

Second Handbook of Research on Mathematics Teaching and Learning

Edited by
Frank K. Lester, Jr.



NATIONAL COUNCIL OF
TEACHERS OF MATHEMATICS





Second Handbook of Research on Mathematics Teaching and Learning



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*A Project of the
National Council of Teachers of Mathematics*

Frank K. Lester, Jr.

EDITOR



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*This handbook is dedicated to the memory of four eminent researchers who were authors
of chapters in the first handbook, but who have since died —*

Merlyn J. Behr 1932–1995

Robert B. Davis 1923–1997

James J. Kaput 1943–2005

Alba G. Thompson 1946–1996

*With their passing, the mathematics education community lost talented, dedicated researchers;
those of us who knew them lost valued friends.*

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PREFACE

In 1965, a bit more than 40 years ago, in response to growing interest in mathematics education research, the Board of Directors of NCTM appointed a Committee on Research in Mathematics Education (Johnson, Romberg, & Scandura, 1994). One result of the committee's deliberations was the publication in 1967 of a slim volume arguing for the need for more high-quality research (Scandura, 1967). In the preface to this volume, the editor observed that "leading citizens believe in research as never before. . . . [And] many teachers are asking for evidence to support or deny the current crop of claims demanding changes in curriculum and pedagogy. . . . Recommendations for change should be based on research" (Scandura, 1967, p. iii). In the same year, another prominent American mathematics education researcher, Robert Davis, wondered "in a society which has modernized agriculture, medicine, industrial production, communication, transportation, and even warfare as ours has done, it is compelling to ask why we have experienced such difficulty in making more satisfactory improvements in education" (Davis, 1967, p. 53).

The situation is not so different today, but the criticism has become more strident. Presently, in the middle of the first decade of the 21st century, diverse individuals and groups have been promoting a variety of old and new instructional approaches, programs, and policies for mathematics education. Researchers are being exhorted to gather and analyze data for evaluating the efficacy of various instructional approaches and curricula as never before. Individuals both within and outside of the mathematics education research community have begun to promote specific agendas, some largely for political reasons. Thus, a pressing challenge for researchers has been to reach out to its constituents by providing research-based advice about questions that concern that constituent community, and to do so in an unbiased, rational, and convincing manner. Members of the research community have discovered that the general public understands and appreciates very little about *what* researchers do or *why* they do it. They find themselves in the position of having to defend their work. Many factions—from teachers to politicians—are calling for mathematics educators to find out "what

works" in classrooms at all levels (see, for example, U.S. Department of Education, What Works Clearinghouse, www.whatworks.ed.gov, retrieved August 6, 2006). This emphasis on studies to document what works presents mathematics education researchers with a dilemma: How can they provide research results showing "what works" and at the same time advance the theoretical and model-building goals that are so important to scientific inquiry in every field?

The purpose of this Handbook is not to answer "what works" questions, although answers to many of these sorts of questions can be found in this volume. Instead, the primary purposes are to coalesce the research that has been done, to provide (perhaps) new conceptualizations of research problems, and to suggest possible research programs to move the field forward. It is only by occasionally stepping back and looking at where mathematics education research has been and suggesting where it should be going that the field is likely to ever be able to answer the questions that practitioners, policy makers, and politicians are asking.

HISTORY OF THE DEVELOPMENT OF THIS HANDBOOK

In May 2002, the Educational Materials Committee (EMC) of NCTM sent the NCTM Board of Directors a request to authorize and fund the development of a new edition of the *Handbook of Research in Mathematics Teaching and Learning*, which had been published in 1992. This request followed a review by the EMC of a proposal from Douglas Grouws and the NCTM Research Advisory Committee (RAC) regarding the revision of the Handbook. Grouws was the editor of the original Handbook and he and the RAC were convinced that, after 10 years, it was time to begin work on a new edition. In July of that year, the Board of Directors approved the EMC's request, and by September NCTM president Johnny Lott had convinced me to serve as editor. In November 2002, I had a very productive discussion with Harry Tunis, Director of Publications at NCTM and George Johnson, President of Information Age Publishing, Inc. (IAP). The

result of that discussion was an agreement between NCTM and IAP to publish the new Handbook. A few days later, I met for the first and only time with the advisory board to conceptualize the new Handbook and begin to identify potential authors. The members of the advisory board included some of the most prominent members of the mathematics education community: Doug Grouws, University of Missouri; James Hiebert, University of Delaware; Carolyn Kieran, Université du Québec à Montréal; Judith Sowder, San Diego State University; and past-President of NCTM Lee Stiff, North Carolina State University. My colleague at Indiana University, Paul Kehle, was to serve as assistant editor for the project. At the end of our two-day meeting we had sketched out the overall structure of the Handbook, including identification of chapter topics to emphasize or de-emphasize, and possible authors. Although the topics remained essentially the same throughout the development of the Handbook, some of the authors changed over time for various reasons. During the ensuing four-plus years, the members of the advisory board were also available to give much-needed advice and, in general, to help out in various other ways. During the development process, authors and author teams submitted outlines of their chapters for review by at least two outside reviewers. Having received the reviews, authors proceeded to prepare an initial draft that was closely scrutinized by the same reviewers, who provided valuable constructive criticism of the drafts. Of course, I, as editor, also reviewed each draft and, when appropriate, provided additional commentary. The authors then used the reviewers' feedback to prepare a second draft. I reviewed the second drafts to determine if the authors had adequately addressed the concerns raised by the reviewers. In a few cases, additional work was called for. Once I decided that a draft was acceptable, the editorial staff at Indiana University commenced to prepare it to be sent to the publisher. On average, the process from submission of an outline to submission to the publisher took two full years.

AUDIENCE

The audience remains much the same as for the 1992 Handbook, namely, mathematics education researchers and other scholars conducting work in mathematics education.

This group includes college and university faculty, graduate students, investigators in research and development centers, and staff members at federal, state, and local agencies that conduct and use research

within the discipline of mathematics. The intent of the authors of this volume was to provide useful perspectives as well as pertinent information for conducting investigations that are informed by previous work. The Handbook should also be a useful textbook for graduate research seminars. (Grouws, 1992, p. ix)

In addition to the audience mentioned above, the present Handbook contains chapters that should be relevant to four other groups: teacher educators, curriculum developers, state and national policy makers, and test developers and others involved with assessment. Taken as a whole, the chapters reflect the mathematics education research community's willingness to accept the challenge of helping the public understand what mathematics education research is all about and what the relevance of their research findings might be for those outside their immediate community.

SCOPE

More than 30 years ago, in an effort to define the scope of a reference volume on research in mathematics education, the Publications Committee of NCTM asked mathematics educators to answer two questions about the kind of

reference on research in mathematics education which you would insist be on your shelf, the shelf of every doctoral student in mathematics education, and the shelf of every person responsible for research in mathematics education. What would you want such a reference to contain? What do you think such a reference need not contain? (Shumway, 1980, p. v).

The advisory board for the present Handbook considered the same two questions. A quick perusal of the previous research compendia published by NCTM (i.e., those published in 1980 [Shumway], and 1992 [Grouws]) shows that the scope of mathematics education research has expanded with each new publication. Recognizing a need "to introduce the beginner to some of the issues and problems in the research process" (Shumway, 1980, p. v), the editorial board of the 1980 volume decided to devote nearly one-third of the volume to this topic, with the balance dealing with the identification of critical, productive problems for researchers, dealing primarily with learning, teaching and curriculum.

Not only was the 1992 handbook considerably longer than Shumway's volume, its scope expanded dramatically. In addition to chapters dealing with specific aspects of mathematics learning (i.e., chapters on research on addition and subtraction, multiplication and division, rational numbers, problem solving, esti-

mation and number sense, algebra, geometry and spatial reasoning, probability and statistics, and advanced mathematical thinking), new avenues of active inquiry appeared. For example, attention to instructional issues was quite prominent (chapters on learning and teaching with understanding, mathematics teaching practices, and grouping for instruction) and research on teachers and teacher education had become part of the mainstream (chapters on teachers' beliefs and conceptions, teachers' knowledge, teacher professionalism, and becoming a mathematics teacher). The topics of the remaining chapters of that handbook indicated that new emphasis had been placed on technology; on assessment and international comparisons; and on philosophical, social, personal, and cultural dimensions of mathematics education (one chapter each on the nature of mathematics, the culture of the mathematics classroom, ethnomathematics, affect, gender, and race, ethnicity, social class, language, and achievement in mathematics).

The current Handbook demonstrates that the research community has sustained its focus on problems of learning, teaching, teacher education, assessment, technology, and social and cultural aspects of mathematics education. But, in addition, some new areas of interest have emerged or been expanded.

The present volume has six sections. The first section, *Foundations*, contains a chapter devoted to two areas not specifically addressed in the 1992 handbook: philosophy (Cobb) and theory (Silver & Herbst). These two chapters plus a chapter on research methods (Schoenfeld) illustrate that the mathematics education research community has taken seriously the importance of thinking hard about the underpinnings of its work.

The second section, *Teachers and Teaching*, contains four chapters focusing on: (1) what knowledge mathematics teachers should have (Hill, Sleep, Lewis, & Ball), (2) how teachers are prepared and the kinds of professional development they receive (J. Sowder), (3) what teachers do in their classrooms and the culture within which teaching takes place (Franke, Kazemi & Battey), and (4) teachers' beliefs and affects (Philipp). Teachers and teaching received considerable attention in the 1992 handbook, but the chapters in section I of the new Handbook demonstrate that new perspectives and new lines of inquiry have emerged.

Section III, *Influences on Student Outcomes*, is also made up of four chapters, one of which focuses on how teaching influences what students learn (Hiebert & Grouws). Among the other three chapters in this section, one focuses on a topic that was not present in the 1992 handbook: how curricula influence student learning (Stein, Remillard, & Smith). The other two chapters comprising the balance of this section—one deal-

ing with the notions of ethnomathematics and everyday cognition (Presmeg), the other with issues of race, class, gender, language, culture, and power (Diversity in Mathematics Education Research Center)—illustrate the ever-expanding scope of research in the field.

More than one-third of the volume (11 chapters) is contained in section IV, *Students and Learning*. Several of the chapters address topics present in the previous handbook: whole number concepts and operations (Verschaffel, Greer, & DeCorte), rational numbers and proportional reasoning (Lamon), algebra in the middle school through college levels (Kieran), problem solving (Lesh & Zawojewski), geometric and spatial thinking (Battista), and post-secondary mathematics thinking and learning (Artigue, Batanero, & Kent). Research on probability and statistics, a single chapter in 1992, has been split into two chapters—one on probability (Jones, Langrall, & Mooney), the other on statistics (Shaughnessy). There also are three chapters representing new areas of research that have burgeoned since the 1992 volume: early childhood (i.e., pre-Grade 1) mathematics learning (Clements & Sarama), early algebra and algebraic reasoning (i.e., pre-middle school; Carraher & Schliemann), and the learning and teaching of proof (Harel & L. Sowder).

The ever-increasing role of assessment in mathematics education made it appropriate to devote an entire section to this topic. The first chapter of Section V, *Assessment*, considers how classroom assessment can actually support learning goals, rather than just measure them (Wiliam). The second chapter (Wilson) deals with high-stakes, standardized tests and considers the influence these tests have had on educational practices, and the third chapter (de Lange) considers issues associated with large-scale, international assessments.

The sixth and final section contains six chapters dealing with a wide range of topics and issues. One chapter (Bishop & Forgasz) focuses on the enduring issues and perspectives that relate to equity and access, another with issues and research associated efforts to understand the impact of computer technologies on learning, teaching, and curriculum (Zbiek, Heid, Blume, & Dick), and a third describes an approach for specifying the content for any assessment of students' achievement (Webb). Two chapters consider issues associated with an area of study not previously within the purview of mathematics education research, namely educational policy. The authors of one of the policy chapters (Ferrini-Mundy & Floden) discuss studies that illustrate the formulation, status, implementation, and effects of educational policy that relate to mathematics education. The authors of the other policy chapter (Tate & Rousseau) provide a review of mathematics education research and relevant policy literature that informs school leaders who are endeavor-

ing to build effective school learning environments for traditionally underserved students. In the final chapter of the volume, the author, Mogens Niss, reflects on the state of mathematics education research, where it seems to be going, and what sort of research would need to be done to get us to the utopian state of being able to provide final, conclusive answers to all the important questions in mathematics education.

ACKNOWLEDGEMENTS

A large number of dedicated individuals helped in the development and publication of this Handbook. As I have already noted, the members of the advisory board (Doug Grouws, James Hiebert, Carolyn Kieran, Judith Sowder, and Lee Stiff) were especially helpful in conceptualizing the volume, providing timely advice, and, on occasion, helping me deal with the inevitable frustrations involved in keeping the project on course and schedule. On behalf of the entire mathematics education community, I wish to express my sincere gratitude to them. I also extend my thanks to Paul Kehle, who, unfortunately, was not able to continue to assist me after the first year—he left Indiana University for another position. He provided valuable help during the early stages of the project. Harry Tunis, Director of Publication Services at NCTM was always helpful and supportive of the project at every stage—even when I had to tell him that we would not be able to stick to our timeline! A special thanks is due to the many reviewers who were willing to spend countless hours poring over drafts of chapters, making insightful and always helpful comments. Indeed, the Handbook exists to a great extent because of their tireless, thoughtful, largely unrecognized efforts on behalf of mathematics education.

A colleague suggested to me that when the Handbook is complete I will be better informed about contemporary mathematics education research than anyone else in the U.S. Perhaps; it is at least arguable! To the extent that it is true, I owe it to the authors and reviewers who worked so hard to make this Handbook the source authority on research in the field. I learned a tremendous amount from reading chapter drafts, considering reviewers' comments, and discussing the focus and organization of chapters. I also gained even more appreciation for my colleagues than I had before—and I already had tremendous respect for them. They are such busy folks—I really don't know how they were able to put together such thoughtful and wonderfully insightful chapters. So, to all authors and reviewers, I extend my sincerest, Thank You!

Closer to home, Indiana University doctoral students Paula Stickles and Andrea McCloskey were absolutely indispensable to the success of the project. Paula helped with the thankless job of formatting the final drafts to make them conform to APA standards. Andrea had an even more tedious task in addition to assisting with formatting: checking every reference of every chapter for correctness and completeness. I am convinced that she now knows as much or more about APA reference conventions as anyone! Both Paula and Andrea assisted me with the tedious job of checking page proofs, as did Zac Rutledge, another very able doctoral student. One other person who played a very important behind-the-scenes role in putting drafts into final form was Cheryl Burkey. Cheryl, an accomplished and very experienced copy editor, knows the *APA Publications Manual* extremely well. She copy edited almost all of the drafts in preparation for sending them to the IAP editors for final production. Thank you Paula, Andrea, and Cheryl—your professionalism, competence, and cheerful dispositions made my job much easier and rewarding. Thanks are also due to the staff of Information Age Publishing. In particular, IAP president, George Johnson, deserves special acknowledgement for his patience and understanding as the Handbook slowly became a reality. I wish also to express my thanks to Scott Suckling of MetroVoice Publishing Services for his work on page design and composition and Ann Salinger who created the indices.

Finally, I wish to thank my wife, Diana, who sometimes wondered if her often-cranky husband would ever bring the Handbook to fruition. Now we can begin to enjoy some time together when I will not have some Handbook issue in the back of my mind.

—Frank K. Lester, Jr.
Bloomington, IN

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■ ■ P A R T I ■ ■

Foundations

PUTTING PHILOSOPHY TO WORK

Coping with Multiple Theoretical Perspectives

Paul Cobb

VANDERBILT UNIVERSITY

In inviting me to write this chapter on philosophical issues in mathematics education, the editor has given me the leeway to present a personal perspective rather than to develop a comprehensive overview of currently influential philosophical positions as they relate to mathematics education. I invoke this privilege by taking as my primary focus an issue that has been the subject of considerable debate in both mathematics education and the broader educational research community, that of coping with multiple and frequently conflicting theoretical perspectives. The theoretical perspectives currently on offer include radical constructivism, sociocultural theory, symbolic interactionism, distributed cognition, information-processing psychology, situated cognition, critical theory, critical race theory, and discourse theory. To add to the mix, experimental psychology has emerged with a renewed vigor in the last few years. Proponents of various perspectives frequently advocate their viewpoint with what can only be described as ideological fervor, generating more heat than light in the process. In the face of this sometimes bewildering array of theoretical alternatives, the issue I seek to address in this chapter is that of how we might make and justify our decision to adopt one theoretical perspective rather than another. In doing so, I put philosophy to work by drawing on the analyses of a number of thinkers who have grappled with the thorny problem of making reasoned decisions about competing theoretical perspectives.

In the first section of the chapter, I follow Schön (1983) in challenging what he termed the positivist epistemology of practice wherein practical reason is construed as the application of theory. My goal in doing so is to question the repeated attempts that have been made in mathematics education to derive instructional prescriptions directly from background theoretical perspectives. I argue that it is instead more productive to compare and contrast various perspectives by using as a criterion the manner in which they orient and constrain the types of questions that are asked about the learning and teaching of mathematics, the nature of the phenomena that are investigated, and the forms of knowledge that are produced.

In the second section of the chapter, I argue that mathematics education can be productively construed as a design science, the collective mission of which involves developing, testing, and revising conjectured designs for supporting envisioned learning processes. As will become apparent, my intent in developing this viewpoint is inclusive rather than exclusive. I therefore construe designs broadly and clarify that the learning processes might, for example, be those of individual students or individual teachers, classroom communities or professional teaching communities, or indeed of schools or school districts as organizations. This perspective on our collective activity as mathematics educators gives rise to a second criterion for comparing and

contrasting background theoretical positions, namely that of how they might contribute to the enterprise of formulating, testing, and revising conjectured designs for supporting mathematical learning. This second criterion is therefore concerned with the nature of the potentially useful work that different theoretical positions might do.

In section three, I draw both on the work of John Dewey and on the more recent contributions of neo-Pragmatist philosophers to question the relevance of a number of philosophical distinctions that have featured prominently in the mathematics education research literature. Foremost among these is the central problem of traditional epistemology, that of the opposition between philosophical realism and constructivist positions that deny that ontological reality is knowable. In doing so, I outline an alternative position that Putnam (1987) terms pragmatic realism and clarify its relevance to the issue of comparing and contrasting different theoretical perspectives in mathematics education research. Against this background in section four and five, I then sharpen the two criteria that I propose for comparing background theoretical positions. The first criterion concerns the nature of the phenomena that are investigated and is often framed in terms of whether a particular perspective treats activity as being primarily individual or social in character. I argue that this dichotomy is misleading in that it assumes that what is meant by *the individual* is self-evident and theory neutral. As an alternative, I propose to compare and contrast different theoretical positions in terms of how they characterize individuals, be they students, teachers, or administrators. The second criterion concerns the potentially useful work that different theoretical positions might do. In addressing this issue, I differentiate Dewey's sophisticated account of pragmatic justification and his related analysis of verification and truth from a purely instrumental approach in which methods or strategies that enable us to reach our goals are deemed to be true, and those that do not are treated as false.

In the subsequent sections of the chapter, I use these two criteria as my primary points of reference as I discuss four broad theoretical positions: experimental psychology, cognitive psychology, sociocultural theory, and distributed cognition. Experimental psychology refers to the psychological research tradition in which the primary methods employed involve experimental and quasi-experimental designs, preferably with the random assignment of subjects. The cognitive psychology tradition on which I focus involves accounting for teachers' and students' inferred interpretations and understandings in terms of internal cognitive structures and processes. So-

ciocultural theory and distributed cognition provide points of contrast with this psychological tradition in that people's activity and learning are considered to be situated with respect to the social and cultural practices in which they participate. As I clarify, sociocultural theory developed largely independently of western cognitive psychology by drawing directly on the writings of Vygotsky and Leont'ev and is concerned with people's induction into and participation in relatively broad cultural practices. In contrast, distributed cognition emerged in reaction to mainstream cognitive psychology and tends to focus on people's activity in their immediate physical, symbolic, and social environments.

I conclude from the comparison of the four perspectives that each has limitations in terms of the extent to which it can contribute to the enterprise of formulating, testing, and revising designs for supporting learning. As part of the analysis, I outline the historical origins of each perspective in order to understand adequately the types of phenomena that are investigated and the forms of knowledge produced. I note that the concerns and interests that motivated the development of these four perspectives differ from those of mathematics educators. In light of this conclusion, I propose and illustrate how we can view the various theoretical perspectives as sources of ideas to be appropriated and adapted to our purposes as mathematics educators. In the final section of the chapter, I then step back to consider the issue of competing, incommensurable theories more broadly. In doing so, I locate the approach I have taken within a philosophical tradition that seeks to transcend the longstanding dichotomy between the quest for a neutral framework for comparing theoretical perspectives on the one hand and the view that we cannot reasonably compare perspectives on the other hand.

THE POSITIVIST EPISTEMOLOGY OF PRACTICE

In his influential book *The Reflective Practitioner*, Donald Schön (1983) explicitly targets what he terms the positivist epistemology of practice wherein practical reasoning is accounted for in terms of the application of abstract theoretical principles to specific cases. As Schön (1983, 1986, 1987) notes, this epistemology is apparent in teacher education programs that (a) initially emphasize the acquisition of theoretical principles derived from fields such as psychology and sociology, (b) then focus on the process of applying these principles in subject matter

methods courses before (c) allowing future teachers to engage in this application process in field experiences. This epistemology is also implicit in instructional approaches that separate students' initial acquisition of mathematical concepts, strategies, or procedures from their subsequent application of those concepts and methods. In the research literature, the positivist epistemology of practice is central to what De Corte, Greer, and Verschaffel (1996) term the first wave of the cognitive revolution, the goal of which was to formulate internal information-processing mechanisms that account for observed relations between the external stimulus environment and documented responses. Anderson, Reder, and Simon (1997) exemplified this position in an exchange with Greeno (1997) when they argued that the contributions of cognitive analyses of this type stem from the analytic power they provide for extracting principles that can generalize from one setting to another. Anderson et al. suggested that teachers should base their instruction on these principles while relying on common sense and professional experience when issues arise in the classroom that psychologists cannot yet answer. Most importantly for the purposes of this chapter, the positivist epistemology of practice is also apparent in debates about the implications of background theoretical positions for mathematics learning and teaching.

The most prominent case in which attempts have been made to derive instructional implications directly from a background theory is that of the development of the general pedagogical approach known as *constructivist teaching*. This pedagogy claims to translate the theoretical contention that learning is a constructive activity directly into instructional recommendations. As Noddings (1990) and Ball and Chazan (1994) observe, it is closely associated with the dubious assertion that "telling is bad" because it deprives students of the opportunity to construct knowledge for themselves. For his part, J. P. Smith (1996) clarifies that adherents of the pedagogy tend to frame teachers' proactive efforts to support their students' learning as interfering with students' attempts to construct meaning for themselves. As Smith demonstrates, in emphasizing what the teacher does not do compared with traditional instructional practices, the teacher's role in supporting learning is cast in relatively passive terms, thereby resulting in a sense of loss of efficacy.

It is important to note that constructivism does not have a monopoly on questionable reasoning of this type in which background theoretical contentions that are *descriptive* in nature are translated directly into instructional maxims that are *prescriptive*. For example, an advocate of symbolic interactionism might argue

that as learning involves the negotiation of meaning, students should be encouraged to continually discuss their differing interpretations. Similarly, a devotee of the distributed view of intelligence might argue that as cognition is stretched over individuals, tools and social contexts, it is important to ensure that students' mathematical activity involves the use of computers and other tools. In each of these cases, the difficulty is not with the background theory but with the relation that is assumed to hold between theory and instructional practice. As I have argued elsewhere (P. Cobb, 2002), pedagogical proposals developed in this manner involve a category error wherein the central tenets of a descriptive theoretical perspective are transformed directly into instructional prescriptions. It is noteworthy that similar instructional recommendations (e.g., the importance of small group work) are frequently derived from contrasting theoretical positions (e.g., constructivism, symbolic interactionism, and distributed intelligence). The resulting pedagogies are underspecified and are based on ideology (in the disparaging sense of the term) rather than empirical analyses of the process of students' learning and the means of supporting it in specific domains.

Later in this chapter, I will build on this critique of the positivist epistemology of practice by proposing an alternative view of the relation between theory and practice. In doing so, I will question the assumption that theory consists of decontextualized propositions, statements, or assertions that are elevated above and stand apart from the activities of practitioners. For the present, it suffices to note that work in the philosophy, sociology, and history of science, sparked by the publication of Thomas Kuhn's (1962) landmark book, *The Structure of Scientific Revolutions*, has challenged this view of theoretical reasoning in the natural sciences. In this and subsequent publications (Kuhn, 1970, 1977), Kuhn developed an analysis of both the processes by which scientists develop theory within an established research tradition and the processes by which they choose between competing research traditions. In the case of theory development within an established research tradition, he followed Michael Polanyi (1958) in questioning the assumption that scientists fully explicate the bases of their reasoning by presenting analyses of a number of historical cases. His goal in doing so was to demonstrate that the development and use of theory necessarily involves tacit suppositions and assumptions that scientists learn in the course of their induction into their chosen specialties. Kuhn went on to clarify that these implicit aspects of scientific reasoning serve as a primary means by which scientists distinguish between insiders and

outsiders in the process of establishing the boundaries of scientific communities.

Kuhn (1962) extended this argument about the tacit aspects of scientific reasoning when he considered how scientists choose between competing research traditions by arguing that “there is no neutral algorithm of theory-choice, no systematic decision procedure which, properly applied, must lead each individual in the group to the same decision” (p. 200). Not surprisingly, Kuhn was accused of claiming that the process by which scientists resolve disputes is irrational. It is therefore important to stress that he was not in fact questioning the rationality of scientists. Instead, he was challenging the dominant view that scientific reasoning could be modeled as a process of applying general rules and procedures to specific cases. As Bernstein (1983) observed, “Kuhn always intended to distinguish forms of rational persuasion and argumentation that take place in scientific communities from those irrational forms of persuasion that he has been accused of endorsing” (p. 53). In responding to his critics, Kuhn (1970) subsequently clarified that

what I am denying is neither the existence of good reasons [for choosing one theory over another] nor that these reasons are of the sort usually described. I am, however, insisting that such reasons constitute values to be used in making choices rather than rules of choice. Scientists who share them may nevertheless make different choices in the same concrete situation... [In concrete cases, scientific values such as] simplicity, scope, fruitfulness, and even accuracy can be judged quite differently (which is not to say that they can be judged arbitrarily) by different people. Again, they may differ in their conclusions without violating any accepted rule. (p. 262)

Bernstein (1983) summarized the debate incited by Kuhn’s analysis of scientific reasoning by cautioning that

one must be sensitive to and acknowledge the important differences between the nature of scientific knowledge and other forms of knowledge. But the more closely we examine the nature of this scientific knowledge that has become the paradigm of theoretical knowledge, the more we realize that the character of rationality in the sciences, especially in matters of theory-choice, is closer to those features of rationality that have been characteristic of the tradition of practical philosophy than to many modern images of what is supposed to be the character of genuine *episteme* [i.e., the application of general, decontextualized methods to specific cases]. (p. 47)

In speaking of practical philosophy, Bernstein was referring specifically to the work of Gadamer (1975). The specific phenomenon that Gadamer analyzed when he developed his philosophy of practical activity was the process by which people interpret and understand texts, particularly religious texts. In developing his position, Gadamer responded to earlier work in this area that instantiated the positivist epistemology of practice by differentiating between general methods of interpretation and understanding on the one hand, and the process of applying them to specific texts on the other hand. Gadamer rejected this distinction, arguing that every act of understanding involves interpretation, and all interpretation involves application. On this basis, he concluded that the characterization of theoretical reasoning as the application of general, decontextualized methods serves both to mystify science and to degrade practical reasoning to technical control.

The philosophical positions that Kuhn, Gadamer, and others developed to challenge the dominance of positivist epistemology of practice foreshadowed a number of more recent developments that are familiar to mathematics education researchers. For example, their argument that reasoning in all domains, including those that are typically characterized as formal and abstract, has much in common with practical reasoning anticipated the view proposed by a number of educational researchers that people’s reasoning is situated with respect to their participation in specific activities or practices (e.g., Beach, 1999; J. S. Brown, Collins, & Duguid, 1989; P. Cobb & Bowers, 1999; Forman, 2003; Greeno & The Middle School Mathematics Through Applications Project Group, 1998; Saxe, 1991; Sfard, 1998). Relatedly, their characterization of people as members of intellectual communities anticipated investigations in which the community of practice has been taken as a primary unit of analysis (e.g., Lave & Wenger, 1991; Rogoff, 1990; Stein, Silver, & Smith, 1998; Wenger, 1998).

Kuhn’s and Gadamer’s arguments are also directly relevant to the issues that I seek to address in this chapter. For example, they call into question attempts to develop a so-called *philosophy of mathematics education* if its intent is to be normative and prescriptive by providing foundational principles for the activity of mathematics education researchers and practitioners. Instead, their arguments orient us to see research, theorizing, and indeed philosophizing as distinct forms of practice rather than activities whose products provide a viable foundation for the activities of practitioners. It is for this reason that I shy away from the approach of first surveying currently fashionable philosophical positions and then deriving

implications for mathematics education research and practice from them. Instead, I take as my starting point several background theoretical positions that are already currently influential in mathematics education research: experimental psychology, cognitive psychology, sociocultural theory, and distributed cognition. At the most immediate level, my goal in discussing these theoretical positions is descriptive rather than prescriptive in that I explicate their basic tenets and consider their potential contributions to mathematics education research. This first goal gives rise to a second that is inherently philosophical in nature, that of moving beyond the realization that there is no neutral algorithm of theory choice by considering how we might sensibly compare and contrast different theoretical perspectives given our concerns and interests as mathematics educators. In my view, this latter (meta)issue is of the greatest importance given the profusion of perspectives that has become emblematic of educational research in general, and of mathematics education research in particular.

In order to address this (meta)issue, it is necessary not merely to be explicit about the criteria of comparison, but to justify them. My goal in doing so is not to formulate a general method or procedure for choosing between these and other theoretical perspectives, but to initiate a conversation in which we, as mathematics education researchers, can begin to work through the challenges posed by a proliferation of perspectives. As the neo-pragmatist philosopher Rorty (1979) clarified in his much cited book, *Philosophy and the Mirror of Nature*, philosophy so construed is therapeutic in that it does not presume to tell people how they should act in particular types of situations. Instead, its goal is to enable people themselves to cope with the complexities, tensions, and ambiguities that characterize the settings in which they act and interact. As will become apparent, such an approach seeks to avoid both the unbridled relativism evident in the contention that one cannot sensibly weigh the potential contributions of different theoretical perspectives and the unhealthy fanaticism inherent in the all-too-common claim that one particular perspective gets the world of teaching and learning right.

As an initial criterion, I propose comparing and contrasting the different theoretical perspectives in terms of the manner in which they orient and constrain the types of questions that are asked about the learning and teaching of mathematics, and thus the nature of the phenomena that are investigated and the forms of knowledge produced. Adherents to a research tradition view themselves as making progress

to the extent that they are able to address questions that they judge to be important. The content of a research tradition, including the types of phenomena that are considered to be significant, therefore represents solutions to previously posed questions. However, as Jardine (1991) demonstrates by means of historical examples, the questions that are posed within one research tradition frequently seem unreasonable and, at times, unintelligible from the perspective of another tradition. In Lakatos' (1970) formulation, a research tradition comprises positive and negative heuristics that constrain (but do not predetermine) the types of questions that can be asked and those that cannot, and thus the overall direction of the research agenda. In my view, delineating the types of phenomena that can be investigated within different research traditions is a key step in the process of comparing and contrasting them. As Hacking (2000) observed, while the specific questions posed and the ways of addressing them are visible to researchers working within a given research tradition, the constraints on what is thinkable and possible are typically invisible. This initial criterion is therapeutic in Rorty's (1979) sense in that the process of comparing and contrasting perspectives provides a means both of deepening our understanding of the research traditions in which we work, and of enabling us to de-center and develop a basis for communication with colleagues whose work is grounded in different research traditions.

MATHEMATICS EDUCATION AS A DESIGN SCIENCE

The second criterion that I propose for comparing different theoretical perspectives focuses on their potential usefulness given our concerns and interests as mathematics educators. My treatment of this criterion is premised on the argument that mathematics education can be productively viewed as a design science, the collective mission of which involves developing, testing, and revising conjectured designs for supporting envisioned learning processes. As an illustration of this collective mission, the first two *Standards* documents developed by the National Council of Teachers of Mathematics (1989, 1991) can be viewed as specifying an initial design for the reform of mathematics teaching and learning in North America. The more recent *Principles and Standards for School Mathematics* (National Council of Teachers of Mathematics, 2000) propose a revised design that was formulated in response to developments in the field, many of which were related to attempts to realize the

initial design. Significantly, as part of this revision process, the designers found it necessary to substantiate their proposals by developing a research companion volume (Kilpatrick, Martin, & Schifter, 2003). As a second illustration, the functions of leadership in mathematics education at the level of schools and school districts include mobilizing teachers and other stakeholders to notice, face, and take on the task of improving mathematics instruction (Spillane, 2000). Ideally, this task involves the iterative development and revision of designs for improvement as informed by ongoing documentation of teachers' instructional practices and students' learning (Confrey, Castro-Filho, & Wilhelm, 2000; Fishman, Marx, Blumenfeld, & Krajcik, 2004; Hill, 2001; McLaughlin & Mitra, 2004, April). At the classroom level, the design aspect of teaching is particularly evident in Stigler and Hiebert's (1999) description of the process by which Japanese mathematics teachers collaborate to develop and revise the design of lessons. This aspect of teaching is also readily apparent in analyses that researchers such as Ball (1993), Lampert (2001), and Simon (1995) have developed of their own instructional practices.

Although the three illustrations focus on developing, testing, and revising designs at the level of a national educational system, a school or school district, and a classroom respectively, the ultimate goal in each case is to support the improvement of students' mathematical learning. As a point of clarification, I should stress that the contention that mathematics education can be productively viewed as a design science is *not* an argument in favor of one particular research methodology such as design experiments. Rather than being narrowly methodological, the claim focuses on our collective mission and thus on the concerns and interests inherent in our work. This framing of mathematics education research therefore acknowledges that a wide spectrum of research methods ranging from experimental and quasi-experimental designs to surveys and ethnographies can contribute to this collective enterprise. Furthermore, this framing recognizes the value of theoretical analyses that result in the development of conceptual tools that can contribute to this enterprise. The fruitfulness of the framing stems from the manner in which it delineates core aspects of mathematics education research as a disciplinary activity. These core aspects include:

- Specifying and clarifying the prospective endpoints of learning
- Formulating and testing conjectures about both the nature of learning processes that aim towards those prospective endpoints and the specific means of supporting their realization

(Confrey & Lachance, 2000; Gravemeijer, 1994a; Simon, 1995)

This formulation is intended to be inclusive in that the learning processes of interest could be those of individual students or teachers, of classroom or professional teaching communities, or indeed of schools or school districts viewed as organizations.

A discussion of the relative merits of various goals or envisioned endpoints for students', teachers', and school districts' learning is beyond the scope of this chapter. It is, however, worth noting that the choice of goals and thus of what counts as improvement is not a matter of mere subjective whim or taste. Instead, our choices involve judgments that are eminently discussable in that we are expected to support them by giving reasons. As an illustration, I outline my view of the primary issues that should be considered when formulating endpoints for students' learning. I propose that, at a minimum, the process of formulating instructional goals should involve delineating central mathematical ideas in particular mathematical domains and clarifying more encompassing activities such as mathematical argumentation and modeling (Lehrer & Lesh, 2003). In addition, I contend that it is important to justify the proposed endpoints in terms of the types of future activities to which they might give students access. One important consideration is therefore the extent to which the envisioned forms of mathematical reasoning have what Bruner (1986) termed *clout* by enabling students to participate in significant out-of-school practices in relatively substantial ways. For example, it is apparent that public policy discourse increasingly involves the formulation and critique of data-based arguments. Students' development of the relatively sophisticated forms of statistical reasoning that are implicated in such arguments therefore has *clout* in that it enables students to participate in a type of discourse that is central to what Delpit (1988) termed the culture of power (cf. G. W. Cobb, 1997).

In my view, it is also important to take explicit account of the function of schools in differentiating between and sorting students when gauging the extent to which envisioned forms of mathematical reasoning have *clout* (cf. Secada, 1995). As an example, Moses and C. Cobb (2001) clarify that the goal of the Algebra Project is to make it possible for *all* students to have access to and to succeed in high school algebra courses that function as gatekeepers to college preparatory tracks, and thus to future educational and economic opportunities. As Moses and Cobb also indicate, cultivating students' development of mathematical interests such that they come to view classroom

mathematical activities as worthy of their engagement should be an important goal in its own right. Although many students will not pursue career trajectories beyond school that involve direct engagement in mathematical activity, it is nonetheless reasonable to propose that mathematical literacy include an empathy for and sense of affiliation with mathematics together with the desire and capability to learn more about mathematics when the opportunity arises.

The importance of this latter goal becomes even more apparent when we note that an increasing appreciation for mathematics inherent in the development of mathematical interests is compatible with what D'Amato (1992) refers to as *situational significance* wherein students come to view engagement in mathematical activities as a means of gaining experiences of mastery and accomplishment, and of maintaining valued relationships with teachers and peers. D'Amato contrasts this general way in which learning mathematics in school comes to have value for students with a second that he terms *structural significance*. In this second case, students come to view achievement in mathematics as a means of attaining other ends such as entry to college and high-status careers, or acceptance and approval in household and other social networks. As he notes, not all students have access to a structural rationale for learning mathematics in school. Gutiérrez (2004) observes, for example, that many urban students do not see themselves going to college, hold activist stances, have more pressing daily concerns (e.g., housing, safety, healthcare), or do not believe that hard work and effort will be rewarded in terms of future educational and economic opportunities. As D'Amato (1992), Erickson (1992), and Mehan, Hubbard, & Villanueva (1994) all document, students' access to a structural rationale varies as a consequence of family history, race or ethnic history, class structure, and caste structure within society. Their analyses indicate that failure to give all students access to a situational rationale for learning mathematics by framing the cultivation of mathematical interests as an explicit goal of design and teaching will result in what Nicholls (1989) termed inequities in motivation.

The issues on which I have focused when discussing the formulation of goals for students' learning clearly reflect my personal view. However, the fact that they are open to critique and challenge indicates that I cannot choose goals on the basis of unarticulated whim or fancy. For example, it might be argued that in focusing on students' participation in significant out-of-school practices, I failed to consider the practices of mathematicians as my primary point of reference (cf. J. S. Brown et al., 1989). Alternatively,

it might be argued that my concern for equity in students' access to significant mathematical ideas is inadequate in that I failed to adopt social justice agenda (cf. Frankenstein, 2002; Gutstein, in press). In responding to both critiques, I would be obliged to give additional justifications to back up the types of goals I have proposed. The process of specifying and clarifying prospective endpoints for students' learning inherent in such exchanges is analogous to the type of reasoning that Kuhn (1962) argued is involved in choices between competing research traditions. In both cases, the type of rationality involved "is a judgmental activity requiring imagination, interpretation, the weighing of alternatives, and application of criteria that are essentially open" (Bernstein, 1983, p. 56). Kuhn contrasted subjectivity with the process of developing supporting reasons or warrants that satisfy communal standards of rationality. Furthermore, he argued that these standards are, as Rorty (1979) put it, "hammered out" by members of an intellectual community as they pursue their collective enterprise. Viewed in these terms, it is quite possible for mathematics educators to make rational decisions about instructional goals while acknowledging that what counts as improvement in students' mathematical learning is itself open to revision.

The view that I have proposed of mathematics education as a design science gives rise to a second criterion for comparing and contrasting background theoretical perspectives that concerns their usefulness. Stated as directly as possible, the usefulness criterion focuses on the manner in which different theoretical perspectives might contribute to the collective enterprise of developing, testing, and revising designs for supporting learning. The illustrations I gave of design at the levels of a national educational system, a school or school district, and a classroom indicate that it will be important to consider for whom particular perspectives might do useful work when comparing perspectives with respect to this criterion. For example, a perspective that does useful work for a classroom teacher responsible for supporting the mathematical learning of specific groups of students might not necessarily be valuable for a school or district administrator whose primary concern is with the learning outcomes of a relatively large student population. This second criterion reflects the view that the choice of theoretical perspective requires pragmatic justification. In a later section of this chapter, I sharpen this criterion by drawing on Dewey's account of pragmatic justification. First, however, I pause to consider traditional epistemological distinctions that have plagued discussions of philosophy in mathematics education. In doing so, I outline an

alternative position that Putnam (1987) terms internal or pragmatic realism.

PRAGMATIC REALISM

Discussions of philosophical issues in mathematics education have focused primarily on epistemology in recent years. Epistemology is concerned with inquiries into both the nature of knowledge and the process by which we come to know. It therefore addresses questions that center on the origin, nature, limits, methods, and justification of human knowledge (Hofer, 2002). These discussions were initially sparked in large part by the increasing influence of radical constructivism in mathematics education research in the 1980s. The general *psychological* contention that learning is an active, constructive process is common to numerous variants of constructivism and has become widely accepted. A key distinguishing feature of radical constructivism is the manner in which it supplements this psychological contention with the *epistemological* assertion that it is impossible to check whether our ideas and concepts correspond to external reality (von Glasersfeld, 1984). It is important to emphasize that radical constructivists do not deny the existence of a pre-given external reality. Their central epistemological claim is instead that this reality is unknowable precisely because it exists independently of human thought and action. Radical constructivists therefore speak of knowledge that has proven viable as fitting with, rather than matching, external reality in that it satisfies the constraints of that reality as much as a key satisfies the constraints of a particular type of lock (von Glasersfeld & Cobb, 1984).

Viewed within a broader philosophical context, radical constructivism is a recent variant of a long line of skeptical epistemologies that can be traced back to the sophists of ancient Greece. Radical constructivism, like other skeptical positions, challenges the traditional project of epistemology, that of identifying a universal method for determining whether a particular theory or conceptual scheme matches or corresponds with external reality. The controversy that radical constructivism has evoked within the mathematics education community stems from the manner in which it, as a skeptical position, plays on the fear that unless we overcome human fallibility and achieve finality in our knowledge claims, we have achieved nothing (Bernstein, 1983; Latour, 2000). The pragmatic realist position that I will outline also seeks to end the traditional epistemological project. However, whereas radical constructivism contends

that it is impossible to bridge the gulf between knowledge and reality, pragmatic realism questions a fundamental assumption common to both traditional epistemology and to skeptical positions such as radical constructivism. I can best clarify these two contrasting approaches by focusing on the notion of *reality* that underpins the arguments of both philosophical skeptics and their realist counterparts.

The reality on which realists and skeptics both focus their attention is not populated with tables and chairs, students and teachers, or differential equations and geometry proofs. It is instead an imagined, or perhaps better, an imaginary realm that has been the center of philosophical debate since the time of Plato. Putnam (1987) refers to it as Reality with a capital “R” to distinguish it from the world in which our lives take on significance and meaning. As Bernstein (1983) observes, although skeptics seek an end to the traditional epistemological project, they are also more than willing to play the traditional epistemological game. For all their differences, realists and skeptics agree on the basic image of people as knowers separated from Reality (with a capital “R”). Radical constructivists, for example, question neither the relevance of this notion of an imaginary, ahistorical Reality nor the value of debating whether it is knowable. Putnam (1987), in contrast, follows John Dewey in challenging the dichotomy between a putative external Reality on the one hand, and the concepts and ideas that people use to think about and discuss it on the other. In contrast to radical constructivism, his goal is not to offer new solutions to traditional epistemological problems, but to question the problems themselves. In his view, realists and skeptics are both enthralled by what Dewey (1910/1976) termed “the alleged discipline of epistemology.” The appropriate response to both realists and skeptics is therefore not to join them in their debate, but to challenge their exchange as an academic exercise of limited relevance.

For Putnam and Dewey, and indeed for a number of influential twentieth century philosophers including Goodman (1978), Quine (1992), and Taylor (1995), the world is not a screen shutting off nature but a path into it. As Dewey (1929/1958) put it, “experience is *of* as well as *in* nature. It is not experience that is experienced, but nature” (p. 12, italics in the original). In speaking of people being in nature, Dewey displaced the traditional preoccupation with Reality with a focus on people’s activities in the realities in which they actually live their lives. As Putnam clarifies, the quest for certainty is given up once this focus is adopted in favor of understanding how people are able to produce fallible truths and achieve relative security in their knowledge claims.

In Putnam's view, the traditional epistemological goal of prescribing to scientists and non-scientists alike how they ought to reason is unjustifiable self-aggrandizement. Like Dewey, his primary concern is instead with processes of inquiry as they are enacted by flesh-and-blood people. Putnam's contribution to this pragmatic line of thought is particularly relevant to my concerns in this chapter because he addresses the issue of how people cope with the multiplicity of perspectives that characterize various domains and social contexts.

As Putnam observes, contemporary science has taken

away foundations without providing a replacement. Whether we want to be there or not, science has put us in a position of having to live without foundations. . . . That there are ways of describing what are (in some way) the "same facts" which are (in some way) "equivalent" but also (in some way) "incompatible" is a strikingly non-classical phenomenon. (p. 29)

Putnam goes on to note that scientists in many fields switch flexibly from one perspective to another and treat each set of facts as real when they do so. On this basis, he concludes that these scientists *act* as conceptual relativists who treat what counts as relevant facts and as legitimate ways of describing them as relative to the background theoretical perspective that they adopt (cf. Sford, 1998). In making this claim, he does not focus on scientists' commentaries on the practices of their chosen disciplines. Instead, he uses the pragmatic criterion employed by Dewey, Charles Sanders Peirce, and particularly by William James, namely that the indicator of what people actually believe is not what they say about their activity (i.e., espoused belief) but the suppositions and assumptions on which they risk acting. In the case at hand, scientists risk acting on the assumption that the differing constellations of phenomena that they investigate when they adopt differing theoretical perspectives are each real (with a small "r"). In taking this stance, Putnam rejects as irrelevant the traditional epistemological project of determining which of these constellations of phenomena correspond to an imaginary Reality. He terms his position *pragmatic realism* in that he takes at face value the realities that scientists investigate. In his view, questions concerning the existence of abstract mathematical and scientific entities should be addressed not by making claims about Reality (with a capital "R") but by examining disciplinary practices. In this regard, he notes approvingly that Quine (1953)

urges us to accept the existence of abstract entities on the ground that these are indispensable in mathematics, and of microparticles and space-time points on the ground that these are indispensable in physics; and what better justification is there for accepting an ontology than its indispensability in our scientific practice? (Putnam, 1987, p. 21)

Putnam makes it clear that pragmatic realism is not limited to mathematics and science.

[It] is, at bottom, just the insistence that realism is *not* inconsistent with conceptual relativity. One can be *both* a realist and a conceptual relativist. Realism (with a small "r") . . . is a view that takes our familiar common sense [or everyday] scheme, as well as our artistic and scientific and other schemes, at face value, without helping itself to the thing 'in itself' [i.e., Reality with a capital "R"]. (p. 17, italics in the original)

He goes on to clarify that the world looks both familiar and different when we reject the dichotomy between Reality and people's ideas about it.

It looks familiar, insofar as we no longer try to divide up mundane reality into a 'scientific image' and a 'manifest [everyday] image' (or our evolving doctrine into a 'first-class' and a 'second-class' conceptual system). Tables and chairs . . . exist just as much as quarks and gravitational fields. . . . The idea that most of mundane reality is an illusion (an idea that has haunted Western philosophy since Plato . . .) is given up once and for all. But mundane reality looks different, in that we are forced to acknowledge that many of our familiar descriptions reflect our interests and choices. (Putnam, 1987, p. 37)

Putnam takes care to differentiate *conceptual relativism*, the notion that the phenomena that are treated as real differ from one perspective to another, from the view that every perspective is as good as every other. "Conceptual relativity sounds like 'relativism', but it has none of the 'there is no truth to be found . . . "true" is just a name for what a bunch of people can agree on' implications of relativism" (1987, pp. 17–18). Putnam instead characterizes truths as fallible, historically contingent, human productions that are subject to correction. His goal in adopting this view is to rehabilitate the notion of truth while simultaneously rejecting the view that Truth should be ascertained in terms of correspondence with Reality. In his view, it is imperative to preserve the notion of truth given that people make and risk acting on the basis of their truth claims in both mundane and scientific realities.

Putnam demonstrates convincingly that scientists and the proverbial person in the street both *act* as

conceptual relativists. However, in the cases that he considers, the various perspectives are well established and are not subject to dispute. They therefore contrast sharply with the fluid situation in mathematics education research where theoretical perspectives continue to proliferate and are openly contested. Pragmatic realism is nonetheless relevant in that it provides an initial orientation as we begin to address the meta-issue of how we might compare and contrast different perspectives. For example, it leads us to question claims made by adherents to a particular perspective that their viewpoint gets the world of teaching and learning right—that the phenomena that they take as real correspond to Reality. Arguments of this type involve what Putnam terms the fiction that we can usefully imagine a God’s Eye point of view from which we can decide which perspective matches Reality. Pragmatic realism instead orients us to acknowledge that we make choices when we adopt a particular theoretical perspective, and that these choices reflect particular interests and concerns. In addition, pragmatic realism alerts us to the danger of inferring from the conclusion that there is no neutral algorithm of theory choice that any theoretical perspective is as good as any other. In Putnam’s view, we can counter this “anything goes” claim by making the criteria that we use when justifying our theoretical choices an explicit focus of scrutiny and discussion. It is to contribute to such a discussion that I have proposed two potentially revisable criteria for comparing and contrasting background theoretical perspectives in mathematical education research. The first criterion is concerned with the nature of the realities that are investigated by researchers who adopt different theoretical perspectives whereas the second focuses on the extent to which research conducted within a particular perspective can contribute to the enterprise of formulating, testing, and revising conjectured designs for supporting envisioned learning processes. In the next two sections of this chapter, I sharpen each of these criteria in turn.

THE NOTION OF THE INDIVIDUAL AS CONCEPTUALLY RELATIVE

Historically, mathematics education researchers looked to cognitive psychology as a primary source of theoretical insight. However, during the last fifteen years, a number of theoretical perspectives that treat individual cognition as socially and culturally situated have become increasingly prominent. In their historical overview of the field, De Corte et al.

(1996) speak of first wave and second wave theories to distinguish between these two types of theories. The goal of first-wave theories is to model teachers’ and students’ individual knowledge and beliefs by positing internal cognitive structures and processes that account for their observed activity. They portray the emergence of second-wave theories as a response to the limited emphasis on affect, context, and culture in first-wave research. Theories of this type typically treat teachers’ and students’ cognitions as situated with respect to their participation in particular social and cultural practices.

The distinction that De Corte et al. (1996) draw between cognitive and situated perspectives has become institutionalized within the mathematics education research community and serves as the primary way in which we compare and contrast theoretical perspectives. These comparisons focus on the extent to which particular perspectives take individual cognitive processes or collective social and cultural processes as primary. Although these comparisons throw important differences between perspectives into sharp relief, they are misleading in one important respect. As I will illustrate when I compare the four background theories of experimental psychology, cognitive psychology, sociocultural theory, and distributed cognition, the notion of the individual is conceptually relative. Adherents to these different perspectives conceptualize the individual in fundamentally different ways. Furthermore, these differences are central to the types of questions that adherents to the four perspectives ask, the nature of the phenomena that they investigate, and the forms of knowledge they produce. I will therefore concretize the first of the two criteria that I have proposed for comparing and contrasting theoretical perspectives by teasing out these differences.

I can best exemplify the difficulties that arise when the notion of the individual is taken as self evident by previewing my discussion of distributed theories of intelligence. The term distributed intelligence is perhaps most closely associated with Pea (1985; 1987; 1993). A central assumption of this theoretical perspective is that intelligence is distributed “across minds, persons, and symbolic and physical environments, both natural and artificial” (Pea, 1993, p. 47). Dörfler (1993) clarified the relevance of this theoretical perspective for design and research in mathematics education when he argued that thinking

is no longer considered to be located exclusively within the human subject. The whole system made up of the subject and the available cognitive tools and

aids realizes the thinking process. . . . Mathematical thinking for instance not only *uses* those cognitive tools as a separate means but they form a constitutive and systematic part of the thinking process. The cognitive models and symbol systems, the sign systems, are not merely means for expressing a qualitatively distinct and purely mental thinking process. The latter realizes itself and consists in the usage and development of the various cognitive technologies. (p. 164)

This theoretical orientation is consistent with the basic Vygotskian insight that students' use of symbols and other tools profoundly influences both the process of their mathematical development and its products, increasingly sophisticated mathematical ways of knowing (Dörfler, 2000; Hall & Rubin, 1998; Kaput, 1994; Lehrer & Schauble, 2000; Meira, 1998; van Oers, 2000).

In developing his distributed perspective, Pea (1993) has been outspoken in delegitimizing analyses that take the individual as a unit of analysis. In his view, the functional system consisting of the individual, tools, and social contexts is the appropriate unit. Not surprisingly, Pea's admonition has been controversial. For example, Solomon (1993) responded by arguing that, in distributed accounts of intelligence, "the individual has been dismissed from theoretical consideration, possibly as an antithesis to the excessive emphasis on the individual by traditional psychology and educational approaches. But as a result the theory is truncated and conceptually unsatisfactory" (p. 111). It is worth pausing before adopting one or the other of these opposing positions to clarify what Pea might mean when he speaks of the individual. His arguments indicate that individuals, as he conceptualizes them, do not use tools and do not take account of context as they act and interact. They appear to be very much like individuals as portrayed by mainstream cognitive science who produce observed behaviors by creating and manipulating internal symbolic representations of the external environment. Pea's proposal is to equip these encoders and processors of information with cultural tools and place them in social context. This proposition is less contentious than his apparent claim that all approaches that focus on the quality of individuals' reasoning should be rejected regardless of how the individual is conceptualized. The relevant issue once we clarify what we are talking about is not whether it is legitimate to focus on individual teachers' and students' reasoning. Instead, it is how we might usefully conceptualize the individual given our concerns and interests as mathematics educators.

USEFULNESS AND TRUTH

A concern for the usefulness of different theoretical perspectives carries with it the implication that they can be viewed as conceptual tools. This metaphor fits well with the characterization of mathematics education as a design science and orients us away from making forced choices between one perspective and another. However, for this criterion to itself do useful work, we have to clarify what it means for a theoretical perspective to contribute to the collective enterprise of mathematics education. Prawat's (1995) discussion of three types of pragmatic justification identified by Pepper (1942) is relevant in this regard.

The first type of pragmatic justification is purely instrumental in that actions are judged to be true if they enable the achievement of goals. Prawat (1995) indicates the limitations of this formulation when he observes that "a rat navigating a maze has as much claim on truth, according to this approach, as the most clear-headed scientist" (p. 19). In the second type of pragmatic justification,

it is not a successful act that is true, but the hypothesis that leads to the successful act. When one entertains a hypothesis, Pepper (1942) points out, it is in anticipation of a specific outcome. When that outcome occurs, the hypothesis is verified or judged true. (Prawat, 1995, p. 19)

As Prawat goes on to note, this is a minimalist view in that hypotheses are treated as tools for the control of nature. This portrayal sits uncomfortably with Tolumin's (1963) demonstration that the primary goal of science is to develop insight and understanding into the phenomena under investigation, and that instrumental control is a by-product.

The third type of pragmatic justification, which Pepper (1942) calls the qualitative confirmation test of truth, brings the development of insight and understanding to the fore. This type of justification builds on Dewey's analysis of the function of thought and the process of verification. Dewey viewed ideas as potentially revisable plans for action and argued that their truth is judged in terms of the extent to which they lead to a satisfactory resolution to problematic situations (Westbrook, 1991). In taking this stance, he equated experience with physical and conceptual action in a socially and culturally organized reality, and maintained that it is characterized by a future-oriented projection that involves an attempt to change the given situation (Sleeper, 1986; J. E. Smith, 1978). He therefore contended that foresight and understanding

are integral to thought, the primary function of which is to project future possibilities and to prepare us to come to grips with novel, unanticipated occurrences.

These aspects of thought are central to Dewey's analysis of verification as a process in which the phenomena under investigation talk back, giving rise to surprises and inconsistencies. He clarified that some of these surprises can be accounted for relatively easily by elaborating underlying ideas, whereas others constitute conceptual impasses that typically precipitate either the reworking of ideas or their eventual rejection (Prawat, 1995). The crucial point to note is that, for Dewey, verification constitutes a context in which theoretical ideas are elaborated and modified. As Pepper (1942) put it, ideas judged to be true are those that give *insight* into what he termed the texture and quality of the phenomena that serve to verify them. In the case of mathematics education, for example, ideas are potentially useful to the extent that they give rise to conjectures about envisioned learning processes and the specific means of supporting them. However, these ideas are not simply either confirmed unchanged or rejected during the process of testing and revising designs. Instead, Dewey drew attention to the process by which ideas that are confirmed evolve as they are verified. In his formulation, *the truth of fallible, potentially revisable ideas is justified primarily in terms of the insight and understanding they give into learning processes and the means of supporting their realization*. This is the criterion that I will use when I focus on the potentially useful work that experimental psychology, cognitive psychology, sociocultural theory, and distributed cognition might do in contributing to the enterprise of formulating, testing, and revising conjectured designs for supporting envisioned learning processes.

As a point of clarification, it is important to note that Dewey's intent in analyzing verification processes was descriptive rather than prescriptive. He was not attempting to specify rules or norms for verification to which scientists ought to adhere. Instead, he sought to understand verification as it is actually enacted in the course of inquiry. He claimed that regardless of the pronouncements of philosophers who strive to identify the Method for determining Truth, verification as it is enacted by scientists is an interpretive process in which they modify and elaborate theoretical ideas. Dewey's analysis has far reaching implications in that it challenges a distinction central to the traditional project of epistemology, that between the context of discovery and the context of justification. As Bernstein (1983) explains,

while no contemporary philosopher of science has wanted to claim that there is a determinate decision procedure or method for advancing scientific discovery, many have been firmly convinced that

there are (or ought to be) permanent procedures for testing and evaluating rival theories. This is the basis of the "orthodox" distinction between the context of discovery and the context of justification. The latter has been taken to be the proper domain of the philosopher of science. His or her task is to discover, specify and reconstruct the criteria. (p. 70)

Popper's (1972) contention that to be scientific, hypotheses and theory have to be open to falsification is the most widely cited proposal of this type. As a rough rule of thumb, Dewey would probably not have disputed Popper's claim. However, his analysis of verification indicates that he would add that the application of this rule necessarily involves wisdom and judgment. In this regard, he would concur with Bernstein's observation that

Popper frequently writes as if we always know in advance what will count as a good argument or criticism against a conjecture. The basic idea behind the appeal to falsification as a demarcation criterion between science and nonscience is that there are clear criteria for determining under what conditions a conjecture or hypothesis is to be rejected. (p. 70)

As Dewey pointed out and as Kuhn (1962; 1977), Lakatos (1970), and Feyerabend (1975) subsequently underscored, the situation becomes less straightforward than Popper's proposal implies when we focus on the actual practices of scientists. Although the phenomena under investigation talk back, they do not serve as a jury that tell us unambiguously whether a theoretical idea should be accepted or rejected.

We frequently do not know, in a concrete scientific situation, whether we are confronted with an obstacle to be overcome, a counterinstance that can be tolerated because of the enormous success of the theory, or with evidence that should be taken as falsifying our claim. Data or evidence do not come marked "falsification"; in part, it is we who decide what is to count as a falsification or refutation. (Bernstein, 1983, p. 71)

Furthermore, as Dewey in particular emphasized, the ideas under scrutiny are moving targets that evolve as scientists, in effect, engage in a dialogue with the phenomena under investigation.

Latour and Wollgar's (1979) analysis of activity in an organic chemistry laboratory and Pickering's (1984) of a high energy physics laboratory substantiate Dewey's analysis of verification, particularly as the work carried out in both laboratories resulted in Nobel prizes. The case studies reveal that the scientists did not simply formulate conjectures, conduct experiments to test them, and then passively let the

resulting data determine which were confirmed and which were false. To be sure, the scientists had theoretical ideas and conjectures expressed in terms of those ideas. However, they also had views about how the experimental apparatus functioned, how it could be used, and so forth. As Latour and Wollgar, and Pickering both document, it was typical for the apparatus to initially not behave as the scientists expected. In these situations, the scientists attempted to adjust to the perceived anomaly in a number of ways that included revising the theory under investigation, revising their views of how the apparatus functioned, or tinkering with and rebuilding the apparatus itself (Traweek, 1988). Their goal in doing so was to achieve what Pickering (1984) terms a robust fit between these different aspects of the experimental situation. Furthermore, the scientists judged the theoretical ideas that they successfully revised and elaborated in the course of this problem-solving process to be true because they gave them insight into the phenomena they were investigating. Latour and Wollgar's and Pickering's analyses of the process by which scientists produce truths as they engage in a dialogue with nature are consistent with Dewey's contention that truths are made through the process by which they are verified (Westbrook, 1991). This pragmatic notion of truth and, relatedly, of usefulness underpins my comparison of different theoretical perspectives in terms of their potential to contribute to the enterprise of formulating, testing, and revising conjectured designs for supporting envisioned learning processes.

COMPARING THEORETICAL PERSPECTIVES

My purpose in comparing and contrasting the four theoretical perspectives is to illustrate the relevance of the two criteria I have proposed: (a) How the individual is conceptualized in the differing perspectives, and (b) the potential of the perspectives to contribute to our understanding of learning processes and the means of supporting their realization. I therefore make no pretense at providing a comprehensive overview of theoretical perspectives in mathematics education. For example, I largely ignore both symbolic interactionism (Bauersfeld, 1980, 1988; P. Cobb, Wood, Yackel, & McNeal, 1992; Voigt, 1985, 1996) and discourse and communicational perspectives (Ernest, 1994; Pimm, 1987, 1995; Rotman, 1988, 1994; Sfard, 2000a, 2000b). Experimental psychology refers to the psychological research tradition in which the primary methods employed involve experimental and quasi-experimental designs, preferably with the

random assignment of subjects. My decision to focus on this perspective has been influenced by the strong advocacy for research designs of this type by personnel in the Institute of the Educational Sciences and other U.S. government funding agencies.

My discussion of cognitive psychology is limited to theoretical orientations that involve what MacKay (1969) termed the actor's viewpoint. The goal of psychologies of this type is to account not merely for teachers' and students' observed behaviors but for their inferred interpretations and understandings in terms of internal cognitive structures and processes. This restricted focus is premised on the observation that the relationships we establish with teachers and students in the course of our work as mathematics educators frequently involve collaboration and mutual engagement. As Rommetveit (1992) and Schutz (1962) both illustrate, relationships of this type involve communicative interactions characterized by a reciprocity of perspectives typical of the actor's viewpoint. I will therefore have little to say about cognitive approaches that involve the observer's viewpoint (MacKay, 1969) and that focus on internal cognitive processes that intervene between an observed stimulus environment and observed response activity. Sociocultural theory and distributed cognition are both second wave perspectives as described by De Corte et al. (1996) and provide points of contrast with cognitive psychology by viewing individual activity as situated with respect to social and cultural practices. As I will clarify, sociocultural theory has developed largely independently of western cognitive psychology by drawing inspiration directly from the writings of Vygotsky and Leont'ev whereas distributed cognition has emerged in reaction to information-processing psychology and incorporates aspects of the Soviet work.

Experimental Psychology

In speaking of experimental psychology, I refer to the psychological research tradition whose primary contributions to mathematics education have involved the development of assessment instruments, particularly norm-referenced tests, and the findings of studies that have assessed the relative effectiveness of alternative curricular and instructional approaches. This perspective merits attention given the far reaching implication of a recent legislative initiative in the United States known as the No Child Left Behind (U.S. Congress, 2001). As Slavin (2002) observes, the Act "mentions 'scientifically based research' 110 times. It defines 'scientifically based research' as 'rigorous, systematic and objective procedures to obtain valid

knowledge,' which includes research that 'is evaluated using experimental or quasi-experimental designs,' preferably with random assignment" (p. 15). Slavin goes on to note that the recently established Institute for the Educational Sciences, a major federal funding agency, is organized "to focus resources on randomized and rigorously matched experimental research on programs and policies that are central to the education of large numbers of children" (p. 15). Whitehurst (2003, April), the Institute's Director, addresses this point when he states that although "interpretations of the results of randomized trials can be enhanced with results from other methods" (p. 9), "randomized trials are the only sure method for determining the effectiveness of education programs and practices" (p. 6). In a nutshell, "randomized trials are the gold standard for determining what works" (p. 8). As Whitehurst's reference to "what works" indicates, the emphasis on a particular research method profoundly influences both the nature of the questions asked and the forms of knowledge produced. In this regard, Slavin (2004) clarifies that well-designed studies of this type are not limited to x versus y comparisons but can "also characterize the conditions under which x works better or worse than y , the identity of the students for whom x works better or worse than y , and often produce rich qualitative information to supplement the quantitative comparisons" (p. 27).

Knowledge claims of this type are premised on a particular conception of the individual. In teasing out this conception and thus the nature of the reality that is the focus of investigation, it is important to distinguish between the abstract, collective individual to whom knowledge claims refer and the individual students who participate in experiments. This collective individual is a statistical aggregate that is constructed by combining measures of psychological attributes of the participating students (e.g., measures of mathematical competence as measured by achievement test scores). As Danziger (1990) demonstrates in his seminal historical analysis of experimental psychology, the purpose of experimentation is to make predictions about how certain variations in instructional conditions affect the performance of this abstract individual as assessed by aggregating the scores of individual students. This statistically constructed individual is abstract in the sense that it need not correspond to any particular student. The construction of this collective individual enables investigators who work in this tradition to avoid the issue of individual differences and the challenges involved in accounting for the reasoning and learning of specific students (Danziger, 1990). This is accomplished by treating differences in measures of students' performance as

error variance. Students are then characterized by the extent to which their performance deviates from group norms that measure the performance of the collective individual.

This methodological approach of investigating the performance of the abstract, collective subject rests on two underlying assumptions. The first is that students are composed of discrete, isolatable attributes or qualities that vary only in degree from one student to another (Danziger, 1990). This assumption makes it legitimate to combine measures of students' performance. Students so portrayed are therefore limited to possessing larger or smaller measurable amounts of these psychological attributes. Second, the environments in which students acquire these capacities are composed of independent features that the investigator can manipulate and control directly. The implicit ontology is that of environmental settings made up of separate independent variables and students composed of collections of dependent psychological attributes. Together, these two theoretical suppositions ground investigations that seek to discern causal relationships between the manipulation of instructional conditions on the performance of the collective, abstract individual. This theoretical underpinning indicates that experimental psychology should be treated as a theoretical perspective rather than merely as a set of methodological prescriptions.

The forcefulness with which adherents of this perspective have attempted to advance their viewpoint has elicited a number of responses. One line of critique claims that the approach of casting students as objects in studies that focus on their responses to environmental manipulations is ethically dubious and potentially dehumanizing. As these critiques target the morality of investigators working in this tradition, it is worth noting with Porter (1996) that

social quantification means studying people in classes, abstracting away their individuality. This is not unambiguously evil, though of late it has been much criticized. Much, probably most, statistical study of human populations has aimed to improve the condition of working people, children, beggars, criminals, women, or racial and ethnic minorities. (p. 77)

A second line of attack develops the argument that experimental psychology lacks theoretical depth. For example, Danziger (1990) contends that this analytic approach both separates people from the social contexts in which their actions take on significance, and eschews the study of the interpretations that they make in those contexts. Criticisms of this type miss the mark in my view because they use the

theoretical commitments of alternative theoretical perspectives as criteria against which to assess experimental psychology. Judgments of theoretical depth do not transcend research traditions but are instead conceptually relative to the norms and values of particular research communities. Experimental psychologists, for example, make judgments of theoretical depth by gauging the extent to which the findings of investigations contribute to their collective understanding of the psychology of the abstract, statistically constructed individual. More generally, exchanges between experimental psychologists and their opponents typically involve people talking past each other as they point to features of the different realities that they investigate. In my view, it is more productive to assess the value of experimental psychology in terms of its potential contributions to a specific enterprise such as that of formulating, testing, and revising conjectured designs for supporting mathematical learning. There is, however, one issue that requires additional scrutiny before we consider the potential usefulness of this theoretical perspective.

Advocates of experimental psychology frequently argue with considerable vehemence that their perspective and its associated research methods are scientific, and that all other perspectives and methodological approaches are not. As the case studies conducted by Pickering (1984) and by Latour and Woolgar (1979) indicate, this claim is not based on an examination of the practices of scientists whose work is widely acknowledged to be of the highest caliber. The claim instead capitalizes on folk beliefs about what it means to be scientific and objective. In the history of the discipline, what was important

was the widespread acceptance of a set of firm convictions about the nature of science. To be socially effective, it was not necessary that these convictions actually reflected the essence of successful scientific practice. In fact, most popular beliefs in this area were based on external and unanalyzed features of certain practices in the most prestigious parts of science. Such beliefs belong to the rhetoric of science rather than its substance. . . . Such unquestioned emblems of scientific status included features like quantification, experimentation, and the search for universal (i.e., ahistorical) truths. (Danziger, 1990, p. 120)

It is instructive to compare the features of scientific practice that advocates of experimental psychology deem to be critical with those identified by the National Research Council (2002):

Scientific Principle 1

Pose Significant Questions That Can Be Investigated Empirically

Scientific Principle 2

Link Research to Relevant Theory

Scientific Principle 3

Use Methods That Permit Direct Investigation of the Question

Scientific Principle 4

Provide a Coherent and Explicit Chain of Reasoning

Scientific Principle 5

Replicate and Generalize Across Studies

Scientific Principle 6

Disclose Research to Encourage Professional Scrutiny and Critique (pp. 3–5)

Eisenhart and Towne (2003) clarify that these principles were identified by “reviewing *actual* research programs—both basic and applied—in natural science, social science, education, medicine, and agriculture” (p. 33, italics in the original). As they note, scientifically based research is best defined not by the employment of particular research methods but by characteristics that cut across a range of methods. I would only add that the use of these or any other set of principles to assess specific research programs necessarily involves interpretations and judgments of type that all scientists including experimental psychologists make in the course of their practice.

It is ironic that the claim that experimental psychology has hegemony over what is regarded as scientific lacks empirical grounding and is contradicted by the available evidence. This claim is ideological in the pernicious sense of the term. It rests on the indefensible proposition that the methods of experimental psychology are theory neutral and constitute the only guaranteed means of gaining access to Reality with a capital “R”. As I have demonstrated, experimental psychologists’ use of these methods is theory laden and reflects theoretical suppositions and assumptions about the nature of the reality that they are investigating. In using these methods, they attempt to gain insights into the psychology of the statistically constructed collective individual, a character that is composed of a set of isolatable psychological characteristics and that inhabits a world made up of manipulable independent variables. This observation does not threaten the credibility of experimental psychology as a viable research tradition: The research practices that any research community establishes as

normative necessarily entail theoretical commitments. The observations, however, do undermine the claim that experimental psychologists have found the Method for discerning Truth. The failure of many of their most forceful advocates to acknowledge that research in this tradition is conducted from a theoretical perspective closes down debate and, in my view, highlights the political nature of boundary disputes between science and non-science. Adherents to experimental psychology frequently state that their primary motivation in conducting investigations is to improve the students' intellectual and moral welfare. I accept these statements at face value but question the implication that insights developed in any other perspective can make at best marginal contributions to students' intellectual and moral welfare. This brings us to the issue of usefulness.

Adherents of experimental psychology repeatedly emphasize their commitment to conduct research that is pragmatically relevant. For example, Whitehurst (2003, April) stresses that "the primary focus for the Institute [of Educational Sciences] will be on work that has high consideration of use, that is practical, that is applied, that is relevant to practitioners and policy makers" (p. 4). In addressing this and similar claims, it is important to consider for whom the forms of knowledge produced by experimental psychology might be useful. Danziger's (1990) historical analysis is pertinent in this regard. He reports that

after the turn of the [twentieth] century, psychologists' relations with teachers became increasingly overshadowed by a new professional alliance which was consummated through the medium of a new set of investigative practices. The group of educators with whom psychologists now began to establish an important and beneficial professional alliance consisted of a new generation of professional educational administrators. This group took control of a process of educational rationalization that adapted education to the changed social order of corporate industrialism. The interests of the new breed of educational administrator had little in common with those of the classroom teacher. Not only were the administrators not directly concerned with the process of classroom teaching, they were actually determined to separate their professional concerns as much as possible from those of the lowly army of frontline teachers. In this context, they emphasized scientific research as a basis for the rationalized educational system of which they were the chief architects. In the United States the needs of educational administration provided the first significant external market for the products of psychological research in the years immediately preceding World War I. (p. 103)

Danziger goes on to clarify that knowledge about the responses of the statistically constructed collective individual to different instructional conditions served the needs of administrators who managed institutions in which instruction is carried out in groups. Furthermore, the conception of learning environments as composed of manipulable independent variables was directly relevant to administrators who sought to both distance themselves from and manage classroom instructional processes. Danziger also demonstrates that the consequences of this alliance were at least as profound for experimental psychology as they were for education. For example, it was during this period that the goal of gaining insight into individual mental processes was displaced by that of putting psychological prediction at the service of administrative needs. In addition, the high value that experimental psychologists place on quantification was influenced by the usefulness of statistical constructions based on group data to administrators. On the basis of these observations, Danziger concludes that experimental psychology was transformed in large measure into what he terms an administrative science in that it investigates the reality that administrators seek to manage.

In terms of the metaphor of a theoretical perspective as a conceptual tool, Danziger's analysis indicates that experimental psychology is a tool that has been fashioned to produce forms of knowledge that fulfill administrative concerns. The resulting forms of knowledge enable administrators who are removed from the classroom and who do have little if any specialized knowledge of teaching and learning in particular content domains to make informed decisions about the curricula and instructional strategies that teachers should use. Although these forms of knowledge can be of some relevance to teachers, they are less well suited to the contingencies of classroom teaching. This is because the knowledge produced of individual students is, as Danziger (1990, p. 165) puts it, "a knowledge of strangers" who are known only through their standing in the group. It does not therefore touch on the challenges, dilemmas, and uncertainties that arise in the classroom as teachers attempt to achieve a mathematical agenda while simultaneously taking account of their students' proficiencies, interests, and needs (Ball, 1993; Davis, 1997; Fennema, Franke, & Carpenter, 1993; Lampert, 1990, 2001). I therefore question the frequent assertion made by proponents of experimental psychology that the forms of knowledge it produces are of equal relevance to teachers and administrators. It is significant that the examples that proponents cite to illustrate the need for these forms of knowl-

edge almost invariably concern administrators and policymakers but not classroom teachers. For example, Whitehurst (2003) does not mention classroom teachers when he explains that personnel at the Institute of Educational Sciences “recently completed a survey of a purposive sample of our customers to determine what they think we ought to be doing to serve their needs. The sample included school superintendents and principals, chief state school officers, and legislative policy makers” (p. 5).

In summary, the arguments that adherents of experimental psychology advance in support of their perspective are based on the claims that it has hegemony over the production of scientific knowledge about teaching and learning, and that these forms of knowledge are pragmatically useful. I have suggested that the first of these claims is primarily ideological but that the second has merit provided we clarify for whom the resulting forms of knowledge are useful. Proponents of this perspective offer us a bold vision in which studies involving experimental and quasi-experimental designs constitute the primary basis for an ongoing process of educational improvement. It is worth noting that this vision is based on belief rather than evidence and is not supported by the historical record. There was an initial period of enthusiasm for such a vision in the early part of the last century, but this enthusiasm has since waned.

In spite of these promising beginnings, the history of treatment-group methodology in research in education and educational psychology was not exactly a march of triumph. . . . In due course . . . a certain pessimism about the prospects of experimental research in education began to set in. The claims made on behalf of the quantitative and experimental method had undoubtedly been wildly unrealistic, and in the light of changing priorities the illusions of the early years were unable to survive. (Danziger, 1990, p. 115)

In my judgment, the claim that sustained improvements in mathematics teaching and learning can be made by relying *almost exclusively* on the findings of treatment-group studies is untenable. To be sure well-designed studies of this type can, in all probability, make an important contribution. However, advocates of experimental psychology should, in my view, moderate their rhetoric lest the history of high hopes and dashed expectations repeat itself.

Cognitive Psychology

In comparing and contrasting cognitive psychology with other theoretical perspectives, I restrict my

focus to theoretical approaches that seek to account for teachers’ and students’ inferred interpretations and understandings in terms of internal cognitive structures and processes. Cognitive approaches of this type take on a challenge sidestepped by experimental psychology, that of accounting for specific students’ and teachers’ mathematical reasoning and learning. It is useful to distinguish between two general types of theories developed within this tradition. The first are theories of the *process* of mathematical learning that are intended to offer insights into students’ learning in any mathematical domain, whereas the second are theories of the development of students’ reasoning in specific mathematical domains. Pirie and Kieren’s (1994) recursive theory of mathematical understanding serves to illustrate theories of the first type. They differentiate a sequence of levels of mathematical reasoning and model mathematical understanding as a recursive phenomenon that occurs as thinking moves between levels of sophistication. In their theoretical scheme, students who are novices to a particular mathematical domain initially make images of either their situation-specific activity or its results. The first significant development occurs when students can take such images as givens and do not have to create them anew each time. Later developments involve students noticing properties of their mathematical images and subsequently taking these properties as givens that can be formalized. One of the notable features of Pirie and Kieren’s theory of the growth of mathematical understanding is its broad scope in tracing development from the creation of images to formalization and axiomatization.

Additional examples of theories of the process of mathematical learning include Dubinsky’s (1991) theory of encapsulation, Sfard’s (1991; Sfard & Linchevski, 1994) theory of reification, Dörfler’s (1989) analysis of protocols of action, and Vergnaud’s (1982) analysis of the process by which students gradually explicate their initial theorems-in-action. Each of these theorists consider the primary source of increasingly sophisticated forms of mathematical reasoning to be students’ activity of interpreting and attempting to complete instructional activities, not the instructional activities themselves. Furthermore, they each characterize mathematical learning as a process in which operational or process conceptions evolve into what Sfard (1991) terms object-like structural conceptions. Proficiency in a particular mathematical domain is therefore seen to involve the conceptual manipulation of mathematical objects whose reality is taken for granted. Greeno (1991) captured this aspect of mathematical proficiency when he introduced the metaphor of acting in a mathematical environment

in which tools and resources are ready at hand to characterize number sense. For her part, Sfard (2000b) describes mathematical discourse as a virtual reality discourse to highlight the parallels between this discourse and the ways in which we talk about physical reality. The intent of each of the theoretical schemes I have referenced is to provide a conceptual framework or toolkit that can be used to develop accounts of the process of specific students' mathematical learning (Thompson & Saldanha, 2000).

In contrast to these general conceptual frameworks, theories of the second type focus on the development of students' reasoning in specific mathematical domains. Examples include analyses of early number reasoning (Carpenter & Moser, 1984; Fuson, 1992; Steffe & Cobb, 1988), multiplicative reasoning (Confrey & Smith, 1995; Streefland, 1991; Thompson, 1994; Vergnaud, 1994), geometric reasoning (Clements & Battista, 1992; van Hiele, 1986), algebraic reasoning (Filloy & Rojano, 1984; Kaput, 1999; Sfard & Linchevski, 1994), and statistical reasoning (Konold & Higgins, in press; Mokros & Russell, 1995; Saldanha & Thompson, 2001). It is important to note that these domain specific frameworks do not focus on the mathematical development of any particular student but are instead, like theories of the first type, concerned with the learning of an idealized student that Thompson and Saldanha (2000) refer to as the *epistemic individual*. Researchers working in this cognitive tradition account for variations in specific students' reasoning by using the constructs that comprise their framework to develop explanatory accounts of each student's mathematical activity. This approach enables the researchers to both compare and contrast the quality of specific students' reasoning and to consider the possibilities for their mathematical development.

The common element that ties both types of theories together is their portrayal of both the epistemic individual and of specific students as active constructors of increasingly sophisticated forms of mathematical reasoning. The metaphor of learning as a process of construction can be traced to the eighteenth century Italian philosopher Giambattista Vico who was the first to advance an explicitly constructivist position when he argued that "the known is the made" (Berlin, 1976). Vico's arguments anticipated several of the major claims that the German philosopher Immanuel Kant made 80 years later. In his treatise *A Critique of Pure Reason* (Kant, 1998), Kant contended that our perceptions are always in the form of objects because our minds have a priori structures or intuitions of space and time. He also argued that in addition to being perceived in space and time, objects are experienced through four

a priori categories of understanding: quantity (e.g., plurality, totality), quality (e.g., negation), relation (e.g., causality and dependence), and modality (e.g., possibility and necessity).

Kant's philosophical analysis provided the backdrop against which the most significant contributor to constructivism in cognitive psychology, Jean Piaget, conducted his research (Fabricius, 1979). Although Piaget is typically viewed as a psychologist, he described himself as a genetic epistemologist (Piaget, 1970). In this context, the term genetic denotes genesis and encompasses both the origins and development of forms of knowledge. Piaget explained that, ideally, he would have studied the origins and subsequent evolution of foundational categories of understanding as they occurred in the history of humankind. However, as this was impossible, he attempted to gain insight into epistemological issues by studying the development of these notions in children. Significantly, the concepts that he focused on in his investigations correspond almost exactly to the fundamental categories of thought proposed by Kant 150 years earlier. His intent was to account for the development of these categories by relying on constructs such as assimilation, accommodation, and equilibration thereby showing that there is no need to posit that they are *a priori*.

The aspect of Piaget's work that has gained most attention among both psychologists and educators is his claim to have identified a sequence of invariant stages through which children's thinking progresses. The cognitive theorists I have referenced have, in contrast, focused primarily on the process aspects of Piaget's theory. Piaget drew on his early training as a biologist to characterize intellectual development as an adaptive process in the course of which children reorganize their sensory-motor and conceptual activity (Piaget, 1980). In appropriating Piaget's general constructivist orientation to the process of development, cognitive theorists have necessarily had to adapt his theoretical constructs because their concern is to gain insight into the process of students' mathematical learning rather than to address problems of genetic epistemology. The resulting characterizations that they propose of the individual differ significantly from that of experimental psychology.

As we have seen, experimental psychology focuses on how variations in instructional conditions affect the performance of the statistically constructed collective subject. In contrast, cognitive psychology focuses on how the epistemic individual successively reorganizes its activity and comes to act in a mathematical environment. Thus, whereas experimental psychologists direct their attention to an

environment composed of manipulable independent variables, cognitive psychologists seek to delineate how the world of meaning and significance in which the epistemic individual acts changes in the course of development. We have also seen that experimental psychology characterizes specific students in terms of the deviation of their performance from that of the collective subject. In contrast, cognitive psychology characterizes specific students in terms of the nature or quality of their mathematical reasoning. Thus, whereas experimental psychology is premised on the assumption that students possess large or small amounts of psychological attributes, cognitive psychology is premised on the assumption that students' development involves *qualitative* changes in their mathematical reasoning.

In cases such as this and other cases where characterizations of the individual contrast sharply, it is tempting to try and determine which perspective gets people right. Following Putnam (1987), I have argued that this quest is misguided and that it is more productive to compare and contrast differing theoretical perspectives in terms of their potential usefulness. In this regard, I have suggested that the forms of knowledge produced by experimental psychology are particularly useful to administrators who are responsible for managing educational systems at some distance removed from the classroom and who have little grounding in classroom teaching and learning processes in particular content domains. Given their administrative concerns and interests, it is unlikely that they would see value in the forms of knowledge produced by cognitive psychologists working in the tradition on which I have focused. Conversely, having conducted studies that involve experimental designs (P. Cobb et al., 1991; P. Cobb, Wood, Yackel, & Perlwitz, 1992), I can attest that the resulting forms of knowledge are not well suited to the demands of instructional design at the classroom level. A fundamental difficulty is that studies involving experimental designs do not produce the detailed kinds of data that are needed to guide the often subtle refinements made when improving an instructional design at this level.

Proponents of the cognitive tradition that I have discussed frequently assume that the types of explanatory frameworks they produce constitute an adequate basis for both classroom instructional design and pedagogical decision making. I have contributed to the development of such a framework (Steffe & Cobb, 1988; Steffe, von Glasersfeld, Richards, & Cobb, 1983) and have attempted to use it to guide the development and refinement of classroom instructional designs. On the basis of this

experience, I question the claim that such frameworks are, by themselves, sufficient. The primary difficulty is precisely that the forms of knowledge produced are cognitive rather than instructional and do not involve positive heuristics for design (Gravemeijer, 1994b). In my view, researchers working in this tradition have inherited perspectives and associated methodologies from cognitive psychology but have not always reflected on the relevance of the forms of knowledge produced to the collective enterprise of mathematics education. The issue of how these perspectives and methodologies might be adapted so that they can better contribute to the concerns and interests of mathematics education has rarely arisen because their sufficiency has been assumed almost as an article of faith (see Thompson, 2002, for a rare discussion of these issues).

Given that cognitive theories of mathematical learning do not, by themselves, constitute a sufficient basis for design, the question of clarifying the contributions that they can make remains. Drawing on my experience of developing and revising designs at the classroom level, I can identify three contributions of theories of development in particular mathematical domains that I do not claim are exhaustive. First, domain-specific theories typically include analyses of the forms of reasoning that we want students to develop. These analyses are non-trivial accomplishments and can serve to specify the "big mathematical ideas," thereby giving an overall orientation to the instructional design effort. Second, domain-specific frameworks can alert the designer to major shifts in students' mathematical reasoning that the design should support. Third, the designer's or teacher's use of a domain-specific framework to gain insight into specific students' mathematical reasoning can inform the design of instructional activities intended to support subsequent learning. This use of a cognitive framework is particularly evident in the successful Cognitively Guided Instruction program developed by Carpenter, Fennema, and colleagues (Carpenter & Fennema, 1992; Carpenter, Fennema, Peterson, Chiang, & Loef, 1989; Fennema et al., 1993). As I will clarify when I discuss distributed cognition, I consider this emphasis on instructional activities as the primary means of supporting students' mathematical learning to be overly restrictive. It is also worth noting that domain-specific cognitive frameworks do not orient designers to consider issues of equity in students' access to significant mathematical ideas even though the resulting designs can, on occasion, make a significant contribution in this regard (Carey, Fennema, Carpenter, & Franke, 1995; Silver, Smith, & Nelson, 1995).

In summary, general and domain-specific theories of mathematical learning are both concerned with the learning of an idealized student, the epistemic individual. Variations in specific students' reasoning are accounted for by using the constructs central to theories of this type to develop explanations of their mathematical activity. Although there are substantive differences between the various cognitive theories I have referenced, they are tied together by the characterization of mathematical learning as a constructive process in the course of which students successively reorganize their sensory-motor and conceptual activity. From the perspective of experimental psychology, this multiplicity of theories is frequently taken as evidence that the cognitive tradition on which I have focused is not scientific. It is therefore worth reiterating that experimental psychologists have achieved unanimity only by eschewing a concern for cognitive processes and structures that account for observable performance and by treating variation in performance as error variance. Whereas the forms of knowledge produced by experimental psychology are well suited to the concerns of administrators, cognitive theories can contribute to the development and improvement of classroom instructional designs.

Sociocultural Theory

At their core, the forms of knowledge produced by the cognitive theorists on whose work I have focused concern the process by which the epistemic individual successively reorganizes its activity. In contrast, the forms of knowledge produced by sociocultural theorists concern the process by which people develop particular forms of reasoning as they participate in established cultural practices. This theoretical perspective treats intellectual development and the process by which people become increasingly substantial participants in various cultural practices as aspects of a single process. Consequently, whereas cognitive theorists investigate the activity of the epistemic individual, sociocultural theorists investigate the participation of the *individual-in-cultural-practice*. In a very real sense, the two groups of researchers are attempting to understand different realities. It is necessary to clarify the origins of sociocultural theory in order to understand the reality into which it gives insight.

Contemporary sociocultural theory draws directly on the writings of Vygotsky and Leont'ev. Vygotsky (1962; 1978; 1981) made his foundational contributions to this perspective during the period of intellectual ferment and social change that followed the

Russian revolution. In doing so, he was profoundly influenced by Marx's argument that it is the making and use of tools that serves to differentiate humans from other animal species. For Vygotsky, human history is the history of artifacts such as language, counting systems, and writing that are not invented anew by each generation but are instead passed on and constitute the intellectual bequest of one generation to the next. In formulating his theory of intellectual development, Vygotsky developed an analogy between the use of physical tools and the use of intellectual tools such as sign systems (Kozulin, 1990; van der Veer & Valsiner, 1991). His central claim was that just as the use of a physical tool serves to reorganize activity by making new goals possible, so the use of sign systems serves to reorganize thought. He viewed culture as a repository of sign systems and other artifacts that are appropriated by children in the course of their intellectual development (Vygotsky, 1978). It is important to stress that for Vygotsky, children's mastery of a counting system does not merely enhance or amplify an already existing cognitive capability. He instead argued that children's ability to reason numerically is created as they appropriate the counting systems of their culture. This example illustrates Vygotsky's more general contention that children's minds are formed as they appropriate sign systems and other artifacts.

In the most well known series of investigations that he conducted, Vygotsky attempted to demonstrate the crucial role of face-to-face interactions in which an adult or more knowledgeable peer supports the child's use of an intellectual tool such as a counting system (Vygotsky, 1981). He concluded from these studies that the use of sign systems initially appears in children's cognitive development on what he termed the "intermental" plane of social interaction and that, over time, the child eventually becomes able to carry out what was previously a joint activity on his or her own. On this basis, he argued that the child's mind is created via a process of internalization from the intermental plane of social interaction to the "intramental" plane of individual thought.

The central role that Vygotsky attributed to social interactions with more knowledgeable others usually features prominently in accounts of his work. However, there is some indication that shortly before his premature death in 1934, he began to view the relation between social interaction and cognitive development as a special case of a more general relation between cultural practices and cognitive development (Davydov & Radzikhovskii, 1985; Minick, 1987). This aspect of sociocultural theory was developed more fully after Vygotsky's death by a group of Soviet psychologists, the most prominent of whom was Alexei Leont'ev.

Although Leont'ev (1978; 1981) acknowledged the importance of face-to-face interactions, he saw the encompassing cultural practices in which the child participates as constituting the broader context of his or her development. For example, Leont'ev might have viewed interactions in which a parent engages a child in activities that involve counting as an instance of the child's initial, supported participation in cultural practices that involve dealing with quantities. He argued that the child's progressive participation in specific cultural practices underlies the development of his or her thinking. Intellectual development was, for him, synonymous with the process by which the child becomes a full participant in particular cultural practices. Because he considered the cognitive capabilities that a child develops to be inseparable from the cultural practices that constitute the context of their development, he viewed those capabilities to be characteristics of the child-in-culture-practice rather than the child *per se*.

The second contribution that Leont'ev made to sociocultural theory concerns his analysis of material objects and events. Although Vygotsky brought sign systems and other cultural tools to the fore, he gave less attention to material reality. In building on Vygotsky's ideas, Leont'ev argued that material objects as they come to be experienced by the developing child are defined by the cultural practices in which he or she participates. For example, a pen becomes a writing instrument rather than a brute material object for the child as he or she participates in literacy practices. In Leont'ev's view, the child does not come into contact with material reality directly, but is instead oriented to this reality as he or she participates in cultural practices. He therefore concluded that the meanings that material objects come to have are a product of their inclusion in specific practices. This thesis serves to underscore his argument that the individual-in-cultural-practice constitutes the appropriate analytical unit.

To clarify the distinction between the individual as characterized by cognitive and by sociocultural theorists, it is useful to contrast the role attributed to tools and social interactions in the two perspectives. Sociocultural theorists have sometimes accused proponents of the cognitive tradition on which I have focused of adopting a so-called Robinson Crusoe perspective in which they study the reasoning of socially and culturally isolated people. It is therefore important to emphasize that adherents of the cognitive tradition readily acknowledge that students' reasoning is influenced by both the tools that they use to accomplish goals and by their ongoing social interactions with others. However, tools and others' actions

are considered to be external to students' reasoning, and the focus is on documenting how students interpret them. Following Vygotsky, sociocultural theorists question the assumption that social processes can be clearly partitioned off from cognitive processes and treated as external conditions for them. These theorists instead view cognition as extending out into the world and as being inherently social. They therefore attempt to break down a distinction that is basic to the cognitive perspective and indeed to experimental psychology, that between the reasoner and the world reasoned about. Furthermore, following Leont'ev, sociocultural theorists typically situate tool use and face-to-face interactions within encompassing cultural practices. From this perspective, students' actions are viewed as elements of a system of cultural practices and students are viewed as participating in cultural practices even when they are in physical isolation from others.

In considering the potential usefulness of sociocultural theory, I should acknowledge that the instructional designs that I and my colleagues have developed in recent years to support students' mathematical learning are broadly compatible with some of the basic tenets of this theoretical perspective (P. Cobb & McClain, 2002). Nonetheless, I contend that sociocultural theory is of limited utility when actually formulating designs at the classroom level. The contributions of Davydov (1988a, 1988b), notwithstanding, it is in fact difficult to identify instances of influential designs whose development has been primarily informed by sociocultural theory. This becomes understandable once we note that the notion of cultural practices employed by sociocultural theorists typically refers to ways of talking and reasoning that have emerged during extended periods of human history. This construct makes it possible to characterize mathematics as a complex human activity rather than as disembodied subject matter (van Oers, 1996). The task facing both the teacher and the instructional designer is therefore framed as that of supporting and organizing students' induction into practices that have emerged during the discipline's intellectual history. While the importance of the goals inherent in this framing is indisputable, they provide only the most global orientation for design. A key difficulty is that the disciplinary practices that are taken as the primary point of reference exist prior to and independently of the activities of teachers and their students. In contrast, the ways of reasoning and communicating that are actually established in the classroom do not exist independently of the teacher's and students' activity, but are instead constituted by them in the course of their ongoing interactions

(Bauersfeld, 1980; Beach, 1999; Boaler, 2000; P. Cobb, 2000). A central challenge of design is to develop, test, and refine conjectures about both the classroom processes in which students might participate and the nature of their mathematical learning as they do so. Sociocultural theory provides only limited guidance because the classroom processes on which design focuses are emergent phenomena rather than already-established practices into which students are inducted.

Extending our purview beyond the classroom, there are two areas where, in my judgment, sociocultural theory can make significant contributions. I introduce each by first discussing recent bodies of scholarship developed within this tradition that are relevant to mathematics educators. The first body of scholarship is exemplified by investigations that have compared mathematical reasoning in school with that in various out-of-school settings such as grocery shopping (Lave, 1988), packing crates in a dairy (Scribner, 1984), selling candies on the street (Nunes, Schliemann, & Carraher, 1993; Saxe, 1991), playing dominoes and basketball (Nasir, 2002), laying carpet (Masingila, 1994), woodworking (Millroy, 1992), and sugar cane farming (de Abreu, 1995). These studies document that people develop significantly different forms of mathematical reasoning as they participate in different cultural practices that involve the use of different tools and sign systems, and that are organized by different overall motives (e.g., learning mathematics as an end in itself in school versus doing arithmetical calculations while selling candies on the street in order to survive economically). This approach of contrasting the forms of reasoning inherent in different cultural practices bears directly on issues of equity in students' access to significant mathematical ideas.

Although an adequate treatment of cultural diversity and equity is beyond the scope of this chapter, it is worth noting that a number of investigators have documented that the out-of-school practices in which students participate can involve differing norms of participation, language, and communication, some of which might be in conflict with those that the teacher seeks to establish in the mathematics classroom (Civil & Andrade, 2002; Ladson-Billings, 1998; Zevenbergen, 2000). An emerging line of research in mathematics education draws on sociocultural theory to document such conflicts and to understand the tensions that students experience (Boaler & Greeno, 2000; Gutiérrez, 2002; Martin, 2000; Moschkovich, 2002). The value of work of this type is that it enables us to view students' activity in the classroom as situated not merely with respect to the immediate learning environment, but with the respect to students' history

of participation in the practices of particular out-of-school groups and communities. It therefore has the potential to inform the development of designs in which the diversity in the out-of-school practices in which students participate is treated as an instructional resource rather than an obstacle to be overcome (Bouillion & Gomez, 2002; Civil, 2002; Gutstein, 2002; Warren, Ballenger, Ogonowski, Rosebery, & Hudicourt-Barnes, 2001).

The second relevant body of sociocultural research centers on the notion of a community of practice. In their overview of this line of research, Lave and Wenger (1991) clarify that they consider learning to be synonymous with the changes that occur in people's activity as they move from relatively peripheral participation to increasingly substantial participation in the practices of established communities. In doing so, they also argue that the tools used by community members carry a substantial portion of a community's intellectual heritage. Franke and Kazemi (2001) and Stein et al. (1998) have used Wenger's (1998) more recent formulation of these ideas to analyze teachers' learning as they are inducted into the practices of an established professional teaching community. However, mathematics education researchers are yet to exploit the full potential of the notion of community of practice in my view. This becomes apparent when we follow Wenger (1998) in noting that this notion brings together (a) theories of social structure that give primacy to institutions, norms and rules, and (b) theories of situated experience that give primacy to the dynamics of everyday existence and the local construction of interpersonal events. These two types of theories correspond to a dichotomy in the teacher education literature between analyses that focus on the structural or organizational features of schools and analyses that focus on the role of professional development in supporting teachers' reorganization of their instructional practices (Engestrom, 1998; Franke, Carpenter, Levi, & Fennema, 2001).

As Engestrom (1998) observes, the notion of community of practice has the potential to transcend this dichotomy in the literature by providing a unit of analysis that captures social structures that are within the scope of teachers' engagement as they develop and refine their instructional practices. Elsewhere, I and my colleagues have drawn heavily on this notion to develop an analytic approach for locating mathematics teachers' instructional practices within the institutional settings of the schools in which they work (P. Cobb, McClain, Lamberg, & Dean, 2003). The goal of research of this type is to gain insight into the processes by which teachers' instructional practices are partially constituted by

the institutional setting of the schools in which they work. The potential contribution of such work is indicated by the substantiated finding that teachers' instructional practices are profoundly influenced by the institutional constraints that they attempt to satisfy, the formal and informal sources of assistance on which they draw, and the materials and resources that they use in their classroom practice (Ball & Cohen, 1996; C. A. Brown, Stein, & Forman, 1996; Nelson, 1999; Price & Ball, 1997; Senger, 1999; Stein & Brown, 1997). Analyses that document these affordances and constraints can inform the development of designs for supporting teachers' learning. In particular, they orient researchers and teacher educators to consider whether their collaborations with teachers should involve concerted attempts to bring about change in the institutional settings in which the teachers have developed and revised their instructional practices.

In summary, sociocultural theory characterizes the individual as a participant in established, historically evolving cultural practices. Thus, whereas the cognitive tradition that I discussed accounts for learning in terms of the epistemic individual's reorganization of its activity, sociocultural theory does so by documenting process by which people become increasingly substantial participants in various cultural practices. I have questioned the relevance of sociocultural theory to the development of designs at the classroom level but identified two other areas in which it can potentially make contributions. The first concerns equity in students' access to significant mathematical ideas and involves analyzing the out-of-school practices in which they participate. The second involves analyses of the institutional settings in which teachers develop and refine their instructional practices that can inform the development of designs for supporting their learning. In both these areas, the central issue is that of understanding how people deal with the tensions that they experience when different practices in which they participate are in conflict. In the first case, the conflicts are between the out-of-school practice in which students participate and those established in the mathematics classroom, whereas in the second case, the conflicts are between the practices of the schools in which teachers work and those established in a professional teaching community.

Distributed Cognition

As I noted, sociocultural theory has developed largely independently of western psychology and draws inspiration directly from the writings of Vygotsky and Leont'ev. Distributed cognition, in contrast, has developed in reaction to mainstream cognitive science

and incorporates aspects of the Soviet work. Mainstream cognitive science should be differentiated from the cognitive tradition on which I have focused in this chapter in that it involves the observer's rather than the actor's viewpoint and posits what Anderson (1983) terms cognitive behaviors that intervene between stimulus and observed response activity. Several of the most important contributors to distributed cognition such as John Seeley Brown (1989), Alan Collins (1992), and James Greeno (1997) in fact achieved initial prominence as mainstream cognitive scientists before substantially modifying their theoretical commitments. Whereas sociocultural theorists usually frame people's reasoning as acts of participation in relatively broad systems of cultural practices, distributed cognition theorists typically restrict their focus to the immediate physical, social, and symbolic environment. Empirical studies conducted within the distributed tradition therefore tend to involve detailed analysis of either specific people's or a small group's activity rather than analyses of people's participation in established cultural practices.

As I indicated when clarifying the claim that the notion of the individual is conceptually relative, the term distributed cognition is most closely associated with Roy Pea. Pea (1985, 1993) coined this term to emphasize that, in his view, cognition is distributed across minds, persons, and symbolic and physical environments. As he and other distributed cognition theorists make clear in their writings, this perspective directly challenges a foundational assumption of both mainstream cognitive science and of the cognitive tradition that I have discussed. This is the assumption that cognition is bounded by the skin and can be adequately accounted for solely in terms of internal processes. Distributed cognition theorists instead see cognition as extending out into the immediate environment such that the environment becomes a resource for reasoning. As a consequence, the individual is characterized in this tradition as an element of a reasoning system.

In developing this position, distributed cognition theorists have been influenced by a number of studies conducted by sociocultural researchers, particularly those that compare people's reasoning in different settings. In one of the most frequently cited investigations, Scribner (1984) analyzed the reasoning of workers in a dairy as they filled orders by packing products into crates of different sizes. Her analysis revealed that the loaders did not perform purely mental calculations but instead used the structure of the crates as a resource in their reasoning. For example, if an order called for ten units of a particular product and six units were already in a crate that held

twelve units, experienced loaders rarely subtracted six from ten to find how many additional units they needed. Instead, they might realize that an order of ten units would leave two slots in the crate empty and just know immediately from looking at the partially filled crate that four additional units are needed. As part of her analysis, Scribner demonstrated that the loaders developed strategies of this type as they went about their daily business of filling orders. For distributed cognition theorists, this indicates that the system that did the thinking was the loader in interaction with a crate. From this perspective, the loaders' reasoning is therefore treated as emergent relations between them and the immediate environment in which they worked.

Part of the reason that distributed cognition theorists attribute such significance to Scribner's study and to other investigations conducted by sociocultural researchers is that they capture what Hutchins (1995) refers to as cognition in the wild. This focus on people's reasoning as they engage in both everyday and workplace activities contrasts sharply with the traditional school-like tasks that are often used to investigate cognition. In addition to questioning whether people's reasoning on school-like tasks constitutes a viable set of cases from which to develop adequate accounts of cognition, several distributed cognition theorists have also critiqued current school instruction. In doing so, they have broadened their focus beyond mainstream cognitive science's traditional emphasis on the structure of particular tasks by drawing attention to the nature of the classroom activities within which the tasks take on meaning and significance for students.

J. S. Brown et al. (1989) developed one such critique by observing that school instruction typically aims to teach students abstract concepts and general skills on the assumption that students will be able to apply them directly in a wide range of settings. In challenging this assumption, they argue that the appropriate use of a concept or skill requires engagement in activities similar to those in which the concept or skill was developed and is actually used. In their view, the well-documented finding that most students do not develop widely applicable concepts and skills in school is attributable to the radical differences between classroom activities and those of both the disciplines and of everyday, out-of-school settings. They contend that successful students learn to meet the teacher's expectations by relying on specific features of classroom activities that are alien to activities in the other settings. In developing this explanation, Brown et al. treat the concepts and skills that students actually develop in school as relations

between students and the material, social, and symbolic resources of the classroom environment.

It might be concluded from the two examples given thus far, those of the dairy workers and of students relying on what might be termed superficial cues in the classroom, that distributed cognition theorists do not address more sophisticated types of reasoning. Researchers working in this tradition have in fact analyzed a number of highly technical, work-related activities. The most noteworthy of these studies is, perhaps, Hutchins' (1995) analysis of the navigation team of a naval vessel as they brought their ship into harbor. In line with other investigations of this type, Hutchins argues that the entire navigation team and the artifacts it used constitutes the appropriate unit for a cognitive analysis. From the distributed perspective, it is this system of people and artifacts that did the navigating and over which cognition was distributed. In developing his analysis, Hutchins pays particular attention to the role of the artifacts as elements of this cognitive system. He argues, for example, that the cartographer has done much of the reasoning for the navigator who uses a map. This observation is characteristic of distributed analyses and implies that to understand a cognitive process, it is essential to understand how parts of that process have, in effect, been sedimented in tools and artifacts. Distributed cognition theorists therefore contend the environments of human thinking are thoroughly artificial. In their view, the cognitive resources that people exercise in particular environments are partially constituted by the cognitive resources with which they have populated those environments. As a consequence, the claim that artifacts do not merely serve to amplify cognitive process but instead reorganize them is a core tenet of the distributed cognition perspective (Dörfler, 1993; Pea, 1993).

In contrast to sociocultural theory, distributed cognition treats classroom processes as emergent phenomena rather than already-established practices into which students are inducted. For example, distributed theorists consider that aspects of the classroom learning environment such as classroom norms, discourse, and ways of using tools are constituted collectively by the teacher and students in the course of their ongoing interactions. In my judgment, the distributed perspective therefore has greater potential than sociocultural theory to inform the formulation of designs at the classroom level. A number of design research studies have in fact been conducted from this perspective (Bowers, Cobb, & McClain, 1999; Confrey & Smith, 1995; Fishman et al., 2004; Hershkowitz & Schwarz, 1999; Lehrer, Strom, & Confrey, 2002). In investigations of this

type, researchers both develop designs to “engineer” novel forms of mathematical reasoning, and analyze the process of students’ learning in these designed learning environments together with the means by which that learning is supported (P. Cobb, Confrey, diSessa, Lehrer, & Schauble, 2003; Collins, Joseph, & Bielaczyc, 2004; Confrey & Lachance, 2000; Design-Based Research Collaborative, 2003; Edelson, 2002; Gravemeijer, 1994b). I noted that designs developed from the cognitive perspective on which I have focused typically emphasize the development of instructional activities. In contrast, distributed cognition theorists’ characterization of students as elements of reasoning systems orients them to construe the means of supporting students’ mathematical learning more broadly. The means of support that are incorporated into designs are usually not limited to instructional activities but also encompass classroom norms, the nature of discourse, and the ways in which notations and other types of tools are used. Thus, whereas design from the cognitive perspective involves developing instructional activities as informed by analyses of specific students’ reasoning, design from the distributed perspective focuses on the physical, social, and symbolic classroom environment that constitutes the immediate situation of the students’ mathematical learning.

Given my generally positive assessment of the usefulness of the distributed perspective in informing the development, testing, and revision of designs, it is also important to note two potential limitations. The first limitation concerns the scant attention typically given to issues of equity. For example, the focus of researchers who develop and refine designs at the classroom level usually centers on students’ individual and collective development of particular forms of mathematical reasoning. Pragmatically, it is essential that students come to see classroom activities as worthy of their engagement if the designs are to be effective. However, the process of supporting students’ engagement by cultivating their mathematical interests is rarely an explicit focus of inquiry (for exceptions, see diSessa, 2001; Eisenhart & Edwards, 2001, April). As a consequence, differences in students’ engagement that might reflect differential access to the instructional activities used and to the types of discourse established in the classroom can easily escape notice. In my view, this limitation stems from an almost exclusive focus on the classroom as the immediate context of students’ learning. This focus precludes a consideration of tensions that some students might experience between aspects of this social context and the out-of-school practices in which they participate. Adherents to this perspective

can address this limitation by coordinating their viewpoint with a sociocultural perspective that enables them to see students’ classroom activity as situated not merely with respect to the immediate learning environment, but also with respect to their history of participation in the practices of out-of-school groups and communities.

The second limitation concerns the manner in which the characterization of the individual as an element of a reasoning system is taken as delegitimizing cognitive analysis of specific students’ reasoning. As I indicated earlier in this chapter, this view of the individual reflects the evolution of the distributed perspective from mainstream cognitive science in that it involves equipping the individual as portrayed by this latter perspective with cultural tools and placing it in social context. I consider this restriction to be a limitation of the distributed perspective because it fails to acknowledge the contributions that cognitive analyses of specific students’ reasoning can make to the process of adjusting and modifying an instructional design (see P. Cobb, 1998, for a detailed discussion of this point). Furthermore, in the hands of a skillful teacher, the diversity in students’ reasoning is a primary resource on which he or she can draw to support sustained classroom discussions that focus on substantive mathematical issues. Earlier in this chapter, I followed Putnam (1987) in endorsing Quine’s (1953) argument that we should accept the existence of abstract mathematical entities on the grounds that this ontology is indispensable to the mathematics viewed as a practice. Similarly, I contend that we should accept the existence of specific students’ mathematical reasoning because this ontology is indispensable to design and teaching in mathematics education. It is therefore necessary, in my view, to resist theoretical arguments that delegitimize this ontology. In the next section of this chapter, I indicate how it might be possible to capitalize on the potential contributions of the distributed perspective while circumventing this limitation.

In summary, the distributed perspective emerged in response to the limited attention given to context, culture, and affect by mainstream cognitive science (De Corte et al., 1996). The individual is characterized as an element of a reasoning system that also includes aspects of the immediate physical, social, and symbolic environment. In contrast to sociocultural theorists’ focus on people’s participation in established cultural practices, distributed theorists usually conduct detailed analyses of reasoning processes that are stretched over people and the aspects of their immediate environment that they use as cognitive resources. In my view, the distributed perspective has greater

potential than sociocultural theory to contribute to the formulation of designs at the classroom level because it treats classroom processes as emergent phenomena. However, I tempered this positive appraisal by noting two limitations. These concern the limited attention given to issues of equity and the disavowal of cognitive analyses of specific students' mathematical reasoning. Taken together, these two limitations indicate the value of attempting to capitalize on the ways that multiple perspectives can contribute to the enterprise of developing, testing, and revising designs for supporting learning. I address this issue shortly.

REFLECTION

The contrasts I have drawn between the four theoretical perspectives are summarized in Table 1. The development of these contrasts involved the use of two criteria of comparison. The first criterion concerns how each perspective characterizes the individual whereas the second focus on the potential of each to contribute to our understanding of learning processes and the means of supporting their realization. Following Kuhn (1962; 1970), it is important to acknowledge that these criteria are not neutral standards but are instead values that I propose should be considered when coming to terms with the multiplicity of theoretical perspectives that characterize mathematics education research. As I indicated, these criteria reflect commitments and interests inherent in the view that mathematics education is a design science and, for this reason, are eminently debatable and are open to critique and revision. Furthermore, the process of using

them to compare theoretical perspectives necessarily involves interpretation and judgment. Consequently, mathematics educators who use these same criteria “may nevertheless make different choices in the same concrete situation” (Kuhn, 1970, p. 262). My intent in proposing criteria and illustrating how they might be used has therefore not been to shut down debate in the face of competing theoretical perspectives, but to move the debate to a meta-level at which we are obliged to give good reasons for our theoretical choices. At this meta-level, the ideological fervor with which particular perspectives are sometimes promoted is no substitute for justifications that articulate the choice criteria or values together with the interests and concerns that they reflect. The pragmatic realist stance that I have taken to the issue of multiple theoretical perspectives will have achieved what Rorty (1979) refers to as its therapeutic purpose to the extent that justifications of this sort become commonplace.

THEORIZING AS BRICOLAGE

As part of the process of discussing the four theoretical perspectives, I briefly outlined the historical origins of each. I noted, for example, that experimental psychology has been profoundly shaped by its alliance with educational administrators. In contrast, the origins of the cognitive tradition on which I focused can be traced to Piaget's interest in problems of genetic epistemology. Sociocultural theory for its part reflects Vygotsky's and Leont'ev's commitment to the notion of the “the new socialist man” and to the view of education as a primary means of bringing about this change in Soviet society (van der Veer & Valsiner, 1991).

Table 1.1 Contrasts Between Four Theoretical Perspectives

Theoretical perspective	Characterization of the individual	Usefulness	Limitations
Experimental Psychology	Statistically constructed collective individual	Administration of educational systems	Limited relevance to design at classroom level
Cognitive Psychology	Epistemic individual as reorganizer of activity	Specification of “big ideas” Design of instructional activities	Means of supporting learning limited to instructional tasks
Sociocultural Theory	Individual as participant in cultural practices	Designs that take account of students' out-of-school practices Designs that take account of institutional setting of teaching and learning	Limited relevance to design at classroom level
Distributed Cognition	Individual element of a reasoning system	Design of classroom learning environments including norms, discourse, and tools	Delegitimizes cognitive analyses of specific students' reasoning

Finally, distributed cognition emerged in reaction to perceived shortcomings of mainstream cognitive science. The concerns and interests that motivated the development of each of these perspectives differ significantly from those of mathematics educators. In terms of the metaphor of theoretical perspectives as conceptual tools that I introduced earlier in this chapter, each of these perspectives is a tool that has been fashioned while addressing problems that are not of immediate concern to most mathematics educators. It is therefore unreasonable to expect that any one of these perspectives is ready-made for the collective enterprise of developing, testing, and revising designs. Given the limitations that I have discussed of each perspective with respect to this enterprise, the question that arises is not that of how to choose between the various perspectives. Instead, it is how they can be adapted to the concerns and interests of mathematics educators.

In addressing this question, I propose to view the four perspectives as sources of ideas that we can appropriate and modify for our purposes as mathematics educators. This process of developing conceptual tools for mathematics education research parallels that of instructional design as described by Gravemeijer (1994b).

[Design] resembles the thinking process that Lawler (1985) characterizes by the French word *bricolage*, a metaphor taken from Claude Lévi-Strauss. A *bricoleur* is a handy man who invents pragmatic solutions in practical situations. . . . [T]he *bricoleur* has become adept at using whatever is available. The *bricoleur*'s tools and materials are very heterogeneous: Some remain from earlier jobs, others have been collected with a certain project in mind. (p. 447)

Similarly, I suggest that rather than adhering to one particular theoretical perspective, we act as *bricoleurs* by adapting ideas from a range of theoretical sources.

To illustrate this approach, I take as a case an interpretive framework that several colleagues and I developed over a number of years while addressing concrete problems and issues that arose while working in classrooms (P. Cobb, Stephan, McClain, & Gravemeijer, 2001; P. Cobb & Yackel, 1996). The intent of the framework is to locate students' mathematical reasoning in the social context of the classroom in a manner that can feed back to inform instructional design and teaching. For my current purposes, it suffices to note that the framework involves the coordination of two distinct perspectives on classroom activity. One is a social perspective that is concerned with ways of acting, reasoning, and arguing that have been established as normative in a classroom community. From this per-

spective, an individual student's reasoning is framed as an act of participation in these normative activities. The other is a cognitive perspective that focuses squarely on the nature of individual students' reasoning or, in other words, on their specific ways of participating in communal classroom activities. Analyses developed by using the framework bring the diversity in students' mathematical reasoning to the fore while situating that diversity in the social context of their participation in communal activities.

We take the relation between the social and cognitive perspectives to be one of reflexivity. This is an extremely strong relationship that does not merely mean that the two perspectives are interdependent. Instead, it implies that neither exists without the other in that each perspective constitutes the background against which mathematical activity is interpreted from the other perspective (Mehan & Wood, 1975). For example, the collective activities of the classroom community (social perspective) emerge and are continually regenerated by the teacher and students as they interpret and respond to each other's actions (cognitive perspective). Conversely, the teacher's and students' interpretations and actions in the classroom (cognitive perspective) are not seen to exist apart from their participation in communal classroom practices (social perspective). The coordination is therefore not between individual students and the classroom community viewed as separate entities but between two alternative ways of looking at and making sense of what is going on in classrooms.

This interpretive framework is a *bricolage* in that the social perspective draws on sociocultural theory (Cole, 1996; Lave, 1991; Rogoff, 1990) and the cognitive perspective draws on both cognitive psychology (Piaget, 1970; Steffe & Kieren, 1994; Thompson, 1991), and distributed accounts of cognition (Hutchins, 1995; Pea, 1993; Wertsch, 1998, 2002). One of the key theoretical constructs that we use when we take a social perspective, that of a classroom mathematical practice, serves to illustrate how we have appropriated ideas to our agenda as mathematics educators. We developed this construct by adapting sociocultural theorists' notion of a cultural practice. As I indicated when discussing sociocultural theory, this idea is attractive because it makes it possible to characterize mathematics as a complex human activity rather than as disembodied subject matter. However, I also noted that the notion of a practice as framed in sociocultural theory is problematic from the point of view of design because practices are typically characterized as existing prior to and independently of the teacher's and students' activity. We therefore modified this notion by explicitly defining a classroom

mathematical practice as an emergent phenomenon that is established jointly by the teacher and students in the course of their ongoing interactions. This is a non-trivial adaptation in that students are then seen to contribute to the development of the classroom norms and practices that constitute the social situation of their mathematical learning.

We also made modifications when fashioning a cognitive perspective appropriate for our purposes by adapting ideas from the cognitive perspective I have discussed and from distributed cognition. For example, we took from the cognitive perspective the notion of learning as a process of reorganizing activity. However, influenced by distributed theories of intelligence, we found it important to broaden our view of activity so that it is not restricted solely to solo sensory-motor and conceptual activity but instead reaches out into the world and includes the use of tools and symbols. The rationale for this modification is at least in part pragmatic in that our work as instructional designers involves developing notation systems, physical tools, and computer-based tools for students to use. We therefore needed an analytic approach that can take account both of the diverse ways in which students reason with tools and symbols, and of how those ways of reasoning evolve over time.

Although we saw value in distributed accounts of intelligence, we could not accept this theoretical orientation ready-made given its rejection of analytical approaches that focus explicitly on the nature of specific students' reasoning. The adaptation we made was to modify how the individual is characterized. As I have indicated, distributed theorists appear to have accepted the portrayal of the individual offered by mainstream cognitive science and propose equipping this character with cultural tools and locating it in social context. The primary adaptation we made was to characterize the individual not as needing to be placed in its immediate physical, social, and symbolic environment, but as already acting in that environment. The tools and symbols that students use are then viewed as constituent parts of their activity rather than as standing apart from or outside their activity.

Once this modification is made, what is viewed as a student-tool system from the perspective of distributed cognition becomes, in the psychological perspective we take, an individual student engaging in mathematical activity that involves *reasoning with* tools and symbols. Thus, although the focus of this psychological perspective is explicitly on the quality of individual students' reasoning, its emphasis on tools is generally consistent with the notion of mediated action as discussed by sociocultural theorists such

as Wertsch (2002). Further, as I have indicated, the remaining component of the functional system posited by distributed theorists, the classroom social context, becomes an explicit focus of attention when this psychological perspective is coordinated with the social perspective. The interpretive framework therefore characterizes students as reasoning with tools while participating in and contributing to the development of communal practices. With regard to usefulness, the framework yields analyses of students' mathematical learning that are tied to the classroom social setting in which that learning actually occurs. As a consequence, these analyses enable us to tease out aspects of this setting that served to support the development of students' reasoning. This, in turn, makes it possible to develop testable conjectures about ways in which those means of support and thus the instructional design can be improved.

In keeping with the critical stance I took to the four theoretical perspectives, I should acknowledge that the interpretive framework has at least two major limitations. These concern the isolation of classroom learning environments from the institutional settings of the schools in which they are located, and failure to address issues of cultural diversity and equity in students' access to significant mathematical ideas. As a consequence, these two areas have become a focus of our research in recent years (P. Cobb & Hodge, 2002; P. Cobb, McClain et al., 2003). It is also worth noting that the framework does not draw on the theoretical perspective of experimental psychology. This is primarily because experimental psychology views the classroom learning environment from an administrative viewpoint and characterizes it as composed of independent variables that can be manipulated from the outside. In contrast, the framework I have outlined characterizes the classroom learning environment from the inside as jointly constituted by the teacher and students. Analyses developed when using this framework will therefore be of little value to most administrators. The experimental psychology perspective might be useful when the goal is to contribute to public policy discourse about mathematics teaching and learning.

My purpose in discussing the interpretive framework has been to illustrate the process of adapting and modifying ideas appropriated from a range of theoretical sources. The pragmatic spirit of the bricolage metaphor indicates that the goal in doing so is to fashion conceptual tools that are useful for our purposes as mathematics educators. The metaphor therefore serves to differentiate relatively modest efforts of the type that I have illustrated from more ambitious projects that aim to develop

theoretical cosmologies (Shotter, 1995). For example, our goal in developing the framework was to craft a tool that would enable us to make sense of what is happening in mathematics classrooms rather than to produce a grand synthesis of cognitive psychology, sociocultural theory, and distributed cognition. In my view, theorizing as a modest process of bricolage offers a better prospect of mathematics education research developing an intellectual identity distinct from the various perspectives on which it draws than does the attempt to formulate all-encompassing theoretical schemes.

INCOMMENSURABILITY

The approach I have illustrated for comparing theoretical perspectives is best viewed as a proposal about how we might cope with competing theoretical perspectives in mathematics education research. The account that I have given of the relations between different perspectives can be contrasted with analyses that portray a historical sequence of theoretical perspectives, each of which overcomes the limitations of its predecessors. Accounts of this latter type depict the history of a field as an ordered progression that typically culminates with the perspective to which the writer subscribes. As Guerra (1998) notes, these narratives are based on the implicit metaphor of theoretical developments as a *relentless march of progress*. For example, adherents of distributed cognition frequently portray their perspective as overcoming the limitations of mainstream cognitive science. Proponents of perspectives that have supposedly been superseded typically employ a different type of narrative that is based on the metaphor of *potential redemption*. For example, experimental psychologists frequently characterize the increasing prominence of competing perspectives as a period in which the issue of what counts as credible evidence has been largely ignored, and portray their perspective as offering redemption by making educational research a prestigious scientific enterprise.

The account I have given reflects an alternative metaphor, that of *co-existence and conflict*. The tension between the march of progress and potential redemption narratives indicates the relevance of this metaphor. I fleshed out this metaphor by proposing two criteria for comparing and contrasting competing perspectives. The first focused on how each perspective characterized the individual, thereby delineating the types of phenomena that proponents of the different perspectives are investigating. The second focused

on the potential contributions of each perspective to the collective enterprise of formulating, testing, and revising designs for supporting learning. In developing this second criterion, I followed Dewey (1890/1969), Pepper (1942), and Prawat (1995) in arguing that a theoretical perspective is a useful conceptual tool for mathematics educators to the extent that it gives insight and understanding into learning processes and the specific means of supporting their realization. In using these two criteria, I noted that the goals for which each of the four perspectives was originally developed differ from those of mathematics educators. On this basis, I argued that we should view the various co-existing perspectives as sources of ideas to be adapted to our purposes.

The contrasting ways in which the different perspectives characterize the individual indicate that they are incommensurable. In terms of Putnam's (1987) pragmatic realism, adherents to the differing perspectives ask different types of questions and produce different forms of knowledge as they attempt to develop insights into different realities. I followed Putnam (1987) and Kuhn (1962, 1977) in arguing that the realities that researchers investigate are conceptually relative to their particular theoretical perspectives, but rejected the claim that any of these realities is as good as any other. The approach I took is consistent with Feyerabend's (1975) claim that we cope with incommensurability both in research and in other areas of life by drawing comparisons and contrasts in the course of which we delineate similarities and differences. Feyerabend also argued that there is no single ultimate grid for comparing theoretical perspectives, and demonstrated that they can be compared in multiple ways. The primary challenge posed by incommensurability is to develop a way of comparing and understanding different perspectives. It was for this reason that I discussed the two criteria I used in some detail.

As Bernstein (1983) observed, the process of comparing incommensurable perspectives has parallels with anthropology in that the goal is to figure out what the "natives" think they are doing (Geertz, 1973). Kuhn (1977) described how he attempts to avoid merely imposing his viewpoint when he discussed his work as a philosopher and historian of science.

[The] plasticity of texts does not place all ways of reading on a par, for some of them (ultimately, one hopes, only one) possess a plausibility and coherence absent from others. Trying to transmit such lessons to students, I offer them a maxim: When reading the works of an [historically] important thinker, look first for the apparent absurdities in the text and ask yourself how a sensible person could have written them. When

you find an answer, I continue, when those passages make sense, then you may find that more central passages, ones you previously thought you understood, have changed their meaning. (p. xii)

The openness inherent in this stance to incommensurability has the benefit that in coming to understand what adherents to an alternative perspective think they are doing, we develop a more sensitive and critical understanding of some of the taken-for-granted aspects of our own perspective. This understanding is critical to the justification of theoretical choices in that these justifications should, in my view, involve the specification of both the choice criteria used and the interests and concerns that they reflect.

As Geertz (1973) emphasizes, the absence of abstract, cross-cultural universals does not condemn anthropology to absolute relativism.

If we want to discover what man amounts to, we can only find it in what men are, and what men are, above all other things, is various. It is in understanding that variousness—its range, its nature, its basis, and its implications—that we shall come to construct a concept of human nature that more than a statistical shadow and less than a primitive dream has both substance and truth. (p. 52)

Similarly, the absence of a neutral framework for comparing incommensurable theoretical perspectives does not condemn us to absolute relativism in which theoretical decisions amount to nothing more than personal whim or taste. A primary purpose of philosophy for Dewey, Gadamer, Kuhn, Putnam, and Rorty is to transcend the dichotomy between an unobtainable neutral framework on the one hand and an absolute, “anything goes” brand of relativism on the other. Philosophy as they conceive it is a discourse about incommensurable perspectives and discourses. Its intent is both edifying and therapeutic in that it aims to support conversation both about and between various perspectives. My purpose in this chapter has been to advance a conversation of this type in the mathematics education research community.

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Correspondence concerning this article should be addressed to Paul Cobb, Vanderbilt University, Peabody College Box 330, Nashville, TN 37203. E-mail: paul.cobb@vanderbilt.edu

THEORY IN MATHEMATICS EDUCATION SCHOLARSHIP

Edward A. Silver and Patricio G. Herbst

UNIVERSITY OF MICHIGAN

This chapter addresses the use of theory in mathematics education scholarship.¹ We approach our task as one of characterizing how theory is used in our field, taking a practice-oriented perspective rather than a normative one. Yet we do not advocate a passive stance toward theory development and use. Rather, we intend that the review we offer here, and the embedded framework for considering theory as central to the relationships between problems, research, and practice, will stimulate reflection, growth, and improvement in the way that theory is viewed, developed, and used in our field.

As Mason and Waywood (1996) have noted, “theory is a value-laden term with a long and convoluted history” (p. 1055). A synopsis of that history may be seen in contemporary usage of the word *theory*. The Oxford English Dictionary (OED) lists the following as nonobsolete meanings of the word:²

3. A conception or mental scheme of something to be done, or of the method of doing it; a systematic statement of rules or principles to be followed.

4. **a.** A scheme or system of ideas or statements held as an explanation or account of a group of facts or

phenomena; a hypothesis that has been confirmed or established by observation or experiment, and is propounded or accepted as accounting for the known facts; a statement of what are held to be the general laws, principles, or causes of something known or observed.

b. That department of an art or technical subject which consists in the knowledge or statement of the facts on which it depends, or of its principles or methods, as distinguished from the *practice* of it.

c. A systematic statement of the general principles or laws of some branch of mathematics; a set of theorems forming a connected system: as *the theory of equations, of functions, of numbers, of probabilities*.

5. In the abstract (without article): Systematic conception or statement of the principles of something; abstract knowledge, or the formulation of it: often used as implying more or less unsupported hypothesis (cf. 6): distinguished from or opposed to *practice* (cf. 4b). *in theory* (formerly *in the theory*): according to theory, theoretically (opp. to *in practice* or *in fact*).

6. In loose or general sense: A hypothesis proposed as an explanation; hence, a mere hypothesis, speculation, conjecture; an idea or set of ideas about something; an individual view or notion. Cf. 4.

¹ No single chapter could cover the extensive territory that our charge encompasses, so we have made choices about how to focus the chapter in ways that we hope will complement other sources that a reader may find helpful on this topic. One excellent resource is Mason and Waywood (1996). Moreover, given the intimate connection of theory to philosophical perspectives, on the one hand, and to research methods and the nature of evidence, on the other hand, we urge readers interested in theory to also read Paul Cobb’s chapter on philosophical perspectives and Alan Schoenfeld’s chapter on research methods, both of which appear in this volume. Other chapters in this volume will explicate theory as it is used and developed in specific areas of research in our field.

² <http://dictionary.oed.com>, retrieved May 16, 2005.

The attentive reader of this chapter will note the use of most, perhaps all, of these senses of the word theory in this chapter.

A fundamental assumption of this chapter is that theory appears in our scholarship, especially in research in mathematics education, in many guises and at many levels. Our goal here is to describe, illustrate, and discuss a variety of these ways. But we also intend to bring some order to this diversity, both by pointing to characteristics of theory that appear to be common across the many uses and by providing a unified framework that locates theory as a central matter in the research endeavor in our field. Along the way we suggest several scopes of work where theories of different kinds can be useful. To accomplish our task we use many examples, some short, some long, which have been chosen from the literature that we know best to make key points. Although we treat many examples in this chapter, we do not profess to be comprehensive. Our omission of some theoretical perspective should not be interpreted as implied critique, nor should inclusion be seen as overt endorsement. Our examples are chosen only to help with the larger task undertaken here.

We prepared this chapter against the backdrop of two seemingly contradictory trends in mathematics education scholarship. On the one hand, there has been a growing concern for increasing the relevance and usefulness of research in our field. The press for relevance has come not only from policymakers and practitioners but also from within the community of scholars itself. On the other hand, mathematics education scholarship has become increasingly attentive to theory in recent years. Whereas the role of theory in mathematics education scholarship was usually tacit (at best) 30 years ago, it has become much more visible and important in contemporary scholarship.

What makes these two trends appear to be contradictory is an ambient skepticism in the field of mathematics education about the possibility that theory-driven research can also be relevant to educational practice. It is not uncommon to hear teachers and many other mathematics education professionals complain that mathematics education scholarship is “too theoretical.” This declaration appears to rest on OED meanings 5 or 6 (above); the sense in which “true in theory” is euphemism for “not true in practice.” Moreover, one can often hear researchers deride work for being “too applied,” thereby identifying research that is closely tied

to educational practice as insufficiently theoretical. Although one can debate the validity of such statements and views, there can be little debate that these views are deeply and widely held.

There can be little argument that there is a legitimate tension between some of the meanings evoked by the words *theory* and *practice* when these words are pitted against each other. In his analysis of some of the challenges faced by teachers who enter research-oriented doctoral programs in education, David Labaree has deftly captured the essence of this tension in characterizing the challenging transition that individuals make if they leave teaching careers and enter research-oriented doctoral programs: “The shift from K–12 teaching to educational research often asks students to transform their cultural orientation from normative to analytic, from personal to intellectual, from the particular to the universal, and from the experiential to the theoretical” (2003, p. 16). Nevertheless, despite the tensions that are evident when one pits the words *theory* and *practice* against each other, we contend that mathematics education scholarship can be both invested in theory and relevant to educational practice.³ The essence of our argument is straightforward. Viewed historically, mathematics education scholarship is itself a practice—an institutional practice, done by scholars who belong to disciplines and to organizations and are thus disposed to address certain realities, to perceive certain puzzles, and to employ particular kinds of tools. As participants in a scholarly field, not only do we researchers examine educational practice, but also many in our midst come from that practice or are active reformers of that practice. Thus, we assert that theory, research, and practice in mathematics education should exist in a more harmonic relation than has been the case to date.

Although scholarship in the field of mathematics education has become more conversant with theory (more theory-friendly) over the past 35 years or so, it has done so in a variety of ways; that is, the use of theory in scholarship has become more common, yet ways of using theory are quite diverse. We offer a framework that can account for the diverse ways in which scholarship has made room for theory and how scholarly pursuits might be *improved* by being more self-conscious about matters of theory.

Given that theory has become much more evident in scholarship in our field in the past decade or so, the moment seems propitious for a serious examination of the role that theory plays and could play in the

³ For another excellent treatment of the tension between theory and practice in mathematics education, see Malara and Zan (2002). Perceptive commentary on the potential connection between research and practice has also been offered by Lester and Wiliam (2002).

formulation of problems, in the design and methods employed, and in the interpretation of findings in education research. But, given increased pressures for relevance, we think that such an examination should not take an *insider* perspective only—focusing on research questions, research methods, and research findings. To be sure, theory has much to offer in regard to all three of these—helping to shape research questions, suggest research methods, and explain research findings—but we think a broader view is needed to span the perceived chasm between scholarship and practice in mathematics education. Through our consideration of the place of theory in mathematics education scholarship in this chapter, we hope to offer the field some new ways to think about the relationship between scholarship and educational practice.

Many in the policy and practice communities would argue that the growth in attention to theory in recent decades has driven the field away from central concerns of educational practice. As noted above, many view theory as a likely culprit in the perceived gap between the work done in educational research and the problems and needs of educational practice. In contrast, Sfard (2005) argued that researchers' attention to theory is not the culprit in the separation of research and practice; rather, it is researchers' careless use of theory that inhibits both the explanation and communication of findings and their implications. We hope to offer a view of theory use in mathematics education scholarship that will narrow the gulf between research and practice.

In this chapter we describe the emergence of theory (in its many forms) in mathematics education scholarship, and we offer examples of ways in which theory has contributed to, or might contribute to, our individual and collective endeavors. In our view, a careful examination of the diverse ways in which research in mathematics education has become and can continue to grow more theoretical should promote continued improvements in the formulation, examination, and justification of research problems. This examination could in turn transform debates over educational research, educational policy, and educational practice into collective deliberations on the ways in which it is or might be productive to give theory a more central role in our field.

THE EMERGENCE OF THEORY IN MATHEMATICS EDUCATION SCHOLARSHIP

In 1980 the National Council of Teachers of Mathematics (NCTM) published the first book that might legitimately

be dubbed a “handbook of research in mathematics education” in the United States. Aptly titled *Research in Mathematics Education* (Shumway, 1980), this was one volume in a series of professional reference books. A close examination of that volume offers a glimpse at the state of research, and research critique, in our field at that time. Of particular relevance to this chapter are commentaries in the 1980 volume that address the status and role of theory in research in mathematics education. For example, in his chapter on “the research process,” David Johnson (who had served a decade earlier as the first editor of the *Journal for Research in Mathematics Education*) decried the lack of theoretical grounding for much of the research in mathematics education at that time: “Perhaps the most significant shortcoming of much contemporary research [is] the lack of adherence to any type of theoretical or conceptual framework to direct the research efforts and to provide a basis for interpreting the results of the experimentation” (1980, p. 31).

There can be little argument that the situation has changed. Although many legitimate questions and concerns can and should be raised about the “state of the art” in using theory in our field, a perusal of articles in major research journals in our field reveals that theory is alive and well in recent work, as evidenced by the frequent appearance of the words *theory*, *theoretical framework*, and *theorizing*. Further support for this claim comes from an interesting analysis by Lerman and Tsatsaroni (2004) of articles published in two major research journals in our field (*Educational Studies in Mathematics* [ESM] and the *Journal for Research in Mathematics Education* [JRME]). They noted a sharp decline between 1990–1995 and 1996–2001 in the number of published articles that they judged to have no evident theoretical framing; the decline was from 10% to 5% for ESM and from 24% to 11% for JRME.

This shift in emphasis and visibility is likely due in part to the fact that the importance of theory in relation to education research in general has received considerable emphasis over the past quarter century. But other factors have likely also contributed to the growth in attention to theory in research in mathematics education.

Disciplinary Connections

An essential characteristic of the field of mathematics education is that its questions and concerns are deeply tied to matters related to the teaching and learning of mathematics. As such, the need for ties to other academic disciplines is obvious. For example, research in mathematics education is

concerned with mathematics, hence its ties to such disciplines as mathematics itself and to epistemology.

In interviews conducted by Silver and Kilpatrick (1994) with leading scholars in the field of mathematics education, many researchers commented on the need for mathematics education to maintain strong ties to well-established disciplines. Some expressed a desire to tie research in mathematics education even more closely to its “mother discipline” of mathematics by considering the historical-epistemological dimension of research on student learning or by examining the learning of advanced topics in school mathematics.⁴

For those among us who consider scholarship in mathematics education as a form of development of scholarship in mathematics, the meaning given to the word *theory* could have, at least in a professional deformation sense, some roots in the discipline of mathematics. OED definition 4c (above) refers to the use of the word *theory* in the discipline of mathematics; the Wikipedia⁵ expands on its usage in mathematics as follows:

In mathematics, the word **theory** is used informally to refer to certain distinct bodies of knowledge about mathematics. This knowledge consists of axioms, definitions, theorems and computational techniques, all related in some way by tradition or practice. Examples include group theory, set theory, Lebesgue integration theory, and field theory.

The term **theory** also has a precise technical usage in mathematics, particularly in mathematical logic and model theory. A theory in this sense is a set of statements in a formal language, which is closed upon application of certain procedures called rules of inference. A special case of this, an axiomatic theory, consists of axioms (or axiom schemata) and rules of inference. A theorem is a statement which can be derived from those axioms by application of these rules of inference. Theories used in applications are abstractions of observed phenomena and the resulting theorems provide solutions to real-world problems. Obvious examples include arithmetic (abstracting concepts of number), geometry (concepts of space), and probability (concepts of randomness and likelihood).

Although a direct mapping of theory from mathematics to mathematics education is not possible, definitions of *theory* associated with mathematics and other disciplines that mattered in the inception of our field may nevertheless serve as useful heuristics that help uncover what our own theories are or might be.

Other researchers in the Silver and Kilpatrick (1994) study pointed to the very fruitful connections that have been formed over many years with the field of psychology—connections that have helped illuminate many aspects of mathematics learning and performance. Yet others noted the importance of the social dimensions and organizational contexts of teaching and learning interactions and events, which underscores a need to develop ties to other disciplines such as anthropology, sociology, and linguistics.

Steiner (1985) and other scholars have pointed out that the field of mathematics education, and especially research in this field, occupies a space at the intersection of many other academic disciplines. Sierpinska et al. (1993) put it very well when they noted that our field lies at the intellectual “crossroads of many well-established domains such as mathematics, psychology, sociology, epistemology, cognitive science, semiotics, and economics” (p. 276). Although this positioning of our field sometimes leads to disputes over the “ownership” of certain problems or the “legitimacy” of certain claims and approaches, it also promotes giving attention to theoretical issues. Scholars in most social science disciplines (e.g., psychology, sociology) typically justify their research investigations on grounds of developing understanding by building or testing theories. More generally, some researchers have argued that educational research should emulate scientific research “the long-term goal of which . . . [is] to generate theories that can offer stable explanations of phenomena that generalize beyond the particular” (Shavelson & Towne, 2002, p. 3).

Diverse Perspectives

When David Johnson wrote his chapter nearly 3 decades ago, psychology was virtually alone as a source of theory for mathematics education researchers. Reflecting the prevailing view of his time, Johnson (1980) referred exclusively to “psychological theory,” which he described thus,

First, a theory serves as a means to explain observed phenomena. Second, it enables the researcher to make predictions about as yet unobserved relationships. Third, and possibly most important, a theory structures the conduct of inquiry by guiding the researcher in the process of asking questions, formulating hypotheses, and determining what key variables and relationships to investigate. (p. 31)

⁴ For more on the historical development of mathematics education as a scholarly field, and its early ties to mathematics and psychology, see Kilpatrick (1992).

⁵ <http://en.wikipedia.org/wiki/Theory#Mathematics>, retrieved May 16, 2006.

Johnson urged researchers in mathematics education to “*first investigate the adaptability of various psychological theories . . . to the learning and teaching of mathematics, and [only] in the event such adaptation is not feasible, move to the creation of a new theory*” (p. 32). Although psychological theories still have considerable influence in contemporary research in mathematics education, one can also see many other disciplinary roots for the theoretical perspectives taken by researchers in the field.

In their study of articles published in *ESM* and *JRME*, Lerman and Tsatsaroni (2004) noted growth during the period 1990–2001 in attention to sociological and sociocultural theories, as well as increase in the use of linguistics and semiotics. Clearly, mathematics education researchers have had opportunities in recent years to draw on a broad array of theoretical positions to support their research investigations beyond the ones available when David Johnson wrote his chapter.

The diversification of theoretical perspectives in the field is closely tied to the emergence of modes of inquiry that are more qualitative and less quantitative in nature, and to the emergence of paradigms for experimental research alternative to the experimental paradigms in psychology—such as didactical engineering (e.g., Artigue & Perrin-Glorian, 1991) and design experiments (e.g., Cobb, Confrey, diSessa, Lehrer, & Schauble, 2003). The quantitative research methods, adapted from other fields such as agriculture, that dominated education research in the 1950s and 1960s relied on experimental and quasi-experimental research designs and statistical theories as warrants for generalizable claims concerning teaching and learning. Unfortunately, though such research occasionally *demonstrated* some important phenomenon or relationship, it rarely *explained* the basis for the relationship or why a relationship existed. As researchers in mathematics education became dissatisfied with the results of such research and frustrated by the myriad questions that such research tended to leave unanswered, they began to turn away from quantitative approaches and toward more qualitative approaches. This turn toward the qualitative demanded increased attention to theory. It was through the careful use of theory that case studies and detailed analyses of small-scale interventions could generate generalizable claims.

Academic Politics

Pragmatics and politics have also undoubtedly played at least some small role in increasing attentiveness to theory. A *theoretical perspective* (as

opposed to a *practical perspective*) currently dominates the process of scholarly publication in mathematics education (Hanna, 1998). Thus, it is common for research journals that publish empirical studies examining some aspect of mathematics teaching and learning to require that the papers accepted for publication provide an explicit theoretical framework. In fact, manuscripts are often rejected for journal publication if they are deemed atheoretical or if their theoretical foundations are insufficiently explicated.

It can be argued that university and academic politics may also contribute to the growth of attention to theory in our field. Most researchers in mathematics education work in university settings, holding faculty appointments in mathematics departments or in schools of education. Given that research in mathematics education often deals with the teaching or learning of precollege mathematics or content found in introductory courses in college-level mathematics, rather than with “cutting edge” mathematical ideas, university-based researchers find it useful to establish the academic quality and intellectual rigor of their work by connecting it to disciplines that have higher status within the academic community. As Lagemann (2000) and other commentators have written, this is a feature of educational research more broadly, and it has deep roots in the historical development of educational scholarship and attempts to increase its status within the academic community.

For these reasons alone, it is not surprising that mathematics educators have in recent decades formed diverse and strong disciplinary allegiances and connections to theories and methods from other disciplines and used them to undergird their own research. Some of the disciplinary connections, such as those to psychology and to mathematics itself, are longstanding; others, such as those to anthropology and sociology, are of a more recent vintage. Moreover, because research in mathematics education is a relatively young enterprise, its practitioners have wisely looked to the theories and methods from longer established disciplines as a basis for framing their work. And the proliferation of disciplinary connections has led in recent years to much greater diversity in the paradigms undergirding research reported in the leading research journals in the field. Yet another factor influencing the growth of attention to theory in mathematics education is the increasing internationalization of the field.

Internationalization

Research in mathematics education can be viewed as an international endeavor even more than a

national enterprise. In fact, as has been noted by several commentators (e.g., Bishop, 1992; Fischbein, 1990; Kilpatrick, 1993), a vital international community of mathematics education researchers has developed and continues to flourish. A discussion document prepared in advance of a meeting entitled “What Is Research in Mathematics Education and What Are Its Results,” which was sponsored by the International Commission on Mathematics Instruction (ICMI), pointed to the existence of diverse voices and perspectives among members of our community (Sierpiska et al., 1993). The growth of an international community of mathematics education researchers and scholars has fostered consideration of a broad array of theoretical perspectives, many of which grew naturally within the confines of the cultural and historical contexts of particular countries and then captured the attention of others in the international community as both the theories and the community itself matured.

Contemporary conversations within the international mathematics education research community often focus on matters of theory and epistemology. The contents of proceedings for conferences of the International Group for the Psychology of Mathematics Education (PME) reveal this strong tendency. Moreover, in a book intended to summarize and highlight important features of research conducted by members of the PME (Nesher & Kilpatrick, 1990), four of the seven chapters deal almost exclusively with issues of learning topics and processes in particular mathematics content domains, and the other three deal primarily with theoretical issues or a blend of epistemological and content issues. In fact, considerable progress has been made internationally in studying the learning and teaching of mathematical content and processes from a variety of interesting and important theoretical perspectives.

(RE)CONSIDERING THE PLACE OF THEORY IN MATHEMATICS EDUCATION SCHOLARSHIP

One way to begin a consideration of the role of theory in mathematics education scholarship would be to ask what the field would be like if we had no theories, or if our theories were left implicit and kept from view. Some might argue that we can easily conduct this thought experiment by simply examining the historical record of scholarship in our field.

The earlier quote from David Johnson is only one of many that could be offered as evidence of the frequent calls for greater attention to theory in mathematics education scholarship over the years.

It is not difficult to find commentary on the state of mathematics education scholarship that characterizes the field as atheoretical. One way of visualizing this state of affairs is to refer to the scholarship triangle shown in Figure 2.1.

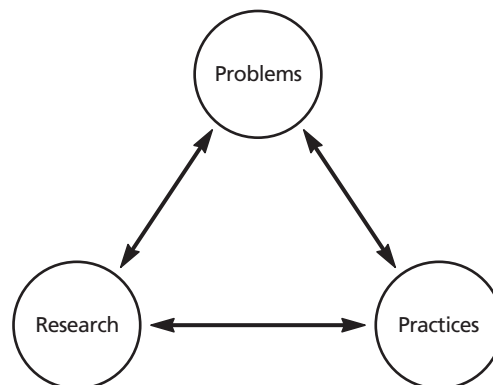


Figure 2.1 The scholarship triangle.

This triangle describes a view of scholarship that is constituted by relations between mathematics education *practices* and *research*, each of which is connected to a set of *problems*. Note that the connection between problems and practices is bidirectional, with stable practices attesting to problems and with perceptions of problems suggesting new practices. For example, certain established practices (e.g., activities in which children or adult engage with mathematics, algebra instruction in eighth-grade classrooms, children’s play with wooden blocks) can pose certain kinds of problems (e.g., puzzlements about states of affairs that are different than what might be desired or surmised, undesirable differences in achievement across groups of students, nonoptimal use of resources, low yield in student achievement or motivation). The reverse can also hold. Practices may exist in response to perceived problems, such as when practitioners engage in innovative practices oriented to solving problems of certain kinds (e.g., the problem of ensuring that teachers know the mathematics they teach has inspired the practice of teaching mathematics to teachers in professional development programs, the problem of making access to higher education equitable motivates special programs for low-SES students).

Scholarly discourse about the connections between problems and practices has long been a part of mathematics education scholarship, though it has not always made explicit use of theory. In fact, an atheoretical approach is often quite appealing, offering a straightforward, commonsensical characterization of practices and problems deemed sufficient to guide

research, which is the third vertex of the scholarship triangle shown in Figure 2.1.

In our view, research in which the role of theory is unspecified and apparently absent is not always lacking in merit. It often addresses problems deemed important by the public and points to critical issues. Moreover, such research can be influential and important. One need only consider how many citations of results derived from the National Assessment of Educational Progress [NAEP] (e.g., Silver & Kenney, 2000) or the trends in the Third International Mathematics and Science Study [TIMSS] (e.g., Mullis, Martin, González, & Chrostowski, 2004) are included in research articles to see the influence that research that is not theory-based can have on the field.

Such research is not restricted to large-scale surveys of national or international scope. Consider, for example, George Levine's (1972) study of the attitudes of elementary students and their parents toward mathematics and other subjects. He asked students and their parents to rate four school subjects (one of which was mathematics) in regard to their importance relative to 9 statements such as "when I was a student I did best in this subject" (p. 52, parent questionnaire) and "I do my best work in this subject" (p. 53, pupil questionnaire). He looked into the relative placement of mathematics in the ratings and found that students and their parents had a high regard for mathematics.

In terms of the scholarship triangle, Levine's (1972) research can be described as addressing the practice of studying elementary mathematics and the problem of increasing students' positive attitudes. Levine had a hunch that in addition to increasing teachers' positive attitudes toward mathematics, another driver to increase students' attitudes toward mathematics could be parental attitudes. Other than this hunch, supported to some extent by prior research, no theory was explicitly involved that justified why attitudes were important, why parents' attitudes might be important to look at, or what was the warrant for looking at mathematics in comparison with other school subjects as opposed to mathematics in comparison with other activities students might like to do.

Another interesting example to consider is the recently published study by Judson & Nishimori (2005) on the differences between American and Japanese high school students of calculus, with respect to their conceptual understanding of the derivative and their

ability to use algebra in solving derivative problems. The investigators found no significant differences between the two groups of students in conceptual understanding, but they noted higher fluency on the part of the Japanese students. In terms of our triangle, we could describe Judson and Nishimori's (2005) study as addressing the practice of thinking and problem solving in calculus and the problem of describing and sizing up the gap between two school systems of economically competing nations. Prior cross-national studies like TIMSS and their lack of data for high school served as warrant to do this study, in which the authors relied on rather intuitive (though mathematically sophisticated) ideas of "conceptual understanding" and "algebraic skills" to code students' responses to problems. Additionally, no theory other than mathematics itself justified the items included in the testing, so students were compared in their understanding of the fundamental theorem of calculus or of Riemann sums without much information about what the authors considered these topics and items to be cases of, nor consideration of why particular versions of test questions were used and not others.

These two examples of interesting research are offered to underscore the observation that scholarship in our field includes some research⁶ that can be legitimately justified on the basis of connections between practice and problems. Research in our field has long been done in this way—responding to problems derived from practice. That is, research has been a tool to produce specific and trustable information about problematic aspects of practice. Research has also been done to systematically describe practices originated in response to problems. In this way, research has been a tool to monitor or evaluate the impact of new practices. Research that looks into problems derived from practice has taken a variety of forms, such as comparing alternative manifestations of a given practice (e.g., when curriculum experts have done content analyses of materials being used in instruction; or when evaluators have inspected the differential achievement by comparably instructed students of different race, class, or gender). Research that looks into practices oriented to solving problems may look among other things like evaluating the practical outcomes of a problem-solving attempt (e.g., when evaluation specialists compare instructional treatments on the basis of students' achievement, when researchers have inspected the added value of novel practices such as professional

⁶ In order to deal precisely with the role of theory in scholarship, we use the word *research* here in a rather restricted sense—as a particular research study done within certain timeframe, in response to a problem or question, and addressing a particular patch of reality. We recognize and endorse a broader use of the word *research* that is commonly used to refer to practices in which researchers also conceive problems, identify practices, and formulate theories; those activities may also be viewed as research. Our use of the word *scholarship* here is our way of capturing that larger sense of practices that might be dubbed research.

development or new materials such as manipulative materials or technology).

The arrows pointing from practices to research and from problems to research in Figure 2.1 are bidirectional because problems and practices have not just directed research but also at times drawn on research for useful or applicable results or for records of how problems have been addressed. Thus, practices may have looked into prior research in order to make policies of certain scope (e.g., as when differential achievement gains of two instructional treatments have been used to recommend policy decisions as to how to shape practice). In identifying and handling problems, the field has also looked to research to help inform decisions about the appropriate grain size for further inquiry into a problem (e.g., when reviews of the research literature have illuminated the importance to control for SES in studying differential achievement). Mathematics educators have for a long time been interested in questions like those, yet they have often proceeded with little overt attention to or investment in theory.

One could argue that neither example given above was truly atheoretical scholarship. Tacitly, each investigation relied on an organization of ideas that helped the authors make distinctions and decisions along the way. This is what Mason and Waywood (1996) referred to as *background theory*, “because every act of teaching and of research can be seen as based on a theory of or about mathematics education” (p. 1056). Though we recognize the sense in which this argument may be considered valid, we prefer to focus our attention on what Mason and Waywood dubbed *foreground theory*, in which theoretical considerations are made explicit.

Toward that end we propose the insertion of theory at the center of the scholarship triangle (formed by practice, problems, and research) as a way to capture our conception of how theory has provided or may provide added value to mathematics education scholarship (see Figure 2.2).

In Figure 2.2 theory appears as the center of gravity of the triangle, and it mediates relationships among the three vertices of the scholarship triangle shown earlier in Figure 2.1. We think that this framework can be used to describe an emerging state of affairs in mathematics education scholarship, in which theory is becoming more and more of an explicit feature in the work of mathematics education scholars, enabling and mediating connections between pairs of vertices in the triangle, often suggesting connections that may have not been visible before. In the remainder of this chapter, we will use this frame to describe and

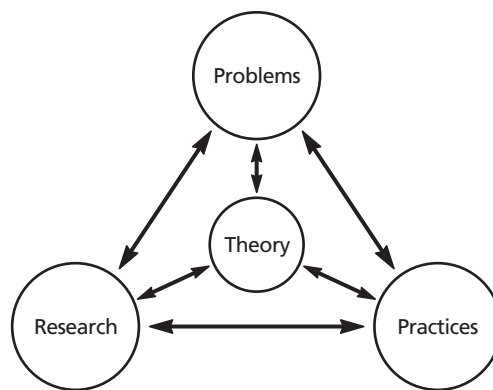


Figure 2.2 A theory-centered scholarship triangle.

exemplify the roles that theory plays in the work done in our field.

MAKING THEORY USEFUL: THEORY AS A VITAL MEDIATOR IN OUR SCHOLARSHIP

Whereas the role of theory in mathematics education scholarship was largely tacit in work published 30 years ago in our field, it has become more prominent in current scholarship. In this section we discuss how scholarship in mathematics education has related emerging uses of theory to each of the vertices of the triangle shown in Figure 2.2.

In positing that theory mediates the connection between each pair of vertices of the triangle we are proposing that one might ask: What could theory be doing there? In each of the next three subsections, we consider how theory might serve as a mediator of the dyadic interaction between vertices of the triangle in Figure 2.2 as a way to sketch possible answers to that question.

Theory As Mediator of Connections Between Research and Problems

What are some ways in which the connection between problems and research has been mediated by theory? As we note above the activity of scholarly problem posing has looked into prior research in order to inform the grain size in which further inquiry into a problem might proceed. Alternatively, the activity of addressing problems might have looked into public commentary or into theory. Here we attend to the roles that theory has played helping research get a handle on problems (see Figure 2.3).

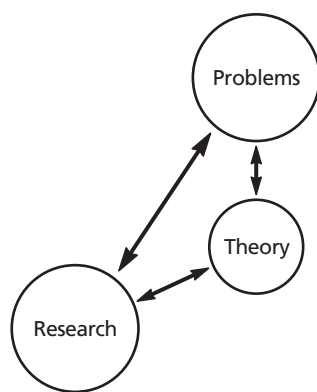


Figure 2.3 Theory mediating connections between research and problems.

Recalling the earlier quotes from David Johnson and the discussion of the close historical ties between psychology and mathematics education, we begin by noting that one way that theory has mediated the way problems appeal to research and research informs problems has often occurred through the use of intellectual formulations imported from other disciplines, such as psychology or sociology, *to interpret (or give meaning to) the results of research* in studies where these results had been produced from analysis, data, and problem formulation, which are independent of that theory. Consider for example John Clement's (1982) landmark study of college students' responses to algebra word problems in which they were asked to write equations to express relationships between two quantities, such as the following: "There are six times as many students as there are professors." Clement found that, when students were asked to write an equation that expressed the relation between S (the number of students) and P (the number of professors), many students who were otherwise competent in algebraic symbol manipulation mistakenly wrote $6S = P$. He further probed their thinking and found that the same individual who had modeled the statement using $6S = P$ might relate to the same situation in the correct way ($S = 6P$). In trying to explain what his results might mean Clement made an appeal to theory:

We call this evidence for "shifting between approaches" and hypothesize that it reflects an unobservable internal process of shifting between cognitive schemes used to deal with the problem. This provides evidence for the notion that human cognition is not always based on consistent processes—schemes that lead to contradictory results can apparently exist fairly autonomously and independently in the same individual. . . . There is an important implication of the theory that contradictory schemes for viewing problems can exist in people. The theory implies that students who are successfully taught a standard

method for a mathematical skill may still possess intuitive, nonstandard methods that compete for control. (Clement, 1982, p. 28)

Thus this theory that individuals can be "of two minds" helped Clement interpret the empirical data displaying contradictory ways in which a student modeled the "students and professors" problem.

A theory, however, need not come in handy only to interpret the results of research in regard to a problem. It could also be imported from somewhere else as *a lens to look at data gathered or coded* and to proceed with the analysis. This role of theory may still presume no relationship between the nature of the lens and the nature of the problem. In work on teacher beliefs, Tom Cooney and his colleagues (see Cooney, 1985, 1994; Cooney, Shealy, & Arvold, 1998) used theories of individual intellectual development by Perry (1970) and by Belenky, Clinchy, Goldberger, and Tarule (1986) as lenses to look at interviews, observations, and artifacts pertaining to a problem that was somewhat distant from the one examined by Perry and Belenky: the individual development of beginning teachers. The notion that this development in this population was a case of the more general intellectual development in young adults described in the theories of Perry and Belenky et al. is symptomatic of this use of theory in our field: It presumes that the problems investigated are particular cases of more general problems addressed by theories generated in other social or behavioral sciences and that mathematics teaching and learning is a context of application of those theories. In one way or another, all our scholarship can display this relationship to theory, nesting our scholarship to larger fields of human inquiry, but not all our scholarship relates to theory as an imported tool to look at data collected from a homegrown problem.

Theory could also appear yet a bit earlier in the research work, as *a tool that helps give closure to the corpus of data, pointing to relevant sources of data* that complement the sources already identified by a problem, or to relevant categories that might not be represented in the data. Again, it can do so without necessarily requiring theory to play a role in the definition of the problem. Some researchers have thus used theories that describe a population of interest in order to make sure their sample to study a problem is comprehensive. At times these theories are unlikely to be identified as theories in the strict sense; they are more or less shared ways of talking about a population or a kind of data. An example of this use of theory is provided in the study by Cooney and Henderson (1972, p. 21) on the "ways mathematics teachers help

students organize knowledge.” The researchers set out to develop a taxonomy of the instructional moves that good teachers used in “assisting students in organizing knowledge” (p. 22). They analyzed the transcripts of 44 observed lessons in grade 7–12 classrooms, identified key instructional moves that appeared to support students’ organization of knowledge, and proposed a scheme consisting of nine elements. To explain how they developed this taxonomy, the authors noted

The taxonomy was developed through an interaction of logical considerations and analyses of the transcripts. It was felt that basically an item of knowledge is organized by incorporating it in some sort of structure. This is done by establishing one or more relations between the item and other knowledge within the structure. Hence the concept of a relation as a set of ordered pairs could be used to explicate each of the relations identified. Theoretical relations were constructed on the basis of logical considerations and the previous writings of Henderson, Smith, Ryle and others. Analysis of the transcripts suggested that modifications of the theoretical relations were necessary in order to adequately describe the classroom discourse. The present taxonomy then, was conceived in part from theoretical considerations and in part from empirical evidence gained through analyses of the transcripts. (pp. 22–23)

Although Cooney and Henderson derived their taxonomy through empirical analysis, it is interesting and important to note that the table in which they presented their data contained several cells with 0 observations at a particular grade level, and one relation (abstracting) was not observed at any grade level. Had they developed their taxonomy merely as a description of their data corpus, they would have come up with a shorter list, certainly one that did not contain abstracting. But they also used theory to guide their development of the taxonomy, particularly the theory of the logic of teaching developed by B. O. Smith and Henderson. As a result of working with both the theory and the data, and the resident tensions between the relations predicted by the theory and those observed in the data, Cooney and Henderson were able to propose and argue forcefully for a taxonomy of nine relations that could be used to characterize how teachers assist students to organize their knowledge.

A theory could also appear even earlier in mediating the connection between research and problems. A theory could provide ways of examining and transforming a problem that had initially been formulated through common sense, turning the problem into a researchable problem. The research done by Hill and Ball (2004) characterizing and

measuring teachers’ knowledge of mathematics for teaching provides an example. The research is grounded in the problem of gauging teacher knowledge of mathematics that makes a difference in their teaching, particularly as it regards the evaluation of professional development opportunities for teachers. The transit from that problem to the design of instruments and collection of data for the research is mediated by the development of a theory of the object of inquiry—teachers’ knowledge of mathematics. The authors recounted the argument that the surprisingly little evidence of a relationship between teacher knowledge and student achievement is symptomatic of the field not having attended to the particular form of teacher knowledge that might make a difference. They built on Shulman’s (1986) notion of *pedagogical content knowledge* to argue that, “at least in mathematics, how teachers hold knowledge may matter more than how much knowledge they hold” (p. 332). And they referred to Ball’s notion of *knowledge of mathematics for teaching* (a particular kind of specialized knowledge of content; see Ball, Lubienski, & Mewborn, 2001) as the form of knowledge of mathematics that a teacher needs to have to be able to do the activities of teaching (e.g., explaining what an idea means, appraising students’ methods for solving problems, determining whether invented student methods can be used in other problems). This theoretical framing of what can be meant by *teacher knowledge of mathematics* directed their pursuit of the research. Thus, among the assessment items that the researchers developed to measure teacher knowledge one finds, for example, an item that reads, “Imagine that you are working with your class on multiplying large numbers. Among your students’ papers, you notice that some have displayed their work in the following ways” (p. 350). The item shows three different ways in which students worked out 35×25 and asks “which of these students would you judge to be using a method that could be used to multiply any two numbers?” Other items also demonstrate how the theoretical conceptualization of knowledge of mathematics for teaching directs the collection of data informing teacher knowledge by way of designing items that involve teachers in doing the mathematical work that they need to do while teaching.

A theory could also function like *a machine that produces problems as questions framed using the constructs and relations found within the theory*—problems that might or might not be readily seen as applicable or relevant to some concern in educational practice. Nevertheless, research that is spawned from theoretically generated questions can eventually help address a practical problem. An example of this use

of theory can be found in Steffe's (2004) study of his own interactions with two children in the context of a constructivist teaching experiment on fractions.

Steffe's investigation was prompted by the theoretically important issue of reconciling the notion of a teacher's *hypothetical learning trajectory* (Simon, 1995) with the model of individual knowing proposed by radical constructivism (von Glasersfeld, 1995). Through a comparison between what the teacher-researcher had hypothesized to be the difference between the fractional schemes of two children and a retrospective investigation of how both children arrived at the notion of commensurate fractions, Steffe (2004) argued that a teacher's hypothetical learning trajectory regarding children's learning of fractions cannot be accounted for solely in terms of their progress in acquiring various fractional models, but rather that "learning trajectories of children must be constructed by teacher-researchers who participate first-hand in children's constructive activity" (p. 155). Steffe's inquiry into how the two children constructed the notion of commensurate fractions in interaction with the teacher-researcher emphasizes that the hypothetical learning trajectories are schemes that are themselves being adapted in the constructive activity of the teacher-researcher. Thus a theoretically generated question led to a study of teaching (outside of a classroom) in which a researcher demonstrated that hypothetical learning trajectories are less of a prediction based on existing schemes and forthcoming tasks and more of an emergent product of the interaction among individuals operating in the same milieu with different schemes.

At the risk of oversimplifying Steffe's investigation and findings to offer an example of this role of theory, we note that what appeared to be a completely theoretically generated research question seems to speak to a quite pressing pragmatic problem, that of whether any conclusive judgment as to the developmentally appropriate nature of a task can be made *a priori* for a child based solely on the child's current schemes. Thus, a question about a feature of a theory (a prediction, a necessary notion) may motivate the research, but this research may eventually translate into information on one or more publicly owned problems.

A final example of a way in which problem formulation has used theoretical considerations to draw on research that provides information is to consider the logical and conceptual structure of reviews or syntheses of past research on a problem. The background of a problem is often established on the basis of *an organized totality of the previous studies*

on related problems. Not the studies themselves but the conceptual scheme that assembles the studies into a totality is a scholarly artifact that we suggest is a kind of theory in our field. To the point that the way a summary of the available research is organized shapes the extent to which some things about a problem are "known" and others are unknown and to the point that considerations such as what has or has not been sufficiently studied often drive decisions as to what else should be studied, we can assert that the organizational scheme of a review of research has functioned in mathematics education scholarship like theories function in more mature disciplines.

Some literature-organizing efforts have explicit theoretical aims, such as Anna Sfard's deft synthesis of major trends and perspectives in research on mathematics learning (Sfard, 1998) that contrasted approaches that reflected the acquisition metaphor and those that were captured by the participation metaphor. Her explicit aim was to raise theoretical issues and concerns through her analysis of prior work. Sfard's work offers an important example of a role for theory in the formulation of a problem through the organization and synthesis of prior work.

But such undertakings are not always explicitly oriented by or toward theory. Consider, for example, Begle's (1979) account of the research on mathematics teachers. He organized the empirical literature to seek insights into the relationship between teacher effectiveness and two large categories of teacher characteristics—teacher attitudes and teacher knowledge. Although Begle identified constructs, categories, and relationships, he was not explicit about the theory that connected them, perhaps because he viewed the relationships as ones to be accepted at face value: "There is no doubt that teachers play an important part in the learning of mathematics by their students. However, the specific ways in which teachers' understanding, attitudes, and characteristics affect their students are not widely understood" (p. 27).

Although Begle did not explicitly portray his literature review as an exercise in theory, and some might think it pretentious to apply the label *theory* to a summary of research, theory appeared to lurk implicitly in his review, in the form of assumptions of what research to look for and how to weigh it. For example, Begle argued the importance of the question of what role teachers played in students' learning, by considering teachers to be resources or assets for students' learning and by assuming that their differential effectiveness in promoting student achievement could be explained in relation to differences in their attitudes toward mathematics or in their knowledge of mathematics.

In the sense that a thorough, problem-oriented review of research can serve as a bridge between problems and research, insofar as a review of research posits an organization of the extant research knowledge and makes visible certain problems—and the associated categories, constructs, and relationships—a review can be argued to be a theoretical artifact, to function as a theory. Any such organization groups some things together and separates others, highlights some problems and leaves others—and the associated categories, constructs, and relationships—less evident. In the case of Begle’s review, for example, the actual activity of teaching mathematics, and the differential ways in which the learning resource represented by teachers in relation to students and the mathematics to be learned might function, was at that time a minor area of research; Begle’s organization of the literature did nothing to make it visible (see Ball et al., 2001; Cohen, Raudenbush, & Ball, 2003). Thus, although we have called attention to the (often implicit) theoretical nature of the organization of a literature review, unless the theoretical aspects are made explicit as in the case of Sfard’s article, it may be a weak use of theory if the goal is to advance the research enterprise.

In summary, in this subsection we have described the bidirectional relationship between problems and research as one in which problems call for research to be done, and research produces information on problems. We have argued that one of the roles of theory in our scholarship is that of articulating or mediating that bidirectional relationship between research and problems. And we have identified and exemplified at least six distinct ways in which theory (or “theory”) plays the role of mediator between problems and research:

- Theory as intellectual instrument imported from other disciplines to interpret results of research on a problem
- Theory as intellectual instrument imported from other disciplines as a lens to analyze data and produce results of research on a problem
- Theory as a tool that helps give closure to the corpus of data to study a problem, complementing the sources identified by the problem
- Theory as the means to transform a commonsensical problem into a researchable problem

- Theory as a generator of researchable problems that may later be translated in commonsensical terms
- (Often implicit/background) theory as systematic organization of a corpus of research on a problem

As a by-product of these different manifestations of theory we can identify a more nuanced notion of *problem*. On the one hand there are problems in the sense of undesirable or inconvenient happenstances to be fixed or managed (and research is expected to do that). On the other hand there are problems in the sense of puzzling, compelling phenomena that provoke interest and curiosity to find out more about them, to understand them (and research is expected to do that). The two notions of *problem*, the one active and the other one contemplative, are not disjoint, but the extent to which their correspondence can be worked out seems to owe to the involvement of theory in mediating the connection between problems and research. Figure 2.4 shows how the notion of a scholarship triangle accommodates this dual notion of problem.⁷

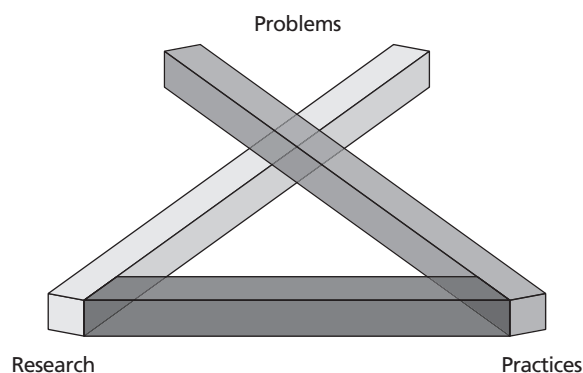


Figure 2.4 An open trilateral as a model of scholarship on two kinds of problems.

We propose those various ways in which theory mediates the connection between research and problems as a set of different entry points that have at various times been used by scholars in our field. The examples also show that these different roles of theory mediating the connection between problems and research are not neatly sorted in a historical development process. Research practices in our field display different kinds of engagements with theory as it regards posing and investigating researchable problems.

⁷ We acknowledge Nicolas Balacheff’s suggestion of this diagram and of the need to elaborate on the different nature of the problems.

Theory Mediating Connections Between Research and Practice

What are some ways in which the connection between research and practice has been mediated by theory? As we note above, scholarship in our field has addressed various practices: Individual mathematical thinking, teaching and learning in classrooms, and the development of mathematics teachers are three distinct practices of mainstream interest in our community (along with a number of others). Furthermore, new practices have developed in response to some research work: New curricula, pedagogical tools, or practices, even new institutions (such as summer institutes for mathematics teacher educators) have been developed that respond in some way to research. Another role of theory is that of mediating the connection between research and the practices with which our community is concerned (see Figure 2.5).

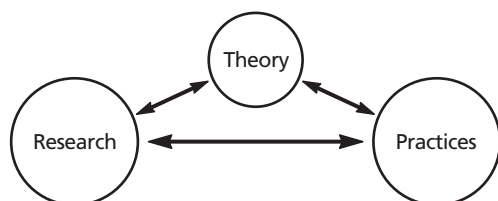


Figure 2.5 Theory mediating the connections between research and practice.

A widely held view of the role that theory plays in relating research to practice is judgmental: The more theoretical the research is, the less practical and applicable it is assumed or asserted to be. For some, perhaps many, teachers, teacher educators, and curriculum developers in mathematics education, research that is invested in theory seems to be irrelevant to practice. Although researchers might argue with this view by critiquing the critique (e.g., Sfard, 2005) or by arguing that many aspects of research have potential relevance to practice (e.g., Silver, 1990), we think it might be more productive in this chapter to look at what this view criticizes and to ask, What are the roles that theory plays and could play in connecting research to practice? What are the practices that our research connects to, for better or worse, through the use of theory? How does or could theory articulate research and practice in our scholarly community?

The word *theory* is often used to label sets of *prescriptive statements that could be used to guide the deliberate enactment of a practice*. Unlike physics, which concerns natural phenomena, teaching and learning are social phenomena. As Goffman (1974) pointed out, we humans tend to frame social events

as “guided doings”; that is, we assume that actors of the social world are in some significant sense willful agents of practices that they can significantly shape. Accordingly, the word *theory* as it relates to a social phenomenon such as educational practice is at times biased by such framing. In particular, the word *theory* is often used to indicate what, according to some vision, the agents *should* be doing—*theory* takes its everyday meaning, as an ideal or desired way of conducting social events. In that scheme of things, *practice* is used to refer to what social actors happen to or eventually are able to do, and *theory* is used to refer to what they should, or should be able to, do. We refer to these theories as prescriptive, noting that we make no distinction between forceful prescriptions and friendly suggestions, between progressive or reactionary ones—all of those “theories,” no matter their force or their intention, relate to practice in similar, prescriptive or normative ways. We submit that when *theory* is understood in that prescriptive way, the relationship between research and practice is mediated by theory in at least two possible ways.

On the one hand research uses (prescriptive) theory as a norm against which to evaluate practices. An easy example of that is how mathematics itself has operated as a theory in curriculum research on the content deployed in textbooks (see for example Herbst, 1997). In such cases, mathematical knowledge as represented in texts and practices of the discipline has provided the theory—what should be the definition of a concept, what theorems should be proved, and so on. Textbooks have been looked upon as empirical material that attests to the practice of curriculum development, and the research has been one of describing, perhaps assessing the gap between theory and practice. Note also that a second example of this kind of involvement of prescriptive theories in mediating the relationship between research and practice is the use that some researchers have given to policy documents such as the NCTM *Standards* (NCTM, 1989, 2000) or derived ones in the evaluation of instructional programs.

On the other hand, practice has used research to shape prescriptive theories of the research-based “best practices” kind. Examples of this include the use that consumers of research might give to cognitive research on learning in developing a notion of best practice in teaching or, more frequently, the ways in which research often translates into “implications for practice.” In both of these cases, the key theoretical operation is the assumption that relationships among constructs described or explained by research correspond to actual or desirable states of affairs in the real world that agents of a practice can willfully decide

to strive towards. One could dispute the warrant of conceiving human action as amenable to voluntaristic theorizing, yet it is undeniable that the social world in which the practices of mathematics education exist (the world of schools, for example) frames those practices as guided doings. Thus agents of practices are under institutional obligations of trying harder and being better in whatever socially accepted definition of those terms, and the more specific those terms are the more they are identified as theories to be aimed to and the more amenable practices are to be certified as good by way of research. Theory as a conception of what is desirable thus helps research assist practice.

Diametrically opposed to the notion that theory's role is to prescribe or guide practice is the notion that social practices, such as mathematics teaching and learning, are complex social systems whose dynamics depends only partially on what their actors are willing to do. Rather, the conditions in which educational practices exist allow them to happen in similar ways as natural phenomena happen: Even when key agents may be unknown and some levers may not be controllable, the phenomena a practice produces are amenable for description and explanation. In that perspective, theory, rather than being the means for research to legitimize practice, is *a tool for research to understand practice*.

In particular, *theories can be languages to encode and read, that is to describe, a practice* so that researchers can examine practice according to such reading. There are many examples of descriptive theories in mathematics education research, particularly as regards the description of the practices of thinking and problem solving. The theory of conceptual fields developed by Gerard Vergnaud, for example, has been used to describe different kinds of conceptions tied to a common conceptual field. Examples worked out are those of additive structures—which include conceptions such as putting together, comparing states, or comparing transformations—and multiplicative structures—which include conceptions such as repeated addition and ratio. The theory of conceptual fields provides the means to describe the manifestations of those conceptions in the observation of children's mathematical work by requiring the observer to identify the problems, operational invariants, and signifiers involved. Much of the empirical work done by scholars like Carpenter, Fuson, Behr, and their associates concurs with Vergnaud's observations (see chapters 13 and 14, this volume, for more on this).

But theories that help research understand practice may not only aim at describing, they might also aim at explaining certain practices. It may be useful to borrow a distinction between causal and rational explanations

from physical science. Likewise in mathematics education scholarship, among explanatory theories some may help research understand practice by disclosing the *causes* for practices, as when one looks for what has created a certain phenomenon. The recognition of a widespread prevalence of errors in arithmetic calculation has led researchers at various times to rely on theory to offer causal explanations for students' errors. For example, in a landmark analysis of children's subtraction errors, Brown and VanLehn (1980) proposed cognitive *repair theory* as a theoretical explanation for common errors based on children's incorrect generalization from correctly learned procedures and their attempts to perform local repairs on global procedures when problems were encountered. This gave rise to other explanatory theories that were proposed across a variety of errors (e.g., Resnick et al., 1989) to reconcile the apparent paradox that students could consciously do things that turned out to be incorrect in response to knowledge they acquired by way of doing things correctly.

Other explanatory theories may help research understand practice by providing rational arguments for why a certain phenomenon should not be surprising, or why it is plausible. An example of an explanatory theory of the rational kind can be found in the constructivist theory of learning (von Glasersfeld, 1995), which explains changes in a learner's action in terms of adaptations of schemes to fit what the learner perceives to be the situation in which she is operating. Another example may be found in Schoenfeld's (1988) article, "When Good Teaching Leads to Bad Results." In his analysis of one teacher's classroom, Schoenfeld theorized that an educational system that valued a particular kind of instruction—oriented toward having students master the skills needed to pass a test—could also encourage the relative isolation of intrinsically connected mathematical practices such as geometric construction and geometric proof. He provided a rational explanation for a practice of instruction that appeared to be dysfunctional for the development of students' mathematical dispositions. Researchers in France relied on similar observations to make the point that epistemological obstacles (i.e., items of knowledge, such as 'decimal numbers are like natural numbers but with a dot,' which may be true within a certain domain of practices but false beyond that domain) exist that could have a didactical (or instructional) origin (Brousseau, 1997). That is, these obstacles might exist because they help teacher and student get their classroom work done.

Finally, a particular case of theories that provide rational explanation is that of statistical theory, which helps research understand practice by demonstrating

correlations between variables at play in a phenomenon. This last variety of theories is common in educational research at large and is sometimes (but not always) connected to theories of one or the other kind (causal or rational). Ethington and Wolffe's (1984) study of gender differences in mathematics achievement and Hembree's (1990) study of mathematics anxiety are examples of how statistical theory has been used to create explanations.

In addition to describing or explaining practice, theories that help research explain practice may also aim at predicting aspects of practice. A theory may help predicting what instances of a phenomenon may look like. For example, the repair theory of arithmetic bugs (Brown & VanLehn, 1980, 1982) not only provided a rational explanation of subtraction errors by proposing that students would engage in local repairs of a partially forgotten procedure. The theory also had the capacity to predict which repairs would more likely be done for which kind of problems (provided that students had forgotten the correct procedure), thus predicting which errors would likely be observed.

Another way in which theories may enable research to predict practice is by enabling researchers to construct probes that would perturb the stability of practices and elicit some of the norms that undergird their variability. Herbst and Brach (2006) used a theory of instructional situations proposing that mathematical work in classrooms is framed within customary systems of norms that regulate division of labor between student and teacher to propose a range of possible tasks to students. The theory allowed researchers to predict that students would react to the presumption that they be responsible for doing those tasks in their class by commenting on (and often amending) the share of labor that their teacher should be responsible for.

The distinction between theories that prescribe practice on the one hand and theories to understand practice is not as sharp as it may seem. Inasmuch as every observation is an experiment of sorts, eventually the extent to which the observed exists in a natural state is very much a construction itself. Without a scholar looking at it there would be no *phenomenon* per se, there would be people here and there doing more or less similar things (see Margolinas, 1998). That is, even the most descriptive approaches of research to practice include a prescription of what that practice should be that allows it to be visible and isolated from

the rest of experience. Yet this is a different kind of prescription: Rather than prescribing what the practice should be like, it prescribes what the researcher will consider to be an instance of the practice, without necessarily expecting that agents of a practice should strive to behave as the theory says. A theory could therefore prescribe what is to be seen (how to observe something) whereas (possibly another) theory might also serve to understand what is seen.

To illustrate this use of theory we turn to Brousseau's (1997, especially pp. 137–144, 261, and chapter 5) notion of the *didactical contract*. To understand its origins, consider a scenario that might have been observed in elementary classrooms during the times of the so-called New Math. In concert with the goal of bringing school mathematics closer in content and aims to the mathematics of mathematicians, abstract mathematical ideas, particularly algebraic structures, had found their way into the prescribed curriculum for elementary school. Also, Piagetian research on the psychogenesis of knowledge had brought attention to the role of action in the formation of cognitive schemes. Into that mix were added prescriptive pedagogical theories, such as those proposed, for example, by Zoltan Dienes, that outlined how the teaching of abstract structures could be accomplished by having students actively play with different kinds of concrete materials (for example, different games), all of which had a common structure underlying the actions the child was meant to do.

From a mathematical perspective, the abstract structure to be taught could be thought of as the equivalence class of the games or structured materials that the student would play with—none of them per se being isomorphic to the structure itself (see Schoenfeld, 1987). From a psychological viewpoint, the claim could be made that the abstract structure was a viable way to describe the (second-order) model that an observer could make of the cognition of an individual student who was successfully playing any one of those games (*viability* in the sense that the model would not be falsified as long as the play was successful).⁸ Pedagogical theories of the time, however, turned those games, which could be appropriate places where someone who knew the structure could see a representations of it, or the play of those games, that could be appropriate contexts for the observation of knowledge-in-action and learning, into methods of teaching. To teach a given structure to students who did not know the structure, teachers were to have

⁸ Mathematics can certainly provide a language for an observer to describe what a student is doing—as long as the observer is clear not to equate the ideational system that the observer points to with that language with the ideational system present for the student. That is, stating that everything happens “as if” the student is thinking *T* is not the same as saying that the student is thinking *T*.

them play one game, then another one, and another one, until the students would see the structure. To characterize such instruction, Brousseau referred to the *Jourdain effect*: The student moves a set of cups of yogurt and the teacher says, “You have discovered the Klein group.”⁹

For Brousseau, the observation of the Jourdain effect was much more than an amusing interlude. He saw it as a symptom of a possible breach of expectations in instruction. To characterize those expectations, Brousseau brought in the notion of the didactical contract that he had also used to explain other puzzling happenstances in student-teacher interactions (see Brousseau & Warfield, 1999).

Brousseau’s accomplishment in naming the Jourdain effect a consequence of the didactical contract was not just to point at something gone wrong but also to say how it was, in a way, reasonable. After all, the student has played the game with the yogurt cups, not necessarily because it was fun or interesting but most surely because it was assigned by the teacher in school. Hence the student could expect the teacher to care to see what she has done and to give an official meaning to what she has done. Furthermore, because they are in a mathematics class, what they are doing has little chance of being officially meaningful just in its character of play; it must also be mathematical in some official way, no matter what it has meant for the individuals themselves.

The teacher knows that she has the responsibility to sanction what the action is worth, or at least she knows that the student thinks she has that responsibility. Furthermore the teacher knows that such responsibility is predicated on her supposedly different, greater knowledge of the subject, which, again, she knows she has or at least she knows that the student thinks she has. To be under the curricular prescription of having to teach abstract structures by concrete methods imposes extra constraints on the teacher that make the teacher’s actions perplexing and yet also plausible, reasonable, predictable.

As this example illustrates, the point of a theory that describes and explains is not to say that something is wrong (or right for that matter) but to describe what it is and explain why it happens, that is, to shift attention away from simpleminded blame or handyman troubleshooting. One could say that the teacher’s actions breach the contract at least in regard to how valid it is to label those ordinary actions after such a serious mathematical idea, but

one understands that the teacher has to provide a label for the play of the game and that she has to find indications of knowledge of the algebraic structure in students’ actions. It is not unavoidable for a teacher to do those things, but if she does them one can understand that to be a predictable outcome of the didactical game the student, the teacher, and the knowledge at stake are summoned to play. In that sense our example illustrates more than just how a theory can produce a meaningful research result out of the observation of an undesirable happenstance in practice. It also illustrates how a theory can transform a commonsensical problem into a researchable problem: Being attuned to what makes the original observation meaningful enables one to look for places where other related observations could be made, and even to create experimental lessons where those phenomena could be reproduced and studied.

What theory does in this case is give the grounds to restore reason to what happened or, as Gaston Bachelard would say it, make the real appear as something that one should have thought of.¹⁰ Theory does that without asking one to forget what happened: Theory allows one to see something as *odd* (or singular, surprising, special, funny, amusing, worthy of attention, pick your word) and as *reasonable* (plausible, expectable, predictable) at the same time. Ideologically one could continue to be disgusted by what happened and unmoved by the explanation. As we say above, if we were thinking from our positions as actors in the educational system (in addition to being researchers, both of us are parents and teacher educators) we might like classroom practices to be different, better for the sake of children, for the sake of mathematics, and for the sake of the society that receives them eventually. But theory bestows the mission to react rationally rather than ideologically; theory allows one to *address the system* that produces problems, not just attack the symptoms of those problems. Like the high fever that comes with an infection, the aberration called Jourdain effect might be less of a problem than the pedagogical theories or the curriculum on whose reaction the aberration appeared. But again, one can only ponder the merits of such assertion if one has a theory to stand on. We suggest that insofar as it affords grounds for civil discourse, insofar as it provides a way to see that people (not only children) do odd things for a reason, theory is useful. All of this is visible in the

⁹ Guy Brousseau named the Jourdain effect after a character in an episode from the *Bourgeois Gentilhomme* by Molière. In the play, after an evidently nonpoetic verbiage by the rich and coarse Monsieur Jourdain, his private tutor tells him that what he has produced is actually prose. In naming the Jourdain effect, Brousseau reminds us that a structure imposed by an observer of a scenario may not be shared by the actor(s) in the scenario.

¹⁰ “Le réel n’est jamais ‘ce qu’on pourrait croire’ mais il est toujours ce qu’on aurait dû penser” (Bachelard, 1938, p. 13).

example of the Jourdain effect, but there is more to the role of theory in mathematics education scholarship.

Another way in which theory mediates the relationship between research and practice is by providing means to find out what is generalizable or what is useful from research to practice—especially when these are not the things officially allowed by the traditional way of arguing generalizability in empirical research. Thus theories help practice read research by suggesting ways in which the results of research might orient policy-making (even if research cannot *validate* those policies). Theories may also help practice find by-products of research that practitioners may use at their own risk.

Consider, for example, the ways in which notions regarding the negotiation of task demands in classrooms have moved into the work of teachers and teacher educators. One strand of this body of scholarship begins with the work of Walter Doyle (1983, 1988) on academic tasks and flows into the work of Mary Kay Stein and her colleagues in the QUASAR project (Silver & Stein, 1996). Stein borrowed Doyle's conception of the centrality of academic tasks in creating learning opportunities for students, and she adapted and particularized it to the mathematics classroom at about the time of the introduction of the NCTM *Standards*. Although Stein did not refer to the notion of didactical contract, the theoretical and empirical contributions of this work can be seen as illuminating the origins of some contractual difficulties and entanglements.

Central to the work on academic tasks done by Stein and colleagues (e.g., Henningsen & Stein, 1997; Stein, Grover, & Henningsen, 1996; Stein, Smith, Henningsen, & Silver, 2000) has been the Mathematical Task Framework (MTF). The MTF (see Figure 2.6) underscores the important role that mathematical tasks play in influencing students' learning opportunities in the way they unfold during classroom instruction.

The MTF portrays a mathematical task as passing through phases, from the task as found in curricular materials, to the task as it is set up by the teacher in the classroom, to the task as enacted by students and their teacher interacting with one another and with the task

during the lesson. The MTF underscores the important role that mathematical tasks play in influencing students' learning opportunities. Moreover, it points to the centrality of the work that teachers do with and around mathematical tasks. Teachers' decisions and actions influence the nature and extent of student engagement with challenging tasks and ultimately affect students' opportunities to learn from their work on such tasks. The MTF and associated notions have contributed to theoretical understanding of the centrality of tasks in the academic learning that occurs in classrooms, the changes in cognitive demands that may emerge as students and teachers work on mathematics tasks in the classroom, and how changes (especially reductions) in cognitive demand can affect (usually constrain) students' opportunities to learn. The MTF was used as the basis for analyses of empirical data on the extent to which teachers succeed in maintaining the cognitive demand of challenging tasks in the classroom (Stein et al., 1996), factors that influence their success (Henningsen & Stein, 1997), and the relationship between teachers' success in maintaining the cognitive demand of tasks and the mathematics learning of their students (Stein & Lane, 1996).

The MTF has also been transformed into a tool with which to consider some of the challenges that inhere in teachers' use of complex tasks in the mathematics classroom. Recognizing that the MTF research suggests that teachers need to learn to orchestrate the work of the students while resisting the persistent urge to tell students precisely what to do, thereby removing the opportunity for thoughtful engagement, by responding to student queries and requests for information in ways that support student thinking rather than replace it, Stein and colleagues (Stein et al., 2000) constructed a set of narrative cases to illustrate observed patterns of classroom activity in which high-level task demands were maintained (or declined) during a lesson. These cases, and a second generation of narrative cases (e.g., Smith, Silver, & Stein, 2005) have been used widely by teachers in teacher professional development projects and in teacher preparation programs. This example raises to visibility the potentially important role that tools,

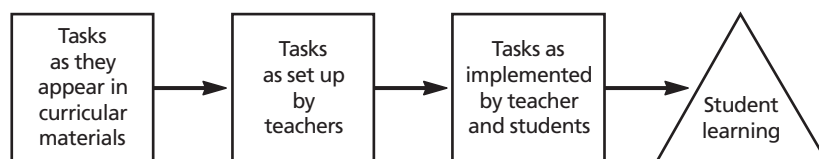


Figure 2.6 The Mathematical Tasks Framework.

such as frameworks, can play in the work of theory mediating connections between research and practice. One can think of other examples, such as the framework used in Cognitively Guided Instruction (CGI) to organize addition and subtraction problems not by operation but rather by underlying structure (Carpenter, Fennema, Franke, Levi, & Empson, 1999); this framework has had extensive use with and impact on teachers, teacher educators, and curriculum developers.

In this subsection we have argued that one of the roles of theory in scholarship is that of articulating or mediating the bidirectional relationship between research and practices. We have identified and exemplified at least four distinct ways in which theory (or “theory”) plays the role of mediator between research and practice and made some distinctions within those ways:

Prescription, and elaborated that in this sense theory can be

- a system of prescriptions of what an educational practice should be like and used by research to evaluate the merit of an actual practice
- a system of “best practices” derived from research that has inspected an actual practice (usually with a criterion such as student achievement)
- a definition of what that practice consists of that allows it to be visible for the purposes of analysis and inquiry

Understanding, and elaborated that in this sense theory can be

- a language of description of an educational practice
- a conceptual system that may explain the causes of an educational practice
- a conceptual system that may establish the grounds on which the existence of an educational practice is plausible or reasonable
- a system of correlated statistical variables concurrent in an educational practice

Prediction, and elaborated that in this sense theory can be

- a system of conjectures that foretell instances of a phenomenon
- a system of assumptions that enable researchers to construct probes that perturb the stability of practices so as to elicit some of the norms that undergird their variability

Generalization, and elaborated that in this sense theory can be

- an account of which results of research might orient policy-making and why
- an organization of by-products of research that practitioners might use

Through this examination a more nuanced notion of practice also emerged. We demonstrated that researchers in mathematics education are concerned with real educational practices. But they engage in other kinds of practices, though these are often embedded in activity and thus less obvious. One is the practice of doing research itself; another is the practice of deriving implications from research (see Schoenfeld, this volume, for more on these practices). The connection between research and those practices benefit from the use of theories just as much the connection between research and educational practices does.

Theory Mediating Connections Between Practice and Problems

We initiated our consideration of the scholarship triangle (Figures 2.1 and 2.2) by arguing that scholarship inquires on problems posed by educational practices. We argued that it makes sense to deal with at least two meanings of *problem*: problem as undesirable happenstance in a practice and problem as provoking puzzle of a practice. Likewise we argued that scholarship also focuses on practices that respond to problems. We noted that new practices are often developed precisely to curb undesirable happenstances (as when a new program is established to remedy underachievement) or to capitalize on the potential of untapped resources (as when a new program is developed to teach mathematics with technology). Further, scholarship relates to other practices in addition to the educational practices that we study: The practice of doing research and the practice of shaping policies were provided as key examples. We now elaborate on how the investment of theory might help mediate the relationships between problems and practice (see Figure 2.7).

How does one know what makes sense to investigate, what is puzzling about a practice? How does one know what is undesirable or problematic about a practice or what needs improvement? How does one know what is the practice that produces a given problem? How does one know what elements need to be put together to develop a new practice? Theory seems to play a role in helping answer those questions. Theory can allow one to identify the practices that pose problems and

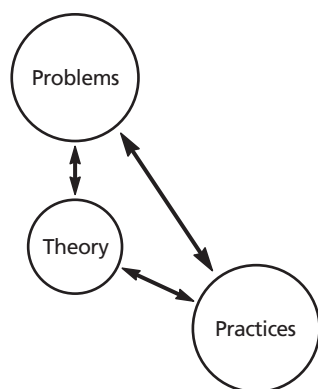


Figure 2.7 Theory mediating the connection between problems and practice.

to imagine possibilities in practice that can lead into researchable problems. In elaborating on this role of theory we attend to the more nuanced notions of problem and practice presented above.

One of the senses of *problem* is that of an inconvenience or an undesirable circumstance. All the practices that scholarship is concerned with, the educational practices studied, the novel practices studied experimentally, the practices carried out while involved in doing research, and those carried out when communicating research to policymakers and practitioners could all be problematic in that sense of the word *problem*. We suggest that theories may help each of those practices deal with those kinds of problems.

Consider first the case of the problems posed by educational practices, be those achievement gaps in K–12 education or myriad others. Even if problems were of the kind, “How can we fix this? How can we improve this?,” there could be a role for theory in helping *propose a solution to a problem*. Note for example that this was the role that van Hiele’s theory of levels of geometric thinking played in helping researchers come to grips with the problem of large numbers of students failing to learn the content of traditional high school courses in geometry focused on formal proof (Senk, 1983, 1989). With the assistance of the theory, scholars understood the problem as one of student readiness and sought to solve it by providing different, developmentally appropriate curricular options for students who had to take high school geometry but whose assessments did not show capacity to benefit from a course in formal geometry. The course on “informal geometry” was one of those curricular options, offered to solve a problem (see Cox, 1985; González & Herbst, 2006; Hamlin, 2006; Hoffer, 1981).

But theories also help one understand some calls for improvement as problematic. The role of theory

in such cases is not to tell how to improve doing something practical but rather to help understand what the terms involved in the call for improvement mean, to analyze and characterize what it might mean to do it better, and to support attempts to explain why one or another attempted intervention succeeds or fails. For example, the increasingly well-articulated theory of mathematical knowledge for teaching (Ball & Bass, 2000; Ball, Hill, & Bass, 2005; Ball et al., 2001; Hill & Ball, 2004) offers many reasons why a simplistic call to improve mathematics teaching by having teachers take more mathematics courses is unlikely to succeed unless those courses are very carefully designed to reflect the conception of mathematics knowledge for teaching that is embodied in the theory.

Theory is pivotal in identifying practices and delimiting their scope in order to address a problem. For example, the recognition that the mathematical thinking that is of interest to the betterment of mathematics education is manifest in practices that involve communication with others (peers, teacher, or an observer), activities (such as problems), and artifacts (inscriptions, tools) has led many researchers to theorize the object of study not any more as an autonomous cognitive system but rather as a larger communicational or activity system, a discourse (see Sfard & Lavie, 2005). Thus the theoretical notion that language is not just a means for somebody to express what she thinks but actually a tool for her to think with, enlarges the practice being considered (to include, for example, the interactions between subjects and interviewer, as shown in Sfard & Lavie, 2005) and adds possible problems (e.g., understanding the register or the modality that enables or inhibits communication between interviewer and subject).

Also, theory can play an important role when one wishes to define and propose as important a problem of practice that needs improvement. Consider, for example, the subtle shift in the argument for instructional change in the NCTM *Standards* documents over a decade or so. The initial rationale for proposed changes was based on allegations of low instructional expectations, in documents such as *A Nation at Risk* (National Commission on Excellence in Education, 1983). This led initially to calls for higher standards for student achievement. But the nature of the problem was further elaborated and informed over a decade or so of discussion and argument by theoretical considerations that went beyond the policy of establishing higher standards for student achievement. Consideration of what mathematics is and what school mathematics could be about, which were nurtured by theoretical accounts of the history of mathematics and of mathematical problem solving,

helped inform the posing of more complex, nuanced problems regarding notions like *opportunity to learn* and *instructional quality*. Theory and evidence played an important role in shaping the emerging argument for change, and the spotlight shifted from a focus on low expectations for students to a consideration of the ways in which schools and teachers support or inhibit student learning. As this example suggests, theory can be the ally of practitioners who wish to define problems in ways that honor their complexity and point to the inadequacy of simplistic solutions that might be suggested.

In our prior discussion of problems we noted that the word *problem* refers not only to inconveniences that need fixing but also to puzzles that appeal to the intellect. The relationship between those problems and practice is another place where there is a role for theory to play. Specifically, theory can help establish the way in which puzzling questions that matter to scholars might also matter to practice and to practitioners. For example, why is it that mathematics lessons sometimes go smoothly, exactly as planned, but often they do not? Why is it that students have little difficulty learning to add and subtract decimal fractions, but many struggle to learn to multiply and divide them?

We noted before that the practices that we are concerned with include not just the existing practices that call for scholarship to begin with, but also the new practices that result from scholarship. Some of those practices, such as the development of learning environments enhanced by various kinds of technologies, actually create opportunities to pose problems of both kinds noted above. And theory can play a role in mediating the posing of those problems as well. The emergence of dynamic geometry software has not only provided a useful tool to do constructions in classrooms but also motivated concerns such as how to maintain a role for proving in spite of the software's capacity to inspect a large number of cases. A theoretical reconsideration of the relationship between proof and perception such as that offered by Michael Otte (1998) has been instrumental not only in reaffirming the importance of proof but also in providing heuristics for the design of dynamic geometry-based tasks that create the need for proof. On the other hand, whereas all sorts of technologically enhanced learning environments may have been proposed and accepted as tools to improve practice, without necessarily posing problems that need fixing, scholars have also found ways of puzzling about them. Theory has played a role in that as well.

One such problem-as-puzzle that has been posed is that of whether and how technology changes the nature of the mathematics to be learned rather than

just the way in which the mathematics is learned. In proposing the notion that technology might operate a transposition of knowledge, a computational transposition, Balacheff (1994; see also Balacheff & Kaput, 1996) was banking on the existence of a theoretical notion of didactical transposition (Chevallard, 1985) that had argued for the role of the medium (text in particular) in shaping the message for the case of disciplinary knowledge turned into academically learnable content over time.

In fact one could turn the previous observation around and notice that the development of new practices, including the development of technology-enhanced educational environments of many kinds, has allocated a role for theory in relating to the problems that call for that development. A classic example is the development of the LOGO programming language and the many interactive environments based on turtle graphics. A theory of cognition that allocated an important role to the interplay between on the one hand the analytical, differential-geometry-kind of thinking called for by the programming interface and on the other hand the synthetic, Euclidean-geometry-kind of thinking called for by the visual interface was key in the design of the language. This theory was instrumental to deploy into practice the notion that young children should exploit their intuitive knowledge of drawing to learn the properties of geometric figures.

Finally, the practices we are concerned with include not only the practices studied or created, but the very practice of doing research. Like a craftsman, the mathematics education scholar is often involved in making tools (to make tools, to make tools) to study practices. In many cases that aspect of the practice of research has been simplified by importing tools and methods from other disciplines, such as the use of experimental and quasi-experimental designs from psychology and the analyses of variance and covariance from statistics. Yet even in those cases implicit theories about the participants and about the treatments have been beneath the practice of choosing designs, tools, and methods. The emergence of case-study designs and qualitative methods for data analysis have been only the most visible evidence of a much more important development in our field. This development could be characterized as the inclusion of a more deliberate theoretical stance regarding our practice of research, by using theory to justify choices in study design, data-collection instruments, and data-analysis procedures. In particular, mathematics education researchers have developed research designs that use the word *experiment* but that do not defer to the psychological experiment for characterization. One of those is the *experimental*

epistemology (also called *didactical engineering*) that Guy Brousseau and his colleagues in France used to carry out research within the theory of didactical situations, as illustrated in Brousseau's (1997, ch. 3) studies of the emergence of rational numbers and decimal numbers in fourth-grade mathematics. The qualitative-quantitative distinction is ill prepared to describe this kind of research. Brousseau's objective was to find a set of situations that could create the conditions for the emergence of specific ideas about fraction (such as commensurability). The object of study was thus not the individual cognition of a child but the system of relationships between a cognitive being and the elements of its environment relevant to the mathematical ideas at stake. In terms of design for data collection, however, the research implemented the situation under study in a real classroom.

Underlying this use of an implementation paradigm was a theory of experimentation very different from that of experimental psychology: Unlike in the experimental psychology paradigm where the implemented situation would be seen as treatment and the yield on the individual students would be seen as outcome, Brousseau's experimental research took the real classroom as the treatment or experimental probe on the situation under study and saw the yield in the robustness of the situation (in terms of its capacity to actually produce the ideas that it sought to produce). For Brousseau what was potentially variable was the *treatment*, and the possible cause of the variation in those outcomes was the variability induced by actual students impersonating the cognitive agent addressed by the situation. A theory of observation was also at play that permitted attributing the actions of each and every real child to an empirical aggregate that could illuminate the extent to which the (theoretical) student might or might not take the situation to different places.

Another example of how a theoretical stance about what research practice should be has facilitated the doing of that practice has been common in the United States: the constructivist "teaching experiment" and its generalization, the "design experiment." This form of theory use is dealt with extensively by Cobb (this volume); Cobb et al. (2003); and Kelly and Lesh (2000), so we direct the reader to these sources for more details.

In this subsection we have argued that one of the roles of theory in scholarship is that of articulating or mediating the bidirectional relationship between problems and practice. We have elaborated on this relationship by building on the multiple meanings of the words *problem* and *practice* that are relevant to our scholarship. We have proposed that theories help

relate problems to practice in at least the following eight ways:

- A theory is a proposed solution to a problem of practice.
- A theory establishes the criterion according to which one state of a problem in a practice can be compared to another, so as to judge whether there has been any improvement (i.e., it establishes the meaning of *improvement*).
- A theory identifies the boundaries of the practices that pose problems and how those practices contribute to the information about the problem.
- A theory helps argue that a certain state of affairs in a practice is actually a problem (either as a circumstance that needs improvement or as a puzzling contingency that needs explanation).
- A theory helps establish the way in which puzzling questions that are problems for scholars might also be problems to practitioners or create inconveniences to the pursuit of practice.
- A theory can help envision the inconveniences and puzzles that might result from the unfolding of new practices.
- A theory can help design new practices.
- A theory can help make and justify choices (e.g., design, sampling, coding) in scholarly practices such as research, consulting, or communication.

MAKING USEFUL THEORY: THEORY MAKING AS A PART OF SCHOLARSHIP

The foregoing discussion was predicated on the premise that theory is instrumental to the relationships among practice, problems, and research. This approach seemed justified given the historical emergence of mathematics education as a field of practice and reflection on practice. Other social sciences (such as psychology and economics) also seem to have followed a similar historical developmental trajectory: Economics, for example, was first the art of doing multiple things with limited resources and only later the science of the transformations of goods and services. Mathematics education seems to be at a reasonably early stage in a similar kind of development. But it may be time to prepare for and initiate the next stage of development.

Mature social sciences grow not only in response to external demands and responses, but also from within. Their own dynamics tend to build them as solid fields of inquiry, in addition to fields of practice. And theory plays a critical role in strengthening that development from within. In fact, inward-focused development often fuels the specification of new theories and theory-based tools that offer new possibilities for application in the ways described in the previous section.

In this concluding section of the chapter we consider briefly the role of theory not so much as a mediator of relationships among practice, problems, and research, as we have in the bulk of this chapter, but rather (or also) as the collector, beneficiary, or target of that interplay in a fundamentally academic theory-making exercise. What roles could theory play when doing the job of strengthening the field as an endeavor of academic scholarship?

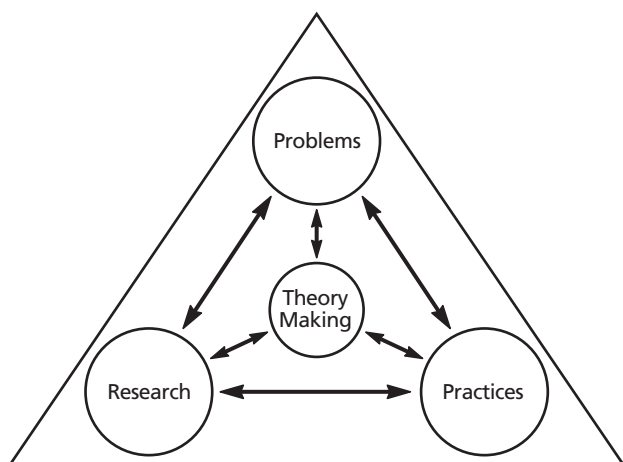


Figure 2.8 Scholarship as a theory-making endeavor.

In addition to the local theories that help mediate specific connections among the three vertices of the scholarship triangle, there are at least two kinds of theories that we scholars should aspire to build. The first could be dubbed *grand theories* of mathematics education. This type of theory responds to a need for broad schemes of thought that can help us organize the field and relate our field to other fields, much in the same way as evolutionary theory has produced a complete reorganization of biological sciences. It can also be seen as a means to aggregate scholarly production within the field. Notwithstanding the famous critique by the sociologist C. Wright Mills (1959) of such approaches in the social sciences, this has long been the goal of some pioneers in the field of mathematics education (e.g., Steiner, 1985).

According to Skinner (1985), a grand theory is any theory that tries to provide a comprehensive explanation of social life, history, or human experience. The development of a grand theory of mathematics education could be useful in providing warrants for our field's identity and intellectual autonomy within apparently broader fields such as education, psychology, or mathematics. In that sense, *a grand theory could be helpful to organize the field*, imposing something like a grand translational or relational scheme that allows a large number of people to see phenomena and constructs in places where others only see people, words, and things. A grand theory of the field of mathematics education could seek to spell out what is singular (if anything) of *mathematics education as an institutional field* or perhaps seek to spell out connections with other fields that may not be so immediately related and that establish the field as one among many contributors to an academic discipline.

This seems to have been the aim of Yves Chevallard's TAD (*Theorie Anthropologique du Didactique*, or Anthropological Theory of Didactics; see Chevallard, 1991; Chevallard, Bosch, & Gascón, 1997). In this approach, scientific production of mathematical knowledge and its communication within and across institutions (including schools, teacher preparation centers, and mathematics departments in universities), as well as the internal divisions of labor that facilitate production and communication, are turned into objects of study. Mathematics education writ large, and likely including most of what the research community does, becomes an object of study, which the TAD theory attempts to describe and explain. Its goal might be to push the limits of the theoretical approach: For example, can an anthropological framework provide an account of all the problems that the field generates and of the theoretical problems generated in addressing those?

Reciprocally, a theory that organizes the field could also take the form of *a theory of mathematics, a mathematical interpretation of the practice of mathematics education that took that practice as a repository of mathematical ideas* just as the historical documents of mathematics constitute another repository of mathematical ideas and just as (as viewed by the Piagetian studies of genetic epistemology) individual development is a repository for the mathematical ideas (such as space, time, causality, and moral judgment) from an ontogenetic perspective. The problems posed could relate to the evolution of mathematical ideas, and the theories made or tested could be epistemological ones, cultivated in educational settings much like Piagetian epistemology was an epistemological theory that used child development as the medium in which to

observe ontogenesis of knowledge. Important threads that inspire this theoretical approach are found in the works of Guy Brousseau (particularly his seminal idea that for each piece of mathematical knowledge there is an *adidactical* situation, a set of relationships between a subject and an antagonist milieu, which creates the meaning of the former; see Brousseau, 1997, chapter 1).

Embodiments of mathematics are present in human actions in school and out of school. The actions in which a teacher and her students use the Pythagorean theorem are as much a cultural representation of the Pythagorean theorem as Proposition I.47 in Book I of Euclid is. It seems that the more that our culture becomes conversant with the idea of cultural embodiments of ideas and with the technologies that can capture them, the more likely we are to witness the development of an empirical epistemology of mathematics. Balacheff's (1995) proposal to identify mathematical conceptions in coherent spheres of mathematical practice provides the beginning of a theory of mathematics in practice.

A grand theory could also be a statement of the field's concerns—a statement that identifies the field. In this case, scholarship does its job with relative freedom by way of naming its components in some taken-as-shared vocabulary as we do when we explain what mathematics education is to scholars from outside our community. So, for example, we use expressions like “mathematical thinking,” “mathematical knowledge,” “mathematical communication,” or “learning of mathematics” to identify objects that we take for granted to exist. Yet their theoretical nature, their constructed being, may be questioned or even denounced when we have to explain what we do to others.

One way to resolve this tension might seem to be a return to David Johnson's long-ago suggestion that we first borrow theory from other social science fields, and only if we can find nothing useful, to invent our own theories within mathematics education. But there is no guarantee that this approach will solve the problem, though it may appear to finesse it. Consider, for example, what we mean by the expression *mathematics learning*. We are mindful that students might be learning mathematical ideas in the mathematics class, but we also are aware that they might very well be learning something that is not mathematics, such as learning how to adapt to their teacher, how to get a good grade, or how to speak English. If the capacity to talk about mathematics learning were dependent on taking learning theories from elsewhere and applying them to mathematics, we might be better integrated with our colleagues in educational research yet limited in our capacity to

understand what we care about. A general theory of learning, such as one based on *communities of practice* and *legitimate peripheral participation* (Lave & Wenger, 1991), offers a compelling framework to analyze student learning as a public activity in classrooms or teacher learning as a communal activity in collegial settings such as schools (Stein, Silver, & Smith, 1998). Yet, without fundamental reconsiderations, this theory might be insufficient to distinguish, for example, the sense in which student learning is becoming a central participant in the game of school mathematics from the sense to which student learning is becoming a central participant in a game of mathematical ideas in school.

Our capacity to see the teaching and learning of mathematics where others just see teaching and learning, and yet others see only social interaction, attests to the different purposes for which individuals and organized fields construct theories. Having a grand theory of mathematics education that distinguishes constructs like *mathematics learning from learning*, beyond the mainstream distinction between genus and species, would be useful in making scholars in our field more resourceful and autonomous. With better theories of our own, mathematics educators will be well positioned to say things of interest to a wide variety of other fields from which we now freely borrow their theories, such as communication studies, information science, cognitive anthropology, cognitive psychology, or organizational studies.

A second type of theory grows from the need to inform a discrete variety of practices, including individual mathematical thinking, teaching and learning in classrooms, or mathematics teacher education. That is, we need what the sociologist Robert K. Merton dubbed *middle-range theories*, comparable in scope to mathematical theories (e.g., Lie theory, spectral theory), which identify subfields of study and the specific methodologies used to study those. We in mathematics education would be better able to inform outsiders about those specific practices if our scholarship contributed to the building of theories that aspired to inform, both in the sense of help understand and help improve, these practices.

We need considerable attention to middle-range theories that conceptualize patches of reality and guide empirical research, bridging the gap between grand theorizing and empiricism. Although many scholars will be attracted to the goals of grand theory outlined above, we think that more fertile ground is likely to be found if we till the soil for middle-range theories. According to Mjøset (1999), Merton studied the evolution of the social sciences and

claimed that social science was still too young. When they matured, as physics already had, the body of middle range theories would converge into a system of universal laws on the model of experimental natural sciences. Until social science matured, he concluded, it should give priority to middle range theories, deliberately avoiding universal theory.¹¹

Accordingly, a possible expectation for theory making in mathematics education could be not to develop a grand theory of the field but to parse the field into patches where middle-range theories could be developed as a result of the process of doing scholarship. In particular, these theories could help expose and identify things that are invisible without it, producing constructs that are operationally defined but only visible indirectly or by their consequences. Most modern science is of this kind; mathematics education scholarship has been making strides in that direction through the more extensive involvement of theory. We suspect that the building blocks for much of this kind of theory-building can be found in the other chapters in this volume.

To illustrate one example of such a middle-range theory, consider the case of mathematics instruction augmented by Brousseau's notion of *didactical contract*, which was introduced above. This idea, the didactical contract, appears to be comparable in nature to others like the *social contract* of Jean-Jacques Rousseau or *social capital* in the scholarship of Pierre Bourdieu. They are not as much precise concepts that can be defined operationally, but rather omnibus concepts that designate a perspective from which to be attuned to notice some things while looking at the real in search for phenomena and to develop finer grained systems of operational concepts that explain them.¹² We argue however that the notion of didactical contract has turned out to be a foundational contribution to the development of instruction as a theory of a middle range.

We commented above about the initial uses of didactical contract, for example pointing to the significance of the Jourdain effect, when illustrating the role that theory can play while connecting research to practices or problems. The hypothesis that a didactical contract was in place was useful to explain what was odd in the observations labeled after the Jourdain effect. And this had to do with recognizing a number of things that regulate classroom mathematical work in situations where learning a new idea was at stake. But

the notion of didactical contract itself, albeit useful to explain in general why some things happened, was not what Brousseau was interested in developing a theory of.

The key problematique Brousseau pursued asked on what conditions it was possible to communicate the ideas of the discipline of mathematics while at the same time making them function in (even emerge from) the activity of children's thinking about problems. The theory of *didactical situations*, within which the didactical contract is only a subsidiary notion, is a theory of the conditions under which the knowledge that has to be taught can emerge as a by-product of the interaction between the learner and an associated mathematically relevant environment (the milieu). This rendering stresses the affinity of Brousseau's work and that done in the United States by teacher-researchers like Deborah Ball (1993) and Magdalene Lampert (1990). One might conjecture that insofar as those independent lines of work pursued agendas for research on the enacted curriculum, both were probes on Bruner's provocative assertion that any subject can be taught to anyone at any age in "some" intellectually honest way (Bruner, 1960). Both agendas were different, however, in how they accounted for the work of the teacher and how they contributed to the development of a middle-range theory of mathematics instruction. Specifically, the contribution of those who asked the question of what it takes to teach mathematics in an intellectually honest way has been the notion that the teacher of mathematics is a player of a complex game who must select moves and strategies depending on the mathematical work that has to be accomplished, balancing commitments, some of which are unconscious and which come to the fore in the form of tensions or dilemmas that need management (Ball, 1993; Chazan, 2000; Chazan & Ball, 1999; Heaton, 1994; Lampert, 1985, 1990, 2001; Smith, 2000).

We bring this brief review up to suggest that as far as their usefulness to explain the actions of the teacher in instruction there is complementarity between the notion that a didactical contract provides entitlements and imposes responsibilities on the teacher and the notion that a teacher experiences her work as one of managing tensions and dilemmas. The role of the didactical contract in the theory of didactical situations was to identify a source of relatively ancillary regulations that needed to be managed as a precondition to engage in the experimental study

¹¹ Note that because this quote is from an online publication we cannot provide a page number.

¹² Many mathematical concepts function as omnibus concepts. For example, parallelism or area are typically not captured by precise operational definitions. Rather, they designate systems of related phenomena amenable to theories within which less important but more precise concepts can be defined (e.g., alternate interior angles, sequence of nested polygons).

of the emergence of knowledge and in order to make epistemological claims that validly referred to classroom mathematical work. In that role, the notion of contract could serve to explain why the teacher or the student might at times get out of the roles they were expected to play in the knowledge game associated with a task. Specifically what structure the didactical contract imposed on the ordinary work of a teacher and what sorts of tensions and contradictions a teacher would have to manage by way of playing the role that the didactical contract prescribes on her were not a principal part of the theoretical work in the theory of didactical situations.

The work of the teacher in the classroom has however been the object of theorization in the United States. In the 1960s Kenneth Henderson developed a relational conception of teaching into a research program that challenged then-current ways of studying teaching in educational research (which focused on personal characteristics of the teacher or on teacher behaviors). Henderson (1965) proposed a sophisticated and abstract conception of teaching as a set of relationships between and among elements of a triplet: the teacher, the student, and the knowledge at stake. Later scholars have named that relationship many things, including the “I, Thou, It, triangular relationship” (Hawkins, 1974), the *didactic system* (Chevallard, 1985), and the *instructional triangle* (Cohen et al., 2003). The important point is likely not what each chooses to call it, but rather the common, collectively acquired capacity to identify and recognize the set of relationships among those three elements and within a school institution as an object of inquiry. The notion of didactical contract has turned out to be a particularly helpful notion to turn a descriptive theory of instruction based on that relational conception of teaching into an explanatory theory (see, e.g., Herbst, 2003, 2006). The complementarity between the hypothetical obligations that the didactical contract places on any teacher and the dilemmas and tensions that individual teachers perceive is currently being exploited to study empirically the practical rationality that underlies the work of teaching (Herbst & Chazan, 2003, in press). Overall we can describe this work as one of trying to flesh out a theory of the didactical contract that can contribute a way of crafting rational explanations for instructional phenomena, hence contributing to the development of a theory of instruction of the middle-range kind.

With this theoretical insight in hand, the field is now able to grasp the idea that interaction in any mathematics classroom is not merely the projection of what an individual teacher, individual students, or the discipline of mathematics itself bring to the

encounter, but rather the joint construction of a viable response to the conditions and constraints that make possible as well as affect their encounter. The emergent explanatory theory of instruction based on a relational conception of teaching is a possible example of how theory making can help sustain the identity of mathematics education as a field of inquiry. It does so by way of giving the means to stake a claim on a perspective on an object of study, by directing where some efforts should go, and by giving the means to relate to other disciplines inside and outside educational research.

The development of the notion of the didactical contract from being an omnibus concept into being a system of constructs that can afford explanation and facilitate experimentation exemplifies one way in which a relational conception of instruction has been developed. The hypothesis that a didactical contract articulates teacher and student mutual responsibilities vis-à-vis the knowledge at stake provides a way to explain how what participants actually do in classrooms is as much the joint construction of a response to the conditions in which they work together and for each other as it is an expression of who they are. What use does that have? In our opinion it allows the same kind of evolution in thinking mathematics instruction that the notion that errors have a conceptual basis had for the way researchers think about individual cognition. In the case of cognition, theory helped scholarship evolve from seeing students’ mistakes as indications of a deficit to the recognition that errors are an indication of the having of a different knowledge, which is adapted to cope with a range of situations. That theoretical recasting of what errors are has served to restore rationality to the child; this has at least potentially helped promote a more respectful and empathetic treatment of learners by teachers. This has not meant that educators ignore error, any more than medicine ignores disease, but the evolving understanding of the *logic of errors* has helped support the design of better instructional treatments, in much the same way that the evolving understanding of the *logic of diseases* has helped the design of better medical treatments. Like that understanding of the nature of errors, the understanding of the nature of instructional actions provided by an explanatory theory of instruction constitutes a particular achievement that mathematics education scholarship has made and that contributes to the scholarly world writ large. Mathematical cognition in its instructional context and mathematics instruction in institutional context are two areas in which the work of our field has developed key contributions towards theories of the middle range. These contributions say things about cognition,

communication, and mathematics to the educated world that no other academic field can say.

Advancing the field of mathematics education—both its scholarship and its practice—requires theoretical advances of this type. Adapting the oft-cited motto of “If you want to get ahead, get a theory” (Karmiloff-Smith & Inhelder, 1975), originally offered to capture some aspects of children’s cognitive developmental trajectory, to our purposes here and applying the maxim to mathematics education professionals as a collective, we can think of no better precept both to end this chapter and to guide our future endeavors: If we want to get ahead, we need to develop theories and use them wisely!

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METHOD

Alan H. Schoenfeld

UNIVERSITY OF CALIFORNIA, BERKELEY

This chapter is concerned with research methods in mathematics education and, more broadly, with research methods in education writ large. As explained below, space constraints do not allow for the detailed consideration of individual methods, or even classes of methods. Hence I have chosen to address broad metatheoretical issues in much of the chapter, which is divided into three main parts. Part 1 provides an overview of the process of conducting and reflecting on empirical research. It examines the major phases of empirical research and some of the issues researchers must confront as they conduct their studies. A main thesis underlying the discussion in Part 1 is that there is a close relationship between theory and method. I describe the process of conducting empirical research and elaborate on how researchers' theoretical assumptions, whether tacit or explicit, shape what they choose to examine, what they see and represent in the data, and the conclusions they draw from them. Part 2 presents a framework for evaluating the quality of research. In it I argue that research must be judged by at least the following three criteria: trustworthiness, generality, and importance. A range of examples is given to elaborate on the issues discussed in Parts 1 and 2. In Part 3 I try to bring together the general arguments from the first two parts of the chapter by focusing methodologically on a topic of current interest and long-term importance. As this *Handbook* is being produced, there is great pressure on educational researchers and curriculum developers in the U.S. to employ randomized controlled trials as the primary if

not sole means of evaluating educational interventions. In an attempt to move forward methodologically, I propose and discuss an educational analog of medical "clinical trials": the structured development and evaluation of instructional interventions. Part 3 offers a description of how the development and refinement of educational interventions might be conducted in meaningful ways, beginning with exploratory empirical/theoretical studies that reside squarely in "Pasteur's quadrant" (Stokes, 1997) and concluding with appropriately designed large-scale studies.

Before proceeding, I should comment about what this chapter is and is not. It is not a survey of research methods or (with the exception, in some sense, of Part 3) a "how to" guide to research. Such an approach would require a volume as large as this *Handbook* itself. Moreover, it would be largely redundant. There exist numerous handbooks of research methods in education, many weighing in at close to 1000 pages (see, e.g., Bruning & Kintz, 1987; Conrad & Serlin, 2005; Denzin & Lincoln, 2005; Green, Camilli, & Elmore, 2006; Keesee, 1997; Kelley, & Lesh, 2000; LeCompte, Millroy & Preissle, 1992; Riley, 1990; Tashakkori & Teddlie, 2002). To give just one example of the extent of the methodological domain, the *Handbook of Complementary Methods in Education Research* (Green, Camilli & Elmore, 2006) contains chapters on 35 different research methods. The methods that begin with the letters C and D alone include: case studies: individual and multiple; cross-case analysis; curriculum assessment; data modeling;

structural equation modeling; definition and analysis of data from videotape; some research procedures and their rationales; design experiments; developmental research; theory, method, design and statistical analysis; and discourse-in-use. It should be clear that even a cursory coverage of methods, much less a “how to,” is beyond what can be done in this chapter.

What can and will be done is to take a bird’s eye view of the terrain—to examine some overarching issues regarding the conduct of empirical research. It should be noted that from this perspective mathematics education is both special and not special. Mathematics education is special in that it is the focus of this *Handbook* and one of the best-mined fields of empirical research. All of the examples discussed in this chapter come from or serve to illuminate issues in mathematics education. At the same time, however, the issues addressed by these examples—What processes are involved in making sense of thinking, learning, and teaching? What are the attributes of high quality empirical research? How might one characterize a rigorous development and testing process for instructional interventions?—are general. The discussions in this chapter apply to all empirical research in education; indeed, to all empirical research.

PART 1 ON THE RELATIONSHIP BETWEEN THEORY AND METHOD; ON QUALITATIVE AND QUANTITATIVE METHODS; AND A FRAMEWORK FOR EXAMINING FUNDAMENTAL ISSUES RELATED TO EMPIRICAL INQUIRY

*There is no empirical method without
speculative concepts and systems; and there is no
speculative thinking whose concepts do not reveal,
on closer investigation, the empirical material
from which they stem.*

—Albert Einstein

All empirical research is concerned with observation and interpretation. This is the case when one is crafting “rich, thick” descriptions (Geertz, 1975) of classrooms or of aboriginal cultures; it is also the case when one is conducting randomized controlled trials of rats running mazes after being subjected to different training regimes or of students taking mathematics assessments after being taught from different curricula.

What may be less obvious, but is equally essential, is that all empirical research is concerned with and deeply grounded in (at times tacit but nevertheless strong) theoretical assumptions. Even the simplest observations or data gathering are conducted under the umbrella of either implicit or explicit theoretical assumptions, which shape the interpretation of the information that has been gathered. Failure to recognize this fact and to act appropriately on it can render research worthless or misleading.

In this opening part of this chapter I focus on issues of theory and method. First, I provide some examples to make the point that theory and method are deeply intertwined—that, as the quotation from Einstein attests, there are no data without theory and there is no theory without data. Then I proceed to put some flesh on the bare bones of this assertion. I offer a framework for conducting and examining empirical research. Readers are taken on two “tours” of this framework, one describing an example of qualitative research and one describing an example of quantitative research. A main point of the discussions is to show that divisions between the two types of research are artificial—that the same theoretical and empirical concerns apply to both.

On Framing Questions, Data Gathering, and Questions of Values

From the onset of a study, the questions that one chooses to ask and the data that one chooses to gather have a fundamental impact on the conclusions that can be drawn. Lurking behind the framing of any study is the question of what is valued by the investigators, and what is privileged in the inquiry.

For example, a recurrent issue in college level mathematics is typically posed as follows: “Is there evidence that small classes (e.g., recitation sections with thirty or fewer students) are more effective than large lecture classes?” What must be understood is that the way this question is operationalized and the choice of evidence that will be used to inform a decision are consequential.

One way to judge course effectiveness is to examine student scores on a uniform end-of-term examination. For reasons of efficiency, students in large lecture classes are often tested using skills-oriented multiple choice tests. Thus, one might decide to give such tests to students in both small and large calculus classes, and look for differences in scores.¹ It might

¹ How well a skills-oriented test might actually reflect what a group of students has learned, and what conclusions can be drawn from such using such tests, are serious matters. Those issues are considered in the discussion of Ridgway, Crust, Burkhardt, Wilcox, Fisher and Foster (2000) later in this chapter.

well be the case that on such a test there would be no statistically significant differences between the scores of students in large and small classes. On the basis of this evidence, the two forms of instruction could be judged equivalent. Once that judgment has been made, cost might be used as the deciding factor. The institution might opt to offer lecture classes with large enrollments.

An alternative way to evaluate course effectiveness is to look at the percentage of students in each instructional format who enroll in subsequent mathematics courses or who become mathematics majors. With that form of evaluation, small classes might produce better results. On the basis of such evidence, the institution might decide (cost factors permitting) to offer classes with small enrollments.

The point of this example is that both test scores and subsequent enrollment rates are legitimate measures of the outcomes of instruction. Each can be quantified objectively and used as the justification for policy decisions. Yet, the two measures might lead to different conclusions. A decision to use one measure or the other, or a combination of both, is a reflection of one's values—a reflection of what one considers to be important about the students' experience. In this sense, even simple quantitative data gathering and analysis are value-laden. The same is the case for qualitative analyses. Historians, for example, will decide that certain pieces of evidence in the historical record are relevant to their framing of an historical issue while others are not. These acts of selection/rejection are consequential for the subsequent representation and analysis of those data.²

On the Relationship Between Theory and Data

In recent years “evidence-based medicine” (see, e.g., the Cochrane Collaboration at <http://www.cochrane.org/index0.htm>) has been advocated by some, notably by federal administration figures such as Grover Whitehurst, director of the U. S. Department of Education's Institute for Education Sciences, as a model for how to conduct empirical research in education (see, e.g., Whitehurst, 2003). For this reason I have selected as cases in point for this discussion of the relationship between theory and data some uses of the experimental paradigm in medical research.³ Direct connections to research

in mathematics education will be drawn after the examples have been presented.

Consider as an example the use of the “male norm” in clinical studies (Muldoon, Manuck, & Matthews, 1990; National Research Council, 1994; World Health Organization, 1998; Wysowski, Kennedy, & Gross, 1990), in which the results of male-only studies have been assumed to apply to both men and women. The March 2005 issue of the *New England Journal of Medicine* reported the results of a 10-year study of women's use of low-dose aspirin to combat heart disease. Among the findings are the following. In contrast to the situation with men, for whom taking low-dose aspirin on a daily basis has consistently been shown to lower the likelihood of heart attacks, taking a low daily dose of aspirin did not, overall, reduce the likelihood of a first heart attack or death from cardiovascular disease for women. However, there were age-specific results: Aspirin did substantially reduce the likelihood of heart attacks in women over the age of 65. Similarly, recent medical research indicates that there are differential risks of diabetes for different subpopulations of the general population.

There is a sampling point here: assuming that the results of a study (no matter how well executed) that is conducted on a subpopulation will apply to the population as a whole is not necessarily warranted. Selecting an appropriate sample is a subtle art, and unexamined assumptions may skew a sample badly. Conversely, studies that average results over an entire population may fail to reveal important information about specific sub-populations—that is, averages may mask important effects. (See, e.g., Siegler, 1987, and the discussion of Bhattachargee, 2005, below.)

This example also makes an equally important point regarding the researchers' underlying conceptual models. When “male norm” studies were paradigmatic, the assumption was that a random selection of males was a random selection of people—that gender didn't matter. That is, the experimenters did not consider gender to be a relevant variable in their experiments. This failure to conceptualize gender as a variable rendered the studies of questionable value.

In sum: Whether it is tacit or explicit, one's conceptual model of a situation, including one's view of what counts as a relevant variable in that situation, shapes data-gathering—and it shapes the nature of the conclusions that can be drawn from the data that are gathered. As will be discussed later in

² N.B. Historians, and social scientists in general, often make their cases via narrative. One must understand that narrative is a form of representation; my comments about representations apply to narrative work as well.

³ Throughout this chapter I discuss examples of significant current interest such as controversies over randomized controlled trials as the “gold standard” for educational research. In doing so I am attempting to achieve simultaneously the dual goal of addressing enduring points of concern and clarifying current issues.

this chapter, issues such as the characteristics of the student population (e.g., what percentage of students are second-language learners?) or of the environment (e.g., is the school capable of implementing a curriculum as intended?) can be fundamental factors shaping what takes place in a learning environment. Whether and how those factors are taken into account in formulating a study and gathering data for it will shape how that study's findings can be interpreted.

A second issue, touched on in the class size example discussed above, concerns the experimenter's selection of outcomes (dependent variables) and the selection of measures to document those outcomes. To give a medical example: By the 1990s hormone replacement therapy (HRT) had become a commonly recommended treatment for some of the symptoms of menopause. When subsequent research examined an expanded set of outcomes such as the incidence of heart disease, breast cancer, and strokes, the value of HRT was called into question (see Medline Plus, 2005, for an overview). Delayed or unexpected consequences are also an issue. The devastating impact of thalidomide was not discovered until some years after the drug had been in common use.

It may seem quite a leap to compare the results of such medical studies with the results of educational interventions. However, there are direct analogues. Like medical interventions, educational interventions can have unintended and often long-term consequences. For example, a body of research in the 1970s and 1980s, which included the qualitative documentation of classroom interactions and results, documented the results of students' school mathematics experiences. These were summarized by Lampert (1990) as follows:

Commonly, mathematics is associated with certainty; knowing it, with being able to get the right answer, quickly (Ball, 1988; Schoenfeld, 1985b; Stodolsky, 1985). These cultural assumptions are shaped by school experience, in which doing mathematics means following the rules laid down by the teacher; knowing mathematics means remembering and applying the correct rule when the teacher asks a question; and mathematical truth is determined when the answer is ratified by the teacher. Beliefs about how to do mathematics and what it means to know it in school are acquired through years of watching, listening, and practicing. (p. 32)

Let me reframe this summary in terms of contemporary discussions. As Lampert indicated, years of learning mathematics passively result in a population that tends to be mathematically passive. That population may be able, on demand, to perform

some mathematical procedures—but it tends not to possess conceptual understanding, strategic competency, or productive mathematical dispositions. If the measures and descriptions of educational outcomes that are employed in empirical research fail to take into account such centrally important classes of outcomes (e.g., conceptual understanding as well as procedural competency; the ability to apply one's knowledge to novel concepts; problem-solving ability; beliefs and dispositions; drop-out rates), then researchers, teachers, and policymakers who wish to make judgments on the basis of those outcomes are potentially misinformed about the likely consequences of their decisions.

A Framework for Conducting and Examining Empirical Research

The preceding metatheoretical comments frame what is to come in this section. In what follows I set forth a framework for conceptualizing empirical work, whether that research is qualitative or quantitative, in *any* field. Figure 3.1 (modified from Schoenfeld, 2002, with permission) offers a framework within which to consider issues of method. After an introductory caveat, I briefly introduce the framework. Then I work through it in some detail.

Caveat

The discussion of Figure 3.1 proceeds in an ostensibly straightforward manner, from the “beginning” of the research process (conception and formulation of problems) to its “end” (drawing conclusions). However, the linear nature of the exposition and the representation in Figure 3.1 belie the complexity of the process, which is decidedly non-linear as it plays out in practice. Research proceeds in cycles, in which one considers and then reconsiders every aspect of the process. Even within cycles, insights (including those caused by failure or chance observation) may cause a reformulation of underlying perspective, or of what are considered salient phenomena; they may result in new representations, alternative data gathering or new ways of thinking about data that have already been gathered; and new conclusions. Specifically, Figure 3.1 is not to be taken as a linear prescription for research.

In simplest terms, empirical research is concerned with making observations of and drawing conclusions about some “real world” situation. Data are gathered and interpreted, and conclusions are drawn. That process is represented by the dotted line from Box A to Box F in Figure 3.1. The conclusions drawn are sometimes just about the situation itself (“I observed

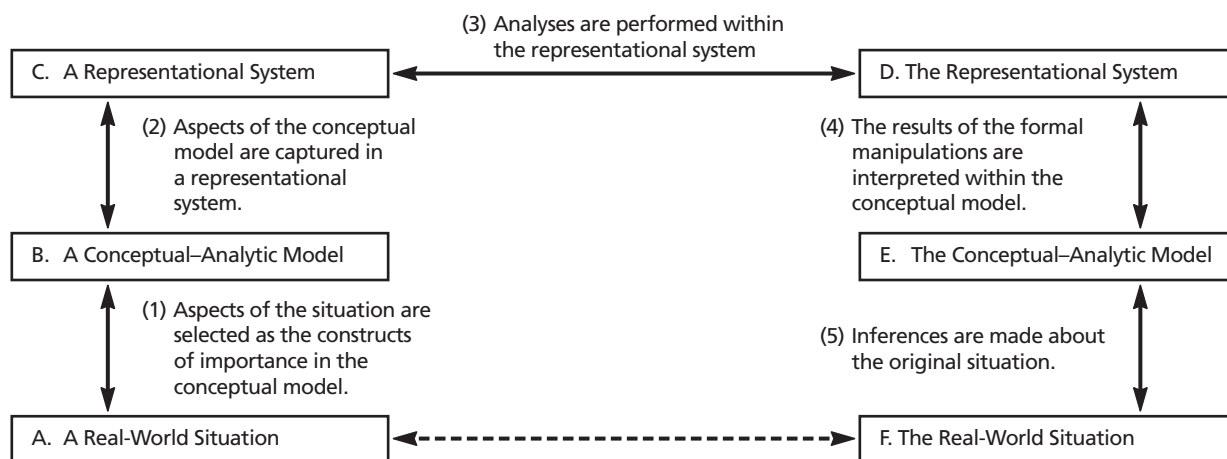


Figure 3.1 A schematic representation of the process of conducting empirical research.

the following...”), but more typically they are drawn with intimations of generality (“What happened here is likely to be the case in circumstances that resemble those described here.”) and importance (“This information should shape the ways we think about X, Y, and Z.”). The main purpose of Figure 3.1 is to indicate that the pathway from observations to conclusions is not as simple as it might seem.

In line with Figure 3.1, I claim that all empirical research involves the following processes:

- conceptualization, in which the situation to be analyzed is seen and understood in certain (often consequential) ways;
- the creation, use, or refinement of a conceptual-analytic framework or model, in which specific aspects of the situation are singled out for attention (and, typically, relationships among them are hypothesized);
- the creation, use, or refinement of a representational/analytic system, in which aspects of the situation singled out for attention are selected, represented and analyzed;
- the interpretation of the analysis within the conceptual-analytic framework or model; and
- attributions and interpretations from the preceding analytic process to the situation of interest (and possibly beyond).

To illustrate the main points above I consider at some length one primarily qualitative example and one primarily quantitative example.

A First Example

As a first qualitative example I discuss the decision I made, nearly 30 years ago, to explore aspects of

students’ metacognitive behavior during problem solving. (Extensive detail regarding this work can be found in my 1985 book *Mathematical Problem Solving*.) The starting place for this work seemed simple. I brought students (either by themselves or in pairs) into a room near my office (my “laboratory”) and asked them to solve a series of problems out loud. I was in the vicinity while they worked on the problems, and I occasionally intervened if a long time had passed without their saying something audible. I videotaped the students’ solution attempts and saved their written work.

The primary sources of data for analysis were the videotapes I made of their problem-solving attempts and the written work they produced while working the problems. On the basis of those data I drew inferences about the students’ decision making during problem solving and its impact on their success or failure at problem solving. I also drew inferences about the frequency and import of the students’ “executive decision making” in general.

To illustrate the issues involved, I start with Box A at the lower left of Figure 3.1, and make a circuit of the figure by following the arrows up, around, and down to Box F. To begin, it should be clear that I was making a fair number of assumptions about the “real world situation” examined here—students solving problems in the laboratory. Two major assumptions were that (a) the students’ problem solving behavior in the laboratory bore some relation to their problem solving behavior in other contexts; and (b) the students’ overt actions bore some relation to their internal cognitive processes.

Both of these assumptions were and are controversial to some degree. Regarding (a), for example, over the years some researchers have questioned the value of

laboratory studies, saying that the artificial behavior induced in the laboratory renders laboratory studies of little or no value in understanding the kinds of interactions that take place amidst (for example) the blooming complexity of the classroom. Regarding (b), for quite some time there have been controversies over the role of verbal reports as data. Retrospective reports of thought processes were roundly discredited in the early years of the 20th century, and for some years *any* reports of thought processes were deemed illegitimate. (Indeed, behaviorists banished the notion of thought processes from “scientific” explanations of human behavior.) In the 1980s Nobel prize winner Herbert A. Simon and colleague K. Anders Ericsson wrote a review (Ericsson & Simon, 1980) for *Psychological Review* and then a book (Ericsson & Simon, 1984) entitled *Verbal Reports As Data*, trying to make the case that although post hoc reports of thought processes could not be taken as veridical, “on the spot” verbalizations of what one was doing could be taken as data suggestive of the individuals’ thought processes.

One could say a great deal more about assumptions (a) and (b)—teasing out what “some relation” means in each of them is a nontrivial exercise! What matters here is something simpler. Wherever one comes down with regard to assumptions (a) and (b), the fact is that they *are* assumptions, and one’s stance toward them shapes how one considers the data gathered. What should be clear is that a form of naïve realism—that the videotapes and written record *directly* capture (some of) what people were thinking as they worked on the problems—is not warranted. Equally clear is that I began my work with specific assumptions about what “out loud” problem-solving protocols could reveal; I entered into the work with a set of underlying assumptions about the nature of cognition that framed the way I saw what was in the tapes. Someone who was not a cognitive scientist, or whose orientation to cognition was different, would not look for or see the same things.

When I began examining the videotapes, I knew there was *something* important about students’ decision making during problem solving—something that was a factor in success or failure—but I did not know what it might be. My earlier work had focused on teaching an explicit decision making strategy, to help students use their problem-solving knowledge effectively. Now I was looking at videotapes made before the instruction, trying to identify causes of success or failure. I was looking at the tapes “from scratch” in part because the fine-grained coding schemes I had found in the literature had not seemed informative.

My research assistants and I watched a fair number of tapes, trying to figure out how to capture

events of importance in a coding scheme. We started in a somewhat systematic way, looking for what we called “reasons to stop the tapes.” These occurred at places in the videotapes where we saw students acting in ways that seemed to bear on the success or failure of their problem solving attempts. We made a list of such events and composed for each event a series of questions designed to trace its impact on the problem solution. This was a prototype analytic scheme. And after polishing it a bit I asked my students to try to analyze the data using it.

When we reconvened, my research assistants were unhappy. They said that the scheme we had developed was impossible to use. Far too many of our questions, which had seemed to make sense when we looked at one tape, seemed irrelevant on another. Our system had so many reasons to stop a tape, and so many unanswerable or irrelevant questions when we did, that whatever was truly important about the problem-solving episode was lost among the huge collection of questions and answers.

Confronted with this failure, I decided to begin again. I chose to look at an “interesting” tape—a tape in which it seemed that the students “should have” solved the problem but did not. My assistants and I tossed the coding scheme aside and looked at the tape afresh. As we did, I noticed one particular decision that the students in the videotape had made. They had chosen, without much deliberation, to perform a particular computation. As the solution unfolded, they spent a great deal of time on the computation, which I realized would not help them to solve the problem. As I watched them persevere in the computation, things clicked. That single decision to perform the computation, unless reversed, could result in the expenditure of so much time and energy in an unprofitable direction that the students were essentially guaranteed to fail to solve the problem.

I had the feeling I was on the trail of something important. My assistants and I looked at more tapes, this time searching for consequential “make-or-break” decisions. It turned out that these were of two kinds: paths wrongly taken and opportunities missed. These make-or-break decisions were consequential in more than half of our tapes. With this understanding, we had a new perspective on what counts as a major factor in problem solving. This new conceptual/analytic perspective oriented us differently toward the tapes and changed our subsequent data analyses. At this point, with a conceptual model in place, we were in Box B of Figure 3.1.

[Before proceeding, I must stress that not every study involves a new conceptual model; most studies involve the use or refinement of well-established

conceptual models. The point of this particular example is that any conceptual model highlights some things and obscures or ignores others; it takes some things into account and does not consider others. For example, my analyses of the videotapes of students solving problems did not, at that point, include a focus on issues of affect or belief. They did not include the detailed examination of student knowledge or knowledge organization, save for the fact that I had been careful to have the students work problems for which I had evidence that they possessed adequate knowledge to obtain a solution. (It is of little theoretical interest when a student fails to solve a problem simply because he or she lacks the knowledge that is essential to solve it.) Hence, as I was examining the problem-solving tapes, I was viewing them through a particular theoretical lens, one that focused on the impact of a particular kind of decision making. The videotapes might well have supported different kinds of analyses, but other aspects of the students' solutions were not to be seen in our analyses (and, equally important, ceased to be salient to us as we analyzed the tapes). I also note that this example demonstrates the dialectic between representational/analytic schemes and conceptual frameworks, thus illustrating the non-linear character of Figure 3.1.]

Once my research assistants and I had a first-order conceptual-analytic framework, we needed a representational scheme to capture and analyze our data. In simplest terms, we decided to parse problem-solving sessions into major chunks called "episodes," periods of consistent goal-oriented activity on the part of the problem solver. The notion of an episode was a useful device for identifying the loci of consequential decisions. The places where the direction of a solution changed were the natural boundaries between episodes, and they were often the sites of consequential decisions. It also turned out that, at a gross level, there were relatively few kinds of episodes: reading the problem, working in a structured way to analyze the problem, planning, implementing a plan, and working in a somewhat random or ill-thought-out way ("exploration").

Over time we refined this representational scheme, which was later supplemented by a more compact and informative time-line representation suggested by Don Woods (Figure 3.2, below, is an example). With a representational scheme in place, we were able to code the data. Things were straightforward. The idea was to represent the contents of a videotape (typically a 20-minute problem-solving session) by an episode diagram, which identified and labeled the episodes and the consequential decisions in a problem session.

At this point we were comfortably ensconced in Box C, working within a particular representational system. It is important to observe that the representational system reified our conceptual model. Events that did not fit into the representational scheme were not captured in the representation, and thus were not fodder for data analysis.

My research assistants and I worked together on a series of tapes, developing a communal understanding of the meanings and types of episodes, and of consequential decision making (including the failure to act on a relevant piece of information). We then coded some tapes independently. Our codings matched more than 90% of the time.

With the consistency of coding established, we were working within Box D—performing analysis with and within the representational system. Coding the sessions was still a matter of interpretation, but with practice it became a relatively straightforward task, as indicated by the high interrater reliability. The hard work had been done in the conceptualization of the scheme. Once the tapes were represented in the coding scheme, data analysis was simply a matter of counting. More than half of the students' problem-solving attempts were represented by the following schematic diagram: an episode of reading the problem followed by an episode of exploration (and failure). That is, the bar graph in Figure 3.2 represented more than half of the problem-solving sessions we coded.

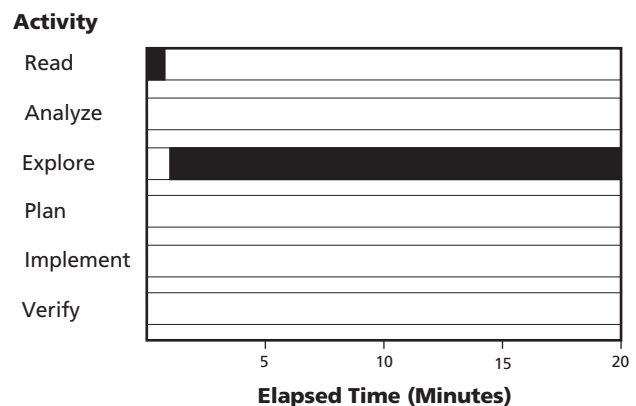


Figure 3.2 A timeline representation of a typical student solution attempt.

These data had a straightforward interpretation. More than half the time, the students—who had the knowledge required to solve the given problems—failed to solve the problems because of their poor choice of initial solution direction. As long as the students lacked effective mechanisms for reflecting on and undoing those initial decisions, they did not

have the opportunity to use what they did know to solve the problems.

This interpretation placed us in Box E, and the extrapolation to Box F (and beyond) was also straightforward. We recognized, of course, that factors other than monitoring and self-regulation affected problem solving success—knowledge and problem-solving strategies among them. In our work, we had controlled for these by giving the students problems that they had the subject matter knowledge to solve; thus we saw the devastating effects of poor metacognitive decision making. We asserted that ineffective monitoring and self-regulation were significant causes of problem-solving failure, both in the laboratory and beyond it.

As noted, there were many assumptions made here, in the attribution of causality both in laboratory problem solving and in the extrapolation to more general problem solving. At this point, given the small number of students we had videotaped, our ideas about the importance of metacognitive decision making were suggestive but not yet validated. Further studies led to substantiation of those ideas. The study of accomplished problem solvers (in the laboratory) documented the ways that effective monitoring and self-regulation can be a productive force in problem solving. Subsequent research showed the robustness of the phenomena of monitoring and self-regulation, and their importance. (See, e.g., Brown, 1987; deCorte, Green, & Verschaffel, 1996; Lester, 1994.)

A second example

I now examine the issues raised by a prototypical quantitative study (Bhattachargee, 2005). Again, I focus on the kinds of choices that are made at various points in the process described in Figure 3.1 and the impact they have on the conclusions that can be drawn. Consider the task of determining whether students learn more from Curriculum X or from Curriculum Y. As many people see it, this is as straightforward as you can get: All one has to do is perform some sort of randomized controlled trial in which half the student population is randomly assigned each treatment, and then see what differences emerge on an appropriate outcome measure. Would that life were so simple.

The complexities begin in the real-world context, Box A. Even before articulating a conceptual model, there are choices at the pragmatic level. Does one randomize the instructional treatment at the student level (with two adjacent students perhaps studying different curricular materials)? At the classroom level? At the school level? Such considerations are often driven by practicalities. But then, when one goes to

the conceptual level (Box B), choices get much more complex—and consequential.

At least two kinds of conceptual issues are fundamental in shaping how one does research on curricular outcomes. The first is methodological, the second subject-matter related. A fundamental conceptual issue related to curriculum implementation is what one considers an “implemented curriculum” to be. One perspective is as follows.

Perspective 1. A curriculum is the set of instructional materials and preparation to use them that teachers are given. Whatever the teachers do with those materials in the classroom is the “implemented curriculum.”

In this case, what counts as the simplest measure of the curriculum’s effectiveness is the average performance of all those students who were in classrooms where that curriculum was used.

Another perspective is as follows.

Perspective 2. There is a strong degree of interaction between curriculum and context. Given different contexts or different degrees of support, there may be more or less fidelity of curriculum implementation. “Degree of fidelity of implementation” (in conformity with the intention of the designers) matters and should be taken into account in analyses of curriculum impact.

Questions of interest to people with this orientation include the following. What kinds and levels of support are necessary, in what kinds of contexts, to guarantee some level of fidelity of implementation for a particular curriculum? What is the curricular impact (measured in terms of student outcomes) when there is some degree of curriculum fidelity? When one is considering a choice between two curricula, what kinds of outcomes can one expect for each, given the resources that one has to put into implementing them?

This distinction in framing has significant consequences. Here is an example, taken from a recent issue of *Science* (Bhattachargee, 2005). In a randomized trial in a school district, three schools used Curriculum X and three used Curriculum Y. The schools were roughly matched on demographics. When one looked at overall curriculum implementation—that is, the complete sets of scores from students who had worked through Curriculum X and Curriculum Y—no statistically significant differences between outcomes were found. Is one to conclude, then, that the two curricula are equally good?

The answer depends on one's conceptual model. For those who adhere to perspective 1 as described above, the situation is straightforward. Given that the schools were randomly assigned to the treatments and the data showed no differences, it follows (within perspective 1) that the curricula are equally effective. But for those who hold perspective 2, there might be a world of difference between the two curricula.

It turns out that of the three schools that used Curriculum X, one school embraced the curriculum and implemented it in a way consistent with the designers' intentions. Students at that school outperformed those who used Curriculum Y. At a second school the implementation of Curriculum X was uneven. There, scores were not statistically different from overall scores on Curriculum Y. In a third school Curriculum X was poorly implemented, and students did poorly in comparison to Curriculum Y.

"On average" the two curricula were equally effective. The averages are uninformative, however. Another way to look at the data is as follows. When Curriculum X is implemented as intended, outcomes are superior to outcomes from Curriculum Y. Under those conditions, Curriculum X is preferable. But when Curriculum X is not implemented effectively, students would do better with Curriculum Y. Hence instructional leadership should assess the capacity of the staff at each site to implement Curriculum X—either now or with professional development—and decide on that basis whether to use it at that site. From perspective 2, then, the decision as to whether to use Curriculum X or Curriculum Y is context-dependent, depending on the school staff's current or potential capacity to implement either curriculum with some degree of fidelity. Note that this is a very different kind of conclusion than the kind of conclusion drawn by those with the "curriculum is context-independent" perspective.

Here is a relevant analogy. Suppose there are two surgical treatments for a particular condition. Treatment A returns patients to full functioning *if* they undergo a full regimen of physical therapy for a year, but the results are unsatisfactory if a patient does not. Treatment B is reasonably but not completely effective, regardless of whether the patient undergoes physical therapy. Suppose that, on average, not that many people follow through on physical therapy. On average, then, people do slightly better with Treatment B than with Treatment A.

Put yourself in the position of a patient who is facing surgery for that condition. Would you want your doctor to recommend Treatment B on the basis

of the statistical average? Or would you rather have the doctor explain that Treatment A might be an option for you, but only if you commit yourself to a serious regimen of physical therapy afterward? Both statements represent legitimate interpretations of the data, within the frames of particular conceptual models. Those models make a difference. As a patient, I would much rather be offered the second choice. There is no reason to settle for the statistical average if there are reliable ways to beat that average. (One should settle for it, however, if one does not have the wherewithal to follow through with physical therapy.)

To return to the curricular example: one's conception of what is meant by "curriculum implementation" has tremendous implications for the ways that findings are reported and interpreted. One can report on data of the type discussed by Bhattachargee (2005) either by saying

- (a) "There were no significant differences between Curriculum X and Curriculum Y" or
- (b) "Curriculum X is superior to Curriculum Y under certain well-specified conditions; Curriculum X and Curriculum Y produce equivalent test scores under a different set of well-specified conditions; and Curriculum Y is superior to Curriculum X under yet another set of well-specified conditions."

The possibilities for *acting* on the information in (a) and (b) differ substantially.⁴ I now consider conceptual models related to subject matter.

Just what does it mean to know (to have learned) some particular body of mathematics? This is not only a philosophical issue, but a practical one as well: Different conceptual models of mathematical understanding lie at the heart of the "math wars" (see Schoenfeld, 2004). One point of view, which underlies much of the "traditional" curricula and standardized assessments, is that knowledge of mathematics consists of the mastery of a body of facts, procedures, and concepts. A more current perspective, grounded in contemporary research, is that mathematical knowledge is more complex. The "cognitive revolution" (see, e.g., Gardner, 1985) produced a fundamental epistemological shift regarding the nature of mathematical understanding. Aspects of mathematical competency are now seen to include not only the knowledge base, but also the ability to implement problem-solving strategies, to be able to use what one knows effectively and efficiently, and more (deCorte, Greer, & Verschaffel, 1996; Lester,

⁴ This idea is not new: see, e.g., Brownell, 1947.

1994; Schoenfeld, 1985a, 1985b, 1992). In elementary arithmetic, for example, the National Research Council volume *Adding It Up* (2001) described five interwoven strands of mathematical proficiency:

- *conceptual understanding*—comprehension of mathematical concepts, operations, and relations
- *procedural fluency*—skill in carrying out procedures flexibly, accurately, efficiently, and appropriately
- *strategic competence*—ability to formulate, represent, and solve mathematical problems
- *adaptive reasoning*—capacity for logical thought, reflection, explanation, and justification
- *productive disposition*—habitual inclination to see mathematics as sensible, useful and worthwhile, coupled with a belief in diligence and one's own efficacy. (p. 5)

Fine-grained analyses of proficiency tend to be aligned with the content and process delineations found in the National Council of Teachers of Mathematics' (NCTM, 2000) *Principles and Standards for School Mathematics*:

Content: Number and Operations; Algebra; Geometry; Measurement; Data Analysis and Probability;

Process: Problem Solving; Reasoning and Proof; Making Connections; Oral and Written Communication; Uses of Mathematical Representation.

These views of proficiency extend far beyond what is captured by traditional content-oriented conceptual frameworks.

In the experimental paradigm, one's view of domain competency is typically instantiated in the tests that are used as outcome measures. What view of mathematical proficiency one holds, and how that view is instantiated in the outcome measures one uses for educational interventions, can make a tremendous difference.

The issues at stake are as follows. Traditional assessments tend to focus on procedural competency, while assessments grounded in broad sets of standards such as NCTM's *Curriculum and Evaluation Standards* (1989) or *Principles and Standards* (2000) include procedural (skills) components but also assess conceptual understanding and problem solving. In a rough sense, the traditional assessments can be seen as addressing a subset of content of the more comprehensive standards-based assessments. Hence a

choice of one assessment instead of another represents a value choice—an indication of which aspects of mathematical competency will be privileged when students are declared to be proficient on the basis of test scores. As the following example shows, these choices are consequential.

Ridgway, Crust, Burkhardt, Wilcox, Fisher, and Foster (2000) compared students' performance at Grades 3, 5, and 7 on two examinations. The first was a standardized high-stakes, skills-oriented test—California's STAR test, primarily the SAT-9 examination. The second was the Balanced Assessment test produced by the Mathematics Assessment Resource Service, known as MARS. The MARS tests cover a broad range of skills, concepts, and problem solving. For purposes of simplicity in what follows, scores on both tests are collapsed into two simple categories. Student who took both tests are reported below as being either "proficient" or "not proficient" as indicated by their scores on each of the examinations. More than 16,000 students took both tests. The score distribution is given in Table 3.1.

Table 3.1 Comparison of Students' Performance on Two Examinations

MARS	SAT-9	
	Not Proficient	Proficient
Grade 3 (N = 6136)		
Not proficient	27%	21%
Proficient	6%	46%
Grade 5 (N = 5247)		
Not proficient	28%	18%
Proficient	5%	49%
Grade 7 (N = 5037)		
Not proficient	32%	28%
Proficient	2%	38%

Unsurprisingly, there is a substantial overlap in test performance: Overall 73%, 77%, and 70% of the students at Grades 3, 5, and 7, respectively, either passed both tests or failed both tests. The interesting statistics, however, concern the students who were rated as proficient on one test but not the other.

For each grade, consider the row of Table 3.1 that reports the SAT-9 scores for those students rated "proficient" on the MARS test. At Grades 3, 5, and 7 respectively, 88%, 91%, and 95% of those students were rated proficient on the SAT-9. Thus being rated proficient on the MARS test yields a very high probability

of being rated proficient on the SAT-9. That is: being declared proficient on the MARS exam virtually assures having the procedural skills required for the SAT-9.

The converse is not true. Consider the final column of Table 3.1, which indicates the MARS ratings of the students who were rated proficient on the SAT-9. Approximately 31% of the third graders, 27% of the fifth graders, and 42% of the fifth graders who were declared proficient by the SAT-9 were declared not proficient on the MARS exam. That is, possessing procedural fluency as certified by the SAT-9 is clearly *not* a guarantee that the student will possess conceptual understanding or problem-solving skills, as measured by the MARS test. Indeed, the students who were declared proficient on the SAT-9 but not the MARS test—roughly 1/3 of those declared proficient on the SAT-9—can be seen as false positives, who have inappropriately been deemed proficient on the basis of a narrow, skills-oriented examination.

Once an assessment has been given, the die has been cast in terms of data collection. One is now in Box C in Figure 3.1, where there exist standard techniques for scoring tests and representing test data. The pathway from Box C to Box D in Figure 3.1 is relatively straightforward, as are analyses within Box D. This, after all, is the province of standard statistical analysis. However, *interpretation*—the pathway to Boxes E and F—is anything but straightforward, for it depends on the conceptual models being employed.

There is suggestive, though hardly definitive, evidence (see, e.g., Senk & Thompson, 2003) that nearly all of the National Science Foundation-supported standards-based curricula have the following property. When the test scores of students who have studied from the NSF-supported curricula are compared with test scores of students who have studied from more traditional skills-oriented curricula, there tend to be no statistically significant differences between the two groups in performance on skills-oriented tests (or the skills components of broader tests). However, there tend to be large and significant differences favoring the students from the NSF-supported curricula on measures of conceptual understanding and problem solving. Thus, if appropriately broad assessments are used, comparison studies will tend to produce statistically significant differences favoring the performance of students in these standards-based curricula over the performance of students from more traditional comparison curricula. However, if skills-oriented assessments are used, no significant differences will be found. Hence at the curriculum level, the use of measures that focus on skills can result in curricular false negatives—the tests will fail to show the real differences that exist.

The fundamental point to be taken from the preceding discussion is that the specific contents of any given assessment matter a great deal. One can draw meaningful conclusions about the relative efficacy of two curricula on the basis of a particular assessment only when one knows what the assessment really assesses (that is, when a content analysis of that assessment has been done). Without a content analysis, it is impossible to interpret a finding of “no significant differences.” Such a finding might occur because both curricula are equally effective. Or, it might occur because an inappropriately narrow assessment failed to pick up what are indeed significant differences in impact. For this reason, a report of a randomized controlled trial that does not contain a content analysis of the assessment employed is of no value. Indeed, the conclusions drawn from it may be false or misleading.

Ironically, this is the mistake made by the nation’s most ambitious attempt to provide information about curricular effectiveness, the What Works Clearinghouse (WWC). WWC (<http://www.whatworks.ed.gov/>) does not conduct research itself. Rather, it was created to review and report findings from the literature. WWC searches the literature for studies that meet stringent methodological criteria. Studies that qualify for vetting by WWC must be of one of the following three types: randomized controlled trials, quasi-experiments that use equating procedures, or studies that use regression discontinuity designs. These are vetted for technical proficiency and empirical flaws. Only studies that make it through WWC’s methodological filter are reported.

WWC committed the fundamental error identified above in reporting one of the few studies that did make it through its methodological filter. In a report (What Works Clearinghouse, 2004), WWC gave part of the statistical analyses in the study it examined (a quasi-experimental design with matching reported in 2001 by C. Kerstyn) full marks. Here is what WWC (2004) said about its choice of that part of the study:

The fifth outcome is the Florida Comprehensive Assessment Test (FCAT), which was administered in February 2001. The author does not present the reliability information for this test; however, this information is available in a technical report written by the Florida Department of Education (2002). This WWC Study Report focuses only on the FCAT measures, because this assessment was taken by all students and is the only assessment with independently documented reliability and validity information.

Note that reliability and validity are psychometric properties of an assessment: They do not provide a characterization of the actual content of the examination. Neither Kerstyn nor WWC conducted content analyses of the FCAT exam. For all one knows, it could be as narrow as the SAT-9 examination discussed by Ridgway et al. (2000). The Kerstyn study reported “no significant differences”—but why? Was it because there were none, or because the narrowness of the measure used failed to reveal a significant difference that actually existed? Because of the lack of information provided by WWC, it is impossible to know. Given that WWC failed to conduct a content analysis of the FCAT, the findings reported in the WWC report are at best worthless and at worst misleading. In addition, WWC’s unwillingness to conduct content analyses of the measures used in the randomized controlled trials of mathematics studies makes it impossible for WWC to achieve its core mission. WWC was created with the intention of conducting meta-analyses of the literature—to sort out through analytical means the impact of various curricula. Properly conducted, the analyses and meta-analyses are intended to reveal information such as the following: “Curriculum X tends to be strong on procedural skills and on conceptual understanding, but not especially strong on problem solving. Students tend to do well on tests of geometry, measurement, and number, but they do less well on tests of algebra and data analysis.” Given that WWC has refused to conduct content analyses,⁵ WWC can offer no insights of this type. Once again, what is attended to, both in conceptual models and in assessments, is highly consequential.

In sum, although one must be proficient in the application of quantitative and qualitative methods on their own (specifically, the pathway from Box C to Box D in Figure 3.1), such proficiency is no guarantee that the interpretation of the results will be meaningful or useful. A meaningful report must respect all of the pathways from Box A to Box F in Figure 3.1.

Discussion

In this section I have focused on some fundamental issues of theory and method. First, I argued that theory and method are deeply intertwined. Every empirical

act of representation, analysis, and interpretation is done in the context of a (sometimes explicit, sometimes implicit) conceptual and theoretical model. The character of such models shapes the conclusions that are produced by subsequent analysis and interpretation. Second, I have presented a framework (Figure 3.1) that highlights major aspects of empirical research including conceptualization, representation, analysis, and interpretation. I remind the reader that although the figure and the linearity of prose as a medium may suggest that the process is linear, it is not: the process is cyclical, and there can be substantial give-and-take between all of the aspects of research reflected in Figure 3.1 during each cycle of research. The extensive discussion of Figure 3.1 highlighted the complexity of the process and the ways in which conceptual models can affect what one captures in data and how those data are interpreted. Third, I deliberately chose to work through one prototypically qualitative and one prototypically quantitative example to indicate that the fundamental issues of focus, data gathering, data analysis, and interpretation of findings are the same whether one is conducting qualitative or quantitative research.⁶ The serious question to be considered is not, “is this research of one type or another” but “what assumptions are being made, and how strong is the warrant for the claims being made?”

Finally, I want to point to the fact that the framework outlined in Figure 3.1 can be used reflectively, both as one conducts research and as one examines research conducted by others. Each of the pathways between the boxes in Figure 3.1, and each of the boxes, represents a series of decisions made by the researcher. Thus, for example, the pathway from Box A to Box B indicated by Arrow 1 (“aspects of the situation are selected as the constructs of importance in the conceptual model”) offers a reminder that any choice of focal phenomena represents a set of theoretical commitments. This provides the opportunity to reflect on the choice and implications of the conceptual model that is being (even if tacitly) employed. For example, which phenomena are not taken into account by this perspective? Which are given significant emphasis? How are those theoretical biases likely to shape the interpretation of the situation?

⁵ I served as the Senior Content Advisor for WWC’s mathematics studies (at first for middle school mathematics, then for all mathematics reports) from WWC’s beginnings. I resigned in early 2005 when WWC refused to correct the flaws identified above and reneged on a commitment to publish an article in which I had discussed such issues. For details see Schoenfeld (2006a, 2006b).

⁶ If space permitted I would include a third example. Suppose one wanted to conduct an ethnographic study of classrooms using different curricula, with a focus on (say) discourse structures and their impact. It is left as an exercise for the reader to work through Figure 3.1, with regard to issues such as unit of analysis, selection and form of data, outcome measures (e.g., test scores, or discussions of identity), and interpretation. All of the issues that arose in the quantitative example arise here as well.

Similarly, the pathway between Boxes B and C indicated by Arrow 2 (“aspects of the conceptual model are captured in a representational system”) represents an act of *data selection and reduction* as well as representation. In historical studies, for example, whose voices are selected and heard? Or, suppose one is conducting classroom observations. Does one take field notes or make videotapes? If one tapes, what is the focus of the camera? If one takes notes, are they structured according to a predetermined system (in which case they reflect an explicit focus on particular aspects of the situation) or are they somewhat open (in which case the selection is tacit)? For example, data-gathering during the days of the process-product paradigm typically consisted of tallying certain kinds of behavior (teacher actions, student actions) and looking for correlations with educational outcomes (e.g., test scores). In contrast, many current studies of classroom discourse focus on the character of student and teacher interactions, and the results in terms of community norms, beliefs, and knowledge. Each act of data selection, reduction, and representation will have the potential to illuminate certain aspects of a situation, and to obscure others (or even render them invisible). Even if the selection is done with great fidelity to the theoretical model, an act of sampling is taking place.

The third arrow, “analyses are performed within the representational system,” is deceptively simple. The key questions to ask are, What is meaningful within the representational scheme? What can be said about the quality of the inferences drawn? It should be obvious that great care must be taken in subjective analyses. But it is equally important to take comparable care in the case of ostensibly objective quantitative analyses. The results of data analyses will be no better than the quality of the data that are subjected to analysis. For example, there may be a statistically significant difference in the performance levels of two classes on an outcome measure. But is the cause a difference in the two instructional treatments, the fact that they were taught by different teachers, or (if the same teacher taught both) either the enthusiasm of the teacher for one treatment over the other or the fact that one course was taught in the morning and the other right after lunch? Many of the variables that affect performance go unmeasured in statistical analyses. I shall review the issue of *trustworthiness* of analyses in the next section.

The fourth arrow is the mirror image of the second. Just as the passage from a conceptual model to a representational system involves data selection and reduction, the return from the representational system to the conceptual model involves significant

acts of interpretation. A difference in two measures might be statistically significant, for example—but is it meaningful or consequential? If so, along what lines? Or, to take a qualitative example, suppose the representational system involves coding student-to-student dialogue in classroom interactions. If the coding scheme focuses on the frequency of interactions and dialogic “take-up,” one might, for example, get a picture of a highly collaborative working group. But what was the collaboration about? An analysis of the content of the interactions might or might not indicate that the group was focused productively on important mathematical issues. Thus the extrapolation from representational system to the conceptual system must be made with care.

Finally, there is the return from the conceptual model (Box E) to the “real world” situation—the original Box A, now Box F. Here too there is at least interpretation, and perhaps extrapolation. or example, what are test scores taken to mean? History has made clear the consequences of confusing test scores such as IQ tests with the traits they ostensibly represent, such as “intelligence.” Likewise, whether one attributes mathematical proficiency to a good score on the SAT-9 or the Balanced Assessment tests can make a big difference. And, saying “students from Curriculum X did (or did not) outperform students from Curriculum Y on this test” is a very different thing than saying “Curriculum A is (or is not) better than Curriculum B.” I address the idea of *generality* in the next section.

PART 2

ASPECTS OF RESEARCH—ISSUES OF TRUSTWORTHINESS, GENERALITY, AND IMPORTANCE

In this section I discuss three fundamental issues related to all research studies. Those issues can be posed as questions that can be asked about any study:

- Why should one believe what the author says? (the issue of trustworthiness)
- What situations or contexts does the research really apply to? (the issue of generality, or scope)
- Why should one care? (the issue of importance)

The following diagram (Figure 3.3) may be useful in thinking about the ultimate contributions made by various studies or bodies of studies:

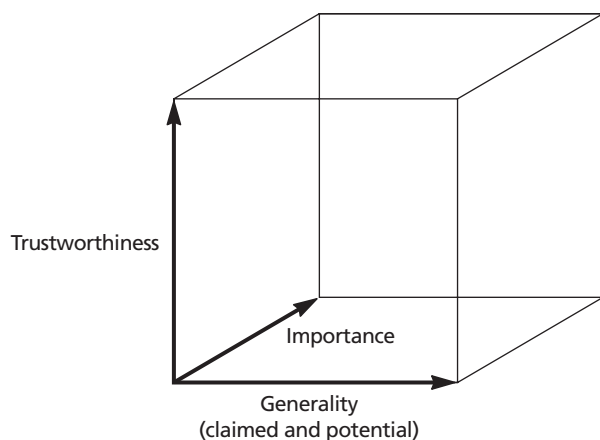


Figure 3.3 Three important dimensions along which studies can be characterized. (Reprinted, with permission, from Schoenfeld, 2002, p. 450)

As an example, a superbly written and insightful autobiographical account might score high on the trustworthiness and importance dimensions, while low on generality—although it might, by virtue of elaborating what might be seen as common experience, have intimations of generality in some ways. The same could be said of some rich descriptions of rather singular events, such as Fawcett’s (1938) description of a 2-year-long geometry course. Fawcett’s course served as an important and well-documented existence proof: It *is* possible to teach a course in which students develop certain kinds of understandings of the mathematical enterprise. A technically proficient comparison of two instructional treatments might rank reasonably well on the trustworthiness dimension. Such a study might or might not be important, and might or might not have potential generality, depending on the particulars of the situation. Myriad dissertation studies with conclusions of the form “students using materials that I developed scored better than students using a standard text” score low on both. However, individual and collective evaluations of some of the newer and widely used mathematics curricula begin to suggest generality and, as the findings mount, some importance (see, e.g., Senk & Thompson, 2003). Studies that are technically proficient but flawed along the vertical pathways illustrated in Figure 3.1 are *not* trustworthy. By considering the three dimensions in Figure 3.3 one can get a sense of what studies can make as their contributions. In what follows I address the three dimensions one at a time.⁷

Trustworthiness

*What did the President know,
and when did he know it?*

—United States Senator Sam Ervin,
during the Watergate hearings

The United States Senate’s impeachment hearings of President Richard Nixon, known as “the Watergate hearings,” were one of the defining moments of the American presidency. Richard Nixon’s presidency was at stake. Time and time again, Senator Sam Ervin boiled things down to their evidentiary core. The answer to his oft-repeated question “What did the President know, and when did he know it?” would, once arrived at with a significant degree of certainty, determine Nixon’s fate as president.

What mattered in Nixon’s impeachment trial—what matters in all legal proceedings—is the idea of “a significant degree of certainty.” Legal matters, unlike matters of mathematics, are not axiomatic. Typically, one does not resolve complex legal issues with absolute certainty; rather, the standard is whether particular claims have been substantiated “beyond a reasonable doubt.” The underlying evidentiary issues are: What warrants are given for the claims being made? How believable and trustworthy are they? How robust are the conclusions being drawn from them?

Thus it is in mathematics education as well. As indicated in the previous section, once one has developed a theoretical orientation toward a situation, the core processes of empirical research are the gathering, representation, manipulation, and interpretation of data. A core question regarding the quality of the conclusions drawn from the research is, how *trustworthy* are each of those processes?

In what follows I elaborate on a number of criteria that are useful for examining the quality of empirical and theoretical research. Aspects of trustworthiness include the following, which are discussed below:

- Descriptive and explanatory power
- Prediction and falsification
- Rigor and specificity
- Replicability
- Triangulation

⁷ The three aspects of research I have called “dimensions” are not truly independent, of course. What I am offering here is a heuristic frame and an argument that attention to all three aspects of research is essential.

Descriptive and Explanatory Power

Descriptive power denotes the capacity of theories or models to represent “what counts” in ways that seem faithful to the phenomena being described. Descriptions need not be veridical but they must focus on what is consequential for the analysis. To give a classical mathematical example, consider a typical related-rates problem that involves a ladder sliding down the side of a building. The building is assumed to be (or is explicitly stated to be) vertical and the ground horizontal. In the diagram representing the situation, the ladder, the building, and the ground are represented as *lines* that comprise parts of a right triangle. What matters for purposes of the desired analysis are their lengths, and the way the ladder is moving. That information, properly represented and analyzed, enables one to solve the given problem; that information and nothing else is represented in the diagram and the equations one derives from it. What does not matter (indeed, what would be distracting in this context) includes how many rungs are on the ladder or how much weight it might support. In a different context, of course, such things would matter quite a bit. The issue of descriptive power, then, is, does the research focus on what is essential for the analysis, in a way that is clear and compelling?⁸

Explanatory power denotes the degree to which a characterization of some phenomenon explains how and why the phenomenon functions the way it does. Before getting down to educational specifics, I note that explanatory power, like many of the properties described in this section of this chapter, is an issue in all analytic fields. In mathematics, many people will prefer a constructive proof to an existence argument that employs proof by contradiction. The latter says that something exists, but not how to build or find it; the former, in providing a “blueprint” for finding or building it, provides more detailed guidance regarding how and why it exists.

A fundamental issue here is the distinction between correlation—the fact that X and Y tend to co-occur—and what I shall call explanation at a level of mechanism, an attempt to say how and why X and Y are linked. (Note that causal or constraint-based explanations are the traditional forms of explanation at a level of mechanism, but that statistical or probabilistic models also provide such explanations.)

As one example, consider Mendelian genetics as an expression of a theory of heritability. Surely, before Mendel, the notion that traits appear to be passed from generation to generation had been observed. Indeed, the observation had some power—but that power was limited. With Mendel came a suggestion of mechanism—the idea that (at least in the case of specific traits of relatively simple plants such as peas) genes determine the characteristics of those traits. And with that suggestion came the possibility of experimentation and the refinement of the underlying theoretical notions.

Productive, data-based explanations need not be right in all detail—but they do need to be taken seriously in order for the field to progress. For example, Galileo (at great personal cost) advanced a solar-centric theory of planetary motion, which was more accountable to data and explanation than the faith-based earth-centric model it supplanted. Centuries later this work was established on a much more solid footing when Newton proposed a theory of gravitation that had the potential to take things further. His theory explained, for example, why planetary orbits are elliptical rather than circular. (To anticipate arguments made later in this section, it should also be noted that issues of prediction and falsifiability make a big difference. They allow for the evaluation of competing theories, and the refinement of explanations. And, to emphasize a point made earlier, increased explanatory power may come from a shift in conceptual-analytic frameworks.)

One example within mathematics education concerns the relationship between teachers’ classroom practices and student learning. In the 1970s and 1980s a dominant approach in classroom research was the process-product paradigm, in which the data gathered focused on (a) tallies of specific classroom practices (e.g., time on task, worksheet use, asking certain kinds of questions); (b) student outcomes, typically as measured on standardized tests; and (c) the statistical relationships between (a) and (b). The results of such analyses were typically of the form “students did better when teachers did X more frequently,” with the implication that it would be good for teachers to do X more often. Of course, researchers had ideas as to why some practices were more productive than others—

⁸ As I stress in Part 1 of this chapter, a great deal of what one attends to in empirical research depends on the conceptual-analytic frameworks that orient one to the phenomena at hand. Thus, when I say that descriptive power denotes the capacity of theories or models to represent what counts, I am making a contextual statement: a characterization or representation that has substantial descriptive power has that power relative to the conceptual-analytic framework being employed. Finding a “better description” may entail finding an alternative conceptualization.

but the research methods themselves did not explore how or why they worked.⁹

In contrast, more recent studies of classroom activities focus on notions such as discourse communities, practices, and classroom norms and their impact—students’ participation structures, their sense of the mathematics, and their understanding as reflected on a variety of assessments. Boaler (2002), for example, described in detail the philosophy and teaching practices at two schools, and the beliefs and understandings they are likely to engender. She interviewed students about their understandings regarding the nature of mathematics and their sense of the in-school enterprise. She examined their performance on standardized tests and on problem-solving tests of her own construction. As a result, readers understand that there are differences—and they have been provided evidence that substantiates a plausible story about how and why those differences came to be.

I should stress that I do not wish to privilege any form of explanation in this discussion. In a field that draws from the social sciences, the sciences, and the humanities, there will be various ways to try to explain how and why things happen. What is important is that the attempt be made and that claims must be held accountable to data. Thus, if one claims that teachers’ decision making is based on their knowledge, goals, and beliefs (e.g., Schoenfeld, 1998), one should offer models of that decision making and enough evidence to warrant such claims; if one claims that a particular kind of action or sequence of actions on the part of a teacher supports the development of an “oppositional culture” in the classroom, one should offer evidence of the growth of opposition and link that growth in plausible ways to the teacher’s actions.

Prediction and Falsification

I first describe issues of prediction and falsification in general, then with regard to educational research.

In the physical and other sciences, prediction has been the name of the game, and the potential for falsification a theoretical essential. Physics is the archetype, of course (and, alas, the source of much

inappropriate positivism in education). Newton’s laws, for example, say that under certain conditions, certain things will take place. Those predictions, and myriad others, serve both practice and theory. In practical terms, they allow people to build things that work consistently and reliably. Predictions are a mechanism for theoretical progress. As noted above, for example, a solar-centric explanation of planetary motion ultimately triumphed over the earth-centric view because the former explanation fit the data better than the latter. A theory of gravitation based on Newtonian mechanics and the inverse square law of gravitational attraction provides close descriptions and predictions of planetary motion, and allows for fine-grained predictions of planetary movement. The Newtonian view prevailed for centuries, but with some known anomalies—some of its predictions were not quite right. In one of the more famous incidents in the history of science, Einstein’s theory of relativity predicted that under the conditions of a solar eclipse, the planet Mercury would appear to be in a different location than Newtonian theory predicted. It took some years before the eclipse took place—and Einstein’s view was substantiated.

The point, as noted by Popper (1963), is that Einstein could have been proved wrong. Had Mercury not been where Einstein predicted, there would have been evidence of problems with Einstein’s approach. That is, Einstein’s theory was *falsifiable*. Evidence could be gathered that substantiated it, but also that cast it into doubt. According to Popper:

- Every “good” scientific theory is a prohibition: it forbids certain things to happen.
- A theory which is not refutable by any conceivable event is non-scientific.
- Every genuine test of a theory is an attempt to falsify it, or to refute it. Testability is falsifiability; ...
- Confirming evidence should not count except when it is the result of a genuine test of the theory; and this means that it can be presented as a serious but unsuccessful attempt to falsify the theory;

⁹ Indeed, subsequent studies showed some of the limitations of this kind of approach. In one study comparing highly effective teachers with other teachers, Leinhardt (1990) showed that the highly effective teachers (defined as those teachers whose students performed in the top performance range for their demographic group—whether “learning disabled,” “gifted,” or anything in between) uniformly established clear disciplinary routines for their classes at the beginning of the school year. In a companion study, however, Leinhardt et al. (1991) documented the limited utility of such findings. It turned out that many of the teachers had rather shaky mathematical knowledge, extending only a little bit beyond that of the curriculum; and that the standardized tests used to assess students (and teachers) were rather narrow and procedural. Thus, the other work could be reinterpreted as indicating that a high level of discipline is effective in helping teachers remain within their “comfort zone” and tends to produce students who have mastered procedures but may have little or no conceptual understanding.

- One can sum up all this by saying that the criterion of the scientific status of a theory is its falsifiability, or refutability, or testability. (Popper, 1963, p. 36)

The kind of predictions made in classical physics represent only one type of prediction, which has given rise to many misconceptions about predictions in the social sciences. Although “absolutist” arguments in domains such as physics may have face validity, theorists such as Toulmin (1958) argue that they do not apply in more complex, contextual situations in which human actions are involved. Toulmin seeks ways to mediate between absolutism on the one hand and relativism on the other; theorists such as Pickering (1995) replace notions of absolutism with concepts of scientific *practices* that are bound to the histories of the scientific communities in which they emerge. Simply put, the notions of prediction and falsification are unsettled. For that reason, I outline a range of prediction in the sciences and then education, while trying to preserve what I can of the notion of theory-testing.

As noted, the physical sciences sometimes support predictions of the type “under these conditions, the following will take place.” All of the traditional laws in the sciences afford such predictions. For example, the ideal gas law $PV = nRT$ says that under certain conditions, the values of three of the variables P , V , n , and T determine the value of the fourth. Likewise, the creation of the periodic table as a theoretical characterization of atomic structure supported predictions about the existence of elements that had not yet been found. These kinds of predictions are deterministic.

The life sciences often entail predictions that are not deterministic in the sense above, but which still describe an expected state. Consider, for example, predator-prey models of animal populations. In simplest terms, predators flourish when they are few in number and there are many prey, but they diminish in number when they are densely crowded and few prey remain as food. Prey flourish when they are few in number and predators are few in number, but they diminish in number when they are overcrowded or the number of predators is large. All of these states, and population change rates, can be quantified, at which point the predator-prey model will predict changes in the sizes of both populations. The fates of individual animals are not determined in such models,

but trends are. In this sense, predictions are not absolute: A theory is not “true” or “false” in the sense that it proposes a universal, and one counterexample serves to invalidate it. Nonetheless, the theory does give rise to models, and the accuracy of the models (and the theory that generated them) can be judged by their fidelity to actual data. Similarly, Mendelian genetics predicts the percentages of offspring that will have specific traits, but not (except in the case when $p = 1$) the traits of individuals; it yields probability distributions regarding the traits of individuals. Yet, at least cumulatively, this is a strong form of prediction. (And the predictions led to refinements of the theory—for example, data that did not conform with theory led to the uncovering of linked traits.)

A weaker form of prediction has to do with *constraints*. Here, evolutionary theory is a primary example. Under ordinary circumstances, evolutionary theory cannot specify how an organism will evolve—just that it will be responsive to its environment.¹⁰ However, evolutionary theory does impose constraints about the ways on which organisms change over time: There is, for example, greater complexity and differentiation. Thus, according to the theory, certain evolutionary sequences are plausible and others are implausible (if not impossible). Thus the theory can be challenged by empirical evidence. Every exploration of geological strata offers potential disconfirmation.

And then, of course, there is weather prediction, which is not yet a science. However, various models of climatic behavior can be assessed and refined. The main point here is not that those in the business of building climatic models “have it right”—far from it. But, because each model allows for predictions, and variations in the models support different predictions, the careful examination of the predictions made and their relation to the theory can help to improve prediction and the underlying theory.

I now turn to educational issues. Of course, only some empirical work in the social sciences or in education involves prediction; substantial bodies of empirical work in education (e.g., autobiographical reports and descriptive studies) involve no claims beyond those made about the evidence discussed. However, I note that a great deal of descriptive work contains implicit claims of generality, and thus of prediction. As soon as there is the implication that “in similar circumstances, similar things happen,” one is,

¹⁰ There are exceptions in simple cases of natural selection, where the relationship between certain animal traits and the environment is clear. In one classic case, for example, it was possible to predict that as pollution darkened local trees, the population of moths would darken because darker moths on trees were less visible to predators than lighter moths. (And, when pollution was reversed, the population balance changed back in the other direction.)

at least tacitly, making a prediction (see the section on *generality* below).

Within educational research, as in the examples from the sciences discussed above, there is a wide range of prediction. Randomized controlled trials offer one kind of prediction: The assumption underlying experimentation is that under conditions similar to the circumstances of experimentation, results similar to the results of experimentation will be obtained. This is not unique to statistically oriented studies, however: The same is often true of “rich, thick” anthropological descriptions. A main purpose of descriptions of productive classroom discourse structures is to explain not only how things took place (descriptive power) but why students learned what they did (explanatory power), thus enabling others to try similar things in the hope of obtaining similar results (prediction). The more that such descriptions and claims can be made rigorous, the more likely they are to have a productive impact on practice and to serve theory refinement.

Here are some illustrative examples. Brown and Burton’s (1978) study of children’s arithmetic “bugs” described the authors’ analyses of children’s errors in base-ten subtraction. Brown and Burton found that children’s patterns of errors were so systematic (and rule-based) that, after giving children a relatively short diagnostic test, they could predict with some regularity the incorrect answers that those students would produce on new problems. Brown, Burton, and colleagues (Brown & Burton, 1978; Brown & VanLehn, 1982; VanLehn, Brown, & Greeno, 1984) provided well-grounded explanations of why students made the mistakes they did. But prediction added a great deal to their work. First, the data provided clear evidence of the power of cognitive models: If you can predict the incorrect answers a student will produce on a wide range of problems before the student works them, you must have a good idea of what is going on inside the student’s head! Second, prediction played a role in theory refinement: If predictions do not work, then one has reason to look for alternative explanations. Third, a fact often overlooked by those who view the “buggy” work as overly simplistic and mechanistic is that this work provided clear empirical evidence of the constructivist perspective. In 1978, many people believed in the simplistic idea that one teaches something (perhaps in multiple ways) until the student “gets it,” and that nothing has been “gotten” until the student has learned whatever it was to be learned. Brown and Burton showed that students had indeed “gotten” something: They had developed/learned an incorrect interpretation of what they had been taught and used it with consistency. That is, what they did

was a function of what they perceived, not simply what they had been shown. If that isn’t data in favor of the constructivist perspective, I don’t know what is. (See also Smith, diSessa, & Roschelle, 1993/1994.)

Another famous study, predating Brown and Burton, is George Miller’s 1956 article “The Magical Number Seven, Plus or Minus Two.” After numerous observations in different intellectual domains, Miller hypothesized that humans have the following kind of short-term memory limitation: We can only keep between (roughly) five and nine things in working memory at the same time. That hypothesis gives rise to simple and replicable predictions. For example, carrying out the multiplication

$$\begin{array}{r} 634 \\ \times 857 \\ \hline \end{array}$$

requires keeping track of far more than nine pieces of information. According to Miller, it would be nearly impossible to look at these numbers, close your eyes, and compute their product. This has been shown time and time again—once at a talk I gave to more than a thousand mathematicians, many of whom, at least, can do their arithmetic. (It *is* possible to do such products if one knows certain arithmetic shortcuts, or if one rehearses some of the subtotals so that they become “chunked” and only use up one “slot” in short-term memory. But such instances are rare.) Note that the element of falsifiability of Miller’s work is critically important. If a nontrivial fraction of the people given tasks like this succeeded at them, then Miller’s claim would have to be rejected or modified.

A third class of examples consists of process models of cognitive phenomena. In physics, diSessa’s models of *phenomenological primitives* and the ways they develop and shape cognition provide predictions of the ways people will interpret physical phenomena, and tests of the adequacy of the theory. In a different domain, the Teacher Model Group at Berkeley (see, e.g., Schoenfeld, 1998) constructed detailed models of a range of teachers’ decision making. One theoretical claim made by the group is that a teacher’s in-the-moment actions can be modeled as a function of the teacher’s knowledge, goals, beliefs, and a straightforward decision procedure. Each new case explored (an inexperienced high school mathematics teacher conducting a more or less traditional lesson, an experienced high school teacher conducting an innovative lesson of his own design that largely went “according to plan,” and an experienced third-grade teacher conducting a lesson in which the agenda was coconstructed with the students) tested the scope and adequacy of the theory. Again, the idea is straightforward. The more that theoretical claims can

be examined and tested by data, the more there is the potential for refinement and validation.

Rigor and Specificity

At this point, little needs to be said about the need for rigor and specificity in conducting empirical research. The more careful one is, the better one's work will be. And the more carefully one describes both theoretical notions and empirical actions (including methods and data), the more likely one's readers will be able to understand and use them productively, in ways consistent with one's intentions.

Precision is essential; a lack of specificity causes problems. One historical example is the potentially useful construct of *advance organizer* introduced by Ausubel (see, e.g., Ausubel, 1960). Over the years, a large number of studies indicated that the use of advance organizers (roughly speaking, top-level introductions to the content of a body of text) improved reading comprehension. But an equally large number of studies indicated that the use of advance organizers did not make a difference. Why? On closer examination, the construct was so loosely defined that the advance organizers used in various studies varied substantially in their characteristics. Thus the results did as well. Similarly, a theory that involves terms such as *action*, *process*, *object*, and *schema* needs to say what those terms are, in clear ways. And, of course, appropriate detail and warrants need to be provided in discussing those terms. One example of a productively used term is the *didactical contract* as used by the French (see, e.g., Brousseau, 1997). The term has a specific meaning, which provides a useful backdrop for many such studies. In contrast, the generic use of the term *constructivist* (for example, in the meaningless term *constructivist teaching*) has not been helpful.

Note that rigor does not simply mean rigor in the use of one's data analyses (the path denoted by Arrow 3 in Figure 3.1). It means attending carefully to all of the pathways from Box A to Box F in Figure 3.1.

Replicability

Every person is different; every classroom is different. How can one possibly speak of replication in education? The idea seems strange. One might replicate experiments of some types—but how many educational researchers do experimental work? And, if every classroom is different, what does it mean to replicate someone else's research study?

One way to think about the issue of replicability is to think about generality. As noted above and as will be elaborated below, there is a tacit if not explicit aspect of generality to most studies: the expectation is that the lessons learned from a study will apply, in some way, to other situations. To that degree, some key aspects of those studies are assumed to be replicable. The issue, then, is, how does one characterize those aspects of a study—in enough detail so that readers can profit from the work in the right ways, and so that they can refine the ideas in it as well? Thus, for example, studies like those referred to in the section on prediction (Brown and Burton's studies of arithmetic bugs, Miller's article on the magic number seven plus or minus two, and the Teacher Model Group's article on teachers' decision making) all involve prediction; hence they should be potentially replicable as well.

Multiple Sources of Evidence (Triangulation)

In a well-known article published in 1962, Martin Orne introduced the concept of the *demand characteristics* of an environment via this personal anecdote:

A number of casual acquaintances were asked whether they would do the experimenter a favor; on their acquiescence, they were asked to perform five push-ups. Their response tended to be amazement, incredulity and the question 'Why?' Another similar group of individuals were asked whether they would take part in an experiment of brief duration. When they agreed to do so, they too were asked to perform five push-ups. Their typical response was "Where?"

The idea, later abstracted as *context effects*, is that context makes a difference: People will do things in some circumstances that they might do differently (or not at all) in other circumstances. Some behavior is purely artifactual, *caused by* the particulars of the circumstances.

For this reason, the use of multiple lenses on the same phenomena is essential. In some cases, that means employing multiple methods to look at the same phenomena. Thus, observations, questionnaires, and interviews can all be used to challenge, confirm, or expand the information gathered from each other. Similarly, behavior that manifests itself in some contexts may or may not manifest itself in other contexts.¹¹ One should constantly be on guard with regard to such issues.

¹¹ *Context* is meant to be taken very broadly here. Orne's example is a case in point: a question asked of a friend is very different from a question asked of a voluntary participant in an experiment. Similarly, a question asked of individual experimental participants may be treated very differently by *pairs* of subjects.

Generality and Importance

A central issue with regard to any research is its scope, or generality—the question being, how widely does this idea, or theory, or finding, actually apply? A second, equally critical issue is importance. A study may or may not be warranted to apply widely. But whether it is or is not, one must ask: Does it matter? Just what is its contribution to theory and practice?

The concept of generality is slippery, because in many papers a few instances of a phenomenon are taken as an indication of the more widespread existence (or potential existence) of that phenomenon. Thus, it may be useful to distinguish the following kinds of generality:

- The *claimed generality* of a body of research is the set of circumstances in which the author of that work claims that the findings of the research apply.
- The *implied generality* of the work is the set of circumstances in which the authors of that work appear to suggest that the findings of the research apply.
- The *potential generality* of the work is the set of circumstances in which the results of the research might reasonably be expected to apply.
- The *warranted generality* of the work is the set of circumstances for which the authors have provided trustworthy evidence that the findings do apply.

There is often a significant gulf between these. And there is often slippage between evidence-based claims (warranted using the constructs within the conceptual-analytic system—the path from Box C to Box D in Figure 3.1) and explicit or implicit claims of generality (the path from Box A to Box F in Figure 3.1). Thus, for example, an experimental study may compare outcomes of a new instructional method with an unspecified control method. If the study is done by the author using a measure that he or she developed for the purpose of this study, the warranted conclusion may be “students in this context, taught by the instructor who developed the materials and tested using an assessment similar in kind to the materials, did better on that assessment than students who took a somewhat different course.” The warranted generality is thus quite low. The claimed generality might be “students in the experimental group outperformed control students,” with the implied generality being that other students who experience the new instructional method will similarly outperform other students. The potential generality is actually

unknown—it depends on the contexts, the character of instruction in experimental and control groups, the attributes of the assessment, and more.

A key point is that the warranted generality of a study and its importance are not necessarily linked. For example, Wilbur and Orville Wright demonstrated (with substantial trustworthiness!) on December 17, 1903, that a heavier-than-air machine could take flight and sustain that flight for 59 seconds. That was the warranted generality. However, that flight was critically important as an existence proof. It demonstrated that something *could* be done and opened up fertile new territory as a result. Likewise, many studies in mathematics education can be seen as existence proofs. They may demonstrate the existence of a phenomenon worthy of investigation (e.g., early studies of metacognition or beliefs) or of instructional possibilities (e.g., early studies of Cognitively Guided Instruction). The findings of studies with existence proofs are not yet general—but there may be the potential for them to be.

Conversely, a body of research that has broad generality can turn out to be theoretically sterile and to have little practical value. One such example is a spate of studies conducted in the 1980s, which showed that a substantial proportion of those examined produced the equation $P = 6S$ instead of $S = 6P$ in the problem

Using the letter S to represent the number of students at this university and the letter P to represent the number professors, write an equation that summarizes the following sentence: “There are six times as many students as professors at this university.”

(Clement, 1982; Clement, Lochhead, & Monk, 1981; Rosnick & Clement, 1980). The phenomenon was documented repeatedly, and various attempts were made to get people to do better at such problems. However, after all was said and done, the field had no real understanding of why people made this mistake, and no effective mechanisms for preventing or fixing it. Thus—given that this body of work was general, but did not produce significant understandings or applications—we see that generality and importance are at least somewhat independent dimensions.

In what follows I briefly describe the (warranted-to-potential) generality of a number of studies and discuss the importance of those studies. Broadly speaking, the studies are clustered by their degree of generality. For the most part I have chosen studies whose trustworthiness is well established. Judgments of importance reflect the author’s perspective but are also included as a reminder that importance is an essential dimension to take into account.

Studies of Limited Warranted Generality

There is a large class of studies for which there is limited warranted generality, but which are worth noting because of their current or potential importance. Such studies may be of interest because they offer existence proofs, bring important issues to the attention of the field, make theoretical contributions, or have the potential to catalyze productive new lines of inquiry.

An autobiographical report may be of interest because it is motivational, or, like other historical documents, it illuminates certain historical decisions or contexts. (For example, what was the historical context for the *Brown v. Board of Education* decision?¹²) In cases such as these there are no warranted conclusions beyond those described, but there may be lessons to be learned nonetheless. (Note that there is an implied aspect of generality to historical studies in Santayana's oft-quoted statement that "those who cannot learn from history are doomed to repeat it.")

Another singular event is the existence proof—a study that shows that something is possible and may elaborate on the means by which it became possible. For example, Harold Fawcett's classic 1938 volume *The Nature of Proof* demonstrated that it was possible to create a classroom mathematical community in which students engage meaningfully in many of the activities in which mathematicians engage (e.g., making definitions and deriving results logically from them). Early papers on Cognitively Guided Instruction (CGI) showed that it is possible to capitalize on young students' informal models and situational understandings to help them produce a wide range of meaningful solutions to word problems (see, e.g., Carpenter, Fennema, & Franke, 1996). Moll, Amanti, Neff, and Gonzalez (1992) showed how it is possible to develop instruction that builds on the kinds of cultural knowledge and traditions that students have in their out-of-school lives. Gutstein (2003) showed that it is possible to teach a mathematics course in which students do well on traditional measures of mathematical performance and become engaged as social activists as well. Boaler (in press) showed that it is possible to create a discourse community in a high school mathematics classroom with a large percentage of low-SES and ESL students in which there are high mathematical standards—and the classroom accountability structures are such that the students hold each other accountable for producing meaningful and coherent mathematical explanations.

Each of these studies had a high level of trustworthiness, in that it satisfied many of the criteria discussed in the previous section of this chapter. Each had relatively low warranted generality, in that it made the case that something had happened in specific circumstances, with a small number of students. But each showed that something could be done, opening up a previously undocumented space of possibilities. In that sense, they rate high on the potential importance scale (and may, indeed, pave the way to phenomena that become more general). Over time, some (e.g., Fawcett's study) have languished as inspirational examples not taken up on a large scale; some (e.g., Cognitively Guided Instruction) have had increasing impact.

Other studies bring important phenomena to readers' attention. They may suggest as-yet-undocumented generality, opening up arenas for investigation. Thus, for example, Cooney's (1985) case study of "Fred" showed that a teacher may espouse a set of values that sounds compatible with a particular pedagogical direction (in this case, "problem solving") but may interpret those terms in contradictory ways and thus act in ways contrary to (the normative interpretation of) those values. Cohen (1990) showed that a teacher may adopt the rhetoric of a particular pedagogical approach and believe that she is implementing that approach, while in fact assimilating (in the Piagetian sense) many of the surface aspects of that approach to her long-established pedagogical practices. Eisenhart et al. (1993) showed that a particular teacher, subject to the pressures of her environment, wound up teaching in a way differently than she intended. Each of these studies made a well-documented case that something important was happening in one specific set of circumstances. But each study also made a plausibility case that the phenomenon under discussion was more widespread.

A classic example of this genre is Heinrich Bauersfeld's 1980 article "Hidden Dimensions in the So-Called Reality of a Mathematics Classroom." Bauersfeld revisited the data from a dissertation by Shirk (1972), which had focused largely on the content and pedagogical goals of beginning teachers. Examining the same data, Bauersfeld focused on the social dimensions of the classroom. His analysis addressed four areas of classroom research that had hitherto received a negligible amount of attention: "the constitution of meaning through human interaction, the impact of institutional settings, the development of personality, and the process of reducing classroom

¹² In a 1954 decision that had far-reaching implications, the United States Supreme Court declared that the establishment of segregated public schools was unconstitutional.

complexity” (Bauersfeld, 1980, p. 109). Although the warranted generality of the findings was small, the face-value typicality of the classroom he explored made at least a plausibility case that the phenomena under investigation were relatively widespread. The phenomena were not (yet) claimed to be general but were seen as worthy of investigation.

In a similar way my 1988 article “When Good Teaching Leads to Bad Results: The Disasters of Well Taught Mathematics Classes” presented a trustworthy account of one classroom, in which students had developed a series of counterproductive beliefs about the nature of mathematics as a result of receiving well-intentioned but narrow instruction that focused on preparing the students with particular procedural skills to do well on a high-stakes test. By virtue of the typicality of the instruction in the focal class, and the causal explanation of how the classroom practices resulted in specific outcomes, the article had substantial potential generality.

Other papers may be important because of their theoretical or methodological contributions. For example, diSessa’s 1983 chapter on *phenomenological primitives* introduced a new way to conceptualize the development of conceptual structures and reframed the theoretical debate on the nature of conceptual understanding. Brown and colleagues (e.g., Brown, 1992; Brown & Campione, 1996) introduced and elaborated on the notion of design experiments, describing nonstandard instructional practices and a novel way to think about data gathering and analysis in the context of such instruction. Yackel and Cobb’s (1996) discussion of sociomathematical norms provided a theoretical tool (which will have widespread application) for examining the ways in which “taken-as-shared” classroom practices can shape individual cognition. Similarly, Cobb and Hodge’s (2002) characterization of diversity as a function of differences in cultural practices rather than a measure of “spread” of some particular demographic variable, although not yet widely used, has the potential to reframe studies of diversity in productive ways. Therein lies its importance.

Studies Where Some Degree of Generality is Claimed and/or Warranted

The simplest claims of generality are statistical. Consider, for example, the following datum from Artigue (1999):

More than 40% of students entering French universities consider that if two numbers A and B are closer than $1/N$ for every positive N, then they are not necessarily equal, just infinitely close. (p. 1379)

In and of itself, this statement (whose generality is warranted in statistical terms) may or may not seem important. However, this and other data (e.g., that a significant number of such students believe that the decimal number 0.9999... is infinitely close to but less than 1) help to establish realistic expectations regarding the knowledge of entering calculus students, and expectations about necessary focal points of instruction.

In a similar way, some findings from the U.S. National Assessment of Educational Progress (NAEP) pointed to serious across-the-board issues in American mathematics instruction. Perhaps the best-known single item from NAEP is the following (Carpenter, Lindquist, Matthews, & Silver, 1983):

An army bus holds 36 soldiers. If 1128 soldiers are being bussed to their training site, how many buses are needed?

The calculation is straightforward: 36 goes into 1128 thirty-one times, with a remainder of 12. Hence 32 buses are needed. Here is the frequency of responses from the stratified nationwide sample of students who worked the problem:

23%	32
29%	31R12
18%	31
30%	other

The vast majority of students who worked the problem performed the necessary computation correctly. But then, more than a third of the students who did the computation correctly wrote down an impossible answer (the number of buses needed must be an integer, after all!), and a substantial number of others rounded down instead of up, which makes no sense in the given context.

Once again, the statistics attest to the generality of the finding. What made this finding *important* was the way it fit into an emerging body of research findings. In the early 1980s, researchers were coming to understand the importance of beliefs (Schoenfeld, 1983), both in terms of their impact on students’ mathematical performance and in terms of their origins in classroom practices. Many of the arguments in support of beliefs had been local: In a specific class, students experienced these specific things; they developed the certain understandings of the mathematical enterprise as a result; and they then acted in accord with those understandings. In small-scale studies the argument had been made that students had come to experience word problems not

as meaningful descriptions of mathematical situations, but as “cover stories” for arithmetic operations; they understood their task as students as uncovering the relevant mathematical numbers and operations in the cover story, performing the operations, and writing the answer down. In this view, the ostensible reality described in the cover story played no role in the problem once the numbers and operations had been extracted from it. The data on the NAEP provided evidence that the problem was a national one. Therein resides the importance of the research.

In a similar way, I first explored the role of metacognition in mathematical problem solving in the research described in part 1 of this chapter. The data in the original experiment were suggestive: more than half of the videotapes I examined were of the type shown in Figure 3.2. This finding was potentially important. The question was whether it would turn out to be general. The number of cases I had examined was small, but *a priori* there was nothing to suggest that they might be atypical. When the kinds of effects demonstrated in my early studies were replicated elsewhere and tied to emerging theories of the importance of metacognition in other domains (e.g., Brown, 1987), the generality and importance of the phenomenon became increasingly clear. Note that this kind of sequence is typical: Individual studies point to aspects of a potentially interesting or general phenomenon, and an expanding body of studies refine the idea over time.

A set of ideas that is in its early stages, suggesting generality but not yet fully explored or validated, has to do with the attributes of particular kinds “learner-centered” learning environments. Engle and Conant have suggested that

productive disciplinary engagement can be fostered by designing learning environments that support (a) problematizing subject matter, (b) giving students authority to address such problems, (c) holding students accountable to others and to shared disciplinary norms, and (d) providing students with relevant resources. (2002, p. 399)

The authors provided a detailed analysis of a series of discussions in a “Fostering a Community of Learners” classroom (Brown & Campione, 1996), demonstrating how each of the four themes identified above played out in that classroom. They then briefly reexamined discourse patterns in two other types of instruction known for engaging students productively

with disciplinary content: Hypothesis-Experiment-Instruction Method Classrooms (Hatano & Inagaki, 1991; Inagaki, 1981; Inagaki, Hatano, & Morita, 1998) and the Chèche Konnen project (National Science Foundation, 1997; Rosebery, Warren, & Conant, 1992). This evidence suggests the potential generality of the findings. Readers’ familiarity with other such environments—e.g., Scardamalia and Bereiter’s “Knowledge Forum” classrooms (Scardamalia, 2002; Scardamalia & Bereiter, 1991; Scardamalia, Bereiter, & Lamon, 1994) or my problem-solving courses (Schoenfeld, 1985) may add to the sense of generality and importance of the findings. Time will tell.

A final kind of argument with some generality and potential importance, which is a bit further along than the Engle and Conant study, is done by aggregating over a collection of studies, each of which offers consistent data. Such an argument is made in Sharon Senk and Denisse Thompson’s 2003 volume, *Standards-Based School Mathematics Curricula: What are They? What do Students Learn?* Senk and Thompson present the data from studies evaluating a dozen *Standards*-based curricula. A careful reader could find reasons to quibble with a number of the studies. For example, many of the assessments employed in the studies were locally developed.¹³ Hence one could argue that some of the advantages demonstrated by the *Standards*-based curricula were due in part to assessments tailored to those curricula. In addition the conditions of implementation for the curricula (e.g., the preparation that teachers received before teaching the curriculum) may have been higher than one might expect in “ordinary” curriculum adoptions. Nonetheless, the pattern of results was compelling.

The book is divided into sections assessing curricula at the elementary, middle, and high school levels. Putnam (2003) provided the following summary of the evaluations of *Math Trailblazers*, *Everyday Mathematics*, *Investigations*, and *Number Power* (all elementary curricula):

The four curricula ... all focus in various ways on helping students develop conceptually powerful and useful knowledge of mathematics while avoiding the learning of computational procedures as rote symbolic manipulations.

The first striking thing to note about [them] is the overall similarity in their findings.

Students in these new curricula generally perform as well as other students on traditional measures of

¹³ A concern about locally developed measures is that there is the potential for bias (giving an unfair advantage to the “experimental” curriculum) whenever the developers of that curriculum are closely tied to the developers of an assessment used in the evaluation of that curriculum. Whether advertently or not, items on the assessment could be designed in ways that favor the experimental curriculum.

mathematical achievement, including computational skill, and generally do better on formal and informal assessments of conceptual understanding and ability to use mathematics to solve problems. These chapters demonstrate that “reform-based” mathematics curricula can work. (p. 161)

Chapell (2003) summarized the evaluations of *Connected Mathematics*, *Mathematics in Context*, and *Middle Grades MATH Thematics: The STEM Project* (all middle grades curricula) as follows:

Collectively, the evaluation results provide converging evidence that *Standards*-based curricula may positively affect middle-school students’ mathematical achievement, both in conceptual and procedural understanding. . . . They reveal that the curricula can indeed push students beyond the “basics” to more in-depth problem-oriented mathematical thinking without jeopardizing their thinking in either area (pp. 290–291)

Swafford (2003) examined the evaluations of the following high school curricula: *Core-Plus Mathematics Project*, *Math Connections*, the *Interactive Mathematics Program (IMP)*, the *SIMMS Integrated Mathematics Project*, and the *UCSMP Secondary School Mathematics Program*. She concluded that:

Taken as a group, these studies offer overwhelming evidence that the reform curricula can have a positive impact on high school mathematics achievement. It is not that students in these curricula learn traditional content better but that they develop other skills and understandings while not falling behind on traditional content. (p. 468)

Although many of the individual studies in Senk and Thompson (2003) may be problematic in some regard, the cumulative weight of the evidence in favor of the impact of the *Standards*-based curricula—and the absence from the literature of any comparable evaluations favoring the more traditional, skills-oriented curricula—is such that the findings, although preliminary, acquire a nontrivial degree of trustworthiness. There is a degree of generality in that all of the curricula examined, although differing substantially in style and content, all emphasized conceptual understanding, problem solving, and deeper engagement with mathematical concepts than the “traditional” curricula with which they were compared. And if students learn more, there is a *prima facie* case for importance.

A final example with a somewhat different kind of warrant is the claim made by Stigler and Hiebert that “teaching is a cultural activity (1999, p. 85).” As Stigler and Hiebert noted, the Third International

Mathematics and Science Study (TIMSS) video study performed a random sampling of classrooms selected from the larger TIMSS study, which was carefully constructed; the final video study sample included 100 German, 50 Japanese, and 81 U.S. classrooms that the authors claimed “approximated, in their totality, the mathematics instruction to which students in the three countries were exposed” (p. 19). Classroom videos were then coded for various kinds of detail, and the codings were analyzed. The data revealed that there was much less within-country variation than across-country variation—that is, that there were relatively consistent practices in each nation that differed substantially from the practices in the other nations. It goes without saying that one might code for things other than those captured by the coding scheme developed for the TIMSS study. But the main finding, backed up by descriptions of lesson content, appears trustworthy and (by virtue of the sampling) general across the three countries in the study. Subsequent replication studies are extending the findings. But, do the results matter? In this author’s opinion, the answer is a clear *yes*. Cross-national comparisons reveal what turn out to be cultural assumptions that one would not see if one kept one’s eyes within one nation’s borders. They reveal practices that challenge one’s notion of what may be possible and thus challenge one to think about the premises underlying instruction.

Studies Where Significant Generality, if Not Universality, is Claimed or Implied

On the one hand, as Henry Pollak once said, “there are no theorems in mathematics education.” On the other hand, there are some well-warranted results of significant generality. Many of these have to do with (attributions of) cognitive structures. Thus, for example, although one can certainly quarrel with major aspects of Piaget’s work, phenomena such as children’s development of *object permanence* and *conservation of number* are well established and well documented. Similarly, replication and follow-up studies of phenomena such as “the magic number seven plus or minus two” (Miller, 1956) have established the limitations of short-term memory as essentially universal. Cumulative bodies of literature with regard to the role and existence of schemata or their equivalents as forms of knowledge organization have established the utility of those constructs for describing people’s knowledge structures, likewise for studies of beliefs and metacognition as shapers of cognition. Cumulatively, these studies provide documentation of phenomena that are robust (hence

the research is trustworthy); that are essentially universal; that provide an understanding of individual cognition; and are thus clearly important.

Robust findings can, of course, be misused in applications to mathematics education. For example, the fact that people who are fluent in secondary school mathematics possess a large number of schemata for solving word problems does not imply that students should be directly taught a large set of those schemata. (See the exchange between Mayer, 1985, and Sowder, 1985, for a discussion of this issue.) Nor should a focus on cognitive structures in the preceding examples be taken in any way as suggesting that cognitive structures provide complete and coherent explanations of thinking, teaching, and learning. As noted in previous sections, many ideas from sociocultural theory have great promise in helping to unravel the complex interactive processes in which all humans engage. The arena is important, and many individual findings are trustworthy, but the field is still in its genesis. One expects to see many examples with a sociocultural emphasis—studies of discourse patterns, identity formation, etc.—in this category of studies when the third edition of this *Handbook* is published.

Finally, I note another category of potential universals: large-scale comparative studies of various types. As examples, cross-national studies such as TIMSS (see, e.g., Mullis et al., 1998; Mullis et al., 2000) and PISA (see, e.g., Lemke et al., 2004), provide trustworthy comparative information at nationwide levels about student performance. However, it should be remembered that any assessment is only as good as the items it uses (cf. Part 1 of this chapter). Moreover, any assessment reflects the mathematical values of its developers. Specifically, PISA and TIMSS emphasize different aspects of mathematical competence (PISA being more “applied”). Nations that do well on one assessment do not necessarily do well on the other. Other comparative studies (e.g., Lee, 2002) reveal trends in similarities and differences in the mathematical performance of various subgroups of larger populations. The same caveats apply.

Discussion

A major challenge in the conduct of educational research is the tension between the desire to make progress¹⁴ and the dangers of positivism and reductivism. The point of Part 1 of this chapter was that

reductivism, at least, comes with the territory: One’s explicit or implicit theoretical biases frame what one looks at, how one characterizes it, how one analyzes it, and how one interprets what one has analyzed. It is easy to slip into positivism as well.

It is in that spirit that I refer to the ideas in Part 2. The question addressed here is, what makes for good empirical work? I have argued that research must be examined along three somewhat independent dimensions. The first is trustworthiness—the degree of believability of the claims made in a piece of research. The core issue is: if claims are made, do the warrants for them ring true? As discussed above, there are various criteria for the trustworthiness of empirical research: a study’s descriptive power; its explanatory power; whether the claims made are in some sense falsifiable; whether the study makes predictions and, if so, how well those predictions fare; how rigorous and detailed the work is; whether the work has been described in ways that allow for attempts at replication and, if so, whether the findings are duplicated or extended; and whether the study offers multiple lenses on the same phenomena and multiple lines of evidence that converge on the same conclusions. Of course, not every criterion is relevant for every study; the nature of a study will determine how each of the relevant criteria is applied. But trustworthiness is an essential quality of good research, and attending to the criteria discussed in Part 2 will help one to make informed judgments about it.

Trustworthiness is not enough, however. Simply put, a study may be trustworthy (and thus publishable in the sense that it meets the appropriate methodological criteria) but trivial, along one or both of the two other dimensions: generality (scope) and importance. Assessing the generality of a result is often a delicate matter. Typically authors imply the generality of a phenomenon by tacitly or explicitly suggesting the typicality of the circumstances discussed in the study. Implying generality is one thing, however, and providing solid evidence for it is another. For that reason I have introduced the notions of *claimed*, *implied*, *potential*, and *warranted generality* as ways to think about the scope or generality of a study.

Importance is, of course, a value judgment. But it is an essential one, to be made reflectively.

¹⁴ If one defines *progress* in educational research as either clarifying/adding to the field’s knowledge or producing information or materials that help people do something more effectively, then just about all research is aimed at progress in some way. I intend this broad a definition.

PART 3
FROM IDEAS TO IMPLEMENTATION:
A RECONSIDERATION OF THE CONCEPT
OF “CLINICAL TRIALS” IN EDUCATIONAL
RESEARCH AND DEVELOPMENT

My purpose in Part 3 of this chapter is to describe a sequence of research and development activities employing both qualitative and quantitative methods that is intended to serve as a mechanism for the improved development and effectiveness of educational interventions. This effort is motivated both by a need, in general, for mechanisms to improve the curriculum design and evaluation process (Burkhardt, 2006; Burkhardt & Schoenfeld, 2003) and the wish to bring together the statistical and mathematics education communities in profitable ways (Scheaffer, 2006). I shall try to draw upon the best of three traditions that are often seen as in conflict with one another: educational research and design as craft, fundamental research in mathematics education, and calls for the scientific validation of curricular effectiveness through experimental means.

This part of this chapter differs substantially from the two that preceded it. Earlier in this chapter, my goal was generality. Part 1 offered a scheme for characterizing and reflecting on all empirical research, and Part 2 offered a framework that can be used to assess the quality of all empirical research studies. In a sense, the issues addressed in Parts 1 and 2 are timeless—questions of how to conduct research of increasingly high quality will always be with us. Here my focus is more narrow. In a chapter that addresses issues similar to those addressed here, Clements (2002) poses this question: “Why does curriculum development in the United States not improve?” (p. 599) One of my goals is to help solve this practical problem. The problem is timely for practical and political reasons, as will be discussed below. It is far from ephemeral, however. The general issue of experimentalism in education, and the scientism that bedevils it, have been problematic since long before I entered the field.

I begin with some caveats, in the hope of avoiding misinterpretations of what follows. I am *not* proposing that curriculum development is the solution to instructional problems in mathematics education; nor am I suggesting that it is the only (or even the primary) way in which mathematics instruction can be improved. A curriculum is a tool, and a tool is only effective in the hands of those who can wield it effectively. The fact that I do not focus here on issues of professional development, or on strategies for

improving instruction via professional development, should not be taken as an indication that I underestimate the value of professional development as a vehicle for improving instruction. In theoretical terms, I do not propose that curriculum development is the solution to the problem of “travel”—the ability of materials developed in one context to be used profitably in another (National Academy of Education, 1999). The more narrow issue that I address is, how can we refine the curriculum development process, so that the curricular tools that are developed are more likely to be effective (when used in the right ways)? The approach I take is to try to develop a serious educational analog to the idea of clinical trials in the development of medicines and medical treatments. The general approach outlined here should be applicable to all educational interventions.

I note by way of preface that I have been engaged in educational research for long enough to see a number of pendulum swings in what appears to be the eternal (and erroneous) dialectic between emphases on qualitative and quantitative methods. In the 1960s and 1970s, an emphasis on experimental and statistical methods was stifling the field. The problem was often a misapplication of good statistical methods. In many cases a researcher employed a “Treatment A *versus* treatment B” experimental design to compare curricular materials developed by the researcher with some form of control treatment. The assessment measures employed were typically standardized tests or measures developed by the author, each of which have their own problems (recall the discussion in Part 1 of this chapter). Sometimes the experimenter taught both the experimental and control classes, in which case the most significant (and unmeasured!) variable may have been teacher enthusiasm for the experimental materials, or the fact that one course was taught before lunch and one after. Sometimes different teachers taught the experimental and control classes, in which case teacher variation (again unmeasured) may have been the most important variable shaping outcomes. A number of scholars, including myself (see also Kilpatrick, 1978), have argued that there was as much scientism as science in educators’ attempts to adopt scientific methods in educational research—for example, I called for a moratorium on the use of factor analyses on “mathematical abilities” until researchers could explain what the factors determined by statistical methods actually meant.

Over the 1980s and 1990s a flowering of the cognitive and sociocultural perspectives led to a wide range of methods and approaches. Attention to these, combined with a reaction against the sterility of the earlier decades of “scientific” work, led to a

general abandonment of quantitative research in the field. This too is problematic, unless there is a firm commitment on the part of researchers to provide the warrants that allow research to be deemed truly trustworthy (cf. Part 2 of this chapter). In a column leading up to the American Educational Research Association annual program in 1999, program chair Geoffrey Saxe and I (1998) reproduced the following quote from a letter I had received:

At [Annual Meetings] we had a hard time finding rigorous research that reported actual conclusions. Perhaps we should rename the Association the American Educational Discussion Association. . . . This is a serious problem. We serve a profession that has little regard for knowledge as the basis for decision making. By encouraging anything that passes for inquiry to be a valid way of discovering answers to complex questions, we support a culture of intuition and artistry rather than building reliable research bases and robust theories. Incidentally, theory was even harder to find than good research. (p. 33)

At the federal level, at least, the pendulum has swung back in the direction of experimentalism in recent years. Here is an excerpt from the U.S. Department of Education's (2002) Strategic Plan for 2002–2007:

Strategic Goal 4: Transform education into an evidence-based field

Unlike medicine, agriculture and industrial production, the field of education operates largely on the basis of ideology and professional consensus. As such, it is subject to fads and is incapable of the cumulative progress that follows from the application of the scientific method and from the systematic collection and use of objective information in policy making. We will change education to make it an evidence-based field. We will accomplish this goal by dramatically improving the quality and relevance of research funded or conducted by the Department, by providing policy makers, educators, parents, and other concerned citizens with ready access to syntheses of research and objective information that allow more informed and effective decisions, and by encouraging the use of this knowledge. (U.S. Department of Education, 2002, p. 51)

Grover Whitehurst, Director of the Department of Education's Institute of Education Sciences, has followed through on that agenda. IES has funded the

What Works Clearinghouse, some of whose limitations are discussed in Part 1 and in Schoenfeld (2006a, 2006b). Early in his tenure, Whitehurst made his agenda clear in a presentation (2002) to the American Educational Research Association. Referring to Donald Stokes's argument in *Pasteur's Quadrant* (Stokes, 1997; see below), Whitehurst argued strongly in favor of his Institute's mission to support fundamentally applied work, in "Edison's quadrant":

Yes, the world needs basic research in disciplines related to education, such as economics, psychology, and management. But education won't be transformed by applications of research until someone engineers systems and approaches and packages that work in the settings in which they will be deployed. For my example of massed versus distributed practice, we need curricula that administrators will select and that teachers will follow that distributes and sequences content appropriately. Likewise, for other existing knowledge or new breakthroughs, we need effective delivery systems. The model that Edison provides of an invention factory that moves from inspiration through lab research to trials of effectiveness to promotion and finally to distribution and product support is particularly applicable to education.

In summary, the Institute's statutory mission, as well as the conceptual model I've just outlined, points the Institute toward applied research, Edison's quadrant. (Whitehurst, 2002; unpaginated.)

While paying homage to other forms of research, Whitehurst goes on to state his stance in clear terms: "Randomized trials are the only sure method for determining the effectiveness of education programs and practices."¹⁵ (Whitehurst, 2002)

But *sure* in what sense? As noted in the section of Part 1 called "a second example," how one interprets curricular effectiveness depends on how (if at all) one defines and attends to context. And where and when does one move from investigative studies to documentation of effectiveness? Must they be separate? In this section I address these issues.

Because much of the move toward evidence-based research in education is based on comparable work in the health sciences (see, e.g., the work of the Cochrane Collaboration, at <http://www.cochrane.org/>), I shall make my proposal by way of analogy to the model of clinical trials favored in medical and pharmaceutical research. It should be understood, however, that in using this analogy I am not accepting the reductive

¹⁵ Whitehurst goes on to say that randomized controlled trials (RCTs) are not appropriate for all research questions, and that other methods are appropriate to supplement the findings of RCTs. However, it is clear that "questions of what works are paramount for practitioners; hence randomized trials are of high priority at the Institute."

metaphor of curricular “treatments” in education as being anything like “simple” drug treatments for medical conditions. In what follows I wish to honor the complexity of the human and social processes that characterize the educational enterprise. What I wish to show is that the pharmaceutical enterprise is more complex than many of those who would analogize to education would have it. There is much to learn from the systematic study of proposed medical interventions, *if* the proper analogies are made, fully respecting the complexity of the educational process. If they are not, however—if steps in the process are skipped, or if proper attention is not paid to the complexities of educational research discussed in Part 1 of this chapter—then there is the potential for reductivism or scientism to do more harm than good.

I begin, with some prefatory comments about *Pasteur’s quadrant*. Stokes (1997) observed that, traditionally, basic and applied research have been seen as being at opposite ends of the research spectrum. This is captured by the one-dimensional picture in Figure 3.4.



Figure 3.4 Basic and applied research viewed as polar opposites.

Moreover, basic research has normally been thought of as preceding applied research, which precedes large-scale applications in practice. Stokes argued, however, that the basic/applied dichotomy was too simple. Although some research is almost exclusively basic (the work of physicist Niels Bohr is a classic example) and some is almost exclusively applied (Thomas Edison’s being a case in point), some critically important research has been framed in ways that allowed it to simultaneously make fundamental contributions to theory *and* to the solution of important practical problems. Pasteur’s research, which advanced the field of microbiology while helping to cure problems of food spoilage and disease, is the generic example of such work. Stokes proposed replacing the scheme in Figure 3.4 with a two-dimensional version, shown in Figure 3.5.

Whereas Whitehurst argued that an emphasis on Edison’s quadrant—in essence, focusing on “What Works” without necessarily asking why—is essential for making progress on curricular issues in mathematics education and other fields, I shall argue that a more balanced approach, with much more significant attention to work in Pasteur’s quadrant, will have a much larger payoff. To do this I shall make a direct analogy with clinical studies in evidence-

Research is inspired by:

		Consideration of Use?	
		No	Yes
Quest for Fundamental Understanding?	Yes	Pure Basic Research (Bohr)	Use-Inspired Basic Research (Pasteur)
	No		Pure Applied Research (Edison)

Figure 3.5 A 2-D representation of “basic” and “applied” considerations for research (Stokes, 1997, p. 73).

based medicine. The arguments made here have been influenced by a long-term collaboration with Hugh Burkhardt. An approach parallel to the one discussed here has been called the “engineering approach” to education (see Burkhardt, 2006; Burkhardt & Schoenfeld, 2003).

The U.S. Food and Drug Administration (FDA, 2005) described four “steps from test tube to new drug application review”: preclinical research and three phases of clinical studies (see <http://www.fda.gov/cder/handbook/>). In preclinical drug (or indeed, any medical treatment) research, a wide range of experimentation is done with animals, in order to determine that proposed interventions are likely to be useful and cause no harm to humans. The FDA then defines Phase 1, 2, and 3 studies as follows:

Phase 1 Clinical Studies. Phase 1 includes the initial introduction of an investigational new drug into humans. These studies are closely monitored and may be conducted in patients, but are usually conducted in healthy volunteer subjects. These studies are designed to determine the metabolic and pharmacologic actions of the drug in humans, the side effects associated with increasing doses, and, if possible, to gain early evidence on effectiveness.... The total number of subjects included in Phase 1 studies varies with the drug, but is generally in the range of twenty to eighty. (<http://www.fda.gov/cder/handbook/phase1.htm>, August 14, 2004)

Phase 2 Clinical Studies. Phase 2 includes the early controlled clinical studies conducted to obtain some preliminary data on the effectiveness of the drug for a particular indication or indications in patients with the disease or condition. This phase of testing also helps determine the common short-term side effects and risks associated with the drug. Phase 2 studies are typically well-controlled, closely monitored, and conducted in a relatively small number of patients, usually involving several hundred people. (<http://www.fda.gov/cder/handbook/phase2.htm>, August 14, 2004)

Phase 3 Clinical Studies. Phase 3 studies are expanded controlled and uncontrolled trials. They are performed after preliminary evidence suggesting effectiveness of the drug has been obtained in Phase 2, and are intended to gather the additional information about effectiveness and safety that is needed to evaluate the overall benefit-risk relationship of the drug. Phase 3 studies also provide an adequate basis for extrapolating the results to the general population and transmitting that information in the physician labeling. Phase 3 studies usually include several hundred to several thousand people. (<http://www.fda.gov/cder/handbook/phase3.htm>, August 14, 2004)

I have abstracted the general approach to phases 1, 2, and 3 on the left-hand side of Figure 3.6. My argument is that there are analogues of such studies in education, and that it would be useful to follow through with them. The educational analogues are given on the right-hand side of Figure 3.6.

In what follows I flesh out the details of Figure 3.6.

Preliminary and Phase 1 Studies in Educational R&D

Research in each of Bohr's, Edison's, and Pasteur's quadrants makes significant contributions as preliminary and Phase 1 studies in educational R&D.

Although the emphasis here is on curricular design, it is worth recalling that many curriculum ideas have their origins in Bohr's quadrant (research with a focus on fundamental understanding)—and in addition that such research encompassed a wide range of methods, from experimental studies in the laboratory to classroom ethnographies. Basic research in the 1970s and 1980s resulted in a fundamental reconceptualization of the nature of mathematical thinking, along multiple dimensions. Theoretical considerations in the field at large resulted in an epistemological shift away from an acquisitionist theory of knowing to a constructivist view, laying the foundation for a different conception of student engagement with mathematics. Laboratory studies revealed the mechanisms by which problem-solving strategies could be elaborated and learned, and demonstrated that poor metacognition could hamper problem solving. Classroom observations and laboratory studies demonstrated the negative impact of counterproductive beliefs on mathematical performance, and classroom studies suggested the origins of such beliefs. All of this work informed NCTM's (1989) *Curriculum and Evaluation Standards for School Mathematics*, which served as a catalyst for the National Science Foundation to issue a series of requests for proposals (RFPs) for curriculum development in the early 1990s. Hence basic research

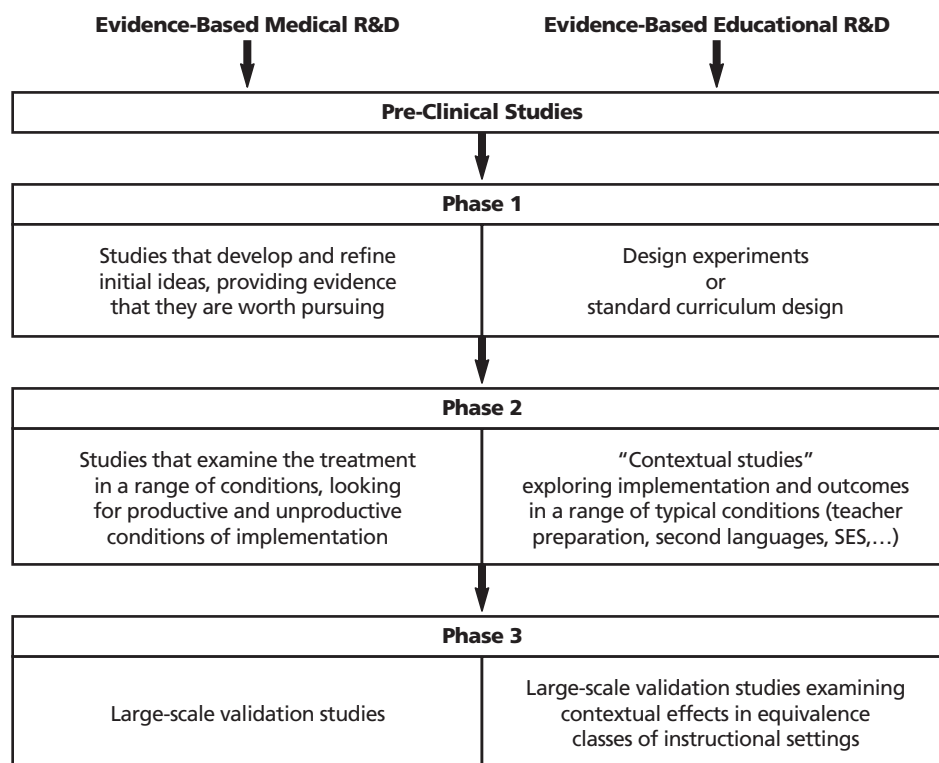


Figure 3.6 Potential parallels between evidence-based research in medicine and education.

played a fundamental role in shaping curriculum development over the final quarter of the 20th century, and in shaping the criteria by which curricula should be evaluated.

Once the goals and theoretical orientation had been established, fundamentally applied work played a major role in curriculum development. The teams that produced curricula in response to the National Science Foundation's curriculum RFPs differed in character, with some being university-based and some housed in organizations known for the production of instructional materials. However, all shared the property that they drew on whatever sources they could—knowledge from research where available, but equally if not more important given the state of the art, prior experience in curriculum development and, more generally, what has been called the wisdom of practice (see, e.g., Shulman, 2004). Given the timetable for curriculum development and refinement—typically, the grant for the development of an n -year curriculum had a term of n years!—the process of curriculum development and refinement necessarily resided squarely in Edison's quadrant.

Over the same time period, work in Pasteur's quadrant—work that simultaneously addressed theoretical issues and pressing issues of practice in fundamental ways—was beginning to become a productive reality. Early exemplars, before the concept of design experiment had been named, were called *apprenticeship environments* (Collins, Brown, & Newman, 1989). These included my (1985a, 1985b) courses in mathematical problem solving, Palincsar and Brown's (1984) work on reciprocal teaching, and Scardamalia and Bereiter's (1985) work on facilitating the writing process. Related concepts within the mathematics education community have included teaching experiments (Cobb, 2000, 2001; Steffe & Thompson, 2000), classroom-based research (Ball,

2000; Ball & Lampert, 1999; Lampert, 2001), and curriculum research as design (Battista & Clements, 2000; Roschelle & Jackiw, 2000).

The term *design experiment* was introduced into the literature by Ann Brown (1992) and Allan Collins (1992). The authors intended for both of the terms in the name to be taken with great seriousness. Design experiments were exercises in instructional *design*: they involved the creation of (novel) instructional environments (contexts and materials) that played out as real instruction for real students. They were also true *experiments*—not necessarily in the statistical sense, though data were often gathered and analyzed statistically, but in the scientific sense, in that theory-based hypotheses were made, measurement tools were established, a deliberate procedure was rigorously employed, outcomes were measured and analyzed, and theory was revisited in light of the data.

Brown (1992) encapsulated this complexity as shown in Figure 3.7. It is worth noting that the double arrow between “contributions to learning theory” and “engineering a learning environment” represents the dualism inherent in working in Pasteur's quadrant.

The meaning of design experiments has not been settled in the literature. Related but clearly distinct formulations may be found in Collins (1999), the Design-Based Research Collaborative (2003), and Cobb, Confrey, diSessa, Lehrer, and Schauble (2003). Collins provided a contrast between traditional psychological methods and the context and methods of typical design experiments. In the latter case one gives up a significant degree of control and straightforward experimental prediction for a much more complex set of human interactions, and the need to develop rich characterizations of those interactions. The Design-Based Research Collaborative noted the interaction of theoretical and empirical goals, and the cyclic nature of design, enactment, analysis, and redesign;

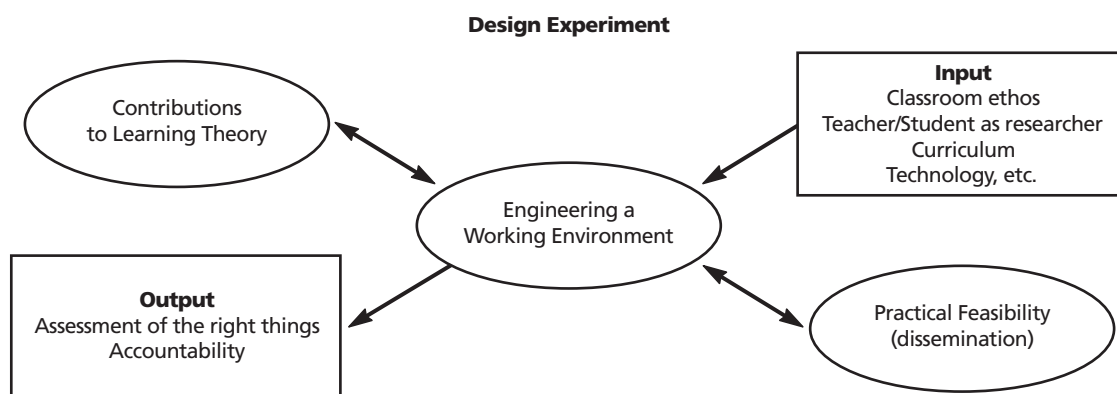


Figure 3.7 The complex features of design experiments. (From Brown, 1992, p. 142. With permission)

they called for methods to “document and connect processes of enactment to outcomes of interest.” Cobb et al. identified a wide range of studies that involve design and fit squarely into Pasteur’s quadrant: small-scale versions of a learning ecology that enable detailed studies of how and why they function the way they do (e.g., Steffe & Thompson, 2000); collaborations between a teacher and a research team to construct, conduct, and assess instruction (e.g., Cobb, 2000, 2001; Confrey & Lachance, 2000; Gravemeijer, 1994); preservice teacher development experiments (Simon, 2000); collaborations between researchers and in-service teachers to support the development of a professional community (Lehrer & Schauble, 2000; Silver, 1996; Stein, Silver, & Smith, 1998); and systemic interventions such as in New York City’s Community School District #2 (High Performance Learning Communities, 2003). They then delineated the following common features of those efforts:

- Purpose: “to develop a class of theories about both the process of learning and the means that are designed to support that learning.”
- “Highly interventionist methodology: Design studies are typically test beds for innovation.”
- Theory testing: “Design experiments create the conditions for developing theories yet must place those theories in harm’s way.”
- “Iterative design.”
- “Pragmatic roots: theories developed during the process of experiment are humble not merely in the sense that they are concerned with domain-specific learning experiences, but also because they are accountable to the activity of design.” (Cobb et al., 2003, pp. 9–10)

Despite the differences in emphasis among those who theorize about design experiments, there is certainly an agreement that one desirable outcome of design experiments is a “road tested” design for an instructional environment or curriculum that shows promise in terms of student learning and does not have any major downside risks.

However one arrives at them—inspired by basic research (origins in Bohr’s quadrant), drafted by skilled designers on the basis of their intuitions or produced by a commercial process (designed or refined in Edison’s quadrant), or produced by a process that attends to both theory and materials development (Pasteur’s quadrant)—the road-tested design described in the previous paragraph has all of the desired properties that a Phase 1 product should have: promise, feasibility, and no obvious risks. So: What next?

One possibility, consistent with much of the current political zeitgeist, is to move directly to randomized controlled trials. After all, if you have a product that you think “works,” why not “prove it,” using what has been called the gold standard of experimental methods? In what follows I shall argue that such an approach, although well intended, represents a mistaken parallel to evidence-based medicine. The next step in clinical trials is to move to Phase 2, mid-scale exploratory studies. There is an appropriate educational analogue, which has not been explored by the field at large.

Phase 2 Studies in Educational R&D

I begin with a story that illustrates the main point underlying the approach suggested in this section. I had been following the use of a standards-based curriculum in local classrooms. Implementation was uneven at times, but what I saw in local middle schools, across the boards, confirmed the impression I had from the literature. In comparison with students who experienced the “traditional” curriculum (represented by two state-adopted textbooks in California), students who experienced the standards-based curriculum performed respectably with regard to skills, and they were better off when it came to concept acquisition and problem solving. At that point, I was leaning toward a blanket recommendation that the newer standards-based curricula be adopted. It appeared that the floor for such curricula was higher than the floor for traditional curricula.

Then I visited an off-track algebra class in a local high school. This was a second semester algebra course being taught in the first semester. All of the students in the class were there because they had had some difficulty with algebra—they were either repeating the course or had entered it from a tracked course that had covered one semester of algebra over the course of a year. By and large, the students were disaffected and saw themselves as poor mathematics learners.

In this school as in many urban schools, privilege played a role in course assignments. The “best” teaching assignments, such as calculus, went to those with the most seniority and clout; the most problematic assignments, of which this was one, went to those who had little choice and sometimes little motivation to teach them. The person assigned to teach this course was a computer science teacher, and the students were assigned to his room—which had computers bolted to each desk.

The curriculum calls for a great deal of group work. Roughly speaking, students are expected to work 10 problems a day in class, largely in groups; this work

prepares them for the 10 problems they are assigned each day for homework. The in-class work is essential, because the homework from this curriculum looks somewhat different from the work in the traditional curriculum. Parents or other caregivers are less likely than usual to be able to help with homework.

Simply put, the class was dysfunctional. The teacher assigned students to groups but provided neither monitoring nor assistance; some students listened to headphones, some drummed basketballs or otherwise indicated their disaffectedness. With computers bolted to their desks, even those students who wanted to work found themselves in an environment that was crowded, noisy, and inhospitable to such work. (The teacher reacted to an early complaint about the level of chaos by saying that the disruptive students would soon drop out, and that the classroom would then be quieter.) Generally speaking, students were either unwilling to work in class or found it nearly impossible to do so. This caused serious problems for those who tried. Unable to work in class, they needed to do 20 problems (the in-class assignment *and* the homework assignment) each night, often without support, in order to keep up.

The fact is that (short of replacing the teacher with a more attentive and competent one, which was not likely to happen) these students would have been better off taking a course that used the traditional curriculum, for the following reasons. Being given straightforward presentations, even those aimed at procedural learning, would have been better than receiving no instruction at all. Seatwork on standard problems might have been doable for a larger percentage of the students. And many of them could have received homework help from parents or caregivers. In sum, in this somewhat pathological context, using a *Standards*-based curriculum actually lowered the floor. In this kind of context the students would not have been well served, but they would have been better served, by the traditional curriculum.

This kind of story will come as no surprise to those who are familiar with urban classrooms. However, it illustrates an essential point that seems to have been missed by many. A curriculum is not a “thing” that is “given” to students, with consistent effects. A curriculum plays out differently in different contexts. The same instructional materials are unlikely to result in the same patterns of learning in inner-city schools that have broken-down facilities and a low proportion of credentialed staff that they produce in suburban schools that have modern facilities and a well-credentialed staff. Some curricula will work well with, or be easily adaptable to, a student body that has a high proportion of English language learners;

others, because of their high threshold of formal or academic language, may prove difficult for such students. Some may require a significant amount of teacher preparation before implementation can be successful; some may not. If one is in the position of choosing a curriculum, one wants to be aware of these *contextual effects* and choose a curriculum that has the potential to provide the best results given the real or potential affordances of the present context.

This is precisely the kind of information provided by Phase 2 studies in medical research. The pharmaceutical industry has long understood, for example, that drugs may have differential impacts (some positive, some negative) depending on: the health of the patient and what other conditions he or she may have; on what other medicines the patient may be taking; and on whether the medicine is taken on a full or empty stomach, or with particular foods. Thus, for example, Medline Plus, a web server made available by the U.S. National Library of Medicine and the National Institutes of Health at <http://www.nlm.nih.gov/medlineplus/druginformation.html>, has standard categories of information related to various medical treatments. A typical page describing a drug or treatment regimen includes categories such as the following:

- *Before Using This Medicine*
This category includes information about allergies, special situations such as pregnancy or breast feeding, appropriateness for particular populations, and interactions with other conditions and/or medicines.
- *Proper Use of This Medicine*
This category includes information about necessary conditions for proper use. This includes a drug or physical therapy regimen, but also information such as “to be taken on a full stomach” or even as specific as “do not drink grapefruit juice with this medicine.”
- *Precautions While Using This Medicine*
This category includes descriptions of downside risks to taking the medicine, possible complications, and things that should be avoided (typically, taking alcohol or other medicines that might interact with the one being considered).

In pharmacology, the point of Phase 2 studies is to uncover this kind of information, *before* proceeding to large-scale testing via randomized controlled trials. In broad terms, the issues are, What needs to be in place before the proposed approach stands a decent chance of being effective? Which outcomes should

be tracked, in what ways? What makes effectiveness unlikely? What kinds of collateral benefits or costs might one encounter, with what frequency, for what populations? Only after such questions have been answered—when one has a decent sense of where and when something works—should one set about the large-scale documentation *that* it works in those circumstances.

In what follows I explore the educational analogue. Lest I be accused of reductiveness at the start, I should point out that public health may be a better analogy than pharmacology for instruction-in-context. The rough parallel is that curricula play a role in the educational system (where there is room for anything from almost total neglect to faithful implementation of what is proposed) that is analogous to the role that medicines play in public health implementation (where there is room for anything from almost total neglect to faithful implementation of what is proposed). What I am proposing here is somewhat speculative and (like all work in Pasteur's quadrant) will require a good deal of adaptation and midcourse correction to be fully effective.

Suppose, then, that one has developed some instruction,¹⁶ possibly as part of a larger learning environment. Assume that the instruction extends over a substantial period of time—perhaps a semester, a year, or some number of years. The path toward the development of the instruction may have been any of those described in the section “Preliminary and Phase 1 Studies”—it may have been inspired by research, may have emerged from a team of skilled designers, or may be the product of one or more design experiments. The instruction has presumably been used with a relatively small number of students and found promising. What next?

It seems clear that the instruction should be implemented and examined in a range of contexts that are representative of the wide range of circumstances in which it might be adopted. (These might include suburban schools, inner city schools, schools with low and high proportions of English language learners, schools with low and high proportions of credentialed mathematics teachers, and so on.) This implementation and analysis should be systematized, perhaps as follows.

Although no two schools are the same, it might be possible to identify a collection of “Phase 2 schools” that, in some way, represent equivalence classes of

schools. That is, suppose one selects a collection of schools such that each school has a collection of demographic and perhaps administrative attributes that typify a significant number of schools. It is not unreasonable to expect that the implementation in a Phase 2 school that represents an equivalence class would be at least somewhat similar to implementation in other schools in that equivalence class, with somewhat similar results. Hence implementation over a range of schools selected from different equivalence classes would suggest what might happen if the instruction were implemented on a wider scale. It would also indicate which kinds of contexts are likely to experience beneficial results, and which are not.

In each of the Phase 2 schools, one would investigate questions such as the following:

- Under what conditions can the instruction be made to work (and what does one mean by work)?
- What measures, both of process (e.g., classroom dynamics, and discourse patterns) and product (mathematical or other outcomes) can be used or developed to help understand and calibrate the impact of such instruction?
- How does the instruction get operationalized in different contexts, and with different kinds of support systems? What are the patterns of uptake, of interaction?
- How does implementation in different contexts reflect on the theoretical underpinnings of the design?
- How might the curriculum be refined in light of what is revealed by the attempts to implement it?

In the early stages of Phase 2 work in these varied contexts (Phase 2a), the work would resemble a family of coordinated design experiments. Understanding and describing instructional impact in different contexts would still be a contribution to fundamental research at that point; a wide range of methods would still be appropriate to explore the implementation, and to find out what support structures, in different environments, improved outcomes. This kind of work could provide “user's guides” to the curricula. (If a district has a particular profile, it might expect certain patterns of outcomes. Thus it might want to consider

¹⁶ In what follows I use the term *instruction* because it is somewhat generic. I intend a very broad interpretation in what follows. If, for example, the intended instruction requires certain classroom configurations (e.g., the kind of “jigsawing” used by Brown and Campione, 1996, in their FCL work) or certain technology (as in Scardamalia and Bereiter's, 2002, Knowledge Forum work), those are considered to be part of the instruction.

modifying certain administrative features of the environment and providing professional development along certain lines to its staff.)

One would, of course, expect some aspects of this kind of work to be more efficient than in a first design experiment: Some of the key phenomena would have been noted in prior R&D, and some of the student assessment tools would have been developed. Indeed, in the interests of efficiency and with an eye toward later studies, one would want to have or develop similar observational and assessment tools for as many of these contexts as possible.

Once the basic research in the different contexts has been done, efforts turn toward systematization (Phase 2b). Are there features of implementation fidelity that are essential to produce certain kinds of student outcomes? Looking across the various contexts in which the instruction has been implemented, are there features of those contexts that seem consistently to produce certain kinds of effects? Are there certain contextual variables that, having been identified in the more detailed studies, can be captured more efficiently in summary fashion once one knows to look for them? (For example, early studies of accountability—to what and to whom are students accountable, and how does that play out in terms of learning and identity?—can be expected to be highly descriptive and detailed. After some time, however, one can imagine that the character of accountability structures could be codified in straightforward ways.)

A major goal of the work in Phase 2b is to develop the instrumentation that supports the meaningful conduct of Phase 3 studies. This includes the ability to capture central aspects of the context as described in the previous paragraph. It includes broader contextual variables found to have an impact on implementation or outcomes. (These might, for example, include larger aspects of the context, such as whether the state or district has high-stakes accountability tests of particular types. Given that many teachers and schools depart from planned curricula for substantial periods of the school year in order to focus on test preparation, the presence of such tests might have a significant impact and should be included as a variable.) It includes characterizations of implementation fidelity. And it includes a range of outcome measures. Content-related measures would, in the best of circumstances, be independent of the curriculum and standards-based. (Recall that a major

problem with early curricular evaluations such as those found in Senk and Thompson, 2003, is that many of the student assessments were locally developed, and thus not comparable to other such measures.) Other measures might well include characterizations of the learning community, of student beliefs, and of other aspects of the students' mathematical identities. Some comparative studies would be conducted during either Phase 2a or 2b, in order to assess the relative impact of the instruction and gauge the plausibility of the claim that the instruction is indeed worth adopting. To be explicit in summary, much of Phase 2 research as imagined here would reside squarely in Pasteur's quadrant—contributing to basic understandings, but also laying the foundations for the kind of applied work done in Phase 3.

Phase 3 Studies in Educational R&D

The main focus of Phase 2 was to understand how and why instruction “works,” to revise and improve that instruction, to identify relevant variables, and to develop relevant instrumentation. With that established, Phase 3 can be both straightforward and informative. One can imagine two kinds of Phase 3 studies: (1) expanded studies replicating the effects discovered in Phase 2, and (2) comparative in-depth evaluations of any number of promising curricula. The two kinds of studies can be done independently, or studies of type 1 (which are in essence comparative studies of two treatments, the instruction and a control) can be done as subsets of studies of type 2.¹⁷

A key aspect of such studies is the random assignment of students to instructional treatments. That said, the rest of the work should be reasonably straightforward, in conceptual terms.¹⁸ That is, it should be, *if* the proper work has been done in Phase 2, and the right perspective is taken toward the analysis of the data. As noted above, major goals for research in Phase 2 are to identify contextual variables and be able to code for them, and to use appropriately independent and meaningful assessments of student performance. With such tools at one's disposal, the relevant issue becomes one of interpretation.

Recall the second example discussed in Part 1, which illustrated two different ways to think about the results of randomized controlled trials. In this author's opinion, aggregating across all contexts—“on average, students who studied from Curriculum A

¹⁷ For a broad discussion of comparative studies, as part of the “engineering approach,” see Burkhardt (2006). For an extended treatment of some of the relevant statistical issues, see Scheaffer (2006).

¹⁸ Of course, the devil is in the details. To say statistical analyses are conceptually straightforward is one thing; to carry them out correctly can be something else altogether. Even issues of the appropriate unit of analysis are non-trivial.

outperformed students who studied from Curriculum B”—is of little value. Such an approach provides evaluative information in summary, but it provides little useful information for anyone faced with a choice of instructional materials. (What if, for example, Curriculum A outperformed Curriculum B in general, but Curriculum B outperformed Curriculum A by a significant margin in schools that are similar to the ones for which instructional materials must be chosen? In that case Curriculum B is the right choice, even though Curriculum A is better on average. Or, what if the research reveals that Curriculum B is superior under present circumstances, but that a program of professional development will equip your staff to employ Curriculum A to better effect?) In sum, the most beneficial role of randomized controlled trials in curricular evaluation would be to investigate the hypotheses raised in Phase 2 studies—to determine which contextual variables have which effects on implementation fidelity of a wide range of curricula, and what the effects of those curricula are in those conditions. That kind of information would be truly useful for decision makers.

CODA

I began doing research in mathematics education in the mid-1970s. At the time, the field's primary research methods (in the United States, at least) were statistical, but their use was often unsophisticated, and the field as a whole suffered somewhat from a reductive form of what has been called “science envy.” The field of cognitive science had not yet coalesced (the first annual meeting of the Cognitive Science Society was held in 1978), and many of the fields that would ultimately contribute both methods and perspectives to our current understanding of mathematical thinking and learning (among them anthropology, artificial intelligence, linguistics, philosophy, psychology, and sociology) were at best peripheral to the enterprise. When I began my research on problem solving, the cutting edge of research consisted of attempts to make sense of factors that shaped the success or failure of individuals working problems in isolation in the laboratory. Major constructs that have now come to be cornerstones of our understanding (the very notion of cognitive modeling; the roles of metacognition and beliefs; the concepts of identity and of communities of practice) and the methods that help to elaborate them were only beginning to emerge, if they were on the horizon at all.

The progress that has been made since that time is nothing short of phenomenal. In little more than a quarter century there have been major epistemological shifts, accompanied by a flourishing of tools, techniques, and the theoretical perspectives that underlie them. The cognitive sciences and sociocultural research within mathematics education have both matured and become increasingly robust; what at first seemed almost like thesis and antithesis have, over the past decade or so, begun a synthesis that seems increasingly promising in terms of its possibility to help explain issues of (mathematical) thinking, teaching, and learning. The same is the case for the artificial distinction between quantitative and qualitative methods, which becomes less and less important as one begins to ask these central research questions: “What assumptions are being made? What claims are being made? What warrants are being offered for those claims?” This is remarkable progress, and one hopes and expects to see more.

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■ ■ P A R T I I ■ ■

Teachers and Teaching

ASSESSING TEACHERS' MATHEMATICAL KNOWLEDGE

What Knowledge Matters and What Evidence Counts?

*Heather C. Hill, Laurie Sleep, Jennifer M. Lewis,
and Deborah Loewenberg Ball*

UNIVERSITY OF MICHIGAN

For more than two hundred years, teachers—and what they know—have been objects of scrutiny. Teachers have been tested, studied, analyzed, lauded, and criticized. In short, both they and their performance have been assessed to an extent rare in other professions. Both the purposes and methods for assessing teachers have varied. With some assessments, the goal has been to evaluate individuals' qualifications for the work of teaching, while with others, the focus has been to analyze the knowledge teachers use to do that work. Most assessments have been evaluative, either explicitly or implicitly, seeking to appraise the adequacy of individual teachers' knowledge or the quality of their performance. Some assessments have contributed to building evidence about the knowledge needed for teaching, thus helping to establish criteria for professional qualifications and the methods for certifying them. What is assessed differs across approaches, and how it is assessed varies: Some are indistinguishable from a test that could be given to students, while others pose tasks special to the work of teaching. Teachers have been interviewed and observed; they have

been queried, given tasks, and asked to construct portfolios representing their work. The variation in purposes, test content, and assessment methods has increased over the past 30 years, as teacher testing has become more commonplace and the number of teacher tests has multiplied.

Developing a more coherent approach to the assessment of teachers' knowledge, and in particular their knowledge of mathematics, is now both possible and necessary. Three important contemporary pressures call out for a system of assessment that is not only rigorous, but professionally relevant and broadly credible. The first is a political environment that demands that students be taught by "highly qualified" teachers. Too many students, especially those in under-resourced schools, currently are assigned to teachers who are not certified to teach mathematics (Darling-Hammond & Sykes, 2003; Ingersoll, 1999). Still, what counts as "qualified" and how this could be measured and certified remains a significant problem. A second pressure is the growing need to establish evidence on the effects of teacher education on teachers' capacity, and of teachers' knowledge and skill on their

taking an exam, usually offered at the local or county level, that certified those with passing scores as eligible for classroom work (Angus, 2001; Haney, Madaus, & Kreitzer, 1987). It was only at the turn of the twentieth century that the completion of professional education programs eclipsed certification exams as a pathway into teaching, though by the late twentieth century, certification exams had returned full force. In this section, we examine the earliest exams, describe their decline and subsequent rebirth in the 1980s. In doing so, we focus closely on the content of these assessments in order to examine the mathematical knowledge thought to be required for teaching.

Early teaching exams were notable for what they included—not only the three “R’s”, but also questions designed to assess the moral fiber, ability to manage a classroom, and, possibly, religious affiliations of those tested (Angus, 2001). Over the course of the nineteenth century, exams lengthened to include a wider array of subject matter and, by the 1850s, some pedagogical questions as well. This period was also marked by a change in exam format, from an oral exam often administered by a local district board to written exams administered by county or state officials.

The mathematics sections of these written assessments reflected closely the curriculum of the day. The state of Michigan, for instance, maintained a vigorous teacher examination system through the early twentieth century, issuing tests in physics, literature, geography, arithmetic, history, civics, botany, physiology, and the “theory and art” of teaching, among other topics. An 1895 exam² contained the following arithmetic problems:

1. A pole 63 feet long was broken in two unequal pieces, and $\frac{3}{5}$ of the longer piece equaled $\frac{3}{4}$ of the shorter. What was the length of each piece? Give a good solution.
2. Two men hire a pasture for \$20. The one puts in 9 horses, and the other puts in 48 sheep. If 18 sheep eat as much as three horses, what must each man pay?
3. A boat whose rate of sailing in still water is 14 miles an hour, was accelerated $3\frac{1}{2}$ miles per hour in going down stream, and retarded the same distance per hour in coming up. It was five hours longer in coming up a certain distance than in going down. What was the distance?
4. (a) How do you read any decimal? (b) How do you express decimally any common fraction?

(Michigan Department of Public Instruction, 1896, pp. 297–298)

The state exam also contained the following algebra problems:

1. There is a number consisting of three digits, the first of which is to the second as the second is to the third; the number itself is to the sum of its digits as 124 to 7; and if 594 be added to the number, the digits will be reversed. What is the number?

2. Find the value of x :

$$\frac{\sqrt{3x+1}+3}{\sqrt{3x+1}-3} = \frac{\sqrt{7x+8}+4\frac{4}{5}}{\sqrt{7x+8}-4\frac{4}{5}}$$

(Michigan Department of Public Instruction, 1896, p. 297)

Candidates answering these and other problems were granted a state certificate for teaching any grade. Michigan also maintained, through this period, a separate county examination system in which graded certificates were available to candidates in rural areas. This system reflected the tradition of local control over such exams, but more important, also accommodated the realities of the rural teacher labor market. Candidates in rural areas could choose to sit for an exam leading to a general certificate, which authorized its holder to teach at any grade level in the county. These general certification exams were substantially similar in length and content to the state exam excerpted above. Records show that the vast majority of rural teacher candidates, however, chose instead to sit for exams leading to primary (K–4) certificates. These exams did not have a formal algebra section, and occasionally had less computationally intensive arithmetic problems. County education commissioners and legislators argued that requiring more difficult tests for the rural population would have led to fewer certified teachers, a chronic problem in underpopulated counties. In fact, even the primary-grade exams boasted a 50% failure rate in some counties (Michigan Department of Public Instruction, 1897, p. 152). One observer noted that, “The chief cause for failure is a want of knowledge of the subject; and this is due to poor teaching” (Michigan Department of Public Instruction, 1897, p. 153).

Beyond illustrating how mathematics was imagined to be used in daily life at the turn of the twentieth century, these exams and their attendant materials reveal the views of teacher knowledge held during this period. To start, many believed teachers should know more difficult mathematics than they were expected to teach. On the typical primary certification exam, for instance, candidates were asked to solve classic middle school arithmetic problems—rates, proportions, and percents:

² Throughout the chapter, the item numbering is our own, and does not necessarily reflect the item numbering on the actual tests.

1. (a) Explain as to a class the difference between the simple and local value of figures.
(b) Write in figures fourteen million, one thousand and five hundredths; three trillion, two hundred one and one thousand seventy billionths; also write in words 7504306.0405214.
2. I own a horse and a farm; one-fourth the value of the farm is four times the value of the horse. Both taken together are worth \$1,700. Find the value of each. Write out a complete analysis.
3. A man sold a lot for \$84 and by so doing gained $\frac{1}{5}$ of what it cost. What % would he have gained if he had sold for \$100? Analyze.
4. A field of 5 acres in form of a square is to be surrounded by a fence $4\frac{1}{2}$ feet high, to be built of boards 6 inches wide, placed horizontally. The lower board is to be four inches above the ground, and there is to be a space of 5 inches between the boards. What will be the cost of the boards required at \$18 per M?
5. A merchant gets 500 barrels of flour insured for 75% of its cost, at $2\frac{1}{2}\%$, paying \$80.85 premium. For how much per barrel must he sell the flour to make 20% upon cost price?
6. A man devotes 40% of his income for household purposes, 35% for the education of his children, 16% of the remainder to charitable purposes, and saves the remainder, which is \$705.60. What is his income?
7. One-half of a stack of hay will keep a cow for 20 weeks, and $\frac{3}{4}$ of the stack will keep a horse 120 days. How many weeks will the whole stack of hay keep both the cow and the horse?

(Michigan Department of Public Instruction, 1896, p. 315)

Although it is possible that these topics were taught in the primary grades during this era, this is probably not why this content was included. In their annual reports, state superintendents of the time asserted that the “idea that a teacher should know very much more than he is expected to teach is so generally agreed upon that no discussion here is necessary” (Michigan Department of Public Instruction, 1898, p. 10). Mathematics certification exams throughout this period were also marked by the use of numerically complex examples. Whereas the examiners could have chosen mathematics problems that yielded easily to mental computation or shortcuts, few did. This implies that successful teacher candidates of this time would not only understand the general principles for solving such problems, but also have a solid foundation in facts, procedures, and patience.

According to state documents, the difficulty of these exams was in part a political move, the state bureaucracy’s attempt to ratchet up the quality of rural teachers by making the test more difficult and removing the possibility of political favoritism. Defending an increase in the difficulty of test questions, the Superintendent of Public Instruction Henry R. Pattengill wrote in 1896: “The rural schools have received the benefit of a better prepared, more mature, and more broadly

educated class of teachers. . . . Sympathy, politics, sect, ‘pull’, should play no part in the choice of teacher. It is not understood that the teacher need be examined year after year in the same studies, but every teacher should forever be a student” (p. 7). Another state report of the time presaged current political slogans: “Better qualified teachers in the country schools should be our motto” (Michigan Department of Public Instruction, 1897, p. 10).

The exams and their materials reveal other key assumptions about teacher knowledge. In Michigan, nearly all the arithmetic problems were what would today be called “real-world,” ones that asked teachers to engage in the practical calculations facing businesses, farms, and banks. Problems of a purely mathematical nature—for instance, those that explored the conceptual underpinnings of arithmetic—were not included. The problems included were based on the mathematics curriculum that teachers were expected to transmit to their all-male students (Michalowicz & Howard, 2003), and suggested that test writers viewed teachers’ mathematical knowledge as the knowledge they were responsible for instilling in future business owners, farmers, and bankers.

And, despite the applied focus of the exams, some Michigan state superintendents advocated what might today be called conceptual knowledge. In 1896, for instance, graders were cautioned that “in arithmetic, a knowledge of principles and general accuracy in method shall be considered not less than three times as important as obtaining a correct answer” (Michigan Department of Public Instruction, 1896, p. 313). At an 1896 conference on teacher testing, one county commissioner stated, “Our examinations should be of such a character as to demand less of the memory and more of the reason; less rote work, more sequence, more analysis, and more simplicity” (Michigan Department of Public Instruction, 1897, p. 409). Without evidence from the actual grading of exams, it is difficult to determine the extent to which this sentiment was implemented. However, the existence of these admonitions foreshadows the debates in the twentieth century over procedural fluency and conceptual understanding.

Finally, these tests also illuminate nascent political struggles about what should be known in order to teach. These struggles continue into the present; in fact, the words Pattengill wrote in 1894 could well have been written today:

The question of teachers’ examinations has always been a perplexing one. No matter how excellent the questions for such examinations may be, nor how thoroughly and honestly conducted, it will still

students' learning. Many question the effectiveness of professional training and argue that it is unnecessary; by this logic, all that is needed to be qualified to teach is a mathematics major and experience. Evaluating this claim requires empirical evidence about teachers' knowledge and skill. This, in turn, necessitates a method for appraising such capacities. Third is the need to distinguish what makes teaching professional—that is, a domain of professional knowledge and skill not possessed by just any educated adult. Is the knowledge mastered by someone who majors in mathematics sufficient content knowledge for teaching? If not, then what exactly distinguishes mathematical knowledge for teaching and how could this distinction be established? Given the confluence of these three forces, the field has an opportunity to advance both the tools to assess teachers' mathematical knowledge and our collective understanding of the notion of mathematical knowledge *for teaching*. A set of agreed-upon reliable and valid methods of assessing teachers' mathematical knowledge would afford the capacity to gather and analyze the sorts of evidence needed to make progress on these issues.

Toward that end, this chapter reviews and seeks to appraise the myriad ways in which U.S. teachers' knowledge of mathematics has been assessed. It asks: What is measured on tests of teachers' mathematical knowledge? What *should* be measured? *How* should it be measured? And how has the evolution of both assessment methods and scholarly thinking about teachers' mathematical knowledge influenced these tests? Answering these questions has taken us into territories that have often remained disconnected. On one hand, the history of teacher testing consists largely of assessments designed to *certify* teachers, that is, to attest to the adequacy of teachers' knowledge to teach mathematics. On the other hand, the history of research on teaching and teacher education reveals a stream of systematic efforts to *investigate* what teachers know, and to associate that knowledge with their professional training and their instructional effectiveness. Although these two lines of work—certification and investigation—have both focused on the assessment of teachers' mathematical knowledge, they have proceeded almost independently of one another. They differ in the knowledge they seek to assess, the methods they use to do so, and the conclusions they aim to make. In this chapter, we assemble these lines of work under a common light in order to consider how teachers' knowledge might be responsibly assessed. Important to note, however, is that the assessment

of teachers is hotly contested terrain. Our goal is to contribute resources that might help move the debate from one of argument and opinion to one of professional responsibility and evidence.

The chapter is organized in four parts. First we review the history of U.S. teacher certification testing, beginning in the nineteenth century, and trace its evolution and re-emergence in the 1980s. In the second section, we examine early *scholarly* research involving the measurement of teachers' mathematical knowledge. Both early certification exams and scholarly studies were constructed in the absence of any elaborated theory about the elements of mathematical knowledge for teaching, a fact that led to important limitations on the interpretations of this early work. We then show, through extended example, that in contrast to the views of teaching mathematics implicit in both early teacher certification tests and early scholarly research, teaching requires knowing more than simply how to solve the problems a student might solve. In the third section, we describe measurement methods in studies that explore the mathematical knowledge as it is *used* in teaching. This section provides a review of the tools and instruments developed to study teachers' mathematical knowledge. In the fourth section, we return once again to teacher certification exams, considering the extent to which modern exams have taken up the ideas and methods that appear in scholars' study of mathematical knowledge for teaching.

Throughout, we observe that the mathematical assessments used in each period are linked both to contemporary methods of assessment and ideas about the mathematical knowledge teachers need for success with students. We do not explore ideas about teachers' mathematical knowledge in their own right, but instead describe how these ideas have been instantiated in and influenced the development of assessments.¹ We also limit our analysis to U.S. teacher assessments, as reviewing international teacher assessments is beyond the scope of a single chapter.

TESTING TEACHERS: HISTORICAL APPROACHES TO ASSESSING TEACHERS' MATHEMATICAL KNOWLEDGE

The history of assessing teachers' mathematical knowledge begins with the history of teacher examinations for certification. For the majority of U.S. history, in fact, the main route to teaching was through

¹ For a fuller treatment of mathematical knowledge for teaching, see Ball, Lubienski, and Mewborn (2001).

be true that examinations alone are an inadequate test of a candidate's ability to teach school. The qualifications requisite for good school teaching cannot be ascertained by any set of questions whether oral or written; and yet, where the teachers must be employed for a large extent of territory and for many schools, no other plan has been suggested better than the plan of examinations combined by the supervision exercised by the country school commissioner.

The first essential element in a teacher is good scholarship. No amount of tact, or method, or skill in the use of devices will make up for the deficiency in scholarship. In thus emphasizing the value of this factor, we by no means overlook the value of tact and good method. If, however, we are to do without one of these, we think we could more safely risk the teacher with scholarship than the one with method and poor scholarship. (Michigan Department of Public Instruction, 1894, pp. 1–2)

In Pattengill's view, and in the exams of the early 1890s, content knowledge was separate from knowledge of the "theory and art" of teaching. Which is more important to student learning remains both an unanswered question and a topic of hot debate.

In contrast to the business-focused mathematics on the exams of the 1890s, the period after Pattengill's tenure saw an increase in teaching-specific mathematics questions on these exams. For instance, while half the 1900 state exam contained problems similar to those found in previous years, the arithmetic section also carried the following:

1. In what arithmetical processes is drill the essential element?
In what operation is memory a factor?
2. When may the algebraic equation be used to advantage in arithmetic?
Is it advisable to teach its transformations in eighth grade arithmetic?
3. How do you present the table of Surveyors' Long Measure so that link and chain stand for ideas in the mind of a child?
Write the table.
4. Outline a first lesson on the subject of ratio.

(Michigan Department of Public Instruction, 1901, p. 70)

Without information on the grading of specific answers, it is difficult to tell exactly what types of knowledge test developers desired to assess; however, it seems likely that these state assessments were shaped by Michigan's growing class of professional teacher educators, some of whom perhaps called attention to the mathematical work teachers needed to do once inside classrooms. In fact, in 1900 the State Board of Education passed control of the state-level certification exams to the presi-

dent of the normal school system (Michigan Department of Public Instruction, 1901, p. 69).

These results refer to the exams in one state; without similar and detailed analyses of teacher exams in other state archives, it would be difficult to tell whether Michigan's tests were typical. However, we think it likely that they were, and that the major issues regarding test content were common to other states, as well. It is especially striking that these major issues remain unresolved today: how difficult the exams should be—and by extension, how many prospective teachers should be excluded on the basis of lack of knowledge; whether it is content knowledge or knowledge of methods that make a good teacher; whether conceptual or procedural knowledge should be assessed. The passage of 125 years has done little to bring closure to these important questions.

These exams are also revealing in how they *measured* teachers' knowledge. Nineteenth-century teacher exams were designed to draw inferences about *individuals* and their capacity to teach children. Yet it would be nearly a half century before modern psychometrics, with its concerns for reliability and validity, began to influence the development of teacher certification exams. Meanwhile these assessments were used to make high-stakes decisions about individuals' careers without actual evidence of their effectiveness in this regard.

In the interim, however, the use of teacher exams shrank rapidly. As reported in Angus (2001), the late nineteenth century saw a growing class of professional teacher educators begin to criticize these exams, primarily on the grounds that they were too easy, thus allowing incompetent teachers to serve. Histories of teacher certification also note that local control of these exams encouraged favoritism and left open the possibility that examiners would know less than the teachers they tested. In response, teacher educators successfully convinced state bureaucracies and legislatures to approve what was then an alternative form of certification: completion of professional training, usually at a normal school or teachers' college. By 1937, 28 states had abolished teacher testing (Wilson & Youngs, 2005), requiring instead the completion of a professional preparation program. These programs, according to Wilson and Youngs, came to include courses in educational history, psychology, educational foundations, teaching methods, and assessment. Although teacher educators would soon face criticism from other academic disciplines, the first half of the twentieth century saw an increasing claim to professional knowledge in education, and

the assent of state legislatures to allow teacher preparation to occur in professional schools.

The early history of teacher testing, in fact, might help explain the aversion of many teacher educators to it. Angus (2001) describes this aversion as a byproduct of professionalization: teacher educators sought to legitimize their product and accumulate resources by claiming a codified body of knowledge and basing entry to their profession on the completion of coursework designed to cover that body of knowledge. Teacher tests like Michigan's, by virtue of being "too easy" and a "back door" into teaching, threatened teacher preparation programs. These tests were also largely controlled not by educational professionals but by either bureaucrats or elected officials. These twin problems—the perceived ease of teacher tests and the lack of control over testing by educational professionals—would afflict the profession in even greater ways later in the century, after teacher certification exams regained political favor.

Enabled by the development of standardized tests and test firms, and in response to teacher surpluses and urban school officials' concerns about how to hire the "best" candidates, certification exams reemerged in the second half of the 1900s. The National Teacher Examination (NTE), established in 1940, eventually became the most widely administered—and most studied—teacher examination in the U.S. The NTE saw three major growth periods. The first two were fueled less by concerns about underqualified teachers than by Southern white attempts to block efforts to reduce racial discrimination. As Baker (2001) describes, school districts in the South typically paid comparably educated African American teachers less than whites through the 1940s. Challenged by NAACP litigation, Southern districts eliminated this practice from official policy. To continue a de facto policy of discriminatory pay, many turned to using the early NTE, on which African Americans tended to score lower than whites. Ben Wood, the major author of the NTE, aggressively promoted the test for this purpose to Southern school officials, touring the South to spread the word. As Baker recounts, "When school leaders asked about how white and black teachers might score on the NTE, [Wood] informed them that on previous administrations of the test the average score of blacks was 'at the lower fifth percentile' of whites" (p. 326). During the second period of NTE expansion in the 1960s, Arthur L. Benson, director of teacher testing at the Educational Testing Service (ETS), which assumed control of the NTE during the 1940s, repeated Wood's strategy. This time, the issue was the desegregation of

schools; Benson sold the NTE to Southern states by again pointing out that African American and white teachers performed differently on the exam. By 1968, nine Southern states required the exam of at least some teacher candidates. Criticism of this practice led ETS in 1971 to issue guidelines on the proper use of the NTE and to attempt to eliminate test bias, but African Americans continued to score lower—and thus be more likely to fail the certification tests—through the 1980s.

The third wave of state adoptions of the NTE occurred in the 1980s in response to *A Nation at Risk* (National Commission on Excellence in Education, 1983). This wave of adoptions was, by all reports, driven by public suspicion about teacher quality. The most popular among these tests were basic skills assessments (Haney et al., 1987). As Wilson and Youngs (2005) report, 37 states had adopted basic skills testing as a requirement for certification by 2002. Over the last two decades of the twentieth century, the percentage of teachers taking one of these assessments prior to entry into either preservice education programs or teaching increased dramatically. As in earlier periods, professional educators and professional education associations (e.g., the National Educational Association) remained opposed to the tests, and perhaps with good reason, as pressure mounted from non-education quarters, mainly economists, political scientists, and policy-makers who questioned the value of professional education in teaching (Hess, 2002; Walsh, 2001). The solution many non-educators propose is to return to a system in which passing an exam certifies a teacher to work in classrooms, regardless of the candidate's level of formal professional training or experience. Such proposals have met with strenuous objection from some involved in teacher education (see, for example, Berliner, 2005; Darling-Hammond & Youngs, 2002).

In this debate, much hinges on the quality of the test itself—what it assesses, and whether what it assesses is reasonably related to successful classroom teaching performance. The NTE tested mathematical knowledge among elementary and secondary teachers. All teachers took mathematics "items," as mathematics problems are referred to in modern assessment terms, as part of the "Core Battery" test, which assessed communication skills, general knowledge (including mathematics), and professional knowledge. Elementary teachers also took mathematics items as part of the "Elementary School Specialty Area Test," an examination that measured subject area knowledge and pedagogical knowledge across the elementary school curriculum.

Secondary teachers intending to be certified to teach mathematics took a stand-alone mathematics test.

Several items from an elementary-level core battery test illustrate the nature of this assessment:

1. An elevator operator is not allowed to carry more than 10 passengers in the elevator at one time. If there are 35 people waiting on the ground floor, what is the minimum number of trips up that the elevator must make in order to transport these 35 people?
a) 3 b) $3\frac{1}{2}$ c) 4 d) 5 e) $5\frac{1}{2}$
2. Statement: "The product of any two numbers is always greater than or equal to either of those numbers."
Which of the following examples proves the statement above FALSE?
a) $1 \times 1 = 1$ b) $3 \times 4 = 12$ c) $5 \times 1 = 5$ d) $\frac{5}{2} \times 4 = 10$ e) $\frac{1}{2} \times 4 = 2$
3. Which of these is NOT a correct way to find 75% of 40?
a) 75.0×40 b) $(75 \times 40) \div 100$ c) $\frac{75}{100} \times 40$ d) $\frac{3}{4} \times 40$ e) 0.75×40
4. A pattern requires $1\frac{7}{8}$ yards of a certain fabric. If five remnants of the fabric are on sale and their lengths are as follows, which of these lengths will provide enough fabric and result in the least waste?
a) $1\frac{1}{2}$ yd b) $1\frac{3}{4}$ yd c) $1\frac{15}{16}$ yd d) 2 yd e) $2\frac{1}{16}$ yd
5. Written as a percent, 2 =
a) 0.02% b) 0.2% c) 2% d) 20% e) 200%

(ETS, 1984, pp. 68–73)

Two observations about this assessment stand out. In contrast to the early teacher tests we uncovered, the NTE Core Battery contains a mixture of basics (e.g., item #5 above), disciplinary knowledge (e.g., item #2 above, where candidates must recognize a counterexample), and word problems (e.g., #4 above) that reflect, like the early tests, the ways mathematics was imagined to be used in contemporary society. This new mix of items reflected, perhaps, the changing nature of mathematical knowledge in society.

Second, the items are arguably easier than those on both nineteenth century tests and contemporary tests. Unlike the historical tests, for instance, few items extend beyond the formal elementary school curriculum. And few problems are, like those on the nineteenth century assessments, numerically complex; in many cases, teacher candidates likely did not have to put pencil to paper to correctly answer the item. Further, quite a number of the items are extremely basic. Item #2, for instance, might assess mathematical reasoning and proof from one per-

spective, but from another, it simply asks candidates to recognize a case that violates the statement made in the problem situation. One reason for the easier assessments might be the methods adopted for testing teacher candidates, described below. Another might be that these were conceived in many quarters as basic skills tests. Nevertheless, the items contained on this assessment constitute a very different theory about the substance of teacher knowledge than earlier teacher certification exams, and from the certification exams to come.

Interestingly, some items from the NTE Elementary School Specialty Area test in the 1980s seemed to tap into what today we might call "mathematical knowledge for teaching." In fact, taken as a whole, these elementary exams presage what Shulman would soon name as "pedagogical content knowledge" (Shulman, 1986): the items convey a broad sense that teachers need a specialized knowledge, something beyond what other educated adults know. The following is an example of one such item (ETS, 1987, p. 19):

$$\begin{array}{r} 521 \\ -386 \\ \hline 245 \end{array} \quad \begin{array}{r} 348 \\ -187 \\ \hline 261 \end{array} \quad \begin{array}{r} 863 \\ -37 \\ \hline 836 \end{array} \quad \begin{array}{r} 508 \\ -43 \\ \hline 545 \end{array} \quad \begin{array}{r} 108 \\ -26 \\ \hline 182 \end{array}$$

A pupil's work on five subtraction problems is shown above. Which of the following is the most appropriate diagnosis and suggestion about the pupil's work in arithmetic?

- a) The pupil has not had enough practice in doing this type of problem. The teacher should assign the pupil at least ten such problems each day for a few weeks.
- b) The pupil shows no recognizable error pattern in the problems shown; the poor performance is probably due to lack of interest. The teacher should seek some means of motivating the pupil to give more attention to class work.
- c) The pupil does not understand place value as it relates to subtraction problems. The teacher should have the pupil use concrete materials to explore this concept.
- d) The pupil has no understanding of the subtraction concept. The teacher should have the pupil review basic subtraction facts with the help of concrete materials.
- e) The pupil probably understands the subtraction concept but has not memorized the basic subtraction facts. The teacher should use drill activities and games to help the pupil memorize those facts.

Answering correctly by choosing option (c) requires knowledge of subtraction and the standard procedure for subtracting multi-digit numbers. It also emulates the reasoning that teachers do *in practice*—sizing up student work and deciding what to do next.

Another item focuses on a common mistake that students make when learning about the multiplication of rational numbers:

Ralph has worked with decimal representations of numbers in class. In response to a decimal exercise posed by his teacher, Ralph wrote $.2 \times .4 = .8$ on his paper. Ralph's work is best evaluated as

- a) correct, and this fact can be verified by multiplication of fractions
- b) correct, but this fact cannot be verified by multiplication of fractions
- c) incorrect, and multiplication of fractions can be used to show why it is incorrect
- d) incorrect, and decimal addition can be used to derive the correct answer
- e) incorrect, but neither multiplication of fractions nor decimal addition can be used to demonstrate why it is incorrect

It takes only basic mathematical knowledge to know that answer choices (a) and (b) are incorrect, because $.2 \times .4$ is $.08$. It would be possible to distinguish between answer choices (c), (d), and (e) using purely mathematical knowledge as well, but most people other than teachers are unlikely to have done such work or have occasion to do so. One would need to know that the model of multiplication as repeated addition (2×4 is the same as $4 + 4$) does not apply here: $.2 \times .4$ is not the same as $.4 + .4$. That in fact would yield $.8$, which is incorrect. This knowledge is not the sole province of teachers, but teachers need to know the meanings of operations for teaching in a way that the common educated person does not. The answer (c) is correct, because one could show that

$$\frac{2}{10} \times \frac{4}{10} = \frac{8}{100}$$

by multiplying across numerators and denominators. Again, this is not knowledge unavailable to adults other than teachers, but teachers would have the most facility working with such ideas about meanings of operations and how they can be modeled.

The next item is similarly constructed to gauge mathematical knowledge that is used in teaching multiplication (ETS, 1987, p. 23):

In developing concepts of multiplication, the following types of exercises should be included:

- (1) 2×23 (3) 4×13
- (2) 3×20 (4) 5×35

Which of the following is the most appropriate sequencing of these exercises?

- a) 1, 2, 4, 3
- b) 1, 3, 4, 2
- c) 2, 1, 3, 4
- d) 2, 3, 4, 1
- e) 3, 4, 1, 2

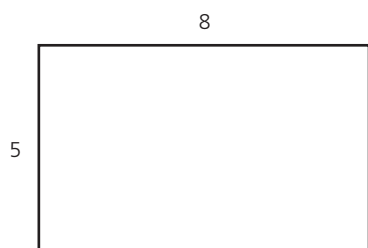
This item draws upon pure mathematical knowledge (multiplication of a one-digit by a two-digit number), but having such knowledge is not sufficient to identify the correct answer. This item also demands knowledge of how this topic is learned by children—the logical sequencing, the typical errors children make and the obstacles likely to be encountered by a child who is new to learning this skill. Thus, it encompasses basic mathematical competence but goes far beyond into mathematical territory that teachers traverse in their work with children.

However, test items that tap mathematical knowledge for teaching can be difficult to construct. As later generations of item writers discovered, particularly as scholars moved into measuring knowledge in this domain, items tend to suffer from three types of problems. One is the construction of an unambiguously correct answer. Teaching is heavily contextualized—teachers work with particular materials, and with particular children. It is also highly variable: textbooks have different approaches to teaching specific topics, and different developments of those topics over time. And students have unique learning and instructional histories. This contextualization and variability work against writing items in this domain. For instance, in the item above, arguments can be made for a sequence that starts with 3×20 , as it yields easily to mental mathematics and modeling via a number line. Also, a teacher who plans to develop related number facts (3×2 , 3×10 , 3×20) to logically reason about the problem might choose one of the options that begins with 3×20 (b or c). However, a teacher who plans to begin with a partial product method might choose 23×2 over 20×3 :

$\begin{array}{r} 20 \\ \times 3 \\ \hline 0 \\ 60 \\ \hline 60 \end{array}$	$\begin{array}{r} 23 \\ \times 2 \\ \hline 6 \\ 46 \\ \hline 46 \end{array}$
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As one can see, 20×3 yields a 0 in the first partial product, which might be confusing to children just learning this method. Items without clear correct answers were not uncommon occurrences in the tests we examined.

The following item illustrates a second problem associated with these tests—items that appear to be contextualized in teaching but that, in fact, do not require special professional knowledge (ETS, 1987, p. 28):



To find the perimeter of the rectangle above, a child can add 5 and 8 and multiply by 2 [$2(5 + 8)$] or multiply each number by 2 and then add [$(2 \times 5) + (2 \times 8)$]. This example illustrates an application of which of the following properties?

- a) Distributive property of multiplication over addition
- b) Commutative property of addition
- c) Commutative property of multiplication
- d) Associative property of addition
- e) Associative property of multiplication

Although this is important knowledge to have, and a kind of mathematical problem teachers are likely to come across in their work with children, this item could be solved simply by knowing the mathematics (properties of addition and multiplication). The school learning context makes no further knowledge demands for determining the correct answer to the item; it is, for testing purposes, window dressing. Note this contrast with the previous two items, in which specialized knowledge of children and learning in the domain of mathematics was required to determine the correct answer.

Third, some attempts to measure the mathematical knowledge used in teaching produced items that that depended more on ideology than professional skill. For instance (ETS, 1987, p. 15):

In teaching a third-grade class regrouping with two-digit numbers for the first time, which of the following would provide the best means for a teacher to demonstrate the concept?

- a) Manipulatives
- b) Illustrations drawn on the chalkboard
- c) A chart
- d) Examples from an appropriate mathematics textbook
- e) Examples from a local newspaper

The test materials identify (a) as the correct answer. Yet we might argue that several of these possibilities are plausible, and that it is not clear on what evidence any one answer is best. For example, a diagram (b) that showed the regrouping of tens and ones and linked it to the symbolic form might help pupils to link the representation to the written procedure more clearly than with manipulatives. It is unlikely that there is any one best approach in this example, and

as we pointed out above, good teaching would likely depend on the particulars of the situation. Here ideology about teaching—that concrete materials are always best—seems to override judgment.

These items demonstrate how an earlier version of teacher tests had an underlying sensibility about specialized knowledge for teaching, one that predates the formalized naming of “pedagogical content knowledge.” Second, the items show the difficulty of constructing measures that tap into this knowledge domain with precision. We return to both issues below.

In both NTE tests, mathematical knowledge at the elementary level is subsumed under “general knowledge”—it is one of several areas (literature, fine arts, social science) in which the successful teacher candidate was expected to be conversant. Scores were not returned for subjects separately. In this view of teacher knowledge, general proficiency across a wide range of topics is needed for successful teaching.

In addition, the methods for measuring teacher knowledge underwent a significant shift in the years since our nineteenth-century examples. With the development of more sophisticated methods for scaling items and scoring responses, test developers could better assure candidates and education officials of at least the reliability of the measures. These new methods—and related technology, such as computer-scannable response sheets—changed the face of the tests. Multiple choice items came into vogue. More items were put on the assessments to increase the accuracy of individual-level scores. And, perhaps to increase the precision of measurement among less knowledgeable teachers, easier items were added.

One area of trouble for the NTE, however, was its predictive validity—that is, whether teachers’ performance on the assessment correlated with the quality of their classroom instruction or their students’ learning. As Wilson and Youngs (2005) describe, six studies addressed the relationship between NTE score and student learning; all but one failed to find a relationship. Official documents make clear that ETS never claimed predictive validity, however, instead proffering their test construction process as proof of validity. The tests were written at the request of educators, to specifications drafted and reviewed by educators, and based on an analysis identifying what teacher candidates should know in a particular area, the tests must represent the knowledge needed to teach well.

The claims to validity made by ETS and other large teacher testing firms lead back to the issue of control over these professional tests. One hallmark of a profession, in fact, is control of barriers to entry by members of the profession itself (Abbott, 1988).

Yet through the 1980s, and even today, there has been only a weak claim to professional control over teacher assessments (Haney et al., 1987). Even after control of teacher testing shifted from local and state bureaucracies to large testing firms, the representation of professionals—in this case, mathematics educators, teachers, and mathematicians—on key decision-making panels remained more nominal than real. Certainly testing firms have included teachers, school officials, disciplinary experts, and others in the test construction process. On the whole, however, the leaders in the field have not tended to play a role or to be consulted in this work. Instead, control over the tests remains to a large extent in the hands of psychometricians, other testing experts, and officials from the state departments of education that mandate the assessments.

Throughout U.S. history, then, teacher tests have been largely controlled by those outside the education profession: local officials, state legislators and bureaucrats, and testing firms. By many accounts (Berliner, 2005; Haney et al., 1987; Wilson & Youngs, 2005), these tests have tended to capture basic skills rather than more complex and job-specific competencies associated with helping students learn.

At the same time, rather than contributing expertise to the process, many educators condemn any form of standardized testing. In a 2002 issue of *English Education*, nearly all articles railed against the very idea of standardized testing for teachers:

Virtually all of the criticisms leveled against testing in schools also apply to the quick and dirty attempt to demand accountability in testing teachers. Timed tests given to children are really evaluating speed rather than thoughtfulness, and the same is true when they're given to adults. Multiple choice tests and contrived open response items are not meaningful ways of assessing how much students understand, and neither are they particularly effective in telling us how well educators can educate. (Kohn, quoted in Appleman & Thompson, 2002, p. 96)

Others take the view that educators need to assume control over licensing exams, moving toward new forms of assessment and focusing on professionally relevant knowledge and skill. We review some of these new assessments in the last section of this chapter. First, however, we review *scholarly* approaches to *studying* teachers' mathematical knowledge—studies that measure teacher knowledge not for high-stakes decisions, but instead for the purpose of uncovering what teachers do know, understanding the relationship between teacher knowledge and student learning, and,

eventually, for understanding the nature of knowledge and knowing in teaching. Insights from this body of work have opened new possibilities for the assessment of teachers' mathematical knowledge.

EARLY SCHOLARLY RESEARCH INVOLVING THE MEASUREMENT OF TEACHERS' MATHEMATICAL KNOWLEDGE

Studies measuring teachers' knowledge date to the 1960s, originating in what is now known as the "educational production function" literature. The main goal of this research program was to predict student achievement on standardized tests as a function of the resources held by students, teachers, schools, and others. Key resources were seen to include students' family background and socioeconomic status, district financial commitments to teacher salaries, teacher-pupil ratios, other material resources, and teacher and classroom characteristics (Hanushek, 1981; Greenwald, Hedges, & Laine, 1996). Published mainly in economic journals, these studies have been influential in shaping American public opinion and policy on how best to foster school improvement. In particular, educational economists' studies of teachers and their knowledge and skills have been especially influential.

One approach taken in this literature involves using data on teachers' preparation, coursework, and experience to predict student achievement. Key measures included overall teacher education level, certification status, number of post-secondary subject matter courses taken, number of teaching methods courses taken, and years of experience in classrooms. By using such measures, researchers assumed a connection between formal schooling and employment experiences and the more proximate aspects of teachers' knowledge and performance that produce student outcomes. This makes sense, given teacher educators' assertion that formal preparation provides the knowledge needed for teaching. However, reviews of educational production function studies have disputed the extent to which variables like teacher preparation and experience in fact contribute to student achievement (Begle, 1972, 1979; Greenwald et al., 1996; Hanushek, 1981, 1996), with conflicting interpretations resting on the samples of studies and methods used for conducting meta-analyses. Wayne and Youngs (2003) argue that when studies use subject-matter specific markers of teacher preparation—for instance, certification in mathematics rather than general certification—results are a bit

more positive. This increasing subject-matter specificity is one characteristic of studies conducted in the educational production function tradition.

A second approach measures teacher knowledge by looking at teachers' performance on certification exams or other tests of subject-matter competence. The first study to do so was the Coleman report, *Equality of Educational Opportunity*, completed in 1966. Coleman and colleagues measured teacher knowledge via a multiple-choice questionnaire, then used teachers' scores to predict student achievement in both reading and mathematics. They found that teacher scores did indeed predict student achievement in both subjects, and that this relationship grows stronger in the higher grades. Notably, however, none of the items on their measure focused specifically on mathematics; instead, the questionnaire asks teachers to complete a "short test of verbal facility" with items such as:

Dick apparently had little _____ in his own ideas, for he desperately feared being laughed at.

- a) interest
- b) depth
- c) confidence
- d) difficulty
- e) continuity

Thus the first study to identify a positive relationship between teacher knowledge and student mathematics achievement did not actually measure teachers' mathematical knowledge at all. Instead, it measured teacher knowledge via a vocabulary test—close, some would argue, to a test of general intelligence. Similar studies have taken advantage of available data from the NTE to create composite measures of teacher knowledge to compare to student achievement scores (Strauss & Sawyer, 1986; Summers & Wolfe, 1977). And more recently, Ferguson (1991) found that the mix of reading skills and professional knowledge captured on the Texas Examination of Current Administrators and Teachers, an exam used in the 1980s, was positively related to student achievement.

By the 1990s, some studies began to focus on how teachers' *mathematical* knowledge related to student gains in mathematics achievement (Harbison & Hanushek, 1992; Mullens, Murnane, & Willett, 1996; Rowan, Chiang, & Miller, 1997). This move toward subject-matter-specific measures coincided with increasing indications that teachers' effectiveness was related to their knowledge of subject matter rather than their general or pedagogical knowledge. Rather than use proxy measures such as degrees or mathematics coursework, however,

the strategy was to administer (or obtain teachers' scores on) problems they might assign students. Harbison and Hanushek (1992), for instance, administered a fourth grade student assessment to both teachers and students, using scores from the first group to predict performance among the second. Mullens et al. (1996) used teachers' scores recorded on the Belize National Selection Exam, a primary-school leaving exam administered to all students seeking access to secondary school. In both cases, the authors provided little information about the actual content of the assessment; Harbison and Hanushek (1992) listed the mathematical topics tested (number recognition, measurement, multiplication and division, rational numbers, unit measures, four operations, story problems) but did not include any actual items. Because these articles described assessments given to students, however, it seems probable that both studies measured teachers' competency with basic computation and mathematical procedures, rather than knowledge of teaching those topics.

Rowan et al. (1997) employed a somewhat different approach, one that began an increased effort to capture the mathematical knowledge used in classrooms, a trend we will take up in more detail in the next section of the paper. In this study, the authors used a one-item assessment to explore how teacher knowledge related to high-school student gains in the National Education Longitudinal Study. The item, developed by the Teacher Education and Learning to Teach study (Kennedy, Ball, & McDiarmid, 1993), reads:

Your students have been learning how to write math statements expressing proportions. Last night you assigned the following:

A one-pound bag contains 50 percent more tan M&Ms than green ones. Write a mathematical statement that represents the relationship between the tan(t) and green(g) M&Ms, using t and g to stand for the number of tan and green M&Ms.

Here are some responses you get from students:

Kelly: $1.5t = g$
 Lee: $.50t = g$
 Pat: $.5g = t$
 Sandy: $g = 1/2g = t$

Which of the students has represented the relationship best? (Mark ONE)

- ☐ All of them
- ☐ Kelly
- ☐ Lee
- ☐ Pat
- ☐ Sandy
- ☐ None of them. It should be _____
- ☐ Don't know.

When stripped of the teaching context, this might easily be a problem that would appear on an end-of-chap-

ter test or a student exam. The problem is different in that it is situated in a common task of teaching. However, like items on the NTE test, the teaching situation can be considered window-dressing.

Studies that directly measure teachers' verbal or mathematical knowledge typically show a positive relationship to student mathematics achievement (e.g., Boardman, Davis, & Sanday, 1977; Ferguson, 1991; Hanushek, 1972; Harbison & Hanushek, 1992; Mullens et al., 1996; Rowan et al., 1997; Strauss & Sawyer, 1986; Tatto, Nielsen, Cummings, Kularatna, & Dharmadasa, 1993; for an exception, see Summers & Wolfe, 1977; for reviews, see Greenwald et al., 1996; Hanushek, 1996; Wayne & Youngs, 2003). By using such measures, these studies assume a relationship between teacher content knowledge as measured by such assessments and the kinds of teaching performances that produce improved student achievement. Yet there is little evidence in this literature for what is actually measured by these assessments, and how what is measured relates to classroom performance. These tests were not developed, for the most part, by specifying domains of teacher knowledge and developing an assessment from this map. Instead, the claim to validity is that these assessments covered material in the K–12 curriculum, and which should be known by the students themselves at the end of schooling. This is true—but may miss important elements of the knowledge that makes teachers successful. And using these theoretically impoverished tools also limits the conclusions that could be drawn from these studies. Is it that teachers who perform better on these student-level tests simply make fewer mistakes in classroom teaching? Or, do teachers who perform better offer a qualitatively different kind of mathematics instruction? Measuring quality *teachers* through performance on tests of verbal or mathematics ability may overlook key elements in what produces quality *teaching*.

The conjecture that teachers may need to know subject matter differently than their students or non-teachers has become the subject of intensive research over the last two decades. In his presidential address to the American Educational Research Association, Shulman (1986) argued for the centrality of subject matter in teaching, drawing attention to the particular ways that teachers must know and use content knowledge in their work. He introduced the term “pedagogical content knowledge,” as a special kind of teacher knowledge that intertwines content and pedagogy. According to Shulman (1986, 1987) and colleagues (Wilson, Shulman, & Richert, 1987), pedagogical content knowledge includes understanding which topics students find interesting or difficult, the common

misconceptions that students have, and what forms of representation are useful for teaching particular topics. More recently, scholars who focus on studies of mathematics teachers and teaching have hypothesized that in addition to topic-specific knowledge of students and teaching, teachers might in fact use subject-matter knowledge in ways unique to teaching. Our research group, for instance, has developed the notion that teachers have mathematical knowledge that is “specialized” for the work of teaching. This specialized knowledge of the content can be seen in the demands of providing explanations for mathematical procedures or ideas, representing mathematical phenomena, and working flexibly with non-standard solution methods (Ball & Bass, 2003; Ball, Thames, & Phelps, 2005; Hill, Schilling, & Ball, 2004). In her study of Chinese teachers, Ma (1999) points to the depth and coherence of these teachers' knowledge both over the range of mathematics topics and over time, a kind of knowledge that she labeled “profound understanding of fundamental mathematics.” And Ferrini-Mundy, Floden, McCrory, Burrill, and Sadow (2005) argue that teaching algebra requires teaching-specific mathematical practices, among them “bridging” between what students know and disciplinary knowledge, and “trimming” disciplinary knowledge to levels that are understandable by students, but that are also still mathematically accurate.

To illustrate what it means to know mathematics *for teaching*, not just to know mathematics, we turn to an example from a topic in the upper elementary curriculum: division of fractions. By considering the ways teachers might need to know the mathematics behind this topic, we offer conceptual proof that there is more to be known than has been captured to date in most of the educational production function studies described above. This conceptual proof also serves as an outline of the mathematics-specific teaching knowledge used by teachers. Following the example, we then turn in the next section to measurement methods used in the literature that uncovered this knowledge.

We begin with the following problem:

$$\frac{5}{6} \div \frac{1}{3}$$

A common way to calculate the answer is using the infamous “invert-and-multiply” algorithm:

$$\frac{5}{6} \div \frac{1}{3} = \frac{5}{6} \times \frac{3}{1} = \frac{15}{6} = \frac{5}{2} = 2\frac{1}{2}$$

To teach division of fractions to students, teachers must of course be able to carry out the procedure correctly themselves. Thus one aspect of knowing mathematics for teaching is being able to do the mathematics that one is teaching one's students. Yet being able to divide fractions oneself is far from sufficient. After all, teaching mathematics is not simply "knowing" in front of students. Teaching requires making the content accessible, interpreting students' questions and productions, and being able to explain or represent ideas and procedures in multiple ways. For example, a student might ask a teacher why it works to invert and multiply, or why the second fraction is "flipped over" rather than the first. Or, a student might be perplexed because the answer, $2\frac{1}{2}$, is larger than both $\frac{5}{6}$ and $\frac{1}{3}$ —isn't dividing supposed to make the answer smaller? To respond to these questions, teachers must understand and be able to explain why the algorithm works, and be able to explain why and in what cases the quotient is larger than both the dividend and the divisor.

Or, instead of questioning why the standard procedure works, a student might claim that she doesn't even need to learn how to invert and multiply because she has found an easier way to divide fractions—namely, to just divide the numerators and divide the denominators:

$$\frac{5}{6} \div \frac{1}{3} = \frac{5 \div 1}{6 \div 3} = \frac{5}{2} = 2\frac{1}{2}$$

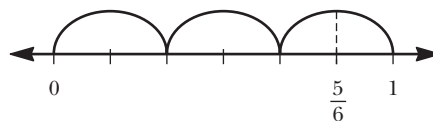
This situation is not uncommon; students often develop their own methods of computation—some valid and some not—and teachers must be able to size up these alternative approaches. In this case, the teacher would have to be able to judge whether the student's method for dividing fractions is mathematically valid or whether the correct answer was simply a coincidence. And, if the method is valid for this particular problem, would it work in general to divide any two fractions, or does it only work in special cases? Generating and testing examples and counterexamples in teaching involves mathematical reasoning. In fact, even the disposition to consider a method's generalizability and efficiency is in itself an example of the type of mathematical knowledge used in teaching.

What else arises in teaching division of fractions? Students, of course, do not always solve problems correctly. For example, suppose a teacher sees the following calculation on a student's paper:

$$\frac{5}{6} \div \frac{1}{3} = \frac{6}{15} = \frac{3}{5}$$

Spotting this as an incorrect answer is insufficient. A teacher must be able to figure out what steps the student might have taken to produce this error, as well as the reasons it might have been made, and then, in light of this mathematical analysis, determine an appropriate response.

After analyzing the above error, for example, a teacher might decide to pose a word problem that corresponds to the calculation to encourage the student to think about whether $\frac{3}{5}$ is a reasonable answer. Or, the teacher might decide to use a number line to represent the calculation:



Choosing representations such as story problems or diagrams involves mathematical reasoning and skill. For example, to use the number line representation above, a teacher needs to be able to map each element of the computation to the representation: Where is the $\frac{5}{6}$, the $\frac{1}{3}$, and the $2\frac{1}{2}$? Why does this representation model division? In this case, the representation is using a measurement interpretation of division (i.e., How many $\frac{1}{3}$ s are in $\frac{5}{6}$?). Five-sixths is marked on the number line and each loop represents "measuring off" $\frac{1}{3}$. Together, the loops represent the number of $\frac{1}{3}$ s that "go into" $\frac{5}{6}$: Two full loops and a half a loop, or two and a half $\frac{1}{3}$ s. In addition to making these correspondences, explaining this representation requires an understanding of central mathematical ideas such as the meaning of division and attention to the unit. The teacher must also recognize that, although this representation models division with fractions and shows why the quotient is greater than the dividend and the divisor, it does not readily explain the invert-and-multiply algorithm.

For this purpose, a teacher might rely on the inverse relationship between multiplication and division, noting that dividing by a number is equivalent to multiplying by its reciprocal. Alternatively, a teacher could appeal to students' knowledge that multiplying by 1 does not change the value of a number. In this case, the complex fraction

$$\frac{\frac{5}{6}}{\frac{1}{3}}$$

is multiplied by a convenient form of 1,

$$\frac{\frac{3}{1}}{\frac{1}{3}}$$

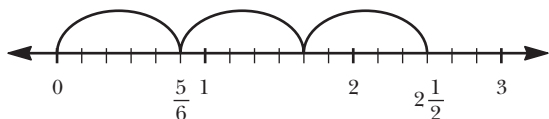
to show that $\frac{5}{6} \div \frac{1}{3} = \frac{5}{6} \times \frac{3}{1}$:

$$\frac{5}{6} \div \frac{1}{3} = \frac{\frac{5}{6}}{\frac{1}{3}} = \frac{\frac{5}{6} \times 1}{\frac{1}{3} \times 1} = \frac{\frac{5}{6} \times \frac{3}{1}}{\frac{1}{3} \times \frac{3}{1}} = \frac{\frac{5}{6} \times \frac{3}{1}}{1} = \frac{5}{6} \times \frac{3}{1}$$

Another means for explaining the invert-and-multiply algorithm again turns to the meaning of division, but this time using a partitive interpretation. In the case of

$$\frac{5}{6} \div \frac{1}{3},$$

the partitive interpretation asks: What number is $\frac{5}{6}$ one-third of? Like the measurement interpretation, the partitive interpretation can also be represented on a number line. If $\frac{5}{6}$ is one-third of the length of the whole, then the length of whole can be found by putting together three lengths of $\frac{5}{6}$, in other words, multiplying $\frac{5}{6}$ by 3, which is equivalent to $\frac{5}{6} \times \frac{3}{1}$, which is precisely the invert-and-multiply step:



Understanding these subtleties requires a robust understanding of the different meanings of operations and the ways they can be represented. And, as this example shows, teachers not only need to be able to map between a representation and the problem or concept being modeled, they must also determine which representation or explanation would be most appropriate for a particular instructional goal. Thus, and significantly for the assessment of teachers' mathematical knowledge, an item on a test for teachers that asks for the correct answer to

$$\frac{5}{6} \div \frac{1}{3}$$

might reveal knowledge of basic mathematics, but not *all* the possible types of knowledge needed for teaching this topic.

Clearly, global and verbal assessments do not capture the sort of mathematical knowledge and reasoning required to navigate the situations of teaching sketched in the example above. At the time when most of the education production function studies were conducted, however, few alternatives existed; scholars needed to make inferences about teachers' mathematical knowledge based on proxies and indirect indicators. Aligning teachers' and students'

scores on tests was a good bet. Yet these studies can only make inferences about the value of the specific knowledge domains that appear on the assessments: teachers' knowledge of the content they are directly responsible for teaching. The same can be said of the teacher certification exams reviewed to this point: although teaching-specific mathematics questions crept in and out of the tests of the nineteenth and twentieth centuries, they never captured much real estate. Why is this important? Without broader conceptualizations of teacher knowledge and without assessments that reflect this broader conceptualization, we lack precision about the kinds of teacher knowledge that matter. Is it, as some have concluded from analyzing the educational production function literature, simple competency in the topics teachers teach students? Or do teachers need to know "advanced" knowledge, a view common in the 1800s and again today (Education Trust, 1999)? Or, do teachers need specialized knowledge of the topics they teach, illustrated by the teaching tasks in our example? If we had assessments that reliably and validly measured these different conceptions of mathematical knowledge for teaching, and could construct models of student achievement using teachers' scores on such assessments, we could specify more precisely the nature of the knowledge that makes a difference for instructional quality and student learning. Without this precise specification, we cannot design preservice and in-service learning experiences that equip teachers with this knowledge.

Through the 1980s and 1990s, researchers began to approach these questions from another perspective: studying teachers' knowledge by studying teaching and teachers. Through this work, researchers began to hypothesize about the discipline-specific knowledge of content, students, and instruction required for the work of teaching. The studies they conducted focused on teachers' knowledge in practice situations; nearly all involved measuring or making inferences about teacher knowledge in some way. We turn next to a discussion of the strategies used by researchers engaged in this line of research.

METHODS FOR MEASURING PROFESSIONALLY SITUATED MATHEMATICAL KNOWLEDGE

Sparked in part by Shulman and colleagues' conceptualization of pedagogical content knowledge, and by a perceived need to assess the content knowledge needed for teaching as a basis for certification and licensure decisions, researchers in the 1980s

and 1990s sought to identify what teachers know (or should know) in teaching classroom mathematics. From the start, these studies suggested that teaching requires more than the ability to do the mathematics in the school curriculum. Teaching mathematics, in this view, is not the same as standing at the board and doing mathematics in front of students; it entails additional mathematical knowledge, competencies, and skills—what we call “mathematical knowledge for teaching” (Ball & Bass, 2003). This knowledge, as shown in the division of fractions example above, is multifaceted, including not only teachers’ ability to solve the problems their students are expected to solve, but also to understand the content in the particular ways needed for teaching it, to understand what students are likely to make of the content, and to craft instruction that takes into account both students and the mathematics.

In this section of the chapter, we discuss the methods used to study—and later measure—mathematical knowledge for teaching. To do so, we shift from making inferences about conceptions of teachers’ mathematical knowledge based on analyses of assessments, as we did with historical teacher exams and the NTE, to exploring the tools used to develop a theory of mathematical knowledge for teaching. We do not provide results from these studies, or chronicle the growth of this concept, because others have done so (Ball et al., 2001). Instead, we focus on the methods used, in part to provide readers with a sense for the array of methods and instruments that can measure mathematical knowledge for teaching. We also discuss the inferences that can be made from each type of research and consider each method’s advantages and disadvantages.

We have organized this section of the chapter both by method and chronology. We first discuss observations of teaching practice, beginning with efforts in the 1980s to uncover aspects of mathematical knowledge for teaching and later, post-2000, as they began to be used as a method for quantifying the mathematical characteristics of classroom instruction. We next discuss written tasks and interviews, linked methods used in the 1980s and beyond to assess the extent to which teachers held mathematical knowledge in forms useful to instruction. As concern over the level of teachers’ mathematical knowledge grew, so did programs designed to improve such knowledge—and measures to evaluate the results of these programs. We then describe some of the results of efforts to develop such evaluation tools—multiple-choice and other methods designed to assess teacher learning.

The assessment methods used in (and lessons learned from) the studies described below are related

only distantly to the methods used in early teacher certification exams and to the assessments given to teachers in the educational production function studies. While they are related more closely to the newer teacher certification exams described in the next section, development of the methods and content for both types of assessment have proceeded, to some degree, on separate tracks. One result is that lessons from one field (e.g., large-scale certification testing) do not easily penetrate another field (e.g., close studies of teacher knowledge). We observe here and in the next section the extent to which the separation of these paths has inhibited the development of stronger measures.

Uncovering Mathematical Knowledge for Teaching: Observations of Teaching Practice

Perhaps the earliest and most widely adopted technique for measuring teachers’ mathematical knowledge is one that, on its face, does not appear to be a measurement strategy at all: direct or videotaped observations of teachers’ mathematical instruction. However, because scholars followed these observations by careful analysis and explication of the quality and characteristics of the mathematics delivered to students, and because scholars often made inferences about teachers’ knowledge from such evidence, we argue that this strategy verged on measurement. We describe some of these early efforts below.

Early observational research focused on teachers’ mathematical knowledge had some common elements. Researchers typically collected tens, if not hundreds, of hours of observations or videotapes; published research, though, typically focused on a tiny fraction of the data, often even just a few minutes. Analysis that appeared in print was primarily qualitative, with researchers using methods and coding systems tailored specifically to the mathematical topics and questions at hand. Scholars typically combined observational records with other sources of data to gain insight into mathematical knowledge for teaching.

One classic example of a study in this tradition is Leinhardt and Smith’s (1985) study of expertise in mathematics instruction. To explore the relationship between teacher behavior and subject matter knowledge, the authors studied eight teachers intensively, collecting three months of observational field notes from their mathematics lessons, ten hours of videotaped lessons, interviews about the videotaped lessons and other topics, and responses to a card sort task. They used the classroom observations of instruction to construct a measure, ranking teachers’ knowledge as high, medium, or low based on “in-class discussions

over 3 years and by considering their presentations and explanations as well as their errors” (p. 251). This strategy—sorting teachers by their actual in-class mathematical performance—is rare in the literature, and, unfortunately, not well explicated in their published work. The authors then examined teachers’ knowledge in light of performance on interview tasks and examined three teachers’ teaching of fractions in much more depth by intensive description of single lessons on simplifying fractions. This method—thick description (Geertz, 1973)—would prove a mainstay in probing the mathematical knowledge needed for teaching.

Borko et al. (1992) provide another example of the observational method for measuring teachers’ knowledge of mathematics. Their article focuses on a few minutes of an hour-long review lesson, moments in which a student teacher was asked by a student to explain the division of fractions algorithm. Audio-tapes of the lesson were augmented by fieldnotes taken by live observers, and allowed the construction of a dependent variable of sorts: an assessment of this teacher’s capacity to provide a conceptually-based justification for the standard algorithm. The authors showed that this teacher’s capability in this area was poor—she used a concrete model that actually represented multiplication rather than division. They then used data from interviews, her performance on open-ended mathematics problems, and records from this teacher’s teacher education program to explain her in-class performance.

These early observational studies began to record elements of mathematical knowledge for teaching: presentations, representations, explanations, linking between key mathematical elements in instruction, flexibility, and other uses of content knowledge that could be seen in observations of teachers’ work. As a result, many of these studies were largely descriptive; rather than inferring any *individual’s* overall level of knowledge, classroom practice, or learning, these studies served as explorations of the *territory* of mathematical knowledge for teaching. Borko et al., for instance, did not make global generalizations about their teacher’s level of mathematical knowledge for teaching or mathematics instruction per se, or attempt to use their observations to inform any high-stakes (e.g., licensure) decision. This is entirely appropriate, given the limitations of the method.

More recently, several standardized protocols for observing the mathematical quality of classroom instruction (or videotaped records) have emerged in response

to the need to study teacher performance, learning, and change. These protocols are designed to make more generalizable inferences about particular teachers’ classroom mathematical work, and thus differ from the early observational studies in the sense that they are intended to evaluate, with some degree of confidence, how well a teacher or group of teachers can teach mathematics. Leaving aside instruments that focus solely on the pedagogical aspects of teaching mathematics—e.g., the degree to which students work in groups, work on extended investigations, or answer questions—a number of instruments combine descriptions of the nature of classroom work with estimates of teachers’ skill and knowledge in teaching: the *Reformed Teaching Observation Protocol* (RTOP) (Sawada & Pilburn, 2000), *Inside the Classroom Observation and Analytic Protocol* (Horizon Research, 2000), and the *Learning Mathematics for Teaching: Quality of Mathematics in Instruction* (LMT-QMI) instrument (LMT, 2006a).³ These instruments vary in the extent to which they infer the *quality of the mathematics in instruction*, as opposed to the *quality of mathematics instruction*. All three instruments, for instance, ask for ratings of the extent to which content is presented accurately. All three ask whether the content presented to students is mathematically worthwhile. And all three ask about some elements of what some consider “rich” instruction—the use of representations, explanations, and abstractions, for instance. As such, all three contain elements that some might use to infer the *quality of the mathematics in instruction*, or the accuracy of content, richness of representation and explanation, and connectedness of classroom tasks to mathematical principles. We argue that while not a measure of teacher knowledge per se, the quality of the mathematics in instruction is a product of, and thus closely related to, teachers’ mathematical knowledge for teaching.

RTOP and Horizon’s instruments, however, both embed the ratings of teacher knowledge in larger scales intended to measure the extent to which classroom instruction aligns with the National Council for Teachers of Mathematics standards. For instance, RTOP has the following prompts:

- In this lesson, student exploration preceded formal presentation.
- This lesson encouraged students to seek and value alternative modes of investigation or of problem-solving.
- Connections with other content disciplines and/or real world phenomena were explored and valued.

³ We also surveyed the TIMSS-R Video Math Coding Manual (2003). The coding effort here appears more focused on the quality of the problems presented and enacted in classrooms, rather than making inferences about teacher knowledge.

- Students were actively engaged in thought-provoking activity that often involved the critical assessment of procedures.

The issue here is not whether these are desirable lesson characteristics, but instead what is being measured. These two instruments are designed, according to their materials, to measure the quality of mathematics instruction, where quality is defined as both the richness and correctness of mathematical content *and* the way material is conveyed to students. In these instruments, no direct measure of teacher knowledge is available. Instead, teacher knowledge is estimated as a component of how mathematical material is presented to students. The RTOP and Horizon instruments are also designed for rating both science and mathematics lessons, which limits the specificity with which they can ask about particular mathematical practices.

The LMT-QMI instrument was intended to serve as a means to validate multiple-choice measures of teachers' knowledge for teaching mathematics. As such, it is catholic with respect to teaching style, but places heavy emphasis on ways in which teachers' mathematical knowledge might appear in instruction. It includes many of the elements common to the three instruments, but at a finer level of specification. The accuracy of mathematical content, for instance, is assessed over a broad domain of prompts, including use of mathematical language, computational errors, explanations, and notation. The rubric also asks about the presence and accuracy of mathematically rich elements of instruction, including representations, links between multiple representations, explanations, justifications, and the explicit development of mathematical practices. As such, the LMT-QMI measures the quality of the mathematics in instruction.

Observations of teaching have high validity, if the goal is measuring mathematical knowledge for teaching, because classrooms are where mathematical knowledge for teaching is expressed—in teachers' classroom moves, explanations, representations, and computations. However, all three instruments suffer from two main problems. The first involves language and interpretation. Put simply, prompts such as “the mathematics/science content was significant and worthwhile” are just a collection of words; their interpretation will be shaped by the coders' knowledge and views of mathematics itself, and will vary significantly across individuals. In our own work using LMT-QMI, for instance, we have found that the rating of an explanation as accurate or inaccurate—or even as an explanation at all—is shaped by observers' knowledge of the specific mathematical topic, knowledge

of what constitutes explanation within mathematics, knowledge of mathematics education, and interpretation of the instrument. The three instruments attempt to eliminate this problem in various ways: Horizon researchers, for instance, spent two days in training and were given an annotated coding manual prior to conducting actual observations. LMT coders spent nearly two years coming to agreement on the coding scheme itself, and on how particular video clips should be rated. And RTOP offers online training for those interested in using their instrument.

The second problem relates to generalizability and the logistics of measurement in the context of research projects: one must observe teachers for multiple lessons before reaching any conclusions about their level of mathematical knowledge for teaching (Rowan, Harrison, & Hayes, 2004; Stein, Baxter, & Leinhardt, 1990). On any particular day, a teacher might be working on a topic with which she is unfamiliar, making a generalization from that day's performance to her overall level of knowledge invalid; there is also natural variation in teachers' knowledge across content areas (number, operations, geometry) that may bias results from a small sample of lessons. Clearly, making many observations over a large sample of teachers participating in preservice or in-service coursework would be a significant burden on any study; the hundreds of teachers required to model statistically student achievement effectively prohibits this strategy. Thus while classroom and videotape observations can be generative for answering fine-grained questions about particular teachers working with students around particular topics, this technique is not amenable to studies that formally test the effects of preservice or teacher development programs, or that link different conceptualizations of teacher knowledge to student achievement.

Exploring Teacher Knowledge: Mathematical Interviews and Tasks

A second method used widely for investigating mathematical knowledge for teaching is mathematical tasks and interviews. These mathematical tasks are different from mathematics assessments, which we consider below, by virtue of the fact that they are not designed to yield generalizable inferences about individual participants' knowledge, but to help scholars understand the nature and extent of teachers' knowledge. In other words, the focus of inference is a group of teachers' performance on the task itself, often to extend our understanding of the task or of teacher knowledge very generally, rather than the performance of individual teachers (e.g., drawing inferences

about mathematical knowledge as one component of suitability for teaching). Tasks can be paper-and-pencil problems or included as components of interviews exploring teachers' mathematical knowledge (Borko et al., 1992; Ma, 1999; Simon, 1993; Tirosh & Graeber, 1990) and have typically been designed to study the nature of teachers' mathematical knowledge in specific content areas such as multiplication, rational numbers, division, geometry, and functions (Ball et al., 2001). Some of the tasks that have been used to study teachers' knowledge, such as those derived from elementary textbooks or from studies of children's thinking, could also be used to assess students' mathematical knowledge. Other tasks have been explicitly designed to investigate mathematical knowledge beyond what students would be expected to be able to do, such as explaining what misconception led to a student's alternative method or generating a representation that could be used to teach a particular concept. These tasks are often based on situations that arise in teaching such as analyzing a student's error or answering a student's question. Providing a detailed analysis of all of the mathematical tasks and interview probes that have been used to study teachers' mathematical knowledge for teaching is beyond the scope of this chapter; however, we describe below some of the tasks that have been used in one content area, division, as a representative sample of the types of tasks that have been used.

Some tasks used on written assessments and interviews were, literally, the same tasks given to students. For instance, Graeber, Tirosh, and Glover (1989) investigated the extent to which prospective elementary teachers held misconceptions about multiplication and division similar to those commonly held by children—namely, that multiplication always yields larger numbers and division always yields smaller numbers. They used a written test constructed by slightly modifying 26 problems that had been administered by Fischbein, Deri, Nello, and Marino (1985) in their study of adolescents' misconceptions about operations used to solve multiplication and division word problems. The test presented various word problems and asked respondents not to perform the calculation, but “to write an expression in the form of ‘a number, an operation, and a number’ that would lead to the solution of the problem” (Graeber et al., 1989, p. 96). It was administered to 129 prospective elementary teachers; 33 follow-up interviews were conducted. Interviewees were given a problem similar to one that had been answered incorrectly on their written form. If they still answered incorrectly, they were asked to explain their answer and show how they would check their solution. Upon realizing (or being told) that the original

expression was incorrect, interviewees were asked to explain what they thought might have caused them to write an incorrect expression. This example shows the ways tasks and interviews might be combined. It also demonstrates one major strength of this method: questions can be tailored to the answers given by specific respondents, often on the spot, to probe the reasons for misconceptions or the support for understanding. It also illustrates a central finding of the studies that used this method: some prospective teachers had serious gaps in their understanding of the mathematics they would be expected to teach.

In another study, Tirosh and Graeber (1989) investigated whether prospective teachers' misconceptions about multiplication and division were explicitly held or just implicitly influenced their calculations. A written instrument was administered to 136 prospective elementary teachers, and then 71 were interviewed. The test asked respondents to label each of the following statements as “true” or “false” and to provide a justification for their response:

- A. In a multiplication problem, the product is greater than either factor.
- B. The product of $.45 \times 90$ is less than 90.
- C. In a division problem, the quotient must be less than the dividend.
- D. In a division problem, the divisor must be a whole number.
- E. The quotient for the problem $60/.65$ is greater than 60.
- F. The quotient for the problem $70 \div \frac{1}{2}$ is less than 70.

There were also questions that asked respondents to write expressions for word problems, as well as perform calculations that served as counterexamples to the statements above. This example demonstrates a class of problems slightly removed from actual classroom teaching itself, but that is not knowledge used exclusively in teaching. Few teachers would give the above examples A–F directly to students; yet a firm foundation in operations with rational numbers, and in mathematical proof and reasoning, is necessary for teaching children successfully. Building on this work, Tirosh and Graeber (1990) then explored whether cognitive conflict could be introduced to prompt prospective teachers to confront their misconceptions, in particular, the misconception that in division the quotient must be less than the dividend. Twenty-one respondents who correctly computed $3.75 \div .75$ but agreed that “the quotient must be less than the dividend,” were selected for interviews. The interview protocol was designed to point out the conflict between these two responses and to consider the source of this inconsistency. This study reinforced the predominant

findings of other research—i.e., that some teacher candidates do not adequately know the mathematics they will teach—but also used tasks and interviews as a means of studying teacher learning.

With the work of Ball (1990), the field turned more directly toward examining knowledge of mathematics that is specialized to teaching. Ball studied elementary and secondary prospective teachers' knowledge of division in three contexts: division with fractions, division by zero, and division with algebraic equations. The following example tasks required respondents to generate and explain representations, and interview probes were designed to gather information about their notions of what counts as a mathematical explanation:

1. People have different approaches to solving problems involving division with fractions. How would you solve this one:

$$1\frac{3}{4} \div \frac{1}{2}$$

2. Sometimes teachers try to come up with real-world situations or story problems to show the meaning or application of some particular piece of content. This can be pretty challenging to do. What would you say would be a good situation or story for $1\frac{3}{4} \div \frac{1}{2}$ —something real for which $1\frac{3}{4} \div \frac{1}{2}$ is the appropriate mathematical formulation?
3. Suppose that a student asks you what 7 divided by 0 is. How would you respond? Why is that what you'd want to say?
4. Suppose that one of your students asks you for help with the following:

$$\text{If } \frac{x}{0.2} = 5, \text{ then } x =$$

How would you respond? Why is that what you'd do?

These four items illustrate different aspects of what Ball and her colleagues (Ball & Bass, 2003; Ball, Thames, et al., 2005) call specialized mathematical knowledge. Using fractions to calculate the answer to real-life problems is not uncommon among mathematically literate adults; however, constructing real-life problems to illustrate fractional computation is likely limited to teachers. Responding to students' questions about division by zero requires more than the knowledge that “this cannot be done”—it requires mathematical knowledge of the reason this is undefined. And the fourth item requires insight into how to make the procedures of algebraic manipulations comprehensible to learners, a different kind of understanding from simply being able to do the procedure oneself.

These items, written by Ball and others working on the Teacher Education and Learning to Teach project (TELT) (Kennedy et al., 1993), have been widely used in the field. Ma (1999), for instance, compared U.S. teachers in the TELT sample with Chinese teachers,

analyzing responses to the items. She found that many Chinese teachers had a depth of understanding, for example how mathematical topics are related/connected, not present in U.S. teachers.

Another example of an instrument that taps specialized knowledge for teaching is described by Simon (1993), who investigated prospective teachers' knowledge of division using written open-response items and individual interviews. The questions were designed to assess the connectedness of prospective teachers' knowledge and their understanding of units:

1. Write three different story problems that would be solved by dividing 51 by 4 and for which the answers would be, respectively:

a) $12\frac{3}{4}$ b) 13 c) 12

2. Write a story problem for which $\frac{3}{4}$ divided by $\frac{1}{4}$ would represent the operation used to solve the problem.
3. In the long division carried out as in the example below [the actual item shows the problem 715 divided by 12 being calculated with the standard long division algorithm], the sequence *divide, multiply, subtract, bring down* is repeated. Explain what information the *multiply* step and the *subtract* step provide and how they contribute to arriving at the answer.

Like Ball's items, these ask respondents to perform tasks that are unique to teaching: writing story problems that take into account the different interpretations of remainders, and explaining how portions of the standard long division algorithm work. The entire written instrument was administered to 33 prospective elementary teachers, and another eight were interviewed about a subset of the problems. Interviewees were asked to “think aloud as they solved the problem” and their comments were probed by the interviewer. Simon's results echoed previous studies, finding that, in general, prospective teachers' knowledge was fragmented and procedural.

The division tasks described above are examples of the types of written questions and interview prompts that have been used to better understand teachers' knowledge of mathematics. In developing the tasks, researchers had to grapple with questions about what mathematical knowledge is needed for teaching and how it might be assessed. It makes sense that the source of many such tasks were problems students encounter in K–12 schools, for teachers certainly must be able to solve the problems they give to their students. But the tasks also try to tap into mathematical knowledge needed for teaching beyond what students need to know, for example, knowing not only how to perform an algorithm, but also why it works and how it can be represented to students. Thus, designing these tasks

required researchers to reconceptualize what it means to know elementary mathematics in ways that are central to teaching and enabled scholars to develop a more refined conception of the mathematical knowledge needed for teaching.

The use of these tasks helped explicate what knowledge for teaching might look like—in essence, making the case for specialized, professional knowledge. These studies have also brought attention to problems with the quality of mathematical instruction delivered in the U.S. and can be used to inform the content and methods of teacher education courses. And, from a measurement standpoint, there are certain benefits to using these types of measures. Once tasks are designed, they are fairly low cost and easy to implement. There is also high face validity, at least as compared to standard mathematics tests and/or multiple-choice exams.

However, their use is also constrained in ways similar to direct observations. Analyzing or grading teachers' responses is, at scale, a lengthy and expensive process. Further, these tasks and interviews cannot be used to make inferences about particular individuals' (or even groups of individuals, we argue) level of knowledge, for they have seldom been studied to provide information regarding their validity, reliability, and generalizability. Contrary to some beliefs in mathematics education research, using an open-ended task or interview format does not free researchers from an obligation to examine the measurement qualities of the assessments used. Instruments with only a few mathematics problems, or with a very narrowly defined band of mathematics content, encounter the same generalizability problem that using only a handful of observations engenders, that of making a broad claim about knowledge from only a few samples of teachers' work. Open-ended tasks and interview probes are not always reliable, either; teachers' performance on one open-ended task designed by Ball—asking teachers to provide a story that represents $1\frac{3}{4}$ divided by $\frac{1}{2}$ —has been used as evidence that U.S. teachers lag in mathematical knowledge for teaching. A pilot of the multiple-choice version of this measure, however, suggests that part of the item fails to discriminate reliably between highly knowledgeable and less knowledgeable teachers.

Moreover, although these tasks are often situated in teaching contexts, they do not necessarily indicate how a teacher uses mathematical knowledge in practice. A teacher may perform well on the tasks in a clinical or instructional setting but not be able to access this knowledge during a lesson (Borko et al., 1992). Observations of teaching have revealed differences in teachers' mathematical knowledge that

were not evident in interview tasks, such as their skill in selecting and using representations and their emphasis of key concepts during lessons (Leinhardt & Smith, 1985).

Assessing Teacher and Student Learning: A New Generation of Teacher Assessments

As researchers uncovered elements of mathematics knowledge unique to teaching, scholars and evaluators faced new measurement needs. One concerned the development of tools to test the relationship between mathematical knowledge for teaching and student achievement; the other involved the development of efficient tools to track change in teachers' mathematical knowledge over time. In both cases, researchers required tools that could be used to make valid and reliable inferences about individuals' or groups' mathematical knowledge for teaching. And researchers needed these tools at scale. Although they are not used for teacher certification, they are often used to deliver information about dozens if not hundreds of teachers, often at multiple time points.

These issues spurred the development of what many consider to be an unusual choice given the history of antipathy between professional educators and teacher assessments: pencil-and-paper tests. In some cases, such as the Study of Instructional Improvement/Learning Mathematics for Teaching (SII/LMT) measures (LMT, 2006b) and the SimCalc rate and proportionality teaching survey (Shechtman et al., 2006), these tests come in multiple-choice format. In other cases, such as the Knowledge for Algebra Teaching (KAT) measures (Knowing Mathematics for Teaching Algebra Project, 2006) and the Diagnostic Teacher Assessments in Mathematics and Science (DTAMS) measures (Bush, Ronau, Brown, & Myers, 2006) the instruments contain both multiple-choice and short open responses—often simply the answer to a computation problem, but sometimes short explanations for problems or procedures. All of the above tests, however, emphasize the ability to measure teachers' knowledge at scale, and with known reliability and validity. Teachers' answers to the items are marked as correct or incorrect, and a score is calculated, typically in standard deviation units. Thus these assessments are easily scored, with the ease of scoring a major factor in the selection of this format.

One reason for this selection was the scope of the research projects and intended uses of these measures. Having new insight into mathematical knowledge for teaching, scholars now turned to modeling its growth and contributions to the educational production function. For instance, the SII/LMT measures grew out of

a study designed to examine and compare the performance of elementary schools, teachers, and students participating in one of three whole-school reforms. Teacher learning was one potential target of these reforms, and also a mediator of student outcomes (Hill, Rowan, & Ball, 2005). Later, the instrument's authors received continued funding from the National Science Foundation to pilot additional items and extend the measures upwards to middle school. The SimCalc project began as a curriculum development project, one aimed at introducing the mathematics of change and variation in the early grades, typically with some technological component. The SimCalc instrument grew out of interest in teachers' learning as a result of engaging with their professional development and curriculum, and the desire to model student achievement as a function of teacher learning (Shechtman et al., 2006). KAT began as a project to explore the nature of knowledge used in teaching algebra and progressed to instrument development. Instrument developers hope to use these measures in models of student achievement. DTAMS is solely a measures development project, but has designed measures to be used not only to diagnose the status of middle school teachers' mathematical knowledge, but to track teacher learning and student achievement as a function of teacher knowledge.

This new generation of pencil-and-paper tests shares some common attributes. All, for instance, contain items intended to represent the mathematics problems encountered in teaching, rather than only "common" content knowledge, or knowledge of mathematics that is common across professions and available in the public domain. As such, items focus on teachers' grasp of representations of content, unusual student work or mathematical methods, student errors, mathematical explanations, teaching moves, and other raw materials of classroom mathematics. This sets these instruments apart from historical teacher tests, which focused only on asking teachers to solve the mathematics problems they would be teaching to students, or in some cases, mathematics problems somewhat more advanced than the curriculum they would be teaching. The new instruments also focus on relatively narrow mathematical domains, rather than returning an overall score across all mathematical content. And, critically in our view, all these tests have been developed by experts in mathematics, in the teaching of mathematics, and in psychometrics. This collaboration has led to explicit interest in the multidimensionality of the measures. Rather than attempting to write measures that represent "pure" mathematical or pedagogical knowledge, as many test development firms do for technical reasons, these

measures embrace the idea that teachers' knowledge has many facets that must be included in any instrument. Project psychometricians, for their part, have been willing to work with this specification and, in turn, lent generous expertise in gauging the reliability and validity of these instruments. Reliability and validity analyses, which we shall discuss in more detail below, is relatively rare anywhere in the educational assessment enterprise (Messick, 1989).

One major difference among these pencil-and-paper assessments is in instrument goals and, by extension, the interpretation of scores. Two projects, KAT and DTAMS, criterion referenced their exams—DTAMS to nine policy documents describing both middle school student *and* teacher knowledge and KAT to a rubric that outlines the categories of teacher knowledge and the content involved in teaching algebra. By choosing to criterion reference and assuming defensible cutoffs for different performance levels, scores on this measure can be concretely interpreted vis-à-vis this standard—e.g., teacher Y knew X% of the mathematics needed according to standards. A third project, SII/LMT, initially designed its measures to discriminate accurately among teachers, in essence ordering them as correctly as possible relative to one another and to the underlying trait being assessed, mathematical knowledge for teaching. Although this design allows users to measure change over time as teachers learn, raw frequencies have no meaning relative to any criterion, such as "the mathematical knowledge teachers need to know." Finally, the fourth project, SimCalc, designed its measures to gauge curriculum-specific mastery of content—whether teachers grasp key principles in rate and proportionality, principles needed to teach using the SimCalc curriculum.

It is in the arena of underlying theory, however, that these instruments differ most. Despite claiming to cover roughly the same terrain, these projects have strikingly different approaches to specifying domains for measurement—in essence, different approaches to organizing what is "in" mathematical knowledge for teaching. SII/LMT has four knowledge domains; SRI has two; DTAMS has four; and KAT's conceptual framework crosses three aspects of knowledge for teaching and four mathematical domains. None of these conceptual maps match one another exactly, and none match Shulman's original formulation of pedagogical content knowledge. From one view, these differences might reflect ways the content itself (algebra, number) influences the knowledge needed for teaching. But at least to those interested in building theoretical coherence around mathematical knowledge for teaching, the variety of approaches is distressing.

Some examples from the actual assessments help elaborate these theoretical differences. Three-quarters of DTAMS' items measure important types of mathematical knowledge that teachers are charged with developing in students:

1. *Memorized facts and skills*: Which of the following numbers is the least common denominator of $5/9$ and $7/12$? (a: 108; b: 36; c: 72; d: 3)
2. *Conceptual understanding*: Name two numbers between 1.35 and 1.36. (a: $1.3\overline{5}$ and $1.\overline{35}$; b: 0.351 and 0.352; c: 1.345 and 1.354; d: There are no numbers between them.)
3. *High-order thinking*: Mr. Short is three paper clips in height. Mr. Tall is five buttons in height. Two buttons laid end-to-end are the same height as three paper clips laid end-to-end. Find the height of Mr. Tall in paper clips. Justify your solution.

These are relatively common problems in the “public domain,” so to speak, of mathematical knowledge. Other than the fact that teachers are more likely than the general population to be solving such problems, none represent knowledge specific to teaching. One quarter of the items measure mathematically situated pedagogical knowledge:

1. A student said, “Whenever you add two non-zero integers the sum is always between the two integers.” Explain why this is incorrect. Explain how you would use a diagram or model to help the student understand she is incorrect.
2. A student claims that all squares are congruent to each other because they all have four right angles. Why is this claim incorrect? Explain how you would help the student understand the error in her thinking.

These pedagogical items, which like the higher-order thinking items are in short-answer format and hand-scored, mimic the mathematical judgments teachers must make during classroom work—understanding common mathematical errors and designing instruction to correct them.

The SII/LMT items, by contrast, are in multiple-choice format and cover three domains of teacher knowledge, with one domain sub-divided into two constituent parts:

Content Knowledge:

Common content knowledge, or the mathematical knowledge teachers are responsible for developing in students:

Ms. Dominguez was working with a new textbook and she noticed that it gave more attention to the number 0 than her old book. She came across a page that asked students to determine if a few statements about 0 were true or false. Intrigued, she showed them to her sister who is also a teacher, and asked her what she thought.

Which statement(s) should the sisters select as being true? (Mark YES, NO, or I'M NOT SURE for each item below.)

	Yes	No	I'm not sure
0 is an even number	1	2	3
0 is not really a number. It is a placeholder in writing big numbers.	1	2	3
The number 8 can be written as 008.	1	2	3

Items in this category are similar to DTAMS' memorized facts and skills and conceptual understanding. They tap knowledge that is in the public domain, including that used in other professions (e.g., writing 8 as 008 as in computer science).

Specialized content knowledge, or mathematical knowledge that is used in teaching, but not directly taught to students:

Imagine that you are working with your class on multiplying large numbers. Among your students' papers, you notice that some have displayed their work in the following ways:

Student A	Student B	Student C
$\begin{array}{r} 35 \\ \times 25 \\ \hline 125 \\ +75 \\ \hline 875 \end{array}$	$\begin{array}{r} 35 \\ \times 25 \\ \hline 175 \\ +700 \\ \hline 875 \end{array}$	$\begin{array}{r} 35 \\ \times 25 \\ \hline 25 \\ 150 \\ 100 \\ +600 \\ \hline 875 \end{array}$

Which of these students would you judge to be using a method that could be used to multiply any two whole numbers?

	Method would work for all whole numbers	Method would <i>not</i> work for all whole numbers	I'm not sure
Method A	1	2	3
Method B	1	2	3
Method C	1	2	3

Here, the teacher must inspect the three student methods to determine what, in fact, is occurring. Student A, for instance, has multiplied 5×25 and then 30×25 ; student B is using a version of the standard U.S. algorithm; and student C has used a partial product method. If the teacher understands these explanations for the students' methods, she must then make a determination about whether each method generalizes to the multiplication

of other whole numbers, perhaps by referencing the commutative or distributive properties of multiplication. This work is entirely mathematical, but it is not mathematical work done by many non-teaching adults.

Knowledge of Content and Students

Knowledge of content and students, or the amalgamated knowledge that teachers possess about how students learn content:

Takeem's teacher asks him to make a drawing to compare $\frac{3}{4}$ and $\frac{5}{6}$. He draws the following:



and claims that $\frac{3}{4}$ and $\frac{5}{6}$ are the same amount.

What is the most likely explanation for Takeem's answer? (Mark ONE answer.)

- a) Takeem is noticing that each figure leaves one square unshaded.
- b) Takeem has not yet learned the procedure for finding common denominators.
- c) Takeem is adding 2 to both the numerator and denominator of $\frac{3}{4}$, and he sees that that equals $\frac{5}{6}$.
- d) All of the above are equally likely.

Here, a teacher must look beyond the standard method for comparing fractions (finding common denominators) to see that Takeem has focused on the amount that is missing from each whole. This knowledge is likely held by teachers who have worked with children learning to compare fractions; it also corresponds to Shulman's "common student misconceptions" portion of pedagogical content knowledge.

Knowledge of Content and Teaching

Knowledge of content and teaching, or mathematical knowledge of the design of instruction, includes how to choose examples and representations, and how to guide student discussions toward accurate mathematical ideas.

While planning an introductory lesson on primes and composites, Mr. Rubenstein is considering what numbers to use as initial examples. He is concerned because he knows that choosing poor examples can mislead students about these important ideas. Of the choices below, which set of numbers would be best for introducing primes as composites? (Mark one answer.)

- | | Primes | Composites |
|----|--|------------|
| a) | 3, 5, 11 | 6, 30, 44 |
| b) | 2, 5, 17 | 8, 14, 32 |
| c) | 3, 7, 11 | 4, 16, 25 |
| d) | 2, 7, 13 | 9, 24, 40 |
| e) | All of these would work equally well to introduce prime and composite numbers. | |

In this item, the teacher must first know that students are likely to think that all prime numbers are odd; including 2 is important given this misconception. Students are also likely to think that odd numbers cannot be composite—making choice (d), which tests both misconceptions, the best of the four options. This SII/LMT category is the furthest from pure mathematical knowledge, as it involves reasoning about both students and teaching. It corresponds with another component of Shulman's pedagogical content knowledge (choosing the "best representation") and to DTAMS' pedagogical knowledge category.

The schematic for item development used by the Knowledge of Algebra for Teaching is more detailed, consisting of 24 separate cells that are the product of two algebraic content areas, three types of algebra knowledge for teaching, and four domains of mathematical knowledge. The first side of this three-dimensional matrix names two central algebraic topics: expressions, equations, and inequalities; and functions and their properties. The second side names algebra knowledge for teaching, which in this view consists of knowledge of school algebra, or the mathematics in the intended curriculum that *students* should learn; advanced knowledge, or college-level mathematics that gives teachers perspectives on the trajectory of mathematics beyond middle or high school; and teaching knowledge, or mathematical knowledge beyond that directly taught to students, but which is useful in instruction. This last category includes the mathematics involved with student misconceptions, knowledge of materials and texts, and ways to develop mathematics within and across lessons with particular goals in mind. Finally, the third side of the matrix encompasses four domains of mathematical knowledge, including "core" content knowledge, or declarative or substantive knowledge; representations of mathematical content such as number lines, tables, graphs, and area models as well as the materials that can be used to represent content during instruction (e.g., algebra tiles); applications and contexts, such as situations that can be modeled by linear functions; and reasoning and proof, or knowledge of how truth is established in the discipline.

Four items illustrate aspects of this domain specification. The first item comes from the category of school algebra, or the topics that teachers would directly teach students. It illustrates an "application" problem from KAT's four domains of mathematical knowledge, modeling the ways in which exponential functions might be used in real-life situations:

Which of the following situations can be modeled using an exponential function?

- i. The height h of a ball t seconds after it is thrown into the air.
- ii. The population P of a community after t years with an increase of n people annually.
- iii. The value V of a car after t years if it depreciates $d\%$ per year.

- A. i only
- B. ii only
- C. iii only
- D. i and ii only
- E. ii and iii only

This problem is similar, perhaps, to those found in high school or college textbooks. This item would fall in the SII/LMT common content knowledge category, and in DTAMS' conceptual knowledge category.

The next KAT item is from the category of advanced knowledge or knowledge beyond the actual middle or high school curriculum, and focuses on functions and their properties:

For which of the following sets S is the following statement true?

For all a and b in S , if $ab = 0$, then either $a = 0$ or $b = 0$.

- i. the set of real numbers
- ii. the set of complex numbers
- iii. the set of integers mod 6
- iv. the set of integers mod 5
- v. the set of 2×2 matrices with real number entries

- A. i only
- B. i and ii only
- C. i, ii, and iv only
- D. i, ii, iii, and iv only
- E. i, ii, iii, iv, and v

Answering this item properly requires an important type of mathematical reasoning. Teachers must be sensitive to the fact that familiar properties from algebra are not necessarily true in general, but, rather, depend on the set being referenced. In the item above, the statement *if $ab = 0$, then either $a = 0$ or $b = 0$* is true when working with all the choices save for the set of real 2×2 matrices and integers mod 6.⁴ So although integers mod 6 is not in the K–12 curriculum, teachers must not lead high school students to believe that the original statement is universally true. This category is unique to the KAT measures and represents one

very traditional line of thinking about the mathematical knowledge teachers need—that a teacher should know “very much more than he is expected to teach” (Michigan Department of Public Instruction, 1898, p. 10; see also Education Trust, 1999, p. 4).

The third item is open-ended, tapping “teaching knowledge” of student errors, and is also classified as “core content knowledge” by the KAT project:

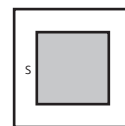
A student solved the equation $3(n - 7) = 4 - n$ and obtained the solution $n = 2.75$.

What might the student have done wrong?

To answer correctly, teachers must know how to correctly solve this equation for themselves, and they must also know a common student error, namely that students will forget to distribute the 3 over the 7 as well as the n . This corresponds to the SII/LMT category named knowledge of students and content.

The last item is also open-ended and draws from the “teaching knowledge” category, focusing on representations of expressions:

Hot tubs and swimming pools are sometimes surrounded by borders of tiles. The drawing at the right shows a square hot tub with sides of length s feet. This tub is surrounded by a border of 1 foot by 1 foot square tiles.



How many 1-foot square tiles will be needed for the border of this pool?

- a. Paul wrote the following expression:

$$2s + 2(s + 2)$$

Explain how Paul might have come up with his expression.

- b. Bill found the following expression:

$$(s + 2)^2 - s^2$$

Explain how Bill might have found his expression.

- c. How would you convince the students in your class that the two expressions above are equivalent?

Here, teachers must “see” how the same representation might yield very different expressions, mapping closely between the representation and the construction of each expression. Teachers must also make a judgment about how best to prove these two expressions are equivalent, a question that grapples with the complex intersection between content, students, and teaching. This item would fall, in the SII/LMT scheme, into the “specialized” and “knowledge of content and teaching” categories.

⁴ For example, when working with the set of integers mod 6, 4 times 3 equals 12, which is 0 mod 6, but neither 4 nor 3 is equal to 0 mod 6.

Finally, the SimCalc materials have no similar domain map, instead using a domain map for *student* learning of their “mathematics of change” curriculum materials. Nevertheless, their items do reveal a theory of teacher knowledge. The item below, for instance, asks teachers to analyze an unusual solution method—and in so doing, to draw on underlying knowledge of proportional relationships as represented in such a table:

Students are working on creating tables of proportional relationships. During the lesson, Mr. Lewis has the following table up on the board.

x	y
3	12
2	8
4	16
8	32

He asks the students to come up with another (x,y) pair that could fill in the blank cells. One student says that you can create a new (x,y) pair by taking the averages of the last two rows in the table. That is, the new x is

$$\frac{4+8}{2} = 6,$$

and the new y is

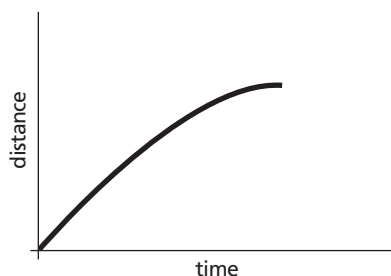
$$\frac{16+32}{2} = 24.$$

Mr. Lewis had hoped that the students would pick a value for x , then multiply by 4 to get a value of y . He is intrigued by this student's solution to the problem, however, and wonders whether it will always work.

This method will produce a valid x,y pair: (Mark ALL that apply.)

- a) Only when x and y are always whole numbers.
- b) Only when the last two rows have even numbers.
- c) Only when the last row is exactly double the next-to-last row.
- d) Always, as long as the table represents a proportional relationship.
- e) Always, as long as the table represents a linear function.

In addition, because this novel curriculum asks students to explain, interpret, and represent linear functions, proportional relationships, and rates, teachers are often asked to do the same:



Ms. Chopra asks her students to explain why the graph above is NOT proportional. Her students' explanations are given below.

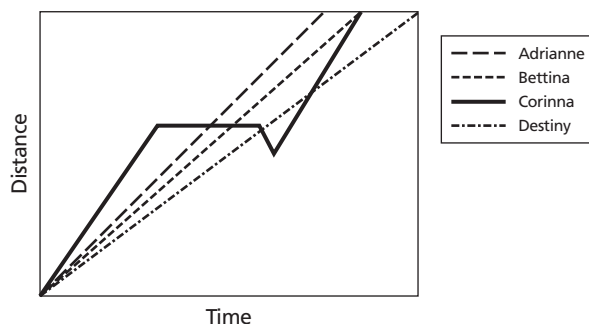
Which of the following student statements explains why the graph is NOT proportional? (Mark ALL that apply.)

- a) You can't have a curve in a graph.
- b) The relationship between the distance and time is not always constant.
- c) The graph is not labeled correctly.
- d) The ratio of rise over run is different at different points on the curve.
- e) At every point on the curve, the d/t is not always the same ratio.
- f) None of the above.

This item blurs the distinction between common and specialized knowledge set out by the SII/LMT instrument developers; here, both students and teachers are asked to know mathematics in ways that probe the underlying meanings and to provide mathematical explanations to counter a common misconception.

However, many more of the SimCalc items ask teachers to solve the same types of problems they would be asking students to solve:

Mr. Aneesh presents the following graph to his class. The graph represents the entire race run by Adrianne, Bettina, Corinna, and Destiny.



Mr. Aneesh asks the students to describe what is happening during the race.

Which of the following student statements are FALSE? (Mark ALL that apply.)

- a) Corinna and Bettina finished at the same time.
- b) All runners finished the race at the same time.
- c) Adrianne had the highest average speed.
- d) Adrianne led the race from start to finish.

These four projects have progressed the furthest in instrument development, in the sense that all have taken care to specify domain maps, completed pilot work with large groups of teachers, conducted psychometric analyses, and begun work on validation of the measures. However, many other pencil-and-paper tests exist. Most faculty responsible for teaching mathematics content and methods courses to preservice teachers, for instance, write end-of-unit or end-of-course assessments. Research projects interested in studying

the development of teacher knowledge design their own instruments, tailored to their own program or purpose. Schweingruber and Nease (2000) describe a content knowledge assessment used to gauge learning in a teacher professional development program at Rice University. Phillip et al. (in press) describe and make available a test that combines an assessment of preservice elementary teachers' content knowledge with their knowledge of students. In it, preservice teachers are presented with student work, asked whether that work is correct and, if not, to explain the student error. Some of the correct student work presents unusual mathematical solutions or representations, such as non-standard algorithms or unusual representations of number and content. Many of the student errors are drawn from the literature on student learning and error analysis.

As this review shows, the field has responded strongly to the need for pencil-and-paper assessments that can be used to study teacher and student learning. Many build on the types of items and mathematics described in the initial work on teachers' knowledge identified in the beginning of this section, and thus come closer to tests of professional knowledge than have past assessments, particularly those used in the educational production function literature. Although development is not complete on any of these assessments, and such assessments are far from perfect vehicles for this work, significant progress has been made. One reason for this is the interest on the part of funders, particularly the federal government, in developing such measures; three of the projects described in detail above were funded by the National Science Foundation, and one by the U.S. Department of Education. Measures development is expensive, and a commitment on the part of major funding agencies is critical. Another reason might be a climate of increased accountability, not only for districts, schools and students, but also for the professional development providers who serve these educators.

Like any form of assessment, the instruments reviewed here have significant drawbacks. To start, few would agree that instruments that use multiple choice or short-answer responses can represent completely the mathematical knowledge teachers use in their work. First, the construct of mathematical knowledge for teaching is still emergent. Without better theoretical mapping of this domain, no instrument can hope to fully capture the knowledge and reasoning skills teachers possess. Second, teaching any subject matter involves complex professional judgments, and such

judgments are not always amenable to testing in a format with single correct answers. While there may be some regularities in student errors, for instance, the errors of any particular student are shaped by his or her prior learning experiences and the task at hand; either may lead to idiosyncrasies. And while base ten materials may generally work well to represent multiplication of decimals, whether they will work well in a particular situation depends in part on the teacher's skill in using them, students' exposure to the use of base ten materials to model whole number operations, and so forth. Teachers' reasoning about these situations is not well reflected in right/wrong formats. Problems that draw only on mathematics avoid some of these complications, but also fail to gauge other aspects of knowledge and reasoning that are key to classroom teaching.

A third problem with multiple-choice/short-answer assessments involves teachers' reactions to such instruments. Several of the projects described here note teacher resistance to assessments and, in particular, to multiple-choice forms of assessments. Teachers, whose work is about helping others learn, are well aware of the problems inherent in assessing complex learning outcomes. They know that many important achievements are difficult to measure, and that multiple-choice formats are often inadequate in capturing learning. Using multiple-choice formats to assess teachers' own knowledge suggests a limited view of the complexity of professional knowledge and thus conflicts with professionals' reasonable experience with and distrust of test questions framed in this form.⁵

Fourth, validity work on these tests has generally lagged behind assessment development itself. Test developers are most likely to have conducted standard psychometric analyses, such as construct identification via factor analyses and reliability analyses. Reliabilities for these assessments meet or exceed industry standards for the specific test goals listed, such as diagnosing teacher knowledge or tracking groups of teachers over time. However, for the most part, validation work has been limited. The DTAMS and the SII/LMT projects both conducted content validity checks, ensuring that the test reflected key elements of the mathematics curriculum for students, and elements of teacher knowledge suggested by the literature described above. KAT is currently conducting a discriminant validity study, giving their assessment to college mathematics majors, for instance, to see whether this group scores high on advanced mathematics knowledge but low, as expected, on teaching mathematics

⁵ That most multiple-choice tests are disappointing in their capacity does not rule out the possibility of designing new types of items that take advantage of the format rather than distorting the nature of the knowledge being assessed.

knowledge. Construct identification, however, is only one form of validity. Another important form is predictive—whether teachers' performance on the assessment is related to the mathematical quality of instruction, and to student learning. Currently, only the SII/LMT test developers have performed this work, and even there, more work remains to be done.

Finally, using multiple-choice/short answer assessments as part of an evaluation or study requires at least basic statistical expertise—a t-test or ANOVA to examine teacher gains, for instance. In some cases, such as the SII/LMT measures, scores are returned not in raw frequencies but in standard deviation units. Users also have the option of creating their own forms from the item pool. This contrasts with the analytic demands of many of the other methods described here. Eventually, many of these pencil-and-paper assessments may be put online, a condition under which these assessments will likely be scored automatically by computer.

Other Methods

New technologies may also allow improvements on pencil-and-paper assessments, at least for small-to-moderate scale research projects. Kersting (2004) and associates at LessonLab have developed an online measure that gauges teacher knowledge by examining responses to ten short video clips of teaching. Teachers' responses are credited according to the amount they discuss features of the mathematics, student thinking and understanding, teaching strategies, and according to the amount that the discussion of the clips embeds these observations in a cause/effect framework. Like interviews or open-ended tasks, this method may be cumbersome to score. However, the technical properties of the instrument appear promising, and the assessment format itself holds three appealing features. First, it complements current professional development methods, particularly the use of videotape for teacher analysis and reflection. Second, an analysis of mathematics teaching may be less threatening to teachers than a pencil-and-paper assessment. Finally, the assessment can be embedded in the flow of teachers' work in a video-based professional development setting.

Other embedded assessment techniques also exist. For instance, Sowder, Phillip, Armstrong, and Shappelle (1998) effectively used discourse analysis—or studying what teachers say and how they talk about mathematics—to trace growth in teachers' knowledge of division of fractions. One benefit of this method is that it requires no statistical expertise. Discourse analysis, however, does require close and specialized train-

ing. It is a promising approach because teachers' use of mathematical language is both an indicator and target of growth in knowledge and skill; teachers need, in classrooms, to be able to use mathematical terms accurately and precisely. But more, they also need to be able to “translate” between children's home language, informal mathematical language, and disciplinary language (Ball, Masters-Goffney, & Bass, 2005; Pimm, 1987). Our own observations suggest that this linguistic capacity may be held variably in today's teaching population.

The drawbacks to discourse analysis include a lack of generalizability past the particular professional development (or other) setting that participants are in and difficulties separating individuals' skill with language from group skill with language. Further, such measurements do not scale easily, and they require finely calibrated frameworks for interpreting and analyzing discourse patterns. That said, this remains an intriguing possibility, one that should be investigated more formally.

Finally, some key studies have used self-reports of improvements in knowledge and skill as proxy measures for actual growth (e.g., Garet, Porter, Desimone, Birman, & Yoon, 2001; Horizon Research, 2002; NCES, 1999). However, the degree to which these are related to actual teacher knowledge growth is largely unknown.

In sum, scholars have developed a variety of methods for studying mathematical knowledge for teaching, from those intended to help understand the construct itself to those designed to provide evidence of teachers' learning in this domain. Our review of these methods has taught us several things. First, developers in this field report that there is in fact considerable interplay between assessment and theoretical development. Writing test items, for instance, helps illuminate aspects of mathematical knowledge specific to teaching; conducting a discourse analysis can help illuminate the understandings teachers have of students and mathematics. Second, every method has both advantages and drawbacks; there is no one perfect method, although there may be better and worse methods for particular research questions. In fact, more diversity in assessments and assessment formats should be encouraged, to allow better matching between the subjects of inquiry and instruments used.

Third, these assessments should be widely available, and widely used. This requires broader support than exists now in the mathematics education community. For instance, there is currently no clearinghouse through which scholars “shopping” for a measure can collect and compare information about each measure's content, design, and technical features.

Also, there is no systematic method for disseminating measures or for training individuals in their use. Instead, scholars and evaluators sit at the mercy of the research projects and agencies developing these measures, whose capacity to disseminate varies according to funding and the availability of personnel. But broader use of these measures in the study of teacher knowledge and learning would surely benefit the field, as results from different research projects would be more directly comparable and, we hope, more rigorously evaluated.

Finally, each of the efforts described here have made an attempt to bring the measurement of teachers' mathematical knowledge closer to the actual practice of teaching itself. In many cases, the developers of these assessments explicitly state their desire to measure the mathematical knowledge used in teaching, although the frameworks for what those measures might be vary. One would expect that this improves the validity of the assessments, although as we noted above, most projects do little predictive validity work on their measures. A similar effort to move the assessment of mathematical knowledge closer to actual teaching has been occurring in the field of teacher certification testing, a topic to which we now turn.

CONTEMPORARY APPROACHES TO TESTING TEACHERS FOR CERTIFICATION

In the last decades of the twentieth and into the twenty-first centuries, three elements converged to spur the development of new forms of teacher certification tests. First, policy-makers and the public made increasing demands for accountability on the part of teachers and teacher educators; as we have seen, the amount of required teacher testing grew markedly over the period 1980–2000. Second, mathematics educators began to understand the nature of mathematical knowledge for teaching, investing heavily in the idea that beyond knowing content, teachers must also have profession-specific mathematical knowledge about content, students, and teaching. Finally, this era saw emergent interest on the part of professional educators in testing teachers; although opposition to teacher testing still runs high in some quarters, at least two national professional organizations have been established to promote new forms of teacher assessment. In this way, the historic antipathy between teaching and teacher educator organizations and testing agencies has begun to crumble. Whether the professional organizations

can gain control over widespread teacher testing, and whether the tests produced are satisfactory, remains to be seen.

In this section, we describe in detail the mathematical elements of the three most prominent of these efforts. Two of the assessments are for beginning teachers: the Praxis Series (the successor to the NTE), and the Interstate New Teacher Assessment and Support Consortium's (INTASC) portfolio assessment system. Although INTASC is not now formally in use for certification in any state, Connecticut's second-stage licensure system uses a portfolio assessment based on INTASC principles, and INTASC is looked to as a leader in the field. The third assessment is an endorsement for experienced practitioners: the National Board for Professional Teaching Standards (NBPTS) certification. We examine each of these assessments along four dimensions: its history and development; a description of the test and how it is used; an analysis of sample mathematics items; and, based on our analysis of available items, some conjectures about the test's implicit views of teaching, of mathematics, and of the mathematics needed for teaching. We also briefly review other tests used in the U.S., including the California Subject Examinations for Teachers (CSET), and the tests developed by the American Board for Certification of Teacher Excellence (ABCTE). Each assessment plays a unique role in the current debates over how to certify teachers.

Before beginning, it is important to note what these modern certification tests are designed to do. Unlike the mathematical assessments common in the scholarly literature, the tests discussed below are designed to allow the user to draw inferences about a candidate's suitability for teaching. As this implies, there are often great stakes attached to teachers' performance on these exams: pay raises, at a minimum, and more often the ability to compete for jobs in one's chosen profession. The high-stakes nature of many of these assessments helps explain why there has been so much debate over them and so much effort devoted to ensuring their accuracy. We begin with Praxis, the most commonly used teacher certification assessment in the U.S.

The Praxis Series

The Praxis Series began to replace the National Teacher Examination (NTE) in 1993. Currently, 43 states use at least one portion of the series for beginning teacher certification (ETS, 2006a). In the wake of increasing calls for teacher testing, and building on

new advances in psychology and research on teaching, Praxis was to take a more robust view of teaching and the knowledge needed to teach. Praxis takes a three-stage approach to teacher certification that includes tests of prerequisite content knowledge, subject-specific content knowledge and pedagogical knowledge, and interactive teaching skills. Each construct in Praxis would be measured separately, and at different stages in a new teacher's career.

In many states, teachers begin by taking Praxis I, a multiple-choice test of general academic knowledge, before admission to teacher education programs. This test is similar to the SAT and in some states the tests are used interchangeably. If the study guide questions are representative, the mathematics questions in Praxis I are similar to the SII/LMT "common content knowledge" category, meaning mathematics that is common across professions and, in many cases, in the public domain. This makes sense, given that this assessment is administered prior to the start of formal teacher education programs.

Praxis II is a range of multiple-choice and constructed-answer assessments that test general academic and pedagogical knowledge as well as subject-specific knowledge for teaching. Praxis II is generally taken at the completion of a teacher education program. Elementary teachers might see mathematics questions on one of three separate tests: the content knowledge test for elementary teaching, a widely used assessment, containing roughly 30 multiple-choice items on mathematics; the curriculum, instruction, and assessment test, where roughly 20% of 110 multiple choice items ask about mathematics; and the content area exercises, a test that includes as one of its four prompts a problem that asks teachers to design mathematics instruction or respond to a student error. At the secondary level, ETS maintains several mathematics-specific Praxis II assessments, including content knowledge; proofs, models, and problems; general mathematics; and pedagogy. The mathematics pedagogy assessment is similar to the elementary content area exercise test, in that it asks teachers to respond to prompts about how best to design or implement mathematics instruction.

Praxis III is an inventory of teaching skills that is conducted by observing the beginning teacher in her classroom. Only two states are currently using Praxis III, and none require it for initial licensure. Praxis III is used instead in some districts and states for professional development.

Each state that uses Praxis requires its own combination of tests, and each state determines its own

passing rates for licensure. So, for example, Idaho requires that its teacher candidates take a single subject-area Praxis II test for licensure; Hawaii requires the Praxis I test and two Praxis II tests, the Principles of Learning and Teaching as well as a subject-matter area test. Because of the prevalence of Praxis II, and its claim to measure subject-matter-specific disciplinary knowledge, we focus on it for much of the discussion below.

Praxis II, like other modern teacher certification tests, stakes its claim to validity on the idea that it captures the knowledge needed to successfully teach. Two aspects of assessment design buttress this claim. First, the Praxis II tests were developed through a process designed to include key stakeholders in the education and certification of teachers. In mathematics, this included subject-matter specialists (mathematicians, mathematics educators), teachers, mathematics coordinators in school districts, psychometricians, and psychologists. Second, the items on the assessment were developed following what the test developers call a "job analysis," or a review of the mathematical knowledge teachers need in their jobs. This "job analysis" is not an ethnographic study of a mathematics classroom with special focus paid to the teacher and the knowledge employed in teaching. Rather, by "job analysis," the developers mean that subject-matter specialists examined earlier tests such as the NTE, textbooks, the NCTM Standards documents, curricular documents, then drew up lists of topics that teachers would need to know for teaching mathematics. This list was then shared with teachers and other school personnel who made suggestions and changes and, when approved, translated into assessment items.

Because ETS does not release active Praxis items, it is difficult to characterize the results from this process with any degree of certainty. If Praxis study materials are any guide, however, the problems prospective teachers face on the content area tests are largely dependent on the Praxis II assessment their state requires. For both elementary and secondary content knowledge tests, among the most commonly used, the items resemble those that have comprised teacher assessments for generations. The following items are taken from study guides and test preparation materials (ETS, 2005b, p. 11; ETS, 2006c, p. 4):

1. $15(4 + 3) = 15 \times 4 + 15 \times 3$

The equation above demonstrates which of the following?

- The distributive property of multiplication over addition
 - The commutative property of multiplication
 - The associative property of multiplication
 - Additive inverse and additive identity
2. Riding on a school bus are 20 students in 9th grade, 10 in 10th grade, 9 in 11th grade, and 7 in 12th grade. Approximately what percent of the students on the bus are in 9th grade?
- 23%
 - 43%
 - 46%
 - 76%
3. Which of the following is equal to 8^4 ?
- 4,032
 - 4,064
 - 4,096
 - 4,128
4. For which of the following values of k does the equation $x^4 - 4x^2 + x + k = 0$ have four distinct real roots?
- 2
 - 1
 - 3
- II only
 - III only
 - II and III only
 - I, II, and III

Items 1–3 are from elementary materials; item 4 is from secondary test preparation materials. These items ask teachers to solve problems that contain mathematical content common to the public and other professions, including mathematics as a discipline. And they are not far from the types of items found in most teacher certification exams for the past two centuries, with changes in mathematical content and item design to allow the tests to stay current with the content taught in elementary and secondary schools.

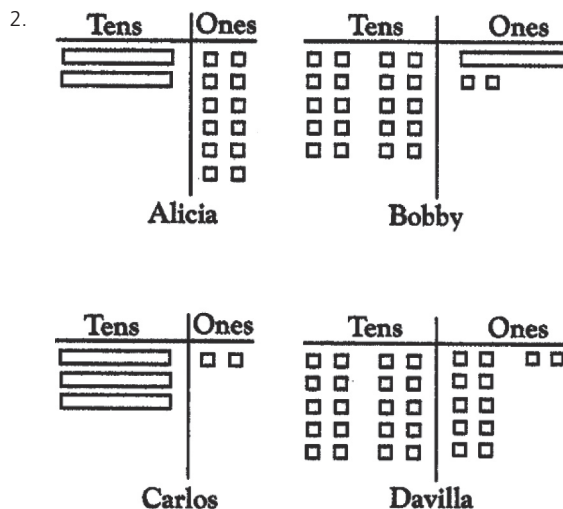
By contrast, the small number of mathematics items on the elementary curriculum, instruction, and assessment test appear aimed at assessing prospective teachers' ability to work with the mathematics in instruction (ETS, 2005a, pp. 5–6):

1.

$$\begin{array}{r} \frac{4}{16} \\ - \frac{1}{8} \\ \hline \frac{3}{8} \end{array} \quad \begin{array}{r} \frac{5}{9} \\ - \frac{1}{2} \\ \hline \frac{4}{7} \end{array} \quad \begin{array}{r} \frac{7}{16} \\ - \frac{1}{5} \\ \hline \frac{6}{11} \end{array}$$

The examples above are representative of a student's work. If the error pattern indicated in these examples continues, the student's answer to the problem $9/11 - 1/7$ will most likely be

- $\frac{10}{4}$
- $\frac{8}{7}$
- $\frac{8}{4}$
- $\frac{9}{18}$



The illustrations above show how four students, Alicia, Bobby, Carlos, and Davilla, used base ten blocks to represent the number 32. Which of the students used the blocks to represent the number 32 in a way that reveals an understanding of the underlying concepts of a numeration system based on powers of ten?

- Alicia
- Bobby
- Carlos
- Davilla

The first item asks teachers to recognize or reason through students' errors; the second asks about how materials represent concepts. Taken together, these items seem to draw from the kinds of thinking described in Shulman's "pedagogical content knowledge" and the finer subdivisions of this construct by modern academic test developers (e.g., KAT's "teaching knowledge" or SII/LMT's "knowledge of content and students" and "knowledge of content and teaching"). These items are also the modern version of those on the NTE's elementary school specialist area test.

Note that the second item again illustrates the difficulty of writing defensible multiple-choice items in this arena. According to test materials, Carlos' representation is best. However, Alicia might also have a solid understanding of base ten numeration, as evidenced by the fact that she showed 2 tens and 12 ones—a representation of 32 commonly used for subtraction with regrouping. In fact, many teachers use base ten materials to explicitly teach the equivalence of Carlos' and Alicia's representations of 32, and we hazard that some educators might go as far as to argue that Alicia has a better understanding than Carlos for particular purposes (e.g., subtraction with regrouping). Although the presence of this item on Praxis released materials might reflect a simple mistake, it may also indicate a significant flaw in the construction of this assessment, in that we cannot imagine many mathematics educators would approve of such

an item. If mathematics educators do not have a voice in the construction of this assessment, it again calls into question the degree of professional control over the assessment. It also harkens to the precursor to the Praxis II, the National Teacher Examination, which carried other such ambiguous items.

Moving further toward practice, two Praxis II tests (elementary content area exercises and secondary mathematics pedagogy assessment) ask teachers to answer essay questions involving mathematical scenarios like the following:

You are teaching a unit on solving quadratic equations. You have already taught the students how to solve quadratics by taking square roots and by factoring. In your next lesson, you plan to teach the students how to solve quadratic equations by completing the square.

Design a homework assignment for your students to complete after the lesson on solving quadratic equations by completing the square. The homework assignment should consist of 5 problems that review previously taught skills and concepts while also providing practice in the newly introduced material.

Briefly explain your rationale for including the skills and concepts that the problems illustrate.

(ETS, 2003, p. 106)

Here, and in every other sample secondary-level prompt available to us, teachers must engage in lesson design. Other prompts include “Describe an investigation you would have your students make to discover the difference between ‘regular’ and ‘equilateral’ [polygons]. Your investigation may involve the use of any type of manipulative or any software package” (ETS, 2003, p. 114). And “Describe a strategy, using pictures or manipulatives, that you could use to help foster the students’ conceptual understanding of equivalent fractions” (p. 110). On their face, these tasks do appear to tap mathematical knowledge as it is used in teaching; responses are scored based on not only demonstrated understanding of the mathematics, but also on the basis that the instruction described therein is “very likely to achieve the desired goals” (ETS, 2005, p. 2). However, it does so in a particular way, assuming that good teaching equates with constructing lessons from scratch. Left out in this type of assessment is the kind of mathematical knowledge used when adapting curriculum materials, or using these materials well.

Finally, like some of the professional organizations we describe below, the Praxis has moved into the realm of measuring teaching in the classroom. The Praxis III assessment, now used in two states, evaluates teachers during their first and second years in the

classroom via observations and interviews by trained assessors. It assesses knowledge in 19 criteria in four interrelated domains: organizing content knowledge for student learning; creating an environment for student learning; teaching for student learning; and teacher professionalism. Although not specific to mathematics instruction, the prominence of content knowledge is noteworthy. The description of this domain begins with the following: “Knowledge of the content to be taught underlies all aspects of good instruction. Domain A focuses on how teachers use their understanding of students and subject matter to decide on learning goals, to design or select appropriate activities and instructional materials, to sequence instruction in ways that will help students to meet short- and long-term curricular goals, and to design or select informative evaluation strategies” (Dwyer, 1998, p. 182). Nonetheless, the specifications for content knowledge observed in Praxis III are generic, so how and whether teachers’ mathematical knowledge for teaching is observed and appraised remains an open question. Whether mathematical knowledge is closely evaluated also depends in large part upon whether the observer is knowledgeable in this area.

Interstate New Teacher Assessment and Support Consortium (INTASC) Portfolio Assessments

Another effort to reformulate teacher certification has taken place under the auspices of the Interstate New Teacher Assessment and Support Consortium (INTASC), a consortium of state education agencies and national educational organizations sponsored by the Council of Chief State School Officers. INTASC was created in 1987 to provide a forum for states to learn collaboratively about and develop new accountability requirements for teacher preparation programs, new assessments for teacher licensing and evaluation, and new professional development programs. Following the publication of its *Model Standards in Mathematics for Beginning Teacher Licensing & Development: A Resource for State Dialogue* (1995), INTASC began to develop standards-based licensing tests for teachers. On the view that no single test could adequately assess a prospective teacher’s ability to teach, INTASC recommended three types of licensing tests: 1) a test of content knowledge; 2) a test of teaching knowledge (pedagogy); and 3) an assessment of actual teaching. The first two types of tests—content knowledge and teaching knowledge—are intended for the conclusion of teacher preparation programs as a requirement for a provisional licensure.

To qualify for a permanent license, teachers would complete a third, performance-based test in the initial years of teaching. INTASC used a portfolio format for its assessment, “a collection of documents that tell the story of a candidate’s teaching as it develops over a period of time” (INTASC, 2006). Portfolios had been growing in use as K–12 assessments through the 1980s, and their extension to teacher assessments made sense: they can provide evidence of the development of a teacher’s practice and include instructional materials, student work samples, video records of teaching, and written commentaries. Like their model standards, INTASC’s portfolio assessments are intended to be a resource that can be adopted or adapted, for example, by schools of education as a requirement for program completion, or by states as a requirement for certification or completion during the induction years. We focus on INTASC’s portfolio assessment both because of its novel format and its view of teacher knowledge.

The INTASC mathematics teaching portfolio documents approximately 8 to 12 hours of a beginning teacher’s practice, organized around a central mathematical topic. Portfolio entries include a description of the teaching context; a sequence of lesson plans highlighting central features of the lessons such as the mathematics content, tasks, and accommodations for three focal students; two featured lessons each documented with 20–30 minutes of video, assessments, and student work samples; a cumulative student assessment and scoring criteria, accompanied by student work marked with feedback; and an analysis of teaching and professional growth. Along with these records of practice, each entry also requires a written commentary with a rationale for and evaluation of the teacher’s instructional choices. The following excerpt describes part of the commentary required for the series of lessons:

The Mathematics: Identify your goals and expectations for student learning across this series of lessons. Begin your commentary with a statement of these goals. Continue with a description of the broad mathematical concept(s) or idea(s) that unifies the lessons. Why is this concept or idea important to the study of mathematics? How do the lessons you have designed build a cohesive set of plans to address this concept or idea? Explain the mathematical connections across the lessons. Provide one or two examples of how the series of lessons builds on mathematics the students have already learned. Give one or two examples of how the series of lessons serves as a foundation for the mathematics students will learn later.

Provide a general description of your goals and expectations for students to reason mathematically, solve problems, communicate mathematically, and see connections within mathematics,

connections to other disciplines, and connections to the real world across these lessons. Provide specific examples of how your lessons will promote reasoning, problem solving, communication, and connections through the mathematical concept or idea that is the focus.

(INTASC, 1996, p. 25)

Completed portfolios are evaluated based on five interpretive categories—tasks, discourse, learning environment, analysis of learning, and analysis of teaching. Guiding questions for each category illustrate aspects of teachers’ performance that could be considered in the collection and analysis of evidence.

The INTASC portfolio assessment is thus very different from prior teacher certification exams in that teacher knowledge of mathematics is assessed through the tasks, discourse, and analysis of student learning presented in the portfolio. Three of the evaluation framework’s guiding questions are identified as specifically capturing facets of mathematical knowledge used in teaching: the appropriateness of the tasks for the instructional goals; the teacher’s use of notation, language, and representation; and the accuracy of the teacher’s interpretation and evaluation of information about students. Other areas of the portfolio might also demonstrate a teacher’s use of mathematical knowledge in teaching, for example, in making decisions about when to probe students or in the ability to maintain the cognitive demand of a task during its implementation. Thus, the INTASC assessment has the potential for capturing the type of professionally situated knowledge identified by researchers—mathematical knowledge as it is deployed in the work of teaching. However, recognizing a teacher’s skillful task selection in mathematics or imprecise use of mathematical language requires portfolio readers with significant mathematical knowledge. Thus, although the INTASC portfolio has the possibility of capturing mathematical knowledge as it is used, or misused, in teaching, its ability to do so would depend on the reader.

It is also worth noting that INTASC materials embrace the particular vision of mathematics instruction that is articulated in the INTASC standards and NCTM’s *Professional Standards for the Teaching of Mathematics* (1991). Although the assessment handbook says that “there is no singular right way to teach mathematics” (INTASC, 1996, p. 36), teachers are required to demonstrate evidence of particular instructional formats and types of mathematics tasks. For example, one of the featured videotaped lessons must show students engaged in “mathematical problem solving and reasoning” where problem solving is defined as “students exploring new ideas and trying to pull prior

learning together to address non-routine problems, not simply applying an algorithm to a new context,” and reasoning is defined as “students making and testing conjectures, constructing arguments, judging the validity of arguments, formulating counterexamples, etc.” (INTASC, 1996, p. 27). In addition to problem solving and reasoning, the featured lessons must, between them, capture other aspects of instruction such as both whole group and non-whole group formats, student-to-student discourse, and the introduction or development of a mathematical concept. Whereas historical teacher tests kept knowledge of pedagogical techniques and knowledge of content separate, they are here intertwined; teachers must exhibit specific instructional skills. While this is part of a larger story about changes in “pedagogical” knowledge over the last two centuries, it warrants noting here because of its implications for the orientations and skills novice teachers must possess.

National Board for Professional Teaching Standards (NBPTS) Certification

In contrast to both the Praxis and INTASC assessments, the National Board for Professional Teaching Standards (NBPTS) certification is a voluntary endorsement for teachers who already possess the minimum state requirements for teacher certification and who have at least three years of experience. Certification is currently available in 24 areas (NBPTS, 2006b)—including Early Childhood/Generalist, Middle Childhood/Generalist, and Early Adolescence/Mathematics—and is valid for ten years, after which teachers can apply for renewal. There are specific standards for each certificate that describe “what an accomplished teacher should know and be able to do” (NBPTS, 2006a). In fact, the National Board Standards informed the development of INTASC standards, and, like INTASC, the NBPTS is endorsed both by governance (e.g., National Governors’ Association; National School Boards Association) and teacher education associations (e.g., NCATE); NBPTS is also backed by the National Education Association.

NBPTS certification has two components: assessment center exercises and portfolio entries. These two types of performance-based assessments are designed to measure not only teachers’ content knowledge, but also the skills and judgment they must deploy routinely in practice. Although the specifics and standards vary, the process is the same across different types of certificates.

Assessment center exercises focus primarily on content knowledge and are designed to measure what accomplished teachers should *know*, rather than ques-

tions that can be studied for. Teachers complete six open-ended response exercises at designated testing centers, located across the nation. Candidates are given up to 30 minutes to respond to each exercise. Organized around “challenging teaching issues,” exercises require teachers to use their content knowledge in pedagogical situations, for example, to identify student misconceptions and plan an appropriate instructional strategy, or to describe a learning experience based on a foundational concept or in response to a student’s inquiry. NBPTS contracted with ETS, the same organization that publishes Praxis, to develop its scoring systems.

All six assessment center exercises for the Early Adolescence/Mathematics certificate are focused on mathematics topics: algebra and functions, connections, data analysis, geometry, number and operation sense, and technology and manipulatives. For both the Early and Middle Childhood/Generalist certificates, however, only one of the six exercises explicitly involves mathematics. The Early Childhood/Generalist scoring guide describes such an exercise:

Candidates will demonstrate their ability to identify mathematical misconceptions or difficulty in a student’s work, to state the fundamental prerequisites needed by this student in order to learn this particular mathematical concept, and to plan an instructional strategy based on real-world applications. They will also be asked to choose the materials and to provide a rationale for their choice of these materials that will be used to teach these prerequisites. (NBPTS, 2005b, p. 9)

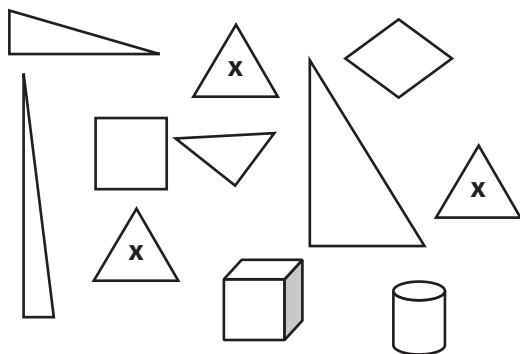
The scoring guide provides the example in Figure 4.1.

In this example, candidates must identify and explain a common student misunderstanding—failing to recognize non-equilateral triangles as triangles—then design instruction to remedy this problem. This exemplar can be characterized as “pedagogical content knowledge,” for it asks teachers to solve mathematics-specific problems involving student misunderstandings and teaching strategies. However, because there is only one mathematics-specific task on the generalist certification assessments, it is unlikely that *mathematical* pedagogical content knowledge is the construct that is contained in the overall score teachers receive. Instead, teachers’ scores on all of the assessment center exercises and portfolio entries, across content areas, are weighted and combined to make a final assessment about their level of expertise.

The NBPTS portfolio allows for in-depth examination of practice. It requires four entries, three of which document specific aspects of a teacher’s practice

Stimulus

A kindergarten student is having difficulty with a mathematical concept:



The student was asked to mark an X on all of the triangles.

Prompts

1. Identify the mathematical misconception/difficulty of this student work sample.
2. What fundamental concepts are prerequisites for this student at this grade level in order to learn this skill?
3. Based on real-world applications, state your goal for an instructional strategy or learning experience to help this student. Plan a learning experience or instructional strategy based on this goal that would further the student's understanding of this mathematical concept.
4. What materials would you use to teach this mathematical concept to this child? Provide a rationale for your choice of materials.

Figure 4.1 Example in the Early Childhood/Generalist scoring guide. (Reprinted with permission from page 84 of the National Board for Professional Teaching Standards, Early Childhood/Generalist Scoring Guide, 2005. All rights reserved.)

using records such as videotapes, student work samples, and related instructional tasks and classroom assessments.⁶ In addition to creating these records, teachers write reflective commentaries that describe, analyze, and evaluate their practice. Similar to the assessment center exercises, the Early and Middle Childhood/Generalist certificates require only one of the four portfolio entries to focus on mathematics. And in fact, this entry requires teachers to document their integration of mathematics and science, as described in the 308-page portfolio instructions document for the Middle Childhood/Generalist certificate:

In this entry you will demonstrate how you help students better understand a “big idea” in science using relevant science and mathematics knowledge. You will engage students in the discovery, exploration, and implementation of these science and mathematics concepts, procedures, and processes by integrating these two disciplinary areas. This entry is designed for

you to provide evidence of your ability to plan, describe, illustrate, assess, and reflect on your teaching practice.

For this entry, you must submit the following:

- Written Commentary (14 pages maximum) that contextualizes, analyzes, and evaluates the teaching and integration of the math and science instruction.
- One Video Recording (15 minutes maximum) that demonstrates how you engage two small groups of students, with at least three students per group, in a science lesson that integrates mathematics.
- Three Instructional Materials (no more than a total of 6 pages) related to the lesson featured on the video recording that will help assessors understand what occurred during the lesson. (NBPTS, 2006d, p. 169)

Each of the portfolio entries is evaluated independently according to a detailed rubric addressing selected NBPTS standards. For example, the portfolio entry described above is scored for standards such as knowledge of students, knowledge of content and curriculum, meaningful applications of knowledge, and reflection. For the two mathematics-specific certificates (Early Adolescence/Mathematics and Adolescence and Young Adulthood/Mathematics), the four portfolio entries each focus on a different aspect of mathematics teaching: developing and assessing mathematical thinking and reasoning, whole class mathematical discourse, small group mathematical collaborations, and contributions to student learning.

Our inspection of the NBPTS documents suggests several stances toward the mathematical knowledge needed for teaching, and toward the measurement of this construct. First, the NBPTS assessments do offer the opportunity to capture how teachers use their knowledge of mathematics in practice. For example, the mathematics-related assessment center exercise for the Middle Childhood/Generalist certificate describes a high scoring response as follows:

The response offers clear, consistent, and convincing evidence of the ability to demonstrate pedagogical and content knowledge of math by accurately identifying the math misconception/error, providing an instructional strategy to assist student understanding of concepts/skills needed to accurately solve the mathematical problem, and providing a rationale that supports the instructional strategy. (NBPTS, 2005c, p. 53)

Like the INTASC standards, the NBPTS assessment presumes that teachers need intertwined knowledge

⁶ The fourth focuses on accomplishments outside the classroom.

of content, students, and instructional design—and fluency with that knowledge, as evidenced by the short span of time allotted for teachers to answer the assessment center exercises. In this way, both the assessment center exercises and the portfolio entries reflect two aspects of teachers' mathematical work in classrooms: using integrated knowledge, and using it in the moment, rather than in a context more removed from actual classroom work. Thus, the NBPTS assessments better measure professionally situated mathematical knowledge than did past teacher certification exams, but, like the INTASC portfolios, are dependent on the samples of practice the teacher submits, as well as the evaluator's ability to recognize a teacher's use of mathematical knowledge in practice.

Second, there are a number of measurement-related concerns, some more pressing than others. While both INTASC and NBPTS have provided evidence for the validity of their measures, evidence on their reliability and other technical properties is more difficult to come by. Such evidence should be broadly available. Also, the small number of mathematics/science questions within K–8 exercises and prompts raises concerns about the degree to which the sampled tasks generalize to teachers' overall level of expertise in teaching. This is complicated by the fact that teachers can choose lessons for portfolio entries, and those lessons may not represent their everyday practice (Porter, Youngs & Odden, 2001). And this is even further complicated in the NBPTS generalist certificates by the combination of mathematics and science in the same portfolio task. Because NBPTS is interested in measuring accomplished general practice, they chose not to design separate measures of proficiency in teaching these two content areas. But accomplished practice might be discipline-specific. This leaves under-assessed the specific mathematical qualities of a teacher's instructional practice.

Third, both the NBPTS and INTASC portfolios seem as much designed to capture teachers' *reflection* on practice as the practice itself. For instance, the written commentary guidelines for the NBPTS include prompts such as: *What evidence of inquiry, intellectual engagement, discussion, and content is demonstrated in your video recording? How did you further students' knowledge and skills and engage them intellectually? How does the discussion and/or activity featured on the video recording reveal students' reasoning and understanding? Describe a specific example from this lesson as seen on the video recording that shows how you ensure fairness, equity, and access for students in your class* (NBPTS, 2006d, p. 173–174). This is a view of teaching that is heavily tailored toward ideas about teachers as “reflective practitioners” (Schön,

1995). It is worth considering, however, whether accomplished reflection on teaching is required for accomplished teaching; if it is not, then NBPTS might be missing a class of proficient educators. Moreover, the relationship of articulate reflection to effective instruction is not clearly established; yet the NBPTS makes this a central demand of its assessment strategy. Are teachers who can describe, explain, and reflect on their work better teachers than those who are less able to articulate their designs, purposes, or analyses? Possibly they are, for it may be that the ability to articulate one's practice is an indicator of deliberateness, and that the ability to write cogent reflections an indicator of analytic capacity, both of which may predict student achievement. But the nature of the evidence for this remains unexplained.

Finally, like INTASC, with its very specific view of mathematics instruction, and Praxis, with its instructional design tasks, the view of instruction presented in the NBPTS precludes some types of teachers from being recognized as accomplished—for instance, those who follow scripted curriculum materials, or those who do not teach mathematics using the inquiry/discussion/engagement methods suggested above. In the scoring guide, for instance, teachers whose portfolio is rated poorly are urged to consider whether it “provide(s) evidence that you have created a classroom environment that promotes active learning by all your students” and “provide(s) evidence that you were able to engage students in an effective classroom discussion or inquiry appropriate to the goals of your teaching” (NBPTS, 2005a, p. 15). Thus, there are examples of competent teaching that might not be recognized by NBPTS. Should a teacher who controls most classroom talk, but who also gives clear explanations of mathematical procedures or careful demonstrations of proofs be recognized as accomplished? A read of NBPTS documents suggests not: the organization bases its scoring system on what we have called the *quality of mathematics instruction*, or teachers' mathematical knowledge and pedagogical choices, considered together. However, other stances are also possible.

Other Certification Tests

Several other certification tests affect a large number of prospective teachers.

American Board for Certification of Teacher Excellence (ABCTE)

The certification program designed by the ABCTE is meant to attract career-changing adults into teaching, allowing these individuals to circumvent teacher

education programs and other state requirements for entry into the profession. As part of ABCTE certification, teachers must possess a bachelor's degree, pass "rigorous examinations of subject area and professional teaching knowledge" (ABCTE, 2006a), and submit materials for a federal background check. This process brings to mind certification routes from the eighteenth and nineteenth centuries, in which teacher candidates possessed some minimum level of schooling, passed an exam, and were assessed for moral fiber. As we shall see below, many of the debates evoked by the ABCTE program echo those from the late nineteenth century.

In mathematics, as in most other ABCTE subjects, the exam is entirely multiple-choice and administered via computer. Test content is based on ABCTE standards, which were developed in turn by reviewing state and professional standards. Standards for elementary teachers suggest that test developers expect teachers to know not only the content in the K–6 curriculum, but also content their students will encounter in middle or high school. For instance, in the area of algebra, teachers certified via the elementary exam must know the symbolic language of mathematics, "including applying proportional reasoning, defining the inverse function and performing arithmetic operations on functions, solving systems of linear and quadratic equations and inequalities, and extending and recognizing linear patterns" (ABCTE, 2006b). Many of these topics are typically taught in middle or high school. The middle/high school standards extend into content often taught in post-secondary education, such as least squares regression, matrix algebra, and calculus topics such as limits, derivatives, integrals, and convergence of sums. Secondary test items reflect this emphasis:

1. Twenty gallons of solution A, which is 35% acid, is mixed with 30 gallons of solution B. The resulting mixture contains 20% acid.

What percent of solution B is acid?

- a) 5% b) 10% c) 12.5% d) 15%

2. The following list of ordered pairs represents a functional relationship:

(1,1) (2,4) (3,9) (4,16)

What principle of functional relationships supports this statement?

- a) The x values are all different.
 b) The relationship between each pair of numbers is reciprocal.
 c) The pairs are listed in consecutive order.
 d) The relationship between each pair of numbers is the same.

3. What is $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$?

- a) 0 b) 4 c) 8 d) 12

The first item is perhaps a modern-day equivalent of the types of problems that appeared on the 1890s Michigan teacher exams; the second taps more conceptual knowledge, investigating teachers' understanding of the definition of functions; the third item taps a topic typically not taught until college. Overall, this is a mathematically demanding assessment, with items that are computationally intensive, designed to push on candidates' misconceptions, and to push into territory beyond that typically taught in high school.

To date, the ABCTE certification process has not been widely used. Although the program is funded by two U.S. Department of Education grants totaling \$40 million, it certified only 135 individuals between its founding in 2001 and April of 2006 (Schimmel, personal communication, April 19, 2006). One reason for the slow start is that the ABCTE certification must be recognized by states before teachers can gain employment; in mid-2006, only five states recognized the ABCTE certification. Another reason might be the proliferation of state and district-based alternative certification programs, which reduce incentives for teachers to seek the ABCTE stamp of approval, and also for states to recognize the organization's certification process.

The tepid response by states and teachers has not dulled the uproar over the ABCTE certification process. For the most part, critiques center around the assumption that teachers can be certified via a multiple-choice exam, absent professional training and field experiences. As Berliner (2005) writes of the ABCTE and similar tests, "it is a near impossibility to adequately assess quality teaching through pencil-and-paper tests of professional knowledge. . . . These tests fail in part because of the complexity of classroom environments and the near impossibility of capturing that reality in pencil-and-paper formats" (p. 211). Educators also assert that in other professions, such as law, medicine and architecture, the certification process includes not only an exam but also extended professional learning and apprenticeship.

Thus the ABCTE has reopened the debate over who is qualified to teach, and how best to assess individuals' skills and capacity for this work. Teacher educators, many of whom oppose any form of alternative certification, have singled out this test-based certification method for particular opprobrium, much as educators at the turn of the twentieth century railed against the teacher tests administered by local school boards and officials. Many issues are the same. Teacher educators claim there is a professional body of knowledge learned primarily through preservice preparation and apprenticeship, and that tests provide a "back door" for under-qualified individuals to enter teaching. ABCTE support-

ers argue that teachers' subject matter knowledge is of the foremost importance, and that there needs to be a method to identify knowledgeable individuals to help alleviate teacher shortages, shortages caused in part by the bureaucratic restrictions most states have placed on the entry to teaching. ABCTE supporters also take the view that teachers need to know more advanced content than they are expected to teach students, much as the Michigan superintendent (and test promoter) Henry R. Pattengill did over 100 years earlier. Teacher educators, for their part, tend to focus more on promoting deep knowledge of the specific curriculum teachers will teach students.

California Subject Examinations for Teachers

The California Subject Examinations for Teachers (CSET) are reviewed here briefly as an example of both a state-specific test, and as an example of a customized exam developed by National Evaluation Systems (NES, 2006) specifically for one state's use. NES is the second-largest supplier of teacher certification assessments, providing tests to not only California but also Texas and a number of smaller states. In California, NES customized tests to align with California curriculum frameworks and student standards.

Two sample items from the elementary-level (multiple subjects) exam provide some insight into the composition of the test:

1. If the number 360 is written as a product of its prime factors in the form a^3b^2c , what is the numerical value of $a + b + c$?
a) 10 b) 16 c) 17 d) 22
2. The problem below shows steps in finding the product of two two-digit numbers using this standard multiplication algorithm. The missing digits in the problem are represented by the symbol $\square$.

$$\begin{array}{r} \square 9 \\ \times 36 \\ \hline 29\square \\ + \square 4\square\square \\ \hline \square\square\square\square \end{array}$$

What is the hundreds digit in the product of the two numbers?

- a) 1 b) 4 c) 6 d) 7

These and other items assess upper-elementary content (e.g., slope; proportionality). They also appear designed to draw on both candidates' basic skills and on their ability to reason through novel mathematics problems. On this test, candidates' mathematics and science ability is reported as a separate subtest score, allowing test users to gauge individual competency in both areas combined. Praxis II, in contrast, only reports overall scores to employers.

Candidates for secondary mathematics positions must take two mathematics examinations, one focusing on number theory and algebra and the other focusing on geometry and probability and statistics. To teach advanced placement classes or classes in analysis and calculus, a candidate must take a third mathematics examination covering the history of mathematics and calculus. The examinations were designed by NES to follow the mathematics content standards for California Public Schools, and these are expressed as lists of topics to know for teaching. This parallels the development of Praxis, where items were developed to reflect mathematical knowledge represented by lists of topics. The example below shows the kind of background CSET provides to explain the undergirding of its mathematics items:

1.4 Linear Algebra

- a. Understand and apply the geometric interpretation and basic operations of vectors in two and three dimensions, including their scalar multiples and scalar (dot) and cross products.
- b. Prove the basic properties of vectors (e.g., perpendicular vectors have zero dot product)
- c. Understand and apply the basic properties and operations of matrices and determinants (e.g., to determine the solvability of linear systems of equations)

(Mathematics Content Standards for California Public Schools,
Algebra I: 9.0; Algebra II: 2.0; Mathematical Analysis: 1.0;
Linear Algebra: 1.0–12.0)

One can see the direct connection between this policy statement and the item below:

If vectors $\vec{a} = (a_1, a_2)$ and $\vec{b} = (b_1, b_2)$ are perpendicular,

$$A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix},$$

and if we identify vectors with column matrices in the usual manner, then show that the vectors $A\vec{a}$ and $A\vec{b}$ are perpendicular for all values of θ .

Like the elementary exam, the secondary exam appears to emphasize reasoning and proof, at least in its constructed response items. The exam also appears quite challenging; the following items draw from post-secondary content:

1. Which of the following statements refutes the claim that $GL_{\mathbb{R}}(3)$, the set of 3×3 invertible matrices over the real numbers, is a field?
 - a) There exist elements A and B of $GL_{\mathbb{R}}(3)$ such that $AB \neq BA$.
 - b) There exist elements A and B of $GL_{\mathbb{R}}(3)$ such that $\det(AB) = \det(A)\det(B)$.
 - c) If A is an element of $GL_{\mathbb{R}}(3)$, then there exists a matrix A^{-1} such that $A^{-1}A = I$.
 - d) If A is an element of $GL_{\mathbb{R}}(3)$, then there exists a matrix A such that $\det(A) \neq 1$.
2. Show that the subset of complex numbers of the form $a + bi$ with a and b rational numbers satisfies the axioms of a field under the operations of addition and multiplication of complex numbers.

In all, however, many of the items on the CSET, ABCTE, and secondary Praxis II look quite similar. This overlap is one characteristic of current secondary teacher certification tests: aside from different content emphases and more or less difficult items, there is less variation than one would think, given the proliferation of current tests.

The Future of Certifying Teachers of Mathematics

At no time in this nation's history have there been so many routes to becoming a certified teacher; teachers can do everything from taking multiple-choice content exams to constructing elaborate records of their teaching practice. Where does this leave those interested in moving teacher certification toward an assessment of more job-relevant, discipline-specific professional skills? We begin by discussing the benefits and drawbacks of current certification exams, then in our conclusion consider some of the issues that might lead to the next generation of certification tests.

Clearly, most new forms of teacher licensure testing have significant benefits as compared to past forms and, in some cases, even as compared to scholarly methods for assessing mathematical knowledge for teaching. To start, unlike their historical counterparts, most of the current tests attend carefully to issues of reliability and, increasingly, validity. For instance, a number of studies comparing NBPTS teacher candidates who have received certification and those who have not have revealed measurable differences between NBPTS-certified and regularly certified teachers and their impact on student learning and growth (Bond, Smith, Baker, & Hattie, 2000; Cavalluzzo, 2004; Goldhaber, Perry & Anthony, 2004; Vandevort, Amrein-Beardsley, & Berliner, 2004; for a counterexample, see Sanders, Ashton, & Wright, 2005). This is a key question centering on predictive validity: Does the assessment provide information about the candidate's actual effectiveness in the classroom? If not, the assessment is of little value.

Second, most current certification tests explicitly intend to assess professionally specific knowledge of mathematics, such as pedagogical content knowledge. And, in the case of performance-based assessments—such as observations, portfolios, the Praxis content exercises, or the NBPTS assessment exercises—not only does the content of the assessments more accurately reflect ways that teachers use mathematics in practice,

the assessments themselves emerge from or are closely tied to the actual work teachers do. Given that sociologists typically define professions by the presence of a codified body of knowledge with professionally controlled barriers to entry for new practitioners, this is a significant advance. In fact, efforts such as INTASC and NBPTS actually contribute to this development, for their standards attempt to articulate a coherent and comprehensive vision of the knowledge, skills, and dispositions needed for teaching.

Another benefit of performance-based assessments is they may actually stimulate teacher reflection and learning, thus providing an opportunity for professional development. For instance, individuals who seek ABCTE certification take a self-assessment and, if necessary, work with a learning advisor to refresh and/or learn content areas prior to taking the actual exam. Teachers have reported that the NBPTS certification process inspires growth and self-evaluation (Rotberg, Futrell, & Lieberman, 1998) and encourages teachers to become more reflective about their practice (Haynes, 1995; Letofsky, 1999; Sumner, 1997; Tracz et al., 1995).

However, new forms of teacher certification also have drawbacks. To start, the closer to practice the assessment, the less likely it is to be used. The secondary content knowledge Praxis II, for instance, was taken by six times more examinees than the secondary mathematics pedagogy exam (ETS, 2006b, p. 11). Only 2 of the 43 states that use Praxis I and/or II for certification require Praxis III. And only small numbers of teachers seek NBPTS certification; one recent survey found that only 2% of middle grades teachers and 3% of high school teachers had applied for certification in the past three years (Horizon Research, 2002).

A second issue centers on equity: Research routinely suggests certification testing has adverse impacts on teachers from minority groups, who for instance have historically scored lower on both the NTE and Praxis.⁷ Minority candidates may successfully complete teacher education coursework and internships but not meet their states' requirements for a passing score on Praxis II (Albers, 2002). This raises questions about the construction of the Praxis tests themselves. Some suggest the content and the format privilege the knowledge base of certain groups: for example, that Praxis tests in English favor white literature over minority literatures, and therefore disadvantage minority teacher education students. It has also been suggested that the curricula and style of learning predominant

⁷ See Haney et al. (1987) for a discussion of the NTE. For Praxis, Wakefield (2003) examined statistics released from the Georgia Professional Standards Commission. These statistics revealed that in 2001, European Americans had a 93% pass rate on Praxis II, while African Americans had a 71% pass rate (p. 383).

in historically black universities and colleges is not reflected in standardized tests in general and in Praxis in particular (Albers, 2002).

There also have been equity concerns about NBPTS certification. Some believe that the process creates an unnecessary hierarchy within the field of teaching. Even though a growing array of grants, scholarships, and no-cost loans are now available to cover the cost of participation in the certification process, some argue that the \$2,500 fee may still limit who can participate in the process (King, 1994; Marshall, 1996). Others critique the lack of inclusion of “culturally sensitive pedagogies” in the framework (Hamsa, 1998; Irvine & Fraser, 1998). Furthermore, while the number of minority applicants is proportionate to their representation in the teaching work force (17%), their achievement of certification is lower (only 11% of NBPTS-certified teachers are minority). NBPTS and other researchers are critically examining the role of cultural bias in these results as well as support mechanisms that might address this situation (Kraft, 2001; SRI, 2004; Serafini, 2002).

A third concern involves reliability. How accurately can the assessments determine how much mathematical knowledge for teaching an individual has? Standardized assessments, such as Praxis, return indices of reliability that can be used to assess the accuracy of the test. For more novel formats, such as portfolios and constructed response tasks, the determination of the degree of accuracy is more complex—dependent on the selection of specific tasks, scoring rubrics, and also on the ability of the person scoring the assessment. Reliability between scorers must be reached, but there must also be some assurance, in our view, that if one took another sample of a teacher’s mathematical practice, the same overall judgment regarding certification would be reached. With any stakes attached to the assessment, this is a critical issue.

Yet another issue is the lack of evidence on the validity of many of these assessments, or the degree to which these certification tests predict what teachers can do in classrooms to help students learn. Aside from the NBPTS and some of the scholarly measures, few assessments have been validated by anything more than a review of the literature and signatures from the practicing teachers and mathematics educators who serve on standards development and assessment review panels. The need for more evidence on predictive validity in particular—that is, whether teachers who pass specific certification tests produce better-educated students than those who do not—is dire.

Fourth, it is interesting to note how the mathematical topics to be tested have changed over time. Today there is an increasing emphasis on algebra in

some exams: fully 40% of the CSET items for mathematics certification in California are algebra problems, and the Praxis II examination for secondary mathematics teachers has 16–18% algebra items, up from 12–14% on the NTE. Interest in geometry has grown as well: in the days of the NTE, geometry items represented 16–18% of the entire pool of mathematics items, and the Praxis II exams, which replaced the NTE in most states, now have 22–26% geometry out of all math items. What counts as mathematically important knowledge for teaching seems to vary over time and by test.

So where does this leave us? Clearly, there is currently no ideal certification assessment, at least in mathematics. However, we do stand in a better position than just twenty years ago. Basic research into mathematical knowledge for teaching has supplied a vision for what might be measured on these assessments, one that contrasts with the “teachers should know a little more than their students” view taken during most of U.S. history. And advances in assessment theory—both psychometrics and test theory in general—have encouraged the adoption of better formats and improved test characteristics. Yet much remains to be done. Despite the alternative vision of what should be measured, we do not know whether this is the mathematics that actually matters to student learning. Pattengill’s old question—whether teachers need to know content, or teaching methods, or both—survives today, although in slightly different form. And the items on this list are complicated by the fluid nature of professional knowledge, and by debates over mathematics teaching more broadly. Some of what counted as profession-specific mathematical knowledge on the 1900 certification exams does not count as knowledge today; what counts as knowledge today might not count as knowledge tomorrow.

CONCLUSION

Teacher examinations have a complicated history and pose a number of serious contemporary dilemmas. Even 150 years after the first written teacher tests emerged in the U.S., there is little agreement on what, whom, and how to measure, and for what purpose. Pressing questions—such as the balance of knowledge of content *and* knowledge of pedagogy, the nature of content knowledge useful for teaching, and the “content” of pedagogical knowledge—have not been answered. Meanwhile, the number of teacher exams continues to increase. And pressures for more accountability, both among the programs

that prepare teachers and among teachers themselves, continue to mount.

Given these pressures, it is not likely that teacher testing will subside. The challenge, perhaps, is to create the best tests possible. Measuring teacher knowledge, even using standardized modes of assessment, *can* be done in ways that honor and define the work of teaching, ratify teachers' expertise, and help to ensure that every child has a qualified teacher. Doing so requires carefully constructed instruments that take seriously the work of teaching and that can be used at scale. In this concluding section, we turn to consider possible ways to contribute to the improvement of teacher assessment, both standardized and other forms. Our remarks take the form of proposed directions and foci, based on our review of practice in this domain.

Measure Mathematical Knowledge for Teaching

It is no surprise that we argue for measuring the mathematical knowledge used in teaching. Valid teacher assessments should not be remote from what teachers are asked to do in classrooms, with real students, materials, and content. The extended example of division of fractions in the middle of this chapter illustrates some of our own bets about what kind of knowledge matters; others have different bets, and there will no doubt be some evolution toward the "correct" mix of problem types, tasks, and items that should be contained on teacher assessments. With effort, there can also be an evolution toward appropriate formats for the assessment, through in-person observations, pencil-and-paper assessments, portfolios, or other formats. These directions can bridge more effectively the scholarly work on teacher knowledge and the policy-related work on assessing teacher quality.

However, we are worried about two trends. One is this field's tendency to conflate teachers' knowledge of mathematics for teaching with other types of knowledge or skill. For instance, many of the assessments we studied conflated knowledge of instructional approaches (e.g., using manipulatives; involving students in meaning-making; teaching in line with the NCTM standards) with knowledge of mathematics. This is reminiscent of our distinction between the *quality of mathematics instruction*, which we defined as inclusive of instructional approaches, and the *quality of the mathematics in instruction*, which we defined as focused specifically on the actual mathematics deployed in the course of a lesson. Other assessments of mathematical knowledge for teaching required teachers to design lessons, a possibly separate skill. And still other assess-

ments appear to be scored based on a teacher's ability to reflect about her instruction. Such assessments are not necessarily invalid; however, these other elements of teacher knowledge should be recognized as a crucial component of what is being measured, and test scores should be interpreted appropriately.

Mathematical knowledge for teaching should also not be represented as only one element in a composite score. Mathematics teaching is important in its own right, and students deserve teachers who have qualified for their jobs based on knowing mathematics in the ways it should be known for teaching—not demonstrating preparedness in many areas, including mathematics. This is especially true, in our analysis, for elementary-school certification exams, where mathematics is likely to be combined with other areas (e.g., science, in the elementary CSET and NBPTS) or with all subjects (e.g., elementary content area assessment of the Praxis II).

Measure with Care

No test is perfect; as we have shown, every format, whether multiple choice, observational, or performance based, has advantages and disadvantages. Recognizing this, the task facing educators and policy-makers is to carefully and responsibly link the research/policy question being investigated to the most appropriate form of assessment.

Two other concerns are worthy of mention. Assessments can send unintended messages about their purpose. The format of the assessment may serve, inadvertently, as a model of assessment that teachers believe should then be used in assessing their students. Multiple-choice exams are vulnerable to many criticisms, yet their use for the testing of teachers may suggest to teachers that this is a preferred assessment method. Even when the content of the multiple choice questions is focused on robust expressions of what teachers know to teach, the format may speak louder and have greater implications than intended by the developers. Similarly, mathematics educators have worked hard to counter the misconception that mathematical competence is demonstrated by quick solutions to routine mathematical problems. Certain testing formats, multiple-choice in particular, may solidify this misconception among teachers.

This field also needs to recognize how the technical requirements for standardized testing affect test content. In his scathing critique of standardized testing in history, Wineburg (2004) notes that the vagaries of modern psychometrics conspire to hide what students know about history rather than measure what they *do* know. Wineburg describes how analytic techniques

such as biserial correlation disadvantage those whose knowledge base is different than the mainstream, and that only items that produce a good “discrimination index” (that is, that differentiate between high achievers and low achievers) are included in multiple-choice exams, regardless of the intrinsic value of their content. He writes, “If ETS statisticians determined during pilot testing that most students could identify George Washington, ‘The Star-Spangled Banner,’ Rosa Parks, the dropping of the bomb on Hiroshima, slavery as a main cause of the Civil War, the purpose of Auschwitz, Babe Ruth, Harriet Tubman, the civil rights movement, the ‘I Have a Dream’ speech, all those items would be eliminated from the test, for such questions fail to discriminate among students” (2004, p. 1409). Measures of mathematical knowledge for teaching are vulnerable to the same dilemma.

Use Multiple Approaches

One way to effect well-considered improvements in this enterprise is to avoid repeating the mistakes of the past. One of the problems with assessing professional knowledge has been reliance on traditional test formats. In this chapter, we have examined both standard assessments formats and more novel forms of assessment, including discourse analysis, video analysis, and portfolio assessment. The most widely used teacher test today is the Praxis series, which offers multiple-choice and open-ended items in the paper-and-pencil tests that measure inert knowledge for teaching, as well as real-time observation of teaching that is designed to assess knowledge in action. Only two states are currently using the observation-based Praxis III for certification, but we think this combination of testing formats is promising. New uses of multimedia for documenting teaching offer other avenues for comprehensive appraisals of teaching work, as discussed earlier in the chapter.

Meet Professional Standards of Rigor in Assessment

Another important step is to confront the challenge of rigor, and what “rigor” might appropriately mean. New assessments should be held to more rigorous standards than old assessments, which were amazingly often validated only with reference to the content they covered, despite their use in making inferences about practice and its quality. New assessments should be rooted in theory, and scores should be validated using a number of methods. In our own work, paper-and-pencil measures of the mathematical knowledge for teaching are validated by close analyses

of the mathematical work of teachers that we videotaped. Is the mathematical knowledge that teachers demonstrate on a paper-and-pencil test expressed in their teaching? Can we show that students of highly qualified teachers, as defined by this paper-and-pencil test, learn more mathematics? If all mathematics assessments for teachers answered such questions, tests could bring real value to our efforts to improve educational quality. Rigorous standards for assessment already exist in the form of American Educational Research Association/American Psychological Association guidelines (1999) and the field can readily use them.

Learn from Other Measurement Methods

In the introduction to this chapter, we noted that seldom do lessons from one line of work measuring teachers' mathematical knowledge “travel” to other lines of work. Qualitative researchers have much to learn from large-scale test developers; large-scale test developers need lessons learned in qualitative research to succeed. More crossover needs to occur between these separate enterprises.

Attend to Issues of Equity

Attention to equity is crucial. As we saw, exams have a long and disturbing history of excluding people of color from the teaching profession. That these assessments regularly discriminated among candidates on the basis of characteristics unrelated to the knowledge and skill needed for teaching was a serious problem, and a threat to the validity of the exams themselves. Doing well on the tests must be related to success in teaching, not demographic characteristics. When patterns of performance are predicted by race, examinations must be scrutinized to uncover what it is about the test questions that may be producing this result. Tests that are invalid across groups in such ways must be challenged publicly; moreover, their use also serves to inhibit the development of a diverse teaching force.

Investigate the Relationship among Mathematical Knowledge for Teaching, Other Domains of Teaching Knowledge, and Student Learning

Much work remains to be done to specify what we are calling “mathematical knowledge for teaching.” Important to understand is how this domain intersects with other domains of knowledge for teaching, and how these together affect student

learning. We know, for example, that mathematical knowledge for teaching intersects knowledge for equitable teaching. Being explicit about mathematical practices makes mathematics learning accessible for wider populations. Mathematical knowledge for teaching may also act upon (or be acted upon by) teachers' ability to motivate students to learn, to organize classrooms for productive instruction, and to produce learning gains.

Increase Professional Role and Control

The history of U.S. teacher testing in mathematics has been one in which the tests have largely been developed by those outside the profession: local officials, bureaucrats, and later, professional test developers. In the coming years, professionals in education and in mathematics must develop the earned authority to control the identification of professional knowledge and the means of certifying it reliably. The judgments involved require specialized insight and perspective, rooted in evidence that is tied to instructional effectiveness—that is, instructional practices that reliably produce student achievement. Improvement in assessing teachers depends on close involvement of experts in mathematics instruction.

Where does the tour on which we have been lead? We have seen that teacher assessment was originally narrow, closely bound to the content of the curriculum. Teacher qualification was judged by teachers' ability to solve difficult word problems of the sort they might teach their students. Over the last 100 years, approaches to assessment have multiplied. What is assessed has broadened dramatically. How it is assessed has similarly expanded. Even within the same testing firm, different assessments are developed. The field has moved from a narrow and underspecified conception of the knowledge necessary for teaching to a broadband approach which lacks the precision and common agreement about requirements that one would expect of a profession. The agenda for assessing teacher knowledge is clear: The need to prepare and certify knowledgeable and skilled teachers who can be reliably effective with a wide range of students has perhaps never been greater. With the expansion of interest in and approaches to assessment, the field needs to move toward clearer precision, specification and agreement of measures, and broader skilled use of reliable and valid tools for appraising teachers' knowledge, skill, and performance. Research on teaching and teaching effectiveness is ready for this step; the current state of policy and practice demands it.

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THE MATHEMATICAL EDUCATION AND DEVELOPMENT OF TEACHERS

Judith T. Sowder

SAN DIEGO STATE UNIVERSITY

The nation can adopt rigorous standards, set forth a visionary scenario, compile the best research about how students learn, change textbooks and assessment, promote teaching strategies that have been successful with a wide range of students, and change all the other elements involved in systemic reform—But without professional development, school reform and improved achievement for all students will not happen. Unless the classroom teacher understands and is committed to standards-based reform and knows how to make it happen, the dream will not be realized.

—Principles for Professional Development, American Federation of Teachers, 2002, p. 2

Teachers matter. A variety of recently published documents support the notion that the key to increasing students' mathematical knowledge and to closing the achievement gap is to put knowledgeable teachers in every classroom. For example, in a survey designed by Lou Harris and undertaken in 1998 by the Recruiting New Teachers organization, a carefully chosen sample of a cross-section of U.S. adults was questioned about how to improve America's schools (Haselkorn & Harris, 1998). Of the 2525 people interviewed, an overwhelming majority agreed that "improving the quality of teaching is the most important way to improve public education" (p. 1). Other documents essentially repeating this message include the 1998 report *Every Child Mathematically Proficient: An Action Plan* from the Learning First Alliance, the 2000 report *Before It's Too Late: A Report to the Nation* from the National Commission on Mathematics and Science Teaching for the 21st Century as requested by the U.S. Department

of Education, the 1996 and 2003 reports *What Matters Most: Teaching for America's Future* and *No Dream Denied* from the National Commission on Teaching and America's Future, the 2000 report *The Mathematical Education of Teachers* (MET) from the Conference Board of the Mathematical Sciences, the 2001 act *No Child Left Behind* from the U.S. Department of Education (2002), and the 2003 report *Mathematical Proficiency for All Students* from the RAND Mathematics Study Panel.

The purpose of this chapter is to address what it means to prepare teachers of mathematics and to provide them, while they progress in their careers, with the professional learning opportunities they need to lead their students to succeed in learning mathematics.

In this chapter I attempt to answer, as much as is possible on the basis of theory and research evidence, the following questions:

1. Why has professional development become a priority in realizing current goals for mathematics education?
2. What are the goals of professional development?
3. What principles can be used to guide the design of professional development?
4. How do teachers learn what they need to know for teaching mathematics?
5. How do teachers learn from their professional communities about teaching mathematics?
6. What can teachers learn by investigating their own teaching of mathematics?
7. What can be learned from research on teacher change?
8. What can be learned from research about the preparation of teachers to teach mathematics?
9. What do researchers know about issues that affect professional development, such as support and accountability, preparing teacher leaders, evaluating professional development, and policy issues?
10. Where do we stand now, and what comes next?

In the ten major sections of this chapter, I address these ten questions.

Before beginning to address these questions, I raise six caveats that will clarify some decisions I made in organizing and writing this chapter. First, because this volume also contains chapters on teacher knowledge, teaching, and teacher beliefs, I address these topics only to the extent needed to provide a basis for much of the ensuing discussion. Second, the more general literature on professional development and teacher education provides a context from which to interpret the literature within mathematics education. Thus, the two bodies of literature are intermingled in this chapter. Third, many of the claims made by authors and quoted here are based on reviews of literature, and, thus, for understandable reasons, the original literature on which they base their claims is not usually included here. Fourth, much of the research discussed here could easily fit under more than one heading; that is, the categories I use are not disjoint. My placement decisions are apt to be different from those of some readers. Fifth, some authors make distinctions among teacher education, professional development, and teacher change. For example, Lerman (2001) and Ponte (2001) associated teacher education with preservice teacher preparation, that is, courses and experiences designed for students prepar-

ing to be teachers, whereas professional development and teacher change are linked to in-service programs and teachers' changing roles and could occur in taking courses, completing projects, reading, reflecting, sharing experiences, and so on. That distinction is not always made here because much of what is true for professional development is also true for teacher preparation. For example, an inquiry approach can be used in both; strategies such as focusing on children's thinking or studying cases can be used in both; changes of beliefs about mathematics and about learning mathematics can be expected from both, even though these approaches are discussed in sections on professional development. Likewise, topics such as mentoring are discussed within the section on teacher preparation. Indeed, the National Science Foundation now uses the term *professional development* to refer both to teacher preparation and to the development of practicing teachers. That said, some issues, such as the effects of methods courses and field experiences, relate to teacher preparation but not to professional development of practicing teachers. For this reason, I devote a section to only the preparation of teachers. And, sixth, although the term *reform* has been both overused and often abused, it so thoroughly permeates the literature that I realized that I could not completely avoid its use here. When possible, however, I use the term *principled knowledge* in the manner articulated by Spillane (2000) as a way of thinking about the goals of school mathematics reform:

Reformers want principled mathematical knowledge, as distinct from procedural knowledge, to receive more attention in school work. Whereas *procedural knowledge* centers on computational procedures and involves memorizing and following predetermined steps to compute answers, *principled knowledge* focuses on the mathematical ideas and concepts that undergird mathematical procedures. Procedural knowledge has dominated the K through 12 curriculum. Reformers also propose that students develop a more sophisticated appreciation for doing mathematics including framing and solving mathematical problems, articulating conjectures, and reasoning with others about mathematical ideas: Students need to appreciate mathematical activity as more than computation. (p. 144)

In this chapter I focus on what researchers can tell us about teaching mathematics for principled knowledge and the implications this information has for the design of programs of teacher education and development. Changing the focus from procedural

to principled knowledge is a new frontier for many teachers, administrators, and policy makers. The knowledge demanded is not an add-on; for many, this change can mean a complete reconceptualization of such fundamental questions as: What does it mean to teach mathematics? What is mathematics? What mathematics is important for teachers to know and why? What does teaching mathematics entail? How does one acquire knowledge? How can teachers productively assess that knowledge and use assessment data to plan instruction? The answers to all these questions change from those that underlie teaching for procedural knowledge.

**ADDRESSING QUESTION 1:
WHY HAS PROFESSIONAL DEVELOPMENT
BECOME A PRIORITY IN REALIZING CURRENT
GOALS FOR MATHEMATICS EDUCATION?**

The standards movement of the 1990s coupled with a call for school reform, both set against a backdrop of performance-based accountability demands that include high-stakes testing, conspire to change the ways in which schools operate. The standards movement was heralded by the publication of the *Curriculum and Evaluation Standards for School Mathematics* by the National Council of Teachers of Mathematics (NCTM) in 1989; this document set the stage for standards in other K–12 disciplines. NCTM, too, continued to publish standards for teaching mathematics (1991) and assessing mathematics learning (1995) and in 2000, published the *Principles and Standards for School Mathematics*. States have also developed standards for teaching mathematics, some based on the NCTM Standards and some not.

The accountability movement is more recent and results from recognition that all students have the ability to learn—and to learn more than they are currently learning—even while the student population becomes more diverse. To some, *accountability* means that all should know their basic skills, but others recognize that “the future of education clearly presses beyond the basics to more demanding forms of academic learning” (Sykes, 1999, p. 153). The accountability movement is perhaps best represented in the United States by the 2001 *No Child Left Behind* Act, which, by law, calls for vast changes that are intended to result in all children’s becoming proficient in key areas.

Education reformers realize that instructional improvement and increased student achievement

depend on the professional development of teachers and administrators (e.g., Ball & Cohen, 1999; Elmore & Burney, 1999; Nelson & Hammerman, 1996; Sykes, 1999; C. L. Thompson & Zeuli, 1999). Hawley and Valli (1999) provided evidence from a number of sources that “one of the most persistent findings from research on school improvement is, in fact, the symbiotic relation between professional development and school improvement efforts” (p. 129). Sykes (1999) noted that the view that professional development can act as a “significant lever for education improvement” (p. 151) is fairly recent and has occurred as the result of three major societal changes: systemic reform efforts, new research knowledge about instruction, and the escalation of the commitment to quality education for all. Systemic reform has been a major policy influence.

Recognition of the need to change the way in which mathematics is taught and learned is international in scope—“Across the world, . . . mathematics teacher education programmes are preparing teachers to work with and promote reform in the practice of school mathematics” (Adler, 2000, p. 205). International researchers view teachers as the key figures on which the success of current reforms depends (Ponte, 1994). In the United States, the escalation of the commitment to providing quality education to all students, at a time when the student population grows more diverse, has led to higher demands on teachers, who in turn need the support provided through appropriate professional development. In mathematics education, “mathematics for all” has been a foundation for all volumes of NCTM *Standards* (1989, 1991, 1995, 2000).

Even with increased recognition that teacher professional development must be a priority, the professional development offered in mathematics often does not meet teachers’ needs. According to Ball and Cohen (1999), most professional development funding is spent on

sessions and workshops that are often intellectually superficial, disconnected from deep issues of curriculum and learning, fragmented, and noncumulative (Cohen and Hill, 1997; Little, 1994). Rarely do these in-services seem based on a curricular view of teachers’ learning. Teachers are thought to need updating rather than opportunities for serious and sustained learning of curriculum, students, and teaching. (pp. 3–4)

In many school districts a “one size fits all” mentality limits the type of professional assistance teachers receive (Lord, 1994). Some frame the issue even more

dramatically—Miles (1995) called current professional development a “joke”—“pedagogically naive, a demeaning exercise that often leaves its participants more cynical and no more knowledgeable, skilled, or committed than before” (p. vii). Others (e.g., Elmore, 2002a; Hargreaves, 1995; Middleton, Sawada, Judson, Bloom, & Turley, 2002) have been milder in stating their criticism of current professional development efforts but, nonetheless, have found them to be ineffective in terms of current reform and accountability demands.

The confirmation of the belief that much conventional professional development is ineffective and wasteful was found by Hawley and Valli (1999) to be one of four converging developments that appeared to provide a consensus among stakeholders that major changes in how professional development is conceptualized and conducted are needed. A second development they noted was the body of research that showed how change is related to professional development. Schools cannot change unless teaching is improved. And schools will not be improved for children unless schools also become places for teachers to learn. A third development was a “growing agreement that students should be expected to achieve much higher standards of performance, which include a capacity for complex and collaborative problem solving” (p. 128). The view of teaching as transmission of knowledge is at odds with the type of performance now expected of children. The fourth development they noted was that the research on how people learn leads to strategies for teaching and assessment that differ from those most commonly used. Many of teachers’ core beliefs need to be challenged before change can occur.

Case studies have become a useful way to understand and evaluate the effectiveness of professional development or intervention into the teacher preparation process. Large-scale research projects in which professional development has been a major factor often use case studies as a means to study the effects of the professional development on individual teachers. The results can be dismaying, as exemplified in a case study by D. K. Cohen (1990). His portrayal of Mrs. Oublier has attained the status of a study that can be used as a generic description of a class of teachers who have misinterpreted the principles underlying the professional development they received. Mrs. Oublier, an elementary school teacher in California, willingly participated in professional development that included familiarization with the 1985 *Mathematics Framework for California Public Schools* (California Department of Education), a framework much different from California’s current framework. She eagerly adopted new

curriculum materials and activities introduced to her during professional development in which she participated. She believed that she had revolutionized her practice. Cohen noted that she had simply adapted these new materials and activities to her traditional teaching style. She continued to teach mathematics as though there were only right and wrong answers, and she discouraged exploration that would reveal students’ understanding. Cohen observed many tensions that lay beneath the surface of the work in Mrs. Oublier’s classroom. Because her own understanding of mathematics was still superficial, she could not have been successful with the give and take of open discourse. Cohen reflected on his observations of Mrs. Oublier, stating that the kind of change professional developers believe may be taking place is illusory.

The teachers and students who try to carry out such change are historical beings. They cannot simply shed their old ideas and practices like a shabby coat, and slip on something new. . . . As they reach out to embrace or invent a new instruction, they reach out with their old professional selves, including all the ideas and practices comprised therein. The past is their path to the future. Some sorts of mixed practice, and many confusions, therefore seem inevitable. (p. 339)

Mrs. Oublier had little opportunity for sustained guidance and support. She had much to unlearn, but no one to help her do this unlearning. The lessons here for the need for sustained professional development and mentoring are significant.

These developments and stories show that the quality of preparation and professional development of teachers must improve well beyond current levels if teachers are to become “capable of far more sophisticated forms of practice” (Sykes, 1999, p. 153). Exploration of how this improvement can come about permeates the literature of teacher education.

ADDRESSING QUESTION 2: WHAT ARE THE GOALS OF PROFESSIONAL DEVELOPMENT?

To teach mathematics for principled knowledge, teachers need an extensive knowledge base, technical teaching skills, and epistemological stances that should all be goals of teacher preparation and professional development programs (Elmore, 2002a; Sykes, 1999). To meet these goals, professional development must provide opportunities for professional growth on the part of teachers and motivate them to develop the knowledge, skills, and dispositions they need to

teach mathematics well. Related outcomes of professional development include changing teachers' understanding of how students learn mathematics and of the nature of mathematics, of mathematical knowledge, and of teaching mathematics well. Ideally, this new understanding will result in mathematics teachers equipped to choose appropriate curricula, plan instruction, and organize their classrooms to promote and support learning for *all* the students they teach. Professional growth is thus marked by change in teachers' knowledge, beliefs, and instructional strategies. Professional development should be designed to address the needs of teachers so that they can teach for principled knowledge in a manner envisioned by researchers and educators. Several educators have addressed the issue of what these needs are. Ball and Cohen (1999) suggested that teachers need to become serious learners of practice rather than learners of strategies and activities. For teachers to become such learners would entail their coming to understand well the mathematics they teach and what it means to reason mathematically. They need to learn to attend to their students in insightful ways, a skill that "requires expertise beyond what one gathers from one's own experience" (p. 9). They need to "develop and expand their ideas about learning, [which would require that] longstanding beliefs and assumptions about learning would need to be examined" (p. 9). They would need to better understand pedagogy and how to establish a classroom culture that supports learning goals. Ball and Cohen noted that even when these needs are addressed, the complexity of interactions and situations in classrooms also affect the ways in which teachers teach.

Borasi and Fonzi (2002) also identified several needs of teachers that could serve as a focus for professional development. In addition to the needs identified by Ball and Cohen, they included the needs to develop a vision and commitment to school mathematics reform, to become familiar with exemplary curriculum materials, to understand equity issues and their implications for the classroom, to cope with the emotional aspects of engaging in reform, and to develop an attitude of inquiry towards one's practice.

In this section I address these teacher needs by first grouping them into six, sometimes overlapping, goals and then considering literature that informs and supports each goal. These goals are to develop (a) a shared vision for mathematics teaching and learning, (b) a sound understanding of mathematics for the level taught, (c) an understanding of how students learn mathematics, (d) deep pedagogical content knowledge, (e) an understanding of the role of

equity in school mathematics, and (f) a sense of self as a mathematics teacher.

Goal 1. Developing a Shared Vision

When professional development providers plan, they themselves are guided by a vision of what they want teachers to know and be able to do. These visions will differ according to the providers' views of the role of school mathematics, how children learn, and how instruction should be designed to best help children learn. Those who believe that the goal of school mathematics is to lead students to develop principled knowledge of the mathematics they need to function in today's society will formulate their goals and structure their programs in ways that reflect this vision.

Several documents published over the past few decades have contributed to the formation of a vision of what teaching and learning mathematics encompasses. The NCTM Standards documents of 1989, 1991, 1995, and 2000 have provided images of classroom teaching and learning that are far different from those familiar to most teachers. For example, the *Principles and Standards for School Mathematics* (NCTM, 2000) provided a vision of the mathematics all students ought to know and be able to do. The Teaching Principle is a recognition of the role of teachers in attaining this vision: "Effective mathematics teaching requires understanding what students know and need to learn and then challenging and supporting them to learn it well" (p. 16). Accordingly,

teachers need to know and use "mathematics for teaching" that combines mathematical knowledge and pedagogical knowledge. They must be information providers, planners, consultants, and explorers of uncharted mathematical territory. They must adjust their practices and extend their knowledge to reflect changing curricula and technologies and to incorporate new knowledge about how students learn mathematics. They must also be able to describe and explain why they are aiming for particular goals. (p. 370)

Other sources such as *Adding It Up* (National Research Council [NRC], 2001a), with its intertwined strands of abilities and dispositions representing mathematical proficiency; *Making Sense: Teaching and Learning Mathematics With Understanding* (Hiebert et al., 1997), with its descriptions of what learning with understanding entails; *Reconstructing Mathematics Education* (Schifter & Fosnot, 1993) and *What's Happening in Math Class* (Schifter, 1995b), with their stories of teachers who transformed their instruction; *Relearning to Teach Arithmetic*, with its videotapes showing children's approaches to multiplication and division;

the *Integrating Mathematics and Pedagogy* (Philipp, Cabral, & Schappelle, 2004, 2005) with its rich sets of videos of children's thinking, organized by mathematics topic (within whole number and rational number areas), grade level, and other categories; *Learning and Teaching Linear Functions* (Seago, Mumme, & Branca, 2004), with its video cases of real classroom lessons; *Fostering Algebraic Thinking* (Driscoll, 1999), with its grounded assistance for teaching algebra beginning in Grade 6; *Mathematics Teaching Cases* (Barnett, Goldstein, & Jackson, 1994) and *Windows on Teaching Math* (Merseeth, 2003), with their cases of classroom teaching prepared for discussion by teachers—these and many other sources reflect a common vision of teaching and learning mathematics and provide professional developers with assistance in their planning and materials for presentations and workshops. Some of this literature was based on research; some, on shared values and beliefs. Cooney (1994) reminded us that teachers' beliefs about teaching "are shaped by social situations and therefore can only be reshaped by social situations" (p. 109). Professional development can provide the social situations teachers need to develop and clarify their values and beliefs while they learn new content and skills.

Ferrini-Mundy (1997) quoted from Fullan in her discussion of vision as a process rather than a product: "Shared vision, which is essential for success, must evolve through the dynamic interaction of organizational members and leaders" (Fullan, 1993, p. 28). In her project, designed to initiate reform of mathematics at many schools, Ferrini-Mundy learned that vision is "evolutionary and open-ended" (p. 124) and that in her project, teachers and administrators "became more and more articulate and definite about the goals and nature of the mathematics changes as they accumulated experience and evidence from their practice" (p. 124).

Including a shared vision as a goal of professional development does not necessarily require the visible focus one maintains for the remaining goals discussed here. A teacher's vision of learning and teaching mathematics unfolds while the other goals are being addressed.

Goal 2. Developing Mathematical Content Knowledge

Mathematics continues to be an essential component in the work of professionals in a variety of disciplines, some of which have existed only for the past few decades and many of which, though not new, have changed the ways in which they use mathematics. Technology has forever changed the way math-

ematics is used not only by professionals but also by workers in many fields. Accordingly, expectations for school mathematics are also changing—students will need to know more and different mathematics to be successful. And, in turn, expectations for teachers of school mathematics are different than they were in the 20th century. In the RAND report (2003), these expectations are stated in terms of urgency—"as both a matter of national interest and a moral imperative, the overall level of mathematical proficiency must be raised, and the differences in proficiency among societal groups must be eliminated" (p. xi).

Yet many policy makers underestimate the knowledge required for teaching mathematics. More than a decade ago, Ball and Wilson (1990) noted and countered two common but mistaken assumptions about obtaining knowledge needed for teaching: that traditional mathematics content needed by teachers can be best obtained through traditional undergraduate mathematics courses and that content knowledge is the only professional knowledge teachers need. Many policy makers continue to exhibit these assumptions, even though many educators believe that what mathematics knowledge is needed to teach well remains uncertain (Ball, Lubienski, & Mewborn, 2001). However, welcome attempts have been made (and continue to be made) to determine the content knowledge needed to teach mathematics, and these needs have been expressed in terms of a deep understanding that requires at least some specially designed coursework or programs. The quality of the mathematical knowledge also needs to be addressed. At the 2003 Department of Education Secretary's Summit on Mathematics, Ball, in her presentation, said that the qualities of mathematics knowledge needed for effective instruction included being "respectful of the integrity of the discipline, able to be extended and opened up for learners—*unpacked*, justified and reasoned, connected within and across domains, building on earlier ideas and anticipating more advanced topics, and organized psychologically as well as logically."

The need for teachers to know mathematics differently than mathematicians do has been recognized by educators for a very long while. More than a century ago Dewey addressed this issue:

Every study or subject thus has two aspects: one for the scientist as a scientist; the other for the teacher as a teacher. These two aspects are in no sense opposed or conflicting. But neither are they immediately identical. For the scientist, the subject matter represents simply a given body of truth to be employed in locating new problems, instituting new researches, and carrying them through to a verified outcome. . . . The problem of the teacher is a different one. . . . What

concerns him as teacher is the ways in which that subject may become part of experience, what there is in the child's present that is usable with reference to it; how such elements are to be used; how his own knowledge of the subject-matter may assist in interpreting the child's needs and doings, and determine the medium in which the child should be placed in order that his growth may be properly directed. He is concerned, not with the subject-matter as such, but with the subject-matter as a related factor in a total and growing experience. (Dewey 1902, pp. 29–30)

The mathematics community is only now beginning to understand the need for teachers to know appropriate mathematics, as described by Dewey. Recent documents from mathematical organizations have addressed the mathematical needs of teachers. *The Mathematical Education of Teachers*, published by the Conference Board of the Mathematical Sciences (CBMS, 2000), provided detailed information about the appropriate mathematics teachers need to know to teach at elementary, middle, and secondary school levels. The work of the writing committee was influenced by the then recent publication by Ma (1999). Her study of Chinese and American elementary school teachers' knowledge, in which she described the "Profound Understanding of Elementary Mathematics (PUFM)" needed to teach elementary grades, led many mathematicians to understand that the mathematics of elementary grades is not trivial and that teachers need more and different preparation than has been common. Additionally, the writers recognized that the ways present and future teachers need to know mathematics differs from the ways other college mathematics students need to know mathematics.

A second document that includes the U.S. mathematics community's content recommendations for the preparation of teachers is the *Undergraduate Programs and Courses in the Mathematical Sciences* (Mathematical Association of America [MAA], 2003), prepared by the Committee on the Undergraduate Programs in Mathematics (CUPM), a standing committee of the MAA. The teacher education recommendations in this document are informed by the MET recommendations and are embedded in a set of recommendations for the mathematical education of all undergraduates.

Substantial evidence indicates that too few classroom teachers have the mathematical knowledge described above (Borko & Putnam, 1995; RAND, 2003). Preservice teachers at all levels have been found to know rules and procedures but lack knowledge of concepts and reasoning skills (Wilson, Floden, & Ferrini-Mundy, 2001). The current efforts to focus school mathematics instruction on students' developing principled knowledge of mathematics must go hand-

in-hand with teachers' developing principled knowledge of mathematics also. Expanding principled knowledge calls for a shift from content as a focus to "content in relation to student learning and context" (Elmore, 2001, p. 11).

Professional development provides an opportunity for teachers to learn more mathematics, even when the focus is on student thinking or curriculum or classroom events. The projects described later in this chapter resulted in increased teacher knowledge of mathematics even though knowledge development was not their primary focus.

Goal 3. Developing an Understanding of How Students Think About and Learn Mathematics

Schifter (2001) has reminded the mathematics education community that additional mathematical skills needed for teaching may evolve not from a focus on mathematical content but from

attending to the mathematics in what one's students are saying and doing, assessing the mathematical validity of their ideas, listening for the sense in children's mathematical thinking even when something is amiss, and identifying the conceptual issues on which they are working. (p. 131)

Attending to students' reasoning has become a focus of many professional development programs. The Cognitively Guided Instruction (CGI) work is but one example of using student thinking in professional development. The literature on CGI is extensive; several studies (e.g., Carpenter & Fennema, 1992; Carpenter, Fennema, Peterson, & Carey, 1988; Carpenter, Fennema, Peterson, Chiang, & Loef, 1989; Fennema, Carpenter, Franke, & Carey, 1992; Fennema et al., 1996; Franke, Carpenter, Levi, & Fennema, 2001; Franke, Carpenter, Fennema, Ansell, & Behrend, 1998; Franke, Fennema, & Carpenter, 1997; Jacobson & Lehrer, 2000) over a number of years pointed to teachers' changes of instructional practice that led to substantial gains in student achievement. This research is based on the hypothesis that if teachers listen to children, understand their reasoning, and teach in a manner that reflects this knowledge, they can and will provide children with a mathematics education better than if they did not have this knowledge. To this end, the researchers developed a framework of children's thinking about early number problems—problem types and solution strategies—and used this framework as the core of a professional development program. This research has provided evidence of the

efficacy of focusing professional development on student thinking and learning.

Developing an understanding of how students think about mathematics is a goal of professional development—it is also a strategy used in professional development. When teachers examine work of their own students, they gain insight into their students' thinking and understanding. When teachers examine the work of other students, in company with other teachers, they discuss the kinds of strategies used and hypothesize about the students' knowledge, the kind of instruction received, and the instructional changes needed to improve understanding and achievement.

Student thinking can also be thought of as an interpretive lens that helps teachers think about their students, the mathematics they are learning, the tasks that are appropriate for the learning of that mathematics, and the questions that need to be asked to lead them to better understanding. As with any interpretive lens, development takes place over time, with many experiences with individual children and with whole classes of students. Professional development that has as a goal teachers' coming to understand how children think about and learn mathematics can awaken teachers to the power of understanding children's thinking and could lead them to develop a listening capacity that deepens over time.

A later section of this chapter is devoted to the exploration of projects that have focused on using student thinking as an approach to teacher learning. These studies confirm the powerful effect that the study of student reasoning can have on the ways in which teachers think about mathematics learning and gear their own instruction to account for this new knowledge.

Goal 4. Developing Pedagogical Content Knowledge

The grade-band content recommendations described in *The Mathematical Education of Teachers* (CBMS, 2000) are buttressed with descriptions of mathematical content as it relates to pedagogy. They focus on providing teachers with robust knowledge of mathematics that is conceptual in nature and appropriate for teaching. Teachers

need to see the topics they teach as embedded in rich networks of interrelated concepts, know where, within those networks, to situate the tasks they set their students and the ideas these tasks elicit; . . . they must be able to appraise and select appropriate activities, and choose representations that will bring into focus the mathematics on the agenda. (p. 55)

The knowledge described in this document aligns well with the notion of *pedagogical content knowledge*, that is, the knowledge of a subject one needs to teach it, which Shulman (1986) described as including

the most useful forms of representations of those ideas, the most powerful analogies, illustrations, examples, explanations, and demonstrations—in a word, ways of representing and formulating the subject that make it comprehensible to others. [It] also includes understanding of what makes the learning of specific topics easy or difficult; the conceptions and preconceptions that students of different ages and backgrounds bring with them to the learning of those most frequently taught topics and lessons. (p. 9)

This description is reminiscent of the quote from Dewey earlier in this section—the knowledge of the scientist (or mathematician, in this case) and the knowledge of the teacher are quite different because their goals are so very different.

Grossman's (1990) description of the four central components of pedagogical content knowledge is helpful to those developing teacher education programs and professional development opportunities for mathematics teachers. Her four categories (here changed to refer to mathematics) are (a) an overarching knowledge and belief about the purposes for teaching (mathematics); (b) knowledge of students' understandings, conceptions, and potential misunderstandings (in mathematics); (c) knowledge of (mathematics) curriculum and curricular materials; and (d) knowledge of the instructional strategies and representations for teaching particular topics (in mathematics).

The categories are not independent of one another. According to Borko and Putnam (1995), having an overarching conception of mathematics teaching (the first category above)

serves as a “conceptual map” for instructional decision making, as the basis for judgments about classroom objectives, instructional strategies and student assignments, textbooks and curricular materials, and the evaluation of student learning. . . . Teachers' overarching conceptions are a particularly salient component of the professional knowledge base. (p. 47)

Effective teachers can often predict what mathematics students will understand, how they will understand it, and some of the potential for misunderstandings (Grossman's second category above). They know, for example, that children's work with whole numbers often leads them to believe that multiplication “makes bigger,” which is untrue when their world of numbers is extended to include fractions and decimal numbers.

These teachers can plan more effectively because they can anticipate students' difficulties. They know what prior knowledge must be present to understand something new. They know how to scaffold knowledge to assist students in developing understanding. They know how to listen to students. Much of this knowledge comes from practice, but teachers who themselves have poor understanding of mathematics are unlikely to develop this type of knowledge, particularly when the mathematics in the curriculum becomes more sophisticated, such as it does when moving from additive to multiplicative situations or from Cartesian coordinates to polar coordinates.

Effective mathematics teachers think about mathematics curricula (Grossman's third category above) in terms of *big ideas*, such as proportional reasoning or the mathematics of change, around which to structure instruction. According to Kennedy (1997), attending to the central ideas rather than to minutiae is one way to define *conceptual understanding*. Effective teachers recognize the particular strengths and weaknesses of the textbooks and materials they are using. For example, a sixth-grade teacher studied by Philipp, Flores, Sowder, and Schappelle (1994) expressed astonishment that the mathematics text his district had adopted "allocated only three pages to perimeter and area, which indicated to him that the authors thought these difficult concepts could be learned by looking at a few textbook examples of regular figures and then learning formulas" (p. 165). Many effective teachers have (and often own) an array of materials they use when teaching mathematics. Another teacher interviewed by Philipp et al. (1994) said, "I have been known for a long time as having an incredible resource library. It's too bad I didn't buy stock in Dale Seymour and Creative Publications"¹ (p. 173). This teacher and others interviewed and observed by these researchers were able to use manipulatives and other materials effectively because they knew exactly what mathematics children would learn from any particular manipulatives when appropriate scaffolding was provided.

Knowledge of the strategies and representations for teaching particular topics (Grossman's fourth category above) can be characterized as "an extensive repertoire of powerful representations and the ability to adapt these representations in multiple ways in order to meet specific goals for specific sets of learners" (Borko & Putnam, 1995, pp. 48–49). Grouws and Schultz's (1996) description of *pedagogical content knowledge* in mathematics as including, but not limited to, "useful representations, unifying ideas, clarifying

examples and counterexamples, helpful analogies, important relationships, and connections among ideas" (p. 444) embodied this knowledge component. The work of Sowder et al. (1998), on the analysis of the mathematical content knowledge necessary to teach multiplicative structures in the middle grades also fits with this fourth component. Many teachers fail to develop this pedagogical knowledge of mathematics because they lack the fundamental mathematical knowledge that undergirds such pedagogical content knowledge. According to Borko and Putman (1995), teachers come to understand the various messages about changing instruction through their existing knowledge and practices, and teachers' knowledge and beliefs serve as "powerful filters through which learning takes place" (p. 60).

The teacher knowledge characterized by Grossman as combining many types of knowledge is, of course, the sort of knowledge teachers need, and the knowledge components must be integrated in practice, even though researchers at times, rightly or wrongly, separate them theoretically so that they might be more easily studied.

Simon (1997), too, characterized the knowledge teachers need as the combination of aspects of knowledge. The eight areas (overlapping and often interdependent) he identified illustrate this intertwining of facets of knowledge needed by teachers.

1. Knowledge of and about mathematics . . . [that] defines what is worth learning and the activities that are appropriate to the enterprise.
2. A personally meaningful model of mathematics learning . . . [that includes a] teacher's concepts of how mathematical knowledge is developed.
3. Knowledge of students' development of relevant concepts [that allows teachers] to make sense of the students' mathematics.
4. Relationship to students' mathematics . . . [that] encompasses commitment to understanding his or her students' mathematics, ability to elicit and probe their mathematics, and ability to analyze students' mathematical activity and form useful models of their knowledge.
5. A personally meaningful model of mathematics teaching . . . is itself recursively embedded . . . [and] is constantly in a state of construction and renovation.

¹ These companies sold, among other things, manipulative materials for teaching mathematics.

6. Ability to define appropriate learning goals for students, . . . [and] longer range goals as well as goals for spontaneous interventions, [implying] the ability to identify key mathematical ideas of the mathematics being considered, as well as to make use of knowledge of students' concept development and knowledge of that teacher's own students' mathematics.
7. Ability to anticipate how students' learning might ensue, . . . learning to anticipate how student learning might progress
8. Ability to construct lessons consistent with one's model of teaching . . . [by] constituting new forms of practice both by adapting new techniques and by rethinking the role of familiar ones. (pp. 81–82)

Although progress has been made, “the nature and development of [teachers'] knowledge is only beginning to be understood by the present generations of researchers in teaching and teacher education” (Munby, Russell, & Martin, 2001, p. 877). Limited knowledge of key mathematical ideas can lead to “missed opportunities for fostering meaningful connections between key concepts and representations” (Borko & Putnam, 1995, p. 44). Fundamental changes are needed in what we consider to be the nature of teacher knowledge. We now have at least some knowledge about what teachers need to know to be effective teachers, and this knowledge should guide us in developing programs for teacher education and professional development.

Goal 5. Developing an Understanding of the Role of Equity in School Mathematics

The NCTM Standards documents (1989, 1991, 1995, 2000) advocated exemplary mathematics instruction for *all* students. In fact, in the 2000 Standards the Equity Principle was quite deliberately listed as the first of the six principles to describe a high-quality mathematics education: “Excellence in mathematics education requires equity—high expectations and strong support for all students” (NCTM, 2000, p. 11). This principle was the core element of the vision of the Standards and required that “all students, regardless of their personal characteristics, backgrounds, or physical challenges, must have the opportunities to study—and support to learn—mathematics” (p. 12). This sentiment can be found in other documents; for example, the Holmes Group (1990), an influential consortium of deans of the nation's colleges of educa-

tion, claimed that “schools fail students if they deny them a passport to the mainstream; they also fail if students graduate with a sense that their own culture is worthless” (p. 22).

But what exactly is *equity*? And how is it incorporated into the mathematical education of teachers? Weissglass (1997) had defined it thus:

Equity is the ongoing process (not a product) of increasing our own and society's capacity and commitment to

- completely respect individuals as complex thinking and feeling humans with different social, cultural, gender, and class backgrounds and values, and
- provide the necessary resources to assist people in learning. This includes overcoming the effects of any mistreatment on the ability to learn, whether it be at the hands of individuals or institutions. (pp. 120–121)

Equity must be a goal of professional development because teachers teach in school settings marked by diversity, schools in which people of different racial, ethnic, language, and socioeconomic backgrounds coexist. “Professional development must attend to teachers' unique circumstances, particularly in those contexts where social justice and educational equity are most needed” (Quiroz & Secada, 2003, p. 104). The difficulty with dealing with everyone equitably is that we are many times unaware that we are not doing so. Or, as Delpit (1995) has so aptly stated: “We all interpret behaviors, information, and situations through our own cultural lenses; these lenses operate involuntarily, below the level of conscious awareness, making it seem that our own view is simply ‘the way it is’ (p. 151).

Reviews of research on teacher knowledge (Tate, 1994) have not focused on teachers' understanding of mathematics as it relates to “society's democratic and economic processes” (p. 56). Rather, “research on teachers' subject matter knowledge has been built on the assumption that mathematics is a neutral, objective, abstract, culture-free discipline” (p. 56). Yet according to Secada (1989), an equitable curriculum for teachers would include

real contexts that reflect the lived realities of people who are members of equity groups, and unless those contexts are as rich in the sorts of mathematics which can be drawn from them as from others, we are likely to stereotype mathematics as knowledge that belongs to a few privileged groups. (p. 49)

In Sleeter's (1997) review of literature of mathematics, multicultural education, and teaching, she noted many connections between multicultural education and mathematics. However, she found that professional development in multicultural education does not usually help teachers make this connection. But professional development on these issues

can help them grapple with some concepts and issues that are fundamental to multicultural mathematics, such as culturally relevant pedagogy, pedagogy for language differences, a deep-level understanding of culture, and the operation of institutional discrimination both inside and outside schools. (p. 693)

Professional developers and teachers have to avoid what Cahnmann and Remillard (2002) called the pitfall of relying on "token symbols of so-called culture that do not of themselves enhance students' learning opportunities. Embracing culturally contextualized practices is not simply a matter of incorporating ethnic symbols and artifacts into the mathematics classroom" (p. 199).

Tate (1994) reviewed other literature to make the point that teachers need to be prepared to effectively teach diverse populations. He described two multicultural approaches to teacher education. The ultimate goal for the first, the Teaching the Exceptional and Culturally Different (TECD) approach, is the assimilation of all students into the American mainstream so that they can compete in the job market. Teaching is viewed as a static process of passing on information; mathematics is viewed as a set of facts and procedures to be learned. In the second, the Education that is MultiCultural and Social Reconstructionist (EMCSR) approach, the preparation of students to critically analyze social conditions is advocated. Tate, borrowing from Dewey (1920) and Ernest (1991), described this approach as based on three interconnected sets of beliefs: "All knowledge is tentative and fallible; . . . education should prepare students for democracy; . . . and education should be built on the child's interests and experiences" (p. 61). Those using this approach address diversity by "promoting critical thinking by situating students in realistic problem contexts found in our multicultural society and having them analyze and solve problems" (p. 61). Teachers using this approach would encourage development of a community of learners and must "facilitate genuine discussion between the students and the teacher" (p. 61).

One particular area in which attention to equity sometimes fails is with the type of learning expected of students in diverse classrooms. Teachers of students from low-income families or second-language families

often expect too little of their students, teach mathematics procedurally, and focus on basics. Yet a large body of research indicates that when teachers of students in diverse classrooms focus on meaning-making and on mathematics content, students do learn more and better mathematics than when they are taught only procedures and skills (e.g., Boaler, 2002; Knapp et al., 1995; Ladson-Billings, 1994; Silver, Smith, & Nelson, 1995; Zeichner, 1992). This is the good news.

The bad news is that few teachers in these schools can teach in this manner. "Teaching in urban schools demands a different set of skills and abilities and requires people who themselves are committed to protecting learners and learning" (Ladson-Billings, 1999a, p. 233). Many teachers lack these skills. Further, many teachers in urban schools are the youngest and least skilled, ready to move on when they have achieved enough seniority. Sykes (1999) turned to teacher preparation for an answer: "One clear implication is that teachers in the future must be capable of far more sophisticated forms of practice than in any prior era. The stakes around their learning are now much higher" (p. 153).

Thus we have a problem—teachers of the future are unlikely to be capable of these forms of practice without major changes in our teacher preparation and professional development programs. Zeichner (1992) pointed out that demographic composition of the teaching corps is unlikely to change significantly; thus "the problem of educating teachers for diversity will continue to be one of educating white, monolingual, mostly female teacher education students" (p. 1) to teach an increasingly diverse population with backgrounds and life experiences very different from their own. According to Ladson-Billings (1999b), applicants to teacher preparation programs who are most apt to have the background and skills that would make them good urban teachers are too often screened out of programs. In short, the education community is only beginning to acknowledge and deal with the challenges presented by the NCTM Equity Principle.

Goal 6. Developing a Sense of Self As a Teacher of Mathematics

This particular goal, like the goal of developing a shared vision, can be a byproduct of sound, ongoing professional development. Dewey (1916/1944) has told us that the self is "something that is in continuous formation through choice of actions" (p. 351). A knowledge of self develops when teachers regularly engage "in reflection, in, on, and about their values, purposes, emotions, and relationships" (Day & Sachs, 2004, p. 9). Dewey.1916/1944, p. 351) This goal was

addressed in the NCTM *Professional Standards for Teaching Mathematics* (1991) as part of its standard on developing as a teacher of mathematics:

Being a teacher of mathematics means developing a sense of self [as] a teacher. Such an identity grows over time. It is built from many different experiences with teaching and learning. Further, it is reinforced by feedback from students that indicates they are learning mathematics, from colleagues who demonstrate professional respect and acceptance, and from a variety of external sources that demonstrate recognition of teaching as a valued profession. Confident teachers of mathematics exhibit flexibility and comfort with mathematical knowledge and commitment to their own professional development within the larger community of mathematics educators. (p. 161)

A sense of self as a confident teacher of mathematics is particularly difficult for elementary teachers to develop because of the mathematical anxiety many of them experience. Much of this anxiety is due to their lack of understanding of the mathematics they teach—in fact, many have had negative experiences with mathematics from the time they were in elementary school, in part because their own teachers were probably “math anxious” and did not understand well the mathematics they were teaching. Professional development that allows teachers to experience learning as an intellectual endeavor in which mathematics is explored together with others can lead them to have confidence in their abilities to develop understanding of other mathematical content (Nelson & Hammerman, 1996; Sowder, Philipp, Armstrong, & Schappelle, 1998). That is, developing the ability to explore mathematics intellectually can be generative. Strong, ongoing professional development can lead teachers to this point, and, further, teachers should be helped to understand that this type of learning can also be generative for their students. “It literally creates understanding in the mind of the thinker” (C. L. Thompson & Zeuli, 1999, p. 346).

Shedding anxiety about the teaching of mathematics can lead to a sense of empowerment. Research reviewed by Hargreaves (1995) showed that when teachers become empowered, they

work collaboratively rather than individually, take more risks (Little, 1987), commit to continuous rather than episodic improvement (Rosenholtz, 1989), tend to be more caring with students and colleagues alike (Nias, Southworth, & Yeomans, 1989; Taafaki, 1992), have stronger senses of teaching efficacy (Ashton & Webb, 1986), are more assertive in relation to external pressures and demands (A. Hargreaves, 1994), experience greater opportunities to learn and improve

from one another (Woods, 1990), and have access to more feedback (Lortie, 1975) and opportunities for reflection (Grimmett & Crehan, 1991. (p. 19)

In spite of this empowerment, teachers are not entirely ready to cast off old practices and beliefs. Like teachers in other fields, teachers of mathematics need to be convinced that new teaching practices work before they are willing to adopt them (Guskey, 1995). Contrary beliefs are sometimes held simultaneously. Only when the inconsistencies of certain beliefs, such as a view of teaching exclusively as presentation versus teaching as mutual exploring, are brought to the forefront can teachers address them. And, as is to be expected, new anxieties can result in such exposures.

Other evolving aspects of a teacher’s sense of self as a mathematics teacher are the deep emotions and passions that are “often at the very heart of teacher commitment and desire” (Hargreaves, 1995, p. 24). Teachers who share the vision of mathematics as a place in which students can become intellectually alive are likely to be passionate about teaching mathematics. They are also more likely to want to share this passion with their colleagues, to feel a commitment toward empowering others to teach principled mathematics. They come to think that their roles as mathematics teachers have a moral component. In a chapter on conceptualizing teacher development Cooney (2001) said: “I suggest that there remains a philosophical perspective that suggests that reform-oriented classroom is more consistent with the kind of society most of us would embrace. I do so under the assumption that the teaching of mathematics, or any subject for that matter, is ultimately a moral undertaking” (p. 11).

Coping with change is certain to have emotional aspects. Professional development providers need to recognize this aspect and help teachers address it. One tends to think of change as an outcome of professional development, but new teachers can also benefit from attending to the emotional aspect of teaching during their teacher preparation. For them, “learning how to handle the emotional responses . . . was as important as learning how to conduct tasks, meet new experiences, make judgments, build relationships, or assimilate new knowledge (Tickle, 1991, p. 320).

A sense of self as a mathematics teacher not only will differ among teachers but also will change for individual teachers while they progress in their profession. Day and Sachs (2004) described five career phases through which teachers pass:

1. Launching a career: initial commitment (easy or painful beginnings).

2. Stabilization: find commitment (consolidation, emancipation, integration into peer group).
3. New challenges, new concerns (experimentation, responsibility, consternation).
4. Reaching a professional plateau (sense of mortality, stop striving for promotion, enjoy or stagnate).
5. The final phase (increased concern with pupil learning and increasing pursuit of outside interests, disenchantment, contraction of professional activity and interest). (p. 11)

Finally, a sense of self can be thought about in terms of *identity formation*, which Wenger (1998) reminded readers is not only or always an individual process but is a process that can involve the communities in which teachers practice.

Identity formation is a lifelong process whose phases and rhythms change as the world changes. From this perspective, we need to think about education not merely in terms of an initial period of socialization into a culture, but more fundamentally in terms of rhythms by which communities and individuals continually renew themselves. Education thus becomes a mutual developmental process between communities and individuals, one that goes beyond mere socialization. It is an investment of a community in its own future, not as a reproduction of the past through cultural transmission, but as a formation of new identities that can take its history of learning forward. (pp. 263–264)

In the discussion of various types of professional development in the remainder of this chapter, the effects of the career stages of individuals are important to consider.

Summary and Discussion of Goals for Professional Development

The six goals stated and discussed here are not independent of one another. For example, if mathematics learning is to be useful in the classroom, it cannot be separated from the learning of pedagogical knowledge. Knowing mathematics and having the associated pedagogical knowledge to teach mathematics will lead teachers to develop a sense of self as a mathematics teacher. Learning about children's thinking about mathematics will affect teachers' visions of mathematics teaching (D. K. Cohen & Hill, 2001).

The goals listed here will not be achieved if teachers are simply told what to do or how to teach differently. Rather,

Teachers themselves must make the desired changes. To do so, they must acquire richer knowledge of subject matter, pedagogy, and subject-specific pedagogy; and they must come to hold new beliefs in these domains. Successful professional development efforts are those that help teachers to acquire or develop new ways of thinking about learning, learners, and subject matter, thus constructing a professional knowledge base that will enable them to teach students in more powerful and meaningful ways. (Borko & Putnam, 1995, p. 60)

ADDRESSING QUESTION 3: WHAT PRINCIPLES CAN BE USED TO GUIDE THE DESIGN OF EFFECTIVE PROFESSIONAL DEVELOPMENT?

The quality and effectiveness of professional development programs are, as noted in the introduction of this chapter, of interest not only to educators and policy makers but also to the public at large. Various educators have prepared lists of principles of effective professional development, some based on reviews of research and some based on years of success as a professional developer. Success is usually measured by teacher changes in knowledge of mathematics, beliefs about mathematics, and instructional practice and by increased student learning. These four aspects of success are somewhat easier to measure, but no more important, than other goals of professional development described earlier: a vision for mathematics teaching and learning, an understanding of the role of equity in the mathematics classroom, and a sense of self as a mathematical teacher.

Basing their synthesis of research on professional development that successfully contributed to student learning, Hawley and Valli (1999) identified eight characteristics that both embody this research and take into account recent "calls for action" (p. 137). A similar cut on the literature on successful professional development, that is, professional development that improves student learning, has been provided by Elmore (2002a). Elmore called his list a consensus view based on professional literature and academic research. Both syntheses apply more generally to professional development and both are based on literature about professional development that leads to student success. Clarke (1994) has compiled 10 important principles of mathematics professional development on the basis of both mathematics education research literature and his own years of providing professional development.

Table 5.1 A Comparison of Three Lists of Elements of Successful Professional Development

Hawley and Valli (1999, p. 138)	Elmore (2002a, p. 7)	Clarke (1994, p. 38)
Driven, fundamentally, by analyses of the differences between (a) goals and standards for student learning and (b) student performance.	Focuses on a well-articulated mission or purpose anchored in student learning of core disciplines and skills; derives from analysis of student learning of specific content in a specific setting.	Addresses issues of concern and interest, largely (but not exclusively) identified by the teachers themselves.
Involves learners (such as teachers) in the identification of their learning needs and, when possible, the development of the learning opportunity and the process to be used.	Focuses on specific issues of curriculum and pedagogy derived from research and exemplary practice; connected with specific issues of instruction and student learning of academic disciplines and skills in the context of actual classrooms.	Solicits teachers' conscious commitment to participate actively; involves a degree of choice for participants.
Is primarily school based and integral to school operations.	Involves active participation of school leaders and staff.	Involves groups of teachers from a number of schools; enlists support from administrators, students, parents, and broader community.
Provides learning opportunities that relate to individual needs but for the most part are organized around collaborative problem solving.	Develops, reinforces, and sustains group work using collaborative practice within schools and networks across schools.	Enables participating teachers to gain a substantial degree of ownership by their involvement in decision making and by being regarded as true partners in the change process.
Is continuous and ongoing, involving follow-up and support for further learning, including support from sources external to the school.	Sustains focus over time—projects an expectation of continuous improvement.	Recognizes that change is a gradual, difficult, and often painful process and affords opportunities for ongoing support. Recognizes that changes in teachers' beliefs about teaching and learning derive largely from classroom practice and, thus, such changes will follow the opportunity to validate, through observing positive student learning, information supplied by professional development programs.
Incorporates evaluation of multiple sources of information on outcomes for students and processes involved in implementing the lessons learned through professional development.	Uses assessment and evaluation for active monitoring of student learning and providing feedback on teacher learning and practice.	Encourages participants to set further goals for their professional growth.
Provides opportunities to develop a theoretical understanding of the knowledge and skills to be learned.	Embodies a clearly articulated theory or model of adult learning.	Allows time and opportunities to report successes and failures and to share the wisdom of practice.
Is integrated with a comprehensive change process that deals with the full range of impediments to and facilitators of student learning.		Recognizes and address the many impediments to teachers' growth at the individual, school, and district levels.
	Models effective practice delivered in schools and classrooms and is consistent with the message.	Models desired classroom approaches to project a clearer vision of the proposed changes.

As can be seen in Table 5.1, Hawley and Valli's (1999), Elmore's (2002a), and Clarke's (1994) lists of what makes professional development successful have considerable overlap. The first two included recognition of the importance of a theoretical base for professional development. Other researchers have agreed that however professional development is designed, it

will be ineffective unless it is grounded in sound theories of learning, particularly adult learning (Knapp, 2003; Knight, 2002; Mewborn, 2003).

Clarke's (1994) list lacks the emphasis on theory found in the other two lists, but his list addressed issues in more practical ways. All three lists provide guidance for those planning professional develop-

ment and could be used as a means for considering the success of a professional development program. (The fit of different elements of each list into a table is somewhat forced but allows for some comparisons.)

Primary commonalities include the role of determining the purpose of a professional development program, the role of teachers in deciding on foci (or, at least, a focus on issues relating to teachers' needs), the need to have support from other constituencies (e.g., administrators, peers, parents) to undertake changes in instruction, the important role of collaborative problem solving, the need for continuity over time, the necessity of modeling the type of instruction expected, and the need for assessment that provides teachers with feedback they need to grow. Hawley & Valli's (1999) list seems to have been developed with a focus on the needs of teachers (see, in particular, the second component); the second (Elmore, 2002a) was more focused on student learning. Both are the result of literature reviews, in contrast to Clarke's (1994) list, which was based on experience with long-term professional development of mathematics teachers in Australia and is more pragmatic in nature. Even with these differences, the three lists have a great deal in common.

Others who study professional development have made some of the points found in the first three lists and some new ones. For example, Borasi and Fonzi (2002), in their monograph on professional development prepared at the request of the National Science Foundation (NSF), reviewed successful professional development programs in mathematics education and, on the basis of their review, recommended that programs be sustained and intensive, be informed by how people learn best, focus on the critical activities of teaching and learning rather than on abstractions and generalities, offer a rich set of diverse experiences, and foster collaboration. Also, in a report of work in NSF Teacher Enhancement projects, Friel and Bright's (1997) recommendations for professional development in mathematics included many of the elements above; new in their list were a focus on the need to understand children's thinking, the link between curriculum and teachers' decision making, the need to develop teacher leadership, the consideration of ways to deal with the tensions of change experienced by teachers, and the important role of becoming mathematics learners themselves and developing an understanding of the mathematics they are expected to teach. Knapp's (2003) review of literature on professional development that "is likely to deepen teachers' knowledge and skill *as well as* prompting application of this knowledge and skill in the classroom" led to recommendations including opportunities that

(a) "challenge teachers intellectually and are built around powerful images of teaching and learning;" (b) engage teachers as active learners with "concrete images of what high-quality practice looks like;" (pp. 120–121), (c) provide reinforcement through long-term exposure to ideas and interactions with other teachers; and (d) address specific problems of practice. This set of recommendations reinforces the need for teachers to be challenged by powerful images of good practice.

Ball and Cohen's (1999) examination of the needs of teachers were discussed previously in this chapter.

Missing in all but Friel and Bright's (1997) and Ball and Cohen's (1999) recommendations is a focus on developing a strong content-knowledge base through professional development. Perhaps this need was assumed; also, content knowledge is viewed as less important in some areas (e.g., reading) than in other areas (e.g., mathematics). Wilson and Berne (1999), in their review of high-quality professional development programs in mathematics, found that providers in these programs "acknowledged that professional teaching knowledge might include, at the very least, knowledge of subject matter, of individual students, of cultural differences across groups of students, of learning, and of pedagogy" (p. 177). All the projects they reviewed involved communities of collegial learners who were redefining their teaching practice and teacher learning, leading to their reconceptualizing both teaching and professional development. Wilson and Berne also noted that teachers often undertake professional development with preconceived ideas of what they will find helpful and relevant and that these ideas undergo, at times, radical revision, a point to consider in the light of the recommendations of others (above) that teachers should identify their learning needs and participate in planning for their professional development.

Professional developers might do well to also consider, as did Hargreaves (1995) in a literature review of professional development programs, reasons some professional development is *not* effective: Teachers are likely to reject knowledge and skill requirements when (a) the requirements are imposed or encountered in the context of multiple, contradictory, and overwhelming innovations; (b) teachers (except for those selected for design teams) are excluded from the development; (c) professional development is packaged in off-site courses or one-shot workshops that are alien to the purposes and contexts of teachers' work; or (d) teachers experience them alone and are afraid of being criticized by colleagues or of being seen as elevating themselves on pedestals above them. Hargreaves concluded that

The reason that knowledge about how to improve teaching is often not well utilized by teachers is not just that it is bad knowledge (though sometimes it is), or even badly communicated and disseminated knowledge. Rather, it does not acknowledge or address the personal identities and moral purposes of teachers, nor the cultures and contexts in which they work. (p. 14)

Brown and Borko (1992) noted that because learning to teach requires the acquisition of both a knowledge base and observable teaching behavior, cognitive psychologists are drawn to research on teachers' knowledge and their development of teaching skills. Indeed, much of the literature on professional development is produced by cognitive psychologists. Borko and Putnam (1995, 1996; also Brown & Borko, 1992) reviewed the literature on professional development programs, much of it in mathematics education, and concluded that there was substantial evidence showing that teachers in these programs did experience significant changes in their instructional practices. These changes depended on having "opportunities for teachers to construct knowledge of subject matter and pedagogy in an environment that supports and encourages risk taking and reflection" (Borko & Putnam, 1995, p. 59). Thus, in their view teachers can be considered to be both the objects and agents of change.

The above reviews on what makes professional development effective or ineffective were based on examinations of many studies. A final research study discussed here is one in which effective professional development was investigated through a large-scale, empirical data-collection and analysis. Garet, Porter, Desimone, Birman, and Yoon (2001) examined a national probability sample of more than a thousand mathematics and science teachers who had participated in professional development sponsored by the Eisenhower Professional Development Program. The dimensions of the study included the forms of professional development activity; the duration; the collective participation of groups of teachers from the same department, school, or district; and the core features, including the content focus, the opportunities for active learning, and the degree to which the activities were coherent and aligned with teachers' goals, state standards, and assessments. The researchers found that sustained and intensive professional development was more likely to be effective, as reported by teachers, than was shorter professional development. Their results also indicated that professional development that was focused on academic subject matter (content), was based on providing teachers opportunities for "hands-on" work (active

learning), and was integrated into the daily life of the school (coherence) was more likely to produce enhanced knowledge and skills than programs lacking these features. Thus they confirmed the importance of professional development that is focused on mathematics content, and their results support literature that said that coherence and collective participation were related to improvements in teacher knowledge and practice.

In one way or another, these reviews of the literature (except the last one) on successful professional development used student learning as a criterion for success. But strengthening the connection between teacher learning and student learning is difficult because reform efforts do not always match the system in which the reform is to take place. Moreover, teacher professional development (TPD) that

pursues more ambitious learning goals, requiring, for example, conceptual understanding of subject matter or application of knowledge to novel problems and new situations of use, is unlikely to exert much influence even if the TPD itself is well designed and supported. State and district curriculum frameworks, instructional materials, tests, and not least, longstanding traditions will combine to subvert teachers' efforts to change instructional practice. . . . [These portrayals] yield a paradox. If TPD is to be successful, it must fit with the regularities in place, but if it fits, it is unlikely to exert much beneficial influence on teacher or student learning. (Sykes, 1999, p. 160)

Sykes (1999) did provide suggestions for ways to strengthen the teacher-student learning connection. First, this connection should be used as a criterion for the selection and design of teacher professional development. Planners must have a theory about how the learning opportunities can affect the eventual learning by students. Second, professional development should be embedded in the content of the student curriculum. Teachers need opportunities to develop a deeper understanding of the curriculum they teach; they need to develop ways of presenting the content to their students; and they need to understand how students learn the curriculum. Third, teachers need to better understand student learning and the multiple ways to examine that learning. Fourth, teachers need opportunities to fully understand new implementations, whether they be changes in curriculum, in testing, or in instructional practice. "Relatively few schoolwide innovations have been validated on the basis of their impact on student learning" (p. 168). And fifth, an evaluation process that has student learning as an outcome is needed to develop better accountability.

A SEGUE TO DISCUSSING THE ACQUISITION OF PROFESSIONAL KNOWLEDGE

Professional development is an umbrella term for many types of activities and settings. Loucks-Horsley and her colleagues (Loucks-Horsley, Love, Stiles, Mundry, & Hewson, 2003) listed several: (a) aligning and implementing curriculum including curriculum-replacement units and selecting instructional materials; (b) participating in collaborative structures such as professional networks, study groups, and partnerships with mathematicians in business, industry, and universities; (c) examining teaching and learning through collaboration and action research, case discussions, examination of student work and thinking, scoring of assessments, and lesson study; (d) immersion in inquiry in mathematics and into the world of mathematicians; (e) practice teaching, including coaching, demonstration lessons, and mentoring; and (f) participating in workshops, institutes, courses, and seminars that can include developing professional developers and using technology for professional development. My organization of professional development activities and settings in this chapter is different from the organization of Loucks-Horsley et al., but, overall, I discuss most of the types they listed.

In the following three sections (and subsections), I focus on various types of professional development in which teachers' knowledge needed for teaching well is considered important, under the premise that without well-prepared teachers, students will not achieve to their capacities. Because teacher knowledge can be acquired in many ways and because the literature on professional development is vast and diverse, I have organized this literature in a way that helps me discuss it and that readers can, I hope, follow. I found the types of teacher knowledge suggested by Cochran-Smith and Lytle (1999), *knowledge-for-practice*, *knowledge-in-practice*, and *knowledge-of-practice*, to be a useful organizational tool. These authors described *knowledge-for-practice* as acquired by learning from formal professional development programs and university coursework. The emphasis here is that this is shared knowledge, already known by others such as those who provide teacher education and professional development experiences. *Knowledge-in-practice* is the practical knowledge of teaching known by competent teachers "as it is embedded in practice and in teachers' reflections on practice" (p. 250). Here teaching is treated as a *craft*. Teachers acquire knowledge-in-practice when they deliberate on their practice, "consciously reflecting on the flow of classroom action and invention of knowledge in action in order to take note of new

situations, intentionally and introspectively examining those situations, and consciously enhancing and articulating what is tacit or implicit" (p. 268). This kind of learning is the "stuff" of communities of practice, of lesson study. Finally, when teachers use their own classroom and school sites to investigate learning, knowledge, and theory, they generate *knowledge-of-practice*; they are learning from practice. "Teachers learn when they generate local knowledge of practice by working within the contexts of inquiry communities to theorize and construct their work and to connect it to larger social, cultural, and political issues" (p. 250). This kind of knowledge is acquired when teachers practice inquiry of teaching, whether or not they undertake research in a more formal sense.

Certainly this method of categorizing the ways in which teachers acquire knowledge is not a perfect one. The primary disadvantage of using this organizational tool is that, in reality, the examples of each are not clear cut, and in many cases the examples I have selected to characterize a particular type of professional development could also be used to characterize other types of professional development. But the purpose of this categorization is to provide examples of the different types of professional development (many of which are also relevant to teacher preparation) rather than to fully describe each project that serves as an example.

ADDRESSING QUESTION 4: HOW DO TEACHERS LEARN WHAT THEY NEED TO KNOW TO TEACH MATHEMATICS? THAT IS, HOW DO THEY ACQUIRE KNOWLEDGE-FOR-PRACTICE?

The literature on formal professional development, that is, on the contexts and strategies that provide the knowledge needed for the practice of teaching mathematics, are considered here within four categories: approaches that focus on student thinking; approaches that focus on curriculum, whether student textbooks or curriculum designed expressly for professional development; approaches that focus on classroom activities and artifacts; and, finally, approaches that make use of formal coursework focused on the knowledge needed for teaching well. I next discuss each of these approaches and provide examples of professional development programs and projects within each of these categories. The projects cited are not, by any means, exhaustive of those that have been successful in providing teachers with the professional knowledge they need to teach mathematics well. Those that are

included have been the subject of research reports, although not even all of those can be discussed here. My purpose is rather to provide examples of an array of ways that each approach can be used.

A Focus on Student Thinking As an Approach to Teacher Learning

A review of research by staff members at the Eisenhower National Clearinghouse (2000) on examining student work and thinking led to suggestions for using this strategy productively. Although the strategy appears easy to apply, professional developers need to ensure that the tasks examined are appropriate and that student work varies in terms of the types of responses and the quality of explanations. Teachers' motivation for studying student work can vary from curiosity about disappointing student performance on standardized tests to the desire to develop a rubric for examining student work on particular problems (e.g., on a performance-based evaluation of learning), to the desire to design instruction that is based on what students know. Teachers can examine student work on their own, but having someone with better pedagogical knowledge of the mathematics being examined can frequently lead teachers to a deeper understanding of children's thinking and reasoning processes.

Little (2004) has described three complementary ways in which written student work can be used in professional development. First, it can be used as a resource for strengthening teacher knowledge, providing evidence of "what works" in the classroom, and developing the teachers' confidence in their own instructional practices. Second, student work can catalyze a spirit of inquiry and provide a basis for developing a professional community. Third, student work can provide evidence of student learning to be used for external accountability, thus assuring that instruction is aligned with standards.

It is precisely this constellation of purposes that makes "looking at student work" of particular interest in the landscape of teachers' continuing professional development (CPD). Embedded in the arguments for looking closely and collectively at student work, and embodied in the conception and design of specific approaches, these purposes reflect certain prevailing interests and emerging tensions in teachers' work and teacher development more generally. In this regard, "looking at student work" is a case emblematic of contemporary professional development in the USA. (p. 95)

Warfield (2001) found that elementary school teachers who have a research-based knowledge of children's thinking and deep understanding of the mathematics they teach "can extend what [they] learn about the mathematical thinking of their children and the ways in which they use what they learn to inform their practice" (p. 137). She described several ways this extension can take place. These teachers could

- (a) pose questions that go beyond asking children to describe their solution strategies to asking them to think more deeply about the mathematics underlying those strategies; (b) understand students' mathematical thinking that differs from what might be expected based on the research-based information on children's thinking; (c) critically examine that thinking to determine whether it is mathematically valid; and (d) use what they learn about their students' thinking to create tasks that enable students to extend their understanding. (p. 137)

Some examples of projects that use student thinking as a means of developing teacher knowledge are next discussed.

Example: Cognitively Guided Instruction (CGI)

This primary-grades program was briefly described in the Goals section as an example of meeting the goal of developing understanding of how students think and learn about mathematics. An extensive body of literature on the effectiveness of using student thinking to guide instructional decision making was cited. This body of research is based on the hypothesis that if teachers listen to children, understand their reasoning, and teach in a manner that reflects this knowledge, they can and will provide children with a better mathematics education than if they did not have this knowledge. The research began in the late 1980s and continues to generate new research studies. In the first publications of their work (e.g., Carpenter et al., 1988; Carpenter et al., 1989), the CGI researchers reported on their study of 40 first-grade teachers—20 randomly assigned to a treatment group and 20 to a control group. The treatment consisted of a month-long workshop during which the participants studied children's thinking about addition and subtraction problems. Earlier work by Carpenter and Moser (1983) provided a detailed analysis of this domain of addition and subtraction problems and was used to familiarize the treatment-group teachers with the processes children use to solve problems. The treatment consisted of a summer workshop followed by a year of teaching after the workshop. During the second year, the teachers were observed teaching, and treatment teachers were found to listen to their stu-

dents while the students were solving problems and to encourage children to solve problems significantly more than control teachers. The treatment teachers overcame their reluctance to place a greater emphasis on problem solving and found that their students achieved better than students of the control teachers.

In other reports (e.g., Carpenter & Fennema, 1992; Fennema, Carpenter, Franke, & Carey, 1992), the researchers provided detail on the teachers' use of knowledge of student thinking and the changes in teachers' beliefs and knowledge about children's abilities to solve problems. They undertook case studies of individual teachers to illustrate the CGI child-focused instruction and noted how two teachers who had access to the principled knowledge offered through CGI differed in how they used this knowledge in teaching. They studied the extent to which teachers in the workshop internalized the knowledge of children's thinking and then adapted instruction to meet student's needs, which varied from teacher to teacher.

In a longitudinal study (Fennema et al., 1996) of using children's mathematical thinking in primary classrooms with 21 CGI teachers, a pattern of increasingly greater gains was found for students who had been in CGI classrooms for more than one year. Overall, the gains in students' concepts and ability to solve problems were substantial. The descriptions of teacher gains and of these researchers' further study (Franke et al., 1997) of the self-sustaining, generative change that resulted from teachers' work with CGI are more appropriately discussed in later sections.

Examples: Teaching to the Big Ideas (TBI) and Developing Mathematical Ideas (DMI)

In the Teaching to the Big Ideas project, Schifter and colleagues (Schifter 1998; Schifter, Russell, & Bastable, 1999) examined the role of student thinking in elementary school together with exploration of mathematical content as avenues for teacher development, reflecting their belief that an analysis of student thinking provided a powerful site for continued mathematics learning. They acknowledged the difficulties inherent in moving from a presentation type of instructional practice to teaching as engaging students in mathematical discourse through understanding the student thinking that prompts students' questions and responses during classroom discussions of mathematics. "In a practice in which student thinking takes center stage, classroom practices become much harder to manage and much less predictable" (Schifter, 1998, p. 56). The elementary school teachers in the TBI project participated as mathematics learners in sessions with project staff. The lessons addressed the content of elementary school mathematics and were

directed toward mathematical issues that would challenge adult learners. The teachers in the classes struggled with the ideas. "Though most adults, including most elementary teachers, learned years before how to operate with fractions, few developed an articulated sense of the conceptual distinctions between whole numbers and fractions" (p. 78). They also dealt with these ideas in their own classrooms, which became resources for learning about children's thinking.

An important part of the work of the TBI seminar involved learning how to use children's perplexities to make such deep conceptual issues visible. The TBI teachers began their investigations into children's mathematical thinking by analyzing other teachers' students, studying videotapes of clinical interviews and classroom discourse as well as written materials illustrating student work. . . . However, the teaching practice TBI was working to help these teachers construct requires a facility that goes beyond an understanding of basic principles of children's mathematical thinking in general. Rather, it involves developing a new ear, one that is attuned to the mathematical ideas of one's own students. (pp. 78–79)

In addition to helping the participants work through mathematical topics by engaging in the mathematics and by examining student thinking, the providers also intended for teachers to develop a "disposition to inquiry. . . . [They] must come not only to expect, but to seek, situations in their own teaching in which they can view the mathematics in new ways, especially through the perspective that their students bring to the work" (Schifter, 1998, p. 84).

The TBI was a professional development and research project that led the authors to develop curriculum materials for teachers of Grades K–6—the Developing Mathematics Ideas (DMI) materials. The seven volumes of DMI, each of which focuses on a different mathematical topic, are being used at many sites for professional development seminars. The materials are heavily based on children's thinking. S. Cohen (2004) described this use of the materials:

During seminar meetings teachers analyze the mathematical thinking of the children depicted in the cases (both video and print), listening to the students for the logic of their thinking, the parts of their reasoning and their understanding that are strong, and the mathematical issues that the children are still working on. (p. 32)

Three aspects of DMI that, together, "help describe DMI's particular place within the new genre of professional development" (S. Cohen, 2004, p. 34) are "(1) solidity and complexity of the mathematics under study, (2) the concurrent examination of teach-

ers' and students' mathematics; and (3) the parallel between the seminar's pedagogy and elementary classroom pedagogy as envisioned by both national Standards and the DMI designers" (pp. 33–34).

S. Cohen (2004) reported on a research project she undertook to examine teachers' development of understanding both of mathematics and of children's thinking about mathematics over the course of a year of participation in DMI seminars. On the basis of teacher reports, she described the benefits to children:

Children began to experience themselves, the subject matter, and the classroom community differently. The teachers see increased student engagement in the classroom, and it shows up in four ways. (1) The students enjoy mathematics, (2) they feel like they can do it, (3) they go further, more deeply into the subject matter than the teachers had thought was possible, and (4) a new kind of community is built. (p. 245)

Cohen found that these changes, described by the teachers, paralleled the changes she and the other researchers had found among the teachers. She noted the supportiveness, agenda, and rigor of the seminar community. "Professional development [groups] such as these are vibrant, rigorous, respectful intellectual communities that support teacher learning in all of its complexity" (p. 293).

Example: Integrating Mathematics Assessment (IMA)

This IMA project (Gearhart & Saxe, 2004; Saxe, Gearhart, & Nasir, 2001) was also designed to help elementary school teachers understand children's mathematical thinking and learn to use the knowledge gained to guide children to a deeper understanding of the mathematics. The researchers worked with a group of teachers who were using the *Seeing Fractions* "replacement unit" that had been designed for California teachers for use in place of textbook pages on fractions. The teachers were participants in the researchers' Integrating Mathematics Assessment program. This program was built on the assumptions that (a) when children solve problems, they make use of what they already know to create opportunities for developing new understandings, and (b) for an instructional approach to be effective, it had to help children build new knowledge based on prior knowledge. Thus, a curriculum alone was not necessarily sufficient for constructing understandings; the role of the teacher was to support students' development. Ongoing assessment was a critical factor. The professional development with these teachers consisted of cycles of activity; during each cycle, the first focus was on the teachers' understanding of the mathematics

(with the teacher as learner), then on understanding children's mathematics (with the teacher as researcher), and then on implementation of the lesson (with the teacher as educator). Assessment was critical to the third phase, and several ways to assess children's understanding were practiced.

The researchers compared the students of the IMA teachers to both students of a second group of teachers who had participated in professional development not focused on children's understanding of mathematics and students of a third group of teachers who had not participated in professional development. The IMA students had better conceptual understanding than students from other groups and had as much or more procedural proficiency. The researchers concluded that the replacement curriculum was not sufficient for students' improved learning—teachers needed to build activities and assessments to move these students toward understanding. In this project, the curriculum-replacement unit was used in the professional development to help teachers understand the mathematics and to learn what to expect from students using that curriculum.

Summary

Providing opportunities for teachers to examine student work has been quite productive, as indicated by these projects. The major focus of each was teachers' coming to understand student thinking about mathematics at their grade levels so that their instructional practice might improve and student learning might be enhanced. Learning the mathematics of the grades taught and how students learned this mathematics was essential to the success of all projects. The teachers had opportunities to "practice" on their own students, and the students' reactions were powerful reinforcers of the value of teaching for understanding.

A Focus on Curriculum As an Approach to Teacher Learning

Many researchers and professional developers believe that school curriculum materials should be designed with both teachers and students in mind. For example, Schifter, Russell, and Bastable (1999) argued that "innovative materials can aid teachers in rethinking their mathematics programs; indeed, we are convinced that curriculum must be thought of as a vehicle supporting ongoing teacher development, especially with regard to teachers' mathematical understanding" (p. 31).

The authors of *Adding It Up* (NRC, 2001a) also suggested that instructional material, including textbooks, could assist teachers in understanding math-

emational concepts and student errors and in learning about effective pedagogy. Ball and Cohen (1996), too, spoke of the usefulness of curriculum materials, when integrated into professional development, in generating learning. The belief that student curriculum materials can be useful in a professional development environment is not new. Indeed, in the preface of the 1977 edition of *The Process of Education* (1960/1977), Bruner wrote, "A curriculum is more for teachers than it is for pupils. If it cannot change, move, perturb, inform teachers, it will have no effect on those whom they teach. It must first and foremost be a curriculum for teachers" (p. xv). According to Brown, Smith, and Stein (1996), research has shown that well-designed curricula can provide strong professional development for teachers of mathematics.

Many efforts over the past decade or so have been aimed at providing well-designed curricula for school mathematics. Soon after the publication of the *Curriculum and Evaluation Standards for School Mathematics* by NCTM in 1989, the National Science Foundation funded several school mathematics curriculum projects, for example, *Investigations in Number, Data, and Space* (Russell, Tierney, Mokros, & Economopoulos, 2004) at the elementary level; *Connected Mathematics* (Fey, Fitzgerald, Friel, Lappan, & Phillips, 2006) at the middle school level; and *Core-Plus Mathematics Project* (Coxford et al., 2003) at the secondary level. Each has spawned an industry of professional development workshops and conferences focused on helping teachers prepare to use the materials in their classrooms. The National Science Foundation has funded centers to study of the effects on the elementary curricula (ARC Center in Lexington, MA, <http://www.comap.com/elementary/projects/arctest/>), the middle school curricula (Show-Me Center at University of Missouri, <http://showmecenter.missouri.edu>), and the secondary curricula (COMPASS at Ithaca College <http://www.ithaca.edu/compass/>). Each center's website Home Page cites studies on the effectiveness of the various curricula, related primarily to student achievement but also to teacher learning.

In this section both children's textbooks and curricula designed for professional development are explored.

Example: Studying the Manner in Which Teachers Used Adopted Textbooks for Professional Development

Remillard and her colleagues (e.g., 1999, 2000; Remillard & Bryans, 2004; Remillard & Geist, 2002) have studied the effectiveness of using curriculum materials for professional development. In a 1999 study, Remillard described two levels of curriculum

development. The first is what curriculum writers do when they produce curriculum materials for use in the classroom. The second is what teachers do as they "alter, adapt, or translate textbook offerings to make them appropriate for their students" (p. 318). In this research report, Remillard reviewed literature on recent teacher development projects, concluded that teachers need to have well-designed curriculum guidance, and suggested that researchers need to be more in tune with the role textbooks play in teachers' larger curricular agendas. In a 2000 study, she addressed the question of whether curriculum materials can support teachers' learning. By observing the manner in which two 4th-grade teachers "constructed curriculum" in their classrooms, she analyzed the nature and contexts of their learning over an academic year when they used a newly adopted reform-oriented textbook. Using her data, she was able to consider the manner in which teachers *read* the textbook, with *reading* referring "to a constructive and dynamic process of meaning making through engaging written text, such as would be found in a textbook" (p. 335). She also found and interpreted the manner in which teachers read both students and the tasks with which the students engage. She found that substantial learning occurred during these three types of reading. The teachers reexamined their beliefs and understandings during this process and, in so doing, influenced the curriculum. Remillard (2000) concluded that new textbooks alone will not affect teaching unless teachers are provided support to learn to construct curriculum from text in good ways. "To promote productive use of curriculum materials, professional development opportunities need to foster teachers' reading and decision making [and] deepen and broaden their mathematical knowledge" (Remillard, 2000, p. 347). Even when professional development was provided, however, Remillard and Bryans (2004) found that the perspectives and orientations teachers bring to their teaching of a particular curriculum can greatly affect the *enacted* curriculum in their classrooms. But when curriculum is well designed, it can be useful in professional development for many teachers and can contribute to larger instructional agendas.

Curriculum developers have also recognized the need for appropriate professional development and have suggested that good curriculum materials,

when developed through careful, extended work with diverse students and teachers, when based on sound mathematics and on what we know about how people learn mathematics, are a tool that allows the teacher to do her best work with students. (Russell, 1997, p. 248)

Russell advocated using innovative curriculum materials with both preservice and in-service teachers and suggested that the curriculum can provide a forum for thinking about both mathematics content and students' mathematical thinking. When teachers can explore the depth of the mathematics and the changes in instruction called for in some of the new curriculum materials, such opportunities can lead to a commitment to the curriculum and to changes in teaching practices needed to be successful with the curriculum (Loucks-Horsley et al., 2003). Schoen, Cebulla, Finn, and Fi (2003) also found that professional development of secondary school teachers specifically aimed at preparing to teach the Core-Plus curriculum was important for teachers and led to higher student achievement.

A curriculum might exist not in traditional textbook format but rather as a computer microworld in which students can learn a narrow set of related concepts. Bowers and Doerr (2001) presented a microworld focused on concepts related to mathematical change to secondary preservice and in-service mathematics teachers in classes on two campuses. As students, the participants found value in conceptual explanations and agreed that "teaching with technology involves rethinking the format of activities and what counts as an acceptable explanation and solution" (p. 126) and they recognized "tradeoffs in having exploration before or after symbolization" (p. 126). As teachers, they learned "the value of building on students' incorrect explanations" (p. 130) and the "influence of hidden supports and constraints of technology on students' mathematical activities" (p. 131). The mathematical insights were all related to their increased conceptual understanding of mathematical change.

Another way in which curriculum has been extended is the deliberate development of instructional units to replace part of an adopted curriculum that falls short of expectations. In the IMA project described above, the curriculum-replacement unit was used in the professional development to help teachers understand the mathematics in the unit and to anticipate what to expect from students using that curriculum.

In professional development, the curriculum can sometimes be a set of materials or experiences designed to provide teachers with specific experiences and ways of learning. The two projects described next fit within this category.

Example: Introducing Mathematics Teachers to Inquiry

In this project, Borasi and Fonzi (1999) developed a set of materials that could be used to offer professional development or to provide experiences valu-

able to prospective teachers, primarily at the middle school level. The goals included helping teachers in "becoming mathematical problem solvers/inquirers, understanding the nature of mathematics and the 'big ideas' in mathematics, and developing mathematical confidence" (p. 45). Their package of materials, to be used by professional providers, incorporated an inquiry approach both to learning and to teaching. Their framework included a sequence of activities:

beginning to develop a community of learners, experiencing inquiry as learners, reflecting on the inquiry experience, developing images of inquiry-based mathematics classes, articulating the principles of an inquiry approach, scaffolding a first "experiences as a teacher" of an inquiry approach, moving towards teaching mathematics through inquiry throughout the curriculum, and looking back and ahead. (p. 63)

(Each of these components was explained in more detail than provided here and was accompanied with a list of suggested activities.) The supporting print and electronic media materials provided professional developers with a detailed plan for implementing their program.

Borasi and her colleagues (Borasi, Fonzi, Smith, & Rose, 1999) described using the three mathematical-inquiry units they had designed with two groups of teachers of heterogeneous backgrounds. Working on the instructional units, which provided scaffolding, the teachers experienced inquiry learning of mathematics that they could take back to their classrooms. The mathematics topics (e.g., tessellations) could be used at various grade levels. The teachers' backgrounds varied—some were teaching special education classes. Participants were positive about the experience, and many found that the help they had received through their own inquiry-based learning led them to want to teach other mathematics using this approach.

Borasi and Fonzi (2002) described their units and the ways they can be used. Their analysis of results of their program led them to claim that it was "successful in initiating a long-term process of rethinking one's pedagogical beliefs and practices, as well as in promoting some immediate instructional change" (pp. 196–197). They found that follow-up was a key to success.

Example: Relating Mathematical Knowledge and Teaching

In this 3-year research study Sowder and colleagues (Sowder, Philipp et al., 1998; Sowder & Schappelle, 1995) investigated the relation of middle grade teachers' understanding of mathematics at their grade levels to their teaching practices. The researchers and selected teachers met weekly for a year, then once or

twice a month during the second year. The 3-hour meetings focused on rational number, quantity, and proportional reasoning, with prolonged discussions about the concepts involved—an informal curriculum designed around teachers' content needs. Teachers were observed several times over the 2-year period. The results of the study were presented in terms of the effects of the seminars on teacher knowledge, case studies of the changes in teachers' instructional practices over the 2 years, and evidence of change in student learning during the second year. The teachers' conversations at the beginning of the first year were profoundly different from the conversations at the end of the first year and during the second year. Although the researchers chose initial topics and mathematical problems, the teachers controlled the agenda during the class period while they mulled over and persisted in trying to fully understand the mathematics. The positive effects of the seminars on the teachers were measured by pretests and posttests on content and by written comments from teachers.

The first year was difficult for these teachers. They struggled to understand while they came to realize that the mathematics they taught was not trivial; in fact, it was quite complex. But the teachers' changing understanding of content and their comfort level with content grew noticeably stronger, sometimes dramatically so, and influenced mathematics instruction in their classrooms. Student learning in their classes during the second year was measurably better than during the first year. The teachers changed their expectations of their students' capabilities when they learned that students could understand mathematics in ways they themselves had not understood before. Their views about the role of curriculum materials changed as they came to recognize the difficulty of teaching good mathematics without access to good instructional materials. The manner of classroom discourse also changed. Teachers began to probe for understanding and were delighted to find that students wanted to share their solutions with others. When asked why they now focused on questioning, even though questioning had never been a topic of discussion during the seminars, the teachers said that they had come to understand the importance of questioning when reflecting on the seminars and the manner in which the researchers' questioning them had helped them in developing their own mathematical understanding. The case studies of these teachers illustrated the need for support over an extended period of time for teachers to make meaningful change.

Example: The NCTM Academy

For approximately 4 years, The National Council of Teachers of Mathematics offered a series of

professional development workshops through their Academy for Professional Development. The primary focus was the content of the *NCTM Principles and Standards of School Mathematics* (2000), as portrayed in the *Navigations* series published by NCTM. Each year several long-weekend academies were offered in different parts of the country, for elementary, middle, and secondary school teachers. Instructors were carefully selected and teachers reported that they found the workshops useful for implementing Standards-based instruction in their classrooms. The NCTM Academy was quite different from the other projects in that each academy offered short, concentrated programs on one topic.

Summary

All the research projects described in this subsection focused on developing, through the use of curricular materials, teachers' understanding of the content the teachers taught. But the approaches were quite varied. Borasi and Fonzi (1999, 2002) introduced mathematics to teachers through a set of carefully selected problems that were conducive to an inquiry approach to learning mathematics and were materials that the teachers would use in their own classrooms. Remillard and her colleagues (e.g., 1999, 2000; Remillard & Bryans, 2004; Remillard & Geist, 2002) studied the use of actual elementary school curricula in professional development and concluded that teachers need to develop curricula from textbooks, and to do so, they had to understand the mathematics in the textbooks. Sowder et al. (1998) led teachers to explore mathematical content in ways that introduced them to the challenges of the mathematics they taught, using a somewhat Socratic approach. The mathematics of elementary school is challenging (Ma, 1999) and needs to be a major focus of professional development. The fundamental message of the first three projects is that teachers need assistance in coming to understand well the mathematics they teach but that when they become confident of their own abilities to learn, the need for outside assistance diminishes. The purpose of the NCTM Academies was somewhat different, but they were attractive to teachers because they provided opportunities to explore Standards-based instruction by focusing on limited content (for example, geometry in Grades 3–5) and were of short duration.

A Focus on Case Studies As an Approach to Teacher Learning

Studying cases as a way of developing knowledge within a profession has long been used in law and medicine. Shulman, in his 1985 American Educational Re-

search Association Presidential Address (published in 1986), made the case that education could also profit from the study of cases. He described cases as providing “knowledge of specific, well-documented, and richly described events” (p. 11). In 1992 he advocated that the study of cases be included in postgraduate programs much as they are in areas such as law and business. Cases are now frequently used in education to assist teachers in examining their practice and their students’ reasoning and understanding.

When cases are used in formal professional development, a facilitator usually guides the case discussion and can influence the focus, the progress, and the outcomes of the discussion. A facilitator can play the role of a devil’s advocate to encourage teachers to consider their claims in a careful fashion and to model the manner in which one can respectfully challenge the assumptions and ideas of others in the group (Barnett, 1998). Levin (1999) has noted other research that supported the important role of a facilitator in challenging misinterpretations and assuring that the discussions are fruitful, particularly with preservice teachers. She noted, “Recurring evidence that case discussions held in small groups followed by processing in the larger group may mean more active involvement and more attention by the participants than cases read but never discussed, or discussed without a facilitator” (p. 112).

In this section I consider how teachers can come to hear about, read about, and sometimes see how other teachers teach and the effects these experiences have on their professional development. Three types of cases are discussed: written cases, videocases, and multimedia cases.

Examples: Studying Written Cases of Classroom Events

Richert (1991) noted three components of cases that are important for teacher learning: They are descriptive in nature (even though they may contain analytical elements); they describe teaching practice; and they are “situated in a way that is significant for thinking of them as ‘texts’ for teacher learning (Brown, Collins, and Duguid, 1989), they are about particular students in a particular setting, at a particular time, for a particular purpose” (p. 116) and represent knowledge “as it is embedded in the practice of teachers’ work” (p. 121). Furthermore:

[Cases] present the dilemmas of teaching, the tradeoffs, the uncertainties. They capture teacher actions as they exist in the uncertain context where those actions occur. . . . [Cases] provide the potential for connecting the act of teaching with the cognitions and feelings that both motivate and explain that act. They offer a vehicle for making the tacit explicit. (p. 117)

Others have noted these and other qualities of case studies. Lee and Yarger (1996) found the merit in case studies to be that they provided rich detail and were explicit about contextual factors.

Cases provide opportunities for teachers to make judgments about what is worthwhile, to develop critical analysis of teaching and learning that is student centered, to analyze situations and weigh the effectiveness of various alternatives, to exchange perspectives with peers, to reflect on their own practices, and, in so doing, to extend their pedagogical content knowledge and become empowered in ways that lead to changes in beliefs about teaching (Barnett, 1991, 1998; Barnett & Friedman, 1997; Merseth, 1996, 2003). Teachers can examine what other teachers think and do, together with their rationales, then compare these findings with how they themselves think and what they do, together with their own rationales (Richert, 1991). In addition to the opportunities to explore teachers’ and students’ thinking and coming to understand mathematics in new ways, cases offer a safe environment for teacher inquiry because they provide teachers opportunities to explore and discuss instruction that is not their own or that of peers (Merseth, 2003).

Merseth (1996), in her review of research on cases and case methods, found that studying cases showed that skillful teachers do not operate “from a set of principles or theories, but rather build, through experience in contextualized situations, multiple strategies for practice” (p. 724). Studying cases, by helping teachers become more reflective about their own practices, allows them to develop this skill. Merseth (2003) has used case methods with secondary school mathematics teachers. Barnett, whose work has focused on case studies in mathematics with elementary school teachers, reviewed research that showed that case studies, as a method of professional development, can bring about changes in practices, beliefs, and awareness of student learning (Barnett & Friedman, 1997). Barnett’s work is based on the theoretical framework of Spiro and colleagues (Spiro, 1993) in which a case-based curriculum

that is effective in helping people acquire advanced knowledge in complex, context-dependent, and ill-structured domains . . . presents opportunities to reexamine a variety of ideas from different vantage points and in new contexts, thus offering the possibility of establishing multiple connections among cases or experiences that on the surface may seem dissimilar. (Barnett & Friedman, 1997, p. 391)

In later work, Barnett (1998) used aspects of cases and case discussions to illustrate four areas of development. The case studies seemed to be used suc-

cessfully to promote a deeper understanding of mathematics. First, transcripts and conversations with the teachers illustrated how the discussions help teachers “extend and resolve their own understandings of the mathematics” (p. 87). Second, teachers learned to “consider student thinking as a feedback loop that is used to continually reevaluate, revise, and learn from teaching experiences” (p. 88); Barnett believed that this process also influenced the ways in which teachers selected tasks for their students. Third, the case studies helped teachers to understand that concepts they thought were difficult were actually quite easy, and, just the opposite, mathematics they thought was easy was often more complex than they had realized. Fourth, the opportunities to critically examine alternative views helped them reconsider their own understandings and beliefs. Barnett concluded by noting two hypotheses: that case methods such as these

demonstrate the extraordinary value of collective inquiry and critical reflection [and that] an individual teacher’s decision to pursue an idea that has been carefully examined through the case discussion process is more likely to endure the tests of classroom practice. (p. 92)

Four pivotal experiences “help move teachers beyond their own perceptions and beliefs” while participating in case studies (Barnett & Friedman, 1997, pp. 389–390): First, the cases are written by other classroom teachers and provide opportunities for peers to collaborate in formulating approaches to the case. Second, the discussions model a way of teaching; the facilitator is a guide, not an expert. Third, the issues are usually unresolved and leave teachers with a desire to continue the inquiry. Fourth, teachers have opportunities to see classroom experiences through the eyes of students. Barnett and Friedman argued that “reflection about specific student thinking in the context of classroom situations is perhaps the most potent experience a teacher can have” (p. 390).

Stein, Smith, Henningsen, and Silver (2000) referred to cases in work such as that of Barnett and Merseeth as *dilemma-driven* because each case focuses on a pedagogical problem, thus helping teachers realize that teaching is inherently dilemma-driven and that they face trade-offs when selecting a course of action in the classroom. In contrast, Stein et al. called their own cases *paradigm* cases because they “embody certain principles or ideas related to the teaching and learning of mathematics” (p. 33); they are designed to help teachers understand tasks and the associated cognitive demands, and they require that teachers critically reflect on their own practice. They claimed

that teachers often do not know how to think about their own practices. By learning to critically examine and reflect on their practices, using the Mathematical Tasks Framework the authors designed, teachers can “bring meaning to the myriad actions and interactions that constitute classroom activity” (p. 34). This framework focused on student learning as the outcome of mathematical tasks (a) as they appear in curricular materials, (b) as they are set up by teachers, and (c) as they are implemented by students. Stein et al. used *cases*, that is, episodes of classroom practice, as a way to situate the abstract ideas of their Tasks Framework. They explained, “Once teachers begin to view our cases as *cases* of various patterns of task enactment, they can begin to reflect on their own practices through the lens of the cognitive demands of tasks and the Mathematical Tasks Framework” (p. 34). The Mathematical Tasks Framework and the use of tasks and cases themselves were developed as part of the Quantitative Understanding: Amplifying Student Achievement and Reasoning (QUASAR) project.

Other researchers have also used cases successfully as a form of professional development that supports teacher change. For example, Walen and Williams (2000) designed a research study that was based on their belief that “reflection on the practice of teaching is vital to professional growth [and] as a data source in the study of teaching” (p. 5). Their review of research on case methods led them to conclude that cases used in group settings can lead to a community of learners because of the power of shared inquiry. They collected data on two cases used during a 2-day teacher meeting with 115 teachers. Teachers began the first case by passively accepting their situation and ended by challenging the system. “As individuals became aware of similarities of their ideas, teachers expressed confidence in their ability to find ways to change the system. Teachers genuinely enjoyed the opportunity that this case provided for discussion” (p. 15). The authors situated their work within Merseeth’s (1996) conceptual framework—but it differs from other studies because of the manner in which Walen and Williams viewed the role of reflection:

We see reflection not as one more among many psychological processes, subject to development and training, but as a fundamental ontological fact of human existence: Reflection is a movement among modes of being, a movement from the unreflective ready-to-hand mode to the more abstract unready-to-hand or present-to-hand modes. For us, reflection is intimately tied to individual concerns and is always a movement from within a context of concern. Thus, we see reflection as bringing out for each teacher what is of individual concern, allowing those concerns to be

examined and made sense of within the context of their own lived experience.” (pp. 22–23)

Case-based studies have been reported in other countries. For example, constructing cases as part of school-based professional development in Taiwan helped teachers to become more aware of and help students who had difficulties learning mathematics (Lin, 2002).

Examples: Video Case Studies

The study of videos of teachers teaching is another form of professional development using cases. Video cases offer an opportunity for teachers to share a common experience of teaching practice and can act as a scaffold for developing theory from practice and applying theory to practice.

The potential that video has for making practice accessible in manageable sized chunks, and the opportunity video offers to pause, rewind, and replay practice is worthy of attention. It has the possibility of portraying the complexity of practice and at the same time enables a focus into the mass of action. (LeFevre, 2002, p. 13)

But developing video cases is expensive, and they are more difficult to use than written cases because of the need for appropriate and dependable technology, both drawbacks that are disappearing, however. Thus, more of these cases are expected in the future. The video-cases studies I discuss here are only a sample of such studies.

Classroom video of American, German, and Japanese teachers taped as part of the Third International Mathematics and Science Study (TIMSS) have been widely used in professional development because of their power to draw teachers into deep discussions of instructional practice (U.S. Department of Education, 1997). The videos clearly illustrate very different teaching styles unique to each country (Stigler & Hiebert, 1997); they provoke thoughtful analysis of the differences and of how these instructional differences might be related to achievement differences.

Seago and Mumme (2002; Mumme & Seago, 2002), in their Videocases for Mathematics Professional Development project (VCMPD) have developed videocases as tools for K–12 mathematics teacher development. They deliberately avoided creating the image that the teachers in the videos were exemplary teachers because they were not convinced that exemplary images would help teachers understand and reason about the practice unless they could relate to the class and teacher being observed. They believed that “video provides an opportunity to study an instance of

teaching that is emotionally distant enough to create a safe place to scrutinize practice carefully” (Seago & Mumme, 2002, p. 5). They hypothesized four ways in which teachers could learn from videocases:

1. “Videocases afford teacher opportunities to learn mathematics embedded in teaching” (p. 11); that is, teachers can learn *useable* mathematics within discussions of the videocases.
2. “The cross-analysis of multiple and varied video segments can provide teachers the opportunity to generalize mathematical and pedagogical ideas” (p. 13).
3. “Videocases afford the opportunity for teachers to develop a more complex view of teaching” (p. 15). Seeing real teaching that is “messy” and complex provides teachers opportunities for identifying and extracting what is relevant and important.
4. “Videocases can provide teachers opportunities to develop new norms of professional discourse and a more precise language of practice” (p. 17). Teachers need ways to critically analyze teaching practice and to make claims that they can support; they need to learn accepted ways of describing what they are seeing and thinking (e.g., terms such as *classroom culture* and *teacher directed* become part of their professional language).

The VCMPD videos of cases through which to study classrooms are not unique. The CGI and DMI projects (both discussed earlier) have video cases to be used in professional development; both of these sets of videos can also be useful in teacher preparation courses. The difficulty is not so much in locating good video to use in professional development; it is rather that the video should provide needed lessons for teachers and that the provider using the videos should have the background to help teachers profit from the viewing of video.

Some video case studies described here focused on the learning of preservice teachers whereas others focused on practicing teachers. The manner in which these two groups engaged in cases, whether video or not, differed. Richardson and Kille (1999) pointed out that preservice teachers “do not have the cognitive schema or practical knowledge that are acquired from teaching experience to be able to ask appropriate questions and reconstruct their own sense of theory and practice” (pp. 121–122). This difference carries over to many types of professional experiences educators provide for both groups. Sowder, Philipp

et al. (1998) concluded from their work that teachers contextualized, in terms of their own students, almost every mathematical task the group undertook, whereas preservice teachers were unable to think in terms of what children can and cannot do. Videocases do, however, provide a context through which preservice teachers can explore classroom situations.

Examples: Multimedia Case Studies

These case studies usually include some video but complement the video with other types of media. In an intensive year-long study, Lampert and Ball (1998) used videotapes of themselves teaching third graders (Ball) and fifth graders (Lampert) to create records of teacher practices that could communicate the complexity of teaching. The focus was on instruction over an entire school year rather than on isolated lessons or even units of mathematics. In addition to the videos of most lessons and some audiotapes of the remaining lessons, they collected field notes written by observers of every lesson, their own written reflections after each lesson taught, and records of student work. A curriculum was developed around these materials and then used in their teacher preparation program. The researchers' aim was to

put beginning teachers in a position where they have the capacity to knit the resources of theory with those of practice as they explore "big ideas" in knowing teaching. Just as place value and functions are big ideas in knowing mathematics, discourse, curriculum, the teacher's role, and classroom culture are all big ideas in knowing teaching. (p. 79)

Prospective teachers carried out investigations on a topic or question. An investigation might have included examples of student work, excerpts from the teacher's journal, and segments of video. Lampert and Ball analyzed 68 investigations; most analyses focused closely on the teaching. For example, one project focused on "What does Deborah do when a student gives the wrong answer?" (p. 96). The analyses led the investigators "to think that the multimedia environment offered significant potential as a site for a new pedagogy of teacher education" (p. 109) but that challenges remained to explore in future projects: The investigators did not know what the teacher education students actually took from their work in these classes, how to manage the interplay between big ideas and the contexts and particulars of practice, and what types of policies or programs would be most appropriate for supporting teacher education programs.

The difficulties in finding a proper field placement for preservice teachers and the challenges of trying to use student thinking in instructional practice

led Masingila and Doerr (2002) to design multimedia case studies that could be used to explore "the complexities of teaching and the complexities of teaching *about* teaching" (p. 238). They used a case study of a middle-school teacher teaching a four-day sequence of lessons on the use of data; the case included a video overview of the school setting, the teacher's lesson plans, video of the class lessons, students' written work, the teacher's reflections and anticipations on each lesson in a video journal, and the transcripts of all the included video. They found that this rich video source served as a site in which the preservice teachers could investigate, analyze, and reflect on the practice of the teacher and the effects of her instruction on student learning.

Van Es and Sherin (2002), building on multimedia case studies, developed and successfully used a software tool, Video Analysis Support Tool (VAST), to support secondary intern teachers in developing abilities to "notice and interpret aspects of classroom practice that are important to reform pedagogy" (p. 8). However, they did not investigate how the participants' teaching might have been affected by the experience.

The possibility of interactive media increases the usefulness of technology in teacher development. Sullivan and Mousley (2001) suggested that interactive media can be a tool used to develop both "student teachers' and teachers' understandings of the complexities of mathematics classrooms as well as about the many roles that they can play. It allows moments that have the potential for active classroom decision-making to be 'frozen,' discussed, and explored" (p. 162).

Summary

Case-study methods include the study of written cases, the study of video cases, and the study of multimedia. The study of classroom events and activities is formal in the sense that the events and activities studied are selected by a facilitator who then guides discussion by teachers who have read about the events or watched videos of the events. At times, teachers' writings about events in their own classrooms are shared with the group, and usually the discussion is still coordinated by a facilitator. Using case studies such as these, teachers can talk about instruction in a way that is nonthreatening and rewarding. Cases also preserve the complexity of teaching while providing opportunities to examine it. The mathematics inherent in these events provides opportunities to increase content knowledge. During discussions of pedagogy teachers examine the practices of others and reflect on the cases and take away lessons, both mathemati-

cal and pedagogical, that they can apply to their own instruction.

Formal Course Work

Many teachers seek out professional development opportunities in continued university coursework, such as master's degree programs, certificate programs, or focused courses on mathematical topics they feel unprepared to teach. Through this type of professional development, teachers earn university credit toward credential programs, fulfill an obligation for continuing education, obtain higher salaries, and participate in structured learning experiences that are otherwise unavailable to them. At the elementary level, degree programs are sometimes offered for cohort groups, thus allowing teachers to form communities of support. But frequently teachers, particularly secondary teachers, are not in cohorts and have classes with, for example, master's-degree students who are from many school sites or who are not even educators but rather students needing mathematics for some other purpose. Teachers in such situations are less likely than those in cohort groups to form communities of support.

Courses offered in programs for teachers should be based on sound principles of professional development (Loucks-Horsley et al., 2003), but often they are not. For example, graduate-level courses in mathematics are often taught in a lecture format only. Some institutions offer teachers separate courses that focus on developing pedagogical knowledge together with content knowledge. The disadvantage of separating teachers from other mathematics students, particularly at the secondary level, is that some mathematicians believe that the mathematics taught to prospective or practicing teachers is less rigorous than the mathematics in regularly offered courses and are more likely to consider these students to be second-class citizens in the department.

Acquiring Knowledge-for-Practice: Discussion

Acquiring *knowledge-for-practice*, that is, knowledge that belongs to someone else who shares it with teachers, is a common form of professional development and is what most people think of when the term *professional development* is used. Professional development provided by school districts or funding agencies is usually offered in one or more of the forms discussed here.

As noted in the introduction to this section, the categorization scheme (using student thinking as an approach to teacher learning, using curriculum

materials as an approach to teacher learning, using classroom activities and artifacts as an approach to teacher learning, and formal courses) used to organize the many projects and studies that fit under *acquiring knowledge-for-practice* can lead to a false implication that any particular project was narrowly focused in terms of its delivery of professional development. For example, the study of written cases of classroom events is placed in the subsection on using classroom activities and artifacts, but the study of written cases is bound to focus on the student thinking that is exemplified in the case. Thus this work could have been described as using student thinking as an approach to teacher learning.

Some Common Features

There were commonalities across projects; two in particular deserve attention. First, all the professional development projects described here provided teachers with opportunities to themselves learn mathematics. For example, in the DMI project (Schifter 1998; Schifter, Russell, & Bastable, 1999) the learning of mathematics by the teachers was overt and taken seriously. And in the IMA project (Gearhart & Saxe, 2004; Saxe, Gearhart, & Nasir, 2001) the first focus was on the teachers' understanding of the mathematics, with the teacher as learner. In fact, one might question the success of any mathematics professional development in which the teachers do not themselves learn mathematics, whether or not their mathematics learning is a stated goal.

A second commonality was that all the projects described in this section fostered an *inquiry approach* to learning and teaching; that is, the providers emphasized teachers' "engagement in problem-solving and theory-building about important situations and concepts" (Lloyd, 2002, p. 1). For example, Borasi and Fonzi (1999), in their *Introducing Math Teachers to Inquiry* project, consciously set out to provide teachers with opportunities not only to learn mathematics but to do so by experiencing learning themselves. The teachers were placed in situations in which they worked through the instructional units they would be using, an experience that required them to adopt an inquiry stance toward learning, and eventually, to teaching. As with the opportunities to learn mathematics, one might question whether professional development that does not foster inquiry as a necessary component of mathematics learning could be considered successful.

Mention of such commonalities serves as a reminder that projects selected as examples of a particular approach could have been selected differently, because most projects used more than one approach.

Thus the examples should not be construed as being limited to the particular focus under which they are placed. The placement here is simply one means of looking at how different approaches were realized in different projects and studies.

**ADDRESSING QUESTION 5:
HOW DO TEACHERS LEARN FROM THEIR
PROFESSIONAL COMMUNITIES ABOUT
TEACHING MATHEMATICS? THAT IS, HOW DO
THEY ACQUIRE KNOWLEDGE-IN-PRACTICE?**

Recall that the second category of knowledge described by Cochran-Smith and Lytle (1999) was *knowledge-in-practice*, that is, the practical knowledge of teaching “embedded in practice and in teachers’ reflections on practice” (p. 250). Lesson study (e.g., Lewis, 2000), which could be considered as a special type of case study, is discussed here as an example of learning-in-practice because in lesson study teachers deliberate on the practices they observe with others. Communities of practice and professional development schools also provide settings for learning from practice. Learning often takes place in settings in which teachers join with other teachers—making learning a communal process. These settings can be thought of as sites of practice that “seem promising as contexts through which teachers can learn mathematics, in particular the mathematics it takes to teach well” (Ferrini-Mundy, 2001, p. 125).

Professional Communities

Many professional communities into which teachers might be drawn to participate do not necessarily use the strategies described in the previous section, but teachers profit from them professionally. Many of these communities are informal—a teacher might meet occasionally with other teachers who are teaching the same grade level within their school to develop an instructional plan that meets state standards. Sometimes pinpointing exactly what teachers are learning is difficult—at times teachers exchange stories that might seem like gossip but could be regarded as “learning about the moral principles that guide one another’s work” (Hargreaves, 1995, p. 22).

The professional communities in which teachers seek membership are often outside their school districts. We have much yet to learn about the effects of participation in professional associations such as state and national organizations (Little, 1993), although we know that effects are real. For many teachers, attend-

ing a national NCTM conference, for example, is a heady experience. These and other teacher members subscribe to journals and website offerings, purchase books and instructional materials, and attend conferences and professional academies of NCTM. State organizations of mathematics teachers often have strong memberships and offer conferences that are well received by teachers. Thus, teachers do participate in and take advantage of opportunities offered by professional societies, but this type of professional activity is nebulous and thus its effect on teachers is more difficult to determine.

A professional community shares a common base of specialized knowledge, and that specialized knowledge in a community of teachers is of the types discussed in earlier sections of this chapter—content and pedagogical knowledge and understanding how students think. The formation of teacher communities is inherently more difficult than community formation in other professions because participation is voluntary: Teachers most often experience teaching as individuals behind closed doors and do not feel the need for outside assistance. Also, the work of these communities often takes place primarily outside the workday and thus are seen by some as taking away from time with family or other personal pursuits.

Yet many teachers feel the need to participate in a professional community; they seek to find or establish such a community because, instead of being extraneous to their lives within a school, “constructing and participating in such professional communities (Little & McLaughlin, 1993) in schools is itself a vibrant form of teacher and school development [in which the] members are constantly searching for ways to improve practice” (Hargreaves, 1995, p. 19).

In a review of literature on teacher communities, Grossman, Wineburg, and Woolworth (2001) found that “a key rationale for teacher community is that it provides an ongoing venue for teacher learning” (p. 947). Although teacher learning is an important outcome, it is not the only incentive for teachers to join together in communities of practice. Wenger (1998) argued that “issues of education should be addressed first and foremost in terms of identities and modes of belonging, and only secondarily in terms of skills and information” (p. 263). Members of a true community share a sense of purpose, usually with an understanding that they are responsible to one another for achieving their goals; they coordinate their efforts to assure student learning, learn together to improve their practice, and share responsibilities for decision making about matters pertaining to their group (Secada & Adajian, 1997). Lord (1994) has listed seven other characteristics that help define professional

communities of teachers; some overlap with those of Secada and Adajian:

1. Teacher ownership or, at the very least, increased partnership for teachers in decisions regarding professional development activities.
2. A collective commitment to acquiring and using new knowledge in the subject areas, especially knowledge that could be characterized as 'cutting edge.'
3. A reliable connection to resource-rich institutions, organizations, or associations independent of the school or school district, e.g., university education, liberal arts, and science departments, as well as libraries, museums, theaters, businesses and industries, and civic agencies.
4. Intensive, and in some cases long-term, professional relationships among participants.
5. A perspective on the profession of teaching that extends beyond the four walls of the school and beyond the duration of individual teachers' careers.
6. A greater commitment to 'lateral accountability' within the teaching profession, i.e., the critical review of teachers' practices by other teachers.
7. High levels of teacher involvement in the reform of system wide structures. (p. 199)

Lord (1994) advocated working with communities beyond the school because they provide access to intellectual resources that often do not exist in a small community at the school level, even though external communities commonly "operate at the margins of local school district life" (p. 201). Some teachers seek out communities in which they can find these intellectual resources and establish professional identities. Philipp et al. (1994) found that a common element in the lives of four extraordinary teachers whom they studied was long-term, active participation in professional communities outside their schools, which provided them with identities as professional mathematics teachers.

From another viewpoint, Grossman et al. (2001) suggested that summer workshops and professional days away from the school are often viewed as activities that take place during free time rather than as part of an ongoing professional life. They argued that professional communities located in the school offer "the possibility of individual transformation as well as the transformation of the social settings in which indi-

viduals work" (p. 948), noting that the expectation of a teacher's teaching for 9 months then learning for 2 weeks in the summer is like "having a marathoner train all week long and eat only on weekends" (p. 993). But the ways in which professional communities within a school can affect the instructional process should not be underestimated—they can have a profound effect on classroom practices and student learning.

A community of practice is considered by some to be more than the professional communities just described. Communities of practice "create, expand, and exchange knowledge about their practice" (Gallucci, 2003, p. 15). They vary in terms of their strength and their openness to engage with new ideas about their practices. These variations can "affect teachers' interpretations of standards-based reform policies, their engagement with learning opportunities, and, subsequently, the kinds of changes they make in classroom practice" (p. 14). They are more likely to remain intact if the membership is stable and if the members think that they are profiting professionally by their participation.

Mathematics education researchers have, in recent years, begun to study professional communities of teachers and communities of practice in mathematics. Some examples follow.

Example: Communities of Practice among Mathematics Specialists

Nickerson and Moriarty (2005) worked with teachers in eight low-performing, urban elementary schools in which the district had placed mathematics specialists who had 60–90 minutes each day of professional development time together. The teachers were all taking mathematics and pedagogy classes to earn a mathematics-specialist certificate. In some schools, teachers took advantage of their time together to develop a community of practice within their school. They developed a shared sense of purpose, and their discussions focused on how to provide students the best opportunities possible to learn mathematics. But at other schools, the teachers used the time for grading, individual lesson planning, and discussion of their students' shortcomings. Where teachers formed a community of practice, they were more apt to have good relationships with other teachers in the school and with the principal; they respected and accessed the particular knowledge of one another; their mathematical content knowledge was strong enough that they felt comfortable in their classrooms; and they were familiar with the culture and language of their students.

The next two examples provide instances in which researchers have taken theoretical stances that helped them study communities of teachers in new ways.

Example: QUASAR

The QUASAR project researchers set out to explore the possibility of changing student achievement in low-performing middle schools at several locations around the United States. The project was focused on providing assistance to the staff of the middle schools and then evaluating the effects, using the site as the unit of analysis of change. Instead of focusing on one teacher at a time, the researchers (Stein & Brown, 1997) studied the “process that takes advantage of the synergy, support, and motivation supplied when a ‘critical mass’ of teachers undertakes reform for all students in a given school” (p. 156). By changing the focus of attention from the individual to the group, they defined *learning* at each site by a transformation in the ways teachers participated in the collaborative activities and in the community that was being formed.

In a QUASAR study of teachers in one school, Stein and Brown (1997) used the *framework of change* to define, as well as measure, *teacher learning* as movement from the peripheral to a fuller form of participation. The researchers visited the school a number of times over several years and so had to consider changes in teaching staff along with other data collected. They decided to examine teachers’ participation by measuring *learning* as breadth of participation in all the activities in which members participated. That is, a long-time staff member was expected to participate in more activities than a newcomer, and each teacher was expected to increase his or her range of participation over the course of the project—thus both breadth and depth of activity were measured. These measures proved to be useful indices of participation and, thus, of learning.

This learning encompassed the gradual appropriation of a complex set of knowledge and skills related to valued classroom instructional practices as well as professional work associated with building, sustaining, and explaining a reform program of mathematics instruction. Moreover, teachers were developing the attitudes, values, and orientations that underlie the central tenets of the reform movement. The learning that occurred did so without the teachers being explicitly taught in any sort of structured manner, but rather through their ongoing exposure to old-timers and the work practices of the community.” (p. 170)

The teachers were motivated to work collaboratively on projects they valued. Each teacher’s contribution depended on the level of expertise of that teacher, and all were able to move forward and to learn.

In a second study, in a different school, the researchers (Stein & Brown, 1997) found that the

framework of the study had to be altered because reform goals and joint productive activity had not yet taken hold. They turned to work by Tharp and Gallimore (1988), for whom *learning* was viewed as assisted performance, which “within the social organization of schools helps to explain the organizational and interpersonal factors necessary for teacher learning to occur and the obstacles that must be overcome in most schools in order for reform communities to become established” (Stein & Brown, 1997, p. 171). *Learning*, though again viewed as a process of transformation of participation, was defined in terms of the *zone of proximal development* (ZPD, a term used by Vygotsky, 1978), and the learning of teachers was studied while these teachers moved from assisted performance to unassisted performance.

The framework Tharp and Gallimore (1988) used had four stages. In the first stage, performance was assisted by others; in the second, performance was assisted by oneself; in the third, performance was independent and automated; in the fourth, a stage of review and revision, the first three stages were revisited. The researchers examined lines of influence, which they called *chains of assessment*, and the manner in which they became *chains of assistance*. They were led to study the larger social system in which the teachers worked. Principals arranged for expert and experienced people to help the teachers and worked alongside the teachers during these sessions.

In the beginning of this study, resource partners were, together with administrators, sources of assistance. (The school had committed to using resource partners as part of the agreement with the QUASAR project.) The district, however, held to its schedule of testing, and consequently the district administrators, the principal, and teachers at this middle school had conflicting goals. Also, the common planning time funded by QUASAR was not used productively. Thus, little progress was made during the first two years. In the third and fourth years, however, the district provided more leeway to the school mathematics teachers, and they progressed toward the QUASAR goals of reformed teaching of mathematics. The common planning time was used as an “effective activity setting for teachers learning through assisted performance” (Stein & Brown, 1997, p. 184). The researchers began to see movement through the ZPD while teachers worked toward unassisted performance.

Stein and Brown (1997) noted that their two studies were sociocultural in nature and “channeled attention away from the cognitive attributes and instructional practices of individual teachers toward the collaborative interactions that occur as teachers (and others) attempt to develop and improve their school’s

mathematics instructional programs” (p. 185). Yet this channeling of attention played out differently in the two schools:

In the communities-of-practice framework, this transformation of participation is characterized as movement from *peripheral to fuller* forms of participation in the overall work activities of the community. In the framework advocated by Tharp and Gallimore, the transformation is characterized as movement from *assisted to unassisted* performance in specific activity settings. (p. 185)

In another study, Stein, Smith, and Silver (1999) provided detailed examples of how professional development evolved over time at two QUASAR sites and the manner in which, at the first site, the teachers took from the university partners the lead in designing professional development. The evolution of the work at this site led teachers to form a community of teachers with a shared vision of the mathematical competence they wanted their students to attain. These teachers decided that they no longer needed the expertise provided by the outside developers. Apparently because these two outside providers never taught in or even observed their classes, the teachers concluded that the outsiders did not really understand the work of the teachers. A lesson to be learned from this experience is that when mathematicians become involved in providing professional development for teachers, it is imperative that they visit classrooms and that they do so as learners. Only then will they understand the constraints under which teachers work, the teachers’ students, and the type of help teachers need.

Example: Situating Work in Instructional Practices

To study teachers’ instructional practices, Cobb, McClain, Lamberg, and Dean (2003) also situated their work in the instructional setting. The authors developed “an interpretive perspective that seeks to situate teachers’ instructional practices with respect to the affordances and constraints of the schools and districts in which they work” (p. 13). They noted that most research on teachers has focused, on the one hand, on the role of professional development that leads to teachers’ reorganizations of their teaching practices and, on the other hand and independently of the first, around the structural and organizational features of schools. They viewed the school in which they worked as a “lived organization” consisting of interconnected communities of practice, and they developed what they called an *analytic* approach to research that they have used to study teaching as a distributed activity and to situate teachers’ instructional practices “within the institutional settings of the schools and

school districts in which they work” (p. 13). The notion of interconnected communities of practice led them to distinguish among three types of interconnections: (a) *boundary encounters* occurred when the teachers’ community joined in activities with another community (e.g., the district mathematics leaders); (b) interaction with *brokers* who could “bridge activities of different communities by facilitating the translation, coordination, and alignment of perspectives and meanings (Wenger 1998)” (p. 19); and (c) *boundary objects*, interconnections among communities of practice based on *reification* (a term also attributed to Wenger) that “play a significant role in enabling members of different communities to coordinate their activities even when they are used differently and have different meanings” (p. 19). They used the example of a district pacing guide, used by the teachers, as a boundary object.

Three general aspects of the Cobb et al. (2003) analytic approach, taken together, provide a general method of understanding specific situations and thus make a useful way to study teachers. First, using their approach provides a way to account for different interpretations held by different communities to “acknowledge and account for the frustrations and antagonism that they experience” (p. 21). Second, people’s actions, as situated within a school “as a lived organization, constitute resources that might enable members of particular communities to move beyond viewing members of other communities merely as impediments to their agendas” (p. 21). Third, this type of analysis can “support the formulation of strategies for institutional change that involve the creation of new tools as prospective boundary objects as well as the orchestration of boundary encounters and the development of brokers” (p. 21). Together, these three aspects constituted a “*general* method for understanding the specific settings in which particular groups of teachers work” (p. 21) and allowed the research group to more effectively collaborate with teachers.

In the next part of this section I turn to another type of professional community connecting schools and universities.

Professional Development Schools (PDSs)

Some school sites link with universities to form professional communities. PDSs are an outgrowth of the work of the Holmes Group, a consortium of research universities dedicated to improving teacher education. The Holmes Group prepared three sets of goals: In 1986, they published goals for tomorrow’s teachers; in 1990, goals for tomorrow’s schools; and in 1995, goals for tomorrow’s schools of education.

One goal stated in the first report was “to connect our own institutions to schools” (Holmes Group, 1986, p. 4). The second report introduced Professional Development Schools to provide “continuing learning by teachers, teacher educators, and administrators” (Holmes Group, 1990, p. 7) with a major commitment of “overcoming the educational and social barriers raised by an unequal society” (p. 7). A PDS would “devise for itself a different kind of organizational structure, supported over time by enduring alliances of all the institutions with a stake in better professional preparation for school faculty” (p. 7). In the third report the Holmes Group called for connecting “professional schools of education with professionals directly responsible for elementary and secondary education at local, state, regional, and national levels to coalesce around higher standards” (Holmes Group, 1995, p. 20) . . . so that a PDS would function “as a place where prospective and practicing educators from the school and the university immerse themselves in a sea of inquiry in pursuit of ever more effective learning” (p. 20). By 1999, some 250 professional development schools were located across the United States. (Cochran-Smith & Lytle, 1999).

Lanier (1994) referred to PDSs as places of “ongoing invention and discovery; places where school and university faculty together carry on the applied study and demonstration of the good practice and policy the profession needs to improve learning for young students and prospective educators” (p. ix). These schools provide opportunities for school faculty and university faculty to join together in research and in rethinking instructional practice (Darling-Hammond, 1994a). PDSs, from the beginning, have had equity as a central focus (Valli, Cooper, & Frankes, 1997). They are viewed as places that produce professionals. Because “professionalism starts from the proposition that thoughtful and ethical use of knowledge must inform practice,” PDSs, by focusing on “knowledge and expertise, ethical commitment to clients, and responsibility for setting standards” (Darling-Hammond, 1994a, p. 4), are professionalizing the field of teaching.

The critical connection between teacher education and professional development, beginning with teacher induction, has been recognized by school districts, universities, and states, and PDSs are recognized as a formal way of acknowledging and addressing this need (NRC, 2001b). PDSs have been established at a large number of sites, and the movement has gained support from the American Federation of Teachers, the National Education Association, and the American Association of Colleges for Teacher Education (Book, 1996; Darling-Hammond, 1994b). In fact, Maryland state law requires that teacher education include a

one-year internship in a PDS. The NRC (2001b) study of PDSs was a literature review in which more than 600 PDS sites were identified. By far the majority of PDSs studied reported having teachers and university faculty working together to improve teacher education. Teachers who participated in a PDS showed greater understanding of diversity and the needs of children than teachers who had not had this experience.

Mathematics learning and teaching are rarely singled out in descriptions of PDSs. Exceptions are the PDS described by Whitford (1994), in which the increased cooperation led teachers to reorganize their instruction around the NCTM *Standards*, and the PDS described by Valli, Cooper, and Frankes (1997), in which a university professor guided a group of teachers in implementing the NCTM *Standards*. PDSs, because of their comprehensive natures, often include mathematics, a core subject taught in K–12 schools. The NRC Committee on Science and Mathematics Teacher Preparation (NRC, 2001b) considered PDSs to be an exemplary approach to teacher education, places where “teachers, students, and university faculty create new knowledge and experiment with, evaluate, and revise practices” (p. 76).

Establishing a PDS is not an easy task. Neither a university nor a school can begin this type of collaboration unless it builds on prior work together. “Building coherence requires building shared understandings, which takes dialogue, time, and compromise” (Arends & Winitzky, 1996, p. 551). Financing a PDS is also problematic. Schools rarely allocate much funding for professional development, and universities are often unwilling to expand budgets for teacher education to include the type of collaboration necessary for a PDS partnership to exist. A third problem is the existing reward structure for university faculty. Work in a PDS is often viewed as service, which counts little toward promotion, tenure, or merit pay. “For PDSs to survive, universities will have to take seriously this kind of research and count it in the reward structure for faculty” (Book, 1996, p. 206). Only when issues of how collaboration is undertaken, when all those involved benefit, when trust and respect exist, when the focus is clear to all involved, and when the agenda is manageable can PDSs hope to succeed as a means of furthering the professionalization of teaching.

Lesson Study

Hiebert, Gallimore, and Stigler (2002) claimed that U.S. “teachers rarely draw from a shared knowledge base to improve their practice” (p. 3) and that specific research information has had little effect. Literature they reviewed indicated that the type of *craft*

knowledge needed for teaching does not follow from the work of educational researchers. This finding led them to consider requirements for building a professional knowledge base for teaching from practitioner knowledge and to reflect on whether this approach would be more successful than trying to build a professional knowledge base from research. "Observation and replication across multiple trials can produce dependable knowledge as well [as experimental methods can]" (p. 12). They considered lesson study to be a large-scale system that could facilitate this type of transformation.

Lesson study, a recent phenomenon in this country, is a basic practice in Japanese schools at all levels and in all subject areas. Recognizing what U.S. educators could learn from Japanese lesson study led to a U.S.-Japan Teacher Development Workshop following the 2000 ICME-9 (International Congress on Mathematics Education) conference in Japan. The conference proceedings were published by National Academy Press (Bass, Usiskin, & Burrill, 2002). Chapters by Shimizu and by Yoshida provided background on Japanese lesson study, and these chapters are the basis of the information in the next paragraph.

Japanese educators at this conference explained how lesson studies, which are of different levels and different types, operate in Japan (Shimizu 2002; Yoshida, 2002). Intraschool lessons are held a few times a year within a particular school. Lesson studies also occur at the city level, the school-district level, and even the national level. The three basic parts of a lesson study consist of choosing a particular topic and designing a lesson around that topic, teaching the lesson to students with teachers observing the particular lesson, and discussing the lesson with the observing teachers. The discussion, usually conducted by the lesson teacher, begins with comments. The process is cyclic, and each lesson study serves as preparation for the next. Lesson studies are the primary means of education for novice teachers and of professional development for experienced teachers. An obvious outcome is that participating teachers learn much about the skill of teaching. A less obvious, but important, outcome is that the participating teachers learn a great deal of content knowledge through lesson-study discussions.

Studied lessons are often called *research lessons*. They are set apart from everyday lessons in that "they are observed by other teachers; . . . they are carefully planned, usually in collaboration with one or more colleagues; . . . they are focused; . . . they are recorded; . . . (and) they are discussed" (Lewis & Tsuchida, 1998, pp. 12–14). National research lessons are disseminated all over the country. At the national

level, thousands of teachers might participate in a lesson study.

The first two steps of lesson study—preparing and implementing a lesson—are not new to U.S. teachers (although these activities rarely involve other teachers), but the third step—analyzing and evaluating the lesson—is uncommon. Ball (2002) listed three skills needed to undertake lesson study. First, a teacher must have the ability to pay attention to and teach every student in the class, knowing what each child understands or does not understand. Second, the teacher must know mathematics well enough to be able to organize the content, create appropriate activities, and make adjustments as needed during class. Third, a teacher must be able to work with others on developing knowledge for teaching. Hiebert and Stigler (2000) have also described lesson study:

The knowledge being shared through lesson study is not just collections of lesson plans that teachers can pull off the shelves and use. The goal of lesson study is not just to produce lessons that can be copied but to produce knowledge about teaching upon which colleagues can build (see Ball & Cohen, 1996). Such a knowledge base grows as a teacher reflects on and improves what others have done, working to understand the basis for the improvements. (p. 12)

The impact of research lessons can be substantial. Lewis (2000) listed nine ways in which these lessons contribute to Japanese instruction: They (a) provide professional development; (b) help teachers understand student thinking (or "see children"), (c) spread knowledge of new content and new approaches, (d) help individual teachers connect their practices to school goals and broader goals, (e) allow competing views to be heard, (f) create a demand for improved instruction, (g) shape national educational policy, and, last but not least, (h) honor the central role of teachers in the development of instruction. Lewis (2002) has worked with teachers in American schools to implement lesson study. To be successful, she advocated that American educators must answer the following questions: "What are the essential features of lesson study that must be honored when lesson study is conducted in the US? How do educators improve instruction through lesson study? What supports will be needed for lesson study in the US, given its system and culture?" (pp. 6–7). Lewis then provided tentative answers to these questions and noted that the educational systems in the United States and Japan differ greatly and that lesson study must be thoughtfully adapted rather than "borrowed."

Acquiring Knowledge-In-Practice: Summary and Discussion

The process of acquiring knowledge-in-practice can take many forms. Communities of practice established within a school can have a powerful influence on the mathematics instruction and student learning in that school. However, establishing such a community is not at all easy, and maintaining the community is difficult, particularly because teachers move in and out of the school. Several positive characteristics of communities of practice were listed, and examples of communities of practice were described. Lesson study, originating in Japan, is being studied in this country as a form of practice in which teachers can reflect on and learn from their own practices.

Professional development schools continue to be a source for learning, but, for many reasons, they are difficult to establish. Where they exist, they provide professional development for both teachers and university colleagues. PDSs serve as venues for forming collaborations, but other groups have collaborated successfully without actually establishing a PDS. Clark et al. (1996) provided an example of professional development consisting of collaboration in the form of dialogue in which teachers and researchers engage. *Dialogue* was considered to be

a means of achieving parity in collaboration while facilitating mutual reflection, growth, and change. Instead of teachers and researchers trying to take on the *work* of one another as their own—an approach which may be particularly *disempowering* for teachers—reciprocity is achieved dialogically, in part by talking through the hidden assumptions which teachers and researchers bring about the “other.” (p. 228)

These collaborations centered on evaluation of portfolios in English and science classrooms but might also be a useful technique in mathematics, particular as a way of introducing mathematicians into the work of teachers.

ADDRESSING QUESTION 6: WHAT CAN TEACHERS LEARN BY INVESTIGATING THEIR OWN TEACHING OF MATHEMATICS? THAT IS, HOW DO THEY ACQUIRE KNOWLEDGE-OF-PRACTICE?

Cochran-Smith and Lytle (1999) called their third type of teacher knowledge *knowledge-of-practice*. This type of knowledge can be generated when teachers investigate learning and teaching in their own classrooms and school sites. Their work is located within

the “contexts of inquiry communities to theorize and construct their work and to connect it to larger social, cultural, and political issues” (p. 250). Stated more formally, *knowledge-of-practice* is

constructed in the context of use, intimately connected to the knower, and, although relevant to immediate situations, also inevitably a process of theorizing. From this perspective, knowledge is not bound by the instrumental imperative that it be used in or applied to an immediate situation; it may also shape the conceptual and interpretive frameworks teachers develop to make judgments, theorize practice, and connect their efforts to larger intellectual, social, and political issues as well as to the work of other teachers, researchers, and communities. (pp. 272–273)

Practitioner Research

In the 1940s Kurt Lewin introduced research that teachers undertake to better understand their own teaching and its effects. It has evolved into “an ongoing process of systematic study in which teachers examine their own teaching and students’ learning through descriptive reporting, purposeful conversation, collegial sharing, and reflection for the purpose of improving classroom practice” (Eisenhower National Clearinghouse, 2000, p. 18). According to Holly (1991), “Action research, at root, constitutes participative learning for teachers. Who better to analyze teaching and learning than the teachers themselves through the agency of action research?” (p. 135). Hiebert et al. (2002) advocated the development of a knowledge base for teaching, beginning with practitioner knowledge:

The teacher-as-researcher movement has oriented teachers to studying their own practice, thereby making it more public and testing its effectiveness (Berthoff, 1997; Burnaford, Fisher, and Hobson, 1996; Cochran-Smith and Lytle, 1993; 1999). During the same time that the movement has been increasing educators’ awareness of the richness of teachers’ personal knowledge, it also has focused attention on the kind of teacher learning that is required to teach more effectively. (p. 11)

Much of the early work in teacher research in mathematics, commonly referred to as *action research*, took place in Australia and South Africa and has been described by Crawford and Adler (1996), who viewed research activities on the part of teachers as a prerequisite for changes in the quality of mathematics education. They claimed that teachers who act in “generative research-like ways . . . may learn about the

teaching/learning process, and about mathematics, in ways that empower them to better meet the needs of their students” (p. 1187). Borasi and Fonzi (2002) considered action research to offer “an ideal way for teachers to learn more about teaching and learning mathematics and to apply the results immediately to their own practice” (p. 100). Middleton et al. (2002) described research that moved from the study of systems to a new model of teacher as professional: “Under such a model, teachers are seen as knowledge producers, curriculum designers, and policy analysts with a unique configuration of knowledge, skills, and practices that have merit in the larger order of knowledge production, curriculum design, and policy analysis” (p. 411).

Action research has multiple forms—teachers might work alone to pursue a research interest, they might work together in inquiry teams, or they might work with university researchers. They might produce their own data or undertake a study of existing data on a question of interest. The defining factors are that teachers themselves formulate or contribute to the formulation of the research questions; they collect data to answer the questions; they use an action-research cycle of planning, acting, observing, and reflecting; they work collaboratively when possible; they have access to outside sources of knowledge and stimulation; and they document and share their research (Loucks-Horsley et al., 2003). According to Cochran-Smith and Lytle (1999), teacher research, particularly when collaboratively undertaken, has the potential to

profoundly alter the cultures of teaching. . . . In work of this kind, the image of knowledge as collectively constructed is particularly striking; knowledge emerges from the conjoined understandings of teachers and others committed to long-term highly systematic observation and documentation of learners and their sense making. To generate knowledge that accounts for multiple layers of context and multiple meaning perspectives, teachers draw on a wide range of experiences and their whole intellectual histories in and out of schools. (p. 275)

Teachers who undertake research are likely to use the results, and they gain power over decision making that guides their practice. “The change process—with all its emotional undertow—is internalized and personalized to the point where the teacher action researcher becomes ‘hooked’ on his or her change agenda” (Holly, 1991, p. 153). This process was exemplified in a project in Arizona in which a group of teachers worked with researchers studying children’s reasoning using the “model of teacher-as-researcher as

a mechanism for supporting their unique needs and questions of interest but also for disseminating their understandings across a larger audience” (Middleton et al., 2002, p. 411).

Some concerns have been raised about the validity of action research. Jaworski (1998), working with a group of teacher researchers focusing on the ways research interfaced with reflective practice, had the following to say about this matter:

In making public the outcomes of their research, teachers contribute to research rigour. It is hard to validate the research in terms of substantive outcomes. For example, what Julie [a teacher engaged in action research] learned about mathematical talk must be seen in the context of Julie’s classroom and personal teaching perspective. However, in offering the outcomes of her research to other teachers, she shares her own insights and invites responses. Validity rests in the possibility of others seeing and making sense of Julie’s work and using it to illuminate their own work and thinking. This is a rather different way of regarding research rigour, but nevertheless important to the use of research as a vehicle for developing teaching. (p. 27)

Obstacles need to be overcome in the process of planning and undertaking teacher research (ENC, 2000). Undertaking research always demands a significant amount of time and is not always recognized as a form of professional development. When possible, teachers arrange to have this work count toward district-required professional development. Also, teachers have different goals and priorities, and not all want to become involved in research. If teachers are, for example, teaching from a new curriculum or participating in other professional development activities, they often cannot, at the same time, undertake research that requires additional time and effort. For others, however, action research provides an avenue to develop professionally. The ENC (2000) report provided an example of a teacher from New York who wanted to study how interactions with computers affected learning in her mathematics class. Her research led to a collaborative effort with another teacher and a university researcher and to a sizable grant from the National Science Foundation. She remarked, “Action research lends itself to classroom use through professionally inquisitive teachers. When this happens, the classroom becomes a learning environment for the teacher as well as the student. It involves doing what comes naturally to teachers, questioning classroom practices” (p. 20).

Knowledge-Building Schools

Whereas *action research* often refers to one or a few teachers in a school undertaking research in their classrooms, *knowledge-building schools* are schools in which research is embedded into the school structure. Groundwater-Smith and Dadds (2004) described a school-improvement project in which teachers from seven schools in Australia “sat together and discussed the possible formation of a coalition of knowledge-building schools . . . constructed and examined in ways which illuminate understanding rather than [serving] as a means of proving a particular case” (p. 247). The participants wanted to embed inquiry practices into their teaching and develop a workplace-learning culture, believing that “professional learning is not an exclusively individualistic enterprise but that learning and growth can take place at the organizational or corporate level” (p. 248).

The Inquiry of Teaching, Or Practical Inquiry

The *inquiry of teaching*, that is, “the ongoing everyday inquiry into their students’ learning that teachers are expected to conduct” (Hammer & Schifter, 2001, p. 471) is different from research on teaching and learning as practiced by researchers, whether or not the researchers are teachers. Inquiry of teaching could be considered a form of research, and, in fact, the boundary between inquiry of teaching and teacher research is indistinct. Hammer and Schifter have examined this border by distinguishing between the inquiry of teaching and the inquiry of research: “Both . . . inquire into students’ knowledge and reasoning, but they do so under different conditions and with different opportunities and constraints” (pp. 475–476). Some teachers are uninterested in research but do want to know more about their teaching and their students and undertake daily analysis of their students’ thinking. Their study is distinguished from research in terms of the audience, the scope and purpose of the inquiry, and the manner in which teachers identify themselves, whether as researchers or as teachers who study their students’ thinking. Inquiry of teaching is often prompted by the use of innovative textbooks and other instructional tools that lead to teachers’ experimenting with instructional practices. “Curriculum reform requires teacher inquiry into students’ knowledge, reasoning, and participation” (p. 442).

Cognitively Guided Instruction researchers have used a different label, *practical inquiry*, for what seems closely related to inquiry of teaching. These researchers have recently turned their attention to the under-

standing of teachers’ self-sustaining, generative change in the context of professional development (Franke et al., 1998). The authors claimed that for change to occur, teachers must be engaged in *practical inquiry*, that is, inquiry that helps the teachers understand their students, the contexts in which they work, and the practices of the community in which the teachers work. It can be viewed as a teacher’s “questioning and reflecting about his/her practice with a specific focus. The focus of a teacher’s practical inquiry determines what a teacher sees as critical, and what constitutes an opportunity for reflection” (p. 68). In undergoing this type of change, teachers are not learning new procedures but, instead, are making basic epistemological changes in their views of learning and teaching.

In the first of two levels of reflecting on practice, one’s aim is to see “what works.” Knowing some practices that can be used successfully can encourage the teacher to try to find other such activities or practices. This type of reflecting on practice is self-sustaining because it supports continuation of using these practices, but at this first level, teachers do not know *why* a practice is effective. When teachers also engage in detailed analysis of their practices, but now in relation to their own thinking, they come to understand the “principled ideas that drive their practice and their continued practical inquiry” (Franke et al., 1998, p. 68). This second level is both self-sustaining and *generative* because when teachers understand the relation between their practices and student learning, they have a basis for generating new insights about effective instruction. Thinking in terms of these two levels provides fodder for those individuals who provide professional development for teachers.

A concern for self-sustaining, generative change shifts the focus of professional development from the factors that initiate change to principles that make it possible for teachers to continue to grow and learn. . . . In order to understand the effects of a teacher development program, it is necessary to understand how the participants in the program construe and implement the principles. Teachers, as well as students, do not simply assimilate knowledge. Teachers in teacher development programs may construct interpretations of a program’s first principles that are quite different from the principles intended by the designers of the program. (p. 68)

Acquiring Knowledge-of-Practice: Summary and Discussion

“The knowledge-of-practice conception turns on the assumption that the knowledge teachers need to

teach well emanates from systematic inquiries about teaching, learners and learning, subject matter and curriculum, and schools and schooling” (Cochran-Smith & Lytle, 1999, p. 274). Whether this inquiry is viewed by the teacher as research or as practical inquiry depends on whether the inquiry has some sort of research question and employs research methods.

One way to distinguish knowledge-of-practice from knowledge-in-practice is to consider the knowledge gained in lesson study, described in the previous section. From the standpoint of the teachers participating in the lesson study as observers and discussants, they are acquiring knowledge-in-practice by engaging in a community discussion of the lesson. But the person who teaches the lesson and participates in discussions about the class is more likely to be acquiring knowledge-of-practice. (Recall that the second category of knowledge described by Cochran-Smith and Lytle [1999] was *knowledge-in-practice*, that is, the practical knowledge of teaching “embedded in practice and in teachers’ reflections on practice” [p. 250].)

ADDRESSING QUESTION 7: WHAT CAN BE LEARNED FROM RESEARCH ON TEACHER CHANGE?

Recent research and thinking about teaching has led to a “radically revised picture of what kind of instruction should take place in the classroom” (Goldsmith & Schifter, 1997, p. 20). A climate of inquiry and sense making, a change in the role of the teacher as the sole authority, and a focus on reasoning and problem solving call for significant changes in a teachers’ beliefs about mathematics, about children’s learning, and about teaching. On the part of the teacher these changes call for a fundamental reconceptualization of his or her role in the educational process—not an easy change to make. According to Nelson and Hammerman (1996), researchers must come to understand

what it is in innovative professional development projects that makes this process of change possible for teachers, and why it works. . . . We must specify the nature of the change that is at issue and advance a theory of the process by which teachers make these changes. (p. 4)

Instructional change can occur in several ways (Richardson & Placier, 2001). Change that is mandated rather than initiated by teachers may be negative and, more often than not, leads to teachers’ becoming recalcitrant and resistant to implementation of changes they feel forced to make. Such change, usually minimal and

superficial, is the type referred to by Loucks-Horsley and Stiegelbauer (1991) in the following:

Too often policy makers, administrators, even teachers assume that change results from a policy mandate, an administrative requirement, adoption of new curricula, or a revised procedure. They assume that a teacher will put aside tried-and-true practice and immediately use a new one with great facility. In the most absurd situations, there is the companion assumption that the new program can soon be evaluated for its impact on students. In reality, change takes time and is accomplished only in stages. (p. 17)

Because teacher change is a process rather than an event, it must be considered in terms of continuous growth over time. Darling-Hammond (1990) said that the reason change is slow and difficult is that in addition to perseverance, “it requires investments in those things that allow teachers, as change-agents, to grapple with transformations of ideas and behavior: time for learning about, looking at, discussing, struggling with, trying out, constructing and reconstructing new ways of thinking and teaching” (p. 240).

One type of experience necessary for teachers to undertake instructional change is the opportunity to see how innovative teaching practices can enhance their students’ learning, particularly when they see that their students become more confident and more engaged in learning, with resulting higher achievement. “Then, and perhaps only then, a significant change in teachers’ attitudes and perceptions is likely to occur” (Guskey, 1989, p. 446).

Ways of Describing Teacher Change

Researchers have tried, in several ways, to describe the change process in terms of differences in instructional practice. For example, the manner in which teachers question their students can provide insight into their orientations toward learning and instruction (A. G. Thompson, Philipp, P. W. Thompson, & Boyd, 1994), and the study of the manner in which teachers question students can become a way to measure change. These authors claimed that teachers who have a *calculational orientation* are focused on procedures for obtaining answers. Such teachers speak almost exclusively in the language of numbers and operations. Teachers who have a *conceptual orientation* focus students’ attention on the “rich conception of situations, ideas, and relationships among ideas. [They] strive for conceptual coherence, both in their pedagogical actions and in students’ conceptions” (p. 86). Teachers’ implementation of new curricula is particularly influenced by their orientations toward mathematics teaching.

Other researchers have described stages that occur in teacher growth. On the basis of studies of teachers in professional development programs designed to help teachers change their teaching practices, Schifter (1995a) proposed four stages of conceptions of mathematics teaching:

- (1) an ad hoc accumulation of facts, definitions, and computational routines;
- (2) student-centered activity, but with little or no systematic inquiry into issues of mathematical structure and validity;
- (3) student-centered activity directed toward systemic inquiry into issues of mathematical structure and validity;
- (4) systematic mathematical inquiry organized around investigation of “big” mathematical ideas. (p. 18)

Each of these broadly defined stages “entails an understanding of what counts as ‘doing mathematics,’ of the extent to which mathematical results are interconnected, and where mathematical authority resides and how it is established” (Schifter, 1995a, p. 18). The model was developed to assess change on the part of program participants, but Schifter and her colleagues found it more useful as a pedagogical heuristic that could clarify goals, interpret practice, and guide the design of professional development programs. These stages are similar to ones she and Fosnot used in their SummerMath program (Schifter & Fosnot, 1993).

The Cognitively Guided Instruction researchers also developed a way of documenting growth on the part of teachers (Fennema et al., 1996). They categorized teachers as belonging to one of four levels, similar but not identical to those proposed by Schifter (1995a). In Level 1, instruction focused on the learning of procedures and was guided by an adopted textbook. In Level 2, instruction was similar to that of teachers in Level 1 but was enriched by problems similar to those studied during the teacher workshops. Teachers were exploring new ways of teaching, but within a model of fixed routines. Classrooms of teachers in Level 3 were characterized as those in which children were engaged in solving and reporting solutions to problems that were not found in standard textbooks. The teachers talked about specific children’s thinking. By Level 4, teachers based instructional decisions (to a greater extent than those at lower levels) on the kinds of problems children were capable of solving, the strategies children use, and the kinds of communication children are capable of using. At the end of their longitudinal study, 90% of the teachers were at Level 3 or 4; Level 4 was the researchers’ category for

the most cognitively guided instruction they observed. They concluded, “Starting with an explicit, robust, research-based model of children’s thinking, as we did, enabled almost all teachers to gain knowledge, change their beliefs about teaching and learning, and improve their mathematics teaching and their students’ mathematics learning” (Fennema et al., 1996, p. 433). The levels and patterns of change were further discussed in a book chapter (Franke et al., 1997) in which the authors noted that teachers at lower levels were able to change instructional practice without substantially changing beliefs, but that for progress from Level 3 to Level 4, changes in teachers’ beliefs were essential.

Schifter’s (1995a) and Franke et al.’s (1997) descriptions of stages of development were based on their work with teachers. Goldsmith and Schifter (1997) used theories of cognitive development in addition to personal observations to provide descriptions of the understandings of individuals over time, while teachers’ mental organizations were being reconstructed to allow for more complex thought and action. They believed that studying these stages could be fruitful avenues for research on teachers and teaching. Their first stage of development is characterized by instruction that is organized around the transfer of knowledge; thus teachers’ instruction at this stage was focused on providing opportunities for children to automatize and generalize the demonstrated procedures, as happens in the majority of mathematics classrooms. In subsequent stages, teachers become “less intent on helping students acquire facts and procedures, and more involved in building on what (and how) their students understand” (p. 23). The final stage is one in which teachers’ instruction is aligned with that envisioned in the NCTM Standards. Although this last stage has different interpretations, one constant is that teachers focus on understanding what their children are capable of achieving and take this information into account when planning instruction. The authors spoke of stages as providing important information about the process of change—noting that the development of pedagogical content knowledge can be seen as orderly but that it usually is not. Thus, research on models of mathematics teaching might profitably focus on the orderliness of change. “Research directed toward mapping the issues teachers confront as they enact their new beliefs and understandings in the classroom will help to create a fuller picture of how teachers move through the terrain of creating a reformed mathematics practice” (p. 38).

Goldsmith and Schifter (1997) encouraged attention to both psychological and sociocultural mechanisms in research on teacher development and suggested that each view of learning will lead to a different

way of viewing teacher change. They emphasized the importance of language in the process of “becoming acculturated to new forms of mathematics teaching and new ways of communicating those thoughts and ideas” (p. 45). Teachers often need new vocabulary to talk about the changes in their instruction and in student learning and new images of good instruction:

Teachers’ efforts to construct new forms of mathematics teaching can be affected considerably by the images they have of good teaching, the nature of collegial relationships, the criteria and procedures for job evaluations, and the kinds of professional development that prevail in their schools and districts. (p. 45)

Goldsmith and Schifter also suggested that interactions with students affect teachers’ development and that research centered around these interactions could prove useful. Finally, they said that still to come is “a well-developed account of what sustains people as they undertake the often difficult and frustrating task of restructuring their understanding” (p. 46). Their earlier work indicated that “courage is an important ingredient in the motivation equation” (p. 46).

The difficulties of moving through these stages were acknowledged by Goldsmith and Schifter (1997). They spoke of teachers’ having “one foot in each of two conflicting paradigms” (p. 28) when they move from one set of beliefs to another, conflicts that they resolve over time. Even while attitudes and perceptions are changing, teachers experience a certain amount of anxiety and reluctance to change instruction until they are certain that the changes will be successful in their classrooms (Borasi & Fonzi, 2002; Guskey, 1995). At times teachers’ changing perceptions can bring about a belief that their past teaching practices actually harmed their students, a belief that can bring with it a great deal of pain on the part of a conscientious teacher (Sowder, Philipp et al., 1998). Schifter and Fosnot (1993) found that for their SummerMath teachers the process of transformation was “highly affect-laden” (p. 195). Teachers who have transformed their mathematics teaching and want to help others sometimes encounter professional jealousy, which, in turn, can cause tension and pain (Koch, 1997). Professional developers must be aware of and able to help teachers deal with the affective issues related to change.

Psychological and Sociological Frameworks for Studying Teacher Change

Studies of mathematics teachers’ development have generally followed two broad frameworks within

which to develop and explore theories about teacher change. In the first, based on a psychological perspective, individual teachers are themselves the objects of study—how they learn, how they teach, and how they transform their practices, usually as a result of some professional development intervention. The Goldsmith and Schifter (1997) stages discussed above are based on a psychological perspective. Case studies of individual teachers—and there are many such studies—fit within this genre of research. The bulk of the studies reviewed here fit within this framework.

The second type of framework is based on a sociocultural perspective in which teachers’ learning is conceptualized as a process of *transformation of participation* in a community of practice (Stein & Brown, 1997). Whereas in the psychological framework the study of teacher change is focused on changes in the individual teachers, their knowledge, beliefs and practices, one using a sociological perspective instead focuses on the learning processes of teachers by looking “beyond formally structured events to other times and places in which individuals learn; . . . the focus on analysis shifts away from pedagogical activity toward an analysis of the community’s activities and learning resources” (Stein & Brown, 1997, p. 162).

Stein and Brown (1997) further delineated this sociocultural perspective into two frameworks. One is based on Lave and Wenger’s (1991) “theory of learning through legitimate peripheral participation in communities of practice” (Stein & Brown, 1997, p. 155) and the other on Tharp and Gallimore’s (1988) “model of learning as movement from assisted performance to unassisted performance through a Zone of Proximal Development” (Stein & Brown, 1997, p. 155).

Although these are the two most commonly used frameworks, other frameworks, too, provide ways to study teacher change. For example, Drake, Spillane, and Hufferd-Ackles (2001) used the construct of *identities* as a framework for studying teacher change. They claimed that a teacher’s sense of self, or sense of identity, reveals the way that a teacher knows himself and his life and that teachers’ stories “can serve as a lens through which teachers understand themselves personally and professionally and through which they view the content and context of their work, including any attempts at instructional innovation” (p. 2). Using this framework, Drake et al. identified three types of elementary mathematics teachers: *Turning-point teachers*, who had had mostly negative experiences with mathematics learning but had had a recent positive experience and who focused on using manipulatives and on the affective components of reform; *failing teachers*, who had only negative experiences with learning and teaching mathematics, were traditional

in their teaching, and focused only on superficial aspects of reform; and *roller-coaster teachers*, who had had both positive and negative mathematics learning experiences and now focused on both the content and affective components of reform. Drake et al. noted, “It is not surprising to find that individual teachers exposed to identical reform programmes will respond differently depending in part on the dispositions and beliefs which are embedded in their identities as teachers and as learners” (p. 3). The authors claimed that a better understanding of teachers’ positions in terms of their mathematics identities could be used to “inform the design of future policies, curricula, and teacher development activities intended to foster instructional innovation” (p. 17).

Another framework that has been used to study professional development of elementary school mathematics teachers is the *Diffusions of Innovations* framework of Rogers (1995). Diffusion research can be used to understand how innovations are shared over time among the members of a social system. Jacobs and Raynes (2001, 2002) used this framework to study professional development being offered by teacher leaders in a small school district. They found four diffusion principles useful in guiding their work. First, the characteristics of an innovation can help determine how quickly it will be adopted. “Innovations that do not require radical changes but have highly visible benefits over current practices can have an inherent advantage” (Jacobs & Raynes, 2002, p. 4). Second, an adoption of an innovation is determined by information gathering and decision making, both of which require time and opportunities to explore the innovation. Third, those who communicate about the innovations are more effective if they have “personal or social characteristics that make them similar to but also respected by those considering adoption” (p. 5). Fourth, “the social system in which an innovation would be used has structures and norms that encourage its members to act in particular ways . . . [and that can] increase or decrease the rate of diffusion” (p. 5).

In yet another way to understand how teachers develop their practices, Simon (2001) intertwined two lines of study, the empirical work of conducting research on mathematics teacher development and the theoretical work of elaborating theory of mathematics teaching and learning, as the basis for his work on the Mathematics Teacher Development project. Teachers’ development “can be thought of generally as evolution from more traditional practices toward practices more consistent with current mathematics education reforms” (pp. 160–161). Simon and his research group developed two constructs, a perception-based perspective and a conception-based perspective,

which they used to describe the mathematical experiences of the students with whom they worked. The *perception-based perspective* can be thought of as “best achieved by students examining for themselves mathematical situations in which the relevant mathematics can be seen; that is, they learn best through first-hand experience with the mathematics” (p. 163). Simon viewed the teacher’s role as a guide to “afford students’ direct apprehension of the mathematics” (p. 163). The *conception-based perspective* refers to “an array of perspectives that share the following assumptions:

1. Mathematics is a product of human activity. Humans have no access to mathematics that is independent of their ways of knowing.
2. What individuals see, understand, and learn are afforded and constrained by their current conceptions.
3. Mathematics learning is a process of transformation of learners’ knowing and ways of acting. (p. 164)

Someone with a perception-based perspective assumes “that everyone is looking at the same mathematics; what is seen is assumed to be unvarying from person to person” (p. 164). Someone with a conception-based perspective makes “no assumption that what is seen matches with what others see” (p. 164) in any particular situation. Simon argued that theoretical frameworks of mathematics teaching and learning can promote discourse and advance understanding of our field.

A Role for Researchers in the Study of Teaching and Teacher Change

Steps for professional development design and research were suggested by Borko (2004) in her AERA presidential address, in which she called upon researchers “to play a leadership role in providing high-quality professional development for all teachers” (p. 13) by addressing two major questions: “What do we know about professional development programs and their impact on teacher learning? (and) What are important directions and strategies for extending our knowledge?” (p. 3). She noted that in her discussion of successful research programs on teaching and teacher change her choices were constrained by the limited availability of such programs. She suggested design experiments to bring researchers and professional developers together to adapt existing successful professional development to new subject areas by considering a successful program and engaging “in multiple design/research cycles to refine the program and

study its impact on the development of professional community and the learning of individual teachers” (p. 12). Just as educational innovations are changed by users, so is professional development program changed by the manner in which it is implemented. She thus suggested research “investigating the balances and tradeoffs between fidelity and adaptation, and consider which elements of a program must be preserved to ensure the integrity of its underlying goals and principles” (p. 13). Finally, she noted the need to develop instruments to study “change over time in teachers’ subject matter knowledge for teaching and instructional practices, and analytic tools that can separate out the influences of various program, school, and individual factors on teacher and student learning” (p. 13).

Several successful professional development programs have been described in this chapter. Borko’s suggestions lay out terrain for conducting research on professional development and teacher change in mathematics.

This discussion of teacher change is based on a vision of mathematics teaching as described in my first of six goals and found in the above descriptions of final stages of change (Fennema et al., 1996; Goldsmith & Schifter, 1997; Schifter, 1995a). An enhanced vision of mathematics teaching and learning develops hand-in-hand with a developing knowledge base for teaching.

A Few Words About the Role of Reflection in Teacher Change

Throughout the research discussed in this chapter, the topic of reflection and its role in furthering teachers’ professional development and education is a constant in the discussions. The importance of teachers’ reflecting on their practice is a recurring theme in research on teaching, and the degree and kind of reflection is often used to describe teacher change. Schön (1983, 1987) is most often credited for this focus on reflection, although recognition of its importance can be found in Dewey’s work (e.g., 1910/1977). In Mewborn’s (1999) research on reflective thinking among preservice teachers, she noted that the term *reflection* is used in different ways but that three features are common in all uses. First, reflection is always considered to be qualitatively different from recollection or rationalization. Because the object of reflection when applied to teaching is problematic in nature, the teaching itself is recognized as problematic. Second, “action is an integral part of the reflective process. . . . Reflection and action together are seen as a bridge across the chasm between educational theory and practice” (p.

317). Third, reflection requires introspection but also requires outside probing; thus, it is “both an individual and [a] shared experience” (p. 317).

The ability to reflect on practice is considered a necessity for effective instruction.

Teachers need to be able to analyze and reflect on their practice, to assess the effects of their teaching, and to refine and improve their instruction. They must continuously evaluate what students are thinking and understanding and reshape their plans to take account of what they have discovered. (Darling-Hammond, 1998, p. 8)

The notion of teacher inquiry is at the heart of reflective practice and plays a central role in “many of the prevailing conceptions of teacher learning including critical reflection, reflection in and on action, personal and pedagogical theorizing, narrative inquiry, action research, and teacher research” (Barnett, 1998, p. 81). In their work with four “extraordinary” teachers, Philipp et al. (1994) were struck by the quality of the reflectiveness of these teachers and their tendencies to, as A. G. Thompson (1992) said, “think about their actions, vis-à-vis their beliefs, their students, the subject matter, and the specific context of instruction” (p. 139).

But important issues need to be considered when we ask or expect teachers to become reflective about their practices. First, Hargreaves (1995) noted that although

reflection is central to teacher development, the mirror of reflection does not capture all there is to see in a teacher. It tends to miss what lies deep inside teachers, what motivates them most about their work. However conscientiously it is done, the reflective glance can never quite get to the emotional heart of teaching. (p. 21)

Second, time is needed for developing the ability and habit of reflecting on practice. Reflection rarely occurs when time is not a resource available to teachers (Pugach & Johnson, 1990). Third, “like many enduring educational concepts, [reflection’s] rhetorical appeal has outstripped the evidence. . . . Reason only has limited access to that which drives our actions, which rather limits the promise of reflection for improving practice” (Knight, 2002, p. 232). Within the classroom setting, teachers rarely have opportunities to reflect—they must constantly be observing students and making on-the-spot inferences and decisions. Only unusual teachers will carve out the time needed to reflect on their own instruction and students’ learn-

ing unless support is given in terms of time and strong collegial relationships.

ADDRESSING QUESTION 8: WHAT CAN BE LEARNED FROM RESEARCH ABOUT THE PREPARATION OF TEACHERS TO TEACH MATHEMATICS?

Many of the goals, types, and strategies we have described for professional development are also applicable to teacher preparation. For example, teacher educators use videos that illustrate student thinking; school curriculum materials; case studies; and, most certainly, formal course work in their teacher preparation programs. Still, some aspects of teacher preparation need to be considered apart from professional development of practicing teachers. Here I discuss the development of a new frame of reference for teaching, ways of improving teacher preparation, and the particular role of mathematicians in preparing teachers.

Developing a New Frame of Reference for Teaching

A prospective teacher undergoes a lengthy *apprenticeship of observation* (Ball & Cohen, 1999; Ebby, 2000, attributed to Lortie, 1975; Kennedy, 1999) during his or her K–12 education. In mathematics, this apprenticeship can be described as experiencing mathematics classes as an immutable sequence beginning with checking homework, asking questions about the homework, watching the teacher demonstrate how to do new problems, then receiving an assignment to work problems similar to those demonstrated (Romberg & Carpenter, 1986). University course experiences are likely to reinforce this framework. Thus to improve the way in which teachers are prepared to teach mathematics, the mathematics education community must attend to changing this familiar model of a mathematics class. “The kind of teaching that reformers envision requires teachers to shift their thinking so that they have different ideas about what they should be trying to accomplish, interpret classroom situations differently, and generate different ideas about how they might respond to these situations” (Kennedy, 1999, p. 56). According to Kennedy (1999), unless this shift is initiated during teacher preparation, teachers develop habits of teaching based on their pre-preparation experiences and solidify these practices, making change all the more difficult. The demands of the job are intense

when teachers begin teaching, so to find that new teachers often fall back on practices they know best is not surprising. Dewey (1904) expressed this concern a century ago:

The student adjusts his actual methods of teaching, not to the principles which he is acquiring, but to what he sees succeed and fail in an empirical way from moment to moment; to what he sees other teachers doing who are more experienced and successful in keeping order than he is; and to the injunctions and directions given him by others. In this way the controlling habits of the teaching finally get fixed with comparatively little reference to principles in the psychology, logic, and history of education. (p. 24)

The influence of the apprenticeship-of-observation framework is more prevalent at the secondary level than at the elementary or middle school level. Secondary school teachers have experienced many more mathematics classes, most, if not all, taught in a traditional fashion. They are likely to have taken far fewer courses dealing with education or pedagogy. “Teachers need to know and use ‘mathematics for teaching’ that combines mathematical knowledge and pedagogical knowledge” (NCTM, 2000, p. 370). Providing secondary school teachers with a new frame of reference for how mathematics should be taught is particularly challenging.

Improving Mathematics Teacher Preparation

What makes teacher preparation effective? In this section I explore what many researchers and educators have described as components of a strong teacher preparation program and changes needed to make programs more effective.

A concern for better preparation of teachers of mathematics and science led the National Research Council, in 1998, to appoint a committee to study teacher preparation. The committee was comprised of mathematicians and scientists, mathematics and science educators including K–12 teachers, and a business representative. The committee’s report, *Educating Teachers of Science, Mathematics, and Technology* (NRC, 2001b), was written to identify “critical issues in existing practices and policies for K–12 teacher preparation in mathematics and science” (p. xiii). Several recommendations were made about the characteristics that teacher education programs in mathematics and science should exhibit. The recommendations are broad and call for interactions among all groups concerned with K–12 teacher preparation. They recommend that programs should have the following features:

- Be collaborative endeavors developed and conducted by mathematicians,² education faculty, and practicing K–12 teachers with assistance from members of professional organizations and mathematics-rich businesses and industries . . . ;
- Help prospective teacher to know well, understand deeply, and use effectively and creatively the fundamental content and concepts of the disciplines that they will teach . . . ;
- Unify, coordinate, and connect content courses in mathematics with methods courses and field experiences . . . ;
- Teach content through the perspectives and methods of inquiry and problem solving . . . ;
- Present content in ways that allow students to appreciate the applications of mathematics . . . ;
- Provide learning experiences in which mathematics is related to and integrated with students' interests, community concerns, and societal issues . . . ;
- Integrate education theory with actual teaching practice, and knowledge from mathematics teaching experience with research on how people learn mathematics;
- Provide opportunities for prospective teachers to learn about and practice teaching in a variety of school contexts and with diverse groups of children . . . ;
- Encourage reflective inquiry into teaching through individual and collaborative study, discussion, assessment, analysis, [and] classroom-based research and practice; and
- Welcome students into the professional community of educators and promote a professional vision of teaching. (pp. 69–71)

Howey (1996) also addressed the elements needed to make teacher preparation programs effective. Though many of his suggestions were too general to be appropriate for this chapter, he did suggest eight understandings, abilities, and dispositions that should “permeate a preservice program in the form of inter-related pedagogical activities designed to promote them” (p. 163) and that could profitably be the focus of a discussion by those who prepare mathematics teachers. He offered a different slant from that of the National Research Council on ways to make teacher preparation effective.

Understand and celebrate individual and cultural diversity; understand the subject matter to be taught and be able to represent it in multiple ways pedagogically; reflect on the moral and ethical consequences of policy and classroom practice; analyze and justify teaching practice; engage in teaching as a shared responsibility; monitor student understanding and foster conceptual learning; engage learners in active, self-monitoring learning tasks; and relate experiences in school to critical issues in society. (Howey, 1996, p. 163)

Carrying out these recommendations requires new ways of thinking about teacher preparation. Wilson and Ball (1996) suggested three ways that teacher education programs can provide models of teaching different from those the majority of entering teachers have experienced. One way is to sort out what needs to be preserved and what needs to be changed. Being able to manage a large group of students will always be something teachers are required to do. But new components of teacher education might need to be added—how to elicit student talk, how to ask good questions, how to undertake classroom discourse. A second way to provide models of teaching is to create opportunities for prospective teachers to view the sort of instruction that is needed to teach for understanding. Observing good teachers can be productive, but probably only if sufficient guidance is provided. For example, watching a teacher work with groups in a classroom should not leave the impression that the teacher is not teaching. Models might exist as videos that can be discussed in a teacher education class. Third, new teachers will need to be more prepared to “reason in the face of a host of dilemmas of practice” (p. 133). Reasoning in and about practice requires new ways of watching and hearing children. It means learning to reason about children’s development, about instructional tasks, and about the interaction between the two.

Lampert and Ball (1999) also suggested changes in teacher preparation that might result in teachers better prepared for teaching mathematics. They suggested orienting teacher education around investigations of practices of teaching and learning instead of focusing teacher education programs on providing knowledge and skills for teaching. Making this change would require a focus on “inquiring into concrete phenomena of practice” (p. 43).

Investigations draw on but are not restricted to current theories and formal knowledge about teaching and learning in general or a particular kind of teach-

² Throughout the recommendations, science and technology were included in the original document but were omitted here for the purposes of this chapter.

ing and learning. Instructors help teacher education students access and use ideas of the field, and they also encourage students to make novel conjectures and advance possible novel interpretations. Engaging with situated questions of practice can support a context for the intertwining of knowledge use and knowledge construction. (Lampert & Ball, 1999, p. 44)

Few teacher preparation programs currently focus on investigations, although individual professors might make investigations a focus in individual courses. Programs that are associated with a Professional Development School might more readily accept this challenge.

The Role Children's Thinking Can Play in Teacher Preparation

Preparing teachers to teach in new ways requires that they be provided opportunities to reason in and about practice and to learn to listen to, hear, and watch students. The opportunity to observe and study children thinking about mathematics has proved to be a powerful component of professional development, as noted earlier, but the study of children's thinking can also be a productive component of teacher preparation. For example, the Cognitively Guided Instruction model most often used in professional development of practicing teachers has also been used successfully with preservice teachers (Philipp, Armstrong, & Bezuk, 1993; Vacc & Bright, 1999). The Developing Mathematical Ideas (DMI) program, in which teachers examine how children develop understanding of the big ideas of elementary school mathematics, has been successfully used with preservice teachers (Schifter, Russell, & Bastable, 1999). Studying mathematics cases such as those of Barnett et al. (1994) for elementary and middle grades and Merseeth (2003) for secondary school can also provide insight into children's ways of understanding mathematics, together with new ways to think about teaching.

The Integrating Mathematics and Pedagogy (IMAP) project (Philipp et al., in press) also focused on children's thinking during teacher preparation. Prospective teachers in their first mathematics content course who volunteered to become involved in the project were randomly assigned (as much as was possible) to one of five treatments: two that provided opportunities to study children's thinking through interviewing children or through watching videos of interviews of children, two that visited classrooms either of exemplary teachers or of teachers unknown to the researchers, and a control group. Belief changes were measured using a project-developed, web-based beliefs-assessment instrument on which students whose beliefs reflected better understanding of children's

thinking and of the nature of school mathematics had higher scores. The treatment groups who studied children's mathematical thinking had significantly higher gain scores on the beliefs-assessment instrument than either the group who observed traditional teachers or the control group, but not than the group who had observed weekly in classes taught by carefully selected teachers.

Others have also provided opportunities for prospective teachers to work with children and observe their thinking. Nicol (1998) used "prospective teachers' weekly interactions with students as springboards for investigations of mathematics, teaching, and learning" (p. 45). Using a framework of questioning, listening, and responding, she found that the prospective teachers began to pose questions and listened and responded in ways they had not at the beginning of the project.

At universities with large numbers of prospective teachers, opportunities for preservice teachers to work with K-12 students one-on-one are difficult to arrange. Videos of children doing and explaining mathematics provide a different, but also effective, way to study children's mathematical reasoning if instructors help teachers to understand the children's thinking.

Methods Courses, Field Experiences, Induction, and Mentoring

Most prospective teachers of mathematics take a course (or more) on methods of teaching mathematics. This course is the most likely place that Lampert and Ball's (1999) suggested changes begin. Mewborn (1999, 2000) studied four preservice teachers during a field-based mathematics methods class so that she could examine what these future teachers found problematic and how they resolved their problems. She found that they viewed the knowledge they gained in professional education classes as rules or prescriptions to apply to classrooms, and they became disillusioned when these rules were ineffective. Thus she encouraged the prospective teachers to analyze their experiences in a way that would lead to reflective judgments rather than evaluative judgments. She concluded that "teacher education that is conducted in a setting that promotes investigation and inquiry into the problems of mathematics teaching seems to hold promise for assisting preservice teachers in becoming inquiring, reflective mathematics teachers" (Mewborn, 1999, p. 339). Ebby (2000), too, studied ways of thinking about relating coursework and fieldwork. Her study of the very different ways prospective teachers make connections between their coursework in teacher preparation and their fieldwork led her to conclude that teacher educators need to rethink how

methods courses are organized and delivered. Rather than focusing only on developing new knowledge and beliefs, methods courses should help future teachers develop habits of mind so that they can learn from the classroom—"methods courses need to be explicitly oriented towards learning from fieldwork" (Ebby, 2000, p. 94). Howey's (1996) previously described recommendations might be useful to consider in light of Ebby's research.

In the Mewborn (1999) and Ebby (2000) studies, methods courses had field-experience components. However, such is not always the case. When separated, the field experience can be either a positive or negative experience. A review of research on teacher preparation commissioned by the U.S. Department of Education (Wilson, Floden, & Ferrini-Mundy, 2001) addressed issues of school experiences as part of teacher preparation programs. Both experienced and newly certified teachers considered clinical experiences, such as student teaching, as the most powerful element in their preparation to teach. Research showed significant shifts in attitudes during clinical experiences, but unclear is whether these changes enhanced the quality of the experiences. Too often school experiences, particularly student teaching, are disconnected from university-based components of teacher preparation. Problems with appropriate placement are common, and this problem is magnified by the fact that classroom teachers have a powerful influence on prospective teachers. Across several studies "one theme that emerges is that field experiences led to more significant learning when activities are focused and well structured" (p. 19). Teachers who had fifth-year induction internships were more satisfied with being a teacher than teachers who lacked this induction experience.

According to Darling-Hammond, Berry, Haselkorn, and Fidler (1999), research indicates that 30% of new teachers leave within 5 years, that beginning teachers of mathematics leave the field at an even higher rate, and that secondary school teachers leave at a higher rate than elementary or middle school teachers. Higher salaries in other professions are one inducement to leave teaching. Inadequate preparation is another. "The disproportionate placement of underprepared teachers in disadvantaged schools is especially problematic given the mounting research evidence that fully qualified teachers . . . are more effective than those who do not possess this knowledge and skill; especially in contexts where skillful teaching is most needed" (p. 189). The importance of strong teacher preparation followed by induction that includes mentoring and further professional development can reduce attrition and, at the same time,

strengthen teachers' abilities to be effective. Darling-Hammond and her colleagues suggested that all our reform efforts will be at risk if our teaching force is not prepared to raise student achievement. The need for new programs that better prepare teachers is a call to

envision the professional teacher as one who learns from teaching rather than as one who has finished learning to teach, and the job of teacher education as developing the capacity to inquire systematically and sensitively into the nature of learning and the effects of teaching. [This approach to knowledge production is] one that empowers teachers with greater understanding of complex situations rather than seeking to control them with simplistic formulas or cookie-cutter routines. (Darling-Hammond, 1998, p. 9)

Mentoring new teachers is considered by many to be a crucial element in reducing attrition. The role of a mentor was well described by authors of *Mentoring in Mathematics Teaching* (Jaworski & Watson, 1994):

The mentoring process offers an intensive opportunity to develop the philosophy and practice for all concerned. Mathematics teaching is a particularly interesting field in which to mentor because schools can exhibit vastly different styles of teaching and make contrasting demands on teachers. (p. 10)

A great deal of knowledge-in-practice is acquired during the first years of teaching. A master-teacher mentor who has content knowledge, knowledge of teaching, and knowledge of how to reflect on practice (Cochran-Smith & Lytle, 1999), together with time for the beginning teacher to learn from the mentor teacher, can effectively influence the knowledge-in-practice gained by the new teacher.

As educators study the manner in which mathematics teachers are prepared, consideration must also be given to alternative ways of obtaining certification to teach. Wilson, Floden, and Ferrini-Mundy (2001), in their study of teacher preparation programs, also considered the success of high-quality, alternative certification programs. They found that a more diverse pool of teachers has entered these programs but that the programs have a mixed record in terms of the quality of teachers recruited and trained.

What Teachers Carry from the University into the Classroom

The move from the university to the classroom was the focus of studies by researchers in the *Learning to Teach Mathematics* project (Borko et al., 1992, Eisenhart et al., 1993). These studies carry important messages for those trying to reform teacher prepa-

ration. The researchers examined the progress of a small number of prospective middle school teachers through their final year of teacher preparation and their first year of teaching. The researchers' goal was to "describe and understand the novice teachers' knowledge, beliefs, thinking, and actions related to the teaching of mathematics" (Eisenhart et al., 1993, p. 11). The 1992 report focused on a classroom lesson in which the novice teacher, Ms. Daniels, was unsuccessful in providing meaningful justification for the division-of-fractions algorithm, although she had the most extensive mathematics background in the group and believed that good mathematics teaching included making mathematics relevant and meaningful for students. In interviews at the beginning of her student teaching, she was unable to give meaning to division of fractions. She developed some knowledge about division of fractions during student teaching but was unable to use this knowledge to help students understand division of fractions. The methods course did not challenge her beliefs, and, in fact, she was disappointed in this class because it lacked the examples and explanations she had expected to learn.

The authors of the first study (Borko et al., 1992) noted two major implications of the study. "First, prospective teachers must be given the opportunity in their university course work to strengthen their subject matter knowledge (p. 219)," not just by increasing courses but through improvement in the curriculum and instruction offered in university mathematics departments. Second, university course work must provide prospective teachers opportunities to develop the "concepts and language to draw connections between representations and applications on the one hand and algorithms and procedures on the other" (p. 220). Prospective teachers' fundamental beliefs about teaching and learning must be challenged; they need more opportunities to strengthen their pedagogical content knowledge.

The second study in this project (Eisenhart et al., 1993) focused on the complex process of teaching for understanding. The data collected for the study were again the interviews of eight prospective teachers when they completed their professional coursework and began student teaching. Ms. Daniels was again the focus of the study. The researchers noted that "teaching for conceptual knowledge was a major theme of the mathematics education course work, an expressed commitment of the placement schools, and a part of the student teachers' own beliefs about good teaching" (p. 10). Yet although the teachers and teacher educators they observed "struggled to teach for conceptual knowledge, they often appeared to emphasize procedural knowledge instead" (p. 10).

Eisenhart et al. (1993) examined the tensions created over learning to teach for conceptual understanding in their university courses and the procedural knowledge required for their daily responsibilities in their placement classrooms. "Tensions felt by the student teachers seemed to blind them to the conceptual knowledge, and ways of teaching for conceptual knowledge, that the instructor was trying to help them learn" (p. 29). Additional tension came from the fact that administrators appeared to want teachers to teach for both conceptual understanding and procedural knowledge yet held teachers accountable only for the procedural knowledge. As a result, teachers in the schools taught primarily for procedural knowledge. Ms. Daniels received no support for the ideas she had learned in her methods course.

The researchers (Eisenhart et al., 1993) concluded by offering two recommendations. First, student teachers must be placed with teachers who will reinforce the importance of teaching for understanding. Second, the university must find a way to provide time and opportunities for the prospective teachers and the instructors "to explore and develop approaches to teaching that are considered desirable" (p. 38). This study illustrates the need to find ways for prospective teachers to develop a framework that will be strong enough to combat the apprenticeship-of-observation framework that not only is brought with them into teacher education but also is sometimes reinforced by what they observe even while in a teacher education program.

When the same questions are examined in more than one country, the results can be surprisingly similar. For example, Oldham, van der Valk, Broekman, and Berenson (1998) analyzed lesson plans of prospective teachers from four countries. All the prospective teachers had been told to prepare a lesson on area, and all were interviewed about the lesson. The researchers found that all lesson plans had some procedural approaches, and some also used conceptual approaches. Many plans were completely oriented around the teacher as the leader, and a smaller number had some student activities. The researchers believed that this model provided them with robust frameworks and insights across cultures concerning the role of the teacher.

A Model for Learning to Teach

Some researchers are proposing new ways to think about teacher preparation. The *experiment model* for learning to teach, developed by Hiebert, Morris, and Glass (2003), is a way of thinking about the objectives of coursework during teacher preparation. The authors first acknowledged and described the

tools that teachers need to become effective mathematics teachers—personal mathematical proficiency and the knowledge, competencies, and dispositions to learn to teach for mathematical proficiency. They then proposed that “as a way of making some aspects of teachers’ routine, natural activity more systematic and intensive” (p. 207), teachers need to learn to treat lessons as experiments. When designing a lesson, a teacher should be guided by the learning goals for the lesson and by hypotheses about the instructional activities planned to achieve the learning goals. “Lessons-as-experiments require, on the one hand, constructing local theories regarding the relationships between teaching and learning and, on the other hand, the tying of the theories to *this* learning goal in *this* context” (p. 209). Data are collected and analyzed in terms of the learning goals and are used to make informed decisions about future lessons. Following this model leads prospective teachers to acquire “the identity and dispositions of a professional teacher” (p. 211) with the obligation to draw from and contribute to a shared knowledge base. The authors suggested that “the principles and processes that are proposed for the generation of knowledge needed to improve teaching are the same as those that are proposed for the generation of the knowledge needed to improve teacher preparation” (p. 213).

Mathematicians’ Role in Teacher Preparation

The mathematics community has, for a number of years, actively promoted more involvement by mathematicians in the preparation of teachers. At the time the first NCTM Standards document was released in 1989, the National Research Council released *Everybody Counts: A Report to the Nation on the Future of Mathematics Education* (1989). *Everybody Counts* was a consensus document of the Mathematical Sciences Education Board, the Conference Board of the Mathematical Sciences, and the Committee on the Mathematical Sciences in the Year 2000. Mathematicians were well represented on these three boards. The document authors spoke forcefully about the role of mathematicians in the education of teachers:

Undergraduate mathematics is the linchpin for revitalization of mathematics education. Not only do the sciences depend on strong undergraduate mathematics, but also all students who prepare to teach mathematics acquire attitudes about mathematics, styles of teaching, and knowledge of content from their undergraduate experience. No reform of mathematics education is possible unless it begins with revitalization of undergraduate mathematics in both curriculum and teaching style. (NRC, 1989, p. 39)

In 1991 the Committee on the Mathematical Education of Teachers, representing the Mathematical Association of America (MAA), published *A Call for Change: Recommendations for the Mathematical Preparation of Teachers of Mathematics* (Leitzel, 1991). The authors believed that the substantive changes in mathematics required corresponding changes in the preparation of teachers. They emphasized that “although skills in mathematics must be acquired, processes, concepts, and understanding should take precedence” (p. 39) and called for a rethinking of the collegiate curriculum in mathematics that would allow future teachers to grapple with their own learning of mathematics. The most recent version of the report from the MAA’s (2003) Committee on the Undergraduate Program in Mathematics (CUPM) and the Conference Board of the Mathematical Sciences’ report (2000) *The Mathematical Education of Teachers* (discussed in the earlier section on what mathematics teachers need to know) both addressed the mathematical and pedagogical preparation of teachers and the role of the mathematics community in this work.

Another example of mathematicians’ recognition of their role in preparing teachers is the MAA initiative Preparing Mathematicians to Educate Teachers (PMET). Funding was received from the National Science Foundation for three summers of workshops for college and university mathematics and mathematics education faculty to prepare them to offer better courses for prospective teachers at the elementary, middle, and secondary levels.

The NCTM’s 2000 version of standards, *Principles and Standards for School Mathematics*, was developed with a great deal of advice from professional associations of mathematicians in the form of recommendations from various Association Review Groups that included groups of mathematicians. The considerable influence of faculty in 2-year and 4-year colleges and universities was apparent and was welcomed. Hyman Bass, in his role as the president of the American Mathematical Society, said in an interview (2001) that “mathematicians and scientists have a special responsibility that extends their traditional roles in research and education at the university level to concerns for K–12.”

ADDRESSING QUESTION 9: WHAT DO RESEARCHERS KNOW ABOUT ISSUES THAT AFFECT PROFESSIONAL DEVELOPMENT?

The people who provide professional development play, of course, a key role in whether the six previously discussed goals of professional development are

met. Who are these providers? How do they become providers? What professional development and other experiences do *they* need and receive? What role is played by school site and district administrators? What resources do they provide? If they have a vision of what should be happening in their classrooms that is different from that of teachers, how is this difference resolved? How does it affect teacher evaluation? What is the effect of district, state, and national policies on professional development? How is the success of professional development, at any level, determined? These issues are explored in this section.

Developing Professional Developers

Those responsible for providing professional development fall into several categories. In one category are university educators, particularly those in schools and colleges of education, who are frequently engaged with professional development and who may or may not receive university recognition for this work. Most mathematics departments do not reward work with teachers; thus, mathematicians who work in professional development and teacher preparation are usually established and find themselves interested in school mathematics, often because of their children's experiences in mathematics or because they are approached to teach mathematics courses for teachers. To be effective, these providers must spend time in classrooms at the level of the teachers whom they teach. In another category are those whose careers are built around professional development, such as professional providers associated with the Education Development Center or Marilyn Burns Associates. In a third category of providers are those who become teacher leaders within a school or district—they may act as resource teachers, mentors, coaches, consultants, curriculum developers, and professional development providers. In some cases they fill this role by choice—a teacher who wants more challenges might enter a master's program that provides them with knowledge and skills needed to lead. Some teachers may be recognized for their potential by administrators, other teachers, or outside collaborators and may be asked to take on additional responsibilities with coaching, mentoring, providing workshops, or working as resource teachers. Teachers as professional development providers need their own professional development (Fonzi, 2003; Sugrue, 2004). Spencer, Moon, Miller, and Elko (2000) have developed a casebook to support the development of teachers who wish to become teacher leaders.

Loucks-Horsley et al. (2003) identified three key elements for professional developers, each of which must be obtained through their own professional de-

velopment opportunities. These elements (from pp. 230–231) are summarized here:

1. *Effective professional developers must have diverse knowledge and skills*, including a deep understanding of the appropriate level of school mathematics; knowledge of how both children and adults learn; knowledge of school organization, structures, and culture; self-awareness and the ability to be self-critical; willingness and ability to learn from mistakes and successes; and an understanding of the process of implementing and evaluating changes.
2. *Professional developers use a repertoire of professional development strategies*, and the strategies needed depend partly on the role played. If a new curriculum is being implemented, the role of the professional developer might include workshops focusing on the curriculum. If a professional developer is charged with providing assistance to new teachers, mentoring and coaching skills will be required. If the content knowledge of teachers within a school is weak, the professional developer might need to focus on developing understanding of the mathematics being taught.
3. *Professional developers have their own learner community* with whom to collaborate and share. They need access to many resources and information. They need their own professional development through courses, workshops, and attendance at state and national meetings. They should not be expected to work in isolation.

The knowledge and skills needed can be accessed in a variety of ways. Colleges and universities offer courses on leadership and management. Workshops offered by, for example, the Association for Supervision and Curriculum Development (ASCD), the American Federation of Teachers (AFT), and the National Council of Teachers of Mathematics (NCTM) can provide teachers leaders with the kinds of information and skills they need. Sometimes state funding, such as for the California Mathematics Project, provides opportunities for teachers both to develop as leaders and to participate in a professional community. In large school districts, special provisions and opportunities are offered to potential leaders. In small school districts, a teacher might become a leader by being mentored by a recognized teacher leader.

Potential leaders are often recognized through their interest in mathematics, their voluntary participation in professional development opportunities of many kinds, their interest in their students' reasoning and thinking, and their abilities to work with others. "Leadership is not a personality trait but an attribute of self-development in social relationships (Grossman et al., 2001, p. 996).

Schifter and Lester (2002) studied the role of facilitators of a professional development seminar (DMI), asking whether facilitators should play an active role and, if so, what they should do and what they must understand to be able to intervene appropriately. They built on the work of Remillard and Geist (2002), who had observed DMI seminars and were drawn to study the instances that "required facilitators to make judgments about how to guide the discourse. [These moments, they argued,] arose from conflicts among the goals and commitments of the facilitators, the expectations of the participants, and the agenda of the curriculum" (Schifter & Lester, 2002, p. 6). These moments were ultimately referred to as *openings in the curriculum*.

Schifter and Lester (2002) argued for an active role by facilitators of professional development (in this case, of DMI) and said that if "instances of discontinuity between participants' ideas or beliefs and the goals of the curriculum are to provide fruitful opportunities for learning, then the facilitator must take determined action to exploit them" (p. 15). The authors described the knowledge facilitators must have to be able to exploit such moments and the type of support DMI facilitators are provided.

Responding to openings for teacher learning is not just a matter of having the right cognitive dispositions. Just as important, effective facilitation requires courage—courage to challenge the thinking of other adults; courage to redirect a discussion that is moving in an unproductive direction; and courage to face the agitation, and sometimes even the tears, that result when firmly held ideas begin to crack. This form of facilitation also demands a stance of respect for and commitment to the teachers being supported and the ideas to be explored. Perhaps this disposition is best reflected in one facilitator's injunction to herself and her colleagues: "We can do better—go deeper—than where we are now." (p. 21)

A story of teachers' becoming professional development leaders can be found in the work of Jacobs and Raynes (2001, 2002). These researchers worked with a group of primary-grades teachers in schools in low-socioeconomic communities. The professional development offered was voluntarily undertaken by

teachers who were not satisfied with their own mathematics instruction. Even though some did not even like mathematics when they began, they found that the focus on listening to children and using student thinking as a basis for instruction was compelling. During the third year of the project, teachers took on leadership roles and began to offer professional development to other teachers in their schools.

The one underlying principle that all of the teacher leaders embraced was that participation in the learning opportunities they designed needed to be voluntary. They recognized that real change is not always smooth. It requires risk taking and a willingness to critically examine one's own practices—often practices that have been taken for granted for many years. They were convinced that this type of professional development could only succeed if a teacher wanted to change. Therefore, they targeted their support toward teachers who volunteered to participate and work to improve their mathematics instruction. On the other hand, the teacher leaders also made sure that their learning opportunities did not preclude other teachers from initially participating peripherally and then becoming more deeply involved at a later date. (Jacobs & Raynes, 2001, pp. 4–5)

Teachers become leaders through a number of routes (Fonzi, 2003). Many begin simply, by honing their own practice or presenting at a local or state meeting for teachers of mathematics. When a teacher first presents at a meeting or conference, she or he has probably been prompted to do so by someone such as a university professor or project director. Some teachers are asked to coach new teachers or act as supervisors of prospective teachers while they are concurrently themselves teaching. Others become engaged in action research and want to share their results. When teachers become engaged in projects and work with other teacher leaders, they gain self-confidence while they are gaining more knowledge and are being asked to provide leadership in many ways. Some will become full-time leaders, working with a school district as a mathematics coordinator or as a resource teacher for a number of schools.

University educators also need to develop new ways of providing professional development. Many of those ways have been described in various sections of this chapter.

Not only will professional developers need to build their repertoires beyond workshops and courses, they will also need to learn how to manage these repertoires in relation to their goals and the context within which they are working. Not only will professional developers need to develop teachers as individuals, they

will also need to learn how to develop whole communities of practice. And, finally, not only will professional developers be held accountable for adding new techniques or skills to teachers' repertoires, but also for teachers' successful enactments of valid practices that raise student achievement. Never before have the stakes been higher or the learning demands greater. (Stein et al., 1999, pp. 267–268)

Supporting Professional Development

Professional development is the key to improving education and effecting changes in student achievement (e.g., Adler, 2000; Guskey, 1995), but only when it is offered in a form that is relevant to teachers and addresses their specific needs and concerns (Guskey, 1995). In addition, effective professional development must “provide regular opportunities for participants to share perspectives and seek solutions to common problems in an atmosphere of collegiality and professional respect” (Guskey, 1995, p. 121). The authors of *Adding It Up* (NRC, 2001a) also advocated that when teachers come together to work on practice, they should be provided with opportunities that will result in serious and substantive professional development and that these opportunities can result only when those providing professional development have strong mathematical backgrounds and understand students' thinking, students' learning, and teachers' thinking about both mathematics and their own instructional practices.

Resources for and of Professional Development

Professional development, particularly in mathematics and science, can lead to changes that result in teaching for understanding; professional development can also be viewed as a resource for change. “Schools can support teacher change by responding to teacher learning” (Gamoran, 2003, p. 66). Gamoran and his colleagues studied six schools in which teachers and researchers collaborated to investigate factors that brought about teaching for understanding. They found evidence that resources that support teaching for understanding include “time and other material conditions, expertise from outsiders and insiders, and communities of teachers within schools that facilitated the professional development collaboratives” (p. 68). They also found that teaching for understanding in mathematics and science can itself generate resources that can help sustain such teaching. The social and human resources generated included the development of professional communities and the increased knowledge of teachers. Material resources generated in their study included curriculum materials and addi-

tional resources that could be used to facilitate teachers' work in the classroom.

For professional development to be “high quality, sustained, and systematically designed and deployed to help all students develop mathematical proficiency” (NRC, 2001a, pp. 409–410), teachers need the time and resources to engage in sustained efforts to improve their practices.

The Roles of Administrators

Administrators have a major role in bringing about change that can foster teaching for understanding.

Leaders who wish to support teaching for understanding must not only allocate resources, but also foster new ways of generating resources; and they must deal with not only material resources (time, materials, and compensation), but also human resources (teachers' knowledge, skills, and dispositions) and social resources (relations of trust and collaboration among educators). Although educators in positions of authority (e.g., principals and curriculum coordinators) have special roles to play in responding to these tasks, they cannot provide all the leadership necessary. Instead, leadership must be distributed throughout the organization, including teachers who take on leadership roles that support their learning and that of their colleagues. Developing capacity to support teaching for understanding involves recognizing new leadership demands and engaging in a distributed response. (Gamoran, Anderson, & Ashmann, 2003, p. 107)

Principals in the Gamoran et al. (2003) study did not play a major role in articulating a vision for the mathematics and science teaching in their schools beyond fostering such general visions as “all students can learn.” Administrators who played a role in articulating this vision were more often district specialists and outside researchers. Teachers themselves usually began professional development with visions of what they intended to gain, but the tempering of these visions by what they learned and by their work with other teachers led to clarification of how to enact what had been learned. “Thus the professional development groups developed their own visions of what teaching for understanding meant in their particular settings” (p. 115).

Nelson (1998), in her *Lens on Learning* project, found that many administrators view mathematical knowledge as hierarchical, with facts and operational facility coming first for students and with the possibility of students' being taught and learning concepts by melding them with practice of math facts. Thus, parents' demands for instruction on skills and facts are seen as reasonable. When faced with calls for reforming mathematics instruction, administra-

tors do not understand that teachers need deeper understanding of mathematics; rather, they think that the purpose of professional development is to teach teachers new techniques and that their role is to support teachers in trying these new techniques. Other administrators, who view mathematics as a way of knowing and thinking, can explain to parents that mathematics focused on understanding might seem less efficient but is more enduring. These administrators understand that beliefs must change before practice can change, that fundamental instructional changes are called for, and that this process takes time. They recognize that professional development is a process rather than an event.

In Nelson's (1998) study, administrators "developed views of what a professional development project that aimed at reconceptualization might look like" (p. 204). They came to understand that the reforms being introduced in schools would require a deep examination of beliefs and ideas about learning, teaching, and the nature of mathematics and mathematical knowledge. Just as teachers and children need opportunities to investigate new ideas and restructure their conceptual understanding, so do administrators. Some of Nelson's work is reflected in the statements made in the NCTM (2000) Standards' comments on the role of administrators:

Administrators at every level have responsibilities for sharing the instructional mission in their jurisdictions, providing for the professional development of teachers, designing and implementing policies, and allocating resources. . . . When administrators themselves become part of the mathematics learning community, they develop deeper understandings of the goals of mathematics instruction. They can understand better what they are seeing in mathematics classrooms (Nelson, 1999) and can make more informed judgments about the curricular, technological, and pedagogical resources teachers need. (p. 376)

Knapp (2003), too, has noted that administrators need professional development to fully understand what improving learning, teaching, and instructional development entails.

The Challenges Faced

Challenges to change go hand-in-hand with support needed for change to take place. Gallucci (2003) advocated that schools be organized so that vision setting and professional learning are encouraged and teachers are involved in communities of practice. He recognized that the most serious barriers to the formation and longevity of such communities include time and other institutional structures. Overcoming these

barriers invariably translates into money. The major challenge is then, to no one's surprise, cost (Garet et al., 2001).

The nature of challenges that faced school professionals in the Gamoran et al. (2003) study were described in the chapter by Anderson (2003):

- Providing resources for classroom teaching. For mathematics these resources include tools and teaching materials, ability to understand and respond to students' reasoning, understanding of mathematics and how children learn mathematics, understanding of differences among students, and ability to develop in classroom learning communities social norms that promote learning;
- Aligning purposes, perceptions, and commitments. Teachers, parents, and students often value traditional schools because they are safe, predictable, and do not make excessive demands. "Yet it seems unlikely that teaching for understanding can become a widespread practice without changes in professional cultures" (p. 18). Complete transformation of schools is unlikely; the alternatives are conflict over resources, standards, and practice; *compromise*, which is accommodation instead of transformation, or *coexistence*, which is described as accommodation together with transformation.
- Sustaining teaching for understanding. To sustain such teaching requires both collective and individual efforts that depend on working interdependently rather than dependently, with leadership from professional communities.

Additionally, even teachers and administrators who support teaching for understanding often do not share researchers' notions of what teaching for understanding entails.

Finally, a seemingly trivial but an actually widespread and serious challenge was noted by Guskey (1995)—that many teachers think any reform is once again a fad that will pass, and they share this view with younger colleagues who express concerns over implementing new programs.

Evaluating Professional Development

Although teachers' satisfaction with professional development is desirable, more information is needed to judge how much teachers have learned, whether teachers are likely to change their instructional prac-

tices, and whether student achievement is likely to be affected (Guskey, 1995; Mumme & Seago, 2002; Wilson & Berne, 1999). One effective way to evaluate professional development is to consider what elements research has shown to be successful—for example, Table 5.1 from earlier in this chapter provides three such lists (one by Hawley and Valli, 1999; a second by Elmore, 2002a; and a third by Clarke, 1994) that contain similar elements expressed in different ways.

The most comprehensive guidelines for evaluating professional development (Guskey, 2000) are quite recent. Understanding Guskey's models requires understanding his view that professional development must be systemic to be effective:

Experience has taught us that you cannot be successful in any endeavor designed to improve student learning by focusing only on professional development (Sparks, 1999). Too many other things in the system affect student learning, such as the curriculum, assessment, school organization, materials, support, and leadership. The dynamic interaction of these elements requires a systemic approach (p. 69)

Guskey presented two evaluation models, one focused on improvement of student learning and the other on the five critical levels in professional development evaluations.

The need to clarify the “precise nature of the relationship” (Guskey, 2000, p. 72) between professional development and student achievement has become more pressing, particularly because government funding agencies consider this relationship imperative in their consideration of future funding. Guskey's first model includes three factors that influence the quality of professional development: content characteristics, process variables, and context considerations. The first two of these factors have been discussed in previous sections of this chapter. The context characteristics “refer to the ‘who,’ ‘when,’ ‘where,’ and ‘why’ of professional development” (p. 74). All three factors are important to the success of professional development, primarily because “most content and processes must be adapted, at least in part, to the unique characteristics of the setting (Firestone & Corbett, 1987; Fullan, 1985; Huberman & Miles, 1984)” (Guskey, 2000, p. 74) but also because if any one of these three factors is neglected, the effectiveness of the entire experience will be reduced.

Guskey's (2000) second model consists of five levels of information that address a broad range of *what* questions related to professional development but do not provide much information that contributes to learning *why*. Still, because each higher level builds on the previous levels, this model can be particular-

ly useful in evaluating professional development in mathematics. At Level 1, participants' reactions are considered. Collecting information at this level is relatively easy and usually consists of asking questions about comfort level, such as the temperature of the room. Guskey referred to these reactions as *happiness quotients*. At Level 2, the focus changes to measuring the knowledge, skills, and activities in which participants engaged and, at times, the attitudes changed by the experiences. Level 3 focuses on the organization—what support is provided and what is changed. For example, professional development may not result in desired change because of organizational characteristics. Gathering such information is not easy but is necessary to inform future change initiatives. Level 4 focuses on participants' use of new knowledge and skills. The central question at this level is whether what participants learned made any difference in their professional practices. “The key to gathering relevant information at this level rests in the clear specification of indicators that reveal both the degree and quality of implementation. In other words, how can you tell if what participants learned is being used and being used well?” (p. 84). At Level 5, the level of most sophisticated teaching, student learning outcomes are measured. Both intended and unintended learning are important to consider, and, thus, multiple measures are necessary. Measures from student records should be augmented by affective indicators, and information on factors such as “student self-concepts, study habits, school attendance, homework completions rates, and classroom behaviors” (p. 86) could be included.

Guskey (2000) warned that “evaluation at any of these five levels can be done well or poorly, convincingly or laughably” (p. 86) and that effectiveness at one level tells no story about effectiveness at the next level.

Evaluators of professional development in mathematics can also profit from Borasi and Fonzi's (2002) identification of nine learning needs of teachers. The authors, in an NSF-requested review of professional development that supports school mathematics, analyzed promising professional development experiences by discussing each experience in terms of each of these needs, some of which were identified earlier but are listed in full here because they can be used as an inventory for evaluating professional development at both a formative and a summative level.

1. Developing a vision and commitment to school mathematics reform,
2. Strengthening one's knowledge of mathematics,
3. Understanding the pedagogical theories that underlie school mathematics reform,

4. Understanding students' mathematical thinking,
5. Learning to use effective teaching and assessment strategies,
6. Becoming familiar with exemplary instructional materials and resources,
7. Understanding equity issues and their implications for the classroom,
8. Coping with the emotional aspects of engaging in reform, and
9. Developing an attitude of inquiry toward one's practice.

No single model emerged from their analysis of successful professional development programs—many programs met these nine needs, but in different ways.

Policy Issues

Policy matters such as how schools are structured, how curriculum is adopted and implemented, how teachers are recruited, how teachers are evaluated, and how testing influences what and how mathematics is taught are relevant, in some manner, to the professional development of teachers and to teacher preparation. But *policy matters* can also refer to the fact that policy *matters*—schools and universities cannot provide professional development and teacher education independent of local, state, and national policies. Many of us in mathematics education have for too long ignored the policy implications of what we do.

Teachers teach from what they know. If policymakers want to change teaching, they must pay attention to teacher knowledge. And if they are to attend to teacher knowledge, they must look beyond curriculum policies to those policies that control teacher education and certification, as well as ongoing professional development, supervision, and evaluation. (Darling-Hammond, 1990, p. 240)

Policy makers have the power to create an infrastructure to support professional development in ways that involve both school-level staff as well as external groups (Knapp, 2003). Small districts in particular often lack personnel to undertake meaningful, long-term professional development. In California, during a time of budget surplus, the state funded a Professional Development Institute (PDI) with sites at many areas across the state where teachers could participate in summer and year-long professional development, primarily offered by universities and primarily focusing on increasing content knowledge and pedagogical content knowledge. Teachers received attractive sti-

pendents and in many cases also earned university credit. The PDI was intended to continue indefinitely but was abandoned in 2004 when the budget surplus vanished. Before that time many hundreds of teachers took advantage of the professional development offered by PDI. This initiative serves as an example of how policy makers at the state level can provide a large-scale infrastructure for professional development.

District- and School-Level Policies

Teachers work in policy environments, whether or not they think about their work in this way. Policy makers at the district level play an important role in professional development by “creating a variety of venues and occasions for learning, helping to establish a focus and incentives for professional learning, and enabling teachers to connect their learning experiences over time” (Knapp, 2003, p. 147). At the site level, policy decisions can affect the allocation of resources, the scheduling of classes, the extra-curricular duties of teachers, and other such matters. Hargreaves (1995) suggested that some site-level professional development focus on political agendas and concerns: “Teachers who have been introduced to the micro-politics of schooling in relation to their own institutions can achieve breakthroughs in insight, action, and effectiveness that help them secure support and resources for their students (Goodson & Fliesser, 1992)” (p. 18). The Urban Mathematics Collaboratives funded by the Ford Foundation in the early 1990s serve as an example of teachers' becoming empowered after coming to understand how they could affect decisions regarding their work.

At the district level, administrators make decisions regarding curriculum, school structures, assessment, and opportunities for professional development. Teachers are expected to make changes in their instructional practices to align with state and national standards, but if they have few opportunities and few incentives to learn how to change practice, they are less likely to make the changes dictated by policy (Spillane, 2000). Policies that call for changing instruction require far more than mandates.

Coming to know involves the reconstruction of existing knowledge rather than the passive absorption of knowledge (Anderson & Smith, 1987; Confrey, 1990). Moreover, sense making is influenced by the social, physical, and cultural contexts of the sense maker. (J. Brown, Collins, & Duguid, 1989; Resnick, 1991). Applying these ideas to policy implementation, we notice that implementers construct what it is policy asks of them, and it is these understandings they respond to in implementing policy. Thus, the policy stimulus is not all that matters: Implementers' beliefs, knowl-

edge, and experience, as well as their situation, also influence the ideas they come to understand from policy. (Spillane, 2000, p. 146)

If policies are to lead to the making of needed changes, those policies must be aligned with one another. Teachers and administrators might ask questions such as

Does a new curriculum framework stress “implementation of texts,” thereby describing a passive teacher and student role, or are teachers assumed to participate in the construction of practices that start with the students’ experiences and needs and aim at student outcome standards? Does an assessment system evaluate student understanding, or does it test for rote recall of disconnected facts? Do teacher evaluation systems look for teaching behaviors aimed at keeping students quiet or for practices that engage students actively in their learning? Do administrator licensing standards require that principals know how students learn and how teachers teach for understanding, or do they stress noninstructional matters? Do school accountability requirements enforce current, highly fragmented bureaucratic structures and uses of time, or do they allow for more integrated and student-centered forms of staffing and fund allocations? (Darling-Hammond & McLaughlin, 1996, p. 213)

If high quality professional development is taken seriously as the most important mechanism for improving teachers, then schools must make a major financial commitment to providing it. But schools can rarely afford to provide high-quality professional development to all the teachers in a district, and choices need to be made. In a report of a study by Garet et al. (2001), the researchers were blunt about their findings:

A major challenge to providing . . . high-quality professional development is cost. Schools and districts understandably feel a responsibility to reach large numbers of teachers. But a focus on breadth in terms of number of teachers served comes at the expense of depth in terms of quality of experience. Our results indicate a clear direction for schools and districts: in order to provide useful and effective professional development that has a meaningful effect on teacher learning and fosters improvement in classroom practice, funds should be focused on providing high-quality professional development experiences. This would require schools and districts either to focus resources on fewer teachers, or to invest sufficient resources so that more teachers can benefit from high-quality professional development. (p. 937)

One way to think about this challenge is to consider seriously, at least in Grades 5 and up, that teach-

ers specialize, that they receive high-quality professional development in one or two areas but are not expected to be experts in all the subject areas in the elementary school curriculum. Hiring of mathematics specialists has been frequently advocated in elementary school for teachers of Grades 5 and above (CBMS, 2000; Learning First Alliance, 1998; NCTM, 2000; NRC, 1989).

Evaluating Teachers

The manner in which teachers are evaluated is usually a site-based matter undertaken with some guidelines from the district. All too often, the on-site evaluation of teachers works against the very reforms that a district is attempting to make (Spillane, 2000). Spillane distinguished *form-focused understanding*, which refers to “pedagogical forms of learning activities, including students’ work, instructional materials, and grouping arrangements” (p. 154), from *functional understandings*, which focus on “what counts as mathematical knowledge, doing mathematics, and learning and knowing mathematics” (p. 154). When teachers receive professional development that leads them to attempt to teach for functional understanding but are then evaluated by administrators who focus on form, the mismatch can lead to discouragement and probably to their abandoning teaching for functional understanding. Of course, the situation could be reversed—a teacher who is form-focused may be evaluated on her students’ success in achieving functional understanding, confusing the teacher who does not know how to teach for functional understanding. Darling-Hammond and McLaughlin (1996) extended these ideas to consider administrative leadership:

To support teaching for understanding and the professional development necessary to it, new forms of teacher evaluation will need to emphasize the *appropriateness* of teaching decisions to the goals and contexts of instruction and the needs of students, rather than focusing on teachers’ adherence in prescribed routines. Evaluation must be conceived not as a discrete annual event conducted by supervisors making brief visits bearing checklists but as a constant feature of organizational and classroom life for practitioners. . . . A critical role for administrative leadership involves creating and sustaining settings in which teachers feel safe to disclose aspects of their teaching, admit mistakes, try and possibly fail, and continue to seek out advice and counsel. (p. 215)

State- and National-Level Policies

A great deal of variance is found in the ways that state policies affect teachers. If the policies regard-

ing professional development, teacher preparation, teaching expectations, curriculum, and testing are aligned, achievement scores are likely to increase. For example, Wilson, Darling-Hammond, and Berry (2001) undertook a case study of a 15-year policy in Connecticut to recruit, prepare, and support teachers; this policy, combined with strong and consistent leadership and investment in research and development, led to success in creating effective teachers and increasing achievement. The intention of the Connecticut State Department of Education was to move the state's educational system toward a "conception of teaching grounded in strong disciplinary and pedagogical content knowledge as well as in a broad research base about teaching and learning" (p. 25). The Department staff realized that continued school improvement depended on a coherent system of continued educator improvement, which required aligning teacher preparation, teacher evaluation, and professional development with the state's student and teaching standards. The primary focus of all professional development is to provide teachers with the skills, knowledge, and abilities to improve student learning. One factor in the success in Connecticut has been the focus on improving teacher preparation and supporting beginning teachers. Licensing requirements were changed to strengthen clinical experiences for beginning teachers. All teacher preparation programs were required to demonstrate that their graduates understood the state's student standards and teaching standards. A strong support system for new teachers was put into place. "By systematically educating mentors who support and evaluate novice teachers, the support system has increased capacity in beginners and veterans alike" (p. 4). The system has high standards for teachers, with strong competition for openings. Only top-quality teachers are hired.

Another large-scale study of state education reform in mathematics was undertaken by D. K. Cohen and Hill (2001) in California. In 1985 a new state framework for mathematics turned from a focus on basics to a focus on teaching for understanding. The reforms advocated in this framework were continued by the state department of education when they contracted outside agencies to develop replacement units that could be used for particular curriculum topics, such as fractions. A state adoption of textbooks forced publishers to conform (as much as was possible) to the 1985 framework. Large-scale professional development was undertaken. Cohen and Hill studied this reform beginning in the early 1990s by focusing on professional development as a way to examine the relations between policy and practice. They found that only a small portion of teachers were affected by the

reform, partly because of the fragmentation of educational governance and partly because of missed opportunities on the part of the reformers. Their analysis did show that when assessment was consistent with professional development, "teachers' opportunities to learn pay off in students' math performance . . . [showing] the significant role that the education of professionals can play in efforts to improve public education" (p. 148). They pointed out that the opportunities to study policy as they had done are rare because states and districts are often unwilling to mandate curriculum and thus are unable to focus professional development on the curriculum. Unfortunately, the policy makers and reformers in California

made no attempt to learn systematically about how the reforms played out in schools and classrooms. As a result, they had no evidence to inform either their work or public debate, and when skeptics charged that the reforms were diluting instruction to basic math skills, state officials and professionals had no basis on which to investigate or respond. (p. 187)

One could disagree with this analysis. California state officials and professionals did attempt to establish ways of recording learning by developing first the California Assessment Program (CAP), which suffered a line-item veto by the governor, then the California Learning Assessment System (CLAS), with items in reading and mathematics that led to a public outcry before also being abandoned. Both of these state examinations were based on the state framework, which had many similarities with the NCTM Standards, and the tests were held in high regard by most mathematics educators who knew about them. New policy makers in California believed that mathematics reform had gone too far, and much has changed on the educational scene. However, the need for professional development continued to be recognized, and in 2000 the state legislature funded the Mathematics Professional Development Institutes previously described. Hill and Ball (2004) evaluated these Institutes by developing pretests and posttests to measure teacher learning; these tests were administered at most sites. Their report provided evidence that statewide, large-scale, funded professional development is beneficial in terms of teachers' learning of both content knowledge and pedagogical content knowledge.

Today, throughout the country, a major factor influencing what teachers teach is the student-testing program within the state, further complicated by the federal rulings regarding the testing for Academic Yearly Progress (AYP) as mandated by the No Child Left Behind legislation and the threat of sanctions for schools that do not meet the required increases in

scores. These new demands have been met with criticism—severe in some cases. The federal requirements of upward change each year ignore much of what is known about testing. For example, Linn, Baker, and Betebenner (2002) noted that “interpretations of the law also should recognize the volatility of school-level results from year to year and provide states with latitude to identify ways of reducing that volatility” (p. 16). After examining the psychometric work pertaining to national testing, Tucker (2004) had this to say.

A central tenet of the No Child Left Behind Act is that educational improvement at a school can be measured by comparing student scores on standards-based tests from one year to the next. An important question about such a strategy that has gotten surprisingly little attention, is this: How accurate are such year-to-year comparisons? The answer is that they are much less accurate than people assume—in some cases, wildly inaccurate. . . . Most standards-based tests are based on a technical psychometric methodology called Item Response Theory, or IRT. Item response theory makes many critical assumptions, both of a practical nature—in getting all the technical details of test development right—and of a theoretical nature—in its one-dimensional model for assessing student’s knowledge. Few states have the resources to implement IRT-based tests with the attention to detail they require. Such tests should be reliable for assessing standard procedural skills, such as solving a quadratic equation. Unfortunately, the more thoughtful and thus unpredictable a test, the more likely equating methods will misperform.

In addition to the psychometric problems of large-scale testing, equity is an issue, primarily because low-performing schools are most likely to have second-language students and students of color. Elmore (2002b) argued that testing is not the way to solve problems of equity.

The threat of such measures is supposed to motivate students and schools to ever higher levels of achievement. In fact, this is a naïve view of what it takes to improve student learning. Fundamentally, *internal* accountability must precede *external* accountability. . . . Giving test results to an incoherent, atomized, badly run school doesn’t automatically make it a better school. A school’s ability to make improvements has to do with the beliefs and practices that people in the organization share, not with the kind of information they receive about their performance. *Low-performing schools aren’t coherent enough to respond to external demands for accountability.* (p. 4)

Beyond the problems of psychometrics and equity, the phenomenon of tests driving what is taught

is not necessarily bad. Instead, testing can be one way to drive desired changes that are not otherwise taken seriously.

D. K. Cohen and Hill (2001) advocated taking more seriously the need for better research about education policy and practice: “Such work could contribute to efforts to improve instruction and to understanding improvement. Research of this sort could not determine decisions, but it can inform understanding, debate, and, in some cases, decisions” (p. 189).

ADDRESSING QUESTION 10: WHERE DO WE STAND NOW, AND WHAT COMES NEXT? PERSONAL THOUGHTS

The recognition that mathematics-teacher change is a process, rather than an event, is being successfully used in some places to justify the need for resources for multiyear, coherent programs of professional development. Policy makers, including district and state personnel, are recognizing that workshops lacking continuity and direction have little effect on teachers. Yet large numbers of teachers remain untouched. The efforts described in this chapter provide evidence that good professional development *can* make a difference. However, “if we are to move past these and other ‘pockets of wonderfulness,’ we must advance to another level of sophistication as mathematics educators and change facilitators” (C. Seeley, personal e-mail, September 6, 2004). Scale-up of successful professional development programs remains the greatest challenge we educators face.

I once thought of teacher preparation as a solution to this problem. Teachers who are well prepared at universities, I believed, could lessen the need for professional development, which seemed almost too enormous a task to undertake. With so many teachers close to retirement, the education system would have new, better prepared teachers to replace them. I came to realize, of course, that this solution was untenable. Preparing teachers to teach mathematics in four short years, often beginning with students who have little real understanding of mathematics, is not possible. Rather, our goal should be to prepare them for future learning, in part because at the university we can focus only on learning-for-practice (and not enough of that), and we know they have much more to learn *in* practice while teaching. Even the arena of preparation that belongs to the university is an undertaking made doubly difficult by transient graduate-student instructors, by mathematicians who fail to model good instruction, by methods courses taught as though all subjects are

taught in the same way, and by what sometimes seems to be random placement for student teaching. Even one of these problems can have a significant influence on a teacher's preparation to teach. Few future teachers have an idea of what understanding mathematics for teaching entails, and, unfortunately, few have sufficient opportunities during their university coursework to change this state of knowledge of mathematics. The career stages described by Day and Sachs (2004) earlier in this chapter need to have a zero stage before the first stage—it is “Preparing to become learners of teaching practice.”

Unfortunately, professional developers, whether coordinators, university professors, or private providers, tend to reinvent the wheel time and time again. Yet according to Elmore (2001),

We know enough about teacher learning, its connection to practice, and the connection of practice to student performance to design some fairly powerful interventions that would reduce variability in practice and increase the consistency with which teachers influence student learning in powerful [ways]. The main constraint is not knowledge; [it is] the *social construction of teaching*. That is, most of our theories about student learning can't be deployed because the people who teach don't believe them—in fact, actively oppose them—and the organizations in which teaching occurs—schools and school systems—are designed in ways that make it virtually impossible to use them. (p. 10)

Why do teachers not believe theories about student learning? As researchers, we have not taken seriously our job to relate research to practice, to work with teachers on classroom research that builds on and tests the theories about student learning.

Research on teachers' knowledge, beliefs, preparation, and professional development is a fairly new phenomenon. Nelson (1997) noted that the mental life of teachers has been a topic of study only since the mid 1970s. For many years, research on teaching used a process-product paradigm that focused on teacher behaviors rather than on teachers' thinking. The 1991 Professional Standards from NCTM was not a research document, but it provided researchers with a clear agenda for undertaking research that could confirm or disconfirm the ideas in that document, and it moved researchers beyond process-product research into new paradigms for professional development, many of which have been discussed in this chapter.

One paradigm in particular illustrates ways that professional development has evolved to encompass features quite different from the traditional workshop type of professional development (Stein et al.,

1999). These features (summarized from pp. 239–241) include

- *Teacher assistance embedded in or directly related to the work of teaching.* Teachers need assistance that focuses on their day-to-day efforts to teach in these new and demanding ways. Such assistance is best provided by support directly focused on an individual teacher's practice . . . or by guided group discussions surrounding selected authentic artifacts from practice, such as student work, instructional tasks, or cases of instructional practice written by and for teachers (Ball and Cohen, 1999).
- *Teacher assistance grounded in the content of teaching and learning.* A characteristic of new calls for reform focus on meeting high standards associated with the major school subjects (Little, 1993). . . . Teachers frequently need to encounter the discipline as learners themselves before grappling with how to teach it (Ball, 1991). Professional development should give them the opportunity to do so.
- *Development of teacher communities of professional practice.* The new paradigm for professional development encourages collegiality among teachers to counter the isolation typical of teaching. . . . A new set of goals has been placed under the purview of professional development, including community building; the development of teachers' capacities to explain, challenge, and critique the work of peers (Lord, 1994); and the development of teacher leaders (Lieberman, Saxl, & Miles, 1988).
- *Collaboration with experts outside the teaching community.* Teachers cannot be expected to be knowledgeable about all aspects of school reform, subject-matter standards, or professional practice. . . . Outside experts—often university-based educators—bring fresh perspectives, ideas about what has proven successful elsewhere, and an analytic stance toward the school-improvement process (Little, 1993).
- *Consideration of organizational context.* Teachers perform their work within multiple contexts, including schools, districts, communities, and states; the values and established procedures each have an impact on classroom practice. Professional developers must carefully analyze the constraints and alternatives offered by each of these contexts, ranging from the unwritten cultural norms to explicit regulations and policies.

This emerging paradigm moves from providing workshops and focusing on techniques to building teachers' capacities, including strong content knowledge and a variety of ways to build practice on children's thinking and clear learning goals. The new paradigm requires "a new conceptual grounding, one that also incorporates theories of how teachers learn within organizations and through interactions with others (Stein & Brown, 1997)" (Stein et al., 1999, p. 265).

Too little has been said here about connecting teachers with research. Some action research was discussed, but here I mean something more. How do teachers come to know about relevant research? What is the responsibility of researchers in sharing their research? What is the responsibility of the researcher in undertaking research that will be relevant to teachers? What is the responsibility of teachers in seeking out research that can help them be better teachers? What is the responsibility of teachers in sharing with researchers their issues and concerns and questions that could be addressed by researchers? What is the responsibility of researchers *and* teachers in mutually identifying and undertaking meaningful research? In a 2003 *Journal for Research in Mathematics Education* editorial, Silver used the metaphor of border crossing to describe the work of researchers and teachers—each traveling across a border to meet the other, with the values of each group as the currency to be exchanged.

In the research community, the valued currency is theory. . . . Work that contributes to the development or refinement of theory is highly valued. In contrast, across the border in the land of educational practice, the valued currency is practical application. Work has value in this community to the extent that it can be directly applied to the improvement of some important domain of practice. (p. 183)

The metaphor of border crossing can be thought of as a cyclic activity, wherein additional knowledge is created when additional conversations are held between researchers and practitioners. Ruthven (2002) described these cycles in phases.

In one phase of this cycle, scholarly knowledge is (re)contextualized and activated within teaching, stimulating (re)construction of craft knowledge. In the complementary phase, craft knowledge is elicited and codified through researching, stimulating (re)construction of the scholarly knowledge. In both phases, conversation involves the filtering and reformulating of knowledge: only certain derivatives of scholarly knowledge will prove capable of being productively incorporated within craft knowledge;

equally, only some derivatives of craft knowledge will prove able to be fruitfully appropriated as scholarly knowledge. (p. 595)

Situations that facilitate these conversations are needed. At times, a researcher seeks access to classrooms to undertake research. At other times, teachers approach a researcher for assistance in answering some question. These are not ideal situations—in the first, the teachers have probably had little or no input into the research; in the second, the particular question asked is not a research question. Both sides must be open and seek opportunities for conversations. At times, outside agencies bring teachers and researchers together and provide opportunities for sharing.

On a final note, in the introduction to this chapter I focused on the national need for more and better teacher education. Education, however, is a global enterprise, being shaped by government policy universally (Day & Sachs, 2004). Although different countries have different educational systems, the recognition that teachers must be well prepared is universal. At the time I began to write this final section of this chapter, I received notice of the Fifteenth International Congress on Mathematics Instruction (ICMI, 2004) Study, The Professional Education and Development of Teachers of Mathematics. The study focus was selected because "the continued development of teachers is key to students' opportunities to learn mathematics" (p. 1). The call for participants noted that the past decade "has seen substantial increase in scholarship on mathematics education and development" (p. 3), but much remains to be learned about the "knowledge, skills, personal qualities and sensibilities that teaching mathematics entails, and about how such professional resources are acquired" (p. 3). This invitation to participate provided lists of topics and questions that could serve as a research agenda throughout the world. A list of topics dealing with teacher preparation included the structure of teacher preparation, recruitment and retention, curriculum of teacher preparation, the early years of teaching, the most pressing problems of preparing teachers, and the history and change in teacher preparation. In the strand on professional development questions rather than topics are proposed:

What sorts of learning seem to emerge from the study of practice? In what ways are practices of teaching and learning mathematics made available for study? What kinds of collaboration are practiced in different countries? What kinds of leadership help support teachers in learning from the practice of mathematics teaching? What are crucial practices of learning from

practice? How does language play a role in learning from practice? (p. 7).

The answers are often universal, and we in the United States are too often insular and not open to answers found in other countries. Thus, more than one border needs to be crossed while we, researchers, administrators, policy makers, and teachers, seek to better educate teachers for the purpose of improving student learning.

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MATHEMATICS TEACHING AND CLASSROOM PRACTICE

Megan Loef Franke

UCLA

Elham Kazemi

UNIVERSITY OF WASHINGTON

Daniel Battey

ARIZONA STATE UNIVERSITY

Ms. Michaels began her third-grade mathematics lesson by posing a problem for all her students to solve. The problem drew on an idea students raised the previous day regarding the equal sign. Ms. Michaels continued to pose a series of problems intending to build and extend students' ideas. She sequenced the problems carefully; each one built mathematically on the previous one. Ms. Michaels reminded the students about what they had discussed the previous day and tried to get them re-engaged with the same mathematical ideas. As the lesson went on, she asked students general questions about how they would solve the problems and why. Students shared their ideas, usually with the whole class, while Ms. Michaels focused them on the important mathematical ideas.

In another classroom, at the very same school, Ms. Jimenez's third-grade class was working on a similar set of mathematical ideas. Ms. Jimenez also created a series of problems to pose. The sequence of problems she created did not develop or build on

a coherent mathematical idea. The students were asked to solve the first problem and discuss it with their partner to see if they agreed and could explain their solutions. As Ms. Jimenez circulated the room and listened to students explain to each other, she asked questions that focused on prompting students to further detail their thinking. She asked students to share publicly, and again she and her students asked questions about detail and about why a student had taken a particular approach.

Student participation varied greatly in these two classrooms. As Ms. Michaels worked through the sequence of problems, an increasing number of students stopped participating in both solving and sharing, while in Ms. Jimenez's class, a majority of students continued to participate throughout the sequence, both in solving the problems and in the sharing and discussion that followed.

Advances in theory and research would characterize aspects of the classroom practice depicted in both

cases as productive. The teachers listened to students' mathematical thinking and tried to use it to encourage conversation that revolved around the mathematical ideas in the sequenced problems. Yet, these brief vignettes only depict the skeleton of what occurred in the classrooms. They convey little about what shaped student participation, how students' interactions arose from the social space of the classroom, and how the many discursive storylines fed into the way students engaged with one another and the teacher. Clearly, teaching is not just about starting with mathematically rich problems, even ones connected to what students are thinking. And it is also not just about listening to students and asking them to describe their thinking. But these are central parts of the work, and although researchers in the field are beginning to agree on what constitutes central features of classroom practice, we have much to do to elaborate the details. In particular, we need to consider the relationship between particular classroom practice and opportunities for students to engage. How is student participation negotiated in the classroom? In what ways are students able to participate across various classroom practices? This work is critical because students' ways of being and interacting in classrooms impact not only their mathematical thinking but also their own sense of their ability to do and persist with mathematics, the way they are viewed as competent in mathematics, and their ability to perform successfully in school.

In reviewing the research on classroom practice in mathematics, we have a dual focus. We attend to what teachers like Ms. Michaels and Ms. Jimenez do to structure opportunities for students to understand and learn mathematics. In the vignettes show, the focus is on the teacher and how we understand the teacher's role in classroom mathematical work. Our intent is to also consider classroom practice as experienced by students, attending to what students bring and take from their participation in the social context of the classroom.¹ In maintaining this dual focus, we will summarize what we do know about teaching and classroom practice while attending to what remains underdeveloped or underspecified.

OVERVIEW OF CHAPTER

This chapter begins with making explicit the views of teaching that guided our review. We discuss in detail the

theoretical perspectives about teaching and learning that frame our views of classroom practice. We then use what we know from the research literature about mathematics teaching and classroom practice to situate our review of the three features of classroom practice we see as currently most central in helping us come to understand the teaching and learning mathematics: (a) creating mathematical classroom discourse, (b) developing classroom norms that support opportunities for mathematical learning, and (c) building relationships that support mathematical learning.

CONCEPTUALIZING CLASSROOM PRACTICE

Teaching Mathematics

What constitutes good teaching² is consistently controversial and will remain controversial. Our goal in this chapter is not to resolve the controversy but rather focus on what research and theory can currently tell us about classroom instruction with the intent of (a) making explicit current findings (drawing widely on both research and theory); (b) highlighting where future research can contribute to our ongoing understanding of classroom practice; (c) attending to how research takes into account or raises questions about supporting student engagement, participation, and learning in classroom practice; and (d) characterizing how current findings and conceptions of classroom practice relate to issues of equity.

Teaching mathematics in classrooms, no matter what the level, engages students, teachers, administrators, and schools in contexts that vary from day to day in ways that make it difficult to create a formula, a set of guidelines, or even a set of practices that all teachers should engage in. We can lay out a theory that can help provide a base for thinking about mathematics classrooms and a set of principles that we can link to outcomes under particular conditions (so we can provide some research-based evidence about what would happen if someone else were to engage under similar circumstances). Our intention is that as researchers and teachers read this chapter, they will finish with the feeling they know something about effective mathematics classroom practices and

¹ The students' perspective is not evident in either vignette and often not evident in research on classroom practice.

² We consider the act of teaching as involving a number of different practices, with different people, which go beyond classrooms. Teachers teach when they work with parents, talk with the principal, and engage in professional development. This chapter is about one aspect of teacher work, that which occurs as teachers work with students in the context of the classroom. In this chapter, we focus on teaching and classroom practice, recognizing it is not all of teachers' work, nor separate from the other aspects of the work, but it is a place where opportunities for student and teacher learning are created and developed.

the kinds of questions to ask to help shape their teaching or studies of teaching.

As we consider classroom practices here, we are interested in practices that have been shown to positively impact both students and teachers. And although we recognize that we must focus on classroom practices that impact students' mathematical understanding, we must also be clear that mathematical understanding involves students' relation to the mathematics—how they see themselves as doers of mathematics. We also recognize that productive classroom practices create ongoing opportunities for teacher and student learning, where students and teachers see themselves as comfortable, confident, and knowledgeable in their abilities to engage in mathematics.

We also examine the work on teaching and mathematics classroom practice through the lens of equity. We present research that considers the opportunity for each student to become successful, to become a part of ongoing engagement, to be a full participant, where assumptions about particular reasons for a student's participation and success or failure are challenged. We recognize that focusing on individual children and creating opportunities in relation to individuals is not enough to ensure equity. We consider how teaching and classroom practice can create opportunities that support and inspire changes in long-standing classroom, school, and district practices that lead to inequities; challenge assumptions on which instruction is consistently built; and support the development of students' identities, particularly disenfranchised students, as competent doers of mathematics.

Claims about Teaching

We want to recognize up front that making claims about teaching and classroom practice in general is difficult. Classrooms involve people, the lives of people over time, people who work in social, cultural, and political contexts that shape how they do their work and how that work gets interpreted. So as we put forward claims about what is known about teaching and classroom practice, we must remember that any aspect of classroom practice may evolve differently depending on the classroom, the teacher, the student, and the broader social, cultural, and political context. However, we do not want these concerns to so override the interpretations of research that we become unable to make progress in our understanding. We ask that the reader consistently consider that the statements we make must be taken as interpretable within

particular local contexts (and as always there exists variability; see Goldenberg & Gallimore, 1991).

A Conception of Teaching³

Teaching is relational. Teachers, students, and subject matter can only be understood in relation to one another. The teacher works to orchestrate the content, representations of the content, and the people in the classroom in relation to one another. Students' ways of being, their forms of participation, and their learning emerge out of these mutually constitutive relationships. Teaching is also multidimensional. Within research on classroom practice, several images of this multidimensionality emerge, images that overlap with one another but that are often emphasized in one line of work. Teaching is creating an environment for learning mathematics, orchestrating participation so that students relate to representations of subject matter and to one another in particular ways (Ball & Bass, 2000a, 2000b; Civil, 2002; Cobb, 2000; Lampert, 2001; Moschkovich, 2002). Teaching is eliciting and interpreting what students do and know. It is detailing students' learning trajectories (Fennema et al., 1996; Kazemi & Franke, 2004; McClain, 2000; Simon, 1997; Simon, Tzur, Heinz, & Kinzel, 2004). Teaching is principled decision-making that emerges from complex interactions between teachers' knowledge, beliefs, and goals (Schoenfeld, 1998, 1999a, 2000, in press). Teaching is learning in and from practice in ways that continually supports students (Franke, Carpenter, Levi, & Fennema, 2000; Kazemi & Franke, 2004; Lave, 1996).

Specifying a View of Teaching

Lampert has written extensively and in detail about the teaching of mathematics. She makes explicit her view of teaching and the theory and research underlying that view. We draw on her work here to provide the foundation for understanding classroom practice in mathematics. Our hope is that the portrait she provides will allow those who share her view as well as those who may differ to see where the similarities and differences exist and how a particular view of teaching can be used to make sense of the literature around classroom practice.

Lampert views teaching practice not as a collection of actions that the individual teacher completes but as the evolution of relationships. "Teaching practice is what teachers do, but it is more than how teachers behave with students or the actions of individual

³ Throughout the chapter we draw on a range of theoretical perspectives. We draw on cognitive and sociocultural theories of learning as well as critical theory and critical race theory. Our intention is to draw on work from these theoretical perspectives that taken together can help shape our understanding of teaching and classroom practice in mathematics.

teachers; action is behavior with meaning, and practice is action informed by a particular organizational context” (2004, p. 2). For Lampert, then, teaching is about building relationships—between students and the teacher and among students themselves around mathematics—and engaging together in constructing mathematical meaning. Teaching involves orchestrating the content, the representation, and the people in relation to one another. It is about making decisions in the moment that serve the individuals and the collective. It is about understanding the students, who they are as individuals and as a group, and continuing to learn more about that as one engages in the process of learning. It is about working together to negotiate meaning.

Supporting learning comes from knowing the students, the situation, and the content and then making decisions that support interaction that productively engages students in moving their ideas forward. Teaching is about knowing about the various tools and the meanings students make of them and how they relate to the content and the development of students’ thinking as well as knowing how this plays out in the context in which one teaches. Teaching is deliberate work, but it is deliberate work that takes into account the interaction among people and ideas and context. Teaching requires adaptation and learning for teachers and students.

Within this view the complexity and layers of teaching emerge. Rather than teaching being a summation of individual acts, it becomes multifaceted in that relationships among people and content are constantly negotiated. As this negotiation takes place, teachers must learn and adjust to the changing nature of individual and collective learning.

Conceptions of Student and Teacher Learning

Classrooms need to be places where teachers and students are engaged in rigorous mathematics in ways that both parties learn. Opportunities must exist in classrooms for teachers to fine-tune their skills and knowledge as they learn what works, for whom, and when. Opportunities must also exist for teachers to engage in sense-making about their own practice in relation to the mathematics and the students. We have chosen in this chapter to review the work that enables us to discuss mathematics classroom practice that affords learning for teachers and students—classroom practice that makes possible ways for teachers and students to participate together without marginalization.

Emerging Consensus on Student Learning

The field appears to be moving toward some consensus about the focus of student learning (as evidenced by *Adding it Up*; see other chapters in this volume). This emerging consensus includes the idea that all students need opportunities to develop both concepts and skills, to develop flexibility in their abilities to engage with mathematical ideas, and to engage in what some may call higher order or critical thinking. Developing these ideas is about rigor in both depth and breadth and is how we frame understanding in this chapter.

The authors of *Adding it Up* (National Research Council, 2002) defined *understanding* as consisting of five interwoven strands of mathematical proficiency: conceptual understanding, procedural fluency, strategic competence, adaptive reasoning, and productive disposition. We find it quite important that these five strands include one that focuses on the disposition of the learner, highlighting that understanding is more than a set of knowledge and skills; understanding includes one’s perception of oneself in relation to the mathematics. In the words of Dutro and colleagues,

Students learn to locate themselves within the dichotomies of schooling. They narrate themselves and are narrated by others into storylines of success and failure, competence and incompetence, participant and non-participant, included and excluded, etc. In addition, students bring other storylines from home, the media, and from their own particular . . . ways of negotiating the world. (Dutro, Kazemi, & Balf, 2006, pp. 25–26)

These storylines, as they emerge and become reinforced through classroom practice, shape students’ ways of being in the classroom and their stances towards learning and doing mathematics.

In *Adding it Up*, the focus on the interwoven strands highlights that these ideas about oneself and the acquisition of knowledge and skills are not developed in a linear fashion. One does not, the authors claimed, develop mathematical skill first then the others follow or develop conceptual understanding and then the others follow but rather all of the aspects of understanding must be addressed together over time.

These perspectives on understanding mathematics are shaped and further defined as one relates them to views of learning. Learning here is seen as social and shared, where teachers and students bring histories and identities to the interactions, where participation is the focus. Knowledge and skills help shape who we are and how we see ourselves (Greeno & MMAP, 1998; Lave & Wenger, 1991; Rogoff, 1990; Wenger, 1998; Wertsch, 1991). They help shape the ways we interact with others in new and existing settings. Language and tools mediate these interactions. Learning

is not about receiving information; it is about engaging in sense-making as we participate together.

Connecting Views of Learning Mathematics with the Challenge for Teaching

Within the field of mathematics education, researchers seem to agree in principle that classrooms that support mathematical proficiency would be places where students are encouraged to be curious about mathematical ideas, where they can develop their mathematical intuition and analytic capabilities, where they learn to talk about and with mathematical expertise. Difficulties arise as we have left these ideas underspecified. In particular, helping teachers, who have not experienced such learning environments themselves, to adapt their classroom practice in significant ways to meet these goals for all students requires helping teachers understand the details of what it would mean to engage in this kind of mathematical work and how to support students in coming to participate in this way.⁴ We attempt in this chapter to provide some of that specificity and give direction to future inquiry in the field.

We know from studies like the Third International Mathematics and Science Study (TIMSS) that, in the United States, most mathematics instruction is not consistent with current reform ideas (Hiebert et al., 2003; Hiebert & Stigler, 2000; Stigler & Hiebert, 1997, 1999). Most U.S. mathematics classrooms maintain an initiation-response-evaluation (IRE) interaction pattern, where the evaluation move on the part of the teacher focuses on students' answers rather than the strategies they use to arrive at them. The teacher assumes responsibility for solving the problem while student participation often involves providing the next step in a procedure. TIMSS has further reported that U.S. students had little opportunity to discuss connections among mathematical ideas and reason about mathematical concepts. A number of studies support the TIMSS report, and agreement exists that currently many mathematics classrooms do not provide sufficient opportunities for students to develop mathematical understanding.

Researchers also know that, even in classrooms where teachers are attempting to teach for understanding as specified by the NCTM *Standards* (and other reform documents), teachers often maintain the IRE pattern and maintain many existing mathematics practices inconsistent with the reform (Hiebert & Stigler, 2000). This was even the case when the teachers were using conceptually rich curricula (RAND Math-

ematics Study Panel, 2002; Spillane & Zeuli, 1999). Changing existing mathematical classroom practice has proven to be challenging, particularly inasmuch as the change requires not simply adopting a set of practices or curricula but must include a change in how teachers engage around the practice with their students. For instance, simply using manipulatives, putting students in cooperative groups, or asking higher order questions does not lead to classrooms that support the development of mathematical understanding (Stigler & Hiebert, 1997, 1999; Webb, Nember, & Ing, 2006). How teachers and students engage with higher order questions, engage students in cooperative groups, or use manipulatives matters.

A growing body of research is beginning to provide insight into mathematical classroom practice that supports mathematical proficiency. Many mathematics reform projects found that practicing procedures and asking students to solve many problems does not support the development of understanding (Carpenter, Ansell, Franke, Fennema, & Weisbeck, 1993; Carpenter, Franke, Jacobs, & Fennema, 1998; Fennema et al., 1996; Hiebert & Wearne, 1992). Decreasing time on conventional procedural skill does not decrease students' proficiency on routine problems (Carpenter, Fennema, Peterson, Chiang, & Loef, 1989; Hiebert & Wearne, 1992). Spending more time on a problem supports a greater degree of reflective and analytic thought (Hiebert & Wearne, 1992) but the nature of discourse around a problem is critical (Cobb, Wood, & Yackel, 1993; Cobb et al., 1991; Hiebert & Wearne, 1992, 1993; Stein, Grover, & Henningsen, 1996; Yackel, Cobb, Wood, Wheatley, & Merkel, 1990). Teachers who know about their students' mathematical thinking can support the development of mathematical proficiency. Knowing about students' mathematical thinking supports opportunities for question asking linked to students' ideas, eliciting multiple strategies, drawing connections across strategies, and so on. When teachers ask more questions and ask for more than recall, providing students' opportunities to express their ideas and actions verbally, students develop understanding. When students are required to describe their strategies in detail and why they work, they develop understanding. These initial studies, as they supported the development of aspects of reform-based classroom practice, provided knowledge about aspects of teachers' work that researchers can leverage to better understand classroom practice.

⁴ We draw attention throughout the chapter to "detailing" practice. Details matter. Describing practices in our research and our work in classrooms allows for unpacking, supports conversations about meaning, and helps us be explicit about agreement and disagreement.

Summary of Stance Towards Classroom Practice Organizing this Chapter

Maintaining our dual focus on what teachers do and how that is experienced by students; keeping equity central; drawing on a current, coherent conception of teaching and learning; and attending to the limits and strengths of the reform studies leads us to a particular approach to our review of classroom practice. Research suggests the value of listening to students' mathematical ideas and building on them, that attending to the details of student thinking and the mathematics (particularly the mathematical task) supports student learning, and creating opportunities for a range of ideas to be a part of the classroom discourse matters for students and teachers alike. Lampert's conception of teaching parallels these ideas and suggests that knowing the development of students' mathematical thinking, understanding the mathematical tasks, and orchestrating conversation around the particulars of student thinking and mathematics are all a part of teaching for understanding. In addition, Lampert's conception would require that teachers know who their students are within the social environment of the classroom (and potentially outside) and who the students are as an evolving intellectual community. Our equity focus requires we attend to the multiple aspects of who students are and how they participate. These understandings lead us to elaborate three features of classroom practice (a) shaping classroom mathematical discourse, (b) developing classroom norms that support engagement around the mathematical ideas, and (c) developing relationships with students and the class in a way that supports opportunities for participation in the classroom's mathematical work.⁵ We now turn to an extended discussion of each of these features of classroom practice.

SUPPORTING DISCOURSE FOR DOING AND LEARNING MATHEMATICS

Shaping mathematical discourse is a significant aspect of a teacher's work. How teachers and students talk with one another in the social context of the classroom is critical to what students learn about mathematics and about themselves as doers of mathematics. What it means to do and learn mathematics is enacted through the discursive practices that form in the classroom. As we argued at the beginning of this chapter,

teaching is relational. Through classroom discourse, one can see how students, the teacher, and subject matter interact and what the consequences are for students. Developing mathematical understanding requires that students have the opportunity to present problem solutions, make conjectures, talk about a variety of mathematical representations, explain their solution processes, prove why solutions work, and make explicit generalizations. A number of researchers have been engaged in research to better understand the discourse practices that support the development of students' mathematical understanding. Although in its relative infancy, this work has already made a number of contributions to what researchers know about mathematics classroom practice.

In this section of the chapter, we contrast the predominant discourse structure of initiation–response–evaluation, or IRE, with structures that enable discussion and deliberation of mathematical ideas. We review how researchers have conceptualized and documented these alternative discursive structures and discuss both the role of the teacher in managing and enabling such structures and how these structures position students in relation to one another, the teacher, and the subject matter. Issues of equity, diversity, and social justice are again both implicitly and explicitly central in understanding the impact of classroom discourse practices. Mathematics classrooms are necessarily cultural and social spaces that can either perpetuate social inequities by privileging certain forms of discourse and ways of reasoning or reorganize them by positioning multiple and hybrid forms of learning and knowing as “having clout” (Cobb & Hodge, 2002; Gutierrez, Baquedano-Lopez, & Tajeda, 1999; Gutstein, *in press*; Lee, 2002). Although researchers know that the role the teacher plays in supporting discourse practices is critical to their success and consequences, our review claims that we know little about what teachers need to do to best support classroom discourse in a way that opens participation and supports the development of students' knowledge and identities. For instance, we do not know whether argumentation, where students are aligned in opposition to one another, is productive for all students' development of mathematical understanding.

The IRE Pattern and Its Alternatives

Communication in the traditional mathematics classroom can be characterized by teacher talk: teach-

⁵ All of this mathematical work within classrooms requires teachers understand the key mathematical ideas of the work they do with students and a particular stance towards that engagement. Chapters in this volume delineate that in detail. Here we focus on the classroom teaching work, not what is supporting it.

ers explaining procedures, giving directions, explaining mistakes in ways that require very little student-to-student or even student-to-teacher talk (Hiebert et al., 2003; Porter, 1989; Silver & Smith, 1996; Stigler & Hiebert, 1997). The most dominant classroom discourse pattern has been the IRE, where the lesson follows a teacher-dominated pattern of teacher-initiated question, student *response*, and teacher *evaluation* (Cazden, 2001; Doyle, 1985; Mehan, 1985). The IRE pattern is a well-documented part of mathematics classrooms in the United States and elsewhere, especially for students from economically disadvantaged backgrounds (Silver, Smith, & Nelson, 1995). In mathematics classrooms students are typically asked to listen and remember what the teacher said. Usually, little emphasis has been placed on students' explaining their thinking, working publicly through an incorrect idea, making a conjecture, or coming to consensus about a mathematical idea. Even in classrooms where teachers are attempting to teach for understanding they often maintain this pattern. Spillane and Zeuli (1999) found in their study of reform-minded teachers that they predominately engaged in procedure-bound discourse, rarely asked students to do more than provide the correct answer, focused on procedural rather than conceptual knowledge, and engaged students in memorizing procedures to calculate answers (see also RAND Mathematics Study Panel, 2002).

Over the last 15 years, a growing body of research has documented and theorized alternatives to the IRE discourse structure. New conceptions of classroom discourse focus on how conversations foster mathematical argument so students can come to understand the different forms of mathematical explanation, create a public knowledge base that can be used by the class as a resource, align students with one another and the content, develop students' mathematical identities, and generally foster higher level thinking (Ball & Bass, 2000b; Boaler & Greeno, 2000; Cobb et al., 1993; Lampert, 2001; Yackel et al., 1990). O'Connor and Michaels (1993) described classroom conversations as providing a site for aligning students with one another and the content of the academic work while also socializing them into ways of engaging in mathematics. In addition teachers, through classroom discourse, can draw on a range of student resources, particularly for English language learners (ELL), and build on them in a way that attends to the strengths of students, not their deficits (Moschkovich, 1999, 2000). Creating these opportunities takes more than simply posing the problem and asking the children how to solve it within a non-threatening environment.

Although the kind of discourse we have just described seems like such a necessary and productive part of classroom practice, what teachers must do to support such opportunities is complex and not well characterized in the literature. The way teachers support mathematical discourse matters. Even in what may seem like the simplest form of classroom discourse—large-group discussion to foster students' participation in thinking through a problem—the teacher must attend to many issues. The teacher must attend to who is participating, how they are participating, the mathematical ideas being pursued, the students' linguistic and mathematical backgrounds, the students' current understandings, and the attitudes and identities the students bring to the conversation (Lampert, 2001). The teacher must give each student the opportunity to participate in working through the problem while simultaneously encouraging each student to attend to the solution paths of others, in ways that she can orchestrate opportunities for students to build one another's thinking. While attending to this, the teacher must also actively take a role in making certain that the class gets to the implicit and explicit goals. She needs to make judgments about what to avoid, navigate through solution paths that do not always work, respond to incorrect statements, and watch out for those not participating. She must also find a way to make explicit the underlying mathematical similarities and differences in the solutions in a way that makes sense to her students. All these actions and decisions must of course fit within the given period of time of a lesson, a unit, and a school year.

Teachers are expected to pose problems but not provide answers (Lampert, 1990), stop or slow down the discussion to provide access to more students (Rittenhouse, 1998), model the academic discourse for the students (Ball, 1993; Lampert, 1990; Rittenhouse, 1998), comment and elaborate on student ideas (Rittenhouse, 1998), and question student reasoning so as to foster certain habits of mind (Lampert, 1990; Lampert, Rittenhouse, & Crumbaugh, 1996; Rittenhouse, 1998). Thus, as Ball (1993) pointed out, the teacher is responsible for the students' learning of mathematical content and, at the same time, for fostering a discourse environment that both supports students and helps to create, among them, new identities that include a favorable disposition towards mathematics. It is no wonder IRE remains prevalent.

Conception of Discourse in Mathematics Education

Researchers and theorists alike are calling for a focus on mathematical communication. How commu-

nication is defined and why we need it in classrooms varies. However, all perspectives include social engagement, through talk and shared representation, with a focus on developing mathematical ideas. Some researchers highlight communication as creating publicly shared knowledge that can serve as a future resource; others see the opportunities for multiple ways to participate with the content, while still others see communication as a way to further develop the practices of mathematics: explanation, argument, and so on. Some researchers focus on students' sharing their solutions and ideas surrounding those solutions, while others focus on developing mathematical arguments. Some see mathematical conversation as another opportunity to develop further mathematical knowledge, while others look to the social and individual benefits.

So although researchers engaged in understanding mathematical conversations and their benefits may think differently about what to look for in a conversation and what the benefits may be, what seems critical here is that to engage in the type of mathematics required by such documents as *Adding it Up* (NRC, 2002) and the NCTM *Standards* (1989, 2000) in a way that makes use of what is known about learning in a social context, mathematical conversations are necessary. Yet, just talking is not enough. Cazden (2001) and Nemirovsky, Barros, Noble, Schnepf, and Solomon (2005) argued that teachers need a repertoire of ways to engage students in mathematical conversation with the understanding of what a particular conversational move may afford or constrain and how to support it in the classroom with a range of students. Questions then become about what types of conversational moves support student participation and learning in mathematical conversations. How can teachers support opportunities for engagement in mathematical conversations in ways that are productive for students?

Engaging in Classroom Conversations

Much of our initial learning about engaging students in mathematical conversations comes out of the research on cooperative groups. Increasing the level of discourse in cooperative groups often produces greater student learning; explaining to other students is positively related to achievement outcomes, even when controlling for prior achievement (Brown & Palincsar, 1989; King, 1992; Peterson, Janicki, & Swing, 1981; Saxe, Gearhart, Note, & Paduano, 1993; Webb, 1991; Yackel et al., 1990). However, just getting students to talk was not enough; what they talked about mattered. Often students in talking with a peer shared a procedure with little discussion about con-

cepts or strategies (Webb, 1991; Webb et al., 2004). Webb and others have found that most often in these cases the students are mirroring what teachers do as they engage in conversations with students. However, although talking could often be productive for the student sharing, it is typically less productive for the student hearing it. The learning of the students hearing an explanation is dependent on the teacher and the support the teacher provides that allows students to learn together.

Consistent throughout the research on classroom talk, whether from the cooperative learning work or from work on classroom discourse, is the fact that details matter (Webb et al., 2004). Sfard and Kieran (2001a) found that students were neither precise nor explicit in their talk (see also Nathan & Knuth, 2003). Often the language students used to describe their ideas was not taken-as-shared by those listening, referents students made were unclear, and they generally did not provide enough detail. This is much like the cooperative group literature that differentiates non-elaborated from elaborated help or specific questions about a problem solution from questions about following a procedure. Elaborated help and specific questions, both more detailed, led to more student learning. Kieran (2002) stated

it is the way we make our thoughts available that is critical, it is not just about making them 'available,' it is how...utterances that were neither complete nor ever expanded upon seemed much less conducive to the emergence of mathematical thought for both participants. (p. 219)

So one of the most powerful pedagogical moves a teacher can make is one that supports making detail explicit in mathematical talk, in both explanations given and questions asked.

The most prominent finding about creating opportunities for mathematical conversations in classrooms is that what the teacher does to structure these opportunities matters. Teachers need to scaffold, monitor, and facilitate discourse around the mathematical ideas in ways that support student learning (Kieran & Dreyfus, 1998). Wood (1998) contrasted discourse patterns in which teachers' questioning *funneled* student responses to discourse patterns in which teachers' questioning *focused* student mathematical thinking. In the former, the teacher's questions funnel the conversation so that the teacher actually does the bulk of the intellectual work. In the latter, the teacher's questions focus student attention on important mathematical ideas but place the responsibility of the intellectual work on students. Sherin (2002) described how the

teacher used a “filtering approach” to focus students’ attention on particular mathematical ideas. After soliciting multiple solutions from students during which they listened and evaluated one another’s ideas, the teacher intentionally filters the ideas, choosing which ones to pursue with the whole class in order to advance particular instructional goals. Although researchers have long recognized the *potential* of teacher practices to foster meaningful conversation and student learning in classrooms (e.g., Cazden, 1986; Forman, 1989, 1993), researchers have only begun to study ways of changing classroom discourse from traditional recitation patterns in which the teacher dominates classroom exchanges to more balanced and student-centered communication in which students take a more active role in classroom discourse (Carpenter, Fennema, Franke, Levi, & Empson, 1999; Cobb & Bauersfeld, 1995; Forman & Ansell, 2002; Hufferd-Ackles, Fuson, & Sherin, 2004; Leinhardt & Steele, 2005; Martino & Maher, 1999; Nathan & Knuth, 2003; Sherin, 2002; Wood, Williams, & McNeal, 2006; Yackel, 2002).

In working to create opportunities for rich conversation a number of researchers have found ways teachers can provide support. From Palinscar and Brown’s (1984) *reciprocal teaching* to Hogan and Pressley’s (1997) *instructional scaffolding*, researchers have developed structured ways for teachers to engage with students that have proven to produce more conceptually based conversation as well as productive student outcomes.

Researchers have in mathematical classrooms also learned something about the general ways teachers can support rich mathematical conversations. Teachers have found certain strategies to be effective in supporting student engagement and discourse. Yackel, Cobb, and Wood (Yackel, Cobb, & Wood, 1991; Yackel et al., 1990) found two strategies that teachers used to develop norms around students’ explaining their methods. First, teachers used specific situations that arose spontaneously during group work as a springboard for whole-class discussion, and second, the teacher invented specific situations that they could use to discuss participation with the class. These teachers also intervened in small-group work directly to help renegotiate student participation. It took significant attention on the teachers’ part to support the development of different forms of participation.

Silver and Smith (1996; Silver, 1996) found through their QUASAR work that teachers need to (a) give students sufficient time to generate and explore their own ideas and discuss their solutions, (b) value understanding solutions by pressing for justification and explanation, (c) refrain from stepping in and providing a solution strategy for the student, and (d) create op-

portunities for competent performance in sharing to be modeled by a capable student (see also Stein et al., 1996). Forman, Larreamendy-Joerns, Stein, and Brown (1998) drew on a study of a teacher engaged in creating opportunities for argumentation. They found that the teacher orchestrated discourse by recruiting attention and participation from students, aligning students with one another on the basis of their mathematical ideas, highlighting positions through repetition, and pointing out implicit but important aspects of an explanation as she expanded on what was shared. These findings are similar to those of Strom, Kemeny, Lehrer, and Forman (2001). In addition, they highlight the teacher’s stance and excitement toward the collective work. She made her investment in the work explicit for the students. In all of these cases teachers’ moment-to-moment participation acts to structure the conversations in ways that allow for all students to participate in meaningful mathematical work.

Beyond these general ideas about mathematical discourse a few promising core ideas and practices are emerging. The four ideas we touch on briefly highlight the detail at which we can come to understand mathematical discourse and the direction we can go in developing additional knowledge to support teachers and students as they engage in this work. We look at the research around revoicing as it represents an area where there is accumulating evidence about the role teachers play in the particulars of how they respond to students within a mathematical conversation. Second, we look at the work discussed earlier on the ways tasks play a role in the development of mathematical discourse. Third, we look particularly at mathematical discourse for English language learners. Fourth, we propose that a discourse practice drawn from science classrooms, interrogating meaning, as defined by Rosebery and Warren, can further efforts to make equity central in thinking about mathematical discourse and participation. We end with emerging work about students’ own experiences of mathematical discourse and participation.

Revoicing

Revoicing, most simply, is reuttering of someone’s talk. This reuttering can involve repetition, expansion, rephrasing, or reporting what a student says (O’Connor & Michaels, 1993, 1996, drew on the work of Goffman in their elaboration of revoicing). Revoicing can accomplish a range of goals and can either support or limit productive discourse. Revoicing serves to clarify or amplify an idea and allows the teacher to substitute mathematical vocabulary for everyday words or redirect the conversation. Beyond these goals, re-

voicing can communicate a way of thinking about doing mathematics, a respect for student ideas, and an encouragement of students' developing mathematical voice. Here revoicing can frame the content discussion and the social roles. In addition, most recently researchers have investigated the use of revoicing as a pedagogical move to position students in relation to one another and to the mathematical argument (Forman et al., 1998). The teacher can use revoicing as a way to align students to a particular argument or way of thinking about the mathematics. Revoicing then is a way of orchestrating the conversation.

Studies of teachers in the mathematics classroom that examine both the types of revoicing that occur and the impact of the revoicing on students have found that often revoicing plays a supportive role in terms of both mathematical support for ideas and the development of students' identities around the learning of mathematics (Forman et al., 1998; O'Connor & Michaels, 1993, 1996; Strom et al., 2001). These case studies of teacher and classrooms provide support for the ways revoicing structures student participation as students work together and connects students to the mathematics. Work like that of Forman and her colleagues or Strom and her colleagues is also beginning to clarify the particulars around how revoicing can support student learning and how that revoicing brings together the mathematics, forms of notation and representation, and the students.

However, this type of language socialization into the discourse practices of academic argumentation is not without its difficulties. First, it creates teaching dilemmas. For example, conflicts can arise between the teacher's content objectives and a pedagogy that honors the intellectual process, particularly when it means leaving incorrect mathematical ideas on the table (Ball, 1993; Lampert, 1990). Second, because academic argumentation can easily deteriorate into oppositional or confrontational talk, there is a risk of interpersonal conflicts that spill over the intellectual content or even the boundaries of the classroom (Lampert et al., 1996; O'Connor, 1998). Thus, it is particularly important for teachers to attend to creating a set of social norms suitable for classroom debate alongside the sociomathematical norm about what is accepted as an adequate mathematical answer or justification (Horn, in press; Yackel & Cobb, 1996). Finally, these discourse practices may be aligned with, or in conflict with, practices from the students' home or the practices from other communities that they participate in outside of school (Gallego, Cole, & The Laboratory of Comparative Human Cognition, 2001; O'Connor, 1998; Walkerdine, 1990).

Although the work on revoicing has stimulated detailed look at supporting classroom discourse in math-

ematics and there is evidence that revoicing occurs and can productively support student engagement and learning, researchers do not yet know when particular aspects of revoicing are productive and when they might lead to disengagement or learning. There is enough evidence to argue that the teacher's role in orchestrating through revoicing and other means is crucial, and that continued research in this area can support both the development of classroom practice and the learning of teachers.

The Role of Tasks

Silver and his colleagues in their focus on mathematical tasks describe how worthwhile mathematical problems can "give students something to talk about" (Silver, 1996; Stein & Lane, 1996). These worthwhile tasks from their perspective are ones that engage students in thinking and reasoning about important mathematical ideas. They drew on the NCTM *Standards* and further described worthwhile tasks that can be solved in multiple ways, involve multiple representations, and require students to justify, conjecture, and interpret (see also Engle & Conant, 2002, for discussions of productive disciplinary engagement).

Stein and Lane (1996), drawing on the work of the QUASAR project, reported that the highest learning gains on a mathematics performance assessment were related to the extent to which tasks were set up and implanted to engage students in high-level cognitive thinking and reasoning. Selecting and setting up a high-level task, though, does not guarantee students' high-level engagement. However, starting with a good task is necessary for providing opportunities to engage students in high-level thinking. So starting with a cognitively demanding task allows the teacher to engage students in sharing their thinking, comparing different approaches, making conjectures, and generalizing (Silver & Smith, 1996).

Oftentimes as teachers begin to think about engaging their students in mathematical discourse they draw upon the tasks they have been using, and these tasks often lend themselves to using a single procedure and do not lend themselves to the types of conversations that stimulate rich discourse. Supporting productive discourse would be easier for teachers if they worked from mathematical tasks that allowed for multiple strategies, connected core mathematical ideas, and were of interest to the students.

Classroom Discourse for English Language Learners

There is significant evidence that students bring to the classroom ways of participating in conversations;

language socialization patterns develop in the home environment. The teacher, through classroom discourse practices, plays a role in helping to socialize students with potentially varying notions about participating in mathematical conversations. Revoicing is a way to begin to understand this, but the challenge becomes even greater when the students do not speak English as their native language (O'Connor, 1998). However, to date, much of this research has been conducted in monolingual classrooms, led by exemplary teachers.

Moschkovich (2002a, 2000b) advocated a perspective for ELL that focuses on *participation in mathematical discourse*. The focus on participation expands and complicates the typical explanations of how ELL students learn mathematics and what the difficulties are. According to Moschkovich (2002b), the focus on participation in mathematical discourse starts with the assumption that knowing and doing mathematics is not merely an issue of vocabulary or semantics. Instead, it leads educators to identify the resources that ELL students can bring to mathematical discussions and the ways to build on these strengths to construct meaningful mathematical understandings (Gutierrez, 2002b). This implies that ELL students should be allowed to use their primary language (particularly in small groups) to talk about mathematics. With the help of the teacher, who can explicitly link these ideas to how they can be expressed in the English mathematical register, students will be able to bridge their mathematical knowledge and their mathematical language proficiency in English (Moschkovich, 2002b).

Nemirovsky et al. (2005) in their studies of high school mathematics classrooms built on the work of Moschkovich (1999) and found that the relationship between everyday language and the mathematical register is one of continuity. This continuity implies that the mathematical meanings are normally expressed in the combined use of everyday language and the specialized register and that, far from being “dangerous,” the open discussion of multiple word usages, including those that tend to be mistaken from a mathematical point of view, enriches mathematics learning.

Revoicing can be one such strategy that supports the participation of students with limited academic English proficiency in mathematical discussions (Moschkovich, 1999). Rather than adopting the typical perspective that sees students’ limited English proficiency as a barrier to understanding, teachers can use revoicing to frame students’ first language as a resource for the construction of meaning (Moschkovich, 2000). Through revoicing, teachers acknowledge and demonstrate respect for students’ contributions, while also potentially elaborating, clarifying, or formalizing students’ thinking. These types of friendly

uptakes of student voices may lead to a greater degree of student participation in the discourse of mathematics. Therefore, revoicing contributes to a new model for classrooms, in which students learn academic English and mathematics at the same time.

Enyedy and his colleagues (Enyedy, Rubel, Castellon, Mukhopadhyay, & Esmond, 2006) studied revoicing in a multilingual high school algebra classroom. They found that revoicing was not being used to initiate arguments between students (as described by Forman et al., 1998). However, revoicing was a frequently used tool. The teacher in this classroom used revoicing to support elaboration, clarification, and positioning with respect to mathematical tasks, goals, and roles. Even though the students did in this class engage in making mathematical claims and justifying them, the teacher did not use revoicing as a means to create alignments and oppositions in an argument. So, little debate occurred. Enyedy and his colleagues (2006) argued then that it may be more productive to conceptualize revoicing as a “flexible rhetorical form that can take on many functions” beyond facilitating student debate.

Interrogating Meaning

An additional discourse practice with promise for supporting mathematical learning and providing opportunities for students from different backgrounds to participate is raised in the work of Rosebery, Warren, and their colleagues (Rosebery, Warren, Ballenger, & Ogonowski, 2005; Rosebery, Warren, & Conant, 1992). In attempting to create opportunities for students to engage in discussion around science inquiry, Rosebery and Warren have developed an approach they term *interrogating meaning* (Rosebery et al., 2005). Here students interrogate situated meanings, familiar and unfamiliar in light of one another, and in the process make explicit the assumptions of use, purpose, and context. Engaging in interrogating situated meaning engages students in the ways language and symbolic practices are used more generally in a discipline and allows the teacher to be explicit that this is not separate from engagement with the knowledge of that discipline. Interrogating meaning engages students in an analytic stance toward the ways of seeing. In Rosebery and Warren’s work students were encouraged by the teacher to interrogate (ask questions of, challenge, wonder about) “deceptively familiar experiences, such as motion down a ramp and ideas such as ‘faster and faster’, ‘speed’ or ‘darkness’ in relation to situations of use and perspective” (p. 66). As students and teachers engaged with the often-conflicting ways students made sense of the ideas, the students

made explicit to themselves and to the teacher their conceptualizations and stances toward those conceptualizations in ways that supported ongoing learning as students made connections between what they knew and what they were learning.

What is appealing about Rosebery and Warren's conception of interrogating meaning is the opportunities this move provided for participation. Interrogating meaning creates a norm around the expected nature of questioning and challenging one's ideas. The focus is not then on who was wrong, or whose explanation was incomplete, but on unpacking every response in ways that support learning.

Although we know little to date about interrogating meaning in a mathematics classroom, the conceptualization of the pedagogical move and the research in science education by Rosebery and Warren leads us to see promise in promoting both the focus on mathematics we hope to support and the opening of participation across students.

Considering Students' Experiences of Classroom Discourse

As our review of the classroom discourse literature indicates, there has been growing attention to the role teachers play in managing classroom discourse. A set of studies is also emerging that considers the experiences students have in classroom discussions. Some of this work includes students' own accounts of these experiences, whereas others are interpretations of students' experiences. Attention to both ways of considering students' experience of classroom discursive practices is clearly a fertile area for research. Importantly, this body of work has often focused on marginalized students. We summarize a number of articles here that reflect this promise.

Empson (2003) presented an in-depth analysis of two of the lowest performing students in a first-grade unit on fractions. Her examination of classroom interactions provided new insights into the learning gains made by these students. In particular, the teacher leveraged certain interactional mechanisms to support the students to learn mathematics and enhance their dispositions towards doing mathematics: (a) Tasks were used to elicit students' understanding, not just their confusions, and (b) teachers positioned the students, through interaction, as mathematically competent. The teacher often supported the students to work through problematic reasoning and then positioned them as authorities on these strategies to the group. In instances when the two students' problematic reasoning was resolved during group discussions, the teacher most often used revoicing strategies to repair

their claims or positioned the students as evaluators of other students' comments or alternative strategies. Rarely did the group conversations cause the focal students to opt out. As Empson's work has continued, she is working with colleagues to use these observations of students to create a set of conjectures about how teachers can take proactive roles to use discourse strategies and participation structures to position students in ways that allow them to *display* understanding and competence (Empson, Turner, Dominguez, & Maldonado, 2006).

The teacher's ability to draw out, connect with, and frame student achievement and competence is clearly linked to her ability to know students, cultivate norms for generative interactions, and build relationship with students, ideas that we will discuss in depth in the major sections of this chapter. By analyzing student participation Moschkovich (2002) found that students brought different ways of talking about mathematical objects and mathematical situations to the discussions. None of these ways of talking should be seen as privileged over the other; instead each way of talking can contribute in its own way to the mathematical discussion and bring resources to the conversation (p. 12). Instead of concluding that the students were wrong or lacked vocabulary, Moschkovich underscored that they were bringing a different point of view or using different language than the teacher expected. Moschkovich concluded that taking a discourse approach to mathematical learning means considering the different ways of talking and different points of view students bring to the discussion.

In much of her work, Boaler (Boaler, 2003; Boaler & Greeno, 2000) intentionally elicits the feelings and interpretations students have of their own classroom experiences. The results have been striking and have repeatedly shown that the practices of learning mathematics enacted in classrooms defined the mathematical knowledge that students learned and their orientations and dispositions toward mathematics.

Lubienski (2002) found in her case study of a seventh-grade class that the high SES and low SES students experienced mathematical discussions differently. Whereas the higher SES students claimed to feel confident in contributing to and in sorting out the ideas presented, the lower SES students found discussions frustrating and wanted the teacher to show them how to do the problems. Lubienski's case study stands in contrast to much work, including the other two studies reviewed above, that have documented the benefits of engaging students in collective mathematical discussions. This case and others (see, e.g., Civil, Baxter, Woodward, & Olson, 2001) begin to open up the complications that can arise in

classroom practice that centers on mathematical discussions. Much research remains in understanding how classroom practices are negotiated so that they do not have differentially negative impacts on any particular group of students.

Summary

The nature of mathematical discourse is a central feature of classroom practice. If we take seriously that teachers need opportunities to learn from their practice, developing mathematical conversations allows teachers to continually learn from their students. Mathematical conversations that center on students' ideas can provide teachers a window into students' thinking in ways that students' individual work cannot do alone. Mathematical conversations provide opportunities for teachers to hear regularly from their students and to learn about the range of ideas students have about a particular mathematical idea, the details supporting students' ideas, the values students attach to those ideas, and the language students use to express those ideas. The knowledge teachers gain from engaging with their students in conversations is essential for teaching for understanding. This is a place where teachers can learn and hone their skills.

Courtney Cazden (2001) in her second edition of *Classroom Discourse* pointed out that teachers are being asked more often to add nontraditional discussions to their repertoires to better support the development of students' higher level thinking. She also pointed out that the "challenges of deciding, planning, and acting together across differences of race, ethnicity and religion are growing" (p. 5). So more than ever educators need to pay attention to who speaks and how opportunities are provided for varied participation. Cazden suggested, as do other educators, that when engaging in different forms of discourse the rules and practices for students be made explicit. Cazden claimed that educational purpose and equitable opportunity to learn remain the most important design principles for classroom discourse. This remains true within mathematics education. The classroom discourse literature consistently finds that (a) many forms of discourse support student participation and learning, (b) the teacher plays a pivotal role in supporting productive discourse and participation, (c) details within the discourse matter, and (d) issues of equity remain.

So much of the work we reviewed addresses the concerns raised by Cazden around both enriching repertoires of different opportunities for talk as well as considering issues of equity. Revoicing, interrogating meaning, and attending to detail all provide opportunities

to continue to challenge the community of teachers to examine how we support mathematical discourse, how the way we structure talk supports the development of students' mathematical identities in productive ways, and how our practices as teachers make room for only limited opportunities for participation.

The limited citations here clearly indicate that the research in these areas has only just begun. What is reassuring is that the work being done is providing productive ways of detailing classroom practice in ways that attend to teacher learning. In addition this work provides a way of thinking about how to continue to engage in research around classroom discourse. It also, taken together with the studies of cooperative group work, demonstrates that we have evidence about how to support student discourse and that these forms of discourse are productive for student learning.

ESTABLISHING NORMS FOR DOING AND LEARNING MATHEMATICS

As the research on managing classroom discourse shows, much of teachers' classroom work involves negotiating—with students—the context for learning, or in other words, supporting the development of classroom norms. Research has shown mathematics classrooms to be particular kinds of social contexts where the structures of activity within them afford and constrain what is learned, how it is learned, and which students learn it (Boaler, 1997; Yackel & Cobb, 1996). As Lampert et al., (1996) argued, for children to learn to formulate mathematical claims in school, the terms and structure of discourse need to be reconfigured. And beyond that, a sense of what counts as mathematical work in school needs to be reconfigured. Lampert (2001) described it as "establishing and maintaining norms of action and interaction within which the teacher can teach and students can study." (p. 51) The teacher plays a critical role in nurturing the mathematical climate of the classroom. Wood, Cobb, and Yackel (1991) describe the teacher's role as guiding mathematical dialogue among students to describe their solutions. The teacher uses these contributions to ask more questions and support the learning of the individual and the group.

In this chapter, we will describe the relationship between managing classroom discourse and the demands it places on the teacher to establish, maintain, and continually negotiate with students the norms that govern classroom interactions and mathematical work. Quoting Goffman (1974), Wood defined norms as the "interlocking networks of obligations and ex-

pectations that exist for both the teacher and students [that] influence the regularities by which students and teacher interact and create opportunities for communication to occur between the participants” (Wood, 1998, p. 170). These “hidden regularities guide the actions of participants in the classroom . . . [and] become the taken-for-granted ways of interacting that constitute the culture of the classroom” (Wood, 1998, p. 170). Key to the developing understanding of the centrality of norms is Yackel and Cobb’s work on distinguishing social norms from sociomathematical norms. They have drawn researchers’ attention not only to what is generally valued in classroom interaction but also to how the teacher and students engage with one another mathematically (Yackel & Cobb, 1996). Several other bodies of work have provided further contours to this work. Lampert, we will show, theorized extensively about her moves to create a particular set of norms related to doing mathematics in the classroom. Work by Civil, Boaler, and others has also underscored the challenges teachers face in supporting the cultivation of particular norms, especially in light of the complex ways students’ social and schooling histories position them in relation to one another as well as to mathematics.

While teachers create opportunities that structure participation, they do not do this alone. Schools and communities have long cultural histories that shape what happens in classrooms. Families and students bring their own histories and identities that shape participation. Yet, together, the teacher, students, and families can shape who learns and how. Who gets to talk, when, about what, who participates and how, the kinds of solutions and questions considered acceptable, and so on are determined by those participating and the experiences they bring to the classroom. Shaping participation is not always accomplished explicitly but can be driven by implicit goals, beliefs, and identities of the teacher, school, and community. Boaler and Greeno (2000) reported that the practices that constitute mathematics learning in one classroom may be vastly different from one another and foster different understandings of mathematics. Shaping learning opportunities in the kinds of classrooms we are advocating requires explicit attention to who gets to participate and how. The ways norms are shaped influences which students learn, what they learn, and how they learn it.

Relating Classroom Norms to Doing Mathematics

Students come to mathematics classrooms with definite notions about what it means to “do” mathematics.

In an early study of first-grade classrooms, Franke and Carey (1996) found students to be quite articulate about what it meant to do mathematics and do it successfully in their classrooms. The first graders clearly said they knew their answer was wrong if the teacher asked them a question about it. They explained either that they had to be fast at math to be good at it or that everyone who had a strategy was good at math. And student responses within classrooms were quite consistent.

Students, like these first graders, develop definite notions of what constitutes participation in a mathematics classroom, and often they come to mathematics classrooms expecting participation that includes working in textbooks, solving pages of problems, listening to teachers explaining, finishing, and finishing quickly. Students typically do not see explaining their thinking as a part of doing mathematics, and they certainly do not typically consider justification of their mathematical ideas as part of the process. This makes creating classroom cultures that support the development of understanding through ongoing mathematical discourse challenging, particularly for those students who are already in middle and high school with a great deal of traditional school experiences (Silver, Kilpatrick, & Schlesinger, 1990; Silver & Smith, 1996).

Depending on one’s theoretical perspective, the discussion of creating opportunities for participation can focus on rule-setting, creating a context for learning, or developing a set of classroom norms. In each case, the concern is with shaping the learning environment in a way that supports student learning and with developing certain dispositions toward doing mathematics. However, given the goals of creating a particular type of mathematics classroom where sense-making and discourse are central and where different forms of participation are acceptable and expected (so each student is included), detailing the development of norms is critical. Cobb et al. (1993) emphasized that it is essential to adopt a perspective that focuses on the culture of the classroom community even if one’s primary concern is to understand individual students’ construction of mathematical knowledge, as individual mathematical learning is influenced by both the mathematical practices and the norms negotiated by the classroom community.

Distinguishing Social From Sociomathematical Norms

In considering norms for classrooms that support mathematical understanding, one must consider the regularities in classroom social interactions that constitute the grammar of classroom life. These general

classroom norms apply to any subject matter area and are not unique to mathematics but sustain classroom cultures characterized by explanation, justification, and argumentation (Cobb, Yackel, & Wood, 1989; Yackel et al., 1991). The classroom norms, interactively negotiated by teachers and students, set guidelines for acceptable participation and constrain individual actions. From the perspective of general social norms, these regularities include how participants engage together—how participants use tools, work together, listen, respect one another's ideas, and determine language to support participation.

Adding it Up describes particular norms for creating mathematics classroom communities. The norms proposed, based on theory and research, include (a) value placed on ideas and methods, (b) student autonomy in choosing and sharing their problem solving methods, (c) appreciation of the value of mistakes as sites for learning for everyone, and (d) renegotiation of authority for whether something is correct or sensible so that authority lies in the logic and structure of the subject and not with the status of the teacher or popularity of the person making the argument. These norms shape opportunities for students to share their thinking, engage in discussion around ideas, value and learn different ways of thinking about problems, and build mathematical arguments.

Lampert's (2001) description of evolving norms, although similar to those put forth in *Adding it Up*, are more specific. She focused on what it means to investigate meaning in public. Lampert (2001) argued that names accord importance and, by labeling them, make particular ideas, skills, and habits usable. She articulated three explicit activities related to mathematical problem solving that she names and works to establish with students: (a) finding and articulating the "conditions" or assumptions in problem situations that must be taken into account in making a judgment about whether a solution strategy is appropriate, (b) producing "conjectures" about elements of the problem situation including the solution, and (c) revising conjectures based on mathematical evidence and the identification of conditions (Lampert, 2001, p. 66). She describes what it means to display alternative legitimate responses and how to disagree. She works on how to use tools, specifically the use of journals. She works at developing norms for small-group as well as large-group interaction and for written communication (Lampert, 2001). Central to all of these norms is mathematics and mathematical work.

Yackel and Cobb (1996) argued that within mathematics classrooms norms are developed that are particular to the mathematical activity and guide the mathematical work, norms that go beyond the general

norms for managing discourse. These norms would be more closely aligned with Lampert's ideas of what it means to engage in mathematical disagreement or to revise a mathematical explanation. Yackel and Cobb termed these *sociomathematical norms*. Knowing that one is expected to explain one's thinking is a social norm; knowing what counts as an acceptable mathematical explanation is a sociomathematical norm. Sociomathematical norms include such norms as those around what counts as mathematically different, mathematically sophisticated, mathematically efficient, or mathematically elegant explanations.

Considering the Consequences of Norms for Student Learning and Dispositions

Attention to sociomathematical norms is critical because it allows one to explicitly address the mathematical aspects of teachers' and students' work in the mathematics classroom. In particular, it allows one to make explicit the implicit ways in which the mathematical work is being driven by particular ways of considering sophistication, elegance, and difference in mathematical ideas. Sociomathematical norms can also be a way to make explicit the teacher's role in relation to both the students and the mathematics. Moreover, attention to norms enables one to think about the values that are being enacted as students engage in the work of mathematics with one another.

Yackel and her colleagues have pointed out that implicit in the inquiry approach to mathematics are social norms that can foster students' development of social autonomy (Cobb et al., 1991; Cobb et al., 1989; Kamii, 1985; Nicholls, Cobb, Wood, Yackel, & Patashnick, 1990). The sociomathematical norms that are negotiated within inquiry-based mathematics classrooms can foster the development of intellectual autonomy: "students who are intellectually autonomous in mathematics are aware of, and draw on, their own intellectual capabilities when making mathematical decisions and judgments as they participate in these (classroom) practices" (Yackel & Cobb, 1996, p. 473).

Lampert (2001) also discussed the broader goals she has for developing norms for articulating conditions, making conjectures, and revising ideas. She described these activities as central to the work of doing mathematics in the classroom, but she also discussed at length her efforts to see that these practices also help students to become "the kinds of people who study in the classroom and who expect others to do so as well" (p. 265) and "who are academic resources for themselves and for one another" (p. 266). She commits herself to helping students learn to practice Polya's *intellectual virtues*, those of intellectual courage,

intellectual honesty, and wise restraint. For Lampert, developing norms for what constitutes mathematical reasoning (developing and revising conjectures and conditions) should also lead students to learn certain intellectual virtues.

Studying the Emergence and Negotiation of Norms

Classroom norms develop through ongoing negotiation among the participants. Mathematics is continually negotiated and institutionalized by a community of knowers. Mathematical activity in the classroom occurs within mathematical practices that have been institutionalized by the classroom community and are taken-as-shared by its members. According to Cobb et al., (1993), “the background for mathematical activity and dialogue is constituted by the results of prior negotiations of mathematical meanings as well as by current social norms” (p. 112). As Lampert (2001) asserted, developing classroom mathematical norms happens at the beginning of the year but continues throughout the year as norms are continually renegotiated.

Creating norms does not occur solely as a result of the teacher’s statements or actions. As teachers and students work together on mathematics, ways of participating in those practices, and the language supporting them, develop. Regularities in these interactions shape the norms and support learning. Thus, the development of norms can be constantly renegotiated. The cause for renegotiation, according to Much and Shweder (1978), often occurs when either the teacher’s or child’s expectations are not fulfilled or when there is a perceived breach of a social norm.

Cobb, Yackel, and Wood studied the norms that developed in the classrooms in their elementary school project. They documented four types of norms, norms around (a) working together, (b) students’ thinking through ideas for themselves, (c) who decides on acceptable explanations, and (d) what types of mathematical reasons are acceptable. They found that acceptable explanations were mathematics-based and not status-based. Acceptability of explanations was not decided by the teachers’ response; students were expected to show mathematically why a solution was correct. In fact, project students were significantly more likely to believe that one succeeds in mathematics by attempting to make sense of things and significantly less likely to believe that success stems from conforming to the teacher’s or other children’s solution methods, from attempting to be superior to classmates, or from being lucky, neat, or quiet (Nicholls et al., 1990; Yackel et al., 1991). In addition, rather

than providing a procedural explanation of the steps to a solution without reference to the quantities or the mathematical ideas, students expected to reference the mathematics.

McClain and Cobb (2001) studied the process through which sociomathematical norms of acceptable mathematical solutions, mathematical difference, and mathematically efficient solutions were established in a first-grade classroom. They focused particularly on the teacher’s proactive role. They found that one of the teacher’s primary challenges was to overcome the dichotomy between providing all the direction and structure for students when discussing solutions and following the students’ own interests and interpretations. In addition, they found a relationship between the emergence of particular sociomathematical norms. As the class worked on what counted as a mathematically different solution, the teacher leveraged those ideas to support the development of what counted as an efficient or sophisticated solution. Further, the norms for what counted as an efficient solution gave direction to students’ learning about arithmetic and supported the development of flexible reasoning, in particular towards algebraic understandings of properties of operations.

Significant in McClain and Cobb’s (2001) study of the emergence of sociomathematical norms was the role teacher’s notation played in supporting the development of norms of mathematical efficiency and sophistication (see also McClain, 2000). This finding adds to the understanding that talk is not the only contributor to the development of sociomathematical norms. For example, how the teacher chose to represent and notate students’ strategies provided lasting records of ideas students used as they made judgments about efficient solutions. In middle school classrooms, Hershkowitz and Schwarz (1999) also noted that the use of nonverbal interactions between students and tools, in this case multi-representational software and multiphase activities, contributed to the development of sociomathematical norms. In their study, how students engaged with the tools themselves, as well as with the activity structure, supported sociomathematical norms of what counts as evidence and what counts as a good hypothesis.

Kazemi and Stipek (2001) found four sociomathematical norms distinguished classrooms in which there was a press for conceptual learning and understanding: (a) explanations were supported by mathematical reasons, (b) mistakes created opportunities to engage further with mathematical ideas, (c) students drew mathematical connections between strategies, and (d) consensus was reached through mathematical argumentation. They concluded that when teachers

created a high press for conceptual thinking, resulting in the development of these four types of norms, mathematics drove not only the activity but the students' explanations as well, thus increasing student understanding.

Silver et al., (1995) found that creating an atmosphere of trust and mutual respect in QUASAR middle school classrooms was critical for the development of mathematical discourse communities. Specifically, they reported that in supporting students to share alternative explanations and question each other's ideas, it was important to avoid criticizing and blaming and to focus on respecting each other's ideas (Silver et al., 1995). They pointed to the importance of norms developed around attention to language. Here they talked about supporting and attending to the language of the students, developing a shared vocabulary, respecting a range of ways of using language to share ideas, and supporting the sharing of information in a form everyone can understand (Silver, 1996). They found it challenging to get students to participate together in ways that challenged one another's thinking and justified their ideas. They suggested that developing the social norms might first need to occur prior to developing those norms focused particularly on the mathematics (Silver & Smith, 1996).

Emerging Complications in the Negotiation of Norms

A number of studies have also drawn attention to the impact that traditional mathematics instruction has on students and the challenges created for teachers who attempt to create new classroom norms based on mathematical reasoning. These studies point to the need to attend to the dispositions and values that students learn as classroom norms are negotiated.

Boaler (2001) found that if students encountered textbook situations that departed from their expectations, they became confused not only because of the limited extent of their mathematics knowledge but also because of the regularities of the mathematics classrooms to which they had become accustomed (Greeno & MMAP, 1998). When students engaged in assessments that asked questions in a different way than in their classrooms, they tried to draw upon their classroom practices and found that these practices were limited in helping them respond to the task. They could not (or did not) use the mathematics they learned. Students' practices were highly specific to the mathematics classrooms, and when they were in different situations, even exams, the students became confused because they tried to follow the cues they had learned in class and found that those practices

did not help. Boaler (2001) concluded that students learn a set of practices "and these come to define their knowledge" (p. 9). In class, students who adapted and applied the mathematical methods and who discussed ideas and situations with different people could use a variety of practices and more readily found ones that could be used in the "real world."

In interviewing students Boaler and Greeno (2000) found that many of the students who talked about disliking mathematics felt so not because of the cognitive demands of the mathematics or because they were unsuccessful (often they were quite successful) but rather because they wanted to pursue subjects they felt offered opportunities for expression, interpretation, and agency (they saw mathematics as about receiving knowledge). These findings underscore the sometimes invisible impact of classroom practices that promote traditional approaches to mathematics as a discipline. The findings depart markedly from virtues promoted by Lampert, Cobb, Yackel, and others of developing students' intellectual autonomy and virtues, discussed above, of the importance of intellectual autonomy, intellectual courage, intellectual honesty, and wise restraint.

Civil and Planas (2004) documented the challenges to creating sociomathematical norms for mathematical deliberation in classrooms, particularly for low-income students of color. Students in the classes studied were not accustomed to analyzing one another's work, especially within the mathematics class (Civil, 2002). As teachers worked to establish a set of sociomathematical norms (involving students engaging together around mathematical thinking and exchanging ideas), the researchers found that forms of participation were influenced by the organizational structures at school and the emerging memberships they created. Civil and Planas reported that when they tried to open up the patterns of participation in the classroom, power and status structures were deeply engrained. Students who were athletes or from a gifted class had the status and power in the class that made it difficult for others to participate. The athletes and "gifted" students were listened to and their ideas given credence over others regardless of the mathematical argument put forth. Other students, without this status, did not see it their place, or worth their time, to participate. These existing forms of participation, often created by structures outside the classroom, made creating particular types of norms challenging at best. Civil's work (1995, 1998, 2002) leads to a consideration of how to support teachers as they attempt to recognize and address the complexity involved in negotiating such norms.

Developing Norms and Teacher Learning

We have argued throughout the chapter that one aspect of designing classroom practice is to consider teachers' learning through practice. Cobb and his colleagues have pointed to how teachers in their project learned through their interaction with their students and how this learning was shaped also by the norms that were negotiated (Cobb et al., 1989; Wood et al., 1991). This is particularly true for the development of sociomathematical norms. If teachers are expected to facilitate discussion, teachers need the opportunity to make sense of a wide array of student solutions. This need returns us to the importance of teachers' developing deep knowledge of students' thinking and subject matter (Yackel, 2002). Teachers must learn how to identify increasingly sophisticated solutions and know how to support students as they detail them and justify them through conversations. The kind of sociomathematical norms that teachers attempt to develop in the classroom will reflect how they are able to make the mathematics central as they attend to the sharing, make sense of a justification, or decide about the sophistication of a solution. Thus, teachers' learning opportunities are shaped by the sociomathematical norms that are developed in the classroom. If only social norms are developed, such as sharing solutions so that ideas are on the table, and the mathematical differences among strategies are not consistently part of the conversation, teachers do not have to listen for them, focus on them, or learn from them.

Summary

In this section, we have argued that as teachers plan and carry out instruction, they must attend to the norms that govern interactions in general and interactions around mathematics in particular. The social and sociomathematical norms that emerge in classrooms have consequences for what students learn about particular mathematical ideas as well as what it means to do mathematics. As many researchers have also argued, the norms that define the classroom culture also impact the kinds of intellectual virtues that students learn. Negotiating norms for engaging in mathematical deliberation together is not without its challenges and complications. In this section, we again drew on research of students' experiences in classrooms to raise questions about how the negotiation of norms positions students with one another and with the subject matter. Negotiating sociomathematical

norms demands, then, that the teacher know not only how to manage discourse but also how to draw on subject-matter knowledge and knowledge of students' mathematics, how to use tools strategically, and how to build relationships with students. These activities also imply that teachers need to learn from their practice as they engage in this work.

BUILDING RELATIONSHIPS FOR DOING AND LEARNING MATHEMATICS

Developing norms that open opportunities for participation in ways that support the development of mathematical discourse can be challenging (Civil & Planas, 2004; Silver, 1996; Silver & Smith, 1996) and requires building relationships with students. Building relationships with students allows teachers to challenge assumptions about who students are and what they bring to the mathematics classroom in a way that supports creating opportunities for participation and mathematical learning. Building relationships with students involves understanding their identities and the experiences that have shaped them as well as allowing the students to get to know one another and the teacher. Attending to understanding mathematics requires that teachers attend to the details of the students' work, the various identities students develop in relation to the mathematics and the classroom community, and the cultural histories that shape students' identities and participation.

Within mathematics education, significant work has focused on getting to know students' mathematical thinking. Knowing about students' mathematical thinking involves knowing the details of how one's students are making sense of particular mathematical ideas and at the same time knowing in general how students develop in relation to a particular mathematical idea. Knowing students' mathematical thinking provides the substance for the mathematical conversations. The benefits of knowing students' mathematical thinking and the trajectories that describe particular mathematical development is well documented (National Research Council, 2002).⁶

Simon (1997; Simon et al., 2004) emphasized the detailed knowledge of student trajectories teachers need to support their classroom practice. The work of the teacher is embedded in the decisions teachers make around students' trajectories. Teachers' knowledge of the cognitive learning trajectories of each student in

⁶ We cannot stress enough how critical this aspect of teachers' work is. We do not detail research on knowing students' mathematical thinking here as it is detailed both in other chapters in this volume and in a number of content-based reviews.

the class provides the basis for choosing tasks that support students' reflective abstraction. These trajectories provide a way to think more clearly about the mechanisms underlying students' acquisition of particular knowledge. Mechanisms are critical here. Understanding the mechanisms enables teachers to choose activities or tasks that provide opportunity for learning.

Cognitively Guided Instruction: A Detailed Example

Cognitively Guided Instruction has much to say about classroom practice that supports the development of students' mathematical understanding. Although the focus of the CGI work was not detailing instruction, much can be drawn from the body of work that can inform classroom practice. The work focused on supporting teachers and learning with teachers as they engaged with research-based information about students' trajectories within particular whole number domains. The data from the experimental study of first-grade classrooms, the qualitative studies of six teachers over a year, the longitudinal study of first-through third-grade teachers, and the follow-up study to the longitudinal studies are quite consistent (Carpenter, Fennema & Franke, 1997; Carpenter et al., 1999; Carpenter et al., 1989; Carpenter et al., 1998; Fennema et al., 1996; Fennema, Carpenter, Franke, & Carey, 1993; Franke et al., 2000). Together these studies provide links between teacher learning and student outcomes, and between teacher learning and changes in classroom practice. Teachers who listened to their students' explanations and asked for multiple strategies to problems had higher achieving students on written measures of skill and problem solving. Spending twice as much time learning number facts did not mean students could recall their number facts better; actually, students in classrooms where teachers spent more time on problem solving and half as much time learning their number facts were more successful using recall to solve problems. Teachers who could report accurately and in detail about their students' mathematical thinking had higher achieving students. These teachers were also ones who continued to learn through their practice.

In the CGI longitudinal study of first-through third-grade teachers and their students and in the follow-up study, 4 years after the professional development work ended, CGI researchers found that although many teachers figured out how to ask their students about their thinking and share strategies it was more difficult for teachers to engage in discussions that required students to compare strategies (Fennema et al., 1996; Franke et al., 2000). They also

found that teachers who were able to detail the development of student's mathematical thinking, who asked students how they solved problems and made use of what they heard from students in their instruction, were those who saw themselves as continuing to learn from their students. These teachers, engaged in what CGI researchers termed *generative growth*, believed that they had a way to make sense of what they learned from interacting with their students, a way of making sense mathematically of what they heard from students. Here teachers had both knowledge to support their practice and a stance toward that knowledge, a stance that they could add to and create ongoing understanding for themselves and their students.

CGI provides evidence that teachers' classroom practice that (a) includes eliciting and making public student thinking, (b) involves eliciting multiple strategies, (c) focuses on solving word problems, and (d) uses what is heard from students to make instructional decisions led to the development of student understanding. Further, teachers whose classroom practice drew on their (a) detailed knowledge of the development of students' mathematical thinking within a domain, (b) organization of student thinking in relation to the mathematical content, and (c) notions that they could continue to learn from their practice contributed to the development of student learning. Clearly, this could be interpreted simply and pronouncements could be made about "just get teachers to ask their students how they solved a problem," but it is important to note that factors about the teachers themselves and their classroom practice evolve together, and it is that coming together that drives how teachers engage in their classroom practices.

Beyond Knowing Children's Thinking

Research then provides substantial evidence that building relationships with students around the specifics of the mathematical work is critical to developing understanding. However, we argue that relationships based on getting to know students' mathematical thinking alone are limited. Teaching to open opportunities for participation requires getting to know students' identities, histories, and cultural and school experiences, all in relation to the mathematical work.

Educators need to know what happens in the world of the children with whom they work. They need to know the universe of their dreams, the language with which they skillfully defend themselves from the aggressiveness of their world, what they know independently of the school and how they know it. (Freire, 1998, p. 72)

When the classroom environment and pedagogical practice are brought into line with the lived experiences of students, student performance is positively influenced (see reviews by Dilworth & Brown, 2001 and Gallego, Cole & Laboratory of Comparative Human Cognition, 2001). In addition, without attention to students' cultural and racial backgrounds, mathematics classrooms will continue to play a role in producing and maintaining the inequities that exist in society (e.g., Gutstein, 2003; Moses & Cobb, 2001; Oakes, Muir, & Joseph, 2003).

We purposely focus this section on building relationships as we intend to push beyond seeing teachers' work as eliciting students' thinking in mathematics and consider what it means for teachers and students to get to know one another in ways that lead to different opportunities for participation in mathematics for teachers and students, challenge the status quo, and allow for the possibility of the distribution of student and teacher voices.

Drawing on Ideas of Identity and Culture⁷

Embedded in current definitions of mathematical understanding are notions of students' perceptions of themselves in relation to the mathematics. Notions of identity are included in understanding not only because educators want students to like mathematics for the purposes of engagement and motivation over time but also because developing a particular stance towards the mathematics is part of understanding the mathematics (National Research Council, 2002). The relationship students develop with mathematics—how they see themselves in relation to the mathematics—influences how they participate, how they make sense of the mathematics, and the different ways they persist. This is all a part of understanding mathematics. This places one's mathematical identity not as something peripheral to doing mathematics but as central.

Gee (2001) described identity as "being recognized as a certain 'kind of person' in a given context." Wenger (1998) described identity as the "pivot between the social and the individual" (p. 145). Both would argue that identities are constantly being formed and re-formed in the dialectic between social structures and individuals' lived experiences. Identities reflect histories of engagement in sociocultural communities, institutions, and practices. As Stuart

Hall (1996) explained, "[This view] accepts that identities are never unified and, in late modern times, increasingly fragmented and fractured; never singular but multiply constructed across different, often intersecting and antagonistic, discourses, practices, and positions" (p. 4).

These conceptions of identity illustrate the importance of thinking about classrooms as places where identities are constructed and reconstructed through discourse. In many mathematics classrooms the identities constructed through discourse practices perpetuate beliefs about who is mathematically literate, and often this construction differentiates students on racial lines (Boaler, 2002; Martin, 2000).⁸ A number of researchers have begun to explore how understanding identities in the context of a classroom's discursive practices can help us understand students in a new way and see how to challenge existing inequities. For example, students are often located and locate themselves within dichotomous identities of achievement: successful versus struggling. Students' own sense of struggle may be hidden behind what they find appropriate to share and divulge about themselves to their peers or their teacher (see, e.g., Dutro et al., in press).

Researchers have recently begun to examine how individuals are positioned and position themselves within the mathematics classroom context (Gresalfi, 2004; Martin, 2000; Nasir, & Saxe, 2003). Just as we researchers have learned a great deal about the different ways that males and females may participate in and with mathematics, we are now just beginning to understand how students from different races and cultural backgrounds may also have different participation histories that influence how they engage in mathematics. Often minority students believe that they must choose between a strong academic and a positive ethnic identity (Nasir & Saxe, 2003). How the tensions continue to play out matters for how students engage in school and in mathematics. Martin (2000, 2003, 2006, in press) studies the co-construction of African American and mathematics identities, highlighting the challenges that many African Americans face in negotiating positive identities as mathematics learners. Martin has shown that mathematically successful African American students not only maintain their sense of self as African Americans and as students but also construct a number of other positive identities;

⁷ We draw heavily here on work outside of mathematics education. This work has been reviewed in a number of places (see Dilworth & Brown, 2001) and not only serves as a basis for conceptualizing what can happen in mathematics classrooms but also provides details we see useful in mathematics classrooms.

⁸ We recognize that race is not the only way students are differentiated (e.g., language, gender, sexual orientation). However we focus here specifically on issues of race as we see the need for mathematics classrooms to attend to race more explicitly. Research on race and mathematics teaching is limited. See equity chapter in this volume for a more detailed examination of these issues.

they are not “acting white.” He frames mathematics teaching and learning as *racialized forms of experience*—where meanings for race and racism emerge in the day-to-day struggles for mathematics literacy among the African American adolescents and adults he has studied. In all of their stories of mathematics teaching and learning, race was salient. Martin (2006, in press) connects the identities developed by the African Americans in his study to the sociohistorical and structural forces that shaped the kinds of mathematics practices in which students participated. Martin (in press) argued for the need to

acknowledge that mathematics learning and participation, like many other areas in life can be viewed as *racialized forms of experience*; that is, as experiences where race and the *meanings for race* emerge as highly salient in structuring (a) the way that mathematical experiences and opportunities to learn unfold and are interpreted and (b) who is perceived to be literate, and who is not.

The complexity of issues of identity for students of color in the United States highlights the importance of understanding how minority students structure and manage emerging tensions as they construct and negotiate ethnic and academic identities in the course of their everyday activities.

Drawing on Cultural Practices

Identities are shaped, constructed, and negotiated through cultural practices (Holland, Lachoitte, Skinner, & Cain, 1998; Lave & Wenger, 1991; Martin, 2000; Moll & Gonzalez, 2004; Nasir, 2002; Wenger, 1998). Understanding cultural practices becomes essential to understanding students’ evolving identities, both the cultural practices students engage in outside of school and those of the classroom.

Leacock’s (1971) definition of culture

refers to the totality of a group’s learned norms for behavior and manifestations of this behavior. This includes the technological and economic mechanisms through which a group adapts to its environment, its related social and political institutions and the values, goals, definitions, prescriptions, and assumptions which define and rationalize individual motivation and participation (p. 35).

So individuals’ ways of participating are shaped by the norms, values, and goals of the group. As teachers and students together create a new set of norms they must understand the existing norms for doing and talking to one another. Understanding the cultures—the

groups students have participated in—can help the teacher build relationships based on a depth of understanding and not on assumptions or stereotypes.

A challenge to developing relationships with students that draw on their histories and cultural experiences involves the deficit views of families that frequently exist. These deficit views often permeate mathematics classrooms (Oakes et al., 2003; Spencer, 2006). A number of researchers argue for the need to change deficit views of families and communities and look to draw on the strengths that exist (Brice-Heath, 1982; Civil, 2002; Cole, Griffin, & LCHC, 1987; Moll & Gonzalez, 2004). Researchers drawing on the *cultural difference* perspective rather than the *cultural deficit* perspective argue that teachers’ work requires that they find ways to reduce the cultural mismatch between the home and classroom cultures. The goal of reducing the mismatch is to use what the children already know, along with the associated cultural practices, as resources for understanding in the classroom (Moll & Greenberg, 1990, cited in Gallego et al., 2001). Many researchers have investigated and developed strategies to reduce mismatch, and Gallego and her colleagues in analyzing this work stated that, “these examples make it clear that a strategy of local accommodations of school culture to home culture can be educationally productive” (p. 979).

Moll and his colleagues documented how households contain accumulated *funds of knowledge*, the skills, abilities, and practices that support the functioning of the household. Moll and Gonzalez (1997) pressed teachers to come to know the cultures of their students. They argued that teachers need to become qualitative researchers in some ways as they come to understand students in their home environments, detailing the strengths and skills students use in the home that can be used to understand participation (including language use) in school. This work explicitly rejects the notion that the problem of underachievement either is located within the students or is due to students’ culture, and it shifts responsibility for underachievement to understanding what students do bring to school and how schools can draw on the wealth of knowledge and experience that the student brings (Moll, 1992).

Civil has taken these ideas into the mathematics classroom through the BRIDGE Project. The explicit goal of the project was the development of mathematics teaching that builds on students’ backgrounds and experiences (Civil, 2002; González, Andrade, Civil, & Moll, 2001). Although this work points to the potential of bringing parents and families into schools so teachers see students’ strengths, it also makes explicit the challenges of actually building on the everyday

mathematics knowledge and experiences of students in schools. Civil found that often the mathematics in everyday activity was hidden: People did not see themselves as doing mathematics and often rejected the idea that what they were doing was mathematics (Gonzalez et al., 2001). And although this work is challenging it does begin to point to the fact that students' competencies identified outside of the classroom are often not used as a resource in school.

Hand (2003) analyzed classroom dynamics in two reform-driven algebra classrooms in an urban high school. She found what she has termed a *participation gap* between the two classrooms. In one classroom any activity that was not directly related to the lesson was treated as a potential distraction to students' learning and thus prohibited. In contrast, in the other classroom, the teacher searched for links between mathematics and the knowledge and practices that the children brought with them to the classroom. In the class where the teacher validated students' contributions, greater participation was evident. Erickson and Mohatt (1982) found that when teachers used language patterns that approximated the students' home cultural patterns or balanced these patterns with the Anglo instructional patterns typically used in school, they were more successful in improving student academic performance. Here the goal was to include student culture in the classroom as authorized knowledge. In work with African American students researchers have found that teachers have successfully built on forms drawn from African American churches. The teachers built relationships in their classrooms that were marked by social equality, egalitarianism, and mutuality stemming from a group rather than an individual ethos. The focus was on collective and personal responsibility (Henry, 1992; Hollins, 1982; Foster, 1993, all as cited in Dilworth & Brown, 2001).

The work of Lipka and the Alaskan Natives, a group of indigenous people of Alaska whose language and culture has been marginalized within the educational system by its emphasis on English-only language and middle-class White American cultural practices (Lipka, 1994, 2002, 2005; Lipka, Wildfeuer, Wahlberg, George, & Ezran, 2001), draws upon the knowledge of the Yup'ik elders in making curricular decisions. Yup'ik elders, teachers, bilingual aides, students, school-board members, and university faculty created a working environment where both insider and outsider knowledge was valued. In this way the teaching of mathematics became a community project that drew on and respected both local knowledge and more traditional mathematics. The researchers have documented that the achievement of students whose teachers used the curriculum after receiving

professional development surpassed that of comparable groups of students (Lipka, 1994, 2002; Lipka et al., 2005; Lipka, Wildfeuer, Wahlberg, George, & Ezran, 2001).

Moschkovich (2002), through her work on discourse and language, has pushed for making the mathematical practices of different groups accessible to students in classrooms. She has stated that it is not the using of everyday practices that is itself important but rather making connections between familiar practices and academic mathematical practices. She suggested that there needs to be a balance between the everyday and academic mathematics so students can be encouraged to engage in the study of mathematics while also providing them the discursive practices necessary for pursuing mathematics. The goal of the work that draws on students' cultural experiences is to understand and use the resources students bring to the classroom, thereby allowing students to accept and affirm their cultural identity while at the same time developing a critical perspective that can challenge the existing accepted practices of schooling. This work points to the value of teachers' understanding students' identities and cultural practices in building classroom relationships. Bringing these identities within the classroom walls legitimizes different forms of participation and allows for more diverse mathematical knowledge.

Teaching As Building Relationships Around Identity and Culture

Although we can make a theoretical argument and provide a number of existence proofs that attending to both identity and cultural practices in building relationships with students makes a difference in students' participation, the field continues to try to identify what exactly it means to attend to identity and cultural practices in mathematics classrooms. We argue here that educators can learn from culturally relevant pedagogy (and related literatures) and the critical stance it takes in providing opportunities for students of color and use it as a way to think about the types of relationships that can be built with students, families, and communities to shape the teaching and learning of mathematics.

Culturally relevant teaching, from Ladson-Billings's (1995) perspective, must develop students academically, support and nurture cultural competence, and develop a sociohistorical or critical consciousness. Culturally relevant teachers organize their classrooms and interactions with their students in ways that are guided by the following principles: (a) Treat students as competent and they will likely demonstrate com-

petence, (b) provide instructional scaffolding so students can move from what they know to what they need to know, (c) focus the classroom on instruction, (d) extend students' thinking and abilities, (e) gain in-depth knowledge of both students and the subject matter, and (f) link student understanding to meaningful cultural referents. How these principles are instantiated matters greatly. Although some may argue that these principles signify good mathematics teaching, what differs here is the focus on the relationships with students, and how students are respected for who they are and what they bring to the classroom (Ladson-Billings, 1995). Ladson-Billings has argued that culturally relevant teaching fosters student achievement and provides ways for students to maintain cultural integrity, develop academic success, and recognize, understand, and critique current social inequities (Ladson-Billings, 1995).

Many scholars have pursued the notion of culturally relevant teaching (see reviews by Gallego et al., 2001; Mercado, 2001). And a growing body of research shows how African American, Latino, and Native American students draw on their cultural and community knowledge to help them succeed in mathematics (Gutstein, 1997, 2003; Martin, 2000; Nasir, 2002).

Gutstein, Lipman, Hernandez, and de los Reyes (1997), in a multiyear study of mathematics teaching in a Mexican-American community, focused on three components they saw as critical to culturally relevant teaching: thinking critically in general and about mathematics, building on both students' informal math knowledge and their cultural and experiential knowledge, and gearing teachers' orientations to students' cultures. In their observations of mathematics lessons, the researchers watched teachers encourage students to explain and justify their answers, push students for multiple interpretations, and encourage mathematical communication (p. 720). They observed teachers' using students' out-of-school knowledge to help students understand mathematical concepts. And teachers came to understand and acknowledge students' cultures by building relationships with students and families.

Gutstein (2003) continued this work through a 2-year study in an urban, Latino seventh/eighth grade mathematics classroom where he taught using these principles of culturally relevant pedagogy along with reform curricula. Gutstein worked to help students develop sociopolitical consciousness, a sense of agency, and positive social and cultural identities while also supporting them to "read" the world using mathematics, developing students' mathematical power, and changing their disposition towards mathematics. He

found that, although it was challenging to merge his various goals, attending to both sets of goals supported student participation and learning.

Like Ladson-Billings and Gutstein, Frankenstein (1990, 1995, 1997) has looked for ways to connect to students' lived experiences so that students develop mathematical understanding while learning to challenge social inequities. Frankenstein operated from the position that effective curricular choices build upon students' existing funds of knowledge. She found that students were often reluctant to acknowledge—despite the gap in their formal mathematical knowledge base—that they are already quite capable logical thinkers in their everyday lives as consumers and workers. In addition, Frankenstein found that several of her students had a very particular vision of what counted as legitimate school knowledge, and as such she faced some resistance to addressing social justice issues through mathematics. For Frankenstein, mathematics can be a tool for grappling with social issues. She argued that the incorporation of ethnomathematics into mathematics curricula can be both culturally and intellectually empowering for students and can lead to increased participation in mathematics.

Most of this work studies exceptional teachers trying to challenge social inequities. However, Gau Bartell engaged more typical teachers in learning to teach mathematics for social justice. Gau Bartell (2005) engaged eight secondary mathematics teachers through the creation, implementation, observation, and revision of a mathematics lesson that incorporated social justice goals. She found that the teachers struggled in balancing the mathematics and social justice. In one group, mathematics "trumped the social justice," as teachers never addressed social justice issues, and for the other group, the mathematics was often not connected with social justice. In this group, teachers were so focused on guiding students to a particular idea that they supported conclusions that the mathematics did not. Although this work speaks to the difficulty for teachers to change their practice to incorporate social justice issues, researchers are just beginning to develop these ideas. It represents a fruitful area for future inquiry.

Summary

Within this section on building relationships the theory and literature point to the importance of understanding students beyond how they think about mathematics. Although understanding how students think about mathematics is critical and has been shown to make a difference for teachers' and student learning, understanding students in terms of their

race, cultural histories, and previous experiences enables teachers to build relationships that challenge assumptions and open opportunities.⁹

Just as knowing how individual students' thinking is supported by knowing trajectories of students within particular content domains, knowing students' identities in relation to school and mathematics is supported by knowing the communities in which they have participated. If teaching involves orchestrating and negotiating participation, as we have argued here, teachers need to understand students' participation histories. Teachers need to recognize the forces inside and outside that shape the multiple identities for students as they engage in mathematics.¹⁰ Teachers need to know how to draw students' identities into the mathematical work, support them to evolve in how they participate, honor different forms of participation, and structure opportunities that allow for different participation forms. We have described some work that begins to document what this can look like in a mathematics classroom. Yet researchers still know little; we particularly know little in terms of the type of critical stance some would argue for in creating classroom cultures and supporting participation. Teachers clearly, though, hold many assumptions about how certain groups of students do or should participate that can narrow forms of participation (often due to their own experiences), and opportunities are needed within the context of teaching to learn to challenge those assumptions in ongoing ways so that each student is challenged to participate in and understand the mathematics.

"Bringing the lived experiences" of students into the mathematics classroom is both challenging and controversial. It is challenging in that researchers who have begun to try to accomplish this have found resistance from a variety of sources, including the students themselves. It is controversial because including lived experiences can be accomplished in a way that focuses on deficit and not strength. The challenging and controversial nature of the work only points to the need for building relationships in classrooms that allow for a multidimensional understanding of each other and the mathematical work.

CONCLUSION

We began this chapter with two brief scenarios drawn from the mathematical teaching practices of Ms. Mi-

chaels and Ms. Jimenez. Both Ms. Michaels and Ms. Jimenez engaged in some aspects of practice supported by research findings that emerged from large-scale reform projects. They posed a sequenced set of problems, elicited student thinking, asked follow-up questions, facilitated classroom discussion around mathematical ideas, and used what they knew about students' mathematical thinking. They each engaged in these practices to different degrees and in somewhat different ways, and those differences had an impact on student participation. What is unknown from the short description of a single lesson in these teachers' classrooms are the details about how the teachers orchestrated mathematical conversation, the norms established in each of the classrooms around what it means to do mathematics, and the kinds of relationships teachers had developed with their students. Knowing more about each of these would enable us to have a better sense of how and why the practices in these two classrooms played out in the way they did and how they shaped student participation. We argued throughout this chapter that there have been important advances in research on classroom practice since the publication of the last *Handbook*. We can articulate and detail some critical features of classroom practices. We know what to begin to look for in Ms. Michaels's and Ms. Jimenez's classrooms. We recognize the need to know more about what surrounds those classroom practices, and this is illustrated by what we do not know about Ms. Michaels's and Ms. Jimenez's classrooms, no matter how much detail we provide about the interaction from one day. We believe that the field is in a productive place to move understanding of classroom practice forward by attending more closely to the relationship between the already articulated critical features of classroom practices and the surrounding discourse, norms, and relationships.

Three Features of Classroom Practice

We chose to focus the chapter on three features of teachers' classroom practice: discourse, norms, and building relationships. We chose to focus on these three aspects of mathematics classroom practice because consensus is building that students need opportunities in classrooms to share their mathematical thinking, discuss alternative approaches to solve problems, use mathematical tools flexibly, and so on. Providing students these types of opportunities requires

⁹ One needs to be careful here in avoiding essentializing by drawing on narrow preconceived notions of cultures. Notions and understandings about cultural histories need to be continually challenged to support the development of a complex set of understandings about culture and the student in relation to his or her culture.

¹⁰ We see this as part of what Ball, Goffney, and Bass (2005) called for in having teachers bring an *equity awareness* and willingness to act on that awareness to the mathematics classroom.

that teachers and students develop new ways to engage together around mathematical ideas—specifically, it requires new forms of mathematical discourse. Creating opportunities for discourse demands attention to the negotiation of both social and sociomathematical norms. The negotiation of these norms to support discourse requires building relationships between teachers and students that include attention to the identities students bring, their participation histories, and the norms and cultural practices of their communities. All of this is needed so that assumptions are continually challenged and renegotiated around what it means to participate and be literate in mathematics. And within this, issues of race, class, and gender need to be made salient.

The research on classroom discourse and specifically mathematics discourse has been accumulating over a number of years, and educators are in the position of beginning to understand the kinds of foci that can facilitate understanding mathematics classroom practice. The work on revoicing, interrogating meaning, and supporting language provides details around what teachers can attend to in managing discourse. And although researchers are accumulating evidence about the details of when and how to use these ideas to manage discourse, further research is needed, particularly to relate how teachers manage discourse to the norms that govern discourse and the teacher-student relationships that are built and enacted.

We also have evidence that classroom norms that attend to the details of working through mathematical ideas are critical for creating classroom cultures where students can develop mathematical understanding. The work on sociomathematical norms has pushed the field to consider the kinds of norms needed to support mathematical work and mathematical reasoning. We also have evidence that creating the norms that expect students to share their mathematical thinking, challenge and test ideas publicly, and work to make sense of one another's ideas is not easy; it involves issues of power and positionality as well as issues of identity that develop both inside and outside the classroom and are shaped by the societal, community, and school structures. This all makes building relationships with students critical to the success of developing classroom norms that support the development of mathematical understanding.

Evidence clearly exists to support the need for developing relationships with students, relationships built upon an understanding of students' past experiences, their cultural histories, their race, and their participation histories. Teachers who attend to the cultural backgrounds of their students, who build relationships with students as well as their families

and communities, make a difference in the success of students in their classrooms. Assumptions need to be challenged about who can do mathematics, what counts as mathematical argument, and what it means to participate in mathematical work. These assumptions are constructed around long-standing personal and societal beliefs about "groups" that shape interactions with individuals. Developing relationships with students that push teachers to get to know the multiple identities students bring, how those identities have been shaped, and the communities that they learn from provides teachers with a different way to see students and opens opportunities for what it means to participate in the mathematics classroom. We recognize that building this type of relationship with students will not resolve all of the equity issues in mathematics classrooms. We recognize the societal, structural basis for much inequity. However, we (as do others) see that building different kinds of relationships and opening different opportunities for participation and practice can lead to using mathematics to help transform what happens for students of color, English language learners, students living in poverty, and other marginalized groups.

Advancing Research on Classroom Practice

In the conclusion to this chapter, we would like to propose the direction needed in order to advance research on classroom practice. To do this, we draw on Lampert's (2005) and Graziani's (2005) recent work studying Italian language instruction in Rome. Following their lead, we argue that the three features of classroom practice we have discussed, although important in helping us think about the work of teachers and the experiences of students, still need to be assembled together and studied as they emerge, unfold, and lead to student learning and empowerment. We propose that the field needs to identify routines of practice and study how the three features of classroom practice discussed in this chapter work together in such routines. By *routines of practice* we mean that we need to agree on core activities (within each mathematical domain and at appropriate grade levels) that could and should occur regularly in the teaching of mathematics. We propose that such routines could become the central hub for teachers' practice and should become the focus of the study of classroom practice. Research would clarify how these routines of practice support teacher and student learning in different contexts with different teachers and students.

To connect to Simon's writing about learning trajectories (1997; Simon et al., 2004), these core

routines of practice should be set within an explicit framework for understanding the development of students' mathematical competence in any particular domain. In addition to our knowledge of students' thinking, we also want to underscore the role of mathematics. As seen through the other chapters in this volume, researchers are accumulating significant knowledge about particular domains of mathematics and how the content itself makes a difference in how ideas evolve in classrooms and in work with teachers (see Franke, Carpenter, & Battey, in press). Mathematics itself needs to be one of the fundamental considerations in both identifying the core practices and analyzing the surround—in how to consider how these core practices play out in classrooms. We cannot ignore that orchestrating mathematical discourse around algebraic thinking may need to look different than orchestrating discourse around addition of whole numbers. So although we argue for focus on routines of practice, we want to be careful and explicit in detailing how these routines of practice are embedded within particular mathematical domains. It is with this detail that we feel we will gain the information we need to support teachers and advance research.

In recent writings, we see advances in specifying routines. We would like to consider a few of these in order to begin to compare the level of specificity in which researchers are beginning to tackle this issue and to lay some of the groundwork for what work lies ahead. What Lampert (2005) and Graziani (2005) described in their analyses is that the Italian language school has a sophisticated framework for understanding linguistic competency. Within this framework, key instructional activities or routines of practice have been identified and in fact extensively specified. As teachers learn these routines of practice, they engage simultaneously in understanding teachers' instructional moves and students' contributions and understandings. Set within the framework for understanding linguistic competence, the entire set of instructional routines allows teachers to develop their knowledge and pedagogy in a principled way. The spare set of routines also enables teachers to develop their practices around common objects of study. We can imagine that researchers in mathematics education, within various mathematical domains, could also marry the research on student thinking with classroom practice by beginning to identify core practices that teachers could follow. The routines of practice serve as a set of focal points (all conceptually connected) in which to embed our developing understanding of the interrelations of the features of classroom practice discussed in this chapter.

Stein and colleagues, in their current project, have built one possible model for framing the work of teachers when planning and enacting lessons for the purpose of “moderating the degree of improvisation required by the teacher during a discussion,” (Stein et al., 2006). They propose five practices:

1. anticipating likely student responses to rich mathematical tasks
2. monitoring students' responses to the tasks during the explore phase
3. selecting particular students' responses to present their mathematical responses during the discuss-and-summarize phase
4. purposefully sequencing the students' responses that will be displayed
5. helping the class make mathematical connections between different students' responses. (pp. 7–8)

As Stein and colleagues discuss in their paper, teachers must make many decisions within each of these practices, and clearly the work that is summarized in this chapter about the work of teaching is implicated in each of these five practices. However, drawing teachers' attention to the goals and routines for bringing mathematical reasoning into the public space of the classroom and helping students engage with it would advance knowledge of classroom practices that support students' intellectual and dispositional growth. We see the promise of connecting the fourth and fifth practices proposed by Stein and colleagues to a recent close examination of interactional patterns in different types of reform-oriented classes studied by Wood et al., (2006). In *strategy-reporting* classrooms, children present different strategies. “Children presenting may be asked to provide more information about *how* they solved the problem by the teacher but rarely by other student listeners” (p. 224). In *inquiry/argument* classrooms, children also share strategies, but the goal of sharing is for other listeners (both teachers and students) to ask questions for further clarification and understanding. These discussions, “often include a challenge or disagreement from student listeners or teachers, which initiates an exchange that in turn prompts the thinking of justification in support of children's ideas” (p. 224). Their extensive analysis of the patterns of interaction in these classrooms provides nuanced images of discourse patterns that enable participants to (a) pull mathematical ideas together, (b) identify and explain mathematical errors, and (c) develop robust mathematical arguments and reasoning.

What the field needs is to define these routines of practice so that the research and teacher education communities can engage with them, unpack them, and study them.¹¹ This first step in identifying routines would involve both conceptual and empirical work. How should we define a *routine*? What important mathematical ideas do students work with in particular mathematical domains that would help us converge on particular routines? To what degree should we specify the routine? How do we make visible teachers' work in managing discourse, establishing norms, and building relationships with students as we specify and study the enactment of routines? The routines of practice should be examined in relation to the critical features of classroom practice raised throughout this chapter, connecting the details of practice to the surround in which they are embedded. Understanding the relationships in the details of practice, discourse, norms, and personal relationships allows researchers to look both within and outside of classrooms. This would potentially serve to connect what happens in classrooms to the social and political structures that influence what occurs. As with Civil's work (Civil & Planas, 2004), we would begin to see how the structures of schools and the social and political nature of communities shape what happens for teachers and students in mathematics classrooms.

We recognize that the research directions we propose have already been initiated by many of those cited throughout this chapter who are looking closely at classroom practice and how teachers and students together shape what occurs. We also recognize that this type of research is challenging for a number of reasons. First, the work is challenging because one needs to either find or create classrooms in which the identified routines of practice exist. Second, the data needed to understand the routines of practice in relation to the surround is complicated—it requires detailed descriptions of classroom interactions over time, interviews with teachers and students, and an understanding of the school culture and students' communities. It also requires attention to race, class, gender, and language, and other ways students and teachers may be marginalized within the school and community. Despite these complications, we see tremendous potential for advancing our understanding of classroom practices if researchers take on such challenging work and create a richer, synthetic view of how discourse, sociomathematical norms, and student-teacher relationships contribute to the enactment of classroom practices that support student learning and empowerment.

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¹¹ There may be different routines of practice across elementary, middle, and high school mathematics.

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MATHEMATICS TEACHERS' BELIEFS AND AFFECT

Randolph A. Philipp

SAN DIEGO STATE UNIVERSITY

Be careful how you interpret the world: It is like that.

—Erich Heller (Also attributed to Helen Keller)

A colleague once complained to me that after his daughter had successfully completed her last required mathematics course from a highly regarded university, she reported to him that as a result of that course she hated mathematics. My colleague was particularly upset because his daughter had enjoyed mathematics up to that point, and he was worried that the negative effects of her last mathematics class might linger long into her future. For many students studying mathematics in school, the beliefs or feelings that they carry away *about* the subject are at least as important as the knowledge they learn *of* the subject. A 2005 Associated Press poll (AP–AOL News, 2005) showed that nearly 40% of the adults surveyed said that they had hated mathematics in school, and although those polled also acknowledged hating other subjects, twice as many people said that they hated mathematics as said that about any other subject. While students are learning mathematics, they are also learning lessons about what mathematics is, what value it has, how it is learned, who should learn it, and what engagement in mathematical reasoning entails.

To understand students' experiences with school mathematics, one must understand a central factor in their experience: mathematics teachers. Two de-

cades have passed since Lee Shulman (1986) rejected George Bernard Shaw's infamous statement "He who can, does. He who cannot, teaches" as overly simplistic because it was not reflective of the highly specialized knowledge required of teachers. He introduced the now famous term *pedagogical content knowledge* to refer to the complex knowledge that lies at the intersection of content and pedagogy and that teachers must possess to make the curriculum accessible to their students. Shulman ended his landmark article by suggesting that a more apropos saying would be "Those who can, do. Those who understand, teach" (p. 14). Researchers studying teachers' knowledge, beliefs, and affect related to mathematics teaching and learning are still trying to tease out the relationships among these constructs and to determine how teachers' knowledge, beliefs, and affect relate to their instruction.

The focus of this chapter is a consideration of what researchers have to say about teachers' beliefs and affect. A saying by Heller, "Be careful how you interpret the world: It is like that," indicates that the way one makes sense of his or her world not only defines that person for the world but also defines the world for that person. Beliefs might be thought of

as lenses through which one looks when interpreting the world, and affect might be thought of as a disposition or tendency one takes toward some aspect of his or her world; as such, the beliefs and affect one holds surely affect the way one interacts with his or her world. Although few researchers have examined the relationship between mathematics teachers' affect and their instruction, the existing research shows that the feelings teachers experienced as learners carry forward to their adult lives, and these feelings are important factors in the ways teachers interpret their mathematical worlds.

ORGANIZATION OF CHAPTER

I begin this chapter with a brief review of two chapters from the first *Handbook of Research on Mathematics Teaching and Learning* (Grouws, 1992): one on teachers' beliefs, written by Alba G. Thompson, and a general chapter on affect, written by Doug McLeod. After summarizing the state of research on mathematics teachers' beliefs and affect at the time of publication of the first handbook chapter, I review the research conducted on teachers' beliefs since that time in two sections. In the first section, I consider what beliefs are, how they are measured, what stances are taken on the role of inconsistent beliefs, and how they are changed. In the second section, I review emerging areas of research related to mathematics teachers' beliefs by looking to research on teachers' beliefs about students' mathematical thinking, teachers' beliefs about curriculum, teachers' beliefs about technology, and teachers' beliefs about gender. I then review the research on teachers' affect. I begin with a brief review of the relationship between affect and achievement and then summarize studies that address teachers' affect. Because little research specifically addresses the topic of mathematics teachers' affect and several researchers have proposed frameworks on students' affect that might be helpful for considering teachers' affect, I follow the research on teachers' affect with a review of some of the student-focused frameworks. After providing a review of the research on teachers' beliefs and affect, I consider researchers who have taken a broader look at the construct of beliefs by considering such constructs as *teachers' orientations*, *teachers' perceptions*, and *teacher identity*, and I highlight how studying identity provides one promising path by which to integrate teachers' beliefs and affect. I end with some final comments.

WORKING DEFINITIONS/DESCRIPTIONS OF TERMS

Many of the terms in this chapter are not used in the literature in a uniform way. However, I recognize that readers can become confused about the relationships among terms, and so I have attempted to distill meanings that capture distinctions that emerge in usage by researchers, and I list them in Figure 7.1. I call these *definitions/descriptions* because they are based upon a combination of the literature usage and the dictionary definitions. These definitions/descriptions are not intended to stand alone as definitions, but instead I provide them to support the reader in drawing distinctions among the commonly used meanings of terms in this chapter. Each of these terms is discussed in the chapter, but I provide them at the beginning (see Figure 7.1) so that the reader can refer to them as needed.

THE STATE OF RESEARCH ON MATHEMATICS TEACHERS' BELIEFS AND AFFECT AT THE PUBLICATION OF THE FIRST HANDBOOK OF RESEARCH ON MATHEMATICS TEACHING AND LEARNING

Brief Summary of A. G. Thompson's 1992 Handbook Chapter on Teachers' Beliefs and Conceptions

Twice in her review of the literature on teachers' beliefs and conceptions, A. G. Thompson (1992) noted the importance for researchers studying mathematics teachers' beliefs to make explicit to themselves and to others the perspectives they hold about teaching, learning, and the nature of mathematics, because these perspectives greatly affect researchers' approaches to and interpretations of their work. In keeping with her own advice, A. G. Thompson stated her stance that researchers must consider the discipline of mathematics and the relationship between what a teacher thinks about mathematics and how the teacher teaches. She embraced a view of mathematics "as a kind of mental activity, a social construction involving conjectures, proofs, and refutations, whose results are subject to revolutionary change and whose validity, therefore, must be judged in relation to a social and cultural setting" (p. 127). The conception of mathematics teaching associated with this view of mathematics and reflected in several documents A. G. Thompson cited from the 1980s was "one in which

Affect—a disposition or tendency or an emotion or feeling attached to an idea or object. Affect is comprised of *emotions*, *attitudes*, and *beliefs*.

Emotions—feelings or states of consciousness, distinguished from cognition. Emotions change more rapidly and are felt more intensely than attitudes and beliefs. Emotions may be positive (e.g., the feeling of “aha”) or negative (e.g., the feeling of panic). Emotions are less cognitive than attitudes.

Attitudes—manners of acting, feeling, or thinking that show one's disposition or opinion. Attitudes change more slowly than emotions, but they change more quickly than beliefs. Attitudes, like emotions, may involve positive or negative feelings, and they are felt with less intensity than emotions. Attitudes are more cognitive than emotion but less cognitive than beliefs.

Beliefs—Psychologically held understandings, premises, or propositions about the world that are thought to be true. Beliefs are more cognitive, are felt less intensely, and are harder to change than attitudes. Beliefs might be thought of as lenses that affect one's view of some aspect of the world or as dispositions toward action. Beliefs, unlike knowledge, may be held with varying degrees of conviction and are not consensual. Beliefs are more cognitive than emotions and attitudes. (I do not indent this definition under affect because, although beliefs are considered a component of *affect* by those studying affect, they are not seen in this way by most who study teachers' beliefs.)

Beliefs System—a metaphor for describing the manner in which one's beliefs are organized in a cluster, generally around a particular idea or object. Beliefs systems are associated with three aspects: (a) Beliefs within a beliefs system may be primary or derivative; (b) beliefs within a beliefs system may be central or peripheral; (c) beliefs are never held in isolation and might be thought of as existing in clusters.

Conception—a general notion or mental structure encompassing beliefs, meanings, concepts, propositions, rules, mental images, and preferences.

Identity—the embodiment of an individual's knowledge, beliefs, values, commitments, intentions, and affect as they relate to one's participation within a particular community of practice; the ways one has learned to think, act, and interact.

Knowledge—beliefs held with certainty or justified true belief. What is knowledge for one person may be belief for another, depending upon whether one holds the conception as beyond question.

Value—the worth of something. A belief one holds deeply, even to the point of cherishing, and acts upon. Whereas beliefs are associated with a true/false dichotomy, values are associated with a desirable/undesirable dichotomy. Values are less context-specific than beliefs.

Figure 7.1 Working definitions/descriptions of terms (listed alphabetically, except that *emotion*, *attitude*, and *belief* are listed from least to most cognitive component of *affect*. I accept sole responsibility for these definitions while crediting many).

students engage in purposeful activities that grow out of problem situations, requiring reasoning and creative thinking, gathering and applying information, discovering, inventing, and communicating ideas, and testing those ideas through critical reflection and argumentation” (p. 128). The role of teachers' mathematical conceptions was a recurring theme underlying her 1992 review chapter, no surprise to readers familiar with her research (e.g., A. G. Thompson, 1984; A. G. Thompson, Philipp, Thompson, & Boyd, 1994; P. W. Thompson & Thompson, 1994).

The term *belief* is so popular in the education literature today that many who write about beliefs do so without defining the term. A. G. Thompson (1992) found, “For the most part, researchers have assumed that readers know what beliefs are” (p. 129). Further, many educators contend that distinguishing between knowledge and belief is unimportant for research, but investigating how, if at all, teachers' beliefs and knowledge affected their experience is important. Although Thompson used both *beliefs* and *conceptions* in the title of her chapter, in most of her chapter, she seemed to

think of beliefs as a subset of conceptions, and her definition of *conceptions* included *beliefs*. And yet, at times she seemed to use the terms interchangeably. She referred to teachers' conceptions “as a more general mental structure, encompassing beliefs, meanings, concepts, propositions, rules, mental images, preferences, and the like” (p. 130). When using the term *conceptions*, Thompson, recognizing the important relationship between knowledge and beliefs, seemed less interested in drawing distinctions between these terms, and she stated, “To look at research on mathematics teachers' beliefs and conceptions in isolation from research on mathematics teachers' knowledge will necessarily result in an incomplete picture” (p. 131). However, early in her chapter she recognized two distinctions between beliefs and knowledge, *conviction* and *consensuality*; these distinctions are generally referred to in the field.

First, beliefs can be held with varying degrees of conviction, whereas knowledge is generally not thought of in this way. For example, whereas one might say that he or she believed something strongly, one is

less likely to speak of knowing a fact strongly. Second, beliefs are not consensual, whereas knowledge is. One is generally aware that others may believe differently and that their stances cannot be disproved, whereas with respect to knowledge, one finds “general agreement about procedures for evaluating and judging its validity” (A. G. Thompson, 1992, p. 130).

A. G. Thompson defined a *belief system* as “a metaphor for examining and describing how an individual’s beliefs are organized” (1992, p. 130). Drawing upon the work of Green (1971) and Rokeach (1960, cited in A. G. Thompson, 1992), she highlighted three aspects of beliefs systems identified by Green, all related to the notion that because a belief is never held in total isolation from other beliefs, considering how beliefs are held in relation to one another is useful. First, some beliefs serve as the foundation for other beliefs in a quasi-logical structure, meaning that some beliefs might be thought of as *primary beliefs* whereas others serve as *derivative beliefs*. Thompson’s example of a primary belief was a teacher’s belief that clearly presenting mathematics to students is important, and an associated derivative belief that might follow is that teachers should be able to readily answer any questions asked by students. A second dimension of beliefs systems is that beliefs can be either *central*, which means strongly held, or *peripheral*, which means less strongly held and more susceptible to change. Green contended that primary beliefs might not necessarily be more central than the associated derivative beliefs. In the example above, the derivative belief that a teacher must be prepared to answer questions may be more central (strongly held) to the teacher than the primary belief that a lesson must be clear. A teacher might even, perhaps as a result of a professional development opportunity, change her primary belief about presenting clear, sequential lessons without affecting the derivative, more central, belief about being able to answer any questions posed. Green’s third dimension of beliefs systems is that beliefs are held in *clusters* that are more or less in isolation from other clusters. One outcome of holding beliefs in this manner is that people can avoid confrontations between belief structures. Another outcome is that these beliefs systems may appear contradictory or inconsistent from the point of view of an observer.

The major portion of her chapter was a review of the literature about topics of study that, at that time, A. G. Thompson (1992) considered relatively new: teachers’ conceptions of mathematics, their conceptions of mathematics teaching and learning, and the relationship between these conceptions and instructional practice. She ended the section by discussing

changing teachers’ conceptions. I briefly summarize these topics below.

She first defined a *teacher’s conception of the nature of mathematics* as “that teacher’s conscious or sub-conscious beliefs, concepts, meanings, rules, mental images, and preferences concerning the discipline of mathematics” (A. G. Thompson, 1992, p. 132). Her examples of ways that scholars have highlighted teachers’ conceptions included Ernest’s (1988, cited in A. G. Thompson, 1992) three conceptions of mathematics as (a) a dynamic, problem-driven discipline, (b) a static, unified body of knowledge, or (c) a bag of tools. She described Lerman’s (1983, cited in A. G. Thompson, 1992) *absolutist* view of mathematics as universal, absolute, certain, value-free mathematical knowledge and the *fallibilist* view of mathematics as developing through conjectures, proofs, and refutations. She described Perry’s (1970, cited in A. G. Thompson, 1992) scheme of intellectual and moral development and Copes’s (1979, cited in A. G. Thompson, 1992) attempt to adapt this general scheme to mathematical knowledge. She also described Skemp’s (1978) ideas of instrumental and relational understanding. Her look at teachers’ conceptions of mathematics teaching and learning focused upon models of mathematics teaching, and she drew upon the work of Kuhs and Ball (1986, cited in A. G. Thompson, 1992) to highlight four views: learner-focused, content-focused with an emphasis on conceptual understanding, content-focused with an emphasis on performance, and classroom-focused. In considering the relationship between teachers’ conceptions and their instructional practices, Thompson noted that although some researchers found a strong relationship whereas others found more variability and inconsistency, in general, a higher degree of consistency was reported by researchers studying the relationship between teachers’ conceptions of mathematics and their instructional practices than by researchers studying teachers’ conceptions about teaching and learning and their instructional practices. She cautioned that inconsistencies between professed beliefs and instructional practice raise a methodological concern related to how beliefs or conceptions are measured, and she suggested that researchers must go beyond teachers’ professed beliefs and at least “examine teachers’ verbal data along with observational data of their instructional practice or mathematical behavior” (A. G. Thompson, 1992, p. 135).

In the last major section of her chapter, on changing teachers’ conceptions, A. G. Thompson (1992) reported studies of preservice teachers who showed little change, and she noted that teachers often assimilate new ideas to fit their existing schemata

instead of accommodating their existing schemata to internalize new ideas. She suggested that studies providing in-depth, detailed analyses would be required to better explicate teachers' difficulties in accommodating new ideas into their existing schemata. She described a promising paradigm and indicated that teachers' beliefs and practices underwent large changes when teachers learned about children's mathematical thinking (Carpenter, Fennema, Peterson, Chiang, & Loef, 1989).

A. G. Thompson (1992) ended her chapter with a few suggestions. First, she was concerned that too many researchers thought of beliefs systems as static entities to be uncovered whereas she viewed beliefs systems as dynamic mental structures that were susceptible to change in light of experience. She challenged researchers to look less at how beliefs affected practice and more at the dialectic between beliefs and practice. She reiterated the position put forth by some researchers that distinguishing between teachers' knowledge and teachers' beliefs is not useful and that, instead, researchers should investigate teachers' conceptions encompassing both beliefs and any relevant knowledge—including meanings, concepts, propositions, rules, or mental images—that bears on the experience. She also noted that virtually no researchers had investigated the extent to which teachers' and students' conceptions interact during instruction.

Brief Summary of McLeod's 1992 Handbook Chapter on Affect

McLeod defined the *affective domain* as referring "to a wide range of beliefs, feelings, and moods that are generally regarded as going beyond the domain of cognition" (McLeod, 1992, p. 576), and he identified several specific terms that make up the affective domain. *Emotions*, *attitudes*, and *beliefs* were the three terms to which he devoted most attention. Emotions change more rapidly than attitudes and beliefs; they are also less cognitive and are felt more intensely than attitudes and beliefs. An example of a negative emotion is the feeling of panic experienced by some students when doing, or even thinking about doing, mathematics, whereas an example of a positive emotion is the satisfactory feeling of "Aha!" experienced by students who have an insight during mathematical problem solving. McLeod noted that emotional reactions had not been major factors in research on affect in mathematics education, and he conjectured that the reason for this lack of attention was that emotions, being prone to changing quickly, lack the stability necessary for researchers to think that they can reliably gather questionnaire data about them.

Attitudes refer to "affective responses that involve positive or negative feelings of moderate intensity" (McLeod, 1992, p. 581). Attitudes are more cognitive in nature and more stable than emotions; they are felt less intensely. Examples of attitudes students might experience toward, for example, geometry are liking, disliking, being curious about, or being bored by the subject. One connection between emotions and attitudes is that repeated emotional reaction to an experience related to mathematics can result in automatizing that emotion into an attitude toward that experience. McLeod provided the example of a student who has repeated negative experiences with geometry proofs. The initial emotional response the student experiences will, with time, become more automatic and less physiologically arousing and will eventually lead to the student's forming a more stable response, an attitude, that can be measured by use of a questionnaire. For example, a student who is struggling in a geometry class may feel discomfort, or even illness, when faced with devising geometry proofs, and with time these feelings may result in the more general attitude that the student dislikes geometry, and, eventually, mathematics.

Beliefs are more cognitive in nature than attitudes (and, hence, also than emotions), are generally stable, and are experienced with a lower level of intensity than emotions or attitudes. McLeod concluded that beliefs tend to develop gradually and that cultural factors play a key role in their development. Four categories of beliefs addressed in McLeod's chapter are beliefs about mathematics, beliefs about self, beliefs about mathematics teaching, and beliefs about the social context. These discussions were almost exclusively about students' beliefs, not teachers' beliefs.

Although his major focus on affect was on emotions, attitudes, and beliefs, McLeod addressed several other affect-related concepts, such as confidence, self-concept, mathematics anxiety, and learned helplessness. He also addressed concepts more closely related to cognition, such as autonomy, intuition, metacognition, and social context.

A major theme identified by McLeod in his 1992 chapter was that "all research in mathematics education can be strengthened if researchers will integrate affective issues into studies of cognition and instruction" (p. 575), and he drew heavily from Mandler's theory that affective factors arise out of emotional responses to interrupted plans. According to Mandler, one approaches a task with a schema for how the task will be completed, and if the anticipated sequence of actions cannot be completed, the result is a physiological response. For Mandler, these affective factors were connected to one's knowledge and beliefs, because

interpretations of interruptions vary from individual to individual. McLeod offered the example of a sixth-grade student solving a story problem. If the student believed that all mathematics problems should be solvable in a couple minutes but the student was unable to solve the problem in that period of time, the student might experience an arousal that he or she would interpret as negative. If these experiences were repeated frequently, the student might develop a negative attitude toward story problems, and, in many cases, such negative attitudes toward one aspect of mathematics generalize to negative attitudes toward mathematics in general or, perhaps even worse, toward the student's view of himself or herself as a mathematical learner. If, however, students believe that story problems can challenge even good problem solvers and require a longer period of time to solve, then arousal at an inability to quickly solve the problem might not be interpreted as negative. The student's interpretation of the experience, not the experience itself, determines the outcome.

In the upcoming section on affect, I relate the research since the publication of McLeod's chapter to two conclusions he drew. First, little research in mathematics education integrated cognitive and affective factors, and McLeod called for more integration, which, unfortunately, is not generally reflected in the recent research. At the end of the chapter, however, I describe some promising directions for such research.

Second, although McLeod addressed affect generally, he noted that teacher affect in mathematics education had been studied little, a trend that seems to have continued. However, the emotional responses felt by students learning mathematics and the associated attitudes that develop are believed to linger into adulthood and may have important implications for teachers. I look to some research on comparison of children's and adults' affective responses, for example, research showing that mathematics test anxiety can equally affect children and prospective teachers, and I consider implications.

Summary of the State of Research on Beliefs and Affect at the Publication of the First Handbook

A. G. Thompson (1992) found that most research on teachers' beliefs and conceptions was interpretive in nature, employed qualitative methods of analysis, and was comprised of in-depth case studies with small numbers of subjects. She suggested that more such in-depth studies were needed to explain why teachers did not accommodate new conceptions into their

existing schemata. Methodological approaches used at the time for these in-depth studies included "interviews, classroom observations, stimulated-recall interviews, linguistic analysis of teacher talk, paragraph-completion tests, responses to simulation materials such as vignettes describing hypothetical students in classroom situations, and concept generation and mapping exercises such as the Kelly Repertory Grid Technique" (p. 131). Those who conducted experimental or pseudoexperimental studies of beliefs used Likert-type instruments, and they tended to measure beliefs isolated from knowledge.

McLeod, (1992) in his chapter on affect, addressed beliefs as a cognitive component of affect and called for affect to be included in the study of cognition and, therefore, of beliefs but found that little research had been conducted on teachers' affect. A. G. Thompson (1992), in her chapter on teacher beliefs, addressed the importance of considering beliefs together with knowledge and referred to this construct as *teachers' conceptions*. Thompson did not address affect. This lack of attention to affect was common throughout the first handbook. With one exception, all index references to *affect* from the first National Council of Teachers of Mathematics (NCTM) *Handbook of Research on Mathematics Teaching and Learning* (Grouws, 1992) were to the McLeod chapter on affect in mathematics education. The one exception was Schoenfeld's (1992) mention, in his chapter on learning to think mathematically, that space constraints precluded his addressing the topic of affect and that readers should refer to the work by McLeod for "authoritative starting points" (p. 358). Absent from both chapters was the topic of technology, either as a tool to support the study of teachers' beliefs and affect or as an object of study about teacher's beliefs or affect with respect to technology.

Setting the Context for Mathematics Teacher Education Research Since 1992

Much has changed since 1992, when the first handbook was published. I note five occurrences that, at least indirectly, have affected research in mathematics education in the United States and, to some extent, abroad. These five occurrences are the general acceptance and infusion of ideas from the NCTM Standards' documents (NCTM, 1989, 1991, 1995) into the educational arena; the increased number of outlets for publishing research in mathematics education; the increased politicization of United States education in general and educational research in particular; technological advances affecting the manner in which we obtain and report research and capture,

edit, and post video for use when assessing teachers' beliefs or affect; and the emergence of sociocultural and participatory theories of learning.

Occurrence 1: The Acceptance and Infusion of Ideas From the NCTM Standards' Documents Into the Educational Arena

Between 1989 and 1995 the National Council of Teachers of Mathematics published three important documents: *Curriculum and Evaluation Standards for School Mathematics* (1989), *Professional Standards for Teaching Mathematics* (1991), and *Assessment Standards for School Mathematics* (1995). (Although NCTM subsequently published the *Principles and Standards for School Mathematics* [2000], this document appeared too late to influence much of the research reviewed for this chapter.) The *Curriculum and Evaluation Standards* document, which emphasized mathematics as problem solving, communication, reasoning, and making connections, particularly affected curriculum development in the United States and, as a result, also affected research that was conducted (Hiebert, 1999). Although this document had been published before the first *Handbook* (Grouws, 1992), it had had no substantial influence on either practice or research by that time. However, in the next decade, that situation was to change, with publication of hundreds of research studies about mathematics education reform (Ross, McDougall, & Hogaboam-Gray, 2002). One might conclude from the abundance of studies on reform that schools were engaged in important and fundamental change. However, a peek into randomly selected American classrooms has led to the conclusion that the reform movement in the United States has not led to widespread change in mathematics instruction (Hiebert et al., 2005; Stigler & Hiebert, 1999). Whereas reform documents have had an effect on mathematics education research, they have had a much smaller effect on what takes place in American schools. When Lerman (2002) wrote that "a reform along the lines of that in the USA could not take place in the UK because of the current dominance of the official pedagogical field" (p. 237), he could, to some extent, also have been writing about the reform's effect in U.S. schools (Becker & Jacob, 2000; Gregg, 1995; Smith, 1996). One reason that schools change so little is that teaching is cultural (Stevenson & Stigler, 1992), and major commitments are required to enact meaningful school-based change. I am reminded of the response of a mathematics curriculum specialist in the mid 1990s to my question about whether the reform movement had affected the teaching of mathematics in his district: He had ensured that every school received a copy of the *Standards* (NCTM, 1989). But

even if major reform initiatives designed to implement innovative curriculum were funded, meaningful and lasting changes in schools would not occur without sustained professional development designed to change teachers' beliefs. In their review of research studies published about mathematics education reform, Ross, McDougall, and Hogaboam-Gray (2002) concluded that the main obstacle to implementation was teachers' beliefs about mathematics teaching. During the past 10 years, many other researchers have studied teachers' beliefs as they relate to reform (e.g., Lloyd & Wilson, 1998; Steele, 2001; Sztajn, 2003), and some have developed instruments to measure teachers' beliefs as they relate to reform (e.g., Ross, McDougall, Hogaboam-Gray, & LeSage, 2003). Although others studying teachers' beliefs may not explicitly mention reform, their focus reflects beliefs that are in the spirit of the reform.

Occurrence 2: The Increased Number of Publishing Outlets

A second notable occurrence to affect research in mathematics education over the past decade has been the increased number of outlets for publishing research in mathematics education in general and research on mathematics teachers in particular. Journals that have begun publishing since publication of the 1992 *Handbook* (Grouws) include *Hiroshima Journal of Mathematics Education*; *International Journal for Mathematics Teaching and Learning*; *The International Journal for Technology in Mathematics Education*; *Canadian Journal of Science, Mathematics, & Technology Education*; *International Journal of Science & Math Education*; *Journal of Mathematics Teacher Education*; *Mathematics Education Review*; *Nordic Studies in Mathematics Education*; and *Zentralblatt Für Didaktik Der Mathematik (International Reviews on Mathematical Education)*. To indicate the effect such journals have had on research on teachers' beliefs and affect, I consider more closely the *Journal of Mathematics Teacher Education*, first published in 1998 by Kluwer Academic Publishers. This journal is devoted to publication of research, critical analyses, and critiques related to mathematics teachers' learning and development at all stages of their professional development. The journal provided a new outlet for research on teachers' affect and beliefs, and many studies on these topics have been published in this journal. In his final editorial as the first editor for this journal, Cooney (2001), after reviewing the first four volumes of the journal, wondered whether the articles reflected a field that is generating subsequent research questions but failing to provide the necessary foundation for substantial progress. In acknowledging that mathematics teacher education is

inherently a field of practice, Cooney was concerned that we teacher educators not allow the practical orientation of our work to preclude the development of powerful and useful new constructs:

The problems we study may be inherently practical, but surely their solutions (in the sense of greater wisdom) lie in our ability to see teaching and teacher education outside the confines of the acts themselves. This makes life difficult. It is relatively easy for us to see teaching from a more abstract level if we are not part of the act. In teacher education, however, teacher educators are indeed part of the act. So how is it that we can educate ourselves to engage in Dewey's notion of reflective thinking or to follow von Glasersfeld's notion of reflection which requires us to step out of ourselves and see our actions from a vantage point beyond our actions themselves? It is only from this meta-vantage point that we can begin developing schemes and constructs that capture the webs in which we are all entangled. Simply put, we cannot engage in an analysis of our own activity using only the language of that activity itself. (pp. 256–257)

Cooney emphasized the need to engage in theoretical case-study research that goes beyond reporting anecdotes to develop constructs that might be applied to help make sense of teaching and learning environments.

Occurrence 3: The Increased Politicization of United States Education and Educational Research

Questions about the quality of research in mathematics education have not been isolated to concerns about mathematics-teacher education. Even the broader field of educational research has come into question. Some contend that educational research tends to lack influence or usefulness, partially because it does not lead to practical advances that can be readily applied by practitioners (Burkhardt & Schoenfeld, 2003). By comparing education to other applied fields, Burkhardt and Schoenfeld suggested that the research-based development of tools and processes used by practitioners found in, say, the medical field are lacking in education. Mathematics educators who raise questions about research in our field often take as given that qualitative research is the appropriate means by which to conduct research, and their questions are about the methodological, analytical, or interpretive issues related to qualitative research (Simon, 2004). However, with the political agendas of granting agencies often driving the research that is conducted (Silver & Kilpatrick, 1994), a call for “scientifically based research” has affected funding agencies (Eisenhart & Towne, 2003). Sadly, during the 1990s, many states in the United States experienced a “Math

War” (Becker & Jacob, 2000) that pitted mathematics educators calling for innovation and reform against those who believed that the best way to proceed was through a “back to basics” approach. Although published research has generally not yet reflected this recent politicization of educational research, it is a third notable occurrence since 1992 to influence subsequent research on teachers’ beliefs and affect.

Occurrence 4: Technological Advances That Support Obtaining and Reporting Research, and Capturing, Editing, and Posting Video for Use When Assessing Teachers’ Beliefs and Affect

Many researchers conducting literature reviews today begin the task with their computers. Many articles are available through online journals or online libraries. Furthermore, authors routinely post presentations and papers on their websites or send pdf versions of papers upon request. These changes speed the process of conducting research but do not fundamentally change it. However, other technological advances have changed the process of assessing teachers’ beliefs and affect. For example, one group of researchers developed a web-based survey with embedded video requiring open-ended responses to be used when assessing elementary school teachers’ beliefs (Integrating Mathematics and Pedagogy, 2003), together with a user’s manual also available on-line (Integrating Mathematics and Pedagogy, 2004). Those completing the survey respond online, and their data are downloaded to a spreadsheet. Use of this survey enabled the researchers to conduct a large-scale experimental study (Philipp et al., 2005) that would have been impossible 20 years ago.

Occurrence 5: The Emergence of Sociocultural and Participatory Theories of Learning

A fifth development to affect the research on teachers’ beliefs and affect has been the increasing adoption by researchers of emerging theories of learning. By 1992, when the first handbook was published, theoreticians (e.g., Brown, Collins, & Duguid, 1989; Lave, 1988) had already raised questions about psychological theories for explaining the complexities associated with learning, and during the 1990s, sociocultural (Cobb, 1995) and participatory (Wenger, 1998) frameworks were increasingly adopted by mathematics education researchers. The researchers investigating teachers’ beliefs and affect since 1992 have increasingly adopted these theoretical frameworks.

RESEARCH ON TEACHERS' BELIEFS CONDUCTED SINCE 1992

I address, in the first part of the section on mathematics teachers' beliefs, four important areas: defining *teachers' beliefs*, measuring them, considering inconsistencies in beliefs, and changing beliefs. In the second part of the beliefs section, I review emerging areas of research related to mathematics teachers' beliefs and how their beliefs are changed by looking to beliefs about students' mathematical thinking, beliefs about curriculum, beliefs about technology, and beliefs about gender. Although all these areas were studied before publication of the first handbook chapter, they have received substantially more attention from researchers since that time.

What Are Beliefs?

The construct *belief* is of great interest to those attempting to understand mathematics teaching and learning. An ERIC-database search for 1990s' articles based on the terms *teacher* and *belief* located 3,105 documents, and although the numbers of documents for the 1980s (1,382) and the 1970s (745) were smaller, clearly the subject of teacher beliefs has been popular.¹ When the term *mathematics* was added to the terms *teacher* and *belief*, the 23, 123, and 407 items located for the 1970s, 1980s, and 1990s, respectively, though far fewer than the number of generic teacher-beliefs documents, still reflect an area of growing importance in our field. In all these studies, the term *belief* was used. However, as popular as the term *belief* is in education research, it is often undefined by those who study it (Pajares, 1992; A. G. Thompson, 1992). Since publication of A. G. Thompson's 1992 *Handbook* chapter, the amount of research on mathematics teachers' beliefs has increased, and the research has included efforts to define *belief*. However, in general, no clear agreement about the definition has been reached. Two constructs often closely related to beliefs are values and knowledge.

Belief Versus Values

If you must tell me your opinions, tell me what you believe in. I have plenty of doubts of my own.

—Johann Wolfgang von Goethe

In the 1950s, Edward R. Murrow hosted a radio show titled "This I Believe," for which Americans wrote about the "rules they live by, the things they have found to be the basic values in their lives" (Murrow, 1951/2005). National Public Radio recently revived this show with modern-day Americans from all walks of life reading short essays about the core beliefs guiding their daily lives. Topics included belief in the power of love to transform and to heal; belief in one's personal responsibility to positively affect society; belief in empathy; and belief that the presence of a person who often feels perplexed, torn by issues, and unsure of his or her beliefs can be an asset, especially "in periods of crisis, when passions are high and certainty runs rabid" (Gup, 2005). These statements of belief are examples of creeds people believe *in*. Murrow (1951/2005) referred to these as "basic values," and I take that stance: A belief *that* is about beliefs, but a belief *in* is about values.

In a review of the literature on values in mathematics teaching, Bishop, Seah, and Chin (2003) addressed the differences and the similarities found in the ways researchers think about beliefs and values. One difference they identified is that beliefs tend to be associated with a true/false dichotomy whereas values are often associated with a desirable/undesirable dichotomy. They contended that beliefs are, therefore, more context-dependent than values, because whereas a true/false judgment must be made in reference to some object, desirable/undesirable dichotomies are associated with more general, less context-dependent, attributes. They provided the example of the belief that mathematics is fun, a true/false judgment made about a particular subject. This judgment must be made in context, because holding this belief does not imply that one finds all subjects, or even all mathematics, fun. One who views fun as a personally desirable quality in a more universal way is likely to seek fun throughout his or her life. For a person who values fun, the belief that mathematics is fun whereas literature is not fun would be more pertinent than for a person who values something else, say usefulness, more than he values fun. To summarize this distinction between beliefs and values, beliefs are true/false statements about constructs whereas the choice of the particular constructs one finds desirable or undesirable represents one's more context-independent values. These more context-independent values are often viewed as more internalized than beliefs and, hence, harder to change.

¹ This search was conducted in 2001, before the United States Department of Education closed the ERIC database in an effort to restructure the search engine to include only documents that met a particular standard of "evidence-based" research.

In another approach to distinguishing between beliefs and values, some researchers view values as a subset of beliefs. Rokeach (1973, cited in Bishop et al., 2003) viewed values as enduring beliefs, and Clarkson and Bishop (1999, cited in Bishop et al., 2003) viewed values as beliefs in action. Raths, Harmin, and Simon (1987, cited in Clement, 1999) presented several attributes of beliefs that, if present, would constitute a value. They viewed values as beliefs that one chooses freely from among alternatives after reflection and that one cherishes, affirms, and acts upon. Because life situations may invoke incompatible values, many researchers relax the constraint that one must act upon one's values and instead suggest that so long as one is committed to a particular belief, it might be said that that belief is a value for the person (Clement, 1999). As such, values influence, rather than determine, the choice of possible actions available (Bishop et al., 2003).

The similarities in researchers' views on values and beliefs may be greater than the differences, and the two terms are often used interchangeably by mathematics educators (Bishop et al., 2003). Just as people hold incompatible values, so too do they hold beliefs that may conflict. Values exist within more complex values systems, and individual values alone seldom determine one's decisions, actions, and responses (Bishop et al., 2003). Because beliefs also exist within systems (A. G. Thompson, 1992), in assessing a teacher's values or beliefs, one must recognize the role played by a person's incompatible or conflicting values and beliefs. Consider, for example, a teacher who values students' development of mathematical proficiency. She will want to support their developing a combination of conceptual understanding, procedural fluency, and problem-representing and -solving approaches, all built upon the capacities to reason logically about relationships and justify their reasoning and upon the disposition to view mathematics as useful, worthwhile, and attainable (National Research Council, 2001). If this teacher also values raising her students' test scores as measured by a standardized test focused primarily upon procedural fluency, then she is likely to find herself in a quandary. If this teacher also teaches in a low-performing school, subject to strict local or federal guidelines (like those currently applicable in U.S. schools), then the requirement to immediately improve students' test scores is likely to cause the teacher to feel pressured into pursuing a course of action different from the one she believes will lead her students to develop mathematical proficiency. This teacher will have to prioritize among competing values. These priorities will vary across situations and contexts (Bishop et al., 2003); for example, for some time before the standardized test, the teacher may feel more con-

strained to alter her usual curricular and pedagogical approaches to focus upon developing procedural fluency. Simply observing this teacher's performance is unlikely to provide one an accurate assessment of her values or beliefs. Furthermore, this teacher may be unaware of how these conflicting forces are affecting her curricular and pedagogical decisions; thus, teasing out the relationships among her values and beliefs and her thoughts and actions would be difficult, even in conversation.

Belief Versus Knowledge

Not . . . what opinions are held, but . . . how they are held; instead of being held dogmatically, [liberal] opinions are held tentatively, and with a consciousness that new evidence may at any moment lead to their abandonment.

—Bertrand Russell, *Unpopular Essays*, 1950

Across disciplines of anthropology, social psychology, and philosophy, one finds general agreement in viewing *beliefs* as “psychologically held understandings, premises, or propositions about the world that are felt to be true” (Richardson, 1996, p. 103). The notion that a belief is thought to be true raises one of the more common distinctions drawn between belief and knowledge (Pajares, 1992; A. G. Thompson, 1992), with researchers often viewing *knowledge* as “belief with certainty” (Clement, 1999). This definition is similar to one that is more than 2,300 years old, Plato's definition of *knowledge* as “justified true belief” (McDowell, 1987, cited in Furinghetti & Pehkonen, 2002, p. 42), but with one important difference. By using the term *true*, Plato implied the existence of an external reality of which one could know and be certain, and von Glasersfeld's observation that for many people today “there reigns the conviction that knowledge is knowledge only if it reflects the worlds as it is” (von Glasersfeld, 1984, p. 20) captures the sense that the view of knowledge put forth by Plato continues to prevail.

The notion that knowledge must be associated with truth has influenced many definitions of *knowledge* and, therefore, the distinction between knowledge and belief. For example, Scheffler (1965, cited in M. S. Wilson & Cooney, 2002) viewed both *belief* and *knowledge* as cognitive constructs but viewed knowledge as a stronger condition than belief. For Scheffler, one knows *Q* if three conditions hold: One believes *Q*; one has the right to believe *Q*; and *Q* is, in fact, true. Under this definition, a belief that is true is not knowledge for an individual unless one has a warrant for believing it, that is, unless one's believing is based upon evidence. Fur-

thermore, under this definition, a belief is not knowledge unless it is true. This truth requirement alarms many, because the means by which truth is determined is debatable, and the history of science indicates that what was taken as true at one point in time was generally modified or subsumed into more encompassing theories at other times. In von Glasersfeld's (1993) view of constructivism, the notion that knowledge is a representation of reality is rejected. Instead, the most one can say about ability to predict physical phenomena is that knowledge has proved to be viable under particular circumstances. Cobb, Yackel, and Wood (1992) noted important implications for teachers' holding the position that to know is to accurately represent what is outside the mind. Teachers who hold this view are likely to believe that they can use materials as a means for presenting readily apprehensible mathematical relationships as if there is a direct mapping between the materials and the mathematics, and all that is necessary for students to construct this mapping is for a teacher to clearly present the mathematics by using relevant materials (Cobb et al.). For example, a teacher might believe that the place-value relationships that students so often struggle to comprehend may be obvious if the students use base-ten blocks.

Some are troubled by the view of radical constructivists, who reject the notion that one has access to truth. For example, Goldin (2002b), who thought that the radical constructivist epistemology was deeply flawed and was "entering the realm of *passé*" (p. 205), feared that defining *knowledge* as justified true belief is a grave mistake made by "cultural relativists" (Goldin, 2002a, p. 64), because doing so "leaves no convenient word to distinguish beliefs that are *in fact* true, correct, good approximations, valid, insightful, rational, or veridical, from those that are *in fact* false, incorrect, poor approximations, invalid, mistaken, irrational, or illusionary" (Goldin, 2002a, p. 65).

This debate about the relationship between knowledge and belief is unlikely to cease, and I suspect that researchers will continue to disagree about the affordances and constraints of various relationships between the two. More important for researchers is to take clear stances on how they are viewing beliefs, stances that are reflected in their operationalization of the construct. For example, M. S. Wilson and Cooney (2002) dropped the truth condition and modified Scheffler's second condition so that they suggested that *X* knows *Q* if *X* believes *Q* and *X* has reasonable evidence to support *Q*. M. S. Wilson and Cooney acknowledged that although in using this definition, they dodged the claim to know reality, the major question of knowing what constitutes evidence is unresolved. Using this definition, researchers can,

however, acknowledge that what is evidence for one may not be evidence for another.

Educational researchers attempting to understand teachers' knowledge and beliefs can benefit by recognizing how teachers look at these constructs and by drawing distinctions between knowledge and belief according to the world view of the subjects holding the knowledge or belief. The important question for researchers studying mathematics teachers' beliefs and knowledge is not whether some conception is true in an ontological way but how a teacher views the conception. As a researcher, I have found the following stance useful when I attempt to understand how a person holds a particular conception: A conception is a *belief* for an individual if he or she could respect a position that is in disagreement with the conception as reasonable and intelligent, and it is *knowledge* for that individual if he or she could not respect a disagreeing position with the conception as reasonable or intelligent. By this definition, agreement upon what constitutes "a reasonable, intelligent position" is unnecessary; instead, the principles are left for the subject to apply and for the researcher to reveal. Under this definition, one person's belief may be another person's knowledge. For researchers to know *how* one holds a notion may be as important as knowing *what* one holds as the notion. Two people who hold contradictory *beliefs* about something may have more in common with each other than with a person who holds one of those conceptions as *knowledge*. For example, consider the following three people and the notions they hold about students' learning of concepts and procedures: Person A holds as *belief* the notion that students are better served by being taught concepts before procedures; Person B holds as *belief* the notion that students are better served by being taught procedures before concepts; Person C holds as *knowledge* the notion that students are better served by being taught concepts before procedures. On the one hand, Persons A and B disagree about the beliefs they hold, but because they hold the notions as beliefs, they may find important points of agreement in a discussion, because they could imagine reasonable, disagreeing positions. On the other hand, Persons A and C, who hold the same notion but one questions it and the other does not, may experience unexpected difficulty when holding a meaningful discussion about their positions. The real roadblocks to meaningful dialogue are created when at least one of party of a disagreement holds a notion as knowledge and, hence, does not respect the position of those who disagree as reasonable or intelligent. Unfortunately, in mathematics education, this lack of ability to find common ground for discussion

has fueled the fires in what have come to be known as The Math Wars (Becker & Jacob, 2000).

Summary

General distinctions have been drawn among values, beliefs, and knowledge, with values and knowledge often defined in terms of beliefs. Values are generally viewed as a type of belief to which one is deeply committed. Values are also viewed as preferences that are not associated with truth values, whereas beliefs are held by an individual as true/false dichotomies with the understanding that others may disagree. If one takes the ontological view that truth exists and people have access to it, then knowledge might be viewed as true belief. If one takes a view that truth, though it may exist, is not accessible to humans and instead the best one can hope for is viability, then knowledge is belief with certainty. Although for researchers to understand *what* beliefs teachers hold is important, perhaps equally important is understanding *how* teachers hold these beliefs.

A Comment About Terminology

A. G. Thompson, in her handbook chapter (1992), used the terms *conception* and *belief*, and although at times she seemed to use the terms interchangeably, more often she used *conception* to refer to a general construct that included beliefs as a component. Other researchers have used the term *conception* in their work, but with little agreement on its meaning. For example, Tirosh (2000) used *conception* as interchangeable with knowledge whereas Knuth (2002) used the term to represent knowledge and belief “in tandem” (p. 85). Andrews and Hatch (2000) noted that for some the term *conception* is essentially cognitive whereas for others it is affective. Although attempts to unify the definitions of constructs might seem critical for moving the field forward, I am not convinced of this need, because the constructs that researchers apply when studying the integration of teachers’ beliefs, knowledge, and practices are operationalized as much by the research methodologies used as by a particular definition. Throughout this chapter, I use the terms as the researchers used them in their studies and attempt to tease out similarities and differences in their approaches.

Measuring Beliefs

However one chooses to define (or not define) *beliefs*, when studying the beliefs of others, researchers must draw inferences from what people do or say (Pajares, 1992). Mathematics education researchers have typically approached the study of teachers’ beliefs in

one of two ways: by using case-study methodology or by using beliefs-assessment instruments. Perhaps the more common approach has been to use case-study methodology to provide detailed descriptions of the beliefs of a small number of teachers by relying upon rich data sets that include some combination of classroom observations, interviews, surveys, stimulated-recall interviews, concept mapping, responses to vignettes or videotapes, and linguistic analyses. These data are often collected over a period of time and triangulated. These rich data sets are important for theory building, inasmuch as they enable researchers to consider interrelationships in the complex world of teachers. Such studies also enable researchers to meet the challenge put forth by A. G. Thompson (1992) to investigate the dialectic relationship between teachers’ conceptions, including their beliefs and knowledge, and teachers’ practices. Researchers in most studies I examined took this case-study approach to studying beliefs.

Likert-Scale Surveys

The case-study approach to studying teachers is powerful for building theory, but testing theory often requires tools for measuring the beliefs of larger groups of teachers. Typically, beliefs of large numbers of mathematics teachers are measured using Likert scales, and I present four examples of Likert scales used for measuring mathematics teachers’ beliefs.

As part of their long-term study of teachers’ use of knowledge about children’s mathematical thinking, referred to as Cognitively Guided Instruction, Carpenter, Fennema, and colleagues developed a beliefs survey, the Mathematics Belief Scale, comprised of 48 Likert-scale items with four subscales to assess Role of the Learner, Relationship Between Skills and Understanding, Sequencing of Topics, and Role of the Teacher (Fennema, Carpenter, & Loef, 1990; Peterson, Fennema, Carpenter, & Loef, 1989). The Role of the Learner subscale measured the belief that children are able to construct their own knowledge instead of being receivers of knowledge. The Relationship Between Skills and Understanding subscale measured the belief that skills should be taught in relation to understanding and problem solving rather than in isolation. The Sequencing of Topics subscale measured the belief that children’s natural development, rather than the logical structure of formal mathematics, should guide the sequencing of topics. The Role of Teacher subscale measured the belief that instruction should facilitate children’s construction of knowledge rather than consist of teachers’ presenting materials. An example of an item from The Role of the Learner subscale is “It is important for a child to be a good lis-

tener in order to learn how to do mathematics." This beliefs survey has been used by researchers working with prospective teachers (Vacc & Bright, 1999) and with practicing teachers (Fennema et al, 1996).

Zollman and Mason (1992) developed a beliefs instrument based upon the National Council of Teachers of Mathematics (1989) *Curriculum and Evaluation Standards for School Mathematics*. Their instrument was designed to assess teachers' beliefs about the *Standards*, and they selected 16 items to reflect 16 of the 54 standards included in the NCTM document. The items were stated as nearly direct quotes or the inverse of direct quotes from the *Standards* document, and each item met the criteria that the answer should not be "intuitively obvious" (Zollman & Mason, 1992, p. 359) and the item could be incorporated into a single sentence easily stated in a positive or negative manner. During their pilot work developing the survey, the authors found that respondents appeared to emphasize aspects of the items that were distracting, and so to ensure that respondents focused attention on the intent of the item, select words were capitalized. An example of a positively stated item is "Students should share their problem-solving thinking and approaches WITH OTHER STUDENTS"; a negatively stated item is "Children should be encouraged to justify their solutions, thinking, and conjectures in a SINGLE way" (Zollman & Mason, 1992).

Researchers seeking to use Likert-scale beliefs surveys to study mathematics teachers' beliefs use existing surveys, as Vacc and Bright (1999) used the survey developed by Fennema et al. (1990), develop a new survey, as Zollman and Mason (1992) did, or create a hybrid, drawing upon an existing survey. Hart (2002) administered a three-part beliefs' survey to prospective elementary school teachers before and after they had completed an integrated content/methods course; the first two parts were adapted from existing instruments, and the third part was Hart's creation. The first part was a form of Zollman and Mason's (1992) *Standards'* Belief Instrument; the second was adapted from Schoenfeld's (1989, cited in Hart, 2002) Problem-Solving Project. A positively stated item from this second part, which Hart used to assess change in teachers' beliefs about teaching and learning mathematics within and outside the school setting is "Good mathematics teachers show students lots of different ways to look at the same question"; an item that was negatively stated is "Good math teachers show you the exact way to answer the math question you will be tested on." Hart's third part was comprised of two items on teacher efficacy, with the first stating, "I am very good at learning mathemat-

ics," and the second stating, "I think I will be very good at teaching mathematics."

Enochs, Smith, and Huinker (2000) developed a Mathematics Teaching Efficacy Belief Instrument designed to be used with prospective teachers. Their beliefs survey was comprised of 21 Likert-scale items, 13 on the Personal Mathematics Teaching Efficacy subscale and 8 on the Mathematics Teaching Outcome Expectancy subscale. An item on the former subscale was "Even if I try very hard, I will not teach mathematics as well as I will most subjects," and one from the latter was "The mathematics achievement of some students cannot generally be blamed on their teachers."

One concern about self-report surveys is whether teachers' reports are accurate. Although Ross et al. (2003) provided evidence that the self-report data for elementary school teachers' degrees of implementation of standards-based mathematics teaching in Ontario, Canada, were reliable and valid, whether teachers' self-reports of their practices are different from their self-reports of their beliefs remains an open question. Is a Likert-scale survey a valid measure on which to base inferences about teachers' beliefs? In the next section, I present an example from a group of researchers whose concern about the validity of Likert-scale beliefs surveys led them to develop an alternative survey that can be administered to large numbers of subjects.

An Alternative to Likert-Scale Beliefs Surveys for Large-Scale Data Collection

A group of researchers engaged in a large-scale research and development project referred to as IMAP (Integrating Mathematics and Pedagogy, 2003; see, also, Ambrose, Clement, Philipp, & Chauvot, 2004) developed a web-based survey requiring open-ended responses to overcome three problems they had identified with use of Likert-scale surveys. They noted, first, that inferring how a respondent interprets the words in Likert-scale items is difficult. For example, in the item "It is important for a child to be a good listener in order to learn how to do mathematics" (Fennema et al., 1990), how do researchers know how the respondent interprets the idea of *good listener*? Might a respondent think differently about this statement depending upon whether she was thinking of a child listening to a teacher demonstrating a procedure, a teacher presenting a problem situation, or a child sharing a solution strategy? Individual respondents are not able to explain their responses for Likert items, so researchers can only infer what considerations may have guided the respondents. A second difficulty the IMAP researchers identified with Likert items is that

responses provide no information for determining the importance of the issue to respondents (Ambrose et al., 2004). McGuire (1969) stated, "When asked, people are usually willing to give an opinion even on matters about which they have never previously thought" (p. 151). For example, although teachers who build their instruction around having children solve problems and share solutions with other students are likely to agree strongly with the statement "Students should share their problem-solving thinking and approaches WITH OTHER STUDENTS," teachers who have not thought much about basing instruction upon their students' sharing their thinking with other students may, when reading this statement, think that it makes sense and may agree or even strongly agree. The third concern the IMAP researchers raised is that Likert scales tend to provide little, and often no, context. For example, Collier (1972) developed a Likert-scale survey containing the item "In mathematics, perhaps more than in other fields, one can find set routines and procedures." Respondents might interpret such a comment in multiple ways, and their responses give no indication of their interpretations. Some respondents may differentiate between middle-school algebra and primary-grade arithmetic and believe that one has set routines and procedures and the other does not. Because no context is associated with the item, respondents may feel compelled to consider mathematics in general when, in fact, they view different levels of mathematics in different ways.

The IMAP team set out to create a beliefs survey by identifying four characteristics of beliefs identified in the literature as accounting for the critical role that beliefs play in teaching and learning and, thus, were important for the way the researchers attempted to measure the beliefs. First, beliefs influence perception (Pajares, 1992). That is, beliefs serve to filter some complexity of a situation to make it comprehensible, shaping individuals' interpretations of events (Grant, Hiebert, & Wearne, 1998). Beliefs might be thought of as serving as a type of constructed model (von Glasersfeld, 1993) or theory (Nisbett & Ross, 1980) *through* which one looks, so that beliefs affect what one sees or notices (Mason, 2002). The IMAP team addressed this issue in their web-based survey by providing respondents with complex situations to interpret (Ambrose et al., 2004). Second, beliefs are viewed as drawing one toward a position or, at least, predisposing one in a particular direction. Allport (cited in McGuire, 1969) considered a belief as a readiness to respond, and Rokeach (1968) thought of a belief as a disposition to action. The notion that beliefs are dispositions toward action has played a key role in some important research on beliefs (e.g., Cooney, Shealy, & Arvold,

1998), because one's beliefs can be inferred by attending to the manner in which one is disposed to act in a particular situation. Because the IMAP team intended their beliefs survey to be used by practicing and prospective teachers, they took into account the fact that many prospective teachers are not yet positioned to act and thus many means available to researchers for measuring practicing teachers' dispositions toward action are unavailable for use with prospective teachers. They addressed this issue in their beliefs survey by providing respondents with scenarios in which they were asked to make teaching decisions. The prospective and practicing teachers' dispositions to act in these situations provided evidence from which to infer their beliefs. A third characteristic the IMAP team identified as pertinent for their work was that beliefs are not all-or-nothing entities; they are, instead, held with differing intensities (Pajares, 1992, citing Rokeach, 1968). To address this characteristic in their survey, they provided tasks with multiple interpretation points; the order in which questions were posed was carefully constructed to elicit views that were important to the respondents before "giving away" the "preferred" answers. For example, in one task, after being shown a video clip, respondents were asked, "What stands out to you in this video?" before they were asked, "What are weaknesses of the teaching in the video?" Those who identified weaknesses *before* being prompted for weaknesses were scored as holding the designated belief more strongly than those who did not mention weaknesses until asked to do so. Furthermore, because of the electronic administration of the survey, respondents could not return to change their responses to previous items. To allow for the differing intensities with which individuals hold beliefs, the team devised the scoring rubrics to differentiate among *strong evidence*, *evidence*, *weak evidence*, and *no evidence* for a respondent's holding a belief (Integrating Mathematics and Pedagogy, 2004). Furthermore, they were careful not to claim that an individual lacked a particular belief but instead stated that they found no evidence for the belief in the responses the individual provided. Fourth, beliefs tend to be context specific, arising in situations with specific features (Cooney et al., 1998). The IMAP team addressed this issue by situating survey segments in contexts, and they inferred a respondent's belief on the basis of his or her interpretation of the context (Ambrose et al., 2004).

This beliefs survey was used in an experimental study of prospective elementary school teachers enrolled in a mathematics course and randomly assigned to (a) concurrently learn about children's mathematical thinking by watching children on video or working directly with children, (b) concurrently visit elemen-

tary school classrooms of conveniently chosen or specially selected teachers, or (c) a control group that had no relevant additional instructional activities outside the mathematical course (Philipp et al., 2005). Those who studied children's mathematical thinking while learning mathematics developed more sophisticated beliefs about mathematics, teaching, and learning and improved their mathematical content knowledge more than those who did not study children's thinking. Furthermore, beliefs of those who observed in conveniently chosen classrooms underwent less change than the beliefs of those in the other groups, including those in the control group. The researchers concluded that by developing a context-based open-ended beliefs survey, they were able to measure prospective elementary school teachers' beliefs related to mathematics and to mathematics learning and teaching years before these prospective teachers were in the position to act as teachers. Beliefs of PSTs could be changed over the course of a semester, although the authors did not speculate whether these changes would remain with time (Philipp et al., 2005).

Summary

The predominant approach used by researchers to measure beliefs is to employ qualitative measures, including teacher interviews, classroom observations, responses to simulated materials, concept generation and mapping, paragraph completion, or Kelly Repertory Grid techniques. Although these qualitative approaches provide rich data sets, they are expensive to employ across large numbers of participants; researchers who need to collect large data sets often rely upon less expensive approaches, such as Likert-scale surveys. Some have identified problems with Likert-scales surveys, and one example was provided of an alternative beliefs survey that was developed and successfully used to survey large numbers of prospective elementary school teachers without relying upon Likert-scale responses.

Inconsistent Beliefs and the Role of Context

Because of the complexity of teachers' beliefs systems, researchers may find that teachers hold beliefs that appear to be inconsistent with their teaching practices. One approach taken by researchers to explain inconsistencies is to examine whether particular beliefs within a beliefs system are more central or primary, and hence play a greater role in influencing practice, than other beliefs. Another approach is to study whether a teacher's perspective on his or her practice might help explain the apparent contradiction. In this section I share examples of both ap-

proaches and conclude by considering the stances researchers take regarding inconsistent beliefs.

In a study of six novice elementary school teachers, Raymond (1997) investigated the inconsistency between one teacher's mathematics beliefs and teaching practice. For each teacher, data, collected during the 10-month period between March of their first year of full-time teaching and December of their second year, included an introductory phone interview, six 1-hour audiotaped interviews, five monthly classroom observations, an analysis of several samples of lesson planning, a concept-mapping activity in which teachers presented their views of the relationships between mathematics beliefs and practice, and a take-home questionnaire on mathematics beliefs and factors that influence teaching practice. Raymond analyzed these data within four categories—teachers' beliefs about the nature of mathematics, teachers' beliefs about learning mathematics, teachers' beliefs about teaching mathematics, and teachers' mathematics-teaching practices—on a 5-level scale ranging from *traditional* to *nontraditional* (traditional, primarily traditional, even mix of traditional and nontraditional, primarily nontraditional, and nontraditional). The focus of her case study was a fourth-grade teacher, Joanna, selected, first, because her expressed beliefs were similar to those of the other five teachers and, second, because her case was dramatic in that although her beliefs about the nature of mathematics were fairly traditional, her beliefs about learning and teaching mathematics were placed at the *nontraditional* end of the scale.

Joanna expressed the view that students should discover mathematics without being shown and that students learn mathematics better when solving problems. Joanna also stated the beliefs that teachers could be effective without following a textbook and that they should use various activities from various sources; she was a strong advocate of using manipulatives to achieve hands-on learning. Raymond noted a major inconsistency between Joanna's nontraditional beliefs about mathematics teaching and learning and her traditional practices. Joanna's class was run in an orderly way with her students' working exclusively at their desks in a quiet atmosphere, reflecting strict discipline. Joanna established herself as the authority, presenting information as teacher-directed instruction; although some teacher-student dialogue took place, on no occasion was student-to-student dialogue observed. Joanna rigidly followed her mathematics textbook, and the students worked quietly on problems from the textbook without access to manipulatives.

Joanna's beliefs about mathematics were categorized as traditional—at the opposite end of the continuum from her beliefs about mathematics teaching

and learning but on the same end as her practice. Joanna identified her own negative experiences as a mathematical learner as the primary influence on her beliefs about the nature of mathematics but her own teaching practice as the main influence on her beliefs about teaching and learning. Despite her negative view of mathematics as a student, and perhaps because of her negative experiences, Joanna wanted her own students to experience mathematics in a more positive light than she had experienced, a desire that may account for her nontraditional beliefs about learning and teaching mathematics. But if Joanna held these nontraditional beliefs she espoused, why was her practice traditional?

Raymond (1997) concluded that this inconsistency arose from the effects on Joanna's teaching of time constraints, scarcity of resources, concerns over standardized tests, and students' behavior. As a novice teacher, Joanna was particularly concerned with classroom management and discipline. Raymond reported also that, in at least one case, Joanna seemed to view her mathematics-teaching practice in terms of what she wanted to do, or thought she should do, rather than what she accomplished. She claimed that her belief in teaching mathematics through manipulatives led to her using them, but the only time Joanna actually used manipulatives was during a class she described as chaotic.

Raymond (1997) concluded, further, that her study provided evidence that the teacher's beliefs and practice were not wholly consistent and that her practice was more closely related to her beliefs about mathematics content than to her beliefs about mathematics learning and teaching. She also suggested that during teacher education, beliefs of prospective elementary school teachers might be more explicitly addressed so that teachers become aware of the beliefs with which they enter and attend to how these beliefs begin to change during this important period of growth.

Hoyles (1992) reflected upon how her changing view of beliefs affected her thinking about inconsistent beliefs. During her work in the early 1980s on a research project designed to identify characteristics of good secondary-school teaching practice, she and her colleagues approached teachers' beliefs as a construct that researchers might access from outside the classroom. Assuming that beliefs could be assessed in a decontextualized manner and that a linear relationship existed between teachers' beliefs and their classroom practices, the researchers first assessed teachers' beliefs and then observed teachers in the classroom. By the early 1990s, after observing the changes teachers experienced while interacting with a classroom innovation involving computers, Hoyles drew upon work in situat-

ed cognition (Brown et al., 1989) and subscribed to the notion that situations coproduce beliefs through activities. She came to view beliefs as situated and dialectically constructed from the relationships among activity, context, and culture and, thus, teachers' beliefs about mathematics or mathematics teaching and learning as being affected by factors such as grade level, students' level, textbook, or computer use.

For Hoyles (1992), viewing all beliefs as situated reconciled the apparent inconsistencies between teachers' beliefs and actions reported by researchers. Once the embodied nature of beliefs is recognized, the notion of inconsistent beliefs is an irrelevant consideration that can be replaced by considering the circumstances and constraints within settings. When Hoyles reconsidered data from the 1980s study with this new lens, she noted "the remarkable harmony between what the teacher wished to 'deliver' and 'what the students wanted'—an example of beliefs constructed in practice" (p. 40). Hoyles continued, "Such harmony exemplifies a good practice—a different blend of variables in the setting might produce alternative pictures of success. The notion of situated beliefs allows us to cope with this diversity" (p. 40).

Skott (2001) also considered the influence of context on teachers' beliefs, and although his conclusion about context differed from Hoyles's, he shared her conclusion about inconsistencies. He investigated the relationship between teachers' beliefs about mathematics and mathematics teaching and learning and the teachers' classroom practices. Skott introduced the term *school-mathematics images* to "describe teachers' idiosyncratic priorities in relation to mathematics, mathematics as a school subject and the teaching and learning of mathematics in schools" (p. 6). On the basis of responses to a questionnaire comprised of open and closed questions, administered to Danish student teachers two months before their graduation, Skott selected 11 student teachers, representing a variety of school-mathematics images, to interview after graduation. Four teachers who presented school-mathematics images inspired by current reform efforts in school mathematics were videotaped teaching their mathematics classes for 2 to 3 weeks during their first 18 months of teaching; they participated in informal discussions after each lesson and in a comprehensive interview after the videotaped lessons were completed. Skott selected one of the four teachers, Christopher, as the subject of a research study.

Christopher's responses to the questionnaire and the first interview indicated that his school mathematics images were in contrast to those he remembered experiencing as a high school student in the Danish

gymnasium, in which teachers lectured at the board. He favored process aspects of mathematics; he viewed his role as initiating and supporting investigative activities so that his students could assume responsibility for their own learning. Christopher was committed to school-mathematics reform but also to his students' developing broad educational, nonmathematical skills: learning to plan, reflecting on their own learning, working cooperatively, relating critically to information, and independently finding solution strategies and models.

After graduation, as a music and mathematics teacher in a typical Danish *folkeskole* (a municipal school for Grades 1–10), Christopher enjoyed freedom in his choice of teaching methods and curricula. He was observed teaching his Grade 6 mathematics class, in which he had introduced a textbook different from the traditional book used by the previous teacher. Because as a novice teacher Christopher felt overwhelmed with lesson preparation and grading, he relied upon the textbook for the aims, the content, and the tasks for instruction, but he strongly influenced the flow of the classroom by the methodology he chose. He lectured little and addressed the whole class primarily to initiate small-group or individual work. He sought student explanations and elaborations, and he encouraged students to explore in the noisy, but productive, classroom.

Skott (2001) described two visits to Christopher's classroom. The first visit took place during the second day of a unit on area; the students were cutting congruent triangles to make parallelograms, cutting parallelograms to make rectangles, and measuring areas of desks, blackboards, and other items. When two students called for help with a problem, Christopher listened to the weaker student's approach and asked the student questions, although he recognized that the approach would be nonproductive, before he asked whether they could approach the problem another way. He then pursued the stronger student's approach with the two students. Skott viewed this episode as being consistent with Christopher's school-mathematics image, because he attempted to initiate and support the students' learning by letting them devise suggestions and by refraining from deflecting suggestions that might be wrong or insufficient.

The second visit took place during the next lesson. When two other boys asked for help to find the areas of rectangular lawns drawn with different scales, Christopher intervened and led the two students through a series of computationally oriented questions in such a way as to deplete the task of any mathematical challenge and replace it with carrying out simple, prescribed computations. Skott

(2001) viewed this episode as being inconsistent with Christopher's school-mathematics image and Christopher's actions as replacing his general reformist school-mathematics priorities with traditional values in mathematics education.

When confronted with the roles he had played in the two episodes, Christopher, who was often critical of his own teaching, was satisfied with both. Skott (2001) suggested that although Christopher's second interaction appeared to reflect inconsistencies both with his action during the first class and with his stated beliefs, the relationship between Christopher's school-mathematics images and his actions could explain his seemingly inconsistent behavior. Christopher's evaluation of each episode encompassed aspects of his approach to teaching beyond students' mathematical learning, and Skott identified three such aspects that became apparent in the two episodes and in the interview. First, Christopher consistently mentioned the need to think of the children as learners generally, not solely as mathematical learners, for example, as needing to develop self-confidence in mathematics and in general. Second, Christopher encouraged and expected his students to interpret and solve tasks on their own and to discuss strategies and results among themselves. In keeping with this goal, he provided minimal support before they started working, and as a result, the students were in constant need of assistance from the teacher during group work. Third, Christopher's constructivist view of learning did not lead to a particular method for teaching, and he spoke about the need to play multiple roles when teaching. Understanding these three aspects helped Skott understand Christopher's actions. In the first episode, he delved into the first student's suggestion because he believed that the student's self-confidence was an important consideration. In the second episode, he ensured that the students arrived at the correct answer because of his concern for the general management of the class, realizing that one of the two boys was extremely demanding and would require excessive attention unless the two boys believed that they had solved the problem. Skott concluded that teachers have "multiple and sometimes conflicting educational priorities, and that in this case the priorities related to Christopher's SMIs [school-mathematics images] were dominated by others concerned with managing the classroom with broader educational issues" (p. 18).

Skott (2001) concluded that Christopher's school-mathematics image shaped and filtered the objects of his reflection and framed his interpretation of what he saw but was insufficient for explaining all his actions, sometimes being overshadowed by the more general educational priorities of building students'

confidence and managing the class. Skott also noted how the teacher's classroom actions tended to replicate themselves. Christopher used the same calculational approach (A. G. Thompson et al., 1994), seemingly incompatible with his school mathematics image, that he had used to move the class along by appeasing these two students with a number of other groups of students, even when the motives for which it was originally used, building student confidence and managing the class, no longer were issues. The way the teacher coped with the original problem became a "prototypical action type" (Skott, 2001, p. 22) that was applied even when the original reasons for applying the action type were no longer present.

Skott's (2001) conclusion that the relationship between the teacher's school-mathematics images and his classroom practices varied according to the classroom situation seems to support Hoyle's (1992) notion that beliefs are situated. Although both Skott and Hoyle considered context critical to assessment of teachers' beliefs, they differed in their views on the situativity of beliefs. Skott contended that Christopher's beliefs did not change with the situation but that Christopher continued to believe in the importance of mathematical processes and the teacher's role in initiating and monitoring student activity to facilitate both the processes and the products. The change was in his specific goals during the two episodes—from mathematical learning of the two students in the first to class management in the second. Skott (2001) elaborated upon how this view differed from that of Hoyle (1992):

It is not the situation that produces a new set of school mathematical priorities. Situations that are outwardly different—for example, a research interview as opposed to a classroom setting—do not necessarily create or radically change a teacher's beliefs. Rather, the situations, that is, very specific situations each concerned with a specific interaction between student(s) and teacher, create or co-produce competing objects and motives of the teacher's activity. It is these competing objects and motives that form the basis of what apparently, but only apparently, is a new set of priorities produced under new and slightly different circumstances. In other words, the contextual embeddedness of Christopher's activity did not necessarily lead to a similar contextualisation of his school mathematical priorities. (Skott, 2001, pp. 24–25)

Skott acknowledged one general conclusion *about* beliefs research: No general conclusions should be expected *in* beliefs research. Furthermore, because teachers and the contexts in which they work differ, the relationship between classroom practices and teachers' beliefs may require more detailed, long-

term investigations of teachers' approaches in specific classrooms over several studies to begin to generate preliminary understandings of theoretical constructs.

Sztajn (2003) also considered the relationship between teachers' beliefs and practices as mediated by the contexts in which teachers were working, but instead of analyzing one teacher in two contexts as Skott had done, she analyzed two teachers who held similar beliefs but taught in different contexts. She observed each of two elementary school teachers a week at a time for four weeks spaced throughout the semester and conducted five semistructured interviews with each teacher. When visiting the teachers, she observed throughout the day, including during lunch, breaks, and staff meetings. Other classroom data included the teachers' lesson plans, grading books, students' report cards, mathematics worksheets and completed assignments, and materials from activities in other subjects. To triangulate her data related to the class context, Sztajn interviewed the principal, other teachers, and parents from each school.

Because she set out to study beliefs that go beyond mathematics and mathematics teaching and learning experiences, Sztajn (2003) initially coded the data using seven descriptive categories: the teacher, the students, the classroom, the school, the parents, general educational goals, and mathematics. The two public-school teachers she investigated taught classes in small, midwestern towns, one a third-grade and the other a fourth-grade class; each class was comprised only of Caucasian children. The teachers held similar beliefs about mathematics, believing that problem solving and basic skills were important but that basic skills had to be mastered before children could engage in problem solving. They believed that some children bring basic-skills knowledge from home but that the others lack both basic and social skills. In spite of the similarity of their beliefs about mathematics, their instruction differed; to explain the differences, Sztajn turned to their beliefs that go beyond mathematics.

Sztajn (2003) found that the concept of students' needs, which included teachers' beliefs about children, society, and education, emerged as an important factor in the teaching of both teachers and accounted for the differences in their instruction, because the two schools differed with respect to the manner in which the children were viewed. The students of the third-grade teacher, Theresa, were at a lower socioeconomic level than the students of the fourth-grade teacher, Julie. At Theresa's school, 40% of the students received free or reduced-cost lunches, compared to only 10% of the students at Julie's school. The educational levels of the parents, as estimated by the principals and teachers, differed; most

parents of the students at Theresa's school held low-income, manual-labor jobs whereas in Julie's school, many middle-income parents were doctors, lawyers, and university professors. Although Theresa believed in the importance of students' engaging in problem solving and developing higher order thinking skills, because she saw her students as coming from unstable, chaotic homes—poor environments for children—basic facts, drill, and practice were at the core of what she thought her students needed to prepare for their roles in the workplace. Most important for Theresa, whose teaching goal was to form responsible citizens, was that “learning and following rules in a responsible and organized way is what her students need in order to find their place in society” (p. 64). The fourth-grade teacher, Julie, held beliefs about mathematics and about students' needs that were similar to Theresa's, but because she thought that she had to deal with few problems and worried little about teaching rote basics, she structured her classroom around problem solving and projects. In Julie's class, even for routine mental-computation tasks, students shared various solution strategies. Although at the time of the study Julie and Theresa approached teaching differently, the approach Julie described having used previously, at a school in which children were more like Theresa's students, was very similar to Theresa's approach. Both teachers believed that teaching social skills was a main role of school. For example, Julie said that responsibility was important to teach, but because she thought that her students learned responsibility at home, they needed less practice on social skills and could explore more sophisticated mathematical content.

Sztajn (2003) contended that both teachers

are the heroes of the stories presented—never the villains. They are not, in a Machiavellian way, trying to hold students back, lessen their chances of going to college, or ensure that they will remain where they are in the socioeconomic spectrum. Quite the contrary, both of them use their best judgments when defining what their students need, and once they form this concept, they direct their work and all their efforts toward meeting these needs. (p. 71)

Sztajn concluded that because these students came from different socioeconomic backgrounds, they were learning different mathematics. It was the teachers' beliefs about children, society, and education, not their beliefs about mathematics, that accounted for the differences in instruction.

Summary

Researchers who have investigated apparent inconsistencies between teachers' beliefs and practices

have attempted to explain them. Raymond found two factors that accounted for such inconsistencies. First, a teacher's practices were more in line with her beliefs about mathematics than her beliefs about mathematics teaching and learning. Second, as important as beliefs about mathematics were, observed inconsistencies between a teacher's espoused beliefs and her practice were explained by general educational issues, such as time constraints, resources, standardized tests, and students' behavior.

Hoyle (1992) found that when she viewed beliefs as decontextualized, she observed inconsistencies, but when her view of beliefs became more contextualized and situated, she could explain apparent inconsistencies by considering the circumstances and constraints within settings.

Skott (2001) found that a teacher's beliefs about mathematics and about mathematics teaching and learning were important for understanding the teacher's practice but that these beliefs were often overshadowed by the more general educational priorities of building students' confidence and managing the class. But unlike Hoyle, who concluded that beliefs were situated, Skott concluded that not the teacher's beliefs but the teacher's goal for the particular activity changed, and after understanding the goal, the researcher could explain apparent contradictions between the teacher's beliefs and actions.

Sztajn (2003) studied two teachers who held similar beliefs about mathematics but taught in very different contexts and differed in their teaching. She found that attending to the teachers' beliefs about mathematics was insufficient for explaining the teachers' practices; only after considering the teachers' beliefs about children, society, and education was she able to account for the differences in the teachers' practices.

A Stance on Inconsistent Beliefs

What are we mathematics education researchers to make of inconsistencies? Is resolving inconsistencies important for the work we do? Within mathematics, consistency is so central that, historically, when mathematicians have faced a new way to consider some aspect of mathematics that seemed to them unimaginable or even untrue, they would, often after much resistance, accept this new way so long as no inconsistencies arose. As a classic example, 19th-century mathematicians realized that rejecting Euclid's fifth postulate led to consistent, albeit previously unimaginable, possibilities, resulting in mathematicians' accepting the findings and inventing new non-Euclidean geometries that led to productive ways for considering the physical world. Other examples are the

expansion of the number systems to include irrational numbers, negative numbers, and complex numbers and Cantor's theory of transfinities, which engendered new ways of thinking about infinity. Although inconsistencies are unacceptable in logical domains, people, in general, have come to accept that inconsistencies abound in human affairs. So what are the implications for how we choose to consider inconsistencies when studying teachers' beliefs?

Most researchers cited in this section found that the inconsistencies that arose when documenting teachers' beliefs and actions ceased to exist after they better understood the teachers' thinking about some aspects of their contexts. I propose that as a research stance in studying teachers and their beliefs, we researchers assume that contradictions do not exist. Taking this stance when we observe apparent contradictions, we would assume that the inconsistencies exist only in our minds, not within the teachers, and would strive to understand the teachers' perspectives to resolve the inconsistencies. Inconsistencies should still present problems, but for researchers instead of teachers. I do not propose that this position is ontological; that is, I do not suggest that inconsistencies are nonexistent or can all be explained away. I suggest, instead, that researchers who assume that inconsistencies do not exist will attempt to better understand teachers' beliefs systems and that the circumstances surrounding the teachers' practices will often, as for the researchers cited in this section, lead to resolution of the inconsistencies.

Changing Beliefs

Two general approaches are considered for the relationship between the change in beliefs and the change in behavior. One notion, supported by research, is that to change behavior, one must first change one's beliefs, because beliefs act as filters that affect what one sees (Pajares, 1992) and people have difficulty seeing what they do not already believe. However, if beliefs must change before behavior, then how do beliefs change? Can one change his or her beliefs merely by reflecting upon them? Or might reflection and action form a dialectic that can contribute to teachers' learning and growth? If Socrates was right that the unexamined life is not worth living, perhaps it is equally true that the un-lived life is not worth examining.

Guskey (1986) presented an alternative model predicated on the notion that "significant change in teachers' beliefs and attitudes is likely to take place only *after* changes in student learning outcomes are evidenced" (p. 7). He offered a linear model, pro-

posing that teachers who implement an instructional change that leads to students' success will then alter their related beliefs. In 1993, von Glasersfeld, in stating, "If one succeeds in getting teachers to make a serious effort to apply some of the constructivist methodology, even if they don't believe in it, they become enthralled after five or six weeks" (p. 29), also observed that changes in teachers' beliefs follow changes in their instruction.

In his research on teacher change, Guskey (1986) examined classroom teachers' success in implementing what he described as a relatively minor alteration in their practices. Grant, Hiebert, and Wearne (1998) wondered how Guskey's model might hold for substantive changes to teachers' practices, such as those called for in the effective-schools literature. They noted that implementers of past programs had often incorrectly assumed that simply *telling* teachers how they should teach would lead to the desired changes in practice and wondered whether, instead of telling, *showing* teachers models of teaching would be effective? What would teachers notice when observing reform-oriented instruction applied with their own students, and would what the teachers noticed be related to the beliefs they held? To investigate these questions, nine teachers observed reform-oriented lessons for 6–12 weeks. The teachers' beliefs about what mathematics should be learned and how to teach mathematics were placed on a continuum (from *skills/teacher-responsibility* end to the *process/student-responsibility* end). The researchers then compared the positions of the nine observing teachers' beliefs along the continuum with their evaluations of the alternative instruction, and they discovered a clear relationship between the teachers' beliefs and what they observed during the instruction. Beliefs of four of the nine teachers were on the skills/teacher-responsibility end of the beliefs continuum, and these four teachers tended to focus on one particular aspect of the reform-oriented lesson or to notice features that the researchers believed were tangential, not the intended point of the instruction. For example, whereas the intended purpose for using manipulatives was to provide opportunities for teachers to promote multiple strategies that might link representations for the students, the teachers on the skills/teacher-responsibility end of the continuum thought that their purpose was to present the *only* way to solve problems.

Three of the nine observing teachers, whose beliefs were placed in the middle of the beliefs continuum, recognized some features of the instruction but did not connect these features with the larger intended goals. For example, one teacher noted the importance of explaining strategies, doing story prob-

lems, and engaging in critical thinking, but she juxtaposed these against her view of the teacher's role as the provider and demonstrator of strategies. This teacher seemed to be driven by an image of a teacher as one who removes obstacles from her students' paths. When working with story problems, she tried to support her students' success by helping the students focus on key words. One teacher who was classified in the *middle-ground* category later taught the place-value unit she had observed. She explained that except for making a little change, she "pretty well stuck to their lesson plans." The change she made was that instead of encouraging students to solve the problems as they chose, she explained the rule of solving by adding from the one's column.

Beliefs of two of the nine observing teachers were placed at the process/student-responsibility end of the beliefs continuum; these teachers recognized the features of instruction important to the researchers and understood the underlying goals of the instruction. For example, even after hearing other teachers say that the approach was based on use of manipulatives, one of these two teachers noted that the approach was less about manipulatives and more about having children describe their thinking processes and solve problems in different ways.

The authors concluded that their study provided evidence for the position that the beliefs teachers hold filter what they see and, consequently, what they internalize. Teachers whose beliefs are at odds with a particular instructional approach may not change their thinking or their practices solely by observing the approach. The researchers suggested that observations combined with other activities designed to support the teachers' reflecting upon the experience, such as collegial discussions of the instruction, are more likely to lead to change.

In a study of 14 third-grade teachers undergoing professional development, Borko, Mayfield, Marion, Flexer, and Cumbo (1997) found a similar result. When teachers' beliefs were incompatible with the goals of the staff developers, the teachers generally either ignored the new ideas or inappropriately assimilated the ideas into their existing practices. Borko et al. concluded that beliefs served as filters through which ideas were perceived, and teachers needed to be challenged to reflect upon their beliefs.

Although Benbow (1995) acknowledged that beliefs act as filters through which teachers interpret their school experiences and influence their classroom practices, he rejected the linear model—that beliefs change must precede change in practice—in investigating the relationship between prospective elementary school teachers' mathematical beliefs and

their classroom teaching in the context of an early field experience. On the basis of data from three measures of beliefs, two Likert-type surveys and one open-ended questionnaire, he concluded that the preexisting beliefs held by the preservice teachers played key roles in their planning and implementation decisions for their mathematics lessons. At the same time, the preservice teachers' interpretations of the instructional outcomes affected their beliefs and, therefore, their subsequent practices. The early field experience had little effect on the preservice teachers' beliefs about mathematics, but it did affect their beliefs about pedagogical issues, such as their becoming more open to alternative points of view. The most dramatic belief change measured by the researcher was in the preservice teachers' *personal mathematics-teaching efficacy*, that is, their confidence in their own abilities to help students learn mathematics. Benbow concluded that the preservice teachers' core beliefs underwent little change and that, overall, the early field experience was primarily a confirming experience for the participants.

Reflection and Changing Beliefs

Cooney et al. (1998) also found that reflection played an important role in the growth of prospective secondary-school teachers over their last year in an undergraduate teacher preparation program. They applied a rich theoretical perspective, drawing heavily from the work of Green (1971, cited in Cooney et al., 1998) to attend not only to what beliefs the prospective teachers held but also to the ways in which the beliefs were held. They considered the three aspects of Green's belief structures: Beliefs can be held as primary or derivative beliefs; they then can be held with differing degrees of strength; and they can be held more centrally or more peripherally in relation to other beliefs. Another important factor for Green was the role that evidence plays in the ways one holds beliefs, because nonevidentially held beliefs are impervious to change through reason or evidence whereas beliefs held evidentially can be changed through teacher reflection. A related construct is *teacher authority*, because for one who holds beliefs nonevidentially, the role of an outside authority figure is important for the construction of beliefs. Cooney et al. applied two frameworks for considering authority, one based upon Perry's (1970, cited in Cooney et al., 1998) scheme, which was developed by investigating how college males think and come to know, and the other based upon women's ways of knowing (Belenky, Clinchy, Goldberger, & Tarule, 1986, cited in Cooney et al., 1998). Cooney et al. "were interested in capturing the meanings each of the teachers ascribed to his or her

teacher-education experiences” (p. 315), and doing so required them to assess each preservice teacher’s beliefs structures by collecting multiple data sources and searching for disconfirming evidence across the data. Beginning with a written survey administered to all the students in the class, the researchers referred to the students’ responses to the survey along with classroom-observation field notes and written assignments over the first few class sessions to select four students who initially held a range of positions toward mathematics teaching and learning. The survey questions, some rather unconventional and designed to assess the preservice teachers’ views of mathematics, included this item:

Consider the following similes: A mathematics teacher is like a news broadcaster; entertainer; doctor; orchestra conductor; gardener; coach. Choose the simile that you believe best describes a mathematics teacher and explain your choice. Choose a simile that you believe does not describe a mathematics teacher and explain your choice. (p. 314)

They administered a similar written survey later in the year and conducted individual interviews at five points over the course of the year: The initial interview was conducted in November during the integrated content and pedagogy course, and the fifth interview was in August near the end of the post-student-teaching seminar. Other data sources included observation field notes that were discussed weekly by the research team. Their results were published as four separate case studies in which relationships were drawn between the types of beliefs changes each student experienced and the manners in which the students held their beliefs. On the one hand, Greg’s beliefs changed over the course of two quarters; he had begun the program “adamantly opposed to using technology in the classroom” (p. 317) but later stated that in looking for a job, he would look for a school with a Mac lab. Henry, on the other hand, was initially opposed to using technology in education, and his belief did not change over the same period of time. Cooney et al. (1998) attributed Greg’s shift to his own involvement in technologically rich learning environments, but these were the same learning environments in which Henry was involved. The authors suggested that not the beliefs the two students began with but the ways they held their beliefs had made the difference.

Greg was open to others’ perspectives and enjoyed the exchange of diverse opinions, whereas Henry tended to either accept or reject an opinion, holding tightly to his own, and felt frustrated when his beliefs were challenged. He held a more *authoritarian view* in which assimilation, but not accommodation, may be

possible. Greg held a *reflective stance* toward the world whereby he was able to hold onto his core ideas while reformulating his core beliefs when faced with perturbations. The two students who changed the most were the two who were most reflective and who were open to teaching as a problematic activity.

The authors presented four characterizations for how the preservice teachers held their beliefs, characterizations that were further elaborated by Cooney (1999): isolationist, naive idealist, naive connectionist, and reflective connectionist. Henry was classified as an *isolationist*, one who

tends to have beliefs structured in such a way that beliefs remain separated or clustered away from others. Accommodation is not a theme that characterizes an isolationist. For whatever reason, the isolationist tends to reject the beliefs of others at least as they pertain to his/her own situation. (Cooney, 1999, p. 172)

The *naive idealist* “tends to be a received knower in that, unlike the isolationist, he/she absorbs what others believe to be the case but often without analysis of what he/she believes” (Cooney, 1999, p. 172). The last two characterizations are considered to be connectionist positions that emphasize reflection and attention to the beliefs of others as compared to one’s own. “The naive connectionist fails to resolve conflict or differences in beliefs whereas the reflective connectionist resolves conflict through reflective thinking” (Cooney, 1999, p. 172). Cooney et al. (1998) considered Greg a *reflective connectionist*, and they suggested that this position set the stage for the preservice teachers to emerge into reflective practitioners (Schön, 1983). The researchers concluded that a goal of teacher education was to produce reflective connectionists, but they could not conceive of some of the students in their study moving in that direction. They suggested that understanding the belief structures of students might help teacher educators make decisions about effective activities:

We submit that understanding the belief structures as evidenced by these four young teachers can provide a basis for conceptualizing the nature and activities of teacher education programs. The inculcation of doubt and the posing of perplexing situations would seem to be central to the promotion of movement from being a naive idealist or even isolationist to being a connectionist. Inciting doubt and making the previously unproblematic problematic can have significant impact on a person’s world and lead to varied and perhaps unsettling responses. It is not enough to make mathematics and teaching problematic for teachers. We need to understand the effect of this inculcation of doubt and also understand the kind of

support teachers need to make sense of it. (Cooney et al., 1998, pp. 330–331)

The authors ended by suggesting that the research community needs to develop a better understanding of the linkages between what teacher educators provide to preservice teachers and the effect these activities have on the preservice teachers' beliefs systems.

Mewborn (1999) investigated reflection and its effect on four preservice elementary school teachers in a field-based mathematics methodology course. She noted that reflection and action must go together, and she used the terms *verbalism* and *activism* to refer to reflection without action and action without reflection, respectively. Mewborn created a community of learners among four preservice elementary school teachers, herself, and a fourth-grade teacher in whose class they observed and worked. Over the course of the 11-week experience, the preservice teachers observed and discussed lessons in the fourth-grade class, observed and discussed videotaped examples of children's thinking, interviewed children and discussed the experience, and taught small groups of children and discussed the experience. Mewborn studied what the four prospective teachers found problematic, and she classified the concerns into four somewhat-ordered categories of issues related to classroom management and the classroom context; issues related to teaching mathematics; issues related to children's mathematical thinking; and, to a lesser extent, mathematics content. She found that the preservice teachers were able to engage reflectively, but they needed support in learning to observe mathematics teaching and learning environments and, in particular, in developing an internal locus of authority for pedagogical ideas. Initially, the students held an external locus of authority whereby they would turn to the classroom teacher or to Mewborn for answers. Later, they moved to a second stage in which the locus of authority was both internal and external, and during this stage, they moved beyond asking questions and began to state problems and generate hypotheses. During the third stage, which was reached while the students immersed themselves with children's mathematical thinking, the locus of authority was internal, and the preservice teachers became fully reflective when they not only presented hypotheses but also searched for evidence by which to test their hypotheses. Mewborn identified five elements of the design of the field experience that she considered critical to successfully helping the students become reflective about teaching and learning mathematics: (a) The field experience was approached from an inquiry perspective; (b) the students, teacher, and teacher educator participated

as a community of learners; (c) the community was nonevaluative; (d) the preservice teachers were given time to reflect; and (e) the field experience was subject specific.

Many have noted the need to support teachers in their becoming reflective practitioners, but we educators must be careful lest we think that this support, alone, will be sufficient to create major change in our schools. The organization and structure of schools present a formidable challenge to the kind of change called for in the reform, even if we create reflective preservice teachers. Gregg (1995), in a case study of a beginning high school teacher's acculturation into the school mathematics tradition, concluded that encouraging teachers to view their current practices as problematic may not be sufficient to promote reform, because the school culture has provided explanations designed to institutionalize the problems as matters to be coped with, not resolved. Gregg noted, for example, that the school mathematics tradition has separated the acts of teaching and learning so that a teacher can be considered to have met the obligation to teach even though the students may not have met their obligation to learn. This view is perpetuated by the commonly accepted conception of ability as capacity, because the teacher cannot be accountable for failing to successfully teach students who are unable to learn. Gregg provided the example of designating a test as "too hard" or certain questions as "unfair" because many students scored poorly on them, so that teachers can blame the test, even if they constructed it, instead of blaming their instruction; blaming their instruction might lead to questioning other of their fundamental beliefs and practices related to the teaching and learning of mathematics. Gregg doubted that "a teacher in the school mathematics tradition would question or reflect on the taken-as-shared beliefs and practices of this tradition as a result of students' poor test performance" (p. 463). He concluded that to be successful, a reform movement "must challenge the classroom, school, and societal obligations that characterize teachers' roles in the school mathematics tradition" (p. 463). In other words, Gregg raised the prospect that looking to change teachers' beliefs and knowledge within the current school mathematics tradition will, for the vast majority of teachers, fail.

Two Obstacles to Changing Beliefs: Teachers' Caring and Teachers' Belief That Teaching is Telling

Cooney (1999) suggested that to help teachers change their beliefs, teacher educators must find ways to support teachers to become more reflective while they unravel their notions about teaching and rebuild them in a rational way. But he identified two

major roadblocks to attaining this effect: teachers' overwhelming propensities to be caring teachers, and teachers' orientations toward teaching-as-telling. Cooney recommended supporting preservice teachers while they transform their notions of caring and telling into notions that encourage attention to context and reflection by helping them move beyond their concerns for children's personal comfort levels to consider students' intellectual needs and by integrating content and pedagogy in mathematics teacher education to support the transition of teachers toward reflection. These two obstacles are related, because the challenges teachers face when they change their role in the class affect their perceived sense of their effectiveness in meeting the needs of their students.

Nearly all mathematics educators seem to agree that effective teaching is more than telling, a view reflected in the U.S. national reform documents (NCTM, 1989, 2000). However, even expert instructors grapple with the difficulties of developing and maintaining a view of teaching as more than telling. In a study during which a researcher and a teacher collaborated on the curriculum that they each taught to separate ninth-grade mathematics classes, Romagnano (1994), the researcher, described difficulties he experienced when he encouraged his students to solve problems independently without first being told what to do, whereas the teacher with whom he was working avoided such difficulties by being more directive with the class. Neither Romagnano's nor the other teacher's approach was without challenges, because whereas the teacher's approach removed much of the conceptual mathematics for the students, Romagnano's approach often led his students to frustration so that they avoided participating in classroom activities. He concluded that the differences between his and the teacher's views of mathematics, of how it is learned, and of the role of the teacher in the process led to his facing the Ask Them or Tell Them dilemma, whereas the teacher faced no such dilemma.

Chazan and Ball (1999) described challenges they experienced when trying not to "tell" while teaching a high school algebra lesson and a third-grade class, respectively.

Chazan described how in the algebra lesson he had taught to a lower track high school class the students disagreed on whether, when determining the average raise given to employees of a company, they should include a person who received no raise. When the discussion became heated and several students became excited, Chazan found himself wondering how to strike a balance between his desire for his students to develop greater confidence in their abilities to discuss their way through a mathematical problem and

his desire for the students to reach some common ground from which they could stand back and reflect upon the mathematics. Chazan wondered whether, or how, he should intervene. During a lesson Ball taught to third graders, the students seemed to agree on an incorrect way of labeling the number line, and when her attempts to induce disequilibrium were unsuccessful, she was left wondering how best to support the students' learning. Whereas Chazan was concerned that his students might have engaged in an unproductive disagreement, Ball was concerned that her students were engaged in an unproductive agreement. Chazan and Ball were aware that telling can take different forms and that the type of telling must be considered with the context. For example, a teacher's telling through demonstration of a mathematical procedure for the students to practice is different from a teacher's telling by attaching conventional mathematical terminology to a distinction the students are already making. In spite of this knowledge and their vast experience teaching without simply telling, Chazan and Ball, two expert teachers, faced difficult challenges in their lessons. They warned that the commonly accepted reform exhortation not to tell is a statement about what *not* to do, but it contributes nothing toward supporting teachers while they consider what they might do. They described three factors they balanced in considering how to proceed during the class discussion: the mathematics under consideration, the nature and direction of the class discussion, and the social and emotional climate of the class.

If expert teachers face difficulties with teaching without telling, one could expect that many teachers struggle even more. Smith (1996) suggested that a *teacher's efficacy*, originally defined as "the extent to which the teacher believes he or she has the capacity to affect student performance" (Tschannen-Moran, Hoy, & Hoy, 1998, p. 202), is being challenged by a reform movement that calls for teachers to approach teaching from an entirely different perspective. Smith pointed to the existence of a core set of beliefs that characterize how most mathematics teachers in the United States view mathematics, mathematics learning, and mathematics teaching. For most teachers, school mathematics is a fixed set of facts and procedures for determining answers, and the authority for school mathematics resides in the textbooks, with the teacher serving as the intermediary authority between the textbooks and the students. For them, teaching mathematics requires telling, or providing clear, step-by-step demonstrations of these procedures, and students learn by listening to the teachers' demonstrations and practicing these procedures. These beliefs support teachers' senses of efficacy, because the con-

ception of mathematics as a fixed set of facts and procedures restricts the content teachers must know; thus they can think that they have mastered the necessary content. The notion of teaching-as-telling provides a detailed but attainable model that teachers can hope to master. Telling students how to perform procedures also supports teachers' senses of efficacy, because the conventional nature of procedures is such that students cannot be expected to know them until the teacher shows them, and so the students' successes in mastering the procedures can be attributed to the teacher. Furthermore, the belief that students learn by listening, watching, and practicing clearly specifies to the students and the teacher what the students should be doing in class.

Teachers of mathematics, like all teachers, need to believe that their teaching actions have significant causal impact on their students' learning. Telling, irrespective of its pedagogical strengths and weaknesses, provides a clear model for teachers of mathematics to develop a sense of efficacy. (Smith, 1996, p. 393)

Smith went on to suggest that the reform movement has substantially changed what teachers must know and do to succeed in helping their students learn. How can teachers who have learned in a telling environment achieve a sense of efficacy consistent with the reform? Two things must happen. First, teachers must learn about the reforms and attempt to change their practices accordingly; a growing body of research on teachers' knowledge, beliefs, and practices in the context of reform has addressed this need. Second, according to Smith, teachers' success in making the changes to their practices must bring about a reconceptualization of their senses of efficacy, and this transformation has received no explicit focus. Now, nearly a decade since Smith's appeal, I think that the movement to consider teachers' identity, to be discussed later in the chapter, might address this call for research into teacher efficacy.

Summary

Researchers, teacher educators, and professional developers are generally not interested in just measuring teachers' beliefs; they want to change beliefs. The research indicates that the beliefs teachers hold affect their views of the instruction they observe, complicating attempts to answer the question of whether beliefs' change precedes or follows change in instruction. Determining which changes first is less important than supporting teachers to change their beliefs and practices in tandem, and reflection is the critical factor for supporting teachers' changing beliefs and practices. Through

reflection, teachers learn new ways to make sense of what they observe, enabling them to see differently those things that they had been seeing while developing the ability to see things previously unnoticed. While teachers are learning to see differently, they challenge their existing beliefs, leading to associated beliefs change. Change in teachers' beliefs may not lead to change in their practices, or vice versa, but I conjecture that the most lasting change will result from professional development experiences that provide teachers with opportunities to coordinate incremental change in beliefs with corresponding change in practice. Teacher educators and professional developers must better understand not only *what* beliefs teachers hold but also *how* they hold them, because the ways that teachers hold their beliefs affect the extent to which existing beliefs can be challenged. Two impediments to changing teachers' beliefs are concern for the well-being of children that often inhibits teachers' willingness to challenge students and difficulty in overcoming the classroom challenges that derive from moving beyond their role as the teacher as one whose job it is to tell students how to be successful.

TEACHERS' BELIEFS, PART II: FOUR AREAS OF RESEARCH

Three major areas of research on teachers' beliefs are their beliefs about students' mathematical thinking, about curriculum, and about technology. These three areas are important because of their potential role in changing teachers' beliefs. One other area I briefly review in this section is teachers' beliefs about gender.

Teachers' Beliefs Related to Students' Mathematical Thinking

For more than 25 years, mathematics educators have gained substantial knowledge about children's mathematical thinking within specific content domains, and this research has served as an important base for supporting professional development of prospective and practicing teachers (Grouws, 1992; National Research Council, 2001). Prospective elementary school teachers care fundamentally about children, but not necessarily about mathematics (Darling-Hammond & Sclan, 1996), so helping prospective teachers learn about children's mathematical thinking has been an important aspect of teacher development. Likewise, professional development based on children's thinking can help teachers create rich instructional environments that promote mathematical inquiry and understanding, leading to docu-

mented improvement in student performance (S. M. Wilson & Berne, 1999). I now turn to recent research on teachers' beliefs about children and the relationship of teachers' beliefs about mathematics, teaching, and learning to teachers' understanding of children's mathematical thinking.

Elementary School Teachers' Beliefs Related to Students' Mathematical Thinking

Fennema et al. (1996) conducted a longitudinal study of 21 primary-grade teachers over a 4-year period during which the teachers participated in a Cognitively Guided Instruction (CGI) (Carpenter, Fennema, Franke, Levi, & Empson, 1999) professional development program focused on helping teachers understand the development of children's mathematical thinking by interacting with a specific research-based model. They studied changes in the beliefs and in the instruction of the teachers and growth in the children's learning. Data sources for measuring teachers' beliefs were comprised of audiotape transcriptions of classroom observations, interviews, Mathematics Belief Scale (Fennema et al., 1990) scores, and field notes from many informal interactions with the teachers. As noted previously, the beliefs survey was comprised of 48 Likert-scale items assessing four subscales: Role of the Learner, Relationship Between Skills and Understanding, Sequencing of Topics, and Role of the Teacher. The Role of the Learner subscale measures the belief that children are able to construct their own knowledge instead of being merely receivers of knowledge. The Relationship Between Skills and Understanding subscale measures the belief that skills should be taught in relation to understanding and problem solving rather than in isolation. The Sequencing of Topics subscale measures the belief that children's natural development, rather than the logical structure of formal mathematics, should guide the sequencing of topics. The Role of the Teacher subscale measures the belief that instruction should facilitate children's construction of knowledge rather than consist of teachers' presenting materials.

The researchers engaged in an extensive 2-year process to define levels of teachers' instruction and levels of teachers' beliefs. These two multilevel scales (that later were collapsed into one) were scored on a scale of 1, 2, 3, 4-A, and 4-B. The belief levels were based on the extent to which teachers held the belief that children can solve problems without instruction and the belief that what the teachers know about their students' thinking should inform the teacher's curricular decisions. Level 1, the lowest level, reflected the belief that children cannot solve problems without instruction. Level 2 teachers were struggling with the belief that children can solve problems without instruction.

Level 3 reflected the belief that children can solve problems without instruction, but the teacher believed only in a limited way that his or her students' thinking should be used to make instructional decisions. The highest level was divided into Levels 4-A and 4-B; Level 4-A reflected the belief that children can solve problems without instruction in specific domains and that teachers should use knowledge of their students to guide interactions with them; Level 4-B reflected the belief that children can solve problems without instruction across mathematics content domains and that teachers should use knowledge of their individual children's thinking to inform their decision making, regarding both interactions with the students and curriculum design.

The researchers used the same Levels (1, 2, 3, 4A, and 4B) to characterize levels of cognitively guided instruction. Level 1, the lowest level of cognitively guided instruction, was characterized by the teacher's providing few or no opportunities for students to engage in problem solving and share their thinking. The highest level, Level 4-B, was characterized by the teacher's providing opportunities for children to engage in various problem-solving activities and eliciting children's thinking, attending to the thinking children shared, and adapting instruction according to what was shared. That is, at Level 4B, instruction was driven by the teacher's knowledge of individual children.

Results (Fennema et al., 1996) indicated that over the course of the longitudinal study, 18 of the 21 teachers increased their levels in terms of sophistication of their beliefs, and 18 of the 21 teachers increased their levels in terms of cognitively guided instruction. Overall, 17 teachers increased on both levels. At the beginning of the study, 2 teachers were categorized as holding beliefs at Level 4 and 2 teachers were categorized for instruction at Level 4 (I do not know whether these were the same teachers), and by the end of the study, 11 of the 21 teachers were categorized as holding beliefs at Level 4, and 7 of the 21 teachers were characterized for instruction at Level 4. The authors noted that in spite of the obvious relation between levels of instruction and levels of belief, they were unable to compare relationships, because a teacher's beliefs and instruction were not always categorized at the same level, and they found no overall pattern as to whether a teacher's level was higher for beliefs or for instruction. In short, the researchers found no consistent relationship between change in beliefs and change in instruction. Regarding the 17 teachers whose final ratings were higher than their initial ratings on both beliefs and instruction, 6 teachers' beliefs changed first, 5 teachers' instruction changed first, and the change

occurred simultaneously for 6 teachers. The authors stated that the exploration of the relationships between change in beliefs and change in instruction was complex and could be understood only in terms of specific teachers. In stating their conclusions, the authors noted that although many factors contributed to the teachers' change, the two that seemed most critical were that the teachers learned the specific research-based model about children's thinking and used that model in their classrooms.

One other important finding in this study (Fennema et al., 1996) was that "gains in students' concepts and problem-solving performance appeared to be directly related to changes in teachers' instruction" (p. 430). The authors attributed the changes in the students' achievement to the facts that teachers provided the students more opportunities to engage in problem solving, encouraged their students to share their thinking, and adapted instruction to the problem-solving abilities of the students. Although the authors expected that more problem-solving emphases by the teachers would lead to an increase in problem-solving performance by their students, they found striking that the shift in emphasis from skills to concepts and problem solving did not result in a decline in students' computational-skill performance.

In a follow-up to the original CGI study (Peterson, Fennema, Carpenter, & Loef, 1989), Knapp and Peterson (1995) conducted phone interviews with 20 of the original 40 CGI teachers 3–4 years after their initial introduction to CGI; they found that all but one of the teachers reported using CGI in some way in their current classroom practices. They identified three patterns of CGI use: (a) Ten teachers reported "steadily, if often gradually, developing their use of CGI" (p. 47) and currently using it as the main basis for their instruction; (b) 6 teachers reported having used CGI more extensively when they first learned about it but currently using it only supplementally or occasionally; and (c) 4 teachers reported having never used CGI more than supplementally or occasionally. Knapp and Peterson found that "the relationship between changes in teachers' beliefs and their classroom practices was often interactive, and in many cases, beliefs seem to have followed practice" (p. 61). They reported that the teachers' beliefs in CGI principles began to develop during the workshop but deepened over time when their students generated solutions to complex mathematical problems.

In a longitudinal study of 496 German elementary school students in 27 classrooms, Staub and Stern (2002) also found a relationship between teachers' beliefs as measured by the CGI Belief Scale (Fennema et al., 1990) and students' achievement. Staub and

Stern found that students of third-grade teachers who scored higher on the CGI Belief Scale displayed higher achievement in solving mathematical word problems than did students of third-grade teachers who scored lower on the CGI Belief Scale. Furthermore, as in the study by Fennema et al. (1996), they found that the students of teachers who scored higher on the CGI Belief Scale did no worse on mathematics-facts achievement than the students of teachers who scored lower on the survey. In fact, the authors concluded, "Teachers with a cognitive constructivist view of learning tend to be more successful in fostering math fact achievement" (Staub & Stern, 2002, p. 354).

Preservice Elementary School Teachers' Beliefs Related to Students' Mathematical Thinking

Vacc and Bright (1999) investigated the effects that introducing Cognitively Guided Instruction had on preservice elementary school teachers' changing beliefs about teaching and learning and on their instructional use. Thirty-four preservice teachers enrolled in a 2-year undergraduate cohort for preservice teachers participated in the 2-year study starting with the beginning of the students' junior year when they began the professional development program and continuing through the end of student teaching at the end of their senior year. On four occasions all students completed the CGI Belief Scale (Fennema et al., 1990); Vacc and Bright (1999) also conducted eight on-site, formal observations of each prospective teacher. In-depth case studies of two students were conducted, and data for these case studies comprised reflective journal entries written during the mathematics methods course and student teaching, videotapes of four mathematics lessons conducted during the student teaching, and three open-ended interviews. Vacc and Bright applied the Levels of Instruction framework and Levels of Belief framework used by Fennema et al. (1996) for determining the levels of the two student teachers.

Results of the study (Vacc & Bright, 1999) indicated that although the preservice teachers' beliefs-scale scores changed little during the first year of the 2-year program of professional course work, which included a 10-hour/week internship in a professional development school, the scores increased significantly during the third semester during their enrollment in the mathematics methodology course, about one third of which was devoted to instruction related to CGI. The students' belief-scale scores continued to increase significantly during the fourth semester during their student-teaching experiences. On the basis of their analysis of the case-study data, Vacc and Bright categorized one of the two preservice teachers at the end of

student teaching at Level 3 for belief and Level 3 for instruction; that is, they found that she believed that children can solve problems without instruction and, in a limited way, that children's thinking should be used to make instructional decisions; also, she was putting these beliefs into practice. The other teacher was categorized at between Levels 2 and 3 for belief and at Level 2 for instruction; therefore, she was providing limited opportunities for her children to engage in problem solving or share their thinking, and she was, in a limited way, eliciting and attending to children's thinking. The authors concluded that counter to previous research indicating that preservice teachers' beliefs are resistant to change, this study provided evidence that preservice teachers' beliefs did change. The researchers were cautious about drawing conclusions regarding either the depth of the changes in beliefs or the contributions of various factors to the change. They concluded, "The data indicate the possibility that intensity of experience and a focus on children's thinking in the mathematics methods course may be keys for helping preservice teachers change their views" (p. 108).

Ambrose (2004) studied the beliefs of prospective elementary school teachers engaged, in Southern California, in an early field experience linked to a mathematics course. The purpose of the early field experience, described by Philipp, Thanheiser, and Clement (2002), was to motivate students to recognize the importance of learning mathematics by exploring the complexities involved when interviewing and tutoring elementary school children. In her theoretical framework for beliefs, Ambrose (2004) considered two primary sources for beliefs: emotion-packed experiences and cultural transmission. An example of an emotion-packed experience, described later in this chapter in the section on affect, is the prospective elementary school teacher Jo's negative childhood experiences with timed multiplication tests (Walen & Williams, 2002). The second source of beliefs mentioned by Ambrose (2004), cultural transmissions, might be thought of as the hidden curriculum in which people are engaged every day; beliefs created as a result of these experiences are held at a subconscious level. Examples of hidden curricula at the secondary school level were provided by Schoenfeld (1988): He found that a "well-taught" high school geometry course perpetuated the views that (a) all problems can be solved in just a few minutes; (b) the answer is what counts; (c) students are passive consumers of other people's mathematics; and (d) the world of deductive geometry is separate from the world of constructive geometry. Ambrose (2004) noted that beliefs have a filtering effect on people's thinking, so that, through time,

beliefs become more resilient. Ambrose provided the example shared by a colleague, who, in an effort to help her prospective elementary school teachers understand that children come to school with a great deal of knowledge on which teachers can build, had them interview kindergarten children during the second week of school. These preservice teachers were amazed at some of the problem-solving skills the students demonstrated, but instead of taking these skills as evidence that students bring much informal knowledge to school, some mentioned how impressed they were with how much the teacher had taught the children during the first week of school! The students' belief that one's mathematical knowledge is a result of school learning colored their view of the experience in such a way that they misinterpreted it.

Ambrose (2004) presented four mechanisms by which students' beliefs systems might change; the first two of these were mentioned as mechanisms for creating new beliefs:

- (a) they can have emotion-packed, vivid experiences that leave an impression; (b) they can become immersed in a community such that they become enculturated into new beliefs through cultural transmissions; (c) they can reflect on their beliefs so that hidden beliefs become overt; (d) they can have experiences or reflections that help them to connect beliefs to one another and, thus, to develop more elaborated attitudes. (p. 95)

Ambrose (like Cooney, 1999, mentioned earlier) noted that the two beliefs central to prospective teachers related to teaching as explaining and teaching as caring about students; the study was designed to provide opportunities for the prospective teachers to challenge their belief about teaching-as-telling by building upon their caring of children.

The 15 prospective elementary school teachers who participated in the study were simultaneously enrolled in a mathematics course and in a Children's Mathematical Thinking Experience. Data sources were surveys, interviews, prospective teachers' written work, field notes, and a computerized beliefs survey completed at the beginning and end of the semester. Ambrose (2004) conducted an intense analysis of one student, Donna, and compared the emergent themes from Donna's case with experiences of the other 14 students. Ambrose learned that for Donna the most powerful experiences were her three sessions working with a fifth-grade student on fractions, assessing the child during the first session and trying to expand the child's fraction understanding during the second and third sessions. The second session was a high point for most of the prospective teachers, because they felt

confident that they had successfully taught the children something about fractions, but during the third session, most were surprised to discover that the children with whom they had previously worked experienced difficulty with the same fractions topic the prospective teachers thought they had taught during the previous lesson. The language the preservice teachers used captured the intensity of the experience; they described feelings of being *stoked* or *excited* by the events of the second session, whereas words they used to describe the third session included *shocked*, *disappointed*, and *aggravating*. Ambrose concluded that for many of the prospective teachers, their initial beliefs were undifferentiated and were thus held in an unreflective manner. The intense experiences in working with the students resulted in their beliefs' becoming more salient to them. Whereas many teacher educators hold belief reversal as the goal for work with prospective teachers, Ambrose found that instead of letting go of their old beliefs, the prospective teachers in her study held onto their old beliefs while forming new ones, and hence the prospective teachers engaged in actions with their children that reflected a combination of beliefs about mathematics teaching and learning. Ambrose suggested that the students' experiencing a "failed teaching experiment" may have been an important component in their learning experience, and she concluded by stating, "Providing prospective teachers with intense experiences that involve them intimately with children poses a promising avenue for belief change" (p. 117).

D'Ambrosio and Campos (1992) studied five preservice teachers enrolled in their senior year of the mathematics program at the Catholic University of São Paulo, Brazil, and, like Ambrose (2004), they found that providing preservice teachers with opportunities to learn about children's mathematical thinking led to their reflecting upon conflicting situations that arose. These situations created a state of disequilibrium for the preservice teachers, leading them to question normally accepted instructional practices and generally to develop characteristics of reflective practitioners (Schön, 1983).

Secondary School Teachers' Beliefs Related to Students' Mathematical Thinking

Most researchers investigating the relationship between teachers' beliefs and students' thinking have focused at the elementary school level. Nathan and Koedinger (2000b), however, studied 67 secondary school mathematics teachers from the United States and 35 mathematics education researchers to determine their predictions of problem-solving difficulty for a set of arithmetic and algebra problems. Nathan

and Koedinger constructed six types of problems using three presentation formats (story problems, word-equation problems, and symbolic-equation problems), and for each format, they wrote two problems, one with the start unknown and the other with the result unknown. Their previous research had indicated that students found result-unknown story problems and result-unknown word equations to be the easiest; result-unknown equations, start-unknown story problems, and start-unknown word equations of easy-to-medium difficulty; and start-unknown equations to be the most difficult of the problems. On the one hand, Nathan and Koedinger suggested that this research supported the view that the development of students' verbal reasoning precedes the development of their symbolic reasoning, and they developed a model they named a *Verbal-Precedence Model* to reflect the student data. On the other hand, the teachers incorrectly predicted that the students' symbolic-reasoning skills would develop first, with word-problem-solving ability developing later, and the researchers referred to the model they developed to reflect the teachers' view as the *Symbolic-Precedence Model*. On the basis of their data, they concluded, "High school mathematics teachers hold beliefs that cause them to systematically misjudge students' symbolic- and verbal-reasoning abilities" (Nathan & Koedinger, 2000a, p. 212). The researchers found that in two commonly used mathematics-textbook series the development seemed to match the teachers' *Symbolic-Precedence Model*, and they suggested that through repeated exposure to textbooks, "teachers internalize the symbolic precedence view as a basis for their predictions of problem difficulty for students" (Nathan & Koedinger, 2000b, p. 181).

In a follow-up study, Nathan and Koedinger (2000a) asked 107 Grade K–12 teachers to rank order the six mathematics problems used in the previous study (Nathan & Koedinger, 2000b); each teacher also completed a 47-item Likert-scale assessment designed to assess each of six constructs related to teachers' views of mathematics, mathematics instruction, and student learning:

1. Algebraic procedures are "best" for effectively solving mathematical problems.
2. Invented solution methods are effective for solving problems.
3. Arithmetic problems are easier than algebra problems and should be presented first, and mathematics problems presented in words are most difficult and need to appear later in the curriculum (symbolic-precedence view).

4. Students may enter the classroom with valid ways of reasoning, and teachers should encourage invented strategies.
5. Correct answers are more important than reasoning processes.
6. Alternative solution strategies (such as arithmetic, guess-and-check, and other nonsymbolic methods) demonstrate gaps in students' knowledge.

Results indicated that across grade levels, the teachers reflected recent mathematical reform views, and they tended to disagree with views that challenged reform-based views. The elementary school teachers were most likely and high school teachers were least likely to agree with reform-oriented views. Teachers across all grade levels correctly predicted that result-unknown problems would be easier than start-unknown problems. High school teachers incorrectly ranked symbolic equations as easier than verbal problems, and the rank orders of the elementary school teachers were similar to those of the high school teachers, although elementary school teachers were more likely to rank problems on the basis of algebraic and arithmetic structure than on presentation format. Middle school teachers stood out as a group because their rank orderings were most consistent with the research on students' performance. That is, "middle school teachers were most accurate in predicting students' problem-solving performance in contrast to the view held by many high school teachers that symbolically presented problems were easier to solve than verbally presented problems (Nathan & Koedinger, 2000a, p. 226). In their discussion, the authors suggested that middle school teachers, who work with students who have not typically received formal algebra training, may have occasions to observe students' use of more informal and invented methods, and thereby the teachers may have more opportunities to observe the transition from arithmetic to algebraic reasoning. They suggested, further, that elementary school teachers, who held the strongest reform-based views of learning and teaching, are relatively unfamiliar with students' algebraic reasoning and may have therefore expected that students would operate with a traditional vision of algebra curriculum and instruction. High school teachers were least aware of the efficacy of students' invented algebra solution strategies, and the authors suggested that because high school teachers have greater expertise in their content areas, they are personally more distant from their novice students' difficulties. The authors introduced the term *expert blind spot*, a feature they attributed to teachers with high levels of content knowledge but low levels of

awareness of alternative interpretations for symbolic equations.

Nathan and Petrosino (2003) studied 48 prospective secondary school teachers enrolled in a "nationally acclaimed teacher education program" (p. 911) in the United States to test their *expert-blind-spot hypothesis*, stipulated as follows:

Educators with advanced subject-matter knowledge of a scholarly discipline tend to use the powerful organizing principles, formalisms, and methods of analysis that serve as the foundation of that discipline as guiding principles for their students' conceptual development and instruction, rather than being guided by knowledge of the learning needs and developmental profiles of novices. (p. 906)

Participants were categorized in one of three ways, on the basis of their subject-matter knowledge: (a) 13 were placed in a "basic math" category because they had not completed a precalculus course; (b) 16 from the general population of teacher education students who were pursuing licensure to teach elementary school but who had completed at least one calculus course were placed into the HiMath knowledge group; (c) 19 who were enrolled in a specialized program for mathematics and science majors and had completed many higher level mathematics courses were placed in the MathSci category. Using problems similar to those used in previous studies (Nathan & Koedinger, 2000a, 2000b), Nathan and Petrosino (2003) asked all participants to rank order six problems as they would predict beginning-level algebra students would experience ease or difficulty in solving them. The participants also responded to the 47-item beliefs survey used in the previous study (Nathan & Koedinger, 2000a). Results of the predictions of problem difficulty (Nathan & Petrosino, 2003) indicated that the preservice teachers with more advanced mathematics knowledge (the MathSci and HiMath groups) were far more likely to follow the symbolic-precedence view of algebra development than the preservice teachers with less mathematics knowledge (the Basic Math group). Results of the beliefs survey supported this finding, with a greater percentage of those preservice teachers in the two higher mathematics knowledge groups agreeing that using algebraic formalisms is best for solving complex problems, whereas students in the Basic Math group were more likely to believe that instruction should build on students' intuitions and invented methods. Nathan and Petrosino (2003) conjectured,

Educators with greater subject-matter knowledge tend to view student development through a domain-centric lens and consequently tend to make judgments

about student problem-solving performance and mathematical development that differ from actual performance patterns in predictable ways. (p. 918)

Summary

Researchers measuring teachers' beliefs about children's mathematical thinking have correlated these beliefs to teachers' instruction and to students' learning. The Cognitively Guided Instruction group (Carpenter et al., 1999) developed an instrument to measure teachers' beliefs about children's abilities to solve problems without instruction and about teachers' beliefs about the role that knowledge of children's thinking should play when one makes instructional decisions, and they used this instrument in a study in which they found that teachers' beliefs were related to instruction reflecting a focus on children's mathematical thinking, and, furthermore, teachers' instructional changes were related to gains in students' understanding and problem-solving performance (Fennema et al., 1996). Knapp and Peterson (1995) found that most of the original CGI teachers who had made instructional changes to focus upon children's mathematical thinking continued to apply this focus several years later. Staub and Stern (2002) used the same beliefs survey and found a direct link between teachers' beliefs about children's mathematical thinking and students' achievement. Preservice elementary school teachers provided with opportunities to learn about children's mathematical thinking can change their beliefs, and the intensity of the experience associated with focusing upon children's thinking was identified as a cause for these changes (Ambrose, 2004; Vacc & Bright, 1999). Although teachers' beliefs about children's mathematical thinking have been studied more extensively at the elementary school level than at the secondary school level, Nathan and his colleagues have investigated secondary school mathematics teachers' beliefs about students' mathematical thinking. Nathan and Koedinger (2000b) found that the development of students' verbal reasoning skills preceded the development of their symbolic reasoning but that high school mathematics teachers believed just the opposite, and they coined the term *expert blind spot* (Nathan & Koedinger, 2000a) to refer to the relationship between teachers' higher levels of content knowledge and lower levels of awareness of students' understanding. Nathan and Petrosino (2003) found that the expert-blind-spot characterization also applied to prospective secondary school teachers. Although researchers studying teachers' beliefs about students' mathematical reasoning have noted the complex relationship between beliefs and practices, several studies provide evidence

that teachers' beliefs often change as a result of observing children's mathematical reasoning (Ambrose, 2004; D'Ambrosio & Campos, 1992; Knapp & Peterson, 1995; Vacc & Bright, 1999).

Teachers' Beliefs Related to or Changed by Use of Mathematics Curricula

Any mathematics class has two curricula: the intended curriculum and the enacted curriculum. The intended curriculum, that which is stated or explicit, is comprised of the materials of a class, including textbooks, curriculum guides, and course descriptions. Although many think of these materials as defining the curriculum, one cannot understand the experience students have in a class without attending to how materials are enacted by a teacher, with students, in a particular context (Ball & Cohen, 1996). Just as the knowledge and beliefs teachers hold about mathematics, teaching, and learning affect the ways in which they enact curricula, so too should we expect that teachers might learn from the curricula they use. That is, teachers not only adapt and change curricula but also are changed by the curricula they use. The relationships between teachers' beliefs and the innovative curricula they use are addressed in this section.

Lloyd (2002) highlighted two aspects of reform-oriented curricula designed to support teacher learning. First, by focusing upon the exploration of solutions to real-world problems, reform-curricula designers emphasize a more inquiry-oriented approach to mathematics than is found in traditional curricula, and the materials are formatted to support these mathematical and pedagogical differences. Second, reform-oriented curricula include more extensive information for teachers than traditional curricula. For example, they provide historical information, details about different representations, and examples of students' reasoning. Lloyd (1999, 2002) found that using the reform-oriented Core-Plus curriculum over several years supported a high-school teacher's learning or relearning of mathematical subject matter. Furthermore, as a result of becoming more familiar with the curriculum and interacting with students, this teacher's beliefs about the graphical representations of functions became richer and more nuanced.

Collopy (2003) reported on a study designed to investigate two elementary school teachers' changes in beliefs related to teaching and learning mathematics and their changes in instructional practices resulting from their first year's use of *Investigations in Number, Data, and Space*, a reform-oriented curriculum. The two teachers were experienced elementary school teachers working in similar contexts, schools

comprised of a high percentage of at-risk students described as “below average in mathematics.” Both teachers sat on their district mathematics committees, both had recently used a district-mandated traditional Addison-Wesley mathematics textbook, and both had volunteered to pilot the *Investigations* curriculum.

Collopy (2003) collected data about the teachers’ beliefs and practices in three stages. During the first stage, she collected data on the teachers’ backgrounds and baseline data on their beliefs and knowledge in an extended formal interview and on their initial use of the curriculum through 7 or 8 classroom observations. During the second stage, she focused on investigating the teachers’ decisions about curriculum use and changes in beliefs and practices through six days of observations, in two-day sets, a month apart. During the third stage of data collection, Collopy collected data on teachers’ reflections; use of the curriculum; and beliefs, knowledge, and instructional practices by observing each class for two consecutive weeks during the final weeks of the school year and conducting a final interview of each teacher. Collopy conducted two data analyses: a thematic analysis and a segment analysis. In the thematic analysis of the interviews and field notes, she developed categories of codes to explore the stability, changes, and relationships among the teachers’ beliefs about mathematics, students, pedagogy, curriculum, and themselves as learners and teachers. She presented her interpretations during each final interview; both teachers concurred with her interpretations. In the segment analysis, Collopy tracked the format and focus of each teacher’s practice across the year by coding each observed lesson into segments and coding each segment for length in minutes, instructional format, the teacher’s role and focus, student behavior, and teacher’s expectations for students’ thinking.

From her detailed case studies, Collopy (2003) concluded that the two teachers differed dramatically in what they had learned from the materials and ways they engaged with the materials as a support for learning. Ms. Clark held a tightly integrated set of beliefs about mathematics, student learning, the teacher’s role, the purposes of mathematics instruction, and her own mathematical efficacy, and her instructional practice was consistent with these beliefs. She believed that computational speed and accuracy distinguished successful students from unsuccessful students, that her students needed to learn the rules of mathematics, and that mathematics topics should be presented systematically from easier to harder. Ms. Clark agreed with the emphasis in the *Investigations* materials on students’ understanding mathematical concepts but not with the definition of *understanding* as “familiarity with

the magnitude of numbers, mathematical relations, and the meaning of mathematical operations and situations” (p. 296); Ms. Clark defined *understanding* as the memorization and correct execution of standard algorithms. Eventually she became frustrated with the *Investigations* materials, in which speed and rote memorization were downplayed, and she adapted lessons to compensate, changing their essence by omitting alternative problem-solving strategies and mathematics discussions she considered distractions; her frustration level rose until January, when she shelved the *Investigations* curriculum and returned to her more traditional curriculum. Collopy’s segment analysis indicated that in Ms. Clark’s use of the *Investigations* curriculum, 61.8% of the observed time was devoted to procedures and correctness, 35.4% to organizational and management issues, and 2.8% (consisting of 24 minutes during one lesson) to conceptual understanding.

The other teacher, Ms. Ross, approached the *Investigations* curriculum differently than Ms. Clark, significantly changing her approach to instruction in shifting from her early focus on procedures and correctness (94% of the early observations to 21% for November through May) to conceptual understanding and mathematical reasoning (from no early focus to more than 75% in later observations). Unlike Ms. Clark, whose actions were driven by beliefs about mathematics as a set of facts and procedures that one learns through exposure to direct and clear instruction, Ms. Ross held beliefs about the importance of developing students’ confidence for learning mathematics by building upon their prior knowledge. Ms. Ross saw no clear structure for mathematics, so she relied upon her curriculum materials and objectives for the state’s standardized tests to direct the content of and emphasis in her instruction. Ms. Ross followed the curriculum carefully, reading and using the student notes, and she learned mathematics while challenging her beliefs. For example, in the process of asking students to demonstrate, write, and discuss their ideas, she came, for the first time, to see that problems can be solved in many ways.

Although the two teachers in this study had similar backgrounds, taught in similar situations, and willingly piloted the same reform-oriented curriculum, their enactment of the curriculum provided different opportunities for them to learn, differences Collopy (2003) explained in terms of the beliefs most integral to the teachers’ identities. Ms. Clark’s identity as a teacher and learner of mathematics was centered in her belief that mathematics is a set of rules and procedures that students learn by being shown, incompatible with the beliefs underlying the curriculum, so in

her curriculum adaptation, she lost most opportunities to learn from the materials. The beliefs most integral to Ms. Ross's identity as a teacher were not about mathematics but, instead, were about developing students' confidence for learning mathematics and were compatible with those espoused in the curriculum. Thus, she was able to use the materials as they were intended and was, thereby, poised to learn much from their use. Collopy's findings show that while considering how to build opportunities for teachers to learn from their curriculum, writers must also consider the beliefs held by teachers using the materials.

Remillard and Bryans (2004) extended the work of Collopy (2003) by studying the way the reform-oriented *Investigations* curriculum supported the learning of urban elementary school teachers in one school over several years. In this study, seven of the eight teachers who volunteered to participate and attend monthly study-group meetings and related research activities attended for two years. During the first year, each teacher was observed 2–4 times and interviewed at least twice; during the second year, six of the seven teachers were observed 7–8 times and interviewed three times. Two of the teachers' classrooms were designated as focus classrooms, which were observed daily over 2-week periods four times during the second year; each focus teacher was interviewed approximately five times. Data were comprised of audiotapes of the lessons and written field notes, which were later used to complete a predesigned observation instrument. The researchers conducted segment analyses of these data, documenting the time devoted to each task, the teacher's aim and focus of the task, and ways the students engaged with the task.

Three broad categories of teachers' use of the *Investigations* curriculum were identified by Remillard and Bryans (2004): *intermittent and narrow*, *adopting and adapting*, and *thorough piloting*. Teachers whose use was *intermittent and narrow* used the materials minimally and relied primarily on their own teaching routines and other resources to guide their curricula over the year. *Adopting and adapting* described use of the materials as a guide for determining what topics to teach and how to sequence them, but with teachers' adapting the materials to fit their familiar strategies and approaches to teaching. *Thorough piloting* referred to use of all parts of the curriculum guides. Although teachers whose use was in this category continued to draw upon activities and strategies from other sources, when they used the *Investigations* curriculum, these teachers followed the lessons as suggested in the guide and studied and grappled with most of the information provided for the teacher.

To address each teacher's relationship with the *Investigations* curriculum during the first 2 years of use, Remillard and Bryans (2004) introduced the construct *orientation toward curriculum*,

a set of perspectives and dispositions about mathematics, teaching, learning, and curriculum that together influence how a teacher engages and interacts with a particular set of curriculum materials and consequently the curriculum enacted in the classroom and the subsequent opportunities for student and teacher learning. (p. 364)

They presented a dynamic model designed to capture the interactive relationships among teachers' perspectives and beliefs, the use of curriculum materials, the enacted curriculum, and consequent opportunities for teachers to learn.

To illustrate orientation toward curriculum, Remillard and Bryans (2004) described three veteran teachers, Jackson, Reston, and Kitcher, who first used the *Investigations* curriculum during the study. Jackson held beliefs about mathematics teaching and learning incompatible with those underlying the curriculum, and his curriculum use was categorized as intermittent and narrow. Jackson not only seemed intent upon using his own materials to emphasize procedural skills but also, in his use of the curriculum, failed to use the suggestions to facilitate students' work and to have students explain their reasoning or discuss their strategies. The other two veteran teachers, Reston and Kitcher, shared views about mathematics teaching and learning that seemed compatible with the curriculum, and so one might have expected their use of the materials to differ from Jackson's. Kitcher's use was as intended (thorough piloting), but Reston's use, in spite of her views of mathematics, teaching, and learning, was similar to Jackson's (intermittent and narrow). Remillard and Bryans found that Reston's "general mistrust of published materials trumped any potential compatibility between her beliefs about mathematics and those represented in the book" (p. 366). That is, her orientation toward curriculum, not her beliefs about mathematics, teaching, and learning, explained her skeptical approach to the curriculum and led her to use the curriculum only as a collection of useful activities.

Use by four of the eight teachers in the study was categorized as thorough piloting of the curriculum, and although these four teachers shared beliefs about mathematics, teaching, and learning that were consistent with those reflected in the *Investigations* curriculum, others in the study did as well, leading the authors to conclude that their shared ideas about the role of curriculum materials, not their beliefs, were

the most significant factor in shaping their orientations toward the curriculum. The teachers viewed curriculum resources as guides or partners in their teaching, and they were all committed to learning about the curriculum and giving it a fair trial. They viewed curriculum as more than a source of activities for students: They saw it as a resource for themselves.

The authors (Remillard & Bryans, 2004) noted that, except for Kitcher (with 25 years' experience), those thoroughly piloting the curriculum were the teachers with the least teaching experience—Graves, Larson, and Schwarz (4, 3, and 3 years, respectively)—who, unlike the other five teachers, lacked well-established routines in their practices and appreciated the structure and learning opportunities available in the curriculum. Graves and Larson used the curriculum to develop deeper insights into students' thinking, and they used those insights to reconstruct their roles as teachers to create more opportunities to attend to their students' thinking. Larson and Schwarz placed substantial emphasis on exploring the mathematical ideas embedded in the units to extend their mathematical knowledge. The authors concluded,

Standards-based curricula are not a panacea, but they can play an important role in fostering reform-based practices. However, without additional support for teachers, the impact of these curriculum materials is likely to be unpredictable and varied. Teachers using curricula like Investigations would benefit from opportunities to explore the content of the materials and have conversations with others about how they use them. (p. 386)

Clearly, curricula may be an important site of learning for practicing teachers. What of prospective teachers? Spielman and Lloyd (2004) attempted to use reform-oriented curricula to change prospective elementary school teachers' beliefs. They arranged for two classes of mathematics for elementary school to be taught by the same instructor using different curriculum models. The *textbook section* was taught using a popular mathematics textbook and pedagogical strategies the researchers believed reflected the philosophy of the textbook authors. The class met weekly for 3 hours, and a typical lesson began with a homework review during which groups of students (who were prospective teachers) were assigned to discuss problems then present solutions to the class, followed by the instructor's comments on the homework review and lecture on the new material. Following the lecture, the instructor showed a short video or students worked, individually or in groups, on a problem or activity at their tables, often with manipulatives available. Student volunteers shared outcomes of their work with the class until a consensus

was formed. During the final hour of class, after the instructor made final comments or summary statements, students worked alone or in groups on homework, with the teacher available to answer questions.

The *curriculum-materials section* used units from two reform-oriented middle school curricula, *Mathematics in Context* and *Connected Mathematics Project*, selected to correspond to the mathematical topics in the textbook section (Spielman & Lloyd, 2004). Whenever possible, instructional decisions were made to support the intent of the authors of the curricula. For example, the curricula materials rarely offered rules or solutions to sample problems, and the instructor refrained from providing rules or solutions. A typical lesson began, as in the textbook section, with the students' reviewing homework by working in groups and sharing their solutions. However, instead of providing solutions or explanations when students had questions, the instructor referred the students to other students for support, often redirecting or rephrasing questions. During the lively homework discussions, students responded to one another's questions without being called upon, and although the students expressed frustration with the instructor's refusal to offer rules, procedures, or solutions, "the students appeared to become accustomed to this type of classroom" (p. 35). After the homework review, the students worked together at their tables on problems or activities, and they would occasionally watch video. Although the videos were the same for both classes, the problems and activities differed both in type and presentation. In the curriculum materials section, no instructor presentation or lecture preceded work on problems, and the instructor's role was to clarify and pose questions. During the last hour of the class students worked on their homework, but instead of answering students' questions, the instructor posed other questions designed to facilitate students' finding their own solutions.

Students completed a Likert-scale teaching-beliefs survey and a mathematics-content instrument as pretests and posttests. No between-group differences were found on the content test, but differences were found on the beliefs survey. The students in the textbook section saw the instructor as more of a classroom authority than did the students in the curriculum-materials section. When asked whether the textbook, other students, or the instructor had been most beneficial to learning, only about 10% of the students in either group selected the textbook, but more than 68% of the students in the textbook section selected the instructor whereas more than 82% of the students in the curriculum-materials section selected other students. A paired-sample *t*-test comparing students'

pretest and posttest responses on the percentage of time students thought should be devoted to teacher lecture and explanation or to group work and discussions showed significant differences for the students in the curriculum-materials group, but no significant differences were found for students in the textbook section. Over the course of the semester, the amount of time students in the curriculum-materials group thought should be devoted to teacher lecture and explanation decreased (from 30.6% to 22.2%) whereas the amount of time they thought should be devoted to group work and discussion increased (from 54.1% to 63.7%). Spielman and Lloyd (2004) concluded that prospective elementary school teachers' beliefs about textbooks, teaching, and learning changed as a result of using reform-oriented middle school textbooks to learn mathematics.

Summary

Several mathematics curricula have been designed to promote a reform-oriented approach (NCTM, 2000; NRC, 2001) to mathematics teaching and learning. The curriculum designers of these materials recognized the role that curriculum might play in supporting teachers' learning, and they included instructor-support materials designed to provide guidance for classroom instruction. These curricula are said to speak *to* teachers, rather than *through* them (Remillard, 2000). As important as curriculum materials are for supporting teachers in the classroom, assuming that teacher-proof curriculum can, or should, be used would oversimplify the matter; the research indicates that if teachers' beliefs about mathematics, teaching, and learning are not consistent with the beliefs that serve as the foundation of the reform-oriented curriculum, the teachers do not use the materials as they were intended (Collopy, 2003; Remillard & Bryans, 2004). Holding particular reform-oriented beliefs is a necessary, but not sufficient, condition for teachers to use a curriculum as intended, because Remillard and Bryans (2004) found that another factor, teachers' orientation toward curriculum, played a significant role in curriculum use even for teachers who held reform-oriented beliefs about mathematics, teaching, and learning. According to Collopy (2003), a teacher who viewed mathematics as a set of procedures and rules was less open to implementing the curriculum as intended than was the teacher who lacked such a clear structure for mathematics. The less experienced teachers who had yet to develop classroom routines for teaching mathematics were more likely than the experienced teachers to read and learn from the supportive materials in a study by Remillard and Bryans, indicating that providing these curricular materials to teachers early in their careers may be a promising means

of supporting their growth as teachers. Spielman and Lloyd (2004), studying prospective teachers enrolled in a course of mathematics for elementary school, found that prospective teachers need not yet be in their own classrooms to accrue the educational benefits of reform-oriented curricula; beliefs of students in a course based on a reform-oriented curricula changed more than the beliefs of students who used a conceptually oriented college textbook.

Teachers' Beliefs About Technology

Prospective teachers believe that computers are important in education, and they want to use them in their preservice programs, but except at a few institutions, students do not learn to use technology effectively (Willis & Mehlinger, 1996). This failure to prepare teachers is reflected in the fact that elementary school teachers generally do not like to teach with computers, and when they do use them, they most commonly do so for drill and practice activities (Cummings, 1998). Using a Likert-scale instrument to survey 33 elementary school teachers about their uses of technology, Cummings found that although most teachers thought that they possessed expertise about using computers, they cited time spent preparing for computer instruction as the greatest barrier to computer use.

Schmidt (1998) investigated beliefs of teachers in Grades 4–6 about the use of calculators and concluded that they reflected their more traditional views about mathematics. In a follow-up study, Schmidt (1999) provided the teachers with an in-service during which they spent one week using calculators to explore fifth- and sixth-grade concepts and applications. After the in-service, during which teachers kept reflective journals; evaluated teaching materials; read and discussed journal articles about a variety of topics, including research about calculators; and participated in and led discussions, the teachers continued their professional development by engaging in action-research projects and sharing the results with colleagues, attending the state mathematics conference, and cohosting a conference for teachers and principals about using calculators. Before and after the in-service, 32 teachers completed a two-part questionnaire: In the first part, they gave personal and professional characteristics and answered three open-ended questions related to what would facilitate the integration of calculators into mathematics, what most hindered the use of calculators in mathematics, and how students in their classes currently used calculators. In Part 2, comprised of 29 Likert-items, respondents provided one of six re-

sponses (strongly agree, agree, slightly agree, slightly disagree, disagree, and strongly disagree) to items such as “If calculators are used in school mathematics programs, students no longer need to know paper-pencil computing techniques” (Schmidt, 1999, p. 24), a statement with which the teachers strongly *disagreed* on both administrations of the questionnaire. On another item, “Calculators are tools that allow students to focus more attention on mathematics concept development and understanding” (p. 25), the teachers strongly *agreed* on both the pretest and posttest. After finding little change in the teachers’ beliefs about calculator use as a result of the in-service, Schmidt suggested that the “teachers’ perspective and philosophy of teaching and learning mathematics constrained their beliefs about calculators” (1999, p. 32).

Wiegel and Bell (1996) investigated the integration of computers into six mathematics content classes for prospective elementary school teachers—three sections each of the first and second courses. Data for the first course consisted of precourse surveys and a topic-specific survey, part of a required but ungraded assignment, for which students wrote about their attitudes and beliefs about mathematics and the integration of computer activities into the mathematics content course. Data for the second course consisted of one essay, weekly reflections from all students, and anecdotal comments collected during the computer labs. Used in the first mathematics course were two interactive microworlds, one for geometry and the other for fractions, created as part of a research grant directed by Les Steffe and John Olive. The students in the second mathematics course used a spreadsheet, a probability microworld, LOGO, and Geometer’s Sketchpad. The authors concluded that the students and instructors all rated the overall experience positively, and that the classroom atmosphere, which was tense at the beginning of the semester, seemed to relax when students used computers, working in small groups and communicating with one another. The students reported appreciating the opportunity to use the computer, an activity they came to see as a welcome change in the class routine. Although the authors of this mid-1990s’ study found that many of the students needed to learn to operate computers and manipulate programs and that often the students’ attention to the basic mechanical operation of the computer detracted from their learning the mathematics, college students today would have more expertise with computers. Also, because their computer work was to be ungraded, the students wanted to stop using the computers toward the end of the semester to prepare

for the paper-and-pencil items they expected on the final examination.

Tharp, Fitzsimmons, and Ayers (1997) conducted a study designed to determine whether providing teachers with instruction on the use of graphing calculators would have an effect on the teachers’ beliefs about computers or on the teachers’ instruction. Five 3-hour, monthly in-services designed to help teachers integrate graphing calculators into their instruction were provided to 261 Grades 6–12 teachers (168 mathematics teachers, 72 science teachers, and 21 teachers of other subjects). Topics covered during these in-services included solving linear equations; studying mathematical modeling, exponential growth, projectile motion, and parametric equations; and connecting multiple (graphic, tabular, and numeric) representations. Teachers watched videotaped lessons offered by experienced teachers using calculators in their classes. All the teachers were required to provide instruction using graphing calculators in their own classrooms. Data included pre and post questionnaires to assess how teachers’ attitudes toward graphing calculators and mathematics shifted because of the instruction and the use of technology. In journals, teachers described their initial use of the graphing calculators during instruction. Likert-scale items, such as “Calculators should ‘only’ be used to check work,” and “When doing mathematics, it is only important to know how to do a process and not why it works,” were used to measure the teachers’ views of calculator use and mathematics learning, teachers’ access to calculators, teachers’ efficacy, student efficacy, and administrative support. Results indicated that teachers’ beliefs about the use of calculators changed significantly to favor using calculators as an integral part of instruction. A significant positive correlation was found between teachers’ views of mathematics and their views on the use of calculators in the classroom, with those teachers holding more rule-based views of mathematics tending to hold the view that calculators do not enhance instruction and those with less rule-based views of mathematics more likely to view calculators as integral for mathematics and science instruction. After coding and analyzing all the teachers’ journal entries, the authors found that although all the teachers attempted to integrate the use of the graphing calculator into their instruction, the rule-based teachers tended to quickly return to limiting the amount and type of calculator use in their classrooms. Whereas the rule-based teachers tended to focus upon the students’ emotional reactions, the non-rule-based teachers were more likely to concern themselves with the students’ conceptual understanding and thinking. The authors concluded

that technology can be used to support changes in teachers' instruction, followed by changes in teachers' beliefs.

Fleener (1995) used a Likert scale to measure the attitudes toward the use of technology held by 233 elementary, intermediate, and high school classroom teachers and by 78 preservice teachers enrolled in a mathematics methodology course for K–3 teachers. Items on the 29-item scale included “Calculators make mathematics fun” and “Students should not be allowed to use calculators until they have mastered the concept.” Fleener’s previous research had shown that a critical issue (which she labeled the *mastery issue*) that divided teachers related to whether they believed that students should be allowed to use calculators prior to having mastered the concept. In the 1995 study, Fleener investigated whether teachers with different views on the mastery issue also differed with respect to their responses to other items on the survey. Results showed that 55% of the preservice teachers were in the *mastery group* (i.e., believed that students should attain conceptual mastery before being allowed to use calculators), and Fleener found no differences between the mastery views of preservice and practicing teachers. She did, however, find differences in the ways both preservice and in-service teachers in the mastery group tended to respond to other items compared with teachers not in the mastery group. Finally, she concluded that the mastery issue may be key for affecting teachers’ beliefs about using calculators.

Walen, Williams, and Garner (2003) investigated the relationship between students’ use of calculators in college mathematics courses for prospective elementary school teachers and their views of appropriate use of calculators in elementary school classrooms. Although technology has been identified as an essential component of teaching and learning mathematics (NCTM, 2000), Walen et al. (2003) reported that many college mathematics faculty limit the use of calculators for fear that their students will grow to depend upon them or because calculators enable students to “get answers that they don’t ‘know’” (p. 447). The researchers contended that this position, held by many faculty, reflects a view of mathematics as primarily comprised of a set of rule-governed calculations and procedures. They studied four college mathematics classes in which calculators were incorporated into a collaborative setting dedicated to students’ learning with understanding (NCTM, 2000). They hoped that because calculators were readily available for students, they would use them “naturally and unreflectively to accomplish mathematical tasks” (Walen et al., p. 449). To determine whether college students who learned mathematics in such an environment developed be-

liefs about how calculators should be used in elementary school classrooms different from the beliefs of others, Walen et al. (2003) collected informal participant-observation notes, classroom artifacts, and questionnaire responses from students in four sections of a mathematics content course designed for prospective elementary school teachers. They also conducted four case studies of students enrolled in the course. Their results indicated that when asked to solve computational tasks, the students used more mental approaches or paper-and-pencil approaches than calculator solutions. In other words, their students did not overuse the calculator by using it on simple problems. When analyzing the students’ responses to three open-ended questions, the researchers found that the three most common responses to the question “When do you use a calculator?” were when accuracy or confidence in the answer was needed, when the problem was large or complex, or when trying to save time, with each answer given by about one third of the respondents (although a person may have given more than one answer). Overall the researchers concluded that their students reported positive experiences using the calculators in their mathematics classes.

The researchers (Walen et al., 2003) had hoped that the classroom experiences provided to their students would positively influence their attitudes toward calculator use in elementary classrooms and their decisions to model the experiences by using calculators with their subsequent students. However, they were disappointed to see that their “success towards this goal was limited” (p. 457). The second open-ended question was “When do you think students should use a calculator?” The most common answer, given by about half the students (32 of 66), was that students should use the calculator only after they knew the operations; 8 students responded that children should never use calculators because they will handicap students’ learning. The most common response (by 42 of 66) to the third open-ended question—“Is there a time when a calculator shouldn’t be used?”—was that calculators should not be used while students are learning the basics. The researchers concluded that although some students’ responses regarding their subsequent teaching with calculators paralleled their descriptions of their positive experiences using calculators, many students’ responses did not. Instead, the prospective teachers in this study held “seemingly contrasting views that it is acceptable for them to use a calculator to do an arithmetic problem, but it is not acceptable for their students” (p. 459). The researchers suggested that their students found themselves in two different worlds, one in which, as doers of mathematics, they should use calculators whenever the

need arises and the other in which, as teachers helping children learn mathematics, they juxtapose the use of calculators against a set of beliefs about mathematics teaching: “that basic arithmetic skills must be learned; that the task of their future students was not to *do* mathematics, but to *learn* mathematics”; and the calculator was neither an integral nor welcome part of this context. The students’ view of mathematics as something that must first be learned and then done caused them to view calculators as inappropriate for children before they learn the mathematics.

Summary of Teachers’ Beliefs About Technology

Teachers need support to effectively use computers or calculators in their classes, but even when teachers are comfortable using computers or calculators for their own learning, they may not believe that using technology with their students is appropriate. Teachers’ beliefs about appropriate use of technology for children are constrained by their beliefs about mathematics (Tharp et al., 1997; Walen et al., 2003) and by their beliefs about teaching and learning mathematics (Fleener, 1995; Schmidt, 1999). One issue that appeared in several studies was the belief that calculators should be used only after students have learned the mathematics (Walen et al., 2003) or have mastered the concepts (Fleener, 1995).

Mathematics Teachers’ Beliefs Related to Gender

Results of the Third International Mathematics and Science Study (TIMSS) indicated that, although performance of boys and girls worldwide is comparable at the 4th- and 8th-grade levels, by the last year of senior high school, in 18 of 21 countries participating in the testing, boys perform significantly better on mathematics assessments than girls, and even in the remaining 3 countries (including the United States), the differences favored males (Mullis, Martin, Fierros, Goldberg, & Stemler, 2000). In the United States, males and females take similar mathematics classes at the K–12 level, but males participate in mathematics more than females after high school (Levi, 2000). Although boys and girls at the early elementary school level have differed little in achievement-test scores, when researchers extended the focus of their investigations to the types of strategies students used, gender differences between boys and girls appeared as early as first grade, with boys consistently using more abstract strategies than girls used to solve problems (Fennema, Carpenter, Jacobs, Franke, & Levi, 1998). Such early differences may explain why major differences appear at the senior high school level, because students who

learn to make sense of mathematics early on by virtue of applying mental computation and estimation strategies are demonstrating the kind of understanding of numbers and operations that will help them be more successful when studying more advanced mathematics (Sowder, 1998).

Some proposed reasons boys consistently outperform girls in mathematics at the senior high school level and some differences favoring boys that appear even much earlier relate to the views that students have about mathematics. For example, both female and male students in the United States, Australia, and Japan have stereotyped mathematics as a male domain, and the males held this stereotype more than the females (Keller, 2001). Might these students’ beliefs be affected by teachers’ beliefs? Although much remains to be learned about how teachers’ verbal and nonverbal behaviors affect students, the finding that teacher expectancy affects pupils’ performance, sometimes known as the *educational self-fulfilling prophecy*, has been clearly established (Rosenthal, 2002). Less clear, however, is what the research on teachers’ beliefs about boys, girls, and mathematics has shown.

Li (1999), in a review of literature on teachers’ beliefs related to gender, found that although results were inconclusive, the literature indicated that teachers had different beliefs about male and female students and that they tended to stereotype mathematics as a male domain. Teachers tended to overrate male students’ mathematics capabilities and had more positive attitudes about male students. Li found no differences between male and female teachers’ beliefs.

Helwig, Anderson, and Tindal (2001) found little evidence of a relationship between third- and fifth-grade teachers’ ratings of their students’ mathematics skills and gender. They also agreed with an earlier conclusion by Brophy (1983, cited in Helwig et al., 2001) that teachers’ perceptions of their students were accurate.

In urban and rural Kansas, Leedy, LaLonde, and Runk (2003) used a survey to assess mathematics attitudes of 4th-, 6th-, and 8th-grade girls and boys participating in a regional mathematics contest; their parents; and their teachers. The students were selected on the basis of their interest in the mathematics competition and their mathematical abilities. Using 12 Likert-type items drawn from the Fennema-Sherman Scales (Fennema & Sherman, 1976), the authors assessed the students on six factors; they modified the items to assess parents’ and teachers’ views of mathematics as a male domain and usefulness of mathematics. The teachers were comprised of two groups, those mathematics teachers who served as coaches for the mathematics teams and those who did not. Results indicated

that the boys, fathers, parents of sons, and noncoaching mathematics teachers more strongly supported the belief that mathematics is a male domain than the girls, mothers, parents of daughters, and mathematics coaches. The most strongly endorsed individual statement related to gender bias was "Men are naturally better at mathematics than women," whereas the most strongly endorsed nonbiased statement was "It is just as appropriate for women to study mathematics as for men" (p. 289). The authors questioned whether agreement with the first statement is an indication of gender bias or of "real, inherent differences between girls and boys, men and women, in how they learn and think about mathematics" (p. 289).

The belief that mathematics is a male domain was also found among teachers in Switzerland. Keller (2001) drew from TIMSS data and used a Likert-type scale she designed to assess teachers' and 6th-, 7th-, and 8th-grade students' aptitude concepts, such as mathematics as a male domain. Results indicated that students and teachers stereotyped mathematics as a male domain, and evidence showed that teachers' views affected the students' stereotyping. Keller concluded that the assumption, put forth by Fennema, "that teachers transmit their views of mathematics during instruction and that students adopt such views is more plausible than the assumption that teachers adopt the views of students" (p. 171).

Tiedemann (2000) studied the beliefs of Grades 3 and 4 German elementary school students and their teachers. The teachers were each asked to name three girls and three boys from their classes, one of each gender performing in mathematics at the low, medium, and high levels, and then the teachers completed a questionnaire for each identified child. The 21-item questionnaire addressed six categories, including estimation of the child's competence, attribution of improvement, and presumed self-concept of the child. The results indicated that even for equally achieving, average boys and girls, teachers rated mathematics as more difficult for the girls than the boys. Teachers attributed unexpected failures more to low ability and less to lack of effort for girls than for boys. Tiedemann also reported similar results for third- and fourth-grade Russian teachers, who thought of mathematics as more a male domain than a female domain. Tiedemann concluded, "If teachers' beliefs are important influences on how they interact with and teach students, these teachers' beliefs could be seen as an influence on the development of gender differences in mathematics" (p. 205).

Summary of Teachers' Beliefs Related to Gender

Researchers raise more questions than they answer related to teachers' beliefs about boys, girls, and mathematics and the effects of these beliefs on children's performance. Apparently, teachers continue to hold the stereotype that mathematics is a male domain, and evidence indicates that this teacher belief affects students' beliefs. Most research in this area has been conducted using Likert scales, although case-study research has also proven to be useful.

RESEARCH ON TEACHERS' AFFECT SINCE 1992

One can better understand the state of the research on teacher affect in 1992 by considering the intersection of McLeod's handbook chapter on affect and A. G. Thompson's handbook chapter on teachers' beliefs and conceptions. Neither author wrote about teachers' affect. In writing about affect, McLeod focused on students' affect when solving mathematics problems, and in her chapter, Thompson focused on teachers' beliefs and conceptions but not on their affect. McLeod called for researchers to begin to incorporate affect into research on cognition, but most of those responding have not focused specifically on teachers.

I begin by sharing some of the little research that has addressed teachers' affect as it relates to mathematics teaching and learning and then turn to work that addresses either affect in general or affect-related frameworks developed to describe children's experiences, because I contend that the frameworks may be applicable for work with prospective and practicing teachers learning or engaged in mathematics. But first, I address a more general question: Does affect matter in the learning of mathematics? I report two reviews of research that clearly indicate that, at least for students, affect is related to achievement.

Relationship Between Affect and Achievement

The relationship between affective factors and achievement appears to be important for researchers in mathematics education. Ma and Kishor (1997) conducted a meta-analysis of 113 studies to investigate this relationship between students' attitudes toward mathematics and their achievement in mathematics. Their results indicated an overall mean effect size of 0.12, which they concluded was "statistically significant but not strong for educational practice" (p. 39). They found that gender differences were too weak to

have practical implications and that whereas the relationship between attitude toward mathematics and achievement in mathematics was not strong at the elementary school level, it “may be strong enough for practical considerations at the secondary school level” (p. 40). They concluded that the junior high school level may be the most important period during which students shape their attitudes toward mathematics as these attitudes relate to students’ mathematics achievement and that by high school, many students’ attitudes are fixed or stable, which may result in their having less effect on or being less affected by achievement in mathematics. They also found a relationship between attitudes toward mathematics and the means by which researchers measured achievement in mathematics. The researchers found that attitudes toward mathematics could predict achievement measured using complex conceptual and procedural mathematics tasks, such as applications and problem solving, better than achievement measured using less mathematically complex tasks. They concluded that researchers should work toward substantially refining their measures for affect. They also called for studies in which researchers differentiate among students’ ability levels, take into account students’ grade levels and ethnicities, and examine the relationships between varying school characteristics (e.g., school size, school mean socioeconomic status, school policies) and mathematics achievement.

Ma (1999) conducted a meta-analysis of 26 studies narrowly focused on the relationship between anxiety toward mathematics and achievement in mathematics and reported a significant correlation of $-.27$. Although several definitions for *mathematics anxiety* were used by the authors of the studies analyzed, Ma drew upon McLeod’s (1992) definitions to conclude that *affect* was a general construct and *mathematics anxiety* was a distinct construct generally associated with the more intense feelings students exhibit in mathematics classrooms, including tension, helplessness, dislike, worry, and fear. Ma also summarized theoretical models put forth for the negative relationship between mathematics anxiety and mathematics achievement. According to one model, when arousal states are either below or above an optimal state, performance decreases. Under this inverted-U-curve model depicting a curvilinear relationship between anxiety and performance, some arousal is beneficial to performance, but when the arousal level increases too much, performance drops. Although this model is popular with arousal theorists, Ma found that most researchers tended to consider the relationship between arousal and achievement to be linear, with increased arousal associated with decreased achievement. Ma cited researchers who have

identified this linear relationship among children at the elementary and secondary level, among adults in general, and particularly among college students. A primary reason posited for the negative relationship between mathematics anxiety and mathematics achievement stems from the theory of test anxiety. Researchers who adhere to this notion regard mathematics anxiety “as a kind of subject-specific test anxiety” (Ma, 1999, p. 522). In his meta-analysis, Ma quantified the potential improvement in mathematics achievement when anxiety toward mathematics is reduced, reporting that when an average student who is highly anxious about mathematics is able to decrease that level of anxiety, improvement from the 50th to the 71st percentile in mathematics achievement may occur. Ma thought that this meta-analysis resulted in more definitive results than the study of the relationship between attitude and achievement (Ma & Kishor, 1997) because the instruments used to measure mathematics anxiety are more effective than those used to measure the more general construct of attitude, perhaps because “anxiety toward mathematics is easier to measure than attitude toward mathematics in that it is more operationally definable for researchers and more verbally expressible for students” (Ma, 1999, pp. 533–534). Although Ma identified several measures of mathematics anxiety, almost half the studies (12 of 26) used the same one—the Mathematics Anxiety Rating Scale (MARS) (Richardson & Suinn, 1972); he found that the research results were similar regardless of the anxiety-measurement instrument. As in the 1997 study, Ma found that the type of instrument the researchers used to measure mathematics achievement made a difference. Researchers who used standardized achievement tests tended to report weaker relationships than those who used researcher-made achievement tests or mathematics teachers’ grades. No gender differences were found in this study.

Mathematics Teachers’ Affect

The meta-analyses described previously indicated that affect in general and mathematics anxiety in particular impede students’ mathematical learning. Clearly, this is an important finding about students, but if the anxiety dissipates by the time these students enroll in college, we need not concern ourselves with mathematics anxiety of prospective or practicing teachers. Unfortunately, this dissipation does not occur, as the following studies indicated.

Harper and Daane (1998) investigated the causes of mathematics anxiety among prospective elementary school teachers enrolled in a U.S. mid-sized southeastern university before and after they completed

their mathematics methodology courses. Fifty-three prospective teachers enrolled in three methodology courses completed three assessments. Mathematics anxiety was assessed using the 98-item Mathematics Anxiety Rating Scale (MARS) (Richardson & Suinn, 1972) test as a pretest and a posttest. The MARS test is a self-rating Likert-scale comprised of items the respondent rates in terms of the degree of anxiety aroused. Items include "Thinking about beginning a math assignment," "Figuring out your monthly budget," and "Doing a word problem in algebra" (Bessant, 1995, pp. 334–335). Factors influencing mathematics anxiety (FIMA) were assessed with a 26-item checklist designed by the authors and used only as a pretest. Respondents completing the FIMA assessment were asked to either agree or disagree with each item on the assessment, and if they agreed, they then indicated whether they believed that the factor had caused them to experience any mathematics anxiety. Examples of items on this survey are "There was an emphasis on drill and practice," "I lacked an interest in math," and "I had embarrassing or negative experiences in past math classes." To further assess the influence of the methods course, the authors developed a 7-item methods-course reflection prospective teachers completed at the end of the course; it was designed to assess effects on students of (a) working with a partner, (b) working in cooperative groups, (c) working with small groups or in centers, (d) using manipulatives, (e) working on problem-solving activities, (f) writing about mathematics in journals, and (g) doing fieldwork in local schools. Finally, 11 of the 53 students were interviewed. Results indicated that mathematics anxiety persists in prospective elementary school teachers and that, often, the anxiety originated in elementary school. Causes for these students' mathematics anxiety included an emphasis on right answers and the right method, fear of making mistakes, insufficient time, and word problems and problem solving. The mathematics methods course led to decreased levels of mathematics anxiety for 44 of the 53 prospective teachers, but 9 of them experienced increased levels of mathematics anxiety during the course.

Mapolelo (1998) conducted a case study of one Botswanian preservice teacher's beliefs and attitudes toward mathematics. In Botswana, mathematics teaching was, at that time, procedurally oriented, and many students had difficulty with it in elementary school. These difficulties continued into teacher training to the extent that more than one third of the 490 prospective teachers at the teacher-training colleges failed the mathematics final examination. Mapolelo selected the case-study student because she expressed encountering difficulty in mathemat-

ics and had previous teaching experience. Dudu was 22 years old and an average student with "less than impressive overall class performance, particularly in mathematics" (p. 339) and two years of primary-level teaching experience in a school in a remote area. Dudu, like most of the preservice teachers, reported that she had had unpleasant experiences with mathematics at the junior and senior secondary levels. Results indicated that Dudu had been frustrated by an educational system in which memorizing facts and procedures without understanding the memorized procedures was emphasized. She came to view mathematics as something that was right or wrong, and she saw little place for reasoning when engaged in mathematics. Dudu was obsessed with failing examinations and attributed passing any examination to luck. Mapolelo concluded that to develop the better mathematics teachers needed in Botswana, mathematics educators needed to attend to the beliefs and affect of preservice teachers.

Philippou and Christou (1998) reported that Greek prospective teachers "brought very negative attitudes to Teacher Education" (p. 196). For example, 24% of the prospective teachers agreed with the statement "I detest mathematics and avoid using it at all times," and 62% agreed with the statement "I do not think mathematics is fun, but I have no real dislike of it" (p. 197). Efforts to change these attitudes were successful over the course of the program, during which prospective teachers completed two mathematics content courses based on the history of mathematics (with particular attention paid to the contributions of the Greek culture to mathematics) and a methods course. Attitudes that changed were the prospective teachers' satisfaction from mathematics and their views of its usefulness. The prospective teachers' deeply rooted anxieties about mathematics did not seem to change, and the researchers concluded that students' emotions are resistant to change.

Bessant (1995) studied attitudes of 173 Canadian university students enrolled in the introductory statistics courses in the mathematics, psychology, or sociology department. The author administered a 35-item mathematics-attitude scale the first week of class, followed a week later by a slightly reduced 80-item version of the Mathematics Anxiety Rating Scale (MARS) (Richardson & Suinn, 1972). He also administered a questionnaire designed to identify students' learning approaches. Using factor analysis, the author identified six factors that accounted for 54.9% of the total variance: general evaluation anxiety, everyday numerical anxiety, passive-observation anxiety, performance anxiety, mathematics-test anxiety, and problem-solving anxiety. The most im-

portant factors were general evaluation anxiety, referring to students' apprehension over the completion of mathematics-related tasks, and mathematics-test anxiety from thinking about a test one day, one hour, or five minutes before it began and waiting for the return of the test. The other factors were "statistically less significant, but they are of theoretical value" (p. 335).

In their study designed to investigate students' emotional responses to timed tests, Walen and Williams (2002) provided a deeper look into issues that might explain Bessant's findings related to the role of general test anxiety. In the only study that I found to establish a relationship between affect experienced by children and adults, Walen and Williams conducted two case studies of college students and one of a third-grade child. They built their work on Mandler's (1989) framework of emotion as resulting from cognitive analysis and physiological response. In particular, visceral arousals can be produced by some discrepancy or the interruption of some ongoing action, and these emotional responses can substantially affect cognitive functioning. One experiencing such stress tends to focus on aspects of a situation deemed important by that individual, but Walen and Williams (2002) pointed out that although this focus may appear to be helpful, often "what an individual considers important in a given context may not lead to a resolution of the perceived stress" (p. 364). Timed mathematics tests are, for some people, such a context.

Walen and Williams juxtaposed this notion of stress induced by interruptions or blockage of activity, an occurrence that varies from individual to individual, with the more culturally based view that people have of time and temporality. Whereas in the Western culture time is seen as "objective, continuous, universal, linear, and infinitely reducible" (2002, p. 364), many in *polychronic* cultures view time as more subjective and variable in speed. For such people, "time is not something life is measured against, but rather something that conforms to life" (p. 364). Walen and Williams cited the American Indian culture as including a view of time that often clashes with the Western view. For example, whereas in the American Indian culture, careful, slow, well-considered responses are rewarded, this approach does not serve well for one required to provide quick answers on timed tests.

Walen and Williams (2002) reported case studies of Pat and Jo, two women enrolled during different semesters in the same mathematics course for elementary school teachers, and a case study of Em, a third grader. Pat, of Asian descent, and Jo, of American Indian descent, were nontraditional students returning to school after spending time as mothers and home-

makers. Pat reported past experiences of failure that caused her to feel anxious and insecure about doing mathematics, but she also reported that she enjoyed mathematics. This seemingly contradictory comment left the instructor (the first author) unsure of Pat's meaning. During problem-solving, small-group activities, the instructor came to see Pat as confident and articulate and not at all anxious or insecure. The instructor was convinced that Pat was learning the mathematics well and so was baffled when Pat failed the first 1-hour exam. As a matter of practice, the instructor informally interviewed each student after each test; during her interview, Pat explained that she had felt that she was progressing satisfactorily on the test until she realized that she had only 20 minutes left to complete the exam and panicked. She explained, "My heart pounded. I saw spots. For me, the test was over. I turned in the exam and left." The instructor offered Pat the opportunity to take as much time as she needed on the subsequent test; Pat responded with great relief to the offer. On the next exam, Pat required an extra 30 minutes to complete the exam; the instructor noted, however, that she saw Pat begin to panic when she saw the first student turn in the exam. On the remaining exams, Pat needed no additional time, and she told the instructor, "When I have all the time I need, I don't even need the whole hour" (p. 369).

Jo's story is almost identical to Pat's. She appeared to understand in class, but during the first exam, the instructor noted that Jo "paled, then flushed, then broke out with beads of sweat" (Walen & Williams, 2002, p. 369). Although Jo assured the instructor that she was okay, the instructor did not believe her. Jo failed the exam, and she reflected later on her experiences taking tests in elementary school: "It was those tests . . . those tests that we took on multiplication. I'm so slow. I just couldn't do it. I understand, but I just can't do it when there's *time*" (p. 369). The instructor arranged for Jo to have extra time on subsequent tests, but Jo, like Pat, did not need it. When turning in the next exam, "she smiled and said, 'I did it. I did it in the hour. I don't need more time. Because I could have more, I didn't need it. I didn't even feel a bit scared'" (p. 370).

Em's story provides a glimpse into the possible development of Pat's and Jo's feelings of anxiety and fear about mathematics tests. Em was described as a bright, articulate third-grade girl who was "particularly successful in mathematics" (Walen & Williams, 2002, p. 370). She was mathematically curious, and she seemed conceptually oriented. However, she had difficulty memorizing her multiplication facts and responding to the 100 facts in 5 minutes, and this frustration left her "dreading every single Friday" (p. 370).

She seemed aware of other children in the class who also struggled with these multiplication timed tests, and she distinguished between one student who was not very good in mathematics and other students who, like her, were. Her teacher had told Em that she was aware that Em knew her facts but that these timed tests had to be administered because the other classes were taking them. The culminating experience for Em was a Friday on which only the students who had mastered all their tables could attend a banana-split party. Em described feeling humiliated by the prospect of being excluded and stayed home to have her own banana-split party. She said that had she gone to school that day but not attended the party, the other students would think "that I wasn't very good at mathematics" (p. 374).

In their analysis of these three case studies, Walen and Williams (2002) noted that these students experienced neither math anxiety nor even test anxiety; instead, their enemy was time. All three students were comfortable doing mathematics, and they

had trouble with timed mathematics tests precisely because they were willing to view mathematics as we wanted them to—as personal, meaningful, and contextual. It was not the mathematics, but rather the imposition of a foreign temporal structure over their mathematical knowing, that led to their discomfort and eventual paralysis in mathematical testing situations. (p. 375)

This research is particularly important for thinking about teachers' affect, not only because it provides two rich case studies of prospective teachers' affect but also because the researchers draw connections between the anxiety experienced by a child doing mathematics and that experienced by adults. Pat and Jo were both studying to be teachers. I am left wondering how many adult students like Pat and Jo are teaching in the classroom now? Worse yet, how many of them feel constrained to perpetuate practices that they recognize as harmful to students because that is what they believe is expected of them?

Summary of Studies of Elementary School Teachers' Mathematics Anxiety

Results of these studies indicate that mathematics anxiety among prospective elementary school teachers is of worldwide concern. Causes of mathematics anxiety have been identified as fear of failure, general test anxiety, and emphasis on right answers and right methods instead of on developing ways of reasoning about mathematics. Efforts to help prospective elementary school teachers reduce their mathematics anxiety are mixed, depending upon the approaches

taken. Simply providing more time during test taking was sufficient for some prospective elementary school teachers to overcome serious anxiety experienced during exams. Coursework promoting mathematical sense making in less rigid environments supported reduced mathematics anxiety among prospective elementary school teachers, although in one case the changes that occurred supported the students' sense of the usefulness of mathematics but not their deeply rooted anxieties toward mathematics.

Perhaps the most compelling finding is that the causes of negative affect toward mathematics or mathematics learning tend to go to prospective teachers' experiences as learners of mathematics. The process of preparing mathematics teachers starts long before prospective teachers take college courses; it begins when students enter kindergarten and first begin to learn mathematics in school. Therefore, I turn now to several studies that addressed affect, but not teacher affect. I consider these studies important because understanding the affect of K–12 students may help us understand the affect of our prospective and practicing teachers. Further, these studies all extend Mandler's widely used theory of affect.

Frameworks for Considering Affect: Extending Mandler's Framework

Hannula (2002) extended the work of Mandler (1989) by incorporating the influence of less intensive emotional states not addressed in Mandler's theory. Whereas Mandler focused upon interruptions that occur only while students are in the process of engaging in mathematics, Hannula argued that he considered students' reactions when they are thinking about mathematics, both generally and specifically. He presented a framework in which the mathematics-related emotions students experience were classified into four evaluative processes: (a) associations when generally thinking about the concept *mathematics*, (b) expectations when thinking about engaging in mathematics, (c) situational emotions when actually doing mathematics, and (d) values when considering the role of mathematics in relationship to other goals. For Hannula, each is a process that produces an expression of an evaluation of mathematics. The first evaluation is a nearly automatic association students make when, for example, they are asked how they feel about mathematics; it depends solely upon their prior experiences. Because it is not related to a particular context, it is an evaluation commonly assessed using questionnaires. The second evaluation is the most cognitive of the four; it goes deeper than the first because whereas for the first, students react generally to their image of

mathematics, for the second, students imagine a specific mathematical situation and the consequences involving some emotions that would follow. The third evaluation occurs when a student, while engaged in doing mathematics, engages in a continuous, often unconscious, evaluation of the situation with respect to her or his personal goals. This evaluation is represented as an emotion. The fourth evaluation is based on, often unconscious, cognitive analysis in which a student evaluates his or her life and the value given to different goals in it.

In a study of Finnish students in Grades 7–9, Hannula (2002) applied his framework to the case of Rita. At the beginning of the year, Rita held negative associations about mathematics in general and about word problems in particular. In spite of these negative associations and negative expectations, Rita exerted much effort when studying mathematics, a fact Hannula interpreted as a reflection of Rita's values related to mathematics. Hannula described experiences Rita had while learning mathematics in a group of three students; two goals she held during the first task were the cognitive goal of understanding the solution of the task and the social goal of effectively interacting with her two peers. Rita experienced frustration with the mathematics and with her feelings of being left out of the group discussions. Later when Rita began to think that she understood the mathematics, her attitude toward mathematics improved. Hannula found that Rita's evaluation of mathematics changed for the better "regardless which of the four evaluations (emotion, expectation, association, values) we consider" (p. 41), and he concluded that the proposed framework provided a useful way to describe attitudes and their change. For Hannula, this case study indicated that attitudes can change over a relatively short period of time and, finally, that Rita's negative attitude toward mathematics provided a "successful defence strategy of a positive self-concept" (p. 42).

Gómez-Chacón (2000) presented a model for the study of the interaction between cognition and affect in mathematics. She subdivided affect related to mathematics into two constructs, *local affect*, which referred to the states of change of feelings or emotional reactions one experiences while engaged in mathematical activity, and *global affect*, which is the concept of self and the beliefs about mathematics and learning one holds. The global affect results from "routes followed in the local affect" (p. 151); that is, global affect might be thought of as the beliefs and identity that one develops as a result of the feelings and emotional reactions one experiences while engaged in mathematics within the individual's sociocultural context. Fear of mathematics is an example of global affect. Gómez-

Chacón developed instruments for measuring the cognition-affect interaction, and she applied those instruments in a study of 23 cabinetmaking students in their late teens enrolled in a job-training workshop in Spain.

One instrument Gómez-Chacón (2000) developed was a *problem mood map*, in which codes are used by students to reflect their emotional reactions while engaged in mathematics. At the end of each mathematical activity, the students were asked to select from a list of emotions they might have experienced while working on the activity. The 14 emotions and descriptors offered, selected as a consensus of the most relevant emotional reactions the students in this study exhibited in the classrooms, were curiosity, cheerful, despair, calm, hurry, boredom, bewilderment/confused/puzzlement, brain teaser, liking, indifference, amusement, confidence, blocked, and *De abuty*, which is a Spanish term meaning "just great." Gómez-Chacón contended that students' coding of their moods while engaged in mathematics could foster awareness of their emotional reactions, and, as a result, they might feel more in control of their emotions; also, teachers could gather information about their students' affective reactions at different stages of a lesson.

In applying her framework to a student named Adrián, Gómez-Chacón (2000) concluded that using her instrument to assess both local and global affect was a valuable way to describe Adrián's affect. For example, although Adrián exhibited the global affective trait of fear toward mathematics, he exhibited a variety of local affective traits, including anxiety, satisfaction, and surprise, while solving problems. She also found that Adrián's negative emotions were associated with a lack of understanding of the mathematics. Gómez-Chacón concluded that although the model for affect put forth by Mandler (1989), whereby students experience affect because of interruptions in problem-solving situations, was valid, the model needed to be enhanced to account for both local and global affect.

Hannula (2002) and Gómez-Chacón (2000), like Mandler (1989), considered affective factors experienced by students while engaged in mathematical problem solving. However, they both believed that Mandler's framework needed to be expanded to consider additional factors. Hannula extended Mandler's framework by considering associations, expectations, and values one holds toward mathematics, and Gómez-Chacón extended the framework by considering the global affective characteristics. Both also found in their case studies that a student's affect was inversely correlated with the student's mathematical understanding.

In her dissertation, Malmivuori (2001) presented a theoretical analysis centered on affect but integrated with cognition and beliefs around particular social environments. She addressed, in addition to affect, beliefs, metacognition, self-regulation and self-perceptions, motivation, and the influence of context on all these factors; she built her framework upon constructivist and sociocognitive theories, and I contend that she attempts to synthesize constructs in such a way as to address teacher identity. Although considering all these elements simultaneously is difficult, rather than isolate these constructs, she consistently drew connections between and among them in an extraordinarily ambitious review, analysis, and reconceptualization based upon 600 references! She wrote that the main point of her study was

to constitute the theoretical and dynamic linkages between the often applied constructs and educational research results, as well as of the mathematics education research results with affect, that would also apply to and clarify self-regulated learning processes or the dynamic interplay of affect and cognition more generally. (p. 299)

Envisioning a study that utilizes her rich theoretical framework is difficult; still, her work is important for those rethinking affect, and Malmivuori's work is an important contribution to the future of research on affect.

Mathematical Intimacy, Integrity, and Meta-Affect

Goldin (2002a) expressed concern that research in mathematics education has focused primarily on cognition and far less on affect; he attributed this situation, in part, to the popular myth that mathematics is a purely intellectual endeavor in which emotion plays little role. Yet, he noted that, contrary to this myth, negative emotion is widespread in mathematics, and he recommended that further attention "be devoted to the psychology of developing effective affect in students" (Goldin, 1998, p. 154). According to Goldin, "When individuals are doing mathematics, the affective system is not merely auxiliary to cognition—it is central" (2002a, p. 60). He, like McLeod, subcategorized affect as emotions, attitudes, and beliefs, but he included a fourth category comprised of values, ethics, and morals. *Emotions* are rapidly changing states of feeling usually embedded in context. *Attitudes* are moderately stable predispositions toward ways of feeling in classes of situations that involve a balance of affect and cognition. *Beliefs* are internal representations to which the holder attributes truth,

validity, or applicability, and these representations are usually stable, highly cognitive, and highly structured. Also like McLeod, Goldin (2002a) viewed emotions as rapidly changing and highly affective; he viewed attitudes as more stable and cognitive than emotions, and he viewed beliefs as the most stable and cognitive of the three. The *values, ethics, and morals* comprising his fourth category are deeply held preferences that might be characterized as personal truths, are stable, may be highly structured, and are highly affective and highly cognitive.

Goldin, working with DeBellis, identified several other affective constructs related to mathematics; two of these constructs are *mathematical intimacy* and *mathematical integrity*. *Mathematical intimacy* is used to describe a psychological relationship between an individual and mathematics that connects with the individual's sense of and value of self. DeBellis and Goldin (1999) provided the example of a child, Jerome, experiencing difficulty solving a problem in which he moved 10 jelly beans from Jar 1 containing 100 green jelly beans into Jar 2 containing 100 orange jelly beans, and then, after mixing up the jelly beans, he moved 10 jelly beans from Jar 2 (now containing 100 orange and 10 green jelly beans) back to Jar 1. The child was puzzled to discover that there were the same number of the orange jelly beans in Jar 1 as green jelly beans in Jar 2, and this relationship stayed constant regardless of how many jelly beans he moved from Jar 1 if he moved the same number back to Jar 1. The mathematical intimacy of the child was reflected in the child's concern about, and involvement in, the problem.

Jerome's interactions, from which we infer intimate engagement, include his close proximity to the jelly beans when performing the experiments, his raised voice, his deep breaths, the gesture of brushing his hand through his hair, his shrugging of shoulders, his smiling, and the silent pauses. He sits back in his chair as if to push himself away from the experiment, to distance himself when the outcome contradicts his expectation. (p. 253)

Mathematical intimacy might be thought of as the extent to which a person working on a problem is engaged with the problem.

Mathematical integrity "describes an individual's affective psychological posture in relation to when mathematics is 'right,' when a problem is solved satisfactorily, when the learner's understanding is sufficient, or when mathematical achievement is deserving of respect or commendation" (DeBellis & Goldin, 1999, p. 253). Mathematical integrity is associated with a learner's desire to want to understand, with one's dis-

comfort toward inconsistencies, and with a focus on justifying one's reasoning. Often students confronted with tests or other evaluations are encouraged, by being awarded partial credit for giving responses even though they know that they do not understand, to bluff. This behavior, though it may lead to a higher test score, compromises the student's mathematical integrity because instead of focusing on whether the submitted work makes sense, the student focuses on the grade that results from bluffing.

DeBellis and Goldin (1999) contended that a problem solver who possesses both mathematical intimacy and mathematical integrity will experience more mathematical power and perseverance than one who lacks these attributes. They noted that a relationship exists between mathematical intimacy and mathematical integrity. Absence of integrity is a major obstacle to intimacy, because the experience of pretending or bluffing one's way through a problem not only blocks the individual's understanding of the problem but also prevents one from experiencing intimacy in relation to the mathematics. Conversely, if one lacks mathematical intimacy, the need for mathematical integrity is reduced.

Goldin (2002a) also proposed the construct of *meta-affect* to refer to "affect about affect, affect about and within cognition that may again be about affect, the monitoring of affect, and affect itself as monitoring" (p. 62). Goldin suggested that most affective experiences exist in a context that contributes to the interpretation of the affective experience and provided the example of a person riding a roller coaster because he or she seeks the experience of fear, which, in that context, is pleasurable. "The cognition that the person is 'really safe' on the roller coaster permits the fear to occur in a meta-affective context of excitement and joy" (p. 62). However, the experience can change entirely if, for example, the rider hears an unfamiliar noise and thinks that the roller coaster may be malfunctioning. The change in the rider's meta-affect creates a different experience for the rider, whether or not anything has actually changed. Goldin wrote, "It might seem that the 'cognitive' belief, that the ride is in fact safe, is the main essential to the joyful meta-affect. In this sense, the belief stabilizes the meta-affect" (p. 62). He explained that other beliefs and values play stabilizing roles for meta-affect. Applying this notion to mathematics might help explain how a particular experience, such as difficulty solving a mathematics problem, might be interpreted in different ways depending upon the beliefs and values held by the problem solver. Whereas one student might interpret the difficulty as an indication that he or she is a failure, another might view the difficulty

with anticipation for a feeling of satisfaction at the expected success. The ways students view the context surrounding the task, together with the beliefs or values held by the student, are integrally related to the interpretation of the experience. "Powerful affective representation that fosters mathematical success inheres not so much in the surface-level affect, as it does in the meta-affect" (p. 63).

Summary of Teachers' Affect

Because so little literature focuses on the intersection between teachers and affect, I have included studies of student affect. Goldin and DeBellis introduced the constructs of *mathematics intimacy*, *mathematical integrity* (DeBellis & Goldin, 1999), and *meta-affect* (Goldin, 2002a), and Hannula (2002) and Gómez-Chacón (2000) extended Mandler's theory by including *less intensive states* and *global states*, respectively. Malmivuori (2001) presented a framework centered on affect but integrated with cognition, metacognition, beliefs, self-regulation and self-perceptions, motivation, and the influence of context on all these factors.

Research indicates that attitudes are less amenable to change among students in high school (Ma & Kishor, 1997) and college (Hembree, 1990) than among those in elementary and middle grades. One might, therefore, reasonably assume that prospective and practicing teachers' mathematics affect or anxiety may have been formed when the teachers were precollege students, and the research supports this assumption. Although the view that the affect of students provides insight into the affect of teachers is a conjecture, evidence shows that similarities exist between adults' and children's feelings about mathematics.

I know of no research linking teachers' affect about mathematics and mathematics learning to their classroom instructional decisions, but the prospect that the two may be linked is sufficient reason to think more carefully about what can be done to support prospective and practicing teachers in developing more positive affect toward mathematics and mathematics teaching and learning. In an age of accountability, can one reasonably assume that teachers who enjoy mathematics will teach more of it, or teach it with a deeper sense of intimacy and integrity (DeBellis & Goldin, 1999)? Will those who have more positive affect toward mathematics be more positive in their approaches to the teaching of mathematics? If teachers are able to share enthusiasm for mathematics, will their students develop more positive impressions of mathematics?

Consideration of the relationship between fear of failure and mathematical learning is important in

the preparation of mathematics teachers. In-service work with practicing teachers generally does not include evaluative components, and some in-service providers attribute their success to the nonevaluative aspect of the experience. But prospective elementary school teachers generally must complete mathematics courses at the university level, and these courses are difficult for some students. The tension between the gatekeeper role of such courses and the affect promoted as a result of the students' concern seems to be worth additional consideration. For example, although we educators may believe that an important component of understanding mathematics is knowing when we do not understand something, students may feel compelled to hide their lack of understanding lest they receive an unsatisfactory grade and fail to meet the requirements for becoming teachers; students may learn that the best way to pass their courses is to hide their lack of understanding from the teacher, and possibly from themselves, thereby compromising the development of mathematical integrity (DeBellis & Goldin, 1999). Or, perhaps worse, if we help prospective teachers learn the mathematics that we consider important for them to learn but they feel that the experience was unpleasant and prefer to avoid mathematics, as practicing teachers, they may teach mathematics in a way that is consistent with their negative affect. Can a position be found between the evaluative, gatekeeper model generally applied with preservice teachers and the nonevaluative, collaborative model generally applied with practicing teachers, a position that would maintain the academic integrity required by universities but would also incorporate the long-range benefits valued by those engaged with in-service teachers' professional development? Even though providing professional development for practicing teachers and preparing preservice teachers raise different sets of issues, turning to professional development collaboratives that take as their primary goal the teaching of mathematics to practicing teachers may be useful as a context for considering innovative approaches to teaching mathematics to prospective teachers in ways that also foster prospective teachers' development of positive affect (see Sowder, this volume).

When we researchers in the field of mathematics education strive to make the field a scientific discipline, we draw heavily from other disciplines, such as psychology and sociology, gaining more theoretical perspectives in our field (*Educational Studies in Mathematics* [ESM], 2002). One result of a young and healthy discipline is that "it has become the norm rather than the exception for researchers to propose their own conceptual framework rather than adopt-

ing or refining an existing one in an explicit and disciplined way" (ESM, 2002 p. 253). The editors of *Educational Studies in Mathematics* (2002) and the *Journal of Mathematics Teacher Education* (Cooney, 2001) raised the possibility that the existence of these journals may have contributed to the proliferation of local theories that may not sufficiently push the field forward. This tendency to develop new conceptual frameworks seems strong in the area of research on affect in mathematics education; each study seems to introduce a new theory about affect. Mandler's theory provided a much-needed starting point around which others could build, and most researchers studying affect in mathematics education at least reference Mandler. We should keep in mind Mandler's (1989) caution that although a search for consensus on definitions of *affect* may be futile, we should take care in how we use the term. Defining our terms is certainly not a problem limited to the study of affect, as we saw in research on teachers' beliefs.

BEYOND BELIEF AND AFFECT?

Some researchers contend that teachers' beliefs should not be isolated but, instead, should be considered a component of more encompassing constructs. In this section, I look to such constructs, teachers' *orientations* and teachers' *perspectives*, and to a new area of research that holds possibilities for integrating research on teachers' cognitions, beliefs, values, and affect: communities of practice and teachers' identities.

Teachers' Orientations

In her handbook chapter, A. Thompson (1992) argued for the position that researchers ought not attempt to separate the study of teachers' beliefs from teachers' knowledge, and she used the term *conception* to refer to both. Later, she and colleagues (A. G. Thompson et al., 1994) used the term *orientation* to refer to teachers' views of mathematics and mathematics teaching. They introduced the terms *conceptual orientation* and *calculational orientation* to refer to the images teachers hold toward pedagogical tasks and the goals they served. Although teachers with conceptual and calculational orientations may agree upon the long-term goal of helping students develop problem-solving skills, their approaches differ in important ways. Actions of a teacher with a conceptual orientation are driven by an image of a system of ideas and ways of thinking she intends her students to develop; an image of how these ideas and ways of thinking can be

developed; ideas about features of materials, activities, and expositions and the students' engagement with them that can orient students' attention in productive ways; and an expectation and insistence that students be intellectually engaged in tasks and activities. Although a teacher with a calculational orientation may share the general view that solving problems is important, the actions of such a teacher are driven by a fundamental image of mathematics as the application of calculations and procedures for deriving numerical results. Associated with a calculational orientation is a tendency to speak exclusively in the language of number and numerical operations, a predisposition to cast solving a problem as producing a numerical solution, an emphasis on identifying and performing procedures, and a tendency to disregard context and to calculate upon any occasion to do so. Whereas a conceptually oriented teacher can identify the important concepts embedded in a problem, a calculationally oriented teacher tends to treat problem solving as flat; that is, nothing about the problem stands out, except that the answer is most important.

Conceptual and calculational orientations lead to differences in classroom discussions. Consider the problem "Susan drives 240 miles in 5 hours. What is the average speed of her trip?" The question "How did you solve this?" usually elicits from students the calculations they performed: "I divided 240 by 5, and the answer is 48." A teacher with a calculational orientation may consider this an appropriate answer, but a teacher with a conceptual orientation might question further to direct students toward the underlying conceptual ideas, the quantities in this problem, and the relationships among those quantities: "When you divided 240 by 5 and got 48, what is 48 a number of? That is, to what does 48 refer in this situation?" Whereas a calculationally oriented teacher might ask, "What did you do?" and mean "What calculation did you perform?", a conceptually oriented teacher might look for an explanation of the students' reasoning, not for the students' calculations, and ask, "What were you thinking?" or "What are you trying to find when you do this calculation (in the situation as you currently understand it)?"

Teachers who wish to become more conceptually oriented toward mathematics and mathematics teaching must reflect deeply about their goals for, and image of, mathematics and mathematics teaching. To shift their fundamental image from mathematics as doing things to mathematics as reasoning in particular ways is difficult. Furthermore, teachers who embark on such a journey frequently encounter their old patterns and goals, for example, in finishing a conceptual lesson by presenting the standard procedure

instead of having students use notation to represent the conceptual methods they produced for solving one or several problems. Too little textbook support for conceptual teaching is provided, and far too little support is available in terms of state frameworks for mathematics instruction.

A. G. Thompson and P. W. Thompson (1994, 1996) applied the constructs of conceptual and calculational orientations in analyzing 2 one-on-one teaching experiments in which different instructors worked with the same child. The first instructor, a mathematics teacher named Bill who possessed a strong and elaborate understanding of the concept of rate but who experienced difficulty in teaching the concept of rate to one of the stronger students in his sixth-grade class, acted like other good quantitative reasoners who are unable to use arithmetic to reason quantitatively.

[Bill had] come to use arithmetic in two ways simultaneously—as a representational system and as a formulaic system to express an evaluation. What we did not foresee was the shortcoming of this development in regard to teaching. Bill's quantitative conceptualizations appeared to be encapsulated in the language of numbers, operations, and procedures. He thus had no other means outside the language of mathematical symbolism and operations to express his conceptualizations. The language of arithmetic served him well as a personal representational system, or as a system for communicating with other competent quantitative reasoners. Yet, . . . that language served him poorly when trying to communicate with children who knew the tokens of his language but had not constructed the meanings and images that Bill had constructed to go along with them. (P. W. Thompson & Thompson, 1994, p. 300)

In the follow-up article, A. G. Thompson and Thompson (1996) described a subsequent lesson in which the same child worked with an instructor who had what the authors referred to as *knowledge for conceptual teaching*, comprised of "clear images of understanding a mathematical idea conceptually, how those images might be expressed in discourse, and what benefits might accrue to students by addressing the conceptual sources of their difficulties" (p. 3). In less than 30 minutes, the instructor had supported the child in her initial development of the concept of *speed* as the "mutual accrual of distance and time." The teacher who had struggled with the child on the previous day observed the lesson but failed to notice that a key aspect of the successful lesson was helping the child attend to the proportionality of corresponding quantities, distance and time, by attending to the segments representing each of these two quantities—even though he reasoned proportionally

to solve these problems himself. Teacher knowledge was not enough to support the child's reasoning, and only the second instructor's ability to coordinate his understanding of the content and the child's thinking and to use language that oriented the child toward the important underlying concepts enabled the child to develop the concept of rate.

The construct *orientation* as used by A. G. Thompson et al. (1994) incorporated teachers' knowledge, beliefs, and values about mathematics and mathematics teaching. In using the term, they described teachers' images, views, intentions, goals, and tendencies, and they operationalized the construct by attending to teachers' language and actions. Although they addressed more than beliefs, they did not explicitly address affective issues. In the next section I describe an attempt to expand the investigation of teachers' beliefs to include affective factors.

Teachers' Perspectives

When studying teachers, researchers apply their conceptual lenses to interpret teachers' knowledge, beliefs, and practices. Simon and Tzur (1999) highlighted two approaches taken by researchers studying teachers. Some researchers apply their conceptual frameworks, which encompass the current knowledge from the field, to tell their own stories about teachers, providing, for example, deficit studies focused primarily on what teachers lack; the teachers' perspectives, aspects of the teachers' practices that are critical to the teacher, are generally missing from the story. In the other approach, the researcher tells the story from the teacher's perspective, but the researcher's conceptual framework may be missing, and consequently the story may not add to the theoretical development of the field. Simon and Tzur (1999), in their methodology for studying mathematics teacher development, focus upon the teacher's approach by using their conceptual framework to provide accounts of the teacher's perspective. A key principle guiding their work is that "every teacher's approach is rational and coherent from his or her perspective" (p. 261) and that the job of the researcher is to understand the teacher's underlying perspectives that might coherently account for the approach without separating aspects of that teacher's perspectives into disparate components, such as beliefs, knowledge, and methods of questioning. They have contended that the term *teacher's practice* includes not only everything the teacher does that contributes to his or her teaching, including planning, assessing, or interacting with students, but also the teacher's values, skills, intuitions, and feelings about those aspects of the practice. These

epistemological commitments are associated with particular methodological commitments, including the view that one cannot assess the relationships among teachers' beliefs, knowledge, values, intuitions, feelings, and practices without gathering rich data. They refer to the unit of analysis for their qualitative work as a *set*, comprised of at least two consecutive related mathematics lessons and interviews before, between, and after the lessons.

Applying this methodological framework to study a group of teachers engaged in mathematics education reforms, they (Simon, Tzur, Heinz, Kinzel, & Smith, 2000; Tzur, Simon, Heinz, & Kinzel, 2001) considered these teachers in transition in terms of two perspectives commonly referred to in the literature: conception-based perspectives and traditional school-mathematics perspectives. The authors referred to *conception-based perspectives* as emergent and constructivist perspectives based on three assumptions: (a) Mathematics is created through human activity, and humans have no access to a mathematics that is independent of their ways of knowing; (b) individuals' currently held conceptions constrain and afford what they see, understand, and learn; and (c) mathematical learning is a process of transforming one's knowing and ways of acting. They contrasted this perspective with the *traditional school-mathematics perspective*, in which mathematics is viewed as existing independently from human experience and students are believed to passively receive mathematical knowledge by listening to and watching others. Teachers holding the perception-based perspective behave as if the mathematical relationships were properties of the objects being considered instead of as a function of the knowledge of the perceiver. Simon et al. (2000) provided a metaphor to capture the difference between the perception-based and conception-based perspectives:

We think of *perception* as looking through a lens. The lens represents what the perceivers bring to the situation, which structures their perception. Perceivers who hold a perception-based perspective do not consider that there are lenses affecting what they see. Rather, they assume that what they see is what is "out there," that what is out there enters *as is* through their senses. . . . In contrast, perceivers who hold a conception-based perspective consider that they can see only that that results from looking through particular lenses and that they have no way to compare their perceptions (through their lenses) with lens-free perceptions. As a consequence, they make no claims as to what is out there, and they attempt to understand what is perceived as a contribution to understanding the perceiver. The analogy serves to emphasize that those who we infer hold a perception-based perspective are not rejecting a

conception-based perspective. They have never considered such a perspective. (p. 594)

The researchers found that neither perspective adequately characterized the views of these teachers in transition because the teachers did not view children as passively receiving mathematical knowledge, but, although they viewed mathematics as an interconnected and understandable body of knowledge, they also viewed mathematics as existing independently from human activity, and therefore “accessible *as is* to all learners” (Simon et al., 2000, p. 593).

Communities of Practice and Teacher Identity

When learning theorists evolved during the twentieth century beyond the stimulus-response of behaviorism to consider the contents of people’s minds, their research focus changed from teachers’ actions to understanding of teachers’ thought processes, including their planning, their interactive thoughts and decisions, and their theories and beliefs (Clark & Peterson, 1986). The preponderance of research on mathematics teachers’ beliefs is focused upon understanding teachers’ beliefs, investigating the relationship between teachers’ beliefs and practices, and changing teachers’ beliefs. However, even before the publication of the first NCTM *Handbook* (Grouws, 1992), researchers on learning questioned a focus solely upon the contents of a person’s mind without attending to the contexts in which learning occurred (Brown et al., 1989; Lave, 1988). By the early 1990s, during further evolution, some mathematics educators had adopted sociocultural perspectives, and a related theoretical framework that was to affect research on teachers’ professional development, *communities of practice* (see Sowder, this volume), emerged.

Lave and Wenger (1991) suggested that by shifting their focus from the individual learner to groups of learners, they could account for the learning that results from changing the way one participates with others who share common goals. Wenger (1998) further elaborated the relationship between a social community’s practices and individuals’ identity construction resulting from participating in these communities. Many researchers studying teacher professional development have adopted the *communities-of-practice* framework and the construct of *identity*; I provide three examples of research in which mathematics educators applied this theoretical lens to their work. Two examples are drawn from major professional development projects based around reform-oriented approaches to teaching and learning mathematics. In

the CGI project (Carpenter et al., 1999) elementary school teachers learned about children’s mathematical thinking as a means for changing their beliefs and their practices; the QUASAR project (Silver & Stein, 1996) focused upon inner-city middle school reform through professional development of teachers. The third example is work of Forman and Ansell (2001) with a single teacher.

In work related to the CGI project, Kazemi and Franke (2004) applied a transformation-of-participation framework to describe the learning of a group of teachers engaged in professional development focused on children’s mathematical thinking. Building upon Wenger’s (1998) framework, they studied ways in which the participation of the group of teachers changed over time, because “shifts in participation do not merely mark changes in activity or behavior. Shifts in participation involved a transformation of roles and the crafting of new identities” (Kazemi & Franke, 2004, p. 205). Their contribution in writing about this work was “to provide an analytic frame for understanding teacher learning as shifts in participation” (p. 206). They met with 10 cross-grade teachers at an inner-city elementary school, after school once a month throughout the year, to discuss students’ mathematical thinking. Teachers adapted common problems for their students and brought student-work examples to discuss at each meeting. The study covered the first year of the teachers’ participation; data analyzed were seven workgroup-meeting transcripts from audio recordings, written teacher reflections, copies of student work shared by the teachers, and end-of-the-year teacher interviews. The authors analyzed their notes and the transcripts to understand how the teachers talked about student work, noting the mathematical and pedagogical issues that were raised, and identified other descriptive themes that consistently emerged across the year, such as teachers’ sharing successes from their practices or teachers’ generating questions about their practices.

Kazemi and Franke (2004) noted two major shifts in teachers’ workgroup participation. First, early in the year, teachers new at eliciting children’s thinking tended to underestimate the children’s abilities and reported students’ unsuccessful attempts to solve problems, but by the third session, they had begun to interact with their students about their strategies and expressed amazement at their students’ innovative strategies. The teachers helped one another by suggesting ways they might select students to share strategies so as to observe something identified as interesting.

Second, the teachers began to notice mathematical issues related to place value in the children’s

strategies; these issues led them to reconsider their mathematical goals for their students. Some teachers came to redefine place value less in terms of students' memorizing the names of units (ones, tens, hundreds) and more related to their using groups of 10 to solve problems (e.g., to find the difference between 48 and 111, one teacher shared that two of her students reasoned that $48 + 60$ is 108, and 3 more is 111, so the answer is 63.)

Kazemi and Franke (2004), although noting the importance of traditional components of studies of teacher change, explained that the framework they had applied enabled them to focus upon shifts in the participation of the study-group teachers:

We do not argue that examining individual teachers' developing knowledge and beliefs is unimportant. In fact, these are key resources for a developing community. However, we believe that by attending to shifts in participation, we can understand the following aspects of teacher learning: (a) how teachers working together supported the development of each other's thinking and the practices they used in their classrooms; (b) how and when teachers asked each other for help and contributed to discussions in the workgroup because of their own experimentation in the classroom; and (c) how teachers looked at the strategies students in other classrooms used and then used those as markers for what to expect from their students. (p. 231)

Stein and Brown (1997) drew upon sociocultural theories to explain teacher learning in the QUASAR project, suggesting that the "location of the phenomenon of learning" (p. 159) changed from being "located in the cognitive structures and mental representations of individual teachers" (p. 159) to being "situated in the fields of social interactions between and among individuals" (p. 159). Stein and Brown explained that when the unit of analysis shifted from the individual to the group, teacher change was defined with respect to the changing roles of the teachers as transformation of participation. For them, the communities-of-practice framework, found useful by many, lacked specificity as an analytic tool. To apply the framework in their study, Stein and Brown (1997) examined the range of the community's work practices and determined that the breadth of teachers' participation increased to include working on portfolio systems, planning curriculum, involvement with parents, making presentations to others, and working on articulation issues across other programs. Further they observed how teachers' depth of participation, as measured by movement from peripheral to more central roles, changed. For example, participants moved from peripheral involve-

ment in "old-timer" colleagues' presentations to giving mini-presentations in the local community or copresenting to external audiences to the point of giving presentations on their own. Stein and Brown concluded that the community-of-practice framework provided a lens for viewing collaboration as a source for teacher learning and development.

In the same paper, Stein and Brown (1997) reported on a QUASAR site where a reform community with common goals and joint productive activity had not become established. The community-of-practice perspective was inappropriate for explaining teacher learning in this situation, so the authors relied instead upon Tharp and Gallimore's (1988, cited in Stein & Brown, 1997) *assisted-performance perspective*.

Stein and Brown (1997) concluded that "studies of teacher change in mathematics tend to overlook noncognitive variables such as affect or motivation for change" (p. 186). They found that the sociocultural frameworks they applied enabled them to "situate the learning of new knowledge and skills in the context of the larger meaning-making capacities and affective characteristics of individuals" (pp. 186–187) and, thus, to "integrate the study of cognitive change with the examination of goals, motivations, and identity" (p. 187).

Forman and Ansell (2001), like the authors of the previous two studies, drew from the communities-of-practice framework to view learning as a social activity, but their focus was on one teacher, Mrs. Porter, not a group of teachers. To support their theoretical framework, they adopted a methodological approach in which their unit of analysis bridged the individual and the social. Initially, in a top-down approach, they sampled major events in a third-grade mathematics classroom over several months before making a more fine-grained analysis of classroom-discourse patterns. They were surprised to find that the nature of the talk when the class was discussing standard strategies differed from their talk about student-invented strategies. During discussions of student-invented strategies, the teacher "emphasized sense making, risk-taking, persistence, being logical and finding increasingly efficient strategies" (p. 131), but in discussions of standard algorithms, she spoke in a different voice, emphasizing the confused and complicated nature of standard algorithms. Forman and Ansell (2001) traced these contrasting voices to personal experiences of Mrs. Porter. Notice the central role of teacher affect in their comments:

The voice of her earliest experiences (both as a student and a teacher) spoke of alienation, low self-confidence, passive acceptance of authority, lack of initia-

tive, and a sense of incompetence and confusion, especially in mathematics. This first voice seemed to be connected, in part, to an instructional approach that relied upon students memorizing algorithms without understanding why they work . . . [and] that ignored or discounted students' own attempts to make sense of their strategies for solving problems. The voice of her more recent experiences spoke of excitement, the enjoyment of learning from others through discussion, risk-taking, a passionate commitment to fostering students' sense-making, and an intense involvement in helping her students express their ideas and reflect on them. This voice seemed associated with an instructional approach that emphasized careful attention to students' different ways of solving problems and their informal knowledge base. Both of these voices also seemed to be linked to the broader social and institutional contexts of her classroom. (p. 134)

What is an *identity*? Collopy (2003) wrote, "A teacher's identity is the constellation of interconnected beliefs and knowledge about subject matter, teaching, and learning as well as personal self-efficacy and orientation toward work and change" (p. 289). Van Zoest and Bohl (2005) also viewed *teachers' identities* as embodying their knowledge, beliefs, commitments, and intentions, but they included, also, all the ways teachers have learned to think, act, and interact. For Van Zoest and Bohl, a teacher may hold many identities, but a given identity is not context specific—individuals carry it with them when they move from context to context. Wenger's (1998) view of identity cannot be captured in a brief definition; he devoted one third of his book to discussing it, so I state what identity is not for Wenger instead of what it is:

[Identity] is not equivalent to a self-image; it is not, in its essence, discursive or reflective. We often think about our identities as self-images because we talk about ourselves and each other—and even think about ourselves and each other—in words. These words are important, no doubt, but they are not the full, lived experience of engagement in practice. (p. 151)

Sfard and Prusak (2005) took issue with Wenger's notion that "there is something beyond one's actions that stays the same when the actions occur, and also that there is a thing beyond discourse that remains unchanged, whoever is talking about it" (p. 16). They sought a definition of *identity* that is "operational, immune to undesirable connotations, and in tune with the claim about identities as man-made and collectively shaped rather than given" (p. 16). They thought of identity as narratives that are told by an author, about an identified person, to a recipient; the author, identified person, and recipient may be different people or

the same person. They distinguished between *actual identities*, consisting of stories about the actual state of affairs (e.g., "I am an army officer" or "I am a good driver"), and *designated identities*, consisting of narratives presenting an anticipated state of affairs (e.g., "I want to be a doctor" or "I have to be a better person"). Sfard and Prusak argued that identities are crucial to learning because they may act as self-fulfilling prophecies, and they suggested that this perspective on identity might serve as the missing link between learning and sociocultural context.

Summary

Researchers have considered constructs related to beliefs, with *orientation* (A. G. Thompson et al., 1994), *perspective* (Simon & Tzur, 1999), and *identity* (Wenger, 1998) serving as three such terms. An important issue for the future of research on teachers' beliefs is how researchers negotiate between constructivist and sociocultural perspectives. For example, Rogoff (1997) argued that the assumptions underlying the theoretical perspectives that explain how individuals acquire knowledge are fundamentally different from the assumptions underlying the transformation-of-participation perspective associated with focusing upon groups of people, and she argued that because no individual exists in isolation or out of cultural context, all learning, even reading a book, is social and should be seen from this perspective. Cobb (1995) presented the position that psychological and sociological perspectives should be viewed together:

It is tempting to respond to the conflicting assumptions of the sociocultural and constructivist perspectives by claiming that one side or the other has got things right. However, I will instead argue that the two perspectives evolved to address different problems and issues and they are complementary in several respects. (p. 379)

P. Thompson (in P. Thompson & Cobb, 1998) agreed that psychological and social perspectives depend upon each other, but he chose to view the psychological perspective as more fundamental because "it aligns more explicitly with what I take as our fundamental goal of making a positive, lasting difference in students' lives after they leave our classrooms" (p. 19).

Lerman (1998, cited in Van Zoest & Bohl, 2005) suggested that in choosing a unit of analysis for educational research, one must be able to zoom in and out, changing focus to account for the full spectrum of locations of cognitive development. P. Thompson and Cobb (1998) applied the same metaphor, arguing, "To achieve a unification of psychological and social perspectives would mean that we become able

to 'zoom' out or in with respect to a set of problems without losing sight of where we started" (p. 4).

Researchers who take a psychological approach by focusing on an individual teacher's beliefs will continue to inform the field as will those who take a sociocultural approach, by, for example, focusing on communities of practice. We in the field must balance these approaches to gather data useful for answering the types of questions driving our work. Furthermore, when we become comfortable with both approaches, the questions we consider asking will change.

SOME THEMES AND FINAL THOUGHTS

Changing Beliefs, Infusing Affect

One noteworthy difference between research on teachers' beliefs and affect is that whereas research on teachers' beliefs has been extensive and subsumed into almost all areas of research on mathematics teaching and learning, the study of teachers' affect has not. Mathematics educators generally agree on what beliefs are; we now face a greater challenge than defining *beliefs*: how to change teachers' beliefs. If beliefs are lenses through which we humans view the world, then the beliefs we hold filter what we see; yet what we see also affects our beliefs—creating a quandary: How do mathematics educators change teachers' beliefs by providing practice-based evidence if teachers cannot see what they do not already believe? The essential ingredient for solving this conundrum is reflection upon practice. When practicing teachers have opportunities to reflect upon innovative reform-oriented curricula they are using, upon their own students' mathematical thinking, or upon other aspects of their practices, their beliefs and practices change. Furthermore, although prospective teachers seldom are embedded in practice-based environments, when they are provided opportunities to learn about students' mathematical thinking and reflect upon the experiences, their beliefs change.

Researchers have found that for some teachers, beliefs change before practice, whereas for others, changes in practice precede changes in belief. I suspect that ultimately research will show that the most meaningful changes take place when teachers' beliefs and practices change together, but additional research in this area will lead to a better understanding of the relationship between teachers' changing beliefs and practices.

Judging by the attention to teachers' affect in the mathematics education literature, one might con-

clude that teachers' affect is not nearly as important as teachers' beliefs. Yet teachers' affect is critically important! If prospective or practicing teachers are to develop deeper content knowledge and richer beliefs about mathematics, teaching, and learning, then positive affect must be considered. I have asked several groups of prospective secondary mathematics teachers, the majority of whom had previously completed their undergraduate mathematics major, to define *mathematical proficiency*. Generally they talk about learning procedures, understanding concepts, and problem solving; a smaller number mention proofs or reasoning. Few students mention affective issues. The definition of *mathematical proficiency* presented in the consensus document *Adding It Up* (NRC, 2001) includes five interrelated strands that, together, comprise proficiency. The first four are *conceptual understanding*, *procedural fluency*, *strategic competence* (the ability to formulate, represent, and solve mathematical problems), and *adaptive reasoning* (the capacity to think logically and to informally and formally justify one's reasoning). The fifth strand, *productive disposition*, is "the tendency to see sense in mathematics, to perceive it as both useful and worthwhile, to believe that steady effort in learning mathematics pays off, and to see oneself as an effective learner and doer of mathematics" (NRC, 2001, p. 131). The inclusion of confidence in the ability to learn mathematics as a component of mathematical proficiency is important as a statement that proficiency in mathematics has affective aspects. Teachers make important decisions about the manner in which they teach mathematics, and elementary school teachers often decide how much time to devote each day to mathematics. If the mathematics courses we educators offer to prospective teachers address the first four strands of mathematical proficiency without addressing mathematical disposition, I suspect that we will continue to produce many teachers who lack the positive dispositions associated with creating positive mathematics-learning experiences for students. Although secondary school teachers generally have more positive affect than elementary school teachers toward mathematics, we must help secondary school teachers to consider not only what mathematics they are teaching but also the experiences they create for their students.

We can support prospective and practicing teachers by helping them attend to their own affect toward mathematics. Do they suffer from mathematics anxiety? Are they affected by timed tests? When they experience what Mandler referred to as *interrupted plans*, how do they interpret the experiences? Do they have experiences of mathematical intimacy? Do they strive for *mathematical integrity*, that is, the ability to

know whether they understand—or have years of being falsely assured that they understand mathematics undermined their abilities to view their own mathematical understanding objectively? When they publicly reflect upon their own experiences, they may also reflect upon the experiences of their colleagues, and in so doing, they may begin to understand the wide range of experience represented by their colleagues and, thus, by the students they one day will teach.

Constructs, Measurement, and to the Future

As researchers, we do not study beliefs or affect in general; we study them in context. Researchers studying teachers' beliefs or affect are generally careful to make explicit what beliefs or affect they are studying, and this practice has facilitated convergence upon common definitions in the field. Many researchers view *knowledge* as belief with certainty and *values* as deeply held, even cherished, beliefs. Researchers studying affect tend to include beliefs as a component of affect, but the majority of research on teachers' beliefs has not included affect.

Ma (1999) found more definitive results in a meta-analysis about mathematics anxiety than he and a colleague found in a meta-analysis about attitudes toward mathematics (Ma & Kishor, 1997), and he attributed the difference to the fact that researchers have more effectively operationalized instruments for measuring mathematics anxiety than for measuring the general construct of attitude (Ma, 1999). Clearly, a field of research is extended by researchers' applying careful definitions of their constructs. But so too is a field extended when researchers consider emergent constructs that encapsulate new ways to consider old constructs. The epistemological commitments entailed by considering teachers' conceptions, perspectives, orientations, identities, or other general constructs that subsume beliefs or affect are associated with particular methodological commitments. For example, Simon and Tzur (1999) argued that studying teachers' perspectives involves the investigation of the relationships among teachers' beliefs, knowledge, values, intuitions, feelings, and practices, and to gather such rich data required a unit of analysis referred to as a *set*, comprised of observation of at least two consecutive related mathematics lessons and interviews conducted before, between, and after the lessons. In-depth qualitative research, in which data sources are triangulated using a variety of tools, is important for theory building, but for theory testing, researchers need methodologies that can be used with large numbers of subjects. Self-report data are widely used for measuring affect and beliefs of prospective and

practicing teachers because such measures are easy to administer and score. However, because people may, at times, be unaware of their beliefs (Furinghetti & Pehkonen, 2002) and may offer opinions on matters about which they have given little thought (McGuire, 1969), Likert scales are of limited use. One promising alternative for assessing beliefs and affect is to employ internet technologies, such as video streaming and data downloading, in developing surveys that require open-ended responses to provide data richer than those gathered in Likert-scale instruments (Integrating Mathematics and Pedagogy, 2003).

We mathematics education researchers must continue to pursue fundamental theoretical questions that move our field forward, often through qualitative research. However, education is a political endeavor, and the past 10 years have found the mathematics education community embroiled in a political struggle for relevance. Until we produce more quantitative studies designed to test theories, too many people in positions of power will continue to ignore our research.

I accepted the task of reviewing the literature on teachers' beliefs and affect for this volume, and although teachers' beliefs and affect are closely related, I was unable to integrate the review of these two areas. Perhaps in the future, teachers' beliefs and affect will converge in such a way as to make an integration more natural. To review all the research on teachers' beliefs and affect since the publication of the last NCTM *Handbook* (Grouws, 1992) is beyond the scope of a single chapter, and with the increases in the size of the field and the number of publication outlets, I expect that for a third handbook, the amount of research to summarize will be far greater than the current body. What would such a volume contain—separate chapters on teachers' knowledge, teachers' beliefs and affect, teacher education and professional development, and teachers and teaching? Might all these chapters be integrated into one chapter under a common theme, say, teacher identity?

Teachers know; teachers believe; teachers feel; teachers participate; teachers belong. The choice researchers make about which constructs to apply in their work is one of the most important research decisions made. With new constructs come new questions and new methodologies. One way to think about how new constructs emerge and fit into existing constructs is to turn to Bernstein (1999, cited in Lerman, 2002), who distinguished vertical knowledge structures from horizontal knowledge structures. Science provides an example of a *vertical knowledge structure* in which each new theory can subsume its predecessors, whereas mathematics education is an example of a *horizontal knowledge structure* because new theories sit alongside

their predecessors. New theories come with their own language and epistemological commitments, and, as such, they cannot be used to refute other theories, but instead they introduce different questions and new methodologies for answering these questions. The research on teachers' beliefs and affect has grown from psychological theories of learning that focused upon the contents of the minds of individuals, and the field has now adopted sociocultural theories through which researchers look at the world anew. I expect that the newer theories will neither replace nor subsume the old theories but, instead, will sit alongside them. Researchers will continue to study emerging sociocultural constructs, such as teacher identity, that include teachers' knowledge, beliefs, affect, and more. But researchers will also continue to study these components in isolation, and I think that this work will continue to be important.

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■ ■ P A R T I I I ■ ■

Influences on Student Outcomes

HOW CURRICULUM INFLUENCES STUDENT LEARNING

Mary Kay Stein

UNIVERSITY OF PITTSBURGH

Janine Remillard

UNIVERSITY OF PENNSYLVANIA

Margaret S. Smith

UNIVERSITY OF PITTSBURGH

The purpose of this chapter is to review research on how mathematics curriculum¹ influences student learning. Interestingly, there was not a similarly named chapter in the 1992 *Handbook of Research on Mathematics Teaching and Learning* (Grouws, 1992), suggesting that the mathematics curricula of the '70s and '80s and their relationship to student learning were not viewed as a significant object of scholarly investigation at the time. Most likely, this stance reflected the era more broadly, a time during which arithmetic—primarily computation—comprised the lion's share of work in the elementary years, and algebra—principally procedures for manipulating symbolic expressions—formed the majority of high school work in mathematics. In this milieu, textbooks were viewed primarily as a resource for problem sets:

“Very few had references to mathematical concepts or principles and virtually none had problems from everyday life or from other fields” (Senk & Thompson, 2003 p. 9). The absence of a chapter on curriculum in the 1992 *Handbook* may also reflect the period in time—a time during which reformers focused on helping teachers create lessons based on activities found *outside* of commercial textbooks.² This tendency may have reflected a general disillusionment among mathematics educators with the textbooks available at the time most of which were developed by publishing companies and did not draw on research on how students learn mathematics (Ball & Feiman-Nemser, 1988).

Current interest in how curriculum materials influence student learning has been fomented by sev-

¹ As described later, we use the term curriculum broadly to include mathematics curriculum materials and textbooks, curriculum goals as intended by teachers, and the curriculum that is enacted in the classroom.

² Examples include Cognitively Guided Instruction (Carpenter, Fennema, Peterson, Chiang, & Loef, 1989), replacement units, and manipulative-based programs.

eral policy shifts in mathematics education and educational research and practice more generally. First, the field has seen a sharp rise in curriculum development activity since the writing of the 1992 Handbook chapter—much of it in response to publication of the *Curriculum and Evaluation Standards for School Mathematics* (National Council of Teachers of Mathematics, 1989). As states adopted standards that reflected the NCTM's vision, the publishing industry moved quickly to make adaptations to their textbooks. At the same time, the National Science Foundation (NSF) funded extensive curriculum development projects that produced an entirely new set of curriculum materials that entered the market in the mid- to late-nineties (referred to as “standards-based curricula”). This activity dwarfed even the explosion of curriculum development that occurred during the new math era of the late '50s and '60s (Usiskin, 1997).

These materials have generated particular interest of researchers because they embody an approach to mathematics teaching and learning that is qualitatively different from textbooks or instructional resources previously available. The standards-based curricula had a common set of design specifications that included alignment with the *Curriculum and Evaluations Standards for School Mathematics* (NCTM, 1989), heavy use of non-numeric representations (e.g., diagrams, manipulatives), an expanded content base (e.g., in elementary curricula, topics that go beyond traditional arithmetic to include statistics and graphing, geometry, and pre-cursors to algebraic reasoning), and extensive use of calculators. Moreover, in order to meet new goals for student learning (i.e., mathematical thinking, reasoning, problem solving, connecting, communicating, seeking evidence, and constructing arguments to make predictions and support conclusions), these new curricula de-emphasized paper-and-pencil skills and focused on students' active construction of and communication about solutions to challenging problems. A subset of these curricula also incorporated technologies (e.g., intelligent tutors) that were not previously available.

The kind of instructional experience supported by the standards-based curricula represented a substantial departure from conventional practice in the U.S. As summarized by Fey (1979) from three studies commissioned by the National Science Foundation (Stake & Easley, 1978; Suydam & Osborne, 1977; Weiss, 1978) and corroborated by the recent TIMSS study (NCES, 2003; Stigler & Hiebert, 2004), the majority of mathematics lessons in the U.S. involved extensive, teacher-directed explanation of new material followed by student seatwork on paper-and-pencil assignments with little or no discussion or exploration of concepts. As

such, reformers cast these new curricula in the role of change agent with the hope that they would help teachers to transform their instruction from a focus on basic skills to conceptually based problems that require thinking, reasoning, problem solving and communication (Senk & Thompson, 2003). To others, however, these new curricula represented a swing of the pendulum too far toward an emphasis on the processes of doing mathematics to the detriment of time spent developing computational efficiency (Wu, 1997).

The standards-based era of curriculum development has been associated with increased vigor in the mathematics education community and an upsurge in research activity. Because the new curriculum materials and the standards documents on which they were based, offered a radically different vision of what it means to learn mathematics and consequently how it should be taught in the classroom, they prompted a substantial amount of research on the role of teachers in the standards-based classroom, as well as research on the relationship between teachers' beliefs and knowledge and their instructional practice. All of these studies are critical to the central argument of this chapter and our response to its driving question—How do curriculum materials influence student learning? As we detail below, the influence of curriculum materials is mediated by teachers and students interacting with those materials in classroom contexts.

A second policy shift, occurring in the late 1990s and early 2000s, catalyzed another research emphasis that raised different kinds of questions about the influence of curriculum materials on student learning: “Do these new materials work?” Because the passage of the *No Child Left Behind Act* restricted the use of federal monies to those programs backed by scientific evidence of student learning (NCLB, 2002), many curriculum developers became eager to “prove” the effectiveness of their materials. In addition, harsh criticism of education research and educational practices as not scientifically based led to calls for research on the effectiveness of educational programs in general, including innovative curricula such as the NSF-sponsored materials (Mosteller & Boruch, 2002; NRC, 2002; 2004). Finally, in response to perceived needs of practitioners to distinguish effective from non-effective programs, the US Department of Education established the *What Works Clearinghouse*. This Internet-based website features the results of research, that qualifies as scientifically based, on the outcomes of educational programs and curriculum materials, as measured by gains in student achievement. This climate of accountability led to a substantial number of studies that aimed to discern what students who were exposed to different kinds of curriculum materials learned. In

this chapter, we review both these effectiveness studies and the broader field of research on how teachers and students use curricula.

CONCEPTUAL ISSUES, DEFINITIONS, AND BOUNDARIES

The term *curriculum* has different meanings in different contexts. Therefore, we begin with a brief discussion of the term and its varied applications in order to clarify how we are using it in this chapter. In so doing, we present the framework that structures our discussion. We also discuss how the construct of *curriculum materials* or *textbooks* has evolved and shifted in educational discourse and, particularly in mathematics education. Finally, we specify the boundaries of our review.

Multiple Meanings of Curriculum

Very broadly, curriculum refers to the substance or content of teaching and learning—the “what” of teaching and learning (as distinguished from the “how” of teaching). Those who study curriculum frequently examine planned and unplanned components of what is taught or experienced in classrooms (Jackson, 1992). However, among educational decision makers, the term curriculum is often used to refer to expectations for instruction laid out in policy documents or frameworks. Currently, and in direct response to the recent flurry of curriculum material development, many mathematics education researchers and practitioners use the term *curriculum* to refer to the material resources designed to be used by teachers in the classroom, such as “standards-based curricula.”

Research on teaching and curriculum, however, has revealed that a substantial difference exists between the curriculum as represented in instructional materials and the curriculum as enacted in the classroom by teachers and students. Curriculum theorists use a number of terms to distinguish between the curriculum outlined in a guide or set of materials and that enacted in the classroom. *Formal* (Doyle, 1992) or planned curriculum (Gehrke, Knapp, & Sirotnik, 1992), for example, refer to the goals and activities outlined by school policies or designed in textbooks or by teachers. The objectives set out in curriculum frameworks or state standards as well as those specified in scope and sequence charts in textbooks also represent formal curricula, sometimes referred to as

the *institutional* or *intended* curriculum. The *enacted* curriculum refers to what actually takes place in the classroom (Gehrke et al., 1992). In order to identify the impact that the enacted curriculum has on students, researchers use terms such as the *experienced* (Gehrke et al., 1992) or *attained* (Valverde, Bianchi, Wolfe, Schmidt, & Houang, 2002) curriculum. It is worth noting that distinctions are infrequently made between the curriculum as outlined by policy makers or curriculum designers and the curriculum interpreted or intended by the teacher.

In our effort to examine the influence that curriculum materials have on student learning, we have found all these meanings of curriculum to be significant and interrelated, yet uniquely important. In short, the influence of curriculum materials on student learning is not straightforward and cannot be understood without examining the curriculum as designed by teachers and as enacted in the classroom. Drawing on our earlier work (Remillard, 1999; Stein, Grover, & Henningsen, 1996), we have conceptualized these various meanings of curriculum as unfolding in a series of temporal phases from the printed page (*the written curriculum*), to the teachers’ plans for instruction (*the intended curriculum*),³ to the actual implementation of curricular-based tasks in the classroom (*the enacted curriculum*). (See Figure 8.1.)

Within and between certain of these phases, interpretative and interactive processes transform the curriculum. *Between* the written and intended phases, teachers bring their prior understandings, beliefs, and goals to bear on the written curriculum and, in the process, transform it into a form that they believe will be workable in the classroom. *Within* the enactment phase, the teacher and the students, in interaction with each other, bring the curriculum to life and, in the process, create something different than what could exist on the pages of the book or in the teacher’s mind or lesson plan. While all of the phases have a bearing on student learning (the final triangle), the classroom activities that occur during the enactment phase most directly influence how students experience mathematics and what they learn (Carpenter & Fenemba, 1988; Wittrock, 1986). We have added return arrows from the enacted curriculum and student learning to the transformation processes between the written and intended curriculum to indicate that the enacted curriculum and teachers influence teachers’ future interactions with written curriculum.

The oval in Figure 8.1 identifies possible factors that mediate the interpretive and interactive processes

³ Our use of the term, “intended” differs from TIMSS’ use of the term. TIMSS used “intended” to refer to what we call the “written” curriculum.

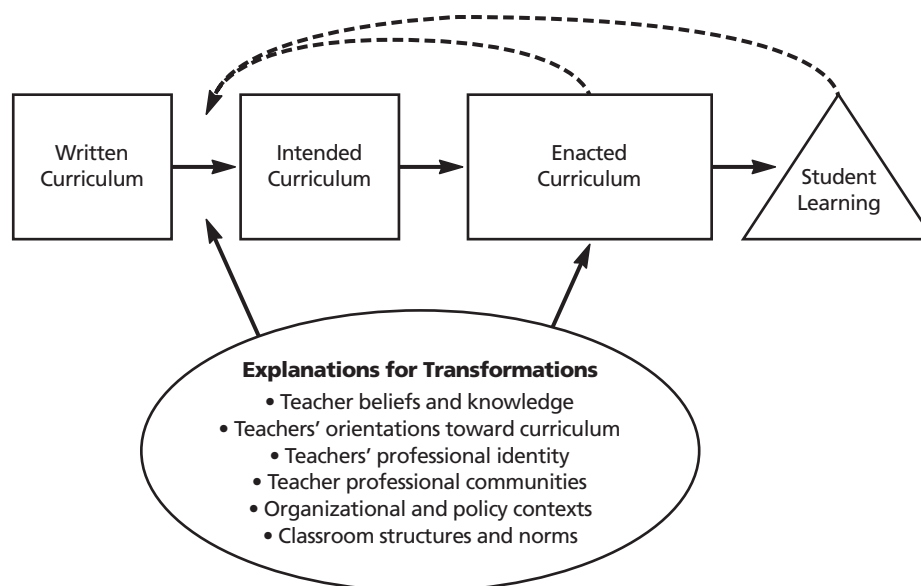


Figure 8.1 Temporal phases of curriculum use.

that occur within and between phases. A variety of studies, some specifically focused on curriculum use, others on the implementation of reform in general, have identified one or more of these factors as influencing how curriculum use unfolds. A number of researchers, for example, have explored the manner in which individual teacher characteristics and capacities, such as subject matter knowledge (Brophy, 1991, 2001; Stein, Baxter, & Leinhardt, 1990), beliefs (Cohen, 1990; Jennings, 1996; Putnam, 1992; Remillard, 1992; Spillane, 1999; Spillane & Jennings, 1997), and professional identity (Remillard & Bryans, 2004) influence how teachers understand and enact reform. Increasingly, studies—inside and outside of mathematics reform—have begun to suggest explanations for transformations that are external to the individual teacher. For example, research suggests that strong professional communities can help teachers to learn complex new ways of teaching (Cobb, McClain, de Silva Lamberg & Dean, 2003; Little & McLaughlin, 1993; Louis, Marks, & Kruse, 1996; Stein, Silver, & Smith, 1998) whereas teachers working in isolation are apt to continue to practice in the manner they always have. In addition, implementation of new programs and practices at the school level is influenced by organizational and policy contexts such as, the level of support from the principal (Berends, Kirby, Naftel, & McKelvey, 2001; Bodilly, 1998; Datnow, Hubbard, & Mehan, 2002; Fullan, 1991; Kirby, Behrends, & Naftel, 2001). Finally, factors central to the social organization of the school and classroom, including students, classroom structures, and norms (Doyle, 1983) can influence the curriculum transformation process.

The framework depicted in Figure 8.1 provides an organizing structure for the studies reviewed in this chapter. In *Section One*, we review research that has addressed the impact of curriculum materials on student learning. Many scholars and policymakers have (implicitly) conceptualized these studies as testing a (causal) relationship between the written curriculum (the first box in Figure 8.1) and student learning (the final triangle in Figure 8.1). The driving equation is, “Does curriculum X work?” with an emphasis on the written materials (not their enactment) as the intervention and a focus on student outcomes (usually achievement scores) as the indicator of what it means to work. After reviewing the findings of this research, we highlight the pros and cons of viewing the relationship between curriculum and student learning in this way; we then turn to the mediating relationships depicted in the figure to identify *how* curriculum influences student learning.

The studies reviewed in the next two sections begin to unpack how curriculum influences student learning by focusing on the transformations written curricula undergo as teachers engage with and use them in the classroom. In *Section Two* we review research on how teachers interpret curriculum materials which can be viewed as a discussion of the relationship between the written curriculum and the intended curriculum (the first and second boxes in Figure 8.1). In *Section Three* we review studies that have focused on what happens when curricular tasks are unleashed inside the classrooms; as such, this section focuses on studies falling within the box labeled enacted curriculum. The research reviewed in *Section Four* deals with studies that

aim to identify reasons for the variations uncovered in Sections Two and Three (i.e., why do teachers interpret and enact curriculum in different ways?) and thus can be seen as a discussion of the explanations that appear in the oval. Finally, in *Section Five* we review research on the relationship between the enacted curriculum and student learning (the final box and the final triangle of Figure 8.1). As such, Section Five also examines the question, “Does curriculum X work?” but this time the intervention is broadly conceived to include how teachers and students engage with the curriculum resource as well as the curriculum itself.

When considered together, the research conducted on the phases of curriculum use paints a complex picture of how curriculum influences student learning. More specifically, it points to the fallacy of assuming that the materials themselves are the primary agent in shaping opportunities for student learning and instead uncovers the important role played by the interpretive and interactive influences of teachers and students. Because they often represent the starting point for instruction, the materials are important and their influence merits substantial investigation; thus, they remain a focal point of this chapter. However, tracing *how* they are important necessarily takes one into the realm of teachers, students, and classrooms. Consequently, the reader will find overlaps between the research examined in this chapter and the chapter by Hiebert and Grouws on classroom teaching.

Curriculum Materials: An Evolving Construct

Because curriculum materials are a focal point of this chapter, we begin with a brief examination of what mathematics textbooks or curriculum materials are and how they and the way they are perceived have changed throughout the history of U.S. public education. Over the years texts and curriculum materials have varied in focus, style, philosophy and degree of comprehensiveness and have taken on different roles in the minds of teachers and administrators. While we use the terms *curriculum materials* and *textbook* (along with instructional resources and guides) somewhat interchangeably to refer to printed or electronic, often published, materials designed for use by teachers and students before, during, and after mathematics instruction, many teachers and mathematics educators draw sharp delineations between the two.

During the latter part of the 20th century, mathematics *textbooks* common to classrooms in the U.S.

were viewed as a source of explanations and exercises for students to complete. The textbook played a central role in most mathematics classrooms and instruction was aimed at teaching students what they needed to know in order to complete the exercises and answer the questions in their books (Jackson, 1968).

Unlike the term mathematics textbook, the term mathematics *curriculum materials* often refers to instructional guides that place substantial emphasis on both pedagogy (the how of teaching) and mathematics (the what of teaching). To many, the term curriculum materials was used to connote something akin to an “anti-textbook” because these resources offered programs of instruction that rejected the notion that learning mathematics involved completing decontextualized exercises in a book. In contrast to textbooks, which were developed and marketed by commercial publishing companies, curriculum materials tended to be designed by mathematics experts and mathematics education researchers and, prior to the late 1990s, were sold independently to a fairly small market. For most standards-based curriculum materials, students’ work during instruction involves investigative projects instead of exercises found on the pages of a “student textbook.” Student textbooks are replaced by thin, often consumable, student workbooks that are designed to support students’ investigative work. The centerpiece of most lessons is the thinking that is required to grapple with the investigative task; student work books are designed to support that thinking by providing a basis for recording, summarizing or reflecting on one’s actions and thinking.

Curriculum material development played a central role in the “modern mathematics” period of the late 1950s and early ’60s. During this time, numerous scientists and mathematicians were solicited to design instructional materials that would prepare the next generation of citizens to be scientifically competitive. The NSF played a major role in funding the development of the majority of these programs, often referred to as the “New Math.”⁴ The curriculum products of these content experts sought to place equal emphasis on “computational skill” and “understanding the basic concepts of mathematics and of their interrelationships, i.e., the structures of mathematics” (Begle, 1970, p. 1). Like texts of the past, these materials were designed with students (specifically talented students), not teachers, in mind. The consideration these developers gave to the teacher’s role is less clear, but it is understood that they grossly “overestimated

⁴ Specifically, the term New Math refers to the curriculum materials developed in the 1960s by the School Mathematics Study Group (MSG). It is often used to refer to all reform-oriented mathematics materials produced during the period of modern mathematics reform.

the materials' independent educative power" and underestimated the influence of the teacher in their use (Cohen & Barnes, 1993, p. 215). As a result, the curriculum materials developed during the modern math period are frequently labeled as attempts to be "teacher proof."

Like the New Math materials, curriculum materials developed during the period of standards-based reform initiated in the early 1990's are also seen as a potential means for shifting instructional practices. Unlike the New Math, however, the role of the teacher is generally explicitly acknowledged in these materials; most include detailed pedagogical guidance for teachers and some include resources to support teacher learning. For example, *Investigations in Data, Number and Space*, an NSF-supported elementary curriculum, includes "Notes to Teachers" that explain the mathematical significance of the ideas that are featured in particular lessons and how students might think about those ideas and "Dialogue Boxes" that illustrate how a student discussion surrounding those ideas might unfold in the classroom. As such, these materials can be viewed as being written for teachers and their learning, rather than being written for direct consumption by students. Remillard (2000) refers to this approach to curriculum design as curricula speaking *to*, as opposed to speaking *through*, the teacher.

The expectation by NSF that curriculum developers distribute their materials through established commercial publishers has had far-reaching influences. Most standards-based materials have experienced unprecedented visibility and success in the commercial markets. At the same time, the final products reflect compromises reached between developers and publishers. And, ironically, curriculum materials developed independently are marketed and sold along side, and sometimes as extensions of, instructional materials and textbooks developed "in house." Thus, another result of the way standards-based curriculum materials have been marketed may be a blurring of distinct lines that—at one time—clearly differentiated them from conventional texts.

The role of curriculum materials has also been conceptualized differently—even within the same period of time. Some individuals view curriculum materials as a direct blueprint for instruction—a plan to be unerringly implemented. According to this view, material that appears on the written page of the curriculum is meant for faithful execution by teachers and students. Others view curriculum materials as a

resource to be drawn upon by teachers as they construct lessons and units. This view suggests that complete fidelity of implementation is impossible because teachers will always bring their own frames of understanding and their knowledge of the local context to bear on how they use curricular materials. Inevitably, this view argues, teachers and students construct their own unique version of the curriculum—a version that necessarily springs from their goals, needs, and understandings (Remillard, 2005).

Literature Selection and Boundaries of this Review

In preparing this chapter, we reviewed research on the effects of mathematics curriculum materials or textbooks, both standards-based and conventional, on students and teachers available in peer-reviewed journals in addition to other sources such as research reports and edited volumes. A substantial portion of the research reviewed in this chapter was spurred by the wave of curriculum development supported by NSF and initiated after the publication of the NCTM *Standards* (1989). As noted earlier, the introduction of these curriculum materials to the public catalyzed substantial research activity—activity that was not matched by studies of conventional textbooks. The marked differences in the number of studies conducted on NSF-supported in contrast to conventional textbooks can be attributed to at least three reasons: a) NSF-supported curriculum developers were required by the funder to conduct evaluations and were sometimes provided with the funds to do so; b) the majority of NSF-supported curriculum developers were university faculty or researchers whose graduate students and colleagues conducted studies on the use of the new materials; and c) the research conducted by commercial publishing companies tended to be market driven, focusing on purchasing decisions rather than on measures of student outcomes or teacher use (NRC, 2004, pp. 25–28).

In an effort to orient the reader to the range of curriculum materials available in the U.S. during the time this chapter was written, we have identified the most well-known curriculum materials in Table 8.1. The first column identifies the full name of the text; the second column identifies the grade band for which the materials were developed. In the third column we identify the funder (if any and if known).

In the final column we indicate the category or group to which we have assigned the curriculum for

Table 8.1 Mathematics Curricula Commonly Used In U.S. Public Schools

Curriculum	Level	Funder (if any and if known)	Standards-based or Conventional
* <i>Everyday Mathematics</i> ; SRA/McGraw Hill	Elementary	NSF	Standards-based
* <i>Investigations in Number, Data, and Space</i> ; Scott Foresman	Elementary	NSF	Standards-based
* <i>Math Advantage K–6</i> ; Harcourt Brace	Elementary		Conventional
<i>Math in My World</i> ; McGraw Hill	Elementary		Conventional
* <i>Math K–5</i> ; Scott Foresman/Addison Wesley	Elementary		Conventional
<i>Math Land</i>	Elementary		Standards-based
* <i>Math Trailblazers</i> ; Kendall/Hunt	Elementary	NSF	Standards-based
<i>Number Power</i> (supplemental)	Elementary	NSF	Standards-based
<i>Opening Eyes to Mathematics</i> ; Visual Mathematics	Elementary		Standards-based
<i>Saxon Math</i>	Elementary		Conventional
<i>Silver Burdett Ginn Math</i>	Elementary		Conventional
<i>SRA Math</i> ; McGraw Hill	Elementary		Conventional
* <i>Connected Mathematics Project</i> (CMP); Prentice Hall	Middle School	NSF	Standards-based
<i>Heath Mathematics Connections</i> ; DC Heath & Co.	Middle School		Conventional
<i>Math Advantage</i> ; Harcourt Brace	Middle School		Conventional
<i>Math Alive</i> ; Visual Mathematics	Middle School		Standards-based
* <i>Mathematics: Applications & Connections</i> ; Glencoe/McGraw Hill	Middle School		Conventional
* <i>Mathematics in Context</i> (MiC); Holt, Rinehart, & Winston	Middle School	NSF	Standards-based
<i>Mathematics Plus</i> ; Harcourt Brace & Co.	Middle School		Conventional
* <i>MathScape: Seeing and Thinking Mathematically</i> ; Glencoe/McGraw Hill	Middle School	NSF	Standards-based
* <i>MathThematics</i> (STEM); McDougal Littell	Middle School	NSF	Standards-based
<i>Middle Grades Math</i> ; Prentice Hall	Middle School		Conventional
<i>Middle School Math</i> ; Scott Foresman/Addison Wesley	Middle School		Conventional
* <i>Middle School Mathematics Through Applications Project</i> (MMAP; unpublished)	Middle School	NSF	Standards-based
<i>Passport Series</i> ; McDougal-Littell	Middle School		Conventional
<i>Pre-Algebra: An Integrated Transition to Algebra and Geometry</i> ; Glencoe/McGraw Hill	Middle School		
* <i>Saxon Math</i>	Middle School		Conventional
<i>Cognitive Tutor</i>	Middle/High School		Standards-based
<i>College Preparatory Mathematics</i> (CPM)	High School		Standards-based
* <i>Contemporary Mathematics in Context</i> (Core-Plus)	High School	NSF	Standards-based
* <i>Interactive Mathematics Program</i> (IMP); Key Curriculum Press	High School	NSF	Standards-based
* <i>Larson Series</i> , Grades 9–12; Houghton Mifflin/McDougal Littell	High School		Conventional
* <i>MATH Connections: A Secondary Mathematics Core Curriculum</i> Grades 9–12; IT'S ABOUT TIME, Inc.	High School	NSF?	Standards-based
* <i>Mathematics: Modeling Our World</i> (MMOW/ARISE); W.H. Freeman & Co.	9–12	NSF	Standards-based
* <i>Systemic Initiative for Montana Mathematics and Science</i> (SIMMS) <i>Integrated Mathematics</i> ; Kendall/Hunt	High School	NSF	Standards-based
* <i>University of Chicago School Mathematics Project Integrated Mathematics, Grades 7–12</i> ; Prentice Hall	Middle/High School	NSF, Amoco Foundation, Carnegie Corp. of New York, General Electric Foundation	Standards-based

*= Curriculum included in NRC panel report, *On Evaluating Curricular Effectiveness*.

the purposes of this chapter. Curriculum materials are identified as either “standards-based” or “conventional.” *Standards-based curriculum materials* include the NSF-funded materials, curriculum materials that were inspired by documents that *preceded* the 1989 *Curriculum and Evaluation Standards* (e.g., AAAS’s *Benchmarks for Scientific Literacy*; and NRC’s *Everybody Counts*, and the draft version of the *Curriculum and Evaluation Standards for School Mathematics* that circulated in the mid-eighties), and, finally, curriculum materials that were driven by the ideas in the NCTM *Standards*, but not funded by the National Science Foundation (e.g., the *Cognitive Tutor*). *Conventional curriculum materials* include commercially developed textbooks with earlier editions that had been on the market prior to the release of the *Curriculum and Evaluation Standards*, editions that were not influenced by the earlier reform documents mentioned above. It should be noted that many of these publishers attempted to align the content of subsequent editions with the NCTM *Standards* after they appeared in 1989; however, this alignment was primarily accomplished through backward-mapping as opposed to using the *Standards* as a design template from the start, thereby giving these textbooks a substantially different “look and feel” than those materials that we label standards-based.

When we cite studies that explicitly involve one or more of the materials that appear in Table 8.1, we identify the curriculum by name. When overall statements are made regarding the characteristics or efficacy of various kinds of materials, we refer to the two general categories of standards-based or conventional.

Finally, the focus of our examination of the influence of curriculum on students is on classrooms—teachers and students. Thus, our discussion of curriculum materials centers on the resources and materials that teachers typically interact with as they plan and carry out their lessons. We have not included policy documents or the curriculum frameworks developed by districts or states in our review. In taking this approach, we do not, however, ignore the role that policy and organizational designs play in curriculum use. We recognize that state and local policy shape the manner in which teachers think about and use curriculum materials. We account for these contextual features primarily in our discussion of the explanations of how the curriculum is transformed between and within phases of enactment (see oval in Figure 8.1).

SECTION ONE: RESEARCH ON CURRICULUM MATERIALS AND STUDENT LEARNING

The *Curriculum and Evaluation Standards for School Mathematics* (NCTM, 1989) were created based on a broad consensus of professionals (teachers, mathematics educators, mathematicians, users of mathematics, etc.) and were framed in broad, easy-to-agree-with terms (EEPA, 1990). The curricular resources that were developed to embody them, however, offered much more detailed specification regarding what should be taught and how it should be taught, thereby providing concrete targets for critics of the Standards. As these new standards-based curriculum materials began to penetrate the market in the mid- to late-nineties, critics charged that they watered down important mathematical concepts, relied too heavily on real-life problems at the expense of pure mathematical problem solving and discovery, provided too little attention to paper-and-pencil calculations, and were not suited for college-level preparation (Wu, 1997).

Despite the fact that conventional textbooks used at the time had little or no evidence of their effectiveness, the release of these new curriculum materials was soon followed by demands for “proof” that they were helping, and not harming, students. Kilpatrick (2003) noted that these demands should not have been surprising:

Anyone proposing a new school mathematics curriculum faces the task of justifying its adoption. Teachers, parents, and students themselves may be dissatisfied with the current situation, but that does not mean they necessarily welcome change of the sort the new curriculum might demand. Curriculum developers are always in the position of “selling” their product by convincing their potential clients that the change it entails is both manageable and for the better. Research showing improved performance, or at least performance that is no worse than at present, is often used to bolster arguments being made on other grounds. (p. 477)

The fact that the standards-based curriculum materials challenged the status quo by embodying a radically different set of goals for student learning, however, meant that they had a particularly steep hurdle to overcome (Romberg, 1992). Moreover, the demands for “proof” were fueled by intense controversy, embodied by disagreements between two opposing camps (composed primarily, though not entirely, of mathematics educators in one group and university-based mathematicians in the other) that became so strident

that they were dubbed the “math wars” (Schoen, Fey, Hirsch, & Coxford, 1999). As this controversy played out in newspaper editorials, Internet discussions, and articles in popular periodicals, its polarized nature often made it difficult for the general public to distinguish ideology from empirical fact.

In this section, two kinds of studies are reviewed: content-based studies of curriculum materials and evaluations of what and how students learn from curriculum materials. These two kinds of studies share the underlying assumption that curriculum materials—in and of themselves—matter. Although some authors acknowledge the fact that curriculum materials are not self-enacting, these studies do not highlight the enactment process but rather focus on the materials themselves or on the relationship between the adoption of written materials and student outcomes. That is, most studies do not attempt to examine how teachers interpret curriculum materials or how teachers and students interact with the materials in the classroom. Returning to Figure 8.1, the reader might imagine a solid arrow drawn from “Written Curriculum” to “Student Learning” as a way to characterize the studies cited in this section.

Research on Content of Curriculum Materials

The majority of mathematics teachers rely on curriculum materials as their primary tool for teaching mathematics (Grouws, Smith, & Sztajn, 2004). If a topic is not included in their curriculum materials, there is a good chance that teachers will not cover it.⁵ And, as noted by Hiebert and Grouws (this volume), one of the best substantiated findings in the literature on classroom teaching and student learning is that students do not learn content to which they are not exposed. Thus, the identification of *what* mathematical topics are covered in a given set of curriculum materials is of fundamental importance.

Others argue, however, that “the mere presence of content in a textbook does not ensure that students will learn that content. For real learning to take place, textbooks must focus sound instructional strategies on the ideas and skills that students are intended to learn” (Project 2061, n.d., part 2). The myriad decisions that curriculum developers make regarding issues such as how material should be presented reflect the developers’ theory of how students learn mathematics. These

dimensions, which we discuss under the label of *how* content is presented (as opposed to *what* content is presented), are important because they set into motion different pedagogical approaches and different opportunities for student learning.

Finally, some curriculum materials aim to promote teacher learning in addition to student learning. If one believes that students’ opportunities to learn are the product of ongoing interactions among the text, the student and the teacher, then what students learn will be influenced by their teachers’ understanding of and presentation of the material in the text. Given the plethora of findings about the impoverishing effect that limited content knowledge has on teachers’ interpretation and use curriculum materials, especially novel and highly demanding materials such as standards-based curriculum materials (see studies reviewed in Section Four), we argue that a third dimension worthy of investigation is the extent to which the curricular materials are educative (Davis & Krajcik, 2005) for teachers. Each of these approaches to examining students’ opportunities to learn—What Content is covered? How is Content Presented? And with what support for teacher learning?—is reviewed below.

What Content is Covered?

Content analysts normally compare selected curriculum materials against a set of external criteria to determine depth and/or breadth of coverage. Analysts in the United States typically use standards, frameworks, or other countries’ curricula as their external criteria (NRC, 2004). Over the past decade, the external criteria used by researchers to analyze curriculum materials have varied widely, reflecting the various values held by the individuals who have conducted them.

Not surprisingly, these variations have produced different results. In their review of 36 content analyses, the NRC (2004) found that the ratings of many curricular programs vacillated from strong to weak depending on who had reviewed them and according to what criteria. Differences in values have not and cannot be decided by empirical analyses. However, once one’s values are clear and learning goals compatible with those values have been specified, questions regarding what curriculum is most effective (for achieving those goals) can be answered empirically (Hiebert, 1999). As concluded by the NRC panel (2004), curriculum decision makers for schools and districts should select

⁵ In recent years, the chances of students not being exposed to a topic if it is not in the curriculum have been somewhat mitigated by state standards. Most districts check their curricula for alignment to state standards. If a topic is found to be in the standards but not in their textbooks, they frequently find supplementary materials to address the gap.

reviews whose underlying dimensions are compatible with their values and learning goals.

It is in this spirit that we report the approaches and findings from three of the most prominent content analyses performed over the past decade: Project 2061 of the American Association for the Advancement of Science (AAAS), the ratings of curricula by the US Department of Education, and the work of Mathematically Correct.

Project 2061. One of the first, systematically documented analyses of the mathematical content of curricula was conducted in the late nineties by the American Association for the Advancement of Science [AAAS]) of middle school (and subsequently of algebra) textbooks. According to their website, the emergence of the NCTM Standards in 1989 along with AAAS's benchmarks for scientific literacy (which were closely aligned with the NCTM Standards) provided—for the first time—a solid, widely acknowledged conceptual basis for evaluating textbooks based on what students should learn. In their evaluation of 13 middle school curricula, they used a “relatively small but carefully chosen set of benchmarks to identify the strengths and weaknesses of the curricula.” These benchmarks, they claimed, “deal with concepts and skills that nearly everyone would agree are important for middle school students to achieve.” Another important aspect of their methodology was the decision to conduct in-depth examinations of a relatively small—but important—number of topics as opposed to examining more topics in a more superficial manner.⁶

The benchmarks used for their review of middle-school curricula included the following:

Number Concepts: The expression a/b can mean different things: a parts of size $1/b$ each, a divided by b , or a compared to b .

Number Skills: Use, interpret, and compare numbers in several equivalent forms such as integers, fractions, decimals, and percents.

Geometry Concepts: Some shapes have special properties: Triangular shapes tend to make structures rigid, and round shapes give the least possible boundary for a given amount of interior area. Shapes can match exactly or have the same shape in different sizes.

Geometry Skills: Calculate the circumferences and areas of rectangles, triangles, and circles, and the volumes of rectangular solids.

Algebra Graph Concepts: Graphs can show a variety of possible relationships between two variables. As one variable increases uniformly, the other may do one of the following: increase or decrease steadily, increase or decrease faster and faster, get closer and closer to some limiting value, reach some intermediate maximum or minimum, alternately increase and decrease indefinitely, increase or decrease in steps, or do something different from any of these.

Algebra Equation Concepts: Symbolic equations can be used to summarize how the quantity of something changes over time or in response to other changes. (Project 2061, n.d., part 1)

Results are presented in a matrix form for each of the 13 curricula that were reviewed. These matrices provide a content rating for each of the above criteria at one of three levels: most content, partial content, and minimal content. Page numbers and sections that provide evidence for meeting each of the content levels are provided.

The results of this review have been summarized as follows:

Most middle grades textbooks do a credible job addressing number benchmarks. However, only the best ones develop meanings for fractions, for example, by having students measure, build models, use number lines, and compare fractions to acquire a full understanding. Almost all textbooks present the formulas in geometry. But even some of the best ones fail to relate geometry skills to real-life ideas, such as the triangular structures used in bridges or the relationship between the distance around a city park and the area enclosed. . . . Few textbooks do a good job teaching how graphs show relationships, and instead focus on simple, linear graphs. The best series involve students with data collection in situations like a bicycle tour, giving them first-hand experience connecting concepts, such as time and distance, with tables and graphs. With this solid foundation, variables and equations are used naturally and with understanding (Kulm, 1999).

Only four of the thirteen middle-grades curricula—all standards-based—were rated satisfactory; that is, high enough to be confident that students would learn the content of the selected benchmarks. These curricula were, ranked in order, *Connected Mathematics*, *Mathematics in Context*, *MathScape*, and *Middle*

⁶ This speaks to the criticism of American curricula and programs as being a “mile wide and an inch deep” (Schmidt, McKnight, & Raizen, 1997) by focusing on the identification and treatment of “big ideas,” the understanding of which should place students in good stead for future mastery of more advanced mathematical content.

Grades Math Thematics. None of the conventional curricula were rated as satisfactory. These textbooks were judged to be lacking in their coverage of important mathematics, and to provide little development in sophistication from grades 6 to 8. (See Appendix A for ratings of various curricula by the AAAS's Project 2061, the Department of Education, and Mathematically Correct.)

U.S. Department of Education. In 1999, the U.S. Department of Education conducted a review of mathematics curricula to identify promising and exemplary curricula. According to the NRC panel (2004), this evaluation was guided by eight criteria structured in the form of questions (page 68):

1. Are the program's learning goals challenging, clear, and appropriate; is its content aligned with its learning goals?
2. Is it accurate and appropriate for the intended audience?
3. Is the instructional design engaging and motivating for the intended student population?
4. Is the system of assessment appropriate and designed to guide teachers' instructional decision making?
5. Can it be successfully implemented, adopted, or adapted in multiple educational settings?
6. Do its learning goals reflect the vision promoted in national standards in mathematics?
7. Does it address important individual and societal needs?
8. Does the program make a measurable difference in student learning?

Although *The Curriculum and Evaluation Standards* of the NCTM are not specifically mentioned, they were used (and assumed to be used) as the national standards referred to in question #6. Not surprisingly, once again, the standards-based curricula fared better under these criteria than did more conventional curricula, with the *Connected Mathematics Project* (CMP), the *Middle School Mathematics through Applications Project* (MMAAP), the *Cognitive Tutor*, *College Preparatory Mathematics* (CPM), *Contemporary Mathematics in Context* (Core-Plus), and the *Interactive Mathematics Program* (IMP) being named exemplary and *Everyday Mathematics*, *MathLand*, *Number Power*, and the *University of Chicago School Mathematics Program* (UCSMP) *Integrated Mathematics 7–12* being labeled promising.

Shortly after these evaluations became public, the criteria used by the Department of Education came under attack by opponents of NCTM *Standards* and Standards-based reform. For example, Richard Askey, a professor of mathematics at the University of Wisconsin and one of the authors of an open letter (signed by 200 professional mathematicians) sent to prominent national newspapers and to *Education Week* stated, "Some of the recommended programs are very careless with the mathematical core." Furthermore, he questioned the role of the NCTM *Standards*, saying, "To put in as one of the criteria that curriculum materials should be aligned with the NCTM Standards is inappropriate. We want good mathematics programs, and these do not necessarily have to be based on the NCTM Standards." Other prominent mathematicians defined their concerns with specific features of the "exemplary" curriculum materials, such as their light treatment of standard algorithms and heavy use of calculators (e.g., Klein, 2000).

Mathematically Correct. The organization of concerned mathematics professionals and parents known as Mathematically Correct have provided the third set of noteworthy content reviews in the past decade. Their reviews of second, fifth and seventh-grade mathematics textbooks (both standards-based and conventional textbooks) are guided by the concept of preparing all students for pre-algebra by eighth grade and by various sets of standards. In the seventh grade review (presented here for comparison to the criteria used by AAAS in their review of middle school textbooks), this organization evaluated "a set of key topics, representing an array of algebra readiness concepts, knowledge, skills, and problem-solving applications. Key benchmarks within these areas were drawn from various standards documents and used as the content basis for the review of each mathematical topic" (Mathematically Correct, n.d.a, p.1) The content areas were: Properties, order of operations; Exponents, squares, roots; Fractions; Decimals; Percents; Proportions; Expressions and equations-simplifying and solving; Expressions and equations-writing; graphing; Shapes, objects, angles, similarity, congruence; Area, volume, perimeter, distance.

Under each of these topic areas, detailed specifications were provided regarding what students should learn. Compared to the AAAS criteria for middle school curricula, the Mathematically Correct criteria focus more heavily on mastering procedural manipulations and rules. For example, under the first topic area, "Properties, order of operations" they state:

At this level students should master the rules of order of operations and the properties of the real number system (commutative, associative, distributive, identity and inverse) and be able to apply them, with justification, with all four operations in calculations involving fractions, decimals, percents, positive and negative numbers, and in the simplification of simple roots and powers. This should include the simplification of numerical expressions, the evaluation of expressions with substitution of numerical values and the solution of simple one and two step algebraic equations (Mathematically Correct, n.d.a, p.2).

The concern of Mathematically Correct is mastery of what they term, the mathematical core, the essential elements of knowledge and skills that students need to succeed in more advanced mathematics. Their criteria for mastering this core include procedural fluency, conceptual understanding and the ability to apply newly learned information to novel situations—criteria that most subscribers to the NCTM Standards would endorse as well. As we shall see in the next section, one of the main differences between NCTM Standards subscribers and Mathematically Correct members can be found in their respective notions of how—and in what order—students should learn the mathematical core.

Under the Mathematically Correct evaluations, conventional textbooks fared better than did standards-based curricula. (Appendix A identifies the curricula that have been rated by Mathematically Correct and the overall “grade” they received.) The highest “grade” received by a standards-based curriculum was a “C” for *Everyday Mathematics*; all others received a “D” or “F.” Meanwhile, all but two of the conventional curricula received an “A” or “B” (Mathematically Correct, n.d.b).

In sum, the three content reviews cited in this section make clear the need for transparency regarding the criteria used for content review of curriculum materials. Using very different sets of criteria, AAAS and the US Department of Education reviews produced recommendations that were nearly the opposite of the recommendations produced by reviews conducted by Mathematically Correct. This suggests that consumers and decision makers must first define what *they* value in terms of student learning and then select a review that encompasses their values as closely as possible (Hiebert, 1999).

Another way of framing the issue is in terms of tradeoffs. For example, one feature that distinguishes the AAAS review from the Mathematically Correct review is the extent to which each focused on depth versus breadth. Whereas AAAS examined a few topics in great depth (reflecting their belief that students

need to develop multi-faceted, in-depth understanding of a few, salient mathematical topics or ideas), Mathematically Correct’s reviews reflected the comprehensive range of skills that they believe students need to succeed in advanced mathematics courses. Each approach can be seen as having both virtues and drawbacks. Being aware of the criteria used in content reviews allows decision makers to surface and articulate this and other tradeoffs (some of which will be illuminated in the next section) and make informed decisions regarding what best fits with their K–12 instructional program, their goals for student learning, and their institutional resources and constraints.

How is Content Presented?

The above analyses primarily focused on *what* is included (and excluded) in various curriculum materials. We turn now to a discussion of *how* that content is presented, that is, analyses that more sharply focus on pedagogical intent. Similar to the analyses of content coverage discussed above, the criteria that researchers have used to make judgments about the pedagogical intent of various curricula are necessarily related to personally held views regarding the nature of mathematics and how students learn it. If one believes that mathematics is best learned through student construction based on active exploration, one set of criteria will be selected to guide one’s analyses. On the other hand, if one believes that mathematics is best learned through direct instruction and skills practice, then review criteria will be of a different sort.

A host of curricular features have the potential to influence how teachers use and thus how students experience curriculum materials. These include pretests to assess prior student knowledge, embedded assessments, suggestions for tailoring the material for students at different levels, the inclusion of group work activities, and the encouragement of student discussion of ideas. In this section, we begin with a discussion of several overarching design features that can have potentially far-reaching effects on how students learn mathematics. We then briefly review how two organizations, AAAS and Mathematically Correct rated curriculum materials with respect to how they presented content.

Overarching features of curricula. Most standards-based and conventional curricula intend for students to learn concepts, skills, applications, problem-solving and efficient procedures. They differ, however, with regards to the order and manner in which these elements are presented, the balance that is struck among various elements, and organizational style. Each of these is discussed below.

1. *Order and manner of presentation.* Conventional curricula tend to rely on direct explication of the to-be-learned material as well as careful sequencing and accumulation of lower-level skills before presenting students with the opportunity to engage in higher-order thinking, reasoning and problem solving with those skills. Students typically are expected to master definitions and standard algorithmic procedures *before* they are exposed to opportunities to apply their knowledge. In contrast, standards-based materials rarely explicate the to-be-learned concepts *for* the students; instead they rely on student engagement with well-designed tasks to surface the concepts. Active exploration of the concept-to-be-learned is viewed as allowing students to induce important facets of the concept based on their own reasoning and thinking. After the concept has been surfaced and its features explored by students, the curriculum and teacher step in to apply definitions, standard labels, and (sometimes) standard procedural techniques related to the concept.

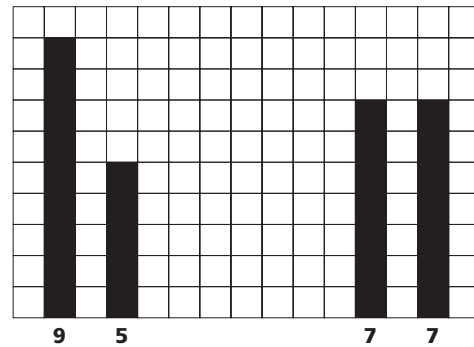
To illustrate the above contrast, consider two ways in which curricula might approach the concept of the mathematical average. Conventional curricula would begin with the provision of an explicit definition of the concept of average (i.e., a measure of central tendency) and would then demonstrate the conventional algorithm for finding the average of a set of numbers (taking the sum of the numbers in the set and dividing the sum by the number of addends). After practicing this procedure on several sets of numbers, students would be asked to apply what they have learned by finding the average in the context of word problems. In contrast, standards-based curricula would not begin with an explicit definition of the concept of average but rather would immerse students in a task that requires that they experience what an average means, for example, as is done with the task shown in Figure 8.2 (found in *Visual Mathematics/Math Alive*).

As students grapple with this task, they would be approaching the concept of average through a frame of “leveling” and thus be encouraged to recognize what an average does, i.e., it evens out the extremes of a given population, a recognition that is important when deciding which measure of central tendency might be the most valid representation of a population given a particular question. Students could also be drawn to recognize that the average can be a number that is not represented in the population, as is the case in this problem. Only after exploring these more qualitative, conceptual features of the mathematical average would students be introduced to the conventional procedure for finding the average.

The theory of knowledge and of how students learn is radically different in these two approaches.

The pairs of numbers in a–d below represent the heights of stacks of cubes to be leveled off. On grid paper, sketch the front views of columns of cubes with these heights before and after they are leveled off. Write a statement under the sketches that explains how your method of leveling off is related to finding the average of the two numbers.

a) 9 and 5 b) 16 and 7 c) 7 and 12 d) 13 and 15



By taking 2 blocks off the first stack and giving them to the second stack, I've made the two stacks the same. So the total # of cubes is now distributed into 2 columns of equal height. And that is what average means.

[Taken from *Visual Mathematics*, (Bennett & Foreman, 1989)]

Figure 8.2 Task used in *Visual Mathematics* to develop students' understanding of the concept of average.

In the more conventional approach, knowledge is viewed as a Gagne-like (1977) hierarchy of skills and learning is viewed as proceeding from lower-level to higher-level skills. Without the prerequisite lower-level skills, this approach argues, students will not be able to master higher-order skills. In the standards-based curricula, on the other hand, learning is viewed as the development of understanding through the construction of increasingly detailed relationships between concepts. In this perspective on learning, students are viewed as making meaning of new information by relating it to their prior knowledge; learning is said to have occurred when students develop an understanding of the relationships between and among concepts and between concepts and procedures (Hiebert et al., 1997).

2. *Balance.* Curriculum materials also differ with respect to the balance that they strike among a number of elements including a) between concepts and procedures, b) between calculator-aided vs unaided computation, and c) among various kinds of representations including numerical data, manipulatives, tables, graphs, and equations. Again, the particular balance that is struck reflects the developers' philosophy regarding how students learn mathematics. For example, in summarizing reviews of algebra curricula, Clopton, McKeown, McKeown, and Clopton from Mathematically Correct (1998) expressed concern re-

garding the impact that calculators have on the development of computational fluency and tended to discourage their use. Many researchers favorable to the NCTM Standards, on the other hand, applaud the introduction and use of calculators, because of their ubiquity in present-day society and their belief that off-loading computational burdens from students “frees up” cognitive space for higher order thinking and reasoning. With regard to the balance between concepts and procedures, not surprisingly, conventional materials tend toward a balance that weighs heavily toward practice on procedures while standards-based curriculum materials are weighted more toward the development of concepts and problem solving.

3. *Organizational style.* Finally, conventional versus standards-based curriculum materials differ in terms of their overall organization. Most conventional textbooks are organized into units and chapters that follow topics that have been identified as appropriate for each of the particular grade levels. Many are also organized as a “spiral,” meaning that specific topics are introduced and re-introduced throughout a set of grades with increasing levels of sophistication each time that students encounter them. Students are not expected to master the topic the first time it is introduced, but rather are expected to deepen their understanding and attain eventual mastery over time as the topic is revisited. Stein and Kim (2006) refer to spiral and other tightly sequenced curricula as *integral* curricula. In integral curricula, knowledge and skills are tightly woven into the fabric of the curriculum (i.e., they cannot be easily separated out) and they must be taught in a specified sequence over the years.

Integral curriculum materials can be contrasted with *modular* approaches in which the subcomponents can be separated and even recombined into new and varied configurations with little loss to the overall effectiveness in attaining curricular goals (Stein & Kim, 2006). Many standards-based curriculum materials are modularized with each module designed to represent a big idea or integrating conceptual theme. For example, *Bits and Pieces*, a sixth grade module from *Connected Mathematics*, subtitled “Understanding Rational Numbers” focuses on moving between and among fractions, decimals, and percents. Typically, students complete 6–12 modules per school year.

By and large, research has been silent on the trade-offs associated with integral versus modular forms of curricular organization. Stein and colleagues (Stein & Kim, 2006) have proposed that integral curricula place more demands on the social cohesion of a school’s

faculty (which they conceptualize as social capital) because they require that all teachers buy into one curriculum and dutifully carry out their responsibilities with respect to teaching the requisite skills at each grade level. Modular approaches, on the other hand, can afford to be taught by some teachers, but not all. Such approaches can also be used in variable sequences (within a given grade level or sometimes across two consecutive grades). Moreover, because the to-be-learned knowledge and skills are more easily identifiable and viewed as separable from the curricular fabric, modular curriculum materials are more amenable to progressive implementation within a reform effort (i.e., implementing only a partial set of modules for the first year, a few more the second year, and so forth).

In sum, the above discussion of design features illustrates the range of decisions that developers make (either implicitly or explicitly) as they design materials for student consumption. Moreover, the features discussed above—order and manner of presentation, balance, and organizational style—are not features that typically appear on textbook review checklists. Rather they are subtle attributes whose presence becomes felt over time as teachers use the materials and observe their impact on how students and their colleagues engage with the materials. Surfacing these distinctions and their potential implications for how students learn mathematics would be a significant move forward for decision makers in most schools and districts.

Reviews of how content is presented. The reviews conducted by AAAS and Mathematically Correct illustrate the selection of two very different sets of dimensions for judging the pedagogical quality of mathematics curricula. AAAS claims to have based their review of middle school curricula on criteria “derived from research on learning and teaching and on the craft knowledge of experienced teachers” (Project 2061, n.d., part 1, p. 3). Reviewers were trained to rate the extent to which textbooks explicitly included information with respect to 24 instructional criteria ranging from the specification of prerequisite knowledge to the guiding of interpretation and reasoning. The 24 criteria were organized into seven categories (identifying a sense of purpose, building on student ideas about mathematics, engaging students in mathematics, developing mathematical ideas, promoting student thinking about mathematics, assessing student progress in mathematics, and enhancing the mathematics learning environment), each of which was justified by findings from research on learning and teaching.⁷ The criteria

⁷ Sources that are cited include *Science for All Americans* (AAAS, 1990), *Benchmarks for Science Literacy* (AAAS, 1993) and the *Handbook of Research on Mathematics Teaching and Learning* (Grouws, 1992).

and organizational categories are also cited as being compatible with NCTM's 1989 Standards.

One example of an organization category will illustrate the view of teaching and learning embodied by the AAAS approach. Category IV, labeled "Developing Mathematical Ideas" is justified as such:

Mathematics literacy requires that students see the link between concepts and skills, see mathematics itself as logical and useful, and become skillful at using mathematics. Six criteria are used to determine whether the material justifies the importance of benchmark ideas, introduces terms and procedures only as needed, represents ideas accurately, connects benchmark ideas, demonstrates/models procedures and provides practice.

The above rationale cites the need for practice and accuracy, but also stresses the importance of students viewing mathematics as meaningful and useful; moreover, much of the emphasis is on the making of connections and the introduction of terms and procedures *as needed*. As such, the view of teaching and learning appears to be closely aligned with a constructivist, meaning-making approach to mathematics.

AAAS's analysis of instructional strategies revealed stark contrasts among the textbooks with respect to the adequacy of instruction for the specific benchmarks. Four curricula are highly rated for their engagement of students, development of mathematics concepts, and support of teachers, the same four that were rated high for content: *Connected Mathematics*, *Mathematics in Context*, *MathScape*, and *Middle Grades Math Themes*. The remaining curricula (all conventional) were judged to be unsatisfactory with respect to instructional support. They were judged to be particularly unsatisfactory in offering a purpose for learning mathematics, taking account of student ideas, and promoting student thinking.

Mathematically Correct based their reviews of pedagogical worthiness on examination of two main dimensions: "quality of presentation" and "quality of student work." For "quality of presentation," the following were examined: "clarity of objectives within each lesson and within a topic as a whole, clarity of presentation, a sufficient number and appropriate examples, reasonable guided practice or scaffolding of the early stages of implementation of new knowledge, and a general sense that students would have a high probability of mastering the material at the level at which it was taught" (Mathematically Correct, n.d.a, p. 5). For "quality of student work," they examined the following: "Is there enough chance to practice what is taught, is the work well designed, and is the work at an appropriately high level? Importantly, a

range depth and scope of student work should be provided, appropriately building from the simpler, less abstract cases to the more difficult ones" (Mathematically Correct, n.d.a, p. 5).

Using these criteria for their review of seventh grade curricula, the Mathematically Correct reviewers gave the two standards-based curricula (*Connected Mathematics* and *Math Themes*) the lowest scores. According to their website (Mathematically Correct, n.d.c, p. 5), "these books fall notably short in content, have ill-defined goals for each lesson, as least in the eyes of the reviewers, and have insufficient or poorly designed student work." The curricula that we have called conventional fared much better, notably those published by Saxon. Unlike the other conventional curricula which the reviewers found to be compromised by a "fusion" of teacher-directed lessons and activities based on "discovery learning and the constructivist theory of knowledge," (Mathematically Correct, n.d.c, p. 4) the Saxon curricula were found to have clarity of focus, regular practice on important topics, and to be concise enough to discourage teacher variation with regards to decisions regarding coverage.

The reader can begin to get a sense of how these two sets of evaluations privilege different approaches to teaching and learning mathematics. Not surprisingly—as with the content analyses—the same curriculum fares very differently depending on which set of criteria are applied. Again, decisions can be viewed as tradeoffs set against the backdrop of consumers' beliefs and values, combined with an analysis of their institutional resources and constraints. Potential consumers need to pose to themselves and answer questions such as the following: What kinds of knowledge and skills will prepare our students for their most likely futures? Do our students need to become procedurally sharp by building on their successes through repetition and diligent practice? Or is our ultimate goal students who recognize and begin to deal with complexity—students who, nonetheless, may have to deal with frustration along the way because they will not always arrive at a correct solution in a neat and timely manner? Are our teachers up to the task of teaching in more complex ways and/or following an integral curriculum to assure a coordinated instructional program across grade levels? The posing and answering of these questions depends on awareness of one's values, an accurate picture of one's institutional context, and the ability to assess curricula according to criteria that are transparent to consumers.

The Support of Teacher Learning

As noted earlier, the standards-based curricula tend to embed much of the mathematics learning

implicitly within the tasks themselves. Students are viewed as learning mathematics by “doing mathematics” as opposed to having mathematical concepts and procedures explicitly and directly taught to them by the curricular materials (and the teacher). This student-centered approach to the learning of mathematics places considerable demand on teachers who must organize the physical and normative environment to be conducive to student inquiry, scaffold student actions and thinking as they grapple with cognitively challenging tasks, deal with student ideas that are off-track, and, finally, bring closure to a lesson that is rooted in many different student-designed methods of approaching a given task and a variety of levels and kinds of understanding of the to-be-learned concept(s).

Recently, researchers have begun to identify ways in which curricular materials might be designed so as to assist teachers to learn subject matter themselves—a prerequisite for this kind of teaching—as well as to learn how to better assist student learning of subject matter (Ball & Cohen, 1996; Davis & Krajcik, 2005). These features comprise elements that are added on top of the “base curriculum,” that part of the curricular materials that is designed for direct student consumption. Educative curricula incorporate elements that are meant specifically for *teacher* learning and are distinguished from typical teacher guides that steer teachers’ actions but not their thinking.

This is a relatively new area of research but is becoming increasingly important as student-centered materials gain a foothold in the marketplace and as teachers struggle to learn a new way of teaching mathematics. The most developed framework for judging how well curricula promote teacher learning identifies five high-level guidelines for educative curricula (Davis & Krajcik, 2005, pp. 5–6):

1. Educative curricula could help teachers learn how to anticipate and interpret what learners may think about or do in response to instructional activities.
2. Educative curricula could support teachers’ learning of subject matter.
3. Educative curricula could help teachers consider ways to relate units during the year.
4. Educative curricula could make visible curriculum developers’ pedagogical judgments.

5. Educative curricula could promote teachers’ pedagogical design capacities or their ability to use personal resources and the supports embedded in the materials to adapt curriculum to achieve productive instructional ends (as opposed to performing “lethal mutations”) (Brown & Campione, 1996, p. 291).

Building on this work, Stein and Kim (in press) analyzed two standards-based curricula (*Everyday Mathematics* and *Investigations*) in order to assess the extent to which they (a) made visible the curriculum developers’ rationales for specific instructional tasks and/or for particular learning pathways advanced in the base curricula; and (b) helped teachers to anticipate how students might approach these tasks. Referring to the visibility of developers’ judgments as “transparency,” Stein and Kim reasoned that teachers who had a clear idea of where a particular task was headed mathematically would be better equipped to guide the classroom enactment of the task, especially when it did not go as planned. The ability to anticipate student thinking in response to the task, they reasoned, would allow teachers to plan ahead for how they might deal with the diversity of student responses that open-ended tasks often generate. They found differences between the two curricula with *Investigations* offering more rationales regarding the mathematical significance of tasks and more support for teachers to anticipate student thinking. Preliminary evidence regarding the impact of these differences on teacher implementation of these two curricula suggests that the open-ended tasks of *Investigations* can be enacted with considerable fidelity when teachers make use of the teacher support material whereas the relatively more constrained tasks of *Everyday Mathematics*⁸ were nonetheless difficult to implement with fidelity when teachers were unsure of the mathematical purpose of the tasks or how students might respond to them (Stein, Kim, & Seeley, 2006).

Brown (in press) distinguishes two different principles that can be used to organize teacher materials: resource-centric and procedure-centric. A resource-centric approach to the design of teacher materials emphasizes the key building blocks of lessons and tries to make visible the pedagogical affordances of the building blocks, but leaves up to the teacher how, when, and whether to implement the various building blocks. A procedure-centric approach, on the other hand, focuses on the actions involved in carrying out the lessons. Stein and Kim (in press) view

⁸ The majority of *Investigations* tasks were coded as “doing mathematics” (open-ended, unstructured), whereas the majority of *Everyday Mathematics* tasks were coded as “procedures with connections to underlying concepts and meaning.” See Stein et al. (1996) for an explanation of these levels of cognitive demand.

the teacher materials that accompany *Everyday Mathematics* as procedure-centric and the teacher materials that accompany *Investigations* as resource-centric. They stress that how and where to implement each depends on the nature of the base curriculum and the institutional context.

In this sub-section (Research on Content of Curriculum Materials), we have reviewed research that has focused directly on curriculum materials, framing our review in terms of the opportunities for student learning afforded by different curricula. Our basic finding is that students' opportunity to learn is in the eye of the beholder. An examination of Appendix A illustrates what may now seem to be predictable disagreements between Mathematically Correct and AAAS/Department of Education reviewers (see, for example, the disparate ratings for *Connected Mathematics*). If one adopts the view of teaching and learning espoused by the NCTM standards (that served as a design blueprint for most standards-based curricula) and consults reviews that base their criteria on the NCTM Standards and like-minded sources, one finds that the standards-based curriculum materials are highly rated. On the other hand, if one believes that mathematics is better taught and learned through more explicit, direct instruction and practice of what is taught and consults reviews that build their criteria from sources such as Mathematically Correct, one finds that the standards-based curriculum materials come up considerably short.

A second finding, which is obvious but nevertheless worth noting, is that standards-based and conventional curriculum materials really do appear to provide very different opportunities for student learning. Although the various reviewers used different criteria, they used them consistently suggesting that standards-based curriculum materials provide more opportunities for students to learn how to think and reason and fewer opportunities to learn and practice basic skills. Conventional textbooks, on the other hand, appear to provide more systematic opportunities to learn definitions and practice demonstrated skills but fewer opportunities to learn when and how to apply those skills.

Finally, our review suggests that standards-based curriculum materials differ with respect to the kind of support materials that they provide *for teachers*. We also suggest that this issue is more complex than discovering that one approach (e.g., the resource-centric approach) is more effective than another. Rather, the kind of teacher-support materials that are most helpful will most likely depend on the nature of the base curricula, as well as the experience level and instructional capacities of the teachers who will be implementing the materials. The issue of teacher learning from curriculum materials opens up an entire new set

of design considerations that developers must attend to and that researchers must learn to evaluate.

Our examination of curriculum materials with respect to students' *opportunities to learn* represents only the first step—albeit an important first step—to investigating curricula's actual impact on student learning. In the next sub-section, we turn to studies that collected data on what students learned when they were exposed to these curriculum materials in the classroom.

Examination of Student Learning from Mathematics Curriculum Materials

All of the NSF-supported curriculum projects were required to conduct both formative and summative evaluations of their materials (NSF, 1989). Virtually all developers conducted pilot tests of their materials as various units were under construction or in different stages of completion. A key element of these formative assessments was to uncover what students were able to do and understand as a result of their exposure to the curricula's tasks. Developers also conducted summative evaluations, many of which are reported in an edited volume by Senk and Thompson (2003) entitled, *Standards-Based Mathematics Curricula: What Are They? What Do Students Learn?* Some developers also received additional funding once their materials had been completed to conduct additional studies of their impact.

On the whole, the early evaluations of standards-based curricula as reported in the Senk and Thompson volume were promising. After reviewing syntheses of findings for the four elementary curricula, Putnam (2003) stated:

The first striking thing to note . . . is the overall similarity of their findings. Students in these new curricula generally perform as well as other students on traditional measures of mathematics achievement, including computational skill, and they generally do better on formal and informal assessments of conceptual understanding and ability to use mathematics to solve problems (p. 161).

Similarly, Chappell (2003) notes that, collectively, the evaluation results provided for three middle school curricula provide “converging evidence that standards-based curricula may positively affect middle school students' mathematics achievement, in both conceptual and procedural understanding.” (p. 290). And, finally, Swafford (2003) reviewed syntheses of findings across five high school curricula and found that students in the reform programs performed as well as students in traditional programs on tests designed to measure traditional content and that they

outperformed them on tasks designed to measure the content and processes emphasized in their materials (pp. 458, 460). Given the very different goals and approaches of the new standards-based curricula, these initial reports suggesting that students taught with these materials demonstrated at least comparable levels of computational learning as students taught with conventional curricula and superior understanding of concepts and problem solving were greeted with enthusiasm by their proponents.

That said, none of the approaches or methodologies used in the studies reported in the Senk and Thompson volume was without flaws. One of the biggest concerns with the studies reported in this volume was the fact that curriculum designers were the primary researchers on most of the studies which inevitably raises conflict-of-interest issues. On the positive side, however, developers who also conduct research on their curriculum materials are in the position to design assessment tasks that tap the knowledge that students should acquire from exposure to their curriculum due to their deep understanding of the material. For example, Mokros (2003) reports on a developer-designed assessment comprised partially of individual interviews that assessed the quality of explanations and use of representations and tools, prominent learning goals associated with the *Investigations* curriculum. These kinds of assessments should yield better information on the extent to which students have learned from the specific curriculum than will standardized tests.⁹

Overall, however, the curriculum designers did not appear to take full advantage of their “insider knowledge” and design such assessments. In one of the concluding chapters of the Senk and Thompson volume, Kilpatrick (2003) noted that many of the most distinguishing features of the curricula that were studied and reported on in this volume were not assessed in terms of the student learning outcomes that would be expected. Rather, the most common instruments used were standardized tests, items from national and international assessments, and the PSAT or SAT. According to Kilpatrick, it appeared as though these measures were being used “more to reassure parents and the public that the new curricula were not harming students than to assess common learning goals” (p. 479).

Other concerns that have been raised about drawing conclusions based on these early studies as reported in the Senk and Thompson volume include the often limited variability in conditions of

the studies. For example, the study of the *Trailblazers* curriculum included primarily teachers who had received professional development from the curriculum developers themselves. Also, in many of the early studies, the materials were not always in their final forms, and, in some cases, the curriculum materials had not been used long enough to establish stable implementations. Finally, issues can be raised about the comparability of the groups to which some of the curricula were compared. This leaves one with limited ability to make strong conclusions based on any one of the above studies, although the number of studies and the diversity of approaches represented in the Senk and Thompson volume might—for some readers—increase confidence in the general patterns that emerged.

The above studies were undertaken during the early phases of the dissemination of standards-based curricula. As such, the studies reported in the Senk and Thompson volume might be best understood as “proof of concept” studies, indicating that the standards-based curricula “are working in classrooms in ways their designers intended for them to work” (Kilpatrick, 2003, p. 472). It should also be noted, however, that these studies did not satisfy skeptics who demanded that independent researchers develop more rigorous methodologies with which to examine student learning, a discussion to which we now turn.

Comparative Studies Conducted by External Researchers

Although it may seem simple to evaluate whether or not a new curriculum is better, worse or no different from existing approaches, designing a study to do so is fraught with pitfalls and complications, leading an NRC panel convened to examine evaluations of mathematics curricula to conclude that “comparative evaluation study is an evolving methodology” (2004, p. 96). To start, credible outcomes measures must be identified or created. This can be particularly problematic with standards-based curricula that focus on conceptual understanding, thinking, reasoning, and problem solving—outcomes for which the field has limited measures. Two accessible measures that assess conceptual understanding, reasoning and problem solving, the New Standards Reference Exam and Balanced Assessment, have not been widely used (Schoenfeld, 2006). Many researchers use their own measures whose psychometric properties have not been established (Kilpatrick, 2003).

⁹ On the other hand, a limitation of most of the items that were developed to measure student learning by curricular developers is that their psychometric properties were not reported, that is, no reliabilities were reported and often curricular validity was the only type established.

Another challenge involves coming to grips with the issue of the differing goals of standards-based vs. conventional curricula. As noted earlier, both would agree to the need for developing some conceptual understanding and some procedural competence; however, the emphasis placed on each varies dramatically between the two kinds of curricula, as does the amount of emphasis placed on reasoning, problem solving, and communication. In addition, there are differences in the topics covered, with, for example, elementary standards-based curricula covering topics not typically found in conventional textbooks including geometry, statistics, and algebraic reasoning. In a curricular evaluation, the question thus becomes: Does one assess only the goals and topics that the two kinds of curricula have in common or does one assess all students on all goals and topics, regardless of which curriculum they were exposed to?

Third, despite being at the heart of this methodology, the creation of comparison groups can be surprisingly difficult. For example, comparison groups are most often selected based on comparability of student samples. Despite the fact that the comparability of teachers is often not considered, experience suggests that teachers in experimental vs. control groups often differ in their levels of enthusiasm, recent professional development experiences, and in their ability to attract a certain kind of student. The NRC report (2004) raises another common concern: studies that compare classrooms using standards-based curriculum with classrooms whose instruction represents a potpourri of instructional styles and approaches. Even if the two sets of classrooms (i.e., the experimental and control groups) are comparable on student background characteristics, the cleaner, more deliberate instructional approach represented by the standards-based classrooms almost always “wins.”

A final issue is determining how the curriculum materials were enacted in the classroom. One cannot say that a curriculum is or is not associated with a learning outcome unless one can be reasonably certain that it was implemented as intended by the curriculum developers. Moreover, there is ample evidence that teachers vary in how they implement the same curricular task (Stein et al., 1996; Tarr, Chávez, Reys, & Reys, in press). Assessing implementation quality using observational methods is extremely resource intensive, especially in large-scale studies, and credible self-report of implementation on survey measures is notoriously difficult to achieve (Ball & Rowan, 2004). Nevertheless, claims about the effectiveness of one curriculum over another are commonly made citing student achievement data, but providing little or no data on degree of implementation.

Given the limitations associated with comparative studies, the question naturally arises: What claims can be made about the impact on student learning of various curricula? This question sits at the heart of the charge given to the NRC panel and their conclusion was:

... the corpus of evaluation studies as a whole across the 19 programs studied does not permit one to determine the effectiveness of individual programs with a high degree of certainty, due to the restricted number of studies for any particular curriculum, limitations in the array of methods used, and the uneven quality of the studies (2004, p. 3).

The report goes on to caution that the inconclusive finding of the panel should not be interpreted to mean that these curricula are ineffective either, but rather that problems with the data and/or study designs prevented the panel from making confident judgments about their effectiveness. (The curricula marked with an asterisk in Table 8.1 were included in the NRC review).

Incontrovertible evidence of curricular effectiveness can be provided only by studies that randomly assign students to standards-based vs. conventional curricula, the so-called “gold standard” of the randomized controlled trial (Mosteller & Boruch, 2002; National Research Council, 2002). Although the NRC panel identified no randomized controlled designs for the 19 programs that they studied, the What Works Clearinghouse (http://www.whatworks.ed.gov/PDF/Topic/math_topic_report.html) has produced a Topic Report on middle school curriculum-based interventions that identifies four randomized controlled studies. Two of these studies yielded significant effects: a study by Morgan and Ritter (2002) that examined the effectiveness of the *Cognitive Tutor Algebra I* and a study by Kirby (2004) on the effectiveness of *I CAN Learn*. Both curricula are standards-based and computer-mediated. The study by Kirby (2004) consisted of one teacher who taught both *I CAN Learn* and a conventional curriculum and used the Georgia Criterion-Referenced Competency Test, a state test, as the outcome measure. The study by Morgan and Ritter (2002) involved nearly 400 students and 18 teachers across five schools and used the Educational Testing Service’s (ETS) Algebra I End-of-Course Assessment as the primary outcome measure. In both studies, students who used the standards-based curriculum outperformed the comparison group that was taught with a conventional curriculum (McDougal Littell’s *Heath Algebra* in the case of the *Cognitive Tutor* and an unspecified “traditional” curriculum in the case of *I CAN Learn*). The other two randomized controlled studies cited by the

WWC failed to find significant differences in student learning favoring either the standards-based (the *Expert Mathematician/UCSMP*) or conventional curricula (*Transition Math* or *Saxon*).

Turning to studies that do not meet the gold standard of the randomized controlled trial, one finds less-certain evidence for any one specific curriculum, but, nevertheless interesting patterns across findings from several large-scale studies that compared achievement in classrooms using a variety of standards-based curricula to achievement in classrooms using conventional curricula. Despite the differences in grades and curriculum studied, and the methodologies employed, many of these studies have produced fairly consistent findings. The first is that students taught using standards-based curriculum, compared with those taught using conventional curriculum, generally exhibited greater conceptual understanding and performed at higher levels with respect to problem solving (e.g., Boaler, 1997; Huntley, Rasmussen, Villarubi, Sangtong, & Fey, 2000; Thompson & Senk, 2001). Second, these gains did not appear to come at the expense of those aspects of mathematics measured on more traditional standardized tests. Compared with students taught using conventional curricula, students who were taught using standards-based curricula generally performed at approximately the same level on standardized tests that assess mathematical skills and procedures (e.g., Riordan & Noyce, 2001; Thompson & Senk, 2001). The differences that occurred were usually not significant, and some show the “standards-based curriculum” students doing slightly better, while others show the “conventional curriculum” students doing slightly better. For example, students in the Core Plus Mathematics Project (CPMP) (a standards-based curriculum) outperformed others on tests of algebraic concepts set in real world contexts, but the students taught using more traditional texts outperformed those in CPMP on tests of algebraic skills set in questions without contexts that did not allow calculators (Huntley et al., 2000). Unsurprisingly, students tend to do well on tests that match the approaches through which they have learned.

It is striking to note the similarity between the patterns of findings reported in the Senk and Thompson volume and the findings of the larger-scale comparative studies conducted by external reviewers. In both cases, students taught using standards-based curricula tended to hold their own on tests of computational skills and to outperform students taught with conventional curricula on tests of thinking, reasoning, and conceptual understanding. This pattern of findings—not the findings of any one study—has prompt-

ed some to point to the overall efficacy of standards based curricula (e.g., Schoenfeld, 2002).

But, efficacy for what? It is important to note that students tended to perform best on tests that aligned with the approaches by which they had learned, repeating the well-worn finding that students learn what they are taught. Combined with the findings from the analyses of curriculum materials cited earlier, the research examined here suggests that students taught using conventional curricula will learn to compute better, but not to think and reason better than students taught using standards-based curricula. Students taught using standards-based curricula, on the other hand, will learn to think and reason better and (maybe) to compute as well as their counterparts.

Finally, readers must look beyond sound-bite findings in order to ascertain the level of learning actually achieved by students—both in conventional and standards-based interventions. For example, what does it mean to perform “as well as” one’s counterparts? In some studies cited above that meant scoring in the lower percentiles of mathematics achievement (e.g., see Morgan & Ritter, 2002; Kirby, 2004). And, studies finding “no difference” do not always mean that students performed dismally. For example, the What Works Clearinghouse reported “no effect” for a standards-based program (the *Expert Mathematician*) because there were no significant student achievement differences between students exposed to it vs. a more conventional curriculum. Nevertheless, students made learning gains in both groups (Viadero, 2004 as reported by Schoenfeld, 2006).

The above discussion harkens back, once again, to values. Should schools and teachers primarily be held accountable for developing students’ capacities for basic proficiencies or for the development of students’ abilities to think, reason and problem solve? If both, how should the two be combined?

Methodologically, we acknowledge the superior power of randomized controlled studies to determine the effectiveness of one curriculum over another and we join the National Research Council in calling for more experimental studies. At the same time, we feel that it is important to acknowledge the questions that randomized controlled studies cannot answer and to make a plea for other kinds of studies as well. Returning to the framework depicted in Figure 8.1, the majority of the comparative studies cited above did not collect data on how the various curricula were enacted in the classroom (the third box in our framework—see Figure 8.1); none examined the way in which teachers engaged with the materials to create the intended curriculum (the second box in our framework). When implementation measures

are included, the findings can become less clear. For example, Boaler and Staples (in press) studied algebra classes across three high schools, approximately half of which used conventional methods, and half of which used standards-based methods. In tests of algebra at the end of their first year of high school, students in the two groups performed at the same level, both in terms of skills and conceptual understanding. However, this was not an indication that the curricular approach did not matter, only that there was a need to look beyond curriculum to the ways in which teachers implemented them. Indeed the most significant factor in comparisons of algebra achievement in this study was the teacher, with large variations between classes *within* the same curriculum. The importance of individual teachers and their particular teaching decisions was also reported by Huntley et al. (2000) who found that different classes in the same curriculum varied to a large degree in classroom implementation and student achievement. And, finally, using multiple, non-observational measures of curricular implementation,¹⁰ Balfanz, Mac Iver, and Byrnes (2006) found differences in levels of curriculum implementation that were associated with differences in student achievement with classrooms that exhibited higher levels of implementation averaging higher achievement gains.

As noted earlier, many scholars and policymakers have (implicitly) conceptualized the question behind student-outcome studies as the testing of a (causal) relationship between curricular materials (the written curriculum—see first box of Figure 8.1) and student learning (the final triangle in Figure 8.1). Our framework, together with findings associated with most of the above studies, suggests, however, that such a conceptualization—although useful for some reasons—is incomplete. Although the above studies that focused on student outcomes can reveal *whether* a particular curriculum achieved superior outcomes, they cannot shed light on *how*. However, knowledge of how an effect was achieved is crucial, both for enhancing the field's understanding of teaching and learning mathematics as well as for others who wish to implement the curriculum and for designers hoping to improve the curriculum. Thus, a review of *only* these studies is insufficient. Without accompanying knowledge of how the various curricula were implemented, students' opportunities to learn remain shrouded. In the best case, this limits our un-

derstanding of how desirable effects were achieved; in the worst case, it raises questions about the extent to which the achieved effects on student learning can be attributed to a particular curriculum. Thus, we call for studies that are designed to include both large-scale tests of curricular effectiveness and smaller, but embedded, observational studies of instructional practice.

There are too few studies of this kind, two notable exceptions notwithstanding. Tarr, Reys, et al. (in press) compared classrooms that used either standards-based or conventional curricula, collecting both achievement data and observational data on the extent to which the classrooms exhibited a "standards-based learning environment (SBLE)." These observations documented the extent to which teachers created learning environments consistent with the tenets of the NCTM Standards and were not tied to any particular curriculum. They found that students in the classrooms using standards-based curricula that also had high levels of SBLE implementation outperformed students who were in classrooms using standards-based curricula but having lower levels of SBLE implementation. In the aforementioned longitudinal study of three California high schools, Boaler and Staples (in press) utilized a range of qualitative and quantitative methods including student assessments, interviews, and questionnaires, and over 600 hours of classroom observations. As such, they were able to report not only that students in Rail-side high school (where instruction was built around standards-based curricula) achieved at higher levels,¹¹ but also the ways that the teachers brought this about through their responses to student thinking, their questioning, and the allocation of time. More studies like this—that attend to both student learning outcomes and the classroom-based mechanisms that appear to relate to those outcomes—are needed.

Despite few studies that examine student learning and instructional practice together, there are many research studies that have investigated the mechanisms and processes by which teachers engage curricula, both in their planning and during instruction. These studies provide an important complement to comparative studies that primarily focus on student outcomes. This brings us to the second area of research that will be reviewed: what happens when teachers engage with and interpret curriculum materials.

¹⁰ The intervention consisted of the University of Chicago School Mathematics Program middle school curriculum supported by professional development, coaching and a surrounding whole-school reform program.

¹¹ The Railside students achieved at higher levels on all measures (researcher-designed and district examinations) except the standardized test used by the state of California.

SECTION TWO: HOW TEACHERS ENGAGE WITH AND INTERPRET CURRICULAR MATERIALS

This section examines teachers' use of curriculum materials as a factor in student learning from curriculum. As our framework suggests, teachers have a substantial role in interpreting and using written curriculum materials and enacting them in the classroom. In this section, we focus on the first step in this process that is labeled in our framework as the relationship between the written curriculum and the intended curriculum.

As discussed earlier, the written or formal curriculum refers to the curriculum represented in curriculum materials or other teaching resources. The intended curriculum refers to the teacher's intentions, that is, the curriculum the teacher plans to enact. As our model suggests, the intended curriculum is often a derivative of the written curriculum; however, there are distinct differences between the two. On a conceptual level, written and intended curricula are different kinds of representations of curriculum. The former is a set of written guidelines and resources aimed at leading and supporting teachers to enact what curriculum developers imagine taking place in the classroom. The latter is comprised of a set of ideas and plans held in the teacher's head. In this sense, the intended curriculum is more closely aligned with an image of classroom practice than the written curriculum.

A second distinction between the two is that the intended curriculum represents the written curriculum as transformed by the teacher. In other words, the process of reading and using curriculum materials necessarily involves interpretation and meaning making on the part of the teacher. In this way teachers do not merely read and follow written curriculum. Just as reading itself is a process of interpretation (Mailloux, 1982), teachers draw on their own experiences and knowledge to make meaning of what they read in curriculum guides. In this sense, they transform the written curriculum from generic, written description and guidelines to both an image of classroom practice and something that has specific meaning to them.

Scholarly interest in this transformation process and the factors that influence it is motivated, in part, by questions about the extent to which curriculum materials can influence teachers and, subsequently, impact students. Research that examines teachers' use of cur-

ricula assumes that teachers play a mediating role in how materials influence student learning and seeks to provide clarity about a) the processes by which teachers use, interpret, and transform curricula, b) the factors that influence it, and c) how conditions and contexts can impact these processes.¹² In this section, we explore how researchers have framed and studied the relationship between the written and intended curriculum and the stances toward conceptualizing the teacher-curriculum relationship represented in the literature. The factors and conditions that influence how teachers engage with curricular materials (b and c above) are explored in Section Four.

Framing of the Relationship between Written and Intended Curriculum

The framework guiding this chapter assumes that written and intended curriculum differ in significant, although sometimes, subtle ways. This view, however, has not always been embraced by curricular scholars. Historical studies of school curricula, for instance, relied heavily on textbooks of the period to reconstruct the contents of classroom practice (Love & Pimm, 1996; Walker, 1976). By highlighting ways that teachers adapted or resisted unfamiliar curriculum programs developed during the period of curriculum reform in the late 1950s and 1960s, researchers such as Sarason (1982), Stake and Easley (1978), and Berman and McLaughlin (1973, 1978) challenged conventional assumptions that teachers merely followed their textbooks and that texts alone represented classroom instruction. These studies frame teachers as active users of curriculum, rather than conduits of it and opened the door to questions about the various ways that teachers make use of curriculum materials in shaping the curriculum they intend to teach. The following discussion examines the variety of ways researchers have described and categorized teachers' approaches to curriculum use.

Content Coverage

As part of a year-long study of how content is determined in fourth grade classrooms, Freeman, Belli, Porter, Floden, Schmidt, and Schulle (1983) undertook case studies of seven teachers using mathematics textbooks published in the 1970s. Their analysis produced what the authors identified as four styles of textbook use that focus on the mathematical content

¹² There are only a handful of studies that examine the processes by which teachers interpret curriculum exclusively; typically, such research is incorporated into studies of how the curriculum is enacted in the classroom. As a result, some of the studies cited in this section are also cited in Section Three.

covered. They used the term *textbook bound* to refer to an approach to textbook use in which the teacher followed the guide lesson by lesson, beginning at the first chapter and, more than likely, not completing the last few chapters before the close of the academic year. The *selective omission* style, was similar to textbook bound, however, the teacher skipped some chapters entirely. Teachers who were identified as *basics focused*, placed priority on teaching the four operations and used their textbook selectively to assist them in doing so. Freeman and colleagues also identified a subcategory of these teachers who included the topics of measurement as a basic skill. The final style of use, identified as *management by objectives*, involved following a set of objectives imposed by district and school policy and using the textbooks as tools to attend to the goals set out by these policies.

It is likely evident to most readers that the categories proposed by Freeman et al. (1983) reflect the conventional mathematics curricula that dominated the educational scene at the time. However, what stands out about the approach of these researchers is their focus on content coverage over pedagogical strategies or other dimensions of textbook use.

In their examination of 39 middle school teachers' use of both standards-based and conventional textbooks, Tarr, Reys, et al. (2006) also focused on mathematical content coverage. These researchers were interested in determining how closely teachers followed the content offered in their mathematics curriculum materials, particularly in terms of the distribution of mathematical topics. By comparing the proportion of each program devoted to number and operations, algebra, geometry and measurement, and data analysis and probability to the proportion of lessons taught devoted to each topic, they found that all teachers made both omissions and additions to the material found in the text in favor of number and operations. Lessons on data analysis and probability were omitted most frequently.

Components of the Curriculum

Other ways of categorizing curriculum material use place primary emphasis on differences in *how* teachers use the same lesson. Remillard and Bryans (2004), for example, studied 8 elementary teachers during the first two years of using *Investigations*, a standards-based curriculum program. In their analysis, which drew on interviews and observations, they characterized teachers with respect to the extent to and thoroughness with which they used the structure of the curriculum, its mathematical content, and the pedagogical suggestions and information offered.

The two teachers who used the curriculum *narrowly and intermittently* relied primarily on their own teaching routines and other resources to guide their curriculum choices over the year. When they did use the standards-based curriculum program adopted in their school, they tended to use it by selecting tasks that seemed familiar and using the repertoires they had developed over years of teaching when enacting them in the classroom. The two teachers identified as *adopting and adapting* the curriculum used the materials as a guide for the general structure and content of their mathematics curriculum. They regularly adopted mathematical tasks from the curriculum guides, but drew on their own strategies and approaches to enact them in the classroom, adapting them to fit their familiar approaches to teaching. The authors identified four of the eight teachers as *thorough piloting* the curriculum. These teachers used the materials as their primary guide in teaching mathematics, which included structuring the curriculum, selecting tasks, and facilitating students' work on those tasks. They tended to read and use all parts of the curriculum guides and sought to follow the lessons as suggested.

An important distinction made in Remillard and Bryans' (2004) categorization system is between teachers who read and follow pedagogical guidance offered by the curriculum and those who relied on their own repertoires when using tasks provided by the curriculum. In other words, the same tasks can be used differently by different teachers. Moreover, because the most recent generation of curriculum materials places substantial emphasis on pedagogical practices, considering how and whether teachers use pedagogical information in the materials is critical to understanding the relationship between the written and intended curriculum.

In his study of middle school science teachers using a standards-based science curriculum, Brown (2002) also focused on *how* teachers used the curriculum guide, emphasizing the shifting nature of this relationship. He identified three different ways that teachers used the materials—offloading, adapting, or improvising. *Offloading* refers to when teachers offload curriculum design responsibilities onto the curriculum and follow it fully. *Adapting* refers to when teachers use suggestions offered in the materials but adapt them to suit their particular needs or preferences. *Improvising* refers to when teachers move away from the curriculum suggestions substantially and design their own.

These types of curriculum use are similar to those offered by Remillard and Bryans (2004) in that they focus on the extent to which the teacher used the

guidance offered by the authors. Brown, however, suggests that teachers tend to engage in different kinds of use at different times. The same teacher might offload in one instance, allowing the curriculum to carry the weight of curriculum design. In another instance, she might improvise, drawing on the curriculum minimally. Although his study was of middle school science teachers, Brown's finding that teachers' approach to curriculum use can vary from lesson to lesson depending on a number of factors contributes to what is understood about the relationships between the written and intended curriculum.

Program Philosophy

Lambdin and Preston (1995) studied middle school teachers using *Connected Mathematics* (CMP), a standards-based curricula. Using interviews and observations, these researchers developed three classifications of teachers to represent the patterns they observed across teachers studied. These classifications highlight the teacher's stance in relation to the underlying philosophy of the program. *Frustrated methodologists*, for example, were teachers whose established classroom routines were in conflict with the goals of the curriculum and were reluctant to change. These teachers' use of the curriculum was mechanical in nature and "characterized by disjointed often superficial use of curriculum materials" (p. 135). The frustrated methodologists described by Lambdin and Preston were often torn between two philosophies of teaching, theirs and that underlying the curriculum. The *teacher on the grow*, describes a group of teachers "open to change and anxious to learn" (p. 136) in order to use the new curriculum more successfully. These teachers often experienced initial reluctance or doubt about the intent of the new curriculum, but, overtime, developed appreciation for its intent and approach. Lambdin and Preston described the final group of teachers as *standards bearers*. These teachers had established classroom routines and pedagogical approaches closely aligned with the philosophy of the program.

Lambdin and Preston's (1995) categories for teachers in relation to curriculum materials focus on the degree of match between the teacher's philosophy of teaching mathematics and the pedagogical stance represented in the curriculum. These researchers found that confidence in one's mathematical knowledge was critical in determining how teachers were classified. Of equal importance, however, was their understanding of the program's epistemological stance or philosophy. In fact, they found more evidence of teachers increasing mathematical knowledge and gaining confidence in their knowledge through use of the curriculum than

they found of teachers coming to understand and appreciate new pedagogical approaches.

Looking across these differing schemes for categorizing how teachers use curriculum materials, we see significant interest in and reason for identifying and examining the relationship between the curriculum as written and as intended by the teacher. Indeed, there is overwhelming evidence that teachers use and enact curriculum in substantially different ways. The varieties of kinds of use examined hint at the complexity of the relationship between written curriculum materials and teachers' uses of them. Moreover, these varied categorization schemes represent some of the possible ways that curriculum use can differ (e.g., in content coverage, in extent and nature of reliance on various curriculum components, in epistemological and philosophical stance). An examination of the kinds of use identified in these studies as a set can offer insight into the dimensions along which curriculum use can vary. As a field, we ought to take as much interest in these dimensions as we do in the variations within each study. Considering these dimensions as a set can shed light on the complexity of curriculum use. While most studies tend to focus on a single dimension, such as philosophical stance, attending to a set of them can offer a fuller understanding of what matters in teachers' curriculum use. Further, attending to these dimensions raises questions for further research. Are there other dimensions along which variation in use occurs that matter? What are the differential consequences of these kinds of variation for classroom activity? Do some variations matter more than others?

A second reason to examine the dimensions of variation in material use identified by researchers is to uncover points of convergence across studies. The different foci taken by researchers along with the introduction of new categorical schemes and terms to label kinds of use points to a need for clarity in how curriculum use is examined and identified. Given the reality that the predominant methods for studying curriculum use involve intensive observation over extended time frames, it is critical that the field establishes structures that enable aggregation of findings across studies.

A final point to consider when examining studies of curriculum material use by teachers is the extent to which the dimensions of use identified parallel the dimensions used to frame analyses of the curriculum materials themselves. The studies above that examined content coverage by teachers (e.g., Freeman et al., 1983, and Tarr, Chavez et al., in pres) can, to some extent, be viewed as mirroring analyses of curriculum materials that focus primarily on content and topic coverage (see pages 327–330 in this chapter). On the other hand, the research cited above by Remillard and

Bryans (2004), Brown (2002), and Lambdin and Preston (1995) that focused on teachers' use of pedagogical and philosophical dimensions of curriculum materials align with studies of curriculum materials that focus on how the content is presented, both to students and teachers (see pages 330–335 in this chapter). We believe that the knowledge base associated with how curriculum impacts student learning would be advanced if the field established common structures for examining both the materials themselves and the dimensions along which researchers examine teachers' use of those materials.

We now turn to different conceptualizations of curriculum use in the literature and how these stances shed light on the manner in which teachers interact with curriculum materials in shaping the intended curriculum.

Conceptualizations of Curriculum Use

In an analysis of the literature on teachers' use of mathematics curriculum materials Remillard (2005) identified several different ways curriculum use was studied in the literature; three of these approaches represent differing stances on what it means to use curriculum. As a result, they reveal different dimensions of the relationship between the written and intended or enacted curriculum. These three stances on curriculum use include: use as following or subverting, use as interpretation, and use as participating with. The following descriptions are accompanied by illustrative examples from the literature.

Curriculum Use as Following or Subverting

Studies that frame curriculum use as following or subverting view the written curriculum as embodying discernible and complete images of practice and examine the degree to which teachers follow these guidelines with fidelity. These studies implicitly contend that the degree of match between the written curriculum materials and enacted curriculum is measurable. As a result, researchers undertaking such studies concern themselves with how curriculum writers and others might achieve greater clarity and closer guidance for the teacher. These studies that seek to capture "treatment integrity," as called for by the NRC (2004) report, are critical to assessing the impact of curriculum materials on student learning.

This perspective is evident in studies of teachers using conventional curricula as well as in studies of teachers using standards-based curricula. In these studies, researchers found many more cases of subverting the curriculum than of close following. As a result, they tended to make recommendations that might

result in greater fidelity, such as increased specificity within curriculum materials and enhanced professional development. It is worth noting that in almost all cases, researchers found that teachers' deviations from the written curriculum reflected tendencies toward conventional pedagogical approaches and representations of mathematics.

Studies by Freeman et al. (1983), Freeman and Porter (1989), Stodolsky (1989), and Sosniak and Stodolsky (1993) all provide examples of analyses of teachers' use of conventional curricula. Their findings revealed that teachers used texts selectively, focusing on basic skills instruction and core mathematical topics, and demonstrated, in Freeman and Porter's words, that the conviction that textbooks determined the mathematics curriculum reflected "a narrow view of teacher decision making" (p. 404).

Studies of teachers using innovative curriculum materials since the 1980s (e.g., Donovan, 1983; Manouchehri & Goodman, 1998; Stephens, 1982; Tarr, Chavez, et al., in press) examined teachers using curriculum materials that reflect unfamiliar views of mathematics and approaches to mathematics teaching. These researchers all found that teachers tended to use mathematical topics and tasks from the curriculum guides, but made alterations as well, often subverting the designers' intentions, transforming the tasks to fit conventional approaches to teaching and classroom management structures.

In a study of 39 teachers from six different states, 17 using standards-based and 22 using commercially produced texts, Tarr, Reys et al. (in press) examined textbook fidelity and its relationship to student outcomes. They measured fidelity by considering: (a) the extent to which the teacher covered topics and lessons in the textbook, (b) the amount of time spent on topics in the textbook, (c) in the case of standards-based curriculum, the extent to which the classroom activities reflected a "standards-based learning environment." They found that, on average, teachers used 60 to 70% of the lessons in the curriculum guides, regardless of the type of curriculum. In fact, the greatest influence on whether lessons were omitted was their topic. Teachers of both kinds of curriculum materials were more likely to omit lessons on data and probability than on number and operations. While the use of standards-based curriculum materials was not a predictor of the presence of a standards-based learning environment (SBLE), measured on a ten point scale, only one classroom using commercially generated curriculum materials had a SBLE rating over 5. Half of the classrooms using standards-based curriculum materials were rated between 6 and 10 on the SBLE scale.

As a group, studies that examine the closeness of match between a written curriculum and actual practice shed light on the aspects of materials teachers tended to follow as well as the variety of ways they adapt and subvert suggestions in these materials. While teachers are likely to follow the mathematical topics offered in curriculum materials, they do so differentially, favoring topics such as number and operations over probability and statistics. Moreover, they are much less likely to follow the pedagogical recommendations or embrace the philosophical stance of the materials.

Curriculum Use as Interpretation

Research grounded in a view of curriculum use as interpretation assumes that fidelity between written plans in a teacher's guide and classroom action is impossible (Remillard, 2005). This view holds that teachers bring their own beliefs and experiences to their encounters with curriculum to create their own meanings, and that by using curriculum materials teachers interpret the intentions of the authors. In her book on teachers' encounters with curriculum materials, Ben-Peretz (1990) argues that teachers draw on personal knowledge and experience to "assign meaning to the curriculum materials they use daily in their classrooms" (p. 71). Research from this point of view investigates the nature of teachers' interpretations, the factors that influence them, and the resulting classroom practices.

Researchers who view curriculum use as a process of interpretation have applied this view to a range of initiatives intended to influence teaching, including educational policy. In their studies of the relationships among state-level policy, district and classroom practices, researchers with the Educational Policy and Practice Study (EPPS)¹³ maintained that policy and its multiple instantiations, such as textbooks and tests, are open to interpretation. Beginning in 1988, EPPS used case-study methods to consider how elementary teachers in California learned about, understood, and acted on state-level mathematics policy in the late 1980s. The state's initial efforts to alter mathematics instruction placed heavy emphasis on approved textbooks to communicate its message of reform. Hence, how teachers interpreted and used their new textbooks was central to this work (e.g., Ball, 1990; Cohen, 1990; Heaton, 1992; Putnam, 1992; Remillard, 1992; Weimers-Jennings, 1990; S. M. Wilson, 1990).

A particularly striking finding from this research with respect to the interpretation of policy and curriculum materials was the conviction with which all of the participating teachers believed that their teaching reflected the ideas of the new policy's reform intentions as a result of their faithful use of a particular textbook or curriculum program. However, their interpretations of the goals of the reform initiative and their uses of their texts varied tremendously. Variations included subtle and not so subtle adaptations of the plans suggested in the text as well as diverging interpretations of what it means to engage students in problem solving or discuss their solution strategies.

A number of researchers studying teachers using standards-based curriculum materials have focused their analyses on teachers' interpretations of the unfamiliar curricula as well as the NCTM *Standards* themselves. Collopy (2003) studied two teachers using the *Investigations* curriculum for the elementary grades and found significant differences in their uses. The most extreme case of this kind of variation was in how the teachers used the illustrative dialogues provided in the curriculum. These scripted conversations, presented in the teacher's guide like the dialogue in a play, illustrated an example of a discussion a class might have about a focal concept or phenomenon. One teacher read them in order to anticipate the ideas that might come up during a class discussion while the other teacher used them as scripts and had students read the different student parts aloud. Collopy attributed these dissimilar interpretations to the teachers' contrasting views of curriculum and the degree to which they had firmly established pedagogical repertoires and curriculum structures.

Like those that frame curriculum use as following or subverting, these studies also highlight variations in how teachers use curriculum resources. They also go a step further in illuminating the process by which these variations occur. By framing curriculum use as an interpretive process, researchers make visible the ways that teachers' meaning making processes shape the intended curriculum. The factors that influence teachers meaning making are discussed in Section Four of the chapter.

Curriculum Use as Participating With

A third perspective taken by researchers studying teachers and curriculum materials focuses on the teacher-text relationship or the activity of using the text. This perspective treats curriculum material use

¹³ This group began its work in 1988 on a study of the relationship between state-level mathematics policy and classroom practice in California. Later, in 1992, as it expanded its focus to include three states and policy in two subjects (mathematics and reading), the group took the title EPPS.

as collaboration with the materials (Remillard, 2005). Central to this perspective is the assumption that teachers and curriculum materials are engaged in a dynamic interrelationship that involves participation on the parts of both the teacher and the text. While there is significant overlap between this perspective and the stance that focuses on use as interpretation, the core difference is the focus of the analyses. Indeed, researchers in either category might view curriculum use as a process of interpretation. However, the research in this group seeks to study and explain the nature of the participatory relationship. In other words, the distinguishing characteristic of this perspective is its focus on the activity of using or participating with the curriculum resource and on the dynamic relationship between the teacher and curriculum. Studies taking this perspective consider the ways that particular features of the curriculum resource, as well as the teacher's interpretations of them, work together to shape the intended curriculum.

The view that curriculum material use involves a dynamic interchange between teacher and curriculum is reflected in Lloyd's (1999) study of two high school teachers using *CORE Plus*, a standards-based curriculum. The study investigated "how and why two teachers encountered particular successes and difficulties as each implemented a set of novel curriculum materials for the first time" (p. 229). The author examined the teachers' conception of the curriculum and of key ideas central to it, including exploration and cooperation, and the teachers' resulting mathematics teaching. She argued that "curriculum implementation consists of a dynamic relation between teachers and particular curricular features" (p. 244), but also suggested that this relationship can be strained by tensions between the structure outlined in the curriculum guide and the teachers' need to construct curriculum that is responsive to students.

Remillard's (1999, 2000) study of two teachers using a textbook that represented a commercial publisher's response to the NCTM *Standards* (1989) also examined teachers' participation with curriculum. The study illustrates how two fourth grade teachers interacted with the same text in different ways to construct contrasting opportunities for student learning. The analysis highlights the ways the teachers read the textbook and explores the factors that contributed to different approaches to reading. Not only did the two teachers read entirely separate features of the text (exercises for students versus supplementary activities), they also read for different purposes (potential activities and assignments versus big ideas to guide planning). These contrasting uses of the textbook resulted in different plans from the same lesson in the text.

Studies like these that frame and examine curriculum use as an interactive process not only capture how teachers interpret curriculum materials, but focus attention on the role the materials play in this process. They illustrate the complexity of transformation from the written to the intended curriculum.

While highlighting different aspects of the teacher-curriculum relationship, the above variations in the way curriculum use is conceptualized in the literature also point to fundamental differences in theoretical and conceptual assumptions underlying the research. These differences, when not made explicit, further complicate efforts to examine findings across studies and illustrate the need for greater clarity in how these constructs are conceptualized.

The relationship between the written and intended curriculum (the first and second boxes of Figure 8.1) has been neglected in most studies of how curriculum influences student learning. Yet the set of studies reviewed here point to important variations in curriculum use that are introduced even before the lesson begins. These studies are important because they challenge the assumption that curriculum materials tend to be implemented uniformly across teachers, schools, and districts. If research suggests that the curriculum is altered the minute that a teacher begins to read it, then the assumption that curriculum materials are single-handedly—or even primarily—responsible for student performance becomes questionable and the need to broaden the logic to include a discussion of how teachers interpret the curriculum becomes apparent. Thus, we see this set of studies as laying the first stone in the foundation of a chain of reasoning, evidence and argument that must underlie future studies of how curriculum influences student learning.

Throughout this chapter, we call for this chain to be broader, more inclusive, better-articulated, and more internally coherent than what currently exists. As we have already noted, the studies reviewed in Sections One and Two reveal the need for greater consistency with respect to the dimensions that are examined when analyzing the curriculum materials (phase one of the framework) and curriculum interpretations (phase two of the framework). Another point raised for us by these studies concerns the consistency with which the term *fidelity of implementation* is used (fidelity is an important concept in the "following vs. subverting" approach to studying curriculum use). More specifically, might "fidelity" mean something different when one is discussing a standards-based versus conventional curriculum materials? If conventional materials—as described earlier—present content and exercises in a straightforward manner, might not the attainment of fidelity be, not only easier to achieve, but also easier to measure? One would

examine the extent to which teachers provided the correct definitions, examples, and procedures and whether they monitored student practice accurately. Standards-based curricula, as noted earlier, tend to present information indirectly, relying on tasks—and students' work on those tasks—to expose the concepts to be learned, thereby placing considerable demand and discretion on teachers to help students correctly surface and explicate concepts and desirable ways of thinking and reasoning. Here, one would need to examine the extent to which teachers' ways of assisting students was compatible with this indirect style of teaching, the extent to which the teacher listens to and "pulls from" students important ideas, and, if and how the students correctly surface the intended material. In the first case, one appears to be measuring fidelity to a given set of teacher procedures, in the second, the fidelity is to an underlying philosophy and theory of learning.

SECTION THREE: THE ENACTMENT OF CURRICULA IN CLASSROOMS

This section examines the ways in which curricular tasks are set up and enacted during instruction. Here our focus moves beyond teachers' intentions to the examination of what actually occurs as curriculum materials are enacted in the classroom—the third phase of curricular use shown in Figure 8.1. We begin this section by discussing the various ways in which curriculum enactment has been studied and providing a rationale for the approach taken herein—a focus on mathematics tasks. We continue by exploring the source and nature of mathematical tasks that are used as the basis of instruction in our nation's classrooms. We then turn our attention to research that illuminates the ways in which mathematical tasks are set up and enacted by classroom teachers and their students and the processes that influence the ways in which tasks actually unfold in the classroom.

Ways in Which Curriculum Enactment Has Been Studied

Studies that can shed light on the enactment of curriculum in the classroom vary with respect to the aspects of instructional practice to which they have attended. Over the past decade, researchers have conducted studies focused on classroom discourse (e.g., O'Connor, 2001), questioning (e.g., Boaler & Brodie, 2004), intellectual authority (e.g., Wilson & Lloyd, 2000) and on normative practices (e.g., Boaler & Sta-

ples, in press) in classrooms in which standards-based and conventional curricula were being used. While not focusing directly on curricular enactment, these studies provide insight into the challenges and possibilities of orchestrating teaching and learning interactions around standards-based curricular tasks.

In this section, we organize our discussion of these and other studies through the conceptual lens of mathematical instructional tasks. We have made this decision for several reasons. First, classroom instruction is generally organized and orchestrated around mathematical instructional tasks. That is, students' day-to-day work in mathematics classrooms consists of working on a tasks, activities, or problems. For example, an analysis of 100 eighth grade lessons revealed that the delivery of content "was accomplished primarily by working through problems" (NCES, 2003, p. 144). Second, the tasks with which students engage determines what they learn about mathematics and how they learn it. According to Doyle (1983, p.161), "tasks influence learners by directing their attention to particular aspects of content and by specifying ways of processing information." Finally, tasks are a way of framing how researchers have looked at the enactment of curriculum (e.g., Doyle 1983 1988). And, because a discussion of tasks typically implicates many of the dimensions discussed above (i.e., discourse, questioning, authority structures, normative practices), we are able to discuss many of these attributes of implementation under the overall umbrella of task.

The Source and Nature of Mathematical Tasks

We define mathematical tasks as classroom activities the purpose of which is to focus students' attention on a particular mathematical idea. Tasks include expectations regarding what students are expected to produce, how they are expected to produce it, and the resources available for so doing (Doyle, 1988; Stein et al., 1996). Curricular materials or textbooks are the main source of mathematical tasks used by teachers for classroom instruction. An analysis of responses to teacher questionnaires, administered under the auspices of the National Assessment of Educational Progress in 2000, revealed that more than two-thirds of students at grades 4 and 8 had teachers who reported that students worked on problems from the textbook on a daily basis (Grouws et al., 2004). An additional 20 percent of students at both grades 4 and 8 had teachers who reported using tasks from the textbook once or twice a week. Although these percentages represented a significant decrease in textbook use from 1992 (Grouws et al., 2004), these data continue to

Task A Martha's Carpeting Task	Task B The Fencing Task
Martha was recarpeting her bedroom, which was 15 feet long and 10 feet wide. How many square feet of carpeting will she need to purchase?	Ms. Brown's class will raise rabbits for their spring science fair. They have 24 feet of fencing with which to build a rectangular pen to keep the rabbits. - If Ms. Brown's students want their rabbits to have as much room as possible, how long would each of the sides of the pen have to be? - How long would each of the sides be if they had only 16 feet of fencing? - How would you go about determining the pen with the most room for any amount of fencing? Organize your work so that someone else who reads it will understand it.

Figure 8.3 Two different mathematical tasks that involve the calculation of area (Stein et al., 2000, pp. 1–2)

make salient the centrality of textbooks in classroom instruction. Data from the most recent TIMSS video study provide additional evidence of the centrality of textbooks. Specifically, 98% of the lessons from the U.S. analyzed in this study used a textbook or a worksheet (NCES, 2003).

As we have discussed earlier in this chapter, curricular materials vary considerably with respect to the nature of the mathematical tasks that are found within them. For example, in comparison to conventional curricula, standards-based curricula have more contextual problems, more tasks that can be solved by using a range of strategies, and fewer exercises that require only application of procedures (Senk & Thompson, 2003). Making distinctions between tasks is important because all tasks do not provide the same opportunities for students' thinking and learning. Tasks that ask students to perform a memorized procedure in a routine manner lead to one type of opportunity for student thinking; tasks that demand engagement with concepts and that stimulate students to make connections lead to a different set of opportunities for student thinking.

Consider, for example, the two tasks shown in Figure 8.3. Task A (Martha's Carpeting Task) requires the simple application of the formula for finding the area of a rectangle (i.e., $A = l \times w$, where l and w refer to the length and width of a rectangle respectively). There is no ambiguity about what needs to be done in order to solve this problem or how one should go about doing it. Therefore it can be viewed as a task with low-level cognitive demands. By contrast in order to solve Task B (The Fencing Task), students must do more than apply a formula in order to be successful. They must generate and test different pen configurations in order to identify the pen that yields the maximum area using two different lengths of fencing (first 24 and

then 16 feet). Students must then generalize beyond the two cases in order to determine the dimensions of the pen that will yield the largest area for *any* amount of fence. As a result, Task B can be viewed as a task with high-level cognitive demands. Students must construct their own approach for solving the task as none is obvious or prescribed. Although the area formula may be used to determine the area of various pen configurations, the task also requires students to engage in additional mathematical thinking and reasoning that includes looking for patterns, making and testing conjectures, and finding a generalization that can be explained and justified.

Hence, the tasks with which students become engaged in the classroom form the basis of their opportunities to learn *what* mathematics is and *how* one does it (Doyle, 1983, 1988). What students can learn from problems such as Martha's Carpeting Task is significantly different from what they learn from the Fencing Task. Over time, the cumulative effect of classroom-based tasks is students' implicit development of ideas about the nature of mathematics—about whether mathematics is something that they personally can make sense of, and how long and how hard they should have to work to do so.

Since the introduction of the *Curriculum and Evaluation Standards for School Mathematics* (NCTM, 1989) there has been an attempt to describe the types of tasks that promote and develop students' understanding of mathematical concepts and ideas. Although these descriptions differ with respect to the actual language used to describe the nature of the activities in which students should engage, they share a common focus on developing students' capacity for non-algorithmic thinking. For example, *worthwhile mathematical tasks* (NCTM, 1991) are described as "ones that do not separate mathematical thinking from mathemati-

cal concepts or skills, that capture students' curiosity, and that invite them to speculate and to pursue their hunches (p. 25)." Hiebert and his colleagues (Hiebert et al., 1997) have argued that in order for students to build mathematical understandings, the tasks with which they engage "must allow the students to treat the situations as problematic, as something they need to think about rather than as a prescription they need to follow" (p. 18). Such experiences, in their view, leave behind important insights into the structure of mathematics and strategies for solving classes of problems. More recently, Horn (2005, p. 22) has used the term "groupworthy" to describe tasks that "illustrate important mathematical concepts, allow for multiple representations, include tasks that draw effectively on the collective resources of a groups, and have several possible solution paths." While similar in some ways to the task descriptions provided by NCTM and Hiebert and his colleagues, Horn's notion of groupworthy tasks also makes salient the need to provide students who are working collaboratively—an organizational structure often found in classrooms that use standards-based curricula—with tasks that allow for multiple entry points so that group members who bring different strengths to the table can contribute to solving the task at hand.

Stein and her colleagues (Henningsen & Stein, 1997; Smith & Stein, 1998; Stein et al., 2000; Stein et al., 1996; Stein & Lane, 1996; Stein & Smith, 1998) have provided a taxonomy of mathematical tasks based on the kind and level of thinking required to solve them. From this perspective, mathematical tasks are viewed as placing *high-level* cognitive demands on students when they appear to engage students in the processes of active inquiry and validation (what Stein and colleagues refer to as "doing mathematics") or encourage them to use procedures (broadly defined to include standard and nonstandard procedures, formulas, and algorithms) in ways that are meaningfully connected to concepts or understanding. Tasks that encourage students to use procedures, formulas, or algorithms in ways that are not actively linked to meaning; or that consist primarily of memorization or the reproduction of previously memorized facts are viewed as placing *lower-level* cognitive demands on students. Examples of tasks that would be classified at each of the four levels of cognitive demand in the Stein et al. taxonomy are shown in Figure 8.4. In the TIMSS Video Study (NCES, 2003; Stigler & Hiebert, 2004) a similar distinction was made between low-level tasks that can be solved with basic computation and procedures (*using procedures*) and high-level tasks that focus on concepts and connections among mathematical ideas (*making connections*).

In addition to providing a taxonomy for classifying mathematical tasks, Stein and her colleagues

also provide a framework for tracking the cognitive demands of mathematical tasks as they unfold during instruction and for exploring the connection between instruction and student learning. The Mathematical Tasks Framework (MTF) distinguishes three phases through which tasks pass as they unfold during a lesson (Stein et al., 1996): First, as they appear in curricular or instructional materials (i.e., on the printed pages of textbooks, ancillary materials, or as created by teachers); next, as they are set up or announced by the teacher; and finally, as they are actually enacted by students and the teacher in the classroom—in other words, the way in which students actually go about working on the task. All of these, but especially the enactment phase, are viewed as important influences on what students actually learn.

This research suggests that the cognitive demands of a task can change as the task unfolds. For example, the task that appears in curricular materials is not always identical to the task that is set up by the teacher in the classroom (reflecting the interpretive processes discussed in Section Two) and, this task, in turn, is not always identical to the task that the students actually do. Between the set-up and implementation phases, tasks can transform for a variety of reasons, many of which have to do with the press of the classroom environment. It is interesting to note, however, that tasks that are set-up at a low-level are nearly always implemented as intended, a point to which we will return (Stein et al., 1996).

In the remainder of this section, we use the Mathematical Tasks Framework as an organizing frame to discuss large and small-scale studies that provide insights into the ways in which mathematics tasks are enacted in the classroom.

Setting Up and Implementing Mathematical Tasks

Research has shown that maintaining high-level cognitive demands during implementation is much more difficult than selecting and setting up high-level tasks. In a study involving 144 lessons selected to be representative of instruction at four middle schools that participated in the QUASAR Project (Stein et al., 1996; Henningsen & Stein, 1997), three-quarters of the tasks were set-up in ways that demanded that students either engage in the active processes of "doing mathematics" or use procedures with connections to concepts, meaning, or understanding (Stein et al., 1996). The study also showed, however, that the higher the demands that a task placed on students at the set-up phase, the less likely it was for the task to be carried out at that level during the implementa-

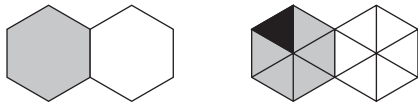
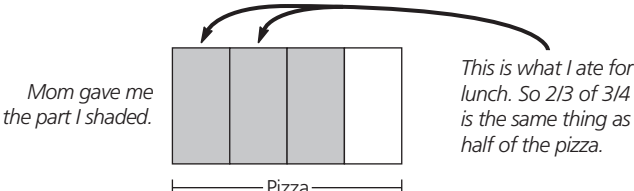
Lower-Level Demands	Higher-Level Demands
Memorization What is the rule for multiplying fractions? <i>Expected Student Response:</i> You multiply the numerator times the numerator and the denominator times the denominator OR You multiply the two top numbers and then the two bottom numbers.	Procedures with Connections Find $1/6$ of $1/2$. Use pattern blocks. Draw your answer and explain your solution. <i>Expected Student Response:</i>  First you take half of the whole which would be one hexagon. Then you take one-sixth of the half. So I divided the hexagon into six pieces which would be six triangles. I only needed one-sixth so that would be one triangle. Then I needed to figure out what part of the two hexagons one triangle was and it was 1 out of 12. So $1/6$ of $1/2$ is $1/12$.
Procedures without Connections Multiply: $\frac{2}{3} \times \frac{3}{4}$ $\frac{5}{6} \times \frac{7}{8}$ $\frac{4}{9} \times \frac{3}{5}$ <i>Expected Student Response:</i> $\frac{2}{3} \times \frac{3}{4} = \frac{2 \times 3}{3 \times 4} = \frac{6}{12}$ $\frac{5}{6} \times \frac{7}{8} = \frac{5 \times 7}{6 \times 8} = \frac{35}{48}$ $\frac{4}{9} \times \frac{3}{5} = \frac{4 \times 3}{9 \times 5} = \frac{12}{45}$	Doing Mathematics Create a real-world situation for the following problem: $\frac{2}{3} \times \frac{3}{4}$ Solve the problem you have created without using the rule and explain your solution. <i>One Possible Student Response:</i> For lunch, Mom gave me three-fourths of the pizza that we ordered. I could only finish two-thirds of what she gave me. How much of the whole pizza did I eat? I drew a rectangle to show the whole pizza. Then I cut it into fourths and shaded three of them to show the part Mom gave me. Since I only ate two thirds of what she gave me, that would be only two of the shaded sections. 

Figure 8.4. Examples of tasks at each of the four levels of cognitive demand.

tion phase. Indeed the kind of tasks that reformers have suggested as most essential for building students' capacities to think and reason mathematically (high-level tasks) appear to be the very tasks that students and teachers have the most difficulty carrying out in a consistent manner. Overall, tasks that were set up to require a high level of cognitive engagement were found to decline into less demanding student engagement more than half of the time (in 62 out of 107 tasks or 58%).

The difficulty teachers in the United States have enacting tasks at a high level is echoed in the results of the recent TIMMS video study. In this study, a random sample of 100 8th grade mathematics classes from each of seven countries was videotaped during the 1999 school year. Although 17 percent of the problems used by teachers in the U.S. were coded as high level (*making connections*), none of these problems was implemented as intended (NCES, 2003; Stigler

& Hiebert, 2004). Instead, most of the making-connections problems were transformed into procedural exercises. Hence, the authors concluded, U.S. 8th grade students spend most of their time in mathematics classrooms practicing procedures regardless of the nature of the tasks they are given.

The *Inside the Classroom* study, conducted by Horizon Research, provides additional insight into the nature of mathematics instruction in the United States (Weiss & Pasley, 2004). In a study of 364 lessons, drawn from representative K–12 mathematics and science classrooms, only 15 percent of the lessons “were structured and implemented in a manner that engaged students with important mathematics or science concepts” and were viewed as “very likely to enhance student understanding of these concepts and to develop their capacity to do mathematics or science successfully” (p. 25). The majority of lessons (59%) were rated as low quality and were deemed unlikely

to enhance students' understanding of mathematics or science concepts. Interestingly, the authors also report that the quality of lessons was not associated with a particular approach to teaching—both reform oriented and traditional lessons were found in each of the three categories (i.e., low quality, medium quality, high quality).

Case studies of teachers attempting to use challenging tasks and curricula in their classrooms provide insights into what actually occurs as these tasks are enacted by teachers and students during instruction. In their study of the early efforts of teachers to implement mathematics instructional reform in California, EPPS researchers observed an odd combination of old ways and new practices coexisting (Cohen et al., 1990). This was most evident in the case of Mrs. O who on the surface appeared to embrace reform, but a closer look revealed that not much had changed (Cohen, 1990). While Mrs. O used innovative instructional materials and activities that were designed to help students make sense of mathematics, she used them in a way “that conveyed a sense of mathematics as a fixed body of knowledge of right answers rather than as a field of inquiry” (Cohen, 1991, p. 19). Students were not asked to explain correct or incorrect solutions or the strategies used to derive them and there was no communication among students.

The *mélange* of old methods and new materials that characterized Mrs. O's practice highlights the challenges faced by teachers as they have endeavored to use new tasks and curricula in their classrooms and to take on new roles in the classroom. Other studies provide insights into the nature of the challenges faced by teachers who have attempted to use new tasks in new ways. For example, teachers struggle with relinquishing authority in the classroom (Wilson & Goldenberg, 1998; Wilson & Lloyd, 2000; Wood, Cobb, & Yackel 1991), ensuring that students feel successful as they work on more challenging mathematical tasks (Smith, 2000), knowing when to ask questions and when to provide information (Romagnano, 1994), and providing an appropriate amount of support and structure (Lloyd, 1999). These studies and others make it clear that taking on these new roles and responsibilities is not easy for teachers. As EPPS researchers (Cohen et al., 1990, p.163) have concluded, “changing one's teaching is not like changing one's socks”—but rather it requires a deep-seated change in belief about what it means to teach and learn mathematics.” In Section Four, we discuss the role that teachers' beliefs and other mediating factors play in influencing the transformation of the written curriculum to the intended curriculum and the intended curriculum to the enacted curriculum.

Investigating Processes Involved in Task Implementation

Here we turn our attention to classroom processes that influence what teachers and students actually *do* during classroom instruction. What students do during instruction depends, to some extent, on the normative practices that have been agreed upon either explicitly or implicitly by the students and teacher. According to Carpenter and Lehrer (1999, p. 26), “although the selection of appropriate tasks and tools can facilitate the development of understanding, the normative practices of a class determine whether they will be used for that purpose.” Hence we confine our discussion here to what happens during task implementation—the actions and interactions of teachers and students as they work on mathematical tasks. In Section Four, we discuss the factors (shown in the oval in Figure 8.1) that influence these interpretive processes and in so doing provide plausible explanations for the transformations that often occur during the task implementation phase.

In the previously cited study of QUASAR Project middle schools (Henningsen & Stein, 1997; Stein et al., 1996), a handful of patterns emerged that capture characteristic ways in which high-level tasks unfold in the classrooms of teachers who are attempting to reform their instruction so as to be more compatible with the *Standards* (NCTM, 1989, 2000). These patterns and the processes associated with them are described below.

Some tasks that were set up to place high levels of cognitive demand on student thinking were indeed implemented in such a way that students thought and reasoned in complex and meaningful ways. When this happened, teachers generally drew on a number of processes in order to support students' engagement in a way that maintained the demands of the task. As shown in the second column of Table 8.2, these processes include scaffolding of student thinking (i.e., assisting student thinking in a manner that preserves task complexity), modeling of high-level thinking and reasoning by the teacher or more capable peers, pressing for explanation and meaning, and selecting tasks that built on students' prior knowledge.

Other tasks that were set up to place high levels of cognitive demand on students' thinking, however, exhibited declines in terms of how students actually went about working on them. When tasks declined, a different set of processes tended to be operating in the classroom environment (see first column in Table 8.2). These processes involved a variety of teacher, student, and task-related conditions, actions, and norms. Although tasks can decline in several different ways,

Table 8.2 Processes Associated with Maintenance and Decline of High-Level Cognitive Demands

Processes Associated with the Decline of High-Level Cognitive Demands	Processes Associated with the Maintenance of High-Level Cognitive Demands
Routinizing problematic aspects of the task	Scaffolding of student thinking and reasoning
Shifting the emphasis from meaning, concepts, or understanding to the correctness or completeness of the answer	Providing a means by which students can monitor their own progress
Providing insufficient time to wrestle with the demanding aspects of the task or so much time that students drift into off-task behavior	Modeling of high-level performance by teacher or capable students
Engaging in high-level cognitive activities is prevented due to classroom management problems	Pressing for justifications, explanations, and/or meaning through questioning, comments, and/or feedback
Selecting a task that is inappropriate for a given group of students	Selecting tasks that build on students' prior knowledge
Failing to hold students accountable for high-level products or processes	Drawing frequent conceptual connections
	Providing sufficient time to explore

one common pattern is into tasks in which students use procedures without connections to meaning. (See Henningsen & Stein, 1997 for a examples of patterns of maintenance and decline of high level tasks.)

Instead of engaging deeply and meaningfully with the mathematics, students ended up utilizing a more procedural, often mechanical and shallow approach to the task. In this type of decline, one of the most prevalent factors operating in the environment was teachers' "taking over" and doing the challenging aspects of the tasks for the students. High-level tasks tend to be less structured, more difficult, and longer than the kinds of tasks to which students are typically exposed. Students often perceive these types of tasks as ambiguous and/or risky because it is not apparent what they should do, how they should do it, and how their work will be evaluated (Doyle 1988; Romagnano, 1994). In order to deal with the discomfort that surrounds this uncertainty, students often urge teachers to make these types of tasks more explicit by breaking them down into smaller steps, specifying exact procedures to be followed, or actually doing parts of the task for them. Should the teacher succumb to such requests, the challenging, sense-making aspects of the task are reduced or eliminated, thereby robbing students of the opportunity to develop thinking and reasoning skills and meaningful mathematical understandings.

A review of the processes in Table 8.2 suggests the central role of teachers' questions to the outcome of a lesson. Asking questions that scaffold or support students' continued engagement with a task and that press students to explain and justify their thinking are key to sustaining the cognitive demands of mathematical tasks. Alternatively, failure to ask questions that go beyond the correct answer or that communicate

accountability for thinking and reasoning are associated with the decline of high level tasks. The nature of teachers' questioning and its impact on students' opportunities to learn are highlighted in both large and small-scale studies. For example, in the study of 350 representative mathematics and science lessons, Weiss and Pasley (2004) indicated that effective questioning was key in helping students develop an understanding of mathematical concepts and in making connections, yet rare in the observed classes. According to the authors, such scaffolding allows teachers to monitor students' understanding and encourages students to thinking more deeply and critically. The study suggests, however, that teachers more frequently asked questions that focus on getting the correct answer rather than on helping students make sense of mathematics concepts.

In a comparative study of mathematics teachers in three high schools using standards-based vs. conventional curricula, Boaler and Brodie (2004) found that teachers using standards-based curricula asked a broader range of questions than those who used a conventional curricula. In particular, nearly all of the questions (95%) asked by the teachers using conventional curricula fell into the category of procedural questions. While teachers using standards-based curricula also asked many procedural questions (60–75% of the questions asked were categorized as such) they also asked a large percentage of questions that were classified as conceptual and probing. Boaler and Brodie (2004, p. 780) argue that teachers' questions play an important role "in shaping the nature of classroom environments and the mathematical terrain that is traversed . . . and the cognitive opportunities offered to students."

Several case studies also point to the critical relationship between questioning and maintaining the integrity of the task (e.g., Smith, 2000; Wood et al., 1991). For example, in a study of a second grade teacher who was implementing standards-based mathematics materials, Wood et al. (1991) note that initially the teacher was unsure how to deal with the various procedures, solutions, and incorrect answers that students came up with during their explorations. The teacher's initial attempt to deal with this challenge resulted in beginning with a student's incorrect response and leading the student through a series of questions that would ultimately lead to the answer the teacher had in mind. According to Wood et al. (1991), "as the teacher reflected on the use of this approach later, she recognized that attempting to guide students in this manner was just another way for the teacher to ensure students get the right answer, whether they understand how to or not" (p. 604).

Related to the issue of what *type* of question to ask is the challenge of determining *when* to let students struggle without teacher input, *when* to ask leading questions, and *when* to tell students something directly (NCTM, 1991). These decisions are at the heart of what Romagnano (1994) refers to as "The Ask Them or Tell Them Dilemma." This dilemma grew out of the pull and tug of when to withhold and when to provide information to students. Romagnano found that it was the frustration and lack of clarity resulting from both the nondirective questions and the use of more complex problems that caused students to disengage. While consistent with the findings regarding the implementation of novel tasks (Doyle, 1988), this presented a dilemma to Romagnano, the teacher with whom he worked, and to many other teachers who are engaged in such work. Romagnano indicated that, "being directive was a problem because I knew that when I was that way, I reduced the chances that any significant mathematics would be learned" (1994, p. 123).

These studies raise important issues about the teacher's role in "discussion-intensive teaching" (Chazan & Ball, 1999) that is often intertwined with the use of the kind of high-level tasks found in standards-based curricula. Although teachers are told *not* to tell, they are given no explicit direction regarding what they could or should do instead. This leaves teachers struggling to determine *how* to support students without taking away the challenge *before* they completely disengage from activity in frustration. Chazan and Ball (1999, p. 9) argue that for teachers to direct a discussion productively, they must have "a repertoire of ways to add, stir, slow, redirect the class's work."

In their study of teachers at Railside, Boaler and Staples (in press) indicate that a feature of the curriculum—one that could not be seen in a review of the curricular materials themselves—was the follow-up questions that teachers asked students. According to the authors, these questions "shaped the course of implementation" (p. 19). The questions, written prior to teaching the lesson, provided teachers with specific ways of focusing students on the key mathematical ideas in the lesson leaving less "in the moment" decision-making regarding the next instructional move.

In this section, we have reviewed the changes that often occur once curricular tasks are unleashed in real classroom settings. By adding new dimensions to our understanding of how curricula are transformed over time, they provide further building blocks in our evolving chain of logic regarding how students learn from curriculum. The studies reviewed in this section suggest, once again, that the introduction of human interaction with curriculum materials brings variation to the information and styles of learning to which students will be exposed. In the previous section, the variation was introduced by the interpretive processes of the teacher "reading" the curriculum materials. In this section, variation is introduced by the flux of the classroom life—a space comprised of students as well as teachers. Instead of just one person interpreting the curriculum (the teacher as he or she plans), there is now a roomful of students interpreting the curriculum tasks and materials and bouncing off of each others' interpretations. As the teacher responds to these interpretations, the task is often transformed into a learning opportunity that is quite different from what was intended by the curriculum developers.

Interestingly, although not the specific focus on any study of which we are aware, the conceptual ideas and empirical findings presented herein suggest that one might expect less variation in the classroom-based enactment of conventional as opposed to standards-based curricula. Most of the studies cited above were conducted in classrooms in which teachers were using standards-based curricula or similarly inspired reform materials. When teachers set up instructional tasks in the way outlined by the materials, the stage was set for the study of how teachers and students enact high-level task (since these kinds of tasks comprise the majority of activities in standards-based materials). However, as noted earlier, low level tasks tended to implemented as intended (Stein et al., 1996). This is because there is much less ambiguity regarding what to do and how to do it. Thus, one would expect less variation in how teachers enact tasks that are found in conventional curricula (since these kinds of tasks—procedural and

definitional activities—comprise the majority of conventional curricula).

This observation reinforces the point made at the end of the previous section regarding the concept of fidelity. There, we noted that the implementation of curricular tasks with fidelity would be expected to be more attainable and measurable with conventional as opposed to standards based curricula. Here we have extended the idea to encompass the notion that enactment of standards-based curricula can logically be expected to lead to greater variability than the enactment of conventional curricula. This is important because—as noted throughout this chapter—it is not the materials themselves but rather how students' experience the materials in the classroom that determines what students will learn. If standards-based classrooms have more variable implementation, then predicting what students will learn from them becomes more difficult.

SECTION FOUR: EXPLAINING TRANSFORMATIONS WITHIN AND BETWEEN DIFFERENT PHASES OF CURRICULUM USE

The preceding sections detail some of the transformations that occur between the written and intended curriculum and within the enactment phase (see Figure 8.1). Research on mathematics teaching and teachers' use of curriculum materials highlights the ways teachers read and interpret written curriculum materials differently (Section Two) and planned tasks and instructional designs are reconstructed by students and teachers as they unfold in real-time lessons (Section Three). In addition to identifying curricular transformations, much of the research seeks to explain them by identifying factors that influence transformations. The teacher figures most prominently in these explanations. However, much of this literature recognizes that the reasons underlying the ways teachers and students transform curriculum are multi-faceted and complex. In addition to individual teacher characteristics, scholars have identified students, the teaching context, and the curriculum itself to be influential in these processes. In the following sections, we discuss the kinds of factors that have been identified in the research. Rather than providing an extensive review of this research, our aim is to delineate those factors that appear influential across the body of studies. The specific studies cited are offered as illustrations.

The Teacher Matters

Studies that examine the way that teacher-related factors influence how teachers use materials in planning and enacting curriculum are most prominent in the literature. The factors most studied include teachers' beliefs about mathematics, teaching, and learning; their knowledge of mathematics; and their experiences as teachers and as students. More recently researchers have begun to conceptualize additional influencing factors, including teacher identity and their stance or orientation toward curriculum materials.

Beliefs and Knowledge

The research that focuses on the role that teachers' beliefs and knowledge play in teachers' curriculum use is part of a larger body of research on the influences of beliefs and knowledge on teaching practices more generally. In fact, the 1992 *Handbook* devoted an entire chapter to the role of teachers' beliefs in mathematics teaching (Thompson, 1992) and a second chapter to the role of teacher knowledge (Fennema & Franke, 1992). The research reviewed in these two chapters demonstrates the critical role teachers' beliefs about mathematics, teaching, and learning and their knowledge of mathematics and students' learning play in influencing their pedagogical and curricular decisions. We limit our discussion here to the portion of this research that is specific to curriculum use. A substantial portion of this research focuses on a) teachers' beliefs about the nature of mathematics and the nature of knowledge, b) teachers' goals for students and their beliefs about purposes of school, and c) teachers' mathematical knowledge. In most cases, these beliefs are tightly intertwined with one another and cannot be examined in isolation (Putnam, Heaton, Prawat, & Remillard, 1992). In the following sections we discuss examples of studies that illustrate how these beliefs influence curriculum use.

Examinations of teachers' beliefs about mathematics, how it is learned, and the nature of mathematical knowledge have figured prominently in studies of teachers' use of standards-based curriculum because these materials offer views of mathematics that conflict with those typically held in mainstream culture. Indeed, much of this research illustrates the ways that teachers' beliefs about mathematics and how it is learned influence how they interpret and use curriculum materials. This influence is particularly evident when conflicts exist between teachers' beliefs and the ideas embraced by the curriculum designers. A number of researchers have pointed to the deeply rooted nature of teachers' beliefs about mathematics (Wilson & Goldenberg, 1998) as an explanation for both sub-

stantial departures from and resistance to suggestions offered by standards-based curricula (Chavez, 2003; Lloyd & Wilson, 1998; Romberg, 1997). Other studies have attributed different teachers' uses of the same curriculum to substantial differences in the teachers' beliefs about mathematics and how students learn it (e.g., Collopy, 2003; Lloyd, 1999; Remillard, 1999).

While discussions of teachers' beliefs about mathematics permeate research on teachers' use of curriculum resources, there is general agreement that they should not be viewed in isolation. Putnam et al., (1992) used case studies of elementary teachers' responses to new curricula and new state policy, to illustrate how teachers' beliefs about mathematics were tightly intertwined with their beliefs about knowledge, teaching, and learning more generally. Other researchers have identified additional aspects of curriculum materials that conflict with beliefs commonly held amongst teachers. Stephens (1982) and Donovan (1983), for instance, illustrated how conflicts between teachers' beliefs about classroom control and management and purposes of school and those ideas represented in an innovative curriculum led teachers to ignore or change key features in the curriculum.

Limits in teachers' mathematical knowledge are frequently used to explain the ways that teachers use curriculum materials, particularly materials designed to foster understandings of concepts and relationships underlying standard mathematical procedures. Heaton (1992) and Cohen (1990) provide two examples of elementary teachers failing to appreciate the mathematical complexity underlying activities suggested in curriculum. Both teachers they report on used interactive mathematics tasks drawn from standards-based curriculum programs without developing the underlying mathematical ideas. Similarly, in their two-year study of two seventh grade mathematics teachers using a standards-based, middle-school curriculum, Manouchehri and Goodman (1998) found the teachers' mathematical knowledge to be the greatest influence on how they used the curriculum.

Orientation

Factors other than teachers' beliefs about mathematics teaching and learning have been found to influence their use of materials. Some studies indicate that teachers' perceptions of curriculum materials or textbooks as a particular genre of teacher resource influence how they engage and use them. In a study of two teacher education programs, Ball and Fei-

man-Nemser (1988) found that the majority of pre-service teachers tended to associate textbooks with traditional and undesirable teaching practices. As a result, they were reluctant to draw on these resources when planning mathematics lessons. Remillard and Bryans (2004) used the term "orientation" to refer to a teacher's stance toward curriculum materials in teaching. These researchers found this orientation to be influential in how and whether teachers used the curriculum regardless of the match or mismatch of the ideas in the curriculum to the teacher's view about mathematics teaching.

Professional Identity

Researchers have also found that teacher's professional identity, or how they see themselves and their roles as teachers, influences curriculum use. Spillane (2000) refers to teachers' identity as "who they are, their sense of self, and their habits of mind . . . an individual's way of understanding and being in the world" (p. 308). The concept of identity builds on and extends the large body of work that considers how teachers' knowledge and beliefs influence their curricular decision making and practice. Identity, as a construct, integrates what one knows, feels and is inclined to do in a particular context into an inseparable whole, making activity, rather than knowledge, central. While knowledge is often viewed as static, identity is active. Further, the notion of identity highlights the contextual and social nature of knowing and being. Identity is a social construct that develops in relation to others and particular contexts.¹⁴ "Identities develop in and through social practices" (Boaler & Greeno, 2000, p. 173). That is, teachers' professional identities are produced in relation to their work as teachers in classrooms with students. And, as Spillane (2000) and Drake and Sherin (2006) suggest, identities are subject-matter specific. The role the particular subject matter plays in a teacher's professional identity is, perhaps, best illustrated by Spillane's case of an elementary teacher who routinely facilitated inquiry-oriented discussions amongst her fifth graders in the area of reading, writing, and grammar. Her mathematics instruction, on the other hand, tended to focus on directing students to memorize and follow procedural rules.

A significant component of teachers' professional identity is how they construct their roles in relation to students. Many teachers, for instance, associate good teaching with telling and showing students what to do,

¹⁴ For organizational purposes, we have placed professional identity under the heading of the teacher. However, as this discussion suggests, identity is a social and context-specific phenomenon and, thus, embraces influential factors discussed in the following sections, including students, context, and curriculum materials.

whereas, standards-based curriculum materials outline a much less didactic role for the teacher. J. Smith's (1996) analysis of the fundamental conflicts between factors that tend to positively influence teachers' sense of their own efficacy, as outlined by self-efficacy research, and the view of teaching promoted by the NCTM Standards helps to explain teachers' curriculum transformations. Frykholm (2004) found that using curriculum materials that conflicted with their ideas about good teaching resulted in considerable discomfort for teachers. He argued that using standards-based curriculum materials required teachers to develop tolerance for this discomfort. Lloyd (1999) observed a different kind of conflict between a teacher's identity and standards-based curriculum materials. One of the teachers in her study of teachers using a high school curriculum viewed herself as an innovative, project-based mathematics teacher and, consequently, found the curriculum overly restricting and difficult to follow closely.

Students Matter

A number of studies have revealed that students' responses to standards-based curricula appear to be influential in how teachers use new materials. Teachers tend to be concerned about the high level of independent thought, problem solving, and self monitoring demanded of students by the tasks found in standards-based curricula, expect that students cannot manage these demands, and consequently restructure or adapt the lessons to make them less complex and more readily accessible to students. As described previously, Stein et al. (1996) observed teachers reducing the complexity of tasks during instruction in response to students' struggles with high demand tasks. M. Smith (2000) associated the tendency to reduce task demand to teachers' concerns about student success. As the previous discussion on self-efficacy indicated, teachers' reluctance to allow students to struggle is associated with how they define and assess their role as teacher.

Some researchers have suggested that teachers' responses to challenging curriculum materials can be associated with perceived rather than actual concerns about students' difficulties. Wilson and Lloyd (2000) also observed that teachers' curricular decisions were often influenced by the "invisible hand of students" (Borasi, 1990). However, they found no evidence that the students in the classrooms they observed were having the kind of difficulties identified by the teachers. They suggest that "the invisible hand was teacher *perception* of student resistance, not student resistance *per se*" (p. 28).

A number of researchers have identified the challenges teachers face in sharing intellectual author-

ity with students as another influential factor in how teachers use curriculum materials. As indicated earlier, standards-based curriculum materials include tasks and teaching recommendations that offer students a degree of autonomy and decision making that is not typical in traditional teaching practices in the United States. Many teachers, however, are not comfortable sharing authority with students and implement curricular suggestions in ways that are more structured and directive than recommended (Wilson & Goldenberg, 1998; Wilson & Lloyd, 2000) and in ways that provide a level of support and structure they deem necessary (Lloyd, 1999; Stein et al., 1996).

The Context Matters

Many studies have also identified the teaching context as an important factor in teachers' use of curriculum materials. Most of these studies highlight the ways features of the context constrain teachers from using the curriculum as intended by the developers or as fully as they would like. Some studies demonstrate the ways that elements of the context enable curriculum use. Some contextual features that figure prominently in this body of research include time, local cultures, and the extent and nature of teacher support.

Time

Limits on time for planning and to devote to mathematics instruction have both been identified as factors that influence how teachers use curriculum materials. Many of the standards-based curricula assume that a significant amount of time will be devoted to mathematics lesson—90 minutes or more. Moreover, the complexity of the mathematics and the instructional recommendations require that teachers spend more time preparing than they are accustomed to. In a study of middle school teachers using *CMP*, Keiser and Lambdin (1996) identify a number of issues related to time that influence how teachers use their materials. They identify conflicts between the time available for instruction and planning and the amount of time required. They also examined how teachers used time both in planning and during instruction. They argue that using standards-based curricula as intended requires flexibility in class scheduling and timing, a luxury infrequently afforded teachers.

Local Cultures

A number of studies have pointed to the ways in which the departmental, district, school, or community culture influences teachers' use of curriculum. This is particularly the case for standards-based curricula where a mismatch with the local norms is likely. As

mentioned above, these conflicts emerge around students' and teachers' roles during mathematics class, sharing authority with students, and norms for behavior (Floden et al., 1981; Lloyd, 1999; Manoucheri & Goodman, 1998; Wilson & Lloyd, 2000; Wood et al., 1991; Cobb, & Yackel 1991; Floden, et al., 1981).

Some researchers have focused attention on the norms of the school as a workplace for teacher learning. As noted earlier, the introduction of standards-based materials challenges what teachers know and how they think about mathematics, as well as their thinking regarding how students learn mathematics and what their role as teachers should be in the classroom. Research on teacher change suggests that teachers are more apt to change their approach to instruction when supported by a group of colleagues who are also struggling to learn innovative approaches to instruction (Coburn, 2001; Spillane, 1999). Some researchers who have explored the role of teacher-teacher interaction in supporting teacher reform in mathematics have utilized sociocultural concepts such as "communities of practice" to explain the ways in which groups of teachers influence each others' learning and adaptation to new forms of teaching (Cobb et al., 2003; Franke & Kazemi, 2001; Stein et al., 1998). More recently, Stein, Coburn and colleagues have utilized social capital as a theoretical construct to gain purchase on how teacher-teacher interaction influences teachers' experiences with the implementation of innovative curricula (Stein & Coburn, 2005).

Finally, since the introduction of NCLB, various policy-related factors have impacted what gets taught and how. The consequences associated with performing poorly on state-mandated tests, for example, have led to the introduction of pacing guides and other methods that McNeil along with others have suggested "narrow" the curriculum (McNeill, 2002). In addition, many districts have either begun with the reform of literacy instruction or have invested the greatest proportion of their resources in literacy. As such, the improvement of mathematics instruction often takes a back seat in overall school and district operations (Stein, Toure, Acquerelli, & Ferguson, 2003).

Teacher Support

A number of studies of teachers using standards-based curriculum materials report on the role of professional support in enabling and influencing teachers' work with the curriculum (Davenport, 2000; Remillard, 2000; Van Zoest & Bohl, 2002). Most of these studies align with what is known about the features of effective professional development in general. For example, effective professional development is content-based, promotes active learning, and is perceived by

teachers to be part of an overall coherent program of teacher learning (Garet, Porter, Desimone, Birman, & Yoon, 2001). An additional feature that is often cited as critical in mathematics teacher professional development is the opportunity for teachers to engage in mathematics thinking and reasoning as learners, often exploring the same or similar tasks to what appears in the curriculum materials for students (Smith, 2001).

The Curriculum Matters

One factor that has the potential to influence transformations between written, intended, and enacted curriculum that has received minimal attention in the literature is the curriculum itself. How do characteristics or features of the particular curriculum influence how a teacher uses it? In order to consider this question, researchers must begin with analyses of curriculum materials themselves. This is an underdeveloped area of research. However, some of the ways curriculum materials have been characterized for the purpose of analysis include a) broad types, such as conventional versus standards-based, b) particular features or characteristics of the materials, and c) how they are designed to communicate with and educate teachers.

Conventional versus Standards-based Curricula

There is some evidence that teachers use standards-based curriculum materials differently than they use traditional curriculum, although there are few studies that compare use of two different curricula by the same teacher. Herbel-Eisenmann, Lubieniski, and Id Deen (in press) undertook a comparative case study of a middle school math teacher using *Connected Mathematics*, a standards-based curriculum, with one class of students and a conventional textbook with another class. They found that the nature of the instruction was notably different across these two classes with respect to classroom organization and the types of tasks students were asked to do. However, these differences seemed closely related to the teacher's beliefs that students and their parents should have a say in the kind of instruction they received. In this setting, students opted into a traditional or reform track. Respecting their choices, this teacher adjusted her teaching practices to fit the curriculum she was using.

Many teachers are less flexible in their practices and find it difficult to use a curriculum that conflicts with their beliefs about mathematics and how it should be taught. In his analysis of seven studies of teachers using standards-based curriculum, Romberg (1997) noted that the mismatch between the teachers' underlying beliefs about mathematics and learning and

those implicit in the curriculum was a primary factor in how teachers used the curriculum. In his study of 53 teachers using either standards-based or conventional curricula, Chavez (2003) found that teachers using standards-based curricula felt less free to make adaptations and were less likely to see themselves as able to determine course goals and objectives.

Curriculum Features

Recently, scholars have questioned the value of applying such general classifications (standards-based or conventional) to analysis of influential factors in curriculum use, arguing that they mask substantial differences across curricula within categories (NRC, 2004)¹⁵ and fail to provide reliable insights into how particular curricular features influence teachers' use (Remillard, 2005). Brown (2002) argues that understanding how a teacher uses curriculum resources and the resulting classroom practices requires an integrated analysis of the particular curriculum resources, the resources the teacher brings, and how they interact. Little research has been undertaken that examines these interactions, however the evidence provided here suggests this to be a fruitful area for future research.

The importance of taking into account the interactions between teachers and curriculum features is illustrated by Lloyd's (1999) findings from a study of two high school teachers using Core-Plus. One teacher found the curriculum too unstructured and struggled to navigate a curriculum where students' ideas were central. As a result, he tended to provide more support for students and more structure in the lesson than the curriculum suggested. The other teacher found the curriculum too structured for her taste.

Another feature frequently considered to be influential in how teachers use curriculum materials is sequencing—the way the material is ordered in the book. A commonly held assumption is that lessons near the end were more likely to be eliminated than those placed earlier in the book. To consider this hypothesis, Tarr et al., (2006) compared the lessons taught and omitted by 9 teachers using the same conventional textbook, *Glencoe*, and found that placement in the text was less influential in omission decisions than the particular topic of the lesson. Teachers omitted lessons throughout the book. Moreover, they omitted lessons on data analysis and probability at a significantly greater rate than lessons on number and operations or algebra.

Beyond content and sequencing, curriculum designers make a number of critical decisions about how

the materials they produce are structured and communicated to the teacher. Many of the standards-based materials are designed to look substantially different from the conventional texts which were a mainstay in most classrooms in the U.S. We know little about how teachers engage these varied offerings.

Educative Curriculum

One characteristic common to the design of a number of the standards-based curriculum materials is their intent to communicate directly with teachers. Because these materials are relatively new, we know little about how teachers respond to or use these features. There are a few examples in the literature of teachers using these features to enhance their understanding of the purpose of particular lessons and to anticipate students' responses (Davenport, 2000; Rodriguez, 2000). There are also examples of teachers ignoring these features (Remillard & Bryans, 2004; Sherin & Drake, n.d.) or using them in different ways than were intended by the authors (Collopy, 2003). Drawing on research on teacher learning, Davis and Krajcik (2005) offer a set of design principles that might guide the development of educative curriculum materials. However, these principles are yet to be tested in practice.

The work cited in this section suggests possible explanations for why teachers interpret and enact curriculum in particular ways. Briefly, support can be found for factors that reside within the individual teacher, within students (or teachers' perceptions of students), within the context, and within the curriculum itself. These studies and their findings are important because they suggest additional pieces of the conceptual landscape that might be incorporated into both large and small-scale studies of curricular effectiveness.

These studies are also important from a practical perspective. The explanations that are offered for how teachers interpret and enact curriculum have primarily been generated from studies in which teachers have been attempting to adjust their instruction to be aligned with standards-based approaches to teaching and learning. Thus, if variations in curricular effectiveness are related to these factors, they suggest potential leverage points for interventions aimed at assisting teachers to enact standards-based curricula in ways that are more aligned with developers' intentions.

Finally, we believe that the study of curriculum materials and the manner in which they interact with the resources that teachers bring to the table (Brown,

¹⁵ See Stein and Kim (2006) for an exception; this study investigates similarities and differences between two standards-based curricula.

2002) deserves more attention. In this perspective, curriculum materials are viewed as tools that can enable or constrain teachers' thinking and actions (Wertsch, 1998). Research conducted from this perspective in science education has uncovered properties of curriculum materials that appear to influence instruction in important ways (Brown, 2002; Schneider & Krajcik, 2002). Stein and her colleagues (in press) have adopted this perspective in their research on elementary curriculum materials and are exploring how particular features of the materials interact with certain teacher characteristics (e.g., knowledge of mathematics for teaching), and with characteristics of the school environment (e.g., amount of teacher turnover; level and kind of social capital to which teachers have access). In doing so, their work attempts to change the question from, "Does Curriculum X work?" to the question: "Which curriculum works best under which conditions?"

SECTION FIVE: HOW THE ENACTED CURRICULUM INFLUENCES STUDENT LEARNING

In this section we examine the impact of curriculum on student learning by looking at the relationship between the enacted curriculum (the third box in Figure 8.1) and what students appear to learn from their instructional experiences (the final triangle in Figure 8.1). We have limited our discussion to studies that involve observation as a measure of instructional practices rather than those that rely solely on self-report data. Using surveys alone to provide insights into classroom practices is problematic for several reasons. As Ball and Rowan point out (2004, p.4), "key descriptions of practice used in survey instruments are seldom understood uniformly by respondents." In addition, it is difficult for teachers to accurately remember classroom events and interactions (NCES, 2003), and this can lead to their inadvertently misrepresenting their practice (Ball & Rowan, 2004). This issue is made salient by the discrepancy between teachers' responses to questionnaire items and actual classroom observations in the TIMSS video study (NCES, 2003). For example, 86% of the U.S. eighth grade teachers surveyed reported that the videotaped lesson was consistent with current ideas about teaching and learning mathematics yet the observations suggest that the goal of the observed lessons was to learn and use procedures (NCES, 2003).

Although there have been many studies in the past 15 years (discussed earlier in this chapter) that

have linked the use of particular curricula to student achievement or have analyzed the ways in which teachers and their students have enacted mathematical tasks or curricula, few studies have connected the curriculum (or tasks) *as enacted* with student learning or achievement. Three studies cited below provide evidence that the cognitive demands of the mathematics tasks in which students engage are related to student learning.

Evidence gathered across scores of middle school classrooms in four QUASAR middle schools has shown that students who performed the best on project-based measures of reasoning and problem solving were in classrooms in which tasks were more likely to be set up and implemented at high levels of cognitive demand (Stein & Lane, 1996; Stein, Lane, & Silver, 1996). In these classrooms a teacher's success in enacting tasks in ways that maintained the rigor of the tasks was related to the processes identified in the right-hand column of Table 8.2. For students in these classrooms, having the opportunity to work on challenging tasks in a supportive classroom environment translated into substantial learning gains on an instrument specially designed to measure high level thinking and reasoning processes. It is worth noting that the school that achieved the highest student learning gains (Site A), and the greatest percentage of tasks that were set-up and enacted at a high level, was using the standards-based *Visual Mathematics* curriculum. Results from QUASAR also show that students who had the lowest performance on project assessments were in classrooms where they had limited exposure to tasks that required thinking and reasoning (Stein & Lane, 1996).

The results of the 1999 TIMSS video study provide additional evidence of the relationship between the cognitive demands of mathematical tasks and student achievement. In this study, a random sample of 100 8th grade mathematics classes from each of six countries (Australia, the Czech Republic, Hong Kong, Japan, the Netherlands, Switzerland) and the United States, were videotaped during the 1999 school year. The six countries were selected because each performed significantly higher than the U.S. on the TIMSS 1995 mathematics achievement test for eighth grade (Stigler & Hiebert, 2004). The 1999 study revealed that the higher-achieving countries implemented a greater percentage of *making connections* tasks in ways that maintained the demands of the task. With the exception of Japan, higher-achieving countries did not use a greater percentage of high-level tasks than in the U.S. All other countries were, however, more successful in not reducing these tasks into procedural exercises. Hence, the key dis-

tinguishing feature between instruction in the U.S. and instruction in high achieving countries is that students in U.S. classrooms “rarely spend time engaged in the serious study of mathematical concepts” (Stigler & Hiebert, 2004, p. 16).

In a recent study Boaler and Staples (in press) report the results of a five-year longitudinal study of 700 students in three high schools. Students at one high school, Railside, used a standards-based curriculum designed by teachers around key concepts (e.g., What is a linear function?) and featuring groupworthy tasks drawn from curricula such as *College Preparatory Mathematics* (CPM) and the *Interactive Mathematics Program* (IMP) and a textbook of activities that use algebra manipulatives. The students at the other two high schools used conventional curricula. The researchers report that students at Railside achieved at higher levels than those at other schools. In particular, by the end of the second year Railside students significantly outperformed all other students in a test of algebra and geometry.

Boaler and Staples (in press) indicate that one factor contributing to the success of students at Railside was the high cognitive demand of the curriculum and the teachers’ ability to maintain the level of demand during enactment through questioning. According to the authors, “the support that teachers gave to students did not serve to reduce the cognitive demand of the work, even when students were showing signs of frustration. . . . At Railside, teachers were highly effective in interacting with students in ways that supported their continued thinking and engagement in the core mathematics of the problems” (p. 27). In addition, the authors point to other sources of the Railside success story that include the strong commitment of teachers to the advancement of equity, the development of a curriculum that enhances success for *all* students by including problems that value a range of abilities, and the accountability placed on students for their own learning and that of their peers. Hence, Boaler and Staples conclude that although the curriculum played a part of the success of Railside students, “at the heart of this system is the work of the teachers, and the numerous different equitable practices in which they engaged” (p. 31).

The results of a study of middle school curriculum being undertaken by the Center for the Study of Mathematics Curriculum¹⁶ provides additional evidence of the importance of the learning environment in student achievement. In this study, Tarr et al. (in press) investigated the impact of three factors on student achievement: curriculum type (conventional vs. standards-based), fidelity of curriculum implementation, and the nature of the learning environment.¹⁷ The study included more than 4200 students in grades 6–8 from 11 middle schools across 6 states. The curricula included both standards-based curricula (i.e., *Connected Mathematics*, *Mathematics in Context*, and *Math Themes*) and conventional curricula from commercial publishers (e.g., Addison Wesley, Glencoe, Harcourt Brace).

The study’s findings suggest that student achievement in mathematics cannot be predicted solely by the type of curriculum used or by the fidelity of implementation of a curriculum. However, student achievement in mathematics could be predicted by the nature of the classroom environment. Specifically, a standards-based learning environment was associated with higher performance on an assessment of thinking, reasoning and problem solving regardless of the curriculum being used. A standards-based learning environment, however, was more frequently found in classrooms that used standards-based curricula. Particularly interesting is the finding that achievement was highest among students who experienced a standards-based curriculum in a standards-based learning environment over two consecutive years.

The findings of this study suggest that the learning environment is a critical factor in student learning and that standards-based curricula are most effective when the normative practices are in place that promote understanding, that is, learning is viewed as problem solving, alternative strategies and perspectives are discussed publicly, and explanations are given to support conjectures and approaches.

In addition to the above studies that have linked observed instructional practice and student achievement across many classrooms, there have been several smaller scale studies, focused on students’ performance in arithmetic in the primary grades, that provide additional evidence of the connection between in-

¹⁶ The Center for the Study of Mathematics Curriculum (CSMC) is one of the Centers for Learning and Teaching Initiative funded by the National Science Foundation. The goal of this initiative was to encourage cross-institutional collaboration on the preparation of future K–12 STEM leaders and on research on critical issues in the field. CSMC is a collaboration of the University of Missouri, Michigan State University, University of Chicago, Horizon Research, Columbia Public Schools, Grand Ledge Public Schools, and Kalamazoo Public Schools.

¹⁷ The authors use the term “Standards-Based Learning Environment” (SBLE) to describe classrooms where students make conjectures and explain responses and teachers use students’ thinking as the basis for instruction and encourage multiple perspectives and strategies.

struction and learning (Carpenter et al., 1989; Hiebert & Wearne, 1993; Wood et al., 1991). These studies show that students in classrooms that used standards-based curricula that included on-going opportunities to engage with high-level tasks outperform students who are exposed to more conventional curricula and instructional approaches. For example, in a study of instruction and learning in second grade classrooms, Hiebert and Wearne (1993) report that students in classrooms that used an alternative to the more conventional textbook program (i.e., a standards-based curricula) showed higher levels of performance than students in classrooms that used a conventional curricula. The alternative program used contextualized problems, encouraged the use of different representational forms and different solution strategies, and allowed for a public discussion and sharing of students solution strategies (Hiebert & Wearne, 1993). Thus the authors conclude that, “instructional tasks and classroom discourse mediate the relationships between teaching and learning” (p. 420).

One question raised by these studies is ‘how do mathematical tasks and discussions around those tasks influence learning?’ One explanation is that students learn mathematics by solving problems and by listening to what the teacher and their peers tell them. More likely, however, is that teaching influences students’ cognitive processes of thinking which in turn influences their learning (Carpenter & Fennema, 1988; Hiebert & Wearne, 1993; Wittrock, 1986). According to Hiebert and Wearne (1993, p. 421) “certain kinds of instructional tasks and discourse encourage more productive ways of thinking.”

The studies reviewed in Section One and this section can be seen as complementing one another. By focusing on tasks, discourse, and students’ opportunities to learn, the studies reviewed in this section help to uncover the mechanisms by which curricula might improve student performance. In this way these studies begin to explain why two different classrooms using the same curriculum might result in different levels of student performance. The larger scale studies in Section One, on the other hand, are useful for suggesting what teachers on average might be expected to accomplish using the newer curricula and thus we learn something about the feasibility of implementing them at scale.

SUMMARY AND CONCLUSIONS

We begin by summarizing and in some cases elaborating on the major points that were surfaced in Sections One–Five.

Curricula Differ in Significant Ways

In Section One, we concluded that the analyses of curriculum materials completed to date suggest that the mathematics curriculum materials available today differ significantly from each another, with the most consistent differences found between those that we have labeled standards-based vs. conventional. Standards-based curricula embody an approach to learning that focuses on the students’ active construction of important ideas and concepts while conventional based curricula, by and large, present content directly and expect the teacher to explicitly teach students the skills, concepts and procedures that are the goal of the lesson. Our review also pointed to the need for evaluators to clearly articulate the criteria by which they judge the quality of curricula. As shown by the ratings in Appendix A, the same curriculum fares very differently depending on the criteria by which it is judged.

These Differences Impact Student Learning

The pattern of findings associated with the comparative, mostly quantitative, research discussed in Section One suggests that the above differences matter, at least with respect to how much computational and conceptual knowledge students gain relative to each other when one group is taught using standards-based and the other using conventional curricula. Students taught using standards-based curricula tend to “keep up” with their conventionally taught counterparts with respect to computational knowledge but to surpass them in conceptual knowledge and their ability to solve non-routine problems. However, all students tend to do best on tests that align with the way they were taught, giving conventionally taught students a slight edge on traditional standardized tests and standards-taught students a considerable edge on measures of thinking, reasoning, and problem solving.

Kilpatrick (2003) has suggested, and we agree, that findings such as these appear to be oriented toward reassuring the public that the new standards-based curricula are not harming students more than assessing what students truly appear to be learning from them. Regardless, the findings that curricula differ in significant ways and that these differences impact student learning point to the role that healthy discussions of values (what kind of mathematics should students be learning?) must play in decisions to adopt one curriculum over another.

From a research—or knowledge-building—perspective, these findings feel somewhat unsatisfy-

ing. Knowing how much students learn may be less important than knowing what and how they learn. For example, an emerging insight from some of the studies cited in Section One that included qualitative data gathering (e.g., Huntley et al., 2000) is that students who learn and practice symbolic skill manipulation under the auspices of a standards-based curriculum may learn such skills in a qualitatively different way than do students who are taught skills in a direct, sequenced, and structured manner. We need investigations to understand more about the forms of knowledge and understanding that students develop in standards-based versus conventional curricular approaches, and about how the particular approaches support this development. Students most likely do not simply develop more or less knowledge but rather acquire knowledge, beliefs and understandings that differ in important ways including how they become available for use at later points.

No Curriculum is Self-Enacting

The studies reviewed in Sections Two and Three reveal the variation to the written curriculum that is introduced once human interaction enters the picture. Section Two focused on the interpretive processes by which teachers “read” curriculum; Section Three focused on the transformative forces that are unleashed once curricular materials are actually used in the classroom. The research and conceptual ideas introduced in both sections suggest that—while not self-enacting—curricula that focus on the development of specified skills and procedures offer less room for interpretation and transformation than do curricula that focus on reasoning, problem solving, and strategy development and depend on the actions of both students and teachers to bring their designs to fruition.

Counterarguments that have or might be made to this statement include the observation by Mathematically Correct that some of the so-called conventional curricula have become so overlaid with suggestions for problem solving activities and group work that teachers now have to make decisions regarding what to cover and how to prioritize their time between teacher-directed instruction and these “constructivist”-based activities. On the other hand, some might argue that technology-enhanced standards-based curricula such as the *Cognitive Tutor* offer a narrower berth for teacher interpretation because of the large amount of instructional time that occurs directly between the student and the computer.

Standards-Based Curricula are Challenging to Enact Well

Along with their unstructured learning space and room for variable enactment, standards-based curricula offer more enactment challenges to the average teacher. The studies cited in Section Three demonstrate the ways in which tasks from standards-based curricula can be distorted such that they no longer embody the vision or goals that the developer intended. Using these curricula in ways that unlock the potential that their developers envisioned requires considerable knowledge and time on the part of the teacher, as well as a philosophical orientation toward teaching and learning that aligns with that of the developers. Other factors that influence the level of success that can be expected from using standards-based curricula are identified and discussed in Section Four.

The high degree of variation associated with the enactment of standards-based curricula point to the need to interpret studies of student learning from such curricula with caution. Given the high degree of “slippage” associated with the enactment of high-level tasks, it is unclear whether limited student learning should be attributed to the materials or the manner in which they were implemented in the classroom. This is less often the case with interpreting studies of conventional curricula.

The Success of Standards-Based Curricula is Influenced by Multiple Factors

Sections Four and Five identify factors within and outside the classroom that impact the ease with which a successful implementation of a standards-based curriculum can be achieved. The studies in these sections are important to teachers because they uncover the variety of contextual factors that often accompany successful implementation of standards-based curricula but that are sometimes not overtly articulated as supporting factors. The studies that looked closely at particular teaching interactions in the classroom provide a level of detail that is needed by teachers who wish to implement tasks from standards-based curricula. From a research perspective, more fine-grained studies of classroom implementation provide a clearer, more complete picture of how and under what conditions standards-based curriculum lead to improvements in students’ learning.

* * *

We began this chapter with the introduction of a framework that illustrated various phases of cur-

riculum use. Uncovering how curriculum influences student learning, we argued, would not be possible without reviewing what is known about how the curriculum is mediated before and in the process of making its way to students. By reviewing and summarizing the implications of studies that lay at various markers along this temporal pathway, we have identified and described—not only whether—but also *how* curricula influence student learning. This review of the research on how curriculum influences student learning has illuminated the vast conceptual territory that lies between the curriculum as a designed object and student learning.

Our claims rely on studies that we have patchworked together to form a longer chain of reasoning regarding the phases through which curricula traverse. In the patchwork process, we have had to gloss over inconsistencies in conceptualizations, definitions of terms, and points of focus that existed across the various studies. In the chapter we call for the field to establish more agreement regarding what to focus on and how so that studies can be accumulated and build toward a solid understanding of how curricula are transformed and eventually influence student learning. Studies that are conducted to examine the entire chain—from materials analyses to teacher interpretation to curricular enactment to student learning—would also add considerably to our knowledge base.

Looking toward the future—including studies that might help curriculum developers improve their materials—perhaps the most pressing question is how to combine the best of both conventional and standards-based curricula into a more unified and balanced approach. How can teachers effectively teach procedural skills in the context of the open-ended, problem-based forms of instruction that characterize standards-based curricula? Standards-based lessons that encourage a variety of ideas, solution strategies, and forms of representation have the advantage of engaging students in thinking and learning to justify and reason. However, they also have the disadvantage of not converging neatly into a common core of knowledge that all students share and know—especially not in any predetermined time frame, be it a lesson, a grading period, a year, or in time for a high-stakes test. Making sure that all students eventually reach clarity and closure on important mathematics ideas and procedures is not trivial in this form of instruction (National Research Council, 2001).

The limitations of conventional curricula that focus too narrowly on procedural skill have been well documented; yet there are proficiencies that students who are exposed to those curricula gain. Learning how to incorporate elements of that learning into the

curricula of tomorrow will be an important task. Given the ambition of the standards-based curricula—both in terms of the depth of student learning for which they aim and the numbers of teachers they hope to influence—we need to get much smarter about how we design studies to help the continuing evolution of those curricula.

Although we feel as though this chapter brings some conceptual order to the territory that lies between curriculum materials and student learning, the territory of the factors represented in the oval in Figure 8.1 is much less well conceptualized. As discussed in Section Four of this chapter, these factors have been supported by a host of studies, however, most studies looked at one, or at most two, factors in isolation from the others. Missing is research that systematically relates the factors to one another in a conceptually coherent manner. For example, we know that teachers' knowledge and beliefs influence how they interpret curricular materials and how they enact them in the classroom. However, we know little about how this knowledge and beliefs are influenced by context, including the informal communities in which teachers talk about mathematics and, most importantly, negotiate the meaning of new curricula. Moreover, we know little about how different features of curricula (how transparent their goals are; how much information they provided about possible student responses to their tasks) are related to various levels and kinds of teacher knowledge.

The research reviewed herein points to the challenges that most teachers face in learning to implement standards-based curricula well. There is beginning to be a body of research that examines *how teachers learn* through implementing these new curricula. We have not reviewed this research as it did not directly bear on the question of how curricula influences student learning. Nevertheless, we feel that it is an important component of the conceptual landscape that sits between the design of materials and student learning. If teachers are the mechanism through which curriculum becomes transformed and goes on to influence students, well-designed studies on what and how teachers learn from using these materials would form an important part of the overall picture.

On a related note, because these new standards-based curricula have been placed into the role of change agent, another kind of conceptual work is suggested as well: How teacher learning from curricular materials might be expected to happen at scale within schools and districts. Cobb et al. (2003) have pointed to a bifurcation of the research literature in this regard with one body of literature on professional development and another body on organizational/structural

facilitators and constraints to instructional improvement. With few studies that bridge the two, as a field we are left with decontextualized knowledge about effective professional development and knowledge about organizational supports for change that fails to make contact with the particular needs of teacher learning in mathematics. Thus, we need more studies that systematically integrate schools and districts as settings for curriculum implementation and what we know about the kind of teacher learning required to enact these curricula well.

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Appendix A

Ratings of Various Curricula by the U.S. Department of Education, AAAS, and Mathematically Correct

DOE: Curricula identified for recognition as Promising or Exemplary

AAAS: Middle school curricula identified as Satisfactory or Unsatisfactory

MC: Overall ratings of curricula

Curriculum	Level	Standards-Based (SB) or Conventional (C)	DOE	AAAS	MC
<i>Everyday Mathematics</i> ; SRA/McGraw Hill	El	SB	Promising		2nd grade: C 5th grade: C–
<i>Investigations in Number, Data, and Space</i> ; Scott Foresman	El	SB			2nd grade: F 5th grade: F
<i>Math Advantage K–6</i> ; Harcourt Brace	El	C			2nd grade: B 5th grade: B–
<i>Math in My World</i> ; McGraw Hill	El	C			2nd grade: B+ 5th grade: B–
<i>Math K–5</i> ; Scott Foresman/Addison Wesley	El	C			2nd grade: B+ 5th grade: B–
<i>MathLand</i>	El		Promising		
<i>Number Power</i> (supplemental)	El	SB	Promising		
<i>Saxon Math</i>	El	C			2nd grade: B 5th grade: B+
<i>Silver Burdett Ginn Math</i>	El	C			2nd grade: B 5th grade: B
<i>SRA Math</i> ; McGraw Hill	El	C			2nd grade: A 5th grade: A–
<i>Connected Mathematics Project (CMP)</i> ; Prentice Hall	MS	SB	Exemplary	Satisfactory	F
<i>Heath Mathematics Connections</i> ; DC Heath & Co.	MS	C		Unsatisfactory	
<i>Math Advantage</i> ; Harcourt Brace	MS	C		Unsatisfactory	B
<i>Mathematics: Applications & Connections</i> ; Glencoe/McGraw Hill	MS	C		Unsatisfactory	B
<i>Mathematics in Context (MiC)</i> ; Holt, Rinehart, & Winston	MS	SB		Satisfactory	
<i>Mathematics Plus</i> ; Harcourt Brace & Co.	MS	C		Unsatisfactory	
<i>MathScape: Seeing and Thinking Mathematically</i> ; Glencoe/McGraw Hill	MS	SB		Satisfactory	
<i>MathThematics (STEM)</i> ; McDougal Littell	MS	SB		Satisfactory	D+
<i>Middle Grades Math</i> ; Prentice Hall	MS	C		Unsatisfactory	B
<i>Middle School Math</i> ; Scott Foresman/Addison Wesley	MS	C		Unsatisfactory	B
<i>Middle School Mathematics Through Applications Project (MMAP)</i> ; unpublished	MS	SB	Exemplary		

Curriculum	Level	Standards-Based (SB) or Conventional (C)	DOE	AAAS	MC
<i>Passport Series</i> ; McDougal-Littell	MS	C		Unsatisfactory	Passport to Algebra & Geometry: A Passport to Math: C
<i>Saxon Math</i>	MS	C		Unsatisfactory	Alg ½: A Math 87: C
<i>Transition Mathematics</i> (Scott Foresman)	MS			Unsatisfactory	
<i>Cognitive Tutor</i>	MS/HS	SB	Exemplary		
<i>College Preparatory Mathematics</i> (CPM)		SB	Exemplary		
<i>Contemporary Mathematics in Context</i> (Core-Plus)	HS	SB	Exemplary		
<i>Interactive Mathematics Program</i> (IMP); Key Curriculum Press	HS	SB	Exemplary		
<i>University of Chicago School Mathematics Project Integrated Mathematics</i> , Grades 7–12; Prentice Hall	HS	SB	Promising		

THE EFFECTS OF CLASSROOM MATHEMATICS TEACHING ON STUDENTS' LEARNING

James Hiebert

UNIVERSITY OF DELAWARE

Douglas A. Grouws

UNIVERSITY OF MISSOURI

TEACHING MATTERS . . . BUT *HOW* IS NOT EASY TO DOCUMENT

We begin with the following claim: The nature of classroom mathematics teaching significantly affects the nature and level of students' learning. Such a claim seems obvious. Everyone can remember teachers who were especially effective. Surely, teaching makes a difference. Systematically collected data support most people's personal experience by showing that different teachers produce different levels of student achievement gains (Nye, Konstantopoulos, & Hedges, 2004; Sanders & Rivers, 1996; Sanders, Saxton, & Horn, 1997). In fact, the cumulative effect over several years of effective teachers is substantial. Having good teachers really does make a difference.

But what makes mathematics teachers effective? This question does not have an obvious or easy answer. The answer is not found by searching the reports cited above on teacher effectiveness. Many reasons could explain why some of the teachers in these

studies were more effective than others. Classrooms are filled with complex dynamics, and many factors could be responsible for increased student learning. This is not one of those pedantic educational issues that researchers get tangled up in by splitting hairs and that, in the end, simply demonstrate what everyone already knows. This is a very central and difficult question to answer. It is, in a real sense, the core question of education. What is it, exactly, about teaching that influences students' learning? Although laypersons in the street often have quick answers based on recollections of their favorite teachers, documenting particular features of teaching that are consistently effective for students' learning has proven to be one of the great research challenges in education.

Our aim in this chapter is to tackle directly the issue of teaching effectiveness—why it has been so hard to document, what is known about it, and how the mathematics education community can learn more. We first examine why it has been so difficult to establish robust links between teaching and learning, then we present a few claims that organize and

structure the literature on teaching effects in what we hope are helpful ways, and finally we outline a set of goals and strategies to guide future work in this central and urgent research domain. Although we believe the claims we make about links between mathematics teaching and learning are well supported, we do not present an exhaustive review of the research literature to substantiate them. Rather, we develop arguments to support our claims, select a few studies that illustrate them, and refer the reader to other, more extensive reviews of relevant research.

Before considering some of the difficulties facing researchers who try to document the effects of teaching, we need to define what we mean by *teaching*. Simple definitions, presented in some classic treatments of teaching, are a good starting point. Thorndike (1906) defined *teaching* as the methods used to help students achieve the learning goals valued by society. Gage (1978) defined *teaching* as “any activity on the part of one person intended to facilitate learning by another” (p. 14). These definitions are incomplete because, although they contain much of what we mean by *teaching*, they treat teaching as a one-way relationship—teachers acting on students. In reality, teaching is influenced by students and has a bidirectional quality. Cohen, Raudenbush, and Ball (2003) captured this quality by saying that “instruction consists of interactions among teachers and students around content” (p. 122). If we use Cohen et al.’s definition but add back the goal-directedness explicit in Thorndike’s definition and restrict the environments to school classrooms, we have a working definition that sets the boundaries for this chapter: Teaching consists of classroom interactions among teachers and students around content directed toward facilitating students’ achievement of learning goals.

Why has it been so difficult to identify features of teaching, as defined here, that show consistent and strong effects on students’ learning? Examining this question sets the context for both the claims we make about links between mathematics teaching and learning and the strategies we propose to advance this central research domain in mathematics education. Because the issues surrounding the difficulty of linking teaching with learning are not unique to mathematics, and because they have been addressed in the broader literature, we examine the question both within this broader context and within contexts specific to mathematics.

A Claim That Appears Obvious is Strikingly Difficult to Specify

Despite more than 100 years of work on specifying effective teaching, the field still is searching for

reliable, demonstrable recommendations of teaching methods. This is not to say that recommendations have not been proposed. The education literature is filled with suggestions, recommendations, and sure-to-succeed descriptions of teaching. Pestalozzi, one of the great educators of the past, was said to have claimed discovery of “the ultimate method of effective instruction” in the early 1800s (Reusser, 2001, p. 1). A perusal of more recent analyses indicates that Pestalozzi’s optimism is alive but not widespread.

Reviews of the teaching-learning literature indicate considerable differences of opinion, not just about which teaching methods are effective, but even about how much is known about whether any methods are particularly effective. Some researchers portray a field that is making good progress in documenting links between teaching and learning, but other researchers see less progress. The third edition of the *Handbook of Research on Teaching* includes two chapters that review, with considerable confidence, the connections between particular teaching behaviors and students’ learning (Brophy & Good, 1986; Rosenshine & Stevens, 1986). More recently, Brophy (1999) reinforced this optimism by identifying 12 principles of effective teaching based on accumulated research.

In contrast to these optimistic claims, Romberg and Carpenter (1986) argued that research on teaching and research on learning had constituted two separate fields in mathematics education with few attempts to connect learning and teaching. Consequently, there was little to say about specific links between the two. Romberg and Carpenter’s less optimistic view is found in the same third edition of the *Handbook* that contains the positive treatments by Brophy and Good (1986) and Rosenshine and Stevens (1986).

Observations about the absence of evidence-based connections between teaching and learning can be traced back to at least the middle of the last century. A report released by the American Educational Research Association Committee on the Criteria of Teacher Effectiveness concluded:

The simple fact of the matter is that, after forty years of research on teacher effectiveness during which a vast number of studies have been carried out, one can point to few outcomes that a superintendent of schools can safely employ in hiring a teacher or granting him [*sic*] tenure, that an agency can employ in certifying teachers, or that a teacher education faculty can employ in planning or improving teacher education programs. (Barr et al., 1952, cited in Duffy, 1981, p. 113)

Two decades later, several reviews suggested that efforts to link particular teaching features with students’

learning still had produced few results (Berliner, 1976; Heath & Nielson, 1974).

Even with continuing attention to the effects of classroom teaching on learning since these early reviews, researchers still are not uniformly persuaded by the evidence (Shuell, 1996). Countering Brophy's (1999) optimistic view, the two chapters in the most recent fourth edition of the *Handbook of Research on Teaching* that deal explicitly with the effects of teaching on learning are decidedly cautious about claims that link teaching with learning (Floden, 2001; Oser & Baeriswyl, 2001).

The uncertainty in the research community about what is known regarding the effects of teaching on learning can be explained by some of the challenges facing researchers engaged in this work and the ways in which the challenges have (and have not) been met. Before examining these challenges, we must point out that this uncertainty in the field has its costs. Among them is the perception by key U.S. policymakers that research does not inform classroom practice:

Education has not relied very much on evidence, whether in regard to how to train teachers, what sort of curriculum to use or what sort of teaching methods to use. The decisions have been based on professional wisdom or the spirit of the moment rather than on research (Grover Whitehurst, U.S. Assistant Secretary of Education, as quoted by J. Traub in the *New York Times*, Nov. 10, 2002).

Pressures are increasing to provide evidence-based descriptions of effective mathematics teaching. Consequently, the focal issue of this chapter is not only fundamental, from a research perspective, but urgent, from a policy perspective.

The first step in making further progress in establishing connections between teaching and learning is understanding why this is so hard to do. In the following sections, we outline a number of difficulties that face researchers who try to connect teaching with learning. These are not excuses; they are severe obstacles that must be resolved. Indeed, after studying the following challenges, we expect that readers might change their question from "Why is it so hard to connect teaching with learning?" to "Is it even possible to establish reliable connections between particular features or methods of teaching with students' learning?" Anticipating this discouragement, we continue the chapter by developing several claims about features of teaching that affect learning, claims that we believe have accumulated considerable empirical support in spite of the challenges.

Useful Theories of Teaching Are in Short Supply

Making progress in any scientific field is difficult without explicit theories. In fact, some researchers would argue that fields are scientific to the extent that empirical research is guided by theory (National Research Council, 2002). Theories are useful because they direct researchers' attention to particular relationships, provide meaning for the phenomena being studied, rate the relative importance of the research questions being asked, and place findings from individual studies within a larger context. Theories suggest where to look when formulating the next research questions and provide an organizational scheme, or a story line, within which to accumulate and fit together individual sets of results. Karmiloff-Smith and Inhelder (1974) titled a provocative article, "If You Want to Get Ahead, Get a Theory." They were describing children's efforts to develop their understanding of physical phenomena, but the analogy holds equally well for researchers developing an understanding of connections between teaching and learning.

Robust, useful theories of classroom teaching do not yet exist. Theories that consider connections between classroom teaching and students' learning are even less developed (Floden, 2001; Oser & Baeriswyl, 2001). Within mathematics, theories of learning have been more clearly articulated than theories of teaching. Although theories of learning provide some guidance for research on teaching, they do not translate directly into theories of teaching and cannot be used, by themselves, to stimulate or coordinate research agendas on the effects of teaching on learning. Components or features of teaching that are likely to play important roles in useful working theories have been identified, and beginnings of theories of mathematics teaching have been proposed (e.g., Brousseau, 1997; Freudenthal, 1973; Gravemeijer, 1994; Leinhardt & Greeno, 1986; National Research Council, 2001; Schoenfeld, 1998; Simon, Tzur, Heinz, & Kinzel, 2004). These efforts signal important progress. But theories that specify the ways in which the key components of teaching fit together to form an interactive, dynamic system for achieving particular learning goals have not been sufficiently developed to guide research efforts that can build over time.

Without guiding theories, researchers are feeling their way through murky waters. Studies that have focused on teaching-learning relationships often are chosen because they align with the personal interests of researchers. When researchers pursue different research questions and use different measures of teaching and of learning, their findings often remain

isolated from one another. The data do not easily accumulate to reveal patterns and general principles. To surmount these obstacles and move ahead, researchers need to construct useful theories. Those who attempt to develop theories face at least three major conceptual challenges.

Challenge 1: Different Teaching Methods Might Be Effective for Different Learning Goals

There is no reason to believe, based on empirical findings or theoretical arguments, that a single method of teaching is the most effective for achieving all types of learning goals. Perhaps some methods of teaching are more effective for, say, memorizing number facts whereas other methods of teaching are more effective for deepening conceptual understanding and still other methods are more effective for acquiring smooth execution of complex procedures. In addition, some methods of teaching might be especially effective for showing gains in the short-term whereas others might be better for retaining and increasing gains in the long-term. Because a range of goals might be included in a single lesson, and almost certainly in a multi-lesson unit, the best or most effective teaching method might be a mix of methods, with timely and nimble shifting among them.

One consequence of recognizing that different teaching methods might be effective for different learning goals is the fact that empirical studies that compare one teaching method to another using blunt learning measures are difficult to interpret. For what kinds of learning are the teaching methods claimed to be effective?

Challenge 2: Teaching is a System of Interacting Features

A teaching method consists of multiple features that interact with one another in almost countless ways (Cohen et al., 2003; Design-Based Research Collective, 2003; Stigler & Hiebert, 1999). Theories of teaching must account for this fact by treating teaching as a system of interacting features rather than a collection of independent and interchangeable features. This means that the effects of features of teaching on students' learning cannot be measured independently of the system in which they operate. The effects of a particular feature on students' learning likely depend on its interactions with other features in the system.

Two simple examples illustrate the system nature of teaching. One of the striking results from the 1995 Video Study of the Third International Mathematics and Science Study (TIMSS) was the difference in teaching eighth-grade mathematics found between Japan and the United States. In general, lessons in Japan challenged students to solve difficult mathemat-

ics problems whereas lessons in the United States emphasized the practice of lower level skills (Stigler & Hiebert, 1999). Casual observation of the lessons suggests that Japanese teachers asked more higher level questions than U.S. teachers, but a closer look showed that the major difference was in the *kinds* of higher order questions that were asked and *when* the questions were asked (Kawanaka & Stigler, 1999). Japanese teachers asked most of their higher order questions about student-presented solution methods for solving problems whereas U.S. teachers asked most of their higher order questions about methods they, themselves, were demonstrating. Japanese teaching was distinguished not just by the existence of high-level questions but in how the questions fit within and reinforced the system of teaching being employed.

A second example of the system nature of teaching comes from research on a feature of teaching often associated with current recommendations—asking students to work in small groups. Clearly whether small groups function productively to help students achieve learning goals depends on many surrounding factors, including the knowledge and skill students acquire for working collaboratively and the kinds of tasks they are assigned (Good, Mulryan, & McCaslin, 1992; Webb, 1991; Webb, Troper, & Fall, 1995). While the first author of this chapter was reviewing videotaped lessons as part of the TIMSS, he observed one eighth-grade teacher asking students to break into their small groups and work together to answer the question “What is the name for a 12-sided object?” Students quizzed each other quickly on whether anyone knew the name and then visited for the remaining time about non-mathematical topics. This task did not lend itself to collaborative investigation. The effect of small-group work is shaped by the system of which it is a part.

A consequence of the system nature of teaching is that the effects on students' learning of individual features of teaching, like higher-order questions or the use of small groups, are difficult to isolate. Systems of teaching, not single features of teaching, affect students' learning. This is not to say that all features within a system are equally responsible for their effects on student learning. Particular features, or combinations of features, might account for more of the system's effects than other features. But the effect of any particular feature is influenced by related and interacting features. Theories of teaching must account for this fundamental complexity.

Challenge 3: The Influence of Mediating Variables

The effects of teaching on learning are complicated, not just by the features of teaching interacting

with each other, but by factors that mediate the effects of the system. In a classic analysis, Wittrock (1986) argued that teaching does not affect learning directly. The effects of teaching are mediated by students' thinking—their attentiveness during instruction, their interpretations of the teacher's presentations and of the tasks they are assigned, their entry knowledge and skills, and so on. Wittrock was not the first to comment on the important role that students play in determining the effects of teaching on learning (Berliner, 1976; Doyle, 1978; Winne & Marx, 1980). Berliner (1976), for example, cited "a student's active time-on-task" (p. 10) as a complicating factor in connecting teaching and learning. More recently, Weinert, Schrader, and Helmke (1989) proposed an even stronger claim by saying that "the activity and cognitive orientation of the learners are as important as the activities and the behaviors of the teachers" (p. 899) when specifying the relationships between teaching and learning.

The set of variables related to students' thinking is not the only factor that mediates the effects of teaching on learning. A large number of contextual conditions undoubtedly influence the nature and level of students' learning (Berliner, 1976; Dunkin & Biddle, 1974). These include conditions both inside and outside of the classroom—number of students in class, length of daily lessons, home support for completing homework, and so on.

Some of the variables that mediate the effects of teaching on learning exert a stronger influence on the relationship than simply redirecting or moderating the impact of teaching on learning. These variables can reverse the causal flow so that they change the nature of teaching. For example, students can push back and alter the teacher's intended approach in both subtle and not-so-subtle ways (Cohen et al., 2003; Cooney, 1985). This is why we adjusted Thorndike's and Gage's definitions of *teaching*, as described earlier, to include two-way interactions rather than just one-way interactions from the teacher to the students.

Developing theories of teaching and, in particular, theories that connect teaching with learning is clearly a challenging task. The lack of robust widely-accepted theories is not surprising. Although the development of such theories would enable a more quickly accumulating research base, researchers would not be wise to wait for such theories to pursue empirical studies. Theoretical and empirical work can boost each other. Unfortunately, as we argue in the next section, conducting empirical studies on teaching-learning connections is fraught with a host of additional challenges. As the reader will notice, however, not all of these challenges are unique to empirical research; some are

related to challenges we have already identified that confront theory development.

Documenting the Effects of Teaching on Learning Is Methodologically Difficult

Collecting trustworthy data on the effects of teaching on learning is hampered by a number of methodological difficulties. We place three kinds of difficulties at the top of the list for posing critical threats to research efforts and accounting for the challenge of linking teaching and learning. In Gage's (1978) terms, the first considers what variables to study and the second asks how to measure them. The third kind of difficulty—individual differences among students—has reappeared in various forms throughout the history of research on teaching and learning and remains a significant issue.

Accounting for Relevant Factors

Many, many factors, inside and outside of school, influence what and how well students learn. In addition to quality of teaching, influencing factors include things like the level of home support (health care, education values and resources, and so forth), peer pressures and social relationships, students' motivation, curricular materials, and assessments. To determine the effects of teaching in the midst of all the competing forces, researchers need to account for these forces in some way. Ideally, researchers would conduct experiments in which all of these forces are equalized across comparison groups, or at least are randomly distributed across the groups. Given the nature of school constraints, randomly assigning students to treatments for sufficiently long periods of time is often impossible. The consequence is that the results of studies can have many alternative interpretations. Numerous factors, other than teaching methods or in combinations with them, can explain the learning outcomes.

If researchers are able to conduct studies with random assignment of students to teaching methods, a problem remains of implementing the same method in a consistent way across classrooms. As we have noted, teaching interacts with students (as well as other contextual variables, such as the teacher's experience and expertise) and is likely to change somewhat from classroom to classroom. It can be difficult to know whether the critical features of teaching are being implemented in the same way across all sites. Limiting the research design to a single classroom or a single teacher does not solve the problem because it leaves to later replications the question of whether the findings will apply to other classrooms and teachers.

A final thorny issue arises when testing the relative benefits of particular features of teaching. As noted earlier, teaching is a system of interacting elements. Isolating particular features of the system is difficult and can even distort the effects of teaching being compared. As an example, consider the question of whether concrete manipulative materials aid students' learning.¹ First, one would need to decide how students' learning would be measured. This, in itself, is a challenging issue because different kinds of learning might be possible with manipulative materials than without such materials. Suppose this issue was resolved and treatments were designed with and without materials. Should the treatments contain the same teaching features except for the use of materials? Or should two different treatments be designed—one to provide the most effective possible approach with materials and another without materials? The first choice might lead to rather contrived treatments for one or both conditions because, under normal conditions, it is unlikely that teaching methods would be identical except for the presence or absence of materials. The second choice would lead to results that would be difficult to interpret—between-group differences might be due to the use of materials or, instead, to other differences between the methods that arose as a consequence of changing other features in the teaching system.

Accounting for all the relevant factors when designing studies or interpreting results on relationships between teaching and learning is a serious methodological challenge. As we will argue later, this does not mean that planned, comparative research is impossible, but it does suggest that the simple isolation of variables within teaching systems is unrealistic.

Creating Appropriate Measures

Developing instruments to measure students' learning and instruments to describe classroom teaching that are reliable *and* valid is a continuing methodological challenge (Ball & Rowan, 2004; Berliner, 1976), especially when using the instruments with large samples (Hamilton et al., 2003; Rowan, Correnti, & Miller, 2002). Describing teaching is, in many ways, more challenging than measuring students' learning due, in part, to its bewildering complexity and, in part, to the relatively less attention it has received.

Attempts to describe classroom teaching must consider what to measure and how to measure it. The question of what to measure is tied closely to our earlier discussions of developing theories that point

researchers to important features of teaching and discussions of accounting for the relevant factors in the teaching system. We focus here on the issue of *how* to describe teaching, once the "what" is determined.

Two methods for describing teaching have been used most heavily: classroom observations (Dunkin & Biddle, 1974) and survey questionnaires (Rowan et al., 2002). Both methods suffer from limitations that must be accounted for when designing studies of teaching. Classroom observations can be influenced by biases of the observer. Establishing and maintaining inter-observer reliability on subtle but potentially important features of teaching is difficult. Observation protocols can be worked out in advance to reduce observer bias, but this requires deciding in advance what to observe. Deciding what to observe, in turn, presumes theories (at least hypotheses) of relationships between teaching and learning. Teacher questionnaires have even more serious problems, especially when the questionnaires are administered on a single occasion and ask teachers to make judgments about teaching approaches and emphases (Hiebert & Stigler, 2000; Malara & Zan, 2002; Mayer, 1999; Wubbels, Brekelmans, & Hooymayers, 1992). Teachers have a tendency to over-report features of teaching that they believe are expected. In other words, self-report surveys might indicate how teachers think they should be teaching, but the validity of these reports is suspect for describing teaching practices.

Alternative instruments to describe teaching are being developed. Video provides numerous advantages over observations and questionnaires (Stigler, Gallimore, & Hiebert, 2000) both in terms of reliability and validity. In addition, it allows teaching to be analyzed by different researchers using different approaches over long periods of time, and it affords serendipitous discoveries. When researchers can afford the expense of collecting and analyzing videotape samples of teaching, it offers many methodological benefits. Less expensive alternatives that can be used routinely on a large scale include redesigned and carefully targeted questionnaires (Ross, McDougall, Hogaboam-Gray, & LeSage, 2003; Rowan et al., 2002), combinations of measures such as questionnaires and case studies (Stecher & Borko, 2002), teacher logs that include frequent diary-like notes by teachers (Porter, Floden, Freeman, Schmidt, & Schwillie, 1988; Rowan, Harrison, & Hayes, 2004), and classroom artifacts that include information regarding teachers' use of instructional materials and teaching methods, classroom activities, and students' work (Borko, Stecher,

¹ See Hiebert (2003) for a similar analysis with respect to the effects of using calculators on students' computational skill.

Alonzo, Moncure, & McClam, 2003). Many of these instruments are currently in the development and testing phase, so it is too early to evaluate their ultimate usefulness.

One methodological issue that cuts across questions of what to measure and how to measure it is the unit of analysis. What constitutes a meaningful unit of teaching? Much of the research that focused on connections between specific teaching features and student outcomes, often referred to as process-product research (Dunkin & Biddle, 1974), used relatively narrow or local units, such as the type of questions a teacher asked or the amount of time spent on each instructional task. Because we believe that teaching is best viewed as a system of interacting features, we propose using broader units that preserve the potentially important interactions. The typical daily lesson is one such unit (Stigler & Hiebert, 1999). The daily lesson has the advantage of being large enough to include key interactions among teaching features and small enough to be thoroughly analyzable. A disadvantage of using the daily lesson as the unit is that students' learning does not occur in lesson chunks. Sequences of lessons defined by full treatment of particular concepts or topics constitute other possible units of teaching, but these can become too large to digest and analyze in detail. Clearly, the tradeoffs must be weighed carefully.

Teaching Often Gets Interpreted as Teachers

To conclude the discussion of why the obvious sounding claim of "teaching affects learning" is so difficult to specify, we focus on a conceptual issue. Questions of how teaching affects learning often get muddled with questions of how teachers affect learning. Some might view this distinction as less substantive and more pedantic than the theoretical and methodological challenges posed in the earlier sections, but we believe the confusion between teachers and teaching is important because it can interfere with theoretical and methodological efforts to establish clearer connections between teaching and learning. If the differences between teachers and teaching are not clarified, research findings are misinterpreted and misunderstood.

To reiterate, we focus on *teaching* in this chapter—classroom interactions among teachers and students around content directed toward facilitating students' achievement of learning goals. This is different from *teachers*—their beliefs, subject matter knowledge, years of experience, personality traits, and so on. Characteristics of teachers surely can influence their teaching, but these characteristics *do*

not determine their teaching. Teachers with different characteristics can teach in essentially the same way and vice versa. No doubt this a major reason for the absence of clear links between teachers' personal and background characteristics and student achievement (Dunkin & Biddle, 1974).

We chose to focus on teaching in this chapter, with a targeted and constrained definition, not only to make our task manageable but also because the teaching we define is largely under the control of the teacher. Some characteristics of teachers, such as personality traits, are quite unalterable, but the methods used to teach can be changed. This is not to say that changing teaching is easy, but it is something that teachers' control. Teaching is something that can be altered under supportive conditions and that can benefit from research-based information.

The distinction between teachers and teaching has been recognized by researchers for some time, but the distinction often disappears. In their classic 1974 book, Dunkin and Biddle recognized the distinction and even gave teacher characteristics a special name—*presage* variables. But the distinction has not always been made, perhaps because of a mistaken assumption that teacher characteristics determine, or nearly determine, the nature of teaching. The confusion is especially apparent, and nearly pervasive, in the public press and in many policy documents. Most news reports and policy statements, for example, that critique education and call for better teaching point to characteristics of teachers, not methods of instruction. The annual report on the condition of U.S. education sponsored by the Pew Charitable Trust has a section on "Teaching Quality" but reports no information on teaching methods, only characteristics of teachers such as certification programs and years of experience (Olson, 1997). In the first two sentences of the 2004 annual report of the U.S. Secretary of Education, *teacher* and *teaching* are equated and then used interchangeably with improved teaching apparently assumed to result from changing teacher characteristics (Paige, 2004). The policy document *Before It's Too Late* produced by the highly visible commission chaired by U.S. Senator John Glenn includes four key points in the foreword about improving mathematics and science teaching in the 21st century (National Commission on Mathematics and Science Teaching for the 21st Century, 2000). The third of the four points actually focuses on teaching, and suggests that improving teaching is a key to improving students' learning. But, consistent with Secretary Paige's report, the point is developed by recommending upgrading the quality of teachers as measured by professional and personal

characteristics, not by recommending upgrading the quality of teaching methods.

One explanation for blurring the distinction between teachers and teaching is the absence of a shared, accepted knowledge base linking teaching with learning. If Secretary Paige or the Commission chaired by Senator Glenn wanted to recommend improved teaching methods, what would they say? Perhaps discussions of teaching quickly shift to teachers because not enough is known about teaching methods that support particular kinds of learning. At the same time, we believe that ignoring or blurring the conceptual distinction between teachers and teaching hinders progress in building a knowledge base. Confusion between teachers and teaching undermines research efforts to link teaching with learning.

Despite the Difficulties, Research-Based Claims Are Emerging

Given the challenges we have outlined, it now might seem surprising that empirically supported claims linking teaching with learning exist at all. However, despite the difficulties, researchers have constructed a variety of local hypotheses to guide their efforts and have used a variety of methods to investigate connections between teaching and learning. Out of this array of theoretical and empirical work, a few general patterns that link features of teaching with particular kinds of learning can be detected. This is a bold claim because it says there is sufficient empirical data, not just plausible arguments, to link some features of teaching with some types of learning.

Because the goal of this chapter is to describe empirically based connections between teaching and learning, we reviewed the literature with two primary criteria in mind. First, we wanted to select learning outcomes that were educationally significant across grade levels and specific mathematical topics. We were not concerned with detailing the kinds of teaching that promote learning of particular competencies, like solving quadratic equations or subtracting whole numbers with regrouping. Rather, we were interested in more general outcomes, like skill efficiency or problem-solving ability or conceptual understanding. If we could find links between teaching and more general outcomes, the findings would have wider applicability.

The second criterion we used to guide the review was to select only studies that described classroom teaching with some detail, that included well-described measures of students' learning, and that were designed to examine connections between the nature of the teaching and what and how well students learned. Studies that addressed well at least some of the concep-

tual and methodological challenges reviewed earlier were of special interest. Note that requiring studies to describe the nature of the classroom teaching eliminates a number of important areas of research on students' learning. For example, a well-established principle of learning is that the extent to which new material is learned well is influenced by what is already known (National Research Council, 1999). But few studies in mathematics education have examined these effects in the classroom by varying systematically the extent to which teaching connected with students' current knowledge or ways of thinking. Although the principle still has important implications for teaching mathematics, its empirical support does not come primarily from classroom studies of teaching and learning mathematics. As another example of excluded literatures, a rapidly increasing set of studies exists on the effects of different curricula on students' learning (National Research Council, 2004). At this point, however, most of these studies do not include detailed descriptions of classroom teaching so they are not included here. The interested reader will find the literature on the effects of curricula on students' learning reviewed by Stein, Remillard, and Smith (this volume).

In the next sections of the chapter, we present several patterns that connect mathematics teaching and learning. We begin by identifying one of the most firmly established but most general connections between teaching and learning. Commonly referred to as "opportunity to learn," this claim says that students learn best what they have the most opportunity to learn. We then increase the lens magnification slightly and identify two educationally significant learning outcomes in mathematics that have attracted considerable attention from researchers studying teaching effectiveness: skill efficiency and conceptual understanding. We identify the kinds of teaching that are empirically related to these learning outcomes.

The patterns we describe were induced by looking across many studies with an eye toward identifying convergent findings. We believe they are the patterns that enjoy the strongest empirical support at this point in our research history. Other patterns are emerging but do not yet have as much support at this point. For example, considerable work has been conducted on classroom discourse and its effects on learning (see, for example, Forman, 2003; Lampert & Cobb, 2003), but, in our judgment, the empirical links between particular kinds of discourse and particular kinds of student learning are not yet as developed as the patterns we describe. Parenthetically, we include in our review some studies that report data on classroom discourse, but we interpret the results to support a kind of teaching that includes more than just a particular kind of discourse.

Our reading of the literature indicates that the patterns we identify are sufficiently documented that they warrant action. That is, we believe they can be accepted with high enough levels of confidence that they can be used to inform policy and influence classroom practice. Saying this yet another way, they provide substantive empirically-based connections between teaching and learning that Secretary Paige or the Glenn Commission on Teaching could have used to formulate recommendations regarding improved teaching. In addition, the patterns point to fertile areas for future research. They suggest ingredients for modest theories of relationships between teaching and learning, and they map out a territory for promising empirical studies.

OPPORTUNITY TO LEARN: STILL THE KEY ENABLING CONDITION

“Opportunity to learn is widely considered the single most important predictor of student achievement” (National Research Council, 2001, p. 334). Defined by the National Research Council (2001) as “circumstances that allow students to engage in and spend time on academic tasks . . .” (p. 333), the concept of opportunity to learn has been proposed for some time as an explanation for differences in achievement among students (Floden, 2002). It has been used to account for differences in mathematics learning across a variety of contexts, including between-country differences on international comparisons. Fletcher (1971), for example, in a reanalysis of mathematics achievement data from the First International Study of Mathematics Achievement (Husen, 1967) found opportunity to learn a key explanatory construct. He summarized, “the inescapable conclusion, therefore, seems to be that reported ‘achievement’ is virtually synonymous with ‘coverage’ across countries (p. 145).” Carroll’s (1963) influential model for school learning included time as well as topic coverage as an important variable and promoted the concept’s popularity. Researchers from inside and outside of mathematics education continue to include opportunity to learn on the short list of well-documented findings that link teaching and learning (Brophy, 1999; Grouws, 2004; National Research Council, 2001).

Opportunity to learn is not entirely a function of teaching or entirely under the control of the teacher. The curriculum the teacher is required to use, for example, surely influences students’ opportunity to learn (Stein et al., this volume). But teaching, as we have defined it, plays a major role in shaping students’

learning opportunities. The emphasis teachers place on different learning goals and different topics, the expectations for learning that they set, the time they allocate for particular topics, the kinds of tasks they pose, the kinds of questions they ask and responses they accept, the nature of the discussions they lead—all are part of teaching and all influence the opportunities students have to learn.

Opportunity to learn as a concept that links teaching and learning is best viewed as something more nuanced and complex than simply exposure to subject matter. Put simply, students who are exposed to a topic obviously have a better chance of learning it than students who are not. But, opportunity to learn can mean something more interesting and useful. Consider first graders exposed to a lesson on calculus. Do they have an opportunity to learn calculus? Only in an overly simplistic sense. In reality, first graders would not have a good opportunity to learn calculus because they are not sufficiently prepared to learn the material and cannot engage in the tasks and discussions that would support such learning. They might learn something by sitting through the lesson, like how to sit quietly and listen to things they do not understand. Why? Because that is what the conditions afford. It might not be the intended learning, but it is, in a real sense, what they have an opportunity to learn. So, “opportunity to learn” is not the same as “being taught.” Opportunity to learn includes considerations of students’ entry knowledge, the nature and purpose of the tasks and activities, the likelihood of engagement, and so on.

Vygotsky’s (1978) well-known concept of the *zone of proximal development* captures the general principles behind the example of first graders being taught calculus. The zone of proximal development refers to the space within which learning can be expected, under supportive conditions, given the person’s current level of functioning. It describes what is possible, given appropriate opportunities.

Opportunity to learn can be a powerful concept that, if traced carefully through to its implications, provides a useful guide to both explain the effects of particular kinds of teaching on particular kinds of learning and improve the alignment of teaching methods with learning goals. If conceptual understanding, for example, is a valued learning goal, then students will need opportunities to develop such understanding. Similar statements can be made for other learning goals, such as smooth execution of procedures, problem-solving skill, and so on. These kinds of learning do not happen automatically; students need continuing opportunities to achieve them.

Although the concept is useful, the fact that opportunity to learn still is the best link between teaching and learning signals that the research community has not advanced very far in teasing out more specific connections. In the next section, we take a small step in identifying more specific connections between features of mathematics teaching and students' learning. The two patterns we describe can be thought of as instances of opportunities to learn: opportunities to develop skill efficiency and opportunities to develop conceptual understanding.

TWO PATTERNS: TEACHING FOR SKILL EFFICIENCY AND TEACHING FOR CONCEPTUAL UNDERSTANDING

Two of the most valued learning goals in school mathematics are, and have been for some time, skill efficiency and conceptual understanding (Resnick & Ford, 1981). By *skill efficiency*, we mean the accurate, smooth, and rapid execution of mathematical procedures (see, for example, *skill learning* in Gagne, 1985). We do not include the flexible use of skills or their adaptation to fit new situations. By *conceptual understanding*, we mean mental connections among mathematical facts, procedures, and ideas (Brownell, 1935; Davis, 1984; Hiebert & Carpenter, 1992). This definition suggests that conceptual understanding grows as mental connections become richer and more widespread. As we mentioned earlier, considerable research has been conducted to identify the kinds of teaching that facilitate students' achievement of these two learning goals.

An initial and rather obvious finding is that some kinds of teaching support skill efficiency and other kinds of teaching support conceptual understanding. That different kinds of teaching facilitate different kinds of learning is a direct consequence of opportunity to learn. Different kinds of teaching provide different opportunities to learn that, in turn, yield different kinds of learning. This notion has been expressed for some time. Here is what Edward Thorndike had to say in 1912, "It is possible, by focusing attention upon immediate facility, to choose methods of teaching that are excellent for that, but defective for the more important service of arousing in a pupil the desire and power to educate himself [*sic*]" (p. 197). Gagne (1985) refined the notion by detailing different instructional techniques appropriate for different learning outcomes. And Brophy (1988) reminded educators that the work he reviewed linking teaching behaviors to students' learning mostly had used measures of routine

skill learning. Other teaching features might have been identified if other learning outcomes had been used.

The story gets more complicated when one notices that neither theory nor empirical data indicate a simple correspondence between one method of teaching and skill efficiency and between another method of teaching and conceptual understanding. The best way to express current knowledge in the field is to describe some features of teaching that facilitate skill efficiency and some features of teaching that facilitate conceptual understanding, and to indicate where these features overlap or intersect.

At this point we should recognize an important historic distinction that is related to but different from the distinction in learning goals we are making here. In a classic book on learning and teaching, Ausubel (1963) proposed a 2×2 matrix with rote versus meaningful learning on one axis and discovery versus expository teaching on the other axis. Ausubel contended that these dimensions were independent. Expository teaching, said Ausubel, does not necessarily produce rote learning, and discovery teaching does not necessarily produce meaningful learning. We agree with Ausubel and raise the distinction here for two reasons. First, we want to alert readers that we are not necessarily linking skill efficiency with expository teaching. Based on our definition, skills might be executed efficiently under rote or meaningful learning conditions. Conceptual understanding, on the other hand, would require meaningful learning.

A second reason for raising Ausubel's (1963) distinction is that we do not expect the features of teaching that facilitate skill efficiency and conceptual understanding to fall neatly into categories such as "expository" or "discovery." In fact, the features of teaching we describe do not fit easily into any of the categories frequently used to describe teaching: direct instruction versus inquiry-based teaching, student-centered versus teacher-centered teaching, traditional versus reform-based teaching, and so on. Although these categories and labels have been useful for some purposes because they capture constellations of features and treat teaching as systems of interacting components, they also can be misleading because they group together features in ill-defined ways and connote different kinds of teaching to different people. For our purposes, avoiding past connotations is important. In fact, we will argue that most of these categories, distinctions, and labels are now more confusing than helpful, and further advances in research as well as clarity of policy recommendations will benefit from abandoning these labels.

But abandoning the usual labels for methods or systems of instruction in favor of identifying features

of teaching that link with learning appears to violate a principle we espoused earlier—teaching is a system of interacting elements, and the effects of an individual feature are determined by the system of which it is a part. Can these two points of view be reconciled? Can researchers single out features of teaching that facilitate learning and still believe that the specific effects of individual features depend on the systems of teaching in which they operate? We think so, for two reasons. First, by looking for patterns across multiple studies, we are searching for features that replicate their effects through different implementations of similar systems and even through different systems. We suspect that the exact effects of target features vary somewhat from system to system, but we expect some of these features to emerge as sufficiently robust across systems to appear as detectable patterns. Second, we assume that when we identify specific features of teaching we actually are identifying clusters of features. The features we single out in the following sections probably operate within a supportive set of less apparent features. The regularity that can be detected in their effects on students' learning might be the result of clusters of features that function well within a range of systems of teaching. We single out specific features of teaching because the empirical data do not yet allow a more precise interpretation. But the hypothesis of systems of teaching predicts that these features are simply the most salient aspects of surrounding clusters. Perhaps the development of sharper research tools will afford more precise descriptions in the future.

Becoming Efficient in Executing Skills

No empirical studies that we are aware of set out to examine which features of teaching support skill efficiency and which support conceptual understanding. Studies have contrasted methods of teaching that emphasize skills versus understanding (e.g., Brownell & Moser, 1949; Cobb et al., 1991; Heid, 1988; Hiebert & Wearne, 1993), but no single study or set of studies aimed to optimize these two learning outcomes and sort out the features of teaching associated with each. In addition, different researchers used different measures of teaching and learning, so patterns across studies must be abstracted by interpreting designs and results and inducing consistencies, not simply by adding up the findings.

The large set of studies conducted within the process-product paradigm provides the best evidence

from which to induce patterns for links between teaching and skill efficiency. The literature includes studies in mathematics and in other subjects, and, for the most part, the features of teaching that correlate with skill efficiency are similar across subjects. The most problematic aspect of interpreting this work is characterizing the nature of the dependent measures. Many studies used standardized achievement tests to assess students' learning. Although these tests often are composites of a range of items, they include a heavy skill component, require relatively quick responses, and are restricted to closed-ended, multiple choice formats. Consequently, we believe we can appropriately infer that skill efficiency, as we defined it earlier, is captured by these outcome measures.

Recall that process-product studies attempt to identify relationships between what teachers do in the classroom (the process) and what students learn as a result of this instruction (the product). To provide trustworthy data, these studies must include large sample sizes, use reliable classroom observation instruments to measure classroom processes, and assess student learning in reliable ways. We review several studies that meet these criteria and provide insights into one set of teaching strategies that is related to mathematical skill learning as measured by standardized achievement tests. The interested reader can examine a comprehensive summary of this line of research and its implications across many subject matter fields in the work of Brophy and Good (1986).

In a naturalistic study of mathematics teaching, Good and Grouws (1977) examined the teaching performance of more than 100 third- and fourth-grade teachers over a 2-year period. An initial finding was that the impact of many teachers on students' mathematics learning, as measured on a standardized achievement test, was not very stable over time. This finding suggests that their teaching methods might have changed from year to year, perhaps due to the changing composition of their classes. The authors were able, however, to identify a group of teachers who consistently facilitated better than statistically predicted achievement results over 3 consecutive school years² and teachers who were less successful than would have been expected during the same time period. Results from the study indicated that teaching effectiveness was associated with the following behavioral clusters: whole-class instruction³; general clarity of instruction; a nonevaluative, task-focused en-

² The teachers were identified by using achievement data over a two year period and their success was confirmed for the third year during the course of an observational study.

³ Whole class instruction was the organizational mode in the high classrooms (and low classrooms) with teachers employing instruction via groups in the middle of the distribution.

vironment; higher achievement expectations (more homework, fast pace); student initiated behavior; and classrooms relatively free of behavioral problems. The faster pace finding was also identified in a study of fifth-grade classrooms conducted by McDonald (1976) and singled out by Leinhardt (1986) as a characteristic of teaching associated with standardized achievement gains. Other common findings in the literature are illustrated in the next process-product study we examine.

Using the process-product paradigm, Evertson, Anderson, Anderson, and Brophy (1980) studied junior high school teachers of mathematics ($N = 29$) and English ($N = 39$). They used both low-inference and high-inference observational coding schemes and measured student learning using the California Achievement Test (CAT). Effective mathematics teachers asked more questions than did less effective teachers with most of the questions lower-order product questions. The authors summarized that “the most effective [mathematics] teachers were active, well organized, and strongly academically oriented” (p. 58). Further, as in the previous described study, the effective teachers tended to emphasize whole-class instruction with some time devoted to seatwork. In general, they actively managed their class efficiently. Although the terminology used in this study was different from that in the Good and Grouws (1977) study, some obvious parallels exist in their findings.

A series of studies conducted by researchers at the Far West Regional Laboratory for Educational Research and Development during the 1970s focused on linking teacher behavior to student achievement in mathematics and reading. This research became known as the Beginning Teacher Evaluation Study (BTES), although in practice the studies involved teachers with at least 3 years of experience and concentrated on research rather than evaluation. Interestingly, none of the teacher behaviors measured was a significant predictor of student learning in both mathematics and reading. This finding lends credence to the position that some important teaching actions are content specific.

The results from the BTES about mathematics teaching and learning tend to agree with the previously cited studies but add some interesting findings when data are examined at the individual student level. The studies found a positive relationship between student achievement on a standardized achievement test and high success rates when engaged in mathematical work during classroom lessons (correct answers except for occasional errors). Conversely, there was a negative relationship when students were engaged in activities with low success rates. The researchers

considered this an important enough finding to refine the concept of opportunity to learn to mean “the amount of time a student spends engaged in an academic task [related to the intended learning] that s/he can perform with high success” (Fisher et al., 1980, p. 8). Interestingly, the attention given to high success rates in these studies provided the rationale for Good and Grouws emphasizing the importance of students being well prepared for seatwork before working independently. An overview of the studies in the BTES can be found in Powell (1980) with a detailed account provided in the five volume final report by McDonald and Elias (1976).

The features of teaching that emerge from the studies just reviewed can be summarized as follows: teaching that facilitates skill efficiency is rapidly paced, includes teacher modeling with many teacher-directed product-type questions, and displays a smooth transition from demonstration to substantial amounts of error free practice. Noteworthy in this set of features is the central role played by the teacher in organizing, pacing, and presenting information to meet well-defined learning goals.

The features of teaching identified above facilitate students’ skill efficiency as measured by standardized tests. What effects these features have on students’ conceptual development is unclear. Because conceptual understanding is of great interest in mathematics education and infuses the recently conceived goal of mathematical proficiency (National Research Council, 2001), we devote considerable space in the next section to examining links between teaching and students’ conceptual understanding.

Developing Conceptual Understanding

We defined conceptual understanding earlier as the mental connections among mathematical facts, procedures, and ideas (Brownell, 1935; Davis, 1984; Hiebert & Carpenter, 1992). This simple definition works well for our purposes but is not without controversy. Some researchers argue that this definition places too much emphasis on the internal or mental representations of individuals and that social/cultural factors should be considered (Lave, 1988; Wenger, 1998). From this sociocultural view, understanding can be seen as an activity of participating in communities of people who are becoming competent in the practices of the trade. In our case, this would mean participating in a community of people who are becoming adept at doing and making sense of mathematics as well as coming to value such activity. We believe that both the mental connections and the cultural participation views of understanding can be useful (Cobb,

1994; Sfard, 1998) and that the claims we make in the following sections are reasonable under either definition. However, because the research examples we review were conducted within the then-prevailing mental connections view, this definition should be kept in mind when reading the next sections.

Teaching That Promotes Conceptual Development: Two Key Features

Educators always have been interested in promoting conceptual understanding, at least in their rhetoric, but there has been a recent explosion of interest in the topic. On the day we drafted this paragraph, amazon.com found 90,448 book entries for the terms: teaching for understanding. Among the best-selling titles were *Teaching for Musical Understanding*, *Strategies That Work: Teaching Comprehension to Enhance Understanding*, and *Shakespeare Without Fear: Teaching for Understanding*. Two well-known books that contain material relevant for this chapter are *Teaching for Understanding* edited by Cohen, McLaughlin, and Talbert (1993) and *Mathematics Classrooms That Promote Understanding* edited by Fennema and Romberg (1999). Clearly, the topic does not lack for analyses and recommendations.

The research base that links features of teaching with students' understanding is much thinner than the current attention might suggest. Looking across the landscape of research connecting teaching and learning, we agree with Brophy's (1997) observation that the research conducted provides less insight into how students construct understanding during instruction than in how they develop skill efficiency. This is due, in part, to the challenge of creating measures that assess conceptual understanding and the fact that much past work has used standardized tests and other similar measures that are more skill oriented. Nevertheless, we can identify two features that emerge from the literature as keys to promoting conceptual understanding. We view our presentations of both features as plausible conclusions drawn from past research *and* as descriptions of domains in which future research could be especially rich and useful.

Feature 1: Teachers and Students Attend Explicitly to Concepts

A clear pattern across a range of empirical studies is that students can acquire conceptual understandings of mathematics *if* teaching attends explicitly to concepts—to connections among mathematical facts, procedures, and ideas (Gamoran, 2001; Hiebert, 2003; National Research Council, 2001). By *attending to concepts* we mean treating mathematical connections in

an explicit and public way. Brophy (1999) described such teaching as infused with coherent, structured, and connected discussions of the key ideas of mathematics. This could include discussing the mathematical meaning underlying procedures, asking questions about how different solution strategies are similar to and different from each other, considering the ways in which mathematical problems build on each other or are special (or general) cases of each other, attending to the relationships among mathematical ideas, and reminding students about the main point of the lesson and how this point fits within the current sequence of lessons and ideas.

To elaborate our claim and to illustrate the findings that support it, we will review several studies that typify those upon which the claim is based. More extensive reviews of this literature can be found in *Adding It Up* (Chapters 6, 7, and 8, National Research Council, 2001), in *A Research Companion to Principles and Standards for School Mathematics* (Kilpatrick, Martin, & Schifter, 2003), in *Mathematics Classrooms That Promote Understanding* (Fennema & Romberg, 1999), and in a chapter by Hiebert and Carpenter (1992) in a previous handbook for teaching and learning mathematics.

In many ways, the claim that students acquire conceptual understanding of mathematics if teaching attends explicitly to mathematical concepts is a restatement of the general observation that students learn what they have the best opportunity to learn. At a surface level, the claim seems rather obvious and, perhaps, not even very interesting. But it becomes more interesting when one discovers that the claim is supported across a wide range of research designs and holds true across different instructional treatments or systems. The studies we review, and the larger literature from which these studies are drawn, display a remarkable variation. Variability across studies can be described in the following ways:

- The effects appear in studies of varying designs—from tightly controlled studies of shorter duration to more loosely designed studies of longer duration;
- Both teacher-centered and student-centered teaching systems or methods (using the old labels) that have attended to conceptual development have shown higher levels of students' conceptual understanding than similar methods that have not attended to conceptual development as directly;
- The ways in which concepts are developed in classrooms can vary—from teachers actively directing classroom activity to teachers taking

less active roles, from methods of teaching that highlight special tasks and/or materials to those that highlight special forms of classroom discourse to those that highlight student invention of solution strategies.

We believe the evidence does not justify a single or “best” method of instruction to facilitate conceptual understanding. But we believe the data do support a feature of instruction that might be part of many methods: explicit attention to conceptual development of the mathematics.

To support and clarify the claim, we chose the following first set of studies to review because they focused on the same mathematical topic—multidigit addition and subtraction in the primary grades—but implemented quite different treatments using different research designs. Fuson and Briars (1990) investigated the effects of a particular conceptual instruction approach on students’ acquisition of the standard addition and subtraction algorithms. Specially designed instruction on addition was given to 169 first and second graders, and 75 of the second graders also received special instruction on subtraction. Instruction included careful demonstrations of the algorithm using base-10 blocks, with special attention to the pattern of the regrouping as the numbers increased to three and four digits. Connections were explicitly drawn between the physical and written representations. Some teachers used whole-class formats with teacher demonstrations and class discussions, and other teachers asked students to work in small groups and complete some problems on their own. In all classrooms, the focus was on understanding the standard algorithm, not inventing new strategies. Before instruction, and then after 3–6 weeks of instruction on place value and addition, plus an additional 2–4 weeks for those receiving instruction on subtraction, students were assessed using a traditional written skills test, a specially designed written test on underlying concepts, and individual interviews. As might be expected for students this young, pretest performance was quite low. For example, only 9 of the 169 students correctly regrouped when completing a multidigit addition task. In contrast, performance was uniformly high on the posttest. For example, 160 of the 169 students correctly regrouped on a similar posttest item. Even more striking were the ceiling levels of performance on conceptually based items, including those on the interviews asking students to explain why and how regrouping works in the multidigit algorithm. Given the high levels of performance across the full sample, the particular ways in which the teachers implemented the instruction (e.g., whole

class demonstrations versus small group discussions) apparently did not substantially affect the outcomes. The key seemed to be the explicit attention to the conceptual underpinnings of the algorithm through connecting written and physical representations.

The absence of a control group in the Fuson and Briars (1990) study leaves many questions unanswered. These are answered, in part, as the authors point out, by the fact that the level of performance by the first graders and the (lower achieving) second graders in their study was higher than that of many third and fourth graders on similar items reported on the National Assessment of Educational Progress. This comparatively high performance supports claims about the effectiveness of the instruction. In addition, the findings were replicated by the authors (reported in the same article), with only slightly lower levels of performance, with 2,723 second graders in the Pittsburgh City Schools. Clearly, the effects of this instruction were not an artifact of a special sample. In sum, the findings can be viewed as an existence proof that attending to conceptual underpinnings—to connections among mathematical ideas and representations—can facilitate students’ conceptual understanding. The results also show strong skill learning, a finding that we will explore in a later section of the chapter.

A second study, conducted 40 years earlier by Brownell and Moser (1949), employed control groups and fills in another part of the picture on the effects of instruction that attends explicitly to conceptual development. Brownell and Moser studied the effects of teaching multidigit subtraction algorithms “meaningfully” versus “mechanically.” About 1,400 third graders in 41 schools were involved in the study. Two different algorithms, equal additions (adding equal amounts to the minuend and subtrahend to keep the difference the same) and decomposition (the current standard U.S. algorithm of regrouping), were crossed with two teaching approaches (meaningful and mechanical) to create four conditions. Teachers in the 41 schools who volunteered to participate indicated which of the approaches they preferred. Classrooms were assigned to conditions by taking into account teachers’ preferences and by trying to equalize students’ entry abilities across conditions. Each lesson for each condition during the 15-day experimental period was prescribed, and all lessons were quite teacher-directed. The meaningful or conceptual approach included the use of physical objects (bundles of sticks) to illustrate grouping by tens, writing numerals in expanded notation and connecting written and physical representations, and delaying the step-by-step algorithm until after examining the meaning of the number ma-

nipulations. The mechanical approach began with a demonstration of the step-by-step algorithm and did not include a rationale for the procedure. The time saved by this approach was used for practicing the procedures. Students' learning was assessed immediately after the instruction and 6 weeks later using written computation tests and individual interviews. Items measured speed, accuracy, explanations of procedures, and transfer to new kinds of subtraction problems. The results are complicated, but, in general, the following conclusions can be drawn. On the immediate measures of speed and accuracy, the mechanical approach produced higher scores. On measures of retention and transfer, the meaningful approach produced higher scores. The meaningful approach worked better for the decomposition algorithm than the equal additions algorithm and for students who had not already memorized a subtraction algorithm before they began instruction. In general, if good explanations, retention, and transfer are valued learning goals (Brownell, 1948), then the meaningful approach showed the most promise, regardless of which algorithm was taught.

Critics might argue that the results reported by Brownell and Moser (1949) are not conclusive because the mechanical approach was highly constrained and the large sample of teachers, without classroom observations to suggest otherwise, could have implemented the approaches in quite different ways. The criticisms are appropriate and should be considered when interpreting the results. Still, we can reasonably conclude that explicit attention to conceptual underpinnings resulted in higher levels of conceptual understanding. This reinforces the conclusion drawn from Fuson and Briars (1990) using a much different research design and separated in time by more than 40 years.

A third study on multidigit addition and subtraction fills in another part of the picture. Hiebert and Wearne (1993) examined the effects of a mixed teacher- and student-centered form of conceptually-based instruction on conceptual understanding (and procedural skill). The 147 second graders in the six classrooms of one school were assigned to one of two treatment conditions for the 12 weeks normally devoted to multidigit addition and subtraction by the teachers. Conceptually-based instruction focused on building a foundation of place value understanding by connecting written and physical (base-10 blocks) representations of numbers and then asked students to invent their own procedures for solving addition and subtraction problems. Teachers demonstrated the standard algorithms if they were not presented by students. Traditional instruction followed the textbook as ordinarily used by the teachers. The focus in

these classrooms was on demonstrating and practicing the standard algorithms. Four classrooms (one high achieving and three average-low achieving) received the conceptually based instruction taught by specially trained teachers and two classrooms (one high achieving and one average-low achieving) received the traditional instruction taught by the classroom teachers. Weekly observations and audiotapes recorded the tasks used during the lessons and the talk by teachers and students in all the classrooms. Instruction, as implemented, showed differences between the number and kind of tasks presented and the nature of the classroom discourse. In general, students receiving the conceptually based instruction spent a greater percentage of time working on the rationale for procedures and examining the legitimacy of invented procedures. Students receiving the traditional instruction spent a greater percentage of time practicing taught procedures. On the basis of beginning-of-the-year and end-of-the-year tests, students in the conceptually based classrooms gained more on skill items and, especially, on conceptual items such as explaining the rationale for computation procedures and transferring them to new kinds of problems. By the end of the year, the performance profile of the lower achieving classroom that received the conceptually based instruction looked much like the higher achieving traditionally instructed classroom.

The use of specially trained teachers for the conceptually based instruction and the relatively small sample size in Hiebert and Wearne (1993) lead to less-than-conclusive results but for different reasons than in Fuson and Briars (1990) or in Brownell and Moser (1949). Each study contains some design deficiencies, but the deficiencies lie in different places. That is why we avoid drawing conclusions about the links between the specific kinds of instruction on multidigit addition and subtraction described in each study and the acquisition of specific skills and understandings. But we do see a pattern across these studies, and others like them, in the general feature of attending to conceptual development. We conclude that when teaching attends explicitly and directly to the important conceptual issues students are more likely to develop important conceptual understandings. This conclusion is reinforced further by considering a few additional and very different kinds of studies.

The evidentiary base supporting conceptual teaching is not restricted to research in the computational skills domain. In a summary report of three separate experimental studies of mathematics learning defined broadly in classrooms at Grades 4, 6, and 8, respectively, Good, Grouws, and Ebmeier (1983) found substantial support for an instructional program that was

teacher directed, tightly controlled, and emphasized attention to conceptual development. To define (conceptual) development, the authors drew on the early work of Brownell (1947) on meaningful learning and the subsequent recapitulation of this work, termed *relational understanding*, by Skemp (1976). They summarized their use of the term *development* as “that part of a lesson devoted to increasing comprehension of skills, concepts, and other facets of the mathematics curriculum” (p. 203).

Common to the set of studies reported by Good et al. (1983), and quite different from the preceding studies reviewed in this section, was the use of a treatment that asked teachers to follow a prescribed, highly structured daily teaching plan that employed whole class instruction and included a brief introductory segment (attend to homework, short review, mental computation practice), a development component (about one half the remaining class period), seatwork, lesson closure, and assignment of homework. In the studies, intact classrooms were randomly assigned to treatment and control conditions. Control group teachers were asked to teach using their usual teaching methods.

A number of unique features of these studies are worth noting when interpreting the findings. First, total test scores and subtest scores of the mathematics portions of standardized achievement tests were the main dependent measures used to determine student learning, although as will be mentioned later, supplementary content measures were used in some studies. Second, analysis of variance on residualized gain scores were used to examine the pre-post test differences on each measure. Third, class means were used as the unit of analysis in the statistical tests employed. Finally, interpretation of results took account of treatment fidelity using data from various combinations of classroom observations, teacher diaries, and self-report questionnaires.

In the fourth-grade study of 40 classroom teachers, the treatment teachers exhibited substantially more of each of the requested teaching features than did the control teachers. Furthermore, the scores of the students of the treatment teachers showed significant improvement from pretest to posttest moving from the 27th percentile to the 57th percentile on national norms, significantly higher than the student scores from the control group ($p = .003$) (Good & Grouws, 1979). On a specially designed content test developed by an outside consultant to measure specific content in the curriculum including skills, concepts, and verbal problem solving, the treatment group again outperformed the control group, but the statistical difference was more modest ($p = .10$) with only small differ-

ences between the treatment group and the control group on the verbal problem-solving items.

In the sixth-grade study involving 36 teachers and in the eighth-grade junior-high study involving 19 teachers (41 classes), the treatment was modified to give regular, systematic attention to verbal problem solving. In both studies, the treatment students outperformed the control students on the standardized achievement test and on a research-constructed measure of verbal problem solving. The differences on both measures, however, were not as large as in the fourth-grade study. The authors hypothesized that the smaller effects in the sixth-grade study might have been due to smaller differences between teaching methods in the treatment and control classrooms than in the original study. They attributed the lack of larger teaching differences to the possibility of sixth-grade teachers visiting with and learning from the fourth-grade treatment teachers who participated in the study the previous school year. The somewhat diminished fidelity of Grade 8 teachers to the teaching expectations requested in the treatment might be due, according to the authors, to a shorter training period than in previous studies and to a possible treatment request overload—the teachers were asked to implement all the teaching requests in the previous study plus the requests associated with increased attention to verbal problem solving. In spite of less fidelity to treatment than in the previous studies, the researchers did find a statistically significant difference between treatment and control classrooms on the problem-solving measure at the sixth-grade level when 7 poorly implementing teachers were removed from the analysis.

The researchers are quick to point out that they “are *not* suggesting that the instructional program used in the study is the best approach to take for facilitating the mathematics achievement of students” (Good et al., 1983, p. 91), but rather that the ideas within the program including more attention to conceptual development, systematic attention to review, and utilization of appropriate homework assignments might be of considerable value to teachers who use a whole-class organizational pattern for teaching mathematics. The important message from these studies for the current discussion is the recurring link between explicit attention to the development of mathematical connections among ideas, facts, and procedures and the increased conceptual learning by students. The studies not only support this contention but also seem to show that such instruction facilitates the development of important mathematical skills beyond those associated with computational skills, a point to which we will return in a later section.

In contrast to the rather tightly controlled and precisely targeted designs of the studies reviewed above, Fawcett (1938) and Boaler (1998) reported multi-year studies on the cumulative effects of instruction at the secondary level. Fawcett described a 2-year experimental course in high-school geometry for which instruction was redesigned to begin with the reasoning of students and build toward conventional deductive forms of reasoning through, in part, analyzing the logic of real-life arguments. Conceptual aspects of this instruction are apparent in the explicit attention to connections among mathematical objects—definitions, arguments, proofs, claims, and so on. Performances of the 25 students in this course were compared to those of a similar number of students in a more traditional (2-year) course in geometry. On conventional measures of geometric content, on retention of content, and on transfer measures involving the analysis of reasoning in nonmathematical situations, students in the experimental group outperformed students in the control group. Although many design features undermine the comparative claims (e.g., the absence of information about the instruction provided in the more traditional course), a fact acknowledged explicitly by Fawcett (1938), it does seem clear that students in the experimental group engaged with important mathematical ideas and developed important understandings.

Boaler's (1998) study, 50 years later, addressed some of the design deficiencies of Fawcett's (1938) study and reached similar conclusions. Boaler followed a cohort of students at two different secondary schools for 3 years. The schools were chosen because they claimed to be implementing different forms of instruction. After observing regularly lessons in both schools, Boaler described the instruction in one school as an efficient and well-ordered textbook-based approach to procedural competence. Teachers demonstrated procedures and students practiced them. Instruction in the other school was directed toward more conceptual learning goals and revolved around open-ended projects. Students often worked cooperatively, sharing and explaining solution strategies. Entry measures showed that the approximately 200 students in the first school and the approximately 100 students in the second school performed similarly when they began. But, after 3 years, students in the second school, emphasizing conceptual problems and discussions, performed better than their peers on both specially designed conceptual measures and on standardized tests.

In summary, the findings from these studies typify those found in much of the literature that assesses the effects of instruction that explicitly attends to conceptual development of mathematics. Students receiving

such instruction develop conceptual understandings, and do so to a greater extent than students receiving instruction with less conceptual content. As noted earlier, this claim might not be surprising because it is a straightforward instantiation of the principle of opportunity to learn. It becomes more interesting, however, when one notes that conceptual development of the mathematics can take many pedagogical forms. Concepts can be developed through teacher-centered and highly structured formats as in some of Fuson and Briar's (1990) classrooms and in Good et al.'s (1983) classrooms or through student-centered and less structured formats as in Fawcett's (1938) classroom. The claim becomes even more interesting and important when making two additional observations that will be developed later in the chapter. First, as we have noted, the findings from several of the studies we reviewed suggest that instruction emphasizing conceptual development facilitated skill learning as well as conceptual understanding. Second, although instructional attention to conceptual development might seem like an obvious feature of "good" teaching and might be expected in most classrooms, it is a feature consistently absent in descriptions of U.S. mathematics classrooms. Before exploring these two issues, we present the second feature of instruction that is associated consistently with increased conceptual understanding.

Feature 2: Students Struggle with Important Mathematics

Our interpretation of the literature on teaching for conceptual understanding points to a second feature of teaching that consistently facilitates students' conceptual understanding: the engagement of students in struggling or wrestling with important mathematical ideas. Unlike the first feature, this second feature might not be obvious to readers, so we first clarify what we mean by *struggle*, then elaborate the theoretical connection between struggling with and understanding mathematics, and finally review a small sample of empirical studies from this perspective.

We use the word *struggle* to mean that students expend effort to make sense of mathematics, to figure something out that is not immediately apparent. We do *not* use *struggle* to mean needless frustration or extreme levels of challenge created by nonsensical or overly difficult problems. We do not mean the feelings of despair that some students can experience when little of the material makes sense. The struggle we have in mind comes from solving problems that are within reach and grappling with key mathematical ideas that are comprehensible but not yet well formed (Hiebert et al., 1996). By struggling with important mathematics we mean the opposite of simply being presented

information to be memorized or being asked only to practice what has been demonstrated.

Dewey (1910, 1926, 1929) devoted a good deal of attention to elaborating the processes and consequences of this kind of struggle. For Dewey, the process of struggle was essential for constructing deep understandings. The process begins, said Dewey (1910) with “some perplexity, confusion, or doubt (p. 12).” It continues as students try to fit things together to make sense of them, to work out methods for resolving the dilemma. Traditional school instruction, noted Dewey (1926), is plagued by a press for quick answers. This short-circuits opportunities to think about the conceptual aspects of the problem. Deep knowledge of the subject, said Dewey (1929), “is the fruit of the undertakings that transform a problematic situation into a resolved one” (p. 242–243).

Other cognitive theorists have developed similar arguments that connect the process of struggling to make sense of the subject with deeper understandings of the subject. Festinger’s (1957) theory of cognitive dissonance developed the notion of cognitive perplexity as a central impetus for cognitive growth. In a more targeted way, Hatano (1988) identified cognitive incongruity as the critical trigger for facilitating the development of reasoning skills that display conceptual understanding. For Hatano, cognitive incongruity places the student in the situation described by Dewey as filled with perplexity, confusion, or doubt. Applying a similar theoretical construct to school classrooms, Brownell and Sims (1946) contrasted the repetitiveness of most school tasks with the processes they argued are essential for developing understanding—students must feel a need to resolve a problematic situation and then be allowed “to muddle through” (p. 40). What kinds of tasks or problems pose appropriate problematic situations? Vygotsky’s (1978) zone of proximal development provides a guide for answering this question because it can be interpreted as the space within which a student’s struggle is likely to be productive. That is, novel problems near the boundaries of a student’s zone of proximal development are problems that are within reach but that present enough challenge, so there is something new to figure out.

Mathematicians and others who are concerned about how students interact with the discipline of mathematics have alluded to struggling with key mathematical ideas as a natural part of doing mathematics. In his classic book on problem-solving heuristics, Polya (1957) echoed themes of Dewey (1910) by opening the preface to the book’s first printing with the following:

A great discovery solves a great problem but there is a grain of discovery in the solution of any problem. Your problem may be modest; but if it challenges your curiosity and brings into play your inventive faculties, and if you solve it by your own means, you may experience the tension and enjoy the triumph of discovery (p. v).

Polya continued by noting that experimentation and struggle are an integral part of developing the discipline of mathematics and suggested that students should experience this side of mathematics as well as the systematic, deductive side. Brown (1993) explicitly brought together Dewey’s and Polya’s views around the notion of struggling with mathematics, a phenomenon Brown called “confusion.” He suggested that struggling with real problems is quite different than the tidy curriculum students usually receive, and he argued that appropriate struggle (or confusion) reveals to students a key aspect of mathematics. “We seem incapable of appreciating what it means to be confused, what virtues may follow from it and what kind of pedagogy may make use of it” (Brown, 1993, p. 108).

The logic that connects struggle with understanding, the learning goal of interest in this section, can be expressed using either of the definitions of *understanding* presented earlier. If understanding is defined as the mental connections among mathematical facts, ideas, and procedures, then struggling is viewed as a process that reconfigures these things. Relationships among facts, ideas, and procedures are re-formed when new information cannot easily be assimilated or when the old relationships are found to be inadequate to make sense of a new problem (Piaget, 1960; Skemp, 1971). Struggle results in restructuring one’s mental connections in more powerful ways. If understanding is defined as participating in a community of people who are becoming adept at doing and making sense of mathematics, then struggling is vital because it can be an essential aspect of personal meaning-making within the community (Handa, 2003). It is central to what mathematicians do (Brown, 1993; Polya, 1957). In summary, struggling with important mathematics is implicated in both definitions of understanding by identifying the common processes hypothesized to develop understanding.

Although little, if any, research has tried to isolate and test the effects of struggle on the development of students’ conceptual understandings, many findings suggest that some form of struggle is a key ingredient in students’ conceptual learning. In a provocative review of cognitive training studies (outside of mathematics education), Robert Bjork (1994) pointed out a curious feature in many of the findings: When trainees experi-

enced difficulties and could not smoothly master the targeted skills, they developed deeper or more useful competencies in the long run. The difficulties were introduced either intentionally or accidentally into the training procedure and took different forms—changing the context of problems, withholding feedback, introducing interfering distractions, and so on. The key condition was that trainees faced some obstacles along the way. In these cases, the trainees generalized their learning more appropriately and transferred it more effectively to novel tasks. Apparently the trainees had to think harder about what they were learning. Bjork (1994) concluded:

Whatever the exact mixture of manipulations that might turn out to be optimal [for learning], one general characteristic of that mixture seems clear: It would introduce many more difficulties and challenges for the learner than are present in typical training routines (p. 189).

This conclusion is not surprising, said Bjork, because it simply affirms the observation, now widely accepted in cognitive psychology, that learners construct their own interpretations and memories of activities rather than simply absorbing them from the environment. When students struggle (within reason), they must work more actively and effortfully to make sense of the situation, which, in turn, leads them to construct interpretations more connected to what they already know and/or to reexamine and restructure what they already know. This yields content and skills learned more deeply.

No series of classroom studies parallels those of the laboratory studies in the training literature reviewed by Bjork (1994), but, when viewed through the lens of students' struggle, the findings from a number of classroom studies are suggestive. Again, we review only a few studies here to illustrate the nature of these findings. First, in a study outside of mathematics, Capon and Kuhn (2004) found that students in a graduate MBA course who learned a new business concept through attempting to solve problems rather than listening during a lecture and discussion could apply what they learned to more effectively explain a related concept. The authors accounted for the results by arguing that working to solve problems promoted more sense making and integrating of information, in other words more conceptual understanding.

We now review several studies in mathematics education that support the claim that some features of teaching can facilitate students' opportunities to struggle, in the sense that we have defined it, and when this happens students develop greater conceptual under-

standing. Although Carpenter, Fennema, Peterson, Chiang, and Loef (1989) did not interpret their data from this perspective, they reported findings that are consistent with our claim. The authors compared the teaching and learning in 40 first-grade classrooms. Half of the teachers from these classrooms were randomly selected to receive a 4-week summer course (called Cognitively Guided Instruction, or CGI) that focused on young children's knowledge and skills of simple addition and subtraction. Classroom observers recorded the kinds of teaching that occurred for addition and subtraction the following year, and the researchers administered a variety of student achievement measures. Teachers who had received the 4-week course spent a higher percentage of instruction time on word problems (often thought to be more challenging for students) whereas the control teachers spent a higher percentage of instructional time on number facts problems and on review. In addition, the CGI teachers posed more arithmetic problems to the whole class, listened more often to the solution procedures presented by students, and gave feedback less often on whether students' answers were correct. All these features of teaching are consistent with the hypothesis that students in the classrooms of CGI teachers had more opportunities to wrestle with the mathematics involved in adding and subtracting whole numbers. On many of the achievement measures, there were no significant differences between students in the two groups. However, students in the classes taught by CGI teachers performed better on the "complex addition and subtraction problems," problems requiring more than routine knowledge of arithmetic. Interestingly, students in these same classrooms also demonstrated significantly higher recall of number facts during individual interviews than students in the control classrooms. Although only suggestive, the results reported by Carpenter and colleagues (1989) are consistent with our primary claim in this section.

A study that addresses more directly the claim that when teaching promotes constructive struggle with mathematics students' understanding increases was conducted as part of the QUASAR Project (Silver & Stein, 1996; Stein & Lane, 1996). In particular, the results of the study suggest that mathematical tasks that make higher cognitive demands on students facilitate greater conceptual development (Stein, Grover, & Henningsen, 1996; Stein & Lane, 1996). The mathematical tasks presented to middle-school students were examined for the cognitive effort, or "thinking processes," the tasks were likely to elicit. The most challenging tasks were presumed to require reasoning mathematically (problem solving, pattern finding, conjecturing) and connecting proce-

dures with concepts. The least challenging tasks were presumed to require executing procedures without considering connections to other ideas, memorizing facts and rules, and exploring unimportant mathematics or exploring in a haphazard, unpurposeful way. The level of challenge of a sample of 144 tasks presented in 3 classrooms in each of 4 middle schools over 3 years was correlated with the achievement gains by sixth, seventh, and eighth graders tested at the beginning and end of each school year. Achievement was measured using a specially designed pool of 36 open-ended mathematics problems that allowed tracing changes in students' performance over several years. Results are quite complicated but clearly indicate that students attending schools in which teachers presented *and* faithfully implemented more challenging problems were more likely to develop increased conceptual understanding of the mathematics.

The study by Hiebert and Wearne (1993), reviewed earlier, supplements the findings of the QUASAR studies because teachers in the classrooms that facilitated greater understanding asked a higher percentage of questions that required explanations and analyses, and students in these classrooms responded with longer utterances. These forms of discourse require more intellectual work from students and can reveal students who are struggling to make sense of mathematics and to express their emerging understandings.

That students can make sense of classroom discourse that displays genuine cognitive and mathematical struggle was documented nicely by Inagaki, Hatano, and Morita (1998). Fourth and fifth graders in 11 classrooms were observed during discussions of alternative methods for adding fractions with unlike denominators. Although the discussions were filled with confusions and incorrect conjectures along with correct analyses, most students made sense of the discussions and improved their understanding as demonstrated through both their verbal statements and written work.

Case studies of classrooms in which teachers engage students in struggling with important mathematics add support to the claim that struggle is a key enabler for conceptual understanding. Detailed descriptions of classrooms in which authors reflect on their own experiences as teachers are especially compelling (Ball, 1993; Fawcett, 1938; Heaton, 2000; Lampert, 2001; Schoenfeld, 1985). Classrooms in which this kind of struggle is noticeably absent and, in turn, students do not improve their understanding (Schoenfeld, 1988) complement the positive cases.

A number of questions regarding classrooms in which students are engaged in constructive struggle

with mathematics remain unanswered. In contrast to the first feature of explicit attention to concepts, whether struggle can be incorporated into a wide range of pedagogical forms is unclear. Struggle is usually associated with student-centered or student inquiry approaches. But we can imagine teacher-centered approaches that provide targeted and highly structured activities during which students are asked to solve challenging problems and work through challenging ideas. In fact, it seems plausible that students' struggle should be sufficiently bounded and directed so that it centers on the important mathematical ideas. This requires some level of teacher guidance. There are few principles, however, from theory or data, that speak to the degree of structure needed to facilitate (or undermine) the productive effects of students' struggle. Parenthetically, the possibility that appropriate struggle can be built into teacher-centered approaches is one reason why we think the old labels of student-centered versus teacher-centered instruction can be so misleading.

A striking contrast should be apparent between teaching that encourages students to struggle and teaching for skill efficiency, described earlier, that ensures high success rates. Although these are not completely contradictory features, they would likely be found in quite different systems of teaching. On the one hand, this is consistent with what one might expect—different features and clusters of features are effective for promoting skill efficiency than are effective for conceptual understanding. But what explanation can account for the fact that skill learning was relatively high in classrooms where features of teaching associated with conceptual development were being implemented? We turn next to consider this question.

Teaching Features that Promote Conceptual Understanding also Promote Skill Fluency

We noted earlier that no simple correspondence exists between features of teaching and learning outcomes. One place in which the complex nature of teaching and learning becomes apparent is in the effects of conceptually oriented teaching on skill learning. Many of the reports on the conceptual development of students also indicate that their skills increased at a level equal to or greater than students in the control groups (Boaler, 1998; Fawcett, 1938; Fuson & Briars, 1990; Good et al., 1983; Hiebert & Wearne, 1993; Stein & Lane, 1996). Apparently, it is not the case that only one set of teaching features facilitates skill learning and another set facilitates conceptual learning. In this case, two quite different kinds of features both seem to promote skill learning.

One way to explain this observation is that the *nature* of skill learning might be somewhat different under the two instructional conditions. Under one condition, instruction is quickly paced, teachers ask short-answer targeted questions, and students complete relatively large numbers of problems during the lesson with high success rates. Under the other condition, instruction is more slowly paced, teachers ask questions that require longer responses, and students complete relatively few problems per lesson. Using Wittrock's (1986) argument, the cognitive mechanisms that students are likely to use under these different conditions would be different, leading to different skill competencies. In fact, Bjork's (1994) review seemed to show exactly that: Trainees who mastered skills under conditions that stimulated more conceptual work acquired different competencies—they were better able to adapt their skills to solve new kinds of tasks. Following the authors of *Adding It Up* (National Research Council, 2001), we use the word *fluent* (rather than *efficient*) to describe this kind of competence.

The foregoing analysis produces the following conjecture. The two features we identified for developing conceptual understanding are promising candidates for facilitating a more ambitious learning goal—mathematical proficiency (National Research Council, 2001). *Mathematical proficiency* is the proficiency achieved through integrating five kinds of competencies, including conceptual understanding and procedural fluency.⁴ Because mathematical proficiency has not been defined long enough to serve as the outcome measure for empirical studies, no unambiguous data are available to test the relationships between teaching and students' mathematical proficiency. But the findings of improved conceptual understanding and procedural fluency related to the two features of teaching we described above make plausible the conjecture that these two features can facilitate the development of mathematical proficiency.

Summary

We believe that two features of classroom mathematics teaching facilitate students' conceptual development (and perhaps mathematical proficiency)—explicit attention to connections among ideas, facts, and procedures, and engagement of students in struggling

with important mathematics. We believe both features are justified by not only the theoretical arguments but also the empirical data. We have not identified all of the teaching actions that might call students' attention to relevant connections or engage them in productive struggle, but we have described a sample of strategies (e.g., posing problems that require making connections and then working out these problems in ways that make the connections visible for students). The centrality of the teacher was noted when reviewing the teaching features that facilitate the acquisition of skill efficiency. On the basis of the current literature, systems of teaching that facilitate the development of understanding seem to operate with the teacher playing both more and less active roles.

Our reading of the literature suggests that other features of teaching that sometimes are associated with conceptual development (e.g., use of concrete materials, asking higher order questions) are too specific and tied too closely to particular classroom conditions to support claims that apply across classrooms. That is, not enough evidence exists to conclude that these more specific features, which work well under certain conditions or within particular systems of teaching, will work well under other conditions or in other systems. The two features we identified are more general and seem to operate effectively across a range of contexts and teaching systems.

We have not commented on whether both teaching features—attending to connections and struggling with important mathematics—are necessary for promoting conceptual understanding, whether either one is sufficient, whether there are interactions between the two, and so on. We also have not speculated about how these features might function as part of clusters of features. Perhaps, for example, each of these two features is part of a cluster that travels, as a cluster, across different systems of teaching. Increased conceptual understanding might be due to a cluster of teaching features rather than just the single highlighted feature. We believe these issues, together with the specific teaching actions that are required and the mechanisms that are responsible for the effects of these teaching features, define promising research agendas for the future. We also believe that they warrant investigation using outcome measures of the re-

⁴ In brief, mathematical proficiency is the simultaneous and integrated acquisition of five kinds of mathematical competencies, or "strands":

- *conceptual understanding*—comprehension of mathematical concepts, operations, and relations
- *procedural fluency*—skill in carrying out procedures flexibly, accurately, efficiently, and appropriately
- *strategic competence*—ability to formulate, represent, and solve mathematical problems
- *adaptive reasoning*—capacity for logical thought, reflection, explanation, and justification
- *productive disposition*—habitual inclination to see mathematics as sensible, useful, and worthwhile, coupled with a belief in diligence and one's own efficacy" (National Research Council, 2001, p. 116).

cently defined and more ambitious learning goal of mathematical proficiency (National Research Council, 2001). After commenting about the state of current classroom teaching with respect to these features, we pick up on this claim by identifying a number of principles to guide research agendas that would advance our knowledge of key relationships between teaching and learning.

Absence of Features That Support Conceptual Development from U.S. Mathematics Teaching

Before leaving the section on “Developing Conceptual Understanding,” we feel compelled to point out that U.S. mathematics classrooms show a striking absence of the very two features we identified as facilitating conceptual understanding. Ever since U.S. classroom practices have been studied systematically, these two features rarely have been found. Hoetker and Ahlbrand (1969) surveyed early descriptions of classroom teaching, across all subject fields, and reported that, since records have been kept, U.S. school teaching often has focused on lower level skills using a tightly controlled and curtailed question-and-answer routine called *recitation*. A series of studies in the 1970s confirmed this general observation for mathematics classrooms. The authors described classrooms focused on developing routine skills with a basic pattern of teaching that was quite consistent across classrooms (Fey, 1979; National Advisory Committee on Mathematics Education, 1975; Stake & Easley, 1978). Welch (1978) described a typical mathematics lesson this way:

First, answers were given for the previous day's assignment. A brief explanation, sometimes none at all, was given of the new material, and problems were assigned for the next day. The remainder of the class was devoted to students working independently on the homework while the teacher moved about the room answering questions. The most noticeable thing about math classes was the repetition of this routine (pp. 5–6).

Recent surveys have shown that this lesson pattern continues today and that instruction rarely includes explicit attention to conceptual development or engages students in struggling with key mathematical ideas (Hiebert et al., 2003; Rowan et al., 2004; Stigler,

Gonzales, Kawanaka, Knoll, & Serrano, 1999; Weiss, Pasley, Smith, Banilower, & Heck, 2003).

The mathematics Video Study of the 1999 TIMSS provides recent and direct information on the nature of U.S. mathematics teaching (Hiebert et al., 2003). To compare eighth-grade mathematics teaching in the United States with six higher achieving countries in Europe and Asia (Australia, Czech Republic, Hong Kong, Japan, Netherlands, and Switzerland), the 1999 Video Study collected a nationally representative sample of about 100 videotaped lessons from each country. An international team of researchers developed more than 75 codes to analyze the structure and organization of lessons, the mathematical content of the lessons, and the way in which content was worked on during the lessons. Only codes that could be applied with at least 85% interrater agreement were used.

One code from the 1999 Video Study that is especially instructive is the nature of the mathematics problems presented and the way in which the problems were worked on with the students.⁵ On the basis of the written materials or the oral presentation by the teacher, all problem statements could be classified into one of three categories defined by the mathematical processes that were apparently intended: using procedures, stating concepts, and making connections. “Making connections” is of most interest here because making connections (among ideas, facts, and procedures) lies at the heart of our definition of conceptual development. A making connections problem could be “Graph the equations $y = 2x + 3$, $2y = x - 2$, and $y = -4x$, and examine the role played by the numbers in determining the position and the slope of the associated lines.” Problem statements classified as stating concepts asked students to recall or illustrate definitions or properties. These problems seldom involved conceptual development because they simply asked students to recall or repeat concepts or propositions. A stating concepts problem could be “Show the point (3, 2) on the coordinate plane.” Using procedures problems asked students to use apparently known procedures to solve problems, for example, “Solve $2x + 3 = 5$.” Results showed that 17% of the problems in an average U.S. lesson indicated an intent to make connections; this was within the range of 15% to 24% evident in all other countries except Japan (Japan was an outlier with 54%).

The picture changes, however, when considering the way in which teachers worked with students on problems. All problems that had some public discussion or visible work were coded a second time into one

⁵ This code was developed by Margaret Smith (2000) and built upon earlier work by Mary Kay Stein and colleagues (Stein, Grover, & Henningsen, 1996).

of the same three categories. A fourth category, giving only the answer, was needed for those cases when no mathematical work was done with students except reading the answer. A question of special interest is, "What happened to the making connections problems when they were implemented in the classroom?" In all countries except for Australia and the United States, 37% to 52% of these problems were worked on so that the connections implied by the problem statement were made explicit with the students—through examining the problem, comparing solution methods, justifying why the solution methods worked, and so on. In Australia, 8% of the making connection problems were worked on in this way. In the United States, so few making connection problems were worked on in this way that the percentage rounded to 0% (Hiebert et al., 2003). This nationally representative sample of eighth-grade mathematics lessons shows that students in the United States are not receiving instruction that supports conceptual development.

Before speculating about the reasons for the absence of these teaching features from U.S. classrooms, we point out the degree to which researchers in the Video Study had to probe the actual work of teachers in these classrooms to detect the differences among countries. If coding had stopped with the kinds of mathematics problems presented to students, U.S. classrooms would have looked similar to several other countries. Describing teaching—how students and teachers interact about content—required moving beyond the curriculum to describing the details of how teachers use the curriculum. Documenting connections between teaching and learning requires describing teaching with considerable detail and precision. Notice that this fact calls into question findings that connect opportunity to learn with student achievement when opportunity to learn is defined solely in terms of content coverage (Fletcher, 1971; Gamoran, Porter, Smithson, & White, 1997; Porter et al., 1988). To the extent that teachers transform problems, as they did in the TIMSS 1999 Video Study, the opportunity to learn indicated by an analysis of the curriculum might not be the opportunity to learn experienced by the students. This becomes especially critical if one hopes to understand the mechanism lying behind statistical relationships and develop policy recommendations based on the findings. Simply put, curriculum changes are unlikely to lead directly to changes in students' opportunity to learn.

A number of factors might explain the longstanding absence from U.S. mathematics classrooms of teaching features that support conceptual understanding. These include the difficulty of changing practices that are embedded in a culture, the lack of

subject matter and pedagogical knowledge that teachers need in order to teach in different ways, the challenge of enacting complex skills even when they are known by teachers, the limiting nature of common professional development activities, the pressure of district- and state-mandated tests, and the absence of a useful and accessible knowledge base for teachers to improve their practices. We will not develop the arguments that justify each of these explanations, but we will use the last of the explanations, the absence of a knowledge base, as the springboard for the final section of the chapter. The question we tackle in the final section concerns the kind of research agendas that could stimulate further progress in linking teaching and learning. What kind of information will be most helpful and how can it be produced?

FUTURE DIRECTIONS FOR RESEARCH THAT CONNECTS TEACHING AND LEARNING

In this section, we identify several principles that we believe will elevate the quality of research that aims to link mathematics teaching and learning. Some of these principles can be applied immediately, and some will require development themselves. We believe that attending to each of the principles, even if they are not fully developed, will increase the chances that research findings will contribute to a growing, coherent, and useful base of knowledge for improving classroom teaching.

Be Explicit About Learning Goals

The distinction we elaborated earlier, between teaching features that support skill efficiency and those that support conceptual understanding, foreshadows the first principle of conducting research on teaching/learning relationships: Researchers must be clear and explicit about the kinds of learning they will study. Talking about teaching, in general terms, as "effective" or "not effective" is no longer helpful. Teaching is effective (or not) for helping students achieve particular kinds of learning goals. Teaching is effective *for* something and the something must be described and measured as precisely as possible.

We are not saying that a simple correspondence exists between specific features of teaching and specific learning outcomes. This empirical question has yet to be answered. Of course, the data we reviewed earlier indicate already that the correspondence will not be straightforward. The features of teaching that support conceptual understanding might also sup-

port skill fluency. Regardless of how these particular relationships are resolved, one lesson from past work is that further progress in understanding connections between teaching and learning depends on being clear about the learning outcomes to be measured and on developing assessments that measure these outcomes directly.

We recommend that the learning goal of mathematical proficiency (National Research Council, 2001) become part of some research agendas in the future. This ambitious goal, which is defined as the integration of five strands of knowledge, has not yet been operationalized through instrument development. Considerable effort will be needed to construct tasks that measure the integration of these strands. Measuring each strand individually misses the point of the construct. It is the integration of the strands that characterizes this promising goal. How to handle these conceptual and methodological challenges presents an immediate task worth pursuing.

Build and Use Theories

Once learning goals are chosen and instruments developed to measure them, theories must guide the research process. Ideally, theories will guide key aspects of the *entire* process: (1) the generation of testable hypotheses about the connections between teaching and learning, (2) the selection and modification of methodological designs that are appropriate for the questions, and (3) the analyses and interpretation of the data. Earlier, we identified the lack of theories of teaching and learning relationships as a major cause for the absence of better claims. Now, we remind the reader that the existence of theories, even local theories, has great benefits for guiding the design of individual studies and for connecting the findings from individual studies to build a useful knowledge base for teaching.

We underscore one benefit of working from theories—they allow researchers to *understand* what they are studying. Theories of teaching provide a framework within which to understand the interactions among features of teaching and the reasons that features of teaching might be more or less effective, within different systems, for facilitating particular kinds of learning. Unless researchers, and teachers, understand why teaching influences learning in a particular way, the knowledge is of little use (Rowan et al., 2002). Understanding the reasons for the effects of teaching allows teachers to adapt the features of interest to their own setting without fatal changes (Brown & Campione, 1996). This makes knowledge useful. When connections between teaching and learning are not under-

stood, isolated findings of “what works” can become rules or prescriptions for practice that are applied too rigidly or changed inappropriately. Users need to know how to modify or adjust the prescriptions to fit the expected differences from classroom to classroom. This benefit of working from theories becomes critical for our discussion of policy in the concluding section of the chapter.

The process through which theories of mathematics teaching and learning relationships are best developed is not yet clear. Ruthven (2002) described two different processes for building theories of teaching, attributed to French and Italian researchers, respectively. A first approach is characterized by top-down activities in which researchers specify the hypotheses to be tested and teachers implement the tests as prescribed. The second approach is characterized by bottom-up activities in which the teachers experiment with various instructional strategies and, together with researchers, detect more general principles that gradually emerge. We believe it is too early in the history of mathematics education research to tell whether one or the other, or a third, approach will be most productive for generating useful theories. All approaches should be considered.

A call to build theories is, fundamentally, a call for researchers to be clear about the hypotheses they use when conducting empirical studies. Hypotheses are researchers’ expressions of how well they understand the phenomena under investigation. Explicit statements of hypotheses provide a measure for how the field is advancing in its collective understanding of the phenomena. Thus, hypotheses and the theories they generate not only aid researchers in conducting empirical studies, they provide an essential way for researchers to communicate with one another regarding their own understandings of the teaching-learning relationships. As specific, local hypotheses are made explicit, are refined, and are accumulated, theories can emerge that tie together individual studies and enable the construction of a useful knowledge base.

Set Realistic Expectations for What a Knowledge Base Can Do

A third principle that can inform future research agendas is recognition of the appropriate role of research in setting policy and recommending practice. To define an appropriate role for research on relationships between teaching and learning, consider first Gage’s (1978) observations. Gage argued that to develop a science of teaching defined by high predictability and control would be unrealistic. The reasons are found in the challenges presented earlier in docu-

menting straightforward connections between teaching and learning. But, said Gage, a scientific basis for the art of teaching is realistic. It is realistic to expect an increasingly sound theoretical and empirical basis for making informed (although still uncertain) decisions. Gage quoted an 1891 observation by the philosopher Josiah Royce:

It is vain that the inadequacy of science is made a sufficient excuse for knowing nothing of it. The more inadequate science is when alone, the more need of using it as a beginning when we set about our task. (p. 20)

A realistic expectation, in our view, is that a knowledge base can inform policy and practice and can be used to make decisions with some level of confidence. The level of confidence rises as the knowledge base becomes more detailed, richer, and more coherent. Because research is further advanced in some areas than others, decisions about teaching practices that support some learning goals will be made with more confidence than decisions about achieving other learning goals.

The uncertainty of research claims is quite similar to that found in many other professional fields that involve the complexities of human behavior. Consider the field of health and nutrition. Scientific research on the effects of exercise, diet, and drugs informs recommendations regarding healthy lifestyles. Fruits and vegetables are good for you in a variety of ways. A small daily dose of aspirin reduces the risk of stroke. A high-fiber diet reduces the risk of colon cancer. But, like in education, most of these recommendations are made with some level of confidence, not with certainty. And as the specificity of the recommendation increases, the confidence level declines. Fruits and vegetables are good, but which ones are best, and exactly how many servings should everyone have? Should everyone take a small dose of aspirin each day? What about young people, and what about those who have mild stomach reactions to aspirin? Like education, the problem is that so many interactions influence the effects of any particular treatment—physiology of the individual, amount of exercise, weight, age, and so on. Specifying optimal diet, drug treatments, and lifestyle with certainty is impossible. But a knowledge base that informs decisions with increasing levels of confidence can be built.

One final parallel between decisions about physical health and about education is important. Readers surely have noticed that, as new information about nutrition and drug treatments becomes available, recommendations change, sometimes dramatically.

Should you eat butter or margarine? For years, margarine was the healthy alternative to butter. With new knowledge about trans-fatty acids, butter might be the better choice. Recommendations regarding more serious issues, like hormone treatments and breast cancer, also have witnessed considerable changes, even reversals. This does not mean that the medical profession is being silly or lazy or is conducting thoughtless and flawed research. Rather, the problems are extremely complicated, and sorting out all of the interactions and conditions under which treatments affect outcomes is enormously difficult. It also is noteworthy that people generally believe they benefit from the latest and currently best information that the health and medical professions can provide, even if it will change in the future. The same should be true in education with respect to relationships between teaching and learning.

One implication of this analysis is that educators, and the public, should not expect critical experiments in education. They should not expect single studies, regardless of how large or how well conceived, to answer the question of what kind of teaching is best for achieving particular learning goals. Well-designed studies that yield clear results can increase the level of confidence about which features of teaching to recommend, but the recommendations always are subject to change when better information becomes available or when conditions change.

We conclude this section by reminding the reader that not all decisions in education are, or should be, based on research. Learning goals, for example, are statements about what society most values and should be selected, in part, through public debates about values (Hiebert, 2003). How important is conceptual understanding? Should it be a priority or are other competencies more important? Research can inform society about what is possible but not about what is most valued. Once learning goals have been selected and made explicit, based at least partly on value judgments, then researchers can examine the features of teaching that facilitate the achievement of such goals.

Account for the Costs and Benefits of Different Research Approaches

The fourth principle we propose to elevate the quality of research on classroom teaching and learning relationships is to carefully consider the advantages and disadvantages of different research designs and strategies when conducting empirical studies and interpreting data. We elaborate this principle by considering the issues related to three particular contexts: research designs that compare the effects of

instructional methods; research designs that correlate features of teaching with student learning; and the trade-offs between small-scale qualitative studies and large-scale quantitative studies.

Comparing the Effects of Different Instructional Methods

Research designs that compare students' learning under two or more instructional methods often are quasi-experiments (Campbell & Stanley, 1963). Some of the methodological features are consistent with usual conventions of scientific experiments (controlling some variables while studying differences among others) but some are not (lack of random assignment of students to instructional method). One advantage of these comparative studies is that they often have some ecological validity. That is, they are conducted in real classrooms with all of the usual complexities. Instructional methods are implemented as systems, with all of the usual interactions. A second advantage is that these studies have the opportunity to control some of the most obvious alternative explanations for the results. For example, pretests can show similar entry knowledge and skills by students across treatment groups, content can be similar across groups except in ways dictated by the different teaching methods, and classroom observations can ensure faithful implementation of methods.

Some of the disadvantages of the quasi-experiments that compare teaching methods are inherent in the design and are difficult for researchers to control. The ecological validity that assures a realistic implementation of a teaching method also guarantees that the method will not be exactly replicable in other settings. As noted earlier, many classroom conditions, including the students, influence the way in which methods are implemented and, consequently, influence the learning outcomes. The challenge for researchers is to determine whether robust features of teaching within the system yield particular kinds of outcomes regardless of the naturally occurring changes in the system from classroom to classroom. This requires multiple replications, a strategy not frequently employed by educational researchers. But replication is one of the most powerful tools available to sort out which features of teaching make a difference across different implementations of a system of teaching, and even across different systems.

Another disadvantage of these comparative studies of teaching is easier for researchers to address. Reports of comparative studies often provide insufficient descriptions of the teaching features or methods studied. Without full descriptions of the teaching features

being compared, the results cannot be interpreted in ways that connect particular kinds of teaching with particular kinds of learning. And without these kinds of interpretations, it is impossible to *understand* the relationships that are documented. Superior learning reported during a thinly described "student-centered" approach compared with a thinly described "direct teaching" approach is of no value. There is no way to develop more specific hypotheses about what, exactly, in the student-centered approach facilitated students' learning. And it is these ill-understood connections that can lead to "lethal mutations" when implementing the teaching features in other conditions (Brown & Campione, 1996).

Thinly described teaching methods often are special deficiencies in reports of studies that include a control condition. The experimental method usually is described in some detail, but the control method can be described as cryptically as "traditional instruction was provided" or "the textbook was followed closely." Thinly described control methods undermine the educational significance of a claim that the experimental method yielded better results. This is because the power of such a claim depends as much on the quality of the control method as the quality of the experimental method. When benefits are claimed for one teaching method over another, regardless of the methods being compared, the claim is only as educationally significant as the quality of the weaker method. If the weaker method is extremely weak, or not sufficiently described to judge its quality, then results showing that the alternative method yielded better results are of little importance. To evaluate comparative claims between teaching methods, all the methods must be fully described.

We conclude this section with a suggestion about research questions. History shows that hypotheses in social science research are almost always confirmed (Greenwald, Pratkanis, Lieppe, & Baumgardner, 1986). This is especially true for predictions regarding the superiority of an experimental condition over a control condition. Several well-known research biases, such as the Hawthorne effect, probably explain the frequent confirmation of hypotheses. But, as Greenwald and colleagues (1986) argued, this tendency can and should be countered by changing the nature of the research question. Rather than asking "Is Method A better than Method B?" researchers should ask "Under what conditions is Method A better than Method B?" Translating the suggestion into the situation of interest here, we recommend asking more questions of the kind "Under what conditions will teaching features of Set A facilitate achievement of particular mathematics learning goals more effectively than teaching features

of Set B?" Indeed, this is exactly the characteristic of Brownell and Moser's (1949) study that received applause from Cronbach (1986). Cronbach echoed Greenwald and colleagues' (1986) recommendation by noting the benefits of documenting the conditions under which particular instruction approaches are effective rather than claiming one approach is superior to another.

Correlating Features of Teaching with Students' Learning

A common research approach to connecting teaching with learning has been to measure a range of specific features of teaching and gains in students' achievement and then, through correlation techniques, to identify the features of teaching that correspond best with students' learning. Known as *process-product research* (Brophy & Good, 1986; Dunkin & Biddle, 1974), this approach often focuses on counting the frequency of teacher behaviors and correlating these with student gains on standardized achievement tests. The approach was primarily responsible for the patterns described earlier between features of teaching and students' skill efficiency.

Advantages of the correlation approach center around the ability to sort out features of teaching that might be key facilitators of learning and those that are part of the implementation but might not be critical. For example, teaching might vary along dimensions of asking students to present alternative solution methods and time spent working in groups developing shared solution methods. One of these features (or an interaction between the two) might be more important than the other for students' conceptual development. Well-designed correlation research could provide useful information. Schoen, Cebulla, Finn, and Fi (2003) presented an example of how correlations can be used to sort out key instructional variables in classrooms using reform curricula. Although the traditional process-product approach selected a specific kind of teacher behavior and used standardized tests to measure learning, the correlational approach is not limited to these independent and dependent variables. The approach is especially useful to create initial maps of the terrain when few hypotheses exist to suggest which features of teaching might facilitate which kinds of learning.

A potential danger of the correlation approach is the flip-side of its greatest advantage—the assumption that teaching consists of a collection of interchangeable features and that the effects of individual features can be measured independently of their interactions with other features in the system. Although many re-

searchers have been explicit that “any teaching act is meaningful only in relationship to other instructional behaviors that occur in the classroom” (Good et al., 1983, p. 20), their warnings have often been overlooked in summaries and syntheses of research and when research findings are incorporated into professional development activities for teachers. Without considering the systems of which features are a part, recommendations from process-product relationships can be as simple and misguided as “Teachers should engage in process [X] more often.” Thus, as we noted earlier, when discussing the challenges faced in developing adequate theories of teaching and learning relationships, teaching must be conceived as a system of interacting features and overly simplistic notions of teaching must be avoided.

In spite of the danger, we believe that correlation approaches have a role to play in linking teaching with learning. They can help to identify features of teaching that might have major effects on learning and deserve further study, and they can provide descriptive data to inform the development of hypotheses regarding teaching-learning relationships. Equally important, especially in the complex classroom environment where school learning takes place, correlational studies can assist with decisions on which variables will likely play major roles and which will play minor roles in particular learning situations and in this way help develop testable hypotheses and assist with the design of manageable experiments.

Balancing the Benefits of Small-Scale Qualitative Studies and Large-Scale Quantitative Studies

In the current research climate, there are increasing pressures to quantify regularities through large-scale experimental studies that use random assignment and seek to sort out cause and effect between classroom practice and student learning (Coalition for Evidence-Based Policy, 2003). Partly in response to these pressures, the National Research Council sponsored a report that proposed guidelines for what counts as “scientific” research in education (National Research Council, 2002). A number of researchers have argued that a balanced approach between qualitative and quantitative approaches has the best chance of moving the field forward (see, for example, Maxwell, 2004). We believe that small-scale qualitative approaches and large-scale quantitative approaches and mixes between the two all have useful roles to play in advancing our understanding of the relationships between teaching and learning. Gage's (1978) simple observation is a good starting point for outlining the contributions

of different approaches: qualitative approaches reveal what is possible and quantitative approaches demonstrate what is probable. But we can be more specific.

Qualitative and small-scale studies can provide a number of critical ingredients for the research enterprise. Such studies are essential for developing and testing measuring instruments for teaching and learning, including the high-quality measures of student understanding of complex mathematical ideas called for in the National Research Council (2004) report on the effectiveness of the evaluations of mathematics reform curricula. Qualitative and small-scale studies also can support the thorough development of instructional treatments before large-scale studies test their effects with larger samples of students. Additionally, although case studies are not frequently undertaken to test hypotheses, they can, as Yin (2000) argued, rule out important alternate hypotheses if the studies are carefully designed.

Finally, and perhaps most importantly, qualitative smaller scale studies can provide data that deepen our understanding of how systems of teaching work to facilitate learning (Maxwell, 2004). Another brief story from the medical field is instructive. As reported during the December 7, 2004 National Public Radio program “Morning Edition” (Knox, 2004), a very small-scale study had recently tested the effect of a vaccine targeting a pre-leukemia condition called myelodysplasia. The vaccine not only induced immune responses in about 60% of the cases but unexpectedly created remissions in several cancer cases. “Now the challenge,” according to Richard Knox, “is to figure out why the vaccine works.” Until the researchers understand the reasons for the vaccine’s effect, they cannot adapt it to trials under other conditions or on large scales. There are good reasons to apply a similar logic to research on teaching and learning. Qualitative and small-scale studies can detect the existence of teaching effects and then unpack the system of interacting teaching features to explore the reasons for the effects. Understanding why particular features of teaching facilitate particular kinds of learning is essential for creating more refined hypotheses, for adapting the teaching system to other conditions, and for testing its effects on a large scale.

Large-scale quantitative studies play an equally critical role in building a useful knowledge base for teaching. Because the links between teaching and learning are mediated by so many contextual variables, large-scale studies are needed to gauge the size of these effects and estimate the likelihood of finding similar links between teaching and learning across multiple classrooms. Researchers cannot control all of the classroom influences on students’ learning and all of the interaction effects among the teaching features

of interest and the systems in which they operate, so replication over large numbers of classrooms and students remains one of the best strategies for detecting links between teaching and learning (Hiebert, Gallimore, & Stigler, 2002). We recommend taking seriously the call for large-scale cause-effect studies that aim to test hypotheses about relationships between teaching and learning. Along with Cohen et al. (2003) and others before them (e.g., Brownell & Moser, 1949), we recommend studying the effects of implementing systems of teaching under different conditions with clear learning goals in mind. To the extent that conditions not of interest can be controlled (e.g., instruction implemented similarly across classrooms, random assignment of students to treatments), the results can be interpreted with clarity and confidence. As we argued earlier, full control is unlikely but need not preclude efforts to design studies with appropriate controls.

Smaller scale qualitative studies and larger scale quantitative studies are essential for building a theoretically driven and practically useful knowledge base for teaching. Neither is sufficient; both are necessary. The confidence with which conclusions can be drawn about the effectiveness of features or systems of teaching increases with the convergence toward similar conclusions from studies using different methods. We agree with the National Research Council (2002), “If a research conjecture or hypothesis can withstand the scrutiny of multiple methods its credibility is enhanced greatly” (p. 64). The issue is not one of choosing between approaches but rather one of using both approaches in a mutually beneficial way.

FINAL THOUGHTS

We began the chapter with the goal of examining directly the issue of teaching effectiveness—why it has been so hard to document, what is known about it, and how the mathematics education community can learn more about the relationships between teaching and learning. We contended that these questions get to the heart of the educational research enterprise because (1) the questions lie at the foundation of education activity and are fundamental research issues and (2) the improvement of classroom practice depends on the answers. This places researchers in a difficult position. The issues they are dealing with are fundamental and complex, but researchers are under pressure from policy and government agencies to produce simple, easy-to-understand evidence-based answers and to produce them quickly. What constitutes an appropriate response to this challenging dilemma?

One response to the challenge, currently popular in the United States, is to address the simple question “What works?” The “What Works Clearinghouse” launched in 2004 (Viadero, 2004) is the latest in a series of efforts by the U.S. government (U.S. Department of Education, 1987), state governments (e.g., Dixon et al., 1998), and professional organizations (e.g., Cawelti, 2004) to provide practitioners with straightforward evidence-based advice for improving classroom practice. The What Works Clearinghouse plans to roll out a series of recommendations on a range of educational issues by conducting reviews of intervention programs, practices, and products. Similar to the review procedure used by Dixon et al. (1998) to identify best practices in mathematics classrooms, the What Works Clearinghouse will screen out most studies for review using a set of selection criteria. In particular, all studies that do not include a cause-effect analysis through the use of comparison groups will be excluded. The studies remaining constitute one segment of the large-scale quantitative studies described earlier. These studies are an important part of the picture. They provide essential information regarding teaching-learning relationships. But, usually, these large-scale comparative studies do not help consumers understand why the documented relationships exist or exactly how particular kinds of teaching facilitated particular kinds of learning. As we have argued, users must understand why particular features of teaching facilitate students’ learning in order to deal appropriately with the modifications that inevitably are needed for their own classrooms. Some level of understanding is essential to prevent lethal mutations in teaching (Brown & Campione, 1996).

We applaud the concept of conducting and reviewing research to answer questions of what works. Educators need trustworthy information to inform classroom decisions. Our concerns about the limitations of the process engaged by the What Works Clearinghouse in no way lessen the importance or urgency of the goal toward which the Clearinghouse is working. In fact, we believe this goal must be aggressively pursued by researchers in the mathematics education community. Mathematics education researchers are in the best position to create and sustain a better alternative.

We conclude this chapter by proposing a richer, more useful, but longer term alternative to “what works.” We propose the coherent and systematic construction of a knowledge base that documents robust links between teaching and learning *and* provides insights into the mechanisms that are responsible for such links. Researchers will need to participate in efforts to generate knowledge about teaching-learning

relationships that have direct relevance for classroom teachers and to regularly review available evidence to provide recommendations accompanied with estimates of confidence. We expect that these efforts will follow a range of different models for researcher-teacher collaboration (Hiebert et al., 2002; Ruthven, 2002) but will be characterized by the principles outlined in this chapter. Effective teaching will be studied in relation to clearly specified and measured learning outcomes so that conclusions will be phrased, not just as “effective teaching” but as “effective teaching under conditions [X] for learning [Y].” Hypotheses and theories will guide all aspects of the research process so that conjectures can be offered about why and how the targeted teaching features or systems facilitated (or did not facilitate) students’ learning. The conditions under which features were effective will be detailed so that users can interpret the conclusions in light of their own local situations. Tests of hypotheses about teaching-learning relationships will be replicated in a variety of contexts to study the influence of contextual factors so that conclusions can be drawn with increasing specificity and confidence. Reviews will look for patterns of consistent findings from studies using different research methods to increase the confidence with which conclusions can be offered.

To build the richer knowledge base we envision, debates about research methods must shift from debates about which methods are appropriate to debates about how to increase the quality of the methods employed. The principle that multiple methods are better than one method must be accepted, and researchers will need to pursue the goal of gathering data with higher levels of trustworthiness, regardless of method. Discussions like those initiated by Hanna (1998), Kilpatrick (1993), Lester and Lambdin (1998), Nissen and Blomhøj (1993), and Simon (2004) on the quality of research in mathematics education should continue and be extended.

A knowledge base of the kind we envision cannot be built without resources. Conducting the large-scale comparisons of teaching that provide critical kinds of information are especially expensive. Assembling networks of researchers who can launch coordinated series of studies and build on one another’s work requires long-term investments. No research enterprise that produces important advances is cheaply funded. U.S. mathematics education research funding comes primarily from the National Science Foundation and the U.S. Department of Education, and, as a Rand-conducted study of funding for mathematics education research from 1996 to 2001 pointed out, neither agency has clear strategic long-range planning for mathematics education R&D funding. In fact, fund-

ing for research and development in both agencies is a miniscule part of their total education budgets (Lacampagne, Newton, & Markham, 2001). Apparently high-quality evidence-based information on teaching-learning relationships in the mathematics classroom will cost much more than the United States currently is willing to spend.

The knowledge base we envision that links mathematics teaching and learning is beginning to take shape, even with limited resources. But it is difficult to find. It is hidden in the archives of libraries and scattered across time and among countless materials. Teachers (and researchers) must work hard to access it. Recent reviews have succeeded in pulling together some information (English, 2002; Kilpatrick et al., 2003; National Research Council, 2001; this volume), and we hope this chapter extends and contributes to these efforts. But most of this work is post hoc, requiring reviewers to detect patterns across work that was conducted independently and without connections. Much can be gained by researchers launching research agendas that are intentionally and explicitly connected. Through theory development, planned replications, and other research activities that connect research communities, a coherent, rich, accessible, and useful knowledge base can be built. Questions of "What works?" can then be answered with the richness and insight teachers deserve.

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AUTHOR NOTE

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CULTURE, RACE, POWER, AND MATHEMATICS EDUCATION

Diversity in Mathematics Education Center for Learning and Teaching

UNIVERSITY OF CALIFORNIA, LOS ANGELES

UNIVERSITY OF WISCONSIN, MADISON

UNIVERSITY OF CALIFORNIA, BERKELEY

For their tireless efforts toward equity and justice, we dedicate this chapter to

Rosa Parks

&

Coretta Scott King

INTRODUCTION

Considerable attention in mathematics education research has been paid to understanding and confronting differential mathematics achievement. Such research is complex, as it is situated within and framed by broader educational, social, and political contexts. Embedded in these contexts are issues of race, class, gender, language, culture, and power. Mathematics education researchers not only have to grapple with issues of mathematics teaching and learning, but also come to understand the ways in which these broader educational and cultural contexts shape opportunities for all students to learn mathematics (Cobb & Nasir, 2002; Martin, 2000; Moschkovich, 2002a). Moreover, such research brings into question the role of mathematics education by situating it within these broader contexts (Apple, 1992; Martin, 2003).

Race is one of the most, if not the most, salient framing characteristics for differential achievement in the discourse that surrounds the “achievement

gap” (Schoenfeld, 2002; Tate, 1997). Additionally we, the authors of this chapter, were aware of the significance of race in our everyday work with teachers and students and noted that its impact in mathematics education went beyond issues of differential achievement. Thus, in this chapter we specifically examine research aimed at understanding the various intersections between issues of race and mathematics teaching and learning. We follow the field in rejecting perspectives of race as biologically determined and adopted a perspective of race as constructed, contested, and reified through social activity. Additionally, our synthesis of the research around race and mathematics education provides a context, as Perry (Perry, Steele, & Hilliard, 2003) suggested, to “grapple with the notion that not all racial minorities occupy the same political position in this society” (p. 9). This acknowledgment brings relations of power into our analysis. More specifically, as discussed later in this chapter, the racialization of mathematics education is intertwined with issues of power and authority, making it difficult, if not impossible, to examine issues

of race without also considering the impact of power dynamics on mathematics teaching and learning.

We also note that the field often treats culture as race and vice versa (see special issue of *Educational Researcher*, 2004). Some work treats cultural characteristics of people from a racial group as homogenous and immutable; in this model, a static view of race determines culture. Other work opts to deal with local cultural practices as a way of avoiding seemingly static racial categories, often missing broader patterns of behavior that socially construct racial experience. Both of these trends speak to the difficult intersection between race and culture. Thus, we expanded our work to include research examining issues of culture and mathematics teaching and learning. We focus on research in mathematics education that has conceptualized culture not as a static individual characteristic, but as constantly negotiated by individuals in their everyday activities. Conceptualizing culture in this way means that cultural knowledge is an integrated system of learned patterns of behavior, ideas, and products with a focus on people's everyday lived practices (K. Gutiérrez, 2002; Moll & Gonzalez, 2004; Rogoff, 2003). This chapter attempts to highlight and suggest areas for future research that explores the complexity in students' cultural activity, racial experience, and mathematics education.

Individual Mathematical Performance

Before turning to a discussion of future research directions, we begin with a brief examination of the concept of individual mathematical performance as it relates to differential achievement and solutions to inequitable educational situations. Researchers and the media often report achievement gaps in mathematics across broad groups of students. Explanations for these gaps in mathematics achievement both structure larger agendas for addressing equity in schools and affect individual teacher-student interactions. Many of the explanations, however, both in the media and in research, explicitly and implicitly frame the problem as one of individual student learning, asserting that some students "just don't try hard enough" or "fail because they are disadvantaged." Although these explanations often stem from research that examines the success or failure of individual students, they are also mistakenly generalized to the differential patterns of achievement for *groups* of students.

Research that focuses on the mathematical characteristics and performance of individual students frames equity as an issue of equal access to high-quality mathematics education that allows *all* students to

succeed. This notion of equity as equality suggests that equity can be realized when students have equal access to resources, high-quality teachers, and appropriate instructional support regardless of race, class, gender, and so on. However, researchers concerned with issues of social justice and longstanding social inequities, or what Ladson-Billings (2006) referred to in her American Educational Research Association Presidential Address as the "Education Debt," argue that "equality" in educational opportunities does not necessarily address the histories of groups and how those histories have shaped the social structures and beliefs at the base of opportunities in and outside of school.

Rochelle Gutierrez (2002) described the goal of equity in mathematics education as "being unable to predict student patterns (e.g., achievement, participation, and the ability to critically analyze data or society) based solely on characteristics such as race, class, ethnicity, sex, beliefs and creeds, and proficiency in the dominant language" (p. 153). She proposed a two-fold approach to achieving this condition: first, providing students with high-quality mathematics education, or what she called *dominant* mathematics, and second, supporting students in using mathematics to perceive and confront inequitable situations in their lives, or *critical* mathematics. In this chapter, we take up Gutierrez's call to the field, to explore the processes involved in differential mathematics achievement and to consider mathematics education as a means of and for a socially just society.

In support of this task, this chapter reviews research within and beyond the field of mathematics education that conjoins accounts of individual-level processes with critical social and cultural processes involved in learning and teaching (Allexsaht-Snider & Hart, 2001; Boaler, 2002; Greeno, 2004; Martin, 2000; Moschkovich, 2002a). Further, instead of making comparisons of individuals on the basis of what has been considered "normative" or "adaptive" (Baratz & Baratz, 1970; Leacock, 1971; Nieto, 2004; Secada, 1992), the research reviewed is concerned with elaborating and deepening conceptions of learning to account for differences and regularities both within and across social and racial groups as well as social and cultural contexts. As Secada (1992) and others noted, over the last 20 years, group comparisons have typically been developed within research with a majority of White children, primarily by White researchers, whose perspectives were shaped by their sociocultural experiences in the dominant culture. A charge to the field of mathematics education, then, is to conduct research within non-dominant populations of students who experience marginalization, and to attend to the positioning of these groups vis-à-vis their White coun-

terparts and to broader sociopolitical structures and forces. *White* as a racial category also needs explication as it entails certain privileges and status within current educational systems (Lee, 2004). For these reasons, we focus this chapter on research that conceptualizes larger constructs such as culture, power, and race as ways to understand issues in mathematics education.

Organization of the Chapter

To help make explicit current understandings, research, and theory around culture, race, power, and mathematics education, we first review the literature that uses culture as a way of conceptualizing achievement differences within and between groups. This review focuses on literature that portrays culture as fluid, socially negotiated, and “lived in the everyday.” This review attempts to understand the influence of what we summarize as *theories of cultural activity* on perspectives of learning, development, and culture and the relations between them with respect to differences in opportunities to learn mathematics for non-dominant students. Then we turn to mathematics education research that suggests it is not enough to examine culture without attending to broader issues of power and race. This section reviews research that takes into consideration the constructs of power and race at the classroom level and in larger contexts. This literature stresses the need to consider these constructs in order to fully understand and reconceptualize differential achievement in mathematics education. Finally, we discuss issues around and implications of researching culture, race, and power in mathematics education.

CULTURAL ACCOUNTS OF MATHEMATICS TEACHING AND LEARNING

Culture has long been a focus of research on individual differences in cognition, learning, and development (Lave, Murtaugh, & de la Rocha, 1984; Nunes, Schliemann, & Carraher, 1993; Rogoff, 2003; Saxe, 1988a, 1988b). However, over the past 10 years studies in this area have gained prominence and offered alternative perspectives on the role of culture in learning, in terms of both what counts as learning and who has access to it. One reason for this heightened interest is that research drawn from various theories of cultural activity has afforded an understanding of knowing and learning as a function of what an indi-

vidual accomplishes over time and across the various communities and practices in which he or she participates.¹ These theories encompass such theoretical approaches as situated cognition, activity theory, cultural historical activity theory, and sociocultural theory. Findings across these lines of research point to the fact that mathematics classrooms are necessarily cultural and social spaces that can perpetuate social inequities by privileging certain forms of discourse and ways of reasoning or reorganize them by positioning multiple forms of learning and knowing as “having clout” (Cobb & Hodge, 2002; Gutierrez, Baquedano-Lopez, & Tajeda, 1999; Gutstein, in press-a).

Understanding the cultural entailments of mathematics learning requires complicated analyses of how people live and learn culturally both within and outside of the mathematics classroom. This chapter reports on (1) the links that researchers have uncovered between the diverse ways that students from different backgrounds negotiate learning within and outside of mathematics classrooms, (2) the structures for mathematics learning within mathematics classrooms, and (3) the relations students develop with the domain of mathematics. All told, this work suggests that differences in mathematical achievement among groups do not rest solely upon students’ cultural/mathematical backgrounds, but also in the sociopolitical organization of mathematics classrooms (and mathematics education in general). Moreover, although culture is commonly characterized as synonymous with group membership, this perspective is broadened by an emphasis in this work on how individuals create, contest, and reconfigure roles and relationships within the communities in which they participate.

Research that takes a cultural activity perspective aimed wholly or in part at persistent inequities in mathematics education has found traction in focusing on the mathematics learning of nondominant students both inside and outside of the mathematics classroom. We examine contributions of this research to the field of mathematics education to improving understanding of the processes through which students negotiate mathematics learning and doing across a variety of contexts. One major contribution of this research is the extensive accounts of the participation of students and teachers in various *communities* of mathematics classrooms, the curricular and participation structures within these communities that guide the class’s joint work together, and the relation of these structures to the identities students develop as

¹ *Participation* in this case refers broadly to the way that individuals interact within a community of practice—whether they play a central role in perpetuating what goes on inside of it or actively contest or challenge from the margins.

learners and doers of mathematics (see, e.g., Cobb & Hodge, 2002; Martin, 2000; Nasir, 2002).

The recognition of the mathematics classroom as a functioning community where teacher and student activity in it is shaped by (and shapes) a set of norms and practices for learning mathematics highlights the importance of issues such as competence, ownership, and alignment in engaging in this community. In particular, alignment between the practices and identities of home and school has implications for whether students negotiate ways of participating that serve their individual goals, as well as the goals of the classroom community (Cobb & Hodge, 2002; Hand, 2003).

A second and growing contribution of this research is documentation of the wide variety of mathematical practices and identities that students bring to the classroom from their home and local communities, which has expanded conceptions of competent classroom participation (see, e.g., Carraher, Carraher, & Schliemann, 1985; Lipka, 1994; Moll, Amanti, Neff, & Gonzalez, 1992; Nasir, 2002; Taylor, 2004). Students' practices and ways of reasoning are often marginalized in mathematics classrooms where teachers rely on traditional scripts of, and formats for, classroom instruction. Broadening mathematical activity to recognize and value the multiple ways that students participate in mathematics can draw in students who may normally be sidelined. Finally, this research offers promising models of classroom learning environments that begin to address issues of race and power in the mathematics classroom by focusing squarely on issues of cultural relevancy and social justice (see, e.g., Gutstein, 1997, 2003, in press-b; Ladson-Billings, 1994; Moses & Cobb, 2001a).

We begin the next section by explicating the key constructs of participation and identity that undergird the research from a cultural activity perspective. Next, we examine how these constructs have been employed to recenter analyses of student achievement to consider differences in students' *opportunities to learn* mathematics both in classrooms and more broadly. Examining opportunities to learn in mathematics classrooms for diverse groups of students has prompted new ways to think about differential access to engagement in mathematics in relation to the social and cultural resources that students bring to the learning environment. As a framework, opportunity to learn orients the review of the literature by highlighting particular themes that run across research with a perspective of culture as activity. These themes include (a) creating equitable opportunities to learn in the classroom, (b) broadening classroom discourse practices, (c) expanding conceptions of mathematical competence, (d) bridging in-school and out-of-school

mathematics knowledge, and (e) reforming the culture of mathematics.

Participation

Participation has become an important construct in situated and sociocultural analyses of students' school and classroom experiences (Boaler, 1999; Boaler & Greeno, 2000; Cobb & Hodge, 2002; Cobb, Stephan, McClain, & Gravemeijer, 2001; Engle & Conant, 2002; Gonzalez, Andrade, Civil, & Moll, 2001; Lave & Wenger, 1991; Martin, 2000, in press-b; Moschkovich, 2002a). As Sfard (1998) has observed, analytical frameworks that embrace the *participation metaphor* regard learning as a constant process of doing and becoming within a context populated by practices and activities in which individuals engage. An analytic focus on participation in studies of learning can highlight in what practices and activities students engage and how, as well as recenter analyses of student achievement and performance on the actual practices and activities in which students, teachers, and mathematics interact. Several researchers concerned with issues of equity in education broadly have achieved this recentering in different ways.

Cobb and Hodge (2002) focused on practice and participation in their analytical perspective on diversity and equity in mathematics education. They proposed a *relational perspective* that highlights "the relations between the specifically mathematical practices in which students participate in the classroom and the practices of the out-of-school communities of which students are members" (p. 251). They emphasized the continuities and discontinuities between students' ways of reasoning, talking, and interacting rooted in their out-of-school communities and those of their mathematics classroom. They argued that these relations, as they play out in classroom interactions, are the locus of the successes and inequities that arise in mathematics classrooms. In this formulation, what students are motivated to do in mathematics, how they engage in doing it, and, hence, what they are learning are understood through investigating patterns of students' participation in and across different contexts, in relation to the normative practices of the mathematics classroom. This analytical frame locates student achievement in the relationships between students' ways of participating in mathematics—which may be shaped by students' histories of engagement in their out-of-school communities—and the norms and valued practices of their mathematics classrooms.

Gutierrez and Rogoff (2003) used practice and participation to understand culturally related approaches to learning. They argued that although re-

search on cultural learning styles was an important move away from deficit-model explanations, it led to an “overly static and categorical” approach to the relation between culture and individual participation and engagement in school-related practices (p. 19). They proposed that instead of attributing cultural differences to individual or group traits, researchers concerned with understanding how students’ culture and communities influence their school achievement should also focus on “variations in individuals’ and groups’ histories of engagement in cultural practices” (p. 19). By distinguishing between membership in a particular group and participation in practices of cultural communities, researchers can attend both to regularities in the practices and organization of cultural communities as well as to how students differently engage in and make meaning of those practices.

A number of researchers have explored the implications of situations in which students’ everyday practices are misaligned with classroom practices for students’ social and academic positioning in the classroom (Diamondstone, 2002; Hand, 2003; Nasir, 2004). For example, Hand (2003) found that the *a priori* distinctions between social and intellectual activity that often get organized in mathematics classrooms can provoke tension between the ways of being, talking, and reasoning that students from diverse backgrounds bring to the mathematics classroom and the normative aspects of the mathematics classroom. She argued that “open” participation structures, or those that afford negotiation around the framing and positioning of participation, are more likely to encourage broad-based participation among a range of students, versus “closed” or rigid participation structures, which tend to foster student resistance. This perspective attends not only to the relation of students’ participation practices in and out of the classroom, but also to the relation of this alignment to students’ deep engagement in or opposition to classroom mathematical activity. Both Cobb and Hodge (2002) and Hand (2003) pointed to the relation of students’ participation and engagement in school and mathematics to ongoing negotiation and development of who they are and who they want to become—in other words, students’ developing and constantly negotiated identities.

Identity

The notion of identity is of central concern in studies of participation as it is intimately linked to social practice and is concerned with both what is made available to individuals in the various social and cultural communities they inhabit and how they enact their participation across them. Although the field of

mathematics education (and education research more broadly) has yet to agree upon a working definition of *identity* (Sfard & Prusak, 2005), the theories of cultural activity that frame this review generally follow Wenger (1998) in proposing that, “identity serves as the pivot between the social and the individual” (p. 145). The working definition that the field eventually arrives at will need to attend to multiple aspects of identity formation and negotiation. For example, the concept of identity must account for the perceptions that individuals hold about themselves and for those held by others about them, and the relation of these multiple perceptions to an individual’s social positioning in interaction. This definition will also need to acknowledge multiple identities (or dimensions of identity), such as being an African American and being a mathematics learner (Martin, in press-b), that individuals manage within and across contexts. What a perspective of identity of this kind theorizes is an *identity in practice* that is constantly being formed and reformed in dialectic between social structures and individual lived experiences. In this way, identity captures both the histories of engagement of various sociocultural communities and institutions and the practices they develop over time, as well as the trajectories of individuals as they negotiate and adapt these practices in and across local communities.

A number of researchers have examined identity negotiation and development within and beyond the mathematics classroom, particularly for nondominant students (Boaler & Greeno, 2000; Cobb & Hodge, 2002; Gresalfi, 2004; Martin, 2000, 2006, in press-a, in press-b; Nasir, 2002; Nasir & Hand, 2006; Sfard & Prusak, 2005). Martin’s (in press-b) research, for example, focuses on the challenge many African Americans face in negotiating positive identities as mathematics learners both inside and outside of school. He argues that “any analysis of identity construction and students’ becoming doers of mathematics must simultaneously consider African American identities as well” (p. 6). This is because mathematics education is shaped by a master narrative within society that positions African Americans as less capable in mathematics than their White peers. In response to and in contestation of the master narrative, Martin focuses on the success of some African American mathematics learners in overcoming structural and cultural impediments to mathematical achievement in order to forge new identities as Black mathematics learners. Martin’s significant contributions to our understanding of racialized mathematical identity are explored further in the section on race and power.

Nasir has also examined the relation between identity and learning among African American youth

but has explored this connection across multiple activity settings, including dominoes, basketball (Nasir, 2002), and the mathematics classroom (Nasir & Hand, 2006). Nasir studies the organization of social and intellectual resources for students' identity development and learning in relation to students' goals and strategies for participation in various social practices. Drawing on Wenger's *community of practice* model of learning, she has observed a relationship among students' goals, identities, and learning. Identity formation shapes and is shaped by learning. Similar relationships are found between goals and learning and between goals and identity. For example, in the case of elementary and secondary school students gaining competence in the game of dominoes, shifts in goals and identities accompanied age-related shifts in practice. More specifically, older, more experienced, and skilled players employed more complex strategies and had increasingly complex mathematical goals. Nasir also explored the notion of a *practice-linked identity* to capture the sense of connection that high school basketball players developed to the practice of basketball, in contrast to mathematics. In this study, players incorporated more of themselves into the practice of basketball as they drew heavily on a range of available resources for participation to accomplish their play. Nasir's research in this area has shown how identities involve aspects of both community (e.g., relationships) and learning (e.g., mastery), which both affect and reflect identity.

The literature on participation and identity illustrates the complexity underlying individual activity in any given social context, including a mathematics classroom. It also emphasizes the importance of examining the processes by which individual stakeholders in mathematics education (e.g., students, teachers, parents, policymakers, mathematicians) help to *construct* what "counts" as mathematics learning and what it means to be a mathematics learner, both within the local context of the mathematics classroom and in the broader system of mathematics education.

Assessing Opportunities to Learn

The consideration of participation and identity in research on mathematics learning communities shifts the focus from assessing what an individual student knows to discerning what she or he has an opportunity to learn within classroom mathematical practices (Greeno & Gresalfi, 2006). The construct of *opportunities to learn* (or OTL) initially referred to whether certain content appeared in curriculum materials or was included in the implemented curriculum. Recently the construct has been revived by a number of

researchers to challenge a common assumption that learning is dictated by the curriculum, instead of being constructed within a classroom system of teachers, students, mathematical tasks, and material and ideological resources (Greeno, 2003; Lampert, 2001). This perspective highlights the enactment of mathematical content within a classroom culture, which affords particular opportunities for engaging with mathematical practices and ideas.

For example, it has been argued that mathematics instruction that prioritizes finding solutions or performing rapid computation over making sense of mathematical ideas and the connections between them can serve to constrain students' opportunities to engage deeply in mathematics (Boaler, 1998; Bransford, Brown, & Cocking, 1999; Carpenter, Franke, & Levi, 2003; Hiebert et al., 1997; Jacobs, 2002; Kilpatrick, Swafford, & Findell, 2001). Although this finding does not necessarily speak directly to the gap in achievement scores, researchers like Boaler (1997, in press) have linked meaningful participation in inquiry-based mathematics classrooms to greater affiliation with, persistence, and achievement in mathematics for women and students of color. There is general consensus in the field of mathematics education that mathematical practices that are guided by principles of mathematics reform, if implemented with fidelity to the curricular goals and intentions, can support powerful engagement in rich mathematics for *all* students (Kilpatrick et al., 2001; National Council of Teachers of Mathematics, 2000; Schoenfeld, 2002).

Opportunities to learn are related to identity in recognizing the diverse ways that opportunities can be *taken up* by students as a function of their cultural and mathematical histories. Student participation in the mathematics classroom is related to affiliation with and membership in local and broader communities (Nasir, 2002) as well as negotiation in moment-to-moment interaction about positioning oneself and being positioned by others (Wortham, 2006) with respect to these communities and the local classroom practices. Thus, differences in students' perceptions of and participation practices with opportunities to learn ultimately shape the nature of their mathematical experiences (Gresalfi, 2004).

To date, the research on opportunities to learn mathematics has focused primarily on comparing the experiences of dominant and nondominant students in classrooms, with less attention to the broader social and political structures shaping these learning environments. Issues of equity, however, run across multiple levels and social contexts and thus suggest that analyses of opportunities to learn be expanded beyond current research efforts.

Creating Equitable Opportunities to Learn

Researchers in mathematics education have investigated the relations between opportunities to learn mathematics for dominant (i.e., Whites and Asian Americans) versus nondominant students (i.e., African Americans, Latino/as, and women) in a variety of ways. The work of Jo Boaler and her colleagues, in particular, has figured prominently in this effort by identifying curricular strategies and classroom mathematical practices that serve to level the playing field in classrooms of students from diverse backgrounds.

Boaler's (1997, 1998, 2002, 2003, 2006) longitudinal studies of traditional versus reform mathematics curricula in the United Kingdom and the United States have focused on the nature of classroom mathematical practices under these different approaches, focusing specifically on what these curricula afford for the mathematical understandings that students develop and the views students construct of mathematics and themselves as mathematics learners. In her early work, Boaler (1997) found that reform mathematics classrooms were more likely to engage students in practices that focused on mathematical problem solving, developing a variety of solution paths, and providing justification for their mathematics work, and that those practices inherently offered opportunities for more students to participate in them. The students in the reform classrooms developed a conceptual understanding rather than just procedural knowledge, enabling them to apply this knowledge to a variety of tasks. Boaler (2002) has taken these findings further, demonstrating how reform mathematics can promote equity. Boaler and Greeno (2000) found that students taking Calculus developed significantly different perspectives on the role of learning mathematics depending on the nature of the classroom's mathematical activity (e.g., discussion-based format versus lecture-based format). Although these findings do not necessarily address issues of equity directly, Boaler and Greeno contended that students who developed a more "connected" relation to the learning of mathematics—meaning that they view mathematics as a meaningful aspect of their lives and important to who they see themselves becoming—were more apt to pursue mathematics over the long run. Thus, the nature of a classroom's mathematical activity could provide opportunities for more students to see themselves as mathematics learners.

Boaler (2003) has continued to pursue these issues in her work and has recently reported a variety of findings in which issues of equity are foregrounded. She highlights the work of the Railside mathematics department, which was highly successful at producing strong mathematics learners in a diverse urban high school.² Boaler argued that the success of this department was largely dependent on the ability of the teachers to foster what she calls *relational equity* among all students in their classrooms. She described relational equity as focusing on how "students learn to treat each other and the respect they learn for people from different circumstances to their own" (Boaler, in press, p. 5). This perspective of equity links students' mathematical engagement directly with the deep sense of commitment and respect students develop for one another within the classroom mathematics community. To promote relational equity among students, the teachers at this school drew on strategies of complex instruction (Cohen & Lotan, 1997), which minimizes status differences among students and promotes group accountability (Boaler & Staples, in press). They also utilized an internally developed reform mathematics curriculum, which offered a rich set of group-worthy mathematics tasks for students to grapple with together. Boaler's work over the years has helped paint a nuanced picture of mathematics classrooms that focuses on meaningful and respectful mathematical engagement, where all students' mathematical ideas are respected and their mathematical identities are cultivated.

In-depth case studies of mathematics learning within classrooms and across schools, like the ones presented by Jo Boaler and her colleagues, offer critical perspectives on how teachers, departments, and school administrations create conditions for relational equity among their students. Rich and comprehensive analysis of the practices of teachers and departments that play a role in fostering these relations can be described as design principles for learning environments. Although these principles can help guide the development of better opportunities for students in mathematics within their local schools, they do not necessarily address the systematic and structural aspects of inequity in mathematics education. Additional research needs to be conducted on the application of these principles across a range of schools and classrooms, and the broader social and structural barriers that shape, support, and constrain their proper implementation.

² At the time of Boaler's (2002) study, Railside student demographics were approximately 38% Latino/a, 23% African American, 20% White, 16% Asian or Pacific Islanders, and 3% other groups.

Broadening Classroom Discourse Practices

One aspect of classrooms that support relational equity is that they utilize a discussion-based format, where students work in groups and as a whole class to engage deeply with mathematical concepts and procedures. Some researchers concerned with English language learners (ELL) and students from nondominant backgrounds suggest that organizing mathematics classrooms around inquiry-based discussion may sometimes serve to perpetuate inequities among students if language and status differences are not taken into account. For example, in her work as a teacher-researcher, Lubienski (2000, 2002) found that her high and low socioeconomic-status (SES) students experienced mathematical discussions differently. While the higher SES students claimed to feel confident in contributing to and sorting out the ideas presented, the lower SES students found discussions frustrating and wanted the teacher to show them how to do problems. Using examples from her classroom, Lubienski (2002) argued that one cannot assume that particular practices, such as open-ended discussions, will necessarily work for all students. Although she does not suggest an alternative practice, a possible interpretation of her finding is that lower SES students will be more successful if taught by traditional methods. This conclusion is inconsistent with Boaler's findings. A more nuanced interpretation of Lubienski's findings is that the practices she studied did not take into account the everyday discursive practices or ways of reasoning of the low SES students. For example, others have found that when students' diverse ways of communicating are taken up as part of classroom practices, all students involved benefit (Brice-Heath, 1982; Staples & Hand, 2004; Warren, Rosebery, & Conant, 1994).

Research that closely examines the role of language in mathematics classrooms has yielded significant insights into the linguistic resources that are often missed in classrooms with students from diverse cultural backgrounds. Moschkovich (1999, 2002a), in studying the participation of ELL students, has examined how classroom practices create opportunities for students to demonstrate their competence in mathematics. She has argued that teachers and researchers should not focus on what students lack (in this case mathematics vocabulary) but instead focus on how students participate in everyday and mathematical discourses as well as on how they draw on multiple resources to communicate mathematically. Specifically, by analyzing students' participation, Moschkovich (1999) found that students brought different ways of talking about mathematical objects (e.g., narrative, predictive, and argumentative) and points of view of

mathematics situations to the discussions (e.g., standard definition of parallelogram versus dynamic view of trapezoid as half of a parallelogram). None of these ways of talking is privileged over the other; instead, each way of talking "can contribute in its own way to the mathematical discussion and bring resources to the conversation" (p. 12). Also, by examining the point of view that the students brought to the problem, Moschkovich illustrated how instead of concluding that the students were wrong or lacked vocabulary, they were simply bringing a different point of view than the teacher expected. Moschkovich concluded that taking a discourse approach to mathematical learning means considering the different ways of talking and different points of view students bring to discussions. This kind of approach shifts the focus of math instruction for ELL students away from vocabulary development toward mathematical content. In concluding her analysis of middle school math discussions, Moschkovich (2002a) provided further implications for mathematics instruction for bilingual and ELL students:

- Instruction should support engagement in conversations that go beyond vocabulary translation, and involve students in communicating about concepts.
- Teachers should support ALL students in participating in discussions. They can move toward this goal by providing opportunities for bilingual students to participate in discussions and by learning to recognize the resources that these students use.
- Instruction should also support students' use of resources from the everyday register, as well as resources such as gestures, objects, and students' first languages.
- Assessment must consider more than vocabulary, expanding to include how students use the situation, the everyday register, and their first languages as resources.
- Determining if a student's error is a conceptual misunderstanding or a language problem is not as important as listening to students and uncovering their competence, which requires a complex perspective. (Moschkovich, 2002a, p. 207)

Issues of language and discursive practices have implications for a large group of students. Mathematical and scientific language, with its particular precision, must be learned by all students. This is often accomplished as students are apprenticed into the community of mathematics and science during their elementary and secondary schooling. The discourse

of the mathematical community seems to align with the discourse patterns of the dominant society (R. Gutiérrez, 2002), privileging students from that socioeconomic and language group. In advocating for a mathematics community that is more inclusive, attention may need to be focused in two directions: (a) to apprenticing students from nondominant groups into the mathematics community so that they develop the necessary discourse proficiency at the same time that the resources they bring to the classroom to explore concepts are validated and valued; and (b) to opening the domains of science and mathematics to reflect more of the diversity of the United States and the world. This may be a task that requires more attention from teachers who come from discourse communities that are different from those of the students. The work of Moschkovich (2002a), Lubienski (2002), and others has begun to define the issues and suggests questions that need to be addressed.

Expanding Conceptions of Mathematical Competence

Examining opportunities to learn in mathematics classrooms for diverse groups of students also prompts new ways to think about student achievement in mathematics. The recentering of analyses of student learning on student experience and practice allows for the identification of various student competencies not easily captured by traditional forms of assessment or, for that matter, those that were not previously valued as mathematical competence (Cohen & Lotan, 1997). Departing from traditional notions of *competence*, competence from a cultural perspective is recognized as being coconstructed by teachers and students in relation to classroom opportunities to learn and to what students are held accountable. This analytic shift from *achievement* to *competence* distinguishes a culturally biased deficit approach to assessment from a more culturally inclusive notion of mathematical understanding. Focusing on what students *are* doing opens conceptions of mathematical competence to the possibility of accommodating the diversity in students' ways of knowing and paths to learning.

One approach to expanding what it means to be mathematically competent is to investigate students' mathematical activities outside of school, challenging existing beliefs that pathologize or construct as deficient the cultural practices of nondominant populations. Since Gay and Cole's (1967) landmark ethnography on mathematics learning in a Liberian Kpelle society that focused on community-based out-of-school activities, sociocultural studies of mathematics learning have proliferated (see, e.g., Lave, 1988;

Saxe, 1991; Scribner, 1984). Important conclusions of researchers comparing the everyday use of mathematics (e.g., selling candy or fruit, purchasing items at a local store) with performance in school mathematics contexts not only found students to be competent in mathematics in out-of-school contexts, but also suggest the importance of analyzing and drawing on children's informal strategies into the community of practice of the mathematics classroom (Carraher et al., 1985; Lave et al., 1984; Saxe, 1988a, 1988b; Taylor, 2004).

In the first of many studies of young Brazilian street vendors, Carraher et al. (1985) analyzed the everyday use of mathematics by children selling fruit on the streets of Recife, Brazil. The results of the study indicated that students competent in out-of-school settings could not reach equally accurate solutions to the same problem when posed as an in-school task, suggesting that the context in which problem solving is happening cannot be separated from the problem and the act of problem solving itself. Similarly, Lave et al. (1984), in a study comparing the problem solving done by grocery shoppers at the supermarket to their performance on paper-and-pencil tests, concluded that the supports present in the supermarkets could not be divorced from the problem solving process of making purchases. In other words, the environment of the supermarket provided supports for calculations that did not exist when the task was one given in a school-like format. The importance of the setting to the problem-solving process is indicated by this study as "people and settings together create problems and solution shapes, and moreover, they do so simultaneously" (p. 94). In both of these studies not only were people found to be competent in mathematics in out-of-school contexts, but this research also suggests that the environment in which problem solving takes place contributes to mathematical accuracy.

In studying how school mathematics learning is related to out-of-school mathematical practices, Saxe (1988a) found no statistically significant difference between schooled and nonschooled children for problems of currency arithmetic and ratio comparison in out-of-school contexts. On the other hand, in looking at differences in performance on school arithmetic problems, he observed that the children who were sellers did solve more problems drawing on the informal strategies developed in the marketplace than did the nonsellers. Saxe argued for drawing on informal strategies such as these in classroom practice.

In a study of how school mathematics learning might be supported by out-of-school mathematical practices, Taylor (2004) explored the purchasing

done by children before and after school at local convenience stores in a northern California city. He documented conventions, artifacts, and other supports that assisted children in making purchases in these real-life situations. Taylor found that accessing the mathematics necessary to complete purchases required children to draw on knowledgeable others, such as older children and store employees. Taylor also found currency to be a powerful context to draw on in school. In working with young children on noncontextualized addition and modeling problems in a school context, Taylor has shown that the use of currency as a base-ten model provided more support in arriving at correct solutions to problems than did base-ten blocks. This study illustrated how students' everyday practices may support the learning of fundamental mathematics concepts in mathematics classrooms. It is important to point out that although these studies illustrate the mathematical fluency with which children engage everyday tasks, they do not suggest that these youth are incapable of success in school mathematics. Rather, studies of students' out-of-school mathematical activities expose a misalignment between the opportunities to learn in school mathematics and students' mathematical practices developed outside of the classroom.

The work of both Saxe (1988a) and Taylor (2004) suggests that expanding notions of mathematical competence to include the valuation of students' informal strategies developed in out-of-school contexts can support students' mathematical knowledge development within school contexts. Uncovering students' mathematical competency both in and out of school broadens the field's understanding of student mathematics achievement and begins to change conceptions of mathematical competence. However, broadening conceptions of mathematical competence by looking outside of the classroom provokes the question of how these insights can be applied generatively to classroom mathematics curriculum and instruction. Research examining the features of out-of-school mathematics activities in comparison to school practices may help bridge students' participation in mathematical activity, and mathematical activity that supports this engagement.

Bridging In-School and Out-of-School Mathematics Knowledge

One project that attempts to forge links between in-school and out-of-school knowledge is the Yup'ik project. The Yup'iks are a group of indigenous people of Alaska whose language and culture has been marginalized within the educational system by its emphasis on English-only language and middle-class,

White American cultural practices (Lipka, 1994, 2002; Lipka, Wildfeuer, Wahlberg, George, & Ezran, 2001). The Yup'ik Project is a collaboration among university researchers, teachers (both indigenous and White) and Yup'ik elders to explore and describe the traditional number system used by the Yup'ik as well as the mathematics inherent in Yup'ik cultural practices. The project has begun to publish a culturally based curriculum (*Math in a Cultural Context*, Detselig Enterprises, Calgary, Canada) and has documented that the achievement of students whose teachers use the curriculum after receiving professional development surpasses that of comparable groups of students (Civil, 2001; Lipka, 1998, 2005; Webster, Wiles, Civil, & Clark, 2005).

Moll and his colleagues have conducted seminal work on bridging the in-school and out-of-school knowledge of students from households in working-class Mexican communities (Gonzalez, Moll, & Amandi, 2005; Moll et al. 1992; Moll & Greenberg, 1990). These researchers developed a conceptual framework known as *funds-of-knowledge* to begin to account for the nature and structure of knowledge and skills organized in and across these households. Funds-of-knowledge are "historically accumulated and culturally developed bodies of knowledge and skills essential for household or individual functioning and well-being" (Moll et al., 1992, p. 133). The funds-of-knowledge work explicitly rejects the notion that the problem of underachievement either is located within the students or is attributable to their cultures and communities. Instead the locus of responsibility for underachievement shifts to the school and acknowledges the complexities of students' lives in and out of school, seeking to support the students' scholastic achievement by drawing on the wealth of their home and community experiences. Moll (1992) argued "that these families and their funds-of-knowledge represent a *potential* major social and intellectual resource for the schools" (p. 22).

Whereas the initial funds-of-knowledge work of Moll and his colleagues focused on language and literacy, the BRIDGE project extends the work to mathematics (Civil, 1995a, 1995b). An explicit goal of the project was "the development of mathematics teaching that builds on these students' backgrounds and experiences" (Civil, 1995a, p. 2), and it incorporated the same three basic components of the original project: (a) ethnographic household visits, (b) teacher-researcher study groups, and (c) classroom implementation and curriculum development (Civil, 1994). Views of the families as "somehow disorganized socially and deficient intellectually" were exploded during the data gathering and study-group analysis in the BRIDGE project (Moll et al., 1992, p. 134). When faced with caring and interested parents

on the other side of the home interview table, teachers had to confront their notions of the homes these children came from as dysfunctional and begin to examine the potential that existed there (Moll et al., 1992). Civil (1998) suggested that this type of direct connection made between parent and teacher could have an impact on the teacher's thinking and could ultimately influence his or her teaching. The incorporation of mathematical funds-of-knowledge into school mathematics presented particular challenges, however, within this project. A question that evolved from the study of household funds-of-knowledge was the extent to which it was possible for schools to build on everyday mathematical situations (Civil, 1995b). Researchers found, for example, that frequently the mathematics in everyday activities was hidden, such that people engaging in the activity did not acknowledge what they were doing as mathematics and at times even directly rejected it being characterized as math (Gonzalez et al., 2001).

Although the BRIDGE project does seem to have positively affected teacher and student attitudes about school, and although learning clearly was happening in the BRIDGE classrooms, it is not as clear what mathematics the children were learning. As early as 1995, Civil was raising issues regarding the depth of mathematics the children were learning, stating, "unless we have a clearer picture of the mathematical opportunities in a module and on how to push for these, the modules risk to present only surface applications of mathematics, often not challenging enough for the students" (1995a, p. 15). Civil's (2001) recent work documents the explicit connections that students made between a gardening project that she designed with an intermediate grades mathematics teacher and area measurement. The activity was designed to foreground the mathematics by "looking for opportunities in which mathematics will occur naturally" (p. 404).

Moschkovich (2002b) summarized much of the work done in the area of integrating everyday mathematical practices into school mathematics in the interest of making "the mathematical practices of different groups accessible to more students" (p. 8). Her own recommendations for implementing such practices, however, carry a caveat about the overuse of everyday mathematics at the expense of academic mathematics. She is particularly concerned about the emphasis on practical mathematics being implemented or interpreted as part of a tracking policy or as a disguise for vocational education. "It is not merely using everyday mathematics that is important but making connections between the familiar practices of everyday activities and academic mathematical practices" (Moschkovich, 2002b, p. 9). In the end, Moschkovich is asking for a balance between everyday and academ-

ic mathematics that will not only motivate students to engage in the study of mathematics but will also provide them with the discursive practices they will need to pursue more advanced studies such as documenting and constructing narratives about their solution processes and reflecting on the "efficiency or generality of different approaches to a problem" (Moschkovich, 2002b, p. 9).

Reforming the "Culture of Mathematics"

Balancing the need for individuals to draw on practices, identities, and realities of their everyday, lived experience with the goal to master formal, domain-related practices and identities within the context of schooling has been a reoccurring theme in the field of education (Dewey, 1902, 1938). R. Gutiérrez (2002) framed this longstanding tension in a particular way with respect to equity in mathematics. Progress on equity cannot be made, she argued, as long as deficit perspectives continue to permeate research and practice, and the field of mathematics perpetuates a culture that excludes individuals with different perspectives (such as minorities and women). To create conditions for equitable mathematics education, she proposed that the field coordinate two different approaches to mathematics instruction. One is mathematics that "reflects the status quo in society, that gets valued in high-stakes and credentialing, that privileges a static formalism in mathematics, and is involved in making sense of a world that favors views and perspectives of a relatively elite group" (p. 151), or what Gutierrez called *dominant mathematics* (in which she included "reform" mathematics). She contrasted this with "mathematics that squarely acknowledges students are members of a society rife with issues of power and domination" (p. 151), or *critical mathematics*, that empowers students to challenge the structures, perspectives, and processes through which they are marginalized. The combination of *dominant* and *critical* mathematics, she argued, "serve as an entrance for students to critically analyze the world with mathematics, and being able to critically analyze the world with mathematics may be an entrance to engage dominant mathematics" (p. 152).

The contention that mathematics education is embedded in a broader "culture of mathematics" that is privileged and privileges a certain few is echoed by researchers who take a critical stance on mathematics education (Delpit, 1988; Ladson-Billings, 1997; Tate, 1994). This critical perspective is serving to push mathematics education research to take seriously issues of power, race/racism, and "White privilege" in mathematics education. In the section that follows, we highlight some of the contributions that the theories

of cultural activity have made to the conceptualization and consideration of differences in opportunities to learn between dominant and nondominant students in mathematics education.

Summary

Researchers who study mathematics education with a perspective of learning as cultural activity have made significant strides in disentangling the issues involved in differential achievement—reconceptualized as differences in opportunities to learn. Due to the space constraints of this chapter, we have reviewed the research of only a select group of researchers who are recognized within the field as pioneering this work. This review is by no means exhaustive, and the research has suffered in general from complicated methodological and political challenges. However, the research reviewed here offers important insights into the processes by which a *participation gap* (Hand, 2003) between students from diverse social, cultural, and racial backgrounds can arise within mathematics classrooms and how classrooms can be structured to better afford opportunities to participate in mathematics by a wider range of students.

From a theoretical perspective, research based on a perspective of culture as constituted by and through activity provides new constructs for examining the interplay of culture and the individual. By focusing on participation and identity as key aspects of individual and joint social practice, this perspective acknowledges the role of both individual agency (as individuals choose, adapt, and reject practices) and sociocultural processes (as broader social, political, and racial practices and identities are remade in local classrooms) in shaping the learning experience. Thus, while acknowledging that students bring perspectives, values, and routines to the mathematics classroom from their home and local communities, this perspective emphasizes the processes of negotiation, reconciliation, and rejection that students manage with respect to their activities across multiple communities.

The complex nature of examining these processes within and beyond the mathematics classroom has been aided by the development of complementary methodological tools and techniques. Tools such as video interaction analysis and qualitative data analysis software and techniques such as documenting trajectories of participation and repertoires of practice help researchers capture, analyze, and identify features of learning communities and various aspects of participation within them. These features and aspects can then be linked back to broader cultural practices, discourses, tools and artifacts, and identities, which implicate issues of power

and access in the classroom. Locating practices, scripts, and norms found in the local context of the mathematics classrooms within global social and cultural hierarchies can reveal the danger of “neutral” mathematics instruction in perpetuating inequitable processes that marginalize nondominant students.

Attending to the variety of ways individuals engage in mathematical tasks and other activities in different contexts, this perspective has also expanded the notion of what it means to do mathematics and to be a learner and doer of mathematics. On the one hand, definitions of mathematical thinking and reasoning have been widened to include the processes by which people solve mathematical problems informally with the material and social resources around them. Thus, part of calculating the statistics of a pro basketball player involves being part of a community of players and spectators who find this activity meaningful. On the other hand, learning mathematics in school and identifying oneself as a mathematics learner is bound up in social and cultural discourses that position particular mathematical practices as being “competent” or “rigorous” and certain individuals as being “smart” and “capable.” Thus, what this research has revealed is how mathematics learning itself is organized within a cultural practice that can serve to either enfranchise or marginalize different groups of individuals.

Finally, one common thread across the research presented in these sections is the rejection of deficit and cultural-deficit thinking. Culture is acknowledged as a critical element in trying to understand the academic disparities between economically advantaged populations and impoverished ones. But instead of viewing the cultures of these families and communities as pathological and the source of academic school failure, culture is viewed as an area of students’ lives that can contribute to academic success if appropriately understood. At the same time, however, this research has tended to shy away from making explicit links between opportunities to learn found in mathematics classrooms, cultural practices both in and out of the mathematics classroom, and persistent racial inequities (Cobb & Nasir, 2002; Martin, in press-b).

A potential tension in research aimed at identifying and valuing the cultural practices and identities of nondominant groups of students is the possibility of reifying what are basically essentialist accounts of racial and ethnic communities. This way of thinking, or essentializing of traits of groups of students, has potentially negative consequences for students subject to its application.

The obvious problem with essentializing is that, by treating cultural behaviors, values, and practices as fixed and immutable, it slips perilously close to com-

mitting at least two types of error that were observed with cultural deficit thinking. First, in associating and universalizing cultural practices and patterns to all members of a group, the space for the member to be treated as an individual shrinks. Essentializing contributes to a form of *cognitive reductionism* where cognitive complexity is simplified and diminished (Gutierrez & Rogoff, 2003). A person, as an individual, is rendered secondary in the process of teaching and learning. Second, as in the case of cultural deficit thinking, with essentializing there is a tendency to reify culture, that is, to see it as isolated, immune from impact of other cultures and the dominant social and economic forces. If the role of culture is going to be given a central place in educational research and practice, researchers must learn how to think and talk about cultural patterns and practices and student learning without slipping into essentializing and cultural-deficit thinking (Gutierrez & Rogoff, 2003). In order for culture to remain a viable explanatory construct for understanding and improving the persistent underachievement of certain student populations, much care and attention must be exercised in developing theories and pedagogies that do not essentialize cultural traits.

Much of the research reviewed in this section has focused on the conditions of individual classrooms or schools, out-of-school practices, or communities, with only limited attention to how historically based processes and structures that involve race and power have shaped these local contexts and practices over time. Thus, perhaps one of the most important challenges to arise out of research based on a cultural activity perspective is the need to better understand the relation between race and culture. On the one hand, research that isolates the individual from the cultural processes in which she develops has merged the two constructs, treating race and culture as if they were one and the same. In this way, the characteristics of all people from a racial group are treated as being homogenous. In this model race determines culture. On the other hand, the research just reviewed has tended to completely divorce the two and focus only on culture in its most apolitical sense. In this model, culture is all that matters. However, there are a growing number of researchers who have critiqued a focus on culture that excludes consideration of race and power.

CONSIDERATIONS OF RACE AND POWER AT THE CLASSROOM LEVEL

Thus, we turn to mathematics education research that has considered explicitly issues of culture *and* issues

of race and power as they intersect with mathematics teaching and learning. Specifically, this section highlights research that provides models of classroom learning environments that focus squarely on issues of cultural relevancy while also taking into consideration issues of race and power (see, e.g., Frankenstein, 1990, 1995, 1997; Gutstein, 2003, 2006; Ladson-Billings, 1994, 1995, 1997; Moses & Cobb, 2001a, 2001b), thus making race and power central constructs in confronting differential achievement in mathematics education.

The Algebra Project

The Algebra Project, for example, is a mathematics literacy effort with a grassroots implementation process (Moses & Cobb, 2001a, 2001b; Moses, Kamii, Swap, & Howard, 1989; Silva, Moses, Rivers, & Johnson, 1990). The program is designed to make Algebra available to all seventh- and eighth-grade students despite their previous levels of academic achievement. The curriculum is created to develop algebraic thinking using projects that engage students with concrete experiences supported by a culture of mutual inquiry. These experiences are drawn from or build on practices of the communities in which students live. Addition and subtraction of negative numbers, for example, is taught within the context of trips. But the subway context that is used for inner-city kids in Boston is exchanged for a bus trip for students living in rural areas of Mississippi. The curriculum explicitly acknowledges that the mathematics must connect with the lived experiences of students.

The work of Moses and colleagues also makes central issues of race and power, equating the need for mathematical literacy in today's society with the need for Black registered voters in Mississippi in 1961. Moses and Cobb (2001a) have argued that differential access to algebra, which disproportionately excludes African Americans, Latinos, and poor White students from college preparatory mathematics classes, is serving as a form of structural discrimination resembling the use of literacy tests in the '60s. Moreover, algebra is the forum where students learn the symbolism necessary for developing technological knowledge demanded in today's high-tech job market. Thus, mathematical literacy is not just needed for access to college preparatory mathematics classes but is also necessary to meaningfully participate, with economic viability, in today's society. Mathematical literacy, then, is the key to citizenship; it becomes a civil-rights issue, and a necessary component in promoting economic and civic equality.

More recent work of the Algebra Project, the Young People's Project, encourages students to take up this right (Kirkland, 2002). Their work organizes young people to work to change their education and the way they relate to it, that is, encouraging students to "demand to understand" in an effort to challenge and transform their marginalization. Specifically, the Young People's Project is dedicated to the creation of mathematically literate communities through the recruitment and training of core high school and college Math Literacy Workers. These Math Literacy Workers come from the communities the Algebra Project seeks to serve, and they focus on developing their knowledge capacity so that they are able to organize and manage math literacy work independently. This work might consist of mentoring middle and elementary school students, providing ongoing after-school workshops for younger students, providing community events for families and community members, and facilitating team organizing for mathematics competitions. In the process, the Math Literacy Workers collectively contribute to the development of ever-expanding networks of mathematically literate young people. For this to occur, though, young people themselves must demand the right to receive a quality education, and the Young People's Project is working to create that demand.

The work of the Algebra Project, and the extended work of the Young People's Project, focuses squarely on issues of cultural relevancy by building a curriculum that explicitly connects with students' lived experiences. Additionally, this work considers the effects of power structures on mathematics teaching and learning by acknowledging the gate-keeping role of mathematics. Their work explicitly confronts this gate-keeping role of mathematics education by helping *all* students, particularly those traditionally denied opportunities, pass through the gates. It has the potential to confront differential mathematics achievement and provide quality mathematics instruction to all students. More research is needed that looks at the effects of the Algebra Project and the Young People's Project on student achievement in mathematics.

Culturally Relevant Pedagogy

Other compelling models of classroom learning environments consider both the multiple constructs of culture, race, and power, and how students can use their knowledge (of mathematics and other subjects) to challenge current oppressive and inequitable structures (see, e.g., Frankenstein, 1995, 1997; Gutstein, 2003, 2006; Ladson-Billings, 1994, 1995, 2001; Tate, 1994, 1995). Ladson-Billing's (1994, 1995, 1997,

2001) work, although not specific to mathematics education, provides a theory of culturally relevant pedagogy that speaks to issues in mathematics education, and as such is important to consider here. From her study of eight successful teachers of African American students, Ladson-Billings developed a theory of culturally relevant pedagogy. Specifically, she argued for the importance of a three-pronged approach to culturally relevant teaching that proposes to (a) produce students who can achieve academically where achievement is not limited to standardized assessment, (b) produce students who demonstrate cultural competence, and (c) develop students who can both understand and critique the existing social order. Two of these three components will be expanded upon here: cultural competence and understanding and critiquing the existing social order (or developing a sociopolitical consciousness).

As a pioneer of culturally relevant teaching, Ladson-Billings argued that teachers should develop in students a cultural competence. That is, students should be provided with "a way to maintain their cultural integrity while succeeding academically" (Ladson-Billings, 1995, p. 476). In classrooms that promotes cultural competence, teachers must first be aware of their own culture and its role in their lives (Ladson-Billings, 2002). They can then work to effectively respond to students from different cultures and classes while valuing and preserving the dignity of cultural differences and similarities between individuals, families, and communities. A classroom that promotes cultural competence acknowledges that everyone has a cultural history that shapes their identity. Moreover, in such a classroom, the academic and cultural assets students bring to the classroom are seen as enriching the community, and the students and teacher together continuously strive to learn about one another and the assets each person brings. Additionally, Ladson-Billings (2001) argued that in classrooms that promote cultural competence in students, the teacher "uses culture as a basis for learning" (p. 98). That is, teachers aim to capitalize on students' prior knowledge, and they view students' culture as a means through which they can acquire new knowledge.

In addition to promoting students' cultural competence, teachers enacting a culturally relevant pedagogy provide students with the tools they need to understand the social structures around them, see how those social structures (such as institutional racism) may affect their lives, and teach students how to challenge those structures. In classrooms where teachers employ culturally relevant pedagogy, teachers and students together create knowledge "in conjunction with the ability (and the need) to be critical of con-

tent” (Ladson-Billings, 1994, p. 93). To be critical of content means culturally relevant teachers attempt to make knowledge problematic, challenging students to view knowledge as a means for transforming the world in which they live. For example, in a more contextualized examination of two classroom teachers in her study, Ladson-Billings (1994) noted that both teachers held exceptionally high expectations (demonstrated through both words and actions) of their students while simultaneously helping students understand that societal expectations for them are generally low. In this way, students recognized the teacher’s act of holding high expectations and their own efforts to meet those expectations as acts to challenge and defy prevailing societal beliefs.

Although Ladson-Billings is not referring specifically to the teaching of mathematics, mathematics education researchers argue similarly, noting that mathematics teaching entails a shift from thinking of preparing students to live within the world, as it currently exists, to thinking about how to prepare students to restructure “those social systems . . . in order to remove barriers that women, minorities, and others experience” (Secada, 1989, p. 47). Mathematics educators need to work toward using mathematical knowledge to empower students to work for social justice and to confront issues of unequal power relationships that exist in the world in which we live (Martin, 2003). Thus, mathematics education faces a twofold imperative: It needs to provide students with mathematics instruction that includes the mathematics deemed necessary for success in the current system (a similar component to Ladson-Billings *academic achievement*) while simultaneously providing students an opportunity to use mathematics to expose and confront obstacles to their success (Apple, 1992; R. Gutierrez, 2002; Gutstein, 2003, 2006; Martin, 2003; Secada, 1989; Tate, 1994, 1995).

Teaching Mathematics for Social Justice

Teaching mathematics for social justice addresses both of these components. Math teachers employing social-justice pedagogies address the first imperative by recognizing the necessity of mathematical knowledge and including mathematics-specific goals for their students (see, e.g., Frankenstein, 1990, 1997; Gutstein, 2003, in press-a). They address the second imperative by engaging students in using mathematics to analyze their world critically in an effort to ultimately promote a democratic society in which all have an opportunity to participate fully (see, e.g., Frankenstein, 1995; Gutstein, 2003; Skovsmose, 1994).

Frankenstein (1990, 1995, 1997), for example, described her attempts to teach mathematics for social justice with working class adults in basic college mathematics courses. In these courses, Frankenstein employed a critical mathematics pedagogy focused explicitly on using statistical tools to analyze social issues critically (e.g., income data, wealth distribution, home mortgage distribution, the tax system). Her critical mathematics pedagogy involved promoting “the ability to ask basic statistical questions in order to deepen one’s appreciation of particular issues [and] it also involved the ability to present data to change people’s perceptions of those issues” (Frankenstein, 1990, p. 336). The goal of this critical analysis was to prompt students to question their assumptions about how society is structured and to enable them to act from a more informed position on social structures and processes.

Gutstein (2003, 2006, in press-a, in press-b) also described his enactment of a social-justice pedagogy in a Chicago public middle school with a predominately low-SES Mexican and Mexican American population. His work further demonstrates how such a pedagogy addresses issues of culture, race, and power and works to promote students’ use of mathematics to transform oppressive structures. Gutstein (in press-b) conceptualized the pedagogy of teaching mathematics for social justice as developing four main components: “a) academic ‘success’ (i.e., both mathematical power and what is needed to pass gate keeping tests); b) sociopolitical consciousness; c) a sense of social agency; and d) positive social and cultural identities” (p. 8).

To achieve his goals of teaching for social justice, Gutstein created 17 real-world mathematics projects that connected to students’ lives (e.g., examining wealth distribution; analyzing SAT and ACT exam scores by race, class, and gender; questioning whether racism is a factor in mortgage loan opportunities). To achieve the mathematics-related objectives in the classroom, Gutstein used *Mathematics in Context*, a curriculum developed by the National Center for Research in Mathematical Sciences Education and the Freudenthal Institute (1997-1998). More specifically, Gutstein employed a *pedagogy of questioning* in his classroom. He created a classroom environment where students posed their own meaningful questions, engaged in understanding their own realities in sociopolitical context, discussed interrelationships and complexities among questions, engaged and analyzed multiple perspectives, and interacted with questions that connect to actions and social movements. Mathematics played a central role in this pedagogy of questioning. Students used mathematics to develop sociopolitical

understanding of their life conditions and broader society. For example, students used mathematics to uncover what the money for one B-2 bomber would mean in terms of college education for thousands of Latino/Latinas. In this project, students

used mathematics to investigate and calculate the various costs for different ways to use public money; they considered ramifications, shared their views with others, and wrote about them; put their mathematical analyses into sociopolitical, historical context; and built community, a shared sense of purpose, and a dramatically different orientation toward mathematics and its use in understanding reality. (Gutstein, in press-b, p. 24)

The results of Frankenstein's and Gutstein's work are promising. Frankenstein (1997) found that the use of critical mathematics pedagogy changed her students' perceptions of mathematics and their ability to understand mathematics, both important factors in helping students reach the mathematics-specific goals of the course. At the same time, she engaged her students in using mathematics to analyze their world. For Gutstein's students, many developed mathematical power, and the cocreation of a classroom environment where students discussed significant issues of justice and equity, and where students actively and consistently raised their own questions, seemed to foster agency among students.

Although teaching mathematics for social justice holds potential for addressing differential achievement in mathematics because of its simultaneous consideration of issues of culture, race, and power and their intersection with mathematics teaching and learning, more research is needed that examines the effects of the implementation of such pedagogy on students' mathematical learning. It is important, however, to note that when determining whether this pedagogy is effective in terms of students' learning of mathematics, researchers must also consider the purpose of mathematics education. In the cases described here, the purpose of mathematics education is not functional literacy, or "at best, to generate a few more individual successes" (Gutstein, 2006, p. 211); rather, the goal is to conceive of mathematics knowledge as the ability to use mathematics to critique and transform oppressive structures—math literacy is "knowledge for liberation from oppression" (Gutstein, 2006, p. 211).

Frankenstein's and Gutstein's work describes the practices of teacher researchers who are committed to teaching for social justice. The next question is how to build on that work to study what it means for the "average" teacher to learn to teach mathe-

matics for social justice. Gau (2005), in a study of eight secondary mathematics teachers engaged in learning to teach for social justice through the creation, implementation, observation, and revision of a mathematics lesson that incorporated social-justice goals, found that teachers struggled to find a balance between the mathematics and the social justice. For one group, this meant that the mathematics took priority over the social justice, and in four separate lesson implementations they never addressed the social justice goals of their lesson. For the other group of teachers, this meant that mathematics need not be tied in all the time. Instead, they focused on developing the social-justice goal of their lesson, to the detriment of the mathematics. In fact, in this latter case, teachers seemed to be so focused on having students come to a particular conclusion, that they inappropriately interpreted data, suggesting to students that the data supported conclusions that were, in fact, not founded by the data.

Additionally, although teachers in both groups conceptualized teaching mathematics for social justice similarly to that expressed in the literature, they did not ever express that it include the goal of students' learning mathematics. Rather, they conceptualized it as students' using known mathematics to analyze and confront inequities, suggesting that the social-justice component is always an "add on" to the curriculum. More work is needed in this area to see what teachers struggle with as they learn to teach mathematics for social justice, and the implications this has for the potential of such a pedagogy to affect both students' learning of mathematics and students' ability to use mathematics to critique and transform oppressive structures.

Researchers who work in the area of mathematics teaching for social justice (see, e.g., Esmonde, 2006; Gau, 2005; Gutstein, 2003, 2006) also note that such teaching is not just a collection of supplemental projects that could be "dropped into" any context. Rather, this pedagogy permeates all aspects of the classroom. It is a pedagogy that is "forged *with*, not *for*" students and is continually negotiated as students' understandings of their own sociopolitical contexts grow and as the questions they wish to explore evolve (Freire, 1970/1993, p. 30). Additional research on the implications of teaching mathematics for social justice on achievement is still needed, but this work, along with the work of Moses and colleagues in the Algebra Project and Ladson-Billing's theory of culturally relevant pedagogy serves to highlight work that places culture, race, and power at its center.

RACE AND POWER

In the previous section, we examined research that places culture, race,³ and power⁴ at the center in understanding issues of mathematics teaching and learning. Here that examination continues. However, whereas the previous section examined these constructs within the context of the classroom, this section will be situated within a broader, structural context. More closely, this section highlights larger policies and mandates that shape and structure mathematics education. The focus is on two questions. First, how do issues of race and power manifest themselves within the structures that provide mathematics education to students? Second, how might an analysis of race and power within this broader context help in understanding differential achievement in mathematics?

It is important to mention that two issues focus the discussion in this section. First, presently little research centers itself within mathematics education and provides a structural analysis of race and power. Where this literature does exist, we have sought to cite it and its contribution to the field. However, because of the limited amount of such research, in some instances we cite researchers outside of mathematics education that examine race and power. A second reason flows directly from the first. Because there is limited research within the field, this section raises issues and questions with respect to race and power in relation to mathematics education. Many of the issues raised here draw on the experiences and work of the authors in schools and districts that serve poor and minoritized communities. Such communities provide a rich context within which to explore the impact of racialized experiences and power dynamics on differential achievement in mathematics.

Structuring Opportunities to Learn Mathematics

Orfield, Frankenberg, and Lee (2003) stated that the level of segregation of schools is worse now than in 1968. Students of color and Whites are increasingly not in the same schools. Moreover, only 15% of the intensely segregated White schools have populations in which more than half are poor enough to receive free and reduced lunches. For Black and Latino students the percentage is 86%. Schools in communities predominantly consisting of Blacks and Latinos

are poorer, and they generally have fewer AP courses, fewer credentialed teachers (Darling-Hammond & Sykes, 2003), more out-of-field teachers (Ingersoll, 1999; Rogers, Jellison-Holmes, & Silver, 2005), and buildings in worse conditions (Oakes & Saunders, 2002). Kozol (2005) and others (Frankenberg, Lee & Orfield, 2003; Hunter & Donahoo, 2003) have ascribed this situation to a new form of apartheid in the U.S. school system, where low-income public schools have become hypersegregated with populations of up to 99% students of color.

Along with the material conditions in “apartheid” schools, Kozol (2005) notes that in urban schools there are another set of conditions around how we talk about students and the ways they are expected to participate. Kozol points out that he has heard hypnotic slogans like “I’m smart! I know that I’m smart,” repeated everyday, “but rarely in suburban schools where potential is assumed” (p. 36). These non-material conditions shape the opportunities of students of color—often blaming them for their own failure. At the same time that these students are blamed for their failure, the system of mathematics education continually fails them. As such, even if the material conditions were equitable, the non-material normative aspects of schools would still construct failure for students of color (Oakes & Lipton, 1999).

The literature on access and opportunity to learn mathematics documents how experiences differ along racial lines. Overall, segregated minority schools offer less access to upper-level math and science courses, many not offering courses beyond Algebra II. Oakes, Muir, and Joseph (2000) wrote that,

A student can only take a high level class in science and mathematics if his or her school offers such classes or if his or her school opens up access to these courses to all students. In other words, how far a student can go down either the mathematics or science pipeline depends on his or her access to particular courses. (p. 12)

On the basis of a student’s race, he or she can expect to experience mathematics education differently (Hunter & Donahoo, 2003). Students of color often experience a lesser form of education, in mathematics and otherwise. In contrast, adequate mathematics course offerings (Lee, Burkham, Chow-Hoy, Smerdon, & Geverdt, 1998), qualified mathematics instructors (Rogers et al., 2005), quality mathematics curricula, and mathematics teachers who respect their culture

³ By *race*, we mean the very real ways in which a student’s skin color (and the social significance that society attaches to that skin color) frames that student’s educational opportunities and experiences.

⁴ In terms of *power*, we refer to the set of relationships and hierarchies that frame interactions.

and hold high expectations of them is the minimum expectation for White children. Whereas a Latina student can expect to attend a school with a large number of courses below Algebra, a White student can expect, as a matter of course, that the classes necessary for her to prepare herself for college will be present at her school. In this sense, statistically speaking, whiteness has a higher property or currency value. With whiteness comes advantage, more valued cultural practices as well as property, educational buildings in better neighborhoods that draw higher taxes, and therefore more funding.

As described in earlier sections, White culture often determines what is “normal” and also constructs the dialogue or ideology for understanding the “other.” This dialogue is constructed and reinforced in mathematics education, for example, when achievement scores are reported in terms of race, and lower test scores are ascribed to race (ignoring the fact that “White” is also a racial category). Educators fail to ask how the racial and cultural entailments of whiteness provide opportunities for large groups of White students to be consistently ahead of their Black and Latino/a counterparts. Instead, the success or failure of a White student often gets framed as an individual act, acclaiming or pathologizing the individual rather than the race.

Ideologies are embedded within language and ways of talking that perpetuate stereotypes of the “other.” These broad Discourses, as Gee (1990) and colleagues call them, structure the ways of talking about children of color, communities of color, and structure our individual actions (Gee, 1990; Gee, Hull, & Lankshear, 1996). Gee (1990) calls this dialogue Discourse with a big *D* because it contains ideologies, beliefs, practices, and ways of being that further the power of the dominant culture. There is more going on in individual success or failure, or individual interaction, than what is actually seen in front of us. Individual interaction sits inside of a historical reality; it sits within history, within a context, and within a relation of power. The stories embedded in these Discourses limit the ways of talking and thinking about people of color and can limit how one thinks about their intelligence and abilities, quality of family life, and cultural resources (Kana’iaupuni, 2005; Ryan, 1971; Warren, 2005).

Counterstorytelling is a method that “aims to cast doubt on the validity of accepted premises or myths, especially ones held by the majority” (Delgado & Stefancic, 2001, p. 144). Counterstories or counternarratives challenge the privileged discourse, or way of talking and relating, giving voice to people of color and critiquing racialized stereotypes. These counternarratives construct alternative Discourses to the

mainstream, counteracting the essentializing that exists within the dominant narrative. The stories that researchers like Kozol (2005) and Martin (2003, 2006, in press-a, in press-b) bring to the education community serve as counternarratives to the myth that schooling and mathematics education are neutral and color blind. In the following sections, race and power are central in creating counternarratives about the neutrality and color blindness of policies both inside and outside mathematics education.

The increasing segregation, decreasing access, and pervading Discourses place race and educational structures as central in educational opportunity. We see these areas as important for the field to explore in relation to mathematics education in particular. Though much of this work has been done outside mathematics education, we think the work on counternarratives provides an area for future research to challenge prevalent Discourses and to open access to those not in power.

Mathematics for All

In the 1980s, several national reports were released that called attention to serious problems in mathematics and science education; one of them was *A Nation at Risk* (National Commission on Excellence in Education, 1983). Among the findings reported in *A Nation at Risk* were that from 1963 to 1983 average mathematics SAT scores dropped nearly 40 points, only one third of 17 year-olds could solve a mathematics problem requiring several steps, and between 1975 and 1980 remedial mathematics courses in public 4-year colleges increased by 72%. The business and industry sector also provided impetus for improving mathematics education by demanding an improvement in workplace proficiency. Criticism had been directed at public education because employees failed to demonstrate, beyond the use of computational algorithms, proficiency levels in reasoning and problem solving (Vandegrift & Dickey, 1993).

In response, national organizations produced documents that advocated changes in mathematics curricula. The National Council of Teachers of Mathematics (NCTM) produced the *Curriculum and Evaluation Standards* (NCTM, 1989), and the Mathematics Sciences Education Board (MSEB) published *Everybody Counts* (MSEB, 1989) and *Reshaping School Mathematics* (MSEB, 1990). These documents described the goals for mathematics education as problem solving, mathematical power, access to technology, and constructivist learning (Huetinck, Munshin, & Murray-Ward, 1995). These documents were part of a national

movement reconstructing how educators think about the teaching and learning of mathematics.

The NCTM *Standards* (1989) took a particular stand on equity as well:

- In developing the standards, we considered the content appropriate for *all* students.
- The mathematical content outlined in the standards is what we believe *all* students will need if they are to be productive citizens of the twenty-first century. (*italics added*)
- We believe that *all* students should have an opportunity to learn the important ideas of mathematics expressed in these standards. (*italics added*) (p. 9)

The statement made by these excerpts is that all students should have the opportunity to learn high-level mathematics and all students need to learn mathematics. There is an implicit counter-Discourse within this movement; namely that African-American, Latino, and poor children can and should learn mathematics. This national movement intended to bring about greater levels of mathematics achievement for all students. It was born out of a desire to increase the level of mathematics literacy of Americans and to help prepare more students for mathematics-dependent fields such as engineering and computer technology.

The 2000 NCTM *Principles and Standards* pushed notions of equity in mathematics education further by having one of six principles focus specifically on equity. The equity principle states,

*All students, regardless of their personal characteristics, backgrounds, or physical challenges, must have opportunities to study—and support to learn—mathematics. . . . Equity does not mean that every student should receive identical instruction. . . . All students need access each year to a coherent, challenging mathematics curriculum taught by competent and well-supported mathematics teachers. . . . Well-documented examples demonstrate that all children, including those who have been traditionally underserved, can learn mathematics when they have access to high-quality instructional programs that support their learning. (*italics added*, pp. 11–13)*

The 2000 *Principles and Standards* show a marked increase in attention to equity in comparison to a decade earlier. Again, the idea embedded in the *Standards* is that everyone can learn mathematics, though this is spelled out in more specificity in the 2000 *Standards*. Both sets of standards, though there is a shift in attention, frame the goal of mathematics education as a problem “for all.”

The mathematics education community has taken several steps in moving schools towards the goal of greater equity in the mathematics education of students. Such efforts include comprehensive reviews of the mathematics achievement progress of traditionally underserved students such as English language learners, girls, and African Americans (Lee, 2002; Reyes & Stanic, 1988; Schoenfeld, 2002; Secada, 1992; Tate, 1997) and research studies that have revealed gross inequities in the course-availability, course-placement, and learning opportunities of these underserved students (Oakes, 1990; Oakes, Quartz, Ryan, & Lipton, 2000; Paul, 2003). The field of mathematics education is deeply indebted to the work of these scholars—many of whom are outside of the field of mathematics education research. However, we want to argue for increased efforts—within the mathematics education community—that attend to issues around the structures of schools in general and to issues of race, racism and the racialized experiences of students of color that are enforced through these structures. We argue that the lack of attention to these issues may account for the lack of progress towards equity. We posit that one cannot overcome the inequity experienced by students of color and in schools that serve these students without addressing how and why those inequities came to be and are held in place. Race and the Discourses around race continue to bring inequities into being and hold them into place. Furthermore, power—the set of relationships and hierarchies that frame interactions—is often ignored in the efforts at reform in mathematics education. As with race, power both brings into being and holds into place the inequities that we presently see in schools. By focusing our attention on equity, we deal only with effects while ignoring the causes of the inequity that we see. The issue is not only that underserved children have access to far fewer rigorous mathematics courses, but that they attend schools where such disparities are not questioned or critiqued. The realities of race (the real ways in which students’ skin color and the social significance assigned to that skin color) and power (implicit and explicit ways that larger structures, institutions, and normative ways of talking and thinking shape the access, opportunity, and experiences of individuals and groups) hold such inequities in place—making them accepted, acceptable, and normal.

Dismantling the inequities that these scholars bring to the fore such as underfunding certain schools (Kozol, 2005; Oakes, Rogers, Silver, Horng, & Goode, 2004; Rothstein, 2000), a shortage of certified mathematics teachers (Darling-Hammond & Sykes, 2003; Ingersoll, 1999), tracking (Oakes, 1985), cultural conflicts, and standardization (Rogers et al., 2005), we

believe, is a matter of wrestling with and confronting Discourses, practices and the ideologies (such as racism) which hold these structures in place. We use this next section to raise issues and questions as to how the mathematics education community deals with equity, race, and systematic power relationships.

Moving from Rhetoric to Reality

A number of researchers have offered critiques of the NCTM standards and the “Mathematics for All” movement, asking whether the call to provide quality mathematics instruction *for all* is simply rhetoric (Apple, 1992/1999; Martin, 2003). Martin (2003) cited several events that have shaped and tested the mathematics education community’s commitment to equity including specific calls for greater equity (Reyes & Stanic, 1988) and seminal documents such as the NCTM *Standards*. After nearly 2 decades of equity-minded reform, he argued there have been very few appreciable outcomes for African American, Latino, and Native American students (see, e.g., Lee, 2002; Schoenfeld, 2002; Tate, 1997). The color-blind discourse prevalent in NCTM’s seminal documents (1989, 2000) may be a contributing factor in the mathematics education community’s inability to make more headway towards equity.

The 1989 standards use equality rather than equity as a frame. Such a framing does not recognize that students have different needs and that the same instruction will not necessarily produce equitable results. The 2000 standards show a shift from equality to equity. The document notes specifically that there is no one-size-fits-all program for students. (“Equity does not mean that every student should receive identical instruction. . . . All students need access each year to a coherent, challenging mathematics curriculum taught by competent and well-supported mathematics teachers,” p. 11.) However, this document still makes no mention of race or power and the ways in which these factors make access to mathematics education inequitable (Apple, 1992/1999). Instead this latter document frames inequity as an issue of personal characteristics, background, and instruction only (“All students, regardless of their personal characteristics, backgrounds, or physical challenges, must have opportunities to study—and support to learn—mathematics. . . . all children, including those who have been traditionally underserved, can learn mathematics when they have access to high-quality instructional programs that support their learning,” pp. 11–13). Such a framing makes this document, and its consumers, vulnerable to ideologies of defi-

ciency (Cuban & Tyack, 1988; Hull, Rose, Fraser, & Castellano, 1991). Furthermore, this lack of attention to race, gender, and SES allows historical, social, and economic reasons for underachievement to be cast as individual deficiencies (Cuban & Tyack, 1988; Hull et al., 1991).

Martin (2003) has argued that the color-blind discourse (i.e., “for all”) found in these policies and documents glosses over the complexities of race and power so present in schools and school systems. For instance, Martin asked who are the students currently not receiving quality mathematics instruction (i.e., who are the *all* spoken of in these documents)? Why are *all* of these students currently not receiving quality mathematics instruction? For whom do we traditionally not consider this content appropriate? What backgrounds keep students from access to high-quality mathematics? These questions bring texture and complexity to the current efforts at equity and force educators to confront the great effort that will be required to achieve it.

Grappling with the realities posed in the previous questions focuses attention on the racialization of mathematics education. More closely, these questions bring light to how the opportunities of students within mathematics are distributed differentially based upon their race (and the social significance assigned to their race). These opportunities are not only technical (e.g., course offerings), but normative such as the conceptualizations of, the talk about, and beliefs around non-White students, their families, and their communities. It is often these norms (that embed themselves in schooling structures, opportunities, and differential resources) that make the attainment of equity so problematic. Although the call for “Mathematics for All” is noble, if it does not address these normative ways of talking and thinking, it will not change the realities of education for students of color.

Next, we present two examples from our work in schools to raise structural and policy issues for mathematics educators to address and document in future research. We situate both cases within the contexts that they arise and use them to encourage different kinds of questions for mathematics educators to research.

Algebra for All⁵

In California, the call for “Mathematics for All” has played out specifically in policies that require all of its eighth-grade students to take a 1st-year Algebra course. The California mandate, “Algebra for All,”

⁵ Details about this study can be found in Spencer (2006).

serves as an instantiation of the larger “Mathematics for All” movement, and we use this mandate to draw out questions that the mathematics education community needs to address in more depth to better understand issues of equity, race, and power in schools. The structures of schools are reified.

In recent years, Algebra has been declared a necessity for all students (Moses & Cobb, 2001a; Paul, 2003). Part of the outcry for greater access to Algebra is based upon the gate-keeping power of this course to college preparatory mathematics (Moses & Cobb, 2001a). Students who finish Algebra in middle school are positioned to take mathematics courses in high school that are necessary for 4-year college and university admissions. Research revealed that a large number of African American, Latino, and Native American students were not engaging in this course in middle school (Oakes, Joseph, & Muir, 2003). As a result, many in the mathematics education community supported and endorsed a middle-school requirement of Algebra. California took up this decree, requiring its districts to enroll all of its eighth graders in Algebra. On the surface, this decree appears quite egalitarian. Yet, the implementation of such a decree has had significant problems.

The same beliefs, conceptions, labels, and Discourses about urban students that kept them enrolled in non-Algebra courses before the mandate have kept them enrolled in inferior “Algebra” courses after the mandate. Mandates and decrees constantly get interpreted through existing lenses and Discourses within schooling practice. The more the field can understand the lenses, Discourses, and beliefs embedded in the structure of schooling, the more researchers can document and develop successful future reforms to address inequities. These Discourses, in the passing on of ideologies and beliefs, are a form of power acting on the educational opportunity of students of color. In the following section, we pose how using the lens of Discourse can help us research race and power in relation to the “Algebra for All” mandate.

Labels: Recurring Ideologies

Oakes et al. (2003) report on the implementation of the “Algebra for All” mandate in urban districts in California. The mandate, adopted in 2000, has led to all eighth-grade students being enrolled in “Algebra.” Subsequent to the mandate, Paul (2003) documented the proliferation of Algebra and pseudo-Algebra courses. Courses were titled with names such as pre-Algebra, 2-year Algebra, 1-year double-dose Algebra, Math Essentials, and Honors Algebra. Typically, only one of these courses, for example Honors Algebra,

actually fulfilled the Algebra requirement that makes students eligible for the college-preparatory mathematics track in high school. Because only a small number of students are allowed to enroll in Honors Algebra, these iterations of Algebra courses work to recreate the inequity that they profess to correct.

This re-naming of courses serves as one example of how a potential change in structure gets reified because the people and their ideologies around learning mathematics have not changed. Despite the changes in course titles, the dispositions towards students changed little (Oakes et al., 2003). Prior to this decree, teachers’ conceptions of “remedial” students focused their instruction on math facts because of perceived mathematical ability associated with lower track classes (Raudenbush, Rowan, & Cheong, 1993). After the decree, teachers’ conceptions remained the same, however, this time, they were referring to their “Algebra Essentials” students (Spencer, 2006). This Discourse about “remedial” students keeps teachers from engaging students in complex or nonprocedural work, and maintains talk about homework not being turned in and a general lack of parental involvement. Algebra students were given a host of intellectually stimulating, complex, and real-world mathematics problems to engage with (Raudenbush, Rowan, & Cheong, 1993). In opposition to the remedial students, the Discourse in reference to Algebra students constructs them as capable, hardworking, well-raised, and advanced. In addition to reinforcing ideological norms, labels reinforce structural inequities. The new titles further disenfranchised students—giving them the illusion that they were actually engaging in a college-preparatory mathematics course—without changing the actual content and goal of their courses (Oakes et al., 2003; Spencer, 2006). The labels have power in that they route students into or away from college-bound trajectories. Furthermore, these routes shape students’ relationships to mathematics. Put another way, how students see themselves as thinkers and learners of mathematics.

Historically, the same rationales given for why particular students could not think, learn, or achieve 50 years ago are the same rationales presently given for why they cannot think, learn, or achieve (Cuban & Tyack, 1988; Hull et al., 1991; Spencer, 2006). Labels such as Algebra Essentials, lazy, low skilled and remedial serve as a proxy for African Americans and Latinos without dealing with racial implications. The labels signify underlying ideologies about students of color (Cuban & Tyack, 1988; Hull et al., 1991; Spencer, 2006). Not explicitly attending to race allows these ideologies to remain with superficial changes, no matter how well thought-out the policy.

This perpetuation of inequity and deficit views suggests that the work of mathematics education is to find ways to maintain concern with mathematics content, teaching, and learning, while attending to race and power to understand how the arrangement of mathematical opportunities inside and outside of school interact and contribute to current and historical inequities (Atweh, Forgasz, & Nebres, 2001; Gutstein, 2003; Martin, 2000; Oakes, 1990). In addition, mathematics educators need to develop meaningful interventions, inside and outside of school, to empower marginalized students with mathematics so that they can change the conditions that contribute to inequities (Abraham & Bibby, 1988; Anderson, 1990; Apple, 1992/1999; D'Ambrosio, 1990; de Arbreu, 1995; Frankenstein, 1990, 1994; Gutstein, 2002, 2003; Martin, 2000; Moses & Cobb, 2001a).

For example, instead of the state imposing a blanket eighth-grade Algebra mandate, what if the large urban district or a researcher posed the question: Why are more African American and Latino/a students not enrolled in college-preparatory mathematics courses? Educators could start to answer this question by interviewing African American and Latino/a students, asking them about their experiences in mathematics, the reasons for their course choices, and their future goals. Next, researchers may interview African American and Latino/a parents and ask about the goals that they have for their children. Finally, observing mathematics classrooms and talking with teachers about why they do not recommend more of their Black and Latino/a students for college-preparatory mathematics courses could inform the field about how to design interventions to change classroom practices to open opportunities for students of color. After this work, which is only the beginning, mathematics educators would have more knowledge to inform curricular, administrative, and policy decisions as to how mathematics instruction can better serve African American and Latino/a students and teachers of these students.

Such an approach demystifies the *all* and places a face on those students whom the district is currently not serving. It also gives credence to the lives and experiences of minoritized children and communities, and it forces schools and teachers to confront dynamics of power and race that exist within their school. This approach shares power with students, parents, and communities rather than subjecting them to the mandates of politicians and administrators. Instead, the current "Algebra for All" mandate reifies the stratified system of education existent in these schools. It misrepresents the reality to students and community members who believe that their children are engaged in high-quality Algebra coursework.

Standardized Labels: The Continued Process of Racialization

Many of the same issues previously discussed have manifested themselves in national and local movements towards greater standardization. Policies that establish accountability systems, high stakes standardized tests, and mandated curricula also work to reify labels and more importantly ideologies about children and communities of color.

The culture of standardization places students in groups based on judgments about their academic abilities. These judgments often fall into deficit modes of thinking and are solidified in the schooling institution through state-, district-, and school-sanctioned labels. In California, these labels evolved into terms like *far below basic*, *below basic*, *strategic*, or *intensive*. Grouping students by test performance sanctions these labels, making them more acceptable, and creates an institutional Discourse of deficit thinking tied to linking ability with individual standardized test scores (Perry et al., 2003; Rose, 1988). This deficit Discourse in schools, as before, is more often associated with students of color.

Labeling is one way of grouping by race while never making explicit that these labels construct how students of color are seen. Put another way, these labels become proxies for speaking about students of color without referring to race (Perry et al., 2003; Pollock, 2004; Rose, 1988). For example, whereas a teacher or administrator may be reticent to say that their African American boys have trouble sitting still in class, they might state that their "kinesthetic" students have trouble keeping still in class. Taken a step further, they might add that their kinesthetic students need "hands on" lessons to learn. *Kinesthetic*, unlike *African American*, is an ahistorical and seemingly neutral term. It does not carry with it centuries of oppression and denied rights. It therefore fits with a color-blind ideology (DeCuir & Dixson, 2004) and is an acceptable way of talking about students of color. Similar terms such as *Attention Deficit Disorder (ADD)*, *hyperactive*, *remedial*, and more recently *far below basic* and *math essentials* operate in the same manner. Such labels avoid and cover up the complex and real issues of racialized experience that we as educators need to begin to grapple with. By not documenting, understanding, or deconstructing these Discourses we do not produce work that adequately counters long standing racial and power differentials and the ways that they shape the mathematical experiences of minoritized students.

Summary

Our aim in this section on race and power was to illustrate how not using power and race as central

analytical tools in the work of mathematics education reproduces current inequitable practices. Although these cases are preliminary and the descriptions of them largely based on the experiences of the authors involved in this work, they help to illustrate how Discourses and schooling practices are taken for granted, normal, and neutral because they are part of the schooling institution. Better understanding and unpacking the social systems, policies, and narratives that structure classroom learning can help mathematics educators implement new reforms, navigate the political system, and develop policies that work for, rather than against, African Americans, Latinos, and the poor. This would require us to develop new understandings about how power employed through policy, states, districts, Discourses and Whiteness influences the learning that goes on in classrooms. It also means listening and taking up the concerns of those not empowered by the current system in researching mathematics education.

CONCLUSION

Over the last 10 years, there has been a growing concern within education and mathematics education that we need to examine what is happening for groups of students. Although there is recognition that teaching matters, research suggests that as we consider teaching, we cannot ignore cultural histories or essentialize them, and we cannot ignore that learning occurs within schools and communities that are shaped by cultural histories. Research is accumulating that demonstrates that the structures of schooling and society serve to both support and limit student opportunities. However, as we found throughout this chapter, we still know little about the details around how cultural histories and social structures shape and are shaped by communities and their histories. In mathematics education research, we know even less about these details.

The lack of available literature in mathematics education suggests that race, racism, and power remain undertheorized in the field. In other words, the literature does not sufficiently address how race interacts with the experiences of students of color or White students in mathematics education. The high value placed on mathematics education, the career opportunities open to those with strong mathematics backgrounds, and the qualifications for entrance into elite universities situate mathematics in a different way than many other fields. Therefore, it is possible that how racism and the normative ways of talking about students of color play out is different with re-

spect to mathematics from other disciplines. The ways that these normative Discourses that build themselves into the institution of mathematics education through course names such as Math Essentials and labels like *far below basic*, are one example of enacting power on students in systematic ways. The literature does little to make explicit how the schooling institution enacts power, in fairly implicit ways, on students of color. Hopefully, this chapter raises issues for mathematics education researchers to further explore how issues of race, racism, and power structure opportunities and experiences for all students so the field can find ways to counteract inequities.

Standardized tests serve as an example of how social structures and cultural histories shape access, opportunities, and experiences of groups of students. This review suggests that these tests often hold cultural, racial, and social-class biases, making them better assessors of students' cultural practices or their SES, revealing little about students' mathematical understanding. Moreover, standardized-test results have deep consequences for students of color, as the results are often used to inappropriately sort students. As long as differential achievement is associated predominantly with standardized-test scores, students of color will continually be harmed by an inequitable distribution of individual sanctions due to a limited understanding of achievement and neglect of the history of underserving them. Additionally, such sanctions place responsibility on the students for overcoming disparities in achievement, funding, and opportunity to learn that society-at-large and the institution of schools create and perpetuate.

The characterization of inequities in mathematics education synthesized throughout this chapter underscores the benefits of and necessity for (a) researching the concerns of those who have been disenfranchised, (b) questioning the privilege of the powerful, (c) recognizing classrooms as racialized spaces, even all-White classrooms, and (d) looking outside the field of mathematics education for theoretical perspectives and methodologies that can contribute to our understanding of culture, race, power, and mathematics teaching and learning.

Researching Culture, Race, and Power in Mathematics Education

The examination of race, culture, and power with respect to student achievement and learning in mathematics raises different questions for our current system of mathematics educators. Research questions are needed that can help guide studies of mathematics education in both untangling and challenging pro-

cesses that perpetuate current inequity and injustice in mathematics education and in society writ large. A few researchers have already proposed research questions that attempt to address issues of culture, race, and power in mathematics education (see, e.g., R. Gutierrez, 2002; Gutstein, 2006; Martin, in press). These questions stem from the perspectives that these researchers hold about the relations of race, culture, and power in mathematics, the nature of mathematics teaching and learning, as well as the role of mathematics education in society. These researchers share a concern with analyzing what counts as mathematics learning, in whose eyes, and how these culturally bound distinctions afford and constrain opportunities for students of color to have access to mathematical trajectories in school and beyond. The questions offered by Martin, Gutstein, and Gutierrez share a common theme in observing that mathematics education, and assumptions within it about mathematics teaching and learning, have historical, cultural, and political underpinnings, that have privileged White, middle- and upper-class students over students of color.

We have also embedded questions within the chapter in an effort to reveal subtle power relations, racialized experiences, and implicit ideologies and practices that constrain the development of a more equitable system of mathematics education. This is also an attempt to challenge the field to move beyond studies of mathematics teaching and learning that reduce culture to race, race to non-White students, and that inadvertently strengthen narratives and Discourses that marginalize and oppress. In proposing new areas of research, we do not seek to minimize the work that has been done to date to bring about greater equity in mathematics education. We also recognize the difficulty of doing such research in a contentious social and political climate, and simultaneously with the development of new methodological tools and frameworks.

However, different kinds of questions often require different methodologies. Asking questions about systematic inequities leads to methodologies that allow the researcher to look at multiple levels simultaneously. This means that mathematics education research should take a multifaceted approach, aimed at multiple levels from the classroom to broader social structures, within a variety of contexts both in and out of school, and at a broad span of relationships including researcher to study participants, teachers to schools, schools to districts, and districts to national policy. It is important, then, that researchers understand that policies *do* play out as well as *the ways in which* they play out at the classroom, school, district, and state levels.

Methodologies that capture the relationships between individuals, groups, classrooms, schools, communities, and social structures are needed. In other words, research that avoids looking at these elements in isolation has much more potential for informing the field and policy decisions about culture, race, and power in relation to teaching and learning mathematics. For instance, Spencer (2006) provides a multi-level framework for studying the mathematics achievement of African American students. In this framework, Black student participation and identity, the mathematics content present in classrooms, teacher's dispositions towards and explanations about the participation of their African American students, schools' responses to the needs of African American students, and the historical context within which the school is seated each take a central role. The potential of such a framework is in its ability to address the complex and relational nature of a phenomenon such as "student achievement." For example, such an analysis could help answer how the historical significance assigned to a particular group of students impacts the relation of a school to that group as well as the relation of that group to the school. More central to our work, such an analysis may help us understand how a teacher's perceptions of students as doers of mathematics impacts how that group views themselves in the enterprise of learning mathematics.

Both qualitative and quantitative research have strengths and limitations. Work using narratives, ethnographies, and historical analyses allow research to speak to multiple levels of practice in order to see nuanced details. Although these methods are sometimes discredited, not counted as research, or not given the same respect as other forms, their multilevel nature situates them as particularly powerful in understanding the details of relationships discussed in this chapter. Standards of quality and thoroughness can be achieved just as in quantitative analyses. Similarly, quantitative methods such as multilevel modeling can uncover systematic issues of inequity. Although multilevel modeling such as Hierarchical Linear Modeling (HLM) is already respected as a form of research, when using such techniques we must be just as thoughtful and careful that we are actually measuring what we intend. The measures must be sensitive enough to allow for the subtle ways that culture, race, and power can influence teaching and learning in mathematics classrooms. When this is the case, multilevel modeling allows researchers to understand complex causal relationships that can uncover power dynamics within social structures that shape the experiences of groups. This quantitative work allows researchers to understand complex relationships on much broader scales

than by using qualitative methods. We recognize that quality research of this sort is difficult and challenging to do. However, if we are to take equity seriously, we cannot avoid engaging with these issues in new ways.

In addition to different kinds of questions and methodologies, this work will push the field to develop new ways of understanding results. We attempt to provide some different frameworks for understanding the relationships between culture, race, and learning in this chapter. This is not an argument for a particular framework; rather, multiple lenses and theoretical perspectives will be needed to understand our work in relation to social structures and cultural and racial histories. If mathematics education researchers were all to challenge our assumptions about why we have the results we do, it might open up new insights into the complex relationships discussed in this chapter. New frameworks for understanding the interactions between culture, race, and power would shape how we discuss and understand the work of mathematics education.

The seemingly color-blind, neutral policies in mathematics education and in education in general, and discourses about students and student achievement, serve to privilege some students and cultural practices over others. Our discussions in this chapter reveal that not paying explicit attention to the ways race and power manifest themselves in such policies can lead to a masking of the racialization of mathematics education and of the damaging effects of recurring ideologies for students of color. These challenges considered, if the nation is to take seriously current inequities in math education, then it is unacceptable for the field of mathematics education to ignore something that so obviously defines the relationships and realities of many students in the United States.

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THE ROLE OF CULTURE IN TEACHING AND LEARNING MATHEMATICS

Norma Presmeg

ILLINOIS STATE UNIVERSITY

Mathematics education has experienced a major revolution in perceptions (cf. Kuhn, 1970) comparable to the Copernican revolution that no longer placed the earth at the center of the universe. This change has implicated beliefs about *the role of culture* in the historical development of mathematics (Eves, 1990), in the practices of mathematicians (Civil, 2002; Sfard, 1997), in its political aspects (Powell & Frankenstein, 1997), and hence necessarily in its teaching and learning (Bishop, 1988a; Bishop & Abreu, 1991; Bishop & Pompeu, 1991; Nickson, 1992). The change has also influenced methodologies that are used in mathematics education research (Pinxten, 1994). Researchers now increasingly concede that mathematics, long considered value- and culture-free, is indeed a cultural product, and hence that the role of culture—with all its complexities and contestations—is an important aspect of mathematics education.

Diverse aspects are implicated in a cultural formulation of mathematics teaching and learning. Other chapters in this volume focus more centrally on some of these aspects, for instance the following:

- the role of language and communication in mathematics education;
- equity, diversity, and learning in multicultural mathematics classrooms.

These topics will thus not be a focus in this chapter, although some issues from these fields will enter spontaneously. Topics that are central in this chapter are those arising from and extending the notions of

ethnomathematics and everyday cognition (Nunes, 1992, 1993). Various broad theoretical fields are relevant in addressing these topics. Some of the theoretical notions that are apposite are rooted in—but not confined to—situated cognition (Kirshner & Whitson, 1997; Lave & Wenger, 1991; Watson, 1998), cultural models (Holland & Quinn, 1987), notions of cultural capital (Bourdieu, 1995), didactical phenomenology (Freudenthal, 1973, 1983), and Peircean semiotics (Peirce, 1992, 1998). From this partial list, the breadth of this developing field may be recognized. This chapter does not attempt to treat the general theories in detail: The interested reader is referred to the original authors. Instead, from these fields this chapter highlights some key notions that have explanatory power or usefulness in the central focus, which is *the role of culture in learning and teaching mathematics*. The seminal work of Ascher (1991, 2002), Bishop (1988a, 1988b), D'Ambrosio (1985, 1990) and Gerdes (1986, 1988a, 1988b, 1998) on ethnomathematics is still centrally relevant and thus is treated in some detail in later sections.

In the last decade, the field of research into the role of culture in mathematics education has evolved from “ethnomathematics and everyday cognition” (Nunes, 1992), although both ethnomathematics and everyday cognition are still important topics of investigation. The developments have rather consisted in a broadening of the field, clarification and evolution of definitions, recognition of the complexity of the constructs and issues, and inclusion of social,

critical, and political dimensions as well as those from cultural psychology involving valorization, identity, and agency (Abreu, Bishop, & Presmeg, 2002). This chapter in its scope cannot do full justice to political and critical views of mathematics education (see Mellin-Olsen, 1987; Skovsmose, 1994; Vithal, 2003, for a full treatment), but some of the *landscapes* from these fields—as Vithal calls them—are used in this chapter to deepen and problematize aspects of the treatment of culture in mathematics education.

This chapter has four sections. The first addresses an introduction to issues and definitions of key notions involving culture in mathematics education. The organization of the second and third sections uses the research framework of Brown and Dowling (1998). In this framework, resonating theoretical and empirical fields surround and enclose the central research topic, and the description involves layers of increasing specificity as it zooms in on details of the problematic and problems of the research issues, the empirical settings, and the results of the studies, only to zoom out again at the end in order to survey the issues in a broader field, informed by the results of the research studies examined, in order to see where further research on culture might be headed. Thus section 2 addresses theoretical fields that incorporate culture specifically in mathematics education; section 3 addresses salient empirical fields, settings, and some details of results of research on culture in mathematics education, and their implications for the teaching and learning of mathematics. Issues relating to technology and the culture of mathematics and its teaching and learning are included in this section. Using a broader perspective, the fourth (final) section collects and elaborates suggestions for possible directions for future research on culture in mathematics education.

DEFINITIONS AND SIGNIFICANCE OF CULTURE IN MATHEMATICS EDUCATION

Why is it important to address definitions carefully in considering the role of culture in teaching and learning mathematics? As in other areas of reported research in mathematics education, different authors use terminology in different ways. Particularly notions such as *culture*, *ethnomathematics*, and *everyday mathematics* have been controversial; they have been contested and given varied and sometimes competing interpretations in the literature. Thus these constructs must be problematized, not only for the sake of clarification, but more importantly for the roles they have played, and for their further potential to be major

focal points in mathematics education research and practices. Further, especially in attempts to bridge the gap between the formal mathematics taught in classrooms and that used out-of-school in various cultural practices, *what counts as mathematics* assumes central importance (Civil, 1995, 2002; Presmeg, 1998a). Thus ontological aspects of the nature of mathematics itself must also be addressed.

Culture

In 1988, when Bishop published his book, *Mathematical Enculturation: A Cultural Perspective on Mathematics Education* (1988a) and an article that summarized these ideas in *Educational Studies in Mathematics* (1988b, now reprinted as a classic in Carpenter, Dossey, & Koehler, 2004), the prevailing view of mathematics was that it was the one subject in the school curriculum that was value- and culture-free, notwithstanding a few research studies that suggested the contrary (e.g., Gay & Cole, 1967; Zaslavsky, 1973). Along with those of a few other authors (notably, D'Ambrosio), Bishop's ideas have been seminal in the recognition that culture plays a pivotal role in the teaching and learning of mathematics, and his insights are introduced repeatedly in this chapter.

Whole books have been written about definitions of *culture* (Kroeber & Kluckhohn, 1952). Grappling with the ubiquity yet elusiveness of culture, Lerman (1994) confronted the need for a definition but could not find one that was entirely satisfactory. Yet, as he pointed out, culture is “ordinary. It is something that we all possess and that possesses us” (p. 1). Bishop favored the following definition of culture in analyzing its role in mathematics education:

Culture consists of a complex of shared understandings which serves as a medium through which individual human minds interact in communication with one another. (1988a, p. 5, citing Stenhouse, 1967, p. 16)

This definition highlights the communicative function of culture that is particularly relevant in teaching and learning. However, it does not focus on the continuous renewal of culture, the dynamic aspect that results in cultural change over time. This dynamic aspect caused Taylor (1996) to choose the “potentially transformative view” (p. 151) of cultural anthropologist Clifford Geertz (1973), for whom culture consists of “webs of significance” (p. 5) that we ourselves have spun.

This potentially transformative view assumes particular importance in the light of the necessity for negotiating social norms and sociomathematical norms in mathematics classrooms (Cobb & Yackel, 1995).

The culture of the mathematics classroom, which was brought to our attention as being significant (Nickson, 1992), is not monolithic or static but continuously evolving, and different in different classrooms as these norms become negotiated. The mathematics classroom itself is one arena in which culture is contested, negotiated, and manifested (Vithal, 2003), but there are various levels of scale. Bishop's (1988a) view of mathematics education as a social process resonates with a transformative, dynamic notion of culture. He suggested that five significant levels of scale are involved in the social aspects of mathematics education. These are the cultural, the societal, the institutional, the pedagogical, and the individual aspects (1988a, p. 14). Culture here is viewed as an all-encompassing umbrella construct that enters into all the activities of humans in their communicative and social enterprises. In addition to a view of culture in this macroscopic aspect, as in these levels of scale, culture as webs of significance may be central also in the societal, institutional, and pedagogical aspects of mathematics education considered as a social process. Thus researchers may speak of the culture of a society, of a school, or of an actual classroom. Culture in all of these levels of scale impinges on the mathematical learning of individual students.

Notwithstanding these general definitions of culture, the word with its various characterizations does not have meaning in itself. Vithal and Skovsmose (1997) illustrated this point starkly by pointing out that interpretations of *culture* (and by implication also anthropology) were used in South Africa to justify the practices of apartheid. They extended the negative connotations still attached to this word in that context to suggest that ethnomathematics is also suspect (as suggested in the title of their article: "The End of Innocence: A Critique of 'Ethnomathematics'"). Some aspects of their critique are taken up in later sections.

How are *culture* and *society* interrelated? The words are not interchangeable (Lerman, 1994), although connections exist between these constructs:

One would perhaps think of gender stereotypes as cultural, but of 'gender' as socially constructed. One would talk of the culture of the community of mathematicians, treating it as monolithic for a moment, but one would also talk, for example, of the social outcomes of being a member of that group. (p. 2)

A. J. Bishop stated succinctly, on several occasions (personal communication, e.g., July, 1985), that *society involves various groups of people, and culture is the glue that binds them together*. This informal characterization resonates with Stenhouse's definition of *culture* as a

complex of shared understandings, and also with that of Geertz, as webs of significance that we ourselves have spun. Considering the cultures of mathematics classrooms, Nickson (1994) wrote of the "invisible and apparently shared meanings that teachers and pupils bring to the mathematics classroom and that govern their interaction in it" (p. 8). Values, beliefs, and meanings are implicated in these "shared invisibles" (p. 18) in the classroom. Nickson saw socialization as a universal process, and culture as the *content* of the socialization process, which differs from one society to another, and indeed, from one classroom to another.

Part of culture as webs of significance, taken at various levels, are the prevailing notions of what counts as mathematics.

Mathematics

As Nickson (1994) pointed out, "one of the major shifts in thinking in relation to the teaching and learning of mathematics in recent years has been with respect to the adoption of differing views of the nature of mathematics as a discipline" (p. 10). Nickson characterized this cultural shift as moving from a *formalist* tradition in which mathematics is absolute—consisting of "immutable truths and unquestionable certainty" (p. 11) without a human face, to one of *growth and change*, under persuasive influences such as Lakatos's (1976) argument that "objective knowledge" is subject to proofs and refutations and thus that mathematical knowledge has a strong social component. That this shift is complex and that both views of mathematics are held simultaneously by many mathematicians was argued by Davis and Hersh (1981). The formalist and socially mediated views of mathematics resonate with the two categories of absolutist and fallibilist conceptions discussed by Ernest (1991). Contributing the notion of mathematics as problem solving, the Platonist, the problem-solving, and the fallibilist conceptions are categories reminiscent of the teachers' conceptions of mathematics that Alba Thompson, already in 1984, gave evidence were related to instructional practices in mathematics classrooms.

Both Civil (1990, 1995, 2002; Civil & Andrade, 2002) in her "Funds of Knowledge" project and in her later research with colleagues into ways of linking home and school mathematical practices, and Presmeg (1998a, 1998b, 2002b) in her use of ethnomathematics in teacher education and research into semiotic chaining as a means of building bridges between cultural practices and the teaching and learning of mathematics in school, described the necessity of broadening conceptions of the nature of mathematics in these endeavors. Without such broadened

definitions, high-school and university students alike are naturally inclined to characterize mathematics according to what they have experienced in learning *institutional* mathematics—more often than not as “a bunch of numbers” (Presmeg, 2002b). On the one hand, such limited views of the nature of mathematics inhibit the recognition of mathematical ideas in out-of-school practices. On the other hand, if definitions of what counts as mathematics are too broad, then the “everything is mathematics” notion may trivialize mathematics itself, rendering the definition useless. In examining the mathematical practices of a group of carpenters in Cape Town, South Africa, Millroy (1992) expressed this tension well, as follows:

[It] became clear to me that in order to proceed with the exploration of the mathematics of an unfamiliar culture, I would have to navigate a passage between two dangerous areas. The foundering point on the left represents the overwhelming notion that ‘everything is mathematics’ (like being swept away by a tidal wave!) while the foundering point on the right represents the constricting notion that ‘formal academic mathematics is the only valid representation of people’s mathematical ideas’ (like being stranded on a desert island!). Part of the way in which to ensure a safe passage seemed to be to openly acknowledge that when I examined the mathematizing engaged in by the carpenters there would be examples of mathematical ideas and practices that I would recognize and that I would be able to describe in terms of the vocabulary of conventional Western mathematics. However, it was likely that there would also be mathematics that I could not recognize and for which I would have no familiar descriptive words. (pp. 11–13)

Some definitions of mathematics that achieve a balance between these two extremes are as follows. *Mathematics* is “the language and science of patterns” (Steen, 1990, p. iii). Steen’s definition has been taken up widely in reform literature in the USA (National Council of Teachers of Mathematics [NCTM], 1989, 2000). Opening the gate to recognizing a human origin of mathematics, Saunders MacLane called mathematics “the study of formal abstract structures arising from human experience” (as cited in Lakoff, 1987, p. 361). According to Ada Lovelace, mathematics is the systematization of relationships (as described by Noss, 1997). All of these definitions strike some sort of balance between the human face of mathematics and its formal aspects. Going beyond Steen’s well-known pattern definition in the direction of stressing abstraction, in a critique of ethnomathematics, Thomas (1996) defined mathematics as “the science of detachable relational insights” (p. 17). He suggested a useful distinction between *real mathematics* (as

characterized in his definition), and *proto-mathematics* (the category in which he placed ethnomathematics). In the next section, Barton’s (1996) characterization of ethnomathematics, which resolves many of these issues and clarifies this dualism, is presented along with some evolving definitions of ethnomathematics. (For details, the reader should consult Barton’s original article.)

Ethnomathematics

What is *ethnomathematics*? In his illuminating article, Barton (1996) wrote as follows:

In the last decade, there has been a growing literature dealing with the relationship between culture and mathematics, and describing examples of mathematics in cultural contexts. What is not so well-recognized is the level at which contradictions exist within this literature: contradictions about the meaning of the term ‘ethnomathematics’ in particular, and also about its relationship to mathematics as an international discipline. (p. 201)

Barton pointed out that difficulties in defining ethnomathematics relate to three categories: epistemological confusion, “problems with the meanings of words used to explain ideas about culture and mathematics” (p. 201); philosophical confusion, the extent to which mathematics is regarded as universal; and confusion about the nature of mathematics. The nature of mathematics is part of its ontology, and because both ontology and epistemology are branches of philosophy, all of these categories may be regarded as philosophical difficulties. The strength of Barton’s resolution of the difficulties lies in his creation of a preliminary framework (he admitted that it might need revision) whereby the differing views can be seen in relation to each other. His triadic framework is an “Intentional Map” (p. 204) with the three broad headings of *mathematics*, *mathematics education*, and *society* (cf. the whole day of sessions dedicated to these broad areas at the 6th International Congress on Mathematical Education held in Budapest, Hungary, in 1988). The seminal writers whose definitions of ethnomathematics he considered in detail and placed in relation to this framework were Ubiratan D’Ambrosio in Brazil, Paulus Gerdes in Mozambique, and Marcia Ascher in the USA.

As Barton (1996) pointed out, D’Ambrosio’s prolific writings on the subject of ethnomathematics have influenced the majority of writers in this area. Thus on the Intentional Map, although D’Ambrosio’s work (starting with his 1984 publication) falls predominantly in the socio-anthropological dimension between *society* and *mathematics*, some aspects of his concerns can

be found in all of the dimensions. In his later work, he increasingly used his model to analyze “the way in which mathematical knowledge is colonized and how it rationalizes social divisions within societies and between societies” (Barton, 1996, p. 205). In his early writing, D’Ambrosio (1984) defined ethnomathematics as the way different cultural groups mathematize—count, measure, relate, classify, and infer. His definition evolved over the years, to include a changing form of knowledge manifest in practices that change over time. In 1985, he defined ethnomathematics as “the mathematics which is practiced among identifiable cultural groups” (p. 18). Later, in 1987, his definition of ethnomathematics was “the codification which allows a cultural group to describe, manage, and understand reality” (Barton, 1996, p. 207).

D’Ambrosio’s (1991) well-known etymological definition of ethnomathematics is given in full in the following passage.

The main ideas focus on the concept of *ethnomathematics* in the sense that follows. Let me clarify at the beginning that this term comes from an etymological abuse. I use *mathema(ta)* as the action of explaining and understanding in order to transcend and of managing and coping with reality in order to survive. Man has developed throughout each one’s own life history and throughout the history of mankind *technē*s (or *tics*) of *mathema* in very different and diversified cultural environments, i.e., in the diverse *ethno*’s. So, in order to satisfy the drive towards survival and transcendence in diverse cultural environments, man has developed and continuously develops, in every new experience, **ethno-mathema-tics**. These are *communicated* vertically and horizontally in time, respectively throughout history and through conviviality and education, relying on memory and on sharing experiences and knowledges. For the reasons of being more or less effective, more or less powerful and sometimes even for political reasons, some of these different *tics* have lasted and spread (ex.: counting, measuring) while others have disappeared or been confined to restricted groups. This synthesizes my approach to the history of ideas. (p. 3)

As in some of his other writings (1985, 1987, 1990), D’Ambrosio is in this definition characterizing ethnomathematics as a dynamic, evolving system of knowledge—the “process of knowledge-making” (Barton, 1996, p. 208), as well as a research program that encompasses the history of mathematics.

Returning to Barton’s Intentional Map, the work of Paulus Gerdes is “practical, and politically explicit,” concentrated in the *mathematics education* area of the Map (Barton, 1996, p. 205). Gerdes’s definition of ethnomathematics evolved from the mathematics implic-

it or “frozen” in the cultural practices of Southern Africa (1986), to that of a mathematical movement that involves research and anthropological reconstruction (1994). The work of mathematician Marcia Ascher (1991, 1995, 2002), while overlapping with that of Gerdes to some extent, falls closer to the *mathematics* area on the Map, concerned as it is with cultural mathematics. Her definition is that ethnomathematics is “the study and presentation of the mathematical ideas of traditional peoples” (1991, p. 188). When Ascher (1991) worked out the kinship relations of the Warlpiri, say, in mathematical terms, she acknowledged that she was using her familiar “Western” mathematics. In that sense her ethnomathematics is subjective: The Warlpiri would be unlikely to view their kinship system through her lenses. Referring to mathematics and ethnomathematics, she stated, “They are both important, but they are different. And they are linked” (Ascher & D’Ambrosio, 1994, p. 38). In this view, there is no need to view ethnomathematics as “protomathematics” (Thomas, 1996), because it exists in its own right.

Finally, Barton (1996) found a useful metaphor to sum up the similarities and differences between the views of ethnomathematics held by these three proponents: “For D’Ambrosio it is a window on knowledge itself; for Gerdes it is a cultural window on mathematics; and for Ascher it is the mathematical window on other cultures” (p. 213). These three windows are distinguished by the standpoint of the viewer, and by what is being viewed, in each case. Although not eliminating the duality of ethnomathematics as opposed to mathematics (of mathematicians), these three distinct windows represent approaches each of which has something to offer. Taken together, they contribute a broadened lens on the role of culture in teaching and learning mathematics.

Several other writers in the field of ethnomathematics have acknowledged the need and attempted to define ethnomathematics. Scott (1985) regarded ethnomathematics as lying at the confluence of mathematics and cultural anthropology, “mathematics in the environment or community,” or “the way that specific cultural groups go about the tasks of classifying, ordering, counting, and measuring” (p. 2). Several definitions of ethnomathematics highlight some of the “environmental activities” that Bishop (1988a) viewed as universal, and also “necessary and sufficient for the development of mathematical knowledge” (p. 182), namely counting, locating, measuring, designing, playing, and explaining. One further definition brings back the problem, hinted at in the foregoing account, of *ownership* of ethnomathematics. Whose mathematics is it?

Ethnomathematics refers to any form of cultural knowledge or social activity characteristic of a social and/or cultural group, that can be recognized by other groups such as 'Western' anthropologists, but not necessarily by the group of origin, as mathematical knowledge or mathematical activity. (Pompeu, 1994, p. 3)

This definition resonates with Ascher's, without fully solving the problem of ownership. The same problem appears in definitions of everyday mathematics, considered next.

Everyday Mathematics

Following on from the description of *everyday cognition* (Nunes, 1992, 1993) and important early studies that examined the use of mathematics in various practices, such as mathematical cognition of candy sellers in Brazil (Carraher, Carraher, & Schliemann, 1985; Saxe, 1991), constructs and issues are still being questioned. In this area, too, clarification of definitions is being sought, along with deeper consideration of the scope of the issues and their potential and significance for the classroom learning of mathematics.

Brenner and Moschkovich (2002) raised the following questions.

What do we mean by *everyday mathematics*? How is everyday mathematics related to *academic mathematics*? What particular everyday practices are being brought into mathematics classrooms? What impact do different everyday practices actually have in classroom practices?" (p. v)

In a similar vein, and with the benefit of two decades of research experience in this area, Carraher and Schliemann (2002) examined how their perceptions had evolved, as they explored the topic of their chapter, "Is Everyday Mathematics Truly Relevant to Mathematics Education?"

All the authors of chapters in the monograph edited by Brenner and Moschkovich (2002) in one way or another set out to explore these and related questions. Several of these authors pointed out that it is problematic to oppose everyday and academic mathematics, for several reasons. For one thing, for mathematicians academic mathematics *is* an everyday practice (Civil, 2002; Moschkovich, 2002a). For another, studies of everyday mathematical practices in workplaces reveal a complex interplay with socio-cultural and technological issues (FitzSimons, 2002). In the automobile production industry, variations in the mathematical cognition required of workers have less to do with the job itself than with the decisions of management concerning production procedures

and organization of the workplace. Highly skilled machinists display spatial and geometric knowledge that goes beyond what is commonly taught in school: In contrast, assembly-line workers and some machine operators find few if any mathematical demands in their work, which is deliberately stripped of the need for decisions involving knowledge of mathematics beyond elementary counting (Smith, 2002). The complex relationship between use of technology and the demand for mathematical thinking in the workplace is a theme that is explored in a later section of this chapter.

Another aspect that is again apparent in all the chapters of Brenner and Moschkovich's monograph is the importance of perceptions and beliefs about the nature of mathematics, both in the microculture of classroom practices (Brenner, 2002; Masingila, 2002) and in the broader endeavor to bridge the gap between mathematical thinking in and out of school (Arcavi, 2002; Civil, 2002; Moschkovich, 2002a). An essential element in all of these studies is the concern to connect knowledge of mathematics in and out of school. (This issue is revisited later in this chapter.) Because of the difficulties surrounding the construct *everyday mathematics* the terminology that will be adopted in this chapter follows Masingila (2002), who referred to in-school and out-of-school mathematics practices (p. 38).

The developments described in this section parallel the genesis of the movement away from purely psychological cognitive and behavioral frameworks for research in mathematics education, towards cultural frameworks that embrace sociology, anthropology, and related fields, including political and critical perspectives. The following section introduces some relevant theoretical issues and lenses that have been used to examine some of these developments.

THEORETICAL FIELDS THAT INCORPORATE CULTURE IN MATHEMATICS EDUCATION

The notion of theoretical and empirical fields is drawn from Brown and Dowling (1998) and provides a useful framework for characterizing components of research. This section addresses some theoretical fields pertinent to culture in the teaching and learning of mathematics. Their instantiation in empirical studies is described in the next section. The reader is reminded again to consult the original authors for a full treatment of theoretical fields that are introduced in this section, which has as its purpose a wide but by no means exhaustive view of the scope of theories that are available for work in this area.

As suggested in Barton's (1996) sense-making article introduced in the previous section, in the last 2 decades there has already been considerable movement in theoretical fields regarding the interplay of culture and mathematics. One such movement is discernible in the definitions of ethnomathematics given by D'Ambrosio, Gerdes, and Ascher, as their theoretical formulations moved from more static definitions of ethnomathematics as the mathematics of different cultural groups, to characterizations of this field as an anthropological research program that embraces not only the history of mathematical ideas of marginalized populations, but the history of mathematical knowledge itself (see previous section). D'Ambrosio (2000, p. 83) called this enterprise *historiography*.

Historiography

Moving beyond earlier theoretical formulations of ethnomathematics, its importance as a catalyst for further theoretical developments has been noted (Barton, 1996). D'Ambrosio played a large role and served as advisory editor in the enterprise that resulted in Helaine Selin's (2000) edited book, *Mathematics Across Cultures: The History of Non-Western Mathematics*. The chapters in this book are global in scope and record the mathematical thinking of cultures ranging from those of Iraq, Egypt, and other predominantly Islamic countries; through the Hebrew mathematical tradition; to that of the Incas, the Sioux of North America, Pacific cultures, Australian Aborigines, mathematical traditions of Central and Southern Africa; and those of Asia as represented by India, China, Japan, and Korea. As can be gleaned from the scope of this work, D'Ambrosio's original concern to valorize the mathematics of colonized and marginalized people (cf. Paolo Freire's *Pedagogy of the Oppressed* in 1970/1997) has broadened to encompass a movement that is both archeological and historical in nature, based on the theoretical field of "historiography" and visions of world knowledge through the "sociology of mathematics" (D'Ambrosio, 2000, pp. 85–87).

Although these antedated D'Ambrosio's program, earlier studies such as Claudia Zaslavsky's (1973) report on the counting systems of Africa and Glendon Lean's (1986) categorization of those of Papua New Guinea (see also Lancy, 1983), could also be thought of as historiography, as could anthropological research such as that of Pinxten, van Dooren, and Harvey (1983) who documented Navajo conceptions of space. Also in the cultural anthropology tradition, Crump's (1994) research on the anthropology of numbers is another fascinating example of historiography. More recent studies such as some of those

collected as *Ethnomathematics* in the book by Powell and Frankenstein (1997) and the work of Marcia Ascher (1991, 1995, 2002) also fall into this category. Many of the cultural anthropological studies of various mathematical aspects of African practices, such as work on African fractals (Eglash, 1999); *lusona* of Africa (Gerdes, 1997); and women, art, and geometry of Africa (Gerdes, 1998), may also be regarded as historiography. As part of ethnomathematics conceived as a research program, this ambitious undertaking of historiography is designed to address lacunae in the literature on the history of mathematical thought through the ages. Much of the work of members of the International Study Group on Ethnomathematics (founded in 1985), and of the North American Study Group on Ethnomathematics (founded in 2003)—including research by Lawrence Shirley, Daniel Orey, and many others—could be placed in the category of historiography (see the list of some of the available web sites following the references).

Because historiography addresses some of the world's mathematical systems that have been ignored or undervalued in mathematics classrooms, it reminds us that there is *cultural capital* involved in the power relations associated with access in school mathematics. Some relevant theories are outlined in the following paragraphs.

Cultural Capital and Habitus

The usefulness of the theoretical field outlined by Bourdieu (1995) for issues of culture in the learning of mathematics has been indicated in research on learner's transitions between mathematical contexts (Presmeg, 2002a). Resonating with D'Ambrosio's original issues of concern (although D'Ambrosio did not use this framework), Bourdieu's work belongs in the field of sociology. The relevance of this work consists in "the innumerable and subtle strategies by which words can be used as instruments of coercion and constraint, as tools of intimidation and abuse, as signs of politeness, condescension, and contempt" (Bourdieu, 1995, p. 1, editor's introduction). This theoretical field serves as a useful lens in examining empirical issues related to the social inequalities and dilemmas faced by mathematics learners as they move between different cultural contexts, for example in the transitions experienced by immigrant children learning mathematics in new cultural settings (Presmeg, 2002a). This field embraces Bourdieu's notions of *cultural capital*, *linguistic capital*, and *habitus*. Bourdieu (1995) used the ancient Aristotelian term *habitus* to refer to "a set of dispositions which incline agents to act and react in certain ways" (p.12). Such dispositions are part

of culture viewed as a set of shared understandings. Various forms of capital are *economic capital* (material wealth), *cultural capital* (knowledge, skills, and other cultural acquisitions, as exemplified by educational or technical qualifications), and *symbolic capital* (accumulated prestige or honor) (p. 14). *Linguistic capital* is not only the capacity to produce expressions that are appropriate in a certain social context, but it is also the expression of the “correct” accent, grammar, and vocabulary. The *symbolic power* associated with possession of cultural, symbolic, and linguistic capital has a counterpart in the *symbolic violence* experienced by individuals whose cultural capital is devalued. Symbolic violence is a sociological construct. In that capacity it is a powerful lens with which to examine actions of a group and ways in which certain types of knowledge are included or excluded in what the group counts as knowledge (for examples embracing the learning of mathematics, see the chapters in Abreu, Bishop, & Presmeg, 2002).

Borderland Discourses

The notion of symbolic violence leaves a possible theoretical gap relating to the ways in which individuals choose to construct, or choose *not* to construct, particular knowledge—mathematical or otherwise. Bishop (2002a) gave examples both of immigrant learners of mathematics in Australia who chose not to accept the view of themselves as constructed by their peers or their teacher—and of others who chose to accept these constructions. One student “shouted back” when peers in the mathematics class shouted derogatory names.

Discourse is a construct that is wider than mere use of language in conversation (Philips, 1993; Wood, 1998): It embraces all the aspects of social interaction that come into play when human beings interact with one another (Dörfler, 2000; Sfard, 2000). The notion of *Discourses* formulated by Gee (1992, his capitalization), and in particular his extension of the construct to *borderland Discourses*, those “community-based secondary Discourses” situated in the “borderland” between home and school knowledge (p. 146), are in line with Bourdieu’s ideas of habitus and symbolic violence. Borderland Discourses take place in the borderlands between primary (e.g., home) and secondary (e.g., school) cultures of the diverse participants in social interactions. In situations where the secondary culture (e.g., that of the school) is conceived as threatening because of the possibility of symbolic violence there, the borderland may be a place of solidarity with others who may share a certain habitus. These ideas go some way towards closing the theoretical gap

in Bourdieu’s characterization of symbolic violence by raising some issues of individuals’ choices, because individuals choose the extent to which they will participate in various forms of these Discourses (see also Bishop’s, 2002b, use of Gee’s constructs).

Gee’s work was in the context of second language learning but is also useful in the analysis of meanings given to various experiences by mathematics learners in cultural transition situations. Bishop (2002a) used this theoretical field in moving from the notion of cultural conflict to that of cultural mediation, in analyzing these experiences of learners of mathematics. If one considers the primary Discourse of school mathematics learners to be the home-based practices and conversations that contributed to their socialization and enculturation (forming their habitus) in their early years, and continuing to a greater or lesser extent in their present home experiences, then the secondary Discourse, for the purpose of learning mathematics, could be designated as the formal mathematical Discourse of the established discipline of mathematics. The teacher is more familiar with this Discourse than are the students and thus has the responsibility of introducing students to this secondary Discourse. As students become familiar with this field, their language and practices may approximate more closely those of the teacher. But in this transition the borderland Discourse of interactive classroom practices provides an important mediating space.

The enculturating role of the mathematics teacher was suggested in the foregoing account. However, as Bishop (2002b) pointed out, the learning of school mathematics is frequently more of an acculturation experience than an enculturation. The difference between these anthropological terms is as follows. *Enculturation* is the induction, by the cultural group, of young people into their own culture. In contrast, *acculturation* is “the modification of one’s culture through continuous contact with another” (Wolcott, 1974, p. 136, as quoted in Bishop, 2002b, p. 193). The degree to which the culture of mathematics, as portrayed in the mathematics classroom, is viewed as their own or as a foreign culture by learners would determine whether their experiences there would be of enculturation or acculturation.

Cultural Models

Allied with Gee’s (1992) notion of different Discourses is his construct of *cultural models*. This construct, defined as “‘first thoughts’ or taken for granted assumptions about what is ‘typical’ or ‘normal’” (Gee, 1999, p. 60, quoted in Setati, 2003a, p. 153), was used by Setati (2003a) as an illuminating theoret-

ical lens in her research on language use in South African multilingual mathematics classrooms. Already in 1987, D'Andrade defined a cultural model—which he also called a *folk model*—as “a cognitive schema that is intersubjectively shared by a social group,” and he elaborated, “One result of intersubjective sharing is that interpretations made about the world on the basis of the folk model are treated as if they were obvious facts about the world” (p. 112). The transparency of cultural models may help to explain why mathematics was for so long considered to be value- and culture-free. The well-known creativity principle of making the familiar strange and the strange familiar (e.g., De Bono, 1975) is necessary for participants to become aware of their implicit cultural beliefs and values, which is why the anthropologist is in a position to identify the beliefs that are invisible to many who are within the culture.

In the context of mathematics learning in multilingual classrooms, Adler (1998, 2001) pointed out aspects of the use of language as a cultural resource that relate to the transparency of cultural models. Particularly in classrooms where the language of instruction is an *additional language*—not their first language—for many of the learners (Adler, 2001), teachers must at times focus on the language itself, in which case the artifact of language no longer serves as a “window” of transparent glass through which to view the mathematical ideas (Lave & Wenger, 1991). In this case the language used is no longer *invisible*, and the focus on the language itself may detract from the conceptual learning of the mathematical content (Adler, 2001).

Valorization in Mathematical Practices

If transparency of culture necessitates making the familiar strange before those sharing that culture become aware of the lenses through which they are viewing their world, then this principle points to a reason for the neglect of issues of valorization in the mathematics education research community until Abreu's (1993, 1995) research brought the topic to the fore. The *value* of formal mathematics as an academic subject was for so long taken for granted that it became a *given* notion that was not culturally questioned. Especially in its role as a gatekeeper to higher education, this status in education is likely to continue. But if ethnomathematics as a research program is to have a legitimate place in broadening notions both of what counts as mathematics and of which people have originated these forms of knowledge, then issues of valorization assume paramount importance.

Working from the theoretical fields of cultural psychology and sociocultural theory, Abreu and col-

leagues investigated the effects of valorization of various mathematical practices on Portuguese children (Abreu, Bishop, & Pompeu, 1997), Brazilian children (Abreu, 1993, 1995), and British children from Anglo and Asian backgrounds (Abreu, Cline, & Shamsi, 2002). As confirmed also in the research of Gorgorio, Planas, and Vilella (2002), many of these children denied the existence of, or devalued, mathematics as used in practices that they associated with their home- or out-of-school settings.

Valorization, the social process of assigning more value to certain practices than to others, is closely allied to Wertsch's (1998) notion of *privileging*, defined as “the fact that one mediational means, such as social language, is viewed as being more appropriate or efficacious than another in a particular social setting” (Wertsch, 1998, p. 64, in Abreu, 2002, p. 183). Abreu (2002) elaborated as follows.

From this perspective, cultural practices become associated with particular social groups, which occupy certain positions in the structure of society. Groups can be seen as mainstream or as marginalised. In a similar vein individuals who participate in the practices will be given, or come to construct, identities associated with certain positions in these groups. The social representation enables the individual and social group to have access to a ‘social code’ that establishes relations between practices and social identities. (p. 184)

Thus Abreu argued strongly that in the learning of mathematics, valorization operates not only on the societal plane but also on the personal plane, because it impacts the construction of social identities. At this psychological level, the construction of mathematical knowledge may be subordinated to the construction of social identity by the individual learner in cases of cultural conflict, as suggested by Presmeg (2002a).

Abreu's ideas are embedded in the field of cultural psychology. Also taking the individual and society into account, another field that has grown in influence in mathematics education research in recent decades is that of situated cognition.

Situated Cognition

The theoretical field of situated cognition explores related aspects of the interplay of knowledge on the societal and psychological planes. Hence it has been a useful lens in research that takes culture into account in mathematical thinking and learning (Watson, 1998). As Ubiratan D'Ambrosio by his writings founded and influenced the field of ethnomathematics, so Jean Lave in analyzing and reporting her anthropological research has influenced the theoretical

field of situated cognition. From her early theorizing following ethnography with tailor's apprentices in Liberia (Lave, 1988) to her more recent writings, following research with grocery shoppers and weight-watchers (Lave, 1997), the notion of transfer of knowledge through abstraction in one context, and subsequent use in a new context, was questioned and problematized. In collaboration with Etienne Wenger, her theorizing led to the notion of cognitive apprenticeship and *legitimate peripheral participation* (Lave & Wenger, 1991; Wenger, 1998). In this view, learning consists in a centripetal movement of the apprentice from the periphery to the center of a practice, under the guidance of those who are already masters of the practice. This theory was not originally developed in the context of or for the purpose of informing mathematics education. However, in its challenge to the cognitive position that abstract learning of mathematics facilitates transfer and that this knowledge may be readily applied in other situations than the one in which it was learned (not born out by empirical research), the theory has been powerful and influential. The research studies inspired by Lave and reported by Watson (1998) bear witness to this strong influence.

In her later writings, Lave attempted to bring the theory of socially situated knowledge to bear on the classroom teaching and learning of mathematics. But issues of intentionality and recontextualization separate apprenticeship and classroom situations, although enough commonality exists in the two situations for both legitimately to be called *practices* (Lerman, 1998). Mathematics learners in school are not necessarily aiming to become either mathematicians or mathematics teachers. As Lerman pointed out in connection with the issue of voluntary and nonvoluntary participation, students' presence in the classroom may be nonvoluntary, creating a very different situation from that of apprenticeship learning, and calling into question the assumption of a goal of movement from the periphery to the center. At the same time, the teacher has the intention to teach her students mathematics, notwithstanding Lave's claim that teaching is not a precondition of learning. The learning of mathematics is the goal of the enterprise, and teaching is the teacher's job. In contrast, in the apprenticeship situation the learning is not a goal in itself, for example, in the case of the tailors the goal is to make garments efficiently. Thus, as Adler (1998) suggests, "It is in the understanding of the aims of school education that Lave and Wenger's seamless web of practices entailed in moving from peripheral to full participation in a community of practice is problematic" (p. 174).

Although the theory of legitimate peripheral participation may not translate easily into the classroom teaching and learning of mathematics, the view of learning as a social practice has powerful implications for this learning and has been an influence in changes that have taken place in the practice of teaching mathematics, such as an increased emphasis on communication (NCTM, 1989, 2000) in the mathematics classroom. Even more, situated cognition as a theory has given a warrant to attempts to bridge the gap between in- and out-of-school mathematical practices.

Use of Semiotics in Linking Out-of-School and In-School Mathematics

In the USA, the *Principles and Standards for School Mathematics* (NCTM, 2000) continued the earlier call (NCTM, 1989) for teachers to make connections, in particular between the everyday practices of their students and the mathematical concepts that are taught in the classroom. But various theoretical lenses of situated cognition (Kirshner & Whitson, 1997; Lave & Wenger, 1991) remind us that these connections can be problematic. There are at least three ways in which the activities of out-of-school practices differ from the mathematical activities of school classrooms (Walkerdine, 1988), as follows.

- The goals of activities in the two settings differ radically.
- Discourse patterns of the classroom do not mirror those of everyday practices.
- Mathematical terminology and symbolism have a specificity that differs markedly from the useful qualities of ambiguity and indexicality (interpretation according to context) of terms in everyday conversation.

A semiotic framework that uses chains of signification (Kirshner & Whitson, 1997) has the potential to bridge this apparent gap through a process of chaining of signifiers in which each sign "slides under" the subsequent signifier. In this process, goals, discourse patterns, and use of terms and symbols all move towards that of classroom mathematical practices in a way that has the potential to preserve essential structure and some of the meanings of the original activity.

This theoretical framework resonates with that of *Realistic Mathematics Education* (RME) developed by Freudenthal, Streefland, and colleagues at the Freudenthal Institute (Treffers, 1993). *Realistic* in this sense does not necessarily mean out-of-school in the real world: The term refers to problem situations that learners can imagine (van den Heuvel-

Panhuizen, 2003). A theory of semiotic chaining in mathematics education resonates with RME in that the starting points for the learning are realistic in this sense. But more specifically the chaining model is a useful tool for linking out-of-school mathematical practices with the formal mathematics of school classrooms (an example of such use is presented in the next paragraph).

In brief, the theory of semiotic chaining used in mathematics education research, as it was initially presented by Whitson (1994, 1997), Cobb, Gravemeijer, Yackel, McClain, & Whitenack (1997), and Presmeg (1997, 1998b), followed the usage of Walkerdine (1988). Although she was working in a poststructural paradigm, Walkerdine found the Swiss structural approach of Saussure, as modified by Lacan, useful in building chains of signifiers to link a home practice, such as that of a daughter and her mother pouring drinks for guests, with more formal learning of mathematics, in this case the system of whole numbers. Walkerdine's chain had the following structure.

Signified 1: the actual people coming to visit;

Signifier 1 (*standing for* signified 1): the names of the people.

Signifier 1 and signified 1 form **sign 1**.

The daughter is asked to raise a finger for each name.

Signified 2: the names of the guests;

Signifier 2: the raised fingers.

Signifier 2 and signified 2 form **sign 2**, the second link.

The daughter is asked to count her raised fingers.

Signified 3: the raised fingers;

Signifier 3: the numerals, one two, three, four, five.

Signifier 3 and signified 3 form **sign 3**, the third link.

In this process, signifier 3 actually comes to stand for all of the preceding links in the chain: Five guests are coming to tea. Note that each signifier in turn becomes the next signified, and that at any time any of the links in the chain may be revisited conceptually. As mother and daughter move through the links of the chain, the discourse shifts successively from actual people to their names, to the fingers of one hand, and finally to the more abstract discourse of the numerals of the whole number counting system.

In her distinctive style, Adler (1998) characterized the associated recontextualizing process as that of crossing a discursive bridge:

That there is a bridge to cross between everyday and educated discourses is at the heart of Walkerdine's (1988) argument for "good teaching" entailing chains of signification in the classroom where everyday notions have to be prised out of their discursive practice and situated in a new and different discursive practice. (p. 174)

The theory may be summarized as follows.

In his *semiology* Saussure defined the sign as a combination of a "signified" together with its "signifier" (Saussure, 1959; Whitson, 1994, 1997). Lacan inverted Saussure's model, which gave priority to the *signified* over the signifier, to stress the *signifier* over the signified, and thus to recognize "far ranging autonomy for a dynamic and continuously productive play of signifiers that was not so easily recognized when it was assumed tacitly that a signifier was somehow constrained under domination by the signified" (Whitson, 1994, p. 40). This formulation allows for the chaining process in which a signifier in a previous sign-combination becomes the signified in a new sign-combination, and so on. The chain has at its final link some mathematical concept that may be taught in school.

Using this process, a teacher can use the chain as an instructional model that develops a mathematical concept starting with an out-of-school situation and linking it in a number of steps with formal school mathematics (Hall, 2000; Hall & Presmeg, 2000). Building on the work of Presmeg (1997, 1998b) in his dissertation research, Hall taught three 4th-grade teachers to build semiotic chains appropriate to the practices of their students and the instructional needs of their individual classrooms. Using a similar semiotic chaining model, Cobb et al. (1997) reported on the emergence (using their *emergent* theoretical perspective) of chains of signification in one 1st-grade classroom. In addition to these sources, examples of mathematical chaining at elementary school, high school, and college levels may be found in Presmeg's (1997, 1998b, 2006) writings. In later research on building bridges between out-of-school and in-school mathematics, Presmeg (2006) preferred to use a nested triadic model based on the writings of Charles Sanders Peirce (1992, 1998). In addition to an *object* (which could be an abstract concept) and a *representamen* that stands for the object in some way, each nested sign has a third component that Peirce called the *interpretant*. This triadic model explicitly allows for learners' individual construction of meaning (through the interpretant) in a way that linear chains of signification can do only implicitly.

In closing this section, three more of Peirce's constructs that are relevant to the role of culture and historiography in mathematics education are introduced,

for the purpose of showing the potential of these theoretical notions to provide a foundation for ethnomathematics characterized as a research program.

**PEIRCE'S NOTIONS OF SYNECHISM,
COMMUNITY, AND COMMENS
IN RECOGNIZING THE ROLE OF CULTURE
IN MATHEMATICS EDUCATION**

In this final part of the section, I want to return to ideas in the beginning of this section by introducing some Peircean constructs that have relevance in the *historiography* of D'Ambrosio (2000) with which this section started. Peirce's (1992, 1998) philosophical writings are useful in attending to the historical development of mathematical thought and thus are relevant to the role of culture in mathematics education (Presmeg, 2003). In particular, the construct of *community*—which he used without definition—and his definitions of *commens* and *synechism*, including his “law of mind,” are apposite to explorations into the place of culture in mathematical historiography. In the light of these constructs (discussed in the following paragraphs), the role of community in the public institution of mathematics education is an issue of fundamental practical importance. The significance of the community of thinkers in the evolution of mathematical knowledge is indicated in Peirce's (1992) somewhat negative designation of the individual—uninformed by the sociocultural milieu—as ignorant and in error: “The individual man, since his separate existence is manifested only by ignorance and error, so far as he is anything apart from his fellows, and from what he and they are to be, is only a negation” (p. 55).

With regard to the genesis and evolution of mathematics, a point that has relevance in Peirce's epistemology is the continuity of past, present, and future. Continuity is central in Peirce's definition of *synechism* as “the tendency to regard continuity . . . as an idea of prime importance in philosophy” (Peirce, 1992, p. 313). *Synechism* involves the startling notion that knowledge in its real essence depends on *future* thought and how it will evolve in the community of thinkers:

Finally, as what anything really is, is what it may finally be come to be known to be in the ideal state of complete information, so that reality depends on the ultimate decision of the community; so thought is what it is, only by virtue of its addressing a future thought which is in its value as thought identical with it, though more developed. In this way, the existence of thought now, depends on what is to be hereafter;

so that it has only a potential existence, dependent on the future thought of the community. (Peirce, 1992, pp. 54–55)

Whether “the ideal state of complete information” is ever an attainable goal is a matter of doubt, but the relevance of *synechism* for the history of mathematics lies in the role attributed to future generations of thinkers in assessing the achievements of the past and present. The notion of *synechism* is further explicated in connecting individual and community ideation through the role of convention in semiosis (activity with signs). Because the semiosis of the individual is mediated by the community through the adoption of certain ways of thinking and representing ideas as conventional, the growth of (mathematical) knowledge manifests continuity. Peirce (1992) cast further light on what he meant by continuity in his *law of mind*:

Logical analysis applied to mental phenomena shows that there is but one law of mind, namely, that ideas tend to spread continuously and to affect certain others which stand to them in a peculiar relation of affectability. In this spreading they lose intensity, and especially the power of affecting others, but gain generality and become welded with other ideas. (p. 313)

Because of the importance of personal interpretations in forging a community of thinkers with its conventions, and thus in the continuity of ideas, Peirce formulated three kinds of interpretant in his semiotic model. Accordingly, he used triads not only in his semiotic model including object, representamen (sometimes called the sign), and interpretant, but also in the types of each of these components. This model includes the need for expression or communication: “Expression is a kind of representation or signification. A sign is a third mediating between the mind addressed and the object represented” (Peirce, 1992, p. 281). In an act of communication, then, there are three kinds of interpretant, as follows:

- the “*Intentional* Interpretant, which is a determination of the mind of the utterer”;
- the “*Effectual* Interpretant, which is a determination of the mind of the interpreter”;
- and
- the “*Communicational* Interpretant, or say the *Cominterpretant*, which is a determination of that mind into which the minds of utterer and interpreter have to be fused in order that any communication should take place.” (Peirce, 1998, p. 478)

It is the latter fused mind that Peirce designated the *commens*. There is a clear resonance of the commens with culture defined as a set of shared understandings (in the previous section).

For the continuity of mathematical ideas and their evolution in the history of mathematics, a central requirement is a community of thinkers who share a “fused mind” sufficiently to communicate effectively with one another—and with posterity through their artifacts—through this commens. Both the intentional and the effectual interpretants are important for communication (and have the implicit potential for miscommunication). But the third member of this triad, the interpretant generated by the commens, leads to the adoption of conventional signs by an intellectual community.

D'Ambrosio and others following the tradition of historiography that he has inaugurated (Selin, 2000) have introduced a different commens. Future generations, through synechism, will judge the impact of this historiography on the ontology and epistemology of mathematics in general, and of mathematics education in particular.

EMPIRICAL FIELDS THAT INCORPORATE CULTURE IN MATHEMATICS EDUCATION

The last section dealt with some theoretical fields that have been useful, or have the potential to be useful, in research on the role of culture in mathematics education. In the current section, some associated empirical fields are discussed. Empirical fields entail broad methodologies of empirical research, but in addition to these, in this section specific methods of research and also the participants, data, and results of selected studies are introduced. These details of participants, time, and place in the research are termed the *empirical settings* (Brown & Dowling, 1998). Thus appropriate empirical fields and settings are the focus of this section, along with some results of research on the role of culture in mathematics education.

Because ethnography is the special province of holistic cultural anthropological research (Eisenhart, 1988), which has as a broad goal the understanding of various cultures, it is natural to expect that ethnography (sometimes in modified form) would be used by researchers who are interested in the role of culture in teaching and learning mathematics. This is indeed the case. However, researchers in mathematics education in general sometimes call their studies ethnography when they have not spent long enough in the field to warrant that term for their research (Eisen-

hart, 1988). For instance, valuable as studies such as that of Millroy (1992) are for mathematics educators to learn more about the use of mathematical ideas in out-of-school settings such as that of Millroy's carpenters in Cape Town, South Africa, the 5½ month duration of her fieldwork might be considered brief for an ethnography, and most ethnographic studies in mathematics education have a shorter duration than Millroy's. For this reason, the field of such research is often more appropriately called *case studies* (Merriam, 1998), because they satisfy the criterion of being bounded in particular ways. If such studies are investigating a particular topic, such as ways of bringing out-of-school mathematics into the classroom, and using several cases to do so, they could be called *instrumental case studies* (Stake, 2000). Where the individual cases themselves are the focus, case studies are *intrinsic* (ibid.).

Most of the studies mentioned in this section may be regarded as instrumental case studies in the sense explained in the foregoing. Research that concerns in-school and out-of-school mathematical practices falls into two main categories. In the first category are studies that attempt to build bridges between informal or nonformal mathematical practices (Bishop, Mellin-Olsen, & van Dormolen, 1991) and those of formal school mathematics. In the second category are studies that link with formal mathematics education less directly but with the potential to deepen understanding of this education, by exploring the culture of mathematics in various workplaces. The studies in both of these categories are too numerous for a summary chapter such as this one to do full justice to the aims, research questions, methodologies and methods for data collection, and the subtleties unearthed in the results of such research. (Again, the reader is reminded to read the original literature for depth of coverage.) What follows is an introduction to the range of research in each of these two categories, with discussion of some of the empirical settings, a sampling of issues addressed, and some of the results.

Linking Mathematics Learning Out-of-School and In-School

In the first category, studies that specifically investigated linking out-of-school and in-school mathematics are of several types. Some researchers interviewed parents of minority learners and their teachers to investigate relationships between home and school learning of mathematics. In England, Abreu and colleagues conducted such interviews for the purpose of investigating how the parents of immigrant learners are able to support their children in their transi-

tions to learning mathematics in a new school culture (Abreu, Cline & Shamsi, 2002). Civil and colleagues in the USA interviewed such parents for the purpose of finding out which home practices of immigrant families might be suitable for pedagogical purposes in mathematics classrooms (Civil, 1990, 1995, 2002; Civil & Andrade, 2002). Related empirical studies in which researchers investigated the transitions experienced by immigrant children in their learning of mathematics have been reported by Bishop (2002a) in Australia and by Gorgorio, Planas, and Vilella (2002) in Catalonia, Spain.

In the next type of study in this category, research was conducted in mathematics classrooms to investigate the effects of bringing out-of-school practices into that arena. Using video recording as a data collection method, Brenner (2002) investigated how four teachers and their junior high school students went about an activity in which the students were required to decide cooperatively which of two fictitious pizza companies should be given the contract to supply pizza to the school cafeteria. In a different classroom video study, Moschkovich's (2002b) research addressed the mathematical activities of seventh-grade students during an architectural design project: Students working collaboratively tried to design working and living quarters for a team of scientists in Antarctica in such a way that space would be maximized while still taking into account the heating costs of various designs. These studies emphasized the point that through their pedagogy and instructional decisions in the classroom, teachers are an important component of the success (or lack of success) of attempts to use out-of-school practices effectively in mathematics classrooms. Beliefs of teachers about the nature of mathematics, what culture is, and its role in the classroom learning of mathematics are strong factors in teachers' decisions in this regard (Civil, 2002; Presmeg, 2002b).

Like the researchers in the preceding paragraphs, Presmeg (1997, 1998b, 2002b, 2006), Hall (2000), and Adeyemi (2004) recognized the lack of congruence of out-of-school cultural practices and those "same" practices when brought into the school mathematics classroom (Walkerdine, 1988, 1990, 1997). Their research used semiotic chaining as a theoretical tool (see previous section) in an attempt to bridge the gaps, not only between these practices in- and out-of-school, but also between mathematical ideas implicit in these activities and the formal mathematics of the syllabi that teachers are expected to use in their practices. The fourth-grade teachers with whom Hall worked built chains of signifiers that they implemented with their classes, starting with a cultural practice of at least some of their students and aimed at linking mathematical ideas in

this practice in a number of steps with a formal school mathematics topic. Chains constructed by the teachers were designated as either intercultural—bridging two or more cultures, or intracultural—having a chain that remained within a single culture. Examples of the first type involved number of children in students' families, pizzas, coins, and measurement of students' hands, linking in a series of steps with classroom mathematical concepts. These are intercultural because the cultures and discourse of students' homes or activities are linked with the different discourse and culture of classroom mathematics, for instance, the making of bar graphs. Manipulatives were frequently used as intermediate links in these chains. The intracultural type, involving chains that were developed within the culture of a single activity, was evidenced in a chain involving baseball team statistics. The movement along the chain could be summarized as follows:

Baseball Game → Hits vs. At Bats → Success Fraction
→ Batting Average

It was not the activity that was preserved throughout the chain, but merely the culture of baseball (at least as it was imported into the classroom practice) within which the chain was developed. The need for interpretation at each link in the chain led to further theoretical developments using a triadic nested model that was a better lens for interpreting the results (Presmeg, 2006).

One study that is ethnographic in its scope and methodology has less explicit claims to link out-of-school and in-school mathematics, although implicit ties between the two contexts are present. This study is a thorough investigation of mathematical elements in learning the practice of selling newspapers in the streets by young boys (called in Portuguese *ardinas*) on the island of Cabo Verde in the Atlantic ocean (Santos & Matos, 2002). Using Lave and Wenger's (1991) theoretical framework of legitimate peripheral participation, Santos and Matos described the goal of their research as follows.

Our goal was to look into the ways (mathematics) learning relates to forms of participation in social practice in an environment where mathematics is present but that escapes the characteristics of the school environment. Because we believe that culture is an unavoidable fact that shapes our way of seeing and analyzing things, we decided to look at a culturally distinct practice and that constituted a really strange domain for us: the practice of the *ardinas* at Cabo Verde islands in Africa. (p. 81)

In addition to interesting explicit and implicit mathematical aspects of the changing practice of the *ardinas* as they moved from being newcomers to full participants, methodological difficulties in this kind of research were foregrounded by Santos and Matos (2002):

The fact that the research is studying a phenomena [*sic*] which was almost totally strange to us in most of its aspects, led us to realize that we had to go through a process which should involve, to a certain extent, our participation in the (*ardinas*) practice with the explicit (for us and for them) goal of learning it but not in order to be a full member of that community of practice. This starting point (more in terms of knowing that there are more things that we don't know than that we know) opened that community of practice to us but also gave us consciousness that methodological issues were central in this research. (p. 120)

Many of the difficulties that Santos and Matos described with sensitivity and vividness are common to most anthropological research and thus also impact investigations of cultural practices in the field of mathematics education. They were required to become part of the culture sufficiently to be able to interpret it to others who are not participants, but at the same time not to become so immersed in the culture that it would be transparent—a lens that is not the focus of attention because one looks *through* it. They faced the difficulty that entering the group as an outsider might in fact change the culture of that group to a greater or lesser extent. On several occasions there was evidence that the mathematical reporting by key informants among the *ardinas* was influenced by the fact that the fieldworker was a Portuguese-speaking woman, whom these boys might have associated with their schoolteachers. Finally, the researchers recognized the not inconsiderable difficulties associated with practical matters concerned with collecting data in the street.

The investigation of the *ardinas*' mathematical thinking could be classified as an ethnographic workplace study. Thus this short account of this research provides a transition to the second aspect of this empirical section of the chapter, mathematics in the workplace.

Culture of Mathematics in the Workplace

In the second category of studies in this section, the culture of mathematics in various workplaces was investigated intensively by FitzSimons (2002), by Noss and colleagues (Noss & Hoyles, 1996a; Noss, Hoyles & Pozzi, 2000; Noss, Pozzi & Hoyles, 1999; Pozzi, Noss, and Hoyles, 1998), and by several other researchers

(see later). FitzSimons investigated the culture and epistemology of mathematics in Technical and Further Education in Australia, and how this education related or failed to relate to the cultures of mathematics as used in workplaces. Noss, Hoyles, and Pozzi examined workplace mathematics in more specific detail: *Inter alia* their studies addressed mathematical aspects of banking and nursing practices. A common thread in these studies is the *demathematization* of the workplace: "As the seminal work of researchers Richard Noss and Celia Hoyles among others indicates, mathematics actually used in the workplace is contingent and rarely utilizes 'school mathematics' algorithms in their entirety, if at all, or necessarily correctly" (FitzSimons, 2002, p. 147).

The same point was brought out strongly, but contingently, in an investigation of mathematical activity in automobile production work in the USA (Smith, 2002). One result of Smith's study was that "the organizational structure and management of automobile production workplaces directly influenced the level of mathematics expected of production workers" (p. 112). This level varied from a minimal expectation on assembly lines—designed to be "worker-proof"—to "a surprisingly high level of spatial and geometric competence, which outstripped the preparation that most K–12 curricula provide" (p. 112), e.g., as manifested by skilled machinists who translated between two- and three-dimensional space with sophistication and accuracy in creating products that were sometimes one-of-a-kind. Organization that used "lean manufacturing principles" (p. 124) to move away from assembly lines and give more autonomy to workers was more likely to enhance and require these highly developed forms of mathematical thinking. Workers are not usually allowed to decide what level of mathematics will be required in their work:

The fact that the organization of production systems and work practices mediates and in some cases limits workers' mathematical expectations and their access to workplace mathematics is a reminder that the everyday mathematics of work is inseparable from issues of power and authority. In large measure, someone other than the production workers themselves decides when, how often, and how deeply they will be called on to think mathematically. (Smith, 2002, p. 130)

Other sources of insights into the mathematical ideas involved in practices of various workplaces are detailed in FitzSimons's (2002) book. These include the following:

operators in the light metals industry (Buckingham, 1997); front-desk motel and airline staff (Kanes, 1997a,

1997b); landless peasants in Brazil... (Knijnik, 1996, 1997, 1998); carpet layers (Masingila, 1993); commercial pilots (Noss, Hoyles, & Pozzi, 2000); ... draughtspersons (Strässer, 1998); semi-skilled operators (Wedeg, 1998b, 2000a, 2000b) and swimming pool construction workers (Zevenbergen, 1996). Collectively, these reports highlight not only the breadth and depth of mathematical concepts encountered in the workplace, but underline the complex levels of interactions in the broad range of professional competencies as outlined above, where mathematical knowledge can come into play—when permitted by (or in spite of) management. (p. 72)

In addition to this broad range of reported research, in an early study Mary Harris (1987) investigated “women’s work,” challenging her readers to “derive a general expression for [knitting] the heel of a sock” (p. 28). The more recent mathematics education conferences have included presentations on the topic of the mathematics of the workplace (e.g., Strässer, 1998). Strässer and Williams (2001) organized a Discussion group on “Work-Related Mathematics Education” at the 25th conference of the International Group for the Psychology of Mathematics Education. They presented the aims of this Discussion Group in the following terms.

Discussions in the group should aim at better understanding the contradictory trends of hiding or revealing workplace maths by means of artefacts and discourses, and identify problems and potentials for teaching and learning work-related mathematics. Current research into the use of mathematics at work (with the spectrum from traditional statistics to ethnomethodology) seems to favour case studies in a participatory style, while different ways of “stimulated recall” are also in use. (p. 265)

From the foregoing, the breadth of this field, the variety of practices that have been investigated from a mathematical point of view, and the scope of methodologies employed are apparent. Many of these studies may be characterized as investigations in ethnomathematics, for example, those of Masingila (1993, 1994) with the art of carpet laying, and Harris (1987). This overlap in classification also applies to research on mathematics in the world of work that is concerned with its artifacts and tools, and with its technology. The confluence of technology and culture in mathematics education is the topic of the last part of this section.

Influences of Technology

A study that stands at the intersection of workplace mathematics studies and those that are con-

cerned with the role and influence of technology in mathematics education is that by Magajna and Monaghan (2003). In their research they investigated both the mathematical elements and the use of computer aided design (CAD) and manufacture (CAM) in the work of six skilled technicians who designed and produced moulds for glass factories. The technicians specialized in “moulds for containers of intricate shapes” (p. 102), such as bottles in the form of twisted pyramids, stars, or guitars. The study is interesting because the mathematics used by the CAD/CAM technicians in various stages of their work (for instance, calculating the interior volume of a mould) was not elementary, and in theory school-learned mathematics could have been used. However, the researchers reported as follows.

Although the technicians did not consider their activity was related to school mathematics there is evidence that in making sense of their practice they resorted to (a form of) school mathematics. The role of technology in technicians’ mathematical activity was crucial: not only were the mathematical procedures they used shaped by the technology they used but the mathematics was a means to achieve technological results. Further to this the mathematics employed by the technicians must be interpreted within the goal-oriented behaviour of workers who ‘live’ the imperatives and constraints of the factory’s production cycle. (p. 101)

An implication of results such as these, resonating with the conclusions of other workplace studies, is that there is no direct path from school mathematics to the mathematics used, sometimes indirectly, in various occupations, where the specific practices are learned on the job. It may not be possible then to gear a mathematics curriculum, even in vocational education (FitzSimons, 2002), to the specific requirements of a number of vocations simultaneously. However, as technology has developed in the last 6 decades, its influence has been felt in mathematics curricula of various time periods, as illustrated by Kelly (2003) and described next.

Each of the mathematics curriculum movements of the last century felt the impact of the state of the current technology of that period. Building on the assertion that “the tool defines the skill,” Kelly (2003, p. 1041) described the influence of technological developments on mathematics curricula in four different time periods, overlapping with the following crucial years:

- 1942—the mainframe computer;
- 1967—the first four-function calculator;
- 1978—the personal computer (microcomputer);
- 1985—the graphing calculator.

In a broad sense, then, technology of various types has influenced considerably the culture of mathematics education taken as a complex of shared understandings (Stenhouse, 1967) about the nature of mathematics and its pedagogy. However, this influence does not imply unanimity about *what* technology should be brought into classrooms, or *how* it should be used. Contestation and conflict accompanied the growing use of computers and calculators of increasing sophistication in mathematics education, as Kelly's account showed vividly. He pointed out that the influence has not only permeated the culture of mathematics education through various curricula, but that as the technology changed and became more interactive through graphical user interfaces, the potential for a different kind of learning of mathematics became possible, and hence the need arose for a different kind of instruction. However, two impediments to the incorporation of new technology into mathematics classrooms have been the inaccessibility of the technology—exacerbated particularly in developing countries (Setati, 2003b)—and a lack of teacher readiness (Kelly, 2003).

Kelly's (2003) portrayal also suggests the importance of the avenues opened up by the graphing calculator, especially in—but not limited to—secondary and post-secondary schooling. Use of graphing calculators and computers impacts the nature of mathematics learned at all levels. The research in this growing field is beyond the scope of this chapter, but two examples highlight the potential of graphing calculators and computers to change the culture of learning mathematics. First, the research of Ricardo Nemirovsky (2002) and his colleagues (Nemirovsky, Tierney, & Wright, 1998) demonstrates how the use of motion detectors and associated technology changes the culture of learning from one of fostering *formal* generalizations in mathematics (e.g., “all x are y ”), to one of constructing *situated* generalizations. The latter kind of generalization is “embedded in how people relate to and participate in tasks, events, and conversations” (Nemirovsky, 2002, p. 250)—and it is “loaded” with the values of “grasping the circumstances and transforming aspects that appear to be just ordinary or incidental into objects of reflection and significance” (p. 251). Nemirovsky illustrated this kind of generalization in the informal mathematical constructions of Clio, an 11-year-old girl working with the problem of trying to predict, based on the graph created on a computer screen by a motion detector, where a toy train is situated in a tunnel.

A second example is from the doctoral research of Paul Yu (2004), who investigated middle grades

students' learning of geometry via the interactive geometric program called Shape Makers (Battista, 1998). Yu's research demonstrated how use of an interactive computer program such as this one has the potential to change the order of acquisition of the van Hiele (1986) levels of learning geometry. For instance, for these learners, a trapezoid was *what the trapezoid maker makes* in comparison with other shapes, and only later would they focus on individual appearance and properties of particular shapes.

The influence of dynamic geometry systems has been considerable, and research on the impact of such interactive software on the culture of school learning of geometry is important and ongoing. As Kelly (2003) pointed out, not only does the tool define the skill, but the development of new tools changes the culture and practices of mathematics itself, for instance in the impact of computer algebra systems on mathematics, or in changes made possible by the ease with which the computer can perform large numbers of routines that are beyond human capabilities.

In summary, then, technology, by entering all avenues of life, influences not only the mathematics of the curriculum and the ontology of mathematics itself, but also the culture of the future, through its children.

This section has examined in broad detail some of the empirical research relating to the role of culture in teaching and learning mathematics. Aspects that were considered included the linking of out-of-school and in-school mathematics learning, both from the point of view of comparing these different cultural practices and from the point of view of building bridges between cultural practices and the classroom learning of mathematics—of “bringing in the world” (Zaslavsky, 1996). A related strand that was outlined was the culture of mathematics in the workplace, and finally, some aspects of the cultural influence of technology on the learning of mathematics were suggested. Clearly in all of these strands ongoing research is needed and in progress. However, perhaps just as significantly, research is needed that will increase understanding of and highlight how these strands are related amongst themselves. For instance, as the use of technology both in the workplace and in the mathematics classroom changes the culture of learning mathematics in these broad arenas, hints of avenues to be explored in future work appear, for example in whether and how technology has the potential to bridge or decrease the gap between formal and informal mathematical knowledge. The theme of technology and culture in future research is elaborated, along with other significant areas, in the final section.

FUTURE DIRECTIONS FOR RESEARCH ON CULTURE IN MATHEMATICS EDUCATION

The final section uses a wider lens to zoom out and consider areas of research on culture in mathematics education that require development or are likely to assume increasing importance in the years ahead. By its nature, this section is speculative, but trends are already apparent that are likely to continue. The influence of technology is one such trend (Morgan, 1994; Noss, 1994).

Technology

Already in 1988, Noss pointed out the cultural entailments of the computer in mathematics education, as follows: “Making sense of the advent of the computer into the mathematics classroom entails a cultural perspective, not least because of the ways in which children are developing the computer culture by appropriating the technology for their own ends” (p. 251). He elaborated, “The key point is that children see computer screens as ‘theirs,’ as a part of a predominantly adult culture which they can appropriate and use for their own ends” (p. 257). As these children become adults with a facility with technology beyond that of their parents (Margaret Mead’s *prefigurative enculturation* comes into play as children *teach* their parents), the culture of mathematics education in all its aspects is likely to change in fundamental ways. Noss (1988) and Noss and Hoyles (1996b) put forward a vision for the role that computer microworlds have played and might play in the future of mathematics education. With accelerating changes in platforms, designs, and software, research must keep pace and inform resulting changes in the cultures of teaching and learning of mathematics in schools. An aspect of potential use for the computer stems from its ability to mimic reality in a world of virtual reality. Noting the complex relationship between formal and informal mathematics (ethnomathematics), Noss (1988) suggested, “I propose that the technology itself—specifically the computer—can be the instrument for bridging the gap between the two” (p. 252). This area remains one in which research is needed, both in its own right and in ways that the change of context of bringing the virtual reality of computer images into mathematics classrooms changes the culture of teaching and learning in those classrooms.

Bridging Mathematics In-School and Out-of-School

Resonating with the Realistic Mathematics Education viewpoint that *real* contexts are not confined

to those concrete situations with which learners are familiar (van den Heuvel-Panhuizen, 2003), Carraher and Schliemann (2002) made a useful distinction between *realism* and *meaningfulness*:

What makes everyday mathematics powerful is not the concreteness of the objects or the everyday realism of the situations, but the meaning attached to the problems under consideration (Schliemann, 1995). In addition meaningfulness must be distinguished from realism (D. W. Carraher & Schliemann, 1991). It is true that engaging in everyday activities such as buying and selling, sharing, or betting may help students establish links between their experience and intuitions already acquired and topics to be learned in school. However, we believe it would be a fundamental mistake that schools attempt to emulate out-of-school institutions. After all, the goals and purposes of schools are not the same as those of other institutions. (p. 137)

The dilemma, then, is *how* to incorporate out-of-school practices in school mathematics classrooms in ways that are meaningful to students and that do not trivialize the mathematical ideas inherent in those practices. This issue remains a significant one for mathematics education research.

Reported research (Civil, 2002; Masingila, 2002; Presmeg, 2002b) has shown that learners’ beliefs about the nature of mathematics affect what mathematics they identify *as* mathematics in out-of-school settings. Students’ and teachers’ conceptions of mathematics as decontextualized and abstract may limit what can be accomplished in terms of meaningfulness derived from out-of-school practices (Presmeg, 2002b). And yet, abstract thinking is not necessarily antagonistic to the idea of reasoning in particular contexts, as Carraher and Schliemann (2002) pointed out, which led them to propose the construct of *situated generalization*. They summarized these issues as follows.

Research sorely needs to find theoretical room for contexts that are not reducible to physical settings or social structures to which the student is passively subjected. Contexts can be imagined, alluded to, insinuated, explicitly created on the fly, or carefully constructed over long periods of time by teachers and students. Much of the work in developing flexible mathematical knowledge depends on our ability to recontextualize problems—to see them from diverse and fresh points of view and to draw upon our former experience, including formal mathematical learning. Mathematization is not to be opposed to contextualization, since it always involves thinking in contexts. Even the apparently context-free activity of applying syntax transformation rules to algebraic expressions can involve meaningful contexts, particularly for experienced mathematicians. (p. 147)

They mentioned the irony that the mechanical following of algorithms characterizes the approaches of both highly unsuccessful and highly successful mathematical thinkers.

Taking into account both the need for meaningful contexts in the learning of mathematics and the necessity of developing mathematical ideas in the direction of abstraction and generalization (in the flexible sense, not to be confused with decontextualization), at least two extant fields of research have the potential to address these issues in significant ways. The notions of *horizontal* and *vertical* mathematizing that have informed Realistic Mathematics Education research for several decades (Treffers, 1993) could resolve a seeming conflict between abstraction and context. In harmony with these ideas, recent attempts to use semiotic theories in linking out-of-school and in-school mathematics also have the potential for further development (Hall & Presmeg, 2000; Presmeg, 2006; Yackel, Stephan, Rasmussen, & Underwood, 2003).

As Moschkovich (1995) pointed out, a tension exists between educators' attempts to engage learners in "real world" mathematics in classrooms and movements to make mathematics classrooms reflect the practices of mathematicians. Multicultural mathematics materials for use in classrooms have been available for some time (Krause, 1983; Zaslavsky, 1991, 1996). However, the tensions, the contradictions, and the complexity of trying to incorporate practices for which "making change" serves as a metonymy at the same time that students are "making mathematics" (Moschkovich, 1995) will engage researchers in mathematics education for some time to come. Bibliographies such as that compiled by Wilson and Mosquera (1991) will continue to be necessary, to inform both researchers and practitioners what has already been accomplished as the field of culture in mathematics education continues to grow.

Teacher Education

In all of the foregoing areas of potential cultural research in mathematics education, the role of the teacher remains important. Noting that concrete and abstract domains in mathematical thinking are not necessarily disparate, Noss (1988) suggested,

The key idea is that of focusing attention on the important relationships involved, a role in which—as Weir (1987) points out—the computer is rather well cast; but not without the conscious intervention of educators, and the careful development of an ambient learning culture. (p. 263)

Bishop (1988a, 1988b) was also intensely aware of the role of the *mathematical enculturators* in personifying the values that are inherent in the teaching and learning of mathematics. In the final chapter of his seminal book (1988a) he suggested requirements for the education—rather than the more restricted notion of "training"—of those who will be mathematics teachers, at both the elementary and secondary levels. He did not distinguish between these levels, for teachers at both elementary school and secondary school have important mathematical enculturation roles. Bishop (1988a) summarized the necessary criteria as follows.

I propose, then, these four criteria for the selection of suitable Mathematical enculturators:

- ability to "personify" the mathematical culture
- commitment to the Mathematics enculturation process
- ability to communicate Mathematical ideas and values
- acceptance of accountability to the Mathematical cultural group. (p. 168)

These ideas still seem timely; in fact the literature on discourse and communication has broadened in the decades since these words were written, to suggest that communication amongst all involved in the negotiation of the cultures of mathematics classrooms (teacher and learners) plays a significant role in the learning of mathematics in those arenas (Cobb et al., 1997; Dörfler, 2000; Sfard, 2000). *Language and Communication in Mathematics Education* was the title of a Topic Study Group at the 10th International Congress on Mathematical Education (Copenhagen, 2004), and the literature in this field is already extensive. But *how* to educate future teachers of mathematics to satisfy Bishop's four criteria is still an open field of research.

As Arcavi (2002) acknowledged, much has already been accomplished in curriculum development, research on teachers' beliefs and practices, and "the development of a classroom culture that functions in ways inspired by everyday practices of academic mathematics" (p. 27). However, open questions still exist concerning ways of using the recognition that the transition from out-of-school mathematical practices to those within school is sometimes not straightforward, in order to inform the practices of mathematics teaching. Arcavi (2002) gave examples of such questions.

However, much remains to be researched. For example, is it always possible to smooth the transition between familiar and everyday contexts, in which students use ad hoc strategies to solve problems, and academic contexts in which more general, formal, and decontextualized mathematics is to be learned?

Are there breaking points? If so, what is their nature? Studies in everyday mathematics and in ethnomathematics are very important contributions, not only because of their inherent value but also because of the reflection they provoke in the mathematics education community at large. There is much to be gained from those contributions. (p. 28)

Clearly ethnomathematics conceived as a research program (D'Ambrosio, 2000) has already permeated the cultures of mathematics education research and practice in various ways. The influence, and the need for research that addresses the complexities of the issues involved, are ongoing.

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Useful Web Sites on Culture and the Learning and Teaching of Mathematics:

<http://www.csus.edu/indiv/o/oreyd/once/once.htm>
http://www.geometry.net/pure_and_applied_math/ethnomathematics.html
<http://www.dm.unipi.it/~jama/ethno/>
<http://phoenix.sce.fct.unl.pt/GEPEm/>
<http://www.fe.unb.br/etnomatematica/>
<http://www2.fe.usp.br/~etnomat>
<http://web.nmsu.edu/~pscott/spanish.htm>
<http://etnomatematica.univalle.edu.co>
<http://www.rpi.edu/~eglash/isgem.htm>
<http://chronicle.com/colloquy/2000/ethnomath/ethnomath.htm>
<http://chronicle.com/free/v47/i06/06a01601.htm>
<http://chronicle.com/colloquy/2000/ethnomath/re.htm>
<http://www.ecsu.ctstateu.edu/depts/edu/projects/ethnomath.html>

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■ ■ P A R T I V ■ ■

Students and Learning

EARLY CHILDHOOD MATHEMATICS LEARNING

Douglas H. Clements and Julie Sarama

UNIVERSITY AT BUFFALO, STATE UNIVERSITY OF NEW YORK

It seems probable that little is gained by using any of the child's time for arithmetic before grade 2, though there are many arithmetic facts that he [sic] can learn in grade 1.

—Thorndike, 1922, p. 198

Children have their own preschool arithmetic, which only myopic psychologists could ignore.

—Vygotsky, 1978/1935, p. 84

For over a century, views of young children's mathematics have differed widely. The recent turn of the century has seen a dramatic increase in attention to the mathematics education of young children. Although mathematics researchers have studied young children's learning for some time, much of the research corpus originated in fields such as developmental psychology. Our goal is to synthesize relevant research on the learning of mathematics before first grade from multiple fields. The synthesis reveals a nascent field that nonetheless is a potential harbinger of theoretical advancement. For example, genetic epistemology, as the study of the origins of knowledge itself, has never been more deeply developed or empirically tested than in recent research on early mathematics learning. Thus, this body of research illuminates foundational issues on the learning of mathematics with theoretical implications for all ages.

We begin with a brief overview of mathematics in early childhood and young children's learning of mathematics. In the bulk of the review, we address children's learning of ideas within the five topical domains determined to be significant for young

children's learning (Clements & Conference Working Group, 2004; NCTM, 2000), because it is most fruitful to focus on the big ideas of mathematics (Bowman, Donovan, & Burns, 2001; Clements, 2004; Fuson, 2004; Griffin, Case, & Capodilupo, 1995; Tibbals, 2000; Weiss, 2002). We then briefly discuss issues of mathematical processes and learning and teaching contexts, including early childhood school settings and equity issues. Space constraints limit discussion of educational implications to those with direct empirical evidence (but see other sources, Clements, Sarama, & DiBiase, 2004).

MATHEMATICS IN EARLY CHILDHOOD

There are at least seven reasons for the recent surge of attention to mathematics in early childhood. First, increasing numbers of children attend early care and education programs. In 1999, 70% of 4-year-olds and 93% of 5-year-olds were enrolled in preprimary education, up from 62 and 90%, respectively, in 1991 (U.S. Department of Education, 2000, p. 7). Several

states are instituting universal pre-K,¹ with about 1 million children enrolled in 1999, and that number is increasing (Hinkle, 2000). In 2001, about two thirds of all 4-year-olds were enrolled, with that ratio increasing (Loeb, Bridges, Bassok, Fuller, & Rumberger, in press; Magnuson, Meyers, Rathbun, & West, 2004). Various government agencies, federal and state, provide financial support for pre-K programs designed to facilitate academic achievement, particularly in low-income children.

Second, there is an increased recognition of the importance of mathematics (Kilpatrick, Swafford, & Findell, 2001). In a global economy with the vast majority of jobs requiring more sophisticated skills than in the past, American educators and business leaders have expressed strong concern about students' mathematics achievement. These concerns are echoed in international comparisons of mathematics achievement (Mullis et al., 1997).

Third, the mathematics achievement of American students compares unfavorably with the achievement of students from several other nations, even as early as kindergarten. Some cross-national differences in informal mathematics knowledge appear as early as 4 to 5 years of age (Starkey et al., 1999; Yuzawa, Bart, Kinne, Sukemune, & Kataoka, 1999).

Fourth, the knowledge gap is most pronounced in the performance of U.S. children living in economically deprived urban communities (Griffin, Case, & Siegler, 1994; Saxe, Guberman, & Gearhart, 1987; Siegler, 1993). That is, differences are not just between nations, but also between socioeconomic groups within countries.

Fifth, researchers have changed from a position that young children have little or no knowledge of or capacity to learn mathematics (e.g., Piaget, Inhelder, & Szeminska, 1960; Piaget & Szeminska, 1952; Thorndike, 1922) to theories that posit competencies that are either innate or develop in the first years of life (Baroody, Lai, & Mix, 2006; Clements, Sarama, & DiBiase, 2004; R. Gelman & Gallistel, 1978; Perry & Dockett, 2002). We will discuss these issues in more depth; here, it suffices to say that young children clearly can engage with substantive mathematical ideas.

Sixth, specific quantitative and numerical knowledge in the years before first grade has been found to

be a stronger predictor of later mathematics achievement than tests of intelligence or memory abilities (Krajewski, 2005).

Seventh, research indicates that knowledge gaps appear in large part due to the lack of connection between children's informal and intuitive knowledge (Ginsburg & Russell, 1981; Hiebert, 1986) and school mathematics. This is especially detrimental when this informal knowledge is poorly developed (Baroody, 1987; Griffin, Case, & Siegler, 1994).

For these reasons, there has been much recent interest in, and attention to, the learning and teaching of mathematics before first grade, the focus of this chapter.

YOUNG CHILDREN AND MATHEMATICS LEARNING

Young children possess an informal knowledge of mathematics that is surprisingly broad, complex, and sophisticated (Baroody, 2004; Clements, Swaminathan, Hannibal, & Sarama, 1999; Fuson, 2004; Geary, 1994; Ginsburg, 1977; Kilpatrick, Swafford, & Findell, 2001; Piaget & Inhelder, 1967; Piaget, Inhelder, & Szeminska, 1960; Steffe, 2004). For example, preschoolers engage in substantial amounts of foundational free play.² They explore patterns, shapes, and spatial relations; compare magnitudes; and count objects. Importantly, this is true for children regardless of income level and gender (Seo & Ginsburg, 2004). They engage in mathematical thinking and reasoning in many contexts, especially if they have sufficient knowledge about the materials they are using (e.g., toys), if the task is understandable and motivating, and if the context is familiar and comfortable (Alexander, White, & Daugherty, 1997).

In a similar vein, most entering kindergartners, and even entering preschoolers, show a surprisingly high level of mathematical skills, with much individual variation. For example, in 1999, 94% of entering U.S. kindergartners could count to 10 and recognize numerals and shapes, and 58% could also read numerals, count beyond 10, make patterns, and use non-

¹ Terms for the preprimary years are not used consistently. We use *pre-K* for prekindergarten, the year before kindergarten entrance (in the United States pre-K children are usually 4 years of age); *pre-K children* and *preschoolers* are often used interchangeably, with plural phrases such as *the preschool years* explicitly meaning the pre-K year and the year before this year (3-year-olds); *toddlers* refers to 1- to 2-year-olds and infants to children below 1 year of age.

² Such everyday foundational experiences form the intuitive, implicit conceptual foundation for later mathematics. Later, children represent and elaborate these ideas—creating models of an everyday activity with mathematical objects, such as numbers and shapes; mathematical actions, such as counting or transforming shapes; and their structural relations. We call this process mathematization. A distinction between foundational and mathematized experiences is necessary to avoid confusion about the type of activity in which children are engaged (Kronholz, 2000).

standard units of length to compare objects (National Center for Education Statistics [NCES], 2000). Similarly, about 68% of a sample of low-income children entering preschool in two states in the United States could count verbally to 5, 44% to 10 (Clements, Sarama, & Gerber, 2005). From 37 to 45% could count small groups of objects (2 to 7). However, only 2 to 16% of the children could perform various nonverbal addition and subtraction problems and 2 to 12% could solve verbal problems with sums of 5 or less. Between 75 and 86% of the children could identify prototypical examples of squares, triangles, and rectangles, and 39 to 74% could identify palpable nonexamples (palpable distractors) or categorize nonprototypical members and nonpalpable distractors (26–59%).

Such findings indicate that foundational mathematical knowledge begins during infancy and undergoes extensive development over the first 5 years of life. It is just as natural for young children to think premathematically and then mathematically as it is for them to use language, because “humans are born with a fundamental sense of quantity” (Geary, 1994, p. 1), as well as spatial sense, and a propensity to search for patterns.

In summary, young children have the interests and ability to engage in significant mathematical thinking and learning. Their abilities extend beyond what is introduced in most programs (Aubrey, 1997; Clements, 1984c; Geary, 1994; Griffin & Case, 1997; A. Klein & Starkey, 2004).

Most of this review is organized around the *big ideas of mathematics* (Bowman, Donovan, & Burns, 2001; Clements, 2004; Fuson, 2004; Griffin, Case, & Capodilupo, 1995; Tibbals, 2000; Weiss, 2002), the overarching concepts that are mathematically central and coherent, consistent with children’s thinking, and generative of future learning (Clements & Conference Working Group, 2004). This organization reflects the idea that children’s early competencies are organized around several large conceptual areas.

In addition, in developing each of these big ideas, children often pass through a sequence of levels of thinking. These developmental progressions can underlie hypothetical learning trajectories. Learning trajectories have three parts: a goal, a learning path or trajectory through which children move through levels of thinking, and instruction that helps them move along that path. Formally, learning trajectories are descriptions of children’s thinking as they learn to achieve specific goals in a mathematical domain and a related, conjectured route through a set of instructional tasks designed to engender those mental processes or actions hypothesized to move children through a developmental progression of levels of thinking (Clements & Sarama, 2004c).

Learning trajectories are useful pedagogical as well as theoretical constructs (Bredekamp, 2004; Clements & Sarama, 2004b; M. A. Simon, 1995). In one study, the few teachers who actually led in-depth discussions in reform mathematics classrooms saw themselves not as moving through a curriculum, but as helping students move through levels of understanding (Fuson, Carroll, & Drueck, 2000). Further, researchers suggest that professional development focused on developmental progressions increases not only teachers’ professional knowledge but also their students’ motivation and achievement (D. M. Clarke et al., 2001; D. M. Clarke et al., 2002; Fennema et al., 1996; G. Thomas & Ward, 2001; R. J. Wright, Martland, Stafford, & Stanger, 2002). A focus on both big ideas and the “conceptual storylines” of curricula in the form of hypothetical learning trajectories is supported by research on systemic reform initiatives (Heck, Weiss, Boyd, & Howard, 2002). For these reasons, this review attempts to explicate the developmental progressions of the big ideas.

This emphasis on content is not meant to deny the importance of processes, which we discuss throughout and focus on in the second last section.

As important as mathematical content are general mathematical processes such as problem solving, reasoning and proof, communication, connections, and representation; specific mathematical processes such as organizing information, patterning, and composing, and habits of mind such as curiosity, imagination, inventiveness, persistence, willingness to experiment, and sensitivity to patterns. All should be involved in a high-quality early childhood mathematics program. (Clements & Conference Working Group, 2004, p. 57)

This structure of goals is consistent both with recommendations of the National Council of Teachers of Mathematics (NCTM, 2000) and with research on young children’s development of a network of logical and mathematical relations (Kamii, Miyakawa, & Kato, 2004). Finally, it is consistent with the conclusions of another review that “the overriding premise of our work is that throughout the grades from pre-K through 8 all students can and should be mathematically proficient” (Kilpatrick, Swafford, & Findell, 2001, p. 10), including conceptual understanding, procedural fluency, strategic competence, adaptive reasoning (capacity for logical thought, reflection, explanation, and justification), and a productive disposition.

Theoretical Frameworks

Three main theoretical frameworks for understanding young children’s mathematical thinking are

empiricism, (neo)nativism, and interactionalism. The introductory quote by Thorndike illustrates an empiricist framework. In traditional empiricism, the child is seen as a “blank slate,” truth lies in correspondences between children’s knowledge and reality, and knowledge is received by the learner via social transmission or abstracted from repeated experience with a separate ontological reality. An extension, traditional information processing theory, uses the computer as a metaphor for the mind and moves slightly toward an interactionalist perspective.

In contrast, nativist theories, in the traditional of philosophical rationalism (e.g., Plato and Kant), emphasize the inborn or early developing capabilities of the child. For example, quantitative or spatial cognitive structures present in infancy support the development of later mathematics, and thus innate structures are fundamental to mathematical development. Many theorists build on Gelman and Gallistel’s *privileged domains hypothesis* (Rittle-Johnson & Siegler, 1998). In this view, a small number of innate or early-developing mathematical competencies are privileged and easy to learn. These are hypothesized to have evolutionary significance and be acquired or displayed by children in diverse cultures at approximately the same age. This is in contrast to other perspectives that would explain relative competence as resulting from frequency of experience. As we shall show in more detail, neither the empiricist nor nativist position fully explains children’s learning and development. An intermediate position seems warranted, such as interactionalist theories that recognize the interacting roles of nature and nurture (Newcombe, 2002).

In interactionalist, constructivist theories, children actively and recursively create knowledge. Structure and content of this knowledge are intertwined and each structure constitutes the organization and components from which the child builds the next, more sophisticated structure. In comparison to nativism’s initial representational cognition, children’s early structures are prerepresentational.

Constructivist theories take several forms. In *trivial constructivism*, children are not passive receivers of knowledge, but knowledge is seen as learning about an objective reality. In contrast, in *radical constructivism* cognition is adaptive and serves to help children organize their experiential world, rather than uncover or discover a separate, objective ontology. Radical constructivism often uses an evolutionary view of cognitive development, in which children generate cognitive schemes to solve perceived problems and test them to see how well they fit the experiential world. *Social constructivism*, building on Vygotsky’s seminal work, holds that individuals and the society are inter-

connected in fundamental ways; there is no isolated cognition (Ernest, 1995).

We believe that the research reviewed here supports a synthesis of aspects of previous theoretical frameworks that we call *hierarchic interactionalism*. The term indicates the influence and interaction of global and local (domain specific) cognitive levels and the interactions of innate competencies, internal resources, and experience (e.g., cultural tools and teaching). Mathematical ideas are represented intuitively, then with language, then metacognitively, with the last indicating that the child possesses an understanding of the topic and can access and operate on those understandings. The tenets of hierarchic interactionism follow; research supporting these tenets will be developed throughout the chapter.

1. *Developmental progression.* Most content knowledge is acquired along developmental progressions of levels of thinking. These progressions play a special role in children’s cognition and learning because they are particularly consistent with children’s intuitive knowledge and patterns of thinking and learning at various levels of development (at least in a particular culture, but guided in all cultures by “initial bootstraps”—see below), with each level characterized by specific mental objects (e.g., concepts) and actions (processes), (e.g., Clements, Wilson, & Sarama, 2004; Steffe & Cobb, 1988).
2. *Domain specific progression.* These developmental progressions often are most propitiously characterized within a specific mathematical domain or topic (see also Karmiloff-Smith, 1992; cf. Resnick’s “conceptual rationalism,” 1994). Children’s knowledge (i.e., the objects and actions they have developed with that domain) are the *main determinant of the thinking within each progression, although hierarchic interactions occur at multiple levels* within and between topics, as well as with general cognitive processes (e.g., metacognitive processes, potentialities for general reasoning and learning-to-learn skills).
3. *Hierarchic development.* This key tenet contains two ideas. First, development is less about the emergence of entirely new processes and products and more an interactive interplay among specific components of knowledge and processes (both general and specific; Minsky, 1986). Second, each level builds hierarchically on the concepts and processes of the previous levels. Levels are coherent and

- often characterized by increased sophistication, complexity, abstraction, power, and generality. However, the learning process is more often incremental and gradually integrative than intermittent and tumultuous (i.e., occurring mainly between stable levels). Various models and types of thinking grow in tandem to a degree, but a critical mass of ideas from each level must be constructed before thinking characteristic of the subsequent level becomes ascendant in the child's thinking and behavior (Clements, Battista, & Sarama, 2001). Successful application leads to the increasing use of a particular level. However, under conditions of increased task complexity, stress, or failure this probability level decreases, and an earlier level serves as a fallback position (Hershkowitz & Dreyfus, 1991; Siegler, 1986). No level of thinking is deleted from memory (Davis, 1984). That is, rerecording a mental representation at a more explicit level does not erase the earlier representation. (Indeed, replacing or deleting any successful representations would eliminate "fall-back" strategies that arguably are essential when new, untested knowledge is being formed, cf. Minsky, 1986; Vurpillot, 1976). The schemes that constitute the representation are instantiated dynamically (i.e., their form is interactively shaped by the situational context, including the task demands), and they can be modified at this time (or new schemes can be formed), although schemes that are basic subschemes of many other schemes become increasingly resistant to alteration because of the disruptive effect alteration would have on cognitive functioning. The continued existence of earlier levels and the role of intentionality and social influences in their instantiation explain why in certain situations even adults fall back to earlier levels, for example, failing to conserve in some contexts.
4. *Cyclic concretization.* Development progressions often proceed from sensory-concrete levels at which perceptual concrete supports are necessary and reasoning is restricted to limited cases (such as small numbers) to verbally based (or enhanced) generalizations and abstractions that are tenuous to integrated-concrete understandings relying on internalized mental representations that serve as mental models for operations and abstractions that are increasingly sophisticated and powerful.
 5. *Co-mutual development of concepts and skills.* Concepts constrain procedures, and concepts and skills develop in constant interaction (Baroody, Lai, & Mix, 2005; Greeno, Riley, & Gelman, 1984). (In imbalanced cultural or educational environments, one may take precedence, to an extent that can be harmful.) Effective instruction often places initial priority on conceptual understanding, including children's creations of solution procedures (e.g., Carpenter, Franke, Jacobs, Fennema, & Empson, 1998), the importance of autonomy (Kamii & Housman, 1999), and mathematical dispositions and beliefs ("Once students have learned procedures without understanding, it can be difficult to get them to engage in activities to help them understand the reasons underlying the procedure," Kilpatrick, Swafford, & Findell, 2001, p. 122). The constructs of concepts and skills include symbolic representations, utilization competence (Greeno, Riley, & Gelman, 1984), and general cognitive skills.
 6. *Initial bootstraps.* Children have important but often inchoate premathematical and general cognitive competencies and predispositions at birth or soon thereafter that support and constrain but do not absolutely direct subsequent development of mathematics knowledge. These have been called *experience-expectant processes* (Greenough, Black, & Wallace, 1987), in which universal experiences lead to an interaction of inborn capabilities and environmental inputs that guide development in similar ways across cultures and individuals (cf. Karmiloff-Smith, 1992).
 7. *Different developmental courses.* Different developmental courses are possible within those constraints, depending on individual and social confluences (Clements, Battista, & Sarama, 2001). Within any developmental course, at each level of development, children have a variety of cognitive tools—concepts, strategies, and skills—that coexist.
 8. *Progressive hierarchization.* Within and across developmental progressions, children gradually make connections between various mathematically relevant concepts and procedures, weaving ever more robust understandings that are hierarchical (i.e., they employ generalizations while maintaining differentiations). These generalizations and metacognitive abilities eventually connect

to form logical-mathematical structures that virtually compel children toward decisions in certain domains, such as those on traditional Piagetian conservation of number tasks, that are resistant to confounding via misleading perceptual cues. Children provided with high-quality educational experiences build similar structures across a wide variety of mathematical domains (again, in contrast to instrumental knowledge).

9. *Environment and culture.* Environment and culture affect the pace and direction of the developmental courses. For example, the degree of experience children have to observe and use number and other mathematical notions and to compare these uses will affect the rate and depth of their learning along the developmental progressions. The degree to which children learn mathematical words, exposure to which varies greatly across cultural groups (Hart & Risley, 1995), affects developmental courses. Words alert children to the class of related words to be learned and to specific mathematical properties, laying the foundation for learning mathematical concepts and language (cf. Sandhofer & Smith, 1999) by providing a nexus on which to build their nascent constructs (Vygotsky, 1934/1986).
10. *Consistency of developmental progressions and instruction.* Instruction based on learning consistent with natural developmental progressions is more effective, efficient, and generative for the child than learning that does not follow these paths.
11. *Learning trajectories.* An implication of the tenets to this point is that a particularly fruitful instructional approach is based on hypothetical learning trajectories (Clements & Sarama, 2004c). On the basis of the hypothesized specific mental constructions (mental actions-on-objects) and patterns of thinking that constitute children's thinking, curriculum developers design instructional tasks that include external objects and actions that mirror the hypothesized mathematical activity of children as closely as possible. These tasks are sequenced, with each corresponding to a level of the developmental progressions, to complete the hypothesized learning trajectory. Such tasks will theoretically constitute a particularly efficacious educational program; however, there is no implication that the task sequence is the only

path for learning and teaching, only that it is hypothesized to be one fecund route.

12. *Instantiation of hypothetical learning trajectories.* Hypothetical learning trajectories must be interpreted by teachers and are only realized through the social interaction of teachers and children around instructional tasks. Societally determined values and goals are substantive components of any curriculum (Confrey, 1996; Hiebert, 1999; National Research Council [NRC], 2002; R. W. Tyler, 1949); research cannot ignore or determine these components (cf. Lester & Wiliam, 2002). There is no understanding without reflection, and reflection is an activity students have to carry out themselves. No one else can do it for them. Yet a teacher who has some inkling as to where a particular student is in his or her conceptual development has a better chance of fostering a further reflective abstraction than one who merely follows the sequence of a preestablished curriculum. (von Glasersfeld, 1995)

NUMBER AND QUANTITATIVE THINKING

For early childhood, number and operations is arguably the most important area of mathematics learning. In addition, learning of this area may be one of the best developed domains in mathematics research, especially in the early primary grades (Baroody, 2004; Fuson, 2004; Kilpatrick, Swafford, & Findell, 2001; Steffe, 2004). Preschool mathematics education, although an emergent area, benefits from a wealth of psychological and early childhood studies.

We discuss children's numerical concepts and operations separately, although they and their components are highly interrelated. We do not limit "operations" to standard arithmetic operations of adding, subtracting, multiplying, and dividing but include counting, comparing, unitizing, grouping, partitioning, and composing. The importance of these ideas is highlighted by their close correspondence to the components of number sense, including composing and decomposing numbers, recognizing the relative magnitude of numbers, dealing with the absolute magnitude of numbers, using benchmarks, linking representations, understanding the effects of arithmetic operations, inventing strategies, estimating, and possessing a disposition toward making sense of numbers (Sowder, 1992). The competencies rest on early quantitative reasoning that begins to develop as early

as the first 2 years of life. Preceding cognitive psychology research concerning early numbers in infants by half a century, Vygotsky said

The first stage is formed by the natural arithmetical endowment of the child, i.e., his operation of quantities before he knows how to count. We include here the immediate conception of quantity, the comparison of greater and smaller groups, the recognition of some quantitative group, the distribution into single objects where it is necessary to divide, etc. (Vygotsky, 1929/1994, p. 67)

Introduction to the Early Development of Quantity

In 1917, Warren McCulloch's academic advisor asked him to describe his research interests. He replied, "What is number, that a man [*sic*] may know it, and a man that he may know number?" His advisor responded, "Friend, thee will be busy as long as thee lives" (McCulloch, 1963, p. 1).

Over more than 100 years, research on early number knowledge has passed through four broad phases (cf. Clements, 1984b). After a prescient analysis of early number by Dewey (1898), researchers studied subitizing (the direct and rapid perceptual apprehension of the numerosity of a group, from the Latin "to arrive suddenly," named by Kaufman, Lord, Reese, & Volkman, 1949, who referred to verbal naming of the numerosity), counting, and the relationship between subitizing and counting (e.g., Douglass, 1925; Freeman, 1912).

In the second phase, the wide influence of Piagetian theory redirected theoretical and empirical study, as it explained the development of number concepts based on underlying logical operations.

Our hypothesis is that the construction of number goes hand-in-hand with the development of logic, and that a pre-numerical period corresponds with the pre-logical level. Our results do, in fact, show that number is organized, stage after stage, in close connection with the gradual elaboration of systems of inclusions (hierarchy of logical classes) and systems of asymmetrical relations (qualitative seriations), the sequence of numbers thus resulting from an operational synthesis of classification and seriation. In our view, logical and arithmetical operations therefore constitute a single system that is psychologically natural, the second resulting from generalization and fusion of the first, under the two complementary headings of inclusion of classes and seriation of relations, quality being disregarded (Piaget & Szeminska, 1952, p. viii).

Counting was viewed as ineffectual, with "no connection between the acquired ability to count and the actual operations of which the child is capable" (Piaget & Szeminska, 1952, p. 61), with the child *capable* of meaningful counting only upon reaching the level of reversible operations (p. 184). Studies in this tradition tended to disregard subitizing "the intuitive numbers" as well as counting, which had been the major focus of earlier studies of number concepts. Instead, Piagetians focused on number conservation, which they considered the sine qua non of all rational (i.e., based on reason) number activity. Number conservation is the ability to state quickly and assuredly that the numerosity of a collection above subitizing range (e.g., more than 7) had not changed after a change in the relationship among its elements, such as increasing the distance between them. In the Piagetian view, children do not acquire a notion of quantity and then conserve it; they discover true quantification only when they become capable of conservation. This develops in three stages: Stage 1, gross quantity, in which children make global perceptual judgments without one-to-one correspondence; Stage 2, intensive quantity, in which they can make one-to-one correspondences perceptually; and Stage 3, in which they construct the notion of unit and numerical correspondence and they understand that inverse changes in length and density can compensate each other and thus that changes are reversible. Spatial qualities no longer determine number. The elements become equivalent and interchangeable units (i.e., equivalent members of a class), differing only by their relative order (i.e., seriation). Each successive element creates a category containing all previous classes (i.e., hierarchical inclusion).

The result was a broad belief, in research and educational practice arenas, that children could not reason logically about quantities until the early elementary school years. This view seemed to be supported by hundreds of investigations of the validity of the Piagetian stages. Studies generally supported the existence of the stages (Clements, 1984a). For example, children scoring lower on tests of conservation scored significantly lower on tests of problem solving (Harper & Steffe, 1968; Steffe, 1966).

The third phase began when new perspectives and research methodologies challenged critical aspects of the Piagetian view. Researchers questioned whether the Piagetian tasks measured number knowledge or other competencies, such as attending to relevant attributes (R. Gelman, 1969). Others doubted the Piaget position that without logical operations there is no possibility of meaningful quantitative reasoning and therefore attempting to develop children's number abilities, such as counting, was indefensible.

Supporting the new perspective, seminal research showed numerical and arithmetic competencies in young children who were not operational in Piagetian assessments (R. Gelman, 1969; Harper & Steffe, 1968). For example, in the “magic” experiments, two plates were shown, one with, for example, 2 mice, designated “the loser,” and the other with three mice, designated “the winner.” After a series of identification tasks, the experimenter surreptitiously altered the winner, changing the spatial arrangement in some experiments, altering the identity of the items or size of the collection in others. The result we emphasize here is that children as young as 3 and sometimes 2.5 years seem to know that transformations involving displacements do not change the numerical value of a display—an early form of conservation of number (R. Gelman & Gallistel, 1978). Children preferred to make such decisions based on equivalence or non-equivalence of verbal numbers, rather than on the 1-to-1 object correspondences emphasized in Piagetian studies. Such studies formed the basis for a nativist view of early number development.

Other researchers built number skills-integration models. Taking an information-processing approach, these empiricist-oriented models used task and scalogram analyses to describe sequences of learning numerals, counting and numeration of small before larger collections, and independence of counting and one-to-one correspondence (M. Wang, Resnick, & Boozer, 1971). Other models postulated hierarchic skills-integration sequences (Klahr & Wallace, 1976; Schaeffer, Eggleston, & Scott, 1974). Numerous studies supported the contention of a multi-dimensional concept of number in young children (Clements, 1984a).

The fourth phase is an extension of the third phrase. Debates continue about the meaning of early (and lack of) competence (Baroody, Lai, & Mix, 2006; Cordes & Gelman, 2005; Mix, Huttenlocher, & Levine, 2002; T. J. Simon, 1997; Spelke, 2003; Uller, Carey, Huntley-Fenner, & Klatt, 1999). Some researchers have attempted to synthesize components of the various theoretical perspectives or create intermediate positions. Nativist, empiricist, and interactionalist accounts have been evaluated both philosophically and empirically. Number knowledge similarly has been subjected to analysis, resulting in more sophisticated conceptualizations. For example, number and quantitative knowledge apparently develops substantially earlier than Piagetian logical operations but nonetheless appears to be built upon inchoate foundations, such as unitizing or creating an image of a collection of objects (classification) and tracking and ordering (seriation).

Issues concerning nativist, empiricist, and interactionalist accounts arise in each of the following main sections. In the remainder of this section, we illustrate these issues by focusing on children’s initial competence with number. For example, using new techniques, some researchers claimed that infants possessed understanding of cardinality. Many of these researchers built on Gelman and Gallistel’s nativist position, including the privileged domains of whole-number counting and simple arithmetic, which necessitated an underlying foundation of both cardinality and ordinality. The hypothesis predicts that conceptual understanding precedes skilled execution of the relevant procedures in these domains (R. Gelman & Gallistel, 1978). In contrast, the empiricist skills-integration approach would predict that exposure frequency determines development, and thus procedural skill would emerge before conceptual development.

Providing evidence for their position, nativists reported that babies in the first 6 months of life can discriminate one object from two, and two objects from three (Antell & Keating, 1983; Starkey, Spelke, & Gelman, 1990). This was determined via a habituation paradigm in which infants “lose interest” in a series of displays that differ in size, density, color, and so forth, but have the same number of objects. Renewed interest when shown a display with a different number of objects provides evidence that they are sensitive to number (Wynn, Bloom, & Chiang, 2002). Thus, infants can discriminate among and match small configurations (1–3) of objects. The experiments also indicate that children’s discrimination is limited to such small numbers. Children do not discriminate four objects from five or six until the age of 3 or 4 years (Starkey & Cooper, 1980). Some researchers have therefore suggested that infants use subitizing, defined as an automatic perceptual process that all people, including adults, can apply only to small collections up to around four objects (Chi & Klahr, 1975). Children can also construct quantitative equivalence relations by 6 to 8 months of age through exchange operations (substituting or commuting objects; Langer, Rivera, Schlesinger, & Wakeley, 2003).

Some initial competence is generally accepted. What this competence *means*, however, remains in dispute. Recent findings suggest that infants in “number” experiments may be responding to overall contour length, area, mass, or density rather than discrete number (Feigenson, Carey, & Spelke, 2002; Tan & Bryant, 2000). For example, infants dishabituated to changes in contour length when the number of objects was held constant, but they did not dishabituate to changes in number when contour length was held constant (Clearfield & Mix, 1999). One model pro-

poses that infants encode object files that store data on each object's properties, so that situations can be processed in terms of individuation or analog properties of these objects such as contour, depending on the situation (Feigenson, Carey, & Spelke, 2002). For example, parallel-processed individuation is used for very small collections, with continuous extent used when storage for individuation is exceeded.

The strongest claim that infants are processing number, rather than perceptual information, is based on two types of evidence. First is the claim that infants discriminate not only collections of objects, but also temporal sequences such as sounds or events like puppet jumps as early as 6 months of age (Wynn, 1998). Second is the claim that infants show cross-modal number abilities; that is, they can match visual representations of certain numbers and auditory sequences consisting of the same number of sounds (Starkey, Spelke, & Gelman, 1990). Such evidence has led to strong nativist claims. More recent evidence, however, has called these results and claims into question. Even if children do sometimes make these connections, general, nonnumerical processes of comparing inputs from different modalities (e.g., from sounds and their locations) may be used (T. J. Simon, 1997). More important, one team found opposing results—either no effect of auditory information and merely a tendency to prefer the larger contour (consistent with several previous studies), or longer looking at *noncorresponding* displays (Moore, Benenson, Reznick, Peterson, & Kagan, 1987). In another study, children as old as 3 years performed at chance on an auditory-visual equivalence test (Mix, Huttenlocher, & Levine, 1996). There is no theoretical reason to justify why children more than 3 years old would have difficulty with tasks on which neonates ostensibly show competence. Consistent with these findings, more recent data found no evidence that infants notice quantitative equivalence between auditory sequences and visual displays (Mix, Levine, & Huttenlocher, 1997). Thus, there is little reason to believe that infants have the ability to deal with cross-modal quantitative correspondence or even discrete quantity (i.e., number).

This is not to say that empiricist positions are supported. Early quantitative competencies have been demonstrated. However, many, if not all, of the earliest competencies may be explicable through other frameworks. Further, how infants may use a variety of processes and how they may be weighted or combined, as well as how they may be evoked differently in various situations, have not been thoroughly studied (Baroody, Lai, & Mix, 2005). Researchers also need to learn more about what role such processes, even if they are numerical, play in children's early cognition

or development of later numerical activity (Nunes & Bryant, 1996).

Part of the problem is, of course, that infant studies are particularly difficult to perform and interpret. For example, the frequently employed habituation methodology, although useful, was developed to study perceptual, not conceptual issues, leaving open the question of what results mean (Haith & Benson, 1998). Nevertheless, research suggests that children discriminate collections on some quantitative bases from birth. Furthermore, most accounts suggest that these limited capabilities, with as yet undetermined contributions of maturation and experience, form a foundation for later connection to culturally based cognitive tools such as number words and the number word sequence, to develop exact and extended concepts and skills in number. We elaborate these issues in each of several areas of numerical and quantitative competence.

Subitizing

Early Work on Subitizing

In the first half of the 20th century, researchers believed that counting did not imply a true understanding of number but subitizing did (e.g., Douglass, 1925). Many saw the role of subitizing as a developmental prerequisite to counting. Freeman (1912) suggested that whereas measurement focused on the whole and counting focused on the unit item, only subitizing focused on both the whole and the unit—thus, subitizing underlies number ideas. Carper (1942) agreed subitizing was more accurate than counting and more effective in abstract situations. (Recall that the term “subitizing” was not created until later, Kaufman, Lord, Reese, & Volkman, 1949.)

In the second half of the century, educators developed several models of subitizing and counting. Some models were based on the notion that subitizing was a more “basic” skill than counting (Klahr & Wallace, 1976; Schaeffer, Eggleston, & Scott, 1974). One reason was that children can subitize directly through interactions with the environment, without social interactions. Supporting this position, Fitzhugh (1978) found that some children could subitize sets of one or two but were unable to count them. None of these very young children, however, were able to count any sets that they could not subitize. She concluded that subitizing is a necessary precursor to counting. Research with infants similarly suggested that young children possess and spontaneously use subitizing to represent the number contained in small sets and that subitizing emerges before counting (A. Klein & Starkey, 1988).

As logical as this position seems, counterarguments exist. In 1924, Beckmann found that young children used counting rather than subitizing (cited in Solter, 1976). Others agreed that children develop subitizing later, as a shortcut to counting (Beckwith & Restle, 1966; Brownell, 1928; Silverman & Rose, 1980). In this view, subitizing is a form of rapid counting (R. Gelman & Gallistel, 1978).

Evidence for subitizing as a separate process from “fast counting” includes reaction time data showing that recognition of 1–3 objects is faster and more accurate than larger numerosities. Reaction times increase with the number of items in a set to be quantified; it takes adults about 250 to 350 ms longer to quantify a set of 7 than 6, or 6 than 5. In contrast, the increase is slight when numbers are small, about 40 to 100 ms longer to respond to 3 than 2, or 2 than 1, and this may be response choice time rather than time to process the stimuli. At least some data indicate that adults can process 3 within the same exposure time as 2 or 1 (Trick & Pylyshyn, 1994).

Nature of the Subitizing Process and the Early Development of Quantity

Researchers still debate the basis for subitizing ability. Some claim that there are two distinct types of enumeration: numerical subitizing and nonnumerical (or figural) subitizing. In addition, some models assume that subitizing is possible only for sets that are simultaneously displayed, whereas others allow for sequential enumeration (Canfield & Smith, 1996).

Recognition of spatial patterns, such as triangular arrays, and attentional mechanisms are the main explanations for those who posit an underlying non-numerical process (Chi & Klahr, 1975; Klahr & Wallace, 1976; Mandler & Shebo, 1982; von Glasersfeld, 1982). For Mandler and Shebo, these abstract geometric patterns are mapped to specific numerosities using processes like those for recognizing colors. For von Glasersfeld, the patterns are constituted by motion, either physical or attentional, forming scanpaths. They become numerical after reflective abstraction raises them to “pure” abstraction. In another model, objects are first individuated by a limited but parallel processing preattentive mechanism associated with object tracking, and then markers are enumerated in a serial process (Trick & Pylyshyn, 1994). There is a neural mechanism that distinguishes single, dual, or triple incidences of a given evident or object, based on inherent temporal parameters.

Other models consider subitizing to be a numerical process. In the Meck and Church model (1983) subitizing is a numerical process enabled by the avail-

ability of the functional equivalent of a number line in the brain that operates on both simultaneous and sequential items. A pacemaker emits equivalent pulses at a constant rate. When a unitized item is encountered, a pulse is allowed to pass through a gate, entering an accumulator. The gradations on the accumulator estimate the number in the collection of units, similar to height indicating numerosity in a stack of blocks. An alternative model postulates an evolutionarily based, abstract mental component dedicated to number perception; that is, the child perceives numbers directly (Dehaene, 1997). This counting-like process guides development of whole number counting, hypothesized to be a privileged domain. Findings from both humans and nonhuman animals support this position (Gallistel & Gelman, 2005).

Research using different methodologies confirms that infants rapidly recognize and use the number of sequentially presented pictures to predict the location of the next picture in the sequence (Canfield & Smith, 1996). There was no evidence they had to reaccumulate items by scanning working memory, supporting subitizing rather than fast-counting models. (G. A. Miller, 1956, agreed that his famous working-memory limit was not related to human subitizing limits.) Thus, the researchers (Canfield & Smith, 1996) judged the Meck and Church model (1983) most consistent with their results. Because infants never saw more than a single stimulus at any moment during the session and the items were presented in a manner that was neither rhythmic nor temporally predictable, the process of combining sequential events was viewed as numerical.

Recent reviews (Baroody, Lai, & Mix, 2005; Mix, Huttenlocher, & Levine, 2002) have concluded that children younger than 3 years tend not to represent any numbers except 1 and 2 precisely. For example, 3-day-old to 5-month-old infants could discriminate between collections of one and two, two and three but not between collections of three and four (Antell & Keating, 1983), and 10- to 16-month-olds successfully discriminated between 2 and 3 but not between 3 and 4 or even 2 and 4 or 3 and 6 (Feigenson, Carey, & Hauser, 2002). Even in studies that did show success, results are equivocal; for example, 10–12-month-olds could discriminate between 2 and 3 items, but not 4 and 5, with mixed results for 3 vs. 4 (Strauss & Curtis, 1984); performance of even 2-year-olds drops off on tasks involving 3 items, with above-chance levels not necessarily indicating exact representations (Starkey & Cooper, 1995).

Developmental differences may exist in the processes that underlie subitizing (cf. Resnick & Singer, 1993). The mental models view (Huttenlocher, Jor-

dan, & Levine, 1994; Mix, Huttenlocher, & Levine, 2002) postulates that children represent numbers nonverbally and approximately, then nonverbally but exactly, and eventually via verbal, counting-based processes. (This in contrast to the accumulator model, which is seen as unable to account for the greater difficulty of sequential presentations, use of overall amount in early quantification, and representation of cardinality, Mix, Huttenlocher, & Levine, 2002.) In the mental models view, children cannot initially differentiate between discrete and continuous quantities but represent both approximately using one or more perceptual cues such as contour length (Mix, Huttenlocher, & Levine, 2002). Children develop the ability to individuate objects, providing the ability to build notions of discrete number. About the age of 2 years, they develop representational, or symbolic, competence (such as shown in symbolic play), allowing them to create mental models of collections, which they can retain, manipulate (move), add to or subtract from, and so forth (although the model does not adequately describe how cardinality is ultimately cognized and how comparisons are made). Early nonverbal capabilities then provide a basis for the development of verbally based numerical and arithmetic knowledge (young children are more successful on nonverbal than verbal versions of number and arithmetic tasks, Huttenlocher, Jordan, & Levine, 1994; Jordan, Hanich, & Uberti, 2003; Jordan, Huttenlocher, & Levine, 1992; Jordan, Huttenlocher, & Levine, 1994; Levine, Jordan, & Huttenlocher, 1992). (One difficulty with this argument is that calling both “arithmetic tasks” may unintentionally hide potential differences in children’s schemes.) In this view, there is no reason to consider early quantitative development solely a number competence, much less assume that number is a privileged domain (Mix, Huttenlocher, & Levine, 2002).

A recent version of this theory (Baroody, Lai, & Mix, 2005) suggests that during the time when children are gaining representational precision, they are also moving toward more generalized but deeper (more interconnected) concepts. Approximate mental models serve as a transition between number based on perceptual cues and one based on an exact, abstract, mental model. Implicit distinctions between discrete and continuous quantities lie on a continuum from clearly distinct for collections of one to indistinct in the case of “many”—more than three or four. Meaningful learning of number words (in contrast to symbolic ability) causes the transition to exact numerical representations. Such development may apply to certain numbers at different times, even among very small numbers. The child can then begin to verbally represent numbers (beyond one and two, which may

be represented nonverbally and exactly and verbally at about the same time). This may provide the basis for understanding cardinality and other counting principles, as well as arithmetic ideas (Baroody, Lai, & Mix, 2006). Anywhere from 3 to 6 years, depending on the home and preschool environments, children also make the transition to using written representations, which help them develop numerical or abstract reasoning.

Research reviewed here is generally consistent with this position, with only a few modifications and additions. What infants quantify are collections of rigid objects; that is, sequences of sounds and events, or materials that are nonrigid and noncohesive (e.g., water), are not quantified (Huntley-Fenner, Carey, & Solimando, 2002). Such quantifications, including number, begin as an undifferentiated, innate notion of amount of objects. Object individuation, which occurs early in preattentive processing (and is a general, not numerical, process), helps lay the groundwork for differentiating discrete from continuous quantity. Multiple systems are employed, including an object-file system that stores information about the objects, some or all of which is used depending on the situation, and an estimator mechanism that stores analogue quantitative information only (Feigenson, Carey, & Spelke, 2002; Gordon, 2004; Johnson-Pynn, Ready, & Beran, 2005). This estimator may be a set of filters (a cognitive scheme detects the numerosity of a group), each tuned to an approximate number of objects (e.g., 3) but overlapping (Nieder, Freedman, & Miller, 2002). The child encountering small sets opens object-files for each in parallel. If the situation elicits quantitative comparisons, continuous extent is retrieved and used except in extreme circumstances. For example, by about 6 months of age, infants may represent very small numbers (1 or 2) as individuated objects. Conversely, large numbers in which continuous extent varies or is otherwise unreliable (McCrink & Wynn, 2004) may be processed by the analog estimator as a collection of binary impulses (as are event sequences later in development), but not by exact enumeration (Shuman & Spelke, 2005).

To compare quantities, correspondences are processed. Initially, these are inexact estimates comparing the results of two estimators, depending on the ratio between the sets (Johnson-Pynn, Ready, & Beran, 2005). Once the child can represent objects mentally, that child can also make exact correspondences between these nonverbal representations and eventually develop a quantitative notion of that comparison (Baroody, Lai, & Mix, 2005).

Even these correspondences, however, do not imply a cardinal representation of the collection (a

representation of the collection qua a *numerosity* of a group of items). That is, one still must distinguish between noncardinal representations of a collection and an explicit cardinal representation. Only for the latter does the individual apply an integration operation to create a composite with some numerical index. Some claim that the accumulator yields a cardinal output; however, it may be quantitative, constituted as an indexing of a collection using an abstract, cross-modality system for numerical magnitude (Shuman & Spelke, 2005) without an explicit cardinality. Comparisons, such as correspondence mapping, might still be performed, but only at an implicit level (cf. Sandhofer & Smith, 1999). In this view, only with experience representing and naming collections is an explicit cardinal representation created. This is a prolonged process. Children may initially make word-word mappings between requests for counting or numbers (e.g., “how many?”) to number words until they have learned several (Sandhofer & Smith, 1999). Then they label some (small number) cardinal situations with the corresponding number word, that is, map the number word to the numerosity property of the collection. They begin this phase even before 2 years of age, but for some time this applies mainly to the word *two*, a bit less to *one*, and with considerable less frequency, *three* and *four* (Fuson, 1992). Only after many such experiences do children abstract the numerosities from the specific situations and begin to understand that the situations named by *three* correspond; that is, they begin to establish what adults would term a numerical equivalence class. Counting-based verbal systems are then more heavily used and integrated, as described in the following section, eventually leading to explicit, verbal, mathematical abstractions.

In summary, early quantitative abilities exist, but they may not initially constitute systems that can be said to have an explicit number concept. Instead, they may be premathematical, foundational abilities (cf. Clements, Sarama, & DiBiase, 2004) that develop and integrate slowly, in a piecemeal fashion (Baroody, Benson, & Lai, 2003). For example, object individuation must be stripped of perceptual characteristics and understood as a perceptual unit item through abstracting and unitizing to be mathematical (Steffe & Cobb, 1988), and these items must be considered simultaneously as individual units and members of a collection whose numerosity has a cardinal representation to be numerical, even at the lowest levels. The explicit, cultural, numeral-based sense of number develops in interaction with, but does not replace (indeed, may always be based on, Gallistel & Gelman, 2005), the analog sense of number. Regardless of its origins in continuous or discontinuous processing, number rather

than amount of substances does achieve a core status in quantitative reasoning for preschoolers to adults (K. F. Miller, 1995). The human facility with language probably plays a central role in linking relations between different representations and thus making early premathematical cognition numeric (Wiese, 2003).

Types of Subitizing

Regardless of the precise mental processes, subitizing appears to be phenomenologically distinct from counting and other means of quantification and deserves differentiated educational consideration. *Perceptual subitizing* (Clements, 1999b) is closest to the original definition of subitizing: recognizing a number without consciously using other mental or mathematical processes and then naming it. Perceptual subitizing employs a preattentive quantitative process but adds an intentional numerical process; that is, infant sensitivity to number is not (yet) perceptual subitizing. The term *perceptual* applies only to the quantification mechanism as phenomenologically experienced by the person; the intentional numerical labeling, of course, makes the complete cognitive act conceptual.

Perceptual subitizing also plays the primitive role of *unitizing*, or making single “things” to count from the stream of perceptual sensations (von Glasersfeld, 1995). “Cutting out” pieces of experience, keeping them separate, and eventually coordinating them with number words are not trivial tasks for young children. For example, a toddler, to recognize the existence of a plurality, must focus on the items such as apples and repeatedly apply a template for an apple *and* attend to the *repetition* of the template application.

How is it that people see an 8-dot domino and “just know” the total number, when evidence indicates that this lies above the limits of perceptual subitizing? They are using the second type of subitizing (a distinction for which there is empirical evidence, Trick & Pylyshyn, 1994). *Conceptual subitizing* (Clements, 1999b) plays an advanced organizing role. People who “just know” the domino’s number recognize the number pattern as a composite of parts and as a whole. They see each side of the domino as composed of four individual dots and as “one four.” They see the domino as composed of two groups of four and as “one eight.” These people are capable of viewing number and number patterns as units of units (Steffe & Cobb, 1988). (Some research suggests that only the smallest numbers, perhaps up to 3, are actually perceptually recognized; thus, sets of 1 to 3 may be perceptually recognized, sets of 3 to about 6 may be decomposed and recomposed without the person being aware of the process. Conceptual subitizing as we use the term

here refers to recognition in which the person consciously uses such partitioning strategies.)

Spatial patterns such as those on dominoes are just one kind. Other patterns are temporal and kinesthetic, including finger patterns, rhythmic patterns, and spatial-auditory patterns. Creating and using these patterns through conceptual subitizing helps children develop abstract number and arithmetic strategies (Steffe & Cobb, 1988). For example, children use temporal patterns when counting on. "I knew there were 3 more so I just said, nine . . . *ten, eleven, twelve*" (rhythmically gesturing three times, one "beat" with each count). They use finger patterns to figure out addition problems. Children who cannot subitize conceptually are handicapped in learning such arithmetic processes. Children who can, may only do so with small numbers at first. Such actions, however, can be "stepping stones" to the construction of more sophisticated procedures with larger numbers.

Factors Affecting Difficulty of Subitizing Tasks

In contrast to what might be expected from a view of innate subitizing ability, subitizing ability *develops*, in a stepwise fashion. That is, in laboratory settings, children can initially differentiate "one" from "more than one" at about 33 months of age (Wynn, 1992b). Between 35 and 37 months, they differentiate between one and two, but not larger numbers. A few months later, at 38 to 40 months, they identify three as well. After about 42 months, they can identify all numbers that they can count, four and higher, at about the same time. However, research in natural, child-initiated settings shows that the development of these abilities occurs much earlier, with children working on one and two around their second birthdays or earlier (Mix, Sandhofer, & Baroody, 2005). Further, some children may begin with "two" rather than "one." These studies intimate that language and social interactions interact with internal factors in development, as well as showing that number knowledge develops in levels, over time. Most studies suggest that children begin recognizing "one," then "one" and "two," then "three" and then "four," whereupon they learn to count and know other numbers (see R. Gelman & Butterworth, 2005, for an opposing view concerning the role of language; Le Corre, Van de Walle, Brannon, & Carey, 2005).

The spatial arrangement of sets also influences how difficult they are to subitize. Children usually find rectangular arrangements easiest, followed by linear, circular, and scrambled arrangements (Beckwith & Restle, 1966; M. Wang, Resnick, & Boozer, 1971). This is true for students from kindergarten to college.

Certain arrangements are easier for specific numbers. Arrangements yielding a better "fit" for a given number are easier (Brownell, 1928). Children make fewer errors for 10 dots than for 8 with the "domino five" arrangement, but fewer errors for 8 dots for the "domino four" arrangement.

For young children, however, neither of these arrangements is easier for any number of dots. Indeed, children 2- to 4-years-old show no differences between any arrangements of four or fewer items (Potter & Levy, 1968). For larger numbers, the linear arrangements are easier than rectangular arrangements. It may be that many preschool children do not use decomposing (conceptual subitizing). They can learn to conceptually subitize, though older research indicated that first graders' limit for subitizing scrambled arrangements is about four or five (Dawson, 1953).

Textbooks often present sets that discourage subitizing. Their pictures combine many inhibiting factors, including complex embedding, different units with poor form (e.g., birds that are not compact as opposed to squares), lack of symmetry, and irregular arrangements (Carper, 1942; Dawson, 1953). Such complexity hinders conceptual subitizing, increases errors, and encourages simple one-by-one counting.

Role of Spontaneous Subitizing in Early Mathematics Development

A series of five studies indicated that the child's tendency to spontaneously focus on numerosity is a distinct, mathematically significant process (Hannula, 2005). Such tendency at 3 years predicted development of cardinality knowledge a year later. Focusing on numerosity was also related to counting and arithmetic skills, even when nonverbal IQ and verbal comprehension were controlled. Results suggested that spontaneous focus builds subitizing ability, which in turn supported the development of counting and arithmetic skills. Thus, children low in the tendency to spontaneously focus on numerosity in the early years are at risk for later mathematical failure (Hannula, 2005).

A quasi-experimental study (Hannula, 2005) showed that it is possible to enhance 3-year-old children's spontaneous focusing on numerosity and thus catalyze children's deliberate practice in numerical skills (cf. Ericsson, Krampe, & Tesch-Römer, 1993). Children in the experimental group showed increased tendency to focus on numerosity and develop cardinality competencies on a delayed posttest compared to children in the control group.

The most effective technique may be to begin naming very small collections with numbers after children have established names and categories for some physical properties such as shape and color (Sand-

hofer & Smith, 1999). Numerous experiences naming such collections help children build connections between quantity terms (number, how many) and number words, then build word-cardinality connections (•• is “two”) and finally build connections among the representations of a given number.

In summary, subitizing small numbers appears to proceed and support the development of counting ability (Le Corre, Van de Walle, Brannon, & Carey, 2005). Thus, it forms a foundation for all learning of number. Across development, numerical knowledge initially develops qualitatively and becomes increasingly mathematical. In subitizing, children’s ability to “see small collections” grows from preattentive but quantitative, to attentive perceptual subitizing, to imagery-based subitizing, to conceptual subitizing (Clements, 1999b; Steffe, 1992). Perceptual patterns are those the child can, and must, immediately see or hear, such as domino patterns, finger patterns, or auditory patterns (e.g., three beats). A significant advance is a child’s focusing on the exact number in these patterns, attending to the cardinality. Finally, children develop conceptual patterns, which they can operate on, as when they can mentally decompose a five pattern into two and three and then put them back together to make five again. These types of patterns may “look the same” on the surface but are qualitatively different. All can support mathematical growth and thinking, but conceptual patterns are the most powerful.

Verbal and Object Counting

As previously described, historically some (Dewey, 1898; Thorndike, 1922) have argued that initial mathematics education should emphasize counting, whereas others (e.g., Piaget, Inhelder, & Szeminska, 1960; Piaget & Szeminska, 1952; B. Russell, 1919) contended that counting was a rote skill until logical foundations were acquired. Even recent accounts treated counting, at least verbal counting, as a rote skill (Fuson, 1992; Ginsburg, 1977). However, number words can be meaningful in some contexts and can orient children to numerical meanings (Mix, Huttenlocher, & Levine, 2002). Further, without language, development of number appears to be severely restricted. For example, members of a tribe with only a “one-two-many” system of counting had remarkably poor performance on number tasks above three (Gordon, 2004). Even with language, without a *verbal counting system*, exact naming of and operations on number do not appear (Pica, Lemer, Izard, & Dehaene, 2004).

Verbal Counting

By 24 months of age, many toddlers have learned their first number word (typically “two”). Words for

larger collections usually appear after children use verbal counting. Depending on their early environment, children begin to try to count using verbal number names at age 2 or 3 years. Important developments in counting continue during the preschool years. Children from ages 2 to 5 years learn more of the system of number words (“one, two, three, ...”) due to a desire to count larger collections and a curiosity about the number word system itself (Baroody, 2004; Fuson, 1988, 2004; Griffin, 2004; Steffe, 2004). In this developmental progression, children learn verbal counting, reproducing the forward number word sequence.

We do not use the common phrase “rote counting” for verbal counting, because children have to learn the principles and patterns in the number system as coded in their natural language, at least for numbers words above 20 (Baroody, 1987; Fuson, 1992). Supporting evidence for this position is that children who could count when given a new point from which to start performed better on all numerical tasks, suggesting that fluent verbal counting does not depend primarily on rote factors, but rather on the recognition that the system is rule-governed (Pollio & Whitacre, 1970). That is, children learn that numbers derive order and meaning from their embeddedness in a system, and they learn a set of rules that allows the generation, not recall, of the appropriate sequence.

Further, cross-cultural studies indicate that learning of counting varies with the language in which the number system is learned (Ginsburg, Choi, Lopez, Netley, & Chi, 1997; Han & Ginsburg, 2001; K. F. Miller, Major, Shu, & Zhang, 2000; K. F. Miller, Smith, Zhu, & Zhang, 1995). For example, Chinese, like many East Asian languages, has a more regular sequence of number words than does English. In both English and Chinese, the numbers 1 through 10 are arbitrary and the numbers after 20 follow a regular pattern of naming the decade name and then the digit name (e.g., “twenty-one”). However, in Chinese (and in many Asian languages rooted in ancient Chinese), there are two important differences. The tens numbers directly mirror the single-digit number names (“two-tens” rather than “twenty”; “three-tens” rather than “thirty”), and the numbers from 11 to 20 also follow a regular pattern (comparable to “ten-one,” “ten-two,” etc.) instead of the obscure “eleven, twelve. . . .” Through 3 years of age, children in the various cultures learn 1 through 10 similarly; however, those learning English learn the “teens” more slowly and with more errors. Further, Asian number words can be pronounced more quickly, providing another significant cognitive advantage (Geary & Liu, 1996).

Overall, children learning Asian languages show a substantial advantage in learning verbal counting.

Support for the notion that this advantage is a direct consequence of language is provided by parallel findings that Asian children do *not* differ in other aspects of number knowledge, including counting small sets and solving simple numerical problems (Fuson, 1992; K. F. Miller, Major, Shu, & Zhang, 2000). One group of researchers abandoned studying U.S. kindergartners because they could not count consistently and accurately from 1 to 50, but Korean kindergartners had no such problems (Miura & Okamoto, 2003). Only U.S. children made errors such as “twenty-nine, twenty-ten, twenty-eleven...”; Chinese children do not make that kind of mistake (K. F. Miller, Major, Shu, & Zhang, 2000). Although all children tend to make more mistakes at decade boundaries, for Chinese-speaking children, these are largely limited to infrequent mistakes with numbers above 60. Conclusions must be made with caution, however, as other cultural factors, such as Asian parents’ emphasize on counting in the early years, may affect these results (Towse & Saxton, 1998; J. Wang & Lin, 2005).

The deleterious effects of less mathematically coherent languages are not limited to counting larger numbers but hamper children’s development of place value, multidigit arithmetic, and other concepts (Fuson, 1992). The effects are detrimental in surprising ways: Even the counting words from 1 to 9 are learned better by Chinese children because they practice these exact numbers when learning to count from 11 to 19 (Miura & Okamoto, 2003).

Fuson and colleagues traced the development of the number word sequence from its beginnings at age 2 to its general extension to the notion of base-ten systems at about the age of 8 (Fuson & Hall, 1982; Fuson, Richards, & Briars, 1982). Most middle-class children less than 3.5 years of age are working on learning the sequence to 10 (Fuson, 1992; Saxe, Guberman, & Gearhart, 1987). For the next year, they develop the sequence from 10 to 20. From 4.5 to 6 years, they still make errors in the teens, but most also develop the decades to 70, although a substantial number count to 100 and higher (Bell & Bell, 1988; Fuson, Richards, & Briars, 1982).

Fuson and colleagues also reported two distinct, overlapping phases: an acquisition phase of learning the conventional sequence of the number words, and an elaboration phase, during which this sequence is decomposed into separate words and relations upon these words are established. During the first phase, the sequence is a single, connected serial whole from which interior words cannot be produced singly. The most common overall structure of the sequences is (a) stable conventional, an initial group that is the beginning of the conventional sequence; (b) stable noncon-

ventional, a group that deviates from convention but is produced with some consistency, and (c) nonstable, a final group with little consistency. The acquisition of longer sequences consists of the extension of the sequence and the consolidation of the extension so that it is produced reliably.

At first, children can only produce the sequence by starting at the beginning. In the elaboration phase, there is a gradual process of constructing relations among these words. Five levels were differentiated and empirically supported (Fuson & Hall, 1982; Fuson, Richards, & Briars, 1982): (a) string level—words are not objects of thought nor “heard” as separate words, (b) unbreakable list level—words are heard and become objects of thought, (c) breakable chain level—parts of the chain can be produced starting from an arbitrary number (enabling certain counting on strategies), (d) numerical chain level—words become units that themselves can be counted, and (e) bidirectional chain level—words can be produced in either direction and the unitized seriated embedded numerical sequence allows part-whole relations and a variety of flexible strategies to be employed. These developments precede and enable changes in addition and subtraction solution strategies (Fuson, 1992).

Fuson and colleagues (1982) also claimed that backward verbal counting is learned in a slow and laborious manner, based on an already-mastered forward sequence. However, specific instruction may enable children to develop backward counting to almost the same extent as forward counting (B. Wright, 1991).

Object Counting

To count objects, children learn to coordinate the production of counting words with indicator actions, such as pointing to or moving objects (Dewey, 1898; Fuson, 1988; Ginsburg, 1977; Judd, 1927). More elaborately, a fully developed object counting scheme has four components. First, there is a situation that is recognized, say a collection of countable items. Second is a goal, to find out how many. Third is an activity, counting, and fourth, the result, a unitary whole of counted items (Steffe & Cobb, 1988).

The easiest type of collection for 3-year-olds to count has only a few objects arranged in a straight line and can be touched as children proceed with their counting. Between 3 and 5 years of age, children acquire more skill as they practice counting, and most become able to cope with numerically larger collections in different arrangements, without needing to touch or move objects during the act of counting.

As discussed previously, experiences in which the configurations are labeled with a number word by adults or older children (“Here are two blocks.”) en-

able children to build meaning for number words as telling how many items in a configuration or collection of items. The capstone of early numerical knowledge, and the necessary building block for all further work with number and operations, is connecting the counting of objects in a collection to the number of objects in that collection. Initially, children may not know how many objects there are in a collection after counting them. If asked how many are there, they typically count again, as if the “how many?” question is a directive to count rather than a request for how many items are in the collection. Children must learn that the last number word they say when counting refers to how many items have been counted. Many 3-year-olds do learn this result of counting: “One, two, three, four. There are four olives.”

How do such competencies develop? Gelman and Gallistel (1978) hypothesized that the ease and rapidity that even very young children display in learning to count indicated that such development was guided by knowledge of counting principles. In particular, they hypothesized that young children know five principles. The three how-to-count principles included the stable order principle (always assign the numbers in the same order), one-one principle (assign one and only one number word to each object), and cardinal principle (the last count indicates the number of objects in the set). The two what-to-count principles were the order irrelevance principle (the order in which objects are counted is irrelevant) and the abstraction principle (all the other principles can apply to any collection of objects). The researchers presented evidence indicating that children understand—explicitly or implicitly—all these principles by age 5, and many by age 3 years. For example, in the “magic” experiments described previously, all three how-to-count principles were followed (not perfectly, as the question was competence, not performance) by children as young as 2.5 years, at least with homogeneous, linearly arranged collections of no more than five items (R. Gelman & Gallistel, 1978). The one-one errors tended to be over- or undercounting on all set sizes, or skipping or double counting on the larger of the set sizing. Almost no children even used the same number word more than once.

As another example, even when children make mistakes, they show knowledge of the principle. For example, if they do not know the standard order of number words, the idiosyncratic order they *do* use is usually stable. Or, they may make mistakes in executing the one-one rule (Fuson, 1988) but assign exactly one number word to most of the objects (about 75% of the objects even for 2- and 3-year-olds). Thus, these might reflect performance errors, not conceptual lim-

itations. Preschoolers give evidence of using the cardinal principle by repeating or emphasizing the last number. They count events, collections that include different categories of objects, and even “missing” objects (e.g., eggs not in a carton). The order irrelevance principle is not demonstrated as widely, but most 5-year-olds will start counting in the middle of a row of objects and count each object.

The principles were expanded and implemented within a computational model in which it was assumed that implicit understanding of principles was characterized as conceptual competence, hypothesized to be a set of action schemata corresponding to those principles (Greeno, Riley, & Gelman, 1984). Theoretically, a main thrust of the nativist argument is that such principles guide children’s acquisition of counting skills and understandings. This was in contrast to Piagetians (e.g., Elkind, 1964; Kamii & DeVries, 1993), who maintained that counting errors occurred because the child lacked fundamental understandings, for example, recognizing the logical need to order objects.

Other researchers gave more importance to repeated, often massive, experience and demonstrations, modeling, or scaffolding from adults in learning counting competencies (e.g., Fuson, 1988). Supporting this view are findings that suggest children count skillfully before they understand the principles that underlie counting (Bermejo, 1996; Briars & Siegler, 1984; Frye, Braisby, Lowe, Maroudas, & Nicholls, 1989; Wynn, 1992b). From this perspective, children may abstract features that are common to all acts of counting (e.g., counting each object exactly once) and distinguish them from those that are incidental (e.g., starting at the end of a row). Another possibility is that children build on their verbal number recognition (subitizing) skill to abstract the counting principles from their own counting experience (Le Corre, Van de Walle, Brannon, & Carey, 2005) or to make sense of adult efforts to model counting procedures (Baroody, Lai, & Mix, 2006).

Additional research has clarified these issues, especially the main principles. Children improve on counting in a “stable order” but do not show signs of understanding the principle (Fuson & Hall, 1982; Fuson, Richards, & Briars, 1982). Researchers agree that children as young as 3 years usually display one-to-one correspondences in counting (Fuson, 1988). However, they do not do so in all situations. There are actually several related but distinct competencies, including coordinating number words with objects and keeping track of counted objects (Alibali & DiRusso, 1999). Even 5-year-olds show difficulty sequencing large disorganized collections. Further, coordination

of words with objects actually involves a chain of two types of correspondence, that between the production of a number word and an indicating act such as pointing, and another between that act and each item to be counted. At age 3 or 4, children violate the first type of correspondence in two main ways, by pointing to an object without saying a word and by pointing to an object and saying more than one number word. They violate the second by skipping over objects or by double-counting (see Fuson, 1988, for an elaboration of many less frequent errors). Gesturing helps children maintain both types of correspondences, keeping track and coordinating number words with objects (Alibali & DiRusso, 1999). The rhythmic, physical motions may focus children's attention on the individual items, aiding segmentation. Further, the use of touching may lessen the working memory demands of counting by externally representing some of the contents of working memory, such as marking the child's place in the set of objects. In summary, for these two principles, evidence does not provide strong support for the nativist view but does not disprove it; children could understand the principle and have difficulty carrying out the procedures.

Evidence regarding the cardinality principle is perhaps most inconsistent with a strong nativist view. Cardinality does not appear to be a component of many children's early counting in most situations (Fuson, 1988; Ginsburg & Russell, 1981; Linnell & Fluck, 2001; Schaeffer, Eggleston, & Scott, 1974). As previously stated, to these children, the purpose of counting is the action of enumeration (Fluck, 1995; Fluck & Henderson, 1996). Showing an even more limited understanding, some children, asked why they count, seem perplexed by the notion of counting as a purposeful activity (Munn, 1997). Fuson and colleagues posited that repeating the last number of a counting act is initially merely the last step in a chain of responses that children learn to meet adult requests (Fuson & Hall, 1982; Fuson, Richards, & Briars, 1982). Even when children respond correctly to the cardinality question, they may not fully understand it (Sophian, 1988). For example, asked to indicate which object(s) to which their last-number-word response referred, children often indicate only the last object. As another example, children give the last-number-word response even when it is incorrect (e.g., when they were asked to begin counting with a number other than one, Bermejo, 1996). Children may have to represent, or "redescribe," the final number word at an explicit level before it is available to serve the purpose of establishing a relationship between the counting act and cardinality (Karmiloff-Smith, 1992). This theory emphasizes internal processes, including visuo-spatial abilities, which play a greater role than language ability

in the development of cardinality in normally developing young children. However, social interaction probably aids this development (Linnell & Fluck, 2001).

This does not mean that cardinality is not understood, only that it is not an innate notion in counting. Instead, young children's earliest meaning for number words may be rooted in recognition or subitizing (Le Corre, Van de Walle, Brannon, & Carey, 2005). This may be generalized to counting, so that by 3.5 years, many children begin to consciously use and understand the cardinal notion in counting (Wynn, 1990, 1995), at least for small numbers, and most 5-year-olds do so consistently in simple counting situations. However, the idea still undergoes development. For example, younger children may not maintain cardinal and ordinal meanings in Piagetian conservation of number tasks in which the child builds a row of objects equivalent to a given row of objects. However, different tasks reveal greater competence. Many preschoolers, told the cardinal value of each of two sets, can determine whether the items would be put in one-to-one correspondence (and the inverse), providing evidence that they understand the relation between the number words and quantity and can use number words to reason numerically about one-to-one correspondences that are not perceptually available (Becker, 1989; Sophian, 1988). This provides evidence that preschoolers have at least initial integration of the cardinality of the collection as a whole and the individual items in the collection. If asked to count, many 4- and 5-year-olds count and use the information to correctly judge equivalence (Fuson, 1988; Fuson, Secada, & Hall, 1983; Michie, 1984a). Thus, counting can be a meaningful quantifier for children.

The abstraction principle addresses what units children can count. Research indicates that a significant development is to understand fully the unit that one is counting (K. F. Miller, 1989). For example, asked to count whole forks when shown some intact forks and some broken in half, children 4 to 5 years of age were reluctant to count two halves as a single item (Shipley & Shepperson, 1990). Although performance improves on such tasks from 4 to 7 years, a substantial portion of each age had such difficulties (Sophian & Kailihiwa, 1998). Thus, the abstraction principle appears only partially valid.

The nativist position is therefore neither definitively supported nor disproved. Addressing similar issues, a review of research generally supported the skills-before-concepts position, although there was a lack of studies with a strong design (Rittle-Johnson & Siegler, 1998). The argument and evidence against a strictly nativist (privileged domains) position is more compelling than evidence in support of the skills-be-

fore-concepts position. This is not to say the concepts-before-skills position is supported, but rather that the evidence reviewed may provide the most support for an interactionist position, in which growth in each of concepts and skills supports growth in the other (Baroody, 1992; Fuson, 1988). For example, a skills-first perspective may suggest that very early use of number or counting is mathematically meaningless (Ginsburg, 1977; von Glasersfeld, 1982); however, early developing quantitative and numerical concepts may imbue these words and procedures with meaning (Baroody, Lai, & Mix, 2005; Baroody & Tiilikainen, 2003), and the words may help explicate and differentiate the initially syncretic quantities (Clements, Battista, & Sarama, 2001). Further, humans are sufficiently flexible to permit different paths to learning (e.g., mostly skills or concepts first), even if they are differentially generative of future learning (as hierarchical interactionism posits, see also Clements, Battista, & Sarama, 2001). Finally, the extant literature offers many cases in which conceptual knowledge leads to construction of procedures (i.e., concepts result in skills). Moreover, that literature often does not examine the types of experiences that have been offered to children; we contend that especially in school, but also at home, simplistic and reductionistic views of mathematics and mathematics learning have biased research examining relationships between concepts and skills. Researchers need better evidence, including solid causal evidence (Rittle-Johnson & Siegler, 1998), on these issues.

Zero and Infinity

Situations similar to those that engender counting may also lead to children's initial encounters with zero and infinity. Because touching objects is important to young children's initial counting, zero is often not encountered in the act of counting. (Resistance to the ideas is not merely a developmental limitation, of course. Historically, societies have taken considerable time to confront, represent, and incorporate ideas of zero and infinity.) Kindergartners solve problems with zero well, although there is a separation between their knowledge of zero and their knowledge of the counting numbers. In one study, for example, they showed no evidence of applying counting strategies to zero problems and performed better on zero problems than on problems involving one or two, suggesting that such problems were solved differently (Evans, 1983). Preschoolers often have a more limited understanding of zero and infinity (Evans, 1983; R. Gelman & Meck, 1983). However, even 3-year-olds represent zero as the absences of objects more than half the time, and by 4 years, children did as well with zero as

with small whole numbers (Bialystok & Codd, 2000; similar results are reported by Hughes, 1986).

Counting played a larger role in children's early ideas related to infinity (Evans, 1983). Some children showed little or no knowledge of numbers greater than one hundred or the structure of counting. Other children at level showed knowledge of this structure, including generating new, larger numbers. The most advanced children understood that there was no greatest number because it was always possible to create a larger number.

Summary

Accurate, effortless, meaningful, and strategic counting is an essential early numerical competence. Initial level of achievement and subsequent growth in number and arithmetic from preschool into the elementary years are both predicted better by early counting ability than by other abilities, such as visual attention, metacognitive knowledge, and listening comprehension (Aunola, Leskinen, Lerkkanen, & Nurmi, 2004).

Synthesizing across this and the previous sections, we summarize that early numerical knowledge includes four *interrelated* aspects (as well as others): recognizing and naming how many items are in a small configuration (small number recognition and, when done quickly, subitizing), learning the names and eventually the ordered list of number words to ten and beyond, enumerating objects (i.e., saying number words in correspondence with objects), and understanding that the last number word said when counting refers to how many items have been counted. Children learn these aspects, often separately by different kinds of experiences, but gradually connect them during the preschool years (cf. Linnell & Fluck, 2001). Each of the aspects begins with the smallest numbers and gradually includes larger numbers. In addition, each includes significant developmental levels. For example, small number recognition moves from nonverbal recognition of one or two objects, to quick recognition and discrimination of one to four objects, to conceptual subitizing of larger (composed) groups. Development shares similarity across these aspects.

This describes the progression from sensory-concrete to integrated-concrete knowledge. Sensory-concrete refers to knowledge that demands the support of concrete objects and children's knowledge of manipulating these objects (Clements, 1999a). *Integrated-concrete* refers to concepts that are "concrete" at a higher level because they are connected to other knowledge, both physical knowledge that has been abstracted and thus distanced from concrete objects and abstract knowledge of a variety of types (Clements, 1999a). Ultimately,

these are descriptions of changes in the configuration of knowledge as children develop. Consistent with other theoreticians (Anderson, 1993), this distinction assumes that there are not fundamentally different, incommensurable types of knowledge, such as “concrete” versus “abstract.”

Comparing and Ordering

Comparing and Equivalence

Infants begin to construct equivalence relations between sets by establishing correspondences (spatial, temporal, or numerical) as early as 10 months and at most by 24 months of age (Langer, Rivera, Schlesinger, & Wakeley, 2003). Studies of early discrimination and conservation of number, previously discussed, also address children’s initial abilities to compare quantities. To briefly highlight relevant findings, children appear to base comparisons on number and may use nonverbal representations of cardinality (Mix, Huttenlocher, & Levine, 2002). For example, they may share with monkeys a set of mental filters, each tuned to an approximate number of objects (e.g., 3), but overlapping with adjacent filters (e.g., that detect 2 or 4), which explains why discrimination improves with greater numerical distance (Nieder, Freedman, & Miller, 2002). At 3 years of age, children can identify as equivalent or nonequivalent static (simultaneously presented) collections consisting of a few (1 to about 4) highly similar items (e.g., Huttenlocher, Jordan, & Levine, 1994; Mix, 1999). For instance, they can identify  and  as equal and different from  or  (the researchers describe this as a nonverbal competence, but children may have subitized the arrays). At 3.5 years, they can match different homogeneous visual sets and match sequential and static sets that contain highly similar items. At 4.5, they can nonverbally match equivalent collections of random objects and dots—a heterogeneous collection and a collection of dissimilar items.

Children’s success with sequentially presented objects emerges later than with simultaneously presented objects, and sequential events are the most difficult (Mix, 1999). These abilities seem to depend on counting abilities, with sequential events requiring a mastery of counting.

Both infants (R. G. Cooper, Jr., 1984) and preschoolers (Sophian, 1988) do better comparing sets of equal, rather than unequal, number, presumably because there are many ways for collections to be unequal. In brief, children show competence in comparing simultaneously present, equivalent collections as early as 2 or 3 years of age in everyday, spontaneous situations but show only the beginnings of such com-

petence on clinical tasks at 2.5 to 3.5 years of age, with success across a wide range of tasks only appearing at ages suggested by Piaget (Baroody, Lai, & Mix, 2005; Mix, Huttenlocher, & Levine, 2002).

Clinical studies of equivalence also highlight specific difficulties in number comparison. For example, some children will count the sets, recognize the number is the same, but still maintain that one set has more (Piaget & Szeminska, 1952). Children from 4 to 5 years of age do compare set sizes on the basis of misleading length cues, even when the situation is set up to encourage counting (Fuson, 1988).

Conversely, children cannot always infer number from that of an equivalent set. In one study, fair sharing was established by dealing out items to two puppets, and the experimenter counted one set aloud and asked children how many the other puppet had. No 4-year-old children made the correct inference that the number was the same; instead, they tried to count the second set. The experimenter stopped them and asked again, but less than half made the correct inference (Frydman & Bryant, 1988).

Researchers have posited several explanations for preschoolers’ reluctance or inability to use counting (Curtis, Okamoto, & Weckbacher, 2005). Some prefer an explanation that relies on available memory resources (Case & Okamoto, 1996; Pascual-Leone, 1976). However, others state that children believe that counting-based strategies are too difficult or unreliable (Cowan, 1987; Michie, 1984b; Siegler, 1995; Sophian, 1988). Finally, others say that counting is an activity embedded in situated tasks, and that use of it in other tasks develops only over time (Nunes & Bryant, 1996; Steffe, Cobb, von Glasersfeld, & Richards, 1983), as children come to trust the results of counting (e.g., relative to conflicting perceptions of length, Cowan & Daniel, 1989). Most agree that preschoolers do not often use counting to compare numerosities of collections.

However, presented different tasks, preschoolers can perform such inferences, such as from matching-based to counting-based equivalence. Recall studies showed that, told the cardinal value of each of two sets, preschoolers determined whether the items would be put in one-to-one correspondence and the inverse (Becker, 1989; Sophian, 1988). Thus, success is dependent on the task, with success appearing first when one-to-one correspondence (rather than sharing) is used to establish equivalence. Between the ages of 3 and 5, children develop from counting only single collections to being able to use counting to compare the results of counting two collections (Saxe, Guberman, & Gearhart, 1987) and reason about these comparisons across different situations.

Further, counting is not without positive influence. Even when children use misleading length cues, from 70 to 80% turned to counting when asked to “do something” to justify their response (Fuson, 1988). Further, although encouraging 4.5- to 5-year-old children to match and count increases their correct judgments of equivalence, children only spontaneously adopt counting for subsequent problems of the same type (Fuson, Secada, & Hall, 1983). When the counting is performed for them, preschoolers are more likely to base their comparison decisions on the results of counting (Curtis, Okamoto, & Weckbacher, 2005), which may imply that counting imposes a large processing load on children of this age, or that they do not appreciate the relevance of the counting strategy.

Apparently children have to learn that the same number implies the same numerosity and that different numbers imply (and necessitate) different numerosities (Cowan, 1987). That is, many children up to first grade need to learn about the significance of the results of counting in different situations, such as comparing sets or producing equivalent sets (Nunes & Bryant, 1996). Further, they may not have a concept of space that is sufficiently articulated and unitized to reconcile the perceptual evidence and numerical interpretation of the situation (Becker, 1989).

Thus, as we previously concluded, counting can be a meaningful quantifier for children before they reach the Piagetian levels of operational thought about number conservation. However, children’s ability to solve problems using counting develops slowly, as they learn counting skills and about the application of counting to various tasks.

Ordering and Ordinal Numbers

When the topic of *ordinality* is discussed, many authors and some researchers, often assume that all ordinal notions must involve the terms *first*, *second*..., and so forth. This is a limited view. As was previously discussed, numbers have an ordinal property in that they are sequenced. A number may be first or second in an order without considering cardinality, or it may be considered to have a greater or lesser magnitude than another number. We first address young children’s ability to order collections by number, then their ability to deal explicitly with traditional ordinal numbers.

In most studies of conservation discussed previously, children needed only to decide whether collections were equivalent or not, but not to explicitly order them (even determining which contained more). Evidence indicates that human beings can make perceptual judgments of relative quantities early, but somewhat later than they evince other quantitative abilities. As an example, 16- to 18-month-olds were reinforced

for selecting a square containing two dots but not another square containing one dot, regardless of which side it was on and regardless of how big or bright the dots were (Strauss & Curtis, 1984). Then, presented two new squares with three and four dots, respectively, the infants more often chose the square with more dots, indicating sensitivity to the ordinal concept of “more numerous.” Other studies confirm that children differentiate between “less than” and “greater than” relations by 14 months of age (R. G. Cooper, Jr., 1984; Feigenson, 1999; Haith & Benson, 1998), and possible earlier, at least with collections no larger than two (Sophian & Adams, 1987). Even 11-, but not 9-month-olds, can distinguish between ascending and descending sequences of quantities (Brannon, 2002). That is, following habituation to three-item sequences of decreasing numerical displays, in which area was not confounded with number, only 11-month-olds could then distinguish between new displays of decreasing vs. increasing sequences.

Seminal research on equivalence, Gelman’s “magic experiments,” showed that children explicitly recognized which of two sets was the “winner” even if the length, density, or even properties (color) of one set were altered. However, when the change involved an addition or subtraction that created equivalent sets, children correctly responded that there was no winner (R. Gelman & Gallistel, 1978). These experiments also revealed that children about 3 years of age are sensitive to ordinal relations. Shown that 2 was the winner compared to 1, children selected 4 as the winner compared to 3. In both experiments, this competence was just emerging at 2 to 3 years of age. By 3 years of age, children show knowledge of order in comparing collections, separate from their counting skills (Huntley-Fenner & Cannon, 2000; Mix, Huttenlocher, & Levine, 2002).

As with cardinal number, children develop the ability to order numbers over several years by learning the cultural tools of subitizing (Baroody, Lai, & Mix, 2006), matching, and counting (Fuson, Secada, & Hall, 1983). For example, children can answer questions such as “Which is more, 6 or 4?” only by age 4 or 5 years (Siegler & Robinson, 1982). Tasks involving smaller numbers and numbers farther apart are easier (Cowan & Daniel, 1989), suggesting that counting skills are relevant with numbers larger than 4. Unlike middle-income children, low-income 5- and 6-year-olds were unable to tell which of two numbers, such as 6 or 8, is bigger, or which number, 6 or 2, is closer to 5 (Griffin, Case, & Siegler, 1994). They may not have developed the “mental number line” representation of numbers as well as their more advantaged peers. These representations have several aspects, such as

two additional principles of counting. The “plus and minus one” principle involves understanding the generative pattern that relates adjacent numerical values (e.g., a collection of 4 is a collection of 3 with one added). The related comparison principle involves understanding the consequences of each successive number representing a collection that contains more objects (Griffin, Case, & Siegler, 1994). Children who use and understand these principles can reason that if the counts of two collections are 9 and 7, the collection with 9 has more because 9 comes later in the counting sequence than 7.

Finding out *how many more* (or fewer) there are in one collection than another is more demanding than simply comparing two collections to find which has more. Children have to understand that the number of elements in the collection with fewer items is contained in the number of items in the collection with more items. That is, they have to mentally construct a “part” of the larger collection (equivalent to the smaller collection) that is not visually present. They then have to determine the “other part” or the larger collection and find out how many elements are in this “left-over amount.”

Ordinal numbers, usually (but not necessarily) involving the words *first*, *second* . . . , indicate position in a series or ordering. As such, they have different features (e.g., their meaning is connected to the series they describe). Of children entering their first year of school in Australia, 29 and 20% could identify the third and fifth items in a series (B. A. Clarke & Clarke, 2004). In the U.S., performance for first was 60 and 72%, and for third was 9 and 13% for low- and middle-income groups, respectively (Clements, Sarama, & Gerber, 2005).

As in other areas, different languages differentially affect children’s acquisition and use of ordinal words. Differences in ordinal words are even more distinct across languages. Beginning right with *first* and *second* compared to *one* and *two*, English ordinal and counting words often have little or no relation. In comparison, Chinese speakers form ordinal words simply by adding a predicate (*dì*) to the cardinal number name. English-, compared to Chinese-, speaking children show dramatically lower performance on naming ordinal words from the first words and from the earliest ages, persisting into elementary school (K. F. Miller, Major, Shu, & Zhang, 2000). A precipitous decline is between 19th and 21st, with only 30% of the English-speaking sample counting to 21st, versus nearly perfect performance in Chinese-speaking children. However, this research also reveals complexities in the relationships between concepts and language. Although few children understood the unique

features of ordinal numbers, English speakers seem to understand the distinctions between ordinal and cardinal number concepts sooner, perhaps because struggles with the different names engendered reflection on these differences (K. F. Miller, Major, Shu, & Zhang, 2000).

Beginning Addition and Subtraction, and Arithmetic Counting Strategies

For most people in the United States, arithmetic is arguably the most salient topic in elementary mathematics education. However, most people would also argue that this was an inappropriate topic in preprimary education. We begin our discussion considering research on the age at which knowledge of arithmetic seems to emerge.

The Earliest Arithmetic

Educational practice, past and present, often takes a skills hierarchy view that arithmetic follows counting and other simple work with number and therefore is beyond children’s grasp until about first grade. However, decades ago, researchers illustrated the possibility of arithmetic competence in children before kindergarten (R. Gelman & Gallistel, 1978; Groen & Resnick, 1977; Hughes, 1981).

More recently, nativist researchers claimed that even infants have knowledge of simple arithmetic. For example, there is evidence that infants and toddlers notice the effects of increasing or decreasing small collections by one item. For example, after seeing a screen hide 1 or 2 dolls, then a hand place another doll behind the screen, then the screen retracted, 5-month-olds look longer when an incorrect result is revealed (a violation-of-expectations procedure, Wynn, 1992a).

In another study, children placed balls in a box placed high enough that they could not see into the box. They then watched as balls were added or removed (Starkey, 1992). Finally, they were asked to remove all the balls. The question was whether the number of times children reached was the same as the number of balls they placed. To prevent children from feeling whether or not any remained in the box, the balls were secretly removed and replaced one-by-one before each reach. Children did reach the same number of times above chance. However, a flaw in the design was that these removals might have inadvertently kept children from reaching in again when they would have (mistakenly) done so (Mix, Huttenlocher, & Levine, 2002). This would favor tasks with an answer of one. Note that children were evaluated as correct on 64% of the 4 – 3 trials but only 14% on the 4 – 1

trials, even though the latter would be considered the easier of the two tasks.

Despite such problems, authors of the original studies have made strong nativist claims, such as “infants can discriminate between different small numbers . . . determine numerical equivalence across perceptual modalities,” and “calculate the results of simple arithmetical operations on small numbers,” all of which indicates that they “possess true numerical concepts, and suggests that humans are innately endowed with arithmetic abilities” (Wynn, 1992a, p. 749). They have “access to the ordering of and numerical relationships between small numbers, and can manipulate these concepts in numerically meaningful ways” (p. 750). A recent study grounded in that perspective suggested that very young children, viewing dots representing $5 + 5$ and $10 - 5$ (controlled for area and contour length), could discriminate between outcomes of 5 and 10 (McCrink & Wynn, 2004).

Possible alternative explanations have been addressed. For example, it is possible that infants merely track locations, but evidence indicates they can track objects or number of objects (Koechlin, Dehaene, & Mehler, 1997). A different possible confound is that every outcome in Wynn’s work that was arithmetically impossible was also physically impossible; however, evidence indicated that 3- to 5-month-old babies were more upset if an arithmetically impossible result occurred than if one doll changed to another (T. J. Simon, Hespos, & Rochat, 1995).

Still, there are concerns about the nativist claims. In some studies, using standardized procedures, infants were not found to discriminate between correct and incorrect results of addition and subtraction (Wakeley, Rivera, & Langer, 2000). In addition, multiple studies have found that 3- but not 2-year-olds are successful on nonverbal addition and subtraction tasks (Houdé, 1997; Huttenlocher, Jordan, & Levine, 1994). If toddlers do not have the ability, it may be unreasonable to suggest that infants do (see Huttenlocher, Jordan, & Levine, 1994, for a discussion of these and other problems with research claiming that infants perform exact arithmetic).

Further, early behaviors often are explicable through other frameworks (Haith & Benson, 1998). For example, the children might be processing continuous amounts. Using Wynn’s tasks but different size puppets, infants were more likely to attend to the display with the unexpected change in amount and the expected number than the opposite (Feigenson, Carey, & Spelke, 2002; Mix, Huttenlocher, & Levine, 2002). Children’s success with larger number arithmetic (e.g., $10 - 5$, McCrink & Wynn, 2004) may involve their use of an estimator such as in the “large number

exception” situation discussed in the subitizing section. This is supported by cultures that have number words only up to five but can compare and add large *approximate* numbers beyond that range (Pica, Lemer, Izard, & Dehaene, 2004). It is also consistent with the view that infants represent small collections as individual objects but not as groups and large numbers as groups but not as individual objects (Spelke, 2003). Thus, analogue estimators may be innate or early developing and may mediate (Gallistel & Gelman, 2005), but not directly determine, later developing explicit, accurate arithmetic.

Another possible framework for smaller number problems postulates that infants are tracking objects via “mental tokens” using general processes of simple categorization (an early “object concept”) and comparison (Koechlin, Dehaene, & Mehler, 1997; T. J. Simon, 1997; T. J. Simon, Hespos, & Rochat, 1995). Children may possess neural traces of visual information that are not consistent with the final display, and thus their “representations” could be literal and knowledge-based. That is, these babies may *not* be *quantifying* a collection of objects but instead individuating and tracking separate objects, reacting if one disappears or appears mysteriously (Koechlin, Dehaene, & Mehler, 1997; Sophian, 1998). For example, infants’ surprising greater sensitivity to subtraction than addition across multiple studies (Koechlin, Dehaene, & Mehler, 1997; Wynn, 1992a, 1995) may indicate that the smaller the result, the more likely the child will track accurately. Further, experiments directly comparing two models, numeric/symbolic and object file, were consistent with the object file model’s predictions (Uller, Carey, Huntley-Fenner, & Klatt, 1999). For example, infants are much more likely to succeed if the first object in a $1 + 1$ task is seen on the floor before the screen is introduced, compared to tasks in which the screen is introduced first and thus the objects must be imagined there (Uller, Carey, Huntley-Fenner, & Klatt, 1999). Consistent results from other experiments (Uller, Carey, Huntley-Fenner, & Klatt, 1999) and the mixed results of various researchers imply that adults should remain aware that we often attribute numerosity and arithmetic operations to reactions to changes in number, but that may be our inability to decenter and understand that this is our projection of number onto the situation, deeply embedded in our adult cognition (cf. von Glasersfeld, 1982). This does not mean competencies do not exist, or that they are irrelevant to the foundations of numbers development; in contrast, relevant competencies are supported in most studies. However, even comparing via an early version of one-to-one correspondence may be at the service of general goals and not be numerical *for the infant in*

the sense that it would be for adults. Probably the most we can confidently say is that infants react to situations that older children and adults would experience as arithmetical.

Semantic analysis is consistent with this latter interpretation (von Glasersfeld, 1982). For example, children may discover, perhaps by putting up two fingers on each hand, that they can make a configuration of four fingers, but perhaps only at the sensory-motor level. Through application of subitizing or counting and reflective abstraction on the attentional patterns whose unconscious “automatic” application to sensory material made the figurative compositions possible, children eventually rebuild these experiences as arithmetical relationships.

Supporting this constructivist approach is evidence that young children’s active sensorimotor production may develop before their reactive perceptual discrimination of adding and subtracting. For example, when exposed to a standardized violation-of-expectations procedure (Wynn’s were not standardized), 11-, 16-, and 21-month-olds did not look longer at results for $1 + 1$ or $2 - 1$ that were incorrect than those that were correct (Langer, Gillette, & Arriaga, 2003). However, 21-month-olds did reach into a box correctly in about three fourths of the trials. Thus, pragmatic knowledge of what “works” with objects may precede “understanding how” or “what.”

When do children explicitly understand the order relations in arithmetic? An early study showed that children from 3 to 6 years show sensitivity to the effects of adding or subtracting one or two marbles to one or more containers, initially established to hold an equal (but uncounted) number of marbles, when exact computations were not required (Brush, 1978). Children 14 to 28 months showed similar sensitivity to insertions and deletions from hidden sets of numerosity no more than 2, with a caveat that results were affected by children’s bias toward choosing the transformed set (Sophian & Adams, 1987). Thus, as early as 14 to 24 months of age, children appear to understand some sense of addition increasing, and subtraction decreasing, the size of collections, before their counting skills are well established (Bisanz, Sherman, Rasmussen, & Ho, 2005; Cowan, 2003; Mix et al., 2002).

Preschoolers develop in reasoning about the effects of increasing or decreasing the items of two collections of objects. When two collections are created simultaneously by placing items one-for-one in separate locations, many preschoolers correctly judge that the items in the two collections are equal even though they do not know exactly how many objects are in the collections. If items of one collection are then increased or decreased, children as young as

age 3 correctly judge that the collection added to contains more or the collection subtracted from contains fewer than the collection that was not changed. However, problems involving two collections that are initially unequal present a difficulty for 3-year-olds. For example, if two collections initially differ by two objects, and one object is added to the smaller collection, many 3-year-olds will incorrectly say that the collection to which the object was added has more. In contrast, some 5-year-olds know that this collection still has fewer. They know that both the addition and the initial inequality must be taken into account in reasoning about the effect of the addition on the collection (R. G. Cooper, Jr., 1984).

Children’s accuracy is limited until about 3 years of age. Children are initially more accurate with small numbers (operands from 1 to 3) and nonverbal problems (Hughes, 1981; see also Huttenlocher, Jordan, & Levine, 1994). For example, children solve nonverbal arithmetic problems at about 3 years of age. In such tasks, the experimenter might show a number of disks, move them behind a screen, add more disks to the now-hidden collection, and ask the child to make a collection showing the number hidden (Huttenlocher, Jordan, & Levine, 1994). Even when preschoolers develop this ability, their competencies are approximate and are strongly related to their general intellectual competence. This is inconsistent with a perspective that such arithmetical competence is innate. Addition is easier than subtraction for larger problems (operands 5 to 8; (Hughes, 1981). Verbal-based addition develops later, as counting becomes the key basis for understanding verbal computation above small numbers (Cowan, 2003).

Thus, most agree that children have built an initial, explicit understanding of addition and subtraction situations by about 3 years of age. However, it is not until 4 years of age that children can solve addition problems involving even slightly larger numbers with accuracy (Huttenlocher, Jordan, & Levine, 1994).

Most children do not solve larger number problems without the support of concrete objects until 5.5 years of age (Levine, Jordan, & Huttenlocher, 1992). They have, apparently, not only learned the counting sequence and the cardinal principle (usually about 3.5 years) but also have developed the ability to convert verbal number words to quantitative meaning (cf. the ordinal-to-cardinal shift in Fuson, 1992).

However, this limitation is not so much developmental as experiential. With experience, preschoolers and kindergartners can learn “count all” and even “count on” strategies (Clements & Sarama, in press; Groen & Resnick, 1977; Hughes, 1986), a point to which we will return.

Arithmetic Strategies

Once children show dependable signs of understanding addition and subtraction tasks, they also show a range of strategies for solving them. Most people of any age can show diversity in solution strategies, as was recognized for young children more than 50 years ago (Ilg & Ames, 1951). Solutions of children as young as preschool (perhaps because school has not yet negatively affected them, cf. Kamii & Lewis, 1993) are notably creative and diverse (Bisanz et al., 2005; Geary, Bow-Thomas, Fan, & Siegler, 1993). For example, preschool to first-grade children use a variety of covert and overt strategies, including counting fingers, finger patterns, verbal counting, retrieval, derived facts, and covert strategies, some slower and some faster (Siegler & Jenkins, 1989; Siegler & Shrager, 1984; Steffe, 1983). Children are flexible strategists, using different strategies on problems they perceive to be easier or harder. Preschoolers can invent surprisingly sophisticated arithmetic strategies without instruction (Groen & Resnick, 1977).

Strategies may emerge from children's modeling the problem situation. Kindergartners can solve a wide range of addition and subtraction problem types when they represent the objects, actions, and relationships in the situations (Carpenter, Ansell, Franke, Fennema, & Weisbeck, 1993). About 90% used a valid strategy even for the basic subtraction and multiplication problems, with half the children using a valid strategy on every problem. About 62% solved more than three fourths of the problems accurately.

Although researchers cannot always ascertain their strategies, children from all income levels are first able to respond to the simplest tasks, nonverbal addition and subtraction problems with small numbers (no number greater than 4; Huttenlocher, Jordan, & Levine, 1994; Jordan, Huttenlocher, & Levine, 1994). Middle-income children can also respond verbally to these problems; low-income children find this, and conventional skills such as counting, more difficult. Thus, early solutions may use manipulation of nonverbal images when possible, and conventional skills may be requisite for wider types of response types and solution procedures, especially for problems that involve larger numbers. Automatic (fast and accurate) counting skills, such as naming the counting number immediately after a given number, predicts accuracy of arithmetic performance, which predicts overall mathematics achievement for kindergartners (Penner-Wilger et al., 2005). An open question is whether the distinction between nonverbal and verbal responses is mainly a difference in language competencies or also includes the children's possession of a figural collection (which may not

have a cardinal representation) versus a numerosity or protonumerosity (Steffe & Cobb, 1988).

Confirming cultural or experiential influences, Chinese kindergartners answer three times as many addition problems as U.S. children (Geary, Bow-Thomas, Fan, & Siegler, 1993). Language, again, appears to be a substantial factor. Quicker pronunciation of Chinese number words allows a greater memory span for numbers that in turn supports more sophisticated verbal strategies (compared to use of fingers) that in turn support learning number combinations (because addends and sum are simultaneously in working memory).

The structure and abstruseness of English (compared to Asian) counting words slows early learning of counting, as shown earlier. Given the importance of counting in learning subsequent skills such as addition and subtraction, the complexity of the number system may slow learning in those areas as well (Fuson, 1992; Miller, Smith, Zhu, & Zhang, 1995). In addition, languages such as Japanese add coherence to problem statements by integrating items into a set using numeral classifiers (Miura & Okamoto, 2003). A word problem such as "Joe has 6 marbles, 2 more than Tom. How many does Tom have?" in Japanese would be "Joe has 6-round-small-things, 2-round-small-things more than Tom. How many round-small-things does Tom have?" This may help children build images and keep the notion of the sets of objects explicit.

Average 4-year-olds in the United States integrate counting with their arithmetic knowledge and, in doing so, invent counting strategies to compute the effects of addition and subtraction operations on sets of objects. They often initially use a "count all" procedure. For example, given a situation of $5 + 2$, they count objects for the initial collection of 5 items, count 2 more items, and then count the items of the two collections together and report "7." These children naturally use such counting methods to solve story situations as long as they understand the language in the story. After children develop such methods, they eventually curtail them (Fuson, 1992). In an early study (Groen & Resnick, 1977), researchers were surprised that, given only practice on addition problems and demonstrations of the count-all strategy, about half of the 5-year-olds moved from a counting-all strategy to a counting-on-from-larger strategy by themselves. That is, given $2 + 4$, they might start with 4 and count 2 more. This illustrates that young children can invent efficient strategies without direct instruction, or even despite being taught alternative methods (see also Siegler & Jenkins, 1989). They follow different developmental paths, some illustrating awareness of the new strategies, others not, and most using a variety of more and less advanced strategies at any time. This

research also illustrates that failing to solve tasks is not requisite to inventing new strategies; instead, success is often the catalyst (DeLoache, Miller, & Pierroutsakos, 1998; Karmiloff-Smith, 1984, 1986), especially if one is not changing a conception but only changing to a more efficient or elegant approach to producing the same answer. After concentrated experience, 4- to 5-year-olds showed a variety of strategies, often the counting-on-from-larger (also called “min”) strategy, but their successes, rather than failures, appeared to catalyze the invention of new strategies. Transitional strategies were important, such as the “shortcut-sum” strategy, which appears similar to a count-all strategy but involves only one count; for example, to solve $4 + 3$, 1, 2, 3, 4, 5, 6, 7 and answer 7. This often preceded the min strategy, which was preferred by most children once they invented it, especially on problems such as $2 + 23$ where it saved the most work (Siegler & Jenkins, 1989).

Children continue developing and curtailing their counting methods. For example, given a task in which two collections are hidden from view but are said to contain 6 and 2 items, respectively, children may put up fingers sequentially while saying, “1, 2, 3, 4, 5, 6” and then continue on, putting up two more fingers, “7, 8. Eight!” Such counting on is a landmark in children’s numerical development. It is not a rote step, as is true of all such curtailments of processes. It requires conceptually embedding the 6 inside the total (Steffe & Cobb, 1988; Steffe, Cobb, von Glasersfeld, & Richards, 1983).

Counting on when increasing collections and the corresponding counting-back-from when decreasing collections are powerful numerical strategies for children. However, they are only beginning strategies. In the case where the amount of increase is unknown, children count-up-to to find the unknown amount. If six items are increased so that there are now nine items, children may find the amount of increase by counting, “Six; 7, 8, 9. Three.” And if nine items are decreased so that six remain, children may count from nine down to six to find the unknown decrease, as follows: “Nine; 8, 7, 6. Three.” However, counting backwards, especially more than three counts, is difficult for most children (Fuson, Smith, & Lo Cicero, 1997).

When children fully realize that they can find the amount of decrease by putting the items back with the six and counting from six up to nine, they establish that subtraction is the inversion of addition and can use addition instead of subtraction. This understanding develops over several years. There is little sign that children as young as 2 years use or understand the principle (Vilette, 2002), but 3- and 4-year-old children use procedures consistent with the inversion

principle (J. S. Klein & Bisanz, 2000; Rasmussen, Ho, & Bisanz, 2003), showing use of arithmetical (quantitative, not just qualitative) principles before formal schooling. This use is inconsistent, however, perhaps reflecting children’s creative tendency to try out a variety of procedures (Shrager & Siegler, 1998), limits on their information-processing capacities (J. S. Klein & Bisanz, 2000), their nascent understanding of the concept (Vilette, 2002), or some combination. Given a problem with larger numbers, no 4-, some 5-, and most 6-year-olds used the addition-subtraction inverse principle (Baroody & Lai, 2003). In summary, only in the primary grades do many children understand and use the inversion principle in a conscious and inter-related fashion (P. Bryant, Christie, & Rendu, 1999). At that time, they often use addition to solve certain subtraction problems. However, many kindergartners do use this mathematically significant principle consistently in a quantitative manner in some situations.

Summary

Early in life, babies are sensitive to some situations that older people view as arithmetic. They may be using an innate subitizing ability that is limited to very small numbers, such as $2 + 1$. Alternatively, they may be individuating and tracking individual objects. In any case, they possess the beginnings of invariance and transformations and thus have far richer foundations for arithmetic than traditional Piagetian accounts suggested.

The contents of minds can be categorized as data, information, knowledge, understanding, and wisdom (Ackoff, 1989). Infants appear to have data, and perhaps information, but knowledge and understanding develop over considerable time. As for the other areas involving number, calculation first emerges with very small numbers, then extends to larger numbers. Exact calculation is preceded by a period of approximations that are more accurate than random guessing. Only years later do they extend their abilities to larger collections that, although still small (e.g., $3 + 2$), are amenable to solution with other methods. These other methods, usually using concrete objects and based on subitizing and counting, play a critical developmental role, as the sophisticated counting and composition strategies that develop later are all curtailments of these early solution strategies (Carpenter & Moser, 1984; Fuson, 1992). That is, children learn to count from a given number (rather than starting only from one), generate the number before or after another number, and eventually embed one number sequence inside another. They think about the number sequence, rather than just saying it (Fuson, 1992). Such reflection empowers counting to be an

effective and efficient representational tool for problem solving. Thus, educators must study the processes children use as well as the problems they can solve to understand both their strengths and limitations at various ages.

Education

Children can learn arithmetic from 3 years of age and, in some situations, even earlier. Yet most pre-K teachers do not believe arithmetic is appropriate (Sarama, 2002; Sarama & DiBiase, 2004). In multiple countries, professionals in multiple educational roles vastly underestimate beginning students' abilities (Heuvel-Panhuizen, 1990).

Given the abilities of even preschool children to directly model different types of problems using concrete objects, fingers, and other strategies (Carpenter, Ansell, Franke, Fennema, & Weisbeck, 1993), these restricted educational beliefs and strategies are not only unnecessary but also deleterious. That is, children unfortunately constrain the symbols + and – to these limited interpretations and lose the flexibility of their intuition-based solution strategies (Carpenter, 1985), and their confidence in themselves as problem solvers decreases. Textbooks also present symbols and number sentences too soon and in the wrong ways (i.e., before meaningful situations); do little with subitizing or counting, automatization of which aids arithmetical reasoning; and de-emphasize counting strategies (Fuson, 1992). The younger the children, the more problematic these approaches become. No wonder that American schooling has a positive effect on children's accuracy on arithmetic but an inconsistent effect on their use of strategies (Bisanz et al., 2005; Naito & Miura, 2001).

In addition, textbooks offer an inadequate presentation of problems with anything but small numbers. In one kindergarten text, only 17 of the 100 addition combinations were presented, and each of these only a small numbers of times (Hamann & Ashcraft, 1986). Primary students encountered combinations with larger numbers, but less frequently than small-number combinations. Correlational evidence supported that this accounts for difficulty students of all ages have with addition with larger sums (Hamann & Ashcraft, 1986).

Several benefits stem from children's invention and discussion of diverse strategies for more demanding arithmetic problems. The number of different strategies children show predicts their later learning (Siegler, 1995). For example, when learning to solve problems involving mathematical equivalence, children were most successful when they had passed through a stage of considering multiple solution strat-

egies (Alibali & Goldin-Meadow, 1993; Siegler, 1995). Also, it is too seldom recognized that these processes are themselves worthwhile mathematical goals (Steffe & Cobb, 1988). Teachers might encourage children to advance in their sophistication, but effective advances usually do not involve replacing initial strategies with school-based algorithms but instead helping children curtail and adapt their early creations. When asked to explain a concept they are acquiring, children often convey one procedure in speech and a different procedure in gesture. Such "discordant" children are in a transitional state and are particularly receptive to instruction (Alibali & Goldin-Meadow, 1993; Goldin-Meadow, Alibali, & Church, 1993).

For what period are manipulatives necessary? For children at any age, they are necessary at certain stages of development, such as counters of perceptual unit items (Steffe & Cobb, 1988). Preschoolers can learn nonverbal and counting strategies for addition and subtraction (Ashcraft, 1982; Clements & Sarama, *in press*; Groen & Resnick, 1977), but they need concrete objects to give meaning to the task, the count words, and the ordinal meanings embedded in the situations.

However, once children have established that level of thinking, they can often solve simple arithmetic tasks without explicitly provided manipulatives. In one study of kindergartners, there were no significant differences between those given and not given manipulatives in accuracy or in the discovery of arithmetic strategies (Grupe & Bray, 1999). The similarities go on: Children without manipulatives used their fingers on 30% of all trials, whereas children with manipulatives used the bears on 9% of the trials but used their fingers on 19% of trials for a combined total of 28%. Finally, children stopped using external aids approximately halfway through the 12-week study.

Composing and Decomposing

Composing and decomposing numbers is another approach to addition and subtraction, one that often overlaps with counting strategies. Phenomenologically it can be experienced similarly to subitizing; indeed, conceptual subitizing, previously discussed, is a special case of composition of number. Composing and decomposing numbers also contributes to developing part-whole relations, one of the most important accomplishments in arithmetic (Kilpatrick, Swafford, & Findell, 2001).

Composing and decomposing are combining and separating operations that allow children to build concepts of parts and wholes. Toddlers may first learn that wholes consist of parts, then that a whole is more than

its parts and that parts make a larger whole. Next, they may learn that nonverbally represented parts make a specific whole (e.g., ●● and ●● make ●●●●). Later, they may learn that “two” and “two” make “four” (Baroody, Lai, & Mix, 2005). At that point, children can develop the ability to recognize that the numbers 2 and 3 are “hiding inside” 5, as are the numbers 4 and 1. Such thinking continues to develop. As previously shown, most preschoolers can “see” that 2 items and 1 item make 3 items; even 3-year-olds can solve problems such as one and one more, at least nonverbally. However, 4-year-olds do not appreciate the part-whole structure of initial-unknown change problems (Sophian & McCorgray, 1994). They would choose a small number as the answer for both addition and subtraction problems. In contrast, children a year or two older responded with a number that was larger than that given for the final set more often on addition than on subtraction problems, and they responded with a smaller number than that given for the final set more often on subtraction than on addition problems. This is not to say that children always apply such understandings in finding precise answers to arithmetic tasks. The findings are less pessimistic than the view that this level of part-whole thinking is not accessible to children until the primary grades. For example, Piagetian theory might be interpreted as excluding missing-addend tasks until after first or even second grade (Kamii, 1985). However, children appear to understand the part-whole relationships of tasks by kindergarten, although they may not know how or think to apply it to all arithmetic tasks (Sophian & McCorgray, 1994).

At least by 5 years of age, children are ready to engage in tasks that put a substantive demand on their explicit understanding of part-whole relationships, such as initial-unknown change problems. However, they may not apply these understandings in all relevant arithmetic tasks. One study suggests that teachers may need to help children see the relevance in and apply their understandings of part-whole relationships to these types of problems (Sophian & McCorgray, 1994).

Language, Numerals, and Other Symbols

As the brief history of research on early number knowledge indicated, the hypothesized role of language has changed through the four historical phases described previously. In the first phase, counting and number words played a definite, if underspecified, role. The second, Piagetian, phase severely limited the role of language. In the theories of the nativists and skills-integration theorists of the third phase, language was important in building more sophisticated

number concepts but was not central to development per se, especially in its earliest foundations. In the fourth phase, several theorists proposed that language plays a critical role in developing the concept of cardinal numbers. For example, they claimed that language sensitizes children to cardinality and helps them move from approximate to exact representations of number (Spelke, 2002; Van de Walle, Carey, & Prevor, 2000). That is, the word *three* may enable children to create an explicitly cardinal representation that applies to various triads of objects (Baroody, Lai, & Mix, 2006).

Empirical evidence indicates that basic verbal number competencies precede some nonverbal tasks, contrary to the mental-models view (Baroody, Benson, & Lai, 2003). This supports the view that learning number words facilitates the development of exact representations. Thus, as stated previously, human language plays an important role in connecting different representations, making early premathematical cognition numeric (Wiese, 2003). Much of the research discussed in this section supports this view, as does research on learning other concepts, such as color words (Sandhofer & Smith, 1999). Not all language involves the term for the concept. Often, introduction of the term is better delayed until concepts are developed with informal language and experiences (Lansdell, 1999). Further, there is still an open debate about the exact role of language, with some arguing that it is critical for the development of number concepts above four, and others citing evidence of competence in cultures with limited counting words and other evidence that language is not the cause of the development of these concepts (R. Gelman & Butterworth, 2005). For example, there is evidence that when verbal counting is too slow to satisfy time constraints, nonverbal mental magnitudes mediate the production of a number word that approximates that numerosity of a set. It also mediates the ordering of symbolic numbers (Gallistel & Gelman, 2005). In general, research indicates that the interactions between language on the one hand, and number concepts and skills on the other, are more bidirectional, fluid, and interactive than previous accounts allowed (Mix, Sandhofer, & Baroody, 2005).

Children come to elementary school, and often pre-K, with initial knowledge of reading and even writing numerals (Bell & Bell, 1988; Bialystok & Codd, 2000; Fuson, 1992). For example, kindergartners from a range of populations were generally competent with the numerals from 1 to 20, and many from 1 to 100. The early autumn percentages for reading and writing were for 3, reading, 95%, writing, 80%; for 9, 74% and 53%. By the end of the year, the percentages for 3 and 9 for both reading and writing were above 93%. Reversing the digits for numbers in the teens

was a frequent error, due to the reversal in English pronunciation. (Reversals in writing digits, e.g., writing 3 backwards, were ignored in scoring as common errors that children outgrow.)

Children's ability to read or produce numerals or other representations does not imply a commensurate understanding of what they represent (Bialystok & Codd, 2000). In one study, about a fourth of the children entering school were unaware of the function of numerals (Munn, 1998). When asked to represent how many objects there were, children may produce idiosyncratic, pictographic (representing the appearance of objects as well as their numerosity), iconic (one-to-one correspondence between objects and marks), or symbolic (numerals) responses (Ewers-Rogers & Cowan, 1996; Hughes, 1986; Munn, 1998). Similarly, in other studies preschoolers showed an understanding that written marks on paper can preserve and communicate information about quantity (Ewers-Rogers & Cowan). For example, 3- and 4-year-olds invented informal marks on paper, such as tally marks and diagrams, to show how many objects are in a set (Allardice, White, & Daugherty, 1977). However, children this age did not notice when numerals were missing from pictures of such objects and rarely said number words in giving fast-food orders; thus, understanding preceded use of numerals (Ewers-Rogers & Cowan, 1996).

Just as important as learning to read and write representations of number is learning to use such representations functionally. Except for the idiosyncratic representations, children use their symbols to remember the amount in collections, one level of functional use. Children continue to develop competence in functional use of numerals. For example, children were shown several collections labeled with symbols. They were told a teddy bear added one object to only one collection and asked to figure out which one had the extra object. Only some young children could use the symbol to complete the task (Munn, 1998). Functional use is more likely when children use numerals rather than pictographic or iconic representations, and nonconventional representations may not be important precursors to conventional symbol use.

Final Words

Research on number and quantity supports the tenets of the *hierarchical interactionism* framework. For example, cognitive progressions through levels in the specific domains of number, such as counting, comparing, and arithmetic, are evident, building on initial bootstraps, yet undergoing significant development through the early years. There are different courses

for such development. Two examples, emphasizing experiences in the social environment, briefly illustrate these possibilities. First, children from low-income homes may engage in informal premathematical activity and possible premathematical knowledge (Ginsburg, Inoue, & Seo, 1999) but, due to different experiences, may still lack components of the conceptual structures possessed by children from more affluent homes (Griffin, Case, & Siegler, 1994). Second, educational emphases on skills, for example, can lead to instrumental knowledge of, and beliefs about, mathematics (Kamii & Housman, 1999; Skemp, 1976). More specifically, experiences with sophisticated counting strategies, composition/decomposition, and other imagistic approaches lead children to develop different strategies and understandings of number than, for example, experiences limited to simple counting strategies and memorization. Relevant here is the notion, perhaps expressed best by Wittgenstein (1953/1967), but expressed previously by several others (e.g., Douglass, 1925; von Glasersfeld, 1982), of weaving together various threads to form number concepts, in which various different weavings create different number concepts of various strength and applicability. The tenet of progressive hierarchization is illustrated by the connections children come to make between various mathematically relevant concepts and procedures (e.g., connecting experiences with nonverbal subitizing, hearing number words applied to particular situations, and counting to learn cardinality, Klahr & Wallace, 1976; Sophian & Adams, 1987), creating more robust understandings that are hierarchical in that they employ generalizations (e.g., of additive composition) while maintaining differentiations (e.g., discrete vs. continuous compositions).

The domains of number and geometry share commonalities that emerge from the loosely differentiated competencies of the young child. For example, Mix, Huttenlocher, and Levine suggested that "the quantification of infants and young children could be accurately termed 'spatial quantification'" (Mix, Huttenlocher, & Levine, 2002, p. 139). In this view, spatial thinking is essential in its role in the development of number and quantification. It is important for many other reasons.

GEOMETRY AND SPATIAL THINKING

Geometry and spatial thinking are important because they involve "grasping . . . that space in which the child lives, breathes and moves . . . that space that the child must learn to know, explore, conquer,

in order to live, breathe and move better in it" (Freudenthal in NCTM, 1989, p. 48). In this section, we begin with just that space in which the child lives, then examine how the child learns to know it better, and then turn to issues of smaller scale geometric shape, including composition and transformation of shapes. The separation of these two basic geometric domains is based on distinct systems in the primate brain for object perception, or recognizing *what* an object is (inferior temporal cortex, ventral pathway), and for spatial perception, or *where* an object is (posterior parietal cortex, ventral stream; Stiles, 2001; Ungerleider & Mishkin, 1982).

Spatial Thinking

Spatial thinking is important because it is an essential human ability that contributes to mathematical ability. Spatial reasoning is also a process that is distinct from verbal reasoning (Shepard & Cooper, 1982) and functions in distinct areas of the brain (Newcombe & Huttenlocher, 2000). Further, mathematics achievement is related to spatial abilities (Ansari et al., 2003; Fennema & Sherman, 1977; Fennema & Sherman, 1978; Guay & McDaniel, 1977; Lean & Clements, 1981; Stewart, Leeson, & Wright, 1997; Wheatley, 1990). Although researchers do not fully understand why and how, children who have strong spatial sense do better at mathematics. This relationship, however, is not straightforward. Sometimes, "visual thinking" is "good," but sometimes it is not. For example, many studies have shown that children with specific spatial abilities are more mathematically competent. However, other research indicates that students who process mathematical information by verbal-logical means outperform students who process information visually (for a review, see Clements & Battista, 1992).

Similarly, limited imagery in mathematical thinking can cause difficulties. An idea can be *too* closely tied to a single image. For example, connecting the idea of *triangles* to a single image such as an equilateral triangle with a horizontal base restricts young children's thinking. Therefore, spatial ability is important in learning many topics of mathematics. The role it plays, however, is elusive and, even in geometry, complex. Two major abilities are spatial orientation and spatial visualization (Bishop, 1980; Harris, 1981; McGee, 1979). We first discuss spatial orientation, which involves an extensive body of research, then spatial visualization and imagery.

Spatial Orientation

Spatial orientation is knowing where you are and how to get around in the world, that is, understand-

ing and operating on relationships between different positions in space, at first with respect to your own position and your movement through it, and eventually from a more abstract perspective that includes maps and coordinates at various scales. Like number, spatial orientation has been postulated as a core domain, for which competencies, including the ability to actively and selectively seek out pertinent information and certain interpretations of ambiguous information, are present from birth (R. Gelman & Williams, 1997). Infants focus their eyes on objects and then begin to follow moving objects (Leushina, 1974/1991). Toddlers ignore other cues and instead use geometric information about the overall shape of their environment to solve location tasks (Hermer & Spelke, 1996). Again, however, evidence supports the interaction of inborn endowments, (possibly) maturation, experience, and sociocultural influences. What, then, can young children understand and represent about spatial relationships and navigation? When can they represent and ultimately mathematize this knowledge?

Piaget maintained that children are born without knowledge of space, or even permanent objects (Piaget & Inhelder, 1967). His topological primacy thesis posited that they move through stages of egocentric spatial constructions (e.g., objects within reach, or those in front of the child, including topological relations of connectedness, enclosure, and continuity) to allocentric constructions (e.g., objects farther away, including having relationships to one another). Studies can be interpreted in that light. For example, if 6- to 11-month-old children are placed in a maze and repeatedly find a toy by crawling straight and then turning left at an intersection, then are moved to another end, most will incorrectly turn in the same direction that originally led to the toy (Acredolo, 1978). At 16 months of age, children correctly compensate for the change in their position. Piaget claimed that children's first notions are of topological space (e.g., understanding closure and connectedness) and later projective (relations between the child and objects, establishing a "point of view") and Euclidean, or coordinate space.

Research supports Piaget's prediction about development of near space before far space (see Haith & Benson, 1998, for a review). However, Piaget's topological primacy thesis seems of limited usefulness. In addition, research suggests that young children are more, and adults less, competent than the Piagetian position indicated, although substantial development does occur. Young children can reason about spatial perspectives and spatial distances, although their abilities develop considerably throughout the school years. In the first year of life, infants can perceive the

shape and size of objects and can represent the location of objects in a three-dimensional space (Haith & Benson, 1998; Kellman & Banks, 1998). As another example, “egocentrism” is not displayed if landmarks provide cues (Rieser, 1979). Between 5 and 9 months, they develop a geometric mechanism that allows them to identify an object to which another person is pointing (Butterworth, 1991). Older students and adults still display biases and errors in spatial reasoning (Fischbein, 1987; Uttal & Wellman, 1989) and do not always perform successfully on tasks designed by Piaget to assess an underlying Euclidean conceptual system (Liben, 1978; Mackay, Brazendale, & Wilson, 1972; H. Thomas & Jamison, 1975). Thus, both the topological primacy thesis and the traditional egocentric-to-allocentric theory should be replaced. However, Piaget’s constructivist and interactionist positions remain viable, especially that the representation of space is not a perceptual “reading off” of the spatial environment but is built up from active manipulation of that environment (Piaget & Inhelder, 1967). We review research on spatial orientation in four categories: spatial location and intuitive navigation, spatial thought, models and maps, and coordinates and spatial structuring.

Spatial Location and Intuitive Navigation

What kind of “mental maps” do young children possess? Neither children nor adults actually have “maps in their heads”—that is, their mental maps are not like a mental picture of a paper map. Instead, they consist of private knowledge and idiosyncrasies. Children learn about space by developing two types of self-based reference systems and two types of external-based reference systems (Newcombe & Huttenlocher, 2000). The younger the child, the more loosely linked these systems are.

Self- and external-based systems. Self-based spatial systems are related to the Piagetian construct of egocentric space and encode the position of the moving self. The most primitive—both in its early emergence and its limited power—is response learning, or sensorimotor coding. The child records a location or route to a location by a pattern of movements that have been associated with a goal (e.g., looking to the left from a high chair to see a parent cooking). The second is path integration, in which locations are

coded based on the distance and direction of one’s own movement. One’s location is continually updated based on input regarding movement (as well as from landmarks encountered—integrated with external-based systems). Such automatic processes could serve as the foundation for explicit mathematics (Newcombe & Huttenlocher, 2000).³

External-based reference systems are based on environmental structures and landmarks. The landmarks are usually salient, familiar, or important objects. Cue learning associates an object with a coincident landmark, such as a toy on a couch. As with response learning, cue learning is the more limited and less powerful of two systems. The more powerful, place learning, comes closest to people’s intuition of mental maps (albeit, as we shall show, this is a limited metaphor), as it builds locations from distances and directions among environmental landmarks. One example of place learning, taking the edges of a region or walls of a room as a frame of reference, illustrates a possible early, implicit foundation for later learning of coordinate systems.

Finally, when information from these four systems is combined, it is combined hierarchically (e.g., chair in a room, school, city . . .), with different precisions at each level. This provides a best estimate of locations but can also introduce systematic biases (Newcombe & Huttenlocher, 2000). For example, knowing an object was dropped and lost in the left half of the back yard might be combined with a memory of a specific location via path integration, but it may bias the estimate toward the center of the region.

When do these systems emerge? Once they do, do they develop? Development consists of two interrelated aspects. First, children learn through experience which coding systems are more effective and accurate, and in which situations. (Following Newcombe & Huttenlocher, 2000, we use the term coding to mean a memory trace of some type in the information-processing system that supports action in the environment without any commitment to a cognitively accessible “representation” that will, eventually, support thinking about space.) Secondly, each of these coding systems becomes more effective, although the simple systems of response and cue learning only extend the situations to which they apply. We begin by describing the latter development.

³ We use R. F. Wang and Spelke’s (2002) term rather than Newcombe and Huttenlocher’s (2000) term *dead reckoning* because the root of the latter is “deductive reasoning.” With magnetic compasses, sailors could take compass bearings, or headings. Bearings on two landmarks were used to construct two intersecting lines, which determined a location. This starting point was then iteratively updated based on movement—direction and distance (speed times time). In both its implicit beginnings and navigational application, the difficulty is “drift”—small errors lead to increasingly inaccurate calculations. The common written abbreviation of deductive reckoning, “ded. reckoning,” was misread by an early mariner and the name stuck as *dead reckoning*. As a term, *path integration* has no implication of conscious deduction.

Response learning and cue learning. In the first year of life, stationary infants rely on response learning to locate objects in their environment. An example is children who incorrectly turn in the same direction that originally led to a toy even after being physically reoriented (Acredolo, 1978; see also Acredolo & Evans, 1980).

In addition to response learning, two other systems, cue learning and path integration, emerge by at least 6 months of age (Newcombe & Huttenlocher, 2000). As an example of cue learning, 7-month-olds can remember which of two containers holds an object, even after a minute's delay filled with distractions (McDonough, 1999). Landmarks can help children depress an incorrect response after they are rotated, but 6-month-olds were uncertain of the correct choice; 9- and 11-month-olds used the cues successfully (Acredolo & Evans, 1980). As another example, infants associate objects as being adjacent to a parent (Presson, 1982; Presson & Somerville, 1985) but cannot associate objects to distant landmarks. By the age of 1 year, they can use a different colored cushion among an array of cushions to locate a toy (Bushnell, McKenzie, Lawrence, & Com, 1995). Toddlers and 3-year-olds can place objects in prespecified locations near distant landmarks but "lose" locations that are not specified ahead of time once they move. R. F. Wang and Spelke (2002) described a *view-dependent place recognition* system that operates similarly in animals such as ants and bees and argued that in humans as well, determining location and navigating are based on view-specific representations.

Path integration ("dead reckoning"). Infants use response and cue learning when they are stationary. They ignore movement when they are carried (Acredolo, Adams, & Goodwyn, 1984) but use path integration as early as 6 months of age when they have actively moved themselves (Newcombe, Huttenlocher, Drummey, & Wiley, 1998). By 1 year of age, they can encode both distance and direction with some degree of accuracy during self-movement (Bushnell, McKenzie, Lawrence, & Com, 1995). Thus, self-movement appears important, although infants as young as 6 but not 4 months of age can demonstrate above-chance performance after training (to localize a fixed location from two orientations; see also Haith & Benson, 1998, for a review and statement of methodological challenges; D. Tyler & McKenzie, 1990). They can internally represent the amount of rotational movement about their own axis by at least 8 months and code amount of movement along a straight line by at least 9 months. By 16 months, children are likely to perform path integration following movements involving both translation and rotation. They sometimes use this sys-

tem, even more than other systems for certain tasks (Bremner, Knowles, & Andreasen, 1994). A general point is that spatial systems that produce motor activity and spatial representations are intimately connected (Rieser, Garing, & Young, 1994). Path integration has been identified in insects, birds, and mammals, and findings support that aspects of it are fundamental in-born endowments in humans as well (R. F. Wang & Spelke, 2002). The fine calibration of the system, and the ability to ignore distracting visual information, improves from 4 years of age to adulthood, probably on the basis of fine-tuning from the senses, including proprioception and kinesthesia (Newcombe & Huttenlocher, 2000).

Place learning. During their second year, children develop the ability to code locations using objects in their external environment. They also become capable of spatial reasoning, in that they can solve problems with that information. They continue to grow in their abilities in spatial coding, reasoning, and symbolizing through their elementary school years, as they develop spatial visualization abilities such as maintaining and operating on mental images (e.g., mental rotation) and learn to use such tools as language and maps.

The first of these abilities is place learning, which involves creating a frame of reference by coding objects' positions with respect to perceptually available landmarks, using information of their distances and directions. In contrast to Piaget's topological primacy thesis, in which children build a "Euclidean system" only by age 9 or 10, research indicates that toddlers are able to code distance information and use that to locate objects. For example, infants as young as 5 months use spatiotemporal information to track and even individuate objects and from 12 months of age can code distance to support the search for hidden objects (Bushnell, McKenzie, Lawrence, & Com, 1995; Newcombe & Huttenlocher, 2000); however, in one of these studies they could not use separate indirect landmarks at 1 year of age (Bushnell, McKenzie, Lawrence, & Com, 1995). They can also use simple geometric properties of a room to guide a search; for example, looking more often in two corners of a rectangular room with the long wall on the right and the short wall on the left (Hermer & Spelke, 1996).

Place learning using multiple objects may appear after 21 months of age (Newcombe, Huttenlocher, Drummey, & Wiley, 1998). For example, after shown an object hidden in a sandbox, children were taken around the other side of it. Children 21 months or younger were not helped by the availability of landmarks around the sandbox (that is, they did no worse when these landmarks were hidden by a curtain), but older children were significantly more accurate with

the visible landmarks. By 5 years of age, children can represent an object's position relative to multiple landmarks, such as an object midway between two other objects (Newcombe, 1989). From 5 to 7 years, children increase their ability to keep track of their locations in mazes or open areas.

Functional use of such spatial knowledge for searching, which requires coding spatial information and forming and utilizing spatial relationships, develops over the toddler and preschool years (Newcombe & Sluzenski, 2004). Children show significant growth in the ability to search for multiple objects between 18 and 24 months of life and an increase in the ability to use relations among objects between 24 and 42 months (Newcombe & Sluzenski, 2004). In addition, toddlers can systematically check an array by exhaustively searching within groupings at three sites. Preschoolers can plan comprehensive searches in a small area, including memory of sites that have been checked, often using a circular search path (Cornell, Heth, Broda, & Butterfield, 1987).

Selection of systems. As mentioned, the second aspect of development involves learning to select the coding systems that are more effective and accurate in different situations. Evidence of early appearance of all systems and gradual growth favor the view that children change the relative importance they assign to different types of spatial information when these types provide conflicting information (Newcombe & Huttenlocher, 2000), possibly through the formation of metacognitive strategies (Minsky, 1986). Infants' choices among conflicting spatial systems depend on a combination of cue salience, complexity of movement, whether response learning has been recently reinforced, and whether the infant is emotionally secure or under stress (Newcombe & Huttenlocher, 2000). Further development depends on experience. For example, when children attain self-mobility by crawling between 6 and 9 months of age, they can experience failures in response learning. Infants who crawl, and even those with extensive experience with a walker, succeed more often in locating objects' spatial positions (Bertenthal, Campos, & Kermoian, 1994). The longer they have been moving themselves, the greater the advantage, probably because they learn to attend to relevant environmental information and update their spatial codings as they move. This is supported by research showing that when 12-month-old children walk to the other side of a layout and have the opportunity to look at all times, they both look more than children who are guided and also subsequently do better in turning toward the object from the new position (Acredolo, Adams, & Goodwyn, 1984). There is great variability on the onset of crawling (Bertenthal,

Campos, & Kermoian, 1994), which, along with failures in the codes from response learning, probably leads to developmental advances via the variation-and-selection process.

This perspective does not deny the possibility of an apparently qualitative, or general, shift in default propensities (Newcombe & Huttenlocher, 2000), which may be the result of newly available experiences (e.g., crawling) or of the gradual growth in the effectiveness of a new system that has achieved ascendance in the overlapping stages of development theory (Minsky, 1986, see also a later but more developed version in Siegler, 1996). Such shifts may also result from biological maturation, although some argue that experience may be the cause of the development of the nervous system (Thelen & Smith, 1994).

Hierarchical combination of systems. Before their second birthday, at about 16 months, children show the beginnings of a hierarchical combination of spatial reference frames, including categorical coding of regions and fine-grained information, although the categories and combinations of categories develop considerably after that age (Newcombe & Huttenlocher, 2000). For example, children form categories that are not physically demarcated (e.g., the left half of a field), beginning in a limited way at 4–6 years of age (in small rectangles) and expanding to larger regions at about 10 years. Second, they develop the ability to code hierarchically along two dimensions simultaneously. For example, to recall the location of a point in a circular region, people 10 years of age and older code locations for both the distance from the center of a circle and angle, as well as categorical information about the quadrant. Children younger than 7 years do not code the categorical information about quadrants in such tasks, whereas 7-year-olds do if angle information is requested. In general, as they age, children divide regions into smaller and more abstract categorically coded regions. Such changes, rather than a qualitative shift to Euclidean space, appear to explain changes in performance. In summary, hierarchical coding begins at 16 months, but not until 10 years of age do children reach sophisticated levels (Newcombe & Huttenlocher, 2000).

Summary. Children are born with potential abilities in response learning, cue learning, path integration, and place learning. During the 1st year of life, these systems are further integrated, based on feedback from experiences with their physical environment. During the 2nd year, starting at about 21 months, children develop place learning proper; that is, they begin to use distance information from multiple environmental landmarks to define location. Development of symbolic skills may play an important role. From that

point forward, the path integration and place learning systems develop in effectiveness and accuracy. Path integration improves from 4 years on. By 5 years of age, children can represent a location in terms of multiple landmarks, and from 5 to 7 years, develop in their ability to maintain locations in challenging circumstances such as open areas. All four systems are further reconfigured in their application and integration. Finally, there are changes in the size and nature of categories used in hierarchical spatial coding, but these remain limited in the preschool years.

Further, the systems are used with increasing efficiency. At 6 months of age, children are more likely to rely on response learning, but older children, with more visual and especially self-produced movement experiences, rely more on cue learning and path integration (Newcombe & Huttenlocher, 2000). This variation-and-selection approach is similar to Siegler's (1996) approach, as previously discussed regarding the learning of number. Again, the phenomenon of increased variability leads eventually to greater stability and generalizability.

We now turn to what Piaget and Inhelder's (1967) research focused upon, representation that would support reflection. They separated representation that would support action in the world (what we called *spatial coding*) and that would support reflection about the world (*spatial thought*).

Spatial Thought

As children gain the capacity for symbolic thought in their 2nd year, they begin to gain access to their spatial knowledge and thus build upon spatial codings to create accessible representations, supporting the emergence of spatial thought. We first address the question of whether spatial reasoning is an innate or developed ability and then consider spatial perspective taking, navigation through large-scale environments, and the language of space.

Development of spatial thought. Some have argued that abilities such as spatial inference are innate, on the basis of, for example, the finding that children who are blind can infer paths that they have not been taught (Landau, Gleitman, & Spelke, 1981). However, blind children performed less accurately in the key aspect of the task (accuracy at final position) than age-matched sighted but blindfolded children performed in similar tasks. This is noteworthy given that the blindfolding created an artificial task for the latter children (Morrongiello, Timney, Humphrey, Anderson, & Skory, 1995). Thus, at least some visual experience appears important for full development of spatial knowledge (Arditi, Holtzman, & Kosslyn, 1988; Morrongiello, Timney, Humphrey, Anderson,

& Skory, 1995; Newcombe & Huttenlocher, 2000). Nevertheless, spatial representations are spatial, not strictly visual. Visual input is important for full development.

Spatial perspective taking. A component of Piaget's topological primacy thesis was that "projective relations" did not develop until elementary school age. When they developed, children considered figures and locations in terms of a "point of view." As an example, the concept of the straight line results from the child's act of "taking aim" or "sighting." Children perceive a straight line from their earliest years, but they initially cannot independently place objects along a straight path. They realize, based on perception, that the line is not straight but cannot construct an adequate conceptual representation to make it so. At about 7 years of age, children spontaneously "aim" or sight along a trajectory to construct straight paths.

The Piagetians confirmed these theoretical claims with other experiments, such as the "three mountains" task in which children had to construct a scene from the perspective of a doll. For each new position of the doll, young children methodically went about their task of re-creating the appropriate viewpoint, but it always turned out to be from the same perspective—their own! Thus, Piaget and Inhelder inferred that children must construct systems of reference, not from familiarity born of experience, but from operational linking and coordination of all possible viewpoints, each of which they are conscious. They concluded that such global coordination of viewpoints is the basic prerequisite in constructing simple projective relations. For although such relations are dependent upon a given viewpoint, nevertheless a single point of view cannot exist in an isolated fashion but necessarily entails the construction of a complete system linking together all points of view.

However, other studies have shown conflicting results. For example, young children do recognize that other observers see something different and develop in their ability to construct those viewpoints (Pillow & Flavell, 1986). They may use a "line-of-sight" idea, in which they reason that people can see any object for which one can imagine an unobstructed line between their eyes and the object. Some ability to coordinate perspectives mentally is present in children as young as 18 months. For example, children performed better than chance in retrieving a reward after it was hidden randomly in one of two identical left-right locations on a turntable, then its location reversed via either a 180-degree rotation of the turntable by the experimenter or a move by the child to the opposite side of the table (Benson & Bogartz, 1990).

In summary, the development of projective space may involve not just the coordination of viewpoints

but also the establishment of an external framework. That is, a key to solving the three mountains task may be, again, the conflict between frames of reference, experienced and imagined. Full competence in dealing with such conflicts is achieved at approximately the ages Piaget originally claimed. Simultaneously, children are developing additional knowledge of projective relations. At 4 years of age, but not 3, children begin to understand that moving objects nearer or farther increases and decreases its apparent size; they can also indicate how a circular object should be rotated to make it appear circular or elliptical (Pillow & Flavell, 1986).

Navigation through large-scale environments. Navigation in large environments requires integrated representations, because one can see only some landmarks at any given point. Some researchers believe that people learn to navigate using landmarks; then routes, or connected series of landmarks; then scaled routes; and finally survey knowledge, a kind of mental map that combines many routes and locations (Siegel & White, 1975). Only older preschoolers learn scaled routes for familiar paths; that is, they know about the relative distances between landmarks (Anooshian, Pascal, & McCreath, 1984). Even young children, however, can put different locations along a route into some relationship, at least in certain situations. For example, they can point to one location from another even though they never walked a path that connected the two (Uttal & Wellman, 1989).

Children as young as 3.5 years were able, like adults, to accurately walk along a path that replicated the route between their seat and the teacher's desk in their classroom (Rieser, Garing, & Young, 1994). Self-produced movement is again important. Kindergarten could neither imagine similar movements nor point to various objects accurately, but they could recreate the route and point accurately when they actually walked and turned. Thus, children can build imagery of locations and use it, but they must physically move to show their competence.

Comparing routes, as in finding the shortest of several routes, is a difficult task. Children as young as 1 or 2 years can plan shortest routes only in simple situations. For example, 1.5-year-olds will choose the shortest route around a wall separating them from their mothers (Lockman & Pick, 1984). Such early planning may depend on direct sighting of the goal and on an obvious choice of a shorter or more direct route to the goal (Wellman, Fabricius, & Sophian, 1985). Children 3.5 to 4.5 years of age show a mixture of sighting and planning, that is, considering extended courses of action taking into account the overall distances of competing routes, probably on the basis of qualitative dis-

tance-relevant aspects of routes, such as the necessity of backtracking (Wellman, Fabricius, & Sophian, 1985). There is some evidence that young children can compute shortest routes even when the goal is not visible (Newcombe & Huttenlocher, 2000). A significant proportion (40%) of 4-year-olds can not only identify that a direct and indirect route to a given location are not the same distance but can explain why the direct route was shorter (Fabricius & Wellman, 1993). However, ability to plan routes in situations in which the optimal route involves locations that are not close or in sight (when others are) appears at 5–6 years of age.

Considering these results in the context of the broad research on spatial orientation, it is unsurprising that children grow over a number of years in their ability to form integrated spatial representations. However, evidence does not support a simple, qualitative shift from landmark to route to integrated, 2-dimensional representations (Newcombe & Huttenlocher, 2000). Instead, development results from substantive refinements in the effectiveness and connectedness of already-existing representational systems, including hierarchical categorization (e.g., children create more precise embedded categories as they age). Young children undoubtedly rely on cues (landmarks) and partially connected landmark-and-route representations when they have limited experience in an environment and when information-processing demands, such as those on memory; use of strategies (e.g., turning around to visualize how a route will look on the return trip); and inferences required to combine spatial knowledge from different systems overwhelm their cognitive abilities. They develop abilities such as recognition-in-context memory that facilitate acquisition of landmark knowledge (Allen & Ondracek, 1995).

The language of space. The development of both geometric domains, the “what” and “where” systems (Ungerleider & Mishkin, 1982), begins early in life as children represent objects at a detailed level of shape, and simply as a set of axes (a representation of an object allowing a region in front and a region in back; Landau, 1996). However, spatial relations are not perceived “automatically”; they require attention (Logan, 1994; Regier & Carlson, 2002). Children learning English show strong biases to ignore fine-grained shape when learning novel spatial terms such as “on” or “in front of” or when interpreting known spatial terms, and equally strong biases to attend to fine-grained shape when learning novel object names. For example, 3-year-olds will ignore shape and generalize primarily on the basis of an object's demonstrated location when shown an unusual object and told, “This is acorp my box” [sic] (Landau, 1996).

Children represent objects in terms of axes by the second year of life; contrary to Piagetian theory, the development of regions appears to begin with the axis and broaden afterward. However, spatial terms are acquired in a consistent order, even across different languages (Bowerman, 1996), one of the few sequences that is consistent with the Piagetian topological primacy thesis. The first terms acquired are *in*, *on*, and *under*, along with such vertical directionality terms as *up* and *down*. These initially refer to transformations of one spatial relationship into other (“on” not as a smaller object on top of another, but only as making an object become physically attached to another, Gopnik & Meltzoff, 1986). Second, children learn words of proximity, such as *beside* and *between*. Third, children learn words referring to frames of reference such as *in front of*, *behind*. The words *left* and *right* are learned much later and are the source of confusion for many years.

The consistency of acquisition order, along with early learning of space and the reliance on spatial understanding in learning new words (e.g., predicting what new words mean and extending their use to new situations) argue that spatial language builds upon already-constructed spatial concepts (Bowerman, 1996; Regier & Carlson, 2002; Spelke, 2002). However, that does not mean that children merely map spatial words directly onto extant spatial concepts. The concepts may be just forming, and develop from attempts to change spatial relationships of objects (*down as get me down*) to references to all such changes, and finally to static spatial relationships. The words children use encode concepts that are problematic for them—that they are developing—and are used as a cognitive tool to support that development. In addition, adult language helps children consolidate their emerging understandings (Gopnik & Meltzoff, 1986). Further, semantic organization of language appears to influence children’s development of spatial concepts (Bowerman, 1996). For example, while English uses *on* for contact with and support by a surface (*on a table* and *handle on a door*) and *in* for containment, Finnish categorizes the handle on a door and apple in a bowl together; the horizontal support of “on a table” requires a different construction. In Dutch, the hanging attachment of “handle on a door” is a separate construct (Bowerman, 1996). Consistent with the notion that these differences affect children’s special learning, cross-cultural studies show that, for example, children learning English acquire concepts referring to verticality faster than children learning a language such as Korean for which those terms are less central, whereas Korean children learn different meanings for terms that mean *in* as in *fit tightly* and as in *inside a larger container* whereas English-speaking

children are slower to learn that differentiation, with these differences occurring both in production (Choi & Bowerman, 1991) and in comprehension (Choi, McDonough, Bowerman, & Mandler, 1999) as early as 18 to 23 months. Thus, language appears to affect conceptual growth by affecting what kind of spatial relationships and categories children attend to and build.

By 2 years of age, children have considerable spatial competence on which language might be based. Moreover, the use of even a single-word utterance by a 19-month-old, such as “in” may reflect more spatial competence than it first appears when the contexts differ widely, such as saying “in” when about to climb into the child seat of a shopping cart and saying “in” when looking under couch cushions for coins she just put in the crack between cushions (Bowerman, 1996). However, notions such as *left* and *right*, whose relative understanding may require mental rotation (Roberts & Aman, 1993), may not be fully understood until about 6 to 8 years of age. Between these ages, children also learn to analyze what others need to hear to follow a route through a space. To a large degree, however, development past pre-K depends on sociocultural influences on children’s understanding of conventions, such as negotiating which frame of reference is used (Newcombe & Huttenlocher, 2000). Such influences have strong effects. For example, achieving flexible spatial performance is correlated and may be caused by acquisition-appropriate spatial vocabulary (R. F. Wang & Spelke, 2002). Indeed, it may be the unique way that people combine concepts from different inborn spatial systems into mature and flexible spatial understandings. Finally, although language supports simple representations such as cue learning, place learning is difficult to capture verbally because of the multiple simultaneous relationships. This leads us to more apropos external representations, such as models and maps.

Models and Maps

Young children can represent and, to an extent, mathematize their experiences with navigation. They begin to build mental representations of their spatial environments and can use and create simple maps. Children as young as 2 years of age can connect oblique and eye-level views of the same space, finding their mother behind a barrier after observing the situation from above (Rieser, Doxsey, McCarrell, & Brooks, 1982). In another study, 2.5- but not 2-year-olds could locate a toy shown in a picture of the space, even when 2-year-olds were given help (DeLoache, 1987; DeLoache & Burns, 1994).

To make sense of maps, children have to create relational, geometric correspondences between ele-

ments, as these vary in scale and perspective (Newcombe & Huttenlocher, 2000). By 3 years, children may be able to build simple but meaningful models with landscape toys such as houses, cars, and trees (Blaut & Stea, 1974), although this ability is limited through the age of 6 years (Blades, Spencer, Plester, & Desmond, 2004). However, researchers know less about what specific abilities and strategies they use to do so. For example, kindergarten children making models of their classroom cluster furniture correctly (e.g., they put the furniture for a dramatic play center together) but may not relate the clusters to each other (Siegel & Schadler, 1977).

Thus, preschoolers have some impressive initial competencies, but these are just starting to develop. We begin a more in-depth look at this development by considering children's use, rather than production, of models and maps. One study confirms that children can use both models and maps by 2.5 to 3 years of age, but with a twist. Children were shown a location on a scale model of a room then asked to find same object in the actual room (DeLoache, 1987). For example, a miniature dog was hidden behind a small couch in the model, and the child was asked to find a larger stuffed dog hidden behind a full-sized couch. Interestingly, raising a point to which we shall return, 3- but not 2.5-year-olds could find the corresponding object (the authors do not state, but we assume, that the two were in alignment). However, both ages were successful with line drawings or photographs of the room. It may be that the younger children saw the model as an interesting object, but not as a symbol for another space, leading to the counterintuitive result that the more "concrete" model was less useful to them. In support of this notion, having these children play with the model decreased their success using it as a symbol in the search task, and eliminating any interaction increased their success. The 3s were successful with either, revealing cognitive flexibility in their use of the model, as an object per se and a symbol for another space.

In a similar vein, beginning about 3 and more so at 4 years of age, children can interpret arbitrary symbols on maps, such as a blue rectangle standing for blue couch, or "x marks the spot" (Dalke, 1998). Their abilities lack sophistication; for example, preschoolers recognized roads on a map but suggested that the tennis courts were doors (Liben & Downs, 1989). In another study, 4- and 5-year-olds criticized symbols that lacked features (e.g., tables without legs) but could recognize a plane view of their classroom, so findings such as these may be the result of children merely voicing preferences. All could distinguish between representational (in room) and nonrepresentational (outside of room) paper space (Liben &

Yekel, 1996). In any case, by age 5 or 6, children can consistently interpret the arbitrary symbolic relationships used in maps (Newcombe & Huttenlocher, 2000). Yet, children may understand that symbols on maps represent objects but have limited understanding of the geometric correspondence between maps and the referent space. As we shall show, both understandings are developing but have far to go by the end of the preschool years (Liben & Yekel, 1996).

Shortly after 3 years of age, children are able to scale distance across simple spatial representations, a fundamental competence (Huttenlocher, Newcombe, & Vasilyeva, 1999). However, they perform better with symmetric than asymmetric configurations (Uttal, 1996). Children also understand that a map represents space (Liben & Yekel, 1996). By 4 years of age, they can build upon these abilities and begin interpreting maps, planning navigation, reasoning about maps, and learning from maps, at least in simple situations (Newcombe & Huttenlocher, 2000). For example, 4-year-olds benefit from maps and can use them to guide navigation (i.e., follow a route) in simple situations (Scholnick, Fein, & Campbell, 1990). In a similar study, 4- to 7-year-olds had to learn a route through a six-room playhouse with a clear starting point. Children who examined a map beforehand learned a route more quickly than those who did not (Uttal & Wellman, 1989). Children younger than 6, however, have trouble knowing where they are in the space; therefore, they have difficulty using information available from a map relevant to their own present position (Uttal & Wellman, 1989). Preschoolers also have difficulty aligning maps to the referent space, a skill that improves by age 5 (Liben & Yekel, 1996). Competencies in geometric distances and scaling are underway by age 6 or 7, and primary grade children can recognize features on aerial photographs and large-scale plans of the same area (Blades, Spencer, Plester, & Desmond, 2004; Boardman, 1990), but these abilities continue to improve into adulthood. However, even adults do not attain perfect competence.

The ability to use a map to plan routes is more challenging than following specified routes. This ability is forming at about age 5, although the spaces researchers use are usually simple and rectangular. By 6 years, children can plan routes in more complex environments with multiple alternatives, using distance information. For example, by 5 to 6 years of age, children can use maps to navigate their way around a school but are less successful navigating complex streets or a cave (Jovignot, 1995). More research is needed on naturalistic spaces, as well as on children's ability to plan efficient routes (Newcombe & Huttenlocher, 2000).

Young preschoolers show some ability to create models of spaces such as their classrooms (Blaut & Stea, 1974; Siegel & Schadler, 1977). Preschoolers, like older people, could preserve the configuration of objects when reconstructing a room depicted on a map. However, preschoolers placed objects far from correct locations and performed worse with asymmetric than symmetric configurations (Uttal, 1996). Most 4-year-olds can locate clusters of model furniture items in a scale model of their classroom but get confused when they must position the items, getting only about half the items correct (Golbeck, Rand, & Soundy, 1986; Liben, Moore, & Golbeck, 1982). Much of the difficulty may be not in coding and producing locations, but rather in scaling distances, especially as that difficulty is compounded with multiple elements (Newcombe & Huttenlocher, 2000).

There are individual differences in such abilities. In one study, most preschoolers rebuilt a room better using real furniture than toy models. For some children, however, the difference was slight. Others placed real furniture correctly but grouped the toy models only around the perimeter. Some children placed the models and real furniture randomly, showing few capabilities (Liben, 1988). Even children with similar mental representations may produce quite different maps due to differences in drawing and map-building skills (Uttal & Wellman, 1989). Nevertheless, by the primary grades, most children are able to draw simple sketch-maps of the area around their home from memory (Boardman, 1990).

What accounts for differences and age-related changes? Maturation and development are significant. Children need mental processing capacity to update directions and location. The older they get, the more spatial memories they can store and transformations they can perform. Such increase in processing capacity, along with general experience, determines how a space is represented more than the amount of experience with the particular space (Anooshian, Pascal, & McCreath, 1984). Learning is also important, as will be discussed in a following section.

Fundamental is the connection of primary to secondary uses of maps (Presson, 1987). Even young children can form primary relations to spaces on maps, once they see them as representing a space at about 2.5 to 3 years of age (DeLoache, 1987). They must learn to treat the spatial relations as separate from their immediate environment. These secondary meanings require people to take the perspective of an abstract frame of reference ("as if you were there") that conflicts with the primary meaning. You no longer imagine yourself "inside" but rather must see yourself at a distance, or "outside," the information. Show-

ing children several models and explicitly comparing them using language, and possibly visual highlights, can help (Loewenstein & Gentner, 2001), probably because it helps children notice common relationships on subsequent tasks. Such meanings of maps challenge people from preschool into adulthood, especially when the map is not aligned with the part of the world it represents (Uttal & Wellman, 1989). For example, successful map use and mental rotation abilities are correlated in 4- to 6-year-olds (Scholnick, Fein, & Campbell, 1990). Ability to use misaligned maps, especially those 180° misaligned, shows considerable improvement up to about 8 years of age. In addition, children may learn other strategies to deal with the basic problem of conflicting frames of reference (Newcombe & Huttenlocher, 2000). These findings re-emphasize that researchers must be careful how they interpret the phrase *mental (or cognitive) map*. Spatial information may be different when it is garnered from primary and secondary sources such as maps.

Coordinates and Spatial Structuring

As stated, young children can learn to relate various reference frames, and they appear to use, implicitly, two coordinates in remembering direction, either polar or Cartesian. This seems inconsistent with the Piagetian account of the development of two-dimensional space, in which only in later years do children come to "see" objects as located in a two-dimensional frame of reference. That is, Piaget and Inhelder (1967) challenged the claim that there is an innate tendency or ability to organize objects in a two- or three-dimensional reference frame. Spatial awareness does not begin with such an organization; rather, the frame itself is a culminating point of the development of Euclidean space.

To test their theory in the case of horizontality, Piaget and Inhelder showed children jars half-filled with colored water and asked them to predict the spatial orientation of the water level when the jar was tilted. For verticality, a plumb line was suspended inside an empty jar, which was similarly tilted, or children were asked to draw trees on a hillside. Children initially were incapable of representing planes; a scribble, for example, represented water in a tilted jar. At the next state, the level of the water was always drawn perpendicular to the sides of the jar, regardless of tilt. Satisfaction with such drawings was in no way undermined even when an actual water-filled tilted jar was placed next to the drawing. Striking is "how poorly commonly perceived events are recorded in the absence of a schema within which they may be organized" (1967, p. 388). Sometimes, sensing that the water moves towards the mouth of the jar, children

raised the level of the water, still keeping the surface perpendicular to the sides. Only at the final stage—at about 9 years of age—did children ostensibly draw upon the larger spatial frame of reference (e.g., tabletop) for ascertaining the horizontal.

Ultimately, the frame of reference constituting Euclidean space is analogical to a container, made up of a network of sites or positions. Objects within this container may be mobile, but the positions are stationary. From the simultaneous organization of all possible positions in three dimensions emerges the Euclidean coordinate system. This organization is rooted in the preceding construction of the concept of straight line (as the maintenance of a constant direction of travel), parallels, and angles, followed by the coordination of their orientations and inclinations. This leads to a gradual replacement of relations of order and distance between objects by similar relations between the positions themselves. It is as if a space were emptied of objects to organize the space itself. Thus, intuition of space is not a “reading” or innate apprehension of the properties of objects, but a system of relationships borne in actions performed on these objects.

Young children’s grasp of Euclidean spatial relationships is apparently more adequate than the theory posits. Very young children can orient a horizontal or vertical line in space (Rosser, Horan, Mattson, & Mazzeo, 1984). Similarly, in very simple situations, 4- to 6-year-old children (a) can extrapolate lines from positions on both axes and determine where they intersect, (b) are equally successful going from point to coordinate as going from coordinate to point, and (c) extrapolate as well with or without grid lines (Somerville & Bryant, 1985). Piagetian theory seems correct in postulating that the coordination of relations develops after such early abilities. Young children fail on double-axis orientation tasks even when misleading perceptual cues are eliminated (Rosser, Horan, Mattson, & Mazzeo, 1984). Similarly, the greatest difficulty is in coordinating two extrapolations, which has its developmental origins at the 3- to 4-year-old level, with the ability to extrapolate those lines developing as much as a year earlier (Somerville, Bryant, Mazzocco, & Johnson, 1987).

These results suggest an initial inability to utilize a conceptual coordinate system as an organizing spatial framework (Liben & Yekel, 1996). Some 4-year-olds can use a coordinate reference system, whereas most 6-year-olds can (Blades & Spencer, 1989), at least in scaffolded situations. However, most 4-year-olds can coordinate dimensions if the task is set in a meaningful, guided context in which the orthogonal dimensions are cued by the lines of sight of imaginary people (Bremner, Andreasen, Kendall, & Adams, 1993).

Conceptual integration of coordinates is not limited to two orthogonal dimensions. Children as young as 5 years can metrically represent spatial information in a polar coordinate task, using the same two dimensions as adults, radius and angle, although children do not use categorizations of those dimensions until age 9 (Sandberg & Huttenlocher, 1996).

However, performance on coordinate tasks is influenced by a variety of factors at all ages. Performance on horizontality and verticality tasks may reflect bias toward the perpendicular in copying angles, possibly because this reference is learned early (Ibbotson & Bryant, 1976). Representations of figures also are distorted either locally by angle bisection, or by increasing symmetry of the figure as a whole (Bremner & Taylor, 1982). Finally, performance on these Piagetian spatial tasks correlates with disembedding as well as with general spatial abilities (Liben, 1978). Such results indicate a general tendency to produce symmetry or simplicity in constructions that confound the traditional Piagetian interpretation (Bremner & Taylor, 1982; Mackay, Brazendale, & Wilson, 1972).

Thus, young children have nascent abilities to structure two-dimensional space (which they often have to be prompted to use), but older students often have not developed firm conceptual grounding in grid and coordinate reference systems. To consciously structure space with such systems requires considerable conceptual work. Spatial structuring is the mental operation of constructing an organization or form for an object or set of objects in space (Battista, Clements, Arnoff, Battista, & Borrow, 1998). Structuring is a form of abstraction, the process of selecting, coordinating, unifying, and registering in memory a set of mental objects and actions. Structuring takes previously abstracted items as content and integrates them to form new structures. Spatial structuring precedes meaningful use of grids and coordinate systems. On the one hand, grids that are provided to children may aid their structuring of space; however, children face additional hurdles in understanding grid and coordinate systems. The grid itself may be viewed as a collection of square regions, rather than as sets of perpendicular lines. In addition, order and distance relationships within the grid must be constructed and coordinated across the two dimensions. Labels must be related to grid lines and, in the form of ordered pairs of coordinates, to points on the grid, and eventually integrated with the grid’s order and distance relationships so that they constitute mathematical objects and ultimately can be operated upon.

In summary, even young children can use coordinates that adults provide for them. However, when facing traditional tasks, they and their older peers may

not yet be able or predisposed to spontaneously make and use coordinates for themselves.

As a final note, we argue that the term *spatial structuring* be reserved for this specific construct of organizing such two- or three-dimensional concepts, the context in which the term was created (Battista & Clements, 1996; Battista, Clements, Arnoff, Battista, & Borrow, 1998; Sarama, Clements, Swaminathan, McMillen, & González Gómez, 2003). Although one could think of all geometric and spatial activity discussed in this section as “structuring space,” this would enervate the construct. Furthermore, better, specific local theories exist for other areas of geometry and spatial thinking. Building stronger and more detailed local theories is a superior approach for psychology and education (Newcombe & Huttenlocher, 2000).

Spatial Visualization and Imagery

Spatial visualization is the ability to generate and manipulate images, which are internally experienced, holistic representations of objects that are to a degree isomorphic to their referents. Kosslyn (1983) defined four classes of image processes: generating an image, inspecting an image to answer questions about it, maintaining an image in the service of some other mental operation, and transforming and operating on an image. Thus, spatial visualization involves understanding and performing imagined movements of two- and three-dimensional objects. To do this, one must be able to create a mental image and manipulate it. An image is not a “picture in the head,” although the mental processes are similar to those that underlie the perception of objects (Shepard, 1978). For example, they are integrated and can be scanned or rotated as one would do to perceptually available objects, with transitional images and times in proportion to those of perceptual activity (Eliot, 1987; Kosslyn, Reiser, & Ball, 1978). Images are more abstract, more malleable, and less crisp than pictures. They are often segmented into parts and represent relationships among those parts (Shepard, 1978). Some images can cause difficulties, especially if they are too inflexible, vague, or filled with irrelevant details.

People’s first images are static, not dynamic. They can be mentally re-created and even examined but not transformed. Having a dynamic image allows one to, for example, mentally “move” the image of one shape (such as a book) to another place (such as a bookcase, to see if it will fit) or mentally move (slide) and turn an image of one shape to compare that shape to another one.

In Piagetian theory (Piaget & Inhelder, 1967, 1971), children up to 4 years of age cannot construct an entire image of a two-dimensional shape after only

tactile-kinesthetic experience (visual experiences were thought to rely overly on perceptual thinking) because preschoolers are too passive, touching one part of a shape only. Children aged 4 to 7 would touch another part and regulate their actions by establishing relations among them, building a more accurate representation of the shape. Such processes allow them to accurately distinguish between rectilinear (e.g., a triangle) and curvilinear (e.g., circle) shapes and build images of simple shapes.

Although Piaget argued that most children cannot perform full dynamic motions of images until the primary grades (Piaget & Inhelder, 1967, 1971), pre-K children show initial transformational abilities. Some researchers have reported that second graders learned only manual procedures for producing transformation images but could not mentally perform such transformations (Williford, 1972). In contrast, other studies indicate that even young children can learn something about these motions and appear to internalize them, as indicated by increases on spatial-ability tests (Clements, Battista, Sarama, & Swaminathan, 1997; Del Grande, 1986). Slides appear to be the easiest motions for children, then flips and turns (Perham, 1978); however, the direction of transformation may affect the relative difficulty of turn and flip (Schultz & Austin, 1983). Results depend on specific tasks, of course; even 4- to 5-year-olds can do turns if they have simple tasks and orientation cues (Rosser, Ensing, Glider, & Lane, 1984). For children of ages 4 to 8, there were no significant effects for slides and flips (with the trend being a negative effect for younger children) but a dramatic beneficial effect for turns. A slide task was at least as easy as a flip, and turns were most difficult (Moyer, 1978).

Transformations and perspective taking appear to follow the development of perceptual and imagistic reproduction. There is a hierarchical developmental sequence of reproduction of geometric figures requiring only encoding (i.e., building a matching configuration of shapes, with the original constantly in sight), reproduction requiring memory (building a matching configuration from recall), and transformation involving rotation and visual perspective-taking (building a matching configuration either from recall after a rotation or from another’s perspective), with pre-K children able to perform at only the first two levels (Rosser, Lane, & Mazzeo, 1988). In a similar vein, a framework of imagery for early spatial mathematics learning that is generally consistent with the research reviewed here has been proposed by Owens (1999). In each of three categories, orientation and motion, part-whole relationships, and classification and language, children develop strategies in five sub-

categories: emergent (begin to attend, manipulate, and explore), perceptual (attend to features and make comparisons, rely on what they can see or do), pictorial (use mental images and standard language), pattern and dynamic imagery (understand conceptual relationships), and efficient strategies. Preliminary evidence suggests the validity and usefulness of the framework, for researchers and teachers. Using a similar framework, researchers found that 11% of Australian kindergartners were unable to visualize simple shapes at the beginning of the year (2% at the end of the year in the experimental group); 70% were at the level of forming static, pictorial images in conjunction with models or manipulatives (37%), with only 19% at the level of visualizing effects of motions (52%) and 1% using dynamic imagery (10%; D. M. Clarke et al., 2002). Thus, there is much room for growth in the earliest year, but helping teachers understand developmental progressions and learning trajectories can promote that growth.

Education

Little is known, in the spatial sphere, about specific cultural and educational experiences and their impact on these capabilities. Experience-expectant processes (Greenough, Black, & Wallace, 1987) seem to account for much of children's development. Universal experiences, such as the physical world provides, lead to an interaction of inborn capabilities and environmental inputs that guide development in similar ways across cultures and individuals. However, other competencies, such as spatial reasoning and the use and creation of external spatial representations like language, models, and maps, probably develop via experience-dependent processes (Greenough, Black, & Wallace, 1987), and thus capability differs across cultures and individuals. For example, preschool teachers spend more time with boys than girls and usually interact with boys in the block, construction, sand play, and climbing areas and with girls in the dramatic play area (Ebbeck, 1984). Boys engage in spatial activities more than girls at home do, both alone and with caretakers (N. Newcombe & Sanderson, 1993). Such differences may interact with biology to account for early spatial skill advantages for boys (note that some studies find no gender differences, e.g., Brosnan, 1998; Ehrlich, Levine, & Goldin-Meadow, 2005; Jordan, Kaplan, Oláh, & Locuniak, 2005; Levine, Huttenlocher, Taylor, & Langrock, 1999; Rosser, Horan, Mattson, & Mazzeo, 1984). Thus, there are important questions not only for cognitive psychology, but also for mathematics education research.

The psychological research once again indicates that active experiences, here emphasizing both physi-

cal and mental activity, are appropriate and, in some cases, critical for children, especially as they are just developing a skill. For example, 1-year-olds who walk themselves around a display are more active observers and better locators than those who are carried (Acredolo, Adams, & Goodwyn, 1984). Similarly, self-directed movement at 5 years of age led to superior recall of distances in a spatial layout of a room (Poag, Cohen, & Weatherford, 1983). As children develop, they become able to perform well under a greater variety of conditions (e.g., outcomes did not differ among self- or other-directed movement and viewing conditions for 7-year-olds). These and other studies (Benson & Bogartz, 1990; Newcombe & Huttenlocher, 2000; Rieser, Garing, & Young, 1994) emphasize the importance of self-produced movement for success in spatial tasks and suggest the benefit of maximizing such experience for all young children.

Given the early competence in foundational spatial representational systems, there is every reason to assume that rich environments will contribute to spatial competence at the intuitive and explicit levels. For example, adults responsible for young children know that they learn practical navigation early. Channeling that experience is valuable. For example, when nursery-school children tutor others in guided environments, they build geometrical concepts (Filippaki & Papamichael, 1997). Such environments might include interesting layouts inside and outside classrooms, incidental and planned experiences with landmarks and routes, and frequent discussion about spatial relations on all scales, including distinguishing parts of their bodies (Leushina, 1974/1991) and spatial movements (forward, back), finding a missing object ("under the table that's next to the door"), putting objects away, and finding the way back home from an excursion. Verbal interaction is important. For example, parental scaffolding of spatial communication helped both 3- and 4-year-olds perform direction-giving tasks in which they had to disambiguate by using a second landmark ("it's in the bag *on the table*"), which children are more likely to do the older they are. Both age groups benefited from directive prompts, but 4-year-olds benefited more quickly than younger children did from nondirective prompts (Plumert & Nichols-Whitehead, 1996). Control children never disambiguated, showing that interaction and feedback from others is critical to certain spatial communication tasks.

Representing spatial environments with models and maps, as well as implicitly or verbally, is less likely to be spontaneously developed by children. Specific teaching strategies may therefore yield as yet largely unexplored benefits. Models and maps are sociocultural tools (Gauvain, 1991), but ones whose develop-

ment is not well supported in the U.S. culture. For example, school experiences are limited and fail to connect map skills with other curriculum areas, including mathematics (Muir & Cheek, 1986). Most students do not become competent users of maps even beyond their early childhood years.

Such deficits may have negative ramifications past map use. Using and thinking about maps may contribute to spatial development by helping children acquire abstract concepts of space, the ability to think systematically about spatial relationships that they have not experienced directly, and the ability to consider multiple spatial relations among multiple locations (Uttal, 2000). As we have documented, children benefit from exposure to maps (Uttal & Wellman, 1989) and to overhead views (Rieser, Doxsey, McCarrell, & Brooks, 1982), at least when children use the information (cf. Liben & Yekel, 1996). Indeed, using maps to teach children locations before they entered the space helped children to identify rooms out of the learned sequence, thus improving their mental representation of that space (Uttal & Wellman, 1989).

Instruction on spatial ability, symbolization, and metacognitive skills (consciously self-regulated map-reading behavior through strategic map referral) can increase 4- to 6-year-olds' competence with reading route maps, although it does not overcome age-related differences (Frank, 1987). Using oblique maps (e.g., tables are shown with legs) aids preschoolers' subsequent performance on plan ("bird's-eye view") maps, perhaps because symbolic understanding was developed (Liben & Yekel, 1996). Telling very young children that a model was the result of putting a room in a "shrinking machine" helped the children see the model as a symbolic representation of that space (DeLoache, Miller, Rosengren, & Bryant, 1997). Using structured maps that help preschoolers match the elements on the map with the corresponding elements in the space facilitated their use of maps (DeLoache, Miller, & Pierroustakos, 1998). These may be good beginnings, but models and maps should eventually move beyond overly simple iconic picture maps and challenge children to use geometric correspondences. Teachers need to help children connect the abstract and concrete meanings of map symbols.

According to some research, navigation activities that combine physical movement, paper-and-pencil, and computer work can facilitate learning of mathematics and map skills. Such spatial learning can be particularly meaningful because it can be consistent with young children's ways of moving their bodies (Papert, 1980). For example, young children can abstract and generalize directions and other map concepts by working with the Logo turtle (Borer, 1993; Clements,

Battista, Sarama, Swaminathan, & McMillen, 1997; Clements & Meredith, 1994; Goodrow, Clements, Battista, Sarama, & Akers, 1997; Kull, 1986; Try, 1989; Watson, Lange, & Brinkley, 1992; Weaver, 1991), although results are not guaranteed (Howell, Scott, & Diamond, 1987). The interface must be appropriate, and activities must be well planned (Watson & Brinkley, 1990/91). By giving the turtle directions such as forward 10 steps, right turn, forward 5 steps, they learn orientation, direction, and perspective concepts, among others. Walking paths and then recreating those paths on the computer help them abstract, generalize, and symbolize their experiences navigating. For example, one kindergartner abstracted the geometric notion of *path* saying, "A path is like the trail a bug leaves after it walks through purple paint" (Clements, Battista, & Sarama, 2001). Logo can also control a floor turtle robot, which may have special benefits for certain populations. For example, blind and partially sighted children using a computer-guided floor turtle developed spatial concepts such as right and left and accurate facing movements (Gay, 1989). Other simple (non-Logo) navigational programs may have similar benefits. For example, using such software (with on-screen navigation) has shown to increase kindergartners' understanding of the concepts of left and right (Carlson & White, 1998).

As early as the preschool years, through to first grade, U.S. children perform lower than children in countries such as Japan and China on spatial visualization and imagery tasks (Starkey et al., 1999; Stigler, Lee, & Stevenson, 1990). There is more cultural support for activities in these other countries (e.g., using visual representations, expecting competency in drawing), and, similarly, within cultural groups in the United States, and within higher SES families in all countries (Starkey et al., 1999; Thirumurthy, 2003). Research shows that even pre-K and kindergarten children show initial transformational abilities in certain settings (Clements, Battista, Sarama, & Swaminathan, 1997; Del Grande, 1986). While incomplete, research suggests that all children benefit from developing their ability to create, maintain, and represent mental images of mathematical objects.

Shape

Shape is a fundamental construct in cognitive development in and beyond geometry. For example, young children form artifact categories characterized by similarity among instances in shape (Jones & Smith, 2002). Even very young children show strong biases to attend to fine-grained shape when learning novel object names, at least when directed to a rigid object,

contrary to the topological primacy thesis of Piagetian theory (Landau, 1996). How do children come to understand shape? We first introduce several theoretical perspectives, then present the hierarchic interactionist position that is consistent with extant evidence.

Theories of Young Children's Perception and Knowledge of Shape

Piaget and Inhelder (1967) claimed that young children initially discriminate objects on the basis of "topological" features, such as being closed or otherwise topologically equivalent, especially when given only tactile, rather than visual, perceptions of the shapes. Only older children could discriminate rectilinear from curvilinear forms and, finally, among rectilinear closed shapes, such as squares and diamonds, via systematic and coordinated explorations. However, as previously discussed with regard to spatial thinking, support for this view has decreased. Decades ago, researchers reported that even at the earliest ages (2–3 years), children can distinguish between curvilinear and rectilinear shapes, contrary to the theory (Lovell, 1959; Page, 1959).

A difficulty in designing such experiments is in quantifying the degree of equivalence of the shapes. Geeslin and Shar (1979) modeled figures via a finite set of points on a grid. Degree of distortion was defined as the sum of displacements of these points. The authors postulated that children compare two figures in terms of the amount of distortion necessary to transform one figure into another, after an attempt at superimposition using rigid motions and dilations. This model received strong support; however it is more predictive than explanatory.

An overarching criticism has been that Piaget and Inhelder's use of the terms such as *topological*, *separation*, *proximity*, and *Euclidean*, as well as the application of these and related concepts to the design of their studies, are not mathematically accurate (Darke, 1982; Kapadia, 1974; Martin, 1976). For example, proximity, in Piaget and Inhelder's interpretation of "nearby-ness," is not topological, for it involves distance. Piaget and Inhelder also claimed that only by the stage of formal operations (11 or 12 years) do children synthesize notions of proximity, separation, order, and enclosure to form the notion of continuity. However, continuity is not the synthesis of these concepts; it is a defining concept of topology. If it does not develop until a late stage, the argument for the primacy of topological concepts is weakened (Darke, 1982; Kapadia, 1974). In a similar vein, classifying figures as topological or Euclidean is problematic, as all figures possess both these characteristics.

In sum, results of many of the Piagetian studies may be an artifact of the particular shapes chosen and the abilities of young children to identify and name these shapes (G. H. Fisher, 1965). This does not support a strong version of the topological primacy thesis. The class of topological properties may not enable young children to identify certain shapes. Visually salient properties such as holes, curves, and corners; simplicity; and familiarity—rather than topological versus Euclidean properties—may underlie children's discrimination.

Although the topological primacy thesis is not supported, Piaget and Inhelder's theory included a second theme. They claimed that children's representation of space is not a perceptual "reading off" of their spatial environment but is constructed from prior active manipulation of that environment. From this perspective, abstraction of shape is the result of a coordination of children's actions. Children

can only "abstract" the idea of such a relation as equality on the basis of an action of equalization, the idea of a straight line from the action of following by hand or eye without changing direction, and the idea of an angle from two intersecting movements. (p. 43)

The Piagetian's tactile-kinesthetic experiments appeared to support this view.

In contrast, Gibson (1969) stated that motor activity plays at most an indirect role in perception. She also disagreed with the empiricists who believed that perception was a matter of association. Instead, she claimed that perception involves both learning and development, increasing the person's ability to (selectively) extract information from the stimulation in the environment. Her theory is similar to Gestalt theory, which emphasizes (a) wholes, irreducible to parts, that drive toward "best structures" and (b) developmental processes including articulation and differentiation, rather than accretion by association. However, Gibson emphasized learning in sensory reorganization through processes of filtering and abstraction. Perception begins as only crudely differentiated and grossly selected. With growth and exposure, perception becomes better differentiated and more precise, as the person learns detection of properties, patterns, and distinctive features. Gibson suggested that learning is facilitated by instructors' emphasizing distinctive features and by beginning with broad differences in those features and moving toward finer distinctions.

From this perspective, perception is active, adaptive, internally directed, and self-regulated. In contrast to a radical constructivist orientation, Gibson assumed an external reality, including structure and

information already existing in the stimulus, without the need for cognitive mediation. Consistent with the constructivist interpretation, other research indicates that, although components such as edge-lines may already be discrete, and basic perceptual competencies such as size constancy (correcting for variations in distance) inborn (Granrud, 1987; Slater, Mattock, & Brown, 1990), one cannot claim that ontological reality is divided into invariants (E. Wright, 1983). There is no “information” in the environment separate from the individual.

Also in the constructivist tradition, the theory of Pierre and Dina van Hiele posits that students progress through qualitatively distinct levels of thought in geometry (van Hiele, 1986; van Hiele-Geldof, 1984). At Level 1, the visual level, students can only recognize shapes as wholes and cannot form mental images of them. A given figure is a rectangle, for example, because “it looks like a door.” They do not think about the defining attributes, or properties,⁴ of shapes. At Level 2, descriptive/analytic, students recognize and characterize shapes by their properties. For instance, a child might think of a square as a figure that has four equal sides and four right angles. Properties are established experimentally by observing, measuring, drawing, and model making.

The hierarchic interactionalism perspective builds most directly on elaborations and revisions of the most educationally relevant theory, that of the van Hieles, introducing other theoretical and empirical contributions. To begin, although both theories posit domain-specific developmental progressions, hierarchic interactionalism hypothesizes several revisions of the van Hiele levels (Clements & Battista, 1992). These revisions have been subsequently supported (Clements, Battista, & Sarama, 2001; Clements, Swaminathan, Hannibal, & Sarama, 1999; Yin, 2003). A level antecedent to the visual level is required to describe the empirical corpus. At Level 0, pre-recognition, children cannot reliably distinguish circles, triangles, and squares from nonexamples of those classes. Children at this level are just starting to form visual schemes for the shapes. These early, unconscious schemes are formed through several initial bootstrap competencies. For example, pattern matching through some type of feature analysis (Anderson, 2000; Gibson, Gibson, Pick, & Osser, 1962) is conducted after the visual image of the shape is transformed by heuristics built into the visual system that imposes an intrinsic frame of reference on the shape, possibly using symmetry (Palmer, 1989).

Research indicates that even infants can perceive wholes as well as parts of geometric patterns (Bornstein & Krinsky, 1985) and that children as young as 3 years engage in active spatial analysis in both construction (Tada & Stiles, 1996) and perception (Feeney & Stiles, 1996), a process that changes with development. That is, children will draw a + using 4 separate line segments, treating intersections as junctions of separate parts and will even choose 4 small segments as being most like the goal object. Older children produce one long vertical and two short horizontal segments, and adults two long segments. As another example, when copying a square divided into fourths (all lines parallel or perpendicular), some young children use all short segments, and others use multiple closed forms (e.g., four squares drawn inside the larger square). As early as 4.5 years, most children’s constructions are similar to those of adults. Thus, the youngest children parse out simple, well-formed, spatially independent parts and use simple combination rules such as seriation and adjacency, then organization around a central point, to connect these parts. Older children relate parts across intersections and parse out continuous unsegmented parts, coordinating relations across boundaries (Tada & Stiles, 1996). These first units may be inborn, well-defined, rigid, unarticulated primary structures that may not follow Gestalt principles. Indeed, Gestalt processes themselves, and what adults perceive, may be a result of a developmental process. Development includes both components of *spatial analysis*—identification of the parts of a geometric form and integration of those parts into a coherent whole. Whereas 4-year-olds use fragmented strategies, such as drawing segmented forms radiating around a central point, children as young as 6 years of age possess multiple spatial-analytic strategies and can use strategies like those of adults, at least in simple situations. The strategy they use is a function of both their capabilities and the complexity of the pattern they are copying (Akshoomoff & Stiles, 1995). Throughout development, children process more parts and more difficult sets of relations, such as intersections and oblique segments, in increasingly higher order hierarchical units (Akshoomoff & Stiles, 1995).

Simple, closed shapes may initially tend to form undifferentiated, cohesive units in children’s phenomenological perceptual experiences (c.f. Smith, 1989). (Research indicates that shapes are perceived holistically with a separate subsystem; Ganel & Goolale, 2003.) For example, contour is a salient characteristic for young children (e.g., Tada & Stiles, 1996)

⁴ We reserve the term *property* for those attributes that indicate a relationship between parts, or components, of shapes. Thus, parallel sides, or equal sides, are properties. We use *attributes* and *features* interchangeably to indicate any characteristic of a shape, including properties, other defining characteristics (e.g., straight sides) and nondefining characteristics (e.g., “right-side up”).

that, when suggestive of a spatially continuous object, may direct the perception of the shape as a whole unit. Nascent schemes may ascertain the presence of the features of closed and “rounded” to match circles, “pointy” features along a figural scan-path to match triangles (without necessarily attaching numerical significance to this path, cf. Piaget & Szeminska, 1952; von Glasersfeld, 1982), four near-equal sides with approximately right angles to match squares, and approximate parallelism of opposite “long” sides to match rectangles.

As they develop triangle schemes (cognitive networks of relationships connecting geometric concepts and processes in specific patterns), children develop, tacitly at first, the ability to “see” both the parts *and* the whole so that they do not focus so much on the angles (“pointy parts”) that they consider an angle (e.g., upside-down V) to be a triangle. Similarly, they learn not to focus only on the enclosing, simple contour of the shape that initially led them to consider shapes such as chevrons to be triangles (Owens, 1999). Typical “geometry deprived” (Fuys, Geddes, & Tischler, 1988) young children have a limited number of imagistic schemes for triangle, for example, an equilateral triangle and a right triangle, both with a horizontal base (Hershkowitz et al., 1990; Vinner & Hershkowitz, 1980). These prototype schemes are not rigid, but they have constraints. For example, the more the lengths of the legs of a right triangle differ in length, the less likely it will be assimilated to that prototype. Such prototypes can be thought of as having multivariate distributions of possible values (e.g., for the relationships between the side lengths and for degree of the base’s rotation from the horizontal) in which the nearer to the mode of the distribution the perceived figure is (equal for the side lengths and 0° rotation of the base for this example), the more likely it will be assimilated to that prototype (parallel distributed processing, or PDP, networks and other structural theories model this type of system, see Clements & Battista, 1992; McClelland, Rumelhart, & the PDP Research Group, 1986).

What is the role of language in this process? As the initial schemes are developing, so do mappings similar to those of number, color, and other properties (Sandhofer & Smith, 1999). That is, children first learn that the question “what shape?” is answered by words such as *circle* and *square*. They then map these shape words to a few concrete examples. Next, they combine these abilities to produce correct shape names for prototypical examples of common shapes. Only after these experiences do they build shape categories (probably because the PDP-type networks of each of these examples contains much more information than geometric shape and thus property-to-prop-

erty matching and shape category creation is a slower developmental process, cf. Sandhofer & Smith, 1999). Thus, shape words and names help organize and direct attention to the relevant features of objects.

This analysis reveals that the nature of the van Hiele levels also requires clarification. The “visual” level includes visual/imagistic and verbal declarative knowledge (“knowing what”) about shapes. That is, it is not viable to conceptualize a purely visual level (1), followed and replaced by a basically verbal descriptive level (2) of geometric thinking—a common interpretation. Instead, different types of reasoning—those characterizing different levels—coexist in an individual and can be developing simultaneously but at different rates. Consistent with the tenet of *progressive hierarchization*, levels do not consist of unadulterated knowledge of only one type. This view is consistent with literature from Piagetian and cognitive traditions (e.g., Minsky, 1986; Siegler, 1996; Snyder & Feldman, 1984), as well as reinterpretations of van Hiele theory (Clements & Battista, 1992; Gutiérrez, Jaime, & Fortuny, 1991; Lehrer, Jenkins, & Osana, 1998; Pegg & Davey, 1998).

Certain conditions, such as a request to explain one’s decision, may prompt children, even at early levels, to abstract, attend to, and describe certain features (“this is pointy”). They might also consciously attend to a subset of the shape’s visual characteristics and use such a subset to identify geometric shapes. Thus, their descriptions of shapes may include a variety of terms and attributes (Clements, Battista, & Sarama, 2001; Lehrer, Jenkins, & Osana, 1998). However, these are “centrations” in the Piagetian sense, not integrated with other components of the shape. This supports the recognition of both comparison-to-prototype and attention-to-attributes in young children’s geometric categorization (Clements, Swaminathan, Hannibal, & Sarama, 1999; see also Lehrer, Jenkins, & Osana, 1998). Conditions also affect how such features and prototype matching are processed. For example, young children’s overall acceptance rate of both examples and nonexamples increases with the inclusion of palpable distractors (Hannibal & Clements, 2000).

Thus, even at a very early age children attend to some attribute of their imagistic scheme, even if it is only its “prickliness” for triangles (c.f. Lehrer, Jenkins, & Osana, 1998). Then, as the scheme becomes better formed (and as the child gains the ability and predisposition to give increased selective attention to single dimensions when comparing objects, Smith, 1989), the child is able to discern more attributes (both defining and nondefining) that s/he uses to construct her or his definition of a triangle. Children can match and identify many shapes (most

reliably, prototypical forms determined by both biology and culture). However, they often attend only to a proper subset of a shape's visual characteristics and are unable to identify many common shapes. In tactile contexts, they can distinguish between figures that are curvilinear and those that are rectilinear but not among figures within those classes. Even in visual contexts, they may not be able to construct an image of shapes, or a re-presentation of the image. They are unable to rotate shapes and place them into part-whole relationships (Wheatley & Cobb, 1990). Thus, before Level 1, children lack the ability to construct and manipulate visual images of geometric figures. This example of *cyclic concretization* is consistent with the Piagetian tradition of the construction of geometric objects on the "perceptual plane" before a reconstruction on the "representational" or "imaginal plane" (Piaget & Inhelder, 1967).

Later in development, additional visual-spatial elements, such as the right angles of squares, are incorporated into these schemes, and thus traditional prototypes may be produced. Further, older children can attend to these features separately, whereas younger children are not able or predisposed to focus on single features analytically (Smith, 1989; Vurpillot, 1976). Therefore, younger children can produce a prototype in identifying rectangles without necessarily attending to the components or specific features that constitute these prototypes. For children of all ages, the prototypes may be overgeneralized or undergeneralized compared to mathematical categorization, of course, depending on the examples and nonexamples and teaching acts children experience. Also, progress to strong Level 1 and eventually to Level 2 understanding is protracted. For example, primary grades continue to apply many different types of cognitive actions to shapes, from detection of features like fat or thin, to comparison to prototypical forms, to the action-based embodiment of pushing or pulling on one form to transform it into another (Clements, Battista, & Sarama, 2001; Lehrer, Jenkins, & Osana, 1998).

Thus, young children operating at Levels 0 and 1 show evidence of recognition of components and attributes of shapes, although these features may not be clearly defined (e.g., sides and corners). Some children operating at Level 1 appear to use both matching to a visual prototype (via feature analysis) and reasoning about components and properties to solve these selection tasks. Thus, Level 1 geometric thinking as proposed by the van Hieles is more *syncretic* (a fusion of differing systems) than *visual*, as Clements (1992) suggested. That is, this level is a synthesis of verbal declarative and imagistic knowledge, each interacting with and enhancing the other. We therefore suggest

the term *syncretic level* (indicating a fusion of different perspectives), rather than *visual level*, because we wish to signify a global combination of verbal and imagistic knowledge without an analysis of the specific components and properties of figures. At the *syncretic level*, children more easily use declarative knowledge to explain why a particular figure is not a member of a class, because the contrast between the figure and the visual prototype provokes descriptions of differences (Gibson, 1985).

Children making the transition to the next level sometimes experience conflict between the two parts of the combination (prototype matching vs. component and property analysis), leading to mathematically incorrect and inconsistent task performance. For example, one girl started pre-K with a stable concept, a scheme of the triangle as a visual whole. However, when introduced instructionally to the attributes of triangles, she formed a separate scheme for a "three-sided shape." For some time thereafter, she held complex and unstable ideas about triangles, especially when the two schemes conflicted (Spitler, Sarama, & Clements, 2003). As another example, many young children call a figure a square because it "just looked like one," a typical holistic, visual response. However, some attend to relevant attributes; for them, a square has "four sides the same and four points." Because they have not yet abstracted perpendicularity as another relevant and critical attribute, some accept certain rhombi as squares (Clements, Battista, & Sarama, 2001). That is, even if their prototype has features of perpendicularity (or aspect ratio near 1), young children base judgments on similarity (in this case, near perpendicularity) rather than on identity (perpendicularity), and therefore they accept shapes that are "close enough" (Smith, 1989). The young child's neglect of such relevant (identity) attributes or reliance on irrelevant attributes leads to categorization errors. This is consistent with early findings that preschoolers show a slow development of skills, sudden insight, and regression (Fuson & Murray, 1978).

Mervis and Rosch (1981) theorized that generalizations based on similarity to highly representative exemplars will be the most accurate. This theory would account for the higher number of correct categorizations by those children who appeared to be making categorization decisions based on comparison to a visual prototype without attention to irrelevant attributes. Finally, strong feature-based schemes and integrated declarative knowledge, along with other visual skills, may be necessary for high performance, especially in complex, embedded configurations. To form useful declarative knowledge, especially robust knowledge supporting transition to Level 2, children must

construct and consciously attend to the components and properties of geometric shapes as cognitive objects (a learning process that requires mediation and is probably aided by physical construction tasks using manipulatives as well as reflection often prompted by discussion, points to which we shall return).

Thus, the *hierarchical interactionism* theory predicts that children are developing stronger imagistic prototypes and gradually gaining verbal declarative knowledge. Those figures that are more symmetric and have fewer possible imagistic prototypes (circles and squares) are more amenable to the development of imagistic prototypes and thus show a straightforward improvement of identification accuracy. Rectangles and triangles have more prototypes that are possible. Rectangle identification may improve only over substantial periods. Similarly, shapes such as triangles, the least definable by imagistic prototypes discussed here, may show complex patterns of development as the scheme widens to accept more forms, overwidens, and then must be further constrained.

Children's variegated responses (some visual, some verbal declarative) may be another manifestation of this syncretic level. Further, they substantiate Clements's (1992) claim that geometric levels of thinking coexist, as previously discussed. Progress through such levels is determined more by social influences, specifically instruction, than by age-linked development. Although each higher level builds on the knowledge that constitutes lower levels, the nature of the levels does not preclude the instantiation and application of earlier levels in certain contexts (not necessarily limited to especially demanding or stressful contexts). For each level, there exists a probability for evocation for each of numerous different sets of circumstances, but this process is codetermined by conscious metacognitive control, control that increases as one moves up through the levels, so people have increasing choice to override the default probabilities (*progressive hierarchization*). The use of different levels is environmentally adaptive; thus, the adjective *higher* indicates a higher level of abstraction and generality, without implying either inherent superiority or the abandonment of lower levels as a consequence of the development of higher levels of thinking. Nevertheless, the levels would represent veridical qualitative changes in behavior, especially the construction of mathematical representations (i.e., construction of geometric objects) from action.

In summary, geometric knowledge at every level maintains nonverbal, imagistic components; that is, every mental geometric object includes one or more image schemes—recurrent, dynamic patterns of kinesthetic and visual actions (Johnson, 1987). Imagis-

tic knowledge is not left untransformed and merely "pushed into the background" by higher levels of thinking. Imagery has a number of psychological layers, from more primitive to more sophisticated (each connected to a different level of geometric thinking), which play different (but always critical) roles in thinking depending on which layer is activated. Even at the highest levels, geometric relationships are intertwined with images, though these may be abstract images. Thus, images change over development. The essence of Level 2 thinking, for example, lies in the integration and synthesis of properties of shapes, not merely in their recognition. Through the process of *progressive hierarchization*, children at this level have transcended the perceptual and have constructed the properties as singular mental geometric objects that can be acted upon, not merely as descriptions of visual perceptions or images (cf. Steffe & Cobb, 1988). Ideally, however, these objects are "neither words nor pictures" (Davis, 1984), but a synthesis of verbal declarative and rich imagistic knowledge, each interacting with and supporting the other. The question, therefore, should not be whether geometric thinking is visual or not visual but rather whether imagery is limited to unanalyzed, global visual patterns or includes flexible, dynamic, abstract, manipulable imagistic knowledge (Clements, Swaminathan, Hannibal, & Sarama, 1999).

Thinking and Learning About Specific Shapes

With these processes in mind, it is useful to consider children's learning about specific shapes. Infants may be born with a tendency to form certain mental prototypes. When people in a Stone Age culture with no explicit geometric concepts were asked to choose a "best example" of a group of shapes, such as a group of quadrilaterals and near-quadrilaterals, they chose a square and circle more often, even when close variants were in the group (Rosch, 1975). For example, the group with squares included square-like shapes that were not closed, had curved sides, and had nonright angles. So, people might have innate preferences for closed, symmetric shapes (c.f. Bornstein, Ferdinandsen, & Gross, 1981). Further, symmetry affects shape perception in two ways, on the global level (e.g., preference for symmetry about the vertical, and to lesser extent, horizontal axis) and local level (symmetries within a shape even if the axes are not vertical or horizontal). For example, global symmetries support the recognition of a prototypical "diamond" (with the long diagonal vertical), but the local symmetry still affects the recognition of any rhombus, even if oriented with a horizontal side (Palmer, 1985).

Culture influences these preferences. We conducted an extensive examination of materials that teach

children about shapes from books, toy stores, teacher supply stores, and catalogs. With few exceptions (and with signs that this is changing in recent years), these materials introduce children to triangles, rectangles, and squares in rigid ways. Triangles are usually equilateral or isosceles and have horizontal bases. Most rectangles are horizontal, elongated shapes about twice as long as they are wide. No wonder so many children, even throughout elementary school, say that a square turned is “not a square anymore, it’s a diamond” (Clements, Swaminathan, Hannibal, & Sarama, 1999; Lehrer, Jenkins, & Osana, 1998). In one study, 4- to 5-year-olds considered rotated squares no longer the same shape or even size, 6- to 7-year-olds retained its characteristics but lost its category and name—it was no longer a square, and 8- to 9-year-olds achieved invariance (Vurpillot, 1976). This may reflect systematic bias for horizontal and vertical lines and a need for perceptual learning and flexibility, but restricted experiences exacerbate such limitations. Research indicates that such rigid visual prototypes can rule children’s thinking throughout their lives (Burger & Shaughnessy, 1986; N. D. Fisher, 1978; Fuys, Geddes, & Tischler, 1988; Kabanova-Meller, 1970; Vinner & Hershkowitz, 1980; Zykova, 1969).

Specifically, what visual prototypes and ideas do young children form about common shapes? Decades ago, Fuson and Murray (1978) reported that by 3 years of age over 60% of children could name a circle, square, and triangle. More recently, A. Klein, Starkey, and Wakeley (1999) reported shape-naming accuracy of middle-income 5-year-olds: circle, 85%; triangle, 80% (note all were close to isosceles); square, 78%; rectangle, 44% (note squares were not scored as correct choices).

A study of young children used the same line drawings previously used with elementary students for comparison purposes (Clements, Swaminathan, Hannibal, & Sarama, 1999); replication studies have been conducted in Singapore (Yin, 2003) and Turkey (Aslan, 2004). Children identified circles quite accurately: 92%, 96%, and 99% for 4-, 5-, and 6-year-olds, respectively, in the United States. Only a few of the youngest children chose the ellipse and curved shape. Most children described circles as “round,” if they described them at all. Thus, the circle was easily recognized but relatively difficult to describe for these children. Evidence suggests that they matched the shapes to a visual prototype. Turkish children showed the same pattern (Aslan, 2004).

Children also identified squares fairly well: 82%, 86%, and 91% for 4-, 5-, and 6-year-olds, respectively. Younger children tended to mistakenly choose non-square rhombi; 25% of 6-year-olds and 5% of 7-year-

olds did so in Singapore. However, they were no less accurate in classifying squares without horizontal sides. In Singapore, 7-year-olds were less likely to correctly identify these as squares than were 6-year-olds (Yin, 2003). Children in all three countries were more likely to be accurate in their square identification when their justifications for selection were based on the shape’s attributes (e.g., number and length of sides).

Children were less accurate at recognizing triangles and rectangles (except in Turkey, where children identified these shapes only slightly less accurately, 68% and 71%, than squares, 73%, Aslan, 2004). However, their scores were not low; about 60% correct for triangles. Although ages of children in the studies differed, both the U.S. and Singapore data revealed a phase in which children chose more triangle examples and distractors, then “tightened” their criteria to omit some distractors but also some examples. The children’s visual prototype seems to be of an isosceles triangle.

Young children tended to accept “long” parallelograms or right trapezoids as rectangles. Thus, children’s visual prototype of a rectangle seems to be a four-sided figure with two long parallel sides and “close to” square corners.

Additionally interesting are cross-cultural and longitudinal comparisons. Similarly aged children in the United States performed slightly better than Singapore children on circles and squares, but the reverse was true for triangles, rectangles, and embedded figures (Yin, 2003)—the more complex tasks. However, these differences lacked statistical significance.

In the follow-up study, children ages 3 to 6 were asked to sort a variety of manipulative forms (Hannibal & Clements, 2000). Certain mathematically irrelevant characteristics affected children’s categorizations: skewness, aspect ratio, and, for certain situations, orientation. With these manipulatives, orientation had the least effect. Most children accepted triangles even if their base was not horizontal, although a few protested. Skewness, or lack of symmetry, was more important. Many rejected triangles because “the point on top is not in the middle.” For rectangles, on the other hand, many children accepted nonright parallelograms and right trapezoids. Also important was aspect ratio, the ratio of height to base. Children preferred an aspect ratio near one for triangles, that is, about the same height as width. Other forms were “too pointy” or “too flat.” Children rejected both triangles and rectangles that were “too skinny” or “not wide enough.” These same factors, with one additional factor of size, similarly affected children’s judgments in Turkey (Aslan, 2004).

Another study addressed recognition of squares at various orientations. The percentage of primary

children recognizing squares was 96.1 for squares with a horizontal base, and 93.2, 84.5, and 73.8 for rotated squares, the last 45° from the horizontal (Kerlake, 1979). Not until the age of 8 did children begin to generalize the concept of square. Again, the explanation for such phenomena may be that shapes are perceived relative to a reference-frame structure in which the orientation of the axes is taken as the descriptive standard. The visual system has heuristics for assigning the frame, which usually, but not always, allows the detection of shape equivalence. However, the reference orientation results from an interaction of intrinsic structure (e.g., symmetry) and orientation relative to the environment (verticality or gravity) and the observer (Palmer, 1989).

A study not designed to test the van Hiele theory also provides evidence on classification schemes. Children of 3, 4, 5, 7, and 9 years of age, and adults, were asked to sort shapes including exemplars, variants, and distractors (Satlow & Newcombe, 1998). A substantial change occurred between 4 and especially 5 years of age to 7 years, consistent with Keil's description of characteristic-to-defining shifts, in which older children relied more on rule-based definitions and less on perceptual similarity than younger children. Younger children were more likely to accept distractors with characteristic features and reject variants. Development regarding recognition of variants was incremental, but identification of distractors showed sudden improvement. Consistent with research discussed, shapes with multiple variants, such as triangles, were more difficult. The authors stated that this evidence disconfirms theories of general development, including Piaget's cognitive and van Hiele's mathematical ability. However, the shift itself is consistent with our *hierarchical interactionalism* reinterpretations of van Hiele theory (Clements, Battista, & Sarama, 2001).

In their play, children's behaviors were coded as "pattern and shape" more frequently than six other categories (Seo & Ginsburg, 2004). About 47% of these behaviors involved recognizing, sorting, or naming shapes. However, children's capabilities, in play and other situations, exceed naming and sorting shapes. This continues into the primary grades, as we have shown (Lehrer, Jenkins, & Osana, 1998). As a final example from the U.K., 7-year-olds were given a set of shapes in random order and asked to match and name each. More than 90% easily matched the shapes, but the percentage naming each was circle, 97.4; square, 96.4; equilateral triangle, 92.8; rectangle, 78.1; regular hexagon, 55.3; and regular pentagon, 31.1. They were also less confident about drawing shapes they had more difficulty naming. The percentage able to name properties of the shapes varied

from 92 for the square to 80 for the pentagon. These data substantiate the conclusion, consistent across the studies reviewed here, that children can easily distinguish the shapes but are exposed to a limited number of shape names. Although their performance is lower for unfamiliar shapes, these children can still name properties, further evidence that they are in "geometrically deprived" environments. We now turn to additional aspects of children's knowledge of shape and spatial structure.

3-D Figures

There is a paucity of research on very young children's knowledge of 3-D figures. Babies only 1 or 2 days old can maintain object size despite changes in distance (and thus change in size of the retinal image, Slater, Mattock, & Brown, 1990). That is, they habituate in looking at a sphere of constant size that changes in distance from the newborn, but not when both this distance and the size of the sphere is changed so that the sphere subtends the same angle on the retina (Granrud, 1987). In addition, infants can perceive 3-D shapes; this is limited, however, to continuously moving objects, rather than single or even multiple static views of the same object (Humphrey & Humphrey, 1995).

Two related studies required children to match solids with their nets. Kindergartners had reasonable success when the solids and nets both were made from the same interlocking materials (Leeson, 1995). An advanced kindergartner had more difficulty with drawings of the nets (Leeson, Stewart, & Wright, 1997), possibly because he was unable to visualize the relationship with the more abstract materials.

Congruence, Symmetry, and Transformations

Young children develop beginning ideas not just about shapes, but also about symmetry, congruence, and transformations. The ability to detect symmetry develops early (Vurpillot, 1976). Infants as young as 4 months dishabituate more quickly to symmetric figures than asymmetric figures, at least for vertical symmetry (Bornstein, Ferdinandsen, & Gross, 1981; Bornstein & Krinsky, 1985; Ferguson, Aminoff, & Gentner, 1998; C. B. Fisher, Ferdinandsen, & Bornstein, 1981; Humphrey & Humphrey, 1995). A preference for vertical symmetry seems to develop between 4 and 12 months of age, and vertical bilateral symmetry remains easier for children than horizontal symmetry, which in turn is easier than diagonal symmetries (Genkins, 1975; Palmer, 1985; Palmer & Hemenway, 1978). Extreme spatial separation of components of the pattern caused infants to lose the advantage for vertical symmetry (Bornstein & Krinsky, 1985). Further, they do

not dishabituate to horizontal symmetry or different vertical patterns (C. B. Fisher, Ferdinandsen, & Bornstein, 1981), so initial competence is limited to “goodness of organization.”

There appear to be two phases in recognition of symmetry (Palmer & Hemenway, 1978). The system first selects potential axes of symmetry defined by symmetric components via a crude but rapid analysis of symmetry in all orientations simultaneously. It maps alignment relations, using detection of closed loops, (non)perpendicularity, intersections, and protrusions, to produce structured representations (i.e., structurally consistent matches between identical attributes and relations of the objects in each). Vertical and, to a lesser extent, horizontal axes are preferred, because orientation relationships such as *above* and *beside* develop early; vertical is preferred because *beside* is bidirectional (Ferguson, Aminoff, & Gentner, 1998). The system then evaluates specific axes sequentially in a detailed comparison (Palmer & Hemenway, 1978), which has a significant influence on shape recognition (Palmer, 1989).

Children often use and refer to rotational symmetry as much as they do line symmetry in working with pattern blocks, such as remarking that an equilateral triangle was “special, because when you turn it a little it fits back on itself” (Sarama, Clements, & Vukelic, 1996). They also produce symmetry in their play (Seo & Ginsburg, 2004). For example, preschooler Jose puts a double unit block on the rug, two unit blocks on the double unit block, and a triangle unit on the middle, building a symmetrical structure. Play in this category, “pattern and shape,” was the most frequent of the six categories coded. For people of all ages, symmetric shapes are detected faster, discriminated more accurately, and often remembered better than asymmetrical ones (Bornstein, Ferdinandsen, & Gross, 1981). However, many explicit concepts of symmetry are not firmly established before 12 years of age (Genkins, 1975).

Many young children judge congruence (Are these two shapes “the same?”) based on whether they are, overall, more similar than different (Vurpillot, 1976). As with symmetry, the comparison of two figures may evoke a pair of structured representations, and the comparison is represented as a mapping between sets of relations between components of the representations (Ferguson, Aminoff, & Gentner, 1998). Children younger than 5.5 years may not do an exhaustive comparison and until about 7 years of age may not attend to the spatial relationships of all the parts of complex figures (Vurpillot, 1976). Shown pairs of figures, some congruent, but all rotated, kindergartners tend

to judge all pairs as “different,” considering orientation a significant feature (Rosser, 1994).

With guidance, however, even 4-year-olds and some younger children can generate strategies for verifying congruence for some tasks. Preschoolers often try to judge congruence using an edge-matching strategy, although only about 50% can do it successfully (Beilin, 1984; Beilin, Klein, & Whitehurst, 1982). They gradually develop a greater awareness of the types of differences between figures that are considered relevant and move from considering various parts of shapes to considering the spatial relationships of these parts (Vurpillot, 1976). Thus, strategies supersede one another in development (e.g., motion-based superposition), becoming more powerful, sophisticated, geometrical, and accurate (recall that rigid motions were discussed in the section on spatial thinking).

The origins of the symmetry and equivalence concepts, however, may lie in early actions. Evidence indicates that children as young as 18 months will pick up two blocks and bang them together and are more likely to choose equivalent blocks than other available blocks. Children may progress through levels, from experiencing similar input between two hands (symmetry in action), to bilateral banging, to relating the block forms (at about 2 years of age), stacking equivalent blocks (an early explicit attempt to create a static expression of equivalence), placing blocks side by side, and eventually spacing equivalent blocks (Forman, 1982). Thus, human sensitivity to symmetry and equivalences may be prefigured in the bilateral symmetry of our anatomy, prefigured in the sense that particular types of object manipulatives are more likely to occur as a result of our having hands bilaterally opposed, and the feedback that is thus pleasing, as having identical objects in two hands (Forman, 1982), perhaps reflecting perception of the equivalence of our own bodies and even neurological symmetries.

Under the right conditions, children of all ages can apply similarity transformations to shapes. Even 4- and 5-year-olds can identify similar shapes in some circumstances (Sophian & Crosby, 1998). The coordination of height and width information to perceive the proportional shape of a rectangle (fat vs. skinny, wide or tall) might be a basic way of accessing proportionality information. This may serve as a foundation for other types of proportionality, especially fractions.

Education

Research indicates the importance of educational experiences. If the examples and nonexamples children experience are rigid, not representing the range of the shape category, so will be their concepts. For example, many children learn to accept only isosceles tri-

angles, often only with a horizontal base, as “triangles.” Others learn richer concepts, even at a young age; for example, one of the youngest 3-year-olds scored higher than every 6-year-old on shape-recognition tasks (Clements, Swaminathan, Hannibal, & Sarama, 1999). Concepts of two-dimensional shapes begin forming in the pre-K years and stabilize as early as age 6 (Gagatsis & Patronis, 1990; Hannibal & Clements, 2000). It is therefore critical that children be provided better opportunities to learn about geometric figures between 3 and 6 years of age. This learning will be more effective if it includes a full range of examples and distractors to build valid and strong concept images, including dynamic and flexible imagery (Owens, 1999).

Unfortunately, U.S. educational practice does not reflect these recommendations. Children often know as much about shapes entering school as their geometry curriculum “teaches” them in the early grades. This is due to teachers’ and curriculum writers’ assumptions that children in early childhood classrooms have little or no knowledge of geometric figures. Further, teachers have had few experiences with geometry in their own education or in their professional development. Thus, it is unsurprising that most classrooms exhibit limited geometry instruction. One early study found that kindergarten children had a great deal of knowledge about shapes and matching shapes before instruction began. Their teacher tended to elicit and verify this prior knowledge but did not add content or develop new knowledge. That is, about two thirds of the interactions had children repeat what they already knew in a repetitious format as in the following exchange: Teacher: “Could you tell us what type of shape that is?” Children: “A square.” Teacher: “Okay. It’s a square” (B. Thomas, 1982).

A more recent study confirmed that current practices in the primary grades also promote little conceptual change: First-grade students in one study were more likely than older children to differentiate one polygon from another by counting sides or vertices (Lehrer, Jenkins, & Osana, 1998). Over time, children were less likely to notice these attributes, given conventional instruction of geometry.

Such comparisons may be present even among preschoolers in various countries (Starkey et al., 1999). On a geometry assessment, 4-year-olds from America scored 55% compared to those from China at 84%. Thus, cultural supports are lacking from the earliest years in the United States. Computer environments can catalyze children’s thinking about squares and rectangles for young children. In one large study (Clements, Battista, & Sarama, 2001), some kindergartners formed their own concept (e.g., “it’s a square rectangle”) in response to their work with Logo mi-

croworlds. This concept was applied only in certain situations: Squares were still squares, and rectangles were still rectangles, unless one formed a square while working with procedures—on the computer or in drawing—that were designed to produce rectangles. The concept was strongly visual in nature, and no logical classification *per se*, such as class inclusion processes, could be inferred. The creation, application, and discussion of the concept, however, were arguably a valuable intellectual exercise.

Less is known about teaching concepts of 3-D shapes. Certainly, consistent experience with building materials such as building blocks seems warranted (Leeb-Lundberg, 1996), especially as children engage in considerable premathematical play with these materials (Seo & Ginsburg, 2004) and can build general reasoning skills in this geometric context (Kamii, Miyakawa, & Kato, 2004). However, it is critical that teachers mathematize such activity. They can engage in fruitful discussions of blocks and other solids, and using specific terminology for solids, faces, and edges makes such discussion particularly beneficial (Nieuwoudt & van Niekerk, 1997).

Children can develop theories-in-action involving congruence as toddlers, stacking congruent blocks to make towers (Kamii, Miyakawa, & Kato, 2004). Block building can be used to develop such early notions throughout the preschool years. Beginning as early as 4 years of age, children can create and use strategies for judging whether two figures are “the same shape,” and over the next couple of years, they can develop sophisticated and accurate mathematical procedures for determining congruence. Using puzzles and matching games, with task selection and scaffolding adjusted to children’s strategies, caretakers can effectively move children through this learning trajectory.

There is mixed evidence regarding young children’s ability with geometric motions. Pre-K to kindergarten children may be limited in their ability to mentally transform shapes, although there is evidence that even these sophisticated processes are achievable (Ehrlich, Levine, & Goldin-Meadow, 2005; Levine, Huttenlocher, Taylor, & Langrock, 1999). Further, these children can learn to perform rotations on objects (physical or virtual), and a rich curriculum, enhanced by manipulatives and computer tools, may reveal that knowledge and mental processes are valid educational goals for most young children. For example, interventions improve the spatial skills of low-income kindergartners, especially when embedded in a story context (Casey, 2005). Computers are especially helpful, as the screen tools make motions more accessible to reflection, and thus bring them to an explicit

level of awareness for children (Clements & Sarama, 2003; Sarama, Clements, & Vukelic, 1996).

Computer environments also can be helpful in learning congruence and symmetry (Clements, Battista, & Sarama, 2001). Indeed, the effects of Logo microworlds on symmetry were particularly strong for kindergarten children. Writing Logo commands for the creation of symmetric figures, testing symmetry by flipping figures via commands, and discussing these actions apparently encouraged children to build richer and more general images of symmetric relations (with possibly some overgeneralization). Children had to abstract and externally represent their actions in a more explicit and precise fashion in Logo activities than, say, in freehand drawing of symmetric figures. There is undeveloped potential in generating curricula that seriously consider children's intuitions, preference, and interest in symmetry.

Research supports the use of manipulatives in developing geometric and spatial thinking in young children (Clements & McMillen, 1996). Using a greater variety of manipulatives is beneficial (Greeb, 1978). Such tactile-kinesthetic experiences as body movement and manipulating geometric solids help young children learn geometric concepts (Gerhardt, 1973; Prigge, 1978). Children also fare better with solid cutouts than printed forms, the former encouraging the use of more senses (Stevenson & McBee, 1958). Pictures also can be important; even children as young as 5 or 6 years (but not younger) can use information in pictures to build a pyramid, for example (Murphy & Wood, 1981). Thus, pictures can give children an immediate, intuitive grasp of certain geometric ideas. However, pictures need to be sufficiently varied so that children do not form limited ideas. Further, research indicates that pictures are rarely superior to manipulatives. In fact, in some cases, pictures may not differ in effectiveness from instruction with symbols (Sowell, 1989). But the reason may not lie in the "nonconcrete" nature of the pictures as much as it lies in their "nonmanipulability"—that is, that children cannot act on them as flexibly and extensively. Research shows that manipulatives on computers can have real benefit.

Instructional aids help because they are manipulable and meaningful. In providing these features, computers can provide representations that are just as real and helpful to young children as physical manipulatives. In fact, they may have specific advantages (Clements & McMillen, 1996). For example, children and teachers can save and later retrieve any arrangement of computer manipulatives. Similarly, computers allow one to store more than static configurations. They can record and replay *sequences* of actions on ma-

nipulatives. This helps young children form dynamic images. Computers can help children become aware of and mathematize their actions. For example, very young children can move puzzle pieces into place, but they do not think about their actions. Using the computer, however, helps children become aware of and describe these motions (Clements & Battista, 1991; Johnson-Gentile, Clements, & Battista, 1994).

Composition of Shapes

Disembedding Shapes

Children develop over years in learning how to separate structures within embedded figures. Visual discrimination, including figure-ground discrimination, seems to be innate, and visual stimuli are perceptually organized in the 1st year of life (Vurpillot, 1976). The primary perceptual structures operate as rigid, indivisible, unanalyzable, and unarticulated up to about 4 years of age, but between 6 and 8 years they become flexible, decomposable, and composable. Eventually these concepts and operations operate hierarchically (i.e., units can be combined, each one serving as a whole to the smaller units of which it is composed and, at the same time, as a component part of the comprehensive structure). Thus, in this age range, children develop the ability to break down line figures and reassemble them in new forms (as in embedded-figures problems), link up isolated perceptual units by means of imaginary lines to identify the more complex structures of incomplete figures, and pass from one structure to another when these are reversible figures. Gestalt theory leads to the following principles.

1. All the lines of a figure are involved in construction of the primary contour structures (PCS).
2. No line or part can belong to more than one primary structure.
3. A line belongs in its entirety to a single primary contour structure.
4. The PCS are preferably symmetrical or at least as regular as possible.
5. The number of PCS must be the fewest possible.
6. Each area entirely surrounded by the lines of a figure and not crossed by another line constitutes a primary area structure (PAS).

Solving embedded-figures problems requires going beyond the primary structure to create secondary structures, borrowed from one or more PCS.

The preschooler can thus find a figure identical to a given PCS but in other cases will incorrectly designate a PCS as the solution, as empirical findings suggest. The following problems present increasing degrees of difficulty, the first three primary, the second three secondary (in which parts can belong simultaneously to several structures, allowing the construction of new units) levels of organization.

1. The PCS that is the frame, or external contour (e.g., largest square in the 3 by 3 arrangement of squares).
2. A PCS of a complex figure (triangle of Star of David).
3. A PCS other than the frame (e.g., the tic-tac-toe of the arrangement of squares).
4. A figure that is not a PCS but encloses a PAS (e.g., smallest square of the arrangement of squares).
5. Similar, but a sum of several PAS.
6. A nonclosed figure that is not a PCS.

In finding a secondary structure, children less than 6 years of age often identify a PCS instead. In sum, before 6 years of age, what children perceive is organized in a rigid manner into structures whose form is determined by Gestalt principles. Children grow in the flexibility of the perceptual organizations they can create. They eventually integrate parts and use imaginary components, with anticipation. Beyond a certain level of complexity, people of any age cannot perceive secondary structures but must construct them piece-by-piece.

Composition and Decomposition

The ability to describe, use, and visualize the effects of composing and decomposing geometric regions is significant in that the concepts and actions of creating and then iterating units and higher order units in the context of constructing patterns, measuring, and computing are established bases for mathematical understanding and analysis (Clements, Battista, Sarama, & Swaminathan, 1997; Reynolds & Wheatley, 1996; Steffe & Cobb, 1988). Additionally, there is empirical support that this type of composition corresponds with and supports children's ability to compose and decompose numbers (Clements, Sarama, Battista, & Swaminathan, 1996).

Block building provides a view of children's initial abilities to compose 3-D objects (as well as their formation of a systemic of logic, cf. Forman, 1982). As they do with 2-D shapes, children initially build structures from simple components and later explicitly synthesize 3-D shapes into higher order 3-D

shapes. Children either engage in little systematic organization of objects or show little interest in stacking in their 1st year (Forman, 1982; Kamii, Miyakawa, & Kato, 2004; Stiles & Stern, 2001). Stacking begins at 1 year, thus showing use of the spatial relationship of *on* (Kamii, Miyakawa, & Kato, 2004). Occasionally, children balance blocks intuitively, but often they place blocks off center or on an edge of a triangular prism. In the latter case, they could recognize lack of success when the block fell but made no attempt to understand it. The *next-to* relation develops at about 1.5 years (Stiles-Davis, 1988). At 2 years, children place each successive block congruently on or next to the one previously placed (Stiles-Davis, 1988). They appear to recognize that blocks do not fall when so placed. Children begin to reflect (think back) and anticipate (Kamii, Miyakawa, & Kato, 2004). At 3 to 4 years of age, children regularly build vertical and horizontal components within a building (Stiles & Stern, 2001). They can use the relations only in sequence at age 3, creating multicomponent structures, but within a limited range. When asked to build "a tall tower," they use long blocks vertically, because they have added to the goal of making a stable tower, making a stable tall tower, first using only one block in this fashion, then several (Kamii, Miyakawa, & Kato, 2004). At 4 years, they can use multiple spatial relations, extending in multiple directions and with multiple points of contact among components, showing flexibility in how they generate and integrate parts of the structure. A small number of children will build a tower with all blocks; for example, by composing the triangular blocks, making subparts to coordinate with the whole (Kamii, Miyakawa, & Kato, 2004).

This developmental progression was used to study children's processes in relation to task complexity (Stiles & Stern, 2001). Three-year-olds could not produce the more complex constructions, including an enclosure, +, and horizontal corner. Their strategy was simple and unsystematic accretion of parts. Children of 4 or 5 years could use a more sophisticated strategy organized around main construction components, but they regressed to simpler strategies for the most difficult tasks. Children of 6 years did not differ from adults on these tasks. Thus, spatial processing in young children is not qualitatively different from that of older children or adults. However, with age, children produce progressively more elaborate constructions. The way they analyze spatial arrays also changes; they segment out different elements and relational structures. For example, to make a +, adults often complete one long segment and add two short segments, but children build four

components around a point. Finally, as they develop, children choose more advanced strategies, but this is influenced by the task.

In this and other studies, Kamii argued for focusing on children's general reasoning with a Piagetian perspective. From a different perspective, children's performance on a block design task was consistent with Vygotskian theory, predicted by both preschoolers' initial abilities and a dynamic assessment of their learning potential (Day, Engelhardt, Maxwell, & Bolig, 1997). Both these areas were domain-specific, supporting the position that cognition is multidimensional.

A related study required children to make cube buildings from 2-D representations or determine the number of cubes in a pictured cube building. A case study indicated that an advanced kindergartner could do the former, but not the latter. Perhaps he was unable to visualize the shapes and needed manipulatives (Leeson, Stewart, & Wright, 1997). Thus, less advanced children in pre-K and kindergarten may require hands-on materials to help them interpret 2-D representations of 3-D shapes.

These 3-D studies only touched on children's abilities to actually compose shapes. Research on 2-D shape composition provides a more detailed picture. The following is a research-based developmental progression, which approximately spans ages 4 to 8 years (Clements, Sarama, & Wilson, 2001; Sarama, Clements, & Vukelic, 1996). From lack of competence in composing geometric shapes, children gain abilities to combine shapes—initially through trial and error and gradually by attributes—into pictures and finally synthesize combinations of shapes into new shapes (composite shapes). Three sources of evidence support the validity of this theory (Clements, Sarama, & Wilson, 2001; Clements, Wilson, & Sarama, 2004). First, the original hypothetical learning trajectory and the developmental progression underlying it emerged from naturalistic observations of young children composing shapes (Clements, Battista, Sarama, & Swaminathan, 1997; Mansfield & Scott, 1990; Sales, 1994; Sarama, Clements, & Vukelic, 1996; Sarama & Clements, in preparation). Second, the levels of the developmental progression were tested iteratively in formative research that involved researchers and teachers. Their case studies indicated that about four fifths of the children studied evinced behaviors consistent with the developmental progression (using an early version of the instrument). By the end of this phase, all participants believed that the developmental progression and the items retained to measure levels in the progression were valid and that they could reliably classify children as exhibiting thinking on the progression. Third, a summative study employed the

final instrument with 72 randomly selected children from pre-K to grade 2. Analyses revealed that the level scores fit the hypothesized structure in which scores from one level would be more highly correlated with scores immediately adjacent to that level than to scores on levels nonadjacent to the given level. Further, the developmental progression showed development across ages, with children at each grade scoring significantly higher than those at the preceding grade did. In addition, support for this theory lies in its consistency with previously discussed research on children's perceptions of shape. In both, parts are related to wholes, with each part initially playing a single functional role in the pattern structure (cf. Tada & Stiles, 1996). Indeed, children spatially isolate parts at first, then arrange them contiguously, and later combine them in an integrative manner, eventually creating more complex units within different structural layers. In both cases, mature cognition is a result of a developmental process in which parts and wholes are interrelated across hierarchical levels.

Summary and Issues

Theoretical Framework

Our theoretical framework, hierarchic interactionism, is supported by the research on geometric and spatial thinking. Foundational geometric competencies are inborn or developed early, become represented with language, and eventually are reflected upon metacognitively (probably following phases of representational redescription, cf. Clements, Battista, & Sarama, 2001; Karmiloff-Smith, 1984, 1986, 1990). Developmental progressions, along with specific mental objects and actions defining each level, have been identified and, especially in the domain of knowledge of traditional topics such as geometric shapes, have been generally confirmed through a variety of research methods, which describe the *hierarchic development* of those topics separately, as well as their progressive hierarchization (Clements, Battista, & Sarama, 2001; Lewellen, 1992). Further, these progressions are topic specific, as the theory indicates, rather than general (van Hiele, 1986). Environment and culture affect spatial reasoning, although many are experience-expectant processes, but they are particularly important in the learning of geometry, with both children and adults showing low levels of geometric thinking if they are deprived of high-quality education. Learning trajectories appear to be an effective basis for curriculum and teaching (Clements, 2003; Clements, Wilson, & Sarama, 2004).

Although the section "Theories of Young Children's Perception and Knowledge of Shape" elabo-

rated on the hierarchic interactionism framework, a few additional points can be made. There is particular support for hierarchic development and the existence of different developmental courses in geometry (Clements, Battista, & Sarama, 2001). That is, levels of thinking about geometric shapes, for example, do not consist of unadulterated knowledge of one type only. Levels of geometric knowledge are not strictly visual, followed by descriptive/analytic (as described in van Hiele, 1986). Instead, children at both levels possess shape schemes that include visual/imagistic and nonvisual or verbal declarative knowledge (“knowing what”) about shapes.

Research has substantiated that children possess several different prototypes for figures (e.g., a vertically and a horizontally oriented rectangle) without accepting the “middle” case (e.g., an obliquely oriented rectangle). In one study, participants studied a preponderance of rectangles with extreme values and few intermediate values of variables such as size (Neumann, 1977). Participants were presented with test items and asked to rate their confidence that they had already studied that particular item. Interestingly, they rated the extremes (e.g., large or small rectangles) much higher than items created by using the mean of these values (e.g., middle-sized rectangles), showing that they extracted multiple foci of centrality, thereby creating several visual prototypes. Thus, they did not cognitively “average” what they had studied. This finding is consistent with studies on the van Hiele theory (Burger & Shaughnessy, 1986; Clements & Battista, 1992; Clements, Battista, & Sarama, 2001; Fuys, Geddes, & Tischler, 1988).

Initially, children have constrained imagistic prototypes for triangles but also possess verbal declarative knowledge. With repeated exposure to exemplars in the culture, these prototypes grow stronger; to the extent these exemplars are limited (e.g., mostly equilateral for many shape categories), acceptance decreases radically, leading to the child’s rejection of both distractors and variants that do not closely match the visual prototype, which may have rigid constraints regarding aspect ratio, skewness, and orientation. This rejection may be particularly noticeable in situations in which shapes are drawn (or otherwise cannot be manipulated) and are presented in a canonical rectangular frame. Nascent declarative knowledge, while developing, does not gain transcendence in the scheme (consistent with the *hierarchic development* tenet of the theory). (Note this provides an elaboration and mechanism for the construct of concept images—a combination of all the mental pictures and properties that have been associated with the concept, Vinner & Hershkowitz, 1980.) Even people who know a correct verbal description of a concept

and possess a specific visual image (or concept image) associated strongly with the concept may have difficulty applying the verbal description correctly. Eventually, exposure to a wider variety of examples and a strengthening of declarative knowledge (“3 straight sides”) leads to a wider acceptance of varieties of geometric figures while rejecting nonmembers of the class. Each type of knowledge increasingly constrains the other (*co-mutual development of concepts and skills*).

Consistent with the tenet of *different developmental courses* and the evidence that children possess multiples types of geometric knowledge, we suggest that children’s knowledge of geometry might be enhanced in different ways. First, their imagistic prototypes might be vastly elaborated by the presentation of a variety of variants, via the systematic variation of irrelevant and relevant attributes (e.g., through dynamic media such as the computer). Children must actively attend to the examples and connect them as through verbal labeling. Such enhancement is accomplished mainly through a usually unconscious visual-induction process. Second, through the presentation of particular tasks and engagement in dialogue about them, children’s verbal declarative knowledge might be refined to extend, elaborate, and constrain their visual knowledge. Preschoolers exposed to such discussions consciously used and were excited about their ability to tell if any figure was a triangle just by counting to see if it had three straight sides (Clements & Sarama, *in press*).

We hypothesize that each of these two paths toward enhancing knowledge of geometry can be followed separately or together. If separately, knowledge of one type can “substitute” for knowledge of another type on certain tasks, within certain limits (including a performance decrement in accuracy or speed of execution). For example, a rich and varied exposure to various examples, could allow near-perfect performance on triangle discrimination tasks such as those discussed previously. Performance would suffer only if a figure were presented that fell outside the range of any of the multiple imagistic prototypes developed by the children, or if the task demanded reasoning based on properties not supported by such imagistic-oriented schemes (e.g., calculations regarding angle relationships), in which case the schemes would be inadequate to the task.

Similarly, well-developed verbal declarative knowledge, even in the absence of exposure to various examples, could allow perfect performance on such simple discrimination tasks. However, performance would suffer (at least in speed of execution) because a chain of reasoning would be required for every figure that could not be immediately assimilated into the

(hypothesized-limited) range of imagistic prototypes. In each of these cases, the characteristics of the figures would have to be “read off” the figure and compared to those that are held in verbal declarative knowledge as characteristics of the class. This type of verbal knowledge would support more sophisticated analysis of geometric figures. Without the imagistic knowledge, however, the range and flexibility of the application of this knowledge would be limited. Research on the learning of geometry shows that students limit their conceptualizations to studied exemplars and often consider common but nondefining features (e.g., an altitude of a triangle is always located inside the triangle) as essential features of the concept (Burger & Shaughnessy, 1986; N. D. Fisher, 1978; Fuys, Geddes, & Tischler, 1988; Kabanova-Meller, 1970; Zykova, 1969). Further, students who know a correct verbal definition of a concept but also have a limited visual prototype associated with it may have difficulty applying the verbal description correctly (Clements & Battista, 1989; Hershkowitz et al., 1990; Vinner & Hershkowitz, 1980). This is consistent with the theory of simultaneous and sequential processing. In this view, pictorially presented materials may be more likely to evoke the visual/holistic/simultaneous processing, whereas aurally presented materials would be more likely to evoke verbal/sequential/linear processing. The type of presentation may interact with individual differences in abilities in these two domains.

This view is consistent with the work of Karmiloff-Smith (1984; , 1986; , 1990), who postulated a repeating 3-phase cycle of representational redescription. Phases, in contrast to structurally constituted stages or levels, are recurrent, general (across-domain) processes that people work through during development or learning. At Phase 1, the building of mental representations is predominantly driven by interactions of the children’s goal-directed schemes with the environment. The children’s goal is behavioral success, or the reaching of that goal, which sometimes is evaluated by consistency with adult responses and feedback.⁵ Such success leads to the recording of isolated correspondences between environmental situations that do or do not allow or aid the attainment of the goal in a form inaccessible to the system (i.e., input-output correspondences along with their contexts in compiled form). In this form, any relationship between bits of knowledge is, at best, implicit.

When behavioral success is achieved, a Phase 2 metaprocess (a procedure that operates on internal knowledge structures) evaluates the knowledge base. Now, the goal is not to behave successfully, but to gain control over the representational forms (Vygotsky, 1934/1986, similarly stated that the development of thought cannot be derived from the failure of thought and postulated a genetic predisposition to gain control of mental representations). The first operation of Phase 2 is to rerecord Phase 1 representations in a form that can be accessed, though not yet consciously. The implicit representations are analyzed into semanticized components, linking them into a simplified but growing network structure that is predicated on the usefulness of the initial correspondences to goal attainment. The second operation is to form relationships between bits of knowledge. These two new operations place demands on cognitive processing, which, together with the (over)simplified structure and the need to test the mental representation in new situations, often leads to new “errors”—ostensibly a step backward to Phase 1 from an adult’s perspective—that mask the progress in explicating representations of the domain. Finally, these two operations constitute an internalization of relationships and processes that were previously only implicit (cf. Steffe & Cobb, 1988).

Once successful rerepresentation is achieved, children develop Phase 3 control mechanisms that integrate and balance consideration of the external environment and the new internal representational connections forged in Phase 2. At the end of Phase 3, these connections are rerecorded again in abstract symbolic form, the first form accessible to conscious thought. Now, performance improves beyond that which was achieved at either of the two earlier phases.

At the syncretic level of geometric thinking, children are implicitly recognizing the properties of shapes. For example, their schemes for squares and for rectangles both contain patterns for right angles. However, these are patterns in spatial subsystems that emerge when instantiated; they are not conceptual objects (mental entities that can be manipulated or scrutinized, Davis, 1984). Right angles are not represented explicitly, and therefore no relationship is formed between them. In general, this type of representation explains how operation at one level can presuppose knowledge from the succeeding level, without allowing access to knowledge at the higher level because

⁵ This sentence could be misconstrued to mean that students are controlled by the external environment. We assume that students are always sense-making beings; however, during this phase in building a representation, they are actively making sense of their social and physical environments, rather than their representations, of which they are not yet conscious. This active sense-making is critical, in that it allows the students to differentiate between environments that do and do not assist goal attainment.

the form of such knowledge is proceduralized or schematized behaviorally and is, therefore, inaccessible to the rest of the cognitive system.

When shapes are dealt with successfully on the level of behavior, metaprocesses rerecord the mental representations, creating a mental geometric object for the visual image of the right angles and a link between these mental objects for different cases of right angles, including links between those in rectangles and those in squares. Because children in Phase 2 are seeking control over their representations of these geometric forms, some increases in “errors” (from an adult’s point of view) may occur (e.g., maintaining the notion that a nonrectangular parallelogram is a rectangle “because you’re only looking at it from the side”).

Eventually, the properties become conceptual objects that are available as data to conscious processes. That is, visual features become salient in isolation and are linked to a verbal label, and the child becomes capable of reflecting on the visual features and thus can explicitly recognize the shape’s properties. At first, however, the flexibility of application is limited. Indeed, children can be expected not only to think at different levels for different topics (Clements & Battista, 1992) but also to think at different phases for different topics. Problem solving and discussion involving the geometric objects help build connections between the constructed knowledge (e.g., of right angles) and other similarly accessible knowledge (e.g., of parallelism and side lengths of rectangles and squares). Eventually, connections are built between properties of figures such as rectangles and squares and of properties of other geometric objects. In this manner, children’s geometric knowledge can become increasingly abstract, coherent, and integrated, because it is freed from the constraints of compiled, and thus inflexible, mental representations. At each level, the degree of integration increases as the connections span greater numbers of geometric and eventually nongeometric topics (cf. Gutiérrez & Jaime, 1988). Here again, instruction has a strong influence. Ideally, it encourages unification; however, the instruction of isolated bits of knowledge at low levels retards such development. Unfortunately, the latter is pervasive in both curriculum materials (Fuys, Geddes, & Tischler, 1988) and teaching (Clements & Battista, 1992; Porter, 1989; B. Thomas, 1982).

To place instructional implications in a different light, only after the third phase do children become explicitly aware of their geometric conceptualiza-

tions; therefore, it is after Phase 3 that the last three instructional steps in the van Hiele model (explicitation, free orientation, and integration) can begin.⁶ An implication is that short-circuiting this developmental sequence (e.g., by beginning with explicitation) is a serious pedagogical mistake. Deprived of the initial construction of their own mental geometric objects and relationships (images), children construct Phase 1 (behaviorally “correct”) verbal responses based on “rules without reason” (Skemp, 1976). A more viable goal is the construction of mathematical meaning from actions on geometric objects and subsequent reflections on those actions.

In summary, development of geometric properties as conceptual objects leads to pervasive Level 2 thinking. This development represents a reconstruction on the abstract/conscious/verbal plane of those geometric conceptualizations that Piaget and Inhelder (1967) hypothesized were first constructed on the perceptual plane and then reconstructed on the representational/imaginal plane. Thus, Level 2 thinking requires what Piaget called the construction of articulated mental imagery, which develops most fully via the combined enhancement of both imagistic and verbal declarative knowledge, as previously discussed. We posit that the same recurrent phases explain the reconstruction on each new plane (level), but that the role of social interaction and instruction increases in importance with higher levels.

The theory of reiterated phases of rerepresentation applies equally to children and adults, regardless of their overall stage of cognitive development (Karmiloff-Smith, 1990). This is consonant with reports of low van Hiele levels among high school students and preservice teachers (Burger & Shaughnessy, 1986; Denis, 1987; Gutiérrez & Jaime, 1988; Mayberry, 1983; Senk, 1989).

Final Words

As stated in the introduction to this section and supported by the research, spatial and geometric thinking are essential human abilities that contribute to mathematical ability. Their importance is highlighted by findings that infants assign greater weight to spatiotemporal information than color or form in their definition of what an object is (Newcombe & Huttenlocher, 2000). Thereafter, toddlers use the shape of objects as the essential cue in learning the identity and names of objects. For example, training 17-month-olds to attend to shape led them to general-

⁶ The first two steps, Information and guided orientation, of the van Hiele model’s five-step instructional sequence would be, with some modification, consonant with the three phases of development described here. This topic is related to, but different from, the topics addressed here and will not be examined.

ize that objects with similar shapes have the same name and engendered a dramatic increase of 350% in learning new words outside of the laboratory (Smith, Jones, Landau, Gershkoff-Stowe, & Samuelson, 2002). As important is spatial thinking, which supports geometry and creative thought in all mathematics. Given their importance, it is essential that geometry and spatial sense receive greater attention in instruction and in research. Unfortunately, U.S. children's performance in geometry and spatial reasoning is woefully lacking. Educators need to understand more about learning and teaching geometry and spatial sense.

Although far less developed than our knowledge of number, research provides guidelines for developing young children's learning of geometric and spatial abilities. However, researchers do not know the potential of children's learning if a conscientious, sequenced development of spatial thinking and geometry were provided throughout their earliest years. This can also be seen as a caveat. Research on the learning of shapes and certain aspects of visual literacy suggest the inclusion of these topics in the early years. Insufficient evidence exists on the effects (efficacy and efficiency) of including topics such as congruence, similarity, transformations, and angles in curricula and teaching at specific age levels. Such research, and longitudinal research on many such topics, is needed.

Finally, competencies in these first two major realms, spatial/geometric and quantitative/number, are connected, probably at deep levels, throughout development. The earliest competencies may share common perceptual and representational origins (Mix, Huttenlocher, & Levine, 2002). Infants are sensitive to both amount of liquid in a container (Gao, Levine, & Huttenlocher, 2000) and distance a toy is hidden in a long sandbox (Newcombe, Huttenlocher, & Learmonth, 1999). Visual-spatial deficits in early childhood are detrimental to children's development of numerical competencies (Semrud-Clikeman & Hynd, 1990; Spiers, 1987). Other evidence shows specific spatially related learning disabilities in arithmetic, possibly more so for boys than girls (Share, Moffitt, & Silva, 1988). Children with Williams Syndrome, who show impairment on visuo spatial construction tasks, can learn reading and spelling but have difficulty learning mathematics (Bellugi, Lichtenberger, Jones, Lai, & St. George, 2000; Howlin, Davies, & Udwin, 1998). Primary school children's thinking about "units" and "units of units" was found to be consistent in both spatial and numerical problems (Clements, Battista, Sarama, & Swaminathan, 1997). In this and other ways, specific spatial abilities appear to be related to mathematical competencies (D. L. Brown & Wheatley, 1989; Clements & Battista, 1992; Fennema

& Carpenter, 1981; G. H. Wheatley, Brown, & Solano, 1994). Geometric measurement connects the spatial and numeric realms explicitly and is the topic of the following section.

GEOMETRIC MEASUREMENT

Geometric measurement can serve as a bridge between the two critical domains of geometry and number, with each providing conceptual support to the other. Indications are, however, that this potential is usually not realized, as measurement is not taught well. Many children use measurement instruments or count units in a rote fashion and apply formulas to attain answers without meaning (Clements & Battista, 1992). In international comparisons, U.S. students' performance in measurement was low (NCES, 1996).

Children's understanding of measurement has its roots in infancy and the preschool years but grows over many years, as the work of Piaget and his collaborators has shown (Piaget & Inhelder, 1967; Piaget, Inhelder, & Szeminska, 1960). As with number, however, Piagetians underestimated the abilities of the youngest children. For example, shown an object that was then occluded by a drawbridge, infants looked longer when the drawbridge rotated past the point where the object should have stopped it (Baillargeon, 1991). Thus, even infants are sensitive to continuous quantity (Gao, Levine, & Huttenlocher, 2000), and even comparisons (Spelke, 2002) and accumulations (Mix, Huttenlocher, & Levine, 2002) of continuous quantity, at least in some conditions (cf. Huntley-Fenner, Carey, & Solimando, 2002). These studies show that early cognitive foundations of mathematics are not limited or unique to number. As with number, however, these abilities have limits. Infants and toddlers can discriminate between lengths of dowels, but only when a salient standard (a same-length dowel or container) was present; 4-year-olds could discriminate with or without such a standard (Huttenlocher, Duffy, & Levine, 2002). Infants and toddlers may lack the ability to create and maintain a mental image of a length.

As we discussed, some believe that humans may be sensitive to amount of continuous quantity and not discrete number (Clearfield & Mix, 1999). However, they do make distinctions. For example, when a small number of objects or portions of a substance is hidden and then revealed, 12- to 13-month-old infants expect that the former cannot be combined to make a larger object but that the latter can be coalesced into larger portions (Huntley-Fenner, 1999a). Recall that 11- but not 9-month-old infants could discriminate sequences of

number (Brannon, 2002). Noteworthy here is that the 9-month-olds could discriminate sequences of size; therefore, nonnumerical ordinal judgments may develop before the capacity for numerical ordinal judgments.

Preschool children know that continuous attributes such as mass, length, and weight exist, although they cannot quantify or measure them accurately. Children 2 to 4 years of age can use the same three types of standards when judging “big” and “little” that adults use: perceptual (object is compared to another physically present object), normative (object is compared to a class standard stored in memory, such as a chihuahua is small for a dog), and functional (Is this hat the right size for this doll?; Ebeling & Gelman, 1988). Children also can coordinate these, preferring the perceptual to the normative if there is a conflict. For example, when asked if an object is larger than an egg, but then given an even larger egg, children will compare the object to the larger egg. They prefer either the perceptual or normative standards to the functional (which must be brought to the attention of 3 year olds, but not 4 year olds) (S. A. Gelman & Ebeling, 1989). Finally, children can switch within contexts and from a normative context to the others but have difficulty switching to a normative context. The normative context may differ from the perceptual and functional contexts in that stimuli are not physically present and thus may be accessible only when no other context has been recently experienced (Ebeling & Gelman, 1994).

As young as 3 years of age, children know explicitly that if they have some clay and then are given more clay, they have more than they did before. However, preschoolers cannot reliably make judgments about which of two amounts of clay is more; they use perceptual cues such as which is longer. Children do not reliably differentiate between continuous and discrete quantity, for example, basing equal sharing on the number of cookie pieces rather than the amount of substance (K. F. Miller, 1984; Piaget, Inhelder, & Szeminska, 1960).

Further, they have not yet integrated their counting (e.g., of discrete entities) with measurement (counting units of continuous quantity). For example, 4- and 5-year-olds were more likely to count when asked which group had “more glasses” than if asked which had “more sand,” even though in each comparison the same sand was in the same glasses (Huntley-Fenner, 1999b). Younger children responded inaccurately when asked to compare sand that was not in glasses. When children observed cupfuls of sand poured into boxes they again did not use counting and used the rate, not duration, of pouring as the basis of judgment (if researchers hid the pouring act, younger children performed the worst). Thus, children must learn to ap-

ply their counting skills to unitized measures of continuous quantities. They appear to do so first for small numerosities. For example, 6-year-olds were more accurate in addition tasks when red blocks were placed in cylinders instead of liquid, but only for small numerosities (5 or less). To measure, children have to overcome a natural inclination to quantify continuous substances with mental processes that are analogously continuous (i.e., that do *not* involve discrete units and counting). In a striking example, 3- to 5-year-old children were no less successful comparing amounts of sand in piles than when the same amount was shown in 3 vs. 2 discrete glasses (Huntley-Fenner, 2001).

Despite such challenges, young children can be guided to have appropriate measurement experiences. They naturally encounter and discuss quantities (Seo & Ginsburg, 2004). They initially learn to use words that represent quantity or magnitude of a certain attribute. Then they compare two objects directly and recognize equality or inequality (Boulton-Lewis, Wilss, & Mutch, 1996). At age 4–5 years, most children can learn to overcome perceptual cues and make progress in reasoning about and measuring quantities. They are ready to learn to measure, connecting number to the quantity (even though the average U.S. child, with limited measurement experience, exhibits limited understanding of measurement until the end of the primary grades). We next examine this development in more detail for the case of length.

Length Measurement

Length is a characteristic of an object found by quantifying how far it is between the endpoints of the object. *Distance* is often used similarly to quantify how far it is between any two points in space. Measuring length or distance consists of two aspects, identifying a unit of measure and *subdividing* (mentally and physically) the object by that unit and placing that unit end to end (*iterating*) alongside the object. Subdividing and unit iteration are complex mental accomplishments that are too often ignored in traditional measurement curriculum materials and instruction. Therefore, many researchers go beyond the physical act of measuring to investigate children’s understandings of measuring as covering space and quantifying that covering.

We discuss length in the following three sections. First, we identify several key concepts that underlie measuring (Clements & Stephan, 2004; Stephan & Clements, 2003). Second, we discuss early development of some of these concepts. Third, we describe research-based instructional approaches that were de-

signed to help children develop concepts and skills of length measurement.

Concepts in Linear Measurement

At least eight concepts form the foundation of children's understanding of length measurement. These concepts include understanding of the attribute, conservation, transitivity, equal partitioning, iteration of a standard unit, accumulation of distance, origin, and relation to number.

Understanding of the attribute of length includes understanding that lengths span fixed distances ("Euclidean" rather than "topological" conceptions in the Piagetian formulation).

Conservation of length includes understanding that lengths span fixed distances (Euclidean rather than topological conceptions in the Piagetian formulation) and the understanding that as an object is moved, its length does not change. For example, if children are shown two equal-length rods aligned, they usually agree that they are the same length. If one is moved to project beyond the other, children 4.5 to 6 years often state that the projecting rod is longer. At 5 to 7 years, many children hesitate or vacillate; beyond that, they quickly answer correctly. Conservation of length develops as the child learns to measure (Inhelder, Sinclair, & Bovet, 1974).

Transitivity is the understanding that if the length of object X is equal to (or greater/less than) the length of object Y and object Y is the same length as (or greater/less than) object Z, then object X is the same length as (or greater/less than) object Z. A child with this understanding can use an object as a referent by which to compare the heights or lengths of other objects.

Equal partitioning is the mental activity of slicing up an object into the same-sized units. This idea is not obvious to children. It involves mentally seeing the object as something that can be partitioned (or "cut up") before even physically measuring. Asking children what the hash marks on a ruler mean can reveal how they understand partitioning of length (Clements & Barrett, 1996; Lehrer, 2003). Some children, for instance, may understand "five" as a hash mark, not as a space that is cut into five equal-sized units. As children come to understand that units can also be partitioned, they come to grips with the idea that length is continuous (e.g., any unit can itself be further partitioned).

Iteration of a unit is the ability to think of the length of a small unit such as a block as part of the length of the object being measured and to place the smaller block repeatedly along the length of the larger object (Kamii & Clark, 1997; Steffe, 1991), counting these iterations.

Accumulation of distance is the understanding that as you iterate a unit along the length of an object and count the iteration, the number words signify the space covered by all units counted up to that point (Petitto, 1990). Piaget et al. (1960) characterized children's measuring activity as an accumulation of distance when the result of iterating forms nesting relationships to each other. That is, the space covered by three units is nested in or contained in the space covered by four units.

Origin is the notion that any point on a ratio scale can be used as the origin. Young children often begin a measurement with "1" instead of zero.

Relation between number and measurement requires children to reorganize their understanding of the items they are counting to measure continuous units. They make measurement judgments based upon counting ideas, often based on experiences counting discrete objects. For example, Inhelder and others (1974) showed children two rows of matches that were the same length, but each row comprised a different number of matches. Although, from the adult perspective, the lengths of the rows were the same, many children argued that the row with more matches was longer because it had more matches. Note that this context differs from the cardinal situations in that the order-irrelevance principle does not apply and every element (e.g., units on a ruler) should not necessarily be counted (Fuson & Hall, 1982).

Researchers debate the order of the development of these concepts and the ages at which they are developed; perhaps education and experience have a large effect on both. Researchers generally agree that these ideas form the foundation for various aspects of measurement. Traditional measurement instruction is insufficient for helping children build these conceptions.

Early Development of Length Measurement Concepts

The same landmarks that aid children in cue or place learning also can affect their representations of the distances separating objects. Piaget, Inhelder, and Szeminska (1960) reported that after placing a third object between two objects, young children claim that the distance is smaller or larger than before. In another study, children judged that two routes, one direct and one indirect, cover the same distance (i.e., straight path is the shortest distance between two points). Subsequent studies have confirmed that most 4-year-olds and about half of 5- and 6-year-olds show such patterns (Fabricius & Wellman, 1993; K. F. Miller & Baillargeon, 1990). Thus, the Piagetian position is that young children do not possess understanding of distance and length.

Mistakes on tasks may not be due to the misconceptions of space that Piagetian theory assumed. First, children can encode and apply distance information. For example, preschoolers do well at simple distance judgment tasks, with this competence appearing as early as 12 to 16 months of age (Huttenlocher, Newcombe, & Vasilyeva, 1999).

Second, there are inconsistencies in the literature on Piagetian tasks. About 40% of 4-year-olds could avoid errors on direct and indirect routes, including giving correct explanations (Fabricius & Wellman, 1993). Further, children 3.5 to 5 years of age appear to understand both the direct-indirect principle and the same-plus principle (if two routes are the same up to a point, but only one continues, it is longer) in a task modification in which the items were screened, so responses would not be the result of perceptual scanning (Bartsch & Wellman, 1988).

In a variation of the conservation-of-length task, children were asked which of five boxes a stick would fit into (Schiff, 1983). Children's judgments remained consistent after sliding. Thus, they correctly judged that a stick would go into the same box after it was moved (e.g., apparently believing that sliding the stick across the table did not change the physical dimensions of the stick). In another study, children were asked to choose a stick to bridge a gap. They appeared to understand that occlusion of the stick did not affect the length (K. F. Miller & Baillargeon, 1990). Children first understand affordances, such as "will this stick fit here," and later integrate knowledge between length and distance (K. F. Miller, 1984).

Across several experiments, then, there is little empirical support for the notion, such as in the Piagetian topological primacy thesis, that conceptualizations underlying children's reasoning about distance and length differ from those of adults. Preschoolers understand that lengths span fixed distances. Still, some researchers hold that complete conservation is essential for, but not equivalent to, a full conception of measurement. Piaget, Inhelder, and Szeminska (1960) argued that transitivity is impossible for children who do not conserve lengths because once they move a unit, it is possible, in the child's view, for the length of the unit to change. Most researchers agree that children develop the notion of conservation before transitivity (Boulton-Lewis, 1987). Although researchers agree that conservation is essential for a complete understanding of measurement, children do not necessarily need to develop conservation before they can learn some measurement ideas (Boulton-Lewis, 1987; Clements, 1999c; Hiebert, 1981; Petitto, 1990). Two measurement ideas that seem to depend on conservation and transitivity are the inverse relation between

the size of the unit and the number of those units and the need to use equal length units when measuring. However, in several anecdotal reports preschoolers understood the inverse relation in reform curricula contexts, and one study (Sophian, 2002) showed an increase in the understanding of effect of object size on measurement of volume in 3- and 4-year-olds, when children are given the opportunity to compare the result of measurements made with different units.

Most researchers argue that children must reason transitively before they can understand measurement adequately (Boulton-Lewis, 1987). Some researchers conclude that the ruler is useless as a measuring tool if a child cannot yet reason transitively (Kamii & Clark, 1997). As with conservation, this may only be true for some tasks. Further, as we have shown before, understanding is not a dichotomous phenomenon. Children as young as pre-K and kindergarten age use transitivity in simple measurement tasks. For example, given two holes and a marked stick, they can compare the depth of the holes by inserting the stick and comparing the marks (Nunes & Bryant, 1996). Such abilities appear in even younger children on some tasks (K. F. Miller, 1989).

On many tasks that *appear* to require general logical reasoning, children find their own strategy to measure, and they do so correctly. These solution strategies do not necessarily match the structural logic of the task. For example, children use intermediate measurements to compare two lengths without explicitly asking the transitivity question. They move a unit to measure the length of an object and do not worry about whether the length is being conserved. Finally, children of all developmental levels solve simple measurement tasks that do not appear to rely heavily on general reasoning.

In summary, children have an intuitive understanding of length on which to base reasoning about distance and length, but that reasoning develops considerably. They may have difficulty mapping words such as *long* onto the adult concept, instead assuming it means endpoint comparison (Schiff, 1983). They need to learn to coordinate and resolve perceptual and conceptual information when it conflicts. Finally, they need to learn to use measurement, understanding that units of lengths can be iterated along successive distances and these iterations counted to determine length. Thus, young children know early that properties such as length (as well as area, volume, mass, and weight) exist, but they do not initially know how to reason about these attributes or to measure them accurately. Nevertheless, when there are no perceptually conflicting cues, preschoolers can accurately compare objects directly.

Before kindergarten, many children lack measurement rules such as lining up an end when comparing the lengths of two objects (Piaget & Inhelder, 1967; Piaget, Inhelder, & Szeminska, 1960), although they can learn about such ideas. Even 5- to 6-year-olds, given a demarcated ruler, wrote in numerals haphazardly with little regard to the size of the spaces. Few used zero as a starting point, showing a lack of understanding of the origin concept. At age 4–5 years, however, many children can, with opportunities to learn, become less dependent on perceptual cues and thus make progress in reasoning about or measuring quantities. From kindergarten to Grade 2, children significantly improve in measurement knowledge (Ellis, Siegler, & Van Voorhis, 2000). They learn to represent length with a third object, using transitivity to compare the length of two objects that are not compared directly in a wider variety of situations (Hiebert, 1981). They can also use given units to find the length of objects and associate higher counts with longer objects (Hiebert, 1981, 1984). Some 5-year-olds, but most 7-year-olds, can use the concept of unit to make inferences about the relative size of objects; for example, if the numbers of units are the same, but the units are different, the total size is different (Nunes & Bryant, 1996). However, even 7-year-olds found tasks demanding conversion of units challenging.

Even kindergartners are fairly proficient with a conventional ruler and understand quantification in measurement contexts, but their skill decreases when features of the ruler deviate from the convention. Thus, measurement is supported by characteristics of measurement tools, but children still need to develop understanding of key measurement concepts. In one study, all K–2 children understood several measurement concepts. However, there were significant age differences on understanding concepts such as iterating a standard unit and the cardinality principle (Ellis, Siegler, & Van Voorhis, 2000). Children initially may iterate a unit leaving gaps between subsequent units or overlapping adjacent units (Horvath & Lehrer, 2000; Lehrer, 2003); therefore, it is a physical activity of placing units along a path in some manner, not an activity of covering the space/length of the object with no gaps. Furthermore, students often begin counting at the numeral “1” on a ruler (Lehrer, 2003) or, when counting paces heel-to-toe, start their count with the movement of the first foot, missing the first foot and counting the second foot as “one” (Lehrer, 2003; Stephan, Bowers, Cobb, & Gravemeijer, 2003). Students probably are not thinking about measuring as covering space. Rather, the numerals on a ruler (or the placement of a foot) signify when to start counting, not an amount of space that has already been cov-

ered (i.e., “one” is the space from the beginning of the ruler to the hash mark, not the hash mark itself). Many children initially find it necessary to iterate the unit until it “fills up” the length of the object and will not extend the unit past the endpoint of the object they are measuring (Stephan, Bowers, Cobb, & Gravemeijer, 2003). Finally, many children do not understand that units must be of equal size. They will even measure with tools subdivided into different size units and conclude that quantities with more units are larger (Ellis, Siegler, & Van Voorhis, 2000). This may be a deleterious side effect of counting, in which children learn that the size of objects does not affect the result of counting (Mix et al., 2002, although we disagree with the authors’ claim that units are always “given” in counting situations—along with most teachers, Mix et al. do not consider counting situations, such as counting whole toy people constructed in two parts, top and bottom, when some are fastened and some are separated, cf. Sophian & Kailihiwa, 1998).

Children are also learning accumulation of distance. Some, for example, measured the lengths of objects by pacing heel to toe and counting their steps (Stephan, Bowers, Cobb, & Gravemeijer, 2003). As one child paced the length of a rug, the teacher stopped the child midmeasure and asked her what she meant by “8.” Some children claimed that “8” signified the space covered by the eighth foot whereas others argued that it was the space covered from the beginning of the first foot to the end of the eighth. These latter children were measuring by accumulating distances. This type of interpretation may not appear until students are 9 or 10 years old (Clements, 1999c; Kamii & Clark, 1997). However, with meaningful instruction, children as young as 6 years old can learn to measure by accumulating distance (Stephan, Bowers, Cobb, & Gravemeijer, 2003).

Finally, young children are developing the foundational ideas of origin and relation between number and measurement. As Piagetian research indicated, they draw on their counting experiences to interpret their measuring activity, to which the “starting at 1” error may be related. If children understand measuring only as “reading the ruler,” they may not understand this idea (Lehrer, 2003). Children also have to understand and apply counting concepts, including one-to-one correspondence and the cardinality principle.

Thus, significant development occurs in the early childhood years. However, the foundational length ideas are usually not integrated. For example, children may still not understand the importance of, or be able to create, equal size units, even into the primary grades (Clements, Battista, Sarama, Swaminathan, & McMillen, 1997; Lehrer, Jenkins, & Osana, 1998;

K.F. Miller, 1984). This indicates that children have not necessarily differentiated fully between counting discrete objects and measuring.

Education

Young children naturally encounter and discuss quantities in their play (Ginsburg, Inoue, & Seo, 1999). They first learn to use words that represent quantity or magnitude of a certain attribute. Facilitating this language is important not only to develop communication abilities, but for the development of mathematical concepts. Simply using labels such as “Daddy/Mommy/ Baby” and “big/little/tiny” helped children as young as 3 years to represent and apply higher order seriation abilities, even in the face of distracting visual factors, an improvement equivalent to a 2-year gain. Language provides an invitation to form comparisons and a method to remember the newly represented relational structure (Rattermann & Gentner, 1998). Thus, language can modify thought (cf. Vygotsky, 1934/1986). Along with progressive alignment, in which children are presented with easy literal similarity matches before difficult matches, language provides powerful scaffolding potential (Kotovsky & Gentner, 1996).

Next, children compare two objects directly and recognize equality or inequality, for example, of the length of two objects (Boulton-Lewis, Wilss, & Mutch, 1996). Following this, children can learn to measure, connecting number to length. Again, language, such as the differences between counting-based terms (e.g., “a toy” or “two trucks”) and mass terms (e.g., “some sand”), can help children form relationships between counting and continuous measurement (Huntley-Fenner, 2001).

Traditionally, the goal of measurement instruction has been to help children learn the skills necessary to use a conventional ruler. In contrast, research and recent curriculum projects suggest that developing the conceptual foundation for such skills is critical to develop understanding and procedures. Kamii and Clark (1997) argued that comparing lengths is at the heart of developing the notions of conservation, transitivity, and unit iteration, but most textbooks do not have these types of tasks. Textbooks tend to ask questions such as “How many paper clips does the pencil measure?” rather than “How much longer is the blue pencil than the red pencil?” Although Kamii and Clark advocate beginning instruction by comparing lengths with nonstandard or standard units (not a ruler), they caution that such an activity is often done by rote.

Many recent curricula promote a sequence of instruction in which children compare lengths, measure with nonstandard units, incorporate the use of

manipulative standard units, and measure with a ruler (Clements, 1999c; Kamii & Clark, 1997). The basis for this sequence is, explicitly or implicitly, Piaget et al.’s (1960) theory of measurement. The argument is that this approach motivates children to see the need for a standard measuring unit. Challenges to this sequence have arisen from studies suggesting that standard units and even rulers might lay a better foundation for initial understanding of measurement, with nonstandard and different standards units introduced later (Clements, 1999c).

Whatever the specific instructional approach taken, four implications can be abstracted from the research. First, measurement is not a simple skill, but rather a complex combination of concepts and skills that develops over years (Clements, Battista, Sarama, Swaminathan, & McMillen, 1997; Sophian, 2002; Steffe, 1991; Stephan, Bowers, Cobb, & Gravemeijer, 2003). Second, initial informal activities can establish the attribute of length and develop concepts such as *longer*, *shorter*, and *equal in length* and strategies such as direct comparison. Third, emphasis on solving real measurement problems and, in so doing, building and iterating units, as well as units of units, helps children develop strong concepts and skills. Fourth, children need to closely connect the use of manipulative units and rulers. When conducted in this way, measurement tools and procedures become tools for mathematics and tools for thinking about mathematics (Clements, 1999c; K. F. Miller, 1984, 1989). Well before first grade, children have begun the journey toward that end.

Area Measurement

Understanding of area measurement also involves learning and coordinating many ideas (Clements & Stephan, 2004). Eventually, children must understand that decomposing and rearranging shapes does not affect their area.

Some researchers report that preschoolers use only one dimension or one salient aspect of the stimulus (Bausano & Jeffrey, 1975; Maratsos, 1973; Mullet & Paques, 1991; Piaget, Inhelder, & Szeminska, 1960; Raven & Gelman, 1984; J. Russell, 1975; Sena & Smith, 1990). For example, 4- and 5-year-olds may match only one side of figures when attempting to compare their areas. Others claim that children can integrate more than one feature of a region but judge areas with additive combination, for example, making area judgments based on the longest single dimension (Mullet & Paques, 1991) or height + width rules (Cuneo, 1980; Rulence-Paques & Mullet, 1998). Children from 6 to 8 years use a linear extent rule, such as the diagonal of a

rectangle. Only after this age do most children move to explicit use of multiplicative rules. This leads to our next concept.

In most of these studies, children did not interact with the materials. Doing so often changes their strategies. Children as young as 3 years are more likely to make estimates consistent with multiplicative rules when in a problem-solving setting (count out the right number of square tiles to cover a floor and put them in a cup) than when just asked to make a perceptual estimation (K. F. Miller, 1984).

Children of 5 to 6 years of age were more likely to use strategies consistent with multiplicative rules after playing with the stimulus materials (Wolf, 1995). Wolf argued that more complex rules are often used when people are more familiar with the materials involved in a task. This may be so; however, children did better when their manipulation followed their estimating sizes. Thus, small numbers of objects and familiarity with materials may be beneficial, but familiarity with the task (the conceptual goal) and the physical and cognitive actions applied to the materials may also encourage more accurate strategies such as scanning one length through another.

Although some researchers imply multiplicative thinking on the part of the child, we take the conservative position that there is little evidence of true 2-dimensional spatial structuring in these studies, although children may be using linear or additive strategies that are more consistent with the result of accurate multiplicative rules.

Another more accurate strategy is superimposition. Children as young as 3 years have a rudimentary concept of area based on placing regions on top of one another, but it is not until 5 or 6 years that their strategy is accurate and efficient. Asked to manipulate regions, preschoolers used superimposition instead of the less precise strategies of laying objects side-by-side or comparing single sides, both of which use one dimension at best in estimating the area (Yuzawa, Bart, & Yuzawa, 2000). Multiplicative thinking may be a result of internalization of such procedures as placing figures on one another, which may be aided by cultural tools (manipulatives) or scaffolding by adults (cf. Vygotsky, 1934/1986). For example, kindergartners who were given origami practice increased the spontaneous use of the procedure of placing one figure on another for comparing sizes (Yuzawa, Bart, Kinne, Sukemune, & Kataoka, 1999). Because origami practice includes the repeated procedure of folding one sheet into two halves, origami practice might facilitate the development of an area concept, which is related with the spontaneous use of the procedure.

Equal partitioning is the mental act of cutting two-dimensional space with a two-dimensional unit. Teachers often assume that the product of two lengths structures a region into an area of two-dimensional units for older students. However, the construction of a two-dimensional array from linear units is nontrivial. Young children often cannot partition and conserve area and instead use counting as a basis for comparing. For example, when it was determined that one share of pieces of a paper cookie was too little, 6 preschoolers cut one of that share's pieces into two and handed them both back (K. F. Miller, 1984).

Conclusion

Measurement is one of the principal real-world applications of mathematics. It bridges two critical realms of mathematics, geometry and real numbers. Number and operations are essential elements of measurement. The measurement process subdivides continuous quantities such as length to make them countable. Measurement provides a model and an application for both number and arithmetic operations. In this way, measurement helps connect the two realms of number and geometry, each providing conceptual support to the other.

Research indicates that measuring in general is more complex than learning the skills or procedures for determining a measure. Learning specific measurement concepts and skills is intrinsically important and also helps children differentiate between two basic types of quantity, discrete and continuous, which are often confused by young children (K. F. Miller, 1984; Piaget, Inhelder, & Szeminska, 1960). That is, representing exact discrete quantities by counting and more precise continuous quantities by measuring may be instrumental in children's development of the distinction between and attributes of these two basic categories (Mix, Huttenlocher, & Levine, 2002). This gradual integration and differentiation across domains supports the previously described principle of progressive hierarchization, which appears consistently in studies of measurement. For example, recall that children who observed agreement between measurement and direct comparison were more likely to use measurement later than were children who observed both procedures but did not have the opportunity to compare them (P. E. Bryant, 1982). Finally, we repeat the caveat that although young children can develop early ideas in measurement of various attributes, such as area and angle, there is little research on how valuable it would be to invest instructional time in these areas rather than others.

PATTERNS AND ALGEBRAIC THINKING

One key foundational ability for all topics in mathematics is the algebraic insight—the conscious understanding that one thing can represent another. Children develop this key idea at about 3 years of age in the simplest form; for example, as described earlier, understanding that a map or picture can represent another space. This ability develops considerably over the next several years, as children interact with symbols and with other people using these symbols in a variety of situations, from maps to spatial patterns to numerical patterns.

From this perspective, algebraic thinking can permeate much of the instruction of these main areas (and the smaller amount of research done on such thinking should not diminish its importance to the curriculum). One common, albeit often underappreciated, route to algebra begins with a search for patterns. Identifying patterns helps bring order, cohesion, and predictability to seemingly unorganized situations and allows one to generalize beyond the information directly available. Recognition and analysis of patterns are important components of the young child's intellectual development because they provide a foundation for the development of algebraic thinking.

Although pre-K children engage in pattern-related activities and recognize patterns in their everyday environment, research has revealed that an abstract understanding of patterns develops gradually during the early childhood years (B. A. Clarke & Clarke, 2004; A. Klein & Starkey, 2004). Of children entering their first year of school in Australia, 76% could copy a repeating color pattern, but only 31% could extend or explain it (B. A. Clarke & Clarke, 2004). In the pre-K years, children can learn to copy simple patterns. In kindergarten, they can learn to extend and create patterns. Further, children learn to recognize the relationship between patterns with nonidentical objects or between different representations of the same pattern (e.g., between visual and motoric, or movement, patterns). This is a crucial step in using patterns to generalize and to reveal common underlying structures. Through kindergarten and the primary grades, children must learn to identify the core unit (e.g., AB) that either repeats (ABABAB) or “grows” (ABAABAAAB) and then use it to generate both these types of patterns. Note that by the pre-K or kindergarten year, many children can name patterns with conventions such as “ABAB.” This is another step to algebra, as it involves using variable names (letters) to label or identify patterns that involve different physical embodiments. Such naming helps children recognize

that mathematics focuses on underlying structure, not physical appearances. Little else is known, except that pattern identification is one of many elements of teaching visual literacy with positive long-term impact in the Agam program (Razel & Eylon, 1990).

In the later months of kindergarten, it is possible that with guidance children can begin analyzing numerical patterns, just as primary grade students can do. These students learn to find and extend numerical patterns—extending their knowledge of patterns to thinking algebraically about arithmetic (Baroody, 1993). Two central themes are generalizing and using symbols to represent mathematical ideas and to represent and solve problems (Carpenter, Fennema, Franke, Levi, & Empson, 1999). For example, children might generalize that when you add zero to a number the sum is always that number or that when you add three numbers it does not matter which two you add first (Carpenter, Fennema, Franke, Levi, & Empson, 1999). Thus, students in the primary grades can learn to formulate, represent, and reason about generalizations and conjectures, although their justifications do not always adequately validate the conjectures they create.

This body of research on young children's understanding of patterns may be used to establish developmentally appropriate learning trajectories for pattern instruction in early mathematics education. However, the main conclusion of this section is that far too little research has been conducted as a basis for curriculum development in the domain of patterns and algebraic thinking. This is equally true of the final content area, to which we now turn.

DATA ANALYSIS

There is a dearth of research on children's learning of data analysis in the preprimary years. From a small amount of research conducted with older students, we can describe in broad strokes that the developmental continuum for data analysis includes growth in classifying and counting and in data representations. (The development of classification per se, a major consideration in the domain, is described in the following section.) For example, children initially learn to sort objects and quantify their groups. They might sort a collection of buttons into those with one to four holes and count to find out how many they have in each of the four groups. To do this, they focus on and describe the attributes of objects, classifying according to those attributes, and quantify the resulting categories. Children eventually became capable of simultaneously classify-

ing and counting, for example, counting the number of colors in a group of objects, as described previously.

After gathering data to answer questions, children's initial representations often do not use categories. Their interest in data is on the particulars (S. J. Russell, 1991). For example, they might simply list each child in their class and each child's response to a question. They then learn to classify these responses and represent data according to category. Finally, young children can use physical objects to make graphs (objects such as shoes or sneakers, then manipulatives such as connecting cubes), then picture graphs, then line plots, and finally bar graphs that include grid lines to facilitate reading frequencies (Friel, Curcio, & Bright, 2001).

As stated, however, research on data analysis before the primary years is sparse. Most consist of anecdotal reports. They are promising, but limited. There is some evidence that preschoolers can understand discrete graphs as representations of numerosity based on one-to-one correspondence (Solomon, 2003). Similarly, although 5- to 6-year-olds were more accurate than 4- to 5-year-olds on most, but not all graphing tasks, all children could interpret the graphs and use them to solve mathematical problems (J. Cooper, Brenneman, & Gelman, 2005). Children performed better than in previous research, possibly because they were provided examples and given feedback and because they were motivated by the task, graphing their progress toward gathering items for a scavenger hunt. Children performed better with discrete than continuous formats.

Across all the available literature, the consensus of a group attempting to create research-based standards for young children concluded that the main role for data analysis would lie in supporting the areas of number and spatial sense (Clements & Conference Working Group, 2004). For example, gathering data to answer a question or make a decision is potentially an effective means to develop applied problem solving and number sense. To do so, curricula and teachers might focus on one big idea: Classifying, organizing, representing, and using information to ask and answer questions.

MATHEMATICAL PROCESSES

We have woven discussion of mathematical processes throughout this chapter but make a few additional points focused on specific processes in this section.

Reasoning and Problem Solving

Reasoning is not a radically different sort of force operating against habit but the organization and coop-

eration of many habits, thinking facts together. Reasoning is not the negation of ordinary bonds, but the action of many of them, especially of bonds with subtle elements of the situation. . . . Almost everything in arithmetic should be taught as a habit that has connections with habits already acquired and will work in an organization with other habits to come. The use of this organized hierarchy of habits to solve novel problems is reasoning. (Thorndike, 1922, p. 193–194)

Thorndike accurately reflects turn-of-the-century perspectives that learning mathematics is like mental exercising, building the mind like a muscle. This perspective was perhaps most problematic in generalizing that reasoning could be explained as bonds and habits.

In contrast, others have thought that reasoning is merely logic, with uneducated people less knowledgeable of that domain. George Boole considered his research on pure mathematics and symbolic logic a central contribution to psychology, entitling his 1854 book, *An Investigation of the Laws of Thought*. The Piagetians agreed. "In short, reasoning is nothing more than the propositional calculus itself" (Inhelder & Piaget, 1958, p. 305). Instead, as has already been illustrated in previous topically oriented sections, young children's reasoning is a complex cognitive process that depends on local and global knowledge and procedures. Space constraints prevent our doing more than the briefest of overviews of young children's mathematical processes, including reasoning, problem solving, and communication, with the goal of illustrating their complexity, impressive extent, and importance.

As some vignettes throughout this chapter have illustrated, even very young children can reason and solve problems if they have a sufficient knowledge base, if the task is understandable and motivating, and if the context is familiar (Alexander, White, & Daugherty, 1997; DeLoache, Miller, & Pierroutsakos, 1998). Consistent with hierarchic interactionism, there is no cogent evidence that these processes undergo substantive qualitative shifts. Instead, development occurs in interactive interplay among specific components of both general and specific knowledge and processes, together with an increasing effectiveness of these components.

Multiple types of reasoning processes can be identified in young children, although the various types overlap. Perceptual reasoning begins early, as similarity between objects is the initial relation from which children draw inferences, but it remains an important relation throughout development (DeLoache, Miller, & Pierroutsakos, 1998, on which this summary is largely based). Once concepts are formed, children as young as toddlers build structural-similarity relations, even

between perceptually dissimilar objects, giving rise to analogies (Alexander, White, & Daugherty, 1997). Analogies in turn increase knowledge acquisition, which increases information-processing capacity through unitizing and symbolizing. Myriad represented relations give birth to reasoning with rules and symbolic relations, freeing reasoning from concrete experience and encouraging the appropriation of cultural tools that vastly expand the possible realms of reasoning. Again, then, we see the mutual development of concepts and skills, with knowledge catalyzing new processes, which in turn facilitate knowledge growth.

In solving problems, children may often use most of these types of reasoning, highlighting the overlap and even lack of real distinction (except for adult purposes of classification or research) among them. One illustration researchers use is the task, described in the “Models and Maps” section, in which children were shown a location of a toy on a scale model of a room and then asked to find the toy in the actual room (DeLoache, 1987). Perceptual similarity and concepts supported children’s relating the model and room. Symbols were important—indeed, the picture was more useful to the youngest children because it served as a symbol. Children’s inferring the toy’s location was analogical, mapping a relationship in one space onto a similar relationship in the other.

Reasoning is, of course, used in solving problems. There are also additional strategies possessed by young children. Luke, 3 years old, watched his father unsuccessfully looking under the van for a washer that had fallen and suggested, “Why don’t you just roll the car back, so you can find it?” Luke employed means-end analysis better than his father did. This strategy involves determining the difference between the current state and the goal and then taking action that reduces the difference between them, reasoning backwards from the goal to set subgoals. Means-end problem solving may emerge between 6 and 9 months when, for example, children learn to pull on a blanket to bring a toy into their reach.

We discussed in previous sections that even young children have multiple problem-solving strategies at their disposal and the ability to choose among them. Means-ends analysis is a general strategy, as are several others. Children know and prefer cognitively easier strategies. For example, in hill climbing, you reason forward from the current state in the direction of the desired goal (DeLoache, Miller, & Pierroutsakos, 1998). Trial and error, with light cognitive requirements, begins early, with Piagetian circular reactions.

These strategies develop in generality and flexibility during the toddler and preschool years, enabling children to address problems of increasing complex-

ity. For example, recall that kindergartners can solve a wide range of addition, subtraction, multiplication, and division problems when they are encouraged to represent the objects, actions, and relationships in those situations (Carpenter, Ansell, Franke, Fennema, & Weisbeck, 1993). The researchers argued that modeling is a parsimonious and coherent way of thinking about children’s mathematical problem solving that is accessible to teachers and children. An Australian study supported the notion that kindergartners could learn to solve a variety of quite difficult word problems (Outhred & Sardelich, 1998). Children used concrete objects but were also required to draw and explain their representations, including the relationships among the elements. By the end of the year, they could pose and solve complex problems using a variety of strategies to represent aspects of the problem, including showing combining and partitioning groups and using letters and words to label elements of sets.

The ability to choose among alternative problem-solving strategies also emerges early in life; for example, in early spatial planning (Wellman, Fabricius, & Sophian, 1985). At 18 months or earlier, for example, children can use multiple strategies to pull a toy into reach with a third object (DeLoache, Miller, & Pierroutsakos, 1998). As an example in a different domain, recall that young children can make adaptive choices among arithmetic strategies, especially if the situations and strategies are meaningful for them (Siegler, 1993). In another study (Outhred & Sardelich, 1997), by the end of their kindergarten year, all children modeled arithmetic problems using concrete materials and accurately solved them. Their drawings showed a variety of strategies for representing the situations, including displaying properties (e.g., size), separating groups or crossing out items, partitioning sets, drawing lines to indicate sharing relationships, drawing array structures to show equal groups, and using letters and words to label items in collections. They could write their own problems and represent problem situations symbolically (they only struggled to accurately represent multistep problems with symbols). All this was accomplished despite the small amount of time the teacher engaged the children in problem solving.

Problem posing appears to be an effective way for children to express their creativity and integrate their learning (S. I. Brown & Walter, 1990; Kilpatrick, 1987; van Oers, 1994). Few empirical studies have been conducted that verify effects of problem posing, however, and none involved young children.

In summary, considering their minimal experience young children are impressive problem solvers. They are learning to learn and learning the rules of

the “reasoning game.” Research on problem solving and reasoning again reveals that children are more skilled, and adults less skilled, than conventionally thought. Finally, although domain-specific knowledge is essential, researchers should not fail to recognize that reasoning from domain-specific knowledge builds upon the basis of mindful general problem-solving and reasoning abilities that are evident from the earliest years.

Classification and Seriation

The complex relationship between seriation and classification and the development of number concepts (Piaget & Szeminska, 1952) has already been discussed. Piagetian theory held that these operations also underlie logic and reasoning (Piaget, 1964). For example, the claim that the abilities to seriate, construct a correspondence between two series, and insert an element into a series developed synchronically was an argument for the Piagetian operational theory of intelligence. The research literature on these constructs is vast (Clements, 1984a); we will only highlight some relevant findings here.

Although research militates against any simple, direct causal relationship, there is also evidence that these processes do play important roles in the development of mathematical reasoning and learning (cf. Kamii, Rummelsburg, & Kari, 2005; Piaget, 1971/1974). For example, children who do not acquire basic competencies in classification (*oddity*—which one is not like the others), seriation, and conservation by kindergarten do not perform as well in mathematics in later schooling (Ciancio, Rojas, McMahon, & Pasnak, 2001; Lebron-Rodriguez & Pasnak, 1977; Pasnak et al., 1987). Similarly, assessments of Piagetian reasoning tasks in kindergarten were related to children’s mathematics concepts years later (Silliphant, 1983).

At all ages, children classify informally as they intuitively recognize objects or situations as similar in some way (e.g., differentiating between objects they suck and those they do not at 2 weeks of age) and eventually label what adults conceive of as classes. Often, functional relationships (the cup goes with a saucer) are the bases for sorting (Piaget, 1964; Vygotsky, 1934/1986). In addition, even infants place objects that are different (6 months), then alike (12 months), on some attribute together (Langer, Rivera, Schlesinger, & Wakeley, 2003). By 18 months, they form sets in which objects in each set are identical and objects in other set are different and, by 2 years, form sets with objects that are similar on some properties but not necessarily identical. Some 2-year-olds and all

3-year-olds will substitute elements to reconstruct misclassified sets (Langer, Rivera, Schlesinger, & Wakeley, 2003). These are often partial arrangements with fluid criteria; nevertheless, they play an essential role in number, through the unitizing process.

Not until age 3 can most children follow verbal rules for sorting. For example, told two simple rules for sorting pictures, children aged 36 months could sort regardless of the type of category, but children even a few months younger could not (Zelazo & Reznick, 1991). Having relevant knowledge and having fewer memory demands were not significant in this case.

In the preschool ages, many children learn to sort objects according to a given attribute, forming categories, although they may switch attributes during the sorting (Kofsky, 1966; Vygotsky, 1934/1986). The result may appear to reflect adult categorizations but often has a different conceptual basis, such as general resemblance (Vygotsky, 1934/1986, calls these “pseudo-concepts”). Preschoolers appear to encode examples holistically, distributing their attention nonselectively across many stimulus features, and then generalize to new stimuli on the basis of their overall similarity to the stored examples (Krascum & Andrews, 1993).

Not until age 5 or 6 years do children usually sort consistently by a single attribute and reclassify by different attributes. At this point, they can sort consistently and exhaustively by an attribute, given or created, and use the terms *some* and *all* in that context (Kofsky, 1966). Young children also learn seriation from early in life. Preverbal infants are able to make perceptual size comparisons and, from the age of 18 months or so, are able to respond to and use terms such as *big*, *small*, and *more* in ways that show they appreciate quantity differences (Resnick & Singer, 1993). By 2 or 3 years of age, children can compare numbers and number pairs on the basis of a common ordering relation (Bullock & Gelman, 1977). At 3 years, children can make paired comparisons, and 4-year-olds can make small series but do not seriate all objects (Clements, 1984a, Piaget & Inhelder reported that 55% make no attempt at seriation and 47% build partial uncoordinated series, Inhelder & Piaget, 1958; 1967). At age 5, 18% make no seriation, 61% build partial series; and 12% solve the problem, but only by trial and error. In a recent study, 43% of 5-year-olds put six lengths in order by length (A. Klein, Starkey, & Wakeley, 1999, but note that Piaget’s lengths were more difficult to distinguish perceptually). Most 5-year-olds can insert elements into a series.

Children exhibit many strategies in seriation. Some choose the smallest (or largest) object to begin, then continue to select the next smallest (largest). Some place randomly selected objects in place. Others begin

with the largest block, then select a proximate block, accepting it only if order is preserved, switching if it is not. Some analyses have claimed that those with less sophisticated, or even incorrect, strategies are not qualitatively different from seriators; they are just missing one or more rules (such as accepting any monotonic increase, rather than only unit differences).

There is also evidence that children can make transitive, deductive inferences, often considered the most difficult of the seriation tasks (if A is longer than B and B is longer than C, then A is longer than C). For example, 4-year-olds do so if they are trained to code and retrieve the relevant information (Trabasso, 1975). The authors claimed, as we did with reasoning and problem solving previously, that the cognitive processes of even young children are not that much different from those of adults, at least in the nature of the strategies used.

As mentioned, Piaget claimed that seriation and serial correspondence (between two series) and reconstruction of ordinal correspondences (a similarity that has become numerical) are solved at approximately the same time (Piaget & Szeminska, 1952). Young children make global comparisons with seriation or correspondences. They then develop the ability to construct series and correspondences intuitively but often fall back on perceptual correspondences. Only at the stage of concrete operations do they solve all the problems. They understand correspondences numerically; that is, the element n represents both the position n and the cardinal value n . Seriation and multiple seriation follow the development of classification and multiple classification, respectively. Generally, the developmental progressions within a domain (such as seriation) are supported, but few developmental concurrences across domains have been found (Clements, 1984a).

Studies involving both classification and seriation suggest that young children are more likely to abstract a property represented through literal comparisons. For example, both 4- and 6-year-olds can match three circles increasing in size with three squares that similarly increased in size, but only 6-year-olds could match the circles to three circles that increased in color saturation, that is, match seriations across dimensions (Kotovsky & Gentner, 1996). If the 4-year-olds are trained to mastery on the former, they can match across dimensions. Thus, as with counting and comparing number, we see a relational shift, in which children's early reliance on physical similarity develops into the ability to perceive purely relational commonalities (Kotovsky & Gentner, 1996).

Education

A strong argument can be made that developing reasoning and problem-solving abilities be the main

focus in any attempt to help babies and toddlers develop in mathematics (see, e.g., Kamii, Miyakawa, & Kato, 2004). Encouraging language can support the growth of reasoning abilities (Rattermann & Gentner, 1998). Presenting children with easy literal similarity matches also helps them solve difficult analogical matches. Similarly, helping preschoolers learn the properties and relationships in a mathematical situation allows them to solve analogies of various types, such as geometric analogy items (Alexander, White, & Daugherty, 1997).

Problem solving can also be facilitated. Even for the very young, there are substantial benefits of varied situations, encouragement of diverse strategies, discussions, simple justifications, and prompts and hints as needed (DeLoache, Miller, & Pierroutsakos, 1998). Children make progress when they solve many problems over the course of years. Children as young as kindergartners, and perhaps younger, benefit from planned instruction, but not prescribed strategies, from a teacher who believes problem solving is important. They benefit from modeling a wide variety of situations (problem types, including addition, subtraction, and, at least from kindergarten on, multiplication and division) with concrete objects, from drawing a representation to show their thinking, from explaining and discussing their solutions, and from connecting representations (Carpenter, Ansell, Franke, Fennema, & Weisbeck, 1993; Outhred & Sardelich, 1997; van Oers, 1994). A final point is that concrete objects often make an important contribution but are not guaranteed to help (Baroody, 1989; Clements, 1999a). Children must see the structural similarities between any representation and the problem situation to use objects as tools. (These issues are discussed in Clements, 1999a.)

All young children need opportunities to achieve at least a minimal level of competence with classification (e.g., oddity) and seriation before they reach the primary grades. Simple teaching strategies can have a significant effect, especially for children with special needs. For example, Pasknak and colleagues (Ciancio, Rojas, McMahon, & Pasknak, 2001; Kidd & Pasknak, 2005; Lebron-Rodriguez & Pasknak, 1977) have taught seriation and classification via a simple learning set procedure, including demonstration, practice, and feedback with many varied concrete examples to a variety of children. These children, including blind and sighted, at-risk pre-K children and at-risk kindergarten children, and older blind children, increased their scores on instruments measuring intelligence quotient and mathematics achievement. (Other instructional approaches may also have these and other benefits as we shall see, Clements, 1984c; Kamii, Rum-

melsburg, & Kari, 2005.) Similarly, although solving oddity problems is difficult for children below the age of 6, telling the children the oddity rule quickly leads to success (DeLoache, Miller, & Pierroutsakos, 1998). Thus, children do not have difficulty following the rule, but inducing it. Game-like instruction may help children learn to induce simple rules. Again, evidence indicates that giving children clues about a rule or discussing it with them enables them to represent and follow it.

When should such instruction begin? Before 3 years of age, informal, child-centered experiences are indicated. Many 2.5-year-old children know a rule and have relevant conceptual knowledge but fail to use it to regulate their behavior. Seemingly impervious to efforts to improve their rule use, 32-month-olds could not label pictures in terms of appropriate categories, even with varieties of extra help in sorting, including feedback and reinforcement (Zelazo, Reznick, & Piñon, 1995). Improvements in sorting by rules may require emerging control over actions.

Materials to think with, to sort, or to order are important. Again, the meaningfulness of the representations and tasks is more important than the form of the materials; thus, for example, computer materials may be as or more useful than physical materials (Clements, 1999a). In one study, children working with computer manipulatives learned classification and other topics as well as children learning from physical materials, but only the computer group gained significantly on seriation (Kim, 1994). Further, the computer manipulatives provided children with a more interesting learning environment that generated more time on task.

Beyond the youngest ages, many educational approaches have been tried, including direct verbal instruction, contingent feedback, and modeling by peers just one level above the target child's assessed level. Researchers disagreed as to the theoretical position their results supported, but all methods were found to be effective in at least some situations (Clements, 1984a). For example, between 28% and 66% of kindergarten children in a transitional level learned all classification including class inclusion following six training sessions on various classification competencies (P. Miller, 1967). The researcher claimed that the training might have catalyzed cognitive reorganization.

Several educational approaches have also been found effective in teaching seriation, which is often easier to teach than classification or number concepts such as conservation (Clements, 1984a). These approaches include televised modeling; developmentally sequenced lessons; use of Montessori materials; a combination of corrective feedback, attention to task

stimuli, and cuing and cue fading; and discussion of children's own seriation strategies. Some studies reported that only children in Piagetian transition stages make substantial gains.

Focused, teacher-centered activities to develop these processes are thus effective, but there are reasons to believe they should be supplemented with occasions in which the processes are used to solve meaningful everyday problems for the child. As Piaget (1971/1974) stated,

The child may on occasion be interested in seriating for the sake of seriating, in classifying for the sake of classifying, etc., but, in general, it is when events or phenomena must be explained and goals attained through an organization of causes that operations [logico-mathematical knowledge] will be used [and developed] most. (p. 17)

For example, although many types of activities may support the learning of classification, a pedagogical guideline of "classify with good causation" (Forman & Hill, 1984) indicates that children will benefit less from sorting shapes according to a teachers' directions and more from sorting three-dimensional objects to distinguish which will and will not roll down a ramp.

A final issue is whether training in Piagetian foundational areas such as classification and seriation aids the development of number. Evidence is mixed. Several studies show positive effects. For example, teaching classification and seriation to kindergarten children improved their performance on tests of number concepts (Lesh, 1972). Similarly, kindergartners taught classification, seriation, and conservation with simple teaching procedures (Pasnak, 1987) made twice the gains of children receiving "traditional mathematics instruction" on measures of general learning and reasoning ability and matched their gains on reading and mathematics achievement. These gains have held longitudinally (Pasnak, Madden, Malabonga, & Holt, 1996).

Findings similar to those of Lesh and Pasnak were reported for pre-K children. Four-year-old children were randomly assigned to one of three educational conditions for 8 weeks: logical foundations (classification and seriation), number (counting), and control (Clements, 1984c). After engaging in activities teaching classification, multiple classification, and seriation operations, the logical foundations group significantly outperformed the control group both on measures of conservation and on number concepts and skills. However, inconsistent with Piagetian theory, the number group also performed significantly better than the control group on classification, multiple classifica-

tion, and seriation tasks as well as on a wide variety of number tasks. Further, there was no significant difference between the experimental groups on the logical operations test, and the number group significantly outperformed the logical foundations group on the number test. Thus, the transfer effect from number to classification and seriation was stronger than the reverse. The domains of classes, series, and number seem to be interdependent, but experiences in number have priority (Clements, 1984c). Note that some number and logical foundation activities were structurally isomorphic, so children received implicit experience with classification and seriation in number activities. For example, in one activity, children counted the blue cars, the red cars, and all the cars. Thus, in meaningful counting situations, the child may have a familiar cognitive tool with which to construct the logical structures required.

LEARNING AND TEACHING CONTEXTS

Early Childhood Mathematics Education

How much mathematics is done in early childhood settings? Observations of the full day of 3-year-olds' lives, across all settings, revealed remarkably few mathematics activities, lessons, or episodes of play with mathematical objects, with 60% of the children having no experience across 180 observations (Tudge & Doucet, 2004). Factors such as race-ethnicity, SES, and setting (home or child care) did not significantly affect this low frequency. Teacher reports from the large Early Childhood Longitudinal Study (ECLS, Hausken & Rathbun, 2004) indicate that kindergarten teachers spend an average of 39 minutes each session, 4.7 days per week, for a total of 3.1 hours each week of mathematics instruction. This is about half of what they spend on reading. Direct instruction was more commonly used with girls and higher SES, and "constructivist" approaches used more often with children of low ability. Each week teachers engage children in manipulative activities 1–2 times and problem-solving and practice (worksheet) activities 2–3 times. Individual skills were not related to the frequency of exposure to different types of mathematics problems. The authors warned that uniformity of activities across the classrooms may account for these findings. Especially given contrasting data from smaller but more focused studies described later, additional caveats concerning the limitation of post hoc teacher self-report data and the lack of specific information on curricula are warranted.

Research is beginning to describe corresponding preschool practices. The large National Center for Early Development and Learning (NCEDL) studies report that children are not engaged in learning or constructive activities during a large proportion of the pre-K day (Early et al., 2005). About 8% of the day on the average involves mathematics activities. A survey (Sarama, 2002; Sarama & DiBiase, 2004) asked teachers from a range of preschool settings at what age children should start large-group mathematics instruction. Family and group care providers chose ages 2 or 3 most often, whereas the other group felt large-group instruction should not start until age 4. Most teachers professed to use manipulatives (95%), number songs (84%), basic counting (74%), and games (71%); few used software (33%) or workbooks (16%). They preferred children to "explore math activities" and engage in "open-ended free play" rather than participate in "large group lessons" or be "doing math worksheets." When asked about mathematics topics, 67% taught counting, 60% sorting, 51% numeral recognition, 46% patterning, 34% number concepts, 32% spatial relations, 16% making shapes, and 14% measuring. Geometry and measurement concepts were the least popular.

A small observational study of four pre-K teachers from two settings revealed that little mathematics was presented in any of the classrooms, either directly or indirectly (Graham, Nash, & Paul, 1997). Researchers observed only one instance of informal mathematical activity with concrete materials and few instances of informal or formal mathematics teaching. Teachers stated that they believed that mathematics was important and that they engaged in mathematical discussions. Apparently the selection of materials and activities such as puzzles, blocks, games, songs, and finger plays constituted mathematics for these teachers.

Several findings support the traditional emphasis on play and child-centered experiences. In one study, children made more progress overall and specifically on mathematics when they attended child-initiated, compared to strictly academically oriented, programs (Marcon, 1992). There was some evidence that these children's grades were higher at the end of elementary school (6th, but not 5th, grade; Marcon, 2002). This may be consistent with practices in some Asian countries. For example, Japanese pre-K and kindergarten education places emphasis on social-emotional rather than academic goals. Preschoolers engage in free play most of the day. Parents interact with their children in mathematics, usually in real-life, such as counting down elevator numbers. Few mention workbooks (Murata, 2004). However, Macron's studies have been criticized on methodological grounds (Lo-

nigan, 2003) and are correlational. Further, exposure to mathematics instruction explained a substantive portion of the greater gains of young Chinese compared to U.S. children (Geary & Liu, 1996).

Although fewer studies have been conducted specifically of small- or whole-group pedagogical strategies, evidence suggests these approaches too can be effective. Small-group work can significantly increase children's scores on tests aligned with that work (A. Klein & Starkey, 2004; A. Klein, Starkey, & Wakeley, 1999). Small-group activities can also transfer to knowledge and abilities that have not been taught (Clements, 1984c). Combinations of whole-group, small-group, everyday, and computer activities were employed in the *Building Blocks* project (Clements & Sarama, 2004a, in press). Large effect sizes supported the strategy of designing a curriculum built on comprehensive research-based principles, including an emphasis on hypothesized learning trajectories (Clements & Sarama, in press). Finally, observations in countries that use far more whole-group instruction with young children suggest its advantages may be overlooked in the United States. For example, the teacher-directed, whole-class Korean approach provides a positive, nurturing environment that offers children the opportunity to develop essential preacademic skills (French & Song, 1998).

In summary, in this new educational arena, several approaches, if performed in high-quality settings, can be effective. Most successful pedagogical strategies, even those with focused goals, include play or play-like activities. All approaches have a shared core of concern for children's interest and engagement as well as content matched to children's cognitive level. Although some studies support general, play-oriented approaches, learning mathematics seems to be a distinct process, even in preschool (Day, Engelhardt, Maxwell, & Bolig, 1997), and approaches focused on mathematics have been successful. We return to this issue in the following section.

Issues of Equity and Individual Differences

Children who live in poverty and who are members of linguistic and ethnic minority groups demonstrate significantly lower levels of achievement (Bowman, Donovan, & Burns, 2001; Brooks-Gunn, Duncan, & Britto, 1999; P. F. Campbell & Silver, 1999; Denton & West, 2002; Entwisle & Alexander, 1990; Halle, Kurtz-Costes, & Mahoney, 1997; Mullis et al., 2000; Natriello, McDill, & Pallas, 1990; Rouse, Brooks-Gunn, & McLanahan, 2005; Secada, 1992; Sylva, Melhuish, Sammons, Siraj-Blatchford, & Taggart, 2005). Ethnic gaps widened in the 1990s (Lee, 2002). The achievement

gaps have origins in the earliest years, with low-income children possessing less extensive mathematics knowledge than middle-income children of pre-K and kindergarten age (Arnold & Doctoroff, 2003; Denton & West, 2002; Ginsburg & Russell, 1981; Griffin, Case, & Capodilupo, 1995; Jordan, Huttenlocher, & Levine, 1992; Saxe, Guberman, & Gearhart, 1987; Sowder, 1992). The SES gap is broad and encompasses several aspects of mathematical knowledge: numerical, arithmetic, spatial/geometric, patterning, and measurement knowledge (Clements, Sarama, & Gerber, 2005; A. Klein & Starkey, 2004). The reason for this gap seems to be that children from low-income families receive less support for mathematics development in their home and school environments (Blevins-Knabe & Musun-Miller, 1996; Holloway, Rambaud, Fuller, & Eggers-Pierola, 1995; Saxe, Guberman, & Gearhart, 1987; Starkey et al., 1999). Also, public pre-K programs serving low-income, compared to those serving higher income, families provide fewer learning opportunities and supports for mathematical development, including a narrower range of mathematical concepts (D. M. Bryant, Burchinal, Lau, & Sparling, 1994; Farran, Silveri, & Culp, 1991).

Differences in specific aspects of young children's mathematical knowledge have been reported in two types of comparisons. First, there are cross-national differences. Some mathematical knowledge is more developed in East Asian children than in American children (Geary, Bow-Thomas, Fan, & Siegler, 1993; Ginsburg, Choi, Lopez, Netley, & Chi, 1997; K. F. Miller, Smith, Zhu, & Zhang, 1995; Starkey et al., 1999). Second, there are differences related to socioeconomic status. Some mathematical knowledge is more developed in children from middle-income, compared to lower-income, families (Clements, Sarama, & Gerber, 2005; Griffin & Case, 1997; Jordan, Huttenlocher, & Levine, 1992; Kilpatrick, Swafford, & Findell, 2001; Saxe, Guberman, & Gearhart, 1987; Starkey & Klein, 1992). The key factors in one study were the educational level attained by the child's mother and the level of poverty in the child's neighborhood (Lara-Cinisomo, Pebley, Vaiana, & Maggio, 2004). These are distinct factors, with income having a direct effect on the child and an effect mediated by the parents' interaction with children (e.g., higher, compared to lower, income parents providing more support for problem solving, Brooks-Gunn, Duncan, & Britto, 1999; Duncan, Brooks-Gunn, & Klebanov, 1994). Low-income, compared to middle-income parents, believe that mathematics education is the responsibility of the preschool and that children cannot learn aspects of mathematics that research indicates they can learn (Starkey et al., 1999). In addition, low-income fami-

lies more strongly endorsed a skills perspective than middle-income families, and the skills and entertainment perspectives were not predictive of later school achievement, as was the “math in daily living” perspective adopted by more middle-income parents (Sonenschein, Baker, Moyer, & LeFevre, 2005). These deleterious effects are more prevalent and stronger in the United States than other countries and stronger in early childhood than for other age ranges.

As an example of this gap, one large survey of U.S. kindergartners found that 94% of first-time kindergartners passed their Level 1 test (counting to 10 and recognizing numerals and shapes) and 58% passed their Level 2 test (reading numerals, counting beyond 10, sequencing patterns, and using nonstandard units of length to compare objects). However, 79% of children whose mothers had a bachelor’s degree passed the Level 2 test, compared to 32% of those whose mothers had less than a high school degree. Large differences were also found between ethnic groups on the more difficult Level 2 test (NCES, 2000). Differences appear even in the preschool years. Another study showed that mathematical knowledge is greater in 3- and 4-year-old Chinese children than in American middle-class children and greater in American middle-class children than in 3- and 4-year-olds from low-income families (Starkey et al., 1999). As a caveat, what mechanisms account for all these cross-national differences is unknown. Some factors seem to be nationally situated, such as teacher knowledge, formal teaching practice, and curriculum standards. Others are transnational, such as language differences, and yet others may be cultural without reflecting national boundaries, such as family values (J. Wang & Lin, 2005). Japanese kindergartners perform better in mathematics than those from the United States, but neither families nor schools in Japan emphasize academics for this age group (Bacon & Ichikawa, 1988). Their lower but perhaps more realistic expectations and reliance on informal instruction at the child’s level, including eliciting interest and providing examples rather than direct teaching of procedures, may account for their success.

Children from low-income families show specific difficulties. They do not understand the relative magnitudes of numbers and how they relate to the counting sequence (Griffin, Case, & Siegler, 1994). They have more difficulty solving addition and subtraction problems. Working-class children in the U.K. are a year behind in simple addition and subtraction as early as 3 years of age (Hughes, 1981). Similarly, U.S. low-income children begin kindergarten behind middle-income children, and, although they progressed at the same rate on most tasks, they ended behind and made no progress in some tasks. For example, although they per-

formed adequately on nonverbal arithmetic tasks, they made no progress over the entire kindergarten year on arithmetic story problems (Jordan, Kaplan, Oláh, & Locuniak, 2005). Further, lower class children were more likely to show a “flat” growth curve for the year.

Similarly, lower income preschoolers lag behind higher income peers in addition, comparing, and subitizing (Clements, Sarama, & Gerber, 2005), as well as on the earliest form of subitizing, spontaneous recognition of numerosity (Hannula, 2005). Further, the subtests on which the gaps were significant were those that assess more sophisticated mathematical concepts and skills, such as composing numbers or shapes (Clements, Sarama, & Gerber, 2005). They often lack foundational abilities to classify and seriate (Pasnak, 1987). Older children entering first grade showed a smaller effect of familial factors on computation than on mathematics concepts and reasoning. Majority-minority contrasts were small, but parents’ economic and psychological resources were strong influences (Entwisle & Alexander, 1990).

Into kindergarten and the primary grades, lower income children more than middle class children, use less adaptive and even maladaptive strategies, probably revealing a deficit in intuitive knowledge of numbers and different strategies (Griffin, Case, & Siegler, 1994; Siegler, 1993). Most 5- and 6-year-old low-income children are unable to answer the simplest arithmetic problems, where most middle-income kindergartners could do so (Griffin, Case, & Siegler, 1994). In one study, 75% of children in an upper-middle-class kindergarten were capable of judging the relative magnitude of two different numbers and performing simple mental additions, compared to only 7% of lower income children from the same community (Case, Griffin, & Kelly, 1999; Griffin, Case, & Siegler, 1994). As another example, about 72% of high-, 69% of middle- and 14% of low-SES groups can answer the orally presented problem, “If you had 4 chocolate candies and someone gave you 3 more, how many chocolates would you have altogether?” Low-income children often guess or use other maladaptive strategies such as simple counting (e.g., $3 + 4 = 5$). They often do this because they lack knowledge of strategies and understandings of why they work and what goal they achieve (Siegler, 1993). However, given more experience, lower income children use multiple strategies with the same accuracy, speed, and adaptive reasoning as middle-income children.

Thus, there is an early developmental basis for later achievement differences in mathematics: Children from different sociocultural backgrounds are provided different foundational experiences (Starkey et al., 1999). This is important, as knowledge of mathematics

in preschool predicts later school success (Jimerson, Egeland, & Teo, 1999; H. W. Stevenson & Newman, 1986; Young-Loveridge, 1989). Specific quantitative and numerical knowledge is more predictive of later achievement than are tests of intelligence or memory abilities (Krajewski, 2005). Those with low mathematics knowledge in the earliest years fall farther behind each year (Arnold & Doctoroff, 2003; Aunola, Leskinen, Lerkkanen, & Nurmi, 2004; R. J. Wright, Stanger, Cowper, & Dyson, 1994).

Research shows that providing educational support to children at risk results in greater school readiness upon entry into kindergarten (Bowman, Donovan, & Burns, 2001; Magnuson & Waldfogel, 2005; Shonkoff & Phillips, 2000), because such support helps young children develop a foundation of informal mathematics knowledge (Clements, 1984c). Early knowledge has been shown to support later school mathematics achievement, and lack of it places minorities on a path away from engagement in mathematics and science (F. A. Campbell, Pungello, Miller-Johnson, Burchinal, & Ramey, 2001; Oakes, 1990). Longitudinal research indicates that attendance in center-based (but not other types of) care in the pre-K year is associated with higher mathematics scores in kindergarten and (to a lesser extent) in first grade (Turner & Ritter, 2004), and that achievement in preschool is related to differences in elementary school achievement for Hispanic children (Shaw, Nelsen, & Shen, 2001). In another study, African American, Hispanic, and female children who attended an intervention preschool program had a significantly greater probability of achieving high scores in fourth grade than their peers who did not attend school (Roth, Carter, Ariet, Resnick, & Crans, 2000). Childcare in general can help, with greater number of hours in childcare correlating with greater quantitative skills in low-income children (Votruba-Drzal & Chase, 2004). High-quality preschool experience is predictive of later school success in mathematics (Broberg, Wessels, Lamb, & Hwang, 1997; F. A. Campbell, Pungello, Miller-Johnson, Burchinal, & Ramey, 2001; Peisner-Feinberg et al., 2001). However, results are sometimes moderate, and there is not always a significant effect for quality.

In comparison, children living in poverty and those with special needs increase in mathematics achievement after high-quality interventions *focused on mathematics* (P. F. Campbell & Silver, 1999; Fuson, Smith, & Lo Cicero, 1997; Griffin, 2004; Griffin, Case, & Capodilupo, 1995; Ramey & Ramey, 1998), which can be sustained into first (Magnuson, Meyers, Rathbun, & West, 2004) to third grade (Gamel-McCormick & Amsden, 2002). For example, the *Rightstart* (now *Number Worlds*) program (Griffin, Case, & Siegler,

1994), featuring games and active experiences with different models of number, led to substantial improvement in children's knowledge of number. Across five studies, almost all children failed the number pretest, and the majority in the comparison groups failed the posttest as well, whereas the vast majority of those in the program passed the posttest. Children in the program were better able to employ reasonable strategies and were able to solve arithmetic problems even more difficult than those included in the curriculum. Program children also passed five far-transfer tests that were hypothesized to depend on similar cognitive structures (e.g., balance beam, time, money). The foundation these children received supported their learning of new, more complex mathematics through first grade. In a 3-year longitudinal study in which children received consistent experiences through the grades from kindergarten through primary, children gained and surpassed both a second low-SES group and a mixed-SES group who showed a higher initial level of performance and attended a magnet school with an enriched mathematics curriculum. The children also compared favorably with high-SES groups from China and Japan (Case, Griffin, & Kelly, 1999). (A caution is that other research indicates that certain components of the curriculum are difficult to implement, Gersten et al., 2006.)

A series of studies have similarly indicated that the *Building Blocks* curriculum (Clements & Sarama, 2007) significantly and substantially increases the mathematics knowledge of low-SES preschool children. Formative, qualitative research indicated that the curriculum raised achievement in a variety of mathematical topics (Clements & Sarama, 2004a; Sarama & Clements, 2002). Summative, quantitative research confirmed these findings, with effect sizes ranging from .85 (Cohen's *d*) for number to 1.47 for geometry in a small-scale study (Clements & Sarama, in press). In a larger study involving random assignment of 36 classrooms, the *Building Blocks* curriculum increased the quantity and quality of the mathematics environment and teaching and substantially increased scores on a mathematics achievement test. The effect size compared to the control group score was very large ($d = 2.8$), and the effect size compared to a group receiving a different, extensive mathematics curriculum was large ($d = 1.3$). There was no significant interaction by program type (Head Start vs. public preschool).

The *Number Worlds* and *Building Blocks* programs share several characteristics. Both use research to include a comprehensive set of cognitive concepts and processes (albeit *Number Worlds* only does so in the domain of numbers). Both curricula are based on developmentally sequenced activities and help teach-

ers become aware of, assess, and remediate based on those sequences (projects around the world that use research-based developmental progressions help raise achievement of all children, e.g., G. Thomas & Ward, 2001; R. J. Wright, Martland, Stafford, & Stanger, 2002). Both use a mix of instructional methods.

Higher quality programs result in learning benefits upon entering elementary school, including in mathematics (Fuson, 2004; Griffin, 2004; Karoly et al., 1998). Unfortunately, most American children are not in high-quality programs, much less in programs that use research-based mathematics curricula (Hinkle, 2000). Further, children whose mothers had college degrees were nearly twice as likely to be in higher quality, center-based care as those whose mothers had not completed high school (Magnuson, Meyers, Rathbun, & West, 2004). As another example, Hispanic children are less likely to be enrolled in preschool, even though the benefit they receive from attending is double that for non-Hispanic White children in mathematics and prereading (Loeb, Bridges, Bassok, Fuller, & Rumberger, in press). Similarly, children from extremely poor families show a .22 *SD* (compared to .10 *SD* average for all children) advantage in mathematics concepts compared to peers who remain at home. Further, these average gains would be predictably higher if focused mathematics programs were in place. Thus, children from low-income and minority communities are provided fewer educational opportunities. Implications include providing high-quality programs for all children and their families. Even controlling for parents' occupations and education, family practices such as playing with numbers at home have significant impact on children's mathematical development (Sylva, Melhuish, Sammons, Siraj-Blatchford, & Taggart, 2005).

There seems to be a contradiction between two pictures of the mathematical knowledge and competencies of children of different SES groups. On the one hand, the evidence suggests a substantive and widening gap. On the other hand, there are few or no differences between low- and middle-income children in the amount of mathematics they exhibit in their free play (Ginsburg, Ness, & Seo, 2003; Seo & Ginsburg, 2004). The authors often conclude that low-income children are more mathematically competent than expected. This contradiction may have several explanations. Low-income children may not have the same kind of informal opportunities at home (although there is only weak support for this hypothesis, Tudge & Doucet, 2004). Researchers often observe them in a school setting, and there is evidence that low-income families provide less support for mathematical thinking (Thirumurthy, 2003). Thus, it may be that they

evince mathematics in their play in school but are still engaged in such play far less than higher income children. Another explanation is that children have not been provided with the opportunities to reflect on and discuss their premathematical activity. There are huge, meaningful differences in the amount of language in which children from different income levels engage (Hart & Risley, 1995, 1999). Low-income children may engage in premathematical play but be unable to connect this activity to school mathematics because to do so requires the children to bring the ideas to an explicit level of awareness. This is supported by the finding that a main difference between children is not their ability to perform with physical objects, but to solve problems verbally (Jordan, Huttenlocher, & Levine, 1992) or explain their thinking (Sophian, 2002). Consider a child who turned 4 years of age. When asked to solve "how much is ten and one more," she used physical blocks, added 1 to 10, and answered, "eleven." Five minutes later, asked several times using the same wording, "How many is two and one more," the child did not respond and, asked again, said, "fifteen" in a couldn't-care-less tone of voice (Hughes, 1981, pp. 216–217).

In summary, although there is little direct evidence on this, we believe the pattern of results suggest that, although low-income children have premathematical knowledge, they do lack important components of mathematical knowledge. They lack the ability—because they have been provided less support to learn—to connect their informal premathematical knowledge to school mathematics. Children must learn to mathematize their informal experiences, abstracting, representing, and elaborating them mathematically, and using mathematical ideas and symbols to create models of their everyday activities. This includes the ability to generalize, connecting mathematical ideas to different situations and using the ideas and processes adaptively. In all its multifaceted forms, they lack the language of mathematics.

We believe the significance of this conclusion needs to be highlighted. Some authors find that low-income children perform similarly to middle-income children on nonverbal calculation tasks, but significantly worse on verbal calculation tasks (Jordan, Hanich, & Uberti, 2003), consistent with our review. These authors also state their agreement with others (Ginsburg & Russell, 1981) that the differences are associated with language and approaches to problem solving rather than "basic mathematical abilities" (p. 366). We prefer to call these *not* "basic mathematical," but rather foundational abilities. We emphasize that mathematization—including redescribing, reorganizing, abstracting, generalizing, reflecting upon, and giving language to that which is

first understood on an intuitive and informal level—is *requisite to basic mathematical ability*. This distinction goes beyond semantics to involve a definition of the construct of mathematics and then to critical ramifications for such practical decisions as allocation of resources on the basis of equity concerns.

Gender equity also remains a concern. In some cultures, females are socialized to view mathematics as a male domain and themselves as having less ability. Teachers show more concern when boys, rather than girls, struggle. They call on and talk to boys more than girls. Finally, they believe success in mathematics is due to high ability more frequently for boys than girls and view boys as the most successful students in their class. All these may unintentionally undermine girls' achievement motivation (Middleton & Spanias, 1999). In more than one study, boys were more likely than girls to score in the lowest and highest ranges in mathematics (Callahan & Clements, 1984; Rathbun & West, 2004). In addition, there was evidence of a faster growth rate for high-achieving boys (Aunola, Leskinen, Lerkkanen, & Nurmi, 2004). Reasons for this are still unclear, but there are practical ramifications. There are also some indications that boys outperform girls as early as kindergarten on some tasks, such as number sense, estimation, and nonverbal estimation, all of which may have a spatial component (Jordan, Kaplan, Oláh, & Locuniak, 2005). However, in the U.K., preschool girls scored higher than boys (Sylva, Melhuish, Sammons, Siraj-Blatchford, & Taggart, 2005). Thus, the problems are complex, and there are distinct concerns about boys and girls.

Some children show specific learning mathematical difficulties (MD) at young ages. Unfortunately, they are often not identified or categorized broadly with other children as “developmentally delayed.” This is especially unfortunate because focused mathematical interventions at early ages are effective (Dowker, 2004; Lerner, 1997).

Foundational abilities in subitizing, counting and counting strategies, simple arithmetic, and magnitude comparison are critical for young children with MD (Aunola, Leskinen, Lerkkanen, & Nurmi, 2004; Geary, Hoard, & Hamson, 1999; Gersten, Jordan, & Flojo, 2005, note that these studies often ignore mathematics topics other than number). Some children with MD may have difficulty keeping one-to-one correspondence when counting or matching. They may need to physically grasp and move objects, as grasping is an earlier skill than pointing in development (Lerner, 1997). They often understand counting as a rigid, mechanical activity (Geary, Hamson, & Hoard, 2000). These children also may count objects in small sets one-by-one for long after their peers are strategically subitizing these

amounts. Emphasizing their ability to learn to subitize the smallest number, perhaps representing them on their fingers, may be helpful. (Children who have continued difficulty perceiving and distinguishing even small numbers are at risk for severe general mathematical difficulties, Dowker, 2004).

Some children may have difficulty with magnitude comparisons (e.g., knowing which of two digits is larger) and in learning and using more sophisticated counting and arithmetic strategies (Gersten, Jordan, & Flojo, 2005). Their lack of progress in arithmetic, especially in mastering arithmetic combinations, causes consistent problems; thus, early and intensive intervention is indicated.

In another domain, some children have difficulty with spatial organization across a wide range of tasks. Children with certain mathematics learning difficulties may struggle with spatial relationships, visual-motor and visual-perception, and sense of direction (Lerner, 1997). They may not perceive a shape as a complete and integrated entity as children without learning disabilities do. For example, a triangle may appear to them as 3 separate lines, as a rhombus, or even as an undifferentiated closed shape (Lerner, 1997). Children with different brain injuries show different patterns of competence. Those with right hemispheric injuries have difficulty organizing objects into coherent spatial groupings, whereas those with left hemispheric injuries have difficulty with local relations within spatial arrays (Stiles & Nass, 1991). Teaching with learning trajectories based on the developmental sequences described here is even more important for children with learning disabilities, as well as children with other special needs.

Spatial weakness may underlie children's difficulties with numerical magnitudes (e.g., knowing that 5 is a greater than 4, but only by a little, whereas 12 is a lot greater than 4) and rapid retrieval of numeral names and arithmetic combinations (Jordan, Hanich, & Kaplan, 2003). These children may not be able to manipulate visual representations of a number line.

Children diagnosed as autistic need structured interventions from the earliest years. They must be kept engaged with their world, including mathematics. Many children with autism are visually oriented. Manipulatives and pictures can aid children's learning of most topics, in geometry, number, and other areas. Children benefit from illustrating even verbs with dramatizations. About a tenth of children with autism exhibit savant (exceptional) abilities, often spatial in nature, such as art, geometry, or a particular area of arithmetic. These abilities are probably due not to a mysterious talent, but from massive practice, the rea-

son and motivation for which remains unknown (Ericsson, Krampe, & Tesch-Römer, 1993).

In conclusion, there are substantial inequities in mathematics experiences in the early years. Some children not only start behind but also begin a negative and immutable trajectory in mathematics (Case, Griffin, & Kelly, 1999). Low mathematical skills in the earliest years are associated with a *slower growth rate*—children without adequate experiences in mathematics start behind and lose ground every year thereafter (Aunola, Leskinen, Lerkkanen, & Nurmi, 2004). Interventions should start in pre-K and kindergarten (Gersten, Jordan, & Flojo, 2005). There is substantial evidence that this lag can be avoided or ameliorated, but also evidence that our society has not taken the necessary steps to do either. Without such interventions, children in special need are often relegated to a path of failure (Baroody, 1999; Clements & Conference Working Group, 2004; Jordan, Hanich, & Uberti, 2003; B. Wright, 1991; R. J. Wright, Stanger, Cowper, & Dyson, 1996).

FINAL WORDS

Research in early mathematics has and continues to lead the way in investigating fundamental issues in epistemology, psychology, and education. Number and space are common topics of investigations by psychologists engaged in a debate among empiricist, nativist, and interactionalist positions (Haith & Benson, 1998; Spelke, 2000). Further, researchers in mathematics education have emphasized the need to specify children's abilities to learn, and learn to learn, as well as the ecological influences on such learning, from sociocultural background to school learning experiences. There is as yet no consensus about exactly when knowledge begins, what it consists of, how it manifests itself, what causes it to emerge, or how it changes with growth and experience in the earliest years of life. Furthermore, reminiscent of introspective psychology of a century ago (Anderson, 2000), researchers' empirical evidence is consistent with their own theoretical orientation with uncomfortable frequency.⁷ This is another reason why mathematics education research that emphasizes curriculum research will be so valuable (Clements, in press). Because it is result-centered, rather than theory-centered, curricu-

lum research minimizes seductive theory-confirming strategies that tend to insidiously replace the intended theory-testing strategies and maximizes strategies that attempt to produce specified patterns of data and thus mitigate confirmation bias, stimulating creative development of theory (Greenwald, Pratkanis, Leippe, & Baumgardner, 1986). Nevertheless, we already have a growing body of knowledge that is at the least suggestive about early competencies on which to build mathematics learning.

Through more than a century, research has moved from a cautious assessment of the number competencies of children entering school, to a Piagetian position that children were not capable of true numeric thinking, to the discovery of infant sensitivity to mathematical phenomena, to the present debate about the meaning of these contradictions and an attempt to synthesize ostensibly opposing positions. Frequent in the last two of these phases is the paradox of contradictions to Piagetian findings and confirmation of the basic constructivist Piagetian framework, the influence of which has been so fundamental that even substantive new theories were born in reactions to the monumental Piagetian corpus.

Often due to their reactions against certain Piagetian functions and the nature of their methods, researchers tended to create increasingly specialized and local theories. Recently, similarities among those theories have laid the groundwork for new hybrid theories that provide general frameworks but are replete with local detail, including specific innate predispositions as well as competencies, conceptualizations, and strategies along developmental progressions. These progressions are at various levels of detail, both topical (e.g., "number" vs. "counting" vs. "specific counting competencies and errors at the *n*th level of development") and social-psychological (e.g., a broad level for goals, deeper for teachers, deeper for curriculum developers, deeper for researchers).

To develop such theories, researchers need to synthesize psychological and clinical approaches with others, such as those used in mathematics education research. The theoretical framework of *hierarchical interactionalism* that we proposed here is one such attempt. This framework connects *initial bootstraps* with a gradual development of conscious knowledge of systems of mathematical knowledge, depending on educational experiences. Thus, intellectual development results from an interplay between internal and external fac-

⁷ A similar, contributing problem is the theoretical and empirical insularity of the various research communities. Mathematics education researchers are aware of some, but not all, of research from other fields, but those in various branches of psychology are not aware of relevant work in mathematics education research, even making discoveries or inventing "new" research methods that have a long history in mathematics education.

tors, including innate competencies and dispositions, maturation, experience with the physical environment, sociocultural experiences (as opposed to only “social transmission”), and self-regulatory processes (reflective abstraction). To be useful educationally, the roles and interactions of each of these factors must be described in detail within specific domains. The field of early mathematics is fortunate to include several research programs in which the development of such descriptions is underway. Similar proposals and increased clinical, longitudinal, and educational studies of the usefulness of the theories are needed. Studies and specific theories must avoid eclecticism (Newcombe & Huttenlocher, 2000), in which “everything matters,” and garner empirical evidence that details development and the factors that influence it. Evaluating the theories’ usefulness requires well-designed studies that connect specific pedagogical processes and contexts to outcomes to identify moderating and mediating variables and to compare the immediate and long-term outcomes of different approaches. We caution researchers interested in studying young children’s mathematical thinking against restricting their attention to numbers and arithmetic. The mathematical world of young children is much richer.

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WHOLE NUMBER CONCEPTS AND OPERATIONS

Lieven Verschaffel

UNIVERSITY OF LEUVEN, BELGIUM

Brian Greer

PORTLAND STATE UNIVERSITY

Erik De Corte

UNIVERSITY OF LEUVEN, BELGIUM

Whole number concepts and operations constitute, universally, a dominant part of elementary mathematics education. The volume of research devoted to them within mathematics education and other fields such as instructional psychology, experimental psychology, developmental psychology, and (increasingly) neurosciences, is correspondingly vast; hence we have been necessarily selective in constructing this chapter. We intend it to be a systematic and balanced overview, but in the process of selecting and thematically arranging the material, it is impossible not to be influenced by our value systems about mathematics education (research). For example, we consider it relevant to situate the research and theory that we have surveyed within a broader cultural context.

Of course, this chapter overlaps to some extent other chapters, especially that on early childhood mathematics (by Clements and Sarama). To minimize overlap with that chapter, we have narrowed ours to the elementary level (i.e., from about 6 years old), whereas the Clements and Sarama chapter addresses the same topics (as well as others) at the preschool to kindergarten level (i.e., from birth to age 6).

By reference to earlier reviews, we can briefly summarize material that is fully dealt with elsewhere. Thus, we refer the reader to relevant chapters of the previous NCTM *Handbook of Research on Mathematics Teaching and Learning* (Grouws, 1992), in particular those by Fuson, Greer, and Sowder, which we explicitly take as our point of departure (see also De Corte, Greer, & Verschaffel, 1996).

Our review surveys studies and analyses inspired by various theories about the nature of arithmetic cognition and how it develops in students' learning activities. In our characterizations of these underlying theories, we largely follow the terminology used by Greeno, Collins, and Resnick (1996). These authors distinguished three general perspectives that frame the nature of knowing and learning in contrasting and complementary ways, which they labeled behaviorist/empiricist, cognitive/rationalist, and situative/pragmatist-sociohistoric. Representative traditions for the first view, according to which knowing is an organized accumulation of associations and skills, are associationism, behaviorism, and connectionism. Gestalt psychology, symbolic information process-

ing, and constructivism are traditions of research they considered to be illustrative of the second perspective, because of their emphasis on the organization of information in cognitive structures and procedures. Although symbolic information processing and constructivism share the above-mentioned common feature, there are significant differences (of emphasis) between these research traditions. Among the traditions that have contributed to the situative/pragmatist-sociohistoric perspective are ethnography, ecological psychology, and situation theory.

In terms of the trichotomy of Greeno et al. (1996), the first view is scarcely represented here, as it plays little part in contemporary work in the field. However, as pointed out by De Corte et al.

Conceptions of learning as incremental (with errors to be avoided or immediately stamped out), of assessment as appropriately implemented by reference to atomistic behavioral objectives, of teaching as the reinforcement of behavior, of motivation as directly mediated by rewards and punishments, and of mathematics as precise, unambiguous, and yielding uniquely correct answers through the application of specific procedures remain prevalent in folk psychology and, as such, represent the legacy of behaviorism. (1996, p. 492)

Sfard (1998, 2003) suggested a broadly similar contrast between the second and third perspectives as “acquisitionist” versus “participationist.” According to Sfard, the “acquisitionist” grouping consists of those (traditional) cognitivist approaches that explain learning and knowledge in terms of mental entities such as cognitive schemes, tacit models, concept images, and misconceptions, whereas the participationist framework embraces all those relatively new theories that prefer to view learning as a reorganization of activity accompanying the integration of an individual learner within a community of practice. So, the latter framework focuses on the process through which the learner becomes a skillful participant in a given (mathematical) discourse.

For a related, but slightly different, classification of the psychological roots of various theories of (mathematical) cognition and learning, including the important differences between information processing theory and constructivism, see Baroody (2003).

One historical characterization of the contrast between the cognitive/rationalist and situative/pragmatist-sociohistoric views is that they constitute two “waves” of the “Cognitive Revolution” (De Corte et al., 1996), the second wave being a reaction to the perceived limitations of the first (Gardner, 1985; Greer & Verschaffel, 1990). However, as argued by Sfard (1998), the two perspectives cannot be completely

separated, and both are present in almost every current theory (although one perspective is usually much more prominent than the other).

OUTLINE OF THE REST OF THE CHAPTER

The body of the chapter consists of four sections.

In the *first section*, we update ascertaining studies reported in earlier reviews on aspects of individual children’s performance on, and understanding of, a range of key topics, namely:

1. Single-digit computation
2. Multidigit computation
3. Estimation and number sense
4. Word problems
5. The structure of the whole-number system

An overarching theme of earlier research and theoretical analysis, which receives a lot of attention in this first section, was the appropriate balance between procedural competence and conceptual understanding. Particularly since the seminal book edited by Hiebert (1986), the prevailing view has been that procedural and conceptual aspects are not oppositional, but rather complementary and interacting (see also Baroody & Dowker, 2003; Hiebert et al., 1997; Rittle-Johnson & Siegler, 1998; Romberg, Carpenter, & Kwako, 2005; Siegler, 2003). However, as argued by Baroody (2003), some influential thinkers (e.g., Brownell, 1945; Resnick & Ford, 1981; Thorndike, 1922) came to this conclusion earlier.

Section 2 deals with the design, implementation, and evaluation of (experimental) environments for learning and teaching whole number arithmetic. Looking towards the future at the end of her chapter for the Grouws handbook, Fuson (1992, p. 268) described a vision of future mathematics classrooms “as places where children construct meanings for mathematical concepts, words, and written marks and carry out, discuss, and justify solution procedures for mathematical situations.” Further (p. 269), they are places where (a) children engage in mathematical situations that are meaningful and interesting to them; (b) the emphasis is on sustained engagement in mathematical situations, not on rapidly obtaining answers; (c) alternative solution procedures are accepted, discussed, and justified; and (d) errors are just expected way stations on the road to solutions and should be analyzed in order to increase everyone’s understanding.

Exemplary studies reported in the second section reflect that, in the intervening years, substantial

strides have been made towards actualizing this vision through a major development of that period, namely a shift from mainly ascertaining studies of individual children to extensive classroom-based research. The major approaches/topics exemplified in these classroom-based studies are

1. Learning arithmetic as problem solving
2. Teaching arithmetic as inherently algebraic
3. Creating a social-constructivist classroom culture
4. Using emergent models as a design heuristic
5. Teaching word problems from a modeling perspective
6. Changing teachers' knowledge and beliefs

Collectively, these studies represent a richer and more complex view of whole number arithmetic in relation to mathematics as a whole, and of the socially mediated processes within the culture of the elementary mathematics classroom.

In the *third section*, we reflect on the wider contexts of the research—historical, cultural, social, and political. Whole numbers and operations on them are pan-cultural concepts and activities (Bishop, 1988). Within this universality, cultural differences impact teaching and learning, in terms of cognitive tools (most importantly, language) and societal attitudes towards, and beliefs about, both arithmetic and learning/teaching arithmetic. Arithmetic in school constitutes a particular form of activity, a *situated practice* in Lave's (1988) term. In this section, we consider two theoretical perspectives that link with recent important trends that characterize cognition as both situated and distributed. The first is the relatively new field of Ethnomathematics, defined by D'Ambrosio (1985, p. 45) as "the mathematics which is practised among identifiable cultural groups, such as national-tribal societies, labor groups, children of a certain age bracket, professional classes, and so on." The second is Activity Theory, a development of the Russian school of Vygotsky and others. In particular, we consider the characterization by Engestrom (1987, p. 103) of the activity of school-going. These perspectives also focus attention on the lack of connections between arithmetic in school, outside of school, and prior to formal education. Furthermore, it is becoming increasingly obvious—particularly, but by no means exclusively in the United States—that mathematics education is politically situated. Associated fundamental philosophical differences are particularly clearly focused in early arithmetic.

In the concluding *fourth section*, drawing on the research reviewed, we endorse a balanced view of pro-

ficiency as implying intertwined strands of conceptual understanding, strategic competence, adaptive reasoning, productive disposition, and procedural fluency (Kilpatrick, Swafford, & Findell, 2001). This conception clearly implies that students should develop *adaptive expertise*, defined by Hatano (2003, p. xi) as "the ability to apply meaningfully learned procedures flexibly and creatively" rather than merely *routine expertise*, which he defined as "being able to complete school exercises quickly and accurately without understanding." We further propose a vision of early learning of arithmetic as an experience that is foundational in establishing a positive disposition towards mathematics and introduces the young child to defining characteristics of mathematical activity: sense-making, problem-solving, modeling of aspects of the real world, structural analysis, constructive use of symbolic and physical tools, and logical argumentation.

ASCERTAINING STUDIES OF INDIVIDUAL CHILDREN LEARNING AND DOING WHOLE-NUMBER MATHEMATICS

Here we review and illustrate the last decade's developmental studies, expert-novice studies, studies about children with learning problems, neuropsychological research, and so on, on five key topics related to number and arithmetic, always on a necessarily highly selective basis and with reference to existing surveys. Most of the highlighted studies in this section are exemplary for the continuation of the cognitive/rationalist (predominantly information-processing) approach based on experiments or interviews with individual children, with little or no attention to the broader cultural or instructional contexts in which these studies take place. However, at some places, results from classroom-based research and (inter)cultural studies will be discussed insofar as their results are directly relevant.

Single-Digit Computation

Children's Single-Digit Conceptual Structures

The conceptual field of whole numbers is central in elementary school mathematics and foundational for an individual's mathematical education. (We deliberately use Vergnaud's, 1996, term *conceptual field* to draw attention to the complexity of the constellation of ideas relating to whole numbers, rather than the frequently—and, in our view, misleadingly—used term *concept*.) The development of children's understanding and use of natural numbers has attracted an enormous (arguably disproportionate) amount

of literature. For a long time, the view and practice relating to the development of understanding of the natural numbers were heavily influenced by Piaget's "logical operations" framework, including classification, seriation, and conservation as the foundations of the natural number understanding. Accordingly, many educators, aware of Piaget's findings and claims, believed that it was impossible for children to learn about natural numbers in a meaningful way until they had reached the concrete-operational stage, roughly at the age of 6–8, wherein the above-mentioned logical operations become part of their cognitive structures in an integrated way (Kilpatrick et al., 2001; Verschaffel & De Corte, 1996). Research over the last 25 years, however, has questioned the pivotal role of these logical operations in the development of natural number understanding and has shown at the same time the importance of children's declarative and procedural knowledge of counting for this development. In general, this research has convincingly documented that the development of counting ability is intertwined with the development of the understanding of counting principles. Moreover, it has been shown that by the time they enter elementary school most children understand the rules that underlie counting, can perform conventional counting correctly with sets of objects even greater than 10, can use counting to solve some simple mathematical problems, and even know the Arabic symbols for some, if not for most, of the numbers up to 10 (Dehaene, 1993; Kilpatrick et al., 2001; see chapter by Clements and Sarama, this volume).

Besides developmental studies relating to basic concepts of natural number, there are also many analyses of the conceptual structures for addition and subtraction. In an attempt to integrate the work of Steffe and his colleagues on counting and cardinal conceptual units (Steffe & Cobb, 1988), as well as her own work in this area, Fuson (1992) proposed a model of the conceptual structures for addition and subtraction consisting of three levels. At Level 1, children are limited to seeing sums as an aggregate of single-unit items and to seeing only a single presentation of a quantity at a time, with the consequence that they can apply only the most primitive procedures to solve addition and subtraction problems (see below). At Level 2, all three quantities can be simultaneously represented by embedding entities for the addends within the sum, allowing the application of more efficient and more abbreviated counting methods. At Level 3 children have created "ideal chunk items," enabling them to construct simultaneous embedded mental presentations of numbers within a sum. This ability permits recomposition of the addends so that a

problem can be transformed into an easier sum of different addends or be solved using other derived-fact strategies. For an extensive and systematic review of the research about the early conceptual development of numerosity and of the concepts relating to natural numbers, see the chapter by Clements and Sarama in this handbook.

Besides these developmental analyses, a very important area of cognitive-psychological and neuroscientific research describes how people internally represent and process numbers. Among the most influential models for number processing are those of McCloskey, Caramazza, and Basili (1985) and Dehaene and Cohen (1995). The former model assumes that all numerical inputs are initially translated, via notation-specific comprehension modules, into an amodal abstract representation of numbers. The model of Dehaene abandons the idea of a unique number concept and clusters numerical abilities in different groups according to the format in which numbers are manipulated (verbal, Arabic symbols, or quantity). This model posits an analog magnitude code (a kind of mental number line) as the main, and perhaps the only, semantic representation of a number. Dehaene and Cohen also speculated about the anatomical substrates of the different processes involved in their model (see also Delazer, 2003).

Children's Strategies for Single-Digit Computation

As children begin to learn mathematics in elementary school, much of their number activity is designed to help them become proficient with single-digit arithmetic, namely mastery of the sums and products of single-digit numbers and their companion differences and quotients, the necessity of which is not disputed in the current reform documents (Kilpatrick et al., 2001; Thompson, 1999a; Treffers, 2001a, 2001b). However, the view on what it means to have knowledge of these facts and, even more importantly, how these number facts should be learnt and taught has drastically changed in the past decades. Whereas learning single-digit arithmetic was for a long time based mainly on memorizing those facts through drill-and-practice to the point of automatization (with early exceptions such as, notably, Brownell, 1945), current instructional approaches put great emphasis on the gradual development of these number facts from children's invented and informal strategies (Kilpatrick et al., 2001; Verschaffel & De Corte, 1996).

The domain of single-digit addition and subtraction is undoubtedly one of the most frequently investigated areas of numerical cognition and school mathematics. Much work in the domain has been done from a cognitive/rationalist—and, especially, information-

processing—perspective. Numerous older and more recent studies provide detailed descriptions of the progression in children of mastery of orally stated single-digit additions, such as $3 + 4$:

- from the earliest concrete counting-all-with-materials strategy, for example, finding $3 + 4$ by making a set of 1, 2, 3 blocks, then adding 1, 2, 3, 4 blocks to this set of 3 blocks, and then counting the total,
- over several types of progressively more streamlined counting strategies, such as counting-all-without-materials (1, 2, 3, . . . 4, 5, 6, 7), counting-on-from-first (3, . . . 4, 5, 6, 7), and counting-on-from-larger (4, . . . 5, 6, 7), and derived-fact strategies ($3 + 4 = [3 + 3] + 1 = 6 + 1 = 7$), which take advantage of certain arithmetic principles to shorten and simplify the computation,
- to the final state of fact mastery (for extensive reviews of this research, see Baroody & Tiilikainen, 2003; Fuson, 1992; Kilpatrick et al., 2001; Thompson, 1999a).

Similar levels for subtraction have been described, although this developmental sequence is somewhat less clearly defined (Thompson, 1999a).

These and other studies document at the same time how, at any given time during this development, an individual child uses a variety of addition strategies, even within the same session and even for the same item (for an overview see Siegler, 1998). Even older pupils and adults do not always perform at the highest developmental level of fact mastery but still demonstrate a range of different procedures even for simple addition problems (Siegler, 1998).

Proposing and testing models framed in terms of storage in the brain of arithmetic facts in learners having reached the final stage of the above developmental process is a special area of research in numerical cognition (for overviews see Ashcraft, 1995; Dehaene, 1993; Verguts, 2003). Most of this research has been done with adults. Most such models share the notion that in the “expert fact retriever” arithmetic facts such as $2 + 3 = 5$ are memorized in, and (semi) automatically retrieved from, a stored associative network or lexicon (Ashcraft, 1995; Verguts, 2003). Well-known “problem size effects” (i.e., the fact that the time needed to answer single-digit addition problems increases slightly with the size of the operands) and “tie effects” (i.e., the fact that response time for ties such as $2 + 2$ remain constant or increases very little with operand size) are considered in this common view as reflecting the duration and the difficulty of memory

retrieval. According to Ashcraft (1995; Ashcraft & Christy, 1995), both effects faithfully reflect the frequency with which arithmetic facts are acquired and practiced by individuals (but see Baroody, 1994, for a thorough critique of such evidence). However, it is quite generally accepted that not all experts’ knowledge of single-digit arithmetic is mentally represented in separate and independent units. Part of their knowledge about simple addition seems to be stored as rules (e.g., $N + 0 = N$) rather than as isolated facts. A related assumption is that because of relational knowledge not all problems need be represented. For instance, for each commutative pair (e.g., $3 + 5$ and $5 + 3$) there may be only one representational unit in the network.

It is well known that the above-mentioned developmental process from counting to fact retrieval does not proceed smoothly for all children. Single-digit arithmetic among children with mathematical difficulties or learning problems has also attracted a lot of research. Generally speaking, this research shows that children with learning disabilities, and others having difficulty with mathematics, do not use procedures that differ from the progression described above, but they are slower in moving through it (Kilpatrick, Martin, & Schifter, 2003; Torbeyns, Verschaffel, & Ghesquière, 2004). Central but still unresolved research questions are whether these difficulties are associated with general weaknesses in linguistic processes related to representing and retrieving semantic information from long-term memory (Geary, 2003; Jordan, Hanich, & Kaplan, 2003) and how these persistent problems with fact retrieval are related to developmental delays or more fundamental deficits in particular parts of the brain (Geary, 2003). Although according to some authors these persistent retrieval difficulties appear to reflect a highly persistent, perhaps lifelong, deficit (Geary, 2003; Jordan, Hanich, & Kaplan, 2003; Ostad, 1997), others (see, e.g., Baroody, *in press*) argue that poor instruction may be the basis of most of these problems.

Another very important line of research addresses the relationship between the development of these strategies or procedures for doing simple addition and subtraction, and children’s conceptual development. In an attempt to account for this relationship, Fuson (1992) proposed a model of the conceptual structures for addition consisting of three developmental levels, across which both the conceptual structures for quantities change (declarative aspect) and the addition and subtraction strategies become increasingly abbreviated, abstract, internalized (procedural aspect). This model was briefly described in the previous section.

Several other studies have addressed the question about the relationship between declarative and procedural knowledge by investigating whether one kind of knowledge develops before the other. As far as single-digit addition is concerned, this question has focused on the relationship between children's understanding of the commutativity of addition principle and their progression towards more efficient counting strategies (like the counting-on-from-larger strategy) based on this mathematical principle (for an extensive review of this literature see Baroody, Wilkins, & Tiilikainen, 2003; Rittle-Johnson & Siegler, 1998). Rittle-Johnson and Siegler claimed that their findings showed that most children understand the commutativity concept before they generate the procedures based on it, but this claim has been disputed by Baroody (2003). In any case, whereas in previous decades the debate about the relation between conceptual and procedural knowledge was dominated by proponents of two camps (i.e., the "skills-first" versus the "concepts first" view), most researchers now adopt a more nuanced perspective wherein the relationship between procedural and conceptual knowledge is assumed to develop more concurrently and iteratively than suggested in either of the older views, and wherein it is accepted that the nature of this relationship may differ among different mathematical (sub)domains (Baroody, 2003; Baroody & Ginsburg, 1986; Rittle-Johnson & Siegler, 1998).

While many cognitive psychologists continue to investigate the growth of children's procedures for single-digit arithmetic with little or no concern for the impact of the instructional and cultural environments in which this development is situated, a growing number of studies belonging to the situative/pragmatist-sociohistoric perspective make this instructional and broader cultural impact the focus of their work. For instance, several researchers who examined the developmental paths of addition solution methods used by Japanese children have reported that they typically move more quickly from counting-all methods to derived-fact and known-fact methods—without passing through a clearly identifiable stage of more efficient counting strategies—than United States children do. Interestingly, many Japanese children use the number 5 as an intermediate anchor number to think about numbers and to do additions and subtractions before starting to do sums by means of retrieval or using 10 as an anchor in their derived-fact strategies. Arguably, these developmental characteristics of Japanese children are closely related to a number of cultural and instructional supports and practices, such as the emphasis that is put on using groups of 5 in the early arithmetic instruction in general and in abacus instruc-

tion in particular (Hatano, 1988; Kuriyama & Yoshida, 1995; Murata, 2004). Similarly, Torbeyns, Verschaffel, and Ghesquière (2005) found a remarkably quick and good mastery of ties with totals above 10 (such as $8 + 8$) and, correspondingly, an unusually frequent, efficient, and adaptive use of a so-called tie strategy on sums above 10 (i.e., solving almost-tie sums such as $7 + 8$ by means of $[7 + 7] + 1$, rather than by the "decomposition-to-10" strategy $[7 + 3] + 5$) among classes of Flemish children who used a (new) textbook series that put great emphasis on the deliberate and flexible use of multiple solution strategies (rather than on the mastery of the decomposition-to-10 strategy as the only acceptable approach to sums above 10).

Although the research concerning the development of children's strategies for multiplying and dividing single-digit numbers is less extensive than for single-digit addition and subtraction, there is a growing body of studies in this domain (see, e.g., Anghileri, 1989, 1999; Baroody, 1999; Butterworth, Marschesini, & Girelli, 2003; Cooney, Swanson, & Ladd, 1988; Kouba, 1989; LeFevre, Smith-Chant, Hiscock, Daley, & Morris, 2003; Lemaire & Siegler, 1995; Mulligan & Mitchelmore, 1997; Rickard & Bourne, 1996; Steffe & Cobb, 1998). Similarly to the progression for single-digit addition and subtraction, this research documents how, generally speaking, children progress from (material-, fingers-, or paper-based) concrete counting-all strategies, through additive-related calculations (repeated adding and additive doubling), pattern-based (e.g., multiplying $\times 9$ as by $10 - 1$), and derived-fact strategies (e.g., deriving 7×8 from $7 \times 7 = 49$) to a final mastery of learned multiplication products. However, consistency between the names and the characterizations of the different categories is less than for addition and subtraction. As for addition and subtraction, research on multiplication and division has documented how multiplicity and flexibility of strategy use are basic features of people doing simple number combinations, even for older children and for adults (LeFevre et al., 2003). Here, too, the exact features of the organization and the functioning of the postulated multiplication facts store and, more particularly, to what extent (part of) experts' knowledge about the multiplication table is stored as rules ($0 \times N = 0$, $1 \times N = N$, $10 \times N = N0$, etc.) rather than as strengthened associative links between particular mathematical expressions and their correct answers has been, and still is, an important research issue (for a review see Verguts, 2003). On the basis of a recent study with third graders and fifth graders solving multiplication items with the larger operand placed either first (7×3) or second (3×7), Butter-

worth et al. (2003, p. 201) concluded with respect to that issue that

the child learning multiplication facts may not be passive, simply building associative connections between an expression and its answer as a result of practice. Rather, the combinations held in memory may be reorganized in a principled way that takes into account a growing understanding of the operation, including the commutativity principle, and, perhaps, other properties of multiplication.

(See also Baroody, 1999, for a similar conclusion based on a study wherein he found transfer of mastery to unpracticed but commuted versions of practiced multiplication combinations).

Probably the most ambitious and most influential attempt to model this development and this variety of strategy use in single-digit arithmetic from an information-processing perspective has been the model of strategy choice and strategy change in the domain of simple addition developed by Siegler and associates (Siegler & Jenkins, 1989; Siegler & Shipley, 1995; Siegler & Shrager, 1984), which is described and critiqued in the next section.

Strategy Choice and Discovery Simulation (SCADS)

SCADS is a computer model, the latest in a series of models constructed by Siegler and his colleagues (e.g., Shrager & Siegler, 1998). Central to SCADS is a database, with information about problems (in the form of associations between individual problems and potential answers to these problems that differ in strength and the strength of which changes continuously as a result of reinforced practice) and about strategies (including global, featural, problem-specific, and novelty data about each strategy available in the database). Whenever SCADS is presented with a simple addition problem, such as $3 + 6$, it activates the global, featural, and problem-specific data about the speed and accuracy of each of the available strategies, weights these data as a function of the amount of information they reflect and how recently they were generated, and chooses the strategy with the highest outcome generated by this complex weighting process. If a retrieval strategy is chosen, the model specifies the minimum strength of the problem-answer association that must be reached before it will be retrieved as well as the maximum number of attempts that the model will make to retrieve the answer. If this retrieval strategy does not work, a so-called back-up strategy is chosen. The execution of a back-up strategy is represented in the model as a modular sequence of operators. The first time SCADS executes a particular back-up strategy (e.g., counting on from first), the

model devotes its attentional resources to ensuring the correct execution of the strategy, but the more frequently the model executes a strategy, the less attentional resources SCADS needs to monitor this strategy's execution. SCADS allocates these freed attentional resources to the built-in discovery heuristics, which sometimes propose new strategies, some valid, others flawed. SCADS evaluates these proposed new strategies (e.g., counting on from larger) for consistency with conceptual constraints. If the proposed strategy violates these constraints, it is abandoned. If not, SCADS adds it to its strategy repertoire.

According to the developers of SCADS, its performance on single-digit additions and on additions with one addend above 20 is highly consistent with the strategy choice and discovery phenomena that can be observed in 4- and 5-year-old children (as in the microgenetic study by Shrager & Siegler, 1998; see also Siegler, 2001).

Siegler's strategy-choice model has been tested not only for simple addition, but also, although to a much less fine-grained extent, for multiplication. Siegler and Lemaire (1997) reported a longitudinal investigation of French second graders' acquisition of single-digit multiplication skill. Speed, accuracy, and strategy data were assessed three times within the year when children learned multiplication. The data showed improvements in speed and accuracy that reflected four different aspects of strategic changes (namely origin of new strategies, more frequent use of more efficient strategies, more efficient execution of each strategy, and more adaptive choices among available strategies) that generally accompanied learning. According to the authors, these findings supported a number of predictions of the SCADS model.

Although the model of Siegler and his colleagues is considered by many as providing exemplary support for the success of the information-processing paradigm, and although it has influenced and still influences a lot of research in the domain of single-digit arithmetic, the model also has its critics. Starting from a constructivist and social-learning theoretical framework and from a broader data set, Baroody and Tiilikainen (2003) performed a very critical analysis of Siegler's model of early addition performance and its underlying assumptions. Their major criticisms are that the theoretical advance that has resulted from this computer simulation is very limited (because most of what the model does was known and understood already) and that the model in many respects does not behave and learn as real children do. According to these authors, the computer model looks impressively sophisticated, comprehensive, and detailed, but close inspection indicates that it does not provide an ade-

quate account of, and even is at odds with, how children's "expanding web of conceptual and relational knowledge" (Baroody & Tiilikainen, 2003, p. 82) promotes the development of increasingly advanced arithmetical procedures. For instance, SCADS does not model key constructs such as assimilation, integration, and developmental readiness, or key phenomena such as why order of addends initially constrains children's strategy choice, the evolution of children's estimation ability, or their invention of thinking strategies to reason out unknown number combinations (A.J. Baroody, personal communication, May 24, 2004). Moreover, in the model, little or no attention is given to the social, and in particular instructional, context in which the development of arithmetic skills takes place. Indeed, it seems incontrovertible to assume that the occurrence and the frequency, efficiency, and adaptivity with which certain strategies are used by children will depend heavily on the nature of instruction. By instruction, we mean more than the pattern of frequencies of arithmetic facts in textbooks (Ashcraft & Christy, 1995), the number of times a particular item has been shown, or the number of times a child has received positive or negative feedback to a particular item.

Commenting on Baroody and Tiilikainen's (2003) very critical analysis of SCADS and their own "schema-based view" (Baroody & Ginsburg, 1986) that they presented as a more valuable alternative, Bisanz (2003) remarked that, although it is quite clear how SCADS works, this schema-based view, which speaks of "a web of conceptual, procedural, and factual knowledge," is not described in equal detail.

While SCADS exemplifies one of the major virtues claimed for information-processing theories, namely that they encourage precise theorizing and provide a clear basis for evaluating alternative accounts (Cowan, 2003), we certainly do not go as far as many adherents of this approach, such as Klahr and MacWhinney (1998, p. 670), saying that "computational modeling appears to be a necessary condition for advancing our understanding of cognitive development." As we (De Corte & Verschaffel, 1988; Greer & Verschaffel, 1990) and many others (see, e.g., Baroody and Tiilikainen, 2003) have argued, for a variety of reasons, using computer simulation does not *guarantee* clarity or rigor over theories expressed (only) in nonformal language. Even if the model of Baroody and Tiilikainen (2003) lacks the specificity of SCADS, it certainly has pointed to the complex mutual relationship between different kinds of knowledge (conceptual and procedural) in the development of single-digit arithmetic as well as to the crucial role of

the broader sociocultural and instructional contexts within which this development occurs.

Conclusion and Discussion

Although psychologists' and mathematical educators' views of arithmetical expertise and of the method of achieving mastery have changed, the importance of mastery of number combinations has certainly remained highly prominent in elementary school mathematics. Clearly, a child who cannot efficiently produce results for the basic combinations is at a great disadvantage in multidigit written and oral arithmetic (Cowan, 2003; Kilpatrick et al., 2001; Thompson, 1999a).

The available research over the past 10 years has provided further documentation that acquiring proficiency with single-digit computations involves much more than rote memorization. This domain of whole number arithmetic demonstrates (a) how the different components of arithmetic skill (strategies, principles, and number facts) contribute to each other, (b) how children begin with understanding of the meaning of operations and how they gradually develop more efficient methods, and (c) how they choose quite adaptively among different strategies depending on the numbers involved (Kilpatrick et al., 2001). Although researchers have made considerable progress in describing these phenomena and although there are now sophisticated (computer) models that fit (to some extent) with these data, we are still some way from a full understanding of the development of expertise in this subdomain (Cowan, 2003).

A first important focus for future research efforts is the need for increased methodological rigor. For instance, a very quick response or the absence of any conscious awareness of the answering process is no guarantee that an answer is generated by retrieval (Baroody, 1993, 2003; Cowan, 2003; LeFevre et al., 2003). Highly practiced routines can become (quasi-)automatized to a point that they can be carried out very quickly and without any conscious awareness. So, neither self-report nor speed will distinguish automatic retrieval from rapid nonconscious computation or application of a rule. With the development of models of number processing that specify both functional components and neuroanatomical circuits (Dehaene & Cohen, 1995), it might become possible to discriminate retrieval of memorized facts from unconscious computation using event-related potentials or functional magnetic resonance imaging techniques (Baroody, 2003; Cowan, 2003). Evidently, this neuroscientific work may be helpful to further our theorizing. Another important methodological problem is that some of the obtained findings may partly be artifacts of the methodologies used, which have different strengths but also poten-

tial weaknesses. For example, data about the speed and the accuracy of strategies obtained in many studies wherein participants are simply asked to solve the problems (by means of whatever strategy they want) can be seriously biased by selection effects. To control for these effects, researchers have developed the “choice/no-choice method”, wherein strategy data collected in a “(free) choice condition” are compared with those gathered in one or more “no-choice conditions,” wherein participants are (experimentally) forced to solve all items by means of a particular strategy (Siegler & Lemaire, 1997; Torbeyns, Arnaud, Lemaire, & Verschaffel, 2004). However, as argued by Torbeyns, Arnaud et al. (2004), this methodology, like any technique, is not without problems.

Second, although researchers now generally accept that the concepts and skills involved in single-digit arithmetic develop iteratively or simultaneously, rather than in a fixed order (either “concepts-first” or “skills-first,” Baroody, 2003, pp. 10–11), how these two components interact and, more precisely, exactly when and how the development of a particular (sub)skill promotes the discovery of a particular principle (or vice versa) is still a matter of serious debate. As argued convincingly by Siegler and others, further research on this issue requires the application of so-called microgenetic methods, which involve the repeated examination of children’s conceptual and procedural knowledge during the whole learning process (Siegler, 1998; Torbeyns, Arnaud, et al., 2004).

Third, while a lot of research has been done on addition and multiplication, relatively little attention has been given to a *comparative* analysis of the two operations. This raises the question to what extent the role of the different components (namely: strategies, principles, and number facts) in the development of arithmetic skill are the same for single-digit addition and single-digit multiplication. Moreover, compared with the operations of addition and multiplication, subtraction and division have attracted much less research, although clinical and neuroscientific evidence suggests that single-digit arithmetic around these latter operations proceeds quite differently than for their inverse counterparts, especially with respect to performance at the highest level of development (namely the level of fact retrieval).

Finally, although this line of research has unraveled the role of self-invention or self-discovery in the development of addition skill, the impact of cultural and instructional factors is beyond the scope of most of the (cognitive/rationalist) psychological work being discussed in this section. Many of these (computer) models seem to assume that there is a kind of universal taxonomy or developmental sequence of

computational strategies that is fundamentally independent of the nature of instruction or the broader cultural environment. Although it seems plausible that some elements of this development are strongly constrained by general factors other than the instructional and cultural context wherein this development occurs (such as the inherent structure of mathematics, some characteristics of young children’s everyday experiences, the unfolding of certain cognitive capacities in early childhood), others are less constrained and much more dependent on children’s experiences with early mathematics at home and at school (such as the provision of cultural supports and practices as sources to move quickly beyond counting-based methods or the involvement in a [classroom] climate that nurtures and values flexibility).

Multidigit Arithmetic

After they have been introduced to single-digit arithmetic in the first year(s) of elementary school, children spend several years learning multidigit arithmetic. Children in the United States, but also in many other countries, receiving regular classroom instruction have been shown to experience considerable difficulty constructing appropriate concepts of multidigit numeration and appropriate procedures for multidigit arithmetic (Brown, 1999; Fuson, 1992; Rittle-Johnson & Siegler, 1998; Thompson, 1999c).

Illustrations of lack of conceptual understanding are elementary school children’s inability to identify the tens, hundreds, etc. digit in a multidigit number, to relate ungrouped and grouped objects to the different digits in a multidigit number, and to demonstrate or explain 10-for-1 trading with concrete representations (Fuson, 1992; Fuson, Smith, & Lo Cicero, 1997; Fuson, Wearne, et al., 1997; Thompson, 1999c). Lack of procedural knowledge has been evidenced in findings that children of these ages frequently err while solving multidigit arithmetic problems with paper and pencil (see also Fuson, 1992). As a typical example, we refer to the numerous documentations of children applying incorrect variations of the correct procedures involved in the execution of the standard algorithms (“buggy procedures”; see the section *Mental Calculation Strategies Versus Written Algorithms*), because they misremembered one step or reversed two steps of the standard algorithm (Thompson, 1999c).

According to Fuson (1992, p. 263), many errors children in the United States make on numeration indicate that they interpret and treat multidigit numbers “as single-digit numbers placed adjacent to each other, rather than using multidigit meanings for the digits in different positions.” Thus, they seem to be

using a concatenated single-digit conceptual structure for multidigit numbers. This pattern of behavior may stem from classroom experiences

that do not sufficiently support children's construction of multiunit meanings, do require children to add and subtract multidigit numbers in a procedural, rule-directed fashion, and do set expectations that school mathematics activities do not require to think or to access meanings. (Fuson, 1992, p. 263)

A similar analysis of children in the U.K. was provided by Anghileri (1999), who cited the anecdotal example of a child who, after having struggled with the (long) division problem $432 \div 15$ put up his hand and asked: "Could I add the 2 and the 43 because I know that 15 goes into 45 three times?" So, an important focus of the current worldwide reform movement is to help children develop good understanding of the decimal number system as a basis for developing procedural fluency in multidigit arithmetic (Brown, 1999; Kilpatrick et al., 2001).

In this section, we review recent literature on the development of children's conceptual structures and procedures for multidigit arithmetic. First, the research on children's conceptual structures for multidigit numbers will be reviewed. Next, we discuss recent work on the development of children's (invented, informal) strategies for, respectively, multidigit addition and subtraction, and multiplication and division. We categorize these strategies, which typically show up in reform-based classroom environments that support children's invented strategies (discussed later in this chapter), as *mental arithmetic* and contrast them with the standard algorithms for (written) computation. When we use the term "mental arithmetic," we do not use it in the traditional sense of "calculating an exact arithmetic operation without the aid of an external calculation or recording device," but rather in the sense of inventing and applying handy and flexible means of calculation, based on one's understanding of the basic features of the number system and of the arithmetic operations, as well as on a well-developed feeling for numbers and a sound knowledge of the elementary number facts (see Anghileri, 1999; Reys, Reys, Nohda, & Emori, 1995; Thompson, 1999a). A typical characteristic of mental arithmetic thus conceived is that it involves the selection and adaptive use of strategies, possibly idiosyncratic but appropriate for a particular problem and always linked to the conceptual underpinnings. In this view, mental arithmetic does not exclude the use of intermediate notes or "jottings" (as an *aide-mémoire*) if the situation or the problem allows or necessitates it (Anghileri, 1999;

Buy, 2001; Thompson, 1999a). In short, according to this second view, mental arithmetic is "thinking *with* your head" rather than as "thinking *in* your head" (Buy, 2001, p. 122). Thus, it is not the presence or absence of paper and pencil, but rather the nature of the mathematical entities and actions that is crucial in our differentiation between mental arithmetic and (written) algorithms.

Children's Conceptual Structures for Multidigit Numbers

Although the conceptual structures needed to do single-digit arithmetic can, in principle, also be used to solve multidigit problems, these procedures quickly become inefficient when the numbers get larger. Consequently, children must develop conceptual structures that reflect the basic principles underlying our decimal number system as well as the mutual relationships between the different ways in which the numbers of that system can be represented (i.e., the number words, the written marks, and the quantities they represent). So, the fundamental building blocks for children's conceptual structures for multidigit numbers are the single-digit combinations and rules for combining them within the decimal system. Some scholars, like Wu (1999), suggest that the most appropriate way to think about multidigit arithmetic is to regard it as nothing more than a sequence of single-digit arithmetic operations, which is odd, because one needs to understand how to orchestrate the sequence. Wu's view, which fits with an information-processing perspective, may be adequate for routine expertise, but not adaptive expertise as defined by Hatano (2003, and see introduction).

Hereafter we summarize a framework of conceptual structures developed by Fuson, Smith, et al. (1997) and Fuson, Wearne, et al. (1997) for understanding (English) number words for multidigit numbers and multidigit written marks. This framework extends and clarifies the theoretical analysis presented in Fuson (1992) and integrates the theoretical perspectives of several recent projects designed to help children learn number concepts and operations with understanding, which will be reviewed in the second part of this chapter (Carpenter, Franke, Jacobs, Fennema, & Empson, 1997; Fuson & Kwon, 1992; Hiebert & Wearne, 1996; Murray & Olivier, 1989). Fuson, Smith, et al. (1997) and Fuson, Wearne, et al. (1997) called it the UDSSI triad model after the names of the five conceptual structures (unitary, decade, sequences, separate, integrated). According to the authors, several different conceptions may be available to a given child and be used in different situations. Thus, new conceptions are added to, rather than replacing, older conceptions. A sixth conception, the incorrect concatenated single-

digit conception (discussed above) is also included in their framework. Each conception involves a triad of two-way relationships among number words, written number marks, and quantities.

According to the model, the development begins with an established triad structure for single-digit numbers. Multidigit numbers build on and use the unitary single-digit triads of knowledge. Thus, before children can learn about two-digit numbers, they must have learned for 1 to 9 how to read and say number words corresponding to each number mark, write the numeral corresponding to each number word, and count or count out quantities for each mark and number word up to 9. Because the number words for single-digit numbers in English, as in most languages, and the corresponding written marks are arbitrary, most children learn most of the unitary single-digit triads as rote associations.

Unitary multidigit conception. Children begin with a unitary multidigit conception, in which quantities are not differentiated into groupings, and the number word and number marks are not differentiated into parts. So, for 15 doughnuts, for example, the 1 is not related to “teen” in “fifteen,” and the quantities are not meaningfully separable into 10 doughnuts and 5 doughnuts.

Decade-and-ones conception. In the second phase children begin to separate the decade and the ones parts of a number word and start to relate each part separately to the quantity to which it refers, e.g., “fifty” to 50 objects and “three” to 3 objects. They may make the same separation and try to link the decade and the ones parts of a number word to written marks. A typical error made by children in this stage of conceptual development is to write 503 as they hear “fifty-three.”

Sequence-tens-and-ones conception. In the next phase children construct a sequence-tens-and-ones conception, in which each decade is structured into groups of ten. This conception requires the skill of being able to count by tens, but also of “seeing” the groups of ten within a quantity and choosing to count these by tens.

Separate-tens-and-ones conception. In the next stage the child begins to think of a two-digit quantity as comprising two separate kinds of units—units of ten and units of one. When asked to make a two-digit quantity, the child counts the groups of ten using single-digit numbers (e.g., “one ten, two tens, three tens, . . .”) or sometimes even omits the word *tens* while counting, leaving it as understood (“one, two, three, four, five tens”). While using this terminology, the child understands that each ten is composed of 10 ones and can switch to thinking of 10 ones if that approach becomes useful.

Integrated sequence-separate tens conceptions.

In the last stage, children may eventually relate the sequence-tens and separate-tens conceptions of two-digit numbers to each other so that they are able to switch between them rapidly. In this integrated sequence-separate tens conception, bidirectional relationships are established between the tens and the ones component of each of the three parts (number words, marks, quantities) of the sequence-tens and the separate-tens conceptions. This integrated conception allows children considerable flexibility in approaching and solving problems using two-digit numbers.

Concatenated single-digit conception. Even when children have one of the adequate multidigit conceptions and use this conception to add or subtract numbers meaningfully and accurately when these are presented in a word problem or horizontally, they may use a concatenated single-digit conception for the same computation presented vertically and make an error. The vertical presentation elicits an orientation of vertical slots on the multidigit numbers that partitions these numbers into single digits. (The physical appearance of the written multidigit marks as single digits and the nonintuitive use of the left-right position as a signifier may combine to seduce children to use a concatenated single-digit conceptual structure even if they have a more meaningful conception available.)

So far, we have only discussed two-digit numbers, but according to Fuson, Smith, et al. (1997) and Fuson, Wearne, et al. (1997) the conceptual structures for numbers with three or more digits are extensions of these conceptual structures, except that the hundreds and thousands (in English) have more “separate” than “sequence” characteristics because of their more regular named structure (e.g., *three hundred* and *four thousand* by comparison with *thirty* and *forty*).

Fuson, Smith, et al. (1997) acknowledged that the developmental model presented above is deceptively neat in several respects. First, there are qualitative and quantitative differences depending on the language used. Second, children learn the six relationships for a given number (or set of numbers) at different times and may not construct the last triad relationship for all numbers up to 99 for one kind of conception before the first triad relationships for another conception are construed. Third, not all children construct all conceptions; these constructions depend on the conceptual supports experienced by individual children in their classroom and outside of school. Finally, children who have more than one multidigit conception may use different conceptions in different situations or combine parts of different triads in a single situation. So, children’s multiunit conceptions definitively

do not conform to a uniform and stage-like model (Fuson, Smith, et al., 1997).

We now turn to some comments on the above framework. First, the empirical basis of the latest version of the model, as presented above, seems somewhat unclear. Clearly, it is partly based on the ascertaining studies that formed the basis of the earlier version of this developmental model described in Fuson (1992). But, for the latest version, the authors state that it showed up “throughout their work with children in the various design studies around more meaningful, constructive, problem-based learning environments aimed at a better proficiency with (operations with) multidigit numbers” (Fuson, Wearne, et al., 1997, p. 138), without giving much further explication of the precise nature of the empirical underpinnings of the theoretical framework and without much clarification about its generalizability. In other words, it remains unclear what aspects of this development are shaped by specific characteristics of the (innovative) learning environments in which it was observed and what aspects of it are shaped by more general factors that are largely independent of instruction. In an attempt to submit Fuson, Smith, et al.’s (1997) and Fuson, Wearne, et al.’s (1997) UDSSI model to a rigorous empirical test, Collet (2004) has designed a series of tests allowing her to assess every aspect of the triad relationships at each developmental level of the model. This test battery was administered individually to 120 first- and second-grade children from different but traditional learning environments. A provisional analysis of the results suggests that, whatever the instruction received, most (but not all) children construct the unitary conception first. However, once the unitary conception is in construction, the individual pathways seem to become more divergent. Some children construct the decades-and-ones conception before being able to use the sequence-tens-and-ones conceptions (as postulated in the UDSSI model), but some other children can use these latter two conceptions (and are able to count by tens and the tens by ones) without being able to use the decades-and-ones conception. Collet’s (2004) results suggest identification of the aspects of the development of the base-ten system that are independent of instruction (i.e., the unitary conception constructs first) and clarify the developmental aspects that vary from one child to another. Further research is needed to clarify the exact contribution of the learning environments in the diversity of the developmental pathways observed.

Second, Fuson, Wearne, et al.’s (1997) model focuses on one aspect of children’s growing understanding of numbers and number relationships when they

start exploring and operating on multidigit numbers (Fuson, 1992; Jones, Thornton, & Putt, 1994; Treffers, 2001b), namely their base-ten structure. Fuson (1992, p. 265) herself pointed to the fact that besides this “collection-based” interpretation of numbers, there is also a “sequence-based” interpretation wherein the focus is on the number’s position within the number-word sequence. Treffers (2001b) referred to both conceptions as, respectively, the “structuring” and “positioning” representation of numbers. He (Treffers, 2001b, p. 109) defined *structuring* as “seeing numbers as a junction in a network of number relations,” as in decimal splitting (breaking down a number into its compound positional values) or in factorization (relating numbers to each other by multiplication and division), whereas *positioning* is defined as “being able to place whole numbers on an empty number line with a fixed start and end point. . . . Positioning enables students to gain a general idea of the sizes of numbers to be placed” (p. 104). As such, Treffers’s (2001b) positioning interpretation shows some alignment with Dehaene’s theory about how numbers are internally represented in the human mind (and brain), wherein an analog magnitude code (a kind of mental number line) is assumed as the main, if not only, semantic representation of a number (see earlier section on single-digit conceptual structures). Although several mathematics educators working in the domain of multidigit arithmetic give this positioning interpretation an increasingly prominent place in their (experimental) curricula, textbooks, and instructional materials (see, e.g., Beishuizen, 1999; Gravemeijer & Stephan, 2002; Selter, 1998; Treffers, 2001b), we are not aware of any (ascertaining) study documenting the development of this latter aspect of children’s growing conceptual knowledge of numbers (and its relation to the other aspects of multidigit number development) in a broad and systematic way.

Third, the conceptual development described in this section builds on a development that started much earlier, namely at the time of the very first signs of perception of numerosity and understanding of and operating on natural numbers in the early years of childhood (see the chapter on early childhood mathematical thinking), and continues when children start to learn about other numbers that are included in the upper grades of school mathematics (rational numbers, integers, etc.) (e.g., Greer, 2004; Moss & Case, 1999; Vosniadou & Verschaffel, 2004). With respect to this continuation, there is a great deal of agreement that the extension of number concepts beyond the domain of natural numbers is a serious conceptual obstacle requiring drastic conceptual change (Greer, 2004; Vosniadou & Verschaffel, 2004) in the mathe-

mathematical development of children. Making use of their knowledge of natural numbers when interpreting new information about rational numbers gives rise to numerous well-documented misconceptions such as the misbeliefs that “the more digits a number has, the bigger it is” (leading to the misbelief that, for instance, 1.3 is smaller than 1.25) or that “multiplication always makes bigger and division always makes smaller” (leading to the expectation that, for instance, 45 divided by 0.9 will result in a quotient smaller than 45). These misconceptions typically result from the unwarranted overgeneralizations of rules that are (only) valid in the domain of natural numbers (Greer, 1992, 2004; Verschaffel & De Corte, 1996; Vosniadou & Verschaffel, 2004; see also the chapter on rational numbers and proportional reasoning in this volume).

Finally, although we have separated this section from the next ones, it is, of course, difficult to separate the development of base-ten number concepts from learning the procedures for doing multidigit arithmetic. In their review of the relation between conceptual and procedural knowledge of multidigit arithmetic, Rittle-Johnson and Siegler (1998) reported several types of evidence for this relationship. A first type comes from evidence from several large-scale studies that individual children’s success on measures of conceptual knowledge and procedural knowledge are positively correlated. For instance, Hiebert and Wearne (1996) followed children from first grade to fourth grade. They assessed conceptual understanding by asking children to identify the number of tens in a number, to represent the value of each digit in a number with concrete materials, and to make different concrete representations of multidigit numbers. Procedural knowledge was assessed through performance on two-digit addition and subtraction (story) problems, which could be solved either by the standard algorithm or by an invented procedure. The size of the numbers used in the tests differed as the children grew older. Across assessment periods, children who demonstrated higher levels of conceptual understanding also obtained higher scores on the procedural measures. As a second kind of support, Rittle-Johnson and Siegler (1998) referred to another finding from Hiebert and Wearne’s (1996) study, namely that early conceptual understanding predicted not only concurrent, but also future procedural skill. Thirdly, as will be reviewed in later, recent design experiments aimed at improving teaching of multidigit arithmetic by emphasizing concepts of place value and how they relate to steps in a procedure usually led to successful increases in both conceptual and procedural knowledge compared to traditional forms of instruction that are characterized by a gap between the two kinds

of knowledge (Hiebert & Wearne, 1996). At the same time, Rittle-Johnson and Siegler (1998) referred to some research evidence (Resnick & Omanson, 1987) that in conventional instruction that emphasized practicing procedures without linking this practice to conceptual understanding, the links between conceptual and procedural development are much weaker.

Strategies for Multidigit Addition and Subtraction

Whereas existing theory and research offer a rather comprehensive picture of how children learn to add and subtract with small numbers (discussed earlier), the literature about what strategies should be distinguished, and how they develop over time, is much more limited in the number domain beyond 20. At the end of her overview of the research on whole number addition and subtraction, Fuson (1992, p. 269) concluded that, except for research on buggy algorithms, our knowledge of the development of conceptual structures and solution procedures for multidigit addition and subtraction “lags far behind” that for single-digit concepts and skills.

However, since Fuson’s (1992) review, many studies exploring this new domain have been conducted in the United States (Jones et al., 1994), the U.K. (Thompson, 1994, 1999a), Japan (Reys et al., 1995), Australia (Cooper, Heirdsfield, & Irons, 1998), the Netherlands (Beishuizen, 1993, 1999; Beishuizen, Van Putten, & Van Mulken, 1993), Germany (Selter, 1993), Belgium (Torbeyns, Verschaffel, & Ghesquière, 2006; Verschaffel, 1997), and many other countries. These studies have documented the frequent and varied nature of children’s and adults’ use of (informal) strategies for mental computation that depart from the formal written algorithms taught in school.

Fuson, Wearne, et al. (1997) provided a comprehensive analysis of the mental calculation strategies children use to solve multidigit addition and subtraction problems, involving two primary classes of strategies, namely the *decompose-tens-and-ones* method and the *begin-with-one-number* method, as well as a third category of mixed strategies. In the first type of strategy, namely decompose-tens-and-ones, the tens and units of both numbers are split off and handled separately (e.g., $46 + 47$ is determined by taking $40 + 40 = 80$ and $6 + 7 = 13$, answer $80 + 13 = 93$). In the second type of strategy, the begin-with-one-number method, the different values of the second number are counted up or down from the first unsplit number (e.g., $46 + 47$ is determined by taking $46 + 40 = 86$, $86 + 7 = 93$). In the literature, these two methods have a variety of names such as *combining units separately*, *decomposition*, *partitioning* or *the split method* (for the decompose-tens-and-ones method), versus *sequential strategy*, *cumulative*

method or *jump method* (for the begin-with-one-number strategy). Besides these two primary types of strategies (and their mixed versions), some researchers (e.g., Buys, 2001; Treffers, 2001b) have identified a third type, called *compensating* or *varying* strategies, wherein the given numbers are adjusted in a clever and flexible way to simplify the calculation (e.g., $46 + 47 = [45 + 45] + 1 + 2 = 93$ or $46 + 47 = 46 + 50 - 3 = 96 - 3 = 93$), whereas other researchers (such as Fuson, Wearne, et al., 1997) consider such clever strategies as a subcategory of one of their two major categories of solution methods. Evidently, these categorizations apply also for addition and subtraction with multidigit numbers larger than two.

The first two types of strategies (for which we will from now on use the short but telling terms *split* and *jump*) in fact appear to be consistent with the two fundamentally different ways of interpreting two-digit numerals and number words, which Fuson (1992, p. 265) called “collection-based” and “sequence-based” and Treffers (2001b) called “structuring” and “positional” interpretations of numbers (discussed earlier; see also Buys, 2001; Cobb, Gravemeijer, Yackel, McClain, & Whitenack, 1997; Treffers, 2001b). Whereas the idea implicit in the split strategy is that of numbers as collections of items that can be partitioned and recomposed, the view implicit in the jump strategy is that of a counting activity that can be curtailed or chunked. Accordingly, whereas base-ten blocks (or corresponding square/bar/point drawings) have been used typically as concrete referents to teach and to assess the collection-based interpretation and the corresponding split strategy, educators and researchers focusing on the sequence-based interpretation and the jump strategy will often use other representational tools, such as the hundred square or the hundred row and, more recently, the (empty) number line (Beishuizen, 1999; Treffers, 2001b; Wittmann, 2001).

Although these two strategies have a widespread use worldwide, there seem to be great international variations as consequence of differences in the (school) mathematics culture and practice. For instance, in most studies with second- to fourth-grade children in the United States and the U.K. the split method is the dominant one, probably because of an educational tradition that has emphasized written “column arithmetic” and the use of base-ten blocks (sharing with the split method the same conceptual number structure). In most studies involving continental European children, by contrast, the jump method seems more dominant among children who have received instruction in addition and subtraction in the number domain up to 100, probably because of the greater historical attention to mental calcula-

tion in general and the greater instructional focus on the sequential method in textbooks and on materials (such as the hundred board, the number line...) supporting this method (Beishuizen et al., 1993).

Beishuizen’s research program aimed at the characteristics of the solution and learning process of both strategies in the domain 20–100, basically from a viewpoint of cognitive (developmental) psychology (Beishuizen, 1993, 1999; Beishuizen et al., 1993). Like other Dutch researchers, he found that, although Dutch mathematics education in Grade 2 focuses on the jump method, the strategies of Dutch pupils showed a mixed picture. In general, only about half of the third graders systematically used the (explicitly taught) jump method, whereas the other half preferred the split method (as a nontaught alternative). Only a small minority used both strategies in a flexible or adaptive way (e.g., the split strategy for addition and the jump strategy for subtraction problems). However, contrary to what occurs in this early stage, in the long run the split method turns out to be the more complicated and error-prone computation method, whereas the jump method becomes the more efficient and effective one. Especially in subtraction-with-carry problems, pupils using the split method show many errors, such as the well-known “smaller-from-larger bug” (e.g., $63 - 38$ solved as $60 - 30 = 30$ and $8 - 5 = 3$, answer 33). In line with this argument, Beishuizen (1993) observed more errors for the split as compared to the jump strategy when competent and consistent users of either method were compared on mental computation tasks.

Following these exploratory studies on the characteristics of the jump and the split strategies, Beishuizen et al. also conducted a series of intervention studies (Blöte, Van der Burg, & Klein, 2001; Klein, Beishuizen, & Treffers, 1998) wherein they compared the impact of reform-based instruction, focusing on the flexible application of different strategies from the start of the learning process onwards, with traditional instruction, focusing on the mastery of one single strategy, namely the jump strategy. The results of these intervention studies showed that, after instruction, both groups of children applied the jump strategy quite frequently, especially on subtractions (Blöte et al., 2001; Klein et al., 1998). But the children who used the reform-based instruction fitted their strategy choices more often to the number characteristics of the items than the traditionally taught children and relied more frequently on so-called varying strategies. For instance, they solved items with an integer ending with the number 9 (like $35 + 49$) more frequently with a compensation or varying strategy (e.g., $35 + 50 - 1$)

than children who had received traditionally-oriented instruction (Blöte et al., 2001; Klein et al., 1998).

Although the research programs of Beishuizen and his colleagues deepened our understanding of the developmental changes in children's strategy competencies in the number domain up to 100, their results are weakened because of some methodological issues. For example, their operationalization of adaptivity only takes into account item characteristics. As discussed by several authors (Payne, Bettman, & Johnson, 1993; Siegler, 1998), an adaptive strategy choice process takes into account not only item characteristics, but also characteristics of the strategies (e.g., accuracy, speed), the subject (e.g., ability level), and the context (e.g., instruction). In an attempt to overcome these shortcomings, Torbeyns et al. (2006) analyzed the development of children's strategy competencies for doing additions and subtractions up to 100 in terms of the model of strategy change (Lemaire & Siegler, 1995), using the choice/no-choice method (Siegler & Lemaire, 1997). Second graders of high, average, or low mathematical achievement level solved four types of additions and subtractions before and after (traditional) instruction in the jump method. At both measurement times, children answered the items in one choice condition in which they could choose between the jump and the split strategy on each item, and in two no-choice conditions in which they had to solve all items with, respectively, the jump and the split strategy. As in Beishuizen et al.'s studies, at the first time of measurement, most children solved all problems by one strategy, mostly splitting, in the choice condition. Second, although the no-choice data showed that most children became quite accurate in applying the taught jump method, many of them (especially the weaker ones) continued to opt for the split strategy (especially on the addition problems) in the choice condition. Finally, although the children in this study, as in the studies of Beishuizen et al., seemed to adapt their strategy choices to item characteristics (e.g., by using the split strategy much less frequently on subtraction than on addition items), a more comprehensive (but at the same time more demanding) measure of adaptivity, involving also strategy and subject characteristics, revealed that, both before and after instruction, children were remarkably nonadaptive in their strategy choices. Although the reasons for this nonadaptivity remain obscure, it seems evident that this lack of strategy adaptivity is at least partly due to an orientation towards routine rather than adaptive expertise in the instructional environment.

As part of a broader ascertaining study, Reys et al. (1995) assessed mental computation performance, as well as attitude and computational preferences, of

large groups of Japanese students in Grades 2, 4, 6, and 8 on problems involving whole-number additions and subtractions ranging from single-digit to three-digit problems such as $182 + 97$ and $264 - 99$. Problems were presented visually or orally, and students could not make use of paper and pencil. A sample of students in Grades 4 and 8 scoring in the upper and middle quintiles on the mental computation test was interviewed to identify strategies used to mentally compute. A wide range of performance on mental computation was found with respect to the different number types and operations involved at every grade level. The mode of presentation affected performance level, with visual items generally producing higher performance. The most important finding was that the range of strategies used for mental computation was quite narrow, with the most popular approach reflecting a "mental version" of the learned right to left paper-and-pencil algorithm. This tendency to work algorithmically, even in the absence of paper and pencil, was more pronounced for the middle-quintile students than for the high-quintile students.

In the United States, Carpenter et al. (1997) did a longitudinal study investigating the development of children's multidigit addition and subtraction in relation to their understanding of multidigit concepts in Grade 1–3. Students were individually interviewed five times on a variety of tasks involving straightforward, result-unknown addition and subtraction word problems with two-digit numbers for the first three interviews and three-digit numbers in the last two interviews. During the same interviews, children were individually administered five tasks measuring their knowledge of base-ten number concepts, together with a task wherein they had to apply a specific invented strategy to solve another problem and two unfamiliar (missing addend) problems that required some flexibility in calculation. Contrary to the above-mentioned study with Japanese pupils, all students were in classes of teachers who were participating in a 3-year intervention study designed to help teachers understand and build on children's mathematical thinking. The emphasis of this program was on how children's intuitive mathematical ideas emerge to form the basis for the development of more formal concepts and procedures. Teachers learned about how children solve problems using base-ten materials and the various invented strategies children often construct. The researchers identified the following categories of strategies:

- modeling or counting by ones,
- modeling with tens materials,
- sequential strategies (jump strategies),

- combining-units strategies (decomposition or split strategies),
- compensating strategies,
- other invented strategies,
- algorithms (correct as well as buggy ones).

First, the study showed that, under favorable circumstances, children can invent mental calculation strategies for addition and subtraction problems. Second, buggy algorithms occurred more frequently among children who started out working algorithmically early than among children who used invented mental strategies before or at the same time they used standard algorithms. Third, students who used mental calculation strategies before they started using standard algorithms also demonstrated better knowledge of base-ten number concepts and were more successful in extending their knowledge to new situations. Finally, the data suggest that there is no explicit sequence in which the three basic categories of mental calculation strategies (sequential/jump, combining units/split, and compensating/varying) develop for addition. The majority of students applied all three, and the order in which they occurred was mixed. For subtraction, the sequential method was mostly used, but for this kind of problem some compensation strategies were also observed. For other studies reporting similar findings about the development of mental calculation strategies for multidigit addition and subtraction in nonconventional classrooms in close relation to the development of the concepts that support them, we refer to Fuson, Wearne, et al. (1997) and to Hiebert and Wearne (1996).

Strategies for Multidigit Multiplication and Division

Although several researchers have documented that children can also invent strategies for multiplying and dividing multidigit numbers and have described some strategies they use, less progress has been made in characterizing such inventions than for the domain of multidigit addition and subtraction. Here, we summarize the main findings from a recent analysis by Ambrose, Baek, and Carpenter (2003; see also Baek, 1998) of children's invented multidigit multiplication and division procedures (and the concepts and skills on which they depend). We stress again that these constructions did not take place in a vacuum, but in an instructional environment that stimulated children to construct, elaborate, and refine their own mental strategies rather than forcing them to follow a uniform, standardized trajectory for mental or written arithmetic (Ambrose et al., 2003, p. 309). As such, the constructions described by these authors show great similarities with those in Treffers's (1987, Chapter VI)

analysis of how elementary school children gradually build up the algorithms for column multiplication and division starting from their invented informal strategies in an instructional environment based on the principles of realistic mathematics education (RME), and to similar analyses by Anghileri (1999) and Thompson (1999c). For a description of these environments, we refer to the Classroom-Based Research section.

Children's mental calculation strategies for multiplication problems were classified into four categories: direct modeling, complete number strategies, partitioning number strategies, and compensating strategies (Ambrose et al., 2003; Baek, 1998). A child using a *direct modeling strategy* models each of the groups using concrete manipulatives or drawings. A second category, *complete number strategies*, are those based on progressively more efficient techniques for adding and doubling, the most basic one being simply repeated addition. A child using the *partitioning number strategy* will split the multiplicand or multiplier into two or more numbers and create multiple subproblems that are easier to deal with. Finally, a child using a *compensating strategy* will adjust one or both of the multiplicand and multiplier on the basis of special characteristics of the number combination to make the calculation easier. Each of these mental calculation strategies is illustrated in Figure 13.1 (Baek, 1998).

Many children in the study developed their mental calculation strategies for multidigit numbers in a sequence from direct modeling to the complete number strategy to the partitioning numbers strategy (Ambrose et al., 2003; Baek, 1998). Children's strategies for solving multidigit multiplication problems varied with their conceptual knowledge of addition, units, grouping by ten, place value, and properties of the four basic operations.

Ambrose et al. (2003) presented a similar taxonomy for division, distinguishing between the following strategies. First, *Working with one group at a time*, whereby the child repeatedly subtracts the smaller given number from the larger (solving "180 divided into groups of 6" by doing $180 - 6 = 174$, $174 - 6 = 168$, etc.), or repeatedly adds up the smaller given number until the larger one is reached ($0 + 6 = 6$, $6 + 6 = 12$, etc.), and then counts how many subtractions or additions were made. Repeated subtraction or addition occurred regularly for quotitive-division problems, but very rarely for partitive-division problems, because in the latter case children do not know what number to subtract or to add repeatedly. However, these problems can, and are, solved at a comparable basic level by effectively distributing the larger given number in small parts over the different groups, usually with help of

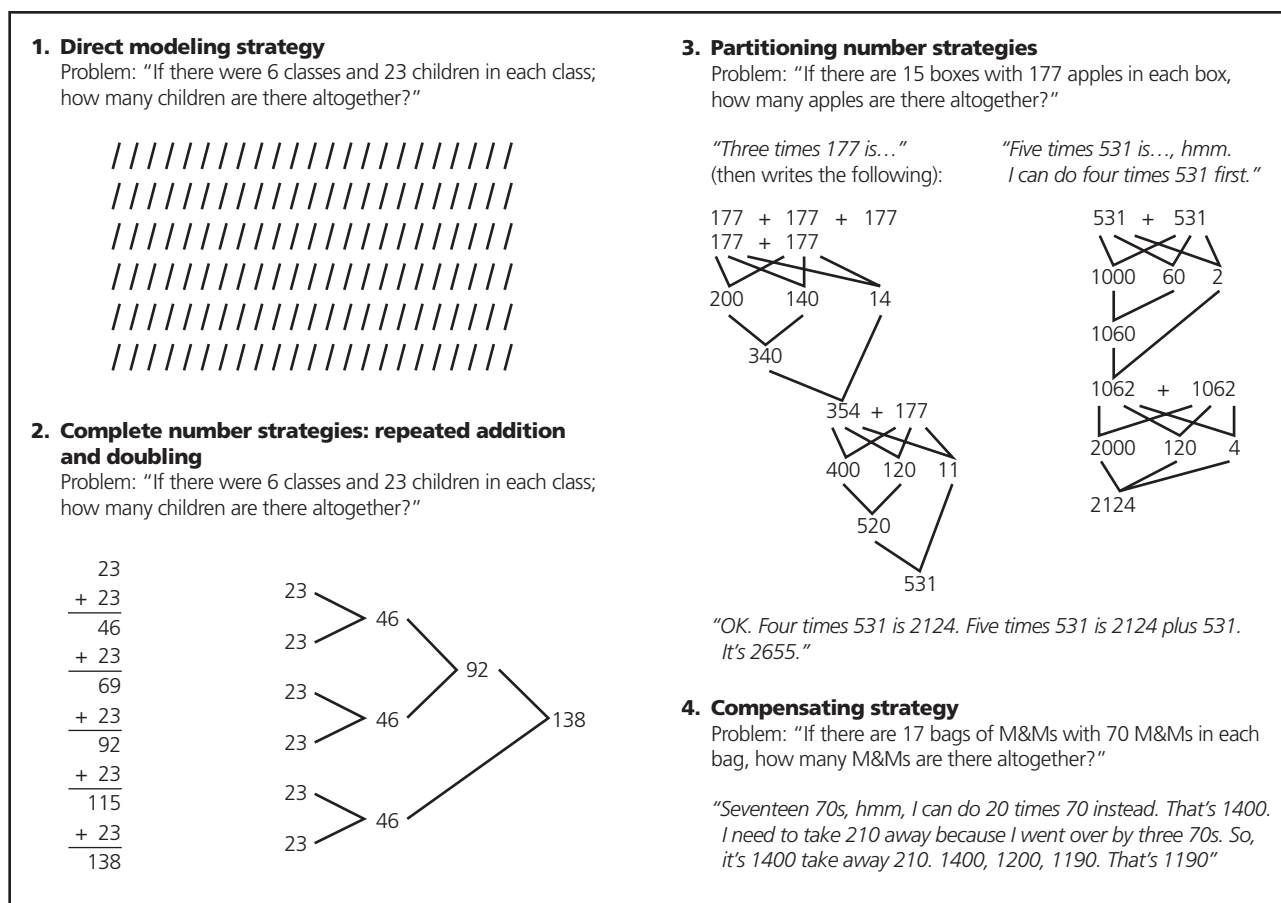


Figure 13.1. Examples of (invented) strategies for multidigit multiplication problems (Baek, 1998, pp. 153–158).

some kind of visual representations (e.g., solving "228 divided into 12 groups" by drawing 12 circles—one for each group—and then distributing the 228 units over the 12 groups by first putting a 10 in each circle, then a 2, then a 5, and finally another 2, while always keeping track how much remains after each distribution round). A second category involves *more abstract strategies that do not involve decomposition of the dividend*, wherein the child subtracts (or divides) in a more efficient and systematic way by relying explicitly on the ten-structure (e.g., solving "544 divided into 17 groups" by subtracting 544 three times by $10 \times 17 = 170$, and, having arrived at 34, by $2 \times 17 = 34$, leading to the solution of $30 + 2 = 32$). A third category consists of *strategies wherein the dividend is split into parts* (e.g., dividing 896 by 35 by dividing first the units [6], then the tens [90] and finally the hundreds from the dividend [800] by 35 while keeping track of the number of times 35 goes into each of these numbers as well as of the remainder of each of these three partial divisions). A final category involves strategies wherein both partitive and quotitive division problems are solved by means of

building up strategies (e.g., solving 544 divided into 17 groups by adding up $170 + 170 = 340$, $340 + 170 = 510$, $510 + 34 = 544$, so there are 32 candies/friend). For similar inventories of informal strategies for multidigit division, see, for example, Anghileri (1999, 2001), Treffers (1987), and, most recently, Van Putten, van den Brom-Snijders, and Beishuizen (2005). Using a 10-category system of strategies and a sophisticated multivariate approach, the latter authors were able to describe in a very detailed and systematic way the evolution of Grade 4 children's strategies for solving multidigit division problems under a realistic mathematics approach in Dutch primary schools. Just after the introduction of division problems with larger numbers, the strategies varied from no-chunking to high-level chunking, but after 5 months, the diversity was reduced, not only for the high-ability pupils but also for pupils with a rather weak mathematics level, to mainly high-level chunking.

Underlying these multidigit mental calculation strategies for multiplication and division are the commutative, associative, and distributive laws, although

they may not be known to pupils by name and may not be applied deliberately (Anghileri, 1999). As such, they constitute what Vergnaud (1996) termed *theorems-in-action*. A possible question for future research is whether making the laws (more) explicit would enhance the adaptivity based on their implicit application.

Mental Calculation Strategies versus Written Algorithms

Besides the kind of (invented) mental calculation strategies described in the two previous sections, there is, of course, another well-known type of procedure for doing multidigit arithmetic, namely the standard algorithm. Algorithms are finite, well-defined step-by-step procedures for accomplishing (familiar) tasks (Plunkett, 1979; Usiskin, 1998). More specifically, in this context, they are recipes for performing calculations with the digits of the numbers in the problem to find the result of an addition, subtraction, multiplication, or division (Thompson, 1999c; Treffers, Nootboom, & De Goeij, 2001). The purpose of any arithmetical algorithm is to reduce a complex calculation to a series of more elementary calculations that can be performed using well-established processes, together with the rules for co-ordinating these smaller calculations. As explained earlier, these standard algorithms are typically done by means of paper and pencil, but they can be, and sometimes are, done mentally.

Standard algorithms have evolved over centuries for efficient, accurate calculation and for the most part are far removed from their conceptual underpinnings. Their compact notation tends to mask the underlying principles that make them work. For instance, the principled basis of the standard algorithm for multiplication and especially for division in many countries is very difficult to unpack. Actually, standard practice in the use of algorithms even demands that one does not think about what the digits represent if one does not want to become confused (Thompson, 1999c). According to the same author, algorithms encourage “cognitive passivity” on behalf of the person using them: “The decision as to how to set out the calculation, where to start, what value to assign to the digits, etc. are all taken out of the individual’s hands” (Thompson, 1999c, p. 173). Taught or invented mental calculation strategies like those discussed in the two previous sections, by contrast, are derived directly and perspicuously from the underlying concepts of base-ten numbers, the fundamental principles of the operations, and the relations among them (Ambrose et al., 2003; Anghileri, 1999; Thompson, 1999a; Treffers, 1987).

Evidently, there are different standard algorithms available for whole-number operations. For instance, Figure 13.2 shows three main written methods for subtraction that have been and are still being taught in

(a)

$$\begin{array}{r} 146 \\ - 29 \\ \hline 117 \end{array}$$

(b)

$$\begin{array}{r} 146 \\ - 29 \\ \hline 117 \end{array}$$

(c)

$$\begin{array}{r} 146 \\ - 29 \\ \hline 117 \end{array}$$

9 + 7 = 16 2 + 1 = 3
3 + 0 = 3 0 + 1 = 1

Figure 13.2 Alternative algorithms for whole-number subtraction.

different Western countries. For an overview of different algorithms for arithmetic, we refer to Morrow and Kenney (1988), Thompson (1999c), and the Web site of Orey (2005).

Ascertaining research on algorithms is rather scarce, except for the extensive (older) work on buggy algorithms by Brown, Burton, and Van Lehn (Brown & Van Lehn, 1982; Burton, 1982) in the early '80s. Drawing on computer-based metaphors, they developed a theory that characterizes observed patterns of errors as “bugs” within correct procedures for computations. A finite collection of such bugs, often operating in combinations, have been shown to account for many of the errors made by pupils. This work has often been held up as an example of the contribution that the information-processing approach can make to mathematics education, being cited by Boden (1988, p. 263) as one of the few examples of a domain modeled in enough detail to be useful to pupils or teachers, by Rissland (1985) as a good example of “principled” modeling of human mathematical behavior, and by Newell (1990, p. 368) as a rare case in which cognitive theory can predict long stretches of behavior. Mathematics educators, by contrast, have been much less enthusiastic (e.g., Cobb, 1990; Fischbein, 1990), a fundamental basis of their criticism being that skill in computation involves much more than procedural

competence. On the one hand, local repair of bugs in children's procedures amounts to treating symptoms rather than the underlying causes of imperfect understanding. On the other hand, it is appropriate to exploit systematicity in errors to diagnose conceptual weaknesses (e.g., Resnick et al., 1989).

Other researchers, particularly mathematics educators, have also studied algorithms. Results show that after having completed time-consuming courses in written algorithms, many students still have not mastered them. For instance, Treffers (1987) reported that after between 50 and 100 lessons, only 2 out of 3 pupils in the Netherlands had mastered the long division algorithm. Moreover, traditionally schooled pupils find it very difficult to relate the algorithm to situations in which it is nevertheless applicable, and they also perform very weakly on test items asking for insight into the algorithm rather than for procedural mastery (Treffers, 1987; see also Verschaffel & De Corte, 1996). Some mathematics educators went further in the '90s and documented the harmful effects of algorithms in elementary school children (Kamii & Dominick, 1998). These manifestations may be interpreted as evidence that teaching with the narrow aim of establishing routine expertise may not be the best way to achieve even that limited goal.

As explained before, algorithmic calculation is given a less prominent place in current educational reform—or rather, to be more precise, its prominence is placed against the background of emphasizing other aspects of proficiency. Although the importance of mental arithmetic and estimation is stressed in all current reform documents, the view is still generally accepted that children should learn in elementary school at least one standard written method of calculation for each of the four arithmetic operations. However, it is also agreed, first, that premature teaching of standard written algorithms should be avoided and that an introduction into written algorithms can only be given after prolonged work with concrete and mental arithmetic. Second, with respect to the way of teaching, there is wide agreement that children should be actively involved in devising the algorithms for written computation, starting from their knowledge about number and single-digit and multidigit operations. Third, more and more people in the mathematics education community accept that there is no absolute need that all pupils reach the highest level of all formal written standard algorithms (Thompson, 1999a, 1999c; Treffers et al., 2001).

Illustrative, and supportive, for this latter reform-based approach are the design experiments realized by the Freudenthal Institute for the development of multiplication and long division, wherein the hypo-

thetical learning trajectories for the four algorithms involve a gradual progression from the kind of informal and invented mental calculation strategies described above through a stage of “column arithmetic” strategies to the culturally-agreed algorithmic calculations (Treffers, 1987; Treffers et al., 2001; see also De Corte et al., 1996; Van Putten et al., 2005; Verschaffel & De Corte, 1996). For similar design experiments done recently in the United States and the U.K., see the Classroom-Based Research section.

Conclusion and Discussion

Whereas in the '70s and '80s research focused on children's solutions of arithmetic problems involving relatively small whole numbers, researchers more recently have paid more attention to problems that involve multidigit calculations and have made significant progress in identifying and characterizing the different conceptual elements and strategies that children construct to calculate with multidigit numbers besides the regularly taught standard algorithms for written computation.

This research has helped us to illustrate another meaning of “mental arithmetic” besides the traditional one wherein mental arithmetic is seen (merely) as doing sums in one's head instead of with paper and pencil. Most classifications of children's procedures for operating on multidigit numbers distinguish among three basic categories of strategies of mental arithmetic (which seem to be closely linked to different conceptions of numbers):

- Jump strategies in which the numbers are seen primarily as objects in the counting row and for which the operations are movements along the counting row—further (+) or back (−) or repeatedly further (×) or repeatedly back (÷).
- Split strategies in which the numbers are seen primarily as objects with a decimal structure and in which operations are performed by splitting and processing the numbers on the basis of this structure.
- Varying strategies based on arithmetic properties in which the numbers are seen as objects that can be structured in all sorts of ways and in which operations take place by exploiting a suitable structure and using the appropriate arithmetic properties (Buys, 2001).

For each of the basic arithmetic operations, each of these three basic forms can be performed at different levels of internalization, abbreviation, abstraction, and formalization (e.g., at a lower level by using models such as the base-ten blocks or the “empty number

line,” at a higher level by noting intermediate steps in formal-arithmetic language, or purely internally).

As in the *Single-Digit Computation* section, the above review of the last decade’s research on multidigit mental arithmetic has pointed to (a) the invented nature of several of these procedures of mental arithmetic; (b) the flexible or adaptive use of different strategies as a basic characteristic of expertise in mental arithmetic (Hatano, 2003); (c) the close relationship between procedural and conceptual knowledge in the domain of multidigit mental arithmetic (Rittle-Johnson & Siegler, 1998); (d) the “dispositional nature” of mental arithmetic, as is convincingly documented, albeit in a negative way, by many traditionally taught children’s inclination to apply their standard algorithms in a stereotyped, inflexible way, even in cases in which adaptive mental arithmetic seems much more appropriate, such as problems like $24,000 \div 6,000$ and $4,002 - 3,998$ (Buys, 2001; Treffers, 1987, 2001b), by their lack of conceptually based “critics” to detect errors in computation, and by their lack of self-confidence to “have a go” and take risks when leaving the safe path of standard algorithms (Thompson, 1999a).

Although we have focused mainly on findings about the development of individual children’s concepts and strategies in multidigit arithmetic and postponed the role of instructional and cultural issues to the later section on classroom-based research, the research reported so far leads already to some important instructional implications. On the one hand, there is a well-documented research tradition on the many kinds of errors (“bugs”) that students make when the algorithmic methods for multidigit addition, subtraction, multiplication, and division are not connected to place value meanings (Kilpatrick et al., 2001), while at the same time there is increasing evidence of the variety of strategies and the gradual progression in these strategies when children are allowed and even stimulated to invent and use their own methods. Typically, these invented strategies can be ordered on a developmental line beginning with concrete methods that model the problem rather directly, towards more efficient methods, which are less transparent and more problem independent (Kilpatrick et al., 2001). So, these informal mental calculation strategies seem to be a perfect stepping-stone to “informal column arithmetic” and, afterwards, to the standard formal algorithms for written computation (Treffers et al., 2001).

Whether these algorithmic methods still need to be mastered perfectly by all children, and (thus) deserve so much instructional time, is a question that elicits hot debates. The idea that drilling students

for long periods on solving problems involving large numbers by means of these written algorithms “seems to be more appropriate to the twentieth century than the twenty-first” (Kilpatrick et al., 2001, p. 214) is gaining ground. Nevertheless, internationally, most elementary school mathematics curricula continue to pay, for mixed reasons, considerable attention to the teaching and learning of these algorithms.

Finally, an important concern, with serious theoretical, methodological, and instructional implications—not only for the research on multidigit arithmetic but also for the previous and the next sections—is what is really meant by “a student is using strategy *x*” or “strategy *x* is being taught.” According to authors such as Gravemeijer (2004), it is inappropriate to think of strategies as ready-made techniques that are available in the repertoire of the children and are consciously or deliberately chosen and applied in a particular situation. Especially with young children, a more plausible way of thinking is that they (initially) base their computations on their familiarity with certain number relations. Take, for instance, a second grader who, when solving the problem $29 + 5$, activates a variety of number relations involving 29 and 5 to solve the problem at hand. One option would be to combine $29 + 1 = 30$ and $4 + 1 = 5$ to conclude that you can take 1 from the 5, add that 1 to 29 to get 30, and add the remaining 4 to the 30 to get 34. As Gravemeijer commented (2004, p. 114),

Then, from the observer’s point of view, it could look like the student is using a building-up-to-10s strategy [or, in our terminology, a jump strategy]. For the student, however, this strategy may not be at the horizon yet. Only after reflecting on substantial experience with similar problems, the student could start to notice a pattern and construe the building-up-to-10s strategy. Even then, it could take a while before the student starts to use this strategy as an *a priori* guidance for choosing a solution procedure.

This distinction recalls Vergnaud’s (1996) term *theorem-in-action*. If this view on (the development of) strategies is true, the instructional goal, then, becomes, according to Gravemeijer (2004, p. 115), “not to teach the students a set of strategies. Instead, our primary goal is for the students to develop a framework of number relations that offers the building blocks for flexible mental computation.”

This idea fits perfectly with the “schema-based view” of arithmetic learning proposed by Baroody and Ginsburg (1986, see also Baroody, Cibulskis, Lai, & Li, 2004).

Estimation and Number Sense

In the lives of both children and adults, many types of questions and situations call for estimation, yet, until recently, curriculum developers have given it short shrift. Before 1980, only isolated voices spoke in support of its role in the mathematics curriculum. Current reform documents, however, call for incorporating estimation activities “into all areas of the program on a regular and sustaining basis” (Sowder, 1992, p. 373; see also Siegler & Booth, 2005). According to these documents, ability to estimate is not only extremely important in many kinds of employment and in the ordinary activities of life, but estimation experience is also claimed to have beneficial effects on many other mathematical abilities such as mental computation, spatial visualization, and measurement (Brade, 2003; Dowker, 2003; Van den Heuvel-Panhuizen, 2001). A third reason why estimation is important is that many types of estimation require going beyond routine application of procedures and applying mathematical knowledge and skills in a flexible way. So it is a privileged vehicle for developing adaptive expertise in mathematics (Siegler & Booth, 2005).

Before these pleas in favor of estimation in curricular documents and recommendations, it had prompted very little research. The scarce research that had been done offered ample evidence that most school-age children are remarkably bad at it and even many adults lacked basic estimation skills (Sowder, 1992), a finding echoed in Siegler and Booth’s (2005) more recent review. In her review of the literature, Sowder (1992, p. 371) referred to three types of estimation: computational, numerosity, and measurement estimation, “each requiring different kinds of understanding and different kinds of skills.” In this section we focus on computational and numerosity estimation. Because of its close relation to estimation, we have decided to review the research on number sense in this section too. As in the other parts concerning ascertaining studies, we pay more attention to the psychological (and psychometric, and neuroscientific) research on estimation and number sense and their development than to studies dealing with the question of how the skill should be addressed in school mathematics. Measurement estimation (and “sense for measures”), the third component of estimation distinguished by Sowder (1992), will not be discussed, as it lies beyond the scope of this chapter. Although numerosity estimation is clearly the first to evolve, we prefer to start our review with computational estimation, as this has, by far, elicited most of the research on estimation.

Computational Estimation

Computational estimation can be defined as “finding an approximate answer to arithmetic problems without actually (or before) computing the exact answer” (Lemaire & Lecacheur, 2002, p. 282). Within computational estimation, a distinction can be drawn between (a) calculating with rounded-off numbers with the intention of finding a global answer to an arithmetic problem given in exact terms (instead of, before, or after making the exact computation), and (b) calculating with estimated values because (some of) the exact data are unavailable (Van den Heuvel-Panhuizen, 2001).

Most of the research on estimation has dealt with children over 10 years old or with adults. This means that estimation has usually been studied after it is at least partially established, rather than charting its emergence (Dowker, 2003). However, some researchers have investigated how estimation abilities develop (in the absence of instruction) before the age of 10 (see Sowder, 1992, for an overview of the studies up to 1992, and Brade, 2003; Dowker, 2003; and Siegler & Booth, 2005, for reviews of the later research).

A comprehensive study on computational estimation done by Reys, Rybolt, Bestgen, and Wyatt (1982) on a large group of pupils from Grades 7 to 12 and out-of-school adults led to the identification among good estimators of three key processes or procedures that are still encountered in recent work, namely reformulation, translation, and compensation.

- Reformulation can be described as the process of altering numerical data to produce a more mentally manageable form while leaving the structure of the problem intact (e.g., changing $[6 \times 347] \div 42$ into $[6 \times 350] \div 42$).
- Translation involves changing the mathematical structure of the problem into a more mentally manageable form (e.g., changing $[347 \times 6] \div 43$ into $347 \times [6 \div 43]$ and then into $350 \div 7$).
- Compensation involves making adjustments to reflect changes that come about as a result of transformation or reformulation of the problem (e.g., $\$21,319,908 \div 26$ is estimated by dividing 26,000,000 by 26, yielding 1,000,000, then compensating downward to $\$850,000$).

The investigators noted that good estimators used a variety of strategies within these three key processes.

Many investigators have characterized good estimators as individuals who, besides having the ability to use these three procedures, “have a good grasp of basic

facts, place value, and arithmetic properties, are skilled at mental computation, are self-confident and tolerant of error, and can use a variety of strategies and switch easily between strategies” (Sowder, 1992, p. 374). Poor estimators, by contrast, seem to be bound, with only slight variations, to one strategy—that of applying algorithms more suitable for finding the exact answer. They have only a vague notion of the nature and purpose of estimation, believe it to be inferior to exact calculation, and equate it with guessing. According to Sowder (1992), Hatano’s distinction between adaptive and routine expertise seems particularly relevant when thinking about how to characterize good and weak estimators and about how to plan instruction on estimation. Indeed, many estimation problems are novel in nature, making it difficult, if not impossible, to delineate a list of knowledge components and skills that, if learned, would constitute a comprehensive knowledge base for computational estimation (Sowder, 1992).

Since Sowder’s (1992) landmark review, interesting work in this domain has been done by LeFevre, Greenham, and Waheed (1993); Dowker (1997, 2003); and Lemaire and Lecacheur (2002). In their cross-sectional study with Canadian participants, ranging from elementary school children to adults, LeFevre et al. (1993) found that until Grade 4 multidigit computational estimation develops very slowly, followed by a leap in estimation ability between Grade 4 and Grade 6, and further improvements beyond Grade 6. Second, using observation of ongoing behavior and immediate retrospective self-reports as their data-gathering techniques, these authors observed a wide strategic variability among children as well as adults, with rounding being the most common approach. Third, both children and adults seemed to adapt their strategy choices to problem characteristics. Finally, the range, the sophistication, and the adaptiveness of the strategy choices increased with age. For instance, a sophisticated strategy like post-compensation (whereby the person corrects for distortion after a computation has been done on rounded or truncated numbers) was used (efficiently) more often by adults than by children. To account for the observed (development in) variety, sophistication, and adaptivity in people’s strategy choices, these authors proposed a process model of computational estimation, consisting of retrieval, calculation, and compensation as the basic strategic components, heavily inspired by Siegler and Shrager’s (1984) model of strategy choices in one-step arithmetic (see *SCADS* section).

Dowker (1997, 2003) investigated the development of arithmetic estimation with special attention to the question how estimation changes in relation to the difficulty level of the items and development of ar-

ithmetical skill. A large group of children between 4–5 and 9–10 years old was divided into five levels according to their performance on a set of mental addition tasks. For instance, a typical Level 2 child could solve $5 + 3$ but not $8 + 6$, whereas a typical Level 5 child could solve $52 + 39$ but not $523 + 168$. Depending on the level at which they were placed, the children then were given one or more sets of estimation items that went beyond the capacities of the child’s arithmetic level. For instance, children categorized in Level 2 received estimation items beyond the capabilities of the addition-to-10 level (e.g., $8 + 6$). The child’s task was not to work out the answers, but to make good guesses about them. Dowker found that estimation performance improved with the calculation performance level of the children and that, for each performance level, estimation accuracy deteriorated as the children moved further and further from the number domain corresponding to their level of arithmetical skill (Dowker, 2003).

In an attempt to overcome some methodological problems of many of the previous studies, Lemaire and Lecacheur (2002) recently investigated people’s strategies in computational estimation of three-by-three-digit addition problems with small units ($263 + 471$) or with large units (e.g., $249 + 743$) using the “choice/no choice” method. Grade 4 and Grade 6 students, and adults, were asked to provide estimates for such calculations either in a choice condition, wherein they could choose between a “round-down” (rounding both operands to the closest smaller decades) and a “round-up” strategy (rounding both operands to the closest larger decades) or in one of two no-choice conditions (either always rounding-down or rounding-up). Results showed that (a) participants of all age groups used the round-down strategy more often and executed it more quickly than the round-up strategy; (b) strategy use and execution were influenced by participants’ age, problem features, and relative strategy performance; and (c) age-related changes in computational estimation included also changes in the relative influences of problem and strategy characteristics on strategy choices.

In their search for factors that might underlie the above-mentioned developments in estimation performance, strategy use, and strategy choice, Siegler and Booth (2005) distinguished three factors: (a) improved conceptual understanding of estimation and its underlying principles, (b) working memory growth, and (c) improved computational skills.

Numerosity Estimation

Of the three types of estimation distinguished by Sowder (1992), numerosity estimation has received

the least research attention. But since the early 1990s, numerosity estimation has received growing attention in developmental and cognitive psychological research. Numerosity estimation is typically described as gauging the cardinal number of a discrete quantity (Brade, 2003), such as beans on a string, people in a football stadium, or candies in a jar.

Subitizing, a process whereby very small numerosities are identified instantaneously, had attracted the attention of experimental psychologists already a long time ago and also leads to the determination of the cardinality of a discrete quantity. However, most of the research on subitizing has been done with children of a very young age, so we will not review this (for an extensive review, see Whalen, Gallistel, & Gelman, 1999).

As for the research on computational estimation, research on numerosity estimation has shown that (a) accuracy of numerosity estimation increases over a prolonged age range, (b) there is a variety in strategies, (c) strategy choices in numerosity estimation are sensitive to problem characteristics, and (d) the age-related differences in accuracy of numerosity estimation stem from the adoption of new, superior strategies, improvement in strategy execution with age, and improvement in the adaptivity of the choices among the existing strategies. Hereafter, we briefly review a few representative studies illustrating the above general findings.

Crites (1992) analyzed the strategies used by third, fifth, and seventh graders who were judged skilled and less skilled in making numerosity estimates on the basis of their results on a paper-and-pencil estimation test. Students' strategies were identified by means of a structured interview. Three strategies accounted for two thirds of the strategies used by the students of all grade levels: (a) benchmark comparison (estimating the numerosity of a new set by application of a known standard), (b) decomposition/recomposition (decomposing the to-be-estimated quantity into small samples and recomposing these small samples to arrive at an estimate), and (c) eyeball (a perceptually based strategy accompanied with an explanation like "looks like there are that many"). Further, the first two kinds of strategies were associated with higher performance scores and the third kind of strategy with lower performance scores. Finally, skilled estimators outperformed less skilled estimators, especially on questions that involved large quantities (greater than 1000).

Brade (2003) analyzed the numerosity estimation strategies and skills in a class of Kindergarten, Grade 1, and Grade 2 children who were involved in an intervention program (consisting mainly of a computer program involving estimation of numerosity tasks arranged in different levels according to the size of the

to-be-estimated quantity) and comparable control classes. All children from each class were pretested and posttested using a free-response task requiring children to estimate the number of objects in drawings containing up to about 80 identical and randomly arranged objects. Using a classification scheme consisting of eight (developmental) levels, each child was scored into one of these levels based on his or her performance on each test. Contrary to his expectations, the author did not find a significant effect of the computer-based intervention. Moreover, a significant relationship between children's ages and their estimation strategies was found, showing, for instance, that most kindergartners did not go beyond a level where they must count at least a portion of the to-be-estimated numerosity before an estimate can be made, whereas Grade 1 and 2 children already started to respond (appropriately) by means of typical small, medium, and large numbers (without ostensive use of counting by ones). However, more sophisticated strategies, like using the above-mentioned benchmark and decomposition/recomposition methods, were almost completely absent. Finally, fairly strong correlations were found between strategy and accuracy data, although some strikingly discrepant cases were observed too.

Camos (2003) examined the development of the strategies people use to determine the numerosity of large sets, from childhood to adulthood. Using Lemaire and Siegler's (1995) framework for describing strategic change, Camos analyzed her data in terms of the repertoire, the distribution, and the efficiency of execution of numerosity judgment strategies. Four main strategies were identified in this range: counting by ones ("1, 2, 3, 4, . . ."), by *ns* (e.g., "5, 10, 15, 20 . . ."), with additions (e.g., "one set of 4, plus a set of 6 and another one of 5 is 15"), and with multiplications (e.g., "6 sets of 5 is 30"). Although counting by ones continued to be present at all ages, counting by *ns* and with additions appeared at 7 years, and counting with multiplications only at 9. The frequency of use of the three new strategies increased with age, and counting by *ns* became the dominant strategy (only) after 11 years. The evaluation of the strategies (in terms of speed, accuracy, and use of manual pointing behavior) showed that all of these strategies became more efficient with age, except for the counting with multiplications (which was the most efficient as of Age 9, but also the least used). Finally, the frequency of use of strategies was examined in relation to the characteristics of the arrays to be counted. Although the manipulated characteristics rendered the arrays more or less difficult to count (by changing size, arrangement, density of subgroups, and so on), none of them

affected the use of strategies until 11 years, suggesting little signs of adaptivity before that age.

In the studies reviewed so far, the objects were arranged randomly. By contrast, in a series of recent studies Verschaffel, De Corte, Lamote, and Dhert (1998), and Luwel, Verschaffel, Onghena, and De Corte (2000, 2003a, 2003b) used tasks wherein the objects were presented within a regularly spaced grid form. Participants (ranging from second graders to university students) were asked to judge numerosities of colored blocks that were randomly arranged within rectangular grid structures (e.g., a grid of 7×7 , 6×9 , or 10×10). Siegler's model of strategic change was used to predict and analyze differences in strategic competence among different age groups and between different tasks. A scrutinized analysis of individuals' response-time and error-rate data, combined with individual interview data, revealed that besides the above-mentioned addition strategies from Camos's classification, many participants applied another strategy, namely a subtraction strategy, in which the number of empty squares was subtracted from the total number of squares in the grid. Second, even from the early years of elementary school on, many participants switched adaptively between these two kinds of strategies. Thus, for the items with few blocks and many empty squares (like a numerosity of 10 blocks in a 10×10 grid) they applied (different versions of) the addition strategy, whereas for items with many blocks and few empty squares (like a numerosity of 90 blocks in a 10×10 grid), most participants applied a subtraction strategy. Third, as expected, the occurrence, frequency, efficiency, and adaptivity with which this subtraction strategy was used increased with age.

All above-mentioned studies investigated people's numerosity estimation strategies in rather artificial contexts. Sowder (1992) referred to a couple of studies dealing with estimation by elementary school children of (large) numerosities embedded in a meaningful context, such as estimating the number of names on a page of a telephone book or how many sugar cubes would fill a matchbox, revealing that the above-mentioned benchmark and decomposition/recomposition methods reported by Camos (2003) were also observed in these realistic contexts. A recent study by Verschaffel, Luwel, and Torbeyns (2006) revealed that the same holds for the so-called subtraction strategy. Groups of second, fourth, and sixth graders were individually confronted with numerosities smaller and larger than 100 embedded in one-, two-, or three-dimensional realistic contexts. Whereas one third of these contexts were unstructured (e.g., a representation of an irregular piece of land with 72 cars unsystematically arranged on it), another third had a clear structure (e.g., a 16×4 rect-

angular parking lot completely filled with 64 cars), and the final third had a semistructure that made the use of the subtraction strategy highly efficient (e.g., a 13×7 rectangular parking lot containing 83 cars). Overall, almost 20% of all identifiable strategies for the semistructured estimation tasks could be classified as subtraction strategies. As expected, this percentage increased significantly with age.

In their review of numerical estimation, Siegler and Booth (2005) also reviewed another line of research, which they called *number line estimation*, which has become the focus of these reviewers' own research attention. In a typical number-line estimation experiment, such as the study by Siegler and Opfer (2003), a person is asked to estimate the position of a set of numbers on a particular scale (e.g., a 0–20, a 0–100, or a 0–1000 scale). Generally speaking, the results of these studies (in terms of the development of estimation performance as well as of strategy variety, efficiency, and adaptiveness) are the same as for the previously reviewed domains of numerical estimation. However, a very interesting specific aspect of this latter study is that it also revealed how people internally represent numerical magnitude. For instance, by means of fine-grained analyses of the patterns of percent absolute error for varied groups of numbers within a particular scale, Siegler and Opfer (2003) found distortions in the numbers used to label collections of objects, due to nonlinear mappings between numbers and their magnitudes.

Number Sense

In many analyses, estimation is connected to number sense (or to numeracy). For instance, Van den Heuvel-Panhuizen (2001, p. 173) started her didactical treatise of estimation as follows: "Estimation is one of the fundamental aspects of numeracy. It is the pre-eminent calculation form in which numeracy manifests itself most explicitly." McIntosh, Reys, and Reys (1992, p. 3) described number sense as follows:

Number sense refers to a person's general understanding of number and operations along with the ability and inclination to use this understanding in flexible ways to make mathematical judgements and to develop useful strategies for handling numbers and operations. It reflects an inclination and an ability to use numbers and quantitative methods as a means of communicating, processing and interpreting information. It results in, and reciprocally derives from, an expectation that numbers are useful and that mathematics has a certain regularity.

Number sense is highlighted in current mathematics education reform documents as it typifies the theme

of learning mathematics as a sense-making activity. Discussions around this “flag term” have resulted in listings of the essential components of number sense (McIntosh et al., 1992; Sowder, 1992) as well as descriptions of students displaying (lack of) number sense (Reys & Yang, 1998).

Probably the most comprehensive and most influential attempt to articulate a structure that clarifies, organizes, and interrelates some of the generally agreed components of basic number sense has been provided by McIntosh et al. (1992). In their model, distinction is made among three areas where number sense plays a key role, namely number concepts, operations with number, and applications of number and operation.

- The first component, knowledge of and facility with numbers, involves subskills such as sense of orderliness of number (e.g., indicate a number on an empty number line, given some benchmarks), multiple representations for numbers (e.g., $3/4 = 0.75$), sense of relative and absolute magnitude of numbers (e.g., have you lived more or less than 1000 days?), and system of benchmarks (e.g., recognizing that the sum of two 2-digit numbers is less than 200).
- The second component, called knowledge of, and facility with, operations, involves understanding the effect of operations (e.g., knowing that multiplication does *not* always make bigger), understanding mathematical properties (such as commutativity, associativity, and distributivity and intuitively applying these properties in inventing procedures for [mental] computation), and understanding the relationships between operations (e.g., the inverse relationship between addition and subtraction, and multiplication and division).
- The third component, applying knowledge and facility with numbers and operations to computational settings, involves subskills such as understanding the relationship between problem contexts and the necessary computation (e.g., understanding that a problem like “Skip spent \$2.88 for apples, \$2.38 for bananas, and \$3.76 for oranges; could Skip pay for this fruit with \$10?” can be solved quickly and confidently by adding the three estimated quantities rather than by exact calculation), awareness that multiple strategies exist for a given problem, and inclination to utilize an efficient representation or method (e.g., not solving

$[375 + 375 + 375 + 375 + 375] / 5$ by first adding the five numbers and then dividing the result by 5), and inclination to review the data and result (having a natural tendency to examine one’s answer in the light of the original problem).

Several researchers have documented the problems children experience with the different aspects of number sense mentioned before. For instance, Reys and Yang (1998) investigated the relationship between computational performance and number sense among sixth-grade and eighth-grade students in Taiwan. Students were interviewed about their knowledge of the different aspects of number sense from the theoretical framework mentioned above. Their performance on number sense was lower than that on similar questions requiring written computation. There was little evidence that identifiable components of number sense, such as use of benchmarks, were naturally used by Taiwanese students in their decision making. This is not surprising because, at least until recently, Taiwanese schools were typically very traditional and computationally driven in their instructional approach (Reys & Yang, 1998).

Given the above conceptualization of number sense, it is evident that, according to many authors, the development of number sense “is not a finite entity that a student has or does not have but rather a process that develops and matures with experience and knowledge” (Reys & Yang, 1998, p. 227) and that this development results from a whole range of activities of mathematics education on a day-by-day basis within each mathematics lesson, rather than a designated subset of specially designed activities (Reys & Yang, 1998).

Conclusion and Discussion

Our review of studies on estimation has revealed how difficult it is for students who receive traditional instruction, with its overemphasis on routine paper-and-pencil calculation, to understand that besides counting precisely and calculating exact answers, there is also something like estimating and developing appropriate procedures and strategies for making appropriate estimations (Kilpatrick et al., 2001; Siegler & Booth, 2005; Van den Heuvel-Panhuizen, 2001). Besides showing that the development of estimation begins rather late and proceeds rather slowly, the available research has documented, as in most other domains, (a) that estimating is a complex activity wherein all strands of mathematical proficiency are involved; (b) that children use several (invented) strategies to find estimates of arithmetic problems,

of numerosities or of positions on a number line; (c) that these estimation strategies vary in frequency of use and efficacy; (d) that estimation strategy choices are influenced by problem characteristics; and (e) that estimation strategy use, execution, and adaptivity improve with age.

At a more general level, the research reviewed in this section has shown that the domain of estimation is, *par excellence*, an area where the recurrent basic features of arithmetical thinking and learning are manifest, notably (a) the interrelationship between declarative and procedural knowledge, (b) flexibility rather than routine skill as the quintessence of expertise in the domain, and (c) the dispositional nature of number sense, involving not only aspects of capacity but also aspects of inclination and sensitivity (see also Dowker, 2003).

The above review also prompts some questions, such as why estimation skill develops so slowly. Besides a number of well-known sources of difficulty, which have been raised in relation to limitations of conceptual understanding, component skills, and working memory, the recent work on number-line estimation suggests an additional source of the slow and incomplete development of estimation skills, namely the nonlinear representations of numerical magnitudes: “If representations of the magnitudes of different numbers are largely indistinguishable, children are unlikely to gain much of a sense of the types of numbers that are plausible answers to numerical problems or that typically accompany real world values” (Siegler & Booth, 2005, p. 211). This idea not only suggests further ascertaining studies on the relationships between number line, computational, and numerosity estimation, but also design experiments aimed at the improvement of estimation accuracy over a wide range of numerical estimation by helping (young) children develop linear representations of a wide range of numerical magnitudes.

We end this section with a conceptual comment on the notion of *number sense*. As suggested already, there is a serious problem with it and, more specifically, with its relation to other aspects of arithmetic competence described in this and other chapters. Number sense seems to be somewhat like *intuition* in the sense that the mathematics education community commonly agrees on its importance, but it is hard to define and even harder to operationalize in order to do research on it. Also, recently there has been a rash of—more or less—related terms such as *numeracy* and *mathematical literacy*, with little definitional precision. Indeed, number sense is, in most cases where we have encountered it, defined so broadly that it includes not only problem solving, but also most, if not all, other

skills and dispositions that have been discussed in the other sections. Although we acknowledge its power in curricular reforms (Thompson, 1999b), we have some doubts about its usefulness for scientific research, unless and until its specific meaning has been defined in a clear and consistent way.

Understanding and Solving Arithmetic Word Problems

Arithmetic word problems constitute an important part of the mathematics program at the elementary school. Historically, they had mainly an application function, that is, they were used to train children to apply the formal-mathematical knowledge and skills (previously) learned at school in real-world situations. Later, word problems were given other functions, in particular as a vehicle for developing students’ general problem-solving skills. According to the current reform documents, word problems should also be mobilized in the earliest stages of the teaching and learning of whole-number calculation to promote a thorough and broad understanding of, and proficiency with, whole-number arithmetic (Kilpatrick et al., 2001; Verschaffel & De Corte, 1997). Indeed, in solving word problems young children get ample opportunities to build a rich and broad understanding of the four basic operations, to display their most advanced levels of counting performance, and to build a gradually more efficient and more abstract repertoire of procedures for whole-number computation.

Word problems have already for a very long time attracted the attention of researchers, both cognitive psychologists and mathematics educators. Here we will review only the work on word problems involving one or more basic arithmetic operations on whole numbers. Moreover, the focus is on studies that have been carried out since the reviews by Fuson (1992) and Greer (1992). Although some issues of instructional and cultural factors will be discussed already here, these issues are mainly dealt with later in this chapter.

Understanding and Solving One-Step Addition and Subtraction Word Problems

Throughout the 1980s and 1990s, concentrated research was carried out on how children learn to do one-step arithmetic word problems. Undoubtedly, the most popular subdomain has been the solution of one-step addition and subtraction problems involving small whole numbers or collections of discrete objects. In the early 1980s a basic distinction emerged that guided much of the subsequent research, distinguishing three classes of problem situations modeled by addition and subtraction. These are situations in-

volving a change from an initial state to a final state through the application of a transformation (change problems), situations involving the combination of two discrete sets or splitting of one set into two discrete sets (combine problems), and situations involving the quantified comparison of two discrete sets of objects (compare problems). Within each of these three major semantic categories, further distinctions were made resulting in 14 different types of one-step addition and subtraction problems (Riley, Greeno, & Heller, 1983; see also Fuson, 1992; Reed, 1999; Verschaffel & De Corte, 1993, 1997).

Numerous empirical studies carried out during this period with children between 5 and 8 demonstrated the psychological significance of this classification scheme, specifically that word problems that can be solved by the same arithmetic operation (i.e., a direct addition or a subtraction with the two given numbers in the problem), but that belong to different semantic problem types, yield different degrees of difficulty, different ways of representing and solving these problems, and different error categories (for reviews of this research, see Fuson, 1992; Reed, 1999; Verschaffel & De Corte, 1997).

On the basis of the results of this research program, researchers have also developed experimental programs for teaching elementary arithmetic word problems, wherein children were taught to materialize or schematize the semantic structure of the addition and subtraction word problems before actually solving them. Some of these experimental programs focused on learning to (re-)represent all 14 semantic problem types in terms of the part-whole structure (Rathmell, 1986; Tamburino, 1982; Wolters, 1983), whereas others are characterized by the use of different material or schematic representations for the distinct basic categories of addition and subtraction word problems (see, e.g., De Corte & Verschaffel, 1985a; Fuson & Willis, 1989; Jaspers & Van Lieshout, 1994). A somewhat different kind of instructional application is the Cognitively Guided Instruction approach, wherein researchers tried to improve in-service teachers' instructional practice and their students' learning outcomes by confronting these teachers with the above-mentioned research-based knowledge of the different types of word problems and of (the development of) children's skills and processes for solving them (Carpenter & Fennema, 1992; see also the section on *Teaching Word Problems*).

Computer simulations were developed concurrently with the empirical research (for a more recent example of the genre, see Leblanc & Weber-Russell, 1996, and for more recent reviews of these computer models, see Fuson, 1992; Reed, 1999; Verschaffel & De

Corte, 1997). Many studies compared the performance, strategy, and error data of students with the behavior of these computer models (for reviews of these empirical studies see Fuson, 1992; Reed, 1999; Verschaffel & De Corte, 1997). For example, empirical support for the assumed central role of part-whole knowledge in children's addition and subtraction word-problem solving comes from the work of Sophian (Sophian & McGorray, 1994; Sophian & Vong, 1995).

Until the early 1990s, the computer models and empirical findings were much more elaborated for change and combine problems than for problems about comparison. Therefore, it is not surprising that in the last decade that last problem category seems to have especially attracted the continued attention of researchers. This research has focused on the most difficult compare problem types, namely compare problems with an unknown reference set, such as "Pete has 9 apples. He has 3 apples more than Connie. How many apples does Connie have?" and "John has 5 marbles. He has 3 fewer marbles than Ann. How many marbles does Ann have?" In an attempt to explain students' difficulties with these types of compare problems, Lewis and Mayer (1987) presented a model that simulates the comprehension processes when solving these problems in quite a different way than according to the hypothetical processes implemented in the above-mentioned computer models. Their model has been tested in several empirical studies with a variety of data-gathering techniques, including the registration of eye movements while reading and solving the problems (Hegarty, Mayer, & Monk, 1995; Verschaffel, De Corte, & Pauwels, 1992), asking participants to retell the problems before answering them (Verschaffel, 1994), asking them to formulate a hypothesis about the required mathematical operation while reading every problem sentence (Kintsch & Lewis, 1993), and doing individual interviews wherein subjects are confronted with a variety of carefully chosen diagnostic tasks (Stern, 1993). Whereas some of the findings support the hypotheses underlying the Lewis and Mayer (1987) model, there are also others for which this model does not provide a proper explanation and for which alternative explanatory frameworks (e.g., Kintsch & Lewis, 1993; Okamoto, 1996) seem more appropriate. In any case, which (combination) of the different available theoretical explanations provides the best explanation for the remarkable difficulty of compare problems with an unknown reference set and for the extra complicating effect of the presence of the "marked" relational term is still a matter of hot debate among (cognitive) researchers.

As with the above-mentioned research programs of Siegler and associates (Shrager & Siegler, 1998;

Siegler, 1998; Siegler & Jenkins, 1989; see *SCADS* section) and Brown, Burton, and Van Lehn (Brown & Van Lehn, 1982; Burton, 1982; see section *Mental Calculation Strategies* section), the cross-fertilization between these empirical studies and computer simulation models is considered, in some circles, as one of the most impressive demonstrations of the cognitive-rationalist, and especially the information-processing, paradigm (De Corte et al., 1996; Greer & Verschaffel, 1990). As a result of the extended joint effort of computer model builders and empirical researchers, Fuson (1992, p. 269) stated that “there is a fairly clear consensus about the development of the skill in solving one-step addition and subtraction word problems over time, both in terms of the underlying conceptual structures and the kind of solution strategies.”

However, it was clear from an early stage that there were still several problematic issues and remaining questions. By no means are all empirical findings consistent with the (computer) models (for systematic comparisons of these empirical data with the predictions of the computer models, see Carpenter & Moser, 1984; De Corte & Verschaffel, 1988; Fuson, 1992). Many of these inconsistencies have to do with the fact that the models do not address important aspects, such as (a) the influence of textual variables on the construction of children’s problem representations and solutions, and (b) the fact that the solution of a word problem does not take place in a sociocultural vacuum but within a particular sociocultural setting, namely a mathematics classroom. To illustrate the first complicating element, we refer to a great number of studies that have convincingly shown how small changes in the wording of the problems may have a dramatic impact on children’s solution processes and skills (Coquin-Viennot & Moreau, 2003; Cummins, Kintsch, Reusser, & Weimer, 1988; De Corte, Verschaffel, & De Win, 1985). Taking into account such findings, some researchers have started to question the basic assumption underlying the computer models that children’s competence to solve word problems depends basically on the development of logico-mathematical knowledge structures, including the general part-whole conceptual schema (= logico-mathematical models), and proposed instead “text-comprehension models” that pay much more attention to the importance of children’s understanding of the statements referring to the situation described in the problem (Cummins et al., 1988). Illustrative for the second complicating factor, namely the impact of the classroom culture on children’s problem-solving processes and skills, are the well-documented examples of children’s errors on arithmetic word problems due to the use of superficial coping strategies they have developed within the particular culture of school

mathematics, rather than to their failure to represent or solve the problem because of the lack of the necessary logico-mathematical or linguistic knowledge (see the next *Conclusion and Discussion* section for a further elaboration of this theme).

A general limitation of the theoretical models that have been developed through a combination of empirical studies and computer simulations is that this work has been almost entirely confined to word problems that can be “unproblematically” modeled by addition or subtraction of small positive whole numbers. In other words, rather than covering the whole range of addition and subtraction situations that (can) occur in the real world, most of this research has instead used a very restricted set of scholastic word problem versions of the real world, without much reflection on the complexities from a sociocultural perspective.

Understanding and Solving Word Problems About Multiplication and Division

Partly because of the complexity of the domain, partly because of the lack of computer models that have tried to provide an integrative picture of the whole representation and solution process, the field of multiplication and division word problems has developed somewhat differently, and with less integration, than that of addition and subtraction word problems. We consider here three prominent lines of research.

First, research on semantic types is concerned with categorizing the situations described in word problems according to how they are or can be schematized (prior to solution). Having reviewed the work done throughout the 1980s and early 1990s in this first line of research, Greer (1992) proposed a synthesis of semantic types—which he termed *models of situations*—for multiplication and division. An important difference from the dominant categorizations for addition and subtraction is that this taxonomy also pays attention to the kind of numbers involved (whole numbers or positive rationals). A distinction is made between situations that are “psychologically commutative” (or “symmetric”) and those that are “psychologically non-commutative” (or “asymmetric”). In the asymmetric classes the multiplier and multiplicand can be distinguished—for example, in the class labeled “equal groups” the number of objects in each group is the multiplicand and the number of groups is the multiplier. By contrast, for Cartesian product (e.g., “How many different combinations can be assembled if you have 3 shirts and 4 trousers?”) the numbers multiplied have symmetric roles. With respect to division for asymmetric cases, two types of division can be distinguished, namely division by the multiplier (partitive division) and division by the multiplicand (quotitive

division). By contrast, for a symmetric class of situations, there is no clear distinction between multiplier and multiplicand, and, consequently, only one type of division. This type of structural analysis provides a valuable framework for research, but it is not very clear to what extent this is useful in considering children's thinking or as a guide to pedagogical strategies (Greer, 1992). Contrary to research on addition and subtraction, in which the categories of problem types also represent hypothetical cognitive structures that are assumed to drive pupils' understanding and solving of particular problems, this does not seem to be so clearly the case for the different classes in the above scheme (Anghileri, 2001; Greer, 1992; Verschaffel & De Corte, 1997). Although this line of research is basically conceptual/epistemological in nature, researchers also point to empirical research (e.g., about levels of difficulty or about typical strategies or errors) to support the validity of certain distinctions or characterizations in their models.

A second line of inquiry that has attracted a lot of research, and that is also extensively reviewed by Greer (1992), concerns the role played by so-called "intuitive models" in the solution of word problems. According to Fischbein, Deri, Nello, and Marino's (1985) theory of primitive models, each arithmetic operation is linked to a primitive, intuitive model, even long after that operation has acquired a formal status. Identification of the operation needed to solve a word problem is mediated by these intuitive models. The model affecting the meaning and use of multiplication is repeated addition, in which a number of collections of the same size are put together. One intuitive model for division is equal sharing of a group of objects (partition), and a second less dominant interpretation relates division to finding the number of equal groups in a given total (the quotitive model). Many research findings, such as the amount and the kind of errors made when confronting students with multiplicative problems involving different kinds of decimals in the role of multiplier or multiplicand (for an extensive overview see Greer, 1992), children's problem solving and reasoning processes in individual interviews around such problems (see also Greer, 1992), and, more recently, the kind of correct as well as incorrect word problems that pupils generate when being asked to construct problems corresponding to particular multiplications and divisions (e.g., 4×1.5 , 0.75×8 , $4 \div 0.5$; De Corte & Verschaffel, 1996) have been successfully interpreted within this theory. However, several questions still exist about how to account for the origin and the further development of these intuitive models (e.g., whether the primitive models reflect natural features of human thinking or rather

how the relevant concepts were taught in school, or both), as well as their precise role in the solution of different types of word problems involving multiplication and division (Greer, 1992; Verschaffel & De Corte, 1997).

The third line of research, not systematically reviewed by Greer (1992), pertains to analyses of computational strategies, that is, the kinds of strategies that children apply to arrive at an answer of a multiplication or division word problem. Generally speaking, this research yielded similar outcomes as for addition and subtraction (see also Verschaffel & De Corte, 1997). First, the results indicate that before pupils have been instructed in multiplication or division, many of them are able to solve problems involving these operations by means of a widespread variety of informal strategies (Kouba, 1989; Levain, 1992; Mulligan & Mitchelmore, 1997; Nunes & Bryant, 1995; Treffers, 1987). An overview of these strategies, ordered in terms of increasing levels of abbreviation, internalization, and formalization, has already been given in the sections on *Single-Digit Computation* and on *Multidigit Arithmetic*. This development towards higher levels of abbreviation, internalization, and formalization proceeds differently for the different categories of problems. For instance, use of derived-fact or known-fact strategies seems to occur later for problems about Cartesian product than for equal group problems. Second, in the early stages of development, children's strategies tend to reflect the action or relationship described in the problem. For instance, they use different material strategies for solving partitive and quotitive division problems that fit very well with the meaning of the problem. Afterwards, children start to apply strategies that are more efficient from a computational point of view but that no longer necessarily match this semantic problem structure. Third, compared to addition and subtraction, the research clearly shows that many children do not progress to secure multiplicative thinking in the primary school years (Anghileri, 2001; Greer, 1992). Or, as Clark and Kamii (1996, p. 48) stated it, "multiplicative thinking appears early but develops slowly." Indeed, in their study wherein a very large group of children in Grades 1–5 were interviewed using a Piagetian task to study the development of their multiplicative thinking, 45% of the second graders demonstrated already evidence of multiplicative thinking, but only 49% of the fifth graders could be classified as solid multiplicative thinkers.

Strategic Aspects of Word Problem Solving

Most of the word problem solving research described above starts from the assumption that a competent problem solver tries to represent the problem

situation and to base the choice of an appropriate solution (whether it be a material or verbal counting action, or a formal-arithmetical operation) on that understanding. However, research has convincingly shown that the process of solving word problems for many pupils is along the lines of another model, whereby the intermediate representation is bypassed and the problem text immediately guides the choice of an operation—a choice that may be based on superficial features such as the presence of certain key words in the text (for instance, the word *less* in the problem text automatically results in a decision to perform a subtraction). The directly evoked operation is then applied to the numbers given in the problem text, and the result of the calculation is found and given as the answer. For a review of these coping strategies we refer to Verschaffel and De Corte (1997) and Verschaffel, Greer, and De Corte (2000).

Moreover, researchers have documented that when confronted with a word problem for which they have no ready-made solution, many pupils do not spontaneously apply valuable heuristic strategies, such as making a sketch or a drawing of the problem, or decomposing the problem into parts, or guessing-and-checking (De Bock, Verschaffel, & Janssens, 1998; De Corte & Somers, 1982; Garofalo & Lester, 1985; Van Essen, 1991). Although many researchers have reported differences between pupils who are good and weak in word problem solving with respect to the use of heuristics, the outcomes of teaching experiments focusing on the development of these heuristic skills have generally been rather disappointing (Schoenfeld, 1992).

These failures of efforts to improve students' problem solving by teaching them heuristics led researchers to study word problem solving from the perspective of metacognition (Garofalo & Lester, 1985; Mevarech, 1999; Schoenfeld, 1992). Several studies have revealed that in many pupils' solution attempts of word problems, moments of metacognitive awareness or self-regulatory activities such as analyzing the problem, monitoring the solution process, and evaluating the outcome are completely absent. A typical approach used by many pupils can be summarized as follows: The student glimpses the problem, quickly deciding what calculations to perform with the numbers given in the problem statement and then proceeds with these calculations without considering any alternatives even if no progress is made at all (Schoenfeld, 1992; Verschaffel & De Corte, 1997). Several researchers reported differences between children who are more and less able in solving arithmetic word problems with respect to self-monitoring and self-control (see, e.g., Garofalo & Lester, 1985;

Nelissen, 1987). Starting from these findings from ascertaining studies, several scholars have started to design and evaluate instructional environments aimed at the development of metacognitive skills for solving word problems (sometimes in combination with heuristic training). In these studies a detailed cognitive model of the heuristic and metacognitive abilities underlying competent problem solving in mathematics is specified and used to guide the development of teacher and student activities that will lead to the acquisition and use of these (meta)cognitive structures. In some studies this training was done on an individual (computer-aided) basis (Teong, 2000), whereas in others the metacognitive instruction was employed in cooperative settings where small groups of 4–6 students studied together. Typical examples of such intervention studies are Lester, Garofalo, and Kroll (1989), Cardelle-Elawar (1995), and Mevarech (1999). Most of the above studies suggest that providing metacognitive training has a significant impact on students' mathematical performance. As will be explained in the section *Emergent Models in Design Heuristics*, in more recent design experiments, the cognitive-rationalist perspective behind these intervention studies has been replaced, or at least enriched, by a more situative orientation, a socioconstructivist orientation, or both, as evidenced by a more critical attitude towards traditional word problems, a greater awareness of the complex relationship between mathematics and reality, and a greater emphasis on authentic mathematical tasks than in the studies belonging to the cognitive-rationalist tradition. The rationale for this shift of attention is given in the next *Conclusion and Discussion* section.

Pupils' Lack of Sense-Making When Doing Word Problems

As already announced before, in recent years the characteristics, use, and rationale of word problems have been critically analyzed from multiple perspectives, including linguistic, cultural, and sociological. In particular, many mathematics educators have argued that the stereotyped and artificial nature of word problems typically represented in mathematics textbooks and the discourse and activity around these problems in traditional mathematics lessons have detrimental effects on children's disposition towards realistic mathematical modeling and sense-making. Many observations have led to the conclusion that pupils answer word problems without taking into account realistic considerations about the situations described in the text, or even whether the question and the answer make sense.

The most dramatic example comes from French researchers who posed (lower) elementary school

children nonsensical questions such as: “There are 26 sheep and 10 goats on a ship. How old is the captain?” It was found that the majority of pupils were prepared to offer an answer to such questions (Baruk, 1985).

Inspired by this and several other striking examples of suspension of sense-making in school mathematics, Greer (1993) in Northern Ireland and Verschaffel, De Corte, and Lasure (1994) in Flanders carried out in parallel a pencil-and-paper study with 11–13-year-old students, using a set of problems including the following:

- A man wants to have a rope long enough to stretch between two poles 12 meters apart, but he only has pieces of rope 1.5 meters long. How many of these would he need to tie together to stretch between the poles?
- Bruce and Alice go to the same school. Bruce lives at a distance of 17 km from the school and Alice at 8 km. How far do Bruce and Alice live from each other?
- John’s best time to run 100 meters is 17 seconds. How long will it take him to run 1 kilometer?

They termed these items *problematic* (P-items, for short) in the sense that a proper answer requires (at least from the researchers’ point of view) the application of judgment based on real-world knowledge and assumptions rather than the routine application of arithmetical operations on the given numbers. A response was classified as a “nonrealistic reaction” if a pupil’s response sheet contained no apparent trace of a judgment based on real-world knowledge. A reaction was scored as a “realistic reaction” if either the answer given revealed that realistic considerations had been taken into account or a nonrealistic answer was accompanied by a comment indicating that the pupil was aware that the problem was not straightforward. Both in Greer’s (1993) and in Verschaffel et al.’s (1994) study, pupils demonstrated a very strong overall tendency to exclude real-world knowledge and realistic considerations when confronted with these P-items. For instance, in Verschaffel et al.’s study, only 17% of all reactions to the P-items were realistic. (The percentages for the three examples of P-items given above were between 0.5% and 3%!)

These studies were replicated in many other countries using the same P-items, the same testing conditions, and the same scoring criteria as in the original studies. Findings have been strikingly consistent with the first results, sometimes to the great surprise and disappointment of these other researchers, who had anticipated that the “disastrous” picture of the Irish

and Flemish pupils would not apply to their students (see Verschaffel et al., 2000, for a review of the earlier replications).

Several follow-up studies have tested the effectiveness of variations in the experimental setting on children’s tendency to solve problems in a realistic way. A first group of studies assessed the effectiveness of making students more alert to consider aspects of reality and to legitimize alternative forms of answering. For instance, Yoshida, Verschaffel, and De Corte (1997) gave half of the pupils an explicit warning at the top of their test sheet (a translation of Verschaffel et al.’s, 1994, test) that some of the problems in the test were problematic and invited them explicitly to write down and explain any lack of clarity or complexities. However, this manipulation did not result in a significant increase in the number of realistic reactions to the P-items in the test. In a different set of studies, one or more categories of P-items were presented in more authentic, performance-based settings. Contrary to the first kind of experimental manipulation, increasing the authenticity of the experimental setting (see, e.g., De Franco & Curcio, 1997) yielded great improvements in students’ inclination to include the real-world knowledge in their problem-solving endeavors (see Palm, 2002, and Verschaffel et al., 2000, for a full account of these studies). However, Van Dooren, De Bock, Janssens, and Verschaffel (in press) showed that the facilitating authenticity effect had only marginal impact on students’ future problem-solving behavior with respect to traditional word problems.

Besides inferring how students think about word problems from their responses to word problems in a paper-and-pencil test, it is, of course, possible to ask them directly. Interviews carried out by Caldwell (1995) and Hidalgo (1997) and Palm (2002) suggest that, although unfamiliarity with the contexts involved in the problems and lack of appropriate heuristic and metacognitive skills may provide contributory explanations, (mis)beliefs about school arithmetic word problems constitute the major reason why so many students solve the P-items in a nonrealistic way. For instance, a 10-year-old interviewed by Caldwell (1995, p. 39) commented as follows in response to the interviewer’s question as to why she had answered a P-item in a nonrealistic way: “I know all these things, but I would never think to include them in a maths problem. Maths isn’t about things like that. It’s about getting sums right and you don’t need to know outside things to get sums right.”

How can the results of some years of mathematics education be the willingness of children to collude in negating their knowledge of reality? This apparent “suspension of sense-making” can be construed as

sense-making of a different sort, namely a strategic decision to play the “word problem game.” As expressed by Schoenfeld (1991, p. 340)

such behavior is sense-making of the deepest kind. In the context of schooling, such behavior represents the construction of a set of behaviors that results in praise for good performance, minimal conflict, fitting in socially etc. What could be more sensible than that?

Students’ strategies and beliefs develop from their perceptions and interpretations of the didactical contract (Brousseau, 1997) or sociomathematical norms (Yackel & Cobb, 1996) that determine—mainly implicitly—how to behave in a mathematics class, how to think, how to communicate with the teacher, and so on. More specifically, this enculturation seems to be strongly influenced by two aspects of current instructional practice, namely the nature of the problems given and the ways in which these problems are conceived and treated by teachers (Verschaffel et al., 2000). Support for the second factor comes from a study by Verschaffel, De Corte, and Borghart (1997), wherein preservice elementary school teachers were asked, first, to solve a set of problems themselves and, second, to evaluate alternative answers from (imaginary) pupils to the same set of problems. The results indicated that these future teachers shared, though in a less extreme form, students’ tendency to suspend sense-making. As a result of these factors, students learn to play the Word Problem Game (De Corte & Verschaffel, 1985b), the rules of which include

- Any problem presented by the teacher or in a textbook is solvable and makes sense.
- There is a single, correct, and precise numerical answer that must be obtained by performing one or more arithmetical operations with numbers given in the text.
- Violations of knowledge about the everyday world may be ignored.

Conclusion and Discussion

In this section we have reviewed research on the development of children’s arithmetic word problem solving processes and skills, focusing on that done since the thorough and extensive reviews by Fuson (1992) and Greer (1992). More precisely, we have described what we see as the continuation of the cognitive-rationalist line of research, although we have, especially in the last section, already touched on a considerable part of the last decade’s research on word problems wherein this theoretical perspective has

been enriched by others, in particular representatives of the situative/pragmatist-sociohistoric perspective.

This research has shown that most children entering school can solve some word problems, not only with an additive but also with a multiplicative structure, by means of solutions based on modeling the actions and relations described in them. Moreover, researchers have documented how children’s proficiency gradually develops in two significant directions. One direction is from informal, external, laborious strategies towards more efficient (i.e., more abbreviated, internalized, formalized) calculation strategies. The other direction is from having a different solution method for each type of problem that is a direct reflection of the problem situation, to a single general method that can be used for classes of problems with a similar underlying mathematical structure (see also Kilpatrick et al., 2001).

Notwithstanding all these accomplishments, there is no shortage of issues and tasks for further research. Besides the specific questions for future research that we raised at the end of each part of this section, there are some broader issues. A first general limitation of past research is that it has largely focused on the initial and middle stages of the development of the concepts and skills needed for solving arithmetic word problems. Therefore, a general research priority should be to investigate as well the upper stages of the development of these concepts and skills, in which the additive and multiplicative concepts extend beyond the domain of positive integers (Greer, 1992, 1994; Vergnaud, 1988). Second, although separate analyses of the conceptual fields of additive and multiplicative structures will doubtless continue, there is a strong need for a comparative analysis between or a synthesis of these two hitherto rather separate bodies of research (as was already pleaded for by Greer, 1992). Such a synthesis should involve both theoretical analyses of the similarities, dissimilarities, and relationships between the conceptual fields of additive and multiplicative structures and empirical studies in which data are simultaneously collected for all four arithmetic operations. Some general comments about differences between addition/subtraction and multiplication/division are appropriate. For the first pair of operations, extension to non-whole-numbers seems to be assumed nonproblematic (there appears to be essentially no research on it)—instead the main problem is extension to negative numbers. By contrast, for the second pair, the extension to positive rational numbers provides major difficulties (e.g., Greer, 1994). Secondly, it may be that by the time students get to multiplication and division, they have already developed, through their work with addition and sub-

traction, some dissociation between calculations and situations, that is to say they have learned that, for the purposes of finding the answer, the calculation, once identified, can be done “off line.”

Finally, there is a strongly growing awareness among researchers that the level of our understanding of how children learn to solve arithmetic application problems drops enormously if we allow the domain to expand beyond the restricted range of traditional school word problems that can be “unproblematically” mapped onto an arithmetic operation with the given numbers and if we are prepared to take seriously into account the complexities of the instructional and cultural context wherein this learning process takes place.

The Structure of the Whole-Number System

Formal Structure of the Natural Numbers and Associated Arithmetic

At the time of “New Math” it was joked that children knew that 6×7 and 7×6 are equal but did not know what they are equal *to*. More seriously, there was a belief, stemming largely from mathematicians, that the structural edifice of mathematics as seen from an advanced standpoint could be communicated directly to children (Kilpatrick, 2003, p. 474). In retrospect, the effort was founded on the error of “replacing the learner’s insight with the adult mathematician’s” (Freudenthal, 1991, p. 112). Perhaps the time is right to reevaluate the role of structural understanding *from the perspective of the learner*.

The name *natural numbers* for the set of positive integers 1, 2, 3, . . . reflects their original conceptualization as Platonic objects and their ubiquitous usefulness for modeling aspects of the real world that afford counting. The four basic operations of arithmetic may be regarded as naturally emergent from reflection on counting acts and manipulations of sets of objects, physical or mental. As a structure, the set of whole numbers, which appends zero to the natural numbers, has been analyzed in great depth and formalized in several variants. Here we deal with this formalization briefly and informally—in standard mathematical texts, the reader can find as much detail and rigor as required.

The ordinal aspects of natural numbers, intuitively grounded in acts of counting, are formalized in the Peano axioms (anticipated by Dedekind) from which the arithmetic operations can be defined, and their properties derived. The cardinal aspects have been formalized in set-theoretic terms, notably by Cantor, Frege, Russell, and Whitehead (Freudenthal, 1983, p. 76). One formulation is to define a whole number, n , as the class of all sets with cardinal n . This formula-

tion corresponds to the idea that sets of 3 sheep, 3 colors, 3 rectangles, and so on, all have “threeness” in common. Alternatively, the whole number, n , can be defined as the set of all the whole numbers (including 0) less than n . As Freudenthal wryly commented (1983, p. 80), “with each profundity we get further away from the phenomenology of number as it is naively expressed.” Nor has history come to an end in this respect, as new formulations continue to appear (e.g., Conway & Guy, 1996).

The set of whole numbers, with the operations of addition and multiplication, forms a structure with properties such as commutativity and associativity of addition and multiplication and distributivity of multiplication over addition (Kilpatrick et al., 2001, Chapter 3). In addition, there are numerous properties relating to order.

It is characteristic of the notion of *number* that it has historically and logically been extended. In particular, the set of whole numbers is extended to the set of integers (positive and negative) and to the set of rationals (all those numbers of the form a/b where $b \neq 0$). An important mathematical motivation for these extensions is the urge for closure. The whole numbers are not closed under either subtraction or division (i.e., it is not always possible to subtract one whole number from another, or divide one whole number by another, and get a whole number as the result), leaving gaps that are closed by the extensions to the integers and to rational numbers, respectively. Although the focus of this chapter is on whole-number arithmetic, we draw attention to the need to anticipate the difficulties students commonly experience in reconceptualizing arithmetic within larger and larger domains as the concept of *number* itself is extended (Lakoff & Nunez, 2000).

Straddling the development of natural numbers as, on the one hand, a practical tool for dealing with calculations on sometimes very large numbers and, on the other hand, the infinite set of natural numbers as a mathematical object is the invention of finite (hence, necessarily generative, like language) representations. A reasonable case can be argued that the decimal system that has emerged as the evolutionarily fittest is optimal insofar as it concatenates polynomial expressions with a constant base; however, the choice of 10 as that base (rather than, say, 8 or 12) appears to be a biological accident, and the specific choice of place-value representation (with the key step that zero could be seen as a number in its own right) is a cultural decision. A great deal of what we call early arithmetic is more precisely described as arithmetic using decimal representation, as one solution to the generative requirement of a finite representation.

Research on Understanding of Structural Properties

There has been considerable research on children's awareness of the structural properties of whole numbers and operations upon them. In accordance with our principled division between this section and the next on *Classroom-Based Research*, here we concentrate on ascertaining studies, which are generally associated with the first level, whereby awareness is inferred from behavior that, in some sense, indicates implicit understanding of the principle—what Vergnaud (1996) termed a *theorem-in-action*. It is unusual for children taught traditionally to spontaneously formulate explicit generalizations and provide general arguments in their support; hence these manifestations are rare in ascertaining studies. Work relating to these higher levels of understanding and the explicitation of structural properties, which is mainly situated within teaching experiments, is dealt with in the second part of the *Classroom-Based Research* section.

At the simplest level, exploitation of properties may be applied in the mastery of the basic number facts. For example, the task of learning or constructing the 100 multiplication facts may be drastically simplified by flexible and strategic use of a wide variety of properties and regularities (Kilpatrick et al., 2001; see earlier section on *Single-Digit Computation*). In the case of multidigit computations, algorithms mostly consist of systematic methods to simplify the computation to a coordination of basic number facts, and these methods depend on several properties, notably the associative and distributive laws in the case of multiplication. As pointed out earlier, the compact notation of standard algorithms tends to mask the underlying principles that make them work (although these principles may be unmasked, as shown by Wu, 1999). On the other hand, other algorithms and taught or invented mental strategies are derived directly from the underlying properties of decimally represented whole numbers and the relations among them (Resnick, 1992).

So-called “smart arithmetic” tasks may be used to probe students' awareness of such networks of relationships. For example, starting from the given equation

$$86 + 57 = 143$$

students may be asked to use it in finding: $86 + 56$, $57 + 86$, $860 + 570$, . . . and so on (Selter, 1996; Van den Heuvel-Panhuizen, 1996). Children's “flexible arithmetic” (Gravemeijer, 1994, p. 37) is based on (often implicit) use of the formal properties of arithmetic, plus others derived therefrom, such as

$$\begin{aligned} a + (a + 1) &= (a + a) + 1 \\ (a + p) - (b + p) &= a - b \end{aligned}$$

Many studies have been carried out that have probed students' understanding of specific properties. As an example of research in this spirit, we consider one of the most basic structural principles, the commutativity of addition and multiplication.

Within a more general theory of the emergence of arithmetical cognition from its rudimentary origins, Resnick (1992) presented a theory of how children's understanding of commutativity of addition develops (see Table 13.1).

Baroody et al. (2003, p. 130) quoted a useful clarification of Resnick's position as follows to the effect that

additive commutativity is reconstructed at each successive level, using both what the child knows from the previous level and new knowledge (e.g., the experience of counting and computing). . . . For a child to understand [for example] that 3 apples + 6 apples = 6 apples + 3 apples [requires] more than just applying an existing [part-whole] schema. . . . It would involve also noticing . . . that in both cases the sum is 9 apples.

They presented evidence that understanding of commutativity is correlated with computational experience and expertise. However, there is a general point that needs to be raised, namely what evidence might

Table 13.1 Developmental Levels of Commutativity (Resnick, 1992)

Level of additive commutativity	Example
Protoquantitative (prenumerical)	apples + oranges = oranges + apples (Part1 + Part2 = Part2 + Part1)
Quantitative (specific numbers in context)	3 apples + 5 apples = 5 apples + 3 apples
Numerical (specific numbers in the abstract)	$3 + 5 = 5 + 3$
Operator (general arithmetic principle)	$a + b = b + a$

we take to show that a child understands commutativity of addition, and in what sense?

- If the child computes the answer to $3 + 5$ by saying “five . . . six, seven, eight” (perhaps accompanied by finger actions), or if the child rapidly and confidently answers the question “do the additions $6 + 4$ and $4 + 6$ have the same answer?”, we infer that the child, in some sense, understands commutativity of addition.
- If the child gives an explicit statement implying that commutativity holds for any two whole numbers, then we infer that the property is understood at some level of generality, at least as the result of empirical observation (“every time I’ve calculated an addition both ways, I’ve got the same result”).
- At the highest level an explicit statement is accompanied by a (more or less) convincing argument, often based on a physical or visual representation (Kilpatrick et al., 2001, p. 77). A key distinction here is that made by Piaget (e.g., Piaget, 2001) between an empirical abstraction, whereby a student notices and generalizes a pattern, and a sense of the logical necessity underlying that pattern, which is the product of reflective abstraction (cf. the comment by Sawyer, 1955, that a mathematician always wants to know the reason for a pattern).

Thus, as with mathematical concepts in general, we suggest it is problematic to speak of understanding of commutativity as something a student either does or does not have.

A key idea Baroody et al. (2003) emphasized is that addition may be treated as a unary or binary operation. Within, say, the whole numbers, addition as a binary operation is a rule that assigns to any two whole numbers another whole number. On the other hand $5 + 3$ can be interpreted as the application of the unary operation $+ 3$ to 5. As discussed below (see previous section on arithmetic word problems), this distinction is reflected in the different classes of situations modeled by addition and subtraction. Moreover, although the binary conception is more conducive to an emergent grasp of commutativity, the unary conception facilitates development of understanding of subtraction as the inverse of addition (Siegler, 2003; though the latter may also be facilitated by experience of working with a part-part-whole binary schema, see also Bryant, Christie, & Rendu, 1999).

Synthesizing a wide range of experimental evidence and theoretical analysis, Baroody et al. (2003,

p. 155) proposed a revised model for development of additive commutativity (Table 13.2).

Table 13.2 Development of Additive Commutativity Understanding (Baroody et al., 2003)

Level 0:	Unary conception + noncommutativity
Level 1:	Unary conception + noncommutativity and binary conception + protocommutativity
Level 2:	Unary conception + protocommutativity and binary conception + part-whole (“true”) commutativity
Level 3:	Unary conception + change-add-to commutativity (“pseudocommutativity”) and binary conception + part-whole (“true”) commutativity

Despite the amount of work surveyed, Baroody et al. (2003) finished by acknowledging that more research is needed on this topic. The fact that the interpretation remains controversial and unclear, despite a considerable effort in experimentation, in itself suggests that there is an inherent messiness arising from individual differences, teaching, differential experience, artefacts of the experimental contract . . . and so on. In a later section the discussion will be resumed in the context of classrooms in which children are encouraged, through strategically chosen tasks and discussions, to enunciate and even explain commutativity as a general principle.

A highly important point, with general relevance, is the suggestion by Baroody et al. (2003) that children’s early understanding of arithmetic should be viewed “in terms of multiple, loosely connected weak schemes” (p. 156). As such, it seems inevitable that a neat linear description of developmental stages will always be complicated by the exigencies of individual differences in cognition and in experience. This reality underlies one major advantage of teaching experiments, in that they afford access to at least partial knowledge of the nature of instruction the child has experienced and how that experience has affected the child’s learning. The model presented in Table 13.2 may therefore be more usefully seen as a framework for interpreting children’s performance.

Understanding of multiplication from a purely computational point of view has also been extensively studied from the aspect of looking for evidence of use of structure, particularly commutativity. For example Baroody and Tiilikainen (2003, p. 105) proposed a possible schema underlying mental multiplication that invokes several rules, such as if 1 and a number are multiplied, the answer is that number, that if 2 and a number are multiplied, the answer is that number

added to itself (see also the earlier section *Children's Strategies for Single-Digit Computation*). An overriding principle is that the order of the numbers multiplied is irrelevant and accordingly may be switched strategically to facilitate the calculation. Interestingly, as Baroody and Tiilikainen pointed out, the research of Butterworth et al. (2003) suggests that Italian children convert small-factor-first combinations to large-factor-first, whereas that of LeFevre et al. (2003) suggests the converse for Chinese experts.

Modeling Different Classes of Situations

The research exemplified thus far is based on counting acts, operations on numbers represented by sets of taken-as-countable objects, or purely symbolic computational tasks. Further complications arise if the task involves the use of arithmetic to model a situation.

For example, consider the awareness of subtraction as the inverse of addition for each of the three main semantic structures: combine, change, and compare. Splitting a superset into subsets as the inverse of combining subsets to form a superset is grounded in actions, as Piaget stressed, and related to his concept of reversibility. So is undoing a change—but in a different way. Conceptualizing “Peter is 4 inches taller than Anne” and “Anne is 4 inches shorter than Peter” is different again, and particularly difficult for young children (Stern, 1993). Inverseness also plays a role in cases in which the operation that yields the answer is incompatible with the “story-line,” as in a change-unknown problem such as “John had 5 apples and found some more. Now he has 9 apples. How many apples did he find?” The computation $9 - 5$ represents an inversion, in a sense, of the direct modeling of the story as $5 + ? = 9$ (note the algebraic nature of this equation).

Likewise, for a simple additive word problem, the arithmetic structure of the operation involved may depend on which class of problems—combine, change, or compare—is involved (De Corte & Verschaffel, 1987). Thus, as mentioned already in the SCADS section, the computationally more efficient counting-on-from-larger strategy, which for a purely numerical task exploits the commutativity of addition, is much more prevalent for a combine problem than for a change problem. The explanation is that the addends in the former case play “symmetrical” functions, whereas in the latter case, the functions are asymmetrical and imply order (a distinction closely related to Weaver’s, 1982, distinction between unary and binary conceptions of addition).

The concept of symmetry also applies in the case of situations modeled by multiplication and division (Greer, 1992). In many classes, it is possible

to distinguish a multiplier that acts upon the multiplicand, implying asymmetric functions, as in the simple case of equal groups, where the multiplicand is the number in each group and the multiplier is the number of groups. As a consequence, there are two types of division, namely partition (division by the multiplier) and quotient (division by the multiplicand) for such situations (Greer, 1992, p. 286; see also elsewhere in this chapter). For this reason, there are interactions between the semantics of modeled situations and computational solutions. For example, Ambrose et al. (2003, pp. 318–319) reported that for an equal groups problem involving 153 bags with 37 balloons in each “[the students] were not inclined to use the commutative property to switch the numbers around, even though it might have made calculation easier” because their thinking was linked to the context. Many observations confirm that, in this sense, an equal-groups-of meaning of multiplication is psychologically non-commutative.

Number Theory

Number theory, in general, is the study of properties of number systems. Originally, its scope was patterns and relations among the natural numbers, to which we restrict ourselves here. Typical topics within this domain are “figurate numbers, whole number patterns and sequences, multiples, factors, divisors, primes, composites, prime decomposition, relatively prime numbers, divisibility, and divisibility rules” (Campbell & Zazkis, 2002b, p. 3). Campbell and Zazkis (2002a) argued for a greater recognition of number theory within the curriculum at all levels. Justifications for this recommendation include the following points:

- The topic is of historical interest.
- It offers a context for fostering structural awareness and algebraic thinking.
- It affords mathematical sense-making, reasoning, argumentation, and proof about familiar mathematical objects.
- It reinforces the idea that mathematics is, at heart, concerned with patterns and relations.

Campbell and Zazkis included examples of studies with preservice teachers, but none with school students. We suggest number theory as a possible area for future research with children, for the reasons cited above. Zazkis (2001) made an interesting suggestion that using problems with specific large numbers could force generalizable thinking that could serve as a bridge to algebra.

CLASSROOM-BASED RESEARCH

In this section, we reflect a major development of the research since the early 1990s, namely a shift from mainly ascertaining studies of individual children to extensive classroom-based intervention research. This methodological shift is, first of all, related to the theoretical shift from the (strictly) cognitive-rationalist analysis of individual children's mathematical thinking and learning, and the task and subject variables influencing these thinking and learning processes, to other (socioconstructivist and sociocultural) frameworks wherein the complex social and cultural context is considered as truly constitutive for what and how children think and learn mathematically. Second, this methodological shift towards classroom-based intervention research is also a consequence of researchers' growing awareness that a general priority for them should be to complement ascertaining studies, which only document and diagnose existing ills, with research efforts aimed towards designing, implementing, and assessing innovative mathematics curricula, programs, teaching/learning activities, and so on, as recommended both by Fuson (1992, p. 269) and by Greer (1992, p. 293). Or, as Selter (2002, p. 221) put it: "mathematics education has a fundamental and clear-cut responsibility to develop practical suggestions for teaching within the capacity of 'average teachers' in a systematic process involving researchers, teachers, designers, textbook authors and children."

This plea for such "design experiments" can be found both in the research field of instructional psychology and in that of mathematics education. As far as the former is concerned, we refer to Greeno et al. (1996) and Schauble and Glaser (1996), who describe design experiments as a new kind of interdisciplinary teamwork that is evolving among practitioners, researchers, teacher educators, and community partners around the design, implementation, and analysis of changes in practice. Results provide case studies that can serve as instructive models about conditions that need to be satisfied for reforms of the same kind to be successful, and about conditions that impede success. Researchers believe that the design of such learning environments and the development of theory must proceed in a mutually supportive fashion. Among the mathematics educators who have pleaded strongly for design experiments as the main type of research in the field are Freudenthal (1991), Rouché (1992), and Wittmann (1995). In his seminal article entitled *Mathematics Education as a Design Science*, Wittmann (1995, p. 356) described the main task of researchers in mathematics education as follows:

Scientific knowledge about the teaching of mathematics, however, cannot be gained by simply combining results from other fields; rather it presupposes a specific didactic approach that integrates different aspects into a coherent and comprehensive picture of mathematics teaching and learning and then transposing it into practical use in a constructive way. . . . In my view, the specific tasks of mathematics education can only be actualized if research and development have specific linkages with practice at their core and if the improvement of practice is merged with the progress of the field as a whole.

According to Wittmann (1995, p. 363), the core of research in mathematics education is "the construction of 'artificial objects,' namely teaching units, sets of coherent teaching units and curricula as well as the investigation of their possible effects in different educational 'ecologies.'" And he continued: "Indeed, the quality of these constructions depends on the theory-based constructive fantasy, the 'ingenium', of the designers, and on systematic evaluation, both typical activities for design sciences." According to Wittmann (1995) both the so-called "developmental research" projects of the Dutch Freudenthal Institute (see, e.g., Gravemeijer, 1994; Treffers, 1987; see also De Corte et al., 1996) and his own team's work on the MATHE 2000 project (see, e.g., Becker & Selter, 1996) are typical examples of this approach.

Design research basically encompasses three phases: developing a preliminary design, conducting a teaching experiment, and carrying out a retrospective analysis. The first phase starts with the formulation of a "conjectured local instruction theory" (Gravemeijer, 2004, p. 109) or a "hypothetical learning trajectory" (Simon, 1995, p. 133) that involves conjectures about (a) the learning goals, (b) the instructional tasks and activities and possible tools that will be used, and (c) the thinking and learning processes in which the students might engage in this instructional environment. Most typically, these conjectures are based on the historical development of mathematics or knowledge about children's informal strategies. These conjectured goals, tasks and activities, and envisaged thinking and learning processes are often not worked out in detail during this first phase. In the second phase, the teaching experiment, those conjectures are put to the test. In the course of the teaching experiment, the hypothetical teaching/learning trajectory is gradually and cyclicly adapted, corrected, refined, and so on on the basis of the input of the students and assessments of their actual understandings. However, we should acknowledge that the nascency and complex nature of design experiments has led to a variety of interpretations and applications, both about the methodologi-

cal requirements of this approach to research and about the constituents of the learning trajectories (for a thorough discussion, see Baroody et al., 2004; Clements & Sarama, 2004b).

Rather than structuring our review of this classroom-based research around the different curricular subdomains of elementary school arithmetic distinguished in the previous section, we judge it more productive to highlight and illustrate some typical features for the design studies realized in these different curricular subdomains during the last decade. The selection of examples represents both a variety of subject-matter content and age levels as well as a diversity in theoretical perspective and background. However, these examples also share some “family resemblances”: (a) the active and constructive view on mathematics learning; (b) the recognition of the crucial role of students’ prior knowledge for future learning; (c) the orientation towards conceptual understanding and problem solving; (d) a reluctance to “impose” a single representational model or solution strategy on all learners and the pursuit of adaptive expertise (Baroody et al., 2004); (e) the importance of social interaction and collaboration in doing and learning mathematics, and, more generally, of conceiving learning mathematics as a community activity (Romberg et al., 2005); (f) the efforts to embed mathematics learning in authentic and meaningful contexts; (g) the seeking of an appropriate balance between “horizontal and vertical mathematization” (Treffers, 1987); (h) the focus on basic principles (or, as Baroody et al., 2004, or Romberg et al., 2005, would say: on “big ideas” that underlie numerous concepts and procedures across topics) while leaving most of the details to students (Paulos, 1991); (i) the belief that the above instructional design principles do not hold only for teaching mathematics to mathematically able children, but also to low achievers; and (j) the attention to the teacher’s role in the learning process. This implies that all these examples represent quite a radical departure from a “traditional skills approach” to mathematics instruction based on the view that mathematics learning is a highly individual activity consisting mainly in absorbing and memorizing a fixed body of decontextualized and fragmented knowledge and procedural skills transmitted by the teacher (Baroody, 2003). But these examples can also be differentiated from a so-called “laissez-faire problem-solving approach,” which was especially influential in the initial period of the current reform movement in countries like the United States (Baroody, 2003; Romberg et al., 2005) and the Netherlands (Van den Heuvel-Panhuizen, 2001). They generally fit with what Baroody (2003) called the investigative approach, that is, a blend of a conceptual and

a problem-solving approach, aiming at the mastery of basic skills, conceptual learning, and mathematical thinking and characterized by both meaningful and inquiry-based instruction and by purposeful learning and practice. From a methodological perspective, the selected examples can be considered as prototypes of the new generation of ecologically valid design experiments (as defined above) rather than the traditional kind of teaching experiments, which focused more on internal rather than external or ecological validity.

Problem-Solving as a Basis for Learning Multidigit Calculation

According to many researchers and policy makers, students can learn about number through problem solving—including both standard aspects of computation, as well as more puzzle-like problems that promote understanding of the structure of the whole-number system, elicit algebraic modes of reasoning prior to the use of formal algebraic notation, and so on. Taking the description of problem solving in NCTM’s *Principles and Standards for School Mathematics* (2000, p. 116) as “a major means for developing mathematical knowledge” as a starting point, a number of related research projects in the United States and South Africa have addressed children’s well-known deficiencies with multidigit numbers and arithmetic by trying new approaches that are hypothesized to support children’s construction of accurate and robust conceptual structures for multidigit numbers and facilitate the use of these conceptual structures in multidigit calculation. All these programs take a so-called problem-solving approach to teaching multidigit number concepts and operations (see Carpenter et al., 1997; Fuson, Wearne, et al., 1997; Hiebert et al., 1996). In a joint manifest in the *Educational Researcher*, Hiebert et al. (1996, p. 12) explained this problem-solving principle as follows:

The principle is this: students should be allowed to make the subject *problematic*. We argue that this single principle captures what is essential for instructional practice. . . . Allowing the subject to be problematic means allowing students to wonder why things are, to inquire, to search for solutions, and to resolve incongruities. It means that both curriculum and instruction should begin with problems, dilemmas, and questions for students. . . . We use “problematic” in the sense that students should be allowed and encouraged to problematize what they study, to define problems that elicit curiosities and sense-making skills.

These authors justified the practice of problematizing the subject by claiming “that it is this activity that

most likely leads to the construction of understanding" (p. 15). Examples of design experiments wherein this basic principle is applied are contained in a large number of case-study descriptions of classrooms by these researchers in which the open and constructive examination of methods of inquiry and instruction is emphasized. For instance, Fuson, Smith, et al. (1997) carried out a year-long classroom teaching experiment in two predominantly Latino low socioeconomic status (SES) urban classrooms (one English-speaking and one Spanish-speaking) that sought to support first graders' thinking of two-digit quantities as tens and ones, starting from the developmental sequence of conceptual structures for two-digit numbers (the so-called UDSSI triad model) described earlier (Fuson, Smith, et al., 1997). By the end of the year, most children could accurately add and subtract two-digit numbers that require trading (regrouping) by using drawings or Dienes blocks and gave answers by using tens and ones on various tasks. Their performance was substantially above that reported in other studies for U.S. first graders of higher SES and for older U.S. children. According to the authors, children's responses looked more like those of East Asian children than of U.S. children in other (ascertaining) studies.

According to Hiebert et al. (1996), the available evidence from design experiments such as the one by Fuson, Smith, et al. (1997) shows that students who are presented with these kinds of tasks and engage in the kinds of discussions described and illustrated above do develop, first of all, deeper structural understandings of the number system than their peers who move through a more traditional skills-based curriculum. Second, two kinds of strategies are produced by working through problematic situations: (a) the particular arithmetic strategies that can be used for solving particular problems and (b) the general metastrategic approaches or ways of thought that are needed to construct these procedures. As far as the specific strategic aspect is concerned, the research evidence from these authors' studies suggests that students who are allowed to problematize arithmetic procedures perform as well on routine tasks as their more traditionally taught peers. With respect to the metastrategic aspect, there is evidence that pupils who have been encouraged to treat situations as problematic and develop their own strategies can adapt them later, or invent new ones, to solve new problems. A third important observation is that when students experience curricula that treat mathematics as problematic, such as those described above, the traditional separation between conceptual knowledge and procedural skill is weakened. Finally, the authors stress that what students take away from these problem-based environments cannot be

adequately described in (meta)cognitive terms only. There is evidence, according to Hiebert et al. (1996), that students who are allowed and stimulated to treat mathematics as problematic develop also appropriate beliefs and positive attitudes towards mathematics that, in turn, influence their orientation toward future activities.

Whereas the experimental program of Fuson, Smith, et al. (1997) aimed at the development of the integration of declarative and procedural knowledge for doing multidigit addition and subtraction from a basically ten-base and decomposition approach, several mathematics educators in the Netherlands and Germany have recently designed and evaluated experimental training programs for mental addition and subtraction up to 100 or up to 1000 wherein the empty number line, and, consequently, the sequential concept and the jump method, play a crucial role (Beishuizen, 1999; Gravemeijer, 2004; Menne, 2001; Selter, 1998). By way of example, we review here Selter's (1998) design experiment.

As in the above-mentioned design experiments from the United States, Selter (1998) started from the idea that powerful instruction should involve a gradual progression from highly contextualized, externalized, less elegant, and inefficient methods toward more general, internalized, more elegant, and efficient methods (vertical mathematization in Treffers's, 1987, terminology), typically (but not necessarily!) starting from the exploration of realistic problems. Pupils' different solution strategies are discussed, explained, and elaborated via pupil-pupil and pupil-teacher interactions. The teacher's role is to encourage the pupils to develop their own ways of working and to reflect on them, much in the same way as in the design experiments mentioned above. During these moments of reflection, the role of the teacher is to act as a mediator between the "inventions" of the pupils and the "conventions" of mathematics (Lampert, 1990).

Selter's (1998) teaching experiment was conducted with 27 German 8- and 9-year-old third graders. The course consisted of 15 lessons on mental addition and subtraction (as defined in the earlier section on arithmetic word problems) in the domain between 100 and 1000. The aim of the study was not to prove the superiority of this instructional approach to other, more traditional ways of teaching mental arithmetic. Rather, it was more modest, namely to describe in a qualitative way how he tried to realize in one class the idea of vertical mathematization for this part of the curriculum and to document the kind of student learning resulting from it.

In the first stage, the children solved word problems (mostly around the context of a big new cine-

ma center recently built in their city) with their own informal methods. Especially during this first stage, children were stimulated to use the empty number line as an informal and approximate way to support both their thinking and their communication. As il-

lustrated in Figure 13.3, Selter observed a wide variety of solutions to problems such as “In cinema 5, 216 persons can be seated. 148 persons are already sitting there. How many empty seats are there?” From the first stage on, all children had a diary wherein they

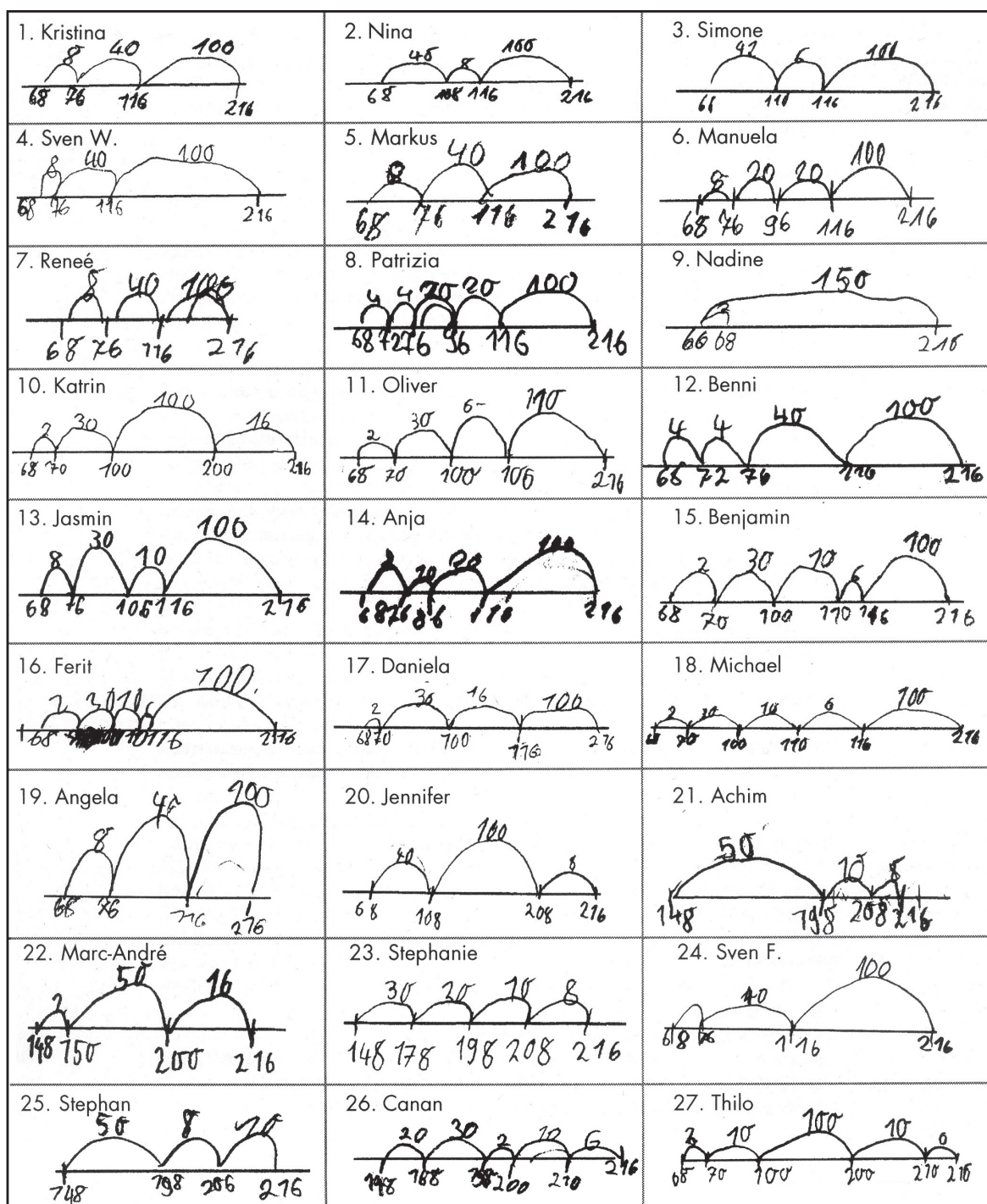


Figure 13.3. All solutions to a subtraction problem (“In cinema 5, 216 persons can be seated. 148 persons are already sitting there. How many empty seats are there?”) (Selter, 1998, p. 16).

had not only to document their strategy by means of the empty number line (on the right half of the page), but also to write a brief verbal description of their solution and, if possible, to put down their method in symbolic expression (on the left half of the page). Many children's descriptions point to a high degree of articulation and reflection.

In the second stage, vertical mathematization was achieved by means of "math conferences" among small groups of pupils as well as whole-class discussions, wherein they explicitly discussed the different methods in terms of elegance and efficiency. An interesting instructional technique that was used in that stage, called "Do it like . . .," involved the explicit instruction to try to solve a given addition or subtraction problem according to a particular strategy that had appeared to be effective during previous lessons. For instance, in a worksheet called "Doing it like Ferit," pupils were asked to apply a compensation strategy (see *Strategies for Multidigit Addition and Subtraction* section) to solve the problem $128 + 96$. Although more elegant and more efficient methods were favored through these activities, children were never forced to use the most efficient methods in their own work (not even at the end of the course). In other words, strategies were never imposed upon children.

During the third part of the course, children focused on the advantages and disadvantages of the different strategies and were stimulated to develop new ones. Selter (1998) documented the generation of several new strategies in this stage, such as the use of a sort of "adding down to" strategy to solve subtraction of close numbers like $312 - 278$.

Finally, children were asked to reflect on the whole course by explaining to other people (i.e., their parents and children from another class) what they had done and learnt in the course.

Looking back, Selter (1998) pointed to a number of important didactical problems that were revealed through this design experiment, such as the handling of the decomposition or split strategies, which are difficult, if not impossible, to visualize by means of the empty number line. Moreover, as the study was basically qualitative and exploratory in nature, he acknowledged the necessity of further, more systematic evaluation.

Laying Foundations for Algebra in Early Arithmetic

One important aspect of algebra, though by no means the only one (Kaput, 1999), is the generalization of arithmetic, and researchers commonly refer to problems student encounter during the transition

from arithmetic to algebra. More recently, however, the emphasis has shifted to recognizing the aspects of algebraic thinking that are inherent within arithmetic and to teaching arithmetic accordingly. Note that by *algebraic thinking* is meant such basic processes as the recognition and exploitation of structure and, above all, generalization. Such processes are not dependent on algebra as the intensive study of the last three letters of the alphabet or, as expressed by Engestrom (1987), making what should be a tool into the object of study itself. Pre-algebra can also entail a focus on analyzing process instead of generating answers.

The refocus on algebra as latent within arithmetic affords complementary advantages, in that it lays foundations for symbolic algebra and at the same time consolidates arithmetical knowledge by exploiting its structural coherence (refer back to the *Structure of the Whole-Number System* section). Furthermore, it exposes children at an early age to one important view of mathematics as "the classification and study of all possible patterns" (Sawyer, 1955, p. 12) and thus contributes to the foundations of a positive mathematical disposition.

Thinking Ahead, Laying Groundwork

By strategic choices in the types of arithmetic activities, the foundations for later symbolic algebra may be established early. Adumbrating the use of letters to denote specific unknowns is simple and familiar—for example, through a question such as "times what equals 20?" The more advanced use of a letter to denote a variable can also be foreshadowed by posing a problem such as "describe how the store owner would figure the amount of refund for any number of returned cans," given that the refund is \$5 per can (Swafford & Langrall, 2000, p. 93). Thus, paralleling the historical development from rhetoric to synco-pated algebra, the groundwork is done for the major conceptual advance whereby generalized quantitative relationships are symbolically expressed—which really begins to pay off when those expressions are conceived of as mathematical entities that can be operated upon. Careful choice of tasks that strain the limits of arithmetical treatment and demonstrate the power and elegance of algebra can provide motivation and rationale for this conceptual advance.

A specific difference between algebra and arithmetic is in the use of the equals sign. Taught without regard to their future, many children come to have a very limited notion of what this symbol means in arithmetic, something like "makes"—which indeed is its major role in arithmetic. This limitation is a natural result of their experience being only of seeing it come between two arithmetic expressions, with that on the right being a single number. As a result, they have no

basis for dealing with arithmetic equations in noncanonical form, with or without unknowns, such as

$$8 + 4 = _ + 5$$

(Carpenter, Franke, & Levi, 2003; Falkner, Levi, & Carpenter, 1999; Seo & Ginsburg, 2003; Van Amerom, 2002). To interpret such equations requires a relational view of the equals sign, namely that the expressions on either side are equal.

Seo and Ginsburg (2003) carried out a study (using several research methods including analysis of materials, participant observations, and clinical interviews) of a second-grade classroom with 29 students in which the teacher was attempting to foster such a relational view. In their study, they specifically addressed the following questions:

Issue 1: How do textbooks present the equals sign?

Issue 2: How did the teacher present the equals sign?

Issue 3: How did context affect the children's interpretation of the equals sign?

In the textbooks used, they found little effort to promote the relational view. The teacher did include a variety of activities promoting it, such as linking the equals sign with the signs for inequality, writing a variety of expressions with the same numerical value, using Cuisenaire rods, currency equivalence (e.g., 1 nickel = 5¢, 1 dime = 10¢), and everyday examples linking "equals" with "same number." Children's interpretation varied widely depending on the context. One student carefully explained that the meaning of the equals sign depended on the mathematical sentence, and most of the others agreed. Also the children made distinctions of meaning that mathematicians would not! They considered that the meaning is different for an addition and a subtraction, also for, for example, $2 + 3 = 5$ (means "answer") and $3 = 3$ (means "the same").

In passing, we note that a lot of the recent work on the equals sign implies that *the* mathematical meaning of the equals sign is equivalence, which is wrong. There are many other meanings encountered later, such as expressing an identity ($a + b = b + a$) or a definition ($f(x) = 3x - 4$) and subtleties such as $17 \div 3 = 5$ remainder 2 and $22 \div 4 = 5$ remainder 2, but $17 \div 3 \neq 22 \div 4$.

Falkner et al. (1999) reported that of 145 sixth-grade students posed the problem

$$8 + 4 = _ + 5$$

every single student thought that either 12 or 17 should go in the box. Such findings serve particularly well to

illustrate the dangers of short-termism. If the aim is to get the children to a high level of proficiency in arithmetic for the next test, then the "makes" meaning will serve its purpose. Unfortunately, such a temporary gain is more than offset by repercussions in the future when students need to work with algebraic uses of the equals sign, notably in equations.

In any case, the relational meaning of the equals sign is needed to express equivalence in the course of exploring the structure of numbers or doing mental arithmetic (in the sense described earlier in the arithmetic word problems section). Thus, according to Falkner et al. (1999, p. 233)

A child who has had many opportunities to express and reflect on such number sentences as $17 - 9 = 17 - 10 + 1$ might be able to use the same mathematical principle to solve more difficult problems, such as $45 - 18$, by expressing $45 - 18 = 45 - 20 + 2$. This example shows the advantages of integrating the teaching of arithmetic with the teaching of algebra. By doing so, teachers can help children increase their understanding of arithmetic at the same time that they learn algebraic concepts.

Structural Insight, Problem-Solving, Argumentation, and Proof

In the *Structure of the Whole-Number System* section, we reviewed evidence that children spontaneously—and mostly implicitly—exploit the structure of arithmetic, in the process demonstrating considerable adaptive expertise. One focus of recent studies is how enquiry-based classrooms can support making the structural properties explicit, through processes of conjecture and refutation, argumentation and (informal) proof. For example, Carpenter, Franke, and Levi (2003, chapter 4) gave several examples of students constructing and testing generalizations about calculations involving 0 and 1, about inverse operations, and about computations with even and odd numbers. There are several accounts of emergent convincing arguments about the commutativity of addition and multiplication. Carpenter and Franke (2001) reported a case study of a sixth-grade class that illustrates the progression of forms of argument used by students to explain commutativity of multiplication, culminating in general argument based on rotating a rectangular array of dots. A similar explanation from a third-grade student was provided by Schifter and Fosnot (2003).

Another set of studies addressed how children tackle puzzle-like situations that depend on insight into structural relationships. An example of an extremely rich problem is "arithmagons" (or "arithmogons"; Becker, Sawada, & Shimizu, 1999; Becker & Selter, 1996; Driscoll, 1999; Mason, 1996; Mason,

Burton, & Stacey, 1982; Wittmann, 2001). The core problem may be stated generally as follows: “A secret number is assigned to each vertex of a triangle. On each side of the triangle is written the sum of the secret numbers at its ends. Given the values on the three sides, find a simple rule for revealing the secret numbers.” For example, the numbers 11, 18, 27 are produced by the secret numbers 1, 10, 17 (see Figure 13.4). A specific problem such as this may be solved by many methods—from undirected trial and error, to systematic trial and improvement, to arguments such as one that begins “the two bottom secret numbers must differ by 7, since added to the top secret number they produce 11 and 18, respectively,” to formal algebra, assigning a , b , c as the secret numbers to produce a system of three equations.

$$a + b = 11 \quad b + c = 27 \quad c + a = 18$$

An insightful step (based on the symmetry of the system of equations) is to add the three equations to find that $(a + b) + (b + c) + (c + a) = 2(a + b + c) = 11 + 27 + 18 = 56$, hence $a + b + c = 28$. Now c can be found by subtracting 11 ($= a + b$) from 28 ($= a + b + c$), and so on.

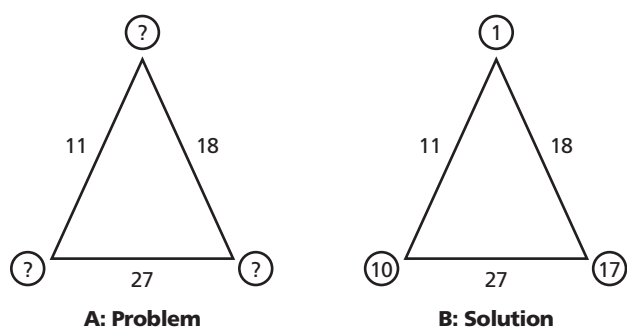


Figure 13.4 Arithmogons (Becker et al., 1999).

Various modifications of the task making it suitable for first-grade students have been used in the MATHE 2000 project (Becker & Selter, 1996, pp. 538–540; Wittmann, 2001).

The problem can also be generalized to other polygons (Mason et al., 1982, p. 160) with interesting results. For any even number of secret numbers, it turns out that the solution is indeterminate, whereas for any odd number of secret numbers, the solution that begins by finding the sum of all the secret numbers by taking half of the sum of all their pairwise sums generalizes. The comparison of odd and even cases leads naturally to a discussion of dependent versus independent systems of equations.

Another rich problem is that of finding which whole numbers can be expressed as the sum of two or more consecutive whole numbers in at least one way

(e.g., $15 = 7 + 8 = 4 + 5 + 6 = 1 + 2 + 3 + 4 + 5$; Driscoll, 1999; Mason et al., 1982). Exploration very naturally leads to the conjecture that all numbers have this property except for powers of 2 (including $2^0 = 1$). As Sawyer (1955, p. 23) commented, “for every pattern that appears, a mathematician feels he [*sic*] ought to know why it appears.” The search for a reason in this case is likely to throw up many conjectures and generalizations, such as that 3 times any number can be expressed as the sum of that number plus the preceding and succeeding numbers, that prime numbers (apart from 2) can be expressed as the sum of two consecutive numbers but not otherwise, and so on. It is perhaps worth pointing out that a lot of practice in computation is a by-product of such investigations.

Blanton and Kaput (2000) described one strand of a year-long teaching experiment with a socioeconomically and ethnically diverse third-grade class in an urban school, part of a district-based project. The focus of the description was generalizing and formalizing the students’ thinking about odd and even numbers, with the class as the grain-size for analysis. The researchers documented progression in complexity of generalizations and supporting justifications about even and odd numbers over the course of the year. Specifically, they found evidence of multiple forms of reasoning (including representational, numerical, and pattern-based), algebraic use of the terms *even* and *odd* as placeholders, use of a generalized referent (e.g., *odd*) to produce a generalization, extension of generalizations across mathematical operations, validation of generalizations through peer argumentation, and construction of sophisticated justifications. For large-scale implementation of an “algebrafication” strategy in early grades, Kaput and Blanton (2001, pp. 344–345) called for classroom-grounded teacher development in three dimensions:

1. Building algebraic reasoning opportunities, especially generalization and progressive formalization opportunities, from available instructional materials, including the algebrafication of existing arithmetic problems by transforming them from one-answer arithmetic problems to opportunities for pattern-building, conjecturing, generalizing, and justifying mathematical facts and relationships.
2. The building of teachers’ “algebra eyes and ears” so that they can spot opportunities for generalization and systematic expression of that generality . . . and then exploit these opportunities as they occur across mathematical topics.

3. The process of creating classroom practice and culture to encourage and support active student generalization and formalization within the context of purposeful conjecture and argument, so that algebraic reasoning opportunities occur frequently and are viable when they do occur.

Creating a Social-Constructivist Classroom Culture

Whereas many researchers focus their analysis of classroom environments on the development of arithmetical thinking and how it can be improved through innovative didactical materials, tools, techniques, and so on, in another influential set of classroom-based studies of the past decade, the focus is on creating, from a very young age, an appropriate classroom culture for arithmetic lessons along lines suggested by a social-constructivist perspective. In reflecting upon their longstanding developmental research project on mathematical learning in the early grades of elementary school, Yackel and Cobb (1996, p. 459) wrote,

we began the project intending to focus on learning primarily from a cognitive perspective, with constructivism as our guiding framework. However, as we attempted to make sense of our experiences in the classroom, it was apparent that we needed to broaden our interpretative stance by developing a socio-cultural perspective on mathematical activity.

According to this sociocultural perspective, the development of individuals' arithmetic concepts and skills cannot be separated from their participation in the interactive constitution of "taken-as-shared" mathematical meanings. An extremely important proactive role in the development of such a classroom microculture is played by the teacher. According to Yackel and Cobb (1996), one of the teacher's major tasks, which has been neglected for too long, is to establish sociomathematical norms. These norms, which are specifically mathematical (and are therefore distinguished from the social norms that relate to classroom actions and interactions in general), regulate classroom discourse and influence the learning opportunities that arise for both the students and the teacher. Typical examples of sociomathematical norms include what counts as a different mathematical solution, a sophisticated mathematical solution, and an efficient mathematical solution. Others relate to what counts as a good mathematical problem, a proper explanation, or an acceptable justification. The importance attributed to sociomathematical norms stems from the contention that students reorganize their specific-

cally mathematical beliefs and values as they participate in and contribute to the establishment of these norms. This claim implies that teachers can support students' development of appropriate dispositions toward mathematics by guiding the development of sociomathematical norms (Yackel & Cobb, 1996).

Starting from Yackel and Cobb's (1996) analysis, McClain and Cobb (2001) collaborated intensively with one teacher to consciously attempt to guide the development of sociomathematical norms and thus influence her students' beliefs about what it means to know and do mathematics, in the context of a teaching experiment. Together with the teacher, they traced the emergence of sociomathematical norms in one classroom over a 4-month period while she was teaching mental computation and estimation with numbers up to 100. The data consist of videotaped recordings from two cameras of about 100 lessons, supplemented with copies of students' written work, field notes, several videotaped clinical interviews with each student and with the teacher. In their article, they documented in great detail the development of sociomathematical norms in this classroom over a 12-week period, wherein the focus was on becoming familiarized with the quantitative environment structured by relationships between numbers up to 20. Ample use was made of the "arithmetic rack," a device with two rows of five white and five colored beads each, which enhances, more than others, multiple approaches to single-digit arithmetic (see Figure 13.5, which illustrates how, with the help of the arithmetic rack, a sum like $6 + 7$ can be solved easily either as $6 + 4 + 3$ or as $6 + 6 + 1$; Treffers, 1991, 2001a). This rich and detailed analysis led to several interesting conclusions and issues for further research.

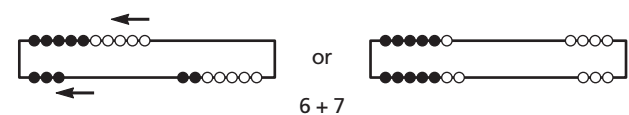


Figure 13.5 Solving a sum like $6 + 7$ in two different ways using the arithmetic rack (Treffers, 2001a, p. 50).

The first issue concerns the complexity and challenges that arise for teachers as they attempt to reorganize their instructional practices along the lines of the curriculum reforms. In the beginning of the experiment, the teacher accepted all students' contributions equally and rejected the view that she should act in any way as a mathematical authority in the class. According to the researchers, she had developed this form of practice as a reaction to traditional instructional practices. In other words, she had developed an instructional approach that could be defined negatively in terms of what she did not do compared with traditional teach-

ing. When she started to guide the development of the norm of mathematical difference at the researchers' urging, she did it in a rather directive and nonargued way. The teacher's learning during the rest of the teaching experiment consisted of improved attempts to cope with the tension between direction and structure, on the one hand, and openness to students' interpretations and interests, on the other.

A second issue concerns the way in which the emergence of the sociomathematical norm of what counted as a different solution formed the foundation for the emergence of the norms of what counted as a sophisticated and as an efficient solution, and the role that notating and symbolizing plays in supporting the emergence of these norms.

The third issue concerns the manner in which the establishment of these norms could directly influence the mathematical agenda. For instance, McClain and Cobb (2001) observed instances in which the students initiated reflective shifts in classroom discussions by conjecturing about the number of different ways in which a task could be solved by grouping, "which appear to involve an initial step in the transition from arithmetic to algebraic reasoning" (p. 264).

One major goal of this section was to demonstrate exemplarily how the teaching and learning of number and arithmetic can, and should, be interwoven with establishing appropriate social and sociomathematical norms from a young age. A second was to show how careful attention to these (subtle) sociocultural aspects of school mathematics can lead to significant learning outcomes related to children's mathematical reasoning and communication skills, and intellectual autonomy, without losing significant learning effects for number and arithmetic "strictu sensu" (see also Gravemeijer, 2004).

Emergent Models as a Design Heuristic

In many approaches to elementary mathematics learning, both older and recent, tactile and visual models play a very important role. However, there are different views about the use of these models. In a traditional skills approach (Baroody, 2003), they are not used because meaningful instruction is not an important goal. In a conceptual approach, the goal of meaningful learning of concepts and skills is typically reached by the imposition of a "ready-made" model (e.g., MAB material, Cuisenaire blocks) by the teacher that concretizes or embodies a given concept or technique. Recent theoretical developments, especially the growing appreciation of social constructivism and sociocultural theories of learning and instruction and of investigative approaches

to mathematics education, have led to strong calls to find an alternative for the teacher-imposed use of tactile and visual models as (unproblematical) "embodiments" of mathematical concepts, as in the above-mentioned conceptual approach.

In a number of (theoretically closely related) recent design experiments dealing with the teaching of certain arithmetic concepts or skills the option of building on symbolizing, modeling, and tool use as personally meaningful activities of students, in the context of social practices, is explored. The traditional metaphor of "transmission of knowledge" with the help of symbols that function as "carriers of meaning" is replaced by the image of students "constructing their own ways of symbolizing as part of their mathematical activity" (Gravemeijer, Lehrer, Van Oers, & Verschaffel, 2002, p. 1; see also Gravemeijer, 2002, 2004). What is called symbolizing and modeling in these studies differs significantly from the use of manipulative materials and visual models that has been common practice for a long time in mathematics education (research). In contrast, within the conception that is paramount in these design experiments, modeling is seen primarily as a form of organizing, in the course of which both the symbolic means and the model itself emerge (Gravemeijer, 2002, 2004; Gravemeijer & Stephan, 2002). This "emergent modeling as an instructional design" approach to the teaching and learning of arithmetic can be illustrated best by referring to several developmental research studies realized by the Freudenthal Institute in the Netherlands (Gravemeijer, 1994; Treffers, 1987), and to other studies conceptualized and realized along similar lines (Cobb et al., 1997; Izsak, 2004). These studies, which deal with the use of models *of* certain meaningful situations transitioning into models *for* arithmetic operations, show how these models emerge as context-specific models that refer to concrete and paradigmatic situations that are "experientially real" for the students. Gradually, the contextual meaning may shift to the background and the visual expression may change, but the idea is that the roots of the model are preserved, and that the students can always fall back on this original meaning (Gravemeijer & Stephan, 2002). Although the proponents of the emergent modeling approach claim that, ideally, the pupils should invent the necessary modeling tools for themselves, they acknowledge (partly based on experiences wherein they have tried to implement this design principle quite radically) that this ideal is mostly not really feasible and that as a reasonable compromise the pupils should at least "experience an involvement in the invention process, even though they do not invent the tools for themselves" (Gravemeijer, 2004, p. 122).

Rather than summarizing Gravemeijer's description of design studies based on the empty number line (Gravemeijer, 2002, 2004; see also Gravemeijer & Stephan, 2002) as a prototypical example of the emergent modeling approach, we refer to a recent study by Iszak (2004), dealing with a different curricular topic. Iszak's (2004) study combined an in-depth analysis of two-digit multiplication instruction in one U.S. fourth-grade classroom with an analysis of learning accomplished by a cross-section of students from the same classroom. More specifically, the study compares how taken-as-shared classroom mathematical practices and individual students used features of rectangular area representations for accomplishing conceptual, procedural, and problem-solving goals. Except for a chapter in Treffers's (1987) book, describing an experimental program for multidigit multiplication based on the principle of "progressive schematization" starting from area representations (but without providing any empirical analysis of classroom interactions with these experimental materials), Iszak (2004) did not find any previous research on the use of rectangular areas to develop students' understanding of multidigit multiplication. Starting from these ideas of Treffers (1987) and the results of ascertaining studies that fourth-grade students have good (intuitive) understanding of arrays, Iszak (2004) designed a learning environment wherein students' understanding of arrays was used as the starting point to develop understandings of multidigit multiplication, as a kind of alternative to the more traditional instructional programs wherein multiplication is introduced as repeated addition (as discussed earlier).

The study was conducted in the context of the Children's Math Worlds (CMW) project, the main objective of which is to make the goals of the U.S. Standards accessible to urban and suburban students and teachers. The CMW two-digit multiplication block builds on another CMW unit aimed at developing connections among single-digit multiplication, equal groups, and areas of rectangles understood as arrays of unit squares. The two-digit multiplication unit scaffolds the development of more efficient strategies with three area representations and a numeric method to break apart factors and products into smaller, easier-to-handle pieces (Figure 13.6): (a) a unit squares representation, (b) a 100s/10s/1s representation, (c) a quadrant representation, and (d) the expanded algorithm representation.

The experimental course consisted of 13 two-digit lessons, spread over a period of 8 weeks, wherein (groups of) pupils' representations and solutions to a small number of two-digit multiplication problems were intensively discussed with the whole class and

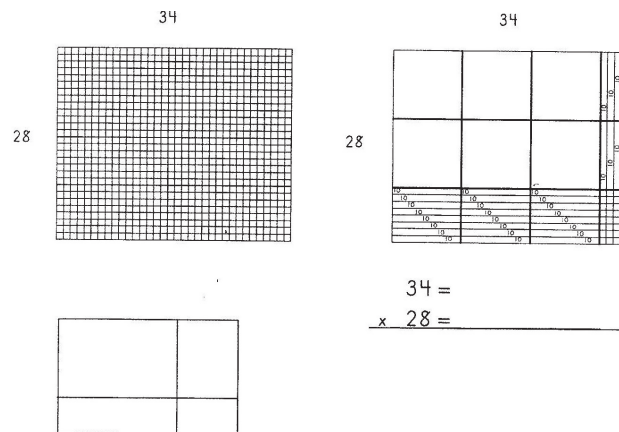


Figure 13.6 The unit squares, 100s/10s/1s, quadrants, and expanded algorithm representations for 28×34 (Iszak, 2004, p. 50).

connected to each other as well as to representations and solutions proposed by the teacher.

A scrutinized analysis of the videotapes of these lessons revealed, first, how taken-as-shared class strategies for finding groups and determining areas in unit squares and 100s/10s/1s representations emerged as the teacher and her students discussed alternatives for problems such as 14×13 and how, afterwards, increasingly more complex problems and accompanying new instructions led to the gradual development of more efficient strategies of the types shown in Figure 13.6. Ultimately, class discussion converged on essentially one multiplication strategy, the expanded algorithm representation shown in Figure 13.6, that coordinated magnitudes of partial products, expanded forms for factors, and the distributive property. As a consequence of this convergence, alternative choices for equal-groups strategies gradually faded from class discussions. The analysis of the whole-class solutions also revealed the challenging decisions that the teacher had to make about the range of strategies to pursue with the students, especially with respect to finding a good balance between providing too little guidance (in which case students might not develop understandings of efficient and general multiplication methods) and too much guidance (leading to too quick removal of alternative methods from the class discussions).

The results on a paper-and-pencil posttest that asked pupils to solve problems ranging between 17×12 and 92×78 and also "to draw any pictures that they need" demonstrated that by the end of the lesson unit a large majority of students could compute correctly with the expanded algorithm, and many could use alternative methods as well.

To gain deeper insight into pupils' strategies, the researcher also conducted videotaped semi-struct-

tured interviews with two pairs of low-, middle- and high-achieving pupils at the end of the unit, wherein they were administered two nonroutine tasks (e.g., a visually presented problem about the size of a 6×17 rectangular pattern and a word problem wherein children had to compute the total number of stores owned by a chain that had 26 stores in each of 38 states). These qualitative data made it clear that the pupils understood class strategies to very different degrees, that some of them did not readily make the required connections, and that more time and support might be necessary if all students are to find an approach to two-digit multiplication based on areas of rectangles. According to the author's analysis, this moderate overall outcome is at least partly due to the sociomathematical norms for justifying solutions applied by the teacher:

Had norms been established in the teacher's classroom that afforded opportunities for students to explain strategies in full, and that required students to take responsibility for asking questions when they did not understand, then the standard for taken-as-shared might have been higher and taken more time to achieve (Iszak, 2004, p. 78).

In a position paper about "developmental research" (unfortunately not available in English), Treffers (1993) extensively discussed the advantages and disadvantages of an instructional approach for multidigit multiplication based on the rectangular area model (as in some older developmental studies from the Freudenthal Institute and as in Iszak's study), and an alternative teaching/learning trajectory based on the repeated addition model (as in the study of Carpenter et al., 1997, described earlier and in other design experiments realized at the Freudenthal Institute; Dekker, Ter Heege, & Treffers, 1982; Treffers, 1987). Although he acknowledged that both approaches have their advantages and disadvantages, he preferred the latter approach based on the repeated addition model, mostly because it fits better with children's intuitive understanding of, and informal strategies for, multiplication. An empirical comparison of the strengths and weaknesses of both instructional approaches to multidigit multiplication seems necessary.

Teaching Word Problems From a Modeling Perspective

As discussed earlier, in recent years, the characteristics, use, and rationale of word problems have been critically analyzed from multiple perspectives, including linguistic, cultural, and sociological (Verschaffel et al., 2000). In particular, many mathematics

educators have argued that the stereotyped and artificial nature of word problems typically represented in mathematics textbooks, and the discourse and activity around these problems in traditional mathematics lessons, have detrimental effects on students' disposition towards mindful and realistic mathematical modeling and problem solving.

One reaction to criticisms of traditional practice surrounding word problems in schools is to undermine the approach that allows students to succeed using superficial strategies based on the "rules of the game." This is done by breaking up the stereotypical nature of the problems posed. For example, by including problems that do not make sense or contain superfluous or insufficient data, students can be guided to interpret word problems critically. A more radical suggestion is to treat word problems as genuine exercises in mathematical modeling.

Without any attempt to be exhaustive, we list a few recent design studies that have been set up recently according to this modeling perspective:

- the Jasper studies of the Cognition and Technology Group at Vanderbilt (1997),
- the numerous design experiments with model-eliciting activities summarized in the recent book by Lesh and Doerr (2003),
- the modeling curriculum for mathematics and science by Lehrer and Schauble (2005),
- English's (1998) design studies on mathematical problem posing,
- the learning environment for mathematical modeling and problem solving in upper elementary school children that Verschaffel et al. (1999) developed and tested a few years ago.

Hereafter we exemplarily describe the last study from that list. Verschaffel et al. (1999) carried out a design experiment aiming at the design and evaluation of a powerful learning environment that elicits in upper primary school children the appropriate learning processes for acquiring the intended competence in mathematical modeling and applied problem solving as well as positive mathematics-related beliefs. The learning environment in the classroom was fundamentally changed, and its design, implementation, and evaluation were done in close cooperation with the teachers of the four participating experimental classrooms and their principals. The intervention consisted of a series of 20 lessons that were taught by the regular classroom teachers (for a more detailed report, see Verschaffel et al., 1999).

The learning environment focused on the acquisition by the pupils of an overall cognitive self-regu-

lation strategy for solving mathematical problems consisting of five stages (representation, planning, execution, interpretation and evaluation, and communicating the answer), and embedding a set of eight heuristic strategies that are especially functional in the first two stages of that strategy (e.g., make a drawing, make a table, use your real-world knowledge, simplify the numbers, make a guess). An additional aim was to foster positive mathematics-related beliefs.

To elicit and support the intended constructive learning processes in *all* pupils, an intervention characterized by the following three basic features was developed. First, a varied set of carefully designed meaningful, complex, and open problems was used differing substantially from traditional textbook tasks. Moreover, these problems were presented in different formats—a text, a newspaper article, a brochure, a comic strip, a table, or a combination of several of these formats. An example is described below.

School Trip Problem

The teacher told the children about a plan for a school trip to visit the Efteling, a well-known amusement park in the Netherlands. If that turned out to be too expensive, one of the other amusement parks might be an alternative.

Each group of four pupils received copies of folders with entrance prices for the different parks. The lists mentioned various prices depending on the period of the year, the age of the visitors, and the kind of party (individuals, families, groups).

In addition, each group received a copy of a fax from a local bus company addressed to the principal of the school. The fax gave information about the prices for buses of different sizes (with a driver) for a 1-day trip.

The first task of the groups was to check whether it was possible to make the school trip to the Efteling given that the maximum price per child was limited to 12.50 euro. After finding out that this was not possible, the groups received a second task: They had to find out which of the other parks could be visited for the maximum amount of 12.50 euro per child.

Second, a learning community was created through the application of a varied set of stimulating and interactive instructional techniques. The basic instructional model for each lesson period consisted of the following sequence of classroom activities: (a) a short whole-class introduction; (b) two group assignments solved in fixed heterogeneous groups of three to four pupils, each of which was followed by a whole-class discussion; (c) an individual task also with a subsequent whole-class discussion. Throughout the whole lesson the teacher encouraged and scaffolded pupils to engage in, and to

reflect upon, the kinds of cognitive and metacognitive activities involved in the model of skilled problem solving. These instructional supports were gradually faded out as pupils became more competent in and aware of their problem-solving activity and, thus, took more responsibility for their own learning and problem-solving processes.

Third, an innovative classroom culture was created through the establishment of new social and sociomathematical norms about learning and teaching problem solving, aiming at fostering positive mathematics-related attitudes and beliefs in children. Typical aspects of this classroom culture were (a) stimulating pupils to articulate and reflect upon their solution strategies, (mis-)conceptions, beliefs, and feelings relating to mathematical problem solving; (b) discussing about what counts as a good problem, a good response, and a good solution procedure (e.g., “there are often different ways to solve a problem”; “for some problems a rough estimate is a better answer than an exact number”); and (c) reconsidering the role of the teacher and the pupils in the mathematics classroom (e.g., “the class as a whole, under the guidance of the teacher, will decide which of the generated solutions is the optimal one after an evaluation of the pros and cons of the different alternatives”).

As mentioned above, the learning program consisted of a series of 20 lessons. One introductory lesson was followed by 15 lessons that systematically addressed each step and each heuristic of the five-step problem-solving model. At the end of each of these 15 lessons pupils completed a sheet for their personal notebook summarizing the essentials about the what, the how, and the why of each step of the model, or of each of the embedded heuristics. At the end of the series four project lessons built around more complex application problems were organized wherein the pupils learnt to use all the elements of the problem-solving model in an integrated and flexible way. The School Trip Problem presented above is an example of such a problem.

To enhance a reliable and powerful implementation of the learning environment, the teachers of the four experimental classes received a great variety of support materials. It is important to mention here that this lesson series as well as the above-mentioned elements of teacher support were elaborated in partnership with the teachers of the participating experimental classes and their principals.

During the period of about 4 months during which the intervention was implemented in the four experimental classes, pupils from seven comparable control classes followed the regular maths curriculum.

Students' progress toward the major goals of the learning environment was assessed using a variety of instruments. Three parallel versions of a written test consisting of 10 difficult nonroutine application problems were used as pretest, posttest and retention test, respectively. A questionnaire on pupils' beliefs and attitudes relating to the teaching and learning of mathematical word-problem solving and a standardized mathematics achievement test that covers the entire mathematics curriculum were both applied as pretest and posttest. In addition, in each of the four experimental classes the solution processes of three pairs of children for five nonroutine problems were videotaped and analyzed before and after the intervention. Finally, in order to assess the implementation of the learning environment by the teachers, a sample of representative lessons was videotaped in each experimental class, and these lessons were analyzed by means of an observation scale. Moreover, the four experimental teachers were interviewed at the end of the whole lesson series about (their personal appreciation of) all aspects of the learning environment.

The results can be summarized as follows. First, whereas no significant difference was found between the experimental and control groups on the word-problem test during the pretest, the former group significantly outperformed the latter during the posttest, and this difference in favor of the experimental group continued to exist on the retention test. Second, the learning environment had also a significant, albeit small, positive impact on children's pleasure and persistence in solving mathematics problems, and on their mathematics-related beliefs and attitudes. Third, the results on a standardized achievement test showed that the experimental classes also performed significantly better than the control classes on the standard achievement test. Fourth, the analysis of pupils' written notes on their response sheets of the word problem test showed that the better results of the experimental group were paralleled by a very substantial increase in the spontaneous use of the heuristic strategies taught in the learning environment. This finding was confirmed by a qualitative analysis of the videotapes of the problem-solving processes of three groups of two children from each experimental class before and after the intervention. Indeed, this revealed that on the posttest these dyads made nearly twice as much spontaneous use of the eight heuristics as during the pretest, and that they also clearly progressed in terms of the spontaneous use of four metacognitive skills (i.e., orientation, planning, monitoring, and evaluation). Fifth, not only the high- and the medium-ability pupils, but also those of low ability benefited significantly—albeit to a lesser degree—from the interven-

tion in all aspects just mentioned. Indeed, there was no significant interaction between the variables experimental group, time of test, and ability level. Finally, the analysis of the videotapes of the lessons of the four teachers indicated that all four experimental teachers implemented the learning environment in (at least) a satisfactory way.

The preceding results show that an intervention, combining a set of carefully designed word problems with highly interactive teaching methods and the introduction of new sociomathematical classroom norms, can lead to the creation of a powerful learning environment that significantly boosts pupils' cognitive and metacognitive competency in solving mathematical word problems. In most of the design experiments in the domain of mathematical modeling, promisingly positive outcomes have been obtained in terms of performance, underlying concept formation, heuristic, metacognitive and communicative skills, and motivational and affective aspects of learning, although these outcomes are sometimes less impressive or convincing than one would have hoped, both in terms of the size of the obtained effects and their impact on all groups of pupils including low-ability children and in terms of the methodological rigor with which the reported effects have been obtained (see also Baroody et al., 2004). However, taken as a whole, the available research evidence shows that, to quote Niss (2001, p. 80), "application and modeling capability can be learnt, and according to the above-mentioned findings has to be learnt, but at a cost, in terms of effort, complexity of task, time consumption, and reduction of syllabus in the traditional sense."

Changing Teachers' Knowledge and Beliefs

Another typical characteristic of the last decade's research in the domain of number and arithmetic is the central place accorded to the teacher's knowledge and beliefs and the general acknowledgment of the importance of the teacher, not only in children's learning but also, for example, in curriculum reform and in research (Sfard, 2005). Researchers generally agree that improvement of mathematics education starts with improvement of the mathematical knowledge and the pedagogical content knowledge of teachers (Ma, 1999).

On the other hand, several studies have documented the weak subject-matter and pedagogical content knowledge of teachers. Subject-matter knowledge includes mastery of the key facts, concepts, principles and explanatory frameworks, procedures, and problem-solving techniques and strategies within the given domain of instruction. Crucial in this respect is also

the level of teachers' understanding of the domain. There is empirical evidence to support the alarming statement about (future) teachers' insufficient mathematical competence. For instance, in the domain of multiplicative structures, Graeber and Tirosh (1988) studied preservice elementary school teachers' knowledge and skills with respect to multiplication and division involving decimals larger and smaller than 1. They found that a considerable number of preservice teachers made the same errors and shared the same misconceptions as observed in 10- to 12-year-olds. In another study that focused on the connectedness rather than the correctness of prospective teachers' knowledge of division, Simon (1993) found that their knowledge base was weak with respect to several types of connections, such as the conceptual underpinnings of the familiar algorithm of division, the relationship between partitive and quotative division and between division and subtraction, and the connection between symbolic division and real-world situations to which it is applicable. Verschaffel, Janssens, and Janssen (2005) conducted a large-scale longitudinal study in which the elementary school mathematical knowledge and skills of a large group of Flemish preservice elementary school teachers from 15 different institutes was assessed by means of a paper-and-pencil test that was administered both at the beginning and at the end of their 3-year training. The 30-items test covered the new standards for mathematics in the elementary school curriculum in Flanders. The test was divided into six subsets differing in terms of the curricular subdomain and of the cognitive operations being addressed by the item. The results revealed that at the beginning of their training preservice elementary school teachers have rather weak mathematical competencies. At the end of their 3-year training, the overall test performance had become substantially better, although there were still reasons to be seriously concerned about the readiness of some student teachers to teach mathematics to elementary school children, especially with respect to their modeling and applied problem-solving skills. Finally, Ma (1999) found that the majority of the U.S. teachers (versus only a minority of the Chinese teachers) involved in her study displayed only procedural knowledge of topics of whole number and arithmetic such as subtraction with regrouping and multidigit multiplication. In fact, the knowledge gap between the U.S. and Chinese teachers paralleled the learning gap between U.S. and Chinese students revealed by scholars like Stevenson and Stigler (1992). According to Ma (1999), this limitation in their knowledge confined their expectations of student learning as well as their capacity to promote conceptual learning in the classroom. Given that the

parallel of the two gaps is, according to Ma (1999), no coincidence, it follows that "while we want to work on improving students' mathematics education, we also need to improve their teachers' knowledge of school mathematics" (p. 144).

The second category of teachers' knowledge—teachers' pedagogical content knowledge—includes several subsystems such as knowledge of mathematics lesson scripts and mathematics teaching routines, knowledge about the kinds of problem types, graphical representations, and so on. that are best suited to introduce particular mathematical notions and skills to pupils, and knowledge of instructional materials available for teaching various mathematical topics. Several studies have documented particular weaknesses in (future) teachers' pedagogical content knowledge, mostly in close relation to weaknesses in their mathematical knowledge base (Ma, 1999; Van Dooren, Verschaffel, & Onghena, 2002; Verschaffel et al., 1997).

Other studies have focused on the impact of teachers' mathematical knowledge on students' learning processes and outcomes. Taken as a whole, the majority of the results are in favor of the old adage, which is also included in the recent publication *Adding It Up* (Kilpatrick et al., 2001) and is echoed in Ma's (1999) book, namely that you cannot properly teach what you do not know yourself.

Given the well-documented relationship between these aspects of teachers' professional knowledge, on the one hand, and students' learning processes and outcomes, on the other, clearly without an improvement in the state of preparedness of the teaching body, calls for teaching with, or for, understanding such as those contained in the U.S. *Standards* (NCTM, 2000) and in similar reform documents are simply doomed. Therefore, researchers have begun to address the question of whether and how (preservice) teachers' cognitions and conceptions can be influenced. These intervention studies differ widely in

- the aspects of (preservice) teachers' cognition addressed (e.g., mathematical content knowledge, pedagogical content knowledge, beliefs about mathematics and mathematics teaching)
- the amount and kind of training provided (e.g., short interventions, interactive video, intensive workshops, workshops followed by classroom visits, and afterschool working sessions)
- the number of preservice teachers involved
- the way training effects were determined (e.g., teacher cognitions, teaching behavior, student outcomes).

In an unusually systematic and sustained intervention study of this kind, Carpenter, Fennema, Peterson, Chiang, and Loeff (1989; see also Carpenter & Fennema, 1992) investigated whether it is possible to improve teachers' instructional practice and their students' learning outcomes by confronting them with the available research-based knowledge of the development of children's skills and processes for solving addition and subtraction (word) problems described in ascertaining studies. They called their approach *cognitively guided instruction* (CGI). Basic ideas underlying the CGI approach are that the teaching/learning process is too complex to specify in advance, and that teaching as problem solving is mediated by teachers' thinking and decision making. Consequently, significant changes in educational practice can best be pursued "by helping teachers to make informed decisions rather than by attempting to train them to perform in a specified way" (Carpenter & Fennema, 1992, p. 460). Twenty first-grade teachers participated in a 1-month long workshop that attempted to familiarize them with the available research findings on learning and the development of addition and subtraction concepts and skills. Teachers learned to classify problems, to distinguish levels of mastery, to identify strategies that young children use to solve different problem types, and to relate those strategies to the mastery levels and problem types in which they are commonly used by children. Afterwards teachers discussed principles of instruction derived from research and designed their own programs of instruction based on those principles. Based on a large set of data-gathering techniques, including classroom observations, measurement of teachers' cognitions and beliefs, and tests of student learning outcomes, the following findings were obtained:

- CGI teachers taught problem solving significantly more and number facts significantly less than control teachers
- CGI teachers encouraged pupils to use a variety of strategies, and they listened to the processes their students used significantly more than did control teachers
- CGI teachers believed that mathematics instruction should build on students' existing knowledge more than did control teachers, and they knew more about individual students' problem-solving processes
- Pupils of CGI teachers outperformed control pupils in number-fact knowledge, problem solving, reported understanding, and reported confidence in problem solving.

Notwithstanding that several unanswered questions remain, these studies demonstrate that sharing with teachers research-based knowledge about students' thinking and problem solving can profoundly affect teachers' cognitions and beliefs about arithmetic classroom learning and instruction, their classroom practices, and, most important, their students' learning outcomes and beliefs. As such, these positive findings provide evidence for the relevance of the constructive conception and the dispositional view of mathematics learning and teaching, which is complementary to one already presented earlier. At the same time, these studies demonstrate the feasibility of this new conception in the setting of a typical elementary mathematics classroom. In light of the extremely time-consuming nature of the knowledge-sharing approach followed in CGI and other similar projects, the question remains whether such an approach can be applied on a large scale. New information technology, such as simulation programs and interactive video, can probably play an important role in helping inservice and preservice teachers interact with CGI or related ideas.

A quite different approach to professional development is followed in the Lesson Study project, the core form of professional development for Japanese teachers (Lewis, 2002). In the Lesson Study cycle, teachers work together to

- formulate goals for student learning and long-term development;
- collaboratively plan a "research lesson" designed to bring to life these goals;
- conduct the lesson, with one team member teaching and others gathering evidence on students' learning and development;
- discuss the evidence gathered during the lesson, using it to improve the lesson, the unit, and instruction more generally;
- teach the revised lesson in another classroom, if desired, and study and improve it again.

Research lessons occur in many settings in Japan. Lesson Study is credited for the shift from "teaching as telling" to "teaching for understanding" in Japanese mathematics education and is highly valued by both Japanese teachers and administrators. However, except for Yoshida's (1999) case study of the impact of the Lesson Study approach on teacher development and improved mathematics instruction in one Japanese school, we know of no research-based evidence (of the type described for CGI) for the impact of this approach to improving instruction through teacher development.

CULTURE, SOCIETY, SCHOOLING, AND ARITHMETIC

Introduction

The purpose of this section is to situate the research summarized in the first two sections in wider human contexts—historical, social, cultural, and political. Although these perspectives are, of course, applicable to mathematics education as a whole, their application to early arithmetic is of particular interest and importance for many reasons, including the following:

- the human (near-)universality of natural numbers, operations on them, and applications
- the importance of early arithmetic in a child's (mathematical) education—not just in the sense that it forms the basis for later mathematics, but also that the child's experience in the early years of formal schooling is foundational for the development of her or his conception of, and disposition towards, mathematics
- the popular perception of arithmetic as the central aspect of mathematics
- the philosophical and ideological debates that are often fought out over this terrain and that influence both mathematics education policy and research on mathematics education.

To begin with, it is appropriate to make some comments from this perspective on the studies summarized in the first two sections of this chapter. In respect of the types of experimental investigations described as ascertaining studies, and as stressed several times during that section, we should avoid any suggestion of a universal developmental trajectory for the development of numerical and arithmetic cognition. As with Piagetian theory, this implies careful acknowledgment that any results are relative to a specific cultural environment, particularly, but not limited to, the nature of instruction in arithmetic and, more generally, schooling and to the cognitive tools (including language) available. This point may seem obvious if, for example, one does a thought experiment on computation with Roman numerals, yet arguably its implications are frequently not given enough weight.

These remarks are clearly applicable to cognitive models such as SCADS, discussed earlier. As commented there, a major shortcoming of this and related models is that “little or no attention is given to the social, especially instructional context in which the development of arithmetic skills takes place.” More-

over, as stated on p. 565, “many of these (computer) models seem to assume that there is a kind of universal taxonomy and/or developmental sequence of computational strategies which is fundamentally independent of the nature of instruction or the broader cultural environment.” In this respect, the model of Baroody and Tiilikainen (2003), although lacking the precision highly valued by many in cognitive science, does acknowledge “the crucial role of the broader socio-cultural and instructional contexts.” As one very straightforward example among several mentioned by Fuson (2003), the emphasis on commutativity in Chinese teaching of multiplicative single-digit combinations is reflected in reaction-time patterns of Chinese adults (LeFevre & Liu, 1997).

Although work within neurosciences is clearly a growing area for the scientific study of mathematical cognition (see Campbell, 2004, for a recent overview of this research), it is limited in several respects. Though there are aspects of the physical environment common to all humans, such as affordances for counting, and although there are common evolutionarily produced phenomena in the development of the brain, the aspects of mathematical cognition that are the products of biological evolution are minimal relative to those that are the products of cultural evolution. The conclusion by Delazer (2003, p. 404) that “recent neuropsychological studies provided clear evidence that different types of knowledge are involved in arithmetical processing and that they are functionally independent and separately implemented in the human brain” may be interesting, but its implications and utility for teaching arithmetic have as yet not been demonstrated. Moreover, the neuropsychological work is largely limited to context-free arithmetic executed with routine expertise, whereas the research on even the addition and subtraction of small whole numbers in context shows the semantic and cognitive complexity of such operations. As Lakoff and Nunez (2000, p. 26) put it, we know something about what parts of the brain are active when we use different arithmetical capacities, and we have some idea of which of these capacities are innate (Butterworth, 1999), but “compared to the huge edifice of mathematics, [that] is almost nothing.”

We may suggest, therefore, that the explanatory power and educational utility of such cognitive and neuroscientific studies would be very significantly increased if they took into account the specifics of instructional environments and aimed to show how those differentially affect mathematical thinking and performance and how they align (or not) with hard-wired cognition. For example, is it possible to relate instruction based on representations of the number line—and the historical record of such representa-

tions and their use—to a theory that number is innately so represented in the brain?

Another inherent limitation of the work from cognitive science is that it is typically limited to the simplest arithmetic, while often making claims that it is—in principle—extendable. Yet it seems obvious that the more complex the arithmetic the more divergence may be expected due to cultural/instructional diversity. By way of example—and still without penetrating at all far into the edifice of mathematics—we may cite the profound effects of linguistic regularity/irregularity for number names from 11–100 documented by Fuson (2003).

A common theme of recent research (as in many of the ascertaining studies reviewed in the first section, such as those of Carpenter et al., 1997) is the invention by children of strategies that they are not directly taught. This phenomenon may be taken as a strong indication of the pervasive influence of cultural context beyond direct instruction, such as the representation of arithmetic operations and applications in children's social worlds outside of school. In the *Classroom-Based Research* section, several examples show how the accumulation of these discoveries has led to increasing consideration of how instruction should take account of, encourage, and be aligned with these inventions.

Generally, in the second section, examples of specific approaches made clear that the nature of the instructional environment is constitutive of mathematical cognition and that what happens in the mathematics classroom is about many more and much more profound things than knowledge of number facts and computational competency, including laying foundations for the student's view of mathematics, as conveyed through sociomathematical norms, in particular (in the third part of that section).

Beyond the physical and cultural environments, it is necessary to consider *designed* environments. The shift from ascertaining to intervention studies provides case studies of precisely how the culture of the mathematics classroom (Seeger, Voigt, & Waschescio, 1998) influences both what mathematics is learned by students and what they learn *about mathematics*. Many of the characteristics common to the representative design experiments discussed in the *Classroom-Based Research* section reflect a cultural perspective, particularly the fostering of mathematical discourse and sociomathematical norms as forms of enculturation and attention to sense-making connected to students' lived experiences outside of school.

Cultural Perspectives on Arithmetic

Sociohistorical Aspects of Number and Arithmetic

Crump (1990, p. 146) summarized the overview of his book *The Anthropology of Numbers* by concluding that “the series of natural numbers, together with the basic arithmetical operations of addition and subtraction, multiplication and division, are a resource open to use in almost any culture.” However, this comment should be clarified by the recognition that the application of the “resource” is highly variable across cultures and throughout history. Many of the uses of arithmetic that are taken for granted in U.S. and European societies are actually reflections of a particular contemporary worldview. In 1985, the Brazilian scholar Ubiratan D'Ambrosio introduced the term and concept of *Ethnomathematics*, a field that he defined as lying on the borderline between the history of mathematics and cultural anthropology. Subsequent work on ethnomathematics has had a variety of foci, notably the demonstration of the pan-cultural nature of mathematics (Ascher, 2002; Bishop, 1988; Selin, 2000), in particular to counterbalance the prevailing Eurocentric view of the development of mathematics (Joseph, 1992; Powell & Frankenstein, 1997).

Although much of ethnomathematical work has been concerned with the mathematics of non-Western peoples, the breadth of the conception of ethnomathematics is clear from the definition given by D'Ambrosio (1985, p. 45): “the mathematics which is practised among identifiable cultural groups, such as national-tribal societies, labor groups, children of a certain age bracket, professional classes, and so on.” Gerdes (1994, p. 20) offered an equally broad characterization as “the field of research that tries to study mathematics (or mathematical ideas) in its (their) relationship to the whole of cultural and social life.” These broad definitions clearly suggest strong theoretical links between ethnomathematics and varieties of sociocultural theory, notably situated cognition and activity theory.

The combination of historical and pan-cultural perspectives offered by ethnomathematics provides a useful corrective to tendencies to assume stability and universality of mathematics discernible in Western mathematics education; however, these issues lie beyond the scope of this chapter. Staying more focused, we find several reasons why the issues raised here are relevant to the teaching of whole-number arithmetic:

First, mathematics is a human activity. The study of arithmetic across time and cultures (Swetz, 1994) and stories about insights into the structure of natural numbers by gifted individuals such as Gauss and Ramanujan (Wells, 1997, p. 44 and p. 77, respectively)

afford an early introduction for children to an understanding of mathematical creativity and mathematics as part of culture.

Second, the historical and pan-cultural lore of number and arithmetic, including representations, algorithms, pattern analysis, games, and puzzles is a rich source for interesting mathematics (e.g., Bazin, Tamez, & The Exploratorium Teacher Institute, 2002; Conway & Guy, 1996; Selter, 2001; Wells, 1997).

Third, an appreciation of the contributions of all cultures to arithmetic (more generally, mathematics) is part of culturally appropriate instruction in countries around the world (including those in a post-colonial phase) that acknowledges the contributions of all peoples (e.g., Stillman & Balatti, 2001; Zaslavsky, 1973).

Fourth, similarly, an appreciation of the contributions of other cultures and cultural variation in doing arithmetic is part of appropriate instruction for the increasingly multicultural classes of the United States, many European countries, and elsewhere (e.g., Orey, 2005; Zaslavsky, 1991).

Arithmetic in School and Out

Arithmetic is not something children do only in school. Before they go to school, children are engaged in many activities with number and arithmetic, which they learn in ways that are in certain respects similar to how they learn their native tongue, namely in context and, very often, with an older person to provide instant feedback. So, by the time a child goes to school, he or she already has a lot of foundational knowledge in place (Clements & Sarama, this volume; Kilpatrick et al., 2001, chapter 5).

Encounters with number and arithmetic in children's lives outside of school are semantically and contextually much richer and varied than the narrow focus of formal education in arithmetic. Consider what a child entering school (in the United States, for the purposes of our example) typically knows about the number 5—what the numeral looks like, how to say the word for 5 in their language, five fingers on each hand, 5 objects, 5 people, 5 inches, what it means to be 5 years old (which is very complex inasmuch as there is “I am five today,” perhaps represented by 5 candles on a cake, as opposed to “I am five,” which probably means between 5 and 5:6, because a child of 5:6 is likely to say “I am five and a half”), five cents/dollars as opposed to a nickel/five dollar bill and their equivalence for some purposes, 5 as part of a date or a time (think about the complexity of interpretations of the number 5 on an analog clock), 5 as the number of a house or a bus, fifth in a series, . . . and on and on. In short, 5 provides a model for a very wide variety of situations familiar to a child.

As formal education proceeds, the connections between school mathematics and children's lived experiences and sense-making become weaker. A dramatic example was discussed in the *Word Problems* section. Children in France were prepared to give answers to questions such as “There are 26 sheep and 10 goats on a ship. How old is the captain?” (Baruk, 1985). Many other examples, in many countries (Verschaffel et al., 2000), point to an apparent effect whereby many children, after some years of formal education in mathematics, are inhibited from saying “that does not make sense” and learn to play the word problem game. Many mathematics educators are making attempts to tackle the perceived lack of connection between math in and out of school; several examples were discussed earlier. Increasing emphasis on data handling, and on modeling and applications, in elementary school is another development contributing to this movement.

Theoretical frameworks have also been brought to bear. Lave (1988, p. 74) used word problems as “a microcosm of theories of learning” in order to suggest “a theory of situated learning as a possible next step in understanding the learning and teaching of mathematics,” She argued (p. 81) that

math in school is situated practice: school is the site of children's everyday activity. If school activities differ from the activities children and adults engage in elsewhere, the view of schooling must be revised accordingly; it is a site of specialized everyday activity—not a privileged site where universal knowledge is transmitted.

A related theoretical framework leads us to consider school mathematics as an activity system (Engestrom, 1987; Engestrom & Miettinen, 1999). Starting from Vygotsky's theory of mediated acts, Engestrom elaborated a triangular structure that he applied to the analysis of many and varied activity systems, in particular that of school-going (Figure 13.7, Engestrom, 1987, p. 103). Although this figure presents an analysis of school-going in general, it applies very appositely to mathematics education in particular. It offers a powerful theoretical lens for analyzing the teaching/learning of elementary arithmetic, including the use of physical and symbolic tools (Cobb, Yackel, & McClain, 2000; Gravemeijer, Lehrer, Van Oers, & Verschaffel, 2002). Moreover, much of the work described in the *Classroom-Based Research* section exactly fits the suggestion by Engestrom and Miettinen (1999, p. 3) that

the type of methodology I have in mind requires that general ideas of activity theory be put to the acid test of practical validity and relevance in interventions

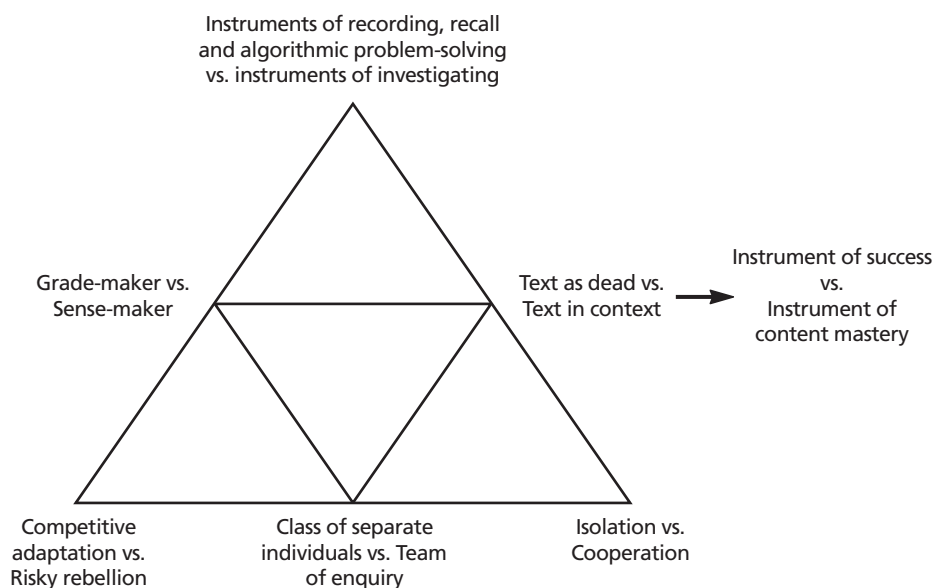


Figure 13.7 The activity of school going (Engestrom, 1987, p. 103).

that aim at the construction of new models of activity jointly with the local participants.

Although as yet only limited research directly uses this theory, it may be suggested as a powerful resource for future work, as a perspective that, for example, “overcomes and transcends the dualism between individual subject and objective societal circumstances” (Engestrom & Miettinen, 1999, p. 3). Moreover, it is in alignment with recent interest in contextual and cultural theories in cognitive science, as shown by the concepts of situated cognition and distributed cognition (Salomon, 1993), and in education in the idea of situated learning in communities of practice (again, as reflected in many of the case studies described in the second section).

Cultural and Educational System Differences

As exemplified at various points in the first two sections of this chapter, many research findings deal with differences in arithmetic teaching and learning that can be attributed to specific or general cultural differences. Pervasive aspects of the culture milieu, such as attitude to learning, apply more broadly to all education, and we will not attempt to cover them here. On the other hand, it is appropriate to deal with language, insofar as there are clear and specific ways in which its influence on early arithmetic teaching and learning can be traced. Then we will turn to aspects of different educational systems, without attempting the complex

task of analyzing in depth the varying extents to which these differences are reflections of cultures *per se*.

Effects of Linguistic Differences on Early Arithmetic

A strong and specific effect on learning and teaching whole-number arithmetic is the observation that languages differ in the extent to which the structure of the whole numbers is mirrored in the system of words used to name them. In particular, Chinese, Japanese, and Korean (all derived from ancient Chinese) systematically name the whole numbers from 11 to 100, whereas European languages do not (Fuson & Kwon, 1992; Miura & Okamoto, 2003). Consequent differences have been noted, as early as kindergarten, in development of verbal counting (Miura & Okamoto, 2003, p. 233). In another comparison, early first-graders (who had no prior experience with base-ten blocks or instruction in using them) were asked to represent 42. The children from the United States, Sweden, and France were most likely to use 42 unit blocks, whereas those from China, Japan, and Korea generally used 4 ten-blocks and 2 units (Miura & Okamoto, 2003, p. 231). It is at least highly plausible that the linguistic differences, at least partially, underlie better understanding of place value by Chinese, Japanese, and Korean first-graders as opposed to those from the United States, Sweden, and France (Miura, Okamoto, Kim, Steere, & Fayol, 1993).

The systematicity of the naming system may be one reason for, and certainly facilitates the use of, early calculating methods using the “make a ten” method. Combined with making the connections between 6, 7,

8, 9 and $5 + 1$, $5 + 2$, etc., this method is taught in first grade in China, Japan, Korea, and Taiwan (Fuson & Kwon, 1992). As a result, most children in these countries, by the end of first grade, can rapidly carry out all single-digit additions. The method is also taught in some European countries, and Fuson strongly advocated using it in first grade in U.S. schools, instead of restricting almost all of first grade to the learning of addition and subtraction below 10. Further strong motivation for adopting this approach is that it lays a strong conceptual foundation for two-digit addition and subtraction in general, and for the understanding of place value in relation to multidigit addition and subtraction. In Liping Ma's (1999) study of Chinese and American elementary teachers, she reported that most of the Chinese teachers who were asked about two-digit subtraction sums referred to the principle of "decomposing a unit of higher value" which is related to abacus use (Ma, 1999, p. 7) and which formed a coherent "knowledge package" (p. 19) relevant in many arithmetical procedures. By contrast, the majority of American teachers showed only procedural knowledge of subtraction with regrouping.

Differences in Educational Systems

As commented by Towse and Saxton (1998), a clear pattern of findings seems to show that children from Asian countries such as Japan, Korea, China, and Singapore outperform those in the United States and Europe (even though the picture is complicated by the methodological quagmire of cross-cultural comparisons and often contradictory results). A considerable amount of research has been undertaken to try to find reasons for these differences and thereby generate ways to improve mathematics education. In addition to specific linguistic factors, possible explanations considered by Towse and Saxton include the influence of preschool education, quality and quantity of teaching, differences in problem-solving strategies, and cultural attitudes to learning mathematics.

However, as remarked by Kilpatrick et al. (2001, p. 31), "In every country, the complex system of school mathematics is situated in a cultural matrix." Thus, as stated by Stigler and Hiebert (1999, p. 103), "teaching is a complex cultural activity that is highly determined by beliefs and habits that work partly outside the realm of consciousness." Awareness of this cultural embedding is needed, accordingly, to temper the temptation to uncritically import aspects of other systems that appear to contribute to success. For example, describing the outcomes of the Third International Mathematics and Science Study (TIMSS), Kilpatrick et al. (2001, p. 32) warned that comparisons of practices, programs, and policies in the United States with those of high-

achieving countries "at best can only be suggestive of the sources of achievement differences."

On the other hand, the example of linguistic factors discussed above, with ramifications spreading upwards to the learning of multidigit computation, suggests that a careful analysis may lead to beneficial results. Another line of research that may well have had tangible benefit is the comparative studies of textbook treatments of whole-number arithmetic, of which an early and influential example was Stigler, Fuson, Ham, and Kim (1986).

An interesting study of the effects of different teaching approaches was reported by Anghileri, Beishuizen, and Van Putten (2002). They explored the written calculation methods for division used by pupils in England and the Netherlands at two points in the same school year. The test involved a series of five calculations and five word problems involving both single-digit and two-digit dividers. Informal strategies were analyzed, and progression towards more structured procedures were identified. Whereas the English approach was characterized by a discontinuity between the formal computational procedure being currently taught and informal solution methods used earlier, the Dutch approach was based on careful progression from informal strategies towards more formal and efficient procedures. Comparison of the methods used by Year 5 pupils in the two countries showed greater success in the Dutch approach on the tests. With respect to strategies, the traditional algorithm was widely used during both tests by English children (with rather low level of success), whereas Dutch children most frequently applied high-level chunking strategies (with rather high levels of success). Although some methodological problems warrant caution, this comparative study can be considered as evidence that learning is more effective when written methods build upon pupils' intuitive understanding and informal strategies in a progressive way rather than by abruptly replacing them with standard procedures.

Political/Ideological Aspects

Towards Evidence-Based Reform: Standards-Based Elementary Curricula

Research on mathematics education, to an unprecedented extent, has become a political issue especially, but by no means exclusively, in the United States (see, e.g., Ginsburg, Klein, & Starkey, 1998). Thus, Kilpatrick (2003, p. 471) pointed out that "research on teaching and learning has become a weapon in the so-called math wars over *Standards*-based school mathematics curricula." As mentioned in the introduction

to this section, elementary arithmetic, for several reasons, is one of the major battlegrounds in that war.

One response to the vociferous criticism in some quarters following publication of the *Curriculum and Evaluation Standards for School Mathematics* (National Council of Teachers of Mathematics, 1989) was to encourage the evaluation of a number of elementary, middle, and high school programs based on the *Standards* (Senk & Thompson, 2003) in the spirit of “evidence-based policy making.” Given the importance of textbooks (Kilpatrick et al., 2001, pp. 36–37), the relative paucity of evaluative research may be considered surprising.

Part II of *Standards-Based School Mathematics Curricula* (Senk & Thompson, 2003) includes evaluations of four elementary grades curriculum projects: *Math Trailblazers*, *Everyday Mathematics*, *Investigations*, and *Number Power*. These evaluations are of textbooks in widespread use, in contrast with most of the examples in the second section of this chapter (with the exception of the work stemming from the Freudenthal Institute). There are inevitably problems in conducting tightly controlled experimentation within the constraints of real schools (Hiebert, 1999). Nevertheless, in an overview of the studies Putnam (2003) concluded that they provided convincing evidence of beneficial effects in that the students in the experimental programs performed at least as well on conventional standardized tests, and better on custom-made tests assessing various other aspects targeted by the projects, such as mental arithmetic and number sense. Moreover, collectively these curricula cover a wider range of topics than has historically been the case, in particular data analysis, modeling, and applications, including in science.

Looking to the future, Kilpatrick (2003, p. 485) identified as major concerns for further development of such evaluative research “a deeper examination of how teachers are using the curriculum, the inclusion of comparisons across new curricula, and the development of more instruments to measure ambitious learning goals.” The difficulties of evaluating curricula have been starkly exposed in a report prepared for the National Research Council (Confrey & Stohl, 2004), which concluded (p. 189) that

The corpus of evaluation studies across the 19 programs studied does not permit one to determine the effectiveness of individual programs with high degree of certainty, due to the restricted number of studies for any particular curriculum, limitations in the array of methods used, and the uneven quality of the studies.

The report makes recommendations that prompted one commentator (Steen, 2004) to remark that “the standards that the committee set for future evaluations to meet the test of ‘scientific validity’ are generally sound, although taken together they pose hurdles that are both administratively and financially impossible to achieve.”

Philosophical Basics: What Does $2 + 2 = 4$ Mean?

$$2 + 2 = 4$$

“There is a mathematically correct answer”

(Masthead for website of Mathematically Correct)

However, it is not so simple. Hersh (1997, p. 16) offered a clarification that we find helpful:

So “two” and “four” have double meanings: as Counting Numbers or as pure numbers. The formula:

$$2 + 2 = 4$$

has a double meaning. It’s about counting—about how discrete, reasonably permanent, noninteracting objects behave. And it’s a theorem in pure arithmetic.

The quotation from “Mathematically Correct” implies that only the second meaning suggested by Hersh counts as part of mathematics. However, even theorems in pure arithmetic are not such a simple matter. Again, we can turn to Hersh (1997, p. 16) for enlightenment:

Fact 1: Mathematical objects are created by humans. Not arbitrarily, but from activity with existing mathematical objects, and from the needs of science and daily life.

Fact 2: Once created, mathematical objects can have properties that are difficult for us to discover.

Thus $2 + 2 = 4$ may be thought of as a property of the whole numbers that is easy to discover, in contrast to, say, Fermat’s last theorem. The set of properties discovered is, of course, time-dependent as are interpretations of those properties (straying outside the positive whole numbers momentarily, the history of interpretations of the equation $-1 \times -1 = 1$ is particularly interesting). Also varying across time are the many different formal interpretations of the system of whole numbers (discussed earlier).

Our preferred description of the first meaning of $2 + 2 = 4$ suggested by Hersh is that it is a mathematical statement that may or may not provide an adequate model for some aspect of reality. In relation to state-

ments about natural numbers used to model aspects of the real world, clearly $2 + 2 = 4$ is *not* an appropriate model for many situations that can be differentiated from situations for which it is an appropriate model only by extramathematical interpretations of the real world (Usiskin, 2004, and see Verschaffel et al., 2000, chapter 7, for more examples), and such interpretations are certainly culturally dependent, as the ethnomathematical literature in particular shows (e.g., Ferreira, 1997). For example, Ascher and Ascher (1997, p. 29) related an oft-told anecdote about an African sheepherder and “someone else variously described as an explorer, trader, scientist, anthropometrist, or ethnologist.” The herder agreed to trade two sticks of tobacco for one sheep but became upset when offered four sticks of tobacco for two sheep. As pointed out by Ascher and Ascher, from one perspective the herder does not understand that $2 + 2 = 4$, whereas from another, the scientist/trader does not understand sheep, which are not standardized units.

The above is not merely an abstract discussion about philosophy; it has implications for the realpolitik of teaching arithmetic, because the beliefs of those who exercise power in educational policy-making and influence public opinion impact what happens in schools. By way of illustration, at the end of his book *Entering the Child's Mind*, Ginsburg (1997, p. 228) quoted an article from the *San Francisco Chronicle* (Saunders, 1994):

If the proposed framework is approved, California's third graders could be taught that even math is relative. [The program] would teach children that there are different “strategies” for solving simple equations. Instead of memorizing $5 + 4 = 9$, the program would tell children to find creative ways to find the answer, such as “I think of $5 + 5$, because I know that $5 + 5 = 10$. And $5 + 4$ is 1 less, so it's 9.” . . .

Maureen DiMarco, Governor Wilson's point person on education and candidate for state schools superintendent, dismisses new-new math as “fuzzy crap.”

The point of view represented here fits clearly the philosophical stance labeled “dualism” by Baroody et al. (2004, p. 228). It is beyond the scope of this chapter to unpack the many implications of the rhetoric (note the title of the article) and the underlying ideologies (of both the reporter and the candidate for state schools superintendent). The reader will see that the dismissal of encouraging *thinking* is incompatible with the views expressed in this chapter, yet the expression and influence of such views are part of the political reality of mathematics education today, and not just in the United States. In his comments, Ginsburg (1997,

p. 228) starkly characterized the choice as “between fundamentalism and modernity.”

SALIENT THEMES AND CHALLENGES

In this final section, we highlight overarching themes of our review, followed by what we see as major challenges facing the field. From many points of view—what kind of research is needed, what it means to have a profound understanding of whole-number arithmetic, how arithmetic fits within the edifice of mathematics, the need to improve understanding of the issues among the general public and those wielding power, and the appropriate preparation and continuing professional development of elementary teachers—the central messages are that the learning and teaching of whole-number arithmetic are extremely complex, culturally embedded, and foundational for every person's mathematical education.

Major Themes of the Research

Complementary Approaches

The contrast between the first two sections of our review of recent research reflects a major development to complement studies predominantly carried out with individual children, largely within the broad family of cognitive perspectives and methodologies, with extended classroom-based action research and design studies within the family of sociocultural theoretical frameworks. Note that the use of the word *complement* is intended to signal our belief in the continuing valuable contribution of cognitive analyses (cf. Greer, 1996).

This contrast corresponds to that between the two broad theoretical orientations, with associated research methodologies, identified by Sfard (2003, p. 355), namely the acquisitionist and the participationist group, that were characterized as follows in the introductory section of this chapter (see p. 558).

The “acquisitionist” group consists of those traditional cognitivist approaches that explain learning and knowledge in terms of such entities as cognitive schemes, tacit models, concept images, or misconceptions. The “participationist” framework embraces all of those relatively new theories that view learning as a reorganization of activity accompanying the integration of an individual learner with a community of practice.

The idea of enculturation into “*the* mathematical community” had a vogue, but, if interpreted too narrowly, is problematic because what children do in

school is very different from what practicing and research mathematicians do. However, “integration with a community of practice” is a more general conception, consistent with D’Ambrosio’s (1985) definition of ethnomathematics as the mathematics of cultural groups, and Lave’s (1992) characterization of school mathematics as a form of situated practice.

Research within the first framework follows the traditional approach that, through systematic enquiry, yields evidence for theoretical descriptions of the development of cognitive processes underlying the identified elements of competence in the domain. The aim and expectation is that this theoretical knowledge can subsequently be applied by teachers, curriculum developers, policymakers and, indeed, those engaged in classroom-based research. Thus it follows what Greeno (2003, p. 312) called the RDDE (research, development, dissemination, evaluation) model in which the components follow a linear temporal trajectory.

By contrast, much of the work surveyed in the second section follows a cyclical or intermeshed pattern of research and development that has been called “developmental research” (Gravemeijer, 1994, chapter 4; Gravemeijer, 2004). Related terms include *design experiments* (Greeno et al., 1996), *interactive research and design* (Greeno & the Middle-School Mathematics Through Applications Project Group, 1998), and *hypothetical learning trajectories* (Clements & Sarama, 2004a). A major advantage of such work is that it greatly reduces the gap between research and practice. However, in most cases the huge task remains of scaling up—from existence proofs with exceptional teachers given strong support—to the generality of teachers, as Selter (2002) commented (see Burkhardt & Schoenfeld, 2003). The work of the Freudenthal Institute stands out in this regard as perhaps the best example of a long-term and large-scale implementation that has substantially impacted mathematics education at a national level.

The contrast between the first and second sections is correlated with divergent aims between psychologists and mathematics educators (De Corte et al., 1996, p. 492). In particular, psychologists who take mathematical cognition as a domain for study tend to accept mathematics education as it is, whereas mathematics educators are likely to want to change it.

The Multifaceted Nature of Proficiency

Perhaps the most salient observation about research on learning/teaching arithmetic in recent years is that it reflects a much more complex and sophisticated view of the domain in many respects. For example, Kilpatrick et al. (2001) characterized mathematical proficiency, with particular emphasis

on whole-number arithmetic, using the metaphor of intertwined strands representing different facets of a complex whole: conceptual understanding, procedural fluency, strategic competence, adaptive reasoning, and productive disposition. They stressed that (p. 116): “the five strands are interwoven and interdependent in the development of proficiency in mathematics.” This conception represents an enrichment of the earlier recognition that conceptual understanding and procedural competence interact in complex ways (Hiebert, 1986)—including the case when understanding is retrospective and facilitated by previous routine use of procedures (Sfard, 2003).

Much of the enrichment, relative to educational policies and goals predominantly restricted to the second strand of procedural fluency, is captured by Hatano’s distinction between routine and adaptive expertise (see earlier in this chapter). In the first section, we reported on numerous observations of children’s spontaneous adaptive expertise—successively streamlined methods for calculations, increasing sophistication of strategy choice, flexible mental arithmetic and number sense, contextual interpretation of outcomes of computational work, awareness and exploitation of structure. . . . Correspondingly, many of the examples in the second section amount to the construction and evaluation of designed instructional environments that foster adaptive expertise, in particular by harnessing the creativity implicit in Hatano’s characterization.

Long-Term Developmental Perspective

Long-term developmental has two meanings with respect to the cultural/historical development (phylogeny) of whole-number arithmetic and with respect to the development for an individual child in an instructional environment and a social environment outside of school (ontogeny).

Using natural numbers and operations upon them is a human activity that is pan-cultural and reaches several millennia back into the past. Although it may be psychologically difficult to avoid a Platonic view of the natural numbers (the name itself is indicative and the mathematician Kronecker famously said that the natural numbers are the work of God, whereas all else is the work of man), the creation of symbol systems to record counted numbers efficiently, and the invention of physical devices and algorithms for calculating with them, represent a cultural activity of constructing and using tools. This cultural-historical perspective is a necessary backdrop when considering development of understanding of arithmetic in today’s student, as Freudenthal, in particular, has argued (1991, p. 47): “Traditionally, mathematics is taught as a ready made subject. Students are given definitions, rules and algorithms, ac-

cording to which they are expected to proceed. Only a small minority learn mathematics this way.”

As we have described, the long arc of children’s experience with whole-number concepts begins before school and continues in their social lives outside of school, in parallel with formal instruction. Even if we restrict attention to whole-number arithmetic, the complexity of this development demands a coherent long-term theoretical framework. Further, when we consider this period in a wider context, a comprehensive conception is needed, such as Vergnaud’s (1996) characterization of the additive and multiplicative conceptual fields, and their synthesis, with a view to the further development of the concept of number. Further elements include the conceptual and not merely procedural extension of the operations to domains beyond the natural numbers (Greer, 1994; Verschaffel & Vosniadou, 2004), acts of generalization that foreshadow formal algebra (Kaput, 1998), and understanding of the mathematical structures in which numbers and operations are embedded.

In terms of the instruction that such a theoretical framework should inform, we have seen, through several examples, that aligning instruction with the child’s spontaneous creations, problem-solving, and sense-making—under guidance, rather than presenting it as “ready-made”—has beneficial results. An example already mentioned is the study by Anghileri et al. (2002) in which Dutch students who were taught in accordance with the principle of progressive schematization outperformed English students who were directly taught standard algorithms.

A common feature of such approaches is that they do not proceed by systematic incremental changes in superficial features such as the size of the numbers. Rather, they operate at the level of larger ideas, in the spirit of the alternative to traditional scope and sequence termed a *distributed curriculum* by Resnick (1992, p. 423), which calls for and fosters well-connected knowledge and adaptive expertise (cf. Ambrose et al., 2003, p. 330, and see the notion of “big ideas” in Baroody et al., 2004).

Sense-Making and Modeling

One kind of sense-making in this context is when doing disembedded calculations where the sense to be made is of the structures that relate numbers and operations upon them, without explicitly modeling aspects of the real world (Hiebert et al., 1997, predominantly addresses this form of sense-making). However, modeling with a different meaning (Greer, 1994, p. 76; Greer, 2004) is often involved in that those structural relationships may be represented using physical

(including symbolic) embodiments, such as blocks (unitary or structured), fingers, number line, abacus (see *Multidigit Arithmetic* section). These embodiments fall into two broad classes, discrete and continuous. The use of collections of uniform, small, hard, permanent objects has the advantage of being immediately accessible to counting expertise, and such a collection becomes internalized as a generalized image for natural numbers and operations on them. Further, the objects can be structured in order to accommodate the decimal system for the whole numbers. On the other hand, the number line has the great advantage that it extends naturally to rational numbers, directed numbers, and real numbers (and eventually, with an orthogonal number line, to both complex numbers and Cartesian graphs). The decimal system has the same design strength in that the extension of computations to rational numbers, decimally represented, is relatively straightforward (but the conceptual restructuring is certainly not; Greer, 1994), although the procedural as well as conceptual problems of extension to rationals represented as fractions are well documented.

As reviewed in the *Single-Digit Computation* section, children first learning to compute, as an extension of counting, are heavily dependent on external devices including their fingers. We have argued and presented evidence for the assertion that the trajectory from that point to the point at which they can do multidigit computations should not require the abandonment of sense-making.

What we might call the mathematician’s approach is exemplified by Wu (1999) who characterized conceptual understanding of multidigit computations in terms of realizing how they are based on laws of arithmetic such as associativity and commutativity. He also showed how those laws themselves can be grasped as intuitively accessible generalizations through the use of arrays of dots. From this perspective, the title of his paper (rightly, in our opinion) labels “basic skills versus conceptual understanding” as “a bogus dichotomy in mathematics education” (Wu, 1999, p. 1).

However, as Wu (1999, p. 5) pointed out, the standard algorithms represent “efficient compression of a valuable piece of *mathematical reasoning* into a compact shorthand” and many mathematics educators believe that a more promising route—with much the same destination in view—is to reach this level of compression more gradually and more in concert with children’s intuitive sense-making. Thus, according to Treffers (1987, p. 267) “The fact that the standard algorithm is taken as the final product of a process of algorithmisation rather than as starting point is, in a way, paralleled by the historical development . . . [through] processes of schematizing and shortening.” As Freud-

enthal (1991, p. 61) pointed out, the process of “progressive schematization” may be highly facilitated by presenting problems as word problems. The work of Ambrose et al. (2003), summarized earlier follows the same principle (and see also Hiebert et al., 1997). For example, a partitive division problem is “Hattie had 544 candies to share with her friends. She gave each of her 17 friends the same amount of candy. How many candies did each friend get?” Here the story provides a simple semantic scaffolding for the logic of the calculation, a kind of “mental manipulative” (Toom, 1999); in particular, this particular example appeals to children’s strong social understanding of fair sharing. At the same time, for the students being studied, this was a genuine problem in that they did not have a routine solution to hand, certainly not the standard algorithm.

The same distinction between sense-making in terms of the structural properties of the system of whole numbers and sense-making in relation to a described real-world or imaginable situation applies also to estimation and number sense.

In what has been described above, word problems are assumed to map unproblematically onto arithmetic operations, reflecting their main purpose, namely to support computation with understanding. We have argued (Verschaffel et al., 2000) that sense-making involves a second aspect whereby the relationship between a description of a situation and the mathematics appropriate for modeling that situation needs to be considered more critically and not always accepted as unproblematic (thereby calling for adaptive expertise—see Van Dooren, Verschaffel, Greer, & De Bock, 2006). Further, as argued by Selter (2002, p. 224), beyond the discussion of a range of particular models is the higher level activity of understanding the nature of modeling activity (Greer & Verschaffel, in press).

A related issue is the common concern about the lack of connection between school mathematics and out-of-school life. The long-term perspective referred to above also extends backwards in time, with a heightened attention to the experience children have prior to formal education, from the counting they have done, from hearing number words and seeing number symbols used in everyday life, and from various experiences in judging, comparing quantities, sharing collections, and so on (see chapter 5 of Kilpatrick et al., 2001). More generally, according to Becker and Selter (1996, p. 516),

The informal knowledge that children possess is valuable and should be used as the basis for mathematics teaching and learning. . . . Connecting or relating formal mathematics to these informal approaches is

not an easy task . . . however, it has potential for making school mathematics more meaningful and less threatening and overwhelming to children.

Challenges Ahead

Conducting Research in a Political Context

In the third section of this chapter, we commented briefly on the ideological and political forces that are, to a considerable extent, shaping mathematics education—most extremely, perhaps, but certainly not only, in the United States (e.g. Dowling & Noss, 1990; Ellerton & Clements, 1994; Thomas, 2001). The same forces are acting on research in mathematics education. The *No Child Left Behind Act* of 2001 calls for the use of “scientifically based research” as the foundation for many education programs and for classroom instruction. On February 6, 2002, a seminar was held at the U.S. Department of Education where “leading experts in the fields of education and science discussed the meaning of scientifically based research and its status across various disciplines” (U.S. Department of Education, 2002). In this seminar, a talk entitled *Math Education and Achievement: Scientifically Based Research* by Russell Gersten, a specialist on reading comprehension and special education, began thus: “This is actually an easy topic to be brief on because there isn’t a lot of scientific research in math [education].”

The validity of research, including a great deal of that reported in this chapter, and the credentials of those in the discipline of mathematics education are, to an unprecedented extent, under fire. Although these comments apply to this volume as a whole, for reasons set out earlier elementary arithmetic is at the epicenter. Yet, educational research is “the hardest science of all” (Berliner, 2002), and research in mathematics education is characterized by “reasonable ineffectiveness” (Kilpatrick, 1981). Recent shifts within the field have led to a proliferation of theories and methodologies that may be regarded as liberating or anarchic, depending on one’s perspective (De Corte et al., 1996, p. 499). (For the anarchic interpretation, see Steen’s, 1999, review of the ICMI Study *Mathematics Education as a Research Domain: A Search for Identity*; Sierpinska & Kilpatrick, 1998).

Consider the following key issue. We have suggested that the teaching of whole-number concepts and operations could be viewed as an opportunity to lay the foundations for many aspects of higher order mathematical thinking—problem-solving, algebraic forms of reasoning, argumentation, symbolizing and representational acts, sense-making, modeling. Examples cited constitute existence proofs demonstrat-

ing the feasibility of classroom teaching that addresses these higher level, long-term aims and the positive results that can be obtained. It hardly seems possible, however, to envisage research that could definitively validate or invalidate the belief of proponents of these approaches that the laying of such early foundations pays off in the long term in older students who are better at problem solving, understanding proof, mathematizing both vertically and horizontally, and so on. The promotion of such approaches is a matter of beliefs and values—just as is the statement in the Mathematics Framework for California Public Schools (California Department of Education, 2000, p. 18 and oft-repeated) that “all students are capable of learning rigorous mathematics and learning it well.” (This example was chosen because the creators of that framework are firmly in the camp of those who call for decision-making based on rigorous scientific evidence.)

Another belief shared by most, if not all, of the mathematics educators cited in this chapter is that, by contrast with traditional teaching that has less ambitious aims of computational fluency, conceptual understanding, and perhaps problem solving (often narrowly conceived), classrooms organized according to the progressive principles exemplified in the *Classroom-Based Research* section will be more intellectually enjoyable and motivating and lay the foundations of positive mathematical dispositions. Again, taking into account the time frame, is this a researchable hypothesis?

In response to the critique of the NCTM-led reform, there have been notable efforts to bring interdisciplinary groups together to try to “balance the unbalanceable” (Sfard, 2003) and to produce consensual interpretations of research. Examples include the report *Adding It up* (Kilpatrick et al., 2001) that concentrates on elementary arithmetic and has been cited at numerous points in this chapter, and *A Research Companion to Principles and Standards for School Mathematics* (Kilpatrick et al., 2003), several relevant chapters of which we have drawn on. Further, in response to criticism of reform-based curricula, an effort has been made to demonstrate empirically their efficacy (Senk & Thompson, 2003) as reviewed in relation to the elementary curricula described earlier (but see the conclusions reached by Confrey & Stohl, 2004). Insofar as these represent a determined effect to improve the interface between research and practice, such reactions may be considered a positive by-product of the ideological conflict.

The Central Importance of Teachers and Instructional Environments

As Selter (2002, p. 222) stated (in commenting on three specific pieces of research, but clearly with

more general relevance), “It is a challenging but necessary task to transform the research and the theoretical background . . . in such a way that it can reach and have an impact on a considerable number of teachers.” As in mathematics education as a whole, there is a wide gap between what can be demonstrated by researchers who have had the benefit of many years of reflection, working in atypically supportive circumstances, and the propagation of the same effective approaches through the ranks of day-by-day teachers. As Carroll and Isaacs (2003, p. 81) put it, “reforms must take account of the working lives of teachers.”

A coherent attempt to improve the teaching of whole-number concepts and operations along the lines suggested above is possible only with a radical change in the preparation and professional development of elementary teachers (Kilpatrick et al., 2001; and see the last part of section 2), in concert with corresponding changes in the nature of curricular materials and assessment. A major obstacle to such reform is the common perception that elementary mathematics and the teaching of elementary mathematics are simple. For this reason and others, the political will to devote the necessary resources is lacking. Thus, although we may indulge here in laying out a utopian solution, it might be more effective to consider what could be achieved in reality.

Building Strong Foundations for Mathematics

If the last 100 years of school mathematics is viewed as a long-running experiment, how much more evidence do we need that radical change is required? The learning of whole-number arithmetic is foundational for a person’s mathematical education. Throughout this chapter, we have sought to emphasize ways in which this is so, including the long-term perspective that takes into account the extensions of the number concept. Further, arithmetic cannot be considered in isolation. We argued, in particular, that reconceptualizing arithmetic as inherently algebraic would serve the complementary aims of laying the groundwork for formal algebra and strengthening the understanding of arithmetic as structured. Arithmetic is foundational for other parts of mathematics—most clearly data handling.

Even more fundamentally, the early years of formal education in mathematics provide the opportunity to establish the forms of discourse, reasoning, and creativity that are definitive of mathematics—appreciation of pattern, problem solving, informal proof, use of symbolic and physical tools, modeling, communication. From this perspective, the importance of procedural competence is in no way diminished but is envisaged as a natural component of a much richer

early mathematical education. In the second section of the chapter, examples have been described that represent promising steps in actualizing such a program. (It should be acknowledged that although our emphasis has been on recent work, there are many noteworthy precursors going back several decades.)

The ideal conception of early learning of arithmetic is as an intellectually and culturally rich experience that connects with children's personal and social lives and lays a firm foundation for mathematical proficiency as defined by Kilpatrick et al. (2001), including a positive and productive disposition towards mathematics as a human activity.

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RATIONAL NUMBERS AND PROPORTIONAL REASONING

Toward a Theoretical Framework for Research

Susan J. Lamon

MARQUETTE UNIVERSITY

Traditionally, ratio and proportion have been considered middle school topics, corresponding to Piaget's theory that children achieve understanding of these topics in adolescence, but when viewed in a larger context, their tight connections with fractions and other multiplicative concepts suggest a body of knowledge that spans the entire curriculum—elementary school through university-level mathematics. Of all the topics in the school curriculum, fractions, ratios, and proportions arguably hold the distinction of being the most protracted in terms of development, the most difficult to teach, the most mathematically complex, the most cognitively challenging, the most essential to success in higher mathematics and science, and one of the most compelling research sites. In the last decade or more, researchers have made little progress in unraveling the complexities of teaching and learning these topics. Worse yet, the number of scholars pursuing long-term research agendas in the field of rational numbers is disproportionate to the mathematical richness of the domain and to the amount of research that needs to be accomplished.

Currently, research in the domain is at a turning point, a propitious time to take stock of where research in this domain has been and where it is going. With the hindsight afforded by the passage of approximately 15 years since the last handbook chapter was begun, I will look back at the major themes and

messages addressed in that chapter, and, taking into account some changes that have occurred since then, reevaluate the research area, identify persistent unresolved issues, and add to the research agenda the new questions that have arisen. This chapter will attempt to explicate the mathematical complexities of the domain so that it is more inviting to researchers who may be interested in studying related topics. I begin by revisiting the three major areas addressed by Behr, Harel, Post, and Lesh (1992): their semantic/mathematical analysis of rational numbers, their principles for qualitative reasoning, and the teaching and learning of rational number concepts.

LOOKING BACK

Behr, Harel, Post, and Lesh wrote the chapter on rational number, ratio, and proportion in the 1992 *Handbook of Research on Mathematics Teaching and Learning* shortly after the Research Agenda Project Conference, at which the leading scholars on rational number topics evaluated the state of the art and set an agenda for future research in middle school mathematics. The consensus of the research community was that “the differences between early and later number concepts need to be recognized and more

adequately understood before instructional programs can be developed to enhance number learning in the middle grades” (Sowder, 1988, p. ix). Rational number construct theory was paramount; multiple attempts were made to capture the essence of rational numbers for pedagogical purposes. Our tradition of teaching for computational ability in the industrial era had left us pedagogically bankrupt for an age that values connections and meaning, and basic questions needed exploration. What does it mean to understand the rational numbers? What are the conceptual hurdles for children as they make the transition from whole number topics to rational number topics? Thus, Behr et al. began with a discussion of the most important work of that era: Behr, Lesh, Post, and Silver (1983), Kieren (1976), and Ohlsson (1987).

These independent analyses represented different perspectives and resulted in different conclusions about how many subconstructs or personalities of the rational numbers show up at the phenomenological level (Freudenthal, 1983), that is, in real situations for which rational numbers are useful. When these analyses were reconciled, *measure*, *quotient*, *ratio*, and *operator* were recognized as distinct subconstructs (Kieren, 1988), and there was some disagreement about whether *part-whole* was distinct from the *measure* subconstruct. Whatever the number of subconstructs, the overriding issue was that current instruction, which addresses only one of them, is inadequate.

Behr et al. (1992) also presented a semantic/mathematical analysis of rational numbers and an accompanying mathematical notation system based on units and the ways in which they may be regrouped to represent the subconstructs of the rational numbers. Their formal microanalysis of the possible unit conversions did not take into account whether these operations have any basis in real problems, whether they represent mental actions that children actually perform, or whether teachers should be advised to teach every possible (reasonable) conversion. Nonetheless, this work served the authors’ stated purpose of emphasizing the critical deficiencies in the mathematics curriculum in terms of the kinds of opportunities it affords students to develop flexibility in unitizing and reunitizing quantities. (*Unitizing* refers to the process of mentally chunking or restructuring a given quantity into familiar or manageable or conveniently sized pieces in order to operate with that quantity. For example, one might think of a case of 24 colas as 24 cans, 2 (12-packs), or 4 (6-packs).)

For several reasons, the Behr et al. (1992) semantic analysis and notational system has not proved useful in ensuing research. The authors stated that “it remains for curriculum development and research to

create problem situations whose solutions would involve these manipulations and to determine whether such solutions can be created by children” (p. 305). The notion of beginning with the mathematics and inventing problems to create a purpose for the mathematics is antithetical to current conceptions of doing mathematics, especially from a mathematical modeling perspective, in which one begins with a real or realistic situation and formulates a mathematical description for the purpose of explaining, predicting, drawing conclusions, or answering questions (e.g., Lamon, Parker, & Houston, 2003).

A second objection relates to the authors’ emphasis on manipulating physical representations. The authors claimed that the manipulation of objects should facilitate the construction of the units given in their analysis (p. 305) and that “a goal of our analysis is to provide a theoretical context that will guide research into the cognition underlying students’ manipulations in transforming physical representations” (p. 315). Building upon Dienes’s (1960) principles of perceptual variability, they advocated the teaching of fractions using a model that entails multiple translations between real-world situations and written symbols, pictures, spoken symbols, and manipulative aids. In fact, Cramer (2003) and Cramer, Post, and delMas (2002) have had success with this translation model in helping students to develop understanding of initial fraction concepts. Nevertheless, Behr et al.’s (1992) formal quantity analysis of rational number constructs deals only with *discrete* representations of rational number and does not address the more important instructional question of how to facilitate children’s conceptual leap to quantities that defy physical representation, those that result from reflective abstraction, and those that do not result from counting and correspondence mechanisms, or direct measurement—the intensive or “per” quantities (Schwartz, 1988). These include such quantities as speed (miles per hour), price (cents per gram), density (grams per cubic centimeter) and tire pressure (pounds per square inch).

The second section of the Behr et al. chapter addressed the intuitive qualitative reasoning that might be applied to rational number and proportional situations, specifically, in determining fraction order or equivalence. Essentially, the concepts of equivalence and ordering of rational numbers by size rely on determining whether a multiplicative relationship varies or remains invariant. In some cases, but not all, qualitative reasoning is sufficient to determine order relationships.

A *quantity* is a measurable quality of an object—whether that quality is actually quantified or not. For example, you can compare the heights of two people

in your family without measuring them. When one is standing beside or near the other, you can tell which is taller. If you are in New York, you can safely say that Philadelphia is closer than Los Angeles without checking a mileage table to get the distance between the cities. Relating quantities that are not quantified is an important kind of reasoning that is easily accessible to children when posed in contexts that they understand (Lamon, 2006). For example:

Yesterday you shared some cookies with some friends.
Today, you share fewer cookies with more friends. Will everyone get more, less, or the same amount as they received yesterday?

Of course, you know that everyone will receive less today. Both the number of people and the number of cookies might be quantified by counting; however, even without quantifying either quantity, the question is easily answered. In fact, the question invites you to determine the direction of change between yesterday's ratio of the number of cookies to the number of people and today's ratio.

Suppose that today fewer people share more cookies. Again, you can tell that everyone would receive more than they did yesterday. Suppose that today more people shared more cookies. In this case, it could be more or less, depending on the numbers chosen.

This type of reasoning is easy for children because it builds on their prior knowledge and experience with sharing. It is, in fact, intuitive. There are nine different situations, only two of which are indeterminate, in which students can think about the change in the size of the ratio of cookies to children without quantifying either quantity and simply by knowing the direction of change in each quantity. Letting + represent an increase, −, a decrease, and ?, an indeterminate change, the nine cases are summarized in the Table 14.1.

Table 14.1 Direction of Change in a Ratio

Change in number of cookies	Change in quantity cookies per person		
	Change in number of people		
	+	−	0
+	?	+	+
−	−	?	−
0	−	+	0

This thinking is useful for some comparisons, but not sufficient for all comparisons. For example, children can rely on their intuitive knowledge to compare

$\frac{4}{7}$ and $\frac{5}{6}$ (today fewer people share more cookies), but other methods are needed to compare, say, $\frac{4}{7}$ and $\frac{5}{8}$.

Behr et al. analyzed a second type of multiplicative task, one that entails the invariance of a product. The example they used was that of the balance-beam task. However, this type of task is more difficult than a simple direct proportion (Vergnaud, 1988). The product of measures is a structure that consists of a Cartesian composition of two measure spaces into a third measure space and is useful in problems involving area, volume, Cartesian products, work, and many other physical concepts. For example, when you multiply (number of men) $\times$ (number of hours they work), you get # of man-hours; when you multiply area of the base of a right prism by height, you get neither area nor height, but rather a new measure, volume. Thus, on several accounts, this type of task is more cognitively demanding: a) It requires the coordination at least three quantities simultaneously; b) knowledge of physical principles is involved; and c) as research has consistently demonstrated (Case, 1992; Inhelder & Piaget, 1958; Marini, 1992; Siegler, 1976), children's understanding of this task and others like it (e.g., projection-of-shadow problems) shows clear developmental trends, both in terms of the M factor (Pascual-Leone, 1969), which limits the amount of information to which a child can attend, and in terms of the level of understanding of the situation.

As is common in neo-Piagetian theories of conceptual development (Case, 1992), it is important to distinguish the types of competence a task requires and the epistemic level of that competence. Any competence can be demonstrated to be present in some form at almost any age (Bruner, 1960). However, the level of functioning with that concept changes noticeably with development; that is, children in certain age groups reach a common developmental ceiling across a variety of tasks (Marini, 1992). Cross-task asynchrony occurs, however, as children solve each task independently, using whatever task-specific background they might have and heavily relying on the contextual factors present in the task.

Although people with a mature understanding of physical situations such as the balance-beam task can determine when the order relation between products is determinate or indeterminate for each of the nine change conditions given in the table above, stating these relations as principles or rules for which initial instruction should help children "construct principles to apply qualitatively to questions of order" (Behr et al., 1992, p. 322) goes well beyond what is typically classified as intuitive knowledge or "knowledge that is self-evident and obvious to the person who has it" (Resnick, 1986, p. 188). Thus, qualitative reasoning is not necessarily intuitive.

In the first handbook chapter, the authors named the clarification of concepts and standardization of vocabulary as prerequisites to progress in research on rational numbers. Their observations remain as compelling today as they were then. Specifically, we still need to “agree on the concepts of fraction and rational number” (Behr et al., 1992, p. 296). Other distinctions such as the difference between ratios and rates, and the difference between reasoning proportionally and understanding proportionality still need clarification. Another persistent issue is the confusion in naming student strategies for determining the equivalence of ratios—the *within* strategy or the *between* strategy—that resulted from the different uses of the term in early research, depending on whether the researcher came from a mathematics background or a science background. Several issues were not addressed in that chapter. What counts as proportional reasoning? What is the connection between proportional reasoning and fractions? What is the connection of proportional reasoning to the proportionality schema (Inhelder & Piaget, 1958)? As work in the field progresses, these questions continue to present themselves for clarification and will be addressed in this chapter.

In the final section of the chapter, Behr et al. discussed their original charge to address research on teaching rational numbers. They noted that, unfortunately, they were “unable to find much research that specifically targeted teaching rational number concepts” (1992, p. 300). The full significance of that statement was that the research domain including fractions, rational numbers, and ratios and proportions had not yet reached a level of maturity from which it could offer empirical propositions for teaching, that is, generalizations that derive directly from empirical findings. Nevertheless, they reported on a different kind of knowledge that informs teaching, a conceptual invention. Conceptual inventions (Shulman, 1986a) do not derive directly from findings in a straightforward way. They are acts of scholarly invention based on theoretical understanding, practical wisdom, and generalizations from empirical work. They explained Lesh’s model for translations between modes of representation for teaching initial fraction concepts and the manner in which it exploits perceptual variability to aid mathematical abstraction. In this chapter, I will report on research that has finally taken rational number concepts into the classroom.

SCOPE AND ORGANIZATION OF THIS CHAPTER

The nature of this chapter is influenced by three important developments. Chief among them is the

fact that six major contributors to rational number research have died since the last handbook chapter (including Merlyn Behr), and several other contributing scholars have retired. These were great thinkers to whom all future researchers owe a debt of gratitude. Unfortunately, new researchers, committed to long-term agendas in this research area, are not stepping up to replace those pioneers. Opportunistic treatment of rational numbers and multiplicative ideas in the course of pursuing other research topics is clearly not sufficient to make progress in this domain. The work that needs to be done is abundant enough to provide career-long research agendas for many scholars.

As the reader will note, the number of references in this chapter predating 1992 is far greater than the number appearing since the last handbook. This crisis—I do not exaggerate to call it that—stems from the complex nature of the mathematics as well as from the difficulty of conducting research because a) teachers are not prepared to teach content other than part-whole fractions; b) long-term commitment is needed because rational number topics are learned over many years; c) the nonlinear development of content does not mesh well with scope and sequence charts currently prescribing mathematics instruction in schools; and d) in comparison to a domain such as early addition and subtraction, little research progress is evident. Indeed, the rational number domain, however critical, has been perceived as a difficult research site. One goal of this chapter is to share recent exciting developments in this research area, to summarize the history and work in this field in understandable terms and to inspire new researchers to adopt personal research agendas based on multiplicative topics.

Second, the diversification of research approaches can drastically change the status of a research domain by making feasible and legitimate work that was previously impossible. In recent years, the design experiment (Brown, 1992, also deceased since the last handbook; Kelly & Lesh, 2000) has come into vogue and has opened the door to new possibilities for research in complex domains such as rational numbers. Brown originally envisioned a dynamic relationship between classroom-based and laboratory-based research, a dialectic in which substantive action aims to make changes in classroom practice but at the same time tries to discover the means to enact those changes most effectively. Just as researchers several decades ago were able to begin research in children’s thinking in addition and subtraction because of the cognitive revolution and the acceptance of Piaget’s clinical interview methods, so has the design experiment provided the same impetus to research in the rational number domain. Short-term and cross-sectional studies have not

been particularly useful in studying a conceptual domain that is subject to developmental processes. Although Piagetian interviews reveal something about the changes underlying children's thought processes, they fail to capture the dynamic processes underlying such change, including some of the fleeting transitional stages, the full range of contexts and decisions that might affect those changes, and the full range of reactions of teachers and students in real classrooms, rather than under highly controlled research conditions. The design experiment allows the researcher to influence, to study, and to adjust the course of the longitudinal study based on the unstaged and serendipitous dynamics of a real classroom.

The third significant development, related to the last, is that research in this field has met a milestone since the last handbook chapter in terms of beginning the longitudinal research in real classrooms (Lamon, in progress). A nonclassical framework for this research was based on top-down analyses of content, bottom-up investigations of student thinking, and neo-Piagetian cognitive psychology, and was implemented in two longitudinal studies using a design-experiment research paradigm. Two 4-year studies with the same children tracked their development from the start of fraction instruction in the third grade until the end of sixth grade, yielding new insights into the teaching and learning of rational numbers. Because the work in this field is reaching a new level of maturity, this is a propitious time to look back and to analyze the research field in terms of the broad stages through which it has progressed. Hopefully taking stock of where we as researchers have been from a vantage point that also allows us to see new directions for future work will give new impetus to research in this domain.

This chapter is divided into three sections, the first of which addresses some persistent language and definition issues, with the goal of promoting better communication. The second section will review the evolution of research in this field, looking at the development of rational number research in the broad stages through which work has progressed. The third section will present a theoretical framework that provided the basis for several longitudinal studies, as well as a summary of the major results of those studies. Research generally raises more questions than it answers. Therefore, the chapter will conclude with research questions that may provide inspiration for those who wish to take up the challenge of research in this field.

To keep this chapter tractable, I will restrict discussion to the content and issues most appropriately addressed in the school years 3–8, spanning the time that children begin fraction instruction until they finish middle school. Understanding fractions, ratios, and

proportions is but a small piece of the huge system of interrelated multiplicative concepts, operations, and associated contexts whose construction extends from the elementary school years through undergraduate and graduate school. However, with an eye toward later usefulness and connections, this chapter will focus on the early years because research into the learning of advanced multiplicative concepts is a moot issue when we educators are failing miserably at teaching the most elementary multiplicative concepts and operations.

Neither will this chapter address fractions and proportional reasoning within the context of inservice or preservice teacher education. Many adults, including middle school teachers (for example, Harel, Behr, Post, & Lesh, 1994; Post, Harel, Behr, & Lesh, 1988) and preservice teachers (e.g., Graeber, Tirosh, & Glover, 1989; Harel et al., 1994; Simon & Blume, 1994), struggle with the same concepts and hold the same primitive ideas and misconceptions as students do. This may be attributed to the fact that the mathematics curriculum has never appropriately addressed the multiplicative conceptual field, and teachers themselves have had the same school experiences as current students receive.

DEFINING TERMS AND CLARIFYING CONSTRUCTS

Several mathematical constructs are ill-defined, and their relationships to one another have become obfuscated as terminology has been used indiscriminately. In other cases, concepts and language need to be periodically updated because meanings have evolved as part of the natural development of a research area. In this section, I will discuss *within* and *between* strategies, the difference between ratios and rates, the difference between fractions and rational numbers, what it means to understand fractions as numbers, the difference between proportional reasoning and proportionality, and the connection between the rational numbers and proportional reasoning.

Within (Internal) and Between (External) Ratios

Freudenthal (1973, 1978) classified a ratio as an internal or within ratio if its constituent magnitudes shared the same measure space, or in Freudenthal's terminology, came from the same "system." Similarly, an external or between ratio is composed of magnitudes from different measure spaces.

The two ratios in the following proportion involve an internal comparison by Freudenthal's definition:

$$4 \text{ people} : 20 \text{ people} = 1 \text{ car} : 5 \text{ cars}$$

whereas this proportion uses external or between ratios:

$$4 \text{ people} : 1 \text{ car} = 20 \text{ people} : 5 \text{ cars}.$$

Confusion arose, however, because research originating in a science tradition (Karplus, Pulos, & Stage, 1983a, 1983b; Noelting 1980a, 1980b) used an alternate definition of a system: a set of interacting elements. According to this alternate conception, the example above shows two systems with people and cars interacting within each of them. The scientist's notion of internal ratio (corresponding to Freudenthal's external ratio) is a comparison of elements within one scientific state or system. An external ratio is one involving elements from two different systems. Thus, the following proportions involve internal and external ratios, respectively, according to the scientist's definition:

$$\begin{aligned} 1 \text{ mile} : 4 \text{ hours} &= 3 \text{ miles} : 12 \text{ hours} \\ 1 \text{ mile} : 3 \text{ miles} &= 4 \text{ hours} : 12 \text{ hours}. \end{aligned}$$

The confusion is easily eliminated by using the terminology "within or between *systems*" or "within or between *measure spaces*." For example, consider the problem:

Richard buys 4 cakes for \$8.95 each.
How much does he have to pay?

You could analyze the problem using, say, Vergnaud's (1988) terminology, *isomorphism of measures* where Measure 1 (M1) refers to numbers of cakes and Measure 2 (M2) refers to costs. Measure spaces usually refer to different sets of objects, different types of quantities, or different units of measure.

M ₁	M ₂
1	\$8.95
4	\$ <i>x</i>

In this case, $1 : 4 = \$8.95 : \x is a proportion whose ratios originate *within measure spaces*.

If you compare the "oranginess" of the liquid in pitchers containing cans of orange concentrate and cans of water, such as the following:

Pitcher 1	Pitcher 2
3 o.j.	5 o.j.
5 H ₂ O	11 H ₂ O

then ratios comparing parts orange concentrate to parts water, such as 3 o.j. : 5 H₂O and 5 o.j. : 11 H₂O are *within-system ratios*.

Ratios and Rates

In the last decade, progress has been made in more clearly specifying the essential differences between ratios and rates. Early definitions of ratio and rate were linked to comparisons *within* and *between* measure spaces. A ratio was considered a comparison between like quantities (e.g., pounds : pounds) and a rate a comparison of unlike quantities (e.g. distance : time), although, as noted by Lesh, Post, and Behr (1988), there was "disagreement about the essential characteristics" that distinguish the two. Ohlsson (1988), for example, defined a ratio as a numerical expression of how much there is of one quantity in relation to another quantity, and a rate as a ratio between a quantity and a period of time.

Schwartz's (1988) distinction between ratio and rate was based upon the nature of the entity whose attribute is described by a multiplicative comparison. If the material is homogeneous (such as a mixture) or a speed is constant, then the quantity (a rate) characterizes the material or the trip as a whole, but if the referent is not homogeneous, the quantity is a more local property of the referent entity and is a ratio. Building on this distinction, Kaput and West (1994) associated ratios with the accumulation of attributes and rates with homogeneity of attributes. A ratio may be iterated in a series of static and independent instances, describing a local property of the referent, so that amounts of each quantity composing it accumulate. An example might be the *building up* strategy often observed in children's work: 2 pencils for \$.10, 4 pencils for \$.20, 6 pencils for \$.30, and so on. In a rate, however, constituent elements always exist in a constant ratio to one another, regardless of how much of the original entity is sampled. For example, if you take samples of a mixture such as paint or orange juice, they are always composed of constituent elements in the same ratio to one another, regardless of the amount of the mixture sampled.

After working extensively with teachers and students on the concept of speed (Thompson & Thompson, 1994, 1996), P. Thompson (1994) suggested that the inherent difference lies not so much in the situations and contexts per se, as in the mental operations by which people apprehend rate and ratio situations. He defined a rate as a "reflectively abstracted constant ratio" (p. 192). This means that one conceives a ratio when he/she conceives two quantities in a multiplicative comparison. However, a ratio *becomes* a rate when that person is able to conceive that the ratio applies

beyond the particular situation in which it was conceived and can think about it as a characteristic of a whole class of covarying quantities. Thompson's distinction between ratio and rate is attractive for those who study children's thinking because, in a neo-Piagetian sense, it applies the words to levels of epistemic competence.

Fractions and Rational Numbers

Often the word *fraction* is used when rational number is intended, and vice versa. The word *fraction* is used in a variety of ways inside the classroom as well as outside the classroom. The many uses of the word are bound to cause confusion. In particular, the word *fraction* has several meanings, not all of which are mathematical. For example, a fraction might be a piece of undeveloped land, whereas in church, it would refer to the breaking of the Eucharistic bread. In the statement "All but a fraction of the townspeople voted in the presidential election," the word *fraction* means *a small part*. When you hear that "the stock rose fractionally," it means *less than one dollar*. In math class, students are disconcerted by the need to learn the technical definition of a part-whole fraction when they already know from their everyday experience of the word *fraction* that it means *any little bit*. When a fraction such as $\frac{4}{3}$ refers to more than one whole unit, the interpretation *a little bit* does not apply very well.

The colloquial usage of the word is complicated enough, but multiple uses of the word in mathematics education are also problematic. Some people use the word *fraction* to refer specifically to a part-whole comparison. It may also refer to any number written in the symbolic form $\frac{a}{b}$. Often the words *fraction* and *rational number* are used interchangeably.

What are fractions? First, *fractions* are bipartite symbols, a certain form for writing numbers: $\frac{a}{b}$. This sense of the word *fraction* refers to a notational system, a symbol, two integers written with a bar between them. Second, *fractions* are non-negative rational numbers. That is, in the set of rational numbers, $\frac{a}{b}$, a and b may be positive or negative integers, with the restriction that $b \neq 0$. Traditionally, because students begin to study fractions long before they are introduced to the integers, a and b are restricted to the set of whole numbers, so that fractions are only a subset of the rational numbers.

I reject the use of the word *fraction* as referring exclusively to one of the interpretations of the rational numbers, namely, part-whole comparisons. Because the part-whole comparison was the only meaning ever used in instruction, it is understandable that *fraction* and *part-whole fraction* became synonymous. However, educators now realize that restricting instruction to

the part-whole interpretation has left students with an impoverished notion of the rational numbers, and increasingly, teachers are becoming aware of the alternate interpretations or subconstructs of the rational numbers and referring to them as *operator*, *measure*, *ratio*, and *quotient*. Part-whole comparisons are on equal ground with the other interpretations and no longer merit the distinction of being synonymous with *fractions*.

Therefore, the terms *fractions* and *rational numbers* are not synonymous and it is more accurate to think of fractions as a subset of the rational numbers. Eventually, it is important for students to understand the full set of rational numbers as well as to grasp the following distinctions:

- All rational numbers may be written in fraction form.

$$\frac{3}{4}, \frac{\sqrt{4}}{3} \text{ (usually written as } \frac{2}{3}\text{),}$$

$$\frac{2.1}{4.1} \text{ (usually written as } \frac{21}{41}\text{) and}$$

$$\frac{1}{\frac{1}{4}} \text{ (usually written as } \frac{4}{1}\text{)}$$

are all fractions and rational numbers.

- All numbers written in fraction form are not rational.
 $\frac{\pi}{2}$ is not a rational number although it is written in fraction form.
- Each fraction does not correspond to a different rational number.

A different rational number does not exist for each of the three fractions $\frac{2}{3}$, $\frac{6}{9}$, and $\frac{10}{15}$. Just as one and the same woman might be addressed as Mrs. Jones, Mom, Mother, Maggie, Dear, Aunt Meg, and Margaret, these fractions are different numerals designating the same rational number. A single rational number underlies all of the equivalent forms of a fraction.

- Rational numbers may be written in fraction form, but they may be written in other forms as well.
- Terminating decimals are rational numbers. Nonterminating, repeating decimals are rational numbers. Percents are rational numbers. Nonterminating, nonrepeating decimals are not rational numbers.

Fractions as Numbers

When I speak of a fraction as a number, I am really referring to the underlying rational number. Understanding a fraction as a number entails realizing, for example, that $\frac{1}{4}$ refers to the same relative amount in each of the following pictures. There is but one rational number underlying all of these relative amounts.

Whether I call it $\frac{1}{4}$, $\frac{4}{16}$, $\frac{3}{12}$, or $\frac{2}{8}$ is not as important as the fact that a single relationship is conveyed. When considering fractions as single numbers, rather than focusing on the two parts used to write the fraction symbol, the focus is on the relative amount conveyed by those symbols (Figure 14.1).

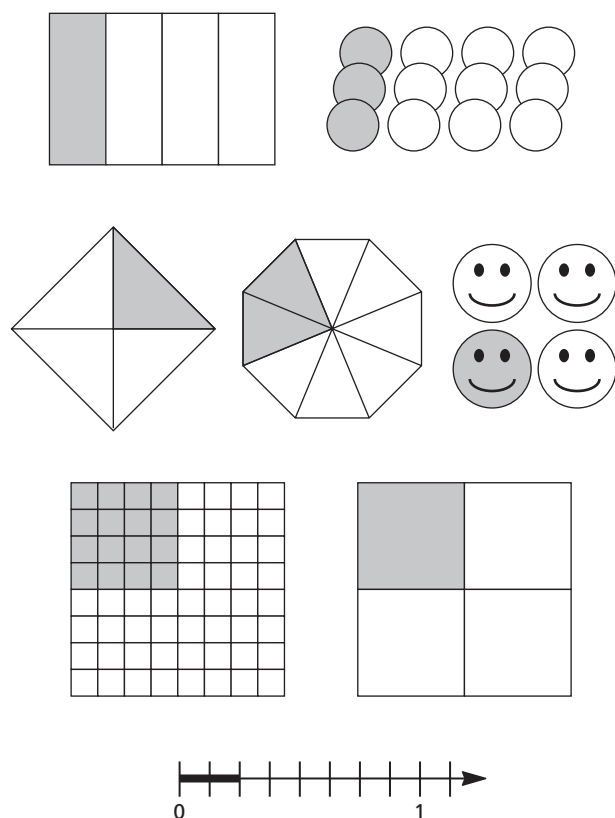


Figure 14.1 Illustrations for a single rational number.

Regardless of the size of the pieces, their color, their shape, their arrangement, or any other physical characteristic, the same relative amount and the same rational number is indicated in each picture. Nevertheless, psychologically, for the purpose of instruction, when fractions are connected with pictorial representations, which fraction name you connect with which picture *is* an important issue. For example, you would not call the first picture $\frac{2}{8}$ or $\frac{4}{16}$.

Understanding Rational Numbers

The rational numbers and proportionality are logical mathematical constructs. These constructs are not terribly useful for instructional and learning purposes. Knowing rational numbers is certainly more than reciting the axioms of an ordered quotient field. Similarly, the definition of *proportionality* as a condition in which two quantities or variables are related in a linear manner offers little information as to what

it might mean to understand proportionality. These two constructs are not disjoint. The rational numbers have a proportional nature; that is, within each equivalence class, each element is a constant multiple of every other. Nevertheless, as expected, understanding the rational numbers and understanding proportionality are not equivalent.

The work of Kieren (1976, 1980, 1983, 1993) and Behr et al. (1983) has highlighted five subconstructs of the rational numbers: part-whole, measure, quotient, operator, and ratio. These subconstructs suggest learning sites from which children might gain insights into the essential nature of rational numbers. Freudenthal (1983) called fractions the “phenomenological source of the rational number” (p. 134). The didactical phenomenology are a set of situations in which important mathematical ideas are grounded in real phenomena; these sites provide the context in which the learning of the mathematics takes place so that both the meaningfulness and the utility of the mathematical ideas are ensured. They are points at which a student may enter into the learning process and reconstruct important ideas for himself because the real phenomena he is exploring beg to be organized by the mathematics of interest. Although much needs to be done in this direction, in recent years progress has been made in building didactical phenomenology (Lamon, 2006).

When educators say that the goal of fraction instruction is to *understand the rational numbers*, we mean that by providing students with experiences with non-negative rational numbers (fractions), we want them to recognize nuances in meaning; to associate each meaning with appropriate situations and operations; and, in general, to develop insight, comfort, and flexibility in dealing with the rational numbers. The goals of fraction instruction are best summarized in the language of number sense (Greeno, 1991; Sowder, 1989). Students who have developed *rational number sense* have an intuitive feel for the relative sizes of rational numbers and the ability to estimate, to think qualitatively and multiplicatively, to solve proportions and to solve problems, to move flexibly between interpretations and representations, to make sense, and to make sound decisions and reasonable judgments. In general, they should be comfortable in reasoning, computing, and problem solving in the domain of rational numbers.

What Is Proportional Reasoning?

Educational researchers have always perceived that something called *proportional reasoning* is a consequence of understanding the nature of rational numbers. Unsurprisingly, for the purposes of textbook in-

struction, large-scale assessment (Lamon, 1988), and for research (Tourniaire & Pulos, 1985) the domain of ratio and proportion has been defined traditionally in terms of two problem types similar to fraction comparison and equivalence tasks: comparison problems and missing-value problems. In a comparison problem, four values are given (a , b , c , and d) and the goal is to determine the order relation between the ratios $\frac{a}{b}$ and $\frac{c}{d}$: $\frac{a}{b} \{<>=\} \frac{c}{d}$. The following example is a typical ratio comparison problem:

John makes lemonade concentrate by using 3 spoonfuls of sugar and 12 spoonfuls of lemon juice. Mary makes concentrate by using 5 spoonfuls of sugar and 20 spoonfuls of lemon juice. Whose lemonade concentrate is sweeter, John's or Mary's, or do they taste the same? (Karplus et al., 1983b, p. 53)

A missing-value problem provides three of the four values in the proportion $\frac{a}{b} = \frac{c}{d}$ and the goal is to find the missing value. This is a typical missing-value problem:

John makes lemonade concentrate by using 3 spoonfuls of sugar and 12 spoonfuls of lemon juice. How much lemon juice would Mary need with 5 spoonfuls of sugar to make her concentrate taste just like John's? (Karplus et al., 1983b, pp. 53–54)

In spite of the implicit definition of the domain of ratio and proportion in terms of these two problem types, the terms *proportionality* and *proportional reasoning* are often used interchangeably (e.g., Karplus et al., 1983b). As a result, both have become ill-defined umbrella terms, referring to anything and everything related to ratio and proportion. In the 1989 Standards, the National Council of Teachers of Mathematics claimed that *reasoning proportionally* “is of such great importance that it merits whatever time and effort must be expended to assure its careful development” (p. 82), and in 2000, the Council claimed that *proportionality* “is an important integrative thread that connects many of the mathematics topics studied in grades 6–8” (p. 217). Although no suggestions were offered for teaching either proportional reasoning or proportionality, these strong statements express a pervasive sense that this reasoning ability is critical to mathematical and scientific thinking, and a sense of urgency about the consistent failure of students and adults to reason proportionally. Even using the basic definition of understanding the structural relationships in comparison and missing-value problems, my own estimate is that more than 90% of adults do not reason proportionally—compelling evidence that this reasoning process entails more than

developmental processes and that instruction must play an active role in its emergence. Yet, without appropriate instructional goals, purposeful teaching for well-structured knowledge is impossible. Clearly, our failure to define terms is a deterrent to progress in this research domain.

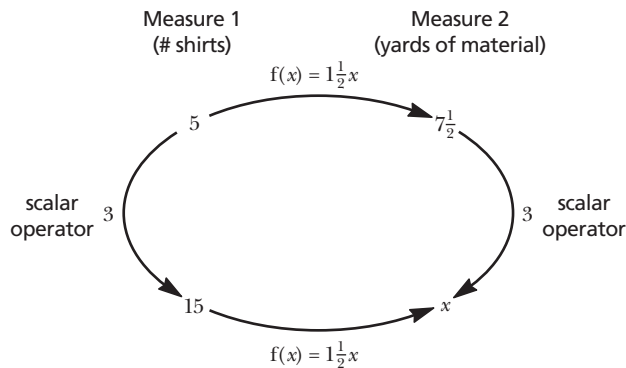
Many scholars share the view that proportional reasoning is a long-term developmental process in which the understanding at one level forms a foundation for higher levels of understanding (Case, 1985, 1992; Hart, 1981, 1984; Inhelder & Piaget, 1958; Kieren, 1988, 1993; Lesh et al., 1988; Noelting, 1980a, 1980b). “The domain represents a critical juncture at which many types of mathematical knowledge are called into play and a point beyond which a student's understanding in the mathematical sciences will be greatly hampered if the conceptual coordination of all the contributing domains is not attained” (Lamon, 1994, p. 90). The ability to solve comparison and missing-value problems is assumed knowledge in the secondary school curriculum (Post, 1986) and as the knowledge of multiplicative topics deepens through the study of algebra, geometry, trigonometry, statistics, and sciences in the secondary curriculum, it becomes the basis for understanding fields of application such as inflation, force, density, the physics of lenses, and the physics of sound (Gray, 1979). This view is consistent with other philosophical perspectives (Bohm & Peat, 1987) and mechanisms described in computer science and psycholinguistics (Pinker, 1997) that view the mind as a recursive processor, constantly embedding knowledge in progressively more complex thoughts.

This perspective suggests that proportional reasoning, defined by one's ability to understand the structural relationships in comparison and missing-value problems, might be conceived as a necessary but not sufficient prerequisite to understanding proportionality. The issue becomes one of where to draw lines (benchmarks) in a long-term process that is not easily segmented; for example, others have referred to some stages of the reasoning process as preproportional reasoning (Inhelder & Piaget, 1958), protoratio reasoning (Resnick & Singer, 1993), ethnomathematical knowledge (Kieren, 1988, 1993), and theorems-in-action (Vergnaud, 1983, 1988). One approach is to define proportional reasoning in terms that present reasonable expectations for students by the end of middle school, keeping in mind that the study of variables, functions, linear equations, vectors, and other topics studied in high school mathematics will continue to broaden and deepen their perspectives on multiplicative relationships. Therefore, I propose that *proportional reasoning* means supplying reasons in

support of claims made about the structural relationships among four quantities, (say a, b, c, d) in a context simultaneously involving covariance of quantities and invariance of ratios or products; this would consist of the ability to discern a multiplicative relationship between two quantities as well as the ability to extend the same relationship to other pairs of quantities.

The critical structural relationships, both scalar and functional, are best illustrated using Vergnaud's (1983) framework for analyzing multiplicative structures. The following example shows a simple direct proportion between two measure spaces. A linear functional relationship exists between corresponding elements of the measure spaces, and a scalar operator transforms quantities of the same type.

If Maria can sew 5 team shirts with $7\frac{1}{2}$ yards of material, how many yards will she need to make a T-shirt for each of the 15 boys on the soccer team?



The ability to give correct answers is no guarantee that proportional reasoning is taking place. Often, proportions may be solved using mechanical knowledge about equivalent fractions or about numerical relationships, or by applying algorithmic procedures (for example, the cross-multiply rule) that circumvent the use of the constant of proportionality.

Which is sweeter, a mixture of 2 teaspoons of sugar with 6 teaspoons of lemon juice, or a mixture of 8 teaspoons of sugar with 24 teaspoons of lemon juice?

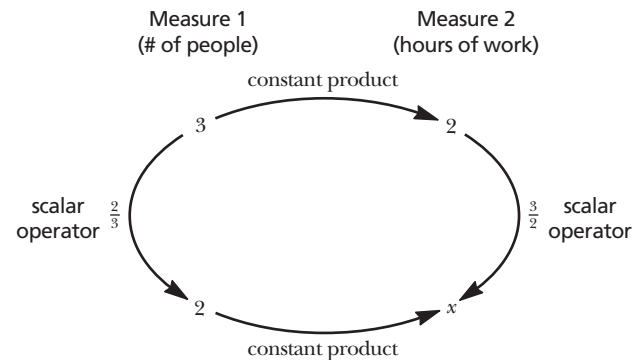
Student: They are the same because 8 goes into 24 three times and 2 goes into 6 three times. (Karplus et al., 1983b, p. 55).

In such contexts, multiplicative explanations have “counted” as proportional reasoning; however an observer cannot say much about these students’ understanding of proportionality. Indeed, most adults do not associate comparison problems with trying to find out which situation has the greater constant of proportionality. As researchers, we must conclude that

proportionality is a much larger construct than proportional reasoning.

The *reasoning* aspect of proportional reasoning entails recognition of the constant ratio between elements of the same measure space and recognition of the functional relationship between measure spaces (recognition of the constant ratio $M_2 : M_1 = 3 : 2$). In the case of a simple inversely proportional relationship, recognizing the structural aspects of the situation entails understanding that there are two scalar operators, one of which is the multiplicative inverse of the other, and that product of corresponding measures is constant.

If 3 people can mow and trim a lawn in 2 hours, how long will it take 2 people to do the same work?



Understanding Proportionality

Proportionality is a mathematical construct referring to the condition or the underlying structure of a situation in which a special invariant (constant) relationship exists between two covarying quantities (quantities that are linked and changing together). In this section, I explore some of the critical aspects of understanding proportionality, both mathematical and psychological, arguing that it is a larger construct than that of proportional reasoning.

The constant of proportionality. The mathematical model for proportional relationships is a linear function of the form $y = kx$, where k is called the constant of proportionality. Thus, y is a constant multiple of x . Equivalently, two quantities are proportional when they vary in such a way that they maintain a constant ratio: $\frac{y}{x} = k$. The constant k plays an essential role in understanding proportionality.

Pedagogically speaking, k is a slippery character, because its guise changes in each particular context and representation involving proportional relationships. It frequently does not appear explicitly in the context in which it plays a role, but rather is a structural element waiting to be discovered beneath the obvious details. In symbols, it is a constant. In a graph, it

is the slope. In a tabular representation, it may be the difference between any entry and the one before it,

# of stacked wooden cubes	1	2	3	4	5	6
height of the stack in inches	3	6	9	12	15	18

or, equivalently, it may be the rate at which one quantity changes with respect to the other expressed as a unit rate.

# of stacked wooden cubes	2	5	9	12	15
height of the stack in inches	6	15	27	36	45

In general, in rate situations, k is the constant rate. In reading maps, it is the scale. In shrinking or enlarging contexts, or in similar figures, it is the scale factor. It may be a percentage if you are discussing sales tax, or a theoretical probability if you are rolling dice. These examples suggest that the interpretation of the constant of proportionality is situation-specific.

Even when a student reasons proportionally, the task of interpreting the constant of proportionality in relatively simple contexts can be challenging. For example, consider this recent discussion that I had with a calculus student (S) as he struggled with many issues related to number versus quantity, working and interpreting within a context, and understanding the constant of proportionality. He was considering the following problem:

Suppose that you are going to take a trip by car on interstate highways. Estimate the amount of gas you will need.

S: I know that the amount of gas I will need (let's say g) depends on the length of the trip I am taking (let's call it d for distance). In fact, if I drive 1000 miles, I will use about twice as much gas as if I drive 500 miles and about three times as much if I drive 1500 miles. So I know that the amount of gas I use (g) is proportional to my distance traveled:

$$g \propto d.$$

This relationship means that g is a constant multiple of d , so I can write

$$g = kd.$$

What is that constant k ? I suspect that it has something to do with my car's fuel economy. I know that on the highway, my car's fuel economy rating is 30 mpg, meaning that it can go about 30 miles per gallon of gas. So I said my equation

would be $g = 30d$. But here is where I get stuck. That equation doesn't make sense because if I plug in 1 mile for my distance, it says I need 30 gallons.

I: Tell me the way you *think* it works.

S: The amount of gas it takes is your miles divided by 30, which is $\frac{1}{30}$ times the number of miles. So I would say $g = \frac{1}{30}d$. So it looks like $k = \frac{1}{30}$. But how was I supposed to know that?

I: Go back to the equation $g = kd$. Solve for k .

S: $k = \frac{g}{d}$

I: What label is on that quantity?

S: Gas over distance—that's gallons over miles. OK! Let me back up. I can write 30 mpg as $\frac{30 \text{ miles}}{1 \text{ gallon}}$. But that's the same as $\frac{1 \text{ gallon}}{30 \text{ miles}}$. So my equation has to be $g = \frac{1}{30}d$.

I: So what does $k = 30$ mpg mean?

S: To drive a distance of 1 mile, I need about $\frac{1}{30}$ gallon of gas.

This scenario suggests that students should be able to

- express the meanings of quantities and variables and the constant of proportionality in the contexts in which they are used.

Additionally, it seems reasonable to expect that an understanding of proportionality entails

- the ability to use proportionality as a mathematical model to organize in appropriate real-world contexts;
- the ability to distinguish situations in which proportionality is not an appropriate mathematical model from situations in which it is useful;
- development and use of the language of proportionality;
- use of functions to express the covariation of 2 quantities;
- the ability to explain the difference between functions of the form $y = mx$ and functions of the form $y = mx + b$. In the latter function, y is not proportional to x , but rather, Δy is proportional to Δx .
- knowing that the graph of a direct proportional situation is a straight line through the origin;
- knowing that the graph of $y = mx + b$ is a straight line intersecting the y axis b units above the origin;
- the ability to distinguish different types of proportionality and to associate each of them with appropriate real-world situations in which they are applicable: direct proportions,

inverse proportions, square proportionalities ($y = kx^2$), and cubic proportionalities ($y = kx^3$);

- knowing that k is the constant ratio between two quantities in a direct proportional situation;
- knowing that k , the constant in an inversely proportional situation, is simply the product of any pair of values of the two quantities;
- knowing that the graph of an inversely proportional situation is a hyperbola.

The Proportionality and Proportional Reasoning Hypothesis

Educators are accustomed to mentioning *rational numbers*, *ratios*, and *proportions* together in the same phrase simply because the terms are so close mathematically and psychologically. Proportional reasoning develops (or should develop) as one studies fractions, a subset of the rational numbers. Thus, proportional reasoning is, to some degree, an indicator of one's rational number sense. However, understanding proportionality is a far more demanding goal than either rational number sense or proportional reasoning. In particular, the ability to use proportionality as a mathematical model of real world situations develops only as one studies higher mathematics and science.

Few would disagree that proportions have always played a key role in quantitative relations in science. Furthermore, recall that Piaget and his colleagues (Inhelder & Piaget, 1958) referred to the "proportionality schema" (p. 173). In general, a schema has always been taken to mean something more complex and encompassing than the phenomenon defined above as *proportional reasoning*. Using clinical interview methods, the Genevans investigated children's development of the proportionality schema in complex situations (e.g., within physical systems such as the balance beam or with an oscillating pendulum). Later, when researchers wished to work in classrooms and test larger numbers of students, they developed tasks amenable to paper-and-pencil large-scale testing (see Karplus, Karplus, Formisano, & Paulsen, 1979; Karplus et al., 1983b). Piaget's tasks were criticized because they required some knowledge of physical principles in addition to proportional reasoning, and tasks such as orange juice and lemonade mixture tasks and Mr. Tall and Mr. Short emerged. Notwithstanding the confusing use of the terms *proportionality* and *proportional reasoning*, implicit in the kinds of tasks that have been devised to study proportional reasoning is an effort to draw the line long before what researchers might call *application problems*.

My guess is that Piaget and his colleagues were correct and that proportionality is a much larger con-

cept than proportional reasoning as we have come to know it. Due to the context-bound interpretations illustrated above, understanding proportionality is tied to investigating and discerning principles and regularities in problem situations before using proportions to express those relationships symbolically, and then interpreting the symbols and what one has learned in terms of the specific context to which they apply (as in the gas consumption problem). Mathematicians recognize this process as mathematical modeling.

My hypothesis is that proportions arise in the study of rational numbers as a natural expression of their equivalence and that as one develops rational number sense through various experiences with many personalities of the rational numbers, one learns to reason proportionally. However, understanding the larger concept of proportionality comes about later, through interaction with mathematical and scientific systems that involve the invariance of a ratio or of a product.

EVOLUTION OF THE FIELD

In any research domain, it is useful to step back periodically and consider the state of the field. In this section, I analyze the evolution of research in the domain of rational numbers and ratio and proportion. In every field of endeavor, research tends to proceed through regular stages, each of which produces distinctive types of research, fueled by beliefs, values, questions, and approaches shared by a scholarly community. Looking at the field in developmental stages provides a useful organization for those who may be considering a research agenda in the field, and at the same time, it foretells "next steps." The following broad stages, named for their characteristic types of research, do not necessarily proceed sequentially; sometimes parallel development occurs: empirical analysis, rational task analysis, clinical interviews, interventions, and longitudinal studies. To borrow a metaphor, this section will not be concerned with rebuilding brick by brick, as much as it will expose the current edifice (Begle & Gibb, 1980). Therefore, rather than summarizing extant literature, only the contributions that have most directly influenced current work will be reviewed.

Empirical Task Analysis: Defining the Problem

Standardized tests, especially those that compare student performance to national and international audiences, are often a stimulus for research and

change. The tests, notwithstanding criticism of their validity and reliability, identify areas in which student performance is weak. Ensuing research examines student performance to better understand the nature and extent of children's preconceptions and misconceptions, often verifying the results in a broad range of contexts with a variety of age groups. In general, the initial focus is on children's incompetencies, on what they *cannot* do. This research tends not to offer explanations, but rather results in conjectures about the variables that affect problem difficulty.

Almost 50 years ago, Inhelder and Piaget (1958) drew attention to proportional reasoning when they designated it the hallmark of their formal operations stage of development. For a long time, research was piecemeal; it was conducted in several different disciplines, including science education, psychology, and mathematics education, and it lacked an underlying framework. Years after Piaget and his associates posited the centrality of proportional reasoning in their developmental theory, a large-scale study of science education in 19 countries (Comber & Keeves, 1973) showed a low success rate on questions involving proportional reasoning, and many researchers began to study the phenomenon (e.g., Karplus et al., 1979; Karplus, Karplus, & Wollman, 1974; Karplus et al., 1983a, 1983b; Noelting, 1980a, 1980b; Wollman, 1974). In 1979, the Rational Number Project (Behr, Lesh, & Post, 1984; Behr, Lesh, Post, & Silver, 1979) began to investigate fractions, ratios, and proportions from a mathematics education perspective.

A large body of research, roughly from 1970 to 1985, produced an extensive inventory of factors that influence the difficulty of proportion problems. Among these are context (Tourniaire, 1983, 1986); how familiar students are with using proportions in a given context (Tourniaire, 1983); the location of the missing element in a proportion in relation to the other three numbers (Bezuk, 1986; Rupley, 1981); whether a problem concerns discrete or continuous quantities (Behr et al., 1983; Pulos, Karplus, & Stage, 1981); whether the number to be found is the largest of the four terms (Abramowitz, 1974; Rupley, 1981); the presence of integral ratios as opposed to nonintegral ratios (Bezuk, 1986; Hart, 1984, 1988; Karplus et al., 1983b; Noelting, 1980a, 1980b; Rupley, 1981); the presence of unit ratios, especially 1:2 (Hart, 1981, 1988; Noelting, 1980a, 1980b); and whether perceptual cues are consistent or inconsistent (Behr et al., 1983; Cramer, Post, & Behr, 1989; Lesh, Landau, & Hamilton, 1983; Novillis, 1976). In addition, researchers noted that children frequently use additive strategies where multiplicative comparisons are required (Hart, 1981, 1988; Karplus et al., 1983b). Many chil-

dren lack understanding of the conventional notation (e.g., 12:1) for expressing a ratio (Silver, 1981), and, in general, they show an overdependence on standard textbook representations when trying to model a given situation (Lesh et al., 1983; Silver, 1981). Schwartz (1987) found that multiplication and division problems involving intensive quantities (for example, "per" quantities) are difficult, even for above-average 12th-grade students. The mode of representation used by students is distinctly problem-specific and in some rational number tasks, when students were encouraged to use concrete aids alone, performance declined (Lesh et al., 1983).

Children experience difficulties linking underlying problem structure and operation when presented with problems in multiplication and division. Faced with problems involving the same content, they change their minds about the operation needed to solve the problem, depending on the numerical data used. Greer (1987) called this phenomenon *nonconservation of operations* (p. 37). A line of research in multiplication and division, the collective work of Bell, Swan, and Taylor (1981), Fischbein, Deri, Nello, and Marino (1985); Bell, Fischbein, and Greer (1984); Luke (1988); and Graeber et al. (1989), hypothesized that the fundamental operations of arithmetic are linked to implicit, unconscious, and primitive intuitive models that constrain students' ability to predict appropriate operations and that these models lead to the assumption that multiplication makes bigger and division makes smaller.

Rational Task Analysis: Understanding the Content

Researchers use empirical analysis to study children's mathematics, that is, children's actual performance on tasks, but they use rational task analysis to examine the content from a mature mathematical perspective, making assumptions about the ways of thinking that are necessary to solve problems. On a theoretical level, rational task analysis identifies the essential mathematical components of the domain, the cognitive objectives of instruction, both conceptual and procedural, with an eye toward the future usefulness of those ideas in higher mathematics. Often, in the process of carefully analyzing the mathematics, implicit processes become explicit for the first time, and a vocabulary develops by which researchers can discuss the various phenomena they are studying. Especially in a domain where the mathematics is so complex this stage is critical for devising a framework by which the research can proceed. That framework provides a rationale that helps the researcher to ask

appropriate questions, to devise activities and problems, and to interpret children's thinking.

By the late 1970s, researchers were convinced of the complexity of the connections among rational numbers, proportional reasoning, and many other mathematical concepts of a multiplicative nature. A number of task analyses examined the huge domain from many different angles and enhanced our understanding of the mathematics and its cognitive demands.

Kieren (1976, 1980) proposed that to understand rational numbers, one must have adequate experience with multiple interpretations, not merely with fractions as objects of computation. For five major interpretations or personalities of rational numbers—part-whole fractions, ratios, operators, quotients, and measures—he analyzed the kinds of mathematical understanding unique to each, the mechanisms such as equivalence and partitioning that help to build competence in each interpretation, and some classroom activities appropriate for building knowledge of each interpretation. Other analyses (e.g., Behr et al, 1983; Ohlsson, 1987, 1988) resulted in different numbers of subconstructs, but, whether one admits five or six interpretations or more, the compelling issue was that the mathematics curriculum has not been providing an adequate foundation for understanding rational numbers.

Freudenthal's (1983) didactical phenomenology helped researchers to relate very complex mathematical ideas (among these, the personalities of rational numbers) to the mental objects, human activities, and real phenomena that are appropriately organized by those mathematical ideas. He suggested many everyday examples accessible to children through which instruction could provide an inroad to the world of rational numbers, from what he described as a "wealth," a veritable "ocean" of examples, "a source that never dries up" (p. 134).

Another contribution was the introductory chapter to a monograph summarizing the presentations at the National Council of Teachers of Mathematics Research Agenda Project Conference on concepts and operations in the middle grades. Hiebert and Behr's (1988) synthesis of the themes of the conference and the status of research on mathematical learning in the middle grades was, in its own right, an expert treatise on the mathematical and psychological dimensions of change in moving from whole numbers to rational numbers.

Schwartz (1976, 1987, 1988) and Kaput (1985) emphasized the role of the mathematics of quantity in multiplicative operations and enhanced researchers' understanding of the cognitive complexity of those operations. Their semantic analysis, which dis-

tinguished extensive and intensive quantities, emphasized the need to always link numbers with their referents and to carefully consider the *relation* between the referents in a problem. An intensive quantity is an abstraction from the comparison of two other quantities. It is not easily attained because its referent is different from that of either entity that composes it, it has no adequate representation, and operations with such quantities do not conform well to dominant cognitive models of multiplication as repeated addition or of (partitive) division as equal sharing.

Vergnaud (1983, 1988) introduced the construct of a *conceptual field*. The multiplicative conceptual field refers to a complex system of interrelated concepts, students' ideas (both competencies and misconceptions), procedures, problems, representations, objects, properties, and relationships that cannot be studied in isolation. The content of the multiplicative conceptual field is extensive, including such topics as multiplication, division, fractions, ratios, simple and multiple proportions, rational numbers, dimensional analysis, and vector spaces, and understanding of this field develops over many years. This construct suggests that learning is not linear, that it is content-based, and that a piece-by-piece conception of research will prove to be inadequate because in a web-like model of the structure of knowledge, contact with one strand will have ramifications across the entire structure. These views are consistent with the conclusions of others (e.g., Bereiter, 1990; Lesh & Lamon, 1992) who have argued that research capable of informing the teaching and learning enterprise in schools (complex systems) must take into account the full complexity of the content, and the classroom, and the children's worlds of experience.

Clinical Interviews: Understanding Students' Thinking

Whereas analyses of rational number knowledge, such as those just discussed, begin with the mathematics and view the content as a prelude to more advanced mathematical concepts, a bottom-up approach to research in the domain begins with children and the knowledge they bring to the task of learning rational numbers. Clinical interviews deeply probe not only children's current understanding of a topic, but how the researcher might affect change in that understanding. Beginning with questions based upon a conceptual analysis, the researcher opportunistically revises or devises questions as the child's primitive conceptions or misconceptions surface and thereby can study the student's constructions in the face of these improvisations. Assuming that the small number

of children who are interviewed demonstrate thinking that is characteristic of the larger population, clinical interviews fuel theories about how certain ideas develop and about how material might be psychologized for instruction. Two kinds of research using clinical interviews have been influential in rational number research: those that study children's construction of knowledge and studies that seek to identify children's competencies, useful intuitive knowledge, and informal strategies.

Steffe and his colleagues study the functioning schemes (repeatable, generalizable actions) that children bring to a specific task and how the children adapt their scheme in response to the specific task at hand. On the basis of a careful study of counting and number sequences, Steffe has articulated a theory concerning the way in which children's formation and use of units is progressively elaborated from early counting through multiplication (Steffe, 1988, 1992; Steffe & Cobb, 1988; Steffe, von Glasersfeld, Richards, & Cobb, 1983). The centrality of the unit in fraction instruction, especially the role of composite units (units of numerosity greater than one) and the fact that ratios and rates may be viewed as complex types of units (Lamon, 1993a, 1994) suggest that unit building may be an important mechanism in accounting for the development of increasingly sophisticated mathematical ideas. Research in the natural development of language hierarchies (Callanan & Markman, 1982; Markman, 1979) substantiates this perspective, suggesting that more sophisticated thinking results when one reframes a situation in terms of a more collective unit. When that happens, a part-whole schema comes in to play and the individual is able to think about both the aggregate and the individual parts that compose it. In turn, the ability to compose and to decompose a unit into its constituent parts adds the flexibility to one's reasoning that is needed in the field of rational numbers.

Olive & Steffe (1994) have embodied these mechanisms by which children build more complex units in a microworld that allows children to experiment with operations on discrete and continuous objects that can help them to understand rational numbers: composing and decomposing, iterating, partitioning, measuring. It further serves as a dynamic medium by which teachers and researchers can investigate the itinerary children take as they construct meaning for rational numbers.

In spite of the fact that many whole number strategies and ways of thinking are not useful for dealing with fractions, research in the last decade has proved that there is not an unbridgeable divide between the two number domains. In particular, Mack (1990, 1995)

demonstrated in her teaching experiments that students' informal and whole-number knowledge can be extended and connected to rational number symbols and procedures. Others have found competencies, even in very young students, that could be exploited in instruction. Student representations of rational numbers and of proportions occur on an informal, qualitative basis long before these students are capable of treating these topics quantitatively (Tourniaire, 1986; Treffers & Goffree, 1985). Children naturally employ a form of mathematical intuition, an informal knowledge system (Kieren, 1983; Streefland, 1984, 1985; Treffers & Goffree, 1985). This informal knowledge includes a visual understanding of ratio and proportion (Steffe & Parr, 1968), especially of congruence and similarity (van den Brink & Streefland, 1979). Some strategies are more primitive and depend on counting, adding, and halving (Hart, 1984, 1988). However, many fairly sophisticated strategies and processes develop independent of instruction (Lo & Watanabe, 1997; Post, Behr, Lesh, & Wachsmuth, 1985; Rupley, 1981; Saenz-Ludlow, 1994).

Partitioning or fair sharing has long been recognized both as an important mechanism for building rational number understanding (Kieren, 1976, 1980; Piaget, Inhelder, & Szeminska, 1960; Pothier & Sawada, 1983; Streefland, 1991) and as a part of students' informal knowledge base beginning in their preschool days. Because the rational numbers are a quotient field, partitioning is an operation that plays an important role in generating the rational number subconstructs (Lamon, 1996) as well as in constructing meaning for operations with the rational numbers (Lamon, 2006; Mack, 2001). Partitioning activities merit a place of greater prominence than they currently receive in the elementary curriculum because children's partitioning strategies develop in efficiency over a long period of time and share important connections with equivalence and the formation of complex unit structures (Lamon, 1996).

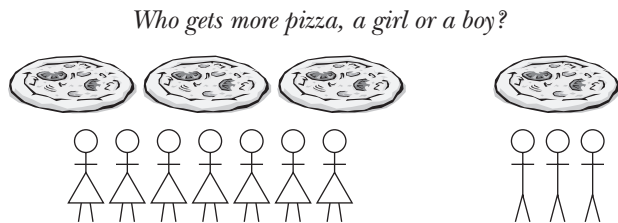
Among the early, correct strategies children use in situations involving proportions is the *building-up* strategy as documented by Piaget, Grize, Szeminska, and Bang (1968), Karplus and Peterson (1970), and Hart (1984). In this strategy, a student establishes a ratio and extends it to a second ratio by addition:

2 pencils cost \$.65
\$.65 for 2 more makes \$1.30
\$.65 for 2 more makes \$1.95

Although it is a primitive strategy by which children reason up to some desired quantity by using pattern recognition and replication, it is useful in solving some

proportion problems. Nevertheless, without additional information, it would not be considered proportional reasoning because it does not take into consideration the constant ratio between the two measure spaces.

Children also demonstrate interesting and productive strategies in ratio comparison problems. Lamon (1989, 1993a, 1993b) found that students are able to use a double counting strategy to successfully solve some problems. For example, consider this comparison task:



Spontaneously, before instruction in rational numbers, children construct complex units when solving problems such as this. They choose one of the ratios (in this case, 1 pizza: 3 boys) and use it to reinterpret the other ratio (3:7). This process is called *norming*, or reinterpreting a situation in terms of some chosen unit. Essentially, it is a double counting process: Measure the ratio 1:3 out of the ratio 3:7 by counting both parts of the pizza-to-boy ratio out of corresponding parts of the pizza-to-girl ratio. It is a fairly easy task for students who use this strategy to represent it in symbols. The ratio arithmetic is

$$(3:7) - 3(1:3) = (3:7) - (3:9) = (0:-2).$$

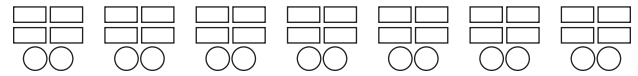
Thus, the student explains, if the pizza were served so that there was always 1 pizza for 3 people, the first group would actually get more because they could have fed 2 more people. Like the building-up strategy, norming is not a good indicator of proportional reasoning because students who use it often fail to recognize all of the structural relationships in a proportion. Although students who use the strategy do not yet conceptualize a ratio as a single number (a divided ratio or an index), the norming strategy appears to be a natural step in the process of conceptualizing a ratio as a unit of units of units (Lamon, 1993a, 1993b, 1994). The child coordinates a) number of pizzas, b) number of people, and further recognizes the new intensive quantity (pizzas per person) composed of a and b. This strategy and the next build upon the unit construction processes identified by Les Steffe and his colleagues (e.g., Steffe, 1988, 1992; Steffe et al., 1983), who documented the role of counting in children's construction of number sequences, and the

mechanisms by which children build up increasingly complex unit structures.

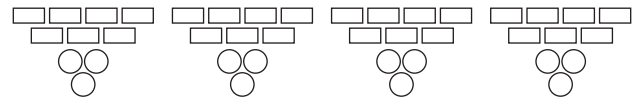
Lo and Watanabe (1997) have documented other variations of a double matching strategy. They asked a fourth-grade student to solve this proportion:

Yesterday I bought 28 candies with 12 quarters. Today, if I go to the same store with 15 quarters, how many candies can I buy?

The child made 7 groups of 4 candies and put quarters in groups of 2:

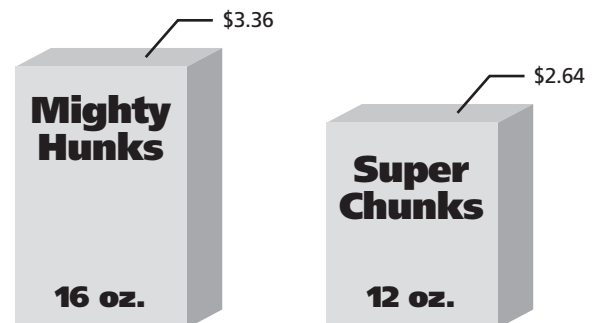


This did not prove satisfactory because there were 6 groups of quarters, not 7. After experimenting with several regroupings of candies and quarters, he arrived at this matching:



Then he was able to verbalize the relationship 3 quarters for 7 candies and imagine one more such matching to arrive at the answer 15 quarters for 35 candies.

Another important intuitive strategy is that of *unitizing*. Recall that unitizing is the cognitive chunking or regrouping of a given quantity into manageable or conveniently sized parcels (Lamon, 1996). One of the most salient differences between those who think proportionally and those who do not is that the proportional reasoners are adept at building and using composite units when the context suggests that it is more efficient to use composite units than to use singleton units (Lamon, 1993a, 1993b). For example, decide which of these cereals is the best buy:



In this case, without pencil or calculator, comparing the prices of 4 ounces of each cereal is more convenient than mentally using a two-digit divisor to compute the cost of 1 ounce of each cereal. Although third-grade students rarely construct such composite

units, fourth and fifth graders begin to develop thinking strategies based on unitizing (Lamon, 1996).

As a whole, the studies in which young children were interviewed before they had had substantial fraction instruction show that powerful, intuitive strategies are available to children at an early age. Even Inhelder and Piaget (1958), speaking of proportional reasoning, observed, "We have seen subjects construct the notion in the experimental situation before they have learned it in school" (p. 314). On the other hand, research with older children who have had 5 or more years of traditional mathematics instruction (Karplus et al., 1983b) suggests that rules and algorithms replace reasoning strategies, and that these often fail the student when integral ratios do not exist. In the Karplus et al. research, a proportion such as one solved by a much younger child in the Lo and Watanabe (1997) study, $\frac{28}{12} = \frac{x}{15}$, would have been one of the most difficult for eighth graders. Researchers must begin to suspect that instruction played a role in the poor performance of the older students. One conjecture is that informal strategies were replaced with rules for creating equivalent fractions (multiply numerator and denominator by the same number) that failed to serve the students well in the absence of integral ratios. Thus, instruction that builds upon children's preinstructional strengths should be a high priority.

Interventions: Correcting Problems

If one suspects that understanding in a particular field has to do both with development and instruction, the question naturally arises as to whether learning can occur more rapidly in older children who, for whatever reason, have misconceptions or deficits in their knowledge. Intervention studies are limited-term teaching experiments that target specific problems with the goal of producing more acceptable outcomes. Unfortunately, in the domain of fractions, ratios, and proportions, these studies have been largely ineffective.

Greer (1987) devised a teaching experiment for children whose problem-solving strategies varied on two juxtaposed problems that differed only in the numbers involved. His goal was to see if they could overcome their nonconservation of operation and recognize the invariance of operation. He concluded that "this conservation can be as difficult to induce as the standard Piagetian conservations" (p. 44).

Hart (1984) described teaching experiments designed to help children overcome the incorrect use of addition in multiplicative situations, only to conclude that "children have misconceptions of a more funda-

mental nature than can be readily cured by a brief series of lessons" (p. 81).

Similar disappointing results came from a number of teaching experiments conducted by the Rational Number Project. In a study of children's understanding of fraction size, Behr, Wachsmuth, and Post (1985) reported that "given the amount of instructional time, the degree of special attention, the extensive use of manipulative aids, and the amount of time devoted to developing rational number concepts," they were surprised that 49% of the students in the study showed "little or no comprehension of fraction addition or rational number size" (p. 128). In a study of children's understanding of order and equivalence, Post, Wachsmuth, Lesh, and Behr (1985) reported, "Even under the conditions of instruction in the teaching experiment—which employed a rich diversity of manipulative aids, placed a heavy emphasis on concept development, and took more time than traditional instruction—many rational number concepts remained exceedingly difficult for these children" (p. 23). Similarly, in an 18-week study in which instruction focused on multiple concrete teaching materials, Behr, Wachsmuth, Post, and Lesh (1984) reported, "Even after extensive instruction, the performance of a significant number of fourth graders demonstrates a substantial lack of understanding" (p. 337). Collectively, these intervention studies indicate that building fraction, ratio, and proportion knowledge will involve a long-term learning process. There will be no quick fixes.

Experimental Materials

In spite of the lack of clear direction from research, experimental curriculum materials have been developed to help students build understanding of rational numbers and other multiplicative content. These materials vary in scope from full middle school curricula to units of study to less comprehensive materials. All have been developed from slightly different perspectives. In the United States, the so-called reform curricula, including Connected Mathematics (Lappan, Fey, Fitzgerald, Friel, & Phillips, 1998) and Mathematics in Context (Romberg & de Lange, 1998), and in South Africa, materials produced by the MALATI Fraction Group at Stellenbosch (Murray, Olivier, & Human, 1996) have incorporated multiple rational number subconstructs (ratio, quotient, measure, operator, and part-whole comparison). In the Netherlands, an extensive study (Streefland, 1991) used materials that developed fractions and ratios simultaneously through realistic problem-centered approaches. Other interesting materials were developed for a study at the Institute of Child Study

at the University of Toronto (Moss & Case, 1999) and in work at Cornell University (Confrey, 1994; Lachance & Confrey, 1995; Scarano, 1996; Smith & Confrey, 1994). The Canadian study, originating in a psychological tradition, began with percents and decimals and used the proportional nature of the rational numbers to help children build number sense. Confrey and her colleagues emphasized a measurement interpretation, using continuous quantities and splitting. All of these materials and approaches were significant in that they took full advantage of children's informal knowledge and encouraged intuitive, sense-making strategies from the onset of instruction, while going well beyond the traditional content of fraction instruction. Unfortunately, the work begun in The Netherlands and in Canada was cut short by the untimely deaths of their chief investigators, Leen Streefland and Robbie Case, respectively.

Questions arise as to whether the new curricula that have incorporated various rational number subconstructs will be effective in increasing students' understanding of the rational numbers and whether students' proportional reasoning ability will be enhanced. The answers to these questions may require long-term follow-up of students who have used the curricula as they study mathematics in high school and college. Questions concerning the effectiveness of the curricula present challenges to the research community to more explicitly define what it means to understand rational numbers and to develop intrinsic measures of understanding. We habitually look to standardized examinations to measure success in spite of the fact that they reflect a different set of values, include more procedural than conceptual content, are often developed for different purposes, and are problematic in that the structure and arrangement of content they assess is inconsistent with the nature and growth of knowledge in the multiplicative conceptual field.

A Temporary Standstill

Research can sometimes seem to come to a halt when, after years of studying a complex domain, scholars have uncovered a complex web of questions and misconceptions and compelling needs, only to realize that they do not have the tools and methods and classrooms needed to answer the challenge. Certainly this was true at the time Behr et al. wrote the first handbook chapter. It was difficult to see where the field was going. Even as late as 1992, Davis, Hunting, and Pearne characterized the field as one that is not producing any definite conclusions, one in which "no real progress is being made" (p. 2). Traditional experimental methods and interventions can fail to

be productive, especially when dealing with complex content in the multifaceted world of the classroom. A research community may need to wait for the evolution and diversification of research paradigms; sometimes they must diverge from classical frameworks and incorporate new perspectives; sometimes, both. In the next section, I discuss a new research framework that incorporates nontraditional perspectives and its use in the new design experiment research paradigm for conducting longitudinal classroom research.

A FRAMEWORK FOR RESEARCH IN THE MULTIPLICATIVE CONCEPTUAL FIELD

In this section, I discuss a framework that is currently being used in longitudinal studies to facilitate rational number understanding and proportional reasoning. This framework rests upon the assumption that pedagogical content knowledge in a complex domain such as the rational numbers does not concern target concepts and operations in the domain, as much as it does core or central cognitive structures, composed of concepts, ways of thinking, and mechanisms for growth that are foundational to a broad range of multiplicative topics. Later in this section the conclusions of that research will be presented, as will questions that might launch additional longitudinal studies using the framework.

Reconsidering Pedagogical Content Knowledge

Pedagogical content knowledge (Shulman, 1986b, 1987) is the kind of knowledge a teacher needs to be able to teach a particular strand of content. One needs to know the students and their current capabilities and, beyond a personal knowledge of mathematics, the teacher needs to know the particular ingredients of understanding that can facilitate children's construction of knowledge about that content. The pertinent issue is *what* content should be psychologized for instruction. Do we in education teach to some goal state, or do we seek some foundational kind of knowledge that can grow to support larger, more complex content?

Historically, rational number education has taken adult knowledge of mathematics and simplified it so that it is palatable for children. As confirmed by a survey reported by Hart (1984), teachers equate understanding of ratio and proportion with the ability to solve problems involving scale drawing, similarity, percentages, trigonometry, probability, enlargement, and so on, or to find equivalent fractions or ratios, or to

solve for missing numbers in a proportion. Nevertheless, teaching definitions, algorithms, and applications of rational number knowledge has not facilitated the development of rational number sense and the ability to reason proportionally.

Rather than teaching to goal states, seeking initial, front-end ideas and processes may be more important. When considered from the perspective of emergent concepts and processes, pedagogical content knowledge in the field of rational numbers and ratios and proportions may not appear to be related to textbook definitions and the problems one uses may not look like the problems or the applications for which proportional reasoning is useful. Therefore, before asking what kinds of problems we educators want students to be able to solve, we might begin by looking for fundamental perspectives on the field, including the cognitive processes, focal points, and big ideas that form the basis of thought and knowledge in rational numbers and related multiplicative topics. This is generally referred to as *conceptual analysis* (Lesh, 1985). Conceptual analyses may be performed at a variety of levels. The previous section in which proportional reasoning and proportionality were discussed is a kind of conceptual analysis. For another example, see Thompson and Saldanha (2003). Identifying student ideas (useful ways of thinking or mechanisms that contribute to the development of more powerful processes) is another form of conceptual analysis (Lesh, 1985). For example, unitizing, norming and partitioning (fair sharing) are primitive student capabilities that underlie rational number ideas. The following analysis, for the purpose of psychologizing critical content in the multiplicative field for instruction, examines additional basic concepts and processes and includes consideration of student thinking about them.

Continuing the Conceptual Analysis

Reasoning: Beyond Mechanization

Proportional *reasoning* refers to detecting, expressing, analyzing, explaining, and providing evidence in support of assertions about proportional relationships. The word *reasoning* further suggests that one uses common sense, good judgment, and a thoughtful approach to problem solving, rather than plucking numbers from word problems and blindly applying rules and operations. Researchers typically do not associate reasoning with rule-driven or mechanized procedures, but rather with mental, free-flowing processes that require conscious analysis of the relationships among quantities. Consider these problems and an eighth grader's approach to them:

- If a bag of topsoil weighs 40 pounds, how much will 3 identical bags weigh?
- If a football player weighs 225 pounds, how much will 3 players weigh?
- If Ed can paint the bedroom by himself in 3 hours, and his friend, Jake, works at the same pace as Ed does, how long will it take to paint the room if the boys work together?
- I was charged \$1.30 for sales tax when I spent \$20. How much sales tax would I pay on a purchase of \$50?
- Bob and Marty like to run laps together because they run at the same pace. Today, Marty started running before Bob came out of the locker room. Marty had run 7 laps by the time that Bob ran 3. How many laps had Marty run by the time that Bob had run 12?

Mason

$$\begin{array}{lll}
 \text{a. } \frac{1 \text{ bag}}{40 \text{ p.}} = \frac{3 \text{ b.}}{? \text{ p.}} & 3 \times 40 = 120 \text{ p.} \\
 \text{b. } \frac{1 \text{ f}}{225 \text{ p.}} = \frac{3 \text{ f}}{? \text{ p.}} & 3 \times 225 = 675 \text{ p.} \\
 \text{c. } \frac{1 \text{ man}}{3 \text{ hrs}} = \frac{2 \text{ men}}{? \text{ hrs}} & 2 \times 3 = 6 \text{ hrs} \\
 \text{d. } \frac{\$1.20 \text{ tax}}{\$20} = \frac{?}{\$50} & \$1.20 \times 50 = 60 \\
 & 60 \div 20 = \$3 \\
 \text{e. } \frac{M}{7 \text{ laps}} = \frac{B}{3 \text{ laps}} & 7 \times 12 = 84 \\
 & 84 \div 3 = 28
 \end{array}$$

Problems a and d can be correctly solved with the student's algorithmic approach. The student's blind application of proportions is a reminder that part of understanding a concept is knowing what it is *not* and when it does *not* apply. Two quantities may be unrelated (as in problem b); one quantity may be related in an inversely proportional way to another quantity, as in problem c (increasing the number of people decreases the time needed to do a job); or the *change* in one quantity might be proportional to the *change* in the other quantity, as in problem e. Many other important relationships do not entail proportional relationships at all, and students must learn to recognize the difference.

Another nuance of the word *reasoning* is that it relies on headwork and logical reasoning patterns to draw conclusions or inferences, rather than written computations.

Colloquially, proportional reasoning is “reasoning up and down.” For example, the following problem can be solved by reasoning up, without using the form $\frac{a}{b} = \frac{c}{d}$ or doing any writing at all.

If a box of detergent contains 80 cups of powder and your washing machine recommends $1\frac{1}{4}$ cups per load, how many loads can you do with one box?

Think: $1\frac{1}{4}$ cups do 1 load
 5 cups do 4 loads
 40 cups do 32 loads
 80 cups do 64 loads

It takes 6 men 4 days to complete a job. How long will it take 8 men to do the same job?

Think: 6 men take 4 days
 1 man doing the work of 6 men all by himself takes 24 days
 8 men, dividing up the work that 1 man did, take 3 days

Reasoning up and down can replace pencil work from the initial stages of fraction instruction. For example, consider the following fraction question, in which the unit is implicit.

These are $\frac{2}{3}$ of Joan’s pennies. How many pennies are $\frac{1}{2}$ of all she has?



The problem gives information about $\frac{2}{3}$ of the unit; from there, one can reason down to $\frac{1}{3}$, then to 1 (because three $\frac{1}{3}$ s make the whole set of pennies), then to $\frac{1}{2}$.

Think: 8 pennies are $\frac{2}{3}$
 4 pennies are $\frac{1}{3}$
 12 pennies are 1 whole set
 6 pennies are $\frac{1}{2}$ of Joan’s pennies

Unfortunately, students rarely reason without a pencil and paper. Yet, from my experience with children, I consider reasoning aloud the most profound

and compelling constructivist activity. Not only does the reasoning aloud enable students in the class to engage in interactive, critical communication in which they influence each other’s interpretations and actions (Cobb, Yackel, & Wood, 1992), but in addition, the student who is reasoning aloud becomes his own best critic. Hearing his own message constitutes instantaneous reflective abstraction and serves to develop his metacognitive skills.

Covariance and Invariance

One of the most ubiquitous ways of thinking and operating in mathematics entails the transformation of quantities or equations in such a way that the mathematical structure is invariant (unchanged). As I have already mentioned, proportional relationships involve one of the simplest forms of covariation. Two quantities are linked to each other in such a way that when one changes, the other one also changes in a precise way with the first quantity. How does one know precisely how the changes in the quantities should be linked? How would one recognize an error in reasoning in either of the problems just solved (laundry detergent and pennies)?

Part of what it means to understand the proportional nature of the rational numbers is to recognize valid and invalid transformations. Two major types of invariance play a role in proportional reasoning. One is invariance of the ratio of two quantities. The second is invariance of the product of two quantities.

In a direct proportion, the direction of change in the related quantities is the same. One says that “ y is directly proportional to x ” or that “ y varies as x .” For example,

For every 2 cups of water in a punch recipe, you add $\frac{1}{2}$ cup of sugar. If you quadruple the recipe so that you are using 8 cups of water, you should use 2 cups of sugar.

The amount of water as compared to the amount of sugar is always the same, no matter how large a batch you make. In any size batch that is made with this recipe, there is always 4 times as much water as there is sugar. The two quantities may be increased or decreased as long as the relationship between the quantities is preserved, that is, as long as their original ratio is maintained. Symbolically,

$$\frac{y}{x} = k, \frac{n_1 y}{n_1 x} = k, \frac{n_2 y}{n_2 x} = k, \frac{n_3 y}{n_3 x} = k,$$

where n_i is a natural number.

For example, in the laundry detergent problem, the ratio of cups to loads was $1\frac{1}{4}:1$. The constant of

proportionality, k , is the divided ratio: $1\frac{1}{4}:1 = 1.25$. That ratio was maintained throughout the reasoning process: $5:4 = 1.25$, $40:32 = 1.25$, and $80:64 = 1.25$. Similarly, in the pennies problem, the ratio (number of pennies):(fractional part of the set) was preserved:

$$8:\frac{2}{3} = 12:1$$

In a second type of invariance two quantities change together, but the direction of change is not the same for both. In the work problem used above, increasing the number of people working on the job should mean that the job can be completed in less time. Conversely, decreasing the number of people working on the job should increase the number of days needed to complete the job. In this situation, “ y is inversely proportional to x ” or “ y varies inversely as x .” The invariance occurs in the product of the two quantities, number of men (m) and number of days (d). Symbolically, $m \cdot d = k$.

# men	# days	# man-days
6	4	24
8	3	24
1	24	24
4	6	24
2	12	24

Multiple layers of development precede the ability to detect the invariant quantity when two quantities are changing together. First of all, students do not always focus on the quantifiable characteristics of a situation. For example, middle school students were asked to think about this situation:

When you get on your bicycle in the front of your house and you ride your bicycle down the street, what changes?

Their answers included the following observations:

- The pedals go up and down
- The trees go by
- I move away from where I started
- My wheels go around
- I pass my friend's house
- I get tired

Students must be encouraged to direct their thinking beyond obvious, surface-level observations toward more significant, quantifiable characteristics such as distance traveled, speed of the bike, height of one's

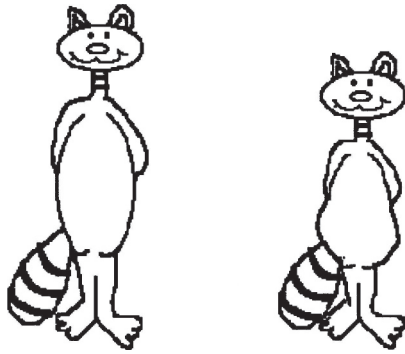
foot above the ground, and so on. If children are not thinking about quantities, it is pointless to talk to them about the ways that quantities change together.

Very young students have a tendency to assume that related quantities change together: If one quantity increases, so does the other to which it is related. For example, in a study of ratio sense, well over 50% of third graders said that if an orchestra can play a composition in 15 minutes, then two orchestras could play it in 30 minutes. Some fourth and most fifth graders observed that 2 orchestras playing would not take longer; it would just be louder. Students often need reminders to think about common sense statements that can help them to think about the way that quantities work together. For example, “If fewer people are working on a task, it should take longer to complete it” or, “If I travel at the same speed but I take a longer bike ride than I did yesterday, then it should take me longer today.”

However, using common sense and recognizing the explicit quantities in a problem are not enough. One must look beyond the given quantities to construct a new quantity. A third (implicit) quantity derives from the relationship of the two changing and connected quantities; this new quantity remains constant. Conceiving of a relationship between two quantities (a ratio) as a quantity is difficult, but an intensive quantity is precisely such an abstraction. It is not explicit in the situation described; it has a different referent than either of the quantities that are present; and it is difficult to represent (Kaput, Luke, Poholsky, & Sayer, 1986).

Multiplicative Reasoning

In the previous problems about the laundry detergent and men working, why did the transformations consist of *multiplying* (or dividing) both quantities by the same whole number? Some degree of mathematical maturity is required to understand the difference between adding and multiplying and contexts in which each operation is appropriate. One of the most difficult tasks for children is to understand the multiplicative nature of the rational numbers. Children who reason additively indiscriminately employ additive transformations, but whether additive reasoning is an invariant stage in the development of proportional reasoning, as Inhelder and Piaget (1958) suspected, is unclear. Other problems, including imprecise colloquial language, often interfere with children's apprehension of mathematical ideas. For example, when asked to compare the sizes of the cats in the following picture, young children will say that the cat on the left “shrunk into” the cat on the right, or that the cat on the right was enlarged. These children simply do not know that shrinking or enlarging entails more than just adding to or subtracting from the height of the cats.



Proportional reasoners are able to differentiate between additive and multiplicative situations and to apply whichever transformation is appropriate. The process of addition is associated with situations that entail adding, joining, subtracting, separating, and removing—actions with which children are familiar because of their experiences with counting and whole number operations. The process of multiplication is associated with situations that involve such processes as shrinking, enlarging, scaling, duplicating, exponentiating, and fair sharing. As students interact with multiplicative situations, analyzing the quantitative relationships, they eventually understand why additive transformations do not work. However, this takes time and experience, and it cannot happen until the student can detect intensive quantities. For example, consider this problem:

For your party, you had planned to purchase 2 pounds of mixed nuts for 8 people, but now 10 people are coming. How many pounds should you purchase?

# People	Pounds of nuts
8	2
10	?

Having 10 people is 2 more than 8 people, but adding 2 pounds of nuts for 2 more people suggests that a whole pound is needed for each person, when originally you were not figuring on a pound per person. Hence, an additive transformation does not preserve the ratio (pounds of nuts : number of people). As this example illustrates, a student is not going to be able to analyze this situation until he “sees” in it the implicit third quantity “pounds per person.”

Abstraction: Beyond Observation and Direct Measurement

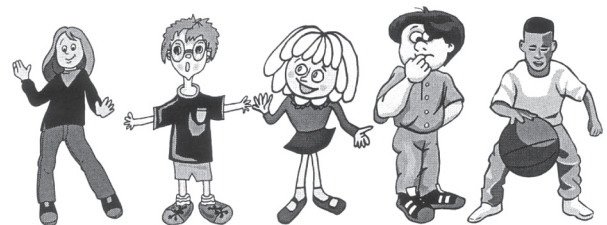
Multiplicative reasoning moves beyond the realm of concrete operations and into the stage of formal reasoning. It deals with abstractions, and Piaget considered it the hallmark of formal reasoning. In this computer age, students are accustomed to a barrage

of sense data; understanding comes from perceptually based data, but in logico-mathematical knowledge, understanding consists in grasping abstractions imposed upon sense data. Abstraction entails imposing a relational structure, a classification scheme apart from the object, event, or image in which it is exemplified. It is not a *perception* as much as it is a *conception*. The construction and use of ratios as measures is one example.

In the previous example, the quantities 8 people and 2 pounds of nuts relied on counting and weighing, two activities with which most people are comfortable. However, the quantity *pounds per person* is both a relationship between the other two quantities, people and pounds, and a new measure derived from the other two measures by considering their ratio. The underlying concept of ratio (a relationship between a pair of quantities) is fundamental to fractions (in arithmetic), to rates (in physics), to similarity transformations (in geometry), and to probabilities (in statistics), just to name a few mathematical applications. Density, slope, and speed, for example, are all called intensive quantities and they differ from extensive quantities (those that tell numerosity, length, height, time, etc.). They are, in fact, not directly observable and not directly measurable. Intensive quantities are relationships (comparisons or ratios) between two other quantities. In short, relative or comparative thinking, as opposed to absolute thinking, is needed to move beyond the sense data.

Relative thinking activities provide the opportunity for students to further expand the range of applicability of certain words that were formerly only associated with additive concepts (Pimm, 1987). For example, the word *more*. Although most children will be familiar with the word as a signal word for addition/subtraction, the word can have a proportional or relational meaning:

Which family has more girls?



The Jones Family

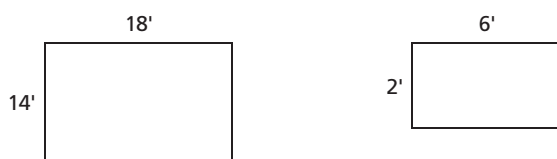


The King Family

Being able to see that 2 girls comprise a larger proportion of the King family ($\frac{1}{2}$ or 50%) than they do of the Jones family ($\frac{2}{5}$ or 40%) is related to understanding percent increase/decrease problems.

Understanding intensive quantities sometimes requires more than intuition and experience; content knowledge sometimes plays a role. In turn, without knowledge of intensive quantities, one cannot reason proportionally. In some cases, mathematical knowledge is needed to quantify some of these characteristics. For example, the property *squareness* relies on a mathematical definition of a square as a quadrilateral with congruent sides and angles.

Which of these rectangles (not drawn to scale) is most square?



Early elementary instruction includes such topics as counting, telling time, making change, measuring length with a ruler, and, perhaps, massing objects. Later topics include perimeter, area, and volume. On more than one account, researchers might argue that this is an impoverished treatment of measurement. Measurement lies at the very heart of human activity; humans have always been preoccupied with measuring their universe, and the units and methods of measurement—which have a long history of development (Klein, 1974)—are essentials of science. Yet, the elementary mathematics curriculum does not address the fact that some quantities cannot be measured directly. Students need time to analyze such characteristics as color intensity, sourness, roundness, the oranginess of a drink, and the crowdedness of an elevator, for example, and to engage in argumentation and justification about how to measure these characteristics and distinguish them from quantities with extensive measures.

In addition, the treatment of measurement in the elementary curriculum does not allow time for students to develop *concepts* of measurement, as opposed to the acts of measuring. For example, by the start of third grade, very few students understand the inverse relationship between the size of a unit of measure and the number of times you can measure it out of some fixed quantity. Very few understand that no matter what the unit of measure, you can break it into smaller and smaller subunits to make your measurement as precise as you need. Unit conversions, in which the same quantity is expressed in terms of different-sized chunks, is related to unitizing. Until measurement

becomes a more reflective process, and one that is connected to other aspects of mathematical thinking, children are inadequately prepared for rational number understanding.

Need for Longitudinal Research

By the time students begin to study fractions, they have had 3 or more years of schooling in addition to 7 or more years of diverse, individual experiences that contribute to their intuitive knowledge. Knowing the path that any particular child has taken in coming to grasp a particular strand of knowledge is impossible because, in fact, many different routes can lead to the same useful understanding. A conceptual field, much like the theorized structure of the human mind (Pinker, 1997), is not linear. Thus, a single best way to reach a goal or to prescribe instruction to a desired end (à la Thorndike) is unlikely to exist. Nor could students possibly be taught each and every context and application that is useful. The process of achieving a flexible number sense in the rational number realm is partly developmental, so that instruction alone is not the full solution. Multiplicative ideas, in particular, fractions, ratios, and proportions, are difficult and develop over a long period of time; brief teaching experiments have had disappointing results. There seems to be no substitute for longitudinal research that takes into account the full complexity of the classroom, the full complexity of students' experiential backgrounds and intuitive knowledge, and the full complexity of the mathematics.

Nevertheless, conducting research has been difficult because the web-like organization of this knowledge defies the linear organization of scope and sequence charts, and the organization of schools into grade levels and terms of fixed length are not conducive to the kind of research that is needed. Time constraints, unstable communities, and other realities of public schools have made longitudinal research very difficult.

Over time, the nature of schools is changing and obstacles to doing long-term research in classrooms are diminishing. Many charter schools, for example, are not bound by traditional rules and organization, and opportunities are increasing for researchers to work intensively over 4 or 5 years with charter schools that are seeking excellence in mathematics. Teacher and student turnover in these schools is not the problem that it is in traditional, urban public schools, and some use mathematics specialists.

A Longitudinal Study

The preceding rational analysis of proportional reasoning, many studies of children's thinking in the field of rational numbers, and a rare bit of serendipity occasioned a longitudinal study of 6 classes of children as they developed rational number meanings and operations from the start of Grade 3 until they finished Grade 6. A glimpse into the frameworks, curriculum, instruments, and results of that study (Lamon, in progress) are offered here with the hope that they can stimulate research in the area in many new directions.

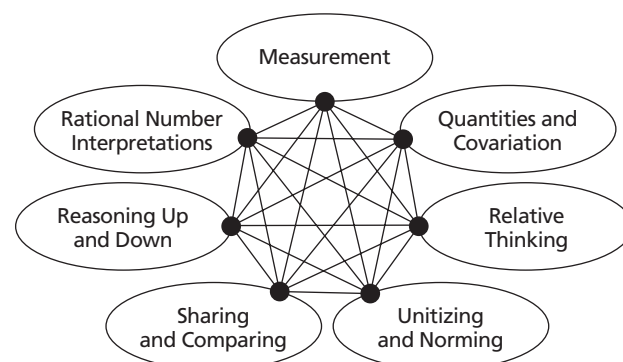
Central Multiplicative Structures

In my longitudinal study, I used a 7-point model for building mathematics instruction that grounded children in one of the subconstructs of rational numbers as well as six other ideas considered to be central to the development of multiplicative thinking. Following Case (1992), I refer to the six topics as *central multiplicative structures*. Case formulated the construct of a *central conceptual structure* in response to the debate about whether a general intelligence results from a developmental process, as proposed by Piaget (e.g., Flavell, 1963), or whether intelligence is modular or domain-specific as proposed by Gardner (1983). Case and his colleagues presented evidence (Case, 1992) that the real case is somewhere in between these two positions. Central conceptual structures are critical systems—built-in and subject to developmental ceilings, and partly nurtured through instruction—including content, action, associations with appropriate contexts, representations, and a web of conceptual relations both within and between them. They are built up by a complex interaction of knowledge and experience over a long period of time and are called *central* because they constitute part of the very backbone of mathematics, ideas and processes and representations that are recurrent, recursive, and of increasing complexity across mathematical and scientific domains. When these structures are sufficiently mature and connected, their owner develops control structures so that they are under her command (reasoning is conscious and deliberate), they enable higher order thinking that was previously impossible, and they affect knowledge and performance across a variety of domains.

Paradigm and Methodology

On the basis of the preceding rational task and conceptual analyses, the following 7 central structures

were hypothesized as critical components of instruction if children are to eventually develop a rich understanding of multiplicative concepts.



For 4 years, in Grades 3 through 6, five cohorts of students in two urban schools in different parts of the country each built their understanding of rational numbers on a different initial interpretation of the symbol $a-babb$: as a part/whole comparison with unitizing, as a measure, as a quotient, as an operator, and as a ratio. A sixth class that received traditional fraction instruction served as a control group. Instruction in all of the classes except the control group entailed elements from the other 6 hypothesized central multiplicative structures. The rationale was this: If these are, indeed, central conceptual structures, then in time (and expectations were that this was going to take a *long* time), children's knowledge built in each of the "nodes" should grow together. As a result, students should demonstrate a usable knowledge of multiple rational number interpretations, not only the one in which their instruction was grounded. Their thinking should be flexible enough to support proportional reasoning in problem-solving situations. This study provided a unique opportunity to compare unconventional approaches with traditional instruction and to document the sequencing and growth of ideas and the breadth and depth of the understanding that developed.

This research was conducted in the paradigm that has become known as *design experiment* research. The chief characteristic of design research (Brown, 1992) is that the product is being designed in a series of systematic trials and adaptations. The data collection, analysis, and refinement of the product being developed (in this case, effective instruction in rational numbers and proportional reasoning) go through an extensive test-and-revise cycle. One's well-researched framework is the starting point, but daily activities are designed and changed in response to classroom obser-

variations to better meet the needs of the students that they are purportedly serving. That is, based on data (e.g., observations, students' written work, classroom discussions), the next day's lessons reflected what the teachers and I learned from that data. Rather than empirically proving that some theory "works," a design experiment actually studies the synergy, the ecology of a fully functioning classroom in all of its complexity. Rather than "controlling" for variables, it allows the researcher and teacher to regulate what they do in response to the diversity they encounter. Simply put, the teachers and I meddled in the research. We brought about innovative forms of learning so that we could study them and tweak them and make them as effective as possible for as many students as possible.

This design research was based upon the following questions:

1. What understanding might develop if fraction instruction were based on a subconstruct other than part-whole comparisons?
2. What sorts of instructional activities support understanding of each of the nontraditional rational number constructs?
3. How long do students take to develop a useful understanding of any single subconstruct of rational number?
4. Because understanding is not an all-or-none affair, what are some of the benchmarks along the path to understanding?
5. Are the subconstructs equally good alternatives for learning concepts and computation?
6. Which subconstructs are powerful enough to connect to other interpretations without direct instruction?
7. Are any developmental processes or mechanisms operating that connect whole number knowledge and rational number knowledge?
8. Will the set of hypothesized central conceptual structures be necessary and sufficient to facilitate proportional reasoning?

Because I was concerned that my results generalize to other public school classrooms, I was careful not to idealize the study in any regard. Results include all students, including learning disabled students, students with behavioral disabilities, one student with a truancy problem, and students for whom English is a second language—all the realities normally found in a mainstream public school classroom. The reality principle applied equally to the teachers. Teachers studied to

learn the material and the teaching methods immediately before applying the content and methods in their own classrooms. Although they were experienced teachers, they were not mathematics specialists.

Children were given the freedom and the encouragement to express their thinking in whatever manner they wished. The two teachers who taught the five classes facilitated learning through the kinds of activities they encouraged and through the problems they posed. None of the groups were taught any rules or operations. Teachers sometimes used whole-class instruction, and occasionally, whole-group discussions of the ideas arose from the activities. However, on most days, the mathematical activity consisted of group problem solving, reporting, and then individually writing and revising solutions for homework.

Three measures were used to compare the five classes with the control group: number of rational number subconstructs the student was able to use by the end of Grade 6, proportional reasoning, and fraction computation ability.

Problem types and children's strategies. A summary of the subconstructs of rational numbers and some of the kinds of classroom activities that supported each of them is given in Table 14.2, using the fraction $\frac{3}{4}$ as an example.

Children have a tremendous capacity to create ingenious solutions when they are sufficiently challenged and when they do not feel expected to follow rules. To give the reader both a sense of the kinds of nontraditional questions explored during instruction and a sense of the kind of student thinking those questions elicited, several examples are given here. More detail about instructional activities can be found in Lamon (in progress).

- Here, a student uses "reasoning up and down" to determine the fractional part of a set when the unit is given implicitly: If 6 books are $\frac{2}{3}$ of all the books on Robert's shelf, figure out how many books are $\frac{5}{9}$ of the books on his shelf.

Jen

$$\begin{aligned} \text{ff } \square\square\square\square\square\square &= \frac{2}{3} \\ \square\square\square &= \frac{1}{3} \\ \square\square\square\square\square\square\square\square &= 1 \\ \square\square\square\square\square &= \frac{5}{9} \end{aligned}$$

Table 14.2 Alternatives to Part-Whole Fraction Instruction

Interpretations of $\frac{3}{4}$	Meaning	Selected Classroom Activities
Part-Whole Comparisons with Unitizing "3 parts out of 4 equal parts"	$\frac{3}{4}$ means three parts out of four equal parts of the unit, with equivalent fractions found by thinking of the parts in terms of larger or smaller chunks. $\frac{3 \text{ (whole pies)}}{4 \text{ (whole pies)}} = \frac{12 \text{ (quarter pies)}}{16 \text{ (quarter pies)}} = \frac{1\frac{1}{2} \text{ (pair of pies)}}{2 \text{ (pair of pies)}}$	Unitizing to produce equivalent fractions and to compare fractions
Measure " $3(\frac{1}{4}$ -units)"	$\frac{3}{4}$ means a distance of $3(\frac{1}{4}$ -units) from 0 on the number line or $3(\frac{1}{4}$ -units) of a given area.	Successive partitioning, reading meters and gauges
Operator " $\frac{3}{4}$ of something"	$\frac{3}{4}$ is a rule that tells how to operate on a unit (or on the result of a previous operation): multiply by 3 and divide the result by 4 or divide by 4 and multiply the result by 3. This results in multiple meanings for $\frac{3}{4}$: $3(\frac{1}{4}$ -units), $1(\frac{3}{4}$ -unit), and $\frac{1}{4}$ (3-unit).	Machines, paperfolding, xeroxing, discounting, area models for multiplication and division
Quotient "3 divided by 4"	$\frac{3}{4}$ is the amount each person receives when 4 people share a 3-unit of something.	Partitioning
Ratios "3 of A are compared to 4 of B"	3:4 is a relationship in which 3 A's are compared, in a multiplicative rather than an additive sense, to 4 B's.	Bi-color chip activities

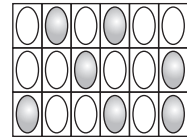
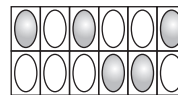
- This problem entails identification of the unit and the notion of operator: Here is one day's supply of juice. You have taken only one drink. How much of your day's supply do you still have left?



Gerry

I measure it out and get $\frac{1}{5}$ 
 but it is $\frac{1}{5}$ of $\frac{1}{2}$ of all my juice that I drank.
 $\frac{1}{5}$ of $\frac{1}{2} = \frac{1}{10}$ drank
 $\frac{9}{10}$ left

- This is a relative thinking question that entails consideration of the how one might measure a quality such as "brownness" of a carton of eggs: Here are two egg cartons, each containing some brown and some white eggs. Which container is more brown?



Bert

I put them into 6 packs like this



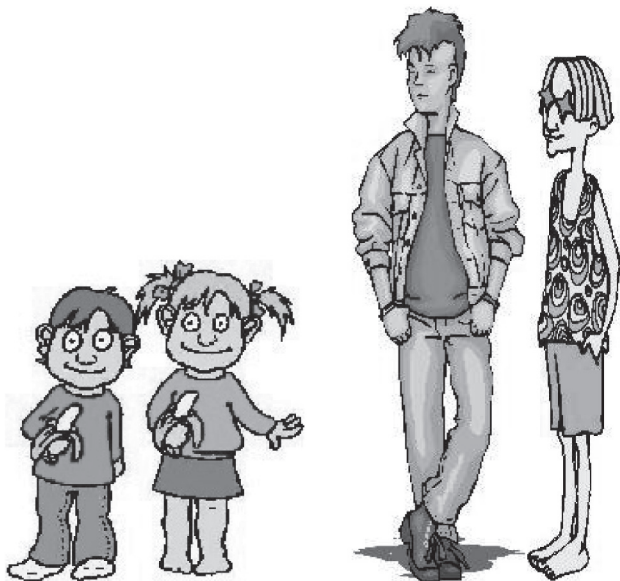
then here I have $\frac{2}{2}$ brown eggs in each



here I have $\frac{2}{3}$ brown eggs in each

$\frac{1}{2}$ is more, so the 12 pack is browner.

- How would you compare rates of growth? The first picture shows Jeb and Sarah Smart when they were younger. The second shows them as they look now. Who grew faster between the first and second pictures?



Grade 5 Student:

Well they both grew taller, but Jeb grew *taller*. Get it? He was shorter then he grew taller. Well, actually, Sarah was shorter then she grew taller, too. But he grew **TALLER** (loud and with emphasis).

(The student realized that his words were not conveying what he meant and showed some frustration in not being able to express himself. There was a long pause as he thought.)

Look. I can show you. (Taking a ruler, he aligned it with the top of Sally's head then and the top of her head now. Then he did the same for Jeb.)

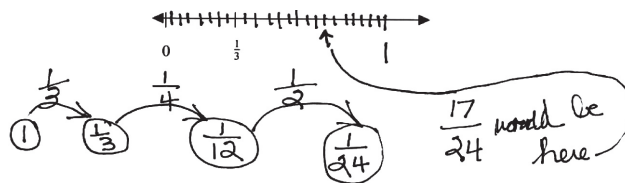
See. The ruler is more slanty for Jeb. That proves he grew faster.

- Rational numbers as measures were introduced using successive partitioning of a number line.

Locate $\frac{17}{24}$ on this number line:



Students were taught to use arrow notation to keep track of the size of the subdivisions after each time they partitioned.



This was a particularly powerful interpretation of rational numbers. After a time, students stopped partitioning the number lines and were able to visualize and reason about the fractions without them.

- Find two fractions between $\frac{1}{8}$ and $\frac{1}{9}$.
Martin, end of Grade 5

$$\frac{1}{9} \quad \frac{8}{8} \div 9 \quad \frac{1}{8} = \frac{9}{8} \div 9$$

$$\frac{1}{8} \div 9$$

$$\frac{1}{9} \quad \frac{10}{90} \quad \frac{11}{90} \quad \frac{12}{90}$$

$$\frac{1}{9} \quad \frac{10}{81} \quad \frac{11}{90} \quad \frac{12}{99} \quad \frac{1}{8}$$

- Find 3 fractions between $\frac{7}{12}$ and $\frac{7}{11}$.
Jonathan, end of Grade 5

$$\frac{7}{11\frac{4}{4}} \quad \frac{7}{11\frac{3}{4}} \quad \frac{7}{11\frac{2}{4}} \quad \frac{7}{11\frac{1}{4}} \quad \frac{7}{11\frac{0}{4}}$$

Annie, Grade 6, midyear

$$\frac{7}{12} \quad \frac{7}{11\frac{1}{2}} \quad \frac{7}{11\frac{1}{4}} \quad \frac{7}{11\frac{1}{5}} \quad \frac{7}{11}$$

$$\frac{7}{23} \quad \frac{7}{45} \quad \frac{7}{56}$$

$$\frac{14}{23} \quad \frac{28}{45} \quad \frac{35}{56}$$

- Next, two students use unitizing to answer the question, "Which is more pizza, $\frac{1}{2}$ pizza or $\frac{3}{5}$ pizza?"

Tommy R.

$$\frac{1 \text{ piz}}{2 \text{ piz}} = \frac{2 (\frac{1}{2} \text{ piz})}{4 (\frac{1}{2} \text{ piz})} = \frac{3 (\frac{1}{3} \text{ piz})}{6 (\frac{1}{3} \text{ piz})} =$$

$$\frac{4 (\frac{1}{4} \text{ piz})}{8 (\frac{1}{4} \text{ piz})} = \frac{5 (\frac{1}{5} \text{ piz})}{10 (\frac{1}{5} \text{ piz})}$$

$$\frac{3 \text{ piz}}{5 \text{ piz}} = \frac{6 (\frac{1}{2} \text{ piz})}{10 (\frac{1}{2} \text{ piz})} = \frac{9 (\frac{1}{3} \text{ piz})}{15 (\frac{1}{3} \text{ piz})}$$

$\frac{3}{5}$ of a pizza is more.

Angela

$$\frac{1 \text{ pizza}}{2 \text{ pizzas}} = \frac{5 (\frac{1}{5} \text{ pizzas})}{10 (\frac{1}{5} \text{ pizzas})}$$

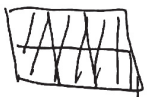
$$\frac{3 \text{ pizzas}}{5 \text{ pizzas}} = \frac{6 (\frac{1}{2} \text{ pizzas})}{10 (\frac{1}{2} \text{ pizzas})}$$

- How would you divide 1 by $\frac{2}{3}$?

Grace

$\frac{2}{3}$ is twice as big as $\frac{1}{3}$ and I know $\frac{1}{3}$ goes into 1 three times. So $\frac{2}{3}$ can go in only half as many times.

Troy



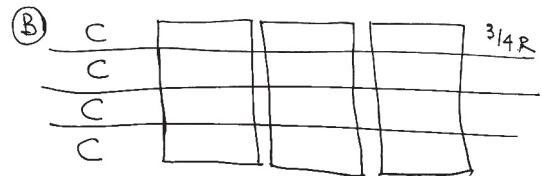
$$1 = 6 (\frac{1}{6} \text{ pieces})$$



$$\frac{2}{3} = 4 (\frac{1}{6} \text{ pieces})$$

After I take out 4 from the 6, I have $2 (\frac{1}{6} \text{ pieces})$ left. This is $\frac{2 (\frac{1}{6} \text{ pieces})}{4 (\frac{1}{6} \text{ pieces})}$

- Here is a question that entails relative thinking and ratio comparison: The 5 people at Table A drank 2 root beers and 3 colas. The 7 people at Table B drank 3 root beers and 4 colas. Which table could be called the cola drinkers?



A, a cola is ~~the~~ matched with $\frac{2}{3}$ of a root beer.

B, a cola is matched with $\frac{3}{4}$ of a root beer.

Cola is stronger at table A because it is matched with less root beer.

- Another ratio comparison question: Which group, the girls or the boys, got the better deal?
2 girls got into the theatre on State Street for \$3.
5 boys got into the theatre on Main Street for \$6.



$$\text{I did } (5:6) - 2(2:3) = (1:0)$$

$$(5:6) - (4:6) = (1:0)$$

The boys price is better because
for the same amount of money
they can get 1 more person in.

Selected Results

After 4 years, all five groups of students had developed deeper understanding of rational numbers as measured by the number of subconstructs they were using than did the students in the control group. The numbers of proportional reasoners far exceeded the number in the control group, and even in computation, achievement was greater in all five groups.

Under different rational number interpretations, children acquired meanings and processes in different sequences, to different depths of understanding, and at different rates. Even children who were taught the same initial interpretation of fractions showed

different learning profiles. The time-honored learning principle of transferability was robust. Not only did children transfer their knowledge to unfamiliar circumstances, but to other interpretations that they were not directly taught. By the end of sixth grade, more than 50% of the children had demonstrated the ability to apply their knowledge to at least two of the rational number subconstructs. Table 14.3 shows the numbers of students from each class who were able to reason proportionally at the end of 4 years, the number from each class who could pass a computation test at the 80% level, and the number from each class who had demonstrated the ability to apply their knowledge to 1, 2, 3, or 4 of the rational number subconstructs.

Student reasoning and problem solving in the five special classes were extraordinary. Their abilities brought into question the conclusion from former (now classic) research studies that number relationships affect the solution of fraction or ratio comparison tasks. For example, consider this comparison task:

Are these fractions equivalent? Show how you figure it out.

$$\frac{3}{5} \quad \frac{7}{11}$$

If you are like most adults, you either used a common denominator or checked the cross products. When elementary school students in the beginning stages of fraction instruction are asked this question, most conclude that the fractions are not equivalent.

Grade 5 student: They are not equivalent because there is no number I can multiply the 3 and the 5 by to get 7 and 11.

Table 14.3 Achievement of the Four-Year Participants from Each Class on Three Criteria

Class	Numbers of Students Achieving Each Criterion						
	Proportional Reasoning	Computation ^a	Number of Rational Number Subconstructs				
			1	2	3	4	5
Unitizing ^b	8	13	19	8	7	1	0
Measures ^c	8	12	17	11	9	3	0
Operators ^d	3	9	15	9	0	0	0
Quotients ^e	2	9	16	7	0	0	0
Ratio/Rate ^f	12	11	18	10	6	0	0
Control ^g	1	6	11	2	0	0	0

^a 80% accuracy or better; ^b $n = 19$; ^c $n = 17$; ^d $n = 19$; ^e $n = 18$; ^f $n = 18$; ^g $n = 20$

Students will, in fact, arrive at the same conclusion for the same reason when asked to compare $\frac{4}{6}$ and $\frac{6}{9}$. They have no notion of equivalence, but they remember their teachers saying that “to get equivalent fractions, you multiply the numerator and the denominator by 1 or 2 or 3, and so on.” Older children who have learned how to reduce or to find common denominators do a little better. However, middle school students generally perform very poorly on this type of comparison task (for example, see Karplus et al., 1983b). Either children do not know how to proceed, or they resort to incorrect additive strategies. Researchers have concluded that the lack of integral multiples between the given ratios and within the given ratios greatly contributes to their difficulty. Students can more easily compare fractions or ratios such as $\frac{3}{12}$ and $\frac{5}{24}$, $\frac{6}{8}$ and $\frac{12}{15}$, or $\frac{5}{9}$ and $\frac{6}{15}$.

Children, who were never given rules, did not perform as the *integral multiples* hypothesis would predict. In fact, when they were given all of the tasks used in the Karplus et al. study, the students made no more errors on comparison tasks without integral multiples than on tasks involving integral relationships. This suggests that the nature of the instruction children receive may have more to do with their ability to compare these fractions than with the kind of numbers involved. The children in the Karplus study were in sixth and eighth grades and presumably had rule-based instruction. When rules and procedures are not learned with meaning, students forget them or do not always realize when to use them. When children are accustomed to thinking and reasoning without rules, which numbers they are given makes little difference.

The question naturally arises as to the kind of learning experiences that will result in a well-rounded understanding of the rational numbers. One way that question is asked is: Should children be taught one interpretation in depth or should instruction include all of the interpretations? This seems to be the wrong question. All of the interpretations do not provide equal access to a deep rational number understanding, and no single interpretation is a panacea. However, interpretations are tightly intertwined, and with the proper attention to the central multiplicative structures, students will naturally develop an understanding of multiple interpretations. Results are briefly summarized by class.

Part-Whole Comparisons with Unitizing

Unitizing added a dynamic aspect to the traditional part-whole interpretation so that children developed a very strong notion of the unit, and of equivalent fractions, which definitely facilitated the operations of addition and subtraction. This interpretation

has strong, natural connections to the measure interpretation, to ratios, and to operators. By the end of sixth grade, some students had a working knowledge of four interpretations.

The part-whole interpretation of rational numbers does not provide a particularly intuitive or insightful path to fraction multiplication. For example, using part-whole language or thinking to multiply $1\frac{1}{3} \cdot 2\frac{1}{2}$ is difficult. Nevertheless, for division, it led to the common denominator algorithm.

$$1\frac{1}{4} \div \frac{2}{3}$$

$$1\frac{1}{4} = \frac{5}{4} = \frac{15}{12} (\frac{1}{3}\text{-parts}) \text{ and } \frac{2}{3} = \frac{8}{12} (\frac{1}{4}\text{-parts})$$

How many times can you measure 8 parts out of 15 parts? $\frac{15}{8}$

$$\text{This tells us that } \frac{5}{4} \div \frac{2}{3} = \frac{15}{12} \div \frac{8}{12} = \frac{15}{8}$$

In this case, both the divisor and the dividend were reinterpreted in terms of equal-sized pieces of the unit so that the division was essentially a whole number division.

Quotients

Quotients shared natural connections with ratios and rates, and there was a great deal of power in being able to interchange quotients and ratios in sharing and comparing contexts. Students could easily choose and use equivalent fractions, compare fractions, and add and subtract fractions. However, the thinking used for addition and subtraction did not prove as easy to use for multiplication and division. The model broke down and became too complicated for multiplication and division, and area models were used to provide a meaningful introduction to multiplication and division. The children were already using the language and ideas of operators, which had evolved during their partitioning activities. Thus, introducing the area models to the children was quite natural and easy.

Operators

Although the operator interpretation provided a useful context for multiplication and division, scaling, and general fraction sense, it did not lead naturally to fraction addition and subtraction. This is because rational numbers have a dual nature. In mathematical terms, they are a field; they obey two sets of group axioms, those for addition and those for multiplication. These operations are defined independently from one another, and in the operator construct, perhaps moreso than in the other interpretations, the separateness of those operations was evident.

However, the representations used for multiplication and division, area models, could be extended quite easily to compare fractions and to build meaning for symbolic addition and subtraction.

$$\text{Add } \frac{2}{3} + \frac{3}{4}$$

Look at $\frac{2}{3}$ of a unit and $\frac{3}{4}$ of the same unit by partitioning a unit area into thirds in one direction and into fourths in the other direction.



It can then be seen that $\frac{2}{3}$ (2 rows) is $\frac{8}{12}$ of that area and that $\frac{3}{4}$ (3 columns) is $\frac{9}{12}$ of that area.

$$\text{So } \frac{3}{4} > \frac{2}{3} \text{ and } \frac{2}{3} + \frac{3}{4} = \frac{17}{12}.$$

Measures

Students developed strong notions of the unit and subintervals, equivalence, the order and density of the rational numbers, and the operations of addition and subtraction. Most children naturally extended their knowledge to the operator interpretation. For example, from their operations on number lines and the supporting use of arrow notation, they already knew that $\frac{1}{2}$ of $\frac{1}{6}$ is $\frac{1}{12}$ and $\frac{1}{3}$ of $\frac{1}{7}$ is $\frac{1}{21}$. They were well prepared for the language of operators and had no difficulty connecting to that division model. Some of these children demonstrated a knowledge of four interpretations by the end of sixth grade.

Ratios

The greatest difference between ratios and the other interpretations of rational numbers is in the way they combine through arithmetic operations. The other interpretations of rational numbers are all different conceptually, but they are indistinguishable once they are written symbolically. They add, subtract, multiply, and divide according to the same rules. However, one does not operate on ratios in the same way as on fractions.

Students who studied ratios and rates as their primary interpretation of rational numbers developed a very strong notion of equivalence classes and of proportional reasoning in general. They easily switched between ratio and part-whole comparisons and had no trouble with fraction addition and subtraction. Most of them developed their own ways of reasoning about

multiplication and division. For example, to multiply $\frac{3}{8} \cdot \frac{2}{3}$, they used proportional reasoning:

$$\frac{1}{3} \text{ of } \frac{3}{8} = \frac{1}{8}, \text{ so } \frac{2}{3} \text{ of } \frac{3}{8} = \frac{2}{8} \text{ or } \frac{1}{4}.$$

Most of the ratio class developed a good working knowledge of ratio, part-whole, and operator.

Conclusions and Implications

The deep connectivity of the rational number subconstructs made teaching only one interpretation impossible. The rational number personalities that I have discussed highlight different and essential characteristics of the rational numbers, but they are inextricably connected. In the end, the measure subconstruct seemed to be the strongest because it connected most naturally with the other subconstructs; in fact, I agree with Kieren that part-whole is not a separate construct, but really a case of the measure subconstruct (Kieren, 1988). Not all of the subconstructs were equally good starting points; operators and quotients were less powerful than the measures, ratios, and part-whole subconstructs.

Children need a starting place in one of the rational number interpretations for their fraction instruction. In this space, they need to develop the ideas of unit and equivalence of fractions. They need to develop techniques for comparison so that they can judge the relative size of fractional numbers. They need sufficient time to work in that interpretation without being given rules so that they develop fraction sense, a comfort and flexibility in fraction thinking. Although the measure subconstruct and the ratio subconstruct were especially powerful, pretesting suggested that informal experiences and intuitive knowledge better prepared some children for being sharers or comparers and others for being concatenators or composers when performing multiple operations. Thus, children were placed into their groups to best take advantage of their preinstructional strengths, and clearly not all children were equally prepared to deal with the measure and ratio interpretations. A more significant factor in overall success, no matter where a child was placed, was the development of the central multiplicative structures.

Will Incorporating More Rational Number Subconstructs Be Effective?

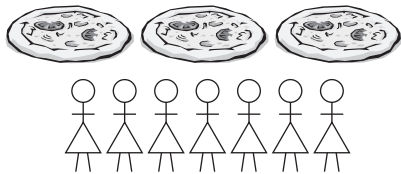
My research suggests that incorporating rational number interpretations other than the traditional part-whole meaning, will, in itself, be insufficient to fa-

cilitate meaningful learning. Children need in-depth grounding in a specific rational number interpretation (part-whole comparisons, ratios, operators, measures, or quotients) to which they can return to center their thinking when they face a new problem. In addition, I question whether studying all of the interpretations will allow students the time to get to know any of the subconstructs in depth.

This observation is supported by evidence, even when working in-depth with a single interpretation, of a long-term learning process. As an example, in another study (Lamon, 1996), many students whose instruction began with partitioning (the quotient subconstruct) took up to 2 years before they could answer the following questions:

If the girls each get an equal share of the pizza, how much pizza will each girl receive?

How much of the total pizza will each girl receive?



Because of all the unnecessary marking and cutting they did, students were unable to reassemble the many pieces that comprised one person's share (3/7 of a pizza) until they had some notion of equivalence and could distribute larger chunks to each person. Understanding the task also demands that one know that when 7 people are sharing something, each share will be 1/7 of the unit, regardless of the unit, whether or not it is composite, and what size the components are. Many adults were instructed long ago that the fraction symbol 3/7 means "3 divided by 7;" however, they fail to realize that that fact has anything to do with the sharing situation presented here, or that 1/7 of the total pizza is 3/7 of a pizza. Thus, partitioning and the quotient interpretation of a rational number are not easily grasped, and a superficial treatment of the process and concepts does not result in understanding. This perspective is further substantiated by the small numbers of children who understand part-whole comparisons in spite of their historical emphasis in instruction (Lesh et al., 1983).

Current part-whole fraction instruction likely could be improved merely by allowing children the time and opportunity to build understanding of the central multiplicative structures and not directly presenting rules and algorithms. In fact, in a 4-year follow-up study in which instruction in the part-whole subcon-

struct was supplemented with materials emphasizing the central multiplicative structures, results were similar to those in the first study. Details concerning the pre- and post-tests and the nature of instruction under each subconstruct are given in Lamon (in progress).

Building meaning took a long time, and during the first 2 years, students in the rational number study were no competition for other students who were already performing fraction computations. However, in the long run, all five classes of students surpassed the rote learners, and their representations showed that they were performing meaningful operations. Of course, this delayed gratification has many implications for teacher accountability, state and national assessment, and many other aspects of schools and schooling.

Developing Research Agendas

Shulman (1987) suggested that not all research sites are of equal importance. Some areas represent more critical sites because they can result in dramatic reconceptions with far-reaching results. As he pointed out, all too often, researchers choose their research sites because of their amenability to available concepts and research methodologies. However, the complexity of the multiplicative conceptual field has been impenetrable with traditional perspectives and research methodologies. Researchers in this area have had to find new ways to think about the content, new vocabulary by which to discuss it, and new methodologies by which to study student thinking (Harel & Confrey, 1994).

Mathematics education has a compelling need for longitudinal studies with the same children in the domain of rational numbers and proportional reasoning. Many of the generalizations about stages and order of learning in mathematics education have been made by taking snapshots of children's thinking. A persistent problem in snapshot research is that a child's actions at any particular time really depend on many causes—constraints and affordances—in his immediate environment. What the child might have done under different conditions, over longer periods of time, goes undiscovered. Particularly in biological research, models suggest that two of the prerequisites for natural selection are enough variation and enough time (Lamon, 2003; Pinker, 1997). This was true for the children in the longitudinal study. For the first couple of years, in comparison to the children who received traditional part-whole fraction instruction, the special classes appeared to make little progress. However, once children found processes that worked and these were internalized, they had more powerful ways

of thinking than the children who had been taught to execute standard algorithms. They had been given the time to develop more diverse perspectives that, in the end, provided them with great flexibility.

Another important consideration is that, given the difficulty and the complexity of the multiplicative conceptual field, progress is unlikely to be made if researchers do not adopt personal research agendas, life-long areas of interest that keep them focused on a significant issue. Researchers are encouraged to carve a niche in which they will focus in-depth on a particular piece of the mathematics and on children's thinking about that mathematics. Research that moves from topic to topic in "opportunity" studies is not as powerful in advancing knowledge in the field.

In the interest of encouraging new researchers to devote their time and talent to this engaging domain, I will conclude with some ideas for building research agendas. Usually studies answer some questions and unearth many others, and this was certainly the case in the longitudinal research reported in this chapter. Some general directions as well as a sampling of more pointed questions are presented.

Adapting the Framework

The framework used for the study reported in this chapter provides a useful starting point and can be used as a springboard for additional studies. It could be used in a number of ways, in whole or in part, for future research.

Research in Microworlds

The work of Les Steffe and his colleagues is an example of conceptual microanalysis. The work of studying in fine detail how children construct units and operate with them is critical in psychologizing mathematical content. Much more work of this nature is needed. In particular, given the importance of measurement ideas to understanding not only rational numbers, but all of mathematics, a microanalysis of the development of measurement could provide another long-term research agenda with broad impact.

The Reversibility of Processes

The concept of multiplicative inverses becomes important when a child enters the world of rational numbers. Children do not always construct operations and inverse operations at the same time. Similarly, Lamon (1994) found that when children first constructed complex units, they were not immediately able to decompose those units into their constituent parts. Researchers know very little about reversibility or about multiplicative operations and inverses, and

these could be subjects for a valuable microanalysis research agenda.

Focus on a Single Central Multiplicative Structure

Another way to conceive of doing research in the domain is to form a research agenda based on one of the central multiplicative structures that seeks answers to the many questions surrounding that single structure. For example, how do children use unitizing in coming to understand each of the rational number subconstructs? Alternatively, building an agenda around *quantities and covariation*, one might study the process by which children recognize the requirements of a ratio-preserving mapping and later, construct such mappings. Very little is also known about children's thinking in inversely proportional contexts.

Focus on Children's Thinking at the Beginning of Third Grade

One might investigate the possibility that different subconstructs might provide better inroads to the rational numbers for different students. Some evidence from pretests given to students in the longitudinal study reported in this chapter, indicated that children who are oldest in their families are more inclined to be sharers (partitioners) and children with older siblings, comparers (ratio thinkers). My hypothesis is that the need for younger siblings to hold their own within their respective families might make them better comparers. Mothers frequently give snacks to the older child with directions to share them with younger brothers or sisters. After being shortchanged a few times, the younger child begins to monitor the division of the goods. In the same vein, some third-grade children were better at concatenating than composing and vice versa. Apparently by third grade, or the beginning of fraction instruction, children may have already developed a propensity for ways of thinking that might make one interpretation of rational numbers a more logical starting point for them in order to build upon their informal knowledge.

Generative Ordering

Kieren's work (e.g., Kieren, 1993) is an excellent example of research that considers the generative nature of knowledge. Intelligence bootstraps itself both by enlarging its domain of effective operation but also by building its own depth or *interiority* by embedding previous thinking and elaborating it into more sophisticated forms. Kieren's work used the recursive nature of knowledge building to try to understand how one can build rational number knowledge from basic concepts and the actions they support. The goal of the framework presented in this chapter was to capture

the germs of the vast system of relationships that interact and eventually build upon one another to facilitate proportional reasoning, but much more generative ordering research (but on a smaller scale than Kieren uses) is needed to more fully understand the power of various elementary structures and mechanisms. For example, beginning with a single idea or action—say, unitizing and norming, or partitioning—use a deep mathematical analysis to determine how far the idea or the process can be pushed to build increasingly sophisticated forms of rational number knowledge. Then, test that theoretical analysis in empirical research to see if or how it corresponds to what children do or can do.

Other Outstanding Questions

Many other researchable questions lie in new areas, or in areas where much fertile ground still exists:

- What are the links between additive and multiplicative structures?
- What mechanisms drive the change from additive to multiplicative reasoning?
- What intuitive (informal) strategies are available to children before fraction instruction?
- What are the limits of a child's intuitive knowledge in building fraction concepts and operations?
- What knowledge beyond fractions is needed to understand rational numbers?
- How does one measure rational number sense?
- What are the developmental ceilings for understanding multiplicative concepts at various ages?
- Which aspects of whole number knowledge are useful in building fraction knowledge?
- How does a child come to understand a rational number as a single quantity as opposed to regarding it as a pair of numbers?
- How is the quantification of motion best taught?
- What do children of various ages understand about distance-rate-time relationships?
- How do children come to understand this triad?
- What is the relationship between the development of increasingly complex unit structures and unit splitting?
- What is the connection of this splitting that leads to exponentiation and the successive partitioning that was used to study the measurement interpretation of rational numbers?
- What are the connections between rational number learning and algebra?
- What are the connections between rational number learning and calculus?
- As children study each of the rational number interpretations, what are the benchmarks by which to judge that their knowledge is moving in a desirable direction?
- What constitutes understanding of each of the rational number subconstructs? Specifically, how is understanding operationalized? (What are the observable effects of understanding a particular rational number subconstruct?)
- How can we assess depth of understanding of the rational numbers?
- How many subconstructs should a student understand before high school algebra? Before calculus?
- Instruction must build upon children's intuitive knowledge but push them beyond it as well. What are the limits of children's intuitive knowledge and qualitative reasoning? How far can that intuitive knowledge be pushed, and what critical mathematical content and ways of thinking will not occur without instruction?
- How can unit analysis help to connect fraction learning with other conceptual areas?
- What are the connections between proportional reasoning (as defined in this chapter) and proportionality?
- What constitutes understanding of proportionality?
- What are some of the benchmarks of understanding between proportional reasoning and proportionality?
- Is it better to teach one rational number subconstruct or all five?
- If one, which should it be?
- What are the conceptual difficulties (learning problems) associated with each rational number subconstruct?
- Do algorithms preclude reasoning, or can older students develop useful knowledge about the central multiplicative structures?

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EARLY ALGEBRA AND ALGEBRAIC REASONING

David W. Carraher

TERC

Analúcia D. Schliemann

TUFTS UNIVERSITY

WHY ALGEBRAIC REASONING?

This chapter provides an overview of research about algebraic reasoning among relatively young students (6–12 years). It focuses on mathematics learning and, to a lesser extent, teaching. Issues related to educational policy, epistemology, and curriculum design provide a backdrop for the discussion.

We will adopt as our working definition of *algebra* a description given by Howe (2005) at a meeting about algebraic reasoning in grades K–12:

[Algebra involves]

- i. Working with variables, and in particular, arithmetic with variables, so the formation of polynomial and rational expressions. This also includes representing, or “modeling” concrete situations with expressions, and setting up equations. It is also often extended to include extracting roots. . . . It also includes manipulating expressions and equations, to simplify, solve and interpret.
- ii. Algebraic structure, primarily as captured in the Rules of Arithmetic (aka, the field axioms). The Rules of Arithmetic encapsulate the legal manipulations on polynomial or rational expressions. If taking rational powers is allowed, they have to be supplemented by the Law of Exponents and the

multiplicativity of fractional powers. These rules, together with the principles for transforming equations (the original techniques which gave rise to the subject known to us as algebra), summarize the basis for algebraic technique, which works on the algebraic expressions described in i), and on equations between them. (p. 1)

The references in Howe’s definition to “arithmetic with variables,” “modeling,” and “Rules of Arithmetic” are especially noteworthy. They reflect the idea that “algebra is inherent to arithmetic.” or, as we have expressed it, arithmetic has an “algebraic character” (Carraher, Schliemann, & Brizuela 2000; Schliemann, Carraher, & Brizuela, 2007). They further suggest that arithmetic and elementary algebra are not fully distinct.

We shall treat *arithmetic* as the science of numbers, quantities, and magnitudes. Students’ understanding of number develops throughout many years, covering the natural or counting numbers, integers, rational numbers (common fractions and decimal fractions) and, to a very limited extent, real and complex numbers. Operations with numbers vary according to the domain of number. This, together with the fact that students continually learn new conventions for representing numbers and operations on them—e.g., place value and rules for computation (Howe,

2006)—means that arithmetic is a moving target, from the student's and from the teacher's perspectives. Some topics, such as the divisibility of integers, are integral to school mathematics but also belong to special sub-domains of mathematics: divisibility for example is part of elementary number theory or "the higher arithmetic" (Davenport, 1968). We will argue that arithmetic entails functions, but the kinds of functions related to operations and properties suitable for elementary and middle school students are yet to be determined and remain a topic of importance for mathematicians and learning specialists to work out.

Algebraic reasoning refers to psychological processes involved in solving problems that mathematicians can easily express using algebraic notation. This is admittedly a difficult and somewhat subjective matter, one that runs the risk of over-attributing competence to students. However, in the same sense that societies solved problems of algebra before the existence of algebraic notation (Harper, 1987), students may be able to work with variables and the Rules of Arithmetic (i.e., the field axioms) before they have been taught algebra. As we shall see, there is also some compelling evidence (e.g., Daydov, 1991) that students can learn to use algebraic notation and techniques themselves.

We will use the expression *early algebra* (EA) to encompass algebraic reasoning and algebra-related instruction among young learners—from approximately 6 to 12 years of age. Such a definition is useful for getting started, but in order to appreciate the puzzles, challenges and controversies of the field one needs to consider ideas and research findings from recent decades. What emerges, we hope, is a more integrated and challenging view about the early mathematics curriculum, especially with regard to the nature and purposes of arithmetic instruction.

RATIONALE AND STRUCTURE OF CHAPTER

In this section we address the rationale and structure of the chapter. Next, in "A Traveler's Guide To Early Algebra," we present terminological and conceptual distinctions useful for getting around in the field. In the three sections on "Entry Points to EA" we consider various approaches and research findings about elementary school children learning EA. In the final sections, "Special Issues in EA" and "Concluding Thoughts," we summarize the main findings and consider the still open issues about EA and their

implications for teacher education, curriculum development and policy.

The idea of writing a chapter on EA may appear unnecessary. After all, handbooks typically do not include chapters on "early calculus," "early geometry," or "early statistics." Besides, textbooks and syllabi from early grades have tended to include relatively little content clearly related to algebra. Furthermore, it is well known that many adolescents have difficulty learning algebra. Could the situation really be more promising for younger students? Would our time and energy not be better spent strengthening the existing early mathematics curriculum? If students had a more solid grounding in arithmetic and elementary geometry, would they not be better prepared when they encountered algebra at its traditional place in the curriculum? These are reasonable questions. However, as we will try to show, there are compelling pragmatic and scientific reasons for paying special attention to the role of algebra in elementary school.

The Recent Focus on Algebraic Reasoning in the Early Grades

Some may react to the idea of introducing algebra in elementary school with puzzlement and skepticism. After all, much prior research highlights the difficulties that middle and high school students have with algebra, showing, for instance, that these students (a) believe that the equals sign¹ only represents a unidirectional operator that produces an output on the right side from the input on the left (Booth, L., 1984; Kieran, 1981; Vergnaud, 1985; Vergnaud, Cortes, A., & Favre-Artigue, P., 1988b); (b) focus on finding particular answers (Booth, 1984); (c) do not recognize the commutative and distributive properties (Boulton-Lewis, 2001; Demana, 1988; MacGregor, 1996); (d) do not use mathematical symbols to express relationships among quantities (Bednarz, 2001; Bednarz & Janvier, 1996; Vergnaud, 1985; Wagner, 1981); (e) do not comprehend the use of letters as generalized numbers or as variables (Booth, L., 1984; Kuchemann, 1981; Vergnaud, 1985); (f) have great difficulty operating on unknowns; and (g) fail to understand that equivalent transformations on both sides of an equation do not alter its truth value (Bednarz, 2001; Bednarz & Janvier, 1996; Filoy & Rojano, 1989; Kieran, 1989; Steinberg, Sleeman, & Ktorza, 1990). Collis (1975), Herscovics & Linchevski (1994), Kuchemann (198), and MacGregor (2001), among others, have attributed students' difficulties to insufficient cognitive

¹ Both "equals sign" and "equal sign" are encountered in the literature. We will adopt the former throughout.

development. And Filoy & Rojano (1989) and Sfard (1995) further relate students' development and their cognitive difficulties with algebra to historical trends in mathematics.

However from time to time in recent decades mathematics educators (Davis, 1967a, 1967b, 1972, 1985, 1989; Davydov, 1991) have discussed the idea of introducing algebra much earlier than its traditional appearance in high school courses. Kaput (1995, 1998) argued that the weaving of algebra throughout the K–12 curriculum could lend coherence, depth, and power to school mathematics, and replace late, abrupt, isolated, and superficial high school algebra courses. Proponents of EA have stressed that the current content of elementary mathematics is not fully distinct from algebra (Mason, 1996; Bass, 1998; Carraher, Schliemann, & Brizuela, 2000, 2005). A deep understanding of arithmetic, for example, requires mathematical generalizations that are algebraic in nature. Some have argued that algebraic notation makes it easier for adults as well as young learners to give expression to such mathematical generalizations (Brizuela & Schliemann, 2004; Carraher, Schliemann, & Brizuela, 2000; 2005; Schliemann, Carraher, & Brizuela, 2007). Bodanskii (1991) suggests that “the algebraic method is the more effective and more ‘natural’ way of solving problems with the aid of equations in mathematics” than the arithmetical method (p. 276). An Algebra Initiative Colloquium Working Group (Lacampagne, 1995; Schoenfeld, 1995) proposed that algebra should pervade the curriculum instead of appearing in isolated courses in middle or high school. And the 12th International Commission on Mathematical Instruction (ICMI) Study Conference on the Future of the Teaching and Learning of Algebra (Chick, Stacey, Vincent, & Vincent, 2001; Lins & Kaput, 2004; Stacey, Chick, & Kendal, 2004) created a special working group on issues related to EA.

A Decisive Moment

Two events capture the current political significance of EA for mathematics educators in the United States.

Event 1: NCTM's Endorsement

The National Council of Teachers of Mathematics (NCTM, 1989; 2000) recommended that an algebra strand be woven deeply throughout the preK–12 curriculum:

Algebra encompasses the relationships among quantities, the use of symbols, the modeling of phenomena, and the mathematical study of change. The word al-

gebra is not commonly heard in elementary school classrooms, but the mathematical investigations and conversations of students in these grades frequently include elements of algebraic reasoning. These experiences present rich contexts for advancing mathematical understanding and are an important precursor to the more formalized study of algebra in the middle and secondary grades. For example, when students in grades 3 through 5 investigate properties of whole numbers, they may find that they can multiply 18 by 14 mentally by computing 18×10 and adding it to 18×4 , thus using the distributive property of multiplication over addition in a way that contributes to algebraic understanding.

As with number, these concepts of algebra are linked to all areas of mathematics. Much of algebra builds on students' extensive experiences with number. Algebra also is closely linked to geometry and to data analysis. The ideas of algebra are a major component of the school mathematics curriculum and help to unify it. A strong foundation in algebra should be in place by the end of eighth grade, and ambitious goals in algebra should be pursued by all high school students. (c.f. <http://standards.nctm.org/document/chapter3/alg.htm>)

In brief, teachers are now expected to introduce concepts from algebra in early mathematics instruction. And, given that NCTM is placed “firmly in the driver's seat of the school mathed reform bus” (Howe, 1998b, p. 243), it is not surprising that textbook publishers have been updating their offerings to comply with NCTM's recommendations.

Event 2: The RAND Mathematics Study Panel Report

A 2003 report commissioned by the Rand Corporation argued that “the initial topical choice for focused and coordinated research and development [in K–12 mathematics] should be algebra” because of its fundamental role “for exploring most areas of mathematics, science, and engineering. . . . [and its] unique and formidable gatekeeper role in K–12 schooling” (RAND Mathematics Study Panel, 2003, p. 47).

Although there is some agreement that algebra has a place in the elementary school curriculum, the research basis needed for integrating algebra into the early mathematics curriculum is still emerging, little known, and far from consolidated.

The NCTM directives are general and ambiguous both in terms of mathematical content (Howe, 1998a) and pedagogical implementation. In a sense, the wheels of change have been set in motion before the destination and outcomes are decided upon. By and large the curriculum materials are yet to be developed. Teacher education programs have only recently begun to respond to the call for EA, often without

having benefited from reflections on prior work of direct relevance. The mathematics education community still needs to work on establishing a solid research basis for such an undertaking.

Five Key Issues

There is another reason for a chapter specifically about EA. Discussions about EA have raised issues related to the nature of mathematics learning, the structure of mathematics, the roles of teachers, and the feasibility of “algebra for all”. Below we mention five major issues in the recurring debate about EA and the mathematics curriculum.

Issue 1: The Relations Between Arithmetic and Algebra

EA goes against the prevailing view in U.S. mathematics education, noted by several (e.g. Chazan, 1996; Gamoran & Hannigan, 2000), that algebra belongs after arithmetic and possibly for only a select minority of students who have the necessary skills and future career needs. Not surprisingly, territorial disputes sometimes arise about the distinctions between arithmetic and algebra. Filoy and Rojano (1984, 1989), for example, claimed there is a clear-cut boundary separating arithmetic from algebra; namely, arithmetical approaches such as “undoing” can handle cases as complex as equations with a variable on one side, but equations with variables placed on each side of the equals sign belong to the realm of algebra.

Whereas EA proponents generally search for continuity between arithmetic and algebra, there have been several well-argued analyses stressing the rupture or discontinuity between the two domains (Balacheff, 2001; Filoy & Rojano, 1984, 1989; Lins, 2001).

Should algebra be treated as a distinct domain with its own methods, objects and perspectives? Or can algebra be developed out of or be ‘grafted’ onto arithmetic? It might seem that this issue could be straightforwardly resolved by consulting research mathematicians about the relations between arithmetic and algebra. But the question also requires the perspectives of mathematical learning researchers, teacher educators, and historians of mathematics. Such specialists typically ask how arithmetic and algebra curricula can take into account student learning, how teachers can learn to conceive anew the curricula, activities, and goals of mathematics in the early grades, how algebra arose historically, and so on.

Issue 2: Process versus Object

Various authors have contrasted mathematics as, on one hand, operation/process/computation/procedure/algorithm and, on the other, as object/

structure/relation (Dubinsky & Harel, 1992a; Otte, 1993; Sfard, 1995; Sfard & Linchevsky, 1994). There is some disagreement over the enduring worth of process approaches. Some authors treat procedural approaches as inherently more primitive and thus in need of replacement by an object orientation; others treat procedural interpretations as different, yet nonetheless desirable even in advanced mathematical thinking. Sfard (1992) claims that students first grasp algebra from an operational perspective and, only later, can develop a structural conception of algebra; therefore, she concludes, the teaching of algebra should start from an operational, instead of from a structural perspective.

Issue 3: The Referential Role of Algebra

For some, algebraic understanding grows out of trying to represent extra-mathematical situations—that is, through modeling (Gravemeijer, 2002; Kirshner, 2001; Lehrer & Schauble, 2000). Others tend to view algebra’s referential role as a source of distraction and interference. This tension comes to a head in discussions about the role of patterns in EA instruction.

Issue 4: Symbolic Representation (narrowly defined)

Substantial differences of opinion can be identified regarding the importance, timing, and meanings associated with conventional algebraic notation. Some argue that algebraic (sometimes referred to as “symbolic”) notation deserves a prominent place in early instruction (Carragher, Schliemann, & Brizuela, 2000, 2005; Schliemann, Carragher, & Brizuela, 2007). Others believe it should be postponed for several years, on the grounds of being developmentally inappropriate or likely to engage students in meaningless symbol manipulation (Linchevsky, 2001).

The papers presented at the Research Forum on Early Algebra that took place in the 25th International Conference for the Psychology of Mathematics Education provide a good sample of the multiplicity of views on EA (Carragher, D. W., Schliemann, & Brizuela, 2001; Linchevsky, 2001; Radford, 2001; Tall, 2001; Teppo, 2001; Warren & Cooper, 2001). Some, like Linchevsky (2001), believe that algebra should continue to be introduced only after students have spent at least a half-dozen years learning the basics of arithmetic. Even proponents of EA (Blanton & Kaput, 2000, 2001; Carpenter, Franke, & Levi, 2003; Carragher, Schliemann, & Schwartz, 2007; Dougherty, in press; Moss, Beatty, McNab, & Eisenband, 2006; Schifter, Monk, Russell, & Bastable, in press), who regard algebra as deserving a prominent place in the early mathematics curriculum, are likely to disagree about issues so basic as the following:

1. What learning tasks and forms of thinking are algebraic?
2. What kinds of evidence are needed to evaluate the presence of algebraic thinking among young students?
3. What pedagogical approaches, teacher education and policy guidelines ought be encouraged?

Issue 5: Symbolic Representation (broadly defined)

One of the fundamental contributions of mathematics learning research has been to highlight the roles of other symbolic systems—particularly tabular, graphical, and natural language systems—that come into play in algebraic reasoning. There is some diversity of opinion regarding the importance of such systems of representation. Many believe they provide crucial entry points for learning algebra; some consider them to be important even after a certain mastery of symbolic reasoning (in the narrow sense) has been achieved. Still others regard them as “pre-algebraic.”

As we will see, in recent years there has been a gradual accumulation of studies of EA (see also Rivera, 2006). One would hope that, as the mathematics education community carefully considers data from empirical studies, beliefs—even ardently held ones—would be reassessed. A conjecture such as, “the more students are confident in their arithmetic competencies, the more they will be reluctant to accept the use of algebra on problems they recognize as pertaining to the field or arithmetic (or computation)” (Balacheff, 2001, p. 252) bears directly on the first issue mentioned above. And it is the sort of conjecture that needs to be assessed in the light of empirical evidence. With these issues in mind, we begin our discussion of the EA research.

A TRAVELER’S GUIDE TO EARLY ALGEBRA

School Algebra and EA

There is a certain agreement in the United States about the algebra students need to know by the end of high school (Bass, 2005). Students are to be familiar with polynomial expressions and certain trigonometric identities. They should be able to solve first order equations in one variable (e.g., $5x - 37 = 2x + 5$), quadratic equations associated with conic sections, and systems of equations in two or more variables. They should be fluent in the use of graphs for representing functions, including elementary trigonometric functions. They should be familiar with mathematical

induction as applied, in particular, to the properties of the binomial coefficients. They should be comfortable handling diverse issues about variation and covariation. They should be able to mathematize word problems, use mathematical models, and apply their conclusions to the original problem contexts. High school algebra represents only a tiny sampling of algebra as understood by research mathematicians, and perhaps not the most essential or fundamental part.

Considerably less agreement can be found in the case of early algebra. Does EA include the same topics as high school algebra? If it does not, what topics are ‘appropriate for minors’? Does it introduce new topics? Does it introduce the same methods (factoring and expansion of polynomial expressions; solving systems of equations, etc.)? Does it require significant departures in the ordering and handling of topics?

A thorough review of algebraic reasoning in the early grades would scrutinize data from nations outside of the United States and western Europe. And some important lessons could be learned from studying how other countries have dealt with contentious issues facing our own educational system. Curricular materials in wide use in Russia and Singapore reveal that students are being exposed to ideas and techniques from algebra much earlier than in the United States, often with striking success (Milgram, 2006b; Davydov, 1991). This is noteworthy and important. Although it is likely that textbooks “make a difference,” we believe that merely adopting course materials from other countries (with translation, of course) is not enough.

Research is gradually providing a sense of young children’s capabilities and of the viability of different approaches to EA and to the teaching needs to successfully implement EA activities. The news is generally more encouraging than it was two decades ago, when studies seemed to suggest that algebra was inherently too difficult for many adolescents. In addition, there is a growing recognition that EA is not simply a subset of the high school syllabus; rather, it is a rich sub-domain of mathematics education where special topics and issues are the focus of research and instruction. Furthermore the field now has a better sense of how representations and situations play roles in students’ mathematical understanding.

One might be tempted to assume that mathematics has an inherent structure that requires learners to follow the same general pathway regardless of the age at which they first begin to study the subject. According to this view, young learners begin with “the basics” and simply move ahead (perhaps) more slowly than older students. Yet for many centuries adolescents learned the basics in Latin by repeatedly practicing

declensions and verb conjugations. It might have appeared obvious to many that relatively young learners would need to learn the basics in like manner—by drilling their declensions and conjugations. But young students' versatility in learning foreign languages does not derive from skills in expressly parsing written sentences. Even though grammar and structural analysis ultimately play important roles in language mastery, they may not constitute the soundest pillars for early language learning. This said, prudence recommends acknowledging the need for a balance between basic skills and conceptual understanding (Wu, 1999).

Competence in algebra requires an ability to derive valid, non-trivial and useful inferences from written forms and their underlying structure. But this in no way suggests that the study of algebraic expressions is the only or even the best point of departure to algebra. As we have stressed before, algebra for young learners may call for somewhat different approaches than those traditionally given prominence in algebra instruction for adolescents, among which are the following (Carraher, Schliemann, & Schwartz, in press):

1. Conventional algebraic notation is not the only vehicle for the expressing algebraic ideas and relations; tables, number sentences, graphs and specialized linguistic structures can also express algebraic ideas.²
2. Context and grounding in physical quantities may be desirable in EA instruction even though they can legitimately be curtailed or downplayed in later instruction.
3. Functions provide opportunities for bringing out the algebraic character of many existing early mathematics topics and activities.

EA Versus Pre-Algebra

One can discern two families of approaches related to mathematical learning and teaching before the traditional onset of algebra instruction.

Pre-Algebra Approaches

Pre-algebra approaches (Filoy & Rojano, 1989; Herscovics & Kieran, 1980; Herscovics & Linchevski, 1994) aim to ease the abrupt transition from arithmetic to algebra. They make efforts to augment or redefine the uses and meanings of mathematical symbols such as $+$, $-$, $\times$, $\div$, and $=$ as employed in algebraic expressions and equations. The rationale is that careful intervention before algebra instruction proper

may mitigate the difficulties that beginning algebra students typically show—difficulties attributed to the inherent differences between arithmetic and algebra. Because young students were presumed to be ill suited for learning algebra, most of the research done in the 1980s, focusing mainly on equations, did not question the idea that algebra should start only in high school.

Herscovics and Kieran (1980) studied the effect of transitional approaches to algebra by attempting to expand seventh and eighth graders' notion of equality and the gradual transformation of arithmetic expressions into algebraic equations. They found that the six participants in the study showed a "clear understanding of arithmetical identities, equations, and algebraic rules" (Herscovics & Kieran, 1980, p. 579). Vergnaud (1985) and Vergnaud, Cortes, & Favre-Artigue (1988b) used a series of activities based on the behavior of a two-plate balance scale to help seventh to ninth grade students at risk of failing in algebra courses. By the end of the study the at-risk students performed better than the control group generating equations from word problems and then solving them through conventional symbolic methods and syntactic rules. Filloy and Rojano (1989) were less successful in their work with geometrical and balance scale models aimed at promoting use of syntactic rules for solving equations. Students in their study required constant intervention by the teacher to adopt algebraic methods.

Over time, as researchers moved away from equation solving as the principal activity of algebra instruction, transitional approaches switched to generalization, number patterns, variables, and functions. Transitional approaches to algebra as the study of variables and functions have been closely associated with computers in mathematics education. Healy, Hoyles, & Sutherland, (1990) and Ursini (1994, 1997, 2001), worked with Logo environments (Papert, 1980), in carefully structured activities to encourage students to move toward general algebraic representations. Despite the gulf between algebraic symbolism and spreadsheet formulas, which refer to values via their grid location, Sutherland and Rojano (1993), Rojano (1996), and Sutherland (1993) found that, through participation in spreadsheet activities, Mexican and U.K. students as young as 10 years of age, performed relatively well on algebraic tasks working with unknown quantities and operating on them as if they were known. A number of software environments specifically address the teaching and learning of variables and functions. Kieran, Boileau, and Garançon (1996)

² Some would treat these auxiliary representations as pre-algebraic (Harper, 1987). Regardless of whether one wishes to confer plenary algebraic status on such representations, they deserve recognition as being vital in students' learning trajectories.

found that software-guided support for students' intermediary representations facilitates the adoption of conventional representations. Heid (1996) found that, as they work with software, students develop algebraic thinking that allows them to model real world situations. Software environments designed by Schwartz (1996a, 1996b) and Schwartz & Yerushalmy (1992b, 1992c) give prominence to multiple representations, such as algebra symbolic notation, numbers, graphs, and natural language, and focus on functions as the basic mathematical object and the core of algebra. In Schwartz's approach, the software is a tool for modeling situations and relationships and a medium for students to represent their understandings and to flexibly move among the different kinds of representations. A functions approach to algebra seems to be extremely valuable. But, as was the case with other transitional or introductory approaches, evaluation of students' learning through the proposed activities is still scarce.

It is interesting to note that, even though many of those who developed transitional approaches to algebra recognize that part of the problem originates from students' previous experiences with arithmetic, they generally do not question the sequence of arithmetic first, algebra later.

EA Approaches

EA approaches recognize that mathematical symbols are employed diversely in arithmetic and algebra (Blanton & Kaput, 2000, 2001; Carpenter, Franke, & Levi, 2003; Carraher, Schliemann, & Schwartz, in press; Dougherty, in press; Moss et al., 2006; Schliemann, Carraher, & Brizuela, 2007; Schifter et al., in press). Likewise, they acknowledge the significant and serious difficulties that many adolescents have with algebra. But EA approaches take a different stance regarding the etiology of adolescents' difficulties with algebra. Without denying that algebra moves towards increasingly abstract mathematical objects and relations and depends on ever more elaborate techniques and representational forms, an EA approach attributes difficulties exhibited by adolescent students of algebra as due in large part to shortcomings in how arithmetic and, more generally, elementary mathematics are introduced. As Booth (1988) suggests, "the difficulties that students experience in algebra are not so much difficulties in algebra itself as problems in arithmetic that remain uncorrected" (p. 29).

We develop this view throughout the present chapter. For the moment an example or two may suffice. Carpenter and colleagues have developed teaching activities for exploring number sentences that support the notion that an equals sign (in cases such

as $5 + 4 = 9$) means more than "yields" or "produces" (Carpenter & Franke, 2001; Carpenter et al., 2003), a point easily conveyed through examples such as $9 = 5 + 4$ and $3 + 6 = 5 + 4$. Number sentences have the additional virtue of being able to express not only the reflexive property ($a = a$) of the equals sign (which students have no difficulty in accepting), but also the symmetric ($a = b \Rightarrow b = a$) and transitive ($a = b$ and $b = c \Rightarrow a = c$) properties. Remarkably, children as young as 8 and 9 years of age appear to be able to adopt and understand these uses of the equals sign when it is introduced to them through carefully constructed activities (Carpenter, Franke, & Levi, 2003).

Number sentence activities can highlight the *continuity* between arithmetic and algebra. In doing so they raise the possibility that the rupture between arithmetic and algebra is a result of a flaw in the way the early mathematics curriculum has been designed and implemented.

The distinction between pre-algebra and EA is not meant to spark a debate about terminology, much less to separate "believers from non-believers." Some of the most distinguished contributions to EA research have come from people not identified as EA researchers. Kieran (1992, 1996, 2006) for example, has provided seminal overviews of research about the teaching and learning of algebra that have been of tremendous importance for EA work. As we noted earlier, Filloy and Rojano (1989) have recognized the relative difficulty of first-order equations in which the variable appears on both sides of the equation as opposed to only one side. We happen to believe they have taken an extreme stance in arguing that these types of equations demarcate a "cognitive gap" between arithmetic and algebra. Our sense is that, although the variable-on-both-sides case is indeed more challenging, there are a number of intermediary cases for bridging the gap; we would further note that undoing, when performed on each side of an equation, is related to legitimate and important applications of inverse operations.

On the Possibilities of EA

Two questions still under debate are: Can young students really deal with algebra? And, can elementary school teachers teach algebra? Concerning the first question, which is the main focus of this chapter, until recently empirical studies on the teaching and learning of algebra in elementary schools, or of what came to be known as EA, were rather scarce. As we have noted elsewhere (Schliemann, Carraher, & Brizuela, 2006), the lack of empirical data in support of EA led researchers, educators, and policy makers to wonder about issues of possibility (Can teachers teach algebra to young students and can young students learn al-

gebra?), implementation (How can the recommendations for EA be put into effect?), and desirability (Is it important or useful for students to learn algebra early or at all?).

Examples of young children's algebraic reasoning and/or use of algebra notation are described by Ainley (1999), Bellisio and Maher (1998), Brito-Lima's (1996, 1997) Lins Lessa's (1995), Slavitt (1999), Smith (2000), Warren (2004), and Warren and Cooper (2005), among others. Adopting different approaches, these studies, as well as the studies described later in this review, show that elementary school children can successfully learn about the rules, principles, and representations of algebra in the early grades.

Regarding the preparation of teachers to teach EA, some progress has been made (see, for instance (Blanton & Kaput, 2001) and (Schifter, 2005)). This however, is beyond the scope of the chapter.

Parsing (Early) Algebra

Authors have parsed algebra and algebraic thinking diversely over the years (see Bednarz & Janvier, 1996; Bednarz, Kieran, & Lee, 1996; Bloody-Vinner, 2001; Boyer & Merzbach, 1989; Harper, 1987; Kaput, 1995, 1998, in press; Kieran, 1996; Kirshner, 2001; van Amerom, 2002). Most have worked on specific dimensions of interest; relatively few have attempted to characterize the field exhaustively. When they have so tried, the categorical structure occasionally exhibits inconsistencies and overlaps. For example, a breakdown of algebra into generalizing, problem solving, modeling, and functions mixes non-disjoint reasoning processes (generalizing and problem-solving) with a topic of mathematics (functions) and another (modeling) (Bednarz, 1996) that can be understood either as a mathematical topic or a set of reasoning processes. This may reflect the fact that the analysis of algebraic thinking is still in its infancy; or it may reflect the fact that instructional approaches differ on so many dimensions that there may not be a way to parse them into disjoint and exhaustive types. We suspect that approaches to EA will never fit comfortably into a Linnaean taxonomy: a certain degree of eclecticism may even be healthy.

The fact that approaches resist pigeonholing does not mean that the field is conceptually intractable. For instance, there is a widespread acceptance of the historical breakdown of algebra into three stages ranging from an initial rhetoric stage (expressing both the problem and its solution in natural language, through a syncopated stage (in which numbers and letters are used to stand for unknowns or variables) and finally to a symbolic stage (where letters are also used to represent *given* or assigned values) (Boyer & Merzbach,

1989; Harper, 1987; Kieran, 1996; van Amerom, 2002). Consistent with this, most EA theorists and practitioners assign considerable importance to natural language representations of algebra relations in the early stages. Likewise, they tend to agree (e.g., Bloody-Vinner, 2001) that the symbolic representation of parameters (e.g. $ax + b$) is inherently more advanced than the representation of expressions for which parameters have been assigned values (e.g. $3x + 9$). Although approaches may be conceptually messy, they provide points of access to significant theoretical and empirical work. In upcoming sections we consider approaches that are overlapping and not exclusive.

Kaput, a pioneer EA researcher, worked for many years trying to organize algebra and algebraic thinking according to general approaches (Kaput, 1995, 1998), eventually settling on three approaches, with two additional overarching strands (Kaput, in press). Kaput's three approaches are the following:

1. Algebra as the study of structures and systems abstracted from computations and relations, including those arising in arithmetic (algebra as generalized arithmetic) and in quantitative reasoning;
2. Algebra as the study of functions, relations, and joint variation;
3. Algebra as a cluster of specific modeling languages to express and support reasoning about the situations being modeled (Kaput, 2006, 2007; Kaput, Blanton, & Moreno, 2007).

Kaput's approaches are understood as *ideal types* in Weber's sense (Weber & Shils, 1949/1917, p. 90). The first approach accentuates the use of algebra to capture mathematical structures. The third approach refers to extra-mathematical modeling, that is, using mathematics to describe real-world data, relations and processes. The two approaches correspond generally to Kirshner's *referential* and *structural* "options" (Kirshner, 2001).

Because functions, relations, and joint variation are prominent in both structural and referential approaches, they do not constitute an alternative to the other two. We will deal with functions shortly, treating them as a special entry point into algebra.

Kaput's (2007) core aspects denote reasoning processes considered to flow through each of his approaches; they include:

- Generalization and the expression of generalizations in increasingly systematic, conventional symbol systems and
- Syntactically guided action on symbols within organized systems of symbols

The syntactic guidance referred to by Kaput conveys a notion that Balacheff (2001) has referred more generally to as a “control structure.” When a student solves a problem by relying on her knowledge of the quantities manipulated, “the control is left in the reference world of the modelling process (p. 252).” This calls to mind Reed & Lave’s (1991) distinction between “manipulation of quantities” and “manipulation of symbols,” made in the context of research with Liberian tailors. Balacheff continues:

The dilemma of controlling the problem-solving process in most situations involving Algebra in the school context is, taking Dettori’s (2001) words, the need for students to be able “to associate meanings to the employed symbol, and to manipulate symbols independently of their meaning.” Boero (2001) may suggest the overcoming of this dilemma by the ability of “a partial ‘suspension of the original meaning’ of the transformed expression” during the transformation process. But all authors insist on the tendency of students to come back to a reference world outside Algebra in order to check either the relevance of the actions they performed or the validity of their solution. (p. 253)

Referring to an analysis by Lins (2001), Balacheff (2001) asserts that, even when literals [algebraic letters] are used “to speak about the actions of objects of a referent world of buckets and tanks” (p. 253), quantities-based thinking falls short of being algebraic because “the control on the meaning of the symbolic expressions

as well as the control of their manipulation depends on an interpretation in this referent world (p. 253). He uses this as the basis for distinguishing between “symbolic arithmetic” on one hand and algebra on the other.

Let us now propose a set of dimensions for characterizing varieties of algebraic reasoning in broad strokes (see Figure 15.1). The Control Dimension comes directly from Balacheff, with semantically driven inferences on one extreme and syntactically driven inferences on the other. The second dimension, Focus, refers to the domain in which the phenomena to be explained lie.

Quadrant 4, in its most extreme form, comprises cases in which algebraic reasoning makes no links to worldly experience or data. Algebra is treated as a self-contained system of structures and symbol manipulation but the symbols do not stand for anything in the real world. Inferences are derived based upon axioms and accepted rules of inference. Cases of modeling that occur within this quadrant are confined to the domain of mathematics itself, for example, when one uses mathematical ideas or representations from a certain mathematical domain to clarify ideas or representations in another mathematical domain. Examples of Quadrant 4 reasoning rely heavily on inferences based on written forms. For example, given the relation, $d = m/v$, where d , m , v are non-negative numbers, one might ask what happens to the value of d , when v is tripled in value.

The extreme reaches of Quadrant 2 comprise cases in which one makes inferences about worldly

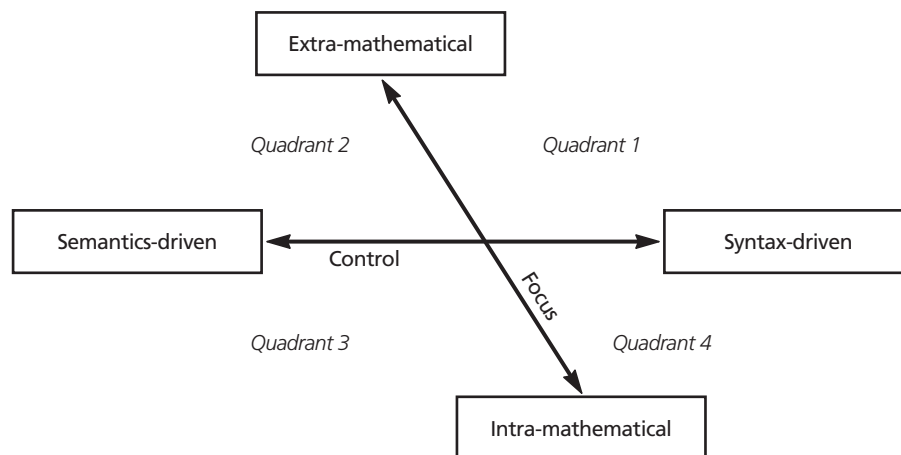


Figure 15.1 Two dimensions underlying approaches to algebra

Note: The *Focus* dimension refers to the domain of phenomena one is attempting to clarify, namely, whether they lie within or outside of mathematics. The *Control* dimension refers to the degree to which inferences are drawn on the basis of the semantics of the problem or the syntax or the representational forms.

³ We do not wish to suggest that light behaves in contradictory fashion. It may simply be that the metaphors we use to understand light (viz., light-as-particle and light-as-wave) are somewhat limited as models.

phenomena based upon information of a non-mathematical nature. For instance, one might attempt to describe how the crowdedness of gas molecules changes when the gas is moved to a container three times the size of the present container.

Quadrant 1 comprises cases in which mathematical representations can influence the drawing of inferences about extra-mathematical phenomena. For example, upon learning that the density of matter can be described as a ratio of mass to volume, a student may use the formula, $d = m/v$ to conclude that tripling the volume reduces the density to one-third of its original value.

Finally, Quadrant 3 cases correspond to those in which one's use of knowledge or intuitions about worldly phenomena help generate the mathematical representations. For example, one may take measurements of the mass and volume of a set of objects made of the same substance, plot measure of mass by volume, and conclude that there is an invariant relation between the mass and volume of items of the same substance. In other words, the student may conclude that $y = kx$ is a reasonable model for the data.

It should be clear from the present analysis that modeling is not associated with any particular region in Figure 15.1. In fact, modeling is a central, unavoidable process in any aspect of mathematics learning. It is true that the sorts of modeling activities associated with Quadrant 4 are the most challenging to young learners. And tremendous obstacles may be required for students to overcome the intrusion of semantic considerations when issues of mathematical syntax deserve prominence. In any case, the models themselves gradually become objects of study and theorization, an idea brought home by Douady's (1986) *outil-objet* (tool-object) analysis. So, for example, one can use mathematics to keep track of changes in humidity and to model how humidity varies as a function of temperature. In such cases one is preoccupied principally with the data. However, as one becomes increasingly involved in large-scale weather forecasting, the models themselves become more and more prominent. Researchers will often find their attention shifting from the data to the mathematical structures for representing data. This still falls short of pure mathematics, where extra-mathematical modeling may play no role at all, but it is still fairly distant from the extra-mathematical extreme of the continuum.

For purposes of exposition, the two complementarities can be considered independent. In actual practice, there is likely to be a substantial association between them. For this reason we have diagrammed the complementarities as non-orthogonal.

The thing about complementarities, such as the wave-particle tension in Physics, is precisely that the poles do not represent alternatives. Any significant theory of light needs to account for its wave-like and particle-like natures, even though they may seem incompatible. We would claim that any attempt to account for EA needs to take into account each pole of the two complementarities, even though each pole pulls in its own direction, so to speak.

Algebra Is Latent in the Existing Early Mathematics Curriculum

In a symposium on "The Nature of Algebra in the K-14 Mathematics Curriculum," Bass presented his view of school algebra (Bass, 1998):

School algebra—and *the root of all algebra*—is about the following:

- The basic *number systems*—the *integers* and the *real numbers* and those derived from them, such as the rational and complex numbers.
- The *arithmetic operations* (+, −, ×, ÷) on these number systems.
- The *linear ordering* and resulting *geometric structure* defined on the real line. By this I mean the notions of size (whether one number is larger or smaller than another) and of distance between numbers.
- The study of the *algebraic equations* that arise naturally in these systems. (p. 9, emphasis in original)

The final component, algebraic equations, is fully expected. The other three ingredients—number systems, the arithmetic operations, and the number line—may at first appear to be non-algebraic because they do not require that one use algebraic notation. Algebra nonetheless underlies arithmetic.

In the next three sections of this chapter we examine how conceptions of arithmetic bear on EA. We will consider specifically the cases where arithmetic is understood as the science of (1) numbers (2) numbers, quantities and magnitudes (3) numbers, quantities, magnitudes and functions.

ARITHMETIC AND NUMERICAL REASONING AS AN ENTRY POINT INTO EA

Arithmetic is conventionally understood to be the science of number. It includes the rational operations (addition, subtraction, multiplication, and division) as well as factorization and extraction of roots.

The Field Axioms and Other Properties of Numbers

The question, “What can one do with numbers?” is tantamount to asking, “What are the operations and properties of the number system in which they reside?” The field axioms in Table 15.1 convey some of these properties. They also hint at the algebraic character of arithmetic. The letters a , b can be understood as placeholders for particular numbers, but they are intended to stand for *any* and *all* numbers in the domain under consideration. Hence the field axioms assert numerical identities, general properties of numbers. One normally expresses the axioms in the formal language of mathematics—using a limited, precise grammar, and a limited set of symbols (such as the symbols for “for all,” “for each,” “there exists,” and so on). This formal language shares much with algebraic notation itself.

Table 15.1 The Field Axioms
(c.f. Weinstein, 1999)

Name	Addition	Multiplication
Commutativity	$a + b = b + a$	$a \cdot b = b \cdot a$
Associativity	$(a + b) + c = a + (b + c)$	$(a \cdot b) \cdot c = a \cdot (b \cdot c)$
Distributivity	$a(b + c) = ab + ac$	$(a + b)c = ac + bc$
Identity	$a + 0 = a = 0 + a$	$a \times 1 = 1 \times a$
Inverses	$a + (-a) = 0 = (-a) + a$	$a \times a^{-1} = 1 = a^{-1} \times a$ if $a \neq 0$

Of course, what is to be learned about number is not captured by the field axioms. For example, the axioms do not reveal how to compute $a + b$, that is, how to express $a + b$ without a plus sign once a and b have been assigned values. The same can be said for multiplication, subtraction and division; the latter two are not even explicitly recognized by the axioms. These rational operations depend upon the particular notational system in which numbers are represented. So, for example, column multiplication as we know it depends upon the structure and conventions for our present base 10, place-value system of notation.⁴ In ancient Egypt multiplication was carried out by a “halving and doubling” procedure (Menninger, 1992) that also works, but in a very different way. It strikes us as important that there be a formal “grammar” of notation systems⁵ akin to the notation-independent grammar of the field axioms.

The field axioms do not even assert that there is a unique value for $a + b$, or that there exist an infinite number of ways to additively compose any number, or

that every positive integer is uniquely factorable. Furthermore, division is closed under both the rational and real numbers, but not under the integers because a/b is not defined if b does not divide a or $b = 0$.

Numbers have additional properties that invite expression in algebraic notation. Consider, for instance, the division identity shown below:

If b is a positive integer and a is a non-negative integer, then there are unique non-negative integers q and r such that,

$$a = qb + r, \quad 0 \leq r < b$$

This means that any non-negative integer, a , can be expressed as an integral multiple of any positive integer, b , plus a non-negative remainder, r , less than the second integer. The division identity proves to be fundamental to Euclid’s algorithm for finding the greatest factor common to two integers, to continued fractions, to the fundamental theorem of arithmetic and to modular arithmetic (Bass, 1998; Carraher, 1990).

These ideas highlight the connections between arithmetic and algebra for those who are already comfortable with algebraic notation and ideas. They indicate that arithmetic has a *potentially algebraic* character. But surely this does not mean that young students who engage in arithmetic are *de facto* engaging in algebra. Or does it? We will need to give this question careful consideration in the ensuing discussion.

Studies That Introduce Algebra Through Generalizations About Numbers

Let us turn our attention for the moment to problems involving number sentences such as those in Table 15.2.

Table 15.2 Number Sentences
from Carpenter et. al. (2003)

$8 - 5 = 3$
$3 \times 4 = 15$
$3 + 5 = 8$
$8 = 3 + 5$
$8 = 8$
$3 + 5 = 3 + 5$
$3 + 5 = 5 + 3$
$3 + 5 = 4 + 4$
$9 + 5 = 14 + 0$
$9 + 5 = 0 + 14$
$9 + 5 = 13 + 1$
$3 + 5 = \square$ or $3 + \square = 8$

⁴ Across countries, there occasionally exist variations in how the computations are notated, particularly in the cases of multiplication and division, but these are of little import so long as they do not lead to different answers.

⁵ There is a similar need for a grammar of symbolic systems that are not strictly notational (Goodman, 1976), such as graphs.

Teachers working in the project led by Carpenter, Franke, & Levi (2003) initiated discussions about the truth or falsity of number sentences as a means of prodding students to think about the structural relations among the numbers. They found that third to fifth grade students in their project produced and justified generalizations such as:

“When you add zero to a number you get the number you started with.” “When you subtract a number from itself, you get zero.” “When multiplying two numbers, you can change the order of the numbers.” (Carpenter et al., 2003)

The first generalization is a natural language formulation of the Additive Identity property. The second statement asserts the property of Additive Inverse and the third remark states the Commutative property of addition. Hence, despite the fact that Carpenter’s problems did not employ the language of algebra and the students did not use algebraic notation in their answers, the students nonetheless were expressing general, algebraic properties of the number system.

Bastable and Schifter (2007) present an intriguing example given by a teacher. The example concerns a 4th grade homework problem in which students were to represent square numbers on graph paper:

Knox [the classroom teacher] explained that Adam [a student] had difficulty articulating his idea “even though he knew exactly how to do it.” But with help, he was able to share his discovery, which the group then began to explore: If you take two consecutive numbers, add the lower number and its square to the higher number, you get the higher number’s square. For example, consider 2 and 3. Adam’s rule says add 2 plus the square of 2 plus 3 to get the square of 3, or $2 + 2^2 + 3 = 2 + 4 + 3 = 9$. Knox writes:

“[The group] had some discussion about whether it would work each time and Fred insisted that it wouldn’t work with a higher number than that, so we tried 7 and 8: $7 \times 7 = 49$; $8 \times 8 = 64$; and $7 + 49 + 8 = 64$.” Adam then raised an additional question: What would you subtract from a square to find the square below the one you have?

That Adam would find it difficult to explain his theorem is not surprising. Natural language is relatively clumsy for expressing algebraic relations. In addition, Adam stumbled upon his theorem in a geometric context, so he was required to shift between drawings of squares and expressions involving numbers. Algebraic notation offers a more succinct formulation of his theorem in ways that preserve the original number and its successor. Once transcribed into algebraic notation, Adam’s reasoning conveys the gist

of the expansion of $(n + 1)^2$: for all natural numbers, n , $(n + 1)^2 = n + n^2 + (n + 1)$

Elsewhere Schifter (1999) identified other examples of implicit algebraic reasoning and generalization among elementary school children in classrooms where reasoning about mathematical relations is the focus of instruction. She concluded that:

Algebraic methods are clearly implicit in the children’s work. . . . As they apply different operations to solve a single word problem, they evidence a sense of how the operations are related. For example, as the children come to see that any missing-addend problem can be solved by subtraction, or that any division problem can be solved by finding the missing factor, they acquire experience with the inverse relationships of addition and subtraction, multiplication and division, and thus with equivalent [number sentences]. And as they develop fluency in a variety of computational strategies, they implicitly apply the laws of commutativity, associativity, and distributivity. (p. 75)

Even when children solve “missing addend problems,” such as “Maria had 7 marbles and won some more, ending up with 15 marbles; how many did she win?” they may be implicitly dealing with the additive inverse axiom. This corresponds to the number sentence $7 + \square = 15$. This sort of problem is known to be difficult for students who think that “winning” (or the plus sign) indicates that one must add in order to find the result. Further subtleties related to addition are discussed in Vergnaud’s (1982) classic treatment of the subject. It is noteworthy that by solving missing addend problems through subtraction one implicitly makes use of the additive inverse axiom.

James Milgram, a reviewer of this chapter, commented that “algebra is a specific TOPIC in mathematics dealing with specific kinds of problems—inverse problems” (Milgram, 2006a). When we asked whether that would include missing addend problems such as the above, he responded:

YES!!! This is a crucial point. For example, ordinary addition is not algebra, but the missing addend problem IS ALGEBRA. This is a student’s first contact with the issues of algebra and it is interesting that in the mathematical instruction in high achieving countries addition and subtraction are introduced at exactly the same time with subtraction being introduced as the inverse problem. In our texts, addition and subtraction are always separate chapters, and the connection between them is not part of typical instruction. This seems to be a part of the reason why algebra is such a hurdle for students in this country.

The above analysis highlights what appears to be an oft-missed digging deeper into arithmetic in order to explore its algebraic nature through the field axioms.

Quasi-Variables

The algebraic character of arithmetic is also witnessed by Fujii and Stephens' (2001) work on *quasi-variables*, a term they use to refer to the implicit variables that students seem to make use of in arithmetical contexts. Quasi-variables, commonly used in Japanese elementary school mathematics, appear "in a number sentence or group of number sentences that indicate an underlying mathematical relationship that remains true *whatever the numbers used are*" (p. 259, emphasis added). For example, a student may note that the expression, $71 - 13$, can be solved by taking away 20 from 71 and then adding 7. It is as if the student reasons that $71 - 13 = 71 - (10 + (10 - 7)) = 71 - (20 - 7) = (71 - 20) + 7 = 51 + 7 = 58$. Now, this could be mere opportunism on the part of the student. The key issue rests on how the three from 13 becomes re-presented as $(10 + (10 - 7))$. But there is the opportunity for the student to gain insight into the structure of the number system. To what extent is the student aware of the fact that she can avoid having to subtract a digit, b , from another, possibly smaller digit, by first subtracting an additional 10 (not difficult) and *adding* the 10-complement, $10-b$? We leave it open for others to decide whether a particular instance is truly algebraic. Suffice it to say here that "quasi-variables" may represent a convenient bridge between arithmetic and algebraic thinking.

Lest the reader think that one cannot have a solid grasp of advanced mathematical ideas until one can articulate them using formal language, consider Thurston's (1994) remarks:

It is common for excellent mathematicians not even to know the standard formal usage of quantities (for all and there exists), yet all mathematicians certainly perform the reasoning that they encode. (pp. 164–165)

This is a nice example of what Vergnaud (1979) refers to as theorems-in-action. One of the most important questions in EA research is how students learn and can be taught to transform theorems-in-action into explicit expressions.

There are many cases in mathematics where particular instances are reasonably construed as examples of more general mathematical objects and relations. One may use a particular drawing of a triangle to represent a triangle in general. One uses a letter to stand for a particular number or a (possibly infinite) set of

numbers. Blanton and Kaput (2000) demonstrated third graders' ability to make robust generalizations and to provide intuitive supporting arguments as they discuss operations on even and odd numbers and consider them as placeholders or as variables. They realized in this analysis that there is a constant shifting back and forth between the specific numbers and the more general classes of odd and even numbers. It is vital that teachers and researchers attempt to discern the scope of validity with which students are effectively operating. It is also helpful to scan current activities for other, possibly broader, contexts. For example, a mathematics teacher can take note of the fact that discussions about even and odd numbers can be construed as examples of modular arithmetic for modulus 2. Students may not appreciate the wider significance of the discussion. And a teacher will need to take into account a host of considerations before deciding whether to explicitly pursue such connections. But if and when she does, some care will be required to distinguish the broadened interpretations from those of her students while encouraging them to partake in the broader view.

The cautious reader will wonder whether we have been "reading algebra into" the cases we have been examining. Is it possible that in our quest for algebra among young students we have been overly zealous in ascribing algebraic interpretations and meanings? In mathematics education, especially with young learners, there is always a risk of over-attributing meaning. But it is a risk one needs to take because of the nature of mathematical reasoning.

Summary: Numerical Reasoning and EA

Several remarks serve to summarize the main points regarding the arithmetic of numbers as an entry point into EA.

1. Arithmetic has an inherently algebraic character and can be usefully regarded as a part of algebra rather than an a domain distinct from algebra;
2. Young students sometimes make algebraic generalizations without using algebraic notation (although natural language is often poorly suited for expressing algebraic relations);
3. Studies of arithmetic as an entry point into algebra are promising, but most of what needs to be known has yet to be investigated.

The first point above may resist appreciation because there are two easily conflated senses of arithmetic.

tic: *arithmetic as currently taught* and *arithmetic as it might be taught*, were its algebraic character given greater prominence. For example, narrow approaches to arithmetic conceal the symmetric and transitive properties of equality. Arithmetic need not be taught in such manner, and many of the difficulties students display in learning algebra at adolescence may reflect the misleading ways that symbols have been introduced.

ARITHMETIC AND QUANTITATIVE REASONING AS AN ENTRY POINT INTO EA

The status of *quantities* and *magnitudes* as mathematical objects depends upon one's perspective on modeling. Mathematicians sometimes prefer to think of them as residing outside mathematics, in the realm of science. However, as Whitney (1968a) noted, quantities can be treated as legitimate mathematical objects:

Let us consider the problem of choosing a model M for masses. An object A has a certain property which we call its "mass"; why not let this property itself be an element of the model? As far as the structure of the model is concerned, we need not theorize on what "mass" really is; we need merely give it certain properties in the model. For instance, if we have distinct objects A and B , with masses m_A and m_B , we may think of the objects as forming a single object C ; its mass m_C should then be $m_A + m_B$. Therefore M should contain an operation of addition, and any further properties we choose. (p. 115)

Because modeling plays such a substantial role in elementary mathematics, there are good reasons for treating quantities and magnitudes as belonging within the domain of arithmetic. This does raise the likelihood that lapses in rigor will occur. But it seems to be an unavoidable part of early mathematics teaching.

In this section we will treat quantities and magnitudes (in addition to numbers) as objects of arithmetic and attempt to characterize how this expanded view of arithmetic provides an entry point into EA.

Quantities, Measures, and Magnitudes

Historically, mathematics arose from human enterprises such as commerce (tallying and weighing), surveying (measuring land), manufacturing, astronomy (measuring and predicting position-time values for heavenly bodies), optics, architecture (building,

perspective drawing), war (trajectories of projectiles), in which physical quantities are prominent. As Atiyah (2005) observed, "Mathematics originates from the physical world . . . but is organized and developed by the human brain" (00:13:37). He later notes: "In some sense . . . the most fundamental notion of mathematics is that of magnitude, size, that some things are 'bigger than others'" (00:19:13). He refers to the specific case of volume, sound, and light, as instances of "an abstract notion of *magnitude*—mathematics starts off with such abstract notions" (00:19:47).

Significant advances in the history of mathematics are constructed upon the abstract notion of magnitude. The Eudoxian theory of proportion (Euclid & Heath, 1956), for instance, is built upon considerations of the relative lengths of line segments. And Newton's method of fluxions for finding a derivative (Newton, 1667/1671) draws on his reflections about motion.

There are admittedly good reasons for distancing mathematics from quantities. As mathematician, Aleksandrov (1989), put it:

We operate on abstract numbers without worrying about how to relate them in each case to concrete objects. In school we study the abstract multiplication table, that is, a table for multiplying one abstract number by another, not a number of boys by a number of apples, or a number of apples by the price of an apple. Similarly, in geometry we consider, for example, straight lines and not stretched threads, the concept of a geometric line being obtained by abstraction from all other properties, excepting only extension in one direction. More generally, the concept of a geometric figure is the result of abstraction from all the properties of actual objects except their spatial form and dimensions. (pp. 1–2)

Let us now put forth some basic distinctions related to quantities, drawing on various authors (Fridman, 1991; Lebesgue & May, 1966; Schwartz, 1996a; Thompson, 1988; Whitney, 1968a, 1968b):

- *Quantities* include countable and measurable properties (mass, length, time), as well as derived properties⁶ (volume, area, velocity, acceleration, density, coefficient of friction).
- *Abstract numbers* are numbers without any intended reference to concrete situations or referents outside of mathematics. Examples are 3, -0.12 , $\frac{3}{4}$, $4i$, π .
- *Concrete numbers* include counts and measures

⁶ We are following conventions of the SI base units here. It is of course sometimes the case that quantities such as area and volume (by SI standards) can be measured directly, in which case they are not derived from the observer's point of view.

- *Counts* express the *cardinality of sets of like objects* through a *natural number quantifier* associated with a *unit*; examples are “3 eggs” and “42 donations.”
- *Measures* are expressions of the magnitude of a selected property of *individual objects*; they are either the result of measurement that assigns a *number* and *unit of measure* to each respective object, or they are *derived* or composed from other measures, as density is derived from mass and volume.
- *Measurable quantities* (Fridman, 1991) of one dimension⁷ are of particular interest; they are by definition
 - *Scalar*.⁸ *Scalar quantities* are those that can be ordered along a single dimension; speed is scalar, but velocity and force are not because they involve direction as well as magnitude
 - *Additive*. *Additive quantities* are scalar quantities for which the operations of addition/subtraction apply
- *Magnitudes* are general representations of size without explicit reference to extra-mathematical quantities (Euclid & Heath, 1956). Line segments, closed regions in the plane, and closed regions in 3-space commonly represent size via length-distance, area, and volume, respectively. Expressions—for instance, A, B, A–B—can also represent magnitudes. Magnitudes can be thought of as generic physical quantities.

Quantitative Thinking and Number Lines

There is a large body of research about the evolution of young children’s understanding of physical quantities, measured as well as unmeasured. The research spans concepts of length, area, volume (Clements & Battista, 1992; Lehrer et al., 1998; Piaget, 1960; Piaget & Inhelder, 1967), mass, time, density, buoyancy and many others (Inhelder & Piaget, 1958; Piaget, 1960, 1970a, 1970b; 1974). However there are two reasons for which it has had a somewhat limited influence on the field. Firstly, given the sparse theoretical groundwork about quantitative thinking, with noteworthy exceptions (e.g., Schwartz, 1996a; Thompson, 1988; Smith & Thompson, in press), it continues to be

challenging to relate investigations about quantities, particularly unmeasured and indeterminate quantities, to studies associated with arithmetic, broadly conceived. Secondly, mathematics educators have tended to avoid studies from developmental psychology that seek out universals and downplay the roles of teaching and particular representational forms on students’ thinking. Although these concerns are justified, there is much that can be appropriated from developmental studies without endorsing the epistemologies of the researchers. And much work remains to be done analyzing the implications of quantitative reasoning for mathematics education.

Some progress has been made with respect to length. This is encouraging because length and linear ordering take on important symbolic roles (representing magnitudes) in number lines, graphs in Cartesian planes, line segment diagrams and so on. So there exist very special reasons for understanding how students make sense of length and distance.

Bruer (1993) provides an overview of the development of young students’ reasoning relating number lines and numerical understanding. This body of work offers important leads for mathematics educators and developers to pursue. It bears directly on Bass’ (1998) appeal for grounding the teaching and learning of real numbers—central to EA—on number line representations:

It is both natural and advantageous to give an early emphasis to the geometric real line model of the real numbers, in which basic arithmetic operations are interpreted geometrically and developed alongside the more algorithmic development rooted in base-10 place-value representation. (p. 15)

The number line is generally compatible with children’s learning about natural numbers and integers, despite the “fence post” issue (focusing on numbers as points when intervals are the relevant invariant) and matters of interpretation related to directed numbers (Peled & Carraher, 2007). But as students’ concept of number develops, it becomes increasingly difficult for them to reconcile operations on numbers with displacements, intervals, and points on the number line; something that has implications for using number lines to discuss scalar quantities. One manifestation of this is the work on students’ belief that “multiplying makes bigger and division makes smaller” (Bell, Fischbein, & Greer, 1984) —a belief that held up well for

⁷ We shall not consider vector quantities and higher-order tensor quantities—forces, velocities, etc.—here even though they are of considerable importance in secondary mathematics and beyond.

⁸ We are uncertain whether there is agreement on the meaning of *scalar*. Some authors seem to treat it as dimensionless, which is to say, not that scalars cannot be ordered, but rather that they are not associated with a particular physical attribute and unit of measure.

the natural numbers but no longer holds for rationals and reals (nor arguably for integers). If $x \times 2/3$ is challenging, it should not be surprising that students will have difficulty in understanding $y = \frac{2}{3}x + b$ as well as its representation as a graph in the Cartesian plane.

Over the past 40 years there have been various attempts to introduce operations on line segments consistent with the Eudoxian theory of proportion (e.g. Braunfeld & Wolfe, 1966; Steiner, 1969). Carraher (1993, 1996) shows how there is room for minimizing the rupture in the transition from natural numbers to rationals by conceiving the numerator of a positive fraction as if it were a natural multiplier and the denominator as if it were a natural divisor.

The work of realistic mathematics education (RME) regarding “empty number lines” (Gravemeijer, 1999; Klein, Beishuizen, & Treffers, 1998) can benefit from and contribute to theory and research about number lines and their relations to EA. This work is predicated on the intriguing notion that there may be advantages to withholding assignment of values to points and line segments so that operations are carried out directly on tokens representing quantities.

Can Students Apply Other Properties of Arithmetic to Magnitudes?

If magnitudes are to be formally treated as objects of arithmetic, they ought to be subject to arithmetical operations and properties, as numbers are. Among other things, this suggests that they should be subject to the field axioms and to some notational grammar. What do expressions such as $A + B$ and $A + (-A) = 0$ mean in the domain of magnitudes? If A and B are line segments, for instance, $A + B$ refers to a particular way of joining them end-to-end. But what does $A + (-A)$ mean for line segments and intervals on the number line? And what about $A + (-B)$? Does it matter if $B < A$ or $B > A$? Multiplication of magnitudes is subject to various representations. $A \times b$, the multiplication of a magnitude times a number is usually depicted as n -folding in a single dimension (Whitney, 1968a, p. 115). When A , B are linear magnitudes, $A \times B$ is generally treated as the area of a rectangle with sides of lengths A and B . But how are students to interpret the product $A \times B$ if A and B stand for quantities of different natures (e.g., an intensive quantity times an extensive quantity)? And how is $A \times b^{-1}$ to be understood on a number line? And what is the multiplicative inverse of a magnitude (or quantity). Furthermore, how is $A \times B^{-1}$ to be interpreted? One cannot simply “ignore the units,” that is, fail to consider the relations to physical quantities. As Confrey and Carrejo (2005) discovered with 5th grade students who were asked to

express, with a fraction, how 3 pizzas were to be shared equally among 10 students, the answers 3, $3/10$, and $1/10$ could all be justified, depending upon the unit assignment to the answers. Even if a consensus were to be achieved—say, around the answer $3/10$ —this might be inconsistent with some students’ conceptualization of the magnitudes and underlying quantities.

Referent-Transforming Properties

Quantities raise numerous issues that do not arise in the domain of abstract numbers. Notable among these is what Schwartz (1996a) refers to as the referent-transforming properties of multiplication and division. Consider the simple task of purchasing bottled juices at a vending machine. Let us imagine that each juice costs \$.75 and that Jonathan wants to purchase 8 juices. How much will he spend? Presuming that one has decided that the problem calls for multiplication (as opposed to, say, repeated addition), there are several ways of construing the multiplication: (a) multiply .75 by 8, obtaining 6.00 and then attaching the unit of measure, \$; (b) multiply \$0.75 by 8, obtaining \$6.00 straightaway; or (c) multiply two quantities, the intensive quantity, \$.75/can, by the extensive quantity, 8 cans, obtaining \$6.00 as the result.

These three approaches may look alike, but entail strikingly different approaches to multiplication. Method (a) is a computation in the realm of abstract numbers, requiring the student to somehow determine the units of the outcome measure. Method (b) involves the multiplication of an extensive quantity by a dimensionless, abstract number; it is somewhat congenial to the repeated addition method. Method (c) involves the multiplication of two quantities, and exemplifies referent-transformation.

The referent transformation character of (c) can be appreciated by considering the juice vending machine as a function machine. Coins are repeatedly inserted but juice comes out. If multiplication were simply a matter of repeated addition, one would expect the inputs and outputs to be of like nature; accordingly, one would expect to put in coins and receive coins instead of cans of juice.

Students will eventually need to deal with problems from science, engineering, or mundane situations in which multiple dimensions are inherent to the situation. The values along the x -axis may be of one nature (e.g., time) whereas the values along the y -axis may be of an entirely different nature (e.g., velocity). At some point the means for converting input values into outputs may require invoking the constant of proportionality of a linear function. For a student to be able to understand how the expression bears

upon the problem at hand, she will need to be cognizant of the units of measure associated with constants and variables. The constant of proportionality, a , in $ax + b$, cannot simply sum x 's repeatedly, because b and the output will be in units of different natures. The constant of proportionality will not act merely on the number associated with x ; it also acts as an exchange function, producing a unit different from that of x and different from a .

Such issues may be too subtle for students who are just beginning to learn about multiplication, but sooner or later they require addressing. How students eventually come to understand the referent-transforming roles of multiplication is an area that deserves careful empirical research.

Referent-transformation is just the tip of the iceberg when it comes to issues students must negotiate when dealing with quantities. The interested reader may wish to consult Smith & Thompson (2007) for further examples of the challenges elementary and middle-school students face in reconciling computational routines with representations of rich problem situations.

EA Studies that Focus On Magnitudes and Measures

The number line research can be regarded as a specific example of the more general claim that “visualization bootstraps algebraic reasoning and algebraic generalization promotes ‘seeing’ new spatial structure” (Boester & Lehrer, 2007). This view is consistent with Davydov’s (1991) approach to EA, with a significant line of work in Japanese mathematics education (Kobayashi, 1988), and also with the Math Workshop approach to elementary mathematics (Goldenberg & Shteingold, in press; Sawyer, 1964; Wirtz, Botel et al, 1965). By not assigning specific values, magnitudes are left indeterminate and even free to vary. As Freudenthal (1973, pp. 296–298) remarked, in a somewhat more advanced context:

[In abstract algebra] a polynomial “ $a_0x + a_1x + \dots + a_nx^n$ ” is not understood in the sense that $a_0, a_1, \dots, a_n$ are real or complex numbers but rather that they are “indeterminates” which then means that in a fixed ground-field P free transcendents $a_0, a_1, \dots, a_n$ have been adjoined and the given expression is considered as a polynomial over the field extension $P(a_0, a_1, \dots, a_n)$. Of course the substitution of elements of P for $a_0, a_1, \dots, a_n$ is allowed, but this is another thing. (p. 298)

One easily appreciates the relevance of indeterminates for introducing variables: only a slight adjustment in thinking would appear to be needed to shift from treating a letter representing a single, indeterminate value to each and every value in the domain. With regard to numbers, this entails a shift in focus from an arbitrary value to all possible values. With regard to length, there would be a shift from an arbitrary length to all possible lengths. These cases are to be contrasted with unknowns, that is, unique values that, for one reason or another, are not yet known.

Given the importance of variables in algebra, there are good reasons for mathematics educators to treat unknown values as indeterminates or variables. This holds even for equations. For instance, in $8 = 5 + x$, x is profitably conceived as a variable (and is thus free to vary) despite the fact that there is only one value, namely 3, for which $8 = 5 + x$ is true. Any other substitution for x is legitimate even though it results in a false statement. Such a framing can help students become familiar with variables from early on rather than having to radically overhaul the meaning of symbols that originally had constricted meanings.

One can imagine a line of research to investigate the conditions under which numerical assignment facilitates or interferes with learning, something from which a learning progression⁹ in number line understanding and EA might eventually be developed.

Why Quantitative Thinking Is Unavoidable in EA

There is a major reason why we cannot conceive of a pedagogically sound approach to EA that does not give considerable attention to quantitative thinking. This reason is well expressed by Fridman (1991):

In scientific theory . . . the concept of number is primary and is introduced and constructed independently of the theory of measurement of quantities; the latter is based on the previously constructed system of real numbers. It is impossible to do this in school, however, because the axiomatic method of building the number system may not be used, at least at the elementary level of instruction. (p. 163)

This argument suggests that, despite the promises of arithmetic of abstract numbers as an entry point into EA, one expects that there will be limitations. It does not seem realistic to first introduce youngsters to the algebra of number and then proceed to problems

⁹ A learning progression is a conceptual framework for the long-term evolution of understanding in a given domain, taking into account what is known about students’ prior understanding as well as the target understanding (c.f. Smith et al, 2006). The approach draws upon learning studies while avoiding the maturationist pitfall common to developmental approaches.

steeped in quantities as “applications” of algebra. One can of course expand the conception of arithmetic to encompass abstract and concrete numbers. But this merely allows arithmetic to save face, so to speak. Issues related to quantities still must be contended with.

The Davydov Approach to EA

In the 1960s, Davydov and colleagues from mathematics and psychology (Davydov, 1991) developed a comprehensive approach to EA where the primary focus was on expressing the basic relationships between explicit and implicit values of quantities. Even before children learned to work with numbers, they were encouraged “[to compare] quantities, to use notation to represent their basic relationships in formulas of equality or inequality (in letter form), and to study the properties of these relationships” (Bodanskii, 1991, p. 293). Bodanskii added:

From the very beginning of instruction, the composition of equations was introduced as a *self-contained* and unique method of solving word problems, and there was no generalization of the arithmetic method. . . . [From the start, children] . . . had to develop an ability to isolate the two equitable quantities and to write down their basic relationship, even before selecting the unknown quantity and searching for other relationships, and then to be guided by their knowledge of this basic relationship. (pp. 289–290)

Table 15.3 provides three examples of the problems children were asked to represent.

Table 15.3 Three EA Problems from Bodanskii (1991).

Problem 1: One brigade built a houses; another brigade built b houses. It is known that a is greater than b . What must be done for the number of houses built by the second brigade to be equal to the number of houses built by the first brigade? (p. 294).

Problem 2: The students in one class made a (38 toys) and in another they made b (29 fewer toys). How many toys did the students in a third class make if c (78) toys altogether were made by the three classes? (p. 295).

Problem 3: In the kindergarten there were 17 more hard chairs than soft ones. When 43 more hard chairs were added there were 5 times more hard chairs than soft. How many hard and soft chairs were there? (p. 302).

Problem 1 requires recognizing that $a - b$ is the missing quantity; problem 2 instantiates the values but is difficult nonetheless, describing the relations among a , b , and c simplifies the problem considerably; problem 3 once again instantiates values but is

even more difficult. Note that the first problem entails algebraic relations in a purely additive context, the letters representing unmeasured quantities. The second problem assigns values to the literals but the real challenge lies in the relational calculus (Vergnaud, 1979, 1982) rather than the computation. The third problem involves an even more *relationally complex situation* (Thompson, 1993) however, once the relations among quantities, known and unknown, are understood, finding the answer is fairly straightforward.

The approach described by Bodanskii encouraged the students to represent the relations between givens and unknown values through several representations: (1) diagrams of line segments configured so as to reflect the part-whole relations of quantities; (2) tables of values; and (3) algebraic notation. Students in the experimental classes used tabular notation and geometrical diagrams (line segments) to express relationships among quantities. Equations were written to register that two quantities were equal, something students would seem to have discovered by their diagrams and notations. Once the equation was written, conventional rules for solving were followed until the answer emerged in the form of a statement “ $x = \dots$ ”.

Although the equation solving among Bodanskii’s students appears to have been largely syntax-driven, the focus the instruction, particularly in the first years of schooling, fell upon examining the relationships among known and unknown quantities. The initial symbolic notation of a problem thus relied extensively upon semantic-driven inferences (see the Control Dimension in Table 15.1).

Bodanskii (1991) reports results showing that children who received instruction in the multiple representation of verbal problems from grades one to four were able to represent and solve equations to find solutions to algebra problems corresponding to first-order equations. Children in the experimental group used algebraic notation to solve the algebraic word problems, performing substantially better than their control peers. The fourth grade students even performed significantly better than sixth and seventh graders in traditional programs (five years of arithmetic, with algebra instruction beginning in grade six).

The Measure Up Project

The *Measure Up* project (Dougherty, 2007) has implemented in an American setting many ideas originating from the Davydov work. In first grade, children learn to compare objects in terms of length, height, area, volume, or mass.

Comparison situations are created for which students must communicate to others the results of

a comparison in the absence of the physical objects. At first children sometimes attempt to use colors to name the physical objects, but this proves unsatisfactory when two objects may have the same color. The teacher guides students to label objects quantities with a letter. Students are also taught to express quantitative relations among quantities through “strips,” natural language (“greater than,” “less than,” “equal to,” “not equal to”) and eventually literals and comparison operators.

The following dialogue takes place between a teacher and two first grade children who are comparing three volumes, D, K, and P:

“I think that volume D is greater than volume K,” said Caylie.

“How do you know that, Caylie? We didn’t directly compare those two volumes,” said Mrs. M.

“Well,” said Caylie, “we found out that volume D is equal to volume P and volume P is greater than volume K, so volume D must be greater than volume K.”

“I agree with Caylie,” said Wendy. “Because volume D and volume P are really the same amount so if volume P is greater than volume K, then volume D also has to be greater than volume K.” (Dougherty, 2007)

We should not assume, of course, that students have a detailed comprehension of volume; for all we know, they may be treating it as a single, orderable quantity (amount of stuff). But even a one-dimensional notion of quantity may be adequate for the case at hand.

As understanding and representation of the reflexive, symmetric, and transitive properties emerge from the comparisons (we noted transitivity in the above example), numbers and operations enter the picture. For example, using part-whole diagrams or line segments, the children learn to explain that given magnitudes A, B and $A > B$, $A - B$ is itself a magnitude that can be added to the lesser quantity (B) or subtracted from the greater one (A) to make the two quantities equal.

As operations are carried out on A, B, there appears to be a shift in what the letters refer to. Whereas the students appear to initially understand labels (A, B, C . . .) as representing particular magnitudes, the magnitudes and operations gradually provide a metric and serve as units of measure (Carraher, 1993). There may also be some foreshadowing of the notion of variable, but one must be cautious about such an inference given the well documented and non-trivial confusion between measures or counts (8P as “8 professors”) and variables (8P as “8 times the number of professors”)—see Clement, Lochhead, & Monk, G. (1981).

Dougherty (2007) described the children’s explanations as follows:

... Given $\frac{B}{E} = 5$ (Equation 1) and $\frac{B}{Y} = 8$ (Equation 2), (read as quantity B measured by unit E is 5 and quantity B measured by unit Y is 5), they conclude that because the quantities B are the same, then unit Y must be smaller than unit E. They justify this with the explanation that unit Y had to be used more times to make B than unit E was used. This means that unit Y is smaller.

Summary: Quantitative Reasoning and EA

Bodanskii’s (1991) results demonstrate that young children can learn to represent and correctly find the unknown in algebra problems corresponding to first-degree equations. However, he does not explore the nature of children’s understanding of algebraic procedures, especially in what concerns the understanding of the rules for transforming equations, functions, and variables. Even though the potential for exploration of variables and functions exists in his approach, children in his study do not deal with variables and functions, but only with unknowns.

The examples described by Dougherty (2007) suggest that a focus on quantities is a fruitful path towards the development of algebraic reasoning, including work with variables and functions, even among very young children.

ARITHMETIC AND FUNCTIONS AS AN ENTRY POINT INTO EA

The concept of function has played an extremely important role in the history of modern mathematics and has rather recently (Dubinsky & Harel, 1992b) become recognized as having a pivotal role to play in mathematics education at the secondary level and beyond. Schwartz, for example, has proposed that functions, rather than equations, be the fundamental object of algebra instruction (Schwartz, 1999; Schwartz & Yerushalmy, 1992a, b, c; see also, Chazan, 1993; Chazan & Yerushalmy, 2003; Yerushalmy & Chazan, 2002). Placing functions at the center of algebra instruction entails conceiving letters as variables, instead of unknowns, to interpret expressions as rules for functions and the Cartesian coordinate system as a space to display the results of calculations, and to take into account the multiple meanings of the equals sign (Chazan & Yerushalmy, 2003). The idea is that functions can serve to “algebrafy” existing content (Kaput, 1998). We have proposed that arithmetical opera-

tions themselves be conceived as functions (Carraher, Schliemann, & Brizuela, 2000, 2005; Schliemann, Carraher, & Brizuela, 2006).

Can Young Students Reason with Functions?

For the purposes of elementary and middle school mathematics,¹⁰ a function is a rule that assigns each element from a domain to a unique element in the co-domain; the domain and co-domains usually are comprised of numbers, but as we suggested in the former section, there are good reasons for treating quantities and magnitudes as fundamental objects; this allows for mappings across sets of like or unlike natures—for example integers to magnitudes or elapsed time values to measures of distance.

By the end of secondary school students are expected to become familiar with a very small part of what is known about functions. What grade school students might be expected to learn is clearly going to be of a much more elementary sort. But we need to ask:

- What evidence exists that young students can reason about functions? What can they understand?
- What evidence, if any, exists suggesting that functions, introduced early, can provide a solid grounding footing for EA?
- And what additional reasons might there be for introducing functions to elementary school students, when functions are known to be challenging even for university level students (Carlson & Oehrtman, 2006)?

Functions As Rules for Generating Collections of Figures

Moss et al. (2006) assessed children's learning about the rules governing visual/geometric and numerical patterns as they participated in an experimental patterning curriculum. In two main studies, one with second graders, the other with fourth graders, in diverse urban settings, Moss and her team of teacher-researchers developed activities aiming at explicitly linking ordinal pattern positions with the number of elements in that position, and to bridge the learning gap between scalar and functional approaches (see Vergnaud, 1983, for the classic analysis of scalar versus functional approaches). The components of Moss' curricula include geometric pattern building with position cards, function ma-

chines, t charts (tables), and activities that integrate these activities. Both geometric patterns and numeric patterns were part of the activities. The activities were designed to foster the integration of students' visual and numeric understandings and to move back and forth across these two types of representations of patterns.

Figure 15.2 shows a sequence of geometric forms from Moss et al.'s (2006) work.

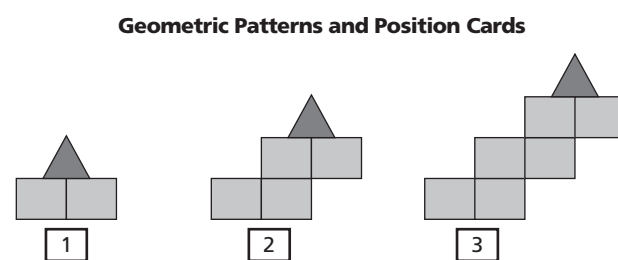


Figure 15.2 A patterns problem from Moss et al (2006). Note the explicit use of the indices corresponding to values of the independent variable.

Note that each step in the pattern sequence is numbered, thus making the order of the produced designs an explicit variable.

The study with second graders was conducted with 78 students in three experimental classrooms in New York and Toronto. On the basis of their results in a number knowledge task, children were assigned to one of three levels (low, medium, high). Twenty-two children in a fourth grade served as a control group. The intervention focused on extending students' prior *numerical* understandings and on helping them integrate their numerical and *visual* knowledge. From this perspective, the merging of the *numerical* and *visual* would provide a new set of powerful insights that would underpin the early learning of a new domain as well as subsequent learning.

A total of 20 lessons were implemented over the period of three months (Moss et al., 2006). Children first worked independently with geometric patterns and numeric patterns. This was followed by lessons designed to foster the integration of visual and numerical understandings. The final set of activities provided opportunities for students to move back and forth across the two types of representations for patterns. The specific activities involved geometric patterns and position cards, multiplicative (number $\times 2$) and composite functions (number $\times 2 + 1$), work with a function machine and tables, among others. The interview administered before and after

¹⁰ We avoid the ordered-pair definition of function that dispenses with a mapping rule. As various authors have pointed out, the ordered-pair approach is likely to be confusing for young students.

the intervention consisted of 10 patterning problems and sub items. Interview results showed that (a) fourth graders in the control group were stronger on number knowledge, (b) the experimental and control groups showed the same rate of improvement on near generalizations, and (c) the experimental group outperformed the control group on far generalizations and explicit reasoning. Children of all ability levels in the experimental group significantly improved from pre to post interview.

Children in the experimental group were able to build geometric patterns based on algebraic representations, to recognize functions from geometric patterns including two-step composite functions, and to use syncopated language to express functions. Even though they had received no formal instruction in multiplication, they displayed understanding of multiplication through use of a variety of invented solution strategies for multiplication problems. They were also more able to apply multiplication than the students in the control group who had received formal instruction in multiplication.

The second study, with fourth graders, involved 50 children in an inner city school and 22 children in a private school, who participated in 18 lessons taught over five weeks and who worked collaboratively over *Knowledge Forum*, an electronic network. Activities included “Guess-my-rule” games, building geometric patterns from rules, deciphering rules from geometric patterns, and word problems involving patterns.

During discussions, all ideas generated by the students were first treated as *improvable*. Discussions took place over a web communication resource (*Knowledge Forum*), where contributions from participants were incorporated into a process of *idea refinement*.

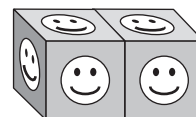
The following is an example of the kind of rules and explanations produced by fourth graders in the experimental group as they discussed the “Cube Sticker” problem (Figure 15.3).

One fourth grader explained her solution as follows:

My theory is that the rule is “times 4 + 2.” I found out the rule right after I made a chart because in the second column there was a pattern going down and the rule was plus 4. Then I thought that the rule was times 4 but when I tried it it didn’t work. So I looked at the chart again and I looked across and I saw that 2 was missing. Then I knew that 2 is constant. Then I got the rule times 4 plus 2. (Moss, Beatty, McNab, & Eisenband, 2005, slide 61)

Cube Sticker

A company makes colored rods by joining cubes in a row and using a sticker machine to put “smiley” stickers on the rods. The machine places exactly 1 sticker on each exposed face of each cube. Every exposed face of each cube has to have a sticker. This rod of length 2 (2 cubes) would need 10 stickers.



How many stickers would you need for:

- A rod of 3 cubes
- A rod of 4 cubes
- A rod of 10 cubes
- A rod of 22 cubes
- A rod of 56 cubes
- What is the rule?

Figure 15.3 A problem discussed by fourth graders in Moss et al. (2006)

In this study, children were assigned to low, medium, or high levels, based on their teacher ratings and report cards. Quantitative analyses on pre- and post-test measures revealed that low, medium, and high achiever students’ scores were improved significantly as the result of the intervention, with an effect size of 3.2 standard deviations.

Moss et al. (2006) concluded that children found rules for patterns, relationships between patterns, and relationships across different representations, thus moving towards understanding the *relationships between the two variables* in the problem. They also demonstrated that they could use mathematical symbolic language and offer proof and justifications for their conjectures. There are important ideas underlying such a statement and they are suitably challenging, so much so that we will devote the next section of our review to some special topics related to patterns and tables. At the end of that section we will be better prepared to deal with functions as an entry point into EA.

Functions Expressed Through Multiple Representations: The Early Algebra, Early Arithmetic Project

Our approach to EA, developed over a series of interview and longitudinal classroom studies implemented over 10 years, is based on the same premise underlying the work of other EA researchers, namely, that arithmetic and algebra are not fully distinct: a deep understanding of arithmetic requires mathematical generalizations and understanding of basic algebraic principles. We focus on algebra as a generalized arithmetic of numbers and quantities and view

the introduction of algebraic activities in elementary school as a move from computations on particular numbers and measures toward thinking about relations among sets of numbers. Specifically, we have proposed (Carraher, Schliemann, & Brizuela, 2000) that the operations of arithmetic should be viewed as functions. Central to our approach is the use of multiple representations, namely, natural language, line segments, function tables, Cartesian graphs, and algebra notation.

In our lessons, problem contexts constitute essential ways to situate and deepen the learning of mathematics and generalizations about quantities and numbers. Contexts are crucial to mathematics educators' concerns since students, particularly young students, learn mathematics through reasoning about various types of situations and activities. A wide range of important research aims to determine how reasoning about situations can foster mathematical understanding (e.g., Moschkovich & Brenner, 2002; T. N. Carraher, Carraher, & Schliemann, 1985, 1987; Nunes, Schliemann, & Carraher, 1993; Schwartz 1996; Smith and Thompson, *in press*; Verschaffel, Greer, & De Corte 2002). Starting from rich problem contexts and situations one hopes that, at some point, students will be able to derive conclusions directly from a written system of equations or an x-y graph drawn in a plane. The main challenge that contexts pose concerns how abstract knowledge about mathematical objects and structures can result from experience and reasoning in particular situations. While contextualized problems and a focus on quantities help in providing meaning for mathematical relations and structures, algebraic knowledge cannot be fully grounded in thinking about quantities. Important questions for EA research underlie the tension between contextualized situations and mathematical structures and relations (see Carraher & Schliemann, 2002; Schliemann & Carraher, 2002).

We have conducted three longitudinal studies with the aim of documenting how children's algebraic thinking evolves as they are introduced to algebra principles and representations over rather long periods of time (one to three years). The first study was comprised of 16 EA lessons developed over the year in a third grade classroom of 18 children (Carraher, Schliemann & Brizuela 2000, 2005; Schliemann, Carraher, & Brizuela, 2006). The second study was implemented in four classrooms (69 children), with one weekly 90 minutes EA lesson from the second semester in second grade to the end of fourth grade (Brizuela & Earnest, *in press*; Carraher, Schliemann, Brizuela, & Earnest, 2006; Carraher, Schliemann, and Schwartz, 2007; Schliemann et al., 2003). The third

study involved 26 students as they progressed from 3rd through 5th grade. In 3rd and 4th grades, the children participated in two 60 minutes lessons per week (total of 50 lessons in 3rd grade and 36 in 4th grade), each one followed the next day by a 20 minute homework review; in 5th grade they participated in one 90 minutes lesson per week (18 lessons total) followed the next day by 30 minutes of homework review. The students in all three studies came mainly from minority and first generation immigrant families located in the Boston area.

In our classroom work we generally begin not with a mathematical representation, such as a number sentence or a graph, but rather with an open-ended problem or a situation. When working with each problem, we seek to generate a teaching and learning environment that is conducive to children's presentation of their own perspectives, ideas, and ways of representing the problem. Children's first verbal reactions to the problem are brainstormed and they are asked to show, on paper, their ideas about the problem and their suggested solutions. The children in the front of the class then share the notations they produced.

For instance, in a lesson implemented at the beginning of third grade, The Candy Boxes Problem (see Carraher, Schliemann, & Schwartz, 2007), the instructor holds a box of candies in each hand and tells the students that the box in his left hand is John's, and all of John's candies are in that box; the box in his right hand is Mary's, and Mary's candies include those in the box as well as three additional candies resting atop the box; each box has exactly the same number of candies inside. Children are then asked to write or to draw something to compare John and Mary's amounts.

To introduce graphs and the Cartesian space (see Schliemann & Carraher, 2002), we start with a statement like: "Maria has twice as much money as Fred." After holding an initial discussion about the situation, students express their ideas in writing. We discuss their representations and introduce a new representation, often foreshadowed in students' own drawings, such as two parallel number lines drawn on the floor, one for Maria's and one for Fred's amounts, represented as points along the lines. We then ask individual children to represent on the parallel lines the places corresponding to different values Maria and Fred could have, as the relation "twice as much" continues to hold. As the number pairs become larger, it becomes impossible for one single child to touch the two points on the two lines, at the same time. We would then rotate one of the number lines by 90 degrees and ask the students to plot themselves as points at the intersections of ordered pairs for Maria and Fred's amounts.

A “Human Graph” is thus built and becomes the object of discussions and the basis for future work on the graphical representation of functions.

Once children have learned to represent a function via tables and graphs, we introduce problems involving the comparison of functions. For instance, in a problem implemented at the beginning of the second semester in fourth grade, *The Wallet Problem* (see Carraher, Schliemann, & Schwartz, 2007), we describe and ask the children to represent the following situation: “Mike and Robin each have some money. Mike has \$8 in his hand and the rest of his money is in his wallet. Robin has altogether exactly three times as much money as Mike has in his wallet. How much money could there be in Mike’s wallet? Who has more money?”

Because the *Wallet Problem* involves the comparison of functions, it becomes appropriate to explore when the two functions are equal, even though the problem, as originally stated, does not suggest an equation. But why did we not already present the problem as an equation, at least by telling the students that Mike and Robin have the same amounts?

To assert from the beginning that Mike and Robin have the same amounts overly constrains the functions and encourages the students to think of the variable as a single value. We leave the functions free to vary in order to encourage students to first explore the variation inherent to each function (total amount as a function of the amount in the wallet).

Only when students are informed that the amounts “happen to be equal” does it become appropriate to use an equation such as $w + 8 = 3 \times w$. Until that moment, an inequality such as $3 \times w > w + 8$ or $3 \times w < w + 8$ may hold.

When one treats equations as the setting equal of two functions, there is no need to treat unknowns and variables as fundamentally different. We prefer to think of an *unknown* as a variable that for some reason or other happens to be constrained to a single value, as when $w + 8$ is set equal to $3 \times w$. In this case, the equation holds only for certain values of w —actually, only one value, but it does not transform w from a variable into a single number or instance.

Through experience with problems of this kind, children begin to deal with more than one function at the same time, analyzing the patterns in the changes in the relationships among quantities, variables, and functions, and finding, on the graph, the value that would make the two functions equal. They also deal with the interaction between the multiple systems of representation they are using (see Brizuela & Earnest, in press).

Once children feel comfortable with the above, we begin to address the representation of equations and

solution of equations through symbolic manipulation. The following are two examples of the problems 5th graders in our third study were asked to discuss, represent in writing as an algebra equation, and to solve using the manipulation rules of algebra:

Problem 1: Anna went to the arcade with some amount of money. She then spent five dollars playing video games. After that, she won a prize where they doubled her money. The same day, Bobby went to the arcade with ten dollars. When he got there, his mother gave him thirty more dollars. Afterwards, he spent half of all of his money playing video games. At the end of the day, Anna and Bobby discovered that they did have the same amount of money. Write an equation showing that Anna and Bobby had the same amount at the end of the day. Solve the equation.

Problem 2: Elizabeth and Darin each have some money. Elizabeth has \$40 in her wallet and the rest of her money is in her piggy bank. Darin has, altogether, exactly five times as much money as Elizabeth has in her piggy bank. Elizabeth’s total amount of money is equal to Darin’s total amount of money. Write an equation showing that Elizabeth’s total amount of money is equal to Darin’s total amount of money. Solve the equation. How much money does Elizabeth have in her piggy bank?

The work with equations represented a considerable challenge for us. As we have discussed in this chapter, previous research had highlighted the difficulties that much older, adolescent, students had in dealing with equations. Moreover, the transition from the semantics of the problem to the syntactic rules of algebra, without loss of meaning is a rather challenging endeavor. However, as we will see next, some degree of success was achieved.

Throughout our three longitudinal studies, we have developed qualitative and quantitative analysis of videotaped classroom discussions, written classroom work produced by the children, and children’s responses to written assessments and interviews.

Our first study, with 3rd graders, constituted a pilot study for us, in the sense that we could then identify major issues that were to be addressed in the next studies. Three main issues we identified in the first study (Carraher, Schliemann, & Brizuela, 2000; 2005; Schliemann, Carraher, & Brizuela, 2001, 2006) should be mentioned here:

First, reasoning about variable quantities and their interrelations would constitute a natural setting for the discussions about variables and functions. However, using realistic situations to model mathematical ideas and relations presents students with challenging issues. On occasion, children were inclined to assign

fixed values to what were meant to be variable quantities, without recognizing their general character. For example, we gave children the following statement: “Tom is 4 inches taller than Maria. Maria is 6 inches shorter than Leslie.” To help them focus on the differences between the unspecified heights, we asked three volunteers to enact the problem in front of the class. When the students agreed about the relative order of heights of the three protagonists, we asked them to draw Tom’s, Maria’s and Leslie’s height and to show what the numbers 4 and 6 referred to in their diagrams. As many as 12 of the 18 students in class assigned particular values to each height, while maintaining the differences between the heights in the story consistent with the information given.

Second, when we introduced function tables relating number of items and price, we found that although the children could correctly fill in the tables, they did not attend to the invariant relationship between the values in the first and second columns. For example, to complete the table in Figure 15.4, they would count by ones to find the values in the first column and then count by threes to find the values in the second column one. When a guess-my-rule game was introduced, students were finally able to break away from the isolated column strategies they had been using.

Mary had a table with the prices for boxes of Girl Scout cookies. But it rained and some numbers were wiped out. Let’s help Mary fill out her table:

Boxes of cookies	Price
	\$ 3.00
2	\$ 6.00
3	
	\$ 12.00
5	
6	
	\$ 21.00
8	
9	
10	\$ 30.00

Figure 15.4 Table to be completed by third graders.

Third, the introduction of letters to denote any value for the first variable in a function table proved to be a powerful tool for children to focus on the general rule relating the two variables.

The activities in our second study, from 2nd to 4th grade, related to addition, subtraction, multiplication,

division, fractions, ratio, proportion, and negative numbers. The project documented, in the classroom and in interviews, how the students worked with variables and functions, using different systems of representation, such as number lines, function tables, graphs, algebraic notation, and equations.

As lessons were implemented during the first year, we witnessed the major role that cognitive development obviously plays in mathematical reasoning: we found, for example, that the idea of “difference,” fundamental to the students’ understanding of additive structures, takes a long time to evolve. It follows directly from the fact that integers comprise a group that any ordered pair of them, a, b , will have a unique difference, $a - b$. This idea takes on subtle differences in meaning across the contexts of number lines, measurement, subtraction, tables, graphs and vector diagrams (Carraher, Brizuela, & Earnest, 2001).

We also witnessed the evolution of children’s written representation for problems involving variables. In their analysis of the 69 children in our second study, Carraher, Schliemann, & Schwartz (2007) showed children’s considerable progress in terms of notation for variables from the beginning of third grade, when they attempted to represent the Candy Boxes Problem, to the middle of fourth grade, when they represented the Wallet Problem (see Figure 15.5).

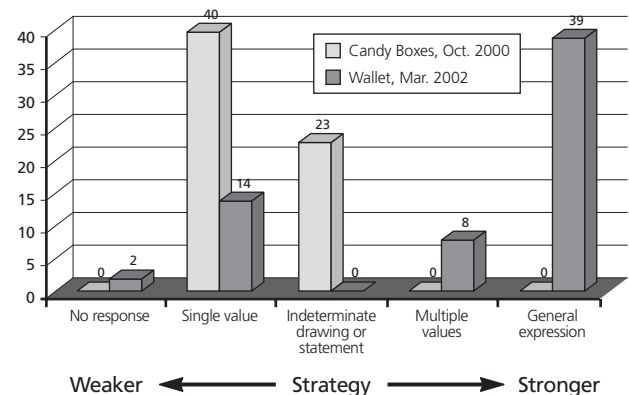


Figure 15.5 Number of children choosing each type of representation for the Candy Boxes problem and the Wallet Problem.

In third grade most children preferred to use a single value for the amount of candies in each box (see Figure 15.6). In contrast, in fourth grade most of them used general notation to represent the amount in the wallet (see Figure 15.7), with only a few still persisting in choosing a specific value.

Concerning the solution of equations, Brizuela and Schliemann (2004) described how, during the

What can you show about John and Mary's candies?
Draw or write something below.



Figure 15.6 A student's representation for the Candy Boxes problem.

Mike	Robin
Mike has \$8 in his hand plus more money in his wallet.  \$8  $N + \\$8 = \square$	Robin has $N \times 3$ money Robin has 3 times as much money as Mike has in his wallet.  $N \times 3 = 3N$

Figure 15.7 A student's representation of the Wallet problem.

implementation of the last lesson in fourth grade, children attempted to represent and solve the following problem:

Two students have the same amount of candies. Briana has one box, two tubes, and seven loose candies. Susan has one box, one tube, and 20 loose candies. If each box has the same amount and each tube has the same amount, can you figure out how much each tube holds? What about each box?

Although some of them still used iconic notations (see Figure 15.8), others generated and solved an equation using letters to represent the unknown amounts (Figure 15.9).

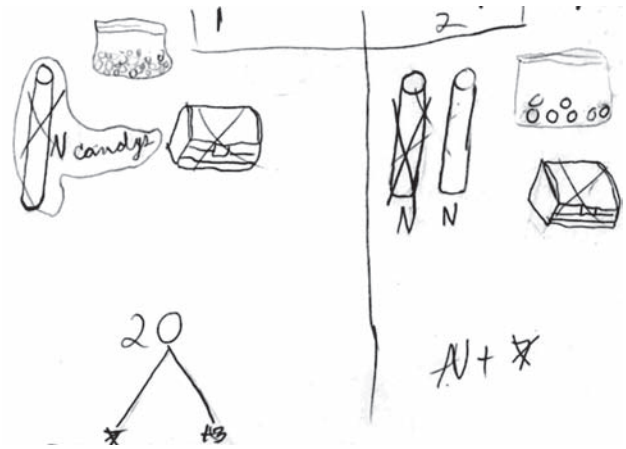


Figure 15.8 An iconic representation of the equation.

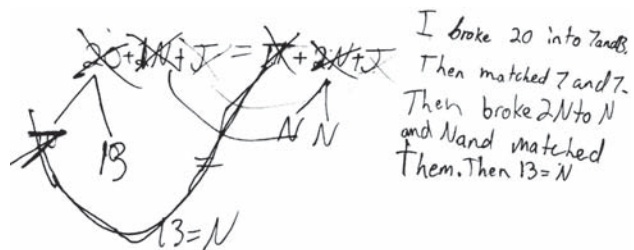


Figure 15.9 An algebraic representation of the equation and its solution.

At the end of 4th grade, three to four weeks after our last class, the students in the experimental group (EG) were individually interviewed. A control group (CG) of 26 5th graders from the same school were also interviewed. We (Schliemann et al., 2003) found that:

- 85% of the 4th graders in the EG correctly stated that the equation $6 + 9 = 7 + 8$ was true while only 65% of the 5th graders in the CG did so.
- To represent that Mary had three times as much money as John, 70% of the EG represented John's amount as N and Mary's as $N \times 3$. In the CG, only 29% of the children did so.
- When asked which of three linear function graphs showed that "Mary has three times as much money as John," 78% of the EG chose the correct line while only 46% in the CG did so. Of those choosing the correct line, 39% in the EG provided general justifications that took into account any possible pair of numbers. In the CG only 25% did so.
- Children were asked to represent in writing and to solve the problem "Harold has some

money. Sally has four times as much money as Harold. Harold earns \$18.00 more dollars. Now he has the same amount as Sally. Can you figure out how much money Harold has altogether? What about Sally?" We found that 56% of children in the EG represented Harold's initial amount as a letter such as N and 49% represented Sally's amount as $N \times 4$. In the EG, 35% of the children wrote $N + 18$ for Harold's amount after earning 18 more dollars, 17% wrote the full equation $N + 18 = N \times 4$ and 27% correctly solved the problem. However, only 6% (four children) systematically used the algebra method to simplify the equation. In the CG, 23% of the children solved the problem but no one used algebra notation or equations to find the solution.

In the third longitudinal study from 3rd to 5th grade we were conducting at the time of writing this chapter, we have found that participants in the project show significantly better results in the EA assessment we have developed throughout grades 3 to 4, in comparison to control group results. In a written assessment given at the beginning of the second semester in 5th grade, 45% of the children correctly solved the equation $n + 6 = 3n$ and explained what the solution meant in relation to the problem at hand (Goodrow & Schliemann, 2006).

One interesting development in this third study was that the classroom teachers, who were present during the lessons and who had run the review of the homework given after each lesson, as well as the other third to fifth grade teachers, decided to implement all the algebra lessons we had developed with a second and third cohort of students. Analysis of the written assessments of the second cohort of students at the end of third and fourth grade, shows that the students who were taught by the teachers perform as well, and sometimes better, than the students who were taught by the researchers (Goodrow & Schliemann, 2006). These assessments included items from the National Assessment of Educational Progress and from the Massachusetts Comprehensive Assessment System, as well as items developed by our project team.

Summary: What Can Young Students Learn About Functions?

Taken as a whole, Moss et al. (2006) results show that transformation rules are accessible to very young students and that patterns activities, if focused on the transformation rules and numerical and geometrical representations, may constitute a meaningful entry

point to EA. Our own results point to the conclusion that young students (8–11 years of age) can benefit from EA activities and learn to (a) think of arithmetical operations as functions rather than as mere computations on particular numbers; (b) learn about negative numbers; (c) grasp the meaning of variables, as opposed to instantiated values; (d) shift from thinking about relations among particular numbers and measures toward thinking about relations among sets of numbers and measures; (e) shift from computing numerical answers to describing and representing relations among variables; (f) build and interpret graphs of linear and non-linear functions; (g) solve algebraic problems using multiple representation systems such as tables, graphs, and written equations; (h) solve equations with variables on both sides of the equality; and (i) are able to inter-relate different systems of representations for functions. We have also found that elementary school teachers can successfully implement the EA activities we have designed. In many of our teacher-researcher meetings, the six teachers we have been working with in our latest study have expressed that (a) they learn mathematics and algebra as they get ready to teach the lessons, (b) the children enjoy and understand the lessons, and (c) the EA lessons help children better deal with other topics in their regular mathematics curriculum. A major research question that we hope to pursue in the next few years concerns the impact of students' EA experiences on their performance in middle and high school mathematics.

CONCLUDING THOUGHTS

Research on EA is still at its infancy. Initial findings are promising, but as we have indicated throughout this review, many basic issues—conceptual and empirical—remain to be clarified and there needs to be a concerted effort by researchers, practitioners, teacher educators and policy makers if the work is to continue productively. The worst possible scenario would be to conclude, "We already know what is needed" and to jump into large-scale implementation. In this final section of the chapter, we share some thoughts about the directions the study of EA and early algebraic reasoning might take in the near future.

What Kinds of Representations Express Algebraic Ideas?

We pointed out earlier that the concept of symbolic representation has undergone a broadening of meaning in recent years. For a long time the concept

has included algebraic notation. Most mathematics educators now agree that even arithmetic notation (number sentences) is symbolic insofar as it contains symbols for representing abstract notions and relations as well as a grammar. Indeed four symbolic or representation systems are currently recognized as fundamental in EA (Janvier, 1987): *arithmetical-algebraic notation, tables, graphs, and natural language*.

It is no coincidence that these four fundamental representation systems are the very ones employed by mathematicians for representing functions. We suggest that the principal means for determining whether representations in these systems are conducive to algebraic reasoning is the extent to which they represent functions and capture functional reasoning. We will illustrate this by considering some of the pitfalls in using patterns and function tables. The basic idea is that that patterns, when not taken as instances of functions, tend to encourage the extension of $f(n)$ to $f(n+1)$ without considering the role of the independent variable, n . When tables are treated merely as patterns, the same issues ensue.

Patterns and Functions

Patterns arise in such a wide variety of everyday life contexts—tailoring, choreography, musical composition, floor tiling, DNA matching, fingerprint identification, numerology, voice detection, architectural design,

poetic rhyme schemes, code-breaking, and so on—that the power of patterns in developing algebraic reasoning can be overlooked. Even in mathematics, a pattern can refer to many things. To put it bluntly, a pattern is not necessarily a well-defined mathematical object with clear connections to mathematical objects and traditions in mathematics. However, it turns out to be immensely important for establishing a coherent research base in EA. And the fact that NCTM (2000) included “understanding patterns” among the Algebra Standards for grades pre-K through 12, makes it especially urgent to decide whether patterns are a suitable topic for introducing young students to ideas of algebra. It is true that NCTM places patterns in the broad context of relations and functions (see Table 15.4). Even so, there is the implicit suggestion that patterns are bona fide mathematical objects, as relations and functions are, and there is no attempt to clarify how patterns may easily fail to establish connections to those objects.

Consider the collection of triangular dots shown in the top row of Figure 15.10. What item comes next? Most people, children and adults alike, will think of something like the arrangement of 15 dots shown in the figure as the “logical” choice.

Technically, any answer can be defended as long as the universe of discourse has not been clarified, (as Bass and colleagues [2005] noted in their criticism of NAEP algebra problems; see also Schwartz, 1975). Many people find it natural to presume that the

Table 15.4 NCTM’s Algebra Standard—Patterns, Relations, & Functions

Grades	Algebra—Understand Patterns, Relations, Functions
Pre-K–2	- Sort, classify, and order objects by size, number, and other properties; - Recognize, describe, and extend patterns such as sequences of sounds and shapes or simple numeric patterns and translate from one representation to another; - Analyze how both repeating and growing patterns are generated.
3–5	- Describe, extend, and make generalizations about geometric and numeric patterns; - Represent and analyze patterns and functions, using words, tables, and graphs.
6–8	- Represent, analyze, and generalize a variety of patterns with tables, graphs, words, and, when possible, symbolic rules; - Relate and compare different forms of representation for a relationship; - Identify functions as linear or nonlinear and contrast their properties from tables, graphs, or equations.
9–12	- Generalize patterns using explicitly defined and recursively defined functions; - Understand relations and functions and select, convert flexibly among, and use various representations for them; - Analyze functions of one variable by investigating rates of change, intercepts, zeros, asymptotes, and local and global behavior; - Understand and perform transformations such as arithmetically combining, composing, and inverting commonly used functions, using technology to perform such operations on more-complicated symbolic expressions; - Understand and compare the properties of classes of functions, including exponential, polynomial, rational, logarithmic, and periodic functions; - Interpret representations of functions of two variables

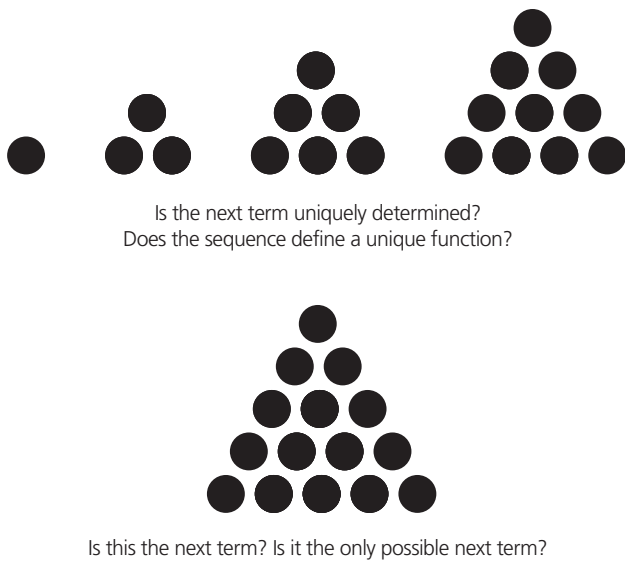


Figure 15.10 An example of a growing pattern and a possible next term.

universe is dots—figures corresponding to triangular number—but nothing requires the next item to consist of dots, or that it even be a geometric figure. In addition, even if one takes the universe of discourse for granted, no rule has been advanced to determine how the next item is created. So the fifth term cannot be deduced from the first four unless further information is supplied.

In classroom settings pattern-discovery tasks can be valuable for getting students to make conjectures about the rule that the creator of the series might have used in order to generate each item. This is in fact the rationale behind the “guess my rule” game that Bob Davis (1967) introduced in fifth grade classrooms four decades ago. Now, it is true that different rules can generate the same finite sequence. And when students discover this they face an important question (Carraher & Earnest, 2003): if two rules generate the same finite sequence, are the two rules equivalent, or are they actually different rules that happen to give the same results over a limited number of cases?

Continuing with the example of triangular numbers, the figures themselves change from instance to instance; so does the number of dots in each figure. These are not quite the same: a student may learn to extend the series without paying particular attention to the number of new dots or the number of total dots. For example, if she is working on a computer she might simply copy the last figure, then copy and paste an additional diagonal edge of dots, and then finally add a single dot. This needs to be taken into account if one wishes to use geometric patterns as an entry point to algebra.

Moreover, the dependent variable or output of the function is presumably the total number of dots. However, this is somewhat arbitrary. If one imagines the drawings in Figure 15.10 superimposed as they are produced in succession, it is easy to think of the change, that is, the increase in dots, compared to the pre-existing case, as the output. This shift in focus corresponds to the distinction between a function and its first derivative. Many miscommunications in the classroom arise around a lack of agreement about what the output is—the total or the change. However, there is another essential yet elusive thing that varies, namely, the position or counter associated with each drawing in Figure 15.10: position 0, 1, and 2 and so forth. This corresponds to the independent variable or input of a function. Students often describe serial patterns as if it were necessary only to pay attention to the dependent variable. That is, they treat sequences of geometric figures as if only one variable were involved. This is often inadvertently encouraged by curriculum materials or by teachers themselves, who may view learning how to extend functions as the primary goal of the activity.

Pattern activities can easily be carried out without students giving explicitly attention to the independent variable. This may not be a reason for concern in the grade band pre-K through 2. However, one hopes that students will eventually be able to express the functional dependency, and this entails referring to the independent and dependent variables and their relation.

Issues Common to Patterns and Tables

The first column of a table generally contains values of the independent variable, while the second column is used to hold the respective values of the dependent variable. This would appear to immunize tables against the problems that often plague patterns activities. However, tables are susceptible to the same shortcomings of patterns activities if one fails to keep in mind their role as expressions of functions. Tables of numbers can serve as function tables, that is, as representations of functions. Whether students view them as such is another matter. In the case of linear functions ($f(x) = ax + b$), students may supply additional values by extending each column independently of the other. They may note for example that the first column “grows by 1’s” while the next grows by 7’s (Schliemann, Carraher, & Brizuela, 2001, 2006). This is not incorrect but it is incomplete. Let us examine why, drawing upon our earlier discussion about serial patterns.

Conceiving a pattern as a “one-variable” sequence restricts attention to how one determines $f(n+1)$ from $f(n)$. In the case of triangular numbers (Fig. 10), one

might count the number of dots along the edge of a triangle in any given case, $f(n)$, then increment that number by one and add it to $f(n)$, yielding $f(n+1)$. The answer is correct although it requires that the student already know the value of $f(n)$. The student will typically say something like “it goes up by 3’s.”

Now, assume that we are not aware of the specific function the student has been working with. Neither the student’s technique nor his explanation is explicit enough for us to know what function he is working on. Even if we assume that the three (in “it goes up by 3’s”) refers to $f(n+1) - f(n)$ and, further, that the independent variable increments by 1 with each row, we would have to examine the given problem in order to determine the precise function underlying the numbers in the table. For all one knows, the function could be any member of the family, $3x + b$; that is, b might be any number.

This failure to fully capture the function is not an inherent shortcoming of “scalar” (Vergnaud, 1983) approaches. Consider the function expressed in closed form as $f(x) = 3x + 7$. The recursive definition of this function is as follows:

$$\begin{aligned} &\text{For all natural numbers, } n, \\ &f(0) = 7 \\ &f(n + 1) = f(n) + 3 \end{aligned}$$

This definition of the function contains: (a) a statement that fixes the value of $f(x)$ at $x = 0$; and (b) a statement describing how to obtain the value of any $f(x + 1)$ from its predecessor, $f(x)$. Students’ scalar methods for filling in and extending function tables show features of iteration and recursion but there are also some significant differences.

It is rarely the case that the student needs to begin with the initial condition. If she wishes to extend the table beyond some $f(x_j)$, already known, she can start there paying no attention to $f(0)$. She simply continues to add the numbers 1 and 3, respectively, to the first and second elements of each ordered pair until she reaches her destination, say, $(x_i, f(x_i))$. The student’s iterative approach is correct; but it is incomplete insofar as it benefits from the initial condition without explicitly recognizing its role in the production of all values in the table. Furthermore, iteration can be inefficient for large values of x : if $x = 100$, it is difficult to determine $f(100)$.

Chances are, the student will never need to consider the initial condition of the function, a condition upon which the whole table is predicated. He can simply act on $f(x)$ to produce $f(x+1)$.

Is a Scalar Approach Valid?

We noted that students’ scalar approaches generally tend not to be fully iterative. This was not to dismiss their approaches. Students notice regularities in patterns and tables of values that are perfectly legitimate. In fact, scalar ways of thinking eventually lead to calculus and differential equations, where great attention is given to Δy and Δx .

Two points must be borne in mind (Cuoco, 1990). First, the closed form and recursive approaches to functions do not share the same domain and co-domain: $f(x) = 3x + 7$ can be defined over real numbers, whereas the recursive definition has a domain of only the natural numbers¹¹ (and an even smaller co-domain). This signifies that a scalar approach will run into difficulties in accommodating the rationals, a stumbling block Kaput & West’s (1994) 10–11 year old students repeatedly came across when using (software-generated) tables to determine missing values.

The second point is that scalar thinking does not easily lead to the closed form representation of a function. A closed form expression, such as $f(x) = 3x + 7$, is far easier to reconcile with an input-output approach to functions, that is, one considers x and an input and $f(x)$ as the output. In the context of tables this amounts to thinking how a value in column one can be operated on to produce the value in column two.

Given this analysis, failures to understand the “algebraic meaning of tables” are easier to understand: students tend to be drawn to sequences as they scan tables in a downward, column-wise fashion. The teacher, on the other hand, may be hoping that students will produce a general rule for obtaining $f(x)$ from x . However, it is extremely challenging to produce a closed form algebraic expression of a function based only on information about changes from row n to row $n+1$. (see Table 15.5 for an illustration of this. Note that it takes 7 steps to find $f(3)$ using the piecewise definition given above.)

What Goals Are Achievable in the Short Term, Mid Term, and Long Term (for Students and Teachers)?

The jury is still out on what algebraic material and approaches make sense in the early mathematics curriculum, and whether EA corresponds to a distinct domain of study or whether it is better integrated into the knowledge base about algebra instruction for older students.

¹¹ We follow Hardy and Wright (1979) in including zero among the natural numbers.

Table 15.5 Steps Required to Determine $f(3)$ for the Function Defined by $f(0) = 7$ and $f(x) = f(n-1) + 3$, for all natural numbers, n

Step		Note
1	$f(3) = f(2) + 3$	From $f(n+1) = f(n) + 3$
2	$f(2) = f(1) + 3$	"
3	$f(1) = f(0) + 3$	"
4	$f(0) = 7$	Given
5	$f(1) = 7 + 3 = 10$	From steps 4 & 3.
6	$f(2) = 10 + 3 = 13$	From steps 5 & 2.
7	$f(3) = 13 + 3 = 16$	From steps 6 & 1.

EA is not just about *when*, *what*, *why*, and *how* are equally as important. It also concerns novel views about arithmetic, algebra, and how they are related. In the United States, arithmetic and algebra have often been treated as distinct subject matters even though certain ideas, techniques, and representations are common to both. Consistent with this view is the notion that arithmetic and algebra need to be *bridged*. The bridging is seen as occurring toward the “end” of arithmetic and the “beginning” of algebra. “Pre-algebra” courses offered one or two years prior to Algebra I in grade 8 or 9 attend to this view of arithmetic and algebra. In these courses, special attention is given to the expanded uses and meanings of symbols from arithmetic. But this *transition* is anything but simple; and part of the reason lies in the unfortunate ways concepts have been treated in the early mathematics curriculum. Consider, for example, the equals sign. In most arithmetic instruction the equals sign has the sense of “yields” or “makes.” Students demonstrate that they subscribe to this notion when they read the sentence, “ $3 + 5 = 8$ ” as “three plus five makes eight.” They may even cheerfully accept expressions such as “ $3 + 5 = 8 + 4 = 12$,” but reject “ $8 = 3 + 5$ ” and “ $3 + 5 = 7 + 1$.” When students begin algebra instruction they are expected to treat the equals sign as a comparison operator expressing an equivalence relation. But, as the preceding examples illustrate, for many students symmetry, and transitivity do not seem to apply to the equals sign.

This makes the difficult task of introducing algebraic equations all the more daunting. Consider, for example, an equation with variables on each side (e.g., $3x = 5x - 14$). Students have to contend with the idea that the expressions on each side of the equals sign can be viewed as functions having variation. The equals sign effectively constrains the values of variable to the solution set: x is free to vary, but the equation is only true for a single value of x . Note that in this

case the expressions on the left and right sides of the equals sign are not interchangeable because they refer to distinct mathematical objects (in the above example, they correspond to the functions $3x$ and $5x - 14$, which are not the same function). This point seems to have gone unnoticed by many mathematics educators, including Freudenthal (1973) who claims that the equals sign means “are the same thing” (p. 302), grounding his point in Leibniz’ idea that two things are equal if they are alike in all respects.

There is an alternative to having students relearn mathematics when they take Algebra I. This alternative rests on a strikingly different view about what arithmetic and elementary mathematics are about. The key idea behind this new view is that arithmetic is a *part* of algebra, namely, that part that deals with number systems, the number line, simple functions and so on. Arithmetic deals with the part of algebra in which particular numbers and measures are treated as instances of more general examples.

Isolated examples can always be treated as instances of something more general. The number 327 is an efficient way to represent $(3 \times 100) + (2 \times 10) + (7 \times 1)$, which is just an instance of the more general expression, $(a \times 100) + (b \times 10) + (c \times 1)$, which can be expressed even more generally for an arbitrary radix or base. Opportunities for generalizing, for thinking about functions and variables, for using algebraic notation, abound in elementary mathematics.

To consider arithmetic as a part of algebra encourages us to view isolated examples and topics as instances of more abstract ideas and concepts. Addition, for example, is a computational method. It is also a function with certain general properties. Likewise multiplication by two is a table of number facts ($1 \times 2 = 2$; $2 \times 2 = 4$; $3 \times 2 = 6$; $4 \times 2 = 8$) and also a function that maps a set of input values to unique output values. The latter idea can be expressed algebraically, for instance, through the mapping notation, $x \rightarrow 2x$, the standard notation for functions, $f(x) = 2x$, or the graph on a Cartesian plane of a relation between x and y corresponding to the equation, $y = 2x$.

It will help to bear these points in mind. It may first appear that the problems we gave children to solve are arithmetical. Upon looking more closely, the reader will note their algebraic character. The categories, *arithmetical* and *algebraic*, are not mutually exclusive.

This does not mean that every idea, concept and technique from arithmetic is manifestly algebraic; however, each is *potentially* algebraic. Concepts such as equivalence can and should be treated early on in ways consistent with their usage in more advanced mathematics unless there are compelling reasons not to.

That arithmetic is a part of algebra is neither obvious nor trivial, especially for those of us who followed the “arithmetic first, algebra much later” route through our own mathematics education.

There is another reason that EA deserves special consideration. The mathematics curriculum tends to be a collection of isolated topics. Yet there are good arguments for treating mathematical (and scientific) concepts—particularly the more abstract concepts—as exceedingly relational. That is, much of their meaning derives from connections to other concepts. For example, one of the important meanings of “fraction” stems from the division of one integer by another. Yet division of integers is often treated in the curriculum without any reference to fractions (and vice-versa). It is thus not surprising that many students do not view the expression, “ $3 \div 4$ ” as related to the fraction $\frac{3}{4}$ (Booth, 1988). In order to correct the fragmented nature of the curriculum it is necessary to take a long-term developmental perspective. In this way, one can more easily see how topics introduced in the early grades foreshadow and set the stage for topics appearing later. Likewise topics appearing relatively late in the curriculum can be more easily understood as building upon, and perhaps departing from, earlier topics.

EA offers a rare opportunity to consider the evolution of mathematical ideas from K–12. That is, it is a special context for examining how early mathematical ideas and methods are related to later ones. This is not to subscribe to the notion that mathematical understanding follows a fixed trajectory, a trajectory laid out entirely by the constraints of psychological development. The course of development depends much on the curricular structure and teaching approaches. That is why we need to ground policy regarding algebra instruction on a careful consideration of the many possible approaches.

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LEARNING AND TEACHING ALGEBRA AT THE MIDDLE SCHOOL THROUGH COLLEGE LEVELS

Building Meaning for Symbols and Their Manipulation

Carolyn Kieran

UNIVERSITÉ DU QUÉBEC À MONTRÉAL

From the days of al-Khwārizmī, and on through Vieta and Euler, algebra was all about procedures and notations. This view of algebra, as a tool for manipulating symbols and for solving problems, has been reflected in school algebra curricula as they developed and took shape throughout the 1800s and into the 1900s. Like the students of today, those of yesteryear no doubt struggled to adapt to the intricacies of this representational and procedural tool. Thus, research conducted during the first half of the 20th century on the learning of algebra focused, for example, on the relative difficulty of solving various kinds of linear equations (Hotz, 1918; Reeve, 1926), the role of practice (Thorndike et al., 1923), and the errors that first-year algebra students make in applying algorithms (Breslich, 1939). During the 1950s and early 1960s, research related to algebra learning was conducted mostly by psychologists with a behaviorist orientation who used the subject area as a vehicle for studying general questions related to skill development, memory, and the like. In contrast, the research that has been conducted at least since the late 1970s when algebra education researchers began to increase in number and coalesce as a community (Wagner & Kieran, 1989) has tended to focus on the kind of meaning students make of their algebra, as well as

on various ways in which to make algebra learning meaningful for students.

It is not surprising that more recent research in algebra learning and teaching has, in general, emphasized pathways to and the development of algebraic meaning. First, Piaget's ideas had begun to circulate among North American researchers during the 1960s and 1970s. The influence of his cognitive development psychology on mathematics education researchers was substantial. Next, studies over several decades had shown that an exclusively skills-based approach to the teaching of algebra did not lead to skilled performance among algebra students (e.g., Monroe, 1915, and the later work of Carry, Lewis, & Bernard, 1980). Nor, according to the ample number of studies of the late 1970s and 1980s, had such approaches led to students' being able to interpret adequately the various ways in which letters are used in algebra (Küchemann, 1981; Matz, 1982), or the structural features of algebraic expressions (Davis, 1975), or equivalence constraints on equations and equation solving (Greeno, 1982). Thirdly, the research on algebra learning since the 1980s has been influenced by socio-democratic forces, such as for example the "Algebra for All" movement in the United States (Chazan, 1996; Edwards, 1990). New ways of

thinking about the content of secondary school algebra, as well as the notion that introducing algebraic activity in primary school (Kaput, 1995) could help in removing some of the obstacles associated with secondary school algebra, began to emerge and be discussed. Lastly, the arrival of computing technology in schools in the 1980s led to reshaping conceptions of what could (and should) be taught in school algebra (Fey, 1989)—ideas that were to a large extent both initiated by algebra learning researchers as well as reflected in their subsequent work.

Whereas much of the 1970s and 1980s research focused on the reporting of primarily empirical findings with respect to students' understanding of and activity in algebra, recent theoretical advances in the field of algebra education research, both in North America and abroad (see, e.g., Lerman, Xu, & Tsatsaroni, 2002; Radford, 2000), have widened the community's perspective on the notion of algebraic meaning and have altered its ways of thinking about how students construct meaning for algebraic objects and processes. The theoretical framework of constructivism with its key tenet that knowledge is actively constructed by the cognizing subject, which had been inspired by Piaget's work and had blossomed in the 1980s, had led algebra researchers to move their attention from, for example, the errors made by students to the ways in which students craft their understandings of algebraic concepts and procedures. Although the focus was still cognitive, it tended to be broader than that suggested by the research analyses carried out just a decade or two earlier. Then, during the 1990s, the sociocultural perspective that had developed outside the mathematics education research community began to be felt within. The earlier constructivist/cognitivist orientation shifted for a large number of algebra researchers toward analyses of social factors affecting algebra learning (see, e.g., Cobb, 1994), with an accompanying interest in the mediating role of cultural tools (e.g., Meira, 1998). This shift also led to an increase in classroom-based studies with a focus on teacher-student and student-student discourse (e.g., Bartolini Bussi, 1995; Sfard, 2001). The increasing use of digital tools in learning, in combination with the sociocultural lens being applied to both the design of learning studies and the analysis of the resulting data, led to new questions about the nature of learning with these tools. Dynamic learning environments prompted the development of new theoretical perspectives in which bodily experience was considered to play a role in the creation of meaning for algebraic objects.

More recent research has thus had not only a stronger theoretical component to frame the empirical work, but also one derived from a more diverse set of theories

than in the past. Shifting views of the content of school algebra have also been exemplified in this research. In order to give adequate consideration to these shifts, the chapter is divided into two parts, the first part of which is devoted to their discussion. This part also includes the presentation of models of algebraic activity and examines various sources of meaning that shape algebra learning. The second part of the chapter deals with the body of research that has been carried out in the learning and the teaching of algebra, primarily since 1990. Although the research studies conducted prior to 1990 have already been discussed in Kieran (1992), this chapter will at times refer back to this earlier body of research to suggest the ways in which more recent work has been informed by the past.

This review draws upon the considerable number of algebra research resources that exist already in the domain, including research syntheses (Bednarz, Kieran, & Lee, 1996; Filloy & Sutherland, 1996; Kieran, 1990, 1992; Mason & Sutherland, 2002; Rojano, 2002; Sutherland, Rojano, Bell, & Lins, 2001; Wagner & Kieran, 1989), related monographs (Kilpatrick, Martin, & Schifter, 2003; National Research Council, 2001; Wagner, 1993), the Proceedings and final volume of the recent ICMI Algebra Study (Chick, Stacey, Vincent, & Vincent, 2001; Stacey, Chick, & Kendal, 2004), and the algebra research reports found in the Proceedings of both the International Group for the Psychology of Mathematics Education (PME) and the North-American chapter of PME (PME-NA).

In addition to the above sources, the journal literature that was extensively surveyed in preparing this chapter includes articles from the *Journal for Research in Mathematics Education*; *Educational Studies in Mathematics*; *Journal of Mathematics Teacher Education*; *Journal of Mathematical Behavior*; *Mathematical Thinking and Learning*; *International Journal of Computers for Mathematics Learning*; *For the Learning of Mathematics*; *Canadian Journal of Science, Mathematics, and Technology Education*; *Recherches en Didactique des Mathématiques*; *Petit x*; *Focus on Learning Problems in Mathematics*; *The Mathematics Educator*; and *Hiroshima Journal of Mathematics Education*. Additional resources, too numerous to mention here but noted in the Reference list, were also consulted.

PART 1. SCHOOL ALGEBRA: BACKGROUND ISSUES AND THEORETICAL CONSIDERATIONS

This part of the chapter deals with some of the issues surrounding current research in algebra learning and teaching, such as the content of school algebra, source-

es of meaning, and models of algebraic activity. It also includes a description of the model that will serve as a framework for the research studies on learning that are presented in the second part of the chapter.

The Current Content of School Algebra: Competing Views

Since the mid-1980s, the content of school algebra has been experiencing a tug of war between traditional and reformist views. The current situation is such that the content varies not only from country to country, but also within a given country. In the United States, for example, algebra classes might be following either reform-oriented or traditional programs, or something in between (see Lester & Ferrini-Mundy, 2004). Traditional (i.e., nonreform) algebra courses still tend to reflect the content and approaches of the texts written in the 1960s and 1970s by Dolciani and her collaborators (e.g., Dolciani, Berman, & Freilich, 1965). These courses, which are typically 1-year courses and have a strong symbolic orientation, include the simplification of expressions and the solving of equations, inequalities, and systems of equations by formal methods, as well as the factoring of polynomial and rational expressions. Interspersed among the various chapters of typical textbooks are word problems of various types that serve to apply the algebraic techniques just covered. Although functions, along with their graphical, tabular, and letter-symbolic representations, are also treated, they are generally accorded a more minor role—the main thrust of the course being polynomial and rational expressions and their manipulation. According to Saul (1998), this emphasis on polynomial and rational expressions is oriented toward recognizing form, which is considered one of the most important aspects of school algebra within this perspective:

This level [of computing with rational expressions] is reached when students learn patterns of factoring, for example. Seeing the factorization $A^2 - B^2 = (A + B)(A - B)$, a student who has mastered certain algebraic concepts can factor $x^4 - y^4$, $4x^2 - 1$, or $\cos^2 x - \sin^2 x$. Mathematically, the student is then letting the variables A and B stand for other rational expressions (or rational trigonometric expressions). (p. 138)

Saul is not alone in his view, one that is shared by several other mathematics educators and mathema-

ticians (e.g., Wheeler, 1996). Pimm (1995) similarly argued that, “algebra is about form and about transformation” (p. 88).

In contrast to the position enunciated above, nontraditional, reformist algebra programs tend to give a great deal of weight to functions, to various ways of representing functional situations, and to the solution of “real-world” problems by methods other than manual symbolic manipulation, such as technology-supported methods. These programs reflect the stance formulated by Fey and Good (1985) that practicing manipulative skills should be replaced with the study of families of elementary functions, a shift that “places the function concept at the heart of the curriculum” (p. 49).

The origins of this changed perspective regarding the core elements of school algebra can be traced back to the 1980s when some mathematicians and mathematics educators promoted the idea that computing technology could have significant effects on the content and emphases of school-level and university-level mathematics (Fey, 1984). Early visionaries soon realized that computing technology could be harnessed to more fully integrate the multiple representations of mathematical objects in mathematics teaching. Soon, a *functional perspective* on algebraic activity could be seen in some research projects, a few of them involving pilot programs in schools.¹

Heid (1988), one of the pioneers of a functional view of algebra (see also Fey et al., 1991; Schwartz & Yerushalmy, 1992), defined it as follows:

The functional approach to the emergence of algebraic thinking . . . suggests a study of algebra that centers on developing experiences with functions and families of functions through encounters with real world situations whose quantitative relationships can be described by those models. (Heid, 1996, p. 239)

The encouraging results emanating from the research program of Fey, Heid, and their collaborators had an impact not only on the developing projects of other researchers at the time but also on the recommendations to be found in the early *Standards* documents² and other related resources published by the National Council of Teachers of Mathematics (NCTM). The movement for function-based approaches to algebra continued throughout the 1990s, becoming more fully elaborated in integrated curricula (see Senk &

¹ This was not the first time that functions became a thread of secondary school mathematics programs. During the 1960s, for example, the modern mathematics movement had included similar proposals (see, e.g., the texts authored by Van Engen et al., 1964). However, at that time, the approach being considered was much more formal, emphasizing sets and distinctions between functions and relations.

² These documents included *Curriculum and Evaluation Standards for School Mathematics* (NCTM, 1989), *Professional Standards for Teaching Mathematics* (NCTM, 1991), and *Assessment Standards for School Mathematics* (NCTM, 1995).

Thompson, 2003, for examples of National Science Foundation-funded, reform-based curricula that were produced). It culminated in the later publication of the *Principles and Standards for School Mathematics*, where the Algebra Standard for grades pre-K–12 emphasized “relationships among quantities, including functions, ways of representing mathematical relationships, and the analysis of change” (NCTM, 2000, p. 37).

If the number of research studies carried out within this perspective is any gauge, a functional perspective on the teaching of algebra became quite widespread in the United States. However, this perspective is not without its critics. For many researchers, algebra does not consist in the study of functions; algebraic expressions can be used without describing functions, and functions can be expressed without using algebra (Lee, 1997). Pimm (1995) has argued strongly against the position that algebra is about functions:

Ironically, technology is being used to insist on screen (graphical) interpretations of algebraic forms. There is a strong presumption that symbolic forms are to be interpreted graphically, rather than dealt with directly. . . . There is currently a rapid process of redefinition of algebra, triggered I feel more by the potentialities of these new [technological] systems and the drawbacks of an over-fragmented mathematics curriculum than by any novel epistemological insight. The view of algebraic objects that is being strongly promoted . . . is of algebraic expressions as functions. (p. 104)

One of the by-products of these competing views of algebra content is the development of hybrid versions of programs of study that attempt to include elements of both traditional (rational expressions and equations) and functional orientations to school algebra. However, such mixed approaches can create additional difficulties for algebra learners. As pointed out by Chazan and Yerushalmy (2003), students become confused regarding distinctions between equations and functions, not being able to sort out, for example, how equivalence of equations is different from equivalence of functions. According to Chazan and Yerushalmy, it is not the case that such questions are unanswerable, but rather that combining functional approaches with more standard treatments of school algebra leads to such dilemmas, and that only rarely are opportunities provided for students to inquire into these questions and to attempt to resolve them.

The various curricular approaches to algebra that are currently available in different parts of the United States, ranging from reform to traditional to in-between, are also to be found in the diversity of algebraic content worldwide (see, e.g., Kendal & Stacey, 2004). Sutherland (2002), who conducted a compara-

tive summary of algebra curricula in several countries, noted surprising differences in the ways in which algebra is specified in these various curricula:

For example, the algebra component of the Australian (Victoria) curriculum is organized around the ideas of “expressing generality,” “equations and inequalities,” and “function,” whereas in the Hungarian curriculum the algebra component refers only to work with systems of equations. . . . The Japanese curriculum is different from the algebra curricula of all the other countries in that there is an emphasis on symbolizing mathematical relationships from elementary school and there is a much more demanding expectation with respect to the use of algebraic symbols for the 15–16-year-old age group than in the other countries studied. . . . The idea of introducing algebra within the context of problem situations is evident within most of the countries studied, although these “problem situations” are sometimes more traditional word problems (e.g., in Italy, Hungary, France, Hong Kong) and are sometimes more “realistic modeling situations.” . . . In general where there is more emphasis on solving “realistic problems” there tends to be less emphasis on symbolic manipulation. (p. 29)

This latter observation by Sutherland regarding the greater the emphasis on realistic problems, the less the emphasis on symbolic manipulation, remains a fundamental issue in the arena of competing ideas on algebraic content. The orientation toward the solving of realistic problems, with the aid of technological tools, allows for an algebraic content that is less manipulation oriented. Such orientations also emphasize multi-representational activity with a shift away from the traditional skills of algebra. The logic underpinning this position is that technological tools can step in and assist with problem solving when students do not have the required manipulative skills. This has led to a certain amount of resistance in the United States to the NSF-funded reform-based curricula at the upper secondary level because the technical aspects of algebra would seem to be minimized. The fundamental issue is really one of values—that is, what it is that mathematicians, mathematics educators, and mathematics education researchers consider essential and important in algebra. The next section on sources of meaning takes a second look at some of the same issues, but this time from an epistemological perspective.

Sources of Meaning

This section examines first the general question of where algebraic meaning comes from and then goes on to describe each of four main sources of meaning: (a) meaning from the algebraic structure itself, involv-

ing the letter-symbolic form; (b) meaning from other mathematical representations, including multiple representations; (c) meaning from the problem context; and (d) meaning derived from that which is exterior to the mathematics/problem context (e.g., linguistic activity, gestures and body language, metaphors, lived experience, image building, etc.).

Algebraic Meaning: Where Does It Come From?

Several researchers have studied the specific question of meaning in school algebra (e.g., Arzarello, Bazzini, & Chiappini, 2001; Filloy, Rojano, & Rubio, 2001; Kaput, 1989; Kirshner, 2001; Lins, 2001; Sackur, Drouhard, Maurel, & Pécal, 1997). The different perspectives that have been described at various times in the literature have recently been reconceptualized by Radford (2004). For Radford, meaning in school algebra is produced in the “crossroads of diverse semiotic mathematical and non-mathematical systems” (p. 163) and is deemed to come from three primary sources: (a) the algebraic “structure” itself, (b) the problem context, and (c) the exterior of the problem context.

Although the third of Radford’s sources of meaning is quite new to algebra research, something akin to it has been discussed by Noss and Hoyles (1996) in their general treatment of “meaning in mathematics education” according to a triple perspective that includes *meanings from mathematical objects*, *meanings from problem solving*, and *meanings constructed by the individual learner*. The parallelism between the three sources of meaning in algebra from Radford and the corresponding view for mathematics at large from Noss and Hoyles has a certain appeal. However, a strictly threefold approach is not broad enough within the context of current views of algebra content in which graphical and other mathematical representations are considered very important. Thus, a second within-mathematical source of meaning in algebra could be added to Radford’s classification, namely *meaning from other mathematical representations, including multiple representations* (see Figure 16.1).

Meaning From the Algebraic Structure Itself, Involving the Letter-Symbolic Form

At the Research Agenda Project conference on algebra in 1987, Booth (1989) argued that

Our ability to manipulate algebraic symbols successfully requires that we first understand the structural properties of mathematical operations and relations which distinguish allowable transformations from those that are not. These structural properties constitute the *semantic* aspects of algebra. . . . The essential feature of algebraic representation and symbol manipulation, then, is that it should *proceed from* an understanding of the semantics or referential meanings that underlie it. (pp. 57–58)

In other words, “the sense of meaningfulness comes with the ability of ‘seeing’ abstract ideas hidden behind the symbols” (Sfard & Linchevski, 1994, p. 224), that is, the symbols become transparent.

This structural source of meaning not only links letter-symbolic representations to their numerical foundations but also provides connections among the symbolic forms of algebra, its equivalences, and its property-based manipulation activity (e.g., Cerulli & Mariotti, 2001). Although the algebra research literature often refers to *the structure of expressions*, the latter phrase both shuns definition (see, e.g., Arcavi, 1994; Linchevski & Livneh, 1999; MacGregor & Price, 1999) and proves difficult for students to grasp (e.g., Demby, 1997). Nevertheless, even though the nature of the meaning that students draw from algebraic structure can be elusive, this source of meaning is considered by many mathematics educators and researchers to be fundamental to algebra learning.

Meaning From Other Mathematical Representations, Including Multiple Representations

The diversity of perspectives underlying the content of school algebra suggests that the question of where algebraic meaning comes from is intimately related to the nature of the algebra courses experienced by students. Courses with a traditional focus on literal expressions, equations, equivalent forms, properties

1. Meaning from within mathematics:
 - 1(a). Meaning from the algebraic structure itself, involving the letter-symbolic form.
 - 1(b). Meaning from other mathematical representations, including multiple representations.
2. Meaning from the problem context.
3. Meaning derived from that which is exterior to the mathematics/problem context (e.g., linguistic activity, gestures and body language, metaphors, lived experience, image building).

Figure 16.1 Sources of meaning in algebra (adapted from Radford, 2004).

and structure will offer quite different meaning-building experiences from those oriented toward functions and the interplay of tabular, graphical, and symbolic representations. The question then is, what role can be played by the tabular and graphical representations, in conjunction with the letter-symbolic, in algebraic meaning making.

Kaput (1989) has argued that the problem of student learning in algebra is compounded by (a) the inherent difficulties in dealing with the highly concise and implicit syntax of formal algebraic symbols and (b) the lack of linkages to other representations that might provide feedback on the appropriate actions taken. As a consequence, he has promoted the kind of mathematical-meaning building that has its source in *translations between mathematical representations systems*. Although tables, graphs, and equations can all be used to display binary relations, a critical difference, according to Kaput, between tables and graphs is that “a graph engages our gestalt-producing ability, which allows us to consolidate a binary (and especially a functional) quantitative relationship into a single graphical entity—a curve or line” (Kaput, 1989, p. 172). The opportunity to coordinate objects and actions within two different representations, such as the graphical and the letter-symbolic, is considered by many to be crucial in creating meaning in algebra (e.g., Fey, 1989; Romberg, Fennema, & Carpenter, 1993; Yerushalmy & Schwartz, 1993).

Meaning From the Problem Context

In contrast to the internal semantics of algebra as a site for meaning making (Booth, 1989), the external semantics of a problem permit the algebra learner to fuse symbols and notations with events and situations, thereby creating an external meaning for certain objects and processes of algebra. A strongly held belief in algebra education is the notion that problem-solving contexts are foundational to the emergence and evolution of algebraic reasoning (e.g., Bednarz & Janvier, 1996; Bell, 1996). This stance is based to a certain extent on historical grounds whereby problem solving made a major contribution to the development of algebra. As well, algebra grew in status to become the privileged tool for expressing general methods for solving whole classes of problems. Additional arguments for the role of problems in algebraic meaning building are related to notions of relevance and purpose (cf., Lee, 1997). Nevertheless, it has been argued that significant epistemological issues are associated with the whole idea of using word problems for creating algebraic meaning making (e.g., Balacheff, 2001; Pirie & Martin, 1997). Therefore, a broader perspective on problem solving has been advanced by Bell (1996):

For introducing and developing algebra, I understand problem solving to refer to the solving of problems by the forming and solving of equations; this is the narrow sense of the term. But the essential mathematical activity is that of exploring problems in an open way, extending and developing them in the search for more results and more general ones. Hence [all algebraic learning] . . . is based on problem explorations. This is the broad sense of the term. (p. 167)

Included within this source of algebraic meaning making are “real problems,” that is, those that involve *modeling* situations. Much of the current modeling activity that occurs in algebra classes uses physical artifacts or technological modeling tools as an integral part of the activity. In these cases, the contextual aspect of the situation is very much intertwined with the use of gestures and body language in the process of constructing a mathematical model. This activity thus combines two sources of meaning, one derived from the problem context and the other that is exterior to the context and that is presented immediately below.

Meaning Derived From That Which Is Exterior to the Mathematics/Problem Context

A recent approach by algebra researchers to thinking about meaning making concerns sources of meaning that are exterior to both the mathematics and the problem situation. (Note that much of this meaning building that is “exterior” to both the mathematics and the problem situation is related to that which is uniquely human, and thus is clearly “interior” in a very real sense.) This approach focuses on students’ processes of meaning production in terms of the way diverse resources such as gestures, bodily movements, words, metaphors, and artifacts become interwoven during mathematical activity (see, e.g., Arzarello & Robutti, 2001; Radford, Demers, Guzmán, & Cerulli, 2003). Past studies of students’ ways of thinking and of their errors in algebra had always suggested that students brought more to bear on their learning of algebra than was accounted for by the theories available at the time. Thus, the explicit focus on bodily activity, language, and past lived experience as a source of meaning would seem a natural evolution in research on the learning of algebra.

On the basis of his studies of students’ production of oral and written signs and the meanings they ascribe to them as they engage in the construction of expressions of mathematical generality, Radford (2000) has observed that

Students, at the very beginning, tend to have recourse to other experiential aspects more accessible to them than the structural one. . . . Novice students bring

meanings from other domains (not necessarily mathematical domains) into the realm of algebra. Hence it seems to us, one of the didactic questions with which to deal is . . . that of the understanding of how those non-algebraic meanings are progressively transformed by the students up to the point to attain the standards of the complex algebraic meanings of contemporary school mathematics. (p. 240)

Radford's closing remark, which implicitly raises the question as to the nature of the algebraic activity whereby students' meanings are gradually transformed and aligned with "the complex algebraic meanings of contemporary school mathematics," leads to the next section dealing with algebraic activity. As will be seen, the section integrates previously elaborated issues of content and sources of meaning in algebra.

Models for Conceptualizing Algebraic Activity

Bell (1996) has described algebra as a means to express generalizations, relations, and formulas; solve problems; denote unknowns; and solve equations. Usiskin (1988) has synthesized four conceptions of algebra: generalized arithmetic, the set of procedures used for solving certain problems, the study of relationships among quantities, and the study of structures. Kaput (1995) has identified five aspects of algebra: generalization and formalization; syntactically guided manipulations; the study of structure; the study of functions, relations, and joint variation; and a modeling language. Other researchers (see, e.g., Gascón, 1994–1995; Lins, 2001; Mason, Graham, & Johnston-Wilder, 2005; Rojano, 2004; Smith, 2003; Star, 2005; Sutherland, 2004) have offered variations on the above, each in accordance with their particular perspectives on algebra, algebraic thinking, or algebraic language.

Participants at an international colloquium on algebra in Montreal in the early 1990s focused on four approaches: generalization of numerical and geometric patterns and of the laws governing numerical relationships; problem solving; functional situations; and modeling of physical and mathematical phenomena (Bednarz et al., 1996). Concurrently, a study by Lee (1997) was in progress, one that adopted an even broader perspective on that which constitutes school algebra. In this study, which explored algebraic understanding, Lee presented the question, "What is algebra?" to a cohort of mathematicians, teachers, students, and mathematics education researchers. Seven themes emerged from her interviews on the question: a school subject, generalized arithmetic, a tool, a language, a culture, a way of thinking, and an activity.

If there was one theme that tended to permeate all others among Lee's interviewees, it was *Algebra is an activity*. Lee (1997) remarked that, "Algebra emerges as an activity, something you do, an area of action, in almost all of the interviews" (p. 187). For example, Pimm, one of Lee's interviewees, is quoted as saying that *action* is the central feature of school algebra: "Algebra . . . is so much more about doing, is actually about action on things, . . . with attention being more on the *transformations* [emphasis added] than on the objects themselves" (p. 187). On the other hand, Kaput, another interviewee, emphasized the alternate side of the coin when he pointed to the importance in algebraic activity of spending a great deal of time in the "building of algebraic objects" (p. 189). And Bell expressed concern about the question of purpose, about students not really "having the experience of what algebra is for" (p. 196).

Building on the idea of algebra as activity (and reflecting the key ideas expressed by Kaput, Pimm, and Bell), Kieran (1996) developed a model that synthesizes the activities of school algebra into three types: *Generational*, *Transformational*, and *Global/meta-level*. This model, hereafter referred to as the GTG model (see Figure 16.2), will serve in Part 2 as the structuring device for presenting findings from the research involving the algebra learner.

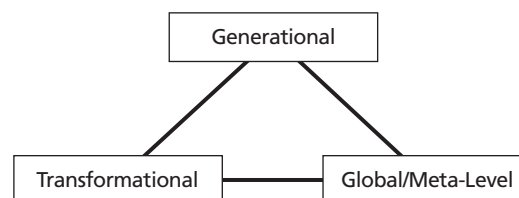


Figure 16.2 GTG model for conceptualizing algebraic activity.

The *Generational* activities of algebra involve the forming of the expressions and equations that are the objects of algebra. Typical examples include (a) equations containing an unknown that represent problem situations (see, e.g., Bell, 1995), (b) expressions of generality arising from geometric patterns or numerical sequences (see, e.g., Mason, 1996), and (c) expressions of the rules governing numerical relationships (see, e.g., Lee & Wheeler, 1987). Much of the meaning building for algebraic objects occurs within the generational activity of algebra. Therefore this activity includes, as well, work with variables, unknowns, and equality; even the notion of that which constitutes the solution to an equation can be initiated within generational activity. Clearly, the meaning that students develop for the notion of, for example, *variable* will

depend on whether the algebra content they have experienced has been derived from a polynomial-equation framework or a functions framework, that is, has tended to emphasize either the letter-symbolic or a combination of mathematical representations. These two frameworks each provide a unique transversal thread to the three types of algebraic activity.

Generational activity is the area where, according to Radford (2001), the role of algebra is that of a language to express meaning, and where the *habit of mind* of those who are in “algebra mode” can find expression (Cuoco, Goldenberg, & Mark, 1996). It may be tempting to equate generational activity with the conceptual aspects of algebra and, similarly, transformational activity with the skill-based aspects of algebra, but this parallelism is not intended. Indeed, as is being proposed in this section and argued throughout Part 2, conceptual work and meaning building occur as well within the transformational activity of algebra.

The second type of algebraic activity—the *Transformational* activities (referred to, by some, as the *rule-based* activities)—includes, for instance, collecting like terms, factoring, expanding, substituting one expression for another, adding and multiplying polynomial expressions, exponentiation with polynomials, solving equations and inequalities, simplifying expressions, substituting numerical values into expressions, working with equivalent expressions and equations, and so on. A great deal of this type of activity is concerned with changing the symbolic form of an expression or equation in order to maintain equivalence. In addition to developing meaning for equivalence, this activity also includes meaning building for the use of properties and axioms in the manipulative processes themselves. Even the technique of factoring has its conceptually meaningful aspects: for example, coming to see that, if the exponent in, say, $x^n - 1$ has several divisors, it can generally be factored in more than one way, and thus that this expression can be viewed in terms of multiple possible structures (e.g., seeing $x^6 - 1$ as a difference of squares and also as a difference of cubes). It is in this way that meaning can be said to be present for the transformational processes of algebra. This factoring example also illustrates that a given task can be both transformational and generational. It is generational to the extent that the learning of a new manipulative transformation, such as $x^6 - 1 = (x - 1)(x^5 + x^4 + x^3 + x^2 + x + 1)$, can induce an extended perspective on the algebraic object itself.

In this sense, transformational activity is not just skill-based work. It is much broader than this in that it can include conceptual/theoretical elements, especially during the period in which certain transformations are being learned. According to Lagrange

(2002), “Technique has a pragmatic role that permits the production of results; but it also plays an *epistemic* [emphasis added] role in that it constitutes understanding of objects and is the source of new questions” (p. 163, translated from the original).

Lastly, there are the *Global/meta-level*, mathematical activities. These are the activities for which algebra is used as a tool but that are not exclusive to algebra. Note that they are not referred to as *global/meta-level* because they involve taking some distance from the mathematical activity at hand and viewing it comprehensively. Rather, they suggest more general mathematical processes and activity. They also provide the context, sense of purpose, and motivation for engaging in the previously described generational and transformational activity. The global/meta-level activities include problem solving, modeling, working with generalizable patterns, justifying and proving, making predictions and conjectures, studying change in functional situations, looking for relationships or structure, and so on—activities that could indeed be engaged in without using any letter-symbolic algebra at all. These are the kinds of activities that provide the surround for engaging in mathematical activity in general, and algebraic activity in particular—and, thus, for constructing and working with algebraic objects and processes.

However, algebra does not equal problem solving, or modeling, or any of the other global/meta-level activities. Although some may find it helpful to think of these global/meta-level activities in terms of “approaches” to the teaching of algebra—such as a generalization approach to algebra or a problem-solving approach to algebra (see, e.g., Bednarz et al., 1996)—algebra includes as well generational and transformational activity, which may not be a necessary component of the designated approach to algebra. On the other hand, aspects of, for example, generalization and problem solving can be present in much of generational and transformational activity of algebra, especially during the early stages of learning. Thus, the global/meta-level activity of algebra is both broader than and at the same time not quite as broad as algebra.

PART 2. SCHOOL ALGEBRA: FINDINGS FROM RESEARCH STUDIES

This second part, which constitutes the main thrust of the chapter, describes the body of research work on the learning and the teaching of school algebra, emphasizing more recent studies but also integrating some

findings from earlier work. The research is presented according to three age ranges: the middle grade and lower secondary level student, the upper secondary and college level student, and the teacher of algebra. Research on the development of algebraic thinking at the primary school level is treated in another chapter (see Carraher & Schliemann, this volume).

Within each of the age-related categories dealing with research on the *learning* of algebra, the GTG model of algebraic activity provides the structural framework. Research where the primary focus is generational activity is treated first, followed by the research dealing with transformational activity, and then that of global/meta-level activity. Because the third type of activity is overarching and cross-cutting, it proved to be more practical, at times, to discuss it within the generational or transformational activity that it served. This was particularly the case for studies involving word problems with the middle grade and lower secondary level student. It is noted that the placement of research studies into the various categories of the GTG model was done according to rather general criteria: For example, if the main emphasis of the study appeared to be transformational activity, then the findings are discussed under that category, even if the study also included some generational activity; if several types of global/meta-level activity were involved, the study is reported in the section related to the major focus. Studies that did not fit this model, such as international assessments, are discussed separately.

Research on the algebra learner occupies the first two sections of this part of the chapter; research that has focused on the algebra teacher and the teaching of algebra is the subject of the third section. The chapter then closes with a fourth section that attempts to draw out that which this body of research on the learning and teaching of school algebra tells us.

The Middle Grade and Lower Secondary Level Student

The research that is reviewed in this section includes those studies carried out with students in the middle grade and lower secondary levels, that is, from Grades 6 to 9 (approximately 11 to 15 years of age). The majority of the research carried out with this age range of pupil has focused on generational activity, with considerably fewer studies dedicated to transformational activity. The studies focusing on global/meta-level activity have been concentrated in the areas of generalizing, proof and proving, and modeling (with problem solving being treated within the context of generational activity).

By way of introduction, this section starts with a brief commentary on results from some recent international assessments. This provides a first glimpse at what students of this age find easy or difficult in algebra. These introductory remarks are followed by the three main sections related to research on the middle grade and lower secondary level student, research categorized according to generational, transformational, and global/meta-level activity. A section on the impact of technology on algebraic activity is then presented, followed by brief concluding remarks.

Background: International Assessments

Two large-scale international studies, the Trends in International Mathematics and Science Study (TIMSS, formerly known as the Third International Mathematics and Science Study) and the Program for International Student Assessment (PISA), announced the results of their 2003 surveys in late 2004 (PISA, 2004; TIMSS, 2004). Forty-eight countries participated in the 2003 TIMSS mathematics achievement assessment for Grade 8 (Gonzales et al., 2004). The number of score points obtained on average in the algebra content area across the 48 countries was 25 out of a possible 53, whereas the U.S. eighth graders in algebra obtained on average 29 score points. Figures 16.3 and 16.4 provide two items from the algebra area of the survey.

The three figures below are divided into small congruent triangles.

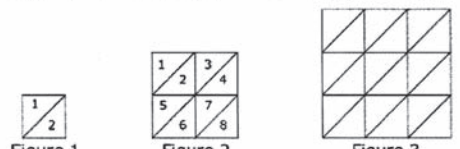


Figure 1 Figure 2 Figure 3

A. Complete the table below. First, fill in how many small triangles make up Figure 3. Then, find the number of small triangles that would be needed for the 4th figure if the sequence of figures is extended.

Figure	Number of Small Triangles
1	2
2	8
3	
4	

B. The sequence of figures is extended to the 7th figure. How many small triangles would be needed for Figure 7?

Answer: _____

C. The sequence of figures is extended to the 50th figure. Explain a way to find the number of small triangles in the 50th figure that does not involve drawing it and counting the number of triangles.

Figure 16.3 One of the released items from the 2003 TIMSS for eighth grade.

Sam wanted to find three consecutive even numbers that add up to 84.

He wrote the equation $k + (k + 2) + (k + 4) = 84$.

What does the letter k represent?

- (A) The least of the three even numbers
- (B) The middle even number
- (C) The greatest of the three even numbers
- (D) The average of the three even numbers

Figure 16.4 Another of the released items from the 2003 TIMSS for eighth grade.

The percentage of correct responses across all TIMSS countries for A, B, and C respectively of the item shown in Figure 16.3 was 36.6%, 22.1%, and 14%. Although the results on this item were universally low, the U.S. students scored higher (50.2%, 29.6%, and 19.4% respectively), perhaps indicating that U.S. students had had more opportunities to examine patterns in this manner than many students internationally. The percentage of correct responses on the item shown in Figure 16.4 (for which the correct answer is A), one with some letter-symbolic content, was 24.7% across all TIMSS countries and 23.1% among U.S. students.

In general, these low results are disturbing—less than 37% correct, worldwide, for each of the four questions shown in the two items of Figures 16.3 and 16.4. Despite the improvement of U.S. eighth graders since the 1995 administration of the test, the fact that only 8 of 48 countries outperformed U.S. eighth graders in algebra in the 2003 assessment suggests that, overall, students are doing quite poorly in algebra. The PISA results, similarly, confirm this situation. The research findings that are presented in this chapter offer a more detailed look at what students know, that is, the nature of the meanings that they attribute to algebraic objects and processes, as well as their approaches to tasks that involve the manipulation of symbols. Some of this research involves the study of alternate learning environments that can lead to more powerful algebraic understandings among students.

Generational Activity: The Middle Grade and Lower Secondary Level Student

The generational activity of algebra involves forming and interpreting the objects of algebra, usually within the context of some global/meta-level activity. This section is one of the larger ones of the chapter dealing as it does with an area where a significant number of research studies on the learning of algebra have been carried out. To render this area more accessible, it has been divided into subthemes according to the source of meaning being invoked in constructing

or interpreting the objects of algebra: (a) primarily a letter-symbolic focus, (b) a multiple-representation perspective, or (c) within the context of word problems. The relatively small number of studies that have focused on the fourth source of meaning, that which is “exterior” to both the mathematics and the problem context, usually included as well one of the other three sources of meaning and are thus treated therein.

Generational Activity with a Primary Focus on the Letter-Symbolic Form. The studies that have focused on the kinds of meanings that students of this age range attribute to the letter-symbolic forms of the objects of algebra have been concentrated in the following areas: variables, expressions, and equations; the minus sign and negative numbers; and the beginnings of structure sense.

Variables, Expressions, and Equations. Whereas past research on the ways in which students of this age range interpret algebraic symbols tended to focus on cognitive levels (e.g., Küchemann, 1981), prior arithmetic experience and methods of thinking (Booth, 1984), and difficulty with notation such as brackets (Kieran, 1979) and the equal sign (Behr, Erlwanger, & Nichols, 1976; Kieran, 1981), more recent work suggests additional factors impinging upon students’ interpretation of algebraic notation: (a) what one is able to perceive and prepared to notice (Sfard & Linchevski, 1994), (b) difficulties with operation signs (Cooper et al., 1997), (c) the nature of the questions asked and the medium in which they are asked (Warren, 1999), (d) the presence of multiple referents and shifts in the meaning of unknowns (Stacey & MacGregor, 1997), and (e) the nature of the instructional activity (Wilson, Ainley, & Bills, 2003).

More specifically, a large-scale study conducted by MacGregor and Stacey (1997) involving 2000 Australian students (aged 11–15) found that students’ interpretations of algebraic notation are based on intuitive assumptions and sensible, pragmatic reasoning about an unfamiliar notation system; analogies with symbol systems used in everyday life, in other parts of mathematics, or in other school subjects; interference from other new learning in mathematics; and poorly designed and misleading teaching materials. The nature of the interpretations brought to bear on algebraic notation by these students suggests the strong role played by the meaning derived from the exterior of the mathematics/problem context. These researchers have argued that recognition of these origins of misunderstanding is necessary for improving the teaching of algebra. In another analysis drawn from the same survey data, MacGregor and Price (1999)

assessed metalinguistic awareness³ and its relation to the use of algebraic notation. The researchers found that very few students with low metalinguistic scores achieved high algebra scores.

The transition from arithmetic to algebra remains an important theme of several studies involving variables, expressions, and equations. Building on earlier research on variables (e.g., Küchemann, 1981; Wagner, 1981), expressions (e.g., Bell, Malone, & Taylor, 1987), and equations (e.g., Kieran, 1988; Whitman, 1976), some of the more recent studies have examined students' abilities to express generalizations and to represent these generalizations in various forms in symbolic expressions (Arzarello, 1992; Lee, 1996; Mason, 1996). Fujii (2003) has introduced younger students to algebraic thinking through generalizable numerical expressions, using numbers as *quasi-variables*—for example, number sentences such as $78 - 49 + 49 = 78$, which are true whatever number is taken away and then added back. Finding that “generalizable numerical expressions can assist students in identifying and discussing algebraic generalizations long before they learn algebraic notation” (p. 59), Fujii claimed that these expressions “allow teachers to build a bridge from existing arithmetic problems to opportunities for thinking algebraically without having to rely on prior knowledge of literal symbolic forms” (p. 62). Other studies (e.g., Graham & Thomas, 2000) have illustrated how calculating technology can be effectively used to develop students' understanding of letters as specific unknowns, generalized number, and variable. In addition, some research studies have focused on the conceptual obstacles involved in moving from simple equations with an unknown on one side to more complex equations with unknowns on both sides and have described phenomena that they have called the *didactic cut* (Filloy & Rojano, 1989) and the *cognitive gap* (Herscovics & Linchevski, 1994).

The Minus Sign and Negative Numbers. In contrast to the findings already accumulated with respect to students' evolving knowledge of the different possible interpretations of the equal sign, we know comparatively little about students' interpretations of the minus sign in algebraic expressions and equations. The subtle distinctions in the ways that the minus sign is used in algebraic expressions and equations were the focus of a study by Vlassis (2004). Of the three possible interpretations of the minus sign, which she described as knowledge of *negativity*, she found that the binary sense predominated among the eighth graders of her

study. She noted a high degree of inflexibility in considering either a unary sense (whereby “subtract 3” means “negative 3”) or a symmetric sense (where, for example, in $-x = 1$, the $-x$ means “the opposite of x ”). Vlassis concluded that the various uses of the minus sign in algebra are counterintuitive to a majority of beginning-algebra pupils and constitute a significant conceptual obstacle in their coming to give meaning to algebraic symbols and processes.

Related research is that involving students' developing notions of negative numbers. Among the few studies in this area is the work of Gallardo (2002; see also Vergnaud, 1989), who applied historical/epistemological analysis to the study of students' extension of the natural-number domain to the integers within word-problem contexts. Gallardo has found that the first three of the four levels of acceptance of negative numbers within historical texts (i.e., subtrahend, relative number, isolated number, and formal negative number) were present among the 12- and 13-year-olds of her study. This suggested to her that students have an intuitive sense of negative numbers long before they are able to formalize this knowledge.

Beginnings of Structure Sense. Linchevski and Livneh (1999) extended some of the earlier research work related to students' structural sense in algebra (e.g., Kieran, 1989; Matz, 1982) in a study that tested whether students' difficulties with aspects of algebraic structure reflect difficulties they already have with numerical structure. They interviewed two classes of end-of-year sixth graders (average age: nearly 12 years) who had not yet learned negative numbers or any algebraic manipulation. They found that, on the whole, students' difficulties with interpreting equations containing several numerical terms and an unknown were a reflection of the same errors that they were making in purely numerical contexts. However, the researchers also noted that some students were inconsistent in that they gave an incorrect answer in one context but a correct one in another. Follow-up interviews with students after they had learned how to operate with negative numbers produced the same results. The researchers suggested that algebra instruction be designed to foster the development of *structure sense* by providing experience with equivalent structures of expressions and with their decomposition and recomposition. They emphasized that more systematic research in this area is needed. One recent thrust in this direction has been taken by Carpenter, Franke, and Levi (2003) whose research with younger students

³ *Metalinguistic awareness* refers, according to the authors, to the ability to reflect on and analyze spoken or written language, for example, being able to pay attention to sounds and spellings of linguistic signs instead of to their meanings. Both symbol awareness and syntax awareness are considered to be components of the metalinguistic awareness necessary for success in learning to use algebraic notation.

has aimed at having students explain and justify the properties they are using as they carry out their arithmetic calculations. Additional research in this area is discussed in the section on transformational activity.

Generational Activity with Multiple Representations.

Most of the approaches to algebra that make use of multiple representations define themselves as *functional approaches*. Tabach and Friedlander (cited in Kieran & Yerushalmy, 2004) suggested that three factors allow for a functional approach in the teaching and learning of algebra: the potential to produce ample numerical tables, the need to use general expressions to express the data in these tables, and the possibility of obtaining a wide variety of corresponding graphs. As an example of such an approach, they referred to the *Compu-Math curriculum* (Hershkowitz et al., 2002), which relies heavily on the use of spreadsheets, in particular. However, not all the research involving multiple representations includes technology environments. In addition, as will be seen, some studies have included multiple representations within word-problem or modeling situations. When the focus of the analysis has been the multi-representational aspect, then the study is included in this section; however, if the word-problem aspect has the higher profile, then the research is treated in the next section. Finally, if the modeling aspect is uppermost, then the study is reserved for the later discussion of global/meta-level activity. This section includes the following themes: tabular representations and the letter-symbolic, graphical representations and the letter-symbolic, and connections among representations.

Tabular Representations and the Letter-Symbolic. Recent studies involving the tabular displays of spreadsheets have investigated whether and how the use of spreadsheet methods lends meaning to algebra. Building on earlier research with spreadsheets as a means to introduce students to algebra-like notation (e.g., Sutherland & Rojano, 1993), Filloy, Rojano, and Rubio (2001) have found that the spreadsheet serves as a bridging tool to algebra because it helps students to create conceptual meaning for algebraic objects and operations, and to move from focusing on a specific example to describing general relationships. For Dettori, Garuti, and Lemut (2001), the functional orientation of spreadsheets allows for their effective use in the investigation of variation but can cause difficulties later when solving equations or inequalities. When these researchers investigated 13- and 14-year-olds working on algebraic problems with spreadsheets, Dettori and colleagues were led to conclude that “spreadsheets can start the journey of learning

algebra, but do not have the tools to complete it” (p. 206). As well, Hershkowitz et al. (2002) have described how students tend, in the spreadsheet environment, to generalize recursively rather than explicitly, which makes it more difficult for them to generate closed-form algebraic rules for patterns.

However, other research work that has included tables of values, but not within a spreadsheet environment, has noted that this representation can be used effectively within generational activity. For example, Cedillo and Kieran (2003) carried out a study with 13- and 14-year-olds in which the pattern of input and output values in the given table motivated the search for a computer “program” (algebraic expression) that would reproduce the pattern. Using TI-92 Plus calculators, the students were able to test the algebraic expressions they were generating to see if their expression would yield the same output values as were displayed in the given table. The researchers pointed out the beneficial role played by the calculator in the development of not only students’ ability to formulate algebraic expressions but also their emerging notions of variable.

Graphical Representations and the Letter-Symbolic.

In a study that focused on the slope-intercept form of a linear function, $y = mx + b$ (Lobato, Ellis, & Muñoz, 2003), the researchers found that students in a first-year reform algebra class developed the notion of slope as a difference (i.e., “it goes up by”) rather than as a ratio—as was intended. By means of the theoretical construct of focusing phenomena, Lobato and colleagues analyzed the regularities in the ways in which the teacher, students, artifacts, and curricular materials acted together to direct student attention toward certain mathematical properties over others. Their analysis revealed the presence of four focusing phenomena: the ambiguity of the “goes up by” language, the use of ordered tables of values, the language used when entering into the graphing calculator both the x-values of a function and the scale of the x-axis, and a lack of coordination of differences in x- and y-values when $x \neq 0$. The researchers suggested that a directed focus of attention on the coordination of covarying quantities would be more likely to lead students to generalize slope as a ratio.

Moschkovich (1998) has pointed to students’ tendencies to ascribe a role to the x-intercept when learning about linear functions of the form $y = mx + b$. A subgroup of 18 students from a ninth-grade pilot first-year algebra course that included the use of the computer graphing utility, *Superplot*, participated in videotaped discussion sessions with a peer of their choice, using the same graphing utility. Although 13 of these students used the x-intercept at least once at the outset

of the group study for either b or m in an equation, or to describe lines as moving along the x -axis as a result of changing b in an equation, their thinking nevertheless evolved over the course of the study. Moschkovich has described the resources used by students as they refined their conceptions of the presumed role of the x -intercept, such as narrowing the problem contexts in which they applied this conception and making connections between conceptions.

Using the graphing facility of the spreadsheet as a potential learning environment, Ainley (1996) explored some of the ways in which this tool could be used to introduce 12-year-olds to the power of generalizing through algebraic notation. Ainley found that spreadsheet activity allowed students to appreciate the purpose and power of the symbolic notation by providing them with a means to generate more data for the problem at hand and thus work towards a solution. In a study by Kieran and Sfard (1999) that used the software, *Algebra Connections*, students became so skilled at graphing linear functions by focusing on the slope and y -intercept that they could mentally approximate the solutions to equations such as $7x + 4 = 5x + 8$ by visualizing the graphs of the two functions and their intersection point. However, in another study, Kieran (2001) found that, for certain nonlinear situations involving rational functions, students experienced considerable difficulty in generating both a graphical and a symbolic representation for the problem situation—the rational function proving to be much more problematic than the linear function.

Researchers have also looked at the ways in which graphical representations are used to give meaning to inequalities. Bazzini, Boero, and Garuti (2001b), who studied the feasibility of a functional approach in the teaching of inequalities to eighth-grade students, have pointed to the positive role that graphical representations can play in helping students to conceptualize the symbolic forms of inequalities, as well as the pitfalls involved in attempting to apply to the solving of inequalities some of the transformational techniques used with equations.

Arzarello and Robutti (2001) have described a study introducing ninth graders to algebra and involving the interpretation of body-motion graphs within an environment that included symbolic-graphing calculators connected to motion sensors (the calculator-based ranger—CBR). Initially students were encouraged to try various running patterns in order to create different graphs. The continuous nature of the CBR graphing allowed students to test conjectures in a direct manner, controllable by their own physical movement. Arzarello and Robutti observed that

students' cognitive activity passes through a complex evolution, which starts in their bodily experience (namely, running in the corridor), goes on with the evocation of the just lived experience through gestures and words, continues connecting it with the data representation, and culminates with the use of algebraic language to write down the relationships between the quantities involved in the experiment. (p. 39)

Connections Among Representations. Although research continues to show that various technological environments can help in developing students' algebraic thinking by building links among symbolic, tabular, and graphical representations (e.g., Friedlander & Tabach, 2001; Koedinger, Anderson, Hadley, & Mark, 1997; see also Balacheff & Kaput, 1996; Ruthven, 1996), such improvements are not automatic. The quality of the tasks (e.g., Hoyles, 2002), the teaching (e.g., Kubinová & Novotná, 2002), and the general learning environment (e.g., Kramarski, 2000) all play significant roles. Furthermore, as pointed out in the review of graphing calculator research carried out between 1985 and 1995 by Penglase and Arnold (1996), graphing technology clearly helped students improve their understanding of function and graphing concepts; however, several of the studies included in their review suggest that students continued to experience difficulties with seeing the relationship between algebraic and graphical representations of functions.

A partial explanation of this phenomenon has been found to lie in the time needed by students to build an understanding for symbolic notation. For example, Nathan, Stephens, Masarik, Alibali, and Koedinger (2002) investigated seventh and eighth graders' ($N = 90$) abilities to solve problems using tabular, graphical, verbal, and symbolic representations (without the aid of technology) and to translate among these representations before and after instruction that involved bridging activities built on students' invented strategies and representations. Before instruction, students experienced more success solving problems using a given representation than translating among representations, with success strongly influenced by the particular representational formats involved. Results suggested that there are significant gaps in students' abilities to comprehend and to produce representations, and that students may attain more fluency with tables and pointwise graphs before symbolic equations and verbal expressions. (See also Koedinger & Nathan, 2004, whose study is described in a later section.)

Yerushalmy (2000) has studied the long-term impact of a problem-based, functional approach to the teaching of algebra, one that was supported by intensive use of graphing technology, and has reported findings similar to those of Nathan and his colleagues.

By means of classroom observations and clinical interviews with students over their 3 years of following the course, Yerushalmy found that the representation of problem situations evolved as follows: from numbers as the only means of modeling, to intensive work with graphs and tables, to the use of more symbolic representations. Students also moved from analyzing patterns of numbers by watching the behavior of increments to more explicit, closed-form formulations of functional relationships involving pairs of numbers. However, Yerushalmy wondered why it took so long for the students to appreciate algebraic symbols, and why they preferred to use the situation as the source for their answers. From the findings of another study she carried out (Yerushalmy & Shternberg, 2001), and long-term observations by Chazan (2000), Yerushalmy (2000) has argued that “a curricular sequence of algebra based on functions might have to address more subtleties and transitions than initially expected . . . and that the complexity of helping students to value algebraic symbols may require more than bridging between representations” (p. 145). In another related study (Gilead & Yerushalmy, 2001), ninth graders who had been taught by a functional approach to the interpretation of equations (comparison of functions) encountered an obstacle when dealing with a certain type of situation not easily modeled by an equation without introducing another variable or parameter. The researchers were led to suggest that graphical tools may be useful as an exploratory support only for canonical problems (i.e., where y -intercept and slope are both given).

In a study of 1 year's duration, Schwarz and Hershkowitz (1999) investigated ninth graders' learning of the concept of mathematical function in an environment that included problem situations, graphing calculators, and multirepresentational software tools, and where students were encouraged to make their own decisions about which representations to use, when and how to link the representations (especially graphs and tables), and in which medium to work. The researchers compared and contrasted the characteristics of students' concept images of functions that emerged in this environment with the concept images described in previous work involving different environments (e.g., Leinhardt, Zaslavsky, & Stein, 1990; Markovits, Eylon, & Bruckheimer, 1986; Schwarz & Dreyfus, 1995). The questionnaire that was presented to students included questions primarily on the interpretation and generation of graphical representations, and to a limited extent on the relation of certain graphs to algebraic representations. The researchers found that the students of the current study often used the

prototypical linear and quadratic functions but did not consider them as exclusive (for persistence of linear models in students' reasoning, see DeBock, Verschaffel, & Janssens, 2002). They suggested further that the numerous manipulative experiences with function examples, which were afforded by the multirepresentational software, influenced positively students' understanding of the functions' attributes. They also claimed that the process of beginning each class activity by small-group work on open-ended situations, followed by full-class discussions that were orchestrated by the teacher, was conducive to students' engaging in a discourse that led to higher levels of mathematical reasoning and conceptual change (see also Yackel & Cobb, 1996).

Generational Activity Within the Context of Word Problems. The solving of word problems has traditionally been considered to involve two phases: the setting-up of an equation to represent the relationships inherent in the word problem, and the actual solving of the equation. Although past studies focused almost exclusively on the equation as the form of problem representation (e.g., Chaiklin, 1989), more recent work has included a much broader range of representations. A further move has been occurring of late that involves an extension of the types of problems being worked on to include the modeling of more open-ended, “real-world” situations. This growing domain of modeling within school mathematics, to the degree that it intersects with the learning of school algebra, is reserved for the later discussion of global/meta-level activity. This particular section is therefore divided into two parts: The first examines the research related to word-problem solving involving multiple representations, and the second, involving the equation representation.

Generating Multiple Representations in the Solving of Word Problems. In a study involving 6 intact classes of seventh and eighth graders at three junior high schools, half of whom were in a treatment group that participated in a representation-based unit on functions and the other half in a comparison group using conventional textbook lessons on algebra that included solving equations, using equations to solve word problems, and a unit on graphing ordered pairs, Brenner et al. (1997) investigated whether instruction on representing word problems by means of multiple representations improved students' abilities to represent and solve problems. The researchers found that instruction on problem-representation skills can work, that is, students learned to create various multiple representations and to use these to find problem solutions. The reform-based instruction was also

found to have the same beneficial effects on language-minority students as on the overall school population of the study. Related research by Koedinger et al. (1997), involving the multirepresentational, problem-solving computer environment, *Cognitive Tutor Algebra I* (Carnegie Learning, 1998), and by Nathan et al. (2002), suggests however that students tend to favor tables of values as problem-solving tools before they gain facility with graphs and equations. Multiple representations of word problems have also been used to introduce students to the form of inequalities. In analyzing some of the videotape data from the TIMSS-R eighth-grade algebra lessons (Stigler et al., 2003), Kieran (2004) noted that Japanese students attempted to solve the inequality situation first by using a variety of representations and methods to solve for the corresponding equality, and then by making adjustments to their “solution” in order to provide an answer for the inequality. (Interestingly, none of the students used a graphical representation.) The teacher then used these various “equality” methods, in particular the table-of-values representation, to help students acquire meaning for the form of the algebraic inequality.

Generating Equations in the Solving of Word Problems. Generating equations to represent the relationships found in typical word problems is well known to be an area of difficulty for algebra students. Nevertheless, word problem situations not only continue to be used as a means for infusing algebraic objects with meaning but have also received increased emphasis in reform programs as vehicles for introducing students to algebra. Research in this area continues to provide evidence of students’ preferences for arithmetic reasoning and their difficulties with the use of equations to solve word problems (e.g., Bednarz & Janvier, 1996; Cortés, 1998; Swafford & Langrall, 2000). For example, Stacey and MacGregor (1999) found that, at every stage of the process of solving problems by algebra, students were deflected from the algebraic path by reverting to thinking grounded in arithmetic problem-solving methods. Another study dealing with the setting up of equations within the activity of word-problem solving has reported that, for sixth and seventh graders, reasoning and symbolizing appear to develop as independent capabilities (van Amerom, 2003). Although some students could write equations to represent problems, they did not use these equations to find the solution, preferring instead to use more informal methods. Some students did, however, use the equations they had generated as a basis for their reasoning about the problems and for finding their solutions. Similarly, Malara (1999) has observed students using numerical substitution within problem representations in order to arrive at solutions (see also

Johanning, 2004). Kutscher and Linchevski (1997) have also noted the beneficial effect of numerical instantiations as mediators in solving word problems. Although contextualized problems are often considered an appropriate avenue for introducing students to algebra (NCTM, 2000), De Bock, Verschaffel, Janssens, and Claes (2000), who explored the influence of authentic and realistic contexts and of self-made drawings on the illusion of linearity in length and area problems, found no beneficial effect of the authenticity factor, nor of the drawing activity, on pupils’ performance. Furthermore, the researchers suggested that realistic problems might, in fact, steer pupils away from the underlying mathematical structure of a problem.

However, according to Nathan and Koedinger (2000a), word problems presented in verbal form are easier for students to solve than comparable questions presented in other formats, such as equations or “word-equations.” These researchers presented a set of 12 problems to a group of 76 high school students in the following formats: Two were in story-problem format (e.g., “When Ted got home from his waiter job, he multiplied his hourly wage by the 6 hours he worked that day. Then he added the \$66 he made in tips and found he earned \$81.90. How much per hour did Ted make?”), two were in equation format (e.g., “Solve for x : $x \cdot 6 + 66 = 81.90$ ”), and two were in word-equation format (e.g., “Starting with some number, if I multiply it by 6 and then add 66, I get 81.90. What did I start with?”). For one problem in each format, the start value was unknown, and for the other, the result was unknown. The equation format was found to be significantly less likely to be correctly solved than either the story-problem or word-equation formats, whereas algebra story problems and algebra word-equation problems were found to be of equal difficulty. In a replication study involving 171 students (Nathan & Koedinger, 2000a), solution success rates for symbolic equations were 25% less than for story problems and nearly 20% less than for word equations. In both of these studies, the start-unknown format was more difficult than the result-unknown problems. Students’ difficulties with problems presented in equation format challenge, according to these researchers, the oft-cited view that story problems are inherently harder than symbolic ones. The many possible entry points, as well as the availability of a variety of solving approaches for story problems, are deemed to be factors accounting for the higher success rates with the verbal format.

In a follow-up study, Koedinger and Nathan (2004) further explored their earlier finding that students are more successful solving simple algebra sto-

ry problems than solving mathematically equivalent equations. They found evidence that this result is not simply a consequence of situated world knowledge facilitating problem-solving performance, but rather a consequence of student difficulties with comprehending the formal symbolic representation of quantitative relations. In fact, students translated story problems into the standard equation format only 5% of the time. According to the researchers, students even after a full year of algebra were particularly challenged by the demands of comprehending the letter symbolic form of the equation: “The language of symbolic algebra presents new demands that are not common in English or in the simpler symbolic arithmetic language of students’ past experience” (p. 149). The researchers suggested that more attention be focused on equation-solving instruction; however, they emphasized that this be done after students have had experience with translating back and forth between English and algebra within story problem situations.

Transformational Activity: The Middle Grade and Lower Secondary Level Student

As noted earlier, the transformational activity of algebra includes collecting like terms, factoring, expanding, substituting one expression for another, adding and multiplying polynomial expressions, exponentiation with polynomials, solving equations and inequalities, simplifying expressions, substituting numerical values into expressions, working with equivalent expressions and equations, and so on. Past research in this area tended to focus on the nature of the manipulative processes used in expression simplification (e.g., Davis, Jockusch, & McKnight, 1978; Sleeman, 1984) and in equation solving (e.g., Bell, O’Brien, & Shiu, 1980; Kieran, 1988; Whitman, 1976). The field has recently experienced a change that includes attention to the theoretical foundations of students’ manipulative work. In other words, algebraic transformations are not viewed simply as procedural, but also as theoretical. Thus, notions of equivalence figure more prominently in some of this later work. Another movement has been the downward shift of symbol-manipulation technology from the college to the secondary school level, even to the lower secondary level. To reflect the current research emphases, this section is divided into the following three parts: theoretical elements of transformational activity, namely equivalence and theoretical control; technical elements of transformational activity, namely the manipulation of expressions and equations; and the use of concrete manipulatives in transformational activity. The reader is reminded that the age range of the student in this section continues to

be approximately 11 to 15 years of age, that is, the sixth to the ninth grader. Comparable research conducted with the older student is reserved for a later section.

Theoretical Elements of Transformational Activity: Equivalence and Theoretical Control. According to Pimm (1995),

One resource of algebra is a rich plurality of symbolic forms; one core notion, that of equivalence. Equivalence and transformation are linked notions, indicating sameness perceived in difference for some purposes, or indifference with respect to others. The existence of multiple expressions ‘for the same thing’ (highlighting ‘naming’) can suggest the very possibility of transforming expressions directly to get from one to another. (p. 89)

Despite the importance of this core idea, only a few of the earlier studies explicitly addressed students’ notions of equivalence (e.g., Greeno, 1982; Kieran, 1984a; Linchevski & Vinner, 1990; Steinberg, Sleeman, & Ktorza, 1990).

An example of some of the later research in this area is the work of Cerulli and Mariotti (2001; see also Nicaud, Delozanne, & Grugeon, 2002). Cerulli and Mariotti have described a teaching experiment on equivalence that involved ninth-grade classes and the algebra microworld, *l’Algebrista*, created by Cerulli (2004). Their aim was to develop in students a theoretical perspective on algebraic manipulation, by basing activity on the concept of equivalence relation, and by distinguishing *proof* from *verification*. The equivalence of two expressions is *proved* if one expression is transformed into the other using the axioms. Students were taught that, with literal expressions, the use of axioms is the only way to prove the equivalence of two expressions, whilst numerical *verification* (i.e., substituting the letters with numbers and computing the expressions) is the main way to show that two expressions are not equivalent. Students worked at changing the form of given numerical and literal expressions by selecting axiom buttons of the computer tool or by creating their own “theorem/transformation buttons.”

Cerulli and Mariotti’s (2001) analysis highlighted two important issues. First, they noted the carryover of the button-icons to students’ paper-and-pencil work. Similar iconic traces of the computer environment have been noted in the written algebra work of the participants of Sutherland’s (1993) spreadsheet study. According to Cerulli and Mariotti, such signs guide the evolution of meaning in students’ algebraic thinking. The second issue concerns the theoretical aspect of students’ activity. Cerulli and Mariotti were able to

provide evidence to support their claim that they had moved students away from a purely procedural interpretation of algebraic manipulation and had inculcated a theoretical perspective by means of the emphasis on axioms and theorems. The importance of theoretical control in algebraic transformational activity has also been articulated by, among others, Balacheff (2001) and Dettori et al. (2001). However, as noted by Demby (1997), students have a great deal of difficulty in identifying the properties they use when they transform algebraic expressions.

Another element of theoretical control that is considered basic to transformational activity is the knowledge that relates the algebraic domain with the arithmetical. For example, numerical substitution activity within expressions and equations can help students make connections between the arithmetical and the algebraic world (see, e.g., Warren & Pierce, 2004). Graham and Thomas (2000) conducted a teaching experiment with nearly two hundred 13- and 14-year-olds in which students used the letter stores of the graphing calculator as a model of a variable and so evaluated different expressions for a variety of inputs. In the case of equivalent expressions, students came to see that different expressions were being used to represent the same process. This activity also had an impact on students' views of expressions and variables, which suggests that a task that is transformational in nature can simultaneously be related to generational activity if it leads to an evolution of students' conceptions of the objects of algebra.

The kinds of errors that students can make in algebraic transformational activity (e.g., Carry et al., 1980; Lemoyne, Conne, & Brun, 1993; Matz, 1982; Sleeman, 1984) have suggested to some researchers (e.g., Kirshner, 1989) that the issue is not an absence of theoretical control but rather a misperception of form. In a study that extended Kirshner's earlier work on the visual syntax of algebra, Kirshner and Awtry (2004) investigated the role of visual salience⁴ in the initial learning of algebra for students in four intact classes of seventh graders (about 12 years old). They found that students did indeed engage with the visual characteristics of the symbol system in their initial learning of algebraic rules: The percentage-correct scores for recognition tasks were significantly higher for visually salient rules than for non-visually-salient rules. Furthermore, visually salient rules turned out to be significantly more difficult to recognize than non-visually-salient rules when presented using tree notation. Similarly, Hewitt (2003), in a study of 40 teachers

and a class of 11- to 12-year-olds, found that the inherent mathematical structure, and the visual impact of the notation itself, had an effect on the way in which equations were manipulated.

Technical Elements of Transformational Activity: Expressions, Equations, and Equation Solving. Among the few recent studies dealing with the transformation of algebraic expressions involving this age group of students are a study by Bazzini, Boero, and Garuti (2001a) and another by Ayers (2000) on bracket expansion errors. A greater amount of research has dealt with equations. For instance, Herscovics and Linchevski (1994) conducted a study involving equations of the type $4 + n - 2 + 5 = 11 + 3 + 5$, with various numerical operations on one or both sides of the equal sign. They found that about 50% of the beginning algebra students whom they tested failed to obtain the correct solution, many of them having equated the left side of the equation with only the first member of the right side—a result consistent with Kieran's (1984b) findings with the same kind of equation. In addition, Herscovics and Linchevski observed that several of the unsuccessful solvers incorrectly grouped the left-hand expression to $4 + n - 7$, an error that they have referred to as the *detachment of a term from the indicated operation*. They noted as well that, with the equation $115 - n + 9 = 61$, some students transformed it to $106 - n = 61$, having committed the error of *jumping off with the posterior operation* (see Linchevski & Herscovics, 1996, for additional errors). Similar findings have been reported in research involving eighth-grade Belgian students (Vlassis, 2001).

In a study that focused on the checking of equation solutions, Perrenet and Wolters (1994) found that eighth graders made the same sorts of errors in their checking that they had made while solving the equation, thereby not detecting that their solutions were erroneous. Analysis of students' erroneous checking behavior disclosed their profound structural misunderstandings, uncertainty, and wishful thinking. In another study related to checking solutions, Pawley (1999) noted that students were more successful in both their equation solving and their checking if they were taught how to check sometime later than they were taught how to translate from sentence to equation.

Up until quite recently, researchers have known very little about the ways in which students of this age range approach the solving of systems of equations and the manner in which they think about its underlying concepts. The recent research of Filloy, Rojano,

⁴ According to the researchers, "visually salient rules have a visual coherence that makes the left- and right-hand sides of the equations appear naturally related to one another" (p. 229). Although $(x^y)^z = x^{yz}$ is considered a visually salient rule, $x^2 - y^2 = (x - y)(x + y)$ is not.

and Solares (2003, 2004) has focused on the spontaneous approaches of 13- and 14-year-olds, who had already been introduced to the solving of one-unknown linear equations, to problems that could be solved by systems of equations. One of the aims of this research was to document the transition from one-unknown representations and manipulations to the representation and manipulation of one unknown given in terms of the other unknown. It was found that students seemed more inclined to make sense of comparison approaches than substitution methods; however, the researchers noted that manipulation difficulties contributed to making the substitution method less accessible. They also observed that the extension of the notion of transitivity of equality from the numeric to the algebraic domain, as well as the idea of substituting one expression with another, was not at all obvious to students.

Further evidence of students' difficulties with the substitution method for solving systems of equations was found by Drijvers (2003). Within a Computer Algebra System (CAS) environment, 14- and 15-year-olds were asked to solve parametric equations, for example, "Solve $ax + b = 5$ for x ." Students experienced difficulty in accepting the expression $(5 - b)/a$ as a solution. According to Drijvers, this required that they conceptualize an expression as an object (Sfard, 1991). Other tasks such as, "Consider the equations $y = a - x$ and $x^2 + y^2 = 10$. Make one equation from these in which y does not appear; you do not need to solve this new equation" (p. 260), which required substituting a variable in one equation by an expression drawn from the other equation, were equally problematic. In some cases, the use of graphical representations helped students to make sense of the equations with which they were working. The obstacles that students encountered in working with these equations led Drijvers to suggest that more attention needs to be paid to the role of paper-and-pencil work throughout CAS activity and to the importance of focused classroom discussions. Friedlander and Stein (2001) found that a significant number of students who could solve equations with the various computer tools at their disposal actually preferred to use paper and pencil.

Within process-object theories (e.g., Sfard, 1991), the transition from process to object tends to be viewed as a leap, a sudden shift in perspective, whereby a process becomes reified as an object. Students who conceive of an expression as a process are considered to be unable to make much sense of, for example, composition of functions (e.g., Ayers, Davis, Dubinsky, & Lewin, 1988), until such time as they make that sudden shift from expression

as process to expression as object. However, recent research by Landa (2003) suggests a more gradual learning process, in that composition of function may be preceded conceptually by composition of argument. Landa, who worked with 14- and 15-year-old students in a spreadsheet environment, found that, before students could make sense of, for example, $g(f(x))$, where $f(x) = 2x + 3$ and $g(x) = x^2$, they first needed to be able to create meaning for replacing the argument in a single function by an expression, as in $g(2x + 3)$. The spreadsheet environment, by means of its numerical calculation of functions with composed arguments, was seen to play a vital role in mediating students' passage toward composition of functions.

The Use of Concrete Manipulatives in Transformational Activity. Research has produced mixed results with respect to the use of concrete models and manipulatives in learning about equations and equation solving. Filloy and Rojano (1989) found that the balance scale and geometric models were not effective for helping students construct meaning for the process of operating on both sides of the equation to solve for the unknown. Boulton-Lewis et al. (1997) reported that the eighth graders of their study did not use the concrete manipulatives (cups, counters, and sticks) that were made available for solving linear equations and suggested that the concrete representations increased processing load. In contrast, advocates of such models (e.g., Brown, Eade, & Wilson, 1999; Linchevski & Williams, 1996; Radford & Grenier, 1996) have argued that the balance scale facilitates the understanding of the operation of eliminating the same term from both sides of an equation. In particular, Vlassis (2002) found that, for the two classes of eighth graders whom she observed over a period of 16 lessons, the balance model was an effective tool for conveying the principles of transformation. According to Vlassis, some errors linked to its usage appeared during the first lesson; however, these tended to disappear over time. Nevertheless, Vlassis emphasized that if students have not already extended their numerical range to encompass the negative integers, cancellation errors will be inevitable.

Global/Meta-Level Activity: The Middle Grade and Lower Secondary Level Student

The global/meta-level activities of algebra include problem solving, modeling, working with generalizable patterns, justifying and proving, making predictions and conjectures, studying change in functional situations, looking for relationships or structure, and so on—activities that could indeed be engaged in

without using any letter-symbolic algebra at all. Although algebraic activity involving problem solving, more particularly word-problem solving, has been integrated into the previous sections on generational and transformational activity, three types of global/meta-level activity for which a significant body of algebra research exists with this age group of learner have been reserved for this section: generalizing, proof and proving, and modeling.

Global/Meta-Level Activity Involving Generalizing.

Generalization activity within algebra has its roots in the use of algebraic notation as a tool for expressing proofs (e.g., Bell, 1976; Fischbein & Kedem, 1982; Mason & Pimm, 1984). The position that generalization is also a route to algebra was developed by Mason, Graham, Pimm, and Gowar (1985; see also Mason, Graham, & Johnston-Wilder, 2005). In some of the pioneering research on the use of algebraic notation as a tool for expressing the general form of geometric and numerical patterns, and for justifying equivalent forms of these patterning relations, Lee (1987; Lee & Wheeler, 1987) found that few students use algebra or appreciate its role in justifying a general statement about numbers. Similar findings have been reported by MacGregor and Stacey (1993), who observed that an additional difficulty was related to students' inability to articulate clearly the structure of a pattern or relationship using ordinary language.

Healy and Hoyles (1999) have pointed out that the visual approaches generated in tasks involving generalization of matchstick patterns can provide strong support for the algebraic representation of sequences and the development of a conceptual framework for functions but emphasized the need to work hard to connect the observed number patterns to symbolic form. Warren (2000), who tested 12- to 15-year-olds ($N = 379$) on tasks related to generalization of patterns, reported that students found this kind of activity to be rather difficult. As a result of follow-up interviews held with some of the students, she noted that the generalization process draws on a variety of skills not traditionally associated with the algebraic domain. In addition, Ainley, Wilson, and Bills (2003), who compared the process of *generalization of the context* with *generalization of the calculation*, found that generalizing the context did not seem to be sufficient to support pupils in moving to a symbolic version of the rule. However, Radford (2000), who has studied the transition from the particular to the general, has argued that such processes take time.

In a study that focused on the role of tabular representations in generalization activity, Sasman, Olivier, and Linchevski (1999) presented eighth grad-

ers (14-year-olds) with tasks in which they varied the representation along several dimensions, namely the type of function, the nature of the numbers, the format of tables, and the structure of pictures. Results showed that varying these dimensions had little effect on students' thinking. According to Mason (1996), a scrutiny of established school practice involving generalization in algebra reveals that often, starting from geometric figures or numeric sequences, the emphasis is on the construction of tables of values from which a closed-form formula is extracted and checked with one or two examples. This approach in effect short-circuits all the richness of the process of generalization. In fact, some researchers (e.g., Moss, 2005) have suggested that tabular representations may actually impede students' recognizing the general relationships underlying patterns and representing them algebraically. Mason suggests several possible investigative approaches that can lead to students' construction of algebraic formulas, including visualization and manipulation of the figure on which the generalizing process is based.

From a study of eighth-grade students and the generalization of patterns, Radford (2003) reported that whereas rhythm and movement play a central role in presymbolic generalizations, symbolic algebraic generalizations require a disembodiment of the mathematical from the spatial-temporal. This researcher noted that the passage from a nonsymbolic to a symbolic algebraic expression of generality involves two ruptures, one with the sensual geometry of the patterns and the other with their numerical features such as rank. Furthermore, these ruptures are, according to Radford (2000),

regulated by a socially established mathematical practice where the teacher plays a central role; this role is that of immersing and initiating the students into the peculiarities of the signs and meanings in which the practice of algebra is grounded. (p. 258)

Global/Meta-Level Activity Involving Proof and Proving.

A great deal of the research on proof and proving with students of this age includes problems drawn from elementary number theory. For example, Edwards (1998), in her study of 14- and 15-year-olds who had not yet been exposed to instruction in proof in their algebra class, found that none of them used standard algebraic notation to justify or explain their true-false decisions about simple statements involving the combining of odd and even numbers. Miyakawa (2002), who reported on a study involving 14-year-old ninth graders working on a proof problem dealing with the sum of two even numbers, found that the difficulty in constructing this kind of mathematical proof

relates not only to algebraic competence and proof conception, but also to general mathematical knowledge. Dreyfus, Hershkowitz, and Schwarz (2001), who observed seventh graders collaboratively constructing an algebraic proof, reported that constructing this proof, which the students were able to do, involved first generalizing the process and then justifying it by means of an algebraic representation.

In contrast to research involving students without prior experience in constructing algebraic proofs, a study by Healy and Hoyles (2000) examined the conceptions of proof held by students who had followed a curriculum that, several years earlier, had mandated activity in formulating and testing conjectures, as well as in explaining and justifying conclusions. The researchers surveyed nearly 2500 high-attaining 14- and 15-year-olds. Consonant with the earlier findings of Lee and Wheeler (1987), Healy and Hoyles found that, when students were asked to pick the argument that they thought would get the best mark, the majority of students chose the ones based on algebraic form. However, students were unlikely to base their own arguments on similar algebraic constructions, feeling that these arguments neither communicated nor illuminated the mathematics involved. Students preferred to give some numerical examples, followed by a narrative explanation—even if they were aware of the limitations of such arguments. According to the researchers, students' responses were influenced mainly by their mathematical competence, but also by curricular factors, their views of proof, and their gender.

Global/Meta-Level Activity Involving Modeling. The majority of the algebra studies involving problem situations have, in general, been carried out with typical word-problem contexts. Nevertheless, a growing number of studies are now including modeling activities that go beyond the traditional problems found in most textbooks. In his plenary address at the closing session of the 12th ICMI Study on The Future of the Teaching and Learning of Algebra, Burkhardt remarked that “modeling as a bridge to algebra” (see below) seemed to turn on its head the idea of algebra as a tool for modeling. However, a number of studies have used real or pseudo-real situations to stimulate algebraic symbolization, for example, the modeling of running with the aid of motion sensors to motivate the development of a symbolic representation of this activity (Arzarello & Robutti, 2001). Stacey and Chick (2004) have commented on the issue of modeling as a bridge to algebra as follows:

Why, at this point in time, is there an emphasis on “modeling as a bridge to algebra” rather than on in-

struction designed to help students use their algebra in modeling the world? It may simply reflect the dominance of interest in beginning rather than advanced algebra. It may reflect a feeling of stalemate in improving students' abilities to set up equations for solving word problems, whether presented traditionally or otherwise, and tackling traditional applications. Alternatively, it may reflect an intention to keep algebra and the real world closer together throughout instruction, so that special lessons to close the gap are irrelevant. In any case, it is important to recall the centrality of Kieran's global/meta-level activity. Proving, modeling, and applying are activities that give purpose to learning algebra. (p. 18)

Since the mid-1990s, there has been a noticeable shift toward interest in modeling activity. However, the term *modeling* is interpreted quite broadly in the research literature to include everything from typical word problems to complex physical situations. For Janvier (1996), modeling, in the true sense of the word, is a rare activity in school algebra (see also Doerr & Tripp, 1999). In line with Burkhardt (1981) and de Lange (1987), Janvier has emphasized that modeling involves two phases: formulation and validation. The formulation phase of modeling comprises first the examining of a phenomenon or situation in order to establish some key relationships between the variables involved. After initial assumptions have been investigated, a more or less complex series of mathematical transformations or operations ultimately lead to a model, often expressed as a symbolic expression. The second phase, that of validation, requires testing the validity of the model by going back to the reality that it is supposed to represent. Similarly, according to Zbiek (1998),

a mathematical model is a combination of one or more mathematical entities and the relationships among them that are chosen to represent aspects of a real-world situation (Niss, 1988); mathematical modeling requires the application of mathematics to unstructured problems in real-life situations (Galbraith & Clatworthy, 1990). (p. 184)

Research in this area includes the work of Lesh, Hoover, Hole, Kelly, and Post (2000) on model-eliciting activities, and Lesh and Doerr (2003) on a theory of models and modeling, as well as, for example, cross-national approaches to research on modeling (Molyneux-Hodgson, Rojano, Sutherland, & Ursini, 1999). (See also the chapter in this volume on problem solving and modeling.) In their research involving both U.S. and Australian middle school students, Doerr and English (2003) have described how students' modeling shifted their focus of thought from finding

a solution to a particular problem to creating a system of relationships that is generalizable and reusable.

Dynamic physical models are often an integral part of modeling studies. For example, research with students in the 11- to 15-years-of-age group has included devices with gears of different sizes (Bartolini Bussi, 1995; Meira, 1998), a screen version of a baby duck swimming to catch up with its mother (Kaput & Roschelle, 1997), a simulation of a moving elevator (Noble, Nemirovsky, Wright, & Tierney, 2004), and motion artifacts such as an electronic walking frog (Schorr, 2003). Winch movement was the focus of a case study by Izsák (2003), who analyzed how a pair of eighth graders developed knowledge for representing with algebra the phenomena produced by this device. Izsák reported that, as these students solved problems about the winch, they learned to distinguish equations that are true for any value of the independent variable from equations that constrain the independent variable to a unique value—distinctions that have been shown to be elusive for many algebra students. Izsák's analysis also demonstrated that students have and can use criteria for evaluating algebraic representations, and that they can develop modeling knowledge by coordinating such criteria with knowledge for generating and using algebraic representations of physical situations. Hines (2001), using a similar winch device in her research, has described the learning trajectory of an eighth grader as he created and interpreted tables, equations, and graphs intended to represent functions originating from explorations with this dynamic physical model.

Among the studies on students' modeling that have involved technology environments is the research of Yerushalmy (1997a). She studied seventh-grade Israeli algebra beginners working in a software environment, *The Algebra Sketchbook* (Yerushalmy & Shternberg, 1993), that allowed them to manipulate visual objects and to use two lexical sets (verbal and graphic) to model temporal phenomena related to functions of a single variable. Yerushalmy's descriptions of the linguistic aspects of students' modeling efforts provide substantive examples of student activity in an environment that does not require the use of algebraic symbols (see also the work of Nemirovsky, 1996, on mathematical narratives). In a related study involving situations in two variables, Yerushalmy (1997b) found that students' ability to produce visual representations and to think about and discuss concepts and processes using graphical terminology was an important factor in their modeling success. Visual considerations were a constant feature of students' analyses, even while they were using tables and talking about variables, and before drawing graphs.

The Impact of Technology on Algebraic Activity With the Middle Grade and Lower Secondary Level Student

Since the publication of the last *Handbook* (Grouws, 1992), research on the learning of algebra has experienced a significant increase in the number of technology-related studies. Although this surge must be balanced with the more modest growth in actual classroom usage of technology (e.g., Mullis et al., 2000, report that of the 23 educational systems participating in the 1999 TIMSS, only 21% of eighth graders reported some degree of computer use in their mathematics classes), the results regarding the effects of technology use on the initial learning of algebra have been quite remarkable. The various tools that have been employed include—as has been pointed out in earlier sections—spreadsheets, graphing calculators, computer algebra systems, calculator-based ranglers, and specially designed software environments. The aim of this section is to synthesize from the results discussed in the preceding sections the role that technology can play in the learning of algebra with students of the middle grade and lower secondary levels.

The greatest impact of the use of technology in algebra learning has been on students' understanding of graphical representations and the way in which these representations allow a significant number of students to visualize in a graphical way the symbolic form of functions. This has thereby also improved students' understanding of certain functions. However, many students continue to experience difficulties with seeing the relationship between algebraic and graphical representations of function (Penglas & Arnold, 1996). The long-term experience of students in problem-based, functional environments supported by intensive use of graphing technology suggests that their representation of problem situations passes through phases, starting with the use of numbers as the only means of modeling, to working with graphs and tables, and finally to using more symbolic representations (e.g., Yerushalmy, 2000). Several studies, involving various technological environments and an assortment of global/meta-level activities to provide context for algebraic work, have in fact pointed out that supporting students in moving to a symbolic version of a rule, a pattern, or a problem situation takes time. Nevertheless, in environments and with activities specially designed for symbolic work, students can and do develop conceptually with respect to thinking with and acting on symbols (e.g., Cerulli, 2004).

The use of technology, and mainly the use of graphing technology, obviously introduces into school algebra other representations besides symbols

and equations. One of the research-tested advantages of this introduction of a variety of representations in technology-supported environments is that such approaches are enabling traditionally unsuccessful students to gain access to the problem-solving aspects of algebra. In fact, “approaches to teaching and learning that emphasize problem solving and exploration, and within which students actively construct and negotiate meaning for the mathematics they encounter, find in this [graphing calculator] technology a natural and mathematically powerful partner” (Penglasé & Arnold, 1996, p. 85). More specifically, Huntley, Rasmussen, Villarubi, Sangtong, and Fey (2000), whose study included ninth graders as well as older students, found that students who were not strong in symbol-manipulation skills, when they were able to use context clues and had graphing calculators available, could outperform symbolically capable students in tasks requiring formulation and interpretation of situations. This suggests that weaker students too are deriving a deeper conceptual knowledge of algebraic objects within technological environments. However, research needs to reveal more about how this happens.

There is a perception among some educators that, when powerful technological tools appear to “do all the work,” there might be little left to teach in an algebra class. However, as the research in this and a later section shows, the opposite is the case. Nevertheless, the nature of the teaching changes, as does that of the tasks that are presented to students. To take advantage of the potential of technology would seem to require not only tasks that are designed to push students beyond the limits of their current algebraic thinking and encourage further development of that thinking, but also approaches to teaching that facilitate such growth in students. However, a great deal more research is needed in the two areas of task design and teaching within technological environments. Further discussion of the role of the teacher in technology-supported algebra classrooms occurs in a later section.

Concluding Remarks on the Algebra Research Involving the Middle Grade and Lower Secondary Level Student

The middle grade and lower secondary level student has received the bulk of the attention of algebra researchers. This would seem appropriate in that algebra is usually introduced with this age range of student. However, a comparison of the earlier studies of the 1980s involving the middle grade and lower secondary level student with the more current work that has been discussed in this particular section of the chapter discloses the presence of a shift in the

nature of the research. Earlier work tended to focus on student difficulties in making the transition from arithmetic to algebra, and on the nature of the algebraic concepts and procedures developed and used by students during their initial attempts at algebra. Later work, while building upon and extending some of this earlier research, has evolved in new directions. Three areas in particular were noted.

One concerns the sources of algebra meaning making. Meaning based on the letter-symbolic form and on the problem situation/context has been extended to include the meaning derived from the use of graphical and tabular representations, as well as the meaning that emerges within dynamic activity. More research, however, needs to be done in this latter area so as to provide further insights into how such meaning develops and becomes disembodied. Another more recent research focus that was signaled in this section, and which will be elaborated more fully in the next section dealing with the upper secondary and college level student, is the role of theorizing within manipulative activity. Although properties and axioms were always an implicit part of symbolic manipulation, they were rarely the theme of algebra research in the 1980s. In contrast, theorizing is now considered to belong not only to the domain of generational activity in algebra, but to transformational activity as well. Finally, the third new direction in algebra learning research is one that was just discussed in the prior section, that of the use of technology. In particular, spreadsheets and graphing technology have played significant roles in this technologically oriented movement in recent algebra research dealing with the middle grade and lower secondary level student.

This section closes with a few additional remarks regarding questions in need of further research. Earlier, it was noted that Radford (2000) had argued that

One of the didactic questions with which to deal is . . . that of the understanding of how . . . meanings are progressively transformed by the students up to the point to attain the standards of the complex algebraic meanings of contemporary school mathematics. (p. 240)

With the broad scope of generational and global/meta-level activity in current research, including functional approaches with their multirepresentational focus, Radford’s question presents a challenge for researchers. School algebra has been taking on so many different forms that one might be led to ask: Are there still commonly accepted “standards of the complex algebraic meanings of contemporary school mathematics”? Although the view of that which con-

stitutes algebraic activity has been considerably enlarged, the answer to this question is, I believe, still yes. And, because algebraic symbolism remains foundational to school algebra, it is strongly recommended that further research related to the learning of the letter-symbolic form be carried out. Areas where the field could benefit from additional study include, for example, the question as to how students can be assisted in (a) becoming aware of structure in patterns and in using symbols to express these patterns, (b) seeing relations between graphical representations and the corresponding letter-symbolic forms, and (c) making connections between their verbal problem-solving activity and the generating of equations. With the increased use of technology in algebra learning, another area where further research is needed is in the interaction between paper-and-pencil and machine work with tasks that have been specially designed to take advantage of this interaction. Further discussion of this issue is offered in the next section of the chapter dealing with the upper secondary and college level student.

The Upper Secondary and College Level Student

This section of the chapter examines the algebra research that has been carried out with the upper secondary and college level student, from the 10th grade onwards (about 15 years of age and up). Investigations that have involved students in teacher training programs at the college level are, however, reserved for the later section on teachers. As with the previous section devoted to the middle grade and lower secondary level student, the first subsection below deals with generational activity, whereas the second treats transformational activity and the third, global/meta-level activity. A fourth section discusses research on the effects of technology use, followed by a fifth that offers concluding remarks. Note that the research studies carried out on algebra learning with the upper secondary and college level student are fewer in number than those conducted with the middle grade and lower secondary level student.

Generational Activity: The Upper Secondary and College Level Student

Creating the objects of algebra, and building meaning for them, is at the core of generational activity, not only for the middle grade and lower secondary level student but also for the upper secondary and college level student. This section examines the research bearing on generational activity accord-

ing to whether the object of study is treated from (a) a primarily letter-symbolic focus or (b) a multiple-representation perspective. The latter of these two sources of meaning is the theme of a larger number of studies than the former, reflecting both the increased attention to functional approaches across reform curricula as well as the traditional emphasis on functions and their various representations with the older student.

Generational Activity With a Primary Focus on the Letter-Symbolic Form. The studies that have focused on generational issues related to the letter-symbolic form among upper secondary and college level students have included research on form and structure, as well as on the kind of meaning building that occurs for the algebraic object, parameter. (A review of the work related to the learning of abstract algebra and of linear algebra is beyond the scope of this chapter.)

Form and structure. Algebra was at one time “all about form,” according to Goldenberg (2003, p. 11). Two examples of form-related questions from a test set for 3rd-year Italian secondary students (16-year-olds) are obtained from a study by Menghini (1994):

Find the value of the following expression:

$$z^4/c^2(b-a) + z^4/c^2(a-b) = \dots$$

For which values of x is the following inequality true?

$$x^2 + x < x^2 + x + 1.$$

Menghini reported that, for the first example, most students carried out very detailed calculations, and for the second, the best answers involved a comparison of the two parabolas represented by the two parts of the inequality. What seemed to be missing from the responses to these tasks, according to the researcher, was the ability to recognize more general laws and forms, or to be able to achieve an abstract understanding independent of the specific numerical context. Similarly, in their study of 11th graders, Hoch and Dreyfus (2004) found that very few students used structure sense and those who did were inconsistent. The more advanced students used it more often than the intermediate ones. The presence of brackets seemed to help students see structure, focusing their attention on like terms and breaking up the long string of symbols.

On parameters. While research with the older student has attended to the issue of shifts in the meanings of literal symbols (Bills, 2001)—including those related to the concept of variable (e.g., Ursini & Trigueros, 1997)—particular interest has been fo-

cused on parameters and their role in algebraic expressions and equations. According to Bloedy-Vinner (1994, 2001), understanding algebraic language related to parameters means (a) understanding from the context which letters are used as parameters, (b) understanding the role of parameters as opposed to the role of unknowns or variables, and (c) understanding the fact that the meaning of a letter as parameter might change throughout the process of solving a problem. In understanding this difference in roles, implicit quantifiers are involved. From a questionnaire designed to test students' understanding of implicit quantifier structures, administered to 82 Israeli students who had taken matriculation exams in mathematics, she found that most of the questions yielded a very low success rate (ranging from 3% to 69%). Furinghetti and Paola (1994), in their study of 199 Italian students, ages 16 and 17, found that only 20 of these students could adequately describe differences among parameters, unknowns, and variables.

Generational Activity with Multiple Representations.

The research interest in multiple representations within functional perspectives is expressed in themes related to the understanding of functions and the interplay between symbolic and graphical representations, with special attention paid to the role of visualization.

Functions and Their Meaning. A function can be regarded as a set of ordered pairs, a correspondence, a graph, a dependent variable, a formula, an action, a process, or an object (Selden & Selden, 1992). When introduced to the formal ordered-pair definition of *function*, students have been found to rely on process-oriented interpretations (Sfard, 1991, 1992) or on their own intuitive concept images (Vinner, 1992; Vinner & Dreyfus, 1989). A significant number of studies have focused on efforts to support students in moving from a process-oriented to an object-oriented view of functions by reifying their process views (see, e.g., Dubinsky & Harel, 1992a). Student misconceptions that have been documented in this area include, for example, identifying functions with just one representation (Vinner, 1992), neglecting domain and range considerations (Sfard, 1992), and relying on the limited information provided by computer screen windows (Goldenberg, 1988). Building on some of the earlier research, more recent studies have explored, for example, the nature of covariational reasoning within the context of exponential functions (e.g., Confrey & Smith, 1994, 1995), the elaboration of a property-oriented view of function that is based on visual aspects of functional growth (Slavit, 1997), the notion of chronicle as a source of error (Janvier, 1998), and

the role of concept maps in assessing conceptual understanding of functions (e.g., Williams, 1998).

Technological environments have been widely used in research on students' conceptual understanding of functions, for example, ISETL (Dubinsky & Harel, 1992b), *DynaGraphs* (Goldenberg, Lewis, & O'Keefe, 1992), and *Visualizing Algebra* (Schwartz & Yerushalmy, 1992). Graphing and graphical representations have been a major theme of the research (e.g., Leinhardt et al., 1990; Romberg et al., 1993), with a particular focus on the establishment of connections between the symbolic and graphical representation of function, often with the use of graphing calculators. This latter research is discussed next.

The Interplay Between Symbolic and Graphical Representations and the Role of Visualization.

The role of visualization as a representational and interpretational tool has been an ongoing area of interest among algebra researchers (e.g., Arcavi, 2003; Dreyfus, 1991). Earlier work produced evidence that even advanced students demonstrate a reluctance to use visual representations (e.g., Eisenberg & Dreyfus, 1991), preferring letter-symbolic over graphical representations, even when the former are more complicated. More recently, drawing on findings by Schoenfeld, Smith, and Arcavi (1993) and Moschkovich, Schoenfeld, and Arcavi (1993), Knuth (2000) examined 9th- to 12th-grade students' understanding of the concept that the coordinates of any point on a line will satisfy the equation of the line, within the context of problems that require the use of this knowledge. Knuth found an overwhelming reliance on letter-symbolic representations, even on tasks for which a graphical representation seemed more appropriate. The findings indicated to Knuth that for familiar routine problems many students have mastered the connections between the letter-symbolic and graphical representations; however, such mastery appeared to be superficial at best. Similar results were found by Slavit (1998) in a study in which graphing calculators were available. According to Slavit, the students had difficulty in coming to view algebra from a multirepresentational perspective and thus tended to favor symbolic modes of work, using the graphing calculator only on graphing tasks. These findings contrast with those reported for younger students, where the use of symbolic modes of work lagged behind those of graphical representations as problem-solving tools. Visualization in advanced problem solving by both college students (the novices of the study) and mathematics professors (the experts) was the focus of a study reported by Stylianou and Silver (2004). The two groups judged visual representations likely to be useful with different sets of problems; furthermore, when it came to actual usage,

the experts constructed visual representations much more often than did the novices.

Dynamic, visual representations were a central feature of a study by Zbiek and Heid (2001), who investigated students' understanding of parameters. They have described an environment involving dynamic control of a slidergraph and provided an example involving a family of functions whose general member is represented by $f(x) = a/(1 + be^{cx}) + d$, where a , b , c , and d are real numbers. At the same time that the slidergraph-control moves along the screen, allowing a selected parameter to take on different values, the graphical representation responds to and reflects the various values that the functional expression takes on. Zbiek and Heid pointed out that the ease with which the slidergraph, unlike static graphing utilities, allows students to vary a single parameter quickly and consistently led to surprising observations and was, in fact, the key to launching student reasoning. This study is an example of one of a growing number related to the use of dynamic environments in algebra in which bodily motion plays a role in meaning building for the functional objects of algebra (see also Kieran & Yerushalmy, 2004).

Zaslavsky, Sela, and Leron (2002) found evidence of much confusion regarding the connection between algebraic and geometric aspects of slope, scale, and angle. Participants, who included 11th-grade students as well as teachers, mathematics educators, and mathematicians, responded to a simple but nonstandard task concerning the behavior of slope under a nonhomogeneous change of scale. Responses revealed two main approaches—analytic and visual—as well as combinations of the two. The researchers recommended that instruction on slope distinguish between the erroneous conception of *visual slope*—the slope of a line (for which the angle is a relevant feature)—and the *analytic slope*—the rate of change of a function. Similar approaches were noted in the Lobato et al. (2003) study involving younger students.

Other related studies have investigated students' interpretations of linear and quadratic graphs as displayed on the graphing screen. Although some studies (e.g., Cavanagh & Mitchelmore, 2000) have suggested that older students continue to experience some of the same difficulties that have been found with younger students, others (e.g., Pierce & Stacey, 2001a) have reported that the use of symbolic, graphical, and numerical representations helps to expand students' previous understandings by allowing them to link visual images to symbolic statements. In particular, Pierce and Stacey observed that college-level students operating in a CAS envi-

ronment experienced little difficulty in moving between graphical and symbolic representations and in fact seemed to prefer these two modes to the movement between tabular and symbolic representations. Furthermore, studies that have investigated the impact of long-term use of graphing calculators on students' conceptual understanding of function and its representations have reported positive results (e.g., Streun, Harskamp, & Suhre, 2000). The time spent in working with multiple representations has been found to be crucial to their learning.

Transformational Activity: The Upper Secondary and College Level Student

Research that has focused on the transformational activity of algebra with the upper secondary and college level student has investigated the following areas: notions of equivalence and meaning building for equivalence transformations, the solving of equations and inequalities, the factoring of expressions, and the integration of graphical and symbolic representations in transformational work. Before examining this research, a few theoretical remarks are offered to provide a context for some of the more recent work that has been carried out in this area.

Theoretical developments by French researchers have led to a rethinking of the nature of technical activity in mathematics. Combining elements from both anthropological theory (Chevallard, 1999) and cognitive ergonomics (Vérillon & Rabardel, 1995), Artigue (2002b) and Lagrange (2000) have elaborated the notion that, as students develop techniques in response to certain tasks, they engage in a process of theory building. In their theoretical approach, which they have applied to research involving Computer Algebra Systems (CAS), activity with technological tools is considered to promote both conceptual and technical growth in mathematics, as long as the technical aspects are not ignored. An increasing number of research studies on the transformational activity of algebra reflect this newly developed perspective.

Related to this perspective, especially with respect to the emergence of theoretical thinking within the context of CAS use, is the work done by Pierce and Stacey (2001b) on the development of algebraic insight. Algebraic insight, which is considered to consist of both algebraic expectation and the ability to link representations, involves a focus on basic properties, structure, key features, and links.

Transformational Activity Related to Notions of Equivalence. Among the research dealing with students' notions of equivalence and equality has been a study carried out by Sackur et al. (1997). These re-

searchers have described how 10th graders' thinking evolved with respect to these notions within a series of "make false"⁵ interviews. The researchers, who used Frege's sense-denotation distinction as their analytical framework, pointed out that,

When we say that a student must know what the expressions denote, we mean that the student must know the following: (a) that an expression like $y(2x + y)$ has a numerical value, (b) that this value depends on the values of x and y , and (c) that this value is not changed by the algebraic transformation rules to which an expression is subjected, for example, that which transforms $y(2x + y)$ into $2xy + y^2$. (p. 47)⁶

Their analysis led them to conclude that, unless students come to realize that algebra is an arena of sense-making and that they can arrive at rules that will permit them to obtain the same results as their teacher or classmates, they will never be able to control their algebraic work.

Research evidence exists that, without interventions of the type conducted by Sackur et al., students do not spontaneously develop an algebraic notion of equivalence, and that the obstacles associated with this notion cut across the age range. For example, Kieran (1984a) found that 8th graders showed little awareness that the equation-solving transformations they were able to execute proficiently preserved solutions. In a study using a larger sample of 8th- and 9th-grade students, and employing tasks similar to those used by Kieran (1984a), Steinberg et al. (1990) noted that although most students knew how to use transformations to solve simple linear equations, many did not spontaneously relate this knowledge to the production of equivalent expressions. Ball, Pierce, and Stacey (2003) developed an instrument to assess students' abilities to quickly recognize equivalent algebraic forms—the "Algebraic Expectation Quiz." On the basis of the performance of 50 students on this test, administered before and after students' progression from 11th to 12th grade and during which time they participated in algebra instruction involving the use of a computer algebra system, the researchers reported that "recognizing equivalence, even in simple cases, is a significant obstacle for students" (p. 15). Pomerantsev and

Korosteleva (2003) presented compelling evidence that difficulties related to understanding algebraic equivalence can extend well beyond the postsecondary level. Their study involved the use of a diagnostic test administered to a large ($N = 416$) sample of students enrolled in different stages of their K–8 teacher preparation program at a major American university. Test items were designed to assess students' abilities to discern and use structural aspects of algebraic expressions; results of this research revealed serious difficulties in doing so, cutting across students in all groups contained in the sample.

The growing increase in the use of CAS in secondary school algebra classes has been accompanied by research interest in the role that this technology might play in the development of students' conceptions of equivalence. Particular attention has been paid to the nature of the tasks that might provoke theoretical growth within the context of transformational activity. For example, Saldanha and Kieran (2005; also Kieran & Saldanha, 2005) reported on a study with 10th graders that involved a combination of CAS- and paper-and-pencil-based tasks on equivalence of expressions. Results suggested that the development of students' theoretical notions of equivalence was guided by the techniques that the tasks invited, and that the most productive learning occurred after the CAS techniques produced an output that conflicted with the students' expectations.

Transformational Activity Related to Equations and Inequalities. While the solving of linear equations has received a great deal of research attention, that of quadratic equations has not. One of the few studies was carried out by Vaiyavutjamai, Ellerton, and Clements (2005). Nearly 500 students from Year 9 classes in Thailand, Year 10 classes in Brunei Darussalam, and 2nd-year university students in the United States attempted to solve the same quadratic equations, all of the form $x^2 = K$ ($K > 0$) and $(x - a)(x - b) = 0$ (where a and b are any real numbers). All students had already learned to solve such equations before participating in the study. The responses to the second type of equation, in particular, suggested the presence of serious gaps in the theoretical thinking underpinning students' work when solving such equations. For example, in solving $(x - 3)(x - 5) = 0$, several students

⁵ A "make false" interview is one that is aimed at having students confront the limits of their own thinking and in which a student is asked to produce an equality that is always false. For example, the researchers presented pupils with tasks such as "Write a false equality for $7x/(7 + x) =$ " and then posed questions related to the responses provided by the pupils.

⁶ Similarly, according to Arzarello, Bazzini, and Chiappini (1994), "the 'denotation' of a symbolic expression in algebra refers to the number set that is represented by the expression; it is determined by the symbolic expression and by the universe in which the expression is considered (for example the equation $x^2 + 1 = 0$ denotes the empty set when it is considered in $\mathbf{R}$ and the set $\{+i, -i\}$ when considered in $\mathbf{C}$)" (p. 42).

who correctly solved the equation checked their solutions by substituting $x = 3$ into $(x - 3)$ and $x = 5$ into $(x - 5)$ and concluded that because $0 \times 0 = 0$ their solutions were correct. Related difficulties were encountered by the U.S. students who were asked to respond True or False to the following statement: *This equation $(x - 3)(x - 5) = 0$ is equivalent to $x^2 - 8x + 15 = 0$, which is a quadratic equation with two solutions. Thus, with $(x - 3)(x - 5) = 0$, the x in the first brackets always equals 3, and the x in the second brackets always equals 5.* Fifty-five percent of respondents answered that this was indeed True. To another question, many were unsure whether the x in the “ x^2 term” represented the same variable as the x in the “ x term” of the equation $x^2 - 8x + 15 = 0$. The finding that most of the students in this study were confused about how the solutions of a quadratic equation related to the equation itself led these researchers to suggest that, if quadratic equations are to remain an important component of middle- and upper-secondary mathematics curricula, then research is needed to guide teachers about how students think about quadratic equations.

Tsamir and Bazzini's (2002) study of 16- and 17-year-old Israeli and Italian students' solutions to algebraic inequalities found that students of both countries intuitively used the balance model to do the same thing to both sides of an inequality and continued to draw on equation-related analogies when dividing both sides by a not necessarily positive value. This reliance on the balance model can be related back to results described in an earlier section of the chapter where the introduction of this model with younger students was found by several researchers to be beneficial, even if some suggested caution. In related research with Israeli students, Tsamir, Almog, and Tirosh (1998) reported that the following methods were used by high school math majors for solving equations and inequalities: algebraic manipulations, drawing a graph, and using the number line. When graphical representations (without technology) were used to solve rational and quadratic inequalities, the solutions were usually correct; however, students often failed to reject the excluded values.

Goodson-Espy (1998) presented 13 college students with a set of eight word problems that involved linear inequalities. Even those participants who were known to be able to solve basic algebraic equations and inequalities when presented in the symbolic format solved the problems using arithmetic forms of reasoning. Similar observations were noted earlier in the research involving younger students who were found to make use of arithmetic forms of reasoning in solving word problems. In all but one case, the participants in the Goodson-Espy study who used algebra to

solve the tasks used equations rather than inequalities. This too echoes the findings reported with younger students who, when introduced to problems involving inequalities, chose to think about and solve for the corresponding equality situation and then make adjustments at the end to produce an answer for the inequality.

Transformational Activity Related to Factoring Expressions. With the emergence of CAS in secondary school mathematics classes, the transformational activity of factoring has been receiving research attention (e.g., Artigue, 2002a; Artigue, Defouad, Duperier, Juge, & Lagrange, 1998). For example, Lagrange (2003) has described the activity of classes of 16- to 18-year-olds (reported initially in Mounier & Aldon, 1996) attempting to find and prove general factorization of $x^n - 1$. The students who were in CAS-equipped classes tried to factor $x^n - 1$ for various values of n on their CAS; the machine produced as output complete factorizations of these expressions. This form of output tended to thwart the mathematical aim of the activity as designed by the teachers, which was to have the students arrive at an awareness of general regularities in the factored forms. A related study by Kieran and Drijvers (2006; see also Kieran & Saldanha, 2007) took advantage of the fact that the CAS produced complete factorizations of $x^n - 1$ in order to provoke a confrontation with the existing, but limited, theoretical thinking of the 10th-grade participants with respect to factoring. This confrontation was found to be very productive for most students, but especially so for those who, upon realizing that they could not generate the same factors as had the CAS, insisted on finding out how to do so either from the teacher or from their group-mates during the working part of the activity, or later during the plenary classroom discussion. These CAS-based encounters resulted in the evolution of not only students' paper-and-pencil factoring techniques but also their theoretical views of the structure of these expressions (e.g., seeing that $x^6 - 1$ could be viewed as a difference of squares, or as a difference of cubes, or as an example of the general rule that they had earlier generated, $x^6 - 1 = (x - 1)(x^5 + x^4 + \dots + x + 1)$).

Another study on factoring is that reported by Nguyen-Xuan, Nicaud, Bastide, and Sander (2002). Two classes of 10th-grade novices (but not beginners) learned factoring techniques in a computer-based tutoring environment for symbol manipulation that provided informative error messages. One class received instruction by means of observing 14 pairs of worked-out examples, each observation being followed by activity on a related example. The second class did not observe any worked-out examples; their activity

consisted only of factoring the 14 pairs that the other class had observed being factored. Both classes had from 4 to 6 hour-long sessions with the given activities. The researchers found that the second class learned more. This finding supported their hypothesis that, although observation may be an effective learning approach for rank beginners, it is less effective than learning by doing for students who are trying to become more expert. The researchers also noted that the time taken to move from novice to expert status was significant—a finding already reported in several other studies.

Transformational Activity Involving the Integration of Graphical and Symbolic Work. Ruthven (1990) conducted a study involving advanced-level mathematics students who had access to graphing calculators while working on double-angle trigonometric identity tasks. The most productive work started, according to the researcher, when students began to compare graphs of the different possibilities and used their manipulative skills to establish which of their different algebraic formulations were equivalent. This finding suggests an interesting approach for using the graphical to motivate the manipulation of the symbolic. In another study, Baker, Hemenway, and Trigueros (2001) noted that college students' work with transformations of functions was assisted by graphical representations when these activities were accompanied by written descriptions of what they were seeing on the screen of the graphing calculator.

The integration of graphical and symbolic representations has also been noted in the transformational work reported in studies on (a) finding the roots of quadratic functions (Mourão, 2002), (b) translating quadratic functions (Zazkis, Liljedahl, & Gadowsky, 2003), (c) working with cubic functions (Gómez & Carulla, 2001), (d) solving linear and quadratic equations and inequalities (Asp, Dowsey, & Stacey, 1993; Tsamir et al., 1998), (e) solving problems involving translations of quadratic graphs and parametric curves within a graphing-calculator environment (Drijvers & Doorman, 1996), and (f) the Newton-Raphson method for approximating the zeros of functions in a CAS environment (Hong & Thomas, 2002). Although the research involving CAS technology has focused on either the exclusive use of the letter-symbolic representation or the interplay between graphical and symbolic representations, some CAS research reports have included only graphing tasks, with no reference to the letter-symbolic (Zbiek, 2003; see also Thomas, Monaghan, & Pierce, 2004).

Traditionally, the study of transformations of functions has belonged to the domain of symbolic representation. However, in some technological environments (e.g., Schwartz & Yerushalmy, 1992, 1996), one graph can be transformed into another through direct actions on the graph. Borba (1993), who worked with 16-year-olds in the *Function Probe* environment developed by Confrey (1991), reported that the software environment influenced the mathematics that was studied and produced. He also argued that the nature of the tasks determined whether the students thought about transformations of functions in a static or dynamic way. Kieran and Yerushalmy (2004) have described other environments permitting dynamic transformation of graphical representations, accompanied by simultaneous change in the corresponding symbolic representation; however, they also noted that research on the impact of directly controlling a change in one representation to effect changes in another remains relatively undeveloped.

Global/Meta-Level Activity: The Upper Secondary and College Level Student

As with the middle grade and lower secondary level student, global/meta-level activity with the older student has included work in problem solving and in proving. However, in several of these studies, the two themes have been integrated and are therefore treated in the same section below. Some studies have also been carried out with the older, more mathematically advanced student in the global/meta-level activity of modeling; this research is briefly presented in the second section that follows.

Global/Meta-Level Activity Involving Problem Solving. Much of the research involving problem situations with the older student, as was the case with the younger student, has included technological tools. An example of such research is a study by Hershkowitz and Kieran (2001), who reported on classes of 16-year-olds exploring a situation involving three families of rectangles that increased in size according to a linear, a quadratic, or an exponential relation. In attempting to respond to the question as to which family of rectangles would eventually surpass the others in terms of area, students were found to go from entering lists to graphical representations by means of the regression tool of the graphing calculator without ever seeing or having to examine the algebraic representation of the situation. All of their problem-solving work involved attention to the data in the lists and the resulting graphs, which led the researchers to question whether the use of such tools within problem-solving situations

can, in fact, lead students away from the use of letter-symbolic representations.

Much of the research on the global/meta-level activity of problem solving with the older student has tended to integrate number-theoretic work and proving. For example, Arzarello et al. (1994) observed a link between expressing the relations of number-theoretic problems in suitable algebraic code and success in determining the solutions to such problems. According to the researchers, appropriate naming is linked to anticipatory aspects of solving and allows students to orient the solution process toward the aims of the problem. They also pointed out that some students who could express the elements of a problem by using natural language were unable to express them using the algebraic language. In contrast, Douek (1999) found that nonstandard representations of number-theoretic problems did not prevent students from producing valid proofs and problem solutions.

However, this may in fact depend on the methods used by students to generate their proofs. Alcock and Weber (2005) reported that students used both referential (involving some numerical instantiation) and syntactic approaches (involving symbolic manipulation) and that students who used referential approaches may have had a more meaningful understanding of their proofs but may not have been able to complete them. In another study with 10th graders who had been given the task of proving their conjecture that $x + 1$ is always a factor of $x^n - 1$ for even values of n , Kieran and Drijvers (2006) found that most students used in their proofs a combination of syntactic and referential approaches, and that the most successful of the proofs were generic in nature.

Global/Meta-Level Activity Involving Modeling. Dynamic physical models have been an integral part of modeling studies with the upper secondary and college level student. Schnepf and Nemirovsky (2001) reported on a study that involved graphing software and hardware that linked a computer to miniature cars on parallel linear tracks and to a miniature stationary bike. The software allowed the system to work in two ways: either by manipulating the cars (or bike) and then modeling this movement by means of graphical representation, or by constructing a graph on the computer that then moved the mechanical device according to the graphical specifications. The latter, which is the opposite of a modeling perspective, was found effective when combined with the former in the development of conceptions of movement, and its representation, among the 12th-grade calculus students who participated in the study. Related work

has been carried out by Rasmussen and Nemirovsky (2003) in an environment involving the physical tool, Water Wheel, with more advanced students.

The Impact of Technology on Algebraic Activity With the Upper Secondary and College Level Student

Just as with the younger algebra student, the presence of technology in the studies involving the upper secondary and college level student has been noteworthy. The main types of technology that have been employed in these studies with older students are the graphing calculator (without symbol-manipulation capabilities) and computer algebra systems, which also include graphing capabilities. Some of the impact of these tools has already been discussed in the individual research studies that have been presented. However, other technology-related studies that cut across the various types of algebraic activity have been reserved for this section. Some of these are large-scale studies, involving meta-analyses or comparison groups; others have focused on the ways in which roles in the algebra classroom change when technology is introduced.

Large-Scale Studies on the Effects of Technology Use.

The studies that are presented in this section are categorized as to whether they deal with calculators in general, graphing calculators, or computer algebra systems. The largest set of studies is that having to do with the impact of graphing calculator usage.

Calculators. The first study to be included herein is a meta-analysis that investigated the effects of calculators (basic, scientific, and graphing) on students' achievement and attitude levels by examining 54 studies, 26 of which dealt with the high school level (Ellington, 2003). It was found that, when calculators were included in instruction but not testing, the ability to select the appropriate problem-solving strategies improved for the participating students. Under these conditions, there were no changes in students' computational and conceptual skills. When calculators were part of *both* testing and instruction, the strategic skills, computational and conceptual skills, and problem-solving skills improved for participating students. Under these conditions, there were no changes in students' ability to select the appropriate problem-solving strategies. Students who used calculators while learning mathematics reported more positive attitudes toward mathematics than their non-calculator-using counterparts on surveys taken at the end of the calculator treatment. Ellington also reported that students received the most benefit when calculators had a pedagogical role in the classroom and were not available just for drill and practice or checking work. However,

she also noted that in only 11% of the studies analyzed had special curriculum materials been designed for calculator use and that this is thus an area in which more research needs to be conducted.

Graphing Calculators. The next two studies involved primarily the use of the graphing calculator within the computer-intensive algebra (CIA) curriculum (Fey et al., 1991). The first is a study by O'Callaghan (1998) that examined the effects of a one-semester computer-intensive algebra and traditional algebra curricula on college students' understanding of the function concept. He found that the CIA students achieved a better overall understanding of functions and were better at the components of modeling, interpreting, and translating. Further, the CIA students showed significant improvements in their attitudes toward mathematics, were less anxious about mathematics, and rated their classes as more interesting.

Hollar and Norwood (1999) tested four classes of intermediate algebra students (two experimental classes and two control classes), with two instructors each teaching one experimental and one control class in their study of the effects of a graphing-approach curriculum that extended O'Callaghan's (1998) CIA study. The control classes covered the same topics as the experimental classes but did not have access to graphing calculators. In the experimental classes, the calculators were used to explore, estimate, and discover graphically, and to approach problems from a multirepresentational perspective. It was found that students in the graphing-approach classes demonstrated significantly better understanding of functions on all four subcomponents of O'Callaghan's function test than did students in the traditional-approach classes. Additionally, no significant differences were found between the graphing-approach and traditional classes either on a final examination of traditional algebra skills or on an assessment of mathematics attitude.

Further evidence of the benefits of graphing-calculator use was obtained from a study that examined the effects of this technology upon students' understanding in elementary college algebra (Shoaf-Grubbs, 1993, as cited in Penglase & Arnold, 1996). The study involved 37 females from an all-women's liberal arts college, divided into an experimental class and a control class, with the only difference between the two groups being the use of the graphing calculator. Statistical analysis of pretest and posttest results showed that the majority of the students in the experimental group achieved significantly better than the control group in the test results, especially in their understanding of graphing concepts. However, Hall (1993), who compared four treatment classes with four control classes on the use of graphing calculators

in a 3-week study of trigonometric functions, found that the graphing calculators had no significant impact on students' achievement.

Another large-scale study involving graphing-calculator use was conducted by Streun et al. (2000) in the Netherlands. These researchers used a three-condition pretest-posttest design to analyze the impact of prolonged use of the graphing calculator throughout the entire school year for all topics of the curriculum (i.e., functions and graphs, change, and exponential and periodic functions). Three experimental classes used the graphing calculator throughout the year; a second set of five experimental classes used the graphing calculator with only one topic for 6 weeks; and four classes, which served as the control group, covered the same subject matter throughout the year but without the graphing calculator. The students who used the calculator throughout the year had enriched solution repertoires and a better understanding of functions. The students who used the graphing calculator for only a short period of time did no better on the posttest than the students in the control group. They merely replaced their symbolic approaches and guess-and-test procedures with graphing methods. Unlike the students who spent more time using the graphing calculator, they had not significantly altered their conceptual understanding of functions. The time factor, which was pivotal in this study, has also been found to be of crucial importance in other technology-related research findings (e.g., Chacon & Soto-Johnson, 1998; Lagrange, Artigue, Laborde, & Trouche, 2001).

Further research involving graphing calculator use is the comparative study of the effects of a 3-year integrated mathematics curriculum for Grades 9–11, the *Core-Plus Mathematics Project*, with more conventional curricula on growth of student understanding, skill, and problem-solving ability in algebra (Huntley et al., 2000). Results indicated that the Core-Plus curriculum was more effective than conventional curricula in developing student ability to solve algebraic problems when those problems were presented in realistic contexts and when students were allowed to use graphing calculators. Conventional curricula were more effective than the Core-Plus curriculum in developing student skills in manipulation of symbolic expressions in algebra when those expressions were presented free of application context and when students were not allowed to use graphing calculators.

Computer Algebra Systems (CAS). The number of large-scale studies involving the use of CAS is much fewer than for the graphing calculator. In one study, Shaw, Jean, and Peck (1997) evaluated the effectiveness of using CAS in an intermediate algebra course by comparing the grade distributions in a follow-up

course of three groups: students who went through the CAS technology-based developmental course ($N=34$), students who went through a traditional developmental course ($N=109$), and students who were not required to take a developmental course ($N=1335$). Students in the developmental program that had the technology-based course did at least as well as the students who did not need to take any developmental course. This was considered a real improvement inasmuch as the students who took the traditional intermediate algebra course did not do as well in subsequent courses as students who took the technology-based course.

Technology and the Changing Roles of Students and Teachers.

A few studies have investigated the work habits of students using technology. For example, Guin and Trouche (1999) categorized students' use of graphic and symbolic calculators into five profiles of behavior: random, mechanical, rational, resourceful, and theoretical. Doerr and Zangor (2000) classified five modes of graphing calculator use: as a computational, transformational, visualizing, verification, and data collection and analysis tool. These studies suggest the potential of technology for broadening the range of mathematical activity engaged in by students. Other studies have noted the changing roles of both students and teachers when technology is integrated into the mathematics classroom. For instance, Farrell (1996) observed that both teachers and students moved toward the roles of consultant and fellow investigator. Goos, Galbraith, Renshaw, and Geiger (2003) analyzed the impact of an assortment of technological tools on classroom interactions in terms of offering opportunities for students to engage constructively and critically with mathematical ideas, that is, as master, servant, partner, and extension of self. The 3-year Goos et al. study showed how technology can facilitate collaborative inquiry during both small-group interactions and whole-class discussions where students use the computer or calculator and screen projection to share and test their mathematical understanding. This finding is in contrast to that reported by Doerr and Zangor (2000), who observed that the graphing calculator as a private device led to a reduction in small-group interactions. However, Goos et al. (2003) have argued that, when calculators and computers are used as partners or extensions of selves and are permitted to become a part of face-to-face discussions, they facilitate communication and sharing of knowledge.

Similarly, Manouchehri (2004) found that the overall complexity of the mathematical discourse increased from before the technology was introduced

to afterward in the senior-level mathematics course that she observed. In particular, she noted that three factors appeared to control the construction of fruitful algebraic classroom discourse: the teacher, the tasks, and the technology. She pointed out that a common tendency in mathematics instruction is to focus on the power of the technology as a presentation device rather than as a discourse participant. In her study, even though the technology served as a powerful information provider, it only became a learning medium when supported by appropriate teacher intervention and tasks. This latter observation has been made time and again by several researchers.

Concluding Remarks on the Algebra Research Involving the Upper Secondary and College Level Student

Although the amount of research conducted with the upper secondary and college level student has been less abundant than that carried out with the middle grade and lower secondary level student, it has produced results that touch upon issues that are both similar to and different from those that emerged from the body of research involving the younger student.

The sources of meaning that have been the object of study in the research with the older student are principally those derived from within mathematics—focusing either on the letter-symbolic or on multiple representations. As was seen, the research involving the letter-symbolic form included studies on structure and on equivalence. With symbol-manipulating technologies being introduced in algebra classes, students' conceptions of equivalence of expressions were a theme of interest. The research involving multirepresentational sources of meaning in algebra, a comparatively larger body of work in relation to that with a letter-symbolic focus, included a broad range of studies. Several of the studies that examined students' notions and usage of symbolic and graphical representations noted students' preference for the symbolic over the graphical. This was in contrast to results from studies involving younger students, which noted their difficulties with using symbolic approaches. Studies dealing with the role of the graphing calculator reported, in general, that sustained use of this technology can provide effective support for multirepresentational thinking among algebra learners. However, the importance of according sufficient time for the conceptual advantages of the technological environment to take hold was noted.

As time has emerged as a key factor in technology-related studies, two other factors have been found to be equally important. One concerns the nature of the

tasks in which students engage; the other is the role played by the teacher in orchestrating the development of algebraic thought by means of appropriate classroom discussion. As is borne out by the results of several studies, the technology by itself is clearly insufficient.

The use of CAS technology in the learning of algebra was found to be more widespread in the research with this age group than with the younger student. Although some of the research with this technology makes use of multirepresentational approaches to algebra learning, a significant amount of this work focuses on the letter-symbolic mode. In line with advances by French researchers in the development of theoretical frameworks for studying CAS use, many of the studies reflect these frameworks by their attention to the emergence of theoretical thought within technical activity. Results from this newly developing sphere of research activity provide evidence that meaning building does indeed occur not only within the generational but also within the transformational activity of algebra. However, a great deal more research is needed in this area.

A related issue that arises with the use of technological environments in general, and with CAS in particular, is the question of the role of paper-and-pencil work in the development of understanding of particular concepts. For example, Asp et al. (1993) claimed that physically constructing tables of values for functions was essential to the development of students' understanding of the relationship between graphs and equations. Warren and Pierce (2004), on the basis of their review of Australasian research in algebra learning and teaching during the years 2000 to 2003, have further suggested that for successful CAS use students need to be able to do the following by hand: solve simple equations, use basic algebraic manipulations to rewrite equations in a form they can use, and be aware of standard techniques used to find solutions. However, they admit that further research is needed with respect to the question of the "by hand" skills that students require when working with CAS, as well as the issue of the place of by-hand work in the learning process (see also Forster, Flynn, Frid, & Sparrow, 2004).

The question regarding the by-hand skills needed by students embarking on CAS activity could usefully be researched by means of studies that carefully document the prior experience and incoming knowledge of students (i.e., their current by-hand skills) and then relate these to the theoretical and technical knowledge that develops (as well as to the obstacles encountered) as a result of the technological experience. Such well-documented research might indeed show that the prior by-hand skills needed by students

for CAS activity vary according to several yet-to-be-determined dimensions.

The second issue regarding the place of by-hand work in the learning process has been emphasized in several studies. For instance, recent research (e.g., Drijvers, 2003; Kieran & Drijvers, 2006) has suggested that it is in the interaction of paper-and-pencil work and CAS activity, particularly in tasks involving confrontation between students' existing theoretical thinking and CAS output, that students develop both technically and theoretically. However, research has just begun to explore this area in, for example, studies related to equivalence of expressions and factoring. Further research focusing on the interaction of paper-and-pencil work and CAS activity, on a variety of topics, is sorely needed.

Before closing this section, a few remarks on needed research that does not necessarily involve technology are offered. With the current curricular interest in problem situations as a means of motivating and inducting a large majority of students into the study of algebra, along with the use of multiple representations for exploring the domain of problem solutions, several issues arise. For example, the obstacles faced by younger secondary students in using letter-symbolic representations as problem-solving tools versus the oft-found reliance of older secondary students on this same mode of representation leads to several questions: What is the nature of the transition between nonusage and usage of algebraic symbols in problem solving? In particular, what are the factors at play when students shift from using non-letter-symbolic (numerical or graphical) to letter-symbolic representations to solve problems? How do students learn to make connections between graphical images and corresponding symbolic representations? More specifically, what is the nature of the connections that students make between graphical and symbolic representations? The mathematical content involved in the study of these questions also needs to move beyond linear situations to encompass, for example, quadratic and trigonometric situations, as well as systems of equations. Research is needed with the upper secondary and college level student on a much wider variety of algebraic content than has up to now been studied.

Algebra Teaching and the Algebra Teacher

In the algebra chapter of the 1992 *Handbook*, Kieran (1992) remarked

Even though the research community knows very little about how algebra teachers teach algebra and what

their conceptions are of their own students' learning, this is not to suggest that there has not been considerable research on new approaches to the teaching of algebra—there has. However, the analysis of results has usually been directed toward detailing learning phenomena. (Kieran, 1992, p. 395)

There are signs that the situation described in the early 1990s has changed in some ways. There has been a noticeable increase in the number of studies related to the teaching or teacher of algebra over the past 15 years or so. Some studies have dealt with the practicing teacher within the algebra classroom; others have been conducted within a context of professional development or inservice training; and yet others have involved the preservice teacher. However, researchers still know relatively little about algebra teaching. As well, there remains a noticeable disconnect between the research on the learning of school algebra and the research on the teaching of school algebra. This latter issue is elaborated throughout the sections that follow.

Research Involving the Practicing Algebra Teacher

One of the first differences one notices in moving from the research focused on the algebra learner to the research focused on the algebra teacher is that the theories used to frame the research tend to be quite different. Whereas studies of algebra learning have included theoretical frameworks based on, for example, constructivism, socioculturalism, semiotic mediation, discourse, embodied cognition, didactical situations, and instrumentation (see also Lerman et al., 2002), the most widely used perspective to frame studies involving the algebra teacher is based on Shulman's (1986) construct of teacher knowledge, which includes content knowledge, general pedagogical knowledge, curriculum knowledge, and pedagogical content knowledge. Shulman identified *pedagogical content knowledge* as the specialized knowledge that teachers need and described it as

the blending of content and pedagogy into an understanding of how particular topics, problems, or issues are organized, represented, and adapted to the diverse interests and abilities of learners, and presented for inspection . . . the category [of knowledge] most likely to distinguish the understanding of the content specialist from that of the pedagogue. (Shulman, 1987, p. 8)

Ball and Bass (2002) have pointed out that this notion of pedagogical content knowledge suggests that

even expert personal knowledge of mathematics often may be inadequate for teaching; knowing mathematics for teaching requires a transcendence of the tacit understanding that characterizes much personal knowledge . . . it also requires a unique understanding that intertwines aspects of teaching and learning with content. (p. 4)

This perspective has led Ball and Bass to focus on what it is that teachers of mathematics *do*, rather than on what teachers need to *know*, and to begin to develop a practice-based theory of mathematical knowledge for teaching. This distinction between what it is that algebra teachers know or need to know versus what they do may be helpful in examining the research that has focused on the teacher and the teaching of algebra.

Since the early 1990s, mathematics education researchers, wishing to characterize the components of Shulman's pedagogical content knowledge for various domains of mathematics, have worked at fleshing out this construct. For the domain of school algebra, the specification of pedagogical content knowledge has involved efforts to explore algebra teachers' knowledge and beliefs, as well as analyses of the practice of algebra teaching. Teachers' preexisting views, expectations, and beliefs are generally considered to act as a filter for both their pedagogical content knowledge and teaching practice, but little research exists on the influence of prior conceptions in the learning processes of algebra teachers. Thus, the research that has involved the practicing teacher of algebra has been concentrated in two areas that are treated sequentially below: the first focusing on knowledge and second on teaching, with the research on beliefs interwoven into both sections.

Algebra Teachers' Knowledge and Beliefs. Regarding the domain of school algebra, and functions in particular, Even (1990) has suggested that pedagogical content knowledge needs to include at least the following: different representations, alternative ways of approaching the concept, strength of conceptual knowledge, a basic repertoire of examples, knowledge and understanding of the concept, and knowledge of mathematics. However, as research has recently disclosed, and which will soon be seen, knowledge of students' ways of thinking about particular aspects of school algebra and functions is an equally important component of pedagogical content knowledge. Even (1993) has applied her framework to research involving prospective secondary level teachers and has found their understandings of the concept of function to be weak and fragile.

More recently, a related framework—the Multidimensional Grid for Professional Competence in elementary Algebra (MGPCA)—has been developed by French researchers (Artigue, Assude, Grugeon, & Lenfant, 2001, as cited in Doerr, 2004) with dimensions similar to Shulman's, but they have been elaborated specifically for the teaching of algebra: epistemological, cognitive, and didactic. The *epistemological* dimension includes knowing the following: the content of algebra, the structure of algebra, the role and place of algebra within mathematics, the nature of valuable algebra tasks for learners, and the connections between algebra and other areas of mathematics and to physical phenomena. The *cognitive* dimension includes knowing the following: the development of students' algebraic thinking, students' interpretations of algebraic concepts and notation, students' misconceptions and difficulties in algebra, different approaches taken by learners, ways to motivate learners, and theories of learning. The *didactic* dimension includes knowing the following: the curriculum, the resources, different instructional representations, different practices and approaches taken by other teachers, the connections across the grade levels, and the nature and development of effective classroom discourse. The MGPCA grid has been used as a tool for collecting data and analyzing preservice teachers' initial relationship with algebra, and its evolution, throughout their 1st year of combined teacher training and classroom teaching experience in the field. In general, Artigue et al. have noted that, with respect to the cognitive dimension, teachers begin to become sensitive to students' learning difficulties during their 1st year of fieldwork, but that the length of the training period is too short for the trainees to develop the means for analyzing these difficulties or for generating remedial strategies.

One of the current themes of the existing research inquiring into algebra teachers' pedagogical content knowledge concerns teachers' knowledge of students' algebraic thinking. Recent research suggests discrepancies between teachers' predictions of students' difficulties and students' actual difficulties. Sixty-seven high school mathematics teachers, as well as 35 mathematics education researchers, participated in a study conducted by Nathan and Koedinger (2000a) on teachers' and researchers' beliefs about the development of algebraic reasoning. This group was asked to rank order 12 mathematics problems from easiest to most difficult. Four were in story-problem format, four were in equation format, and four were in word-equation format. A comparison of their predictions with the actual difficulty as experienced by high school students, which was discussed in an

earlier section on algebra learning research, shows some important differences. The most salient discrepancy was that teachers and researchers predicted that story-problems and word-equation problems would be more difficult than symbol-equation problems, whereas students found symbolically presented problems most difficult. These differences between students' performance and teachers' predictions could have a significant effect on how teachers perceive students' reasoning and learning in the classroom and in how they teach to these perceptions. In a follow-up study involving 105 K–12 teachers and using similar tasks, Nathan and Koedinger (2000b) found that the middle school teachers came closest to predicting actual student performance. Furthermore, Nathan and Petrosino (2003) have provided evidence to support the argument that it was the well-developed subject-matter knowledge of the high school teachers (i.e., the “expert blind spot” syndrome), which underpinned their view of the equation-before-word-problem sequence, that led them to inaccurately predict student problem-solving difficulty.

Another study (Hadjidemetriou & Williams, 2002) addressing the issue of teachers' knowledge of what students find easy/difficult in algebra was one in which teachers' judgments of the difficulty of items dealing with graphical conceptions, and their awareness of errors and misconceptions, were analyzed and contrasted with learners' difficulty hierarchy. Teachers' judgment of what is difficult was found to be structured by the curriculum and by their mathematical knowledge. Also noted was a gap between pupils' difficulties and teachers' perception of these difficulties: Teachers underestimated technical difficulties of graphing and overestimated the difficulty of interpretive work. Similarly, in a study of 8 upper secondary school teachers in Sweden, Bergqvist (2005) found that the teachers tended to underestimate their students' reasoning levels on conjecture tasks and that the teachers erroneously believed that only a small group of students in each class could use higher level reasoning in mathematics.

Research has also begun to address the question of teachers' knowledge of mathematical links between lessons. Even, Tirosh, and Robinson (1993) compared the connectedness of lessons for two novice secondary teachers and an expert teacher on the topic of equivalence of algebraic expressions. They found that only the expert teacher used connections in mathematical content to guide the unfolding of her lessons. In a later study on teachers' knowledge of students' misconceptions, and the teaching approaches that would address these misconceptions, Tirosh, Even, and Robinson (1998) investigated four 7th-grade teachers'

awareness of students' tendency to "finish" algebraic expressions (e.g., to write the expression $2x + 3$ as 5x or 5) and their ability to respond on-the-spot to pupils who showed this tendency in class. Although two of the teachers were aware of this tendency, the means they used to address it in class were only partially successful. The authors argued that,

while one can use the mathematics education literature in order to raise teachers' sensitivity to students' ways of thinking, . . . this literature does not offer enough information and discussion regarding the impact of various approaches and teaching methods related to this conception; . . . it is important for teachers to be acquainted with various teaching methods and to be aware of their pros and cons in different contexts with different teaching aims, and with different students. (pp. 62–63)

Although the interaction between algebra teachers' knowledge and their beliefs is a complex and understudied area, it is widely held that both knowledge and beliefs shape teachers' teaching practice and that teachers' professed beliefs do not always match their instructional practices (Cooney, 1985). It has also been shown that limitations in both content knowledge and pedagogical content knowledge may influence teachers' beliefs and their instructional decisions and actions (Nathan & Koedinger, 2000b). Furthermore, findings by Nathan and Petrosino (2003) have suggested that strong content knowledge in algebra teachers produces beliefs that, in the absence of well-developed pedagogical content knowledge, lead to pedagogical decisions based on the structure of the mathematical domain rather than on the actual ways in which students think. An additional factor that has been found to influence algebra teachers' beliefs is the organization of mathematical material in textbooks (Nathan & Koedinger, 2000b). Cultural factors and related curricular demands also play a role, according to a study by Cai (2004), who found that U.S. and Chinese teachers viewed students' responses involving concrete strategies and visual representations differently. Although both U.S. and Chinese teachers valued responses involving more generalized strategies and symbolic representations equally highly, Chinese teachers expected sixth graders to use the generalized strategies to solve problems whereas U.S. teachers did not.

The existing research on algebra teachers' knowledge and beliefs has, up to now, barely begun to explore the various dimensions of knowledge that teachers need for teaching algebra. If there is one dimension that seems quite pressing, it would be the cognitive dimension of the MGPCA model, that is, the dimension

that includes knowing the following: the development of students' algebraic thinking, students' interpretations of algebraic concepts and notation, students' misconceptions and difficulties in algebra, different approaches taken by learners, ways to motivate learners, and theories of learning. These areas of pedagogical content knowledge are considered crucial to the expertise of the practicing teacher of algebra (Nagasaki & Becker, 1993) yet remain largely unresearched. Furthermore, these areas are exactly those where research on the algebra learner has produced abundant findings. Research on the algebra teacher has yet to tap into this resource and make connections between these two bodies of research.

The Practice of Algebra Teaching and Influences on That Practice. Although the above studies on teachers' pedagogical content knowledge have tended to use Shulman's framework, or derivatives of it, the emerging body of research on the practice of algebra teaching reflects a broader set of theoretical perspectives. For example, Boaler (2003) has drawn from Pickering (1995) to analyze the *dance of agency* occurring in reform algebra classes. Within this framework, she described the ways in which a teacher encouraged the interplay between human and disciplinary agency, that is, taught in such a way that students were able to interweave their own thoughts and ideas with standard methods and procedures of the discipline of mathematics. Atweh, Bleicher, and Cooper (1998) have employed a sociosemiotic perspective developed by Halliday and Hasan (1989) to investigate the social context of two 9th-grade mathematics classrooms that differed in the socioeconomic backgrounds and genders of their students. The researchers found that the students in the observed classes were perceived differently by their teachers and that each teacher conducted his classes in a manner that was consistent with his perceptions of the students' needs and abilities. Coulange (2001), a French researcher, has combined the anthropological approach of Chevallard (1999) and the theory of didactical situations of Brousseau (1997) in her study of the ways in which a 9th-grade teacher supported the approaches of students so as to ensure bringing the lesson to closure and achieving her teaching objectives.

Other frameworks that have been adopted in studies of algebra teaching include, for example, Davis's (1996) theory on evaluative, interpretive, and transformative listening, which was used by Coles (2001) to analyze the change that occurred in a given teacher's class as the listening of both students and teachers became transformative. Additional approaches to analyzing teacher change in the classroom have ranged

from using (a) perspective taking in middle-school mathematical modeling (English & Doerr, 2003), (b) students' ways of thinking as a motivation for recasting the tasks of teaching about functions (Doerr, 2003), and (c) the occasioning of the growth of teachers' mathematical understandings by focusing on students' mathematics within the classroom collective (Simmt, Davis, Gordon, & Towers, 2003). Another example of a framework for analyzing algebra teaching is presented in a study by Solomon and Nemirovsky (1999). They built upon "Chazan and Ball's (1995) idea that 'any discussion holds the potential for discrepant viewpoints' and that it is the teacher's role to 'manage' these views" (Solomon & Nemirovsky, 1999, p. 217) in their development of a tool for describing the complexity involved in managing the wide range of ideas and choices that emerge from the interactions among students and the teacher.

In some studies on the practice of algebra teaching, the focus has been on the beliefs of the algebra teacher and the role that these beliefs play in teaching. For example, Doerr and Zangor (2000) reported the results of a classroom-based study on the interaction between the role and beliefs of the teacher and the patterns of students' use of graphing calculators in support of their learning of precalculus mathematics. This interaction led to the creation and development of a set of ways the tool was used in the classroom and of related mathematical norms. Doerr and Zangor found that the teacher's confidence, flexibility of use, and awareness of the limitations of the technology led to the establishment of a norm that required results to be justified on mathematical grounds and to the devaluing of regression equations or appeals to the calculator as an authority in a mathematical argument. Simmt (1997), in a study of six 11th- and 12th-grade teachers' use of graphing calculators in the classroom, and the ways in which this use reflected their views of mathematics, found that the graphing calculators were primarily used as a device to provide graphical images from which the students were expected to observe and make generalizations about transformations of the quadratic function. The teachers' philosophies were manifested not so much in their choice of activities but in how they followed up those activities with questions and summary notes.

Another factor influencing teaching practice is the teacher's vision of the role of teacher talk in the algebra classroom. From her analysis of traditional and reform-oriented classroom teaching, Boaler (2003) reported that "the teachers in the traditional classes *gave* students a lot of information, while the teachers of the reform classes chose to *draw* information out of students, by presenting problems and asking students

questions" (p. 4). A particularly effective teacher, rather than asking students to present their finished problems, chose to ask students to present their ideas *before* anyone in the class had finished working on the problem, so that students could actively and collaboratively build on each others' ideas. Boaler argued that researchers need to acquire such detailed conceptions of effective teaching if they are to understand and impact practice.

A further example providing detailed descriptions of effective teaching is drawn from the TIMSS-R 1999 video study of eighth-grade algebra lessons around the world (Hiebert et al., 2003). The particular lesson involved a Japanese class (Stigler et al., 2003) where the teacher used a specific problem situation, accompanied by concrete materials to represent it, to help students create meaning for inequalities and for their algebraic form. While the students were working at finding a solution to the problem, the teacher circulated around the class, taking note of the various solution methods being worked at. Afterward, he asked certain students to come to the blackboard to present their solutions in increasing order of mathematical sophistication. The teacher orchestrated the placement of the different student approaches at the board not only so that all the solutions would remain on the board for the class to see and discuss, but also so that he could build upon one method in particular to move ahead with his lesson. The way in which the blackboard was sequentially organized with different representational approaches for the solving of the problem was considered by the teacher to be an important learning tool for the students of his algebra class.

The eclectic nature of the theoretical frameworks of the above studies of algebra teaching reflects the diversity of an emerging field of study where the focus is on what algebra teachers do, rather than on what they know. This diversity can, however, present a challenge in a field where basic knowledge of what constitutes effective algebra teaching is still lacking. The community of researchers of algebra teaching needs to develop a set of appropriate theoretical frameworks for the systematic interpretation of teachers' practice of algebra teaching and their development as learners and teachers of algebra. Although some of the above research has focused on acquiring detailed conceptions of effective teaching practice, the question as to how teachers become effective in their teaching of algebra remains a crucial one. The potential contributing role of professional development and preservice training in this evolutionary process is thus the subject of the next section on research involving algebra teachers in programs of professional or inservice development, as well as in preservice training.

Research Involving the Algebra Teacher Within a Context of Professional Development or Inservice Training

The research that has been conducted within the context of the professional development and inservice training of the algebra teacher has tended to focus on two areas in particular, the integration of new curricula, teaching approaches, and technological environments into the algebra classroom, as well as the nature of the algebra content knowledge and beliefs of teachers involved in these programs.

Integrating New Curricula, Approaches, and Technological Environments Into the Algebra Classroom.

The research that has focused on algebra teachers' integration of new curricula into the classroom has highlighted the multiple dimensions involved in studying such phenomena. For instance, Bartolini Bussi and Bazzini (2003) have reported on projects in Italy that include teacher-researchers cooperating with university mathematics educators and with other experts. These projects have typically involved (a) a deep analysis of the nature of the mathematics that is the starting point of the inquiry, (b) the strong involvement of teachers working in their own classrooms, and (c) the observation of both individual and collective learning processes. This study illustrates how, in Italy, teaching and learning have been studied in conjunction one with the other. A related example is the Italian ArAl project (Malara, 2003), an inservice teacher education project that was devoted to the renewal of the teaching of arithmetic and algebra in the sixth to eighth grades. The research explicitly included the classroom teacher as a researcher and emphasized the development of pedagogical content knowledge, the teacher's role in the teaching of mathematics, the impact of his/her personality, as well as social issues within the class group. Chapman (2001) has also pointed to the role played by mathematical belief structures in long-term teacher change involving implementation of new approaches.

Although the practice of situating teachers' professional development within the context of their own classrooms has just been considered above, few studies have also included preservice teachers as additional participants in the learning process. English (2003) has reported on such research, which was based on the four-tiered collaborative model of Lesh and Kelly (2000) and where middle-school teachers and preservice trainees gained powerful insights into students' mathematical development by observing the students as they engaged in modeling problems. Regular meetings between researchers and teachers, which took place both before and after the implementation of

each set of modeling activities, were thought to be essential to the success of the collaborative research.

A study (Hallagan, 2004) implementing the reform-based *Connected Mathematics Project* focused on the development of new awarenesses among eighth-grade teachers as a result of their involvement in the project. More specifically, the use of "Ways of Thinking" sheets helped teachers recognize the multiple ways in which their students interpret given tasks, as well as the role of visual images in facilitating students' work on equivalent-expressions tasks. Within another reform-based project, the *Core-Plus Mathematics Project*, involving 40 teachers and their 1466 students in 2 schools, Schoen, Cebulla, Finn, and Fi (2003) analyzed teacher variables related to student achievement. Generally, teaching behaviors that were consistent with the Standards' recommendations and that reflected high mathematical expectations were positively related to growth in student achievement. In another study related to the use of *Core-Plus* curricular materials, Lloyd and Wilson (1998) observed and interviewed an experienced high school teacher during his first implementation of a 6-week unit on functions. The researchers found that the teacher's well-articulated ideas about features of a variety of relationships in different representations supported meaningful discussions with students. The teacher communicated deep and integrated conceptions of functions, dominated by graphical representations and covariation notions.

The increasing implementation of multirepresentational and CAS technology, and curricula involving related new approaches to school algebra, has been reflected in research addressing teacher adaptation and teacher change. Some studies have, for example, examined teachers' attempts at implementing reform-oriented curricula, and their beliefs and attitudes towards different types of curricula (e.g., Heid, Blume, Zbiek, & Edwards, 1999; Stacey, Kendal, & Pierce, 2002). Some teachers have been found to experience tension between familiar methods and the use of technology, or between the curriculum and their personal educational conceptions (e.g., Chazan, 1999; Haimes, 1996; Lloyd, 1999; Slavit, 1996). Others have been found to need support in thinking about how to use symbolic calculators effectively in class and how to reorganize their use of class time when students are involved in CAS work (Guin & Trouche, 2000). In a study of three teachers' implementation of CAS in the classroom, Kendal and Stacey (2001) found that the teachers made very different uses of the CAS even though they had all been involved in planning an agreed common program. The teachers privileged technology use in different ways that were

consistent with their beliefs about algebra and about the teaching and learning of algebra. On the other hand, in a 4-year technology-implementation study in Mexico (Rojano, 2003) involving 15 schools of seventh, eighth, and ninth graders, Cedillo and Kieran (2003) observed that, when algebra teachers noticed that the introduction of calculator technology led to improved student learning, as well as a more positive attitude toward mathematics among both stronger and weaker students, they altered their teaching styles to more learner-centered approaches.

The studies in this section, which have focused on integrating new curricula, approaches, and technological environments into the algebra classroom, have presented a diversity of findings that range from descriptions of multilevel support programs that facilitate teacher change to the difficulties experienced by teachers in attempting to implement new curricula and technologies. Several studies have suggested the ways in which teachers' beliefs both shape and are shaped by the implementation process. Although each study has provided its own insights, further research is needed on the long-term impact on algebra teaching practice of the various implementation projects. Few studies have extended beyond the initial implementation to explore the evolution in teachers' practice and its relation to the professional development program that triggered it.

Algebra Content Knowledge and Beliefs. Although most of the professional development studies of implementation described above included some initial indicators regarding teacher change, other research that has involved the professional development or inservice training of algebra teachers has rather focused on algebra teachers' content knowledge. The areas that have been investigated include knowledge of conic sections (Hitt, 1998); the algebraic and geometric aspects of slope, scale, and angle (Zaslavsky et al., 2002); symbolic and graphical representations of rational functions (Arcavi, 2002); and number-theoretic ideas (Barkai, Tsamir, Tirosh, & Dreyfus, 2002; Zazkis, 1998). Zehavi (2004) has reported that the way in which the problems were presented and the prevailing atmosphere of the inservice group influenced the resulting mathematical behaviors and problem-solving products. Another dimension that has been studied concerns teachers' beliefs. Menzel and Clarke (1998) observed that the inservice teachers involved in their study were divided in their views as to whether the generalizable verbal description of a mathematical relationship should be regarded as algebraic—a finding that relates to Nathan and Koedinger's (2000b)

results on teachers' beliefs as to what constitutes algebraic activity.

An additional area of inservice teachers' content knowledge that has been researched is that of proof and proving. Knuth (2002) carried out a study with inservice secondary school mathematics teachers (grades 9–12) who were participating in ongoing professional development programs. By means of semistructured interviews with participants in order to examine their conceptions of proof, Knuth found that, although teachers recognized the variety of roles that proof plays in mathematics, the view of proof as a tool for learning mathematics was noticeably absent. With the increasing interest in proof and proving as a potential global/meta-level activity for motivating the work of students with algebraic symbols, the findings of Knuth's study suggest that further teacher development is needed in this area.

The above brief examination of the research involving the algebra teacher within professional development and inservice programs has disclosed that little attention has been paid thus far to the study and development of algebra teachers' pedagogical content knowledge. No studies were found that were similar to the Cognitively Guided Instruction program for elementary school teachers of mathematics (Carpenter, Fennema, Peterson, Chiang, & Loef, 1989). This professional development program with its focus on researched-based evidence regarding children's actual thinking in mathematics has been shown to be effective in changing elementary school teachers' instructional practices. It is suggested that similar programs that focus on the research-based thinking of algebra students be developed. Ironically, for many teachers the need for such knowledge may not be obvious. In a study of 9 high school teachers and their perspectives on what constitutes good mathematics teaching and how it develops, Wilson, Cooney, and Stinson (2005) found that the teachers thought that good teaching requires a sound knowledge of mathematics, promotes mathematical understanding, engages and motivates students, and requires effective management skills. Experience was considered the primary contributor. Nevertheless, many of the studies cited in this chapter and elsewhere (e.g., Krainer, 2005) maintain that knowledge of students' mathematical thinking, in addition to knowledge of mathematics, are necessary preconditions for good teaching.

Research Involving the Preservice Teacher of Algebra

Research involving the future teacher of algebra, although still a relatively undeveloped area of investigation, has been growing steadily. The complexity of the

process of educating students to become secondary level mathematics teachers is reflected in programs (e.g., Bednarz, 2001) that do not attempt simply to “deliver” methods of instruction; instead, emphasis is placed on involving future teachers in a process that leads them to construct the knowledge they will need to grapple with, design, and carry out classroom interventions. However, research that informs as to the nature of this complexity is still quite rare, especially with respect to the teaching of algebra. Recent research involving the preservice teacher of algebra has tended to focus on the following themes: the relation between future teachers’ subject-matter knowledge and their beliefs, the characteristics of their beliefs and attitudes, and the nature of their subject-matter knowledge.

In a study of the role played by prospective teachers’ background knowledge on their beliefs about algebra learning (Nathan & Petrosino, 2003), 48 candidates from an established research and teaching university participated: 35 of them had completed calculus or above and 13 had not. Of the 35 with advanced mathematical knowledge, 16 were in a program for mathematics and science majors and going on to teach mathematics or science at the secondary level; the remaining 19 were teacher education students going on to teach at the elementary level (as were the 13 who had basic mathematical knowledge). The preservice teacher education program in which they were enrolled included the study of subject matter with the use of modern technology, extensive use of the TIMSS video cases, the study of effective classroom interactions, and the development of models of teaching. Participants first ranked six problems in accordance with their expectations of the ease or difficulty that beginning-algebra students would experience when solving them and then responded to a belief survey that broadly addressed issues in mathematical learning, problem solving, and the role of algebra in complex problem solving. The pattern of results, in combination with findings from their previous research, suggested to the researchers that advanced mathematics knowledge, rather than algebra teaching experience, mediates teachers’ views of algebra students’ development and that highly developed subject-matter knowledge can blind teachers to the actual ways of thinking of algebra students.

Other research related to the subject-matter knowledge of the preservice trainee, as well as the beliefs associated with that knowledge, includes a study by Van Dooren, Verschaffel, and Onghena (2002, 2003). In their investigation of preservice teachers’ preferential strategies for solving arithmetic and algebra word problems, they observed that future secondary school

mathematics teachers use algebraic methods for solving even very easy problems that could have been handled more appropriately with arithmetic methods. Half of the primary school preservice teachers were found to switch flexibly between arithmetic and algebraic methods, whereas the other half had difficulty with algebraic methods. The authors also noted that the methods the teachers used as individuals were strongly correlated with what they would expect their future students to use and by which they would evaluate student work. Similar results have been reported by Schmidt and Bednarz (1997). In a study conducted by Zazkis and Liljedahl (2002) involving a group of preservice elementary school teachers, participants’ attempts to generalize a repeating visual number pattern led to the verbal expression of the generality without necessarily being accompanied by algebraic notation. However, participants perceived that their solutions that did not involve algebraic symbolism, although complete and accurate, were inadequate.

Studies documenting the knowledge of algebra and its teaching that are held by preservice teachers include the research by Even (1998), who presented 152 prospective secondary mathematics teachers with questions related to moving from one functional representation to another. Her results illustrated how knowledge about different representations is interconnected with knowledge about different approaches to functions, knowledge about the context of the presentation, and knowledge of underlying notions. Similarly, Sánchez and Llinares (2003) identified a clear link between preservice teachers’ subject-matter knowledge and their pedagogical reasoning for the topic of function and its representations. However, Even and Tirosh (1995) found preservice teachers’ subject-matter knowledge to be insufficient for understanding and explaining students’ reasoning, as was noted as well in the Nathan and Petrosino (2003) study above.

In a study of the nature of preservice trainees’ beliefs and attitudes, Vermeulen (2000) has reported that 1st-year mathematics education students often enter training programs with established concept images in algebra that are not associated with successful mathematics teaching practice. This study, which presented a portrait of the suggested current teaching approaches for eighth-grade algebra in South African schools, contrasted these goals with the orientation of student teachers, many of whom indicated altogether different views on the teaching of algebra and showed their resistance to change. Cooney, Shealy, and Arvold (1998) have noted that preservice programs can seem to have a random effect on the participating teachers being trained, and that such randomness should be ex-

pected until researchers begin to better understand the relation between teacher-education activities and the impact of these activities on teachers' belief systems. Similarly, Proulx (2003) found that preservice secondary mathematics teachers were differentially impacted by the teacher training program they were following and that the major factor affecting this impact was their personal frame of reference, which filtered their interpretation of the preservice experience.

Stump (1999), who investigated the notions of slope held by a group of preservice and inservice teachers, found that few thought of slope as a rate of change involving two variables. When questioned about students' conceptual difficulties regarding slope, several teachers answered rather in terms of procedural aspects, such as being able to compute the slope formula. In a related study, Stump (2001) observed preservice teachers teaching a basic algebra course after they had completed a secondary mathematics methods course. She found that the students experienced difficulty in presenting the notion of slope within real-world situations in the classes they were teaching and suggested that methods courses provide experiences that challenge students to construct connections among geometric, algebraic, and real-world representations of slope (see also Zbiek, 1998).

Bowers and Doerr (2001), who used computer-based activities to challenge prospective mathematics teachers' procedural orientations, noted in their findings the intertwined development of subject-matter insights and specific views of teaching. In their research, Chazan, Larriva, and Sandow (1999) asked, "What kind of mathematical knowledge supports teaching for 'conceptual understanding'?" The researchers used tasks from the existing literature on the solving of equations, and questions about teaching students to solve linear equations, to bring to the surface preservice teachers' substantive knowledge in this area. The authors' lack of satisfaction with the nature of the information obtained led them to raise the question of whether finer distinctions might be necessary for describing teachers' substantive knowledge of the mathematics they teach.

The majority of the above studies indicate that research involving the preservice teacher has begun to focus on the complex relation between subject-matter knowledge and pedagogical content knowledge. However, much remains to be done. The finding that teachers with developed subject-matter knowledge may attempt to flesh out underspecified pedagogical content knowledge by drawing on subject-area knowledge suggests first of all that the field needs to develop adequate descriptions of what is considered to be ap-

propriate pedagogical content knowledge for the domain of algebra. Once this is achieved, a great deal more research is needed on the interaction between content knowledge and pedagogical content knowledge for algebra, as well as on how algebra teachers develop both.

Concluding Remarks on the Research Focusing on Algebra Teaching and the Algebra Teacher

Although we now know relatively more about the algebra teacher than we did in the early 1990s, a great deal remains to be researched. There are major areas that have only begun to be studied, for example, the practice of algebra teaching and how teachers learn to be effective teachers of algebra. The above review of research related to studies on algebra teaching and the algebra teacher point to four areas in particular as being in need of focused study.

The first concerns the lack of appropriate models for the observation and analysis of the practice of algebra teaching. The eclectic collection of theoretical frameworks currently being used in research on algebra teaching may work against the systematic building up of a body of coherent knowledge in that area. Second, ways need to be found to incorporate the large body of research findings regarding the learner of algebra into the professional development and preservice and inservice training of algebra teachers. Teachers' insufficient knowledge of students' ways of thinking in algebra is, in part, a reflection of the lack of attention paid to the research base that exists regarding the algebra learner in many programs of teacher training and professional development. Research needs to be conducted on the role played by the gradual integration of this research base into the development of the pedagogical content knowledge of algebra teachers and their practice. Third, the interaction between teachers' content knowledge and pedagogical content knowledge, in conjunction with their understanding of how students' subject-matter-specific knowledge develops, requires major research efforts. Related to this suggestion is the recommendation that research attention focus simultaneously on both the teaching and learning of algebra, and the relation between the two. Fourth, with the steady increase in the use of technology as a tool for algebra learning, considerably more research needs to be done on the importance of the role played by the teacher in maximizing its benefits. Several studies have emphasized that even though technology is a powerful addition to the algebra classroom, it can only become a real learning tool when supported by appropriate

teacher intervention. The nature of such intervention remains an understudied area of research.

Two further areas in need of attention have been pointed to by Doerr (2004). One concerns the lack of shared methodologies in the field, which can hinder progress. While the majority of studies use questionnaires, interviews, and observations, and others use concept maps (e.g., Leikin, Chazan, & Yerushalmy, 2001) and varying forms of action research (e.g., Raymond & Leinenbach, 2000), the methods for analyzing the data are almost as varied as the number of researchers. This latter diversity was noted above with respect to theoretical frameworks for the study of algebra teaching. The other point raised by Doerr concerns the fact that

much of the work of teacher educators, in preparing teachers for practice, is not part of the larger body of research, but rather resides as the uncoded collective wisdom of practitioners and is not subject to the scrutiny of research. (2004, p. 283)

Attention to both of these areas could, according to Doerr, result in the establishment of a professional knowledge base for investigating the teaching of algebra.

To close this section, I return to Tirosh et al. (1998), who remarked:

The mathematics education literature does not offer enough information and discussion regarding the impact of various approaches and teaching methods . . . it is important for teachers to be acquainted with various teaching methods and to be aware of their pros and cons in different contexts with different teaching aims, and with different students. (pp. 62–63)

We researchers have built a large base with respect to the learner of algebra and have developed an extensive understanding of the nature of algebra learning. We have yet to develop the same understanding of algebra teaching and of the kinds of practice that are effective in bringing about such learning in algebra students.

IN CONCLUSION: WHAT DOES THIS BODY OF RESEARCH TELL US?

This chapter has focused especially on the more recent research in the learning and the teaching of algebra, dating from the early 1990s onward. Nevertheless, explicit links were made with the research that was carried out prior. Students ranging in age from about 11 years and upward, that is to say, from about the

sixth grade of middle school up through secondary school and into college, were the subject of the learning research that was examined. Studies involving practicing algebra teachers, as well as those following programs of professional development or inservice or preservice courses, were included in the research that was surveyed regarding the algebra teacher.

The chapter was divided into two parts. The first part looked at the two main competing views of school algebra, at the primary sources of meaning from which the algebra student draws in making sense of the objects and processes of algebra, and at the various models for conceptualizing algebraic activity. The GTG model with its threefold categorization according to the generational, transformational, and global/meta-level activity of algebra served as the framework for presenting the research findings on the learning of algebra within the second part of the chapter. Research on the teaching and teacher of algebra completed the second part.

As the vision of school algebra has widened considerably over the decades—moving from a letter-symbolic and symbol-manipulation view to one that encompasses multiple representations, realistic problem settings, and the use of technological tools—so too has the vision of how algebra is learned. The body of research that was surveyed in this chapter has provided compelling evidence that students derive meaning for algebraic objects from multiple sources. A complex interplay occurs within meaning building among the letter-symbolic and graphical forms, problem situations, and linguistic and gestural activity, as well as past lived experience.

With respect to the manipulative/transformational activity of algebra, the research focus has also expanded. Whereas symbol manipulation has in the past been regarded as a primarily rule-based activity, more recent research has been examining the importance of students' coming to view algebraic transformations as a theory-laden activity. The research involving symbol-manipulation technology has similarly been exploring whether the use of this technology can lead students to embrace a conceptual perspective on symbol manipulation, at the same time that they are developing algebraic techniques. Although this research has begun to show some of the ways in which students think about algebra as theory-based activity, the affordances of such approaches require a great deal more study.

A comparison of the research related to the transformational activity of algebra with that related to the generational activity suggests nevertheless a gap between the two. Within generational activity, students draw upon a wide variety of sources of meaning for algebraic symbols, including multiple representations,

word problem situations, and physical and linguistic activity. However, transformational activity narrows the focus primarily to the letter-symbolic. Little is known about how the meaning derived from multiple sources evolves in students and is transformed from embodied and contextual meaning to an understanding of symbols that are subject to formal manipulation. This is an area where further research is especially needed.

Another area where future research might fruitfully be carried out concerns the interaction between functional and nonfunctional approaches to algebra that are included within given programs of instruction. As noted within the chapter, such hybrid programs, which may in fact constitute the majority of algebra programs at the present time, can be a source of confusion to students with respect to the articulation between generational and transformational activity. Teaching experiments designed to assist students in distinguishing function approaches from equation approaches, and in meaningfully integrating them, are strongly recommended.

Research that has investigated the transformational activity of algebra in technological environments such as CAS has signaled the importance of the quality of both the tasks and the classroom discussions during which the teacher attempts to draw out the conceptual aspects underpinning the transformational activity. This research has emphasized that the insertion of such technology into the algebra classroom does not remove the need for paper-and-pencil algebraic techniques; in fact, just the opposite seems true. Further studies that examine the articulation between machine techniques and paper-and-pencil algebraic techniques, as well as the nature of the tasks that enhance the emergence of both conceptual and technical expertise within symbol-manipulation-technology environments, are however needed. CAS research has also noted that the use of this technology as a didactical tool can occasion mathematical discussions that do not normally occur in algebra classrooms. However, for these discussions to be fruitful, the role of the teacher was found to be of crucial importance. A great deal more research is needed concerning the impact of classroom discussion on algebraic growth: what kinds of discussions are most beneficial, how best to encourage students to present their ideas, with what particular kinds of tasks, with which students, at what specific moments, and so on.

The role of word problems in algebra instruction remains somewhat problematic. Although some research reported the advantages of experience with realistic word problems and modeling situations in algebraic activity, others reported the opposite. Furthermore, research on the representation of word-problem

situations by algebraic equations continued to show the difficulties that students have with such activity. However, it is not the case that students cannot solve these problems by a variety of arithmetic methods; in fact, they are more likely to solve correctly word problems than either similar-equation or word-equation formats of questions. Research suggests however that word-problem solving, as it is currently engaged in, may not be the most productive route to the generation of equations, particularly with the younger student.

Although generating symbolic representations of problem situations was shown to be fraught with obstacles, the use of graphical representations for both solving problems and interpreting the letter-symbolic was found in many studies to be an effective approach. This is an area where graphing technology has played a significant role. Research has shown that students sometimes require several years to effectively generate equations to model problem situations, but that when that happens, the letter-symbolic becomes a more preferred mode of representation than the graphical. Research that can help to clarify the factors involved in the successful transition to a symbolic mode of representation for problem situations would be most useful.

Research with the older student, which has tended to focus more on the within-mathematical sources of meaning than on the sources of meaning outside of mathematics, indicates that a majority of students continue to experience difficulty in “seeing structure.” Just as the younger adolescent needed a great deal of time in coming to see how to model situations with equations, so too has the older student experienced difficulty with what has proved to be another time-demanding phase of symbolic learning, that is, seeing form and structure in those symbols. Although the *Principles and Standards for School Mathematics* (NCTM, 2000) recommends that students in Grades 6 to 8 begin to “represent and analyze mathematical situations and structures using algebraic symbols” (p. 222), the existing evidence suggests that few sixth graders in the United States are actually called upon to engage in symbolic work. Yet, in other countries (e.g., China, Russia, Singapore, and South Korea), students begin a formal study of algebra that involves not only the development of algebraic reasoning and generalization, but also the use of algebraic symbols and the solving of equations as early as the fourth grade (Cai et al., 2005). If, as the findings discussed earlier suggest, students need a great deal of time in becoming comfortable with algebraic symbols and in acquiring the fluency and power that symbols can provide, then serious consideration ought to be given to the notion of having students begin the process at an earlier age.

With respect to the use of technology in algebra learning, some have argued that tools such as computer algebra systems can be made to do all the symbolic work, thus removing the need for students to engage in such learning themselves. However, research has shown that the conceptual does not evolve in the algebra learner if the technical aspects are neglected. The use of technology does not lighten the burden of algebraic work if students have not developed machine techniques to obtain what they want from the machine and conceptual tools for interpreting the machine output. Furthermore, these conceptual tools for interpreting machine output seem to emerge from a complex interplay of paper-and-pencil and machine techniques.

The last section of this chapter involved the algebra teacher and the teaching of algebra. This is a field of study that has been growing during the past 20 years. However, researchers still know relatively little about what makes for effective algebra teaching and how algebra teachers learn to develop their craft. Yet, the current body of research has yielded some notable findings that in turn suggest areas in need of further study. Research has shown that teachers' knowledge of students' actual algebraic thinking may be quite limited. Students can reason beyond what many teachers think they are capable of. In addition, some studies have revealed that the weaknesses that exist in teachers' pedagogical content knowledge may be compensated for by more highly developed subject-matter knowledge that inappropriately maps the structure of the algebraic domain onto students' learning trajectories. This suggests the need for research into the ways in which teachers' knowledge of students' thinking can be further developed, by for example tapping into the research base that exists with respect to the learning of algebra. As emphasized by Stacey and MacGregor (2001),

Awareness of how students think about . . . relationships and of what they are likely to perceive in mathematical situations will help teachers to decide what aspects of presentation need special emphasis and what follow-up will be necessary, leading to better teaching and better outcomes. (p. 152)

A related component of algebra teachers' pedagogical content knowledge that has been found to be in need of development, especially among beginning algebra teachers, concerns the knowledge of students' misconceptions and the teaching approaches that could address these misconceptions. Researchers in this area have argued that the existing research literature has yet to evolve regarding descriptions not only

of teaching approaches that work, but also of those that have been found not to work. It has been proposed that research needs to go beyond documenting what students think and how they interpret various objects and processes to include more information on the advantages and disadvantages of different teaching approaches, the contexts in which they were used, and the nature of the students with whom the approaches were tried. This suggests a closer connection between the study of the learning of algebra and the study of the teaching of algebra.

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PROBLEM SOLVING AND MODELING

Richard Lesh

INDIANA UNIVERSITY

Judith Zawojewski

ILLINOIS INSTITUTE OF TECHNOLOGY

During its relatively short history, research on mathematical problem solving has largely drawn on, and evolved from Polya's (1957) description of problem-solving heuristics. A great deal of research in the 1970s and 1980s focused on processes and strategies thought to be used by mathematicians. Begle (1979) reviewed early stages of this research. He concluded that

No clear-cut directions for mathematics education are provided by the findings of these studies. In fact, there are enough indications that problem-solving strategies are both problem- and student-specific often enough to suggest that hopes of finding one (or a few) strategies which should be taught to all (or most) students are far too simplistic. (p. 145)

Similarly, Schoenfeld in his 1992 review of the literature concluded that attempts to teach students to use general problem-solving strategies (e.g., draw a picture, identify the givens and goals, consider a similar problem) generally had not been successful. He recommended that better results might be obtained by developing and teaching more *specific problem-solving strategies* (that link more clearly to classes of problems), by studying how to teach *metacognitive strategies* (so that students learn to effectively deploy their problem-solving strategies and content knowledge), and by developing and studying ways to eliminate students' counterproductive beliefs while enhancing productive *beliefs* (to improve students' views of the nature of mathematics and problem solving). How-

ever, when Lester (1994) compared his own 1980 list of issues in problem-solving research to those of Schoenfeld's in 1992, he concluded that little had changed. In fact, even a decade later when Lester and Kehle (2003) compared a current list of issues to those described by Lester in 1994, they concluded that, still, little progress had been made in problem-solving research and that the literature on problem solving had little to offer to school practice.

The lack of impact and cumulateness in the research on mathematical problem solving is unsurprising, given that this area of research has been criticized over the years for its lack of a theoretical base (e.g., Begle, 1979; Kilpatrick, 1969a, 1969b; Lester, 1994; Lester & Kehle, 2003; Silver, 1985). Further, coinciding with the lack of cumulateness in the research are discouraging cyclic trends in educational policy and practices. In the *Handbook for Research on Mathematics Teaching and Learning*, Schoenfeld (1992) described how, in the United States, the field of mathematics education has been subject to approximately 10-year cycles of pendulum swings between emphases on basic skills and problem solving, and reviews of the literature over these cycles suggest that few knowledge gains have been made from one cycle to the next.

Schoenfeld, whose literature review was written near the end of a cycle that many people considered to be the decade of problem solving, concluded his chapter with some optimism toward a future that would stay focused on problem solving both in school

mathematics and in mathematics education research. However, since the 1992 *Handbook* was published, the United States and many other countries have experienced strong shifts back toward curriculum materials' emphasizing basic skills. This trend has been especially strong due to the worldwide emphasis on high-stakes testing of basic competencies, and due to the alignment of state- and local-level standards and curriculum with the goals of these tests.

Assuming that the pendulum of curriculum change again swings back toward an emphasis on problem solving, what directions can be taken that will break the to-and-fro cycles between an emphasis on problem solving and basic skills? Different directions for research emerge when connecting problem solving and the learning of mathematics, and when studying problem solving as a developmental process—just as in content areas such as early number, proportional reasoning and geometry are studied developmentally in mathematics education. Therefore, this chapter will focus on the following questions: What new perspectives and directions can lead the field to the next level? What approaches or perspectives can be rejected? What has been learned, and what needs to be learned, in order for the next generation of initiatives to succeed where past ones have not? Will the pendulum swing back?

On the one hand, there is evidence that the amount of research on problem solving per se appears to be on the decline (Lester & Kehle, 2003; Stein, Boaler, & Silver, 2003) and that knowledge accumulation on instruction in problem solving is lagging (Lester & Kehle, 2003). For example, the authors counted the number of articles on research in problem solving in the *Journal for Research in Mathematics Education* (JRME) since 1980¹ and found 31 articles in the 1980s, 22 articles in the 1990s, and 4 articles between 2000 and 2003.²

On the other hand, perhaps there are reasons to be optimistic. One indicator that the pendulum of curriculum change may indeed be starting to swing back toward an emphasis on problem solving is at the international level. For example, in a collection of articles in a 2001 issue of the *Journal for Educational Change*, Maclean (2001) described why a number of Asian countries, and especially those that recognize that their prosperity depends on knowledge economies rather than natural resources, have been shifting their curricular emphases toward critical thinking, technological modernity, and

mathematical problem solving—and away from a curriculum that emphasizes mainly skill-level instruction. They are doing this despite the fact that students in many of these Asian countries are already reported to be performing at very high levels on international tests of conventional skills.

Nathan (2001) and Tan (2002) provided a specific instance of this trend, describing initiatives in Singapore as emphasizing creativity and innovation, and developing school environments to support the initiatives. The impetus for such change is an increasing demand for future workers who have developed higher-order skills and abilities. Such change is seemingly imminent in North America and coincides with the goal for *all* students to have access to an education that emphasizes creativity, innovation, and problem solving (National Council of Teachers of Mathematics, 2000).

A second indicator of a pendulum swing back toward problem solving is apparent from recent research that emphasizes how mathematics is used in fields such as engineering, medicine and business management—where professionals tend to be heavy users of mathematics, science, and technology (e.g., Hall 1999b; Lesh, Hamilton, & Kaput, in press; Magajna & Monaghan, 2003; Noss & Hoyles, 1996; Noss, Hoyles, & Pozzi, 2002). Experts in such fields report that the nature of problem solving has changed dramatically during the past 20 years (Lesh, Hamilton, et al., in press; Lesh, Zawojewski, & Carmona, 2003; Rosenstein, in press). Furthermore, advisors to future-oriented programs emphasize that powerful new technologies for conceptualization, communication, and computation are leading to fundamental changes in the levels and types of mathematical/scientific understandings and abilities that provide foundations for success beyond school (e.g., Accreditation Board for Engineering and Technology, 2004).

Among mathematics educators, there is a growing recognition that a serious mismatch exists (and is growing) between the low-level skills emphasized in test-driven curriculum materials and the kind of understanding and abilities that are needed for success beyond school. For example, researchers such as Gainsburg (2003a, 2003b), and Hall (1999a) have explicitly noted the differences between the mathematics used by these professionals and the mathematics encountered in schools. These differences include the knowledge and abilities needed to create and modify

¹ Titles were identified that included references to problem solving, real world, applications, word problems, problem-solving strategies, problem-solving heuristics, problem-centered programs, verbal problems, and mathematical reasoning

² However, the same trend is not evident in *Educational Studies in Mathematics*, since 51 problem-solving articles appeared in the 1980s, 52 articles in the 1990s, and 11 articles between 2000 and 2002.

mathematical models—and to draw upon interdisciplinary knowledge, or knowledge that integrates thinking from a variety of textbook topic areas.

Finally, a third indicator of a pendulum swing back toward problem solving is evident in three current areas of research in mathematics education that are highly relevant to mathematical problem solving. These include research on situated cognition, communities of practice, and representational fluency. The next section of this chapter describes a variety of ways that these three areas provide fresh perspectives that can inform future directions in research in problem solving. Lester and Kehle (2003) observed that “there is general agreement that new perspectives are needed regarding the nature of problem solving and its role in school mathematics” (p. 509).

The purpose of this chapter is to describe trends, perspectives, and key elements of a research agenda that have potential to set a new and different direction for research on mathematical problem solving. A critical characteristic of this agenda for problem-solving research is the clarification of the relationships and connections between the development of mathematical content understandings and the development of problem-solving abilities (Lester & Charles, 2003; Lester & Kehle, 2003; Schoen & Charles, 2003; Silver, 1985; Stein et al., 2003). Clarifying these relationships can inform the development of curriculum and instruction by providing alternatives to treating “problem solving” as an isolated topic, separate from the learning of substantive mathematical concepts. In other words, by clarifying relationships between concept development and the development of problem-solving abilities, our intent is to suggest alternatives to teaching that are based on the assumption that problem-solving abilities develop by first teaching the concepts and procedures, then assigning one-step “story” problems that are designed to provide practice on the content learned, then teaching problem solving as a collection of strategies such as “draw a picture” or “guess and check,” and finally, if time, providing students with applied problems that will require the mathematics learned in the first step. When taught in this way, problem solving (and its strategies) is portrayed as concept- and context-independent processes, isolated from important mathematical ideas.

Our alternative perspective treats problem solving as important to developing an understanding of any given mathematical concept or process. This perspective is also based on the recognition that differences between the problem solving of novices and experts go beyond the observed behaviors (i.e., the things they *do*) to the ways in which they interpret and reinterpret a problem situation (i.e., the things they

see). Therefore, the study of problem solving needs to happen in the context of learning mathematics (and vice versa). Further, research needs to focus on students’ interpretations, representations, and reflections—in addition to the computations that they carry out, the deductive reasoning processes that they employ, the skill-level achievements that they develop, or rules and procedures that they learn to execute. In other words, emphasizing synergistic relationships between learning and problem solving portrays a vision of problem solving and learning as being about *mathematical models and modeling*, where a problem situation is interpreted mathematically, and that interpretation is a mathematical model.

AN OVERVIEW OF PAST RESEARCH ON MATHEMATICAL PROBLEM SOLVING

The purpose of this section is to reflect on how several prevailing lines of research related to problem solving have evolved, to uncover underlying assumptions and critical issues, and to raise questions about what it means to learn problem-solving strategies, metacognitive processes, and beliefs and dispositions related to affect and to the development of a productive problem-solving persona. The discussion will introduce opportunities to consider an alternative perspective for future research on mathematical problem solving, in particular, a *models-and-modeling perspective* (Lesh & Doerr, 2003a).

We begin with a discussion of the major areas of research on mathematical problem solving prior to 1990. In the second part of this section, we describe how subsequent research on higher-order thinking (metacognition, habits of mind, beliefs and affect) evolved from this earlier work. In both of these parts, questions are raised, issues identified, and implications for future research described. Later, in the second half of the chapter, we examine current lines of promising research and further describe ways in which a models-and-modeling perspective can provide a productive framework to guide future research directions.

Early Problem-Solving Research

Three major areas of research in problem solving appearing in the *Journal for Research in Mathematics Education* prior to 1990 were determinants of problem difficulty, distinctions between “good” and “poor” problem solvers, and problem-solving instruction (Lester, 1994; Lester & Kehle, 2003). Each of these categories provides important perspectives and information that

is used here to identify and raise questions about apparent underlying assumptions of each area.

Task Variables and Problem Difficulty

Goldin and McClintock (1979/1984) wrote the seminal book on task variables in problem solving, complementing a large number of studies that appeared in the *Journal for Research in Mathematics Education* (Lester & Kehle, 2003). Lester and Kehle described this genre of research as focusing “almost exclusively on features of the types of problems students were asked to solve in school” (p. 506). They identified “four classes of variables . . . that contribute to problem difficulty: content and context variables, structure variables, syntax variables and heuristic behavior variables” (p. 506).

Kilpatrick’s (1985) description of this early work stated that “problem solving characteristics [were] selected for no special reason other than that they were easily measured and looked like they might have something to do with making a problem tough to solve” (p. 6). Kilpatrick (1985) described how the theme of task variables was revisited in the 1980s by researchers holding an information-processing perspective, emphasizing computer “modeling [of] the processes used in solving problems and predicting the types of errors to be made” (p. 6). He summarized the research on task variables in the following way:

In the past two decades we have clearly come to a much more sophisticated view of the interactions between task characteristics and the characteristics of the problem solver, and this new view seems likely to yield much useful research (although many of my colleagues in mathematics education appear skeptical on this point). (p. 6)

Skepticism is warranted, given that aptitude-treatment-interaction (ATI) and hypothesis-testing paradigms were long ago described by Cronbach and Snow (1977) as not helpful in establishing “best” instructional practices. In ATI studies, the examination of interaction between educational treatment and student aptitudes was originally thought to have potential, because if a particular treatment could be identified as appropriate for a particular type of student, then clear guidance could be provided for curriculum and instruction of students with the targeted characteristics. However, Cronbach and Snow pointed out major problems with ATI and hypothesis-testing studies:

It will be evident in what we have said through the book that we have serious reservations about the dominant style of research in instructional psychology. Insofar as educational reality is shaped by mul-

tiples parameters of student and treatment, traditional experiments cannot be powerful enough to support adequately complex conclusions. Insofar as variance arises out of the history of a particular class, it will be impossible to catch important causes, in the net of statistical inference. . . . In instructional research, hypothesis-testing turns us toward sterility. (p. 519)

Furthermore, from our perspective, some important variables relevant to problem solving research were not even identified, such as the study of student *response* variables. In particular, task variables alone do not account for the notion that each individual interprets a problem situation differently. That is, the problem solver’s interpretation depends not only on external factors (i.e., task variables) but even more so on internal factors (i.e., how one interprets, or “sees,” the mathematics problem). Research that moves beyond the simple identification of task variables to more sophisticated considerations of response variables represents a view shifting from problem solvers as simple processors of information toward problem solvers as interpreters and creators of mathematical models, or ideas—where thinking involves the development and use of complex artifacts and complex conceptual systems.

Another issue concerning early studies on task variables is that the studies rarely looked at problem-solving situations outside of school (Lester & Kehle, 2003). Therefore, recent ethnographic research examining arithmetic used by people in everyday situations, and mathematics used in “heavy use” environments such as engineering and architecture, has been very important to continued work in problem-solving research. Studies of the use of mathematics in everyday situations, such as people making decisions in grocery stores (e.g., Lave, 1988), have revealed that mathematics as taught in school is seldom used. Rather, people generally develop and routinize their own algorithms to accomplish everyday tasks—constructing their knowledge locally and situated to the specific context.

Ethnographers, such as Hall (1999a) and Gainsburg (2003b), who have examined problem solving in workplace environments, have found that professionals making “heavy use” of mathematics (such as architects and engineers) do make use of “school” mathematics, but not necessarily as direct applications of procedures and rules. The work of such professionals has been found to often require the interpretation and mathematization of local problem situations where mathematical models are developed to meet the constraints of the problem at hand, and the interpretation and presentation of mathematical solutions are frequently in multimedia formats (i.e., verbal presentations, charts, graphs, written narratives, written directions, etc.). Therefore, the importance of study-

ing the nature of problem solving outside of conventional school mathematics has become evident.

Expert/Novice Problem-Solver Studies

A second emphasis in early research on problem solving was the study of how the behavior and performance of “good” and “poor” problem solvers differed. Lester and Kehle’s (2003) summary of findings in this area include (a) “Good problem solvers know more than poor problem solvers and what they know, they know differently—their knowledge is well connected and composed of rich schemas”; and (b) “Good problem solvers tend to focus their attention on structural features of problems, poor problem solvers on surface features” (p. 507).

Krutetskii (1976), Kilpatrick (1985), Schoenfeld (1985), Silver (1985), and others also have suggested that experts may perform better not only because they know more mathematics, but also because they know mathematics *differently* than the novices. An analogy may help clarify the distinction. Larry Bird (arguably one of the most successful basketball players in the history of the game) during a television interview was shown a 3-second video clip of an “amazing move” that he had made during one of his recent games. He was asked to comment on why he did what he did. What was amazing about his comment was that, just by looking at this short video clip, he was able to recount for the interviewer where everybody on the basketball court had come from and where each was going. While most television viewers saw individual players and individual movements as “pieces of information,” Bird saw the situation in terms of patterns, organized around his experience in the game. Thus his knowledge was not only more extensive but organized differently. Although his knowledge was situated (in his breadth and depth of personal experience in the game), his knowledge was also flexible and transferable to new situations in new games. Thus, he possessed more than a collection of skills, he possessed a conceptual system of situated, adaptable knowledge that contributed to making him a great player.

Krutetskii’s (1976) descriptions of the problem solving of gifted mathematics students have similarities to the description of Bird’s expertise. His work provides insights about why descriptions of expert past processes may not serve well as prescriptions to guide novices’ next steps during problem solving. Krutetskii reported that gifted mathematics students do not simply have a larger number of ideas and strategies in their repertoire than the novices; they have ideas, strategies, and representations that appear to be organized into a highly sophisticated, elaborate network of knowledge, providing them with power-

ful ways to approach problem-solving situations. The talented students are described as able to generalize broadly and rapidly from a single instance, curtailing normal chains of reasoning by skipping intermediate steps and moving rapidly from problems to solutions, whereas the average student needs to see and process every logical step. Krutetskii found that gifted students tend to perceive the underlying mathematical structure of problem situations very rapidly, whereas others tend to notice and remember relatively superficial problem characteristics. Notice that the characteristics of expert problem solvers identified by Krutetskii cannot be directly taught to the nonexperts. One cannot directly teach novice problem solvers to “generalize broadly,” “identify the underlying structure,” or “look ahead to skip steps,”—especially assuming that novices often view mathematical problem solving as keeping track of and carefully processing pieces of information.

A landmark study of expert medical diagnosis by Elstein, Shulman, and Sprafka (1978) demonstrates how problem solving in medical diagnosis taps abilities that are organized more around the individual’s experiences (including their own speciality and their own practical experiences) than around a collection of diagnostic heuristics that are used across the field. This was a surprising finding at the time:

The most startling and controversial aspects of our results have always been the finding of case specificity and the lack of intraindividual consistency over problems, with the accompanying implication that knowledge of content is more critical than mastery of a generic problem-solving process. (p. 292)

Elstein et al.’s findings illuminated the importance of considering the learning of problem-solving processes as embedded in, and linked to, the content and context of the *situation*, rather than existing as a stand-alone processes or skill. Similarly, more recent studies of people in everyday settings (e.g., Carraher, Carraher, & Schliemann, 1985; Carraher & Schliemann, 2002; Lave, 1988; Lave & Wenger, 1991; Saxe, 1988a, 1988b, 1991), as well as studies of experts in workplace environments requiring heavy use of mathematics (e.g., Gainsburg, 2003b; Hall, 1999a, 1999b) have solidified the notion that the knowledge of both experts and novices tends to be organized more around the mathematics of the *situation* than it is around general problem-solving heuristics or conventional mathematical topics.

However, what is known about experts’ knowledge is still in very early stages of development. As Lester and Kehle (2003) cautioned,

Studying how experts think, make decisions, and solve problems does not guarantee that one is studying the experts at what actually makes them experts. For example, one can certainly give expert mathematicians textbook problems to solve, and compare their strategies and mental representations to those of novices. But expert mathematicians do not solve textbook problems for a living. . . . even if cognitive scientists have studied the right people, they may have studied them doing the wrong tasks. (p. 504)

Future study on the *nature* and *development* of expertise needs to go beyond describing typical behaviors of experts to investigating how experts “see” problematic situations: How do they mathematize problematic situations? describe them? systematize them? dimensionalize them? quantify them? Research on the development of expertise also needs to go beyond an assumption that experts first learn the content, then learn the problem-solving strategies, and then learn ways to appropriately select and apply the already-learned mathematics. Rather, the development of expertise seems to involve the holistic co-development of content, problem-solving strategies, higher-order thinking, and affect—all, to varying degrees, and situated in particular contexts (Zawojewski & Lesh, 2003). Future research questions need to address how the development of expertise evolves within problem-solving episodes and over experiences. Furthermore, research on expertise, itself, needs to be viewed developmentally. In particular, the meanings of problem-solving processes and strategies should be assumed to develop similarly to the way concepts develop for topics like rational numbers or proportional reasoning.

The final section of this chapter will describe the potential of research methodologies in which researchers assume that problem-solving expertise evolves in multi-dimensional ways (mathematically, metacognitively, affectively, etc.)—and that researchers’ own understandings of expertise also are likely to evolve over the course of a study.

Instruction in Problem-Solving Strategies

In mathematics education, Polya-style problem-solving strategies—such as *draw a picture*, *work backwards*, *look for a similar problem*, or *identify the givens and goals*—have long histories of being advocated as important abilities for students to develop. Although experts often use these terms when giving after-the-fact explanations of their own problem-solving behaviors, and researchers find these terms as useful descriptors of the behavior of problem solvers they observe, research has not linked direct instruction in these strategies to improved problem-solving performance. Why do after-the-fact descriptions of past activities not

necessarily provide guidelines for next steps during future problem-solving activities? Perhaps one reason is because using a strategy, such as *draw a picture*, assumes that a student would know what pictures to draw when, under what circumstances, and for which type of problems. Therefore, mastering *draw a picture* (in general) depends on interpretation abilities—not just execution abilities. Thus, developing systems for interpreting problem situations is as important, if not more important, than developing processes for doing particular strategies.

Our interpretation of Polya’s heuristics is that the strategies are intended to help problem solvers think about, reflect on, and interpret problem situations, more than they are intended to help them decide what to *do* when “stuck” during a solution attempt. To convey our perspective on Polya’s heuristics, we first set the stage by considering how the educational use of Polya’s problem-solving heuristics has evolved historically. Consider Schoenfeld’s (1992) described use of Polya’s problem-solving strategies:

There is no doubt that Pólya’s accounts of problem solving have face validity, in that they ring true to people with mathematical sophistication. Nonetheless, through the 1970s there was little empirical evidence to back up the sense that heuristics could be used as a means to enhance problem solving. . . . In short the critique of the strategies listed in *How to Solve It* and its successors is that the characterizations of them were descriptive rather than prescriptive. That is, the characterizations allowed one to recognize the strategies when they were being used. However, Pólya’s characterizations did not provide the amount of detail that would enable people who were not already familiar with the strategies to be able to implement them. (pp. 352–353)

Schoenfeld went on to describe the amount of detail that would be needed to make problem-solving strategies accessible to those not already familiar with them. Using the *examining special cases* problem-solving heuristic, Schoenfeld illustrated its application to three different mathematical problems (determination of the closed form of a formula for a series represented as a summation; given two polynomials whose coefficients are identical but in “backwards” order, find [and prove] the relationship between the roots of the two functions; given a defined sequence, determine whether the sequence converges and, if so, to what value). In the context of these problems, he described how

each of these examples typifies a large class of problems and exemplifies a different special-cases strategy. . . . Needless to say, these three strategies hardly

exhaust 'special cases.' At this level of analysis—the level necessary for implementing the strategies, one could find a dozen more. This is the case for almost all of Pôlya's strategies. (pp. 353–354)

In other words, Schoenfeld claimed that the *examine special cases* problem-solving strategy is, in a sense, “too large” or “too general” a strategy to be useful for prescriptive purposes. He recommended that each conventional heuristic derived from Pôlya's work (e.g., *consider a special case*) should be broken down into longer lists of more specific strategies. An analogy might be made to the process of learning to cook. One can observe and identify when a person is cooking, but learning to cook (as a concept in itself) can be thought of as involving a number of smaller, more specific aspects of cooking (e.g., principles of baking, or making a sauce).

Schoenfeld concluded, describing his own research (1985) and that of others (e.g., Heller & Hungate, 1985; Silver, 1979, 1981), that a knowledge base is available for the careful, prescriptive characterization of problem-solving strategies, which could lead to more detailed specifications of all the problem-solving strategies and heuristics for instruction, which could be linked to various classes of problems. However, with the longer lists of more detailed heuristics to learn and remember comes the need to manage and select from those lists the appropriate problem-solving strategy for a given situation. Therefore “higher order” managerial rules and beliefs would need to be introduced that specify when, why, and how to use the “lower order” prescriptive processes. Thus, Schoenfeld emphasized the need for future research on metacognition and beliefs.

Although Schoenfeld's own classroom-based research (1985, 1987) indicated success using the above-described approach, one needs to keep in mind that the instruction was implemented by a world-class teacher, who was teaching in a complex and lengthy learning environment where many different factors were at work. Thus, even though some indicators of success were achieved, the reasons for success are difficult to sort out. As Silver (1985) pointed out early on, even when a particular problem-solving endeavor has been shown to be successful in improving problem-solving performance, it is not clear *why* performance improved.

The dilemma we see is that, on the one hand, short lists of descriptive processes (i.e., the conventional problem solving strategies) appear to be too

general to be meaningful for instructional purposes; on the other hand, long lists of prescriptive processes (i.e., more detailed and specific problem-solving strategies for classes of problems) tend to become so numerous that knowing when to use them becomes the heart of understanding them. Furthermore, adding more metacognitive rules and beliefs to be acquired as another layer for instruction only compounds the difficulty for classroom application.

In contrast, we suggest a perspective on Polya's heuristics as not only prompting ways of selecting and carrying out procedures and rules (i.e., “doing” mathematics³), but also as a means of developing systems for interpreting and describing situations (i.e., “seeing” mathematically). Recent research on problem solving in complex problematic situations suggests that the abilities involved in “seeing” are as important as abilities involved in “doing” (Lesh & Doerr, 2003b). Consider, for example, the work of structural engineers, described by Gainsburg (2003b). In particular, for engineers, carrying out the procedures contained in industry-established mathematical models (i.e., “doing the mathematics”) is simultaneously both essential and inadequate. The engineers interviewed described how the behavior of the *assumed* buildings (to which the industry-standard models relate) is idealized, whereas actual applications involving more complicated situations and nonideal shapes make the use of conventional models and procedures insufficient for the needed work. Although the industry-standard mathematical models are critical for the engineers' work, the engineers must be aware of the oversimplifications embedded in the models and adapt or combine the models with other models to address the real-world complex problem. As a result the experienced engineer, using what is called *engineering judgment*, draws from many resource areas, including engineering theory, software, experience, and common sense to make judgments about whether to use and trust mathematical methods, or to modify, or to completely override them.

Research on teaching mathematical problem solving needs to move beyond the notion of trying to directly teach what experts apparently “do” to novices, and beyond transforming descriptive information about expert problem-solving behavior into long lists of prescriptive processes for novices to learn. For example, our interpretation of Polya's (1957) heuristics, and the examples that he used to illustrate them, suggests that the heuristics are intended to help stu-

³ *Doing* is portrayed in this chapter as carrying out procedures, manipulations. This is somewhat in contrast to *doing* as described by Schoenfeld, Goldenberg, and others, who reserve the term for *doing* mathematics as professional mathematicians do—meaning the creation of new mathematics.

dents go *beyond* current ways of thinking about a problem, rather than intended only as strategies to help students function better *within* their current ways of thinking (about givens, goals, and possible solutions steps). Rather than providing a specific list of suggestions (draw a picture, identify a similar/simpler problem) to try out when “stuck,” as strategies for plotting specific “next steps,” we propose that Polya’s heuristics can be thought of as providing a language to help problem solvers think back about their problem-solving experiences. By *describing their own processes*, students can use their reflections to develop flexible prototypes of experiences that can be drawn on in future problem solving.

Using this fresh perspective leads to the emergence of new questions concerning teaching and learning of problem solving: To what extent are students able to particularize a small number of descriptive (heuristic) processes, rather than trying to apply strategies that are recalled from an already-learned list? What collection of descriptive processes seem to be most productive? How do understandings of these descriptive processes develop? What instructional strategies (such as promoting reflection) are useful to facilitate students’ development of their descriptive processes?

Research on Higher-Order Thinking

Research and development efforts focused on higher-order thinking in mathematical problem solving have run a course similar to that of research and development efforts focused on problem-solving strategies. Metacognitive behaviors, habits of mind, and dispositions (including affect and beliefs) of expert problem solvers have been described, identified, and labeled and then have been assumed to be productive traits that function across problems, within all stages of solving a given problem, and for all good problem solvers.

Lester and Kehle (2003), in their review of research on problem solving, summarized some of the higher-order thinking characteristics of good problem solvers:

- Good problem solvers are more aware than poor problem solvers of their strengths and weaknesses as problem solvers.
- Good problem solvers are better than poor problem solvers at monitoring and regulating their problem-solving efforts.
- Good problem solvers tend to be more concerned than poor problem solvers about obtaining “elegant” solutions to problems. (p. 507)

Given that characteristics such as these are perceived as accurate and stable, instructional studies in higher-order thinking typically transform such observed metacognitive characteristics of experts directly into instructional goals for novices. However, although students in these studies sometimes seem to gain an awareness of the taught higher-order thinking strategies and dispositions, their knowledge about and attempted use of the strategies has not necessarily led to better problem solving (Lester & Kehle, 2003).

In this section, we break up *higher-order thinking* into three areas: (a) metacognition, (b) habits of mind, and (c) beliefs and dispositions. Each area is used to raise questions and issues about research in higher-order thinking in general, and to pose an alternative perspective. One issue raised concerns the assumption that higher-order-thinking capabilities exist as stable and general traits or capabilities within and across problem situations. Another question raised is whether higher-order thinking exists separately from one’s mathematical interpretation of a local problem situation. Finally, other questions are raised about whether descriptions of experts’ apparent ways of higher-order thinking should be transformed into prescriptions for novices.

Metacognition

The notion of “thinking about one’s own thinking” has been in educational literature at least since Dewey’s (1933) discussion of *reflective thinking*. Similarly, Piaget’s (1970) reference to *reflective abstraction* preceded Flavell’s (1976) introduction of the term *metacognition*, where he stated,

Metacognition refers, among other things, to the active monitoring and consequent regulation and orchestration of these processes in relation to the cognitive objects or data on which they bear, usually in the service of some concrete goal or objective. (p. 232)

Since the time that Flavell formulated his notions of metacognition, however, a variety of different kinds of operational definitions of metacognition have emerged in the field; and this proliferation of meanings led Lester (1994) to refer to it as an “elusive construct” (p. 666). Similarly, Campione, Brown, and Connell (1989) concluded

One of the most salient features about metacognition is that the term means different things to different people, with the result that there is considerable confusion in the literature about what is and what is not metacognitive. This confusion leads to apparently contradictory viewpoints, ranging from claims that the concept is too ill-defined or fuzzy to be the object of scientific inquiry

to assertions that things metacognitive are the driving force of learning, and therefore the major aspects of learning we should be studying. (p. 93)

Although definitions for metacognition have not converged to a single meaning, most researchers have described metacognition and cognition as separate entities, and as being hierarchical in nature. For example, Schoenfeld (1992) described metacognition as “self-regulation, or monitoring and control . . . [of] . . . resource allocation during cognitive activity and problem solving” (p. 354). He referred to metacognitive ability (e.g., “executive control,” p. 355) as the mechanism that enables problem solvers to break up larger and more complex problems into subtasks, prioritize the subtasks, efficiently sequence each subtask, and finally do each subtask. Another example is in Kuhn’s (1999) hierarchical separation, which was apparent in his description: “Metacognitive knowing operates on one’s base of declarative knowledge. . . . What do I know, and how do I know it?” (p. 18).

Partly due to the described theoretical formulations, most empirical studies also have operationally defined metacognitive behavior as being separate from cognition. For example, Pintrich (2002) categorized observations of metacognition into strategic knowledge, knowledge about cognitive tasks, and self-knowledge. Wilson and Clark (2004) parsed metacognition into the awareness individuals have of their own thought processes during the development of a solution, the self-evaluation of those thought processes, and the self-regulation of those thought processes. In fact, Wilson and Clark explicitly addressed the hierarchical separation, noting that “the objects on which metacognition acts are cognitive objects” (p. 33). Wilson and Clark, like Pintrich, were successful at *describing* metacognitive actions during problem solving. Their identification and categorization of behaviors were based on students’ self-reports, ensuring a high level of validity in the interpretation of students’ behaviors. Specifically, students twice watched a videotape of their own problem-solving processes and retroactively described their own awareness, regulation, and evaluation processes (by selecting predetermined statements on cards). However, even though Wilson and Clark obtained conclusive evidence that students did in fact use metacognitive language to describe how they went through cycles of awareness-evaluation-regulation-evaluation during past problem-solving activities, the researchers were unable to relate these metacognitive behaviors directly to improved future problem-solving performance.

A few empirical studies have treated cognition and metacognition as integrated, for example, by em-

bedding the teaching of metacognition into the learning of substantive content. Lester, Garofalo, and Kroll (1989) described the introduction of metacognitive skills in one “regular” and one “advanced” seventh-grade classroom. The teacher served as an external monitor during problem solving within the teaching of a mathematics unit, encouraged discussion of behaviors considered important for internalizing metacognitive skills, and also served as a model of good executive behavior. In their findings, Lester, Garafalo, and Kroll described a dynamic link between the learning of mathematics content and the learning of metacognitive processes. They emphasized that metacognitive processes and skills should be learned in the context of learning substantive mathematics, and that “control processes and awareness of cognitive processes should develop concurrently with the development of an understanding of mathematical concepts” (p. 86). Thus, they hypothesized that “metacognition training is likely to be most effective when it takes place in the context of learning specific mathematical concepts and techniques” (p. 86).

More recently Kramarski, Mevarech, and Arami (2002) also embedded instruction in metacognition within the teaching of specific content—whole and rational number systems. Using a treatment-control design, they taught the same whole and rational number system content to both groups but withheld specific instruction on metacognition from the control group. In the treatment group, the students were taught to generate and answer four types of general questions related to comprehension, connections, strategies, and reflection. The students who were exposed to metacognitive instruction significantly outperformed students who were not on standard tasks (those more customarily used by mathematics teachers—tasks including simplified data and ones that can be solved by applying a straightforward algorithm that was previously taught) and authentic tasks (tasks that portray problem-solving situations, including realistic data, and are not solvable by the straightforward application of a known algorithm). Further, the students in the treatment group were better able to reorganize and process information, and to justify their reasoning.

Why did performance improve in these two cases? Was improvement truly due to learning metacognitive processes? Or did the students learn the mathematics concepts better or differently? Perhaps asking metacognitive questions during instruction on substantive mathematics prompts students to reflect on the content they are learning, and thus superior learning of the content led to better problem-solving performance *in the domain*.

As early as 1985, Silver expressed doubt about the ability to isolate the variables that might lead to improved problem-solving performance and suggested that research would benefit from developing ways of describing problem solving in terms of conceptual systems that influence students' performance (p. 257). Those conceptual systems might be called higher order conceptual systems that are tightly integrated systems of content knowledge, metacognitive capabilities, beliefs, and dispositions. In contrast to a view of metacognition and cognition as hierarchically separate, higher order conceptual systems can be thought of as developing in parallel and interactively with cognition, and as necessarily embedded in the learning of content in problem-based activities.

An assumption that seems to exist in research on metacognition is that the deployment of specific metacognitive behaviors is *always* productive. For example, the research of Schoenfeld (1985, 1987), Kramarski et al. (2002), Pintrich (2002) and Wilson and Clark (2004) applies the targeted metacognitive strategies across problem-solving tasks, across all stages of the problem-solving process, and across the mathematical content in the problems. Schoenfeld's (1992) description of his early studies (1985, 1987) suggests that students can be trained to ask themselves metacognitive questions throughout a problem-solving episode and across problem-solving activities, and that these particular questions are always productive. He described how when small groups of students were engaged in challenging mathematical problems, he (as their teacher) circulated through the classroom asking specific questions: What (exactly) are you doing? (Can you describe it precisely?) Why are you doing it? (How does it fit into the solution?) How does it help you? (What will you do with the outcome when you obtain it?). Initially students were at a loss to answer the questions. However, recognizing that, despite their discomfort, the professor was going to continue asking those questions, students began to defend themselves by discussing the answers to the questions in advance. By the end of the term, Schoenfeld indicated that this behavior became "habitual" (p. 356). He concluded that his students' use of metacognitive activity after this training was more similar to that of experts than previously, although he said "the students' solution is hardly expert-like in the standard sense, since they found the 'right' approach quite late in the problem session" (1992, p. 357).

A different point of view might be that the use of explicit teacher interventions, intended to help students acquire metacognitive habits, may be disruptive for some students at certain times. Furthermore, drawing explicit attention to a metacognitive activity

may be disruptive even in situations where its tacit use might have been productive. For example, consider a metacognitive strategy for problem solving: "Monitor your progress so that you won't go down wrong paths." Problem solvers are indeed likely to explore some paths that are ones they later choose not to take, particularly early in the solution development process. Therefore, verbally cautioning students early in the session to monitor their work in order to avoid going down the wrong path may seem to be a reasonable instructional strategy. However, such a verbal intervention may not only interrupt the problem solvers' thinking processes (whether the problem solver is an individual or a group), but the intervention may be interpreted to mean that the path selected is indeed wrong—simply because of a belief that the teacher would not say anything if one were on the "right" path. An alternative way of thinking is to consider how students can develop an *awareness* that early in complex problem-solving situations, going down "unproductive" paths is common; that something can be learned from the journey down a so-called wrong path; and that part of problem solving is detecting mismatches between the problem situation and trial solutions, leading to iterative cycles of expressing, testing, and revising current ways of thinking. In other words, a productive metacognitive strategy may be more powerful in a tacit and analytical form, rather than in an explicit and deterministic form.

Consider another example of how explicit attention to metacognition can be counterproductive. Human expertise in the performing arts, or in sports such as golf, involves the use of fluidly functioning conceptual or procedural systems. The actions or moves coordinated by the systems are not routine because they are often used to engage in new and unfamiliar situations—much like in problem solving. In such situations, competitors may try to use disruptive techniques to induce "paralysis by analysis" in opponents. For example, John McEnroe, the well-known "bad boy" tennis player, was reported to have won one of his most famous matches by complimenting his opponent on several details about his extraordinary serve (Fein, 2005). Apparently, his opponent started thinking about his own serve, and his explicit metacognitive thinking about his serve started causing him problems.

Lesh, Lester, and Hjalmarson (2003) assumed that the nature of most metacognitive behaviors should be expected to vary in their productivity across problems and over different phases of solving a problem. For example, *brainstorming*, in which group members try to generate a diversity of ideas without being critical of one another's proposals, may be productive in early

stages of work on complex and challenging problems. Later in the problem-solving process, however, progress may depend more on detailed monitoring and assessment, whereas continued brainstorming may prevent progress on the solution. Therefore brainstorming, as a behavior or strategy, should be viewed as neither good nor bad across the board. The desirability of brainstorming is dependent on the function that it is intended to serve for the problem at hand and for the stage in problem solving where it is to be used.

Assuming that a particular metacognitive strategy is not necessarily productive all of the time leads to the emergence of new areas of research. For example, Hamilton, Lesh, Lester, and Yoon (in press) reported that students do indeed know they go through multiple cycles of expressing, testing, and revising their own ways of thinking about problems, and the students reported engaging in a revision process when weaknesses were detected in their trial solutions. Thus a number of research questions seem to emerge: How do problem solvers come to recognize inadequacies of their own interpretations, which then lead to further testing and revision of their ideas? What mechanisms facilitate students' recognition that current ways of thinking are unproductive? What mechanisms help students generate ideas for revising their current ways of thinking? How do these mechanisms evolve or develop within and over problem-solving experiences? Along what dimensions do these mechanisms evolve (e.g., from unstable to stable, from "primitive" to mature, from external to internal, and from concrete to abstract)?

Studies on understanding mechanisms for detecting problems in one's way of thinking need to involve developmental investigations of problem-solving and metacognitive processes, similar to how mathematics educators have investigated the nature of children's development of early number concepts, rational number concepts, geometry concepts, and other domains of concepts. With rare exceptions (e.g., Lesh & Zawojewski, 1992; Zawojewski & Lesh, 2003), such developmental perspectives have not been applied to problem-solving processes, metacognition, and other higher-order-thinking areas, such as beliefs, values, and dispositions. Yet, when a developmental perspective is taken on these areas, a vision of problem-solving processes and higher-order thinking can be developed that has the potential to inform teachers' practice. For example, in the above-named sources, we described students' early conceptions of a common problem-solving strategy, *look for a similar problem*, and identified the pitfalls for students who attempted to acquire the strategy as a habit. We suggested that the problem-solving strategy, as an instructional tool, be recast from a developmental perspective: "Look

at a particular problem from different perspectives." Over a series of experiences where problem solvers explicitly consider problems from various perspectives, a group (and individuals in a group) may begin to use their experiences to serve as powerful "stories," or metaphors, for thinking about other structurally similar situations. Thus, the problem-solving strategy can be viewed developmentally and as evolving over experiences.

Habits of Mind

Cuoco, Goldenberg, and Mark (1996) described habits of mind as "mental habits that allow students to develop a repertoire of general heuristics and approaches that can be applied in many different situations" (p. 378). For example, they suggest that students should develop general habits of mind such as taking ideas apart and putting them back together and making data-driven conjectures, as well as other mathematics-topic-specific habits (e.g., algebraic and geometric habits of mind). Although students aren't expected to understand the *topics* that mathematicians deal with, Cuoco et. al. "would like students to think about mathematics the way mathematicians do" (p. 377). The notion of habits of mind has become popular in mathematics education rhetoric in recent years, particularly in professional development and more recently in research. Thus, the question addressed in this section is the extent to which habits of mind can enhance directions for research on higher-order thinking in mathematical problem solving.

From our perspective, three issues emerge when considering habits of mind as a potential direction for future research in problem solving. The first issue concerns whether emulation of the apparent behaviors (or habits) of professional mathematicians should serve as a basis of instruction in problem solving for nonexperts. The second issue concerns whether anything new and productive is being added to a research agenda in problem solving by investigating habits of mind compared to investigating problem-solving strategies and metacognitive processes or beliefs. The third issue concerns what the term *habits* may inadvertently communicate to students and teachers, such as the expectation that a habit should be productive across all contexts and all stages of problem solving.

The first issue we raise concerns the apparent assumption that teaching habits of mind (and other behaviors and strategies) of professional mathematicians to students will lead to improved problem-solving performance. For example, the assumption is evident in Cuoco, Goldenberg, and Mark's (1996) suggestion that teaching habits of mind can be a means to help students emulate the problem-solving/research

processes of professional mathematicians. Similarly, Schoenfeld (1992) described a number of studies of students engaged in mathematical activity similar to that of mathematicians and as a result suggested that involving students in microcosms of mathematical practice might improve problem-solving performance of students. Little research is available that investigates the direct link between instruction in habits of mind to improved problem-solving performance. We conjecture that although such studies may document that students exhibit desired behaviors (as seen in the section on metacognition), links to improved problem solving will be elusive.

What may explain the lack of empirical support for teaching the behavior of mathematicians to students? Part of the problem may be explained by Lester and Kehle's (2003) observation that little is known about *individual differences* among experts in mathematical problem solving. For example, di Sessa, Hammer, Sherin, and Kolpakowski (1991) found that expert scientists do not all think or work alike: Some work alone, and some in groups, some work from hypothesis to proof, whereas others constantly switch from generalization to specific case perspectives and vice versa. Therefore, a reasonable assumption is that more than one profile of "typical" mathematicians' problem-solving behaviors and habits of mind are likely to exist.

Another questionable aspect of the assumption is the sole use of pure mathematicians as the role models for problem solving. Other professionals, who also engage in creating and modifying mathematics, do so in ways that are different from those of pure mathematicians. For example, Gainsburg (2003b) described how the nature of proof in the work of mathematicians is quite different than the nature of proof in engineering. One difference is in the role of assumptions. For the mathematician, the starting assumptions are assumed to be true, and the work is in the construction of the logic to link the end statement to the assumptions. For the engineer many of the starting assumptions are not identified (e.g., how the design will be simplified, how the design and the environment interact), so identifying the assumptions and justifying their accuracy is a main challenge. Gainsburg (2006) described how in structural engineering there are mathematical methods for analyzing a building with a completely stiff (rigid) floor and another for a completely flexible floor. These extremes describe no real floor; yet the engineer has to choose one (or come up with an entirely different means of analysis) that will satisfy building codes. Selecting the more conservative mathematical method is safe, because of the assumption that the real behavior of the given floor

would not be as severe as an ideal floor. However, such assumptions may be modified when resulting in an overly expensive redesign of a building, or resulting in situations where the given design simply cannot carry the loads assumed. Thus, the "proof" that the structure meets building codes is contained in the mathematical argument linking quantities associated with the building to a final set of forces, whereas carrying out the procedures is easy and often accomplished in spreadsheets. The nature of the mathematics of engineers, then, appears to be quite different than the mathematics of mathematicians.

The second issue we raise concerns the potential for the construct, habits of mind, to contribute to research on problem solving beyond what has been learned from investigations on problem-solving strategies and metacognition. The habits of mind in Table 17.1 were drawn from Levasseur and Cuoco (2003) and Goldenberg, Shteingold, and Feurzeig (2003). Most of these habits of mind are easy to map to conventional problem-solving strategies and metacognitive behaviors and therefore seem to simply be new names for old conceptions of problem-solving strategies, heuristics, and metacognitive behaviors. For example, the *guessing is not necessarily bad* habit of mind is virtually the same as the conventional problem-solving strategy, *guess and test*; and the *seeking and using heuristics to solve problems* habit of mind is virtually the same as using executive control to deploy the cognitive resources at hand.

A few of the habits of mind that were cited by the original authors are not listed in Table 17.1. These include *distinguishing between agreement and logical necessity* (Goldenberg et al., 2004, p. 25) and *think algebraically* (Levasseur & Cuoco, 2003, p. 33). We did not include them in Table 17.1 because they are different than the others. They are more closely linked to the learning of specific content (e.g., geometry or algebra). By focusing on such content-specific strategies, Driscoll and colleagues (Driscoll, 1999; Driscoll, Moyer, & Zawojewski, 1998; Driscoll et al., 2001) created professional development materials that more narrowly specified the *think algebraically* habit of mind by addressing *generalize from computation* and *build rules to describe functions*. This specification process is similar to what Schoenfeld suggested in 1992 when he encouraged the development of more detailed and specific problem-solving strategies (previously described). Thus, for the purpose of this chapter, even these kinds of content-specific habits of mind appear to be renamed compilations of problem-solving strategies and metacognitive strategies that have been investigated in the past research. Consequently, the construct of habits of mind seems to have the same strengths and weaknesses as its tra-

Table 17.1 Habits of Mind and Their Counterparts in Problem Solving or Metacognitive Strategy

Habit of Mind	Problem-Solving (PS) or Metacognitive (M) Strategy
Guessing is not necessarily bad (Levasseur & Cuoco, 2003, p. 28)	Guess and test (PS)
Challenge solutions, even correct ones (Levasseur & Cuoco, 2003, p. 29)	Look back (PS)
Analyzing answers, problems, and methods (Goldenberg et al., 2003, p. 26)	Check the reasonableness of your answer (PS)
Look for patterns (Levasseur & Cuoco, 2003, p. 29)	Look for a pattern (PS)
Conserve memory* (Levasseur & Cuoco, 2003, p. 30)	
Specialize (Levasseur & Cuoco, 2003, p. 31)	Use simpler numbers (PS)
Use alternative representations (Levasseur & Cuoco, 2003, p. 31)	Draw a picture (PS)
Think about word meanings (Goldenberg et al., 2003, p. 16)	Act it out (PS)
	Write an equation (PS)
Carefully classify (Levasseur & Cuoco, 2003, p. 32)	Make a table (PS)
	Organize the data (PS)
Justifying claims and proving conjectures (Goldenberg et al., 2004, p. 21)	What are you doing? (M)
Seeking and using heuristics to solve problems (Goldenberg et al., 2004, p. 28)	Why are you doing it? (M)
	Is it helping? (M)
	What strategy/tactic/principle can be used in order to solve the problem/task? (M)

* By remembering general patterns—authors' explanation

ditionally named counterparts. Therefore, although such habits of mind are perhaps useful in professional development, where the goal is to stimulate teachers to rethink old practices, they seem to offer no new insights or directions to the existing state of research on problem solving.

Finally, we are concerned that the terminology, habits of mind, conveys a notion of mathematical thinking and problem solving that is not productive. Because in normal everyday language, the word *habit* generally refers to rules or procedures that are executed without much conscious thought, it is usually associated with simple condition-action behaviors. Little thought is given to when an individual habit is “good” or “bad”—generally a particular habit is considered stable and one that should be discontinued (e.g., smoking) or adopted (e.g., saying “please” and “thank you”). On the other hand, there are times when habits of mind, such as *carefully classify* and *challenge solutions, even correct ones* serve productive functions and other times when they can be counterproductive. Therefore, the terminology of *habits of mind* seems to emphasize the automated searching for and doing of explicit heuristics and metacognitive strategies, rather than emphasizing the interpretive, reflective, and generative aspects of problem-solving processes that fit with less behavioristic conceptions of competence.

Beliefs and Dispositions

Beliefs and dispositions related to affect have become an important topic of study related to higher-order thinking. For example, by the late 1980s and early 1990s, the role of metacognition was linked to a wide range of noncognitive factors such as those involved in feelings or other affective variables (e.g., Lester, Garofalo, & Kroll, 1989; Schoenfeld, 1987, 1992). The line of reasoning is that if novice problem solvers can learn how to use various problem-solving heuristics, then the ability to self-regulate, monitor, and call upon control mechanisms often are critically influenced (positively or negatively) by such things as beliefs, attitudes, feelings or dispositions. However, Schoenfeld (1992) pointed out that this picture is complicated by the fact that many students (and often their teachers) need to “unlearn” counterproductive beliefs and dispositions they have developed from past classroom experiences in order to learn and apply the needed metacognitive processes and strategies.

Perceptions about the role of beliefs and dispositions in problem solving have changed little since Schoenfeld's 1992 review of the literature, where he focused on Lampert's (1990) list of common beliefs that have strong (often negative) influences on students' mathematical thinking:

- Mathematics problems have one and only one right answer.

- There is only one correct way to solve any mathematics problem—usually the rule the teacher has most recently demonstrated to the class.
- Ordinary students cannot expect to understand mathematics; they expect simply to memorize it and apply what they have learned mechanically without understanding.
- Mathematics is a solitary activity, done by individuals in isolation.
- Students who have understood the mathematics they have studied will be able to solve any assigned problem in five minutes or less.
- The mathematics learned in school has little or nothing to do with the real world.
- Formal proof is irrelevant to the processes of discovery or invention. (Schoenfeld, 1992, p. 359)

Schoenfeld's (1992) review of research on the beliefs and dispositions of students, in part based on work of McLeod and Adams (1989), led him to conclude that students "abstract their beliefs about formal mathematics—their sense of their discipline—in large measure from their experiences in the classroom" and that "students' beliefs shape their behavior in ways that have extraordinarily powerful (and often negative) consequences" (p. 359). He described how the mathematical practice in the classroom shapes and is shaped by beliefs of students (and teachers, and society). The implications are far-reaching, in as much as what happens in the culture of the classroom determines, in part, future courses taken in college by students, intended majors reported by students, and subsequent approaches to mathematics instruction taken by teachers (Amit, 1988).

The notion of stable, trait-like beliefs and dispositions has been a mainstay in the literature. For example, McLeod (1989) described how emotions that are experienced repeatedly over time transform into relatively stable attitudes and beliefs. Consequently, recommendations for classroom practice naturally focus on eliminating what are believed to be negative beliefs and dispositions—by using direct instruction (providing information and examples counter to given stated beliefs and attitudes), by changing teachers' practice (Schoenfeld, 1992), or by changing the curriculum and the culture of the classroom (McLeod, 1989). For example, consider the first belief listed above—that "mathematics problems have one and only one right answer" (Schoenfeld, 1992, p. 359). Students could receive direct instruction informing them that not all

mathematics problems have one and only one right answer, and then they might engage in a collection of activities that result in more than one correct solution (i.e., multiple reasonable solutions). McLeod's suggestion would go even further, encouraging systemic changes in the curriculum and culture by embedding problems with multiple reasonable solutions throughout the curriculum and embedding explicit discussion of the belief whenever possible.

In these described instructional approaches to dealing with debilitating beliefs and dispositions, an implicit message exists that the belief in question (e.g., in this case, the belief that "mathematics problems have only one right answer") is generally "bad." However, students develop most such beliefs precisely because they served some positive function at some point in their school career. For example, the belief that "mathematics problems have only one right answer" tends to be counterproductive in the context of complex problems, where multiple reasonable solution paths and multiple levels and types of solutions exist. But when students take high-stakes tests comprised of timed multiple-choice items (often claiming to include "problem solving" items), then students benefit from the belief that the problems to be encountered do indeed have exactly one correct answer. Even in complex problems, where a real-life situation involves a client who expects only one type of answer as being correct, it would be counterproductive for a student to disregard the notion that "this problem has only one right answer." Therefore, the *problem* of beliefs and dispositions exists not in the trait itself but in students' ability to use the "trait" (or belief) flexibly and appropriately in different circumstances.

When beliefs (or other attributes such as feelings and dispositions) are thought of as all-encompassing rules or simple declarative statements, they may be productive in some situations for some purposes, but they often are counterproductive in others. The goal, therefore, should be to help students recognize the difference and to use such beliefs in ways that are appropriate. The notion of stable trait-like beliefs and dispositions should be abandoned in favor of the notion of developing a productive "problem-solving persona" or "identity" that involves a complex, flexible, and manipulatable profile of attitudes, feelings, dispositions, and beliefs.

The patterns that form a problem-solving identity are complex, involving varied motivational patterns, affective reactions, and cognitive and social engagements in different circumstances both within a given task and across tasks.⁴ For example, Hamilton et al.

⁴ The notion of flexible problem-solving identity described here goes beyond McLeod's (1989) description of locally dynamic emotion that emerges within a problem-solving episode.

(in press) described how productive problem solvers are not those who adopt some fixed problem-solving personae. Instead, productive problem solvers manipulate their problem-solving personae to fit continually changing circumstances. Instead of trying to get students to learn a list of problem-solving or metacognitive strategies, Hamilton et al. designed opportunities for student reflection following their engagement in problem-solving activities in which students repeatedly expressed, tested, and revised their ways of thinking about the solution. As a result of describing their own problem-solving processes, students gradually developed dynamically changing and manipulatable problem-solving personalities—which integrated metacognitive functions, problem-solving strategies, beliefs, attitudes, feelings, and values.

Middleton and Tolum (1999) also have examined students' problem-solving identity profiles as complex entities involving many conventional variables (e.g., motivation, interest). They found that the conventional "traits" (that are considered to be stable across situations) often change as students progress through various modeling cycles within a specific complex problem-solving episode.⁵ They described students' profiles in terms of peaks and valleys of thoughts, feelings, arousal states, and social interactions that played out across the lifespan of the problem-solving activity. For example, in their focus on motivation, they argued that there is no such thing as an "unmotivated child"—where the term *unmotivated* means an absence of will or desire. Middleton and Tolum suggested that what is conventionally perceived as lack of motivation (a stable trait) actually refers to the fact that the child wants to do something other than what the educator desires at a particular moment in time. They argued that research would benefit from a different conception of motivation—motivational models viewed as adaptive systems that interpret and restructure experience in a format that guides future engagement.

More recently, Goldin (in press) also has described affect as being more complex than conventionally viewed. Drawing on the notions of meta-affect previously described in DeBellis and Goldin (1997, 1999) and Goldin (2002), and the previously described constructs of mathematical intimacy and mathematical integrity (Debellis, 1998; Debellis & Goldin, 1997, 1999), Goldin highlighted the need to develop mathematical self-identity, to achieve mathematical integrity, and to experience intimacy with mathematics during problem solving and modeling. He viewed the development of affect as an essential structural part of learning

mathematics—rather than as peripheral to cognitive aspects of learning. That is, when students interpret situations mathematically, they do a great deal more than simply engage concepts that are purely logical or mathematical in nature; their interpretations also involve feelings, values, beliefs, and dispositions. Further, Goldin described how mathematically powerful affect involves positive feelings about mathematics *as well as* ambivalent or negative feelings. For example, in mathematical modeling the process of testing and revising often leads to frustration, but one can learn that the frustration is often associated with the anticipation of eventual learning of something new, or the satisfaction of eventually solving the problem.

Middleton, Lesh, and Heger (2003) have developed methods for revealing students' beliefs and dispositions in research settings. They conducted their research with small groups solving thought-revealing, model-eliciting activities. The activities were carefully designed to elicit mathematical models as "answers" to problems—and to reveal students' intermediate and final models as they collaborate during the group problem-solving process (Lesh, Hoover, Hole, Kelly, & Post, 2000). Middleton et al. described how, using these activities, the relationship between students' modeling of mathematics and the self in relation to mathematics can be observed over reasonably brief periods of time. During the problem-solving episodes, students externalize their mathematical thinking to group members by proposing their varying perspectives, and by working to coordinate, negotiate, and sometimes reject ideas. As a result, students overtly assess and revise their mathematical ways of thinking. Using this approach, Middleton et al. have been able to describe how students *simultaneously* construct a framework for understanding classes of mathematical tasks as embodiments of mathematical structure and also construct a framework for understanding their own identity in relationship to mathematics in the problem.

Given a view of beliefs and disposition as flexible and adaptive parts of problem solvers' conceptual system, new research questions need to be posed. Productive questions would seek to understand how successful individuals learn to manipulate their own problem-solving personae to suit the situation, or how they alter a given situation to some degree to fit their own profiles. Research on students' continually adapting problem-solving personae needs to include studies of how students' dispositional components develop along dimensions, such as from external to internal,

⁵ Students were given problems in which they worked in small groups to create mathematical models for realistic clients who expressed well-specified needs (i.e., model-eliciting activities as described by Lesh, Hoover, et al. 2000).

and from barren and distorted to increasingly coordinated and adaptable.

Similar to the directions for research on metacognition described earlier, investigations on beliefs and dispositions would benefit from studies that investigate the development of relevant beliefs, feelings, values, and dispositions by involving students in activities where they express, test, and revise their own attributes during post-hoc reflections on problem-solving experiences. For example, in the research described by Hamilton et al. (in press), students developed their own personal models of modeling (i.e., context-specific minitheories about their own learning and problem solving experiences) by using different kinds of reflection tools. A variety of reflection tools were designed to investigate ways to develop more effective learning and problem-solving personae—both as individuals and as members of various problem-solving groups or learning communities. Results of these studies suggest that productive problem-solving identities do not consist of rigid profiles of behaviors or attributes that govern students' behaviors across all situations; they are not a single rigid lists of beliefs, behaviors, dispositions, or values that function in the same ways across all situations—and across all stages of learning or problem solving. Instead, productive profiles are those that students themselves manipulate in ways that are productive in a variety of changing situations.

Summary and Reflections on Research in Higher-Order Thinking

A great deal of research has in common a view of higher-order thinking (e.g., metacognition, motivation, beliefs, attitude, disposition) as separate from the cognitive activity of learning content. However, recent suggestions for research in mathematical problem solving emphasize the parallel and interactive development of both mathematic concepts and higher-order thinking—as well as a variety of attributes that constitute a productive learning and problem-solving persona (e.g., Goldin, in press; Lesh, Lester, & Hjalmarson, 2003; Lester and Kehle, 2003; Middleton, Lesh, & Heger, 2003; Stein et al., 2003). Understanding how students *develop* their own ways of thinking about their own conceptual systems needs to be based on the assumptions that individuals have idiosyncratic conceptual systems influenced by their own personal experiences and that individuals' conceptual systems are dynamically evolving both within and across problem-solving activities. Furthermore, no strategy, process, behavior, or characteristic should be expected to always be productive for every problem, nor for every stage in learning or problem solving.

Research on higher-order thinking in mathematical problem solving needs to be based on the assumption that mathematical ideas and higher-order thinking develop interactively, and the work needs to pursue a greater understanding of the nature of activity that provides opportunities for individuals' problem solving persona to develop. The notion of *flow*, as conceptualized by Csikszentmihalyi (1990), offers a promising direction for such research. *Flow* is a metaphor for a type of experience that is most enjoyable: “like being carried away by a current, everything moving smoothly without effort” (Csikszentmihalyi, 1990, p. xiii). People describe the feelings they experience during flow as including intense “concentration, absorption, deep involvement, joy, a sense of accomplishment . . . as the best moment in their lives” (p. 176), and *flow* experiences are more likely to occur when one is engaged in a difficult enterprise that requires complex skills and leads to a challenging goal than in moments of leisure or during receptive entertainment, such as viewing television (Csikszentmihalyi, 1990). Therefore, the notion of flow seems highly relevant for the study of development of a problem-solving personae, and a close analysis of the characteristic dimensions of flow experiences may help identify the types of experiences that will more readily result in students' co-development of mathematics learning, problem solving, metacognition, and dispositions. For example, Csikszentmihalyi described flow as occurring when the situation provides concrete goals and manageable rules, allows for persons to adjust the opportunities for action to their own capabilities, provides information that makes self-assessment of current progress possible, and makes concentration possible (screening out distractions). Further, the model-eliciting activities briefly described above have these characteristics, making them a potentially useful research site in which to conduct studies related to flow.

Research concerning the development of higher-order thinking should include students reflecting on their own solution processes over a series of model-eliciting activities, identifying their own trends and patterns of feelings related to various phases of problem solving. For example, in post-hoc reflection activities, students can be asked to coordinate landmarks in the solution process with whether time seemed to fly by versus when time seemed to drag, or when they felt “in control” versus felt like they were “floundering.” Research questions might include, How are the patterns noticed related to phases of problem solving? How are patterns noticed related to students' problem-solving identity? When students study their own reflections over a series of modeling activities, what personal theories do they develop about what

constitutes mathematical problem solving and modeling? The goal for future research should be to better understand how cognition and higher-order thinking *co-develop* in the context of learning mathematics through problem solving.

NEW DIRECTIONS AND PERSPECTIVES FOR RESEARCH IN MATHEMATICAL PROBLEM SOLVING

Lester (1994) and Lester and Kehle (2003) suggested that mathematical problem solving is more complex than previously thought. Concept development and the development of problem-solving ability are highly interdependent and far more socially constructed and contextually situated than traditional theories have supposed. Further, neither concept development nor the development of problem-solving abilities proceeds in the absence of the development of beliefs, feelings, dispositions, values and other components of a complete problem-solving persona. Problem-solving ability cannot be completely separated and studied apart from all of these aspects. Thus, research on problem solving needs to be framed differently than in the past and needs to be based on a definition of problem solving that embraces its complexity in school and beyond. In particular, Lester and Kehle pointed out that that “far too little is known about students’ learning in mathematically-rich environments” beyond school (p. 510).

Even though the emerging view of problem solving as a complex endeavor may seem daunting and may create the need for researchers to significantly revise many beliefs that have driven past research, the need for a significant change in perspective should not be a surprise. Historically, as disciplines develop they not only evolve through continuous and incremental steps; they often go through major shifts and revolutions. For example, the 19th century shift from Euclidean geometry to non-Euclidean geometries and the introduction of Gödel’s theorem in the 1930s were radical departures from the conventional work of mathematicians in their day. The academic discipline of mathematics, which is more than 300 hundred years old, has gone through a number of such reconceptualizations, and it continues to do so and prospers by doing so.

In contrast, research in mathematics education is very young, and although some of the earliest work was conducted in the mid-1930s (e.g., Brownell, 1970/1935), work directed at mathematical problem solving (in particular, word problems) did not gain

much attention until the 1960s. Further, the first research journal devoted entirely to mathematics education in North America, *Journal for Research in Mathematics Education*, did not begin until 1970, and *Educational Studies in Mathematics* (based in Europe) began about 10 years earlier. The field of mathematics education, due to its relative youth, arguably has yet to experience any major and enduring conceptual shifts, and research that clarifies the need for such shifts should be viewed as a major achievement.

The purpose of the second half of this chapter is to articulate a promising shift in perspective concerning future directions for research on mathematical problem solving. A brief overview of the perspective is provided here, and distinguishing characteristics of this perspective are described in greater detail, and with more illustration, in the final section of this chapter. In particular, a models-and-modeling perspective (Lesh & Doerr, 2003a) is proposed for conducting research and interpreting results. In keeping with the American Pragmatists perspective (Thayer, 1982), the proposed shift is not intended to be a “grand theory” of learning or problem solving. Instead, a models-and-modeling perspective can serve as a framework that encourages the integration of ways of thinking drawn from a variety of practical and theoretical perspectives. On the other hand, the proposed perspective is not theory neutral; rather it is generally aligned with extensions of constructivist and sociocultural philosophies about the nature of mathematical concepts and abilities.

One characteristic of research using a models-and-modeling perspective is that the conceptual models researchers use to study and understand mathematical problem solving are expected to be continually under development, or *under design*. Just as students develop mathematical models as they go through iterative cycles of expressing, testing, and revising their solutions to complex problems, researchers’ models of students’ modeling and problem solving are expected to go through similar cycles. Instead of conducting studies designed to determine whether a particular explanation, predictive model, or hypothesis is “right” or “wrong,” research using a models-and-modeling perspective is grounded in the design and production of tangible tools and products for school-based use by applying the principles portrayed by a proposed conceptual model of learning and teaching.

The second characteristic of a models-and-modeling perspective is that the research activity is organized around the production of products and tools that are tested in classrooms. Because the tools educational researchers develop are often intended to help induce changes in the very “subjects” the researcher is trying to study, the expectation is that their tools

and their corresponding conceptual models will need continuous modification, extension, and refinement. Therefore, the data gathered informs not only the revision of the tools (through an analysis of how well the tools inform the explanations, predictions, and descriptions in the model) but also the development and design of the researcher's conceptual model of learning and teaching. Thus, a models-and-modeling perspective on research in problem solving has the potential to simultaneously improve learning and teaching while also providing concrete, theory-based, and experience-tested results to inform the continued design of the research model for understanding how the learning of problem solving and mathematics co-develop.

A third characteristic of the proposed perspective is an increased emphasis on theory *development*, and a decreased emphasis on theory borrowing. Mathematics education research needs to shift toward a stronger focus on theory building, including the development of the field's own methodologies, theoretical models, and tools (Kelly & Lesh, 2000; Lesh, Kelly, & Yoon, in press). General theories of learning are useful for guiding the design of worthwhile learning activities and theoretical models, but useful educational models for learning and teaching in specific contexts often need to draw on and build from more than a single practical or theoretical perspective. Similar to the development of realistic solutions to realistically complex problems, model development and theory building in research usually involves trade-offs among competing factors. Therefore, research from a models-and-modeling perspective involves the development of specific methodologies, theoretical models, and tools that are designed in response to the problem being investigated.

A fourth characteristic of a models-and-modeling perspective is that it embraces a goal of producing results that need to be generalizable (i.e., sharable and reusable), but also powerful for local contexts—the context of the school, the context of the students and their teacher, etc.—and, most important, for the context of the mathematics being learned (Lesh, 1985; Silver, 1985; Stein et al., 2003). Thus, the innate tension between generalizability and specificity is constantly at the forefront of a modeling perspective.

The fundamental building blocks for the proposed models-and-modeling perspective are addressed in this section. First, a fresh view of mathematical problem solving, critical to this perspective, is discussed. Then, implications for problem solving are gleaned from current areas of research that are particularly relevant to a models-and-modeling perspective: situated learning, communities of practice,

and representational fluency. The chapter ends with an explicit discussion of the critical features and aspects of a models-and-modeling perspective on future directions for problem-solving research.

A Fresh Perspective on Problem Solving

A useful theory of mathematical problem solving probably needs to be sensitive to the fact that the nature of problem solving has not remained unchanged throughout history. For example, in a technology-based *age of information*, powerful tools for computation, conceptualization, and communication are leading to fundamental changes in the levels and types of mathematical understandings and abilities that are needed for success beyond school. As stated by the National Research Council (1999),

Changing technologies will continue to alter skills and eliminate and create jobs at a rapid rate. Although skill requirements for some jobs may be reduced, the net effects of changing technologies are more likely to raise skill requirements and change them in ways that give greater emphasis to cognitive, communications and interactive skills. (p. 71)

The changes in the nature of mathematics needed for problem solving are remarkable compared to just 50 years ago: Traffic jams are mathematically modeled and used in traffic reports; cell-phone-tower placement is based on mathematical models involving three-dimensional topography of the earth, the availability/cost of property for placing the towers, etc.; mathematical models that form the basis for Internet search engines are based on different assumptions, each one leading to new and more efficient ways to conduct searches. In these new situations, the mathematics used is multitopic, draws on innovative and creative representations and media, and often involves mathematical modeling. Yet, little seems to change for the nature of “problems” encountered by students in mathematics classes. In most school mathematics, problem solving involves thinking about things that are countable or measurable, and the favored problems are those that are readily cast within the conventional topics for teaching—whereas other, multitopic mathematical modeling problems are left for special (expendable) days or chapters at the ends of books.

A fresh perspective of problem solving is needed—one that goes beyond current school curricula and state standards. In particular, a forward-looking view of mathematical problem solving needs to tap research that describes people adapting and creating mathematics for use in everyday environments and in workplaces that require heavy use of mathemat-

ics. A fresh view of problem solving needs to view the learning of mathematics and problem solving as integrated, as largely based on modeling activity, and as a construct that is itself continually in need of development. Each of these concerns is addressed in this section, leading to and deriving from a proposed definition of problem solving.

Beyond School Mathematics

In 1989, when Stanic and Kilpatrick reviewed the literature on mathematical problem solving, they identified three main themes—each suggesting a somewhat different perspective on a definition of problem solving. The first theme was “problem solving as context,” in which mathematical problems are used to motivate learning mathematics (e.g., justify teaching a specific topic, motivate intrinsic interest in mathematics, provide the development of new skills through carefully sequenced problems, practice learned techniques). This theme presented problem solving as a means to an end, rather than an end in itself. The second theme was a view of problem solving as a set of skills to be learned in their own right. Schoenfeld (1992) suggested that this second theme characterized the mathematical problem-solving curriculum and implementation of the 1980s, in which problem-solving strategies were to be taught as skills to be learned followed by a set of problems that were conducive to practicing the specified strategy. The third theme described by Stanic and Kilpatrick was the “art of solving problems”—meaning what mathematicians *do* as they solve difficult and perplexing problems, formulate proofs, and articulate the new knowledge to the field of mathematics. This last theme is related to how mathematicians select and deploy their mathematical knowledge to solve problems. However, Stanic and Kilpatrick’s themes did not refer to human adaptation and creation of mathematics and mathematical models in environments where heavy use is made of mathematics. A fresh perspective on problem solving needs to take into account the situations beyond school mathematics, especially those in which evolving technology plays a large role.

Although traditional topics of school mathematics courses are clearly fundamental in the work of engineers, architects, and business administrators, the method in which the mathematics is deployed and often combined and recreated for the situation is not in the formal “mathematical” fashion learned in school. In fact, one might say that the traditional topics serve as good descriptors of the work, but when used to interpret and understand new and dynamic situations, the mathematics generated and deployed by the users is more complex, situated, and multidis-

ciplinary than the conventional topic descriptions imply. Thus, the users may not even recognize the relationship between the mathematics taught in school and the mathematics they use in problem-solving environments. For example, Gainsburg (2003b), in her study of structural engineers at work, described their use of many of the mathematical topics taught in school, ranging from geometry, algebra, measurement, and graphs to dimensional analysis and probability. However, these topics were not used in the same “formal” way that they appear in mathematics classes, and she described how one of the main ways that the engineers used their knowledge was to describe, interpret, and understand the structure of engineering situations for making decisions or predictions. Of particular note was that the engineers, once submerged in the context of their work, found it difficult to identify school-learned mathematics in their daily work. A similar lack of awareness of the school mathematics used by professionals has been documented for technicians (Magajna & Monaghan, 2003), as well as for architects and engineers (Hall, 1999a, 1999b).

Not surprisingly, a similar gap in perception between school mathematics and the problem solving of the profession may exist for students at the preprofessional stages of education. Capobianco (2003) provided a vivid illustration of the gap in her interviews of first-year engineering students at Purdue University who were making the transition from successful high school science and mathematics accomplishments to college-level engineering. When asked about the purpose of their field (i.e., engineering), most of the students were quick to respond, “problem solving,” but when asked to expand, students did not appear to understand very much about the knowledge needed for engineering work and the contexts in which they would need to solve problems.

Identifying and understanding the differences between the mathematics of school and of the workplace is critical to the formulation of a fresh definition for problem solving. A vivid illustration is found in a report by the Mathematics Sciences Education Board ([MSEB], 1998) where they described how employees who make heavy use of mathematics to solve problems effectively must be able to work in dynamic environments with open-ended problems and be able to integrate mathematics with other disciplines and the real world. Such uses of mathematics require that mathematical knowledge be reconstituted or created for the local problem situation, and that useful content knowledge involves the integration of ideas and abilities related to a variety of mathematics topics and other disciplines. For example, Oakes and Rud (2003) described how professionals using technological tools

need to adapt rapidly to continually evolving conceptual systems underlying each new generation of technological tools. Another illustration is in the work of Magajna and Monaghan (2003), who have described technicians as using mathematical procedures to not only solve problems but also shape the problems to adapt them to the technology. They also described how technicians sometimes choose to find a technological solution that avoids underlying mathematics problems associated with existing models or methods used in their fields. Parker (1998) also described fields such as engineering as involving problem situations in which the user of mathematics needs to understand and modify mathematical systems rather than acquire piecewise knowledge of components of the system, which is more characteristic of school mathematics.

One notable finding across studies of professionals who make heavy use of mathematics is that mathematical modeling consistently emerges as one of the most important types of activities that needs to be mastered. Gainsburg (2003b) emphasized the role of mathematical modeling in her observations of structural engineers at work. Cross (1994) also described new technologies and newly emerging fields as increasingly using mathematical modeling as the primary form of design. For instance, the new interdisciplinary field of nanotechnology is heavily technology-based and requires the construction of mathematical models, conventions, and procedures to standardize operations across the field. The newest fields are experiencing an increasing need for problem solvers who can create elementary yet powerful constructs and conceptual systems for use in these fields, create mathematical models to use with the technological tools of the trade, and adapt previously developed mathematical models to new problem situations.

A Definition for Problem Solving

Given that the most difficult aspects of newly emerging problem-solving situations involve the development of useful ways to think mathematically about relevant relationships, patterns and regularities, definitions for *problems* and *problem solving* are needed that are consistent with these characteristics—and that do not separate problem solving from concept development and the ways that these concepts are used in “real life” situations beyond school. Therefore, we propose the following definition:

A task, or goal-directed activity, becomes a problem (or problematic) when the “problem solver” (which may be a collaborating group of specialists) needs to develop a more productive way of thinking about the given situation.

Developing a “productive way of thinking” means that the problem solver needs to engage in a process of interpreting the situation, which in mathematics means modeling. Thus, *problem solving* is defined as the process of interpreting a situation mathematically, which usually involves several iterative cycles of expressing, testing and revising mathematical interpretations—and of sorting out, integrating, modifying, revising, or refining clusters of mathematical concepts from various topics within and beyond mathematics. In contrast to defining problem solving as a search for a procedure to traverse from the “givens” to the “goals” of a problem, the proposed definition views problem solving as iterative cycles of understanding the givens and goals of a problem. Once that understanding is attained, the linking between the two often is almost trivial.

Lester and Kehle (2003) proposed a definition of problem solving similar to the one we pose here. In their discussion, they described mathematics education as moving *away* from a view of “doing mathematics as consisting of performing low-level procedures (i.e., performing skills and routines), recalling facts and formulas, applying conceptual knowledge, and engaging in high-level problem solving (sometimes referred to as critical thinking or creative thinking)” (p. 501) and moving *towards* a view of complex mathematical activity and the mathematics of modeling complex systems. They claimed that such a perspective results in a “more authentic view of students’ cognitions as they exist in busy classrooms and in complex realistic settings” (p. 517). We concur, recognizing that in virtually every field where ethnographic studies have been conducted to investigate similarities and differences between experts and novices, results have shown that not only do experts do mathematics differently, but they also see mathematics differently than novices. Thus, mathematical problem solving is about seeing (interpreting, describing, explaining) situations mathematically, and not simply about executing rules, procedures, or skills expertly. The proposed definition also emphasizes the fact that mathematics is, above all, the study of structure. That is, an interpretation is mathematical if the description or explanation focuses on structural characteristics of this situation.

Learning Problem Solving and Learning Mathematics

Our definition for problem solving also embraces the notion that people learn mathematics *through* problem solving and that they learn problem solving *through* creating mathematics (i.e., mathematical models). Embedded in a models-and-modeling perspective is an emphasis on situations in which the problem solver is expected to *create*, *refine*, or *adapt*

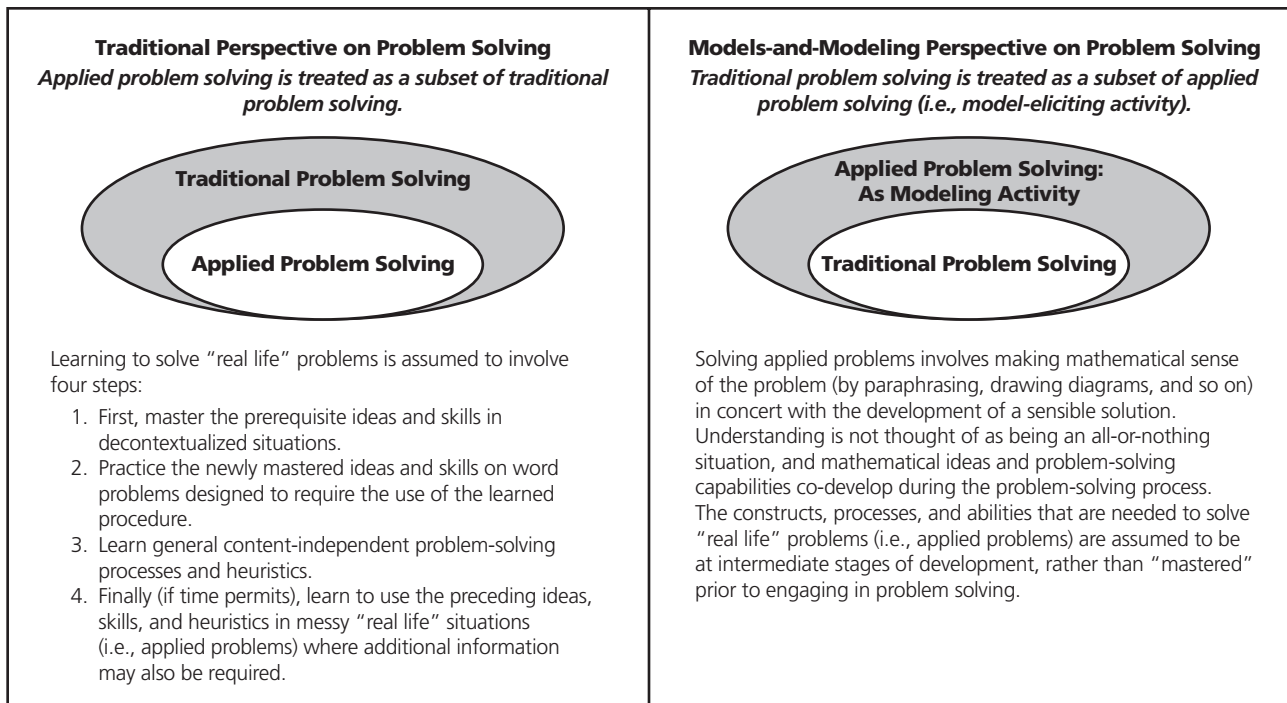


Figure 17.1 Views of problem-solving: traditional versus modeling (Adapted from Lesh & Doerr, 2003b).

mathematical interpretations, or ways of thinking and procedures.⁶ Figure 17.1, adapted from Lesh and Doerr (2003b), compares and contrasts a traditional perspective on problem solving with a models-and-modeling perspective.

As Figure 17.1 suggests, within the boundaries of the traditional school mathematics curriculum, the assumption tends to be made that “real life” applied problems are the most difficult types of problems to solve. Therefore, they are commonly addressed only after computational procedures have been learned, the procedures have been practiced on sets of story problems, and problem-solving strategies have been taught. Thus, only in the final stages of instruction are students engaged in solving realistic and complex applied problems (if time permits). In this traditional perspective, applied problems (requiring mathematical modeling) are a small subset of the problem-solving experiences in which students engage.

From a models-and-modeling perspective, on the other hand, the assumption is that the learning of mathematics takes place *through* modeling. In other words, students *begin* their learning experience by developing conceptual systems (i.e., models) for making sense of

real-life situations where it is necessary to create, revise, or adapt a mathematical way of thinking (i.e., a mathematical model). Given model-eliciting activities (described earlier and illustrated below), students are expected to bring their own personal meaning to bear on a problem, and to test and revise their interpretation over a series of modeling cycles. Students are assumed to simultaneously gain an increasing understanding of both the problem situation and their own mathematization of the problem, which can be described as “local concept development” (Lesh & Harel, 2003; Lesh & Zawojewski, 1992), capturing the integration of learning problem solving and learning mathematics. Therefore, students’ applied problem-solving experiences (i.e., mathematical modeling) drive the learning in the conventional curriculum, and traditional story problems become a subset of the applied problems through which students learn mathematics.

Model-Eliciting Activities

Our definition for problem solving has embedded in it an assumption that, in response to a realistically complex situation, the problem solvers will engage in “mathematical thinking” as they produce, refine, or

⁶ Note the similarity in the process of developing a research model, as described in the introduction to the section, and students’ development of a mathematical model in problem solving. In both cases the problem solver/researcher is required to create a model for dealing with new situations for an expressed purpose, and the development of the model takes place *through* the process of expressing, testing, and revising the model *in context*.

adapt complex artifacts or conceptual tools that are needed for some purpose and by some client. Such activities need to be designed to ensure that the solution (artifact, tool) problem solvers create *embodies* the mathematical process they constructed for the situation, and thus these types of problems are called *model-eliciting activities*.⁷ For example, one model-eliciting activity, called Camp Placement (Lesh & Zawojewski, 1992), required that a procedure (i.e., mathematical model) be produced for a track-and-field camp director to use in the assignment of campers to groups in order to form competitive teams. Using available data about students' capabilities (i.e., track-and-field best scores, comments from the campers' coaches), the procedure created would not only be applied to a current set of campers, but also to all future sets of campers arriving with similar data. Another model-eliciting activity, called Big Foot (illustrated in Lesh & Harel, 2003), required students to develop a procedure (i.e., a model) for police detectives to use to predict a person's height from a shoe imprint in the soil. The students were told to prepare a procedure that could be used by any police officer in future detective work at a variety of crime scenes.

Both of the preceding model-eliciting activities are quite different from conventional story problems. Instead of requiring students to produce short mathematical answers to questions about premathematized situations, students' solutions (i.e., models) are complex artifacts (or tools) that need to be useful for a given client in a given situation *and* those artifacts need to be sharable and reusable in other situations, for other data sets, or by other people. Because solutions to these types of activities typically evolve over several iterative modeling cycles, the student-generated assumptions and constraints also evolve and change. Therefore, assessing progress on a model-eliciting activity is different than "checking an answer" in traditional story problems. An "end-in-view," as described by English and Lesh (2003), who based their description on Archambault (1964), is used to guide solution development as the problem solver keeps in mind the expressed needs of the client, as the trial solution and assumed conditions change.

Model-eliciting activities require students to focus on the structural and systemic characteristics of the "thing" that is being designed, constructed, or modeled. For example, in the Camp Placement Problem (described above) the procedure developed for the camp director may involve a series of rules, sometimes

applied in a stochastic manner, and may be accompanied by graphs, tables, or step-by-step procedures to help communicate how the tool is used. The final solution embodies the system of objects, relationships, and operations that the students thought were important. Thus, the students' work reveals important aspects of their mathematical thinking. For the Big Foot problem, Lesh and Doerr (2003b) reported how one group of students had classmates line up in rank order of height against a wall and then noted something about a "six times" relationship. Although their written response revealed no formal use of proportions, their graph suggested that the group used trends to establish a "six times" relationship between foot length and height of a person. The transcript made from a videotape of the group work confirmed this interpretation of the group's final product; clearly the group used trends, suggested that the foot length be multiplied by six to obtain the height, and did not formally use proportions. Thus, the procedure developed by the students revealed aspects of their thinking not only to the researchers, but also to the teachers and the students themselves.

The iterative modeling cycles that students go through during the solution of model-eliciting problems make it clear how traditional story problems represent only a small part of the modeling process. Figure 17.2 depicts the modeling process as consisting of four steps of interacting processes that do not necessarily occur in any fixed order. *Description* is the creation of a model, which involves mapping from the real world⁸ to the modeled world. *Manipulation* of the mathematical model happens within the modeled world and functions to generate predictions (or actions) related to the original problem-solving situation in the real world. *Prediction* carries relevant results from manipulation back into the real world, and *verification* checks the usefulness of the prediction (or action) in the context of the real world. Problems detected in the verification phase often prompt a cycle of expressing, testing, and revising a trial solution.

When students are engaged in model-eliciting activities that involve more than a single cycle of modeling, their *initial* interpretations of these situations tend to be immature, primitive, or unstable compared with the interpretations that underlie their final solutions (Lesh & Harel, 2003). Furthermore, as students develop solutions they generally do more than simply make modifications to a single existing model. For example, small groups of students generally pose a

⁷ Lesh, Hoover, et al. (2000) and Lesh, Hoover, et al. (1993) described six principles for designing model-eliciting activities.

⁸ Note that a "real world" can also be an imagined world, such as the ones in Camp Placement and Big Foot. The critical feature is that the "real world" has enough context and relevance to the student to facilitate meaningful engagement in the mathematical modeling.

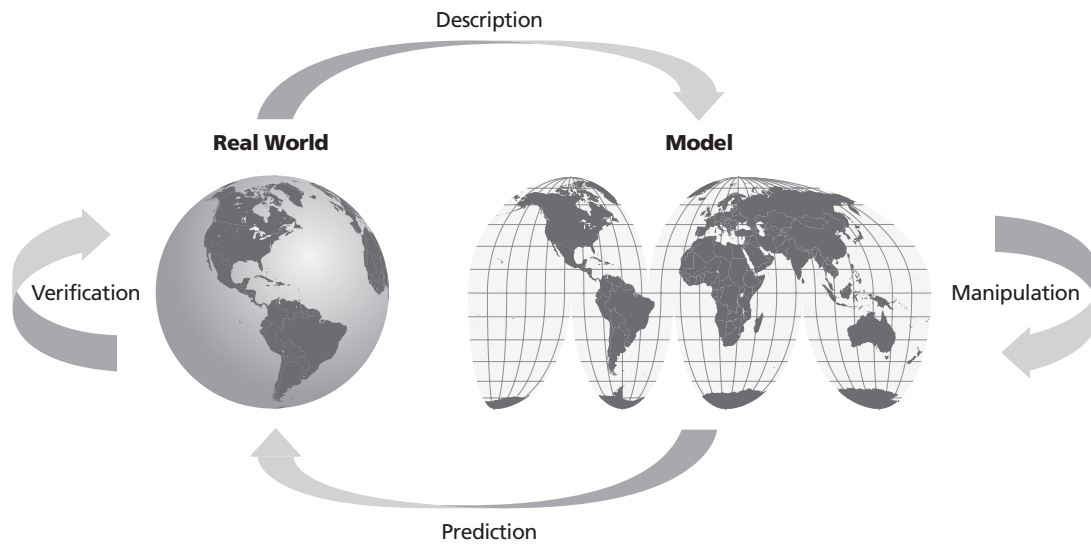


Figure 17.2 Modeling cycles. (Adapted from Lesh & Doerr, 2003b, p. 17).

number of trial solutions that result in the need to sort out, integrate, or discard models (Lesh & Yoon, 2004). Also, trial solutions tend to be expressed using a variety of representational media—including spoken language, written symbols, graphs and graphics, and experience-based metaphors. Each of the representations emphasizes and de-emphasizes somewhat different aspects of the problem-solving situation (Lesh & Doerr, 1998) and therefore also needs to be sorted out, integrated, or discarded. Another source of multiple interpretations results when a variety of technological tools are available, in which multistage solutions tend to require a good deal of planning, monitoring, communicating, and assessing processes (Hamilton et al., in press). Further, when students interpret situations mathematically, their interpretations go beyond logic and mathematics to include feelings, dispositions, values, and beliefs (Goldin, in press), resulting in a need to consider competing models from a dispositional point of view. The breadth and depth of this cyclic process has been described as *making practice mathematical*, which means that the full process of modeling as problem solving is a mathematical practice that needs to be learned through experience. “Making practice mathematical” is in contrast to “making mathematics practical”—which describes the role of traditional story problem in school mathematics (Lesh, Yoon, & Zawojewski, in press).

In contrast to the full modeling situation depicted in Figure 17.2 and described above, traditional story problems generally provide information that is given in prequantified forms. The mathematical question usually requires only short, definitive mathematical

answers, rather than the development of a tool (e.g., a procedure, or algorithm) for making decisions based on mathematical analyses and results. Story problems tend to include words that are carefully selected to make apparent the mathematical procedure required to solve the problem, thus the above description—“making mathematics practical.” Because the “givens” and “goals” in traditional story problems already occur in premathematized forms, the most problematic aspect involves making meaning out of symbolically described situations. (In contrast, model-eliciting activities require students to create mathematical descriptions of meaningful situations.) Thus, traditional story problems begin in the modeled world with word-based symbolic descriptions in which the key characteristics have already been quantified. Then the word-based symbols are translated to a mathematical equation, computation, or procedure, which is manipulated to yield a mathematical (symbolic) answer, still not leaving the modeled world. If and when students “check their answer for reasonableness,” they may map their answer back to the real world. In as much as the problem-solving process begins in the modeled world, upon completing the solution process, students will have only engaged in at most one quarter to one half of a modeling cycle.

An Evolving View of Problem Solving

Just as students’ solutions to model-eliciting activities go through cycles of testing and revision, researchers’ views of problem solving also can be expected to go through cycles of revision over time. For example, Lesh and Doerr (2003b) described the evolution of

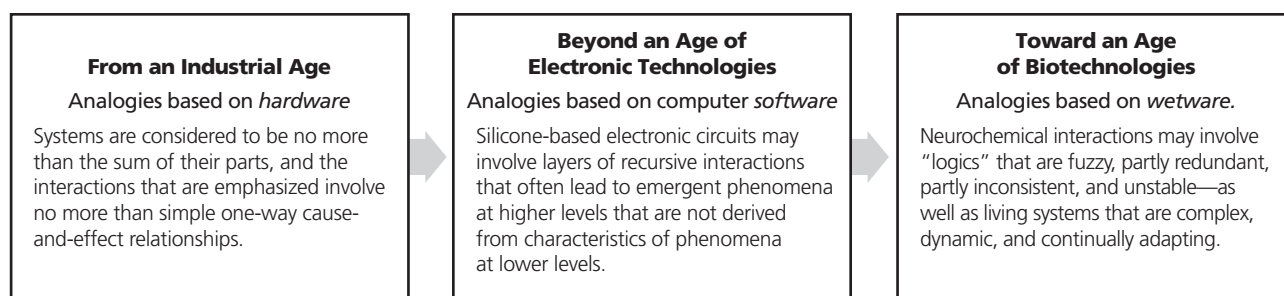


Figure 17.3 Recent transitions in models for making (or making sense of) complex systems. (Adapted from Lesh and Doerr, 2003b, p. 30.)

views of learning and problem solving by using an analogy from industry. Figure 17.3 illustrates how the industrial age, or “hardware” age, was characterized by assembly-line and machine metaphors, reflecting an assumption that the whole is simply a sum of the parts, and that causal relationships are simple one-way connections. Behaviorist models of learning mathematics (e.g., Gagne, 1975; Gagne & Briggs, 1974) reflect these assumptions, in which problem solving is viewed as a linking together of pieces of prerequisite knowledge to solve a well-specified problem.

The subsequent age of electronic technologies, or the “software” age, introduced the assumption that more complex systems involve layers of recursive interactions, which in turn involve information management schemes (related to storage and retrieval) as well as metacognitive processes that operate on the lower order skills. The analogy emphasized learning theories that were based on machine metaphors for learning. For example, Newell and Simon (1972) described their theories of learning and problem solving on the basis of their computer simulations of human problem-solving behavior. The goal of their work was to develop a global theory of problem solving that would account for problem-solving performance independent of the task domain (Lester & Kehle, 2003).

More recent descriptions involve metaphors aligned with biotechnologies, or an age of “wetware” (e.g., neurochemical interactions). These descriptions of learning and problem solving introduce additional complexities: logics that are “fuzzy,” partly redundant, frequently inconsistent, and often unstable. Systems-as-a-whole are assumed to have properties, referred to as *emergent properties* of the system, that cannot be directly derived from the system’s constituent elements. The wetware analogy requires an assumption that the systems involved are highly complex, dynamic, and continually adapting. Therefore scientists (i.e., educational researchers) design situations that lead to the development of living organisms (i.e.,

cognitive systems) that would not have been likely to evolve naturally. Rather than begin with inert “things” from which life is conjured up, the scientists begin with living systems and cultivate new kinds of living systems. When applied to human learning, problem solving can be modeled as conceptual systems that go on living and functioning, in contrast to a hardware analogy that involves condition-action rules in which systems only “run” when they are triggered by specified conditions.

The assumptions underlying the wetware metaphor are highly reminiscent of Papert’s (1991) description of a classroom learning community as a teaming community of conceptual systems (or models for making sense of experience)—each of which is competing for survival in the minds of the resident students, or community of students. In such an environment, survival of the fittest means the survival of models that are useful (for immediate purposes), reusable (for other purposes), sharable (with other people), and durable. Clearly the wetware metaphor suggests the need to consider new definitions, issues, and directions in research on mathematical problem solving and learning.

Promising Lines of Research on Mathematics Learning

Research on mathematical problem solving may be migrating to other areas of research, under different names. An explanation offered by Lester (1994) and Lester and Kehle (2003) was that the introduction of the various *standards documents* by the National Council of Teachers of Mathematics (1989, 1991, 1995, 2000) may have reduced the emphasis on problem-solving research per se by introducing three additional process standards as important for school mathematics: connections, communication, and reasoning. Another explanation offered by Lester (1994) was that constructivism “has replaced problem solving

as the dominant ‘ideology’ driving mathematics education research” (p. 668). He pointed out that some researchers were using the terms *problem solving* and *constructivism* interchangeably and questioned the appropriateness of doing so.

We offer a third possible explanation: a shift from viewing problem solving as a *thing* toward viewing it as related to *a large category of things* that need to be learned. Consider Dark’s (2003) description of the skills that are needed for high-performance workplace environments such as engineering and design—and her conclusion that what people need to know about problem solving goes far beyond lists of rules, strategies, or beliefs. Instead, *models* for describing experience are important goals of mathematics teaching and learning, because in an information age, in knowledge economies, in learning organizations, and in global societies, people need to describe and explain the systems within which they operate. Given a view of problem solving as multidimensional and complex, we have identified three areas of current research that are especially relevant to establishing future research directions for mathematical problem solving advanced in this chapter. These are the areas of situated cognition, communities of practice, and representational fluency. Each of these research areas is discussed in the following three sections as they relate to a models-and-modeling perspective on problem solving.

Research on Situated Cognition

Situated cognition refers to learning and problem solving in context (e.g., Greeno, 1998; Greeno & the Middle-School Mathematics Through Applications Project Group, 1997). Studies investigating situated cognition have ranged from those focusing on everyday tasks to those focusing on high-performance work environments that require heavy use of sophisticated mathematics. In everyday contexts, such as street vending (Saxe, 1988a), grocery store shopping (Lave, 1988; Lave & Wenger, 1991), and others (Boaler, 2000; Carraher et al., 1985; Nunes, Schliemann, & Carraher, 1993; Saxe, 1991), significant types of mathematical thinking emerge in response to local problems, and the mathematics that is created to meet local needs tends to be quite different than the knowledge and abilities learned in school. For example, in school-learned mathematics, students’ abilities and ideas tend to be organized more around abstractions corresponding to the topics emphasized in textbooks, whereas the conceptual systems that people develop to make sense of real-life experiences tend to integrate ideas and experiences drawn from a variety of textbook topic areas and often making use of a vari-

ety of representational media (Greeno, 1991; Noble, Nemirovsky, Wright, & Tierney, 2001).

Similarly, in environments requiring heavy use of substantive mathematics, such as engineering (Gainsburg, 2003a, 2003b, 2006; Parker, 1998), banking (Noss & Hoyles, 1996), nursing (Hoyles, Noss, & Pozzi, 2001; Noss et al., 2002), and other technology-based workplaces (e.g., Magajna & Monaghan, 2003), the users of mathematics tend to organize their mathematical ways of thinking around situations and problem contexts. A main difference from everyday tasks is that these situations often require users to draw on powerful industry-based mathematical models and procedures learned in professional settings (e.g., professional collegiate courses, internships, on the job). However, the problems encountered in these environments frequently do not fit the assumptions underlying the relevant conventional mathematical approaches and therefore are actively adapted or discarded and recreated to meet changing and novel conditions.

Overall, the research from situated cognition consistently demonstrates that nearly all people who are engaged in solving problems in their local contexts are able to develop mathematical concepts and conceptual tools for problem situations that are powerful and reusable. However, these very same people often do not recognize how the mathematics they are using, adapting, or creating is related to the mathematics they learned in school (Gainsburg, 2003b; Lave, 1988). Their perception is that the mathematics used in the real world is significantly different from the corresponding concepts and procedures emphasized in school. Our contention is that the mathematics is indeed different than that learned in school.

While studies in situated cognition have illuminated distinctions between the nature of mathematics needed in “real world” activity and the nature of mathematics typically taught in school, these studies also raise questions about the role of transfer of knowledge to new situations: If people respond to tasks by developing mathematical procedures and models that are situated to the current context, how does one generalize or transfer knowledge? From a models-and-modeling perspective, transfer is based on the idea that in the real world, people seldom take the trouble to develop generalizable conceptual tools if the problem they encounter will only need to be solved once. On the other hand, if the problem requires that a tool be devised to be used more than once or is to be shared with other people, then the problem solver will need to devise a mathematical tool in a way that will move it beyond task-specific knowledge. Consider a problem for which a person needs to develop routines that reduce large classes of tasks to situations that no longer

are problematic. When the routines are initially developed to be adapted for large classes of tasks, transfer takes place as the problem solver revises and tests the earlier procedures to meet the new constraints. However, once the routines are established, tasks that were once problematic become little more than exercises in following rules.

The ability to flexibly adapt existing knowledge to new conditions is especially important in environments where technological tools are advancing so rapidly that “the knowledge students gain in college may be considered obsolete within only a few years” (Oakes & Rud, 2003, p. 224). For example, in fields such as industrial engineering, problem solvers are continually encountering and creating *new* kinds of systems for new kinds of problems in their work environments and therefore need to continually adapt or create tools and models to meet the new challenges (Gainsburg, 2006). Further, emerging fields such as nanotechnology, logistics, biotechnology, and computing are places where professionals need to *anticipate* advances and prepare for the future (National Academy of Engineering, 2004).

Lesh, Cramer, Doerr, Post, and Zawojewski (2003) suggested that engaging students in sequences of structurally related, situated modeling activities can facilitate transfer—when discussions and explorations focus on the structural similarities among related mathematical models. The modeling sequences begin with a model-eliciting activity (described earlier), in which groups of students develop their own mathematical model to meet a client’s needs for a specified purpose. The initial model-eliciting activity is followed by a model-exploration activity in which students are asked to think *about* the model they have produced and other competing models. The final model-adaptation activity provides opportunities for students to adapt their own, or another model recently explored, to a new situation.

An illustration is drawn from the Small Group Mathematical Modeling Project for Improved Gender Equity in Engineering⁹ (SGMM Project; Zawojewski, Diefes-Dux, & Bowman, under development). A modeling sequence was designed using the context of improving the manufacture of the material used to form the surface of artificial hip replacement joints (Diefes-Dux, Hjalmarson, Miller, & Lesh, under development;

Diefes-Dux, Moore, Zawojewski, Imbrie, & Follman, 2004; Hjalmarson, Diefes-Dux, & Moore, under development; Moore & Diefes-Dux, 2004;). The roughness of the surface determines how well the hip replacement moves within the hip socket, providing motivation for the development of a mathematical model for determining a measure of roughness.¹⁰ Students were given images of surfaces of material at an atomic level along with a scale that could be used to estimate the heights of the various molecules pictured. The problem was written to require students to design a procedure for measuring roughness that could be applied to other images of materials and thus required generalization. Teams of 4 students were given 1 hour to complete the problem, and their resulting mathematical model was used in the subsequent model-exploration activity.

The model-exploration activity introduced a few conventional mathematical models for measuring roughness in various disciplines. The model-exploration homework activity required students to think *about* the mathematical models by comparing and contrasting the conventional models to each other and to their own models. They used each model to measure roughness of selected surfaces and then identified the relative trade-offs and constraints involved in each model. The final model-adaptation activity was a major project that required the 4-student teams to recast one of the models for measuring roughness as a computer program that would read digitized images of surfaces (where the gray scale of each pixel was quantified to represent a particular height). As an output, the computer would produce the measure of roughness for that image. Because the computer program developed by the students could actually be run on digitized sample images, students and their instructors were readily able to assess progress and final products by running their trial programs on various nano-scale images of materials, which naturally prompted subsequent cycles of revising and again testing their model (i.e., the computer program).

The theoretical framework for model development sequences draws heavily on Dienes’s (1960) theory of multiple embodiments for developing mathematical concepts. Dienes emphasized that students need to go beyond thinking with a given embodiment (i.e., model) to also thinking *about* it. To do so, stu-

⁹ This was a 4-year NSF-funded project at Purdue University (Award #0120794-HRD) whose goal was to enhance the engineering education experience for both females and males by using mathematical modeling activities. The experiences were designed to address important goals set out by the Accreditation Board for Engineering and Technology ([ABET], 2004) that call for engineering programs to provide more real-world, design-based, team-oriented tasks in order to help students learn what engineering work is like and to help them learn more about the professional work environment of engineering.

¹⁰ This “roughness” task is similar to the “ness” tasks designed by Judah Schwartz and colleagues (Balanced Assessment for the Mathematics Curriculum, 1999) for high school students.

dents need to encounter a concept represented in multiple embodiments (i.e., models) so that they will not solely attend to irrelevant features that are inevitably embedded in specific embodiments. By using multiple embodiments to represent a concept, students begin to recognize the general, abstract concepts that the various embodiments are intended to convey. Similarly, by creating, adapting, and comparing several structurally similar embodiments of a mathematical model, students have opportunities to compare and contrast models, to think about the similarities and differences among them, and to investigate the relationships among alternative models. The generalizing of knowledge from the initial situation to an unfamiliar, yet somewhat close situation is what constitutes *transfer* from a models-and-modeling perspective.

Model development nearly always involves situated cognition: Models are developed for a specific purpose, for specific people, and for specific situations. Nonetheless, model development generally involves much more than task-specific knowledge, because it is seldom worthwhile to develop any significant type of model unless it is intended to be sharable (used by other people) or reusable (in other situations), and therefore generalizable. Situated learning can indeed transfer, when the tasks for which the knowledge is generated require the development of sharable and reusable conceptual tools.

Research on Communities of Practice

Research on *communities of practice* has been closely associated with research on situated cognition. In addition to emphasizing the fact that knowledge is contextually situated, the notion of communities of practice emphasizes the fact that knowledge is also socially situated. Although an emphasis on the social aspects of understanding is not new (Mead, 1962, 1977; Wenger & Snyder, 2000), the resurgence of social perspectives has been fueled, in part, by the increasing expectation in high-performance workplaces, such as engineering, where teams of specialists need to work together in the development of satisfactory solutions to complex problems. Therefore, the ability of individuals to function well within a group is a highly prized attribute of employees (e.g., ABET, 2004), and the characteristics of well-functioning groups also are important to understand and develop. Thus, the “problem solver” is often the *group*, where a diversity of powerful technological and conceptual tools is brought to the solution process, and the trial solutions posed go through cycles of testing and revision.

Another reason for the renewed interest in social perspectives on understanding is that the development of learning theory has increasingly drawn on so-

cial aspects to explain how learning occurs. Whether learners act individually or as a group, the assumption is that an essential mechanism for moving the learner (group or individual) beyond current ways of thinking is through the interaction of a *variety* of alternative conceptual systems that are potentially relevant to the interpretation of a given situation.

Consider early research on problem solving, which was largely based on information-processing learning theories (e.g., Newell & Simon, 1972), which tended to rely on computer simulations to model thinking, which generally focused on the learning of individuals in isolation, and which generally treated the mind as if it were disembodied from context. More recent research on situated cognition, however, has provided the stepping stone back to social perspectives, because explanations of how individuals learn (i.e., how their cognitive systems evolve beyond current ways of thinking) have been shown to depend heavily on both cultural capital and interactions among people (e.g., Boaler & Greeno, 2000; Greeno, 2003; Lave & Wenger, 1991; Wenger, 1998; Yackel & Cobb, 1996; Zawojewski, Lesh, & English, 2003).

Although the work of Mead (1952, 1967) and other early American Pragmatists (Thayer, 1982) established many of the most distinctive characteristics of modern research on sociocultural perspectives, Vygotsky’s (1978) notion of *zone of proximal development* has more often been used to form the basis for modern theories of learning based on social perspectives. Although originally focusing on how a teacher questions the student, Vygotsky’s ideas have been adapted to also consider how student-to-student interactions facilitate learning. Thus, the role of an expert (e.g., a teacher, a tutor, a mentor) has varied in research that draws on a social perspective. One line of research emphasizes learning in an apprenticeship environment (e.g., Boaler & Greeno, 2000; Greeno, 2003; Lave & Wenger, 1991; Wenger, 1998;). Lave and Wenger (1991) coined the phrase “legitimate and peripheral participation” to describe the initial role of apprentices who join a community of practice. In these apprenticeships, the novices do a lot of observing of the experts at work, are given jobs that are “within their zone of proximal development,” and work side-by-side with experts, gradually internalizing what was originally only externally observed.

Another line of research emphasizes peers learning from each other under the guidance of a teacher, tutor, or mentor. In these problem-solving situations, groups of students (pairs, small groups, whole classes) may be viewed as communities of practice that are engaged in a task that is considered to be too challenging for any one individual. Consequently, the role

of the teacher/tutor/mentor is considered critical to guiding the group experience. For example, one role for the teacher in mathematical problem solving is to ensure that the learning community develops productive sociomathematical norms (Yackel & Cobb, 1996) in order to optimize learning via student-to-student interactions. In another example, one of the tutor's roles in medical problem solving is to help members of the team assign responsibilities (e.g., researching a particular topic, preparing a spreadsheet for the group to use) to individuals, which are to be accomplished prior to their next meeting (Koschmann, Glenn, & Conlee, 2000). In both examples, the assumption is that group-based discourse sets up optimal opportunities for individuals' ideas to be challenged within their zone of proximal development, leading to the further development of those ideas. Instead of solely depending on teacher-student interaction to prompt learning, the main learning is thought to occur through student interaction, as the teacher or tutor orchestrates the experience and ensures that the intended learning takes place.

A third line of research focuses on learning and problem solving in fields like business or engineering, where diverse teams of problem solvers often function as communities of practice in which the role of the teacher/tutor/mentor disappears (e.g. Cook & Yanow, 1993; Wenger, 2000; Wenger & Snyder, 2000; Yanow, 2000;). For example, in new innovative industries (e.g., information technologies, nanotechnology), where members of the workplace community represent a diversity of expertise, the major success of the business depends on the organization-as-a-whole's ability to develop expertise as a community—by adapting to new technologies and innovations, new consumer demands, an evolving workforce, and so on. The lack of a “teacher” or “mentor” in these new industries is in contrast to the traditional physical trades (e.g., carpentry, plumbing) where expertise is passed on from senior members to junior. A challenge for these new companies is that management cannot simply “will” diverse communities of practice to work together productively. Traditional “authority figures” can bring together people who have the desired and varied expertise, and who have interest in a given joint enterprise (i.e., a problem), but the authority figure does not orchestrate the work of the group. Instead, as the diverse members of the group work together, needed leadership roles often must emerge flexibly from within in response to the emerging challenges and opportunities.

If a goal of school mathematics is to prepare students for business environments, how does flexible, as-needed leadership come from within a group, rather

than from an external authority figure? How does the role of the classroom teacher change in order to nurture the development of leadership from within the group? How can leadership internal to the group be fostered in classroom settings?

Regardless of the role of a teacher/tutor/mentor, a major contribution made by research in communities of practice is that the learner is not assumed to know *nothing* when encountering a learning or problem-solving situation. Instead, learners generally are thought to bring *some* understanding to the table. Then, interactions among group members provide opportunities for individuals' understandings to be tested, integrated, differentiated, extended, revised, or rejected. Therefore the knowledge (or the solution to the problem) that emerges is viewed as *developing* rather than as being in a state of *learned* versus *not learned*, or *solved* or *not solved*.

Important research questions emerge from this developmental perspective: Along what dimensions does knowledge develop? For example, how does knowledge externally encountered in social problem-solving settings gradually become internalized? How do groups and individuals recognize the need to develop beyond current ways of thinking? What are characteristics of productively functioning groups? How do productive roles of individuals change throughout multistage learning or problem-solving processes?

Some answers to such questions are emerging from transcripts involving small-group problem-solving episodes that are simulations of reasonably complex and realistic decision-making situations. Such studies reveal that successful solutions are typically attained when the group goes through cycles of expressing, testing, and revising their solutions (Lesh & Doerr, 2003a). Furthermore, because significant new ways of thinking often develop within relatively brief periods of time, researchers can often directly observe processes that contribute to local conceptual development (Lesh & Harel, 2003). For example, it certainly is possible to observe students taking on and considering other students' perspectives, comparing and contrasting them to their own, and as a result revising their own ways of thinking about a problem solution. This function of internalizing what was external is one dimension of local concept development. Various other dimensions of knowledge development can also be studied in these contexts, including evolution in thinking from concrete to abstract, from case-specific to general, from simple to complex, or from a collection of uncoordinated, immature ideas to more coordinated, mature, and effective knowledge.

Notice that the focus of the preceding research questions and observations is on the development of

knowledge more than on the development of *individuals or groups*. Shifting beyond studies of human development toward studies of *idea development* has many practical advantages for mathematics educators. In particular, the researcher can study how *mathematical ideas* evolve in the community by observing the evolution of a concept, idea, or model that is externalized during the group interaction (through talking, note-taking, picture-drawing, etc.). Even though three-person teams may sometimes function in ways that are different than isolated individuals, the opportunity to study problem solving rather than problem solvers emphasizes *idea* development over *child* development. In essence, researchers can use studies of problem-solvers-who-are-a-group to help them understand problem-solvers-who-are-isolated-individuals in much the same way that other researchers use studies of experts to help understand novices—or studies of gifted students to understand average ability students. In each case, no assumptions need to be made that problem-solvers-who-are-groups are no different than problem-solvers-who-are-individuals, or that experts or gifted students are no different than novices or average-ability students. The focus is on how mathematical ideas develop, and on mechanisms that prompt rethinking and revision of those ideas.

When Lesh and Yoon (2004) used simulations of complex real-life problem-solving situations to investigate both problem-solvers-who-are-groups and problem-solvers-who-are-isolated-individuals, they found that (a) regardless of whether the problem solver is a group or an individual, the problem solver's thinking often involves a community of alternative ways of thinking, and (b) the evolution of ideas that evolve from this community of minds often resembles the evolution of a community of living, interacting, and evolving biological systems, as described by Dawkins (1990). Regardless of whether the problem solver is a group or an individual, the mathematical model that develops often evolves by sorting out experiences, by adapting to new environments, and by inheriting mathematical traits from multiple lineages (i.e., the various individual ideas of group members). Specifically, for group problem solving, thinking in early phases is based on communities of constructs posed by individuals in the group, which are all most likely at intermediate or immature stages of development. For the model, or *idea*, to develop in productive ways, certain kinds of Darwinian attributes and processes must be present, such as diversity (as described by Minsky, 1987), selection (i.e., survival of stable and productive systems), communication (i.e., spreading ideas throughout the community), and accumulation (i.e., preserving adaptations so they apply to new situations). Applying the wetware anal-

ogy (previously described) to a community of minds in problem solving raises new and interesting perspectives for research on problem solving and modeling. For example, in the classroom setting, the class as a whole might be characterized as an ecological system, where communities of constructs and conceptual systems are complex, dynamic, interacting, and continually adapting systems.

The study of knowledge development conducted with the small group as the unit of study benefits from the natural opportunities for internal ideas to be externalized via various representational media (e.g., talking, writing notes, drawing pictures, producing solutions upon which everyone in the group agrees). This is in contrast to studying idea development within an individual, which usually requires researcher probes and prompts in order to get students to reveal their thinking (and students often respond in an effort to please the researcher). In small-group problem solving, idea development often can be observed in "real time." Furthermore, concerns about the validity of data gathered from think-aloud interviews, retroactive self-reports, and surveys is avoided, because all data about the problem-solvers' thinking is generated naturally during the problem-solving process (using field notes, audiotapes, and videotapes).

Research on Representational Fluency

Representations and the tools to produce them are among the most important artifacts that students project into and encounter in the world (Steffe & Thompson, 2000; Lesh & Doerr, 1998). Researchers' interest in representational capabilities is growing as new kinds of representational forms are constantly being introduced with the ever-increasing use of computers and related devices for computation, conceptualization, and communication (Kaput, Hegedus, & Lesh, in press). In particular, newly developed types of expressive media emphasize the importance of graphics-oriented, dynamic, and creative representations (Johnson & Lesh, 2003), as well as multiple-linked representations (e.g., table, graphs, and equations) (Kaput, 1991). Furthermore, problems themselves often occur in multirepresentational forms and require a solution (i.e., a product or a conceptual tool) to be developed that involves a variety of media for the purpose of explanation, simplification, justification, or prediction (Lesh, 1987). Finally, representational fluency is important for professional uses of mathematics, where the problem solver often is a team of diverse specialists communicating from remote sites, and where resources and expertise tend to be distributed in different geographic locations. Thus, representational fluency is critical to enhancing the

communication capability and conceptual flexibility that are important to the development of solutions to many real-life problem-solving situations.

In most classrooms, students learn to adopt conventional mathematical language and symbol systems introduced by teachers, textbooks, and technology, with little attention to the dynamic and changing use of representations in the technological world. Further, Thompson and Sfard (1994) cautioned that unless special care is taken, the representational systems that students acquire in school may do little more than represent one another without serving as embodiments for students' internal conceptual systems, and without having the power to describe or explain other systems that occur in students' lives. Therefore, the representational media that students encounter in school need to be more than systems that the students look at; the representational media need to be systems that students develop and use, that is, that the systems become models that the students develop and use to predict, describe, and explain.

Supporting Lesh and Doerr's (2003b) claim that, when average-ability students clearly recognize the need for a specific type of conceptual tool, they often are able to develop (or at least significantly refine or revise) concepts that are both powerful and mathematically important, Lehrer and Schauble (2000) conducted a classroom-based study that engaged elementary students in the development of mathematical models that were intended to serve as classification systems. They found that young students were indeed able to develop and represent mathematical models, and they illustrated their findings using examples of first/second- and fourth/fifth-grade students who created their own languages, notation systems, diagrams, experience-based analogies, and other media to express their thoughts. Lehrer and Schauble's study, building on extensive previous work, not only establishes the possibility that young students can create and represent mathematical models, but also provides an example of research designed to produce a trail of documentation of students' modeling and problem-solving processes. Lehrer and Schuable did so by designing their studies to ensure that students externalize their thinking in representations, which in turn allows important insights to be made concerning the students' thinking, as well as the researchers' own evolving conceptual model of modeling and problem-solving capabilities of children of different age levels.

Another promising line of research looks at the local development of representational ability within the context of a problem-solving session. diSessa and colleagues (diSessa, 2002; diSessa, Hammer, Sherin, & Kolpakowski, 1991; diSessa & Sherin, 2000) have

provided a useful description of *metarepresentational competence*: the capabilities students have to construct, critique, and refine external representational forms. Rather than studying the specifically prompted use of standard classroom-taught representations (e.g., the Cartesian coordinate system, or representations of functions), diSessa and colleagues study metarepresentational competence situated in a problem-solving episode where students *need* to develop a representation or *need* to revise a representation in real time—on the spot as they are engaged in the problem. Under these conditions, diSessa et al. (1991) have found that, with minimal intervention from a teacher or other authority, even novice problem solvers reveal local development of metarepresentational competence within the boundaries of the session. Further, they found that this development is most likely to occur if students talk with other students about their representations, which provides opportunities to test representations on peers—usually exposing the weaknesses of early representations. Through continual cycles of expressing, critiquing, and revising representations, students are able to improve their representation until it satisfies a perceived need.

One remarkable finding in this area of research is the lack of relationship between students' mastery of conventional classroom-taught representations and their metarepresentational capability (Izsák, 2003). diSessa et al. (1991) pointed out that much of what students' know about representation appears to exist independent of instruction. Additionally, diSessa (2002) noted that simply presenting one's work or explaining one's thinking has little influence on the development of metarepresentational capability. Instead, the critical feature of classroom activity that contributes to the development of metarepresentational capability is that the activity involves the production, inspection, testing, and revision of representations for a specific purpose. In other words, meta-representational capability involves modeling, and the researchers' descriptions of iterative cycles of expressing, testing, and revising representations are very much in line with the models-and-modeling perspectives posed in this chapter.

Directions for Future Research

Research on situated cognition, communities of practice, and representational fluency all point to new directions for research in not only mathematical problem solving, but other technology-rich domains such as engineering and science education. In particular, Lesh, Hamilton, et al. (in press) have drawn on these areas of research, examined the use of mathematics in technology-rich environments, and as a result proposed a

coherent set of research questions that can provide a basis for discussion about future directions for research on mathematics learning and problem solving:

- What is the nature of problem-solving situations in which some type of elementary-but-powerful mathematical (engineering, scientific) constructs and conceptual systems (i.e., models) are needed for success in a technology-based age of information?
- What levels and types of understandings and abilities are needed for success in these situations?
- How do these understandings and abilities develop? Along what dimensions? How are they integrated in students' conceptual models?
- How can development of these understandings and abilities be facilitated, as well as documented and assessed, in teaching and learning?

Important questions, such as these, need to be investigated through research—not simply resolved through political processes such as those that are used in establishing curriculum standards and standardized-test specifications. These research questions focus on the changing nature of mathematics and situations in which mathematics is used rather than focusing on the nature of students, human minds, human information-processing capabilities, or human development. Therefore, research teams need to include those with broad and deep expertise in mathematics and science to play significant roles in the research. Furthermore, these expert participants should include not only the creators of mathematics (i.e., “pure” mathematicians), but also heavy users of mathematics (e.g., “applied” mathematicians, scientists, engineers).

Addressing these research questions requires fresh perspectives that move away from an information-processing perspective introduced to problem-solving research more than 30 years ago by Newell and Simon (1972)—in which problem solving was described as a search in a problem space for applications of known methods to get from givens to goals. In many ways, this view of problem solving still continues to drive a large share of research on mathematical problem solving. Yet, most early proponents of an information-processing perspective now describe this view as inadequate. For example, Greeno (1998) has advocated a “situated perspective” in which he described the problem solving space is as “emerg[ing] in the process of working on the problem” (p. 7), rather than a search for already known procedures. Some of the

limitations of an information-processing perspective were recognized by Silver (1985) when he suggested that the field of mathematics education needed to go “beyond process-sequence strings and coded protocols” in research methodologies and beyond “simple procedure-based computer models of performance” to developing ways of describing problem solving in terms of conceptual systems that influence students' performance (p. 257).

The prevalent paradigms adopted in past research on mathematical problem solving have been based on an assumption that the way students learn to solve problems is to first acquire the mathematical knowledge needed, then acquire the problem-solving strategies that will help them decide which already known procedure to deploy, then acquire the metacognitive strategies that will trigger the appropriate use of problem-solving strategies and mathematical knowledge, and finally unlearn beliefs and dispositions that prevent effective use of problem-solving and metacognitive strategies, while also developing productive beliefs and affect. This perspective has led to theories of teaching problem solving in which virtually everything to be learned has been reduced to lists of simple rules or declarative statements (e.g., draw a picture; justify claims and prove conjectures; notice that many mathematical problems have multiple reasonable answers; ask yourself, “Why are you doing that?”).

In general, such theories separate concept development from the learning of problem solving by focusing on content-independent problem-solving processes that are assumed to function in relatively invariant ways across diverse sets of tasks and situations. Yet, results from research based on these preceding assumptions have been unimpressive. Consequently, the time seems right to consider alternative assumptions. Many of the more promising recent themes in research—such as those related to situated cognition, communities of practice, and representational fluency—seem to be converging to something that we, and colleagues, are calling a *models-and-modeling perspective* on mathematical learning and problem solving.

Taking a models-and-modeling perspective, and focusing on students' mathematical models and modeling activities, predisposes researchers to take on fresh perspectives on mathematics learning and problem solving. For example,

- Researchers who study students' models and modeling naturally take on integrated approaches to studying the co-development of mathematical concepts, problem-solving processes, metacognitive functions, and beliefs or dispositions.

- Researchers who develop models of students' modeling (thinking and learning) naturally draw on integrated approaches by using a variety of practical and theoretical perspectives—including cognitive and social perspectives, mathematical/epistemological perspectives, developmental psychology perspectives, and so on—to make predictions and explanations.
- Researchers who study how modeling and problem-solving capabilities are learned naturally view problem-solving processes developmentally, in much the same way as those who have investigated the development of mathematical concepts in topic areas such as early number conceptions, rational number concepts, and concepts in geometry, algebra, or calculus.
- Researchers who study students' models and modeling activities naturally gravitate to simulations of "real life" problem-solving situations (e.g., model-eliciting activities).
- Researchers who study the changing nature of mathematics in the real world are drawn to research sites that involve complex multistage projects in which the problem solver is often a diverse team of specialists who have access to powerful conceptual technologies.
- Researchers who study modeling problems assume that mathematical thinking involves creation and interpretation (description, explanation, communication) at least as much as it involves computation, deductive reasoning, and the execution of procedures.

Models-and-modeling perspectives on classroom practice move beyond exclusive reliance on explicitly teaching problem-solving and metacognitive processes that are derived from descriptions of experts' past behaviors. Instead of viewing a novice problem solver as "stuck" and in need of help in the search for a problem space for an appropriate procedure to apply, models-and-modeling perspectives emphasize the fact that students generally are not "stuck" but instead *do* have initial ideas that are relevant, albeit primitive, egocentric, and biased. The goal is to enhance students' ability to use, extend, refine, and develop those mathematical ideas that they *do* bring to bear on the problem they are solving. Thus, other research questions that become important are

- What kind of problem-solving strategies help students use ideas that they *do* have in their initial encounter with a problem-solving situation?

- How can students minimize the debilitating influences associated with the availability of several unstable interpretations of a problem (i.e., conceptual models, trial solutions)?
- How are competing interpretations (i.e., trial solutions, models) differentiated, reconciled, integrated, or discarded altogether?
- How are successively more complex and refined models (i.e., interpretations, trial solutions) gradually constructed?

Zawojewski and Lesh (2003) addressed these types of questions and suggested that the knowledge and use of problem-solving strategies is not black-and-white—learned versus not learned. We argued that, like other forms of mathematical knowledge, students' understanding and use of problem-solving strategies develop and are important components of concept development. Therefore, the role that conventional problem-solving strategies play in instruction needs to be rethought. To illustrate, we described how the problem-solving strategy, "draw a picture," actually plays out in many activities that simulate real-life problem solving. For example, as small groups engage in a model-eliciting activity, group members often use a variety of different forms of representations to explain and defend their early ways of thinking to their teammates. In their early experiences in this type of problem solving, both their interpretations and their representations are often undeveloped and primitive. However, over a series of problem-solving experiences, students often mature in illustrating and explaining their way of thinking to others—and become more proficient at using diagrams and other representations to explore, explain, and converse with others and themselves. They begin to recognize that pictures drawn during early phases of solving a problem may have some inaccuracies, but that with cycles of testing and revising their ideas, the evolving diagrams can lead to enhanced understanding. Therefore, the diagrams themselves are not "right" or "wrong" things to do in order to solve problems; rather, they serve a function of communicating with teammates (or oneself), and of helping to improve the mathematical interpretation as the group engages in cycles of expressing, testing, and revising their thinking. Therefore, the conventional problem-solving strategy, "draw a picture," may be viewed developmentally when considered in the context of communication, and, we propose that the heuristic "draw a picture" can be transformed into "seeking to better understand the situation by explaining your way of thinking to a peer, or a member of a different problem-solving group."

Models-and-modeling perspectives also emphasize the notion that knowledge develops along multiple dimensions. Following the lead of Piaget (e.g., Inhelder & Piaget, 1958; Piaget, 1952; Piaget & Beth, 1966; Piaget & Inhelder, 1975; Piaget, Inhelder & Szeminska, 1960), domain-specific mathematics education research has studied the development in students' understanding of many important mathematical constructs (e.g., early number, rational numbers and proportional reasoning, algebra). As a result, explicit descriptions of stages of concept development have been described for these topics, and have been useful for curriculum and software developers who wish to design textbooks that will guide students towards deeper understandings of important ideas. However, for research and development in problem solving, more complex views are required.

In most of the literature on the development of mathematical knowledge, researchers have studied idea development along only one dimension. For example, Piaget (e.g., Piaget & Beth, 1966) emphasized development from concrete operational systems of thought toward formal operational systems. Vygotsky (1978) emphasized development along the dimensions of external to internal cognitive functions. Bruner (1960) emphasized increasing representational fluency and the development of progressively more powerful representational media in which to express current ways of thinking. Models-and-modeling perspectives adopt more sophisticated conceptions of development based on the observation that when students (or groups) go through a series of modeling cycles in which they integrate, differentiate, revise, and refine their existing relevant ways of thinking, development seldom occurs along a single, one-dimensional, ladder-like sequence. Instead, development occurs along a variety of dimensions, and students' final interpretations often inherit characteristics from a variety of the problem solvers' early interpretations. This multidimensional and integrated view of development requires not only new frameworks for thinking about research on problem solving but also new methodologies and tools for investigating development.

The Need for New Research Tools and Methodologies

Researchers in mature sciences tend to devote significant portions of their energies toward developing tools for their own use. The first hallmark of a high-quality tool is that it does indeed focus on the construct in question; second, the tool is based on assumptions that are consistent with those attributed to the construct being investigated. In research on math-

ematical problem solving, such tools should range from the development of research sites (or criteria for identifying the sites) to tools that can be used for observations or measurements.

In contrast to research in more mature sciences, few tools exist for measuring constructs claimed to be important in mathematical problem solving. Consequently, researchers have often been forced to measure the problem solving and thinking of students, teachers, and other relevant "subjects" using tools based on assumptions and theories that are inconsistent with the theories and assumptions that the researchers consider to be reasonable. For example, Ridgway, Zawojewski, and Hoover (2000) described how norm-referenced standardized tests that are based on psychometric models for studying problem solving can fall far short of capturing students' problem-solving capabilities.

Understandings and abilities related to problem solving and modeling are especially difficult to measure for a variety of reasons. First, problem solving and modeling are about interpreting (e.g., describing, explaining) situations mathematically. Nontrivial problem situations always require more than a single level or type of interpretation. Furthermore, the very act of interpreting a situation tends to cause the problem solvers' relevant interpretation systems to change. Therefore, to engage students' interpretation systems is to change those systems. In addition, the student models that emerge as potential solutions in the problem-solving process are molded and shaped by the structure of the situation (i.e., problem, task) being interpreted, as well as by the conceptual structures that the problem solvers employ. Thus, what researchers can observe are interactions among the structuring abilities of students and the structure of the tasks, and the data consists of auditable trails of documentation that problem solvers generate as they go through multiple cycles of expressing, testing, and revising their ways of thinking. To study significant types of models and modeling, researchers need to examine interactions between how students structure their experience *and* how teachers and other activity designers structure the problem situations.

Such interactions are captured in multitiered design studies (Kelly & Lesh, 2000; Kelly, Lesh & Baek, in press; Lesh & Kelly, 2000; Lesh, Kelly, & Yoon, in press) and particularized by Zawojewski, Chamberlin, Hjalmarson, and Lewis (in press) in their application of design research to teacher development. Multitiered design studies are based on the assumption that every participant in an evolving complex system (the students, the teachers, the professional development provider, the researchers) will be designing and rede-

signing their models (i.e., solution, way of thinking, theory, set of principles) in response to information gathered, documented, and analyzed. The notion of *design* draws on the work of research engineers, who create, test, and revise designs rather than conduct classical scientific experiments to determine whether or not “something works.”¹¹ The notion of a *multitiered* study embraces the complexity of school mathematics environments in which students’ progress cannot be considered in isolation from teachers’ implementation, which in turn cannot be considered without understanding the teachers’ conceptual systems, which in turn cannot be considered without understanding the nature of the professional-development experience, and so forth. Given that a change in one part of a complex system can reverberate throughout the whole system, research on problem solving and modeling needs to involve all participants to fully understand the information gathered.

The use of thought-revealing activity at all levels (the students, the teachers, the professional-development providers, and the researcher) is critical to producing a trail of documentation. Thought-revealing activities are designed to reveal the participants’ thinking naturally, as a part of their own practice. For students, model-eliciting activities are designed to be thought revealing. Similarly, Doerr, and Lesh (2003) have described a number of teacher-level thought-revealing activities, such as producing a lesson plan, or an evaluation rubric.

Data from the student-level and teacher-level thought-revealing activities provides opportunities not only for researcher analysis, but also self-assessment of progress by the participants who subsequently make decisions about next steps, or subsequent experiences. Designing research studies requires a specific plan for producing a trail of documentation from each participant, which ensures that conceptual models brought to bear are revealed (at least partially), and that changes in the models over time can be detected. The result is that insights into the interacting development of student models, teacher models, and researcher models simultaneously informs the research *and* the subsequent development of the learning experiences at all levels.

Consider, as an illustration, Lesh’s multitiered design experiment at Purdue University’s Center for Twenty-First Century Conceptual Tools (TCCT). He organized three- to five-person teams of graduate re-

search assistants and faculty colleagues to work together in semester-long sequences of biweekly meetings in which participants collaborated to design thought-revealing (model-eliciting) activities for students. The activities, designed for middle-school students, were intended to be high-fidelity simulations of “real life” situations where mathematical thinking would be needed for success. As the activities were field tested at diverse sites, and as the designers developed tools to document and assess the mathematical understanding and abilities that were most critical for success in the problem-solving activities, the designers repeatedly expressed, tested, and revised their own current ways of thinking about the nature of mathematical thinking needed for success beyond school. Iterative rounds of field testing with students, consensus building among designers and researchers, and redesign of the tasks were built into the design experiment. The kinds of mathematical knowledge and abilities identified by the participants as emphasized in the activities were quite different from what is typically emphasized in traditional tests and textbooks used by the schools, and even what is emphasized in applied mathematics courses (Lesh, 2003). For example, quantifying qualitative information, sorting and selecting data to use when all data is relevant yet not exactly what is needed, and creating and designing mathematical procedures to process information illustrate the types of knowledge that were found to be critical for solving such problems. Thus, the participants learned that mathematical abilities emphasized in model-eliciting activities represent a larger range than those outlined in district-level, state-level, and even national-level documents.

A number of researchers have begun to use multitiered design studies to examine teachers’ learning in the context of teachers examining and facilitating their own students’ learning. Schorr and Lesh (2003) described a professional-development study in which teachers were asked to interpret their students’ work (on thought-revealing, model-eliciting activities) for the purpose of improving implementation and assessment of problem-solving performance. They found enormous changes in the teachers’ reflections that corresponded with improved performance on the part of students. Clark and Lesh (2003) described the use of concept maps as a reflective tool in a multitiered design study. Teachers were asked to create a concept map for each student-level model-eliciting activity they had implemented in their classroom. The purpose of

¹¹ Conducting a series of studies in an attempt to establish “what works” seldom leads to conclusive results. This is because if the investigation indicates that the something “did not work,” there exist so many alternative explanations for the lack of findings (e.g., poor implementation, student turnover, lack of resources) that the findings are explained away. Thus, the same research question concerning “what works” gets recycled again and again, resulting in a series of conflicting findings leading to inconclusive results.

the concept map was to provide other teachers with a curricular guide that would identify the concepts and skills elicited in student work on the particular activity, and to identify the state and district standards that were aligned with the expected student work on the problem. The development of the concept maps proved to be a nontrivial problem for teachers and was also thought revealing. Over professional-development cycles, teachers' concept maps evolved from literally describing components found in various standards documents to interpretations that grappled with the notion of *big mathematical ideas*, the differences in skills, concepts, and strategies, and eventually to deep discussions that identified concepts and connections between additive and multiplicative reasoning.

Schorr and Koellner-Clark (2003) conducted a case study of one teacher's development in the context of studying his own students' thinking as the students engaged in model-eliciting activities. Schorr and Koellner-Clark were able to avoid a common problem in which professional development results in teachers who can talk about "reform" but do not actually implement what they are talking about. In this case they documented how a teacher actually began to think *differently* about his student's mathematical thinking—which is essential for making permanent changes in perceptions about teaching and learning mathematics.

A number of recent dissertation studies have also been based on multitiered design experiments. For example, Carmona (2004) focused on developing assessment and evaluation tools for describing and scoring students' work. She organized her dissertation as a multitiered design study to document the teachers' and the assessment team's models for interpreting students' work. She found that as teachers and assessment team members worked on ways to document students' understanding, they noticed more and more about students' thinking.

Another example is Richardson (2004), who was interested in the diffusion of students' knowledge among peers, in contrast to the dissemination of teacher knowledge to students. She facilitated teacher development of mechanisms to enhance student diffusion of knowledge during small-group solutions of model-eliciting activities. For example, one of the teachers provided problem-solving teams with a place in the classroom to go to get needed data, providing an opportunity for cross-group interaction. The teacher expected that interactions across groups would result in solutions that converge along one path. Instead, Richardson and the teacher discovered that although students and groups appropriated the ideas of others, they fit the new ideas into their own line of reasoning.

They further developed their own ideas, sometimes eliminating ideas that they found to be useless, and sometimes creating totally new lines of reasoning.

Hjalmarson (2004) also used a multitiered design study in which teachers were motivated to develop "presentation tools" to enhance groups' class presentations of their solutions to model-eliciting activities. Each of the teachers expressed an interest in trying to get students to think more deeply about each other's mathematical models. The teachers each developed their own "tool" (e.g., peer feedback form, planned teacher questions) to enhance the presentation phase of the activity and then revised or changed the tool on the basis of what was learned over a series of model-eliciting activity presentations. The trail of documentation produced by the teachers provided a window into the teachers' ways of thinking about students' learning to present mathematical models to other students.

The main characteristics of multitiered design studies are (a) that activities used by both teachers and their students are designed to be thought-revealing and model-eliciting, (b) that the activities devised require teachers to design a product that is important to them, which results in iterations of expressing, testing, and revising one's own models, and (c) that teachers' models for learning and teaching problem solving are studied as teachers examine the models students design in response to a model-eliciting problem. An assumption underlying this paradigm is that not only will the students and the teachers change as a result of their experience, so too will the professional-development provider and researcher. The role of the latter two participants, then, is to prompt iterative modeling cycles by students and teachers, and *also* to document their own iterative cycles as they revise their own understandings of model development by teachers and students. Clearly, the type of research described is complex and would be enhanced by a cluster of companion studies that use different perspectives, focusing on different participants, and therefore introducing greater diversity of thought into the research process.

An Illustration: The Evolving Models-and-Modeling Perspective

One of the most significant distinguishing characteristics of models-and-modeling perspectives on mathematics learning and problem solving is the assumption that the theory, model, or research perspective being used by the researcher needs to be expressed, tested, and revised systematically as part of the research process—and that these researcher-level models develop along a number of dimensions simi-

larly to the way student-level and teacher-level models develop. In particular, research models should be expected to evolve not only within a particular study, but also over series of studies. For example, the models-and-modeling perspective being proposed in this chapter has gone through a number of major shifts in thinking, and corresponding revisions, based on studies over time. As a close to the chapter, two of the dimensions along which the proposed models-and-modeling perspective has evolved are used to illustrate how the researchers' model is under design while also guiding the work of the research. In particular, we share a glimpse of how the principles for developing model-eliciting activities emerged from early work in understanding the development of rational number ideas and applied problem solving and we share a glimpse of how the researchers' view of students' mathematical knowledge evolved.

Two projects implemented by Lesh and colleagues in the 1980s, the Applied Problem Solving Project (Lesh, Landau, & Hamilton, 1983) and the Rational Numbers Project (Lesh, Post, & Behr, 1985), launched the process of expressing, testing, and revising a perspective on learning and teaching mathematics, which later came to be called a models-and-modeling perspective. The goal of these research projects was to build on the work of Piaget (e.g., Piaget, 1952; Piaget & Inhelder, 1975; Piaget et al., 1960), who studied the evolution of mathematical cognitive structures over long periods of time. Although these projects were designed to do the same, they were designed to look at smaller nuggets—local concept development—over short periods of times, such as in one class period.

In the Applied Problem Solving Project, researchers used complex problem-solving tasks as research sites in which they videotaped small groups of students engaged in collaborative problem solving. One of the early tasks used was the Chicago-to-Tokyo problem, which required students to determine the distance from Chicago (their home town) to Tokyo on different representations of the world (a globe, a Mercator projection map—which is a rectangular map where land masses near the poles look very large, and an Interrupted Mollweide projection map with simplified lobes—which looks like a flattened-out orange peel). The small groups of students who were engaged in solving the problem were usually surprised to find that each map resulted in different distances. Their transcribed conversations were studied and revealed that in successful groups, although students' initial approaches to reconciling the differences could be characterized as a collection of independent, uncoordinated, egocentric ideas, over the course of the problem-solving session, as students tested their ideas

on each other, they began to take into account each other's perspectives. They gradually developed a coherent explanation of the discrepancies and decided together on a distance that made sense to them all. Although the process of expressing, testing, and revising their ideas was the goal of the task designed, the students' final mathematical model could only be inferred from the videotaped conversations. The researchers realized that if the problem-solving activities could be designed to capture the group's mathematical way of thinking as a final product, subsequent research studies would be strengthened. Thus, the first principle for task design emerged: that students' mathematical models would be documented in writing as the "answer" to the problem. (This was later called the "model-documentation" principle.)

As researchers grappled with the desire to have students' final written responses be more than the production of a numerical/unitized answer (e.g., the distance between Tokyo and Chicago), they encountered the challenge of designing activities that revealed students' mathematical models as *answers* to complex problems. As a result, a process of expressing, testing, and revising a full set of task design principles began. For example, given that a mathematical model is seldom required as a solution to an isolated problem situation, the model-eliciting activity design needed to require that the students' solution (i.e., model) be shared with other people or used on other sets of data. Thus, the generalizability principle (later called the sharability and reusability principle) was born and was included in an early set of design principles published in Lesh and Lamon (1992). Later iterations of the six principles for task design have been published by Lesh, Hoover, and Kelly (1993) and Lesh, Hoover, et al. (2000).

Another dimension along which models-and-modeling perspectives developed was how students' knowledge of mathematics was viewed. In both the Applied Problem Solving Project and the Rational Numbers Project, activities were designed to reveal information about students' understanding of conventional mathematical topics (e.g., students' rational number conceptual system, students' proportional reasoning system). However, the task designers learned that the more realistic the model-eliciting activity, the more students brought unexpected and contextually based mathematical ideas to bear on the problem. Hind-sight is always so clear—because realistic modeling problems almost always have more than one reasonable solution students were bringing their own conceptual systems to bear, and these conceptual systems were largely situated in their own experience and were often multitopic with respect to conventional mathe-

matics. As a result, the researchers' interpretations of students' mathematical models became less oriented around conventional mathematics topics and more associated with the *local conceptual models* students developed for the given problematic situation.

The models-and-modeling perspective on mathematical problem solving and learning has evolved along a number of dimensions and has been developed by a cadre of researchers. Organizing the study of mathematical problem solving around mathematics—as the study of systems, and models of those systems—can result in the simultaneous enhancement of learning and instruction while also increasing the field's understanding of what constitutes mathematical problem solving and learning. Future research needs to embark on investigation of the many questions posed in this chapter, to involve an ever-increasing community of experts who can provide the needed conceptual shift in the field, and to continue to grow and develop the notion of a models-and-modeling perspective on research in mathematical problem solving.

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TOWARD COMPREHENSIVE PERSPECTIVES ON THE LEARNING AND TEACHING OF PROOF

Guershon Harel

UNIVERSITY OF CALIFORNIA, SAN DIEGO

Larry Sowder

SAN DIEGO STATE UNIVERSITY

One of the most remarkable gifts human civilization has inherited from ancient Greece is the notion of mathematical proof. The basic scheme of Euclid's Elements has proved astoundingly durable over the millennia and, in spite of numerous revolutionary innovations in mathematics, it still guides the patterns of mathematical communication.

—Babai, 1991, pp. 1–2

This chapter examines proof in mathematics, both informal justifications and the types of justification usually called mathematical proofs. The introductory section below calls for what we label a “comprehensive perspective” toward the examination of the learning and teaching of proof, and identifies the various elements of such a comprehensive perspective. Our viewpoint next centers on students’ outlooks on proof, as described by the “proof schemes” evidenced in students’ work; the second section elaborates on this proof-scheme notion and includes a description of various proof schemes as

well as a listing of the various roles proof can play in mathematics. The third section then gives a brief overview of how the idea of proof in mathematics has evolved historically, and why historical considerations could be a part of educational research on the learning or teaching of proof. The fourth and fifth sections include a look at selected studies¹ dealing with proof, at both the precollege and the college levels, with an effort to show the value of the proof-schemes idea. The final section offers some questions prompted by the earlier sections.

¹ To assure a degree of quality control and for practical reasons, we have restricted our survey to writings that have undergone an external review process, except in the case of a few doctoral dissertations that we examined. Examination of at least one case—the teaching and learning of mathematical induction—was omitted here since it has been treated thoroughly elsewhere (Harel, 2001). We regret that no doubt we have unintentionally overlooked other pieces of research or commentary that could have improved this chapter.

COMPREHENSIVE PERSPECTIVE ON PROOF²

No one questions the importance of proof in mathematics, and in school mathematics (Ball & Bass, 2003; Haimo, 1995; Schoenfeld, 1994). Overall, the performance of students at the secondary and undergraduate levels in proof is weak, as the findings reported in this paper will show. Whether the cause lies in the curriculum, the textbooks, the instruction, the teachers' background, or the students themselves, it is clear that the status quo needs, and has needed, improvement. Earlier reviews of research on the teaching and learning of proof (e.g., Battista & Clements, 1992; Hart, 1994; Tall, 1991; Yackel & Hanna, 2003) have informed and inspired more recent studies of proof learning. This chapter argues for "comprehensive perspectives" on proof learning and teaching and provides an example of such a perspective. A comprehensive perspective on the learning and teaching of proofs is one that incorporates a broad range of factors: mathematical, historical-epistemological, cognitive, sociological, and instructional. A unifying and organizing element of our perspective is the construct of "proof scheme." This construct emerged from a long sequence of studies into the concept of proof (Harel & Sowder, 1998), and, in turn, was utilized to address a cohort of foundational questions concerning these factors:

Mathematical and Historical-Epistemological Factors

1. What is proof and what are its functions?
2. How are proofs constructed, verified, and accepted in the mathematics community?
3. What are some of the critical phases in the development of proof in the history of mathematics?

Cognitive Factors

4. What are students' current conceptions of proof?
5. What are students' difficulties with proof?
6. What accounts for these difficulties?

Instructional-Socio-Cultural Factors

7. Why teach proof?
8. How should proof be taught?
9. How are proofs constructed, verified, and accepted in the classroom?

10. What are the critical phases in the development of proof with the individual student and within the classroom as a community of learners?
11. What classroom environment is conducive to the development of the concept of proof with students?
12. What form of interactions among the students and between the students and the teacher can foster students' conception of proof?
13. What mathematical activities—possibly with the use of technology—can enhance students' conceptions of proof?

This list does not purport to be exhaustive or unique; other researchers may choose a different list of questions or formulate those that appear here differently in their attempt to form their comprehensive perspective to proof. This list, however, represents those questions that we have confronted in a decade of investigations into the learning and teaching proof; hence, these questions constitute an essential part of the *content* of our comprehensive perspective on proof. Nor is the classification of these questions as it is outlined here invariable. Clearly, it is difficult—if not impossible—to classify these questions into pairwise disjoint categories of factors. Our use of the labels indicates a loose association rather than a firm classification. In particular, none of the questions should be viewed as a stand-alone question; rather they are all interrelated, constituting a cohort of questions that have guided the comprehensive perspective on proof offered here as an example. Thus, even though the headings of the third, fourth, and fifth sections of the chapter correspond to the three categories outlined in the above list, the content of each section pertains to the other two.

An important character of our perspective is that of *subjectivity*: While the term "proof" often connotes the relatively precise argumentation given by mathematicians, in our perspective "proof" is interpreted subjectively; a proof is what establishes truth for a person or a community. With this interpretation, "proof" connotes an activity that can permeate the whole mathematics curriculum, from kindergarten on as well as throughout the historical development of mathematics. This subjective notion of "proof" is the most central characteristic of the construct of "proof scheme," which will be defined shortly. When we wish

² All aspects of proof addressed in this paper must always be understood in the context of the learning and teaching of proof. Even when we address mathematical, historical, or philosophical aspects of proof, the goal is to utilize knowledge of these aspects for the purpose of better understanding the processes of learning and teaching of proof. Thus, phrases such as "research on proof," "perspective on proof," and the like should always be understood in the context of mathematics education, not of mathematics, history, or philosophy per se.

to make clear that we mean the mathematically institutionalized notion of proof, we will write “mathematical proof,” otherwise the terms “proof” or “justification” will be used in this subjective sense.

Subjectivity equally applies to how the above questions should be understood. Some of these questions, particularly the first three, might be interpreted as calling for objective, clear-cut answers. Not so. The meaning of proof, its role, and the way it is created, verified, and accepted may vary from person to person and from community to community. One’s answers to these questions are greatly influenced by her or his philosophical orientation to the processes of learning and teaching and would reflect her or his answers to questions such as: What bearing, if any, does the epistemology of proof in the history of mathematics have on the conceptual development of proof with students? What bearing, if any, does the way mathematicians construct proofs have on instructional treatments of proof? What bearing, if any, does everyday justification and argumentation have on students’ proving behaviors in mathematical contexts? Our emphasis on subjectivity—a motive we will repeat a few times in this paper—stems from the well-known recognition that students’ construction of new knowledge is based on what they already know (e.g., Cobb, 1994; Piaget, 1952, 1973a, b, 1978; Vygotsky, 1962, 1978), and hence it is indispensable for teachers to identify students’ current knowledge, regardless of its quality, so as to help them gradually refine it. To avoid unnecessary misunderstanding we note here, and we will repeat later, that subjectivity toward the meaning of proof does not imply ambiguous goals in the teaching of this concept. Ultimately, the goal is to help students gradually develop an understanding of proof that is consistent with that shared and practiced in contemporary mathematics.

Why are Comprehensive Perspectives on Proofs Needed?

Comprehensive perspectives on proof are needed in an effort to understand students’ difficulties, the roots of the difficulties, and the type of instructional interventions needed to advance students’ conceptions of and attitudes toward proof. A single factor usually is not sufficient to account for students’ behaviors with proof. For example, while some of the difficulties students have with proof can be accounted for by the cognitive factor, such as the students’ lack of logical maturity and understanding of the need for proof, research studies conducted in the last two decades have given evidence that these sources are insufficient to provide a full picture of students’ dif-

ficulties with mathematics in general and proof in particular. Balacheff’s (1991) research, for example, has shown that students “have some awareness of the necessity to prove and some logic” (p. 176) and yet they experience difficulty with proof.

Official documents and research papers on the learning and teaching of proof almost never present their theoretical framework as a comprehensive perspective on proof, but stated curricular goals, instructional recommendations, research design, and so forth, may reveal some elements in the scholars’ (potential) perspective on proof. For example, Balacheff (1991) points out that common to mathematics curricula in different parts of the world is the goal of training students in the construction and the formulation of “deductive reasoning,” which is defined as “a careful sequence of steps with each step following logically from an assumed or previously proved statement and from previous steps” (National Council of Teachers of Mathematics [NCTM], 1989, p. 144). Balacheff points out that this characterization of proof, which is close to what a logician would formulate, is almost the same in mathematics curricula all over the world. Remarks in these documents on the conceptual development of proof with students stress that proof “has nothing to do with empirical or experimental verification” and “call attention to the move from concrete to abstract” (Balacheff, 1991, p. 177). This suggests a form-driven perspective, according to which the basis for one’s answer to the question, how should proof be taught, is the form in which mathematics is organized and accordingly presented in scientific papers or books.

Approaching the concept of proof from an encompassing perspective such as the one suggested here is, in our view, a critical research need. Balacheff, a world-recognized pioneer in the area of the learning and teaching of proof, has addressed this need by raising on several occasions (e.g., 2002) the question: “Is there a shared meaning of ‘mathematical proof’ among researchers in mathematics education?” Balacheff was not referring to just the standard, more or less formal definition of mathematical proof. Rather, his question, as he puts it, is “whether beyond the keywords, we had some common understanding” (Balacheff, 2002, p. 23). By “common understanding,” we believe he means agreed-upon parameters in terms of which one can formulate differences among perspectives into research questions. We agree with Balacheff that without such an understanding it is hard to envision real progress in our field. The comprehensive perspective on proof presented in this paper delineates a set of such parameters.

THE CONCEPT OF PROOF SCHEME

Consistent with our characterization of a comprehensive perspective on proof, several critical factors must be considered in addressing the question, What is proof? First, the construction of new knowledge does not take place in a vacuum but is shaped by existing knowledge. What a learner knows now constitutes a basis for what he or she will know in the future. This fundamental, well documented fact has far-reaching instructional implications. When applied to the concept of proof—our concern in this paper—this fact requires that an answer to the above question takes into account the student as a learner, that is, the cognitive aspects involved in the development of the concept of proof with the individual student. Second, one must maintain the integrity of the concept of proof as has been understood and practiced throughout the history of mathematics. Third, since the concept of proof is social—in that what is offered as a convincing argument by one person must be accepted by others—one must take into account the social nature of the proving process. We see, therefore, that one's answer to the above question attends to a range of factors: cognitive, mathematical, epistemological-historical, and social. The approach we provide here takes into consideration these factors. The conceptual framework of our answer has been formed over a long period of extensive work on students' conception of proof, and it incorporates findings reported in the literature as well as from our own studies. The latter included a range of teaching experiments and a three-year longitudinal study (see, for example, Harel & Sowder, 1998; Harel, 2001; Sowder & Harel, 2003) as well as historical, philosophical, and cultural analyses (e.g., Hanna, 1983, Hanna & Jahnke, 1996, Harel, 1999, Kleiner, 1991). The foundational element of this framework is the concept of "proof scheme."

Obviously, learners' knowledge, in general, and that of proof, in particular, is not homogenous; most commonly, a high-school student's conception of proof is different from that of a college mathematics major student, and a contemporary mathematician's conception of proof is different from that of an ancient mathematician, which, in turn, was different from that of a Renaissance mathematician, etc. "Proof scheme" is a term we use to describe one's (or a community's) conception of proof; it will be defined precisely below. In Harel and Sowder (1998) we offer an elaborate taxonomy of students' proof schemes based on seven teaching experiments with a total of 169 mathematics and engineering majors. The experiments were conducted in classes of linear algebra (el-

ementary and advanced), discrete mathematics, geometry, and real analysis. Later, in Harel (2006), this taxonomy was refined and extended to capture more observations of students' conceptions of proof as well as some developments of this concept in the history of mathematics. In this paper we use our taxonomy of proof schemes to describe or interpret

- (a) historical developments and philosophical matters concerning the concept of proof (third section),
- (b) results from other studies concerning students' conception of and performance with proof (fourth section), and
- (c) curricular and instructional issues concerning proof (fifth section).

Definition of "Proof Scheme"

The definition of "proof scheme" is based on three definitions.

1. **Conjecture versus fact.** An assertion can be conceived by an individual either as a *conjecture* or as a *fact*. (A conjecture is an assertion made by an individual who is uncertain of its truth.) The assertion ceases to be a conjecture and becomes a fact in the person's view once he or she becomes certain of its truth.

A critical question, to which we will return shortly, is the following: How in the context of mathematics do students render a conjecture into a fact? That is, how do they become certain about the truth of an assertion in mathematics?

The definitions of "conjecture" and "fact" are the basis for the notion of *proving*.

2. **Proving.** *Proving* is the process employed by an individual (or a community) to remove doubts about the truth of an assertion.

The process of proving includes two subprocesses: *ascertaining* and *persuading*.

3. **Ascertaining versus persuading.** *Ascertaining* is the process an individual (or a community) employs to remove her or his (or its) own doubts about the truth of an assertion. *Persuading* is the process an individual or a community employs to remove others' doubts about the truth of an assertion.

Mathematics as sense-making means that one should not only ascertain oneself that the particular

topic/procedure makes sense, but also that one should be able to convince others through explanation and justification of her or his conclusions. In particular, the convince-others, public side of proof is a social practice (e.g., Bell, 1976), not only for mathematicians but for all students of mathematics. The definition of “persuading” aims at capturing this essential feature of proving. As defined, the process of proving includes two processes: *ascertaining* and *persuading*. Seldom do these processes occur in separation. Among mathematically experienced people and in a classroom environment conducive to intellectual interactions among the students and between the students and the teacher, when one ascertains for oneself, it is most likely that one would consider how to convince others, and vice versa. Thus, proving emerges as a response to cognitive-social needs, rather than exclusively to cognitive needs or social needs—a view consistent with Cobb and Yackel’s *emergent perspective* (1996).

As defined, ascertaining and persuading are entirely subjective, for one’s proving can vary from context to context, from person to person, from civilization to civilization, and from generation to generation within the same civilization (cf. Harel & Sowder, 1998, Kleiner, 1991, Raman, 2003, Weber, 2001). Thus, we offer the following definition:

4. **Proof scheme.** A person’s (or a community’s) proof scheme consists of what constitutes ascertaining and persuading for that person (or community).

We repeat here what we highlighted in Harel and Sowder (1998): Our definitions of the process of proving and proof scheme are deliberately student-centered. Terms such as “to prove,” “to conjecture,” “proof,” “conjecture,” and “fact” must be interpreted in this subjective sense throughout the paper. We emphasize again that despite this subjective definition the goal of instruction must be unambiguous—namely, to gradually refine current students’ proof schemes toward the proof scheme shared and practiced by contemporary mathematicians. This claim is based on the premise that such a shared scheme exists and is part of the ground for scientific advances in mathematics.

Taxonomy of Proof Schemes

The taxonomy of proof schemes consists of three classes: the *external conviction proof schemes class*, the *empirical proof schemes class*, and the *deductive proof schemes class*. Below is a brief description of each of these schemes and some of their subschemes. Further elaboration of the different schemes is provided, as needed, throughout the paper. Relations of our tax-

onomy to other taxonomies and to the functions of proof within mathematics are addressed at the end of this subsection. (For the complete taxonomy, see Harel & Sowder (1998) and Harel (2006).)

External Conviction Proof Schemes

Proving within the *external conviction* proof schemes class depends (a) on an authority such as a teacher or a book, (b) on strictly the appearance of the argument (for example, proofs in geometry must have a two-column format), or (c) on symbol manipulations, with the symbols or the manipulations having no potential coherent system of referents (e.g., quantitative, spatial, etc.) in the eyes of the student (e.g., $(a+b)/(c+b) = (a+\cancel{b})/(c+\cancel{b}) = a/c$). Accordingly, we distinguish among three proof schemes within the *external conviction* proof schemes class:

- External conviction proof schemes class
 - Authoritarian proof scheme
 - Ritual proof scheme
 - Non-referential symbolic proof scheme

Empirical Proof Schemes

Schemes in the *empirical proof scheme* class are marked by their reliance on either (a) evidence from examples (sometimes just one example) of direct measurements of quantities, substitutions of specific numbers in algebraic expressions, and so forth, or (b) perceptions. Hence, we distinguish between two proof schemes within the *empirical proof scheme* class:

- Empirical proof schemes class
 - Inductive proof schemes
 - Perceptual proof schemes

Deductive Proof Schemes

The *deductive proof schemes* class consists of two subcategories, each consisting of various proof schemes:

- Deductive proof schemes class
 - Transformational proof schemes
 - Axiomatic proof schemes

All the *transformational* proof schemes share three essential characteristics: *generality*, *operational thought*, and *logical inference*.

The *generality* characteristic has to do with an individual understanding that the goal is to justify a “for all” argument, not isolated cases and no exception is accepted.

Evidence that *operational thought* is taking place is shown when an individual forms goals and subgoals and attempts to anticipate their outcomes during the evidencing process. Finally, when an individual

understands that justifying in mathematics must ultimately be based on logical inference rules, the *logical inference* characteristic is being employed (see also Harel, 2001).

Unlike the proof schemes in the previous two classes (the external conviction proof schemes class and the empirical proof schemes class), the transformation proof schemes require a more elaborate demonstration. Consider the following two responses (taken from Harel, 2001) to the problem:

Prove that for all positive integers n ,

$$\log(a_1 \cdot a_2 \cdots a_n) = \log a_1 + \log a_2 + \cdots + \log a_n.$$

Response 1

$$\begin{aligned} \log(4 \cdot 3 \cdot 7) &= \log 84 = 1.924 & \log(4 \cdot 3 \cdot 6) &= \log 72 = 1.857 \\ \log 4 + \log 3 + \log 7 &= 1.924 & \log 4 + \log 3 + \log 6 &= 1.857 \end{aligned}$$

Since these work, then

$$\log(a_1 \cdot a_2 \cdots a_n) = \log a_1 + \log a_2 + \cdots + \log a_n.$$

A probe into the reasoning of the students who provide responses of this kind reveals that their conviction stems from the fact that the proposition is shown to be true in a few instances, each with numbers that are *randomly* chosen—a behavior that is a manifestation of the empirical proof scheme.

Response 2

- (1) $\log(a_1 a_2) = \log a_1 + \log a_2$ by definition
- (2) $\log(a_1 a_2 a_3) = \log a_1 + \log a_2 a_3$. Similar to $\log(ax)$ as in step (1), where this time $x = a_2 a_3$.

Then

$$\log(a_1 a_2 a_3) = \log a_1 + \log a_2 + \log a_3$$

- (3) We can see from step (2) any $\log(a_1 a_2 a_3 \cdots a_n)$ can be repeatedly broken down to $\log a_1 + \log a_2 + \cdots + \log a_n, n = 1, 2, 3, \dots$

It is important to point out that in Response 2 the student recognizes that the process employed in the first and second cases constitutes a pattern that recursively applies to the entire sequence of propositions, $\log(a_1 a_2 \cdots a_n) = \log a_1 + \log a_2 + \cdots + \log a_n, n = 1, 2, 3, \dots$

In both responses the generalizations are made from two cases. This may suggest, therefore, that both are empirical. As is explained in Harel (2001), this is not so: Response 2, unlike Response 1, is an expression of the transformational proof scheme. To see why, one needs to examine the two responses against the definitions of the two schemes. While both responses share the first characteristic—i.e., in both the students respond to the “for all” condition in the

log-identity problem statement—they differ in the latter two: whereas the mental operations in Response 1 are incapable of anticipating possible subsequent outcomes in the sequence and are devoid of general principles in the evidencing process, the mental operations in Response 2 correctly predict, on the basis of the general rule, $\log(ax) = \log a + \log x$, that the same outcome will be obtained in each step of the sequence. Further, in Response 1 the inference rule that governs the evidencing process is empirical; namely, $(\exists r \in R)(P(r)) \Rightarrow (\forall r \in R)(P(r))$. In Response 2, on the other hand, it is deductive; namely, it is based on the inference rule $(\forall r \in R)(P(r)) \wedge (w \in R) \Rightarrow P(w)$. (Here r is any pair of real numbers a and x , R is the set of all pairs of real numbers, $P(r)$ is the statement “ $\log(ax) = \log a + \log x$,” and w in step n is a pair of real numbers $a_1 a_2 \cdots a_{n-1}$ and a_n .)

The axiomatic proof scheme too has the three characteristics that define the transformational proof scheme, but it includes others. For now, it is sufficient to define it as a transformational proof scheme by which one understands that in principle any proving process must start from accepted principles (axioms). The situation is more complex, however, as we will show in the section on historical and epistemological considerations (third section). For the purpose of this chapter we will introduce only the Greek axiomatic proof scheme and the modern axiomatic proof scheme—as manifested, for example, in Euclid’s *Elements* and Hilbert’s *Grundlagen*, respectively. The distinction between these two schemes is further discussed in the third section.

Relations to Other Taxonomies

In broad terms, the empirical proof schemes and the deductive proof schemes correspond to what Bell (1976) calls “empirical justification” and “deductive justification,” and Balacheff (1988) calls “pragmatic” justifications and “conceptual” justifications, respectively. Balacheff further divides the pragmatic justifications into three types of justifications: “naïve empiricism” (justification by a few random examples), “crucial experiment” (justification by carefully selected examples), and “generic example” (justification by an example representing salient characteristics of a whole class of cases). “Generic example” in our taxonomy belongs to the deductive proof scheme category, for an analysis similar to that we applied to Response 2 (above) will show that “generic example” satisfies the three characteristics of the transformational proof scheme. Balacheff further classifies conceptual justifications into two types: “thought experiment,” where the justification is disassociated from specific exam-

ples, and “symbolic calculation,” where the justification is based solely on transformation of symbols. In our taxonomy, the latter corresponds to the *referential symbolic* proof scheme. This scheme is the direct opposite of the *non-referential* proof scheme defined above. Recall this is a scheme where neither the symbols nor the operations one performs on them represent a coherent referential reality for the student. Rather, the student thinks and treats symbols and operations on them as if they possess a life of their own without reference to their functional or quantitative meaning. In the referential symbolic proof scheme, to prove or refute an assertion or to solve a problem, one learns to represent the situation algebraically and performs symbol manipulations on the resulting expressions, with the intention to derive information relevant to the problem at hand. We return to this scheme in the next section.

The above definitions and taxonomy are not explicit enough about many critical functions of proof within mathematics. There is a need to point to these functions due to their importance in mathematics in general and to their instructional implications in particular. For this, we point to the work of other scholars in the field, particularly the work by Hanna (1990), Balacheff (1988), Bell (1976), Hersh (1993), and de Villiers (1999). de Villiers, who built on the work of the others mentioned here, raises two important questions about the role of proof: (a) What different functions does proof have within mathematics itself? and (b) How can these functions be effectively utilized in the classroom to make proof a more meaningful activity? (p. 1). According to de Villiers, mathematical proof has six not mutually exclusive roles: verification, explanation, discovery, systematization, intellectual challenge, and communication. At the end of the next section, after the relevant schemes are defined, we show that these functions are describable in terms of the proof scheme construct.

MATHEMATICAL AND HISTORICAL- EPISTEMOLOGICAL FACTORS: SOME PHASES IN THE DEVELOPMENT OF PROOF SCHEMES IN THE HISTORY OF MATHEMATICS³

Deductive reasoning is a mode of thought commonly characterized as a sequence of propositions where one must accept any of the propositions to be true if he or she has accepted the truth of those that preceded it

in the sequence. This mode of thought was conceived by the Greeks more than 20 centuries ago and is still dominant in the mathematics of today. So remarkable is the Greeks' achievement that their mathematics became an historical benchmark to which other kinds of mathematics are compared. We are here particularly interested in analyzing the proof scheme construct across three periods of mathematics: Greek mathematics, post-Greek mathematics (approximately from the 16th to the 19th century), and modern mathematics (from the late 19th century to today). For the sake of completeness we say a few words about pre-Greek mathematics.

Proof schemes are applied to establish assertions in specific contexts. In this respect, therefore, in addressing proof schemes one must attend to the nature of the context about which the assertions are made. Also of importance is the motive—the intellectual need—that might have brought about the conceptual change from one period to the next. Thus, our discussion will center on three interrelated aspects of the historical-epistemological development: (a) the context of proving, (b) the means of proving (i.e., proof schemes), and (c) the motive for conceptual change. Understanding these elements can shed light on some critical aspects of the learning and teaching of proof, as we will see.

Some Fundamental Differences

Practical World versus Ideal Existence: The Emergence of Structure

Generally speaking, pre-Greek mathematics was concerned merely with *actual* physical entities, particularly with quantitative measurement of different objects. Their “formulas” were in the form of prescriptions providing mostly approximations for measured quantities (area, volume, etc.). Ancient geometry developed in an empirical way through phases of trials and errors, where conjectures were proved by means of empirical—inductive or perceptual—evidence. Even Babylonian mathematics, which, according to Kleiner (1991, p. 291), “is the most advanced and sophisticated of pre-Greek mathematics,” lacked the concept of proof as was understood and applied by the Greeks. Accordingly, it is safe to conclude that proving in pre-Greek mathematics is, by and large, governed by the empirical proof schemes.

The Greeks elevated mathematics from the status of practical science to a study of abstract entities. In their mathematics the particular entities under

³ We are restricting our discussion to classical Greek mathematics and the mathematical developments that grew out of it. The development of deductive reasoning in China, India, and other non-Western cultures is not considered in this paper.

investigation are idealizations of experiential spatial realities and so also are the propositions on the relationships among these entities. The difference between pre-Greek mathematics and Greek mathematics, however, is not just in the nature of the entities considered—actual spatial entities versus their idealizations—but also in the reasoning applied to establish truth about the entities. In Greek mathematics, logical deduction came to be central in the reasoning process, and it alone necessitated and cemented the geometric edifice they created. To construct their geometric edifice the Greeks had to create primary terms—terms admitted without definition—and primary propositions—propositions admitted without proof—what the Greeks called axioms or postulates. In contrast, the mathematics of the civilizations that preceded them established their observations on the basis of empirical measurements, and so their mathematics lacked any apparent structure; it consisted merely of prescriptions of how to obtain measurements of certain spatial configurations.

Constant versus Varying Referential Reality

In constructing their geometry, as is depicted in Euclid's *Elements*, the Greeks had only one model in mind—that of imageries of idealized physical reality. This is supported by the fact that the Greeks, in fact, strove to describe their primitive terms (e.g., “a point is that which has no parts”), which indicates that their sole imagery was that of physical space. Hartshorne (2000) points out that this way of thinking did not start with Euclid. About one hundred years before Euclid, Plato spoke of the geometers:

Although they make use of the visible forms and reason about them, they are not thinking of these, but of the ideals which they resemble; not the figures which they draw, but of the absolute square and the absolute diameter, and so on . . . (*The Republic*, Book IV).

From the vantage point of modern mathematics, neither the primitive terms nor the primary propositions in Greek mathematics were variables, but constants referring to a single spatial model (Klein, 1968; Wilder, 1967)—as is expressed in the ideal world of Plato's philosophy. This ultimate bond to a real-world context had an impact on the Greeks' proof scheme. Specifically, in his proofs Euclid often uses arguments that are not logical consequences of his initial assumptions but are rooted in humans' intuitive physical experience (the method of superposition, which allows one to move one triangle so that it lies on top of another triangle, is an example). In this respect, while proving in Greek mathematics was governed by the

deductive proof scheme and can be characterized as an axiomatic proof scheme, it is different from that of contemporary mathematics where every assertion must be, in principle, derivable from clearly stated assumptions. Wilder (1967) points to a crucial difference between Greek mathematics and modern mathematics: Namely, in modern mathematics the primitive terms are treated as variables, not just undefined; they are free of any referent—real or imagined. In Greek mathematics, on the other hand, they are undefined terms referring to humans' idealized physical reality. Treating primitive terms as undefined is fundamentally different from treating them as variables. Wilder quotes Boole (from 1847) to stress this difference:

The validity of the processes of analysis does not depend upon the interpretation of the symbols which are employed, but solely upon the laws of their combination. Every system of interpretations which does not affect the truth of the relations supposed, is equally admissible . . . (in Wilder, 1967, p. 116).

Despite the monumental conceptual difference in the referential imageries between Greek mathematics and modern mathematics, the essential condition in applying deductive reasoning in both is the existence of primary terms and primary propositions (axioms).

Content versus Form

The deductive reasoning—or the axiomatic method, as it often is called—of the Greeks dominated the mathematics of the western world until the late 19th century and in essence is still intact today. The difference in referential imageries between Greek mathematics and modern mathematics entails another critical difference: In Greek mathematics proof is valid by virtue of its *content*, not its *form*. Since the Greeks' concern was about relationships among entities in their physical space, the *form* of the proof cannot be completely detached from the *content* of that space. In this conception, one would not consider, for example, the question of whether there exists a consistent geometric model in which the parallel postulate does not hold. In modern mathematics, on the other hand, proof is valid by virtue of its form alone.

In *Greek Mathematical Thought and the Origin of Algebra*, Klein (1968) adds clarity to this distinction. In Greek science concepts are formed in continual dependence on their “natural” foundations, and their scientific meaning is abstracted from “natural,” prescientific experience. In modern science, on the other hand, what is intended by the concept is not an object of immediate insight. Rather, it is an object whose scientific meaning can be determined only by its connec-

tion to other concepts, by the total edifice to which it belongs, and by its function within this edifice. This difference accounts, for example, for the fact that the Greeks, while interested in idealized entities, were highly selective in their choice of these entities. For instance, what we now call transcendental numbers, like the number π , disturbed the Greeks. “They were unhappy that their ideas of perfection in geometry and arithmetic seemed to be challenged by the existence of such a number” (Ginsburg & Oppen, 1969, p. 214–215). In modern mathematics, on the other hand, entities can be quite arbitrary. Upon encountering transcendental numbers modern mathematics incorporated them into a mathematical structure, that of the real numbers, determined by a set of axioms.

It should be highlighted that the idea that the objects are determined by a set of axioms was a revolutionary way of thinking in the development of mathematics. An important manifestation of this revolution is the distinction between Euclid’s *Elements* and Hilbert’s *Grundlagen*. While the *Elements* is restricted to a single interpretation—namely that its content is a presumed description of human spatial realization—the *Grundlagen* is open to different possible realizations, such as Euclidean space, the surface of a half-sphere, ordered pairs and triples of real numbers, and so forth, including the interpretation that the axioms are meaningless formulas. In other words, the *Grundlagen* characterizes a structure that fits different models. This obviously is not unique to geometry. In algebra, a group or a vector space is defined to be any system of objects satisfying certain axioms that specify the structure under consideration. To reflect this fundamental conceptual difference, we refer to the Greeks’ proving means as the *Greek axiomatic proof scheme* and to the proving means of modern mathematics as the *modern axiomatic proof scheme*.

Operation versus Results of an Operation

Klein (1968) argues that the revival and assimilation of Greek mathematics during the 16th century resulted in fundamental conceptual changes that ultimately defined modern mathematics. This conceptual transformation culminated in Vieta’s work on symbolic algebra, where the distinction between modern mathematics and Greek mathematics began to crystallize.

The use of symbols in the modern sense is due mainly to Vieta, who was followed in this effort by Descartes and Leibniz. Until then, mathematics had evolved for at least three millennia with hardly any symbols. Symbolic notation was the key to a method of demonstration, and Leibniz—more than anyone else in his generation—advanced the role of symbol-

ism in the process of mathematical proof. He was the first to conceive of proof as a sequence of sentences beginning with identities and proceeding by a finite number of steps of logic and rules of *definitional substitution*—by virtue of symbolic notation—until the theorem was proved.

The works of Vieta that led to the creation of algebra and that of Descartes that led the creation of analytic geometry illustrate another important difference between Greek mathematics and modern mathematics: the focus on “operation” versus “results of an operation.” Ginsburg and Oppen (1969) discuss the distinction as described by Boutroux, a mathematician, who analyzed the evolution of mathematical thought. According to Boutroux, the Greeks restricted their attention to attributes of spatial configurations but paid no attention to the operations underlying them. As an example, Ginsburg and Oppen mention the problems of bisecting and trisecting an angle by means of straightedge-and-compass only. As is well known, the Greeks offered a simple way to solve the bisecting problem but the solution of the trisecting problem had to wait two millennia, until Galois, a French mathematician, solved it in the 19th century. The Greeks attended to geometrical objects (rectilinear angles in this case) by investigating their attributes—whether an angle can be bisected or trisected, for example. The 19th century mathematics, on the other hand, investigated the operation themselves—their algebraic representations and structures. Specifically, the standard Euclidean constructions using only a compass and straightedge were translated in terms of constructibility of the real numbers (i.e., a real number is constructible if its absolute value is the distance between two constructible points). This translation leads to an important observation about the structure of constructible numbers; namely, the constructible numbers form a subfield of the field of the real numbers. A deeper investigation into the theory of fields led to a proof of the impossibility of certain geometric constructions, including the impossibility of finding a single method of construction for trisecting any given angle with the classical tools. More importantly, with this theory one can understand why certain constructions are possible whereas others are not. The Greeks had no means to build such an understanding. They did not attend to the nature of the operations underlying the Euclidean construction, and hence were unable to understand the difference between bisecting an angle and trisecting an angle and why they were able to perform the former but not the latter.

Thus, not until the 17th century with the invention of analytic geometry and algebra did mathematicians begin to shift their attention from the result of math-

ematical operations to the operations themselves. By means of analytic geometry, mathematicians realized that all Euclidean geometry problems can be solved by a single approach, that of reducing the problems to equations and applying algebraic techniques to solve them. Euclidean straightedge-and-compass constructions were understood to be equivalent to equations, and hence the solvability of a Euclidean problem became equivalent to the solvability of its corresponding equation(s).

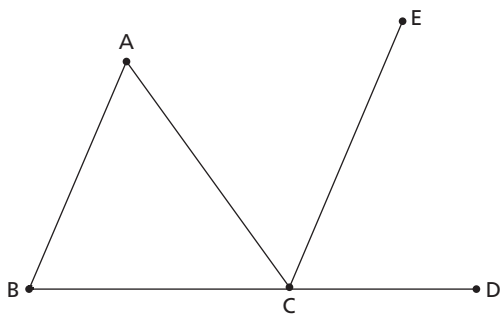
Cause versus Reason

Consider Proposition I.32 and its proof in Euclid's *Elements* (slightly adapted from Heath, 1956, pp. 316–317):

In any triangle, if one of the sides is produced, then the exterior angle equals the sum of the two interior and opposite angles, and the sum of the three interior angles of the triangle equals two right angles.

Proof:

Let ABC be a triangle, and let one side of it BC be produced to D .



I say that the exterior angle ACD equals the sum of the two interior and opposite angles CAB and ABC , and the sum of the three interior angles of the triangle ABC , BCA , and CAB equals two right angles.

Draw CE through the point C parallel to the straight line AB (by Proposition I.31).

Since AB is parallel to CE , and AC falls upon them, therefore the alternate angles BAC and ACE equal one another (by Proposition I.29).

Again, since AB is parallel to CE , and the straight line BD falls upon them, therefore the exterior angle ECD equals the interior and opposite angle ABC (by Proposition I.29).

But the angle ACE was also proved equal to the angle BAC . Therefore the whole angle ACD equals the sum of the two interior and opposite angles BAC and ABC .

Add the angle ACB to each. Then the sum of the angles ACD and ACB equals the sum of the three angles ABC , BCA , and CAB (by Common Notion 2).

But the sum of the angles ACD and ACB equals two right angles. Therefore the sum of the angles ABC , BCA , and CAB also equals two right angles (by Proposition I.13 and Common Notion 1).

This proof appeals to two facts, one about the auxiliary segment CE and the other about the external angle ACD . Note that the property holds whether or not the segment CE is produced and the angle ACD considered. One might then raise the question, what is the true cause of the property proved? This question was a center of debate during the 16th–17th centuries about whether mathematics is a *science*. Philosophers of this period, according to Mancosu (1996), used this proof to demonstrate their argument that mathematics is not a perfect science because “implication” in mathematics is a mere logical consequence rather than a demonstration of the *cause* of the conclusion. Their argument was based on the Aristotelian definition of science, according to which one does not understand something until he or she has grasped the why of it. “We suppose ourselves to possess unqualified scientific knowledge of a thing, . . . , when we think that we know the cause on which the fact depends as the cause of the fact and of no other . . .” (Aristotle, p. 111–112).

Mathematical statements of the form “A if and only if B” provided an additional argument against the scientificity (i.e., causal nature) of mathematics, for—these philosophers claimed—if mathematical proof were scientific (i.e., causal), then such a statement would entail that A is the cause of B and B is the cause of A, which implies A is the cause of itself—an absurdity.

This position entailed rejection of proof by contradiction, for such a proof does not demonstrate the cause of the property that is being argued. When a statement “A implies B” is proved by showing how not B (and A) leads logically to an absurdity, one does not learn anything about the causality relationship between A and B. Nor does one gain any insight into how the result was obtained. Consequently, proof by exhaustion (e.g., Archimedes’ known method of proof for calculating volume, area, and parameter of different objects), which is necessarily based on proof by contradiction, also was unsatisfactory to many mathematicians of the 16th and 17th centuries. They argued that the ancients, who broadly used proof by exhaustion to avoid explicit use of infinity, failed to convey their methods of discovery.

Not all philosophers of the time held this position. According to Mancosu (1996), Barozzi, for example, argued that some parts of mathematics are more scientific (causal) than others; but that a proof by contradiction is not a causal proof, and therefore it

should be eliminated from mathematics. Others, like Barrow, argued that all mathematics proofs are causal, including proof by contradiction:

It seems to me . . . that Demonstrations, though some do outdo others in Brevity, Elegance, Proximity to their first Principles, and the like Excellencies, yet are all alike in Evidence, Certitude, Necessity, and the essential Connection and mutual Dependence of the Terms one with another. Lastly, that Mathematical Ratiocinations are the most perfect Demonstrations. (Quoted in Mancosu, 1996, p. 23)

Of particular interest is the position held by Rivaltus on the issue of causal proof in mathematics. Mancosu (1996) illustrates this position by Rivaltus' commentary on Archimedes' proof for the theorem that the area of the surface of the sphere is four times the area of a great circle of the sphere. In this proof, Archimedes inscribes and circumscribes the sphere with auxiliary solids to show that the surface of the sphere can be neither smaller nor greater than four times the great circle—a typical Archimedean proof by exhaustion, which necessarily involves proof by contradiction. So, there are two issues here: the use of proof by contradiction and the use of the auxiliary solids (as in the case of the proof of Proposition I.32). Each of these two features render, in the eyes of some philosophers of the time, Archimedes' proof non-causal. Rivaltus rejects this possibility on the basis of a distinction between "cause" and "reason":

Ostensive demonstrations in mathematics are not considered more perfect than the ones by contradiction, since in these disciplines it is not made use of the cause of the thing, but of the cause of the knowledge of the thing. . . . The figures drawn are not truly the *cause* [italics added] of that equality but are *reasons* [italics added] from which we know it. From whence it follows that whatever is more fit to knowledge is more appropriate to the mathematician. But we know more easily that absurdities are impossible, false and repugnant by reason that we know the true things. Indeed the truths are concealed and conversely the errors are obvious everywhere. . . . Again it is to be observed that the Geometers do not make use of the cause of a thing, but of the cause from which the thing is known. Indeed it is sufficient to them to show the thing to be so and they do not enquire by which means it is so. (Rivaltus, 1615; quoted in Mancosu, 1996, pp. 26–27)

The Aristotelian theory of science and particularly its appeal to cause and effect manifested itself in another aspect of mathematical practice during the 17th century—that of the use of "genetic definitions." These definitions appeal to the generation of math-

ematical magnitude by motion; Euclid's definition of a sphere as an object generated by the rotation of a semicircle around a segment taken as axis is an example. An example of a non-genetic definition is that of a circle defined as a set of all points in the plane that are equidistant from a given point in that plane. According to Mancosu (1996), although the insistence on the use of genetic definitions was not universal, important mathematicians of the time emphasized their importance because they were viewed as demonstrating cause, and, hence, conform to Aristotle's epistemological position of what constitutes a science. For example, Barrow stated that of all the possible ways of generating a magnitude, the most important is the method of local movements, and he uses motion as a fundamental concept in his work in geometry" (Mancosu, 1996, p. 96), and Hobbes and Spinoza, "emphasize the role of genetic definitions as the only causal definitions, thereby excluding the nongenetic definitions from the realm of science" (p. 99).

Motives

The Greeks' motive for constructing the remarkable geometric edifice that we now call Euclidean Geometry was their desire to create a consistent system that is free from paradoxes. Avoiding paradoxes constituted, in part, an intellectual need for the transition from Greek to modern mathematics. According to Wilder (1967, based, in part, on Freudenthal, 1962), against the customary view that attributes the change to the introduction of non-Euclidean geometry,

there was little evidence of excitement or even interest in the mathematical community regarding the work of Bolyai and Lobachevski, for at least 30 years after its publication. . . . The . . . impact of the admissibility of non-Euclidean geometry into mathematics only promoted an evolution already under way (p. 116).

Hacking (1980) argues that the change in perspective occurred much before the late 19th century as is commonly held, but was pioneered by Leibniz. In fact, Hacking insists that Leibniz was aware of the conceptual change and even explained why it did not develop earlier:

Leibniz himself has a plausible explanation of why the concept of proof emerged at this time. [The idea that proof is independent of its content] is not to be expected when geometry is the standard of rigor. Geometrical demonstrations can appear to rely on their content. Their validity may seem to depend on facts about the very shapes under study, and whose actual

construction is the aim of the traditional Euclidean theorems. (p. 170)

Here, too, Descartes' contribution was crucial: What brought about Leibniz' new way of understanding the concept of proof was Descartes' algebrization of geometry. By its nature algebra is a way of thinking where one can dispose of or reduce overt attention to content. Together with the recognition that the non-Euclidean geometries are just as consistent as the Euclidean, there seems to have been a general feeling that both the Euclidean and the non-Euclidean systems were still only candidates as sciences of space.

The change in proof scheme from Greek mathematics to modern mathematics was necessitated again by the attempt to establish a consistent foundation for mathematics—just as the Greeks did to construct a consistent system—one that is free from paradoxes. The mathematicians who launched major attempts to form a consistent mathematics, like Zermelo and Russell and Russell and Whitehead, were largely motivated by the desire to meet the crisis in the foundations of mathematics caused by contradictions, primarily those resulting from the Cantorian theory of sets:

(The) Greek situation, as the latter appears to us through the haze of centuries, is striking. In both situations, crises had developed which threatened the security of mathematics; and in both cases resort was taken to explicit axiomatic statement of the foundations upon which one hoped to build without fear of further charges of inconsistency. (Wilder, 1967, p. 117)

Wilder (1967) also notes that like the Greek mathematicians, the modern mathematicians who launched major attempts to form a consistent mathematics (e.g., Zermelo and Russell) had only one model in mind, albeit a different model. In the case of Zermelo, he focused on “a portion of set theory sufficient for ordinary mathematical purposes, yet carefully limited to avoid the known contradictions” (p. 117). Russell too, like Euclid and Zermelo, “had only one model in mind, and he aimed at making the conditions surrounding his system so stringent that there could be only one model” (p. 118).

Not everyone in the 19th–20th century gave his or her blessing to the new development in mathematics. Both Poincaré and Weyl, for example, while acknowledging the legitimacy and accomplishment of the axiomatic method, were disturbed by the widespread preoccupation with it. They called for a return to problems of “mathematical substance” and argued that its role should be limited to giving precision to already created mathematical entities (Wilder, 1967). Nor were all mathematicians of the 17th century en-

thusiastic about the emergence of a new context of mathematics. The modern notion of number is a case in point. The emergence of negative numbers raised questions as to the utility of symbols without a concrete referent and especially without a geometrical referent. How is it possible, for example, to subtract a greater quantity from a smaller one, where the mental image of “quantity” is nothing else but a physical amount or a spatial capacity? Moreover, how is it possible to understand such a statement as $(-1)/1 = 1/(-1)$, where the quantity 1 is larger than the quantity -1 , and therefore, the division of 1 by -1 must be greater than the division of -1 by 1? (See Mancosu, 1996.)

Why Are These Historical-Epistemological Factors Relevant?

What does this brief history account tell us about the learning and teaching of the concept of proof? It is still an open question whether the development of a mathematical concept within an individual student or a community of students parallels the development of that concept in the history of mathematics, though cases of parallel developments have been documented (e.g., Sfard, 1995). If this is the case, one would expect that the path of development would vary from culture to culture. Are there common elements or phases to different paths of development across cultures? Did the development of the concept of proof in, for example, China and India follow a similar path to that of the Western world or was there a leap in time from using perceptual proof schemes to modern axiomatic proof schemes? What are some of the salient social and cultural aspects that might impact the trajectory of these paths? Independent of these particular questions, whose treatment goes beyond the scope and goals of this paper, considerations of historical accounts can evoke other research questions whose answers can potentially direct the development and implementation of instructional treatments of mathematical concepts and ideas. In what follows, we will discuss such questions about the concept of mathematical proof evoked by the historical account outlined above.

Pre-Greek Mathematics versus Greek Mathematics

The Greeks had likely constructed their deductive mathematics on the basis of the mathematics of their predecessors, which was mainly empirical. This mere fact evokes a pedagogical question as to the role of the empirical proof schemes in the development of the deductive proof schemes. The empirical proof schemes are inevitable because natural, everyday thinking utilizes examples so much. Moreover, these schemes have value in the doing and the creating of

mathematics. They are even indispensable in enriching one's images by creating examples and non-examples, which, in turn, can help generate ideas and give insights. The question is how to help students utilize their existing proof schemes, largely empirical and external, to help develop deductive proof schemes? As a historian might ask what events—social, cultural, and intellectual—necessitated the transition from pre-Greek mathematics to Greek mathematics, a mathematics educator should ask what instructional interventions can bring students to see a need to refine and alter their existing external and empirical proof schemes into deductive proof schemes.

As was indicated, while pre-Greek mathematics was concerned merely with *actual* physical entities, and its proving was governed by the empirical proof schemes, Greek mathematics dealt with idealizations of spatial and quantitative realities and its proving was deductive. There seems to be a cognitive and epistemological dependency between the nature of the entities considered and the nature of proving applied. To what extent and in what ways is the nature of the entities intertwined with the nature of proving? For example, students' ability to construct an image of a point as a dimensionless geometric entity might impact their ability to develop the Greek axiomatic proof scheme, and vice versa. As far as we know, this interdependency has not been explicitly addressed and its implications for instruction have not been considered.

The motive for the Greeks' construction of their geometric edifice, according to the historical account presented earlier, was their desire to create a consistent system that was free from paradoxes such as those of Zeno. For example, to avoid Zeno's paradoxes the Greeks based their geometric proofs strictly in the context of static concepts. What does this tell us about how to help students see a need for the construction of a geometric structure, particularly that of Euclid? What is the cognitive or social mechanism by which deductive proving can be necessitated for the students?

Greek Mathematics versus Modern Mathematics

As was noted, in constructing their mathematics the Greeks had only one model in mind—that of imageries of idealized physical and quantitative realities. Further, it is these imageries that determined the axioms and postulates on which their geometric and arithmetic edifices stood. In contrast, in modern mathematics entities can be quite arbitrary and one's images of these entities are governed by a set of axioms. This sharp construct is best manifested by Euclid's *Elements*, on the one hand, and Hilbert's *Grundlagen*, on the other: While the former is restricted to a

single interpretation, the *Grundlagen* is open to different realizations. Here too the question of intellectual necessity for the transition from Greek mathematics to modern mathematics is of profound pedagogical importance. Historians like Klein and others characterize this transition as revolutionary: It marks a monumental conceptual change in humans' mathematical ways of thinking. More research is needed to systematically document these difficulties to better understand their nature and their implications for instruction.

Another critical difference between Greek mathematics and modern mathematics has to do with *form* versus *content*. In Greek mathematics, the *form* of the proof could not be completely detached from the *content* of the spatial or quantitative context. In contrast, in modern mathematics proof is valid by virtue of its form alone. Here, too, we know of no studies that document systematically students' mathematical behaviors in relation to these fundamental characteristics of Greek mathematics—when students learn Euclidean geometry, for example, as compared to when they learn finite geometry as a case of modern mathematics.

Post-Greek Mathematics

Symbolic algebra, which began with Vieta's work, seems to have played a critical role in the transition from Greek mathematics to modern mathematics, particularly in relation to the reconceptualization of mathematical proof as a sequence of arguments valid by virtue of their form, not content. In the new concept of proof, one would begin with identities and by virtue of rules of symbolic definitional substitutions proceed through a finite number of steps until the theorem is proved. With symbolic algebra, mathematicians shifted their attention from results of operations (e.g., whether and how an angle can be bisected or trisected) to the operations themselves (e.g., the underlying difference between bisecting and trisecting an angle). A critical outcome of this shift was the discovery that all Euclidean geometry problems can be solved by a single approach, that of reducing the problems to equations and applying algebraic techniques to solve them.

The role of symbolic algebra in the reconceptualization of mathematics in general and of proof in particular raises a critical question about the role of symbolic manipulation skills in students' conceptual development of mathematics, in general, and of proof schemes, in particular. Can students develop the modern conception of proof without computational fluency? And in view of the increasing use of electronic technologies in schools, particularly computer algebra systems, one should also ask: Might these tools deprive students of—or, alternatively, provide students

with—the opportunity to develop algebraic manipulation skills that might be needed for the development of advanced conception of proof? In addressing this question, it is necessary, we believe, to distinguish between two kinds of symbolic proof schemes: non-referential and referential. As we discussed earlier, in the former scheme, neither the symbols nor the operations one performs on them represent a quantitative reality for the students. Rather, students think of symbols and algebraic operations as if they possess a life of their own without reference to their functional or quantitative meaning. By contrast, in the symbolic referential proof scheme, to prove or refute an assertion or to solve a problem, students learn to represent the statement algebraically and perform symbol manipulations on the resulting expressions. The intention in these symbolic representations and manipulations is to derive relevant information that deepens one's understanding of the statement, and that can potentially lead her or him to a proof or refutation of the assertion or to a solution of the problem. In such an activity, one does not necessarily form referential representations for each of the intermediate expressions and relations that occur in the symbolic manipulation process, but has the ability to attempt to do so in any stage in the process. It is only in critical stages—viewed as such by the person who is carrying out the process—that one forms, or attempts to form, such representations. This ability is potential rather than actual because in many cases the attempt to form quantitative representations may not be successful. Nevertheless, a significant feature of the referential symbolic proof scheme, which is absent from the non-referential symbolic proof scheme, is that one possesses the ability to pause at will to probe into the meaning (quantitative or geometric, for example) of the symbols.

Together with the emergence of symbolic algebra, a new conception of mathematical entity, particularly that of number, began to emerge. A mathematical entity, in this conception, is not necessarily dependent on its “natural” pre-scientific experience but on its connection to other entities within a structure and its function within that structure. For example, while the Greeks were highly selective in their choice of numbers—they rejected irrational numbers, for example—the post-Greek mathematicians began to accept them. This conceptual change was not without difficulty. For example, some mathematicians of the 17th century rejected the utility of negative numbers, which they viewed as symbols without real experiential referents. The conceptual attachment to a context—whether it is the context of intuitive Euclidean space or that of R^n —was dubbed the *contextual proof scheme*. In this scheme, general statements, intended for vary-

ing realities, are interpreted and proved in terms of a restricted context. The question of the developmental inevitability of the contextual proof scheme has not been fully addressed in mathematics education research. Some evidence exists to indicate that even students in an advanced stage in their mathematical education have not developed this scheme. For example, it has been shown that many mathematics majors enrolled in advanced geometry courses have major difficulties dealing with any geometric structure except the one corresponding to their spatial imageries, and that mathematics majors enrolled in linear algebra courses interpret and justify general assertions about entities in a general vector space in terms of R^2 and R^3 entities (Harel & Sowder, 1998, Harel, 1999). Such findings have major curricular implications. For example, they raise major doubts as to the wisdom of the practice of starting off college geometry courses with finite geometries or of introducing general vector spaces in the first course of linear algebra.

The debate among philosophers during the Renaissance about the scientificness of mathematics and the mathematics practice that ensued is of particular pedagogical significance. As we have outlined, the question was whether the mathematical practice in which “implication” is a mere logical consequence, rather than a demonstration of the cause of the conclusion, is scientifically acceptable. This, in turn, raises questions about the acceptability of proof by contradiction and proof by exhaustion. Were these issues of marginal concern to the mathematicians of the 16th and 17th centuries, or had they been significantly affected by it? To what extent did the practice of mathematics in the 16th and 17th centuries reflect global epistemological positions that can be traced back to Aristotle's specifications for perfect science? These are important questions, if we are to draw a parallel between the individual's epistemology of mathematics and that of the community. As noted by Mancosu (1996), this debate had a deep and profound impact on the practice of mathematics during the 15th to 18th centuries. For example, the practices of Cavalieri, Guldin, Descartes, and Wallis reflected a deep concern with these issues by, for example, explicitly avoiding proofs by contradiction in order to conform to the Aristotelian position on what constitutes perfect science. This history shows that the modern conception of proof was born out of an intellectual struggle—a struggle in which Aristotelian causality seems to have played a significant role. Is it possible that the development of students' conception of proof includes some of these epistemological obstacles (in the sense of Brousseau, 1997)—obstacles that may be unavoidable, for they

are inherent to the meaning of concepts in relation to humans' current schemes?

We conclude this section on the relevance of history and philosophy of mathematics to the learning and teaching of proof with questions pertaining to the idea of "genetic definitions"—mathematical definitions that utilize motion to generate magnitudes—from the 17th century. As we have indicated, the use of such definitions was viewed by some important mathematicians of the time to conform to the Aristotelian epistemological position on the centrality of causality in science. Can this account for the positive impact that dynamic geometry environments might have on advancing students' proof schemes? What is exactly the conceptual basis for the relationship between motion and causal proofs? In a later section we will report on several studies that have examined this effect.

Functions of Proof

Earlier, in the second section, we described a portion of our taxonomy of proof schemes. The discussion that followed brought up several other schemes that emerged in the history of mathematics. In the rest of this section, we depict all the schemes mentioned in this paper (Table 18.1) and discuss their functions within mathematics. This list is not complete; we only depict those schemes that are needed for the discussion in this paper (for the complete taxonomy, see Harel & Sowder, 1998).

Table 18.1 Proof Schemes

External Conviction	Empirical	Deductive
Authoritative	Inductive	Transformational
Ritual	Perceptual	Causality
Non-referential		Greek axiomatic
		Modern axiomatic

As we indicated earlier (see the second section), de Villiers (1999) built on the work of others scholars—particularly Hanna (1990), Balacheff (1988), Bell (1976), Hersh (1993)—to address important questions about the role of proof. Specifically, what different functions does proof have within mathematics itself and how can these functions be effectively utilized in the classroom to make proof a more meaningful activity? de Villiers suggests that mathematical proof has six not mutually exclusive roles:

- verification
- explanation

- discovery
- systematization
- intellectual challenge
- communication

In what follows, we will show that all of these functions but one (intellectual challenge) are describable in terms of the proof scheme construct. Some of the proof schemes used to interpret these functions appeared in the taxonomy presented above. A description of each—with our additions and modifications—follows.

Verification refers to the role of proof as a means to demonstrate the truth of an assertion according to a predetermined set of rules of logic and premises—the *axiomatic proof scheme*.

Explanation is different from verification in that for a mathematician it is usually insufficient to know only that a statement is true. He or she is likely to seek insight into why the assertion is true. We referred to this as the *causality proof scheme*.

Discovery refers to the situations where through the process of proving, new results may be discovered. For example, one might realize that some of the statement conditions can be relaxed, thereby generalizing the statement to a larger class of cases. Or, conversely, through the proving process, one might discover counterexamples to the assertion, which, in turn, would lead to a refinement of the assertion by adding necessary restrictions that would eliminate counterexamples. Lakatos' (1976) thought experiment on the proof of Euler's theorem for polyhedra best illustrates this process. In some cases one may ask whether a certain axiom is needed to establish a certain result, or what form the result would have if a certain axiom is omitted. We considered this as a case of the *axiomatizing proof scheme*.

Systematization refers to the presentation of verifications in organized forms, where each result is derived sequentially from previously established results, definitions, axioms, and primary terms. This too is a case of the axiomatic proof scheme. The difference between systematization and verification is in the extent of formality.

Communication refers to the social interaction about the meaning, validity, and importance of the mathematical knowledge offered by the proof produced. Communication can be viewed in the context of the two subprocesses that define proving: ascertaining and persuading.

Intellectual challenge refers to the mental state of self-realization and fulfillment one can derive from constructing a proof. As we mentioned earlier, this role does not correspond to any of our proof schemes.

With the notion of proof scheme as an organizing concept—appended with these functions—we will now present selected findings reported in the literature that pertain to students' conceptions of proof. Of course, we are unable to describe without speculation most of these findings in terms of the proof scheme construct, because these studies had not been conceptualized or designed with our proof scheme construct in mind. However, as we will see, much can be said about these research findings in relation to proof schemes.

COGNITIVE FACTORS: A LITERATURE REVIEW OF STATUS STUDIES WITH INTERPRETATIONS IN TERMS OF STUDENTS' PROOF SCHEMES

Examining student performance in an area of mathematics is a natural first step in judging whether curricular and teaching efforts in that area seem to be adequate and not to require any special attention. Hence, we first look at several status studies of proof performance. We interpret the results in terms of the (apparent) proof schemes involved in the results, following the proof schemes focus of this chapter. The first two parts of this section will report, respectively, on precollege and college students' conceptions of proof. The findings provide evidence that the pervasive proof schemes among the two populations of students are those belonging to the external conviction proof scheme class and the empirical proof scheme class.

With the deductive proof scheme playing a significant role in mathematics, there is also relevant information from examining performance in logical inferences—a critical characteristic of this scheme. Any description of mathematical proof, the ultimate in justification, will include some mention of logical inferencing ability. Someone not versed in, for example, the mathematical meaning of an “if . . . , then . . .” statement, or who is not comfortable with reasoning patterns like *modus tollens*, or who does not understand quantified statements and the role of a counterexample, or who cannot deal correctly at some stage with negating “. . . and . . .” and “. . . or . . .” statements, or who cannot explain the logic of an indirect proof, surely cannot carry out or even understand many mathematical proofs (including disproofs). Hence logical inferencing ability is a basic tool for the process of proving in mathematics and likely enters also into many justifications of a less sophisticated sort. But logic is central to the deductive proof schemes. For example, the transformational proof scheme, which constitutes the essence of the proving process in mathematics and is expected to de-

velop with at least college-bound students and mathematics major students, should be present in students' mathematical behavior. As we have discussed earlier, “logical inference” is one of the three essential characteristics of this scheme. A third part of this section will include findings reported in the literature that seem germane to the characteristics of these deductive proof schemes.

Status Studies: Precollege Students

NAEP Studies

A first place to look for data is the periodic National Assessment of Educational Progress in the United States [NAEP], involving, typically, students at ages 9, 13, and 17 (and now reported by grade: 4, 8, and 12). The wide geographic sampling and the large sample sizes, usually many thousands of students, in a NAEP indicates the U.S. picture, which can be further checked with smaller studies and contrasted with studies outside the U.S. For our purposes, however, one limitation of the NAEP has been that only a few items on proof or logic can be included, because of the scope of the tests. Another limitation is that since all the NAEP questions considered here are multiple-choice items, it is difficult to pinpoint the actual reasoning students employed in answering them.

In the planning stage, the first NAEP mathematics assessment (1972–1973) included mathematical proof and logic as objectives to be evaluated, although only a few items tested these areas (Carpenter, Coburn, Reys, & Wilson, 1978, p. 10). Later NAEPs used different designs for setting objectives to be tested. The fourth NAEP (1985–1986), for example, included items testing “mathematical methods,” with a few intended to include “a general understanding of the nature of proof and axiomatic systems, and logic” (Carpenter, 1989, p. 3). Analysts of the results concluded that “most 11th-grade students demonstrated little understanding of the nature and methods of mathematical argumentation and proof” (Silver & Carpenter, 1989, p. 11), citing results on items requiring the recognition of counterexamples (with success rates of 31%–39%—see Figure 18.1 for two items), on items testing understanding of the terms “axiom” and “theorem” (fewer than one-fourth and about half correct, respectively), and on items dealing with an undisclosed but “straightforward” item on indirect proof (about one-third correct) and mathematical induction (similar results) (pp. 17–18). Even when only those students who had taken geometry were considered, results were just slightly better than those for students who had taken mathematics only through first-year algebra. The overall performance led the analysts to con-

clude that “the generally poor performance on these items dealing with proof and proof-related methods suggests the extent to which students’ experiences in school mathematics, even for students in college-preparatory courses, may often fail to acquaint them with the fundamental nature and methods of the discipline” (p. 18).

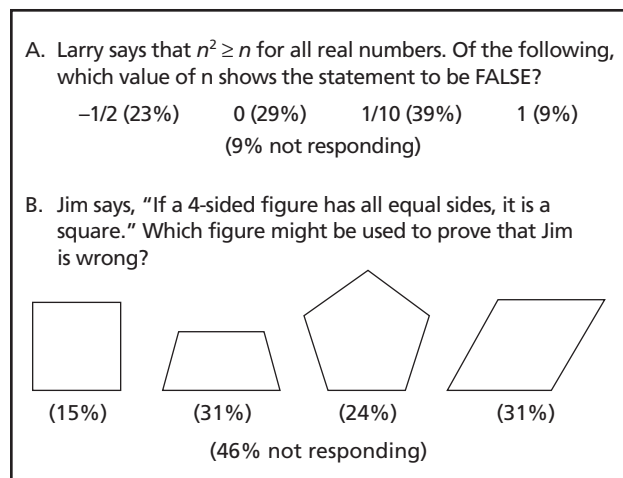


Figure 18.1 Sample NAEP items involving recognition of a counterexample (percents of those responding are in parentheses (after Silver & Carpenter, 1989, p. 17).

Note: Percents may total more than 100 because of rounding.

The test framework for the sixth NAEP (in 1992) did not focus specific attention on proof and logic items. But in commenting on performances on geometry items, the analysts (Strutchens & Blume, 1997, ch. 7) noted that many results could be explained by assuming that the students were basing their responses on the appearances in drawings—the perceptual proof scheme—rather than through reasoning. For example, when asked to choose from a set of triangles one that did not have a particular property, only 21% of the students made the correct choice (p. 180), but when asked whether four given statements applied to an illustrated construction of an angle bisector, 51% were correct on all four, leading the analysts to write, “Often a figure can foster correct reasoning. . . .” (pp. 181–182).

Seventy-seven percent of the Grade 8 students, however, were successful at recognizing a counterexample to a false statement about quadrilaterals. It is interesting to note that the first NAEP (1972–1973) included a multiple-choice item on logic that involved recognizing the logical equivalent of “All good drivers are alert.” Only about one-half of the 17-year-olds, 11th graders, chose the correct alternative (“A person who is not alert is not a good driver”). The analysts for those NAEP results noted, however, “The logic exercises were probably answered as much by the semantic

context of the problems as by any knowledge of logic” (Carpenter et al., 1978, pp. 126–127). This is in line with the earlier observation that proof schemes can vary from context to context.

Overall, then, the limited picture about proof understanding from the NAEPs is that at best only a small percent of high school students are equipped to deal effectively with the deductive proof schemes, with most apparently relying on the empirical proof schemes.

Other Status Studies in the United States

Studies in which justification and proof have been foci are perhaps more telling than the limited messages from the NAEPs. Some have been large-scale studies and hence are particularly significant. For example, Senk’s (1985) study involved 1520 U.S. geometry students in 74 classes in 11 high schools in 5 states within a month of the end of the course. After items in which the student was asked to supply missing reasons or statements, each student was asked to give four proofs, with the first two requiring only one deduction beyond those from the given information, as in the example in Figure 18.2 (72% correct). But only 32% could prove the textbook theorem, The diagonals of a rectangle are congruent.

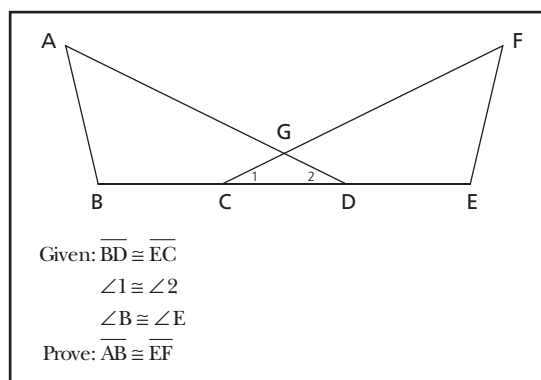


Figure 18.2 A sample geometry proof item from Senk’s study (1985, p. 451).

The overall results were dismayed for the course in the U. S. in which deductive proof schemes should be expected to develop: “[The] data suggest that approximately 30 percent of the students in full-year geometry courses that teach proof reach a 75-percent mastery level in proof writing. . . . 29 percent of the sample could not write a single valid proof” (Senk, 1985, p. 453). At the time, about half of high school graduates took a course in geometry, so it is interesting to speculate about what more recent performances might be, when roughly 80% of U. S. high schoolers take geometry (U.S. Department of Education, 2000, p. 122).

Other, smaller studies in the U.S. have also involved students in geometry or later courses, since virtually all formal work with proof has traditionally been introduced in the geometry course (9th or 10th year is most common). But, for example, the interviewees in Tinto's (1988) study felt that proof was used only to verify facts that they already knew—an antithesis to the discovery or explanation functions of proof discussed earlier.

Thompson's study (1991) is of special interest since her subjects were all advanced students taking the last course of a curriculum targeting university-bound students and emphasizing reasoning and proof as a major strand. Yet Thompson expressed concern about the number of students who "proved" a statement by providing a specific example—a manifestation of the inductive proof scheme, and only about one-third of her subjects could find a counterexample to a number theory statement (For all integers a and b , if a^2 is divisible by b then a is divisible by b), in a "prove or disprove" context. Thompson also referred to the "enormous difficulty that students had with indirect proof" (1991, p. 23), with only 3% able to complete one indirect proof (that the sum of a rational number and an irrational number is an irrational number). Difficulties with indirect proof could well be related to the earlier discussion of the causal proof scheme.

Knuth, Slaughter, Choppin, and Sutherland (2002) found that 70% of roughly 350 students in grades 6–8 used examples (the empirical proof scheme) in justifying the truth of two statements (show that the sum of two consecutive numbers is always an odd number; show that when you add any two even numbers, your answer is always even, p. 1696). Only a few students attempted general arguments. On a more positive note, across grade levels, the students did show an increasing sensitivity to adhering to a given definition.

Since the focus of these studies was proof performance, they provide even more striking evidence than the NAEP studies did, that most U. S. students, even those in college-preparatory programs, do not seem to utilize deductive proof schemes.

Status Studies Outside the United States

Another large-scale study of student performance in justification and proof shows that weak performance and reliance on less mature proof schemes is not solely a U.S. phenomenon. Healy and Hoyles (1998, 2000) conducted a study involving 2459 English and Welsh 14–15 year-olds (finishing ninth graders in 94 classes in 90 schools). That the students were in the top 20%–25% on a national test is noteworthy, as is the fact that the English-Welsh national curriculum gives attention to conjecturing and explaining/justifying conclusions

and is also, in U.S. terms, integrated with attention to both algebra and geometry, so presumably the items were reasonable for those grade 9 students. The students were asked to describe what proof means and what it is for, and to judge given proofs, as well as to decide on the truth or falsity and to construct their own proofs for two algebra and two geometry items. Over a quarter of these able students had little or no sense of the purpose of proof or its meaning (2000, pp. 417–418). The average score on the constructed-proof items was less than half the maximum, with 14%–62% of the students, depending on the item, not even able to start a proof (1998, p. 2). Of those able to start a proof, 28%–56% could proceed only minimally (1998, p. 2). It is comforting that the students were better at selecting correct proofs, favoring general arguments and clear and explanatory arguments that they found convincing (1998, p. 3). When the stated purpose was to get the best mark, however, the students often felt that more formal—e.g., algebraic—arguments might be preferable to their first choices (1998, p. 3). We interpret this last finding as an indication of the authoritarian proof scheme and the ritual proof scheme, in that in the eyes of these students proof must have a certain appearance (ritual) as determined by the teacher (authority).

It is interesting that, in the large, the findings of this study are consistent with those of Coe and Ruthven (1994), who looked at the proof performances of a much smaller group of "advanced level" British students in university preparation schools, on a project for an end-of-course assessment. These students, like those in the Healy and Hoyles study, relied predominantly on examples and techniques for analyzing numerical data, but showed little feel for the purposes of proof. Sample items from Healy and Hoyles' and Coe and Ruthven's studies are presented in Figure 18.3.

An older study by Reynolds (reported in Lovell, 1971) also examined proof ideas held by British students. Most revealing were the responses of 153 6th formers (final year of secondary school) in mathematics, with about 20% endorsing 16 examples as establishing a given statement—a manifestation of the empirical proof scheme. On the other hand, about 70% of these 6th-formers could complete an indirect argument involving a geometric situation (to show the inequality of a pair of alternate interior angles in a drawing with non-parallel lines) that had been started (p. 75), in contrast to the very weak (3%) performance of Thompson's (1991) U. S. students mentioned above.

Still other studies outside the U.S. also indicate difficulties with certain proof ideas. Galbraith (1981) studied the conceptions of proof held by 12–17-year-old Australian students, most of whom were 13–15

Proof or disproof construction (Healy & Hoyles, 1998)	Prove whether the following statements are true or false. Write down your answer in the way that would get you the best mark you can. 1. When you add any 2 odd numbers, your answer is always even. 2. If p and q are any two odd numbers, $(p + q) \times (p - q)$ is always a multiple of 4.
"Starting point" item (Coe & Ruthven, 1994, p. 46)	Any four-digit number is rewritten with its digits arranged in ascending and descending order of size. The smaller is then subtracted from the larger and the process is repeated. Example: 7345 gives $7543 - 3457 = 4086$ 4086 gives $8640 - 0468 = 8172$ Then write $7543 \rightarrow 8640 \rightarrow 8721 \rightarrow \dots$ Complete this chain and then try again with other four-digit numbers. Now try with 3 or 5 or 6 or n digits.

Figure 18.3 Sample proof items (adapted from Healy & Hoyles (1998, 2000) and Coe & Ruthven (1994)).

years old. He found that 28% of 130 responses indicated a lack of understanding of what a given counterexample told one about a given statement. On another problem, only 18% of his 73 students felt that one counterexample was sufficient to disprove a conjecture. In commenting on how many counterexamples are sufficient, one student even suggested "Sometimes one is enough and sometimes it isn't" (p. 19). The confusion about the role of examples in direct proofs versus in indirect proofs versus in disproofs is consistent with our own findings (Harel & Sowder, 1998), a phenomenon we accounted for in terms of students' empirical proof scheme. Many students in Galbraith's study were guided by their mental pictures of what a geometric term suggested rather than a given definition, and relied on just one diagram without considering other possibilities (p. 23)—a clear manifestation of the perceptual proof scheme.

Porteous (1986, 1990) gave questionnaires to about 400 British students, ages 11–16, and interviewed 50 of them. From the questionnaires, he found that more than 40% of the responses endorsed completely a generalization on the basis of examples only, and only about 10% then offered proofs on their own when asked to explain their decisions (1990, p. 591). Furthermore, 83% of those not offering a proof initially claimed to understand a given proof of an assertion, but only 61% of all the students, including the ones offering a proof on their own, were sure that the author was correct about the assertion after seeing a proof (1996, p. 8). From the students' reactions to given proofs, Porteous concluded that "It is only when a pupil devises a proof for himself that he is convinced of the truth of a statement. A proof

provided by a teacher may have some effect, but it is small in comparison with a d(i)scovered proof . . . we need to encourage pupils to investigate relationships for themselves, in order to produce their own reasons, or proof, for general statements. This is not to say, of course, that empirical work has no real part to play in the learning process" (1996, pp. 21–22).

Williams (1980) interviewed 11th grade Canadian students in a college preparatory program and concluded that fewer than 30% showed a grasp of the meaning of mathematical proof, that about half of the students saw no need to prove a statement that they regarded as obvious, roughly 70% did not distinguish between inductive and deductive arguments, and fewer than 20% understood how indirect proof works.

Spanish researchers Recio and Godino (2001) conducted a study involving two groups ($n = 429$ and $n = 193$) of beginning university students. They found that about a third of the 429 students and less than a quarter of the 193 students could prove both of two elementary statements (the difference between the squares of consecutive natural numbers is odd and equal to the sum of the numbers, and the bisectors of adjacent supplementary angles are perpendicular). Also, fewer than half of the group of 429 were successful on each individual proof. Roughly 40% used empirical reasoning: "Empirical inductive [proof] schemes were the spontaneous type of argumentation in a high percentage of students when they were confronted with new problems, in which it was necessary to develop new proof strategies, different from the learned procedures" (p. 91).

In Israel, Fischbein and Kedem (1982) found that their high school students did not appear to un-

derstand that no further examples need be checked, once a proof was given, a finding confirmed by Vinner (1983), who also noted that many high school students (35%), even high attaining ones (39%), seem to regard a given proof as the method to examine and verify a later particular case.

Similar results were found in a recent Japanese study (cited in Fujita & Jones, 2003). The official curriculum of Japan calls for students to “understand the significance and methodology of proof.” However, even though most 14 to 15 year old students are successful at proof writing, “around 70% cannot understand why proofs are needed” (p. 9).

Internationally, overall the most positive conclusion seems to be that the proof glass is not completely empty but that it is by no means even close to full. Even “good” performances may be tainted by little understanding or appreciation of the functions of proof. The prevalence of empirical proof schemes for most students seems to be international.

Status Studies: University Students

One might expect that university students would perform better than secondary students on proof activities. As the Recio and Godino (2001) study indicated, however, performance is disappointingly low in many entering university students. There are only a few formal studies that have examined university students’ later conceptions of proof. In one example, up to 80% of prospective elementary school teachers derived the truth of a general statement from examples (Goetting, 1995), confirming a result by Martin and Harel (1989) about the pervasiveness of the empirical proof scheme among students. Martin and Harel had emphasized proof in their course for prospective elementary school teachers (who had had geometry in high school) and then asked them to judge whether particular arguments were mathematical proofs for a statement covered and proved in the course (If the sum of the digits of a whole number is divisible by 3, then the number is divisible by 3) or an unfamiliar one (If a divides b , and b divides c , then a divides c). The arguments were either inductive (based on specific instances) or deductive (assertions via general statements). More than half of the 101 students considered inductive arguments to be mathematical proofs. As many as half of these college students also accepted a false deductive argument as being a mathematical proof, apparently reacting to the appearance of the argument—indicating the dominance of the ritual proof scheme—and over a third found both inductive and deductive arguments to be acceptable mathematical proofs. Morris (2002) echoed this last

result, with 40% of her pre-service teachers affirming that both inductive and deductive arguments assured certainty and failing to distinguish between the two types of arguments.

But prospective elementary teachers are often not very sophisticated in mathematics, so one might expect majors in technical fields, especially mathematics majors, to perform much better in proof settings. Such is the case, but only for some students, judging from anecdotes from university mathematics faculty. Furthermore, there are studies that point out areas of weakness, usually with small numbers of students. For example, case studies show that although some students may possess the transformational proof scheme, others seem to enter—and leave—university with little or limited abilities with, and views of, proof, mostly demonstrating external and empirical proof schemes (Sowder & Harel, 2003).

For some university students, even recognizing whether a given argument constitutes a proof cannot be taken as a given. In interviews, Selden and Selden (2003) asked their eight mathematics majors to judge whether four given mathematical arguments were valid, with only one of the arguments valid. The students performed only at a 46% correct level with their first judgments but did improve to 81% correct by the end of the interview with their fourth judgments, with the Seldens crediting the improvement to the added experience and reflection during the interview. Although the study did not examine long-term effects, it suggests that repeated validating activities may be a valuable growth opportunity.

Knowledge about what factors are important in devising proofs is limited. In an expert-novice study, with four graduate students in mathematics as the experts and four undergraduates as the novices, Weber (2001) noticed that “an understanding of mathematical proof and a syntactic knowledge of the facts of a domain are not sufficient for one to be a competent theorem prover” (p. 107); experts have what Weber calls “strategic knowledge,” knowledge that undergraduates do not exhibit—knowledge of the domain’s proof techniques (p. 111), knowledge of which theorems are important and when they are useful (p. 112), and knowledge of when, and when not, to use strategies based on symbol manipulation rather than deeper knowledge (p. 113). In a similar vein, Raman’s (2003) interviews of mathematics students and faculty suggested the importance of a “key idea”—“an heuristic idea which one can map to a formal proof with appropriate sense of rigor” (p. 323). She concluded, “For mathematicians, proof is essentially about key ideas; for many students, it is not” (p. 324). In expert-novice studies, one often does not know how the

experts acquired their expertise (or whether there is a selection factor involved), but knowing what differences exist may give ideas for instructional emphases. But in questioning university mathematics faculty about university mathematics majors' proof understanding, a "twice is nice" theme emerged: Exposure to the same material twice allows the student, on the second exposure, to focus on proof methods (Sowder, 2004). Perhaps these second exposures are helpful in attaining other aspects of Weber's strategic knowledge and Raman's key ideas, and in growing in deductive proof schemes. Marty (1991) felt that his explicit attention to proof methods, rather than the common focus on new mathematical content, helped his college students succeed in later mathematics courses.

University students' distinctions among axioms, definitions, and theorems are not sharp. Vinner (1977), for example, found that only about half of a group of Berkeley sophomores and juniors in mathematics could correctly identify all of three statements about exponents as definitions, as opposed to theorems or laws or axioms. Since much of algebra focuses on algorithms and these students had first studied the material in high school, perhaps the lack of such distinctions is expected. Yet, when Brumfiel (1973) questioned a class of University of Michigan juniors and seniors, nearly all of whom had completed a university course in formal geometry, he found that their mastery of similar distinctions were shockingly deficient. For example, collectively the students could recall only one axiom (two points determine a line), about half called a definition of isosceles triangle a theorem, and all were certain that, for two (given) independent postulates about points and lines, one could be deduced from the other. In Israel, Linchevsky, Vinner, and Karsenty (1992) found that only about one-fourth of their university mathematics majors understood that it is possible to have alternate definitions for concepts. These studies speak to about-proof topics. But about-proof topics cannot be mastered without understanding the proof topics themselves. In this case the topics in question involve the meaning and role of axioms and definitions. These studies, then, suggest a weak, or even absent, axiomatic proof scheme among mathematics majors—an observation that is consistent with findings of our study with mathematics majors (Harel & Sowder, 1998).

Findings Germane to the Transformational Proof Scheme

Recall that the deductive proof scheme class consists of two schemes: the transformational proof scheme and the axiomatic proof scheme. The former

is a special case of, and a conceptual prerequisite for, the former. The research findings we have just reported paint a gloomy picture as to the quality of students' proof schemes. The failure of mathematics instruction to help many students—even college-bound students and mathematics majors—construct the transformational proof scheme is evidenced in these reports. In this section we provide a closer look at the characteristics of the transformational proof scheme, particularly the deduction process characteristic.

As was defined earlier, the transformational proof scheme has three characteristics: *generality*, *operational thought*, and *logical inference*. As the language may suggest, our definition of the transformational proof scheme naturally draws one to Piaget's theory of intellectual developmental (for example, Inhelder & Piaget, 1958). Battista and Clements (1992) focused on Piagetian thought in their discussion of proof in an earlier handbook (pp. 439–440), identifying three levels of justification/proof, with the third level being marked by the student being "capable of formal, deductive reasoning based on *any* assumptions" (p. 439)—the generality characteristic—whereas at the second level, the student's reasoning has an empirical reality, with the student not realizing that an argument may cover all cases. The third level of thinking is often identified with Piaget's "formal operational" level—the operational thought characteristic. Although the nominal approximate age for the development of formal operational thought is early adolescence and there are different aspects of formal operational thought (e.g., proportional thinking, ability to control variables, propositional logic), studies suggest that many high school and college students have not reached the formal operational level in some areas important in mathematics. For example, Lawson, Karplus, and Adi (1978) gave tasks in several areas, including propositional logic (one such item is Task 1 in Figure 18.4), to students at grades 6, 8, 10, 12, and 13–14. They found "the lack of clear and substantial improvement with age for the propositional logic items," with percents correct by grade on the Task 1 item being 1.0%, 2.1%, 10.1%, 12.0%, and 16.2% (p. 469). A study of 16-year-old English students found that only 30% tested at a level consistent with formal operational thought (cited in Adey, 1999). It appears that it is risky to assume that logical reasoning at a high level develops automatically in the course of schooling, whether within or outside mathematical contexts.

Psychologists other than Piaget have examined logical thinking for many years, with especially fascinating work done during the last thirty years (for example, Evans, 1982; Rips, 1994; Wason & Johnson-Laird, 1972). More recently, Johnson-Laird (2001) has ob-

served that “deductive reasoning is under intense investigation. The field is fast moving and controversial” (p. 441), with at least some of the controversy coming from different hypothesized mechanisms of reasoning for selected tasks. There does not, however, seem to be controversy about the following findings. Performance on logical inferences involving modus ponens is usually reasonably good, but performance on those tasks involving modus tollens is weak, as is a full understanding of inferences involving if-then statements. Students are too willing to use invalid inference patterns like affirming the consequent (i.e., “if p then q ” and “ q ” yielding “ p ”). In general, humans do not seem to process negative statements as facily as affirmative ones, and context is an extremely important variable in performance on deductive reasoning tasks (Wason & Johnson-Laird, 1972). Figure 18.4, for example, gives logically isomorphic tasks on which the success rates of adults are strikingly different. Only 2 out of 24 (about 8%) of the respondents correctly identified the two cases that must be considered for Task 1, but 88% did so on Task 2 (reported in Wason & Johnson-Laird, 1972). Thus, our discussion here focuses primarily on findings and studies dealing with logic in mathematical settings, calling on known work from psychology when it seems particularly pertinent.

Although our focus is mainly on studies dealing directly with mathematics, one sobering study involving everyday contexts deserves special note, because some of the data come from prospective secondary school mathematics teachers. Using items from Eisenberg and McGinty (1974), Easterday and Galloway (1995) compared the performance of last-year university students planning to teach middle school or high school mathematics with those of 7–8th graders and 12th graders on a variety of reasoning tasks. The tasks were based on everyday contexts that should be non-

suggestive (e.g., “If John is big, then Jane is big. John is big. Is Jane big? Yes/no/maybe”). In particular, on modus tollens tasks, the college students scored 47%, about the same as the 12th grade calculus students but 20% *less* than the 7–8th graders’ 67%. These particular 7–8th graders were studying geometry and may therefore have been exposed to logic, but an earlier comparison in 1986 with 7–8th graders in advanced sections had given a similar result, with the 7–8th graders scoring 62% and the college students only 40% on the modus tollens tasks (Easterday & Galloway, 1995, p. 433). The authors concluded, “College students are barely performing better than children whom they may one day teach” (p. 435), and neither group was by any means topping out in performance.

There seem to be only a few data on students’ abilities with specific ideas from logic and within mathematics. Mentioned earlier were the limited results from NAEP, and there the picture was clouded by the use of a non-mathematical context. The large-scale Longitudinal Proof Project in England has, however, looked at students’ performance on if-then statements involving number theory ideas (Hoyles & Kuchemann, 2002). Among their findings was that 62% of 14-year-olds thought that a given if-then statement (for example, if the product of two numbers is odd, then the sum of the numbers is even) “said the same thing” as its converse. However, the longitudinal nature of the study also allowed Hoyles and Kuchemann to find a 7% improvement on this item over that of the same students at age 13. (In passing, it may be worth noting that so many theorems, especially in geometry, do have true converses, so it is easy to see how students may be insensitive to the logical difference between an if-then statement and its converse.)

Even advanced students have difficulty with quantifiers. Dubinsky and Yiparaki (2000) found that univer-

Task 1. Given four envelopes with a letter on the front side and a number on the back, select just the envelopes definitely needed to be turned over to find out whether they violate the rule. Separate envelopes show on front: D and C, and on back: 5 and 4. Rule to test: If a letter has a D on one side, then it has a 5 on the other side. Percent correct (D, 4): 8%	Task 2. Given four envelopes with a space for a stamp on one side and sealed or not, select just the envelopes definitely need to be turned over to find out whether they violate the rule. Envelopes show (a) back of sealed envelope; (b) unsealed envelope with flap up; (c) front of an envelope with stamp; (d) front of an unstamped envelope. Rule to test: If a letter is sealed, then it has a 5 pence stamp on it. Percent correct (a, d): 88%
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Figure 18.4 Two logically isomorphic tasks with “abstract” [Task 1] and “concrete” [Task 2] contexts (after Wason & Johnson-Laird, 1972, pp. 191–192).

sity students at various levels, including some in an abstract algebra course, had much greater trouble giving the mathematical meaning of a doubly quantified statement when the existential quantifier appeared before the universal quantifier (“There is a positive number b such that for every positive number a , $b \leq a$ ”—19% correct) than when the quantifiers were reversed, universal before existential (“For every positive number a there is a positive number b such that $b < a$ ”—59%). Selden and Selden (1995) noted that university mathematics students may have difficulties in even restating mathematical statements precisely, with their largely third- and fourth-year students often giving incorrect responses when quantifiers were involved. Thus, although there may be areas of apparent strength in the use of logic (for example, the use of modus ponens), there appear to be many areas of weakness as well, and at a wide gamut of levels of schooling.

An important point is that everyday usage of logical expressions may differ considerably from the precise usage in mathematics. Epp (2003) has summarized the differences in a compelling way, and O’Brien, Shapiro, and Reali (1971) have referred to “child’s logic” in describing some of the differences. For example, “or” in everyday usage is most often in the exclusive sense (“I’ll wear my sandals or my tennis shoes”), in contrast with the inclusive convention common in mathematics. An everyday if-then statement (for example, “If you finish your work, then you can watch the game”) often connotes what would be an if-and-only-if statement in mathematics and to many children seems to be an “and” statement. The disparities between everyday usages and mathematical usages are so marked that explicit instruction in logic as used in mathematics would seem to be necessary, with contrasts to the less precise everyday usages pointed out, yet, as Epp contends (2003), perhaps exploiting non-mathematical usages that do reflect the mathematically precise ones, as exemplars for the latter.

Another interesting disparity between everyday usage and mathematical usage is that of indirect proof. According to Freudenthal (1973) indirect proof is a very common activity. Seven to eight year old children used contradiction in game playing and checking conjectures (Reid & Dobbin, 1998). Antonini (2003) even found that indirect argumentations occurred spontaneously by students in his interviews with them about mathematical assertions. Yet research has shown that students experience difficulties with proof by contradiction in mathematics. Leron (1985), for example, observed that despite the simple and elegant form of certain proofs by contradiction, students experience what seem insurmountable difficulties. Lin, Lee, and Wu Yu (2003) see the ability to negate a statement as a prereq-

uisite ability for succeeding at a proof by contradiction. They found that the difficulty levels of students’ negating a statement can be ordered decreasingly as negating statements without quantifiers, negating “some,” negating “all,” and negating “only one.”

In general, then, there are many weak spots in students’ likely grasp of the logical reasoning used in advanced proof schemes. Is it a chicken-egg question, or can logical thinking and proof performance grow together? Later in this section we summarize several studies in which explicit instruction in logical principles was incorporated into high school geometry courses.

INSTRUCTIONAL-SOCIAL-CULTURAL FACTORS: EVIDENCE POINTING TO CURRICULUM AND INSTRUCTION

Students’ instructional history is without question an important variable in studying their performance in justification/proof settings. Hoyles (1997), in particular, makes a compelling case that studies of advanced learners’ performance are relatively meaningless without knowing what curricula (and what teaching emphases) they have received. Two important dimensions of instructional history are the curriculum and the teaching. We begin with studies of teaching and teachers, giving some evidence of the *delivered* curricula. Following this, we review a few studies that highlight the vast differences in curricula, primarily the *intended* curricula as evidenced by national or regional guides. The earlier report of several status studies of students’ performance have given a glimpse of the *learned* curricula. We then attend to some studies that give evidence of what might be achieved through *revised* curricula and teaching, as delivered by teachers with a different perspective on teaching, with the section including a review of a few studies that show the possibilities offered by technology. We conclude with several studies that have focused explicitly on logic as a vital component for what we call the deductive proof schemes.

Current Status

Teachers and Teaching

The emphasis that teachers place on justification and proof no doubt plays an important role in shaping students’ proof schemes. “You get what you teach” is a common aphorism, and the intended curriculum may differ markedly from the delivered curriculum, especially across classrooms. The essentiality of opportunity to learn must be recognized not only at the intended curriculum level but also in the teachers’

enacted curriculum. Yet, a study involving 62 mathematics and science teachers in 18 high schools in six states allowed Porter (1993) to note that for the mathematics teachers, "On average, no instructional time is allocated to students learning to develop proofs, not even in geometry" (p. 4).

Results on items allied to the areas of justification and proof in the 1996 NAEP mathematics assessment indicate not only that students perform much worse on items requiring explanation and justification, but also that at grade 8, the students of teachers who devote more time to "developing reasoning and analytical ability to solve unique problems" have noticeably higher overall scores than other students (Silver & Kenney, 2000). The importance of the role of the teacher (and the curriculum) in fostering justification and proof is further highlighted by the videotape study in the Third International Mathematics and Science Study (TIMSS), in which 30 eighth-grade sessions in each of Germany, Japan, and the U.S. were analyzed. The report reaches the following conclusion.

The most striking finding in this review of 90 classes was the rarity of explicit mathematical reasoning in the classes. The almost total absence of explicit mathematical reasoning in Algebra and Before Algebra courses raises serious questions about the ways in which those subjects are taught. In order for these courses to help introduce students to mathematical ways of knowing, some of the logical foundations of mathematical knowledge should be explicit. Of course, the total absence of any instances of inductive or deductive reasoning in the analyzed United States classes cries out for curriculum developers to address this aspect of learning mathematics. (Manaster, 1998, p. 803)

It should be clear that only an authoritarian proof scheme is likely to be fostered in these classrooms.

Despite Porter's (1993) finding of little attention to proof in high school, one would certainly expect more explicit attention to proof in the mathematics at those grade levels where proof most often is a conscious part of the curriculum. Senk (1985) noted, however, that there were consistent differences across schools in the geometry students' performances on her proof tasks. Tinto (1988) too noted that one of her four teachers seemed markedly different from the others in his approach to geometry. In their large-scale study of proof in British classrooms, Healy and Hoyles (1998) found that students who had been expected to write proofs and who had classes in which proof was taught as a separate topic performed somewhat better on proof items than other students. Thompson (1991), on the other hand, did not notice differences across her nine teachers and schools; her

sample, however, included three private schools and two magnet schools and so was perhaps not representative. Overall, it appears that at least some of the deficiencies in students' acquisition of more sophisticated proof schemes may stem from the lack of opportunity to engage in proof-fostering activities, even in courses where one would expect much attention to proof.

The evidence from the status studies of university students' proof knowledge suggests that some, if not many, precollege teachers are unlikely to teach proof well, perhaps because their own grasp of proof was probably limited in college and may not have grown since then. Knuth (2002a) examined the conceptions of proof of 16 practicing secondary school mathematics teachers, most with backgrounds that would pass a face-validity test for knowledge of mathematics. In interview settings, Knuth asked the teachers to respond to general questions about proof (e.g., What purpose does proof serve in mathematics?), to evaluate given arguments (both proofs and non-proofs), and to identify the arguments that were most convincing. Although all of the teachers endorsed the verification role of proof, none mentioned the explanatory role of proof (see the section on the concept of proof scheme). Six of the 16 thought it might be possible to find contradictory evidence of a (non-specified) statement that had been proved. Four of the 16 tested a statement with a given, endorsed proof with further examples (cf. Fischbein & Kedem, 1982), even though all of the teachers eventually acknowledged that it would not be possible to find a counterexample. Even though the teachers collectively correctly identified 93% of the correct arguments as being proofs, over a third of the non-proof arguments were rated as being proofs! Ten of the 16 accepted the proof of the converse of one statement as a proof of the statement. Thirteen of the 16 teachers found arguments based on examples or visual presentations to be most convincing. Although Knuth felt that their responses may have been directed toward *personally* convincing rather than *mathematically* convincing, that mathematics teachers would be convinced by such arguments more than by a mathematical proof is significant, because it reveals an apparent dominance of the empirical proof schemes among the teachers. Knuth (2002b) further examined these teachers' ideas about proof in the context of school mathematics (versus the earlier just-in-mathematics). In view of the NCTM (2000) recommendation that reasoning and proof be considered fundamental aspects of the study of mathematics at all levels of study, it is disappointing that the teachers in Knuth's study "... tended to view proof as an appropriate goal for the mathematics education of a minority of students" (2002b, p. 83), with 14 not considering

proof to be a central concern in school mathematics. And, to repeat, none even explicitly mentioned the explanatory role of a proof, although seven did mention its verification role—that a proof shows why a statement is true. Thirteen regarded the development of logical thinking or reasoning as the primary role of proof in school mathematics. Yet, the teachers did not completely reject justifications, but were willing to rely on informal “proofs” (e.g., examples and drawings) to support results, a practice that may mislead students into thinking that such are acceptable mathematical “proofs,” and reinforcing the acceptability of their empirical proof schemes.

Curricula

That mathematics curricula differ in their treatments of proof is by no means a recent phenomenon. For example, in his study of students’ proof explanations, Bell (1976) found that proof is the topic that shows the greatest variation in approaches internationally. He noted that this variation can be attributed to the tension between the recognition among teachers that deduction is essential to mathematics but that only the most capable students develop a good understanding of it. There is evidence that this condition remains true today as well. For example, Fujita and Jones (2003) compared the textbook treatments of geometry in lower secondary schools in Japan and Scotland and concluded the following.

Our analysis indicates that . . . Japanese textbooks set out to develop students’ deductive reasoning skills through the explicit teaching of proof in geometry, whereas comparative UK [United Kingdom] textbooks tend, at this level, to concentrate on finding angles, measurement, drawing, and so on, coupled with a modicum of opportunities for conjecturing and inductive reasoning. (p. 1)

It is, perhaps, natural to expect great variation in the treatment of curricular topics within countries that do not have national curricular and educational guidelines. In the U. S., for example, how geometry, the primary locus of proof efforts until recently, should be handled has led to vastly different opinions and occasionally to different approaches or emphases in school geometry (cf., e.g., Hoffer, 1981; NCTM, 1973, 1987; Usiskin, 1987).

Examples of Curricula Aimed toward Enhancing Students’ Proof Schemes

We do not claim that the studies touched on below use our language of proof schemes, but it will be clear that students who have had such treatments should pos-

sess different proof schemes from those that apparently result from traditional teaching. Of particular interest and importance, we think, are the feasibility studies that have involved children in elementary schools.

Earlier Feasibility Studies

Perhaps because of the spirit of the “New Math” times, and the call of the Cambridge Conference Report to move toward more sophisticated mathematics earlier in the curriculum, some studies in the 1970s focused on proof with younger children. To gauge whether and when children seem able to cope with proof or proof-like tasks, Lester (1975) devised a computer-delivered deductive system in a game-like form, with well-defined rules (i.e., postulates) and target configurations (i.e., theorems) to be attained by applying the rules (i.e., with proofs). Hence, the students were dealing implicitly with an axiomatic system.

Lester (1975) sampled 19 students from each of four groups, one group from grades 1–3, a second from grades 4–6, a third from grades 7–9, and the fourth from grades 10–12, gave them practice with the rules, and studied their performances on the target tasks. His grades 7–9 students performed as well as the students from grades 10–12, and the students from grades 4–6 solved about as many tasks as the older students, but took somewhat longer. Lester suggested that “even students in the upper elementary grades can be successful at mathematical activities that are closely related to proof” (p. 23). Indeed, King (1970, 1973) thoroughly developed a 17-day unit dealing with some elementary number theory results (e.g., a number which is a factor of two numbers is also a factor of their sum). He found that a group of 10 above-average sixth graders could reproduce the proofs initially developed with considerable teacher support (in contrast to a non-equivalent control group), but the evidence also suggested that the proofs were given from rote memory.

Fawcett’s study of the late 1930s deserves special mention. The title and sub-title give a good summary: *The Nature of Proof, A Description and Evaluation of Certain Procedures Used in a Senior High School to Develop an Understanding of the Nature of Proof*. Whatever the reason—World War II, the usual inertia in curriculum—this study seemed to have had little impact, even though it was reported in a yearbook of the NCTM (Fawcett, 1938/1995). His approach was surprisingly modern in tone. Fawcett’s summary includes the following.

The theorems [of geometry] are not important in themselves. It is the *method* by which they are established that is important, and in this study geometric

theorems are used only for the purpose of illustrating this method. The procedures used are derived from four basic assumptions:

1. That a senior high school student has reasoned and reasoned accurately before he begins the study of demonstrative geometry.
2. That he should have the opportunity to reason about the subject matter of geometry in his own way.
3. That the logical processes which should guide the development of the work should be those of the student and not those of the teacher.
4. That opportunity be provided for the application of the postulational method to non-mathematical material.

Non-mathematical situations of interest to the pupils were used to introduce them to the importance of definition and to the fact that conclusions depend on assumptions, many of which are often unrecognized. To make definitions and assumptions and to investigate their implications is to have firsthand experience with the method of mathematics . . . (p. 117)

Fawcett's teaching experiment, with a non-equivalent control group, continued through two school years, with the report covering just the first year. As the excerpt above suggests, the students eventually composed, collectively, their lists of undefined terms, definitions, and assumptions. The need for such elements arose in discussing everyday situations, such as the importance of definition in discussing how the governor of Ohio handled a particular bill (pp. 51–52). Of course the teacher played a major role in initiating such discussions and in providing fruitful leads for particular results, but in the large the students were responsible for conjecturing results and then proving them. The evaluation of the experiment was based on a state geometry test and a test of the ability to analyze non-mathematical material. Even though the experimental students, after one year of a two-year treatment, had not covered the usual material in the standard course, their performance (although not reported thoroughly) seemed satisfactory on the 80-point state geometry test: Median 52.0, state median 36.5 (p. 102). More telling was the experimental students' performance on the analysis of non-mathematical material, where they out-performed by far the control group (change score of 7.5 vs. a change score of 1.0; maximum possible not given) (p. 103). Fawcett also quoted the laudatory reactions of visitors and of the students themselves, contrasting the students' final remarks with the largely indifferent attitudes expressed at the beginning of the experiment.

One can only conclude from these studies that upper elementary school children can deal with proof

ideas or actions, and that high school students can develop meaningful understandings of proof if they are taught appropriately.

More Recent Feasibility Studies

Several researchers have emphasized a sociocultural perspective in investigating and enhancing the development of students' proof schemes. According to this perspective, the development of any higher voluntary form of human knowledge cannot be understood apart from the social context in which it occurs. As such, learning is necessarily a product of social interaction. Key to this perspective is Vygotsky's (1978) notion of "zone of proximal development" (ZPD): the difference between what the students can do under adult guidance or in collaboration with more capable peers and what they can do without guidance. A direct and critical implication of this perspective is *instructional scaffolding* (Mercer, 1995), which refers to the provision of guidance and support that is increased or withdrawn in response to the developing competence of the learner. Blanton, Stylianou, and David (2003) further suggest:

Since learning is viewed as a product of interaction, it follows that one's development within the ZPD is affected by the intellectual quality and developmental appropriateness of these interactions (Diaz, Neal, & Amaya-Williams, 1999). In other words, the extent of one's development within the ZPD is predicated in part upon how the more knowing other organizes, or scaffolds, the task at hand. Thus, if we intend to understand development within the ZPD, we must be thinking about if and how tasks can be scaffolded to extend one's learning. (p. 114)

Blanton, Stylianou, and David (2003) have investigated the nature and role of this scaffolding in the learning and teaching of proof. Specifically, they addressed two questions: (a) What is the nature and meaning of instructional scaffolding in the classroom in the development of students' proof ability? and (b) How do different types of scaffolding prompts from the teacher affect students' self-regulatory thinking about proof? Their results suggest

. . . students who engage in whole-class discussions that include metacognitive acts as well as transactive discussions about metacognitive acts make gains in their ability to construct mathematical proofs. Moreover, students' capacity to engage in these types of discussion is a habit of mind that can be scaffolded through the teacher's transactive prompts and facilitative utterances. . . . [This] suggests that students can internalize public argumentation in ways that facilitate private proof construction if instructional scaffolding is appropriately designed to support this. (p. 119)

Some of the most promising work directly or indirectly related to students' growth in elements of proof has been with elementary school children, as early as the primary grades and most often in problem-based settings where questions of proof have come up as intended by-products of the investigations. Evidence exists to indicate that a classroom environment that is conducive to social interactions among students and between the teacher and the students can be productive. Hence, these studies have often incorporated small-group work and whole-class discussions sharing how different children were thinking (that is, justifying their work) rather than just focusing on the numerical result. For example, Zack (2002) had two groups of fifth graders work on the following problem: Find all the squares on an n by n chessboard, first with $n = 4$, then $n = 5$, then $n = 10$. What if it was a 60 by 60 square? Can you prove that you have them all? After working on this problem, her students offered eight different counter-arguments to an answer of 2310 (based on 6 times the established answer of 385 for a 10 by 10). Surely students who experience such instruction will develop different proof schemes than will children whose teachers always judge the correctness of an answer. In some cases (Cobb et al., 1991) researchers worked closely with classroom teachers in designing the tasks and studying the effects of the children's work and their discussions. In other cases (e.g., Carpenter, Franke, Jacobs, Fennema, & Empson, 1998), the teachers were provided with a great deal of background on children's thinking strategies but left to their own in designing and carrying out lessons; however, an earmark of the classroom work was to call for students to give justifications.

Other examples of early appearances of proof ideas come from work aiming toward algebra in the elementary grades (Schifter, Monk, Russell, Bastable, & Earnest, 2003). Some of their instruction deliberately but indirectly deals with such generalizations as commutativity of addition, although that language may not be used in the early exposures. Key are the tasks and teacher questions. Questions such as "Will it always work that way?" or "Why does it work out that way?" are a natural part of the instruction and indeed become an expected part of the lesson. Their work shows that children's conviction about at least some generalizations may follow an interesting path. For example, first graders, when asked "Will it always work that way?" after noting a specific example for commutativity of addition, may be uncertain that the idea will always work. Later, they may be confident that the answer is "Yes" because they have tried lots of examples (an empirical proof scheme), but still later, they may return to uncertainty because they are aware that

there are many untested cases and that they have not tried them all. And, not unexpectedly, some students will over-generalize, accepting commutativity of subtraction as well.

Several Italian studies have also examined the nature of students' justifications as encountered in problem-based lessons (e.g., Boero, Chiappini, Garuti, & Sibilla, 1995; Boero, Garuti, & Mariotti, 1996). One example is their grade 8 students' study of shadows. Using data that the students collected over a period of time about the shadows of two vertical sticks, they examined the question of whether the shadows of a vertical stick and an oblique stick can be parallel, and if so, when. Small-group work, with teacher help, led to clearly stated conjectures (e.g., If the sun's rays belong to the vertical plane of the oblique stick, then the shadows are parallel) that were then examined further, with an eye toward establishing them "in general." In the analysis of these subsequent attempts at general arguments (i.e., proofs), the researchers noticed that in the successful proofs, there were connections with key observations made during the conjecture-forming stage. The researchers' collective work has led to their hypothesis of "cognitive unity," emphasizing the close connection between the reasoning during the formation of a conjecture and the reasoning in an eventual proof:

(D)uring the production of a conjecture, the student progressively works out his/her statement through an intensive argumentative activity functionally intermingling with the justification of the plausibility of his/her choices; during the subsequent statement proving stage, the student links up with this process in a coherent way, organizing some of the justifications ('arguments') produced during the construction of the statements according to a logical chain (Boero, Garuti, Lemut, & Mariotti, 1996, p. 119–120).

They also argue that such a process is followed by many mathematicians—during the conjecturing stage, the mathematician uses arguments that can later be adapted to support his or her mathematical proof—and make the case that much more instruction in mathematics should involve conjecturing.

One series of studies (Maher, 2002; Maher & Martino, 1996; Martino & Maher, 1999) carried out with instruction in a similar problem-based vein, is notable because of its long-term nature (occasional sessions over 14 years, usually separate from the regular mathematics classes) and because of proof behaviors—proof by contradiction, proof by cases, proof by mathematical induction—that arose naturally, if informally, even in elementary school, at least on the part of some students. The nature and flavor of the sessions is communicated by these retrospections of a participant:

Well, we break up into groups . . . like five groups of three, say, and everyone in their own groups would have their own ideas, and you'd argue within your own group, about what you knew, what I thought the answer was, what you thought the answer was and then from there, we'd all get together and present our ideas, and then this group would argue with this group about who was right with this . . . (Maher, 2002, p. 37). . . . You didn't come in and say, "this is what we were learning today and this is how you're going to figure out the problem." We were figuring out how we were going to figure out the problem. We weren't attaching names to that but we could see the commonness between what we were working on there and maybe what we had done in school at some point in time and been able to put those things together and come up with stuff and to do these problems to come up with, what would be our own formulas because we didn't know that other people had done them before. We were just kind of doing our own thing trying to come up with an answer that was legitimate and that no matter how you tried to attack it, we could still answer it . . . (Maher, 2002, p. 32).

As the excerpts illustrate, all of these student-centered, problem-based studies have involved a way of teaching that is in stark contrast to the stereotype of a mathematics class: Students check homework, teacher illustrates something new, students then do seat-work or homework to practice the new material. The didactical contract (Brousseau, 1997) in the experimental classes was obviously quite different from that in the stereotypical one. In particular, the "social norms" were quite different: Students were expected to work together rather than singly, students were to explain their solution methods, and students were to listen carefully and evaluate the explanations of other students—and hence perhaps learn different proof schemes.

Yackel and Cobb (1996) have sharpened the analysis of social norms to identify "sociomathematical norms," those social norms that refer specifically to *mathematical* activity. For example, coming to accept that an explanation is expected might be a general social norm, whereas what constitutes an acceptable mathematical explanation would be a sociomathematical norm (Yackel & Cobb, 1996). Other sociomathematical norms might include norms dealing with when different explanations are mathematically different, or when a justification is acceptable, or when justifications or explanations convey efficiency or elegance. Indeed, Yackel, Rasmussen, and King (2000) focused on the norms that developed during a problem-based undergraduate differential equations course. They found that students "frequently explained their reasoning without prompting, offered alternative expla-

nations and attempted to make sense of other students' reasoning and explanations, despite the fact that their prior experiences were with traditional approaches to mathematics instruction" (p. 276) and, in particular, there was evidence of the development of such sociomathematical norms as, what is an acceptable mathematical justification and what makes up a mathematically different explanation. That the students, who had likely experienced mathematics classrooms with quite different social norms, developed such norms is particularly encouraging. Yackel and Cobb point out that sociomathematical norms should be examined in teaching modes that are different from the inquiry mode in which they have been studied, because the classroom conduct, whether stereotypic or not, will automatically convey what is an acceptable norm. For example, one can wonder whether some students, young or experienced, might be more comfortable with a more directed approach. Dweck (1999) has identified different goal orientations on the part of students. Some students may be guided primarily by performance goals like grades, parent/teacher approval, high marks, or status with others, whereas others have what Dweck calls "learning goals"—their primary interest is in understanding or mastery. A student's extreme preference for performance goals may be an unfortunate aspect of schooling or society as it exists, of course, but it might also result in resistance to sociomathematical norms that do not obviously support perceived performance. It is an interesting question as to whether different sorts of teaching might shape a student's learning goals.

In most of the studies above, outside support for the teacher was particularly important. The question of what is feasible in classrooms without further teacher preparation or researcher involvement is crucial, as, for example, Yackel and Hanna point out (2003). It is nonetheless exciting to envision learners, starting in the primary grades and continuing through high school and college, developing the social and sociomathematical norms about proof, and the proof schemes, that one might wish, but it is daunting to think of the changes needed in curricula and teachers (and testing programs) to support the development of such norms. As an indication of areas of teacher preparation that are important, Martino and Maher (1999) suggest, based on their videotape analysis of their multi-year study, that there is a "strong relationship between (1) [the teacher's] careful monitoring of students' constructions leading to a problem solution, and (2) the posing of a timely question which can challenge learners to advance their understanding" (p. 53). And there may be useful norm-supporting ideas even for classes in which problem-based or

inquiry-based instruction is not the mode. For example, Blanton, Stylianou, and David (2003), based on their work with a college discrete mathematics course and extending the work mentioned earlier (Blanton & Stylianou, 2002), believe that an instructor's attention to likely student understanding (and hence a zone of proximal development) can enable an instructional scaffolding that, along with a careful discussion of (preferably) students' proofs, can support students' understanding of proof.

Again, all of these studies illustrate how appropriate mathematical curricula, together with appropriate teacher intervention, can help students to develop critical elements of deductive reasoning and an openness to the ideas of mathematical proof. As feasibility studies, these efforts have shown that even elementary school children can be remarkably able to make sense of mathematics, if given opportunities. Their explanations not only help them in their communication skills, but the explanations are likely also to help their classmates, to give to their teachers insights into the children's level of understanding, and to engender or foster students' growth in their proof schemes and their grasp of sociomathematical norms.

Use of Technology

The increasing use of dynamic geometry software during the last couple of decades has provoked researchers to look closely into the learning benefits as well as the potential risks of this tool. In this section we focus on studies conducted to investigate the impact of dynamic geometry environments (DGEs) on the learning of proof, primarily because those environments have been involved in several studies.

Generating and measuring many examples, as is now possible and easy with DGEs, would seem only to support the idea that examples prove a result (the empirical proof scheme) and hence interfere with any need for deductive proof. Chazan (1993a,b) noted and corroborated that the ideas that evidence is proof and that proof is merely evidence were widespread. He interviewed 17 students who were taught geometry in a DGE environment (1993b) to see how students fared when they experienced a geometry curriculum that involved both measurement of cases and deductive proof, with the curriculum including attention to the different types of argumentation involved with the two methods. He found both evidence-is-proof (the empirical proof scheme) and proof-is-merely-evidence postures on the part of the students, and hence he noted that the "comparison and contrast of verification and deductive proof certainly deserves explicit attention in mathematics classrooms" (1993b, p. 382), with the teachers involved in the study feeling that

more time should have been spent in dealing with students' doubts about deductive proofs (1993a, p. 109). However, he also noticed that "fewer students considered evidence to be proof . . . whereas more students were skeptical about the limits of applicability of deductive proofs. . . . Some students seemed to become more skeptical about deductive proofs as a result of becoming more skeptical about measurement of examples" (1993a, p. 109). He also suggested, based on interviewee comments, "that the explanatory aspect of proofs is a useful starting point for a discussion of the value of deductive proofs" (1993b, p. 383), that some students have no idea of what deductive proofs are intended to do, and that some students resist the idea that a deductive argument can assure that there cannot be counterexamples.

A special issue of the journal, *Educational Studies in Mathematics* (2000, 44), included four reports on teaching experiments involving DGE. The first four are by Mariotti, by Jones, by Marrades and Gutierrez, and by Hadas, Hershkowitz, and Schwartz. The last paper is by Laborde, and synthesizes the previous four, providing a connecting theme among them by using Brousseau's construct of "milieu." We continue with these studies.

One of the significant results of the four studies is that their findings address a major concern regarding the use of DGEs in the teaching of geometry in school: "the opportunity offered by [DGE] to 'see' mathematical properties so easily might reduce or even kill any need for proof and thus any learning of how to develop a proof" (Laborde, 2000, p. 151).

Jones' study was a teaching experiment with 12-year-old students. The intervention involved students working in pairs or small groups on the classification of quadrilaterals. The instructional activities involved tasks where the students were to reproduce a figure that could not be "messed up" by dragging any of its components (a vertex or segment). More advanced activities included tasks of producing a figure that can be transformed into another specified figure by dragging (e.g., from a rectangle to a square). In each case the students had to explain why their constructed figure was the expected one. Laborde (2000) points out that this type of explanation consists of giving the conditions that imply that the constructed figure is the expected type of quadrilateral, a necessary activity to understand how proof works. From our perspective, these explanations in the context of a DGE involve movement, and, therefore—as we have discussed in earlier in the section—they are causal, and hence deductive. Accordingly, Jones' (2000) study suggests that the DGE does not necessarily eliminate the need for proof in the students' eyes but can enhance students'

deductive proof scheme. The developmental path of students' conception in this study started where students were. Initially, according to Jones, students lacked the capability to describe or explain in precise mathematical language. The instructional emphasis in this stage was on description rather than explanation, where students utilized perception rather than mathematical language to describe their observations. As the emphasis extended to explanations, students' language became more precise but was mediated by the DGE terminology (e.g., the term "dragging"). By the end of the teaching experiment, students' explanations related entirely to the mathematical context.

Mariotti's study was a long-term teaching experiment with 15–16-year-old students. Students were engaged in DGE activities through which the students themselves constructed a geometric system of axioms and theorems as a system of Cabri Geometry commands. Similar to Jones, Mariotti emphasized activities where the task is for pairs of students to construct geometric figures, describe the construction procedure, and justify why the procedure produces the expected figure. The basic conceptual change that Mariotti's (2000) study achieved was in students' status of justification, which transitioned from an "intuitive" geometry—a collection of self-evident properties—to a theoretical geometry—a system of statements validated by proof. The theoretical geometry that Mariotti's students had constructed seems to be more than a deductive system, in that students were not only constructing and proving theorems but also establishing the axioms on which these theorems rest, and thereby laying the foundations for their axiomatic proof scheme.

Hadas, Hershkowitz and Schwarz' (2000) study was done in the context of a geometry course that emphasized the concept of proof. They developed instructional activities involving students making assertions about certain geometric relations and later checking them with a DGE. The choice and sequence of activities were such that upon checking their assertions with the DGE the students would find them to be false—a realization that would make the students curious as to the reason for the falsity of the conjecture. For example, in one of the activities the students began with two tasks. The first task was to measure (with the software) the sum of the interior angles in polygons as the number of sides increases, generalizing their observation, and then explaining their conclusion. The second task was to measure (with the software) the sum of the exterior angles of a quadrilateral. Following this, the students were asked to hypothesize the sum of the exterior angles for polygons as the number of sides increases, and to check their hypothesis by

measuring (with the software) and explain what they found. Hadas, Hershkowitz and Schwartz succeeded in creating in the students a need to find out the cause for their assertion to be untrue. Laborde (2000) points out that such an achievement would have been impossible without the use of a dynamic geometry system, for "the false conjectures came after students were convinced of other properties thanks to the DG system. . . . [The] interplay of conjectures and checks, of certainty and uncertainty, was made possible by the exploration power and checking facilities offered by the DG environment" (p. 154).

Marrades and Gutierrez investigated how DGEs can help secondary-school students (aged 15–16 years) enhance their proof schemes. As in Hadas, Hershkowitz and Schwarz' study, Marrades and Gutierrez showed that a DGE can help students realize the need for formal proofs in mathematics. By interpreting their results in terms of our taxonomy of proof schemes, an important observation reported in their study is that students' transition to deductive proof schemes is very slow; the total teaching experiment lasted 30 weeks, with two 55 minute class per week. Of particular importance is their finding that for this transition to take place instruction must not ignore students' current empirical proof schemes and must institute a didactical contract that attempts to suppress the authoritative proof scheme. Their method was to repeatedly emphasize "the need to organize justifications by using definitions and results (theorems) previously known and accepted by the class" (p. 120). Finally, another significant finding of this study is that the ability to produce deductive proof evolves hand in hand with students' understanding of subject matter: the concepts and properties related to the topic being studied. This is consistent with other findings. Simon and Blume (1996), for example, illustrated that a learner may not fully understand another's proof because of a limited grasp of the concepts addressed in the proof (p. 29). One can argue that such exposure might lead to a disequilibrium and eventually a greater understanding of both the concepts and the proof.

Hence, DGEs are a promising tool, but they do not automatically or easily lead to improved proof schemes. Accomplishing that sort of growth apparently requires a carefully laid-out curriculum (cf. de Villiers, 1999) and considerable adjustment by a teacher accustomed only to telling as the mode of instruction. Lampert (1993), for example, described some of the difficulties encountered by teachers who allowed conjecturing with a DGE. (Making conjectures, itself, may be difficult for students new to the expectation. Koedinger [1998], with an eye toward possible software activities,

noted that with his task of writing a conjecture about kites, given the definition of kite, about a quarter of the roughly 60 geometry students could not come up with a non-trivial conjecture within 20 minutes.) Lampert noted that the change from “sage on the stage” to “guide on the side” required adjustments both for the teacher and the students, since a different sort of didactical contract was involved. Teachers were also concerned about the coverage of a standard body of content (for external testing purposes, or for later courses), as well as the departure from the usual, familiar axiomatic development that often eventuated.

Does Explicit Teaching of Logic Work?

The use of deductive proof schemes, at least implicitly, involves logic. Calls for the explicit teaching of logic are easy to find. More than 40 years ago, consensus at the future-oriented Cambridge Conference on School Mathematics (1963) was that “it is hardly possible to do anything in the direction of mathematical proofs without the vocabulary of logic and explicit recognition of the inference schemes” (p. 39). There have been occasional caveats, as in Suppes’ remarks:

I would not advocate an excessive emphasis on logic as a self-contained discipline. . . . What I do feel is important is that students be taught in an explicit fashion classical rules of logical inference, learn how to use these rules in deriving theorems from given axioms, and come to feel as much at home with simple principles of inference like *modus ponendo ponens* as they do with elementary algorithms of arithmetic” (1966, p. 72).

Studies of the effects of an explicit attention to logic have not, however, indicated that there is then a pay-off in proof-writing. School geometry has long had a goal of improving students’ logical thinking (cf. NCTM, 1970), and several studies have looked at the influence of including an explicit, concentrated treatment of logic in that course (e.g., Deer, 1969; Mueller, 1975; Platt, 1967). Yet, the usual outcome, even when the logic treatment has involved up to four weeks (Platt) and/or a reasonable number of students (Mueller, 146 students; Platt, 12 classes), is a finding of no-significant-differences in proof performance.

It is reasonable to expect that the teacher’s emphasis on logical reasoning, even in the absence of explicit treatment in a curriculum, might influence the students’ use of logic themselves. In a study of the influence of teachers’ language on performance, Gregory and Osborne (1975) explored whether the teachers’ use of logic influenced their students’ performance on a logic test, by examining audiotapes

of five lessons of 20 junior high school mathematics teachers. Although acknowledging some design problems, they found a significant positive correlation between logic test performance for some types of inferences (e.g., when negatives were involved) and the usages of if-then statements by the five teachers with the greatest average use vs. usages of the five teachers with the lowest. Eye-catching, however, was the range in the averages of the number of usages per lesson of if-then sorts of statements: 8.3 to 40.6, with individual lessons giving from 3 to 48 usages (p. 28). Unfortunately, there was no control for content covered (some teachers were teaching number theory, some word problems, and some properties of geometric figures), so the range is only suggestive of what might be an important element in students’ growth in deductive reasoning: the frequency of use of logical statements and logical reasoning in the classroom.

Thus there is much evidence that some important elements of deductive reasoning are not natural parts of students’ repertoires, at a variety of school levels. How to develop these elements so that they are recognized and utilized in mathematical proofs is an open question, and may not be realized through a single, short unit. As the 2000 NCTM Principles and Standards assert, “Reasoning and proof cannot simply be taught in a single unit on logic, for example, or by ‘doing proofs’ in geometry” (NCTM, 2000, p. 56).

REFLECTION

In this chapter we have presented a comprehensive perspective on proof learning—a perspective that addresses mathematical, historical-epistemological, cognitive, sociological, and instructional factors. Comprehensive perspectives on proof are needed, we argued, in order to better understand the nature and roots of students’ difficulties with proof so that effective instructional treatments can be designed and implemented to advance students’ conceptions of and attitudes toward proof. Our perspective grew out of a decade of investigations—empirical as well as theoretical—into students’ conceptions of proof. In various periods and stages of these investigations we have repeatedly confronted questions that collectively address a combination of the five factors mentioned above.

The notion of “proof scheme” serves as the main lens of our comprehensive perspective on proof. Through it, for example, we analyze and interpret students’ proving behavior—in their individual work as well as in their interaction with others—and understand the development of proof in the history of

mathematics. *A proof scheme consists of what constitutes ascertaining and persuading for a person (or community).* This definition was born out of cognitive, epistemological, and instructional considerations. Specifically, a critical observation in our and other scholars' work is that proof schemes vary from person to person and from community to community—in the classroom, with individual students as well as the class as a whole, and throughout history. "Proof," when viewed in this subjective sense, highlights the student as learner. As a result, teachers must take into account what constitutes ascertainment and persuasion for their students and offer, accordingly, instructional activities that can help them gradually refine and modify their proof schemes into desirable ones. This subjective view of proof emerged from our studies and impacted many of the conclusions we drew from them. For example, it influenced our conclusions as to the implications of the epistemology of proof in the history of mathematics to the conceptual development of proof with students, the implications of the way mathematicians construct proofs to instructional treatments of proof, and the implications of the everyday justification and argumentation on students' proving behaviors in mathematical contexts. The subjective notion of proof scheme is not in conflict with our insistence on unambiguous goals in the teaching of proof—namely, to gradually help students develop an understanding of proof that is consistent with that shared and practiced by the mathematicians of today. The question of critical importance is: What instructional interventions can bring students to see an intellectual need to refine and alter their current proof schemes into deductive proof schemes (Harel, 2001)?

The status studies we have reviewed and presented in this paper show the absence of the deductive proof scheme and the pervasiveness of the empirical proof scheme among students at all levels. Students base their responses on the appearances in drawings, and mental pictures alone constitute the meaning of geometric terms. They prove mathematical statements by providing specific examples, not able to distinguish between inductive and deductive arguments. Even more able students may not understand that no further examples are needed, once a proof has been given. Students' preference for proof is ritualistically and authoritatively based. For example, when the stated purpose was to get the best mark, they often felt that more formal—e.g., algebraic—arguments might be preferable to their first choices. These studies also show a lack of understanding of the functions of proof in mathematics, often even among students who had taken geometry and among students for whom the curriculum pays special attention to conjecturing and

explaining or justifying conclusions in both algebra and geometry. Students believe proofs are used only to verify facts that they already know, and have no sense of a purpose of proof or of its meaning. Students have difficulty understanding the role of counterexamples; many do not understand that one counterexample is sufficient to disprove a conjecture. Students do not see any need to prove a mathematical proposition, especially those they consider to be intuitively obvious. This is the case even in a country like Japan where the official curriculum emphasizes proof. They view proof as the method to examine and verify a later particular case. Finally, the studies show that students have difficulty writing valid simple proofs and constructing, or even starting, simple proofs. They have difficulty with indirect proofs, and only a few can complete an indirect proof that has been started.

We believe there is a need for more longitudinal studies regarding students' proof schemes. The most difficult studies to carry out, for both financial and design reasons are longitudinal studies. Yet we cannot gain a solid understanding of the effect on students of continued attention to justification and proof throughout their studies in mathematics, except through longitudinal studies. A one- or two-year exposure (let alone a one-semester treatment) to instruction and curricula attentive to reason-giving can be dwarfed by a multiple-year focus on instruction and curricula that, to use an extreme example, emphasize rote skills likely to be useful in some external testing program. In the latter cases, being asked to give reasons and arguments might well be viewed as aberrations and irrelevant to the perceived "really important" side of mathematics.

The findings from studies of teachers' conceptions of proof do not look much better than those with students. Overall, teachers seem to acknowledge the verification role of proof, yet for many the empirical proof schemes seem to be the most dominant, even in dealing with mathematical statements, and they do not seem to understand other important roles of proof, most noticeably its explanatory role. Some teachers tend to view proof as an appropriate goal only for the mathematics education of a minority of students, not considering proof and justification to be a central concern in school mathematics, as has been repeatedly called for by the mathematics education leadership (e.g., NCTM, 1989, 2000). Studies show that little or no instructional time is allocated to the development of the deductive proof schemes, not even in geometry. In the U.S. explicit mathematical reasoning in mathematics classes is rare, and in algebra and pre-algebra courses it is virtually absent. Many teachers are unlikely to teach proof well, since their

own grasp of proof is limited. It is important to determine better the extent to which teachers are equipped to deliver a curriculum in which proof is central. Results from studies like those of Knuth (2002a) and Manaster (1998), if indeed typical of a widespread performance of mathematics teachers, demand attention both on the part of university mathematics departments, which have a primary responsibility for the preparation of mathematics teachers, and on school districts, which support the continued development of their existing faculty.

The bright side of the findings is that students who receive more instructional time on developing analytical reasoning by solving unique problems fare noticeably better on overall test scores. Likewise, students who have been expected to write proofs and who have had classes that emphasized proof were somewhat better than other students. It also seems possible to establish desirable sociomathematical norms relevant to proof, through careful instruction, often featuring the student role in proof-giving. There has been a concern that the ease with which technology can generate a large number of examples naturally could undercut any student-felt need for deductive proof schemes. Fortunately, several studies have shown that with careful, non-trivial planning and instruction over a period of time, progress toward deductive proof schemes is possible in technology environments, where such desiderata as making conjectures and definitions occur.

An important element in deductive proof schemes is of course the use of logical reasoning. Yet there is evidence that many students, and possibly even many teachers, do not have a good grasp or appreciation of some important principles of logic. Nor is it clear as to how best to devise instruction to improve performance with logic in mathematical (and non-mathematical) contexts. It is unclear to us how best to prepare students to deal with the logical reasoning essential in mathematical proofs and valuable in even informal justifications—osmosis, or explicit attention? And, in particular, the knowledge of teachers of mathematics about logical reasoning may be a matter of concern (e.g., Easterday & Galloway, 1995). We see a need for the incorporation of items on proof or logic (even multiple-choice ones) into the periodic National Assessments of Education Progress. From the practical viewpoint, the NAEPs exist, and they offer a view of performance across the U.S. Even more pleasing would be to see large-scale efforts devoted explicitly to the study of performance in proof and logic, like those in Great Britain (Healy & Hoyles, 1998, 2000; Hoyles & Kuchemann, 2002). A deep look at the students and teachers' knowledge of proof, on the one hand, and at the development of the deductive proof scheme in

the history of mathematics, on the other, has provided us with important insights as to what might account for students' difficulties in constructing this scheme and what instructional approaches can facilitate its construction. In particular, considerations of historical-epistemological developments have led us to new research questions with direct bearing on the learning and teaching of proofs. For example:

1. To what extent and in what ways is the nature of the content intertwined with the nature of proving? In geometry, for example, does students' ability to construct an image of a point as a dimensionless geometric entity impact their ability to develop the Greek axiomatic proof scheme?
2. What are the cognitive and social mechanisms by which deductive proving can be necessitated for the students? The Greek's construction of their geometric edifice seems to have been a result of their desire to create a consistent system that was free from paradoxes. Would paradoxes of the same nature create a similar intellectual need with students?
3. Students encounter difficulties in moving between proof schemes, particularly from the Greek's axiomatic proof scheme (the one they construct in honors high-school geometry, for example) to the modern axiomatic proof scheme (the one they need to succeed in a real analysis course, for example). Exactly what are these difficulties? What role does the emphasis on *form* rather than *content* in modern mathematics (as opposed to Greek mathematics where content is more prominent) play in this transition? Can students develop the modern axiomatic proof without computational fluency? What role does the causality proof scheme play in this transition?

These questions are examples of what we have delineated as important contributions from the history of mathematics to our thinking about students' proof schemes. But there are likely to be other valuable ideas from further study of the growth or development of proof ideas in the history of mathematics. How mathematical proof arose in other cultures—e.g., the basis for the Japanese temple drawings (Rothman & Fukagawa, 1998)—would in itself be fascinating and potentially instructive about how proof ideas might be introduced or developed today. In this respect, our effort to form a comprehensive perspective on proof is an attempt to understand what might be called the

“proving conceptual field,” a term analogous to Vergnaud’s (1983, 1988) “multiplicative conceptual field.” Like the multiplicative conceptual field, the proving conceptual field may be thought of as a set of problems and situations for which closely connected concepts, procedures, and representations are necessary.

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THE DEVELOPMENT OF GEOMETRIC AND SPATIAL THINKING

Michael T. Battista

MICHIGAN STATE UNIVERSITY

The usefulness of axiomatics . . . goes beyond that of demonstration. . . . It permits us to construct simplified models of reality and thus provides the study of the latter with irreplaceable dissecting instruments. . . . [However] axiomatics cannot claim to be the basis of, and still less to replace, its corresponding experiential science. . . . Axiomatic geometry is incapable of teaching us what the space of the real world is like.

—Piaget, 1950, pp. 30–31

Every spatial representation can depend in part on the geometry of the object and in part on the geometry of the subject—and the dosages of each are hard to determine.

—Piaget, 2001, p. 223

In this chapter I examine geometric and spatial reasoning in a broad context. I describe and analyze recent research on the nature and development of students' thinking about geometry, geometric measurement, and aspects of spatial reasoning that are related to geometry. I concentrate on empirical and theoretical research, not curriculum and technological tool design. I focus on key ideas rather than comprehensively surveying the field. Where possible, I discuss the cognitive roots of geometric reasoning.

DEFINING THE SUBJECT: GEOMETRY AND SPATIAL REASONING

Geometry is a complex interconnected network of concepts, ways of reasoning, and representation systems that is used to conceptualize and analyze physical and imagined spatial environments. Geometric reasoning consists, first and foremost, of the inven-

tion and use of formal *conceptual systems* to investigate shape and space (Battista, 2001a, 2001b). For instance, mathematicians employ a property-based conceptual system to analyze and define various types of quadrilaterals and triangles. This system uses concepts such as angle measure, length measure, congruence, and parallelism to conceptualize spatial relationships within and among the shapes. So, defining a square to be a four-sided figure that has four right angles and all sides the same length creates an idealized property-based concept that, with proper spatial grounding, can help people reason more precisely about this special class of shapes.

Underlying most geometric thought is spatial reasoning, which is the ability to “see,” inspect, and reflect on spatial objects, images, relationships, and transformations. Spatial reasoning includes generating images, inspecting images to answer questions about them, transforming and operating on images, and maintaining images in the service of other mental operations (Clements & Battista, 1992; Presmeg,

1997; Wheatley, 1997). Thus, spatial reasoning provides not only the “input” for formal geometric reasoning, but critical cognitive tools for formal geometric analyses.

THEORIES OF GEOMETRY LEARNING PRELIMINARIES: DRAWINGS, DIAGRAMS,¹ FIGURES, SHAPES, AND CONCEPTS— TIPS OF AN ENIGMATIC ICEBERG

Before examining theories on geometry learning, it is important to reflect on and clarify the nature of the various “primitive objects” that students operate on in geometric and spatial reasoning.

The Objects of Geometric Analysis

Researchers often distinguish two types of geometric objects: “Drawing refers to the material entity while figure refers to a theoretical object” (Laborde, 1993, p. 49). Laborde argued that teaching often “confuses drawings and the theoretical geometrical objects that the former represent” (1998, p. 115). She further argued that student difficulties in geometry often arise because students reason about material drawings when they are expected to reason about theoretical geometrical objects (Laborde, 1993). (See Laborde, 1993 for further elaboration of these distinctions.) Similarly, Presmeg observed that, “A picture or a diagram is by its nature one concrete case, yet for any but the most trivial mathematical thinking it is necessary to abstract and to generalize” (1997, p. 305). Presmeg claimed that the “one-case concreteness” of drawings and images is the source of many difficulties in visualization-based mathematical reasoning.

However, the complexity of the issue is only partially captured by the drawing-figure dichotomy. Jones, following Holzl, argued that “learners can get ‘stuck’ somewhere between a drawing and a figure” (2000, p. 58). Furthermore, researchers in psychology and neuroscience have made additional distinctions that illustrate the complexity of identifying the objects of geometric and spatial analysis. For instance, some researchers have characterized the difference between physical and perceptual objects in terms of distal and proximal stimuli: “A distal stimulus is an actual object or event ‘out there’ in the world; a proximal stimulus is the information our sensory receptors receive about that object” (Coren, Ward, & Enns, 1994, p.

485). Other researchers such as Edelman (1992) have distinguished between perceptual and conceptual objects. He argued that perception is nonconscious and operates on signals from the outside world received by the body’s receptors, whereas conception is conscious and operates on the activity of those portions of the brain that record signals from the outside world. Smith (1995) distinguished between a category, which is a group of objects in the world that belong together, and a concept, which is “a mental representation of such a group” (1995, p. 5). Finally, consider the mathematical formalization of an idea: “The concept definition is a form of words used to specify that concept” (Tall & Vinner as quoted in Tall, 1992, p. 496).

Integrating and synthesizing these ideas leads to the identification of five types of basic objects involved in geometric and spatial thought. A *physical object* is an actual physical entity such as a door, box, ball, geoboard figure, picture, drawing, or dynamic draggable computer figure (Coren et al.’s distal stimulus). A *sensory object* is the set of sensory activations evoked when an individual views a physical object (Coren et al.’s proximal stimulus). A *perceptual object* is the mental entity perceived by an individual when viewing a physical object (Edelman’s perception). A *conceptual object* or conceptualization (Edelman’s conception) is the conscious meaning or way of thinking activated by an individual (a) in response to a perceptual object, memory of a perceptual object, or concept definition; or (b) constructed anew from other conceptual objects. A *concept definition* is an explicit formal mathematical verbal/symbolic specification of a conceptual object.²

Another way to distinguish cognitive objects in geometric reasoning is by their use (Battista, in press a). An *object* is a mental entity that is operated on (consciously or unconsciously) during reasoning. A *representation* is something that “stands for” something else (Goldin & Kaput, 1996). In general, in geometric thought, one reasons *about* objects; one reasons *with* representations.

To understand geometric and spatial reasoning, one must understand individuals’ cognitions about the various objects described above, including how these cognitions are interrelated. However, in analyzing these cognitions, one must be aware of several complications.

Complication 1: Conception Affects Perception

What one “sees” is affected by what one knows and conceives. For example, in the size-constancy

¹ Although one might distinguish the terms *drawing* (something created freehand by a student or teacher) and *diagram* (a carefully made graphic given in a textbook or other printed material), the two terms seem to be used synonymously in the literature.

² Two additional types of objects discussed in the research literature—mental models and concept images—will be discussed later.

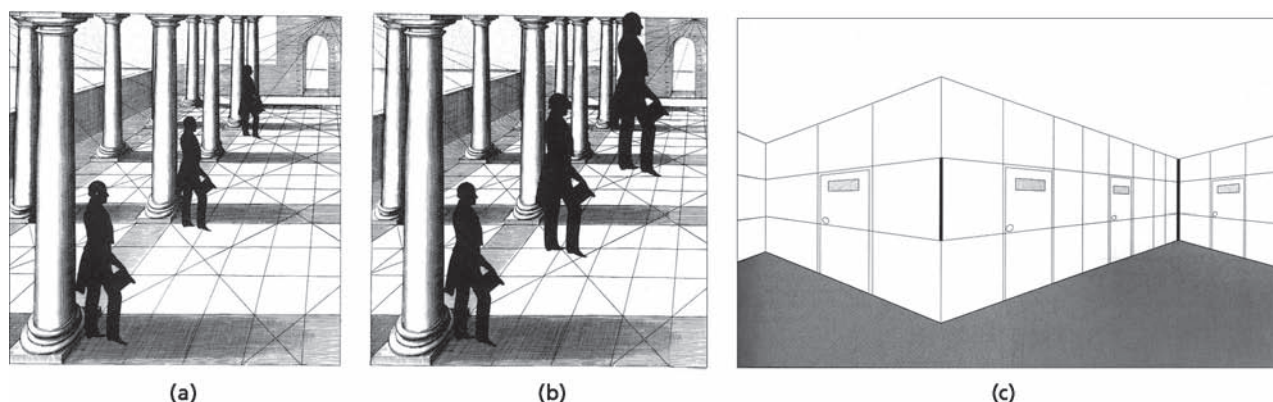


Figure 19.1 Perception of size determined by context (Coren et al., 1994, pp. 488, 497).

phenomenon, the size of an object is conceptualized as constant even though the retinal image it projects varies as the object's distance from the eyes varies; the mind takes into account the apparent distance of the object from the viewer to adjust perceived size (Coren et al., 1994). In Figure 19.1a, the three men appear to be the same size even though the most distant figure is only about a third of the size as the nearest figure. In contrast, in Figure 19.1b, the most distant figure appears much larger than the nearest, even though the actual images are identical. Thus, perception of size is adjusted based on a contextual conceptualization of distance. Figure 19.1c further illustrates how perception is affected by conception and context. Amazingly, the two dark vertical segments are congruent. But our knowledge of the physical world, and the resulting mental models created to represent it, cause us to perceive the two segments as very different in length.

Similarly, verbal labeling can affect conception—an ambiguous figure that is called a broom when presented is drawn to look like a broom, whereas if it is called a rifle when presented, it is drawn as a rifle (Coren et al., 1994). Might a similar phenomenon occur in geometry? Does using a different name for a drawn shape—square, rhombus, diamond—affect a student's perception/conception of the shape? Also, research shows that drawing a figure, in combination with experience and verbal labeling, can make it easier to find when it is embedded in a more complex figure (Coren et al.). Thus again, experience (i.e., current knowledge) affects perception. Finally, research suggests that current knowledge can affect perception of similarities among instances, and thus classification, possibly because such knowledge can influence what stimulus information is attended to (Ross, 1996). For instance, a square with vertical and horizontal sides turned 45° might be perceived as a diamond because previous experience causes a student to attend not

to relationships between parts of the shape but to its position, including implicitly, say, how angles line up vertically and horizontally.

Complication 2: Diagrams as Data or Representations?

Diagrams, or more generally instances of physical objects, play two major roles in geometry. On the one hand, instances of physical objects can be thought of as the *data* for geometric conceptualization—it is through analysis of such objects that geometric shape concepts are derived. On the other hand, instances of physical objects, including diagrams, are used to represent formal geometric concepts. However, both curricula and researchers generally neglect the process of forming concepts from physical objects and instead focus on the “representational” perspective. Consider the following statements by researchers: “Diagrams are intended as models . . . meant to be understood as *representing* a class of objects. Every diagram has characteristics that are individual and not *representative* of the class” (Yerushalmy & Chazan, 1993, p. 25). “Students often attribute characteristics of a drawing to the geometric object it *represents*” (Clements & Battista, 1992, p. 448). Chazan and Yerushalmy (1998) argued that this “representational perspective” on diagrams is rooted in the traditional axiomatic approach to geometry instruction. Geometric “objects” are either undefined or are defined formally in terms of undefined terms; “diagrams are aids for intuition and are not the objects of study themselves” (p. 70). And even though Chazan and Yerushalmy stated that geometry “should more explicitly build on students’ experience of the world around them” (p. 70), they seemed to conceptualize diagrams and figures in representational terms, citing Aleksandrov: “Geometry has as its object the spatial forms and relations of actual bodies, *removed from their other properties and considered from the purely abstract point of view* [italics added]” (p. 70).

The representational perspective focuses on abstract, general mathematical concepts, which are taken as the objects of study. A diagram is a mere indicator of an abstract mathematical concept; so many characteristics of a particular diagram are deemed *irrelevant distractors*. Consequently, most mathematics education research has treated diagrams as imperfect representations of idealized abstract concepts, neglecting the processes by which formal geometric concepts are abstracted from particular instances. Taking a broader perspective can provide additional insights into the processes by which students construct knowledge about shapes and shape classes. Instead of focusing exclusively on diagrams as representations of formal mathematical concepts, researchers must investigate more carefully and explicitly two complementary areas: (a) the process by which students progress from analysis of particular figures to general abstractions about classes of figures (it is from the analysis of particulars that abstract conceptualizations arise) and (b) the mechanisms that enable students to utilize formal abstract geometric concepts to analyze particular figures. Investigating these areas is critical to understanding how students construct understanding of abstract geometric concepts that are applicable not only within mathematics but to real-world problems.

This new perspective reveals several extremely important avenues for investigation. For instance, how does the ability to abstract what is common among a set of shapes (or what is invariant in a dynamic diagram) develop among children, and how can instruction and curricula promote this ability? How are abstract, general conceptualizations produced? Researchers need to describe how students move from the particular to the general. Investigating such abstractive generalization is an important but neglected area in research on geometric thinking. Although mathematics is about abstract concepts, mathematics education research must focus on exactly how individuals derive meaning for abstract concepts from particular instances.

Difficulties With Drawings and Diagrams

Drawings and diagrams are essential for understanding geometric ideas. In traditional use, drawings and diagrams have been used for two major purposes—(a) to represent classes of shapes (e.g., the *set* of rectangles) and (b) to represent geometric relationships (e.g., the bisectors of the angles of a triangle meet at the incenter of the triangle). Students, however, have difficulty with both uses, often because they

do not understand the nature of the objects being considered. In particular, students often attribute irrelevant characteristics of a diagram to the geometric concept it is intended to represent (Clements & Battista, 1992; Yerushalmy & Chazan, 1993). For instance, students might not identify right triangles drawn in nonstandard orientations (having unintentionally abstracted orientation as an attribute). Another difficulty occurs when students use diagrams in proofs (Clements & Battista, 1992; Presmeg, 1997; Yerushalmy & Chazan, 1993). For example, students may assume that sides that look parallel in an accompanying diagram are parallel (adding unintended conditions to hypotheses). Or, students might link a theorem too tightly to the example diagram given with the theorem statement. For instance, if a theorem is originally illustrated by a diagram of an acute triangle, students might believe that the theorem does not apply, or might not think to apply it, to obtuse triangles. Still another type of difficulty can occur when drawings do not capture appropriate geometric relationships—for example, a tangent might be drawn freehand so that it is not perpendicular to the radius it intersects.³ Finally, students might mistake geometric diagrams for pictures of objects, profoundly changing their interpretation. For instance, a bright fourth grader identified nonrectangular parallelograms as rectangles because he interpreted them as pictures of rectangles viewed “sideways” (Clements & Battista, 1992).

THE VAN HIELE LEVELS: RESEARCH AND REVISIONS

A considerable amount of research has established the van Hiele theory as a generally accurate description of the development of students’ geometric thinking (Clements & Battista, 1992). After briefly reviewing the theory, I address recent developments in this important area of research.

The van Hiele Levels of Geometric Thought

According to the van Hiele theory, students progress through discrete, qualitatively different levels of geometric thinking. The levels are sequential and hierarchical, so for students to function adequately at one of the advanced levels, they must have “passed through” the lower levels. Briefly, the levels are as follows (see Clements and Battista, 1992 for a complete

³ Of course, a sloppy drawing might not interfere with reasoning if the individual creating it is aware of the relationships that *should* exist in it.

description). *Level 0 Pre-recognition*: Because immature perceptual activity causes students to attend to only a subset of a shape's characteristics, students are unable to identify many common shapes. *Level 1 Visual*: Students identify and operate on geometric shapes according to their appearance. They recognize figures as visual gestalts, relying heavily on common prototypes. *Level 2 Descriptive/analytic*: Students recognize and can characterize shapes by their properties. *Level 3 Abstract/relational*: Because one property can signal other properties, students form form definitions, distinguish between necessary and sufficient sets of conditions, and understand and sometimes even provide logical arguments. *Level 4 Formal deduction*: Students establish theorems within an axiomatic system. *Level 5 Rigor/metamathematical*: Students reason formally about mathematical systems.

Recent Developments

New developments will be grouped into three categories: extending the level descriptors beyond 2d shapes; reexamining the nature of the levels; and elaborating the levels phenomenologically and psychologically.

Extending the Level Descriptors Beyond 2d Shapes

Early interpretations of the van Hiele levels focused on reasoning about limited aspects of two-dimensional shapes; that is, the theory was investigated primarily for a well-defined but constrained domain. But a number of researchers have extended the levels to other geometric domains. Although difficulties in interpretation and consistency often arise when researchers attempt to generalize the levels to other domains, such studies have increased our understanding of the levels because generalization typically requires reconceptualization and refinement. As examples of these extensions, Gutiérrez and colleagues extended the van Hiele level descriptions to reasoning about 3d shapes (Gutiérrez, 1992; Gutiérrez, Jaime, & Fortuny, 1991). Johnson-Gentile, Clements, and Battista (1994), Lewellen (1992), and Jaime and Gutiérrez (1989) extended the descriptions to motions/transformations. Clements and Battista (1992) summarized researchers' views on the application of the van Hiele levels to deductive/justification reasoning. Aspects of several of the extensions are discussed below.

Gutiérrez et al.'s (1991) extension of the levels to 3d shapes is straightforward and follows the original conception of the levels as described by Clements and Battista (1992). At Level 1, solids are judged and identified visually and holistically, with no explicit consideration of components or properties. At Level 2,

students identify components of solids and informally describe solids using properties, although properties are not logically related. At Level 3, students are able to logically classify solids and understand the logic of definitions. At Level 4, students are able to prove theorems about solids.

In contrast, Gutiérrez and colleagues' extension to 3d visualization is both less straightforward and more ambitious because it attempts to elaborate the relationship between visualization and geometric conceptualization (Gutiérrez, 1992). At Level 1, students compare solids using global perception with no attention given to properties such as angle size, side length, or parallelism. Students are unable to visualize solids, their positions, or motions that they cannot actually see; they manipulate solids by guess and check. At Level 2, students move from global comparisons to visual analysis of components and their properties. Students are able to visualize simple movements from one visible position to another. At Level 3, students compare solids by mathematically analyzing their components. They can visualize movements involving positions that are not visible, and in reasoning about movements, students can match corresponding parts of images and preimages. At Level 4, students can mathematically analyze and formally deduce properties of solids. Visualization is high and is linked to knowledge of properties.

Two observations are relevant to these extensions of the van Hiele levels to 3d figures. First, the descriptors for geometric conceptualization (not visualization) at Levels 3 and 4 in the latter article are a bit less advanced than in the earlier article. One suspects that this may be the case because traditionally there is far less attention given to formal proof in 3d geometry. Second, extending the levels to visualization is complex. On the one hand, visualization is a capability that is not necessarily connected to knowledge of properties. For instance, some students who are not high visualizers may develop analytic (property-based) strategies to help them compensate for a lack of pure visualization skills (Battista, 1990). Other students may possess very high visualization skill well before they develop property-based reasoning about solids. Indeed, some high visualizers can mentally imagine movements of solids so well that, for many problems, they simply have no need to examine components of the solids. On the other hand, Gutiérrez and colleagues have broken new ground by illustrating how visualization and geometric conceptualization can develop hand-in-hand, an idea also proffered by Clements and Battista (2001). However, much more research is needed that clearly describes exactly how geometric conceptualization interacts with visualization during develop-

ment. For instance, at 3d visualization Level 2, how does progression from global comparison to analysis of components and properties enable improvements in visualizing movement, or vice versa?

Reexamining the Nature of the Levels

Originally, the van Hiele levels were conceptualized as periods of development in geometric reasoning characterized not only by qualitatively different thinking, but by different internal knowledge organization and processing. In this view, students are at Level 1 when their overall cognitive organization and processing disposes them to think about geometric shapes in terms of visual wholes; they are at Level 2 when their overall cognitive organization disposes and enables them to think about shapes in terms of their properties. However, difficulties in classifying students' van Hiele level, oscillation of students between levels, and students being at different levels for different concepts have caused many researchers to question whether the levels are discrete. Indeed, one of the major issues that has arisen is, Should students' thinking be characterized as "at" a single level?

Degrees of Acquisition: A Vector Approach

Arguing that students develop several van Hiele levels simultaneously, Gutiérrez et al. (1991) used a vector with four components to represent the degrees of acquisition of each of van Hiele levels 1 through 4. They identified 5 periods in the acquisition of a level: no, low, intermediate, high, and complete acquisition. For example, in the intermediate acquisition period for a given level, students use reasoning characteristic of the level frequently and often accurately. However, students' lack of mastery of the level causes them to fall back on lower level reasoning when they encounter special difficulties. In scoring student responses, Gutiérrez et al. classified each problem solution according to (a) the nature of the reasoning used via the van Hiele level descriptions, and (b) type. Type was quantified by judging degree of attainment of van Hiele level, correctness of answers, and correctness of reasoning (including justification). The numerical score for degree of acquisition of a van Hiele level was the arithmetic average of type weightings for all items that could be answered at that level. Consequently, for example, a student's degree of acquisition vector might be: 96.67% for Level 1, 82.50% for Level 2, 50.00% for Level 3, and 3.75% for Level 4. Using

this vector approach, and focusing on 3d geometry, Gutiérrez et al. described six profiles of level-configurations in students' geometric reasoning. For example, Profile 2 was characterized by complete acquisition of Levels 1 and 2, high acquisition of Level 3, and low acquisition of Level 4.

However, even though level acquisition was described in terms of the vector model, the profiles could easily be interpreted in terms of the original van Hiele levels (as described by Clements and Battista, 1992, with the addition of Level 0). For instance, Profile 2 could be thought of as Level 2 or transition to Level 3. Thus, rather than invalidating the levels, the vector approach merely suggests, as does the research by Lehrer, Jenkins, and Osana (1998), that the original van Hiele levels may need to be elaborated and expanded. Indeed, the profiles suggest a possible decomposition of van Hiele Levels 0–3 into six steps rather than four.⁴ Unfortunately, one difficulty with the profiles approach (and perhaps all current elaborations of the van Hiele theory) is that it lacks the conceptual coherence and clarity of the original van Hiele levels. That is, the original levels have a conceptual-logical coherence and eloquence that not only makes them appealing, but has proved difficult to surpass.⁵

Several other components of the Gutiérrez et al. (1991) research should be mentioned. First, by suggesting that the van Hiele levels consist of configurations of different types of reasoning, this research seemed to encourage other researchers to investigate students' geometric reasoning from a "configuration" approach. Second, Gutiérrez et al. and Clements and Battista (2001) both attempted to deal explicitly with the tricky issue of how to classify reasoning that fit the description of a level but was incorrectly used. That is, if a student clearly is attempting to use properties in identifying shapes, but does so incorrectly, is that student thinking at Level 2? Third, and finally, Gutiérrez et al. have provided a careful and detailed attempt to describe the transition between van Hiele levels.

Elaborating the van Hiele Theory by Merging It With the SOLO Taxonomy

Several researchers have attempted to elaborate and reformulate the van Hiele theory using the SOLO taxonomy (Olive, 1991; Pegg, 1992; Pegg & Davey, 1998; Pegg, Gutiérrez, & Huerta, 1998). Although van Hiele stated, "The most distinctive property of the levels of thinking is their discontinuity, the lack of co-

⁴ These researchers present the profiles as qualitative descriptions of empirical results, not steps in development (A. Gutiérrez, personal communication, 12/14/04).

⁵ Keep in mind that the original levels themselves have had several iterations and interpretations; there have been discrepancies even in the number of these levels.

herence between their networks of relations” (1986, p. 49), Pegg and Davey argued that several groups of researchers have failed to detect these discontinuities and instead have found the levels to be more continuous in nature. Also, similar to Gutiérrez et al. (1991) and Lehrer et al. (1998), Pegg and Davey argued that research suggests that students develop more than one level at a time. They further argued that level discreteness, while appearing valid when the magnification of the analysis is low, seems blurred when the magnification is high. That is, as global descriptions, the levels seem valid, but when one attempts to examine students’ thinking more carefully, the level descriptions may be inadequate.

To deal with the van Hiele theory’s lack of detail in describing students’ progression through the levels, Pegg and Davey (1998) integrated the van Hiele theory with the SOLO theory of Biggs and Collis. In their synthesis of the two theories, Pegg and Davey discussed three major modes of thinking (see also Pegg, 1992). In the iconic mode, students form and operate on mental images of objects with which they have had contact. In the concrete-symbolic mode, students link concepts and operations to written symbols, as long as the context falls within their personal experience. In the formal mode, students are no longer restricted to concrete referents and can systematically consider principles, theories, and ranges of possibilities and constraints; they become capable of formal proof. Each of these modes is further differentiated by progression through levels of complexity, with progression through each mode requiring at least two cycles (Pegg et al., 1998). At the unistructural (U) level, students focus on one isolated aspect of a situation. At the multistructural (M) level, students focus on two or more unrelated aspects of a situation. At the relational (R) level, students interrelate multiple aspects of a situation.

According to Pegg et al. (1998),⁶ van Hiele Level 0 (in the Clements/Battista numbering system) occurs at the iconic mode, in the first UMR cycle (U1, M1, R1). van Hiele level 1 occurs at the iconic mode in the second UMR cycle (U2, M2, R2). At U2, the imaging process focuses on one isolated aspect of a situation, such as the sharpness of a vertex. At M2, several unrelated aspects of a situation are considered. At R2, the individual has full control over the imaging process and is able to identify familiar shapes correctly. van Hiele Level 2 occurs at the concrete-symbolic mode in the second UMR cycle, with Levels U2 (focusing on a single property) and M2 (focusing on many properties). (The earlier U1, M1, R1 concrete-symbolic cycle

is seen as a transition between van Hiele Levels 1 to 2.) van Hiele Level 3 occurs at the R2 Level in the concrete-symbolic mode. Students seem to achieve van Hiele Level 4 as they progress through two UMR level cycles in the formal mode.

The power of integrating the van Hiele theory with the SOLO theory is that it provides a possible framework for discussing finer delineations of the van Hiele levels. However, many issues and questions remain. For example, (a) The fact that the correspondence between van Hiele level and SOLO taxonomy varies, depending on the article read, indicates that researchers are struggling to make this complex correspondence work. The only agreement seems to be that the original van Hiele levels are too coarse to provide adequate distinctions in students’ geometric reasoning. (b) Why should researchers believe that the ability to focus on multiple aspects of a situation is a primary determinant of geometric reasoning? Certainly this ability is important in intellectual development—but is it specific enough to be a primary determinant of geometric reasoning, especially at each van Hiele level? Also, what cognitive mechanisms enable an individual to focus on more than one aspect at a time?

Waves of Acquisition

Several researchers have posited that different types of reasoning characteristic of the van Hiele levels develop simultaneously at different rates, and that at different periods of development, different types or “waves” of reasoning are dominant, depending on the relative competence students exhibit with each type of reasoning (Clements & Battista, 2001; Lehrer et al., 1998; see Figure 19.2). Development depends both on maturation and instruction.

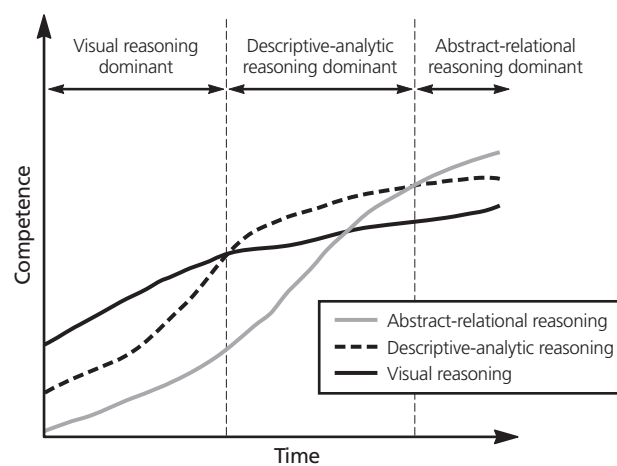


Figure 19.2 Waves of acquisition of van Hiele levels.

⁶ The correspondence between van Hiele levels and SOLO modes and levels varies by research report. See the cited references in this section.

Based on the administration of a single type of task,⁷ Lehrer et al. (1998) argued that “level mixture” was typical for primary-age students, that students’ justifications often jumped across nonadjacent van Hiele levels, and that reasoning varied greatly with task. They argued against portraying geometric development in terms of discontinuities among levels, and instead for it being characterized “by which ‘waves’ or forms of reasoning are most dominant at any single period of time, without necessarily implying the extinction of other forms” (p. 163). They found that children’s geometric thinking was accomplished by a range of mental operations including comparing shapes to visual prototypes, mentally “morphing” one shape into another, and finding a variety of attributes of shapes. They argued that these varied types of reasoning should not be described by the single descriptor “visual.” However, one can argue that the types of reasoning that Lehrer et al. questioned as being visual are, in fact, examples of visual, or Level 1, reasoning, and that Lehrer et al. have provided us with a richer description of such reasoning. That is, the Lehrer et al. research suggests that Level 1 visual reasoning is not as limited and homogeneous as the original van Hiele levels suggest. Furthermore, the characterization of students’ thinking as property-based when they refer to number sides should be questioned (I will return to this issue later in the chapter). And finally, responses the researchers coded as “classification” were not hierarchical classification in the van Hiele sense—students were simply naming figures (it would be a truly exceptional third-grader who used a genuine hierarchical classification). So, as in so many van Hiele studies, the question that arises is whether the “level jumping” was due to students’ reasoning or to quirks in the classification system used by the researchers.

Clements and Battista (2001) proposed the view that the van Hiele levels (seen as types of reasoning) develop simultaneously (albeit at different rates), similar to the “overlapping waves” metaphor suggested by Siegler (1996), and by Lehrer et al. (1998). Visual-holistic knowledge, descriptive verbal knowledge, and, to a lesser extent initially, abstract symbolic knowledge grow simultaneously, as do interconnections between levels. However, although these different types of reasoning grow in tandem to a degree, one level tends to become ascendant or privileged in a child’s orientation toward geometric problems. Which level is privileged is influenced by age, experience, intentions, tasks, and skill in use of the various types of reasoning. In the early years, while visual reasoning (called syncretic to signify combination without coherent

analysis and synthesis) is dominant, descriptive-analytic knowledge begins to emerge and interacts with visual knowledge. The early dominance of visual-holistic reasoning gradually gives way to descriptive-analytic reasoning. As abstract reasoning begins its ascendance, connections among all types of reasoning are strengthened and reformulated. Consistent with this “continuous growth” view, these researchers found little evidence in their own research or that of others of discontinuities between levels as postulated by the van Hiele theory; growth in small increments seemed to be the norm.

Stages or Levels?

Clements and Battista (2001) viewed the van Hiele levels as *levels*, rather than *stages*. They described a stage as a substantive period of time characterized by cognition across a variety of domains that is qualitatively different from that of both the preceding and succeeding stages. In contrast, a level is a period of time of qualitatively distinct cognition, but within a specific domain. In van Hiele research, one of the open questions concerns the size of the “domains” to which the van Hiele levels apply. For instance, Clements and Battista (2001) suggested that many empirical studies indicate that people exhibit behaviors indicative of different levels on different subtopics of geometry (Denis, 1987; Mason, 1989; Mayberry, 1983). This finding, taken together with the observation that students sometimes exhibit different levels on different tasks, suggests that the domains to which levels of thinking apply may be, at most, subtopics within the broad range of geometry.

However, the domain issue may be more complicated than it seems, especially when viewed from a longitudinal rather than a static point of view. For instance, Lewellen (1992) carefully studied 8 students participating in the Logo Geometry Curriculum (Battista & Clements, 1991; Clements & Battista, 2001), which first deals with geometric shapes then with geometric motions. She found that students entering the motions unit all started at Level 1 for motions, but after a short initial learning period, students at Level 2 or in transition to Level 2 for shapes moved up to Level 2 in motions. Lewellen concluded, “These results lend support to the notion that there is a global van Hiele level for each student after they have some initial learning in any new domain” (Lewellen, 1992, p. 279).

In fact, students who entered the motions unit at Level 2 on shapes were able to quickly learn the components and characteristics of objects in this new domain. Rather than limiting themselves to specific

⁷ Detailed discussion of this task will be given in a later section.

values and visual feedback, these students looked for underlying properties, components, characteristics, and rules in problem-solving situations. For example, Kelly developed measure-based strategies for verifying whether or not motions were correct, such as measuring the distance of corresponding points from the flip line for flips and the distance of corresponding points from the turn center for turns. She also knew that a slide moves all points of a figure the same distance and direction. Thus, students like Kelly who were at Level 2 for shapes, having seen the value of analytic conceptions in one domain—for instance, using length and angle measure to describe relationships between components of shapes—seemed to seek out similar conceptions in the new domain of motions. Initially as students worked in the new domain, they did not have such conceptions. The key is that they had developed the ability to use such conceptions, and they understood their power, so they looked for properties in the new domain. Of course, an open question is to describe the degree of acquisition of a level of reasoning—along with enabling mental mechanisms—that is required to attempt to and be able to implement that reasoning in another domain.

An Alternative Route: Elaborating the Original van Hiele Levels

To understand the development of reasoning in geometry, Battista has elaborated the van Hiele levels to carefully trace students' progress in moving from informal intuitive conceptualizations of 2d geometric shapes to the formal property-based conceptual system used by mathematicians. The descriptions for Level 2 given below were developed by Battista in several projects; the description for Level 3 was developed by Borrow (2000) and Battista. This new elaboration considerably expands the van Hiele levels in two places—the development of property-based thinking, and the development of inference about properties.

Level 1: Visual-Holistic Reasoning

Students identify, describe, and reason about shapes and other geometric configurations according to their appearance as visual wholes. They may refer to visual prototypes, saying, for example, that a figure is a rectangle because “it looks like a door,” or they might judge that two figures are the “same shape” because they “look the same.” Students might also say that a square is not a rectangle because rectangles are “long.” Students may justify their responses using imagined visual transformations, saying for instance that a shape is a square because if it is turned it looks

like a square. Orientation of figures may strongly affect Level 1 students' shape identifications. In fact, Level 1 students tend to identify a square only if its sides are horizontal and vertical.

1.1 Pre-recognition. Students are unable to identify many common shapes.

1.2 Recognition. Students correctly identify many common shapes.

Level 2: Analytic-Componential⁸ Reasoning

Students explicitly attend to, conceptualize, and specify shapes by describing their parts and spatial relationships between the parts. However, students' descriptions and conceptualizations vary greatly in sophistication, starting with completely informal and imprecise specifications in Sublevel 2.1 and ending with completely formal geometric specifications in Sublevel 2.3 (which corresponds to original van Hiele Level 2).

Two major interrelated factors contribute to the development of Level 2 reasoning. The first is an increasing ability and inclination to account for the spatial structure of shapes by analyzing their parts and how the parts are related. The second is an increasing ability to understand and apply formal geometric concepts in analyzing relationships between parts of shapes.

2.1 Visual-informal componential reasoning. Students describe parts and properties of shapes informally and imprecisely; they do not possess the formal conceptualizations that enable precise property specifications. Descriptions and conceptualizations are visually based, focusing initially on parts of shapes then on spatial relationships between parts. Examples of references to *parts* are counting the number of sides or angles (usually referred to as “points”) in a shape or saying that a figure has straight sides. Examples of descriptions of *relationships between parts* are specifying that “corners” must be “square” or that “all sides must be even.” In all cases, students describe parts and their relationships using strictly informal language, that is, language typically learned in everyday experience.

Students' informal language ranges greatly in precision and coherence, from using vague and incompletely formulated conceptualizations to informally describing a conceptualization that corresponds to a formal geometric concept. As an example of the latter case, as students examine a tilted rectangle or square, they might informally refer to the property of having all right angles by saying the shape has “square corners” or “straight sides.” Students' descriptions often seem to occur extemporaneously as they inspect shapes (as opposed to describing explicit conceptualizations formulated from previous experiences).

⁸ I use the term *analysis* to refer to the process of understanding objects by resolving them into their components.

2.2 Informal and insufficient-formal componential reasoning. As students begin to acquire formal conceptualizations that can be used to “see” and describe spatial relationships between parts of shapes, they use a combination of informal and formal descriptions of shapes. The formal descriptions utilize standard geometric concepts and terms explicitly taught in mathematics curricula. However, the formal portions of students’ shape descriptions are insufficient to completely specify shapes. For example, a student might explain that a rectangle has “sides across from each other that are equal [formal] and square corners [informal].” Or, a student might say that “a rectangle has sides across from each other that are equal lengths,” making no mention of right angles. Although students often recall properties that they have abstracted for classes of shapes (say, “two long sides and two short sides” for rectangles), their reasoning is still visually based, and most of their descriptions and conceptualizations still seem to occur extemporaneously as they are inspecting shapes.

2.3 Sufficient formal property-based reasoning. Students explicitly and exclusively use formal geometric concepts and language to describe and conceptualize shapes in a way that attends to a sufficient set of properties to specify the shapes. Students have made a decided shift away from visually dominated reasoning because the major criterion for identifying a shape is whether it satisfies a precise set of verbally stated formal properties. So, for example, at Sublevel 2.3, the term *rectangle* refers to a class of shapes that possesses all the properties the student has come to associate with the set of rectangles.

Students can use and formulate formal definitions for classes of shapes. However, their definitions are not minimal because forming minimal definitions requires relating one property to another using some type of inferential reasoning (which occurs at Level 3). Students do not interrelate properties or see that some subset of properties implies other properties. They simply think in terms of unconnected lists of formally described characteristics. The set of properties students give for a class of shapes is a list of all the visual characteristics the student has come to associate with that type of shape, described in terms of formal geometric concepts. (But students seem to recall these property-based specifications rather than discover them on the fly as they inspect shapes.)

Progressing to Sublevel 2.3 requires that formal concepts like side length and angle measure have been raised to a sufficient level of abstraction so that they can be used to form *relational* conceptualizations such as “all sides equal” that describe spatial relationships between shape parts. These relationships, such

as opposite sides equal, must have reached the interiorized level of abstraction, detaching them from their original contexts so they are applicable to new situations and available for analyzing shapes.

Level 3: Relational-Inferential Property-Based Reasoning

Students explicitly interrelate and make inferences about geometric properties of shapes. For example, a student might say “If a shape has property X, it also has property Y.” However, the sophistication of students’ property interrelationships varies greatly, starting with empirical associations (when property X occurs, so does property Y), progressing to componential analyses that explain why one property “causes” another property, then to logically inferring one property from another, and finally to using inference to organize shape conceptualizations into a hierarchical classification system. Because one property can “signal” other properties, students can logically organize sets of properties, correctly specify shapes without naming all their properties, form minimal definitions, and distinguish between necessary and sufficient sets of conditions. At Level 2, students only can use formal properties; at Level 3, students can operate on formal properties.

At Level 3, the spatial relationships described by formal property statements reach the second level of interiorization so that they can be symbolized by the statements, and so that students can reason meaningfully about the statements, in many cases, without having to visually re-present the actual spatial structurings that the statements describe. The verbally-stated properties themselves are interiorized so that they can be meaningfully decomposed, analyzed, and applied to various shapes.

3.1 Empirical relations. Students use empirical evidence to conclude that if a shape has one property, it has another.

3.2 Componential analysis. By analyzing how types of shapes can be built one-component-at-a-time, students conclude that when one property occurs, another property must occur. Students conduct this analysis by making drawings or imagining constructing shapes piece-by-piece. For instance, a student might conclude that if a quadrilateral has four right angles, its opposite sides are equal because if you draw a rectangle by making a sequence of perpendiculars, the sides *must* be equal. (The student explicitly sees visually that making a pair of opposite sides unequal causes some of the angles to *not* be right angles.)

3.3 Logical inference. Students make logical inferences about properties; they mentally operate on property statements, not images. For example, a student might reason that because a square has all sides

equal, it has opposite sides equal. Such reasoning enables students to make the inferences needed for hierarchical classification. For instance, a student whose definition for a rectangle is “4 right angles and opposite sides equal” might infer that a square is a rectangle because “a square has 4 right angles, which a rectangle has to have; and because a square has 4 equal sides, it has opposite sides equal, which a rectangle has to have.” But students do not use this inferencing ability to logically reorganize their conceptual networks about shapes, so they do not adopt a logical hierarchical shape classification system (i.e., they still resist the notion that a square is a rectangle even though they can follow the logic justifying such a statement).

Students’ reasoning is “locally logical” in that they string together logical deductions based on “assumed-true” propositions, that is, propositions that they accept as true based on their experience, intuition, or authority. Thus, students at this level use logic, but they do not question the starting points for their logical analyses.

3.4 Hierarchical shape classification based on logical inference. Students use logical inference *to reorganize* their classification of shapes into a logical hierarchy. They fundamentally restructure their shape classification networks (as opposed to merely making additional connections here and there). It becomes not only clear why a square is a rectangle, but a necessary part of reasoning. Students give logical arguments to justify their hierarchical classifications. Finally, students’ use of logic to draw conclusions provides them with a new way to accumulate knowledge. That is, new knowledge can now be generated not merely through empirical or intuitive means, but through logical deduction.

Level 4: Formal Deductive Proof

Students can understand and construct formal geometric proofs. That is, within an axiomatic system, they can produce a sequence of statements that logically justifies a conclusion as a consequence of the “givens.” They recognize differences among undefined terms, definitions, axioms, and theorems.

Discussion

One of the major criticisms of the van Hiele levels is that students’ progress does not seem to be characterized by jumps from one discrete level to the next, but instead, seems to occur in small incremental steps.⁹ To support this criticism, researchers cite the notion of transition between levels and that students often exhibit characteristics of several levels (e.g., Clements

& Battista, 1992; Lehrer, 1998). Gutiérrez et al. (1991) dealt with this issue by using vectors to indicate the degree of acquisition of various levels. Clements and Battista (2001) and Lehrer et al. (1998) dealt with the issue by positing that several levels may be developing at the same time, but that different types of reasoning become dominant during learning and development. Although each of these proposed revisions has merit and has moved the field forward, they both face two important issues.

Issue 1: Type versus level of reasoning. Throughout the research on the van Hiele theory, there is a lack of distinction between *type of reasoning* and *qualitatively different levels in the development of reasoning*. That is, sometimes the visual-holistic level is used to refer to a type of reasoning that is strictly visual in nature, and sometimes it is used to refer to a period of development of geometric thinking when an individual’s thinking is dominated and characterized by visual-holistic thinking. For instance, Gutiérrez et al. (1991) used vectors to indicate students’ “capacity to use each one of the van Hiele levels” (p. 238). This statement makes sense only if van Hiele levels are taken as types of reasoning, not qualitatively different levels of thought. Similarly, Clements and Battista (2001), along with Lehrer et al. (1998), talked about “waves of acquisition” of types of reasoning defined by the van Hiele levels. To complicate matters further, Pegg and Davey used the SOLO taxonomy to describe cognitive processes that enable van Hiele types of reasoning. Thus, broadly speaking, researchers have intermingled and not yet completely sorted out (a) basic enabling cognitive processes, (b) van Hiele levels as types of reasoning, and (c) van Hiele levels as stages of development of geometric reasoning.

Related to this intermingling, several subissues are apparent. First, exactly what are the basic cognitive processes that enable different types of geometric thought? Are they simply the types of thought described by the van Hiele levels? Or are other, more basic, processes implicated? For instance, research suggests several possibilities for basic processes: (a) visualization/imagery; (b) analysis (decomposing shapes into parts and interrelating those parts); (c) conceptualization in terms of formal concepts; (d) ability to attend to, interrelate, and coordinate various aspects of a situation; and (e) inference/logical deduction. Are there other processes? How do each of these interact and produce the types of reasoning characterized by the van Hiele levels?

Second, is the vector or configuration of activations perspective viable? Certainly there is precedent

⁹ In fact, longitudinal studies often show students making progress within a van Hiele level, but not moving to the next level (Gutiérrez & Jaime, 1998; Clements & Battista, 2001).

for such a view. For instance, a vector model has been used in describing how the brain represents objects in the perceptual field, not by a set of neurons but by a *configuration* of neuronal firing patterns (Churchland, 2002). Is it then reasonable to think that type of reasoning is enabled by activating a *configuration* of enabling processes? Development of geometric reasoning would then occur (a) as the enabling processes become more sophisticated, (b) as new and more useful configurations of processes form and become stable, and (c) as metacognitive processes regulate the use of configurations. But this approach leads back to defining the enabling processes, distinguishing them from the van Hiele levels as types of reasoning, and explaining how the van Hiele levels *as levels* occur in this activation approach. For instance, how exactly does one type of thinking become ascendant or dominant?

Third, the vector-configuration approach runs the risk of losing the compelling conceptual eloquence attained by the original van Hiele levels. In fact, adopting the vector-configuration approach might regress the field into older “structure of intellect” models, with researchers focused so much on defining and measuring enabling processes that describing the nature of different types of reasoning in conceptually coherent ways is lost.

Issue 2: Assessment. Assessment has long been an issue in the van Hiele theory, both because it operationalizes researchers’ conceptions of the levels and because it is necessary for applying the theory to instructional practice. Researchers are still grappling with this issue. For instance, in designing a paper-and-pencil instrument to be used with middle and high school students, Gutiérrez and Jaime (1998) defined four processes (recognition, definition, classification, and proof) and how those processes were implemented by students at different van Hiele levels. *Recognition* refers to recognizing types of geometric figures and identifying components and properties of figures; it includes use of vocabulary to describe properties. *Definition* refers to formulating definitions of concepts during the learning process and using textbook definitions. *Classification* refers to classifying geometric figures into different categories. *Proof* refers to justifying geometric statements. Gutiérrez and Jaime described how each process occurs at several van Hiele levels so that how a student uses a process can be taken as an indicator of the student’s level of reasoning. Assessment tasks required students to describe their reasoning.

However, as always with van Hiele level descriptions and assessments, the devil is in the details. Major questions for researchers are (a) the appropriateness of Gutiérrez and Jaime’s four characterizing processes and (b) how these processes are defined to occur at

each van Hiele level. For instance, their definition of the recognition process focuses on physical attributes at Level 1 but mathematical properties at Level 2, two very different kinds of “recognition” processes that are related but certainly not the same at a basic cognitive level. Thus, it is one thing to devise broad categories of behavioral descriptors; it is another to determine the cognitive processes underlying these categories of behaviors. This has been, and will continue to be, a major challenge facing researchers.

However, despite the difficulty facing researchers attempting to assess students’ van Hiele levels, each serious attempt to do so helps us better understand the nature of the levels and students’ geometric thinking. For instance, in discussing van Hiele assessments, Gutiérrez and Jaime (1998) claimed that even though researchers most often identify students as belonging to Level 3 if they can understand and use hierarchical classifications of shape families, there are other critical attributes of this level. They argued that students at Level 3 can also establish logical connections between geometric properties (a key component in thinking hierarchically and in using logical definitions), and they can accept and identify nonequivalent definitions of geometric shape classes. Do assessments of Level 3 equally assess all of the characteristics of Level 3, and, if not, how does that affect the validity of the assessments? In fact, how do assessments affect researchers’ conceptualizations of the levels?

A totally different approach to assessing van Hiele levels was devised by another group of researchers. In a collaborative effort to find ways to assess elementary students’ acquisition of the van Hiele levels in interview situations, Battista, Clements, and Lehrer developed a triad sorting task, that, with variations, both Clements and Battista (2001) and Lehrer et al. (1998) used in separate research efforts. In this task, students are presented with three polygons, such as those shown in Figure 19.3, and are asked, “Which two are most alike? Why?” Choosing B and C and saying that they “look the same, except that B is bent in” was taken as a Level 1 response. Choosing A and B and saying either that they both have two pairs of congruent sides or that they both have four sides was taken as a Level 2 response. The purpose of this task was to determine the type of reasoning used on a task that students had not seen before (so it was unlikely to elicit instructionally programmed responses).



Figure 19.3 Triad polygon sorting task.

One difficulty with this analysis is that giving the number of sides of a polygon is a “low-level” use of properties. That is, there are different types of property use. The simplest property involves describing the number of components in a shape. For instance, a quadrilateral has four sides; a triangle has three angles. A second, somewhat more sophisticated type of property describes global characteristics. For instance, a quadrilateral is a simple closed curve. Describing the straightness of sides also falls in this category. Finally, and most sophisticated, are properties that describe relationships between parts of shapes. For example, the following statements describe *relational* geometric properties: (a) opposite sides of a parallelogram are parallel and congruent; (b) a rectangle has four right angles; (c) adjacent angles of parallelograms are supplementary; (d) this quadrilateral has one line of symmetry. Examples a–c describe relationships between the sides of shapes, (a) explicitly, (b) and (c) indirectly by specifying angle measures. Example (d) describes a relationship between two “halves” of a quadrilateral, which, with further analysis, could be described in terms of line segments and angles.

The distinction in properties described above suggests that students’ use of number of sides of a polygon is not a very good indicator of Level 2 thinking, which focuses on relational properties. Thus some jumps in levels observed by Lehrer et al. (1998) may have been caused by coding students’ use of number of sides as Level 2. Because a critical factor used in distinguishing van Hiele levels is how students deal with geometric properties, clarifying the meaning of *properties*, as it relates to the van Hiele levels, is important.

Another factor that should be considered with this triad task is that saying Shape B is more like Shape C is not necessarily a less sophisticated response than focusing on number of sides. That is, Shape B is actually more like Shape C if we consider how much movement it takes to transform B into C, compared to B into A. In fact, one could imagine a metric that quantifies the amount of movement required. Thus, the “morphing” response described by Lehrer et al. (1998), and also observed by Clements and Battista (2001), may be an intuitive version of a notion whose mathematization is far beyond the reach of elementary students.

Nevertheless, this being said, the original argument that attending to number of sides is more sophisticated than a visual morphing response may still have some validity. It suggests a shift toward using a more mathematical conceptualization (albeit the primitive one of counting number of sides/vertices).

Another shortcoming of the triad-task approach is pointed out by differences in the ways the research-

ers used the triads. Lehrer et al. (1998) construed each triad task as an indicator of type of reasoning. So students’ use of different types/levels of reasoning on different triads was taken as evidence of jumps in levels. In contrast, Clements and Battista (2001) used a *set* of 9 triad items as an indicator of level of students. To be classified at a given level, a student had to give at least 5 responses at that level. If a student gave 5 responses at one level and at least 3 at a higher level, the student was considered to be in transition to the next higher level. The latter approach, by virtue of its use of multiple items, produces a more reliable indication of level of reasoning. Of course, because it aggregates responses, this approach obscures intertask differences and variability in reasoning. It focuses on determining the ascendant level of students’ reasoning on a particular type of task. (So this issue is related to the type-of-reasoning versus level-of-reasoning dichotomy previously discussed.)

Another difference between the researchers’ approaches is also important. In analyzing students’ reasoning on the triad tasks, Lehrer et al. (1998) classified student responses solely on the basis of the type of reasoning that students employed. In contrast, in classifying students’ van Hiele levels, Clements and Battista (2001) attempted to account for the “quality” of students’ reasoning—each reason for choosing a pair in a triad was assessed to see if it correctly discriminated the pair that was chosen from the third item in the triad. In this scheme, the van Hiele levels for students were determined based on a complicated algorithm that accounted for both type of reasoning used and discrimination score.

However, although discrimination is a reasonable indication of quality of response in the triad context, it does not address another component that might be important in assessing van Hiele levels, the *validity* of reasoning, which involves the accuracy and precision of students’ identifications, descriptions, conceptualizations, explanations, and justifications. That is, a student might validly describe a property of two items in a triad that does not discriminate the items from the third item. Researchers must determine which, if any, of discrimination, validity, or mere use of properties are critical characteristics of van Hiele Level 2.

The last issue in assessment is whether to use individual interviews or paper-and-pencil tests. Because individual interviews provide a more detailed picture of students’ geometric reasoning, they provide a more valid assessment of van Hiele level, but are more time-consuming to administer. In contrast, paper-and-pencil instruments are less time-consuming and are thus more practical for many research projects and classroom teachers who must deal with large numbers

of students. But paper-and-pencil instruments, even those that ask students to write about their reasoning, rarely give as complete a picture of student reasoning as recorded individual interviews. However, no matter which method is used, creating and interpreting tasks that reliably and validly assess van Hiele levels is extremely challenging.

Final thought on the van Hiele theory. Numerous researchers have concluded that the van Hiele levels accurately, if generally, describe the development of students' geometric thinking, especially about shapes. However, these same researchers, in general, seem to find the theory lacking in depth. They would like a more detailed description of this development. I have already described several proposed alternatives and elaborations. But why does the van Hiele theory, though deemed inadequate in detail, also have such a strong ring of validity?

I suggest that the reason for this feeling of validity is that the van Hiele levels describe a progression of thinking that is part and parcel to scientific/mathematical thinking. That is, we can think of both the levels and the "scientific" method as progressing through four phases: perceptualization (resulting in informal, intuitive, surface-level categories and reasoning), conceptualization (resulting in explicit concepts and analysis), organization (resulting in conceptual organization), axiomatization (resulting in formal logical treatments). For highly educated individuals in scientifically advanced civilizations—like mathematics education researchers—moving through these four phases seems like the natural way for individuals to progress from intuitive, everyday reasoning to formal scientific reasoning.

Psychological Mechanisms Underlying the Levels

To understand the cognitive processes underlying geometric thinking, and movement from one level of thinking to the next, one must understand how individuals construct and mentally represent spatial and geometric knowledge. Because such knowledge originates with perception and imagery, I start with these processes.¹⁰

Perception of Shape

Kosslyn (1994) has provided a detailed model of the brain processes involved in perception. This model consists of several subsystems. The preprocessing subsystem accepts sensory input from the atten-

tion window; it outputs both an image and properties that it extracts from the image. The pattern activation subsystem then matches output from the preprocessing subsystem to stored representations of visual patterns that include extracted properties. The pattern activation subsystem is divided into two additional subsystems. The category pattern activation subsystem classifies a stimulus as an element of a category; the exemplar pattern activation subsystem registers specific examples (e.g., Fido, not dogs in general). The two tasks, "recognizing a stimulus as a specific individual and recognizing it as a member of a category, have incompatible requirements; information that is needed to recognize the individual must be discarded to classify it as a member of the category" (Kosslyn, 1994, p. 182). Thus, even at this elemental level, the two essential geometric tasks of apprehending instances and apprehending categories are differentiated.

The process of abstraction seems to be at the root of construction of entities in the category pattern activation subsystem: "To isolate certain sensory properties of an experience and to maintain them as repeatable combinations, i.e., isolating what is needed to recognize further instantiations of, say, apples, undoubtedly constitutes an empirical abstraction" (von Glasersfeld, 1991, p. 57).¹¹ An individual comes to recognize triangles as figural patterns of triangular shapes are abstracted and become linked to the word *triangle*.

Abstraction from the actual sensory material out of which particular configurations or sequences are built up, yields figural patterns that have a certain general applicability and can be semantically associated with specific names. Thus we have, for instance, the notion of 'triangularity' that enables us to *see* spatial configurations as triangles. (von Glasersfeld, 1982, p. 197)

Furthermore, von Glasersfeld argues that perceptual abstraction is active, not passive.

The experiencing subject attends, not to the specific sensory content of experience, but to the operations that combine perceptual and proprioceptive elements into more or less stable patterns. These patterns are constituted by motion, either physical, or attentional, forming 'scan paths' that link particles of sensory experience. (von Glasersfeld, 1982, p. 196)

Individuals form scan paths that enable them to recognize similar perceptual objects (e.g., triangles that look approximately the same). When several different types of triangular perceptual objects are ab-

¹⁰ There seems to be far less relevant literature about cognitive processes underlying thinking at van Hiele levels above Level 2.

¹¹ More will be said about the process of abstraction in a later section.

stracted and associated with the word triangle, an early conceptualization of triangle is created. However, at this early stage of recognition, an individual's "conceptualization" of triangle is usually limited to a small subset of typical prototypes and is holistic in nature. A major problem in placing the van Hiele theory on a sound cognitive foundation is answering the question, How do individuals move to finer, parts-based conceptions of shapes?

From Wholes to Parts

Kosslyn argued that to form an image of a shape requires an *implicit* mental construction in a sequential, part-by-part manner not only of the parts of the shape but how they are related; "to represent such shapes is to extract specific properties that will not change when the object assumes a new configuration or a shape-variant appears. One such invariant is the type of spatial relation between parts" (Kosslyn, 1988, p. 1623). Kosslyn (1994) further argued that in object recognition, global shape is processed first, then parts and characteristics. That is, in perceptual processing, the mind first attempts to match the overall pattern that corresponds to an object; if that does not work, it attempts to match input to representations of individual parts. Thus, even when a shape is processed globally, the perceptual system identifies the shape's parts and computes relatively abstract spatial relations among them. Initially, however, the parts are not explicitly accessible.

Kosslyn's research is consistent with that of Smith (1989). She suggested "that represented objects are built in a bottom-up fashion from a feature level of processing" but "that the represented object is a cohesive unit and is experienced as a unit. Represented objects as wholes are given to conscious experience as unitary entities, so that one cannot get to the represented parts without some work" (p. 127). But in addition to Kosslyn's claim that task demands may cause a holistic-to-parts shift in processing, Smith argued that as children get older, they acquire an increased ability to selectively attend to single dimensions, and thus the characteristics of wholes. Clements and Battista (1992, 2001) described the processes by which parts of shapes gradually become accessible to reflection, but their theory focused on experience, reflection, and learning.

Research on brain processes goes even further, separating the identification of shape parts, the creation of spatial relations between parts, and the creation of the whole. Posner and Raichle (1994) found that pa-

tients suffering brain damage in either the right or left hemispheres possessed very different types of cognitive deficits. When asked to draw target stimuli, right-hemisphere-damaged patients saw the components of the stimuli but were unable to correctly put the components together; left-hemisphere-damaged patients saw the global picture, but not the components of the target stimuli. The patient drawings in Figure 19.4, especially the top row, clearly illustrate the global-versus-parts nature of patients' cognitive deficits. Patients with right hemisphere damage saw the components of the original triangle—the segments made from rectangles—but were unable to correctly put these components together spatially. They did not perceive or record the overall spatial structure of the triangle. In contrast, left-hemisphere damage seemed to cause little deficit in global spatial processing; subjects correctly captured the shape of the triangle.¹² Left-hemisphere damage did, however, cause important details to be omitted from figures. Such omissions can be important geometrically. For instance, Kosslyn (1994) found that the left hemisphere "is critically involved in detecting target figures when they are embedded in other patterns" (p. 202).

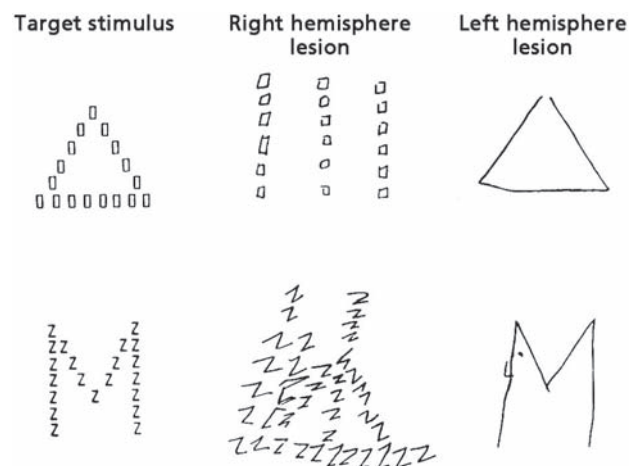


Figure 19.4 Global-versus-parts deficits from hemispheric damage (Posner & Raichle, 1994).

Posner and Raichle (1994) explained these hemispheric differences in terms of attentional scale—zooming in or out: "The right hemisphere is more likely to control large or global scales [zoomed out] while the left does the reverse [zoomed in]" (p. 162). Kosslyn (1994) too argued that a visual object "is represented at multiple scales; at one scale the parts are represented, whereas at another they are embedded

¹² Posner and Raichle (1994) reported, however, some research showing that left-hemisphere damage can cause deficits in assembling parts into overall representations.

in an overall pattern" (p. 161). Furthermore, research has found that observers can selectively focus on *either* global or local features in a visual form. "When we are set to attend to one level of detail, our processing of features at the other level is poorer" (Coren et al., 1994, p. 518). Thus the global-local processing separation is built into the architecture of the brain. However, as individuals learn and become more sophisticated in their reasoning, the separate processes get coordinated and integrated into higher level recognition schemes that are sensitive to the whole, its parts, and relationships between parts. Furthermore, this integration is so automatic in educated adults that it is difficult for them to appreciate the world of the child in which the processes remain separate and un-integrated.

In summary, the psychological/neurological research described above sheds light on several findings in research on geometric thinking. It indicates that perceptual wholes are built up from parts-based representations, but that accessing these parts-based representations is initially impossible and only occurs "with some work." Furthermore, there is a separation in perceiving parts of shapes and relationships between parts, and there is a separation in perceiving global or local features of shapes. These findings help explain three important issues in research on the development of geometry reasoning: (a) why students first attend to whole shapes; (b) why students often notice parts of shapes, but not relationships between parts (Clements & Battista, 1992); and (c) how students shift from focusing on wholes to focusing on relationships between parts, as described in the van Hiele theory.

From Images to Explicit Relationships

How does a child progress from viewing shapes merely as visual images to thinking about them primarily in terms of verbal statements about their parts and relationships between parts? First, students must gain mental access to the parts of shapes, explicitly considered as parts of shapes, so that they can mentally construct spatial relationships between the parts. It is not that before this time students cannot notice the parts of shapes; it is that they do not see these parts as parts of wholes—"parts" are merely seen as self-standing entities. In fact, initial processing likely considers parts and wholes on equal footing—the processing system may not detect whether perceived entities are wholes or parts of wholes (Kosslyn, 1994, p. 168). Thus, a student's description that a shape is "pointy" does not imply that the student is attending to a part of the whole; the student may simply be focusing on a salient entity in the visual field. This lack of an inclu-

sion relation between parts and the wholes to which they belong is characteristic of Level 1 thinking and is similar to a type of conceptualization described by Piaget (Gruber & Voneche, 1977), who claimed that "an arm drawn alongside of a manikin is conceived by the child as 'going with' the manikin not as 'forming part of' his body" (p. 102).

Second, students must mentally construct explicit spatial relationships between the parts of a shape that give it its spatial structure. This requires the mind to "break into" the components of perceptual processing for identifying a shape, abstract structural spatial relationships that characterize the shape, and consciously attempt to conceptualize these relationships. According to the theory of abstraction to be discussed later, breaking into the perceptual processes requires intentional reflection on the nature of the shape, and that the perceptual processes be abstracted at the interiorized level. Once spatial relationships become accessible to reflection, they can be verbally described, refined through empirical testing, and later, linked to interiorized formal conceptualizations.

Several factors can engender the move from images to explicit relationships. First, as has been seen, the ability to break into perceptual processes can be affected by development and experience. Second, Kosslyn discussed an imagery-to-verbalization developmental shift. He claimed that even though children rely predominantly upon imagery in accessing and using information, because processing capacity for verbal knowledge increases with development, children increasingly translate observations derived from images into verbal form so facts can be accessed either imagistically or verbally (Kosslyn, 1980, 1983; Kosslyn, Margolis, & Barrett, 1990). In fact, Kosslyn (1980) suggests that the use of imagistic representation may be reduced as children gain experience and repeatedly access the same ideas. That is, a child "may recode information into propositional form as he or she accesses that information repeatedly" (p. 412). Furthermore, because "certain facts are communicated linguistically and presumably memorized in a propositional format. . . . Retrieval . . . should not require consultation of an image" (p. 412).

Comment

Although a number of theories and studies have been reviewed in an attempt to describe the cognitive processes by which students progress through the early van Hiele levels, this area of research is still in its infancy. This is due in great part because researchers are investigating cognitive processes that cannot be observed. To achieve progress in this domain, it is important for mathematics education researchers

to heed the work of researchers in other fields such as cognitive science and neuroscience. Such research can provide valuable insights into these difficult-to-observe processes. But we must also continue to conduct studies that expose students to instructional treatments and tasks and carefully investigate phenomenologically how students construct meaning for geometric concepts, always attempting to build cognitive models that explain what we observe. It is through the integration of cognitive and phenomenological investigations that we can make true progress in this vital area of research.

OTHER THEORIES RELEVANT TO GEOMETRIC LEARNING

Several other theories seem particularly relevant for understanding geometric reasoning.

The Theory of Abstraction¹³

Learning occurs as individuals recursively cycle through phases of action (physical and mental), reflection, and abstraction in a way that enables them to develop ever more sophisticated mental models. These mental models consist of abstracted objects and abstracted actions that can be performed on those objects. Once objects and actions have been sufficiently abstracted, they themselves become mental objects that can be mentally operated on (e.g., compared, decomposed, connected). The levels of abstraction described below indicate major landmarks in the process of abstracting mathematical objects, affording increasingly powerful reasoning.

Levels of Abstraction

Perceptual abstraction. At the perceptual/recognition level, abstraction isolates an item in the experiential flow and grasps it as an object: “Focused attention picks a chunk of experience, isolates it from what came before and from what follows, and treats it as a closed entity” (von Glasersfeld, 1991, p. 47). When items are perceptually abstracted, we become conscious of them, entering them into working memory. Perceptual abstraction isolates those sensory properties of an experience that are needed to recognize further instantiations of the experience. However, items that have been abstracted only at the perceptual level cannot be re-presented (visualized) and cannot be op-

erated on unless they are physically present. Furthermore, “the abstracted operational pattern necessary to recognize things of a kind does not automatically turn into an image that can be re-presented” (von Glasersfeld, 1995, p. 93).

Internalization. When material has been sufficiently abstracted so that it can be re-presented in the absence of perceptual input, it has been internalized. Internalization is

the process that results either in the ability to re-present a sensory item without relevant sensory signals being available in actual perception or in the ability to reenact a motor activity without the presence of the kinesthetic signals from actual physical movement. Internalization leads to ‘visualization’ in all sensory modalities. (Steffe & Cobb, 1988, p. 337)

According to von Glasersfeld, it is at the internalized level that a concept¹⁴ has been formed, a concept referring “to any structure that has been abstracted from the process of experiential construction as recurrently usable. . . . To be called ‘concept’ these constructs must be stable enough to be re-presented in the absence of perceptual ‘input’” (von Glasersfeld, 1982, p. 194). However, for material that has only been internalized, one cannot reflect upon its re-presentation to consider how it is composed (Steffe & Cobb, 1988). That is, one can only re-present internalized material, not analyze its structure.

Interiorization. “Reflecting upon the results of a re-presentation requires detachment and placing the re-presented activity at a distance in order to analyze its structure and composition” (Steffe & Cobb, 1988, p. 17). To do so requires interiorization, which is “the most general form of abstraction; it leads to the isolation of structure (form), pattern (coordination), and operations (actions) from experiential things and activities” (p. 337). Material has been interiorized when abstraction has disembedded it from its original perceptual context and it can be freely operated on in imagination, including “projecting” it into other perceptual material and utilizing it in novel situations. To use an abstracted item for thinking about a new situation requires that the item be reprocessed, not simply recalled as it has been encountered in the past—reprocessing is the operationalization of interiorization (Steffe, 1998).

Note that researchers should be careful with the notion that abstraction “disembeds” material from its original context. As Hoyles and Healy (1997) have

¹³ Other views of abstraction in geometry can be found in (Hollebrands, 2003; Hoyles & Healy, 1997) and (Mitchelmore & White, 2000).

¹⁴ In this case, von Glasersfeld’s “concept” is the same as my “conceptual object” or “conceptualization.”

noted, abstraction can be *situated* in that, at least at lower levels, it is a process that occurs within, and remains linked to, the context in which it first occurs. That is, at the perceptual and internalized levels of abstraction, context is an inseparable part of an abstraction. At the interiorized level, abstracted material becomes separable from its original context, although links to that context may remain. In fact, these links may be an important part of applying interiorized material in novel situations. Generalized forms of links may form part of the structure of an abstraction that helps individuals recognize new situations in which the abstraction might be applied.

Second level of interiorization. When material reaches the second level of interiorization, operations can be performed on the material without re-presenting it and symbols, acting as “pointers” to the originally abstracted material, can be used to substitute for it.

The Theory of Abstraction Applied to the van Hiele Levels

To apply the theory of abstraction to the van Hiele theory, recall that action-based mental processes enable students to recognize and construct images of shapes—Kosslyn’s sequential part-by-part mental constructions, or von Glasersfeld’s scan paths. During the first van Hiele level, these processes are abstracted at the perceptual and internalized level to recognize and visualize common shapes, respectively. To move up to the second van Hiele level, shape recognition processes must be raised to the interiorized level, allowing these whole-shape recognition processes to be decomposed, and thus the components of shapes to be recognized, operated on, and interrelated. Because interiorization of whole-shape recognition processes permits relational operations to be performed on shape components, relationships between components of shapes can be noticed, starting another cycle of abstraction, focused not on recognition but on relationships. As these relational operations between components of shapes are abstracted at the perceptual and internalized levels, students can describe relationships between components of individual shapes informally. Subsequently, when relational operations become interiorized, they can be applied to sets of shapes so that the student can see that a class of shapes has a set of characterizing properties. At this point a student is capable Level 2, descriptive/analytic, reasoning. General properties, applied to classes of shapes, can be described. To reach the third van Hiele level, relational operations must achieve “symbol” status (von Glasersfeld, 1995)—that is, they must reach the second level of interiorization. At this level, verbal/symbolic statements of relationships can act as substitutes for, or pointers to, these operations, with

individuals able to reason meaningfully about these symbolic statements without having to re-present the actual operations for which they stand.

A Transformational Perspective on Relational Operations

What is the nature of the operations that establish interrelationships between components of shapes? One hypothesis is that the relationships are established with unconscious visual transformations. For example, but differing in detail from Leyton (1992), Battista (in press) posits that in a parallelogram, seeing the relationship that opposite sides are parallel might be based on mentally (but unconsciously) translating one side onto the opposite side. Similarly, seeing that opposite angles in a parallelogram are congruent may be based on an unconscious 180° rotation.

This analysis is consistent with the observation that parallelism is more difficult for students to “see” in typical trapezoids than in rectangles and parallelograms. Seeing parallelism in scalene trapezoids cannot be accomplished using the same mental transformation that establishes this property for rectangles and parallelograms because in a scalene trapezoid, one side is not the translation image of its opposite side. Also consistent with this hypothesis is the common student belief that the property “opposite sides parallel” goes with “opposite sides equal” because the first property is established with a translation that also establishes the second. To disassociate these two properties, parallelism might need to be established, for example, with a translation and a stretch.

Special Types of Abstraction

Three special forms of abstraction are fundamental to geometry learning and reasoning (Battista, 1999a; Battista & Clements, 1996). *Spatial structuring* is the mental act of constructing and abstracting an organization or form for an object or set of objects. It determines an object’s nature or shape by identifying its spatial components, combining components into spatial composites, and establishing interrelationships between and among components and composites. *Mental models* are sets of abstractions that are integrated to form nonverbal mental representations that are activated to interpret and reason about situations (more will be said about mental models in a later section). A *scheme* is an organized sequence of actions or operations that has been abstracted from experience and can be applied in response to similar circumstances. It consists of a mechanism for recognizing a situation, a mental model that is activated to interpret actions within the situation, and a set of expectations (usually embedded in the behavior of the model) about the possible results of those actions.

Three Types of Structuring

Geometry learning can be viewed as involving three types of structuring (Battista, in press b). (a) Spatial structuring constructs a spatial organization or form for an object or set of objects. It determines a perception/conception of an object's nature or shape by identifying the object's spatial components, combining components into spatial composites, and establishing interrelationships between and among components and composites. (b) Geometric structuring describes spatial structurings in terms of formal geometric concepts. That is, in geometrically structuring a spatial situation, a person uses geometric concepts such as angles, slope, parallelism, length, rectangle, coordinate systems, and geometric transformations to conceptualize and operate on the situation. In order for a geometric structuring to make sense to a person, it must evoke an appropriate spatial structuring. (c) Logical structuring formally organizes geometric concepts (i.e., geometric structurings) into a system and specifies that interrelationships must be described and established through logical deduction. To accomplish a logical structuring, individuals must logically organize sets of properties. The construct of structuring, construed more generally, provides another perspective on the van Hiele theory: Spatial structuring is the major focus in van Hiele Level 1 and the transition to Level 2, geometric structuring is the major focus at Level 2 (when it has been broadly accomplished, Level 2 is fully attained), and logical structuring is the focus at Levels 3 and 4 ("axiomatic" structuring is the focus at Level 5).

Construction and Use of Mental Models^{15,16}

An abundant amount of research indicates that many forms of reasoning are accomplished with mental models (Battista, 1994; Calvin, 1996; English & Halford, 1995; Greeno, 1991; Johnson-Laird, 1983; Markovits, 1993). According to the mental model view of the mind, individuals understand or make sense of a situation, including a set of connected verbal propositions describing a situation, when they activate or construct a mental model to represent the situation (Johnson-Laird, 1983). Mental models are nonverbal recall-of-experience-like mental versions of situations that have structures isomorphic to the perceived structures of the situations they represent (Battista, 1994; Greeno, 1991; Johnson-Laird, 1983, 1998). That

is, "parts of the model correspond to the relevant parts of what it represents, and the structural relations between the parts of the model are analogous to the structural relations in the world" (Johnson-Laird, 1998, p. 447). Individuals reason about a situation by activating mental models that enable them to simulate interactions within the situation so that they can explore possible scenarios and solutions to problems. When using a mental model to reason about a situation, a person can mentally move around, move on or into, combine, and transform objects, as well as perform other operations like those that can be performed on objects in the physical world. Individuals reason by manipulating objects in mental models and observing the results. "The behavior of objects in the model is similar to the behavior of objects that they represent, and inferences are based on observing the effects of the operations" (Greeno, 1991, p. 178). Importantly, individuals' use of mental models is constrained by their knowledge and beliefs. That is, much of what happens when we form and manipulate a mental model reflects our underlying knowledge and beliefs about what would happen if we were dealing with the objects they represent. So the properties and behavior of objects in a mental model simulate the properties and behavior we believe the objects they represent possess.

I will now discuss several examples of mental-model-based reasoning in geometry. The first two examples are reformulated from Clements and Battista (2001).

A second-grade student was investigating geometry in a Logo environment. After she had discovered that she needed 90° turns to make a tilted square, she was asked to draw a tilted rectangle. She immediately used 90° turns at the vertices. She was asked, Why?

CA: Because a rectangle is just like a square but just longer, and all the sides are straight. Well, not straight, but not tilted like that (makes an acute angle with her hands). They're all like that (shows a right angle with her hands) and so are the square's.

In later discussions, she also stated that a square is a rectangle.

Obs: Does that make sense to you?

CA: It wouldn't to my [4 year old] sister but it sort of does to me.

¹⁵ Researchers employing the theory of mental models have actually created computer models of mental models for specific situations.

¹⁶ Mathematics education researchers using the notion of concept image have reached conclusions similar to researchers embracing the idea of mental models, specifically that students most often do not use definitions of concepts, but rather concept images—a combination of all the mental pictures and properties that have been associated with the concept—to make decisions. (See Clements & Battista, 1992 for more details) Concept images are similar to mental models.

Obs: How would you explain it to her?

CA: We have these stretchy square bathroom things. And I'd tell her to stretch it out and it would be a rectangle. (Battista & Clements, 1991, p. 18)

It “sort of made sense” that a square could be thought of as a rectangle because CA's mental model of a square enabled her to envision a square being stretched into a rectangle. Because CA had just demonstrated her knowledge that squares and rectangles are similar in having angles made by 90° turns, this response suggests that she may have understood intuitively that all rectangles can be generated from one another by certain transformations, ones that preserve 90° angles. CA's mental models of shapes such as squares and rectangles incorporated allowances and constraints on how they could be acted upon and transformed. In this instance, CA reasoned by performing a simulation of changing a square into a rectangle using a special type of visual transformation, one that incorporated formal geometric constraints—preserving 90° angles—on her mental objects and actions.

Sometimes, however, students' mental models are inadequate to draw valid conclusions. For instance, a second grader was examining her attempt to draw a tilted square in Logo. (She too had successfully taught the turtle to draw squares of various sizes, all of which had sides vertical and horizontal.) Because she did not use 90° turns for the tilted square, her figure was only a crude approximation to a square. In fact, though M had earlier stated that all sides of a square are the same length, in drawing her square, she abandoned this constraint to close the figure (because her angles were not 90°). But M intended to draw a square and concluded that it was a square, reasoning as follows:

Int: How do you know it's a square for sure?

M: It's in a tilt. But it's a square because if you turned it this way it would be a square.

The difference in the validity of CA's and M's reasoning seems to have been due to the degree to which they incorporated relevant knowledge into their mental models. On the basis of a variety of tasks, the researchers found that CA's thinking about shapes was based more on properties than M's, so CA's mental models were more property-based and her imagery more influenced by property-based considerations. M based her thinking on simple visualizations—she, like all students in her class, had seen that squares that are

rotated from the vertical-horizontal-side position often do not look like squares. M's mental-model-based transformation was unconstrained by her knowledge that all sides of a square have the same length. Thus, these episodes raise the question of how students' conceptual knowledge gets incorporated into their mental models and visualization.

A different kind of example is provided by a secondary mathematics teacher attempting to decide on the validity of the following statement: If all the angles in a polygon are equal, then all the sides in the polygon are equal. The teacher said that in thinking about the problem he first imagined a regular pentagon, which he attempted to transform so that it had unequal angles. But he could not imagine how to change some but not all of the sides of the pentagon without changing the angle measures. The mental model this teacher activated to think about this problem made the problem impossible for him to solve. When I asked him to think about a rectangle, he immediately solved the problem. Interestingly, had the teacher originally activated a mental model of a square or regular hexagon, it would have been much easier to implement his transformational strategy. This example illustrates that an important question for researchers is, What determines which mental models are activated during problem solving, and what kinds of instruction guide students to activate the most useful models?

*Mental Models and Imagery*¹⁷

Images are mental representations “that ‘stand in’ for (re-present) the corresponding objects” (Kosslyn, 1994, p. 1). Images can be evoked in the absence of perceptual material (re-presentation or visualization) or by perception (e.g., looking at a diagram). Their formation relies on regions of the brain that support *depictive* representations.

In a depictive representation, each part of an object is represented by a pattern of points, and the spatial relations among these patterns . . . correspond to the spatial relations among the parts themselves. Depictive representations convey meaning via their resemblance to an object, with part of the representation corresponding to parts of the object. (Kosslyn, 1994, p. 5)

A number of researchers have suggested that there are two major types of imagery. Following Lakoff, Wheatley (1998) distinguished between rich images and image schemata. *Rich images* are static, fixed, and contain much visual detail. *Image schemata* repre-

¹⁷ Although I focus on visual imagery, similar discussion can be given for imagery in other modalities such as auditory or kinesthetic.

sent spatial relationships and can be transformed in various ways.¹⁸

The distinction between rich images and image schemata can be explained by the more general theory of mental models. Johnson-Laird and colleagues argued that it is important to distinguish between visual images (which correspond to rich images) and spatial mental models (which correspond to image schemata). Visual images are derived from perception or generated from mental models, with images being views of mental models from particular viewpoints. Images are mentally manipulable when appropriate mental models have been constructed from them (Johnson-Laird, 1998). Thus, in visual reasoning, sometimes people merely construct and scan visual images, and other times they operate on spatial mental models (Knauff, Fangmeier, Ruff, & Johnson-Laird, 2003). Furthermore, sometimes, when aspects of visual images are irrelevant to a task, the evocation of visual images can actually interfere with construction and use of an appropriate mental model and thus impede reasoning (Knauff et al., 2003).

The distinction between images and mental models can also be interpreted in terms of the theory of abstraction. A re-presentation of a visual image results from an internalization of a perception of a spatial object. A mental model of a spatial entity results from interiorization of the mental processes used to conceive of the entity: "The operations that are carried out in reasoning with models . . . are conceptual and semantic" (Johnson-Laird, 1998, p. 457). Thus, to reason geometrically about an image, one must construct a proper mental model of the image, one that captures the image's relevant spatial structure (and the depth of that structure depends on the sophistication of the concepts used in this structuring). That is, geometric reasoning about figures requires a reconstitution of images in terms of appropriately structured mental models. Indeed, if figures are thought about as pictures (using images), not as sets of spatial/geometric relationships (using mental models), students are not thinking about formal geometry (cf. Laborde, 1993).

Concept Learning

Because concept learning, use, and analysis play a significant role in the development of geometric thinking, research on the formation and use of concepts provides insights into that development, especially for understanding thinking at van Hiele Levels 1

through 3. I will summarize some of the relevant findings of that research.

Not only do human minds naturally categorize objects (Pinker, 1997), categorization is essential for intelligent thought and action (Ross, 1996). What the mind gains from forming categories "is inference. Obviously we can't know everything about every object. But we can observe some of its properties, assign it to a category, and from the category predict properties that we have not observed" (Pinker, 1997, p. 307). For example, if, on seeing round, fist-sized reddish-green objects attached to the branches of a tree, you categorize the objects as apples, you can then infer that they are edible and have seeds.

Research suggests that people form two kinds of categories (Pinker, 1997). Fuzzy or natural categories, such as games, are formed in everyday activity. Such categories generally have no clear definitions, have fuzzy boundaries, and are conceptualized mainly in terms of stereotypes and family-like resemblances—some examples are deemed "better" than others. Identification of instances seems to be the major goal. The second type of category consists of formal, explicitly defined categories, such as odd numbers, in which all instances are logically equivalent as representative of the class. Formulating and studying precise definitions is the goal.

Learning geometry involves the formation of both fuzzy and formal categories. Before geometry instruction, and even in primary-grade teaching, as children encounter shapes in the physical world and those shapes are named by adults, children treat shapes as fuzzy categories—shapes are identified but not defined. Subsequent school instruction demands that students conceptualize shapes in terms of verbally stated, property-based definitions. Difficulties often occur as students attempt to reconcile fuzzy categories with formal concepts. The language used in fuzzy and formal concepts may be the same, but the underlying cognitive entities are vastly different. For instance, logically, a square is as much a rectangle as a nonsquare rectangle. But students who possess fuzzy concepts often resist the idea that a square is a rectangle—they think it is only a square. Progressing from the visual to the descriptive/analytic van Hiele level seems very much like moving from fuzzy to formal categories for a set of concepts in a domain.

There seem to be several competing theories for how people develop fuzzy categories. The most prominent seems to be that people develop prototypes—

¹⁸ The distinction between rich images and image schemata seems related to Piaget's notion of figurative knowledge, which results from empirical abstraction, and operative knowledge, which results from reflective abstraction: "Perception yields a series of static, centration-afflicted snapshots. The knowledge we get from perception is figurative, not operative" (Piaget, 2001, p. 21).

mental images or examples that include, usually implicitly, features that are relatively common among members of a category. People decide whether an object belongs to a category by determining if it is sufficiently similar to prototypes (Smith, 1995).

Researchers describe two ways that prototypes might be formed (Krascum & Andrews, 1993). According to the *abstractionist* model, after experiencing a number of category examples, a perceiver abstracts the category features that occur most frequently into a prototype representation. According to the *exemplar* model, category examples are not decomposed into attributes but instead are abstracted to form a representation that consists of a set of discrete, more-or-less intact instances. Krascum and Andrews (1993) stated that many researchers favor the exemplar view because of findings that show that young children are insensitive to component features of objects, show developmental improvement in the ability to selectively attend to single dimensions of stimuli, and perceive objects in global, unanalyzed fashion. However, Krascum and Andrews also cited evidence that is inconsistent with the exemplar model. For instance, they reported that young children do not necessarily fail to decompose objects into their component parts and in fact often fixate on single features. The ability to distribute attention over many stimulus dimensions develops over time. Interestingly, as researchers observe students' learning of geometric shapes, we see many of the behaviors described above—students form prototypes, often limited; they fail to notice important spatial attributes; they sometimes fixate on particular features. Also, Kosslyn's description of perception seems more consistent with the exemplar model and addresses many of the issues raised about it.

In considering how fuzzy categories of (nongeometric) shapes are formed, Gentner and Medina (1998) argued that there is a developmental/experiential shift from perceptual similarity (e.g., recognizing a mobile that is a very close perceptual match to one seen before) to relational similarity (e.g., recognizing the similarity of a mitten covering a hand and a shoe covering a foot). These researchers argued that the perception of similarity is accomplished by a process of "alignment or structure-mapping," and that performing such mappings increases the likelihood of establishing relational rather than purely visual similarities.¹⁹ They hypothesized that giving a common name to shapes in a class can encourage individuals to make comparisons that activate structure mappings that "promote the discovery of deeper

commonalities" (p. 276). Indeed, structure mapping is a central learning mechanism that enables individuals to notice and store abstract relational properties, derive abstract knowledge from instances, and extend that knowledge to new cases. This happens as the relations examined in structure mappings are abstracted to become objects in and of themselves.

Because Gentner and Medina's (1998) theory focuses on *structure* mappings, it has the potential to shed light on the mechanisms by which students formulate properties of geometric shapes when formal definitions are not given. The developmental/experiential shift from perceptual to relational similarity seems particularly relevant in progressing from van Hiele Level 1 to 2 because Level 2 reasoning focuses on relationships between parts of shapes. Gentner and Medina's theory suggests that giving common names to like geometric shapes should cause students to start performing structure mappings that can enable them to see, at least informally, spatial relations common to the shapes. But we do not know how spontaneous these are. Perhaps for most students, the mappings occur only after explicit prompts such as, "How are these shapes alike?" Additional research is needed to examine the potential of the structure-mapping theory in explaining the transition to the early phases of van Hiele Level 2.

Vygotsky's (1986) work sheds further light on the relationship between fuzzy and formal concepts. He argued that children become conscious of their spontaneous (i.e., fuzzy) concepts relatively later, with the ability to define them in words appearing long after they have "acquired" the concepts. In contrast, the development of scientific (formal) concepts usually begins with their verbal definition. However,

though scientific and spontaneous concepts develop in reverse directions, the two processes are closely connected. The development of a spontaneous concept must have reached a certain level for the child to be able to absorb a related scientific concept. . . . In working its slow way upward, an everyday concept clears a path for the scientific concept and its downward development. Scientific concepts, in turn, supply structures for the upward development of the child's spontaneous concepts toward consciousness and deliberate use. Scientific concepts grow downward through spontaneous concepts; spontaneous concepts grow upward through scientific concepts. (p. 194)

Thus, applying this theory to geometric concepts, we see that fuzzy, recognition-based conceptualiza-

¹⁹ A structure mapping is isomorphic-like; it is one-to-one and if two elements are linked in one representation, their images are linked in the corresponding representation.

tions of geometric shapes develop first as the shapes are perceived and named in school and out. For common shapes, and most students, such early experiences “naturally” lead to van Hiele Level 1 reasoning. But getting to Level 2, in which shapes are reasoned about in terms of formal geometric concepts, requires a great deal of work—work that, to be meaningful to students, must connect to the fuzzy concepts developed in Level 1.

Finally, at the formal end of the spectrum on categorization, is the issue of hierarchical classification. According to Jones (2000), following De Villiers, classifications can be hierarchical or partitional. Hierarchical classifications use *inclusive* definitions such as specifying that trapezoid is a quadrilateral with *at least* one pair of sides parallel—which means that a parallelogram is a special type of trapezoid. Partitional classifications use *exclusive* definitions such as specifying that a trapezoid is a quadrilateral with *only* one pair of sides parallel—which *excludes* parallelograms as trapezoids. In general, in mathematics, inclusive definitions (and thus hierarchical classifications) are preferred, although exclusive definitions and partitional classifications are certainly not mathematically incorrect. However, a number of studies show that many students have great difficulty with the hierarchical classification of quadrilaterals (de Villiers, 1994; Jones, 2000). Although research on the van Hiele levels has shed light on this issue, more research is needed that investigates why this difficulty occurs and how to overcome it.

STUDIES OF LEARNING AND TEACHING GEOMETRY

Computer Environments for Learning Geometry²⁰

Early research on the use of computer environments for learning geometry focused primarily on investigating predictions about student learning made by environment designers. More recent research, informed by initial studies and extensive experience with the environments, has investigated more closely the learning processes that occur within the environments and how the environments affect and shape student learning.

I will discuss three major types of computer environments for investigating plane shapes (see Clements and Battista, 1992 for a discussion of other types): Logo-based, Geometric Supposers, and Dynamic Geometry Environments (DGEs).²¹ Two essential features have been noted for such computer environments. First, they require students to provide *explicit* specifications for geometric shapes—using menu selections in Supposers and DGEs, and command lists in Logo. Thus, unlike using paper and pencil, in these computer environments, students cannot make drawings without some level of *conceptual and representational explicitness*. Researchers have argued that this explicitness promotes and supports reflection on and abstraction of geometric concepts, and movement toward van Hiele Level 2. For instance, Clements and Battista (1992), following Papert, claimed that,

Writing a sequence of Logo commands, or a procedure, to draw a rectangle . . . obliges the student to externalize intuitive expectations. When the intuition is translated into a program it becomes more obtrusive and more accessible to reflection. . . . Students must analyze the spatial aspects of the rectangle and reflect on how they can build it from parts. (p. 450)

Similarly, Laborde (2001) claimed that when constructing a square with a given side in a DGE,

With paper and pencil the . . . task is controlled by perception. . . . The same task in Cabri . . . cannot be obtained by eye . . . but uses a circle as a tool for transferring a given distance. The task in Cabri requires more mathematical knowledge about the properties of a square and the characteristic property of a circle. (p. 294)

In the case of DGE research, the amount of explicitness used by students has been, indirectly, a frequent focus of discussion. For instance, in *The Geometer's Sketchpad* (GSP) or Cabri, one can “draw” a rectangle by creating four line segments in a rectangular configuration. Little conceptual explicitness is required, and the drawing does not remain rectangular when its vertices are dragged. Alternately, one can use DGE commands to “construct” a rectangle that preserves its rectangularity with dragging. In this case, it can be argued that each step in the construction requires some conceptual explicitness. The question is, how much? For instance, after creating a segment,

²⁰ Even though research shows that computer environments have great promise for improving geometry instruction, computer use among mathematics teachers is low. Becker (2000) reported that only 1 of 9 secondary mathematics teachers said that a typical student in their classes used computers on more than 20 occasions during a 30-week period.

²¹ Most of the relevant research has been conducted on these three environments. Specialized restricted applets and general drawing programs are not discussed.

students might create a segment perpendicular to it using the perpendicular construction. So they must know, at a minimum, that a specific spatial characteristic of a rectangle is created by the perpendicular construction. Then they must use either perpendicularity or parallelism to construct the remainder of the figure. Because only three right angles are explicitly constructed, how do students know that the last angle must be a right angle? Do they deduce it logically, conduct an informal componential analysis (see the previous description of the Borrow/Battista elaboration of van Hiele Level 3), or merely make a visual assumption? Other properties, such as parallelism or congruence of opposite sides, are not explicitly constructed, so may not be recognized in the construction (but might be visually induced).

Understanding the nature of the conceptual explicitness that occurs in students' constructions of draggable DGE drawings requires much additional research. For instance, are students' conceptualizations of their constructions manifestations of their understanding of abstract mathematical concepts, or are they proceduralized methods for creating visual configurations? That is, when students construct a perpendicular in a DGE, is the perpendicular simply a particular kind of visual configuration, or is it an instantiation of a geometric concept (i.e., two lines intersecting at 90° angles)? Another difficulty with DGEs is that gaining access to commands used in a completed construction is not always straightforward. For instance, to determine how a draggable rectangle was constructed in GSP, one can check all properties of various objects on the rectangle (but the sequence of constructions is not readily available), or one can create a tool for the construction and examine its script. However, if students do not sufficiently understand the syntax of GSP—and learning syntax can be a major barrier for students in DGEs (Laborde, 1992)—students may not be able to use these tools to keep track of what they do, which considerably lessens conceptual explicitness. Similarly, with Logo, although each movement of the turtle requires a command, what the commands mean to students should not be taken for granted. As an example, a student might correctly construct a rectangle, for instance, using trial and error and several turn commands to make each right angle—so there are no explicit commands to make right angles. Without appropriate instructional intervention, the student might not construe the set of RIGHT commands at a vertex as a *set* for which the sum of the inputs determines the size of vertex angle; instead, the commands are taken as separate acts used to visually adjust the turtle's heading. Or, the

shape created may not be a rectangle at all, merely a visual approximation of a rectangle; so even if the set of commands is considered, the properties of the rectangle are not accurately embedded in the commands. Thus, the conceptual explicitness that occurs in computer environments for geometry learning depends on a complex interaction between the commands given in constructing figures, the availability of those commands to student inspection, the reasoning of the student, and instruction.

The second feature attributed to computer environments for facilitating geometry learning is the “repeatability” of drawings (Laborde, 1992). “In this kind of software a necessary condition for a construction to be correct is that it produce several (or an infinity of) drawings which preserve the intended properties when variable elements of the figure are modified” (Laborde, 1992, p. 129). Repeatability is achieved very differently in the different programs. In Logo, repeatability requires the use of procedures, and often variables, adding more syntax requirements. In Supposers, repeatability is discrete and constrained. In a DGE, repeatability is continuous and dynamic. Again, however, a major issue for research is not only what kinds of repeatability engender what kinds of learning, but how students use and, more importantly, conceptualize repeatability. Do students really see draggable constructions in DGEs as *representing* multiple elements in a given class of shapes? Or, are such constructions simply another kind of geometric object that can be studied?

This latter point is critical and returns the focus to diagrams. In paper-and-pencil approaches to studying geometry, there are two major types of entities. First, there are diagrams, mere perceptual objects. Second, there are shape conceptualizations—conscious meanings activated for a class of shapes. DGEs introduce a whole new object—draggable drawings, which researchers often view as representations of shape conceptualizations: “A [draggable] figure captures the relationships between the objects in such a way that the figure [i.e., considered as a formal concept] is invariant when . . . the construction is dragged (in other words, that it passes the drag test)” (Jones, 2000, p. 58).

Furthermore, because draggable drawings are constructed to possess specific geometric properties (and consequently possess additional properties necessitated by the given properties), researchers often see all the properties as “included” in the drawings. However, students may not see any of the properties embedded in draggable drawings and may not even think that the drawings have specific properties. Novice students usually reason about draggable drawings

as geometric entities in their own right, not explicitly as representations. In analyzing draggable drawings, students first notice movement *constraints*, then later, they might conceptualize these constraints in terms of regularities or invariants, and finally, and often only with great effort, these constraints are conceptualized as formal geometric properties. The question of how students interpret DGE drawings, both draggable and not, is not only critical, but in need of much research attention.

Geometric Supposers

Geometric Supposer construction programs allow students to choose a primitive shape, such as a type of triangle or quadrilateral, and perform measurement operations and geometric constructions on it. The programs record the sequence of constructions and can automatically perform it again on other shapes. The focus of *Supposer* programs, therefore, is to facilitate making, testing, and justifying conjectures, based on examples. Most of the research on *Supposer* programs has taken place in the context of guided-inquiry teaching (Wilson, 1993).

Research generally, but not always, shows that use of *Supposer* programs can enhance geometry learning (Clements & Battista, 1992; Yerushalmy, 1993; Yerushalmy & Chazan, 1993). *Supposer* interacting students were better able to identify non-stereotypical examples of shapes, and their learning often extended beyond standard geometry content, for example, to formulating definitions, making conjectures and arguments, posing and solving significant problems, and devising original proofs. After *Supposer* experience, students more frequently used diagrams in their thinking, treated diagrams flexibly, were less often confused by diagrams in nonstandard orientation, used many diagrams to describe classes of figures, often reasoned about a single diagram as a model for a class of shapes, and understood that *Supposer* constructions can include characteristics that are not shared by all members of the class of figures under consideration. *Supposer* activities can also engender movement away from considering measurement evidence as proof (Clements & Battista, 1992). Indeed, *Supposer* work *can* (but does not have to—see below) help students appreciate the need for proof; unlike for textbook theorems, many students consider *Supposer*-generated conjectures as statements that need to be proved before they can be accepted as true.

Yerushalmy (1993) claimed that three major processes are involved in geometric generalization: “formation of samples of examples to serve as a data base for conjectures, manipulations of the samples,

and analysis of ideas in order to form more general ideas” (p. 82). She reported that use of the *Supposer* focused many students’ attention on the nature of the set of examples—what information was needed, the variability of the information, and the need to organize information gathering. Students learned to appreciate the value of generating many examples and of formulating a data-collection strategy that would prevent them from getting lost. Students learned “to use extreme cases, negative examples, and non-stereotypic evidence to back up their conjectures” (p. 82). However, Yerushalmy found that some *Supposer* students did very little genuinely independent work, generating many examples, but with little sense of direction—often asking the teacher what to look for in the examples. Students also struggled with knowing the types of examples they should examine (Yerushalmy, 1993), which suggests that they had naïve ideas about general mathematical statements and their validity.

One can pose many questions about the student learning reported by Yerushalmy. For instance, how deliberate is students’ use of examples? What is the nature of the reasoning that guides students’ choices of examples? How much of the generation of examples is student initiated, and how much teacher generated? If a lot of it is teacher generated, does the generation become interiorized by students, and, if so, how? What do students think as they generate and manipulate examples? What are the elements of students’ analyses of examples—how do conceptualizations of relevant ideas or beliefs affect these analyses?

Chazan (1993b) investigated teachers’ observations that some students using the *Supposer* did not appreciate that measurement evidence could not prove a proposition true for all cases within an infinite domain. Despite the fact that the instructional unit used in the study argued that measurement of examples is not a valid method for verifying the truth of geometrical statements, instructional “activities in favor of deductive proof were not successful with a sizable portion of the students in the study” (p. 109). Also, by the end of the *Supposer* instructional treatment, although “fewer students considered [empirical] evidence to be proof . . . more students were skeptical about the . . . applicability of deductive proofs” (p. 109). However, Chazan argued that the latter finding may not be as negative as it seems. For instance, such skepticism may be warranted (a) because students realized that they were novices at constructing valid deductive proofs, or (b) because, following Lakatos, being skeptical about proof is actually a mathematically healthy behavior. But there is an even deeper issue here. Almost no geometry curricula actually support students’ invention of the deductive method (see Borrow, 2000 for an exception).

That is, even students who are given the opportunity to see the shortcomings of the empirical approach are generally *shown* a “better way” rather than allowed to invent for themselves essential elements of the deductive method. Because students are merely handed the deductive method, not given the opportunity to construct a personal need and meaning for it, they are naturally skeptical of it. Nevertheless, the issue of whether *Supposer* use helps or hinders students’ movement away from justifying conjectures with empirical evidence toward the use of deductive proof remains unresolved.

Logo

Instructional Effectiveness

In general, research supports the use of Logo in geometry instruction (Clements & Battista, 1992, 2001; McCoy, 1996). Students can understand and become competent with representing and reflecting on geometry problems using appropriate Logo environments and instructional activities. For instance, in the *Logo Geometry (LG) Project*, 28 teachers (980 K–6 students) used LG, which provided instructional activities and Logo tools to help students progress through the early van Hiele levels (Battista & Clements, 1991; Clements & Battista, 2001). Students in LG scored significantly higher than control students on a paper-and-pencil test that was not directly connected to LG, making about double the gains of control students. Overall, LG helped students progress from van Hiele Level 1, visual thinking, toward Level 2, descriptive/analytic thinking. Observations of LG students also showed that they moved from (a) non-analytical and/or authoritarian-based notions of knowledge, to (b) more autonomous, empirical bases for establishing truth, and finally to (c) knowledge as reasoning about mathematics as a logical system.

Several general comments should be made about this study. First, students spent 4–6 weeks doing LG. This extensive period of time counterbalanced the “investment-time difficulty” required for learning geometry with Logo. That is, to learn geometry with Logo, students must learn the Logo *language*, which can subtract from the time used to learn geometry per se. However, when such a large amount of time is spent in learning geometry, as with LG, the amount of time learning the Logo language is proportionally smaller compared to the amount of time spent learning geometry. Second, LG not only provided a carefully developed sequence of activities, it incorporated into the Logo language numerous supports that lessened the “learning Logo” load on students. Third, participating teachers were given extensive training on the use of the LG curriculum.

Not all research on Logo has been positive, however. Some studies show no significant differences between Logo and control groups or limited transfer of learning (Clements & Battista, 1992, 2001; McCoy, 1996). For example, ninth-grade Logo students did not differ significantly from control students in high school geometry (Olive, 1991).

Logo as Representational Medium

If one of the major reasons for the instructional efficacy of Logo lies in its representational power (Clements & Battista, 1992), it is important to analyze how this representation facility works for students. Is the representational efficacy of Logo due to its capability to support reflection, to its encouragement of measurement-based analysis and conceptualization, or to the fact that multiple representations enrich conceptualizations by enlarging the web of meanings into which an idea fits? Critical in answering this question is how students command the turtle to make shapes, and, more importantly, how they conceptualize these commands. The following episodes illustrate students using Logo commands to support (a) analytic, measure-based reasoning (Episode 1), (b) visual/spatial and trial-and-error reasoning (Episode 2), and (c) the transition from visual-spatial to analytic reasoning (Episode 3). Some researchers claim that one reason that students sometimes do not learn optimally in Logo environments is that students often reason strictly visually despite the fact that Logo allows more analytic and measure-based reasoning (e.g., McCoy, 1996).

Episode 1. Two fifth graders were drawing a tilted square using equal forward commands and 90° turns. When asked if their figure was a square, the students replied, “Yes, a sideways square. . . . It has equal edges and equal turns.”

Episode 2. Second grader M was examining her attempt to draw a tilted square in a version of Logo that allowed students to “undo” commands (Clements & Battista, 2001). She used a trial-and-error approach, undoing (signified by E for erase) or inserting additional commands when part of the figure looked incorrect. Her commands were: L 35 F 30 R 45 F 30 E E



Figure 19.5 M’s attempts at making (a) a tilted square and (b) a tilted rectangle in Logo.

R 45 R 35 R 35 E F 30 L 35 R 90 R 35 F 30 L 35 R 90 R 35 F 30 F 5 F 5. Because M did not use all 90° turns, her figure was only a crude approximation of a square (see Figure 19.5a). But she concluded that it was a square, reasoning, “It’s in a tilt. But it’s a square because if you turned it this way it would be a square.”

Episode 3. M was then asked to make a tilted rectangle (previously unreported raw data, Battista & Clements, 1987). She entered the following commands:

M: R 45 F 50 R 45 R 10 R 10 R 10 F 20 L 90 Whoa! R 180 F 50 Whoa! It’s supposed to go like this and it goes like this. Because I turned the turtle too far. E E L 25 L 160 F 50 E L 20 R 40 F 50 Phew! R 90 F 20 END [getting the path shown in Figure 19.5b].

Int: Is there anything special about the turns in a rectangle?

M: They’re all 90.

Int: But you didn’t always use 90s.

M: Because I went like this and like this (motions) and then I didn’t turn and I wanted to go like this. And I hadn’t turned 90 and I had to make a couple of guesses and I finally got it right.

Int: Could you teach it to make another tilted rectangle?

M: R 45 F 60 R 90 F 10 L 90 R 180 F 60 R 90 F 10 END

In Episode 2 and the first part of Episode 3, M seemed to take a strictly visual, trial-and-error approach. Her thinking was far different than the fifth graders in Episode 1, who used specific Logo commands to ensure their figure had specific properties. But at the end of Episode 3, M seemed to come to a realization that the turtle needed to make 90° turns to make a rectangle. M’s work brings up a critical issue in the use of Logo, How exactly does Logo move students to higher levels of reasoning?

One possibility is that Logo, by virtue of requiring a more explicit, analytic, and symbolic representation of geometric ideas, supports and encourages reflection and reformulation of those ideas. Clements and Battista (2001) have argued that programming the Logo turtle to make shapes encourages and facilitates students’ reflection on the composition of the shapes. Students analyze the spatial components of shapes and explicitly represent how they can be built from these components, including explicitly conceptualizing how the components are spatially related. This leads to recognition of the shape’s properties.

In Episode 2, however, there was no evidence that giving Logo commands enabled or encouraged M to

understand the nature of the components of a square or how those components were spatially related. There was no evidence that M consolidated the commands into groups that represented the components of the shape—she merely moved the turtle bit-by-bit, seemingly guided only by visual processes. In fact, even though M had previously stated that all the sides in a square are equal, she explicitly violated this notion with her last set of forward commands. M used the same type of visual reasoning in the first part of Episode 3. However, during the second part of Episode 3, a change occurred. The interviewer’s questioning seemed to promote reflection that changed M’s conceptualization of how to make a rectangle—she explicitly used 90° turns. What caused this change? Did M explicitly reflect on the movement commands she used in making her first rectangle and draw some specific conclusions from them—that is, was her realization based on un verbalized, even unconceptualized, abstractions from those experiences? This is possible, but, given the number of commands, it seems unlikely that she conducted a command-by-command analysis. Alternately, did M change her conceptualization by interrelating her current experiences in giving Logo commands for a tilted rectangle to the knowledge she exhibited when she stated that all the turns in a rectangle are 90°? And, what was the basis for her statement that rectangles have 90° turns? Was it her experience in making rectangles that had vertical and horizontal sides—most students in the class had used 90 for these turns? Or, was it the recollection of statements in class, by students and the teacher, that rectangles have 90° turns? What caused and enabled M to connect the concept of 90° turn to a specific type of spatial configuration—a right angle? Whatever it was, it resulted in an interiorized abstraction of this relationship, an abstraction that allowed the relationship to be applied in this new situation of giving commands to make the final rectangle.

When investigating processes that enable discoveries like those of M, it is important to understand how students use and understand sequences of Logo commands. For instance, M twice used the sequence L 90 R 180 to accomplish a R 90 turn (because her initial left turn was the wrong way). Did M explicitly understand the equivalence between the L 90 R 180 sequence and the R 90 command? That is, did she see this sequence as a substitute for R 90, or was the second command in the sequence merely a visual adjustment, with the sequence not being conceptualized for its net result? On the one hand, because this substitution was a common occurrence among students, it is tempting to think that they saw the equivalence in some way. On the other hand, some students give a

sequence of commands such as F 50 F 10 F 10 merely as a way to move the turtle, seemingly unable to integrate the sequence into a length composite equivalent to F 70 (Clements, Battista, Sarama, Swaminathan, & McMillen, 1997). Other students, although able to combine commands when directed to do so, often have difficulty combining commands in more complex situations. For instance, late in an instructional sequence that explicitly attended to encouraging students to combine commands (Clements et al., 1997), a teacher asked if the following commands produced a rectangle:²² F 40 R 90 F 65 R 90 F 20 F 20 R 90 F 50 F 15 R 90. Monica and Nina agreed that it would be a rectangle until they attempted to draw the resulting path on paper—Nina drew a rectangle; Monica did not. Even when Nina explained to Monica that 20 and 20 sum to 40, which matches the other 40, Monica needed to try the path on the computer before she accepted it. She was not convinced by strictly abstract reasoning.

The next episode shows how the Logo medium can help students develop analytic, measurement-based conceptualizations of shapes.

RE: The rectangle is like a square, except that squares aren't long. But on rectangles, they are long.

Int: What does a shape need to be a rectangle?

RE: All of the sides aren't equal. These two [opposite sides] and these two [other opposite] sides have to be equal.

Int: How about 10 on two sides and 9 on the other two? Would that make a square?

RE: Kind of like a rectangle.

Int: Would it be a square too?

RE: [Shaking head negatively] It's not a square. 'Cause if you make a square, you wouldn't go 10 up, then you turn and it would be 9 this way, and turn and 10 this way. That's not a square.

RE's first comment indicated visual reasoning—rectangles are long. On further questioning, RE talked about opposite, but not all, sides being equal—a property-based response. But we cannot attribute this increase in level of reasoning directly to the Logo context—even students not using Logo make such changes in reasoning in response to appropriate questioning. Finally, however, RE reasoned analytically about side lengths using the action-based command context of Logo. The Logo environment seemed to promote,

make accessible, and appropriately support measure-based reasoning for this student.

A final episode further suggests how students' work with Logo can move them toward property-based reasoning about shapes. Prior to the episode, JN had successfully used a variable-input rectangle procedure to draw a tilted rectangle. He was now reflecting on his unsuccessful attempt to use this procedure to make a nonrectangular parallelogram.

JN: Maybe if you used different inputs. [He types in a new initial turn, then stares at the parallelogram on the activity sheet.] No, you can't. Because the lines are slanted. . . .

Tchr: Yes, but this one's slanted [indicating the tilted rectangle that JN had successfully drawn with the Logo procedure].

JN: This one [the parallelogram]—the thing's slanted. This thing [the rectangle] ain't slanted. It looks slanted, but if you put it back [shows how to turn it so that its sides are vertical and horizontal], it wouldn't be slanted. Anyway you move this [the parallelogram], it wouldn't be a rectangle. So, there's no way.

JN was beginning to develop a conceptualization of what is formally captured by the property-based concept that the adjacent sides of a rectangle are perpendicular. In so doing, his reasoning was progressing from the visual reasoning of van Hiele Level 1 to the descriptive-analytic reasoning of Level 2. However, at this time, JN did not possess an interiorized formal concept of perpendicularity (or right angles) that he could apply to the situation.

Zooming in on Conceptual Change

In my final example of Logo research, I summarize a study of Hoyles and Healy (1997), who investigated how one 12-year-old student, Emily, changed her conceptualizations of geometric reflections while working in a Logo-based microworld. On a pretest asking students to draw reflections of figures, two of Emily's answers were correct, three were incorrect. However, in the three incorrect drawings, the reflected images were congruent to their preimages, which is consistent with Emily's description of reflections, "the line . . . has got something on one side and it would be exactly the same on the other." For instance, when drawing the reflection of a vertical line segment through an oblique mirror line, after Emily correctly drew one endpoint of the image, she drew the rest of

²² For consistency, forward commands are always denoted by F and right commands by R. The actual commands given by students varied by microworld.

the image parallel to the preimage. Hoyles and Healy reported that when Emily did reflection problems, she used a ruler placed orthogonally to the mirror line to measure equal distances between some, but not all, corresponding points. Thus, Emily's initial conceptualization of reflection seemed to have two components: First, a reflection consists of a mirror line along with congruent images and preimages; second, some points on the preimages correspond to points on the image via perpendicularity and equidistance (but this idea was vague and inconsistent).

In the first computer task, a blue turtle was supposed to draw the preimage and a red turtle the reflection image about a vertical mirror line. The two turtles were shown with the same position and heading along the mirror. The commands (but no figure) for the blue turtle were shown; the problem for the students was to give the appropriate commands for the red turtle. The initial idea of Emily and her partner Cheryl was to reverse both movement (F, B) and turn commands (R, L), but they quickly decided, without trying their idea, that they only needed to interchange right and left turn commands, which they did, completing the task correctly.

Because the girls' notions of reflections seemed to be tightly connected to Logo commands, Hoyles and Healy next investigated what happened when Logo commands for figures were absent. Students were given two reflection-congruent turtle paths (with the turtles showing), but no Logo commands, and were asked to find the mirror line. Cheryl simultaneously traced the path of both turtles, stopping where she thought they would meet on the mirror line (so she was using a mental model of two linked and simultaneously moving turtles). After using an experimenter-provided tool for getting the turtles to meet on the mirror line, the girls turned the blue turtle through a series of L 10s until it looked as if it were heading along the mirror line. They then turned the red turtle the same angle to the right, and when they saw that it was facing exactly the same direction as the blue, they were convinced that they had found the direction of the mirror line.

Several questions about these first two tasks arise. Precisely how was the girls' left/right reversal strategy connected to or imbedded in their conceptions of reflections? For instance, had the girls previously abstracted that reflection reverses orientation? Or, was turn reversal an element of Logo procedural knowledge that enabled the girls to produce congruent shapes on both sides of a mirror line? How did they discover this knowledge? How exactly did the girls conceptualize the mirror line? From the second task, we might conjecture that they thought that it was

"in the middle" of the two congruent figures and that somehow the turtles drawing the figures made equal angles with the line when they met it.

The next task required students to construct the reflection of a flag when both the flag and mirror line were oblique and no Logo commands were provided. (Colored turtles were used to mark flag vertices so students could find distances between vertices by asking Logo to give distances between specified turtles.) In this task, the girls changed their strategy, trying to set up a correspondence between selected preimage and image points. As shown in Figure 19.6a, Emily projected the endpoints of the flag vertically downward, as if the mirror line were horizontal, making each projected point and its preimage equidistant from the mirror line, in the vertical direction. However, we do not know how the girls arrived at this conceptualization. Is it another manifestation of the vague correspondence idea that Emily used on the pretest?

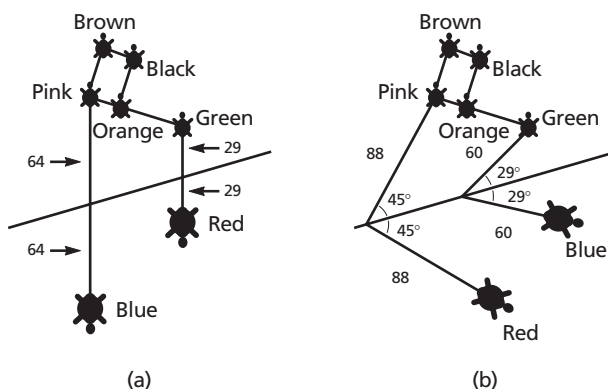


Figure 19.6 Emily and Cheryl's reflection attempts in Logo (Hoyles & Healy, 1997).

When the girls discovered that the distance between the red and blue turtles was unequal to that between the green and pink turtles, they ignored the blue turtle and gave commands to the red turtle. The resulting image flag was congruent to the preimage flag, but when the girls recognized that the image flag's orientation was incorrect, they devised a new strategy for finding the endpoints of the flagpole. They used Logo commands to determine the angle and distance of a turtle on the mirror line to the pink turtle (the preimage turtle); they then turned the turtle the opposite direction from the mirror and went forward the given distance to the image red turtle (see Figure 19.6b). The authors observed,

Across all the [student] pairs and the many different methods observed, our analysis showed how the microworld offered students a *way of talking about* and *operationalising* the angle properties of reflection, by

thinking about turtles turning from a heading along the mirror through the same amount in opposite directions. (Hoyles & Healy, 1997, p. 44)

Finally, to investigate how situated the girls' knowledge was in the Logo context, Emily was asked to reflect a triangle in an oblique mirror using only a ruler and a protractor. She was completely puzzled until she drew a turtle on the reflection line to use as a reference point (see Figure 19.7a). She measured the angle and distance of vertex A to the mirror line, then copied this angle and distance to the other side of the line (locating point A'). As she considered the placement of the image of triangle vertex C, as shown in Figure 19.7a, her interiorization of what she did for A enabled her to progress even further, to a generalizing abstraction.

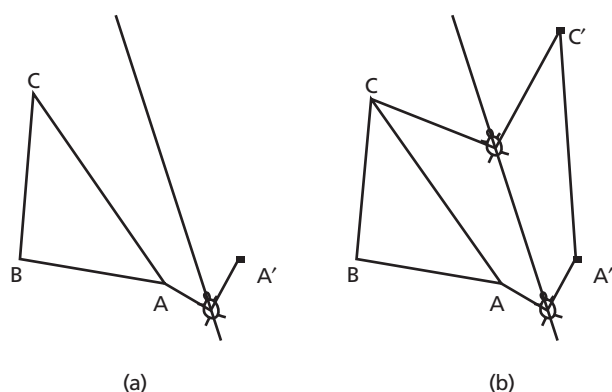


Figure 19.7 Emily's reflection using ruler and protractor and the turtle (Hoyles & Healy, 1997, p. 52).

Emily: That's one [A], now I suppose I have to do the same (indicates C). Can I (she places protractor on turtle) . . . does the turtle have to be there? It's hard to measure to there so, could I . . . hold on . . . I think I've got it. It doesn't matter does it, you can be anywhere as long as the angle and the distance is the same, it doesn't matter where you actually measure from . . . so say there (draws another turtle), the angle is . . . (measures angle and distance to C) so that like (constructs angle and distance to position C') [see Figure 19.7b]. (Hoyles & Healy, 1997, p. 52)

Hoyles and Healy noted that "This last vignette illustrates the extent to which, for Emily, the tools used in her previous explorations had become part-and-parcel of her sense making activities" (1997, p. 53). At first, this non-Logo task presented a major obstacle for Emily because it removed the Logo context in which much of her relevant conceptual knowledge resided. Emily could not think of the problem without turtles for reference points. But when she drew a turtle, two

things happened. First, placing the turtle enabled her to activate her previous turtle-based scheme for dealing with the situation. She was, through the process of interiorization, able to disembed her abstraction from its original Logo context and apply it in this new, paper-and-pencil situation. This interiorized notion, together with her reflection on the arbitrary placement of the turtle, seemed to enable her to develop an explicit notion of the placement arbitrariness—creating a more general and abstract scheme (and her *aha* moment). However, her new, more general conceptualization still seemed tied to the Logo context—she drew a turtle for C. And the effect of the discovery on Emily's orthogonal-correspondence conception of symmetry is unknown. Was it abandoned, or had it been transformed?

Although the data presented by the authors provide an extremely important view of how Emily's conceptualizations evolved, several questions remain unanswered. For instance, in the initial episodes, both Emily and Cheryl seemed to possess a complex interconnected web of knowledge consisting of holistic visual knowledge of reflections as congruent halves on either side of the mirror line, a vague notion of orthogonal corresponding parts in preimage and image, and developing knowledge of producing reflection paths via Logo commands. By the final episode, and as a result of carefully chosen instructional activities, Emily's conceptualization of reflections seemed to change from a turtle-centric path perspective focusing on Logo movement commands to an exocentric view in which the mirror line served as a reference point for measuring angles and distances. This change may have been mediated by her initial vague notion of finding corresponding parts in reflections. Or, it may have been encouraged by the constant demand to translate between the turtle perspective and an exocentric view of reflections (looking "down" on and comparing figures). Somehow Emily's initial vague notion of corresponding parts became connected to and clarified by the concept of "same angle/same distance."

Emily significantly increased the level of sophistication in her thinking about reflections. Her reasoning progressed from visual to descriptive-analytic; from thinking vaguely and visually, to a synthesis of visual, analytic, and measurement-based reasoning, much of it couched in the language of Logo. The Logo environment, including the instructional activities, seemed to provide Emily with tools for making her conceptions of reflections more analytic, precise, and general. Consistent with what Clements and Battista (2001) found in their Logo studies, this micro-world helped Emily and her partner to become more

cognizant of and to reshape their thinking in terms of precise, measurement-based properties.

However, as is often asked of geometry learning in Logo, we might ask how Emily's final conceptualization of reflections compares to traditional mathematical descriptions? That is, Emily constructed a nontraditional conceptualization of reflections—that preimage and image points have equal angles and distances to the mirror line. Although Emily's conceptualization is mathematically correct, it is not as conceptually illuminating as the traditional concept because the traditional concept can more easily be related to the intuitively appealing conception of reflection produced by motion. Indeed, thinking of image and preimage points as endpoints of segments bisected by, and perpendicular to, the mirror line, can be directly related to a (nonrigid) motion, with these segments being the paths of preimage to image points. It would have been interesting to see if and how Emily's final conceptualization could have been evolved into the traditional concept. (For a discussion of students' development of comparable reflection concepts in dynamic geometry, see Hollebrands, 2003.)

Finally, let us return to the question, What characteristics of the Logo environment are essential for encouraging and supporting students' learning? As illustrated in the other studies discussed in this chapter, students in the Hoyles and Healy study seemed motivated to work on problems in the Logo context. But beyond that, was it the Logo-command language per se that supported students' learning, or was it one of several other characteristics of Logo that provided such support? For instance, Logo provides quick, accurate, and interesting visual depictions of students' visual-geometric conceptualizations—a type of uninterpreted feedback. It provides and encourages, through its use of measurement specifications in movement and turn commands, an accessible way to analyze shapes in terms of their components' measures—an important step in moving from pure visual to analytic thought. Thus, it might be conjectured that it is not Logo code per se that supports the development of students' geometric understanding but its provision of (a) appropriate motivation; (b) accurate, quick, and uninterpreted visual feedback; and (c) easy-to-use and precise measurement specifications. These same components also appear in dynamic geometry environments.

Dynamic Geometry Environments

Currently, DGEs seem to be one of the most popular types of software used by mathematics teachers (Becker, 2000) and investigated by researchers. On

the one hand, it has been claimed that DGEs “provide a revolutionary means for developing geometrical understanding. . . . This software seems to make the exploration of geometrical configurations and the identification of meaningful conjectures more accessible to pupils” (Mariotti, 2001, p. 257). On the other hand, skeptics worry that DGEs weaken the role of proof in high school geometry (Mariotti, 2001). The following discussion deals with all aspects of students' learning in DGEs. As in the section on Logo, the discussion will begin with research on students' work at the elementary school level, then move to the middle school and secondary level.

DGE at the Elementary Level

Because DGE software such as Cabri and GSP were originally designed for the secondary level, use of this software in elementary classrooms requires accommodations. For instance, the *Shape Makers* computer microworld is a special add-on to GSP that provides students with geometric shape-making objects (Battista, 1998a). An example is the Parallelogram Maker that can be used to make any desired parallelogram that fits on the computer screen, no matter what its shape, size, or orientation—but only parallelograms. The appearance of a Parallelogram Maker is changed by dragging its vertices with the mouse. In the *Shape Makers* environment, when measurement is introduced, it is done so by displaying relevant measures on the screen automatically; so students do not have to use GSP measurement tools directly. The major focus of *Shape Makers* units is for students to move from van Hiele Levels 0 and 1 to Levels 2 and 3. To illustrate the type of geometric reasoning and learning exhibited by elementary students in DGEs, I describe several episodes with fifth graders' working with the *Shape Makers* microworld (Battista, 1998a, 2001b, in press b) in their regular classrooms.

Episode 1. Three students were investigating the Square Maker at the beginning of their Shape Maker work: MT, “I think maybe you could have made a rectangle.” JD, “No; because when you change one side, they all change.” ER, “All the sides are equal.”

MT, JD, and ER abstracted different things from their Square Maker manipulations. MT noticed the visual similarity between squares and rectangles, causing him to conjecture that the Square Maker could make a rectangle. JD abstracted a holistic movement regularity—when one side changes length, all sides change (thus, he could not get the sides to be different lengths, which he thought was necessary for a rectangle). Only ER conceptualized the movement regularity with complete precision by expressing it in terms of a formal geometric property.

From the van Hiele perspective, one might say that MT's reasoning on this task was at Level 0, JD's at Level 1, and ER's at Level 2. From the structuring perspective, MT and JD had constructed spatial structurings for the Square Maker, with JD's more sophisticated than MT's. ER, in contrast, had constructed a property-based geometric structuring for the Square Maker by applying a previously interiorized concept of equal sides to a spatial structuring similar to that described by JD. Finally, the abstractions made by MT and JD seemed completely situated in the DGE micro-world, whereas that made by ER was not.

Although the above episode illustrates that different levels of reasoning that students can achieve on a task in the Shape Maker environment, it does not show how a student progresses from one level to the next. The following episodes provide insight into this process.

Episode 2. NL was using the seven quadrilateral Shape Makers to make the design shown in Figure 19.8 while a researcher was observing and asking questions.

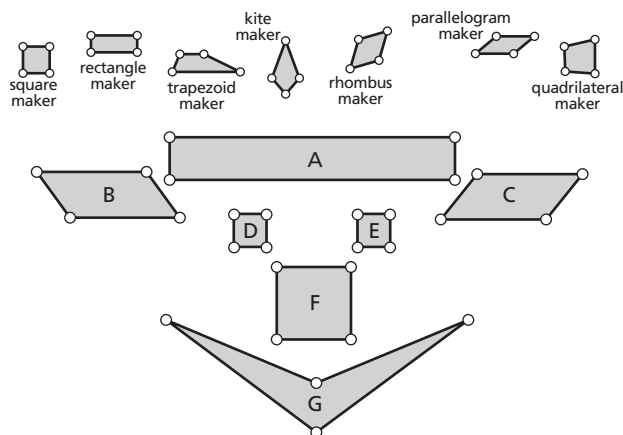


Figure 19.8 Make the design consisting of Shapes A–G with the 7 Shape Makers (Battista, 1998a).

NL: The Rhombus Maker on [Shape] B. It doesn't work. I think I might have to change the Rhombus Maker to [Shape] C.

Res: Why C?

NL: The Rhombus Maker is like leaning to the right. On B, the shape is leaning to the left. I couldn't get the Rhombus Maker to lean to the left, and C leans to the right so I'm going to try it. [After her initial attempts to get the Rhombus Maker to fit exactly on Shape C] I don't think that is going to work.

Res: Why are you thinking that?

NL: When I try to fit it on the shape and I try to make it bigger or smaller, the whole thing

moves. It will never get exactly the right size. [Manipulating the Rhombus Maker] Let's see if I can make the square with this. Here's a square. I guess it could maybe be a square. But I'm not sure if this is exactly a square. It's sort of leaning. The lines are a little diagonal. [Continuing to manipulate the Rhombus Maker] Yeah, I think this is a square maybe. . . .

Res: You said the Rhombus Maker could make the same shape as Shape C, what do you mean by that?

NL: It could make the same shape. It could make this shape, the one with 2 diagonal sides and 2 straight sides that are parallel. It could have been almost that shape and it got so close I thought it was that shape.

NL: [Continuing to manipulate the Rhombus Maker] Oh, I see why it didn't work, because the 4 sides are even and this [Shape C] is more of a rectangle.

Res: How did you just come to that?

NL: All you can do is just move it from side to side and up. But you can't get it to make a rectangle. When you move it this way it is a square and you can't move it up to make a rectangle. And when you move this, it just gets a bigger square.

Res: So what made you just notice that?

NL: Well I was just thinking about it. If it [the Rhombus Maker] was the same shape, then there is no reason it couldn't fit into C. But I saw when I was playing with it to see how you could move it and things like that, that whenever I made it bigger or smaller, it was always like a square, but sometimes it would be leaning up, but the sides are always equal.

This episode clearly shows how a student's manipulation of a Shape Maker and reflection on that manipulation can enable the student to move from thinking holistically to thinking about interrelationships between a shape's parts, that is, about its geometric properties. Indeed, NL began the episode thinking about the Rhombus Maker and shapes holistically and vaguely, saying that she was trying to make the Rhombus Maker "lean to the right" and get "bigger or smaller," and that "the whole thing moves." The fact that NL could not make the nonequilateral parallelogram with the Rhombus Maker evoked a perturbation that caused her to reevaluate the spatial structuring contained in her mental model of the Rhombus Maker. Originally, because her model did not include the constraint "all sides equal," her mental simulations of

changing the shape of the Rhombus Maker included transforming it into nonequilateral parallelograms. Her subsequent attempts to make a nonequilateral parallelogram with the actual Rhombus Maker tested her model, showing her that it was not viable. As she continued to analyze why the Rhombus Maker would not make the parallelogram—why it would not elongate—her attention shifted to its side lengths. This new focus of attention enabled her to abstract the regularity that all the Rhombus Makers' sides were the same length. As she incorporated this abstraction into her mental model for the Rhombus Maker, she was able to infer that the Rhombus Maker could not make Shape C.

This activity encouraged and enabled NL to progress from a spatial structuring that was incomplete, imprecise, and did not explain the Rhombus Maker's movement (van Hiele Levels 0 and 1) to a property-based geometric structuring that did explain that movement (van Hiele Level 2). A combination of two factors may have supported NL's progress. First, during the episode, NL abstracted the "all sides equal" property to a sufficiently high (interiorized) level that enabled her to apply it in various situations. Previous episodes indicated that NL had already concluded that squares have all sides the same length; in Episode 2 she raised this notion to a higher level of abstraction. Second, it is highly likely that NL's conclusions about the Rhombus Maker came about partly because she had previously made a square with it; she viewed the Rhombus Maker as a transformed square. I conjecture that NL's transformation of the Rhombus Maker into a square, a shape that she conceptualized as having all sides equal in length, made a critical connection that eventually enabled her to transfer the concept of equal side lengths to the new shape of Rhombus Maker (via reaching a higher level of abstraction). In a subsequent section, I will discuss how the transformation facility of the *Shape Makers* is a major component of their instructional efficacy.

Episode 2 also illustrates how students using the *Shape Makers* move toward more sophisticated, property-based conceptions of shapes because of the inherent power these conceptions give to their analyses of spatial phenomena. In the current situation, NL developed a property-based conception of the Rhombus Maker because it enabled her to understand why the Rhombus Maker could not make Shape C—something that truly puzzled her. Only when NL conceptualized the movement of the Rhombus Maker in terms of a geometric structuring did she feel that she really understood its movement. NL acquired this property-based conception not because someone told her to learn it, but because it helped her achieve a goal that she was trying

to achieve—understanding why the Rhombus Maker would not make Shapes B or C. (This was a personal goal for NL not only because the problem interested her but because she was a participant in an inquiry-based classroom culture and curriculum.)

Finally, this episode illustrates how geometry learning with the *Shape Makers* is much richer and more powerful than that which occurs in many traditional curricula. First, because NL's learning of properties was purposeful—connected at the outset with attaining a personal goal—it is highly likely that henceforth she would see that knowledge as being applicable and useful. (In fact, NL used this and other properties with increasing regularity in her subsequent work.) Second, because NL constructed her property-based reasoning from her already-existing cognitive structures, the newly constructed knowledge was "well-connected" in the sense that it was firmly anchored in her knowledge web, making it more likely not only to be applied in problem solving but used in further acts of knowledge construction. Third, and finally, it is highly likely that this episode increased NL's overall appreciation for the power of formal geometric reasoning in understanding the environment, again making it more likely that, in the future, NL would seek to use geometry in understanding the world.

Episode 3. The task was to determine which of Shapes 1–7 could be made by the Rectangle Maker (explaining and justifying each conclusion; see Figure 19.9.) Fifth graders M and T predicted that the Rectangle Maker could make shapes 1–3, but not 4–6. They are now checking and discussing their results. (The entire episode took about 25 minutes.)

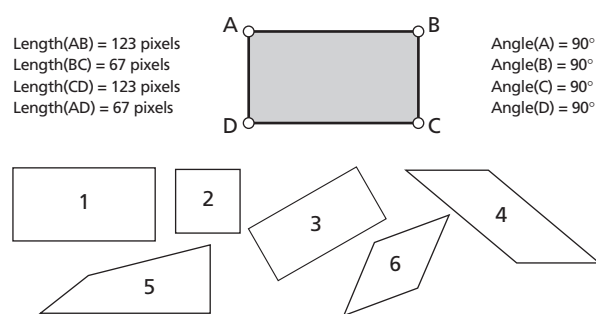


Figure 19.9 Which of Shapes 1–7 can be made by the Rectangle Maker (Battista, 1998a)?

After using the Rectangle Maker to check Shapes 1–3, M and T move on to Shape 4.

T: I'm positive it can't do this one.

M: It [the Rectangle Maker] has no slants. We had enough experience with Number 3, that it can't make a slant.

- T:** Yes it can, it has a slant in the other one [Shape 3]. It [the Rectangle Maker] has a slant right now. . . .
- Tchr:** What do you mean by slant?
- M:** Like this. See how this is shaped like a parallelogram [motioning along the perimeter of Shape 4].
- T:** This is in a slant right now [points at the Rectangle Maker, which is rotated from the horizontal]. [Note that *slant* means nonperpendicular sides for M, but rotated from the horizontal for T.] . . .
- M:** It can't make that kind of a shape [pointing to Shape 4]. . . . It can't make something that has a slant at the top and stuff.
- T:** Do you mean it has to have a *straight* line right here [pointing to Shape 6], like coming across? [T now uses *straight* to mean horizontal.]
- M:** I know they are *straight*, but they are at a slant and it [the Rectangle Maker] always has lines that aren't at a slant. . . .
- Tchr:** [After several failed attempts to unobtrusively get students to reconceptualize the *slant* conception in terms of angles] Keep on really looking at what makes these different. And maybe some of the information on the screen will help you. Watch the numbers up there and see if that will help you. . . .
- T:** [Manipulating the Rectangle Maker after the teacher leaves] Oh, this [the Rectangle Maker] always has to be a 90° angle. And that one [Shape 4] does not have 90° angles. And so this one [Shape 3] has to have a 90° angle, because we made this with that [Rectangle Maker]. So there is one thing different. A 90° angle is a right angle, and this [Shape 4] does not have any right angles.

M and T were trying to find a way to conceptualize and describe the spatial relationship that sophisticated users of geometry would describe as perpendicular. In so doing, and typical of students at van Hiele Level 1, they developed several vague and incomplete spatial structurings, all of which were inadequate. They used familiar terminology and concepts like *slanted* and *straight* that inadequately described the idea with which they were grappling. Progress was finally made, after the teacher left the boys, when T manipulated the Rectangle Maker and focused on its measurements. Through this manipulation, he discovered

and abstracted that the Rectangle Maker always has four 90° or right angles. Furthermore, he abstracted this property sufficiently so that he was able to use it to analyze the differences in Shapes 3 and 4. He subsequently conceptualized that the Rectangle Maker could not make Shapes 4, 5, and 6 because they do not have four right angles. In fact, by the end of the class period, the boys saw that the spatial relationship they were attending to could be described in terms of the formal mathematical concept of right angle. They had constructed a geometric structuring, a van Hiele Level 2 analysis, that enabled them to solve the problem they had embraced.

This episode illustrates how difficult it can be for students to reformulate unrefined spatial structurings of Shape Maker movements in terms of traditional geometric concepts.²³ Only after much guidance, reflection, and experimentation can students construct formal, property-based geometric conceptualizations of Shape Maker movement constraints. However, once students meaningfully move from their initial vague spatial structurings to appropriate geometric structurings, they embrace the latter because of their inherent power.

Similar Results

It is instructive to compare *Shape Makers* results to results from other environments. First, note that M and T's reasoning was similar to, but more advanced, than that of JN in Logo. All three students were struggling with conceptualizing the notion of perpendicularity. In fact, Berthelot and Salin (1998) found that only 50% of 10–11-year-olds used right angles to locate the legs of a rectangular bench if moved. The other students explicitly used only information about length. Also, consistent with the Battista results, Sutherland, Godwin, Olivero, and Peel (2002) found that use of DGE constructions similar to the *Shape Makers* helped elementary students learn about properties of shapes.

DGEs at the Middle and High Levels

In examining instructional DGE use by older students, besides having students investigate geometric shapes by manipulating them on the computer screen, two additional major emphases occur. First, students are asked to use DGE tools to construct draggable figures. Second, students are asked to prove conjectures that arise out of exploration.

²³ The difficulty students have with this process points out just how deficient traditional cursory memorization-based approaches to this topic are. It takes a great deal of thought and appropriate experience for students to use the formal geometric conception-based system with genuine understanding.

To promote understanding of quadrilaterals in junior high students in the UK, Jones (2000) gave a series of DGE tasks in which students were asked to construct *draggable* figures. For instance, students were asked to construct a draggable square and explain why the shape is a square. As students progressed through the instructional sequence, Jones found that, initially, students emphasized description rather than explanation, relied on perception rather than mathematical reasoning, and lacked precise mathematical language. At an interim period in the instruction, students' explanations became more mathematically precise but were intertwined with the operation of DGE software. At the end of the instructional unit, students' explanations were entirely mathematical.

To illustrate, when students were asked to explain why all squares are rectangles, early in instruction one pair wrote: "You can make a rectangle into a square by dragging one side shorter . . . until the sides become equal" (Jones, 2000, p. 76). Later in instruction, students were asked to construct a trapezium that could be modified to make a parallelogram and explain why all parallelograms are trapeziums. The same pair of students wrote, "Trapeziums have one set of parallel lines and parallelograms have two sets of parallel lines" (Jones, 2000, p. 76). The apparent argument in the first example is couched in terms of the DGE—"via dragging, you can see that a square is a special type of rectangle." The second argument is apparently more traditional property-based—"parallelograms have all the properties needed for trapeziums." What we do not know, however, is how students made the transition from DGE-based to formal-geometric conceptualizations. More research is needed that investigates this transition.

In another study, conducted by Noss and Hoyles (1996), two female students, C and M (age 14), were given two flags in Cabri—one the reflection of the other—and asked to find the mirror line. After a short period of dragging points on the flags and noticing the effects on the images, C and M visually located the position of the mirror line. In an effort to be more precise about this location, C had the idea of dragging corresponding points on the flags—preimage and reflection image—so that they met. The girls claimed that such points lie on the mirror line. Subsequently, C and M used two pairs of these overlapping corresponding points to draw the mirror line. With the mirror line drawn, the girls then focused on pairs of corresponding points, which led them to see that the mirror line could be constructed by joining the midpoints of segments between corresponding points.

Similar to Emily in Logo, work with Cabri made C and M's conceptualization of reflections more ana-

lytic. They reformulated their idea of reflections when they saw that the mirror line was formed by joining midpoints of corresponding points. The "analytic" part of this new conceptualization is the geometric description of the reflection concept in terms of constituent parts (corresponding points, segments, midpoints). Also, C and M, like Emily, did not arrive at the traditional view of reflections—they made no mention of orthogonality, so their midpoints notion was incomplete. It would have been useful to see how further instruction could have extended the girls' notion to include orthogonality. It would also have been enlightening to better understand how C and M made transitions through their successive conceptualizations. For instance, we might conjecture that as C and M focused on a pair of corresponding points while manipulating the flags with the mirror line drawn, the two points formed a perceptual unit, bisected by the mirror line, which may have activated an interiorized version of the concept of segment midpoint (either formal, or situated within Cabri).

Artifacts of DGEs versus Geometric Properties

A fundamental open question is if and how students "distinguish fundamental characteristics of geometry from features that are the result of the particular design of the DGE" (Jones, 2000, p. 59). For instance, similar to Holzl, Healy, Hoyles, and Noss (1994), Jones (2000) argued that students need to understand functional DGE dependencies, such as in Cabri, although basic points, points on objects, and points of intersection look identical on the screen, basic points and points on objects are draggable, whereas intersection points are not. Apparently, students have difficulty developing such understanding (Holzl et al., 1994; Jones, 2000). In fact, students often develop non-mathematical conceptualizations of the draggability of figures. For example, asked why her constructed figure could not be "messed up," one student replied, "They stay together because. . . It just glued them together" (Jones, 2000, p. 71). Thus, as Holzl et al., Jones, and others (Goldenberg & Cuoco, 1998) have argued, especially in curricula that make significant use of student exploration, research must investigate how students interpret the construction and behavior of DGE drawings. In essence, students working in DGEs need to see beyond the operational syntax of the environment; the DGE representations must become "transparent." But researchers do not yet know how this happens.

What Happens When Constructing a Draggable Dynamic Figure?

Many instructional treatments involving DGEs introduce "a specific criterion of validation for the solu-

tion of a construction problem: a solution is *valid* if and only if it is *not* possible to ‘mess it up’ by dragging” (Jones, 2000, p. 58). So let us more carefully examine the process of constructing a draggable figure. For instance, two (Year 8) students were attempting to make a draggable rectangle in Cabri (Holzl et al., 1994). After drawing segment AB (see Figure 19.10), they constructed a perpendicular to this segment, passing through B, put point C on the perpendicular and constructed another perpendicular through C, put a point D on that perpendicular and constructed another perpendicular. The students joined points A, B, C, and D and hid the other lines, forming a trapezoid, then dragged A so that the trapezoid looked like a rectangle. The students realized that they had not created a draggable rectangle, but got bogged down in Cabri operations when trying to fix their error. For instance, they attempted to change the nature of point A to be a “point on object [DX].” Finally, when the students were told that this could not be done, they completed their rectangle by creating an intersection point on DX and AB.

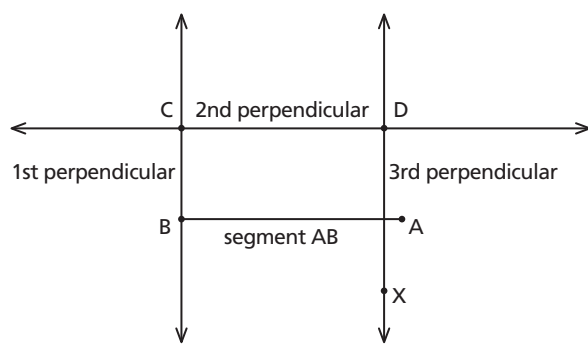


Figure 19.10 Students attempting to make a draggable rectangle in Cabri.

A number of questions must be asked about episodes such as this, which are common in students’ DGE constructions. First, Holzl et al. claimed that the students “were clearly aware of the properties of rectangles and the need to construct perpendicular lines to obtain right angles” (1994, p. 9). But exactly which properties of rectangles did the students know? For instance, was their construction a direct result of knowing that the sides of a rectangle are perpendicular? Or, did they know that rectangles have right angles, and connect the concepts of right angle and perpendicularity to make their construction (perhaps making this connection for the first time)? Also, there was no explicit evidence in their construction that they knew that opposite sides were equal and parallel. Of course, they may have known these properties, but because perpendicularity is sufficient to complete the Cabri

construction, they did not mention them. Second, if the students already knew the properties of perpendicularity, parallelism, and congruence of opposite sides, what geometry did they learn by working on this problem in Cabri? Perhaps no *new* knowledge was acquired, but instead, the students’ knowledge and reasoning were deepened and enriched. The process of applying known concepts in a new situation requires interiorization of the concepts, making them more powerful. Or perhaps connections between properties were newly constructed or extended. Also, if the students noticed parallelism and equality of opposite sides in the constructed figure, they might have seen that perpendicularity implies parallelism and equality of sides (especially if a teacher helped them notice it), moving them toward or reinforcing van Hiele Level 3 thinking. On the flip side of the argument, however, one might ask what the students really understood about the perpendicularity of sides. For instance, the students could have constructed the rectangle visually, using perpendicularity as a procedural tool for making lines meet in a certain way. That is, they knew perpendicularity visually as making things “square,” not a geometrically as involving right angles.

The claim by Holzl et al. (1994) that students working in DGEs “ran up against some fundamental aspects of *its [Cabri] geometry*” (p. 11) raises additional issues. First, does Cabri really have its own geometry? If so, what is it and how is it related to traditional mathematical geometry? For instance, as has already been mentioned, unlike mathematical geometry, Cabri “geometry” has different kinds of points that behave differently. As another example, the idea of “point on object,” as implemented in a DGE, may promote student conceptualizations that are very different from the corresponding mathematical concept. In Birkoff’s formulation of geometry (Moise, 1963), the statement, “let X be a point on segment AB” means that X is any of the points that are elements of the point-set that is segment AB. However, the way the point concept is implemented in a DGE might lead some students to conceptualize a point on a segment as an object “on top of” or separate from the segment, not as part of the segment. In this case, the DGE concept and mathematical concept would be inconsistent. In contrast, in Hilbert’s formulation of geometry (Wallace & West, 1992), *point*, *line*, and *on* are undefined terms, and there is no requirement that individual points be elements of sets of points called lines. In this case, there seems to be no inconsistency between the DGE and mathematical concepts. This recalls the question, What geometric conceptualizations do students develop in DGEs, and how are these conceptualizations related to various formal mathematical concepts?

Moving Toward Justification

At the secondary level, many researchers see the primary purpose for DGE construction activities as a bridge to justification and proof:

The [instructional] approach to the construction problem within the Cabri environment is expected to introduce first, the need to justify a solution, then the need to negotiate the adequacy of this justification. The dragging test has a fundamental role in this process of justification. The fact that students can see for themselves when and which geometrical properties remain constant under dragging can help them to understand when they have produced something that merits justification. . . . In the subsequent mathematical discussions, it will become necessary to negotiate and introduce explicit criteria for the acceptability of the justification itself. (Mariotti, 2001, p. 262) [Presumably “acceptable” means formal proof.]

Consistent with this claim, but counter to suggestions that students who have explored a conjecture extensively in computer environments often feel no need for further justification, de Villiers (1998) reported that he has been able to encourage deductive thought by asking why DGE results are true. A somewhat different perspective was taken by Hershkowitz et al. (2002) who argued that

the dragging operation on a geometrical object enables students to apprehend a whole class of shapes in which the conjectured attribute is invariant, *and hence to convince themselves of its truth* . . . The role of proof is then to provide the means to state the conjecture as a theorem . . . *to explain why it is true*, and to enable further generalizations. (p. 676)

So formal proof (a) helps students to symbolically and precisely encapsulate their empirical discoveries and (b) offers corroboration for and explanation of the validity of the discovery.

However, several questions can be raised about these claims. First, does use of DGEs motivate and promote the learning of proof better than traditional instructional contexts? Second, if students already believe that a proposition is true because of DGE empirical evidence, what precisely is added to *students’* reasoning and learning by formally proving it? Although researchers can see much that might be added, what actually happens with students? This is an empirical question. Third, exactly what are the processes by which DGEs promote the development of proof?

A number of researchers have addressed the latter question. For instance, Mariotti (2001) studied students working on DGE instructional activities as part of the regular curriculum. In the first problem, ninth-grade students were asked to construct the bisector of an angle then describe and justify their solutions. In Alex and Gio’s first solution attempt (Figure 19.11a), they wrote,

We transferred a segment AB . . . [to] segment r1 ($AB = AC$) [despite what they wrote, the students made the two segments AB and AC equal “by eye,” so the transference was not draggable]; we drew two circles . . . center in C and point A and center in B and point A. We joined A and [circle intersection] D . . . [after testing their construction by dragging] Failed! (p. 265)

For their second attempt (Figure 19.11b), Alex and Gio wrote,

We drew a circle . . . [that] gave us the segments AB and AC belonging to r1 and r2, which are equal because they are radii of the same circle. We drew two circles (centre B and C point A); using the intersection . . . of the two circles, we found the point D that we joined with A, determining the angle bisector. (p. 265, 266)

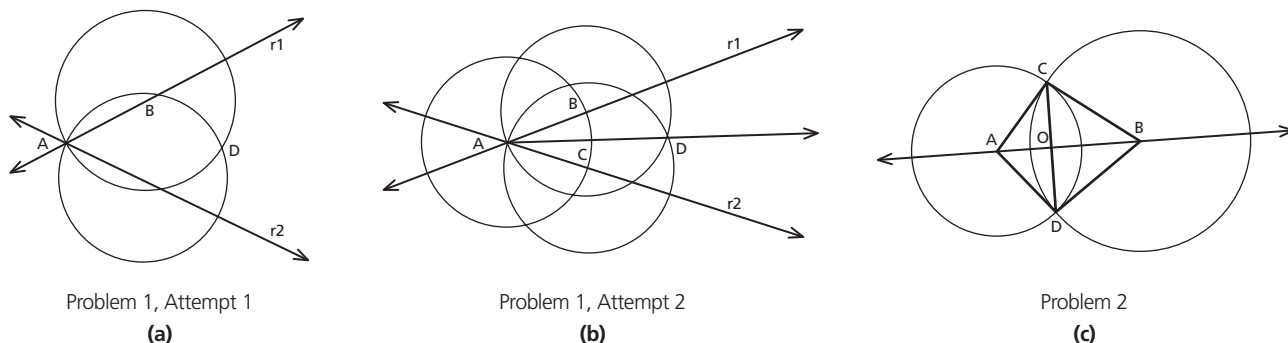


Figure 19.11 Alex and Gio’s attempts to construct the bisector of an angle (Mariotti, 2001).

Note that, other than the one statement, “which are equal because they are radii of the same circle,” the justification provided by the students is an *explanation* of the DGE constructions that they used. Most of what the students wrote is a description of process, not a justification, or even an explanation for why they did what they did.

On a later problem—constructing a perpendicular to a given line through a given point—Alex and Gio drew two circles having centers on the line and passing through the given point C (Figure 19.11c). Their perpendicular was the line through the intersections of the two circles. To prove the validity of their construction, they used the facts that AD and AC are radii of the same circle as well as that BD and BC are radii of the same circle (and the triangle congruence theorems) to show that angles AOC and AOD are congruent and thus right angles. In this case, the students seemed to link the structure of their proof to their DGE construction; in their proof, they took as given that which was produced by their constructions. So here is an example of using a DGE construction to formulate a proof.

It seems that in Alex and Gio’s case, DGE was being used as a problem context—it provided exercises for enriching the proof skills of students who had already been taught formal proof. It helped students learn to use and appreciate geometric proof. It was not being used to evolve students’ informal justifications into formal ones (cf., Borrow, 2000).

However, several issues arise in considering Alex and Gio’s work. First, how did students move from justifying only a small part of their construction in the first problem to the much more sophisticated solution they provided in the second problem? (In order to prove these ideas, students must have already learned about formal proof.) Second, precisely how were students given, and how did they make sense of, a formal theoretical framework for proof? For instance, was the need to prove the validity of constructions generated by students—did they personally feel a need to remove uncertainty—or was it socially motivated by either the classroom culture or demands of the teacher?

A third issue involves the nature of students’ geometric reasoning and how it was supported and furthered by the DGE. For example, what conceptualizations and reasoning (about geometry and Cabri) led students to use equal radii circles in constructing the angle bisector? Were circles seen procedurally as a way of laying off equal segments, or as loci of points in the plane equidistant from a given point? There seems to be evidence of both. Was the students’ approach visual or analytic? That is, did the students use the circles analytically as a way of making point D equi-

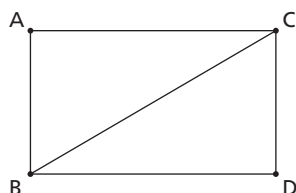
distant from the sides of the given angle; or was their approach strictly visual—they simply tried to find a way to put a point D “in the middle” of the angle? Additional research needs to delve more deeply into students’ thinking processes.

In contrast to the Alex and Gio episodes in which constructing draggable figures seemed to support students’ reasoning, Mariotti (2001) also presented data suggesting a divergence between students’ construction of *draggable* figures and their proofs. For instance, Cathy was unable to make a draggable construction for a perpendicular to line through a point P not on the line. However, she was still able to state and correctly prove how to make such a construction. She proved that if one chooses points S and Q on the line so that $SP = QP$, then the bisector of angle SPQ is perpendicular to line SQ. She derived this proposition by visually manipulating P and observing angle measure changes—her premises, however, were not “constructed” into a draggable drawing. Thus, students can construct valid theorems and proofs even when they cannot construct draggable diagrams. Although this finding is not consistent with claims of the central role that draggability plays in students’ work with proof, it is consistent with the view that DGEs provide an exploratory medium that makes students’ geometric ideas more analytically manipulable and thus more accessible to reflection and refinement.

Research by Healy and Hoyles (2001) provides additional insights into some of these issues. They conducted a study with above average 14–15-year-olds in England who were studying the national curriculum, which emphasizes proof in a range of contexts, not just geometry. However, despite this emphasis, the researchers reported that “only a few students [out of close to 2500 in a nationwide survey] could actually construct a valid deductive argument” (p. 239). In the experimental treatment, which lasted 6 to 8 weeks, after using Cabri to explore congruent triangles and being introduced to formal proof, students were given tasks that the researchers hoped would integrate DGE and proof experiences. (Before participating in this study, none of the students had used Cabri nor had they created sequences of logically justified geometrical statements.) For example, in one task, students were asked to (a) make and describe a Cabri construction of a rectangle, (b) identify the properties of the rectangle that they considered given in their construction, (c) manipulate their constructed figure to discover new properties, and (d) prove one of their discovered properties.

In response to this task, one student wrote the following (p. 243).

(a) Constructed line between 2 points and put perpendicular line through it. Created a parallel line to the perpendicular line and then a parallel line to the first line. (b) All angles equal 90° . Opposite sides parallel. (c) Opposite sides equal. (d) Used a triangle congruence proof to prove that opposite sides are equal. [For the student's proof that the properties in (b) imply the property in (c), see Figure 19.12.]



All angles equal 90°
 $\angle ABD = \angle ACD$. Given.
 $\angle ACB = \angle CBD$. Alternate angles equal.
 $\angle ABC = \angle DCB$. Alternate angles equal.
 $\triangle ABC \approx \triangle BCD$. All angles equal.
 BC is common to both triangles.
 $\therefore$ Both triangles congruent $\rightarrow AB = CD$.

Figure 19.12 Student's proof.

This student was able to construct a draggable rectangle and to use a triangle congruence proof to justify his statement that the opposite sides of a rectangle are congruent. However, consider this student's creation of a draggable rectangle in more detail. After creating the first right angle by using a perpendicular, it is unclear how he conceptualized what he was doing. Did he create parallels because he wanted to give the figure the "opposite sides parallel" property about which he wrote? How did he know, or did he discover, that this would produce right angles (which he did not prove, but assumed in his proof)? Was there any explicit connection between the student's DGE construction and his proof (for instance, did his use of parallelism in constructing the rectangle cause him to focus on parallelism in his proof)?

In the final activity of the Healy and Hoyles teaching experiment, students were asked to construct a quadrilateral in which the angle bisectors of two adjacent angles intersect at right angles. They were to discover, then prove, other properties of this quadrilateral. Tim and Richard created quadrilateral ABCD, then the bisectors of angles ABC and BCD (see Figure 19.13). They measured the angle at the intersection of the two bisectors and dragged the vertices of the quadrilateral until this angle measured 90° . They then noticed that BA was parallel to CD and conjectured that whenever the two angle bisectors were at right angles, BA would be parallel to CD. They dragged the

vertices for further empirical validation of their conjecture. Then, with the angle bisectors intersecting at right angles, the boys used Cabri's "check-property tool" to test whether BA and CD were parallel. The boys were "disappointed and puzzled that the property that the two lines were parallel was declared *not* to be true in general" (p. 244). However, because in the counterexamples presented by Cabri's check-property tool, the angle bisectors did *not* intersect at right angles, the boys decided that the parallel property had been declared invalid because the condition of 90° between the angle bisectors had not been retained by the construction.

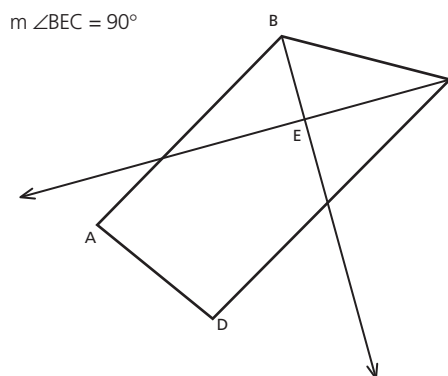


Figure 19.13 Tim and Richard's quadrilateral with angle bisectors of 2 adjacent angles intersecting at right angles.

Note that the boys' realization required sophisticated knowledge about the nature of Cabri constructions and, more importantly, the function of the check-property tool. This latter knowledge is knowledge about Cabri, not geometry. So they had to possess a fairly sophisticated understanding of the nongeometric functioning of Cabri to deal with the instructional task presented to them. Students who do not understand this technology sufficiently would have their geometric investigation thwarted.

Tim and Richard continued investigating their conjecture by constructing a line parallel to CD passing through B. They found that whenever point A was dragged onto this line, the two angle bisectors intersected at right angles, but if A was not on this line, the angle bisectors were no longer perpendicular. They formulated the conjecture, "Whenever the lines are parallel the angle bisectors are perpendicular." In searching for a proof for their conjecture, the boys returned to manipulating the quadrilateral, with A always on the line parallel to segment CD. They measured angles and looked for angles that remained equal when dragging vertices. The boys noticed that the alternate interior angles where the bisector of angle ABC intersected segments BA and CD were always

“the same,” then realized that proof of this equality would show that the two line segments are parallel.

However, before Tim and Richard moved from their Cabri construction to a strictly formal analysis of the situation, they used their construction to further explore their conjectures. But this was complicated by the fact that the angle bisectors did not always intersect the sides of the quadrilateral, so the relevant angles could not always be measured. The issue was resolved when the teacher suggested that they construct lines through the sides of the quadrilateral, which allowed the boys to measure the relevant angles in all draggable positions. At this point, the boys stopped manipulating the figure. The researchers claimed that the boys had “used the figure they had generated with its specific measurements as a generic example, to help them to formulate a general argument” (Healy & Hoyles, 2001, p. 246).

The DGE construction played two important roles in Tim and Richard’s work. First, it provided them with a representation that allowed them to investigate the situation empirically so that they could formulate a valid conjecture about a geometric property. Second, the boys’ manipulations of their DGE representation, including the accurate measurements of angles, helped guide their construction of a proof. This measurement process was not only more accurate, it was more general than that which could have been conducted with paper and pencil because it was maintained during dragging. Thus, in this case, for these students, Cabri seemed to form a bridge between empirical exploration and proof. Dynamic geometry helped these students find reasons why their empirical discovery was true. There was no evidence that the DGE discouraged proof activity.

However, on this same task, many students failed to see the parallel relationship between the sides of the resulting quadrilateral. Also, making a draggable figure with the “givens” was not the route successful students took. Like the two students above, all students who were successful on this problem constructed a general quadrilateral with adjacent bisectors and manipulated it to find when the bisectors were parallel. In fact, students who attempted to construct a quadrilateral with the given bisector-perpendicularity property—as the researchers had intended—encountered two problems. The first was reasoning that the given properties were satisfied by a particular quadrilateral, like a parallelogram. Students using this reasoning either became confused about which properties were given and which were to be deduced or did not see that the solution they derived for a special case was incomplete. The second difficulty was that constructing a general quadrilateral that satisfied the given condi-

tions proved too difficult for students. Because they could not use the Cabri tools available to them to construct the quadrilateral, they became frustrated and failed to solve the problem.

The experimenters’ solution to the second difficulty was that a more flexible, programmable environment is needed.

From all our observations of students’ interaction with Cabri, we have become increasingly aware that the mediation of students’ activities by the software is not necessarily positive for their engagement and for their learning. . . . As we have shown in the work of our less successful students, learners can . . . find themselves in a position where they are unable to use the tools they have in mind, even if they are convinced that their use would make sense mathematically. (Healy & Hoyles, 2001, p. 252)

One can draw several conclusions from the Healy and Hoyles research, though not necessarily those drawn by the authors. First, the DGE apparently helped some students understand and solve posed problems. (Recall, however, that the students who did this seemed to have well-developed proof skills and understanding of associated geometric ideas.) Second, the DGE evidently helped some students develop appropriate proofs.

Just as evident is that other students were stymied in their geometric problem solving. However, it can be argued that the difficulty lies not with the DGE, as claimed by Healy and Hoyles, but with the design of the instructional tasks. For instance, in the last activity, the problem was not chosen so that students could use Cabri to construct, in a reasonably easy way, a draggable figure satisfying the problem hypotheses, even though the researchers, and many students, expected students to do so. If the point of the activity was for students to discover and explain a geometric relationship, there was no reason to expect them to create, *a priori*, a draggable figure that maintained that relationship. It is much more reasonable to expect students to construct a DGE figure that allowed them to reflect on this relationship (as did Tim and Richard). Alternatively, why not give students a draggable figure with the desired properties and let them use it to explore the problem? Perhaps tasks that do not ask students to make their own constructions are better for some students at some times. In general, then, researchers should be asking what students learn by using specific DGE tools, and what knowledge—both geometric and technological—is required to productively use these tools.

Comparisons of student learning with various task designs could be very revealing. For instance, given

the problem of investigating quadrilaterals in which the angle bisectors of two adjacent angles intersect at right angles, researchers might compare the work of students in four different task environments: (a) no DGE, (b) making a draggable figure with the given property, (c) making a draggable figure like Tim and Richard's, and (d) using a preconstructed draggable figure with the given property. The research should compare not only student success rates, but their strategies, required starting knowledge for success, and what they learn from the experience. (Even though the Healy and Hoyles research shows that environment (b) is very difficult for students, it would still be interesting to better understand why it is so difficult.) Related to this suggested research, Arzarello, Michelletti, Olivero, Paola, and Gallirio (1998) attempted to analyze the types of DGE dragging explorations of students during different phases investigation of a geometric problem. Specifically, they investigated types of dragging during empirical activity aimed at trying to understand a problem or generate and explore a conjecture (their "ascending" phase), and activity aimed at building a deductive proof (their "descending" phase). They suggested that different kinds of dragging occur in different phases.

In another study of the use of DGE to support the development of students' proof skills, Marrades and Gutiérrez (2000) described differing levels of sophistication in students' discovery and justification of a theorem about the concurrence of perpendicular bisectors of the sides of a quadrilateral. They stated that "the process of getting conjectures was grounded on the observation of drawings and regularity in the measures of angles" (p. 105). Without prompts for proof, students seemed convinced by empirical validation actions of DGE. However, with teacher prompting, students seemed able to use DGE in concert with proof. When students arrived at a sequence of constructions that enabled them to conceptualize why the theorem was true and their dragging/measuring actions convinced them of the validity of their ideas, they were able to prove the theorem in a way consistent with their construction-based conceptualizations. Thus this research supports the claim that DGE exploration can provide insight and scaffolding for proof, as well as confidence to proceed with a given line of justification (as claimed by de Villiers, 1998, when he cited Polya, "When you have satisfied yourself that the theorem is true, you start proving it").

Unfortunately, many teachers apparently are not taking advantage of DGEs' capability to support the formulation of a proof: "When the students were asked to justify, the teachers did not mention the possibility of using Cabri to find a reason or to elaborate

a proof. It is as if there was no interaction between visualisation and proving" (Laborde, 2001, p. 303).

Final Comments on DGE

Several major issues and areas for further research have arisen from the above discussion.

Issue 1. There are two major theoretical perspectives concerning the use of DGEs to promote the development of students' geometric thinking. In the first perspective, dynamic figures, especially draggable ones, are seen as generating numerous examples. For example, Marrades and Gutiérrez claimed that the main advantage of DGEs "is that students can construct complex figures and can easily perform in real time a very wide range of transformations on those figures, so students have access to a variety of examples that can hardly be matched by non-computational or static computational environments" (2000, p. 95).

In the second theoretical perspective, draggable drawings are seen as interesting, manipulable, visual-mechanical objects that have movement constraints that can be conceptualized and analyzed geometrically (Battista, in press b). This approach is taken in Battista's *Shape Makers* environment and Laborde's "black-box" activities in which students are given a DGE figure without being told how it was constructed and are asked to construct a draggable figure that behaves in the same way (Laborde, 1998). In these draggable drawings, geometric properties are visually and mechanically manifested as constraints and allowances of movement and therefore not only can be seen, but can be felt. Thus, rather than examining *a set of figures* to conceptualize how they are the same, with *Shape Makers* or black-box constructions, students investigate how an interesting manipulable "geometric" object works (which can, with proper instructional guidance, support the construction of meaningful conceptualizations of geometric properties). Seemingly consistent with this view, some researchers argue that DGE should not be conceptualized as a device to merely "enable us to do what we already do faster, better or to a larger extent or degree," but as an environment requiring a fundamentally different cognitive view than has been taken in traditional approaches to geometry instruction (Dörfler, 1993, p. 161).

In considering these theoretical perspectives, which guide both research and curriculum development, several questions arise. First, it is important to test the theories and determine empirically how students conceptualize DGE draggable figures. Do students see the various configurations of a dynamic drawing as examples of a class of geometric shapes, or do they see a dynamic drawing as a geometrically in-

interesting manipulable object? How are students' views on this matter affected by instruction?

Second, because much of geometry is about categories of objects, do students who think about DGE figures as manipulable objects naturally move to thinking about classes of shapes? If not, how can such movement be engendered? Does students' reasoning about DGE manipulable objects transfer to thinking about geometric categories? Battista's *Shape Makers* research suggests that instruction can promote such transfer. Are DGE figures viewed differently by students who have already attained strong conceptualizations of the underlying geometric concepts before using a DGE, as opposed to students who initially construct meaning for these concepts in a DGE? That is, for an expert, a dynamic rectangle maker might indeed be a representation of the class of rectangles, whereas for the novice it might be simply an interesting dynamic object whose movement is constrained in particular ways.

Issue 2. How do students conceptualize particular DGE constructions? For instance, what is involved conceptually when a student learns to construct a parallel to a given line through a point not on that line? What does the construction mean to the student? Is it a procedure for constructing a visual configuration, or is it a representation of a geometric concept? How does using the construction change, supplement, or supplant a student's conceptualization of parallelism? That is, learning changes a student's network of cognitive structures—how does learning to use DGE construction tools change students' related conceptual networks? Conversely, what kinds of conceptual networks enable students to productively employ DGE construction tools? Is the nature of conceptualizations created via use of DGE construction tools the same as those learned through verbal definitions or drawing?

Issue 3. In general, much additional qualitative and quantitative research on the use of DGE in geometry learning is needed. Qualitative research is needed to investigate numerous unanswered questions about the nature of students' construction of geometric knowledge in DGE. Quantitative studies are needed to determine whether use of DGE produces greater learning than use of paper and pencil. However, geometry research, and especially its application to classroom practice, is best served by integrated combinations of qualitative and quantitative studies. For instance, as has been shown previously in this chapter, qualitative research results indicate that high quality geometry learning can occur with use of DGE. But quantitative research is needed to determine the generalizability of these results and whether use of DGE is "better" than use of paper-and-pencil techniques. In

contrast, when quantitative results indicate that one instructional treatment is more effective than another, qualitative methods should be used to determine why such results were obtained.

Issue 4: The allure of draggable figures. Investigating draggable figures in DGE seems to strongly attract the interest not only of students but of researchers in geometry learning. Why is investigating such figures so appealing? One suggestion for the appeal of draggable figures comes from the theory of psychological essentialism. This theory conjectures that people act as if things have essences or underlying natures that make them what they are, and that when an essence is unknown, it may motivate the search for new meanings (Gelman & Diesendruck, 1999). Thus, one reason that DGE draggable figures may be instructionally useful is that students naturally attempt to determine the essences of these objects—how do they move, why do they move the way they do.

Issue 5: Draggability and invariance. A number of researchers have claimed that one of the reasons that DGE draggable figures are useful for helping students develop understanding of geometric properties is that such properties are invariant under dragging movements. For instance, Laborde claimed that the spatial properties of draggable drawings "may emerge as an invariant in the movement whereas this might not be noticeable in one static drawing" (Laborde, 1998, p. 117). Similarly, Battista (in press b) suggested that properties of shapes are more noticeable as invariants of draggable movements than as commonalities across static examples.

Several clues in the psychological literature point toward a possible explanation of this phenomenon. First, the mind's attentional system is especially alert and sensitive to change (Ornstein, 1991). Thus, when viewing a figure as it is dragged, because the figure maintains its identity as a single entity, changes in its shape are naturally attended to. Second, "[the] search for constancy, the tendency toward certain invariants, constitutes a characteristic feature and immanent function of perception" (Cassirer, 1944, p. 21). Third, comparing a set of static figures and comparing various configurations of a dragged figure should both induce structure-mappings (Gentner & Medina, 1998) that can make properties noticeable. However, transforming draggable figures should facilitate this structural analysis because the continuity of the transformation establishes a structural-mapping correspondence between components of compared figures, whereas when comparing static figures, the perceiver must establish the correspondence (which may not be easy). Of course, if students do not attend to the entire dragging transformation, the original figure and

the figure that results from dragging may actually be viewed as two separate entities, losing this facilitative effect. Fourth, a considerable amount of research has shown that objects that move together are seen as part of the same object (e.g. Driver & Baylis, 1989; Regan & Hamstra, 1991). So, dragging motions might create “visual chunking” that facilitates the perception of visual regularities indicative of properties. For instance, because in the Parallelogram Maker the opposite sides move together in a way that maintains their parallelism and congruence, such movement might encourage the formation of two “opposite-sides-parallel” composites or chunks. These composites are a visual manifestation of the geometric properties of opposite sides parallel and congruent (although, in this form, the two properties might not be separable, which is consistent with the observation that children’s initial conceptualization of parallel often includes congruence). Thus these composites might form the spatial structuring needed later for students’ geometric structuring.

RESEARCH ON OTHER GEOMETRIC TOPICS AND NONCOMPUTER ENVIRONMENTS

The National Assessment of Education Progress (NAEP)

Traditionally, the NAEP has been a useful tool for getting an overall picture of U.S. students’ mathematics skills. To illustrate, recent administrations of the NAEP in mathematics have shown some improvement in geometry, but students’ performance in many areas of geometry and spatial sense is still low (Kloosterman et al., in press; Sowder, Wearne, Martin, & Strutchens, 2004). However, it is difficult to reliably integrate recent NAEP results into the research corpus on geometry learning because of two major problems with the NAEP program. First, in recent years, NAEP assessment tasks have not been released to researchers for analysis, making the results almost impossible to interpret seriously. For instance, it is impossible to make much sense out of a vague NAEP item description such as, “Determine the shape formed under certain conditions.”

Second, many NAEP items and scoring rubrics are substantively disconnected from current research on geometry learning. For instance, consider the two tasks shown in Figure 19.14.

Both tasks are intended to assess students’ ability to use properties to compare shapes, a critical goal in

1. In what ways are the figures alike?



2. Explain how shape N is different from shapes P and Q.

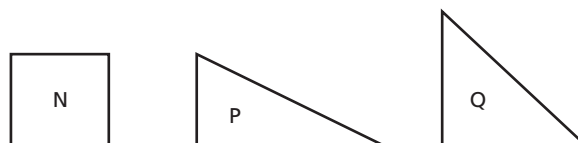


Figure 19.14 NAEP tasks.

the geometry strand that is consistent with van Hiele-based research. However, the property focused on in Task 2 is not, as has been previously discussed, a good indicator of van Hiele Level 2; number of sides does not involve relationships between parts of shapes. In contrast, the properties that discriminate the shapes in Task 1 focus on relationships between parts of shapes and thus are good indicators of van Hiele Level 2. Unfortunately, and inconsistent with research on geometry learning, NAEP administered the low-level Task 2 across the grades (4, 8, 12) and the richer Task 1 only at Grade 4 where one would expect to see only minimal development. Furthermore, the scoring guide for Task 1 is not based on a research-derived classification of justifications. For instance, the scoring guide weighs the following properties equally: having 4 sides or angles, having two sets of equal side lengths, having 4 right angles, having equal heights, having equal areas, and “one is more slanted.” So informal properties, properties dealing with numbers of components, and relational properties are given equal weights. But research suggests that these properties vary considerably in sophistication. Thus, because the NAEP items and scoring are only loosely connected with research on students’ learning of geometry, they provide inadequate information about students’ geometry learning for teaching as well as for curriculum evaluation and development.

Proof and Justification in Geometry²⁴

In 1992, Clements and Battista (1992) concluded that, in geometry, students are extremely unsuccessful with formal proof and that they are deficient in their ability to establish mathematical truth. More recently, Hershkowitz et al. (2002) drew a similar con-

²⁴ See (Harel & Sowder, 1998) for a discussion of the proof schemes students use throughout mathematics, not just in geometry.

clusion: “The teaching of mathematical proof appears to be a failure in almost all countries” (p. 675). In a large study in the United Kingdom, Healy and Hoyles (1998) reported that (a) high-achieving Year 10 students showed poor performance in constructing proofs, with empirical verification the most popular form of justification; (b) students’ proof performance was considerably better in algebra than in geometry; (c) students recognized that once a statement has been proved, no further work is necessary to check its validity; and (d) students were better at recognizing a valid mathematical argument than constructing one themselves. Hoyles and Jones (1998) reported that (a) students do not distinguish between empirical and deductive arguments and prefer empirical arguments; (b) for many students, deductive proof only *adds evidence* for validity; and (c) most students regard proof as an irrelevant add-on and do not understand its purpose. Similarly, Hershkowitz et al. (2002) stated that students rarely see the point of proving. Battista and Clements (1995) argued that although proof is critically important to mathematicians because it enables them to establish the validity of their mathematical thought, students do not conceive of proof as establishing validity, but see it instead as conforming to a set of formal rules that are unconnected to their personal mathematical activity.

Consistent with earlier research, Koedinger (1998) reported that it was rare for high school students to think that evidence beyond examples was needed to support geometric conjectures. He also stated that many students, when considering conjectures, first said that the one example they drew was enough evidence, then, after prompting, that many examples were needed. Most students had to be explicitly asked to write a proof “like they had seen in class” before they attempted to produce one. Also, most students had difficulty formulating proofs. In fact, Koedinger reported that only 10% of students successfully formulated proof problems and began to work on them. Interestingly, students found it especially difficult to state the givens.

McCrone and Martin (2004) found that high school geometry students’ performance was low on almost all items on a proof-construction assessment. Scores were lowest on items in which students had to construct their own proofs, even on items that provided a proof outline (averages ranged from 6–34% correct). Results were a bit better for a fill-in proof (44–50%). Scores were highest, but still low, on an

item that asked students to rewrite a conjecture in “if-then” format and specify the givens and what was to be proved (33–53%). These researchers also constructed a questionnaire that probed students’ beliefs about essential elements of proofs. About 78% of students agreed with “the principle that one purpose of proof is to provide insight into why a statement is true.” About 67% of students agreed with a statement indicating that “knowing that a general statement is true implies that the statement is true for all specific instances.” In contrast, only 22% of students found the logical flaw in a fairly simple two-column proof. On a purported proof that the sum of the angles in any triangle is 180 degrees, students were presented with a table of angle measurements for four scalene triangles—3 obtuse, 1 acute. About half the students claimed that this table constituted a proof, with many students saying so because a range of types of triangles had been tested. Some students claimed that the table did not constitute a proof because it omitted right angles.

Chazan’s (1993a) investigation of high school students’ views of geometry proof elaborates and clarifies some long-standing issues in the research reported above.²⁵ In interviews, he found that 5 out of 17 high school geometry students believed that measuring examples “proved” assertions, 7 did not believe that such empirical evidence proved assertions, and 5 were unclear or changed their minds about this issue during the interview. Some students who believed that empirical evidence is proof focused on the number of examples examined, whereas others focused on the type of examples examined. As an example of the latter approach, in considering a proposition about triangles, one student said, “Right, obtuse, acute and like isosceles or something like that . . . it’s pretty much the same effect as having every one” (p. 369).

Of the students who believed that empirical evidence does not constitute proof, some cited the fact that there may be counterexamples because one cannot examine all examples. Other students cited the inexactness of measurement, a point that raises an important issue. On the one hand, side lengths can be considered as the result of a measuring process (that is, they are obtained by the act of physically measuring and, as such, are only approximations). On the other hand, side lengths can be considered theoretical (that is, they represent the actual length measures of the segments; as such they are exact and without error). In the first case, skepticism about justifications based on lengths is warranted; in the latter

²⁵ First, use of the *Geometric Supposer* was an integral part of Chazan’s students’ courses. But because I focus only on what the research indicates about proof, I discuss this research in this section rather than in the section on the *Supposer*. Second, some students were interviewed in January, some in March—so students had not completed their geometry courses.

case, skepticism is unwarranted. It is important for researchers to determine precisely not only how students think about measures given in proof problems (e.g. approximation versus exact), but how instruction and computer geometry programs deal with the measure/approximation issue and how this affects students' conceptualizations.

In contrast to students who believed that empirical evidence was proof, some students thought that proof was merely evidence (Chazan, 1993a). For instance, some students believed that counterexamples were still possible, even if a deductive proof in a textbook was given. However, Chazan's finding could be due to students not understanding the logic of proof, or it could be a reasonable reaction to the possibility of an incorrectly *implemented* proof (especially if the proof is written by a student²⁶). This brings up an important issue. Theorems presented in school mathematics textbooks, and their accompanying proofs, are generally assumed by teachers and students to be correct because they appear in the book, which is taken as authoritative, rather than because the proofs are critically examined by students and teachers. This is very different from how mathematicians work—usually, proposed new theorems and proofs are greeted with skepticism, and often with attempts to find counterexamples.²⁷ So, for some students, skepticism about deductive proofs should not be viewed as a weakness but rather as justified and healthy.

Another reason that students might be skeptical about the generality of a deductive proof occurs when the proof does not really provide insight into why a proposition is true (recall the McCrone and Martin's, 2004 finding that 78% of students agreed that proofs should provide insight into why statements are true). For instance, one can read a proof and believe in its validity because each step in its argument is valid but still not, in the words of Polya, "see it at a glance." In this case, the proof does not provide enough insight to genuinely make sense of the proposition, so one might still be skeptical. In such instances, it is mathematically healthy and reasonable to further explore the proof by examining examples, perhaps reviewing the proof for some of these examples, even developing alternative proofs (think of how many proofs have been given for the Pythagorean theorem). Thus, instead of assuming that student skepticism about proof must be a negative finding, it is important for researchers to investigate whether students' skepticism is due

to (a) a lack of understanding of the logic of proof, (b) a healthy critical attitude toward mathematical justification (so students are skeptical not about the logic of proof but about particular implementations), or (c) not being exposed to proofs that help them understand why a theorem is true (that is, they do not believe what does not make sense to them).

According to Chazan (1993a), some students believe that deductive proofs apply only to single diagrams or to single types of shapes. In the latter case, a student might believe that a two-column proof had to be written, say, for an acute, obtuse, and a right triangle (and possibly isosceles and equilateral triangles as well). Chazan noted that the notion of multiple proofs for multiple examples might be a carryover from thinking about proof by examples—students want to see multiple examples to convince themselves.²⁸ But there may be more to it than this. Progressing from reasoning about a specific example to reasoning about a deductive proof is similar in some sense to moving from reasoning about specific numbers to reasoning about variables. For instance, for a specific diagram, one might say that each of two sides measures 50 so they are equal in length; in a deductive proof of a general proposition, however, we might say that two sides are congruent—that is, algebraically, the measure of both sides both equal some indeterminate number x . Moving from specific to variable thinking requires a significant jump in abstraction. Thus, the multiple proofs for multiple examples conception might indicate a transitional stage in abstraction in moving from empirical to deductive thought. In any event, it is clear that students possessing the multiple proofs for multiple examples view do not genuinely understand the concept of deductive proof. However, this should not be surprising, given that most instructional treatments do not spend significant time specifically helping students understand the nature of mathematical justification and proof. Instead, these treatments focus mostly on having students master formal methods of proof.

In summary, researchers are still grappling with the basic question: Why do students have so much difficulty with geometric proof? Is it because they see little need for proof, because proof is so abstract that it is very difficult to learn, or is it a combination of the two—that because students see proof as difficult and not useful, they do not expend sufficient intellectual effort to master it? What components of proof are

²⁶ In this research, it is not always clear if the students' statements are about textbook proofs or about student proofs.

²⁷ In fact, analyzing why counterexamples cannot be found often provides insight into why a proposition is true.

²⁸ Of course, there are proofs in mathematics texts that are broken down in terms of cases. But students in school would not have been exposed to many such proofs.

difficult for students, and why? How can proof skills best be developed in students? More research is required on students' development of geometric proof skills and their understanding and beliefs about the nature of proof. As was discussed in the section on DGE, use of DGE has shown promise in encouraging and supporting the development of students' geometric proof. To better understand the nature of the facilitative effect of DGE on proof, however, qualitative research comparing students' proof processes in DGE and paper-and-pencil environments is needed.

Apparent contradictions in the research also need to be resolved. For instance, Healy and Hoyles (1998) reported that students recognized that once a statement has been proved, no further work is necessary to check its validity, but Chazan (1993a) reported that some students believed that counterexamples were still possible, even if a deductive proof in a textbook was given. As another instance, Koedinger (1998) reported that students found it especially difficult to state the givens, whereas McCrone and Martin (2004) reported that scores were highest, but still low, on an item that asked students to rewrite a conjecture in if-then format and specify the givens and what was to be proved. It is likely that such contradictions are due to the fact that, in many cases, small numbers of assessment tasks were examined to draw conclusions, and results are task specific. That is, although there are general processes applicable to most geometric proofs, success on most proofs also depends on understanding specific concepts and situations. Research in this area must balance investigating general versus task-specific components of proof.

Understanding Angles

In an investigation of the development of angle conceptualizations in elementary school students, Mitchelmore and White (2000) describe three stages of abstraction. In Stage 1, similarities between angle-related situations that look alike, involve similar actions, and are experienced in similar circumstances lead to abstraction of *situated angle concepts* (*conceptualizations in my language*). For instance, situations involving hills would be a single situated angle concept. In Stage 2, similarities between different situated angle concepts are abstracted to form *contextual angle concepts*. For instance, students might abstract the similarity between hills and roofs into a contextual angle concept of "slope." In Stage 3, similarities between different contextual angle concepts are abstracted to form *abstract angle concepts*. For instance, students might abstract the similarity between intersections of lines, corners, bends in paths, and slopes. Mitchelmore and White

suggested that construction of a formal mathematical formulation for the standard abstract angle concept of two rays with common endpoint can be considered the fourth stage of angle concept development. They further argued that, when students are forming their first abstract angle concept, they may be in the process of forming additional contextual angle concepts. Furthermore, they claimed that angle concepts are constantly generalized as new situations and contexts are recognized as similar.

Similar to several other researchers in geometry learning, Mitchelmore and White use a theory of abstraction to analyze students' thinking.

For example, students often demonstrate the similarity between a tile and a ramp by physically moving a corner of the tile to fit into the space between the ramp and the horizontal base. . . . The recognition of similarities between different angle contexts is therefore a constructive process requiring reflective abstraction. (2000, p. 216)

Could it be that the critical component of this reflective abstraction is the mental operation of matching corresponding parts of angle contexts, and that this matching is the same type of structural mapping discussed by Gentner and Medina (1998)?

Mitchelmore and White (2000) also suggested that an essential part of the process of mentally constructing a general conceptualization of angle is linking together different angle concepts. This returns the discussion to the issue of abstraction as disembedding/decontextualizing versus linking. In fact, the developmental sequence described by Mitchelmore and White provides a nice example of how the sequence of abstractions that lead to abstract angle concepts involves *both* disembedding and linking. For example, to compare the angle concepts embodied by the corner of a tile and the space between a ramp requires the disembedding operation of interiorization because the original abstractions of these embodiments are applied outside of their initial contexts. But the reflective abstraction that compares the tile-and-ramp-angle concepts can result in a new abstraction that is linked to both ramps and tiles.

Mitchelmore and White (2000) reported three abstract angle concepts: *point* is sometimes used by young children to relate corners and intersections; *a single sloping line* relates sloping and turning objects; and, most common, *two inclined lines meeting at a point*, which they call the *standard angle concept*, can be used to relate all physical angle contexts. Mitchelmore and White found that more than 80% of students in Grades 2–6 could see similarities between a standard angle (as represented by a folding straw) and physical

contexts of scissors, a folding hand fan, a diagram of a road intersection, a four-sided tile, and two intersecting walls. Students had more difficulty relating the standard angle to a wheel or door (turning) and a hill (slope). Less than 30% of students in each of Grades 4, 6, and 8 represented the slope of a hill using a standard angle between the hill and the horizontal, with many, instead, representing the slope as the angle between the vertical and hill. However, this finding must be tempered by the fact that in the representation of the hill used in this study, the angle with the vertical was much more salient (see Figure 19.15).

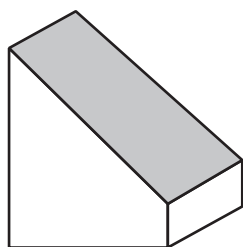


Figure 19.15 Representation of slope of a hill (Mitchelmore & White, 2000, p. 221).

Most importantly, the difficulties that students had relating the standard angle concept to various angle contexts seemed to be directly dependent on the visual availability or salience of structural angle components (the two sides and vertex) in the contexts. For instance, a large portion of students could not identify the two lines that make up a standard angle in the contexts of a wheel or door, and few students saw the standard angle as slope plus horizontal in the hill context illustrated above. Thus, the availability and salience of these structural components seemed to determine students' success in performing the structural mappings required to see similarities between examples from different contexts.

The findings that students have difficulties relating the standard angle concept to inclination and turning are important not only in their own right but because these two contexts are extremely important in postelementary mathematics such as algebra (inclination as slope) and trigonometry (turning). Mitchelmore and White (2000) also suggested that the difficulty that students have in learning to use a protractor might stem from the fact that, on a protractor, several lines can be chosen for the initial side of an angle but the terminal side must be imagined. So again, the cause of students' difficulties seems to be the absence

of structural angle components, which leads to failure in establishing appropriate structural mappings.

Because of the difficulty that students have in seeing standard angles in turning contexts, Mitchelmore and White (2000) claim that defining an angle as an amount of turning (about a point) from one line to another is inappropriate for beginning learners. But the notion that understanding angles in terms of turns is *necessarily* bad for children is contraindicated by some (but not all) Logo research. Indeed, Hoffer claimed that "turns or rotations are natural for young children . . . this dynamic way of working with angles should be started early and carried throughout the students' work" (1988, p. 251). In support of this claim, Clements and Battista's research on elementary students' learning in the *Logo Geometry* curriculum shows that learning about angles in the context of turtle turning and paths can be effective for elementary students, *if the relationship between amount of turn and measure of angle produced is explicitly and carefully taught* (Clements & Battista, 2001). However, continued research is needed on how best to instructionally develop general abstract conceptualizations of angle that are adequate for the many manifestations of this concept in mathematics.

Coordinate Systems and Locations

Locating objects in space is a critical process in geometry and everyday life. Geometry formalizes this process through the use of coordinate systems. In this section, I briefly describe some research issues related to the use of coordinate systems.

Types of Coordinate Systems

Students are exposed to two different types of "coordinate-system" conceptualizations, maps and Cartesian coordinate systems. Maps utilize *nonmetric* ordinal coordinates to specify locations: Labels on axes signify ordered locations without metric relationships. To locate a point in 2d space, one must find specified locations on the axes, then move orthogonally to both axes to find the intersection. The key idea here is not distance, but coordinating order on two intersecting lines.²⁹ On geographic maps, "coordinates" (which can be letters or numbers) often specify intervals, not points. Of course, the difficulty with this conceptualization is dealing with "in-between" locations. In contrast, Cartesian coordinate systems are based on *both* axis-orthogonality and uniform metric axis scales, which ensures that relationships between points are

²⁹ Although axes are usually perpendicular with equally spaced labels, this concept works even if the lines are not orthogonal and labels are not equally spaced.

uniform throughout the coordinate system and allows the system to be used to represent many important geometric properties.

Research is needed to determine which of these two conceptualizations students are developing in their classroom work with coordinate systems. Such research must account for the fact that mathematical coordinate axes have multiple purposes. One use of coordinate axes is simply to locate points in space. This use does not require metric properties of number lines, only order properties. However, full understanding of mathematical coordinate systems affords students more than the ability to locate points. It also allows them to reason about distances between points using coordinates. This is a critical property of Cartesian coordinate systems and one of their major sources of conceptual/reasoning power. To construct a full understanding of Cartesian coordinate systems, merely having students locate points is not enough; students must also analyze distances between points and how those distances can be determined from coordinates (see, for example, Sarama, Clements, Swaminathan, McMillen, & González Gómez, 2003). However, researchers and teachers must be alert to the fact that students' initial conceptualizations of coordinate systems may be skewed by early map-like uses.

An Example of Research on Student Learning of Coordinate Systems

Sarama et al. (2003) found that, when initially working with numerically labeled axes, some fourth graders ignored the labels. For instance, to locate the point for (8, 9), one student, TA, counted 8 lines over and 9 lines up, even though the grid lines were labeled with numbers (so counting was unnecessary). The authors argued that such students did not take the numeric grid labels as symbols or curtailments of the actual counting process. They conjectured that this is similar to students not being able to count on—they cannot operate on the first number in the count as representing a given number of counting acts—they must produce the counting acts. Such students have not raised their counting process to the symbolic, second level of interiorization. The occurrence of the unnecessary counting behavior on axes is also consistent with the conjecture that such students possess an action-oriented meaning for coordinates—coordinates are considered inputs for a counting- or movement-based procedure for locating an object. Such a proceduralized conception of coordinates would be difficult to apply to nondecade coordinates on grids with only multiples of ten labeled (a difficulty that these researchers observed). Another piece of evidence suggesting that students' initial conceptualizations

of coordinates were action based was the finding that some students who did not usually reverse x and y coordinates would do so when the first coordinate was 0. Such students moved horizontally for the second coordinate (e.g., go over 6) as the first step in locating the point, needing a nonzero movement to start their plotting procedure.

Another difficulty observed by Sarama et al. (2003) was students not seeing that coordinates were given with respect to a particular reference point, the origin (0, 0). For instance, some students, after having plotted one point, plotted the next point relative to that point, not the origin. Of course, this error may have occurred because students were working in a Logo environment in which many commands are relative.

So what does it take to construct a fully developed conceptualization of a mathematical coordinate system? Sarama et al. (2003) suggest that students must be able to “distributively coordinate” two orthogonal “conceptual rulers.” This idea is similar to students' conceptualization of 2d arrays of squares (Battista, Clements, Arnoff, Battista, & Van Aukun Borrow, 1998). To construct a properly structured mental model of a 2d array, students must iterate a row composite along the elements of a column. To construct a properly structured mental model of a coordinate plane, one must continuously sweep one coordinate axis (conceptualized as a fully metricized number line) along the orthogonal axis. However, the coordinate axis situation is more complicated than the array situation because arrays are discrete, but coordinate axes are continuous. So one cause of students' problems with coordinates axes may be the difficult shift from discrete to continuous objects. Another difficulty that Sarama et al. (2003) observed is that some students fail to interrelate coordinates and locations of multiple points. For instance, asked to plot (20, 35) and (25, 35), less sophisticated students plotted them separately. In contrast, more advanced students saw immediately that the second point was 5 units to the right of the first point. The advanced students had constructed second-level interiorized versions of plotting schemes and properly structured mental models of frames of reference that allowed them to interrelate locations and coordinates without running through the process of locating each point.

Large- and Small-Scale Spaces

Mathematical coordinate systems function the same whether one is locating a point on a sheet of paper or oneself in a city. However, research in psychology suggests that the cognitive processes used in dealing with these situations differ, which might have important implications for research on the learning

and teaching of coordinate systems. For instance, according to Siegel (1981), large-scale spaces (e.g., a university campus) cannot be perceived from a single viewpoint, and their structure is mentally constructed from a number of observations over time. In contrast, small-scale spaces (e.g., a sheet of paper) can be mentally constructed from a single vantage point. One acts *in* large-scale spaces, but *on* small-scale spaces. Consequently, it might be difficult for students to transfer their thinking about mapping a classroom or playground to coordinate systems shown on a page or computer screen.

In Siegel's (1981) developmental model for the formation of cognitive maps of large-scale spatial environments, in the first phase, as one moves about in an environment, landmarks are noticed and remembered. In the second phase, routes linking landmarks are formed (and with further experience metricized). In the third phase, as coordinated frames of reference are developed, routes are integrated into overall configurational frameworks or survey maps.

In contrast, for small-scale spaces, Piaget's work on students' construction of coordinate systems is still relevant (Piaget, Inhelder, & Szeminska, 1960). For instance, when students were asked to draw a point on one rectangular sheet that was in exactly the same place as a point on another sheet, several levels were observed. At Level 1, students used global visual estimates to locate the point; measuring devices were either not used or used inappropriately. At Level 2, students used only one measurement (say from a rectangle side or vertex); they were satisfied with one measurement because they could not coordinate two. At Level 3, students recognized that two measurements are needed. For instance, they used a single measurement from a corner and tried to visually preserve the slope of the line when using this measurement. Or, they made two oblique measurements. At Level 4, students successfully used two perpendicular measurement dimensions to locate the point. This perpendicularity served as a conceptual device that enabled students to coordinate the measurement dimensions. However, students' use of two coordinated measurements was not yet interiorized; they tended to discover this approach using trial and error instead of immediately using it. Finally, only at Level 5, did students construct an interiorized mathematical coordinate system.

[Level 5] subjects do not begin with a single measurement only to discover the need for a second dimension after a period of trial-and-error. From the outset, they realize the logical necessity to take both dimensions into account and the measurements are straightaway coordinated so as to be at right angles to one another. . . . [This] behavior is operational in nature. Before the facts are given in actual experience, they are engendered by structured anticipatory schema. (p. 169)

Presumably it is at this level that relations of order and distance between *objects* are replaced by similar relations between the *positions* themselves. It is as if a space were emptied of objects so as to organize the space itself. In fact, Piaget argued that

a reference frame is not simply a network composed of relations of order between the various objects themselves. It applies equally to positions within the network as to objects occupying any of these positions and enables the relations between them to be maintained invariant, independent of potential displacement of the objects. (Piaget & Inhelder, 1956, p. 376)

Despite differences in the mental development of large- and small-scale spaces, there are two major commonalities. First, coordination (of actions and objects) is crucial. Second, a frame of reference must be mentally constructed, and actions and objects must be properly integrated into it. Like so many processes in geometry, both the coordination process and the formation of frames of reference depend critically on creating appropriate mental models.

GEOMETRIC MEASUREMENT

The concept of measurement³⁰ is woven throughout the fabric of geometric conceptualization, reasoning, and application. From the historical beginnings of geometry as "earth measure" to Birkhoff's axioms for Euclidean geometry, measurement plays a central role in reasoning about all aspects of our spatial environment. Measurement is critical for understanding the structure of shapes, using coordinate systems to determine locations in space, specifying transformations, and establishing the size of objects. Geometric

³⁰ I use the term *geometric measurement* in the broad and abstract mathematical sense to refer to the concept of assigning numbers to geometric entities in accordance with a set of axioms. At the school level, understanding geometric measurement includes (a) conceptualizing that numbers can be used to quantify the amount of an attribute (e.g., length, area, volume) contained in a geometric object by determining the number of attribute-units that fit in the object and (b) being able to implement procedures for assigning measurements to objects (e.g., iterating units, using a ruler, choosing appropriate measurement units). Thus, understanding measurement requires an integration of procedural and conceptual knowledge.

measurement is also embedded in the graphic representation of functions and algebraic equations.

However, despite the importance of geometric measurement, students' performance on measurement tasks is alarmingly low (Kloosterman et al., 2004; Martin & Strutchens, 2000; Sowder et al., 2004). For instance, according to NAEP results, in the year 2000, less than 25% of 4th graders and only about 60% of 8th graders were able to determine the length of an object pictured above a ruler with its end not aligned to that of the ruler (Kloosterman et al., 2004; Sowder et al., 2004). On a similar item in 1996, the percentages of Grade 4, 8, and 12 students answering this item correctly were 22, 63, and 83, respectively. Thus, almost 1 in 5 seniors in high school seemed to lack understanding of measuring length with a ruler. Performance on area and surface area was worse. In 2000, only about 14% of Grade 8 students could determine the number of square tiles it takes to cover a region of given dimensions, and only about 25% could determine the surface area of a rectangular solid (Sowder et al., 2004). In 1996, only 35% of 12th graders and 27% of 8th graders recognized and could explain why a square, and a right triangle with the same height and twice the width of the square, have the same area (Martin & Strutchens, 2000). And on volume, in 1990, only 55% of 12th graders and 41% of 8th graders knew that a measurement of 48 cubic inches for a rectangular box represented volume (NAEP web site).

Strongly implicated in this low performance is the research finding that for many students—perhaps the majority—there is a basic disconnection between spatial and measure-based numerical reasoning (Barrett & Clements, 2003; Battista, 2001a; Clements et al., 1997). That is, many students do not properly maintain the connection between numerical measurements and the process of unit-measure iteration. For instance, students who incorrectly measure the length of an object when one of its ends is not aligned with “zero” on a ruler do not clearly conceptualize how the ruler's numerical markings indicate the iteration of unit lengths. And most students who correctly use the formulas for the area of a rectangle or volume of a right rectangular prism in standard problem contexts, neither understand why the formulas work nor apply the formulas appropriately in nonstandard contexts. For example, a bright eighth grader who was 3 weeks from completing a standard course in high school geometry—so she was 2 years ahead of schedule for college prep students—responded as follows on the problem shown in Figure 19.16 (Battista, 1998b).

S: It's 45 packages. And the way I found it is I multiplied how many packages could fit in the height by the number in the width, which is 3 times 3 equals 9. Then I took that and multiplied it by the length, which is 5, and came up with 9 times 5, which is 45.

Obs: How do you know that is the right answer?

S: Because the equation for the volume of a box is length times width times height.

Obs: Do you know why that equation works?

S: Because you are covering all three dimensions, I think. I'm not really sure. I just know the equation.

Because of an inappropriate connection between her spatial structuring and the numerical procedure, this student did not understand that the mathematical formula she applied was inappropriate for this problem—a common problem for students. Indeed, only 38% of the students in her geometry class answered the item correctly, despite the fact that all of them had scored at or above the 95th percentile in mathematics on a widely used standardized mathematics test in 5th grade. Battista also found that only 19% of Grade 7 and 8 students in a range of classes in the same school, but not including this geometry class, answered this problem correctly (1999b), and that less than 10% of Grade 3–5 students correctly solved a similar problem in which the box was only one layer high (1998b). Simon and Blume (1994) reported preservice elementary teachers making the same error in an analogous area situation. Similarly, in a study by Reynolds and Wheatley (1996), fourth grader Kristin was trying to determine the number of 3-by-5 cards needed to cover a 15-by-30 rectangle. To solve the problem, she divided 15 into 450. However, the researchers found no evidence that Kristin realized that this quotient would be correct only if the two dimensions of the small card divided, respectively, the two dimensions of the large rectangle. In fact, Kristin seemed suspicious of her procedure, saying that she needed to verify it by drawing. Thus, Kristin was struggling to connect the spatial and numeric aspects of this problem.

In general, students working on these types of nonstandard measurement problems must perform two critical processes: (a) they must construct a proper spatial structuring of the situation; (b) they must coordinate their spatial structuring with an appropriate numerical scheme. Too often, students skip the first process and proceed directly to the second. Also, even when students recognize that they must perform the first process, they often have difficulty doing so. That is, because many traditional curricula prematurely teach numerical procedures for geometric measure-

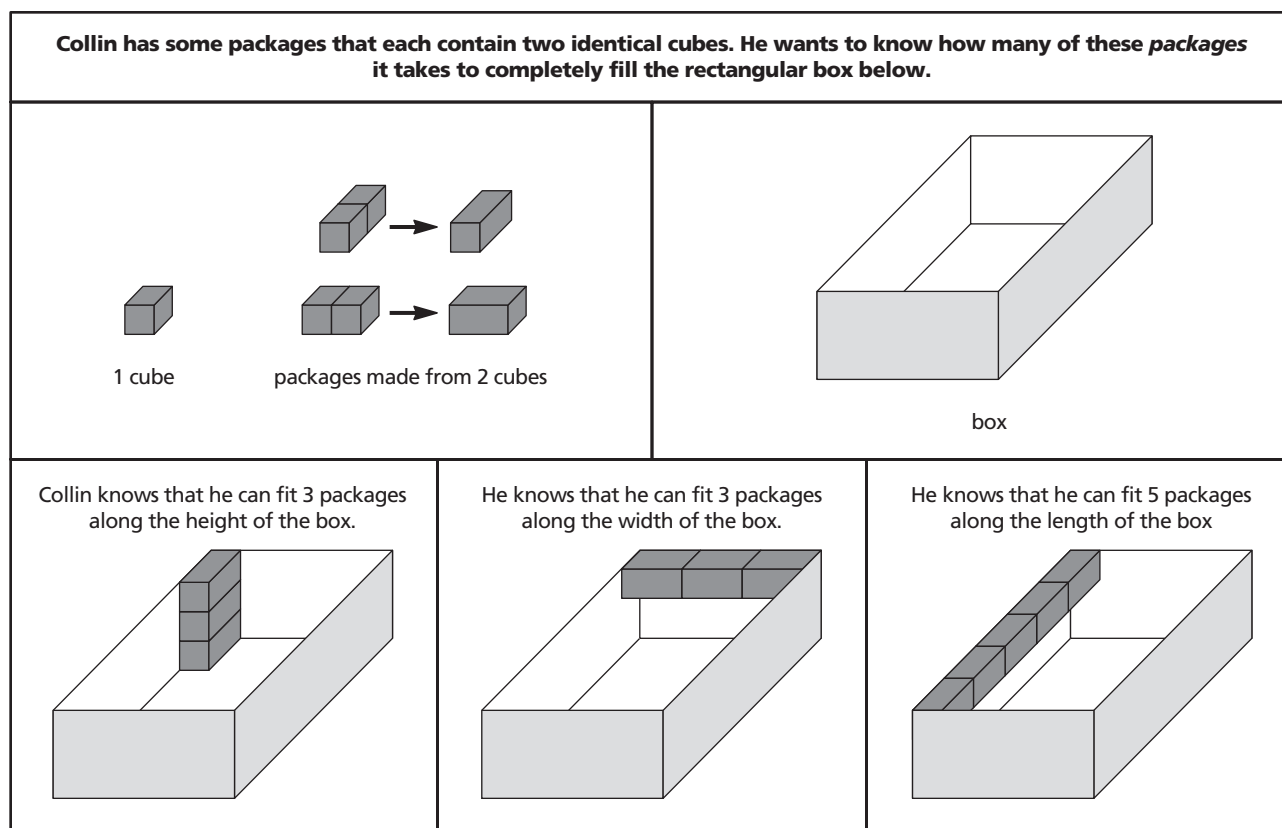


Figure 19.16 Package problem.

ment, students have little opportunity to think about the appropriateness of the numerical procedures they apply, and they have insufficient opportunities to develop skill in spatially structuring arrays of measurement units. In particular, even when students recognize that the volume formula is inappropriate for the Collin problem, they have significant difficulty constructing an appropriate spatial structuring of the packages because they have had so few opportunities to develop their structuring skills.

Research also shows that students commonly interchange measurement units or computational procedures (Chappell & Thompson, 1999; Nunes, Light, & Mason, 1993; Pesek & Kirshner, 2000; Woodward & Byrd, 1983), and that most students have difficulty understanding how the multiplication of two length units produces units of area (Battista, 2004; Kordaki & Potari, 1998; Nunes et al., 1993). *Overall, then, the research indicates that many students' measurement reasoning is superficial, with poorly understood procedures or formulas substituting for deep understanding.* In fact, the traditional premature instructional focus on computational formulas seems to interfere with students' concept development in geometric measurement.

THE DEVELOPMENT OF STUDENTS' REASONING ABOUT LENGTH

In many ways, research on linear measurement has not answered fundamental questions that were asked 2 decades ago. To illustrate, in a still widely cited study, Hiebert (1981) found that many students who failed Piagetian tasks of conservation and transitivity of length apparently still "learned" measurement strategies. He argued that although

conservation and transitivity are logical prerequisites for completing many measurement tasks, children do not seem to use this knowledge when they solve the tasks. . . . They use an intermediate measurement to compare two lengths and do not think to ask the transitivity question; they move a unit to measure the length of an object and do not worry about whether the length is being conserved. Simple skills or techniques apparently allow children to bypass the logical structure of many measurement tasks. (p. 208)

Hiebert conjectured, however, that students who possess a particular Piagetian reasoning ability might

learn related measurement concepts with a deeper level of understanding than those who do not possess this ability—that, for example, students who do not conserve length learn certain measuring strategies only at a rote level. To partially test this hypothesis, Hiebert made a crooked path with rods and asked students to construct a straight path that had just as far to walk on it (the rods used for the straight path were different in length than those used in the crooked path). All children failed this task before instruction. However, whereas 13 of 16 of the students who passed both Piagetian tasks were at least partially successful on this task after instruction, only 2 of 16 of the students who failed the Piagetian tasks were successful. Thus, it seems that conservation, or more likely, the mental operations that enable conservation, are necessary for a conceptually sound completion of this task. Consistent with Hiebert's hypothesis that simple techniques may substitute for deep understanding, Joram, Subrahmanyam, and Gelman (1998) claimed that even though young children can learn some simple measurement skills, actual physical measurement such as determining lengths with paper clips is challenging for many middle-school students. They also reported poor performance on linear measurement estimation even for high school students and adults. Thus, both old and new research suggests that students' knowledge of linear measurement may be more superficial than it appears in most common assessments.

Levels of Sophistication in Students' Reasoning About Length and Length Measure

Several groups of researchers have attempted to describe levels of sophistication in students' development of the concepts of length and length measurement. Because each group has focused on different aspects of length and used different sets of tasks, comparisons are difficult but also revealing.

Clements et al. (1997) observed three levels of sophistication in students' reasoning about path lengths, which were later extended to four levels by Barrett and Clements (2003). At Barrett and Clements' Level 1, students use gross visual comparison of objects. There is a disconnection between numerical measurement and the extent of line segments; students often estimate or guess measurements. At Level 2a, students connect numbers to iterative movements along a path but often have difficulty keeping track of their unit iterations. "Children at this stage

expect that measuring consists in counting, but they are not consistent in their attention to measuring" (p. 55). At Level 2b, students integrate the iteration of a single unit of length into a sequence of units, but restricted to a single direction. So students might correctly iterate a unit along the top of a rectangle but lose track of the unit as they move to an adjacent but perpendicular side. At Level 3, students correctly iterate units around complex multidirectional paths, and they operate on units of units and reason about measurements in the absence of perceptual objects. Number schemes and spatial schemes are properly coordinated.

Battista (2003a, in press c) attempted to integrate previous research and develop a broader characterization of students' construction of meaning for length and length measurement. According to this characterization, there are two fundamentally different types of reasoning about length. *Nonmeasurement reasoning does not use numbers*. Instead it involves visual-spatial inferences based on direct or indirect comparisons, imagined transformations, or geometric properties. *Measurement reasoning involves unit-length iteration*, that is, determining the number of fixed unit lengths that fit end-to-end along the object, with no gaps or overlaps. Measurement reasoning includes not only the process of measuring, but reasoning about numerical measurements (e.g., adding lengths to find the perimeter of a polygon, making inferences about length measurements based on properties of figures). Although students typically develop nonmeasurement strategies before measurement strategies, nonmeasurement reasoning continues to develop in sophistication even after measurement reasoning appears. Furthermore, the most sophisticated reasoning about length involves the integration of nonmeasurement and measurement reasoning.

Nonmeasurement Reasoning

Nonmeasurement Level 0. Students' reasoning about length is appearance-based and holistic. They attend to how things look, focusing on whole shapes rather than systematically on parts within shapes. They often use direct and indirect comparison.

Nonmeasurement Level 1. Students systematically use decomposing/recomposing to compare lengths. At first, students rearrange (physically, by drawing, or in imagination) some or all path pieces, and directly and visually compare the rearranged paths as wholes. Later, students compare two paths by matching, one-by-one,

pairs of pieces that they think are the same length—they do not transform one path into another.

Nonmeasurement Level 2. Students compare path lengths by sliding, turning, and flipping shape parts in ways that allow them to infer, *based on shape and motion properties*, that one transformed shape is congruent to another.

Measurement Reasoning

Measurement Level 0. Students use counting to find lengths; however, their counting does not represent the iteration of a fixed unit length. For instance, students might recite numbers as they continuously move their fingers along a path. Or they might count dots along a path but not as true indicators of unit lengths. (See Sarama et al., 2003, for similar findings.)

Measurement Level 1. Students attempt to iterate what they consider to be a unit length along an object or path. However, because students (a) do not fully understand what a length unit is, or (b) do not properly coordinate iterated units with each other, their iterations contain gaps, overlaps, or different length units, *and are incorrect*.

Measurement Level 2. Students not only understand what a length unit is, when iterating unit-lengths, they properly coordinate the position of each unit with the position of the unit that precedes it so that gaps, overlaps, and variations in unit lengths are eliminated.

Measurement Level 3. Students determine some length measurements without explicitly iterating every unit length. That is, after iterating some length units, students operate on the results of their iterations numerically or logically to find other lengths.

Measurement Level 4. Students numerically and inferentially operate on length measurements *without iterating any unit lengths*. They make complex, property-based visual inferences about measurements using properties of geometric shapes and transformations. At this highest level of measurement reasoning, students fully integrate and apply the processes from Nonmeasurement Level 2 with their measurement reasoning.

Cognitive Processes for the Conceptualization of Length Measurement

Research suggests that students construct meaningful understanding of length measurement as they abstract and reflect on the process of iterating unit lengths (Barrett & Clements, 2003; Carpenter & Lewis, 1976; Hiebert, 1981; Kamii, 1995; Cobb,

Stephan, McClain, & Gravemeijer, 2001; Piaget, Inhelder, & Szeminska, 1960). For instance, Joram et al. (1998) cited research indicating that, in making linear measurement estimates, iterating a unit is the most commonly reported strategy. In one study, 81% of elementary students reported using unit iteration in estimation, with about one third of these students physically marking off units, whereas the other two thirds were observed marking off units with their eyes. Furthermore, consistent with the unit-counting strategy hypothesis, Joram et al. cited one study in which the time it took adults to estimate the length of line segments increased linearly with the length of the segments and another study in which the times taken to estimate the number of 1-cm units in a 6-cm segment were the same as those to estimate the number of 2-cm segments in a 12-cm segment.

But how exactly do the conceptualizations of length and length measurement develop? What are the basic cognitive operations required? Joram et al. (1998) argued that the mind mentally represents counting numbers as magnitudes using a continuous accumulator mechanism (like putting water into a beaker). This mechanism includes a bidirectional mapping between verbal/written number symbols and accumulated mental magnitudes. In this view, counting a set of 5 objects activates the corresponding magnitude on a mental number line (via the accumulator), and, inversely, when a quantity on the mental number line is activated, a portion of the mental number line is subdivided and intervals assigned appropriate numbers.

Joram et al. (1998) discussed how this counting process can be applied to linear measurement estimation and how the process explains why linear estimation is more difficult than counting. According to the accumulator model, the mental representation activated when counting a set of discrete items is a continuous magnitude. During object counting, this continuous magnitude is made discrete by single bursts “of impulses to the accumulator for each item in the set” (p. 422). The perceptual discreteness of the counted objects provides the signals for these bursts. Joram et al. hypothesized that what makes linear measurement estimation more difficult than counting is that, when estimating, there are no perceptual cues to trigger the bursts. Such burst activation requires an individual to mentally segment the linear object into units, which requires (a) recalling and maintaining an image of the length unit, (b) visualizing the iteration of the unit along the object to be estimated, (c) keeping track of where the last unit ended and where the

next one must begin, and (d) maintaining a running count of units while performing tasks a–c. Further complicating matters, the burst interval (interval size) changes, depending on the length unit employed (e.g., inches, feet).

Given that both linear extent and Joram et al.'s (1998) hypothesized mental representation of counting are based on continuous mental models, it is interesting that linear measurement estimation is more difficult to master than counting (Barrett & Clements, 2003; Piaget et al., 1960). In some sense, it would seem that the fact that both length and the counting-accumulator are continuous would make linear quantification easier than discrete quantification. And perhaps, at some level, it is. That is, in some contexts linear extent seems easier to comprehend than counting (thus, for example, the widespread use of bar graphs). However, Joram et al.'s burst-theory explanation suggests that difficulties in applying counting to measuring length reside in the transition from discrete to continuous quantities; individuals must establish an isomorphism between the discrete counting system and linear quantities. The episode below illustrates a student having difficulty with this isomorphism (Battista, in progress; cf., also Clements et al., 1997).

In this episode, when asked how she knew the rectangle she drew was 40 units around, JAK (grade 2) drew irregularly spaced dots along its inside edge, stopping when her count reached 40 (see Figure 19.17). There was no indication that JAK used dots as indicators of fixed unit lengths. She was unable to properly connect the continuous quantity that was the perimeter with the discrete items of the verbal counting sequence.

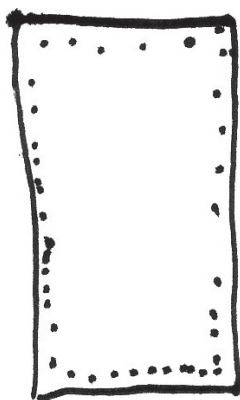


Figure 19.17 JAK's rectangle.

A somewhat different description of the process by which the mind constructs length and length units is given by Steffe (in press), which I will liberally sum-

marize and interpret. According to this theory, individuals abstract continuous experiential items as they move a finger along an item or scan the item with their eyes. Because continuous motion is contained in the abstraction of the object, the perceived object is experienced as continuous. Furthermore, because of the way that an individual becomes aware of a continuous path—the perceptual abstraction contains its endpoints, the *motion* that produces the path, and the duration of the motion—the individual also becomes aware of the path's linear extent. "The property of a segment that we call length is an awareness of the scanning action over an image of the segment along with an awareness of the duration of the scanning action" (p. 3). Extending this analysis, for a segment to be used as a unit in iteration, it must be abstracted at an even higher level. The scanning motion used to perceive the unit length must be interiorized, then applied in steps a–d above.

Cognitive Processes for Conserving Length

What kinds of cognitive processing are required to conserve length? In dealing with conservation of length tasks, Piaget et al. (1960) found that students at the lowest levels (1 and 2a) of sophistication on length conservation failed to simultaneously account for both ends of the sticks, which the researchers saw as an indication that the students were not attending to the intervals of length between the endpoints. So the objects were not compared as continuous entities. Perhaps the reasoning used by students at this level is based on abstraction of experiences with comparisons in which only one set of endpoints are examined, for instance, comparing who is taller when two people are standing next to each other. (Although the *logic* of comparing single endpoints in this context seems obvious to adults, it is complex and may not be at all clear to young children.)

At the next highest level (2b), coordination of parts is clearly evident—student responses indicate "the beginning of a relationship between the two paired extremities (i.e., the four extremities [of the two sticks] taken in pairs" (Piaget et al., 1960, p. 100). This relationship arises as a consequence of students' abstraction and comparison of two scanning motions, one for the extent of each stick. In discussing this level, Piaget et al. made an important observation about operational reasoning. They stated that some students at Stage 2b are close to operational reasoning (which they claimed happens at Stage 3) but not quite there. For instance, one student noted that the sticks were equal when aligned, but when the sticks were moved out of alignment, he had to realign them to convince himself

that they were still equal. Piaget et al. identified this reasoning as empirical, not operationally reversible.

Level 3 is achieved only when students see conservation of length as logically necessary. Piaget et al. (1960) claimed that this happens when objects are viewed in terms of a reference system “which provides a common medium for all objects” (p. 103). Within this medium exists a system of sites that exist independent of positional changes because they are fixed in relation to an invariant backdrop: “When an object undergoes a change of position the empty sites which have previously been occupied are equivalent to sites which were previously empty and are now filled, so that the overall length of the object remains constant” (p. 103). That is, students coordinate the scanning motions performed on objects within a system of possible scannings, which is the “space” in which these motions exist (which suggests that the scannings have reached the second level of interiorization).

In contrast to the Piagetian view, a simpler explanation of the mechanism for length conservation is possible. As children repeatedly compare abstracted scanning motions for nonaligned equal-length objects by placing them in appropriate end-to-end alignments, they abstract a general reversible alignment process. This process, coupled with exposure to adults who interpret the process as indicating that the length of an object does not change when it moves, builds the reasoning structure that supports conservation.³¹

THE DEVELOPMENT OF STUDENTS’ REASONING ABOUT AREA AND VOLUME

Genuine understanding of area/volume measurement requires comprehending (a) what the attribute of area/volume is and how it behaves (i.e., conserving it as it is moved about and decomposed/recomposed), (b) how area/volume is measured by iterating units of area/volume, (c) how numerical processes can be used to determine area/volume measures for special classes of shapes, and (d) how these numerical processes are represented with words and algebra.

A core idea in developing competence with measuring area and volume in standard measurement sys-

tems is understanding how to *meaningfully* enumerate rectangular 2d and 3d arrays of squares and cubes, respectively. Previous research studies have described separate levels of sophistication in students’ enumerating squares and cubes in such arrays (Battista, 1999; Battista & Clements, 1996; Battista et al., 1998). More recently, Battista (2003b, 2004) has integrated the separate models into a more general model for area and volume, which is described below.

Underlying Mental Processes

Battista claims that five basic cognitive processes are essential for meaningful enumeration of arrays of squares and cubes: abstraction, forming and using mental models, spatial structuring, units-locating, and organizing-by-composites (the first three processes have already been discussed).³² The units-locating process locates squares and cubes by coordinating their locations along the dimensions that frame an array. For instance, to understand the location of a square in a rectangular array, an individual must “see” the square in terms of a two-dimensional coordinate-like system—for example, it is in the fourth column and the second row, or it is the fourth unit to the right and the second unit down. The organizing-by-composites process combines an array’s basic spatial units (squares or cubes) into more complicated composite units³³ that can be repeated or iterated to generate the whole array. For instance, in a 2d array, a student might mentally unite the squares in a row to form a spatial composite unit that can be iterated in the direction of a column to generate the array. Students can meaningfully enumerate arrays of squares and cubes only if they have developed properly structured mental models that enable them to correctly locate and organize the squares and cubes.

Levels of Sophistication in Students’ Structuring and Enumeration of Arrays

Level 1: Absence of Units-Locating and Organizing-by-Composites Processes

Students do not organize units into spatial composites, and, because they do not properly coordinate spatial information, their mental models are insuffi-

³¹ Objects actually change length as they are moved from one temperature to another (which was a major problem in establishing stable units of measure in human history). So, when we say that length is preserved, we are implicitly utilizing one particular simplified model of how the world works. Thus, although it might seem that abstracting conservation of length is empirically based, it also depends on social consensus.

³² See Battista (2003b, 2004) for further details. Also, see (Wheatley, 1997) for additional examples of students’ difficulties with these processes in 3d situations.

³³ A *composite unit* is a cognitive entity (an abstraction) that results from mentally uniting and taking as one thing a collection of perceived or re-presented objects.

cient to locate all the units in arrays. Enumeration of squares or cubes seems almost random. The “double-counting” error is ubiquitous.

Level 2: Beginning Use of the Units-Locating and the Organizing-by-Composites Processes

Students not only start to spatially structure arrays in terms of composite units, their emerging development of the units-locating process produces mental models sufficient for them to recognize equivalent composites. For instance, after counting the cubes visible on one side of a building, a student *infers* the number of cubes on the opposite side.

Level 3: Units-Locating Process Sufficiently Coordinated to Eliminate Double-Counting

Students’ units-locating process coordinates single-dimension views (e.g., top, side) into a mental model that is sufficient to recognize the same unit from different views, enabling students to eliminate double-counting errors. However, coordination is still insufficient to build a mental model that properly locates interior squares or cubes.

Level 4: Use of Maximal Composites, But Insufficient Coordination for Iteration

Students structure arrays in terms of maximal composites (rows or columns for area; layers for volume). But, due to insufficient coordination, the student cannot precisely locate these composites, instead estimating their locations.

Level 5: Use of Units-Locating Process Sufficient to Correctly Locate All Units, But Less-Than-Maximal Composites Employed

Students’ mental models correctly locate all squares/cubes in an array. However, although students sometimes get correct answers, because they inefficiently organize arrays into composites, they frequently lose their place in enumeration. Furthermore, students’ structuring and enumeration strategies are not generalizable and are inadequate for large arrays.

Level 6: Complete Development and Coordination of Both the Units-Locating and the Organizing-by-Composites Processes

Students’ interiorized mental models fully incorporate a row-by-column or layer structuring so that they can accurately reflect on and enumerate an array without perceptual or concrete material for the individual units within composites.

Level 7: Numerical Procedures Connected to Spatial Structurings, Generalization

Students’ spatial structuring *and* enumeration schemes reach a level of abstraction at which they can be reflected on and analyzed, thus enabling students to explicitly understand the connection between an enumeration strategy and the spatial structuring on which it is based. Students’ mental models incorporate row-by-column or layer structuring that is abstract and general enough to apply to situations in which the basic units are not cubes (e.g., packages made from two cubes; e.g., see Figure 19.16).

Additional Reasoning

In developing sophisticated reasoning about area and volume measurement, there are additional types of reasoning that students must attain, beyond the levels listed above. For instance, students must generalize the reasoning so that they can reason about fractional units in area and volume measurement, a move that can be difficult for students (Kordaki & Potari, 2002). Students must also generalize their thinking to curved or irregular shapes in which it is not easy to visualize square or cube tilings; in a sense, students must be able to think about deformable units. Students must also generalize their thinking from the small scale spaces in which area/volume are initially studied to large scale spaces such as rooms and buildings. Also, consider the ability to spatially structure the decomposition of, say, a rectangular prism into smaller, non-square units—as might be needed in carpentry (Millroy, 1992). Conceptually, such reasoning may be available at Level 7, but spatially, such reasoning can be extremely demanding and is unlikely to be learned in traditional school settings.

Researchers must carefully investigate students’ attainment of these additional types of reasoning. How, for example, are these types of reasoning related to the seven levels already described? Students must also be able to interrelate area and volume. For instance, advanced thinking about volume (e.g., that needed in calculus) depends on understanding a multiplicative relationship between the height of an object and its cross-sectional area. How do students make the transition from a conceptualization based on iterating layers of cubes to this more sophisticated idea?

Additional Research

Transitions to Abstract Tools

The research seems pretty clear that students need appropriate preparation before learning about the use of standard measuring tools such as rulers and the use of formulas. Students need to develop under-

standing of iteration of appropriate units of length, area, and volume. However, research has not yet clearly described how this transition takes place. Although Battista has described levels of sophistication in the development of students' reasoning about length, area, and volume, other than for volume, he has not followed students during extended periods of instruction. More longitudinal research is needed.

The Connection Between Linear and Area Measurement

Several studies have indicated that students have difficulty properly relating and separating the concepts of length, area, and volume. For instance, preservice and inservice teachers frequently believe that whenever the perimeter of a figure increases, so does its area (Ma, 1999; Tierney, Boyd, & Davis, 1990). Students and teachers also often believe that when the lengths of the sides of a square or cube double, so does the area and volume (e.g., Tierney, Boyd, & Davis). Nunes et al. (1993) found that when 9–10-year-olds were asked to compare the amount of paint needed to cover two rectangles, far more students answered correctly when they used physical squares than when they used rulers. In fact, 26 out of 29 students who used rulers either added the measures of the sides or did not know what to do with the rulers to make an area comparison.

At an even more fundamental level, Outhred and Mitchelmore (2000) conjectured that some elementary students do not relate the number of squares in the rows and columns of a rectangle to the lengths of the sides. Corroboration for this conjecture comes from Battista (1996) who found that some fifth graders, who correctly enumerated the number of cubes in a box when different types of pictures depicting boxes and cubes were shown, were unable to determine the number of cubic centimeters in an unmarked box when they were given a ruler (students were shown a cubic centimeter and told that it measured 1 cm on each edge). However, this idea has not yet been fully and carefully investigated. For instance, Outhred and Mitchelmore's students' errors may have been caused by a lack of spatial structuring of the squares in a rectangular array, or by deficiency in linear measurement. Battista's students presumably were able to structure 3d arrays for cubes correctly, but he did not separately assess their understanding of linear measurement, so there is no way of knowing if the cause of the problem for the students could be attributed to their deficiencies in conceptualizing length. In fact, given the difficulty that many students have with the concept of length measurement, it is not surprising that some of them struggle with building on this concept and properly relating it to area and volume measurement.

Understanding, Making, and Drawing Tilings

A number of researchers have investigated students' understanding of unit-area tilings of larger shapes and the plane, especially as revealed by students' drawings. For instance, Outhred & Mitchelmore (2000) found several levels of sophistication in elementary students' drawings of unit-square coverings of rectangles. At Level 0, the unit squares drawn by students did not completely cover rectangles without gaps or overlaps. At Level 1, unit squares completely covered rectangles without overlap, but the organization was unsystematic; units varied considerably in size and shape. At Level 2, drawings of unit squares exhibited correct array structure, with equal numbers of units in rows, and in columns, but the size of the units was determined visually, not from the dimensions of the rectangle. At Level 3, rectangles' dimensions were used to iterate rows; one dimension was used to find the number of units in a row, while the other dimension was used to determine the number of rows. At Level 4, students did not need to draw units; the numbers of units in rows and columns were used to calculate the total number of squares.

Investigating students' tilings of different kinds of shapes (not just squares) has shed additional light on the cognitive processes underlying tiling. As a case in point, Owens and Outhred (1997) suggested that, to be successful, students have to understand the geometry of the tile and how it affects gaps and overlaps in tiling. Students also have to understand the structure of the pattern of the tiling. Owens and Outhred found that many students were unable to see or draw appropriate tilings. Wheatley and Reynolds (1996) argued that drawing a tessellation of a shape requires students to develop an image of the shape that can be used in drawing it then developing a plan for covering the plane. Wheatley and Reynolds found that some students drew shapes one at a time, seemingly without understanding the overall structure of the tiling. Other students used an overall structure, including the use of composite units, to guide their activity. Thus, tiling a shape requires two critical components. Component 1 consists of constructing a mental model *of the shape* that (a) incorporates critical features of the shape's geometry and (b) can be mentally manipulated. (Some students are able to visualize exactly where a tiled shape should be placed, while other students must use trial and error with physical materials.) Component 2 consists of developing an appropriately structured mental model *of the array* of shapes in the tiling.

Research by Clements, Wilson, and Sarama (2004) that describes levels of sophistication in young children's ability to make a larger shape with pat-

tern blocks helps elaborate these components. For instance, in Clements et al.'s scheme of seven levels, at the second level, children employ trial and error to make simple given shapes, have limited ability to use turns or flips to help them make these shapes, view shapes only as wholes, and see few geometric relationships between shapes. At the third level, shapes are chosen using a gestalt configuration or one component such as side length, vertices may be matched but without a concept of angle size, and turning and flipping are used, usually by trial and error. At the fourth level, children combine shapes with growing intentionality and anticipation, angles are matched by size, and turning and flipping are used with anticipation to select and place shapes. Thus, in relation to Component 1, students gradually come to use the geometry of shapes, moving from holistic to parts-based reasoning, attending to side length before angle measure, and using motion first physically then in imagery, a progression that harkens back to the van Hiele levels and the previous discussion of mental models. The fact that Clements et al.'s last three levels deal with increasing sophistication in the use of composite units is consistent with the notion that students develop Component 2, structuring the tiling array, after Component 1.

Also related to Component 2, Outhred and Mitchelmore (2000) made an important observation about the difference between drawing and using physical shapes to tile the inside of a larger shape. In the case of squares tiling the inside of a rectangle, Outhred and Mitchelmore argued that because of the way that wooden squares fit together, using wooden squares to cover a rectangle “prestructures” the array—“the array structure is inherent in the materials and does not need to be apprehended by the learner” (p. 146). (They also cited research suggesting that such use of squares may not be effective instructionally.) When drawing, however, students must explicitly create an

appropriate structure. Furthermore, “the attempt to represent a covering in a drawing can help children to examine their experience in new ways and lead to new insights” (p. 165).

An episode from Battista et al. (1998) corroborates and extends Outhred and Mitchelmore's claim about material squares pre-structuring arrays. CS was asked to predict how many squares would cover the rectangle shown in Figure 19.18a (making her prediction without drawing). CS pointed and counted as in Figure 19.18b, predicting 30. So clearly she was having difficulty properly structuring the array. When checking her answer with plastic squares, she correctly covered the rectangle with squares. So, consistent with Outhred and Mitchelmore's claim, the placement of the squares was structured by the physical materials. However, CS initially enumerated her correctly placed squares as shown in Figure 19.18c, getting 30, then counted the squares again, first getting 24, then 27. So, although CS was able to correctly place physical squares to cover the rectangle, she was still unable to structure the squares so that they could be correctly enumerated—CS had not mentally constructed the structuring contained in the physical materials.

Let me now address Outhred and Mitchelmore's claim that drawing tilings can help students examine their experience in new ways, leading to new insights into the structure of an array. This claim is consistent with the theory that spatial structuring is a matter of abstracting one's actions on an array, not directly abstracting the array itself. That is, structuring is a reflective not empirical abstraction. However, to gain additional insights from one's drawing, more than abstraction of one's structuring actions is required. For example, typical of young students, CS not only drew an inadequately structured covering as shown in Figure 19.18d, she failed to recognize the inadequacy of her structuring (Battista et al., 1998). Thus, drawing may not promote better structuring than placing

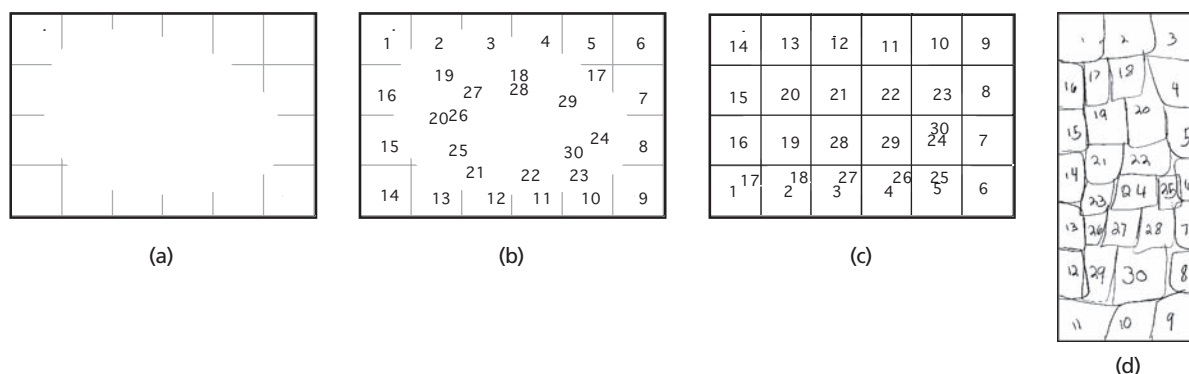


Figure 19.18 CS's structuring of arrays of squares (Battista et al., 1998).

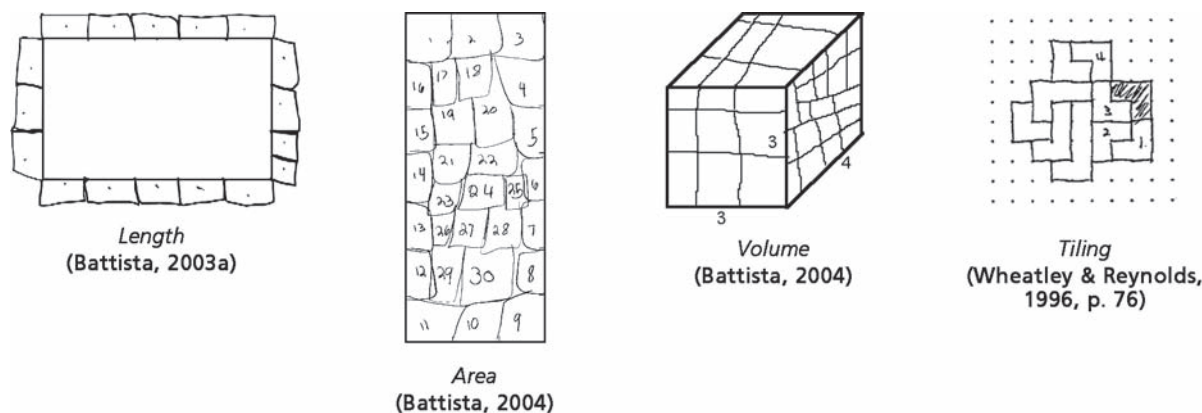


Figure 19.19 Stage 1 of the process of abstracting unit iterations.

physical squares unless students recognize difficulties in their drawings, creating cognitive perturbations. With proper instructional guidance, such perturbations can occur and cause refinements in students' structuring. That is, making drawings and checking drawings using physical squares can cause students to reflect on and refine the structuring contained in their mental models of arrays. In fact, an analogous "predict and check" approach was shown to be very effective in promoting the development of students' structuring and enumeration of 3d arrays of cubes (Battista, 1999). This approach is effective because it focuses students' attention and analysis, not on arrays, but on their *mental models* of arrays.

SYNTHESIS

Cognitive Processes Underlying Geometric Measurement: The Many Faces of Abstraction

Several kinds of abstractions are critical in geometric measurement.

Abstracting the Attribute

First, students must abstract the attribute to be measured (say, length), distinguishing it from all the other spatial attributes possessed by an object. This might happen through reflection initiated by (a) dealing with various kinds of problematic situations (e.g., comparing objects, fitting an object through an opening) or (b) interacting with people who already possess the concept and draw attention to it. Initially,

what students abstract during these experiences may not match the formal concept. However, during these or subsequent experiences, students might notice inadequacies in their conceptualizations, causing perturbations to occur. These perturbations may cause them to refocus their attention and abstract conceptualizations of the attribute that are close enough to the formal attribute that social interactions or problem-solving contexts do not cause further perturbations.

Abstracting Unit Iterations

The process of abstracting unit iterations sufficient to measure various geometric attributes is extremely complex. But, based on current research, we might conjecture the following stages in the abstraction process.³⁴

Stage 1, incomplete abstraction of unit iterations.

Because units and iterations are abstracted at most at the internalized level, iterations are uncoordinated and have gaps, overlaps, and nonequivalence of units. The student work shown in Figure 19.19 illustrates this level for the contexts of length, area, volume, and two-dimensional tiling.

Stage 2, units interiorized and coordinated. To meaningfully and correctly perform unit iteration, students must, at the very least, develop an interiorized mental model *of the unit*. This mental model incorporates relevant elements of the geometry of the unit and permits *coordinated* mental manipulation of the unit so that gaps, overlaps, and nonequivalence are eliminated in the iteration process.

Stage 3, structure of iterations interiorized. Students develop interiorized mental models of the *structure of unit iterations* that make it possible (a) to see a particular unit in relation to the iterated sequence of units, enabling

³⁴ Each step described below must be properly coordinated with numerical operations.

students to understand a unit's location; (b) to structure iterations in terms of composite units; and (c) to establish a part-whole relationship between the iterated units and the whole by reconstituting the whole as the sequence of iterations of the unit (Barrett & Clements, 2003; Steffe, in press; see also Cobb et al. 2001).^{35,36}

Stage 4, numerical measurements become symbols. Mental models of structured unit iterations are raised to the second level of interiorization so that the resulting enumerations become symbols. That is, numbers, construed as measurements, act as symbols for the unactualized process of iteration. This enables students to meaningfully reason about measurements without iterating units.

Integrating Perspectives on Length, Area, and Volume Measurement

Because individual research studies usually have focused on only one type of geometric measurement at a time (length, area, or volume), research has not yet produced a comprehensive theory of geometric measurement. Battista (2003b, 2004) integrated models for the development of area and volume measurement but did not extend this integrated model to length. However, comparing Battista's levels for length to those for area/volume suggests directions and issues for integration. In particular, the question of how nonmeasurement and measurement reasoning are related must be addressed. That is, Battista's levels for area/volume address only one component of understanding these concepts—enumerating units. Nonmeasurement ideas are not addressed, nor is the critical relationship between nonmeasurement and measurement reasoning. A first attempt at developing a general theory of geometric measurement is shown in Table 19.1 (which, at this point, can serve only as a conjecture). In this theory, nonmeasurement and measurement reasoning develop in parallel (though not simultaneously). However, in contrast to this parallel-development hypothesis, Kordaki and Potari (2002) argued that first, students should spatially explore areas; second, they should physically or pictorially transform and compare areas without the use of numbers; third, they should use measurement units; and finally they should use area formulas. Much more research is needed to tease out the interrelationships between nonmeasurement and measurement reasoning.

Final Comments on Measurement

Although students' measurement difficulties are alarming because measure is essential for most real-world applications of geometry, these difficulties may represent only the tip of a huge learning-difficulty iceberg—an iceberg that research has not yet investigated. Indeed, given the pervasiveness of measure in geometric and graphical contexts, poor understanding of measure might be a major cause of learning problems for numerous advanced mathematical concepts.³⁷ For instance, students' learning of high school geometry and analytic geometry is endangered by poor understanding of geometric measure, especially by misunderstandings of length. Furthermore, given students' disconnection between numerical measurements and the process of unit iteration, researchers must question the depth of students' understanding of coordinate systems, function graphing, locus problems, and theorems about lengths or distances.

Specific examples of these endangered topics abound: (a) Comprehending common proofs of the Pythagorean theorem requires understanding length and area and the distinction between the two. (b) Hollebrands (2003) found high school students unable to generalize their conceptualizations of linear magnitude in a way that enabled them to construct the notions of vectors and translations defined by vectors. (c) Almost half of a set of very advanced eighth graders and more than 80% of above average seventh and eighth graders had great difficulty conceptualizing common geometric shapes (rectangles and squares) in terms of measurement-based properties (Battista, 1998a). (d) A number of researchers have documented students' difficulties in understanding the connection between linear equations and their graphs (Dugdale, 1993; Knuth, 2000; Moschkovich, Schoenfeld, & Arcavi, 1993). Knuth, for instance, found that students failed to recognize that the coordinates of graph points used to calculate the slope of a line are solutions to the equation of that line, a finding that could be directly traceable to an insufficiently developed conceptualization of how coordinates are related to lengths. That is, students might see coordinates not as lengths but as markers on a map (just like margin letters on road maps).

³⁵ It is unlikely, however, that a–c occur at the same time.

³⁶ Structuring unit-length iterations increases in complexity when paths are situated in 2d space. For instance, Barrett and Clements (2003) found that maintaining unit lengths was more difficult along nonstraight than straight paths.

³⁷ The notion of poor understanding of geometric measure endangering later mathematics learning was developed in collaboration with Jack Smith.

Table 19.1 Reasoning About Measurable Geometric Quantities

Nonmeasurement Reasoning	Measurement Reasoning
1. Holistic Visual Comparison of Shapes Students' reasoning is appearance-based and holistic. It is appearance-based because students focus strictly on the appearance, or "how things look," of objects. It is holistic because students focus on whole shapes or objects, not parts within shapes. Students' strategies are imprecise and often vague. Conservation and transitivity gradually develop. 2. Visual Comparison of Shapes by Decomposing/Recomposing Students decompose and recompose spatial regions into smaller regions either physically or mentally, then compare the recomposed regions directly. There is no thought about enumeration of units or measured quantities, so no measurement reasoning occurs. When decomposing shapes, students might compare recomposed whole shapes, or they might compare shapes region-by-region. 3. Comparison of Shapes by Property-Preserving Transformations/Decompositions Students compare shapes by decomposing them then sliding, turning, or flipping shape parts in a way that allows them to infer, based on shape or transformation properties, that one transformed shape is congruent to another. Students do not necessarily explicitly mention geometric properties (congruence of shapes, segments, angles) or transformations (slide, flip, or turn), but their reasoning is consistent with such properties and transformations.	1. Use of Numbers Unconnected to Unit Iteration Students use counting to determine measures; however, their counting does not represent the iteration of an appropriate unit. 2. Unit Iteration and Enumeration Students attempt to iterate and enumerate appropriate units of measure to fill the shape being measured. At first, because students do not properly coordinate these units with either each other or the whole object being measured, their iterations contain gaps, overlaps, or different units—students often lose track of units while enumerating. Gradually, students overcome these difficulties. <i>Cognitive Milestones</i> • Units properly maintained, coordinated, and located • Units properly organized into composites • Iterations properly structured • Operating on iterations • Iteration schemes generalized to fractional units • Iteration schemes generalized to non-segment-/square-/cube units 3. Operating on Numerical Measurements Students numerically operate on measurements <i>without iterating units</i>. At this level, iteration seems to have fallen in the background. It is as if iteration is taken as being done.
4. Integrated Measurement and Nonmeasurement Reasoning Students inferentially operate on measurements <i>without iterating units</i>. They can make complex, property-based visual inferences about measurements, which are often accomplished with transformations or by using properties of geometric shapes. At this highest level of measurement reasoning, students fully integrate and apply the processes from Nonmeasurement Level 3 with their measurement reasoning. The difference between Nonmeasurement Level 3 and Measurement Level 4 is that in level 4 students make inferences about numerical measurements, not shapes. For instance, students can develop, meaningfully use, and justify procedures or formulas for finding the areas/volumes of nonrectangular shapes.	

CONCLUSIONS

Despite geometry's importance in mathematical theory and application, students continue to have difficulty learning it with genuine depth. There is much confusion and conceptually shallow, procedural learning. The research described in this chapter discusses in great detail many student difficulties, often suggesting avenues for instructional improvements. But much additional research is needed that delves even more deeply into students' difficulties and carefully examines the effects and effectiveness of alternate instructional approaches.

Moving Beyond the Cognitive Perspective

This chapter has discussed a wide range of research relevant to understanding students' geometric

reasoning. Because of the major research foci in this domain, the research has been both heavily phenomenological—how students construct meaning for geometric concepts—as well as cognitive—what cognitive processes are necessary for students' construction of geometric meaning. Far fewer studies have examined geometry learning from a sociocultural perspective. This perspective would focus on how the geometric meanings students construct are affected by their participation in various communities in which certain mathematical practices, interaction patterns, conceptions, beliefs, and reasoning are shared.

The focus of geometry research on cognitive rather than social processes has been both good and bad for the field. On the positive side, the cognitive approach has produced significant understandings of the development of students' geometric reasoning. Although there is still much to understand about the cognitive processes that underlie geometric reasoning, the cog-

nitive approach not only is critical for adequately investigating these processes, it helps connect the field to important related fields such as psychology. It is critical that this approach continue. However, on the negative side, not enough has been done to understand socio-cultural and affective factors in geometry learning. Researchers do not know nearly enough about how various components of social practice affect students' construction of geometric concepts and reasoning. For instance, what kinds of classroom practice effectively encourage students to personally embrace (rather than mimic) increasingly sophisticated types of justification in geometry, and what social processes enable such progress? (See Borrow, 2000 and Quinn, 1997 for discussions of this issue that combine social and cognitive perspectives.) We also need additional research on how affective factors are interrelated with geometric learning. For instance, how do students "feel" about being required to give formal proofs when they understand neither the need for such proofs nor their logic, and how do these feelings affect students' cognition and learning? Of course, in all but classes dedicated specifically to geometry, classroom practices generally transcend specific mathematical topics; so investigating some of these issues from a geometric perspective may be difficult. However, such investigations are possible (e.g., Quinn, 1997).

Another sociocultural aspect that has been neglected, but not ignored (e.g., Battista, 1999), includes ways that face-to-face interactions in collaborative small-group work affect geometry learning. For instance, how do students make sense of ideas communicated by other students when those ideas are (a) discrepant from their own views or (b) consistent with their views but expressed in different language or reasoning? In class discussions and small-group work, how do vague and inadequate verbal descriptions of spatial ideas (which are inevitable) affect student learning and participation in the classroom culture? What are the social dynamics of small-group work and how do they affect students' geometric thinking and knowledge construction? How does classroom discourse affect students' geometry learning? Thus, although the current cognitive approach to geometry research should continue, numerous important questions should be investigated from other perspectives.

The Attraction of Computer Environments

A great deal of modern research on geometry learning and teaching has focused on the use of computer environments. Why have geometry teaching and research embraced technology more enthusiastically perhaps than any other area of mathematics edu-

cation? First, many geometry computer environments are interesting to students. For instance, elementary students often get really excited when first exposed to Logo or the GSP *Shape Makers*. But the pedagogical and research interest in these environments goes well beyond motivational factors. Environments such as Cabri and GSP are tools that enhance the process of doing geometry for everyone, not just students. That is, DGEs allow individuals to explore geometric ideas in ways that are very different, and arguably better, than explorations with paper and pencil. DGEs significantly extend the ability to examine large sets of precisely drawn examples and to infuse dynamic motion into investigations. So it is no wonder that researchers spend so much time investigating the use of this new and powerful tool in instruction. And, as has been discussed earlier in this chapter, this research is getting progressively more sophisticated and providing major insights in geometry learning and teaching.

However, researchers still need to better understand how these technological enhancements affect students' learning. For instance, even though numerous studies show that DGEs can encourage and support significant learning in students, we do not know the effects of scaling up this use, and we do not know how, in very precise terms, learning in DGEs differs from learning in paper-and-pencil environments. Thus we need more comparison studies, both quantitative, to investigate generality, and qualitative, to investigate differences in cognitive processes.

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RESEARCH IN PROBABILITY

Responding to Classroom Realities

Graham A. Jones

GRIFFITH UNIVERSITY, GOLD COAST CAMPUS

Cynthia W. Langrall and Edward S. Mooney

ILLINOIS STATE UNIVERSITY

There is a broad consensus that the teaching of probability must begin early in order to build sound intuitions and that such efforts must be grounded in experience. ... However, mere experience is not sufficient, as the research on faulty intuitions and fallacious reasoning amply shows, and a major challenge to the field is harvesting what is known from this research to inform teaching.

—Greer & Mukhopadhyay, 2005, pp. 314–315

Greer and Mukhopadhyay's caveat concerning the importance of research in the teaching of probability is timely as we look back more than 16 years since statistics and probability was introduced as a mainstream strand in the school mathematics curriculum. Consistent with curriculum reforms of the period (e.g., National Council of Teachers of Mathematics [NCTM], 1989, 2000), Greer and Mukhopadhyay acknowledge the need to commence probability experientially in the early grades while highlighting the problematic nature of children's probabilistic reasoning—a picture that is replete in the literature. Lastly, their statement inspires educators to look more deeply at the probability research in order to create effective environments for probability learning. This latter suggestion resonates with one of the key intents of this chapter: to inform the teaching and learning of probability by synthesizing research in the field, especially research that has been published since Shaughnessy's (1992) review of stochastics in the first *Hand-*

book of Research on Mathematics Teaching and Learning (Grouws, 1992).

The period since Shaughnessy's review (1992) has been a time of curriculum reform in mathematics for the United States (NCTM, 1989, 2000) and for other countries (e.g., Australian Education Council [AEC], 1991; Department of Education and Science and the Welsh Office [DES], 1991). Among other elements of change, national documents like the *Curriculum and Evaluation Standards for School Mathematics* (NCTM, 1989) harbored the emergence of probability and statistics as a mainstream strand that commenced in the primary grades and continued through the high school years. The pervasive and coherent emergence of statistics and probability in curriculum documents and the potential for it in classrooms at all grade levels has had a profound effect on research in statistics and probability education during this period. The curriculum movements created a refreshing urgency for probability research, and much of the research in

this period has been designed to support perceived needs in classroom teaching and learning or has been stimulated by classroom experiences that manifest the challenges faced by researchers, teachers, and students in dealing with “knowledge of uncertainty.”

Although this *Handbook* devotes separate chapters to research on statistics and probability education, it is critical that we recognize not only the mathematical interrelatedness of the two areas but also the interrelatedness of research in statistics education and research in probability education. Hence we will outline some of the key conceptual bridges between statistics and probability before turning to our main task of reviewing and analyzing research in probability education during the period since Shaughnessy’s (1992) review.

The study of statistics and probability focuses on real-world phenomena that involve uncertainty, that is, cannot be predicted with certainty. For example, a statistician involved in health-related research may be considering the question, “Which of the available medications is preferable for treating the common cold in young children?” When a statistician addresses such a problem, he or she collects, organizes, analyzes, and interprets data (Wild & Pfannkuch, 1999). This data might be the number of days each of the medications takes to remedy the common cold. As such, the data are influenced by both the effect of the medication (causal variation) as well as chance variation or random effects associated with the medications or the subjects. As Moore (1990) explained, “‘Random’ is not a synonym for ‘haphazard,’ but a description of a kind of order different from the deterministic” (p. 98). It is an order that is not attributable to a specific cause or chain of events (determinism) but one that operates in a phenomenon that has uncertain individual outcomes but a regular pattern of outcomes over many repetitions (e.g., the number of days to cure a common cold for many repeats of the same medication on a particular subject).

These random effects or random processes lie at the very heart of statistics and probability. On the one hand, *statistics* uses the characteristics of random processes and probability models of such processes to make inferences about problems involving data; for example, the most preferred medication for the common cold. On the other hand, *probability* focuses directly on describing, quantifying, modeling, and illuminating random processes. For example, one or more approaches to probability measurement may be used to estimate the probability or likelihood of an event such as “medication A cures the common cold within 5 days or less.”

This brief outline of statistics and probability highlights the close connection between the two areas. It also underscores the value of undertaking research in

probability, both because of its complementary role in informing statistics and its special place in describing and quantifying the numerous chance phenomena that operate in our world. Accordingly, in writing this chapter we will concentrate on research into the learning and teaching of probability; however, where appropriate we will make connections between ideas in probability and statistics and timely references to research that has addressed commonalities between the two areas. We begin by briefly revisiting research on probability education in the decades prior to the current period.

SETTING THE STAGE: THE EARLIER REVIEW OF PROBABILITY RESEARCH

In Shaughnessy’s (1992) review he stated that, up to that point, there had not been much involvement by North American mathematics educators in research on the teaching and learning of probability. He observed that, “Since very little probability or statistics has been systematically taught in our schools in the past, there has been little impetus to carry out research on the problems that students have in learning it” (p. 465). Although it would be difficult to take issue with the overall thrust of Shaughnessy’s statement, a number of North American mathematics educators, in the period from the 1950s to the late 1980s, acted as trailblazers for teaching and learning research on probability (e.g., Davies, 1965; Doherty, 1965; Garfield & Ahlgren, 1988; Konold, 1983, 1989; Leake, 1965; Leffin, 1971; Mullenex, 1968; Page, 1959; Shaughnessy, 1977; Shepler, 1970). These North American researchers, like those who charted the way in Europe and other places (e.g., Engel, 1966; Falk, 1981, 1988; Fischbein, 1975; Fischbein & Gazit, 1984; Gillis & Haraud, 1972; Green, 1979, 1983; Jones, 1974; Steinbring, 1984; Varga, 1969a, 1969b) built their research largely on the theoretical models of cognitive-development psychologists like Piaget (e.g., Piaget & Inhelder, 1951/1975) or the work of other psychologists who were interested in people’s learning behavior and decision making when faced with situations involving uncertainty (e.g., Kahneman, Slovic, & Tversky, 1982; Offenbach, 1965; Siegel & Andrews, 1962; Stevenson & Zigler, 1958; Tversky & Kahneman, 1974).

During this time frame, psychologists were interested in probability from a cognitive, epistemological, or decision-making perspective, while mathematics educators harbored strong sentiments that their research had implications for the learning and teaching of probability and an earlier and more extensive inclusion of it in the school mathematics curriculum.

Fischbein, Pampu, and Mînzat (1970) echoed this view in the following conclusion from their research:

In any case, the finding that 9- to 10-year-old Ss [subjects], after a brief instruction, became able to perform chance estimates by comparing numerical ratios and to understand the concept of proportionality might be an argument for starting to teach probabilities while children are still in primary school. (p. 388)

Other early advocates such as Engel (1966), Page (1959), and Varga (1969a, 1969b) reflected similar views to Fischbein's as a result of their own research and their development of exploratory curriculum programs in probability for elementary, intermediate, and secondary students. Steinbring (1984, 1991) even developed an instructional theory that took cognizance of the nature of probability and its different perspectives. However, the efforts of these and other mathematics education researchers in the period prior to Shaughnessy's review (1992) went largely unheralded, as probability received almost no attention in elementary and middle schools and limited exposure in secondary schools. Moreover, when it was taught in secondary schools the focus was largely on "balls-in-urn problems," that is, combinatoric applications.

The protracted gestation finally ended when probability and statistics emerged as a major strand in several national curriculum documents (e.g., AEC, 1991; DES, 1991; NCTM, 1989). Although mathematics educators who had undertaken research on probability thinking and learning in the 3 decades prior to the emergence of these curriculum documents may have felt frustration with the delay, they can take some solace in the fact that knowledge about the teaching and learning of probability was already available before these curriculum programs were introduced across the broad spectrum from kindergarten through secondary grades. There is little doubt that this prior research influenced national curriculum documents in a number of ways: how the aims and "big ideas" in probability were incorporated, how the big ideas were articulated through the grades, and how the design of instruction provided a scaffold for learning.

The research in this prior period also provided an infrastructure and a lighthouse for probability research undertaken during the current period. This infrastructure included a problematique, potential research questions, conceptual frameworks, research designs, assessment instruments, and data-analytic techniques. Notwithstanding the value and significance of this prior research, our thesis is that the research on probability education during the last 15 years has been inescapably driven by the goals and reform directions reflected in

early national curriculum documents (e.g., AEC, 1991; DES, 1991; NCTM, 1989). More specifically, we claim that documents like the *Curriculum and Evaluation Standards for School Mathematics* set in train both general directions for mathematics learning and teaching and particular directions for the content and pedagogy associated with probability and statistics. These reform directions and their influence on schools acted as a catalyst for much of the research on probability education that we will review. In fact, gleanings support from the reform documents' espousal of socioconstructivist (Cobb & Bauersfeld, 1995) orientations to learning and instruction that is grounded in students' intuitions (NCTM, 1989), the research agenda has highlighted the profiling of probabilistic thinking that students bring to the classroom and the development of students' individual and collective probabilistic reasoning during various kinds of instruction.

Because of their importance for probability research in the period since Shaughnessy's 1992 review, an examination of some *national curriculum documents* and their impact on probability research will constitute the first of our five goals in writing this chapter. As our second goal, we will discuss and appraise the key research on probability *learning and teaching* during the period and will make connections where appropriate between the research of this period and that of the previous one. In identifying this learning and teaching research we have endeavored to take an international perspective by capturing research from different parts of the globe. Third, under *curriculum research*, we examine the notion of "probability literacy" and its implications for curriculum. Indeed, if the implementation of probability is to be sustainable, research on probability literacy may well be critical. Fourth, we will present *reflections* that evaluate the research activities of this period against the "wish list" for this period that Shaughnessy foreshadowed in the 1992 review. Fifth and finally, mindful of the refinements espoused for probability in more recent curriculum documents like the *Principles and Standards for School Mathematics* (NCTM, 2000), we will create our own agenda that hopefully will serve as a benchmark for future endeavors in probability research and for the next review.

NATIONAL CURRICULA: A BACKDROP FOR PROBABILITY RESEARCH

As we have observed, the advent of national curriculum documents in the late 1980s and early 1990s spawned statistics and probability as a mainstream strand within the mathematics curriculum. In order to expose this

new mainstream strand and its potential for impacting research in this period, we will examine briefly key elements of probability in the NCTM *Curriculum and Evaluation Standards for School Mathematics* (1989) and in curriculum documents from Australia and the United Kingdom respectively: *A National Statement on Mathematics for Australian Schools* (AEC, 1991) and *National Curriculum: Mathematics for Ages 5–16* (DES, 1989). We chose these latter two curriculum documents because they appeared at almost the same time as the NCTM *Standards* and, with it, seemed to set the scene for much of the curriculum development in probability and statistics worldwide.

In surveying probability in these three curricula documents we will focus on the content and pedagogical directions at the elementary, middle, and high school levels. Although these curriculum documents were revised in the decade following their initial publication (e.g., AEC, 1994; DfEE, 1999; NCTM, 2000), our survey will concentrate on the original documents as they were the first to impact researchers interested in the teaching and learning of probability.

The Probability Curriculum in the Elementary Grades

Table 20.1 presents the “big ideas” in probability that were introduced at the elementary school level

(approximately Grades K through 6) in the three curriculum documents (AEC, 1991; DES, 1989; NCTM, 1989). In all three countries, elementary students were expected to gain an understanding of *chance* (uncertainty) and *random* events through experimentation with simple probability generators (e.g., dice, coins, and spinners) and in some cases through cultural activities (e.g., rainfall, traffic, weather). Students were to use language associated with chance (e.g., *equally likely*, *fair*, and *unfair*) and to recognize the possibility of several outcomes. However, at this level, understanding of randomness did not extend to long-term patterns.

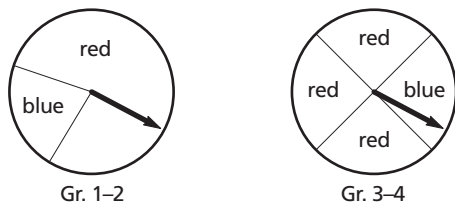
Each of the three countries also incorporated elements of both the *frequentist* (experimental or *a-posteriori*) and the *classical* (theoretical or *a-priori*) approaches to probability measurement. With respect to the frequentist approach (von Mises, 1928/1952), the probability of an event is defined as the ratio of the number of trials favorable to the event to the total number of trials. In the classical approach (Laplace, 1825/1995), the probability of an event is determined by the ratio of the number of favorable outcomes to the total number of outcomes, where outcomes are assumed to be equally likely. Hence, the frequentist approach deals with experimental trials and statistical or *a-posteriori* evidence whereas the classical approach deals with numerical or geometrical measures and *a-priori* assumptions of symmetry or equal like-

Table 20.1 Probability for the Elementary Grades

Country	Big ideas in probability
Australia	• use, with clarity, everyday language associated with chance events (<i>very likely</i>, <i>unlikely</i>, <i>more likely</i>, and <i>equally likely</i>) and use appropriately terms like <i>chance</i> and <i>probability</i> • list systematically possible outcomes for familiar random events including one-stage experiments and some two-stage experiments; deduce the order of probability of the outcomes and test predictions experimentally • place outcomes for familiar events and one-stage experiments in order from those least likely to happen to those most likely to happen (events arising from random generators like the tossing of a coin and from social situations like the weather) • make and interpret empirically based predictions about simple situations (e.g., a drawing pin falling point up or the risk of accidents per 1,000 km traveled for different modes of travel)
United Kingdom	• recognize a degree of uncertainty about the outcomes of some events and that other events are certain or impossible; recognize the possible outcomes of random events (events arising in social situations and from probability generators such as coins and dice) • place events in order of “likelihood” and use appropriate words to identify the chance (of such events); distinguish “fair” and “unfair” • list all the possible outcomes of an event including the outcomes of two combined events that are independent; use a probability scale from 0 to 1; give and justify subjective estimates of probabilities • distinguish between estimates of probabilities based on statistical evidence and those based on assumptions of symmetry; know that different outcomes may result from repeating an experiment and that if each of n events is assumed to be equally likely, the probability of one occurring is $1/n$
United States	• begin to recognize the nature of random processes and explore concepts of chance through games and experiments • use concepts like <i>certain</i>, <i>uncertain</i>, <i>likely</i>, and <i>luck</i> • compare the likelihood of events theoretically and experimentally using both qualitative and quantitative descriptors

likelihood. Although references to these two approaches at the elementary school level are largely nascent, all three curriculum documents make comments like the following: order the outcomes by probability and test predictions experimentally (AEC, 1991), distinguish between estimates of probabilities based on statistical evidence and those based on assumptions of symmetry (DES, 1989) and compare the likelihood of events both theoretically and experimentally (NCTM, 1989).

Not only do the three curriculum documents encourage the use of both frequentist and classical approaches, they also encourage students to make connections between the two. The NCTM *Standards* (1989, p. 56) provided a typical example of the kind of problem that young children could be expected to explore with respect to the two approaches to probability measurement:



Is red or blue more likely? How likely is yellow? How likely is getting either red or blue? If we spin twelve times, how many blues might we expect to get?

The third approach to chance measurement, *subjective probability*, describes probability as a degree of belief, based on personal judgment and information about an outcome. Thus it depends on two things: the event whose uncertainty is contemplated and the knowledge of the person estimating the probability (Batanero, Henry, & Parzys, 2005; Lindley, 1980). In the three elementary curricula children have opportunities to use qualitative (subjective) probability descriptors such as certain, more likely, less likely, and impossible; however, the documents are largely silent on the kind of knowledge base students are expected to use in making such judgments. Some evidence (e.g., Huber & Huber, 1987; Metz, 1998a) suggests that children can use knowledge or beliefs about levels of confidence in dealing with subjective probability.

Although it is not included in Table 20.1, the curriculum documents also focused on teaching aspects of probability. In particular the documents advocated a problem-solving focus, tasks that resonated with children's experiences, and approaches that revealed students' intuitions and confronted their misconceptions. Except for the last approach, the pedagogical directions were largely generic in that they espoused the visions in the mathematics curricula as a whole.

The Probability Curriculum in the Middle Grades

The big ideas in probability for the middle grades (roughly Grades 7 through 9) are presented in Table 20.2. All three curriculum documents focused strongly on having students use numerical probabilities; however, there is still continuing consideration of random processes emphasizing both their short-term unpredictability and their long-term stability. The stability aspect was generally considered in association with experimental (empirical) probability.

The strongest feature in all three middle school curricula is the focus on both the classical and the frequentist approaches to probability. Building on the informal start in the elementary grades, the curricula highlight the determination of theoretical and experimental probabilities and also foreshadow the relationship between the two. For example, the United States document uses the phrase, "by comparing experimental results with mathematical expectations" (NCTM, 1989, p. 109). In dealing with the classical (theoretical) approach to probability, all three documents refer to sample space and the systematic listing of outcomes in simple and multistage experiments. There is also a strong thrust on modeling and simulation of real-world phenomena in the Australian and United States documents. This thrust is illustrated in the following problem (Figure 20.1) taken from the NCTM *Standards* (1989, p. 111). The problem is interesting in that it provides data for modeling and hence makes a nice connection between probability and statistics.

Joan Dyer Data and Questions		
Home runs	9	<i>The table gives the record for Joan Dyer's last 100 times at bat during the softball season. She is now coming up to bat. Use the data to answer the following questions:</i> • What is the probability that Joan will get a home run? • What is the probability that she will get a hit? • How many times can she expect to get a walk in her next 14 times at bat?
Triples	2	
Doubles	16	
Singles	24	
Walks	11	
Outs	38	
Total	100	

Figure 20.1 Softball data problem.

With respect to subjective probability, the Australian and United Kingdom documents relate it to social contexts. For example, students are expected to explore the odds appearing in gambling situations

Table 20.2 Probability for the Middle School Grades

Country	Big ideas in probability
Australia	• understand and explain social uses of chance processes (e.g., investigate the use of probability in insurance, gambling, and games) • construct sample spaces to analyze and explain possible outcomes of simple experiments and calculate probabilities by analysis of equally likely events (including one-, two-, and three-stage experiments) • estimate probability using the long-run relative frequency (i.e., empirical probabilities) • model situations and devise and carry out simulations
United Kingdom	• understand and use relative frequency as an estimate of probability • appreciate, when assigning probabilities, that relative frequency and equally likely considerations may not be appropriate and “subjective” estimates need to be made • understand and apply the addition of probabilities for mutually exclusive events • understand that when dealing with two independent events, the probability of them both happening is less than the probability of either of them happening (unless the probability is 0 or 1) • calculate the probability of a combined event given the probability of each of two independent events and illustrate combined event probabilities of several events using tabulation or tree-diagrams
United States	• model situations by devising and carrying out experiments or simulations to determine probabilities • model situations by using a sample space to determine probabilities • appreciate the power of using a probability model by comparing experimental results with mathematical expectations • make predictions that are based on experimental or theoretical probabilities • develop an appreciation for the pervasive use of probability in the real world

(e.g., horse racing) and to recognize that these odds reflect statements of subjective probability (AEC, 1991, p. 175). Moreover, it is also noted that in some situations only subjective estimates of probability are possible; for example, the probability that a cure for cancer will be found in the next 2 years cannot be based on data or *a-priori* assumptions of equal likelihood (DES, 1989, p. 44).

Once again the three documents make limited reference to pedagogical insights that refer specifically to probability. However, reference is made to the fact that “students have many misconceptions and poor intuitions about probabilistic situations” (NCTM, 1989, p. 110), and hence teachers should encourage students to make conjectures (based on these intuitions) and test these conjectures by experimentation. In addition, the emphasis on real-world modeling and experimentation in the probability sections of the curricula is consistent with the prime place given to problem solving throughout the three curriculum documents.

The Probability Curriculum in the High School

Table 20.3 presents an overview of probability in the three curriculum documents for the high school

years (approximately Grades 10 through 12). It reveals the big ideas intended for students who take *core* programs in mathematics. All documents note that some students will take specialist mathematics courses that may include more rigorous work in probability.

The study of random processes continues in the high school years but is more implicit than explicit in the United Kingdom (DES, 1989) and the United States documents (NCTM, 1989). The fine detail of the Australian document (AEC, 1991, p. 182) actually suggests having students examine “run lengths” as a means of expanding their understanding of randomization. Developing knowledge and understanding of sample space and probability measurement involving compound events is a growing feature of the three curriculum documents at this level. The emphasis is almost totally on classical (theoretical) and frequentist (experimental) approaches to probability measurement; only the Australian document alludes to subjective probability. Modeling real-world problems using simulation to estimate probabilities is a strong feature of the Australian and United States programs.

The major new conceptual developments at this level are the inclusion of random variable and probability distribution. All documents focus on the notion of probability distribution and also consider some

specific probability distributions such as the normal distribution. Although probability distributions play a key role in comparing populations and in making statistical inferences, we will not review research in this area, leaving it to the chapter on statistics (Shaughnessy, this volume).

With respect to pedagogical considerations, the documents recommend that the work should continue to be practical and investigative. Teachers are expected to encourage students to use both theoretical and experimental approaches, especially those involving simulation.

Key Ideas for Research Emanating from the National Curriculum Documents

In undertaking this brief review of three national curriculum documents, we have unashamedly focused on the commonalities of the probability programs rather than their differences. Notwithstanding this perspective, the big ideas in probability in the three curriculum documents are remarkably consistent in scope and evolution from the elementary through the high school grades. We summarize these big ideas below but, in passing, observe that their presence in all three national curriculum documents adds further grist to the argument that researchers across the international arena were influenced in comparable ways. Further, researchers interacted with school classrooms that advocated similar pedagogy.

With respect to probability content, the big ideas that have emerged in the three curriculum programs are the *nature of chance and randomness*, *sample space*, *probability measurement* (classical, frequentist, and subjective), and *probability distributions*. Although many of these conceptual ideas were the subject of research in the period prior to Shaughnessy's review (1992), they became more critical and better articulated during the current period as researchers and teachers grappled with the mainstream introduction of probability and statistics.

With respect to pedagogy, the curriculum documents adopted a number of general reform principles (e.g., problem solving) as well as pedagogical suggestions specific to probability. The major thrust in the pedagogy specific to probability dealt with children's misconceptions about chance and the need for teachers to confront these misconceptions through experimental activity and reflection. On the one hand these new directions highlighted the need for research concerning *teachers' knowledge of probability* and their knowledge about *student cognitions of probability*; on the other hand, they stimulated research on *instructional environments*.

RESEARCH INTO THE LEARNING AND TEACHING OF PROBABILITY: A TIME FOR INFORMING CURRICULUM AND INSTRUCTION

In the wake of our discussion of the national curriculum documents of the late 1980s and early 1990s, we

Table 20.3 Probability for the High School Grades

Country	Big ideas in probability
Australia	• use experimental and theoretical approaches to investigate situations involving chance and determine the likelihood of particular outcomes (including random variables) • generate, use, interpret, and investigate properties of probability models (including normal, binomial, geometric, and Poisson probability distributions) • use simulations to model uncertain events
United Kingdom	• produce a tree-diagram to illustrate the combined probability of several events that are not independent • understand the probability of any two events not happening • consider different shapes of histograms representing distributions (including the <i>normal</i> distribution) with special reference to their mean and dispersion
United States	• use experimental or theoretical probability, as appropriate, to represent and solve problems involving uncertainty • use simulations to estimate probabilities • understand the concept of random variable; create and interpret discrete probability distributions • describe, in general terms, the normal curve and its properties to answer questions about sets of data that are assumed to be normally distributed • apply the concept of random variable to generate and interpret probability distributions including binomial, uniform, normal, and chi square

will review research on the learning and teaching of probability during the current period. As noted in the previous section, with the advent of probability in the school curriculum, researchers and teachers required a better understanding of the probabilistic thinking that students of all school ages brought to the classroom. Moreover, there was an even more critical need for pedagogical knowledge that teachers could use to inform probability instruction.

Our review is organized into two parts. In the first, we present key themes of the research on probability thinking and learning, emphasizing the more recent research of the last decade while recognizing the earlier research on which it was built. In the second part, we highlight research on teachers' probabilistic knowledge and beliefs and research on instructional environments that support probability learning.

Research on Probability Learning

Although Shaughnessy's (1992) review characterized one component of earlier probability research as describing *how* people think, we would argue that the research of that period placed greater emphasis on describing *what* people thought. Reflecting the research of the time, the bulk of Shaughnessy's chapter on probability referred to studies about judgment heuristics, biases, and misconceptions in which most of the data were collected from students' forced-choice responses (p. 476). Apart from the developmental studies on probabilistic thinking and intuitions (e.g., Fischbein, 1975; Piaget & Inhelder, 1951/1975), few studies in that era used methodologies that probed students' reasoning or investigated the meanings students attributed to chance situations. For related reviews and analyses of the probability research of this and earlier periods we refer the reader to Borovcnik and Peard (1996), Jones (2005), and Kapadia and Borovcnik (1991).

Since the 1990s, however, research has followed the lead set by Piaget and Fischbein and has become more intent on addressing the ways in which people think about situations involving chance. The research has expanded beyond describing judgmental heuristics, biases, and misconceptions to include the cognitive development of a wider range of conceptual topics in probability. These conceptual topics are reflected in our analysis of the three national curriculum documents and include students' conceptions of chance and randomness, students' understanding of sample space as it relates to both simple and compound events, and students' thinking about probability measurement in its various forms: classical, frequentist, and subjective.

In the sections that follow, we review research on students' learning of these conceptual ideas, often

linking it back to the research of Piaget, Fischbein, and others of the earlier period. We also analyze several developmental models of probabilistic reasoning that can be used to inform instruction in probability. These models, developed in this period but standing on the seminal work of Piaget and Inhelder (1951/1975) and Fischbein (1975), presage our review of research on the teaching of probability.

Chance and Randomness

This section commences with a review of the extensive research on *chance*. The *Shorter Oxford English Dictionary* (Trumble, 2002) defined chance as "a possibility or probability: as distinct from a certainty" (p. 312). This usage of chance envisages it as a language for describing events that are characterized by unpredictability or possibility rather than certainty. Chance language is also used in measuring the likelihood of some event. For example, the likelihood of obtaining a "3" in the roll of a die can be referred to as a 1-in-6 chance (Gal, 2005). Consequently, chance has become synonymous with probability, albeit in an informal or even colloquial sense.

The Language of Chance

The language that students use to interpret situations involving chance provides insight into the variety of perceptions that this complex construct evokes. Such variety is illustrated by the work of Fischbein, Nello, and Marino (1991) with elementary (Aged 9–11) and junior high school (Aged 11–14) students in Italy. In a written assessment they posed two questions, each of which asked students to indicate whether the event of obtaining a given number was impossible, possible, or certain: (a) when tossing a regular die and (b) when using a uniform random generator with numbers from 1 to 90. The majority of students at both age levels adequately identified impossible, possible, and certain events; however, some students' reasoning revealed nonnormative use of chance language. In the excerpts below, students are identified by grade level (e.g., E4 represents elementary, Grade 4 and M2 represents junior high, Grade 2).

Will one obtain 31 [in (b)]?

"It is impossible", answers the child, "because the probability is very small" (E4).

"It is impossible, because among 90 numbers there is only one 31" (M1).

"It is certain because in the tombola [game] there is a 31" (E5).

Is it possible to get 5 when rolling a die?

"It is impossible . . . because it is only one probability among 6" (M3).

"It is impossible because someone believes that he will win and afterwards it does not come out" (M2). (p. 528)

These excerpts show the gallimaufry of meanings that students attribute to the concepts of possible, impossible, and certain. Green (1983), and more recently Moritz, Watson, and Pereira-Mendoza (1996) and Watson and Moritz (2003b), have described similar misinterpretations and nonnormative conceptions of these fundamental descriptors of chance phenomena.

Working in Israel, Amir and Williams (1999) examined students' conceptions of the term *chance*. They reported that the majority of the students held multiple meanings for the construct. The most common interpretation referred to chance as *something that just happens*. As one student explained, "It's like walking down the road and a stone fell out . . . and you fell over. That would be chance. Just happens by accident" (p. 97). Other interpretations included *unexpected* or *unusual* events—"100 tosses of a coin? It would be chance getting 100 out of 100, but possible to get 55 on one, 45 on the other" (p. 97).

When students quantify chance situations, evidence suggests that they misuse language. The most common example is use of the phrase *50–50* to describe the chance of a particular event. In situations in which an event is *possible*, students frequently use the expression *50–50 chance* to indicate the presence of uncertainty rather than a specific measurement of chance (Watson, 2005). In the excerpt below from Amir and Williams's study (1999, p. 101), observe the student's indiscriminate use of the phrase *50–50* to quantify the chance of any outcome that he considered probable.

I: What is the chance of getting a "4" on a 10-sided die?

J: Probable.

I: Could you give me the chances in a number form?

J: 50–50. Even chance of getting a 6 or a 4 or a 3.

I: What is the chance of getting a number bigger than 6?

J: Probable. 50–50.

I: And of getting an even number?

J: Probable. 50–50.

Tarr (2002) highlighted two distinct ways in which the *50–50* phrase is used inappropriately. One involves

the use of *50–50* chance to describe a set of more than two equally likely outcomes. This is evident in the excerpt above, when the student used *50–50* with an "even chance of getting a 6 or a 4 or a 3." The other misuse of the phrase involves the use of *50–50* chance to describe two outcomes that are not equally likely. In the excerpt above the student might have applied this reasoning by assigning a *50–50* chance to the two events *getting a number bigger than 6* and *getting a number not bigger than 6*. A generalized version of this reasoning, known as the *equiprobability bias*, refers to situations in which all events are treated as equiprobable, even when they are not. Use of the equiprobability bias has been exhibited by students at all levels of schooling and across various cultures (e.g., Batanero, Serrano, & Garfield, 1996; Konold, Pollatsek, Well, Lohmeier, & Lipson, 1993; Lecoutre, 1992; Li & Pereira-Mendoza, 2002; Moritz & Watson, 2000).

Language and beliefs also play an important role when students talk about "fair" probability generators. Watson and Moritz (2003b) used an interview protocol to investigate 108 elementary and middle school students' beliefs about the fairness of dice and how they would determine whether each of a number of given dice (one of which was "loaded") was fair. Watson and Moritz identified four hierarchical levels of student beliefs about the fairness of dice: (a) the dice are *unfair*, based on idiosyncratic reasons; (b) the dice are *fair*, no theoretical or experiential justification; (c) the dice are fair, qualified by referring to the rolling conditions or the physical manufacturing of the dice; and (d) the outcomes are *fair* in the long term, but *short-term variation* is recognized. The researchers also identified four strategies used by students to determine the fairness of dice: (a) relied on *intuitive and idiosyncratic* beliefs (e.g., luck); (b) asserted dice were *fair* but *no test* was necessary; (c) observed the *physical features*, cited *symmetry*, or carried out a few *unsystematic trials*; and (d) performed *systematic trials* of the dice using *small* and *large* samples to varying degrees. Interestingly, Watson and Moritz noted that the association between students' beliefs and their strategies for judging fairness of dice ($r = 0.28$) was statistically significant but not strong. With respect to the longitudinal aspect of their study, they used the same protocol in interviewing 44 of the participants 3 or 4 years later. The results showed that approximately 50% remained at the same level for both beliefs and strategies, almost 50% responded at a higher level, and a very small number responded at a lower level. The association between beliefs and strategies was almost the same as in the initial study. The authors concluded that student beliefs and strategies for determining fairness may produce dilemmas for teachers in using probability generators

like dice and in dealing with language, especially the use of the word *chance* in different contexts.

The diversity of interpretations of chance and the language used to describe it point to the difficulties many students have with the notion of uncertainty. Although there is evidence that children as young as 4 years of age exhibit some understanding of the concept of uncertainty and are able to recognize that some events cannot be predicted (Byrnes & Beilin, 1991; Fay & Klahr, 1996; Fischbein, 1975; Kuzmak & Gelman, 1986), there is considerable empirical data showing that students, as well as adults, overattribute deterministic or causal explanations to situations involving chance (Kahneman & Tversky, 1982; Konold, 1989, 1991b; Metz, 1998a, 1998b; Shaughnessy, 1992). Many misconceptions associated with probabilistic reasoning can be attributed to *causal* or *deterministic* modes of reasoning; that is, to reasoning that assumes that every state of affairs has a cause and that nothing can be attributed to chance or other contingencies (Batanero, Henry, & Parzysz, 2005; Rohmann, 1999).

Causal reasoning is one feature of the *outcome approach* that Konold (1989, 1991b) has introduced to explain the nonstandard reasoning frequently exhibited by school and college students in situations involving chance. For example, when asked to compare the probabilities of a head or tail after a sequence of four heads, students using outcome-oriented reasoning choose “tails” even though they are aware that heads and tails are equally likely. According to Konold, students predict the outcome of the next trial rather than considering probabilities. Moreover, their prediction is often based on a causal or deterministic perspective—“it’s tails’ turn.”

Konold (1989) argued that the use of a causal model for dealing with uncertainty is the most significant difference between novice and expert reasoning in chance situations. He concluded that “as long as students believe that there is some way they can ‘know for sure’ whether a specific hypothesis is correct, the better part of statistical logic and all of probability theory will evade them” (p. 92). This sentiment resonates with Fischbein’s (1975; Greer, 2001) contention that a cultural bias exists toward deterministic thinking. Such a bias may be nurtured by school experiences that emphasize causal explanations and *ipso facto* strengthen intuitions rooted in deterministic modes of thinking.

Characteristics of Random Phenomena

Random phenomena are characterized by having short-term unpredictability and long-term stability. Gal (2005), in considering short-term unpredictabil-

ity, identified two perspectives: (a) randomness is a *property* of an outcome, in the sense that a string of outcomes appears unordered; and (b) randomness is a *process* by which a string of outcomes occur and cannot be predicted according to some basic cause. With respect to the long-term stability of randomness, Moore (1990) highlighted the regular pattern associated with the relative frequencies of outcomes over many repetitions. For example, the event of tossing a coin is a random process because we cannot predict with certainty the properties of either a single toss or a sequence of tosses based on the physics of the coin. However, there is a long-term expectation that 50% of tosses will result in a head and 50% in a tail.

Research on students’ thinking about randomness has occurred over 4 decades and has focused largely on three facets: random mixtures, random distributions, and random selection. Because Piaget and Inhelder (1951/1975) have laid the foundation for research in this area, we present a brief overview of their research on random processes so as to provide a seamless transition into research from the current period. A fuller review of their work has been presented by Jones and Thornton (2005) and Langrall and Mooney (2005).

Piaget and Inhelder’s (1951/1975) research classified students’ conceptions of randomness into three stages of development: preoperational (4 to 7 years), concrete operational (8 to 11 years), and formal operational (beyond 11 years). To examine students’ understanding of *random mixtures* they used the familiar tilt box task; to study *random distributions* they simulated raindrops falling on squares of pavement; and to investigate students’ reasoning on *random selection tasks* they picked objects from a jar, flipped coins, or rotated spinners.

Piaget and Inhelder concluded that students in the *preoperational stage* did not recognize the random mixing of balls in the tilt box. Rather, they predicted regular change patterns and often expected the balls to return to their original position. These students typically placed raindrops evenly on the squares and demonstrated little of the irregularity that is a feature of randomness. On random selection tasks, preoperational students often predicted the most frequently occurring event but exhibited no regard for proportionality. In contrast, students at the *concrete operational stage* predicted a progressive mixing of the balls and also recognized that it was very unlikely that the balls would return to their original positions. Displaying greater understanding of randomness, they constructed irregular distributions that gradually progressed toward regularity as the number of raindrops grew. They also demonstrated more complete quantitative reasoning

in selection tasks. Piaget and Inhelder concluded that these students' actions were based on "qualitative intuitions of proportionality" (p. 54) rather than a sense of the law of large numbers. In the *formal operational stage* students used permutations to deal with random mixing and constructed progressively uniform distributions. They also determined probabilities on the basis of their understanding of proportionality, combinatorics, and the law of large numbers.

Much of the research that built upon and extended the work of Piaget and Inhelder (1951/1975) largely corroborated their findings in relation to students' understanding of various facets of randomness. Two notable exceptions were the studies of Paparistodemou, Noss, and Pratt (2002) on random mixtures and Green (1988) on random distributions.

Paparistodemou et al. (2002) took a contemporary approach to the random mixture task by engaging children (Ages 6–8 years) first in the regular tilt box task and then in a microworld task that allowed them to simulate the random movement of balls in a two-dimensional space. On the regular tilt box task they predicted overregulated arrangements for the balls and exhibited little understanding of randomness, similar to the children in Piaget and Inhelder (1951/1975). By way of contrast, children in the microworld environment reflected an emerging understanding of random mixture manifested in their dealings with haphazard movement, complex movement, symmetry of placement, and size of balls. According to the researchers, children moved beyond describing the random behavior of the balls to actually constructing random behavior.

In a replication of the raindrop task, Green (1988) did not find the extent of growth by age that was reported by Piaget and Inhelder (1951/1975). Subsequently, Green changed the raindrop task to one in which random selections of numbered counters were recorded by marking a cross in the appropriate square on a numbered grid. Using a large sample of students, Aged 7 to 16, Green (1983, 1988) found that children's facilities increased relative to age on the more transparent items representing regular and random patterns. However, on less salient items depicting "semi-random" patterns there was no significant improvement by age. Green's items have been used more recently by Dessart (1995) with middle school (Ages 10 to 13) and college students and by Batanero and Serrano (1999) with secondary students in Spain. In accord with Green's findings, these researchers found no significant differences by age in students' ability to identify random distributions. Clearly the notion of random distribution is a complex one for students of all ages.

Researchers (Batanero & Serrano, 1999; Falk, 1981; Green, 1988; Schilling, 1990; Toohey, 1995) have also studied conceptions of randomness in terms of sequences of outcomes. In a study that typifies the approach and findings of these studies, Batanero and Serrano asked students whether sequences of coin tosses, like the following, were generated by actually tossing a coin or whether the results were contrived:

- H T T T H T T H T T T T T T T H H T T T H T T H T T H T T T T T T T T
- H T H T T H H T H T H H T T H T T H H T T H T H H T T H T H T H T H T H T

The researchers found that "students had greater difficulty in recognizing run properties than frequency properties . . . [and they concluded that] the similarity between the observed and expected frequencies may be more important than run lengths" (p. 562) in students' decisions. For example, most students stated that the second sequence shown above was randomly generated, even though the number of alternations between heads and tails was biased. However, 74% of the students correctly identified the first sequence as nonrandom because the frequency of heads differed from the expected theoretical frequency. To investigate the properties of randomness that students used to make judgments about sequences of outcomes and also random distributions (e.g., modified raindrop task), Batanero and Serrano (1999) asked students to justify their responses. They concluded that students referred to *unpredictability* and *irregular* patterns as indicators of random sequences and distributions and *regular* patterns and *long* runs as indicators of a lack of randomness. Batanero and Serrano interpreted students' difficulties with runs and distributions as an indicator that independence was not intuitive for these students.

Building on Piaget and Inhelder's earlier research, Metz (1998b) investigated randomness from a developmental and nondevelopmental perspective. She examined the reasoning of 5- and 8-year-old children and college students as they confronted classical probability tasks incorporating an experimental component. The experimental component was intended to encourage learning through prediction, experimentation, and reflection. Tasks included the marble tilt box, selection of the spinner most favorable to a target color, and an urn task in which students used sampling to infer the composition of a hidden collection of marbles.

Metz (1998b) found that few kindergarten children (Age 5) were able to interpret random phenomena and those who did failed to do it consistently. She

concluded that kindergarten children had a predisposition for attributing order or regularity to chance situations as evidenced in the children's belief that the marbles in the tilt box would return to their original arrangement and in their assumption that the contents of the urn could be inferred from a small sample of marbles. In contrast, the majority of Grade 3 (Age 8) children exhibited some understanding of uncertainty and randomness. However, task characteristics limited their consistency in interpreting random situations as demonstrated by the fact that about one half of the third graders identified the most favorable spinner whereas none of these children consistently recognized the inherent presence of randomness in the urn sampling task. Although all of the undergraduate students in Metz's (1998b) study interpreted the tasks in terms of random phenomena, they sometimes reverted to deterministic interpretations, and this constrained their ability to recognize uncertainty. Metz concluded that deficiencies in children's performance could not be attributed solely to developmental shortcomings given that undergraduate students were also challenged by the probability tasks in her study.

Given the research findings in this section, it is apparent that the concepts of chance and randomness present an enduring challenge to learners of all ages. By way of summary of the critical elements of the research on chance and randomness for teachers, we draw on Metz's (1998b) synthesis of the special challenges these concepts pose for learners: (a) failure to interpret uncertainty in patterns that emerge over many repetitions of an event; (b) a belief that a person or device (e.g., spinner) can exert control over an event; and (c) a belief that some sort of order, purpose, or reason underlies events.

Sample Space

The concept of sample space is a fundamental part of the process of mathematizing random phenomena. It is an important descriptor of the outcomes of random phenomena and also provides the base for measuring the probability of events. As we have noted earlier, random phenomena are characterized by having multiple outcomes, and the sample space is defined as the set of all those possible outcomes. For example, in the simple random experiment of tossing a coin, the sample space comprises two outcomes: a "head" and a "tail," whereas in a compound random experiment, such as tossing a coin and rolling a die, the sample space can be represented as follows: {(H,1) (H,2) (H,3) (H,4) (H,5) (H,6) (T,1) (T,2) (T,3) (T,4) (T,5) (T,6)}.

Although the concept of sample space appears to be a relatively straightforward aspect of the mathematics of random phenomena, it is more subtle and elusive than it appears. Reflecting on the teaching of probability, Shaughnessy (2003) observed that data from the 1996 National Assessment of Educational Progress (NAEP) indicated that students were weak on the concept of sample space, and he highlighted the value of having students build sample spaces for probability experiments. In this section, we analyze research on the concept of sample space and attempt to interpret and clarify the conflicting results in this research. According to Horvath and Lehrer (1998), understanding and using the notion of sample space requires the coordination of several cognitive skills: (a) recognizing the different ways of obtaining an outcome, (b) being able to systematically and exhaustively generate all possible outcomes, and (c) being able to "map the sample space onto the distribution of outcomes" (p. 123). These cognitive skills will be used as headings to examine and codify the salient features of students' conceptions of sample space.

Recognizing Ways of Obtaining an Outcome

Identifying the set of possible outcomes for even a simple (one-stage) random experiment such as tossing a regular die is nontrivial for many students. Jones, Langrall, Thornton, and Mogill (1999) specifically examined students' (Aged 8 to 9 years) ability and disposition to identify the sample space in random situations. They found that, prior to instruction, 15 of the 37 students in their study did not demonstrate that *all* outcomes could occur in simple event situations. The 15 students exhibited what Jones et al. called the *sample space misconception* in that they either stated a single outcome or eliminated some outcomes because they had occurred on a previous trial. They appeared to think about sample space in a predictive or deterministic way, attributing certainty to situations where it did not exist. A test of the tenacity of this deterministic thinking is reflected in the fact that this misconception remained an intractable problem for 2 of the 15 students in spite of extended instructional experiences with continuous and discrete random generators.

Piaget and Inhelder (1951/1975) did not document sample space as a problematic area for young children. Although subsequent researchers (Borovnik & Bentz, 1991; Green, 1988; Jones, 1974; Jones, Langrall, Thornton, & Mogill, 1997; Schroeder, 1988) raised issues about it, they made relatively little attempt to explain the cognitive mechanisms that students use in recognizing possible outcomes. Jones, Langrall, et al. (1999) revealed some new insights into young chil-

dren's conceptions and dispositions towards sample space, but further research is needed.

Generating All Possible Outcomes

As noted above with simple random experiments, determining the sample space necessitates that one be able to systematically and exhaustively generate all possible outcomes. Generating the outcomes for a compound (two-stage) random situation (e.g., flipping two coins) is even more problematic than for a simple one. Part of the difficulty in determining the sample space for compound random situations has been attributed to students' lack of combinatorial reasoning (Batanero, Navarro-Pelayo, & Godino, 1997; Fischbein & Grossman, 1997), a topic typically under-represented in the school curriculum (English, 2005). Although research on combinatorial reasoning is limited, several studies have produced findings that enrich our knowledge of students' use of combinatorics in dealing with sample space.

In two studies conducted by English (1991, 1993), children (Aged 7 to 12) developed systematic and efficient strategies for generating solutions to meaningful two-stage and even three-stage combinatorial problems. Using the dressing of toy bears to model different combinations, the children in English's studies were able to "develop and modify their solution strategies, detect and correct their errors, and develop generative procedures on their own" (1993, p. 270). The most efficient strategy, coined the *odometer strategy*, was adopted by the majority of children Aged 7 years and older. For a two-stage situation, it involved holding one variable constant (e.g., blue top), matching it with all items of the second variable (e.g., black, white, and blue pants) until all possible combinations had been formed, and then repeating the cycle with a new constant item (e.g., yellow top). In order to develop powerful ideas such as the odometer strategy, English noted that, rather than having direct instruction from the teacher, children needed experiences grappling with meaningful, hands-on combinatorial problems. These kinds of instructional experiences were provided in the Jones, Langrall, et al. (1999) study and likely contributed to the success many students had with regard to compound sample space problems.

Batanero et al. (1997) produced a hierarchical classification of combinational problems after investigating the effects of instruction on the combinatorial reasoning of 14- to 15-year-old Spanish students. This classification, ordered from easiest to most difficult, is presented below.

- *Selection* problems involve determining how many samples of n elements could be ob-

tained from a set of m objects. *Example:* A box contains 10 counters numbered from 1 through 10. A counter is drawn, its numeral is written down and the counter is returned. The process is repeated until a three-digit number is formed. How many three-digit numbers can be obtained using this process?

- *Distribution* problems involve distributing a set of n objects into m cells. *Example:* A toy has three identical red balls and four tubes colored green, yellow, blue, and violet. If only one ball can be placed in each tube, how many different ways can a child place the balls in the tubes?
- *Partition* problems involve splitting a set of n objects in m subsets. *Example:* Sam and Ryan have four cars colored red, orange, blue, and green. How many ways can they share the cars if each gets two?

Selection problems were easiest for students because they could apply a combinatorial formula. Distribution and partition problems were more difficult because students did not recognize how combinatorial operations could be applied to such problems or were unaware of how to translate the problems into the form of a selection problem. Batanero et al. also identified several combinatorial errors in relation to students' construction of sample spaces:

- *Error of order:* Distinguishing the order of elements when it is irrelevant or failing to consider order when it is essential.
- *Error of repetition:* Repeating elements when it is not possible or failing to consider the repeating of elements when it is possible.
- *Nonsystematic listing:* Using trial and error without a recursive procedure to assure identification of all possibilities
- *Faulty interpretation of the tree diagram:* Constructing an inadequate diagram or incorrectly interpreting a diagram.

On the basis of these findings, the researchers suggested that instruction should emphasize "the translation of combinatorial problems into the different models [problem types], recursive reasoning and systematic listing procedures, instead of merely centering on algorithmic aspects and on definitions of combinatorial operations" (p. 196). In essence, the research on combinatorial reasoning underscores the complexities involved in identifying a sample space and the need for instruction that is informed by research-based knowledge of the kinds of errors that

students make in listing sample spaces involving compound events.

Mapping the Sample Space on the Distribution of Outcomes

Although nontrivial, the ability to list the sample space outcomes is of little value unless students recognize the importance of sample space in probability measurement. Yet, the research is replete with evidence that students of all ages do not necessarily consider sample space when determining probabilities or examining outcome frequencies (e.g., Ayres & Way, 2000; Fischbein & Schnarch, 1997; Shaughnessy & Ciancetta, 2002). The work of Fischbein et al. (1991) highlights this point.

Fischbein et al. (1991) examined students' (Aged 9 to 14 years) misconceptions about compound events in specific and general contexts. The *specific* context concerned the tossing of two dice where students were asked about the likelihood of two events, "a 5 with one die and a 6 with the other" or a "6 with both dice." The majority of students responded that the likelihood of these compound events was equal and based their justifications on two modes of reasoning:

- the situation involves chance, hence there is no reason to expect one (compound) outcome over the other
- each simple outcome is equally likely and each toss is independent, hence each (compound) outcome is equally likely.

The first mode of reasoning reflects another aspect of Konold's (1991b) outcome approach and the belief that "anything can happen." The second mode of reasoning is more sophisticated and was characteristic of students who had received instruction in probability. The following response illustrates this mode: "Each die is independent from the other. The probability that with one die, one will obtain a certain number is $1/6$ and it is the same probability that one will obtain the same number with the other" (p. 535). Even though the second mode of reasoning is more sophisticated than the first it fails to take any cognizance of the sample space associated with these compound events. This lack of focus on the sample space was also true for many students who responded correctly for the wrong reason. Instead of considering the sample space composition, they proffered a belief that "same results" (e.g., 6, 6) usually occur less often than "different results" (e.g., 5, 6).

The other context investigated in the Fischbein et al. (1991) study was *general* in the sense that students were asked whether it is more probable, when

tossing two dice, to obtain the same numbers or different numbers. Interestingly, the findings revealed that across these age levels the percentages of correct answers for this problem and a parallel one on the tossing of two coins were greater for the generalized contexts than for the specific contexts. Many of the students, who had provided inappropriate justifications for the specific dice problem, explained their reasoning for the general context in terms of sample space composition. This is illustrated in the following response from a junior high student: "It is more likely to obtain different numbers because, in order to get equal numbers, we have 6 possibilities but for obtaining two different numbers one has 30 possibilities" (pp. 537–538). Although it seems that the more general context motivated students to focus more sharply on sample space, this was not the case for the parallel item involving the tossing of two coins. In the two-coin item, the percentage of correct responses was relatively high, but most students' justifications belied any recognition of the sample space. These baffling differences in students' justifications for the dice and coin problems were attributed by Fischbein et al. (1991) to the richer sample space of the dice context. On the one hand, they claimed that some students exhibit an intuitive capacity to evaluate the sample space in finding the probability of a compound event. On the other hand, they concluded that students do not exhibit a natural understanding that outcomes like (a, b) and (b, a) need to be distinguished and counted separately (p. 547).

As well as providing insights into students' thinking about sample space, the work of Fischbein et al. (1991) and others also highlighted the problematic nature of this research and consequently has seeded many subsequent studies. More recently researchers have begun to observe students' thinking about sample space during teaching experiments and instructional programs.

Horvath and Lehrer (1998) examined the effect of "notational assistance" on Grade 2 students' understanding of the relationship between sample space and distributions. The student tasks involved making predictions with 6-, 8-, and 12-sided dice (e.g., most and least favorable outcomes), experimenting, collecting data, and explaining relationships between their predictions and the data. The study revealed that most children did not relate sample space composition to their experimental dice data, and this was especially true for compound events involving two dice. However, when sample space composition was introduced with a notational system (e.g., bar graphs), the children were able to infer some relationships between sample space composition and experimental data. For

example, in the rolling of two 6-sided dice, they came to realize that a sum of 3 was more likely than a sum of 2 because there are two ways of getting a 3 and only one way of getting a 2. Nevertheless, when their experimentation differed from their predictions, students typically changed their predictions to match past experiences rather than use sample space composition. This preference for experimental data over sample space composition was more pronounced when the context was changed and the notational system was not provided. As part of the same study, Horvath and Lehrer (1998) found similar results with a small sample of fourth- and fifth-grade students. However, after having the benefit of notational assistance, these older students readily incorporated the sample space into their thinking even after the assistance was removed. Moreover, they came to accept experimental variation as characteristic of randomness and based their predictions on sample space composition. In summary, whereas the second-grade students in this study predicted likelihood on the basis of subjective judgments (e.g., my favorite number) and established experimental data as the key criterion for determining likelihood, the older students, albeit only 4 of them, accepted variation in experimental data and seemed to forge an ongoing link between sample space composition and outcome likelihood.

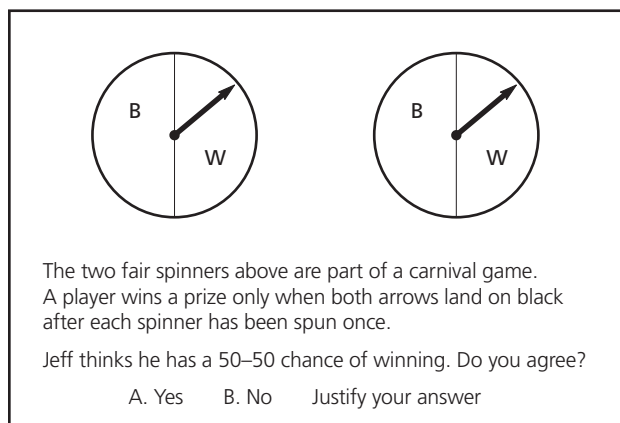


Figure 20.2 Spinner task from the 1996 NAEP (Shaughnessy & Ciancetta, 2002).

Shaughnessy and Ciancetta (2002) also examined students' understanding of sample space in both classical and experimental contexts. Using the task in Figure 20.2, they found that for a large sample of students in Grades 6–12, (a) only 20% of those in Grades 6–8 responded that the chances of winning the game were less than one half, (b) the performance of Grades 9–11 students enrolled in introductory-level mathematics was similar to the NAEP sample of Grade

12 students (28% gave the correct answer and only 8% gave the correct reason), and (c) about 90% of Grades 10–12 students enrolled in upper level courses answered the problem correctly. The researchers observed that a high percentage of students in Grades 6–11 did not consider the sample space when justifying their response. In a follow-up interview with students in Grades 8–12, the researchers investigated whether actually playing the game would encourage students to attend to the sample space. Students were asked to predict the results of 10 rounds of the game, to collect data using the actual spinners, and to reconsider their response. This cycle was repeated before students made a final response to the question. Prior to playing the game, 13 of 28 students responded correctly, but only 5 provided a correct justification. Of these 5 students, 4 listed the sample space and 1 used the multiplication principle to demonstrate that the probability of winning was one fourth. After the game cycles, 11 of the 15 students who had initially responded incorrectly reported that the game was not 50-50, and 8 of them specifically listed the sample space. The authors claimed that, by playing the game and seeing the variation in repeated trials, some students were prompted to construct the sample space for the task. They drew the following conclusion:

The conceptual root of the pedagogical power that we gain from having students conduct simulations is the connection that they can make between the observed variation in data in repeated trials of an experiment, and the outcomes that they expect based on knowledge of the underlying sample space or probability distribution. (p. 5)

The instructional benefits of having students link empirical data with sample space composition are also evident in the studies of Horvath and Lehrer (1998), Jones, Langrall, et al. (1999), and Nisbet, Jones, Langrall, and Thornton (2000).

In a series of cross-cultural teaching experiments in Brazil (Kaufman-Fainguelernt & Bolite-Frant, 1998), Israel (Amit, 1998), and the United States (Maher, Speiser, Friel, & Konold, 1998; Speiser & Walter, 1998), researchers investigated the ways that students (ranging in age from elementary to college) represent and deal with sample space in a two-dice game task. These researchers concluded that the game triggered two sample spaces for students: sample space A with 36 outcomes, the Cartesian product of the set $\{1, 2, 3, 4, 5, 6\}$; and sample space B with 21 outcomes that can be characterized as $\{(1, 1) (1, 2), (1, 3) (2, 3) \dots (6, 6)\}$. In essence, sample space A recognizes pairs like (1, 2) and (2, 1) as *distinguishable*, whereas sample space B does not and includes just one representative, say (1, 2). Nevertheless, students

needed to use sample space A, whose 36 outcomes are equally likely, to compute the unequal probabilities of sample points in B. They did this by essentially using a mapping that is two-to-one for points like (1, 2) and (2, 1) and one-to-one for points like (1, 1). The research suggests that both sample spaces and the map between them are necessary for students to be able to build pervasive solutions for the dice game task. It also highlights that in dealing with compound sample spaces, students need to come to grips with the notion of what constitutes an outcome of a sample space.

From a different cultural perspective, Polaki (2002a) conducted a teaching experiment with fourth- and fifth-grade students (Aged 9 to 10) in Lesotho, Africa, to forge links between probability and sample space composition. The students in Polaki's study were assigned to two instructional groups: one group generated a small set of experimental data in a game situation, then focused on sample space composition to determine the probabilities; the other group examined large-sample computer-generated data prior to considering sample space composition. Both instructional methods positively impacted the students' probabilistic reasoning and their ability to link probability and sample space composition. However, it is interesting to note that there were no significant differences between the two methods, and Polaki conjectured that students might have had difficulty because they were not familiar with computer-generated data.

In contrast to Polaki's study (2002a), students in Pratt's (1998, 2000) microworld studies saw the value of conducting a large number of trials in order to produce a more transparent relationship between sample space composition and relative frequencies. The different findings between the two studies might be attributed to the stronger focus on variation in Pratt's study. Given the importance ascribed to variation by Shaughnessy and Ciancetta (2002), it is not surprising that Pratt's students benefited greatly by observing variation in both small and large samples. In fact, they essentially learned to control variation by preferring large samples over small samples.

Probability modeling presents a different, but related approach to focusing students' attention on sample space. By *modeling* we mean that one "selects a probability generator whose sample space outcomes and their probabilities match with the corresponding outcomes and probabilities of a contextual problem" (Benson & Jones, 1999, p. 2). Benson (2000), using a teaching experiment and tasks like the one in Figure 20.3, found that 6 students in Grades 3 and 4 could construct or select a probability generator that corresponded correctly to the sample space of a contextual problem. All students utilized one-to-one correspon-

dence to make connections between sample space outcomes and their corresponding probabilities; a few also used *n*-to-one or *n*-to-*n* correspondences. In addition, some students recognized when probability generators were *equivalent* and could identify more than one probability generator for a given situation.

Class Draw

Six children, John, Ken, Louis, Sue, Cathy, and Beth, enter a drawing to win a prize. Only one name will be selected. How could you model to find out who wins the drawing?

[Note: Colored bears, dice, two-colored chips, one six-segment and one two-segment spinner were available].

Figure 20.3 Contextual problem on probability modeling (Benson & Jones, 1999).

The modeling tasks in Benson's study seemed to assess students' understanding of sample space more deeply than tasks requiring the listing of all possible outcomes. As she observed,

It is one thing to identify the outcomes and determine the event probabilities associated with a probability generator; it is another to identify or construct a probability generator that will faithfully represent the probability distribution embodied in a contextual task. (p. 3)

Shaughnessy (2003) highlighted the critical need for instructional programs to involve students in building sample spaces for probability experiments. However, as exemplified by the research in this section, teachers need to be aware that the development of students' thinking about sample space takes many twists and turns: (a) an inability or unwillingness on the part of younger children to list the outcomes of even simple experiments, (b) a lack of systematic processing skills to generate or differentiate between outcomes in compound experiments, and (c) a reluctance to link sample space composition and outcome probabilities (probability distributions) in both classical and experimental situations.

Probability Measurement

Batanero, Henry, et al. (2005) provided an insightful historical exposition on approaches to measuring the probability of random events. In particular they underscored three key approaches to probability measurement: classical (Laplace, 1825/1995), frequentist (von Mises, 1928/1952), and subjective (Lindley, 1980). Notwithstanding the importance of

these three approaches, which are described in our earlier section on national curricula, research on probability measurement has focused largely on the classical approach with some recent research beginning to emerge on the frequentist approach. We were not able to locate cognitive research on the subjective approach to probability measurement.

Our review of cognitive research on the classical or theoretical approach to probability measurement has several parts. The first part deals with students' (mostly younger children's) qualitative view of probability in terms of *most likely/least likely events*. Subsequently we examine students' quantitative reasoning on theoretical probability according to the categories: probability of an event, probability comparisons, probability adjustments, and conditional probabilities.

We have already discussed some learning research (e.g., Horvath & Lehrer, 1998; Polaki, 2002a; Pratt, 1998, 2000; Shaughnessy & Ciancetta, 2002) that incorporates to varying degrees a frequentist approach to probability measurement. These studies set the scene for an examination of recent research that focuses more explicitly on experimental probability measurement and the relationship between experimental probability, theoretical probability and sample size.

HT-sequence problem

Part 1. Which of the following is the most likely result of five flips of a fair coin?

- a) HHHTT
- b) THHTH
- c) THTTT
- d) HTHTH
- e) All four sequences are equally likely.

Part 2. Which of the above sequences would be least likely to occur?

Figure 20.4 Coin-tossing problems (Konold et al., 1993, p. 397).

Most Likely/Least Likely Events

Using the task shown in Figure 20.4, Konold et al. (1993) asserted that high school and college-age students perceived *most likely* and *least likely* outcomes very differently. With respect to the HT-sequence problem, most students responded correctly that the *most likely* result was that all four sequences were equally likely. However, in the *least likely* case, only 38% of the students stated that the four sequences were equally likely. Konold et al. argued that this inconsistency for approximately 60% of students could be attributed to reasoning based on the *outcome approach*. That is, students interpreted the problem as asking them to pre-

dict what *would* happen. Hence in predicting equally likely for the *most likely* case, they were not arguing that the sequences were equally likely in the sense of having the same numerical probability; rather, they were using equally likely to mean that "anything could happen" (p. 399). By way of contrast, in the *least likely* case, the majority of students selected item (c) basing their response on the representativeness heuristic (Kahneman & Tversky, 1972) in the sense that this sequence was the least representative.

In a related study, Watson, Collis, and Moritz (1997) asked students in Grades 3, 6, and 9 whether certain events were more likely or equally likely. They found that even when students correctly identified events as more likely or equally likely, their reasoning was not normative. In essence, their findings were similar to those of Konold et al. (1993) in that many students interpreted equally likely as meaning that "anything can happen." The findings of these two studies raise a number of questions about what students of various ages are responding to when they identify the most or least likely event. This caveat is important for studies like those in the next section.

Probability of an Event

A number of studies with younger children (e.g., Acredolo, O'Connor, Banks, & Horobin, 1989; Jones, Langrall, et al., 1997, 1999; Polaki, 2002a) have focused on the construct, probability of an event. In these studies, students were generally asked to identify the most or least favorable outcome usually based on a classical interpretation of probability. In an attempt to avoid the problems raised by Konold et al. (1993) and Watson et al. (1997), they have asked students to justify their responses using both qualitative and quantitative reasoning. The approach taken by Acredolo et al. set the stage for such studies when they examined the reasoning of children in Grades 1, 3, and 5 with respect to the following task: Given a clear plastic bag containing three red jellybeans, two green jellybeans, and one yellow jellybean, indicate the likelihood of randomly drawing a yellow jellybean by sliding a marker along a scale with a sad face at one end and a happy face at the other. They found that students used one of three strategies: (a) a numerator strategy in which they only examined the part of the set that corresponded to the target event, (b) an incomplete denominator strategy in which they examined the complement of the target event, and (c) an integrating strategy in which they related the number of target elements to the total number of elements. More generally, these strategies can be described in terms of part-part and part-whole reasoning as the following studies reveal.

On the basis of their work with students in Grade 3, Jones, Langrall, et al. (1999) concluded that the recognition and use of part-whole relationships, in conjunction with part-part relationships, facilitated students' movement toward quantitative reasoning in probability. Moreover, they found that the part-whole schema, rather than a precise use of fractions, appeared to be a critical element in supporting students' reasoning. A number of other researchers have also identified the importance of part-whole reasoning in younger students' thinking about probability (Jansem, 2005; Metz, 1998a; Polaki, 2002a, 2002b; Ritson, 1998; Tarr, 1997). In fact, Jansem and Polaki have undertaken a detailed analysis of the mechanisms by which elementary school students use different part-part and part-whole schemata in dealing with the probabilities of events.

Probability Comparisons

Some probability measurement situations involve two sets of elements and require students to select the set that is more favorable for a target event (binary-choice task). For example, suppose children are asked which of the following urns is more favorable for red or are they equally favorable: one containing four red marbles and eight green marbles or one containing four red and six green marbles? The consistent finding for this kind of task across several studies (e.g., Falk, 1983; Fischbein et al., 1991; Jones et al., 1997; Piaget & Inhelder, 1951/1975; Way, 1996) is that many students, especially young children, base their choice on idiosyncratic judgments such as "their favorite color" or on restrictive reasoning that focuses on the number of target elements. These students do not consider the number of nontarget elements or the ratio of winning to losing items. Falk, whose studies are seminal in this area, has identified three strategies that students use to identify which set is more favorable: (a) set with more target event, (b) set with less nontarget event, and (c) set with greater difference in favor of target event. Falk observed that as students understood proportionality, they began to use strategies that recognized relationships between the number of winning and losing elements, or between the number of winning/losing elements and the total. In essence, this continuum of strategies resonates with the part-part and part-whole strategies alluded to in the previous section.

The longitudinal study of Watson et al. (1997) illustrates more contemporary research in this area. They surveyed a large sample of students in Grades 3, 6, and 9 on the following problem:

Box A and Box B are filled with red and blue marbles as follows. Each box is shaken. You want to get a blue marble, but you are only allowed to pick out one marble without looking.

Which box would you choose? Please explain your answer.

- (A) Box A (with 6 red and 4 blue)
- (B) Box B (with 60 red and 40 blue)
- (=) It doesn't matter (p. 65)

Their interpretation of students' responses was more complex than that of Falk (1983) and was guided by Biggs and Collis's (1991) development model. Watson et al. described students' responses across four levels for each of two cycles of development that increased in complexity. At the lowest level of the first cycle (*acquisition of the concept*), responses were subjective (pick (=): red and blue are my best colors). At the second level, responses indicated recognition of the colors in the boxes, but no reference to the frequency of colors (pick (=) because there is blue in each). At the third level, responses attended to frequency, but it was inappropriately applied to the problem (pick Box B because there are more marbles to pick from). At the fourth level, responses recognized the number of marbles in each box and compared the two colors (pick (=) because there are more reds in both boxes). In the second cycle (*application of the concept*), more complex explanations were provided. First, the equivalence of the boxes was recognized according to a basic similarity (pick (=) because they are both filled alike). Second, justifications were more mathematical and often used ratios (pick (=) because Box B has ten times the amount in Box A). Finally, responses utilized ratios or percentages for within-box comparisons and then comparisons between boxes (pick (=) because both boxes are 40% blue). Most students in Grades 3 and 6 gave responses at the third level of the first cycle while the majority of Grade 9 students gave responses in the second cycle of development.

The Watson et al. (1997) categorization of responses, based as it is on an extensive age range, has considerable potential for informing probability instruction in both elementary and middle school. It also suggests that students will think differently during the acquisition or building of probability concepts than they do when applying probability concepts.

Probability Adjustments

Citing limitations with tasks typically used in probability of an event and probability comparisons research, Falk and Wilkening (1998) promoted the use of *probability-adjustment tasks* like the one shown in Figure 20.5. In this task students were asked to add

white balls to the second urn so that the likelihood of drawing a white was the same for each urn. Falk and Wilkening claimed that these tasks call for “the assessment of the magnitude of the target probability and at the same time [retain] the need to compare two probabilities” (p. 1341).

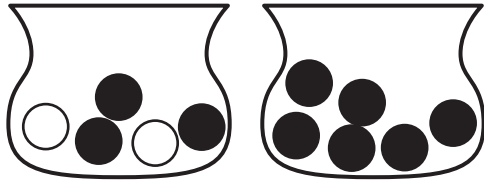


Figure 20.5 Example of a Falk and Wilkening (1998) probability-adjustment task.

The general findings of Falk and Wilkening’s (1998) study indicated that students (Ages 6–7) were unable to generate the target probability using notions of proportionality. Although their strategies were often unsystematic, they did reflect a tendency to focus on the number of either target or nontarget balls in the full urn. Students, Aged 9–10, attempted to integrate the two dimensions but tended to do so by making decisions based on the difference between the number of target and nontarget balls in the full urn. By Age 13, students were utilizing proportionality, albeit at less than a perfect level of performance. Falk and Wilkening claimed that the most notable lesson learned from their research is that “children’s spontaneous indications . . . show that, in a given set-up, binary-choice mastery precedes the ability to adjust probabilities” (p. 1353).

Probability adjustment tasks provide a valuable addition to the network of tasks used to illuminate students’ thinking about probability measurement. This pool acts as a fillip for further research, as do the various caveats of Falk and Wilkening (1998), Konold et al. (1993), and Watson et al. (1997) in relation to *most likely/least likely* probability tasks.

Conditional Probability

Drawing heavily on earlier research of Falk (e.g., 1983, 1988), Shaughnessy (1992) described three salient misconceptions in the literature on conditional probability. The first, referred to as the “Falk phenomenon” or *the fallacy of the time axis*, highlights the difficulty students have in dealing with a conditional probability, $P(A|B)$, where the Event A precedes Event B (see also Batanero & Sanchez, 2005). The second misconception reported by Shaughnessy pertains to the difficulty of identifying the conditioning event (see the card problem in Shaughnessy and the coin problem in Batanero & Sanchez). The third relates to

the difficulty in distinguishing between a conditional statement and its inverse, that is, between events like “I have measles, given that I have a rash” and “I have a rash, given that I have measles” (see Pollatsek, Well, Konold, & Hardiman, 1987). Given the narrow focus of this earlier research, Shaughnessy (1992) called for “systematic investigation of students’ conceptions about conditional probabilities” (p. 475). In the current period, one study specifically addressed this need whereas others extended research on students’ conditional probability reasoning.

In a teaching experiment designed to increase Grade 5 students’ understanding of conditional probability and independence, Tarr (1997) approached conditional probability via a sequence of random tasks that involved *with and without replacement* situations. He detected significant growth in students’ understanding following instruction and characterized aspects of students’ reasoning about conditional probability. One of the key findings in Tarr’s study related to the shift in students’ reasoning from part-part to part-whole comparisons and its impact on their ability to deal with conditional probabilities. Prior to instruction, students relied on part-part comparisons in non-replacement situations and often failed to recognize that the probabilities of *all* events changed. By way of example, consider the following problem:

A candy jar contains an assortment of flavors: 4 grape, 3 cherry, 2 apple, and 1 lemon candies. A grape candy is drawn and eaten. Has the chance of drawing another grape candy from the jar changed or is it the same chance as it was before? Has the chance of drawing a cherry candy changed? An apple candy? A lemon candy? Explain. (Tarr & Lannin, 2005, p. 223)

Students using part-part reasoning responded that the chance of drawing a grape candy had decreased because there was now one less grape candy in the jar, whereas the other flavors had the same chance as before because their numbers had not changed. By way of contrast, students using part-whole reasoning explained that the lemon candy, for example, now had a better chance because there were fewer candies in the jar. Some even stated that the lemon candy now had a probability of 1 out of 9 compared with 1 out of 10 previously. Tarr also claimed that his practice of having students focus on the sample space and the assigning of numerical probabilities after each draw facilitated students’ use of part-whole reasoning.

Many students in Tarr’s (1997) study did not use conventional notation when describing conditional probabilities. Tarr characterized four invented representations, the first three of which appeared mostly before instruction. The first representation was a

verbal description using chance as a unit of measurement. For example in the candy problem, a student might typically respond that drawing a grape candy has “gone down one chance” because one grape candy has been removed. Tarr reported that students who used this representation focused on part-part relationships between the number of target objects and its complement. A second representation involved the use of a frequency comparison that also incorporated part-part reasoning. For example, the chance of drawing a grape candy has decreased because now there is the same number of grape and cherry candies whereas previously there were more grape candies. A third representation, observed for sample space with two outcomes, involved a verbal comparison of the number of objects of the target event (x) with the total number of objects (y) using the phrase “ x of y chance.” The fourth representation, appearing after the instructional program, involved the reporting of probabilities as a ratio of percents. For example in the candy problem, the probability of drawing a cherry candy might be described as 30% out of 90% where 30% represented the percentage of cherry candies initially in the jar and 90% represented the percentage of candies remaining after the removal of a grape candy. Although mathematically incorrect, Tarr reported that this invented notation appeared to make sense to the students.

In this section, we have already made reference to the work of Pollatsek et al. (1987) that revealed college students’ confusion about a conditional statement and its inverse. Watson and Moritz (2002) revisited this work with students in Grades 5 to 11. Using similar items, they examined students’ estimations of probabilities or frequencies for a conditional event and its inverse (e.g., $X|Y$ and $Y|X$). Rather than relying on forced-choice responses, as Pollatsek et al. had, they allowed students to choose the form in which they expressed their estimates:

Please estimate:

- (a) Out of 100 men, how many are left-handed?
- (b) Out of 100 left-handed adults, how many are men?

Please estimate:

- (a) The probability that a woman is a school teacher.
- (b) The probability that a school teacher is a woman. (p. 66)

Watson and Moritz’s results showed that, although the number of correct responses for both items increased by grade level, there were more correct responses for the conditional frequencies item than for the condi-

tional probability item. The researchers stated that “school students rarely had difficulty expressing frequencies but often had difficulties expressing probabilities” (p. 79). They offered two interpretations for the difference in performance on the items: (a) students’ experience with the two contexts, and (b) the fact that the phrase, *out of*, in the frequency item cued students to the relevant conditions. Concurring with the findings of Pollatsek et al. (1987), they claimed that the grammatical structure of the statements contributed most strongly to students’ performance.

Tarr (1997) stressed the importance of having students build conditional probabilities by sorting and counting sample space outcomes, whereas Watson and Moritz (2002) concluded that students handled conditional frequency items better than conditional probability items. In both studies the research is suggesting that “count” or frequency data is important in developing notions of conditional probability. This finding is also consistent with recent psychological research (Gigerenzer, 1994; Hopfensberger, Kranendonk, & Scheaffer, 1999) that highlights the value of students using data (frequencies) to build probabilities. In fact, arguing from a more pervasive pedagogical perspective, Shaughnessy (2003) suggested that “the teaching of probability should actually start with data, and that probability questions should be raised from data sets” (p. 224). This data orientation provides an appropriate backdrop for the next part of our review that deals with data generation and experimental probability measurement.

Experimental Probability, Theoretical Probability and Sample Size

Given the importance of experimental (empirical) probability in national curriculum programs (AEC, 1991; DES, 1989, 1991; DfEE, 1999; NCTM, 1989, 2000), it is surprising that so little research has been undertaken on students’ conceptions of experimental probability, and “the bidirectional relationship between empirical and theoretical probability and the role of sample size in that relationship” (Stohl & Tarr, 2002, p. 314). Studies reviewed previously (Horvath & Lehrer, 1998; Jones, Langrall, et al., 1999; Polaki, 2002a; Shaughnessy and Ciancetta, 2002) have highlighted the difficulty experienced by students of all ages in making links between the composition of the sample space outcomes of a probability generator and the distribution of outcomes the generator produces; in essence, these studies have underscored the cognitive challenge faced by students in making connections between theoretical and empirical probabilities.

In addition to the demanding task of building and analyzing the sample space (Shaughnessy, 2003),

the key element in elucidating the connection between empirical and theoretical probability is the size of the data sample produced by the probability generator. Pratt (2005) claimed that children had a natural proclivity for small samples over large samples in dealing with random phenomena. Nevertheless, the findings of his microworld studies (1998, 2000) suggested that, in an appropriate instructional environment, even 10-year-old students were able to discern the importance of conducting a large rather than a small number of trials in confronting empirical probability and revealing a more transparent relationship between sample space composition and relative frequencies. Studies by Aspinwall and Tarr (2001) and Stohl and Tarr (2002) provide further insights on students' thinking about experimental probability and its relationship to theoretical probability.

During a 5-day instructional program, Aspinwall and Tarr (2001) examined sixth-grade students' conceptions of experimental probability and their understanding of the relationship between experimental probability and sample size. In particular, they investigated whether students understood that, as the number of trials increases, the experimental probability is more likely to reflect the parent population (theoretical probability), and conversely that small samples are more likely to produce nonrepresentative probabilities. Prior to instruction, 5 of their 6 target students showed little or no awareness of the relationship between experimental probability and sample size; typically, they thought that any size sample should reflect the parent distribution. Although students demonstrated a more normative conception of experimental probability and its relationship to sample size following instruction, their understanding was restricted to a realization that events with small probabilities were more likely to occur with a small number of trials and less likely as the number of trials increases. Their responses fell short of a unified understanding of the scope and power of the law of large numbers.

In a related study, Stohl and Tarr (2002) built on earlier research by Stohl (Drier, 2000a, 2000b). They used the *Probability Explorer* (PE) microworld (Stohl, 1999–2002) to explore 2 sixth-grade students' thinking about the relationship between experimental and theoretical probability and the role that sample size plays in that relationship. Using an instructional environment that incorporated the PE tools' facility to perform small- and large-sample simulations and produce multiple data representations (pie graphs, bar graphs, and data tables), the researchers traced students' thinking and reactions to various perturbations associated with the researcher-generated probability tasks. Stohl and Tarr claimed that these two

average students were able to make appropriate inferences based on the data, and, in particular, made valid connections between experimental probabilities and their corresponding theoretical probabilities. Moreover, consistent with Pratt (2000) and Drier (2000a), the students recognized the importance of using large samples when making inferences in probability environments.

Although all of these studies involved only a small number of students, the studies do outline the scope of elementary and middle school students' thinking about experimental probability and its relationship with theoretical probability. Although the evidence is nascent, the research reveals that, with appropriately designed learning resources, students can move beyond their predilection for small samples or a *law of small numbers* (Kahneman et al., 1982), and begin to appreciate the power of the *law of large numbers* in dealing with the duality of experimental and theoretical probabilities.

As we have seen in this section, probability measurement remains the key aspect of any quantitative understanding of random processes. Hence, it is critical that teachers become aware of the salient features of research on probability measurement that have been outlined in this review: (a) the need for concurrent and related development of experimental and theoretical probability, (b) the value of notational assistance and invented language in describing probabilities, (c) the role of simulation and modeling as a means of measuring probabilities, (d) the confounding nature of sample size in measuring experimental probability (e) the need for learners to confront misconceptions, and (f) the importance of integrating various kinds of mathematical reasoning (e.g., numerical and proportional reasoning) in probability measurement experiences.

Cognitive Models of Probabilistic Reasoning

Perhaps the most notable shift in research during this period has been towards the use of neo-Piagetian cognitive models (e.g., Biggs & Collis, 1982, 1991; Case, 1985) to describe different levels of complexity in students' probabilistic reasoning. Most researchers working on cognitive models of probabilistic thinking have drawn upon the general developmental model of Biggs and Collis to profile student growth from naive conceptions to more formal or normative ones.

The Biggs and Collis Model of Cognitive Development

The Biggs and Collis model has been an evolutionary one beginning with the SOLO Taxonomy (Biggs & Collis, 1982) that postulated the existence

of five modes of functioning (*sensorimotor* from birth, *ikonic* from around 18 months, *concrete-symbolic* from around 6 years, *formal* from around 14 years, and *postformal* from around 20 years). According to Biggs and Collis (1991), the emergence of each of the five modes incorporates the continuing development of earlier modes. This multimodal functioning also includes, within each mode, a cycle of learning that has five hierarchical levels. At the *prestructural* (P) level, students engage a task but are distracted by an irrelevant aspect; they actually exhibit reasoning indicative of an earlier mode. For the *unistructural* (U) level, the student focuses on the relevant domain and takes up one aspect of the task. At the *multistructural* (M) level the student identifies several relevant aspects of a task but does not integrate them. In the *relational* (R) level, the student integrates the various aspects and produces a more coherent understanding of the task. Finally, at the *extended abstract* (EA) level, the student generalizes the structure to incorporate more abstract features that represent thinking in a higher mode. Within any mode, the middle three levels are of most importance, because as Biggs and Collis noted prestructural responses belong in the previous mode and extended abstract responses belong in the next. A fuller description of the use of the Biggs and Collis model in stochastic processes is found in Jones, Langrall, Mooney, and Thornton (2004) and Langrall and Mooney (2005).

Jones et al. Probability Framework

In order to present a coherent view of students' probabilistic reasoning, Jones et al. (1997) generated and validated a framework that characterizes elementary students' reasoning on four previously reviewed probability constructs: sample space, probability of an event, probability comparisons, and conditional probability. The framework, based on a synthesis of the probability literature, was supported by the researchers' empirical studies with Grades 1–4 children over a period of 2 years. For each probability construct, the framework describes four levels of reasoning that correspond to four levels of thinking within the ikonic and concrete-symbolic modes of the Biggs and Collis (1991) model. An abstracted version of the framework descriptors is shown in Figure 20.6.

At Level 1, *subjective* thinking, students are narrowly and consistently bound to reasoning that is based on idiosyncratic notions such as a student's favorite color. This level relates to the Biggs and Collis (1991) *prestructural* level within the ikonic mode. The Level 1 descriptors presented in Figure 20.6 illustrate students' proclivity for subjective notions in all constructs, especially sample space.

Students exhibiting Level 2, *transitional* thinking, recognize the significance of quantitative measures, but their responses are often naïve and inflexible. Their thinking is indicative of Biggs and Collis's (1991) *unistructural* level in that they begin to function in the concrete-symbolic mode and focus on one quantitative aspect of a task. The descriptors for Level 2 capture the main thrust of their responses in listing outcomes and quantifying probabilities; however, conflicts in their thinking may result in reversions to subjective judgments.

At Level 3, students' thinking involves the use of *informal quantitative* reasoning. This incorporates aspects of both the *multistructural* and *relational* levels of Biggs and Collis (1991) in that students recognize more than one feature of a task and integrate some features such as attending to the sample space when quantifying the probability of an event. More specifically, as the Level 3 descriptors illustrate, students begin to list outcomes systematically and to use quantitative reasoning and representations to describe probabilities. Finally, at Level 4, students exhibit *numerical* thinking in the sense that they use valid numerical measures to describe probabilities. Moreover, consistent with the characteristics of *relational* thinking (Biggs & Collis), they connect sample space composition and event probabilities.

Although Jones et al. (1997) argued that students' reasoning grows over time, they did not claim that students necessarily follow an ordered progression through the four framework levels. Rather, they saw the framework as presenting a coherent picture of students' reasoning that informs instructional design, implementation, and evaluation (Jones, Thornton, Langrall, & Tarr, 1999). The Jones et al. probability framework has spawned research studies at other grade levels and in other cultures.

Tarr and Jones (1997) expanded the elementary framework into the middle grades, with specific emphasis on conditional probability and independence. Subsequently, a more comprehensive framework (Jones, Thornton, et al., 1999) that merged the elementary and middle school frameworks was published in order to provide teachers with a wider spectrum of students' probabilistic thinking. In a different cultural setting, Polaki, Lefoka, and Jones (2000) used the elementary framework as a springboard to validate a framework that characterized students' probabilistic thinking in Lesotho, Africa. More recently, Volkova (2003) and Skoumpourdi (2004) concluded that the framework validly characterized the probabilistic thinking of elementary students in Russia and Greece, respectively.

	Level 1 (Subjective)	Level 2 (Transitional)	Level 3 (Informal Quantitative)	Level 4 (Numerical)
Sample Space	does not list all outcomes in dealing with <i>simple</i> events	lists outcomes for <i>simple</i> but not for <i>compound</i> events	uses systematic strategies for listing outcomes of compound events	links sample space and event probabilities
Probability of an Event	recognizes certain and impossible events	identifies most or least likely events	uses quantitative reasoning and measures to describe the likelihood of events	assigns numerical probabilities
Probability Comparisons	is unable to distinguish between "fair" and "unfair" situations	- distinguishes between "fair" and "unfair" situations - reverts to subjective judgments	compares likelihood of events using invented numerical representations	uses ratios or numerical measures to compare probabilities
Conditional Probability	uses <i>subjective</i> judgments	recognizes that <i>some</i> event probabilities change in nonreplacement situations	recognizes that <i>all</i> event probabilities change in nonreplacement situations	distinguishes <i>independent</i> and <i>dependent</i> events

Figure 20.6 Some aspects of the Jones et al. (1997) framework.

Watson et al. Model of Chance Measurement

Watson, Collis, and Moritz (1997; Watson & Moritz, 1998) also used the Biggs and Collis (1991) model to characterize the development of students' understanding of chance measurement. Their work differs from that of Jones et al. (1997) with regard to the levels of thinking exhibited by the students in their study. Watson et al. posited that students' thinking could be characterized according to two hierarchical Unistructural-Multistructural-Relational [U-M-R] cycles "associated with the complexity of responses required to successfully complete tasks" (Watson & Moritz, p. 105).

The Watson et al. (1997) model was based on students' (Grades 3, 6, and 9) responses to three classically oriented probability tasks. The first involved comparing the likelihood of one or six when tossing a six-sided die. The second task required students to explain whether a boy's or a girl's name was more likely or equally likely to be drawn from a hat with 13 boys' and 16 girls' names. The third task was a probability comparison situation: Box A contained 6 red and 4 blue marbles, and Box B contained 60 red and 40 blue marbles. Students were asked to explain which box to choose if they wanted a blue marble. The researchers postulated that the first two tasks could be answered within the first cycle of the concrete-symbolic mode

(U_1 - M_1 - R_1) whereas the third task was more cognitively complex and required a second cycle (U_2 - M_2 - R_2). Watson et al. developed the following model to characterize students' responses across the two U-M-R cycles.

P: Decisions are made based on non-mathematical beliefs.

U_1 : At the unistructural level in the first cycle there is the recognition that the outcome of a physical action such as rolling a die is uncertain. This is not seen as equal likelihood or quantified but seems to be based on concrete experiences rather than beliefs or feelings alone.

M_1 : At the multistructural level in the first cycle, the simple uncertainty of the U_1 level becomes qualified in some way; in some contexts a rudimentary numerical comparison may occur.

R_1 : At the relational level in the first cycle, a correct quantification of outcomes (chance) is achieved in a straightforward setting, and a relationship established in a more complex setting.

U_2 : At the unistructural level in the second cycle, consolidation occurs and in a more complex setting, responses offer single constructs without further justification; any potential conflict is not recognized.

- M₂:** At the multistructural level in the second cycle, chance measurement involving multiple settings is sought by using informal ratio notions; the need for resolution of the conflict of settings is realized but not addressed.
- R₂:** At the relational level in the second cycle, a correct mathematical quantitative comparison is made and the result is justified by using ratios and/or percentages. (p. 75)

The researchers reported that their first cycle described the *development* of the concept of chance measurement whereas their second cycle represented *consolidation* of the concept (in terms of using ratio concepts) and *application* of it to more complex problem-solving contexts. Watson and her colleagues have also investigated other probability constructs such as luck (Watson, Collis, & Moritz, 1995) and fairness (Watson & Moritz, 2003b). Similar to Jones et al. (1997), the models they constructed for these latter constructs include only one U-M-R cycle. As researchers continue to build and refine models of development in probabilistic thinking they should remain alert to the possibility of multiple learning cycles within a particular mode of operation. In addition, more research is needed to investigate the formal mode of reasoning about probability.

One further cognitive model on probabilistic thinking (Li & Pereira-Mendoza, 2002) is worthy of mention in that it was built on the written responses of a very large sample of Chinese students ($n = 567$) and the interview responses of 64 of them. The scope of the written questionnaire items was also extensive and incorporated four different categories of probabilistic thinking: identification of impossible, possible, and certain events; interpretation of chance values; chance comparison in one-stage experiments; and chance comparison in two-stage experiments. Li and Pereira-Mendoza's developmental model described students' hierarchical thinking according to one cycle of five SOLO (Biggs & Collis, 1991) levels: prestructural, unistructural, multistructural, relational, and extended abstract. The probability descriptors were similar to those of Jones et al. (1997) and Watson, Collis, and Moritz (1997), even though they related to a wider age range (Grades 6, 8, & 12).

The section on cognitive models of probabilistic reasoning provides a culminating summary of the key research we have discussed on probabilistic thinking and learning: chance and random processes, sample space, and probability measurement. It also provides a timely transition to our review of teaching and instructional environments. Cognitive models of probabilistic reasoning, such as those described above, provide

the theoretical background that is needed to inform teachers' planning, implementation, and assessment of instruction in probability. Moreover, the need for cognitive models of stochastic thinking has been highlighted by several researchers (e.g., Cobb et al., 1991) and by the advent of instructional models like *cognitively guided instruction* ([CGI], Fennema et al., 1996) and *realistic mathematics education* (RME) developed at the Freudenthal Institute (Gravemeijer, 1995).

RESEARCH ON PROBABILITY TEACHING

Stohl (2005) observed that "the success of any probability curriculum for developing students' probabilistic reasoning depends greatly on teachers' understanding of probability as well as a much deeper understanding of issues such as students' misconceptions" (p. 351). Her caveat immediately raises questions about research on *teacher knowledge*, and we note that researchers (e.g., Shulman, 1987) have identified a number of important components of teacher knowledge: mathematical content knowledge, pedagogical content knowledge, and knowledge of student cognitions. These components will guide our review on teachers' probability knowledge and set the stage for examining research on instructional environments in probability.

Mathematical content knowledge refers to knowledge of the concepts, procedures, and processes related to the organization and structure of mathematics. It also refers to the relationships of mathematics with other content areas (Shulman, 1987). Linking this to probability, Steinbring (1991) stated that teachers need to have a simultaneous conceptual and theoretical understanding of probability; further they must be cognizant of probabilistic modeling and the implicit assumptions underlying such models. Kvatinsky and Even (2002) identified three critical areas of teachers' understanding of probability content. First, teachers need to understand the essential features that make probability different from other mathematical fields (e.g., its focus on uncertainty and chance). Second, teachers should understand the aspects of mathematics that support probabilistic thinking and those that inhibit it. Third, teachers must understand the power of probability in dealing with everyday situations. This last understanding resonates with Greer and Mukhopadhyay's (2005) proposition that probability is essential for economic competitiveness in commercial and financial endeavors.

Pedagogical content knowledge is knowledge of effective strategies, representations, models, and examples for presenting mathematical content (Shulman,

1987). According to Steinbring (1991), successful learning of probability is dependent on the representations and activities teachers use. For example, he recommended that teachers use “task systems,” a well-connected series of tasks, to develop probability concepts. Echoing Steinbring’s comments, Kvatinsky and Even (2002) identified various representations and models for probability teaching (e.g., tree diagrams, Venn diagrams, and tables) and advocated that teachers know when each representation or model is appropriate and how they are connected. Kvatinsky and Even also asserted that teachers need a basic repertoire of examples to illustrate probabilistic concepts, properties, and theorems. Such a repertoire should utilize *inter alia* ideas connected to set theory and combinatorics. Finally, Steinbring and Kvatinsky and Even strongly advocated teachers’ use of both the classical and the frequentist approach. In fact, Steinbring recommended the simultaneous use of both approaches in teaching probability concepts.

Knowledge of student cognitions encompasses an understanding of how students think about and learn mathematics. This includes an awareness of the conceptions students have in relation to the content (Shulman, 1987). With respect to probability, knowledge of student cognitions embraces familiarity with different forms of probabilistic understanding including formal, relational, or intuitive thinking (Kvatinsky & Even, 2002). Greer (2001), reflecting on Fischbein’s (1975) work, stressed the importance of teachers’ dealing with students’ probabilistic intuitions, and Kvatinsky and Even warned that “the use of intuitive knowledge in probability [can] lead, in many cases, to wrong answers” (p. 5). In essence, teachers not only need to be familiar with students’ intuitions they need to confront them in the classroom (Steinbring, 1991).

The dialogue in this section reveals that teaching probability goes well beyond “balls in urn” problems. On the one hand, teachers need to understand various theoretical and conceptual aspects of probability; on the other hand, in deciding on the tasks, activities, representations, and technology to be used in instruction, teachers need to understand students’ intuitions and the means by which probabilistic learning can be facilitated. In subsequent sections, we look at what research tells us about teachers’ probabilistic knowledge and instructional environments.

Teachers’ Probabilistic Knowledge

Although the work of Fischbein (1975), Kvatinsky and Even (2002), and Steinbring (1991) has produced insightful theory on the knowledge teachers

need in teaching probability, research on teachers’ mathematical content knowledge, pedagogical content knowledge, and knowledge of student learning is limited. This dearth of research may be due to the fact that the introduction of probability, especially at the elementary level, is still relatively new.

Probabilistic Content Knowledge

In a recent review Stohl (2005) examined four studies that have investigated teachers’ beliefs and content knowledge in probability. Working in New Zealand, Begg and Edwards (1999) found that 22 practicing and 12 preservice elementary teachers had a weak knowledge of probability with only about two thirds understanding equally likely events and even fewer understanding independence. Carnell (1997) investigated 13 preservice middle school teachers’ understanding of conditional probability. He concluded that each of the teachers showed evidence of holding one or more misconceptions described earlier in this chapter: fallacy of the time axis, finding the conditioning event, and confusing conditionality with causality (Falk, 1988). In a broader study in Australia, Watson (2001) developed a profile instrument to gather information about elementary ($n = 15$) and secondary ($n = 28$) teachers’ content and pedagogical knowledge in stochastics. She reported that secondary teachers were significantly more confident than elementary teachers in their ability to teach equally likely outcomes, basic probability measurement, and sampling. In addition, she observed that the majority of teachers had particular difficulty in interpreting 7:2 correctly as “odds” and were unable to transition between part-part odds (7:2) and part-whole probability (7/9 or 2/9). In a related area, Stohl (2005) also noted that mathematics teachers have a *deterministic mindset* in that they experience difficulty in moving mathematically from situations that deal with certainty to situations that deal with uncertainty and nondeterministic reasoning. This finding is corroborated in research on secondary teachers by Nicholson and Darnton (2003) and with elementary teachers by Pereira-Mendoza (2002).

In a study that focused on cultural and language differences, Zaslavsky, Zaslavsky, and Moore (2001) examined 33 Hebrew-speaking preservice teachers’ knowledge of independence and mutually exclusive events. With respect to the 9 tasks given by the researchers, nearly 70% could not explain what it means for events to be mutually exclusive and almost half could not give examples of mutually exclusive events. In addition, few teachers could determine whether two given events were mutually exclusive. In a similar way, 70% of the teachers could not explain when events were independent and more than 40% of them could

not determine whether two events were independent. Not surprisingly, nearly 90% of them did not understand the relationship between independent/dependent events and mutually exclusive events. Despite their inability to deal with independence, about 80% of the prospective teachers could find the probability of selecting consecutive balls from an urn without replacement. Although Zaslavsky et al.'s findings on teachers' content knowledge were largely consistent with studies reviewed by Stohl (2005), their study produced unexpected results when they gave their tasks to Arabic-speaking, English-speaking, and Russian-speaking preservice teachers. Zaslavsky et al. found that English-speaking and Russian-speaking teachers performed better at identifying mutually exclusive events than their Hebrew- and Arabic-speaking counterparts. Proximity of probabilistic language to everyday language seems to act in a perverse or unhelpful way on teachers' probabilistic thinking in the sense that Hebrew and Arabic words for *mutually exclusive events* are used in everyday language whereas equivalent expressions in English and Russian are not.

Working in Mexico, Sanchez (2002) examined 6 secondary teachers' beliefs about using simulation in the classroom after they had experienced the simulation capabilities of Fathom software (Erickson, 2001) over 8 weekly 3-hour sessions. Responding to a follow-up survey, 3 teachers stated that they understood variability better after solving simulation problems with the software. One even mentioned acquiring a better understanding of conditional probability and the law of large numbers. In addition, most teachers, when asked about the usefulness of simulation, focused on ideas associated with a frequentist view of probability. However, not one teacher referred to the value of simulation activities in building students' understanding of randomness, distribution, or the process of validating models of probability.

Although research on teachers' content knowledge in probability is sobering at best, research on teachers' beliefs about the power and value of probability is even more worrisome. Greer and Ritson (1994) surveyed primary and secondary teachers in Northern Ireland vis-a-vis stochastic issues. Most primary teachers considered probability to be relatively unimportant compared to other mathematics topics, and more than 50% of the primary and secondary teachers rarely used probability experiments in their teaching. Similar results were revealed in an Italian study by Gattuso and Pannone (2002). They reported that 91 secondary teachers considered stochastics worthwhile, but the overwhelming majority stated that it took time away from other aspects of the mathematics curriculum.

Pedagogical Content Knowledge and Knowledge of Student Cognitions

Although Stohl (2005) observed that research on teachers' pedagogical content knowledge and their knowledge of student cognition is limited, she did identify some emergent research that is summarized below. Only two of the studies incorporated teachers' classroom practice.

Studies by Haller (1997) and Dugdale (2001) focused to some extent on pedagogical content knowledge and teachers' actions during classroom instruction. Haller observed 4 middle school teachers' probability lessons in a follow-up to a summer program that included experiences with probability content, misconceptions, and pedagogical issues linked to probability instruction. Haller's classroom observations indicated that teachers at the lower end of the probability knowledge spectrum revealed content errors and misconceptions, relied largely on textbooks, and missed opportunities to forge relationships with fractions, decimals, and percents during probability lessons. By way of contrast, teachers with higher probability knowledge made no content errors, enhanced textbook activities, and exploited opportunities to make connections between probability and decimals, fractions and percents. The researcher observed that teaching experience did not appear to have a great impact on the teachers' instruction.

In a study with preservice teachers, Dugdale (2001) used computer simulation to reveal pedagogical content knowledge. During the process of designing a fair game with a pair of dice, teachers came to see the value of considering relative frequencies for a large number of trials. The enabling aspects of the computer software not only convinced them that they had created a fair game, it also motivated them to find theoretical probabilities as a way of verifying computer-generated results. In a real sense, the computer experience enabled teachers to have a salient pedagogical experience that highlighted Steinbring's (1991) advocacy of making connections between frequentist and classical approaches to probability.

The previously mentioned research of Watson (2001) examined teachers' knowledge of student cognitions. She developed a comprehensive protocol to assess among other things elementary and secondary teachers' knowledge of difficulties students might experience with data and chance. In her documentation of student difficulties, only 2 of the 15 elementary teachers mentioned procedural aspects like "finding probabilities," whereas 13 of the 28 secondary teachers referred to calculating probabilities, permutations, and tree diagrams. Although the data as a whole revealed that teachers could identify students' difficul-

ties with probability, their responses, especially those of the secondary teachers, suggested that procedural approaches to teaching tended to dominate. Watson also claimed that there was little evidence that secondary teachers used activity-based approaches like simulation and sampling to reinforce theory. By way of contrast, elementary teachers used activity-based lessons, but their instructional programs in probability seemed to lack coherence.

Although our review in this section has identified some research on teachers' content knowledge in probability, the research on teachers' knowledge of instructional aspects is embryonic at best. This is a serious situation given Greer and Ritson's (1994) finding that almost all the primary and half the secondary teachers in their survey reported not having studied probability in their initial teacher education course. Clearly, researchers need better ways of obtaining data about teachers' pedagogical knowledge, their knowledge of student cognitions, and their instructional practices in probability. Moreover, such research needs to be classroom-based rather than self-reporting data or information on the number of courses or professional development programs in probability they have received (Watson, 2001).

Instructional Environments for Probability

As discussed above, teachers create instructional environments based on (a) their understanding of how students learn, (b) their knowledge of what their students do and do not know, and (c) their repertoire of instructional strategies and tools. With respect to the first two points, we have already highlighted a substantial corpus of research pertaining to cognitive models (e.g., Jones, Langrall et al., 1997, 1999; Watson et al., 1997) that can be used by teachers in developing, implementing, and assessing instructional activities in probability. The research is less robust, however, in studies that have investigated specific instructional strategies for teaching probability (e.g., Fast, 1999) and in the use of calculator and computer environments (e.g., Kissane, 1997a, 1997b; Pratt, 2000; Stohl & Tarr, 2002; Zimmermann, 2002).

Cognitive Frameworks

If we accept the "pedagogic challenge" (Pratt, 2005, p. 175) of designing instruction to build on students' existing views, be they formal knowledge structures or faulty intuitions, cognitive frameworks of students' probabilistic development can serve as useful guides for teachers. In fact, teaching research (Aspinwall & Tarr, 2001; Jones, Langrall, et al., 1999; Polaki, 2002a; Tarr, 1997) has demonstrated that these

frameworks play a key role in instruction because they provide teachers with a coherent overview of the various forms of probabilistic reasoning students exhibit and equip teachers with specific domain knowledge that can be used in the design, implementation, and assessment of instruction in probability.

In terms of the *design of instruction*, cognitive frameworks highlight the complex nature of probability and heighten teachers' awareness of the effects those complexities have on students' reasoning. In turn, this can assist teachers in planning a meaningful sequence of instruction rather than a sequence of potentially disconnected probability activities (Stohl, 2005; Watson, 2001). The use of cognitive frameworks in designing instruction can be amplified through Simon's (1995) notion of *hypothetical learning trajectory*, that is, the formulation of learning goals, learning activities, and a conjectured learning process. Tarr's (1997) teaching experiment on conditional probability and independence serves as an example. His work was informed by domain-specific knowledge drawn from a cognitive framework (Tarr & Jones, 1997), and he used the four-level framework descriptors (ranging from subjective to numerical) to establish learning goals for the students, both individually and as a whole class. With respect to learning activities and the conjectured learning process, Tarr's cognizance of the different levels of students' reasoning guided his selection of instructional tasks and questions. For example, being aware that some students would fail to recognize the changing probabilities in nonreplacement situations, Tarr designed a series of activities that included a task "where one Milky Way bar was drawn without replacement from a bag of 3 Milky Way, 2 Butterfinger, and 1 Snickers candy bars" (Tarr & Lannin, 2005, p. 230). In using this task, Tarr focused discussion on the probability of drawing a Snickers bar, given that the Milky Way bar had been removed from the bag. This drew student attention to the change in the total number of objects in the bag and the consequent changes in probability. The wealth of tasks and instructional activities documented by Tarr have widespread potential for other instructional settings.

With respect to the *implementation of instruction*, frameworks characterize students' conceptual development in a way that helps teachers accommodate the diversity of reasoning reflected by students in their class. The framework serves as a filter for analyzing and classifying students' oral and written responses during instruction and guides teachers in framing questions and written tasks that are accessible to students at different levels of reasoning (see the above illustration for Tarr). Such accommodation and sensitivity on the part of the teacher may enable children

to develop more mature levels of reasoning. Cognitive frameworks can also assist teachers in *assessing* students' performance over time and evaluating the effectiveness of instruction. For example, the candy bar task used by Tarr could just as easily have been used to assess students' progress in relation to the framework descriptors. Teachers can also use such assessment to evaluate their own performance in promoting students' probabilistic learning.

Instructional Strategies

Greer and Mukhopadhyay (2005) depicted instruction in probability as impoverished in that curriculum materials treat probability as the "exposition and routine application of a set of formulas to stereotyped problems" (p. 314). They perceived the contexts of many instructional tasks and assessment items as uninteresting, unrealistic, and often contrived. Although Greer and Mukhopadhyay's (2005) perspective resonates with the "balls in urns" curriculum materials that have dominated much of the teaching of probability, recent research has begun to generate instructional strategies that reflect the visions of the NCTM *Principles and Standards* (2000). In a study that revisited a 1970's teaching experiment in France, Brousseau, Brousseau, and Warfield (2002) examined pedagogy in a primary classroom during 18 lessons on probability. The key strategies that emerged were the need to have children involved in empirical exploration and the collection and analysis of data as a base for building meaning in probability. Even though the data were from another era, these findings set the stage for current research.

Castro (1998) explored the impact of two instructional orientations, *traditional* and *conceptual change*. The traditional instructional orientation focused on a clear linear presentation of mathematical ideas but did not relate to students' misconceptions. The conceptual-change orientation emphasized the eliciting of students' thinking and the encouragement of re-

flection on probabilistic ideas. Castro demonstrated that, compared with the traditional orientation, the conceptual-change orientation significantly improved students' skills in probability calculation and their intuitive reasoning about probability. In discussing this result he highlighted three instructional imperatives: (a) the need to take into account students' preconceptions and beliefs regarding probability, (b) the need to provide an experimental focus to the teaching of probability, as such a focus provoked "the necessary cognitive conflict in the student to perform the transition from the personal model to the scientific model of chance" (p. 252), and (c) the need to incorporate specific training in probability reasoning to challenge misconceptions. This third imperative was consistent with Castro's finding that misconceptions in probability were more resilient among those receiving the traditional instruction, and he attributed this to the fact that the traditional approach did not actually confront students' misconceptions. The notion of confronting misconceptions and engaging students in reflection is not new (e.g., Fischbein, 1975) but is significant in the teaching of probability, as this study and the next two studies reveal.

In a related study, Fast (1999) examined the use of analogies as an instructional strategy for helping students confront probability misconceptions. Forty-one high school seniors were assessed on two parallel sets of tasks. The first set consisted of 10 probabilistic tasks recognized in the research (Kahneman & Tversky, 1972; Konold et al., 1993) as stimulating misconceptions. In the second set, administered immediately after the first set, there were 10 analogous tasks. Figure 20.7 shows an original task and its analogous counterpart. Fast found that students performed better on the analogous tasks (72% success) than on the original tasks (49% success). In interview settings, he observed that many students saw connections between the two versions as indicated by their desire to change their answers to the original tasks. Fast concluded that the use of analogies enabled

Original Task	Analogous Task
In families with five children, which birth order, if either, occurs more often: BGGBG or BBBBB? B = Boy and G = Girl a) BGGBG occurs more often. b) BBBBB occurs more often. c) They occur equally.	In a lottery called Pick 4, a 4-digit number like 2798 is generated. To win, the participant must have chosen the same 4-digit number. Albert has chosen the number 2222 and Bill has chosen the number 2332. Compare their chances of winning. a) Albert has a better chance of winning. b) Bill has a better chance of winning. c) They both have the same chance of winning.

Figure 20.7 Sample task and its analogous task.

students to reconstruct their probabilistic thinking, and he claimed that an analogies approach aids teachers to identify tasks that will help students to face misconceptions. Such an approach aligns with suggestions that teachers should confront students' subjective thinking (Steinbring, 1991) and develop their secondary intuitions (Greer, 2001).

In a European study that utilized simulation and focused on confronting misconceptions, Batanero, Biehler, Maxara, Engel, and Vogel (2005) offered a theoretical position that may have implications for the teaching of probability at all levels. These researchers highlighted the importance of simulation in providing teachers with stochastic experience and knowledge to help them confront probabilistic misconceptions. In addition, they claimed that their study provided teachers with an instructional sequence that incorporated an exemplary modeling approach: (a) introduce a real-world problem involving data analysis, (b) build a simulation model in a technological environment, (c) generate data and build simulation-based inferences, (d) critically evaluate the conclusions in terms of any assumptions, and (e) formally analyze the problem on the basis of mathematical probability and statistics.

Zimmermann (2002) used a similar modeling approach to that of Batanero, Biehler, et al. (2005) in examining 23 high school students' understanding of probability simulation. The students participated in a 12-day teaching experiment that involved whole-class and small-group activities. Following instruction students showed significant improvement in their ability to evaluate and construct a probability simulation and in their understanding of the effect of repeated trials on the empirical probability. However, they did not always understand the implicit assumptions underlying a probability model. Zimmermann's study revealed fresh insights on Steinbring's (1991) caveat concerning the implicit assumptions underlying probability models.

Zimmermann's (2002) study is exemplary in its use of graphing calculators as a simulation tool. Although these calculators have become commonplace in many classrooms, research is limited on the role they play in probability teaching and learning. In contrast, research involving computers has received greater attention even though their use in classrooms is less pervasive.

Computer Environments

During this period of research, Konold (1995b) led the way in exploring the use of computers in the teaching and learning of probability. In particular, his study highlighted the need for software designed to promote sense making and the enrichment of students' intuitions. Konold's work pre-empted the de-

velopment of probability software and subsequent research into its effectiveness. The studies of Pratt (1998, 2000; Pratt & Noss, 2002) and Stohl (Drier, 2000b; Stohl & Rider, 2003; Stohl & Tarr, 2002) are noteworthy in this respect.

Working from the assumption that students hold multiple, even competing, intuitions, Pratt (1998) designed the *Chance-Maker* microworld "in which individuals would meet the consequences of their beliefs" (p. 4). *Chance-Maker* comprises random generator *gadgets* (electronic coins, spinners, dice), some of which are broken in the sense that they produce nonrandom outcomes. Students have access to tools to help them make sense of and edit a gadget's behavior. One tool is the workings box, which lists all possible outcomes for the gadget; other tools include a repeat button to produce large numbers of trials and different options for displaying results numerically and graphically. Pratt found that this computer environment enabled students to generate and examine the long-term behaviors of gadgets vis-à-vis their sample space and data distribution. This in turn supported their understanding of randomness.

Based on similar theoretical assumptions, Stohl's (1999–2002) *Probability Explorer* provides an open-ended learning environment in which students can test their probability intuitions. Probability Explorer allows students to randomly generate a variety of icons (e.g., coins, dice, marbles) that can be represented in graphical or tabular forms. Students determine how much data to generate for the simulation and how to display it. Multiple graphical representations can be presented simultaneously, and numerical results can be presented as odds, probabilities, and decimals. As with *Chance-Maker*, students can change the likelihood of an event and control the sample space and theoretical probabilities for the simulations they construct. Stohl (Drier, 2000a, 2000b) reported that students using Probability Explorer discovered that greater numbers of trials produced distributions that more closely resembled expected theoretical probabilities. She drew the following conclusion:

Simulation tools that give students control over designing experiments, running as many trials as they desire, and viewing graphical representations of results may help in the development of deeper understandings of how theoretical probability, empirical probability and sample size can be used to make inferences. (Stohl & Tarr, 2002, p. 322)

The promise of computer microworlds for probability instruction has yet to be realized in school classrooms; both *Chance-Maker* and *Probability Explorer* have been used mainly in research and development.

It remains to be seen how classroom teachers would incorporate these technology-rich learning environments into their repertoire of instructional strategies.

Pedagogical Guidelines

Although the research on instructional environments is relatively sparse, the literature does offer some guidelines. On the basis of his work with Chance-Maker, Pratt (2005) generated four pedagogical guidelines for probability instruction. Aspects of these guidelines resonate strongly with the theoretical perspectives of Fischbein (1975), Kvatinsky and Even (2002), and Steinbring (1991) that were presented at the beginning of this section.

The first guideline pertains to task design and involves two constructs: *purpose and utility*. A purposeful task is one whose solution holds some meaning for the learner, and a utilitarian task is one whose context enables the learner to see the usefulness of a mathematics concept or skill. As Pratt (2005) contended, “The difficulty in planning lies in linking purpose to utility in such a way that there is a high probability that the learner will stumble across the utility of the mathematical concept as they engage in the purposeful activity” (pp. 182–183). Both Chance-Maker and Probability Explorer were specifically designed to link purpose and utility, and given their success it would seem desirable for this link to be a feature of probabilistic tasks in all environments.

The second guideline, *testing personal conjectures*, relates to Pratt’s (2005) theoretical stance that “old pieces of knowledge coexist with newer pieces of knowledge, either in a connected way or perhaps isolated from each other” (p. 187). From this perspective, instruction is aimed at connecting students’ existing conceptions with new knowledge. This occurs through the use of tasks and questions that encourage students to become aware of the limits (“lack of explanatory power”) of their current ways of thinking in comparison to alternative views. Feedback is the crucial aspect of this process, and Pratt admitted that “the testing of personal conjectures is an especially difficult aim to achieve without the use of technology” (p. 186).

The third guideline, *large-scale experiments*, suggests that students should determine for themselves how many trials to run when collecting data in probability explorations. With that said, tasks should be designed to encourage students to try increasingly greater numbers of trials, and tools should be able to facilitate the collection of large quantities of data. This is another case where the power of technology could be vital as vindicated in the work of Pratt and Stohl.

The final guideline, *systematic variation of the context*, is closely related to testing personal conjectures.

As students experience the lack of explanatory power of their old ideas and come to recognize the power of alternative views, it is important that they experience the reliability of this new knowledge across different contexts. In this way, new knowledge structures are strengthened and eventually supersede earlier intuitions. Thus, students need to be exposed to tasks whose contexts vary while the embedded mathematical structure remains constant and some surface features remain similar to cue the reuse of recently learned knowledge (Pratt, 2005, p. 187).

In this section we have noted that, for the teaching of probability to be effective, teachers need to understand probability from both a theoretical and a conceptual point of view, as well as understand how students’ think (Steinbring, 1991). Having research-based knowledge of students’ intuitions is critical for teachers in deciding on the tasks, representations, and tools to be used in instruction. To this end, several developmental models have been created to describe students’ probabilistic reasoning and to inform instruction (e.g., Tarr & Jones, 1997; Watson et al., 1997). In addition a number of distinctive instructional strategies and environments for the teaching and learning of probability are beginning to emerge (e.g., Fast, 1999; Pratt, 2005).

CURRICULUM RESEARCH

As shown in our discussions of national curriculum documents in Australia, the United Kingdom, and the United States (AEC, 1991; DES, 1989; NCTM, 1989, 2000), data analysis and probability have emerged during this period as a mainstream strand in mathematics curricula. Moreover, these national documents took the bold step of incorporating probability into the mathematics curriculum across all levels of schooling from primary through high school.

Notwithstanding these national visions for data analysis and probability, a very different perspective can be gleaned from the cross-national investigation of curricular intentions in school mathematics conducted as part of the Third International Mathematics and Science Study (Schmidt, McKnight, Valverde, Houang, & Wiley, 1997). The results of this investigation revealed that probability showed the greatest variation among all mathematics topics in terms of when it was introduced in the curriculum: (a) 24 of the 48 countries surveyed introduced probability in Grade 9; (b) 16 countries introduced probability by Grade 6 apparently using an integrated approach from the early grades; and (c) 5 countries “deferred its intro-

duction to a more formal treatment at the end of secondary school” (p. 71). Moreover, although one third claimed that they introduced probability by Grade 6, closer inspection of each country’s data revealed that probability was seldom a significant curricular focus prior to Grade 9.

This international survey suggests that the implementation of probability in schools may not be keeping pace with the expectations of national curriculum documents. Similar evidence is also reflected in research cited earlier where it was noted that primary teachers deemed the teaching of probability unimportant (Greer & Ritson, 1994) and that secondary teachers considered stochastics worthwhile but felt it took time away from other aspects of the mathematics curriculum (Gattuso & Pannone, 2002). There may well be a need for a stronger philosophical orientation to curriculum research in probability, one with the power to materialize H. G. Wells’ prophetic vision: “Statistical thinking will one day be as necessary for efficient citizenship as the ability to read and write” (as cited in Kranzler, 2003, p. 104). Pursuing this orientation, we examine research that focuses on what constitutes *statistical literacy* and more particularly *probability literacy* (Gal, 1994, 2002, 2005; Watson, 1997a; Watson & Callingham, 2003).

Probability Literacy

Literacy and numeracy, although complex to define, have become fundamental concepts of curriculum research (Steen, 1990, 1997, 2001). For example in the United Kingdom, the Numeracy Strategy appears to play a parallel role with the National Curriculum documents (DfEE, 1999) in articulating what mathematics is taught in school. There is also an expanding meaning for domain-specific literacy (e.g., computer literacy, scientific literacy), and it is within the context of domain-specific literacy that research on statistical literacy and probability literacy has found a place and a theoretical base. Gal (2002) took the position that *probability literacy* cannot be conceived of in isolation, which is in accord with the epistemological position of other researchers (Rychen & Salganic, 2003; Stein, 2000) who argue that probabilistic knowledge and dispositions are part of a broader network of skills, knowledge, and beliefs that individuals possess or need to activate in different life settings. In Gal’s more precise terms, probability literacy is closely linked to literacy, numeracy, and his own (Gal, 2002) and Watson’s (1997a; Watson & Callingham, 2003) conceptualization of statistical literacy.

It is helpful to start with a brief discussion of research on *statistical literacy* inasmuch as statistical lit-

eracy includes components that belong to probability literacy. Watson’s (1997a) statistical literacy framework incorporates three increasingly sophisticated tiers: (a) basic understanding of probabilistic and statistical terminology, (b) comprehension of probabilistic and statistical language and concepts when they are embedded in contexts of social communication, and (c) a questioning attitude with respect to claims that are made without proper statistical justification. In an empirical study linked to this framework, Watson and Callingham (2003) used Rasch analysis to show that statistical literacy is a hierarchical construct incorporating six levels of understanding: idiosyncratic, informal, inconsistent, consistent noncritical, critical, and critical mathematical. Watson’s work in statistical literacy not only encompasses chance and data but also aspects of variation. Clearly it has implications for the school probability curriculum.

Gal (2002) argued that statistical literacy involves both cognitive and dispositional components and that some of these components are held in common with literacy and numeracy whereas others are unique to statistical literacy. More specifically, Gal referred to statistical literacy as the knowledge and dispositions that people use to critically evaluate and, where relevant, express their opinions regarding statistical information or data-related arguments. For Gal, the knowledge component of statistical literacy involves five bases: literacy skills, statistical knowledge, mathematical knowledge, contextual knowledge, and a list of critical questions. The dispositional component includes a belief in the power of statistical thinking, a belief in your own capability to undertake statistical thinking, and a belief in the legitimacy of adopting a critical perspective on “information one receives from presumably ‘official’ sources or from experts” (Gal, 2005, p. 45).

Although Gal made only brief reference to probability in his model of statistical literacy, he generated a specific model of probability literacy in his 2005 book chapter. In this chapter, which incorporated a number of allusions to others in literacy and numeracy research (e.g., Moore, 1990; Scheaffer, Watkins, & Landwehr, 1998; Stein, 2000; Utts, 2003), Gal conceptualized *probability literacy* in a similar way to statistical literacy; that is, it contains both cognitive and dispositional components including five classes of knowledge and three dispositional elements. We consider first Gal’s knowledge classes: the big ideas of probability, the figuring of probabilities, the terminology and language of probabilities, the critical questions about reasonableness, and the context within which probability issues are considered.

The notion of *big ideas* was suggested by Moore (1990) and refers to foundational elements that students need in order to understand and interpret probabilistic statements. With respect to probability Gal (2005) identified the big ideas as variation (cf. Metz's, 1997, use of the term *chance variation*), randomness, and independence. Moreover, he argued that these ideas have to be understood in tandem with their complementary "alter egos": stability, regularity, and co-occurrence (p. 47). For Gal, *figuring probabilities* is an important aspect of being able to understand probabilistic statements and communicate with others concerning the likelihood of events. It is also a key element in understanding the classical, frequentist, and subjective approaches to probability. In discussing *terminology*, Gal stressed the need for "probability literate" people to be sensitive to the language of chance in dealing with the many verbal and numerical ways in which probability is communicated (p. 50). Terms like *impossible*, *possible*, and *certain* occur in our everyday lives, and hence it is appropriate that they are incorporated in curriculum documents. Gal's (2005) notion of *critical questions or criticalness* occurs in both statistical and probability literacy. It deals with questions of reasonableness and with being able to make critical responses or to ask critical questions about chance phenomena (cf. Watson, 1997a). More specifically, Utts (2003) and Gal have addressed the need for learners to be aware of flaws, problems, and biases that occur in the derivation, reporting, and interpretation of probability information. This discussion about reasonableness has strong links with previous research on intuitions and misconceptions (e.g., Fischbein, 1975; Kahneman & Tversky, 1972) and unsurprisingly is already represented in international curriculum documents as a caveat on the misconceptions children bring to the classroom. Finally, Gal included, in the knowledge component, an element that deals with *context*. This refers to the importance of understanding the role of probabilistic thinking and language in decisions and dialogue associated with personal and public contexts. For example, people need to understand chance variation in random processes as a means of making sense of media statements about group differences, likelihood of results, significance of differences, and other claims related to statistical inference.

Gal's (2005) dispositional elements of probabilistic literacy include personal sentiments regarding uncertainty and risk, critical stance, and beliefs and attitudes. With respect to *personal sentiments*, Gal noted that people react in different ways when confronted with a need to estimate probabilities of certain events or to deal with risk situations. In essence, emotions like risk aversion or anticipated success affect people's decisions in dealing with chance phenomena in so-

cial, financial, and political situations. With respect to what Gal called *criticalness*, it is one thing to possess the knowledge to adopt a critical approach in probabilistic situations, it is another to actually hold a questioning attitude to messages that may be misleading, one-sided, biased, or incomplete. Children and adults need to be both able and willing to invoke a set of critical questions when faced with arguments that purport to be based on data or empirical research. According to Gal, several *beliefs and attitudes* underlie people's willingness to take risks vis-à-vis critical evaluations. In other words, critical disposition is dependent on people developing a belief in "the need to apply 'critical-response' skills to real world statistical messages" (Gal, 2005, p. 54). Moreover, this belief in the legitimacy of critical action should apply whether the messages come from official or other sources.

It is not surprising that we have noted instances of probability literacy impinging on our earlier overview of national curriculum documents (AEC, 1991; DES, 1989; NCTM, 1989). In the next section we analyze in more depth what research on probability literacy implies for both extant and future curriculum development in probability.

Curriculum Implications From a Probability Literacy Perspective

Both the knowledge elements and the dispositional elements of Gal's (2005) probability literacy model and Watson's (1997a) statistical literacy framework have implications for the probability curriculum. Using their theoretical perspectives we draw implications for the probability curriculum with respect to both content and pedagogical directions.

Content directions. Within the area of *knowledge*, Gal's (2005) model of probability literacy includes big ideas, figuring probabilities, terminology and language, critical questions, and context. Watson (1997a) also identified language and critical questioning as part of her three-tier framework, and her discussion was consistent with Gal in that it revolved around key ideas like chance, probability measures, and risk taking. The writings of Gal and Watson on probability literacy suggest some new content emphases for the probability curriculum and also provide support for much of the content that is already contained in current curriculum documents.

In addition to having students build a robust concept of randomness, Gal (2005) and Watson (1997a; Watson & Callingham, 2003) have called for a strong emphasis on chance variation and independence. We have already noted that national curriculum documents were limited in the extent to which they dealt with chance variation. Even though one Australian

document (AEC, 1994) used the term *chance variation*, the documents, as a whole, fail to build any notion about whether the variation manifest in a set of data has a causal base or whether it is simply due to chance (Metz, 1997; Pfannkuch, 2005). Like Gal and Watson, Moore (1990) contended that it is “crucial for students to grasp the idea that ‘chance variation’ rather than deterministic variation explains many aspects of the world” (p. 99). The notion of independence is incorporated into national curriculum documents but is not given the salience that Gal (2005) advocated. He argued that people cannot fully understand independence without understanding dependence. In a similar way Gal highlighted stability as the complement of chance variation and regularity as the complement of randomness. The addition of these complementary processes (*alter-egos*) to the curriculum via appropriate experiential activities has the potential to greatly enrich key ideas like chance variation, randomness, and independence. Moreover, it means that these key ideas are accessible, albeit through informal experiences, from an early age.

The emphasis in probability literacy on having students find or estimate probabilities is already strongly represented in national curriculum documents. However, I. Gal (personal communication, October 1, 2003) provided fresh insights on how stronger links might be forged among classical, frequentist, and subjective approaches to probability and on how probabilities in the real world can be estimated:

Real world probabilities are usually not computed but *estimated* or *judged* [author’s emphasis], and in ways that do not fit neatly any one of the three views of probability. Usually, information from multiple sources (classical, frequentist) is used, but heavy reliance is also made of a “subjective” or personal judgment process that is affected by a person’s degree of experience in deriving probabilities.

Although the Australian *National Statement* (AEC, 1991) focused on all three approaches to probability measurement and investigated subjective probabilities in social contexts, Gal’s work encourages an even stronger link by using all three approaches to estimate real-world probabilities. Investigations such as the one implied by Gal would also seem timely in future revisions of national curriculum documents.

Gal (2005), Moore (1990), and Watson (1997a) placed strong emphasis on the role of language and terminology in communications about chance. Their research supports the strong emphasis on language that is already present in national curriculum documents. Moreover, in harmony with the theoretical positions espoused by these three writers, curriculum

documents make strong claims about the need for chance language to evolve and be refined as students build up their knowledge of and experience with probability through the grades.

The notion of *critical questions* or the need for people to become familiar with flaws, problems, and biases in stochastic information is a key aspect of Gal’s probability literacy model, Watson’s (1997a) statistical literacy framework, and the writings of Rutherford and Ahlgren (1990) and Utts (2003). In addition, Watson (1997a) and Watson and Moritz (2000) have undertaken valuable research that invited students to challenge claims in media extracts. Watson (1997a) concluded that, at the highest level of statistical thinking, students possessed the confidence to challenge what they read in the media so long as teachers made them “aware of the expectation that they must constantly question conclusions” (p. 110). In the following example, students were asked whether the claim made could be justified on the basis of the statistical processes used.

Some 96 percent of callers to youth radio station Triple J have said marijuana use should be decriminalised in Australia. The phone-in listener poll, which closed yesterday, showed 9,924-out of the 10,000 plus callers-favoured decriminalisation, the station said. Only 389 believed possession of the drug should remain a criminal offence. Many callers stressed they did not smoke marijuana but still believed in decriminalizing its use, a Triple J statement said. (Watson, 1997a, p. 110)

This question involves mostly issues about sampling, and it is true that critical questions about chance are more rare. However, as a result of Watson’s influence on the Tasmanian *Mathematics Guidelines K–8* (Department of Education and the Arts, Tasmania [DEA], 1993), there are several challenge-type questions on chance, one of which is illustrated below.

Peter buys a Tattslotto ticket every week. He never selects 6 consecutive numbers because he believes his chances are better by selecting a number from each line of the entry form. What do you think about this line of reasoning? (DEA, 1993, p. 11)

Watson’s research indicates that appropriate critical questions can be extracted from media, financial, and scientific reports. Moreover, these questions should be introduced in the primary school curriculum, as there is certainly a need for children to begin to question statistical and probabilistic data from an early age.

The final knowledge component deals with context and, in particular, probabilistic issues in social activities, public discourse, and personal decisions. In

essence, Gal (2005) suggested that his earlier-stated knowledge components (big ideas, figuring probabilities, terminology, and criticalness) need to be applied in various social and human contexts. All three national curriculum documents that we visited earlier do this in regard to games of chance and the language of chance. The Australian documents (AEC, 1991, 1994) also incorporated social contexts like gambling where subjective probability, in addition to classical and frequentist probabilities, is needed to interpret the situation. More can be built into the curriculum documents in order to meet the contextual goal that Gal is envisaging. For example, additional activities involving critical questions and the kinds of cultural and media-related tasks generated by Watson (1997a) would bring students nearer to this aspect of probability literacy.

The *dispositional* elements of Gal's (2005) probability literacy model and Watson's (1997a) statistical literacy framework relate more to pedagogical directions than they do to content directions; hence we will discuss these more in the next section. However, dispositional elements such as personal sentiments, critical stance, and beliefs and attitudes also need to be considered in the development of probability content so that learners will have the opportunity to experience situations in which these dispositions are challenged. Sentiments like risk aversion and anticipated success and failure will not generally be felt in decontextualized problems but can occur in games, personal estimates of likelihood, and in problems children pose for themselves. A critical disposition or a willingness to hold a questioning attitude can only be revealed and enhanced by having children from an early age engage in tasks that require them to challenge data or judgments. Similarly, if children are to develop beliefs such as "it is legitimate to be critical about probabilistic messages," these kinds of messages need to be built into the content of tasks and experiences that the students face.

In summary, the knowledge and dispositional elements of Gal's probability literacy model (2005) and Watson's (1997a; Watson & Callingham, 2003) statistical literacy framework have implications for the kind of probability content that should be incorporated in the Pre-K through Grade 12 curriculum. More specifically, notions like chance variation and stability, independence and co-occurrence, figuring likelihood from multiple sources (classical, frequentist, and subjective), language of chance, critical questions, and contexts need to be examined more deeply when future curriculum documents are developed. Curriculum programs also need to take cognizance of children's dispositions towards situations involving chance.

Pedagogical directions. Gal's (2005) probability literacy model and Watson's statistical literacy framework (1997a) are relatively silent about pedagogical issues. However, a number of pedagogical directions can be inferred from their writings. Both of them have a strong thrust on context or the "context of wider social discussion" (p. 108) in Watson's case, and both support a contextual problem-solving approach to probability especially when dealing with critical questions and scrutinizing media statements. This strong contextual orientation is also supported by Rutherford (1997), who noted that the starting point for deciding what constitutes quantitative literacy is less mathematics itself and more the contexts in which people are likely to encounter the need for mathematical skills (p. 62). Hence, pedagogical approaches that focus on problem-solving within cultural, social, and personal contexts are indicative of what Gal, Rutherford and Watson seem to be advocating for the teaching and learning of probability.

Somewhat in contrast with the almost exclusive emphasis of national curriculum programs on classical and frequentist approaches to probability, Gal and Watson emphasize the importance of students using multiple approaches to estimating probability. Even though inserting multiple approaches to probability into curriculum content is relatively easy, it is critical that teachers feel comfortable with the kinds of activities that support the combined use of classical, frequentist, and subjective approaches. The following "Roulette Investigation" from *Activity Book 3* of the Tasmanian Mathematics Guidelines K–8 (DEA, 1993) is indicative of the kind of task that could be used by a teacher to promote all three approaches to probability.

Some people believe that if a particular colour comes up on the [roulette] wheel a number of times in succession then the likelihood of the other colour coming up on the next spin is greater than one-half. For example, if red has come up three times in succession, the chance that the next spin will be black is greater than the chance that the next spin will be red. Investigate this using a toy roulette wheel. (DEA, 1993, p. 11)

In using such a task, teachers can encourage students to estimate and justify their own subjective probabilities as well as to use logical reasoning and data collection to generate theoretical and experimental probabilities, respectively. Moreover, such an investigation affords an opportunity for teachers to confront representativeness (Kahneman & Tversky, 1972).

Our final pedagogical implication from the work on probability literacy deals with Gal's (2005) and Moore's (1990) strong emphasis on the big ideas:

chance variation, randomness, independence, and their complementary elements stability, regularity, and co-occurrence. Contrary to the present pedagogical style in which the teaching of probability seems to focus rapidly on qualitative and numerical likelihood judgments, there is a need for teachers to provide opportunities for students, especially young children, to experience the following: the stability associated with causal action and the variation inherent in chance, the regularity in deterministic phenomena and the short-term unpredictability and long-term predictability of randomness, and the co-occurrence of some events and independence of others. For example, teachers might have students compare and contrast activities like the following:

- the day that follows Monday each week (stability)
- the number that follows a “six” when rolling a die (chance variation)
- the order of the days of the week (deterministic phenomena)
- the percentages of heads and tails when you toss a fair coin a small number of times versus a large number of times (randomness)
- getting a six on your first and second throw of a die (independence)
- throwing a six and an even number when rolling a die (co-occurrence).

Activities such as these, especially if they are located in technology environments, can provide informal experiences for children that involve conjecturing, experimenting, collecting data, building arguments, and discussion of results. As a consequence students may be enabled to build increasingly more sophisticated understanding of chance variation, randomness, and independence and hopefully a firm base for subsequent mathematical approaches to probability.

Our review suggests that curriculum research in probability is in its infancy, not surprising given the brief time that probability has been a mainstream part of the school mathematics curriculum. As current curricula on probability are evaluated, more extensive research is undertaken on the teaching and learning of probability, and further work on probability literacy emerges, the probability strand will undergo further evolution as will the instructional environments that support it.

REFLECTIONS AND FUTURE DIRECTIONS

In his chapter on stochastics in the first *Handbook of Research on Mathematics Teaching and Learning*, Shaugh-

nessy (1992) presented a “wish list” of research for the next decade. The topics in his wish list were directed towards mathematics educators; however, they were also pertinent to researchers in other fields like psychology and sociology. As we near the end of this present review, it is appropriate to reflect on the research presented and appraise it in terms of Shaughnessy’s wish list. We will also generate our own research agenda in probability for the next 10–15 years by suggesting directions and uncharted waters that need to be addressed.

Reflections

Although Shaughnessy’s wish list included research dealing with both statistics and probability, we will concern ourselves with needed research in probability. Shaughnessy’s need areas included the following: the development of assessment instruments, studies on secondary school students’ conceptions and misconceptions, cross-cultural research studies, studies on teachers’ conceptions, studies that examine the effects of instruction on students’ conceptual knowledge and intuitions, studies that examine the effects of computer software on learning and teaching, and the role of metacognition in decision making under uncertainty. For each of these areas, we will evaluate the corpus of research that has been presented in this chapter according to the criteria implicit in Shaughnessy’s wish list.

Development of Assessment Instruments

The key thrust in Shaughnessy’s comments on assessment instruments focused on the need to develop standard and reliable tools, in both written and interview format, that assessed students’ probability conceptions. He also underscored the importance of developing consistent instruments across a wide range of grade levels and a variety of contexts.

As a starting point, it is noteworthy that the current period has seen promising growth in the formulation of *assessment frameworks*. Some of these frameworks (e.g., Garfield, 1994; Jolliffe, 1991), although set in the context of statistics and probability, have focused on more general assessment constructs such as what to assess, the purpose of assessment, who will do the assessment, the method of assessment, and the nature of feedback and follow-up action (see Garfield, 1994; Jolliffe, 2005). Other frameworks (e.g., Jones, Thornton, et al., 1999; Tarr & Jones, 1997; Watson et al., 1997; Watson & Moritz, 2003a, 2003b) have produced characterizations of students’ probabilistic thinking and beliefs in terms of hierarchical levels. Such characterizations, as well as being useful for

monitoring student growth during instruction, have also resulted in the production of written and interview assessments in probability.

These research instruments together with the NAEP assessment material (Shaughnessy & Zawojewski, 1999) have gone some way to addressing Shaughnessy's appeal for standard and reliable probability instruments that cover a wide range of grade levels and incorporate a variety of contexts. In particular, Watson and her colleagues (Watson et al., 1997; Watson & Moritz, 1998, 2003a, 2003b) have produced written survey instruments that measure various aspects of chance measurement, cover a wide range of ages (Grades 3 through 9), and have the advantage that they have been used in longitudinal research and validated on large samples. By way of contrast, Jones and his colleagues (Jones, Langrall, et al., 1997, 1999; Jones, Thornton, et al., 1999; Tarr & Jones, 1997) have generated *interview* assessments. Their instruments cover a range of children's probabilistic reasoning (Grades 1 through 8); however, they have not been subjected to the large-scale population scrutiny that is a powerful feature of Watson's assessments. In addition to these assessment instruments, we underscore Batanero and Serrano's (1999) assessment of randomization and combinatorics with high school students and Amir and Williams's (1999) instruments that examined cultural differences in children's understanding of chance and luck.

Notwithstanding the promising assessment activities adumbrated above, greater coherence is still needed in the way that multiple perspectives of probability (classical, frequentist, and subjective) are assessed. These multiple perspectives of probability have been highlighted for some time as deficit areas in research for both instruction and assessment (Metz, 1997; Steinbring, 1991). We will return to them in discussing our agenda for future research.

Secondary Students' Conceptions and Misconceptions

Shaughnessy (1992) asserted that most of the probability studies have been undertaken with elementary or college students. He called for research on secondary students' thinking about chance, random events, and decision making under uncertainty. In addition, he advocated studies that examine the influence of instruction on secondary students' probabilistic thinking.

This period has seen a much stronger research base on the probabilistic reasoning of secondary students. As described earlier, Batanero and her colleagues have been especially prominent with their

research on the following topics: perception of randomness (Batanero & Serrano, 1999), use of heuristics and biases (Batanero, Serrano, et al., 1996), and combinatorial reasoning (Batanero et al., 1997). Other researchers have also investigated heuristics and misconceptions that secondary students bring to the classroom (e.g., Benson & Jones, 1999; Gras & Totohasina, 1995; Sanchez & Hernandez, 2003; Shaughnessy & Ciancetta, 2002; Zimmermann & Jones, 2002), and, in a related area, Fischbein and his colleagues (e.g., Fischbein et al., 1991; Fischbein & Schnarch, 1997) have examined the evolution of probabilistic intuitions across age ranges that included secondary students. Instructional studies such as Zimmermann's (2002) classroom investigation of secondary students' individual and collective thinking about simulation are beginning to emerge. In fact, the extent of recent research involving secondary students is impressive, as can be readily gleaned from Batanero and Sanchez's (2005) review.

Although Shaughnessy's call for additional studies on secondary students' probabilistic conceptions has been heeded, much of the research has focused on their probabilistic reasoning prior to instruction. There is still a void in the kinds of classroom studies that Shaughnessy advocated, that is, studies that investigate the effect of instruction on secondary students' probability learning.

Absence of Cross-Cultural Studies

Shaughnessy raised concerns about the lack of probability research outside of western countries. He advocated large- and small-scale studies that investigate the influence of different cultures on students' thinking in decision making and probability estimation tasks.

During this period probability research has been undertaken outside of western culture. Jansem (2005) and Lamprianou and Lamprianou (2003) studied the nature of primary students' probabilistic thinking in Thailand and Cyprus, respectively; Li (2000) analyzed probabilistic understanding of middle and high school students in the Republic of China; and Polaki (2002a, 2002b) undertook clinical studies and teaching experiments on the probabilistic reasoning of elementary students in Lesotho. However, as Greer and Mukhopadhyay (2005) noted, these studies relate strongly to English or American research and do not focus on "whether the culture of the participants might have any effect on their way of thinking about probability" (p. 317).

By way of contrast, Amir and Williams (1999) and Peard (1995) have investigated the influence of culture on probabilistic thinking. Probing elementary students' religious beliefs and experiences with random phenomena, Amir and Williams concluded that cultural factors influenced the nature of the students' probabilistic knowledge and in turn their probability learning. Working with high school students, Peard investigated the effect of social background. He reported markedly different probabilistic intuitions for students whose social or family background included extensive familiarity with "track" gambling. Although these studies whet our appetite for research into the influence of culture on probabilistic knowledge, they did not incorporate the strong ethnic influences that seemed to be part of Shaughnessy's agenda.

Teachers' Conceptions of Probability

Given the important role that teachers play in probability learning, Shaughnessy advocated research on the knowledge and beliefs that preservice and practicing teachers exhibit toward probability. He also recommended research and development activities that focus on teachers' knowledge of students' conceptions and misconceptions in probability.

Stohl's (2005) review concluded that there had been limited response to Shaughnessy's call for research on teachers' knowledge and beliefs about probability. Nevertheless, we have seen some research in these areas: Begg and Edwards (1999) and Carnell (1997) have documented primary and middle school teachers' misconceptions, and Watson (2001) has profiled primary and secondary teachers' content and pedagogical knowledge in probability and statistics. This research has been sufficient to raise concerns about teachers' lack of confidence in their knowledge of probability, their anxiety about teaching probability, and their disquiet about the time that probability and statistics take away from other strands in the mathematics curriculum (e.g., Gattuso & Pannone, 2002). With regard to Shaughnessy's appeal for research on teachers' knowledge of students' conceptions of probability, we found only one study (Watson, 2001) that addressed this question. The fact that Watson concluded that both secondary and primary teachers were more familiar with students' procedural knowledge than with their conceptual knowledge highlights the need for teacher development programs in probability. However, Stohl (2005) observed that, although many projects have focused on teachers' understanding of data analysis (e.g., *Teach-Stat*, Friel & Joyner, 1997; *Alabama Quantitative Literacy Workshop*, Yarbrough,

Daane, & Vessel, 1998), there has been little action on programs in probability. Watson's (1997b) program *Chance and Data for Luddites* may be an exception.

Effects of Instruction

Shaughnessy emphasized the need for research on instruction, especially instruction that relates to students' conceptual knowledge of probability. He called for partnerships between researchers and teachers (e.g., through teaching experiments) that would trace changes in students' probabilistic conceptions and intuitions during instruction.

With the emergence of more sophisticated teaching-experiment designs (Cobb, 2000; Lesh & Kelly, 2000; Simon, 2000), there has been a marked increase in instructional research on probability. At the elementary, middle, and high school levels, several studies (e.g., Aspinwall & Tarr, 2001; Jones, Langrall, et al., 1999; Polaki, 2002a, 2002b; Tarr, 1997; Zimmermann, 2002) have traced students' individual and collective thinking in probability during instruction and have identified pedagogical processes that seem to impact learning. In addition, Polaki and Zimmermann have documented and analyzed classroom mathematical norms and practices that may provide valuable pedagogical content knowledge for teachers. Although these teaching experiments involved teacher-researcher partnerships, none of them constituted the kinds of long-term collaborative interventions that Shaughnessy had in mind. The research seems to have reached a stage requiring teaching experiments that use more robust experimental designs and statistical methods to evaluate the effectiveness of instructional interventions.

Effects of Computer Software

This agenda item is clearly related to the previous one. In identifying issues associated with technology, Shaughnessy called for the development of research into computer software that would change students' probabilistic conceptions by taking advantage of the speed, graphics, and simulation potential of microcomputers. He especially underscored the need for researchers and teachers to be able to provide feedback during the developmental phase of such software.

Although there has been manifest support for the use of technology in the teaching and learning of probability (Ben-Zvi, 2000; Biehler, 1991; Dörfler, 1993; Konold, 1991a) and considerable development in probability-oriented computer software and graphics-calculator capability (Edmark Corporation, 1996; Erickson, 2001; NCTM 2002a, 2002b, 2003), research

on the influence of technology on students' probabilistic conceptions is only beginning to emerge. The most promising research involves the use of microworlds and their influence on changing and expanding students' probabilistic thinking (Konold, 1995a; Paparistodemou et al., 2002; Pratt, 1998; Pratt & Noss, 2002; Stohl, 1999–2002). In particular, the work of Pratt and Stohl shows that microworlds can be designed to forge vital links between experimental and theoretical probability and to build cognitive mechanisms that enable children to challenge their own intuitions. There is also evidence of case studies that document the effect of computer software and graphics calculators on probability learning (Kissane, 1997a, 1997b; Konold, 1991a).

The Role of Metacognition in Decision Making Under Uncertainty

Shaughnessy specifically advocated research on metacognitive aspects of probabilistic thinking. He claimed that such research may shed light on students' probabilistic misconceptions and on ways to use metacognitive ideas to challenge such misconceptions.

Regrettably we were unable to locate, in this period, research that has focused on metacognitive aspects of the teaching and learning of probability. The situation in probability seems to reflect Lester's (1994) more general concern about the lack of mathematics education research on metacognition. Perhaps Goos and her colleagues' metacognitive research on mathematical modeling (e.g., Goos, Galbraith, & Renshaw, 2002) might act as a fillip for future research on the role of metacognitive processes in probabilistic modeling.

These reflections on probability research in the last 15 years reveal that gaps still exist in the research outlined in Shaughnessy's (1992) wish list. These extant gaps, together with areas where a need for new research has emerged more recently, will be taken up in the next section when we propose our own agenda for future research.

Future Research Directions in Probability: A Continuing Agenda

Although some aspects of Shaughnessy's (1992) wish list have been addressed during the past 15 years, critical areas remain: developmental research into assessment instruments; long-term teaching experiments, especially in secondary classrooms; studies examining the effects of cultural, social, and ethnic contexts on probabilistic reasoning; research into teachers' knowledge of student conceptions; evaluations of teacher enhancement programs; research into the ef-

fect of microworlds on students' probabilistic reasoning; and classroom research using metacognition to examine students' individual and collective thinking in probability. In generating a new agenda for probability research in the next 15 years, we will revisit these critical areas.

Students' Conceptions of Theoretical and Experimental Probability

Our first agenda item relates to cognitive research that is grounded in both classical and frequentist perspectives of probability. Although there has been considerable research at all age levels on students' conceptions of theoretical probability (e.g., Batanero et al., 1996; Fischbein & Schnarch, 1997; Jones, Langrall, et al., 1997, 1999; Watson & Moritz, 1998), there has been relatively little research on students' thinking about experimental probability and even less on students' understanding of the connections between theoretical and experimental probability (Aspinwall & Tarr, 2001; Stohl & Tarr, 2002).

We believe that there is a need to explore students' thinking about classical and frequentist perspectives of probability across the complete range of ages from early childhood to high school. As an illustration of what we intend, consider the following investigative questions in relation to a probability measurement task like the following: "What color is most likely to come up if a cube with 1 green, 2 red, and 3 blue faces is rolled?"

- How do students address such a task unassisted?
- How do students respond to the suggestion that it might be helpful to experiment by rolling the cube a number of times?
- How do they deal with the experimental data including the size of the sample of data?
- How do they respond to the suggestion that it might be helpful to examine the symmetry of the cube?
- How do students deal with symmetry and its implications for theoretical probability?
- Do students perceive any relationship between likelihood estimates based on experimentation and those based on symmetry?

Research questions and tasks like the ones illustrated above can be used to explore students' thinking during clinical interviews and to trace students' individual and collective thinking during classroom instruction. Although curriculum documents (e.g., AEC, 1994; NCTM, 2000) have stressed the impor-

tance of focusing on experimental probability, and some researchers (e.g., Aspinwall & Tarr, 2001; Garfield & Ahlgren, 1988; Jones, Langrall, et al., 1999; Shaughnessy, 1977; Steinbring, 1991; Stohl & Tarr, 2002) have identified tasks and learning sequences that have incorporated the simultaneous development of both theoretical and experimental probability, only Aspinwall and Tarr and Stohl and Tarr have documented how students responded to these approaches. We believe that cognitive research under the rubric of classical and frequentist probability is a key endeavor for the next period as such research goes well beyond questions of likelihood. It provides a frame for investigating students' conceptions of key ideas such as the following: chance, random phenomena, conditional probability, independence, the law of large numbers, and probability distributions.

The area of subjective probability is more problematic. Even though it is commonly used by people in everyday reasoning, the fact that it is not widely represented in mathematics curricula means that there is less immediate interest on the part of researchers to investigate this aspect of probability measurement. We think it is timely for researchers in mathematics education to examine subjective probability and the way that students conceptualize it.

Adopting a Critical Stance to Stochastic Information

One of the important goals for teaching probability in school mathematics is to prepare students to deal with the statistical and probabilistic information that increasingly impacts their lives. More specifically, criticalness (Gal, 1994), the ability to take an evaluative stance with respect to stochastic flaws and biases contained in media, marketing, and financial information, is of paramount importance in the quest for probability literacy. In spite of its precedence there has been little research on criticalness and little injection of it into current curriculum programs.

Watson's research (e.g., Watson, 1997a, Watson & Moritz, 2000) has made a valuable beginning in this area, especially her studies in which students are challenged to evaluate media claims. Of particular importance for the future was her use of longitudinal designs and large samples to build models of the evolution of students' cognitions on criticalness. Notwithstanding Watson's efforts, the research on criticalness is in its infancy, and there is a need for empirical studies that investigate students' ability to assess social information that involves probabilistic processes like the following: experimental probabilities, the law of large numbers, conditional probability, simulation, sampling, and applications of probability in statistical inference.

The Development of Instructional Theory

Shaughnessy's (1992) wish list included a recommendation for long-term teaching experiments, especially with high school students. We go further by highlighting the need for teaching experiments that investigate diverse approaches to the teaching of probability. In reality, there has not been substantial work on the development of an instructional theory for probability since Steinbring's (1991) groundbreaking work more than a decade ago. Although Steinbring advocated an approach to probability that incorporated both classical and frequentist probabilities (e.g., through *task systems*), he did not attempt to embellish the instructional strategies incorporated in his theory or provide a great deal of empirical data to support his theory.

We claim that there is a critical need for researchers to document instructional strategies for the teaching and learning of probability at all age levels. For example, as noted earlier, recent research (e.g., Gigerenzer, 1994; Hopfsenberger et al., 1999) suggests that probability should be introduced through frequency data tasks rather than probability tasks per se. Studies are needed to build hypothetical learning trajectories (Simon, 1995) for such tasks so as to support teachers' use of the strategy. This data-oriented strategy might also be the basis for forging connections between probability and statistical research. Accordingly, this *frequency* strategy and emerging strategies like those linked to microworld environments (e.g., Pratt, 2000; Pratt & Noss, 2002) need to be trialed and monitored to examine their effects on students' learning and beliefs.

As part of the emphasis on research involving teaching experiments and innovative instructional strategies, we believe that there is a need for studies that use carefully controlled experimental designs and statistical analyses (Kilpatrick, 2001). We are not suggesting a blanket return to scientific positivism but rather a balanced approach that uses mixed designs similar to those suggested by Tashakkori and Teddlie (1998).

The Development of Assessment Instruments

In maintaining assessment as a continuing research endeavor for the next period we wish to propose different directions from those suggested by Shaughnessy (1992). First, in accord with our advocacy of systematic research on students' cognitions vis-à-vis classical and frequentist approaches to probability, we claim that research is needed to develop and validate written and interview instruments that examine students' reasoning about experimental and theoretical probability and their relationship. More-

over, these instruments should be able to examine students' reasoning in experimental and theoretical probability over an extended age range similar to the written instruments used by Watson and her colleagues (Watson, 1997a; Watson et al., 1997; Watson & Moritz, 1998, 2003a, 2003b). Notwithstanding the importance of written instruments, we also underscore Greer and Mukhopadhyay's (2005) caveat that written assessments do not lend themselves to assessing students' understanding of the relationship between theoretical and experimental probability.

Second, in accord with Metz's (1997) call for chance to be assessed along three dimensions: *cognitive* (conceptual understanding), *epistemological* (beliefs), and *cultural* (especially the culture of the classroom), we believe that there is a need for research that evaluates assessment material according to Metz's dimensions. Such assessments need to be used in tandem with research involving teaching experiments. In particular, it is vital that researchers trace students' conceptual understanding of randomness and chance variation and also monitor changes in their "beliefs about the place of chance and uncertainty in the world" (p. 225).

Third, the development of assessment instruments that embody the *cultural* dimension of Metz's (1997) framework have the potential to stimulate sharper research on the effect of different cultures on students' probabilistic reasoning. As Greer and Mukhopadhyay (2005) have noted, there is a need for "a sensitivity to the cultures of others when teaching mathematics to multicultural classes and when teaching a curriculum that is not indigenously grounded" (p. 318). Moreover, with respect to probability, these two researchers declare that there are few studies that bear on the cultural issues endemic to curriculum, instruction, or assessment.

Large-Scale Curriculum and Instructional Research

Investigations that are integrated into a cluster of school-based longitudinal projects in culturally and economically diverse settings have not, for various reasons (e.g., limited awareness of the importance of stochastics), included probability research. Such projects need to include stochastics, and similar to QUASAR (Silver & Stein, 1996), their manifesto should embrace curriculum development, teacher enhancement, computer resources, and assessment materials.

These large-scale research projects will be appropriately situated to use designs such as multitiered teaching experiments (Lesh & Kelly, 2000) and educational development and developmental research (Gravemeijer, 1998). As such they will have the po-

tential to trace the growth of students' probabilistic conceptions and intuitions from elementary through secondary school. They will also generate increased knowledge about students' collective thinking in probability and document sociomathematical norms and classroom mathematical practices (Cobb, 2000) that will help teachers build appropriate learning environments for probability instruction. Moreover, they will provide an ongoing base for teacher support and opportunities to build and monitor teacher knowledge of students' conceptions and misconceptions in probability.

In a related area, this kind of developmental research will be ideally suited to address Shaughnessy's (1992) concerns about the lack of probability research that focuses on metacognition. Given the impetus of research by Goos and her colleagues (Goos & Galbraith, 1996; Goos et al., 2002) that has documented the metacognitive characteristics of successful and unsuccessful problem solvers, there is infrastructure that can be applied to probability research. More specifically, the theoretical base and methodological design associated with Goos's research seems apposite for future metacognitive research in probability, especially instructional research.

CONCLUSION

This chapter presents an examination of research on cognitive, pedagogical, and curriculum aspects of probability in the wake of its introduction as a mainstream topic in the mathematics curricula of various countries (e.g., AEC, 1991, 1994; DES, 1991; NCTM, 1989, 2000). In our analysis of cognitive and pedagogical research in probability, we have compared and contrasted research in this period with research reviewed by Shaughnessy (1992) in terms of philosophical underpinnings, purpose and direction, context, and methodological features. The research in this period reveals a stronger thrust on documenting and characterizing the probabilistic knowledge that students bring to the classroom (e.g., Batanero et al., 1997; Jones et al., 1997; Metz, 1997; Watson et al., 1997) and on tracing students' individual and collective reasoning during instruction in probability (e.g., Batanero et al., 1997; Jones, Langrall, et al., 1999; Polaki, 2002a, 2002b). Researchers in this period have also generated valuable knowledge about learning environments (e.g., Amir & Williams, 1999; Castro, 1998; Horvath & Lehrer, 1998; Jones, Thornton, et al., 1999) including technology-supported environments (e.g., Pratt, 2000; Stohl, 1999–2002; Zimmer-

mann, 2002) that appear to have the facility to change students' probabilistic intuitions and beliefs.

With respect to curriculum research, we have examined the new literature on probability literacy in the wake of evidence suggesting that the implementation of probability in schools may not be as pervasive or as sustainable as national curriculum documents portray. The literature on probability literacy (e.g., Gal, 2002, 2005; Watson, 1997) has identified a need for fresh directions, greater coherence, and an enlargement of ideas that prepare children to deal with the vagaries of variation. In particular, this research advocates a richer understanding of chance variation and randomization; a more unified development of the classical, frequentist, and subjective approaches to probability; and an emphasis on critical thinking that enables children to confront the potential exploitation of media marketing.

We have also generated a research agenda for the next decade or more. In addition to the curriculum research foreshadowed above, this agenda promotes endeavors like the following: the development of instructional theory in probability especially for secondary students, characterizations of student cognitions in relation to experimental and theoretical probability and their connections, investigations into teachers' pedagogical knowledge and knowledge of students' probabilistic cognitions, and explorations of the role of metacognitive processes in the learning of probability especially classroom learning.

In looking back over this period, one of the striking aspects is the emergence of a robust research base on the probability knowledge and conceptions that students bring to the classroom. There is also evidence of a stronger research literature in pedagogical and instructional aspects of probability teaching. Although Greer and Mukhopadhyay's (2005) caveat, "a major challenge to the field is harvesting what is known from this [probability] research to inform teaching" (p. 315), still has currency, the research in this period has made significant contributions to the teaching and learning of probability in schools.

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RESEARCH ON STATISTICS LEARNING AND REASONING

J. Michael Shaughnessy

PORTLAND STATE UNIVERSITY

At the beginning of the 1990s a rather unique professional opportunity came my way when the National Council of Teachers of Mathematics (NCTM) asked me to author a chapter on research on the teaching and learning of probability and statistics for the *Handbook of Research on the Teaching and Learning of Mathematics* (Shaughnessy, 1992). At that time, research on students' understanding of probability and statistics was just beginning to blossom in the United States though research in stochastics had been conducted for several decades in other countries, principally in Europe. During the 1980s, interest in research in probability and statistics in the United States was being fueled by interactions with researchers from other countries at conferences such as the International Conference on Teaching of Statistics (ICOTS) and the International Group for the Psychology of Mathematics Education (PME). Towards the end of the 1990s, the long efforts of a core group of mathematics and statistics educators in the United States to promote an increased emphasis on statistics in school mathematics programs finally took root. Probability and statistics were included among the content standards in NCTM's groundbreaking document, *Curriculum and Evaluation Standards for School Mathematics* (1989). This was the first time that a national organization in the United States, in fact the national organization for mathematics teachers, placed statistics on an equal footing with number sense, algebra, geometry, and measurement as a critical foundation stone for school mathematics. Suddenly statistics became a more attractive area in which to conduct research on student learning. State

curriculum leaders followed the lead of the NCTM *Curriculum Standards* and began to build statistics into their state mathematics frameworks and likewise began to assess students' growth in learning statistical concepts. Consequently, curriculum developers began to pay more than lip service to statistics. Prior to the *Standards* statistics had been a lost stepchild in mathematics curriculum frameworks, the mere frosting on any mathematics program if there was time at the end of the school year. Now statistics is here to stay as a major strand in school mathematics programs in the United States.

This book represents the second effort by NCTM to survey, analyze, and compile research in mathematics education, and I have another unique opportunity to author a chapter, this time focusing on statistics (maybe they are hoping I will get it right this time). However, this time around a synthesis and analysis of the research in data and chance is far more challenging than it was 15 years ago. Research in probability and statistics is no longer a fledgling discipline, and reviewing all of the relevant literature is no longer possible as it was for the first *Handbook*. There has been an amazing boom in research, curriculum development, and assessment in statistics education. A recent review of the research literature in data and chance carried out just in Australasia over a 4-year period from 2000 to 2003 turned up over 150 citations in statistics education (Pfannkuch & Watson, 2005). Compare that to around 150 international references that were in the 1992 *Research Handbook* chapter that covered a 30-year period of research in statistics all over the world, and

the magnitude of the current task becomes apparent. Contributions to the field have come from a very wide range of investigators, including mathematics and statistics educators, statisticians, cognitive psychologists, educational psychologists, and science educators.

Satellite roundtable conferences on statistics education have been held at the last three meetings of International Congress of Mathematics Education (ICME) in Granada (1996), Hiroshima (2000), and Lund (2004), and these have continued to catalyze research in statistics education. There is also now an international group of statistics educators who have been holding semi-annual conferences on Statistics Reasoning, Thinking, and Literacy (SRTL) in Israel (1999), Australia (2001), the United States (2003), and New Zealand (2005). The field is now expanding so rapidly that keeping abreast of everything is quite impossible. However, I do not think that the growth in research in probability and statistics is well known among the general community of mathematics educators. Recently at a conference a famous and very widely read colleague in mathematics education asked me, “Are you still working in statistics? There hasn’t been very much going on there recently, has there?” I was really amazed that he was unaware of what has been going on in the area, and he is probably typical of many mathematics educators.

The sheer volume of research in both probability and statistics has in fact necessitated two separate chapters in this edition of the *Handbook*, one on probability (see Jones, Langrall, & Mooney, this volume) and one on statistics. In one sense, this is rather unfortunate because there are natural connections between statistics concepts and probability concepts, for example in sampling distribution, confidence intervals, and significance tests. Furthermore, researchers must continue to clarify the learning connections between statistics and probability for teachers and students. However, reviews of research are more useful if they focus on particular issues, rather than trying to do everything. In this chapter I will focus primarily on research on students’ learning and reasoning in statistics from the past 15 years or so, making connections to probability along the way. Most of the research discussed in this chapter concerns students’ understanding of descriptive statistics and data analysis, as there has not yet been much research devoted to students’ understanding of inferential statistics—confidence intervals, hypothesis testing, p-values, and so on.

STATISTICS IN THE SCHOOLS TODAY

When the 1989 *Standards* document was updated in NCTM’s *Principles and Standards for School Mathematics*

(2000) (PSSM), statistics continued to hold a prominent position in NCTM’s vision for school mathematics. The two editions of the *Standards* documents (1989, 2000) have had counterparts in other countries that have proclaimed similar messages, touting the important role that statistics plays in the education of our citizens, both as learners of statistics at the school level and as consumers of statistics at the adult level. For example, *A National Statement on Mathematics for Australian Schools* (Australian Educational Council, 1991) and *Mathematics in the New Zealand Curriculum* also called for considerable statistics to be taught as part of school mathematics programs, in part to empower students to critically evaluate data and claims made about data.

National calls for increased attention to statistics have played an important role in catalyzing research in the learning of statistics. However, national calls themselves will not amount to much unless good curriculum materials for students and accompanying guides for classroom teachers are available to implement the suggestions of those calls. Statistics has been embedded in a prominent way in many of the recent curriculum development projects in the United States and Canada throughout the 1990s. These curricula, partly inspired by the development and implementation of the *Standards* documents, have provided another major impetus for research on statistics. The *Standards* documents have influenced curriculum projects such as *Mathematics in Context* (1994), *Core-Plus Mathematics* (1997), *Data Driven Mathematics* (1999), *Connected Mathematics* (1998), and *Investigations Into Number, Data and Space* (1998) just to point out a few examples. The round of curriculum materials in the 1990s systematically wove statistics throughout all their targeted grade levels and incorporated material to help meet the recommendations of PSSM for probability and statistics in K–12 that say that students should:

- Formulate questions that can be addressed with data and collect, organize, and display relevant data to answer them.
- Select and use appropriate statistical methods to analyze data.
- Develop and evaluate inferences and predictions that are based on data.
- Understand and apply basic concepts of probability.

The *Standards* documents and the curriculum development projects of the ‘90s have both helped to increase research interest in how students think about data and chance, and how teachers might address statistical concepts. The mere existence of good

curriculum materials for statistics does not mean that they are being uniformly and faithfully implemented everywhere throughout the United States and Canada. Our teaching force is undernourished in statistical experience, as statistics has not often been a part of many teachers' own school mathematics programs. In many schools there is a tremendous need for professional development in the area of statistics.

At the same time that many teachers need more work with statistics themselves, the number of students taking Advanced Placement (AP) statistics courses in secondary school has greatly increased in the United States. The AP exams, constructed by Educational Testing Service (ETS), provide a mechanism for secondary school students to obtain college-level credit for coursework taken in high school. Growth in the number of students who register for the AP statistics exam in the United States has been the fastest of any AP course in the history of the Advanced Placement Program. In 1997, the first time it was administered, 7,500 students took the AP statistics test. According to the chair of the AP statistics grading committee, the number of students taking the exam grew to 37,000 students in 2000, then to 50,000 students in 2002, and to as many as 65,000 in 2004 (Roxy Peck, personal communication, March, 2004). The AP statistics option has been another catalytic force in the growth of attention to statistics in K–12 schools. Advanced Placement Statistics is a very popular option in secondary schools because most students need to take some sort of statistics course in college, whatever their major. In fact, introductory statistics has recently outgrown calculus for the largest enrollment in any mathematics or statistics class at many colleges and universities.

Spreading the word on the importance of statistics for students and getting good statistics materials adopted and faithfully implemented in K–12 classrooms requires a long-term effort that must be coupled with research on student learning and assessment of student progress. In that regard, perhaps an overall snapshot of student progress in learning statistics is a good starting point for our research journey in this chapter. In the next section I provide a brief overview of students' knowledge of statistical concepts and skills as documented by the *National Assessment of Educational Progress* (NAEP) in the United States.

NATIONAL ASSESSMENT OF EDUCATIONAL PROGRESS: A SHORT OVERVIEW ON STATISTICS

Statistics items have been included in each of the NAEP frameworks since 1973. The variety and depth

of items have continued to grow over the past 30 years in NAEP. The percentage of NAEP items classified as data analysis, statistics and probability has more than tripled at Grade 12 (from 6% in 1986 to 20% in 1996) and almost doubled in Grade 8 (from 8% to 15% over the same time period). In the 1996 administration, the NAEP content strand called Data Analysis, Statistics and Probability was designed to emphasize “the appropriate methods for gathering data, the visual exploration of data, various ways of representing data, and the development and evaluation of arguments based on data analysis.” (National Assessment Governing Board, 1994, p. 77).

In the past, NAEP statistics items have included calculating measures of center (mean, median, mode), as well as making inferences from data, reading and interpreting graphical representations of data, predicting beyond the data given in graphs, identifying which statistic (e.g., mean or median) is more appropriate in various situations, and answering questions about samples and sampling. Information from NAEP tasks can help to identify growth in students' understanding of statistical concepts as well as potential trouble spots for students in understanding statistics. Therefore, NAEP data can be a good source for mathematics educators to formulate researchable questions about student growth and understanding. Some summary highlights of the 1996, and the 2000 and 2003 NAEP findings about students' knowledge of statistical concepts provide a backdrop for the rest of the chapter. In an analysis of the 1996 NAEP statistics items, Zawojewski & Shaughnessy (2000) noted that:

- Over half of 8th-grade students read information from tables, charts, and graphs but experienced difficulty in using the information for other substantive purposes such as drawing conclusions based on data.
- Based on performance on two NAEP items, about three-fourths of students in Grade 12 can successfully read line graphs, yet only 37 percent could read a box plot.
- Comparing 1990 results to 1996 results, there was significant growth in 8th- and 12th-grade students' performance on NAEP items that required they find the mean and median for particular data sets.
- When given a choice, 8th-grade and 12th-grade students tend to select the mean over the median, regardless of the distribution of the data.
- Given their performance on items that assessed the appropriateness of survey samples, over half of the students in Grade

8 appropriately considered the potential for bias and the number of data points.

A subsequent analysis by Tarr and Shaughnessy (in press) of the results on the statistics items from the 2000 and 2003 NAEP administrations found that:

- There was significant growth from 1996 to 2003 in 8th-grade students' performance on NAEP items that required them to determine the mean and median of given data sets. However, when asked to select the most appropriate summary statistic, most selected the mean, regardless of the distribution of the data; only about 1 in 5 students were able to justify their choice of the median.
- Although about 4 in 7 8th-grade students could identify a general formula for the mean, only about 1 in 4 students were able to determine the mean from a grouped frequency distribution, and even fewer demonstrated an understanding of the mean as representative of the entire data set.
- Less than half of the students in Grade 8 considered the potential for bias in sampling, but more than half recognized limitations associated with drawing inferences from small samples.
- Student performance on creating data displays was strong and may have reached ceiling levels in Grade 8 on bar graphs, pictographs, and circle graphs.
- Performance was poor on complex items that involved interpretation or application of information in tables and graphs. There have been little or no gains from 2000 to 2003, and performance on such items may in fact be slightly eroding.

The picture from these selected NAEP highlights is a good news and bad news situation. On the good news side, students' ability to read and understand tables, charts, and graphs, and their success on procedures such as finding the mean and median, are on the rise in the United States. On the bad news side, conceptual abilities to interpret and draw conclusions from graphs, and to make decisions on which calculation is more appropriate (e.g., mean or median) languish behind their procedural counterparts. Student performance on the statistics items on the NAEP has been steadily improving since 1990, although student performance on data and chance items on the NAEP started out at such a rock-bottom level that this improvement might be somewhat misleading. On the other hand, the steady growth of success on NAEP items does suggest that sta-

tistics concepts have been slowly making their way into school mathematics programs. There is still an enormous amount of work to be done to get statistics into every student's school mathematics program throughout the K–12 school years.

PLAN OF THE REST OF THIS CHAPTER

The two principal areas where most of the recent research in statistics education has occurred are students' knowledge and reasoning about statistics and teachers' knowledge of and teaching practices in statistics. To date, far more research has been conducted in students' understanding of statistics concepts than in teachers' knowledge and practices, and so the bulk of the literature reviewed and analyzed in this chapter is on research into student learning and understanding of statistics. A considerable amount of the research on students' understanding has focused on particular concepts or big ideas in statistics, such as centers (averages), variation, comparing data sets, information from samples, students' statistical learning with technology, and students' understanding of graphs. This chapter contains sections on research on students' understanding and reasoning about each of these big ideas. A section on research and development of teachers' understanding of statistics follows the research on student understanding. The chapter will conclude with sections on recommendations for the teaching of statistics, and some recommendations for future research in statistics education.

Before launching into a discussion of the research on student understanding of statistics I wish to begin with an overview and analysis of several models and research frameworks that provide descriptive or interpretive lenses for researchers in statistics education. These models and frameworks include models of *statistical thinking*, models of *statistical literacy*, and cognitive and developmental frameworks that can help to interpret students' *statistical reasoning* as they work on statistical tasks.

Models and Frameworks

For over 3 decades research in probability and statistics has been conducted in a tradition that relies on theoretical frameworks, some built, others borrowed. Earlier frameworks discussed in the first edition of this *Handbook* (Shaughnessy, 1992) included Kahneman and Tversky's theory of heuristics such as representativeness, availability, anchoring, and conjunction in decision making under uncertainty (Kahneman

&Tversky, 1972, 1973a, 1973b; Tversky & Kahneman, 1974, 1983). Other authors (e.g., Shaughnessy, 1977; Konold, 1989; Konold, Pollatsek, Well, Lohmeier, & Lipson, 1993) have applied or extended the work of Kahneman and Tversky. Konold (1989) accounted for types of student reasoning on tasks that did not fall neatly into Kahneman and Tversky's categories of heuristic reasoning. A model for reasoning under uncertainty has turned out to be much more complicated than some of the initial models suggested, as pointed out in several recent interpretive reviews of research in statistics (Konold & Higgins, 2003) and probability (Shaughnessy, 2003a).

Much of the earlier research in probability and statistics was concerned with misconceptions that students had about probability and statistics, a line of research that has persisted even until fairly recently (Fischbein & Schnarck, 1997). However, researchers have begun to pay more careful attention to the details of the development of students' understanding of and thinking about statistics. More mature representations of student thinking have arisen over the past decade. Much of this growth in new models and frameworks has come about from careful documentation and analysis of student thinking by researchers employing a qualitative research methodology while students work on statistical tasks that can elicit an array of responses, from rather naive to somewhat sophisticated responses. As a result, most researchers in statistics (and probability) now hunt for a spectrum of student thinking in their research work, rather than taking an approach that students either do or do not "have it." This alteration of perspectives on student thinking from a misconceptions viewpoint to more of a transitional conceptions position is partly due to an increased attention to and acceptance of a constructivist epistemology that places students at the active center of their learning. It is also partly due to the types of models and frameworks themselves that have been employed in response to a constructivist theory of knowledge.

In reviewing the literature for this second round of the *Handbook*, a new crop of frameworks and models for research in statistics is surfacing. A discussion of some of these models and frameworks is in order early on in this chapter, as they will help provide lenses for discussion and comparison of research results.

STATISTICAL THINKING, STATISTICAL LITERACY, AND STATISTICAL REASONING

The literature on judgment and decision making under uncertainty that was reviewed and analyzed in the prob-

ability and statistics chapter in the first edition of this *Handbook* (Shaughnessy, 1992) included a distinction among *normative*, *prescriptive*, and *descriptive* models that may help to distinguish among the current constructs of statistical thinking, statistical literacy, and statistical reasoning. Garfield (2002) summarized a number of general perspectives on what she called statistical reasoning, including correct and incorrect ways that students reason about statistics, and ways of assessing statistical reasoning. In this chapter I take a perspective that attempts to distinguish between statistical thinking and statistical reasoning, as well as to discuss statistical literacy.

Models of *statistical thinking* help both researchers and teachers to attend to the important concepts and processes in the teaching and learning of statistics. These models reflect what we want learners, consumers, and producers of statistics to know. Thus, models of statistical thinking are primarily *normative* models of what statisticians feel are the important concepts and processes of their discipline.

Models of *statistical literacy* help to identify critical statistical survival skills for both school students and adults. Students are primarily learners of statistics, but also they can be consumers of statistics in making decisions on what to buy, or possibly even producers of statistics if they are working on a research project themselves. Adults are often in job situations in which they are producers of statistics. Models of statistical literacy often have a *prescriptive* tone, suggesting what students and life-long learning adults need to *do* in order to be well informed, or to make good decisions, or to take advantage of the data that are available to them. Statistical literacy may also include recommendations for the development of students' and adults' critical thinking skills, so that claims made with data can be questioned and analyzed.

Finally, cognitive and developmental research frameworks can provide interpretive lenses to help us identify and track students' and adults' *statistical reasoning* and their conceptual development. Models of statistical reasoning are primarily *descriptive* models that help clarify how people are thinking about statistics, what they seem to know and understand, and where they have difficulty. The descriptions of student thinking obtained from models of statistical reasoning can also point out opportunities for scaffolding statistical ideas in the teaching of statistics.

Each of these three important realms, statistical thinking, statistical literacy, and statistical reasoning, while overlapping, are potential focal points for research, teaching, and curriculum development in statistics. They have been the motivating force behind the creation of an ongoing international working group in statistics education that has spawned the SRTL confer-

ences and produced a book on statistical thinking, literacy, and reasoning (Garfield & Ben Zvi, 2004).

A Model of Statistical Thinking

How do statisticians generate statistical questions? How do they decide upon a design for a study, which data to collect, and what to take into account when analyzing the data? Are particular ways of thinking germane to statistics? Wild and Pfannkuch (1999) pre-

sented a four-dimensional model of statistical thinking consisting of two dimensions they called cycles of thinking activity—an Interrogative cycle and an Investigative cycle—and two more dimensions called Types of statistical thinking and Dispositions, respectively (See Figure 21.1).

According to Wild and Pfannkuch when a statistician works on a statistical problem, parts of these four dimensions are continually, and simultaneously, in use (Pfannkuch & Wild, 2000, 2004). They claim that

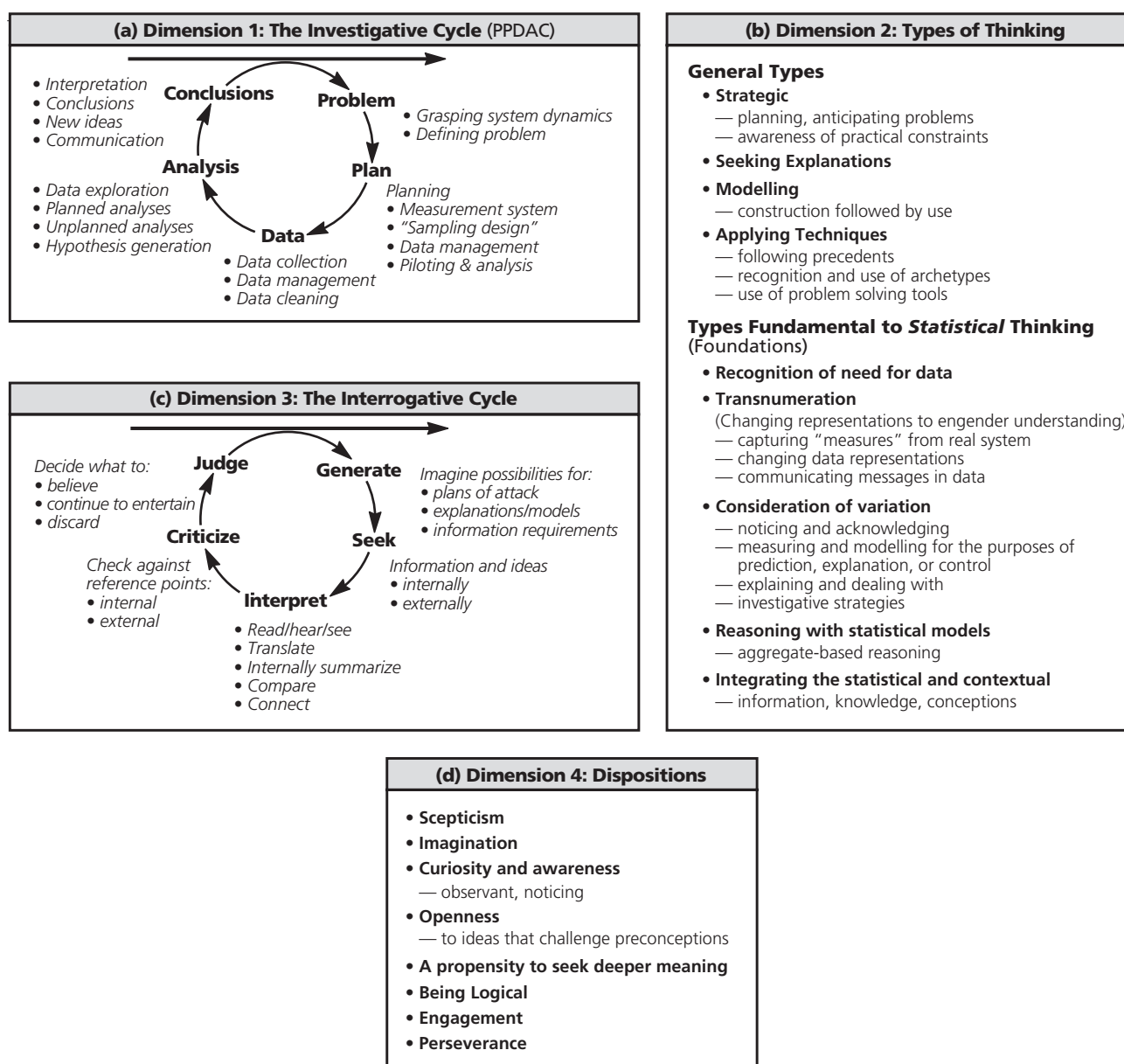


Figure 21.1 A 4-dimensional framework for statistical thinking in empirical enquiry (From "Statistical Thinking in Empirical Enquiry," by C. J. Wild and M. Pfannkuch, 1999, *International Statistical Review*, 67, p. 226. Copyright 1999 by *International Statistical Institute*. Reprinted with permission)

their model of statistical thinking can also be used to analyze student thinking, not just a statistician's thinking, and that such analyses can inform both teaching and curriculum development in statistics. The initial building blocks for their model came from three data sources: students' work on statistical tasks; interviews with student team leaders of statistical projects; and, perhaps most importantly, interviews with six statisticians working in various settings (business, marketing, medicine, etc.). In developing their model Wild and Pfannkuch tapped and expanded upon the writings of previous statistics educators, such as Moore (1990, 1997), who discussed the omnipresence of variability, the need for data and data production strategies, and the measuring and modeling of variability. Statistics educators are likely to resonate with Wild and Pfannkuch's model, whereas mathematics educators who have not delved much into the area of statistics may encounter some thought-provoking and unfamiliar issues. Cobb and Moore (1997) have written eloquently and persuasively about the differences between the disciplines of statistics and mathematics. Some of the differences between solving mathematical problems and solving statistical problems surface in the Wild and Pfannkuch model.

The Investigative cycle, tabbed PPDAC (Problem, Plan, Data, Analysis, Conclusion), is really the mantra of all statistical investigations and is reminiscent of Polya's (1945) seemingly timeless four-step model of mathematical problem solving (Understand, Plan, Execute, Review). However, there are some crucial differences between Polya's model and PPDAC, chiefly lying in the two Ps and the D. Statistical problems are often ill posed at first, as they arise out of messy contexts. Often there is a need for several iterative cycles just between the Problem ↔ Plan phases of the Investigative cycle, in order to adequately formulate a statistical problem from a murky prestatistical situation. The D, for data in the PPDAC cycle of a statistics problem, is very different from the type of data that one might encounter in a mathematical problem. Data for statistics problems are accompanied by excess baggage such as bias, uncontrollable sources of variation, context issues, and so on. Most of the current statistics education in the United States places a heavy emphasis on the DAC parts of the Investigative cycle, but precious little time is devoted in classrooms to the PP parts. If students are given only prepackaged statistics problems, in which the tough decisions of problem formulation, design, and data production have already been made for them, they will encounter an impoverished, three-phase, investigative cycle and will be ill equipped to deal with statistics problems in their early formulation stages.

The dimension on Types of thinking is two-pronged in the Wild and Pfannkuch model. Although some thinking is inherently of a statistical nature, there are also more general types of strategic thinking that remind one of Polya's problem-solving heuristics. The inherently statistical types of thinking highlighted by Wild and Pfannkuch include (a) the need for *data*; (b) attention to *variation*; (c) the use of historical, statistical, and probabilistic *models* (such as those used in inference); (d) the critical importance of *context* knowledge; and (e) *transnumeration*, a word coined by Wild and Pfannkuch. Except for the modeling tools aspect, Wild and Pfannkuch's list is quite different than a typical list of the hallmarks of mathematical thinking, which include processes such as looking for patterns, abstracting, generalizing, specializing, and generating and applying algorithms. In particular, the invented word *transnumeration* needs some further explication.

Wild and Pfannkuch created the word *transnumeration* because sometimes in the data organization and analysis phase, a particular representation of the data can reveal entirely new or different features that were previously hidden. These hidden features may have a major impact on how one interprets the data in that particular context. Wild and Pfannkuch needed a word that went beyond a mere transformation or re-representation of the data, so as to identify instances in which striking features of a context are suddenly revealed. An analogy might be the sudden insight *Eureka!* experience that mathematical problem solvers often speak about.

An example of *transnumeration* occurred in the reanalysis of the O-ring data for the Space Shuttle after the *Challenger* incident (Dalal, Fowlkes, & Hoadley, 1989). It was hypothesized that there might be a relationship between the air temperature at launch time and O-ring failures on the shuttle. The original data set that was being analyzed prior to the launch of the *Challenger* included only information on the launch temperature and number of O-ring failures per launch. No significant trends could be found. However, when data on launch temperatures and the number of O-ring *successes* was included, the augmented data set revealed a cut-off temperature above which O-ring failure had never occurred on the shuttle. The fact that it was probably too cold to launch the *Challenger* was not clearly evident until after this *transnumeration* process.

Although *transnumeration* is more likely to be encountered by statisticians as they investigate a statistical situation, it can also occur during teaching episodes with students. For example, while watching students work with a data set on the wait time between

blasts for the Old Faithful Geyser, Shaughnessy and Pfannkuch (2002) found that when students moved from representing the data in box plots or histograms to creating plots over time, a whole new vista on the data was opened up to them. (The geyser data are discussed in more detail later in this chapter. See for example Figures 2 and 3). Plots over time revealed a short-long cycle of wait times, whereas box plots and other graphical representations of the geyser data masked the nature of the variability. A teaching and learning culture that encourages transnumeration might evolve if teachers and curriculum developers heed the advice of some researchers who have recommended that students have more opportunities to construct their own representations of data rather than working primarily with canned tables and graphs (Cobb, 1999; Lehrer and Romberg, 1996). In this regard, the *Tinkerplots* software (Key Curriculum, 2005a) allows students considerable flexibility in constructing their own visual representations of data and in generating their own hypotheses and conjectures, and thus it encourages the transnumeration process.

The Interrogative cycle in Wild and Pfannkuch's model involves explicit metacognitive activity. There is an intense reflective component in the Interrogative cycle, as statistical problem solvers must always deal with beliefs, emotions, and the danger of their own, or their client's, narrow perspectives. The Interrogative cycle echoes the claim by Shaughnessy and Pfannkuch (2002) that statistical problem solving needs to be done by "data detectives" who continually question and reflect upon the processes of data production and data analysis.

The Dispositions dimension of Wild and Pfannkuch's model has much in common with not only mathematical problem solving, but also problem solving in any arena. All problem solvers need to be curious, be aware, have imagination, be skeptical, be open to alternative interpretations, and seek deeper meaning, as in the look back stage of Polya's model.

Models Focused on Statistical Literacy

"Statistical thinking will one day be as necessary for efficient citizenship as the ability to read and write." This quote from H.G. Wells appeared in Darrell Huff's famous little book *How to Lie with Statistics* (1954). Since then numerous authors have addressed the importance of statistics for an educated and competent citizenry. Consider: "The need for statistical thinking in social decision-making is exemplified every day in the news media" (Watson, 1997, p.107); or, to be statistically literate, one must have "... the ability to interpret, critically evaluate, and express one's

own opinions about statistical information and data-based messages" (Gal, 2003, p.16). There is a clear, overarching importance for students and adults alike to be able to critically read and evaluate information in tables, graphs, and media reports, and to adopt a healthy questioning attitude towards what is presented by sellers and buyers, by scientists and by the government, by politicians and by the news media. Perhaps someday quoting numbers without an adequate basis in fact will be outlawed. Meanwhile everyone is susceptible to those who would stretch the truth, or only tell that part of a data story that suits their purposes. Better that we pay heed to quotes such as those above, than that we fall prey to statements such as this one made by a mid-western congressman several years ago while on the campaign trail: "We will legislate that all children in our state schools will be above average." Although a thorough trek through the literature on statistical literacy is well beyond the scope or intention of this chapter, work in statistical literacy has provided several frameworks that are useful lenses to consider as we examine research in statistics education.

Among the abilities subsumed under statistical literacy are Document Literacy, Prose Literacy, and Quantitative Literacy that comprise the three facets of adult literacy identified by Kirsch, Jungeblut, and Mosenthal (1998). Kirsch et al. presented a five-category framework—*locating, cycling, integrating, generating, and making inferences*—that adults use when they read information from tables and graphs. Mosenthal and Kirsch (1998) illustrated the power of using this framework for understanding the types of skills that adults need in order to read documents by providing a measure of document complexity. This framework for analyzing the graph- and table-reading skills of adults has been used in publications of several large-scale surveys, such as The National Adult Literacy Survey (NALS) and the International Adult Literacy Survey (IALS). The categories in Kirsch et al.'s adult literacy framework are somewhat similar to Curcio's (1989) three levels of reading graphs, "Reading the Graph," "Reading Within the Graph," and "Reading Beyond the Graph."

Reading the Graph corresponds to *locating* in the Kirsch et al. model. *Integrating and generating* are suggestive of Curcio's Reading Within the Graph, and Reading Beyond the Graph includes such activity as *making inferences* in the Kirsch model. The resonance between these two frameworks for graph and table literacy, Kirsch et al.'s and Curcio's, is indicative of what seems to be occurring more and more in research in statistics education: similar statistical reasoning phenomena have been identified independently by separate researchers who are pursuing related but not necessarily identical questions. Although the lan-

guage is different, the types of categories that Curcio and Kirsch et al. developed point to a similar complexity on a scale of graphical literacy. When interpretive frameworks align like this, they cross validate one another in powerful ways.

Gal has been one of the champions in researching and promoting statistical literacy, mainly as part of his interest in general adult literacy (Gal, 2002, 2004). In an article in the *International Statistical Review*¹ (Gal, 2002), Gal claimed that statistically literate behavior by adults depends on their ability to access five knowledge bases: *general literacy* knowledge (like the previously mentioned document, prose and quantitative types of literacy), *statistical* knowledge, *mathematical* knowledge, knowledge of the *context*, and knowledge of how to be *critical* and question claims. Gal also examined how these five knowledge bases can interact with a person's dispositions, beliefs, and attitudes towards data and statistics in general. There is some overlap between Gal's model of statistical literacy and Wild and Pfannkuch's model of statistical thinking discussed above, although they are focused on different constructs: what adults need to do to be statistically literate versus what statisticians do on their job.

In another study Gal (2003) examined six web sites with large databases to analyze the types of knowledge and skills needed to access, process and manipulate the information on those sites. The web sites Gal examined contained information about national statistics, health issues, international education and international economic development. Gal noted that there were press releases, reports, executive summaries, and aggregate data sets available on each of these web sites but they were often in a form primarily intended for policy makers. A considerable amount of context and background knowledge was necessary for someone from the general public to wade through these websites and to make sense of the information that was provided. Gal concluded that the abilities to access, define, locate, extract, and filter information were additional critical skills needed for statistical literacy.

Clearly the type of statistical literacy that Gal described is a different kind of statistical literacy than just reading and evaluating data and graphs. There are many levels and contexts for literacy in Gal's work. He implied that it is important for adults, particularly in their roles as consumers and voters, to become "web-data" literate. However, a person does not just automatically step in and start interpreting and critiquing data and document information on the web. It is challenging to unpack data that appear in reduced formats on the web, especially when the

data come from contexts that are unfamiliar. Gal's reflections on adult literacy should give both researchers and curriculum developers some pause as they think about what pre-cursor skills are necessary for adults to develop this type of web literacy. This type of literacy also should start with students when they are in school. Perhaps students need to begin with digestible chunks from large databases that will help them scaffold from a basic level of statistical literacy (say, reading and interpreting tables and graphs) to the type of literacy that Gal is suggesting. In this regard, the Watson (1997) framework for statistical literacy may provide a roadmap to develop students' statistical literacy skills.

While investigating students' developmental progression in statistical literacy, Watson (1997) identified three Tiers of literacy that arose as students worked on statistical tasks involving the media, newspapers, magazines, news reports, and so on. Tier 1 is an understanding of basic statistical terminology; Tier 2 involves considering and embedding statistical terms within a real context; and Tier 3 encompasses a critical reasoning component. People at Tier 3 are able to question statistical claims and critique media items that involve statistics using their understanding of the basic statistical concepts and their understanding of the context (Tiers 1 and 2).

Watson and her colleagues began to use these Tiers in their research studies to help analyze student thinking on statistical tasks (Watson & Moritz, 1997a, 2000a, 2000b). Much of Watson and her colleagues' subsequent work on statistical literacy arose from the planning, execution, and multiple analyses of the results of a large longitudinal study of the development of school students' knowledge of probability and statistics. About a thousand students in Australia were initially tested on a large set of survey items on statistics and probability, and subsequently many of these students were retested 2 years later and some even 4 years later. Students' understanding of such concepts as means and middles, chance, graphs, variability, sampling, comparing data sets, and their thinking on items from the media involving data and graphs were examined. As a result, over the past decade, Watson has been one of the most active and prolific researchers in statistics education in the world. Her work on students' understandings of particular statistical concepts will be addressed throughout this chapter.

In a reanalysis of data from survey tasks given to students in Grades 3 to 9 over a 7-year period (1993–2000), Watson and Callingham (2003) used Rasch model techniques and combined results on statistical tasks from

¹ *International Statistical Review* is a publication of the International Statistics Institute (ISI).

multiple student surveys from over 3000 students. They found quantitative support for the identification of six levels of development along a uni-dimensional statistical literacy scale, ranging from idiosyncratic to critical mathematical. These six levels are closely associated with Watson's (1997) three Tiers of statistical literacy that initially arose through qualitative analyses of students' responses to the tasks. Watson has thus provided substantial documentation in multiple research studies for the developmental progression of a complex construct called statistical literacy, as well as some of its components (Watson, 1997; Watson & Callingham, 2003; Watson, Kelly, Callingham, & Shaughnessy, 2003). In addition to the research contributions Watson has made in the area of statistical literacy, her three Tier framework—statistical terms, considering terms in context, and justifying statistical claims—provides a useful introductory framework for scaffolding instruction in statistical literacy.

Both Gal and Watson have made it clear that any model of statistical literacy *must* accord a major role to context, as well as the use of statistical terms, tools, and techniques. Furthermore, they both have noted the importance of being able to communicate reactions to statistical information and to critique it (Watson 1997; Gal, 2002). According to Watson and Moritz (1997a) "Judging statistical claims from the media is fundamental to being statistically literate" (p. 129). The research into statistical literacy has unveiled a very deep construct involving a myriad of types of skills and cognitive processes. It is therefore important for researchers to be quite clear about what they mean by statistical literacy when they use the term, as it has many levels of meaning.

Models That Capture Statistical Reasoning

Over the last 15 years a number of researchers have conducted studies that focus on students' reasoning about particular statistics concepts or processes, such as students' notions of average, variation, sampling, comparing data sets, graph sense, and data representations. In most of these studies an explicit, or implicit, conceptual analysis occurs. Researchers attempt to distill and categorize the ways students reason with and think about the concepts, sometimes resulting in the identification of levels of reasoning. In this section I'll first discuss a model that has often been used to identify and assess levels of student reasoning in mathematics and statistics and then point to several emerging models for student reasoning.

The SOLO Model

The SOLO Model is a neo-Piagetian model created to analyze the complexity of student responses to

tasks. SOLO, which stands for Structure of Observed Learning Outcomes, was first developed by Biggs and Collis (1982) as a general model for evaluating learning in any context or environment. It has been heavily used in the past decade particularly by Australian researchers, and often to help code and analyze collections of responses to tasks that arise in clinical interviews. I discuss SOLO here briefly because it has become a prevalent analytical tool in some research circles in statistics education.

SOLO posits five modes of reasoning: sensori motor, ikonic (images), concrete symbolic, formal, and post formal, adding "post formal" to Piaget's original four modes. SOLO also postulates U-M-R (uni-structural, multi-structural, relational) cycles within each mode. Within a given mode, for a particular task, there may be several such U-M-R cycles. These cycles represent increasing orders of complexity while a person functions within a particular mode. Uni-structural responses suggest attention to only one relevant aspect of a task, whereas multi-structural responses involve several disjoint but relevant aspects, and relational responses suggest an understanding that integrates several aspects of a task within a mode. Analyzing student verbal responses is a tricky business, as anyone who has done it is aware. In studies that employ the SOLO methodology, boundaries between the U-M-R levels are often blurred, but that does not necessarily prevent the model from assisting a researcher in identifying a spectrum of complexity in student responses.

A good example of the potential advantages of the SOLO framework can be found in Watson, Collis, Callingham, and Moritz (1995). They introduced the SOLO model and then applied it at the concrete symbolic mode to analyze student responses to a sorting task. Students were given a set of data cards containing information about a set of people, including their eating habits, their TV watching habits, and their weight. The authors originally hypothesized that there would be two U-M-R cycles within the concrete symbolic mode on the data cards task given to students in Grades 6 and 9. Students were first interviewed individually while working on the task, and then groups of students worked together with the data cards and presented a report. In their analysis, the authors outlined a U-M-R cycle with the data cards as follows:

U—Students form images of individual people and invent stories about what the people must be like. For example, they might describe an individual person, who watches a lot of TV, eats a lot of fast food, and seems of higher weight, but no real sorting occurs.

M—Students begin to group cards by aspects, one at a time—perhaps all the high and low TV watchers are grouped in separate piles.

R—Students are sorting by multiple aspects, and making conjectures. For example, people who were high in both TV watching and fast food consumption might be grouped, and conjectures might be made about their weight.

The second cycle that Watson et al. (1995) hypothesized was not found in individual responses but was found when groups of students worked on the card sorting task. This second cycle involved graphical representations of the data (U2, M2), and conjecturing and defending conjectures about the graphs (R2). Groups of students made both histograms and scatterplots of the card data.

SOLO may be a useful research tool for statistics educators, as its framework is designed to assess responses to open-ended complex tasks that elicit a hierarchy of student reasoning. The SOLO model also helps researchers to identify partial success on tasks, so that a spectrum of student reasoning can be observed, rather than an all-or-nothing, right-or-wrong approach. For example, in an investigation of elementary school students' reasoning about data sets, Jones, Thornton, Langrall, Mooney, Perry and Putt (2000) modeled a spectrum of student reasoning using an approach quite similar to SOLO to describe four stages of reasoning: idiosyncratic, transitional, quantitative, and analytical. This four-stage model was subsequently employed by Mooney (2002) to characterize middle school students' statistical reasoning.

Some Emerging Models

The recent development of cognitive and developmental models to interpret student reasoning in statistics is a healthy sign of growing research maturity in the field. A detailed review and analysis of the use of models in statistics education research can be found in Jones, Langrall, Mooney, and Thornton (2004). Among the models they discussed are a series of four-stage, SOLO-like models that they have developed for interpreting and analyzing primary and middle school students' responses to statistics and probability tasks (Jones, Langrall, Thornton, & Mogill, 1999; Jones et al., 2000; Mooney, 2002). In the context of reading and interpreting data in graphs, Jones et al. (2000) superimposed their four stage model of student reasoning (idiosyncratic, transitional, quantitative, and analytical) on Curcio's (1989) three stages of graph sense: reading the data, reading between the data, reading beyond the data. They provided rich descriptions of student reasoning within each cell of the resulting 3×4 (Jones $\times$ Curcio) matrix.

In addition to the models that have emerged from SOLO theory (e.g., Jones et al., 1999, 2000), several groups of researchers, initially working quite independently, have identified and begun to use similar terms to describe types of student thinking. For example, Saldanha and Thompson (2003) referred to "additive" and "multiplicative" thinking when they describe students' work with sampling distributions. Shaughnessy, Ciancetta, & Canada (2004a) identified three levels of student reasoning about variability in a repeated samples environment, "additive, proportional, and distributional." Watson (2002) referred to "additive" and "distributional" reasoning when students are comparing data sets, and Cobb (1999) to "additive" and "multiplicative" thinking when he discussed students' thinking while comparing data sets. The "additive/multiplicative" terminology has been borrowed from other areas of research in mathematics education (Vergnaud, 1983; Doerr, 2000; Thompson & Saldanha, 2003). The terms *proportional* and *distributional* are particularly important for statistics and can have specialized meanings. Examples of the importance of proportional reasoning for handling statistical tasks are presented in Watson and Shaughnessy (2004). Ways of characterizing distributional reasoning are being developed by several groups of researchers, for example, Bakker and Gravemeijer (2004), Shaughnessy et al. (2004a), Shaughnessy, Ciancetta, Best, & Noll (2005). Further discussion of elements of these emerging models is provided in other sections of this chapter.

In this section I have presented some models of statistical thinking, statistical literacy, and statistical reasoning—interpretive and descriptive frameworks that researchers have used when analyzing student responses to statistical tasks. The spectrum of the models and frameworks discussed includes both the particular and the general, from Wild and Pfannkuch's model of statistical thinking that is specifically tailored to statistics, to the SOLO model that is a general framework for analyzing student responses in any content area or discipline. In the next sections I discuss some of the research that has focused on students' understanding of particular statistical concepts and processes, such as centers and average, variability, information from samples, comparison of data sets, and graph sense.

RESEARCH ON STUDENTS' UNDERSTANDINGS OF SOME STATISTICAL CONCEPTS

The research on students' conceptual understanding of statistics has been conducted with a wide range of students from primary, to middle school, to second-

ary, to tertiary. In this regard, there has been a change from the research reported in the first *Handbook of Research in Mathematics Education* (Shaughnessy, 1992) in two ways. First, much of the research reported in the first *Handbook* was conducted with college level students. Second, the stochastics research at the time of the first *Handbook* had concentrated more on probability concepts, or probability distributions, than it did on statistical concepts. This time around a majority of the research in stochastics has been conducted on students' understanding of specific statistics concepts.

Perhaps the overarching goal of statistics education is to enable students (of any age) to read, analyze, critique, and make inferences from distributions of data. The concept of a distribution in statistics is very complicated, and the word is used in different ways. Statisticians talk about *distributions of data*, but they also talk about *sampling distributions* and *probability distributions*. Distributions of data sometimes have an underlying probability distribution, such as the normal distribution or the binomial distribution, or, there may be no apparent normative probability distribution for a data set. The word *sampling distribution* often refers to a finite frequency distribution of repeated observations of some statistic—such as a distribution of means, or sample proportions, or standard deviations—that have been calculated for samples drawn from some population. On the other hand, the word *sampling distribution* may refer to the “distribution of all possible such statistics” for a given population, an infinite, theoretical construct. A more detailed discussion of these various uses of the word *distribution* can be found in Shaughnessy and Chance (2005).

For the purposes of this chapter, an understanding of students' reasoning about distributions is the (oft unstated) goal behind much of the research that has been conducted on students' understandings of particular statistical concepts. The notion of distribution includes concepts like center, shape, and spread. When researchers investigate student thinking about means and middles or about variation, or when they investigate students' thinking while comparing data sets or making decisions or inferences from graphs, they are researching students' understandings of aspects of distributions of data.

Bakker and Gravemeijer (2004) discussed examples of both upwards (from particular data points to entire distributions of data) and downwards (from entire distributions back to particular data points) reasoning by students as they learn to reason about distributions of data. Shaughnessy, Ciancetta, Best, and Noll (2005) presented examples of students' thinking about centers, spread, and shapes as they reasoned about distributions of data.

In the next sections of this chapter, as the research lens focuses on students' understanding of various statistical concepts such as centers, or variation, or information from samples, or on students' graph sense, keep in mind that this division into subsections is one of convenience. The aspects of a distribution—center, shape, and spread—are quite interrelated. Furthermore, distributions are often represented in graphical form so that reasoning about distributions is also confounded with reasoning about the graphs themselves. Though all these concepts and constructs are connected within the field of statistics, research that focuses on a particular concept in statistics can sometimes reveal aspects of student thinking that help to inform the teaching of statistics.

Research on Students' Understanding of Average

Concepts of average—middles and means in particular—are very powerful in statistics because means and other measures of center are used to help summarize information about an entire data set. Furthermore, if the data set is a sample that has been appropriately drawn from a parent population, the sample should mirror aspects of the parent population, and the sample mean should provide an estimate for the mean of the parent population from which the sample was drawn. On a more basic level the mean of a data set or a sample is representative, or in some sense typical, of that sample or data set. “Typical” values of different data sets can provide an efficient, albeit at times potentially misleading, mechanism to compare and contrast those whole data sets. Indeed the mean is a very powerful idea in statistics, and it is fundamental for students' understanding of summary statistics and statistical tests. Thus, there is plenty of motivation for researchers to study students' conceptions of average. It is one of the most important concepts in all of the mathematical sciences. And yet, students' school experiences with concepts of average are often reduced to a computational shell.

What do students do if they are given an opportunity to think intuitively or conceptually about averages? What notions of average, if any, do they employ when making decisions? Are students' notions of average the common statistical notions, such as mean or middles, or do they have other conceptions of what *average* means?

Early Work on Average

Much of the early research on students' understanding of averages was done with tertiary students (e.g., Pollatsek, Lima, & Well, 1981; Mevarech, 1983;

Pollatsek, Konold, Well, & Lima, 1984). Pollatsek et al. (1981) found that when college students were given the population mean (500) for SAT scores and a rather extreme data value from a sample of SAT scores, the students did not take the extreme value into account or adjust their prediction for a sample mean away from 500. Rather, they just predicted that the mean of the sample would also be 500. Students often believe that the mean is always their best bet for any prediction. Sometimes they even believe the mean is the most likely result to occur in a sample, even if the mean itself is not a possible data point. The Pollatsek studies also found that when tertiary students were given means for two unequal-sized samples and asked to find the mean of the combined sample, that the students tended to weight the samples equally and find the midpoint of the two sample means (average of averages—a common mistake by students). Mevarech (1983) referred to this as the “closure misconception,” as she hypothesized that students have a “group structure” in the back of their mind when operating with means or variances in statistics. These early studies on students’ understanding of the mean indicated that many tertiary students have at best a very procedural understanding of the mean, as something to be calculated.

Strauss and Bichler (1988) conducted one of the first studies reported on younger students’ (Age 8–14) conceptions of average. They asked students a series of structured questions in a one-on-one setting to probe students’ understanding of “statistical” or “abstract” properties of average, as well as whether students realized that an average was representative of a set of values. Strauss and Bichler concentrated on computational and measurement properties of the mean, rather than on conceptual properties. For example, they tested to see if students realized that the mean had to be located between the extreme values in a data set. They tested to see if students were aware that the mean is influenced by particular values in a data set. They tested to see if students realized that the average itself did not have to be one of the values in the set, or if students knew that the sum of deviations of data points from the average was zero. They also tested to see if students realized that the mean is in some sense closest to all the values in a data set. (Formally, this would mean that the mean is the value that minimizes the sum of the squared deviations from data points, a fairly advanced concept for students Ages 8 to 14).

In their analysis Strauss and Bichler identified two very different levels of difficulty in their tasks. On the one hand, students were quite aware that the mean was between extremes, and that particular data values can influence the mean. However, more sophisticated

measurement ideas like minimizing deviations, or that a zero data value must also be included and accounted for when calculating the mean, proved extremely difficult for their students. The process of minimizing deviations from the mean normally surfaces in regression analysis in high school or college, and it can be challenging for students at that level, so it is no wonder these younger students had trouble with it. Strauss and Bichler found that children do not think about the concept of the mean in the same ways that statistically mature adults do.

Conceptual Models of Students’ Thinking About Averages

Mokros and Russell (1995) conducted one of the first studies that investigated young students’ conceptual understanding of averages (see also Russell & Mokros, 1996). They interviewed Grade 4, 6, and 8 students in what they called “messy data” situations, using contexts like allowance money and food prices that were familiar to students. The tasks went beyond straightforward algorithmic computations and tried to elicit students’ own developing constructs of average. All of their students had been taught the procedure for finding the arithmetic average, so they had some familiarity with computing means.

The types of tasks Mokros and Russell used tended to ask students to work backwards from a mean to possibilities for a data set that could have that mean. For example, in the Potato Chips problem, students were told that the mean cost of a bag of potato chips was \$1.35, and then they were asked to construct a collection of bag prices that had that mean of \$1.35. The Allowance problem was similar except that students were given several existing data points along with the mean, were asked to create the rest of the collection of data points that would have that mean, *and* were told that they could not use the mean itself as a data point. Mokros and Russell were searching for students’ own preferred strategies when dealing with averages. Their analysis of students’ thinking resulted in the identification of five different mental constructs that students had of average: average as *mode*, average as *algorithm*, average as *reasonable*, average as *midpoint*, and average as *point of balance*.

Mokros and Russell found that students who focused on *modes*, or “mosts,” in data sets had difficulty working backwards from the mean to construct a data set if they were not allowed to use the actual average value itself as a data value. They concluded that modal thinking students do not see the whole data set, the distribution, as an entity in itself. They see only individual data values. About the same time as the Mokros and Russell study, research by Cai (1995) found that although most students could calculate a mean when given all

the data, they had great difficulty working backwards, filling in missing values when given the mean.

Mokros and Russell also discovered that students who had a purely *algorithmic* conception of average were unable to make connections from their computational procedures back to the actual context. For example, in the Potato Chips problem, some students multiplied \$1.35 by 9, and then just divided by 9 again, thus generating a data set that had \$1.35 for an average, but every single value in the data set was also \$1.35. For such students average is something that you *do* with numbers, they have no rich conceptual understanding of average. On the basis of responses gathered during student interviews Mokros and Russell suggested that students who preferred a rule or algorithm for averages may actually have had their own intuitive thinking about average as “typical” interfered with during their schooling.

Students who thought of average as *reasonable* tended to refer to information from their own lives. Perhaps they thought of average as a mathematically reasonable, but not necessarily precise, approximation for a set of numbers. These students thought that no precise answer existed for problems like the Elevator problem, which asked “If there are 6 women who average 120 pounds, and 2 men who average 150 pounds in an elevator, what is the average weight of everyone in the elevator?” Students would say, “But, you don’t know everyone’s exact weight.” For such students, the mean is somewhat representative of a situation, but not always calculable.

Even though students might not formally know what a *median* is, they have a good sense of average as midpoint. Mokros and Russell found that some students worked backwards to a distribution by symmetrically choosing values above and below the average, for example, above and below the \$1.35 value for the Potato Chips problem. Similar to the students who focused on modes, these students also had some trouble when they were not allowed to use the average itself as one of the data points.

Mokros and Russell provided one of the first attempts to build a developmental framework for characterizing students’ thinking about average. In their research they were disappointed to not find more students who had richer conceptions of average, such as average as *balance point*, which might be adapted more naturally to computational algorithms. They concluded that the development of higher-level conceptions of average may need to be scaffolded for students via instructional interactions. They also lamented the fact that some students’ preferred intuitive notions of average, such as average as *modal* or average as *reasonable*, conceptions that do

not provide good underpinnings for connecting to computational algorithms for average.

Foreman and Bennett (1995) make the case that a related intuitive concept of average, average as *fair share* does in fact translate well to computational algorithms. Students represent data in columns of blocks, and then “level” the blocks to produce a visual display of fair shares. This approach starts with whole numbers but can generalize to rational numbers (as stacks of blocks can be sketched and “cut” when necessary). The fair-share approach allows students to generate their own computational algorithms that reflect the physical leveling of stacks of blocks. This “leveling” approach can also lead to the mean as the point of *balance* for a distribution, as it highlights the value of the mean as a representative value of an entire data set. Friel (1998) discussed connections to computational algorithms for both the leveling and the balancing models.

In a longitudinal study of the development of students’ concepts of average in Grades 3 to 9, Watson and Moritz (2000c, 2000d) built upon the work of Mokros and Russell, as well as work of other researchers. Watson and Moritz included questions such as “Have you heard of the word *average*? What does it mean?” and “How do you think they got the average of 3 hours a day for watching TV?” taken from a media report. In another question they told students “On average, Australian families have 2.3 children. What can you tell from this?” In addition to probing for initial understandings of average in media or everyday contexts, Watson and Moritz asked students to work backwards and fill in missing data values when given the mean, and to find the average in weighted mean situations, similar to the tasks of Mokros and Russell.

Watson and Moritz (2000c, 2000d) used the SOLO taxonomy to model student responses on tasks about average and they described six levels of student understanding of average. For example, a response by a student who invented a story about a secret camera that was used to keep an eye on students to see how much television they were viewing instead of doing homework was classified as *pre-structural*. Responses involving colloquial terms for average like “normal” or “alright” were coded *uni-structural*. A reference to a computational algorithm without strong conceptual connections was coded as *multi-structural*. If students’ thinking suggested that they considered the average as a representative for an entire data set, their response would be considered *relational*. Watson and Moritz also later identified two response levels beyond relational with a subset of these same students who successfully dealt with applications of the mean in multiple contexts.

Watson and Moritz have provided strong evidence from a large sample of student interviews (over 90) that students' conceptions of average follow a developmental path that proceeds from "stories" then to "mosts and middles" and finally to the mean as "representative" of a data set. At Grade 3, 71% of the students' responses were either pre-structural or uni-structural. At Grade 5, about 50% of the responses were above the uni-structural level, and by Grade 7 all the students in their study were at least at the multi-structural level. A strong positive association was found between the SOLO levels and student responses by grade level. A partial explanation for this might be that by Grade 7 the three measures of center—mode, median and mean—were to have been covered in the Australian national curriculum recommendations. However, there is also a purely developmental component to the students' understanding of average that is very robust. Watson and Moritz (2000c) conducted follow-up interviews 3 to 4 years later with these same students and found that they had maintained the level of their thinking about average. Once students had developed powerful and flexible conceptions of the mean, including the ability to calculate weighted averages in real contexts, they maintained those conceptions over time.

Watson and Moritz's results suggest that it takes many years for students to develop their concept of the mean to the point where it is a representative of a data set. Whereas Mokros and Russell identified each student's dominant conception of average, Watson and Moritz have documented and analyzed the great variety of conceptions of average that a given student can have. They claimed that, "It would appear that many students hold eclectic ideas associated with average, even when they have not been taught formally" (Watson & Moritz, 2000c, p. 46). They strongly recommended that the teaching of the concept of average should build up from students' initial preferences for "middles" and "mosts" to the more normative conception of mean as representative of a data set. They also recommend delaying the formal introduction of the mean until students' developmental stages have had an opportunity to unfold. Bakker and Gravemeijer (2004) made similar recommendations for delaying the introduction of some of the more formal concepts in statistics, like the mean. They found a wealth of student ideas to build upon, including informal conceptions of center and shape, when students were asked to compare distributions of data, or to generate their own hypothetical distributions of data.

Konold and Pollatsek (2002) added to the literature on the complexity of the *average* concept with a theoretical reflection that extends Mokros and Rus-

sell's earlier work. They postulated four conceptual perspectives for the mean: mean as *typical value*, mean as *fair share*, mean as *data reducer*, and mean as *signal amid noise*. They argued that from a statistical point of view that the mean as "signal amid noise" is the most important and most useful conception of the mean, because they feel that this conception of the mean is the most helpful for comparing two data sets. Furthermore, they recommended that the mean should be introduced to students in the context of comparing data sets. They also argued that inasmuch as conceptions of average like typical or fair share are not as powerful for comparing groups, they shouldn't be emphasized with students. Konold and Pollatsek's conceptions of mean as typical or mean as fair share are more closely tied to a data analysis perspective on statistics, while their conceptions of mean as a data reducer and or mean as a signal are more closely connected to decision-making in statistics. Data reduction is necessary in decision-making in order to locate an informative signal amid the noise of variability.

From a normative point of view (what statisticians are looking for) Konold and Pollatsek may have a good argument. However, on the basis of the work of Mokros and Russell and of Watson and Moritz, mean as fair-share, and subsequently mean as typical value, are perhaps better first introductions to the notion of measures of center, because they build on students' primary intuitions. The mean as data reducer requires more sophistication from students, and a willingness on their part to let go of some pieces of information. This can be difficult for students to do, especially in those instances when *their own* information is included in a data set, for they feel their information will get lost in the data reduction process. The mean as signal amid noise involves yet another level of complexity. It takes considerable experience with data sets to realize that there even is such a thing as "noise" in data. Noise could appear in data just from random variability in samples or in gathering data from probability experiments. Noise could be the result of measurement error, either systematic error or careless error. Noise could be introduced by poor data-production techniques, or biases in sampling procedures. Teachers and students must spend some time focusing on the noise itself before getting too carried away with determining a signal in the data. I am convinced that it is just as important, if not more so, to analyze the variability in data, and to look for both special-cause and common-cause variation in data as it is to compare data sets by comparing their means. Wild and Pfannkuch (1999) agree, and identify variability as the critical component in the development of students' understanding of distributions of data.

Research on Students' Understanding of Variability

Like average, variability is itself a very complex construct. Researchers have often tended to use the terms *variability* and *variation* somewhat interchangeably. I prefer the distinction pointed out in Reading and Shaughnessy (2004), in which variability is the propensity for something to change, and variation is a description of or a measurement of that change. "A survey of various dictionaries demonstrated that *variation* is a noun used to describe the act of varying or changing condition, and *variability* is a noun form of the adjective *variable*, meaning that something is apt or liable to vary or change. The term *variability* will be taken to mean the characteristic of the entity that is observable, and the term *variation* to mean the describing or measuring of that characteristic." (Shaughnessy & Reading, 2004, pp. 201–202). However, I will attempt to remain faithful to the way that particular researchers refer to the concept, variability or variation, when discussing their research.

Within the field of statistics variability arises everywhere. Even in formal statistics variability is a slippery concept. Data vary, samples vary, and distributions vary. Furthermore, variation occurs both within samples and distributions as well as across samples and distributions. A large part of statistical analysis often involves parsing out the relative contributions and locations of sources of variation. The objects within which variability occurs in statistics—in data, in samples, and in distributions—are clearly not independent of one another. Statisticians may be interested in the data in a particular sample, or the distribution of a sample, or the sampling distribution of a statistic, or even variation across several distributions of data.

Similarly, research on students' thinking and understanding about variability could focus on variation in data, or on students' conceptions of variability in samples, or on the variability across several distributions of data that are being compared. Thus, variability occurs within many levels of statistical objects, and our students need to develop their intuition for what is a reasonable or an unreasonable amount of variability in these objects.

It is a bit surprising then, given the overarching importance of variability in statistics (Moore, 1997), that practically no research on students' conceptions of variability was reported prior to 1999. A number of researchers had called for research on the teaching and learning of variability and had raised concerns about the lack of explicit attention to the concept of variability in introductory statistics texts or in school curriculum materials (Green, 1993; Batanero,

Godino, Vallecillos, Green & Holmes, 1994; Loosen, Lioen, & Lacante, 1995; Shaughnessy, 1997). It wasn't until the turn of the millennium that research on and about variability began to appear at professional meetings or in scholarly journals.

Several research efforts have identified a variety of components of variability that indicate the potential depth of students' thinking and understanding about variability. For example, Wild and Pfannkuch (1999) included a number of aspects of variation in their model of statistical thinking, such as *acknowledging*, *measuring*, *explaining*, and *controlling* variation. Reading and Shaughnessy (2004) extended Wild and Pfannkuch's list of aspects of variation to include *describing* and *representing* variability, and Canada (2004) provided a detailed framework for analyzing students' thinking about *noticing*, *describing*, and *attributing causes* of variation. Reading and Shaughnessy also provided evidence from student work on several statistical tasks for both a *description hierarchy* and a *causation hierarchy* for variability. For example, in the description hierarchy lower level responses might be concerned only with outliers or only with middles (uni-structural), whereas a higher level response might mention both middles and extremes (multistructural). At an even higher level a student response might discuss the deviations of data from some fixed value like the mean, thus making connections between the concepts of center and variability (relational).

Some questions come to mind in considering student thinking about variability. Do students acknowledge variability? If so, how do they describe it or talk about it? Are students' conceptions of variability influenced by context? Do they recognize potential sources of variability? Will they attempt to control aspects of an experiment in order to minimize variability? Do students think about variability in a variety of ways, similar to the different conceptions of average that were discussed in the previous section? The next section reviews research that has investigated some of these questions and discusses students' thinking about variability across the spectrum of statistical objects: in data, among samples, and across distributions.

Researchers have now begun to study students' thinking about variability in a number of statistical contexts, such as when reasoning about distributions, dealing with samples and sampling, reasoning about the outcomes of a probability experiment, and comparing data sets (Shaughnessy, Watson, Moritz, & Reading, 1999; Melitou, 2000, 2002; Reading & Shaughnessy, 2000, 2004; Torok, 2000; Torok & Watson, 2000; Watson & Kelly, 2002; Reading, 2003; Ciancetta, Shaughnessy, & Canada, 2003; Shaughnessy, Ciancetta, & Canada, 2003; Watson, Kelly,

Callingham, & Shaughnessy, 2003; Canada, 2004; Shaughnessy, Ciancetta, Best, & Canada, 2004b; Watson & Kelly, 2004). Melitou (2002) presented one of the first reviews of the literature on research on the teaching and learning of variability and described a number of the ways that researchers were beginning to investigate student thinking about the construct. As this chapter was being written, an issue of the *Statistics Education Research Journal* (Vol. 4, No. 1) with a special section on reasoning about variation was in process (J. Garfield & D. Ben-zvi, Eds.). In the next several sections I discuss research on students' thinking about variability, in data, in samples, and in distributions. While this is a convenient approach to discuss student thinking about variability, it is also a somewhat artificial categorization because data, samples of data, and distributions of data all interconnect with one another.

Variability in Data

Data sets tell stories, and the heart of any statistical story is usually contained in the variability in the data. When analyzing data, the role of a student or a statistician is to be a “data detective,” to uncover the stories that are hidden in the data. From a data detective point of view, there are important *signals in the variability* as well as in measures of center. In fact, premature attention to measures of center can result in missing the important trends in the variability in the data. For example, consider the data for a series of wait times for eruptions of the Old Faithful geyser in Yellowstone National Park (Figure 21.2), and the accompanying student graphs of the geyser data (Figure 21.3). This example, taken from Shaughnessy and Pfannkuch (2002), presents data for approximately 3 consecutive days of wait times (in minutes) between eruptions of Old Faithful. Students were asked to work in groups, to analyze the data, and to make a decision on how long they would expect to wait for an eruption of Old Faithful.

Typically, many beginning students first just calculate a mean or determine a median for a day's worth of Old Faithful data and then base their initial prediction on a measure of central tendency. (The mean of First day is 70.1, the median is 70.5; mean of Second day is 79.9, median is 77, and so forth). While the mean

does give a one number summary of the data set, it can also mask important features in the distribution of the data. When these students were subsequently asked to graph the data set, some of their representations uncovered a pattern in the data that was highly bimodal. More importantly, students discovered an alternating short-long pattern in the Old Faithful blasts when they created plots over time, or dot plots or bar graphs (See Figure 21.3).

This oscillating pattern can be completely missed if one just calculates a mean or draws a box-plot for the data. *The signal in the variability is much stronger than the signal in the center in the Old Faithful example.* The variation in the Old Faithful data is not random. There are likely to be some underlying geological causes or relationships for the variation. Shaughnessy and Pfannkuch found that students who attended to the variability in the data were much more likely to predict a range of outcomes or an interval for the wait time for old Faithful, such as “Most of the time you'll wait from 60 to 85 minutes,” than to predict a single number (such as 77 minutes) for the wait time. Means mask what is going on in the Old Faithful data set. The definitive long-short pattern is lost in an average. Means may be critical in making statistical inferences when comparing groups, but variation is often even more important to the data detective.

In the first edition of this *Handbook* I discussed some differences between the points of view of psychologists and of mathematics educators on research in the teaching and learning of statistics and probability. Konold and Pollatsek's (2002) argument on the importance of the mean as the signal in data sets may have arisen from their frustration with students lack of attention to the mean (Konold et al., 1997) as a great tool for making comparisons between data sets, something that psychologists often wish to do in their research work. However, mathematic educators can be equally frustrated when students rush to compute a mean without even considering what the variability in a data set might reveal about the context (Shaughnessy & Pfannkuch, 2002). Students also need to recognize and investigate potential sources of variation within the data and not just rush to looking at centers.

Day 1:	51	82	58	81	49	92	50	88	62	93	56	89	51	79	58	82	52	88
Day 2:	86	78	71	77	76	94	75	50	83	82	72	77	75	65	79	72	78	77
Day 3:	65	89	49	88	51	78	85	65	75	77	69	92	68	87	61	81	55	93

Figure 21.2 Wait times in minutes for the eruption of the Old Faithful Geyser.

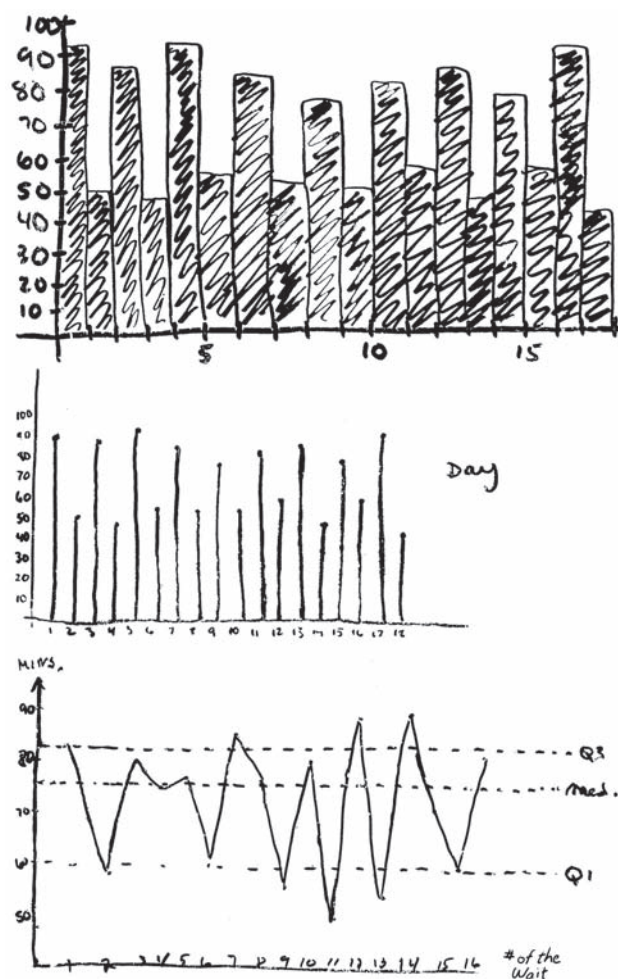


Figure 21.3 Student work samples of graphs of the Old Faithful data.

Variability Among Samples: From Data to Samples

The Old Faithful Geyser data in the previous section provided an example of the importance for students to understand the role of variability in the data itself. Do students acknowledge the variability that can occur from one entire sample to another sample?

A series of studies on students' thinking about how data vary in a sampling environment was precipitated by the Gumball Task in the 1996 NAEP (Figure 21.4).

In a sample of several hundred responses that was pulled from over a 1,000 responses to this item, Zawojewski and Shaughnessy (2000) found that only one student predicted an actual range, from 4–6 reds, rather than a specific number of reds, like 5, or 3, or 10 reds. This may have occurred because students are very accustomed to being asked questions in data and chance that prompt them to respond with single point

value answers. For example: What is the Probability that . . . ? How many . . . would you expect? What is the average of . . . ? The Gumball problem was worded in such a way that it was somewhat natural for students to answer with just one number. Several research issues arose from the Gumball problem. First, what would students do if they were explicitly encouraged to think about a range of possibilities for the data in the Gumball problem? Second, the gumball task was administered only to Grade 4 students in the 1996 NAEP. How would older students respond to the gumball task?

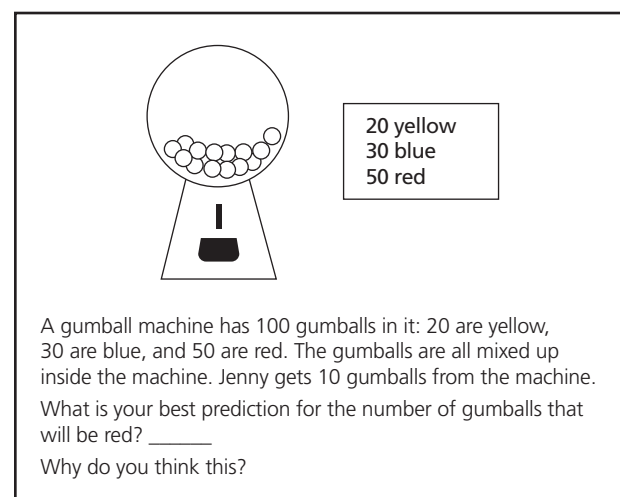


Figure 21.4 The NAEP Gumball Problem (From Zawojewski & Shaughnessy, 2000, p. 259).

In order to explore students' thinking about the variability of the data in a sampling situation, several reformulations of the Gumball problem, called the Lollie² problems, were administered in survey form to over 300 students in Grades 4–6, 9, and 12 in the United States, Australia, and New Zealand (Shaughnessy et al., 1999). Three different versions of the Lollie task (List, Choice, and Range) were given to all the students (See Figure 21.5). In each case, students were asked to supply reasons for their answer.

The Lollie tasks have subsequently been administered to thousands of students in grades 3–12, primarily in Australia and the United States. (Reading & Shaughnessy, 2000; Torok & Watson, 2000; Kelly & Watson, 2002; Shaughnessy et al., 2004a). Responses to the Lollie tasks indicated that there are differences among students on how they acknowledge variability in samples. The types of responses (*high, low, wide, narrow, reasonable*) and the reasons students provide have remained very robust across all these studies. For

² In Australia the word *lollies* is used for hard candies that are wrapped in cellophane or paper.

The Lollie Task

A bowl has 100 wrapped lollies in it. 20 are yellow, 50 are red, and 30 are blue. They are well mixed up in the bowl.



Jenny pulls out a handful of 10 lollies, counts the number of reds, and records it on the board. Then, Jenny puts the lollies back into the bowl, and mixes them all up again.

Four of Jenny's classmates, Jack, Julie, Jason, and Jerry do the same thing. One at a time they pull ten lollies, count the reds, and write down the number of reds, and put the lollies back in the bowl and mix them up again.

What do you think?

(List Version)

1. I think the numbers of reds the students pulled were

I think this because:

2. I think the list for the number of reds is most likely to be (circle one)

A) 8,9,7,10,9

D) 2,4,3,4,3

(Choice Version)

B) 3,7,5,8,5

E) 3,0,9,2,8

C) 5,5,5,5,5

I think this because:

3. I think the numbers of reds went from (a low of) _____ to a high of _____.

I think this because:

(Range Version)

Figure 21.5 The Lollie Task.

example, some students predicted all *high* numbers of reds, like 6, 7, 5, 8, 9, mostly numbers above the expected value of 5 for samples of 10 lollies from this 50% red mixture. These students usually reasoned that there were “a lot of red in there, so it (the red ones) will happen a lot.” Other students, mostly among the Grade 4 students, predicted all *low* numbers (all numbers ≤ 5) and said that there were a lot of “non-reds” in the mixture that would prevent the reds from being pulled very often. Still other students predicted a *wide* list of outcomes, for example, 1, 5, 7, 9, 2 (range ≥ 8)

because “any result could occur, you never know” suggesting that they may be using “outcome approach” reasoning (Konold, 1989) or an equi-probability conception (LeCoutre, 1992). Still other students predicted a very *narrow* list for the numbers of reds, for example, 5, 5, 5, 5, 5, or 5, 6, 5, 5, 6 (range ≤ 1). The “narrow” responses, especially all 5s, occurred more frequently among the older students, Grade 12 in particular. Students who predicted *narrow* reasoned that, “5 is the most likely outcome” or “5 is what you are supposed to get.” Reading and Shaughnessy (2000)

found that the *narrow* predictors were reticent to change their answers, even after they acknowledged that the repeated samples were unlikely to all be identical. Finally, some students' lists were *reasonable* in that they were distributed in a more normative way around 5, such as 3, 7, 5, 6, 5, centered around the expected value within a reasonable range.

There is some evidence that when students are given a chance to actually draw their own samples from a lollie type of mixture, a higher percentage of them will subsequently give *reasonable* predictions for the Lollie task. Shaughnessy et al. (1999) reported an increase from 17% to 55% in *reasonable* responses when a sample of 94 middle school students actually conducted a simulation of the lollie problem with colored cubes in a box. On the other hand, in a study using student interviews, Kelly and Watson (2002) did not find such changes in students' predictions, though they admit there may have been too few trials in their experiment to change students' predictions. The tenacity of beliefs and intuitions about probabilistic and statistical phenomena has been well documented over the years (Kahneman & Tversky, 1972, 1973a, 1973b; Tversky & Kahneman, 1974, 1983; Shaughnessy & Dick, 1991; Shaughnessy & Bergman, 1993). Beliefs and conceptions about data and chance are very difficult to change, and research has suggested that empirical experiments and simulations must be systematically built into instruction over a longer period of time in order to change the patterns of students' intuitive conceptions (Shaughnessy, 1977; Fischbein & Schnark, 1997;).

The emerging conceptual model introduced previously in this chapter has been used by several groups of researchers to describe the progression of student reasoning about variation on the Lollie tasks, from iconic, to additive, to proportional, and finally to distributional. According to Kelly and Watson (2002), some students, particularly younger students, reason "ikonically," by using physical circumstances or personal stories when they predict sample proportions for the lollie task. Students who are reasoning ikonically may say, "They might get more reds because their hand could find them," or "Maybe they are lucky and will get all reds," without any reference to the actual contents or the proportion of colors in the bowl. In a sample of 272 students in Grades 6–12, Shaughnessy et al. (2004a) classified students' reasoning on the Lollie task as predominantly additive, proportional, or distributional. Additive reasoners focus on frequencies, rather than on relative frequencies. Proportional reasoners tend to predict "around 5" for the Lollie problem and defend it with statements such as, "There are 50 red," or "I'd expect 5 red out of the 10 candies." Proportional reasoners explicitly discuss the connections between sam-

ple proportions and population proportions. Distributional reasoners combine both centers and spreads in their reasoning about the Lollie problem. They make comparisons between the sample proportions and the population proportion, and they also *explicitly mention variation about the expected value*. Shaughnessy et al. also discussed transitional phases in students' thinking that occur between proportional reasoning and distributional reasoning. Attention to just one of the aspects of a distribution (e.g., center, shape, or spread) does not necessarily guarantee attention to the other aspects. It takes some time for students to be able to integrate these various aspects of a distribution.

Variability: From Samples to Sampling Distributions

Distributional reasoning involves making connections from populations to samples, and back again. In order to adequately reason about samples pulled from populations, students must have a strong concept of population proportion, what Kahneman and Tversky (1972) have called *the base rate*. The propensity for subjects to ignore the base rate was so prevalent across Kahneman and Tversky's research that recognition of population proportions is clearly an important idea for instructors to emphasize when they first introduce sampling to students. The complexity of the levels of statistical objects involved in sampling, and the proportional nature of the relations among those objects, is highlighted in work by Saldanha and Thompson (2003). They designed a teaching experiment to develop secondary students' understandings of the concept of sampling distribution. In the experiment the students wrestled with a multitude of statistical objects: individual *data values*; *collections of data values* (samples); *statistics* for a sample (like the mean of a sample); and finally, *collections of statistics* for many samples (for example, a distribution of sample means). The concept of population proportion was critical to Saldanha and Thompson's teaching experiment, as was the relationship between sample proportion and population proportion. Saldanha and Thompson emphasized the differences among data, samples, and populations with the students in their teaching experiment. However, they expressed some frustration that even with this explicit emphasis, their students had great difficulty conceiving of and distinguishing among the various levels of statistical objects while working on sampling tasks. Their students "did not have a sense of variability that extended to ideas of distribution" (Saldanha & Thompson, 2003, p. 264).

Saldanha and Thompson's students interacted primarily with a computer environment, generating samples and sampling distributions. One wonders if Saldanha and Thompson's students would have had

more success if they had spent more time experiencing a hands-on approach, doing simulations with objects to generate sampling distributions, rather than in the computer environment. Studies that have involved substantial hands-on simulation activity have shown some success in influencing student thinking in stochastic situations (Shaughnessy, 1977; Shaughnessy et al., 1999). In a second teaching experiment Saldanha (2003) found that the addition of physical, hands-on sampling provided better support for student understanding of sampling distributions than the computer environment alone.

Rubin, Bruce, & Tenney (1991) pointed out that peoples' understanding of a what a sample tells us ranges along a spectrum from, "knowing everything" to "knowing nothing." Knowing everything is the result of overconfidence in how well the sample represents the population from which it is drawn, reminiscent Kahneman and Tversky's (1972) representativeness heuristic. At the other extreme, knowing nothing reflects a belief that a sample is just chance, because absolutely anything can happen in a sample. This other extreme of Rubin's spectrum reflects reasoning similar to Konold's (1989) outcome approach. Another way to think about these two extremes is that at one end people are too fixated on centers—they feel all samples should be perfectly representative—and at the other end they are too fixated on variability, and all the possibilities for individual outcomes. Students need to have a balance between representativeness and variability, between expectation and variation (Watson and Kelly, 2004).

Rubin et al. created several sampling scenarios to investigate how students might attempt to balance sample representativeness with sample variability. For example, in one scenario students were told that packets containing six Gummy Bears were handed out to all the children attending a parade. The packets were assembled from a huge vat of 1 million red and 2 million green Bears that were all mixed up. Then the students were asked how many children out of 100 they thought got packets with exactly 4 green and 2 red bears? Half the students said over 75% would get packets with 4 greens. All but one student said that over 50% of the children would get 4 greens. The wording of this problem focused the students' attention on the population proportion of two-thirds green, and likely tipped the students' thinking towards the representativeness end of Rubin's spectrum. However, when a different version of the problem was given to the students, asking "out of 100 packets, how many children would get 0, 1, 2, . . . , 6 green Bears," they tended to spread their predictions out across all the possible outcomes, tipping their responses toward the variability end of Rubin's spectrum. Rubin et al. found that students had little or no understanding of

the interaction between representativeness and variability on these tasks, and that students had very little intuition for the shape of the sampling distribution for the Gummy Bears packets.

Other researchers have found results similar to Rubin et al. Shaughnessy et al. (2004a) gave a lollies type question to secondary students that asked them to predict a sampling distribution for repeated samples drawn from a lollie jar. They found that many students focused heavily on aspects of variability, predicting overly wide ranges for the outcomes in a sampling distribution that too many extreme results would occur. On the other hand, Saldanha and Thompson (2003) found that students' concept of sampling did not entail much of a sense of variability at all. Their secondary students tended to judge samples purely on representativeness, perhaps relying too heavily on the underlying population proportion.

This series of studies (Rubin et al., 1991; Saldanha & Thompson, 2003; Shaughnessy et al., 2004a, Watson & Kelly, 2004) clearly shows that there is always a tension between representativeness and variability in sampling situations. Furthermore, the framing of the task itself can tip students' thinking in the direction of either an over reliance on the population proportion, or an over reliance on variability. The goal for teachers and curriculum developers is to help students to grow a middle ground between representativeness and variability, between knowing all and knowing nothing, to "knowing something," so that both population proportion and variability are taken into account together. This balance is necessary so that students understand that one can actually know something with a reasonable likelihood in sampling situations. One approach to sampling that may promote a balance for students between expectation and variation involves student-generated confidence intervals (Landwehr, Watkins, & Swift, 1987).

Variability across Distributions: From Informal to Formal Inference

Some research has focused on student thinking about variability across distributions, using tasks in which students are asked to compare several distributions or to make decisions based on a collection of distributions. Gal, Rothschild and Wagner (1989, 1990) found that middle grade students normally did not use the mean when they were asked to compare two data sets. Instead Gal et al. (1989) found that students used statistical strategies, quantitative data summaries, or proto-statistical strategies that focused on incomplete features of the data sets. Some students invented their own stories to compare data sets. Gal et al. (1989) were the among the first researchers to point out the essential role that proportional reasoning can play when

students are asked to compare two data sets. Konold, Pollatsek, Well, and Gagnon (1997) reported similar results with some upper secondary students, who also neglected to use the mean as a comparative measure. Their students often focused on particular data points or individual features of the data rather than on making global comparisons of the distributions. Evidently the power of the mean as a representative measure for making comparisons of data sets is not a primary intuition (Fischbein, 1987), and must be carefully developed within a teaching-learning setting.

In an 8-week design experiment with a fourth grade class Petresino, Lehrer, & Schauble (2003) introduced distributions of measurement data as objects for comparison and decision-making. Students gathered data on the heights of model rockets, some with pointed noses and others with rounded noses and then were asked to decide which was the best design. The students found the median height for both types of rockets, but this did not entirely satisfy them as a comparison basis, because the data for one type of rocket was more inconsistent than the other. They decided to find a way to measure the inconsistency, that is, the variability. This class devised a way to compare the variation for the two rockets by computing differences from rocket height to median rocket height for each type of rocket. Then they created two new distributions from the original distributions, the distributions of height differences from the medians. These fourth graders were thus comparing distributions of residuals—a rather sophisticated concept, but one that arose naturally in this particular design experiment.

The study by Petresino et al. provides a glimpse into classroom dynamics and into the social-conceptual growth of a class of students who attempted to use both centers and variability in their decision making process. It is interesting that the students in the Petresino et al. study looked for a *signal from the variation* when they compared the distributions, rather than a signal from the centers. They felt that the center values for the two types of rockets could be misleading, as one rocket design was more inconsistent than the other in the heights it attained. In this case, it was not just the mean or the median that became the signal to inform a decision (see Konold & Pollatsek, 2002). The variability was the primary *signal* that aided the students' decision. Petresino et al. Also reported that their fourth graders performed very well on the 1996 NAEP Grade 4 statistics tasks that were administered to the class after the 8-week teaching experiment. This suggests that there may be long-term retention benefits when students invent their own approaches to data analysis and the comparison of data sets.

In a series of articles, Watson reported on Grade 3 to 9 students' types of strategies when comparing pairs

of data sets (Watson and Moritz, 1999; Watson, 2001a, 2001b). A version of the data sets used for these studies is presented in Figure 21.6.

In interview settings, Watson and Moritz presented students with the pairs of the data sets from Figure 21.6 (e.g., Yellow-Brown, Pink-Black) and told them that the graphs showed the results of students' test scores in two different classes. They then asked students which class did better in each pair, and why.

In the first study, Watson and Moritz (1999) report that students' strategies fell into two SOLO cycles. The first cycle dealt with equal sized data sets (e.g., Yellow-Brown). Student explanations ranged from a comparison of individual data values, to calculating the total class values (adding up all the values in the data set), to visual comparison strategies ("Yellow goes higher"), and to combinations of these explanations. In the second SOLO cycle, data sets of different sizes were compared (e.g., the Pink-Black data sets in Figure 21.6) and proportional reasoning played a more prominent role. Students' reasoning strategies again included totaling up the scores and visual strategies, but students also calculated the means to compare the two groups. Watson and Moritz thus identified both additive and proportional reasoning strategies among their students. They concluded that students use "bottom line" strategies, such as attending to certain features of the graphs or reasoning about modal clumps, when comparing data sets. Watson and Moritz argued that these informal reasoning strategies can provide opportunities for teachers to introduce the comparison of two data sets prior to calculating means. Bakker and Gravemeijer (2004) came to the same conclusion: Students can, and should, compare data sets using their own intuitive strategies prior to formal statistics. In a follow-up study conducted 3 years later with some of the same students Watson (2001a) found considerable improvement in the level and complexity of students' responses on these tasks.

In a third study Watson (2002) used an interesting methodology to set up cognitive conflict among students and to challenge their thinking. After they had responded to the comparing-data-sets tasks in Figure 21.6, Watson showed students video clips of other students who had compared the data sets using a different strategy. For example, if a student reasoned that, "The Pink Class did better, because they have more students who scored higher," Watson might show this student a clip of a student who computed the means of the Pink and Black class and claimed that Black was better. Or, if a student said that "Yellow did better because they got high numbers" (referring to the height of the 5 column in the yellow data set) she might show a clip of a student who said that there was no difference, because both classes totaled 45, or there was no

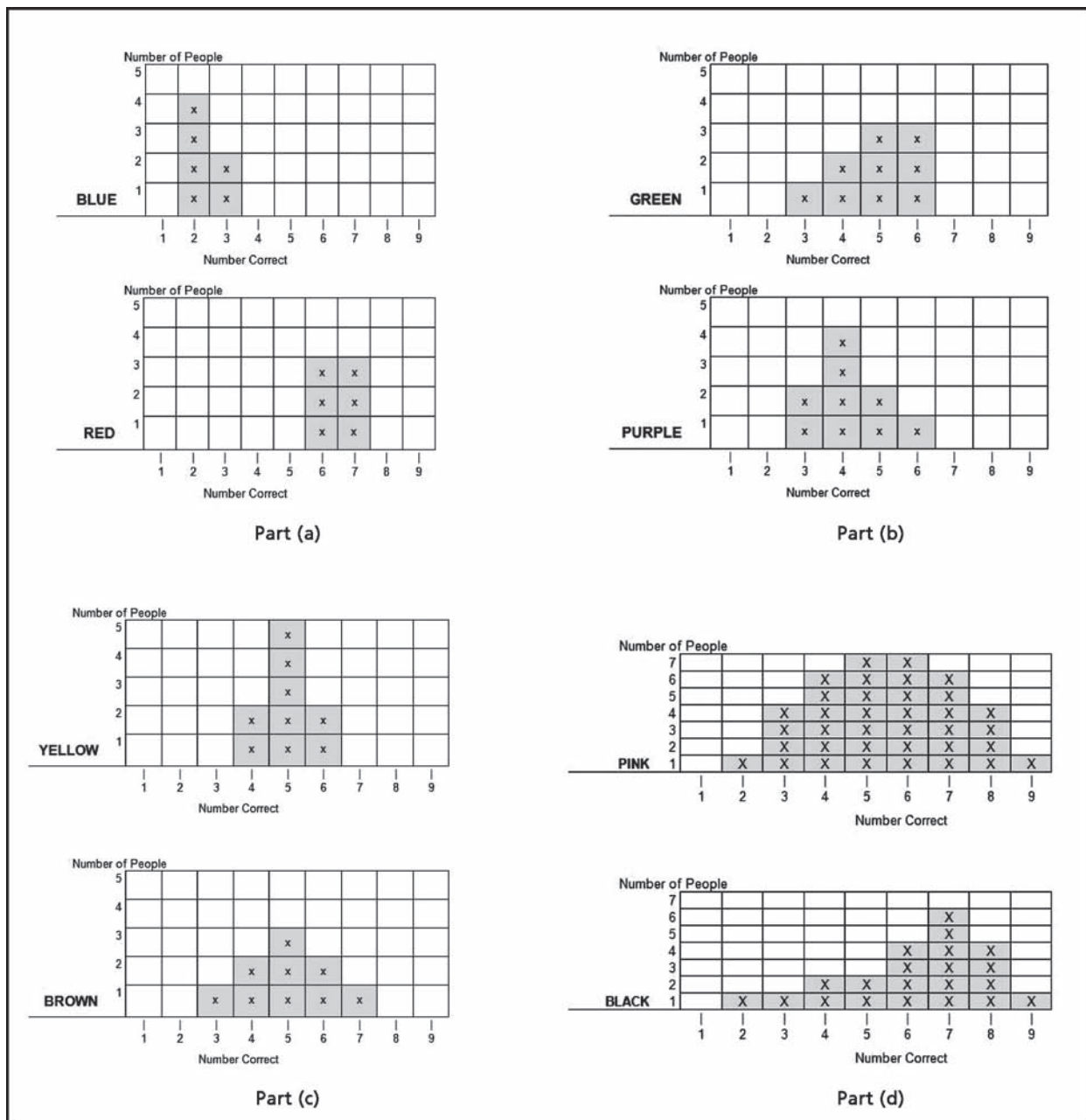


Figure 21.6 Paired data set comparisons. (From Watson & Shaughnessy, 2004. Proportional reasoning: Lessons from research in data and chance. *Mathematics Teaching in the Middle School*, p. 105)

difference because both classes had an average of 15, or that the Brown class did better because it had a 7.

Of those students whose reasoning could improve the second time around (that is, students who didn't reason at the top level the first time), 57% improved on the Yellow-Brown comparison, but only 30% improved on the Black-Pink comparison, after being shown a clip of another student. Students need to reason proportionally to get the Pink-Black comparison correct. Students who only reason additively will usually pick Pink (because

it has higher totals), and it is difficult to change their minds if they are not proportional reasoners. Some students reasoned proportionally on this problem without actually calculating the means. For example, they said, "a higher percentage of the Black class got higher scores." This sort of reasoning is proportional, because students are comparing relative frequencies, not just absolute frequencies. Watson (2002) noted that the improvement rate after the cognitive conflict was nearly identical to improvement rate over 4 years in the (2001a) longitu-

dinal study. Thus cognitive conflict might help accelerate students in their ability to analyze and compare data sets. A question that comes to mind, one acknowledged by Watson, is how stable are the students' responses after the cognitive conflict? What would they have said if asked again 3 or 4 years later?

Shaughnessy (2003b) has used three of these comparing-groups tasks from Figure 21.6 in a survey form with middle and secondary school students. Percentages of types of reasoning, including additive, proportional, and distributional reasoning, on the Brown–Yellow and Pink–Black tasks are shown in Tables 21.1 and 21.2. Although there was some growth across grade levels on the Pink–Black task, comparing unequal sized groups remained a challenge for many students. Nearly three-fourths of the students in the survey said that the Pink class did better. More work with students at comparing unequal sized groups is clearly an important target for teaching and assessment.

Most of the research in this section has dealt with students' informal inference strategies as they compare distributions. Pfannkuch (2005) reviewed the literature on probability and statistical inference, and suggested some approaches to help teachers move from informal inference towards the use of more formal inference tools, such as using simulations in a re-sampling approach to statistical inference with students as early as Grade 12. However, Lipson (2002) found that software capabilities disrupted the transition between empirical and theoretical approaches to inference. Also, del-Mas, Garfield, and Chance (1999) found no substantial evidence that simulations improved their students' conceptual understanding of sampling distributions. Despite these discouraging results Pfannkuch remains optimistic that resampling techniques and simulations can provide a bridge to help students transition from informal inference strategies to a more formal (normative) approach to inference. The concepts surrounding

Table 21.1 Frequencies (%) of Responses for Codes for Yellow–Brown Group Comparisons (Survey Results from NSF Grant No. REC-0207842)

Response	Code	MS (N = 84)	HS (N = 188)	Total (N = 272)
Unclear; Omit	0	19 (23)	56 (30)	75 (28)
Sums	1	53 (63)	76 (40)	129 (47)
Average	2	9 (10)	44 (23)	53 (19)
Variation	3	3 (4)	8 (4)	11 (4)
Variation & Centers	4	0 (0)	4 (2)	4 (1)

Codes: 0—No response, misread, unclear reasoning “they look the same”

1—Using Sums; individual graph characteristics e.g., “there’s more 5’s in yellow,” or “there’s a 7 in brown”

2—Using Average (if computed)

3—Variation explanation; spread or tightness of distributions

4—Sophisticated use of Average and Variation in combination

Table 21.2 Frequencies (%) of Responses for Codes for Pink–Black Group Comparisons (Survey Results from NSF Grant No. REC-0207842)

Response	Code	MS (N = 84)	HS (N = 188)	Total (N = 272)
Pink (mostly)	0	67 (80)	135 (72)	202 (74)
Black—graph characteristics	1	7 (8)	12 (6)	19 (7)
Black—averages	2	10 (12)	31 (16)	41 (15)
Black—Distribution characteristics	3	0 (0)	10 (5)	10 (4)

Codes: 0—No response, or “Pink because of more scores or higher total”; or, black or equal with bad reasoning

1—Black, use of graph characteristics

2—Black, use of averages

3—Black, use of characteristics of the distribution, i.e., “higher proportion of scores are . . .”

statistical inference are very complex, and the transition for students to formal inference is likely to be in process for several years. There is no quick fix to understanding these concepts, anymore than there is a quick fix to understanding the formal concept of a limit in mathematics.

Variability and Random Outcomes in Probability Experiments

Probability is the focal point of another chapter in this volume (Jones, Langrall, & Mooney, this volume), but there are important connections between probability and statistics, particularly when repeated trials of probability experiments generate a distribution of possible outcomes.

Truran (1994) wanted to push students to investigate how far experimental probabilities would have to deviate from expectations before students would consider revising their predictions. Sampling with replacement, Truran asked 32 students in Years 4, 6, 8, and 10 what they would expect in samples of size 9, and samples of size 50 drawn from an urn with 2 green balls and 1 blue ball. Extremes were found to be surprising to most of these students, and some of them thought that their results would tend toward 50–50 if they “did it enough times.” Several students offered personal confidence intervals from 4–8 greens for the small sample, and from 27–40 greens for the large sample. Many of Truran’s students gave responses that involved personal theories about the balls or the urn that did not relate to the actual mixture proportions.

The probability task in Figure 21.7 is an extended constructed response that was given to twelfth graders on the 1996 NAEP. Zawojewski and Shaughnessy (2000) discovered some startling responses to this task.

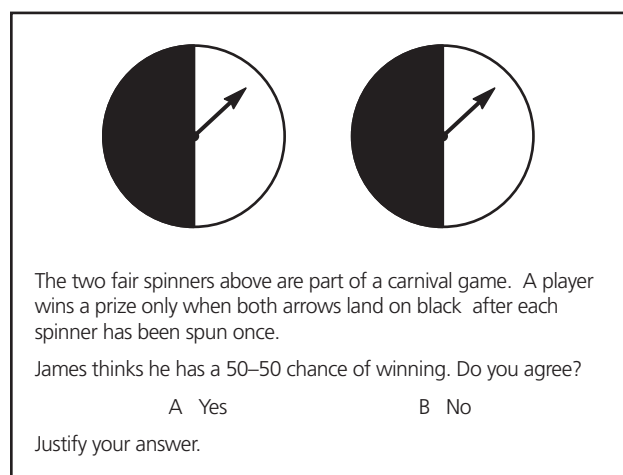


Figure 21.7 The 1996 NAEP Spinner Task (From Zawojewski & Shaughnessy, 2000, p. 263)

Nearly 50% of the 1270 Grade 12 students tested agreed with James, and said that there was a 50–50 chance that both spinners would land black. In a convenience sample of responses ($N = 306$) the reason given by a third of the students in support of a 50–50 chance of both spinners landing black was that since the spinners each were half white and half black there was a 50–50 chance that both of them would land on black. Only 8% of the students in the NAEP sample claimed that the probability was $1/4$ and gave an adequate justification for their answer.

Responses to this item speak volumes about the state of probability instruction in the United States. Simple compound probability problems like this spinner problem are mentioned among the important probability concepts in the *Principles and Standards for School Mathematics* (NCTM, 2000). This spinner problem is not a particularly difficult compound probability problem. In fact, isomorphic versions of this problem appear in many middle school mathematics programs. Shaughnessy and Zawojewski (1999) outlined ways that teachers could use this spinner task, and similar ones, as teaching and assessment tools to explore students’ understanding of probability.

Shaughnessy and Ciancetta (2002) found somewhat more promising results for the spinner task when they surveyed a sample of 652 students in Grades 6–12 and had a subset of students gather actual data. Correct responses initially ranged from 20% in Grades 6–7 to near 90% in classes of students who were studying more advanced mathematics such as precalculus, AP calculus, or AP statistics. Follow-up interviews with 28 of the students from Grade 8–12 were conducted in a setting where students first predicted the results for 10 repeated trials of the spinners tasks and then gathered their own spinner data for 10 actual trials. After conducting their own experimental trials, only 5 students persisted in believing that the chance of both spinners landing black was 50%. As the students experienced the variability in results across several sets of 10 trials and witnessed the actual oscillation in the number of times both spinners landed black, most students realized that the true expectation of winning the spinner game was not 50%, but something much smaller.

Watson and Kelly (2004) assigned SOLO levels to student responses on the 50–50 spinners task. Typical responses in the main three SOLO levels were, “it’s 50–50 because it’s half white” (uni-structural); “the chance that both land black is less than 50% because only one of the three outcomes has both black (multi-structural); and “there are four outcomes, BB, BW, WB, and WW, so the chance is 25% both land black” (relational). Zawojewski and Shaughnessy (2000) found that these same three response categories ac-

counted for most of the student responses to this task on the 1996 NAEP. Watson and Kelly concluded that the concept of independence of the two spinners was not intuitive for their students, and it was especially difficult for the middle school students in their study. This points to a major challenge for teaching and curriculum development in statistics—how do we help students get a better intuitive feel for what a reasonable spread is in experimental probability situations?

Students' recognition of the appropriate variability across the outcomes of a probability experiment is often very dependent on the task. Shaughnessy, Ciancetta, and Canada (2003) compared 84 middle school students' predictions about variability on repeated trials across three task environments: the Lollie task, a task that involved repeated sets of 20 tosses of a regular six sided die, and a single spinner task. The percentage of students' responses that fell within "reasonable" variation guidelines established for these three tasks were 70% for the lollies, 30% for the die, and 49%, for the spinner task. These results suggest that in the mental tug of war that students face between variability (spread) and expectation (centers), that the Lollie problem pulls students more in the direction of acknowledging variability, whereas the die problem pulls students more in the direction of predicting expectation. The spinner problem split the students down the middle into about 50% reasonable and 50% unreasonable predictions for variability across repeated sets of trials. There seems to be an influence from probability on these types repeated trials tasks, especially in the die problem but also somewhat in the spinner problem. Probability instruction might interfere with student thinking about variability, since students may tend to predict 'what should happen, theoretically. In order for students to reason distributionally, in order for them to grow beyond a mere focus on expectation, they must develop their intuition for a reasonable amount of variation around an expected value, not just the expected value itself.

Student Thinking About Association, Covariation, and Correlation

Most of the research studies in the previous sections have dealt with univariate variability. However the most interesting problems in statistics usually involve multivariate situations in which association, dependence, and possible causes of variation lurk, all waiting to be unearthed by a persistent data detective.

In an attempt to see if students would raise questions about a suggested association between two variables, Watson and Moritz (1997a) asked students in Grades 6 and 11 to produce graphical representations

of a nearly perfect relationship between an increase in heart deaths and an increase in motor vehicle use as described in a newspaper article. They also asked students what questions they would raise about such a claim. Using their three-tiered framework for analyzing statistical claims, Watson and Moritz found some students who were reasoning at the third tier because they questioned whether there really was a cause-and-effect relationship between the incidence of heart deaths and motor vehicle use. However, most of their students just accepted the claim about an association between driving a car and having a heart attack without questioning it. Students are often too willing to accept anything that they read in print, and are probably inexperienced in critiquing statistical claims about relationships between variables.

Moritz (2000, 2004) provided an overview and analysis of the research on students' reasoning association and about covariation, and presented bivariate tasks to students in Grades 3 to 9 that involved both positive and negative association. In one task, Moritz gave students the description and graph in Figure 21.8 that portrays a negative association between the number of people in a classroom and the noise level in that classroom. Then, he asked students to explain the graph to someone who could not actually see it.

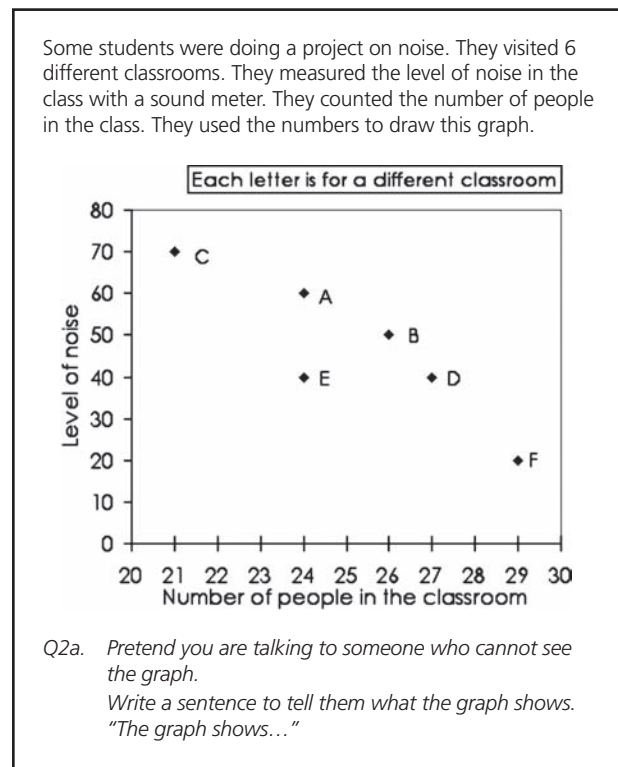


Figure 21.8 Moritz's Covariation Task (Moritz, 2004, p. 238)

Moritz identified four levels in responses to this task: nonstatistical, single aspect, inadequate covariation, and appropriate covariation. Nonstatistical responses did not address anything about covariation, for example “It’s a graph of noise of students.” Single-aspect responders tended to pick out a few data points, perhaps extremes or outliers, and talk about them but they used only a very few data points. If students only talked about one of the two variables, or if they compartmentalized their responses on the two variables and did not talk about how the two were related to one another, their responses were coded as inadequate covariation. Finally, responses that related the changes in the two variables to one another were coded as appropriate covariation.

Moritz (2004) reflected on approaches that might help students to transition through several stages of statistical reasoning, from single data points to the variation within individual variables, to a consideration of how both variables change simultaneously. He agreed with Nemirovsky (1996) who in his work in algebraic thinking suggested that covariation might best be introduced to students by using time as the independent variable. Shaughnessy (2003b) also used graphs over time such as the one in Figure 21.9 in interviews and in classroom teaching episodes with middle and secondary

level students to investigate whether students would look for potential causes of variability in food consumption over time.

Most students did make some conjectures about the variability over time in such per capita food consumption graphs (e.g., per capita milk, coffee, soft drink, & bottled water consumption). Students will even make up something up if they cannot provide a solid contextual explanation for the humps and dips in these food consumption graphs over time. Students have said, “maybe it was inflation,” or “maybe the economy was bad.” They rarely attribute the big jumps or big dips in the food consumption graphs to random variation. Rather, students tend to look for what Wild and Pfannkuch call “special cause” variation. Shaughnessy (2003b) found students who attributed the swings in these food consumption graphs to the baby boom, to improved production and distribution of food, to the Depression, to World War II, to the war in Vietnam, and finally, if all else failed, some students claimed, “It must have been the hippies.” Students do try to make contextual conjectures for why such graphs vary. Nemirovsky (1996) had a point, covariation might be best introduced with time as one of the variables, because students are interested in trends over time, and this type of data connects naturally to topics that are of interest to them.

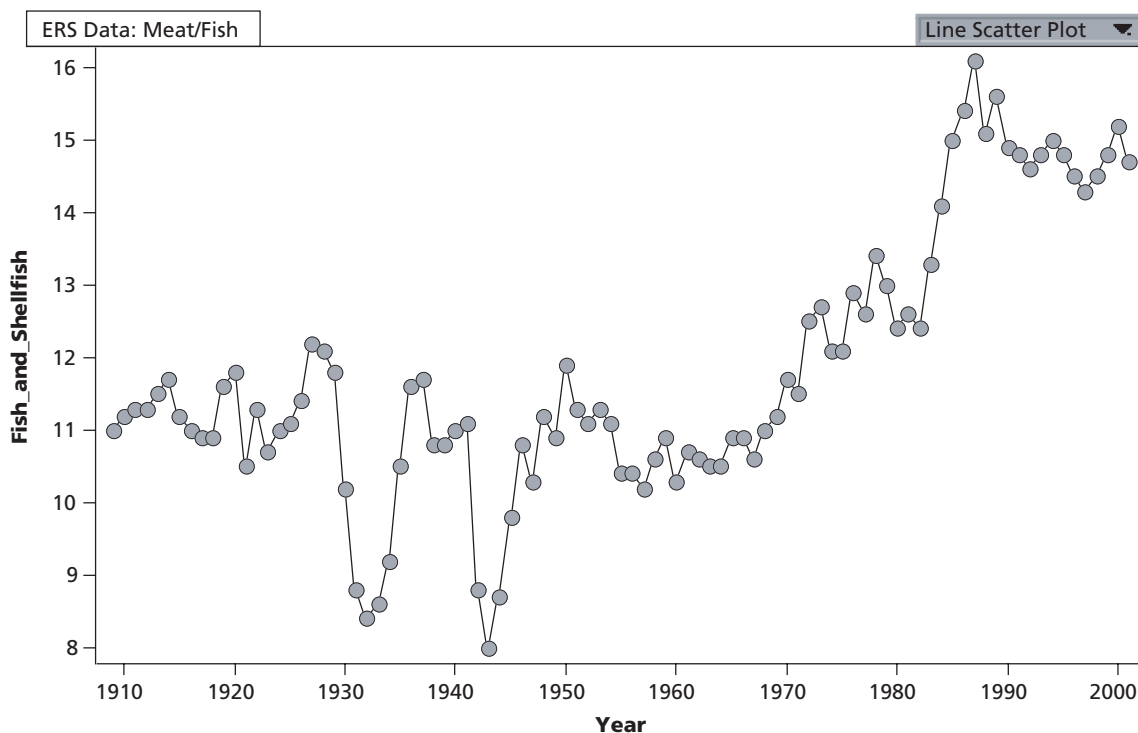


Figure 21.9 Shellfish consumption over time (USDA, 2004).

Whereas Moritz looked at younger student's understanding of covariation, Batanero, Estepa, Godino, & Green (1996) researched beginning tertiary students' reasoning about association of variables in 2×2 and 3×3 contingency tables. Among their findings was the tendency for tertiary students to make decisions about association between two variables on the basis of the frequency in *only one cell* of a contingency table. When reasoning with contingency tables the critical information is usually in the row and column proportions rather than in the cell frequencies. For example the data in Table 21.3 suggest that there may be a relationship between travel class and death rates on the Titanic. The death rate was 38%, 64%, and 76%, respectively, for first, second, and third class travel accommodations (Takis, 1999). The proportion of those who died in each class level is the critical issue, not the absolute frequencies.

Table 21.3 Deaths by Travel Class on the Titanic (From Takis, 1999, p. 483)

Travel Class	Died	Survived	Total
First	122	197	319
Second	167	94	261
Third	476	151	627
Total	765	442	1207

Batanero et al. (1996) found that many students had correct or partially correct strategies when reasoning about association of variables in contingency tables, but they also found three main incorrect reasoning patterns. Some students expected a perfect correspondence between variables, that there should be no exceptions in the data. Batanero et al. called this a *deterministic approach* to association. Other students thought that association could only be in the positive direction and ignored situations in which variables were inversely related, and this they called the *uni-directional* misconception. A third incorrect strategy, called the *localist* approach, students considered only part of the information, such as just one cell, or perhaps just the main diagonal. Batanero et al. concluded that their tertiary students lacked the necessary proportional reasoning skills to reason about contingency tables.

In another study with senior secondary students, Estepa, Batanero, & Sanchez (1999) explored students' ability to make associations between two variables when data sets were presented in two-way tables. Some students used statistical approaches, such as means, totals, percentages, or attempted to compare

the whole distributions. Other students used what Estepa et al. called deterministic strategies, like comparing lowest and highest values, comparing ranges, looking at coincidences, or arguing from their own personal beliefs. Gal (1998) discussed a variety of levels of questions to assess students' understanding of data in two-way tables. Gal noted that percentages are needed to back up opinions or to defend claims that are made about data in two-way tables, because individual cell frequencies are inadequate. Gal encouraged teachers to pose more open-ended, less directive types of questions about two-way tables to promote a higher level of statistical reasoning by students.

Types of Conceptions of Variability

One of the strengths of the research on average noted above has been the spectrum of conceptions of average that researchers have identified, both in investigations of students' thinking and in theoretical conceptual analyses. For example, Mokros and Russell (1995) pointed to average as middle, average as most, average as balance, and average as reasonable. Konold and Pollatsek (2002) mentioned mean as typical, mean as fair share, mean as data reducer, and mean as signal. In a similar way, research efforts on students' understanding of variability are beginning to identify a variety of conceptions of variability, in part because of the variety of statistical objects that can vary, such as data, samples, and distributions. The types of student conceptions of variability identified by recent research include the following:

1. Variability in *particular values*, including *extremes* or *outliers*. In this conception of variability, students focus their attention on particular data values as pointers, often on very large or very small values, or very strange individual values in a graph or in a data set (Konold and Pollatsek, 2002; Konold, Higgins, Russell, & Khalil, unpublished manuscript).
2. Variability as *change over time*. As discussed, this conception of variability may be a good starting point to introduce covariation (Nemirovsky, 1996; Moritz, 2004).
3. Variability as *whole range*—the spread of all possible values. This conception of variability involves the spread of an entire data set or distribution and is closely related to the concept of sample space in probability, the set of *all* the possible outcomes. In this conception students have begun to move away from seeing data only as individual values that

vary, to recognizing that entire samples of data can also vary (Shaughnessy et al., 1999).

4. Variability as the *likely range* of a sample. This conception of variability arises in tasks like the Lollie problem (Reading & Shaughnessy, 2004) or in repeated trials of probability experiments (Shaughnessy et al., 2003). It can lead to statistical tools for representing variability within or across samples, such as box plots or frequency distributions. This conception of variability requires the concept of relative frequency and thus relies on proportional reasoning. It can also lead to the concept of a sampling distribution when applied to the likely range of a distribution of means, or distributions of other sample statistics (Saldanha & Thompson, 2003).
5. Variability as *distance or difference from some fixed point*. This concept involves an actual or a visual measurement, either from an endpoint value (as in a geometric distribution) or from some measure of center (usually the mean or median). Here students are predominantly concerned with the variability of one data point at a time from a center rather than the variability of an entire distribution of data from a center (Moritz, 2004).
6. Variability as the *sum of residuals*. This is a measure of the collective amount a distribution is “off” from some fixed value and provides a measure of the total variability of an entire distribution of data. This is the notion of variability that arose in the design experiment of the Grade 4 class investigated by Petrosino et al. (2003). It provides the foundation for such concepts as standard deviation and regression analysis.
7. Variation as *covariation or association*. This conception of variability involves the interaction of several variables, and how changes in one may correspond to (though not necessarily cause) changes in another. Covariation raises issues about the strength of relationships among variables and poses challenges to parsing out what part of variation is due to chance, and what part may actually be due to cause and effect (Batanero et al., 1996; Moritz, 2004).
8. Variation as *distribution*. Distributions themselves can vary. When the variation between or among a set of distributions is compared, the specter of statistical significance arises. Theoretical probability

distributions emerge to help make decisions about distributions of data, or about sampling distributions (Bakker and Gravemeijer, 2004; Shaughnessy, Ciancetta, Best, & Noll, 2005).

Student Thinking About Information Obtained from Samples and Surveys

In her work with Grades 4 and 5 students, Jacobs (1997, 1999) found that children’s evaluation of survey methods fell into four main categories: *potential for bias*, *fairness*, *practical issues*, and *results*. Some students noted the potential for bias in certain survey methods, but others were more concerned with *fairness* issues. For many students a survey is not fair unless it has representation from all possible subgroups in the survey population. For example, a school survey would have to have a boy and a girl from each class in the school, in order to be “fair.” On the surface, this looks like a sophisticated stratified sampling scheme, but in practice those who favor the fair-sample approach would reject any part of randomization. Jacobs also found that students who were concerned with *results* of surveys disagreed with them if they did not match their own preconceived notions of what should happen. Students also were concerned about how decisive the results of a survey really were. In their minds it was supposed to “be right.”

Several of Jacob’s tasks asked students to evaluate three different survey techniques that she classified as restricted, self-selection, and random. In one task, Jacobs presented students with six different survey scenarios in a school on whether students were interested in conducting a raffle in a school to raise money. Watson et al. (2003) used this task in Grades 3 to 9 as part of their study to measure students’ understanding of sources of variation. Shaughnessy (2003b) used a version of Jacob’s school survey task with secondary school students to see if they could identify important aspects of sampling. The task is presented in Figure 21.10, accompanied by the results from a survey of students in Grades 7–12 in Table 21.4.

Responses to Part 1 of the task were scored 0 to 4, based on the sampling methods described by the students and whether their methods did or did not include explicit reference to (a) sampling, (b) appropriate size (30 to 120 was rated appropriate), (c) stratification, and (d) randomness. Only about a third of the students scored a 3 or 4 on this task, indicating that they had a statistically appropriate sampling plan. Many of the students wanted to survey most, if not all, of the students in the entire school.

The percentage of students who rated each of the three different sampling methods (second part of the

Part 1. A class wanted to raise money for their school trip to Disney World. They could raise money by selling raffle tickets for a Nintendo Game system. But before they decided to have a raffle they wanted to estimate how many students in their whole school would buy a ticket.

So they decided to do a *survey* to find out first. The school has 600 students in grades 7–12 with 100 students in each grade.

How many students would *you* survey and how would *you* choose them? Explain why?

Part 2. Three students in the school suggested different methods to survey the students in the school about buying the raffle tickets.

a) **Shannon** got the names of all 600 children in the school and put them in a hat, and then pulled out 60 of them. What do you think of Shannon's survey?

b) **Raffi** surveyed 60 of his friends. What do you think of Raffi's survey?

c) **Claire** set up a booth outside of the cafeteria. Anyone who wanted to stop and fill out a survey could. She stopped collecting surveys when she got 60 kids to complete them. What do you think of Claire's survey?

(After each of these sampling methods, students were asked to rate the method, and to give a reason for their rating).

☐ GOOD ☐ BAD ☐ NOT SURE

Why?

d) Who do you think has the best survey method? Why?

Figure 21.10 A version of Jacob's Task on Sampling Procedures.

Table 21.4 Percentages Favoring Three Sampling Methods (Survey Results from NSF Grant No. REC-0207842)

Rating/ Method	Shannon (Random)	Raffi (Friends)	Claire (Cafeteria)
Good	24 ^a	37	59
Not sure	9	10	29
Bad	67	53 ^a	12 ^a
Best Method	31 ^a	5	64

^a Indicates the expected response

task) as *good*, *not sure*, or *bad*, is listed in Table 21.4. Shaughnessy's sample of students heavily favored the self-selection approach outside the cafeteria in this task. This is consistent with results found by Jacobs (1997) and Watson and Moritz (2000a). A third of these students felt that asking friends (Raffi's way) was a good way to get a sense of the opinion in the school. Their reasons indicated that they wanted to predetermine the survey results and this was a way to make it happen. For teachers of statistics who promote the importance of random sampling this is a troubling result. So much for democracy!

Another striking result was that only 12% of the students recognized the potential for bias in the self-

selection method outside the cafeteria. For example, some groups of students might not eat lunch in the cafeteria. Only about a third of the students surveyed preferred the random sampling approach (Shannon's way), whereas over half felt that Claire's self selection survey approach was the best, because "everyone has the same chance this way." The fairness criterion for sampling that Jacobs (1997) found so prevalent among younger students is still quite robust among older students.

Watson and Moritz reported on a series of studies to investigate whether students knew what a sample was, whether they would be sensitive to sample size, and whether they would recognize the possibility for bias in real sampling situations from the media (Watson & Moritz, 2000a, 2000b; Watson, 2004). In one longitudinal study they examined students over a 3 to 4-year period, administering the same four tasks three times to students who were in Grades 3 through 11. Students were asked the same questions 2 years later and again 4 years later to see if they had grown in their understanding of a sample as representative of the population from which it was drawn (Watson & Moritz, 2000b). The first question asked was "If you were given a sample, what would you have?" Some students gave examples of samples—samples of food, blood sample, a free sample in the mail, when asked what they would have if they had a sample. Others responded "a little bit," or "a small

portion.” Still others mentioned a “bit of something” or “a part of a group.” Overall, students’ thinking ranged from personal examples, to the notion of a piece, to the idea that a sample should be a representative piece of something larger.

The second question asked students whether they would put more faith in a friend’s recommendation for a car purchase or in the recommendation of Consumer Reports magazine, or if it did not matter one way or the other. Watson & Moritz’s other two questions were based on articles from a newspaper. One article claimed that over 90% of those who phoned in on a survey were in favor of legalizing marijuana, and another article generalized a claim ‘that 6 of every 10 students from a sample in Chicago could easily bring a handgun to school’ to all of the United States. Thus all three of these contexts had the clear potential for bias.

Watson & Moritz describe a progression in thinking about samples in which students (a) do not distinguish between sample and population, then (b) recognize the difference between sample and population but really wanted to sample *everyone*, and finally (c) realize that samples can be used to represent the population and to estimate population parameters. Watson and Moritz found considerable growth in their longitudinal study—50% of the students improved on the four questions after 2 years, and 75% had improved from their initial responses after 4 years. The cumulative school experience and outside world experience seems to have improved these students’ understanding of the concept of a sample.

In another study Watson and Moritz (2000a) interviewed 62 students who had completed the four survey questions in their longitudinal study. In the interviews Watson and Moritz included some more general questions on sampling and sample size such as a “hospital-type” problem in order to investigate whether students would be sensitive to issues of sample size and variability. (The prototype of this task involved the gender percentages of babies born in large or small hospitals. See for example, Kahneman and Tversky, 1972; Shaughnessy, 1977). The hospital-type task went as follows: A sample of 50 students was taken from a large urban school and a sample of 20 students was taken from a small rural school. One of these samples was strange in that it had 80% boys. Which do you think is more likely? (a) The sample is from the small school, (b) it is from the large school, or (c) it could be from the large school or the small school? As part of this study, Watson and Moritz (2000a) identified six different levels of understanding sampling, using *selection criteria* and *sample size* as coding variables. They determined these levels by applying both the SOLO model and their three-tiered

model of statistical literacy—knowing terms, applying terms in context, and asking questions and critiquing the media (Watson, 1997). Their six levels of understanding samples ranged from small samplers with no sampling method to large samplers using random sampling methods that were sensitive to bias. Small samplers were those students who were content to ask just a few people, or to listen to their friend rather than Consumer Reports about the car purchase question. Large samplers recognized the increased power of information when it represented a larger fraction of the population. Students’ own preferred sampling methods ranged from pre-selected or distributed criteria so that “things were fair,” like in the Jacobs (1997) study, to large samplers who wanted to control for bias by pulling random samples.

Watson and Moritz were troubled that so many students failed to recognize the bias in the handgun newspaper article that made a claim about the entire United States on the basis of a survey of just one high school in Chicago. Some students did question the validity of the claims of the Handgun newspaper article, exhibiting Tier 3 reasoning on the literacy scale, but such students were rare. From 55–65% of the students surveyed in each grade level were quite content to believe that those reported interviewed in the Handgun question were representative of everyone in the United States. As for the sample size task, only 9 of 41 students interviewed said that the strange result of 80% boys in the school was more likely to occur at the rural school, because there is more relative variability in small samples. Many of the students who got the School Problem wrong did see the need for and benefits of larger samples on the other questions. They just could not apply what they knew about large/small samples to the School problem. This suggests teachers have their work cut out for them to help students become aware of potential sources of bias in sampling, and to reach the level of critical thinkers at Tier 3 on Watson’s statistical literacy hierarchy. Students can recognize the importance of getting a larger sample as a means to “avoid getting all the same,” or as a means to “avoid too many extreme outcomes” or as a means to “balance” the input more fairly. But they may still not see that larger samples provide a method of tightening relative sample variability. Watson and Moritz sum up what they found about the development of students’ understanding of samples as follows:

Students initially build a concept of sample from experiences with sample products in medical and science related contexts, perhaps associating the term *random* with sampling. As students begin to acknowledge variation in the population, they recognize the

importance of sample selection, at first attempting to ensure representation by predetermined selection but subsequently by realizing that adequate sample size coupled with random or stratified selection is a valid method to obtain samples representing the whole population. (Watson and Moritz, 2000a, p. 63)

Research on Students' Understanding of Graphs

Graphs are critical for data representation, data reduction, and data analysis in statistical thinking and reasoning. A complete review of all the research on reasoning about graphs is well beyond the scope of this chapter. A thorough review of the literature concerning sense-making in graphs is available in Friel, Curcio, & Bright (2001). They define graph comprehension as "the ability of graph readers to derive meaning from graphs created by others or by themselves (p.132)." They discussed the influences of visual perception, the characteristics of graph readers, and the effect that experience with statistics all have on the ability of people to make sense of graphs. Friel et al. (2001) were particularly interested in students' ability to comprehend statistical graphs, as contrasted with graphs of functions in algebra or calculus. It is interesting to compare the references in the Friel et al. review with another article by Roth and Bowen (2001) on reading graphs that appeared in the same issue of that journal. Of the hundreds of references in these two articles, only four or five references appear in both. The overall scope of research on understanding graphs has become enormous. For my purposes, I want to concentrate on a few areas of research on graph sense that are of particular importance to statistical reasoning.

Research on Understanding Particular Types of Statistical Graphs

An analysis of the student results on graph items from the 1996 NAEP indicated that although students performed well when reading information represented in pictographs and stem-and-leaf plots, Grade 12 students ability to read and interpret histograms or box plots lagged behind their performance with other graphical representations (Zawojewski & Shaughnessy, 2000). Performance on histograms, which requires some proportional reasoning, was especially low. Students tested on 1996, 2000 and 2003 NAEP administrations could read graphs fairly well but had trouble interpreting graphs, and even more trouble making predictions based on graphical information. (Zawojewski & Shaughnessy, 2000; Tarr & Shaughnessy, in press). On one extended constructed response task

students were given data for railroad ticket sales and asked which of two different graphical representations of the same data they would use in a presentation on how sales had grown. Shaughnessy and Zawojewski (1999) reported that only 2% of Grade 8 students gave a reason for their choice of graphs that merited the highest score on the NAEP rubric for the rail task. Even though an additional 18% of students gave partially correct answers, the overall performance, especially the number of omits on this NAEP task, suggests that students' graph interpretation skills are weak.

Other research has confirmed some difficulties that students have when reading and interpreting particular types of graphs. Researchers have examined student thinking on bar graphs (Pereira-Mendoza & Mellor, 1991), line graphs (Aberg-Bengtsson & Ottoson, 1995), stem-and-leaf plots (Pereira-Mendoza & Dunkels, 1989; Dunkels, 1994), box-plots (Carr & Begg, 1994), scatter-plots (Estepa & Batanero, 1994), pictographs (Watson & Moritz, 2001), and histograms (Melitiou & Lee, 2002). Pereira-Mendoza (1995) suggested that children should:

1. Explore the assumptions underlying the classification of data and interpretation of the meaning of data.
2. Discuss and explore the possibility of alternative representations.
3. Predict from the data. (Pereira-Mendoza, 1995, p. 6).

His point is that by directing students' attention to alternative representations, teachers can help move students beyond mere drawing and tabulating of data to more critical elements in graph sense.

Carr and Begg (1994) introduced box-plots to 11- and 12-year-old students in order to investigate whether such graphs were appropriate for elementary school students. Their informal observational study of 8 students included a brief instructional component on constructing box-plots, followed by unstructured student interviews to determine if students understood the ideas of center and spread. They concluded that box-plots are an appropriate topic for students in this age group provided that teachers emphasize the understanding and interpretation of the plots, and not just the construction of them. However, not all researchers are in agreement about early introduction of box-plots. Bakker et al (2004) pleaded for delaying the introduction of box-plots due to the difficulties that middle school students have with the proportional reasoning needed to construct and interpret box-plots.

The research on students' understanding of particular graph types suggests that teachers should (a) include a variety of graphical representations and (b) go beyond mere graph construction to discuss the meanings and interpretations of the graphs. Lehrer and Romberg (1996) note that most examples of graphs in textbooks are preprocessed, and they recommend more attention be given to having students construct and share their own graphical representations of data.

A Closer Look at "Bar" Graphs

Given the research on students' understanding of particular graphs, consider the complexity of just one type of graph, the bar graph. Bar graphs are first introduced in the elementary grades to represent frequency counts of categorical or integer data. They follow in a sort of natural way from stacked dot plots, allowing data representation to move from discrete to continuous displays. But this can be a big jump for students, as the exact count that was visible in a dot plot is no longer obvious in a bar and the vertical axis may no longer be integer valued. Bar graphs are first used to represent frequencies, but later on they are used with the bar heights that represent relative frequencies. Still later, when histograms are introduced to represent the frequency or relative frequency of continuous data within continuous intervals, the horizontal axis is no longer categorical or integer valued, it also becomes a continuous scale. Relative frequency distributions are a critical tool for normalizing data in order to compare unequal-sized data sets—but research has shown that this is a very challenging issue for students to grasp.

In their analysis of *document literacy* among adults (discussed previously in this chapter), Kirsch, Jungeblatt, & Mosenthal (1998) and Mosenthal and Kirsch (1998) have done large-scale studies that point out the difficulties that adults have with bar graphs or line graphs that require more than simply reading the data. When integration of information in bar graphs is required, or when adults are asked to summarize in writing the conclusions from a graph, they perform much more poorly than on simple reading-the-graph tasks. These results are consistent with reports about NAEP tasks (Zawojewski & Shaughnessy, 2000; Tarr & Shaughnessy, in press) and with what Gal (2002) discussed about the Grade 12 TIMMS results on graph sense.

The conceptual complexity of bar-type graphs continually increases throughout the teaching and learning process. The values and scales on the horizontal and vertical axes morph from counts, to percentages, and then to percentages over an interval. Watson and Moritz (1997b) discussed the complexity in this transi-

tion among graphical representations (see also Moritz & Watson, 1997). The cognitive transition in bar graphs mirrors the transition from additive to proportional to distributional thinking previously discussed in this chapter, with a heavy reliance on proportional thinking. Without proportional thinking, students see only counts and miss the power of graphs to reduce data and show trends. Proportional thinking is the lynchpin in making sense of statistical graphs.

Research on Components of Graph Sense

Curcio (1987) introduced a framework for graph sense by building on research in the literature on reading. She identified three elements—form, content, and topic—which contributed to the ability of students to construct their own graph-reading schemata. In later work she characterized three levels of reading of graphs: *reading the graph*, *reading within the graph*, and *reading beyond the graph* (Curcio, 1989). These categories help to shed light on the complex nature of understanding graphs. For example consider the graph of fruit juice consumption over time in Figure 21.11. This graph is based on data collected by the United States Department of Agriculture (USDA, 2004) and posted on their website.

In order to be *Reading the graph*, students have to at least understand the scale and the measurement units. This is a graph over time, from 1970 to 2000, and the vertical axis is in gallons per person per year. It is also important for students to realize and understand that "per capita" data are rate data. *Reading within the graph* is especially important for graphs over time, in order to be able to discuss general trends. For example, the timeline shows a fairly steady increase in consumption, with a couple of sudden dips. Why is there a general increase, and why did those dips occur? *Reading beyond the graph*, Curcio's highest level of graph comprehension, includes such skills as projecting into the future and asking questions about the data. For example, one might suggest that fruit juice consumption appears to be dropping at the end of the graph, and that it may level off at some point in the future. Or, one might ask where these data came from, how they were collected, and how the USDA managed to calculate or estimate "gallons of fruit juice per capita" for each of the years.

I have previously argued (Shaughnessy, Garfield, & Greer, 1996) for another level of graph comprehension beyond Curcio's three levels, that is, *reading behind the data or graph*. This involves more than just *reading beyond the graph*. Statistics are within a context. As Wild and Pfannkuch (1999) have noted, one must always search for special causes of variation in the data. The special role of statistics as a scientific dis-

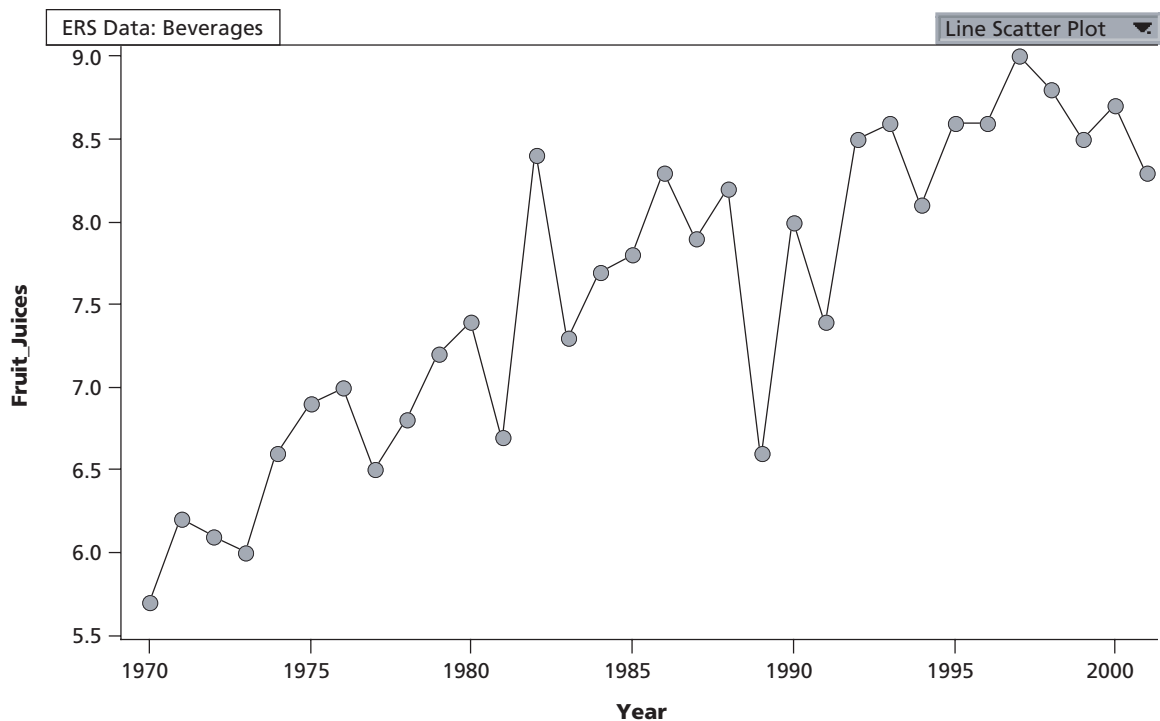


Figure 21.11 Fruit Juice Consumption, Gallons per capita, over time (USDA, 2004).

cipline lies in making connections between the context and the graph, that is, what lies *behind* the graph. For example, in the juice consumption graph one might look at historical, economic, or demographic influences that may have affected juice consumption over the last 40 years. In reading *behind* this graph, one might conjecture that in the mid 1980s, better production of juice came about. Perhaps a switch occurred from smaller orchards to larger centralized locations for collecting and processing fruit for juice. As time went on, a global economy arose so that fruit juice could be produced and shipped more easily all year round from somewhere in the world. This greater availability could contribute to a more rapid growth in fruit juice consumption. Perhaps a killing frost explains the two years, 1981 and 1989, that show a drastic drop in fruit juice consumption. Whatever the cause of the drop *behind* the data in those years, a traumatic event affected juice consumption but was followed by a quick rebound to the status quo in subsequent years. The drop-off during more recent years might be due to increased consumption of other beverages, such as soft drinks or bottled water. These are just a few of the possible special causes that might affect fruit juice consumption over time, which involve looking *behind* the graph, and using some data detective skills to analyze graphical information.

Watson and Moritz (1997a) applied their three tiers of statistical literacy—(a) knowledge of basic statistical terminology, (b) applying statistics in context, and (c) challenging statistical claims—to analyze students' responses to some graphs taken from newspapers. The tiers of statistical literacy match up fairly well with Curcio's three levels of graph sense when one considers statistical literacy about graphs. Watson and Moritz found a lack of attention to scaling issues in the data and graphs they gave their students. Furthermore, only 10% of their students were aware of the complexity of or questioned some of the features of one particularly misleading graph. When students were asked to make some calculations based on information in the graph, many of them ignored the graph altogether and reverted to personal experiences. Watson and Moritz discovered gaps in students' graphical literacy within all three of their levels of statistical literacy.

In another series of studies that investigated students' thinking about pictographs Watson and Moritz (2001) identified three levels of student reasoning. In follow-up studies 3 years later they documented growth in students' thinking about pictographs. They also found that cognitive conflict could improve students' initial reasoning. When a sample of Grade 3 students was shown responses from other children about the pictographs, their reasoning about the graphs tended to improve.

Investigations of students' graph comprehension and graph interpretation skills have been one approach to exploring the components necessary to develop good graph sense. Another approach has involved analyzing student-generated graphs. Following on earlier work in relation to the SOLO model (Watson et al., 1995), Chick and Watson (2001, 2002) used the data card tasks that included information about students' weight, age, favorite activity, and number of fast food meals eaten per week. Students in Grades 5 and 6 were put into groups of three and asked to generate some hypotheses and to construct graphical representations of the information on the case cards. Chick and Watson's students produced three levels of representations of information in the data cards. Some students depicted individual aspects of the data set, such as a table of single values of one variable with no interpretation offered. Other students produced a graph of a single variable, and still others produced a bi-variate representation, such as a scatter graph, indicating a possible relationship between two of the variables. Students' reasoning on the task followed a pattern similar to their graphical representations: looking at individual data points, considering the entire range of values for one variable, or conjecturing cause-and-effect relationships among several variables. Nearly half the students could interpret the data at a higher level than they could represent it graphically, suggesting that students' graphical sense may lag behind their data interpretation skills.

The attention to individual data points exhibited by Chick and Watson's students has been pointed out by other researchers, notably Cobb (1999) and Konold and Higgins (2002, 2003). These researchers have suggested that case-value representations of data are a good starting point to help students develop graph sense and begin making connections to the context. Konold and Higgins (2003) presented case value data in a *value bar* representation and demonstrated a progression of representations from value bars, to end points of the value bars, to stacked dot-plots. This transition path could aid students and teachers to move beyond students' initial preferences for individual case value plots to stacked dot plots that provide a visual representation of the entire distribution of the data, including information about shape, center, and spread.

A summary analysis of the literature on students' understanding of graphs by Friel et al. (2001) identified six behaviors that they considered to be closely associated with graph sense. Each of these six behaviors seems to fit nicely with one of Curcio's three levels of graph reading (read the data, read within the data,

read beyond the data). I also suggest two additional behaviors that fall under the level of reading *behind* the data.

1. Recognizing components of graphs (*Reading the data*).
2. Speaking the language of graphs (*Reading the data*).
3. Understanding relationships among tables, graphs, and data (*Reading within the data*).
4. Making sense of a graph, but avoiding personalization and maintaining an objective stance while talking about the graphs. (*Reading within the data*).
5. Interpreting information in a graph and answering questions about it (*Reading beyond the data*).
6. Recognizing appropriate graphs for a given data set and its context (*Reading beyond the data*).

In addition I would add

7. Looking for possible causes of variation (*Reading behind the data*).
8. Looking for relationships among variables in the data (*Reading behind the data*).

The cumulative results from a number of researchers on graph sense indicate that students have poor graphical interpretation skills and are often unable to reason beyond graphs. Also unless the graph is very straightforward, students may even have trouble just reading a graph at all. In the special case of statistical graphs, reading *behind* the data is critical to making connections between the context and the data. Basic graph sense such as reading, reading within, and reading beyond graphs is critical to statistical thinking, reasoning, and literacy.

Technology and Research on Learning Statistics

My principal research interest in the use of technology in teaching statistics is the development of students' statistical thinking in the conceptual areas discussed in the previous section: averages, variability, information from samples and surveys, and graph sense. The availability of powerful computing tools has led to improved methods of analyzing data and exploring data graphically (Hawkins, Jolliffe, & Glickman, 1992). What sorts of technological environments can enhance student learning of the major concepts in statistics?

Biehler (1993, 1994a, 1994b, 1997) has written extensively on the type of software tools that are necessary to enhance the teaching and learning of statistics. Over a decade ago Biehler described aspects of technology needed to empower students to do interactive exploratory data analysis, using visualization and simulations tools to understand statistical concepts and methods. He proposed the following components for ideal software:

- Student tools for data analysis, for method construction and evaluation, for modeling and for visualization, that can grow and expand along several paths into a professional version, instead of mere technically reduced student versions of professional systems
- A system of coordinated computer experiments, learning environments, and major visualizations that can be adapted to students' and teachers' needs. (Biehler, 1994b, p. 3).

In another early analysis of and report on the use of technology for teaching statistics, Garfield (1990) led a Working Group on Technology and Data of the American Statistical Association (ASA). This Working Group outlined attributes of technological environments that can facilitate the learning of data handling, including:

- Direct access*, which allows students to view and explore data in different forms, including subsets of data and different visual representations.
- Flexibility*, which allows students to experiment with and alter displays of data, change intervals on a graph, and explore different models that may fit the data.
- Connectedness*, so that students are able to access resources on the Internet, as well as to obtain software or data used in the study of other disciplines.
- Representations*, including dynamic ones from which students may choose among different graphs in order to select the best way to interpret and display a data set.

Both Biehler and Garfield have continually recommended that in order to truly enhance statistics education, technology for learning statistics should go way beyond mere statistical packages that carry out procedures. They advocate for technology that puts the design and representation of data structures in the hands of the student, rather than merely allow-

ing selection of procedures from an already existing repertoire. In an analysis of existing and desirable statistical software, Biehler (1997) distinguished among *tools*, *resources*, and *microworlds* for the teaching and learning of statistics with technology. By *tools*, Biehler means the type of software and hardware support that professional statisticians use to practice their trade. For example, *Minitab* (2005) is a software package that consists primarily of statistical *tools* for data analysis. *Resources* would include data sets with references, and background and context information. These types of data sets can often be found at Internet sites. The ERS Web site maintained by the Department of Agriculture (2005) from which the food consumption data (see Figures 9 & 11) were downloaded is one example of what Biehler means by resources. *Microworlds* provide a creative, exploratory environment for students to represent data, and to carry out simulations. Both *Tinkerplots* (Key Curriculum Press, 2005a) and *Fathom* (Key Curriculum Press, 2005b) are examples of microworlds that are flexible and that put a lot of the power in the hands of the students to create a variety of types of graphs and data representations.

Ben-Zvi (2000) presented examples of types of technological tools for learning statistics that are similar to Biehler's categories, such as statistical software packages, microworlds, tutorials, and resources on the Internet. According to Ben-Zvi, tools for statistical learning have been developed to support these areas:

1. Students' active knowledge construction, by "doing" and "seeing" statistics.
2. Opportunities for students to reflect on observed phenomena.
3. The development of students' metacognitive capabilities, that is, knowledge about their own thought processes, self-regulation, and control.
4. The renewal of statistics instruction and curriculum on the basis of strong synergies among content, pedagogy, and technology. (Ben-Zvi, 2000, p. 128)

Ben-Zvi shared a detailed example of some middle school students as they worked with spreadsheets and constructed graphical representations of some data on Olympic records for the 100-meter dash. He documented how even a simple tool like a spreadsheet can be very powerful in the hands of students who are directing their own learning and how a spreadsheet can provide support for the four areas he identified.

Bakker (2002) referred to *Fathom* and *Tinkerplots* as "landscape-type" tools and contrasted them with "route-type" tools with which students have far fewer

choices at their disposal. Examples of route-type tools include the *Mini-tools* discussed by Cobb et al. (1999) and the tools created by the Chance-Plus project such as *Prob-Sim* and *Data-Scope* (Konold & Miller, 1994). Landscape-type tools put students in very open situations, with a variety of powerful choices, including sorting and arranging data into a variety of visual display formats. Bakker expressed concern about starting students with these landscape-type tools, as they might overwhelm students with too many representational choices. He argued that it is better to first introduce students to some more limited route-type tools, such as *Prob-Sim* or *Minitools*, that can provide students with focused opportunities to explore particular concepts. Friel (2007) has provided a detailed review of research on the interaction of technology with the teaching and learning of statistics, including an analysis of the strengths and weaknesses of a variety of types of technological tools for teaching statistics, including both route-type and landscape-type tools.

Research on what students learn about statistics or how their statistical thinking develops while using new statistical tools like *Tinkerplots* and *Fathom* is relatively sparse as of this writing, especially when compared to the general explosion of research in other areas of statistics education. During the development phases of statistical software most of the developers' energy is focused on the tools themselves, or on professional development work with teachers to help them use the new software. Normally very little research is done on student learning or on student conceptual growth in statistics during the developmental phase of software. Finzer (2002) made a good case for the simultaneous integration of research on student learning with software development, so that students' learning issues can inform the software development team in mid-stream, and vice-versa:

In some ideal world the boundary between development of software for use in research on learning and development of software for classroom use would be truly porous so that researchers could easily adapt classroom software for research purposes and classrooms would reap the benefits of educational research. (Finzer, 2002, p. 1)

General overall access for all students and teachers to statistical software is still a problem. Too few teachers and schools are currently using data analysis tools like *Fathom* or *Tinkerplots*. Even AP statistics teachers are more likely to use graphing calculator statistics packages or some of the more traditional data-crunching packages such as *Minitab*. To argue the flip side of Bakker's concern about landscape-type software being too open, in the current teaching of statistics

there may be too much route-type and not enough landscape-type use of technology in the teaching of statistics. The landscape-type software is more likely to help students to become good data detectives. Open-ended micro-worlds give students a lot of autonomy and power in data exploration and analysis, whereas some route-type software might be a bit confining and sometimes overly procedurally oriented.

Some Research on Students' Statistical Thinking with Technology

Bakker and Gravemeijer (2004) presented a detailed analysis of a teaching experiment with seventh Graders that was designed to get students to reason about distributions in informal ways. They noted four critical components of distributions: centers, spreads, density or clustering, and skewness. Their teaching experiment was designed to have students encounter these four components and to move students along from looking at data as individual case values to reasoning about entire distributions of data. Using two computer Mini-tools, a value bar tool and a stacked dot-plot tool, Bakker and Gravemeijer had students investigate two data sets on battery life, data sets that had also been used by other researchers (Cobb, 1999; Cobb, McClain, & Gravemeijer, 2003).

The students first compared the two battery data sets using the value bar tool. During this exploration, students began to use terms like *majority*, *outliers*, *reliability*, and *spread out*. Bakker and Gravemeijer claimed that the value bar tool helped to provide a visual representation of the mean for students. Subsequently the stacked dot-plot tool allowed students to develop qualitative notions of more advanced aspects of distributions such as frequency, classes, spread, quartiles, median, and density (Bakker & Gravemeijer, 2004). The dot plot tool also allowed students to partition data into 2, 4, or more equal-sized groups. This provided an underlying structure for development of concepts such as median, box-plot, density, and eventually histogram. Students were able to reason about clumps of data and compare clumps across the two battery-life distributions, so that the underpinnings of Konold's "signal amid the noise" conception of the mean were available. Cobb (1999) described a similar learning trajectory for a group of middle school students who were exploring the battery data. Cobb's principal lens for the study was classroom discourse and social interaction, which he claims is a necessary component for the type of growth and learning witnessed in his work.

On the basis of the statistical development that they witnessed among the students in their studies, Bakker and Gravemeijer, and Cobb, concluded that

it is important to provide opportunities for students to contribute their own ideas to the statistical learning process and that teachers need to provide a lot of time for discussion and interaction during class explorations of data. They also believe that formal measures to describe distributions, such as median and quartiles, should not be introduced until after students have had opportunities develop their own intuitive notions about distributions.

In reading the Bakker and Gravemeijer study, it is striking how natural the integration of the Mini-tool environment was in the teaching experiment. Furthermore, the process of students' growth from first looking at data as individual data cases, then considering clumps of data, and finally making comparisons and decisions based on clumps and spreads, was clearly supported and enhanced by the Mini-tools technology. Even though these two Mini-tools are more in Bakker and Gravemeijer's route-type software category, the students were able to grow considerably despite the software's limited capabilities. This study was enhanced by the strong alignment between the task environment and the Mini-tool's capability to explore that task environment. Based on these studies on teaching statistical concepts with technology, the need for a close match between the statistical questions that are being investigated and the capability of the software to directly tackle those questions seems paramount. These studies also suggest that the type of classroom discourse that takes place during a statistical exploration also has a major impact on students' conceptual growth in statistics.

Friel (2007) shared a number of visual representations of data from two different types of roller coasters using *Tinkerplots*. Stacked dot plots allow students to partition the two data sets in different ways, to compare spreads, middles, and clumps. Gradually the shape of the two distributions becomes apparent. The roller coaster comparison task is rich in student exploration possibilities, similar to the battery-life task discussed above. The combination of interesting data sets with versatile software tools has made these two environments, the roller coasters and the batteries, excellent candidates for future teaching and research projects.

In reviewing a series of efforts in which researchers have asked students to explore and compare data sets within rich technological environments, Konold et al. (unpublished manuscript) summarized a number of different ways to consider data. Among their research sources were cases of primary teachers who wrote about how their students dealt with data (Russell, Shifter, & Bastable, 2002), some research by Cobb et al. (2003) on how students describe co-

variation in scatter-plots, and some of their own work with secondary school students who explored a large multivariate data set containing information on a number of variables about high school students (Konold, Pollatsek, Well, & Gagnon, 1997). According to Konold et al. (unpublished manuscript) data can be viewed as *pointers*, *case values*, *classifiers*, or *aggregates*. A major issue in statistics education is to find ways to help students' thinking about data to evolve along this continuum. Although Konold has written about the importance of the mean as the "signal amid the noise," indicating that he considers data as aggregate to be the most important way for students view data, in this working paper he has also acknowledged the potential benefits of viewing data in other ways. For example, case values can help to highlight the variability in data, a concept that gets short shrift in school mathematics (Shaughnessy et al., 1999). The types of technological tools that Konold et al. include in their paper support student conceptual growth on centers, as well as on variability.

As noted earlier in this chapter, students have a very difficult time learning about sampling distributions and the accompanying statistical ideas such as the law of large numbers (Saldanha and Thompson, 2003). Chance et al. (2004) shared research on a series of experiments on sampling distributions conducted with college students in introductory statistics courses. This work built upon earlier explorations of tertiary students' understanding of sampling distributions (del Mas, Garfield, & Chance, 1998; Garfield, delMas, & Chance, 1999), and used a software tool developed by del Mas (2001) to run simulations that generate sampling distributions.

In their research Chance et al. identified four prerequisite concepts that students need in order to understand sampling distributions: variability, distribution, sampling, and the concept of the normal distribution. Research reviewed in previous sections of this chapter has already noted the complexity and diversity of student thinking about variability and about samples and sampling, and has noted issues that students deal with when comparing distributions. With the addition of the normal distribution as a prerequisite concept, it is no wonder that Saldanha and Thompson (2003) encountered so much difficulty in their teaching experiment on sampling distributions with secondary students. Chance et al. used the developmental model of statistical thinking that Jones et al. (2000) had introduced when working with elementary and middle school students. Chance et al. uncovered and subsequently validated five different levels of reasoning about sampling distribution among their

college students, namely, idiosyncratic, verbal, transitional, procedural, and integrated types of reasoning.

Even though Chance et al. spent considerable time targeting tasks in which their students ran software simulations to explore sampling distributions, they found only a few students who actually managed to integrate all the statistical concepts involved as they explored properties of sampling distributions.

Part of the problem in developing a complete understanding of sampling distributions appears to be due to students' less than complete understanding of related concepts, such as distribution and standard deviation. We have found our own research progressing backward, studying the instruction of topics earlier in the course and the subsequent effects on students' ability to develop an understanding of sampling distributions. For example, initially we explored student understanding of the effect of sample size on the shape and variability of distributions of sample means. We learned, however, that many students did not fully understand the meanings of *distribution* and *variability*. Thus, we were not able to help them integrate and build on these ideas in the context of sampling distributions until they better understood the earlier terminology and concepts. (Chance et al., 2004, p. 312)

Like Saldanha and Thomspson, they go on to conclude that the "concept of sampling distribution is a difficult concept."

The few pages given in most textbooks, a definition of the Central Limit Theorem, and static demonstrations of sampling distributions are not sufficient to help students develop an integrated understanding of the processes involved, nor to correct the persistent misconceptions many students bring to or develop during a first statistics course. Our research suggests that it is vital for teachers to spend substantial time in their course on concepts related to sampling distributions." (Chance et al., 2004, p. 312)

If the results of the research on student understandings of average, variability, and distribution that were discussed in earlier sections of this chapter can be effective in promoting better and earlier teaching and curriculum development in K–12 statistics, perhaps Chance and others will eventually have a stronger statistical foundation to build upon when they work with tertiary students. In the meantime, tertiary statistics instructors need to heed Chance et al.'s advice and start teaching from where their students actually are in their knowledge of statistics. The research reviewed in this section suggests that technological tools are very important for helping students to transition

from those naive conceptions to richer, more powerful understandings of statistical concepts.

RESEARCH ON AND DEVELOPMENT OF TEACHERS' UNDERSTANDING OF STATISTICS

Much of the research on teachers' understanding of statistical concepts has been derived from professional development work that is intended to extend teachers' knowledge and competence in statistics. This is partly a matter of convenience, combining research with professional development. It is also partly because it is difficult to obtain direct information about teachers' knowledge about statistical concepts. Most K–12 mathematics teachers in the United States have very little background in statistics. The exceptions are those teachers who may have had a concentration in statistics during their masters program for secondary teachers, or middle school teachers who completed one of the few special programs that exist in the United States for middle school mathematics teachers. Teachers know that their statistical understanding is shaky, and so attempts to obtain direct information from them using surveys or interviews can be embarrassing for them. As a result, a good deal of what is known about how teachers think and reason about statistics has tended to be somewhat anecdotal. Sometimes field notes can be taken during observations of classroom statistics activity, or retrospective statements can be obtained from teachers while debriefing trial lessons of new statistics materials.

Research on Teachers' Understanding of Statistics Within Professional Development

Over the course of several curriculum development and professional development projects, Rubin used field notes and reflections after interviews to gather information on teachers' thinking about statistical concepts. In one project several teachers volunteered to trial the Stretchy Histograms and Shifty Lines software from the Elastic project with their students (Rubin & Rosebery, 1988). These two tools were among the very first tools to allow users to click and drag elements of graphs, like bars in histograms, or lines in scatter-plots, and to observe changes in the mean and median of distributions or changes in the equations of regression lines. Rubin and Rosebery found that when teachers asked their students to alter histograms by adding values to the original graph, the

teachers became puzzled. For example, they wondered why did the median not change, when the mean did change? The teachers were unaware that there might be multiple occurrences of the median value in the data set, or that the median remains constant if the same number of data values are added in on either side of it. As a result of Rubin and Rosebery's work, recent software packages like *Tinkerplots* prefer to use stacked dot-plots as the default representation mode rather than histograms. Dot plots preserve every data point, and the median value can easily be highlighted in a dot plot.

In another project, Hammerman and Rubin (2003) worked with middle school and secondary school teachers as they used *Tinkerplots* to clump data into bins, thereby reducing data complexity. This helped the teachers to find ways to make comparisons between groups of unequal sizes. Research discussed earlier in this chapter documented the difficulty that students have when comparing groups of unequal size (e.g., Watson, 2002), and one would expect teachers to have some of those same difficulties. One data set that Hammerman and Rubin gave teachers consisted of T-cell counts for two groups infected with the HIV virus, a control group with $n=186$ and an experimental group with $n=46$. Gender information within the groups was also of unequal size. (The T-cell data set had previously been used by Cobb, Gravemeijer, Doorman, and Bowers, 1999.) *Tinkerplots* allowed the teachers to create bins by T-cell count, then to count the total number of people in each bin, and finally to create pie-graphs of the percentage of men and women in each bin. The pie-graph percentages can disguise the actual bin counts, and decisions made on percentages alone may be misleading because of small sample sizes within some bins. Conversely, actual bin counts can also be misleading if relative sizes are neglected. However, these teachers were simultaneously able to compare both frequencies and relative frequencies for T-cell count by gender, and to use the information to make some conjectures about the relative success of a treatment for HIV for each gender. Hammerman and Rubin (2003) argued that when faced with a lot of complex data, their teachers sought ways to reduce or manage variability, so they could make better judgments.

Several things are striking in the work of Hammerman and Rubin. First, the teachers found ways to use the software to create proportions within bin cells, and to reason proportionally when comparing data sets, rather than purely additively with frequencies. Even though proportional reasoning had its potential pitfalls in this case, the fact that the teachers used proportional comparisons of unequal-sized data

sets is promising. Second, the types of graphical representations that are now available to compare bivariate data (gender $\times$ treatment in this case) have the potential to revolutionize data analysis in classrooms in the future. At the present a typical AP statistics class approach to the T-cell problem might be to put the data into a 2×2 contingency table, sex $\times$ treatment, and examine cell values and margin proportions. Imagine in a few years, if middle school students have already explored bivariate data using powerful visual data-representation tools, the standard numerical statistical procedures for analyzing data might not be their first choice to approach bivariate problems. More of the decision-making power for how the analysis is to proceed could be put directly into the students' hands. The work of Rubin and her colleagues has important implications for the statistical education of teachers.

We believe that one powerful way for teachers to gain expertise in statistical reasoning is to have more experience in "being statisticians" themselves. In the area of exploratory data analysis, teacher education should consist largely of teachers investigating statistical problems that interest them, collecting data, analyzing it, and drawing conclusions in the same way that statisticians would. (Rubin and Rosebery, 1988, p. 17)

In fact, this vision for the statistical education of teachers should be the norm for the statistical education of all of our students.

Makar and Confrey (2002) have also conducted professional development projects in statistics with an imbedded research component. They cleverly merged their teachers' concerns about their students' performance on large-scale state testing in mathematics with an upgrade of the teachers' own understanding of statistics. Makar and Confrey introduced teachers to data analysis with *Fathom* using data from students' performance on the state mathematics test. Teachers were particularly interested in comparing different groups of students on the state test, so Makar and Confrey took advantage of that context. They identified four constructs they wished their teachers to consider as they decided whether differences between groups of students' test scores were really meaningful differences: measurable conjectures, tolerance for variability, context, and an ability to draw conclusions. For example, they pointed out that *tolerance for variability* requires a very different mindset than the deterministic foundations of traditional mathematics courses. Statistics requires decision making under uncertainty. Their teachers discovered a new appreciation for variability within the context of student scores on the state test. What began as a surface-level analysis of just the bare numbers transitioned to a rich discussion re-

lated to context when the teachers were told that the numbers in their data sets were actual state test scores for a set of different classes. In analyzing the teachers' work Makar and Confrey created a framework of five levels of reasoning used by the teachers when they compared data sets: pre-descriptive, descriptive, emerging distributional, transitional, and emerging statistical. Makar and Confrey's levels describe the growth in the teachers' understanding of the role and importance of variability when comparing data sets, and the difficulty of making conclusions about measurable conjectures.

In another study Makar and Confrey (2005) investigated preservice mathematics and science teachers' understanding of variation and distribution during a teacher education course. The teachers were introduced to the *Fathom* software and then given some assessment data to analyze. Makar and Confrey were interested to see if their participants would compare groups just using means, or if they would look at variation and spread in their comparisons. In interviews these preservice teachers made comparisons between groups of students using conventional statistical terms, such as means, measures of spread, discussions of shape, and the proportion of students who improved on the assessment. However, Makar and Confrey also encountered a good deal of non-statistical language among their preservice teachers, such as "clustered" or "spread out" or "bulk of" or "majority" or "clumps" or "big chunk." In their analysis of the responses Makar and Confrey concentrated on "clumps" and "chunks." A *clump* means exactly what it sounds like, including modal clumps of data. A *chunk* is a contiguous subset of a distribution of data, not necessarily a clump. This language is reminiscent of the "bumps" and "gaps" language used by Friel, Mokros, & Russell (1992) in their work with elementary children in the *Used Numbers Project*. It is an intuitive, natural language used to describe a collection of data. Makar and Confrey were capturing the primary intuitions of the student teachers (Fischbein, 1987), the raw material upon which secondary intuitions can be developed through teaching environments. Makar and Confrey warned that the nonstandard terminology that students use should not be ignored, as it might be a rich source of student understanding that could be overlooked.

Heaton and Mickelson have also explored K–6 teachers' understanding of some statistical concepts (Heaton & Mickelson, 2002; Mickelson & Heaton, 2003, 2004). Several of their research efforts have involved a single case study of one third-grade teacher as she integrated statistics investigations into other parts of her curriculum, such as science, literature, or social studies investigations. One report focused

on a teacher's understanding of data and distribution (Mickelson & Heaton, 2004), and another on manifestations of that same teacher's understanding of variability (Mickelson & Heaton, 2003). In some contexts, the teacher exhibited strong statistical reasoning skills, and in other contexts, only much more naive statistical skills. This teacher was acting as both a learner and a teacher in their research project, a dual role that Mickelson and Heaton urge more researchers and professional development leaders to consider, especially with elementary teachers who are trying new ideas out in statistics and data analysis in their classrooms. Although this third-grade teacher had had considerable professional development work in data analysis herself, at times she had difficulty finding an appropriate vehicle to transfer what she herself had learned to the students in her own classroom. Teachers need opportunities to create statistical activities and investigations themselves and to try them out while supported in their classroom by a statistics educator. The teaming of teachers with researchers in the classroom may help to facilitate teachers' abilities to transfer their own insights in statistics to experiences for their students.

Larger Scale Research on Teachers' Understanding of Statistics

Professional development cases, such as those described by Hammerman and Rubin and by Makar and Confrey in the previous section, tend to be with small groups of teachers who are special volunteers. In several studies that involved a larger number of teachers, Bright and Friel (1993) asked primary teachers to create concept maps for probability and statistics. They found that the maps were disconnected and rather sparse in statistical content. Greer and Ritson (1993) surveyed teachers at all levels in Northern Ireland and found a universal need for in-service teacher training in probability and statistics across all grade levels.

Watson (2001b) found a rather clever way to amass a considerable amount of information from teachers on their perceptions of statistics and on their own strengths and weaknesses in the data and chance curriculum in Australia. Using an information profile to assess the need for professional development, Watson asked teachers to rate the importance of certain statistical concepts, to describe their own lesson practices, to estimate their confidence in teaching certain statistical topics, and to suggest two possible student responses to some statistical tasks, one appropriate and the other inappropriate. This proved to be a less threatening methodology for obtaining information on teacher knowledge of statistics than testing them

directly. Watson used Shulman's (1987) knowledge typology to guide the categories of questions on the teacher profile, thereby obtaining information about teachers' content knowledge, knowledge of teaching, pedagogical content knowledge, and other knowledge categories identified by Shulman. Watson was able to administer the profile to over 40 teachers, some in individual interviews, some via written survey, and some via an Internet survey. Of note in Watson's findings were the teachers' lack of confidence in statistics and their lack of knowledge about sampling. Also of note was the need expressed by many of her teachers to further their own professional development in data and chance, as well as their frustration with the lack of support from local authorities to help them in teaching statistical topics.

Watson's profile was designed to provide information about teachers' content knowledge in statistics while assessing their professional development needs. A next step would be to administer such a profile anonymously, directly to teachers, to get a better understanding of their content knowledge. Inasmuch as this is a potentially risky and threatening venture for classroom teachers, it might be easier to investigate the statistical content knowledge of preservice teachers. Canada (2004) conducted a study of preservice elementary teachers' conceptions of variability in three contexts, data and graphs, sampling, and probability. Building upon the aspects of variability identified by Wild and Pfannkuch (1999), Canada began by documenting a variety of types of preservice elementary teachers' thinking about variability. From his analysis of students' responses to survey tasks, an emerging framework evolved consisting of three main aspects: expecting variation, displaying variation, and interpreting variation. Canada subsequently used his framework to compare and contrast the thinking of 6 case study students before and after he imbedded teaching episodes on statistics in a mathematics course for preservice elementary teachers. Canada found growth in all three aspects among the case-study students, including richer conceptions involving expectation of variation in data, more flexibility with displays of variation, and stronger interpretations of variation. Canada's evidence suggests that class interventions can help to strengthen preservice teachers' conceptions of variation.

In response to the tremendous need for professional development in data and chance that a number of researchers have identified, Friel and her colleagues developed some excellent professional development and curricular activities for primary teachers to help strengthen teachers' statistical content knowledge and pedagogical content knowledge (Friel & Bright, 1998;

Teach STAT, 1996a, 1996b;). In project *Teach-STAT* an intensive 3-week in-service course in data analysis for teachers in Grades 1–6 led to the creation of a professional development manual for statistics educators. The in-service materials for teachers were based on the *Used Numbers* curriculum materials that were written for elementary students (Russell & Corwin 1989; Corwin & Friel, 1990; Friel, Mokros, & Corwin 1992).

In Australia, several trials to deliver professional development in statistics for teachers from a wide geographical area were conducted by the Australian Association of Mathematics Teachers (AAMT) (Watson, 1998). The professional development was delivered by regional satellite television, by video-conferencing across the nation, by the preparation of a CD-ROM for teachers, and by the development of a Website with the help of a capital city newspaper. The material delivered was based on workshops and other material to promote data and chance as suggested by the national mathematics statement (Australian Educational Council, 1991). Evaluation by consultants and participants was quite positive (Watson & Moritz, 1997b). However, as with many such technology-based innovations, when the government funding was exhausted the ability to reach teachers was greatly diminished.

A number of other efforts have addressed the professional development needs of teachers in statistics, such as books by Hawkins (1990) and Hawkins et al. (1992) in the United Kingdom, the six books on data and chance of the *Navigations* (2004a, 2004b) series of the National Council of Teachers of Mathematics (see e.g., Burrill, Franklin, Godbold, & Young, 2004; Shaughnessy, Barrett, Billstein, Kranendonk, & Peck, 2004c), and the book *Statistical Questions from the Classroom* (Shaughnessy & Chance, 2005), just to name a few. As documented in a number of studies discussed above, professional development materials for teachers can lead to windows of opportunity for researchers to explore teachers' own understandings of statistical concepts.

SOME RECOMMENDATIONS FOR FUTURE RESEARCH

At the end of my chapter in the first edition of this *Handbook* I put forth some recommendations for future research in the teaching and learning of probability and statistics (Shaughnessy, 1992). One wonders if anyone ever really pays attention to such recommendations, but editors of research handbooks and readers who search through the chapters of such handbooks usually expect a look back over where the research has been, and some suggestions for where

the research could (should?) go next. As an author, it would be fun to have a conversation with readers at this point in the chapter to see what recommendations the readers themselves would make for future research after plowing through this lengthy tome. I do have some thoughts for future research, but first I will take a peek back at my recommendations in the first edition of this *Handbook*. How have they fared?

Research Recommended in the First Edition of the Handbook

As I mentioned in the introduction to this chapter, research in probability and statistics was just beginning to take hold in a number of countries during the decade prior to the arrival of the first *Handbook of Research*. At that time the field was rather naive, and the suggestions I made for future research at that time were probably equally naive. That wish list included: The need for assessment instruments; a request to investigate secondary students' and classroom teachers' conceptions of probability and statistics; a methodological recommendation for more teaching experiments in probability and statistics; and a request to investigate the effects of technology on student learning of statistical concepts. There were also recommendations for cross-cultural studies comparing students' statistical thinking, and a suggestion to investigate the role of metacognition in solving statistical problems, but to my knowledge no one took those two recommendations very seriously.

Progress has been made on some of the recommendations, although in some cases not as much as I would like to have seen. Researchers have investigated secondary students' conceptions of statistics, for example, Saldanha and Thompson (2003) and Shaughnessy et al., (2004a, 2004b, 2005). In addition, many of the studies by Watson and her colleagues reviewed in this chapter have been conducted with students from a range of grade levels, including Grades 9 and 11. However, much more of the research on students' understanding of statistics has been conducted with students from Grades 3 to 8. Perhaps this is because students in these grade levels are actually *doing* some statistics, whereas there is still no guarantee that secondary students, at least in the United States, will study any statistics. Statistics is growing in the United States secondary schools but it is still treated as an optional content area in many schools. It should not be optional.

The research on teachers' understanding of statistics has also begun to blossom with work like that of Hammerman and Rubin (2003), Confrey and Makar (2002), Watson (2001b), and Heaton and Mickelson

(2004). And, there are a few teaching experiments that have been conducted (Saldanha & Thompson, 2003; Saldanha, 2003; Cobb et al., 2003) as well as instances of the more modern version of a teaching experiment, the "design experiment" (Petrosino et. al., 2003), but teaching experiments are still not a frequent occurrence in statistics education. They yield a wealth of information, but they are hard to conduct, very intense, and very time-consuming if done properly. As for research on the influence of technology, even though some interesting "landscape-type" software packages are now available like *Tinkerplots* or *Fathom*, very little research has been conducted on how or what students learn about statistics with these powerful statistical tools. Student reasoning with technology has more often been researched using "route-type" tools (Bakker & Gravemeijer, 2004; Ben-Zvi, 2000).

Concerning the recommendation to develop assessment tools in statistics in the first edition, a good deal of progress has been made. The book *The Assessment Challenge in Statistics Education* (Gal & Garfield, 1997) contains a wealth of assessment information. In one chapter a framework for statistical assessment is presented by Friel et al. (1997). Among the authors who contributed to the development of assessment tools in statistics education are Schau and Mattern (1997) on the use of concept maps, Watson (1997) on using the media, Curcio and Artzt (1997) on small group work, Starkings (1997) and Holmes (1997) on statistical projects, Keeler (1997) on portfolios, Lajoie on using technology in assessment, Lesh, Amit, & Schorr (1997) on using real-life problems, and Jolliffe (1997) on instrument construction. More recently, Garfield, delMas, & Chance (2005) have led a project which has created a website of assessment items for improving statistical thinking.

Recommendations for Future Research: Take Two

This time around my recommendations for future research directions fall under three broad categories: (a) research on conceptual issues in statistics, (b) research on teaching issues in statistics, and (c) some methodological issues for research in statistics.

Conceptual Issues

The notion of distribution needs to be clarified and students' conceptions of the interrelationships of the aspects of a distribution deserve more research attention. Statisticians and statistics educators use the word distribution all the time. The word is used to refer to a single distribution of data, to a sampling distribution of statistics, to a probability distribution. We all

sort of know what we mean, some kind of collection of observations or numbers that is represented in a table or a graph, but it is not very well defined. Probability distributions can be defined in terms of the values that are taken on by random variables (which are themselves a type of function), so clearly the word distribution involves a “range of meanings.” It would be beneficial for statistics educators, present company included, to pin down what we mean when we use the term distribution. Researchers are beginning to talk about “distributional reasoning.” For some researchers, distributional reasoning in data analysis involves the explicit integration of multiple aspects of a distribution, such as centers, shape, and variability (see for example, Shaughnessy et al. 2004a, 2005). A recent meeting of the International Forum on Statistical Reasoning, Thinking, and Literacy held in Auckland, New Zealand (SRTL IV, August, 2005)³ was even entitled “Reasoning about Distributions.” A number of researchers are beginning to articulate their own thinking about distributions and are sharing their research on students’ understanding of distributions in a more prominent way.

More research is needed on tertiary students’ conceptions of statistics. At the time of the first edition of the *Handbook* much of the research had investigated tertiary students’ conceptions of probability and statistics, and very little research had been done with K–12 students. This time it is the other way around. There has been a major influx of studies of K–12 students’ statistical thinking in school mathematics settings, and not so much has been done with tertiary students. What are tertiary students’ conceptions of centers and average? What are their conceptions of variability? How do they reason when comparing data sets? Do tertiary students show any noticeable improvement in reasoning over their K–12 counterparts, or are they stuck at the same levels on the same issues as elementary and secondary students are? Statistics is now required in almost all collegiate majors. Are students learning anything beyond procedures in those courses? Are they learning to be data detectives, or just data crunchers?

The use of student-generated graphs is a very promising area for mining student thinking. Student-generated graphs, an instance of what been called student *inscriptions*, are very powerful tools to investigate student thinking. Student graphs not only provide information about students’ graphical skills, but also information on the level of their thinking about data—cases, clumps, aggregates, and so forth. Inscriptions can also inform us on how students do or do not think about

trends over time, spreads, centers, and shape. Bakker and Gravemeijer (2004), Moritz (2004), and Kelly and Watson (2002) have provided good examples of research that has uncovered a wealth of information from student-generated graphs.

More research is needed on students’ conceptual growth in statistics when they work in technology-rich environments. Too little research has been conducted on the effects of statistical software packages on students’ conceptual growth and thinking in statistics. Studies by Bakker and Gravemeijer (2004), Saldanha and Thompson (2003), Ben-Zvi (2000), and Chance et al. (2004) are among the welcomed exceptions to the dearth of research on the effects of technology on statistical thinking. Currently there is very little research on the effects of “landscape-type” data analysis tools such as *Tinkerplots* or *Fathom*.

More research is needed on teachers’ conceptions of statistics. Teachers have the same difficulties with statistical concepts as the students they teach. Research needs to find ways to help teachers develop in their statistical knowledge and thinking, especially now that statistics has a more prominent role in the K–12 curricula. Some promising inroads into teachers’ understandings of statistics have been made within professional development programs (Watson, 2001b; Heaton & Mickelson, 2002; Hammerman & Rubin, 2003; Makar & Confrey, 2004). Student work samples or the results of student test scores have been used by researchers to catalyze teachers’ reflections and reasoning about statistical issues. This technique is a promising indirect, non-threatening approach to finding out more about what teachers know and do not know about statistics.

Teaching Issues

What is the statistical knowledge necessary for teaching? Ball and Bass (2000) and Ball, Lubienski, and Mewborne (2001) have argued that a special type of knowledge is needed to teach mathematics, that this knowledge is different than just more mathematical content, and that it is more than just what Shulman (1987) called pedagogical content knowledge. For example, a special knowledge is needed for teaching algebra that involves knowing the types of symbolic mistakes and misconceptions that students will have, and how to address them. A special knowledge is needed for geometry that involves the role of examples and counterexamples, and reasoning with proof. Within each content area of mathematics there are special types of knowledge that are critical to effective teaching of that content. What is needed to success-

³ A forthcoming issue of the Statistics Education Research Journal (SERJ) will include edited papers about research on students’ thinking about distribution from the SRTL IV conference (Vol. 5, No. 2, November 2006). There is also a CD of the proceedings available (Makar, 2005).

fully teach statistics? What are the particularly tough concepts and areas for potential misunderstandings in statistics? Examples of statistical knowledge for teaching are addressed in Shaughnessy and Chance (2005) in their discussions of questions that arise from teachers and students in the classroom.

Research is needed on classroom discourse in statistics. Are students being asked to analyze data at a high level? Are statistical tasks posed and discussed in classrooms in ways that promote high level thinking, critical analysis, multiple representations, and thoughtful communication of results and solutions? This type of analysis of classroom processes has been undertaken in research on the teaching of mathematics, but so far no such research has been reported for the teaching of statistics. Analyzing, critiquing, communicating, and representing are critical skills that have been identified by Wild and Pfannkuch (1999) and by Watson (1997) in their models of statistical thinking and statistical literacy, respectively. Are we doing enough to identify and promote the types of teaching that will enhance our students' discourse skills in statistics? How well is statistics taught by mathematics teachers? Statistics is not the same as mathematics (Cobb & Moore, 1997). Are we balancing our classroom discourse between exploratory data analysis (data detective work) and the teaching of statistical concepts and procedures? Discourse analysis in statistics education is a wide-open area for future research.

There has been very little research into students' and teachers' beliefs and attitudes towards statistics. This is another area of research that is wide open for statistics education. There has been research into teacher affect and teachers' beliefs about mathematics (Thompson, 1992) as well as research into student attitudes and beliefs about mathematics (McCleod, 1992), but very little work has been done on students' or teachers' attitudes and beliefs specifically about statistics.

Methodological Issues and Recommendations

The use of multiple research methodologies in research on students' conceptions of statistics has powerful payoffs. There has been a growing trend for researchers to use both quantitative and qualitative methodologies in their research on students' reasoning on statistical tasks. In many studies researchers have gathered and quantified results of surveys on statistical tasks administered to large numbers of students but have also conducted clinical interviews with smaller numbers of students, either in conjunction with survey administration or as follow-up work to the surveys. Hypotheses generated about why students are an-

swering survey questions in particular ways can be validated in detailed clinical interviews. Also, interviews often reveal lines of thought that were in the survey data all the time but that were initially missed by the researchers. I applaud the use of multiple research methodologies and urge my fellow researchers to continue this practice.

The statistics education research community would benefit from a thoughtful discussion and debate about the strengths and limitations of using the SOLO model. The SOLO model has been used by many researchers, particularly in Australia where it was born, to identify levels of student reasoning on a number of constructs. Many research studies reviewed in this chapter employed the SOLO model in an analysis of student responses to statistical tasks. The SOLO model is based on the assumption that development can be represented in hierarchical structures. Is that assumption warranted? One of the criticisms leveled against the SOLO model is that it is not falsifiable, so the validity of any conclusions reached via a SOLO approach cannot be easily challenged. On the other hand, the SOLO model has been genuinely useful in helping to describe student reasoning on a number of concepts in statistics like average, variation, comparison of data sets, and so on. I recommend that the statistics education community engage in healthy debate on the merits and demerits of the SOLO model, and that the debate be published in a public forum.

The statistics education research community would benefit from a thoughtful discussion and debate about the strengths and limitations of using the Rasch model. Recently several statistics educators have begun to use the Rasch model to quantify students' statistical literacy, including tasks involving such concepts as average and variability (see for example, Callingham & Watson, 2005; Watson & Callingham, 2003; Watson et al., 2003). In several instances Rasch measurement has been used in conjunction with the SOLO model in order to scale students' responses to statistical tasks along a SOLO type hierarchy. The results of the Rasch analysis in these studies appear to be very robust, and Rasch provides a way to quantify and measure student constructed responses on surveys and interviews. It is an underpinning assumption of Rasch measurement that the target construct is unidimensional. One wonders if this is really the case with such constructs as average and variability, for which an eclectic variety of types of conceptions have been identified in the literature. Are linear models of complex concepts sufficient for valid measurements? This is a question for debate and discussion within the statistics education community.

Some Implications From Research for the Teaching of Statistics

There are so many recommendations that could be made from research for the teaching and learning of statistics that this section could be a separate chapter just in itself. I will contain myself to just a few.

- *Emphasize variability as one of the primary issues in statistical thinking and statistical analysis.* In the past there has been a tendency to overemphasize centers as the principal concept in statistics, and the important role of variability has been neglected. Students need to integrate the concepts of centers and variability when they investigate data so that they can reason about properties of aggregates of data.
- *Introduce comparison of data sets much earlier on with students, prior to formal statistics.* Bakker and Gravemeijer (2004), Konold and Higgins (2003), and Watson and Moritz (1999) have all found that students can develop their own powerful, intuitive ways to compare data sets prior to the introduction of formal concepts like mean, median, variation, or standard deviation. Have students compare data sets right from the start of their statistical education.
- *Build on students' intuitive notions of center and variability.* Research has uncovered a spectrum of student conceptions about these two important concepts. Students will be in transition from their own colloquial understandings of centers and variability—such as average as “typical” or variability as “things that change over time”—to more statistical understandings of these concepts, such as means or likely ranges. We should start with what students bring to the table on these concepts and build from there.
- *Make the role of proportional reasoning in the connections between populations and samples more explicit.* Although it might seem obvious that carefully chosen samples should be representative of whole populations, a long history of research evidence suggests that people ignore base rates when making inferences from samples or predictions to populations (see for example, Kahneman & Tversky 1972, 1973; Tversky & Kahneman, 1974). Students should have repeated opportunities to actually choose samples themselves, so they have chances to see the proportional relationship firsthand.

- *Remember that there is a difference between statistics and mathematics!* Wild and Pfannkuch's work (1999) and the writings of David Moore (1990, 1997) are signals to everyone who teaches mathematics that there are ways of thinking and analytical tools that are specific to statistics. In particular, statistics is fraught with contextual issues, which is the nature of the discipline, whereas often mathematics strips off the context in order to abstract and generalize.

Concluding Remarks

I had to make choices in developing this chapter so I tried to take to heart the old adage for authors, “Write what you know.” Or, in this case perhaps it is, “Write what you think you know.” There are many opportunities for thoughtful and knowledgeable readers to fill in the many gaps that I left behind. Hopefully the chapter will provide a sufficient network of references so that an interested reader can pursue a particular topic by going directly to the original works and tracing back further among other references. I want to remind the reader that in this chapter you have encountered statistical research through an indirect mode, through the lens of my own personal filter. There is some efficiency in this approach, and that is why people are asked to write reviews and syntheses of the literature. However, this is never any substitute for reading the original sources. I urge all interested readers to do just that, especially in those instances that peak your curiosity. Do not just take my word for it, go and find out for yourself!

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MATHEMATICS THINKING AND LEARNING AT POST-SECONDARY LEVEL

Michèle Artigue

UNIVERSITÉ PARIS 7

Carmen Batanero

UNIVERSIDAD DE GRANADA

Phillip Kent

UNIVERSITY OF LONDON

INTRODUCTION

This chapter deals with research carried out on mathematics thinking and learning at post-secondary level. It tries to point out the evolution of research in this area since the first *Handbook* was published in 1992, its most important advances, their potential and limit for understanding and improving teaching and learning processes at this advanced level.

Synthesizing research advances in a particular area of mathematics education has always been difficult, due to the diversity of educational structures and cultures, and to the diversity of research paradigms. Regarding post-secondary education, the exercise of looking back 10 years could nevertheless seem easier than in other domains of mathematics education. Thanks to the existence of the International Group for the Psychology of Mathematics Education (PME), and its annual conferences, researchers from differ-

ent origins reflecting on these issues established a pattern of regular exchanges and collaborative work. In 1991 this work led to the book edited by Tall: *Advanced Mathematical Thinking*, a good representation of the state of the art¹ at that time. Both the structure and the content of this book show that the predominant concerns then were cognitive ones: identifying cognitive processes underlying the learning of mathematics at advanced levels, investigating the relationships of these processes with respect to those at play at more elementary levels, and understanding students' difficulties with advanced mathematical concepts. Different theoretical constructs supported this research, such as the notions of *concept definition* and *concept image* (Tall & Vinner, 1981), the *process-object duality* (Dubinsky, 1991; Sfard, 1991), or the notion of *epistemological obstacle* due to Bachelard and developed in the didactic field by Brousseau (Brousseau, 1983). However, the book shows an evident common interest in visions of knowledge growth focusing on

¹ There were some significant exceptions: For instance, research carried out in this group did not consider the stochastic domain.

misunderstandings, cognitive conflicts, discontinuities, and hierarchies. Moreover, the research tended to concentrate on just a few mathematical domains: calculus and associated concepts, mathematical rationality, and proof. This vision, unsurprisingly, is reflected in the chapter that refers most directly to post-secondary level mathematics in the 1992 *Handbook*, “The Transition to Advanced Mathematical Thinking: Functions, Limits, Infinity, and Proof” (Tall, 1992).

The situation today is far from being the same, for a lot of reasons, linked both to the internal evolution of mathematics education as a scientific field and to external changes affecting post-secondary education. Research at post-secondary level was firstly influenced by the evolution of the dominant research paradigms from constructivist cognition and cognitive development towards sociocultural and anthropological ones (Lerman & Sierpiska, 1996). Sociocultural and anthropological paradigms were considered in the 1992 *Handbook*—see for instance chapter 20 by Schoenfeld who writes, “This cultural perspective is well grounded anthropologically but it is relatively new to the mathematics education literature” (p. 340). From that time, sociocultural and anthropological approaches have taken an increasing importance, and references to Vygotsky have tended to supplant those to Piaget. Research on mathematical learning at post-secondary level could neither escape this general influence nor ignore the central role these approaches give to the analysis of social and institutional practices in the understanding of knowledge growth. Research was also influenced by the emphasis these approaches put on semiotic mediations, and thus on the semiotic tools of mathematical activity. One can easily understand that such an evolution has a particular resonance in research on advanced topics: in advanced mathematics, students’ relationship with symbolism becomes an essential feature of their relationship with mathematics. More recently, research has also been influenced by the increasing interest in the neuroscientific study of cognition and embodied cognition.

At the same time as these internal changes, the field was affected by a number of external changes, which have been analysed in the ICMI Study on the Teaching and Learning of Mathematics at University Level (Holton, 2001). For example, the following quotation comes from the ICMI discussion document that launched this Study and explained its rationale²:

A number of changes have taken place in recent years which have profoundly affected the teaching of

mathematics at university level. Five changes which are still having considerable influences are (i) the increase in the number of students who are now attending tertiary institutions; (ii) major pedagogical and curriculum changes that have taken place at pre-university level; (iii) the increasing differences between secondary and tertiary mathematics education regarding the purposes, goals, teaching approaches and methods; (iv) the rapid development of technology; and (v) demands on universities to be publicly accountable. Of course, all of these changes are general and have had their influence on other disciplines. However, because of its pivotal position in education generally, and its compulsory nature for many students, it could be argued that these changes have had a greater influence on mathematics than perhaps on any other discipline. . . . As a result of the changing world scene, ICMI feels that there is a need to examine both the current and future states of the teaching and learning of mathematics at university level. The primary aim of this ICMI Study is therefore to pave the way for improvements in the teaching and learning of mathematics at university level for all students. (ICMI, 1997).

All these evolutions have contributed to make the field of research on mathematics thinking and learning in post-secondary education much more diverse today than was the case 10 years ago. Consequently, trying to present a systematic survey of the results obtained in the last 10 years and of the different existing trends could result in a pointillist painting whose driving forces would remain invisible to the reader. Trying to avoid this trap while taking into account the state of the field obliged us to make some choices, and we acknowledge that the vision we present is a personal vision. Our main choice has been to reflect and question some major evolutions we perceive in the field rather than to give a comprehensive view of it. We also decided to approach these evolutions in two different ways: on the one hand by showing how classical research topics such as calculus or linear algebra have been partially renewed in the last 10 years and on the other hand by addressing two emerging research themes: mathematics in engineering courses, and stochastics.

Because this *Handbook* has no chapter devoted to technology, we would additionally like to present our own reflections on technology in this chapter. Consistent with the spirit of the *Handbook*, our choice is not to have a specific section dealing with technology, but rather to integrate the discussion of technology throughout the chapter.

² This discussion document was disseminated through different channels and can be found in the *ICMI Bulletin* number 43 (December 1997) which is accessible on the ICMI website (www.mathunion.org/Organization/ICMI).

The chapter is structured into four main parts. In the first part, we briefly recall the state of the art in the early nineties. In the second part, we consider evolutions that, in our opinion, exemplify how the internal evolution of the theoretical frames in mathematics education has influenced and is influencing research at post-secondary level. For this purpose, we mainly consider research in traditional domains such as calculus and linear algebra. In the third part, we consider evolutions more linked to external factors such as the evolution of the context of post-secondary mathematics education in its institutional, social, cultural, and technological dimensions and the evolution in the relative importance of the different mathematical domains. For this part, which did not have its exact counterpart in the previous *Handbook*, we focus on research carried out about the teaching and learning of mathematics in engineering courses, and research on probability and statistical learning. We hope that this choice helps us question theories and positions as regard mathematical learning and thinking that have generally been established having in mind more or less explicitly the mathematical education of “pure mathematicians” or mathematics teachers and their particular needs, and also that have focused on few and classical mathematical domains. This choice also obliges us to consider the role of technology in learning in a rather different way. Indeed, what is often at stake in these domains is not simply the use of technology for developing usual mathematical knowledge but the way in which technology is changing mathematical activity and understanding, including problem solving, proving, reasoning, modeling, and symbolizing. Mathematical and technological expertise and needs are there much more tightly intertwined.

Finally, in the last part of the chapter, we come back to more general issues, pointing out both the potential and limits of existing research for understanding and for helping improve the current situation of thinking and learning at post-secondary level, pointing out also some evident research needs that are apparent from the literature reviewed in the chapter.

THE EARLY NINETIES

As was mentioned in the introduction, the state of research on mathematical thinking and learning at post-secondary level in the early nineties is rather well represented in the book *Advanced Mathematical Thinking* published in 1991. A main theme of inter-

est for researchers, at that time, was to characterize advanced mathematical thinking (AMT in the following) with respect to more elementary forms of mathematical thinking. It was also to clarify the mental processes that allow students to enter into AMT, and the difficulties students meet in developing such mental processes.

The Nature of AMT and AMT Processes

As pointed out by Dreyfus (1991) in his contribution to the book, characterizing AMT is not something as easy as “there is no sharp distinction between many of the processes of elementary and advanced mathematical thinking” (p. 26). As with more elementary levels, according to Dreyfus, the processes of AMT can be described in terms of representing, visualizing, generalizing, classifying, conjecturing, inducing, analyzing, synthesizing, abstracting, or formalizing. Referring to famous texts by Hadamard (1945) and Poincaré (1913), Dreyfus also pointed out the diversity of mathematical thinking modes, but differentiated essentially two main mathematical styles, according to the relative importance given to visualization and intuition, or to symbolic and analytic approaches.

Beyond these general considerations about AMT, the crucial point, as stressed by Tall in the final chapter of the book, was certainly the acknowledgement of the “thorny nature of the full path of mathematical thinking, so much more demanding and rewarding than the undoubted aesthetic beauty of the final edifice of formal definition, theorem and proof” (p. 251). For the authors, the gap between the logic of the mathematical edifice and the logic of cognitive processes explained the observed inefficiency of university teaching strategies based on the former, for the majority of students. This was evidenced at the time by the high rates of failure in fundamental courses, such as calculus, and also by the limited ability demonstrated later by those students who had passed the fundamental courses. The study of Selden, Mason, and Selden (1989) is from this point of view especially illustrative. These researchers presented students with problems³ that could easily be solved with the techniques at their disposal but were not presented in the usual way. Not one student solved an entire problem correctly, and most of them could not do anything.

The gap mentioned above has been the motivation for several theoretical constructs, and we briefly present the most important ones in the following section.

³ One of these problems was the following: Find at least one solution to the equation $4x^3 - x^4 = 0$ or explain why no such solution exists.

Concept Definition and Concept Image

These notions were introduced by Tall and Vinner (1981) and defined in the following way:

We shall use the term of concept image to describe the total cognitive structure that is associated with the concept, which includes all the mental pictures and associated properties and processes. It is built over the years through experiences of all kinds, changing as the individual meets new stimuli and matures. . . . As the concept image develops it need not be coherent at all times. The brain does not work that way. Sensory input excites certain neuronal pathways and inhibits others. In this way different stimuli can activate different parts of the concept image, developing them in a way which need not make a coherent whole. (p. 152)

One example of this lack of global coherence, often quoted in the literature, is the following: Students are asked first to compare $0.999\ldots$ and 1, then to calculate the sum of the series: $\sum 9/10^n$. Many students answer that $0.999\ldots < 1$ to the first question whereas they correctly answer that the sum is 1 to the second. Several reasons have been given for explaining the first answer. They often rely on the process/object duality we will evoke later on: Students are bound to a process view of the symbolic notation $0.999\ldots$, and this view prevents them from seeing beyond the infinite process whose terms are all less than 1, the object “number 1” that results from it.⁴ When asked the second question, they recognize a geometric series, and activating the formula for its sum they get the correct answer. Notice that this answer generally does not lead students to reconsider their answer to the first question, which is seen as a different one.

The concept image is generally at variance with the concept definition, and this was well evidenced at that time by research carried out on functions. Convergent results showed that many students, even when able to give a correct set-based definition of the notion of function, when asked to say if such or such object given by a discursive, tabular, symbolic, or graphic representation was or was not a function, gave answers at variance with their definition, and answers that could be different for the same object according to the semi-otic representation used.⁵

Epistemological Obstacles

As mentioned in the introduction, other approaches had been developed in the eighties in order to approach the complexity of learning processes at post-secondary level and the distance between these and the current logical organization of mathematical knowledge. One of these approaches, also well represented in the AMT book, relied on the notion of epistemological obstacle, initially due to the philosopher Bachelard, and imported by Brousseau into the educational field. According to Bachelard (1938), scientific knowledge supposes the rejection of common knowledge. In the educational field, this notion has been introduced in order to better understand the status of students' errors and to acknowledge that some of these, generally the most resistant ones, result not from a lack of knowledge but from knowledge that has stabilized because of its efficiency. Let us stress that this knowledge can be social or cultural but that it can also result from school apprenticeship. Some researchers working at post-secondary level soon appropriated this notion in order to understand students' difficulties with advanced mathematical concepts such as the concepts of limit, derivative, and integral. Regarding the limit concept for instance, Cornu and Sierpiska first (Cornu, 1983, 1991; Sierpiska, 1985, 1987) and then Schneider (1991) evidenced the existence of epistemological obstacles related to

The everyday meaning of the word limit, which induces resistant conceptions of the limit as a barrier or as the last term of a process, or tend to restrict convergence to monotonic convergence;

The overgeneralization of properties of finite processes to infinite processes, following the continuity principle stated by Leibniz;

The strength of a geometry of forms which prevents students from clearly identifying the objects involved in the limit process and their underlying topology. This makes it difficult for students to appreciate the subtle interaction between the numerical and geometrical settings in the limit process. (Artigue, 2001, p. 211)

Overgeneralization leads for instance to a belief that the limit of a sequence of strictly positive numbers is strictly positive, or that the limit of a sequence of continuous (resp. differentiable, integrable) functions is contin-

⁴ Edwards (1997) proposed for instance an alternative interpretation based on the difference regularly observed in the treatment of the two equalities: $0.333\ldots = 1/3$ and $0.999\ldots = 1$. This would show that, when manipulating infinite decimal expansions, students do not refer to the definition they have been given but to their practice of division. This practice supports the first equality but cannot make sense of the second one.

⁵ For instance, recognizing a constant function as a function if given by a graphical representation but refusing it this status if given by an algebraic expression without an explicit variable.

uous (resp. differentiable, integrable). The geometrical obstacle leads to a belief that whenever a sequence (F_n) of geometrical objects has, as a limit, an object A, all the magnitudes attached to the F_n have as a limit the corresponding magnitudes for A. As shown by Schneider, this geometrical obstacle often combines with an obstacle she labels the *heterogeneity of dimensions obstacle*, which underlies for instance the following reasoning concerning the computation of the area under a curve: When the size of the subdivision tends towards 0, each of the rectangles tends towards a segment. Thus at the limit, the total area under the curve which is the sum of the areas of the segments is necessarily 0, which is impossible!⁶

Process/Object Duality

Another approach, which was to take increasing importance in the research community, was also already present: that based on the process-object duality. In the early nineties, this approach was associated with some particular names, especially those of Sfard and Dubinsky. In the AMT book, it was especially visible in the chapter written by Dubinsky where he introduced *APOS theory*. According to APOS, conceptualization begins with manipulating previously constructed mental or physical objects to form *actions*; actions are then interiorized to form *processes* that are then encapsulated to form *objects*. Finally, actions, processes and objects are organized more or less coherently in *schemas*, and “a subject’s tendency to invoke a schema in order to understand, deal with, organize, or make sense out of a perceived problem situation is her or his knowledge of an individual concept in mathematics” (Dubinsky, 1991, p. 102).

Dubinsky’s ambition was to isolate small portions in the complex structures of a subject’s schemas and to give an explicit description of these, and especially to develop genetic decompositions of particular concepts such as induction and function, two examples he presented in the same chapter. What was also stressed by Dubinsky is the fact that a learner cannot successfully engage in AMT without developing an object view of the mathematical ideas at stake.

The 1992 Handbook

As mentioned earlier, Tall’s chapter of the *Handbook* closely reflects the AMT book. Tall introduced the different constructs mentioned above together with illustrative examples, and we just refer here to the con-

clusion of the chapter where he gave his own characterization of AMT and proposed an agenda for research. As regards AMT, the crucial point, according to him, is the change in the relationship one develops with mathematical concepts: “To move to more advanced mathematical thinking involves a difficult transition, from a position where concepts have an intuitive basis founded on experience, to one where they are specified by formal definitions and their properties reconstructed through logical deductions” (p. 495), but he also stressed that “In taking students to the transition to advanced mathematical thinking, we should realize that the formalizing and systematizing is the final stage of mathematical thinking, not the total activity.”

According to him, research has thus to investigate and understand the difficult cognitive changes and restructurings that this transition involves. Research evidences the existence of conflicts between students’ intuitive views and formal mathematics; these have to be clarified through clinical interviews, and at the same time formal mathematics has to be itself placed into perspective as a human activity that attempts to organize the complexities of human thought into a logical system. This is the research agenda he proposed.

Beyond the work done under the AMT umbrella, post-secondary research is also present in the first *Handbook* through Schoenfeld’s chapter “Learning to Think Mathematically: Problem Solving, Metacognition and Sense Making in Mathematics”. Synthesizing different approaches to these issues coming from mathematics education and related fields, and the personal work he has carried out in universities on problem solving, reporting on differences between students’ and experts’ problem-solving behaviors, Schoenfeld introduced a framework for the analysis of mathematical cognition. This framework is organized around five dimensions: the knowledge base, problem-solving strategies, monitoring and control, beliefs and affects, practices. We would like to stress here the attention paid by Schoenfeld to the role played by the two last categories: beliefs and practices, and the links he establishes between these, as this attention shows the emergence of the more global and cultural views on cognition whose influence was to increase in the next decade.

Taking into account the specific theme of this chapter, we have focused in this brief summary on research on learning processes and mathematical thinking. Nevertheless we would like to mention that in the early nineties research on post-secondary mathemat-

⁶ Note that recently Gonzalez-Martin and Camacho (2004) evidenced the existence in university students of a variant of this obstacle, when working on students’ conceptions of generalized integrals. It is expressed in the conviction held by students that, for a positive function the integrals $\int_a^{+\infty} f(x)dx$ and $\int_a^{+\infty} f^2(x)dx$ have necessarily the same nature because to a finite area (resp. infinite area) there necessarily corresponds by rotation around an axis (here Ox) a finite volume (resp. infinite volume).

ics education was not restricted to this aspect. Teaching experiments were also developed consistent with these perspectives on learning. For instance, Tall used the notion of generic organizer⁷ to build the software Graphic Calculus for introducing students to calculus concepts; Dubinsky and his colleagues developed the language ISETL and began to use it for teaching functions and algebraic structures; French researchers such as Artigue, Legrand, and Rogalski used the theory of didactic situations and the notion of scientific debate to develop (jointly with physicists) “didactic engineering” for the teaching of differentiation, integration, and differential equations (Artigue, 1991). Many of these developments incorporated technology as a generic organizer, as support for visualization and coordination between semiotic registers, or involved a programming language to support the interiorization and encapsulation of processes.

Thus, if we consider the field of research on mathematics learning at post-secondary level at the beginning of the nineties, there is no doubt that different theoretical constructs had been elaborated and worked out. These structured the didactic reflections of researchers about students’ learning processes and difficulties, allowing researchers to better understand the failure of ordinary educational practices for the majority of students, to design alternatives to these and to test them. Nevertheless, the results obtained were limited to small areas of post-secondary education, both in terms of categories of students and mathematical subjects; the theoretical constructs were not generally integrated into more global didactic structures connecting both learning and teaching phenomena; they remained essentially cognitive constructs and, within this cognitive perspective, relied on the dominant constructivist epistemology. Both the evolution of the field and the evolution of the educational context have produced changes to this situation, and these we present and discuss in the following parts.

EVOLUTIONS DEALING WITH ALREADY-DEVELOPED RESEARCH AREAS AND PERSPECTIVES

In this part we present and discuss evolutions, focusing on those that can be easily connected to research areas and perspectives already mentioned above. First we consider the evolution of ideas about AMT itself.

Second we consider the reinforcement and extension of existing approaches. We discuss the evolution of the process-object oriented approaches through the development of APOS theory, and the development of the proceptual approach. We also consider approaches that, starting from epistemological and/or historical analysis of mathematical knowledge, propose alternative categorizations for approaching the analyses of conceptualization. Third, we discuss the impact of some newer cognitive approaches, focusing mainly on the increasing influence of embodied cognition and of the linguistic approach developed by Lakoff and Nuñez. Fourth, we come to another dimension of the evolution that we see as a consequence of the increasing influence of anthropological and sociocultural approaches in the educational field. Cognition is there seen as something emerging from institutional practices, and understanding learning processes cannot be achieved without analyzing these institutional practices and identifying the norms and values underlying them. We show how these approaches complement the preceding ones for understanding the interaction between the individual and the collective in learning processes and supporting didactic design, and also for understanding issues that have become more and more crucial, such as the secondary/tertiary transition, or the relationships that new generations of students develop regarding mathematics and mathematical activity, and how these affect their learning of mathematics.

Evolution of Ideas About the Nature of Advanced Mathematical Thinking

The publication of the AMT book did not close the discussion about the nature of advanced mathematical thinking and its development. Critical reviews argued about what was referred to by the term *advanced* in the AMT book: mathematics? thinking? both? This was unclear, and the criteria used in the distinction between elementary and advanced mathematical thinking were not really convincing, according to the critics. From that time, the definition of AMT has been a question regularly addressed but unresolved. In the following, we have chosen to illustrate some of the associated discussions by referring to the work carried out since 1998 by a working group of PME-NA titled “The Role of Advanced Mathematical Thinking in Mathematics Education Reform” and the special issue of the journal *Mathematical Thinking and Learning* that

⁷ Tall (1991) defined a *generic organizer* as “an environment that provides the user the facilities of manipulating examples (and, where possible, non-examples) of a concept. The word ‘generic’ means that the learner’s attention is directed to certain aspects of the examples which embody the more abstract concept” (p. 187).

recently emerged from this work (Selden & Selden, 2005). As the editors explained in the introduction to this special issue, the working group “began by discussing such questions as what kinds of earlier experiences might help students make the transition to the kinds of AMT that post-secondary students are often asked to engage” and its interest “rather naturally metamorphosed into efforts at characterizing AMT and looking for seeds thereof that are, or could be, planted early in students’ mathematical careers” (p. 3). The special issue evidenced the diversity of the answers offered within the group. For instance, starting from the fact that manipulating advanced concepts such as the concept of limit does not necessarily require advanced modes of thinking, and that students have difficulty finding referents for these abstract mathematical concepts in their familiar world, Edwards, Dubinsky, and McDonald defined AMT in the following way: AMT is “thinking that requires deductive and rigorous reasoning about mathematical notions that are not entirely accessible to us through our five senses” (2005, pp. 17–18). Harel and Sowder, for their part, defined AMT by referring to the notion of epistemological obstacle. Considering the three conditions imposed by Duroux and Brousseau (Brousseau, 1997) for epistemological obstacles: (a) to have traces in the history of mathematics; (b) to be a piece of knowledge producing valid answers in particular contexts, and invalid responses outside this context; (c) to “withstand both occasional contradictions and the establishment of a better piece of knowledge” (p. 34). Harel and Sowder claimed that “mathematical thinking is advanced, if its development involves at least one of the above three conditions” (2005, pp. 17–18).

These five authors proposed various examples in order to show the pertinence of the perspectives they developed but these examples, in spite of their interest, are insufficient to make the proposed definitions fully convincing. For instance, the first definition relies on the claim that some mathematical objects are directly accessible to our senses, an assertion that can be seriously discussed from an epistemological point of view. This definition also opposes rigorous and deductive reasoning to automatic reasoning or routine, which does not pay attention to the fact that most reasoning processes subtly intertwine routine and rigorous reflection. The separation between what is routine and what needs reflection changes for each learner according to the development of the learning process

and even the academic institution where the learning takes place. Thus the operability of this definition is questionable. As regard the second definition, the separation it artificially introduces between the three complementary conditions imposed by Duroux on the notion of epistemological obstacle, and the fact that these three conditions are given an equivalent role for qualifying AMT, seems difficult to sustain.

More than the attempts at defining AMT, which we do not consider as really successful, what we take from this special issue is the hypothesis made by most of the contributors, and especially Rasmussen, Zandieh, King, and Teppo, that AMT is not specific to a particular level of schooling. As these authors did, we would prefer to speak of *advanced mathematical practices*. Underestimating the relative nature of these, their dependence on institutional context, and linking them too much to formal mathematics can turn an idea that is a priori pertinent for analyzing learning processes into an obstacle. Our ambition in this chapter is to investigate how research carried out up to now at post-secondary level helps us to understand the ways students learn or do not learn, and could better learn mathematics, in the diversity of existing post-secondary institutions, without establishing a priori hierarchies of values among the different mathematics cultures these institutions offer. We need thus to be sensitive to the implicit values carried by the research constructs we use, and our review of AMT research leads us to be very critical in this regard about the construct of advanced mathematical thinking.

The Reinforcement and Development of Existing Approaches

Process–Object Duality

Expanding on the idea of process/object duality introduced above, we would like to focus on two trends: (a) the development of the APOS theory whose influence in research at post-secondary level has strongly increased in the last 10 years, thanks to the collaborative work of several groups of researchers and the existence of institutions such as RUMEC,⁸ and (b) the development of the proceptual approach.⁹

The fundamental base of APOS remains the same as what was extensively presented by Dubinsky (1991). But, since that time, this approach has been used for building genetic decompositions of many different concepts taught at university level, for extending

⁸ Research in Undergraduate Mathematics Education Community.

⁹ We have chosen to focus on these two examples because they nicely illustrate what can be the long-term development of approaches based on the process-object duality, as far as these are widely used. A similar analysis could have been carried out starting from the theory of reification initiated by Sfard.

research based on APOS towards secondary mathematics education, and for elaborating and testing teaching designs based on the theory. APOS theory is a cognitive theory, and this obliges researchers to rely for teaching design on educational approaches complementing what APOS can offer. Thus the idea of cooperative learning has been linked to APOS, leading to a number of different projects (Dubinsky & Schwingendorf, 1997).

The fundamental basis of APOS has not changed, but one evolution is worth mentioning. It resulted from difficulties met by the researchers in satisfactorily explaining all the data they had collected. What was mainly at stake was the schema part of APOS. According to APOS (Dubinsky & McDonald, 2001, p. 277),

a schema for a certain mathematical concept is an individual's collection of actions, processes, objects and other schemas which are linked by some general principles to form a framework in the individual's mind that may be brought to bear upon a problem situation involving that concept. This framework must be coherent in the sense that it gives, explicitly or implicitly, means of determining which phenomena are in the scope of the schema and which are not.

Data obtained in research concerning the chain rule, and then the properties linking the graph of a function and its derivatives, led to a reconsideration of this idea of schema, and to the incorporation of the triad introduced by Piaget and Garcia (1989) in order to better explain the construction of schemas. Incorporating the triad led to the introduction of three different stages in the construction: the Intra, Inter, and Trans stages. As explained in Dubinsky and McDonald (2001, p. 280),

The Intra stage of schema development is characterized by a focus on individual actions, processes, and objects in isolation of other cognitive items of a similar nature. . . . The Inter stage is characterized by the construction of relationships and transformations among these cognitive entities. . . . Finally, at the Trans stage the individual constructs an implicit or explicit underlying structure through which the relationships developed in the Inter stage are understood and which gives the schema a coherence by which the individual can decide what is in the scope of the schema and what is not.

Taking as an example the function concept, one could say that at the Intra level, an individual tends to focus on functions seen as isolated objects and on the activities they perform on these; at the Inter level, he or she begins to make connections between functional objects, and to see how new functional objects can

be created through these connections, to give sense to the idea of transformation of functional objects; at the Trans level, he or she can consider systems of transformations and the mathematical structures that emerge from these.

The second trend in process-object duality we evoke here is that developed by Tall. According to APOS, mathematical learning is achieved through the construction of mental actions, processes and objects, and their organization into schemas. Over the years, the distance between Tall's vision and APOS has progressively increased, and the model he proposes today is something quite different. According to him, cognitive growth of mathematical knowledge presents three main different paths corresponding to three different mathematical worlds (Tall, 2004):

The first grows out of our perceptions of the world and consists of our thinking about things that we perceive and sense, not only in the physical world, but in our own mental world of meaning. By reflection and by the use of increasingly sophisticated language, we can focus on aspects of our sensory experience that enable us to envisage conceptions that no longer exist in the world outside, such as a "line" that is "perfectly straight." (p. 285)

This first world, which Tall names the *embodied world*, applies to the developmental path of Euclidean geometry from our perception of spaces and forms. The second path leads from enactive experiences with quantity and change to numbers, algebra, and calculus. Actions are there "encapsulated as concepts by using symbols that allow us to switch effortlessly from processes to do mathematics to concepts to think about" (p. 285), and Tall names the associated mathematical world the *procept world*. The third world is the *formal world*. Objects are there expressed in terms of formal definitions, and their properties deduced by formal proofs. They belong to mathematical structures defined through axiomatic systems. Geometry becomes axiomatic geometry; calculus becomes formal Analysis. Tall (2001) identified two different ways for the cognitive development of this formal world: natural thinking, which builds from concept imagery towards formalism on the one hand, and formal thinking, which builds from the concept definition, marginalizing imagery and focusing on logical deduction. According to him, individuals generally, according to the context, use one or the other way: "Natural thinking is appropriate for thought experiments that suggest possible theorems; formal thinking is appropriate for establishing formal proofs" (p. 235). But he also stressed that "formal thinking may also lead to 'structure theorems,' whose properties may be used

to develop more subtle visual imagery, now enhanced by the network of formal relationships” (p. 235). Such distinctions are illustrated (Tall, 2001) using examples based on real numbers and infinitesimals. One can understand for instance the transitive property of the real order by relying on the image of the number line; and one can understand it as a consequence of the existence of a subset of positive elements P in a field satisfying the following axioms that make it an ordered field for the relation $x < y$ if $y - x$ belongs to P :

- 1) if x and y belong to P , $x + y$ and xy belong also to P ,
- 2) Any x in the field satisfies one and only one of these properties: x belongs to P , $-x$ belongs to P , $x = 0$.

These two modes of understanding the same property formally expressed are obviously different.

In Tall’s approach, relationships between processes and objects are seen in a much more dialectical way than in APOS, and symbolism contributes in an essential way to this dialectic through the notion of procept. This notion, first introduced by Gray and Tall (1994), expresses the fact that the same symbol can evoke both a process and the concept (i.e., object for Dubinsky) produced by this process; hence the name pro-cept, the condensation of process and concept. According to Tall (1996), “Function, derivative, integral and the fundamental limit notion are all examples of procepts. The theory of functions and Calculus can be summarized in outline as the study of the “doing” and ‘undoing’ of the processes involved” (p. 293).

Learning in this viewpoint means developing a flexible proceptual view. What this exactly covers is not something uniform, and Tall distinguishes between three categories of procepts according to their operational characteristics. Procepts in elementary arithmetic give direct access to the object they represent: The procept “ $4 + 8$ ” represents both the process of adding the number 4 and the number 8, and the number 12 that results from this addition. Algebra is the domain of *template processes*: “ $2x + 3$ ” can be seen as a process and as an object that I can for instance substitute for y in a procept such as “ $2y^2 - 1$ ”. As a process, it is not directly workable: it only tells how to get a numerical value each time a numerical value is given to x ; hence the name template process. Procepts involved in calculus are generally of a third nature as they are associated to symbolizations that do not have the same algorithmic power, even if they can support some operational

work, as for instance change of variables in definite integrals. Interpreting the symbol $\Sigma 1/n^2$ both as a symbol for a process and as a concept does not give means for practically operating with it, and for solving the mathematical problems that can be attached to it. In fact, developing a flexible ability for doing and undoing the processes involved in elementary calculus is certainly a rather complex and long-term construction not only dependent on the development of symbolic techniques and abilities. Tall insists in his most recent work on the role that can be played by enactive experiences and visualization, as mentioned above. The fact that enactive experiences can help students and even young students develop an intuitive sense of calculus concepts such as those of velocity and acceleration has been evidenced by different researchers for a long time (see for instance Kaput, 1992). As regards visualization, the software Graphic Calculus, developed by Tall in the 1980s (Tall, 1986), has provided a paradigmatic example ever since of what can be achieved for developing a first approach to calculus concepts without relying on formal definitions and proofs. It exploits the fact that fundamental notions in calculus such as those of continuity and derivative can be expressed in terms of *local constancy* and *local straightness*, and that the characteristics of computer visualization make it possible to “access” local properties through a finite number of zooms, and to “escape” in some sense the underlying limit process.¹⁰ By working on these visual representations, one can thus explore the properties of mathematical objects that in fact result from an infinite limit process. Connections can be made with symbolic representations, and the interaction between these two different semiotic registers of representation can be used to support conceptualization and the development of procepts.

A major point of interest when thinking about post-secondary mathematics learning is obviously the transition towards formal thinking. Tall has often insisted on the radical change in perspective that this transition requires, and on its difficulty. According to him, only a few students truly enter this world but what is important to stress, and we will have the opportunity to come back to this point later on in the chapter, is that a lot can be achieved through mathematical work in the first two worlds mentioned above, the embodied and the proceptual.

We have summarized above some evolutions of approaches that, in the beginning, focused on the process/object distinction. The distinction between process and object levels of conceptualization is in-

¹⁰ A comparison can be made here with research in stochastics in which simulations serve to create “microworlds” to explore abstract concepts, such as sampling distributions or stochastic processes.

interesting for post-secondary learning because what students are asked to learn in university courses often requires an object level, whereas what they have learnt previously, even if they have developed some familiarity with the notions, does not ensure this object level, except for some limited set of instances in the best cases. In APOS, this initial distinction remained the core of the theory. It aims at representing some *vertical organization* of knowledge whereas the notion of schema takes care of the *horizontal organization* and of the underlying interconnections. But starting from the principle that any form of conceptualization obeys this pattern or, said in another way, that through the process/object duality we approach the epistemological core of knowledge growth in any mathematical domain is a very strong hypothesis that is open to discussion. We will have the opportunity to come back to this point in the last two parts of this chapter. In the research literature moreover, more evidence is emerging that establishing connections plays a fundamental role in knowledge development. In APOS, this dimension is accounted for by the schema (S) part of the model. But this part remains rather undeveloped compared with the APO parts. In our opinion, it is not by chance that researchers met problems with this schema part, which they tried to solve through incorporating the triad of Piaget and Garcia. A new hierarchy has been introduced, but the fundamental problem of the “extension of the concepts” cannot be solved just by introducing a new hierarchy. A fundamental question seems indeed to us the following: What does it mean to have built the concept of function as an object? What does it exactly mean to have an Inter, Intra, or Trans level of the function schema for students, and even for experienced mathematicians? Can we researchers give answers that are independent of the kind of functional objects we deal with, on the kind of operations we are asked to perform on these, on the kind of structures in which these objects are embedded?

The model proposed by Tall does not expose itself exactly to the same criticisms, but some points that are especially important when one thinks about post-secondary education seem to deserve much more research. The notion of procept recognizes the essential symbolic part of mathematical activity. Nevertheless saying that a symbol can flexibly refer to both a process and a concept is not enough for understanding the role played by symbols, and the mathematical work involving them, in conceptualization. This is especially the case in calculus and analysis because, as mentioned above, the operational value of symbols is not necessarily very high.

Symbols are involved in techniques that can be seen as components of mathematical practices. The term *technique* is understood here with the anthropological meaning developed by Chevallard (1992): as a way of doing things, not necessarily something algorithmic. Techniques can be attributed both a pragmatic value linked to their effectiveness in producing results and an epistemic value linked to the way they contribute to the understanding of the mathematical objects they involve. The same can be said of the symbols that support techniques, and thus of procepts. Researchers need certainly to know more about the ways in which work with symbols can help develop the epistemic value of procepts. The second point we would like to make is about the transition towards formal thinking and the reconstructions this requires. More certainly needs to be known about this, and we think that complementary insights can be offered by other approaches, such as those we will describe in the next paragraph.

Epistemological Approaches

The perspectives described in this paragraph can be seen as complementary to the preceding ones. Even if the motivation of research is, as usual, the desire to understand students’ difficulties and to develop didactic strategies helping them to overcome these difficulties, the base of the thinking is here more historical-epistemological than cognitive. This trend seems to us well illustrated by the book synthesizing educational research in linear algebra edited by Dorier (2000), and we will use this example in order to give an account of this position. We will come back then to what can be offered by such approaches to areas other than linear algebra, for instance calculus. As pointed out by Dorier in his epistemological work tracing the origin and evolution of the main concepts of linear algebra such as vector, vector space, linear independence, set of generators, basis, rank, dimension and duality, and linear transformations, the modern theory called *linear algebra* results from a long historical process covering several centuries, from the initial work on linear systems carried out by mathematicians such as Euler and Cramer around 1750 to the axiomatic theory of vector spaces, firstly formulated by Peano in 1880, which became universally accepted only 50 years later. In this sense, “the concept of vector space encapsulates, in a very elaborate product, the result of a long and complex process of generalization and unification” (p. 59).

The importance of this concept did not come just from the fact that it allowed the solution of new

problems¹¹ but more from the fact that it opened new perspectives on old problems and productive connections between problems set up in different mathematical domains once these were recognized as instances of linear problems in adequate vectorial spaces. According to Dorier and his colleagues, the fundamental values of generalization and unification at the core of linear algebra are not easily understood, and the apparent simplicity of axiomatic structures is misleading. Understanding these values requires having already gained sufficient mathematical experience of linear problems in different contexts (geometry, systems of linear equations, sequences, differential equations . . .) and being able to adopt a reflective view in order to connect these different mathematical experiences. When introduced abruptly to the axiomatic structure of vectorial spaces, many students feel understandably overwhelmed with new definitions and vocabulary, immersed in a formalism they cannot really make sense of. Robert and her colleagues termed this the obstacle of formalism (Robert et al., 2000). Making sense of this formalism and understanding abstract linear algebra as more than a pure formal game requires one to establish reflective links with one's previous linear experience.¹² This contributes to make linear algebra, as pointed out by Dorier (2000),

an “explosive compound” of languages, settings and systems of representation. There is the geometric language of lines and planes, the algebraic language of linear equations, n -tuples and matrices, the abstract language of vector spaces and linear transformations. There are the settings of geometry, of algebra, but also of graphical representations which allow a metaphoric use of geometry in higher dimensional spaces. There are the “graphical,” the “tabular” and the “symbolic” registers of the languages of linear algebra. (p. 274)

As evidenced by research, teachers and texts constantly jump among these languages, settings, and semiotic systems as if conversions among these were obvious. To this complexity one must add the diversity of the associated reasoning modes. Sierpiska (2000) for instance distinguished three different reasoning modes that strongly intertwine in linear algebra: the synthetic and geometric in which mathematical objects are, in some way, directly given to the

mind, which tries to grasp and describe them; the analytic-arithmetic mode in which objects are given indirectly by formulas or equations that make calculations with them possible; and the analytic-structural mode in which objects are also given indirectly, but this time through a set of properties. Again, Sierpiska showed that university teachers jump regularly without any precaution from one mode to another, leaving the responsibility for making connections to the student.

This is also an issue of the duality between the Cartesian and parametric points of view. Alvez Diaz (1998) has shown that, even if the conversion between parametric and Cartesian representations of subspaces can be, *a priori*, easily achieved thanks to algorithmic techniques in finite dimensions—those attached to the solving of linear systems—a flexible connection between these two points of view is hardly ever achieved by advanced students. Once more, looking back at history shows that the apparent easiness of such conversions is misleading. Understanding and mastering these took a very long time and was tightly linked to the long-term development of the concept of duality. Once more, an analysis of textbooks and teaching practices evidences the poor sensitivity of university educators to these difficulties.

The research on linear algebra offers views on learning processes that are complementary to the process/object research, emphasizing the specific epistemological status of concepts that have a particular generalizing and unifying role in the mathematical universe and suggesting that specific strategies have to be developed if one wants to have this epistemological status accessible to students. Of course, a more practical initiation to linear algebra can be organized, and this corresponds to an increasing tendency in post-secondary systems facing students' difficulties with abstract linear algebra. Powerful algorithmic techniques in linear algebra can give access to many conceptual ideas, as for instance shown by Uhlig (2002), and one can consider that this level of conceptualization is enough for many students. But research also shows that such an approach necessarily limits the modes of control that students can have on their mathematical work in linear algebra, and that when students are deprived of means of theoretical control on the

¹¹ When students are introduced to calculus concepts, for instance the concept of derivative, they quickly can understand what mathematical power they gain for solving a lot of problems linked to variation and optimization. Finding problems playing a similar role for showing the power gained through introducing linear algebra ideas is not so easy. As shown by the historical development of linear algebra (Dorier, 2000), axiomatic theory took a long time to be recognized as something essential, and this was achieved only when mathematicians used this theory for dealing with infinite nondenumerable spaces in functional analysis, something out of the range of university beginners.

¹² This is the rationale for different didactic strategies presented in the book.

techniques they use, they are easily trapped by “formal skids” as shown by Alvez Diaz (1998).¹³

The research also shows the essential status of the “connective dimension” in learning processes and highlights the diversity of the connections at stake: connections between settings and contexts, between points of view, between languages, between semiotic registers, and between reasoning modes. It thus confirms that cognitive flexibility is a crucial dimension of learning at these advanced levels and shows that this cognitive flexibility requires, to be adequately approached and understood, to go beyond the idea of conversion between different types of semiotic representation—the way in which it has often been analyzed in the research literature.

Coming back to the domain of calculus, one can identify similar phenomena in the transition towards formal analysis. When the fundamental concept of limit was formally defined in the 19th century, the main ambition was to establish the field of differential and integral calculus with solid foundations. This formal concept of limit can be given the status of “proof generated concept” according to the categories introduced by Lakatos (1976). Being sensitive to such mathematical concerns requires some particular mathematical culture, and developing this sensitivity in students certainly requires specific didactic strategies, as is the case for the notion of algebraic structure.

Whatever the way we express it, we touch here on the fact that different forms of conceptualizations can be a priori attached to the same concept, each of these being a coherent whole characterized by the experiences and problems that can be approached and made sense of, the techniques it offers for solving these problems, and a specific vision of mathematical rationality, and based on a specific level of formalization and symbolic manipulation. Conceptualization in any of these forms can a priori reach a high level of sophistication, as evidenced by history, and thus placing these too strictly in a hierarchy is not necessarily the most appropriate way of understanding mathematical thinking and learning, even if this is the general tendency. The idea of hierarchy leads us to see these different forms as corresponding to different cognitive levels and to think of cognitive growth as a transition process between these different levels. This is not necessarily the most appropriate metaphor for cognitive growth, as evidenced by research that shows that in expert practices different forms coexist, that univer-

sity teachers for example jump so naturally from one form to another. If we do not adopt a strict hierarchical perspective, nevertheless other issues immediately emerge: issues of connection, relative importance, and respective role. These are different but not easier to solve, and certainly do not have uniform or universal answers. What researchers can nevertheless infer from both epistemological and educational research carried out up to now is that the connection between different forms of conceptualization is cognitively costly as it obliges to connect different ways of doing and thinking mathematically in a particular area, and to give each of these a particular role, and an importance that of course may vary according to the context. Research in geometry such as that carried out by Houdement and Kuzniak (1999), inspired by Gonseth’s epistemological perspectives, can be considered as a step in this direction. Following Gonseth, they distinguish three main geometrical worlds: the worlds of practical geometry, natural geometry, and axiomatic structures, and they try to describe the characteristics that mathematical work presents in these different worlds. They show that each of these develops its own approach to geometry and rationality, and each has led to the development of sophisticated techniques and results, which are incommensurable between them. They also show that textbooks and elementary courses for preservice teachers jump without any precaution from one world to another (essentially between the first two), and that this fact makes it especially difficult for elementary teachers to understand the relationships between these different geometries, and the consequences for pupils’ difficulties and curricular goals.

The Integration of New Cognitive Approaches

As evidenced by the research literature, for instance by the evolution of the AMT group of PME, educational thinking on learning processes is now influenced by different cognitive approaches. In this part, we consider the influence of neurosciences and theories of embodied cognition. Neurosciences have now informed the investigation of cognition in mathematics with research on perception and geometry (Berthoz, 1998; Longo, 2003) and on numbers and arithmetic, such as those by Dehaene (1996). Thanks to visual brain imaging, researchers can now localize those brain areas actuated in the solving of different

¹³ For instance, Alvez Diaz shows that students customarily manipulate linear objects such as sets of vectors, matrices, linear transformations, linear systems, and determinants through numerical tables. These tables are, a priori, efficient semiotic instruments, but students tend to operate on the lines and rows of these tables automatically, without taking into account the mathematical meaning these operations have or not according to the object the table represents. This behavior generates “formal skids” leading to absurd results or contradictions that students generally do not notice.

mathematical tasks and generate a dynamic image of brain activity. The results obtained show some very interesting facts, for instance that the human brain seems to be genetically equipped with a sort of analog counter that allows humans to globally compare quantities, and moreover that the brain areas involved in this are not the same as those used for counting and exact calculations. Research carried out with people who have brain damage shows also up to what point our numerical abilities are, at brain level, decomposed into a complex of localized processes dependent not only on the task but also on the precise context evoked in the task.

With some exceptions, these results have not concerned up to now advanced learning processes such as those considered in this chapter, but they do renew our attention to the dependence of learning processes on the biological condition of human beings. The increasing influence taken, in the educational world, by theories of embodied cognition reflects this attention. Embodied cognition, which began to develop in cognitive sciences in the eighties (Varela et al., 1991), investigates cognition as a physically embodied phenomenon realized via a process of codetermination between the organism and the medium in which it exists. Within this perspective, abstract understandings are grounded in bodily experience; an example frequently quoted is that of the concept of balance (Johnson, 1987) whose genuine meaning stems from our physical experience of bodily equilibrium and loss of equilibrium. Such bodily experience, according to the theory, gives way to perceptual-conceptual primitives called *image-schemata* that function as patterns and allow for the organization of experience. Their extension to different domains such as art (the “balancing” of colors in a painting), accountancy (“balancing” a budget), or mathematics (“balancing” an equation), which is generally a cultural phenomenon, leads to embodied concepts that are all grounded in this primitive physical experience. These extensions occur through conceptual mappings involving conceptual metaphors, hence the idea of mathematical knowledge as something whose source is metaphorical thinking (Lakoff & Nuñez, 2000).

The article published by Nuñez, Edwards, and Matos in the special issue of *ESM* “Teaching and Learning Mathematics in Context” (Boero, 1999) illustrates quite well how this perspective can be used to reflect on the learning of advanced mathematical concepts, such as the concept of continuity. Starting from the classical distinction between a natural idea

of continuity characterizing a process without gaps (the vision of Euler of a continuous curve as a curve that can be drawn without lifting the pen) and the formalized idea of continuity due to Weierstrass, the authors showed that these two mathematical ideas are grounded in two very different conceptual metaphors. The first one grounds in what they call “the fictive motion metaphor,” which they summarize by the following sentence: “A line is the motion of a traveller tracing that line” (p. 56). According to this cinematic vision, the curve is not a set of points but points can be put on it like milestones along a road, and points can move on it as travellers on a road. Students’ and mathematicians’ language is full of references to this cinematic vision of continuity, as also in everyday discourse. Metaphors can also be associated with the Cauchy-Weierstrass notion but these are radically different: “A line is a set of points”; “Continuity is gaplessness”; “Approaching a limit is preservation of closeness near a point.” Points are in this case constituents of the line, and its continuous character that results from gaplessness is no longer evident. Being continuous for a function means that it preserves gaplessness: The image of a continuous set is a continuous set. The authors use this analysis for criticizing the usual teaching approach that establishes a clear hierarchy between these two conceptions of continuity and presents the $\varepsilon\delta$ definition as the one that captures the essence of the mathematical idea, which is in fact multi-form. These authors reject a vision of learning about continuity that would see it as involving the rejection of the natural and cinematic conception and its replacement by the set-theoretical one, and the teaching strategies that more or less explicitly rely on such a vision. Of course, this does not contradict the fact that the set-theoretic conception, through the mathematical tools it provides, allows the solution of problems inaccessible to the cinematic conception. Even if expressed through another theoretical frame, it is interesting to point out here concerns very close to, for example, those expressed above concerning the different geometrical paradigms.

The influence of embodied cognition is also visible in the way technological issues are addressed today at post-secondary level.¹⁴ In the early nineties, two main approaches in educational research featured at this educational level: the programming approach and the visual and multirepresentational approach. Educational research began to be sensitive to what could be offered by technological devices that simulate movement, thanks to the pioneering work by Kaput

¹⁴ This is for instance visible in the way that Tall (2004) analyzes today the cognitive role of technological tools that he began to develop a long time ago, within other theoretical frames (as mentioned above).

and Nemirovsky with “Math Car” (Kaput, 1992), but this research concerned young students with limited mathematical knowledge. The impressive amount of research work carried out on representations (see the two special issues of the *Journal of Mathematical Behavior* edited by Goldin and Janvier, 1998, for a synthesis) and on the use of multiple representations for learning concepts such as the function concept, leading to what Confrey and Smith (1994) termed an *epistemology of multiple representations*, developed rather independently. As expressed by Borba and Scheffer in a special issue of *ESM* (2004), thanks to the development of technologies like CBR (Calculator-Based Ranger) the context today is changing and new perspectives are being offered to research. Body motion that was only peripheral to the multiple representation discussion becomes an essential component, and research on multiple representations takes another dimension through research concerning embodied cognition.

This connection, and the way it can influence research at post-secondary level is well illustrated in the *ESM* special issue in a paper by Rasmussen et al. (2004). These authors analyzed how three undergraduate students having already completed 3 semesters in calculus and taking a course on differential equations progressively made sense of an unfamiliar tool called the water wheel. Through this analysis, they tried to characterize how bodily activity and tool use can combine in mathematical learning, and how this combination can suggest alternative characteristics of knowing. The water wheel is a complex object. It consists of a clear, circular acrylic disc holding 32 plastic tubes around its perimeter. The disc is mounted on an axle and is free to rotate a full 360° and tilt between 0 and approximately 45° . In the center of the disc are two concentric clear plastic cylinders that contain a variable amount of oil that acts as damping for the system. Water from a bucket with a submersible pump (with adjustable flow rate) falls into several contiguous tubes on the “higher side” of the wheel. Each tube has a small hole at the bottom that allows water to drain out and be directed back into the bucket containing the submersible pump for a continual flow of water into and out of the tubes. When the wheel is tilted, gravity causes the wheel to rotate. When connected to a computer, an optical sensor collects data with real-time displays of angular velocity versus time, angular acceleration versus time, and angular acceleration versus angular velocity. Relying on the work developed by Nemirovsky, Tierney, and Wright (1998) on students’ developing expertise with a motion detector, the authors’ analysis takes into account the evolution of the students’ relationship with the water wheel along several dimensions: becoming aware of how

the tool may respond to certain aspects of one’s kinesthetic engagement and not to others, acquiring an emerging awareness of what the tool does on its own, independent of one’s action; gradually distinguishing which actions may affect the tool and the conditions that need to be in place so that it continues to remain sensitive to the actions; and developing a sense for what outcomes or results are possible, impossible, or difficult to obtain. Analysis along these dimensions helps the authors understand what kind of knowledge can be built through physically interacting with the tool, and how. This leads them to elaborate on a particular type of knowing called *knowing-with*, initially introduced by Broudy (1977). Knowing-with differs from commonly-identified forms of knowing such as knowing-that or knowing-how in that it involves at once the subjective feel and the objective sides of the experience and thus seems to the authors especially appropriate for describing the kind of cognitive construct that is at stake in the interaction with the tool. The language used by the students in this study shows indeed a very strong personal connection with the tool as if they were themselves becoming the tool, and this shapes the knowledge they develop about concepts such as angular velocity and acceleration in this environment.

Without any doubt, research like this attracts attention to learning processes that have been not sufficiently taken into account in post-secondary educational research up to now. The process raises interesting questions, shows intriguing phenomena, but is still in an exploratory phase. What kind of knowledge exactly is built in the sessions with the water wheel? How can students’ utterances that are mainly of a qualitative nature, and often fuzzy, give way to something more analytical? How does knowing-with connect with the other forms of knowledge that develop at post-secondary level? How can it serve to understand and work with other mechanisms, other contexts? These seem today to be mainly open questions.

Investigating how qualitative knowledge that emerges from physical interaction with a calculator can be connected to more analytical forms of knowledge that are aimed at by educational institutions was a focus in the recent doctoral thesis by Maschietto (2002). This thesis addressed the transition between algebra and calculus, and the emergence of the dialectic interplay between, on the one hand, the point-wise or global perspective that characterizes the relationship that students develop with functions in algebra courses, and on the other hand, the local perspective that is attached to calculus and analysis. From a technical point of view, this emergence requires a reconstruction of the relationships with algebraic computa-

tions: The different terms of an algebraic expression are no longer given the same weight; their treatment depends on their respective orders of magnitude and relies on the fundamental idea of relative (or absolute, in nonstandard analysis) negligibility. The mathematical world of algebraic computations is thus deeply affected, at technical level and also in terms of strategies and control. Maschietto explored how these fundamental ideas can develop through adequate interaction with calculators, and the corresponding reconstructions may be promoted by appropriate didactic engineering, without entering the field of formal analysis. Maschietto relies on the potential offered by graphic calculators for visualizing local linearity (a potential that is well recognized today). The influence of embodied cognition is visible in the sensitivity she develops about the students' gestures and discourse as they zoom in and zoom out with the calculator, and about the personal and collective development of the metaphor of microlinearity. But what is added to these perspectives is a careful attention to the mathematical limits of visualization (it shows closeness, but not the order of the approximation that is essential here), a careful attention to the way the microlinearity metaphor can become an operational tool, and to the difficulties met by students in that operationalization. Students are very soon sensitive to the phenomenon of microlinearity that they recognize as an invariant. Moreover the distinction that they make between the two perceptive categories of straight and curved leads them to think that, beyond what they see, there is a more complex process involving infinity (something that is curved cannot become straight through a finite process). But mathematizing the situation in an operational way is not obvious and cannot be left to the students' responsibility. If not carefully dealt with, the microlinearity metaphor easily loses some of its essential attributes and only supports fuzzy discourse and knowledge; the necessary reconstruction of algebraic techniques takes time. Results show that fundamental and deep ideas and techniques of analysis can be developed in such a context rather early, but also that this requires a subtle design of the whole didactic engineering (Artigue, 1988), an adequate sharing of the mathematical responsibilities between the students and the teacher, and evident mathematical and didactic expertise from the teacher.

This research influenced by embodied cognition, by the attention it pays to cognitive processes and the situations in which they develop, and the possible "ecology" of these, leads us into the final section of this part devoted to more global research approaches, inspired by the increasing influence in the educational field of sociocultural and anthropological approaches.

The Integration of More Global Approaches

In this part, we consider approaches to mathematical thinking and learning at post-secondary level in which the consideration of sociocultural and institutional practices plays an essential role. According to the research culture in which it has developed or is developing, this consideration takes different forms and relies on different theoretical frames. These frames are often drawn from outside the field of mathematics education itself, as attested by the increasing number of references to Vygotsky, to Activity Theory, and to frames developed for understanding enculturation processes in various kinds of contexts. It relies also on frames developed inside the educational field. This is the case with the emergent approach developed by Cobb and Yackel (1996) in the USA that aims at connecting psychological constructivist and sociocultural perspectives for the analysis of classroom processes and the design of classroom experiments. This is also the case with the anthropological approach initiated by Chevallard from the early nineties (Chevallard, 1992; Chevallard & Bosch, 1999), which is clearly distinct from constructivist perspectives. The theoretical constructs developed by Godino and his colleagues in Spain (Godino, 2002; Godino & Batanero, 1998; Godino, Batanero & Roa, 2005) or by Cantoral, Farfan, and their colleagues in Mexico (Cantoral & Farfan, 2003) play for other researchers a similar role. These approaches all have their differences in detail, and the distance taken from constructivist and socioconstructivist perspectives may vary, but they share the common view that mathematical objects emerge from human practices, and that these practices are institutional and sociocultural practices. The term *institution* here has a very wide sense and includes any kind of formal or informal structure that organizes or conditions our social and cultural activities. The approaches also share the view that institutions develop with regard to any object that is recognized as a mathematical object a specific idea of what it means to know that object, thus defining a set of institutional norms for knowledge. So, the personal relationship one develops with mathematical objects emerges from the institutional practices and norms one has experienced in the different institutions where this object has been met. Within this perspective, understanding institutional practices and norms is *sine qua non* for the understanding of learning processes. Social and cultural mediations thus play a crucial role, and not by chance do the frames we are discussing pay so much attention to the semiotic dimension of mathematical activity, semiotic tools being an essential channel for these sociocultural mediations. In this part of the chapter, we would like to show the complementary

insights these approaches offer to the understanding of learning processes and students' difficulties. Different examples will help us to illustrate what is offered and to show the diversity of the research developed within these perspectives.

In all the perspectives considered here, individual knowledge emerges from sociocultural practices. Both individual and collective knowledge develop in the classroom, in a subtle intertwining, and one cannot understand individual cognitive development without considering its collective counterpart. Our first example illustrates this dimension, showing one perspective developed within the emergent approach for analyzing the collective construction of knowledge.¹⁵ The research (Stephan & Rasmussen, 2002; Yackel, Rasmussen, & King, 2000) deals with the teaching of differential equations and was carried out in the frame of a 15-week reformed course that tried to benefit from technology for emphasizing graphical and numerical approaches to the topic. From a theoretical point of view, it relies, as mentioned above, on the emergent approach, which considers that learning is both an individual and social accomplishment with neither taking primacy over the other. The design of the instructional sequence was inspired by the theory of realistic mathematics education, initiated by Freudenthal (1991) and further developed by the Freudenthal Institute (Gravemeijer, 1999). According to this, learning trajectories are built, giving students the opportunity to create meaningful mathematical ideas as they engage in challenging mathematical tasks. The development of knowledge is organized around cycles of horizontal and vertical processes of mathematization (Treffers, 1987). Horizontal mathematization refers to "formulating a problem situation in such a way that it is amenable to further mathematical analysis" whereas vertical mathematization "consists of those activities that are grounded in and built on horizontal activity" such as "reasoning about abstract structures, generalizing and formalizing" (Rasmussen et al., 2005, p. 54). Further, in this research the collective development of knowledge is approached through the analysis of argumentation, by using Toulmin's (1969) model of argumentation. According to this model, the core of an argument consists of three parts: the data, claim, and warrant. Briefly speaking, the data provide evidence for the claim, and the warrant explains how the data leads to the claim. Moreover, in case the validity of the warrant is challenged, the presenter must pro-

vide a backing to justify why the warrant, and therefore the core of the argument, is valid. Progression of collective knowledge is associated to the development of taken-as-shared knowledge, and this is achieved through a detailed analysis of the evolution of collective argumentation during the course. The researchers consider that mathematical ideas become taken-as-shared in the classroom community when

either (1) the backing and/or warrants for an argumentation no longer appear in students' explanations and therefore the mathematical idea expressed in the core of the arguments stands as self-evident, or (2) any of the four parts of an argument (data, warrant, claim, backing) shift position (i.e., function) within subsequent arguments and are unchallenged. (Stephan & Rasmussen, 2002, p. 462)

This allows the researchers to develop an interesting analysis of the progression of the classroom knowledge, organized around the emergence and development of six mathematical practices: predicting individual solution functions, refining and comparing individual predictions, creating and structuring a slope field as it relates to prediction, reasoning about the function P (in $P'(t) = f(P)$) as both a variable and a function, creating and organizing collections of solution functions, and reasoning with spaces of solution functions. The researchers stress that the analysis shows a nonsequential development over time of these different practices. For instance, the idea that, for an autonomous differential equation,¹⁶ the phase portrait is invariant by horizontal translation began to emerge in the second session (not of course articulated in the expert language we use here) and went on being discussed until the 11th session. These results are of course linked to the choices made in the design of the instructional sequence regarding the learning trajectory, the precise tasks proposed to students along this trajectory, and the sharing of mathematical responsibilities between teachers and students. The mathematical practices developed here contrast with the practices usually adopted in differential equation courses and offer evidence that teaching designs that are more respectful of the epistemology of the field, both meaningful for students and reasonably ambitious¹⁷ from a mathematical point of view, can exist. The emergent approach, by the attention it pays to the

¹⁵ Another interesting dimension of this research we just mention here deals with the progressive development of sociomathematical norms in this reformed course.

¹⁶ A differential equation $y' = f(y, t)$ is said autonomous if the function f does not depend on t . This reform course only dealt with autonomous differential equations.

¹⁷ The course for instance approaches the dynamics of differential equations depending on parameters and introduces and discusses bifurcation diagrams.

connection between the individual and the collective, helps to understand the essential role played by the classroom community in the individual development of knowledge. By the attention it pays to social norms and sociomathematical norms, it also makes clear that new practices cannot be developed without removing some of the constraints induced by usual norms, and thus raises the issue of the institutional viability of innovative practices.

The second example we present deals with a crucial issue: that of the transition between secondary and tertiary education. This transition is widely known as a serious problem, and the difference between the mathematical cultures of secondary school and college/university is generally admitted as an essential source of it. These two cultures are moreover nowadays generally regarded as two cultures moving on roads increasingly distant, which can only make the transition harder and harder. We are thus interested in looking at this problem from an anthropological or sociocultural perspective. This was the perspective adopted by Praslon (2000) in France, in his thesis devoted to the notion of derivative at the transition between high school and university. Praslon used the anthropological approach of Chevallard and the associated conceptual tools in order to explore the characteristics of the two cultures (limited nevertheless to the case of science-focused students both at high school and university). In this approach, as stated above, knowledge emerges from practices. Practices are analyzed through the notion of *praxeology*, a complex that includes types of tasks, ways for solving tasks called techniques (not necessarily algorithmic techniques in the usual sense), a discourse that explains and justifies techniques called technology, and theories that organize and structure the technological discourse. This led Praslon to characterize the praxeologies involving the notion of derivative in the two different institutions that he considered. This characterization led to interesting results. He showed that, contrary to what is often said, the secondary-tertiary transition in this area is not a transition between informal and formal mathematics, between intuitive and rigorous approaches. Rather the transition is more an accumulation of “micro-breaches,” which affect the balance between the tool and object dimensions of the derivative, the balance between the study of particular objects and objects defined by general conditions, and the balance between algorithmic techniques and techniques that have more the status of general methods to be adapted to each particular case. These shifts also result from the increased autonomy given to students: as regards the choice of appropriate settings, appropriate semiotic registers, and the connections and changes made

between these during the solving process, and more globally as regards the overall development of the solving process (similar tasks often appear in the two institutions, but the number of intermediate questions in a problem is far from being the same). They also result from the incredible diversification of tasks that occurs. From a culture organized around the mastery of a restricted number of tasks that can become reasonably familiar, and in which the progression in complexity is carefully managed, students pass into a culture in which diversity is the norm and the greater number of new notions to be covered in the same time period makes routinization of practices much more difficult. Once more it is evident that cognitive flexibility is an essential criterion for success. In order to make both students and university teachers sensitive to the differences between the two cultures, Praslon has created a set of tasks that, according to his results, are situated today in the “gap” between the two cultures and reveal the main facets of the differences between these. Purposely, none of the tasks requires solution by formal analysis. They are designed to be proposed to the students before their entrance to university or at the beginning of the academic year, and the students’ work discussed with university teachers.

The anthropological framework is also involved in the *instrumental* approach initiated by French researchers for addressing issues linked to the integration of Computer Algebra Systems (CAS) in mathematics education (Guin, Ruthven, & Trouche, 2004). Combining Chevallard’s anthropology with perspectives coming from cognitive ergonomics (Rabardel, 1995), the instrumental approach analyzes how mathematical praxeologies are affected by the use of CAS in their technical and technological components. This approach leads to specific attention on what is called *instrumental genesis*, which is the process in which there is a transformation of a tool (here a CAS) into a mathematical instrument, either for an individual or for an institution. The results obtained (Guin et al., 2004) show the complexity of this process, which for a long time has been underestimated in educational research dealing with technology, and the deep extent to which learning processes intertwine mathematical knowledge and knowledge about the tool itself. The sensitivity to institutional aspects allowed by the anthropological frame leads also to a better understanding of how a teacher’s professional work is affected by the integration of such tools and offers a vision of these changes quite different from what is generally offered by the literature. We will not enter into more details here regarding this approach, whose constructs have been elaborated and used in the context of secondary education up to now (see Guin et al., 2004, for such

details). However, the results obtained on mathematical topics such as calculus that are generally taught primarily at post-secondary level lead us to think that it can be of interest in the future for research involving technology, and especially professional technology¹⁸ as is the case for CAS, at post-secondary level. Its constructs could certainly also be usefully connected to those developed by Nemirovsky and his colleagues mentioned above, as the two approaches share the same fundamental idea: technology *cannot* be considered only as a kind of educational assistant. It deeply shapes what we learn and the way we learn it, and an efficient integration of technology in mathematics education has to take this seriously into consideration, in terms of institutional practices, values, and norms.

In the preceding, we have approached the idea of culture in terms of practices and praxeologies. Other categorizations can be used. Those introduced by Hall (1981) have been used for instance by Sierpiska (1989, 1994) and more recently Nardi (1996). Hall recognizes three types of consciousness, three types of emotional relations to things: the formal, the informal, and the technical. In the context of mathematical culture, the formal level corresponds to beliefs around what is mathematics about, what are the legitimate tools and methods for mathematical work, and so on; the informal level corresponds to schemes of action and thought, unspoken ways of doing things or thinking that result from experience and practice, what is also often called “tacit knowledge”; and the technical level corresponds to the explicit part of knowledge, to the techniques and theories. Approaching learning as an enculturation process leads researchers to try to understand how these three levels and the interactions between these can develop in a given culture, and why. Sierpiska for instance first used this approach in reflecting on the long-term program of research she had developed about the learning of the concept of limit. In this program, the determination of epistemological obstacles, further complemented by the dual notion of “act of understanding,” had played an essential role. Thinking in terms of Hall’s categories, Sierpiska inclined to ask herself what the cultural situation of epistemological obstacles can be. According to her, these can be situated both at formal and informal levels, but not at the technical level.

Nardi (1996) used Hall’s categories in order to understand the tensions between the mathematical culture of students entering the university and that of the university. She also looked at the ways enculturation progresses, taking into account not only the technical

level as is usually done but also the formal and especially the informal level, through the accumulation of experience shared with the expert (here a tutor) and in the process of appropriation by the internalising imitation of the expert’s cultural practices. The data she collected led her to focus on concept image construction and on formalization in order to analyze the enculturation/cognitive processes at stake. She pointed out the tension existing between the informal-intuitive-and-verbal mode of thinking on the one hand, and the formal-abstract-and-symbolic mode on the other, and the difficulties students meet with the mechanics of formal reasoning. These tensions are especially visible in calculus, but Nardi showed that the exercises proposed to students, instead of helping them overcome the epistemological obstacles they face, can reinforce them: for instance when exercises constantly expose students to infinite sums that can be broken up and rearranged as finite sums, reinforcing their belief that infinite sums can simply be treated in the same ways as finite sums.

The two examples just presented focus on the understanding of students’ learning of mathematical concepts. But students’ learning is also affected more generally by the cultural practices they are part of, and by the visions of mathematical activity and mathematical learning these practices induce. As we noted above, such concerns were already present in the first *Handbook*, in the chapter written by Schoenfeld. In our opinion, even if research has remained sensitive to these issues, knowledge in this area did not substantially improve during the last decade. Today, nevertheless, one can think that changes in the context of university teaching—massification of tertiary education and the resulting diversity of the student population, increasing student disaffection for mathematics and science—can stimulate research. Moreover, the development of cultural approaches over the last decade better equips researchers for approaching the difficult questions at stake. From this point of view, the article by Zevenbergen (2001) in the ICMI Study on teaching and learning mathematics at university level is an interesting example. Zevenbergen’s particular concern is about equity and the characteristics of mathematical practices at university that can be seen as excluding some groups of students, preventing them from learning and succeeding. His interest for this issue seems linked to the expansion of the higher education sector and to the resulting increased diversity of students. His approach is cultural and especially relies on sociolinguistics (Halliday, 1988). Language is seen as a form of cultural capital, and the role of language in

¹⁸ That is, a CAS is software created to satisfy the needs of professional users of mathematics, rather than software created for the needs of mathematics learners.

the access to mathematics is seen as something essential. Hence the interest in the analysis of the linguistic practices in mathematics and in the understanding of their potentially excluding role for new categories of students entering the university: those from the working class and those who are not native English speakers. As pointed out by this author, research within these perspectives has been mainly concentrated up to now on elementary and secondary school. Very little has been done at post-secondary level.

Another piece of research that illustrates how the evolution of theoretical approaches can serve to renew research approaches is the recent article by Castela (2004). This research analyzes how tertiary institutions influence students' personal study, and thus learning, by comparing two different types of tertiary institutions that exist in France: specific courses (known as CPGE) that prepare students for entry to the most prestigious business and engineering schools, and university courses. The stimulus for her research was the observed discrepancy between the respective rates of success of students coming from these two types of institutions at the national competition for secondary teachers (CAPES), after 1 year of common preparation. From this, she investigated the characteristics of the mathematical cultures in the two institutions and how these influenced the students' vision of mathematics learning and personal achievement in mathematics, and how this achievement is more or less sufficient for the purposes of entering the CAPES competition. She showed that the type of task proposed at the CAPES (long problems covering extensive mathematical domains) requires specific competences in terms of strategy and structuration of problem solving that are not really taught. Then she tried to match these requirements with the ways that the two categories of students usually work. She for instance showed that CPGE students have a rather operational vision of mathematics and are sensitive to the need for competence in strategy and structuration. University students however perceive mathematics more in terms of content and put more emphasis in their work on lectures. She identified three different approaches to problem-solving in students' practices: a drill-and-practice approach, a reproductive approach, and a transferential approach. The first approach is more representative amongst university students, the last more amongst CPGE students. By deploying the tools provided by didactic anthropology, she then showed how these different attitudes can emerge from the adaptation of students to the institutions in which they study.

We find it interesting to connect these results with those obtained by Lithner, in his observation of students' functioning with textbook exercises (Lith-

ner, 2003). Lithner used three categories in order to qualify the forms of reasoning used by students in task solving:

plausible reasoning (PR) if the argumentation developed

- i) "is founded on intrinsic mathematical properties of the components involved in the reasoning, and
- ii) is meant to guide towards what probably is the truth, without necessarily having to be complete or correct" (p. 32),

established experience (EE) if argumentation

- i) "is founded on notions and procedures established on the basis of the individual's previous experiences from the learning environment,"
- ii) [the same as for plausible reasoning] (p. 34),

identification of similarities (IS), if

- i) "the strategy choice is founded on identifying similar surface properties in an example, theorem, rule, or some other situation described earlier in the text,
- ii) the strategy implementation is carried out through mimicking the procedure from the identified situation." (p. 35)

Even if the academic results of the three categories of students observed by Lithner are quite different, clear similarities appear in their functioning. Almost all of the students' homework time seems to be spent on exercises. IS reasoning dominates strongly, and the evaluation is done by comparing with the textbook solutions section. He also showed that, due to the organization of the textbooks used and to the exercises they propose, most exercises can be solved by IS reasoning. In Lithner's experiment, more elaborated forms of reasoning became necessary when students made errors in the implementation of their IS strategies, but, even in this case, PR remained very limited. What is observed suggests that the forms of work based on memorizing and mimicking can be considered by the students as efficient for passing exams, and he sees these as "unintended by-products of their mathematical instruction" (p. 54).

In this part of the chapter, we have focused on what we perceived as major evolutions of different theoretical approaches linked to both internal and external factors in the field of mathematics education. The choices we made for our presentation helped us structure it, but perhaps we have too strongly opposed the sociocultural and anthropological approaches to the preceding ones. The reality of research is much more complex. Complementarity is certainly a more appropriate term than opposition. Many researchers combine several frames in the same research project.

For instance, Nardi (1996) combined a cultural approach and the use of process/object duality, whilst Praslon (2000) combined an anthropological general approach with different cognitive perspectives used in educational research about calculus. Favoring certain approaches nevertheless shapes the problematics and methodology of the research, and through these the kind of results that one can access and the way they will be expressed. Hence there is an explosion of notions and terms that is not easy to make sense of, but links and partial translations are often possible, as we have also tried to show. In the next part of the chapter, we broaden our discussion to new ideas and new coherences, serving to challenge and complement those already presented, which will provide further evidence of the complexity of the issues at stake, and the fact that any single perspective can only approach this complexity very partially.

EVOLUTION THROUGH THE DEVELOPMENT OF NEW RESEARCH AREAS

In this part, we turn our attention to two areas of research that lie outside the core mathematical domain generally considered by advanced mathematics education: research on the teaching and learning of mathematics in engineering courses, and research on learning probability and statistics. In these two areas we hope to illustrate the effects of the five influences alluded to by the ICMI Study (Holton, 2001):

(1) the increase in the number of students who are now attending tertiary institutions; (2) major pedagogical and curriculum changes that have taken place at pre-university level; (3) the increasing differences between secondary and tertiary mathematics education regarding the purposes, goals, teaching approaches and methods; (4) the rapid development of technology; and (5) demands on universities to be publicly accountable.

Although public accountability is a generally political issue that may seem to lie beyond the interest of researchers, nevertheless it impacts quite directly in several ways, largely due to influence (1), the growing proportion of young people in tertiary education. Public scrutiny of “teaching quality” in universities has become rather common in the last 15 years, as increasing amounts of public money are spent on greater numbers of students. But also there is the case that as student numbers grow then the funding mechanisms have changed in many countries, with students themselves paying growing proportions of

their tuition fees. Thus there is another accountability to be met as students come to expect a good “service” for their money, and perhaps hold stronger opinions than before about the education they expect, setting up a tension with the education that their teachers believe is appropriate for them.

Statistics is one of the most widely taught topics at university level, where many service course students meet advanced stochastic thinking without any prior or concurrent experience of advanced algebra or calculus. At the same time, statistics is separating from mathematics as an academic discipline (e.g., in the training of mathematicians and statisticians; research journals and associations) and is taught by teachers with a variety of backgrounds, rarely by pure mathematicians. As regards secondary education, stochastics is receiving increasing attention in recent curricula (e.g., National Council of Teachers of Mathematics [NCTM], 2000), where it is considered as an essential component of mathematics education, both due to its instrumental role to understand other disciplines and as a vehicle to develop critical reasoning and democratic values.

In the development of mathematics courses for engineering, the main trend comes from the technological influence, particularly stimulated by changes within professional engineering practice. The traditional teaching approach, based on having students spend much time developing fluency with pen-and-paper mathematical techniques, is giving way to students learning how to use computer software (such as spreadsheets or computer algebra systems) to carry out mathematical calculations. On the one hand, this is liberating and empowering, because students can more quickly become engaged in solving realistic engineering problems. On the other hand, mathematicians, and also some engineers, have legitimate concerns that a fundamental understanding of mathematics, as it applies to engineering, could be lost in these changes.

It is, in our opinion, especially insightful to try to understand how research carried out in other areas can contribute to the consideration of mathematical thinking and learning at post-secondary level. It can help us question theories and positions as regard mathematical learning and thinking that have generally been established having in mind more or less explicitly the mathematical education of “pure mathematicians” or mathematics teachers and their particular needs, and also that have focused on a few classical mathematical domains. It also obliges us to consider the role of technology in learning in a rather different way. Indeed, what is often at stake in these domains is not simply the use of technology for developing

usual mathematical knowledge but the way in which technology is changing mathematical activity and understanding, including problem solving, proving, reasoning, modeling, and symbolizing. Mathematical and technological expertise and needs are in these domains much more tightly intertwined.

Learning Advanced Mathematics: The Case of Engineering Courses

This section provides a commentary on recent trends in mathematics education from the viewpoint of “service mathematics,” that is, mathematics as taught to non-mathematics specialists and students studying science, engineering, and other technical subjects. We focus on trends in the United Kingdom and with respect to engineering students (Kent & Noss, 2003). We also consider the development of technology for teaching and learning in service mathematics.

In this section, we move away from a concern with what might be called intra-mathematical structure (and cognition), such as debates on the role of proof in mathematics, or advanced mathematical thinking as discussed earlier, and we will be more concerned with questions of *modeling*—the relationships between mathematics and other knowledge domains, and the ways in which computer technologies are shaping these relationships.

We begin with a sketch of the area of nonspecialist mathematics practice and research. There is in fact rather little tradition of educational research in the nonspecialist area, although arguably this area of research could bring many benefits, given that in terms of the numbers of students involved there are far more than the number of specialist mathematics students, and it is also the area of mathematics education with perhaps the most pressing need for changes in curriculum and teaching approaches, as the “customers” (engineering students and engineering academics) are by many accounts increasingly dissatisfied with the traditional curricula and approaches offered by the mathematicians who teach nonspecialist courses. Up to now, there have been very few studies of the different ways that mathematics and engineering students think about mathematics (see Bingolbali, 2004; Maull & Berry, 2000). One of these few has been Magajna’s interesting and very detailed study of students and their mathematical thinking while studying at vocational college to become engineering technicians (Magajna, 2001; Magajna & Monaghan, 2002).

Innovation in engineering mathematics teaching tends to be practitioner-led, that is, it is carried out by university lecturers themselves, who (in most countries) divide their time between teaching and research. Educational research papers, if written at all, tend to take the form of descriptive reports, not much connected with the research literature of the world of mathematics education. Where a distinct research methodology is followed, the pre-test/intervention/posttest approach is still quite common, which is nowadays out of favor amongst socioculturally influenced mathematics educators. It remains very much an open question what kinds of research can particularly contribute to promoting change amongst higher education teachers (including connecting them to educational research), who have traditionally asserted a high degree of autonomy in their professional lives, including matters of teaching, and often harbor a somewhat negative attitude towards mathematics education (the “math wars” that broke out in the 1990s in the USA illustrate some of the tensions involved here—see Ralston, 2004). There have been some attempts to build a research culture for educational innovation in university teaching. In the UK, for example, the Higher Education Academy Subject Network is a government-funded initiative that aims to support lecturers in all major subject disciplines, including the branches of engineering and science, to carry out educational projects, share their results, and generally network ideas about learning and teaching; one of the significant features of the Network is that it is based on subject-specific support centres, directed and staffed mainly by lecturers from that subject area—the implication being that lecturers tend to feel closer in thinking to their fellow subject professionals than to educational specialists.¹⁹

The last 10 years have been a very interesting period of change for nonspecialist mathematics teaching—for a range of accounts of these developments see the outcomes of the ICMI Study on the Teaching and Learning of Mathematics at University Level, which took place in 1999 and was published in two volumes of papers (Holton, 2000, 2001); see also the proceedings of a regular conference on developments in engineering mathematics (Hibberd & Mustoe, 1997, 2000; Mustoe & Hibberd, 1995). After many decades of stable curricula and teaching methods—typically, substantial lecture courses of up to 100 hours per academic year delivered by mathematics lecturers to large groups of nonspecialist students (and the notion of delivery very much underlines the most common

¹⁹ See, for example, the Mathematics, Statistics and OR Network [<http://www.mathstore.ac.uk>] and the Engineering Subject Centre [www.engsc.ac.uk].

approach, based on transmission of mathematical content, accompanied by students doing extensive exercises in techniques)—recent years have seen some major changes. The impetus for change in mathematics teaching has come from several directions. From the direction of professional engineering comes the enormous change throughout professional practice with the arrival of cheap and ubiquitous computer technology. In many cases (in civil engineering, for example), this has fundamentally changed the technical nature of work, and this is now being reflected in a deep-rooted re-appraisal of undergraduate engineering education. Indeed, new attitudes to “knowledge” are developing in engineering education (Kent & Noss, 2003). More and more knowledge is becoming accessible and potentially relevant to the practicing engineer—not only mathematics and physical science, but knowledge concerned with materials, construction techniques, design methodology, project finance, law, environmental issues, and so on. Broad agreement is emerging amongst engineers that the way to deal with this “knowledge explosion” is to implement a shift in emphasis from teaching built around subject knowledge (i.e., the topics of engineering theory and science, and issues of professional practice) toward teaching about the process of engineering (how knowledge is operationalized), using engineering design as an organizing and motivating principle of an engineering degree.

The other direction of impetus comes from school mathematics: During the 1990s, in the UK (where expansion of university-level education was and remains a key government education policy) and in some other countries, a growing proportion of young people went to university. At the same time, this was accompanied by a widely perceived (but usually officially denied) perception that the level and overall quality of mathematical preparation in schools has declined (Smith, 2004, reports on a major investigation of this phenomenon). To what extent the decline is true or not, certainly whereas in the past a relatively elite group of students would enter university to take degrees in engineering and science, today’s students come from a far more diverse background and have a far broader range of mathematical competence (cf. SEFI, 2002). School mathematics in many countries has also changed radically during the last 20 or so years—a common significant change is the “democratization” of mathematics in schools, a movement that has seen a decline in formal mathematical activities by students (exercises in geometry, algebra, calculus, etc.—mathematics regarded as a body of knowledge to be acquired) and an increase in student-generated investigative activity (projects, etc.—

mathematics regarded as a process in which students can participate).

From the perspective of service mathematics, university teachers ironically seem to be unhappy with a style of mathematics curriculum in schools that emphasizes a way of working that can help to develop students’ abilities in problem solving and applying mathematical ideas—of course, this very much depends on how, and how well, this is done. Yet one can sense a missed opportunity in this situation: Where there seems to be a possibility for making connections, the debate in universities has focused on the emergence of a gap between the expectations of universities about new students and the students’ actual mathematical capabilities; this has been termed the *mathematics problem* in the UK. To a significant extent, however, arguably the notion of “problem” is more a reflection of a dominant conservative viewpoint towards mathematics teaching than a problematic reality. As Steen (2001) describes, through the particular influence of the computer mathematics is becoming more and more relevant to more and more professional workers, including engineers, and therefore a rich *opportunity* is opening up for mathematics education, but this will require mathematicians to be more open to consider changes in the content and approach of mathematics teaching—either that, or to see more and more mathematics teaching being undertaken by the users of mathematics, such as engineering departments, themselves. Here, for example, is a comment recorded (by PK) from a mathematician specializing in engineering mathematics teaching at a large technical university in the UK: “We’re pretty traditional—you would not see much difference between what we do now and 20 years ago, except that the level is lower now.”

The Influence of Technology

Implementation of technology has been a big concern generally in university teaching since the early 1990s, when personal computers became accessible in significant numbers to students in higher education. One of the striking things about this has been the tendency for researchers and especially implementers to regard the higher education setting as distinct from the school setting, and many lessons learnt from the implementation of IT in school level education have noticeably *not* been learnt by those in higher education. For example, technology enthusiasm was particularly rife in the early to mid-1990s, much as in schools about a decade earlier, often in the form of expensive projects that generally proved unsustainable after the special project funding was taken away.

The strength of enthusiasts is that they innovate, they try new ideas, they explore where more cautious colleagues might wait and see. Their weakness is that the outcomes of their enthusiastic endeavours are often poorly contextualised, theorised or evaluated. (Burton et al., 2004, p. 221)

It is pretty much recognized, by mathematics educators at least, that for technology to have a sustainable impact on teaching and curriculum, then there has to be some deep-rooted thinking about the aims and methods of teaching—as the anthropological approach in mathematics education illustrates (see above). Of course, to do this in the complex political atmosphere that surrounds a typical university mathematics programme is no easy task, and even today one is inclined to think, pessimistically, that “university mathematics curricula are virtually immune to change; in mathematical terms, we might say that the mathematics curriculum is invariant under intellectual revolutions” (Steen, 1989).

For all the difficulties inherent in technological innovation in mathematics education, it is evident that those involved in teaching mathematics to nonspecialist students cannot ignore it, inasmuch as computer software is ubiquitous in the professional practice of engineers and is becoming a standard tool for engineering students, for example in the form of specialist Computer-Aided Design packages for the different engineering professions, or structural analysis packages in civil engineering.

The case of structural analysis software is a good illustration of the trends. Structural analysis is essentially a deeply mathematical subject and is central to the professional practice of a civil engineer, and thus also central to his or her education. As part of the design process, the engineer needs to predict the behavior of a proposed structure (the walls of a building, a roof, a bridge, etc.). Before the advent of computers, the working life of an engineer, especially in the early part of his or her career, would be dominated by actually doing structural calculations using pen-and-paper, and a large part of the civil engineering degree was therefore dedicated to giving students an understanding and fluency in a variety of calculational techniques. For the majority of engineers today, all such calculations will be done in practice using computer software. The priority now for education is that students may “do” little mathematics themselves in the form of explicit calculation, yet they are likely to use more mathematics than ever before, implicit within computer software:

No longer do we have to grind through long calculations—the computer will do it for us. The challenge

has changed from the ability to do this to the ability to interpret the meaning of mathematics to engineering and herein lies the challenge and change of emphasis. (Blockley & Woodman, 2002, p. 14)

A typical structural analysis curriculum of 10 years ago would begin with a solid dose of analytical theory—using matrix algebra—followed later (perhaps in the 2nd or 3rd year) by working with computer software. Nowadays, work with computer software is likely to come much earlier, and the place of learning matrix algebra as a preparation for working with software is becoming open to question.

What is the Purpose of Mathematics to an Engineer?

Structural analysis is just one of the analytically based areas of engineering where the curriculum is currently being debated and is evolving into new forms (see, for example, Allen, 2000). In general, one can see a tension between the traditional notion of “pushing” mathematical ideas into the engineering curriculum and of “pulling” ideas from mathematics—that is, the need for a particular piece of mathematics can emerge where the engineering curriculum requires it, rather than being pushed into the student prior to having a meaningful context for it. Of course, this has to be done in a systematic way—mathematics cannot in general be understood in a “just in time” fashion.

In university mathematics education, a strong tendency remains to think that *understanding* of a mathematical technique *must* precede its *application* to the engineering context. In the (precomputer) past, a valid objection to any sort of pull-based mathematics was the uncomfortable notion of anyone trying to use a mathematical idea before knowing the techniques of its application. Mathematicians have tended to regard knowledge of techniques as an essential part of what it means to understand an idea and how to apply it. What users of mathematics such as engineers are wanting however, and it seems that mathematicians will increasingly need to provide, is a form of pull-based mathematics in which the use of mathematical software makes mathematical ideas usable, as in the example of structural analysis above. Carefully designed use of IT can make it possible for students to use mathematical ideas *before* understanding techniques, and to make this part of a genuinely rounded mathematical learning experience (a few examples are given in Kent & Noss, 2003).

It is worth considering at this point the whole question of the purpose of mathematics to a (student) engineer. Kent & Noss (2003) in a UK-based study found widespread agreement across the civil engineering profession about what is the desirable mathematical

competence of a graduate: “Mathematics should instil disciplined thinking and rigour in the development of arguments based on assumption and simplification in modelling, should teach the importance of controlled approximation, and, above all, impress upon students its value as a tool to be invoked when quantitative evidence is needed to underpin assertion, hypothesis, or sheer physical intuition” (Nethercot & Lloyd-Smith, quoted in Kent & Noss, 2003, p. 8). Whilst few would dissent from such a description, there are certainly significant questions about what meanings people attach to the different terms, even, what is “mathematics”? It is notable in talking to experienced engineers that the mathematics that is useful and relevant has in many cases long since come to be thought of as part of engineering, whereas mathematics that is *not* used is regarded as “mathematics.”

Again according to Kent and Noss’s (2003) survey, practicing engineers do not tend to regard mathematics as a problem area—contrary to the sometimes obsessive concern of academics about the “mathematics problem” (see above). Practitioners regard confidence with mathematics as crucial for the majority of engineers, and above all they require a balance of skills across engineering teams. Different employers require different balances, and even specialist analytical design consultancies in civil engineering reported that they only require 10–20% of their engineers to have specialist skills in analysis.

In the past, engineers had to learn a lot of mathematics for practical purposes. At the same time, they could be expected to absorb some understanding of mathematics as a logical way of thinking, and its importance as part of a practicing engineer’s expertise. The availability of computer software for calculations has undone this relationship between practical and theoretical aspects. The teaching of practical mathematics is becoming much more focused on the process of modeling of engineering systems—this results in a decrease in the teaching of calculation techniques, but it does not mean that all manual work with mathematics could be replaced: The right balance must be found. To teach mathematics as a way of thinking, a traditional model exists in the use of formal mathematics—older engineers do look back on courses in Euclidean geometry in their schooldays as having significantly shaped their development. It seems unlikely that formal geometry can re-occupy such a dominant place in the school mathematics curriculum (for one thing, there are too many competing demands on the school curriculum), and new models for this aspect of mathematics learning are not yet widely agreed. Indeed, is a logical way of thinking *only* to be gained through studying “pure”

mathematics? Could the same kinds of analytical problem solving also be developed through, for example, requiring all students to develop substantial skills in (mathematical) computer programming? Certainly in the school context this potential for computer programming has long been advocated, most famously in the case of the language Logo (see Cuoco et al., 1996; Noss & Hoyles, 1996).

Modeling and the Development of “Problem Solving” Curricula

Another emergent factor in engineering mathematics is the role of modeling. However, this should be understood more broadly than it is sometimes interpreted in undergraduate mathematics. Often, the translation between the physical situation and the mathematical analysis is regarded as a small part of the task, whose focus is on the applied mathematics of solving the equations in the model (in the worst cases, the physical situation is but an excuse for students to do a routine exercise in applied mathematics). Yet the process of translation, which involves on the one hand developing a mathematical model of the situation and on the other hand interpreting the mathematical analysis back into the context, is in general a rather complex task, and is intimately related to the issue of learning *how to use* (rather than *how to do*) mathematics, which we suggested above is becoming the focus of engineering mathematics education. See Bissell and Dillon (2000) for an extremely perceptive engineers’ view on the nature of modeling.

As discussed above, an analogous situation to this lies in some aspects of applied mathematics teaching, such as differential equations (see, for example, Stephan & Rasmussen, 2002). As the use of differential equations has been transformed by the availability of mathematical software, both numerical and symbolic (computer algebra systems), both a possibility and a need has opened up to teach about equations differently, less focused on doing techniques of solving differential equations, more focused on the meaning of differential equations, in terms of their geometrical structures and their domains of application.

One of the major trends of recent years in engineering education generally that illustrates the issue of modeling is a growing number of experiments with various forms of *problem-based learning* (PBL). (A few exceptional universities—such as Roskilde University in Denmark—embraced this mode of learning a long time ago, cf. Niss, 2001). The idea behind this is that lecture-based teaching is largely replaced by students working on projects, usually in teams, and being assessed through group-work and continuous assess-

ment. The projects develop increasing complexity as students progress from year to year. Problem-based learning originated in the 1960s in medical education, where it has since become increasingly common. The appropriateness of PBL has been criticized for engineering studies because of its origins in medical education: The argument goes that PBL is good for fact-based knowledge, such as medicine, but is less appropriate for an analytically based subject like engineering (cf. Spencer-Chapman, 2000). Yet making abstract analysis meaningful depends crucially on the student making connections between engineering and mathematics through solving problems (Kent & Noss, 2003); just as in the case of differential equations mentioned above, meaning develops by providing activities for students to develop connections within the subject (e.g., the geometrical structure of differential equations), as well as making connections to other subjects.

Statistics and Probability at Post-Secondary Level

In this section we summarize the research carried out in advanced stochastics, with particular emphasis in the recent emergence of a statistics education research community, which has some links but is almost independent from the advanced mathematics education research community. We reflect on the specific characteristics of advanced stochastics and the potential research challenges that this area sets to educational research.

Many features described for the teaching of “service mathematics” to engineers also apply to the case of statistics. As mentioned above, statistics is one of the most widely taught topics at university level, where it is mainly studied as a tool to solve problems in other fields. However, the variety of previous knowledge and interests of the students involved is even larger than in the case of service calculus, because statistics is taught with almost no exception in all the university majors, from engineering to education, geography, psychology, biology, sociology, journalism, economics, linguistics, and so on. There is also a general dissatisfaction with the mathematically oriented approach to teaching in traditional courses, which often emphasize the teaching of formulas for calculating statistics (e.g., correlation coefficients or confidence intervals) without much concern towards the data context and interpretative activities or simulations that could help students improve their stochastic intuitions. In many cases, the courses are over-mathematized, which involves meeting concepts of advanced stochastic think-

ing for students who do not have any prior or concurrent experience of advanced algebra or calculus.

Technology has also had a major impact in the practice and development of statistics in expanding the range of processes that statisticians and users of statistics can employ to collect, analyze, and interpret data and the amount and type of data they can analyze. This has resulted in the development of new statistical methods, such as exploratory or graphical data analysis, resampling techniques, or data mining and has facilitated the implementation of some techniques that were previously difficult to make use of, such as multivariate or Bayesian analysis. This no doubt has also influenced the increasing demand for statistics education, inasmuch as many people can now apply a method with scarce knowledge of the complex mathematical calculation behind the user-friendly software that is apparently easy to learn. For example, an education or psychology student can easily enter data in a computer, use statistical software to perform a factor analysis in a few minutes, and give an intuitive interpretation of the factors retained in terms of the problem without knowing what an eigenvalue is, how eigenvalues are extracted, or how the different rotations of the components have been obtained.

Modeling and problem solving are also central to the teaching of statistics to professionals. However, statistical problems are often open-ended or ill-defined and have multiple possible solutions. Although model abstraction is common to mathematics and statistics, the context plays a fundamental role in guiding the selection of a statistical model (delMas, 2004), and selecting the model is frequently harder than subsequent mathematical reasoning in working with the model. Algebraic work on the model, for example, is very limited, and although the processes of generalizing, classifying, conjecturing, inducing, analyzing, synthesizing, and abstracting described by Dreyfus (1991) for advanced mathematics still apply to statistics, the meaning of representing, visualizing, or formalizing is quite different. As we will argue later, representing and visualizing apply mainly to the data and are used to create meaning from these data, for example helping to find an adequate model or identifying unexpected sources of variation. Formalizing in statistics applies not just to the mathematical work with models, but to the definitions of units, variables, and categories for data analysis, or designing instruments to measure them, which again are rooted in the context of application. Whereas mathematical practice can be removed from real-world context, statistics practice is highly dependent on the problem context, and this

dependence may lead to reasoning errors that are hard to overcome.²⁰

There is an interest and tradition for educational innovation in the teaching of statistics, and this topic is also taught at advanced level by lecturers with a variety of backgrounds, mostly statisticians, but also economists, health care professionals, engineers, psychologists, or educators. Contrary to the case of engineering, mathematicians or mathematics educators teaching statistics are only a minority. This explains the fact that, until very recently, research in advanced stochastic teaching and learning has not attracted mathematics educators and thus has had a small presence at, for example, the annual Psychology of Mathematics Education Conferences. Of course, this does not mean that the teaching of advanced statistics is without problems or that existing related research on this theme is lacking.

A main difference compared with the teaching of mathematics to engineers is the long tradition of educational research related to the teaching and learning of statistics, and a substantial part of it has been carried out outside the mathematics education community. Research in stochastics thinking, teaching, and learning started during the 1950s with the pioneering work by Piaget and Inhelder (1951) and has always had an interdisciplinary character. As stated by Shaughnessy (1992, p. 465): “*The cross fertilization of research traditions and methodologies in probability and statistics makes it one of the theoretically richest branches in mathematics education.*” Different fields have contributed with various research paradigms and theoretical frameworks, which have been analyzed in different surveys of research, such as Scholz (1991), Shaughnessy (1992), Jones and Thornton (2005), and other chapters in this *Handbook*. Whilst research was still relatively scarce when the first *Handbook* was published, compared with other areas of mathematics it has experienced a notable growth in the past 10 years. However, this has mainly concentrated in elementary and secondary school levels, perhaps unsurprising given the much greater emphasis that recent new curricula for these levels have given to statistics and probability.

Research into advanced stochastic reasoning has interested psychologists for decades. Because psychology is an experimental science that heavily relies on statistics, the efforts to justify the scientific character of this field have led psychologists to examine the va-

lidity of their research paradigms, including the use of statistics in empirical research. An amazing observation is that statistical inference and particularly significance tests were found to be misunderstood and misused by psychologists and experimental researchers at large over 30 years ago, and that the situation still persists in spite of strong debates ever since (Harlow, Mulaik, & Steiger, 1997; Morrison & Henkel, 1970). Moreover, researchers in the field of reasoning under uncertainty have suggested that, even after statistical instruction, students and professionals tend to continue to make erroneous stochastic judgements and decisions (Kahneman, Slovic, & Tversky, 1982). The diffusion of these research results and the increasingly easy access to powerful and user-friendly computers and statistical software, which save teaching time previously devoted to laborious calculations and allow a more intuitive approach to statistics (more real data, active learning, problem solving, and use of technology to illustrate abstract concepts through simulation) have led statistics lecturers to increase their concern towards didactical problems (see, e.g., Moore, 1997). Consequently, a general effort exists to create curricular materials and to evaluate teaching and learning at university level. The influence of the International Association for Statistical Education (IASE) created in 1991 serves to establish links among the different communities interested in statistics education and to support a more systematic research program. Below we summarize these contributions.

Psychological Research on Advanced Stochastic Thinking

Different theoretical approaches in the field of psychology have tried to explain people's poor performance in probability and statistical tasks, which have been widely documented in relation to different concepts that can be considered as part of advanced stochastics, such as randomness, compound probability, association in contingency tables, correlation, conditional probabilities, Bayes problems, sampling, and the test of hypotheses. Traditionally, decisions under uncertainty are defined by incomplete information about the situation, that is, the possible alternatives or their outcomes are only known (at best) in term of probabilities, a feature of many professional tasks (e.g., medical diagnosis, jury verdict, educational assessment). A *fallacy* is the result of a cognitive process that leads to an incorrect conclusion and may consist

²⁰ For example, a very simple rule is computing the probability of the conjunction of two events in terms of simple and conditional probabilities: $P(A \cap B) = P(A) \cdot P(B|A) = P(B) \cdot P(A|B)$. A consequence of this rule is that the probability $P(A \cap B)$ cannot be higher than the probabilities of either of the two individual events. However, when $P(A)$ is very high and $P(B)$ small, people forget the rule and consider $P(A \cap B) > P(B)$. This behavior was first described by Tversky and Kahneman (1983) and termed the *conjunction fallacy*. So, for example students who are asked whether it is more likely that the government will increase the number of grants or that the government will increase the number of grants and the salary of members of parliament will consider more likely the second event.

of the application of an inadequate model or incorrect application of intuitive rules of inference. In trying to explain the mechanisms leading to a fallacy, Kahneman and his collaborators developed the *heuristics and biases* program (Kahneman, Slovic, & Tversky, 1982), which was the dominant paradigm in the early eighties and is still very influential. This assumes that people do not follow the normative mathematical rules that guide formal scientific inference when they make a decision under uncertainty and that, instead, they use simpler judgmental heuristics. Heuristics reduce the complex tasks of assessing probabilities and predicting values to simpler judgmental operations and are in general useful; however they sometimes cause serious and systematic errors and are resistant to change. For example in the *representativeness* heuristics, people tend to estimate the likelihood for an event on the basis of how well it represents some aspects of the parent population. An associated fallacy that has been termed *belief in the law of small numbers* is the belief that even small samples should exactly reflect all the characteristics in the population distribution. Then for example a mother of two boys may incorrectly assume that the likelihood of having a third baby who is a girl is greater than the likelihood of having a baby boy, because a family with 2 boys and a girl is more representative of the general population.

A huge amount of research in stochastics education at both undergraduate and advanced level has been based on the idea of heuristics (see a summary in Shaughnessy, 1992). Large-scale studies carried out by psychologists showed misconceptions related to these heuristics, in which participants assumed that sampling distributions (distribution of the value of the mean and other parameters in repeated samples from the same population) were independent of the sample size, whereas in mathematical theory the spread of these distributions should be inversely related to the sample size. Direct consequences of this for the practice of statistics are that users have an overconfidence in the results of statistical tests, underestimate the width of confidence intervals, and expect a very close result in replication of the experiment, even with a small sample. Recent research on heuristics at advanced level includes studies on: assessment (Garfield, 2003; Hirsch & O'Donnell, 2001), the effect of teaching experiments on the use of heuristics (e.g., Barraqués, 2002; Pfannkuch & Brown, 1996), and misconceptions related to concepts such as randomness (Falk & Konold, 1997) in terms of heuristics or providing alternative explanations for incorrect responses to tasks proposed in research about heuristics. For example, Konold (1991) suggested that students might confuse frequentist probability, which is objective and refers to

the frequency of occurrence of an event in relation to a population, with epistemic (subjective) probability referred only to an isolated event. Then they might confuse the task of assigning a frequentist probability with the task of predicting the next outcome (the *outcome approach*) in a random experiment. This also involves a deterministic view of an uncertain situation, because in the outcome approach, participants tend to rely more on causal explanations rather than accept explanations due to chance and variability.

A different theoretical model in the psychology of decision is the *abstract-rules* framework (Nisbett & Ross, 1980) in which people are assumed to acquire a form of correct statistical reasoning and develop intuitive versions of abstract statistical rules, such as the law of large numbers, which are well adapted to deal with a wide range of problems in everyday life, for example, estimating the average time spent to perform a repetitive task or the approximate cost of the shopping in a supermarket, but the rules fail when applied beyond that range. Such rules are used to solve statistical problems, in which one recognizes some cues in the problematic situation. With respect to training in statistical reasoning the suggestion is that the quality of human inferences and judgments can be improved via statistical instruction: "Many of the inferential principles central to the education we are proposing can be appreciated fully only if one has been exposed to some elementary statistics and probability theory" (Nisbett & Ross, 1980, p. 281), although the suggestion is that statistics teaching should take into account students' intuitive strategies and errors. Moreover, for these authors, even brief formal training in inferential rules may enhance their use for reasoning about chance events; either teaching statistical rules or teaching by having students solve example problems would work. As regards training in conditional logic or conditional probability students need to be trained simultaneously in abstract logical rules and in problem-solving strategies. Problem solving is improved, according to these authors, when the sample space is clearly defined, people recognize the role of chance in the experiment, and the context forces the person to think statistically.

A more recent theoretical framework is the *adaptive algorithms* approach (Cosmides & Tooby, 1996; Gigerenzer, 1994), which supposes that people possess evolutionarily acquired cognitive algorithms that serve to solve complex probability problems, such as conditional probability or problems involving Bayes theorem. Adaptive algorithms serve to solve adaptive problems (such as finding food, avoiding predation, or communicating) and take a long time to be shaped, due to natural selection. Because adaptive algorithms are shaped by natural

environments they are more effective when the tasks are presented in a format close to how data are perceived and remembered in ordinary life. According to this theory people should have little difficulty in solving statistical tasks if the probability data are presented in a natural format of absolute frequencies instead of using rates or percentages (Sedlmeier, 1999). Although the percentage, rate, or frequency representations of probability values in the problem are mathematically equivalent, inasmuch as they can be mapped onto each other, they might be not psychologically equivalent. Gigerenzer (1996) argued that the frequency representation makes some cognitive illusions in statistics disappear and that the cognitive algorithm used to solve the same problem changes with the information representation.²¹

Sedlmeier (1999) analyzed and summarized recent teaching experiments carried out by psychologists that follow either the abstract-rules or adaptive-algorithms theories and involve the use of computers. The results of these experiments favor the adaptive-algorithms approach as an explanation for people's performance in stochastic tasks and suggest that statistical training is effective if students are taught to translate statistical tasks to an adequate format, including tree diagrams and absolute frequencies. These experiments have concerned areas such as compound probabilities, conditional probability, the Bayes theorem, and the impact of sample size on the sampling distributions. However, learning is assessed through participants' performance to tasks that are very close to those used in the training, so evaluating the extent to which students could transfer this knowledge to wider types of problems is difficult. Although this research provides empirical information and potential theoretical explanations for students' difficulties in advanced stochastics, a large amount of work still must be done by mathematics and statistics educators to integrate these results and to use them to design and evaluate teaching sequences in natural settings where students are expected to meet wider curricular requirements.

The Mathematics Education Approach

The approach of mathematics educators is rather different from that taken by psychologists and has fol-

lowed the research paradigms and theoretical frameworks in mathematics education. The mathematical and epistemological analyses reveal that the complexity of concepts, tasks, and students' responses investigated by psychologists is often greater than has been assumed in psychological research and suggests the need for re-analyzing them from a mathematical perspective; for example, centering on isolated types of tasks does not always reveal in depth the students' understanding of a concept, inasmuch as responses are sometimes very dependent on the task variables. Moreover, theoretical constructs taken from mathematics education can contribute to a different perspective for the same phenomena. Take, for example, the concept of correlation, a field that has been extensively studied by psychologists, including Inhelder and Piaget (1955) who considered that the evolutionary development of the concepts of correlation and probability are related and that understanding correlation requires prior comprehension of proportionality, probability, and combinatorics. In these experiments participants are presented two-way tables (two rows and two columns) in which a sample subjects are classified according to the presence-absence of two qualitative attributes, such as eye and hair color (dark or fair hair, brown or blue eyes). Participants are asked to judge if there is a relationship between the two attributes (judgment of correlation) in these 2×2 contingency tables and to justify the procedure to reach their conclusion. In Table 22.1 we describe the data in this type of problem, where a, b, c and d represent absolute frequencies.

Table 22.1 Typical Format for a 2×2 Contingency Table

	B	Not B	Total
A	a	b	a + b
Not A	c	d	c + d
Total	a + c	b + d	a + b + c + d

Piaget and Inhelder found that some adolescents who are able to compute single probabilities only ana-

²¹ For example, in the following Problem 1 (Eddy, 1982) probability data are given in percentages and should be solved with the help of Bayes theorem. Giving the data in a format of absolute frequencies (Problem 2), it transforms to a simpler problem that can be solved by applying the Laplace rule: It is easy to see that out of 107 women with positive test results (8 + 99) only 8 of them will have breast cancer.

Problem 1. The probability of breast cancer is 1% for a woman at age 40 who participates in routine screening. If a woman has breast cancer, the probability is 80% that she will have a positive mammogram. If a woman does not have breast cancer, the probability is 10% that she will still have a positive mammogram. Imagine a woman from this age group with a positive mammogram. What is the probability that she actually has breast cancer?

Problem 2. Ten out of every 1,000 women at age 40 who participate in routine screening have breast cancer. Of these 10 women with breast cancer, 8 will have a positive mammogram. Of the remaining 990 women without breast cancer, 99 will still have a positive mammogram. Imagine a woman from this age group with a positive mammogram. What is the probability that she actually has breast cancer?

lyze the cases in cell [a] in Table 22.1 (presence-presence of the two characters A and B). When they admit that the cases in cell [d] (absence-absence) are also related to the existence of correlation, they do not understand that cells [a] and [d] have the same meaning concerning the association, and they only compare [a] with [b] or [c] with [d] instead. Understanding correlation requires considering quantities $(a + d)$ as favorable to the association and $(b + c)$ as opposed to it. The correct strategy should use the comparison of two probabilities $P(B/A)$ and $P(B/NotA)$, that is

$$\frac{a}{a+b} \text{ with } \frac{c}{c+d}.$$

According to Piaget and Inhelder, recognition of this fact only happens at 15 years of age. Following Piaget and Inhelder, several psychologists have studied the judgment of association in 2×2 contingency tables in adults, using various kinds of tasks. Other psychologists used instead two numerical variables and different numerical, verbal, or graphical representation of data. As a consequence of all this research, it has been noted that participants perform poorly when establishing correlation (see Beyth-Marom, 1982, for a survey) and people frequently use their previous theories about the context of a problem when judging association. The general conclusion is that when data do not coincide with these expectations a cognitive conflict affects accuracy in the perception of correlation (Jennings, Amabile, & Ross, 1982).

Estepa and his collaborators began from these results and, after a mathematical and epistemological analysis of the concept of correlation, designed a questionnaire including a wider range of open-response tasks to give an alternative explanation of pre-university students' strategies and judgments in terms of *misconceptions* regarding correlation (Batanero, Estepa, Godino, & Green, 1996; Estepa, 1993). In the *determinist* conception of association, some students expect that correlated variables should be linked by a mathematical function; students with a *unidirectional* conception of association only perceive direct association (positive sign of the correlation coefficient) and interpret inverse association (negative sign) as independence; *causal* conception consists of assuming that correlation always involves a cause-and-effect relationship between the variables, and in the *local* conception students base their judgment on only part of the data (e.g., they only use one cell in 2×2 contingency tables). These types of conceptions have been confirmed in later research (Morris, 1997, 1999). Another finding was that some mathematical concepts might constitute an *obstacle* when learning correlation, in which case they receive different interpretations. For example, some students

believe a correlation coefficient of -0.7 indicates a smaller degree of dependence than a correlation coefficient of 0.1 because $-0.7 < 0.1$ in the usual ordering of real numbers, whereas in fact -0.7 indicates a stronger dependency than 0.1 .

Estepa and his collaborators organized two teaching experiments based on *computer learning environments* (in the sense of Biehler, 1994, 1997), that is, integrated instructional settings that allow the teacher and a group of students to work with statistical software, data sets, and related problems as well as with a selection of statistical concepts and procedures. The researchers found general improvement in students' strategies and conceptions after teaching, although the unidirectional and causal misconceptions concerning statistical association were harder to eradicate. Using qualitative methods, such as observation, interviews, and the analysis of the students' interaction with the computer, they documented specific *acts of understanding* (following Sierpinska, 1994) of correlation (Batanero, Godino, & Estepa, 1998; Estepa, 1993). For example, students must realize that *the study of the association between two variables has to be made in terms of relative frequencies*; however, in the first teaching session the students tried to solve the problems in terms of absolute frequencies. Although the lecturer pointed out this mistake to them at the end of that session, the same incorrect procedure appeared recurrently for the same students for several sessions, until the students overcame this difficulty. Another example of an act of understanding was realizing that *from the same absolute frequency in a contingency table cell one can compute two different relative conditional frequencies (conditioning by row or column), and the role of the two events in the conditional relative frequency is not interchangeable*. Students' difficulties in discriminating between the probabilities $P(A/B)$ and $P(B/A)$ have previously been described (e.g., Falk, 1986). Again, this was a recurrent difficulty for the students throughout the teaching experiment.

Other researchers have taken mathematics education frameworks to organize research on topics that have scarcely attracted psychologists in complex experimental settings. For example Batanero, Tauber, and Sánchez (2004) used a theoretical framework that incorporates ontological, anthropological, and semiotic ideas (Godino, 2002; Godino & Batanero, 1998) to describe the evolution of students' understanding of the normal distribution in a statistics course based on intensive use of computers. In the theoretical model used, the meaning (understanding) of any mathematical object is conceived as a complex system, that contains different types of interrelated elements: a) problems and situations from which the object emerges, such as fitting a curve to a histogram for empirical

data distributions or approximating the binomial or Poisson distributions; b) representations of data and concepts, for example, histograms or density curves, verbal and algebraic representations of the normal distribution; c) procedures and strategies to solve the problem, such as computing probabilities under the curve, computing standard scores, and critical values; d) definitions and properties, for example, symmetry of the normal distribution, horizontal asymptote, areas above and below the mean, meanings of parameters; and e) arguments and proofs, including deductive and informal arguments. Godino and Batanero (1998) argued that the understanding of a concept emerges from a person's meaningful practices linked to repeated solution of problems that are specific to that concept. Mathematical problems promote and contextualize mathematical activity and, together with actions, constitute the praxemic or phenomenological component of mathematics (*praxis*) as proposed by Chevallard (1997). The three remaining components (concept-definitions, properties, arguments) are produced by reflective practice and constitute the theoretical or discursive component (*logos*). Godino and Batanero also described different dual facets or dimensions of mathematical knowledge, in particular distinguishing between the *personal* and the *institutional* meaning to differentiate between the meaning that for a given concept has been proposed or fixed in a specific institution and the meaning given to the concept by a particular person in the institution.

This theoretical framework was specially suited to describe the complexity of understanding the normal distribution in Batanero, Tauber, and Sánchez's (2004) research, because the understanding of the normal distribution in their experiment was based on the students' previous knowledge about many interrelated concepts such as statistic and random variables, frequency and probability distribution, parameter and statistics, center and spread, symmetry and kurtosis, histogram and density curve, areas under the normal curve, mathematical model and empirical data, their different representations, procedures, and properties. The teaching was based on intensive solving of real problems in which students analyzed different real data sets taken from fields in their areas of interest, which served to gradually suggest to them to reflect about the different elements of meaning of the normal distribution. The idea of different institutional meanings also describes well the changes in meaning and understanding implied by the use of computers: different types of problems (starting from real, multivariate data sets related to a problematic situation that the students should model), representations (wider and quicker availability of interactive

dynamic graphs, tables, and numerical summaries), procedures (knowledge of software options and interpretative abilities substituted for classical probability computations); graphical and iconic language replacing algebraic manipulation, and types of proof (simulation and visualization, instead of deductive proof).

Finally the framework served to picture the general tendencies in the students' *personal meaning* for the normal distribution after the instruction, as well as the variety of these meanings, and to identify main agreements and differences with the intended *institutional meaning*. At the end of the teaching students were given written questionnaires to evaluate specific understanding about particular properties, representations, and procedures and an open-ended task, to be solved with the computer that referred to a data file the students had not seen before. The students were to choose a variable that could be well fitted by a normal distribution, then explain and justify their responses in detail (the file contained data from a real context with 10 different qualitative and quantitative variables, only 2 of which were acceptable solutions to the problem). The students were given freedom to use any resource of the software (statistical graphs or analysis) to support their choice. Quantitative analysis of responses to questionnaires and semiotic analysis of the students' written protocols in the open tasks as well as interviews with a small number of students were used to describe the students' correct and incorrect reasoning about the normal distribution. For example, many students could relate the idea of symmetry to skewness coefficient or to relative position of mean, median, and mode; they were able to compare the empirical histogram and density curve shapes to the theoretical pattern in a normal curve and relate the various graphic representations and data summaries to the geometrical properties of normal distribution. Even when the majority of students learnt to use the software, some of them had difficulties in interpreting areas in frequency histograms produced by the software, in computing areas under the normal curve, or in discriminating between the empirical data and the mathematical model (normal curve).

Influence from Statistics

Another strong influence on research comes from the field of statistics, where interest in education arose especially since the creation in 1949 of an Education Committee by the International Statistical Institute (ISI), through which the ISI promotes the training of official statisticians in developing countries and has organized a series of Round Table Conferences on specific educational problems since 1973 (Vere-Jones, 1997). The International Conferences on Teaching

Statistics (ICOTS) were started in 1982 by the ISI and have continued every 4 years. In 1991 the IASE was created as a separate section of the ISI and took over the organization of these conferences. The journals *Teaching Statistics*, first published in 1979, and *Journal of Statistics Education*, started in 1993, soon became important tools to improve statistics education all over the world. This activity complemented the educational sessions at international and national statistical conferences (e.g., in the biannual Sessions of the International Statistical Institute or in the American Statistical Education Conferences) as well as the Stochastic Thinking Reasoning and Literacy Research Forum started in 1999. With regard to the teaching, learning, and understanding of stochastics at post-secondary level these conference activities promoted a large and varied number of teaching experiences, and many suggestions were published in proceedings and journals. At the same time, some systematic research has been progressively emerging under the influence of the IASE and of several funded projects and initiatives by the American Statistical Association (e.g., the Undergraduate Statistics Education Initiative, USEI,²² and the Consortium for the Advancement of Undergraduate Statistics Education, CAUSE²³).

One example is the research carried out by Garfield, del Mas, and Chance who developed a piece of didactic software (*Sampling Distribution*) and complementary instructional materials (pretest and posttest, instructional objectives, activities) to assess students' previous misconceptions, support discovery and exploration of inferential concepts, and assess change after instruction (Garfield, del Mas, & Chance, 1997). The software allows student to choose among a variety of possible distributions for a continuous variable in a population, including nonstandard models of distributions. In addition it provides different windows where the student can simulate the drawing of samples (for any size and number of samples) from the specified population and visualize the sample outcomes as well as the values of some statistics (e.g., mean, median, variance) in the different samples (sample statistics). From these, there is the possibility to graph the set of different values for these sample statistics to get an empirical sampling distribution for the statistics. Students can add new samples to their previous results little by little, in order to discover the long-run patterns in the empirical sampling distribution and understand the features in the theoretical sampling distribution (distribution of all the possible values of the statistics for a given population and sample size). For

example, they can "discover" for themselves the central limit theorem, according to which the sampling distribution of mean and other statistics will approach to a normal distribution for moderate sample sizes ($n > 30$) even for nonsymmetrical population distributions. They can see how the approximation improves with the sample size, and that as the spread in the sampling distribution decreases the average (value of the sample statistics) approaches to the value of the population parameter.

The same authors carried out a teaching experiment with three groups of students at the university introductory level (delMas, Garfield, & Chance, 1999). The students were supposed to have read a chapter on sampling distribution and the central limit theorem before engaging with hands-on simulation using the software. Class discussions served to compare the shape, center, and spread of empirical sampling distributions for different sample sizes and population distributions and whether empirical results agreed with what was expected in the central limit theorem. The authors were surprised that after the first series of experiments many of their students still showed serious misconceptions. For example, the variability the students expected in the distribution was not consistent with the sample size, and they did not understand that the sampling distribution would resemble a normal distribution with increasing sample sizes. Much better results were obtained when the authors followed a model of learning through conceptual change in which students were asked to test their predictions and confront their misconceptions, following a constructivist model of learning. The authors here concluded that students with misconceptions have to experience contradictory evidence and reflect on this contradiction with their previous expectations before they change their views about random phenomena.

To gather more detailed information about students' reasoning they carried out several guided interviews that suggested a model of developmental stages in students' statistical reasoning: idiosyncratic, verbal, transitional, procedural, and integrated process reasoning (Chance, delMas, & Garfield, 2004), based on work by neo-Piagetian cognitive researchers (e.g., Biggs & Collis, 1991) who have examined the process of cognitive development in everyday and school contexts. In this model students progress from knowing words and symbols without understanding their meaning (idiosyncratic level) to a substantial understanding of the process of sampling and sampling distributions (process level) in a series of stages. Similar mod-

²² <http://www.amstat.org/education/index.cfm?fuseaction=usei>

²³ <http://www.causeweb.org/>

els of developmental stages have also been proposed by some mathematics educators for some elementary stochastics concepts, in research carried out with primary and secondary school students (Jones, Langrall, Mooney, & Thornton, 2004). Finally, we should note this kind of research focusing on the identification on misconceptions and developmental stages is rather classical from a mathematics education point of view.

Although many different theoretical concepts in research carried out by statisticians are taken from psychology or mathematics education, some statisticians are trying to develop specific frameworks to describe statistical thinking as a specific type of thinking that recognizes the variation around us and includes a series of interconnected processes, aimed at identifying, analyzing, quantifying, controlling, and reducing this variation in order to improve or inform decisions and actions in many different fields (Snee, 1990). A notable example is the theoretical framework developed by Wild and Pfannkuch (1999) and Pfannkuch and Wild (2004), who conducted qualitative research on the activities carried out by students and professionals when engaged in statistical investigations from which the authors developed a complex four-dimensional framework including four components. The first component is termed the *statistical investigation cycle PPDAC* (*Problem, Planning, Data, Analysis, Conclusion*) and describes the activities carried out when identifying and solving a statistical problem that is embedded in a wider contextual problem. This cycle includes identifying and setting of the problem, planning for its solution, collecting data, data analysis, and reaching conclusions. Secondly there is an *interrogative cycle*, which is a generic process in statistical problem solving consisting in generating possibilities for causes or explanations, seeking or recalling information, interpreting, translating or comparing information, criticizing this information from both internal and external points of view, and judging the reliability or usefulness of the information. The third component in the model describes the *types of thinking* present in statistical problem solving. In addition to strategic thinking the authors define fundamental modes of stochastic thinking, which include recognizing the need for data, transnumeration (numerical transformation to facilitate understanding, for example, classifying, coding, or representing data), identifying, explaining controlling and measuring variation, use of mathematical models, and synthesis of statistical and contextual knowledge. Finally the model describes a series of *dispositions*, such as curiosity, imagination, or skepticism. Following the publication of the Wild and Pfannkuch paper a number of research projects have focused on teaching or assessing statistical reasoning and its several components as

a whole. For example Pfannkuch, Rubick, and Yoon (2002) described how students use transnumeration or perception of variation to build up recognition and understanding of relationships of variables in small data sets. Another example is the research by Biehler (2005) who analyzed about 60 students' statistical projects and used Wild and Pfannkuch's model to develop an assessment scheme and a project guide to improve the quality of these projects.

The Challenges of Advanced Stochastics

The above summary of research suggests that probability and statistics pose a number of important challenges to research on mathematics thinking and learning at post-secondary level. First, the existing research has been carried out by different scientific communities, and not just by mathematics educators, and for this reason the sources of information are widespread and not always easily available. At the same time, the diversity of research problems and approaches is very wide, something that is also common for undergraduate stochastics. Some research agendas posed by Shaughnessy (1992) and Shaughnessy, Garfield, and Greer (1996), as well as the creation of the *Statistics Education Research Journal* with the specific purpose to promote research show a tendency to unification and linkage of the isolated pieces of research towards developing a more general knowledge about statistical education. However, the influence from philosophical views about randomness, probability, and statistical inference that still continue today is reflected in stochastics teaching and research: "*in the field of probability there continues an ongoing fierce debate on the foundations, even though for pragmatic reasons only, this debate has calmed down in recent times*" (Borovcnik & Peard, 1996, p. 239). These different views influence the role given to probability in the curriculum, from being the central core, to trying to teach statistics without resort to probability (focus on exploratory data analysis only) or favoring classical, Bayesian, or mathematical-abstract approaches to inference.

Secondly, in the case of statistics, the distinction between advanced and elementary topics remains very fuzzy, inasmuch as current curricula in many countries include ideas about association and inference in secondary school alongside situations that require mature stochastic thinking for correct interpretation, such as voting, investment, research planning, or quality control, topics that increasingly form part of the information to which many citizens and professionals are exposed. So, whereas advanced mathematical thinking tends to be used only formally and after some systematic training, advanced stochastic thinking is now being formally or informally used

by many people with little formal training in either mathematics or stochastics. Moreover some apparently simple concepts, such as randomness, that are taken for granted in elementary courses are in fact very complex. For example, judging whether a sequence of outcomes is likely to have been randomly generated involves being able to measure a number of parameters including relative frequency, length and distribution of runs, variation found in subsets, estimating probabilities, and so on. These skills are frequently taught in school; however research in both psychology and education has shown frequent errors and misconceptions related to such simple ideas as probability, average, and distribution. Furthermore, because no random sequence of outcomes exactly fits the expected patterns suggested by probability theory, judging randomness in a particular situation also requires an understanding of hypothesis-testing logic and sampling distribution features. These are both advanced stochastic ideas with which students have many difficulties. Moreover randomness is assumed in building a sampling distribution or in hypothesis testing, so a sound understanding of these two concepts in turn actually rests on the idea of randomness (thus creating a circular situation).

Stochastics is a field in which the need to present advanced concepts to a wide audience of students with varied backgrounds, interests, and capacities is urgent, given the implications of poor stochastic thinking in many fields of human activity. For example, recommendations to substitute or complement statistical tests with confidence intervals (e.g., Wilkinson, 1999) do not take into account the fact that their appealing feature is based on a fundamental misunderstanding (Lecoutre, 1998), which consists of thinking of the parameters as random variables and assuming that confidence intervals contain the parameters with a specified probability. Such interpretation is incorrect in a classical inferential framework, although it is acceptable in Bayesian statistics. For this reason some researchers (e.g., Lecoutre & Lecoutre, 2001) are suggesting to change the practice of statistics towards Bayesian methods and suggest that Bayes's thinking is more intuitive than frequentist probability for students and better reflects students' everyday thinking about uncertainty. Reported results from research focused on teaching Bayesian statistics are limited to a few cases (Albert, 2000), and the decision to change from classical to Bayesian approach is also dependent on researchers' own objective or subjective views of probability. The wide research done on students' and professionals' misunderstanding and misuse of advanced statistics should be complemented by a similar effort in designing and evaluating teaching ex-

periments oriented to help students and researchers overcome these difficulties. In this sense, statistics and probability can be a paradigm for finding ways and approaches to introduce advanced mathematical ideas to wide audiences and to rethink the very meaning of what is advanced mathematical thinking.

CONCLUSION

In this chapter, we have tried to synthesize the evolution of research on mathematics thinking and learning at post-secondary level since the first *Handbook* was published in 1992, and to analyze its most important advances, their potential, and limit for understanding and improving teaching and learning processes at this advanced level. This evolution, which has been fostered by both internal and external factors, has not obeyed a simple dynamic, and the multiplicity of its facets reflects both the intrinsic diversity of educational research and the diversity of the changes that have affected post-secondary education during the last decade. Some of these evolutions were highly predictable, such as the development of the several theories of reification focusing on process-object duality; the increasing attention paid to the semiotic dimension of mathematical activity and to the essential role played by connections between representations, settings, and perspectives in mathematical thinking and learning; and the increasing influence taken by sociocultural and anthropological approaches towards learning processes. Others were less predictable, such as the increasing theoretical influence of current developments in cognitive sciences and embodied cognition, the rapid changes and growth in mathematical practice fostered by information and communication technologies, the reflection of these changes in post-secondary education mathematical curricula that today oblige educational researchers to expand their privileged fields of investigation, the increasing demand of advanced mathematics learning by varied types of students, and the interest in research on teaching and learning mathematics from a variety of disciplines (not just in mathematics education). All these evolutions make the landscape of research on mathematics thinking and learning at post-secondary level today something much more diverse and richer than was the case about 10 years ago.

Looking at what has been achieved, we have the feeling that the research initiated in the Advanced Mathematical Thinking working group of PME in this area, in spite of the diversity of its developments, has avoided a fractionalization of its perspectives and

been able to integrate its previous achievements into complementary and coherent constructs, as we have tried to show in the first parts of this chapter. This certainly provides researchers with a strong and mature basis for addressing the important challenges that research has to face today.

More and more, however, the evolution of post-secondary education obliges us to reconsider the answers we have given to fundamental epistemological issues about the nature of mathematics, and the nature of mathematical learning and thinking. These fundamental issues cannot be discussed without taking into account the current reality of mathematical practices, or by considering only the practice of pure mathematicians working in traditional fields. Post-secondary educational research has from this point of view a specific epistemological role to play in educational research thanks to its proximity with the professional world of mathematics. The increasing importance taken in post-secondary mathematics education by service courses faces us with the necessity of taking a wider perspective. As has been shown in the fourth part of this chapter, we are pushed and questioned by the evolution of mathematical practices in professional fields such as engineering. In the case of statistics, we observe a progressive separation of mathematics from the applications of statistics that originates in the fact that statistics has changed much more rapidly than mathematics, is much more dependent on the context and on information technology, and is usually taught at post-secondary level by lecturers who are seldom mathematicians. At the same time, an increasing amount of research is being carried out in advanced stochastics outside the mathematics education community.

Technology has deeply changed professional practices and mathematical needs and as a consequence changed what has to be learnt and how it can be learnt. As has been argued for the case of stochastics and engineering, these changes oblige us to see in technology more than an educational help, but something constitutive of mathematical practices, and moreover having the effect of changing mathematical practices as well as changing the meaning of mathematical objects. This has an impact on the learning processes and obliges us also to consider forms of and progressions in learning that are different from the usual ones. At the same time these changed practices should support ways for students with a limited mathematical background to make reasonable sense and use of the very sophisticated mathematics that are implemented in the technology they use, thus producing new, more intuitive meanings for these mathematical objects.

From the theoretical point of view, researchers in these new research fields have taken into account only

a very limited number of the theoretical concepts and paradigms developed by the post-secondary mathematics education community. Therefore the study of the extent to which these frameworks can describe the learning of service mathematics or the learning of advanced stochastics or how these frameworks should be complemented with other constructs specific to these fields is still an open challenge for researchers.

Thanks to the global evolution of theoretical frames towards sociocultural and anthropological perspectives, researchers are certainly today better equipped for addressing this issue of mathematical practices, taking into account their different components, both explicit and tacit, in order to analyze their potential cognitive effects. This evolution has also a corollary that any consideration of learning processes is necessarily relative. Knowledge emerges from practices, and the learning processes one can access are those that are made possible by the existing practices. We have thus to be aware that the answers we can provide to the questions at stake are not absolute; they depend on the current state of educational practice. Research has certainly to pay more attention to these dependences than it has done in the past. This makes it necessary to explore more systematically than has been done before what is learnt and how in educational designs different from the traditional ones, as for instance those mentioned in the fourth part of the chapter where the learning of mathematics emerges from the realization of projects and activities in which modeling and technology are given an essential role.

Even if researchers seem today better equipped for approaching the relationships between learning and practices, and sensitive to the cognitive diversity that emerges from their diversity, benefiting today from the different advances is not at all an easy task. We met this difficulty when writing this chapter. For instance, we found it essential to open the chapter to challenging areas such as stochastics or engineering, and our initial project was to end this chapter by some kind of integrated perspective on learning and thinking at post-secondary level. But connecting research in stochastics and research on AMT, two areas of research that have developed in isolated ways and pushed by different logics, turned out to be too difficult. Much more work than what we were able to do within the time devoted to the elaboration of this chapter would have been necessary in order to succeed. What is certainly expressed in this chapter is the specific coherence underlying each of these approaches, and the ways it tends to question the other one, but no more.

We end this chapter with the feeling that research on mathematical learning and thinking at post-second-

ary level is entering now a new and fascinating phase, with difficult and new challenges to face, challenges that will require to be solved to extend interactions and collaborations beyond the traditional community of research in mathematics education at advanced level.

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AUTHOR NOTE

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▣ P A R T V ▣

Assessment

KEEPING LEARNING ON TRACK

Classroom Assessment and the Regulation of Learning

Dylan Wiliam

INSTITUTE OF EDUCATION, UNIVERSITY OF LONDON

INTRODUCTION

When teachers are asked how they assess their students, they are likely to cite tests, quizzes, portfolios, projects, and other more or less formal methods. When, instead, teachers are asked how they know whether their students have learned something, the responses are typically very different (Dorr-Bremme & Herman, 1986). They mention classroom questions, group activities, discussions, posters, concept maps, and even the expressions on the faces of their students. In fact, the origin of the word *assessment* (Latin *assidere*, literally “to sit beside”) is much closer to this more informal meaning. However, the emphasis on assessment as a formal process is pervasive, and mathematics education is no exception.

To a first approximation, then, the research literature on assessment, both generally and in mathematics education, is almost entirely about the formal methods of assessment, particularly tests and examinations. To make matters worse, even when less formal methods of assessments, such as teacher-made

tests, are discussed, the purpose of an assessment is far more likely to be that of making a determination of a student’s existing state of knowledge. Glaser and Silver (1994) observed that, “Aside from teacher-made classroom tests, the integration of assessment and learning as an interacting system has been too little explored” (p. 403). Thus, even when classroom assessment has been studied, the emphasis has tended to be on the concordance of such measures with external measures—in other words, on classroom assessment as an alternative to external assessments. As Kilpatrick, Swafford, and Findell (2001) noted in their survey *Adding It Up: Helping Children Learn Mathematics*, “Even less attention appears to have been paid to how teachers’ assessments might help improve mathematics learning” (p. 40).

The intent of this chapter is to redress the balance by focusing on the role that assessment can play in supporting learning, rather than just measuring it—sometimes described as a distinction between assessment *for* learning and assessment *of* learning¹—but two further qualifications are needed here. The first is that assessment can support learning in a vari-

¹ In the United States, the term *assessment for learning* is often mistakenly attributed to Rick Stiggins (2002), although Stiggins himself has always attributed the term to authors in the United Kingdom. In fact, the earliest use of this term in this sense appears to be a paper given by Mary James at the annual conference of the Association for Supervision and Curriculum Development in New Orleans (James, 1992). Three years later, the phrase was used by Ruth Sutton as the title of a book (Sutton, 1995). However, the first use of the *prepositional permutation* appears to be the third edition of a book entitled *Assessment: A Teacher’s Guide to the Issues* by Caroline Gipps and Gordon Stobart, where the first chapter is entitled *Assessment of Learning* and the second *Assessment for Learning* (Gipps & Stobart, 1997). The distinction was brought to a wider audience by the Assessment Reform Group in 1999 in a guide for policymakers (Broadfoot et al., 1999).

ety of ways such as, for example, when students actually learn something while completing an assessment, as emphasized by Sternberg and Williams (1998, p. 10)—a process that might be termed *assessment as learning*. As Shavelson, Baxter, and Pine (1992) noted, “a good assessment makes a good teaching activity, and a good teaching activity makes a good assessment” (p. 22). Pellegrino, Chudowsky, and Glaser (2001) and Shepard et al. (2005) summarized the research on the design of such formal assessments, and Lester, Lambdin, and Preston (1997) dealt specifically with the possibilities for mathematics.

Accordingly the specific focus of this chapter is not on how teachers and students can use assessment activities to promote learning. The chapter is about assessment as an essentially interactive process, in which the teacher can find out whether what has been taught has been learned, and if not, to do something about it. It is therefore about assessment functioning as a bridge between teaching and learning, helping teachers collect evidence about student achievement in order to adjust instruction to better meet student learning needs, in real time.

The second qualification is that the focus of this chapter is firmly on the learning of mathematics in the mathematics classroom. Effective implementation of the kinds of exemplary practices identified in this chapter will entail careful consideration of a whole range of issues related to classroom management (see, for example, Brookhart, 2004), teacher professional development (William & Thompson, in press) and educational policy (Looney, 2005). Although these issues are clearly important, they are beyond the scope of this chapter.

The importance of the role of assessment in instruction was explicitly recognized in the National Council of Teachers of Mathematics’ six *Assessment Standards for School Mathematics* (NCTM, 1995), the second of which states that “Assessment should enhance mathematics learning” (p. 13). The NCTM’s *Assessment Standards* also make clear that assessment has a role to play not just in making determinations about whether particular teaching activities were successful, but also in teachers’ moment-by-moment decision making (p. 46)—in other words, that teachers should use assessment to “keep learning on track.”

This distinction about the *purpose* of assessment is quite different from the distinction between classroom assessment and external assessments, which is more concerned with where the assessments take place, who sets them, and who scores them (Black & William, 2004a). This is an important qualification,

particularly in the United States, where the term *classroom assessment* is used primarily to mean classroom *summative* assessment.

The next section lays out the reasons for two key assumptions that have been made in the writing of this chapter, namely that assessment is independent both of curriculum and of any particular stance in psychology—in other words, that one can talk about the principles of good assessment without subscribing to a particular view of what should be in the mathematics curriculum, or even to a particular view about what happens when learning takes place. The reader who is prepared to accept these two assertions may comfortably skip the next section without loss of continuity. In subsequent sections, the research on the impact of assessment on learning is reviewed, and I suggest that the effective use of assessment for learning consists of five key strategies:

- (a) Clarifying and sharing learning intentions and criteria for success;
- (b) Engineering effective classroom discussions, questions, and learning tasks that elicit evidence of learning;
- (c) Providing feedback that moves learners forward;
- (d) Activating students as instructional resources for one another; and
- (e) Activating students as the owners of their own learning.

In the final section of the chapter, these five strategies are subsumed within a broader theoretical framework, namely the regulation of learning processes, which allows assessment to be integrated with principles of instructional design.

KEY ASSUMPTIONS

There are two major challenges in synthesizing the research on assessment *for learning* (or *formative assessment* as it is sometimes called). The first is to what extent does an adequate account of assessment for learning require agreement about what mathematics students should learn? The second is to what extent does an adequate account of assessment for learning require agreement about what happens when learning takes place? Each of these is discussed in turn below.

What students should learn in mathematics is a highly contested domain. For example, in discussing the widespread preconception that mathematics

is about learning to compute, Fuson, Kalchman and Bransford (2005) illustrated their discussion with the following question:

What, approximately, is the sum of $8/9$ plus $12/13$?

They pointed out that some people will, sensibly, conclude that the answer is a little less than 2, just by observing that the two numbers to be added are each a little less than 1. Others, however, will attempt to find the smallest common multiple of the denominators of the two fractions. They commented:

The point of this example is not that computation should not be taught or is unimportant; indeed, it is often critical to efficient problem solving. But if one believes that mathematics is about problem solving and that computation is a tool for use to that end when it is helpful, then the above problem is viewed not as a “request for computation,” but as a problem to be solved that may or may not require computation—and in this case, it does not. (Fuson et al., p. 220)

For many people involved in mathematics education, estimation skills are at least as important as, and perhaps more important than, computation skills. Almost 30 years ago, Michael Girling defined *numeracy* as “the ability to use a four-function calculator sensibly” (Girling, 1977, p. 4), suggesting that the skill of being able to assess the reasonableness of an answer was much more important than being able to compute it accurately. Others feel that estimation is much less important than a solid grounding in basic mathematical knowledge and skills (Klein, 2003).

These differences become even clearer in more advanced mathematics. For example, many people need to use the standard formula for solving quadratic equations:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Two points seem important here. The first is that, obviously, this formula needs to be memorized exactly if it is to be of any use. The second is that although some of the people who know this formula would be able to re-create it from scratch by the process of “completing the square,” most would not. Professionals involved in mathematics education would probably agree that the person who could derive this formula from scratch has a deeper understanding than someone who can merely reproduce the formula from memory, but that is not to say that simply knowing this formula is not useful. Indeed, anyone who knows

and can apply this formula accurately has what Richard Skemp called *instrumental understanding*. This is a different kind of understanding from knowing where the formula comes from and how to derive it (what Skemp called *relational understanding*) but it is still a form of understanding (Skemp, 1977). The crucial point here is that different users of mathematics have different needs, and that a well-grounded account of the role of assessment in mathematics education should serve them all.

Building on the work of Raymond Williams (1961), Paul Ernest (1991) identified five broad purposes for mathematics education:

- Acquiring basic mathematical skills, numeracy and social training in obedience;
- Learning basic skills and learning to solve practical problems with mathematics and information technology;
- Achieving understanding and capability in advanced mathematics, with some appreciation of mathematics;
- Gaining confidence, creativity, and self-expression through mathematics;
- Empowerment of learners as critical and mathematically literate citizens in society.

The obvious corollary of the fact that there are differences in people’s perceptions of the purpose of mathematics is that those with different aims will emphasize different aspects of mathematics. For those who see the first of these five broad aims as the major purpose of mathematics education, they will value mathematics curricula that emphasize this purpose. For those who regard the last of the five as most important, they will value very different curricula. The important point here is that although competing groups can construct arguments to justify their claims (Niss, 1993), these are essentially *value* arguments. In particular, there is no way for adherents of one particular view of the purpose of mathematics education to show dissenters that they are wrong. This is why an adequate account of classroom assessment must support any and all of these conflicting views of mathematics education, rather than impose a certain set of views.

Similar arguments apply to the psychology of mathematics education. For the first half of the 20th century, the dominant view about what happens when learning takes place was that the individual creates associations between stimuli and responses. These *associationist* views of learning included the behaviorism of Skinner (for example), as well as a range of other

views. Associationist models of learning explained some aspects of mathematics learning reasonably well but were unable to explain other aspects. For example, an associationist analysis of students' errors in learning multiplication facts would indicate that students' errors should be random—the result of insufficient reinforcement of particular links in chains of stimulus and response—which accords reasonably well with what one observes in practice. However, in other areas of mathematics learning, students' errors are clearly not random; in fact they are highly predictable (see examples in the section on *Eliciting Evidence*). The mounting evidence about the nonrandom nature of students' errors in mathematics led to the development of *constructivist* approaches to the study of learning, where it is acknowledged that students are active rather than passive in the development of their conceptions (see, for example, von Glasersfeld, 1991). Such theories were much better able to account for the systematicity in student errors, but there were some aspects of learning that constructivist views were unable to explain. For example, it was observed that some adults were able to perform calculations in some contexts (e.g., in supermarkets) but not others, such as classrooms (Lave, Murtaugh, & de la Roche, 1984). This idea that learning is often tied to the context in which learning takes place has proved to be particularly powerful in mathematics education. For example, Boaler (2002) found that in some mathematics classrooms, students were able to use the mathematics they had learned outside school, while in others, students were not. This suggests that theories in psychology are not like theories in, say, physics, where new theories include previous theories as special cases (for example in the way that Einstein's relativistic mechanics includes Newton's nonrelativistic mechanics as a special case).

Rather, in psychology, the tendency is for each new theory to be very good at explaining what previous theories did not, but generally not so good at explaining what the previous theories explained well. Many kinds of rote learning are explained well by associationist theories, whereas the regularities in students' errors are better explained by constructivist or information-processing theories, and situated theories explain transfer, or its absence, well. In this sense, each new theory does not replace the preceding theories but rather complements them. For different views of what mathematics students should learn, there may be different views of what happens when learning takes place, although it is important to note that no one view of learning will suffice for even the most specific learning (Sfard, 1998). For example, it might appear at first sight that learning multiplica-

tion facts would be primarily a matter of strengthening associations between stimuli and responses, but research has shown that many students "repair" gaps in their knowledge by using their knowledge of the processes of arithmetic to assemble and combine facts and routines that they can recall (VanLehn, 1990).

For the purpose of this chapter, the important point is that if it is to be useful, an adequate account of classroom assessment cannot dictate what mathematics students should learn nor should it be tied to a single view of what happens when learning takes place.

This chapter will, therefore, as far as possible, avoid putting the assessment cart before either the curriculum or psychology horse. The stance being taken is that assessment is a powerful servant but a bad master. As soon as assessment considerations are allowed to influence either what is to be learnt or what it means to learn, educators are likely to slip from making the important assessable to making the assessable important.

THE PURPOSES OF ASSESSMENT

Educational assessments are conducted in a variety of ways and their outcomes can be used for a variety of purposes. Differences exist in who decides what is to be assessed, who carries out the assessment, where the assessment takes place, how the resulting responses made by students are scored and interpreted, and what happens as a result (Black & Wiliam, 2004a). In particular, each of these can be the responsibility of those who teach the students, or, at the other extreme, all can be carried out by an external agency. Cutting across these differences, there are also differences in the *purposes* that assessments serve. Broadly, educational assessments serve three functions:

- supporting learning (formative)
- certifying the achievements or potential of individuals (summative)
- evaluating the quality of educational programs or institutions (evaluative)

Through a series of historical contingencies, a situation has developed in many countries in which the *circumstances* of the assessments have become conflated with the *purposes* of the assessment (Black & Wiliam, 2004a). So, for example, it is often widely assumed that the role of classroom assessment should be limited to supporting learning and that all assessments with which educational institutions can be held to account must be conducted by an external agency,

even though in some countries, this is not the case (Black & Wiliam, 2005a).

In broad terms, moving from formative through summative to evaluative functions of assessment requires data at increasing levels of aggregation, from the individual to the institution and from specifics of particular skills and weaknesses to generalities about overall levels of performance (although of course evaluative data may still be disaggregated in order to identify specific subgroups in the population that are not making progress, or to identify particular weaknesses in students' performance in specific areas, as is the case in France—see Black & Wiliam, 2005a). However, the different functions that assessments may serve are also clearly in tension. The use of data from assessments to hold schools accountable has, in many cases, because of “teaching to the test,” rendered the data almost useless for attesting to the qualities of individual students (apart, of course, from those qualities that are tested) or for supporting learning.

For similar reasons, some have argued that the uses of assessment to support learning and to certify the achievements of individuals are so fundamentally in tension that the same assessments cannot serve both functions adequately (Torrance, 1993). On the other hand, others have argued that ways must be found to integrate the two (e.g. Shavelson, Black, Wiliam, & Coffey, 2003). For the purposes of this chapter, the crucial point is that the use of assessment should support instruction in *any* assessment regime. Whether the assessment is for purposes of selection and certification, or for evaluation, whether it is conducted through teacher judgment, external assessments, or some combination of the two, classroom assessment must first be designed to support learning (see Black & Wiliam, 2004b, for a more detailed argument on this point). The remainder of this chapter considers further how this might be done.

FORMATIVE ASSESSMENT: ORIGINS AND EXAMPLES

In 1967, Michael Scriven proposed the use of the terms *formative* and *summative* to distinguish between different roles that evaluation² might play. On the one hand, he pointed out that evaluation “may have a role in the on-going improvement of the curriculum” (Scriven, 1967, p. 41), while in another role, evaluation “may serve to enable administrators to decide

whether the entire finished curriculum, refined by use of the evaluation process in its first role, represents a sufficiently significant advance on the available alternatives to justify the expense of adoption by a school system” (pp. 41–42). He then proposed “to use the terms ‘formative’ and ‘summative’ evaluation to qualify evaluation in these roles” (p. 43).

Two years later, Bloom (1969, p. 48) applied the same distinction to classroom tests:

Quite in contrast is the use of “formative evaluation” to provide feedback and correctives at each stage in the teaching-learning process. By formative evaluation we mean evaluation by brief tests used by teachers and students as aids in the learning process. While such tests may be graded and used as part of the judging and classificatory function of evaluation, we see much more effective use of formative evaluation if it is separated from the grading process and used primarily as an aid to teaching.

However, despite Bloom's extension of the term *formative* to apply to the evaluation of individual students (what in this chapter is termed *assessment*) as well as the evaluation of programs or institutions, for the next 30 years, the term *formative* was used almost exclusively in the context of program evaluation. Indeed, the index of the previous NCTM *Handbook of Research on Mathematics Education* lists only one mention of the term *formative* and that is in the section on “Evaluation” in the chapter on Research Methods (Romberg, 1992, p. 58).

Nevertheless, although the term *formative* was rarely used to describe teachers' assessment practices, a number of studies investigated the integration of assessment with instruction, the best known of which is probably Cognitively Guided Instruction (CGI).

In the original CGI project, a group of 21 elementary school teachers participated in a series of workshops over a 4 year period (an introductory 2½-hour workshop and a 2-day workshop before the beginning of the 1st school year, fourteen 3-hour workshops during the first year, four 2½-hour workshops and a 2-day reflection workshop in the 2nd year, and four 3-hour workshops and two 2½-hour review workshops in the 3rd year). During the workshops, the teachers were shown extracts of videotapes selected to illustrate critical aspects of children's thinking. Teachers were then prompted to reflect on what they had seen, by, for example, being challenged to relate the way a child had solved one problem to how they had solved, or might solve, other problems (Fennema et al., 1996,

² Here I am following the convention in American and British English that the term *assessment* applies to individuals, whereas *evaluation* applies to institutions or programs.

p. 407). Throughout the project, the teachers were encouraged to make use of the evidence they had collected about the achievement of their students to adjust their instruction to better meet their students' learning needs.

The teachers in the CGI program taught problem solving significantly more and number facts significantly less than did controls. They also knew more about individual students' problem-solving processes, and their students did better in number fact knowledge, understanding, problem solving, and confidence (Carpenter, Fennema, Peterson, Chiang, & Loef, 1989). More importantly, 4 years after the end of the program, the participating teachers were still implementing the principles of the program (Franke, Carpenter, Levi, & Fennema, 2001).

Another study that showed the substantial benefits of adapting instruction to meet student learning needs was that conducted by Bergan, Sladeczek, Schwarz, and Smith (1991). The performance of 428 Kindergarten students taught by 29 teachers implementing a measurement-and-planning system (MAPS) was compared with that of 410 students taught by 27 teachers who taught their classes as usual. In the MAPS program, teachers, together with an aide and a site manager, assessed their students' readiness for learning in reading and mathematics in the fall, and again in the spring (children were allowed as much time as they needed to complete the items). Teachers in the experimental group were trained on how to interpret the test results and provided with the *Classroom Activity Library*—a series of activities typical of early-grades instruction but keyed specifically to empirically validated developmental progressions—which they could use to individualize instruction, depending on the students' performance in the assessments. At the end of the year, 111 of the students in the control group (27%) were referred for placement in a special education program for the following year, and 80 (20%) were actually placed in special education programs. In the experimental group, only 25 students (6%) were referred, and only 6 students (1.4%) were placed in special education programs. In other words, students in the control group were 4.5 times more likely to be referred for placement in special education, and 14 times more likely actually to be placed in special education programs than students taught by teachers using the MAPS scheme. What is perhaps even more remarkable about this study is that all the schools in the study served districts with considerable socioeconomic disadvantage, and the socioeconomic status of the students in the experimental group was actually lower than that of the control group.

A third example of the use of assessment to improve student learning was a project involving a group of 24 (later expanded to 36) secondary school mathematics and science teachers in six schools in two districts in England (Black, Harrison, Lee, Marshall, & Wiliam, 2003). The work with teachers had two main components. The first was a series of eight workshops over an 18-month period (from February 1999 to June 2000). Seven of the workshops were of 5 hours duration and one was of 3 hours duration. During the workshops, the teachers were introduced to the research basis underlying how assessment can support learning (derived from Black & Wiliam, 1998a, 1998b), had the opportunity to develop their own plans, and, at later meetings, discussed with colleagues the changes they had attempted to make in their practice. The second component of the intervention with the teachers was a series of visits by researchers to the teachers' classrooms, so that the teachers could be observed implementing some of the ideas they had discussed in the workshops, had an opportunity to discuss their ideas, and could plan how they could be put into practice more effectively.

A key feature of the workshops was the development of action plans. As Perrenoud (1998) has pointed out, changing pedagogy requires teachers to re-negotiate the "learning contract" (cf. Brousseau, 1984) that they have evolved with their students, suggesting that radical changes are best effected at the beginning of a new school year. For the first 6 months of the project, therefore, the teachers were encouraged to experiment with some of the strategies and techniques suggested by the research, such as rich questioning, providing feedback to students in the form of comments rather than scores or grades, sharing learning intentions and success criteria with learners, and student peer- and self-assessment (see below). Each teacher was then asked to draw up, and later to refine, an action plan specifying which aspects of formative assessment they wished to develop in their practice and to identify a focal class with whom these strategies would be introduced in September 1999. Most of the teachers' plans contained reference to two or three important areas in their teaching where they were seeking to increase their use of formative assessment, generally followed by details of techniques that would be used to make this happen. In almost all cases the plan was given in some detail, although many teachers used phrases with meanings that differed from teacher to teacher (even within the same school).

Almost every teacher's plan contained some reference to focusing on or improving the teacher's own questioning techniques although only 11 gave details on how they were going to do this (for example us-

ing more open questions, allowing students more time to think of answers, or starting the lesson with a focal question). Others were less precise (for example stating that they intended using more sustained questioning of individuals, or improving questioning techniques in general). Some teachers mentioned planning and recording their questions. Many teachers also mentioned involving students more in setting questions (for homework, or for each other in class). Some teachers also saw existing standardized tests as a source of good questions.

Nearly half the teachers mentioned providing feedback in the form of comments rather than scores or grades, although only 6 of the teachers included it as a specific element in their action plans. Some of the teachers wanted to reduce the use of scores and grades but foresaw problems with this, given school policies on assessment. Four teachers planned for a module test to be taken before the end of the module, thus providing time for remediation.

Sharing the objectives of lessons or topics was mentioned by most of the teachers, through a variety of techniques (using a question that the students should be able to answer at the end of the lesson, stating the objectives clearly at the start of the lesson, getting the students to round up the lesson with an account of what they had learned). About half the plans included references to helping the students understand the rubrics used for investigative or exploratory work, generally using exemplars from the work of students from previous years. Exemplar material was mentioned in other contexts such as having work on display and asking students to assess work using rubrics provided by the teacher.

Almost all the teachers mentioned some form of self-assessment in their plans, ranging from using red, yellow, or green “traffic lights” to indicate the student’s perception of the extent to which a topic or lesson had been understood, to strategies that encouraged self-assessment via targets that placed responsibility on students (e.g., “1 of these 20 answers is wrong: Find it and fix it!”). Traffic lights were mentioned in about half of the plans, and in practically all cases their use was combined with strategies to follow up the cases where the students signaled incomplete understanding. Several teachers mentioned their conviction that group work provided important reinforcement for students, as well as providing the teacher with insights into their students’ understanding of the work.

The other component of the intervention, the visits to the schools, provided an opportunity for researchers to discuss with the teachers what they were doing, and how this related to their efforts to put their action plans into practice. The interactions were in-

tended to be supportive rather than directive, but because researchers were frequently seen as “experts” in either mathematics or science education, there was a tendency sometimes for teachers to invest questions from a member of the project team with a particular significance, and for this reason, these discussions were often more effective when science teachers were observed by mathematics specialists, and vice-versa.

A detailed description of the qualitative changes in teachers’ practices is beyond the scope of this chapter (see Black et al., 2003, for a full account). In quantitative terms, students taught by the teachers developing the use of assessment for learning outscored comparable students in the same schools by approximately 0.3 standard deviations, both on teacher-produced and external state-mandated tests (William, Lee, Harrison, & Black, 2004). Since one year’s growth in mathematics as measured in the Trends in Mathematics and Science Study (TIMSS) is 0.36 standard deviations (Rodriguez, 2004, p.18), the effect of the intervention can be seen to almost double the rate of student learning.

FORMATIVE ASSESSMENT: PREVALENCE AND IMPACT

These studies show that integrating assessment with instruction is both possible and desirable. It is also valued. A sample of 580 principals ranked “Determining what needs to be re-taught after tests” as the most important in a list of 26 assessment competencies desired in their teachers (Marso & Pigge, 1993, pp. 137–138), although teachers rated competence in grading as the most important. However, there is little evidence of such data-driven practices being regularly enacted in classrooms. In a survey first published in 1980, Salmon-Cox stated that “student scores on standardized tests are not very useful to the classroom teacher” and concluded that teachers prefer to rely on their own judgment about student weaknesses and areas of needed help (Salmon-Cox, 1981 p. 631). Stiggins and Bridgford (1985) found that although many teachers created their own assessments, “in at least a third of the structured performance assessments created by these teachers, important assessment procedures appeared not to be followed” (p. 282) and “in an average of 40% of the structured performance assessments, teachers rely on mental record-keeping” (p. 283).

McMorris and Boothroyd (1993) analyzed the quality of tests developed by seventh- and eighth-grade mathematics and science teachers and found that science teachers made greater use of multiple-

choice items whereas mathematics teachers tended to set more computation items. However, for both mathematics and science teachers, the tests were of variable quality, with a significant correlation between test quality and the amount of training in educational measurement the teachers had received.

More recently Senk, Beckman, and Thompson (1997) conducted a survey of assessment practices in 19 mathematics classes in 5 high schools in 3 states. They found that formal tests and quizzes were the most frequently used assessment tools, with a nominal weight across the sample of 77%, whereas written projects and interviews accounted for a further 7%. The tests and quizzes used focused largely on “low-level” aspects of the domains assessed, and the “grading” function of assessment dominated the “assessment” function (58% of all assessments were reported in terms of a “brute grade”). The survey also found that teachers tended to ignore the results of standardized tests in arriving at terminal grades.

In a national survey, Dorr-Bremme and Herman (1986) found that students spent around 12 hours each year taking mathematics tests in elementary school (4th to 6th grade), and approximately twice that in 10th grade, although only about 6 hours was required by the state or district (pp. 16–17), and two substantial review articles, one by Natriello (1987) and the other by Crooks (1988), provided clear evidence that classroom assessment practices had substantial impact on students and their learning, although the impact was rarely beneficial. Natriello’s review used a model of the assessment cycle, beginning with purposes; and moving on to the setting of tasks, criteria, and standards; evaluating performance and providing feedback; and then discussing the impact of these evaluation processes on students. His most significant point was that the vast majority of the research he cited was largely irrelevant because of weak theorization, which resulted in the conflation of key distinctions (e.g., the quality and quantity of feedback). Crooks’s article had a narrower focus—the impact of assessment practices on students. He concluded that the summative function of assessment has been too dominant and that more emphasis should be given to the potential of classroom assessments to assist learning. Most importantly, assessments should emphasize the skills, knowledge, and attitudes regarded as most important, not just those that are easy to assess.

Bangert-Drowns, Kulik, Kulik, and Morgan (1991) reported the results of a meta-analysis of 40 research reports on the effects of feedback in what they called “test-like” events (e.g., evaluation questions in programmed learning materials, review tests at the end of a block of teaching). They found that providing feed-

back in the form of answers to the review questions was effective only when students could not look ahead to the answers before they had attempted the questions themselves (what they called “control for presearch availability,” p. 213). Furthermore feedback was more effective when the feedback gave details of the correct answer, rather than simply indicating whether the student’s answer was correct or incorrect. In the studies in which the students could not look ahead for the answers, and the feedback gave details of the correct answer, the mean effect size was 0.58 standard deviations. Reviews by Dempster (1991, 1992) confirm these findings, as does a review by Elshout-Mohr (1994) that reports many findings not available in English.

The difficulty of reviewing relevant research in this area was highlighted by Black and Wiliam (1998a) in their synthesis of research published since the reviews by Natriello and Crooks. Those earlier two articles had cited 91 and 241 references respectively, and yet only 9 references were common to both articles. In their own research, Black and Wiliam found that electronic searches based on keywords either generated far too many irrelevant sources or omitted key papers. In the end, they resorted to manual searches of each issue between 1987 and 1997 of 76 of the journals considered most likely to contain relevant research. Black and Wiliam’s review (which cited 250 studies) found that effective use of classroom assessment yielded improvements in student achievement between 0.4 and 0.7 standard deviations.

Thirty-five years ago, Bloom suggested that

evaluation in relation to the process of learning and teaching can have strong positive effects on the actual learning of students as well as on their motivation for the learning and their self-concept in relation to school learning. . . . [E]valuation which is directly related to the teaching-learning process as it unfolds can have highly beneficial effects on the learning of students, the instructional process of teachers, and the use of instructional materials by teachers and learners. (Bloom, 1969, p. 50)

At the time, Bloom cited no evidence in support of this claim, but it is probably safe to conclude that the question has now been settled: Attention to formative classroom assessment practices can indeed have a substantial impact on student achievement.

What is less clear is what exactly constitutes *effective* classroom assessment. Although the studies cited above indicate that assessment for learning can improve learning, several studies have found conflicting results. For example, in a study of 32 fifth-grade teachers in Germany, Helmke and Schrader (1987) found that teachers who had an accurate knowledge of their

students (as measured by the teachers' ability to predict achievement test scores) were associated with higher levels of achievement *only* when the teachers also showed a high range of instructional techniques. Students taught by teachers who had a high knowledge of their students' achievement but lacked a range of instructional techniques actually performed worse than students taught by teachers who did not know their students' achievement. This study seems to indicate that collecting data if one cannot do anything with it is counterproductive.

Furthermore, even when teachers do manage to use information about student achievement to adjust or individualize their instruction, teachers may lack the ability to do so effectively. For example, in a 20-week study of 33 teachers in elementary and middle schools, Fuchs, Fuchs, Hamlett and Stecker (1991) found that teachers who received feedback on the achievement of students with learning difficulties in their classes made more adjustments to their teaching programs than teachers not given this information. However, the achievement of these students was improved *only* when this feedback was accompanied by advice from a computerized "expert system", because the teachers not given the feedback from the expert system tended to re-explain how to do problems with the same algorithms that had led to previous failure.

Therefore, if educators are to maximize the potential benefits of formative assessment or assessment for learning, there is a need to understand what, exactly, constitutes effective formative assessment, and this is the focus of the remainder of this chapter.

FORMATIVE ASSESSMENT: THEORETICAL CONSIDERATIONS

In education, the term *feedback* is routinely applied to any information that a student is given about their performance, and the generality of the term obscures that this is, in reality, a frozen metaphor derived from systems engineering. One of the earliest uses of the term was by Norbert Wiener. In 1940, Wiener and his colleagues had been working on automatic range-finders for anti-aircraft guns, which involved mechanisms for predicting the path of airplanes, and realized that the control mechanisms needed for the range-finders were similar to control mechanisms in animals. He realized that purposeful action required the existence of a closed loop allowing the evaluation of the effects of one's actions and the adaptation of future conduct based on past performances (Wiener, 1948). He also realized that there were two kinds of loops: those in-

volving positive feedback and those involving negative feedback. In both kinds of loops, there are inputs to the system and, some time later, outputs from the system. The crucial feature of the loop is that information about the output is fed back to the input side of the system. In positive feedback loops, the feedback serves to drive the system further in the direction it is already going, whereas in negative feedback loops, the feedback acts to oppose the current direction of the system. All positive feedback systems are unstable, driving the system towards either explosion or collapse. Examples of the explosive kind of positive feedback loops are simple population growth in the presence of plentiful supplies of food and in the absence of predators, and inflationary price/wage spirals in economics. Examples of collapse are economic depression, food hoarding in times of shortage, and the loss of tax revenue in urban areas as a result of "middle-class flight" (note that whether the effects are explosion or collapse, both are examples of positive feedback).

In contrast, negative feedback systems tend to produce stability, because they are inherently "self-correcting." One example of negative feedback is population growth with limited food supply, in which the lack of food causes a slowdown in population growth, which in turn, depending on the conditions, produces either an asymptotic approach towards or a damped oscillation about the carrying capacity of the environment. Another example is the domestic thermostat. When the temperature of the room drops below the setting on the thermostat, a signal is sent to turn on the furnace. When the room heats up above the setting on the thermostat, a signal is sent to turn off the furnace.

The foregoing discussion clearly shows that the current uses of the term feedback in education are very different from those in engineering, but, more importantly, the simplistic engineering metaphor may just not be helpful. In systems engineering, negative feedback is good, because it keeps a system under control whereas positive feedback is bad because it leads to explosion or collapse. In educational settings, things are not so clear-cut. Negative feedback may be helpful for correcting learning when it is off-course, but feedback that reinforces learning that is on track is also powerful. And the thermostat does not care how often it is told that the temperature in the room is "wrong" and needs to be corrected, whereas humans are often adversely affected by such information. This is not to say that the insights of systems engineering are irrelevant, but educators should exercise considerable caution in adopting what are essentially metaphors in complex areas like human learning, especially in terms of the relationship between *feedback* and *formative assessment*.

In the United States, the term *formative assessment* is often used to describe assessments that are used to provide information on the likely performance of students on state-mandated tests—a usage that might be described better as “early-warning summative.” In other contexts, the term is used to describe any feedback given to students, no matter what use is made of it, such as telling students which items they got correct and incorrect (sometimes termed *knowledge of results*). These kinds of usages suggest that the distinction between *formative* and *summative* applies to the assessments themselves, but inasmuch as the same assessment can be used both formatively and summatively, these terms are more usefully applied to the *use* to which the information arising from assessments is put.

As Ramaprasad (1983) noted, the defining feature of feedback is that the information generated within the system must have some effect. Information that does not have the capability to change the performance of the system is not feedback: “Feedback is information about the gap between the actual level and the reference level of a system parameter which is used to alter the gap in some way” (Ramaprasad, 1983, p. 4).

Commenting on this, Sadler (1989) noted

An important feature of Ramaprasad’s definition is that information about the gap between actual and reference levels is considered as feedback *only when it is used to alter the gap*. If the information is simply recorded, passed to a third party who lacks either the knowledge or the power to change the outcome, or is too deeply coded (for example, as a summary grade given by the teacher) to lead to appropriate action, the control loop cannot be closed, and “dangling data” substituted for effective feedback. (p. 121)

In this view, formative assessments (or feedback, in Ramaprasad’s terminology) cannot be separated from their instructional consequences, and assessments are formative only to the extent that they impact learning (for an extended discussion on consequences as the key part of the validity of formative assessments, see Wiliam & Black, 1996). Therefore what is important is not the intent behind the assessment, but the function it actually serves, and this provides a useful point of difference between the terms *assessment for learning* and *formative assessment*. Black, Harrison, Lee Marshall, and Wiliam (2004, p. 8) distinguished between the two as follows:

Assessment for learning is any assessment for which the first priority in its design and practice is to serve the purpose of promoting pupils’ learning. It thus differs from assessment designed primarily to serve the pur-

poses of accountability, or of ranking, or of certifying competence. An assessment activity can help learning if it provides information to be used as feedback, by teachers, and by their pupils, in assessing themselves and each other, to modify the teaching and learning activities in which they are engaged. Such assessment becomes “formative assessment” when the evidence is actually used to adapt the teaching work to meet learning needs.

For the purpose of this chapter, then, the qualifier *formative* will refer not to an assessment, nor even to the purpose of an assessment, but the function it actually serves. An assessment is formative to the extent that information from the assessment is fed back within the system and actually used to improve the performance of the system in some way (i.e. that the assessment *forms* the direction of the improvement).

So, for example, if a student is told that she needs to work harder, and does work harder as a result, and consequently does indeed make improvements in her performance, this would *not* be formative. The feedback would be *causal*, in that it did trigger the improvement in performance, but not *formative*, because decisions about *how* to “work harder” were left to the student. Telling students to “give more detail” might be formative, but only if the students knew what “giving more detail” meant (which is unlikely, because if they knew what detail was required, they would probably have provided it on the first occasion). Similarly, a “formative assessment” that predicts which students are likely to fail the forthcoming state-mandated test is not formative unless the information from the test can be used to improve the quality of the learning within the system. To be formative, feedback needs to contain an implicit or explicit recipe for future action.

Another way of thinking about the distinction being made here is in terms of monitoring assessments, diagnostic assessments, and formative assessments. An assessment *monitors* learning to the extent that it provides information about whether the student, class, school, or system is learning or not; it is *diagnostic* to the extent that it provides information about what is going wrong; and it is *formative* to the extent that it provides information about what to do about it. A sporting metaphor may be helpful here. Consider a young fast-pitch softball player who has an earned-run-average of 10 (for readers who know nothing about softball, that is not good). This is the *monitoring* assessment. Analysis of what she is doing shows that she is trying to pitch a rising fastball (i.e., one that actually rises as it gets near the plate, due to the back-spin applied), but that this ball is not rising and therefore becomes an ordinary fastball in the middle of the strike zone, which is very easy for the batter to hit. This is the *diagnostic* as-

essment, but it is of little help to the pitcher, because she already knows that her rising fastball is not rising, which is why she is giving up a lot of runs. However, if a pitching coach is able to see that she is not dropping her shoulder sufficiently to allow her to deliver the pitch from below the knee, then this assessment has the potential to be not just diagnostic, but *formative*. If the athlete is able to use the advice about delivering the pitch from below the knee to make her rising fastball rise, then the feedback given by the coach will, indeed, have been formative. This use of *formative* recalls the original meaning of the term. In the same way that one's formative experiences are the experiences that shape the individual, formative assessments are those that shape learning.

The important point here is that not all diagnoses are *instructionally tractable*—an assessment can accurately diagnose what needs attention without indicating what needs to be done to address the issue. To promote learning, one must collect the correct data in the first place. This is discussed further in the section on *Clarifying and Sharing Learning Intentions* below.

In the examples given above, the action follows quite quickly on from the elicitation of the evidence about student achievement, but the definition of formative assessment given above allows cycles of elicitation, interpretation and action of any length, provided the information is used to form the direction of future learning. For example, in March, a mathematics supervisor may be planning the workshops she will make available to teachers in the summer. By looking at the scores obtained by the students in the district on last year's state-mandated tests, she might discover that, compared to other students in the state, the students are performing relatively poorly on items that assess geometry. As a result, she might plan a series of workshops on teaching geometry for the summer, and, if these workshops are successful, then the students' performance on geometry items should improve. This cycle would be over 2 years in length, given that the supervisor would be using data from tests taken the previous March, and the impact would not be felt until the students took the test the following March (and the results might not be available until July). Furthermore, the data that were used to improve learning in the district were not collected from the students who benefited, but those who were in that grade 2 years earlier. Nevertheless, data from the state tests functioned formatively in this example, since information about student performance was used to make adjustments to instruction (in this case improving the teaching of geometry) that improved the learning of mathematics in the district.

It could be argued that this kind of usage is *evaluation*, rather than *assessment*, because it is the program

that is being improved through the analysis of student data, but there is no clear boundary between the two. A teacher might look through the responses of her students to a trial run of a state test and replan the topics that she is going to teach in the time remaining until the test. Such a test could still be useful as little as a week or two before the state-mandated test, as long as there is time to use the information to redirect the teaching. Again this assessment would be formative as long as the information from the test was actually used to adapt the teaching, and in particular, not only telling the teacher which topics need to be retaught, but also suggesting what kinds of reteaching might produce better results. At this level, both the program and the learning of individual students on whom the data were collected is being impacted by the assessment outcomes, so this example could be thought of as either assessment or evaluation. The crucial point is that the evidence is used to make decisions that could not be made, or at least could not be made as well, without that evidence, with the result that learning is enhanced.

The building-in of time to make use of assessment data is a central feature of much elementary and middle school teaching in Japan. A teaching unit is typically allocated 14 lessons, but the content usually occupies only 10 or 11 of the lessons, allowing time for a short test to be given in the 12th lesson, and for the teacher to use Lessons 13 and 14 to reteach aspects of the unit that were not well understood.

Another example, on an even shorter time-scale, is the use of *exit passes* from a lesson. The idea here is that before leaving a classroom, each student must compose an answer to a question that goes to the heart of the concept being taught at the end of the lesson. On a lesson on probability for example, such a question might be, "Why can't a probability be greater than 1?" Once the students have left, the teacher can look at the students' responses and make appropriate adjustments in the plan for the next period of instruction. The shortest feedback loops are those involved in the day-to-day classroom practices of teachers, where teachers adjust their teaching in light of students' responses to questions or other prompts in real time. The key point in all this is that the length of the feedback loop should be tailored according to the ability of the system to react to the feedback.

All this suggests that the conflicting uses of the term *formative assessment* can be reconciled by recognizing that almost any assessment can be formative provided it is used to make instructional adjustments, but a crucial difference between different assessments is the length of the adjustment cycle. Table 23.1 provides a terminology for the different lengths of cycles.

Table 23.1 Cycle Lengths for Formative Assessment

Type	Focus	Length
Long-cycle	across marking periods, quarters, semesters, years	4 weeks to 1 year
Medium-cycle	within and between instructional units	1 to 4 weeks
Short-cycle	within and between lessons	
day-by-day		24 to 48 hours
minute-by-minute		5 seconds to 2 hours

The foregoing discussion establishes that any assessment can be formative, and that an assessment is formative to the extent that information from the assessment is used to adjust instruction to better meet student learning needs. The adjustment can take place immediately, in time for the next instructional episode, between units, or even between years, and the beneficiaries of the adjustments may or may not be the students on whom information was collected. However, although all these different uses of assessment information may be formative, in that the information is used to adapt instruction to better meet student learning needs, the research evidence cited above suggests that not all are equally effective. To establish what, exactly, constitutes the most effective form of formative assessment requires a deeper look at the research evidence.

An important feature of Ramaprasad's definition of feedback (here *formative assessment*) is that it draws attention to three key instructional processes:

- establishing where the learners are in their learning,
- establishing where they are going,
- establishing what needs to be done to get them there.

Traditionally, this may have been seen as primarily the teacher's role, but one must also take account of

the role that the learners themselves, and their peers, play in these processes. Crossing the three key instructional processes listed above with the three agents involved in the classroom produces the framework shown in Figure 23.1, which provides a way of thinking about the key strategies involved in formative assessment. The subject classroom that is the focus of Figure 23.1 is, of course, itself nested within a school, which in turn is located in a community, and so on. Any adequate account of formative assessment will have to acknowledge these multiple contexts, but they are beyond the scope of this chapter. Furthermore, given that the stance taken in this chapter is that, ultimately, assessment must feed into actions in the classroom in order to affect learning, this simplification seems reasonable, at least as a first-order approximation. For examples of sociocultural approaches to the implementation of formative assessment, see Black and Wiliam (2005b) and Pryor and Crossouard (2005).

This framework suggests that assessment for learning can be conceptualized as consisting of five key strategies and one "big idea" (Wiliam & Thompson, in press). The five key strategies are (a) engineering effective classroom discussions, questions, and learning tasks that elicit evidence of learning; (b) providing feedback that moves learners forward; (c) clarifying and sharing learning intentions and criteria for suc-

	Where the learner is going	Where the learner is right now	How to get there
Teacher	Clarifying learning intentions and sharing and criteria for success	Engineering effective classroom discussions and tasks that elicit evidence of learning	Providing feedback that moves learners forward
Peer	Understanding and sharing learning intentions and criteria for success	Activating students as instructional resources for one another	
Learner	Understanding learning intentions and criteria for success	Activating students as the owners of their own learning	

Figure 23.1 Aspects of assessment for learning.

cess; (d) activating students as the owners of their own learning; and (e) activating students as instructional resources for one another.

The big idea is that evidence about student learning is used to adjust instruction to better meet student needs—in other words that teaching is *adaptive* to the students' learning needs. The following sections describe in more detail these five key strategies and how they fit together within the more general idea of the “regulation of learning processes.”

ENGINEERING EFFECTIVE CLASSROOM DISCUSSIONS, QUESTIONS, AND LEARNING TASKS THAT ELICIT EVIDENCE OF LEARNING

Teachers can use a vast range of strategies to elicit evidence of student learning, from formal testing occasions, through the activities that students routinely undertake in mathematics classrooms, to classroom discussions and informal exchanges with students. In the previous version of this *Handbook*, Webb (1992) outlined a set of principles for assessing mathematics. The chapters by Jan de Lange and Linda Dager Wilson in this volume cover large-scale and high-stakes assessment, respectively, and Clarke (1996) provides an international perspective on contemporary mathematics assessment. In addition, van den Heuvel-Panhuizen and Becker (2003) made proposals for a didactic model of assessment design in mathematics, and an excellent overview of the characteristics of and general principles for the design of “thought-revealing activities” can be found in Lesh, Hoover, Hole, Kelly, and Post (2001). The focus of this section is on classroom discussions and tasks that elicit evidence of learning.

Classroom Discussions

Teacher-led classroom discussion, along with individual seatwork, is one of the staples of mathematics instruction in the United States (and indeed most other countries). In the traditional model of classroom transactions—termed *Initiation-Response-Evaluation* or *I-R-E* by Mehan (1979)—the teacher asks a question, chooses a student to answer the question, and then makes some response to the student's answer. Within this broad structure one can say rather more.

First, it is important to note that the teacher almost invariably dominates such classroom discussions, even in the United States, where it has been shown that American teachers actually talk less than teachers in countries that are more successful in mathematics, at least as measured in international comparisons. For ex-

ample, the 1999 TIMSS video study found that in U.S. classrooms there were 8 teacher words for every student word, whereas in Japan, there were 13, and in Hong Kong 16 teacher words for every student word (Hiebert et al., 2003). Of course, this means that even in a U.S. classroom of 25 students, the teacher speaks 200 times as much as any one student. It also makes clear that the quality of what the teacher is saying is much more important than the quantity, and here the evidence from the TIMSS video studies is compelling.

Although the 1999 TIMSS video study focused on eighth grade, the findings were similar to those of earlier studies (e.g., Weiss, Pasley, Smith, Banilower & Heck, 2003; Rowan, Harrison, & Hayes, 2004) and showed that mathematics teaching in the U.S. “is characterized by frequent review of relatively unchallenging, procedurally oriented mathematics” (Hiebert et al., 2005, p. 125). Why this should be the case is unclear, but the consequences are profound. To see why, it is necessary to discuss briefly some recent work on the nature and development of human abilities.

The term *intelligence* has been used in a variety of ways for at least 100 years, and there is no consensus on its meaning today. The word has been tarnished by its use by the eugenics movement in the first half of the 20th century, especially in the United States (see Selden, 1999, for an extended discussion), and more recently, the term has been used in essentially racist projects such as *The Bell Curve* by Herrnstein and Murray (1994). There is ample evidence that the conclusions reached by such authors is fundamentally flawed (see, e.g., Fish, 2001; Montague, 1999) but the intensity and the politicization of the debate makes it difficult to separate the science from the myth. One particularly unfortunate effect of this debate is that it has led many people to adopt rigid and extreme positions on the issue of intelligence.

At one extreme are those who believe that

- intelligence is determined entirely by one's genes and is fixed for life;
- intelligence tests measure the most important aspects of human thinking;
- intelligence is the most important predictor of success in the workplace; and
- other kinds of ability do not really matter.

At the other extreme, partly in response to the political motives of those who hold the extreme views listed above, are those who reject the concept of intelligence entirely or deny its relevance except in the most limited laboratory studies. For these proponents

- intelligence is determined by the environment and not one's genes;

- intelligence tests measure only the ability to take intelligence tests;
- intelligence does not matter in the real world; and
- there are several different kinds of intelligence, all independent of one another

It turns out that there *is* a high degree of consensus amongst psychologists on the science underlying intelligence, and, predictably, it is somewhere between the two extreme positions described above:

- intelligence is determined by both environment and genetics, and the genetic influence is substantial (Plomin & Petrill, 1997);
- intelligence tests correlate strongly with a range of other measurements of mental capability (Mackintosh, 2000);
- intelligence is strongly associated with success in a wide range of real world activities (Bertua, Anderson, & Salgado, 2006);
- there are several different aspects of intelligence, but most of them are strongly inter-related (Deary, 2000).

In this sense, intelligence seems to be like physical height. Physical height is inherited—tall parents tend to have tall children—but there is also a strong environmental component. The better the standard of nutrition, the taller the child grows. The same is clearly true for intelligence, most clearly evidenced by the *Flynn effect*. Named after James Flynn, this refers to the rise of about one standard deviation that has been observed in IQ scores throughout the developed world over the last 60 years (Flynn, 1984, 1987). Although researchers disagree about the causes of these gains (see, e.g., Neisser, 1998), students' educational experiences are clearly an important element.

Dickens and Flynn (2001) have shown that what is observed about intelligence is best accounted for by the idea that people select, or have selected for them, environments that match their intelligence. People with high intelligence, for example, engage in more of the activities that enhance intelligence, and so become more intelligent whereas people with lower intelligence opt out of or are denied these intelligence-enhancing activities and so lose the opportunities to enhance their intelligence. This suggests that intelligence and environment are mutually constitutive of each other: Environment causes intelligence and intelligence causes environment. However, the model proposed by Dickens and Flynn also suggests that the impact of transient improvements in environment are themselves transient. This explains why compensatory preschool programs have significant effects on intel-

ligence while students are in the program, but the effects diminish when students leave (Barnett & Camilli, 2001), although it should be noted that the improvements in student *achievement* produced by such programs are lasting.

One concrete demonstration on the power of classroom environments to improve student achievement is a study of 191 students in 7 fourth grade classrooms following the "Thinking Together" program (Dawes, Mercer, & Wegerif, 2000) as they learned science. These students outperformed controls in similar schools by 0.74 standard deviations on tests of concept mapping and 0.29 standard deviations on a standardized science achievement test (Mercer, Dawes, Wegerif, & Sams, 2004). Perhaps even more surprisingly, these students outperformed controls by 0.27 standard deviations on a purely spatial intelligence test—Raven's Progressive Matrices (Raven, 1960). Just by increasing the amount of structured talk in the classroom, students improved at performing purely spatial tasks.

The implications of these findings for mathematics classrooms are profound. To maximize mathematics achievement, classrooms must be places where every student is required to engage cognitively with the mathematics she or he is learning, and this appears to be the hallmark of practice both in countries that are successful and exemplary classrooms in the United States (see, e.g., Boaler & Humphreys, 2005). Shulman (2005) has described such cognitively rich pedagogies as *pedagogies of engagement*.

The problem is that, as is clear from the TIMSS video studies (Hiebert et al., 2003), few mathematics classrooms in the United States rigorously employ pedagogies of engagement. In most American mathematics classrooms, the choice about whether to engage is left to the student. In classroom dialogue, the teacher asks a question and then selects a respondent from those who have signaled that they have an answer by raising their hands. Some teachers do try to counter this by occasionally calling on students who have not raised their hands, but this is frequently seen as breaching the terms of the *didactical contract* (Brousseau, 1984). One teacher summed up his predicament thus:

I'd become dissatisfied with the closed Q&A style that my unthinking teaching had fallen into, and I would frequently be lazy in my acceptance of right answers and sometimes even tacit complicity with a class to make sure none of us had to work too hard. . . . They and I knew that if the Q&A wasn't going smoothly, I'd change the question, answer it myself or only seek answers from the "brighter students." There must have been times (still are?) where an outside observer would see my lessons as a small discussion group

surrounded by many sleepy onlookers. (Black et al., 2004, p. 11)

The consequence of this is that some students are deeply engaged in the lesson and, as such, are increasing their capabilities, while others are avoiding engagement and thus forgoing the opportunities to increase their ability.

In other classrooms, participation is not voluntary. Magdalene Lampert (2001) described a lesson she taught in which she made a point of calling on a student who had not raised his hand even though many others had done so. She gave her reasons as follows: "I called on Richard because I wanted to teach him and others in the class that everyone would indeed be asked to explain their thinking publicly. I also wanted to teach everyone that what they said would be expected to be an effort to make mathematical sense" (p. 146).

Leahy, Lyon, Thompson, and Wiliam (2005) described mathematics classrooms in which teachers have gone further and instituted a rule of "no hands up, except to *ask* a question." After posing a question, the teacher decides which student should respond by the use of some randomizing device, such as name cards (see Webb, 2004, p. 175) or a beaker of Popsicle sticks on which the students' names are written. The important point about such classrooms is that mental participation is not optional.

Such a radical change in the "classroom contract" (Brousseau, 1984) may be unwelcome for many students who are accustomed to classrooms where participation is optional, but there is evidence that students' participation practices in mathematics classrooms are malleable (Turner & Patrick, 2004). In particular, there are many strategies that teachers can use to engage students in classroom participation. Where students reply "I don't know," Ellin Keene suggests that teachers can ask, "OK, but if you did know what would you say?" (Carol, 2006), or the teacher can solicit answers from other students and then return to the original student and ask them to select from amongst the answers they have heard. Other possibilities are allowing students to "phone a friend," or, for multiple-choice items, they can "ask the audience" or ask to go "50-50" where two incorrect responses are removed. All these strategies derive their power from the fact that classroom participation is not optional, and even when the student resists, the teacher looks for ways to maintain the student's engagement.

How much time a teacher allows a student to respond before evaluating the response is also important. It is well known that teachers do not allow students much time to answer questions (Rowe, 1974),

and if they do not receive a response quickly, they will "help" the student by providing a clue or weakening the question in some way, or even moving on to another student. However, what is not widely appreciated is that the amount of time between the student providing an answer and the teacher's evaluation of that answer is just as, if not more, important. Of course, where the question is a simple matter of factual recall, then allowing a student time to reflect and expand upon the answer is unlikely to help much. But where the question requires thought, then increasing the time between the end of the student's answer and the teacher's evaluation from the average wait-time of less than a second to 3 seconds produces measurable increases in learning, although according to Tobin (1987) increases beyond 3 seconds have little effect, and may cause lessons to lose pace.

In fact, questions need not always come from the teacher. There is substantial evidence that getting students to generate their own questions enhances their learning. Rosenshine, Meister, and Chapman (1996) found that training students to generate questions while reading increased performance by 0.36 standard deviations on standardized tests, and by 0.86 standard deviations on tests developed by the experimenters. Foos, Mora, and Tkacz (1994) compared four strategies for helping students prepare for a test or examination: (a) tell the students to review the material on which the test is based, (b) provide the students with study methods and materials, (c) tell the students to generate their own study outlines, and (d) tell the students to generate their own study questions, with answers. They found that strategy (d) generated the highest performance, followed, in turn, by (c), (b), and (a). Such student-produced assessments can also be useful to the teacher, because they provide information about what the students think they have been learning, which may not be the same as what the teacher thinks the students have been learning.

Some researchers have gone even further and shown that questions can limit classroom discourse, because they tend to demand a simple answer. There is a substantial body of evidence that classroom learning is enhanced considerably by shifting from asking questions to making statements (Dillon, 1988). For example, instead of asking, "Are all squares rectangles?" which seems to require a simple yes/no answer, the level of classroom discourse (and student learning) is improved considerably by framing the same question as a statement—"All squares are rectangles"—and asking students to discuss this in small groups before presenting a reasoned assessment of the truth of this statement to the class.

Another key feature of classroom questioning is the way that teachers listen to student answers. As many authors have pointed out, when teachers listen to student responses, they attend more to the correctness of the answers rather than what they can learn about the student's understanding (Even & Tirosh, 1995, 2002; Heid, Blume, Zbiek, & Edwards, 1999). In a detailed study, Davis (1997) followed the changes in the practice of one middle school mathematics teacher, focusing in particular on how the teacher responded to student answers. Initially, the teacher's reactions tended to focus on the extent to which the student responses accorded with the teacher's expectations. After sustained reflection and discussion with the researcher over a period of several months, the teacher's reaction placed increasing emphasis on "information-seeking" as opposed to the "response-seeking" that characterized the earlier lessons. Davis termed these two kinds of listening *evaluative listening* and *interpretive listening* respectively. Towards the end of the 2-year period, there was a further shift in the teacher's practice, with a marked move away from clear lesson structures and prespecified learning outcomes, and towards the exploration of potentially rich mathematical situations, in which the teacher is a coparticipant. Most notably, in this third phase, the teacher's own views of the subject matter being taught developed and altered along with that of the students (what Davis termed *hermeneutic listening*). Similar trajectories of change from evaluative to interpretive listening have been observed in other mathematics teachers (English & Doerr, 2004) and in pre-service teachers (Crespo, 2000), as eloquently summarized by a girl in the seventh grade: "When Miss used to ask a question, she used to be interested in the right answer. Now she's interested in what we think" (Hodgen & Wiliam, 2006, p. 16).

What Makes a Good Question?

Careful analysis of students' incorrect responses to standard classroom items can reveal important insights into students' conceptions. For example, DeCorte and Verschaffel (2006) point out that when students are asked to respond to the following item:

$$\underline{\hspace{1cm}} - 12 = 7$$

many students write 18 or 5 in the blank space. Both responses are, of course, incorrect, but in completely different ways, the first probably indicating an arithmetical slip, and the second more likely indicating a lack of understanding of the meaning of the equal sign.

However, understanding the thinking behind students' responses is more straightforward when the questions asked of students have been planned careful-

ly ahead of time expressly for this purpose. Two items used in TIMSS, shown in Figure 23.2 below, illustrate one important feature of the design of good questions. Although apparently quite similar, the success rates on the two items were very different. For example, in Israel, 88% of the students answered the first item correctly, whereas only 46% answered the second correctly, with 39% choosing response (b) (Vinner, 1997). Vinner suggested that the reason for this is that many students, in learning about fractions, develop the naive conception that the largest fraction is the one with the smallest denominator, and the smallest fraction is the one with the largest denominator. This approach leads to the correct answer for the first item but leads to an incorrect response to the second. Further evidence for this interpretation is given by noting that 46% plus 39% is very close to 88%, suggesting that almost half of the students who answered the first item correctly did so with an incorrect strategy. In this sense, the first item is a much weaker item than the second, because students can get it right for the wrong reasons.

Item 1 (success rate 88%)
Which fraction is the smallest?

(a) $\frac{1}{6}$ (b) $\frac{2}{3}$ (c) $\frac{1}{3}$ (d) $\frac{1}{2}$

Item 2 (success rate 46%)
Which fraction is the largest?

(a) $\frac{4}{5}$ (b) $\frac{3}{4}$ (c) $\frac{5}{8}$ (d) $\frac{7}{10}$

Figure 23.2 Two items from the Third International Mathematics and Science Study.

This example illustrates a very general principle in teachers' classroom questioning. By asking questions of students, teachers try to establish whether students have understood what they are meant to be learning. In other words, the teacher is trying to construct a model of the student's thinking. As von Glasersfeld (1987) noted:

Inevitably, that model will be constructed, not out of the child's conceptual elements, but out of the conceptual elements that are the interviewer's own. It is in this context that the epistemological principle of *fit*, rather than *match* is of crucial importance. Just as cognitive organisms can never compare their conceptual organisations of experience with the structure of an independent objective reality, so the interviewer, experimenter, or teacher can never compare the model he or she has constructed of a child's conceptualisations with what actually goes on in the child's head. In the one case as in the other, the best that can

be achieved is a model that remains viable within the range of available experience. (p. 13)

The key phrase here is “within the range of available experience.” If the teacher asks questions that are more like the first TIMSS item above than the second, the range of available experience is narrow, and the range of models that fit, correspondingly large. As Gay and Thomas (1993) pointedly asked, “Just because they got it right, does that mean they know it?” (p. 130).

For teacher questioning to be effective, teachers need to know what kinds of conceptualizations students are likely to have, and need tools to identify them. Consider, for example, the following pair of simultaneous equations:

$$3a = 24$$

$$a + b = 16$$

Many students find this difficult, saying that it cannot be done. The teacher might conclude that they need some more help with equations of this sort, but with this particular pair of equations, a more likely reason for the difficulty is not with mathematical skills but with the student’s *beliefs* (see also Schoenfeld, 1985). If the students are encouraged to talk about their difficulty, they often say things like, “I keep on getting b is 8, but it can’t be because a is.” The reason that many students have developed such a belief is, of course, that before they were introduced to solving equations, they had been practicing substitution of numbers into algebraic formulas, where each letter always did stand for a different number. Although the students will not have been taught that each letter must stand for a different number, they have generalized implicit rules from their previous experience, just as because they are always shown triangles where the lowest side is horizontal, they talk of “upside-down triangles” (Askew & Wiliam, 1995 p. 14).

The important point here is that one would not have known about these unintended conceptions if the second equation had been $a + b = 17$ instead of $a + b = 16$. Items that reveal unintended conceptions—in other words that provide a “window into thinking”—are not easy to generate, but they are crucially important to improve the quality of students’ mathematical learning.

Some people have argued that these unintended conceptions are the result of poor teaching. If only the teacher had phrased the explanation more carefully, had ensured that no unintended features were learned alongside the intended features, then these misconceptions would not arise, but this argument fails to acknowledge two important points. The first is that this kind of overgeneralization is a fundamental feature of human

thinking. When young children say things like “I spend all my money,” they are demonstrating a remarkable feat of generalization. From the huge messiness of the language that they hear around them, they have learned that to create the past tense of a verb, one adds *d* or *ed*. In the same way, if one asks young children what causes the wind, a common answer is “trees.” They have not been taught this but have observed that trees are swaying when the wind is blowing and have inferred (incorrectly in this case) a causal relationship from a correlation.

The second point is that even if one wanted to, one cannot control the student’s environment to the extent necessary for unintended conceptions not to arise. For example, it is well known that many students believe that the result of multiplying 2.3 by 10 is 2.30. It is highly unlikely that they have been taught this. Rather this belief arises as a result of observing regularities in what they see around them. The result of multiplying whole numbers by 10 is just to add a zero, so why should that not work for all numbers? The only way to prevent students from acquiring this (mis)conception would be to introduce decimals before one introduces multiplying single-digit numbers by 10, which is clearly absurd. The important point is that one must acknowledge that what students learn is not necessarily what the teacher intended, and it is essential that teachers explore students’ thinking before assuming that students have “understood” something.

Questions that provide this “window into thinking” are hard to find, but within any school there will be good selection of rich questions in use—the trouble is that each teacher will have her or his stock of good questions, but these questions are not shared within the school and are certainly not seen as central to good teaching. In most Anglophone countries, teachers spend the majority of their lesson preparation time in grading students’ notebooks or assignments, almost invariably doing so alone. In some other countries, the majority of lesson preparation time is spent planning how new topics can be introduced, which contexts and examples will be used, and so on. This is sometimes done individually or with groups of teachers working together. In Japan, however, teachers spend a substantial proportion of their lesson preparation time working together to devise questions to use in order to find out whether their teaching has been successful, in particular through the process known as *lesson study* (Fernandez & Makoto, 2004).

In generating questions, the traditional concerns of reliability and validity do not provide sound guidance as to what makes a good question. For example, many teachers think that the following question, taken from the Chelsea Diagnostic Test for Algebra (Hart, Brown, Kerslake, Küchemann, & Ruddock, 1985), is “unfair”:

Simplify (if possible): $2a + 5b$

This item is deemed unfair because students “know” that in answering test questions, one must do some work, so it must be possible to simplify this expression, otherwise the teacher would not have asked the question. To use this item in a test or an examination where the goal is to determine a student’s achievement would probably not be a good idea. But to find out whether students understand algebra, it is a very good item indeed. If in the context of classroom work, rather than a formal test or exam, a student can be tempted to simplify $2a + 5b$ then the teacher should want to know that, because it means that the student has not yet developed a real sense of what algebra is about.

Asking students which of the following two fractions is larger raises similar issues:

$$\frac{3}{7} \quad \frac{3}{11}$$

In some senses this is a “trick question.” There is no doubt that this is a very hard item, with typically only around one 14-year-old in six able to give the correct answer, compared with around 75% of 14-year-olds being able to select correctly the larger of two “typical” fractions—that is where the numerators and denominators are different, and each less than 12 (Hart, 1981). It may not, therefore, be a very good item to use in a test of students’ achievement. But it is very important for the teacher to know if her students think that $\frac{3}{7}$ is larger than $\frac{3}{11}$. The fact that this item is seen as a trick question shows how deeply the summative function of assessment is ingrained into the practice of most teachers.

A third example, which caused considerable disquiet amongst teachers when it was used in a national test in England in the 1990s, is based on the following item, again taken from one of the Chelsea Diagnostic Tests:

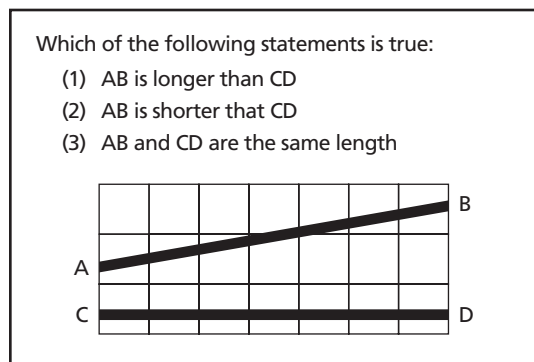


Figure 23.3 Item adapted from the CSMS tests (Hart et al., 1985).

Again, viewed in terms of formal tests and examinations, this may be an unfair item, but in terms of a teacher’s need to establish secure foundations for future learning, it would seem to be entirely appropriate.

All of the questions discussed so far have the potential to elicit evidence of active student conceptions that may hinder learning, and as such, have the potential to support teachers’ instructional decision-making. However, the questions discussed above, and most of the questions used by teachers, are really valuable only when the teacher can ask students to explain or elaborate their answers. Such questions are therefore good as *discussion questions* but less good as *diagnostic questions* (Ciofalo & Wylie, 2006). If it is necessary for each student to explain the reason for her or his answer, then it will take a great deal of time to get around the whole class. Furthermore, those students answering last are more likely to be choosing their justifications from amongst those already given by their peers than thinking hard about their own answers.

Consider the item shown in Figure 23.4. This item has some merit as a discussion item, because it is likely to elicit very quickly the fact that some students in a class think that $a^2 + b^2 = c^2$ for this triangle because it is true for all right triangles. However, as soon as students hear a peer mention that $a^2 + b^2 = c^2$ is true only when c is the hypotenuse, then they may well give the same response.

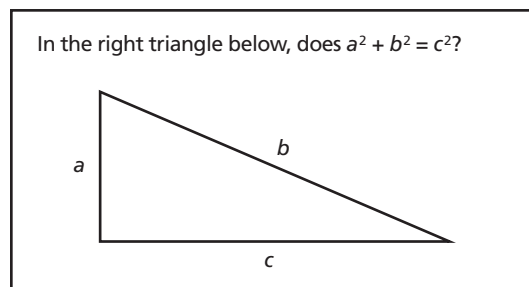


Figure 23.4 Item on Pythagoras’s rule.

As an alternative, consider the item shown in Figure 23.5. For this item, there are 64 possible responses (the statement might be true for each of the six triangles giving a total of 2^6 possibilities), so the chance of a student getting this correct by guessing is less than 2%. Moreover, the item can be used to get a response from every single student in the class at the same time, by giving students a set of six cards labeled A, B, C, D, E, and F. This use of questions with an *all-student response system* (Leahy et al., 2005) provides teachers with much richer data on the level of understanding in a class than is possible with a *single-student response*

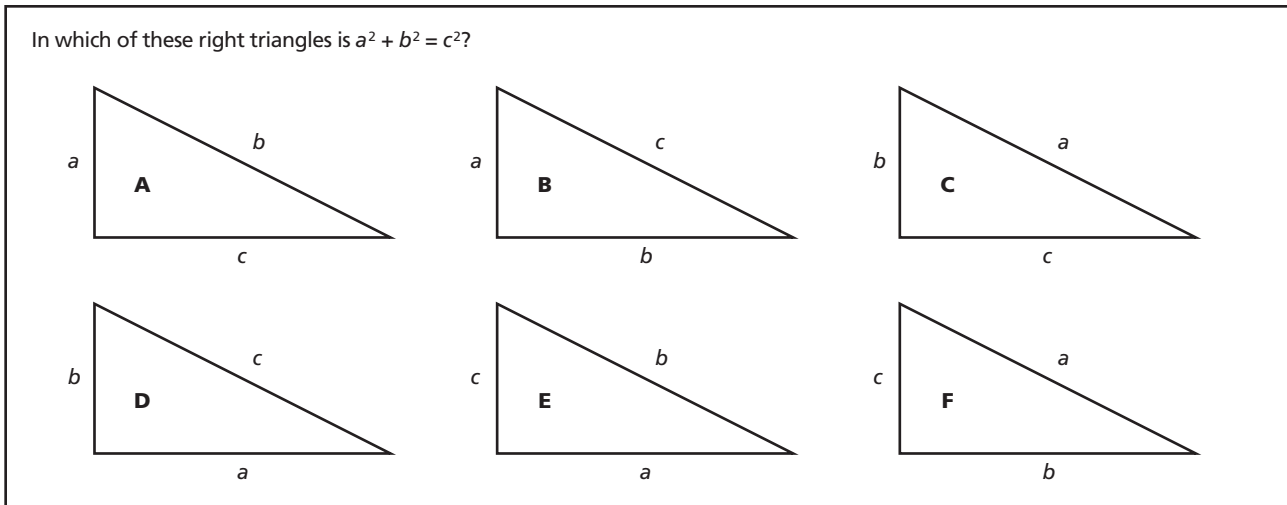


Figure 23.5 Item on Pythagoras's rule for use as an all-student response item.

system such as asking students to raise their hands, or selecting a student at random to answer.

By moving from single-student response systems to all-student response systems, the teacher is able to make more effective instructional decisions in real time by creating hinge-points in the lesson (Leahy et al., 2005). If all students answer correctly, then she can move on. If not, then the teacher has created a “teachable moment.” If no one answered correctly, then the teacher might choose to reteach the material to the whole class using a different approach from the one that she used originally. However, if some of the students answer correctly and some answer incorrectly, this provides an opportunity for a class discussion of the issue. Not only does the teacher know that some students have not understood, but she is able to use the information about the responses of each student to engineer a more effective discussion, by, for example, calling on all those with a particular response to supply their reasons for that response before considering other responses. Of course the aggregation of information from student responses is greatly enhanced by the use of electronic “clickers” (Roschelle, Abrahamson, & Penuel, 2004) but note that most such systems can currently cope with only a single response, so that items such as that in Figure 23.5 would be much more difficult to use. Incorrect responses to these items may reveal “facets” of student thinking (diSessa & Minstrell, 1998) and thus provide important evidence about student learning. In this sense, it is useful if the incorrect responses made by students are interpretable. However, if assuming that students do understand something when they don’t is more damaging than assuming they do not understand something when they do, then a more important re-

quirement for teaching is that the *correct* response is interpretable. In other words, the teacher needs to be sure that the student got the answer right for the right reasons.

Sets of lettered cards work well with multiple-choice items, but not all items can usefully be presented in such a format. For this reason, many teachers have adopted the use of “slates” or “mini white-boards” on which students can write and display an answer for the teacher to see. Such a technique is particularly useful where students’ responses are difficult to predict, such as asking students to write down a fraction between $\frac{1}{6}$ and $\frac{1}{7}$ if they can. Some students will say that it can’t be done, others will generate incorrect solutions, and others will generate correct solutions. The powerful feature of such unstructured response systems is that it allows students to generate substantially correct solutions in nonstandard form, such as

$$\frac{1}{6\frac{1}{2}}$$

Perhaps the most important point in all this is that questions worth asking are not likely to be generated spontaneously in the middle of a lesson. They need careful planning, preferably in collaboration with other teachers, as is practiced in Japanese lesson study (Fernandez & Makoto, 2004).

PROVIDING FEEDBACK THAT MOVES LEARNERS FORWARD

From the reviews of research conducted by Natriello (1987), Crooks (1988), Bangert-Drowns et al. (1991),

and Black and Wiliam (1998a) cited above it is clear that not all kinds of feedback to students about their work are equally effective. For example, Meisels, Atkins-Burnett, Xue, Bickel, and Son (2003) explored the impact of the Work Sample System (WSS)—a system of curriculum-embedded performance assessments—on the achievement of 96 third-grade urban students in reading and mathematics, as measured by the Iowa Test of Basic Skills (ITBS). When compared with a sample of 116 third graders in matched schools, and with students in the remainder of the school district (Pittsburgh, PA), the achievement of WSS students was significantly and substantially higher in reading, but in mathematics, the differences were much smaller and failed to reach statistical significance. The details of the system in use, how it is implemented, and the nature of the feedback provided to students seem to be crucial variables, with small changes often producing large impacts on the effectiveness of the system. Some insight into the characteristics of effective feedback is provided by a study by Nyquist (2003).

The reviews by Natriello, Crooks, and Black and Wiliam, had focused on K–12 education, but Nyquist reported that the findings about the importance of feedback generalize to higher education too. In reviewing the research on the effects of feedback in 185 studies in higher education, he developed the following typology of different kinds of formative assessment:

Weaker feedback only: students are given only the knowledge of their own score or grade, often described as “knowledge of results.”

Feedback only: students are given their own score or grade, together with either clear goals to work towards or feedback on the correct answers to the questions they attempt, often described as “knowledge of correct results.”

Weak formative assessment: students are given information about the correct results, together with some explanation.

Moderate formative assessment: students are given information about the correct results, some explanation, and some specific suggestions for improvement.

Strong formative assessment: students are given information about the correct results, some explanation, and specific activities to undertake in order to improve.

He then calculated the average standardized effect size for the studies for each type of intervention, and the results are given in Table 23.2.

Table 23.2 Standardized Effect Sizes for Different Kinds of Feedback Interventions

	N	Effect
Weaker feedback only	31	0.16
Feedback only	48	0.23
Weaker formative assessment	49	0.30
Moderate formative assessment	41	0.33
Strong formative assessment	16	0.51
Total	185	

Nyquist’s results echo the findings of Bangert-Drowns et al. discussed above. Just giving students feedback about current achievement produces very little benefit, but where feedback engages students in mindful activity, the effects on learning can be profound.

In one study (Elawar & Corno, 1985), 18 sixth-grade teachers in three schools in Venezuela received 7 hours of training on how to give constructive written feedback on the mathematics homework produced by their students (specific comments on errors, suggestions about how to improve, and at least one positive remark). Another group of teachers graded homework as normal (i.e., just scores), and a third group gave constructive feedback to half their classes and just scores to the other half. The students receiving the constructive feedback learned twice as fast as the control-group students (in other words, they learned in 1 week what the others would have taken 2 weeks to learn). Moreover, in classes receiving the constructive feedback, the achievement gap between male and female students was reduced, and attitudes towards mathematics were more positive.

The negative impact of grades was also established in a study of Israeli students conducted by Butler and Nisan (1986). In this study, one group of students received nonthreatening task-related evaluations of their work, one group received normative grades, and one group received no feedback. The level of intrinsic motivation of students given grades was lower than those given the task-related evaluations, and this was also reflected in the level of achievement. Interestingly, those given no feedback also showed lower levels of intrinsic motivation. This suggests that task-related feedback has an important role to play in building intrinsic motivation (see section on *Activating Students as Owners of Their Own Learning* below) and that lack of feedback can, under some circumstances, be as deleterious as the wrong sort of feedback, but the relationship is clearly complex, inasmuch as other studies have not found lack of feed-

back to be consistently associated with lower levels of learning. For example, Grolnick and Ryan (1987) reported a study of 91 fifth-grade children that assessed the effects of different conditions on emotional experience and performance on a learning task. Two directed-learning conditions, one controlling (DLC) and one noncontrolling (DLN), were contrasted with each other and with a third nondirected, spontaneous-learning context (NLC). In both the DLC and DLN conditions, what was to be learned was specified and students knew that there would be an assessment, but in the DLC condition, students were told they would be given grades as they worked, whereas those in the DLN condition were told there would be no grades and that the purpose of the activity was to see what they could learn. In the NLC condition, students were exposed to material without specifications and had no knowledge of the subsequent assessment. On measures of rote learning, both the DLC and DLN groups scored higher than the NLC, but the level of interest and the amount of conceptual learning was lower in the DLC group than in the other two. Furthermore, children in the DLC condition evidenced a greater deterioration in rote learning in a follow-up assessment approximately 1 week later.

In a subsequent study, Butler (1988) investigated the effectiveness of different kinds of feedback in 12 sixth-grade classes in four Israeli schools. For the first lesson, the students in each class were given a booklet containing a range of divergent thinking tasks. At the end of the lesson, their work was collected. Independent scorers then assessed this work. At the beginning of the next lesson, two days later, the students were given feedback on the work they had done in the first lesson. In 2 of the classes students were given scores (which were scaled so as to range from 40 to 99) while in another 4 of the classes, students were given comments, such as "You thought of quite a few interesting ideas; maybe you could think of more ideas." In the other 4 classes, both scores and comments were written in the students' notebooks. Then, the students were asked to attempt some similar tasks and told that they would get the same sort of feedback as they had received for the first lesson's work. Again, the work was collected and assessed.

In order to explore the differences in impact of these treatments on students, the achievement and attitudes toward the subject under study of the highest performing students in each class were compared with the lowest achievers in each class. Those given only scores made no gain from the first lesson to the second. Those who had received high scores in the tests were interested in the work, but those who had received low scores were not. The students given only

comments scored, on average, 30% higher on the work done in the second lesson than on the first, and the interest of all the students in the work was high. However, those given both scores and comments made *no gain* from the first lesson to the second, and those who had received high scores showed high interest whereas those who received low scores did not. The results are summarized in Table 23.3.

Table 23.3 Impact of Feedback Interventions on Achievement and Attitude (Butler, 1988)

	Achievement gain	Attitude toward subject
Scores only	none	High achievers: positive Low achievers: negative
Comments only	30%	All positive
Scores and comments	none	High achievers: positive Low achievers: negative

Far from producing the best effects of both kinds of feedback, giving scores alongside the comments completely washed out the beneficial effects of the comments. When given both scores and comments, most students looked at the score first. The next thing most of these students looked at was the score of their nearest neighbor. In other words, if teachers write careful diagnostic comments on a student's work and then put a score or grade on it, they are likely wasting the time they spend writing comments. The students who get the high scores do not need to read the comments, and the students who get the low scores do not want to. And yet the use of both scores and comments is probably the most widespread form of feedback in the United States, despite the fact that this study (and others like it—see below) shows that it is no more effective than scores alone. The teacher would be better off just giving a score. The students will not learn anything from this, but the teacher will save herself a great deal of time.

A clear indication of the role that ego plays in learning is given by another study by Ruth Butler (1987). In this study, 200 fifth- and sixth-grade students in eight classes spent a lesson working on a variety of divergent thinking tasks. Again, the work was collected and the students were given one of four kinds of feedback on this work at the beginning of the second lesson (again 2 days later):

In two of the classes, students were given written comments;

In two of the classes, students were given grades;

In two of the classes, students were given written praise; and

In two of the classes, students were given no feedback at all.

The quality of the work done in the second lesson was compared to that done in the first. The quality of work of those given comments had improved substantially compared to the first lesson, but those given grades and praise had made no more progress than those given no feedback. At the end of the second lesson, the students completed a questionnaire about the factors that influenced their work that was designed to elicit from the students the reasons for their level of motivation. In particular the questionnaire sought to establish whether the students' effort investment was due to ego-related concerns (such as class rank) or task-related concerns (such as learning the material). The students who had received comments during their work on the topic had high levels of task involvement, but their levels of ego involvement were the same as those given no feedback. However, although those given grades and those given written praise had comparable levels of task involvement to the control group, their levels of ego involvement were substantially higher. The only effect of the grades and the praise, therefore, was to increase the sense of ego involvement without increasing achievement. These findings are consistent with those of Cameron and Pierce (1994), who found that although verbal praise and supportive feedback did increase students' interest in and attitude towards a task, such feedback had little if any effect on performance.

These findings are consistent with the research on praise carried out in the 1970s that showed clearly that praise was not necessarily "a good thing"—in fact the best teachers appear to praise slightly less than average (Good & Grouws, 1975). It is the quality rather than the quantity of praise that is important. In particular, teacher feedback is far more effective when it is infrequent, credible, contingent, specific, and genuine (Brophy, 1981) and focuses on features of the work that are within the students' control (Dweck, 2000; Siero & van Oudenhoven, 1995).

From a young age, students' attitudes to learning are clearly shaped by the feedback they receive. In a year-long study of eight kindergarten and first-grade classrooms in six schools, Tunstall and Gipps (1996a, 1996b) identified a range of roles that feedback played in classrooms in England. Like Torrance and Pryor (1998), they found that a substantial proportion of the feedback given to students focused on socialization: "I'm only helping people who are sitting down with their hands up" (p. 395). Beyond this socialization

role, they identified four types of feedback on academic work, which they labeled A, B, C and D as shown in Table 23.4. In the first type was placed feedback that rewarded or punished the students for their work, for example when students were allowed to leave the classroom for lunch early if they had done good work, or where students were told that if they had not completed assigned tasks, they would have to stay in during lunch to finish their work. Type B feedback was, like Type A, evaluative, but rather than focusing on rewards and sanctions, the feedback indicated the teacher's approval or disapproval (e.g., "I'm very pleased with you" versus "I'm very disappointed in you today").

In addition to the kinds of *evaluative* feedback described above, they identified two types of *descriptive* feedback. Type C feedback focused on academic work as product and emphasized the adequacy of the work with respect to the teacher's criteria for success. At one end of the continuum, feedback focused on the extent to which the work already satisfied the criteria, while at the other end, the feedback focused on what the student needed to do to improve the work. An example of the former is, "This is extremely well explained," pointing out the way that the student had satisfied the teacher's criteria. An example of the latter is, "I want you to go over all of them and write your equals sign in each one"—here the emphasis is on what needs to be done to improve the work.

In contrast to the idea of work as product, Type D feedback focused on the process aspects of work, with the teacher cast in the role of facilitator, rather than judge or coach. As Tunstall and Gipps (1996a) explain, teachers engaged in this kind of feedback "conveyed a sense of work in progress, heightening awareness of what was being undertaken and reflecting on it" (p. 399).

Table 23.4 Typology of Teacher Feedback (Tunstall & Gipps, 1996a)

Evaluative feedback	Type A	Type B
Positive	Rewarding	Approving
Negative	Punishing	Disapproving
Descriptive feedback	Type C	Type D
Achievement feedback	Specifying attainment	Constructing achievement
Improvement feedback	Specifying improvement	Constructing the way forward

The timing of feedback is also crucial. If it is given too early, before students have had a chance to work

on a problem, then they will learn less. Most of this research has been done in the United States, where it goes under the name of “peekability research,” because the important question is whether students are able to “peek” at the answers before they have tried to answer the question. However, a British study undertaken by Simmons and Cope (1993) found similar results. Pairs of students aged between 9 and 11 worked on angle and rotation problems. Some of these worked on the problems using the computer language Logo and some worked on the problems using pencil and paper. The students working in Logo were able to use a “trial and improvement” strategy that enabled them to get a solution with little mental effort. However, for those working with pencil and paper, working out the effect of a single rotation was much more time consuming, and thus the students had an incentive to think carefully, and this greater “mindfulness” led to deeper learning.

The effects of feedback highlighted above might suggest that the more feedback, the better, but this is not necessarily the case. Day and Cordon (1993) looked at the learning of a group of 64 third-grade students on reasoning tasks. Half of the students were given a “scaffolded” response when they got stuck—in other words they were given only as much help as they needed to make progress, whereas the other half were given a complete solution as soon as they got stuck, and then given a new problem to work on. Those given the scaffolded response learned more and retained their learning longer than those given full solutions. In a sense, this is hardly surprising, inasmuch as those given the complete solutions had the opportunity for learning taken away from them. As well as saving time for the teacher, developing skills of “minimal intervention” promotes better learning.

A good example of such feedback is given by Saphier (2005, p. 92):

Teacher: “What part don’t you understand?”

Student: “I just don’t get it.”

Teacher: “Well, the first part is just like the last problem you did. Then we add one more variable. See if you can find out what it is, and I’ll come back in a few minutes.”

Sometimes, the help need not even be related to the subject matter. Often, when a student is given a new task, the student asks for help immediately. When the teacher asks, “What can’t you do?” it is common to hear the reply, “I can’t do any of it.” In such circumstances, the student’s reaction may be caused by anxiety about the unfamiliar nature of the task, and it is frequently possible to support the student by saying

something like “Copy out that table, and I’ll be back in five minutes to help you fill it in.” This is often all the support the student needs. Copying out the table forces the student to look in detail at how the table is laid out, and this “busy work” can provide time for the student to make sense of the task herself (see discussion of performance versus mastery goals below).

The consistency of these messages from research on the effects of feedback extends well beyond school and other educational settings. A review by Kluger and DeNisi (1996) of 131 well-designed studies in educational and workplace settings found that, on average, feedback did improve performance, but this average effect disguised substantial differences between studies. Perhaps most surprisingly, in 38% of the studies, giving feedback had a negative impact on performance. In other words, in 2 out of every 5 carefully controlled scientific studies, giving people feedback on their performance made their performance worse than if they were given no feedback on their performance at all! On further investigation, the researchers found that feedback made performance worse when it was focused on the self-esteem or self-image (as is the case with grades and praise). The use of praise can increase motivation, but then it becomes necessary to use praise all the time to maintain the motivation. In this situation, it is very difficult to maintain praise as genuine and sincere. In contrast, the use of feedback improves performance when it is focused on what needs to be done to improve, and particularly when it gives specific details about *how* to improve (see section on *Activating Students as Owners of Their Own Learning* below).

This suggests that feedback, as the term is used currently in education, is not the same as formative assessment. Feedback is a necessary first step, but feedback is formative *only if the information fed back to the learner is used by the learner in improving performance*. If the information fed back to the learner is intended to be helpful but cannot be used by the learner in improving her own performance, it is not formative. It is rather like telling an unsuccessful comedian to “be funnier.”

CLARIFYING AND SHARING LEARNING INTENTIONS AND SUCCESS CRITERIA WITH LEARNERS

In a chapter entitled “The View From the Student’s Desk” Mary Alice White (1971) suggested that students often had no real idea where they were going in their learning.

The analogy that might make the student’s view more comprehensible to adults is to imagine oneself on a

ship sailing across an unknown sea, to an unknown destination. An adult would be desperate to know where he [*sic*] is going. But a child only knows he is going to school. . . . The chart is neither available nor understandable to him. . . . Very quickly, the daily life on board ship becomes all important. . . . The daily chores, the demands, the inspections, become the reality, not the voyage, nor the destination. (White, 1971, p. 340)

Certainly many authors, notably Bernstein (1975), have shown that not all students share the teacher's understanding of what they are meant to be doing in classrooms. As a simple example, consider the task of identifying the "odd one out" in the following list of objects: knife, fork, hammer, bottle of tomato ketchup. Some students believe that the bottle of tomato ketchup is the odd one out, because the others are all metal tools, whereas other students believe that the hammer is the odd one out because the others appear on the table at meal-times. Neither answer is wrong or right, of course, but as Keddie (1971) pointed out over 35 years ago, schools value some kinds of knowledge more than others, although what precisely is desired or valued is not always made clear.

For example, in a study of 72 students between the ages of 7 and 13, Gray and Tall (1994) found that the reasoning of the higher achieving students was qualitatively different from that of the lower achieving students. This, of course, is not surprising, as the research on expertise (Berliner, 1994) shows that this is a characteristic of expertise. What was surprising in Gray and Tall's study was in the nature of the difference. In particular, the higher achieving students were able to work with unresolved ambiguities about whether mathematical entities are concepts or procedures. In contrast, lower attaining students, by refusing to accept the ambiguities inherent in mathematics are, in fact, attempting a far more difficult form of mathematics, with a far greater cognitive demand.

A simple example may be illustrative here. Consider the number $6\frac{1}{2}$. The mathematical operation between the 6 and the $\frac{1}{2}$ is actually addition. $6\frac{1}{2}$ is a shorthand expression for $6 + \frac{1}{2}$, but when writing an expression in algebra by concatenating two terms, such as $6x$, the implied operation between the 6 and the x is multiplication, not addition. The relationship between the 6 and the 1 in 61 is different again. And yet, very few people who are successful in mathematics are aware of these inconsistencies or differences in mathematical notation. In a very real sense, being successful in mathematics requires knowing what to worry about and what not to worry about. Students who do not understand what is important and what is not important will be at a very real disadvantage.

Quite how big a difference this can make is brought out in a study of seventh-grade science classes conducted by White and Frederiksen (1998). The study involved three teachers, each of whom taught 4 parallel seventh grade classes in two U.S. schools. The average size of the classes was 31 students. In order to assess the representativeness of the sample, all the students in the study were given a basic skills test, and their scores were close to the national average. All 12 classes followed a novel curriculum (called ThinkerTools) for 14 weeks. The curriculum had been designed to promote thinking in the science classroom through a focus on a series of seven scientific investigations (approximately 2 weeks each). Each investigation incorporated a series of evaluation activities. In two of each teacher's four classes, these evaluation episodes took the form of a discussion about what they liked and disliked about the topic. For the other two classes, they engaged in a process of "reflective assessment." Through a series of small-group and individual activities, the students were introduced to the nine assessment criteria (each of which was assessed on a 5-point scale) that the teacher would use in evaluating their work. At the end of each episode within an investigation, the students were asked to assess their performance against two of the criteria. At the end of the investigation, students had to assess their performance against all nine. Whenever they assessed themselves, they had to write a brief statement showing which aspects of their work formed the basis for their rating. At the end of each investigation, students presented their work to the class, and the students used the criteria to give one another feedback.

As well as the students' self-evaluations, the teachers assessed each investigation, scoring both the quality of the presentation and the quality of the written report, each being scored on a 1 to 5 scale. The possible score on each of the seven investigations therefore ranged from 2 to 10. The mean project scores achieved by the students in the two groups over the seven investigations are summarized in Table 23.5, classified according to their score on the basic skills test.

Table 23.5 Mean Project Scores for Students

Group	Score on basic skills test		
	Low	Intermediate	High
Likes and dislikes (control group)	4.6	5.9	6.6
Reflective assessment (treatment group)	6.7	7.2	7.4

Note. The 95% confidence interval for each of these means is approximately 0.5 either side of the mean.

Two features are immediately apparent in these data. The first is that the mean scores are higher for the students doing reflective assessment, when compared with the likes-and-dislikes group—in other words, all students improved their scores when they thought about what it was that counted as good work. However, much more significantly, the difference between the “likes and dislikes” group and the “assessment” group was much greater for students with weak basic skills. This suggests that, at least in part, low achievement in mathematics is exacerbated by students’ not understanding what it is they are meant to be doing.

Although students clearly need to understand the standards against which their work will be assessed, the study by White and Frederiksen shows that the criteria themselves are only the starting point. At the beginning, the words do not have the meaning for the student that they have for the teacher. Just giving “quality criteria” or “success criteria” to students will not work, unless students have a chance to see what this might mean in the context of their own work. Furthermore, as Stiggins (2001, pp. 314–322) pointed out, there is a very real tension between task-specific and generic rubrics. Task-specific rubrics can be written in very clear language and thus can communicate accurately to students what is required, but students need to come to grips with a new rubric for each task. Generic rubrics, on the other hand, can span a number of tasks but are unlikely to provide clear guidance to a student about how to improve her or his work. Another weakness of task-specific rubrics is that, by definition, they focus only on the qualities needed for the specific task and therefore may not contribute to more generalizable skills. Too much attention to one’s performance relative to a detailed rubric may, in fact, be counterproductive (Kohn, 2006). For this reason, Arter and McTighe (2001) suggested that task-specific rubrics are more appropriate for summative assessment, where consistency of scoring is paramount, but that generic rubrics are more appropriate for formative assessment, inasmuch they focus on qualities that transcend the immediate task (for more on the characteristics of good rubrics, see *Relearning by Design*, 2000).

All the features of good rubrics discussed above apply to the use of rubrics for both summative and formative purposes, but if rubrics are to function formatively, additional considerations apply. Effective summative assessment requires those engaged in assessment to share the same construct of quality, and rubrics can play an important role in promulgating common standards (Sadler, 1987). Effective formative assessment requires the student to share the same construct:

The indispensable conditions for improvement are that the student comes to hold a concept of quality roughly similar to that held by the teacher, is continuously able to monitor the quality of what is being produced during the act of production itself, and has a repertoire of alternative moves or strategies from which to draw at any given point. (Sadler, 1989, p. 121)

However, it is possible for a rubric to identify accurately different degrees of quality in a piece of work without explicitly identifying the relationship between the levels. Both a teacher and a student may be able to identify that a particular piece of work merits a 4 on a six-point rubric, but be unclear about what needs to be done to move the piece of work forward to a 5 or a 6, much in the same way as one can tell a softball pitcher that she needs to lower her ERA or get her rising fast-ball to rise without having any idea how to accomplish this. For this reason, it is particularly helpful for formative purposes if rubrics are longitudinal or developmental, so that they are not just descriptions of quality, but effectively *anatomies* of quality—identifying the immediate steps that can be taken to begin improvement.

In clarifying and sharing learning intentions and success criteria with learners, it is also important to be clear about whether the particular aspect of quality sought can be communicated with a rubric at all. Some aspects of quality, such as whether calculations are correct or whether diagrams are drawn in pencil and labeled, can easily be set down in a rubric. On the other hand, some features, such as whether an approach to a mathematical investigation is systematic or not, cannot be so easily described. Using a term like “Adopts a systematic approach” in a rubric may look like a definition of quality, but it is unlikely to be meaningful to those who do not already know what it means to be systematic. In Michael Polanyi’s terms, it is not a rule that can be easily put into effect but is rather a *maxim*:

Maxims cannot be understood, still less applied by anyone not already possessing a good practical knowledge of the art. They derive their interest from our appreciation of the art and cannot themselves either replace or establish that appreciation. (Polanyi, 1958, p. 50)

In such situations, the best we can do is to help students develop what Guy Claxton called a “nose for quality” (Claxton, 1995). Rubrics may have a role to play in this process, as they did in the study by White and Frederiksen. Rubrics were shared with students, but the students were given time to think through, in discussion with others, what this might mean in

practice, applied to their own work. One should not assume that the students will understand these right away, but the criteria will provide a focus for negotiating with students about what counts as quality in the mathematics classroom.

Another way of helping students understand the criteria for success is, before asking the students to embark on (say) an investigation, to get them to look at the work of other students (suitably anonymized) on similar (although not, of course, the same) investigations. In small groups, they can then be asked to decide which pieces of students' work are good investigations, and why. It is not necessary, or even desirable, for the students to come to firm conclusions and can generate a definition of quality—what is crucial is that they have an opportunity to explore notions of *quality* for themselves. Spending time looking at other students' work, rather than producing their own work, may seem like time off-task, but the evidence is that it is a considerable benefit, particularly for students who do not find mathematics easy to learn.

ACTIVATING STUDENTS AS OWNERS OF THEIR OWN LEARNING

The power of getting students to take some ownership of their own work is shown very clearly in an experiment by Fontana and Fernandez (1994). A group of 25 Portuguese primary school teachers met for 2 hours each week over a 20-week period during which they were trained in the use of a structured approach to student self-assessment. The approach to self-assessment involved an exploratory component and a prescriptive component. In the exploratory component, each day, at a set time, students organized and carried out individual plans of work, choosing tasks from a range offered to them by the teacher, and had to evaluate their performance against their plans once each week. The progression within the exploratory component had two strands—over the 20 weeks, the tasks and areas in which the students worked were to take on the students' own ideas more and more, and secondly, the criteria that the students used to assess themselves were to become more objective and precise.

The prescriptive component took the form of a series of activities, organized hierarchically, with the choice of activity made by the teacher on the basis of diagnostic assessments of the students. During the first 2 weeks, children chose from a set of carefully structured tasks and were then asked to assess themselves. For the next 4 weeks, students constructed their own mathematical problems following the patterns

of those used in weeks 1 and 2 and evaluated them as before but were required to identify any problems they had, and whether they had sought appropriate help from the teacher. Over the next 4 weeks, students were given further sets of learning objectives by the teacher and again had to devise problems, but now they were not given examples by the teacher. Finally, in the last 10 weeks, students were allowed to set their own learning objectives, to construct relevant mathematical problems, to select appropriate manipulatives, and to identify suitable self-assessments.

Another 20 teachers, matched in terms of age, qualifications, and experience, using the same curriculum scheme, for the same amount of time, and receiving the same amount of professional development, acted as a control group. The 354 students being taught by the 25 teachers using self-assessment and the 313 students being taught by the 20 teachers acting as a control group were each given the same mathematics test at the beginning of the project, and again at the end of the project 20 weeks later. Over the course of the experiment, the scores of the students taught by the control-group teachers improved by 7.8 points. The scores of the students taught by the teachers developing self-assessment improved by 15 points—almost twice as big an improvement. In other words, the students participating in the self-assessment learned in 20 weeks what the students in the control group classrooms would take 40 weeks to learn—a doubling of the rate of learning.

Whether the students in this study were able to assess their own performance objectively is unclear, and in general, such questions are a matter of heated debate, but very often the debate takes place at cross-purposes. Opponents of self-assessment say that students cannot possibly assess their own performance objectively, but this is an argument about the *summative* function of self-assessment, which is beyond the scope of this chapter. The focus here is whether activating students as owners of their own learning does, indeed, enhance learning, and in this regard, concordance with other, external judgments is a secondary concern. What matters is whether enhancing students' engagement with and ownership of their own learning enhances learning. For this to happen students have to be motivated (literally, moved) to do so, but they also have to possess the necessary cognitive resources.

The key concept here is *self-regulation* (Boekaerts, Maes, & Karoly, 2005), defined as “a multilevel, multicomponent process that targets affect, cognitions, & actions, as well as features of the environment for modulation in the service of one's goals” (Boekaerts, 2006, p. 347). Put simply, students who lack self-regulation skills are unlikely to be able to take control of

and guide their own learning (DeCorte, Verschaffel, & Op't Eynde, 2000). Inevitably, with such a broad area, different researchers have emphasized different aspects of self-regulation.

Winne (1996) emphasized the cognitive aspects of this process, defining self-regulated learning as a “metacognitively governed behavior wherein learners adaptively regulate their use of cognitive tactics and strategies in tasks” (p. 327). Others have observed that many students appear to possess the necessary skills of self-regulation but do not use them in classrooms (and especially in mathematics classrooms), which suggests that the problem is not a lack of skill, but rather a lack of motivation or volition (Corno, 2001). Still others have argued for the integration of different perspectives within sociocultural (Hickey & McCaslin, 2001; McCaslin & Hickey, 2001) or social constructivist (Op't Eynde, DeCorte, & Verschaffel, 2001) perspectives.

These two broad areas—cognition and motivation—have been extremely fertile areas of inquiry over the last quarter century or so, but unfortunately most of the research undertaken has been firmly in one area or the other (Wigfield, Eccles, & Rodriguez, 1998). As Sorrentino and Higgins (1986) pointed out there has been an “almost total exclusion of motivation in research on cognition” (p. 3); Nisbett and Ross (1980), in their influential book on human reasoning, lamented psychology’s “inability to bridge the gap between cognition and behavior” (p. 11). Nevertheless, to realize the potential for self-regulation to improve learning, ways of bridging this gap must be found, because, as Boekaerts (2006) showed, self-regulated learning is both metacognitively governed *and* affectively charged (p. 348).

In the next two sections, the research on cognition and motivation will be briefly reviewed, focusing on the aspects that are most relevant to the idea of activating students as owners of their own learning. The implications of this research for using assessment to activate students as owners of their own learning will then be explored through the use of the dual-processing model of self-regulation proposed by Boekaerts (2006).

Metacognition

It is perhaps not too much of an overstatement to say that everyone agrees that metacognition is important, but no one can agree on what it is, at least in a sufficiently precise definition to put into practice. Even John Flavell, widely credited with inventing the term, acknowledged that it was a “fuzzy concept” (Flavell, 1981 p. 37). Literally, it is “beyond thinking” or thinking about thinking. Flavell (1976) defined the concept thus:

“Metacognition” refers to one’s knowledge concerning one’s own cognitive processes and products or anything related to them, e.g., the learning-relevant properties of information and data. For example I am engaging in metacognition (metamemory, metalearning, metaattention, metalanguage, or whatever) if I notice that I am having more trouble learning A than B; if it strikes me that I should double-check C before accepting it as a fact; if it occurs to me that I had better scrutinise each and every alternative in any multiple-choice type task situation before deciding which is the best one; if I sense that I had better make a note of D because I may forget it; if I think to ask someone about E to see if I have it right. In any kind of cognitive transaction with the human or nonhuman environment, a variety of information processing activities may go on. Metacognition refers, among other things, to the active monitoring and consequent regulation and orchestration of these processes in relation to the cognitive objects or data on which they bear, usually in the service of some concrete goal or objective. (p. 232)

It is thus an “umbrella term [that encompasses all] knowledge and cognition about cognitive phenomena” (Flavell, 1979, p. 906), including knowing what one knows (*metacognitive knowledge*), what one can do (*metacognitive skills*), and what one knows about one’s own cognitive abilities (*metacognitive experience*). Kluwe (1982, p. 202) broadened the term further, by including knowledge about the thinking of others as well as oneself. Brown (1987) identified “four historically separate, but obviously interlinked, problems in psychology that pertain to issues of metacognition” (p. 69):

- whether individuals can have access to their own cognitive processes
- who or what is responsible for executive control of cognition
- what is the role of self-regulation within learning
- what is the role of regulation by others in learning

although there is a continuing debate about whether metacognition has to be a conscious process (Carr, Alexander, & Folds-Bennett, 1994).

Studies have repeatedly shown that students with greater awareness of their own cognitive processes have higher achievement (DeCorte, Verschaffel, & Op't Eynde, 2000), as do students who display more conscious self-regulation (see e.g., Stillman & Galbraith, 1998). Butler and Winne (1995) summarized the situation thus: “Theoreticians seem unanimous—the most effective learners are self-regulating” (p. 245), but this leaves unresolved the question of whether improving metacognitive skills improves achievement, and this is

where the lack of an agreed operational definition of metacognition is most serious.

Many studies have shown that training students in metacognitive strategies improves their performance even in the early school years (e.g., Lodico, Ghatala, Levin, Pressley, & Bell, 1983) and can also generalize metacognitive strategies to new situations (see Hacker, Dunlosky, & Graesser, 1998, for a review), but other studies have failed to find benefits for metacognitive training. For example, in a multilevel study of 444 seventh grade students in 18 heterogeneous classrooms in the Netherlands, Hoek, van den Eeden, and Terwel (1999) found that the achievement of students trained in metacognitive and social strategies simultaneously was no greater than controls just told to work together (although differences in two of the three outcome measures were found in single-level models). Without a clear operational definition of metacognition, this result is difficult to interpret. Those who advocate the benefits of metacognitive training can just claim that the unsuccessful efforts were not “real” metacognition.

What is clear is that many studies *have* found clear benefits for metacognitive interventions in real classroom settings as well as laboratories. In the previous edition of this *Handbook*, Schoenfeld (1992) reviewed the research on metacognition with a specific focus on the learning of mathematics, although most of the studies available at that time had been conducted in laboratory settings or in small-scale trials. Since then, a series of studies have shown that training in self-evaluation improves student achievement in real settings, and across extended periods of time.

For example, Ross, Hogaboam-Gray, and Rolheiser (2002) reported on a study in which 12 fifth- and sixth-grade mathematics teachers were trained in the promotion of systematic student self-evaluation, and the performance of their students ($N = 259$) was compared with students ($N = 257$) taught by teachers who had not been so trained. After a 12-week period, the students undertaking self-evaluation outscored the other students by 0.4 standard deviations on a mathematics task.

Mevarech and her colleagues in Israel have demonstrated that metacognitive training improves student achievement in a range of settings, and across even longer periods of time. Their method is termed *IMPROVE* after the initial letters of the key components:

- Introducing the new concepts;
- Metacognitive questioning in small groups;
- Practicing, Reviewing, and reducing difficulties;
- Obtaining mastery; and
- Verification and Enrichment.

In one year-long study 99 seventh-grade students in three classrooms in Israel were taught using the *IMPROVE* method, and their performance was compared with 148 students from five other classes (Mevarech & Kramarski, 1997). Although the *IMPROVE* students scored slightly lower than the control group students on a pretest, they outscored controls on both an “Introduction to Algebra” test and on a test of mathematical reasoning. The standardized effect size d (Cohen, 1988) of the advantage for the *IMPROVE* students was 0.31 standard deviations for the “Introduction to algebra” test and 0.44 standard deviations on the test of mathematical reasoning. A second study involving 265 seventh graders found a similar result ($d = 0.38$).

Other studies found that the *IMPROVE* method was more effective than just training in metacognitive skills (Kramarski, Mevarech, & Lieberman, 2001) or structured collaborative work (Kramarski, Mevarech, & Arami, 2002), and the benefits extended to both traditional measures of achievement and authentic tasks. More recent work has established that achievement gains are still present after 1 year and are larger for lower achieving students (Mevarech & Kramarski, 2003).

Another research program—Cognitive Acceleration in Mathematics Education or *CAME* (Adhami, Johnson, & Shayer, 1998)—also aims to increase student achievement in mathematics by attention to metacognitive processes. Although it shares many features with *IMPROVE*, the program seems to make somewhat greater demands on teachers and is therefore more difficult to implement than the *IMPROVE* project. Results from 68 classrooms monitored over a 2-year period suggest that the average increase in student achievement was approximately 0.34 standard deviations (Adhami et al., 1997), although the distribution was trimodal, with one peak around 0.8 standard deviations (almost all the classes coming from one school), one peak around 0.5 standard deviations, and another at 0 (which appears to be attributable to schools that did not implement the program as intended).

Other cognitive-behavior interventions have been reviewed by Boekaerts and Corno (2005, pp. 213–221), but the lack of an adequate theorization of metacognition makes comparing different interventions difficult, if not impossible. Nevertheless, the results from *IMPROVE*, *CAME*, and similar interventions suggest that metacognitive interventions can have a substantial impact on student achievement in mathematics, even if the design of such interventions is currently more art than science. It is also no coincidence that the most effective programs for developing metacognitive skills are firmly rooted in specific domains of knowledge (Bransford, Brown, & Cocking, 2000). However, just possessing the skills of being able

to monitor one's own learning is not enough for students to learn mathematics. They also have to want to do so, and, as any mathematics teacher is all too aware, this cannot be taken for granted.

Motivation and Learning

As Edward Deci (1996) noted, "Most people seem to think that the most effective motivation comes from outside the person, that it is something that one skilful person does to another" (p. 9), although there is substantial evidence that the situation is not that simple. In their explanation of Self-Determination Theory (SDT), Deci and Ryan (1985) distinguished between different types of motivation based on the different reasons or goals that give rise to a particular action. "The most basic distinction is between *intrinsic motivation*, which refers to doing something because it is inherently interesting or enjoyable, and *extrinsic motivation*, which refers to doing something because it leads to a separable outcome" (Ryan & Deci, 2000, p. 55).

Because of its power to drive quality learning, the notion of intrinsic motivation has been extensively studied, especially how the actions of teachers and parents can build or undermine it (Connell & Wellborn, 1991; Ryan & Stiller, 1991), and the basic distinction between intrinsic and extrinsic motivation has held up pretty well. However, although the benefits of intrinsic motivation are clear, it is far from clear that extrinsic motivation is antithetical to learning. For example, a meta-analysis of 61 studies that compared a group receiving rewards with a group that did not found that the effects were, on average, very small. Unanticipated verbal rewards increased both attitudes towards tasks and the time that participants chose to spend on them. Anticipated rewards had no impact on attitude but did reduce time spent on tasks, with a much stronger effect when the reward was independent of the quality of performance (Eisenberger & Cameron, 1996). As Ryan and Deci (2000) themselves pointed out, although intrinsic motivation is by definition autonomous, the fact that motivation is extrinsic does not mean that the motivation is necessarily *not* autonomous:

For example, a student who does his homework only because he fears parental sanctions for not doing it is extrinsically motivated because he is doing the work in order to attain the separable outcome of avoiding sanctions. Similarly, a student who does the work because she personally believes it is valuable for her chosen career is also extrinsically motivated because she too is doing it for its instrumental value rather than just because she finds it interesting. (Ryan & Deci, 2000, p. 60)

In these two cases, neither student is intrinsically motivated, but the first is primarily just complying with an external control, whereas the second has embraced the task and is working because she values the goal. In other words, what controls the behavior of the individual involves both whether the motivation itself is internal or external, and whether the value system is internal or external.

When both are external, then behavior is *externally* regulated "by contingencies overtly external to the individual" (Deci & Ryan, 1994, p. 6), whereas *introjected* regulation "refers to behaviours that are motivated by internal prods and pressures such as self-esteem-relevant contingencies" (p. 6), such as the student described above doing his homework under threat of sanctions. *Identified* regulation "results when a behaviour or regulation is adopted by the self as personally important or valuable" (p. 6), as in the case of the student doing the work for her career, whereas *integrated* regulation "results from the integration of identified values and regulations into one's coherent sense of self" (p. 6). These four types of extrinsic motivation—external, introjected, identified, and integrated—form a continuum, in that their intercorrelations have been shown to conform to a simplex pattern in which the largest intercorrelations are closest to the leading diagonal (Ryan & Connell, 1989) and so, together with intrinsic motivation and amotivation (where the individual lacks any intention to act), provide a taxonomy of six kinds of motivation. Although there is undoubtedly a clear hierarchy in terms of the degree of self-determination, this is clearly not a developmental continuum: A student may start out intrinsically motivated but maintain activity only because of external sanctions or instrumental goals, whereas someone may engage in an activity because of the rewards associated with it but then maintain the activity even when rewards are withdrawn (Eisenberger & Cameron, 1996).

The research on internal and external drivers of motivation accounts well for a range of observed student behaviors but pays relatively little attention to why an individual learner might find something interesting or not. Hidi and Harackiewicz (2000) defined *interest* as "an interactive relation between an individual and certain aspects of his or her environment (e.g., objects, events ideas), and is therefore content specific" (p. 152) involving both cognitive and affective components. They pointed out that researchers have placed different emphases on different aspects of interest, with some focusing on specific stimuli that focus attention that may or may not last (generally termed *situational* interest) whereas others have focused on more stable motivational orientations

or personal dispositions towards particular topics or domains (termed *individual* or *personal* interest). As Dewey (1913) noted, catching and holding student interest are two very different things. Mitchell (1993) found that it was possible to generate student interest in mathematics through the use of group work, puzzles, and computer-based activities, but that such interest waned over time. In contrast, meaningfulness and involvement in the tasks built interest that *was* sustained over time.

The approach to motivation discussed above seeks to examine motivation as a cause of engagement in activity. This has been very powerful in understanding why people and, in particular, students in mathematics classes do what they do. However, Csikszentmihalyi (1990) turned this on its head by looking at motivation as an *emergent* property of the interaction between the task in which the individual is engaged and the competences that the individual brings to the task. In this view, motivation is therefore not the cause, but the consequence of engagement. When the task demand is high and the skill levels is low, then the individual experiences anxiety, whereas if the task demand is low and the skill level is high, the individual experiences boredom. But if the task demand is just at the limit of the skill level, then the individual experiences “flow”:

A dancer describes how it feels when a performance is going well: “Your concentration is very complete. Your mind isn’t wandering, you are not thinking of something else; you are totally involved in what you are doing. . . . Your energy is flowing very smoothly. You feel relaxed, comfortable and energetic.”

A rock climber describes how it feels when he is scaling a mountain: “You are so involved in what you are doing [that] you aren’t thinking of yourself as separate from the immediate activity. . . . You don’t see yourself as separate from what you are doing.”

A mother who enjoys the time spent with her small daughter: “Her reading is the one thing she’s really into, and we read together. She reads to me and I read to her, and that’s a time when I sort of lose touch with the rest of the world, I’m totally absorbed in what I’m doing.”

A chess player tells of playing in a tournament: “. . . the concentration is like breathing—you never think of it. The roof could fall in and, if it missed you, you would be unaware of it.” (Csikszentmihalyi, 1990, pp. 53–54)

As well as the drivers of interest that Hidi and Harackiewicz identified, therefore, it is also necessary to attend to matching the level of demand to the skill of the individual.

Of course, the research reviewed above represents somewhat of a “counsel of perfection.” Ideally one would want all students to be intrinsically motivated, and to experience flow in their mathematics classrooms, but this is impossible to achieve in even the best mathematics classrooms all the time. Researchers therefore need to take account of why students do what they do even if they are not intrinsically motivated.

Eccles et al. (1983) identified four components of task value: attainment value, intrinsic value, utility value and cost. *Attainment value* refers to the value to an individual of doing well on a task, particularly in terms of confirming or disconfirming particular aspects of one’s identity (see also Boaler, William, & Zevenbergen, 2000). *Intrinsic value* corresponds to the ideas of intrinsic motivation and flow discussed above. *Utility value* encompasses the value of the task in terms of current and future goals, and is closely related to the identified, introjected and external forms of regulation described above. Finally, Eccles et al. (1983) emphasized the importance of a trade-off between the perceived value (whether it is attainment value, intrinsic value, or utility value) of a task and its cost. Cost includes both the opportunity cost that attempting a task might take, and also the negative consequences such as performance anxiety, risk to one’s view of self if unsuccessful, and so on.

The goals that students actually pursue in mathematics classrooms will therefore depend on a complex calculus of cost and benefit. Bandura (1986) and Schunk (1990, 1991) have shown that students are more motivated to reach goals that are specific, within reach, and offer some degree of challenge, but more recently, researchers have focused on broader issues, and in particular whether the student’s primary orientation is towards *mastery* or *performance*. Ames (1992) defined an *achievement goal* as follows:

An *achievement goal* concerns the purposes of achievement behavior. It defines an integrated pattern of beliefs, attributions, and affect that produces the intentions of behavior and that is represented by different ways of approaching, engaging in, and responding to achievement activities. (p. 261)

Mastery achievement goals are those related to the acquisition of new skills or increasing the level of competence, whereas performance achievement goals are those that are related to the level of performance one is able to produce. Students with mastery goals seek to increase their level of competence, whereas those with performance goals are motivated either to show what they can do (performance approach goals) or to avoid failing (performance avoidance goals). Although mas-

tery goals are clearly beneficial and performance-avoidance goals are clearly detrimental to learning, there is still much debate about the usefulness of performance-approach goals. In a study of a 5-week elementary school math unit, Linnenbrink (2005) found that a combined approach, emphasizing mastery goals within small groups but emphasizing performance goals between groups produced the highest achievement, provided the competition between groups was focused on relative improvement amongst the groups (see also the discussion on *Activating Students as Instructional Resources for One Another* below).

Whether students choose to pursue mastery or performance goals depends, of course, on a range of factors, including the students' beliefs about the likelihood of success, and this, in turn, is influenced by students' ideas about the causes of success in school. Through an extensive research program going back over a quarter of a century, involving questionnaires and interviews with thousands of students, Dweck (2000) has shown that there are substantial differences between students in their beliefs about the causes of success and failure in the classroom. Two dimensions appear to be particularly important. The first is *personalization*: whether success is due to *internal* factors (such as one's own performance) or *external* factors (such as getting a lenient or severe grader). The second is *stability*: whether success is due to *stable* factors (such as one's ability) or *unstable* factors (such as effort or luck).

Dweck and others have found that many if not most students attribute success and failure differently. So for example, the student who says "I got an A" but "She gave me a C" is attributing success internally but attributing failure externally. In fact many studies have found that boys are more likely to attribute their successes to internal, stable causes (such as ability) and their failures to external, unstable causes (such as bad luck or hostile teachers). This would certainly explain the high degree of confidence with which many boys approach tests or examinations for which they are completely unprepared (the important question for such students is not "Have I thoroughly reviewed the material for this test?" but "Do I feel lucky today?"). More controversially, the same research suggests that girls attribute their successes to internal unstable causes (such as effort) and their failures to internal stable causes (such as lack of ability), leading to what has been termed *learned helplessness* (Dweck, 2000). What is clear is that the most adaptive beliefs are that both success and failure are attributable to internal, unstable causes: "it's down to you, and you can do something about it." This is why the strategy of *Clarifying and Sharing Learning Intentions and Success Criteria with Learners* discussed above is so important. When students are clear about the criteria

for success, they are more likely to attribute success and failure to internal unstable causes.

In the 1970s and 1980s, efforts to raise the confidence of students that they could achieve in mathematics focused on self-esteem as a critical factor, but Bandura (1977) has convincingly argued that self-efficacy—a generative capacity to carry one's plans through to completion—is a far more useful focus of inquiry. Students' expectations about their likelihood of success strongly influence their decisions about which tasks to attempt, how much effort to expend, and their persistence in the face of difficulties (Bandura, 1997), and increasing students' self-efficacy has been shown to increase performance in mathematics. Indeed, it appears that self-efficacy may be more important in mathematics than in any other school subject (Pajares, 1996, p. 555).

Students' decisions about whether to pursue mastery or performance goals are also influenced by teachers' practices. In a study of 1571 students in 84 mathematics classrooms from 5th to 12th-grade Deevers (2006) used hierarchical linear modeling to explore the relationship between teachers' formative assessment practices and student attitudes and beliefs about mathematics. He found that although students' self-efficacy beliefs and motivation to learn declined steadily from 5th to 12th grade, students provided with positive constructive feedback were more likely to display mastery orientation, even though teachers gave less of this kind of feedback as students got older.

Closely related to the work on self-efficacy and goal orientation are studies undertaken by Carol Dweck and her colleagues into student perceptions and beliefs about the nature of ability in school (see Dweck, 2000, for a comprehensive summary of this work). Dweck and Leggatt (1986) showed that many students believe that ability is fixed: There are smart people and not-so-smart people. Students holding this "entity" view of ability are likely to adopt a performance orientation to challenging tasks that are set for them. If they are confident in their ability to achieve what is asked of them, then they will attempt the task. However, if their confidence in their ability to carry out their task is low, then, unless they believe that the task is so hard that no one is expected to succeed, they will avoid the challenge. In short, these students are deciding that they would rather be thought lazy than dumb.

In contrast other students see ability as incremental. They see challenging tasks as chances to learn—to get smarter—and therefore in the face of failure will try harder (Mayer, Turner, & Spencer, 1997). What is perhaps most important here is that these views of ability are generally not global—the same students often believe that ability in mathematics is fixed, while at the same time they believe that ability in, for example,

athletics is incremental, in that the more one trains, the more one's ability increases. The crucial thing, therefore, is that teachers inculcate in their students a belief that ability is incremental rather than fixed. Doing so would be advantageous even if it were untrue, because the consequences would be so beneficial, but, as was discussed in the section on *Eliciting Evidence* above, there is overwhelming evidence that "smart is not something you are; it's something you get."

Integrating Motivational and Cognitive Perspectives

A number of ways of bringing together the motivational and cognitive perspectives on self-regulation have been proposed. One of the most promising is the *dual processing* theory developed by Boekaerts (1993). In the model,

It is assumed that students who are invited to participate in a learning activity use three sources of information to form a mental representation of the task-in-context and to appraise it: (1) current perceptions of the task and the physical, social, and instructional context within which it is embedded; (2) activated domain-specific knowledge and (meta)cognitive strategies related to the task; and (3) motivational beliefs, including domain-specific capacity, interest and effort beliefs. (Boekaerts, 2006, p. 349)

As a result of the appraisal of the task, the student begins to act along one of two pathways. If the task appraisal is positive, the student "crosses the Rubicon" (Winne, 2005, pp. 234–236) and begins activity along the "growth pathway" where the goal is to increase competence. In this case, the self-regulation is *top-down* in that the flow of energy is directed by the student. If, on the other hand, the task appraisal is negative, attention shifts away from the learning task and towards the well-being pathway. The student then becomes focused on self-appraisal rather than task appraisal, concentrating on preventing threat, harm or loss. This form of self-regulation is termed *bottom-up* by Boekaerts because it is triggered by cues in the environment, rather than by learning goals. Such bottom-up regulation is not necessarily negative, because by attending to the well-being pathway, the student may find a way to restore well-being, thus allowing a shift of energy and attention back to the growth pathway.

An important feature of the dual regulation model is that it hypothesizes that the balance between top-down and bottom-up pathways of regulation is dynamic, rather than being a dispositional feature of an individual student. Experimental support for this hypothesis was provided by a series of studies summarized in Boekaerts (2001, pp. 24–26) in which it was

found that there was no direct link between domain-specific motivational beliefs and learning intention in any of the mathematics classrooms they studied. Rather, students' decisions about whether to invest effort in a mathematics assignment depended largely on their appraisal of the specific mathematics task in front of them. There is also evidence that interpretations of assessment outcomes from friends and parents mediate the relationship between the assessment outcomes themselves and how they impact students' beliefs (Ross, Rolheiser, & Hogaboam-Gray, 2002).

The dual-processing model allows one to relate the various perspectives on cognition and motivation and learning described above. To simplify somewhat, students who are *personally*, as opposed to *situationally*, interested in a task are likely to engage in activity along the growth pathway, although for students who are not personally interested in a task, a range of factors related to the task-in-context may spark situational interest, thus also triggering activity along the growth pathway. Where personal interest is not the main driver of attention, more concern will be given to considerations of task *value* versus *cost*. The *integrated* and *identified* forms of regulation defined by Deci and Ryan are related to activity along the growth pathway, whereas *external* or *introjected* forms of regulation are related to activity along the well-being pathway. Students who display *mastery* orientation are activating the growth pathway, and those displaying *performance* orientation are activating the well-being pathway.

Self-efficacy beliefs can drive progress along either pathway. Along the growth pathway, self-efficacy drives adaptive (meta)cognitive strategy use, whereas along the well-being pathway, self-efficacy beliefs are likely to steer the learner away from *performance-avoidance goals* and towards *performance-approach goals*. Similarly views of ability as *incremental* help the learner stay on the growth pathway, whereas *entity* views of ability direct activity towards the well-being pathway, where details of the task-in-context, appraised in the light of views of personal capability, will influence decisions about whether to engage in the task.

This extended discussion of the research on metacognition, motivation, and beliefs may seem distant from the central theme of this chapter, but an understanding of these issues is essential in developing an understanding of the role of assessment in learning. Adaptive adjustments to instruction are necessary to maximize learning, so assessment is an essential part of learning, but assessment processes themselves impact the learner's willingness, desire, and capacity to learn (Harlen & Deakin-Crick, 2002). The research reviewed briefly above shows clearly that practices ostensibly designed to support learning actually prevent it from taking place (recall that

the research reviewed by Kluger and DeNisi, 1996, discussed above showed that in 38% of rigorously designed and conducted studies, feedback designed to improve performance actually had the opposite effect).

Although many issues are still unresolved, and a full theoretical synthesis of all the research perspectives is some way off if it is even possible, the existing research on cognition and motivation paints a reasonably coherent picture and provides strong guidance on the design of classroom assessment environments (Brookhart, 1997) that can activate students as owners of their own learning. Feedback to learners should focus on what they need to do to improve, rather than on how well they have done, and should avoid comparison with others. Students who are used to having every piece of work scored or graded will resist this (see, e.g., Smith and Gorrard, 2005), wanting to know whether a particular piece of work is good or not, and in some cases, depending on the situation, the teacher may need to go along with this. In the long term, however, it seems that teachers should aim to reduce the amount of ego-involving feedback given to learners (and with new entrants to the school, perhaps not begin the process at all) and focus on the student's learning needs (Kohn, 1999). Furthermore, feedback should not just tell students to work harder or be "more systematic"—the feedback should contain a recipe for future action, for otherwise it is not formative. Finally, feedback should be designed so as to lead all students to believe that ability—even in mathematics—is incremental. In other words the more students "train" at mathematics, the smarter they get.

Although there is a clear set of priorities for the development of feedback, there is no "one right way" to do this. The feedback routines in each class will need to be thoroughly integrated into the daily work of the class, and so they will look slightly different in every classroom. This means that no one can tell teachers how this should be done—it will be a matter for each teacher to work out how to incorporate some of these ideas into her or his own practice.

ACTIVATING STUDENTS AS INSTRUCTIONAL RESOURCES FOR ONE ANOTHER

As Slavin, Hurley, and Chamberlain (2003) noted, "Research on cooperative learning is one of the greatest

success stories in the history of educational research" (p. 177). However, despite the vast amount of research that has been done in this area and the consensus that cooperative learning does increase student achievement, there is much less agreement on why cooperative learning is effective. Slavin et al. (2003) suggested that there are four major theoretical perspectives on cooperative learning and its effects on achievement.

Two of the perspectives—the *motivational* perspective and the *social cohesion* perspective—focus on the role of motivation. Adherents of the motivational perspective hold that students help their peers to learn because it is in their own interests to do so; the only way they can achieve their personal goals is to ensure that all the individuals in the group are successful. In contrast, theorists within the social cohesion perspective propose that students help their peers because they care about the group. In both of these perspectives, increased student achievement comes primarily from increased motivation and effort.

The other two perspectives—the *developmental perspective* and the *cognitive elaboration* perspective—emphasize the idea that increased learning in cooperative groups stems from the increased cognitive engagement produced in small groups. Empirical work in these two perspectives tends not to emphasize the importance of group goals, which are the hallmark of the motivational perspective, nor the building of team spirit, as required in the social cohesion perspective. Indeed, many writers in this tradition, such as Damon (1984), have explicitly rejected the idea that extrinsic incentives are important in group learning situations (p. 337).

The developmental perspective draws strongly on the work of Piaget and Vygotsky, and in particular the latter, although as Shayer (2003) noted, the differences between these two are much less than is often assumed. The starting point for much of this work is Vygotsky's definition of the *zone of proximal development* (zpd) as "the distance between the actual developmental level as determined by independent problem solving and the level of potential development as determined through problem solving under adult guidance or in collaboration with more capable peers" (Vygotsky, 1978, p. 86).³ Research in the developmental perspective emphasizes the role of more advanced peers in guiding the learning of others.

The fourth perspective—the cognitive elaboration perspective—like the developmental perspective, locates the benefit of cooperative learning in the in-

³ As Chaiklin (2003) made clear, in reading Vygotsky, it is important to understand the cultural and historical context of his work, and in particular, Vygotsky's attempt to distinguish between learning and development. For Vygotsky, development requires changes in the psychological functions that the child can deploy, whereas learning does not. The zone of proximal development (zpd) is not, therefore, just a way of describing what a student can do with support—that could just be learning. Rather the zpd is a description of the *maturing* psychological functions rather than those that already exist, and a focus in instruction on the maturing psychological functions is most likely to produce a transition to the next developmental level.

teraction between students rather than in the motivation for engagement. However, whereas the developmental perspective predicts benefits for the recipient of the support, the cognitive elaboration perspective emphasizes that providers of support also benefit.

Of course, none of these perspectives is contradictory. Slavin et al. (2003) suggested a *logic model* in which group goals lead to both an increased personal motivation to learn as well as a motivation to ensure one's group-mates also learn. This in turn yields peer assessment and correction, with elaborated explanations and peer modeling, resulting in enhanced learning. However, although the four perspectives may be complementary, there are marked differences in the degree of empirical support for each.

Slavin (1995) reviewed studies of cooperative learning in elementary and secondary schools and found 99 that involved interventions of at least 4 weeks' duration in which the performance of students involved in cooperative learning was compared with students who were not. Of the 64 studies that provided *group* rewards that were contingent on the aggregate of the learning of *individual* members, more than 75% (50 out of 64) found significant positive effects on learning, and in none of the 64 studies was the effect of cooperative learning negative. The median effect size in these studies was 0.32 standard deviations, compared with a median effect size of 0.07 in the studies where there were no group rewards or where the group rewards were based on a single group product.

In reviewing the available research on interventions designed to increase social cohesion, Slavin (1995) concluded that interventions that are designed to increase social cohesion but do not provide rewards based on the learning of all group members are no more effective than traditional instruction.

A review of 17 studies looking at the effects of peer interaction on the learning of mathematics, ranging from 2nd grade to 11th grade (Webb, 1991), found that those who provided help benefited little when this help was in the form of answers, procedural information, or managerial information. However, when the feedback was in the form of elaborated explanations, those providing help benefited substantially. The median partial correlation between giving help in the form of elaborated explanations and achievement in mathematics, across the 10 studies on which the relevant information was available, was 0.27. In other words, a 1 standard deviation increase in giving elaborated help resulted in a 0.27 increase in achievement for the help-giver even after controlling for ability. The same study also found a strong negative effect of approximately the same magnitude for students who asked for help but were given only the answer.

The fact that these results were similar across topics and grades suggests that they are likely to generalize to a variety of mathematics classrooms.

Perhaps surprisingly, the effect of peer tutoring can be almost as strong as one-to-one instruction from a teacher and can be stronger than small-group instruction from a teacher, although some studies have found a strong correlation between the mathematical ability of the student giving help, the quality of explanation given, and the amount of learning that results (Fuchs et al., 1996). In a study by Shacter (2000) 109 students drawn from one fourth-grade classroom, one-fifth grade classroom and three combined fifth- and sixth-grade classrooms were randomly assigned to one of four treatments: teacher-student dyad, student-student dyad, same-age peer group with a teacher, and same-age peer group without a teacher. Compared with student-student pairs, the highest achievement was found in teacher-student pairs (standardized effect size of 0.53, 0.86, and 0.55 on concept mapping, essay, and declarative knowledge respectively), but the achievement of students in student-led groups was almost as high (effect sizes 0.50, 0.57, and 0.30), and on average higher than the achievement in teacher-led groups (0.33, 0.67, 0.14). These effects may appear to be larger than some of the results reported above, but the measures in this study were much more sensitive to instruction than is typical for standardized tests.

In fact studies that have explored the effects of group rewards in combination with other interventions have consistently concluded that the "active ingredient" is the establishment of group goals and individual accountability. Fantuzzo, King, and Heller (1992) compared the effects of reward and structure in the context of reciprocal peer teaching in mathematics. In one group, pairs of students were rewarded with a choice of activities if the sum of their scores on a test exceeded a threshold set by the teacher (reward only). In a second group students were trained in a structured method for tutoring each other, taking turns to be tutor and tutee (structure only). A third group of students received both interventions (reward + structure) while a fourth received neither (control). The results showed that reward had a greater effect on mathematics achievement than structure, but the combined effect was greater still. In another study (Fantuzzo, Davis, & Ginsburg, 1995), 72 fourth- and fifth-grade students in urban schools evidencing difficulties in mathematics were assigned randomly to one of three treatments. One-third of the students received a home-based parental involvement program, one third of the students received the parental involvement together with a reciprocal peer-tutoring intervention, and one third of the students were as-

signed to a control group where they just practiced work on which they were having difficulties. Students engaged in both reciprocal peer teaching and parental involvement were more confident and had higher mathematics achievement (on both classroom assessments and standardized tests) than either of the other two groups.

However, despite the clarity of the research evidence about the importance of group goals and individual accountability, it appears that although teachers acknowledge the importance of collaborative learning, few implement it in a way that the research suggests would be necessary for it to be effective. Antil, Jenkins, Wayne, and Vadasy (1998) surveyed 85 elementary school teachers in two districts, and although 93% of the teachers said they employed collaborative learning, follow-up interviews with 21 of the teachers showed that only 5 teachers implemented collaborative learning in such a way as to create group goals together with individual accountability. Furthermore, only 1 of the 21 teachers implemented collaborative learning that satisfied the more complex criteria proposed by Cohen (1994): open-ended tasks that emphasize higher order thinking, group tasks that require input from other members, multiple tasks related to a central intellectual theme, and roles assigned to different group members. Whatever the research says about the benefits of peer collaboration, it seems to be more difficult for teachers to implement than would at first appear.

Whether cooperative learning is more effective for some subgroups of students than others is still a matter of some debate. Although some studies have found greater effects for higher attaining students (e.g., Stevens & Slavin, 1995), others have found more significant effects for lower-attaining students (e.g., Boaler, 2002). Slavin et al. (2003) concluded that most interventions appear to benefit high, medium, and low achievers equally. There is, however, evidence that cooperative learning interventions are particularly beneficial for students of color. A meta-analysis of 37 studies on the effects of small-group collaborative learning on achievement in postsecondary science, mathematics, engineering, and technology (Springer, Stanne, & Donovan, 1999) found a mean effect size of 0.51, with similar impact on persistence ($d = 0.46$) and attitudes ($d = 0.55$). The effects were larger for groups that were held outside class time ($d = 0.65$) than for those held inside class time ($d = 0.44$), and larger in 4-year colleges ($d = 0.54$) than 2-year col-

leges ($d = 0.21$). The effect sizes on achievement were also larger for groups of African-American and Latina/o students ($d = 0.76$) than for white ($d = 0.46$) or heterogeneous groups ($d = 0.42$). The particular value of collaborative learning environments for African-American students has also been emphasized by Boykin and his colleagues (Boykin, Coleman, Lilja, & Tyler, 2004; Boykin, Lilja, & Tyler, 2004).

THE REGULATION OF LEARNING

The preceding sections have discussed in some detail the five key strategies of assessment for learning:

- Clarifying and sharing learning intentions and criteria for success;
- Engineering effective classroom discussions, questions, and learning tasks that elicit evidence of learning;
- Providing feedback that moves learners forward;
- Activating students as instructional resources for one another; and
- Activating students as owners of their own learning.

These five strategies can be integrated within a more general theoretical framework of the *regulation of learning processes* as suggested by Perrenoud (1991, 1998).⁴ Within such a framework, the actions of the teacher, the learners, and the context of the classroom are all evaluated with respect to the extent to which they contribute to guiding the learning towards the intended goal. In some environments, the responsibility for regulating learning, or “keeping learning on track” is with the teacher, in some it is with the student, and in some it is shared (see Vermunt, 2003, for a classification of different kinds of learning environments in terms of the responsibility for regulation).

From this perspective, the task of the teacher is not necessarily to teach, but to create situations in which students learn. This focus emphasizes what it is that students learn, rather than what teachers do, although in an accountability-driven culture, it is hard to maintain this focus. One rather startling example of this is provided by a study of teachers being trained in teaching problem-solving skills to their students that was undertaken by Deci, Spiegel, Ryan, Koestner, and Kauffman (1982). Participating teachers were

⁴ In English, the noun *regulation* has two meanings; one refers to the act of regulating and the other to a rule or law to govern conduct, and so, although the former sense is intended here, the word has the unfortunate connotation of the second. In French, the two senses have separate terms (*régulation* and *règlement*) and so the problem does not arise.

randomly allocated to one of two groups, and both groups were introduced to the important ideas about teaching problem-solving. In both groups teachers were given time to practice the problems and were given both a list of useful hints and the actual solutions to all the problems. The only difference between the two groups was that at the end of the training session, one additional statement was made to the teachers in one group: "Remember, it is your responsibility as a teacher to make sure your students perform up to high standards." Subsequent analysis of recordings of the lessons of the teachers who had been told about the importance of performing up to high standards spent twice as much time talking during the teaching session as the other teachers and made three times as many directives and three times as many controlling statements (e.g., words like *should* and *must*).

Obviously, regulating or controlling the activities in which students engage is only indirectly related to the learning that results (Clarke, 2001) and yet teachers appear to find it difficult to shift from planning activities to planning learning (Franke, Fennema, & Carpenter, 1997). This is especially evident in interviews before lessons in which teachers focus much more on the planned activities than on the resulting learning (e.g., "I'm going to have them do X"). In a way, this is inevitable, because only the activities can be manipulated directly. Nevertheless, in teachers who have developed their formative assessment practices, there is clearly a strong shift in emphasis away from regulating the activities in which students engage, and towards the learning that results (Black et al., 2003). Indeed, from such a perspective, even to describe the task of the teacher as teaching is misleading, since it is rather to "engineer" situations in which student learn.

However, in this context, it is important to note that the *engineering of learning environments* does not guarantee that the learning is proceeding in fruitful ways. Many visual-arts classrooms are *productive*, in that they do lead to significant learning on the part of students, but what any given student might learn is impossible to predict. An emphasis on the regulation of learning processes entails ensuring that the learning that is taking place is as intended.

When the learning environment is well regulated, much of the regulation is proactive, through the setting up of *didactical situations* (Brousseau, 1997). The regulation can be unmediated within such didactical situations when, for example, a teacher "does not intervene in person, but puts in place a 'metacognitive culture', mutual forms of teaching and the organization of regulation of learning processes run by technologies or incorporated into classroom organization and management" (Perrenoud, 1998, p. 100). For

example, a teacher's decision to use realistic contexts in the mathematics classroom can provide a source of proactive regulation, because then students can determine the reasonableness of their answers. If students calculate that the average cost per slice of pizza (say) is \$200, provided they are genuinely engaged in the activity, they will know that this solution is unreasonable, and so the use of realistic settings provides a self-checking mechanism. They are able to keep their learning on track themselves, rather than requiring a teacher's intervention, because they are "metacognitively, motivationally, and behaviorally active participants in their own learning process" (Zimmerman, 1986, p. 308).

On the other hand, the didactical situation may be set up so that the regulation is achieved through the mediation of the teacher or peers. McCaslin and Good (1996) proposed the term *co-regulation* to describe

the process by which the social instructional environment supports or scaffolds the individual via her relationships within the classroom, relationships with teachers and peers, objects and setting, and ultimately the self. Internalization of these supportive relationships empowers the individual to seek new challenges within co-regulated support. (p. 660)

The teacher, in planning the lesson, can create questions, prompts or activities that evoke responses from the students that the teacher can use to determine the progress of the learning (such as the "hinge-point" questions described above) and, if necessary, to make adjustments to the instruction. Examples of such questions are, "Is calculus exact or approximate?" or "Would your mass be the same on the moon?" (In this context it is worth noting that each of these questions is "closed" in that there is only one correct response—their value is that although they are closed, each question is focused on a specific conception.) In classroom discussion, through careful scaffolding of student discussion, teachers can help their students develop the skills of self-regulation (Mayer & Turner, 2002), especially through the establishment and maintenance of sociomathematical norms (Yackel & Cobb, 1996) allowing students to support one another's learning in appropriate ways (McClain & Cobb, 2001; Ross, 1995).

Thoughtful *upstream* planning (before the lesson) therefore creates, *downstream* (during the lesson), the possibility that the learning activities may change course in the light of the students' responses. These *moments of contingency*—points in the instructional sequence when the instruction can proceed in different directions according to the responses of the student—are at the heart of the regulation of learning.

These moments arise continuously in whole-class teaching, where teachers constantly have to make sense of students' responses, interpreting them in terms of learning needs and making appropriate responses. But they also arise when the teacher circulates around the classroom, looking at individual students' work, observing the extent to which the students are on track. In most teaching of mathematics, the regulation of learning will be relatively tight, so that the teacher will attempt to "bring into line" all learners who are not heading towards the particular goal sought by the teacher—in these topics, the goal of learning is generally both highly specific and common to all the students in a class. In contrast, when the class is doing an investigation, the regulation will be much looser. Rather than a single goal, there is likely to be a broad *horizon* of appropriate goals (Marshall, 2004), all of which are acceptable, and the teacher will intervene to bring the learners into line only when the trajectory of the learner is radically different from that intended by the teacher. In this context, it is worth noting that there are significant cultural differences in how to use this information. In the United States, the teacher will typically intervene with individual students where they appear not to be on track whereas in Japan, the teacher is far more likely to observe all the students carefully, while walking round the class and then will select some major issues for discussion with the whole class. This is consistent with what Bromme and Steinbring (1994) discovered in their expert-novice analysis of two mathematics teachers: The novice teacher tended to treat students' questions as being from individual learners, whereas the expert teacher's responses tended to be directed more to a "collective student."

One of the features that makes a lesson *formative*, then, is that the lesson can change course in the light of evidence about the progress of learning. This is in stark contrast to the traditional pattern of classroom interaction, exemplified by the following extract:

"Yesterday we talked about triangles, and we had a special name for triangles with three sides the same. Anyone remember what it was? . . . Begins with E . . . equi- . . ."

In terms of formative assessment, there are two salient points about such an exchange. First, little is contingent on the responses of the students, except how long it takes to get on to the next part of the teacher's script, so there is little scope for downstream regulation. The teacher is interested only in getting to the word *equilateral* in order that she can move on, and so all incorrect answers are treated as equivalent. The only information that the teacher extracts from

the students' responses is whether they can recall the word *equilateral*.

The second point is that the situation that the teacher set up in the first place—the question she chose to ask—has little potential for providing the teacher with useful information about the students' thinking, except, possibly, whether the students can recall the word *equilateral*. This is typical in situations where the questions that the teacher uses in whole-class interaction have not been prepared in advance (in other words, when there is little or no proactive or upstream regulation).

In contrast, the vignettes of mathematics teaching given by Kilpatrick et al. (2001) show how exemplary teachers actually design these teachable moments into their lessons: "Mr. Hernandez and Ms. Kaye have each designed the lesson to afford them critical information about their students' progress. The tasks they frame create a strategic space for students' work and for gaining insights into students' thinking" (p. 349). Similar considerations apply when the teacher collects the students' notebooks and attempts to give helpful feedback to the students in the form of comments on how to improve rather than grades or percentage scores. If sufficient attention has not been given upstream to the design of the tasks given to the students, then the teacher may find that she has nothing useful to say to the students. Ideally, from examining the students' responses to the task, the teacher would be able to judge (a) how to help the learners learn better and (b) what she might do to improve the teaching of this topic. In this way, the assessment could be formative for the students, through the feedback she provides, and formative for the teacher herself, in that appropriate analysis of the students' responses might suggest how the lesson could be improved.

As a concrete illustration of these ideas in action, consider the following account from a research project (Lyon, Leahy, Morris, & Thompson, 2005) of a teacher who was working through a task entitled "Up, Up and Away" (Marquez & Boxley, 2003). The task requires students to complete a table and a graph for a weather balloon that rises at a rate of 8 feet per second. The students are then given a graph of a flare that was set off 2 seconds after the weather balloon and asked three questions:

1. What is the maximum height, in feet, that the flare reaches?
2. From the time the flare is set off, how many seconds will the flare take to reach its maximum height?
3. What is the average speed of the flare, in feet per second, from the ground to its highest point from the ground?

This particular class began to work in groups when the graph of the flare was introduced. The teacher said to the class, “*I’m hearing some problems, especially for the third part. Write your answers to the first part on the dry erase.*” All the answers held up by the students were correct. The teacher then asked the students to hold up their answers to the second question. She said, “*We have 3 different answers: 3, 4, and 5 seconds. It hits the maximum at what time?*” One student answered, “*5 seconds.*” The teacher asked, “*And starts at?*” A student said, “*2 seconds.*”

The teacher asked the students to hold up their answers for the third question and saw that there were still different answers. She said, “*We have different answers. We are going to look at a different problem.*” She then took the students through a similar activity, again looking at average speed, but this time in the context of car journeys where speed was measured in miles per hour. After some discussion, students showed, again by displaying their answers on dry-erase boards, that they understood how to calculate from a distance-time graph the average speed over a specified time interval of a body moving at nonconstant speed.

The students then resumed work on the “Up, Up, and Away” task. After a few moments the teacher asked the groups to hold up their new answers to the third question. Almost all the posted answers were correct, so the teacher asked “*If the average speed is 48 feet per second does that mean that the flare is always traveling at that speed?*” The students answered, “*To have a constant speed the graph would be a straight line.*” In the post-interview the teacher was asked if this was an on-the-fly change to the lesson plan or if the additional problem had been part of her lesson design. The teacher had anticipated that the students might have a problem with the average speed and therefore designed the problem in case there was difficulty during the lesson. She only planned on using the additional problem if it were needed.

This brief extract illustrates several features of the effective regulation of learning. Before the lesson began, the teacher had planned the questions she was going to ask and had provided the students with dry-erase boards, so that she could require responses from all students. She had also developed, within the students, a willingness to be open about their answers, as a step towards the “metacognitive culture” mentioned by Perrenoud above. These are examples of the proactive regulation of learning. As a result of teaching this material previously, she had also identified that the third question might pose particular difficulties—in other words, that this was a hinge point in the lesson—and after some reflection on why this might have occurred, she planned the backup activity. This use of reflection

on past teaching to modify future teaching is an example of *retroactive* regulation of learning. During the lesson, there was also a significant amount of the *interactive* regulation of learning. She listened carefully to the students’ discussions and looked at their work as she walked around the classroom. It was as a result of this that she said, “*I’m hearing some problems*” and decided to collect evidence more systematically about the understanding of her students by asking them to display their responses on dry-erase boards. Because she had evidence from all students, she was in a much better position to make the on-the-fly decision to use the backup activity she had planned. It is also worth noting that the teacher used all-student response systems selectively. At times, she relied on her intuitive perceptions of the class, at times she decided to collect information systematically, and at other times, she allowed students to volunteer contributions. Shulman (2005) describes practices such as these as *pedagogies of uncertainty*. This term accurately reflects the fact that it is impossible to be certain about what students will learn as a result of a particular episode of instruction but the term *uncertainty* suggests that it is impossible to predict anything in advance. As the vignette above shows, in many cases, one might not know what will happen, but one can often reduce the uncertainty by careful planning. In this case, the teacher reduced the possibilities to a simple decision about whether to use the supplementary activity she had prepared. In other cases, the teacher may plan for a more complex range of possibilities, as in the case with the item that asked students for a fraction between $\frac{1}{6}$ and $\frac{1}{7}$, but again, careful consideration of the kinds of responses that students may produce allows teachers to “prefilter” the students’ responses. By anticipating the students’ responses the teacher can simplify the task she is faced with in the classroom. Rather than pedagogies of uncertainty, therefore, it seems more appropriate to describe such pedagogies as *pedagogies of contingency*. The essence of formative assessment is that instruction is contingent on what it is that the students have learned. Teachers can regulate learning proactively by creating moments of contingency, for example by identifying a hinge point in a sequence of instruction as discussed above, and designing a hinge-point question to be used at that time. Teachers can also regulate learning interactively by capitalizing on moments of contingency that may arise spontaneously in their teaching. Through careful reflection after the lesson, they can plan better for the future, both in terms of how to create more moments of contingency, and in terms of how to recognize and take advantage of such moments when they arise.

SUMMARY

In this chapter, I have outlined some of the research that suggests that focusing on the use of day-to-day formative assessment is one of the most powerful ways of improving learning in the mathematics classroom. In other words, even if teachers do not care about deep understanding and instead wish only to increase their students' test scores, then attention to formative assessment appears to be one of, if not the, most powerful way to do this.

To be effective, these strategies must be embedded into the day-to-day life of the classroom and must be integrated into whatever curriculum scheme is being used. That is why there can be no recipe that will work for everyone. Each teacher will have to find a way of incorporating these ideas into her or his own practice, and effective formative assessment will look very different in different classrooms. It will, however, have some distinguishing features. Students will be thinking more often than they are trying to remember something, they will believe that by working hard they get smarter, they will understand what they are working towards, and they will know how they are progressing. The teachers will ensure that students understand what it is that they are meant to be learning, they will be collecting evidence frequently about the extent of students' progress towards the goal, and they will be making frequent adjustments to the instruction to better meet the learning needs of the students.

In some ways, this is an old-fashioned message—indeed, none of the techniques that teachers have used to put these principles into practice in their classrooms is new. What is new is that researchers now have hard empirical evidence that quality learning does lead to higher achievement, even when performance is measured through externally mandated tests. What is also new is the broad theoretical framework of the regulation of learning, which may help teachers to understand how these ideas can be implemented effectively, so that teachers and students can, together, keep the learning of mathematics “on track.”

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AUTHOR NOTE

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HIGH-STAKES TESTING IN MATHEMATICS

Linda Dager Wilson

Test results in schools are often used to make decisions that have serious consequences. The term *high stakes* is used for such tests.¹ For whom are the stakes high? Students are often the constituents who reap both the rewards and the punishments of the tests they take. For example, students may be required to pass a test or a series of tests to earn a high school diploma, or to be promoted to the next grade. Tests might also be used to place students in different courses, or to move them from one stream of the curriculum to another. Students may earn college credit by high scores on advanced placement tests or earn an International Baccalaureate diploma through testing. In Japan, Germany, and France students must pass entrance exams for admission to universities. Thus the stakes associated with tests may be high for students, when either rewards or sanctions are attached to the outcomes.

Other constituents are also affected by the consequences of high-stakes tests. Teachers may be held accountable for their students' test scores, but they may also be required to pass tests themselves in order to earn or maintain their certification. Schools, districts, and states may have accountability measures in place that use the results of student tests to make funding decisions. The No Child Left Behind (NCLB) legislation from the Bush administration is an example of federal funding that is linked, at least in part, to the results of standardized achievement tests at the

state level. Every state has now instituted some sort of state-wide testing program, partially in response to this legislation.

Mathematics as a content area is prevalent in high-stakes testing programs. The NCLB legislation calls for annual testing in reading and mathematics at Grades 3–8 and once in Grades 10–12. Advanced placement tests, such as those administered by the College Board, are offered in statistics and at two levels of calculus. The National Board of Professional Teaching Standards offers national certification for teachers in mathematics, among other content areas. End-of-course exams, required for a high school diploma, are offered in North Carolina, for example, in Algebra 1, Algebra 2, and Geometry. So the teaching and learning of mathematics has the potential to be greatly affected by high-stakes testing.

International high-stakes testing. Although the current political atmosphere in the U.S. seems to call for more and more high-stakes testing in mathematics, as well as other major content areas, some researchers have questioned whether this trend is operating in the reverse in other parts of the world (U.S. Congress, 1992; Kellaghan, Madaus, & Raczek, 1996). Apparently, this is not so. Phelps (2000) looked at data from 31 countries and provinces and concluded that there was, in fact, an overall net increase internationally in all testing from 1974–1999. This included an increase in high-stakes testing during those years as

¹ The term *tests* in this chapter refers to standardized tests. This is not to be confused with multiple-choice, machine-scored, or any other test format, item type, or scoring procedure. Rather, standardized tests are tests that are given under uniform conditions, in which all examinees are asked to respond to the same, or nearly the same, test questions.

well. In the case of large-scale, external testing that was added during those years, most were assessments used for diagnostic or monitoring purposes, and thus not high-stakes. However, the second most frequently added type of test was upper secondary exit exams, followed by university entrance exams and subject-area end-of-course exams. One would expect these three categories to all include mathematics. Phelps also refuted the claim that there is little or no testing in other countries at ages below 16, citing the prevalence of lower secondary exams in such countries as France, Italy, Iceland, Ireland, Denmark, New Zealand, Switzerland, Japan, and Korea (p. 18).

Both commonalities and differences are apparent in the use of high-stakes testing in mathematics in the U.S. and around the world. On the one hand, the use of mathematics exams for promotion (such as a secondary diploma) seems to be similar, as is the use of mathematics exams for entrance to the university. Most U.S. colleges and universities require the mathematics portion of either the Scholastic Achievement Test (SAT) or the American College Test (ACT) for entrance. On the other hand, U. S. students appear to be much more likely to be tested in mathematics in the primary grades.

HISTORICAL CONTEXT OF HIGH-STAKES TESTING IN MATHEMATICS

Much of the origin and development of school testing began in the U.S., starting with the first school examinations in the Boston schools in 1845, which were championed by Horace Mann as a superior way to examine all pupils. Thorndike's (1904) book on mental, social, and educational measurements stimulated the development of the first standardized achievement tests, beginning with arithmetic and English composition, spelling, drawing, and handwriting. Testing as an industry grew rapidly with the use of intelligence testing during World War I. The primary function of such tests was to sort and classify students according to "innate" ability, an illustration of some of the earliest uses of large-scale tests for policy decisions. Psychometrics as a field of study developed in the 1920s as statistical procedures were applied to standardized tests. From their earliest uses, standardized achievement tests had some common features. First, the assumption was made that a single measure or index could be developed to compare individuals on a general, fixed, unidimensional trait (Romberg, 1992). Second, the tests were administered under uniform conditions

and constraints. Third, the tests comprised a collection of items, each having one unambiguous answer. Since those early beginnings, the testing industry in the U.S. has grown into a multi-million dollar business, and the number of tests produced has expanded exponentially.

Today, the use of high-stakes standardized tests in mathematics has the following profile in U.S. schools:

- 26 states currently have or will implement exit exams in mathematics by 2008
- 50 states require mathematics testing in Grades 3–8 and at some year of high school
- 96% of public colleges and universities require an entrance exam in mathematics
- 91% of private colleges and universities require an entrance exam in mathematics

(Breland et al., 2002; Gayler et al., 2003; Potts et al., 2002; U.S. Department of Education NCLB website).

From Grade 3 on, a student in a U.S. public school can expect to be tested at least once each year on a high-stakes test in mathematics. During the high school years the student may be subjected to numerous mathematics tests for a variety of purposes, including state-wide end-of-course exams, an exam for a diploma requirement, and a series of college-entrance and placement exams. Meanwhile that student's teachers have likely been required to take mathematics exams as a certification requirement.

The story of the content and format of large-scale tests in mathematics has not been linear. The early tests established models for the types of instruments that were used in schools throughout the decades, up until the "standards movement" that began in the late 1980s. The first content standards were published by the National Council of Teachers of Mathematics (NCTM) in 1989, and this unleashed efforts in other content areas to do the same (American Association for the Advancement of Science (AAAS), 1993; National Committee on Science Education Standards and Assessment, 1996). The *Curriculum and Evaluation Standards for School Mathematics* (NCTM, 1989) were not only influential in advocating the setting of standards to define specific goals for what students should know and be able to do in mathematics, but the document also called for reform in the way students are assessed. This effort peaked in 1995 with the publication of the *Assessment Standards for School Mathematics* (NCTM, 1995). Both standards documents called for changes in external testing, moving away from the discrete objectives and the sole use of "objective" items

and toward the use of more contextual, constructed-response items that would require human scorers. The middle of the decade of the 1990s saw changes in state testing systems in response to these and other calls for reform. For example, Kentucky instituted an innovative assessment system as part of the Kentucky Education Reform Act of 1990, and Vermont took the bold step towards a portfolio system of assessment in mathematics. Many other states began to supplement their machine-scored, mainly multiple-choice standardized tests with an increased number of constructed-response items. At the same time, nearly every state was developing its own content standards in mathematics, as well as other content areas.

As content standards were developed, interest increased in examining the alignment of standards with assessments (e.g., Romberg, 1995; Romberg & Wilson, 1992). The more simplistic notions of alignment consisted of a simple content match between a test item and a given statement in a set of standards, to determine by examination whether the test items seemed to be measuring the same mathematics called for in the standard statement. In the last decade, alignment analyses have grown in depth, complexity, and scope (AAAS, 2002; Webb, 1997).

In addition to the evolution of content standards and alignment studies, another contextual factor that has affected the nature of high-stakes testing is the growth of technological tools in the teaching and learning of mathematics. Calculators, once nonexistent in the classroom, are now prevalent throughout the grades (10% of fourth-graders, 48% of eighth-graders and 69% of twelfth-graders report using a calculator almost every day to do mathematics in school (National Center for Education Statistics, 2005)). In addition, the type of calculators being used is more sophisticated and their use is at ever earlier grades than before. In addition, the use of computer-assisted technologies has increased (Parsad & Jones, 2005). Standardized tests, in turn, have responded by developing more test items that incorporate the use of calculators, and testing companies are experimenting with ways to situate test questions in a computer setting, rather than with the use of paper and pencil alone (Bennett, 2002; Russell & Haney, 2000).

Currently 24 states have exit exams in mathematics. In all of those states the passing of the exit exam in mathematics either is a diploma requirement or is designed to be a diploma requirement at some future date. Some of these exit exams are described as general mathematics. Some states, such as North Carolina, base the exam on eighth-grade mathematics

standards. Others specify exams that are more directly tied to course content, such as Mississippi's Algebra exam (Center on Education Policy, 2004).

In January, 2002 President Bush signed into law the No Child Left Behind (NCLB) Act, which affects every program authorized under the federal Elementary and Secondary Education Act, first enacted in 1965. Although the federal government allocates only about 7% of its budget to education, which is predominantly the purview of the states, the NCLB Act is having far-reaching consequences for high-stakes testing in mathematics across the country. The law requires that states develop a set of content standards in mathematics, create annual assessments that are aligned with those standards, and administer those tests to nearly all students annually in Grades 3–8, and at least once in Grades 10–12. Further, states must show that all groups of students, disaggregated by poverty, race and ethnicity, disability, and limited English proficiency, reach proficiency in mathematics by 2013–2014. Schools must show evidence of “adequate yearly progress” toward specific annual measurable objectives, and the sanctions associated with not meeting these goals range from mandated changes in instruction to “fundamental restructuring” of “any school that fails to improve over an extended period of time” (U.S. Department of Education, p. 8). The NCLB Act, in effect, legislates that all states will have content standards, will have a set of tests that are aligned with those standards, and will show progress for all students on those instruments, with the threat of serious sanctions if they do not participate. The stakes are high at the school, district, and state levels.

Another piece of the NCLB Act that is having an immediate impact in mathematics assessment is the role of the National Assessment of Educational Progress (NAEP). Since 1972 NAEP has been used as a measure of student achievement in mathematics. Historically, there have been three types of NAEP measures: one for long-term trends, one at the state level, and one at the national level. Since 1973 there have been 9 administrations of the long-term trend instrument in mathematics. These tests are the only consistent gauges of student achievement given over time to a nationally representative sample in the U.S. They are given to students of ages 9, 13, and 17. Although NAEP is not a high-stakes assessment, it is interesting to note here that the average scores in 1999 at all three ages were significantly higher than in 1973 (Campbell, Hombo, & Mazzeo, 2000). The national or “main” NAEP has been administered approximately every 4 years to a nationally representative sample

of students in Grades 4, 8 and 12. The instruments and scales for “main” NAEP are not comparable to the long-term trend NAEP, and in mathematics they have been based on different frameworks over time. The instruments administered in 1990–2003 made use of the 1990 framework (NAEP, 1988), whereas those administered beginning in 2005 will be based on the 2005 framework. In addition to the national results, beginning in 1990 states were given the opportunity to voluntarily participate in main NAEP, in order to be able to see state-by-state comparisons of the results.

The advent of the NCLB legislation has changed the voluntary nature of participation in main NAEP at the state level. Under the law, NAEP would be used as a benchmark against which state results can be compared. To participate in NCLB a state is required to participate in the biennial administrations of NAEP at Grades 4 and 8. States are also required to set proficiency standards in reading and mathematics, and many states have chosen to use the NAEP achievement levels (Below Basic, Basic, Proficient, and Advanced) as the model for those levels. The stated goal of NCLB is for 100% of students in any participating state to attain the level of Proficient in both reading and mathematics by the year 2014. As Linn (2003) has pointed out, if the NAEP achievement levels are used as a guide, this is probably an impossible goal.

This change in the role of NAEP, from a voluntary measure of student achievement to a mandatory benchmark for accountability, was predicted in a congressionally mandated evaluative study of NAEP conducted by the National Research Council (Pellegrino et al., 1999). That committee referred to the increased pressures on NAEP to do “more and more beyond its established purposes” (p. 2). In response to these changing demands, the committee endorsed a second redesign of NAEP.

Ever since the first NAEP mathematics assessment in 1972, the National Council of Teachers of Mathematics has produced “interpretive reports” of the NAEP results (see Kloosterman & Lester, 2004). These reports have examined the results in more depth than the more general reports published by the National Center for Education Statistics (Braswell et. al., 2000). In each report, mathematics educators have looked within specific content areas, such as Number or Geometry, and analyzed the achievement results, often including disaggregated results by student subgroups. When possible, they have also analyzed samples of student work on open-ended items. The reports also include an analysis of survey data on attitudes, beliefs, and instructional context that have been collected from students and classroom teachers in conjunction with the assessment. The audience for these reports

is usually classroom teachers, though they are also intended to be accessible to a more general audience. Examples of statements from the interpretive reports include the following from the Sixth NAEP Mathematics Assessment administered in 1992:

“Performance and participation differences between males and females are disappearing.” (Silver & Kenney, 1997, p. 57)

“Interpreting fractions and decimals as locations on a number line was more difficult for students than interpreting fractions as part of geometric regions.” (p. 115)

As NAEP begins the cycle of biennial testing in the service of No Child Left Behind, the availability of data for the interpretive reports will likely suffer. Beginning with the 2000 administration, researchers for those reports were no longer given access to secure items. So long as NAEP is given this high-stakes role, such a trend will likely continue. Fewer items will be released, researchers will likely be denied access to secure items, and student work will probably not be accessible. In terms of more general usefulness as a data base for the study of student achievement, Johnson argued (2000):

in order to use the data for school improvement or theory building there would need to be more complex NAEP data gathered at the level of the student, the family, the teacher, and school and the state. This emphasis, however, runs counter to the current push to simplify and speed up the data collection and reporting process. (p. 99)

The student achievement data, not to mention the classroom practice data, which have been collected by NAEP over the past 35 years are invaluable to the mathematics education research community. Under the current political pressures to use NAEP as a high-stakes tool with biennial testing, this role for NAEP may be greatly diminished, at the very least.

TECHNICAL ISSUES IN HIGH-STAKES TESTING

When tests are used to make decisions with critical consequences for various constituencies, the technical aspects of the test must be sound. Inasmuch as tests are essentially measures, and all measurement involves error, anyone who is in the position of developing, using, or interpreting a test of student achievement must attend to the potential for error and be mindful of the need to minimize error. In this section

I review recent literature on the possible sources of error in measuring student achievement and how it might be avoided.

Validity

All types of mathematics assessment, whether high-stakes or not, are samples of what students know and can do. This sample data is used to make inferences about certain constructs, such as knowledge, attitudes, or achievement. *Validity* refers to the degree to which the inferences made on the basis of the assessment are meaningful, useful, and appropriate (Brualdi, 2002). Traditional concepts of validity (content-related, criterion-related, and construct-related) have been expanded in recent years, largely through the work of Messick (1989, 1996). He argued that validity should take into account information about actual as well as potential consequences of score interpretations. Validity, according to this view, is not a property of the test, but instead a property of inferences or interpretations made from test scores and is thus essential for all types of assessments, but particularly for high-stakes tests. (See Messick, 1996, for a discussion of the six interdependent aspects of validity that form a general theory).

Two major threats to validity in high-stakes tests are construct underrepresentation and construct-irrelevant variance (Brualdi, 2002). The former refers to a situation in which the concepts or skills that are measured in the assessment fail to include important dimensions or facets of the intended construct (presumably, some content domain in school mathematics). In such a case, the test results are unlikely to reveal a student's true ability regarding that construct. For example, a test of measurement that included only items using the metric system would not be a valid measure of students' ability to solve problems involving nonmetric systems of measure. Construct-irrelevant variance can take two forms. In the first, extraneous clues in the item or the task format may permit some students to respond correctly or appropriately in ways that are irrelevant to the construct being assessed. For example, a student might use "test-wisdom" to ferret out the correct answer without actually knowing or being able to do the mathematics being tested. The other possibility is that extraneous aspects of the task make it irrelevantly difficult for some individuals or groups, such as in the case of a context that is so unfamiliar as to limit access for some students to the mathematics in the task. In either case, the resultant score would not be one upon which to make valid inferences about the student's achievement.

When high-stakes tests are used for accountability purposes, as is proposed in the NCLB legislation, validity issues also arise in the setting of performance standards for the test. Such standards, or levels of achievement, serve the purpose of classifying the test takers into categories, such as "basic," "proficient," "advanced," etc. Performance standard-setting is a judgmental process, often carried out by committees of "experts." To achieve the highest possible level of validity, the process must be carefully planned and executed (see Jaeger, Mullis, Bourque & Shakrani, 1996 for a discussion of the challenges and issues of standard setting, especially for performance assessments). Other aspects of validity are that the distinctions between the levels should be "meaningful and useful" (Stecher, Hamilton, & Gonzalez, 2003, p. 21) for the purposes of the test.

In an analysis of the content validity of selected standardized tests of mathematics achievement that are often used to make decisions about students with disabilities, Parmar et al. (1996) cited numerous technical flaws in the tests. These include inadequate representation of content domains, inappropriate sequencing and placement of items, inappropriate use of age and grade-equivalent scores, and incorrect descriptors assigned to items. One of the conclusions of the study is that "the majority of instruments and procedures used for assessment of students in special education do not reflect important mathematics, nor do they provide information that will drive or guide curriculum or instructional decisions" (p. 128).

Because the content covered in standardized tests is nearly always too narrow to reflect the complexities of teaching and learning, Stake (1995) argued that they are a "flimsy indicator" of the mathematics that students have learned (p. 174). He makes the point that much of the evidence for the learning of mathematics comes from reflective interaction with individual students and with classes of students, so that the complexities of learning can be fully captured. Standardized tests often rely on a single-dimension concept of achievement and thus at best should only be considered as one component in an array of data.

Reliability

No measurement is error-free, and tests are no exception. The reliability of a test refers to the extent to which the measurement is free of random errors, or errors that are not systematic. Reliability is a characteristic of both the population of test takers and the instrument itself, because of the way reliability is computed. Thus a test will show different reliability results for different populations of test takers.

Three potential sources of random error exist in all tests, not just high-stakes tests. Those sources are the test itself, the test takers, and the scoring process. The test may contain random errors that stem from inadequate sampling of the content domain, the ineffectiveness of the distractors (the incorrect options) in multiple-choice items, partially correct distractors, more than one correct distractor, and the difficulty of the items relative to the student's ability, in that a test that is too difficult for a student will have low reliability for that student or population of students. Test takers introduce error by their human nature. Many factors, both external and internal, can affect the consistency of a student's score in a given testing situation, including motivation or illness. An overtired student, for example, may misread the test directions or inadvertently omit sections of the test. Scoring, especially for open-response items, is a third source of random error. Even machine-scored tests can have error if the machine does not correctly read the answers the student intended (an interaction between student error and scoring error). Open-response items introduce more error if scoring rubrics are unclear or if scorers are not highly trained and monitored for high interrater reliability.

Another useful aspect of reliability for high-stakes tests is described by Parmar et al. (1996) as *user reliability*, which they define as the extent to which different users would come to the same diagnostic conclusions when faced with the same results, and the extent to which they would make similar curriculum and instruction recommendations based on that diagnosis. Their analysis of a selection of standardized mathematics tests used to make decisions about students with disabilities cited numerous problems with user reliability of those tests.

Equity

When the results of achievement tests are disaggregated for various subpopulations, one can measure the gaps in achievement among different groups. These achievement gaps among different genders, races, or groups with differing socioeconomic status have changed over the years in mathematics, both for better and for worse. The NAEP data, although not a high-stakes test prior to 2003, shows these differences. For example, in the five NAEP mathematics tests given between 1990 and 2003 (U.S. Department of Education, 2004), the gender gap at Grade 4 has remained small but unchanged (in a meta-analysis of 100 studies on gender differences in mathematics performance, Hyde et al., 1990, found females outperforming males by a negligible amount). However, the gap between Black

and White students, as reported in school records of race/ethnicity, indicates a steady narrowing of the gap over those years. The White/Hispanic gap narrowed from 2000 to 2003, after remaining steady from 1990 to 2000. Finally, the gap between those eligible and not eligible for free or reduced-price lunch also narrowed from 2000 to 2003, when it had not narrowed from 1996 to 2003. Although the news looks more promising in recent years, there is still evidence of disproportionate numbers of minority, low-income, and special-needs students who are failing tests for promotion and graduation (Catalyst Chicago, 2004; Darling-Hammond, 2003).

Aside from the data from high-stakes tests that track gaps among different subpopulations, another issue is the interactions among those groups and the tests themselves. In the literature on testing, the notion of a "fair test" is one that yields comparably valid inferences from person to person and from group to group (National Research Council (NRC), 1999; Pellegrino et al., 2001). In order to avoid bias, test developers must ensure that no task has characteristics that would have different meanings for different subgroups of test takers. This might mean, for example, avoiding contexts or language associated with one particular culture, so that any inferences drawn from those test scores would be comparable across all subgroups. On large-scale tests expert review panels are often set up to review assessment items for this type of bias. Psychometricians might also examine the results of statistical tests for differential item functioning (DIF), which identifies items that produce differing results for members of particular groups after the groups have been matched with regard to the attribute being measured. Some researchers have begun to follow up DIF analyses with cognitive analyses, to try to uncover why certain items function differently across groups. The purpose of such studies is to learn more about how students think about and approach the problems (Lane, Wang, & Magone, 1996).

Many high-stakes tests provide accommodations for certain students who demonstrate special needs. NAEP, for example, began allowing accommodations on the mathematics tests in 1996. These include extra time and bilingual Spanish editions. Accommodations generally fall into four categories: presentation, response, setting, or timing/scheduling (Thompson et al., 2002). Test presentation accommodations may include oral presentations, changes in test content or test format. Response accommodations include allowing students to dictate their answers or write their answers directly into their test booklets. Setting accommodations often involve testing students individually or in a separate room. Finally, timing or scheduling

accommodations include allowing extra time or administering smaller pieces of the test in separate sessions. In a meta-analysis of 150 studies of test accommodations, Sireci et al. (2004) critiqued the studies with respect to an *interaction hypothesis*. The hypothesis states that test accommodations should improve the test scores for targeted groups but not improve the test scores for examinees for whom the accommodations are not intended. Among their conclusions they found that “many accommodations are justified and effective for reducing construct-irrelevant barriers to students’ test performance” (p. 61). In particular, they found that extended time improved the performance of students with disabilities more than it improved the performance of students without disabilities. The authors also argued for the need for future tests to be “constructed and administered more flexibly so that accommodations become unnecessary” (p. 67).

In terms of high-stakes tests, such as those used under the NCLB legislation, Abedi (2004) argued that major issues are involved with measuring adequate yearly progress for students with limited English proficiency (LEP). Among the issues cited are inconsistent LEP classification, the sparse population of LEP students in many states that threatens validity, and the LEP subgroup’s lack of stability. Abedi concluded that NCLB’s mandates may unintentionally place undue pressure on schools with high numbers of LEP students.

Large-scale standardized tests, such as are used in most high-stakes testing situations, are prone to error. In addition to random measurement error, as described above under Validity and Reliability, non-random human errors are associated with tests. Rhoades and Madaus (2003) divided human errors in testing into two types: active and latent. Active errors are those that are committed by specific individuals and are often easy to identify. Examples of active errors would include incorrect scoring of a test or a computer programming error that produces incorrect percentile rankings. Latent errors are systemic in nature and often stem from poor administrative decisions or a broad constellation of circumstances that provide fertile ground for human mistakes. Examples of latent error are state Departments of Education that rely on a single test score to make high-stakes decisions, or an overly demanding testing schedule that puts undue demands on an understaffed or underqualified team to develop, administer, score and report tests. Error can be found in all phases of the testing process, from the development of items to printing of test booklets to the setting of a passing score. Of particular concern is the use of tests to measure change in achievement over time, or trends. A common error is to analyze results from one year to the next of administrations

of the same test, in that the test may have undergone changes that can alter the results. As Beaton (1990) pointed out, “If you want to measure change, do not change the measure” (Beaton et al., 1990, p. 165).

Another issue of particular concern with respect to the NCLB legislation is the determination of adequate yearly progress (AYP). Under this requirement states establish goals for what percentages of students in various subgroups (e.g., low income, minority, limited English proficiency) will meet or exceed proficiency levels on the state’s assessments each year. In order to confirm gains from one measure to the next, or across grade levels, test scores must be comparable from test to test and year to year. *Scaling* is a process used to accomplish this, in which raw scores are converted by means of a mathematical function (which might be either linear or nonlinear) to a common scale. *Vertical scaling* is the process of equating scores of students at different levels. Several assumptions that underlie the process of vertical scaling and equation may be problematic. For example, the tests are assumed to measure the same general content, which is an assumption not easily met by tests that span one or more grade levels. In addition, both the nature of the items and the assessment process may change over grades, in which case vertical equating confounds content changes with method changes. Lissitz and Huynh (2003) have alternative approaches that may resolve some of these issues.

Many testing programs use cut scores to determine the predetermined level at which a student reaches a certain level of achievement. That level might denote the difference between passing and failing, or it might determine categories of achievement, such as “basic,” “proficient,” “advanced,” and so on. The setting of cut scores is used to separate students into different classes of achievement, and it is based upon predetermined concepts of student achievement. Regardless of which method of setting cut scores is used, human judgment is embedded in the process, and therefore the process cannot be regarded as objective. (See Horn et al., 2000, for further discussion of cut scores).

Teachers, as well as students, may be subjected to high-stakes testing, and often the result is not exemplary. In 2004, two thirds of the middle school mathematics teachers in the Philadelphia school district failed the certification test that is used to determine if they meet the qualifications for “highly qualified” under the federal No Child Left Behind Act. These tests are no less immune to error than any other standardized tests.

Many researchers have attempted to find causative links between teacher quality and student achievement. Writing about testing for teacher quality (as in

whether teachers' test scores or observers' ratings are good predictors of professional effectiveness), Ferguson and Brown (2000) described potential sources of error thus:

teacher quality, including teachers' test scores, are measured with error; student quality in a school or district can affect which teachers choose to apply there, creating reverse causation from student performance to teacher quality; particular measures of teacher quality may matter more or less depending on other variables, such as class size, so simple linear models that ignore interactions may produce misleading results; correlations between teacher quality and other inputs such as parental effectiveness can produce biased estimates for the effect of teacher quality if parental variables are omitted from the analysis or measured with considerable error. Further, most studies lack the type of data necessary for sorting out the issues that the advocates of multilevel estimation emphasize. (p. 149)

Goldhaber et al. (2004) studied 3 years of teacher and student data from North Carolina and concluded that teachers who received national certification through the National Board for Professional Teaching Standards were more effective at raising student achievement than teachers who pursued, but failed to obtain, certification. On the one hand, this study looks at one outcome score for students; on the other hand, the national certification process entails an extensive performance-based evaluation, including written exams, lesson portfolios, and classroom videos.

EFFECTS OF HIGH-STAKES TESTING IN MATHEMATICS

When the stakes change from low to high, there are often consequences. DeMars (2000) studied the interactions between test stakes and item format interactions on math and science sections of a high school diploma endorsement test. The findings indicate an interaction between response format and test consequences. Under both multiple-choice and constructed-response formats, students performed better under high stakes (diploma endorsement) than under low stakes (pilot test), but the difference was larger for constructed-response items. Gender and ethnicity did not interact with test stakes; the means of all groups increased when the test had high stakes. Gender interacted with format; boys scored higher than girls on multiple-choice items, girls scored higher than boys on constructed-response items.

Test format (open-ended, multiple-choice, etc) on high-stakes tests also has an influence on instruction. During the school reform movements of the 1990s, much effort was expended on building tests worthy of "teaching to the test" (Resnick & Resnick, 1992). Others questioned at the time whether any one test can be so representative of a desired curriculum that teaching to such a test would equal good instruction (Linn, Baker, & Dunbar, 1991). Under the pressures of No Child Left Behind and other large-scale accountability measures, many of the efforts toward more open, performance-based testing, such as those in Kentucky and Vermont, have been replaced with more traditional tests. When tests use more open-ended items that require writing, teachers place more emphasis on writing in mathematics classes (Taylor et al., 2003). In general, teachers tend to tailor their instructional practices to the format of the tests (Abrams & Madaus, 2003; Pedulla et al., 2003). The format of the test can also have a substantial impact on test results. In a district reporting gains on a high-stakes multiple-choice test, Koretz et al. (1991) gave students a similar test with more open-ended questions and found that student performance dropped off as much as half a standard deviation.

Researchers have documented effects of high-stakes testing on the mathematics curriculum. Taylor et al. (2003) found that teachers in Colorado added new content, such as probability and geometric topics, and placed more emphasis on problem solving and writing in mathematics classes. At the same time, they found that teachers decreased the amount of time spent on other curricular areas, such as science and social studies. Clarke et al. (2003) found that educators in Massachusetts, Kansas and Michigan were more likely to remove content topics than to add them. In those three states, about 10% of educators interviewed thought that the state test had no impact on what was taught. For high school exit exams, Gayler et al. (2004) reviewed existing literature on impact and found that exit tests resulted in broader coverage of state standards, alignment of curriculum with those standards, and targeted course offerings for at-risk students. At the same time, these tests can also narrow the curriculum and force a sequence and pace of instruction that is inappropriate for some students. In a national survey of teachers, Pedulla et al. (2003) found a strong interaction between the level of the stakes in the test and the degree to which it impacted the curriculum; in particular, far more teachers in high-stakes states reported that their own tests reflected the format of the state test than did teachers in low-stakes states. In addition, a substantial majority of teachers at each grade level, but especially at the ele-

mentary level, reported that testing programs in their state have led them to teach in ways that contradicted their ideas of sound instructional practices.

The effects of high-stakes testing are also linked to the grades at which testing is conducted, as documented in a study by Stecher and Barron (2001). Grades in which testing occurs show more profound effects on curriculum and instruction than those grades that are not included in the testing. A more insidious consequence of grade-specific effects is found in a significant nationwide bulge in students enrolled in 9th grade and a tripling of the attrition rate between 9th and 10th grades over the last 30 years (Haney et al., 2004). The change is due, according to their analysis, to the increased practice of holding students back from the pivotal high-stakes exit exams in 10th and 11th grades, as well as to increased course requirements. Frequently, such students drop out of school. These findings are in line with those of Jacob (2001), who reported that high school graduation tests, while they did not have an effect on student achievement in mathematics or reading, did increase the probability of dropping out among the lowest ability students.

STANDARDS FOR APPROPRIATE USE OF HIGH-STAKES TESTS

In light of the effects of high-stakes testing, many educators have called for reforms in the uses to which such tests are put. Many have found fault with the NCLB legislation's emphasis on testing and offered advice on reforms (Abrams & Madaus, 2003; Elmore, 2003; Harvey, 2003; Neill et al., 2004; Stecher et al., 2003). In a broader reaction to the current climate in the U.S. of widespread high-stakes tests for educational policy purposes, the National Commission on Testing and Public Policy (2002) issued a statement that calls for, among other things, less reliance on multiple-choice tests, greater use of multiple sources of evidence for decision-making, critical evaluations of tests for fairness and accuracy, and a separation of testing that is used for instructional purposes and that used for accountability purposes. For more general application of educational testing, the *Standards for Educational and Psychological Testing* (1999) delineates standards for practice in the use of educational tests. It was a joint venture of the American Educational Research Association, the American Psychological Association, and the National Council on Measurement in Education. Among other standards these groups emphasize that when stakes are high, it is particularly

important that the inferences drawn from an assessment be valid, fair, and reliable.

As the research has shown, high-stakes testing in mathematics can have a powerful influence on educational practice, whether for good or ill. It can affect teaching practices, curricular decisions, and the decisions made about schools or individual students. The potential power of testing makes it an alluring policy tool, but a tool that is fraught with tensions. As the National Research Council (1999) pointed out, one of the dilemmas at the heart of this tension is that "policy and public expectations of testing generally exceed the technical capacity of the tests themselves" (p. 30). The primary reason for this is that tests are too often used for purposes other than those for which the tests were designed and validated. It is not uncommon for tests that were developed to produce valid measures of student learning at the aggregate level (e.g., school, state, national) to be used to make decisions about teachers (classroom-level data) or even individual students. Such practices may result in invalid inferences, and thus unfair decisions, being made about those stakeholders.

RESEARCH NEEDS

There are many potential arenas for research in high-stakes testing in mathematics. Many of these were described in the report of the Assessment Working Group of the 2003 Research Catalyst Conference sponsored by the National Council of Teachers of Mathematics (Lester & Ferrini-Mundy, 2003). Though the focus of the conference was on the impact of standards, the assessment group's research agenda covered many components that involve high-stakes tests. The group called for descriptive studies on the evolution of large-scale assessments over the last decade as well as on college entrance exams. They also called for investigations into the construction of assessment instruments that are valid and fair for all students, particularly how test items can be designed to allow greater access to all groups of students. In addition, the agenda designates for high priority studies that investigate how high-stakes tests can be used to improve classroom learning.

The NRC (2002) has noted the need for research on the impact of standards on assessment in mathematics, science, and technology. Among other issues, they emphasize the need for more research into the extent to which teachers have modified their classroom assessment practices to respond to calls for change in standards documents, and to what extent teachers use assessment data to adjust instruction.

They also call for research into college admission tests and the extent to which they are aligned with nationally developed standards.

Because of the important social and political impact of high-stakes testing in mathematics, further research is urgently needed. At the same time, there are potential barriers to the undertaking of research in this area, many of which stem from the nature of high-stakes testing. For example, high-stakes tests are often held in secret, making access to item pools or student data difficult, if not impossible. Further, random assignment and experimental intervention is problematic when students and teachers face serious consequences from the assessment. Yet the very fact that the stakes are high should underscore the critical nature of this research agenda.

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LARGE-SCALE ASSESSMENT AND MATHEMATICS EDUCATION

Jan de Lange

FREUDENTHAL INSTITUTE

INTRODUCTION: IMPORTANCE AND IMPACT

Public View

The science of mathematics education is not a very sexy science if publications in the popular media are taken as an indicator. In many countries one could also use the amount of money spent on this science as a good measure. But sometimes the science of mathematics education hits the media in a big way or becomes even a hot issue that will not disappear from the headlines for years to come.

Recently Germany delivered a prime example when the results of the PISA 2000 study were published. The date of the data release was December 5, 2001, and the cover of *Der Spiegel* (a magazine comparable to the American publications *TIME* and *Newsweek*) of 10 December 2001 tells it all: *PISA-Study: The New Education Catastrophe*. The 16-page-long article in small print makes the message more than clear: Compared internationally the German schools fail. The students are poor readers, are insufficient in arithmetic, and find problem solving completely beyond their possibilities. For Germans PISA is not an Italian city with a leaning tower; Germans are suffering from the PISA “shock.”

The words and adjectives used in the article tell a story of their own: disaster, catastrophe, shock, hopeless, emergency plan. What makes the story extra difficult to accept for Germans is the fact that all neighbors of Germany score significantly better, and that the Germans are at the same level as the U.S.A.

Embarrassing for the Germans, but the Americans do not see any reflection on the PISA results in their *TIME* and *Newsweek*: No media attention at all is probably the fair picture of how the results of PISA 2000 were received in the U.S.A. No news is good news seemed to be prevailing judgment here.

It is worth noting that, according to *Der Spiegel* and many others, PISA had a winner: Finland. The horse race was back, and whatever the organizers of international comparative studies have in mind, it seems inevitable that the horse race component always reaches the front pages. Finland has known it: Flocks of experts went to Finland to see the educational miracle in action. According to *Der Spiegel* the Finns are proud of their comprehensive system. Policy issues are quite often the focus of the interest: the comprehensive system in Finland, the brilliance of the Japanese “average” student in an otherwise miserable system.

There seems to be no interest in the content of the study: There was no discussion of the items that were used, the competencies needed, the quality of the instruments, or the relation to the curricula. And there certainly was no discussion about the whole study being comparative, or norm-referenced, and the fact that it does not indicate any “absolute” quality. It is definitely not criterion referenced. This often-ignored fact can cause serious interpretation problems, especially when considering policy measures. According to the national Dutch report on PISA, the Netherlands was first in Mathematical Literacy (“The Winner”). In the official Organisation for Economic Co-operation and Development (OECD) publication one cannot find the Netherlands because the coun-

try did not meet the statistical standards. The conclusion of the Dutch Ministry of Education was one of being proud of the system, the schools, and the teachers. The system was above all “effective” because the Netherlands is “underspending” on education in the OECD. At the same time math education experts (not excluding the present author) were trying to make a point about the very low level of math education in the Netherlands.

This paradox also runs among experts: In 1997 Lyle Jones argued in his William H. Angoff Memorial Lecture that NAEP (National Assessment of Educational Progress) has raised a high level of controversy because the NAEP report of 1996 shows only 18% of U.S. fourth graders “proficient” and only 2% “advanced” in math (Reese, Miller, Mazzeo, & Dossey, 1997) whereas recent results from TIMSS show that the average math performance for U.S. fourth graders is significantly above the international average (Mullis et al., 1997). Jones asked his audience, “When U.S. fourth-graders perform reasonably well in an international comparison, isn’t it unreasonable that only 20 percent are reported to be ‘proficient’?” Of course the answer is not the seemingly suggested “it is unreasonable,” but there can be very good reasons that this is completely normal—although not always easy to explain to policymakers or journalists.

Mathematics Education View

Not only are international comparative studies popular with the media and policymakers, they are also a subject for much controversy within the math education community itself. One of the key issues is about the preoccupation with competition (the horse race). Another is the assumptions on which comparative studies in mathematics are predicated. Keitel and Kilpatrick (1999) questioned in particular the treatment of the mathematics curriculum as unproblematic. The next is the assumption that a single test can give comparable measures of curriculum effects across countries. Jablonka (2003) discussed the fundamentally situated nature of mathematical literacy (like in PISA, but also in any assessment in context). She argued that contexts will be familiar to some students and not others. And thus any attempt to use a single instrument to assess mathematical literacy beyond the most local context would appear to be self-defeating. Cultural differences exist between countries and will play a major role. Freudenthal found this argument the most important one in discarding international comparative studies. Another problem is the fact that the single scale instrument does not mean that when two countries have (more or less) the same score, the

consequences for policymakers are the same. As an example we analyzed the situation in two neighboring countries, the Netherlands and Belgium, who score almost the same on TIMSS but show big differences in curricula, didactics, culture, and policies (de Lange, 1997a). So if a policymaker wants to learn from these relatively high-scoring countries, where should she go? Probably to Finland.

Politics

One of the concerns that comes up after every large-scale assessment is whether or not such tests are really about measurement. Especially people that look more at the policy side of such studies argue that performance measures are not really about measurement, but about political communication. Political scientists have expressed this view on many occasions, and there is certainly reason to take this view seriously.

Clearly politicians make up their own interpretation of the facts, to their own liking. This can be observed in most countries, but a precondition to make this observation is knowing the facts in some detail. This is most often not the case and beyond what journalists are interested in. If one looks at the large number of national reports that tend to come out parallel with the international report, it is amazing to see how the data were selected. It takes some effort, but it is worth it: large-scale international assessments are indeed about politics, about policy, about political communication.

In the discussions, whether scientific or more public, these politics play a large role. The chain of reasoning is quite often influenced by political views. This may lead to chains of reasoning in the arguments like the ones mentioned earlier that are open to some kind of criticism because of the weak reasoning, not being based on the actual data, or taking the data too seriously. Quite often the points made are mere opinions, not seldom without proper knowledge of the methodology of the study involved. Where appropriate I will deal with a number of controversial issues, hopefully without falling into the trap of weak causal reasoning and mere opinion.

NATURE OF LARGE-SCALE ASSESSMENTS

Large-Scale Assessment As Information

To identify “large-scale assessment” as a subject worthy of inclusion in this book is one thing, to find out what it really is, is quite another. Remarkably,

many of my colleagues seem to agree: They shy away from giving a definition at all, and those who even try come up with rather unsuccessful propositions. In the publication *Reinventing Assessment* (1998) Randy Elliott Bennett proposed the following:

Large-scale educational assessment consists of those tests administered to sizable numbers of people for such purposes as placement, course credit, graduation, educational admissions, and school accountability. It includes group-administered, standardized tests used most often in the secondary through post secondary years. (p. 1)

Bennett is clever enough to use words as “for such purposes as,” leaving room for other purposes. The second sentence seems a bit redundant. But there are few definitions around, so I start with this one and look to improve on this definition, if only for this chapter of the book.

The National Research Council (NRC) publication *Knowing What Students Know* (2001) saw two components in educational assessment: The first context in which educational assessment occurs is the classroom. Here assessment is used by teachers and students mainly to assist learning, but also to gauge students’ summative achievement over the longer term. Second is large-scale assessment, used by policymakers and educational leaders to evaluate programs or obtain information about whether individual students have met learning goals.

The committee adds that large-scale, standardized assessments can communicate across time and place, but by so constraining the content and timelines of the message, they often have limited utility in the classroom. Thus the contrast between classroom and large-scale assessments arises from the different purposes they serve and contexts in which they are used. And large-scale assessments are further removed from instruction but can still benefit learning if well designed and properly used. Finally, large-scale assessments not only serve as a means for reporting on student’s achievement but also reflect aspects of academic competence societies consider worthy of recognition and reward. Thus large-scale assessments can provide worthwhile targets for educators and students to pursue.

The problem with the definition and description just given is that it seems to fit seamlessly to the American situation but ignores the realities in many other countries. International comparative studies may be blooming; international cooperation and recognition still has a long way to go.

Paul Barton made the point of the relatively isolated position in the international large-scale assessment landscape clear in his 1999 publication *Too Much*

Testing. In one area there is very little (large-scale) testing whatsoever in the United States, and a lot in other developed countries: extensive examinations at the exit point for secondary education. In effect, in many countries large-scale testing is quite often identified with these exit-examinations, which are often very closely related to the curriculum. This has a very desirable side-effect that the distinction between classroom testing and large-scale testing, as clearly identified by the committee that is responsible for *Knowing What Students Know*, becomes a fluent connection, especially in those countries where a classroom assessment is part of the large-scale examination. If asked for a definition of large-scale assessments, these countries would probably suggest quite different definitions than the ones above

Large-Scale Assessment as Entrance/Exit

The isolation of the United States in this area has been properly phrased by Eckstein and Noah (1993):

The United States is unique among the countries we have studied (U.S., China, Japan, Germany, England, Wales, France, Sweden, and the former Soviet Union) in having no coordinated, public, national system for assessing student achievement at the end of secondary school. (p. 238)

A more recent study (Stevenson & Lee, 1997) focused on entrance and exit examinations in Japan, Germany, France, and the United Kingdom. The authors observed:

Entrance and exit examinations in these countries are based on a curriculum established by ministries of education at the local, regional or national level. Rather than imposing some arbitrarily defined standards of achievement, the examinations are closely tied to what students have studied in high school. Because teachers are aware of what students are expected to know in examinations, it becomes their responsibility to equip students with the information and skills needed to pass the examination. (p. 47)

This last observation is important because it points to the fact that large-scale assessments need not to be completely different from classroom assessment, as a matter of principle. On the contrary: For students in many countries the exit examination fits seamlessly with their classroom assessment. This fact surely complicates the task to come up with a fitting definition of large-scale assessment. Especially because of the fact that these exit examinations also do not always fit the pencil-and-paper format that people tend to link to

large-scale assessments, not to mention the emphasis on the multiple-choice format.

As Stevenson and Lee observed, the exit examinations typically include open-ended questions that require organization and application of knowledge, and oral examinations that require students to express themselves verbally. Thus exist large-scale assessments that do have more than one measure, use different formats, and try to measure different competencies.

Large-Scale Assessment As Proficiency

Finally I need to mention the developments in large-scale assessments that are the results of information technology. Several examples in existence now illustrate how technology can help infuse ongoing formative assessment into the learning process. The intelligent tutors form a category that makes a difference. *Knowing What Students Know* observes that these intelligent tutoring systems are powerful examples of the use of cognitively based classroom assessment tools blended with instruction. The systems have in common that when the student makes a mistake the system offers effective remediation.

In relation to large-scale assessment caution is necessary when such intelligent tutoring systems are being used nationwide or even made compulsory—as is being considered in the Netherlands. If all schools in a country or state are using a (diagnostic) student tutoring system that also keeps track of student progress over time (maybe over a somewhat narrow band of competencies), can one then conclude that this is large-scale assessment, or should one follow *Knowing What Students Know* and classify this as formative or classroom assessment, which would make it distinct from large-scale assessment?

Definition

Clearly trying to formulate a definition of large-scale assessment, even if only for this paper, is not a simple task. For the scope of this article I will use the following formulation:

Assessment is a process by which educators use students' responses to specially created or naturally occurring stimuli to draw inferences about the students' knowledge and skills (Popham, 2000). Large-scale assessments are assessment instruments intended to be administered to a large number of students for a wide variety of reasons. Three broad purposes can be identified: to assist learning, to measure individual achievement, and to evaluate programs (NRC, 2001). They can be delivered by different media, be individual or teamwork, be formative or summative, use a

variety of formats, can be aligned with a curriculum or not, can be standards based or not, may operationalize competencies (in contrast with a content-oriented curriculum), and can be part of a comprehensive balanced assessment system or not.

STRUCTURING THE LARGE-SCALE ASSESSMENTS DISCUSSION

The definition gives way to a variety of ways to structure the discussion. One class of assessments seems a very natural one, and it is also a very visible one: the class of international comparative studies. In mathematics this class is usually restricted to the IEA (International Association for the Evaluation of Educational Achievement) studies like SIMSS, TIMSS, TIMSS-R and the recent TIMSS—with the meaning of the *T* changed from *Third* to *Trends*—and the OECD studies like PISA 2000, PISA 2003, and so on. But there are numerous other studies, especially in the area of adult mathematical literacy. I will discuss the TIMSS and PISA studies in some detail and make a relation with some of the other studies. All these studies have as a common denominator the international comparative component.

At the national level we run already into problems because of the large spread in formats and uses of large-scale assessments. In many countries exit examinations would be the first assessment that people think of. In others, most notably the United States, the one that comes closest to a national test is the NAEP project. This is a national survey intended to provide policymakers and the public with information about academic achievement of students across the nation. Its goal is of a similar nature as the IEA and OECD studies, albeit on a national level. Trend studies at national level include, for instance, the PPON (Periodieke Peiling van het Onderwijsniveau, Periodic Sounding of the Education Level) study in the Netherlands at elementary level. Of course aside from the NAEP study that has a reputation in regard to the use of standardized tests and the industry that comes with them.

I will discuss in some detail at the national level both national examinations and national trend studies, and to some extent, standardized tests both more traditional and more innovative.

In some countries it is relevant to go down one more level: to that of the states. Of course this is especially relevant in the United States, but other countries have state tests as well, for instance Australia, where Victoria has been very visible when introducing their quite innovative and balanced assessment system. I

will not discuss some consequences of the No Child Left Behind Act in the United States as this issue is addressed in Wilson's chapter (this volume).

It seems reasonable to devote a discussion on the future of large-scale assessment in light of evolving information technology. The trend towards more problem-solving skills and more complex competencies like knowledge organization, problem representation, strategy use, and metacognition in mathematics education seems very challenging, as the underlying assessment principles in these systems are traditional.

The same trend towards more complex competencies in mathematics education also leads to renewed interest in teamwork in real-world problem solving. I will mention some examples that have been in place for quite some time and certainly can be regarded as large-scale assessments.

This leaves a structure with three key tenets: international, national, and state; technology; and assessing real-world problem solving. Of course this is no real taxonomy by any measure but seems a way of clustering that might help the reader to grasp this complex subject. Within each of the five clusters I will discuss some examples, problems and benefits, influence or results, and some suggestions for the future.

REVIEW OF ISSUES IN LARGE-SCALE ASSESSMENTS

International Comparative Studies

TIMSS and PISA

The IEA has conducted studies of cross-national studies since 1959. TIMSS, originally named the Third International Mathematics and Science Study, from 2003 onwards Trends in International Mathematics and Science Study, is the most recent one of the IEA series to measure trends in students' mathematics and science achievement. Additionally the countries that participate are provided with a resource for interpreting the achievement results and to track changes in instructional practices. TIMSS asks students, their teachers, and their school principals to complete questionnaires about the contexts for learning mathematics and science. TIMSS 2003 assesses the mathematics and science achievement of children in two target populations. One target population, sometimes referred to as Population 1, includes children Ages 9 and 10. It is defined as "the upper of the two adjacent grades with the most 9-year-olds." In most countries this is the fourth grade. Population 2 includes children Ages 13 and 14 and

is defined as "the upper of the two adjacent grades with the most 13-year-olds." In most countries this is eighth grade (Mullis et al., 2001b). By assessing these grades using the same target populations as in 1995 and 1999, TIMSS 2003 will provide trend data at three points over an 8-year period. In addition, TIMSS data will complement IEA's Progress in International Reading Literacy Study (PIRLS) being conducted at the fourth grade. According to the authors of the *TIMSS Assessment Frameworks and Specifications 2003* (Mullis et al., 2001b), countries

participating in PIRLS and TIMSS will have information at regular intervals about how well their students read and what they know and can do in mathematics and science. TIMSS also complements another international study of student achievement, the OECD's Programme for International Student Achievement (PISA), which assesses the mathematics and science literacy of 15-year olds. (p. 6)

According to the OECD (2000):

The OECD's Programme for International Student Assessment (PISA) is a collaborative effort among the member countries of the OECD to measure how well young adults, at age 15 and therefore approaching the end of compulsory schooling, are prepared to meet the challenges of today's knowledge societies. The assessment is forward looking, focusing on young people's ability to use their knowledge and skills to meet real-life challenges, rather than on the extent to which they have mastered a specific school curriculum. This orientation reflects a change in the goals and objectives of curricula themselves, which are increasingly concerned with what students can do with what they learn at school, and not merely whether they have learned it. The term 'literacy' is used to encapsulate this broader conception of knowledge and skills. (p. 12)

The first PISA survey was carried out in 2000 in 32 countries, including 28 OECD member countries. Another 13 countries completed PISA 2000 in 2002, and from PISA 2003 onwards more than 45 countries will participate "representing more than one third of the world population" (p. 22). PISA 2000 surveyed reading literacy, mathematical literacy, and scientific literacy, with the primary focus on reading. In 2003 the main focus was on mathematical literacy (published in 2004), and in 2006 scientific literacy will be highlighted.

It will be clear that both studies have a lot of similarities resulting in improper identification of the two series of studies in the media, which is undesirable and confusing. But the descriptions of the organizations that are responsible show that they both claim similar relevance for the studies. It seems safe to say that, for both TIMSS and

PISA, countries participating in their studies will be given information at regular intervals about how well their students read and what they know and can do in mathematics and science. Both studies provide such information and they provide it, methodologically speaking, in a very similar way (based on Item Response Theory, IRT). Even the reporting tables in the respective reports look very similar. We refer to the Technical Aspects Reports for both studies for the methodological aspects (OECD, 2001; Yamamoto & Kulick, 2000).

If there is a problem that both studies share it is the design of the measuring instrument in relation to the validity of the outcomes. Traditionally, validity concerns associated with tests have centered about test content, meaning how the subject domain has been sampled. Typically evidence is collected through expert appraisal of alignment between the content of the assessment tasks and the curriculum standards (in case of TIMSS) and “subject matter” assessment framework (PISA). Nowadays, empirical data are often used before an item is included in a test.

Traditionally validation emphasized consistency with other measures, as well as the search for indirect indicators that can show this consistency statistically. More recently is the recognition that these data should be supplemented with evidence of the cognitive or substantive aspect of validity (Linn, Baker, & Dunbar, 1991; Messick, 1993). Or as *Knowing What Students Know* summarized: “The trustworthiness of the interpretation of test scores should rest in part on empirical evidence that *the assessment tasks actually tap the intended cognitive process.*” (p. 7). One method to do this is a protocol analysis in which students are asked to think aloud as they solve problems; another is an analysis of reasons in which students are asked to provide rationales for their responses; and a third method is an analysis of errors in which one draws inferences about processes from incorrect procedures, concepts, or representations of problems. Although some of these methods are applied only after the test is administered, there is a trend that large-scale assessments like TIMSS and PISA use these methods as well. The use of cognitive laboratories to gauge whether students respond to the items in ways the developers intended has become a new instrument in the developmental process. The use of double-digit coding is another sign of interest in the process of problem solving instead of just judging whether an answer is incorrect or correct. A “correct” or “partly correct” score given not only to each work of the student, but also to which strategy was used or where in the process the students “lost track.”

A point of critique about the validity that remains to be discussed is the way of assessing, and more specifically the choice of item formats. Clearly tension ex-

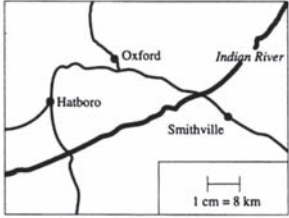
ists between the frameworks (what educators want to measure) and the choice of item format (what educators can afford in a practical and economical way).

Frameworks

The original *Curriculum Frameworks for Mathematics and Science* of TIMSS did not mention item formats (Robitaille et al., 1993). In practice TIMSS relied heavily on the multiple-choice format. That the alignment between the framework and test items leaves some room for improvement may be illustrated by the example shown in Figure 25.1.

EXAMPLE ITEM 3
FRACTIONS & NUMBER SENSE

Distance on map
One centimeter on the map represents 8 kilometers on the land.



About how far apart are Oxford and Smithville on the land?

A. 4 km
B. 16 km
C. 35 km
D. 50 km

Performance Category: Using Complex Procedures

Figure 25.1 TIMSS-item, 1996 (Beaton et al., 1996, p. 63).

This item is from the Population 2 cycle of 1996 (Beaton et al., 1996, p. 63). This rather straightforward problem, in multiple-choice format, is intended to show the use of “complex (mathematical) procedures.” If this is true it shows clearly that TIMSS is not criterion referenced but very normative and that the alignment between framework and test items leaves some room for improvement, especially given the ambitious performance expectations.

The more recent *TIMSS Assessment Frameworks and Specifications 2003* (Mullis et al., 2001b) made major improvements by including a “Question Types and Scoring Procedures” chapter. This made clear that the instrument heavily relies on the multiple-choice formats. The Framework left it to the reader to do the arithmetic to find out how many questions still are multiple-choice:

Two question formats will be used—multiple-choice and constructed response. Each multiple-choice question will be worth one point. Constructed-response questions generally will be worth one, two or three points. However, extended-problem solving and inquiry-items may be worth up to five points. Up to two-thirds of the total number of points will come from multiple choice items. (p. 88)

The wording in the PISA Frameworks is different, and this study seems to be more open to open questions. The PISA 2000 Framework (OECD, 1999) explicitly mentioned the somewhat problematic status of multiple-choice items:

Travers and Westbury (1989) state when discussing the second IEA mathematics study (SIMSS) that: “The construction and selection of multiple choice items was not difficult for the lower levels of cognitive behavior—computation and comprehension.” But they continue: “difficulties were presented at the higher levels.” There is place for the use of multiple-choice formats but only to a limited extent and only for the lowest goals and learning outcomes. For any higher-order goals and more complex processes, other test formats should be preferred, the simplest being open questions. (p. 54)

In practice, PISA 2000 ended up with around 50% of the items being multiple-choice. What is interesting in the 2000 Framework is appendix 2: “Considerations for future survey cycles of OECD/PISA.” In fact it is a plea for a balanced assessment instrument in which students can better show what they can do in a more constructive way: extended-response essay tasks, oral tasks, two-stage tasks, and production items.

For the PISA 2003 Framework, the test instrument consisted of “a combination of items with open-constructed response types, closed constructed-response types and multiple-choice types. About equal numbers of each of these types will be used in constructing the test instruments for OECD/PISA 2003.” (p. 50).

In practice the amount of multiple-choice items is higher than indicated in the Framework. None of the recommendations of the appendix of 2000 were realized.

The validity of the test instrument remains a complex issue. It goes without saying that there is an inherent tension between the traditional choice of item formats, usually with very restricted time (1–2 minutes per item), and the rather ambitious definitions of what the instrument is intended to measure. As Barton (1999) stated:

In the 1990s, there have been constructive attempts to improve the testing enterprise. Serious efforts have been made to broaden tests beyond multiple-choice questions, and to include open-ended questions, per-

formance assessments, and portfolios. However, the assessment reform movement has been slowed over issues of reliability and measurement error. (p. 4)

But not only the concern about “errors” plays an important role in relying so much on multiple-choice, it is also an economic issue: Many countries participating in these large cooperative studies are unwilling or unable to fund much more expensive multiple marker studies, even if such studies have demonstrated their efficacy.

PISA 2003 also had a problem-solving component. Many of the items (Figure 25.2) would fit the

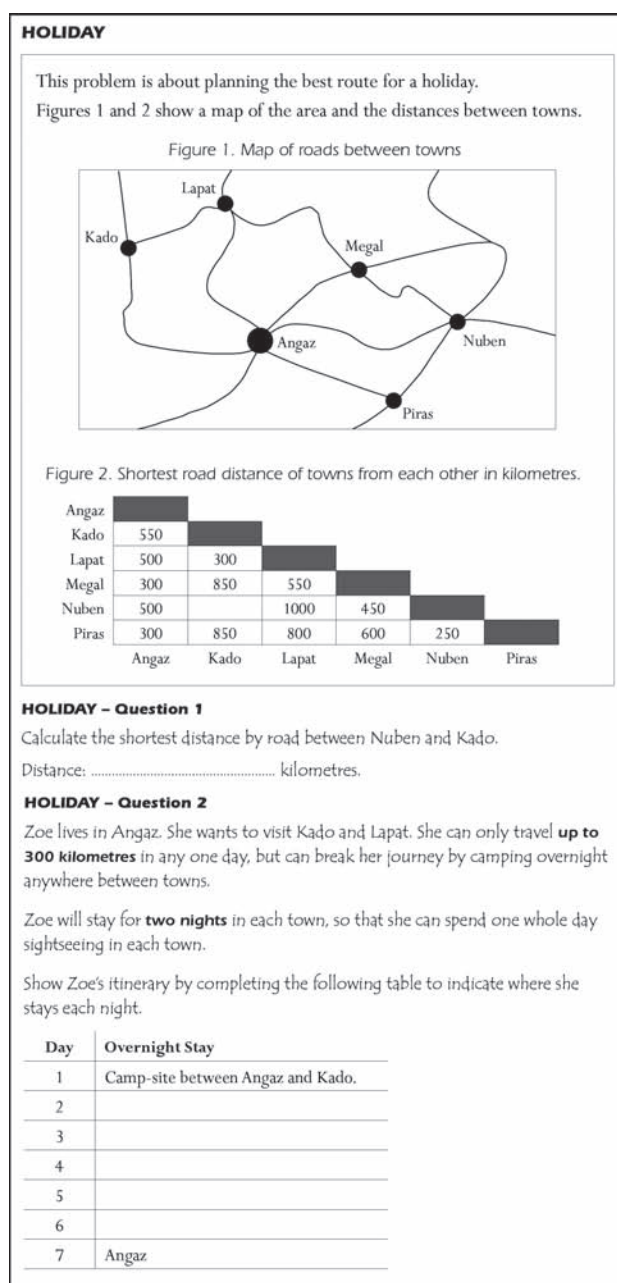


Figure 25.2 PISA-item 2003 (OECD, 2004, pp. 70–72).

mathematics Framework, and given the fact that the instrument for problem solving (PS) had much more open “constructive” items, a study relating the math items and the PS items could be very helpful in advancing the discussion on item instruments and their restrictions in large-scale international studies.

This example (see Figure 25.2) from PISA Problem Solving shows not only the relation to mathematics, but also that students are given the opportunity to engage in a more open problem, like question 2. According to the PISA report on problem solving

[T]he items for problem solving give a first glimpse of what students can do when asked to use their total accumulated knowledge and skills to solve problems in authentic situations that are not associated with a single part of the school curriculum. (p. 18)

One can easily argue that this is always the case in a curriculum: For mathematical literacy, mathematics as taught at school will not suffice. Students need to read, need to interpret tables and graphs (seen by many as belonging to reading literacy), and, indeed, need problem-solving strategies. But seen from the perspective of promising developments on item formats and item quality, the problem-solving component of PISA is interesting, at least. And if TIMSS implements their intent to “place more emphasis on questions and tasks that offer better insights into students’ analytical, problem-solving, and inquiry skills and capabilities,” innovation in large-scale assessments could materialize.

PISA versus TIMSS

The main differences between TIMSS and PISA seem to be the following:

- curricular emphasis for TIMSS versus functional aspect (literacy) for PISA;
- grade-specific structure of TIMSS versus age-specific structure of PISA.

TIMSS uses the curriculum as the major organizational aspect. The TIMSS curriculum model (see Figure 25.3) has three aspects: the intended curriculum, the implemented curriculum, and the attained curriculum. These represent, respectively, the mathematics and science intended for students to learn, and how the education system should be organized to facilitate this learning: what is actually taught in the classrooms, who teaches it, and how is it taught; and finally, what it is that students have learned, and what they think about those subjects.

International curricular diversity was a serious point of concern to the TIMSS study. The goal was

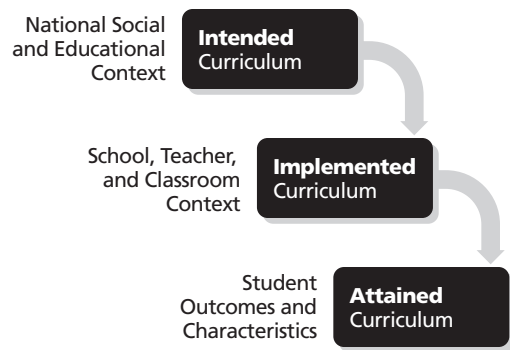


Figure 25.3 The TIMSS curriculum model (Mullis et al., 2001b, p. 3).

to develop an international test that would be equally fair to all participating countries. Therefore subject-matter specialists from all countries were consulted and asked to contribute to the process of test development (Garden & Orpwood, 1996). Most countries participating in TIMSS had an intended mathematics curriculum that matched with more than 90% of the items (Beaton et al., 1996; Kuiper, Bos, & Plomp, 1997). The outliers were the United States and Hungary with 100% matching, and the Netherlands, with 71% matching.

Insiders have discussed the procedure and its validity of this equally unfair analysis. The question not satisfactorily answered is how the mathematics education communities in the different countries were involved, and how representative they were. But if these numbers are accepted, in this context it is worth looking at the minimal matching result of the Netherlands.

As Vos (2002) observed, it was expected that students of other countries would outperform Dutch students. However, contrary to expectations, in 1995 Dutch Grade 8 students performed well on the TIMSS test. Their score was significantly above the international average, just below the four Asian top-scoring countries (Beaton et al., 1996). After some additional research by Beaton it was concluded that somehow the Dutch students were knowledgeable about the 29% of test items that were *remote* from their intended curriculum. In the end it was concluded that the students had the abilities for transfer of their knowledge and skills to items that did not match with their intended curriculum (Kuiper, Bos, & Plomp, 2000). As Heubert and Hauser (1999) explained it can be very appropriate to test students on material they have not been taught, if the test is used to find out whether the schools are doing their job.

PISA takes this point even further: It is based on a dynamic model of lifelong learning in which new knowledge and skills necessary for successful adaptation to a changing world are continuously acquired

throughout life. It focuses on young people's ability to use their knowledge and skills to meet real-life challenges, rather than on the extent to which they have mastered a specific school curriculum.

Given the observation of the TIMSS researcher about the transfer skills of Dutch students, combined with goals of PISA, it may come as no surprise that Dutch students outperformed all other countries in mathematics but did not appear in the official report because "the response rate was too low" (OECD, 2001, p. 236).

The two different approaches can both be critiqued: What does it mean that the Netherlands scored so high with the minimal relation with its curriculum? What does it mean if PISA will not constrain itself to any national curricula? It is clearly *not* true, as Clarke (2003) suggested as a proposition and as Freudenthal supported in 1975 already, that international studies of student achievement may be unintentionally measuring little more than the degree of alignment between the test instrument and the curriculum. What it *does* measure is still a question open to interpretation.

Another indication that shows how difficult it is to make statements that go beyond well-intended opinions can be found in the observation of Westbury in 1992, in relation to SIMSS, when he observed that the lower achievement of the United States is the result of curricula that are not as well matched to the SIMSS test as are the curricula of Japan. But in TIMSS the match was 100% (see earlier), and still the United States did not perform very well.

The Real World

Another concern that the math education community has is about the implicit imposition of a global mathematics curriculum. Keitel and Kilpatrick (1999) have problematized the assumptions on which international comparative studies are predicated. In particular, they questioned the treatment of the mathematics curriculum as unproblematic and the associated assumption that a single test can give comparable measures of curriculum effects across nations. They further suggested that the concept of an "idealized international curriculum" lies behind even the most sophisticated research designs. The problem with this judgment is that no one knows which part of the learning process as demonstrated in the tests can be attributed to schooling and the curriculum.

Without defining what a curriculum is, it is difficult to have a proper discussion. And the question can even be stated whether the whole question about analyzing curricula is a relevant one, given the outcomes of the Netherlands in TIMSS and the PISA conception of learning in society with schools being one

component of the learning process. This of course is a correct observation.

Keitel and Kilpatrick's criticism seems to focus on the underlying assumption of the international "common" curriculum. The TIMSS model seems to invite this kind of criticism, but PISA much less, if at all. However, the problem of creating such an international curriculum seems very much alive as PISA expands to about 70 countries in 2006. Many countries will look at the PISA items as something worth teaching, if alone to get better results next time. This is indeed a very serious problem as many politicians lack the knowledge and skills to realize that PISA only measures part of mathematics, more precisely: literacy. This has led to communication problems in several countries, and by many mathematicians, who tend to see PISA mathematics as representing mathematics as a discipline.

Another concern expressed in Clarke's (2003) article "International Comparative Research in Mathematics Education" is that differences in societies—like Ethiopia and the United States—automatically lead to differences in skills required for effective participation in these societies. This point seems very straightforward, but many of the competencies identified in the PISA Framework or item-collection intend to operationalize competencies that are of some relevance in any society. To state that "to assess the extent to which Ethiopian students possess the mathematical skills required for effective participation in American society seems both futile and uninformative" (p. 153) seems futile itself.

Another point that should be discussed, as it refers to an important aspect given the trend towards more "real-world" problem solving, is the link between mathematical literacy and social practice. Jablonka (2003) argued that even within a single school system there will be contexts familiar to some students and not others. Any attempt to use a single instrument to assess mathematical literacy beyond the most local of contexts would appear to be self-defeating. Again the chain of reasoning is not evidence based. The choice of context in relation to item construction and validation is a very complex one, as shown by many examples from previous experiments in designing curricular materials (e.g., Feijs, 2005). These experiments indicate that one cannot say anything firm about the relationship "context familiarity: success rate." As an example, due to the fear of experts that an item for an international study about the tides of an ocean should not be included because of the participation of land-locked countries, the item was rejected forcefully by the field trials. The influence of contexts should be studied much more systematically than is presently the case, and we researchers should

refrain from strong statements that have proven to be of disputable quality until we have firmer evidence. For the present studies all countries had to agree on the items, including the role of the context.

In a working paper (Nohara, 2001) from the (U.S.) National Center for Education Statistics in which TIMSS-R(epeat), PISA, and NAEP were compared, remarks were made about the inclusion of real-life situations, defined as items not presented strictly in the language of mathematics. According to the paper, this characteristic is significant because connecting mathematics to the world outside of school is a major goal of many mathematics education reform initiatives. It is also significant because it means that students have to choose for themselves the operations and solutions most appropriate for the problem and figure out how they relate to the information provided, thereby “adding to the difficulty of the item.” One may comment here that clearly such contexts require different cognitive demands from the students, but whether one can conclude that this leads to more difficult items is a matter to be studied in more detail, and especially PISA 2003 might offer some direction. The reason for that somewhat optimistic expectation can be found in the same working paper when the author stated that both TIMSS-R and PISA (and NAEP) contained many items situated in real-world contexts: 44% of items for TIMSS-R, 48% for NAEP, and 97% for PISA. The author also noted that in reviewing PISA items, panel members noted that several items in real-life situations presented students with significantly more challenges than others. These contexts were highly unique, that is, not typically encountered in mathematics instruction or textbooks, or required significantly more thought regarding how the nature of the context affects the mathematics involved in the problem. This type of item can be contrasted with the standard word problems typically used in mathematics classes, which can be described as “proxies for reality.” Only a few such items can be found in TIMSS-R or NAEP.

Meaning of Measures

Via the real or perceived influence of context I now come back to the relative meaning of outcomes of large-scale international assessments. The use of context is or could be one factor contributing to the relative difficulty of the assessments. Other factors contributing to the relative difficulty are (Nohara, 2001) the number of items that use the extended-response format, the number that require multistep reasoning, and the amount of computation as “the presence of a computation requirement does not present an ad-

ditional degree of difficulty as it would in an item classified in another content strand” (p. 14).

Looking only at these four factors PISA appears to be the most difficult: It has the highest percentages in all four categories.

If it is true that one can attribute item difficulty to these four factors and conclude that PISA is the most difficult large-scale international assessment, it might give some additional insight into what one can say about the *absolute* level in relation to the *relative* level. In addition to the TIMSS item presented before in this contribution it might be interesting to look at a PISA item that was considered “too difficult” to be included in the final item collection.

In my opinion, Figure 25.4 shows, as part of a wider pattern if one analyzes the items of large-scale assessments, that the “cognitive competency level” operationalized by these tests leaves lots of room for improvement. This implies that the “results” of such studies should be interpreted much more carefully than is usually the case. The horse race outcomes will be differently interpreted if policymakers understand that the metaphor itself is quite inappropriate if the horses do not have to meet a certain qualifying time over a fixed distance but merely are allowed to underperform.

Indeed, as Clarke (2003) summarized the feeling of most of the mathematics education community, it is “the current preoccupation with competition” that always seems to make it into the headlines of the media and always seems to support the present politicians and policymakers. The “Mathematics Olympics” under the banner of PISA 2000 made a media uproar in Germany, as did TIMSS in the United States in 1997.

What is missing quite often is a discussion by somewhat informed mathematics educators about what is being measured from a “content” view and what it means. The quoted working paper from NCES is a laudable attempt to shed some light on this aspect of international comparative studies. Keitel and Kilpatrick (1999) observed correctly the striking fact that almost all of the people with the primary responsibility for conducting the study (TIMSS) have been empirical researchers in education, psychometricians, or experts in data processing. The problem of the content of mathematics education, Keitel and Kilpatrick (1999) argued, has been dealt with as no more than a technical question:

The most monumental and most accurately treated feature of the studies has been the handling of data once they have been collected. Particularly notable has been the fashion in which problems of methodological validity, reliability and quality have been re-

Mathematics Unit 3

HEARTBEAT

For health reasons people should limit their efforts, for instance during sports, in order not to exceed a certain heartbeat frequency.

For years the relationship between a person's recommended maximum heart rate and the person's age was described by the following formula:

$$\text{Recommended maximum heart rate} = 220 - \text{age}$$

Recent research showed that this formula should be modified slightly. The new formula is as follows:

$$\text{Recommended maximum heart rate} = 208 - (0.7 \times \text{age})$$

Mathematics Example 3.1

A newspaper article stated: "A result of using the new formula instead of the old one is that the recommended maximum number of heartbeats per minute for young people decreases slightly and for old people it increases slightly."

From which age onwards does the recommended maximum heart rate increase as a result of the introduction of the new formula? Show your work.

The classification of the situation depends of course on whether or not people are actually interested in data about their own health and body. One can safely argue that this item is somewhat scientific (because of the use of formulas) but many sportsmen and women (joggers, bicyclists, rowers, walkers, etc.) really do measure their heartbeat quite regularly during their exercises. More and more inexpensive instruments using micro-technology have made this aspect of human well-being much more accessible to ordinary people. This explains the situation classification as "Public/Personal."

Because we are really dealing more with modelling and less with trivial problem solving, a classification to the *connections* cluster seems rather straightforward, as well as the overarching idea *change and relationship*.

Comparing two formulas, even though they are merely rules of thumb, that relate to a person's well-being can be an intriguing activity, especially as they are partly presented as "word" formulas.

This usually makes them more accessible to students. Even without asking a question, an initial reaction of students might be to see how their own age will lead to different recommended outcomes. As the PISA students are 15 years of age, the result under the old formula is 205 heartbeats per minute (realizing that the information that the rate is *per minute* is not given), and under the new formula it is 198 (or 197). Thus they might already have found an indication that the statement in the newspaper article seems to be correct.

The example posed is somewhat more complex than this. It requires the students to find out when (at which age) the two formulas give the same result. This can be done by trial and error (a well-established strategy by many students) but the more algebraic way seems more likely: $220 - a = 208 - (0.7 \times a)$, leading to an answer around 40.

From the viewpoint of mathematical literacy as well as from the viewpoint of more curricular-oriented mathematics, that is quite an interesting and relevant problem. We note that the field trial data indicate that 15-year-old students found this problem quite difficult.

Mathematics Example 3.2

The formula *recommended maximum heart rate* = $208 - (0.7 \times \text{age})$ is also used to determine when physical training is most effective. Research has shown that physical training is most effective when the heartbeat is at 80% of the recommended maximum heart rate.

Write down a formula for calculating the heart rate for most effective physical training, expressed in terms of age.

This example *seems* to measure exactly the same competencies as Example 3.1. The correct response rate is almost identical (during field trial). But there is a notable difference: In Example 3.1 students have to compare two formulas and to decide when they give the same result. In Example 3.2 the students are asked to "construct" a formula, something they are not frequently asked to do during their school career, in many countries. From a strictly mathematical viewpoint the question is not difficult at all: just multiply the formula by 0.8—for instance, $\text{Heart Rate} = (208 - 0.7 \times \text{age}) \times 0.8$. It would seem that even such simple manipulation of algebraic expressions, expressed in a practical and realistic context, presents a substantial challenge to many 15-year-olds.

Figure 25.4 PISA item 2003 (OECD, 2003, pp. 64–66).

solved from a purely formal point of view. Questions of content—in all its aspects—have usually been seen as secondary. (p. 245)

I agree with this point and also note that in relation with this point several countries carried out national options, in part because content-related matters gave concern about the validity of the assessment as appropriate for the specific country. An impressive effort in relation to PISA 2000 was carried out by Germany. Although the report (Baumert et al., 2001) was in general positive about PISA 2000, the Germans opted for a National Option because of the

fact that there was a need (in Germany) for more "inner-mathematical problems," for more "basic-skills problems" and "facts." This was done to make the PISA test "closer" to the German curriculum.

A similar action, but then in the opposite direction, was taken by the Netherlands in reaction to TIMSS. As I reported in 1997, the Netherlands (here meaning some influential math educators) concluded that TIMSS did not fit in a satisfying way to the intended/perceived/implemented curriculum, not on the basis of the earlier mentioned relatively low "alignment-quotient," but because of too many multiple-choice items operationalizing too many lower

level thinking skills (de Lange, 1997). Therefore a National Dutch Option was carried out. The international test and the national option had an intersection of carefully selected “anchor items.” Methodologically speaking, the results on the anchor items within the national option or in the international option should be the same. In reality the students did much better on the TIMSS items when they were part of the national option. This outcome supports the arguments to first think what is really being measured and with which instruments before making policy statements regarding the horse race. In fact, the TIMSS score in reality was 541 in Grade 8. But based on the national option as a more reliable measurement, the same TIMSS items would lead to a score of 585. The only

explanation brought up by the national experts, including the IEA group, was that the “national” context made children feel better and more motivated. Or, to put it slightly differently, perhaps the fact that the anchor items were culturally correctly embedded made students perform better.

The content is a point of concern that receives relatively little attention, contrary to the question of the influence of the different cultures. It goes without saying that we measure “cultures.” Bracey’s (1997) observation that “international achievement differences become similarities if we compare the performance of students of Asian cultural affiliation, no matter what school system they attend” is true to a certain extent. One does not have to be a rocket

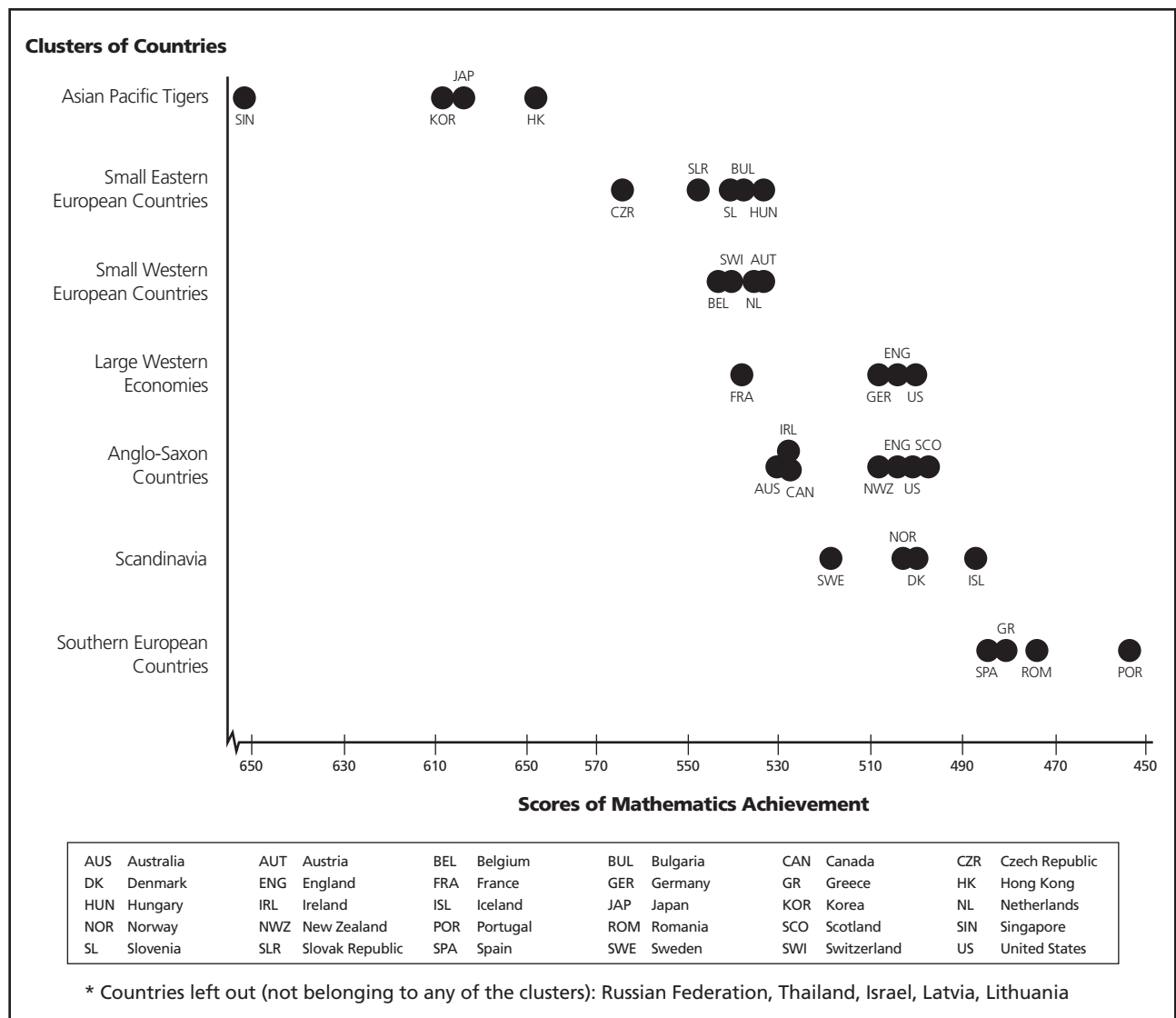


Figure 25.5 Graph showing countries participating in TIMSS 1996 (de Lange, 1997).

scientist, let alone a math education researcher, to see a certain socio-economic-cultural structure in the listing of the horses of TIMSS.

In my opinion one can see a certain clustering of countries, starting with the Pacific Rim countries (referred to as “Asian Pacific Tiger” countries in Figure 25.5), via small European countries, to France, to (former) commonwealth countries, to Germany and the United States, to Scandinavia, and finally Southern Europe (de Lange, 1997). The mathematics education community has the responsibility not only to contribute to such studies (and to influence them in a positive way), but also to critique them when necessary, to draw attention to the content part of such studies, and to inform policymakers what the study might mean beyond the horse race.

Missing Content

In a review of the PISA 2000 report Bonnet (2002) underscored the lack of interest—in content matters—in trying to understand in pedagogical terms how and why pupils gave the responses, or absence of responses, that they actually gave. Instead, he observed, the focus of the report is on trying to analyze results in the light of socioeconomic data and the organization of education systems and schools. One would have wished for more space to be devoted to trying to understand what happens in the classroom in terms of the teaching and learning process. This exactly is what the title of the earlier mentioned IEA/TIMSS promises but fails to materialize. Both studies, TIMSS and PISA, should and could do much better in this respect. It is difficult to point to the real reason behind this phenomenon. Keitel and Kilpatrick (1999) suggested looking at who conducts the study in order to better understand the results.

Taking into account that PISA is organized by the OECD, Bonnet (2002) observed that it is illuminating that chapter 3 of the PISA 2000 report ends at page 91 with tables showing the relationships between the average country performance in the three combined scales (reading, mathematics, and science) and the GDP (Gross Domestic Product) per capita. The true worth of such correlations becomes apparent with the inspiring conclusions that “countries with higher income per capita tend to perform better, on average, but some countries do better or worse than their income would predict” (p. 92). In my opinion, Bonnet was right in concluding that the bias in the report in favor of assessing education from the point of view of economic efficiency is overshadowing another conception of education, prevailing in Europe, whereby education is just as much about social cohesion, personal fulfillment, and cultural development. It is

very frustrating for those who care about children if a minister of education proudly states that the Netherlands has a very efficient educational system, because we outperform most other countries, while spending well below the OECD norm. In this way international comparative studies can be very destructive to the improvement of the teaching and learning process.

Impact of Studies

Consider TIMSS. It is very tempting to use as a source the publication by Robitaille, Beaton, and Plomp for IEA/TIMSS with the title: *The Impact of TIMSS on the Teaching and Learning of Mathematics and Science* (2000). The title of the book is a bit misleading as it deals only in a minor way with the “impact”; the book reports also on national education systems, and the results and findings per country. Although the title of the book stresses—as it should do—the “learning and teaching,” this aspect is difficult to find in the book. This is also to some extent the conclusion of the authors as they state:

the results seem, at least until now, to have made a greater impact on the general public and in political circles through the mass media than they have on educators. On the other hand, it is also true that many of the reports refer to the impact of the study on the process of curriculum reform or on national evaluation programs. Those kind of processes tend to take a long time to bring to fruition, and they are then followed by an equally lengthy and perilous process of implementation. So it is probably unrealistic to expect that the results would have made more of an impact at the classroom level since only three years have elapsed since the publication of the first set of international results. (p. 168)

Without trying to be exhaustive a rather typical example of a specific country’s reaction under the chapter “Impact” will resemble something like the following (based on Switzerland):

TIMSS showed primarily where Switzerland stands in comparison to other countries with respect to achievement in mathematics . . . [and] gave a detailed picture of current levels and patterns of achievement. This is a valuable basis for discussion of needed improvements. TIMSS also gives some indications of possible reasons for such differences in achievement. . . . However, given the way TIMSS was designed, such statements of causality are only valid to a limited extent. It is not easy to deduce directly from TIMSS how improvements might be made. While TIMSS had a direct effect on debate and reflections concerned with educational policy, educational planning, pedagogy, educational science, and teacher preparation, it had only an indirect effect on corresponding practice. (p. 158)

One is tempted to comment that these observations cannot come as a big surprise. Having some experience and involvement with both TIMSS and PISA (see note at end), I find it striking to note the relatively little interest that IEA and OECD have in both content and teachers and students, the key factors in math education. As the studies are usually announced and described as an educational system analysis, and as relations are made between the performance of educational systems and the economy, it is hard to expect much interest from the “real” stakeholders in mathematics education. The introductory remarks for the “impact study” from the executive director of IEA, Hans Wagemaker, make clear where the emphasis lies:

International comparative studies of educational performance . . . can therefore play a central role in assisting policy-makers, curriculum experts, and researchers in judging the extent to which an education system meets national expectations and the extent to which it is likely to contribute to the nation’s overall economic and social well-being. (p. 7)

I now turn my attention to examples of the possible impact of the PISA studies. They are both not about system changes, but about content-oriented matters, taking teachers, children, and content as key variables. In the examples I will discuss two completely different ways the outcomes of PISA 2000 have been used—one of them being by Germany. This cannot come as a surprise after the introduction of this article focusing on the PISA shock; but Germany is also a good starting point for quite another point that I will only mention here, but that is worth further research and analysis. The Germans produced a national PISA report of 550 pages, the international OECD report was 330 pages, and the Dutch report a mere 65 pages. Most countries had something around 150 pages. It is not the statistics that are interesting here, but the message from the report and what has been selected to be included. Even a superficial analysis, which was carried out for this article with the reports mentioned and the one from the United States, makes significant differences visible. As Keitel and Kilpatrick (1999) pointed out very correctly, there is a common myth that “numbers do not lie.” It is now widely accepted that data can be gathered, processed, mathematized, and interpreted in a variety of ways. So a key issue is the question of who influences this process, for what reasons, and through what means. The four studies just mentioned underscore this concern apart from the fact that even numbers can lie.

Back to the very *gründliches* German report. Not only did the German PISA *Konsortium* do an excellent and thoughtful job, it also made recommendations

for immediate improvement, including ones that directly affect the content. According to Blum (2000), the changes should include

- more integration of inner- and outer-mathematical “networks”;
- fewer calculations;
- more thinking activities and student mental “constructions”;
- more reflection;
- more flexible use of schoolbooks.

These goals can be reached when the recommendations that were formulated after TIMSS are implemented:

- development of a different math-problems culture: more open-ended, more “real-world”;
- a new teaching-and-learning culture, with a more exiting cognitive school environment (Klieme, Schümer, & Knoll, 2001);
- more and different professionalization of teachers, emphasizing teamwork.

PISA adds to these recommendations a “very different conceptualization” of mathematical concepts and emphasis of modeling and mathematization, situated in contexts. And, argued the report, the Germans have definitely not reached the optimum in using different representations as a tool to build better conceptual understanding.

Mathematics education is in a state of transition, in part because of the fact that both TIMSS and PISA were taken seriously. Surprisingly the shock and catastrophe that struck Germany as some kind of natural disaster, if one had only the popular media as a resource, has resulted in a government-supported nationwide action-plan with a very strong content part that will result in a different mathematics education culture at schools. Of course, the success of these changes will be measured by PISA 2003, 2006, 2009, and so on. At least in part.

A very different way of using the “very different conceptualization” of PISA was reported by Dekker and Feijs (2005). Classroom teachers (in Philadelphia and Milwaukee, WI) were invited to reflect on their own classroom assessment, initially in their own way using their principles. Teachers were questioned why they asked the questions they asked, why they ordered the questions as they did, what were the presumed learning goals that the items tried to operationalize, and so on.

After getting familiar with the different competency clusters (computation, connections, reflections) and discussing PISA items and other “good” items that

were more of an open kind and needed more student construction, the teachers were again confronted with their own classroom assessments. This Classroom Assessment as a Tool for Teacher Change (CATCH) project was considered powerful because it not only changed the teacher's attitudes towards the teaching-learning-assessing process, but also changed their actual classroom practices—not just their classroom assessments. It seems that using an international comparative study by focusing on the concept of the study from a mathematical point of view, in combination with the available instruments, and by inviting teachers to make this instrument better and more fit for the classroom practice should be strongly encouraged as it can be a powerful tool from the bottom-up in combination with top-down approaches.

In the CATCH project a competency model was used that was quite similar to that of the PISA Framework (2003). The three different competency clusters were visualized in the Assessment Pyramid (de Lange, 1994, 1999) as competency levels. This pyramid turned out to be very effective in changing teachers' attitudes and practices in the classroom. Some of this effect can be attributed to the fact that PISA had adapted a similar model, which is seen as adding validity to the Pyramid.

As indicated before, I would like to close this “international comparative studies” part with a short discussion about other large-scale assessments in mathematics education other than the TIMSS and PISA studies that I have discussed in some detail.

Numeracy, Quantitative, and Mathematical Literacy Assessments

Studies like NAEP and TIMSS are connected to some kind of “idealized” curriculum. PISA is not, as it claims to test mathematical literacy, a term described as

an individual's capacity to identify and understand the role that mathematics plays in the world, to make well-founded mathematical judgments and to engage in mathematics in ways that meet the needs of that individual's current and future life as a constructive, concerned and reflective citizen (OECD, 2001, p. 22).

If the TIMSS publication on the impact of the study expressed some concern about the relative lack of interest in the results, this lack of interest is even more evident in studies about numeracy, quantitative literacy, and mathematical literacy, especially if these are about adults and not, like PISA, for 15-year-olds.

An important event in this respect was the Young Adults Literacy Skills (YALS) project in the United

States, in which young adults (12–15) were assessed on tasks related to three literacy scales, among them quantitative literacy. The description of quantitative literacy was somewhat limited compared to the more recent PISA definition: “The knowledge and skills required to apply arithmetic operations, either alone or sequentially, using numbers embedded in printed material (e.g. balancing a checkbook, completing an order form)” (Dossey, 1997, p. 46; Gal et al., 1999, p.12). The results of this assessment led to conclusions like “performances on the scale were less than what the nation required in human talent for long-term international competitiveness” (Dossey, 1997, p. 48).

The follow-up of YALS was the National Adult Literacy Survey (NALS), carried out in 1992. Gal (1993) concluded that roughly 50% of American adults would either have major difficulty with or be fully unable to handle real world tasks such as

- using a bus schedule to determine departure time;
- identifying a trend on a simple graph about sales figures;
- using a calculator to find the difference between the regular price and the sale price;
- estimating the unit price of a grocery item;
- understanding a table summarizing a school survey;
- calculating interest charges related to a home loan.

In the technical report *Issues and Challenges in Adult Numeracy* (1993) Gal observed that there are two “extreme” views on numeracy. One extreme is to see numeracy as the lower end of mathematics, or whatever math educators attempt to achieve in the early grades (Van Groenestijn, 2002) whereas the other end is viewed as “encompassing a broad set of skills, knowledge, strategies, beliefs and dispositions that people need to autonomously engage in and effectively manage situations involving numbers, quantitative or quantifiable data, or information based on quantitative data” (Gal, 1993, p. 2).

More recently the definition has broadened even more, as in PISA, to reflect that literacy in the area of mathematics goes well beyond the strictly numerical (numeracy) or quantitative aspects (quantitative literacy). Confronted with this problem, I proposed in Madison and Steen's publication on quantitative literacy (2003) to see numeracy as part of quantitative literacy. This includes understanding of and mathematical abilities concerned with certainties (quantities), uncertainties (quantity as well as uncertainty), and relations (types of, recognition of, changes in,

and reasons for those changes) and defines spatial literacy, most likely the most basic and neglected aspect of mathematical literacy, as understanding of the (three dimensional) world in which we live and move (de Lange, 2003). Shaeffer (2003) supported this view because of the indisputable fact that much more in mathematics is useful besides numbers. Indeed, many aspects of statistical thinking (uncertainty) are not about numbers as much as about concepts and habits of mind, he argued.

This discussion is relevant to the problems of large-scale assessments as it is not clear at all what role school curricula play in setting up large-scale literacy assessments, although clearly some are very much tuned to traditional curricula and others are more society oriented. An example is the International Numeracy Survey (INS) in which only a limited number of countries took place: U.K., France, the Netherlands, Sweden, Denmark, Japan, and Australia, organized by The Basic Skills Agency in London in 1997. Two typical examples (out of only 12 problems) are:

The total of 4.25, 6 and 7.74 =
Work out 15% of 700.

Van Groenestijn (2002, p. 34) observed that “though in this study ‘numeracy’ was seen as doing mathematical operations on school-based tasks, it does indicate that the performance on such school-related tasks is disappointing and presumably undermines numerate behavior.” This observation seems somewhat questionable as it directly relates literacy to school mathematics in a restricted basic skills sense and fails to clarify how such skills “undermines” numerate behavior. I cannot but agree on the observation that the performance on the task is disappointing. What can be learned from that is quite another matter.

The NALS was followed by an international study, set up by the OECD: the International Adults Literacy Survey (IALS). This survey was carried out in 1997 in OECD countries. Apart from the United States and Canada 21 countries including Germany, Sweden, Switzerland, and the Netherlands took part. This study defined quantitative literacy quite narrowly, trying to connect traditional arithmetic with the problems of daily life: “To apply arithmetic operations to numbers embedded in printed materials, such as balancing a check-book, figuring out a tip, completing an order form, or determining the amount of interest on a loan from an advertisement” (NRC, 1989, p. 7–8).

Finally I mention the follow-up survey of IALS, the Adult Literacy and Life Skills Survey (ALL) organized by the National Center for Education Statistics (NCES) and Statistics Canada, which uses a more elaborated

numeracy domain. An interesting “Policy Information Report” about this study was published in 2002 (Sum, Kirsch, & Taggart, 2002) for the United States. The report provided a good view of the role of international large-scale assessments as the goal of the report is to call attention to important findings from the IALS and *their implications in terms of educational and workforce development policies*. The report also stated that it is interesting that the results of IALS are becoming available at a time when the role of human capital in influencing the fate of individuals and nations is receiving increased attention. According to a recent report released by the OECD (2001), research shows that the development of human capital is correlated with better health, lower crime, political and community participation, and social cohesion. Some studies, they reported, even *suggest* that the social impacts of acquiring these knowledge and skills could be as large as their impacts on economic productivity. Elsewhere in this article some doubt is cast on this statement, or more precisely on the relation between achievement in education and economic performance in a causal way.

One of the more interesting results is the fact that 46% of the respondents in the United States belonged to Level 1 and 2, “possessing a very-limited to limited mathematical literacy proficiency.” Also noteworthy is the fact that there is no “horse race” ranking in the report: The United States is compared only with all high-income countries, but only as a group.

From the summary I quote the following remarks, clearly identifying the purpose of the report as the title is *Summary of Findings and Policy Recommendations*.

- The U.S. educational system is clearly less productive in raising literacy skills of students per dollar spent.
- The U.S. appears to be living off its past as higher educational investments will inevitably lose ground in the coming decade.
- The nation’s changing demographics will likely exacerbate the literacy skill deficit in coming years.
- Although skills influence labor market success everywhere in the world, this is especially true in the U.S. and other English-speaking countries.
- Mediocre skills and inequality in the distribution of skills may have worked reasonably well in recent years but are not likely to do so in the future.

Given these results it may hardly come as a surprise that mathematics educators pay relatively little attention to these studies, as the report gave no indication

whatsoever about the instruments used and the reasons behind them. It is my opinion that more balanced reports, not only this one, showing the links between policy implications and the assessment instruments are in the interest of everyone.

In 2000 the OECD carried out the earlier discussed PISA study, also with a much wider ranging definition of mathematical literacy. It seems not very reasonable that the difference in definition of literacy can be attributed to the difference in target population: adults or 15-year-olds. Hopefully this change—by OECD—can contribute to newer insights into the importance of mathematics in society and the relevance of seeing mathematics function in the real world.

NATIONAL LARGE-SCALE ASSESSMENTS

Introduction

Two major clusters comprise large-scale assessments on a national level: a discussion of NAEP and the TIMSS Benchmarking Study for the United States at one hand and at the other side the national exit examinations as they exist in many countries around the globe. It goes without saying that, like this whole chapter, I can touch on the subjects discussed and then only in a quite selective way.

NAEP

In 1969 the Carnegie Corporation of New York appointed the Exploratory Committee on Assessing the Progress of Education (ECAPE) with Ralph W. Tyler as chair. According to Tyler (1969) a major purpose of the national assessment program was to provide the lay public with census-like data on the educational achievements of children, youth, and adults—data that furnish dependable background information about educational attainments, the progress being made, and the problems still faced in achieving educational aspirations.

This was the first description for the program that is presently known as the National Assessment of Educational Progress. NAEP is a national survey intended to provide policymakers and the public (definitely lay people) with information about the academic achievement of students across the nation. It serves as a source of information for policymakers, school administrators, and the public for evaluating the quality of their educational programs. Just like TIMSS, NAEP is not tied to a specific curriculum. Whereas TIMSS tried to find a set of instruments that best fit the exist-

ing curricula of all participating countries, NAEP is based on a set of frameworks that describe the knowledge and skills to be assessed in each subject area. The performances assessed are intended to represent the leading edge of what all students are learning.

For over 30 years, the NAEP has served as the United States' only ongoing gauge of student achievement across time. One has to understand that just like with the other large-scale assessments discussed before, the studies do not report or even intend to report on student's individual achievements. This is caused by the multiple-matrix sampling method, the design selected by the founders of NAEP. Not all students will be sampled. On the contrary, a strategically selected sample can support the targeted inferences about groups of students with virtually the same precision as the familiar approach of testing every student. Moreover, like in PISA and TIMSS, not all students are administered all items. It can hardly come as a surprise that one of the current debates is about this aspect of large-scale assessments of this kind: the impossibility to provide student-level reports, something that exit examinations do provide, sometimes in great detail. In NAEP, measurement at the level of individual students is poor, and individuals cannot be ranked, compared, or diagnosed (NRC, 2001, p. 224). Worse even, students who have similar performances on one particular item can have different competencies—something known not only at the student level, but also at country level in TIMSS or PISA.

Silver, Alacaci, and Stylianou (2000) have demonstrated some limitations of scoring methods used by NAEP for capturing the complexities of learning. For a particular item NAEP concluded that 11% of the students gave satisfactory or better responses but can give no further detail about the important differences in the quality of the reasoning demonstrated. In this particular case the reasoning varied from surface-level reasoning to analytic reasoning and to even more sophisticated reasoning.

Dossey, Jones, and Martin (2002) confirmed that indeed a "correct" answer is only part of the story. They did so in the context of the TIMSS study, which used the double-digit coding to find out more information about the strategies used by students in an effective way. In a case that involves three distinct different ways of reasoning one can code a correct answer with 1.1, 1.2, or 1.3, the 1 standing for "correct," the 1, 2, and 3 for the strategy used. In this way important information about the process or strategy of the student can be made visible to a certain extent. But the data offer more for those interested in the actual solution process, in addition to double-digit coding. This can be interesting at the national level, but also at the international level: In

one country most students could be using one—often computational—technique, whereas in another country students use a variety of strategies.

A national analysis of the actual process the students have gone through is done by studying and investigating actual students' work, in combination with double-digit coding and PISA scores—for instance in the Netherlands. Indeed, it seems that stakeholders are increasingly interested in better—which often means more complex—assessments, even on a large scale and even if this means higher costs and less reliability because of intersubjective scoring. I will come back to this issue later, when discussing future developments.

Over the past thirty years (since 1973) three distinct NAEP projects have evolved: the Main NAEP, the Long-Term Trend NAEP, and the State NAEP. The Main NAEP periodically assesses students' achievement in reading, mathematics, science, writing, U.S. history, civics, geography, the arts, and other subjects at Grades 4, 8, and 12.

Trial assessments with voluntary state-by-state assessments were conducted in 1990 and 1992 and were considered so successful that the U.S. Congress authorized regular state assessments starting in 1996 (Allen, Jenkins, Kulick, & Zelenak, 1997).

The content of both the Main and State NAEP programs follows curriculum frameworks developed by the National Assessment Governing Board (College Board, 1996). Kane and Webb (2004) argued that test-item types for the Main and State NAEP assessments are consistent with the current state-of-the-art in achievement testing and have evolved over time with curriculum changes. In contrast, both the students sampling frame and the content of the Long-Term NAEP have remained essentially unchanged.

This Long-Term NAEP, or the Long-Term Trend Mathematics Assessments (LTT Mathematics), has provided a measure of students' achievement over time by administering similar assessments periodically since 1973 (Galindo, Caulfield, Mohr, & McCormick, 2004). It should be noted that the LTT has a computational focus, measuring students' knowledge of basic facts, their ability to carry out paper-and-pencil calculations, their knowledge of basic measurements formulas, and their ability to apply mathematics to daily-life situations such as those dealing with time and money. This focus reflects the school mathematics in the United States in the 1970s. The content areas of the LTT are represented in the Main NAEP. The three types of items used on Main NAEP—multiple-choice, short constructed response, and extended constructed response—are reduced to only multiple-choice and short constructed response in the LTT.

Not only in content and item-type format is the LTT different from the Main NAEP, but also the reporting scales. The Main NAEP is more comparable in many respects with the international comparative studies like TIMSS and PISA. The Main NAEP has three achievement levels: Basic, Proficient, and Advanced. The levels are based on collective judgments of experts about what students are expected to know and do at each grade level (NCES, 2001). The performance is based on a 0 to 500 scale, just like with LTT (and PISA and TIMSS), but in the Trend Study the students' performance is reported in terms of percentages of students, which corresponds to five points on the scale: 150, 200, 250, 300, and 350, with level 150 described as "Simple Arithmetic Facts" to Level 350 as "Multi-step Problem Solving and Algebra." The five LTT performance levels were arbitrarily set at 50-point increments.

Although the results of the LTT study indicate an almost continuous upward—though very slow—trend, the gains document the fact that students' ability to complete traditional mathematics tested by means of a multiple-choice or short-answer format questions has improved across the 30 years, and that the gains were much smaller than the gains on Main NAEP during the same period (Galindo et al., 2004). The "deeper" meaning of this phenomenon is somewhat unclear, as it is unknown how much influence familiarity with the test instrument has on the outcomes. It is a fact that the longer a test exists the more familiar students and other stakeholders become with the nature of the test items. Experts can already recognize a NAEP, TIMSS, or PISA item from a distance, although the items seem to converge, which may be an undesirable development. This familiarity might itself be responsible for the observed almost continuous, but slow, upward trend. Policymakers even go so far as to speak of a TIMSS curriculum, indicating that they think one can identify a framework behind these studies that is reflected in the items and item formats. A familiar request that reaches persons or organizations responsible for these studies is "Please help me to prepare my students for the next study."

Linking the LTT NAEP with the Main NAEP is no easy task because of different content, different scaling, and different item formats. Also linking State NAEP results to those administered by states or local schools is difficult, if not impossible, to achieve (Feuer, Holland, Green, Bertenthal, & Hemphill, 1999); Linn (2000) observed that comparability of state assessments with NAEP scores is sometimes compromised when purposes differ. Add to this the evidence suggested by Kenney and Silver (1997) and Linn, Koretz, Baker, and Burstein (1991) that variation in item for-

mat may result in different estimates of student knowledge and skills across content topics and content dimensions. This point was also made in an explorative study in assessing mathematics problem solving for 16–17-year-olds (de Lange, 1987).

Linking NAEP with TIMSS and PISA is also difficult: The three assessments are targeted towards slightly different student populations, place different emphases on content areas within mathematics, include questions requiring different types of responses and thinking skills, and report results in different ways.

From the content point of view the most commonly addressed NAEP mathematics Content Strand on both NAEP and TIMSS was number sense, properties, and operations, addressed by 32% of NAEP items and 46% of TIMSS-R items, compared to only 9% of PISA items (Nohara, 2001). PISA did pay a lot of attention to Data Analysis, Statistics, and Probability: 31% of the items belonged to this cluster, with 14% in NAEP and 11% in TIMSS. (Note that these data are for PISA 2000; PISA 2003 has a more balanced distribution matrix.) Probably more important than attribution of items to content strands is the fact that one fourth of the items of Data Analysis, Statistics, and Probability were related to the subcategory “read, interpret, and make decisions using tables and graphs,” which may come as no surprise given the fact that PISA aims at measuring mathematical literacy. The more curriculum oriented studies, NAEP and TIMSS, have far fewer items in this subcategory: 4% and 7%, respectively.

subcategories within each of the content strands. This becomes also clear in the percentage of items that present students with real-life situations or scenarios as settings for the problem: NAEP scores 48%, TIMSS 44%, and PISA again is very different: 97%. One should keep in mind that these numbers give some indication of the emphasis placed on real-world situations but do not tell anything about the artificiality, relevance, role, mode of representation, familiarity, and so on, of the context.

A weak spot in any existing large-scale assessment is addressed by many authors and summarized very well in *Knowing What Students Know*. The authors commented on NAEP by noting that NAEP is beginning to be influenced by the call for more cognitively informed assessments of educational programs:

Recent evaluations of NAEP (National Academy of Education, 1997; NRC, 1999) emphasize that the current survey does not adequately capitalize on advances in our understanding of how people learn particular subject matter. These study committees have strongly recommended that NAEP incorporate a broader conceptualization of school achievement to include aspects of learning that are not well specified in the existing NAEP frameworks or well measured by current survey methods. The National Academy of Education panel recommended that particular attention be given to such aspects of student cognition as problem representation, the use of strategies and self-regulatory skills, and the formulation of explanations and interpretations, contending that consideration of these aspects of students achievement is necessary for the NAEP to provide a complete and accurate assessment of achievement in a subject area. (pp. 244–245)

The NRC agreed with this position and added that NAEP should include carefully designed, targeted assessments administered to smaller samples of students that could provide in-depth descriptive information about more complex problem solving over longer periods of time. For instance, smaller data collections could involve observations of solving mathematical problems in groups.

It should be noted that the recommendations suggested in the above publications are not new: Both TIMSS and PISA designers realized this problem and have expressed intentions to address the issues of more complex processes. But the realization of these intentions is no simple task: We have already noted the intentions for more complex assessments that were mentioned in the PISA 2000 framework but have not been realized so far. One can easily guess the reasons: The required “careful design” is extremely difficult, expensive, and not easy to score even if just settling for qualitative descriptions. Time is another big issue.

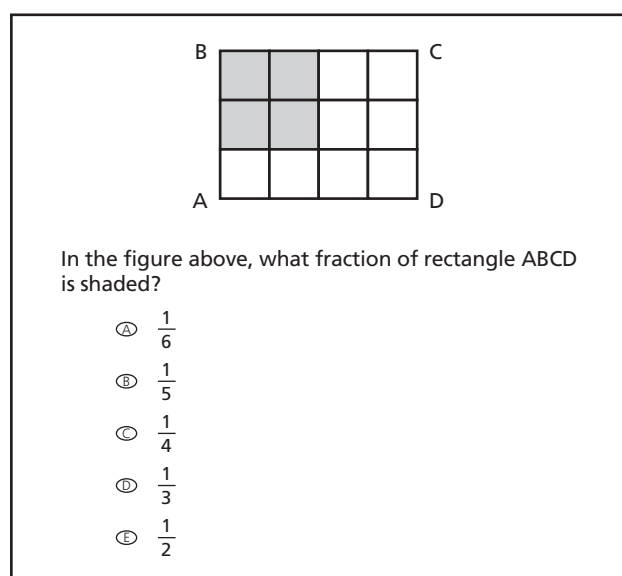


Figure 25.6 Example item from NAEP (NAEP, 2004, p. 29).

Given these observations it may come as no surprise that NAEP and TIMSS address similar sets of

As I will discuss later there are already rather large-scale assessments in existence, even on problem solving in group work, but they take at least one full day to complete. On the other hand, there are also examples of national large-scale (exit) examinations that consist of more than one paper-and-pencil session of 50 minutes or so and do address more complex processes. To discuss these national tests will be the next part of this chapter. But before going to this interesting area I would like to mention an interesting study that tries to connect an international comparative study (TIMSS) with state or local assessments: *The Mathematics Benchmarking Report TIMSS 1999* (Mullis et al., 2001a).

TIMSS 1999, also known as TIMSS-R, focused on mathematics achievement of eighth-grade students. The United States and 37 other countries participated, and important for the United States, 13 states and 14 districts or consortia participated in a voluntary Benchmarking Study. The study provided those states and consortia an opportunity to assess the comparative international standing of their students' achievement and to evaluate their mathematics programs in an international context. And, of course, compare themselves with the score of the United States as a whole.

The Benchmarking Study confirms the extreme importance of looking beyond the averages (horse race) to the range of performance found across the nation. It may come as no surprise that performance across the participating school districts and consortia reflected nearly the full range of achievement internationally: Although achievement was not as high as Singapore, Korea, and Chinese Taipei, the top-performing Benchmarking jurisdictions of Naperville School District and the First in the World Consortium (both Illinois) performed similarly to Hong Kong, Japan, Belgium (Flemish), and the Netherlands. The problem is, as mentioned before, that the international context consists of four very different socioeconomic cultural entities. And one wonders what to do with the finding that the Chicago public schools, the Rochester school district, and the Miami-Dade county public schools are at par with Thailand, Macedonia, and Iran.

This item from the TIMSS Benchmark study (Figure 25.7) illustrates an item from the content area Fractions and Number sense, 8th grade mathematics. Singapore scored 89% correct, the American First in the World Consortium scored 71%, England and the Russian Federation 52%, Maryland 42%, and the Dade County Public Schools scored 20% correct.

Another positive development regarding TIMSS is the inclusion of videotaping in classrooms and Case Studies. In doing this it follows the recommendations of the authors of *Knowing What Students Know* in an unprecedented way. In his overview of the Case

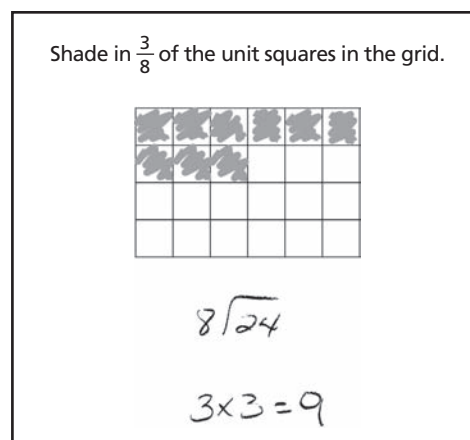


Figure 25.7 TIMSS Benchmark Study-item (Mullis et al., 2001a, p. 72).

Studies, Stevenson (1998) argued that the questionnaires, as used in the main studies of TIMSS and PISA, have both strengths and weaknesses. Their major strength—as with multiple choice—is to cheaply gather large amounts of data on a wide variety of topics from many individuals. On the other hand, and this causes serious problems, respondents may provide answers that are difficult to interpret. Deprived of the opportunity to question the respondents about the meaning of certain replies, to probe for more complete answers, or to ascertain the respondents' understanding of the questions, the investigator can only conjecture about what an answer might have meant. And the opposite is a problem too: What is the real meaning of the question for the respondent?

Case studies, which allow for one-to-one interactions between investigator and respondent, can help to understand the contexts and relationships that lie behind more traditional quantitative surveys of beliefs, attitudes, and practices, especially when seen in combination with the more quantitative data.

The five-volume report from TIMSS on Japan, Germany, and the United States offers examples from standards to homework, from teacher training to response to individual differences. From these examples Stevenson (1998) concluded that the education systems of the different countries are embedded within the culture of each country. As a result of this close relationship, one cannot hope to learn how education systems can be improved or academic achievement increased without understanding the actions, beliefs, and attitudes related to education in that culture. However it is impossible to make causal conclusions about the relation between scores on a test and the culture.

National Exit Examinations, a Personal Experience

Reaching this part of this chapter I am tempted to reflect on this subject in a very personal way. As a student who underwent a national exit examination, as a teacher who was responsible to prepare students for this event, and as a researcher interested in finding out if different tests measure different competencies. As a student I was perfectly aware of the goal of our 11-year-long educational process: to pass the national examination, a national event for all students at exactly the same time, somewhere in May. The final year was merely used to prepare us for the main event: to do three 3-hour long written tests—by no means with multiple-choice but by solving large and complex problems, although not in context. To be prepared was quite easy, to pass difficult. Easy to prepare because there were numerous examples of examination problems from previous years, as all examination test problems are in the public domain after the examination. Passing the exam was difficult because students needed high technical skills and preciseness. Apart from the written tests (9 hours total) most students had to do an oral session as well, under the principle that different cognitive demands could be operationalized under different formats. Some deeper conceptual understanding should be measured in an oral session carried out by the teacher and an outside “expert,” quite often a university mathematician. But if a student had very good grades for the written exams, one did not have to do the orals. And after passing the exam, one had a feeling of absolute freedom: One could enter any university to study an almost unlimited number of disciplines.

As a teacher I felt that the final national examination was a rather threatening event, the moment of truth whether I was a good teacher or not. The only good thing about the last year of schooling was that I had the undivided attention of most students, as they were all very interested in passing the exam. But nightmares for beginning teachers were common: All students failing and the teacher being fired was a dreaded scenario. Of course the teacher was also responsible for scoring the test, but regrettably a randomly appointed colleague had to check the work, and the scores were expected to be within a very narrow band. Reflecting on this practice it is quite surprising that parents completely trusted this process, even today in the twenty-first century. Nowadays the oral has been discontinued because of economical reasons—a problem that again and again overshadows good assessment practices. It was replaced by a school exam, to be designed and scored by the teacher. It accounted for

50% of the total score of the final examination. There was a large degree of freedom for the teacher resulting in many teachers’ trying to copy the “real thing” at the end of the same school year. But more adventurous teachers tried something very different like essays and open-ended problem-solving items.

Although the examinations included quite extensive open-ended problems, the researcher in me was interested in trying to find other ways to operationalize the so-called higher order thinking skills, with expected but still surprising results. I used the liberty of the 50% school examination to find out differences to test out the restricted-time written test, a take-home test, and an oral test. The most striking observation I was able to measure was that the correlation between the different formats was low, indicating that indeed one test cannot measure all cognitive demands or, in this case, that different formats measure different competencies (de Lange, 1987).

Exit Examinations

School exit examinations are not a very popular object of study. One of the few studies, *Secondary School Examinations: International Perspectives on Policies and Practices* by Eckstein and Noah (1993), was announced as filling an informational and conceptual vacuum. The study observed that most (industrialized) countries require students to take extensive examinations at the end of secondary education to determine whether they have satisfactorily met national criteria and are qualified for further education. These examinations are also used to assess individual schools (and teachers, as indicated in my personal note) and a nation’s educational program as a whole. It is remarkable that the United States does not have high school tests at the national level: “The United States is unique among the countries we have studied in having no coordinated, public, national system for assessing student achievement at the end of secondary school” (Eckstein & Noah, 1993, p. 238). In Barton’s paper (1999) the author commented that the examinations in the countries studied by Eckstein and Noah (U.S.A., China, Japan, Germany, England and Wales, France, Sweden, and the former Soviet Union) are closely related to the curriculum. In the United States, he continued, it is hard to conceive of any *national* exam being closely related to the actual curriculum for such a high-stakes examination because of the decentralized control over the curriculum. According to Eckstein and Noah the governmental control of the school curriculum in the United States has been extraordinarily weak.

But this argument of decentralization cannot be used to not introduce decentralized exit examinations. In several countries there are no *national* exit examinations but state examinations (like in Germany and Australia).

Japan, the United Kingdom, France, and Germany were the subjects of a more recent study: *Comparisons of Entrance and Exit Examinations* (Stevenson & Lee, 1997). The authors observed also the relation between curriculum and examination, noting that:

rather than imposing some arbitrarily defined standard of achievement, the examinations are closely tied to what the students have studied in high school. Because teachers are aware of what students are expected to know in examinations, it becomes their responsibility to equip students with the information and skills needed to pass the examination. (p. 47)

It is amazing to see how this fits with the personal experiences described before.

One of the observations in the introduction of the Eckstein and Noah study is the relative lack of systemic, cross-national comparative work, and the fact that, with only a few notable exceptions, the comparative study of examinations has been a largely neglected aspect of comparative education. Although the study is somewhat dated (1993), many observations and conclusions are still very informative if only as sketching possibilities for future development. Quite often things of the past are neglected just for that reason, without looking at the benefits from a strictly educational point of view instead of a practical or economical perspective. A nice example is the disappearance of the oral part of the mathematics examination in the Netherlands: Officially one argument was the lack of validity and reliability with two persons examining a student. In reality it was an economical and logistical argument that did away a format that quite often was highly regarded by students. It is hopeful to observe that the new rules allow for a variety of formats in the school examination part of the exam, bringing back a more balanced assessment system.

The Eckstein-Noah study observed that mathematics is a subject that all must take in some countries, with accommodation for more and less advanced studies. In others, it is differentiated according to specializations in the subject, all at an advanced level, whereas in yet other countries it may not be required at all. Substantial differences among countries in the knowledge required of candidates in mathematics exist to a surprising extent. But what is even more remarkable, especially in light of the (inter)national large-scale assessments, is the observation that format and content are inseparable, an ever-returning point

in any discussion about valid and fair assessment when the designers of large-scale assessments point out that multiple-choice and short-answer questions quite often offer enough possibilities to test the students. It is commonplace that the format of examinations strongly influences both the content and style of teaching. As a consequence, the format of examinations, as much as their content, is a target of the debate over the proper objectives of instruction and over what teachers and schools are supposed to do. The United States and Japan continue to rely heavily on multiple-choice formats. In China as well, in response to the large number of candidates, the high cost of university entrance examinations, and concern about their reliability and objectivity, the examination authorities are increasingly substituting short-answer and multiple-choice questions for the traditional extended-answer format. Concern for reliability and validity thus influences the choice of formats, which in turn influences the content selected for examination, which in turn influences the content and style of classroom instruction.

These remarks are even more remarkable as they were made more than a decade ago, and one assumes that some progress has been made in this area. But this progress fails to show up in the international large-scale assessments to a convincing extent. TIMSS still has a large amount of multiple-choice and short-answer questions, and although PISA may have less than 50% multiple-choice, it still limits itself to very short restricted time-open questions. I noted before that *Knowing What Students Know* recommended strongly that NAEP should include carefully designed, targeted assessments that could provide in-depth descriptive information about more complex activities that occur over longer periods of time (longer than 50 minutes). Having participated in several discussions about the trade-off between “fair-and-valid” assessments and “practical-and-cheap” assessments I have noticed that the decision makers often tend to favor the “practical-and-cheap,” which fits in nicely with the earlier observation that these stakeholders show little interest in the content domain of large-scale assessments.

“Broader” Assessment Systems

That many countries and states have shown that a broader “assessment system” can work in reality has been indicated by the study about the examinations. The variety of formats used for this form of “high-stakes testing” is almost unlimited. An example from the United States that gives indications of future types of large-

scale assessments is the New Standards Project. The choice for this example is not completely at random.

VCE

One of the quite interesting new approaches to something like the New Standards Project was undertaken in the state of Victoria, Australia. One of the principles behind this examination of the Victoria Certificate Assessment Board was that teacher would like to teach to the test. So the test had four different formats, testing very different cognitive demands. I have described this daring approach earlier (de Lange, 1996), and many articles are around by people directly involved. In one of their earlier publications, Money and Stephens stated that applications and modeling are central features of the Mathematics Study Design in the Victorian Certificate of Education (VCE) in the state of Victoria, Australia (Money & Stephens, 1993). All students undertaking mathematics in the final 2 years, that is Years 11 and 12, engage in substantial applications and modeling, with content options up to 50% to provide for the widely varying abilities and interest of students. In my opinion it is fair to say that the new test requirements lead to different practices in teaching (test driven innovation) or, as Money and Stephens saw it: “The new arrangements have had a significant impact on the teaching and learning of mathematics.” (p. 336). Much responsibility is given to the teachers because the Mathematics Study Design for the VCE provides a framework within which *teachers* develop courses, each normally 1 year in length.

Three work requirements, detailed by the teacher but each taking up between 20% and 60% of the course time, provide the basis for determining satisfactory completion and certificate credit for all semester units within VCE Mathematics:

- *Skills Practice and Standard Applications*: the study of aspects of the existing body of mathematical knowledge through learning and practicing mathematical algorithms, routines, and techniques, and using them to find solutions to standard problems.
- *Problem Solving and Modeling*: the creative application of mathematical knowledge and skills to solve problems in unfamiliar situations, including real-life situations.
- *Projects*: extended independent investigations involving the use of mathematics.

For teachers, these work requirements provide the framework for a shift in the balance of instructional patterns and a shift in role from dispenser of information towards learning facilitator, catalyst, and

coach. For students, the work requirements provide an accessible pathway towards formal recognition of the learning processes they have undertaken.

It seems rather trivial to observe that one can only introduce a modeling and applications curriculum when the assessments do really operationalize the goals of such a curriculum (de Lange, 1987), a point repeatedly underlined by many authors (e.g., Blum & Niss, 1991; Money & Stephens, 1993). To achieve this, the Common Assessment Task (CAT) initially consisted of four parts:

- CAT 1, an Investigative Project, is based on a centrally set theme and is intended to represent 15 to 20 hours of students’ work;
- CAT 2, a Challenging Problem, is chosen from four centrally set problems for each course, representing about 6 to 8 hours of students’ work;
- CAT 3, a Facts and Skills Task, is a multiple-choice test of 90 minutes duration, covering the full range of core and optional content;
- CAT 4, an Analysis Task, is a test, also of 90 minutes, consisting of about four structured questions that lead from routine to nonroutine aspects of a problem.

As the reader can judge, my earlier qualification of a “daring” approach seems quite appropriate, especially for real large-scale assessments on a state level.

It cannot come as a surprise that not all CATs survived, but this does not affect the message: To design appropriate large-scale assessments, incorporating the more complex competencies, one is confronted with a daunting and challenging task.

I conclude with some examples from the *Mathematical Methods* Written Examinations 1 and 2 from November 2003. Written Examination 1 (Facts, Skills, and Applications) consists of two parts: Part I and Part II. Part I has 27 multiple-choice items, Part II has 6 short-answer questions. The 27 items of Part I are good for a score of 27 points, the 6 questions for Part II are worth 23 points. Figure 25.8 shows an example of both parts:

Another 1½ hours is given to the students to complete four questions in Written Examination 2: Analysis Task. The four questions have several subquestions (see Figure 25.9), and the total is worth 55 points.

New Standards

As the United States is often seen as “the mother of multiple-choice and other bad habits in assessment,” in part because of the existence of a commercial testing

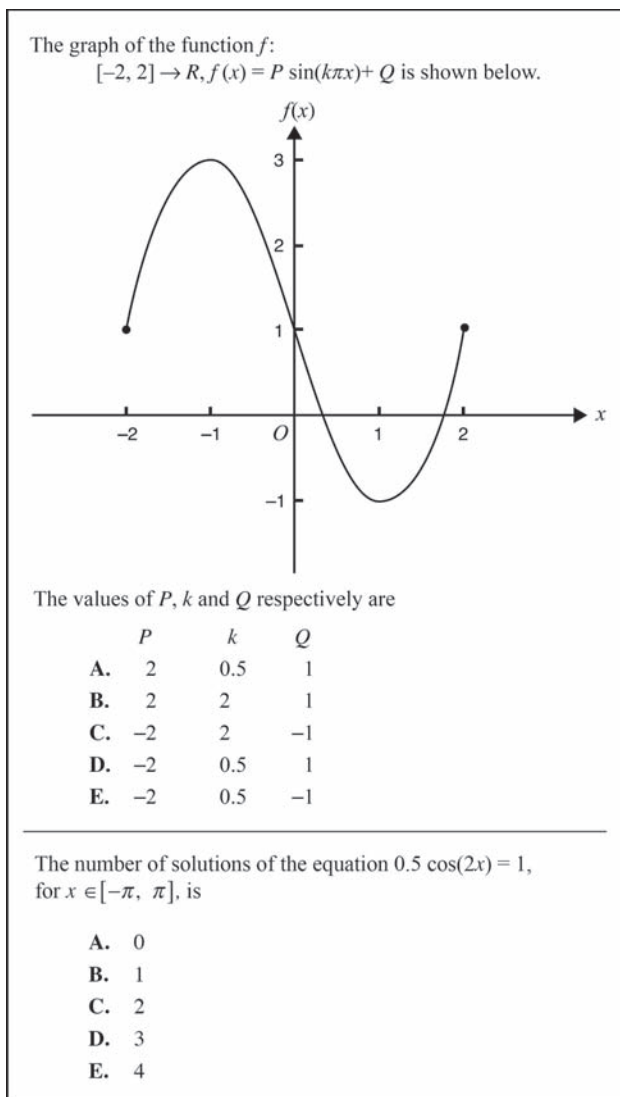
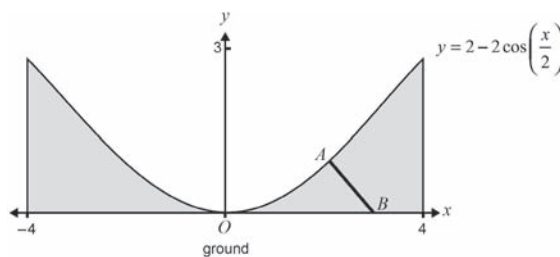


Figure 25.8 Two examples from Part I (VCA, 2003a, pp. 5–6).

industry and its collaborators (Kaplan Test Preparation Courses), it is remarkable that a project like the New Standards Project actually made an impact in the United States. The project illustrates a way to approach a number of issues related to large-scale assessments that differ from the earlier described international and national large-scale assessments.

The program was designed to provide clear goals for learning and assessments that are closely tied to those goals. A combination of on-demand and embedded assessments was to be used to tap a broad range of learning outcomes. Development of the program was through collaboration between the Learning Research and Development Center of the University of Pittsburgh and the National Center on Education and the Economy, in partnership with states and urban

Andrew is making a skateboard ramp. He draws a cross-section diagram with coordinate axes as shown below.



The curve has the equation $y = 2 - 2 \cos\left(\frac{x}{2}\right)$, $-4 \leq x \leq 4$.

All measurements are in meters; the horizontal length of the structure is 8 meters.

- How many meters above the ground is the highest point of the ramp? Give your answer correct to two decimal places.
- Use calculus to show that the gradient of the ramp is always less than or equal to 1.
- Use calculus to find the area of the shaded region, correct to two decimal places.

There is a supporting beam AB on the structure as shown.

A is a point on the curve one meter vertically above the x -axis.

B is a point on the x -axis such that AB is normal to the curve at A .

- Find the exact value of the x -coordinate of A .
- Use calculus to find the exact value of the gradient of the normal to the curve at A .
- Find the exact value of the length of AB .

Figure 25.9 Example from Part II (VCA, 2003b, pp. 4–6).

school districts. Challenging standards were established for student performances at Grades 4, 8, and 10, along with large-scale assessments designed to measure attainment of those standards.

The New Standards Project consists of three inter-related components: performance standards, a portfolio assessment system, and an on-demand exam. The performance standards describe what students should know and the ways they should demonstrate the knowledge and skills they have acquired. The performance standards include samples of student work that illustrate high-quality performances, accompanied by commentary that shows how the work sample reflects the performance standards (New Standards, 1994, 1996, 1997). According to *Knowing What Students Know* they go beyond most content standards by describing how good is good enough, thus providing clear targets to pursue (NRC, 2001, p. 251).

Balanced Assessment Project

Another influential project in the United States that clearly falls under the definition of large-scale assessment is the Balanced Assessment Project. The

collaboration involves not only the U.S. partners (initially Berkeley, Harvard, Michigan State) but also the University of Nottingham, U.K. The project “takes off where multiple-choice ends” (Figure 25.10). Multiple-choice is seen as falling into the range of transforming and manipulating; what the project wants to do is performance assessment that cuts across ranges of mathematical processes, including modeling and formulating, inferring and drawing conclusions, checking, evaluating, and reporting.

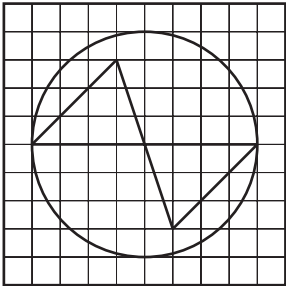
T-Shirt

The aim of this assessment is to provide the opportunity for you to:

- systematically communicate about geometric shapes;
- locate shapes on a grid;
- give a clear set of directions.

The design below, including the 10 by 10 grid, is going to be used on a math team T-shirt. You accidentally took the original design home, and your friend Chris, needs it tonight. Chris has no fax machine, but has a 10 by 10 grid just like yours. You must call Chris on the telephone and tell him very precisely how to draw the design on his grid.

Prepare for the phone call by writing out your directions clearly, ready to read over the telephone.



Write your directions on another sheet of paper.

This is the grid that Chris has in front of him.

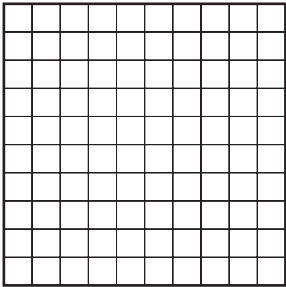


Figure 25.10 Grade 8 sample task from the Balanced Assessment Project (1995).

NCEM

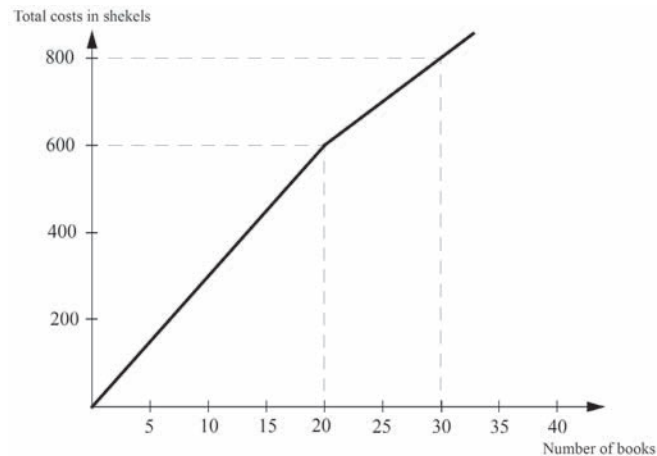
An interesting point that permeates almost all discussions about exit examinations is whether exami-

nations or other large-scale assessments that can promote and stimulate worthwhile educational goals and can function as a high-stakes assessment at the same time. Another point is the fact that in many countries the examination tool becomes a tool for selection of the “best and the brightest.” In their article on reform on the national examination, Amit and Fried (2002) observed the struggle between society’s ambition for high achievement and its simultaneous ambition for mathematical literacy for all. On the one hand, because of the prominent place assumed by mathematics in professional and everyday life, it has become a critical filter, via higher education, for social and economic advancement (Sells, 1978). At the same time, and for the same reason, mathematical literacy is recognized as a basic societal goal, so that good mathematics education, therefore, becomes a basic right for all members of society. In their article Amit and Fried argued that it is possible to align the role of completion examinations in promoting the high quality of advanced studies with the original role as an agent of democracy. They based this conclusion on the new model for the National Completion Examinations in Mathematics (NCEM) in Israel.

The original NCEM consisted of three levels: Ordinary, Advanced, and Highly Advanced. The progression from one level to the next can be characterized by an increasing range of mathematical subjects, but also by an increasing demand for depth in the approach to mathematical ideas. But in the early 1990s Israel’s educational policymakers, mathematics educators, and researchers called for reform of the system because of the rising numbers of students not opting to take the ordinary level examination and therefore not getting the examination certificate.

The answer to these problems came in 1996 when both a new curriculum and a new ordinary level NCEM were introduced. The new examination comprised two separate questionnaires: the basic and the supplementary questionnaire. The first is oriented towards basic mathematical literacy, and the second towards a more complete mastery of mathematical concepts and techniques. The basic part can be taken and retaken independently of the supplementary part. This feature allows schools to develop special tracks geared towards students who will take only the basic questionnaire. It is this basic part that makes the new system work in merging the different objectives. Figure 25.11 gives an example of problems from the basic component (that can be taken when students are in 10th, 11th or 12th grade).

An evaluation by the Ministry of Education showed an increase in the number of students taking the basic examination, with the increase in the basic part being significantly higher than the number of students tak-

**Example 1**

The graph shows the price of books according to the number purchased (see figure above). Answer the questions according to the graph.

- How much do you pay when you purchase 25 books?
- How much does each book among the first 20 cost?
- How much does each book after the 21st book cost?

Example 2

Total 15 people were tested and the following scores were obtained: 8, 7, 7, 9, 8, 8, 9, 5, 7, 6, 6, 4, 6, 10, 8.

- Calculate the average score.
- What is the probability that the score of one of the people chosen at random will be higher than the average?

Example 3

Given a right triangle ABC in which angle D is right. The length of side AD is 40 cm. Angle BAD is 41° . C is a point on BD such that the angle CAD is 38° (see figure below).

- Calculate the length of the leg BD.
- Calculate the length of the segment BC.

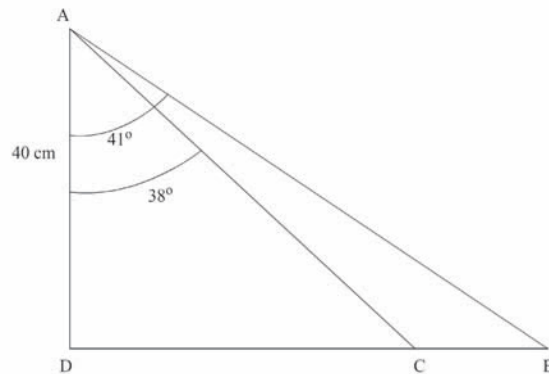


Figure 25.11 Three examples of questions in the Basic Questionnaire (Amit & Fried, 2002, p. 507).

ing the complete examination. Amit and Fried concluded that over the years the gap between the number of examinees sitting for the basic questionnaire and number of examinees sitting for the supplementary part has increased. This indicates that many students take only the basic part and do not continue to complete the requirements for the ordinary level. It is reasonable to assume, based on experience, that students would not have sat for the examination at all for mathematics if this opportunity had not existed. More

students are now enabled to continue to study mathematics throughout high school due to a rather simple intervention: more attention for the functionality of mathematics in the curriculum—and as noted before Israel is certainly not alone in this development—and consequential changes in the assessment system.

Let me make a concluding remark or two before moving on to technology in large-scale assessments. Large-scale and high-stakes assessments clearly can go together in an appropriate way. Regrettably, some

large-scale assessments are “misused” as certain jurisdictions have adopted policies that rely exclusively on achievement test scores to make high-stakes decisions. Many standardized tests do not meet desired criteria, or, as *Knowing What Students Know* stated, current assessment practices are the cumulative product of theories of learning and models of measurement that were developed to fulfill the social and educational needs of a different time. The wish to refashion assessments to meet current and future needs for quality information is recognized beyond doubt, in many countries, and even implemented to a certain extent in some.

LARGE-SCALE ASSESSMENTS, PROBLEM SOLVING, AND TECHNOLOGY

As pointed out before one of the key issues in assessment and also large-scale assessment is not only the need to pay more attention to the mathematical content domain, but also in realizing the idea that assessments are based on modern knowledge of cognition and its measurements. One of the major problems in large-scale assessments is that it should be seen not only as a policy evaluation, but more as a tool for practitioners. Therefore assessment instruments should relate clearly to intended learning outcomes and can be used to improve students’ achievement. These points relate to two major threats to validity in high-stakes tests (Wilson, this volume). The first is *construct underrepresentation* and refers to a situation in which the concepts and skills that are measured in the assessment fail to include important dimensions or facets of the intended construct, in this case the content domain in mathematics. The second is *construct-irrelevant variance* (Brualdi, 2002). This can take the form of extraneous clues in the item or the task format that may permit some students to respond correctly or appropriately in ways that are irrelevant to the construct being assessed. Another possibility is that extraneous aspects of the task make it irrelevantly difficult for some individuals or groups.

Various technologies have been applied to bring greater efficiency, timeliness, and sophistication to multiple aspects of assessment design and implementation. Examples include technologies that generate items; immediately adapt items on the basis of the examinee’s performance; analyze, score, and report assessment data; allow learners to be assessed at different times and in distant locations; enliven assessment tasks with multimedia; and add interactivity to assessment tasks. In many cases, these technology tools have been used to implement conventional theories and methods of assessment, albeit more effectively and

efficiently (NRC, 2001). It can even be argued that access to modern technologies has hindered the development of more valid tasks in respect to the earlier made points. Therefore the focus of the research relating assessments and technologies should shift towards these goals and challenges. As a matter of fact it seems that a precondition to develop valid and interesting tasks is the availability of modern technologies.

Knowing What Students Know provided a number of promising new developments varying from theory-based item generation, developing concepts maps (individual or in groups), complex problem solving in a variety of ways, analysis of complex solution strategies (using complex neural network technology), facilitating formative assessment, technology-based or technology-enhanced learning environments, and more. Many developments seem to be happening at very different levels, and many of the implications are unclear. One of the dilemmas we educators are facing is that we have to reinvent the definition of learning goals in these new environments: Are we abiding with the traditional curricular-based learning goals, or do we see quite new and unexpected horizons? At one side we see successful mathematical modeling competitions in which students work in teams for an extended period of time using technology in a variety of ways on quite a large scale. This might be considered as an existence proof for the merging of large-scale assessments and complex problem solving in groups in a somewhat traditional setting (just curricular mathematics) but where the role of technology is crucial.

An example of large-scale assessment in groups, among others, is the math A-lympiad (see Figure 25.12). This is a modeling competition for students from Germany, Denmark, and the Netherlands (with guests from other countries as well) that has been in existence since 1989 (de Haan & Wijers, 2000). Thousands of students work in many hundreds of teams on an assignment—a very open-ended problem—in which mathematical problem solving and higher order thinking skills must be used to solve a real-world problem. The result is a written report, just like in most other modeling competitions.

The A-lympiad has two rounds: the preliminary round competing at school for a whole day at their own schools, and the international final for the best 16 teams. This final takes most of 2 days. The aforementioned publication has an interesting chapter on the “Assessment of the Papers.” The conclusion is in line with earlier research on problem-solving assessment in mathematics: It takes time, effort and therefore is expensive, but the intersubjective scoring is quite reliable (de Lange, 1987). This is in line with the intermarker reliability found in the PISA study.

Security is an art unto itself

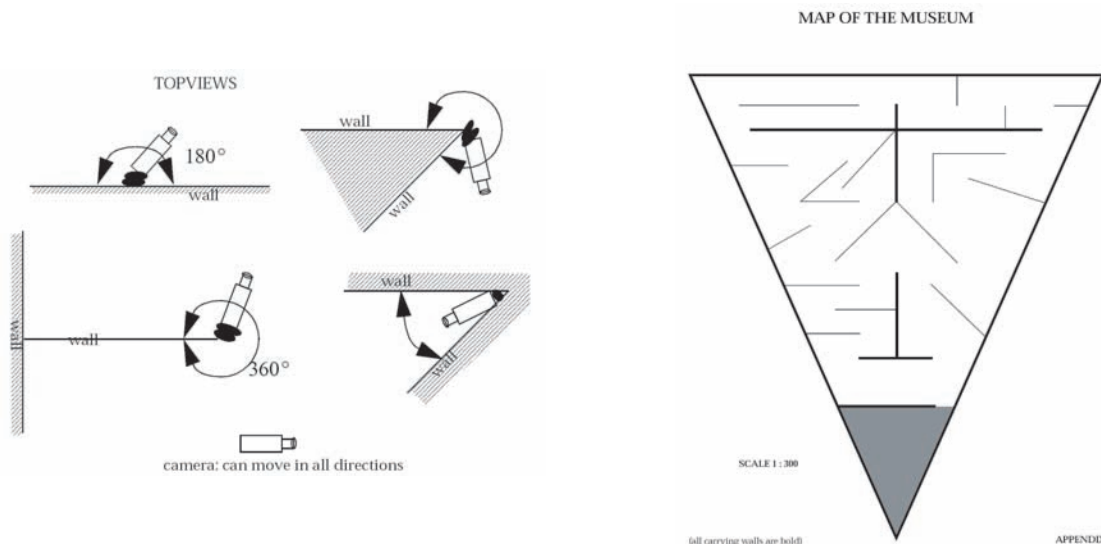
Introduction

A museum for modern art will organize a big exhibition with pieces of a number of great artists in the field of modern art. The organization is busy with preparations for the exposition, and meets with difficulties concerning security.

A new security system

The present security system with video cameras doesn't satisfy the demands of an exhibition this size. The cameras aren't movable enough (to show another corner), but the main objection is, that the present system cannot cover the complete space in the museum. To organize the exposition, it's necessary to get a new security system.

The new type of camera that will be used is already chosen. This camera is so fast in moving (in all directions) and focusing, that one can say that this camera really secures the complete surrounding space (see the figure below).



All walls in the museum go from floor to ceiling.

A disadvantage of this camera is its high price. That's why it's necessary to investigate carefully where the cameras should be placed, for you want the least possible amount of cameras.

Enclosed is a map of the museum. The museum has a triangular formed top view. The grey part contains the entrance, bathrooms, cloakroom and office rooms. It's not necessary to replace the old cameras in this part. The new security system is just needed for the part that is not grey.

Exercise 1

Think of a placement of cameras, in such a way that you use the smallest possible amount of cameras to secure the exhibition room.

Describe the system you used to come to this placement of cameras.

Mark where the cameras should be placed and show that the whole exhibition room will be secured this way.

The exhibition contains a little less than 100 paintings. To show these paintings in a proper way (with enough space in between), at least 280 meters of wall capacity is needed, but: the more wall capacity, the better.

On the map, you can see a number of bold walls. These are carrying walls; the other walls can be removed.

Because the new security system is very expensive, the organization wants to reduce the expenses. To do this, there is the possibility to reduce the number of cameras, but there is also the possibility to remove non-carrying walls. Removal of wall costs f 500, per meter. A video camera costs f 10.000, each.

Exercise 2

How is it possible through removing walls (keep the minimum amounts of capacity you need in mind!) to save expenses on the cameras? In other words: how is it possible, by removing as little wall capacity as possible, to make as many cameras as possible superfluous? And yet, all walls stay secured?

Make a proposal and calculate the savings that this solution will give you.

After this exhibition, the inside of the museum will be changed drastically. First, all non-carrying walls will be removed and just the carrying walls will stay. Because the new inside of the museum will be used for a longer period of time, and for several more exhibitions, an architect agency will be hired to create the new inside. The assignment will be to add 150 meters of wall capacity to the existing carrying walls. Furthermore, 6 cameras must be sufficient for security, yet, there must be an attractive partition of space.

Exercise 3

A number of architect agencies is being asked to hand in proposals in which the conditions are met. Your team works for one of the architect agencies, and your agency really wants the assignment. Of course you won't be a strong rival if you just meet the minimum demands.

Make a proposal for the museum board (including working drawing and short 'covering letter').

Figure 25.12 Assignment in preliminary round of Mathematics A-lympiad, 1996–1997 (de Haan & Wijers, 2000, pp. 59–61).

At the other side is complex problem solving in a less traditional setting: a MashpeeQuest assessment task gives students an opportunity to put complex problem-solving skills to use in a web-based environment that structures their work (Mislevy, Steinberg, Almond, Haertel, & Penuel, 2000). In this example, technology plays at least two roles. The first is conceptual as the information analysis skills to be assessed, and the behaviors that serve as evidence are embedded within the web-based environment. The second is more an operational issue: Because the actions take place in a technological environment, some observations of students' performance can be made automatically. Staying with the area of complex problem solving with technology, some interesting experiments were carried out in Germany by Klieme, Schümer, and Knoll in 2001.

Technology and Assessment: Thinking Ahead (NRC, 2002) argued that technology can help to merge the advances in cognitive sciences and measurement theories, which then should result in a significant leap forward in the science and practice of assessment. In the first paper Gitomer and Bennett (2002) illustrated how researchers at the Educational Testing Service use computer technologies to address a long-standing and very valid criticism of standardized tests. They tend to consist of certain types of traditional test items and have lost sight of the underlying constructs, or cognitive competencies. The authors argued that computer technologies can help "unmask" the constructs underlying traditional assessments, and to make the constructs more visible and explicit in the design of new assessments.

Another paper in this publication by Means and Haertel (2002) described, according to the introduction, the effort to develop computer-based, quality assessments of scientific inquiry. One of the key points they made is that in contrast to standardized tests and the more conventional paper-and-pencil tests used in most (science) classrooms, the new technology-based learning environments reflect the richness and complexity of science inquiry. It is at least interesting to make the observation that the authors and their colleagues saw very exciting opportunities in technology-based assessments in comparison with what they identify as "traditional standardized achievement test," but failed to observe that most of the constructs or competencies can be measured with pencil-and-paper, with or without the use of technology. It is my opinion that the perceived contrast between "Innovative Technology-Supported Assessments" and "Traditional Tests" (p. 15) is a construct of the assessment industry. Technology should not drive innovations in assessments, as has been the case so often in the past. The drive should

come from advances in mathematical cognition, from advances in measurement theory, and should only be facilitated by the use of technology.

Eva Baker (2000) stated that

the previous discussion has focused on the use of technology to make substantial improvements in our ability to assess and understand educational quality, principally measured by student achievement. But there is plenty of resistance to even this level of technology application. (p. 13)

Again, nothing seems new: There has always been plenty of resistance against innovations in assessments—looking back some 30 years shows that many efforts have been made to introduce more valid and fair assessments, with or without technology. Central should be what to measure and how to measure taking the kids and their competencies in the area of mathematics seriously.

Large-scale assessments have committed themselves to measure a wider variety of mathematical competencies. They have indicated that they will include more and more extended formats, and include technology use, starting with the calculator, but that they are also offering technology-based items (like Science in PISA 2006). Finally, high-stakes examinations can be done, although still on a voluntary basis, in a computer environment (Netherlands). There is mention of evaluating or measuring group work as this is regarded widely as an essential competency, and there are working examples in existence. Resistance exists, especially with policymakers, with the people who pay for the large-scale assessment studies. But some progress has been made, and with educators' combined efforts, further advances are around the corner.

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AUTHOR NOTE

In a chapter like this it seems important to make visible some of the author's activities as a professional in this area. I have been on the National Dutch Committee for TIMSS and member of the Subject Matter Item Replacement Committee for TIMSS.

I am Chairman of the International Mathematics Expert Group of OECD/PISA, an advisory body that feels itself philosophically responsible for the mathematics framework. Furthermore I was Principal Investigator of the CATCH-project, and founder of the Math A-lympiad.

I would like to thank my colleague Truus Dekker and the reviewers John Dossey and Ken Travers for their valuable and constructive comments on earlier versions of this article. It is good to know that there are always colleagues willing to spend their time on work of others in order to get a better product.

■ ■ P A R T V I ■ ■

Issues and Perspectives

ISSUES IN ACCESS AND EQUITY IN MATHEMATICS EDUCATION

Alan J. Bishop and Helen J. Forgasz

MONASH UNIVERSITY, MELBOURNE, AUSTRALIA

In the 1992 *Handbook*, two chapters dealt with issues of equity and access. Gilah Leder (1992) contributed the chapter on “Mathematics and Gender: Changing Perspectives,” and Walter Secada’s (1992) chapter was entitled “Race, Ethnicity, Social class, Language and Achievement in Mathematics.” Other chapters alluded to an issue or two, but there was no chapter with the same brief as this current one. Both of the previous authors discussed the problems of terminology and argued for different approaches to research. In Leder’s case she argued for more “in-depth, small sample” research studies, whereas Secada looked for more research within mainstream mathematics education that takes issues of student diversity as a part of the study.

In this chapter, being conscious of the fact that several chapters in this volume focus on students’ learning, and on race, class, and so on, we focus on the issues and perspectives that relate to equity and access. With that in mind we disclaim any attempt to cover the whole research field of equity in mathematics education and instead intend the chapter to clarify and discuss the significant issues that need to be addressed in any current research in this field. We will of course draw on the published research literature in the course of our discussions and clarifications.

High on the list of issue areas is one that must come into this chapter, namely that of terminology. Secada’s (1992) chapter dealt extensively with this topic and raised the significant point that although research on “disadvantaged” or “minority” students is to be welcomed, “it should not preclude inquiry into how the categories of ethnicity, race, social class

and language are constructed and how they are maintained in our society and, by extension, in our research” (p. 623). Who is defining *disadvantage*, why, and from what perspective? Who are the ‘minority’ and do *they* define themselves that way? In South Africa, the Black Africans were referred to as “the minority,” despite being clearly in the majority in terms of numbers.

We are reminded of the study by Becker (1963), who had an interesting perspective on the powerful strategy of labeling, based on his research into deviance in society.

The central fact of deviance is that it is created by society. I do not mean this in the way it is ordinarily understood, in which causes of deviance are located in the social situation of the deviant, or the social factors which prompted his [sic] action. I mean, rather, that social groups create deviants by making the rules whose infraction constitutes deviance, and by applying those rules to particular people and labelling them as outsiders. From this point of view, deviance is not a quality of the act a person commits, but rather a consequence of the application by others of rules and sanctions to an ‘offender’. The deviant is one to whom the label has been successfully applied. Deviant behaviour is behaviour that people so label. (pp. 8, 9)

Deviants are therefore not homogeneous. They share the label and the experiences of being so labelled, and that is all. It is the same with immigrant students, with Black, or Hispanic, or female, or blind, or working-class students. They are not homogeneous

groups, but they share the labels and the experience of being so labelled, as does anyone who is perceived as an outsider to whatever is the mainstream culture. What is important for this chapter is the fact that this labelling affects and indeed controls the educational consequences for those learners so labelled. The provocative research by McDermott (1974, 1996) shows some of these hidden consequences, as can be seen from the two titles “Achieving School Failure...” and “The Acquisition of a Child by a Learning Disability.”

In the case of mathematics education, equity and access have their special problems of definition and interpretation. At one level, we can confidently assert that without *access* to mathematics education there can be no equity; thus in this view *equity* is referring to an outcome whereas *access* is a means of getting there. At another level, we can argue that there are many equities: equity of predisposition, equity of teaching, equity of outcome, as well as equity of access. Equity here is being used as one criterion for judging or evaluating any educational variable, including access. And what precisely does “access to mathematics education” mean? “It is there if you want it.” Or “You can always buy the books.” Or “You have as much chance as anyone of studying mathematics at university!” or “You have the opportunity to learn, what you do with it is up to you.” Where does the responsibility for guaranteeing access lie? And is access, like beauty, in the eyes of the beholder?

The research literature generally indicates that the following student groups have all suffered in some way from conflicts with mainstream mathematics education, largely based on what can be called *Western Mathematics* (Bishop, 1988):

- Girls in “Western” and other societies
- Ethnic minority children in Western societies
- Indigenous “minorities” in Westernized societies (including Black students in South Africa who are numerically in the majority)
- Western “ex-colonial” student groups
- Non-Judeo-Christian religious student groups
- Rural learners, particularly in developing countries
- Learners with physical and mental impairments
- Children from lower class (or lower caste) families

Thus there is no shortage of literatures to help with this chapter; indeed the problem is almost that there is too much to include. However, what this listing, and the consequent surfeit of literature, demon-

strates is that equity and access are not just about students who have physical or mental impairments. Also there are many aspects of commonality between those groups and their problems, which makes an approach based on the different groups not so relevant for this chapter. Instead we shall deal with the common issues and obstacles that all disadvantaged groups face, and the attempts by mathematics educators to research ways that will help them overcome those obstacles.

Firstly there is research at the individual level, dealing with the individual learners and their experiences of inequitable mathematical treatment, which tends to be based mainly on cognitive or affective psychology (see for example, McLeod, 1992; Secada, 1991; Willis, 1992). The construct of *learning disabled* (LD), and, more particularly, of *mathematically disabled* (MD), is one with which cognitive science researchers are familiar (see, for example, Siegler, 2003; Zeleke, 2004). A more recent area of research, cultural psychology, has given further insights into the cultural conditions of learning that inevitably structure the provision of inequitable mathematical treatment and outcomes (see Carraher, Carraher, & Schliemann, 1985; Cole, 1996).

At the second level of the classroom, there are studies of access given or denied to students by virtue of the conditions of pedagogy established by the teacher and the peers (see, for example, Boaler, 1997; Khisty, 1995; Zevenbergen, 2001).

The classroom is part of a larger institution that also structures and constrains the behaviors of teachers and students, and at this institutional level the political and sociological literatures are more active (for example, Apple, 1995; Lubienski, 1997). These literatures also sensitize readers to the fact that it is society at large where the major political battles take place over access, equity, and of course funding, which underlies any decision about educational policies, provision, and resource priorities (Frankenstein, 1995; Knijnik, 1998). In the United States, for example, the influence on equity thinking and practice of the so-called No Child Left Behind Act (The Elementary and Secondary Education Act 2001) is huge and far-reaching, with its emphasis on student outcomes, and on other foci such as the insistence on the use of randomly assigned treatment groups as the only “scientific” kind of research.

Finally, at the broader cultural level are the values, history, and customs that subconsciously but nonetheless effectively shape, constrain, and foster dispositions toward educational equity and access (Keitel, Damerow, Bishop, & Gerdes, 1989). For example, it is not the case that all countries/cultures subscribe

to the aspiration of “Mathematics for all” as shown by D’Ambrosio (1994, 1998).

Thus there are many ways we could have structured this chapter, but we have chosen a relatively simple structuring in that following this introduction, the next section presents some of the rich evidence about inequities in student mathematical performance. The subsequent sections focus on issues arising from studies of policy and intention, regarding equity and access, and on research on practice and policy implementation. The fourth substantive section will consider some of the most significant research issues arising from the two previous sections. We close the chapter by considering some of the research challenges for the future, as a kind of research agenda.

EQUITY IN MATHEMATICS LEARNING OUTCOMES?

One example of the kind of outcomes-based research that makes us aware of inequities in the provision of mathematics education is the mathematics achievement data of approximately 190,000 fourth graders and 153,000 eighth graders from across the United States that were found in the 2003 NAEP results for mathematics (National Center for Education Statistics [NCES], 2003). The results continue a trend in mathematics covering the years 1990, 1992, 1996, 2000, and 2003. The mathematics questions in 2003 covered five content strands: number sense, properties, and operations; measurement; geometry and spatial sense; data analysis, statistics, and probability; and algebra and functions.

According to NCES (2003), the major findings of the national results for students in Grades 4 and 8 included the following:

- for both grades, 2003 average mathematics scores were higher than in all the previous assessment years (e.g., 1990: Grade 4—213, Grade 8—263; 2003: Grade 4—235; Grade 8—278)
- among Grade 4 students in 2003, the percentages performing at or above the Basic level, at or above Proficient, and at Advanced were all higher than in all previous assessment years since 1990.

Of particular interest to us in this chapter are the findings for various subgroups of Grade 4 and Grade 8 students in 2003. The results are shown in Table 26.1 and summarized below.

Table 26.1 Mean Scores for 2003 NAEP Results for Grade 4 and Grade 8 From Various Subgroups

Subgroup	Grade 4	Grade 8
Male	236	278
Female	233	277
White	243	288
Black	216	252
Hispanic	222	259
Asian/Pacific Islander	246	291
American Indian/Alaska Native	233	263
Eligible for free/reduced price school lunch	222	259
Ineligible for free/reduced price school lunch	244	287

Gender

- At both grades, male students scored higher on average than female students.
- At both grades, the average scores for male and female students were higher than in any of the previous assessment years.
- At both grades, the apparent gap in scores was not found to be significantly different than the gap in any of the previous assessment years.

Race and Ethnicity

- At both grades, Asian/Pacific Islander students scored higher on average than did White students. Also, White students and Asian/Pacific Islander students had higher average scores than their Black, Hispanic, and American Indian/Alaska Native peers.
- At both grades, Hispanic students and American Indian/Alaska Native students scored higher on average than Black students.
- At both grades, White, Black, and Hispanic students had higher average scores than in any of the previous assessment years since 1990.
- No significant change was detected in the average score for Asian/Pacific Islander students between 2000 and 2003 at Grade 8. The average scores for Asian/Pacific Islander students were higher in 2003 than in 1990 at both Grades 4 and 8.

Socioeconomic Status (using eligibility for free/reduced-price school lunch as the criterion)

- At both grades, the average mathematics score for students who were eligible for free/reduced-price lunch was lower than for students who were not eligible.
- At both grades, average mathematics scores were higher in 2003 than in any of the previous assessment years for both groups of students.

These kinds of data are frequently found in both national and international studies such as the *Third* (and more recent *Trends in*) *International Mathematics and Science Study* (TIMSS), with gender, race, and class (or socioeconomic status) being the predominant difference variables of interest. The gender variable is arguably the least contentious of the three to be measured in quantitative studies and is thus the most represented in achievement-oriented equity research.

One of the key issues in this research area concerns the extent to which any research is actually trying to study equity and access, or whether the outcomes data happen to show differences between certain defined groups of students. In other words, the differences may be found by accident, so to speak. Thus, for example, in international studies such as the TIMSS, certain sampling decisions were taken in order to get “comparable” samples of students and teaching contexts across countries. This criterion, and those decisions, inevitably meant that any differences found *within* those countries might well have been due to the exigencies and “accidents” of the samples. A classic example is given by the use of the TIMSS tests in Australia to measure the comparative performance of indigenous students. As Lokan, Ford, and Greenwood (1996) reported,

Only about 400 students from the 13 700 or so TIMSS participants identified themselves as Aboriginal or Torres Strait Islander students. To enable more accurate results to be reported for Indigenous students, some additional sampling of schools with relatively high Aboriginal and Torres Strait Islander enrolments was done. . . . Of the 608 students, 330 were in the upper year level, 278 in the lower. . . . The results for the total group are descriptive only, since the additional schools were not randomly chosen. Nevertheless, the very much lower achievement levels, and the smaller differences between lower and upper year levels of the Indigenous students in comparison with the overall Australian results presented in Chapter 3 seem likely to signal real differences. (p. 161)

Another variable, the study of which produces some salutary lessons for equity research, concerns

the student’s language and whether that presents any disadvantage for mathematics achievement. For example, an assumption often made is that students who work and study in their second or third languages are necessarily handicapped in their studies and likely to show underachievement. However, in English-speaking countries, one should not assume that students whose first language is not English are necessarily disadvantaged in mathematics. Among the background data collected in the TIMSS was the main language spoken at home (English or other) and students’ place of birth (English- or non-English- speaking countries) (Lokan, Ford, & Greenwood, 1997). Australian data from Population 1 (9-year-olds) were analyzed by the combination of these two factors. Of the four resulting groups, the lowest achievers were those born in Australia whose families used a language other than English at home (about 4% of the cohort; Lokan et al., 1997). The highest achievers were children living in Australia but born in non-English- speaking countries, whose families had adopted English as the main language used at home.

Huang (2000) analyzed the TIMSS findings of students from five English-speaking countries: England, the United States, Canada, Australia, and New Zealand, confirming the Australian findings. Compared to students who spoke English at home, students whose home language was not English performed substantially less well; this was true whether or not they were immigrants.

Finally, the special education literature is replete with data concerning LD (learning disabled) and MD (mathematically disabled) students and their disadvantages in achieving mathematical understandings and competences, with large age/grade performance differences. As a typical example, Cawley and Miller (1989) reported that 8- and 9- year- old LD children performed at a first- grade level on tasks of computation and application. Zeleke (2004) analyzed and summarized the research and data concerning LD and MD students’ performance on simple arithmetic tasks and concluded, “MD children are characterized by poor strategy choices . . . frequent use of immature strategies . . . less frequent use of retrieval strategies . . . less varied collection of usable strategies . . . and use of significantly fewer problem solving strategies.” (p.12) Clearly there are many implications here, not just for special education and mainstream teachers, but also for policy decisions regarding access and equity.

POLICY ISSUES OF ACCESS AND EQUITY

Given the vast amount of evidence both national and international about systematic differences in math-

emational learning outcomes, we will now explore some of the research and ideas helping to explain the data, and hopefully offering strategies to combat their causes. This section will draw on studies that focus on policy aspects, intended curricula, and other mechanisms, such as examinations, teacher training, inservice aspects, and school structures. It will also give more of the educational and mathematical context of this review.

In the late 1980s and at the turn of the 1990s three international developments had a significant influence on our international and increasingly intercultural field. The first was ICME-6, which took place in Budapest, Hungary, in 1988. At that ICME, and following widespread pressure, a whole day was devoted to the issues and possibilities of Mathematics, Education, and Society. More than 120 people contributed formally to the program that day, and the UNESCO publication that followed (Keitel et al., 1989) drew everyone's attention to the social and cultural aspects that lay behind inequities of policy and practice in many countries. It also built on the groundwork laid at ICME-5 in Adelaide, Australia, where not only was there a whole Theme Group on the topic of "Mathematics for All" (Damerow, Dunkin, Nebres, & Werry, 1984) but also one of the plenary lectures was by D'Ambrosio (1986). In this he introduced the topic of ethnomathematics, thereby changing forever mathematics educators' conceptions of mathematics, and of the mathematics curriculum (see also Ascher, 1981).

The second development took place in 1989, when the National Council of Mathematics Teachers (NCTM) of the United States published *The Curriculum and Evaluation Standards for School Mathematics*, a wide-ranging and far-reaching document that would spearhead the reform of mathematics education across the country. Not only was the aim to change mainstream mathematics education, but particularly to emphasize that this was for *all* students, because all were capable of learning. Later the six principles in the new framework were spelled out in more detail (NCTM, 2000), and one of them is the Equity Principle, which significantly was put first in the order of priorities "Excellence in mathematics education requires equity—high expectations and strong support for all students." (p.12) The NCTM also issued this caveat:

Equity does not mean that every student should receive identical instruction; instead, it demands that reasonable and appropriate accommodations be made as needed to promote access and attainment for all students. . . . Technology can assist in achieving equity and must be accessible to all students. (NCTM, 2000, p. 12)

More reference will be made to this Principle later.

Thirdly, in 1990 UNESCO, together with UNICEF, UNDP, and the World Bank, organized the World Conference on Education for All (WCEFA) involving hundreds of representatives of governments, NGOs, educational bodies, and foundations. The Conference concluded by agreeing on two key sets of recommendations: the "framework for action to meet basic learning needs" and the "world declaration on education for all." They left the educational world in no doubt about their commitment to universal education, including numeracy for all (UNESCO, 1990).

All three of these initiatives highlighted the fact that from the 1990s forward issues of equity and access to mathematics education could not be ignored by those in policy- and decision-making positions. So what are the policy issues and challenges facing those mathematics educators who are seeking to provide an equitable education for all students, and what options does research has to offer?

The Structuring of Mathematics Education by Society, and Vice Versa

The first significant challenge to be faced is that of the valuing and structuring of mathematics by cultures and societies, which have profound effects on the issues of equity and access. Ever since the mathematization of technology in the 18th and 19th centuries, mathematical knowledge has played a strong role in the psyche of modernist and economically focused governments. As Skovsmose (1994) argued in his chapter "A Formatting Power," the formatting of society by mathematics is ever increasing, to which Woodrow (2003) added:

Mathematics has a peculiarly special position in the social and political discourse across the world—to use a new dangerous word it holds global significance. As the 'language of science' it had long assumed power and influence as the terminology of science but during the last half-century it has permeated many of the social sciences, including not only economics but also such social areas of debate as wealth distribution (poverty and affluence) or crime and its causes and consequences. (p. 10)

In every country in the world mathematics now holds a special position, and those who excel at it or its applications also hold a significant position in their societies. The corollary for us in this chapter is that students' mathematical attainment has been used as a selective "filter" for entry to various professions, a fact that mitigates strongly against the ethic of an equitable and accessible mathematics education for all.

Who are the groups controlling this filter effect, and who becomes disadvantaged by it? Professional bodies are the obvious controllers, including those who prescribe the entry requirements for their various professions, but more frequently it is the universities and other educational institutions that act as surrogate filter mechanisms for the professions. Of course universities also want to recruit the best mathematics students for their own purposes, such as to continue and develop the mathematics research programs, and this gives universities a powerful voice in the filtering process. As Woodrow (2003) rightly pointed out, “The ‘cultural capital’ represented by knowing mathematics is ‘gilt edged’ and it has been socially and culturally advantageous to mathematicians to maintain the investment” (p. 19).

In the United States we have seen the developing phenomenon of the “Math Wars” whereby proponents of the NCTM’s reform agenda for mathematics education (see the previous section) ran headlong into the brick walls of the university mathematics establishment, who perceived the reforms as diluting and lowering the high filtering standards they believed they set for their entrants (e.g., Jackson, 1997). The arguments may be cloaked in terms of mathematics content, which is the equivalent of the moral high ground for mathematicians, but the issues are clearly all about equity and access. The debates have spawned web sites (such as <http://mathematicallysane.com>) where evidence of various kinds is disseminated.

The United States is not the only country where these debates take place. In part the arguments in any country are fuelled by the results of international competitive testing, such as TIMSS and its analysts (see Schmidt, McKnight, Valverde, Houang, & Wiley, 1997), where the league tables of results produced by this and similar studies are used by all sides in these debates to support or destroy the arguments for equity. Equity and competition do not sit easily together, and any system that builds its mathematics education on the precepts of competition and rewarding high but narrow achievement will necessarily cast aside arguments for equitable and accessible approaches.

Once again, different countries’ policies regarding mathematics education show certain important variations, all of which have equity implications, due in part to their different histories, cultures, values, and perceived needs. Robitaille (1997) edited a text documenting the various national contexts for mathematics and science in the TIMSS-participating countries, where one can find, for example, that streaming and tracking are not sanctioned everywhere, there are not always policies to integrate students with disabilities in mainstream schools, and private schools do not nec-

essarily get financial or other assistance from governments. Mathematics is not always a compulsory school subject towards the end of formal schooling, and although there are general efforts to encourage girls to continue their mathematics education for longer, in Australia girls already do stay in schooling longer than boys. In some countries the textbooks are mandated by government whereas in others, teachers can choose what texts and teaching materials to use.

However, despite these differences, fundamentally for mathematics educators the most important value that any particular society puts on mathematical education is to a large degree reflected in how the subject is structured in any intended curriculum, and it is to that level that we now turn.

The New Curriculum—Mathematics or Numeracy or Both?

What then are the issues and challenges concerning the nature of the “intended” school mathematics curriculum, and the extent to which it could offer access to all? By saying *intended* here we are using the terminology of the IEA in its international surveys, where they distinguish the curriculum intended by the state or school district, the curriculum implemented by the schools and teachers, and the curriculum attained by the students (Robitaille & Garden, 1989). In this main section of the chapter we are concerned with the intended curriculum and the access issues surrounding its determination. In the following section on “practices” in institutions, we discuss the issues of equity as they relate to the implemented curriculum and the opportunities to learn.

In the 1990s the interpretations of *mathematics* and of *the mathematics curriculum* were debated and continue to be debated. It may seem strange to be debating the nature of mathematics, although philosophers throughout the ages have always done so. But given that mathematics educators approach this problem from the perspective of what a mathematical education is to contribute to all future citizens in a democratic society, this is not an arid philosophical discussion point. It should be at the heart of our teaching, and at the heart of our research.

In democratic societies one must always consider the notion of *democratic education*. Malloy (2002), for example, argued that democratic access to powerful mathematics for all children was a human right and, instead of universal enculturation, “students are educated in a world that concentrates on differences, which consciously or unconsciously separate the rich and poor, educated and non-educated, leaders and followers, and racial and ethnic groups” (p. 18). Povey

(2003) argued in terms of *democratic* mathematics in the context of *citizenship* education. Citizenship was defined in the British context, and the problematic nature of what constituted citizenship and democratic mathematics and what democratic mathematics classrooms might look like were discussed. There was a perceived need for mathematics education to move towards “a deep democracy in which equality and person centredness, engagement, and creativity are recognized as key to effective participation” (p. 61).

In the late 1980s, the construct of ethnomathematics focused the attention of mathematics educators on the “democratic” question of what kind of mathematics should be taught and learned in schools. Various analyses of this construct (Bishop, 1988; Carraher et al., 1985; D’Ambrosio, 1986; Gerdes, 1995; Powell & Frankenstein, 1997) have stretched and generalized it from its roots in the knowledge systems of indigenous communities in third-world countries to encompass any specific and localized forms of mathematical knowledge in any society—carpenters, gamblers, street traders, and so on—who all have their specific knowledge and practices. Bishop (1988) for example wrote about and differentiated between *Mathematics* with an upper-case *M* and *mathematics* with a lower-case *m*, where the second referred to all mathematical knowledges and the first referred to the kind of supposedly culture-neutral and “universal” mathematics that the university mathematicians in Western societies research and teach, and that forms the basis of most mathematics curricula in schools today.

This is a similar position to the argument that the concept of ethnomathematics embraces every form of mathematical knowledge, including the so-called universal mathematics of Western university mathematicians (see, for example, D’Ambrosio & Ascher, 1994), and therefore the educational argument should be about what kind of ethnomathematics to teach. Should one teach *Mathematics* or *mathematics*, including the “specific” ethnomathematical knowledges and practices relevant to the cultural and societal context of the students? And if the answer is “both” then what are the best ways of factoring these different knowledges into a school mathematics curriculum?

This enlarged construct of ethnomathematics does not appear to have been widely accepted as offering a structure for determining the appropriate mathematics curriculum in any given educational context. For example, according to Kassem (2001), despite early hopes and intentions of including cultural referents and contexts, the UK national curriculum has never had a multicultural dimension. The NCTM (2000) *Principles and Standards* document likewise makes no

reference to any cultural contexts for reinforcing or contrasting mathematical practices.

A more widespread and accessible democratizing construct nowadays appears to be that of *numeracy*, despite the fact that different definitions of the term abound (see Jablonka, 2003; Steen, 1990). Gellert, Jablonka, and Keitel (2001) wrote about different conceptions of *mathematical literacy* and their relationships to mathematics, reality, and society, in the context of individuals’ needs for civic participation. Frankenstein (1998) presented four goals (with examples) of a critical mathematical literacy curriculum that would enable teachers to “teach mathematics as a tool to interpret and challenge inequities in our society” (p. 180). The four goals are understanding the mathematics, understanding the mathematics of political knowledge, understanding the politics of mathematical knowledge, and understanding the politics of knowledge.

What the introduction of this term contributes to the equity discussion on school mathematics education is that, despite the different details of the definitions, it focuses attention on the mathematical knowledge deemed necessary to be a competent citizen in a democratic society (see various articles in Kelly, Johnston & Yasukawa, 2003; Steen, 2001). This is surely what equity in mathematics education is all about.

From an equity perspective we would argue that schools certainly *should* give all students access to the so-called universal mathematics knowledge (from here on we shall use the term *Mathematics* with a capital *M*) but schools *must* prepare all students to be competent citizens and should therefore be teaching numeracy to all students.

Numeracy is a widely used term with a myriad of definitions that embrace many of the “mathematics for all” notions described above. According to the Government of Canada (n.d.),

Before the word “numeracy” came into use, there was discussion of terms such as “mathematical literacy” and “quantitative literacy,” which placed the focus on calculations and the ways in which numbers and mathematical concepts were embedded in texts, but which did not take into account the wider practical uses of numbers and mathematics in the workplace and in personal life on an everyday basis.

The Australian Association of Mathematics Teachers (AAMT, 1998) defined what it means to be numerate as being able “to use mathematics effectively to meet the general demands of life at home, in paid work, and for participation in community and civic life” (p. 1). For school education, numeracy was considered to be “a fundamental component of learning, dis-

course and critique across all areas of the curriculum” (AAMT, 1998, p. 1) and involved

the disposition to use, in context, a combination of: underpinning mathematical concepts and skills from across the discipline (numerical, spatial, graphical, statistical and algebraic); mathematical thinking and strategies; general thinking skills; [and] rounded appreciation of context. (AAMT, 1998, p. 1)

The relationships between numeracy and Mathematics seem logically to be of five kinds:

1. Mathematics and numeracy intersect, that is they share aspects but do not include each other.
2. Numeracy is a subset of Mathematics.
3. Mathematics is a subset of numeracy.
4. Numeracy is Mathematics.
5. Mathematics and numeracy are two very different phenomena, having no relationship.

From an equity perspective, choices 4 and 5 do not appear to offer any useful ideas for structuring the intended school curriculum, which satisfy the criteria described in the paragraph above. Nor do many argue for such positions. Only Choices 1, 2, and 3 are supported in the numeracy literature (see Jablonka, 2003; Kelly, Johnston & Yasukawa, 2003), with 2 occupying the least radical position, followed by 1, and then 3, the most radical.

Willis (1996) identified four broad perspectives on the relationships between the mathematics curriculum, disadvantage, and social justice: remedial, non-discriminatory, inclusive, and socially critical. A framework was developed that included statements completing the following sentence stems with respect to each of the four perspectives: “The mathematics curriculum is . . .,” “The problem with ‘disadvantage’ lies with . . .,” “The solution is to . . .,” and “The educational task is to . . .” (Willis, 1996, p. 47). The framework was used with teachers and curriculum writers. Willis (1996) claimed that this enabled them to understand their own positions better and provided “a starting point for developing a more consistent approach to issues of gender justice and the curriculum” (p. 48).

Various curricular approaches have attempted to embed the mathematics in realistic contexts as a way of trying to get the equitable “best of both the mathematical and numeracy worlds,” so to speak. In Holland the most successful approach has been that of the Realistic Mathematics Education (RME) project, based on the educational philosophy of Hans Freudenthal (1983). Under his philosophy the learners

should begin with real, or at least realistic, problem situations from which the teacher can lead them into exploring the relevant and significant mathematical concepts and techniques. From the perspective of the Mathematics in Context project, Romberg and Shafer (2003) described a design study based on RME principles of first-year algebra teaching in a U.S. high school. In this study, all the students had the opportunity to succeed and, in doing so, exceeded the researchers’ expectations. There was also consistent evidence of higher-order thinking and analysis in all of the classes, and not just in the honours class.

Everything depends of course on definitions of *realistic*, *problem-based*, *numeracy*, and so on, but for determining appropriate curriculum structures to be equitable and accessible to all, the key arguments are the same: Will all students study the same curriculum? If so, how and by whom will the pedagogical and micro-curricula decisions be made, in order to involve all students? If not all students will study the same curriculum, how will the choices be made? How will the different curricula be managed and structured? Will different curricular sequences run in parallel, or will the differences relate to the different timing or depth of topics taught?

Mathematics education in schools is thus seen to have a dual function: to prepare students to be mathematically functional as citizens of their societies—arguably provided equitably for all—and to prepare some students to be the future professionals in careers in which mathematics is fundamental, with no one precluded from or denied access to participation along this path. The notion of more than one purpose for mathematics education raises issues of what constitutes an equitable mathematics curriculum and raises questions about the equity implications of systemic, school-based, or classroom-based practices in mathematics education such as tracking/streaming or single-sex settings.

This then moves us into considering the institutions of mathematics education and the research that focuses on issues and challenges at that level.

INSTITUTIONAL STRUCTURING OF MATHEMATICS EDUCATION

Although institutions can be considered in various ways, the most appropriate for this chapter’s focus is the idea of the Opportunity to Learn (OTL). This construct first made its appearance in the studies organized by the International Association for the Evaluation of Educational Achievement (IEA; Keeves,

1995), as the opportunity that students have to learn the content of the items in the tests. Clearly however this definition is limited and could go much wider, particularly if one is interested in equity research.

As an example of the broadening of the OTL concept, in 1994, US President Clinton signed education reform legislation known as *Goals 2000: Educate America Act*, making *opportunity to learn* standards voluntary (NCEO Policy Directions, 1995). In *Goals 2000*, OTL standards were defined as

the criteria for, and the basis of, assessing the sufficiency or quality of the resources, practices, and conditions necessary at each level of the education system (schools, local educational agencies, and States) to provide all students with an opportunity to learn the material in voluntary national content standards or State content standards. (*Goals 2000: Educate America Act*, 1994, Section 3a(7))

The OTL standards addressed the following educational dimensions (NCEO Policy Directions, 1995):

- curricula, instructional materials, and technologies
- teacher capability
- continuous professional development
- alignment of curriculum, instructional practices, and assessments with content standards
- safety and security of the learning environment
- nondiscriminatory policies, curricula, and instructional practices
- other factors that help students receive a fair opportunity to achieve the knowledge and skills in the content standards

Whereas poor and minority students were recognized by proponents of OTL, NCEO Policy Directions (1995) identified the exclusion of students with disabilities.

Oakes (1995) wrote in support of the OTL standards, claiming that they would provide information enabling insights to be gained into why students in particular schools scored as they did in content-standards-related testing—the focus would no longer be exclusively on student performance.

Lynch and Baker (2005) discussed equality in education from the perspective of *condition*, which they defined as “the belief that people should be as equal as possible in relation to the central condition of their lives” (p. 132). They claimed that there were five dimensions of equality: “resources; respect and recognition; love, care and solidarity; power; and working and learning” (p. 137). The OTL characteristics and

Lynch and Baker’s dimensions of equality of condition resonate strongly with mathematics educators’ specific attempts to grapple with definitions of equity, equality, justice, and opportunity to learn within the framework of mathematics learning (see, for example, Hart, 2003). Fennema (1993) defined equity in mathematics education as composed of three parts: “(1) equity as equal educational opportunity, (2) equity as equal educational treatment, and (3) equity as equal educational outcome” (p. 2). She contended that

while there is a semblance of equal opportunity to learn mathematics in today’s world, mere semblance is not enough. . . . Are there less obvious ways in which inequality exists, . . . Legally enforcing equity is a prerequisite to the achievement of equity; but it is not a sufficient condition. . . . Justice in mathematics will not be achieved until the goals of education are met equally by both sexes. (Fennema, 1993, p. 5)

The notions of *equity*, *justice*, and *opportunity to learn* were intertwined in Fennema’s (1993) words. Although Fennema’s definitions and discussion were set in the context of gender, they are clearly applicable to all dimensions of disadvantage and inequality with respect to mathematics education. The work of Allestaht-Snyder and Hart (2001) also extends the notion of equity in mathematics education beyond gender. They contended that “teachers’ work with students in individual classrooms is at the heart of any program designed to attain equity for all students” (p. 93). They recognized that teachers could not do this alone and identified the need for collaborative effort among members of the wider educational community. They used a definition of equity based on two premises:

- that all students, irrespective of race, ethnicity, class, gender, or language proficiency, will learn and use mathematics
- that all associated with children’s education be aware of social, economic, and political factors that hinder or facilitate the mathematics learning of underrepresented students.

For equity in mathematics education to be achieved, Allestaht-Snyder and Hart (2001) maintained that there were three conditions to be met:

1. equitable resource distribution: schools, students, and teachers
2. equitable instructional quality: teachers’ mathematical knowledge, their preparation for mathematics teaching, and their beliefs about and skills in teaching students from diverse backgrounds;

3. equitable student outcomes: achievement, course enrollments, interest, motivation, and valuing of mathematics

Equity would be achieved, they claimed, “when differences among sub-groups of students in these three areas are decreasing or disappearing” (p. 93). Allexsaht-Snider and Hart’s (2001) three conditions for the achievement of equity in mathematics education clearly resonate with Fennema’s (1993) definition of equity in mathematics education. The first two common categories of equity are the organizing parameters for the discussions that follow in this section of the chapter.

Rousseau and Powell (2005) framed their study of equity in reform mathematics on the three variables which they argued shape students’ opportunities to learn mathematics: time, quality, and design. They identified several factors that operated as barriers to reform for one of the two teachers in the study—large class size, high-stakes-standardized tests, high student mobility and absenteeism, and lack of a high-quality curriculum. This female teacher worked in an urban school and her practices were closer to the traditional end of the traditional-reform continuum than were those of the other teacher, a male who taught in a suburban school. According to Rousseau and Powell (2005), the identified barriers to reform noted “were more likely to be present in schools with high proportions of students of color or students living in poverty,” students who were “more likely to be denied access to the kinds of curricula and teaching practices that are associated with mathematics reform” (p. 29).

According to Silver (2003), rural communities worldwide have high poverty levels and low educational attainment levels, and mathematics educational researchers focusing on equity considerations have largely ignored rural students. It is also interesting to note that learning disabilities and physical disabilities are frequently overlooked as categories of inequity in the mathematics education literature. In Australia, van Kraayenoord, Elkins, Palmer, and Rickards (2000) conducted a literature search on literacy, numeracy, and students with disabilities and concluded that there had been greater research emphasis on literacy than on numeracy and that there was a lack of information on all groups of students with disabilities.

Lubienski and Bowen (2000) reported that there were many articles on mathematics and disability published between 1982 and early 1998 to be found in the ERIC database. They claimed that about half of the articles dealt with cognition, most articles were not found in mathematics education journals, and only a handful also included other categories of equity.

It is not clear from their writing, but it appears that Lubienski and Bowen were referring to articles about learning disabilities and not physical disabilities. For this chapter, some research on mathematics learning and physical disabilities, and on mild learning disabilities, have been included in the review.

Equity of Access—Resources: Schools, Students, and Teachers

Oakes (1995) argued that “an adequate and fair distribution of resources, programs, and teachers won’t, by itself, guarantee that disadvantaged students will learn well.” (p. 86). It was noted, however, that not much improvement could be expected from the students without sufficient funding being directed into high-quality instruction. In this section, a selection of research studies illustrating inequities associated with a range of resource issues is presented.

Fenwick (1996) reported that students in poverty (i.e., from low socioeconomic backgrounds) in the United States “are more likely to be taught by inexperienced teachers . . . and by teachers who are either without any certification or not certified to teach the subject they are teaching” (p. 4). This was exacerbated for the poor urban students (mainly Hispanic and African American). These teachers were also likely to have negative attitudes towards the students and were less likely to encourage them.

Secada, Cueto, and Andrade (2002) argued that “group-based inequalities in the distribution of social good,” in their case the opportunity for students to learn mathematics, was “bad social policy from a position of socially enlightened self-interest” (p. 108). Using a wide range of data-gathering methods, their study involved an examination of the opportunities to learn mathematics of Peruvian students from Puno (in southern Peru) and the kinds of schools attended by fourth and fifth graders. The general poverty of the people was reflected in the schools’ resources. Particularly in the regional areas, many schools lacked basic facilities such as water, sewage, and telephones. Good blackboards, however, were found in all schools. The recently reformed mathematics curriculum was described as having a balanced representation of mathematics, yet there had been little change in practices. Arithmetic was the focus, and it was taught by rote. Secada et al. regarded their findings as “showing the crisis in Puno’s overall educational system” (p. 129).

Kitchen (2001) described Guatemala as a country in which political turmoil and student activism were commonplace, where there was inadequate government support for education, teacher professional development was nonexistent, and teachers often had

to work additional jobs to make ends meet. The two teachers in Kitchen's study both worked in inner-city schools attended by an ethnic mix of poor students; one was a school for girls and the other a school for boys. According to one of the teachers, "economic class, not ethnicity, was the great equalizer in Guatemalan society" (Kitchen, 2001, p. 159). Kitchen (2001) maintained that the political and social context in which the two teachers worked was the basis for the conventional pedagogical style they adopted.

In the context of post-apartheid South Africa, Adler's (2001) findings from a 3-year study that focused on learner-centred pedagogy and "mathematical practice that moved beyond 'mathematics as procedures'" (p. 187) challenge notions that equality of resourcing will necessarily bring about equity in opportunities to learn. Classroom observations were conducted in various school settings: urban and rural, primary and secondary, and in a range of relatively well-resourced to impoverished schools. For some students, teachers' use of more resources extended their opportunities for mathematical engagement and growth. For others, particularly in the poorest schools, students' learning opportunities were narrowed. Adler (2001) maintained that the teachers' personal mathematical and professional backgrounds and the learning contexts in which they operated had shaped their changing practices. As a consequence, Adler (2001) contended, it appeared that there had been an exacerbation, rather than amelioration, of the existing inequalities.

Using Fennema's (1993) three components of equity for mathematics learning—access, treatment, and outcomes—equity issues associated with the use of technology for mathematics learning were discussed by Hennesy and Dunham (2002). Inequities with respect to computing technology and the Internet included

- the "global digital divide" between rich and poor countries in terms of numbers of computers per student and Internet access
- inequitable distribution of technology resources between schools with different racial/ethnic profiles and socioeconomic status, and in urban and rural areas
- unreliable technology more likely to be found in disadvantaged schools, and
- gender gaps in use at home and at school.

Access to hand-held technologies (calculators) in schools was considered more equitable; with sophistication and cost increases, inequities were considered likely to increase, however. The extent of technology use was found to be related to the grouping level in which

students are taught when setting/streaming/tracking by "ability" is practiced. Home access to computers was reported as more likely in middle-class and suburban homes. Teachers' attitudes towards and experiences with technology, as well as their biased/stereotyped beliefs and expectations of particular groups of students, were factors that could affect the extent of technology use in their teaching. Inequities in outcomes included achievement and attitudinal differences. Hennesy and Dunham (2002) concluded that "opportunities for access are only a starting point; they may mask physical or psychological restrictions on use and may not translate into equitable use of technology in practice" (p. 158), and they claimed that a radical departure from conventional, inequitable schooling methods, particularly for disadvantaged groups, was needed.

In his book on educating the deaf, Moores (2001) discussed several issues associated with mathematics learning. He stated that "for more than two hundred years, the primary emphasis among educators of the deaf has been communication, with academic achievement receiving secondary attention." (p. 8). Thus, across the curriculum, deaf students were said to spend less time than hearing students on learning academic subjects, including mathematics. At 7 years of age, Moores (2001) claimed, their mathematics achievement scores reflected this difference. Mathematics received "relatively little attention in programs for young deaf children," the time allocated to mathematics was minimal, and "teachers of deaf students tend to underestimate the need for mathematics instruction" (Moores, 2001, p. 336). In preservice courses for teachers of the deaf, communication was central and those teaching mathematics to deaf students from kindergarten through to Grade 12 were relying on "traditional practices such as rote memorization, work sheets, and drill and practice" (Moores, 2001, p. 9). Deficiencies in the preservice and inservice education programs for mathematics teachers of the deaf with respect to mathematics content and the mathematics background of participants were identified. Moores (2001) felt that much was needed "to provide both equity and excellence to . . . deaf students" (p. 10).

Equity and Separated Settings

Segregation has been a common institutional response to the management of differences in education. The degree of segregation varies historically and cross-culturally for different social groups, and often occurs invisibly through broader patterns of residential segregation, selection procedures and parental choice. (Lynch & Baker, 2005, pp. 145–146)

In the educational literature researchers have continued to debate the relative merits of educating particular groups of students in segregated or mixed settings, for example, coeducational or single-sex settings, regular or special classes for those with learning disabilities or physical disabilities. As noted by Lynch and Baker (2005), segregation can result indirectly through circumstance, geography, or choice. Thus, for example, segregation by socioeconomic status (e.g., public versus private education) or by ethnic or racial groupings are found in various contexts. Issues of equity are clearly associated with these issues and are discussed in the literature. Because mathematics serves as a gateway to a number of career options, researchers have shown considerable interest in exploring which educational settings optimize equitable access to, and opportunity for, mathematical learning.

In the area of physical disabilities, the example of hearing-impaired students illustrates the issues. The increasing range and definitions of placement options currently open to students with hearing disabilities in the United States, and the legislative changes that have led to them, were described by Moores (2001). The growth in these options is fairly recent and ranges from residential schools, day schools (only for the deaf), and day classes (classes for the deaf within mainstream schools), to resource rooms (where deaf students return from regular classes to receive special attention) and itinerant programs (students attend regular classes and itinerant teachers assist—schedule of assistance can vary). In Australia, another option is available; students with disabilities (including the deaf) can attend mainstream schools with personal aides (publicly funded) accompanying and assisting them in all lessons. Research on the effects of the different learning settings is sparse. Moores (2001), however, noted that

growing numbers of deaf children are being taught in public school settings. . . . Interest in applying the results of work with hearing children to education of deaf children is growing. Improvements in academic achievement have already been documented, although the gap between scores of deaf and hearing children is still unacceptably large. (p. 337)

In the past, hearing-impaired students' difficulties with mathematics had been thought to be related to the linguistic content of problems or with the students' English-language competence. Kelly, Lang, and Pagliaro (2003) surveyed Grade 6–12 teachers of deaf children in mainstream settings (some in integrated classrooms; others from self-contained classes for the deaf) and in center schools (boarding or day schools for the deaf only). They found that, irrespec-

tive of the setting, deaf students were insufficiently engaged with challenging word problems. Interestingly, teachers in the mainstream classes were found to have higher perceptions of deaf students' problem-solving abilities than the teachers in deaf-only classes. Consistent with Moores's (2001) claim regarding the main focus in their preservice education, the teachers of deaf students attributed the students' low abilities with word problems primarily to their skills and abilities with English.

As another example, one aspect of Tracey's (2002) doctoral study focusing on children with mild intellectual disabilities was also concerned with investigating the effect of regular class placement and special class placement on the children's self-concepts, including mathematics self-concept. It was found that students placed or moved into special classes had higher academic, including mathematics, self-concepts than their counterparts in regular classes, a finding consistent with an earlier study (Burton, 1988). Tracey (2002) claimed that the findings questioned the appropriateness of the continued trend internationally towards the inclusion of students with disabilities into regular classrooms. In a similar vein, a study of students in their first year of schooling by Arnold (1994) found that parents of children with disabilities in the mainstream classes were very satisfied with their social outcomes but less satisfied with their academic progress. Achieving the best balance between those two goals is a major issue for teachers and parents regarding separated educational settings.

Moreover, particular challenges occur when the mainstream curriculum is changed. For example, according to the ERIC Development Team (2002), "many students with disabilities experience difficulties with the [U.S.] reformed math curriculum" (p. 2) including processing and distinguishing relevant information, computational skills, and a lack of reasoning and problem-solving skills. However developments in new information and communication technology can facilitate the learning situation particularly for students with disabilities. For example, in the past, blind students who were unable to read or write the symbols that comprise mathematics had to learn concepts and perform calculations mentally, thus limiting their ability to master the details of mathematics. The development of interactive software was said to help students study the Braille code of mathematics (the Nemeth Code).

Steele's (2004) answer to the dilemma about special versus mainstream teaching was to argue that as the majority of students with learning disabilities (LD) are today being educated in mainstream classrooms, more research attention should be paid to appropri-

ate teaching strategies that mainstream teachers can use. She also argued that such changes could equally facilitate the mathematics learning of the students without disabilities. Her summary of these strategies included the following suggestions:

- Provide advance organizers to introduce the purpose of the lesson.
- Provide additional review of all prerequisites as needed.
- Prioritize, teach, and review major concepts frequently.
- Teach generalization and application to real-life situations.
- Model sequential procedures at a slow pace and with extra clues.
- Present new skills using concrete materials, then pictures, and finally abstract explanations.
- Provide additional practice in small steps with sufficient guidance.
- Be sure directions are clear before starting independent practice.
- Teach students to keep track of their progress with charts and graphs.
- Check for error patterns and related corrections when providing guidance.

Another context in which the issue of segregated or integrated mathematics education is hotly debated concerns single-sex or coeducational settings. The extent of single-sex schooling varies from country to country around the world. Within some countries, gender-segregated schooling is a religious or a cultural norm; in others, it is seen as an alternative for parents to choose for their offspring. Gill (1988) reported that the findings from comparisons of the academic performance and school-related attitudes of students attending single-sex and coeducational schools were inconsistent. In Australia (and the UK), “the distinction between single sex and coeducational schools is interwoven with the division between private and public schooling” (Gill, 1988, p. 3), which, in turn, is closely associated with social class differences. Often, too, the arguments mounted (particularly in the popular press) are shaped by the vested interests of the protagonists, the indicators of success adopted, and factors within the societal context in which the debate is held (Haag, 2000; Leder & Forgasz, 1997).

Single-sex education has been viewed as anachronistic, reflecting times when males and females had different educational needs related to their gendered future roles in society, and for some time, coeducation was considered a means for achieving gender

equality (Leder, 1992). More recently, however, the equity assumptions in support of coeducation have been challenged.

A number of single-sex interventions have addressed gender inequities in mathematics learning outcomes. Leder, Forgasz, and Solar (1996) described a vast range of such interventions that have taken place in different settings, for different lengths of time, for different age groups, and with different aims. The SummerMath program (Morrow & Morrow, 1995) in the United States was one short-term out-of-school intervention that succeeded in assisting 13–18-year-old females to see that they could do mathematics confidently and that mathematics is connected to people and their lives. Several researchers have examined single-sex interventions implemented within regular coeducational secondary school settings (e.g., Forgasz & Leder, 1995; Leder & Forgasz, 1997; Rennie & Parker, 1997). The findings were mixed. Forgasz and Leder (1995), for example, concluded that although “beliefs were that females would benefit most from single-sex classes, there were signs that males derived equal, if not more, benefit from the program than the females” (p. 44). Rennie and Parker (1997) found evidence from teachers that

single-sex classes appeared to hold the most benefit for specific groups of girls who were experiencing a great deal of harassment from boys in mixed-sex classes, and the least benefit for high-achieving girls and boys and for boys in some classes which were particularly difficult to discipline. (Parker & Rennie, 1995, p. 8)

Interestingly, Parker and Rennie (1995) found that the teachers used different strategies in the all-male and all-female classes and that they became more aware of the different needs of girls and boys that may have been missing in coeducational settings.

In the United States, Becker (2001) described the circumstances behind the establishment of public, single-gender academies on a single campus that had previously housed a coeducational high school. Due to a lack of planning and a strong philosophical undertaking for the single gender academies, Becker (2001) felt that the opportunity to experiment with curriculum and instructional approaches to enhance females’ learning opportunities had been wasted, particularly in the mathematics classroom.

The findings discussed above are consistent with the conclusion of Forgasz, Leder, and Vale (2000), who summarized several studies on single-sex mathematics interventions:

Simplistic solutions, such as single sex classes per se, do not appear to have been successful in themselves

in achieving equity and there have been calls for new strategies to be explored. . . . Epistemology, pedagogy and parents' perceptions remain important factors. (p. 323)

Equity and Streaming/Tracking/Setting for Mathematics

The debate on the advantages and disadvantages of streaming/tracking/setting students generally is a long and bitter one (e.g., Edwards, 1998; Loveless, 1998). Over time, mathematics appears to be one of the few disciplines that has attracted continued support for classroom segregation by achievement level, often termed *ability grouping*. In various national and international settings, streaming/tracking/setting of students is practiced to different degrees either systematically or informally at all levels of education. Segregation can occur in several different ways: within-classroom groupings; grade levels within schools with students ranked and divided into homogenous groups, or "topped and tailed" (that is, creaming off the highest and the lowest achievers, with the rest assigned to fairly mixed groups); within-school course selection tracks, for example, academic and technical tracks; within-system tracks, for example, academic (university) bound schools and technical schools.

When streaming/tracking/setting occurs, sometimes the core mathematics curriculum offered is said to be the same, with more time for extending beyond the core for the higher achievers; at other times the curriculum is clearly differentiated with less demanding mathematics offered to the lower streams (Oakes, 1995) and less abstract, theoretical mathematics and more "practical" mathematics offered to those in the technical tracks. The equity-related questions that arise as a consequence of streaming, tracking, or setting, particularly when they occur during the compulsory years of schooling, include the following:

- Who is found in and who teaches the lower streams and technical tracks?
- Are there differences in enrollments in the lower streams and technical tracks by gender, race/ethnicity, social class, or language capabilities?
- Is a differentiated mathematics curriculum equitable?
- Have students' life opportunities been limited by the practices of streaming/tracking/setting?

According to Oakes (1995), minority students are overrepresented in low-track classes and underrepresented in advanced courses and programs for gifted

and talented students. Schools also tend "to place their least qualified teachers with low-ability classes" (p. 87), and expectations of the students in these classes are low. Moody (2001) studied the social and cultural factors in the mathematics learning experiences of two successful female African American students. Both were exposed to tracking at various levels of their schooling. Through tracking, one of the students was isolated from her African American peers who were in lower tracks, many of whom "associated being in high tracks with acting White" (p. 271). Moody concluded that the girls' "experiences exemplify the attributes of tracking that maintain and perpetuate inequalities and inequities in society . . . [and that] tracking seems to *resegregate* our mathematics classrooms, our schools, and our society" (p. 272).

Citing a report from the media, Bartholomew (2002) reported that by the upper secondary years, 94% of students in Britain are taught in setted (streamed) mathematics classes and that

the composition of the different groups is sharply polarised along social class lines, with middle class students concentrated in the higher sets and working class students in the lower sets. . . . In mathematics, it is also the case that boys are frequently over-represented in top set classes. (p. 138)

Bartholomew (2002) also noted that boys continued to outperform girls in top-set mathematics classes.

Even in *high* sets (upper streams), some students have been found to be disadvantaged (Boaler, 1997). Boaler reported that the students' lower-than-expected achievement levels and high anxiety levels, particularly among those from lower socioeconomic backgrounds, were due to the procedural teaching approaches adopted and the fast pace of lessons.

Linchevski and Kutscher (1998) reported findings from three related studies involving Israeli students' learning of mathematics in mixed-ability and same-ability settings. They found that students' mathematics achievements need not be compromised by learning in heterogeneous ability settings. Any increase in the achievement gap due to learning in homogeneous settings was said to be due to the lowered achievements of weaker students rather than the higher achievements of stronger students. There were also positive effects on teachers' attitudes towards mixed-ability mathematics teaching as a consequence of participation in the study.

Equity in Quality of Teaching and Learning

To attain equity in the mathematics classroom, Burton (1996) wrote of a *socially just pedagogy* that in-

volved capitalizing on the heterogeneity in the classroom. Burton saw the need to “celebrate the diversity to be found in classrooms rather than attempting to turn all pupils into carbon copies” (p. 141) or matching “every pupil against a desired norm” (p. 142). Expanding on this, she maintained that “the clash of meanings, interpretations, styles of learning, methods, representations, knowledges, uses for mathematics—all of these, and others I have failed to mention—underpin a rich culture in every classroom” (Burton, 1996, p. 141). For this model of a socially just pedagogy, Burton (1996) claimed that there was a need to change one’s understandings of the nature of mathematics and the conditions supporting its learning, that is a shift from an objective, context-neutral, value-free view of mathematics to one in which mathematics is accepted as a human construct with an emphasis on relationships rather than properties.

Ladson-Billings (1995) defined the notion of an *equity pedagogy* as “the opportunities that *all* children have to benefit from classroom instruction” (p. 130). She described several learning contexts in which inequities in teaching have been identified:

In the single classroom a teacher can deny some students access to educational opportunities while providing special opportunities for others. Sorting, grouping, or tracking students into lower levels where they receive minimal or no instruction from the teacher, while the higher-track students are presented with challenging and intellectually stimulating curriculum and instruction, is an example of how such inequity is structured. (Ladson-Billings, 1995, p. 130)

What can be inferred from the arguments of both Burton and Ladson-Billings is that equity in the mathematics classroom is most likely to be attained in the heterogeneous classroom in which teachers deal successfully with all aspects of the diversity among students—gender, race/ethnicity, language capabilities, social class, special needs, and the full range of mathematics achievement levels. Some researchers exploring equity issues have worked in heterogeneous classrooms. Others have focused on one dimension of equity by examining more homogenous settings including, for example, single-sex classes (homogeneity of gender) and tracked or streamed classes (homogeneity of achievement). One could argue that the Burton and Ladson-Billings models require teachers to have sound mathematical content knowledge and an exceptional range of pedagogical skills. Classroom based research on equity has also involved a focus on pedagogical approaches. In other studies, preservice teacher educa-

tion and ongoing professional education needs have been explored. Representative examples of the range of research undertaken related to mathematics classrooms and equity are described below.

In a year-long ethnographic study, Zevenbergen (2001) analyzed classroom interactions in two Australian classrooms—one in a nongovernment school with a middle-class and elite clientele; the other in a government school serving children of working-class parents. Analyses of episodes with examples of triadic dialogue—teacher asks question, student responds, teacher evaluates response—revealed that the middle-class students participated in the exchanges, whereas the working-class children seemed to resist or failed to recognize the structure of this form of interaction. Zevenbergen (2001) proposed that “one subtle and coercive way that some students are advantaged and others excluded in and through the practices of mathematics is through the dis/continuities between the linguistic habitus¹ of the students and the practices of classroom interactions” (p. 213).

Whereas Zevenbergen focused on social class, Khisty’s (1995) concern in another ethnographic study was with Hispanic students whose mathematics achievements were consistently below the U.S. national average. Rather than focusing on deficit models that assume the need to change students, Khisty (1995) argued for a paradigm shift premised on school failure that involved inadequate and inappropriate instruction given to disadvantaged poor and minority students. Five fully bilingual elementary classroom teachers in schools with high Hispanic populations were observed at various times over a year; 7 to 10 hours of videotape were gathered from each and the classroom discourse analyzed. Two patterns of teacher discourse were identified. The first involved teachers’ efforts in attending to the language of mathematics, and effective and ineffective techniques were noted. The second pattern that emerged concerned the minimal use of Spanish in the mathematics context, with the exception of one teacher. This had occurred even though many students either did not speak or were weak at English. The second pattern is unlikely to have surprised Barwell (2002), who argued that even in mathematics education research, the disadvantage of those who speak less favored languages—and English is often the default, favored language—can be reinforced.

But what happens when students from a range of different language backgrounds are found in the one mathematics classroom and a common language of communication is needed? In the Australian setting, where English is the dominant language, Thomas

¹ Zevenbergen (2001) defined *habitus* as “the embodiment of culture and the lens through which the world is interpreted” (p. 202).

(1997) proposed a range of school-and classroom-based strategies to address the needs of these students: providing more time through extra classes; recognizing that mathematics symbols are not universal, and that in English, mathematical language can have different forms; developing concepts before language; and teaching the four language skills of speaking, listening, reading, and writing within the mathematics classroom.

Carey, Fennema, Carpenter, and Franke (1995) claimed that the development of *diverse curricula for diverse groups* and of *one curriculum for all* were both failed curricular strategies aimed at achieving equity in mathematics education. They maintained that research findings suggested that certain universals in the ways people learn mathematics cut across cultures and that when these were taken into account learning was improved. The *Cognitively Guided Instruction* (CGI) program (see, for example, Carpenter, Fennema, Franke, Levi, & Empson, 1999), they claimed, was one that blends equity concerns with knowledge derived from mathematics education research about how people learn mathematics and contributes to the development of equitable classrooms. Its success was dependent on teachers being concerned about every child and willing to challenge and improve their teaching practices.

Berry (2003) presented findings from three studies in which mathematics classroom practices were consistent with the NCTM (2000) standards and argued that they provided evidence that such classrooms had positive effects on the mathematics achievement of African American students.

Gutstein (2003) described his own attempts, over a 2-year period, to teach mathematics for social justice to a class of Latino students. The three main goals were to help students develop their own social and political consciousness, sense of agency, and social and cultural identities. To achieve these goals, Gutstein (2003) used a curriculum based on the *Mathematics in Context* (Freudenthal Institute) and real-world projects. Gutstein (2003) concluded that although the process was complex, it was possible to teach mathematics so that students could “read the world” with mathematics, develop mathematical power, and change their attitudes towards mathematics. The main factor contributing to the students’ growth, Gutstein (2003) contended, was the classroom environment engendered, in which meaningful issues of justice and equity were discussed.

In another self-study, Lubienski (2000) used a problem-centred curriculum and pedagogy with a focus on socioeconomic backgrounds to see how students reacted to learning mathematics through

problem solving. The lower socioeconomic background students, particularly the females, indicated a strong preference for traditional instruction. They were the most frustrated with the openness of the problems and the teaching approach. Lubienski (2000) was careful not to generalize beyond the sample of one class of students who participated in the study. It was suggested, however, that reform-oriented pedagogy might improve low and high socioeconomic students’ understanding of mathematics while, at the same time, increasing the gap in performance in favor of the high socioeconomic students.

Equity and Affective Factors

The affective domain may not have received as much attention as the cognitive domain among mathematics education researchers. Nonetheless its relevance to mathematics learning is clearly recognized, and the interaction of affect and cognition, for example, was embodied in the research studies reported by McLeod and Adams (1989). With respect to equity, affective factors were clearly identified in the past as contributors to observed gender differences favoring males in mathematics achievement and participation rates (e.g., Fennema & Leder, 1993; Leder, 1992). More recently, research on affective factors associated with mathematics teachers, and with students of mathematics, has been associated with a range of equity dimensions. Findings from some of this work are discussed below.

Zevenbergen (2002) gathered data from 50 Australian teachers on their beliefs about teaching mathematics to socially disadvantaged students. She found that “there was a heavy emphasis on deficit models and a pathologizing of students and families” (p. 148) and that the teachers had low expectations of the students. The literature on beliefs, Zevenbergen (2002) argued, would suggest that the teachers’ practices would reflect their beliefs, with social differences becoming educational ones.

The beliefs of large samples of Australian and U.S. students about the gender stereotyping of mathematics were examined by Forgasz (2001). Data were also gathered on preservice teachers’ views of how students would respond to the same sets of items. For some aspects of mathematics learning, the students’ views were inconsistent with previous research, for example, girls were considered more likely than boys to enjoy mathematics and to be good at it. However, the preservice teachers’ views on how students would respond were consistent with the predicted gendered directions of responses based on earlier findings in the field. Interestingly, in both countries, the views of

the students were similar, as were the views of the pre-service teachers; this was considered partially attributable to the similarities in the mainstream cultural and social dimensions in Australia and the United States. Forgasz (2001) concluded that the preservice teachers' views were "out of touch" with contemporary high school students' views on the gender stereotyping of mathematics.

Atweh, Bleicher, and Cooper (1998) examined the effects of students' socioeconomic backgrounds and gender on teachers' perceptions of their needs and abilities. The teachers and students in two Grade 9 classes participated in the study; one was an all-girls class in a low socioeconomic school, and the other an all-boys class in a high socioeconomic school. Based on interviews and classroom observations, two conclusions were reached. First, the teachers' conduct of their classes was consistent with their perceptions of the students. Second, both teachers were aware of their students' gender and socioeconomic backgrounds. Classroom interactions, it was argued, were consistent with teacher perceptions, and this would "tend to have a self-fulfilling role for teacher expectations" (p. 80).

Ortiz-Franco and Flores (2001) claimed that in the absence of adequate support, many teachers have been successful in providing quality mathematics education for Latino students. Others, however, were said to have "biased views and low expectations for their Latino students that negatively influence their teaching and consequently affect student learning" (p. 245). In their study, two intervention programs had been successful in challenging and changing teachers' attitudes, beliefs, and classroom practices with consequential positive effects on students. Findings from a 3-year study on equity and technology use in secondary mathematics learning indicated that teachers who believed there were differences in the ways boys and girls worked with computers considered girls to be less confident, competent, and interested in using computers than boys (Forgasz, *in press*).

In Britain, deaf children's mathematics achievements are lower than those of hearing children, and there has been no change in that situation in more than two decades (Nunes & Moreno, 2002). An intervention program focused on visual representations was undertaken with deaf pupils aged 7–11 in deaf schools or schools with deaf units. The researchers found that the students in the program had higher achievement levels than a similar group of deaf children from the same schools from the previous year who had not been exposed to the program. Cautious in their conclusions about the effects of the intervention, Nunes and Moreno (2002) attributed the outcomes to a combination of cognitive and motivational

factors; the students were said to have responded positively to the intervention and to have enjoyed working with the booklets. Similar results were found in the research of Barham and Bishop (1991), which used computer-generated visual imagery for teaching key logical connectives.

Rodd (2003) highlighted the role of affect in the mathematics learning of children with "special needs," those with "sense impairments, . . . medical, mobility and developmental conditions, . . . and children who have suffered adverse social conditions" (p. I-127). In her short article, she discussed the relevance of several theoretical frameworks for the analysis of the role of affect in mathematics learning. None, she claimed, included an explanation of the relationship between learning styles and affect. Teaching special-needs students, she maintained, "demands acknowledgment of their specific learning styles otherwise frustration and possibly anger or panic arise" (Rodd, 2003, p. I-129). Although no research evidence was provided, she suggested that autistic children, impaired with respect to their grasp of social situations, "may be more comfortable accessing mathematics via pattern and logic rather than via a (social) context" (p. I-129), and that kinaesthetic activities might suit attention-deficit hyperactive students whose "need to move can be channelled into embodying mathematical relationships" (p. I-129).

ISSUES IN RESEARCHING ACCESS AND EQUITY

In this section we reflect on the research studies mentioned in the previous sections and take a critical view on them. The aim however is not to critique individual studies but to critique generally research in this area, and thereby to help plan worthwhile future studies. Many studies have relevance for the issues of access and equity in mathematics education, and our aim is not to repeat the conclusions from them, but to raise questions that apply to any research in this field, and that we feel have been relatively ignored in the literature.

Research Focus

We must first deal with the issues surrounding the research focus itself. Research is undertaken for various reasons, but in this field of equity in mathematics education, the key motivating force is usually to challenge inequitable practices, and to promote instead equity and access in all their facets. That is, the

research focus is often on issues related to aspects of inequitable practices.

However much research undertaken in mathematics education has generated relevant data and also raised possible implications for questions of access and equity. For example, large-scale international as well as at the local national-level quantitative comparative studies such as TIMSS and NAEP often have data on disadvantaged students and indeed may help to identify specific populations suffering inequities. The problems however are about the focus of that research, and if that focus is not on equity itself, the data may be skewed because of sampling, the tests given, the data collecting procedures, and so on.

We maintain therefore that important questions need to be raised whenever equity is discussed in relation to a particular study:

- What is the research focus?
- What is the background to the research?
- Who is doing the research, and why?
- What is the position of the researcher(s), and what equitable (or inequitable) assumptions lie hidden beneath the study and its data?
- Who is funding the research, and what are the equity assumptions behind the funding agency's stance?

Research Questions

The quality of a research study rests first with the quality of the research questions asked. These questions can be generated from a variety of sources, but hopefully they build on the knowledge gained from previous studies, by extending the range of the earlier studies, by dealing with an aspect revealed by the previous study, or by challenging a finding from a previous study. The questions asked will clearly relate to the research focus referred to above and will themselves be subject to the kinds of assumptions underlying that focus.

Another approach to generating the research questions is to base them overtly on a theoretical model or educational position. In access and equity research that is a common approach and, as we have shown in this chapter, many theoretical positions are available to researchers. There is however the danger that competing theoretical models breed competing research studies, which produce data that are sometimes hard to reconcile or even compare. A good example of this is the difference between psychological and sociocultural models of underachievement that have given rise to very different research studies (see, for example, in gender-related research). Although this issue exists in any field of research, it seems to be

particularly prevalent in equity research, where there are many competing explanations for the presence of inequitable achievements and educational practices.

A third aspect is the extent to which the research study is about access and equity in *mathematics* education. It is true that for most types of educational disadvantage, the equity issues are similar for all school subjects, but it is not true for all types of disadvantage. For example, blind and deaf students face specific challenges related to the subject of mathematics, and gender-related studies have also shown specific aspects of disadvantage because the subject of concern is mathematics. So although nonspecific subject research studies may be of value in interpreting some aspects of nonaccess and inequity within mathematics education, researchers should not necessarily rely on studies of that type to improve access and equity in mathematics education.

Thus some of the questions that need to be raised about any research study, but particularly with any equity-related mathematics research study, are the following:

- What are the research questions based on?
- What previous research has been done in the area?
- What is the theoretical position underlying the research questions?
- How are the questions related to access and equity issues in education?
- How are the questions related to access and equity in *mathematics* education?
- Who is defining these research questions?
- Who is defining equitable and inequitable practices?

Research Participants

If the research study is empirically based, there are many issues concerned with the participants of the research. The field of access and equity is plagued by issues of definition, with the categorizing and labeling of disadvantaged students being the most challenging area. Typically it is the students who are labeled, although the research context is also labeled rather than described, with the consequences of institutionalised self-fulfillment described by McDermott (1974, 1996).

This is the reason for exercising caution in the research process in this area. Whereas it is relatively easy and convenient to use low-inference categories such as *age*, *language*, *place of residence*, and *gender*, researchers must still recognize that the meaning and relevance of those terms are problematic, and that they need to be problematized in any study. They however are rela-

tively easy to deal with compared with research that involves such high-inference categories as *sightedness*, *class*, *indigenous culture*, or even *Afro-American*.

This is also the area where ethical aspects of research studies come to the fore, involving sampling issues, the selection of participants, confidentiality, feedback, and the rights of those participants and their guardians in the research study. The significant questions to be raised in this section are

- Who is the focus of the research?
- Who else is involved but not directly?
- How are they defined, labeled or described?
- What sampling and selection procedures are followed?
- What confidentiality agreements are in place?
- What access do the participants have to the data about them?
- What benefits will accrue to them as a result of the research?

Methodology Issues and Multiple Perspectives

Issues of sampling and definition of the participants are part of the methodology of the research, but it is important to reflect on other specific issues concerning the choice of methodology in research on equity and access. For example, large-scale quantitative studies can certainly point to differences in access and achievement, but they may not reveal a great deal about the reasons for those differences. For that one must turn to more qualitative studies involving case studies and interviews.

In addition, large-scale studies rarely reveal what approaches are beneficial in overcoming inequity and improving access. Small-scale experiments can serve that purpose far better, although once again the involvement of participants in experimental procedures raises ethical issues. Multiple perspectives and approaches are to be recommended to overcome the shortcomings with any one type of research approach.

Experimental work raises another issue of importance that concerns the research site, also involving considerations of the sociocultural context in which the experimental work is done, together with any political pressures that may be applied. Of course these considerations apply equally to any experimental research, but in relation to inequity and access they are far more significant. Whenever there is consideration of unequal opportunities to learn or other inequitable practices, there are histories and contexts to these practices, and there are people who have established and who may be upholding these practices. One has

only to remember the violence associated with the race issue in education in the southern United States to understand the significance of these issues when planning any experimental research into overcoming inequitable practices.

It should also be acknowledged that in some contexts and settings (albeit frequently small-scale models of successful practice) access and equity issues are minimal. Too often, researchers are concerned with identifying problems and ignore the opportunities to research and identify factors that contribute to these success stories.

Thus the significant questions to be raised here are

- What is the sociocultural context of the research?
- What are the histories of the situation or the practice?
- Who are the holders of those histories?
- Who is behind any proposals for change?
- Who are the stakeholders in any future situation?
- Who has the most to lose, or to gain from the research?
- How will the different goals of the stakeholders be balanced in the research?
- What can be learnt from models of successful practice?

FINAL WORDS

In this chapter we have explored access and equity issues in mathematics education from a range of perspectives. Terminology and definitions have been highlighted. Illustrative research findings have been presented that cover a range of issues identified as contributing to inequities including achievement differences, policy and societal factors, the intended and implemented mathematics curriculum, and institutional practices. We concluded the chapter by raising research-related issues that must be considered if research in this field is to provide directions for access and equity to be achieved, and for there to no longer be groups of learners for whom there remain impediments to the learning of mathematics.

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AUTHOR NOTE

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RESEARCH ON TECHNOLOGY IN MATHEMATICS EDUCATION

A Perspective of Constructs

***Rose Mary Zbiek, M. Kathleen Heid,
and Glendon W. Blume***

THE PENNSYLVANIA STATE UNIVERSITY

Thomas P. Dick

OREGON STATE UNIVERSITY

Educators have witnessed explosive growth in the availability of computing technology for mathematics classrooms during the last quarter of a century, and this growth has been accompanied by unbridled enthusiasm for the potential of new technologies in the teaching and learning of mathematics (Fey et al., 1984). A parallel interest in understanding the impact of these technologies on learning, teaching, and curriculum has engulfed the research community. Perhaps the very passion that stimulates research and the growth of new technologies also accounts for why there are many interesting studies and a need for explicit ways to integrate those studies into a solid body of research that both spurs continued scholarship and informs practice. This chapter elaborates on how constructs may be useful tools to further research and enlighten classroom instruction, curriculum matters, and tool development.

The chapter begins with background issues regarding technology in mathematics education and a discussion of what constructs are and how they can be used to further develop, interpret, and apply research. It continues with discussion of constructs

related to several key areas for existing and potential research: technological tools and technology-based mathematical activities, students' behavior in the context of technology, teaching issues related to technology in mathematics education, and effects of technology on mathematical curriculum content.

SETTING THE STAGE

Connecting Past and Future

In the previous *Handbook*, Kaput (1992) documented the continuing growth of electronic technology and predicted, "major limitations of computer use in the coming decades are likely to be less a result of technological limitations than a result of limited human imagination and the constraints of old habits and social structures" (p. 515). At that time, Kaput portrayed structures that conceivably underpinned the user's interaction with a broad range of computing technologies, but the structures were not yet actualized. Rather than predicting where hardware

and software will go in the years to come, we consider where research has been and may go, including a ubiquitous role of technology in work since 1992 and in work that takes us back more than 15 years.

The 14 open questions that Kaput proposed in 1992 have been addressed with varying degrees of interest and of success. One complication is the extent to which research on the teaching and learning of mathematics in technological settings has been premised on the development and accessibility of sophisticated computing tools and appropriate and willing testing sites—both of which take time and resources different from what is required in many areas of mathematics education research. As a consequence, research on technology-intensive mathematics teaching and learning has only recently begun to mature into a well-articulated area of scholarship. As is true of many areas of research, further growth in this arena will depend on the development and articulation of a shared understanding of the phenomena under consideration. As research in this area has developed, researchers have noted, explicitly or implicitly, constructs that describe teaching and learning in these environments. This chapter does not propose a specific agenda for future research on technology in mathematics education but rather submits that scholarly discussion and activity should be framed around constructs that advance collective understanding of past, present, and future work. An attempt to further the development of a shared language of constructs can inform and guide research on the teaching and learning of mathematics in the context of technology.

Technical and Conceptual Activities

Advancing the collective wisdom about the role of technology in mathematics education requires careful distinctions between two different kinds of mathematical activity: technical and conceptual. The technical dimension of mathematical activity is about taking mathematical actions on mathematical objects or on representations of those objects. Procedures can then be built out of sequences of mathematical actions (or out of previously built procedures). Examples of technical mathematical activity include geometric construction and measurement, numerical computation, algebraic manipulation, graphing, graphical transformation, translation between notation systems, solving equations, creating diagrams, displaying, collecting, sorting, and so on. Con-

ceptual mathematical activity involves understanding, communicating, and using mathematical connections, structures, and relationships. Examples of conceptual mathematical activity include finding and describing patterns (inductive reasoning), defining, conjecturing, generalizing, abstracting, connecting representations,¹ predicting, testing, proving, and refuting.

Technical mathematical activity is concerned primarily with tasks of mechanical or procedural performance, whereas conceptual mathematical activity is concerned with tasks of inquiry, articulation, and justification. The potential for student access to technology to influence both the technical and conceptual dimensions of their mathematical activity has been suggested by Borwein (2005) in his description of experimental mathematics (Borwein & Bailey, 2003) in which the computer is used for (a) gaining insight and intuition, (b) discovering new patterns and relationships, (c) graphing to expose mathematical principles, (d) testing and especially falsifying conjectures, (e) exploring a possible result to see whether it merits formal proof, (f) suggesting approaches for formal proof, (g) replacing lengthy hand derivations with tool computations, and (h) confirming analytically derived results.

For researchers interested in the impact of technological tools on mathematics learning and teaching, the distinction between the technical and conceptual dimensions of mathematical activity is both intriguing and problematic. Kaput (1992) noted that “off-loading” routine computations provides a learning efficiency in terms of compacting and enriching experiences. For example, computers or calculators can enable students to examine several graphs in a relatively short amount of time. In turn, this compacted experience allows the student to focus on the common attributes of a family of functions or on the consequences of varying a parameter. The compacted technical activity thus affords an opportunity for enriched conceptual activity. Technology that facilitates connectedness and sharing of results heightens those affordances.

As Artigue (2002) noted, there is a temptation for researchers to explicitly or implicitly treat technical and conceptual dimensions of mathematical activity as though they were fundamentally opposed types of activity. This dualistic epistemological stance tends to view technical activity as entirely mechanical and thus largely devoid of meaning, whereas meaningful learning must be derived from conceptual activity. In terms of its implications for integrating technology

¹ As many who have studied representations have noted (see Goldin & Kaput, 1996, for a discussion of representation), there is a need to distinguish internal representations from external representations. We humans have no direct access to internal representations, and we infer their existence from students' actions on external representations, which might better be termed inscriptions or written artifacts. In this text, when we use the term representation, we mean to indicate an external representation unless stated otherwise.

into mathematics classrooms, such a stance suggests the hypothesis that the primary influence of computing technology is to “free” students from all technical work so they may engage directly in conceptual activity. However, as elaborated by Lagrange (1999) and Hoyles, Noss, and Kent (2004), technical activity using computing technology may involve a combination of routine mechanical actions informed by conceptual reasoning. As such, a line cannot be drawn neatly between technical activity and conceptual activity that distinguishes clearly which is more mathematically meaningful. The underlying and perhaps synergistic relationship between technical and conceptual activities is assumed in this chapter.

Cognitive Technological Tools in Mathematics Education

A discussion of the technical and conceptual dimensions of mathematical activity informs the use of technology in mathematics education, but it does not explain why the potential influence of computer technology may be profoundly different from that of noncomputer tools such as physical manipulatives or drawing instruments. Research questions focused on the influence of technology on mathematics learning and teaching can be sharpened and refined with attention to special differences and contributions technology makes in the mathematical tasks students undertake, the kinds of mathematical activity in which they engage, and the mathematical knowledge and understandings they build.

Pea (1987) used the term “cognitive technologies” for those technologies that help “transcend the limitations of the mind . . . in thinking, learning and problem-solving activities” (p. 91). To facilitate the technical dimension of mathematical activity, a cognitive tool for mathematical activity must allow the user the means to take actions on mathematical objects or representations of those objects. To facilitate the conceptual dimension of mathematical activity, reactive visual feedback becomes critically important (e.g., Hillel, Kieran, & Gurtner, 1989). A cognitive tool must react in response to the user by providing clearly observable evidence of the consequences of the user’s actions “at the surface of the screen” (Balacheff & Kaput, 1996; Balacheff & Sutherland, 1994).

The use of the term cognitive tool in this chapter is intended to include a broad range of technologies used for either the technical or conceptual dimensions of mathematical activity. Distinctions among types of cognitive tools may vary among researchers, and tools having a wide array of uses often do not fit cleanly into any given categorization scheme. For example,

although the term microworld is used extensively in the literature on technology in mathematics education, it is not at all clear that authors share a common definition. Microworlds can vary widely in complexity, ranging from simple systems of virtual manipulatives having a narrow set of objects and actions with limited rules and consequences to exploratory statistical worlds and dynamical geometry systems having a rich array of objects and actions as well as flexible rules that allow for complicated sequences of actions to be captured for repetition or animation.

Balacheff and Kaput (1996) drew a distinction between a microworld and a simulation, the latter of which they described as providing a virtual model of some real-world phenomena, including parameter inputs that can be entered or controlled by the user. The resulting effects of varying the parameter values are then observable by the user. In some cases, a microworld might also be considered a simulation (provided its domain of phenomenology provides a suitable virtual model of some real-world phenomena). The use of the term cognitive tool in this chapter is intended to be inclusive of both microworlds and simulations as well as representational toolkits (Dick, 1992) such as computer algebra systems (CAS), graphing calculators, and spreadsheets that provide a variety of computational tools for dealing with symbolic expressions, graphs, and numeric data. Such tools can include direct access to real-world physical phenomena, either through data collection probes—through interfaces such as the Microcomputer or Calculator Based Laboratory (MBL or CBL)—or through “drivers” of external devices (such as remotely controlled robots). In each case, these tools help transcend the limitations of the mind.

In addition to cognitive tools, other forms of technology are designed to facilitate mathematics learning, teaching, and teacher development. Many of these technologies are cognitive in nature, but they are not mathematical tools. For example, intelligent cognitive tutors, such as the geometry and algebra tutors developed at Carnegie Mellon University (Anderson, Boyle, Corbett, & Lewis, 1990; Ritter, Haverty, Koedinger, Hadley, & Corbett, 2007), provide an alternative learning situation—as tutors, not as tools (even though students have access to a collection of tools within such tutors). Video formats, including the videodisc technology of *The Adventures of Jasper Woodbury* (Zech et al., 1998), allow students to encounter and pursue problems situated in real-world contexts. In these cases, a primary goal of the technology is problem presentation and branching, not use as a problem-solving tool. Online textbooks, online courses, and Internet communities such as the Math Forum (Renninger, Weimar, & Klotz,

1998) offer avenues to serve and connect students and teachers in new ways and in distant settings. Collections of video cases of teaching, including those connected with the Third International Mathematics and Science Study (TIMSS; TIMSS Video Mathematics Research Group, 2003), offer opportunities for teachers and researchers to make teaching an object of study (Cestari, Santagata, & Hood, 2004). Although these and other forms of technology are part of the growing literature in mathematics teaching and learning and could also be used to make the case for constructs, we intentionally drew examples for this chapter from the substantial body of work done regarding cognitive tools to articulate the argument for the usefulness of constructs to conceptualize studies, to connect existing empirical studies, to unfold theories, and to provide a language with which to talk about practice informed by empirical findings and theoretical insights.

What Are Constructs?

What do we mean by *constructs*? *The American Heritage Dictionary* (Pickett, 2000) defines construct as a “concept, model, or schematic idea” or “a concrete image or idea” (pp. 394–395). Our constructs are concepts that have explanatory power regarding technology and the teaching and learning of mathematics. The constructs we have selected for this chapter are ones that have specific applications to mathematics, that have an empirical basis, and that help one understand relationships among tool, activity, students, teacher, and curriculum content.

The constructs on which we will focus are those that are abstracted from research, those that arise from within emerging theories, and those that have arisen from rich descriptions of practice. Although some are

specific to particular types of technology or arose in the data analysis of a particular study, the constructs are essentially explanatory tools that transcend a variety of venues. Each construct shows promise in many different settings, including different types of technology, different mathematical content domains, and different groups of learners. Each construct may start to define or redefine dimensions or stages or capture an underlying component of a larger phenomenon.

Our purpose is to raise configurations of promising constructs to the consciousness of the mathematics education community and to suggest ways in which these constructs have been and can be useful to researchers. Constructs can help frame and influence research and enhance the coherence and connectedness of findings. Use of particular constructs can help refine research questions by drawing attention to structure, relationships, and features related to the roles of technology in the mathematical tasks and activities in which students engage, to the behaviors in which students engage as they work on mathematical tasks in the presence of technology, to the roles of teachers in technology-intensive mathematics education, and to the nature of mathematical content that is learned in technology-intensive curricular worlds. A construct can serve in qualitative and quantitative settings in a variety of ways, including as an independent variable, a dependent variable, a covariate, an explanatory factor, or as the object of study. In addition to their usefulness in research, the chosen constructs are informative for classroom practice and software development.

The rest of this chapter identifies particularly useful constructs in looking at how tools affect aspects of the teaching and learning of mathematics. As suggested by the thick lines added as one moves from the left-hand side of Figure 27.1 to the right-hand side of that

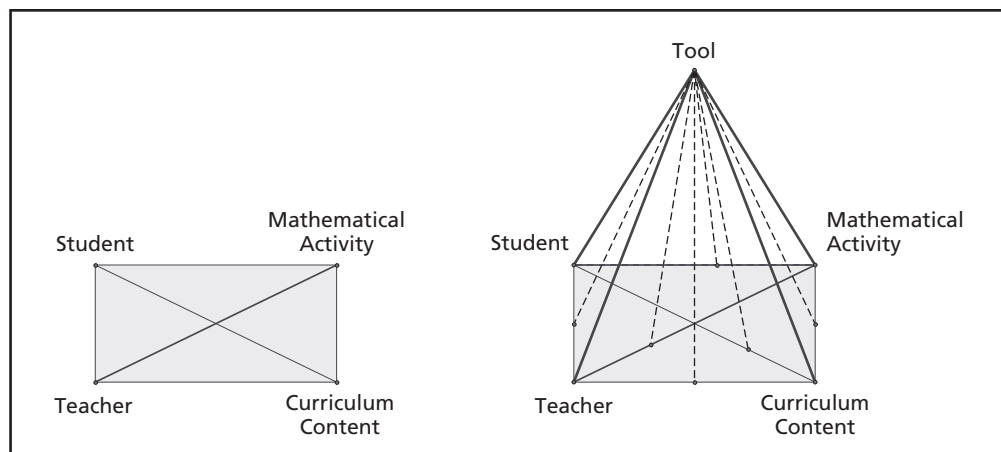


Figure 27.1 Mediation relationships among technology, student, teacher, mathematical activity, and curriculum content.

figure, each of student, mathematical activity, teacher, and curriculum content is mediated by the use of and configuration of the tool. Tool use and configuration also mediate the relationships between any two of the four areas, as denoted by the dotted lines in the right-hand side of Figure 27.1. The remainder of the chapter is organized around the four areas. In the discussion of each area, we identify constructs that address how technology mediates that area and how technology mediates relationships of that area to other aspects of the learning and teaching process. Each section articulates both how these constructs enrich an understanding of existing and potential research and how selected constructs are interconnected. At the end of the chapter, we identify seven specific ways in which constructs inform research on technology in the teaching and learning of mathematics and illustrate each way with constructs discussed in the chapter.

TOOLS AND MATHEMATICAL ACTIVITY

Certain characteristics of cognitive tools can serve as useful constructs for both mathematics education researchers and tool designers. The three constructs discussed here—externalized representation, mathematical fidelity and cognitive fidelity—relate to the ways cognitive tools can provide special opportunities or impediments for learning in the context of mathematical activity. Special attention is drawn to two types of fidelity of the externalized representations of mathematical objects provided by or accessed by the cognitive tool and the technical actions on and linkages to those representations available to the user. Mathematical fidelity refers to the faithfulness of the tool in reflecting the mathematical properties, conventions, and behaviors (as would be understood or expected by the mathematical community). Cognitive fidelity refers to the faithfulness of the tool in reflecting the learner's thought processes or strategic choices while engaged in mathematical activity.

Externalization of Representations— Dynamic Actions and Linkages

Cognitive tools play a special role in mathematical activity by externalizing representations (Heid, 1997). Through externalized, though limited, surrogates for a student's internal mental representations displayed on the surface of the screen, *externalized representations* become visible phenomena that can be shared and discussed with others (e.g., other learners or the teacher). By bringing such representations

literally to the surface, a cognitive tool can allow for unique opportunities for exposing cognitive conflicts. "As mathematical thinking in the face of technology is examined, it will be important not to lose sight of the degree of effectiveness of each application of technology in facilitating students' externalization of representations" (Heid, 1997, p. 8).

Once an external mathematical representation has been created or accessed, the cognitive tool can act as a "user agent" (Kaput, 1992) to perform specific mathematical actions or procedures at the command of the student. Looking at externalization through this idea of user agency suggests research questions that center on a role given to the student—that of tool manager or director. For what kinds of mathematical tasks do students choose to use the technology? How do students use the technology to carry out these tasks? For example, this pair of questions comprised a major strand of a report synthesizing the research findings on handheld graphing technology (Burrill et al., 2002). The authors indicated that this technology was used primarily as a computational and visualization tool and for tasks involving moving between external representational forms of functions, but they also noted that student choices in using the technology were heavily mediated by the teacher. The researchers reported finding few studies (e.g., Ruthven, 1990) that examined what students might do differently because of the access to the handheld graphing technology.

Actions taken on externalized representations with a cognitive tool differ in important ways from actions taken with manipulatives or physical tools. The actions taken with a physical tool are less constrained and are not necessarily as mathematically meaningful as actions on a cognitive tool. For example, a student could stack base-10 blocks vertically to make a toy tower or use a compass point to poke holes in a piece of paper. The extent to which actions with a physical tool are mathematically meaningful is at the discretion of the user. Furthermore, even when an action with a physical tool has consequences with mathematical meaning, that meaning may go unnoticed or be misinterpreted by the student. Physical tools do not automatically react to a user's action with feedback. The student must work to extract feedback from interactions with a physical tool, perhaps with assistance or direction from a teacher.

In its role as a user agent, a cognitive tool can constrain the possible actions on an external representation to be ones that are potentially mathematically meaningful, and it can enforce mathematical rules of behavior of the objects on which it acts. The cognitive tool can also respond to the actions chosen by the student by automatically making potentially mathemati-

cally meaningful consequences of the actions more overtly apparent. Of course, the question of whether the user of a cognitive tool recognizes or correctly interprets an action or its consequences as mathematically meaningful remains, but these constraints and the reactive feedback provided lend greater support for sense-making by the student. Thus, unlike the physical tool, the cognitive tool provides a “constraint-support system” (Kaput, 1992) for mathematical activity. Kaput claimed that the computational contribution of a cognitive tool becomes most apparent in the area of multiple representations when it is used to establish a “hot link,” that is, a dynamic connection between two representations such that an action taken in one representation is automatically reflected in the other. For example, the dilation of a geometric figure in a dynamical geometry program might be accomplished through a “point, click, and drag” action by the user. If there is a hot link to a visually displayed numerical measurement of some attribute of the figure, such as area or perimeter, then changes in the displayed measurement value are dynamically updated continuously as the figure is dilated. Such a hot link achieves a particularly high degree of immediacy in the reactive feedback provided to the user.

Because at least two representations are involved, Kaput (1992) noted that the directionality of a hot link should be considered. In the geometric example mentioned above, the described hot link is unidirectional—the user action is taken on the geometric figure with two related, observable consequences: change in its size and change in the related numerical measurement. A distinctly different hot link is created if, instead, the user action is taken on the numerical measurement (for example, by typing in a specific value in a display field) with the resulting change in value triggering a change in the object by dilation (or some other transformation parameterized by the measurement value). Of course, a hot link could be bidirectional if it offers the option of taking action in either of the two linked representations with the consequences automatically reflected in both. Bidirectionality would, of course, depend on univalence of the mapping in both directions.

In this discussion, we have made the case that cognitive tools provide special environments offering not only access to external representations, but also specific opportunities and “rules of engagement” for the learner’s mathematical activity with those representations. In the section on students and mathematical activity later in this chapter, we introduce research

constructs that draw sharper distinctions among the ways in which a learner may act within and interact with a technological environment. In the present section, our focus is on characteristics of the cognitive tool itself. For the mathematics education researcher to make sense of the learner’s mathematical experiences in such an environment, we suggest that it is important to attend to the faithfulness of those representations to the mathematics accepted by the wider community and to the cognitive intentions of the user.

Mathematical Fidelity

In order to function effectively as a representation of a mathematical “object,” the characteristics of a technology-generated external representation must be faithful to the underlying mathematical properties of that object. The actions that can be taken on that representation, and the consequential behavior resulting from those actions, should reflect accurately the expected mathematical characteristics and behavior. The degree to which this is achieved is a measure of the *mathematical fidelity* of the tool (Dick, 2007).² The desirability for a high degree of mathematical fidelity may seem obvious, but ambiguities of context and technological limitations may make it hard to achieve. The desirability for “user friendliness” may compete with faithfulness to mathematical structure for the tool designer’s priorities.

Dick (2007) used the term mathematical fidelity to point out that the mathematics of the tool does not always represent mathematics as it is understood by the mathematics community. This construct draws attention to the notion that the mathematics students experience in working with technology may not be that which they are intended to experience. There is a range of ways that the disparities accentuated by mathematical fidelity may play out in technology-based mathematics teaching and learning. These ways include differences in the underlying mathematics of seemingly similar computer programs or differences between the mathematics that is intended in a visual display and the possible different interpretations of that display.

Consider, for example, typical calculator graphs of the sine function with increasingly small periods. Figure 27.2 shows a screen capture representation of the graph of such a function. This technology-generated graph of $f(x) = \sin(100x)$ suggests a periodicity and pattern of behavior that does not reflect those of the function rule.

² Later in this chapter, we discuss a different construct of *mathematical concordance* to account for mismatches among the mathematics of the tool, the teacher, the mathematics of the task, and the mathematics of the student.

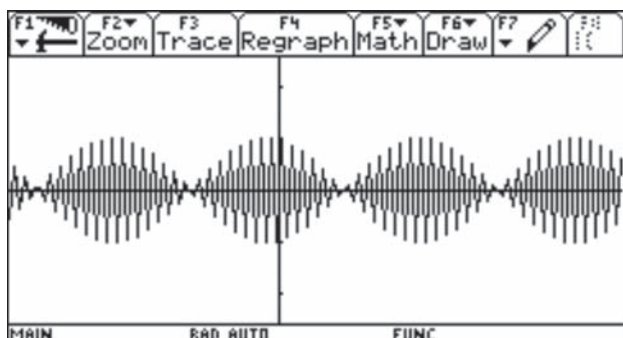


Figure 27.2 Screen capture representation of the graph of the function with rule $f(x) = \sin(100x)$.

While the term itself has not been used widely by researchers to date, the idea behind mathematical fidelity appears repeatedly, particularly in describing phenomena at the interface between the learner and the tool. Dick (2007) noted three areas in which a lack of mathematical fidelity can emerge. First, a lack of mathematical fidelity can emerge through discrepancies between tool and mathematical syntax conventions. For example, the common convention of hierarchy of operations for arithmetic operations may be violated for implied multiplication in the interests of ease of user entry of expressions. Second, lack of mathematical fidelity can emerge through underspecification in mathematical structure. For example, the tool may allow “functions” to be defined by algebraic expressions without specified domains. This can lead to such anomalies as the graph of $f(x) = |\ln(x)|$ (absolute value of the natural logarithm of the real variable x) suggesting a larger domain of real numbers for this function than the domain of real numbers for $f(x) = \ln(x)$. Finally, lack of mathematical fidelity can emerge through limitations in representing continuous phenomena with discrete structures and finite precision numerical computations.

Syntax discrepancies or underspecification of structure have potential for generating confusion for learners and teachers, but it is unclear whether they pose mere nuisances or more serious impediments. When they arise in the midst of classroom activity, it is plausible that the impact on learners may be largely mediated by the teacher’s reactions and interpretations and how such events shape the attitude of the teacher toward technology. Difficulties dealing with the limitations due to discrete structures and finite precision (called “discretisation processes” by Artigue, 2002) have received attention from researchers, especially in the study of function representations and limits. In a study of students exploring asymptotic behavior of rational functions using the Function Supposer, Yerushalmy (1997) noted the “discrepancy be-

tween technology as a support for visual perception and the manipulation of mental objects, but a numerical tool that cannot support the concept of ‘approaching infinity’” (p. 3). Yerushalmy pointed out that the graphical investigation of horizontal asymptotes is inherently thwarted by the impossibility of exploring the entirety of an infinite domain—at the threshold of the precision limitations of the machine, a horizontal asymptote cannot be distinguished from a horizontal tangent. One of the students in Yerushalmy’s study came up with an example that illustrates the difficulty of distinguishing a removable discontinuity from a vertical asymptote when relying on visual evidence: The values of

$$f(x) = \frac{10^{12} - x^2}{10^6 - x}$$

near $x = 10^6$ are very large (visually suggesting a vertical asymptote), but

$$\lim_{x \rightarrow 10^6} f(x) = 2 \cdot 10^6.$$

Even steep portions of continuous function graphs (such as that of $f(x) = x^2$) could visually suggest vertical asymptotes to students.

Hart (1991) conducted extensive task-based interviews with students using graphing technology with a multiple-representation-based calculus curriculum. Several of these tasks were designed specifically to create situations in which a machine-generated representation (usually graphic or numeric) might suggest information contradictory to a student-generated representation (usually algebraic). These tasks included (a) numerical investigations of limits such as

$$\lim_{x \rightarrow 0} (1 + 0.5x)^{1/x} \text{ or } \lim_{x \rightarrow 0} \frac{1}{1 + 2^{1/x}}$$

in which substitution of a sequence of small numerical values for x (0.001, 0.000001, 0.0000000000001) would produce conflicting information due to round-off precision limitations; (b) graphical investigations of functions such as $f(x) = x^3(x-1)^4$ and its derivative $f'(x) = x^2(x-1)^3(7x-3)$, in which the graphs at typical scaling could hide critical point information; and (c) graphical consideration of definite integrals such as $\int_0^{3\pi/4} \sec^2(x) dx$ in which a misapplication of the Fundamental Theorem of Calculus would yield a result conflicting with the graph of the function over the interval of integration. Hart examined how the student arrived at a resolution of these apparent conflicts. In many cases, the source of the contradictory information was essentially a breach of mathematical fidelity in the machine’s representation due to either precision limitations or visual anomalies related to the discrete screen representation of a continuous graph.

Hart found that students' resolutions of such situations were influenced by their confidence in their own algebraic skills, with students having stronger algebra skills less likely to consider a machine-generated graph as authoritatively correct. Interestingly, Hart also found that students with strong algebraic skills using a traditional curriculum without technology were more reluctant to use graphical representations even when the representations provided much more accessible information than a symbolic formula.

One issue centering on mathematical fidelity arises when the mathematics underlying particular tools is not evident to the user. Having to program into the software some behavior not defined by the mathematical system being modeled, software engineers sometimes inadvertently create their own mathematical systems. In dynamical geometry tools, for example, software designers make decisions about the behavior of "arbitrarily placed" points when the figure on which they are placed is dragged (Goldenberg, Scher, & Feurzeig, 2006). Different programmers make different decisions and produce ostensibly similar micro-worlds that act differently in selected cases because they are governed by different rules. The mathematics of the tool can become the object of study resulting in new constructs such as that of *Dynamic Geometry* as an area of mathematics. Colette and Jean-Marie Laborde (2007, p. 44) described activity in this new area of mathematics:

Instead of considering these problems solely as design problems, the researchers involved in the Cabri project recognized these problems as theoretical problems about a new kind of geometry, a dynamical geometry with its own objects and relations. A new field of research about the fundamentals of such a geometry was established. In particular, the theoretical problems pertaining to the axiomatic development of Dynamic Geometry remain under discussion at present. Contributions are being made, some of which were discussed specifically in a vibrant European meeting in Germany in December, 2000. (Gawlick & Henn, 2001, as cited in Laborde & Laborde, 2007)

Dynamical geometry software creates new opportunities to model physical or mechanical motion governed by geometrical constraints (e.g., gear linkages, rolling objects). The simulation of the physical motion can be modeled in a dynamic drawing of the construction as the user drags a point or other element on screen. González-López (2001) studied "the fidelity of the construction by comparing the moving drawings with the real mechanism's performance" (p. 130). González-López echoed the concern of Hölzl (1996, as cited in González-López, 2001) that the de-

mands on dragging (direct manipulation via mouse or stylus device) to provide a faithful simulation go beyond geometry: "The geometrical laws supporting the motion of pictures have no direct link with the physical motion laws nor the continuity of an analytic model" (p. 130).

As technology becomes an increasingly integrated part of school mathematics, careful analysis of issues of mathematical fidelity will be needed. This type of research will necessitate intense collaboration involving mathematicians, computer scientists, and mathematics education researchers.

Cognitive Fidelity

In the context of intelligent tutoring systems, Beeson (1989) has used the term *cognitive fidelity* to refer to the degree to which the computer's method of solution resembles a person's method of solution. Dick (2007) has adopted the term in a somewhat wider context to include the degree to which the (cognitive) tool actions explicitly reflect the user's cognitive actions. Hence, if the external representations afforded by a cognitive tool are meant to provide a glimpse into the mental representations of the learner, then the cognitive fidelity of the tool reflects the faithfulness of the match between the two. In turn, a tool having a high degree of cognitive fidelity holds particular promise to researchers in making visible the thinking of the user.

Olive, Steffe, and others have exploited the idea of a constraint-support system construed to support cognitive fidelity in their research on children's development of rational number concepts through the design of a collection of computer microworlds known collectively as TIMA (Tools for Interactive Mathematical Activity; see Biddlecomb, 1994; Olive, 2000; Olive & Steffe, 1994; Steffe & Olive, 2002). These cognitive tools offer on-screen manipulatives analogous to counters, sticks, and fraction bars. The possible actions on these objects that are afforded to the user are explicitly tied to the development of mental numerical schemes identified in extensive empirical research of children's thinking. These actions include unitizing, uniting, fragmenting, segmenting, partitioning, replicating, iterating, disembedding, and measuring. Hence, the TIMA technology is designed to provide cognitively faithful representations and actions.

Olive and Lobato (2007) made the case that a cognitive tool such as TIMA can allow cognitively faithful actions more easily than physical tools:

One way that technology can enhance the learning of rational number concepts is through the use of com-

puter tools that allow students to enact psychological operations that are difficult to perform with physical materials. In order to establish a relation between a part and a whole in a fractional situation, the child needs to mentally disembed the part from the whole. With physical materials it is not possible to remove a part from the whole without destroying the original whole. With static pictures the part is either embedded in the whole or is drawn separate from the whole. In the former case the child has to mentally unitize one part of the whole while maintaining the unity of the whole and compare these two abstracted units. In the latter case the child has to compare the separate units while imagining that one is embedded in the other. Using a computer tool that provides the child with the ability to dynamically pull a part out of a partitioned whole while leaving the whole intact, the child can enact the disembedding operation that is necessary to make the part-to-whole comparison. This example of one way in which technology can enhance learning draws upon the psychological analysis of rational numbers.

Cognitive fidelity is a particularly important consideration for researchers. By providing action choices to the learner that faithfully reflect potential cognitive choices, tools such as the TIMA technology can provide to researchers more powerful evidence of patterns in children's thinking. In turn, an improved understanding of children's thinking can better inform continuing development of the tools.

An Illustration Involving Parameter Manipulation

Further research into the relationship between cognitive fidelity and the effectiveness of hot links among external representations used as learning aids can offer insights not only to mathematics education researchers but also to instructional designers and engineers of cognitive tools. To appreciate some of the complex issues involved and how the constructs discussed here may be helpful, one might consider a common type of hot-link activity that exploits technology: parameter manipulation with dynamic consequences in another representation. Dynamic virtual simulations of physical phenomena often offer exactly this kind of hot link.

Parameter manipulation is also used widely in investigating families of functions using graphing technology. We will use tool-design decisions about parameter manipulations to illustrate how issues of mathematical and cognitive fidelity can raise research questions. In such an activity, the learner manipulates a representation of the value of a parameter in the symbolic expression for a function—such as the value of the parameter a in the family of functions defined by $f(x) = \sin(ax)$ —and observes the resulting graphi-

cal representation of that function. The learning goal is usually an explicit recognition of the mathematical connection between the parameter and some salient characteristic of the function (such as the period). Although simultaneous visual access to both the symbolic form and the graph of the function seems naturally desirable in a technological implementation of this activity (say with a split screen or two adjacent windows onscreen), there are timing issues related to the parameter changes. Specifically, if a student types in a new parameter value, does the graph change while the student is typing or only after the student signals completion of the change (perhaps by clicking or pressing an ENTER button)? In either case, the screen world may temporarily present conflicting mathematical information to the user (such as an “old” graph simultaneously displayed with a “new” symbolic function formula), a breach of mathematical fidelity.

A tool designer might decide to introduce a “slider bar” as a virtual manipulative controlling the value of the parameter. In this setting, the student need not type in a parameter value. Instead, the student moves the slider to change a number-line representation of the value of the parameter, and the graphical representation is hot-linked to transform continuously with the move of the slider bar. Changing the slider position now directly controls both the shape of the graph and the numerical value of the parameter. The intent behind most slider bars is to enable the user to run through many individual parameter values in a short time, effectively animating the visual consequences of the rapidly changing parameter values. However, it is plausible that the learner's dominant cognitive perception of the user action taken is now more kinesthetic (moving a slider) than mathematical (changing a parameter value). In terms of the cognitive fidelity, the slider bar is not simply a convenient alternative to typing in the parameter value. Rather, it is a third external representation—a virtual manipulative that competes for the user's attention with the symbolic and graphical representations. Does the physical sense of moving a slider obscure rather than enhance the desired cognition of a connection between the parameter value and a salient visual consequence? Such a question is an open one with relevance to both the mathematics education researcher and the tool designer.

The discussion here of some of the issues that arise in this particular case of parameter manipulation is intended to suggest just a few of the complexities of the interplay between a cognitive tool and the learner. The constructs of mathematical fidelity and cognitive fidelity can serve not only to provide a useful vocabulary to cognitive scientists for examining some of those complexities but may also highlight tool characteristics that

provide insights about learning to both instructional designers and tool developers. For the mathematics education researcher, these notions of fidelity can provide a lens for interpreting or making sense of the learner's experiences within the special environment for mathematical activity provided by a cognitive tool. These fidelity constructs are ones in which cognitive-tool designers might have a keen interest, for their engineering and programming decisions bear directly on both. In addition, curriculum developers and instructional designers who assume that learners have access to a cognitive tool also have a special stake in considering the implications these fidelity constructs have for their intended learning outcomes. Researchers are in a unique position to aid these other stakeholders by using fidelity constructs to highlight issues at the interface between the learner and the machine and by more clearly describing what features of the learning environment a cognitive tool affords.³

STUDENT-TOOL RELATIONSHIP

When a student is positioned to engage in mathematical activity in the context of technology a central question is, How will he or she use the technology? A response to this question requires both identification of the point at which the student will choose to bring the technology into play in the process of carrying out the mathematical activity and the determination of the extent to which the technology will function effectively to help the student to accomplish his or her mathematical goal. The constructs of instrumental genesis and technique have productively focused conversation centered on student-tool activities and have concentrated on the relationship between the student and the tool.

Development of Student-Tool Relationships

Instrumental genesis is one of the constructs that have helped frame the conversation about the student-tool relationship. It relates the notion of

"instrument" and derives from ergonomic theories emanating from outside of mainstream mathematics education. Verillon and Rabardel (1995) differentiated between an artifact and an instrument, pointing out the need to consider not only the capacity of the artifacts with which one is working and the tasks in which that person is engaged, but also the relationship between the person and the artifact. The notion of instrument, then, is a psychological one, not a description of a material artifact. The artifact does not automatically assume the role of user-agent (with the computer as a vehicle for accomplishing the purpose of the user, Kaput, 1992); the user needs to develop an understanding of the capacity of the artifact and to develop a relationship with the artifact. As Verillon and Rabardel stated, "The instrument does not exist in itself, it becomes an instrument when the subject has been able to appropriate it for himself and has integrated it with his activity" (p. 84). That is, it is not until this relationship is developed that the artifact becomes what Kaput (1992) described as a user-agent. Artigue (2002) noted the individual and social nature of the development of an instrument:

Thus an instrument is a mixed entity, part artefact, part cognitive schemes which make it an instrument. For a given individual, the artefact at the outset does not have an instrumental value. It becomes an instrument through a process, called instrumental genesis, involving the construction of personal schemes or, more generally, the appropriation of social pre-existing schemes. (p. 250)

As the organizing construct for this conversation about instruments, instrumental genesis is the process of an artifact becoming an instrument (or the process of developing meaningful ways to use the artifact). It is through instrumented action that artifacts evolve into instruments. The question remains, How, specifically, do artifacts such as calculators or computer software become mathematical instruments—tools that the user can employ for his or her own mathematical purposes? This process is not merely a matter of the user becoming acquainted with stan-

³ A relatively new paradigm of educational research methodology may prove especially appropriate in an arena that attracts the interests of these different communities. *Design experiments* (Brown, 1992; Collins, 1992) employ scientifically based research approaches in complex natural learning settings (such as mathematics classrooms), especially those mediated by some technological innovation (such as cognitive tools). The aim of a design experiment is to explore how the technology affects both student learning and educational practice. Design experiments can inform and provide insights into (a) student cognition and learning, (b) the structure of the instructional environment, and (c) design improvements in the technology itself. This kind of research methodology begs for partnerships and collaborations among educational practitioners, researchers, and technological engineers. However, such partnerships are hard to establish, given that the respective goals of these different communities are not necessarily shared, and the tension that exists between the traditions of scientifically based research approaches and those of naturalistic classroom observation (see McCandliss, Kalchman, & Bryant, 2003).

dard uses of the tool; neither the instrument nor the user is unaffected by the development of the instrumental genesis. As described by Guin and Trouche (1999), “the subject has to develop the instrumental genesis and efficient procedures in order to manipulate the artefact. During this interaction process, he or she acquires knowledge which may lead to a different use of it” (p. 201). Thus, instrumental genesis includes both the user shaping the tool for her or his purposes (instrumentalization) and the user’s understanding being shaped by the tool (instrumentation). The core of instrumental genesis in mathematics education is understanding the mathematics of the technology and being able to use it for one’s own purposes.

The construct of instrumental genesis is helpful to researchers in examining the role of technology in learning. It explains how technology does not have the same automatic power for all users and how its intelligent use requires both conceptual and technical knowledge. The construct of instrumental genesis was foundational to Drijvers’ (2003) report of his investigation of how the CAS (in this case, the Texas Instruments TI-89 calculator) could be used to contribute to higher level understanding of the concept of parameter by 9th- and 10th-grade students. Using instrumental genesis, Drijvers accounted for the role of the CAS in students’ learning as a mediator between the physical artifact of the technology and the mental actions of the students. He identified an inventory of obstacles students encountered in developing their relationships with the CAS for the purpose of solving the problems he posed; his analysis of that list of obstacles revealed both conceptual and technical difficulties. For example, one of the obstacles students encountered in the instrumentation process was difficulty in entering expressions into the CAS. Entering expressions requires a conceptual appreciation for the structure of the expressions being entered (e.g., recognizing the need for parentheses in order to avoid entering incorrect expressions) as well as technical knowledge of the notations and conventions of the CAS (e.g., the CAS’s treatment of implicit multiplication, such that ab is interpreted as a variable name rather than the product of a and b , and its convention of responding with “ $\sqrt{}$ ” when the square root key is depressed, requiring the user to enter the ending parenthesis). That the development of full capacity in using a CAS requires both conceptual and technical understanding is not surprising, but the construct of instrumental genesis helped focus attention on this important perspective on learning to use a CAS.

Aspects of Mathematics in Cognitive Tool Use

Attention by researchers to the development of an instrument from an artifact calls for a closer examination of the capacities of the instrument that are to be developed. Thus, concern about instrumental genesis is naturally accompanied by concern about *technique*, the nature of technical capacity that goes beyond rote application of procedures. Examination of this interplay between the constructs of instrumental genesis and technique results in a deeper understanding of both constructs. Guin and Trouche (1999) used the construct of instrumental genesis in their investigation of technique used by 15- to 16-year-olds and 17- to 18-year-olds in CAS environments. Their data helped the researchers identify two phases of instrumental genesis with a CAS calculator. The first phase was a discovery phase in which students were performing actions that helped them detect the function of each key (in this phase students were strongly dependent on the computer, often failing to recognize other relevant information). Students in this phase made almost no use of “understanding tools” (e.g., interpretation, inference, comparison, coordination). In the second phase the students cut down on the number of commands they used, organized their actions using fewer commands, and coordinated their use of those commands. They developed a tendency to seek and use relevant strategies. One of the products of the Guin and Trouche (1999) work is the development of profiles that characterized students’ behavior in using the CAS to solve mathematics problems. With their attention focused on particular aspects of instrumental genesis—instrumentation, instrumentalization, and technique—researchers have been better able to articulate the ways in which students develop in their abilities to use technologies productively (students’ techniques) as well as identify conceptual and technical obstacles to that development. Being able to characterize instrumental genesis, of course, does not guarantee the ability to develop strong capabilities in instrumented action. Having brought the construct of instrumental genesis both into her work with the teachers in an experimental CAS setting and into the design of the curriculum, Artigue (2002) cautioned, “Nevertheless, the students’ instrumental knowledge remained fragile as did the mathematical knowledge they began to build in the field of analysis” (p. 262).

The construct of instrumental genesis can be extended in mathematics education beyond the development of cognitive tools. Although much of the literature on instrumental genesis centers on physically or electronically present tools and technologies,

it is equally viable to think of representations as artifacts that function to a greater or lesser extent as instruments. The questions are parallel. How does a representation become an instrument? Is there a predictable trajectory from representation as artifact to representation as instrument? How students act on multiple technologically available representations is a function of both the stage of their personal instrumental genesis and their understanding of the artifact as a representation of some mathematical object. The core of a student's use of external representations is the extent to which the artifact is not a replacement for the object but serves as a representation to the student of the abstract mathematical object. As Guin and Trouche (1999) pointed out, "It is precisely because mathematical objects are not in the tangible world that the differentiation between a mathematical object and its representation is at the heart of the learning process" (p. 207).

A quintessential and prototypical example that points out the importance of distinguishing between representation as the entity of interest and representation as an encryption that particularizes a mathematical object plays out through *figure* and *drawing*. Used to characterize how students think about geometric displays in dynamical geometry sketches, these constructs, figure and drawing, point out the difference between thinking of the referent (drawing as the visible inscription on the computer screen) and thinking of what is being represented (figure as the theoretical object). This "figure/drawing" distinction is not, of course, limited to technological environments; geometry teachers have long worried about students basing their school geometry proofs on some nonessential feature of the accompanying diagram, and students have long wondered why the distinction matters. The confusion is exacerbated in dynamical geometry environments with the temptation to think of the seemingly endless set of displays under dragging as equivalent to the theoretical object rather than representations of that object. With the powerful range of visual representations available in technological environments, constructs like figure and drawing are helpful in focusing research on mathematical concepts and techniques instead of on less meaningful treatments of procedures. The general constructs of instrumental genesis and technique and the prototype constructs of figure and drawing suggest some overarching perspectives on students' relationships to tools. The next section characterizes students' mathematical activity in the presence of cognitive tools.

STUDENTS AND MATHEMATICAL ACTIVITY

Drawing on theory and research on cognitive mathematical tools, we have identified several constructs that are useful when studying what students do as they use technology in learning mathematics. When students use cognitive tools as they engage in mathematical activity they engage in that activity with goals in mind, and in the course of taking actions in pursuit of those goals, they exhibit certain observable behaviors that can be related to their goals. This section will examine several constructs related to students' use of technology in their mathematical activity: expressive activity and exploratory activity and students' work method when using technology. Expressive activity and exploratory activity are a pair of related constructs that are connected to the nature of students' goals, whereas students' work method when using technology is related to behaviors students exhibit while using technology to engage in mathematical activity.

Goals of Student Activity

Students may have a variety of goals as they work with technology, goals that depend in part on the cognitive demand of the mathematical task. For example, if the cognitive demand of the task is of a purely technical nature, the student's goal in using technology might be to use the tool to carry out a computational algorithm, producing a numeric or symbolic result. If, however, the cognitive demand of the task is of a conceptual nature, the student's goal might be to use the tool to create multiple measures of some attribute from which a generalization might result. So, tasks with quite different cognitive levels might be associated with very different goals. Not surprisingly, students' differing goals often are associated with different types of activity when using technology. For example, if they use cognitive tools to create a representation of a mathematical object (e.g., function, geometric figure) or to construct a game or tool, their mathematical activity may differ substantially from what it would be if they were working with an already-created tool or an already-created external representation of a mathematical object. In particular, two distinct types of activity often are apparent: *exploratory* activity and *expressive* activity. These constructs derive from but also differ somewhat from Bliss and Ogborn's (1989) exploratory mathematical modeling (working with a model created by someone else) and expressive mathematical modeling (creating one's own model). Doerr and Pratt (in press) described how exploratory model-

ing and expressive modeling differ and how they can provide different insights into student learning.⁴

Neither exploratory activity nor expressive activity is a singular entity; each can be thought of as being represented by a continuum. Exploratory activity might range from structured or prescribed activity (e.g., “Press this key. What do you get? Why does that happen?”), to a somewhat less structured exploratory activity (e.g., “Explore this relationship using a triangle, a quadrilateral, a pentagon”), to much less structured activity (e.g., using a specified tool or approach, students create their own examples of a particular mathematical object or instances of a mathematical relationship). Expressive activity might range from students’ selection of a numeric or symbolic computation to be performed and a tool to carry out that computation, to creation of a graph for the solution of an equation, to much more open-ended activity such as creating a game for another student, an activity in which students act as a *bricoleur* or *bricoleuse*—one who builds new structures or objects (Papert, 1993). Expressive activity entails the creation and use of external representations (see the section titled *Externalization of Representations—Dynamic Actions and Linkages*) and therefore may be more conducive than exploratory activity to the development of representational fluency (as described in a subsequent section with that title).

When students are given a procedure to carry out, they are engaging in exploratory activity; however, when students decide which procedures to use they are engaging in expressive activity, albeit a somewhat restricted expressive activity in some instances. It is important to note that both exploratory activity and expressive activity can occur in what often are described as “explorations.” One type of exploratory activity is “guided exploration,” in which the student’s goal is to produce a predetermined result that was chosen by the teacher. In contrast, less guided exploration, in which the student selects some aspect(s) of what to investigate and the approach used to investigate it, is characteristic of expressive activity. When students use technology to help answer a question posed by the teacher using a process suggested by the teacher, their (exploratory) activity is quite different from expressive activity in which they attempt to answer a question

of their choosing with their choice of process. One sees quite different outcomes when studying what students do when they engage in these two forms of “explorations.” One also sees that the nature of students’ exploratory or expressive work depends on both the task and the activity. For example, suppose a teacher, textbook, or peer suggests a task and a particular method to accomplish the task. The student who follows the suggested path to address the given task likely is working in an exploratory mode. The student who creates an original way to address the given task may be working in an expressive mode. The student who follows the suggested path but generates a new task within that path also is working in an expressive way.

One type of setting in which expressive activity results might be one in which a student’s goal is simply to “play”⁵ with the tool, “trying out” various actions that the tool can perform on a particular mathematical object simply to see what kind of results the tool will produce. In an expressive activity such as unstructured play, students can challenge themselves to determine a tool’s capabilities and limitations, and they can begin to develop an understanding of how the tool can accomplish what they intend so they can begin the process of converting the tool into an instrument. From the standpoint of instrumental genesis (Guin & Trouche, 1999)—a construct addressed in depth in the *Student–Tool Relationship* section—this play serves to develop in students an intense, personal, and purposeful relationship with the tool, namely, the psychological component that, when paired with the artifact, comprises the instrument.

The perceived benefits from students’ initial play with a technological tool derive primarily from theory (e.g., the instrumental approach) rather than from empirical studies that explicitly examined play. However, some research studies that have investigated students’ actions when using technology have found that unguided play may not be optimal. Pratt and Ainley (1997) found not only that elementary students produced drawings more often than constructions when given minimal guidance in their introduction to a dynamical geometry tool but also that those drawings did not lead to insights about powerful mathematical ideas that were embedded in the Cabri microworld in

⁴ Doerr (2001) and Doerr and Pratt (in press) made the case that in working with a model created by someone else (exploratory modeling in the sense of Bliss & Ogborn, 1989) students may develop facility with an expanded set of representations and develop conceptual understanding by exploring the consequences of their assumptions. When students create their own models (expressive modeling) there is an opportunity to observe particular aspects of their conceptions, because they make explicit those conceptions as they select objects, represent them, and interpret and validate outcomes from their models. Different insights into learning result from observing what occurs when one does what one is directed to do with a tool as opposed to when one initiates what is to be done with the tool.

⁵ This type of play differs from the play that occurs when playing a game, in which the structure of the game provides direction to the student.

which they were working. Glass and Deckert (2001) suggested that, rather than allowing students' play to be relatively unguided, it be structured by tasks that focus students' attention on relevant mathematical notions. The structuring of students' exploratory activity provided by or facilitated by cognitive tools can focus students on mathematical characteristics and facilitate symbolic descriptions (Clements & Battista, 2001). Play with cognitive tools may involve expressive activity, giving students some freedom to choose tasks, but it also requires some focusing or orchestration by the teacher. Some additional issues related to play that research might address concern the role of play in the instrumentation process; the effects of different amounts and types of structured, exploratory play and unstructured, expressive play; and the effects of different types of guidance that a teacher might offer as the student-tool relationship develops.

Often many paths are available to a student when using a tool in an exploration related to a mathematical task. This multiplicity of paths leads to the "play paradox" described by Hoyles and Noss (1992) and Noss and Hoyles (1996): Because many cognitive tools offer to students such a wide variety of approaches to solving problems, students might not encounter, in the course of their explorations of a problem with the tool, the particular mathematical ideas that were identified as goals by their teacher or by the developers of the curriculum materials. Students often have many possible ways to proceed when using technological tools, but giving students the freedom to select their own approaches raises the possibility that some of the options available to them enable them to bypass the ideas that their teacher intends for them to encounter in the course of their activity. Noss and Hoyles (1996) contended that when one is engaged in play (of a game), one attends to what one experiences in the game, not necessarily to the rules that govern the play of the game, and they noted that

if the teacher brings a new idea to the student's attention from *outside* the activity, then he or she is no longer playing: yet, if it is not imported, he or she might never encounter the idea. Play confers meaning to an activity but blurs the specificity of the intended meaning. (p. 71)

This blurring of specificity might (or might not) result in a change in the level of abstraction. During play the student might engage in a higher level of mathematics than that intended by the teacher, activity that might be related to the student's work method (described in the section on *Students' Technology-Specific Behaviors*).

If the teacher simply brings a new idea to the student's attention during play, that intervention might

substantially change the activity, from being expressive to being exploratory. Noss and Hoyles (1996) suggested somewhat different responses to the play paradox.

Our resolution of this paradox, which is of course only provisional, is that our pedagogical interventions were centered upon encouraging pupils to *reflect* on the task at hand. Of course, this is often quite heavy-handed in comparison to the *laissez-faire* position which some (including, at other times, ourselves) have advocated. (p. 46)

Noss and Hoyles also noted that off-computer discussions were designed to "help children *explicitly* to confront mathematical notions and distinguish them from everyday ones" (p. 54).

Arguments that arise in relation to the play paradox can be viewed as arguments about the exploratory-expressive distinction. At times, the play paradox may result when students treat a task as one involving expressive activity (trying a variety of approaches to see what they produce) while their teacher is expecting exploratory activity (exploring a particular domain using a particular process). This discrepancy can be useful in explaining why outcomes from students' technology use may differ from those that the teacher expects and that the curriculum is structured to promote.

Despite the potential detrimental effects of play to undermine teachers' and curriculum developers' goals as suggested by the play paradox, students' use of technology in a relatively unstructured way also can be quite productive. Harel (1990) and Harel and Papert (1991) provided an example of a very open-ended type of expressive activity that involved students taking on the role of *bricoleur* or *bricoleuse*, in which their goal was to design or produce a tool for other learners. This research centers on the notion of constructionism, which is based on the principle that learning results from a student's construction of artifacts, and that such construction can promote the development of mathematical ideas:

Learners are particularly likely to make new ideas when they are actively engaged in making some type of external artifact—be it a robot, a poem, a sand castle, or a computer program—which they can reflect upon and share with others. (Harel & Papert, 1991, p. 1)

The work of Harel (1990) and Kafai (1995, 1996) suggests that children learn in several ways through designing a game or instructional software for younger students. They learn through design, for example, attempting to integrate the subject matter into a game;

they also learn through programming, which requires reformulation of their knowledge; and they learn through constructing their own mathematical representations. Kafai (1995) identified two types of content integration in the games children created: extrinsic, in which the designer focused more on the game than on the subject matter, and intrinsic, in which the subject matter was more closely integrated with the game. Kafai noted that in several different experiments most children's games employed extrinsic integration, for example, stopping the play at some point to ask the player a content-related question, rather than integrating the content into the game's context. According to Kafai (1995), a challenge for researchers is to determine how to "engage students in more dialogue about making intrinsically integrated educational games" (p. 298). It is tempting for researchers to assume that with sophisticated technology, students could easily create a game in which mathematics is an integral part; however, students might view them as quite separate. Creation of games that intrinsically embody the mathematics requires students to know different mathematics and understand it in a different way from what is required when playing a game and engaging with mathematics are viewed as being sequential, rather than integrated, enterprises.

We next consider how research might profit from the constructs of exploratory activity and expressive activity. Consider research that reports the use of technology to "investigate" some mathematical idea. To better account for outcomes, instead of characterizing students' activity merely as an investigation using technology, researchers need to analyze students' activity when using technology both in terms of the exploratory–expressive distinction as well as on one of the individual continua, either of exploratory activity or expressive activity. This should provide additional explanatory power to studies that previously have considered "exploration," "investigation," or "inquiry" with technology as a singular entity. By using the distinctions that these two constructs provide, researchers can characterize students' activity more accurately

and be able to relate differences in outcomes to the type of activity in which students were engaged, avoiding the possibility of conflicting results that are unexplainable without such a distinction.

Even in response to a particular task, students' goals for using a technological tool may differ greatly, and those goals may be reflected in either exploratory or expressive activity. Researchers who have studied students' use of the dragging feature of dynamical geometry tools have observed that students may have different goals when employing dragging. Arzarello et al. (1998) identified three types of dragging, each with quite different goals (also see Hollebrands, Laborde, & Sträßer, 2006; Rivera, 2005). One type is wandering dragging, a somewhat random type of dragging in which the student's goal is to search for regularities or interesting results that occur when some object is dragged. For example, a student who uses wandering dragging to drag a vertex of a triangle from which an altitude is constructed might observe something about the location of the point representing the foot of the altitude (see Figure 27.3).

In the second type, *lieu muet* dragging (dragging in which the student tacitly or explicitly maintains some condition), the student's goal is to preserve some regularity in the drawing, for example, keeping the measures of one angle and one side of a triangle constant as a vertex is dragged. Figure 27.4 illustrates this type of dragging.

The third type, dragging test (dragging to test a hypothesis), has a different goal, namely, to determine whether a conjecture (the student's or someone else's) is true. For example, in Figure 27.5 a student might test the hypothesis that when $\angle ACB$ is obtuse, the perpendicular from A lies in the exterior of $\triangle ABC$ by dragging vertex A in such a way as to keep $\angle ACB$ obtuse.

Wandering dragging involves a somewhat unconstrained search for results: What interesting result might one notice when examining many examples of some type of figure? *Lieu muet* dragging also involves a search for results, but with some particular condition constraining those results: What is the outcome when

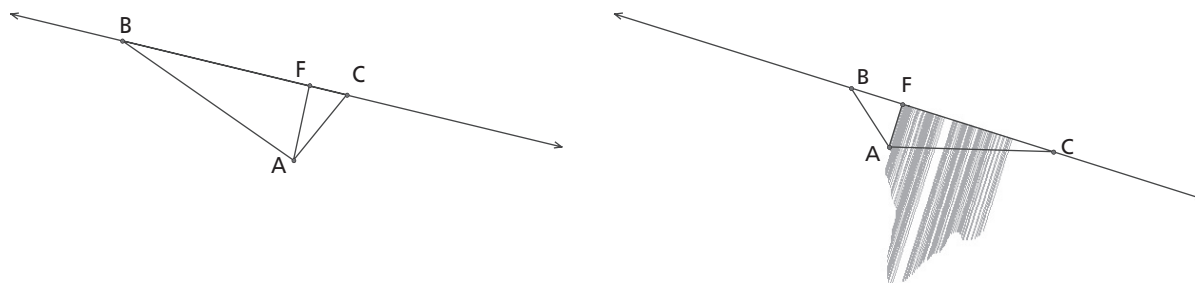


Figure 27.3 A student who drags point A might note what happens to point F.

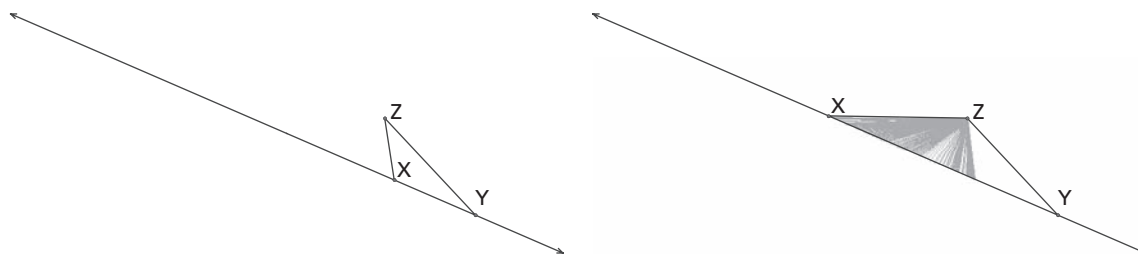


Figure 27.4 A student might drag point X along line XY, keeping $m\angle XYZ$ and the length of segment YZ constant.

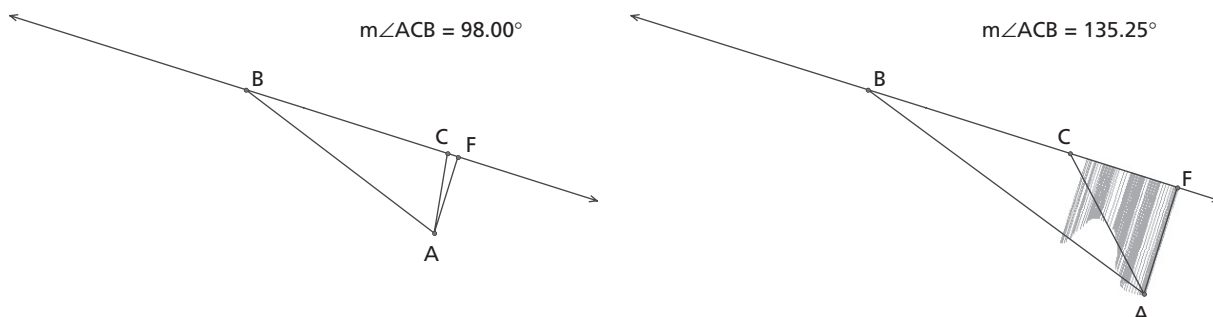


Figure 27.5 A student might drag point A to test a hypothesis about the location of point F.

some aspect of a figure remains invariant? Dragging to test a hypothesis involves a search, not for results, but for confirmation (or disconfirmation) of some particular result: Does this result hold true for many different figures created from a given set of conditions?

These types of dragging can include both exploratory and expressive activity. A student using dragging to test a hypothesis might be doing so in response to a teacher's prompt to drag one vertex of a triangle in a given sketch (e.g., "Try an acute triangle, a right triangle, and an obtuse triangle" to determine whether a particular result holds for each). Such use of technology illustrates exploratory activity. Or, the students' activity might be a much more open-ended wandering dragging that involves the creation of a figure and dragging of its parts to search for some invariant that could lead to a conjecture about such figures. The expressive nature of such activity is reflected in the student's selection of the figure, the dragging to be carried out, and the conclusions that can be drawn. That expressive activity results in part from the task, the curriculum materials, and the teacher's questions and prompts.

These dragging types are important for research because they identify qualitatively different meanings that might underlie a single action made possible by a technological tool and a particular goal. The exploratory-expressive constructs might help a researcher to explain certain learning outcomes that result from

student use of one of these types of dragging rather than another and, again, might help the researcher to explain potentially conflicting results that are masked by the lack of distinction among the types of dragging activities in which students were engaged. Second, relating students' goals and their uses of a tool's capabilities can help to explain very different behaviors that might be observed as students use a tool. Other differences in students' behaviors when using technological tools are the focus of the section that follows.

Students' Technology-Specific Behaviors

When using technological tools, students' goals influence their behaviors, and because not all students' goals are necessarily the same, they do not all exhibit similar behaviors. As observed by Ball and Stacey (2005), data from Pierce (2002, as cited in Ball & Stacey, 2005) suggest that some students are always judicious users of technological tools, some students persist in using technological tools capriciously and inappropriately, and a large number of middle-ground students can be educated to work judiciously. Studying the behaviors students exhibit when working with technological tools can lead to insights about the appropriateness of their use of those tools and about their understanding of mathematics.

A particularly useful idea in characterizing students' behaviors when using technological tools is the

construct of *work method*. Guin and Trouche (1999) identified five work methods, or characterizations of behaviors, exhibited by 17- to 18-year-old students who used symbolic calculators: a random work method, a mechanical work method, a resourceful work method, a rational work method, and a theoretical work method. For each, they indicated the primary source of students' understanding (e.g., investigation, comparison, or inference), the source or tool(s) students used to obtain information (e.g., calculator, paper and pencil, or both calculator and paper and pencil), and students' method(s) of proof (e.g., accumulation of evidence, analogy, or demonstration based on inference). Trouche (2005b) offered several additional components of work method relating to time (e.g., time needed for calculator work) and the student's control of her or his activity (e.g., command process, defined as the student's conscious attitude to consider the information available from all sources and to seek mathematical consistency between those sources). Guin and Trouche also noted that students generally did not operate exclusively with a single work method, and that a student's behaviors often evolved during the instrumentation process, reflecting different work methods at different times.

Trouche (2005b) used a task involving limits of polynomial functions to illustrate contrasting work methods. Students were given the following symbolic rule for a function:

$$P(x) = 0.03x^4 - 300.5003x^3 + 5004.002x^2 - 10009.99x - 100100$$

and were asked to "determine its limit at $+\infty$ " and "determine a calculator window which confirms your result" (Trouche, 2005b, p. 205). A student who used a random (also referred to by Trouche as automatic) work method was unable to analyze the graph that appeared in the standard graphing-calculator window (which did not include the function's rightmost relative minimum) or to determine what might be a more informative window, created a calculator table of values for $P(x)$ for values of x from 1 to 7, and concluded that the limit seemed to be $-\infty$. In demonstrating a mechanical work method—a calculator-restricted work method—a student employed a variety of calculator commands: The Trace command led to location of points that lay outside of the standard window, Zoom commands facilitated searches, and the student obtained the desired result exclusively on the calculator, without reference to its mathematical basis and without recording the work. A student who used a resourceful work method sought coherence when confronted with conflicting results. The student first looked for confirmation of an assertion, based on pre-

viously learned theorems, that $\lim_{x \rightarrow +\infty} P(x) = +\infty$. When the student used a graphic representation of the function, it suggested that the function was decreasing rapidly, even for large values of x . This contradiction led the student to the conclusion that the limit would be $-\infty$, because when there was a great discrepancy between the coefficient of the polynomial's highest degree term and a much larger coefficient of the term with the next-to-largest power, the term with the larger coefficient would dominate. The work of a student using a rational work method entailed factoring out x^4 and finding the limit of each factor using theorems on limit sums and products, then attempting to use first and second derivatives to provide information that would be useful for determining an appropriate calculator window (but not finishing due to lack of time). In contrast to the proof method (demonstration) of the student who used the rational work method, a student who used the theoretical work method established results primarily by analogy, identifying the function as a polynomial function, invoking the theorem concerning its limit at $+\infty$ being that of its highest degree term, and using the overall shape of the graph of a fourth-degree polynomial to make a sketch of the graph.

Students who exhibited the random (or automatic) work method generally employed trial and error, but without verification of results produced by the CAS. These students apparently had a goal that involved finding some calculator operation that would produce a result (not necessarily known to the student to be a correct result) for the task at hand. Other researchers have identified a number of behaviors similar to those of students who exhibited Guin and Trouche's random work method. These behaviors seem to result when students search for some appropriate action with a tool. However, each of these behaviors lacks a mathematical focus. For example, Guin and Trouche (1999) observed in weaker students aged 15 to 16 "avoidance strategies" such as random trials and "zapping" to other commands in the same menu, similar to what Ball and Stacey (2005) reported. Guin and Trouche noted that calculator use does not automatically lead to questioning and reflection:

No doubt this behaviour is different from the activity in the paper/pencil environment where trying something else already requires careful thought. The main difference between the two environments is precisely that which occurs with the calculator, students lose consciousness of the task and there is little mathematical work in their activity. (p. 213)

Trouche (2005a) noted several other behaviors in which a mathematical focus is lacking in students'

work. Trouche reported that Defouad (as cited in Trouche, 2005a) identified zapping, in which students quickly change the graph window without analyzing information from the graph; oscillation between several techniques or strategies; and overchecking that uses more of the calculator's capabilities than is necessary. Trouche (2005a) also reported that in his previous work (Trouche, 1997, as cited in Trouche, 2005a) he noted the phenomenon of automatic transportation, in which students entered all of the problem data into the calculator, with the goal of finding a command that could produce the desired result directly.

Many of these student behaviors involve searching for a command that will generate some result. The goals of these "hunting" behaviors are similar to the goal of curve fitting described by Zbiek (1998)—finding, without using the desired mathematical analysis of the problem situation, some curve that "fits" the data. These goals that entail searching but lack a focus on pertinent mathematical features also appear in Guin and Trouche (1999): "Weaker students often do not question results; activities do not necessarily lead to reflective work, but may instead lead to behaviour she [Artigue, 1995] has called 'fishing behaviour'" (p. 200).

Guin and Trouche's mechanical work method resulted primarily from simple manipulations and led to reasoning based on an accumulation of consistent results from the tool. This seems to reflect a goal that involves finding regularities across multiple instances rather than identifying an underlying mathematical structure. This is similar to what Noss and Hoyles (1996) termed "pattern spotting," when students viewed patterns between numbers or geometric figures as equally valid regardless of those patterns' underlying mathematical structures. Students who engage in wandering dragging may be exhibiting a mechanical work method: They produce a collection of tool-generated results, from which they might note some regularity (without necessarily identifying the mathematical basis for that regularity).

In reference to students' use of spreadsheets, Friedlander and Tabach (2001) found that some students extended a table to thousands of rows to answer a question that could have been answered by using a position formula (specifying the value in a cell in terms of n , its position in the sequence) rather than the recursive formula that they had used to extend the table (specifying the value in one cell in terms of the value in the preceding cell). Despite the efficacy of such an approach, the authors contended that students who used this form of "formula dragging" abused the power of a spreadsheet by constructing large tables in lieu of using what the authors deemed

to be a more efficient symbolic approach (e.g., solving an equation), and they noted that tasks with large numbers brought this issue to the fore for students who wanted to use the symbolic expression rather than the numeric values to obtain a result. Students who use a mechanical work method base their results on an accumulation of tool-generated instances rather than on analysis of underlying mathematical relationships.

In other work methods identified by Guin and Trouche (1999), mathematical analyses played a more prominent role. In the resourceful work method students used a wide range of solution strategies and a variety of sources to generate information (e.g., calculator, paper and pencil, theoretical analyses), sometimes basing results on observations and other times basing them on theoretical analyses. The rational work method entailed little CAS use, and inferences played a primary role in reasoning. In the theoretical work method, understanding derived primarily from interpretation rather than from investigation (as in the mechanical work method) or comparison (as in the resourceful work method); theoretical knowledge, rather than the calculator (as in the mechanical work method) or paper and pencil (as in the rational work method), was the primary tool for information; and reasoning was essentially based on analogy.

Researchers who have attempted to identify productive and unproductive behaviors when students use technology often have focused on inappropriate behaviors similar to those described in conjunction with the random work method, many of which appear to result when students search (sometimes randomly) for appropriate actions. What is needed is research that identifies constructs that are associated with the development of judicious use of technology. An initial step toward identification of those constructs might be empirical studies that provide descriptions of students' successful use of technology (e.g., Dick, 1992; Heid, Hollebrands, & Iseri, 2002), studies that focus on observations of students' behaviors when using technological tools, and studies that identify successful strategies for changing undesirable behaviors. As more is learned about what causes students to use certain work methods, teachers might help students make more judicious use of technology—to use more mathematically based and more productive work methods—causing students whose behaviors are passive, random, or lacking in thoughtfulness to become more judicious users of technology and redirecting them toward behaviors that are based, not so much on the results produced by the tool, but on reflection about the underlying mathematics.

Because the work-method construct consists of a collection of elements that characterize aspects of students' understanding, coordination of information, tool use, and approach to proof, it provides a composite description of the nature of the mathematical basis on which a student's work with technological tools is grounded. This can be useful to researchers in linking changes over time in students' work methods as they use technological tools to the process of instrumentation, in identifying extreme behaviors that may account for unexpected outcomes when using technology, and as a way to relate students' understanding to their behaviors when using technological tools. A study of work methods may also reflect what patterns of privileging occur in the classroom (see the section on the Relationship Between Technology and Practice for additional discussion of the privileging construct).

As noted previously, identifying students' type of activity and behaviors when using technological tools is important, and from a research standpoint it is desirable to use the type of activity in which they engage (exploratory or expressive) and their corresponding behaviors (work method) to infer what students are thinking. An important feature of some technological tools that facilitates this inference is their ability to make students' thinking easier to infer from the actions they take with the tool. In fact, in some instances tools have been designed explicitly to reflect students' thinking by providing choices that mirror students' mental actions. For example, Tools for Interactive Mathematical Activity "provide children with possibilities for enacting their mathematical operations with whole numbers and fractions. They also provide the teacher/researcher with opportunities to provoke perturbations in children's mathematical schemes and observe children's mathematical thinking in action" (Olive & Lobato, in press). In a slightly different way Logo makes students' thinking explicit by enabling students to express geometric ideas in the form of a program that represents an "instantiation of students' expression of geometric ideas" (Clements & Battista, 2001, p. 145). Noss and Hoyles (1996) described programming in general as "a tool for expressing and articulating ideas" (p. 57). They noted that Illich (1973) described such tools as convivial tools—tools that allow users to express their meaning in action.

Students' mathematical behavior is, of course, greatly influenced by the ways in which teachers choose to engage them in mathematical activity. The next section takes up the issue of teaching practice in technological environments.

RELATIONSHIP BETWEEN TECHNOLOGY AND PRACTICE

The relationship between technology and a particular teacher's practice is not always predictable. Two constructs are useful in understanding how teachers involve technology in their work. The degree of match between a particular cognitive technology and a teacher's practice and beliefs underlies the construct of pedagogical fidelity. The construct of privileging centers on a teacher's implicit and explicit choices, including but not limited to choices about technology use.

Match Between Technology and Practice

A teacher's readiness to use a particular form of technology and the nature of how a teacher's use of the technology unfolds center around how the teacher's practice and the nature of that technology align. Dick (2006) assumed that "students learn mathematics by taking mathematical actions . . . on mathematical objects . . . , observing the mathematical consequences of those actions, and reflecting on their meanings" (p. 334). He then described the pedagogical faithfulness of a technological tool as the degree to which the student sees use of the tool as something that furthers this learning. Dick positioned his discussion of pedagogical faithfulness under the more general umbrella of *pedagogical fidelity*. We see pedagogical fidelity more generally as the extent to which teachers (as well as students) believe that a tool allows students to act mathematically in ways that correspond to the nature of mathematical learning that underlies a teacher's practice, regardless of whether the teacher's view of learning is consistent with Dick's assumption that learning requires reflection on observations of the mathematical consequences of mathematical actions on mathematical objects.

Although pedagogical fidelity has not been an explicit research topic, the construct can be used to explain seemingly disconnected or convoluted sets of findings and observations from studies of teachers' thinking, planning, and use of technology in mathematics classrooms. As an example, we consider several studies (Lumb, Monaghan, & Mulligan, 2000; Ruthven & Hennessy, 2002; Zbiek, 1995) that address relationships between mathematical exploration or investigation and classroom use of technology.

In a study of secondary mathematics teachers from seven schools, Ruthven and Hennessy (2002) noted the promotion of investigation as one factor to which teachers attributed successful classroom use of spreadsheets, graphing utilities, geometry programs,

and databases. Their observation might be interpreted as indicating that each of these teachers valued some form of investigation as part of their view of learning and attributed success to the extent to which the tools were faithful facilitators of this valued mathematical activity. In this study, it appears that teachers experienced high pedagogical fidelity because technology supported the investigations they valued for all students and was particularly helpful for learners whom they viewed as “low-attaining students.”

Reports of other studies relate situations in which we wonder whether a key challenge for the teacher was a low degree of pedagogical fidelity. Zbiek (1995) described LeAnne as a teacher who valued a form of investigation in mathematics, but who also wanted students to use particular mathematical procedures in their work and to have a relatively high degree of commonality in how they answered questions. The computer algebra system used in her classroom supported the mathematical investigation she valued, but it also allowed students to have freedom to pursue different methods and ideas in their investigations. The extent to which this freedom resulted in a mismatch between what LeAnne wanted in her classroom and what the tool supported suggests that a low degree of pedagogical fidelity may have been a key factor in LeAnne’s decisions to alter tasks in ways that curtailed students’ opportunities to investigate mathematical phenomena. A similar observation of a teacher seeing the extent to which a computer algebra system allows students many options is found in Lumb et al.’s (2000) description of Steve’s view of materials. As a teacher of Advanced level or A-level 16- and 17-year-old students, Steve began with worksheets to help students learn to use Derive. As students became sufficiently able users of the software, Steve continued to use the worksheets and did not move to using the textbook. Among the many factors that may have played into Steve’s decision to continue with worksheets is pedagogical fidelity. The computer algebra system offered many options and Steve’s decision may have arisen from “a desire to ‘lead’ students to go down a particular route they believe is beneficial to learning” (p. 236).

In these studies and others (e.g., Bottino & Furinghetti, 1996; Mariotti & Bussi, 1998), we see classroom events influenced by teachers’ views of how technology (and curriculum materials) facilitates or impedes mathematical investigations as a form of student mathematical activity that enhances learning, apparently influenced by what happens in instructional settings and how teachers view the appropriateness and usefulness of the technology in their classrooms. The usefulness of pedagogical fidelity in understanding the relationship between technology

and practice extends beyond studies of classroom instruction. Becker and colleagues (Becker & Riel, 2000; Ravitz, Becker, & Wong, 2000) surveyed 4,083 U.S. Grades 4–12 teachers (not all mathematics teachers) chosen by a national probability sample of schools and purposive samples of schools with greatest per-capita computer technology and of schools with substantial participation in educational reform efforts. The research team identified the extent to which each teacher embraced one or some combination of two models of instruction: traditional instruction or constructivist-compatible instruction. Ravitz et al. (1998) carefully articulated what they meant by these two forms of instruction:

Traditional Transmission Instruction is based on a theory of learning that suggests that students will learn facts, concepts, and understandings by absorbing the content of their teacher’s explanations or by reading explanations from a text and answering related questions. Skills (procedural knowledge) are mastered through guided and repetitive practice of each skill in sequence, in a systematic and highly prescribed fashion, and done largely independent of complex applications in which those skills might play some role.

Constructivist-Compatible Instruction is based on a theory of learning that suggests that understanding arises only through prolonged engagement of the learner in relating new ideas and explanations to the learner’s own prior beliefs. A corollary of that assertion is that the capacity to employ procedural knowledge (skills) comes only from experience in working with concrete problems that provide experience in deciding how and when to call upon each of a diverse set of skills. (p. 3)

Teachers were also categorized with respect to the extent to which they reported professional involvement (e.g., working substantially with teachers from other schools, presenting at workshops, attending conferences, publishing, teaching college courses). Based on the self-report data, Becker and colleagues concluded that teachers who were more involved professionally were more likely than other teachers to report constructivist-compatible instruction and greater use of technology. Although these results might suggest that the constructivist-compatible instruction description captures key aspects of practice that suggest high pedagogical fidelity with the technology used, we caution that such a claim requires further investigation. The sample of teachers was not limited to teachers of mathematics, and the technology involved ranged greatly, including but not limited to skill games, electronic mail, browsers, word processors, simulations, and spreadsheets.

The importance of pedagogical fidelity also may be seen in its relationship to other constructs. One example to illustrate this point involves a potential relationship between pedagogical fidelity and the instrumentation and instrumentalization aspects of instrumental genesis. Instrumentation involves the ways in which a student's mathematics is influenced by the technology. In cases in which the tool supports the desired mathematical activities, there may be a high potential for the student to develop rich instrumentation schemes. In contrast, when there is not a high degree of pedagogical fidelity, the student may need to adapt the tool in nontrivial ways to meet the mathematical activity that is consistent with the teacher's view of learning. The student's instrumentalization, or adaptations of the tool to the mathematics, may be well supported and the student may thrive in this situation, or instrumentation may not be supported and the student may struggle to achieve learning goals.

The extent to which instrumentation and instrumentalization are supported in the absence of strong pedagogical fidelity may depend greatly on the teacher's instrumental orchestrations. As described by Trouche (2005b), instrumental orchestration refers "to an organization of the artifactual environment, that an institution (here the schooling institution) designs and puts in place, with the main objective of *assisting the instrumental genesis of individuals* (here students)" (p. 210). Careful attention in research work to the orchestrations that teachers establish and how these orchestrations reflect the pedagogical fidelity of the tools may inform practice as well as help researchers and theoreticians to unravel the complexity of teaching with technology. In addition, articulating orchestrations when describing classroom settings may be one way to add some clarity to what is meant by commonly used but uninformative labels, such as "experimental class," "technology group," and "non-traditional classroom."

When students are engaged in expressive activity or exploratory activity, their teacher might take on different roles (teacher roles are addressed more fully in a subsequent section titled *Incorporating Technology Into Practice*). In exploratory activity students might be involved primarily in doing, with the teacher functioning as manager of the task to be done. When students are engaged in expressive activity, the teacher may function as the manager of reflection, ensuring that students are creating something on which it is worth reflecting and focusing that reflection on the mathematical ideas involved.

Beyond describing pedagogical fidelity in particular cases and using the construct to synthesize studies, describe research settings, and connect constructs,

there is a question of whether and how one might change the degree of pedagogical fidelity. Empirical evidence suggests such changes in teachers' perceptions of the fit between the technology and their view of learning and goals for students' mathematical activity. Graham and Thomas (2000) developed an early-algebra module that used a "press-see-explain" pattern. After pressing keys and seeing results, students routinely were asked to explain the results. The positive reaction to technology by teachers in this study may reflect how the press-see-explain pattern gave these teachers a specific pedagogical strategy to elicit the mathematical explanations they valued. This strategy formed a bridge between technology use by students and evidence of learning as valued by teachers. The teachers seemingly saw this fit between the calculator use and their classroom activity goals, which suggested use of the press-see-explain pattern might have influenced teachers' views of calculator use. Pedagogical fidelity would be enhanced if, for example, teachers moved from seeing calculator use as detracting from learning goals or being good only for checking answers to seeing a press-see-explain form of calculator use that contributed to classroom mathematics goals.

Pedagogical Choices

Beyond the overarching choice to use technology at all, teachers make daily choices about how to use technology in their classrooms. Kendal and Stacey (1999, 2000, 2001) adapted the notion of *privileging* from Wertsch (1990) to describe how teachers, intentionally or unintentionally, use frequently or place a priority on certain things in their practice. Teachers' objects of privileging included types of representation, skills or concepts, and by-hand or by-technology methods. Through written tests and interviews, Kendal and Stacey elaborated a connection between calculus classroom practice and student performance with evidence that students with the same scores on a common calculus test but taught by different teachers differed in their understanding and performance in ways that reflected their teachers' classroom patterns of privileging. One might also ask whether teachers may privilege—or be thought by their students to privilege—certain mathematical activities (e.g., exploration over exposition, production of examples over articulation of definition, illustration over proof) and what relationships, if any, privileging of mathematical activities has to student understanding and performance.

One form of privileging is embodied in Buchberger's (1989) White Box/Black Box Principle for Using Symbolic Computation Software in Math Education. Buchberger argued that some areas of mathematics,

such as common integration rules, are “trivialized” in the sense that “there is a (feasible, efficient, tractable) algorithm that can solve any instance of a problem from this area” (p. 10). He then noted a critical question given the power of symbolic software: “*Should math students learn area X of mathematics when this area has been trivialized?*” (p. 11). His response to this question is the Principle:

I think it is totally inappropriate to answer such a question by a strict “yes” or “no.” Rather, the answer depends on the stage of teaching area X.

- *In the stage where area X is new* to the students, the use of a symbolic software system realizing the algorithms of area X as black boxes would be a disaster. . . . Students have to study the area thoroughly, i.e., they should study problems, basic concepts, theorems, proofs, algorithms based on the theorems, examples, hand calculations.
- *In the stage where area X has been thoroughly studied*, when hand calculations for simple examples become routine and hand calculations for complex examples become intractable, students should be allowed and encouraged to use the respective algorithms available in the symbolic software systems. (Buchberger, 1989, p. 13)

The principle obviously promotes privileging of symbolic software use only when the simple problems are routine and complex problems are intractable. Less obviously, the principle embodies privileging of many facets of mathematics (e.g., problems, concepts, theorems, proofs, algorithms) and of mathematics that not only is going to be introduced with all of these facets but also used beyond its introduction. Coupled with the examples offered by Kendal and Stacey, Buchberger’s White Box/Black Box Principle is a reminder that privileging may have various layers, some of which are less obvious and all of which might influence student learning and achievement.

The white box/black box notion in this principle has been borrowed by many others to convey two extremes regarding whether students are aware of the mathematics they ask the technology to do (white box) or are using the technology without expectation of knowing the mathematical working of the tool (black box). Several authors (e.g., Pimm, 1995) have argued that black-box treatment of technology may be detrimental to students. Lagrange (1999) suggested that students for whom the technical aspects of symbolic work are masked by CAS use may not be able to develop rich technique, a construct discussed earlier in this chapter in *Aspects of Mathematics in Cognitive Tool Use*. In contrast, several authors have contended that some seemingly black-box uses of technology may be

productive. Heid (1988) described how college students were able to engage in mathematical investigations and develop conceptual understanding of calculus topics when manipulation details were delegated primarily to the equivalent of CAS. Her students were able to engage in these investigations without having first mastered by hand the routines that were being executed by the symbolic manipulation program. Dorr and Zangor (2000) noted that the black-box use of a graphing calculator did not impede secondary school students’ encounters with mathematical reasoning. They concluded that the teacher’s questioning drew students’ attention from the workings of the tool to the mathematical justifications of the work, suggesting that this black-box use supported justification activity.

Lurking beneath white-box/black-box debates are potentially different privileging schemes and issues of pedagogical fidelity. For example, where one stands on the white box/black box issue may reflect the extent to which one privileges by-hand computation. If one believes that knowing the details of the manipulations that could be delegated to the tool is critical mathematical activity for students’ learning of mathematics, the privileging patterns intrinsic to the White Box/Black Box Principle strengthen the pedagogical fidelity of the CAS. Privileging, particularly when driven by an adherence to the White Box/Black Box Principle, may also help to explain students’ differing work methods (a construct described in the section on *Students’ Technology-Specific Behaviors*). Interesting research questions arise when one considers how patterns of privileging might unfold differently in two classrooms, both of which claim to adhere to Buchberger’s principle. To what extent might privileging connections among basic concepts, hand calculations, theorems, and proofs correspond to students’ use of resourceful work methods? To what extent might students in classrooms characterized by the absence of privileging these connections but otherwise following the White Box/Black Box Principle be more likely to resort to random work methods or mechanical work methods than students in classrooms that privilege the connections?

We believe privileging also may apply to what technology offers as well as to what teachers do. For example, Goldenberg and Cuoco (1998) discussed geometry environments in terms of the mathematical concept of function. An interesting point they made regards how students may see continuity and dynamic dependency in dynamical geometry versus seeing function as algorithm in a static geometry environment. Dynamical geometry privileges a view of function rooted in dragging, or “continuous real-time

transformation" (Goldenberg & Cuoco, 1998, p. 351). Goldenberg and Cuoco contrasted this view of function in dynamic settings with function as constructive algorithm in static geometry environments (e.g., Geometric Supposer). This observation suggests the need to consider what mathematical views software privileges as well as what teachers (and curriculum) privilege. In short, the match between what the software privileges and what the teacher privileges may be a key aspect of pedagogical fidelity.

Incorporating Technology Into Practice

In addition to constructs to use in articulation of choices teachers make and actions they take, other constructs help to describe other aspects of how teachers go about their work and grow personally and professionally as technology users. One might look at the *teacher role*, a construct that involves describing and contrasting various roles that teachers perform in the classroom. Zbiek and Hollebrands (in press) described 11 roles based on those identified and used by Fraser and colleagues (1987), Farrell (1996), and Heid, Sheets, and Matras (1990). As examples of how teacher role may be useful, we will use two roles (Counselor and Technical Assistant) and their definitions as given by Zbiek and Hollebrands. The Counselor role is one in which the teacher is familiar with the mathematical task at hand and provides mathematical assistance when students ask for it. As a Technical Assistant, the teacher helps students with hardware and software difficulties.

Teacher role is a useful construct to compare technology-present and non-technology-present classroom work of teachers and their students. In particular, several studies have suggested that teachers acted in a Counselor role in technology-using settings more than they or other teachers did in nontechnology settings. On the basis of observations using particular programs rather than mathematics tools, Fraser and colleagues (1987) noted that the technology changed the dynamics of mathematics classrooms with 12- to 14-year-old students by taking on particular roles normally assumed by the teacher while the teacher moved into roles infrequently held when the technology was not in use. In particular, the researchers noted that, when the computer was in use, teachers would join with students in what we call a Counselor role, which seemed to be absent in most teachers' practice. Building from the roles used by Fraser and colleagues, Farrell (1996) compared the roles taken on by six teachers when using graphing calculators in a technology-using precalculus course to the roles taken on by those teachers in the same classes when the technology was not in use. Each 5-minute section of videotaped les-

sons was coded according to whether the teacher ever acted in a particular role and whether any calculator use happened, among other things. On the basis of 283 coded segments, Farrell concluded that teachers functioned as Counselors more frequently when technology was in use (43% of all technology-using 5-minute segments) than when technology was not in use (19% of all non-technology-using segments). The increased prevalence of a Counselor role may not be limited to secondary school settings, and it may be perceived by instructors as well as by observers. When surveyed by Rochowicz (1996), calculus instructors reported taking on a Counselor role when computers or calculators were used in their calculus classrooms in engineering and pre-engineering schools. This set of studies by Rochowicz, Farrell, and Fraser and colleagues suggests that bringing technology into a classroom corresponds to increased instances of the Counselor role.

Teacher role also may be used to describe tensions in a teacher's practice. Despite the seeming association of technology with the likelihood of teachers working in a Counselor role, the Counselor role may be in conflict with teachers' expectations and competing roles. Timmerman (1998/1999) described how Susan, an elementary school teacher, struggled with a desire to take on a Counselor role while using TIMA software (Olive & Steffe, 1994) with Grade 4 students given access to only one computer. Susan was accustomed to all students being engaged in the same task in a teacher-controlled environment. Having two students working together without the teacher's input while the rest of the class was involved in other activity did not allow her control and perhaps did not allow her to be in a Counselor role. Piliero (1994) described a teacher's tension between wanting to serve a Counselor role but more often serving as a Technical Assistant, reacting to students' technical issues as they were using Function Probe (Confrey, 1991, as cited in Piliero, 1994) in a precalculus class. Teacher roles provide a way to capture the tensions that might be useful in identifying differences between teachers' perceptions and practice—differences that might influence or reflect difficulties in implementing technology-intensive curricula.

In addition to ways to describe teachers' roles within the classroom and tensions they feel, ways are needed to describe how technology-rich practice develops. Teacher role may be useful in capturing teachers' experiences as they become fluent users of cognitive tools as pedagogical tools, but they would not explain the total journey of learning to teach with technology. Beaudin and Bowers (1997) introduced the PURIA model for CAS incorporation that Zbiek

and Hollebrands (in press) expanded to account for teachers' growth in the use of any technological tool. The model notes that teachers engage in one or more modes of technology use: personal Play (use without mathematical aims), personal Use (use with clear mathematical goals), Recommendation (recommending or trying technology with small groups of students), Implementation (use of the technology in the classroom), and Assessment (careful attention to the impact of technology use on student learning). Zbiek and Hollebrands argued that PURIA is consistent with evidence found in the reports of prior research. Although it is early to see empirical evidence to support or challenge the value of PURIA as a framework for supporting teachers' growth as practitioners in technology-rich classrooms, the extent to which the classifications provide a way to talk about decades of research on teacher technology use is promising.

PURIA's promise as a way to describe teachers' growth in teaching mathematics with technology includes the extent to which it fits with other constructs related to technology and mathematics teaching and learning. The Play and Use modes allow for teachers' instrumental genesis as they become proficient users of the cognitive tool. As teachers move among Recommend, Implement, and Assess modes, a different type of genesis occurs as teachers grow to see the technology as a pedagogical instrument. Within this genesis of the pedagogical instrument, pedagogical instrumentalization arises from teachers' desires to fit the technology with existing or emerging privileging patterns and teacher roles. The ease with which pedagogical instrument genesis occurs may reflect the degree of pedagogical fidelity.

TECHNOLOGY AND CURRICULUM

The effects of technology on the teaching and learning of mathematics is a function of the curriculum in which it is embedded—how the technology is used, with what mathematical activities students engage, what curriculum materials are used, and upon what content the curriculum is focused. We have identified several central constructs that capture the opportunities for change in curriculum facilitated by technology: representational fluency, mathematical concordance, amplifiers, reorganizers, microprocedures, and macroprocedures.

Representational Fluency

One of the reasons offered for the importance of incorporating technology in mathematics instruc-

tion is the access technology can offer to content that had not previously been included in the school curriculum. This access is facilitated through a range of technological capabilities particularly suited to mathematics instruction. Primary among those technological capabilities is the capacity to accommodate multiple representations, hot-linked and interactive, and a key construct related to the availability of these representations is *representational fluency*. Researchers have used the phrase representational fluency to refer to the interaction between student and representation, usually indicating an ability to move between and among representations, carrying the meaning of an entity from one representation to another and accruing additional information about the entity from the second representation. Entailed in representational fluency is the ability to draw meaning from different representations of the same mathematical entity and to generalize (Becker & Rivera, n.d.) across different representations. Sandoval, Bell, Coleman, Enyedy, and Suthers (2000) provided a comprehensive definition of representational fluency—one that serves well in capturing the phenomenon in the context of multirepresentational techno-mathematical environments:

We view *representational fluency* as being able to interpret and construct various disciplinary representations, and to be able to move between representations appropriately. This includes knowing what particular representations are able to illustrate or explain, and to be able to use representations as justifications for other claims. This also includes an ability to link multiple representations in meaningful ways. (p. 6)

In this chapter, the term representational fluency is construed more broadly than the translation between representations. It includes, as suggested by Sandoval and colleagues, interacting with each of the representations in a meaningful manner. For our purposes, representational fluency includes the ability to translate across representations, the ability to draw meaning about a mathematical entity from different representations of that mathematical entity, and the ability to generalize across different representations. Our operational definition aligns with the notion of representational versatility that Hong and Thomas (2002) took “to include both fluency of translation, and the ability to interact procedurally and conceptually with individual representations” (p. 1002).

Representational Fluency and Mathematics Learning

The study of mathematics centers on the study of abstract concepts, and individuals' access to those

concepts is mediated⁶ through externally present representations of those concepts. These externally present representations are organized in representational systems—sets consisting of external representations, the entities they represent, and the rules that govern transformations of those representations. Each representation system has the potential for revealing certain properties of the represented quantity (and for concealing others), and a set of representations with the potential for foregrounding different aspects of a mathematical entity ostensibly affords students opportunities to consider those entities in a broad range of settings. Availability of representations, however, is insufficient to ensure that students will take advantage of their availability; representational fluency is the issue. Kaput (1989) described the intimate relationship between representational fluency and mathematical meaning, citing translations between mathematical representation systems and translations between mathematical representations and nonmathematical systems (e.g., physical systems, natural language, pictures) as two of four primary sources of mathematical meaning (p. 168). Through the use, interpretation, and coordination of multiple (external) representations teachers and students communicate about mathematical entities; when a broad range of representations is coupled with representational fluency, the opportunity for building rich meaning grows. Technology expands the repertoire of representations available to students for the mathematical entities they study as well as the opportunities to carry meaning between representations; dynamic, interactive technology is fertile ground for the development and use of representational fluency.

The importance of representational fluency is in its capacity to enhance mathematical understanding, and researchers have provided evidence that mathematical understanding emerges when students actively engage in making connections between multiple representational forms. Hall, Kibler, Wenger, and Truxaw (1989) observed how important situational representation was to the success of students who had mastered manipulation of symbolic representations, years later, when they were asked to develop answers to standard word problems. Koedinger and Tabachneck (1994) noted the importance of informal representations in college students' solutions of word problems, and Tabachneck and colleagues (Tabachneck, Koedinger, & Nathan, 1994) observed that merely having access to representations did not ensure student success with word problems. Although their study focused on the strategies students used in solving word problems, it

can be interpreted to suggest that success in solving word problems using multiple representations depends on students' abilities to use the representations interactively, that is, it depends on their representational fluency. Students differ in the extent to which they use representations interactively (that is, in their representational fluency), and these differences are linked to their success in solving verbally stated problems (Tabachneck et al., 1994). Representations are said to be used interactively when an intermediate representation provides feedback that informs the original representation. One with representational fluency can turn to another representation when he or she encounters a local difficulty and return to the original representation once the difficulty is overcome. Researchers (Gluck, 1999/2000; Koedinger, 2001; Koedinger & Anderson, 1998) have identified ways in which technology can encourage the development of representational fluency. For example, coaching students, at opportune times, in appropriate use of the available technological representations in thinking about labels for quantities can assist them in making connections to more concrete situations, and technology-assisted support for thinking about numerical relationships can help students make connections to arithmetic knowledge and facilitate the transition to fluent use of algebraic symbols.

Representational Fluency and Technological Environments

Because technology has the potential for broadening the representational horizon, consideration of representation is fundamental to the development of educational technologies that support problem solving, learning, and learner-centered design issues (Underwood et al., 2005, take this position in their consideration of the development of applets). Researchers (Hoadley & Kirby, 2004) see the thoughtful incorporation of representations as essential to the development of technology:

Thus, representations do a lot of work in the learning context. . . . Seeing the representations that form part of the educational environment and trying to optimize those representations is an important part of the design and instruction. In particular, computers give us unprecedented ability to create, manipulate, and share representations of a wide variety of types. Understanding how they do their work, and how we may design them, is vital. (p. 2)

Enhancement of representational fluency is an affordance, not an automatic result, of the use of tech-

⁶ We consider computer tools to be one of Vygotsky's "psychological tools" or "signs," and hence a means of mediation.

nology. Depending on how it is configured and used, technology can function as cognitive support for the development of representational fluency. Technology like Geometer's Sketchpad or Cabri that dynamically coordinates the actions taken by the user with the representations that appear on the screen has the potential for helping the user to draw meaning about a mathematical entity from its representations if the user reflects on the connection between his or her action and the change in representation. Technology with hot-linked representations that automatically and simultaneously update in reaction to a user action provide the potential for enhancing the user's ability to translate and generalize across representations of the same mathematical entity. The redundancy in information offered by multiple-linked representations has the potential to enhance the user's ability to draw meaning about a mathematical entity from different representations of that mathematical entity. As Kaput (1989) pointed out, "We can have our cake and eat it too, in the sense of being able to trade on the accessibility and strengths of different representations without being limited by the weakness of any particular one" (p. 179). Kaput explained further the potential of technology for enhancing representational fluency but warned that the success of the enhancement depends on the choice of representation and the configuration of the learning environment:

Appropriate experience in a multiple, linked representation environment may provide webs of referential meaning missing from much of school mathematics and may also generate the cognitive control structures required to traverse those webs and tap the real power of mathematics as a personal intellectual resource. However, one must choose such representations carefully and, even more importantly, ensure that the learning environment supports rich sets of actions that will expose underlying invariances and thus enable the student to weave a flexible and enduring web of mathematical meaning. (Kaput, 1989, p. 180)

Gifford (2005) echoed that point of view as he pointed out that technological environments alone are unlikely to enhance representational fluency and that characteristics of curricula that are likely to enhance representational fluency are those characteristics that are inherent in the dynamic interactive representations present in many mathematics-specific technologies:

Research . . . suggest[s] that lessons that afford students access to increasingly complex mathematical representations, or that require students to negotiate their way between and among multiple representations of the same mathematical concepts, procedures, or problem-solving strategies are much more likely to

increase representational fluency than conventional mathematics print-based curricula materials. (p. 3)

Representational Fluency as a Construct for Research

The promise of representational fluency as a construct lies in its potential for providing a focused representational perspective to research on mathematics learning and teaching. In particular, representational fluency provides a lens researchers can productively use in investigations of mathematical activity and of students' mathematical thinking.

Representational fluency as a lens to examine the mathematical activity in classrooms. When viewed as a unit, the ability to translate across representations, the ability to draw meaning about a mathematical entity from different representations of that mathematical entity, and the ability to generalize across different representations can be examined as they affect mathematical activity. Researchers have addressed the effects of representational fluency on mathematical inquiry and on the foci of mathematical curricula.

Research studies have suggested ways in which representational fluency can catalyze mathematical inquiry. For high school precalculus students' investigations of asymptotes in a qualitative technological environment, work in one representation precipitated work in a parallel representation; one student's graphical description of a function with a nonhorizontal slanted linear asymptote ("There is an imaginary line that divides the function in two," Yerushalmy, 1997, p. 18) led her group to a discussion of the form of symbolic function rules that would produce such results. Yerushalmy observed, "The learner's action on an unexplored object presented in one representation (the graph produced by the software) leads to inquiry about the processes underlying that object explored through a parallel representation (the symbolic procedure)" (p. 22). The representational fluency present in the group because of individuals' collective knowledge of representations allowed the group to traverse a path from graphical to symbolic in order to pursue their conjecture. This notion of representational fluency of a group was also studied by Schwartz (1995) who analyzed how students make shared use of multiple representations when they work together. One wonders how the representational fluency of a group relates to the representational fluency of individuals in the group. There may be a connection to expressive activity and exploratory activity (described in the section, Students and Mathematical Activity). In a group, those who contribute most to the discussion may be shaping the group's representational fluency by engaging in expressive activity; those who watch and interpret may be engaged in exploratory

activity. So, representational fluency can be a function of either expressive or exploratory activity, and group members may develop different levels of representational fluency depending on the nature of their expressive or exploratory participation in the group's activity.

The potential that representational fluency offers in the context of multirepresentational settings affords the opportunity for curricula to focus on central mathematical objects and processes and for researchers to study the development of an understanding of these central objects. Chazan (1999) observed that whereas the focus of his teaching in an introductory algebra class in a traditional setting centered on procedures of a litany of solving equations and simplifying expressions, his introductory algebra class in a multirepresentational technological setting focused on central mathematical objects of function and variable. The promise of this functions approach to algebra rests squarely on students' development of representational fluency.

One curriculum constructed as a multirepresentational technology-intensive, functions-based approach to introductory algebra is the Computer-Intensive Algebra (CIA) curriculum (Fey et al., 1991). Analysis of the effects of this newly configured content has been highly influenced by the construct of representational fluency. Studies of implementations of CIA in 8th- and 9th-grade classes demonstrated that a technology-intensive, functions-based introduction to algebra could be constructed to productively enable students to develop an understanding of the concepts of algebra while they used computing tools to move between and among representations and to execute routine procedures (Heid, 1996a). Understanding of fundamental concepts of introductory algebra seemed to be evidenced through representational fluency, and curricula that capitalized on multirepresentational technology provided a platform for activation of that fluency. A range of research studies have documented that in the CIA classes students demonstrated representational fluency as they used computers as tools in solving realistic problems (Heid, Sheets, Matras, & Menasian, 1988; Heid & Zbiek, 1993). That this development of representational fluency may have been a function of the curriculum rather than of the ability level of the students was suggested in a study that compared 9th-grade CIA students with 11th-grade students in an advanced mathematics class. This research reported that students in the 9th-grade CIA classes developed greater flexibility in reasoning with multiple embodiments of mathematical functions than did a group of comparable aptitude that had 2 years of additional exposure in traditional mathematics courses beyond high school algebra (Sheets, 1993).

Representational fluency as a lens to examine how students think about mathematics. Attention to representational fluency has also enabled researchers to observe the ways students call on representations as they develop strategies for solving problems. The availability of multiple representations and the rapidity with which they can be generated provides students with options when they encounter surprising results, and some researchers have suggested that at times those options allow students to avoid interpreting unfamiliar results (Tabachneck et al., 1994).

These situations can easily arise in the context of work with CAS, in that there is a constant need for CAS users to interpret results, and that need elevates particularly when the results of computations are unexpected. In the context of a teaching experiment with 6th- and 7th-grade students in their first formal exposure to work with symbolic algebra (Heid et al., 2002), Kevin, a very bright 7th-grade student, encountered an unexpected result. In the course of creating a function rule to match a given set of characteristics, Kevin noticed that the CAS was giving him a positive value for $|\sqrt{x+5}|$ when $x = -10$. Not having previously encountered

$$|a + bi| = \sqrt{a^2 + b^2}$$

as the modulus of a complex number, Kevin did not venture further on trying to interpret the result and revised his rule to eliminate the expression, $|\sqrt{x+5}|$. Because of the ease of changing the particular symbolic representation and the capability of testing the new rule immediately by graphing it, Kevin was able to quickly generate an acceptable rule. With the multiple representations available in technological environments, students are afforded the opportunity to pursue any of a range of alternative strategies, at times enabling them to steer clear of unfamiliar content. Kevin's encounter with contradictory information was not unlike that of Trouche's (2005b) student whose resourceful work method was described previously in this chapter. Although Kevin's resolution of the conflict led to a correct result, Trouche's student adopted a new (incorrect) theorem to explain the conflict between a previously learned result concerning limits of polynomial functions and the result suggested by the values in a table of function values. Perhaps the availability of multiple representations contributes to students' adoption of a resourceful work method—one that employs multiple approaches—or adoption of a resourceful work method increases the likelihood that students will call on multiple representations.

The availability of multirepresentational technology has opened new curricular horizons in mathematics classrooms, and representational fluency is key to thinking about the effects of those new curricula. Kaput (1994), in his SimCalc project, developed ways to use multirepresentational technology to introduce the ideas of calculus to young learners. Stroup (2002) posited that the cognitive constructs on which younger learners can build as they engage in qualitative calculus experiences differ considerably from the cognitive structures of standard calculus. He noted that the overlap of the standard calculus and the qualitative calculus “allow[s] us to reason powerfully about change and to engage successfully the dynamism of our experience,” and suggested that the technology enables a “*dynamics revolution* in learning” (p. 206). The potential for dynamic, interactive, hot-linked, and connected multirepresentational technology to enable the observation of students’ representational fluency allows researchers to focus on how students (including younger students) can think differently about traditional topics.

Not only has the potential for representational fluency accelerated the introduction of traditional topics, it has allowed researchers to examine students’ (and teachers’) understandings of concepts that have not been a traditional part of the curriculum. In some cases, these studies have revealed general ways of understanding that appear to be critical in these new environments. For example, the dynamic linked-representational statistical tool, Fathom, allows students to simulate random sampling and examine graphs of the resulting distributions of statistics as the samples are generated. A study examining students’ understanding of sampling distribution revealed the complexity of making distinctions among mathematical representations and calling up and using those distinctions at appropriate times (Heid, Perkinson, Peters, & Fratto, 2005). Understanding of sampling distribution required students not only to make careful distinctions involving a sampling distribution, proper subsets of that distribution, a single sample, and a population, but to properly select and apply probabilistic or deterministic reasoning to those concepts—all in the context of representations that appeared very similar. In this case, representational fluency required holding on to the mathematical/statistical meaning of each representation. This ability to make and manage intricately related mathematical distinctions is reflective of representational fluency at a high level of sophistication.

The study of representational fluency can be tightly targeted. Cobb (1999), for example, used purposefully produced tools with constrained capacities

so that he and his colleagues could observe particular facets of students’ reasoning. The minitools focused 7th-grade students’ attention on characteristics of distributions, and the researchers observed students creating a new concept of “hills” to describe features of the data sets students graphed.

The promise of representational fluency as a construct. With a constantly expanding array of newly conceived representations and with current ready access to dynamic, interactive, hot-linked, and connected multirepresentational technology, mathematics teachers and students are becoming increasingly aware of the promise of representations in mathematics teaching and learning. Key to learning, however, is not solely the availability of externally present representations but the actions taken on those representations. That is, it is not so much that the representations exist but that the student interacts in meaningful ways with those representations. Representational fluency as a construct focuses attention on these meaningful mathematical actions.

Representational fluency can be thought of as meaningful and fluent interaction with representations. Researchers can investigate students’ representational fluency as outcomes, as conditions, or as stages of development. A mathematical entity can be viewed through any of its myriad representations, and students’ understandings of an entity are revealed in their interactions with representations. Researchers can use representational fluency in developing an understanding of how students think about traditional school mathematics or about mathematics newly available to schools because of technology. The lens of representational fluency is also an important tool for revealing the nature of students’ understandings of fundamental mathematical ideas. Students who move from one representation of a mathematical entity to another while holding on to the meaning of each of the representations are exhibiting a certain depth of understanding of that mathematical entity. Without the multirepresentational capacity of technology, researchers would not have as broad an access to observing student interaction with representations.

Researchers have offered two somewhat opposing points of view regarding the role of representational fluency in mathematical problem solving. Some (Heid et al., 2002; Tabachneck et al., 1994) have pointed to representational fluency as allowing an end-run for students as they encounter a result they cannot explain. Students with a high degree of representational fluency can quickly generate an alternative representation and avoid having to interpret an unexpected result. Other researchers have credited representational fluency with allowing students to focus on the

mathematical entities being represented instead of on a singular representation of that entity (see Chazan, 1999). Because of the availability and ease of production of multiple technology-based representations of the same mathematical entity, research using this construct can focus on the overall meaning that students credit to a range of representations. By examining how students use fundamental concepts in their work within a range of representations, research using the construct of representational fluency as a lens can provide evidence of the depth of students' understandings of the concepts themselves. Using either perspective on the role of representational fluency, researchers can structure their inquiries into students' mathematical thinking.

Finally, as a construct, representational fluency directs researchers' attention to a capability that may be linked to students' levels of mathematical understanding and allows researchers to examine the extent to which outward manifestations of capabilities in dealing with representations are indicators of deep understanding. A central set of research questions occurs at the interface of representational fluency and other constructs. A productive line of research, for example, might be the relationship between representational fluency and work method or between representational fluency and privileging. Is a student who adopts a resourceful (multiapproach) work method likely to develop a higher level of representational fluency than a student who uses a rational work method (deriving results not from instances but from theorems)? Do particular work methods hide students' representational fluency? Are the students of teachers who privilege multiple representation types or multiple technologies likely to develop more sophisticated representational fluency than the students of teachers who privilege only one representation or technology? These, and a host of other related questions, can help to structure informative empirical studies.

Mathematical Concordance

New content with technological tools can be either intended or unintended. In some cases, technological representations of mathematics are mathematically misleading, and we have referred to the construct of mathematical fidelity to describe this mismatch. But the lack of alignment between the characteristics of a technology-generated representation and the mathematical characteristics and behavior of the represented object are not the only sites for mathematical misalignment. Lack of alignment of mathematics can occur in any of the range of sources for the mathematics that arises in technology-intensive learning en-

vironments, including the teacher's mathematics, the student's mathematics, the text's mathematics, and the tool's mathematics. We capture the compatibility between any two elements of the range of different embodiments of mathematics that arise in technology-intensive environments with the term *mathematical concordance*. We will describe the degree of mathematical concordance as a continuum. Mathematical concordance is different from mathematical fidelity in that mathematical fidelity is about a standard, whereas mathematical concordance is about degree of alignment without reference to a standard.

Researchers have, for example, investigated concordance between the mathematics in which students engage and the mathematics the teacher intended in designing the technological task. At times this low level of concordance is manifested in students' not substantially engaging in the intended mathematical task. In their description of research on the relationship between what was intended to be learned and what was actually learned in the context of 13-year-olds' work on mathematics within a Logo microworld, Hoyles and Noss (1992) observed the task of having the Logo turtle sketch a larger House similar to a given one revert into the task of creating a larger house that looked similar to the given House. Whereas the researchers' and instructors' goal was to engage the students in thinking about proportions, the students' task was to build a bigger House without paying attention to ratios. The students' mathematics circumvented the mathematics intended by the teacher. Incidents like this having drawn their attention to this mathematical discord, the researchers noted in their discussion of the play paradox the lack of mathematical concordance:

The issue we want to pin-point here is what we have termed the *play paradox*. This derives from the following contradictory situation: on the one hand, the computer offers pupils rich and diverse ways of exploring and solving a problem and building upon rather fragile intuitions of the mathematical domain. Yet in exploiting this diversity, pupils are as likely to avoid encountering the mathematical nuggets so carefully planted by their teachers (although of course they may encounter other—unintended—ones). (pp. 45–46)

Dugdale (2007) observed a related phenomenon at the beginning of her work in the development of the program Green Globbs. The goal of Green Globbs was to write equations whose graphs pass through a pseudo-randomly generated set of globbs on a Cartesian plane. Although the program was intended to engage students in thinking about the graphs of different families of functions, students subverted this goal on

an early version of the program when they discovered functions such as $y = 10\sin(10x)$ to be single functions whose compact periodicity ensured that they would hit every glob. In this case there was a low level of concordance between the intended mathematics of the task and the mathematics in which students engaged. Although not fully engaged in the mathematics of the task as intended, the students' mathematics was related to the intended mathematics.

Sutherland and Balacheff (1999) provided a salient example of a low level of mathematical concordance in their report on teaching experiments that revealed "the tension between the student's constructions (which are tied to the phenomenology of the screen, and the student's previous knowledge) and the knowledge which the teacher intends the student to learn" (p. 21). Their example (see Figure 27.6) occurs in the context of a student working on the following problem in a Cabri-Geomètre environment (the authors cited the source of the problem as Capponi & Laborde, 1995, Cabri-classe, Sheet 4-10).

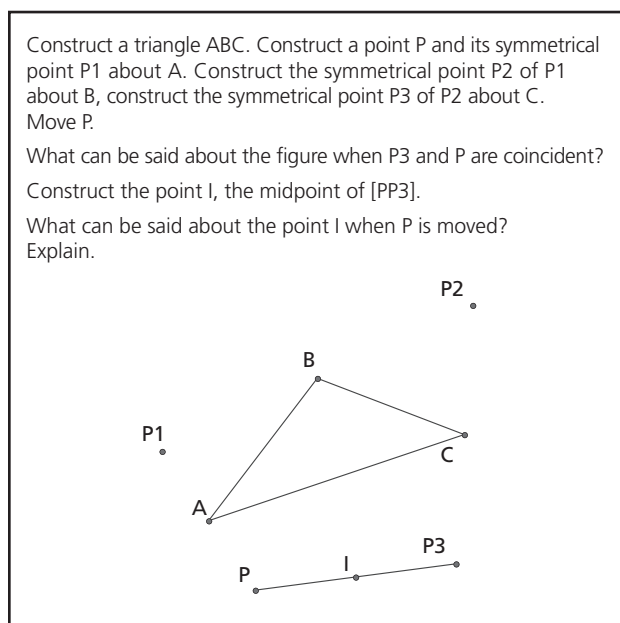


Figure 27.6 Capponi & Laborde's (1995) geometry task.

As intended by the teacher and the task, the student in the study noticed that, as P is moved to different parts of the plane, I is stationary and that ABCI is a parallelogram. The teacher understood that because ABCI is a parallelogram and because A, B, and C are fixed, then I is also fixed. The student focused on establishing that ABCI is a parallelogram but did not see the connection to I being stationary. When the student, with scaffolding provided by the teacher, arrived at a proof that ABCI is a parallelogram, the teacher

was satisfied that the student had solved the problem, despite the fact that the student never established why I is stationary. Identification of the low level of mathematical concordance in this example highlights the importance of the teacher's awareness of the extent to which the students' mathematical goals match those of the teacher. The researchers accentuated the importance of the teacher's recognition of level of mathematical concordance and pointed out that degree of mathematical concordance may be more visible to them in technological environments, arguing that

the teacher is a crucial mediator between individually constructed knowing and socially constructed knowledge. Whatever the medium, students can always construct meanings which are different from those which the teacher intends, the situation is no more complex with computational media than it is with paper and pencil media. However we suggest that with computational media the meanings which students construct are potentially more transparent to teachers if they learn to make sense of the phenomena of the screen. Moreover student/teacher interactions are also potentially more transparent to researchers when students work with computer-based environments. (p. 23)

As the researchers indicated, it is important that the teacher recognized the degree of mathematical concordance and accurately interpreted the student's mathematics in order to steer the student's attention in a fruitful direction.

Although mathematical concordance is certainly not confined to learning in technological environments, the insertion into the student-task-teacher mix of an additional party (the computer) that has its own rules of operation can increase the probability of lack of agreement on mathematical goals. Filtering one's research on mathematics learning in technological environments through the construct of mathematical concordance may provide a more complete account of technology-based learning of mathematics. Of course, divergent mathematical interpretations do not always arise from one of them being incorrect. Sometimes a low level of concordance arises from variant interpretations of the images on the screen. Goldenberg (1988) pointed out how one could provide numerous interpretations for the nature of transformations of graphs as they appear on computer or calculator screens. For example, he pointed out the difficulty of interpreting the transformation of the graph of $f_1(x) = x^2 - 2$ to the graph of $f_2(x) = 2x^2 - 2$; three different possible transformations (horizontal scaling transformation, vertical transformation, and rotational transformation) are suggested in a diagram that appeared in Goldenberg's article (see Figure 27.7).

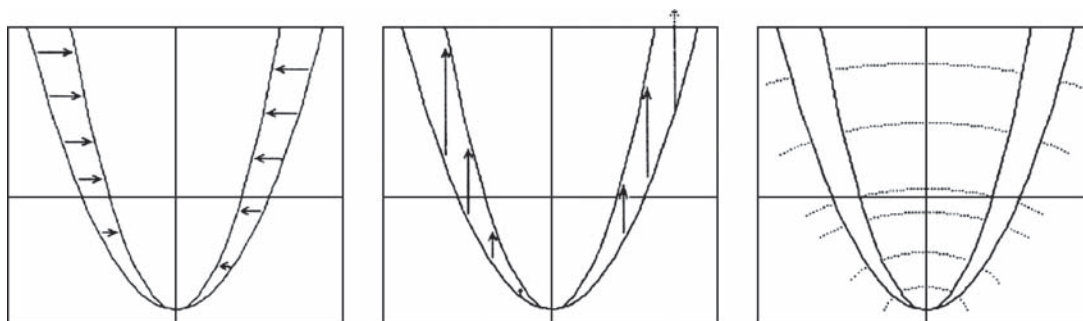


Figure 27.7 Visual descriptions of three different transformations of a graph (Goldenberg, 1988, p. 158). *Note.* Reprinted from *Journal of Mathematical Behavior*, 7, E. P. Goldenberg, Mathematics, metaphors, and human factors: Mathematical, technical, and pedagogical challenges in the educational use of graphical representations of functions, pp. 135–173, copyright 1988, with permission from Elsevier.)

At times the teacher's mathematics will appear to be different partly because the teacher is operating in a new environment and taking some shortcuts. An example of this was reported by Heid and colleagues (Heid, 1996b; Heid, Zbiek, & Blume, 1994) when they recounted how a teacher's (Sara's) weak concept of function notation coupled with her minimal understanding of the computer algebra program she was using led to her systematic misuse and misinterpretation of mathematical notation for function. Within Sara's tasks was the need to evaluate expressions with function notation, such as $f(4)$ given the function f with rule $f(x) = 3x^2 - 10x + 5$. She was using a CAS that required the user to enter in one window an expression to be evaluated and to go to another window to redefine a function. Being somewhat unsteady in her use of function notation, being tentative about the working of the computer program, and being accustomed to using the window to redefine functions, whenever Sara needed to find the value of a function, f , for a given input value, a , she redefined the function by inserting the input value into the function rule in the redefinition window instead of issuing a command to evaluate $f(a)$. For example, to evaluate $f(4)$ given $f(x) = 3x^2 - 10x + 5$, Sara input the expression, $3(4)^2 - 10(4) + 5$, in the redefine window. There was no conflict generated by this action in as much as her action resulted in the numerical value she needed. Her action, however, changed the function rule to a constant function with a value of $f(4)$ and resulted in her losing the potential to use the function rule stored in the computer (now newly defined) in the remainder of the task. Her students followed her strategy with resulting confusion when subsequent use of the function rule stored in the computer yielded unexpected results. What Sara saw as an evaluation tool was designed as a redefinition tool. In this case, the low

level of concordance between the mathematics of the teacher and the mathematics of the tool had unanticipated, unfortunate consequences for her students.

The construct of mathematical concordance can serve to remind researchers that, in analyzing the ways that teachers and students interact with cognitive tools, it is helpful to consider the mathematics of the tool, the mathematics of the teacher, the mathematics the teacher intended through particular technology-based activities, the mathematics that the student engaged in as a result of the technology-based activity arranged by the teacher, and the mathematics that is learned. The construct of mathematical concordance reminds researchers of the importance of the matches and mismatches among these different understandings of mathematics in technological environments.

Amplifiers and Reorganizers: New Emphasis and New Order

In addition to creating technology-intensive curricula that affect the content of school mathematical activity, researchers have developed ways of thinking about technology-based curricula that suggest alternative organizations and emphases in mathematics instruction. Each of these new configurations is accompanied by research on the effects of that vision of curriculum. An overall way to think about and classify these new approaches was suggested by Pea (1985), who used the constructs of *amplifiers* and *reorganizers* to describe possible curricular roles of technology. Pea pointed out that certain uses of technology are amplifiers in that they accept the goals of the current curriculum and work to achieve those goals better. Others are reorganizers in that they change the goals of the curriculum by replacing some things, adding others, and reordering still others. Heid (1997)

found these constructs useful in characterizing and differentiating among the ways in which technology had been used in reports of curriculum research to change the order, emphasis, or content of the mathematics curriculum. Early research on the effects of technology-intensive curricula started in the context of the reform of college calculus courses, and the range of curricular arrangements used in these studies can be characterized by the constructs of amplifier and reorganizer. Technology use in the “calculus reform” projects fell somewhere along a continuum of reorganizers in that, when technology was used, it generally had the effect of changing the content of the calculus curriculum in some significant way. Technology was instrumental in restructuring courses around applications, in making a multirepresentational approach feasible, and in helping focus calculus courses on fundamental concepts. Myriad dissertations and other studies focused on analyzing the effects of these restructured technology-intensive curricula (e.g. Beckmann, 1988/1989; Bookman, 2000; Chappell & Killpatrick, 2003; Cooley, 1996; Cunningham, 1992; Ellison, 1994; Estes, 1990; Girard, 2002/2003; Hart, 1991; Hawker, 1987; Heid, 1988; Judson, 1988, 1990; Meel, 1996; Melin-Conejeros, 1992/1993; Palmiter, 1991; Park, 1993; Park & Travers, 1996; Parks, 1995/1996; Porzio, 1995; Roddick, 1998; Schrock, 1990; Wells, 1996). Similar distinctions can be made about research on curricular approaches to the teaching and learning of algebra at the school level. By taking into account the reorganizer–amplifier distinction, researchers can isolate important structural differences in the uses of technology in the teaching and learning of mathematics.

Sequencing and Emphasis: Microprocedures and Macroprocedures

Technology served as a vehicle to reorganize the curriculum in many of the calculus reform projects of the 1990s, and researchers studying those curricula treated the reorganized curricula as a significant contextual attribute. The advent of symbolic algebra software made possible attempts to have students use computing technology to execute routine procedures, which led to questions about the role of traditional routine procedures without the calculator in mathematics curricula. One category of concerns centered on the sequence of treatments of procedures, examining whether it was better for students to learn the by-hand procedures before using technology to produce the results or whether it made more sense to have students learn to use the technology to produce results before learning by-hand procedures to produce those results.

A pair of constructs that is useful in conceptualizing the use of computer algebra systems in mathematics instruction is that of *microprocedures* and *macroprocedures*. For example, Heid (1988) investigated the effects on students’ conceptual and procedural understandings of a conceptually oriented curriculum that deferred development of routine skills (microprocedures) until the last few weeks of a 15-week applied calculus course and that focused student attention during the first part of the course on the general procedures (macroprocedures) and concepts of calculus. Heid (2003) used the constructs of macroprocedure and microprocedure to describe the design of curricula like these: “Mathematical procedures can be viewed as macroprocedures whose major procedural steps are component microprocedures” (p. 39). Computing technology makes the enlistment of macroprocedures (and the deferment of by-hand execution of microprocedures) possible with the considerable capability of technology to allow the user to chunk mathematical work in packages of various sizes. Lagrange (2000) and Artigue (2002) have raised issues about such resequencing efforts, pointing to the difficulties of what the French have termed the conceptual–technical cut. Ruthven (2002), however, pointed out that the macro-micro strategy in the Heid study deferred only routine skills and not the more general notion of technique promoted by the French researchers:

In its own terms, then, the experimental teaching approach studied by Heid did indeed succeed in ‘resequencing skills and concepts.’ It deferred routinisation of the customarily taught skills of symbolic manipulation until the final phase of the applied calculus course, while the attention given to a broader range of problems and representations in the innovative main phase supported development of a richer conceptual system. Equally, however, in the terms of the French theory, this conceptual development grew out of new techniques constituted in response to this broader range of tasks, and from greater opportunities for the theoretical elaboration of these techniques. At the same time, standard elements emerged from these new tasks, characteristic of the types of problem posed and the forms of representation employed, creating a new corpus of skills distinct from those officially recognised. (p. 284)

The constructs of microprocedures and macroprocedures draw on notions of subprocedures and superprocedures, constructs proposed by Davis (1984) to describe sequences of instructions in mathematics from a cognitive science perspective. Microprocedures and macroprocedures, however, are intended to encompass more than sequences of instructions and are

broad enough to take into account the wider notions of technical expertise and knowledge suggested in the French notion of technique. It is the notion of controlling the grain size of the mathematical activity that is encompassed in the constructs of macroprocedures and microprocedures. Certain configurations of technology put that control in the hands of the teacher and the learner.

Constructs related to technology's effects on the content of school mathematics have potential for affecting research on learning, teaching, and curricular alternatives. Representational fluency provides a lens for understanding mathematical activity and students' mathematical thinking. Mathematical concordance provides a way to think about the extent to which the tool, the teacher, the students, and the curriculum address the same mathematics. The constructs of amplifier and reorganizer provide a global categorization scheme for the curricular content choices that technology enables. The constructs of macroprocedure and microprocedure address the relationship between ways of learning mathematics and ways of organizing mathematics curricula. Representational fluency and mathematical concordance provide targets for attention in research that examines students' thinking and interactions in technological environments. Amplifier, reorganizer, macroprocedure, and microprocedure provide ways for researchers to investigate technology-intensive curricular approaches.

SPECIFIC CONTRIBUTIONS OF CONSTRUCTS

We have focused this chapter on identifying and elaborating on constructs that have promise for informing research on technology in the teaching and learning of mathematics. In that endeavor we have recognized the importance of constructs to refine current directions for research, to suggest new directions, to identify promising variables, to assist research in dealing with the complexity of technology-based mathematics teaching and learning, and to characterize how learning is different (and how it is the same) in technological and non-technological environments.

Refining current directions for research. Consideration of constructs can sharpen one's research focus. It helps researchers deal with complexity by identifying critical features of the phenomena they are observing. For example, the construct of instrumental genesis points to the importance, in examining the effects of technology in mathematics learning, of documenting not only what a student does but the student's pat-

terns of technological action with the tool as well as the student-tool relationship.

Suggesting new directions for research. Constructs help identify targets for new research endeavors as well as ways to organize and interpret existing research. For example, because technology-based approaches to mathematics curricula are remarkably varied, it is helpful in generalizing their effects to look for commonalities. Even though researchers may not have identified their technological approaches as amplifiers or reorganizers, applying these constructs to their reports of curricular research helps in recognizing a critical difference in technology-intensive mathematics curricula. Focus on constructs also raises issues about missing areas of research. Realizing the plethora of constructs relating to inappropriate uses of technology, for example, researchers may be motivated to study and develop characterizations of judicious use of technology.

Identifying promising variables. In mathematics, defining an entity brings it into being. In mathematics education, identifying a construct brings the phenomenon or particular distinctions related to the phenomenon into awareness. Constructs give researchers a finer grained descriptive language for variables in the teaching and learning process—the same variable might function as an independent variable, a dependent variable, or an intervening variable. For example, student achievement could be analyzed as a function of students' dragging behaviors, students' dragging behavior could be analyzed as a function of the type of mathematical activity in which the students are engaged, or students' dragging behavior could be viewed as an intervening variable between the mathematical activity and student achievement.

Dealing with the complexity of technology-based mathematics teaching and learning. Technology mediates (and complicates) the complex interplay among teacher, learner, tasks, and content. The insertion of technological tools (with their own independent rules of operation) as mediators in the teacher–student–task mix increases the likelihood of a mismatch among mathematical or pedagogical goals. Identifying constructs related to these goals helps focus research on important relationships related to these goals. Identifying constructs related to technology use with respect to teachers, students, activity, and content can help researchers handle the complexity. Recognizing the importance of constructs such as pedagogical fidelity and privileging helps to capture important features of teaching in technological contexts. These features can then be examined more closely. Some genres of constructs are particularly helpful in dealing with complexity. Fidelity constructs enable focus on the compatibility of goals. The insertion of an additional

party into the teacher–student–task mix that has its own rules of operations increases the probability of some degree of mismatch among mathematical goals. This highlights the importance of determining the degree of agreement among different understandings of mathematics in technological worlds. It offers the opportunity for a more complete account of technology-based learning of mathematics. Constructs suggest promising configurations of phenomena of interest in research endeavors to explain the impact of technology in the teaching and learning of mathematics. For example, having identified roles of technology, pedagogical fidelity, and instrumental genesis as constructs of importance, one might be drawn to consider elaborating how roles taken by technology or by the teacher in a technological mathematics setting relate to pedagogical fidelity or instrumental orchestration.

Characterizing how learning is different (and how it is the same) in technological and nontechnological environments. Constructs can shed light on the particular facets of learning mathematics that are different in technological contexts. They accentuate the differences technology makes in learning different types of mathematics (conceptual and technical). For example, one might research the role of cognitive tools, perhaps as embodied in intelligent tutors, in learning skills or developing techniques and the role of multirepresentational software in developing an understanding of mathematical concepts. Constructs can identify specific differences in the technology itself that make a difference in mathematics learning. For example, highlighting features of mathematical technology that characterize the representations they offer may lead to research on the effects of those features. Research using the constructs that describe representations can better illuminate the depth of students' understandings of the concepts themselves.

Constructs can promote boundary crossing. Some constructs that arise outside of mathematics education can also be useful in analyzing the teaching and learning of mathematics, and some constructs that arise in technology-related research can also be informative in nontechnological settings. For example, the construct of instrumental genesis, drawn from outside of mathematics education, turns out to be particularly useful in thinking about technology use in mathematics teaching and learning. Having developed the application of instrumental genesis to computing-technology use in mathematics classrooms, it is natural to apply the construct to other uses of tools (say, manipulatives) in mathematics classrooms.

Consideration of constructs can be informative to teachers and tool developers as well as to researchers. Teachers aware of the notion of privileging may have more of a

tendency to examine their own teaching for this bias. Having identified the constructs of mathematical fidelity and pedagogical fidelity, tool developers may have more of a tendency to attend to these features as they build tools, and potential users may have more of a tendency to consider these features in their choice and use of tools.

CONCLUSION

The identification of constructs important to the teaching and learning of mathematics in technological environments paves the way for a comprehensive and connected approach to research in this area. It is our hope that by drawing attention to these constructs we will assist in the formation, framing, and situation of that research. Focusing on the constructs we have identified will assist researchers in explaining not only the effect of technology on the student, teacher, mathematical activity, and curriculum content, but also the effects of technology on the interaction among student, teacher, mathematical activity, and curriculum content in environments that are characterized by intensive use of cognitive mathematics technologies.

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ENGINEERING CHANGE IN MATHEMATICS EDUCATION

Research, Policy, and Practice

William F. Tate

WASHINGTON UNIVERSITY IN ST. LOUIS

Celia Rousseau

UNIVERSITY OF MEMPHIS

In 1985, the Mid-Atlantic Equity Center published *Mathematics and Science: Critical Filters for the Future*, addressing in part mathematics education, school leadership, and academic opportunity for traditionally underserved students. In her monograph, DeAnna Banks Beane argued, “The success of many intervention programs demonstrates that there are no permanent barriers to minority student achievement in science and mathematics. However, the data tell us that the longer we wait to intervene, the more invincible the barriers become” (p. 1). Her remarks are a reminder of the challenges and opportunities in school mathematics requiring clarification and associated strategies for change and improvement. This chapter represents an effort to build upon and extend beyond the literature on school mathematics as discussed in *Mathematics and Science: Critical Filters for the Future*. The purpose of this chapter is to provide a review of mathematics education research and relevant policy literature that can inform school leadership attempting to build effective learning environments for students traditionally underserved in school mathematics.

The political and educational landscape in school mathematics has changed in important ways since

1985. Three significant changes are discussed here. The first change is the introduction of mathematics standards to the education community, specifically the 1989 release of the *Curriculum and Evaluation Standards for School Mathematics* published by the National Council of Teachers of Mathematics (NCTM). This document was a part of a series of mathematics standards documents produced by the Council (NCTM, 1991, 1995, 2000). The role of standards in educational practice and policy-making has gained traction, and today dominates discourse related to school mathematics. The NCTM Standards documents and related educational policy developments have resulted in the rapid evolution of standards-based language. In the post *Curriculum and Evaluation Standards for School Mathematics* era (1990–2002), the word *standard* produced 26,843 documents in the ERIC database. Although the citations were not all directly related to school mathematics, the point here is that standards-based language permeates the education terrain. Most states have standards for school mathematics that signal to local school districts goals for instruction and desired student outcomes.

A second change is a movement calling for educational leadership to more directly address issues of

learning and teaching in schools (Rowan, 1995). Significant changes are taking place in the ways the constructs of teaching and learning are now being defined by researchers, practitioners, and policymakers. During the 1980s and 1990s, cognitive models of teaching and learning were formulated and tested, and many small-scale efforts to transition from the predominant behaviorist models of instructional theory occurred. This research and development has implications for understanding best practice in the design of educational goals, implementation of instructional practices, and development of assessment techniques. Thus, instructional leadership requires a deep understanding of research and development. Many state and federal policies require school district instructional leadership to document the effectiveness or research-base undergirding local change strategies. This represents a new demand on those charged with district-wide and school-level improvements. This chapter is designed for instructional leaders facing today's research-focused managerial demands.

A third and related change in the educational landscape is the *No Child Left Behind Act of 2001* (NCLB). The NCLB Act calls for a new level of Title I accountability by requiring each state to implement accountability systems covering all public schools and public school students. These systems must be based on rigorous state standards in mathematics, annual testing for all students in Grades 3–8, and annual statewide progress objectives ensuring that all groups of students reach established levels of mathematical proficiency within 12 years. Assessment findings and state progress objectives must be broken out by poverty, race, ethnicity, disability, and limited English proficiency to ensure that no group is left behind. School districts and schools that fail to make adequate yearly progress (AYP) toward statewide proficiency goals will, over time, be subject to improvement, corrective action, and restructuring measures aimed at getting them back on course to meet state standards. Schools that meet or exceed AYP objectives or close achievement gaps will be eligible for State Academic Achievement Awards.

Setting rigorous state standards and related accountability models is placing significant pressure on school districts to rethink past practice and to look for effective and sound strategies to support the teaching and learning of mathematics. This chapter is designed to assist teachers, administrators, and community supporters in their efforts to incorporate research-based strategies into the school mathematics program. Additionally, and equally as important, the chapter is an

effort to provide mathematics education researchers with a synthetic review of factors that influence students' learning of mathematics.

ENGINEERING CHANGE: AN OVERVIEW

How can teachers be more productive teachers of school mathematics? How can students' learning of school mathematics be accelerated? These are difficult questions; yet Hamming provided insight into one appropriate response.

Knowledge and productivity are like compound interest. Given two people of approximately the same ability and one person who works 10% more than the other, the latter will more than twice outproduce the former. The more you know, the more you learn; the more you learn, the more you can do; the more you can do, the more the opportunity—it is very much like compound interest. I don't want to give you a rate, but it is a very high rate. Given two people with exactly the same ability, the one person who manages day in and day out to get in one more hour of thinking will be tremendously more productive over a lifetime.¹

The teaching and learning process is embedded in a complex web of schools, school districts, communities, and state governance systems that each play a role in expanding students' opportunity to learn and think about mathematics. Some have criticized the mathematics education community for failing to adequately articulate how access and opportunity to learn mathematics can be expanded to traditionally underserved students (Apple, 1992; Hilliard, 1991; Meyer, 1991). NCTM has recognized this criticism. Recent standards documents produced by the NCTM have called for a focus on equity. For example, the *Principles and Standards for School Mathematics* (NCTM, 2000) stated the following:

The vision of equity in mathematics education challenges a pervasive societal belief in North America that only some students are capable of learning mathematics. This belief, in contrast to the equally pervasive view that all students can and should learn to read and write in English, leads to low expectations for too many students. Low expectations are especially problematic because students who live in poverty, students who are not native speakers of English, students with disabilities, females, and many nonwhite students have traditionally been far more likely than their counterparts in other demographic groups to be the victims of low expectations. Expectations must be

¹ This quote is taken from a transcription of the Bell Communications Research Seminar, March 7, 1986.

raised—mathematics can and must be learned by all students. (pp. 12–13)

High expectations for all students are a new challenge in school mathematics education. Past reform efforts in mathematics education were designed for more select groups. For example, in the post-*Brown* era, the “new math” reform movement sought to improve mathematics education in the United States, as it was thought that good scientific education was a vital component of a strong national defense program and a robust economy (Kliebard, 1987). Initiated in response to the launch of Sputnik by the Soviet Union, this mathematics reform effort designed to address the nation’s scientific crisis did little to address the problems of students of color in urban and rural areas of the United States (Garcia, 1995; Neito, 1995; Tate, 1997). Many responsible for the reform effort stated that their programs should be limited to “college-capable” students (Devault & Weaver, 1970; Kliebard, 1987; NCTM, 1959). The code words *college capable* were a signifier to the educational establishment that only a select few communities and students were appropriate for the reform activities. This is not to say that urban, rural, and poor communities were completely denied opportunities to participate in this reform effort. Instead, these opportunities were limited and insufficient for the curricular and pedagogical changes called for within the reform movement. Thus, for many students—particularly African American and Hispanic students—the late 1950s and 1960s are best characterized as an era of benign neglect with respect to opportunity to learn challenging, high-level mathematics (Tate, 1996).

The focus of mathematics curriculum and pedagogy has evolved in a cyclic fashion. In the late 1960s and early 1970s, a different mathematics movement, “back to basics,” emerged, which focused primarily on elementary and middle schools (NCTM, 1980). This movement was a product of policy directives conceived to address equality of educational opportunity through compensatory education. The back-to-basics effort called for instruction in a narrow set of rudimentary mathematics procedures and facts, often to the exclusion of conceptually rich tasks and advanced mathematical ideas.

Members of the National Council of Supervisors of Mathematics (NCSM) were concerned about the effect this would have on the teaching and learning of mathematics appropriate to the needs in a modern society. A NCSM 1977 position paper urged that we move forward, not “back” to the basics. Not included in the back-to-basics movement were 10 important areas of mathematics students would find essential

as adults: problem solving; applying mathematics to everyday situations; alertness to reasonableness of results; estimation and approximation; appropriate computational skills; geometry; measurement; reading, interpreting, and constructing tables, charts and graphs; using mathematics to predict; and computer literacy. The NCSM position paper was widely influential in school mathematics circles; however, the back-to-basics movement had a pronounced impact on the learning opportunities of low-income urban schools (Strickland & Ascher, 1992).

On the positive side, the basic skills effort resulted in limited gains on narrowly defined aspects of school mathematics for traditionally underserved student demographic groups (Secada, 1992). It served as an existence proof that when teachers and administrators agreed on and supported a common goal in mathematics, students would learn the content. However, as the vision of mathematics education has shifted from largely rudimentary notions to a more challenging standard, the limitations of past pedagogical and school organizational support systems are apparent. The National Research Council (2001) stated the following:

To many people, school mathematics is virtually a phenomenon of nature. It seems timeless, set in stone—hard to change and perhaps not needing to change. But the school mathematics education of yesterday, which had a practical basis, is no longer viable. Rote learning of arithmetic procedures no longer has the clear value it once had. The widespread availability of technological tools for computation means that people are less dependent on their own powers of computation. At the same time, people are much more exposed to numbers and quantitative ideas and so need to deal with mathematics on a higher level than they did just 20 years ago. Too few U.S. students, however, leave elementary and middle school with adequate mathematical knowledge, skill, and confidence for anyone to be satisfied that all is well in school mathematics. (p. 407)

One response to the current state of affairs in school mathematics has been the rapid development and adoption of state-level mathematics standards. Typically, an accountability model is associated with the mathematics content standards to provide indicators of student progress. However, situated in the time between the adoption of mathematics standards and the application of accountability models are important aspects of the educational process. There is an unstated assumption that standards and accountability models are only part of the solution strategy for school improvement in mathematics education. The assumption is that armed with quantitative data, local leadership—teachers, mathematics coordinators

and supervisors, principals, assistant superintendents, superintendents, and school board members—will proactively respond to data. The way in which local school leadership responds to system data has a profound consequence for students' opportunity to learn. Today's school leadership is being called to engineer school improvement and the classroom learning setting. This chapter is written with the hope that it will help the reader understand how research-based strategies can support the engineering of positive change to the structures supporting the teaching and learning of mathematics in educational settings.

The engineer as a metaphor representing a change agent requires a brief explanation. To some, the engineer may seem to be synonymous with the scientist.² The distinction between a scientist and an engineer is partially clarified by examining two activities related to the preparation of each professional—analysis and design. In science classes, students are required to solve problems, observe phenomena in lab settings, record observations, and perform calculations. This process is the essence of analysis. In engineering classes, the instruction often stresses the importance of design. The difference between analysis and design can be described in the following way: If only one solution to a problem exists and discovering it merely entails putting together pieces of discrete information, the activity is probably analysis (Horenstein, 2002). In comparison, if more than one solution exists, and if determining a reasonable path demands being creative, making choices, performing tests, iterating, and evaluating, then the activity is design. Design often includes analysis; however, it also must involve at least one of these latter components. Horenstein (2002) offered the following example to further clarify the difference between analysis and design:

[A] remote-controlled buoy is located off the coast of California and is maintained by the U.S. National Oceanic and Atmospheric Administration (NOAA). It provides 24-hour data to mariners, the Coast Guard, and weather forecasters. Processing the data stream from this buoy, posting it on the Internet, and using information to forecast the weather are examples of analysis. Deciding *how* to build the buoy so that it meets the needs of NOAA is an example of design. (p. 29)

Administrator and teacher leadership charged with addressing our nation's school mathematics challenges must decide how to build effective programs.

Clearly, there is more than one solution to our school mathematics problem, and designing appropriate solutions will require creativity, hard choices, performance tests, iterative action, and evaluation. Like engineers, mathematics educators must study access and opportunity-to-learn issues in great depth and then design an intervention—"learn to build." In contrast, most scientists construct instruments to measure and study phenomena of interest—they "build to learn." This chapter is dedicated to those interested in "learning to build" outstanding school mathematics programs.

The next two sections provide an examination of the challenges facing school mathematics change agents. The second section documents changes in U.S. mathematics achievement by reviewing population trends and national achievement trend studies. A focus of this section is to determine achievement trends of various racial-ethnic and socioeconomic groups. The third section examines opportunity-to-learn (OTL) factors that have the potential to positively influence the learning of mathematics. The intent in this section is to offer possible building blocks to support the engineering of positive change in school mathematics and to review the work of some scholars who have designed school mathematics improvement models based on important OTL factors. The fourth section provides a closer look at research-based cases of successful mathematics programs. This section will highlight both classroom and organizational components that are present in high performing school mathematics programs. The fifth section is a review of two large-scale change efforts in school mathematics and related evaluation information. The sixth and final section is a brief review of the engineering perspective—learning to build—and its importance for school mathematics improvement.

LEARNING TO BUILD: THE PROBLEM DEFINED

Who are the children in the classrooms of today and citizenry of tomorrow? One challenge posed in recent calls for mathematics reform is to improve the academic performance of all students, and particularly students traditionally underserved in mathematics classrooms. For instance, the *National Education Goals Report: Building a Nation of Learners* (National Education Goals Panel, 1995) called for the mathematics performance of all students at the elementary and

² In fact, Hurd (1997) argued that the paradigmatic boundaries of science are shifting toward a science guided by the coaction of science and technology, perceived as an integrated system. Further, he indicated, many in the science community speculate that engineering education may be the best preparation for the natural sciences. Yet, this speculation suggests there are distinct paradigmatic differences.

secondary levels to increase significantly in every quartile and for the distribution of minority students in each quartile to better reflect the student population as a whole. Addressing this challenge requires some insight into the population trends of various demographic groups. A particular focus of this section is population trends over the past 30 years.

The student race/ethnicity population trends have changed dramatically since the 1985 release of *Mathematics and Science: Critical Filters for the Future*. Table 28.1 provides insight into this trend.

Table 28.1 Percentage Distribution of Public School Students in K–12, by Race: Fall 1972 and 2003

Race/Ethnicity	Fall 1972	Fall 2003
White	78%	58%
Black	15%	16%
Hispanic	6%	19%
Other	1%	7%

Note: From U.S. Department of Education, 2005.

The information in Table 28.1 requires additional explanation. The U.S. school-age population declined between 1980 and 1990 but became more diverse. The United States General Accounting Office (GAO, 1993) reported in 1990 about 44.4 million school-age children (ages 5–17), a decline of more than 2.3 million, or 5.8% since 1980. In 1992, the percentages of male and female students 5–18 years old enrolled in school were 51.4% and 48.6%, respectively (National Science Foundation, 1994).

Changes in the racial-ethnic characteristics of the U.S. population have been a part of American life since the first European settlements. However, only in recent decades has the population in the United States become less, rather than more, White. The racial-ethnic diversity of the country is much greater now than at any previous period in history and seems on course to become progressively more diverse for some time to come (Riche & Pollard, 1992; Vernez, 1992). This diversity is reflected in recent trends of school-age children.

During the 1980s, the White school-age population declined by more than 4 million children, or about 12%, and the number of African American children decreased by about 250,000, or about 4%. In contrast, the number of Hispanic school-age children increased by 1.25 million, or 57%, and the number of Asian children rose by over 600,000—an 87% increase. In 1990, White children made up less than 70% of the total school-age children, down from about 75% in 1980 (GAO, 1993).

Similar to the total school-age population, poor children are now more racially and ethnically diverse. On the 1990 census, an individual or family would be categorized as poor if its annual before-tax cash income was below the corresponding poverty threshold for a family of that size. On the 1990 census, the poverty cutoff for a family of four was a 1989 income of \$12,674. During the 1980s, the number of poor school-age children increased by 6% from about 7.2 million to 7.6 million (GAO, 1993). The national poverty rate for school-age children rose from 15.3% to 17.1%. The number of poor Hispanic and Asian children grew by almost 600,000 while the number of poor White children declined, and the African American school-age population in poverty remained relatively constant.

Despite the decline in poor White children, they continued to make up more than 40% of all poor school-age children in 1990, but this percentage changed dramatically by geographic region. White children represented over two thirds of all rural poor children and approximately one third of the urban school-age population in poverty. Yet, regardless of region, African American children experienced the *highest rates* of school-age poverty, from almost 41% in rural areas to 34% in urban areas.

Three other traditionally underserved demographic groups—immigrant households, linguistically isolated (LI) households, and limited English proficiency (LEP) households—each contributed about 5% of all school-age poverty children (GAO, 1993).³ Many of the children were categorized into more than one of these groups. When adjusted for overlap, these three groups totaled nearly 4 million children—more than 9% of all school-age children. More than 30% of these 4 million children were also classified as poor.

³ Some definitions of terms are required here. LI children are in households where no persons aged 14 or older speak “only English” and no persons aged 14 or older who speak a language other than English speak English “very well.” There is no accepted definition of LEP. The term generally refers to students who have difficulty with speaking, writing, or reading English. The GAO (1993) defined LEP children as all persons aged 5 to 17 living in families whose members the Census reported as speaking English “well,” “not well,” or “not at all.” It should be noted that there is considerable variation in actual English-speaking ability among those classified in the “speaks English well” category.

Current demographic trends should be examined in light of mathematics achievement trends. As the demographic context of the United States changes rapidly, how well is our system of education performing in school mathematics across demographic groups?

PROFICIENCY TRENDS IN MATHEMATICS

The purpose of this section is to document changes in mathematics achievement by examining national trend studies to better understand the status of the United States education system. The discussion of national trend data is offered for two related reasons. The first reason is to clearly describe the student achievement problem. The trend studies reviewed in this section are in part a reflection of past practice in school mathematics. Thus, the mathematics trends are linked to limitations of the implemented curriculum, pedagogy, and school organizational strategies. A second reason to discuss national mathematics trend studies is to describe the measures used to determine mathematics achievement and to interpret the findings with a focus on engineering change.

The trend studies should be examined with several concepts in mind. Miller (1995) argued that three intertwined concepts should be taken into consideration when attempting to build effective strategies to accelerate minority student performance on the basis of academic achievement data:

1. Generally, differences in academic achievement patterns among racial/ethnic groups reflect the fact that the variation in family resources is greater than the variation in school resources. His analysis of achievement patterns and resource allocations confirms that most high-SES students receive several times more resources than most low-SES students receive, and much of this resource gap is a function of family resources rather than school resources.
2. Demographic group educational advancement is an intergenerational process. From this perspective, education-related family resources are school resources that have accumulated across multiple generations. On average, investments in the current generation of African American, Hispanic, and American Indian children in the form of intergenerationally accrued education-relevant family resources are significantly less

than comparable investments in White and Asian children.

3. Educational attainment is a function of the quality of education-relevant opportunity structure over several generations. The pace of educational advancement depends on multiple generations of children attending good schools.

Miller (1995) stated the following about these three interrelated concepts:

Current variations in education-relevant family resources are heavily a function of variations in the historical opportunity structure experienced by generations of racial/ethnic groups. At the same time, the quality of the contemporary opportunity structure is crucial to the further evolution of family resource variation patterns. The nation's ability to accelerate the intergenerational advancement process for minorities may be decisively shaped by its capacity to *engineer* [my emphasis] a more favorable opportunity structure for them in the years ahead as well as to supplement family and school resources for those groups at a level commensurate with their actual needs. (pp. 339–340)

The first step in engineering change is problem identification. One goal of recent federal (NCLB Act) legislation and state policy focused on mathematics standards and accountability is to document the achievement level of traditionally underserved students' yearly progress and to provide performance trends at a local level. The theory of action of most standards-based reform initiated at state and federal levels of governance suggests that, armed with quantitative data on how students perform against standards, school leadership will react by making instructional changes required to improve student performance. According to the National Research Council (1999a),

Research on early implementation of standards-based systems shows, however, that many schools lack an understanding of the changes that are needed and lack the capacity to make them. The link between assessment and instruction needs to be made strong and explicit (p. 5).

Why do schools lack an understanding of administrative changes that are needed to improve student performance on specifically designated tests? One or all of the following problems may hamper many school leaders:

- Failure to disaggregate and organize data by race, class, language proficiency, or other relevant demographic variables

- Failure to align local content standards with external performance standards associated with the designated testing system
- Failure to align the testing cycle and fiscal planning

One reason many schools lack the insight to make appropriate instructional changes is related to how they organize and analyze data. Although many states, schools districts, and schools disaggregate data to help provide a more accurate picture of student performance, many educational leaders do not have insight into student mathematical performance by demographic group. This is problematic in that student achievement patterns and trends are potentially overlooked; thus, opportunities for instructional intervention are lost, and future student performance is hampered. Further, lack of clarity about the relationship between content standards and performance standards can result in the implementation of curriculum that is not consistent with outcome measures being employed (NRC, 1999a). Thus, any discussion of achievement trends should be coupled with a clear description of what is being measured. Moreover, the discussion of trends must occur in a time frame that allows for immediate intervention. The timing of tests and the administrative planning cycle further complicate the possibility of intervention. In many states, the test results are produced after fiscal planning has taken place in school districts. This disconnect makes it difficult to plan appropriate interventions for the upcoming school year.

Racial-Ethnic Trends⁴

Rapid growth of the school-age population and changing discourses about racial categories have made it more difficult to classify racial-ethnic, immigrant, and language groups. For example, within the Hispanic, Asian, and African American populations, distinct subgroups have formed, and many have requested unique demographic characterizations. Most national trend analysis of mathematics performance is not conducted at this level of detail. This limitation stated a review of this literature remains instructive for evaluating national trend direction in school mathematics.

NAEP Trends

The National Assessment of Educational Progress (NAEP) trend assessment is largely a basic skills examination. To measure performance trends, subsets of the same items have been a part of successive assessments. Some items have been included in each examination. This practice means that findings from nine NAEP trend assessments provide insight into how students' mathematics proficiency has changed from 1973–1999. NAEP mathematics proficiency scores are available for 1973, 1978, 1982, 1986, 1990, 1992, 1994, 1996, 1999, and 2003.⁵ Tests are administered to a sampling of students across the United States at Ages 9, 13, and 17. The scale scores, which range from 0 to 500, provide a common metric for determining levels of proficiency across assessments and demographic characteristics. NAEP scores reflect student performance at five levels on the scale:

- Level 150—Basic Arithmetic Facts
- Level 200—Beginning Skills and Understanding
- Level 250—Basic Operations and Beginning Problem Solving
- Level 300—Moderately Complex Procedures and Reasoning
- Level 350—Multi-Step Problem Solving and Algebra

The performance-level categories were developed for the 1973 assessment and have continued to be used through the 1999 assessment. However, the language associated with these categories has evolved and changed over this time period. Thus, the “engineer” charged with making decisions about curriculum, teaching, and other relevant educational inputs should be aware that this trend analysis may use language consistent with today's standards-based discourse (NCTM, 1989, 2000). However, the test items may not reflect the problem-solving and reasoning descriptions found in more recent standards documents and state content and performance assessment documents. With this limitation noted, the NAEP trend analysis is a valuable gauge of student performance progress over time. Table 28.2 provides a summary of NAEP racial-ethnic trends in mathematics performance from 1973–1999.

⁴ The trend studies reviewed in this chapter are limited to select national-level analyses that provide insight into student mathematics performance across demographic groups. National studies that did not disaggregate data by demographic group are not included. Moreover, no state-level trend studies or international studies (TIMMS) are included. The period from 1985 to 1999 is a particular focus of this trend analysis summary. This report continues the 1985 effort of DeAnna Banks Beane.

⁵ The 2003 NAEP scores are not included in this discussion. At the time of publication, these findings were not included in the trend study.

Table 28.2 NAEP Trends in Average Mathematics Scale Scores by Race/Ethnicity

Race/Ethnicity	Year	Age 9 Scores	Age 13 Scores	Age 17 Scores
White	1999	238.8 (0.9)	283.1 (0.8)	314.8 (1.1)
	1996	236.9 (1.0)	281.2 (0.9)	313.4 (1.4)
	1994	236.8 (1.0)	280.8 (0.9)	312.3 (1.1)
	1992	235.1 (0.8)*	278.9 (0.9)*	311.9 (0.8)*
	1990	235.2 (0.8)*	276.3 (1.1)*	309.5 (1.0)*
	1986	226.9 (1.1)*	273.3 (1.3)*	307.5 (1.0)*
	1982	224.0 (1.1)*	274.4 (1.0)*	303.7 (0.9)*
	1978	224.1 (0.9)*	271.6 (0.8)*	305.9 (0.9)*
	1973	225.0 (1.0)*	274.0 (0.9)*	310.0 (1.1)*
Black	1999	210.9 (1.6)	251.0 (2.6)	283.3 (1.5)
	1996	211.6 (1.4)	252.1 (1.3)	286.4 (1.7)
	1994	212.1 (1.6)	251.5 (3.5)	285.5 (1.8)
	1992	208.0 (2.0)	250.2 (1.9)	285.8 (2.2)
	1990	208.4 (2.2)	249.1 (2.3)	288.5 (2.8)
	1986	201.6 (1.6)*	249.2 (2.3)	278.6 (2.1)
	1982	194.9 (1.6)*	240.4 (1.6)*	271.8 (1.2)*
	1978	192.4 (1.1)*	229.6 (1.9)*	268.4 (1.3)*
	1973	190.0 (1.8)*	228.0 (1.9)*	270.0 (1.3)*
Hispanic	1999	212.9 (1.9)	259.2 (1.7)	292.7 (2.5)
	1996	214.7 (1.7)	255.7 (1.6)	292.0 (2.1)
	1994	209.9 (2.3)	256.0 (1.9)	290.8 (3.7)
	1992	211.9 (2.3)	259.3 (1.8)	292.2 (2.6)
	1990	213.8 (2.1)	254.6 (1.8)	283.5 (2.9)*
	1986	205.4 (2.1)*	254.3 (2.9)	283.1 (2.9)*
	1982	204.0 (1.3)*	252.4 (1.7)*	276.7 (1.8)*
	1978	202.9 (2.2)*	238.0 (2.0)*	276.3 (2.3)*
	1973	202.0 (2.4)*	239.0 (2.2)*	277.0 (2.2)*

Note: Standard errors of the scale scores appear in parentheses.

* Significantly different from 1999. From NAEP 1999 Trends in Academic Progress, NCES (2000).

The racial-ethnic mathematics scores as measured by the NAEP long-term trend assessment improved for all racial-ethnic subgroups from 1973–1999. The scores for Black and Hispanic students are less consistent than those for White students and demonstrate more abrupt changes. However, the samples of Black and Hispanic students are smaller than that of White students. Smaller samples typically have more variability. Overall, the NAEP trend assessment indicates that all three racial-ethnic groups have experienced positive growth in mathematics proficiency. However, no group by Age 17 was performing on average at the highest student performance level. This finding is a concern given that the performance levels are more closely aligned with a basic skills mathematics curriculum.

NELS Trends

The National Education Longitudinal Study of 1988 (NELS:88) included a nationally representative sample of over 10,000 students, followed from 8th grade (1988) through 12th grade (1992) in nearly 800 high schools nationwide. The schools in the study include public, Catholic, and other private schools and represent a range of enrollment, religious affiliations, geographic settings, school social compositions, as well as various levels of restructuring activity (Newmann & Wehlage, 1995). The NELS:88 mathematics tests were constructed to measure both high-level and low-level skills at three points in time: 1988, 1990, and 1992. Thus, students in the sample were assessed in mathematics at Grades 8, 10, and 12, respectively. The difficulty levels of the first and second follow-up

mathematics tests were adapted to the students' performance levels in the previous administration. Each mathematics test had 40 items. Eighty-one items were used in all forms of the test. The different forms of the test were equated using item response theory (IRT) so the various forms of the test could be equated with a common metric. Units on these tests refer to the number of items answered correctly, after the IRT procedures were used to score the tests and to assign all students on the same scale.

Green, Dugoni, Ingels, and Camburn (1995) reported findings from the NELS:88 second follow-up data set that included mathematics achievement results of high school seniors in 1992. The 1992 NELS:88 second follow-up examination items represent items typically characterized as traditional, basic skills curriculum. Five levels of mathematics proficiency were defined in the study. Green et al. found that African American and Hispanic students were less likely than White and Asian students to demonstrate advanced proficiencies (Levels 4 and 5) on the standardized test of mathematics (12% and 20% compared with 39% and 45%, respectively). Further, 50% of the African American and 42% of Hispanic students were categorized at Level 1 or below. In comparison, 14% of the Asian and 21% of White students performed at Level 1 or below.

Rasinski, Ingels, Rock, and Pollack (1993) compared mathematics scores for sophomores in the 1980 High School and Beyond (HS&B) study and the 1990 NELS:88 (a follow-up study conducted in 1990) by using an IRT scaling procedure that linked the two assessment instruments. The HS&B sophomore cohort mathematics test administered in 1980 consisted of 38 test items and required students to complete the examination in 21 minutes. The test items were quantitative comparisons that required students to mark which of two quantities is greater, indicate their equality, or note a lack of sufficient information to determine a relationship between the quantities. The 1990 NELS:88 first follow-up mathematics test contained 40 items to be completed in 30 minutes. The test items assessed advanced skills of comprehension and simple mathematical application skills. The items included geometric figures, graphs, word problems, and quantitative comparisons (as in the HS&B). Consistent with the HS&B, a multiple-choice format was used in this follow-up test. To compare the performance of the 1980 HS&B sophomore cohort and the 1990 NELS:88 sophomores, 16 quantitative comparisons from the HS&B were included in the 1990 NELS:88 mathematics assessment. Thus, the findings from this study should be viewed as a comparative analysis of a narrow scope of the mathematics content. All racial and ethnic groups with the exception of Asian stu-

dents made statistically significant gains in mathematics performance on the test. In each administration of the test, Asian students on average were the highest performing of the four demographic groups. African American and Hispanic students gained more than Asian and White students in this comparison.

Racial-Ethnic Trend Analysis Summary

The NAEP trend analysis indicates improvement between 1973 and 1999 in all racial-ethnic groups at each age level. During this period, African American and Hispanic students made larger gains than did White students; thus, the performance gap on this assessment between White students and the other two demographic groups closed slightly. The 1980 HS&B and the 1990 NELS:88 sophomore cohort study reported a similar result: African Americans, Hispanics, and Whites made statistically significant gains in mathematics achievement. Further, the gains made by African American and Hispanic students were larger than those of White students.

The NAEP trend analysis and the 1990 NELS:88 sophomore cohort study indicate that the mathematics performance on basic skills items over the past 20 years has improved for the largest racial-ethnic demographic groups in the United States. However, no racial-ethnic demographic group has consistently produced scores that are associated with the highest levels of performance being measured by the NAEP trend analysis.

Socio-Economic Trends

The concept of social class is derived from multiple academic domains and literatures. Most conceptions of social class are linked to the economic roots of class and to varying degrees couple class with political and cultural indicators. The traditional practice in school mathematics achievement data is to organize a hierarchy of classes—working class, lower-middle class, middle class, and so on. This hierarchical framework objectifies high, middle, and low positions on some metric, such as socioeconomic status (SES) in which “Parents’ Education” or “Family Income” is a proxy for class. The limitations of this practice are discussed elsewhere (Grant & Sleeter, 1986; Knapp & Woolverton, 1995; Secada, 1992). However, for the purpose of understanding SES trends in mathematics achievement, a proxy like “Parents’ Education” is instructive. One major limitation of this proxy—and others like it—is that school administrators cannot intervene directly on this variable. However, the importance of family resources must not be ignored in a discussion of achievement (Miller, 1995).

NAEP Trends

From 1978 to 1999, the National Assessment of Educational Progress provided trends in average mathematics proficiency by the highest level of education that students reported for either parent. A summary of the trends in average mathematics scale scores for students at three age levels by parents' highest level of education is provided in Table 28.3.

Students who indicated their parents had less than a high school education have exhibited overall gains in average mathematics proficiency since 1978 across all ages. For students who reported their parents' highest education level was high school graduation, the average proficiency trend has generally improved at Ages 9 and 17. The performance of 13-year-olds in this group was relatively the same during this time period. For students with a parent who graduated from college, only 9-year-olds had an average score in 1999 that was significantly higher than in 1978.

NELS Trends

Rasinski and colleagues' (1993) comparison of sophomore cohorts from the 1980 HS&B study and the 1990 NELS:88 follow-up study documented a consistent pattern of positive gains within SES groups during this period and a difference that is related to student SES. Four SES categories were created by framing the socioeconomic status composite into SES high quartile, SES high middle half, SES low middle half, and SES low quartile. The findings suggest that the highest quartile improved more than the lowest quartile; however, approximately 12% of the lowest quartile in 1990 was missing math test scores, whereas nearly all the 1980 lowest quartile reported mathematics scores. The researchers speculated that the lowest quartile gain could be biased downward as a result of the missing data. The missing data make any interpretation of differential gain between quartiles difficult. However, within each data set—i.e., HS&B 1980 and

Table 28.3 NAEP Trends in Average Mathematics Scale Scores by Parents' Highest Level of Education

Parents' Level of Education	Test Year	Age 9 Scores	Age 13 Scores	Age 17 Scores
Less than High School	1999	213.5 (2.8)	256.2 (2.8)	289.2 (1.8)
	1996	219.8 (3.3)	253.7 (2.4)	280.5 (2.4)*
	1994	210.0 (3.0)	254.5 (2.1)	283.7 (2.4)
	1992	216.7 (2.2)	255.5 (1.0)	285.5 (2.3)
	1990	210.4 (2.3)	253.4 (1.8)	285.4 (2.2)
	1986	200.6 (2.5)*	252.3 (2.3)	279.3 (2.3)*
	1982	199.0 (1.7)*	251.0 (1.4)	279.3 (1.0)*
	1978	200.3 (1.5)*	244.7 (1.2)*	279.6 (1.2)*
Graduated High School	1999	224.4 (1.7)	264.0 (1.1)	299.1 (1.6)
	1996	221.2 (1.7)	266.8 (1.1)	297.3 (2.4)
	1994	225.3 (1.3)	265.7 (1.1)	295.3 (1.1)
	1992	222.0 (1.5)	263.2 (1.2)	297.6 (1.7)
	1990	226.2 (1.2)	262.6 (1.2)	293.7 (0.9)*
	1986	218.4 (1.6)*	262.7 (1.2)	293.1 (1.0)*
	1982	218.3 (1.1)*	262.9 (0.8)	293.4 (0.8)*
	1978	219.2 (1.1)*	263.1 (1.0)	293.9 (0.8)*
Graduated College	1999	239.7 (0.8)	285.8 (1.0)	285.2 (3.9)
	1996	239.7 (1.4)	282.9 (1.2)	287.3 (4.0)
	1994	237.8 (0.8)	284.9 (1.2)	282.7 (3.8)
	1992	236.2 (1.0)*	282.8 (1.0)*	290.2 (3.9)
	1990	237.6 (1.3)	280.4 (1.0)*	276.8 (2.8)
	1986	231.3 (1.1)*	279.9 (1.4)*	280.6 (2.4)
	1982	228.8 (1.5)*	282.3 (1.5)	271.7 (1.8)*
	1978	231.3 (1.1)*	283.8 (1.2)	275.7 (1.9)*

Note: Standard errors of the scale scores appear in parentheses.

* Significantly different from 1999. From NAEP 1999 Trends in Academic Progress, NCES (2000).

1990 NELS:88 Follow-up—SES status is clearly related to mathematics performance.

Green et al. (1995) reported findings from the 1992 NELS:88 second follow-up survey of seniors. In one analysis, Green and colleagues compared achievement across racial and ethnic groups while controlling for SES. The two lowest proficiency levels—below basic and Level 1—and the two highest proficiency levels—Levels 4 and 5—are contrasted. The data indicate that achievement differences exist even when the effects of socioeconomic status are held constant. For example, this study reported that significant differences existed between Whites' and African Americans' test performance within each SES category. Also, there were significant differences between White and Hispanic seniors in the high SES group. The percentage differences among racial and ethnic groups were generally larger in the higher SES groups. There was one exception: Differences in Asian and White seniors' performance were not significant.

SES Trend Analysis Summary

The studies reviewed in this section should be considered with population trends in mind. Clearly, poverty is more severely concentrated among Hispanic and African American children than it is among Whites. Across the various studies of mathematics achievement, a strong relationship between SES and mathematics achievement was present. These studies indicate a need to improve the mathematics achievement of low-SES students as a whole, and even more pressing is the need to raise the mathematics achievement of low-SES minority students. In light of these findings and population trends, the need for intervention in the two geographic regions with the highest poverty levels—urban and rural communities—is apparent.

OPPORTUNITY TO LEARN FACTORS: TIME, QUALITY, AND DESIGN

A close look at the achievement trends reviewed in the prior section suggests that student demographic background is strongly related to mathematics achievement. This is important to know; however, demographic background is out of the control of the teacher, instructional supervisor, school board member, and other school personnel. An educator interested in improved student performance in mathematics

must focus on the variables associated with learning mathematics that can be influenced by specific action and intervention. Yet, as Miller (1995) argued, ignoring family resource variables and SES factors could limit the progress of any reform effort. For students living in poverty, the nature and extent of their in-school experience in many cases must rival the combined in-school and out-of-school academic experiences of resource-rich families, if the goal is closing the achievement gap without penalty to any group. This is not a well-accepted position, however the logic of the argument tacitly undergirds the achievement gap debate. One response to current student underperformance is to examine how opportunity-to-learn variables might inform the design of active intervention on student learning.

Opportunity-to-Learn (OTL) as an important construct influencing—and possibly explaining—the impact of instruction was introduced during the 1960s. Carroll (1963) included OTL as one of five critical constructs in his model of school learning. He defined OTL as the amount of time allocated to the learner for the learning of a specific task. If, for instance, the task assigned a student is to understand the concept of place value, opportunity-to-learn is simply the amount of time the student has available to learn what place value is.

In Carroll's (1963) model, opportunity-to-learn is contrasted with the amount of time the student requires to learn a principle or concept. This latter construct is largely related to the student's aptitude in a concept domain. Thus, whereas teachers have some control over the time available for student learning, they have little control over the time required for student learning. Carroll also contrasted OTL with the amount of time the student actually spends engaged in the learning process. The latter variable, often referred to as time-on-task or engaged time, is thought to be affected by the perseverance of the student and the quality of the teaching. In Carroll's model, OTL represents the maximum value for engaged time.

In contrast to Carroll, Husén (1967) organized OTL in terms of the relationship between the mathematics content taught to the student and mathematics content assessed by achievement tests. In Husén's model, OTL is the overlap of mathematics taught and mathematics tested. Simply stated, the greater the overlap, the greater the opportunity-to-learn.⁶

⁶ Carroll's (1963) and Husén's (1967) opportunity-to-learn models have two important differences. First, whereas Carroll's model describes OTL as an instructional variable (under the control of teachers), Husén's model frames OTL as a measurement variable. Second, Carroll describes OTL as a continuous variable, whereas Husén designed OTL as a dichotomous variable. The most important concern from Carroll's perspective is how much time the student has to learn a specific concept. The most important concern from Husén's perspective is whether or not a student has been provided with quality instruction relative to the concepts included on achievement tests.

Scholars, school leaders, and government agencies have used various combinations of Carroll's and Husén's models to design their own frameworks of opportunity-to-learn (National Governors' Association, 1993; Robitaille & Travers, 1992; Winfield, 1987, 1993). However, Stevens (1993a) identified four variables related to teacher instructional practice and student learning that consistently emerge in these interpretations. In this section, two of the variables are combined; thus, the following three variables form an opportunity-to-learn framework:

1. *Content exposure and coverage variables* measure the amount of time students spend on a topic (time-on-task) and the depth of instruction provided. These variables also measure whether or not students cover critical subject matter for a specific grade or discipline.
2. *Content emphasis variables* affect the selection of topics within the implemented curriculum and the selection of students for basic skills instruction or for higher order skills instruction.
3. *Quality of instructional delivery variables* reveals how classroom pedagogical strategies affect students' academic achievement.

The purpose of these OTL variables is to determine whether or not students are provided sufficient access to learn the mathematics curriculum expected for their grade level and age. According to Stevens (1993b), the OTL variables are "deceptively simple" (p. 234). In general, research in this area examines one variable at a time; however, the OTL conceptual framework developed by Stevens (1993a, 1993b) encourages teachers, administrators, and researchers to

examine the interaction of all three variables simultaneously (see Table 28.4).

This theoretical framework will remain a theory, rather than an active change strategy for most teachers, unless their work is part of a coherent "design" that allows them to take advantage of what is known about opportunity to learn. Two very important variables that emerge from the OTL literature are *time* and *quality*. Time and quality are critical variables because they can be altered with interventions. Thus, time and quality variables derived from the OTL literature form a basis for the construction of school design strategies aimed to improve learning. For purposes of management and leadership, design is critical.

Think of "design" as an innovative portfolio of strategies that will provide students appropriate content exposure, content coverage, content emphasis, and quality instructional delivery. The term *design* is used here to describe how school personnel can construct and package opportunities to learn. Those responsible for the education of children need to be challenged to accept a greater level of responsibility for how teaching and learning is organized. Every educator—teachers, principals, superintendents, and school board members—should have a clear understanding of how the school system and, more specifically, how each school is designed to improve student performance in mathematics. Too many educators fail to see the limitation of long-standing design principles. Still others fail to recognize existing design principles. Some may question the need for a transparent opportunity-to-learn design. However, not having a design is a design for failure. Each state has a measurement system to gauge student performance. These systems are transparent. Similarly, every school and school district should have a learning design that

Table 28.4 Opportunity to Learn: A Theoretical Framework Derived from International Assessments and Research Studies to Examine Students' Access to Intended Curriculum

Variable/Related Study	Definition
Content exposure and coverage (Leinhardt & Seewald, 1981; Leinhardt, 1983; Brophy & Good, 1986; Winfield, 1987, 1993; Suter, 2000)	Teacher arranges class so that there is time-on-task for students. Teacher arranges adequate time for students to learn subject matter and to cover adequately a specific topic. Teacher arranges the curriculum to overlap test content.
Content emphasis (Floden, Porter, Schmidt, Freeman, & Schwille, 1981; LeMahieu & Leinhardt, 1985; McDonnell, Burstein, Catterall, Ormseth, & Moody, 1990; Oakes, 1990; Porter, 1989, 1993; Stevens 1993b; Suter, 2000)	Teacher chooses content from the curriculum to teach. Teacher chooses the dominant level to teach the curriculum (recall, higher order skills). Teacher chooses which skills to teach and which skills to highlight with different groups of students (ability grouping and tracking).
Quality of instructional delivery (Brophy & Good, 1986; Stevens, 1993b; Stevenson & Stigler, 1992)	Teacher uses different pedagogical strategies to meet the learner's needs. Teacher has understanding of the subject matter.

is transparent, open to ongoing monitoring, assessment, and revision.

The appropriate design and management of OTL variables is central to the improvement of school mathematics for many students. The remainder of this section will be devoted to the role of time, quality, and design as they relate to student OTL with traditionally underserved student groups.

Time and School Mathematics

Policies and practices that influence content coverage and time on task in school mathematics are pivotal to the improvement of student performance in the domain. The purpose of studying these opportunity-to-learn variables is to determine whether or not students are provided sufficient time to learn the mathematics curriculum expected for their grade level and age. One very basic principle related to time should be transparent in every classroom. Significant time should be dedicated to mathematics instruction each school day. Further, appropriate time should be allotted to ensure students develop understanding of key concepts and procedures.⁷ Many factors can influence whether or not this basic principle is followed. In this section, several factors related to time and school mathematics will be reviewed.

Course-Taking

Two of the most powerful predictors of school mathematics achievement in large-scale assessments of mathematics have been (a) increased time on task in high-level mathematics and (b) the number of courses taken in mathematics. Generally, these two predictors are interrelated. Evidence indicates that African American, Hispanic, and low-SES students are less likely to be enrolled in higher level mathematics courses than middle-class White students (Secada, 1992). Further, White students on national assessments of mathematics achievement consistently outperform African American and Hispanic students. Thus, it is not shocking that a positive relationship between mathematics achievement and course taking exists across measurement systems (e.g., NAEP, SAT, and ACT).

Course-taking options in the United States are organized in a technology that takes on two forms—curricular and ability tracking. Many comprehensive high schools offer a wide range of mathematics

courses linked to various work-related opportunities. No student could experience all of the coursework, so schools design technologies to regulate the selection process. To this end, students in most high schools are sorted into a curricular track involving a specific course sequence and, ultimately, different opportunities to learn mathematics. Generally, three curricular tracks—college preparation, vocational, and general education—are offered within most traditional high schools. The college preparation track clearly has higher status and provides greater opportunity to learn more demanding mathematics. Curricular tracking has serious implications for student opportunity to learn mathematics.

Similarly, ability tracking is a technology used to sort students into curriculum experiences.⁸ This mechanism for sorting provides different levels of instruction to students across two tracks based on perceived ability. This version of sorting is more difficult to recognize because course labeling can disguise the practice. For example, schools may offer two different courses in geometry. Both may have the same title; however, the mathematics covered in each course may differ in dramatic ways. Another sorting strategy is to offer students different entry points into the college-preparatory coursework at different times (e.g., freshmen year versus junior year). The organizational structure of the school may recognize many tracks or just a few; schools may or may not link tracks to a block of courses or to mathematics only; and schools may have loosely or tightly coupled curricular and ability tracking. Additionally, students may or may not have the option to move across tracks. The opportunity to negotiate new curricular possibilities is an important equity consideration.

Tracking is a serious challenge to mathematics achievement and opportunities to learn mathematics. In theory, tracking as a technology is designed to benefit all students. However, evidence strongly suggests that this goal is not being accomplished (Hoffer, Rasinski, & Moore, 1995). Instead, research studies have indicated that even when tracking systems have positive effects, those effects are more closely associated with those students assigned to high-status tracks (Oakes, 1990; Rock & Pollack, 1995).

One possible solution to the differential opportunities to learn across tracks is to constrain the curriculum options in mathematics at the second-

⁷ Sufficient and appropriate time to learn the mathematics curriculum should be a data-driven decision. Certain mathematical concepts are more difficult to understand. System-wide as well as classroom-based assessments data can inform the process. Both assessment formats are informative with respect to determining the amount of time to devote to a concept.

⁸ We use the term *ability grouping* because this is consistent with the literature on tracking. However, a more appropriate term is *perceived ability*.

ary level. Currently, African Americans and Hispanic students are overrepresented in vocational programs and low-track options. Lee, Croninger, and Smith (1997) found that students learn more mathematics in schools that offer them a narrow curriculum composed of college-preparatory academic courses. This research is suggestive, rather than definitive.

A word of caution: “Course-taking patterns are an important indicator of system quality.” Many students may be enrolled in low-track mathematics courses due in part to prior experiences in elementary and middle school mathematics. Merely mandating a narrower curriculum consisting of college prep mathematics will not address the endemic quality problem of the preK–8 mathematics program. Thus, it is imperative that curriculum constraints toward the college prep model at the secondary level occur in tandem with a close examination of the preK–8 effort.

One state-level change strategy to improve elementary and middle school mathematics is to align the mathematics curriculum with state assessments. This model has implications for time and school mathematics. The next section examines this strategy.

Assessment Practices

The mathematics curriculum in many school districts is aligned with mathematics standards adopted or derived from state or national curricular frameworks. The standards-based reform of mathematics education is often part of a larger systemic change effort that includes academic standards in the core disciplines by grade, holding all students to the same standards, statewide assessments closely linked to the standards, accountability systems with varying levels of consequences for results, computerized feedback systems, and data for continuous improvement (NRC, 1999a). State-level assessment systems and most national testing proposals call for students to be tested in mathematics and reading (NRC, 1999b). This practice has implications for content coverage and time on task in mathematics classrooms in urban school districts and other school systems with large percentages of traditionally underserved students.

Students’ opportunities to learn mathematics are influenced by the assessment policies of the school district. Assessment policy often influences the nature of pedagogy in a classroom. The influence of standardized tests—and, more recently, state-mandated testing—is arguably greater in high-minority classrooms. In a nationwide survey, teachers of high-minority classrooms reported test-specific instructional practices more often than teachers of low-minority classrooms (Madaus, West, Harmon, Lomax, & Viator, 1992). For example, in high-minority classrooms,

about 60% of the teachers reported teaching test-taking skills, teaching topics known to be on the test, increasing emphasis on tested topics, and starting test preparation more than a month before the examination. These practices were reported significantly less often in low-minority classrooms. Moreover, mathematics teachers with high-minority classes indicated more pressure from school district officials to improve test scores than teachers with low-minority classes.

Today, school districts across the country use testing technology as a mechanism to measure school and student progress. However, the role of testing technology is much greater than measurement concerns. Tests do change or at least influence teaching behavior. Many districts are ignoring best practice related to assessment and school mathematics. Two recommendations related to school mathematics and student assessment performance are listed below:

1. Design a curriculum, select quality instructional materials, align curriculum and instructional materials, and then use aligned instructional materials all year. Testing systems are intended to measure the quality of a school’s instructional program. Avoid spending significant time on test preparation. If the combination of the curriculum, instructional materials, and teaching fall short of school district goals, then these factors must be reviewed and improved upon.
2. Use state and classroom assessment data as a way to build a solid instructional program linked directly to student thinking in the content domain.

Fiscal Adequacy

Limited course-taking options and narrow assessment practices are compounded by problems of fiscal inadequacy and resource distribution. The Council of Great City Schools [CGCS] (1992) calculated that the average-per-pupil expenditure in 1990–1991 was \$5,200 in large urban school districts compared with \$6,073 in suburban public school systems. Although both types of school systems allocated about 62% of their budgets to classroom instruction, urban schools spent about \$506 less per child on instruction. Although this study does not use current data related to fiscal resources, it reflects a growing fiscal disparity between urban school systems and some suburban systems and illustrates an important point. How money is spent should be examined carefully. For example, the Commissioner of Education of New York State reported, “The more advantaged districts [in New York State], spend over \$3,000 more per student and

pay their teachers \$20,000 more annually. Students in more advantaged districts are substantially more likely than students in less advantaged districts to perform with distinction on Regents examinations, and they are more than twice as likely to plan to attend four-year colleges” (New York State Department of Education, 2002, pp. vi–vii). The CGCS (2003) calculated that the New York Public Schools would need \$12,537 per pupil to have the resources equivalent to the highest achieving school districts in the state. The fiscal support undergirding instructional practice has implications for meeting new and more challenging demands in mathematics education.

Over the past decade, the average-per-pupil expenditure has constantly increased for urban and suburban school districts. Yet, as Cohen, Raudenbush, and Ball (2003) proposed, rather than focus on fiscal resources as the center of research and policy-making, teaching and learning should be centered and questions of adequate fiscal resources should derive from carefully planned instructional programs. The call for “Mathematics for All” or “Algebra for All” associated with many state content standards proclamations has placed new demands on urban school systems to prepare larger numbers of students in content traditionally reserved for a small percentage of students. Never before has there been a greater need to extend the amount of time students have with mathematics content that is aligned to state curriculum guides and appropriate tests.

Unfortunately, the old saying “time is money” is directly applicable to the implementation of design strategies capable of providing students more time on task in mathematics. Some considerations related to extending time for students in mathematics are listed below:

- Preschool availability;
- Early intervention programs for low performing schools;
- Extended school day opportunities;
- After- and before-school tutorial programs;
- Saturday school;
- Summer school enrichment for all students (not just remediation);
- Community college/university programs;
- Longer school day or expanded year;
- Enrichment and mentoring programs; and
- More individualized or small group instruction.

Each of the strategies listed is integral to a standards-based approach to educational policy-making. These strategies require a sound vision that is direct-

ly linked to fiscal policy. State standards provide an opportunity to plan for success. A simple planning strategy includes (a) adopting a set of mathematics standards, (b) identifying resources needed to achieve the standards (including time-related strategies), (c) formulating a long-term plan that aligns the standards and resources, (d) developing the plan before spending money, and (e) adopting the necessary structural changes to maximize cost-effectiveness. Planning for more engaged time in mathematics is a purposeful act that should be aligned with fiscal management. A school district’s portfolio of mathematics practices and interventions should be clearly aligned to uniform content goals and fiscal management. Too often, districts fail to produce aligned practices and fiscal policy. Yet, a portfolio of aligned practices, interventions, and fiscal policy is the essence of a district’s learning design.

Quality and School Mathematics

Most observers should recognize that the call for more demanding standards in mathematics is a signal for not only what students must know but also what teachers must understand and school systems must support. High standards in school mathematics demand quality instruction and supporting infrastructure. The purpose of this section is to examine quality factors that influence mathematics instruction.

The OTL literature defines *quality* as classroom pedagogical strategies that affect students’ academic achievement. In this case, quality is defined as those pedagogical strategies that positively influence student achievement in school mathematics. Before discussing quality factors, a baseline review detailing what is typical with respect to mathematics pedagogy is helpful.

Traditionally, mathematics pedagogy has emphasized whole-class lectures with teachers modeling one strategy for solving a problem and students passively listening to the explanation. Generally, the lecture is followed by students working alone on a large set of problems that reflect the lecture topic (Fey, 1981; Porter, 1989; Stodolsky, 1988). The purpose of the lecture and problem set is to prepare students to produce correct responses to narrowly defined problems. This pedagogical strategy is often coupled with curricular or ability grouping, with many African American and Hispanic students selected to participate in compensatory mathematics programs that focus on the mastery of low-level computational skills (Strickland & Ascher, 1992). These phenomena are so “normal” in many schools, they have become cultural artifacts. The achievement trends as a result of this model of instruction were reviewed earlier in this section.

In contrast, high-quality mathematics programs generally deviate in important ways from the “normal” approaches to mathematics instruction and classroom practice. A comparison of mathematics teachers in higher and lower performing schools conducted by the North Central Regional Educational Laboratory (NCREL, 2000) revealed important quality factors related to instruction. The findings are summarized in Table 28.5.

Note that the NCREL findings must be understood in light of the centrality of students’ *mathematical reasoning* in higher performing schools. Higher performing schools and teachers provide a learning environment that supports sustained engagement on rigorous mathematical tasks. Teaching as characterized in the higher performing schools is complex and demanding. In contrast, the teaching in lower performing schools is routine and limited with respect to teacher-student discourse patterns. Further, instructional practices in lower performing schools do not center on students’ mathematical understandings and thinking. The characteristics found in teachers working in higher performing schools can be supported in other schools by administrative planning and instructional leadership with the following specific actions:

- Provide professional development that prepares teachers to focus on mathematical understandings and reasoning.
- Provide ongoing professional development focused on content, effective instruction, and student thinking in the content domain.

- Design a curriculum that provides sufficient exposure to difficult concepts.
- Develop programs to address the impact of student/teacher mobility in low-performing schools.

Each of these quality factors will be discussed in greater detail. The focus of the discussion will center on why these factors are key support mechanisms for achieving the quality teaching characteristics indicated in the NCREL study.

Quality Professional Development

What are the “best practices” related to the professional development of mathematics teachers? Every year, school districts sponsor thousands of professional learning opportunities for teachers. There has been a gradual shift in thinking about professional development in many sectors including education and the corporate world (Meister, 1998). A summary of recent shifts in emphasis related to professional development is provided in Table 28.6.

Are shifts in thinking about professional development (as reflected in Table 28.6) consistent with research in mathematics education and teacher learning? What works?

Garet, Porter, Desimone, Birman, and Yoon (2001) conducted the first large-scale empirical comparison of effects of different characteristics of professional development on teachers’ learning. The study used a national probability sample of 1,027 mathematics and science teachers. The results confirm and extend the

Table 28.5 Characteristics of Classrooms and Teachers in Higher and Lower Performing Schools

Higher Performing Schools	Lower Performing Schools
Teachers and students participate in two-way conversations about mathematical ideas.	Conversations tend to be one-way: The teachers tell information to students or look for answers and move on.
Classes exhibit the characteristics of learning communities. There are norms in place so students and teachers are learning together.	Classes have few learning community characteristics. Individuals are more disconnected.
Teachers push for mathematical meaning behind the task.	Teachers lead math tasks; however, meaning-oriented discussion is missing.
Teachers have high expectations that all will learn. They review concepts often, explain ideas thoroughly, invite student thinking, assess student competence, and reteach when necessary.	The expectation is that other sources of help will fill in gaps for struggling students.
Teachers build continuity in the mathematical domain from day to day.	Little continuity is built into mathematical content from day to day.
Students are comfortable with classroom routines and expectations and take initiative in their progress. (They know where to find enrichment materials when finished with an assignment and get started on their own.)	Classroom routines are teacher initiated rather than student initiated. Teachers often remind students of expectations.

Note: From NCREL, 2000, p. 18

Table 28.6 Professional Development Paradigm Shift from Staff Training to Learning

	Old Training Paradigm	Learning Paradigm
Location	Central office	On demand—anywhere
Content	Upgrade math skills	Build core workplace competencies
Methodology	Lecture	Action learning
Audience	Individual teachers	Intact teams of teachers, principals, other staff
Faculty	External university professor/consultants	Internal senior-level district staff and a consortium of university professors and consultants
Frequency	One-time service	Continuous learning process
Goal	Build teacher's inventory of skills	Solve real education issues and improve classroom teaching

Source: Adapted from Meister (1998).

literature on “best practice” in several ways. The study confirms past literature in that the research indicates sustained and intensive professional development is more likely to influence teacher learning, as reported by teachers, than shorter professional development. Also, the research indicates that professional development that focuses on academic work (content), provides teachers opportunities for “hands-on” work (active learning), and is integrated into the daily life of the school (coherence) is more likely to result in enhanced knowledge and skills.

Garet and Associates (2001) extended what is known about professional development and confirmed speculation in the following manner:

Our results provide support for previous speculation about the importance of collective participation and the coherence of professional development activities. Activities that are linked to teachers' other experiences, aligned with other reform efforts, and encouraging of professional communication among teachers appear to support change in teaching practice, even after the effects of enhanced knowledge and skills are taken into account. Such coherence has been hypothesized as important, but with little direct empirical support in the literature to date. Similarly our data provide empirical support that the collective participation of groups of teachers from the same school, subject, or grade is related both to coherence and active learning opportunities, which in turn are related to improvements in teacher knowledge and skill and changes in classroom practice. (p. 936)

This study suggests that if those who are concerned about education are serious about improving the quality of teaching in mathematics classrooms, they need to support and invest in professional learning opportunities for teachers that foster enhanced instructional

practice. A major challenge to the kind of professional development outlined in this study is cost. It is very important that sufficient resources be in place to support a quality professional development model.

Quality Curriculum

The need for a demanding mathematics curriculum aligned with high-quality instructional materials is intuitively obvious. Unfortunately, there is often confusion about the relationship between curriculum and instructional materials. Many systems purchase instructional materials and then treat them like a curriculum. Teachers strive to teach the book from cover to cover with little reflection about the curriculum. In other school districts, curriculum guides are designed, distributed at one-day workshops, placed in school storage, and never used again.

There is serious need for quality district-level curriculum guides in mathematics. In many states, the state-level curriculum framework offers little guidance related to focus; instead, litanies of discrete topics are listed. At the school-district level, curriculum quality can be achieved if the following recommendations for developing and implementing guides are taken seriously:

- Focus on mastery objectives only.
- Reduce the scope of coverage.
- Provide and support the development of more cognitively demanding enrichment materials.
- Allow for variations in completion time and instructional strategy.
- Provide quality instructional materials to schools in a timely fashion.
- Educate principals by focusing their learning opportunities on the relationship between the

curriculum guides, district achievement goals, and test materials.

In the world of high-stakes testing, there is tremendous pressure, real or perceived, to teach to the test. A high-quality curriculum guide that demonstrates an alignment between the instructional materials (including the enrichment materials) and assessment tasks is more likely to result in students' experiencing a coherent and cognitively demanding mathematics classroom than pure reliance on test guidelines.

Mobility and Mathematics

How do schools address the challenges to quality mathematics instruction presented by student and teacher mobility in low-performing schools? High mobility causes a great deal of stress on campus officials attempting to serve these students. In the context of high mobility, a quality curriculum guide that standardizes the curriculum and instructional materials is vital. While schools and classes may deviate on pacing, teachers have a reasonable opportunity to meet individual needs if the curriculum guide has narrowed the coverage and focused on mastery objectives. Further, it is very important that individual student data is transmissible to the new school setting. This will give the teacher an opportunity to construct a data-driven program of study for the student.

Teacher mobility in low-performing schools also is a major problem. Often, new teachers to a system are sent to low-performing schools. The result is not surprising. These teachers either leave the profession or get seniority and transfer to another school. This pattern is consistent and endemic. The result: Low-performing schools are constantly staffed by less experienced teachers and, in many situations, by teachers with emergency teaching certificates.

The time has come for new models of operating in these schools. Questions related to how to retain teachers in low-performing schools require empirical evidence. Here are a couple of speculations on the issue. Retention of teachers in challenging settings may be linked to instructional leadership. Good principals create learning communities that support teachers and students.

Another potential strategy involves the recruitment process. Perhaps with incentives cohorts of well-established teachers can be recruited to low-performing schools. The emphasis is on *cohort*. The goal in this strategy is to embed a core group of excellent teachers in the school setting to influence and mentor the less experienced teachers.

Clearly, it is a disservice to new teachers and students to place novice teachers in the most challenging settings. Remedies to the mobility problem will require major rethinking in the areas of human resource

management and fiscal management. Moreover, mobility issues are often compounded with other system challenges like cultural factors and student language background.

Culture and Mathematics Learning

Today, many calls for equity in mathematics education borrow from opportunity-to-learn constructs found in national and international testing programs. In fact, OTL constructs are foundational in this section. These constructs frame equity largely as the overlap of content taught and content tested. The overlap of content taught and content tested is a serious policy concern. Moreover, opportunity-to-learn constructs have additional explanatory power if aligned with the cultural factors that influence students' mathematics learning (Tate, 1995). Research suggests that developers of equity-related policies in mathematics education should carefully consider incorporating recommendations found in the *Professional Standards for Teaching Mathematics* (NCTM, 1991), which call for mathematics pedagogy to build on (a) how students' linguistic, ethnic, racial, gender, and socioeconomic backgrounds influence their learning; (b) the role of mathematics in society and culture; (c) the contribution of various cultures to the advancement of mathematics; (d) the relationship of school mathematics to other subjects; and (e) the realistic application of mathematics to authentic contexts (see e.g., Ladson-Billings, 1994; Meyerson, 2002; Moses & Cobb, 2001; Nelson-Barber & Estrin, 1995; Rousseau & Tate, 2003; Secada, 1996).

The first NCTM recommendation calls for understanding how demographic group membership may be linked to the learning of mathematics. This recommendation is consistent with *No Child Left Behind* legislation that requires a national accounting of student performance in mathematics by demographic group. However, gathering achievement data by demographic group is very different than reflecting on race-related achievement patterns. Rousseau and Tate (2003) found that mathematics teachers in their study were reluctant to reflect on race and student performance in mathematics. Instead, some teachers in their study indicated they were color-blind and did not notice race or attend to matters of race-related patterns of student achievement. Further, many of the teachers were unwilling to link poor student performance to their teaching or other school-related factors. Teachers tended to blame the students and their families. This kind of practice represents a unique challenge for instructional leadership attempting to engineer school mathematics improvement. The challenge suggests a need for professional practice among teachers and school leaders that differs radically from tradition-

al formats. The need for study groups composed of teachers and school leadership is clear. These groups must foster trust and openly communicate about data and students; in particular, the challenge of discussing race and culture must be met (e.g., Tatum, 1992).

Other cultural factors in school mathematics are related to quality design and school change. Many of these cultural factors largely deal with the aim of school mathematics. More specifically, they deal with the nature and extent to which school mathematics curriculum is linked to the liberal arts tradition of reasoning and inquiry in contexts broader than the problems and concepts found in the discipline of mathematics. Should school mathematics include investigations of how mathematics is used in society and culture? For example, how relevant is political numeracy? Mathematics is part of many aspects of a democracy and can inform the reasoning associated with policy formation and policy analysis. Is this appropriate for middle school or high school students?

Some teachers have embarked on student-led integrated problem-solving investigations that include mathematics, statistics, legal analysis, multimedia techniques, scientific method, and connections to other disciplines (Tate, 1995). The approach is consistent with calls for authenticity in mathematics instruction (Meyerson, 2002). This strategy is designed to build on students' interest and to provide a liberal arts approach to the middle school and high school experience. However, the liberal arts approach may not be consistent with the current demands of high-stakes testing environments. Any disconnect between the liberal arts perspective of schooling and mathematics education is worthy of discussion by teachers, instructional leaders, and policymakers.

Similarly, for some elementary instructional leaders, providing an integrated learning experience that connects mathematics, science, and reading is desirable. This kind of integrated approach has many merits including efficient use of time and building on best practice in early childhood education (Bredekamp & Copple, 1997). If the culture of testing, specifically test preparation activities, substitute for real learning experiences and best practice, then long-term skills like student reasoning ability may be sacrificed.

Language and Mathematics Learning

In a policy analysis of urban students acquiring English and learning mathematics in the context of reform, Secada (1996) raised the following two ques-

tions: "Should their [urban school] efforts at reforming school mathematics specifically address the status of students acquiring English? Or should urban schools assume that these students' needs will be addressed under the broader aegis of reform?" (p. 422). Secada argued that failure to consider the specific learning needs of students acquiring English would be a mistake. He maintains that educators might usefully examine common learning processes that cut across language learning and mathematics learning. Two potential areas of analyses include (a) psychological processes that are common to understanding language and mathematics and (b) sociolinguistic and cultural processes that support the creation of discourse communities in school including how sense making takes place and is validated in these communities (e.g., Kinstch & Greeno, 1985; Lampert, 1990). Secada (1996) described a potential scope of work for educators related to bilingual education and school mathematics:

Newly developing models for teaching mathematics should be scrutinized for their applicability to bilingual learners and adapted as necessary. The limitations of evolving ways to teach mathematics (Lampert, 1990) is a reason to question, but not reject, the developing visions for teaching mathematics (NCTM, 1991). Maybe, with some adjustments—specifically inviting these students to add their thoughts, encouraging them to use their native languages and asking others to translate, slowing down the fast-pace tempo of the classroom, creating an atmosphere in which language variation in the community of discourse is an accepted fact of life—these methods can apply to bilingual learners. (p. 440)

Secada's remarks concerning bilingual learners and school mathematics focused on the importance of modifying instructional time and appropriate instructional accommodations—both critical OTL variables. Time and quality factors permeate the discussion of the research-based cases of the next section.

RESEARCH-BASED CASES OF SCHOOL MATHEMATICS REFORM⁹

In every mathematics reform effort, significant time should be devoted to information gathering and group study of other similar change efforts. Learn as much as possible about related design. The intent of this section is to review a select set of research-based

⁹ Additional information about these cases can be found in Tate and Rousseau (2002) and the January 1996 issue of *Urban Education*. There is some overlap in the evidence presented in this chapter and the Tate and Rousseau article. The cases are presented in this chapter to highlight the importance of time and quality factors.

cases to serve as a model for future information-gathering activities.

The group of studies reviewed in this section was included for three reasons. First, the studies were part of large multi-year projects focused on classroom-based research. These studies provide insight into how time, quality, and design interact to produce positive academic results in school mathematics. In each of the projects reviewed, student performance in mathematics improved over time. Second, each project at some point examined equity-related concerns and looked to intervene in school settings where student proficiency in mathematics was underdeveloped. Finally, each project was included because participants engaged in an effort to reform school mathematics in a manner that was consistent with the teaching practices and curricular goals found in the National Council of Teachers of Mathematics (NCTM) reform documents or state/local mathematics standards. This is not an endorsement of the NCTM Standards but a review of research with the potential to shed some light on the standards debate. The three projects included in this review each have comprehensive research and evaluation components including data on student advancement and other educationally relevant indicators of progress. The research and evaluation aspects of these projects are important because of the rapid advancements of state standards. Massell (1994) reported 41 states have adopted mathematics standards that at least in part are consistent with the NCTM standards series. A brief history of this series is warranted.

In 1980, NCTM, a professional organization of mathematics teachers, supervisors, and college professors, released *An Agenda for Action*, which described a 10-year reform process. A central goal of *An Agenda for Action* was to move the focus of school mathematics from a strictly basic skills curriculum to a more balanced approach that included more demanding mathematics content and appropriate pedagogy to implement this content. Subsequently, but not as a direct result of *An Agenda for Action*, NCTM sponsored the development of the *Curriculum and Evaluation Standards for School Mathematics* (1989), the *Professional Standards for Teaching Mathematics* (1991), the *Assessment Standards for School Mathematics* (1995), and most recently, *Principles and Standards for School Mathematics* (2000). These documents were a product of extensive literature reviews and a series of technical reports that described key themes and ideas in school mathematics. This series of reform documents and the movement to reform school mathematics are important from an equity perspective. Past reform efforts have failed to significantly improve opportunity to learn mathematics for African American, Hispanic, and

low-SES students (Tate, 1996). Thus, a close review of more recent reform efforts is part of the process of learning to build.

Cognitively Guided Instruction (CGI)

Researchers at the University of Wisconsin developed Cognitively Guided Instruction (CGI). The CGI foundation was in part established on Carpenter and Moser's (1983) analysis of young children's learning of addition and subtraction. Subsequently, other research was conducted to understand how teacher knowledge of children's thinking would affect teachers' pedagogical actions and student learning (Carpenter, Fennema, Peterson, & Cary, 1988; Carpenter, Fennema, Peterson, Chiang, & Loeff, 1989). This research suggested that knowledge extracted from studies of learners' thinking can be used by teachers to strategically influence students' learning. The CGI research program supports the argument that knowledge of students' thinking, when integrated, robust, and a part of the established curriculum, can affect the teaching and learning of mathematics (Fennema & Franke, 1992). Carpenter, Fennema, Franke, Levi, and Empson (1999) described the CGI design process and model as follows:

Our research has been cyclic. We started with explicit knowledge about the development of children's mathematical thinking, which we used as a context to study teachers' knowledge of students' mathematical thinking and the way teachers might use knowledge of students' thinking in making instructional decisions. We found that although teachers had a great deal of intuitive knowledge about children's mathematical thinking, it is fragmented and, as a consequence, generally did not play an important role in most teachers' decision making. If teachers were to be expected to plan instruction based on knowledge of students' thinking, they need some coherent basis for making instructional decisions. We designed CGI to help teachers construct conceptual maps of the development of children's mathematical thinking in specific content domains. (p. 105)

CGI is not associated with particular instructional materials. Moreover, CGI does not have an explicit equity component, nor is it targeted at a particular group of students. However, it has been successfully implemented in classroom settings with diverse student groups.

For example, Carey, Fennema, Carpenter, and Franke (1995) described CGI classrooms in a predominantly African American school district. Twenty-two 1st-grade teachers from 11 schools in Prince George's County, Maryland, an urban school setting bordering Washing-

ton, D.C., participated in a research project organized to evaluate the efficacy of CGI with African American students. The student demographics in the classrooms of the study exceeded 70% African American. Further, 7 of the 11 schools participated in Chapter 1, a federally funded program of Title I of the Elementary and Secondary Act, a strong indicator of high concentrations of low-income students in a school. The teachers who participated in the study attended a 2-week summer in-service program that was followed with 5 full-day professional development days offered during the academic year. The researchers documented a change in the teachers' implemented mathematics curriculum, with a greater focus on problem solving beyond that typically associated with the first-grade curriculum. The teachers also displayed an ability to take advantage of student thinking about important mathematical ideas, ultimately building on student understanding to establish new knowledge of school mathematics.

Villasenor and Kepner (1991) reported on the implementation of CGI in a minority context. The study was carried out with 12 treatment classes and 12 control classes in which the percentage of non-White populations ranged from 57% to 99%. The CGI group performed significantly better on a 14-item word problem posttest, an interview on word problems, and an interview on number facts. The CGI students also used advanced strategies significantly more often than non-CGI students on both problem solving and number facts. Peterson, Fennema, and Carpenter (1991) argued that "Villasenor's results are important because they provide concrete evidence for the effectiveness of the CGI approach with a disadvantaged population of students" (p. 78).

The CGI studies suggest that an important set of quality factors related to mathematics instruction is how well teachers (a) understand the structure of a specific mathematical concept, (b) understand students' thinking about the particular mathematical idea, and (c) implement instructional strategies that build on this knowledge of student thinking. As a set, these quality factors are powerful indicators of good instruction with a strong relationship to student learning and performance on outcome measures.

Project IMPACT

Project IMPACT "is a school-based teacher enhancement model for elementary (K–5) mathematics instruction designed to foster student understanding and to support teacher change in predominantly minority schools" (Campbell, 1996, p. 449). Six schools were involved in the original study (three treatment

and three control). The model involved (a) a summer in-service program, (b) an on-site mathematics specialist in each school, (c) manipulative resources for each classroom, and (d) teacher planning and instructional problem solving during a common grade-level planning period each week. The focus of the model was on instructional approaches consistent with a cognitive perspective on learning, emphasizing interaction and collaboration rather than the typical direct instruction approach.

Unlike CGI, Project IMPACT focused specifically on teaching for understanding in urban schools. Thus, content addressing "teaching mathematics in culturally diverse classrooms" was included in the program's summer in-service. Supported by campus-based mathematics specialists, instructional change occurred in most treatment classrooms, particularly where the instructional leadership by the principal encouraged and embraced the reform process. The students in these schools were assessed in the middle and at the end of each school year. Campbell (1996) summarized the results:

The influence of the IMPACT treatment on student achievement was not immediate. The students in the IMPACT treatment schools did not evidence statistically significant higher achievement, as compared to the students in the comparable-site schools, until the middle of second grade; however, once established, this mathematics differential continued through second and third grade. (p. 463)

White (1997), in her dissertation, examined the nature of questioning in four 3rd-grade classrooms both before and after the teachers went through the Project IMPACT summer in-service program. The study documented the question-response pattern, the cognitive level of the question (low or high), and the race and gender of the students who responded. White found that students' educational experiences, as reflected in classroom questioning, differed both between and, in some cases, within classes. Two teachers, Ms. Davis and Ms. Tyler, were fairly equitable in their distribution of questions.¹⁰ "They posed questions to all students across questioning patterns and cognitive levels" (White, 1997, p. 300).

In Ms. Atkins's class, however, the distribution was more skewed. Overall, females answered the majority of the questions. Yet, a look at the different cognitive levels reveals racial patterns as well. White and Asian females answered most of the high-level questions. Black and Hispanic female students were asked a relatively low number of high-level questions. According to

¹⁰ The names were changed to protect the identities of the teachers.

White (1997), the origin of this disparity lies in Ms. Atkins's perceptions of students' academic ability and her own discomfort with mathematics. Ms. Atkins wanted to ask high-level questions, but her own lack of understanding caused her to call only on students whom she thought would give the correct answer. Thus, only the students perceived to be of high ability were selected to answer high-level questions. A similar pattern of focusing only on the students who were perceived to have the greatest mathematical understanding was found in the class of the fourth teacher, Ms. Smith.

This very detailed study of question and response patterns is important for at least two reasons. First, it documents a partial success story for Project IMPACT in terms of improving equity in classrooms. Two of the four teachers appeared to change their practices as a result of their participation in the initial IMPACT summer in-service and ongoing campus-level assistance. Both Ms. Davis and Ms. Tyler were more equitable in their distribution of questions after the in-service than they had been before.

This study is also important because it suggests the need to look closely at teachers' explanations for their actions in order to more fully understand what is happening in the classroom. For example, the case of Ms. Atkins indicates that teachers' inequitable actions can originate from a variety of sources, including inadequate content knowledge.

Project IMPACT suggests that another quality indicator related to mathematics instruction is the relationship among teachers' knowledge of mathematics, teachers' understanding of student thinking about mathematics, and teachers' understanding of race/gender interactions in classroom settings. Project IMPACT is consistent with other research programs that indicate the importance of treating cultural background as a resource for learning (Rousseau & Tate, 2003). For example, Knapp (1995) found that teachers in high-poverty schools who placed the greatest emphasis on meaning in their mathematics instruction made two significant shifts in their thinking about learners. First, they viewed learners as active participants in learning, and second, the teachers used cultural dimensions of instruction to sustain engaged time with academic work. This also is an important lesson from Project IMPACT.

QUASAR

QUASAR is described as "an educational reform project aimed at fostering and studying the development and implementation of enhanced mathematics instructional programs for students attending middle schools in economically disadvantaged communities"

(Silver & Stein, 1996, p. 476). One purpose of the project was to help students develop a meaningful understanding of mathematical ideas through engagement with challenging mathematical tasks. The QUASAR project supported teachers and administrators in six urban middle schools. Each school site worked with a resource partner—typically, mathematics educators from local universities—to improve the school's mathematics instructional program with a focus on mathematical understanding, thinking, reasoning, and problem solving. The site teams operated independently in the design and implementation of its curriculum plan, professional development, and other features of its instructional program. There were regular interactions among representatives from all QUASAR sites. Moreover, each site-based team benefited from financial support, technical assistance, and advice from the QUASAR staff housed at the Learning Research and Development Center at the University of Pittsburgh.

Silver and Stein (1996) described three different analyses used to assess the effectiveness of instruction in QUASAR sites. Unlike the CGI and IMPACT studies, there was no control group in the QUASAR study. One method used to determine the impact of QUASAR was the examination of changes in student performance over time. The results from the first 3 years of the project indicated that "students developed an increased capacity for mathematical reasoning, problem solving, and communication during that time period" (Silver & Stein, 1996, p. 505). A second method of evaluation used a variety of tasks from the National Assessment of Educational Progress (NAEP) as pseudo-control groups (Silver & Lane, 1995). The QUASAR students were given items from the 1992 eighth-grade NAEP. The results were compared to those of NAEP's national sample and disadvantaged urban sample. The findings from the analysis of student performance on the nine open-ended tasks were very informative about the effectiveness of QUASAR. QUASAR students performed at least as well as the national sample on seven of the nine tasks. Silver and Lane (1995) noted that this is an important result, in light of the fact that the national sample had significantly outperformed the disadvantaged urban sample on all nine tasks.

The findings clearly suggest that the mathematics performance gap between more and less affluent students has been significantly reduced for students attending the QUASAR schools. Thus, the performance of QUASAR's students is far greater than would have been expected, given their demographic similarity to NAEP's disadvantaged urban sample, and one can infer that the instruction at QUASAR has a beneficial impact on students' mathematical performance. (p. 62)

A third method of evaluation examined outcomes other than achievement, considering whether QUASAR instruction was linked to increased access and success in algebra coursework. Silver and Stein (1996) reported that students from QUASAR schools were both qualifying for and passing algebra in ninth grade at substantially higher rates than before QUASAR.

The QUASAR project reinforces the importance of students' engaged time with cognitively demanding mathematical concepts. Sustained engaged time with quality mathematics tasks resulted in improved student performance on a wide range of indicators.

Research Case Summary

These research cases have three overlapping themes worthy of note. First, high-quality mathematics was at the center of each effort. More specifically, each program called for mathematics instruction using mathematical tasks not typically associated with the lower educational expectations often found in large urban school districts. In each program, mathematical proficiency consisted of a balance between conceptual understanding and procedural proficiency. Too often instruction is based on extreme positions, rather than a balanced approach with increasing cognitive demand.

A second common feature of each program was the important role of classroom-based assessment designed to better understand student thinking about mathematical ideas. These studies of mathematics teaching support the idea that teachers' knowledge of students' reasoning when integrated with a balanced mathematics curriculum can positively affect the teaching and learning of traditionally underserved students. The assessments used in the research-based cases of mathematics reform were designed to support the learning process and to identify areas in which further instruction is needed. The measures included direct observations of children during classroom activities, evaluation of student work, and asking questions in class. The portfolio of measures in the research-based case studies differed from the measures used to gauge national trends. The latter measures are designed to inform the public about trends in student performance or the effectiveness of large-scale educational programs. The research-based cases included standardized tests, selected NAEP items, and classroom-based activities.

A third common feature of each case was the presence of a strong mathematics professional development program. Two key features of these programs were teacher learning opportunities in the areas of school mathematics content and how children's math-

ematical knowledge develops in the content domains, including what knowledge students were likely to bring with them to school.

These three common themes represent important quality factors. They reinforce how the combination of quality curriculum, cognitive-based assessment tools, and integrated professional development are central to school mathematics design.

TAKING THE DESIGN TO SCALE

The three cases reviewed in the prior section provide important lessons regarding the design of effective programs. However, they are relatively small in scale. Arguably, CGI has had the most widespread impact of the three programs. Hundreds of teachers from various parts of the country have participated in CGI professional development sessions. Although teacher participation in CGI has been linked to student achievement, CGI was not designed to address OTL issues on a systemic basis. Thus, in order to more fully understand how the OTL variables described in the previous sections might come together in the design of a systemic reform effort, it is important to look at examples of design taken to scale. More specifically, each example represents an effort to reform mathematics education in a school district or set of school districts.

The Urban Systemic Initiative

In 1993, the National Science Foundation began the Urban Systemic Initiative (USI) program. The purpose of the program has been to catalyze large-scale systemic change directed toward improving the science and mathematics achievement of students in urban schools. In the 1998–1999 school year, 21 USI sites served over 4.5 million students or 10% of American students in public schools. The students in these school districts represent some of the poorest in the United States, with 69% of the students eligible to receive free or reduced lunch. In addition, 12% are students with limited English proficiency (Kim, Crasco, Blank, & Smithson, 2001).

Six “drivers” serve as the basis for the urban systemic reform program. The drivers have an interdisciplinary philosophical foundation including principles derived from organizational theory, policy studies, ethics, and cognitive science. In practice, the drivers were designed to address mathematics, science, technology, and engineering. However, in this chapter, the focus will be on the role of the drivers in school

mathematics. The six drivers for the National Science Foundation educational system reform are

1. Implement standards-based curricula as represented in instructional practice and student assessment system-wide.
2. Develop a coherent and consistent policy portfolio that supports high-quality mathematics and science education.
3. Converge all resources that are directly or tangentially linked to the support of mathematics and science education into a singular program of upgrades and improvement in mathematics and science for all students.
4. Build broad-based support from parents, policymakers, higher education community, business and commerce, philanthropic organizations, and other segments of the community for the mathematics and science programming and the critical decisions related to system-wide programmatic efforts.
5. Accumulate a broad and comprehensive evidentiary base of student performance indicators—for example, achievement test scores, course-taking patterns, college admissions rates, college majors, and Advanced Placement tests.
6. Improve the achievement of all students, including those traditionally underserved (Kim et al., 2001).

The drivers are consistent with the time and quality variables discussed earlier in this chapter. For example, the USI sites have emphasized implementing quality curriculum and instruction. According to Kim et al. (2001), nearly all of the USI sites have decreased their emphasis on more traditional lecture and demonstration methods and moved towards the types of high-quality instructional practices described earlier. Similarly, most of the USI schools have implemented instructional materials that are aligned with the NCTM Standards documents, including 89% of USI elementary schools and 90% of USI middle schools (Kim et al., 2001). In many cases, these instructional materials were developed with funding from NSF (e.g., *Everyday Mathematics*; *Investigations in Number, Data, and Space*; *Connected Mathematics Project*). In addition, the adoption of these instructional materials has been connected to the revision of district curriculum guides.

Policy changes have also impacted OTL in USI sites. According to Kim et al. (2001), 13 USI sites reported the elimination of tracking and all remedial and non-college preparatory mathematics. In addi-

tion, mathematics requirements for high school graduation have increased in several USI districts since the start of the initiative—changing either from 2 years to 3 years (Chicago, Detroit, Phoenix, Cleveland, Columbus, and Fresno) or from 3 years to 4 years of required high school mathematics (Dallas and New Orleans). Moreover, in 12 of the 21 districts, the graduation requirements for the district exceeded the state graduation requirements (Kim et al., 2001).

Reform efforts in USI sites also have begun to address fiscal adequacy issues through the coordination of multiple funding sources, including Title I, Eisenhower, and other federal, state, and local programs. According to Kim et al. (2001), a total of \$547 million dollars in additional funding was leveraged by the urban systemic districts during the 1999 fiscal year. Moreover, other resources have been brought to the reform effort through partnerships with several different types of entities, including higher education institutions and the private sector. For example, partnerships with higher education institutions have provided the Memphis USI with professional development for its teachers as well as academic support and enrichment activities for students in the Memphis City Schools (Kim et al., 2001).

USI sites have put a significant emphasis on providing quality professional development to their teachers, investing an average of 49% of USI funds on professional development activities. The specific nature of mathematics professional development activities varies across sites, but these activities generally focus on content knowledge, curriculum implementation, assessment, and new methods of teaching. According to Kim et al. (2001), approximately 80 to 90% of USI teachers have been actively involved in professional development, and many report changing their classroom practice as a result.

One of the outcomes of interest with respect to measuring the impact of the USI program is enrollment in gate-keeping and higher level mathematics courses. Changes in enrollment in the USI sites appear to indicate success in getting more students to higher levels of the mathematics pipeline. For example, during the 1993–1994 school year, 158,673 students, or 30% of the total number of 9th–12th-grade students in the first cohort of schools (including Baltimore, Chicago, Dallas, Detroit, El Paso, Miami-Dade, New York, and Phoenix) were enrolled in gate-keeping and higher level mathematics courses. By the 1998–1999 school year, the number of high school students from the first cohort enrolled in these courses had increased to 253,458, or 46% of the total number of students (Kim et al., 2001). In addition, increases in enrollment in Algebra I or above for 8th-grade students have occurred across racial groups.

In the first cohort of sites, African American 8th-grade student enrollment in Algebra I or above increased 58% from the 1993–1994 school year to the 1998–1999 school year. Hispanic 8th-grade student enrollment in Algebra I or above increased 59% during the same time period. For White students, there was a 26% increase in enrollment in 8th-grade Algebra or beyond. In addition to increases in enrollment rates for all groups, the disparities between groups have decreased. In the first cohort of USI sites, there was a 21% decrease in the enrollment disparities between White and African American students in gate-keeping and higher level mathematics courses between the 1993–1994 and 1998–1999 school years. Similar trends can be seen in Hispanic-to-White disparities (Kim et al., 2001).

Another important indicator of the impact of the USI program is student standardized test scores. All seven of the sites in the first cohort have seen an overall improvement in student test scores between the 1993–1994 and 1998–1999 school years. In addition, five of the seven sites in the first cohort have shown reductions in the minority-to-White test score gap. For example, at the start of the USI program in New York, the average passing rate for African American and Hispanic students on the CAT-5 was 26%, compared with 59% for White students. In the 1998–1999 school year, the average passing rate for African American and Hispanic students was 42%, compared with 72% for White students. The scores of both minority and White students increased, and the minority-to-White test score gap decreased (Kim et al., 2001).

Thus, in many ways, the USI program brings together many of the OTL factors described earlier into a design to improve achievement and equity for students who have traditionally been least well-served by public schools. The description of the general USI program drivers and accomplishments gives some indication, in broad strokes, of the possibility of intervention and improvement at a systemic level. However, what this description of the USI program lacks is an understanding of how the various time and quality factors interact to produce positive changes in student achievement. For this reason, we offer another example of the systemic reform of mathematics education.

Mathematics: Application and Reasoning Skills (MARS)

The MARS¹¹ project was a collaboration between the Baltimore City Public School System (BCPSS) and

the University of Maryland. The goal of the project was to “develop, implement, and evaluate a model for systemic reform in elementary mathematics” (Campbell, Bowden, Kramer, & Yakimowski 2003). The project, which lasted from 1996 to 2002, involved 106 elementary schools in BCPSS. Among the several different components of the MARS project, we will focus on three: (a) curriculum, (b) instructional support, and (c) professional development.

One of the key elements of the MARS project involved the redesign of the BCPSS curriculum guide and the subsequent adoption of instructional materials that supported the objectives outlined in the curriculum guide. In addition to this attention to quality curriculum, another element of MARS intervention involved support for teachers as they attempted to change their instructional practices. Exemplary teachers from BCPSS were released from their classroom responsibilities to serve as Instructional Support Teachers (ISTs). The ISTs were generally assigned to one or two schools and provided day-to-day on-site instructional support to teachers. They also designed professional development and organized grade-level planning meetings in their schools to help teachers understand and implement the curriculum. “The IST acted as coach, demonstration teacher, and program catalyst, supporting individual teachers’ efforts to implement intended instructional strategies in the mathematics classroom” (Campbell et al., 2003, p. 4).

A third key element of the MARS project was ongoing professional development for teachers. A variety of professional development activities were provided to teachers. One of these opportunities was the Summer Institute. The Summer Institutes were from 5 to 14 days in duration and focused on (a) mathematical content knowledge of teachers; (b) teaching mathematics for understanding; (c) research on children’s learning of mathematics concepts, organized with a grade-level focus; and (d) teaching mathematics in urban classrooms (Campbell et al., 2003). After the new curriculum guide and instructional materials were introduced, the Summer Institutes also served as an opportunity for teachers to learn how to implement the curriculum. In addition to Summer Institutes, the MARS project involved grade-specific Saturday Workshops during the school year that focused on the mathematics content that would be taught in the upcoming quarter and on the pedagogical strategies and materials appropriate for that content. Other professional development opportunities included af-

¹¹ The MARS project was supported by a Local Systemic Change (LSC) grant from NSF. Although similar to the USI Program in its systemic approach, the LSC program was narrower in scope (in the MARS example, focusing only on mathematics at the elementary grades) and was not specifically targeted at urban areas.

ter-school mini-courses addressing mathematics content and 3-hour graduate courses at the University of Maryland. During the 2001–2002 school year, 2,082 teachers (or 56% of the total number of elementary teachers) from BCPSS participated in some form of elementary mathematics professional development.

Standardized test results demonstrated significant increases in student achievement over the course of the MARS project. For example, between the 1997–1998 and 2000–2001 school year, the second-grade composite scores on the Terra Nova standardized test increased approximately 20 national percentile points. Similar gains were made in Grades 3 and 5. Fourth-grade gains were slightly less and first-grade gains slightly more than the other grades over the same time period (Campbell et al., 2003).

What makes MARS significant for this discussion is not simply the indication that systemic reform can impact student achievement, although that is certainly an important result. Campbell and colleagues (2003) have also examined the impact of the various components of the system, thus allowing greater insight into the influence of the time and quality factors. One of the components of the project that made a difference in student achievement was professional development. Teachers' participation in professional development that was high quality was associated with higher student achievement on the Terra Nova, at levels that were statistically significant. Further, results indicate that the effect was cumulative. For each additional high-quality professional development activity attended corresponding gains in student achievement were evident.

However, not all of the professional development opportunities offered through MARS were determined to be of high quality. Campbell and her colleagues (2003) found that the characteristics of high-quality professional development were (a) the professional development session was taught by either an IST, a well-qualified master teacher with 3 or more years of involvement in the project, or a partner from the University of Maryland; (b) the professional development integrated mathematics content and reform pedagogy; (c) if the professional development occurred following the revision of the curriculum guide, it focused on the guide and adopted textbooks; (d) if the professional development was a workshop, it was organized by grade level; and (e) if the professional development was a workshop, it focused on upcoming objectives in the mathematics curriculum, addressing both mathematics content and pedagogy (Campbell et al., 2003). These characteristics point to the interaction of quality professional development with other quality factors, specifically quality curriculum. For, Campbell and colleagues (2003) found that even

high-quality professional development failed to have a positive effect on student achievement unless it integrated the textbook resources and curriculum guide. Thus, the opportunity for teachers to learn about the curriculum that they were to teach appears to have been an important component of high-quality professional development in the Baltimore case.

Another important element of the MARS intervention was the instructional support provided to the schools. However, as with professional development, not all cases of instructional support were equally effective. Moreover, differences in relative effectiveness seem to have been related to both time and quality factors. A comparison of student achievement in 2000–2001 between schools with on-site instructional support and those without support revealed no significant difference in student achievement on the Terra Nova. Yet, significant differences were evident when high-quality support persons—teachers with a strong understanding of mathematics content and the standards-based curriculum and pedagogy—were assigned to a school for at least 4 years. According to Campbell and associates (2003), “less knowledgeable resources for school-based professional support or knowledgeable resources with less time for teachers to have access to that support were not as effective” (p. 48).

In summary, the MARS example adds to our understanding of systemic reform by highlighting the connections among OTL variables. Although the MARS example points to several of the same characteristics of high-quality professional development as the studies cited previously (e.g., targeted at a grade-level, focused on content and pedagogy), it also indicates a connection between quality curriculum and quality professional development. Moreover, the results suggest that high-quality instructional support of sufficient duration can impact student achievement. Thus, the example of MARS demonstrates how several of the OTL time and quality variables can be brought together to design a systemic program that is successful in raising student achievement.

It's Time to Design

Rigorous mathematics standards preK–12 require thoughtful action and planning. Moreover, the building blocks for engineering a mathematical revolution in any school are time, quality, and design. These three pillars of OTL are foundational for the improvement of the teaching and learning of mathematics in school settings. There are many paths for organizing and implementing change in school mathematics; however, the failure to consider time and quality factors and design issues carefully is a recipe for lost educational

opportunity. Jere Confrey and colleagues at the *Systemic Research Center for Education in Mathematics, Science, and Technology* (SYRCE) designed Figure 28.1 to serve as a conceptual model to inform the organizational design of mathematics teaching and learning in schools.

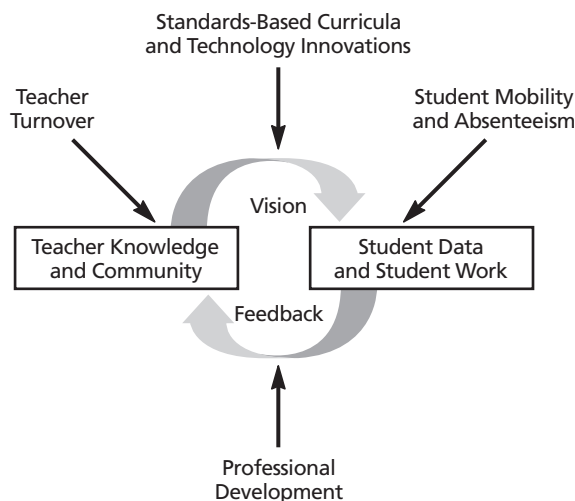


Figure 28.1 SYRCE conceptual model. *Source:* J. Confrey (personal communication, November 2004)

Figure 28.1 is included because it serves as a reminder of key opportunity-to-learn factors and how they interact in our systems of education. As an engineer, it is important to keep in mind a broader conceptual model of school change. This is especially important in light of the day-to-day realities of teaching, administration, and political challenges that face educators. The many events and distractions, such as leadership changes and financial shortfalls, that occur in schools and school districts only become real problems when they influence these three foundational pillars of learning and teaching. Unfortunately, instability and interim leadership are rife everywhere in public schooling. Recall, engineers “learn to build.” Both learning and building require stable, long-term, and insightful leadership. The building process for the engineer is supported by the following model development sequence:

- Model Construction
- Model Exploration
- Model Application
- Model Revision

Models are a language for describing patterns, patterns that can be observed and tested in the real world of schools. Thus, the process of model development emphasizes understanding school factors that influence opportunity to learn mathematics. The model development sequence requires additional clarification.

Model Construction

OTL requires a clear vision and set of learner goals. Many school districts accomplish this part of model construction by adopting all or portions of state mathematics standards and in combination with local system objectives create specific district-level goals for what all students should know and be able to do. District mathematics goals are important quality indicators for teachers, administrators, parents, students, and the broader community. Thus, multiple communication strategies for each constituency should be part of the model’s design. Each constituency should be provided samples of student work. The work should exemplify system-wide expectations and illustrate student work products that meet the district’s mathematics standards. Setting high standards and communicating a clear vision are only part of the model construction process. Ensuring that all students have access to a quality curriculum also is part of the process.

A number of steps are vital to initiating a quality curriculum. Each school should design an academic plan based on local mathematics standards and an associated accountability structure. The process includes

1. Communicating the importance of consistent application of curricular programs and standards.
2. Eliminating courses and academic experiences lacking the rigor of mathematics standards or inconsistent with developmentally appropriate mathematics practice.
3. Providing adequate time in the curricular design to focus on core subjects.
4. Building a timely framework for monitoring student progress using data.
5. Providing teacher and administrator professional development focused on classroom strategies for assessing student learning of mathematics standards.

Vision, curriculum quality, and accountability systems are vital to constructing a model. However, policy and resource alignment also contribute to a comprehensive model design. It is difficult to imagine a sustained change strategy that does not include front-end alignment of school policies and resources to support rigorous standards. For example, ongoing mathematics professional development should be part of the model construction process. In addition, effective student support programs should be prepared and ready to implement in response to updates on student assessments and mobility information. Too often, responses to these types of data are not part of the upfront planning process. Consequently, many school districts find themselves engaged

in triage mode, piecing together programs in real-time, rather than implementing planned data-driven interventions. Resources aligned with an appropriate vision, quality curriculum, and accountability systems are part of the technical core of the model construction process. The technical core is vitally important, but not enough. Ultimately, people explore the model and make decisions.

Model Exploration

Model construction includes planning for reflective examination of student, teacher, and school-level progress. Embedded in the model construction process are key elements of model exploration. The constructed model should include rich opportunities to explore the progress of the organization in relation to its goals. Yet, model exploration is not merely an instrumental activity; instead, exploration is cultural and people driven, with teachers and administrators central to the evaluative process.

As Figure 28.1 indicates, teacher community and knowledge are important components of the feedback loop. Teachers interpret student performance and assessments, and in turn they should rethink their practice and that of the school in conjunction with other instructional leaders. This process is greatly enhanced where community norms are established and teacher turnover is low. Teacher community and norms are linked to the model application process. The importance of teacher collective practice in school mathematics cannot be overstated (see, e.g., Gutierrez, 1996). As Campbell (1996) noted, “it may be unreasonable to expect sustained and reflective reform in isolated classrooms across urban settings. It is not unreasonable to address reform in urban schools where teachers and administrators are working together to develop a shared purpose and meaning” (p. 453).

Both administrators and teachers require training to support their community efforts to build a student-centered, data-driven organization. Classroom strategies for assessing student learning should be central to the common professional development. The community-building and professional-development activities also must be combined with an aligned monitoring process that includes

1. Timely and usable data on student progress.
2. Opportunities for mid-course corrections based on data.
3. Disaggregated student achievement data and mobility information.
4. Recognition and reward for positive results.

Model exploration potentially offers important benefits to schools and school districts. First, model

exploration can provide highly visible evidence related to opportunity to learn and support administrators and teachers seeking successful exemplars of effective mathematics practice (Skrla & Scheurich, 2001). Second, incremental success in student achievement and teacher effectiveness can lead to higher expectations and goals for academic achievement of all demographic groups.

Model Application

One can focus on the design of the model and matters of model exploration without attending to the classroom practices and support systems associated with instruction. Most school systems begin the academic year with some kind of school improvement plan, and although the quality of these plans vary widely, plans for change models exist. Further, it is quite common to find some level of model exploration in most school systems. Although, the quality factors associated with the exploration are often limited. Yet, the most serious challenge facing leadership is model application or implementation. Many instructional programs are adopted, distributed, and discarded each year.

A well-thought-out plan is required to gain any implementation traction. A key factor for success in this area is a fine-tuned curriculum guide that clearly delineates important content aligned to appropriate instructional materials. Further, research-based professional development for teams of teachers and instructional leaders is vital. The work of the teacher collective enterprise and significant time engaged with high-quality professional development are central to the implementation process. And quality student assessments—classroom-based as well as more summative assessments—are linked to collective and individual reflection by members of the teacher community. Quality curriculum, professional development, collective practice, and the other aspects of the model design are a support system for creating a successful teaching and learning process.

Teachers are key to model application. Clearly, mathematics teachers at all levels, kindergarten through college, are central to the improvement of mathematics education. If professional development is to make a difference to students in the classroom, it must be teacher-focused and student-centered. Stigler and Hiebert (1999) wrote:

Improving something as complex and culturally embedded as teaching requires the efforts of all players, including students, parents and politicians. But teachers must be the primary driving force behind change. They are in the best position to understand the prob-

lems that students face and to generate possible solutions. (p. 135)

Successful model application is largely classroom based and includes

1. Teachers and students participating in two-way conversations about mathematics.
2. Teachers pushing for both mathematical meaning behind quality curricular tasks and procedural fluency.
3. Teachers daily building instruction on student thinking and continuity in the mathematics domain.
4. Teachers and administrators organizing classes and support systems to ensure adequate time for students to learn the mathematics content.
5. Teachers and administrators maintaining high expectations for all learners.

These five components of instruction are foundational to model application and opportunity to learn. Although these components are stable in a broader sense, at times the design is not quite adequate to achieve system goals or more ambitious goals require a revised model.

Model Revision

Careful reflection on the progress being made is central to future progress. How is the model helping or hindering teachers and students? Each phase of the model development sequence should be reviewed, and critical discussions about fine-tuning or model abandonment must be held. Revision is not merely an end-of-the-year activity. Model revision is a continuous process that should be part of the design consideration. To aid in both the design and model revision process, the Appendix includes an *Engineering Change Assessment Instrument*. This instrument can be used to help keep track of school and school district progress on key opportunity-to-learn factors.

Taking a model-based approach to the mathematics change process involves constructing a change strategy, exploring the qualities and feasibility of the strategy, applying or implementing the strategy, and based on the data continuously fine-tuning the strategy. A close examination of many school district strategic plans will reveal many goals and targets related to the state accountability systems. Often, what is absent is a coherent change strategy or model for improvement in mathematics that takes seriously the time, quality, and design considerations reviewed in this document. Few school districts explore the qualities or feasibility of their change effort. Rather, many buy into a model

without considering the conditions and constraints that exist in the system. For example, this is evident when major initiatives to improve mathematics, reading, and science at the elementary level are occurring simultaneously. The point is not that three reform strategies cannot occur together; rather, failure to coordinate the changes is a common strategic flaw. Many reasonable change strategies are destroyed as a result of the failure to consider the feasibility of the model.

Engineering Progress: Limiting Conditions

Limiting conditions are conditions that materially affect the appraisal process and, as a consequence, the value conclusion. For example, having no electrical power in a school building is a limiting condition, as it prohibits the use of computers and other electronic equipment. This notion can be applied to mathematics education. What limiting conditions exist that can affect student performance on mathematics outcome measures and, as a consequence, the public's perception of school quality? A brief examination of recent legislation will provide some insight into this question.

Current educational policy and law, more specifically NCLB, calls for educators to carefully examine student achievement by demographic group. Moreover, schools and administrators are being held accountable for improved student performance. This is a radical departure from past educational practice. In fact, this law represents a major shift in federal discourse related to matters of equality. According to Crenshaw (1988), a legal scholar, there are two visions of equality present in anti-discrimination legislation and discourse. One view of equality, which she referred to as the restrictive view, "treats equality as a process, downplaying the significance of actual outcomes" (p. 1341). This section highlighted important processes related to student learning. Although critical to engineering positive progress, the opportunity-to-learn recommendations and other research-based lessons are limited. Specifically, they represent building blocks. However, someone must build the building. Thus, a potential limiting condition is related to people, more specifically to educators willing to do the hard work associated with building mathematical minds.

In the case of school mathematics, the "product" is optimal student learning as measured by state-mandated tests—the intended outcome. Hence, current law as reflected in NCLB is more consistent with another view of equality—the expansive view. Professor Crenshaw (1988) stated this second view of equality, the expansive vision, "stresses equality as a result and looks to real consequences" (p. 1341). The NCLB legislation calls for educators to reflect seriously on

student outcomes by demographic group. This expansive view suggests that equal treatment of students is not equitable if it leads to differential outcomes. This perspective conflicts with the worldview held by many educators that “equal treatment” of all students is optimal. If educators assume a “one-size-fits-all” approach to classroom practice, without careful reflection and planning for individual as well as collective student learning, the product is likely to be unequally distributed opportunities to learn and continued underperformance of traditionally underserved students. In the case of NCLB this “equal treatment” worldview and its associated ideological perspective—color blindness—are limiting conditions.

Many educators who view the “equal treatment” position as a well-meaning and fair perspective assume color blindness as a political or ideological stance. Part of the problem with color blindness is that it ignores students and their performance. Irvine (1990) stated, “by ignoring students’ most obvious physical characteristic, race, . . . teachers are also disregarding students’ unique cultural behaviors, beliefs, and perceptions—important factors that teachers should incorporate, not eliminate, in their instructional strategies” (p. 54). Today, the color-blind method of mathematics education creates barriers to true equality by erecting barriers in school mathematics such as

1. Persistent tracking,
2. Fewer opportunities for African American and Hispanic students to learn from the best qualified teachers,
3. Less access to technology, and
4. Cultural discontinuity between school mathematics and the family life of diverse student groups (National Science Board, 1991; Oakes, 1990; Piller, 1992; Stanic, 1991).

The first three of these barriers to equity, which are quantifiable, are considered acceptable indicators of unequal educational conditions. In contrast, this fourth barrier is subtle and difficult to identify and measure in everyday schooling. The matter of family life and school mathematics warrants additional discussion.

Less frequently, educators have explored the experiences of stakeholders other than teachers in the process of school mathematics reform (see Graue & Smith, 1996). This is a serious limiting condition. In particular, there is limited information available on how parents

- Perceive their children’s mathematics instruction,
- Interpret their children’s performance in light of mathematics standards and state testing, or
- View their role in the mathematics education process.

As Graue and Smith (1996) noted, despite parental presence in many aspects of educational reform rhetoric, researchers of school mathematics practice and design have shown little interest in parents.¹² Ethically, in the context of high-stakes testing and reformed practice, this is a condition that must be addressed by the focused efforts of scholars, school-based educators, and community-based organizations. Matters of ethics represent the final limiting conditions.

The pressures of high-stakes testing, public disclosure of testing results, and the real possibility of job loss or demotions have placed a new and heavy burden on professional educators. This burden will cause some to carefully design change strategies and create exciting learning environments for students. For others, the score-high mentality will create new ethical dilemmas. Reports have emerged of schools and districts removing large numbers of students from opportunities to test, rather than creating appropriate opportunities to learn. The ethical challenges are endless when the stakes are high and very real. As with good engineers, educators must factor ethics into their thinking, everyday planning, and ultimately their design strategy. Failure to do so will endanger the building process.

In pursuit of an engineering strategy in mathematics education, the importance of dedicated educators cannot be underestimated. Many educators are excellent, but some are not. The following questions highlight the traits of those ready to improve school mathematics:

- Does the educator listen to new ideas with an open mind?
- Does the educator consider a variety of solution methods associated with student learning before choosing a design approach?
- Is the educator content with determining a learning design on the basis of trial and error?
- Does the teacher use phrases such as, “I need to understand why students learn mathematics with this approach” and “Let’s consider all possibilities.”

If educators are eager to listen, open to a variety of educational solutions, never content with just trial-

¹² One exception is Family Math (www.lhs.berkeley.edu/equal/FMnetwork.htm).

and-error methods, and pressed to know why a method works with students, they represent the type of teachers and instructional leaders who can engineer changes in mathematics education. These kinds of educators are foundational for "Learning to Build."

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Appendix

Opportunity-to-Learn Building Blocks	Rating Low–High					Description/Areas to Be Addressed
	1	2	3	4	5	
1 Establishing Essential Structures—Organizational Context						
1.1 Set high standards						
1.1.1 Develop clear and rigorous goals for what students should know and be able to do						
1.1.2 Communicate expectations to teachers, administrators, parents, students, community						
1.1.3 Provide samples of student work that meet standards and communicate what is expected to all constituencies						
1.2 Ensure that all students have access to a challenging curriculum						
1.2.1 Implement school-based planning and accountability processes						
1.2.2 Emphasize consistent application of curricular programs and standards						
1.2.3 Eliminate watered-down courses and tracking						
1.2.4 Ensure that curriculum and assignments are aligned with standards						
1.2.5 Focus on core subjects: language arts, reading, mathematics, science, social studies						
1.3 Align standards with assessment and monitor progress						
1.3.1 Provide timely, usable data on student progress						
1.3.2 Encourage mid-course corrections based on data						
1.3.3 Recognize and reward positive results						
1.3.4 Disaggregate student achievement data and carefully monitor minority student data						
1.3.5 Train teachers in classroom strategies for assessing student learning to determine the impact of teaching and classroom activities						
1.4 Develop programs to address the impact of student/teacher mobility in low-performing schools to include						
1.4.1 Standardization of curriculum and materials						
1.4.2 Centralized storage of data						
1.4.3 Transmission of individual student data						
1.4.4 Staff training on impact of mobility and how to mitigate this factor (see Dept. of Defense school model)						
1.4.5 Recruitment and retention of quality teachers at low-performing schools						
1.4.6 Additional use of instructional/curriculum specialists						
1.5 Develop innovative strategies to						
1.5.1 Organize schools into smaller units—small schools have been shown to improve both gap and achievement levels						
1.5.2 Provide quality, sustained professional development and increase its availability of schools and through technology and other innovative approaches						
1.5.3 Encourage collaborative and team approaches						
1.5.4 Foster collegial networks among teachers, principals, schools						
1.5.5 Extend teacher and team planning time						

Opportunity-to-Learn Building Blocks	Rating Low–High					Description/Areas to Be Addressed
	1	2	3	4	5	
1.6 Provide needed policies and resources to support						
1.6.1 Ongoing, quality school site professional development						
1.6.2 Effective, timely student support programs						
1.6.3 Parent/community support initiatives						
1.6.4 Rigorous standards-based teaching and learning for all students						
1.6.5 Governance/organizational alignments necessary to ensure accountability, student progress, and quality teaching and learning for all students						
1.7 Foster research-based characteristics of high-performing schools						
1.7.1 Minimize bureaucracy						
1.7.2 Promote effective leadership at all levels						
1.7.3 Promote access and equity for all students						
1.7.4 Implement rigorous standards-based programs						
1.7.5 Provide significant planning time for teachers						
1.7.6 Orient learning around essential skills, concepts, and inquiry						
1.7.7 Foster parent involvement						
1.7.8 Develop a sense of team, collaboration, and family within the school system or school						
1.7.9 Develop strategies to use research on “bridging the gap” to inform decision-making and to influence the actions of legislators, teacher preparation institutions, educators, community leaders, and parents						
1.7.10 Accountability for quality classroom teaching						
2 Increasing Opportunities to Learn						
2.1 Establish a process for review of subject area achievement goals quarterly, to address needed changes in subject area and grade-level expectations, standards-based teaching/learning practices, and assessment techniques						
2.2 Foster high levels of teacher engagement in “bridging the gap” process through activities that promote involvement with						
2.2.1 School or school system goals for high achievement						
2.2.2 Identification of skills, practices, and knowledge for effective teaching and high achievement						
2.2.3 Teacher training in working with diverse learners						
2.3 Establish a process for monitoring classroom implementation of performance and content standards to include						
2.3.1 Depth of content coverage, extent and time allocated to key concepts, and emphasis on general principles and major concepts						
2.3.2 Quality of instructional delivery and teaching practices such as hands-on activities, use of technology, and student performance activities that include work products						

Opportunity-to-Learn Building Blocks	Rating Low-High					Description/Areas to Be Addressed
	1	2	3	4	5	
2.4 Foster implementation of new approaches to support meaningful ways to make schools/classrooms more compatible with the diverse home backgrounds of students to						
2.4.1 Reduce cultural isolation						
2.4.2 Identify individual student learning styles and build on strengths students bring						
2.4.3 Use modeling and interactive practices and scaffolding to guide students to complete complex tasks						
2.4.4 Enhance use of group learning techniques such as cooperative and team learning, peer tutoring, and reciprocal teaching						
3 Expanding Community Connections						
3.1 Foster increased and meaningful family involvement activities						
3.1.1 Provide specific training to families in tutoring and encouraging students to meet high standards						
3.1.2 Provide community and extended educational opportunities for families at schools, in projects, churches, and community centers						
3.1.3 Develop family resource centers						
3.1.4 Initiate or extend family participation opportunities to foster meaningful roles for families in support of school system/school goals and to encourage family involvement as a team effort in a supportive school environment						
3.1.5 Communicate importance of family support in establishing homework time, study skills, and academic standards in the home						
3.2 Establish necessary collaboration between schools and other community resources to provide needed assistance to						
3.2.1 Ensure availability of support mechanisms within each school						
3.2.2 Establish programs to forge connections between families and community organizations in support of student academic learning and rigorous standards						
3.2.3 Establish opportunities for meaningful proactive roles for business and industry in shaping and supporting rigorous standards at district and school-site levels						
3.2.4 Involve universities in "bridging the gap" initiatives that strengthen the K-16 community connections and programs						
4 Enhancing Teacher Quality and Practice						
4.1 Implement professional development programs that focus on standards-based learning aligned with assessment and with emphasis on practices that promote in-depth content knowledge, key concepts and general principles, real-life hands-on activities, application of technology, and modeling effective teaching strategies						
4.2 Implement quality teaching practices and training programs that utilize strategies for achieving high expectations for all and focus on the need to address and maximize individual students' learning styles						

Opportunity-to-Learn Building Blocks	Rating Low-High					Description/Areas to Be Addressed
	1	2	3	4	5	
4.3 Develop effective classroom assessment practices to enhance						
4.3.1 Use of data to make timely decisions 4.3.2 Use of a variety of classroom assessments including performance tasks and applications, surveys, interval testing, observations, standardized and criterion-referenced tests, student portfolios, teacher logs, course content, modes of instruction, time- allocated key concepts, student activities, etc.						
4.4 Implement teacher training programs that develop skills related to student behavior, especially						
4.4.1 Developing cultural sensitivity 4.4.2 Working with nonmotivated students 4.4.3 Developing effective classroom management skills						
4.5 Provide teachers with powerful, proven techniques successful in “bridging the gap” and in raising student achievement through						
4.5.1 Guiding students to create meaning and solve problems 4.5.2 Providing interactive and collaborative learning activities 4.5.3 Linking academics to community resources 4.5.4 Promoting action research in classrooms						
4.6 Establish personnel practices and programs that promote						
4.6.1 The need to invest in professional development, making available both time and resources 4.6.2 Availability of quality teaches at all schools and in all classrooms 4.6.3 Appropriately prepared and certified teachers 4.6.4 In-house teacher recruitment programs 4.6.5 Degree programs for work-force advancement 4.6.6 Collaborative programs with universities to support pre-service, in-service and community resource needs 4.6.7 Staff stability and quality in low performing schools and reduced teacher mobility 4.6.8 Assignment of high-performing teachers to neediest schools						
4.7 Communicate that research supports three essential characteristics of quality teaching						
4.7.1 In-depth knowledge of content 4.7.2 Effective teaching practices and pedagogy for all students 4.7.3 Higher levels of student achievement						
5 Creating Student Support Networks						
5.1 Create school environments that foster student resilience, provide timely needed academic, social, and psychological support mechanisms						
5.1.1 Develop clear and rigorous goals for what students should know and be able to do 5.1.2 Communicate expectations to teachers, administrators, parents, students, community						

Opportunity-to-Learn Building Blocks	Rating Low–High					Description/Areas to Be Addressed
	1	2	3	4	5	
5.1.3 Provide samples of student work that meet standards and communicate what is expected to all constituencies						
5.2 Extended availability of pre-school and early intervention programs to address needs of low-performing schools and school clusters in critical needs areas, K–12						
5.3 Implement innovative approaches for providing more time on task for students in low-performing schools/school clusters by introducing programs that provide						
5.3.1 Extended-school-day learning opportunities						
5.3.2 After-and before-school tutorial programs						
5.3.3 Saturday school						
5.3.4 Summer school enrichment programs						
5.3.5 Community/adult school extended-day programs						
5.3.6 Community college and university programs						
5.3.7 Longer school day approaches						
5.3.8 Expanded school year approaches						
5.3.9 Enrichment and mentoring programs						
5.3.10 Efficient and productive use of time						
5.4 Increase student support programs to meet needs of students related to increased standards in mathematics through						
5.4.1 Development of supplementary learning activities and programs to promote achievement of more rigorous standards						
5.4.2 Timely support for high-level advanced courses						
5.4.3 Peer tutoring and adult mentoring programs for at-risk students						
5.4.4 Collaborative programs established at universities, libraries, museums, community centers, and with business and industry to support and reinforce academic standards						
5.4.5 Programs that support and increase the number of minority students successfully completing honors/advance-placement courses						
5.5 Closely monitor the availability, enrollment, and content of student support programs to assure access and equity for all students, especially students from low-performing schools/clusters, and to ensure quality of support program content						
5.6 Focus on early identification of students who are at-risk of academic failure in the classroom and implement processes that promote achievement						
5.6.1 Develop early intervention programs for all levels and subjects K–12						
5.6.2 Increase emphasis on reading in early grades						
5.6.3 Implement literacy programs across the curriculum K–12						
5.6.4 Provide special programs for second-language students						
5.6.5 Establish school-linked services/resources to meet multi-ethnic population needs						

Note: The Engineering Change Assessment Instrument is a slightly modified version of the “Bridging the Gap” tool created by the McKenzie Group for urban and rural communities attempting to conduct systemic reform. The McKenzie tool provides a robust and comprehensive framework that can help inform the change process.

EDUCATIONAL POLICY RESEARCH AND MATHEMATICS EDUCATION

Joan Ferrini-Mundy and Robert E. Floden

MICHIGAN STATE UNIVERSITY

In this chapter we review and synthesize research in four areas of educational policy: K–12 mathematics standards and frameworks, K–12 mathematics instructional materials and curriculum, assessment, and the mathematics teaching profession. For each area, we will discuss selected studies that illustrate the formulation, status, implementation, and effects of educational policy that relate to mathematics education. Research on policy *formulation* describes the ways in which politics, practice, and the discipline of mathematics have shaped policy decisions. Studies on the *status* of policy examine and describe variation in policy, primarily at the state level. Research on policy *implementation* has investigated the ways that policies have been interpreted, introduced into practice, and modified as faculty, curriculum developers, and others carry them out. Research on the *effects* of policy looks at the impact policies have had on mathematics teaching and learning. Within each of these four policy areas we discuss both research that has been completed and research that is needed.

We undertook an extensive, though not exhaustive, literature search as the basis for this review. The studies reviewed are eclectic in their origin, with some conducted by those who see themselves primarily as mathematics educators, and others carried out by educational policy experts, assessment experts, or others who have found the domain of mathematics education and its intersection with educational policy a fertile ground for exploration. We hope that our review will encourage and guide further research,

both by mathematics educators and by those working in fields that bear on mathematics education.

THE REVIEW PROCESS

We began the review by selecting search criteria for identifying relevant literature and organizing the studies with a framework that allowed us to address, as appropriate, the four mathematics education policy areas and the four types of policy-related research. The search was based on the broad WordNet® definition of policy as “a plan of action adopted by an individual or social group” (Cognitive Science Laboratory, 2005). In education, such plans for action may be generated from a variety of sources. Examples of major federal policies include the Individuals with Disabilities Education Act (IDEA) of 1997 (U.S. Department of Education, 1997), or more recently, the No Child Left Behind Act (NCLB) of 2001 (U.S. Department of Education, 2002). States are also sources of education policy. For example, many states mandate teacher education and certification procedures or require that schools administer state-approved pupil assessments. States generate or approve instructional standards or curriculum frameworks. Many states and districts also set criteria for the selection of textbooks, uses of technology, and professional development activities for teachers.

With NCLB the federal government has begun to play a greater role in K–12 policy decisions and re-

lated decisions about teacher preparation that affect mathematics education specifically. The increasingly prominent role played by the federal government accompanies the highly visible role that has been played by national professional societies and organizations for the past two decades. In the late 1980s, for example, the National Council of Teachers of Mathematics (NCTM) embarked upon a pioneering effort to create content, teaching, and assessment standards in K–12 mathematics (see NCTM, 1989, 1991, 1995, 2000). This effort by NCTM led to policy recommendations, many of which were adopted or adapted by states and districts. In addition, NCTM's work was taken as a model for the development of standards in other school subjects and influenced a wide range of education policies and education research funding priorities. Hence policies created by professional organizations or national groups such as NCTM or Achieve are also legitimate foci for this review.

Additional sources of policy include funding programs offered by federal and state agencies and private foundations. The programs offered by the National Science Foundation (NSF), such as the Statewide Systemic Initiatives, or the Centers for Learning and Teaching, are also plans of action, and so are considered policies for consideration in our review.

In our literature search, we gave most attention to empirical studies or literature reviews that appeared in refereed sources; we gave less emphasis to the plethora of articles that offered rhetorical (often highly charged) commentary about policy. Empirical studies in mathematics education research have historically focused on core issues of teaching and learning, both in the United States and internationally. This focus is reflected in the first *Handbook of Research on Mathematics Teaching and Learning* (Grouws, 1992), in which 8 of 29 chapters focus on specific subject areas within mathematics. More recent surveys, including that of Sfard and her colleagues for the Tenth International Congress on Mathematics Education (Sfard, 2005) confirm that most mathematics education researchers address issues and problems they consider fundamental to the improvement of mathematics teaching and learning in classrooms, rather than in a broader policy arena. In recent years, this is evident in the predominance of small-scale studies using qualitative methodologies to focus deeply on specific problems about the teaching or learning of mathematics. This emphasis can be corroborated by a simple categorization of articles published in the *Journal for Research in Mathematics Education* over the past three years, where the research mode appearing in a majority of articles is qualitative.

The policies that are devised by local governments, the states, the federal government, professional associations, and funding agencies are critical to the practice of mathematics education. Fundamental research about mathematics teaching and learning alone stands little chance of influencing teaching and learning on a broad scale unless some mathematics education experts become deeply engaged in the policy arena and produce research about policy. Many crucial research questions fall within this domain and have major implications for the health and future of mathematics education. Such questions include:

- In a high accountability policy environment, what is the impact of high-stakes K–12 assessment on mathematics instruction?
- Do teacher certification policies support continued growth in teachers' development of subject matter knowledge for teaching?

These questions do not address what might currently be the most significant policy arena for mathematics education—policies related to subject matter standards at the local, state, and national level. (See Lester & Ferrini-Mundy, 2004, for a summary of a research-agenda-setting conference on this matter.) The need for research addressing policy issues suggests interesting opportunities for mathematics education researchers and for policy researchers who are willing to address questions in this arena.

POLICY AREA ONE: K–12 MATHEMATICS STANDARDS AND FRAMEWORKS

National organizations for mathematics education, especially NCTM, have worked to develop content standards over the past two decades. Prior to that time, despite periodic national reports and general recommendations about curriculum, there was little specific guidance at the national level for K–12 curriculum. International comparative research that has examined curriculum and standards internationally, such as the Third International Mathematics and Science Study (TIMSS), has highlighted the fact that the standards established by other countries are an important influence on mathematics teaching and achievement.

Standards have been a prominent policy instrument in the United States in the past two decades. In most cases, standards are designed to apply to particular categories that are important for policy. These categories include content, curriculum, performance, teaching, and opportunity to learn. Though the labels

have been inconsistently applied, basic descriptions of the categories are as follows: *Content* standards specify the topics that should be covered, often with links to grade levels or bands of grade levels (e.g., content for grades 4–6). *Curriculum* standards, sometimes called a curriculum *framework*, describe the general outlines of the intended curriculum. *Performance* standards specify what students are expected to be able to do. *Teaching* standards describe the instructional approaches teachers should use to help students learn mathematics. *Opportunity to learn* standards, which were discussed but never formally developed in mathematics at the national level, would describe the resources that should be devoted to supporting instruction. (For details see Darling-Hammond, 1994; National Council on Education Standards and Testing, 1992; Oakes, 1989; Smith & O'Day, 1991; Special Study Panel on Education Indicators, 1991.)

State policy makers, with the encouragement of academic policy analysts (e.g., Smith & O'Day, 1991), have seen the formulation of standards as a key step in setting the direction for K–12 mathematics education, both as a way of communicating with practitioners, and as a way to link local and state policy. The U.S. federal government explored the idea of setting official national standards via the Educational Summit in 1989 and the Goals 2000: Educate America Act in 1994 (U.S. Department of Education, 1994), and some policy analysts continue to advocate for national standards (see Ravitch, 2005), but thus far that responsibility has been left to the states.

As noted earlier, in discussing policy research in each area of K–12 mathematics education, we divide the body of the research into four types: research on formulation, status studies, studies of implementation, and studies on policy effects or impact. In the area of standards, research on *formulation* includes studies of the ways that government agencies and other organizations have decided to develop standards and gone about constructing standards and framework documents. *Status* studies provide documentation about the state of standards development or use at the state or local level. We think of studies of *implementation* as works that examine standards documents to determine whether they are coherent, clear, specific, and consistent with other policy documents including other standards or assessments. Some documents suggest that states intended to align their standards with NCTM standards, so studies that look at alignment with NCTM can also be considered implementation studies. Studies of *policy effects* or impact measure the degree to which standards or frameworks have effected a change in teaching practice or student achievement.

Research on the Formulation of K–12 Standards and Frameworks

Policy research can illuminate the ways standards have been developed by looking at the interplay of political, social, and disciplinary factors in the construction of the particular standards adopted by a state agency or national organization, and in the adoption of standards as a policy tool. In the area of standards, where values and beliefs about what should constitute curricular emphases are central, such research can also give glimpses into how values and beliefs influence the development of standards. In the past two decades, most policy research on the formulation of K–12 mathematics education standards has focused on two cases: the series of NCTM standards documents and the evolving California *Curriculum Frameworks and Standards*.

McLeod, Stake, Schappelle, Mellissinos, and Gierl (1996) offered a retrospective account of the development of the NCTM *Curriculum and Evaluation Standards for School Mathematics* (1989). They highlight the role that declining mathematics test scores on U.S. and international assessments played in propelling key actors in the government and in professional organizations toward the creation of systematic plans for improvement. NCTM, influenced in part by the belief that weaknesses in the U.S. curriculum accounted for the poor showing in the Second International Mathematics Study, chose to focus their policy formation efforts on the K–12 curriculum. The actual proposal for creating standards came as a result of discussion at a retreat sponsored by the Conference Board of the Mathematical Sciences (CBMS) focused on a 1983 National Science Board Report (NSB, 1983) that called on professional organizations to take responsibility for improving their fields. In the face of advice from federal policy makers against using the language of standards or of national standards, NCTM decided to design the standards, which were to serve as a guide for the development of higher quality curriculum, instruction, and classroom evaluation. The authors shared a commitment to moving beyond a predominance of “shopkeeper” arithmetic skills” (NCTM, 1989, p. 3) in the U.S. mathematics curriculum, toward an emphasis on problem solving and application.

The Reagan Administration reduced the NSF budget for K–12 mathematics and science education research shortly after the publication of *A Nation at Risk* (National Commission on Excellence in Education, 1983), in part as a result of controversy over federally-funded curriculum efforts (Dow, 1991). In this context, NCTM worked without the aid of government funding programs to produce the 1989 *Curriculum*

lum and Evaluation Standards. The NCTM effort drew upon English-language publications and information produced throughout the international community. Studies from England, the Netherlands, and Australia figured prominently in the conceptualization of the *Standards*. Also prominent was research underpinned by social constructivist views of learning. Concerns about how standards would be received in the U.S. context of local control of education and teacher autonomy were prominent in the development of the document.

Accounts of the development of *Principles and Standards for School Mathematics* (NCTM, 2000) are provided by Ferrini-Mundy and Martin (2003) and Martin and Berk (2001). Drawing on Silver (1990), Ferrini-Mundy and Martin argue that the development of *Principles and Standards* was influenced by theoretical perspectives from research, methodologies from research, and findings from research. They describe the process that NCTM implemented for obtaining input from the field about early drafts of the document, synthesizing and using the results. Martin and Berk (2001) provide an account of the cyclical relationship between research and standards development.

The research that is available about the formulation of NCTM standards documents suggests that the national and local policy context, the availability of research and theoretical perspectives, and the shared commitment by those involved in the development toward a meaningful mathematics education for all students all had impact on the content and form of the documents.

Wilson's research (2003) on the development of the California Mathematics Frameworks highlights the effects of national and state conflicts about the content of K–12 mathematics (also known as the “Math Wars”). That context led to the development of a standards document quite different from the NCTM Standards. Her historical account is based on document analysis and interviews with most of the major figures in the development. It shows how the early version of the Framework was constructed primarily by leaders in the California mathematics education community, working from contemporary ideas in that community about the nature of mathematics, what aspects of mathematics were most important for K–12 pupils, and what approaches to instruction fit best with the intended mathematical learning.

She goes on to describe how the goals and instructional approaches became the subject of controversy, with some research mathematicians joining conservative political actors in criticizing the 1998 Framework and eventually replacing the initial set of mathematics educators as the leaders in the development of later

versions of the Framework. All the key players in this California story accepted the idea that standards, both the Curriculum Framework and later the standards used to guide development of K–12 state assessments, were an important device for influencing mathematics education. The story Wilson tells reveals the ways in which the particular content of the standards was influenced both by content experts (i.e., mathematics educators and research mathematicians) and by others who saw the content as a symbol for broader political issues. The California story is not unique. Similar issues and events play out in other domains where a particular policy is likely to have a high impact and a far reach, especially when the policy calls for a radical departure from the current state of affairs.

As described by Martin and Berk (2001), accountability concerns—in the form of expectations for proven or evidence-based standards and curricula—led to serious questions about the research basis for the 1989 NCTM *Curriculum and Evaluation Standards*. (See Wu, 1998.) In their review, Martin and Berk describe the many ways in which the 1989 Standards generated a flurry of research on the applicability, adaptability, and impact of NCTM's curriculum policy. In addition to the multitude of studies that focused on local efforts to understand and implement the Standards and standards-based curricula, 49 states had developed their own mathematics curriculum standards, claiming that these were based in some sense on NCTM's standards, by the mid-nineties (cf. Council of Chief State School Officers [CCSSO], 1995).

Research on the Status of K–12 Standards and Frameworks

Various types of policy analyses are conducted to summarize the status of standards development by state. Periodic reports by the Council of Chief State School Officers (see CCSSO 1997, 1998, 2001, 2002, 2005) provide current summary information by state about such things as the “status of state content standards in academic subjects.” Similarly, the annual *Quality Counts* reports (Editorial Projects in Education, 2004, 2005) provide information about the status of K–12 standards, frameworks, and curricular expectations, including “grades” by state on standards and accountability. Other organizations also assign “grades” to state standards, according to various frameworks and perspectives developed by organization. Aside from specific details about where individual states are in their standards efforts, the general findings from these types of status reports are that there is great diversity in state standards, and that judgments about their quality are

highly sensitive to the criteria that groups or individuals establish for judging them.

The American Federation of Teachers (AFT) has published reports about that state of state standards (see AFT, 2001, 2003). In their 2001 report, AFT concludes that, although many states were making advances in creating standards, the progress on standards-based reform was being hindered because the assessments were not necessarily aligned with standards.

Another type of status study of standards that has figured prominently in public policy discussions is international work such as that undertaken as part of the 1995 Third International Mathematics and Science Study (TIMSS). William Schmidt and colleagues conducted analyses of the K–12 mathematics standards and textbooks for students in the TIMSS focal grades—grades 4, 8, and 12—from nearly 50 countries (Schmidt, McKnight, Valverde, Houang, & Wiley, 1997). In particular, mathematics standards and textbooks from the U.S. were compared to those from countries whose students performed at the highest levels on the TIMSS eighth-grade assessment. These comparisons revealed that standards and textbooks in the high-achieving countries reflected a kind of curricular coherence not present in U.S. standards and curricula (Schmidt, McKnight, & Raizen, 1997). In particular, Schmidt et al. claim that the mathematics standards and curricula in the high-achieving countries are organized according to the structure of the discipline of mathematics (Schmidt, Wang, & McKnight, 2005; Valverde & Schmidt, 2000). The analysis is summarized in Figure 29.1.

This figure reveals what these high-achieving countries collectively intended to be taught at each grade level as reflected in their national mathematics standards documents and analyzed using the TIMSS Curriculum Framework and document analysis technique (Robitaille et al., 1993; Survey of Mathematics and Science Opportunities [SMSO], 1991, 1995). An analysis of the 1989 NCTM *Curriculum and Evaluation Standards for School Mathematics* compared it with the composite curriculum from the high-achieving TIMSS countries at grade eight (Schmidt, Wang, & McKnight, 2005), and found (given the grade-band structure of the NCTM document) that coverage of all but seven of the 32 topics that are intended in the top-achieving TIMSS countries is recommended for all of grades one through eight. The authors note that in general, based on their analyses of national and state standards in mathematics and science, “Topics enter and linger, so that each grade typically devotes instructional attention to many more topics than is typical of the six high-achieving countries; in addition, each topic stays in the curriculum for more grades than in the high-achieving countries” (p. 525).

Research on the Implementation of K–12 Standards and Frameworks

One type of implementation study is the analysis of state and local standards documents, with attention to clarity, specificity, alignment, and mathematical appropriateness; these features that are contributors to interpretation and implementation. Meeting these criteria is viewed by some as important for ensuring that the policy—that is, standards—have a significant influence on practice.

The literature we identified indicates that the various state standards for K–12 mathematics are highly variable in these features of implementation. Klein et al., (2005), for example, graded the standards on clarity, definiteness, and testability. States receiving the highest overall grades were California, Indiana, and Massachusetts (Grade A) and Alabama, New Mexico, Georgia (Grade B). The national average was D and 29 states received a grade of D or F. In discussing the reasons for the poor grades, the reviewers note such factors as: too much emphasis on calculators, inadequate emphasis on basic number facts and standard algorithms, and insufficient developmental guidance for mathematical reasoning and problem solving. This analysis is the third sponsored by the Fordham Foundation (see also Braden et al., 2000; Raimi & Braden, 1998).

Ongoing analyses of state standards are underway at the NSF-funded Center for the Study of Mathematics Curriculum (CSMC) (see Reys et al., 2006). This status study, rather than grading state standards on specific quality criteria, instead is descriptive. The project was intended to “describe the emphasis and grade-level placement of particular learning expectations as presented in state grade level expectation documents and to document variations across states.” (Reys et al., 2006, p. 1). They report a general lack of consensus in what is included at particular grade levels across the states. In an effort sponsored jointly by NCTM and the Association of State Supervisors of Mathematics (Jackson, 2004), experts reviewed state mathematics standards to examine whether or not there is a “de-facto national curriculum for school mathematics” (p. 1355). In the report of this work, Lott and Nishimura (2005) provide some commentary about what reviewers found to be characteristics of the least and most informative state standards. Standards that used finer grain sizes, used clear language, gave examples, and focused on “big ideas” were seen as more informative, and standards that used imprecise mathematical language, used broad or vague language, were too wordy, or used multiple objectives in single statements were less informative (p. 63). Both of these policy analysis efforts are focused on similarities and differences in standards across states,

Topic	Grade							
	1	2	3	4	5	6	7	8
Whole Number: Meaning	■	■	■	●	●			
Whole Number: Operations	■	■	■	■	●			
Measurement Units	▲	■	■	■	■	■	●	
Common Fractions			▲	■	■	●		
Equations & Formulas			▲	●	●	●	■	■
Data Representation & Analysis			▲	▲	●	●		▲
2-D Geometry: Basics			▲	●	●	●	■	■
2-D Geometry: Polygons & Circles				●	●	●	■	■
Measurement: Perimeter, Area & Volume				●	●	●	●	▲
Rounding & Significant Figures				●	●			
Estimating Computations				●	●	●		
Whole Numbers: Properties of Operations				▲	●			
Estimating Quantity & Size				▲	▲			
Decimal Fractions				●	■	●		
Relation of Common & Decimal Fractions				●	■	●		
Properties of Common & Decimal Fractions					●	●		
Percentages					●	●		
Proportionality Concepts					●	●	●	▲
Proportionality Problems					●	●	■	■
2-D Geometry: Coordinate Geometry					▲	▲	●	●
Geometry: Transformations						●	●	●
Negative Numbers, Integers, & Their Properties						▲	●	
Number Theory							●	▲
Exponents, Roots & Radicals							●	●
Exponents & Orders of Magnitude							▲	▲
Measurement: Estimation & Errors							▲	
Constructions Using Straightedge & Compass							■	▲
3-D Geometry							●	■
Geometry: Congruence & Similarity								■
Rational Numbers & Their Properties								▲
Patterns, Relations & Functions								▲
Proportionality: Slope & Trigonometry								▲

▲ Intended by 4 of 6
 ● Intended by all but one (5 out of 6)
 ■ Intended by all top-achieving countries

Figure 29.1 Mathematics topic coverage in high-achieving countries in TIMSS 1995. Adapted from Schmidt, Wang, and McKnight (2005).

in pursuit of the question of whether there is so much consistency that there is, in fact, a “standard” national curriculum. Both found that there is wide variation and disparity in what individual states require in mathematics at particular grades.

Another class of studies about implementation of standards includes efforts to examine how teachers used standards in their classroom practice. In addition, we have selected some studies that discuss teachers’ interpretation of the meaning of standards, given the relationship of interpretation to implementation,

as well research looking at more general responses from teachers and administrators to the overall standards phenomenon.

Cohen and Ball (1990) examined early implementation of the 1985 California Mathematics Framework, which shared many features with the 1989 NCTM Standards. They found that despite efforts by the California Department of Education to provide teachers only those textbooks that were aligned to the standards and to create state assessments that were also aligned, teachers continued to view mathematics in

traditional ways and struggled to interpret and implement the standards in their classrooms. They tended to interpret the policy as suggestions for topics that they should add to the list rather than as elements useful for various types of mathematical reasoning. (Also see Ferrini-Mundy & Johnson, 1994.) Teachers tended to filter the new policy through the lens of established practice rather than to respond to the new policy by altering their lens.

This notion of filtering the policy through a particular lens is closely related to the matter of interpretation of standards. Researchers in the NCTM Recognizing and Recording Reform in Mathematics Education study (see Ferrini-Mundy & Schram, 1997) found that for teachers, implementation of standards often meant attending only to the surface features of the documents, particularly to advice in the standards that was perceived to be pedagogical, rather than to the mathematical shifts and emphases recommended by NCTM standards. In a study of teachers interpreting state standards in Connecticut, Hill (2001) found that teachers were not familiar with the technical terms that standards writers used in their policy documents, concluding that: “teachers reading these documents imputed more local, and sometimes conventional, definitions to these words. As a result, state standards lost their force.” (p. 289).

Similarly, Wilson and Floden (2001) found that standards-based reform meant different things to different teachers and administrators. For some teachers, the standards-based reform was hardly noticeable, “flowing into a large stream of other reforms.” (p. 213). Other teachers found that standards-based reform provided clarity and language for thinking about their practice. Some teachers who have a history of improving professional practice on their own viewed the standards as stifling. Given the varieties of standards-based reform that teachers encountered, the researchers consistently found teachers and administrators seeking to take a “balanced” approach, which mostly meant a commitment to mixing old with new, and using their professional judgment.

Policy research often finds that implementation is undercut by alignment failure (Webb, 1997). In some cases, such alignment failures are due to assessments that are inconsistent with alignments in curriculum and instruction. In other cases failures are due to deviations in instruction that undermine alignments in curriculum and assessment. In many cases implementation studies find misalignments (of varying degrees of severity) between all three areas: curriculum, instruction, and assessment (La Marca, 2001).

Webb (1999) analyzed mathematics and science standards in four states for their alignment with as-

sessments in those states, using the following criteria: categorical concurrence, range of knowledge concurrence, balance of representation, and depth of knowledge consistency. Although his purpose was primarily to develop and refine methodology for such alignment analysis, he found that the standards and assessments were not consistently well-aligned, signaling that one of the central theoretical assumptions about the standards movement—that is, alignment—is not necessarily in place in implementation. With NCLB requiring this alignment, future status and implementation studies may reveal different trends.

Research on implementation has led to an increased understanding of the role alignment plays in successful reform (Porter, 2002; Smith & O'Day, 1991) and meaningful interpretation of assessment results (Messick, 1989). A knowledge brief produced by WestEd (Ananda, 2003) discusses issues of alignment in the context of NCLB and the attendant emphasis on assessment results as the basis for rewards and sanctions for schools, teachers, and students. The NCLB policy on alignment indicates that “states may select and design their assessment of their own choosing” provided that state assessments are aligned with state academic (i.e., performance) standards and subject-area content standards for each grade level. According to Ananda, the “myriad assessment-and-accountability provisions are pushing states and districts to address many alignment issues that have often received short shrift” (p. 1).

Research on the Effects of K–12 Standards and Frameworks

To think about the effects of the policy tool of K–12 mathematics standards, it is helpful to consider the framework for understanding the influence of standards put forth in the National Research Council (NRC, 2002a) report on that topic. This report notes that standards may influence K–12 practice through three channels: curriculum, teacher development, and assessment and accountability. With respect to curriculum, standards may shape the development of curriculum materials and documents (e.g., district scope and sequence charts), which then influence classroom instruction. With respect to teacher development, standards may be used as a focus for professional development, which may affect teachers' learning by persuading teachers to change their ideas about what is important to teach, or by motivating teachers to pursue opportunities to learn about mathematics content or pedagogy. By leading to changes in teachers' knowledge, skills, and dispositions, standards may thereby have an influence on K–12 teaching and learning. With respect to assess-

ment and accountability, standards may be a basis for such systems, thus giving teachers, students, and others reasons for classroom practice changes that would lead to high performance on the assessments.

Policy studies sometimes focus on a subset of these channels. At other times they look only at the effects on teaching and learning, without trying to understand the channel through which the influence has come. One line of work has focused on standards as part of systemic reform, where assessments and accountability are prominent features but curriculum and teacher development also play a part. Another line of work has focused on the teacher development channel, looking at the ways in which teachers interpret standards. Some investigators have done systematic comparisons of teaching practices with the operating set of standards. The correspondence to standards, or lack thereof, is seen as an indication of the influence of standards.

Central themes that emerged from these studies are that in the early stages of implementing standards, the absence of existing instructional materials that embodied the ideas of the standards led many teachers to adopt superficial aspects of the reforms (Ferrini-Mundy & Johnson, 1997). Teachers may believe they have changed their practice to fit a set of standards, and may talk the language of the standards, without making changes that authors of the standards saw as most important (Spillane & Zeuli, 1999). Those who are most likely to make the changes to practice that are expected by the policy are teachers who have an opportunity to collaborate with colleagues in their efforts to understand the expected changes and enact them in their classrooms (Ferrini-Mundy & Johnson, 1997). The effects of policy are also greater when teachers have deep conceptual understanding of the mathematics they are expected to teach (Lloyd & Wilson, 1998) or participate in professional development activities that promote the development of such understanding (Spillane & Zeuli, 1999). Wilson and Floden (2001) found that, despite the fact that, for some teachers, the clarity of standards and aligned accountability systems proved a catalyst for creating a more coherent practice, most teaching remained more familiar than new, and more ordinary than challenging. Recent work by Jacobs et al. (2006), in their analysis of TIMSS 1999 video study, indicates that “typical mathematics teaching (in the U.S.), in both 1995 and 1999, is more like the kind of traditional teaching reported for most of the past century.” (p. 6).

Observations: Research about K–12 Mathematics Standards and Frameworks

We found very little empirical work that has documented and described the processes of standards de-

velopment and formulation. Given the range of values and commitments that the diverse constituencies (mathematics educators, teachers, mathematicians, administrators) who develop standards bring to the task, and the complexity and differences of viewpoint reported anecdotally by many who have worked on state and national standards, additional research about how standards are created might be useful in assisting new groups coming to the work of creating state, district, or national standards and frameworks. There is much descriptive material available about the status of standards and frameworks, and a trend toward more analytic examinations of state documents. Again, in that work we see the strong influence of differing values and priorities in the choices of criteria for evaluation. Research about implementation and effects of the policies embodied in standards and frameworks is relatively sparse, and does not provide compelling evidence about the impact of standards in mathematics on teachers’ classroom practice. The complexity of undertaking research about implementation and effects is acknowledged in the NRC report about studying the effects of standards (NRC, 2002a), and the framework provided in that document may be a useful tool to guide future efforts in this area.

POLICY AREA TWO: K–12 MATHEMATICS INSTRUCTIONAL MATERIALS AND CURRICULUM

We discuss three areas of policy that relate to instructional materials and curriculum in K–12 mathematics: federally-funded programs to develop curriculum and instructional materials; policies that govern the selection of curriculum materials in states, districts, and schools; and policies about course-taking and curricular opportunities to learn mathematics in schools.

The NSF has played a leading role in supporting the development of mathematics instructional materials through its funding policies, beginning in 1952 with its support of the University of Illinois Committee on School Mathematics. In the “new math” era alone, spanning the period from 1957–1970, NSF invested more than \$500 million in improvement projects for mathematics and science education. Between 1989 and 2002, the NSF invested \$93 million to fund the development of 13 comprehensive mathematics curricula (NRC, 2004, p. 1). Three of these curricula were at the elementary school level, five were at the middle school level, and five were at the secondary school level (NRC, 2004, pp. 19–20). Between 1995

and 2000, several of these major projects received another round of funding to create second editions.

The selection of instructional materials by school districts is governed by a mix of policies. Both at the federal and state level, policies have included the requirement that curricula or instructional materials used in school districts seeking federal funding should be research-based. In some states, such as California, this requirement was present as early as 1999 (cf. <http://www.signetwork.org/SIG2000/California.htm>). Title IX (Section 9101) of NCLB (NCLB, 2001, Title IX, Sec. 9101, No. 37) see <http://www.ed.gov/policy/elsec/leg/esea02/pg107.html#sec9101>) requires the use of materials that are scientifically based, and defines scientifically based research as “research that involves the application of rigorous, systematic, and objective procedures to obtain reliable and valid knowledge relevant to education activities and programs.” In order to receive funding from the U.S. Department of Education’s Comprehensive School Reform (CSR) program for 2002 and beyond, schools must implement a program that “uses proven strategies and methods for learning, teaching, and school management based on scientifically based research and effective practices, and used successfully in multiple schools . . . and that has been found through scientifically based research to significantly improve student academic achievement, or has shown strong evidence that it will.” (<http://www.ed.gov/programs/compreform/legislation.html>).

New policy relevant to instructional materials in mathematics could emerge from the efforts of the recently appointed National Mathematics Panel. The Panel has been charged to “advise the President and the Secretary of Education on the best use of scientifically based research to advance the teaching and learning of mathematics . . . examine and summarize the scientific evidence related to the teaching and learning of mathematics, with a specific focus on preparation for and success in learning algebra . . . issue two reports containing policy recommendations on how to improve mathematics achievement for all students” (<http://www.ed.gov/about/bdscomm/list/mathpanel/factsheet.html>).

In addition to policies at the federal level, a number of states have specific policies about selection of instructional materials (see Seeley, 2003, for a historical summary of mathematics textbook adoption in the U.S.).

Finally, mathematics course-taking policies, such as requiring all students to take algebra in the eighth grade, requiring that algebra two must be completed

for high school graduation, or requiring all students to take four years of mathematics in high school, are now common in states and school districts.

The Advanced Placement (AP) Program, developed in 1955 by the College Board, is one specific area of the high school curriculum that is governed by a national policy-like process. Topic outlines for AP courses are generated largely by surveying colleges and universities. AP course outlines (and assessments) are designed to mirror the content of typical college-level introductory courses. The courses are intended to be an acceptable substitute for the introductory courses offered at more than 2,000 colleges and universities (see <http://apcentral.collegeboard.com/program/0,,150-0-0-0,00.html>, and National Research Council, 2002a).

There has been some policy support for the AP program in the form of U.S. Department of Education funding—the *Advanced Placement Incentive (API) Program* (<http://www.ed.gov/programs/apincent/awards.html>). These grants are intended to help schools comply with the mandates of the Elementary and Secondary Education Act—specifically the “Access to High Standards Act” within NCLB (see <http://www.ed.gov/policy/elsec/leg/esea02/pg14.html> for details). API grants support initiatives to promote pre-AP and AP programs for low-income middle and high school students. Since 2002, the USDE has awarded API program grants to 14 state departments of education as well as to local districts in 10 other states and one U.S. territory. While the California, Texas, and New York Departments of Education have not received API grants, multiple districts within these states and several consortia that include these and other states have also received grants.¹

Research on the Formulation of K–12 Instructional Materials and Curriculum Policies

The formulation of NSF policy to develop K–12 curriculum materials in the post-Sputnik era has been well documented (e.g., Dow, 1991; Lappan & Wanko, 2003). The idea that newer, more modern instructional materials would be crucial to maintaining the nation’s scientific and technological superiority was promulgated by leading mathematics educators such as Edward Begle and Robert Davis, and groups of mathematicians and scientists were at the forefront in the development of the materials. Less has been written about the process of formulating the NSF plans to

¹ The U.S. Department of Education’s full list of *Advanced Placement Incentive (API) Program* grantees is accessible at <http://www.ed.gov/programs/apincent/awards.html>.

fund comprehensive curricula at all three grade levels in the late eighties and early nineties. Language from one of these solicitations indicates the agency's commitment to advancing the standards movement as it had been defined by NCTM:

Because of the focus on problems of mathematics education in our schools, educators have been working toward a plan for improvement. Coming from a long process of deliberation and consensus, the *NCTM Curriculum and Evaluation Standards* and *Everybody Counts* document first steps toward this plan. These and other recent documents address issues such as broadening the content of the current mathematics curriculum, shifting emphasis from computational to problem solving skills, and recognizing the cognitive factors which enter into teaching and learning. They provide a sound philosophical basis for changes in content, method, and assessment. They signal high current interest in mathematics education providing both a foundation and an opportunity for improvement. (NSF, 1989, p. 4)

We were unable to locate any scholarly accounts of the processes of formulating "approved textbook" lists in various states, or about the formulation of K–12 course and credit policies.

Research on the Status of K–12 Instructional Materials and Curriculum Policies

We are aware of one study summarizing funding policies for K–12 instructional materials development. A review of the mathematics education portfolio in NSF's Education and Human Resources directorate (Ferrini-Mundy, Earle, & King, 2005; NSF, 2004) found an imbalance in the cycle of innovation, where there was little evidence of cumulativeness across instructional materials projects, research about learning, and implementation in mathematics education projects resulting from solicitations issued between 1990 and 2003. Of the 133 separate solicitations issued by the Directorate in that period, 10 were for instructional materials development, and the investment in materials development represented only about 5% of the total spending on mathematics education during that period.

Efforts to gauge the quality of instructional materials developed under federal policies can be seen as status studies. The U.S. Department of Education engaged in a process to determine "exemplary" instructional materials (U.S. Department of Education, 1999), and identified five of the NSF-funded materials

as exemplary. These were: *Cognitive Tutor Algebra*, *College Preparatory Mathematics*, *Connected Mathematics*, *Core-Plus Mathematics Project*, and the *Interactive Mathematics Program*. This process resulted in considerable controversy, to be discussed subsequently.

Finn and Ravitch (2004, p. 3) have provided some status research on textbook selection and adoption policies. They report that 21 states across the U.S. have implemented state-wide textbook adoption policies that govern the selection and adoption of textbooks by schools and districts.² Among these, 10 states choose textbooks from which districts must select, 7 states provide an approved list of textbooks from which districts can choose, and 3 states use a combination of these strategies that involves mandating texts for certain grade levels or courses but allowing districts to choose from an adoption list in other grades and courses. Similar textbook adoption policies exist at the district level in many cases where there is no state-wide policy. While the specific procedures for selecting texts for adoption or adoptions lists vary, most districts adopt textbooks on a five to seven year cycle and the process generally includes the use of some kind of committee review process. Some states, such as Texas, generate a list of non-conforming textbooks that districts and schools are expected to avoid. According to Finn and Ravitch (2004) and several other sources, the majority of texts that receive approval are published by four corporations: Pearson, Houghton Mifflin/McDougal Littell, Harcourt/Holt, and McGraw Hill, reflecting recent consolidation of publishing companies into a small set.

Status research is available concerning K–12 curricular and opportunity to learn policies. This includes policy recommendations for mathematics course-taking. NCTM recommends that all students study four years of mathematics in high school (NCTM, 2000, p. 371), increased from a recommendation of three years in the 1989 Standards (NCTM, 1989, p. 124). In 42 states and the District of Columbia, course requirements for high school graduation are set by officials at the state level. In the remaining states, this responsibility is delegated to local school boards.

A number of national reports also have argued for a four-year requirement, with specific advice about the nature of the mathematics that students should study. For instance, Achieve, Inc. recommends that state policy makers "require all students to take a common college and work-preparatory curriculum in math . . . align academic standards in high school with the knowledge and skills required for college and workplace success . . . pursue accelerated options for

² For maps of adoption states see the AAP web site and the Fordham Institute site at <http://www.publishers.org/school/schoolarticle.cfm?SchoolArticleID=32> and <http://www.edexcellence.net/institute/publication/publication.cfm?id=335>, respectively.

earning postsecondary credit while in high school; and monitor results through postsecondary education and use the data to help improve the rigor of course offerings and instruction in high school.” (Achieve, 2004b, p. 3). Increasingly, states and districts are requiring high school exit examinations, and these exams typically include mathematics as a core area.

Achieve (2004b) documented the number and types of mathematics courses that students are required to complete for graduation. There are 13 states that require only two years of mathematics study. Twenty-four states and the District of Columbia require three years, and only five states (Alabama, Arkansas, Mississippi, South Carolina and West Virginia) require all students to complete four mathematics courses for graduation. Nearly half of the states (22) do not specify which mathematics courses students need to take. Of those that do, only Arkansas, Indiana and Texas now or soon will require Algebra I, Geometry, and Algebra II (Achieve, 2004b, pp. 11–12).

Descriptive data about the status of the AP program is made available regularly by the College Board. Originally, the program served only top students from a few high schools. As of 2002, approximately 62 percent of U.S. high schools offered AP courses (NRC, 2002b). Among the eight separate AP courses in mathematics subjects, the most prevalent and popular courses are Calculus AB (first semester college equivalent), Calculus BC (second semester college equivalent), and Statistics. According to the College Board’s summary report on nation-wide AP exam participation, nearly 300,000 eleventh and twelfth graders took one or more exams in a mathematics subject in 2004 (http://www.collegeboard.com/student/testing/ap/exgrd_sum/2004.html).

Research on the Implementation of K-12 Instructional Materials and Curriculum Policies

There has been some research about the implementation of the NSF instructional materials funding policy and its products in the past two decades. Most of these analyses are alignment studies comparing the instructional materials developed under the policy to various frameworks that are either mentioned explicitly in the solicitations or that are of interest in the context of the policy. For example, the American Association for the Advancement of Science (AAAS) Project 2061 (<http://www.project2061.org/>) analyzed middle grades mathematics materials (both NSF-funded and commercial) with respect to the AAAS Benchmarks for Science Literacy (AAAS, 1993). The systematic alignment study process that they implemented involved

examining the materials with respect to six benchmarks: “a concept benchmark dealing with fractions and operations on them; a skill benchmark dealing with equivalent forms of numbers; a concept and a skill benchmark dealing with properties of shapes and computations of circumference, area, and volume; and two concept benchmarks dealing with graphing and equations.” (AAAS, 2000, <http://www.project2061.org/publications/textbook/mgmth/report/part1.htm>). The process also involves looking at how instructional issues are represented in the materials. AAAS concludes “It is clear that reform principles for mathematics education are beginning to have an impact on textbooks and on the developers and publishers who create them.” (AAAS, 2000, <http://www.project2061.org/publications/textbook/mgmth/report/part1b.htm>).

One contribution of such research and evaluation processes has been the design of tools and procedures for measuring alignment, which proves to be a rather difficult process. In TIMSS, Schmidt and his colleagues developed tools and methodologies for comparing instructional materials from the U.S. with those used in other countries. (see Schmidt et al., 1997). In an analysis of several of the NSF-supported mathematics materials, Schmidt et al. (2001) claimed that in many cases these materials lacked the rigor and high standards found in instructional materials from high-achieving nations in TIMSS, and that they also perpetuated the U.S. “mile wide, inch deep” problem of including too many topics in too much of a spiral curriculum. Schmidt argues that a coherent mathematics curriculum should follow the logical structure of the discipline, and that topics should be started and completed, rather than revisited without increase in depth over many years in the school curriculum.

We located systematic accounts by authors of their work in designing instructional materials and consider these descriptions of the implementation of NSF instructional materials policy. Some studies are reported in a special issue of the *American Journal of Education*. Alper, Fendel, Fraser, and Resek (1997) discuss the principles that served as the foundation for their development of the *Interactive Mathematics Program*, one of the earliest of the NSF-funded high school curricula. These principles were: “Students must feel at home in the curriculum; Students must be personally validated as they learn; Students must be actively involved in their learning; Students need a reason for doing problems” (pp. 150–151). They then provide illustrations from their curriculum materials of how each of these principles was instantiated. Usiskin (1997) provides an historical account of the role of application in school mathematics curriculum, including explanation of the ways that applications are

handled in the University of Chicago School Mathematics Project, a program developed simultaneously with the NSF-funded materials. He notes that goals for what students should understand about applications are made as explicit about goals for what students should know about skills. Accounts such as these that provide insights into how developers of instructional materials chose particular ideas from standards documents to frame their work, and are useful as an indicator of how policy goals are interpreted.

There is scholarly activity that relates to the implementation of policies about requirements of using research-based curricula. As states and school districts struggle to find out if the instructional materials they wish to use are research based, and/or have been validated as being effective through research, some resources are emerging. The What Works Clearinghouse (WWC) (<http://www.whatworks.ed.gov/whoware/overview.html>) was established in 2002 by the U.S. Department of Education's Institute of Education Sciences to provide educators, policymakers, researchers, and the public with a central and trusted source of scientific evidence of what works in education. The WWC aims to:

promote informed education decision making through a set of easily accessible databases and user-friendly reports that provide education consumers with high-quality reviews of the effectiveness of replicable educational interventions (programs, products, practices, and policies) that intend to improve student outcomes. (What Works Clearinghouse, What We Do, Overview)

In fall 2004, the WWC released a report on middle grades mathematics materials, which cited only ten studies that met their standards for inclusion. These studies were randomized controlled trials or studies that used quasi-experimental designs. Based on these studies, the report concluded that only five sets of instructional materials had been studied with research designs that met the WWC standards, and the findings about their effectiveness were equivocal.

The only research we located describing the implementation of state textbook adoption policies is focused on the process in California.³ Wilson (2003) reports on a highly politicized and controversial process around California's textbook adoption policy in the 1990s, pitting teachers who supported the reforms against state officials concerned with basic skills and procedures, all surrounded by suspicions about undue influence of the commercial publishing industry. Jacob and Akers (2000) provide an account of the California mathematics educa-

tion policy process during the 1995–1998 period, where a Curriculum Commission, serving in an advisory role to the California Board of Education, chose to enforce a part of the state education code that called for instructional materials to “incorporate principles of instruction reflective of current and confirmed research” (see Jacob & Akers, 2000, p. 4). They argue that this stipulation eliminated a number of “reformist” (NSF-supported) curricula from the approved list.

There is a more robust body of research describing implementation of course-taking and opportunity-to-learn policies. In a recent Council of Chief State School Officers (CCSSO) report (Blank & Langesen, 2005), some data are included regarding algebra in grade eight. NAEP data from 2003 indicate that about 27% of eighth graders across the U.S. report studying algebra in eighth grade, up only slightly from 1996. They also report on course-taking patterns in high school:

As of 2004, 50 percent of U.S. high school graduates took Four Years of high school math. Rates of high school math enrollment vary widely by state. Data as of 2003-04 school year show that nine states had over 50 percent of high school students take Trigonometry or Pre-Calculus by their graduation, which indicates the percent of students taking four years of high school math (with Algebra 1 the first year). In the U.S., 72 percent of graduates took Three Years of high school math. As of the 2003-04 school year, 17 states had over 70 percent of students take Algebra 2 or Integrated Math 3 by graduation, and seven states had over 80 percent at this level. (p. 3). In addition, “In 2003, 27 percent of grade 8 students took Algebra 1 for high school credit, which represents an increase of 3 percentage points from the rate in 1996. In 9 states over 30 percent of students reported taking Algebra 1 in grade 8 as of 2003; and a majority of states had over 25 percent taking Algebra. (p. 17)

Data such as these indicate that policy recommendation calling for higher expectations and earlier introductions of algebra are being implemented.

In another study of high school mathematics course-taking patterns, Kher, Schmidt, Houang, and Wiley (2005) found that the number of routes through high school mathematics course options is surprisingly high and that what constitutes four years of high school mathematics can look radically different for students of similar backgrounds in a single school.

Our search for state and district policies concerning advanced placement courses did not reveal any that require participation in AP mathematics as a condition of high school graduation. However, a

³ The textbook policies of California, Texas, and Florida drive the production of the textbook publishing industry (Finn & Ravitch, 2004).

number of states and districts have engaged in highly organized efforts to substantially increase students' participation in the AP program. One example of this is an initiative in Illinois that promotes AP courses to parents and students, provides support for students' academic achievement, and provides supplies and materials for coursework. Other aspects of the initiative include training teachers and administrators in AP philosophy and methodology, encouraging teaming between high school and middle school teachers, supporting the mentoring of new AP teachers by experienced AP teachers, and promoting teachers' professional development by funding their attendance at institutes conducted by the College Board (see http://www.isbe.net/curriculum/pdf/AP_grant-awards_release.pdf for more information).

To support expanded participation in AP programs, in 2002 the College Board began holding an annual *National Advanced Placement Program® Equity Colloquium* to assist educators in their efforts to ensure that teachers and students across the socioeconomic spectrum have equitable access to AP courses. These supports—and, according to Klopfenstein (2004), many pressures—have led to hundreds of schools (many of them small, rural, or low income) to implement AP programs in the past several years. The Dallas-based AP incentive program successfully uses incentives, including financial supports for teachers and for students based on exam results, reports increased percentages of minority students enrolling in AP courses, taking AP exams, receiving passing grades on AP exams, and applying for college scholarships. (O'Donnell, 2006).

Research on the Effects of K–12 Instructional Materials and Curriculum Policies

The main type of research available about the effects of NSF instructional materials policies is in the form of evaluations of the materials produced with NSF funding. An NRC study committee was charged to evaluate the quality of the 13 mathematics curricula produced with NSF funding beginning in the late 1980s, together with 6 commercially produced curricula, and produced the report, *On Evaluating Curricular Effectiveness: Judging the Quality of K–12 Mathematics Evaluations* (NRC, 2004). After reviewing 147 research and evaluation studies, the committee concluded that

the corpus of evaluation studies as a whole across the 19 programs studied does not permit one to determine the effectiveness of individual programs with a high degree of certainty, due to the restricted number of studies for any particular curriculum, limitations in the array of methods used, and the uneven quality of the studies. (p. 3)

These findings complicate further the recent challenges that curricular choices be based on the available research base about those materials.

The most recent wave of NSF-funded instructional materials also has had another tangible outcome in that it has elicited a strong response from mathematicians and others who are critical of these materials for a variety of reasons. In the fall of 1999 a group of more than 200 mathematicians and others signed an open letter to then-Secretary of Education Richard Riley that was published in the *Washington Post* (November 18, 1999). The letter (see www.mathematicallycorrect.com/riley.htm) called for the U.S. Department of Education to withdraw its endorsement of the ten K–12 mathematics programs—five endorsed as exemplary and five as promising (<http://www.encyc.org/ed/exemplary/>), on the grounds that:

It is not likely that the mainstream views of practicing mathematicians and scientists were shared by those who designed the criteria for selection of “*exemplary*” and “*promising*” mathematics curricula. . . . Even before the endorsements by the Department of Education were announced, mathematicians and scientists from leading universities had already expressed opposition to several of the programs listed above and had pointed out serious mathematical shortcomings in them. . . . [W]e believe that it is premature for the United States Government to recommend these ten mathematics programs to schools throughout the nation. We respectfully urge you to withdraw the entire list of “*exemplary*” and “*promising*” mathematics curricula, for further consideration, and to announce that withdrawal to the public. We further urge you to include well-respected mathematicians in any future evaluation of mathematics curricula conducted by the U.S. Department of Education. Until such a review has been made, we recommend that school districts not take the words “*exemplary*” and “*promising*” in their dictionary meanings, and exercise caution in choosing mathematics programs. (<http://www.mathematicallycorrect.com/riley.htm>)

The endorsement was not withdrawn.

There is a considerable amount of anecdotal discussion about the effects of the textbook adoption policies, but again, we found little actual research on this topic. Representatives of commercial textbook companies respond to these adoption policies by sometimes creating special editions of texts that are designed to meet the standards of a given state (e.g., California, Texas), and have claimed that the reason that U.S. textbooks are “a mile wide and an inch deep” is in part because publishers strive to include the union of all topics in all of the state standards, particularly in the “textbook adoption” states (C. Dillender, personal

communication, Oct. 1, 2003). Delfattore (1992) confirms this view, noting that, because publishers must demonstrate specifically how their textbooks align with the state's standards or curriculum, this requirement "puts publishers on notice that their books had better conform as closely as possible to the curricula of those states" (p. 123).

Similarly, we found little research describing the effects of course-taking and opportunity-to-learn policies. It is early to know the results of the trend toward requiring algebra at the eighth grade, though a number of interesting questions are suggested in RAND Mathematics Study Panel (2003): How do algebra course-taking policies affect the subsequent mathematics course-taking patterns and achievement of urban and minority students? Should algebra be offered at the eighth grade? What is the impact on high school graduation, on placement in college mathematics, and on entry to technical careers in the workforce? Do students who have studied algebra at the eighth grade continue on to reach advanced placement mathematics courses?

We found few studies that looked directly at questions such as these. There is some interest in the impact of participation in AP mathematics courses and high school calculus experience on later course-taking patterns, college placement, or achievement. (e.g., Ferrini-Mundy & Gaudard, 1992; Morgan & Maneckshana, 2000; Morgan & Ramist, 1998; NRC, 2002b). Several studies have indicated that students who take AP courses, even if they do not take or pass the exam, achieve higher college grades than non-AP in the courses into which they place (e.g., Morgan & Crone, 1993; Morgan & Ramist, 1998). However, the advantage of AP participation may diminish as students move to more advanced courses. This is not altogether surprising if one considers the AP experience as acceleration to college-level work. The advantage of early exposure is likely to diminish as students with AP experience enter their second year of college.

Observations: Research about K–12 Mathematics Instructional Materials and Curriculum Policies

It would be very helpful to future policy development to have more research about how policies are formulated, particularly for policy related to mathematics instructional materials development and selection, areas that have emerged as a highly controversial. Such research would likely reveal that the policies guiding these areas are not necessarily grounded in research findings about what might lead to effective instructional materials in mathematics. More research

about formulation also would highlight areas where policy makers are working without an available base of research, and might encourage mathematics education scholars to undertake studies in areas that are central for policy development.

Research about implementation and impact does reveal that, even when policies seem well thought-out and systematic, the resulting implementation and products can lead to substantial disagreement and debate. One implication might be that the policy-making components should include wider diversity and breadth in terms of the stakeholders, with more intentional inclusion of mathematicians, mathematics educators, assessment experts, and K–12 practitioners.

POLICY AREA THREE: ASSESSMENT IN K–12 MATHEMATICS

In the U.S., assessment systems have long been a part of federal and state policy systems. The two most prominent current assessment policies are a cluster of policies associated with the federal No Child Left Behind (NCLB) legislation and periodic national assessments in mathematics, some conducted as part of the National Assessment of Educational Progress (NAEP) and others as part of international comparative studies. Some states and districts have additional assessment policies, including an increasing use of mathematics testing as a requirement for high school graduation.

With the NCLB, the federal government instituted a major new federal assessment policy, requiring states to assess all students in mathematics each year in grades 3–8 (starting in 2005–2006) and once during grades 10–12. Although the law requires testing that produces scores for individual students, the accountability requirements are specified at the school level, rather than at the level of teachers or students. Larger organizational units—districts and states—are also held accountable, with a requirement that they make adequate yearly progress toward the 2013–14 goal of getting all students to the state-determined level of academic proficiency. The requirement for adequately yearly progress also specifies that adequate progress be made for subgroups of students specified by race and income. The assessment initiative builds on state policies already in place.

The intent of the federal NCLB assessment requirements is to improve the achievement of all students by holding schools accountable, including accountability for the achievement of designated student subgroups. The legislation's implicit theory of action is that requiring the publication of assessment

results and tying those results to changes in schools and options for parental choice will lead educators to work toward improving achievement.

In addition to requirements for state assessments, federal policy makers have created a system for national student assessments in mathematics—the National Assessment of Educational Progress (NAEP)—and have supported U.S. participation in international assessments of student achievement, including TIMSS (renamed after TIMSS 1995 as the *Trends in International Mathematics and Science Studies*) and the *Program for International Student Assessment* (PISA). Beginning with the 2002–2003 school year, federal policy for NAEP has strengthened, now making state receipt of Title I funds contingent on participation in the NAEP program component that reports results at the state level.⁴ The goals of NAEP are to compare student achievement in states and other jurisdictions and to track changes in achievement of fourth-, eighth-, and twelfth-graders over time in mathematics, reading, writing, science, and other content domains (National Center for Education Statistics [NCES], 2005).

Similarly, the federal government supports U.S. participation in international assessments, including assessments of mathematics achievement, to provide a basis of comparison—across countries and over time.

With the emergence and growth of the global economy, policymakers and educators have turned to international comparisons to assess how well national systems of education are performing. These comparisons shed light on a host of policy issues, from access to education and equity of resources to the quality of school outputs. They provide policymakers with benchmarks to assess their systems' performances, and to identify potential strategies to improve student achievement and system outputs. (<http://nces.ed.gov/surveys/international/IntIndicators/>)

Other state and district assessment policies vary, with purposes that include providing public information, determining the distribution of resources or interventions, and serving as tools for accountability or for promoting changes in practice.

In short, federal and state policy makers have put in place a variety of required student assessments, intended to provide information on student achievement, with possibilities for comparisons across schools, among countries, and over time. NCLB has required states to link assessments to state standards, a connec-

tion often, but not universally, made previously. The NCLB assessment requirements are, according to the U.S. Department of Education, intended to generate information educators and policy makers can use to improve student achievement.

The tests will give teachers and principals information about how each child is performing and help them to diagnose and meet the needs of each student. They will also give policymakers and leaders at the state and local levels critical information about which schools and school districts are succeeding and why, so this success may be expanded and any failures addressed. (U.S. Department of Education)

Some assessment policies are tied to specific consequences, such as government intervention or parental choice; others are intended to spur change by making assessment results public. Mathematics is a core subject in such assessments.

Research on the Formulation of Policy for K–12 Assessment

As the stated purposes for NCLB testing and NAEP illustrate, assessment policies are established with multiple purposes in mind. Haertel (1999) distinguishes three main purposes for large-scale assessment: providing information for accountability, evaluation, and comparison; focusing public attention; and changing education practices (p. 6). The stated purpose of NCLB assessment highlights the purpose of changing practice, suggesting that change will be spurred by the public attention and accountability provisions built into the overall NCLB legislation.

Policy analysts, drawing on their own experiences as the policy was developed, have described aspects of the development of the assessment policies within NCLB. Goertz (2001) illustrates the accounts given, which indicate that NCLB modified the approach taken in previous federal legislation by requiring states to put in place more uniform and extensive policies for assessment, with federal mandates for improvements in test performance. NCLB mandated that education agencies seeking funds under the legislation (e.g., states and local districts) select at least one assessment in each of mathematics, language arts, and science for common use by all public schools in their purview for measuring the achievement of all students. McDonnell (2005) argues that the NCLB assessment policy

⁴ The NAEP Program includes the Main NAEP and Long-term Trend NAEP. Content tested on the Main NAEP is revised every ten years. Every two years Main NAEP reports mathematics results at the national level years for grades 4, 8 and 12 and results at the state level for grades 4 and 8. The content of the Long-term Trend NAEP has remained "relatively fixed" (NCES, 2005, p. 4), with results to be reported every four years for 9-, 13-, and 17-year olds.

can be seen as part of an evolution in policy over two decades, as research showed that assessments could be an effective way to create the changes in education practice that were needed to move student achievement closer to the high standards that were being set. For the recent standards and accountability movements, the seminal paper by Smith and O'Day (1991) is often cited as a summary of the rationale for using a combination of standards and assessments to promote change in education.

Porter and his colleagues have further developed a framework for specification of assessment policies designed to increase student achievement (e.g., Porter, Chester, & Schlesinger, 2004).

Some accounts, however, see the increasing use of assessments as serving less noble purposes, describing the increased use of testing as misguided or even as a covert ploy for maintaining social stratification (Apple, 2006; Oakes, 1985). These critiques lump assessments of mathematics in with a generally increased policy emphasis on assessment and accountability.

The political history behind recent changes in assessment policy in California has been reported in more detail than is usually found in reports on the formulation of mathematics assessment policies (see Jackson, 1997a, 1997b; Wilson, 2003). In concert with its Mathematics Framework, the California Department of Education had developed an assessment, called the California Learning Assessment System (CLAS), which incorporated open response items in mathematics. As part of the broader controversy about the California approach to mathematics, the assessment came under fire.

Research on the Status of Policy for K–12 Assessment

Documentation about state level assessment policies is included in several periodic reports produced by CCSSO, the AFT, *Education Week*, and the NSF. One aspect of status reports are descriptions of which government units conduct mathematics assessments, at which grade levels, with what frequency. For instance, *Quality Counts 2005* (Editorial Projects in Education, 2005), includes a state-by-state summary of assessments administered by grade and by subject. For example, Arkansas, Georgia, Utah and the District of Columbia report a mathematics assessment in grade one; there are 13 states that do not assess mathematics in grade four.

Sometimes status reports go further, including descriptions of the mathematics content that is assessed. For high school graduation tests, for example, The Center on Education Policy (2004) reported that in 2004, 27 states had a high school exit exam. Achieve,

Inc. (2004a) provides a more substantive analysis in *Do graduation tests measure up: A closer look at state high school exit exams*. The report indicates that about half of the states require such tests as a condition for earning a high school diploma. They have conducted an analysis of the mathematical demands of high school exit exams in the states of Florida, Maryland, Massachusetts, New Jersey, Ohio, and Texas, and draw conclusions such as:

The questions on these exams reflect material that most students study by early in their high school careers. In math, the tests place a heavier emphasis on prealgebra and basic geometry and measurement concepts than on concepts associated with Algebra I and high school geometry.

... The “cut scores” required to pass the tests reflect modest expectations. To pass the math tests, students in these states need to successfully answer questions that, on average, cover material students in most other countries study in 7th or 8th grade. . . .

The tests measure only a fraction of the knowledge and skills that colleges and employers say are essential. Achieve, Inc. (2004a, pp. 1–2)

These analyses of the status of assessment policies show both that the prevalence of assessments is increasing and that the expectations for student performance remain modest. If the intent of mathematics assessment policies is to increase student achievement, the goal seems to raise the floor, rather than to push to high levels.

Research on the Implementation of Policy for K–12 Assessment

If the logic model for assessment policies is that making assessment results public and connecting them to incentives will push the system to bring more students up to the standards, a key aspect of implementation is having the assessments align with the standards. Alignment between assessments and standards has been the subject of frequent study in the last two decades.

In the past decade, several systems have been used for examining alignment between mathematics standards and assessments (e.g., Schmidt et al., 2001; Smithson & Porter, 2004; Webb, 1999). Each of the systems includes dimensions for classifying the content of both assessments and standards, which are used by raters to describe the distribution of content in an assessment or set of standards. These studies often have found a surprising disjuncture between the content specified in the standards and the content in-

cluded on the assessment, even when the standards and assessment were adopted within a single state. In describing example studies of alignment, Porter (2002) notes that a study designed to look at the alignment between state standards and assessments in seventh grade mathematics found that, across four states, assessments were as closely aligned to standards in other states as they were to the state's own standards. Although Porter notes that, because tests represent only a sample of intended content, perfect alignment should not be expected, this study suggests that state standards may not be sufficiently specific to allow an assessment to be tightly aligned with them. "Another possibility is that states have much more work to do to bring their assessments into alignment with their standards (Webb, 1999). This finding seems probable, and it is one about which U.S. Department of Education officials are expressing concern" (p. 6).

The misalignment between standards and assessments is more severe for more challenging content areas. A recent summary of alignment studies produced by the American Educational Research Association (2003) concludes that many tests fail to measure adequately the cognitive complexity or "depth of knowledge" described in state standards and that:

For example, in an analysis of 8th-grade math standards and tests in one state, Porter found that the test questions concentrated on length and perimeter, area and volume, and the performance of routine procedures. But the tests generally neglected more complex topics and cognitive demands included in the standards, such as angles or solving nonroutine problems. Similarly, in an analysis of math standards in three states, Webb found that nearly all of the states failed to measure adequately the "depth of knowledge" described in their standards. (pp. 2, 4)

An alignment study carried out by Achieve concluded that when states use existing norm-referenced tests for their state assessment systems, "these tests do not measure the content of the standards well" (Achieve, Inc., 2002, pp. 8–9).

Research on the Effects of Policy for K–12 Assessment

Research does not yet give a clear picture of the effects of mathematics assessment policies. Varying conclusions are being drawn, with publications sometimes leading to sequences of responses and rejoinders.

Early commentaries on testing policies sometimes criticized the effects that testing programs had on the curriculum, claiming that tests focused on low-level skills, leading teachers to an equally low-level curricu-

lum. But those commentaries seldom cited solid empirical evidence. As measurement experts later noted, research was needed to determine the broad effects of testing (Mehrens, 1998; Porter, Floden, Freeman, Schmidt, & Schwille, 1988).

Subsequent research has shown that effects of testing policies are varied, depending in part on local context and capacity. The conclusion is consistent with research on a broad range of education policies (e.g., Berman & McLaughlin, 1997). Firestone, Schorr, and Monfils (2004) studied the effects of mathematics and science testing policies in New Jersey. The mandated state tests included both multiple-choice and short open-ended questions, intended to measure a range of pupil knowledge, from computational skills to problem solving. Its orientation seemed likely to encourage teaching that included, but went beyond, simple mastery of algorithms.

Firestone and his colleagues found that the influence of the mandated test varied across schools and districts. They saw changes in instruction, many of which might be called, "teaching to the test," but which ran the gamut from drilling on test mechanics to asking probing questions, and trying to get students to explain the reasoning behind their answers. Explanations of differences in effects of testing policies included availability of resources, local political context, and educators' interest in and knowledge about education reform. This research indicates that attempts to determine the typical effect of an assessment policy may be frustrated by variation in what any one policy will produce.

When scholars make broad claims, they often get responses from others who support opposing claims. One recent example is the exchange that began with Amrein and Berliner's (2002) publication of an analysis of the data on the consequences of the implementation of high-stakes state testing policies. Using nationally-normed assessments (e.g., SAT, ACT, NAEP), the investigators looked for changes in student achievement in 18 states which began high-stakes testing policies at varying points in the past two decades. They concluded that, though results on the high stakes exam itself typically increased when higher stakes were attached, the results did not transfer to other measures of academic achievement. Hence, these researchers concluded that apparent positive effects were due to exclusion of students from the high-stakes tests or to narrow forms of training that were unrelated to the intended content of schooling.

Rosenshine (2003) responded with an alternative analysis that led to the conclusion that states with "clear" high-stakes testing policies did have higher scores on the NAEP.

The results in Table 1 show the average NAEP increases in the “clear” high-stakes states were much higher than the increases in the comparison states. In 8th grade mathematics and in 4th grade reading the mean increase for the clear high-stakes states was double the increase for the states without consequences. The effect sizes for the comparisons were .35 for 4th grade math, .79 for 8th grade math, and .61 for 4th grade reading. (<http://epaa.asu.edu/epaa/v11n24/>, paragraph 7)

His analysis differed in comparing states with clear high-stakes testing to those without, rather than looking at individual states across time, as Amrein and Berliner did. Amrein-Beardsley and Berliner (2003) published a rejoinder, finding a positive effect of high-stakes testing in 4th-grade NAEP scores, but not elsewhere.

This exchange illustrates the ongoing debates about the impact of state and national assessment policies. The sources of disagreement are multiple. Some focus on methodology: What comparison group should be used? Can a single setting (e.g., state or district) serve as its own control over time? How can the effects of a testing policy be disentangled from other policies, some changing, some not? What is an appropriate outcome measure? How can controls be built in for changes in the population tested?

Other disagreements come from differences in values attached to outcomes. Is a narrowing of the content taught (to match the test) a desired or disparaged outcome? Are increases in average achievement the most important aspect of improvement, or is narrowing the gap between rich and poor, or Black and White, the most important goal?

A few studies have attempted to consider the impact of high school exit exams on other variables. For instance, Marchant and Paulson (2005) analyzed the relationship of high school graduation exams to graduation rates and SAT scores, through secondary analyses of College Board and NCES data. They found that, “states requiring graduation exams had lower graduation rates and lower SAT scores. Individually, students from states requiring a graduation exam performed more poorly on the SAT than did students from states not requiring an exam.” (p. 1). This result, coupled with the Achieve result about low expectations on these exams, suggests that this policy is not necessarily having the effect of restoring graduation as a valid indicator of preparedness.

Most scholars seem to believe that instituting mathematics testing policies will have effects, but the field seems far from reaching agreement on just what those effects will be, and whether they are desirable.

Observations: Research about Assessment Policy in K–12 Mathematics

Assessment policies in current mathematics education are among the most prominent and prescriptive of all subject areas. The federal policy emphasis on assessment has increased with No Child Left Behind, extending the trajectory along which federal policy makers have been pushing for higher achievement outcomes for two decades. Federal policies have not dictated the specific content to be assessed, but through requirements for broad student participation and connections to NAEP, national policies press states to push achievement higher, and in the general direction represented by the content of NAEP. Analyses of the status and implementation of assessment policies reveal upward movement in expectations, though not yet to levels that match the rhetoric of calling for mastery of challenging content by all students.

Research on the impact of assessment policies shows effects, particularly when these policies are aligned with other policies. Researchers debate whether all the effects observed are desirable, though the general weight of evidence is that assessment is having effects roughly along the lines policy makers are advocating.

POLICY AREA FOUR: THE MATHEMATICS TEACHING PROFESSION

The profession of mathematics teaching is governed and affected by policy conditions of various types at the local, state, and federal levels. In this section we discuss policy relevant to four areas that affect mathematics teaching: entry into the teaching profession (including initial licensure, accreditation, teacher preparation programs); recruitment, retention, and teacher induction; continuing licensure, credentials, and teacher quality requirements; and teacher professional development. Entry into the teaching profession is governed by a complex and largely state-driven set of policies for certification and licensure. These policies include “state-regulated initial certification of new teachers, state-required teacher tests for initial licensing, and voluntary professional accreditation of teacher preparation programs and institutions.” (Cochran-Smith & Zeichner, 2005, p. 26). After a provisional or initial license to teach is granted, continuing licensure depends on meeting additional requirements, such as passing further tests, completing additional education, or getting a favorable rating from an employing school. The specific requirements, tests, and passing scores vary across states. Many states also

require districts to provide some type of induction support to beginning teachers as they enter the career. In addition, there are policies associated with teacher induction, advanced credentialing and teacher professional development that we will discuss in this section.

The education of prospective teachers is governed by state and national accreditation policies that apply at the program and institutional level. The National Council on Accreditation of Teacher Education (NCATE) and the Teacher Education Accreditation Council (TEAC) are the two national accrediting agencies for teacher education programs. NCATE's process includes a specialty program area review component which, in the case of mathematics, is overseen jointly by NCATE and NCTM. State program approval processes vary, but generally involve a self-study and state-administered review.

Other policies address the induction and retention of beginning teachers. The Interstate New Teacher Assessment and Support Consortium (INTASC) is a "consortium of state education agencies and national educational organizations dedicated to the reform of the preparation, licensing, and on-going professional development of teachers." (CCSSO, 2006, p. 1). INTASC has developed core standards for what beginning teachers should know, and also has created model performance assessments for teachers, thus providing policy guidance to state agencies. INTASC has standards specific to mathematics (INTASC, 1995), which call for a view of mathematics teaching consistent with the NCTM Standards. In particular, they note:

They will be expected to provide challenging mathematics to students at very different levels of mathematical understanding and to adequately address the full range of student questions that arise from learning diversity. Teachers of mathematics at every grade level will be expected to teach more mathematics and more challenging mathematics to their students, and to be knowledgeable about the curricula that precede and follow the mathematics they teach. (p. 6)

Local and state policies have been developed about teacher placement in areas of high need (e.g., policies that allow teachers lacking adequate credentials in mathematics to teach mathematics if properly qualified teachers cannot be hired, often tied to district-level alternative certification programs) and policies about teacher retention (e.g., induction programs.) The problem of attrition among beginning

teachers has been well documented (see Grissmer & Kirby, 1987, 1997; Lortie, 1975; Veenman, 1984); these attrition issues may be relatively more serious in mathematics, particularly in urban areas.

Periodically, federal programs have been implemented to increase the size of the mathematics and science teaching force. The most notable is the "Troops to Teachers" (TTT) program, established in 1994 by the U.S. Department of Defense as an alternative certification program.

Among the influential federal policies over the past several decades relative to teaching and teachers (see Earley, 2000), the most current and visible are the policies related to improving teacher quality, recruitment, and preparation in the 1998 Amendments to the Higher Education Act (HEA)⁵ and those related to ensuring elementary and secondary students' access to highly qualified teachers in NCLB. Such federal policies may be influenced by some of the national reports in recent years that have addressed issues about the teaching profession generally (e.g., National Commission on Teaching and America's Future [NCTAF], 1996), as well as issues in mathematics teacher education (e.g., CBMS, 2001; National Commission on Mathematics and Science Teaching for the 21st Century, 2000; Mathematical Sciences Education Board, 1996; NRC, 2001a, 2001b). Such reports typically include strong recommendations for policy. NCTM's 1991 *Professional Standards for Teaching Mathematics* offer specific standards for policymakers, such as

Policymakers in government, business, and industry should take an active role in supporting mathematics education by accepting responsibility for

- supporting decisions made by the mathematics education professional community that set directions for mathematics curriculum, instruction, evaluation, and school practice. (p. 179)

In the mathematics domain, most recent recommendations have focused primarily on the need to strengthen content preparation of teachers.

In response to issues of teacher quality, NCLB requires that all public school teachers of core academic subjects (which includes mathematics) must be "highly qualified" by the end of the 2005-06 school year, with consequences if this is not achieved. (U.S. Department of Education, 2001, pp. 40-41) To be highly qualified, teachers must have a bachelor's degree and full state certification or licensure, and must prove

⁵ HEA was enacted in 1965. The most recent reauthorization by the full Congress was in 1998 (see P.L. 105-244). The 1998 reauthorization was set to expire in 2003 but has been extended several times in response to difficulties achieving agreement on the provisions of the reauthorization.

that they know each subject they teach (U.S. Department of Education, 2001). In addition NCLB requires states to measure the extent to which all students, particularly minority and disadvantaged students, are being taught by highly qualified teachers. States are expected to adopt goals and plans to ensure that all teachers are highly qualified and must publicly report their plans and progress in meeting these goals. NCLB allows states to develop an additional way for current teachers to demonstrate subject-matter competency and meet highly qualified teacher requirements. Proof may consist of a combination of teaching experience, professional development, and knowledge in the subject garnered over time in the profession.

Beyond state credentialing processes, one primary vehicle for advanced teacher credentialing is the National Board for Professional Teaching Standards (NBPTS). Founded in 1993–94, the NBPTS awards advanced certificates for experienced teachers who successfully complete its required assessments. The NBPTS is a non-profit organization, governed by a board of directors, the majority of whom are K–12 teachers. The significance of NBPTS certification varies across states, with some states linking it to pay increases and others simply treating it as information that local districts can use as they see fit. NBPTS offers two certificates in mathematics: adolescence and young adulthood (NBPTS, 2001) and middle childhood through early adolescence (NBPTS, 1998).

In contrast with the strong state policy control, national guidance, and federal control over entry into the profession, teacher preparation programs, induction, and teacher quality, there is very little state-level regulation about the quality and content of the professional development that is expected for the renewal of teaching licenses (Hirsch, Koppich, & Knapp, 2001, cited in Desimone, Garet, Birman, Porter, & Suk Yoon, 2003). School districts, states, professional organizations, universities, and private organizations all are engaged in the business of offering professional development for teachers. The policy environment in this domain is similarly scattered.

The policies that probably have had the most significant impact on teacher professional development in mathematics are federally funded programs for teacher improvement. For example, the National Defense Education Act (NDEA) (PL 85-864) was the main source of funding for institutes for teachers of mathematics and science in the post-Sputnik era, from the mid 1950s until the late 1970s. On the heels of this, the Education for Economic Security Act (EESA) (PL-98-377) was passed in 1984, charging the NSF to provide professional development for teachers of mathematics and science. From 1994–2002, the Eisen-

hower Professional Development Program—a part of Title II of the Elementary and Secondary Education Act of 1994—was “the federal government’s largest investment solely focused on developing the knowledge and skills of classroom teachers in mathematics and science” (Desimone et al., 2003, p. 615).

More recently, the NSF has supported teacher professional development through a variety of programs, including Teacher Enhancement, Systemic Initiatives, and Mathematics and Science Partnerships. And, the U.S. Department of Education provides funding to states for mathematics and science partnerships that offer professional development for teachers.

In summary, the policies affecting the mathematics teaching profession include policies about initial licensure and accreditation of teacher preparation programs; recruitment, retention, and induction; continuing licensure and teacher quality, and professional development. As in other areas, the policies in this arena are generally not specific to mathematics, and there is little mathematics education-specific research that has examined the formulation, status, implementation, and effects of these policies. However, some of the more general research may have important implications for mathematics education. Wilson and Youngs (2005) summarize the historical and political issues surrounding what they posit as a central question in this area: do accountability processes lead to higher teacher quality? (p. 592). They go on to call for rigorous research in this area as a start in addressing the question. We would add that research particular to mathematics education issues, within the more general issues about the teaching profession, would also enlighten the policy discussion.

Research about the Formulation of Policies about the Mathematics Teaching Profession

We found few studies about the formulation of policy relative to the mathematics teaching profession. Although there are historical studies of the overall evolution of requirements (e.g., Sedlak, 1989), and the recommendations for mathematics teacher preparation (e.g., Ferrini-Mundy & Graham, 2003), we do not include those here, because they do not directly address how policies were formed.

There is some available research on policy formulation relative to the teaching profession more generally. Porter, Youngs, and Odden (2001) summarized the processes of developing criteria and standards used in several key teacher assessments, which represent policies governing entry into the profession. As a starting point for the development of Praxis III,

researchers at Educational Testing Service (ETS) established a conceptual guide to describe teaching (Dwyer & Villegas, 1993, cited in Porter et al., 2001). This document included analyses of the job of teaching, reviews of research about teaching, and analyses of state standards for licensure.

Earley (2000) provides an account of some aspects of the formulation of the 1998 Higher Education Act (HEA) legislation. She concludes that the reauthorization was strongly influenced by the 1996 NCTAF report. The policy introduced new accountability and reporting measures for states and institutions of higher education; Earley speculates that the provisions in the HEA were included in part as accountability measures responding to assertions that “teacher education is a culprit in school reform failures.” (p. 33.) As we saw in research about the formulation of standards, it is clear that the context in which policies are constructed can be quite influential in the shaping of the policies.

Porter et al. (2001) contrast the approach used by ETS in developing Praxis III with the approach to the development with that used by INTASC and NBPTS. INTASC also has conducted careful documentation of its development processes (Moss, 1998, referenced in Porter et al., 2001; Porter et al., 2001, p. 276). INTASC convened groups of experienced professionals in specific subject areas to design standards. They provide a detailed description of how the NBPTS standards vetting process occurs, where various layers of committees and experts review the proposed standards and accompanying instruments. The distinction noted by the authors in these two approaches to policy formulation—an analytic, job-demand-based conceptual framework, versus consensus standards developed by practitioners—is interesting. Mathematics teaching-specific policy recommendations generally seem to be in the form of standards developed by an expert practitioner consensus process.

Research about the Status of Policies about the Mathematics Teaching Profession

Various organizations provide periodic information about changes and general features of state-level policies concerning teacher entry, induction, continuing licensure and professional development. It is often difficult to get detailed information about the mathematics-specific elements of the policies.

For many years, the professional association of state officials with responsibility for teacher certification compiled the requirements for every state. This organization, the National Association of State Directors of Teacher Education and Certification (NASDTEC), produces paper copies of these regulations (e.g., NAS-

DTEC, 1997, 2000). A version of the regulations in a uniform format is also published by a private company dedicated to this work (Teacher Certification Publications; <http://home.earthlink.net/%7eteachercertification/index.html>). The Web has eased the compilation of these regulations, which can now be accessed through links at the web sites of the Education Commission of the States (ECS) (<http://www.ecs.org/>) and the National Center for Education Information (NCEI) (<http://www.ncei.com/resources.html>).

Some policy analysts work to pull out what they see as critical features of these descriptions, such as by compiling summaries of recent changes in teacher certification policies, or describing general trends in state policies. Hirsch et al., (2001), for example, write that

Certification and teacher testing are the principal vehicles to date for promoting standards for teaching. Multi-tiered certification systems are becoming increasingly popular ways to demonstrate the growth of teachers' competence over time. While a growing number of states are establishing teacher assessment systems, states are encountering complicated issues regarding the role, validity, and technical quality of teacher assessments. For example, states have yet to evolve cost-effective means of assessing actual classroom performance—or, even more problematic, predicting future performance, in the case of assessments done at the time of initial licensing. (p. 6)

These descriptive studies show some of the common features of teacher certification requirements, such as completion of a bachelor's degree, completing coursework in subjects to be taught, and, for most types of certification, completing some form of practice teaching. Some form of teacher testing is required in 42 states (NASDTEC, 2000, cited in Wilson & Youngs, 2005, p. 596.) Youngs, Odden, and Porter (2003) provide status information about state policy in the area of teacher licensure for the 2001–02 year. They were interested in the extent to which states were using performance assessment, and found that only 18% of states were doing so.

In response to teacher shortages, states often create and implement policy alternatives to university-based teacher certification routes. Feistritzer and Chester (2003) report that most states have some version of an alternative certification program for teachers, and since 1985 about 200,000 people have been certified to teach through alternative routes. Editorial Projects in Education (2004) reports that in 29 states, candidates in alternative routes are required to demonstrate subject matter expertise.

We found little status research or policy analysis describing the status or characteristics of mathemat-

ics teacher education programs. The Association of Mathematics Teacher Educators conducted a survey, (Karp, 2005) to better understand the work of mathematics teacher educators in universities and colleges across the country. They collected survey responses from 427 institutions, and learned that for prospective teachers of grades K–2, most institutions required between 0 and 6 credits of mathematics. In addition, the majority of institutions required between 0 and 3 credits in mathematics teaching methods, and at about 2/3 of the institutions there were 0–3 mathematics education faculty. It is difficult to assemble a comprehensive national picture of the requirements, staffing, and resources involved in the preparation of teachers of K–12 mathematics.

At the state and district levels, beginning teacher induction policies have emerged as a response to the problems of teacher quality and teacher retention. There is no federal policy about teacher induction programs or support for beginning teachers, but Britton, Paine, Pimm, and Raizen (2003) claim that more than thirty states do have policies in this area. Many of these require the mentoring of new teachers. Some school districts also have policies and related programs for new teacher induction in response to state policies. Smith and Ingersoll (2004), on the basis of data from the NCES “Schools and Staffing Survey” (SASS, report increases in the numbers of teachers participating in induction programs, from 51% in 1993–94 to 79% in 1999–2000.

The status studies also show the great variation in the specifics of state requirements. In addition to the “regular” certificate, many states offer others types of certificate, such as “emergency” certificates for cases in which a school is unable to find a certified teacher for some position. These “emergency” certificates have lower requirements and are intended to be valid for only a short time, until a regularly certified teacher can be found.

Research on Implementation of Policies about the Mathematics Teaching Profession

Most studies of policy implementation in the teaching profession are conducted to look at general trends and issues, rather than mathematics-specific matters. Here we highlight some studies that we feel are particularly relevant, and discuss how the trends that they identify may have implications for mathematics.

Some of the status studies described in the previous section characterize the written state policies about what is needed to teach, but as with other poli-

cies, actual practice and implementation may depart from the written policy. Ideally, if the policies requiring a “regular” certificate were thoroughly implemented, the vast majority of teachers would meet those requirements. Policy studies of the certification status of a state’s teaching force give a picture of how well that intention is being met. State teacher certification is not always required for teaching in independent schools. Even within the public schools, however, policy studies have shown that many students are taught by teachers who hold emergency certificates, or whose primary certification is not for the subject area. Darling-Hammond has addressed the prevalence of what she calls “substandard” teacher licenses, noting substantial differences across states. “In Wisconsin and eleven other states, . . . no new elementary or secondary teachers were hired without a license in their field in 1994. By contrast, in Louisiana, 31% of new entrants were unlicensed and another 15% were hired on substandard licenses. At least six other states allowed 20% or more of new public school teachers to be hired without a license in their field” (1999, p. 15).

Some studies look at these issues specifically in mathematics. The National Science Board (2006) reports that 80% of public high school mathematics teachers were fully certified in mathematics in 2002, while 20% were either not fully certified or certified in another field. During the past two decades, there has been a steady problem of recruiting and retaining qualified teachers of mathematics to teach in urban and/or high need districts. In 1999–2000, 69% of U.S. students in grades 5–8, and 31% of students in grades 9–12, were taught mathematics by teachers with no major or certification in the subject (NCES, 2004, p. 10). Situations such as these, which also affect urban and rural youth especially severely, are often a consequence of difficulty recruiting and retaining qualified teachers. Using data from the NCES’ (1999) SASS, Ingersoll (1999b) found that in 1994, 27% of public secondary school teachers assigned to teach mathematics lacked a certificate to teach mathematics. This was a larger percentage than in any other core content area. Out-of-field teaching is associated with school characteristics, with more out-of-field teaching in high-poverty schools and in smaller secondary schools. Ingersoll makes explicit that out-of-field teaching can be seen as a failure to implement certification policy.

Some policy studies have addressed the extent to which the intent of state or national program standards and teacher examinations is implemented in the actual requirements for coursework in programs, and in assessment scoring. For example, some re-

search has examined whether teacher examinations and procedures for determining pass rates actually uphold requirements. Analysts at the Education Trust (Mitchell & Barth, 1999) have examined the subject matter content of teacher certification tests from the two major suppliers, ETS and National Evaluation Systems (NES). They selected a sample of state certification tests in English, mathematics, and science, and engaged their staff and a panel of expert consultants in rating each of the items. For secondary mathematics, this study found that most content on the teacher tests was material from the high school curriculum. The balance of “routine” and more complex items varied across tests, with the test used in Massachusetts having the most challenging items. From this analysis, the authors concluded that the tests were not sufficiently advanced, because “the tests assess mostly tenth to eleventh grade level content. Nothing in either the Praxis II tests or the NES tests probes the intellectual substance of college mathematics that should equip high school teachers with robust backgrounds for dealing with the myriad ideas that will emerge from discussion with students” (Mitchell & Barth, 1999, p. 10). In addition, many states require teachers to get only half the items correct. The authors conclude that implemented policies for teacher certification do not reflect the rigor of the policies’ intentions.

Once teachers are hired, there are challenges in retaining them, in the profession and in particular in high-need schools (Ingersoll, 1999a, 1999b). Retention of teachers of mathematics, particularly in urban settings, is especially difficult. Research reveals a variety of reasons for teacher mobility and attrition; there is some disagreement in the research about how prominently salary figures in this context. Strong factors in teacher attrition include poor working conditions (Ingersoll & Smith, 2003) and issues related to student race and achievement (Hanushek, Kain, & Rivkin, 2001). Neuhaus (2003) reports on a state-funded program in California to increase the proportion of fully certified teachers in low-performing schools. The Teaching as a Priority Program (TAP) involved incentives such as signing bonuses and supplementary funds available for mathematics and literacy instruction. Findings indicate that these kinds of policy incentives might not be effective in inducing teachers to stay in low-performing schools.

In the U.S., one of the most sweeping and heavily resourced statewide induction policy-program combinations is the California Beginning Teacher Support and Assessment System (BTSA), which encourages school districts and other agencies to offer induction programs, and which provides state funding for such

programs. (from <http://www.btsa.ca.gov/>, retrieved January 19, 2005).

In the case of teacher induction, it may be possible to improve the implementation of policies by looking at international practices, where induction policies are in place in a number of countries. The policies tend to be more elaborate and structurally infused within both the national culture and the education system than those in the U.S.; (see Britton et al., 2003, for a report of a study of induction in Shanghai, Switzerland, New Zealand, France, and Japan). In this volume, the researchers describe the experience of the *stagiaire* (first year teacher) in mathematics and the elaborate and highly prescribed induction system in France, which focuses on subject matter. For instance, they recount the centrality for the new teacher of “le cours”—the formal, mathematical content to be organized and taught.” (p. 238). The researchers observed an emphasis on oral fluency in mathematics and careful selection and evaluation of preparatory mathematical tasks as part of the content of the induction experience in mathematics. Induction policies in the U.S. tend not to be subject-specific, and thus their implementation is also general.

Evaluation studies that are aimed specifically at whether funded programs have met the program goals as specified in the “policy” of the funding agency are a type of research that could be construed as studying policy implementation. For example, in a study of district-level implementation of mathematics inservice programs for teachers, as part of the national evaluation of the Eisenhower Professional Development program, Desimone, Porter, Birman, Garet, and Suk Yoon (2002) examined “the degree to which the [PD] activity has a content focus—that is, the degree to which the activity is focused on improving and deepening teachers’ content knowledge in mathematics and science” (p. 1274), a goal of the program. They found that “districts that align professional development with standards and assessments are more likely to offer reform types of activities and are more likely to engage in continuous improvement efforts” (p. 1292).

Some professional development funding programs require or encourage involvement of higher education faculty and disciplinary faculty specifically. Desimone et al. (2002) in the Eisenhower Professional Development Program evaluation, found that “postsecondary institution projects in education departments have significantly more coordination with districts than postsecondary institution projects in mathematics or science departments” (p. 633), and that “projects in mathematics/science departments have high content focus regardless of the type of postsecondary

institution they are in, whereas projects in education and other departments have a high content focus only if they are in research/doctoral universities.” (p. 636). They concluded that the quality of professional development experiences would be strengthened by close coordination between districts and postsecondary institutions. Howey (2000) was also referenced, saying that encouraging K–12 and university faculty to work together will require new policies that offer incentives and working mechanisms for such collaborations.

In summary, studies of policy implementation suggest that there is a distance between the idealized version of the policy and its applied version in practice, particularly in the area of teacher certification. This becomes important in understanding research that examines the effects of policies, because policies are not always implemented with fidelity.

Research on the Effects of Policies Related to the Mathematics Teaching Profession

The fundamental questions in this area center around whether particular policies, especially those related to accountability, credentialing, and program components as specified in state, national, or federal policy, have an impact. The question is complicated not only by the mediating effects of the fidelity of implementation, but also by determining the most suitable outcome variables. These include teacher retention, teacher effectiveness, and ultimately, student performance. However, little research has examined the impact of policy on pupil performance.

In the past decade, scholars have vigorously debated what effect requirements for teacher certification have on the effectiveness of teaching. Sometimes the debates focus on teacher certification *per se*, comparing teachers at the secondary level with regular certification in the subject area to teachers lacking such certification. At other times, the debates focus on the requirements for teacher certification, especially coursework and test performance, looking for associations between those requirements and pupil performance.

Linda Darling-Hammond argues that teacher certification is associated with teaching effectiveness (e.g., Darling-Hammond, 2000a, 2000b; Darling-Hammond, Berry, & Thoreson, 2001). Those on the other side of the debate include Ballou and Podgursky (2000) and Walsh (2001). Wilson and Youngs’ (2005) careful review of rigorous empirical research on the effects of teacher certification concludes that “the trend is toward favoring certified teachers . . . particularly in the area of mathematics” (p. 614). The studies they reviewed used teachers’ learning and knowledge, teach-

ing practice, and/or student learning as outcomes. They note that the great variation in requirements for certification and how researchers define certification make it difficult to draw strong general conclusions about its effects.

Research also has examined the associations between particular aspects of certification and related teacher education programs and the learning of teachers’ pupils. A recent review of the effects of taking subject matter courses found that “Only in secondary school mathematics do the studies provide a clear answer to questions about connection between amount of teachers’ subject matter study and pupil achievement” (Floden & Meniketti, 2005, p. 269). Though research indicates that there is an effect of completing some mathematics coursework, the research does not go so far as to support the requirement for a major in mathematics (or its equivalent), as is built into many state certification requirements and has historically been the norm (see Ferrini-Mundy & Graham, 2003). “For secondary school mathematics, one study showed that the benefit of additional subject matter courses declined after the first five courses, suggesting that, even here, the relationship between subject matter study and student achievement is not as simple as ‘more is better’” (Floden & Meniketti, 2005, pp. 269–270).

Some research also offers support for completing coursework on the teaching of mathematics, another typical part of the requirements for mathematics teacher certification. Monk (1994) found that undergraduate mathematics pedagogy courses contribute more to performance gains of teachers’ pupils than do undergraduate mathematics courses, and that the influence of advanced mathematics courses beyond a threshold set about five may be negligible.

Passing teacher tests is now part of the certification requirements in most states. Little evidence exists, however, about the effects of this requirement on teacher effectiveness. Wilson and Youngs’ literature review (2005) found that much of the research focused on tests that are no longer used and that the literature as a whole gave little information about the connection between performance on teacher tests and pupil achievement.

For advanced certification, however, two recent studies have found an association between passing the NBPTS examination and student achievement. A study in North Carolina (Goldhaber & Anthony, 2004) and one in Florida (Cavalluzzo, 2004) both have found that pupils of NBPTS-certified teachers learn more than pupils of teachers in comparison groups. The North Carolina study found this result for elementary school mathematics (with a smaller, positive effect in

reading) and the Florida study for 9th and 10th grade mathematics. Sanders, Ashton, and Wright (2005) undertook a comparison of National Board Certified Teachers' (NBCT) students' performance growth in mathematics to the performance growth of students of other teachers. They found that "students of NBCTs did not have significantly better rates of academic progress than students of other teachers." (p. 2).

Research on the impact of alternative route programs and induction programs does not typically single out mathematics as a focus, but some of the general findings may have interesting implications for mathematics education. Alternative route programs vary widely in their design and implementation. Those that assume strong subject matter background, such as Troops to Teachers or Teach for America, tend to include preparation that is focused on more general professional practice issues. The more effective alternative route programs are those that include some of the main characteristics of university-based teacher education programs, such as significant support and mentoring, clinical experiences, etc. (see Feistritzer & Chester, 2003; NCTAF, 2003; Wilson, Floden, & Ferrini-Mundy, 2001). Some research has suggested that teachers prepared through alternative routes have reached higher educational achievements than their counterparts who have been certified through traditional processes (Shen, 1999). In his 2003 review of research, Allen concludes that there is only limited support for the conclusion that alternative-route completers can be as effectively as traditionally prepared peers. Thus the research is inconclusive.

Most induction policies require mentoring—that is, systematic support offered by experienced teachers to beginning teachers. (see Feiman-Nemser, Schwillie, Carver, & Yusko, 1999); others also include requirements for participation in professional development. Some researchers are concerned that the focus on mentoring is too narrow and fails to sufficiently problematize the challenges faced by the new teacher. Britton et al. (2003) note that,

"Mentoring" has come to stand for an automatic remedy to a problem that tends to remain unexamined. It is variously seen as insufficient experience of the exigencies of teaching, inadequate information about local practices and customs in the particular school or incomplete knowledge of various sorts required for teaching. . . . Share the common presumption that the novice arrives lacking something. . . the assistance of a mentor—is somehow to make up for this deficiency. (p. 1)

Smith and Ingersoll (2004) found that when mentoring is coupled with other, subject-specific supports, retention

is improved. They comment that "Of all beginners who entered teaching in the 1999–2000 school year, 16% received none of the aforementioned induction or mentoring supports in their first year. Their predicted probability of turnover at the end of the first year was 40%. Twenty-two percent of beginning teachers received three induction components: a helpful mentor from their same field, common planning time with other teachers in their subject area, and regularly scheduled collaboration with other teachers on issues of instruction. Their turnover probability was 28%" (p. 6).

There has been a considerable amount of research on the impact of induction and mentoring programs on teachers' attitudes about their practice, again without specific attention to mathematics. The California Educational Research Cooperative undertook a study of the impact of the BTSA program in 1995–96, reported in Mitchell, Scott, Takahashi, and Hendrick, 1997. About 45% of the teacher respondents reported that their support providers worked in the same subject area and at the same grade level. Overall, the evaluation found that "teachers felt that the program was most beneficial in helping them plan instruction, while support providers, site administrators, and BTSA staff members felt the program was most valuable in helping teachers to organize and manage their classroom" (Porter et al., 2001, p. 284). Bartell and Ownby (1994, summarized in Porter et al., 2001) conducted a study of the implementation of local BTSA programs from 1992–1994, focusing mainly on the nature of the interactions of mentors and teachers, without specific attention to subject matter. BTSA projects at the local levels were charged with taking the lead on this work in 1994. In their account of the various processes of review, the recommendations that came from the field, and the revisions, Porter et al. do not mention any particular focus as having emerged on issues of the place of subject matter, or more specifically, mathematics, in the development of new teachers.

Strong (2005) conducted a review of research about induction programs. He comments:

Of those [studies] that compare different kinds of programs, it appears that comprehensive support such as that advocated by Wong (2004) and identified as having enhanced 'packages' of components by Smith and Ingersoll (2004) are most likely to have an effect on turnover than more basic mentoring programs. (p. 192)

In summary, research that has examined the effects of alternative route and induction programs suggests that programs which are more comprehensive and include subject-specific elements may be more likely to impact teachers' decisions to remain in the

classroom. Johnson and Kardos (2002) have found that new teachers are interested in assistance with curriculum implementation and advice from colleagues in their subject areas. Research that is subject-specific and that makes the link to teaching practice and to student performance is very much needed.

A problem with research about the effects of professional development policy has been the difficulty in linking to student achievement. An analysis released by the U.S. General Accounting Office (GAO) (1984) examined the impact of the NSF Mathematics and Science Teacher Institutes of the '70s and '80s, and found no evidence to connect teachers' participation in the institutes to the subsequent performance of their students in mathematics and science. They report, "as a group, these studies fail to show any relationship between teacher knowledge and the knowledge gain of their students" (GAO, 1984, p. 33, as cited in Earley & Schneider, 1996, p. 308). More recent work suggests that mathematics professional development that is curriculum-based can have an impact on the achievement of teachers' pupils (Cohen & Hill, 2001), as can intervention that is multi-dimensional and involves in-class coaching (Balfanz, MacIver, & Byrnes, 2006). In her review of studies that examined the impact of professional development by considering evidence of student learning, Kennedy (2002) concludes that "programs that focus on subject matter knowledge and on student learning of particular subject matter are likely to have larger positive effects on student learning than are programs that focus mainly on teaching behaviors" (Kennedy, 2002, p. 25).

Observations: Research about Mathematics Teaching Profession Policies

Teachers play a central role in mathematics education, so policy makers and policy researchers are keenly interested in understanding how policies affect teacher recruitment, preparation, selection, retention, and continuing education. In short, they are interested in policies intended to affect the quality of mathematics teachers. Policy research across subject areas has also identified teacher quality policies as an area of high priority and interest.

Given that strong interest, the amount of informative research on mathematics teacher quality policies is disappointing. The status of teacher quality policies across states has been documented, and periodically graded in publications such as *Quality Counts*. But the grades are based more on the beliefs of the report authors than on research about policy implementation and effects. The scholarly journals have published competing analyses about the policies such as requirements for certification,

creation of alternate routes for teacher preparation, and requirements for teacher education program approval. Such studies usually report results for mathematics, along with other subject areas. The scholarly debates remain unsettled, in part due to the methodological challenges in conducting research on these policies.

Those reviewing the teacher quality policy literature (e.g., Cochran-Smith & Zeichner, 2005) offer recommendations for strengthening this area of research. They suggest, for example, building more comprehensive data bases, developing consistent definitions of key terms, paying more attention to impact, and situating research in relevant theoretical frameworks. These recommendations all apply to research on mathematics education teacher quality policies. At present, research in this area remains thin.

SUMMARY AND CONCLUSIONS

This review of research on mathematics education policy has themes that echo those of the broader education policy research literature: variability attributable to the decentralized governance of U.S. education; differences between formal and implemented policies; policy formulation influenced by many factors, with research one among many; and difficulty in tracing changes in pupil learning to particular policies. Policy research in mathematics education does, however, have distinctive features: a longer history with standards and associated assessments; more prominent disputes about desired outcomes; and a relatively large body of research, compared to policy research on other subject areas.

The results for research on policy formulation, status, implementation, and effects reveal the diversity across U.S. states and governmental agencies. Research on mathematics education policy formulation finds policies largely shaped by beliefs and values in the overall education context. In the past quarter-century, for example, standards have been widely promoted. Research studies show how the emphasis on standards has come in large part from the rhetorical appeal of holding high expectations. In mathematics education, research also has shown how important the inter-group (i.e., among mathematics educators, research mathematicians, parents, politicians) differences in value attached to particular areas of mathematics have been in affecting standards development and acceptance.

Research on policy formulation has, though, also documented ways in which research results have shaped policy. Current assessment policies, for example, were shaped by research showing how previous policies failed to produce the changes policy makers

desired. As standards have been revised, research results have also been consulted.

Research on the status of mathematics education policies has been motivated by the variability across political contexts. This research, often sponsored by organizations outside higher education (e.g., the Council of Chief State School Officers, the American Federation of Teachers), is of considerable interest to policy makers who attend to policy trends. Research shows many states or other governments adopting similar policies, with variation in policy details. Anecdotal evidence suggests that policy makers base their own decisions in part on their understandings of what their counterparts are doing. Some status studies add evaluative judgments to the descriptions of policies, giving “grades” to standards or the requirements for certification as a mathematics teacher. Those grades, particularly when published prominently, are also likely to influence continuing policy developments. Such evaluative studies might be considered studies of policy implementation. Once again, differences in values play an important role in the research: a narrowly specified standard might be seen as admirably specific or disastrously confining.

Implementation studies in mathematics education have shown the slippage between intended policy and implemented policy found in most policy research. Some research shows, for example, that assessments intended to be “challenging” include a large proportion of low-level content. Assessment policies may intend to match tests to standards, but the tests are more like each other than like the standards. Nevertheless, the studies often show some effects, generally in the intended direction. Revisions in assessments move them toward greater challenge or alignment, though not as far as policy makers generally intended.

Likewise, the research on effects of mathematics education policies finds policy impact muted by complexity of the phenomena and disputed on value grounds. The effects of policies are documented better for mathematics education than for other school subjects. Effects of teacher quality policies, for example, are usually found primarily in mathematics. Analysts continue to debate what research shows about these policies, due in part to disputes about which effects matter most (e.g., making the curriculum or increasing test scores) and about how key policy variables (e.g., alternate route, full certification) should be defined.

Research shows differences in development and focus across policy areas. Policies around standards were adopted earlier in mathematics than in other subject areas, so research is now examining the second and third revisions of these policies. After ini-

tial general enthusiasm, the initial standards policies were criticized on grounds of form, content, and process. Later versions of policies reflected responses to those criticisms; policy research on mathematics education standards policies also developed, with more refined questions and methods. The result has been a clearer picture of these policies which, though varied, have been shown to have effects.

Policy research on mathematics instructional materials is less well developed. The policies themselves have been less consistent over time and across contexts. The research on these policies is thin, especially in characterizing the intended content and effectiveness in improving student learning. Many of the materials are developed by private companies who may wish to protect their proprietary interests.

In contrast, assessment policies have been widely studied, with mathematics often chosen as the focal subject area. Studies consider assessment policy in conjunction with standards and accountability mechanisms. This policy area shows evidence of research informing further policy development, though other considerations also play a major role in setting policy. Disputes about the desired outcomes lead to disagreements about what direction policy should take, but assessment policy seems likely to grow in importance in the coming years.

Policy makers of every political persuasion agree that teacher quality is a key policy area. Many studies have described, and often judged, the status of teacher quality policies across states. Research on formulation, implementation, and effects is scarcer. Many obstacles impede research on teacher quality policies: key variables are imprecisely defined, myriad factors other than these policies affect their implementation and effects, and the organizations that mediate the policies (e.g., institutions of higher education) are themselves varied and notoriously difficult to influence. The work in mathematics education is, again, more informative than that in other subject areas, but the common conclusion reviewers draw is that the research base is thin.

The literature describes cases, especially around standards and assessments, where research has informed mathematics education policy. Policy will never be based solely on research; values, professional judgment and political expediency will always have important roles, and are not always in line with results of research.

Better connections among research, policy, and practice can be built, however. Such connections can increase the chances that policies achieve their intended effects and that mathematics educators understand the ways in which they are affected by, and might be

able to shape policies. Mathematics educators could also gain a better understanding of how policies and policy research are connected to goals to which various actors aspire.

Policy that affects mathematics education, when it has involved discipline-specific experts, relied primarily on mathematics educators in the period following the New Math, up until the mid 1990s, and thus tended to reflect the shared ideals and values of that community. In the past 10 years there has been a shift to widen the influence to include research mathematicians, whose ideals feature very strong commitment to the centrality of subject matter. Education policy researchers often take mathematics as the subject matter context for their investigations, but sometimes have a limited understanding of the ideals and values of both mathematics educators and research mathematicians. Greater collaboration among mathematics educators, mathematicians, and policy experts would be helpful in building a body of policy-relevant mathematics education research.

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MATHEMATICS CONTENT SPECIFICATION IN THE AGE OF ASSESSMENT

Norman L. Webb

WISCONSIN CENTER FOR EDUCATION RESEARCH

We live in the age of assessment. At no other time in the history of education in the United States, and for that matter in other countries, have so many people, resources, and time been devoted to measuring students' knowledge and performance in mathematics. Assessment has come to be viewed by policy makers and others to be more a lever for reform than merely an indication of the effectiveness of reform. The loosely coupled assessment system pervades every classroom in every school in the nation, at least in grades 3–8 and high school.

The three other assessment chapters (by William de Lange, and Wilson) in this handbook discuss formative assessments at the classroom level, and with regard to large-scale assessments and high stakes assessments. Collectively, their chapters address different purposes for assessments along with a number of other considerations for assessments. This chapter attends to a common issue, not directly addressed by the other assessment authors, that pertains to any purpose of assessment and the use of assessment by researchers. The mathematical content specification for an assessment needs to attend to the purpose for the assessment, but also needs to relate to the qualities of the assessment including its validity for making inferences from the results and its alignment to instruction and the curriculum. This chapter describes an approach for specifying the content for an assessment. The content specification criteria discussed here apply to assessments developed by classroom

teachers, to large-scale assessments, and particularly to assessments used by researchers.

An assessment is one instrument frequently used by researchers to measure student knowledge and learning. The mathematical content measured by an assessment will greatly influence the inferences and conclusions drawn from the results. If the researcher is trying to determine students' knowledge of a mathematical construct, such as proportional reasoning, then the items that are included on the assessment can determine what inferences can be made. If the items only measure students' computation of percentages, then the results would produce information on a very limited part of the proportional reasoning domain. An assessment on proportional reasoning could include items measuring students' computation of rates and ratios, along with reasoning with ratios and proportional relationships (e.g., similar figures and rates). The complexity of the items included on the instrument also could vary, from requiring simple recall of information to solving non-routine problems. How to construct, or select items, for an assessment designed to meet research needs and requirements requires consideration of a number of factors. Content specification is one of the most important. Of course, other factors also are important, including psychometrics—e.g., item difficulty, point-biserial correlations, differential item functioning (DIF) analysis, item format, and the length of the assessment.

Researchers who evaluate curriculum are faced with the issue of using assessments that are fair to all of the mathematics programs included in the analysis. An innovative curriculum can target content areas not normally included in traditional mathematical programs. One strategy is to use an existing instrument, such as a state assessment or national test. However, these instruments will very likely not capture all that students are learning in a given curriculum, including both traditional and innovative programs. Content analysis is one technique that can be used to reveal what content is covered in any assessment and how that coverage compares to the content emphases of the curricula under investigation. Knowing how content has been specified for existing assessment instruments can be very helpful in determining whether the assessment is appropriate for a given application.

In the age of assessment, research is being increasingly reported at the policy level. Student performance on state assessments is being compared to state scores on the National Assessment of Educational Progress (NAEP). Countries and states are being compared on international tests, such as the Third International Mathematics and Science Study (TIMSS) and *Programma for International Student Assessment* (PISA). Achievement gaps between different groups within a population are being studied and reported. Only rarely do such policy studies or other research studies use the mathematical content on the assessments as a basis for drawing conclusions. Sometimes such studies may report on mathematical topics included on the assessments, but do not report on the range of content assessed under these topics, or the complexity of the content measured. A language for content specification can facilitate such a discussion by helping researchers focus on what mathematics is being assessed on different instruments, to what depth that mathematics is being assessed, and how the emphasis on different mathematical topics varies among instruments.

Assessment has assumed increasing importance to schools and teachers with the advent of the *No Child Left Behind Act* in 2003 (see Chapter 24). Assessment results are being used to make decisions on the effectiveness of schools and to identify “failing” schools. And, beginning in 2005–2006, mathematics and reading are the two content areas in which states are to assess every student in grades 3–8 and once in high school each year. These tests are to be aligned to state standards. States that do not meet all of the requirements, as judged by peer review, are subject to the loss of millions of dollars of Title I funding. As a consequence, the annual state mathematics assessments have become very high stakes tests.

Schools faced with sanctions for poor results on the annual state assessments are incorporating benchmarks or other periodic assessments as one means of monitoring students’ progress in learning what is specified in state standards. These benchmark tests, which are administered one to six times during the school year, are to measure whether students learn the materials covered within a six-week period, or the extent to which students are progressing on content tested on the state assessment. Along with monitoring students’ progress, teachers are encouraged to use formative assessments (see Chapter 23) to reveal more about how students are understanding mathematics and what instructional interventions students need to further their understanding.

International comparisons of mathematics performance also have increased in the past decade. Whereas, nearly 20 years elapsed between the First and Second International Mathematics and Science Studies sponsored by the International Association for the Evaluation of Educational Achievement (IEA), assessments for the Third International Mathematics and Science Survey (TIMSS) were administered in grades 4 and 8 three times in eight years (1995, 1999, and 2003). This time span overlapped the administration of the Organisation for Economic Co-operation and Development’s (OECD) *Programma for International Student Assessment* (PISA) administered in 2000 and 2003 in a number of nations (see Chapter 25).

Policy makers and politicians are turning more often to assessments as a means for strengthening education and making schools, teachers, and students more accountable for student learning. Although large-scale assessments can be expensive to develop, administer, and score, the cost to mandate an assessment as a graduation requirement is far less than most of the alternatives—for example, teacher professional development, curriculum development, and reduced teacher/student ratios. But in the haste to comply with federally required timelines or state legislation to have every student be assessed by an end-of-year assessment or that students should pass an assessment to graduate from high school, the timeline to develop such assessments is circumvented or compressed beyond what best practice for assessment development requires. In particular, two years is a minimum time for developing a large-scale test because creation of such a test requires developing a test blueprint, selecting or developing at least twice the number of tasks that will be needed, piloting these tasks, discarding or refining the tasks, creating a test form, field testing the test form, conducting bias and alignment studies, refining the tasks and the test form, and, finally, field testing the test at that time during the school year at which the test will be adminis-

tered. However, in the United States, state assessment directors are being given less than one year to come up with tests that are to be administered statewide. This pressure comes in part from the number of assessments that need to be developed by states because of increasing the number of grades being assessed from three to seven and the number of different groups that are to be assessed including the regular population, English as a second language students, and students with disabilities. This pressure also is imposed by unrealistic expectations of state legislators who too quickly require students to pass a test for graduation with the intent of raising expectations that will lead to improve learning by students.

The push to “politicize” assessments by holding schools and teachers more accountable for the results on the assessments and the urgency to impose such accountability systems on states and schools has resulted in less attention given to general principles that are generally applicable to any form of assessment.¹ A paper-and-pencil test is one of a variety of assessment forms. Others include making observations, analyzing student work samples, and interviewing students. Strong pressures on teachers to meet accountability requirements and to avoid sanctions can result in the assessed curriculum becoming the enacted curriculum (Porter & Smithson, 2001a, 2001b). Under such pressures, teachers can be forced to make their classroom assessments (and even their classroom instruction) conform more with those used for the accountability system (generally, multiple-choice items) and have their instruction restricted to what is measured on the large-scale high stakes assessments. What can happen under such pressure is that teachers may give less attention to how individual students are developing their mathematical understanding, may not use assessment techniques that reveal students’ thinking, may make decisions on too little information (e.g., one or two assessment items), may not be able to rectify different results from different sources of information, and may disregard the mathematics that is being assessed and how this mathematics relates to the mathematics students should be learning. In short, the use of such high-stakes assessments can have very unfortunate effects on teaching practices.

Assessment is a process for making inferences about what students know and can do related to a content domain. In theory, any assessment produces only a sample of what a student knows and can do. Unless the content domain, a body of knowledge, is very narrow, any assessment is too limited to measure all possible ways students may be able to demonstrate

their knowledge of the content domain. Results from an assessment then produce some information about what a student knows. How much confidence an educator can have in this information for inferring what a student knows about a content domain will depend on qualities of the assessment, such as reliability and validity. A clear statement of the content domain is necessary to establish content validity for an assessment. Without specifying the content in some detail, making inferences about what a student knows or can do in relation to a content domain is very difficult. Clear content specifications are an essential component for assessment. In what follows, I discuss specific criteria that can be used in specifying content for an assessment.

CONTENT SPECIFICATION

Central to any form of assessment is specifying what knowledge of mathematics or what mathematical skills should be measured (Webb, 2006). Content specification is an issue at each level of education, from that of a teacher inquiring about what a student knows and can do to that of assessment developers creating a national or international test. At each level, a number of choices have to be made to decide what assessment tasks to include in order to produce both reliable and valid information about a student, or a group of students. Frequently, inferences are made about what students know and can do that are beyond the capacity of the assessment instrument to provide. In part, this is because those making the decisions do not fully understand what content knowledge is being assessed and how the assessment instrument was constructed. An item analysis of a large-scale assessment can result in teachers making instructional decisions about students based solely on their responses to one item, which is too restricted a sample to indicate much beyond the results of the individual item. States also try to make multiple uses of a single assessment, such as measuring the expected standards for K–12 and predicting performance in higher education. How the content is specified for different assessments can be very revealing about those inferences that can and cannot be made from students’ results.

Content specifications need to be shaped by the structure of the content knowledge being assessed. Mathematics has domains of knowledge and inquiry that are interconnected, but generally are referred to as separate areas of study; for example, school math-

¹ Here assessment is used in its broadest sense as gathering information about student learning to make decisions.

ematics is generally broken down into topics such as number, geometry, measurement, algebra, data analysis, probability, and statistics. One starting point for specifying content for a mathematics assessment is to identify what the major mathematical topics are that should be represented on the assessment and what emphasis should be given to each. But these are broad topics that need further delineation. Not only do assessment developers need to know more about what subtopics are to be assessed under each topic, but they need other information, such as the level of complexity that should be represented on the assessment, the form of the assessment tasks (multiple-choice, short-answer, constructed-response), and whether items should measure isolated skills, or assess how students have integrated their understanding across skills. Also, assessment developers should know whether the results from the assessment will be reported by topic, as a total score, or in some other way. Even a teacher who wants to determine what a student knows about a very isolated topic, such as addition of whole numbers, needs to specify in some way the domain of knowledge the student is to know and what the indicators are that the student either has this knowledge, or is progressing toward acquisition of this knowledge. Content specifications then need to be based on the structure of the content and an understanding of how students acquire greater sophistication in mathematics. That is, content specifications should be theory based along with assessment in general (Webb, 1992).

For a large-scale assessment, content specifications take the form of a test blueprint or assessment framework. For classroom assessment, content specifications are presented more informally, but should be tied to the content standards and expectations, as well as to a theory of learning. For both large-scale and classroom assessment, understanding how students learn and how students demonstrate their learning should influence the specification of content for an assessment (National Research Council, 2001).

In the late 1940s, Ralph Tyler introduced into curriculum evaluation the concept of considering multiple dimensions when making a judgment about curriculum effectiveness (Tyler, 1949). In 1956, Benjamin Bloom, a student of Tyler, and his colleagues produced a taxonomy for the purpose of specifying behavioral levels that could be used to evaluate the sophistication of educational goals and assessments (Bloom, Engelhart, Furst, Hill, & Krathwohl, 1956). This was at the time that behavioral psychology was a dominant view of learning and the mathematics education community embraced Bloom's Taxonomy in designing assessments and curriculum. The content-by-behavior assessment matrix of the National Lon-

gitudinal Study of Mathematical Abilities (NLSMA), a project of the School Mathematics Study Group, was greatly influenced by Bloom's work. This matrix listed mathematical topics as the rows (number, geometry, measurement, and algebra) and behavioral levels as columns (computation, knowledge, application, and problem solving). Assessment content is then specified by the percent of the items that should correspond to each cell (number by computation, geometry by application, etc.). But with the rise of cognitive psychology, greater attention began to be given to how students develop relationships among mathematical ideas and how their mental schemas form through mental construction, rather than by rote stimulus and response. Content-by-behavior matrices became less frequently used in structuring assessments. For example, the writers of the National Council of Teachers of Mathematics (NCTM) *Curriculum and Evaluation Standards for School Mathematics* (NCTM, 1989) avoided specifying a content-by-behavior matrix in the assessment component, but rather focused on different types of knowledge. The argument against using a content-by-behavior matrix was that mathematical knowledge should not be pigeonholed into discrete cells (comprehension—number, application—geometry, etc.), but rather should be assessed by activities that extend across cells. That is, an assessment item or task may require a student to use number and geometry, along with recall of facts and strategic thinking. Or, assessments may be structured as a cluster of tasks all building on a single context. Similarly, the National Assessment of Educational Progress framework for the 1990–2003 incorporated the *Curriculum and Evaluation Standards* and used a content-by-abilities matrix in its assessment framework (National Assessment Governing Board, 1996). In this framework five content strands was one dimension while three mathematical abilities served as the second dimension (see Table 30.1).

With the advent of systemic reform in the 1990s and the need to consider the relationships among components of the educational system, the concept of alignment grew in importance. Systemic reform is based on the premise that in order to produce lasting increases in student performance it is necessary to change a multiplicity of components simultaneously, rather than focusing on only one component such as professional development, or curriculum, or assessment. As such, alignment can be defined as the degree to which expectations and assessments are in agreement and serve in conjunction with one another to guide an education system toward students learning what they are expected to know and do (Webb, 1997).

Table 30.1 Content-by-Abilities Matrix Used for the 1990–2003 National Assessment of Educational Progress (NAEP) Mathematics Framework

Content Strands	Conceptual understanding	Procedural knowledge	Problem solving
Number sense, properties, and operations			
Measurement			
Geometry and spatial sense			
Data analysis, statistics, and probability			
Algebra and functions			

The notion of alignment was central to systemic reform and became a critical requirement in the reauthorization of Title I in 1994, requiring states to show that assessments used in monitoring the progress of students served by Title I are aligned with standards. The national *No Child Left Behind Act* requires that all states develop challenging academic content and student achievement standards and tests that are aligned to these standards in the core academic areas of reading/language arts, mathematics, and science (U.S. Department of Education [USDE], 2003). In the establishment of *No Child Left Behind* legislation, alignment of assessments and standards, as judged by a peer-review process, was instituted as one of the key requirements states had to meet to get approval by the USDE.

In addition to the stature it has been given by the USDE, the alignment of expectations for student learning with assessments for measuring students' attainment of these expectations is an essential attribute for an effective standards-based education system. Alignment is a quality of the relationship between expectations and assessments and not an attribute of any one of these two system components. Alignment describes the match between expectations and assessments that can be legitimately improved by changing either student expectations or the assessments. As a relationship between two or more system components, alignment is determined by using the multiple criteria described in detail in a National Institute for Science Education (NISE) research monograph, *Criteria for Alignment of Expectations and Assessments in Mathematics and Science Education* (Webb, 1997).

In standards-based education, alignment criteria are additional considerations for assessment frameworks. Test developers at all levels need to consider the coherence of the system and how the assessments relate to standards, curriculum, and instruction. Five

criteria help specify attributes of assessments that are considered aligned with standards and other system components:

- Categorical Concurrence
- Depth-of-Knowledge Consistency
- Range-of-Knowledge Correspondence
- Balance of Representation
- Structure-of-Knowledge Comparability

Categorical Concurrence

An important aspect of alignment between content standards and assessments is whether both address the same content categories. The categorical-concurrence criterion provides a very general indication of alignment if both documents incorporate the same content. The criterion of categorical concurrence between domains and an assessment is met if the same or consistent categories of content appear in both documents. This criterion is judged by determining whether an assessment includes items measuring content from each content standard. The number of items that should be included on an assessment to produce some information about a student's performance on a content standard or domain of knowledge is primarily a question of reliability. How many items should a test include to produce reliable inferences from the student's responses about the student's proficiency on a content standard or domain of knowledge? Other important factors for consideration, such as breadth of coverage with a sufficient level of complexity, are judged by the other alignment criteria.

The minimum number of items or tasks that should be used to make judgments about a student's knowledge and proficiency of a content standard or domain of knowledge depends on the level of precision of the judgments. In general, the greater the

number of items, the more precise will be the judgments. However, tests and other means for gathering information on student knowledge are constrained by time and other factors. An argument can be made for six items as being the minimum acceptable level for meeting the *Categorical Concurrence* alignment criterion if an assessment is to produce reliable information about what a student knows within a content standard. The number of items, six, is based on estimating the number of items that could produce a reasonably reliable subscale for estimating students' mastery of content on that subscale. Of course, many factors have to be considered in determining what a reasonable number is, including the reliability of the subscale, the mean score, and cutoff score for determining mastery. Using a procedure developed by Subkoviak (1988) and assuming that the cutoff score is the mean and that the reliability of one item is .1, it is estimated that six items would produce an agreement coefficient of at least .63. This indicates that about 63% of the group would be consistently classified as masters (a designated number of items scored correctly as determined by some process to represent a satisfactory level of attainment) or nonmasters, if two equivalent test administrations were employed. The agreement coefficient would increase if the cutoff score were increased to one standard deviation from the mean to .77 and, with a cutoff score of 1.5 standard deviations from the mean, to .88. Usually states do not report student results by content domains, or require students to achieve a specified cutoff score on subscales related to a domain. If a state did do this, then the state would seek a higher agreement coefficient than .63. But, six items are assumed as a minimum for an assessment measuring content knowledge related to a content standard or domain and as a basis for making some decisions about students' knowledge of that domain. If the mean for six items is 3 and one standard deviation is one item, then a cutoff score set at 4 would produce an agreement coefficient of .77. Any fewer items with a mean of one-half of the items would require a cutoff that would only allow a student to miss one item. This would be a very stringent requirement, considering a normal standard error of measurement on the subscale (Downing & Haladyna, 2006).

A minimum number of tasks or amount of information also applies to classroom assessments. Teachers, through probing a student's thinking, can gain insight into the student's conception of a mathematical idea or how the student may be drawing a relationship among mathematical ideas. However, for a teacher to make a generalization about a student's conception to all instances of the mathematical idea or to a domain of knowledge, the teacher will need to observe or in-

teract with the student on a number of occasions and with different examples of the same mathematical idea or ideas to assure that any conclusions reached about the student's knowledge are valid. A teacher will have increasing confidence about a student's mathematical knowledge and performance with additional observations in different settings.

Depth-of-Knowledge Consistency

Content standards and assessments can be aligned not only on the category of content covered by each, but also on the basis of the complexity of knowledge required by each. Depth-of-knowledge consistency between content standards and assessment indicates alignment if what is elicited from students on the assessment is as demanding as what students are expected to know and do as stated in the standards. Four depth-of-knowledge levels are defined to compare the content complexity of content standards with the complexity of an assessment. The definitions were influenced by the Bloom Taxonomy and other schemes for considering content complexity, but are not the same. Bloom's Taxonomy was created at a time when behaviorism was a dominant psychological theory and it strongly reflects this perspective in portraying levels of knowledge. The depth-of-knowledge levels are designed to be used in content analyses and to portray content complexity that is related to cognitive processing (Webb, 1997). The judgment is based on a content analysis rather than on a psychological analysis. Thus, the cognitive processing is inferred from an analysis of the structure of content and the knowledge that students have at a given grade level. For the depth-of-knowledge levels to be verified as cognitive levels would require significant research in cognitive laboratories to determine whether, in fact, these levels represent meaningful degrees in mental functioning. As the definition now exists, there clearly is some relationship between cognition and content complexity, but the main use of the definitions is to compare content descriptions and to use this analysis to make some judgment of the comparability between the complexity of the expectations for student learning and what is required by the assessment.

Interpreting and assigning depth-of-knowledge levels to both standards within domains and assessment tasks is based on the degree to which the standards expect and the tasks require students to process mathematical ideas, from very routine algorithms to non-routine problem solving (Webb, 1999). The descriptions of the levels for mathematics are:

Level 1 (*Recall*) includes the recall of information such as a fact, definition, term, or a simple procedure,

as well as performing a simple algorithm or applying a formula. That is, in mathematics, a one-step, well defined, and straight algorithmic procedure should be included at this lowest level. Other key words that signify Level 1 include “identify,” “recall,” “recognize,” “use,” and “measure.” Verbs such as “describe” and “explain” could be classified at different levels, depending on what is to be described and explained.

Level 2 (*Skill/Concept*) includes the engagement of some mental processing beyond a habitual response. A Level 2 assessment item requires students to make some decisions as to how to approach the problem or activity, whereas Level 1 requires students to demonstrate a rote response, perform a well-known algorithm, follow a set procedure (like a recipe), or perform a clearly defined series of steps. Keywords that generally distinguish a Level 2 item include “classify,” “organize,” “estimate,” “make observations,” “collect and display data,” and “compare data.” These actions imply more than one step. For example, to compare data requires first identifying characteristics of the objects or phenomenon and then grouping or ordering the objects. Some action verbs, such as “explain,” “describe,” or “interpret,” could be classified at different levels depending on the object of the action. For example, interpreting information from a simple graph, requiring reading information from the graph, also is at Level 2. Interpreting information from a complex graph that requires some decisions on what features of the graph need to be considered and how information from the graph can be aggregated is at Level 3. Level 2 activities are not limited only to number skills, but can involve other sorts of skills (e.g., visualization skills and probability skills). Other Level 2 activities include noticing and describing non-trivial patterns, explaining the purpose and use of experimental procedures; carrying out experimental procedures; specifying an algebraic equation and solution; making observations and collecting data; classifying, organizing, and comparing data; determining congruent figures; and organizing and displaying data in tables, graphs, and charts.

Level 3 (*Strategic Thinking*) requires reasoning, planning, using evidence, and a higher level of thinking than the previous two levels. In most instances, requiring students to explain their thinking is at Level 3, as are activities that require students to make conjectures. The cognitive demands at Level 3 are complex and abstract. The complexity does not result from the fact that there are multiple answers, a possibility for both Levels 1 and 2, but because the task requires more demanding reasoning. An activity, however, that has more than one possible answer and requires students to justify the response they give would most like-

ly be at Level 3. Other Level 3 activities include drawing conclusions from observations; citing evidence and developing a logical argument for concepts; explaining phenomena in terms of concepts; and using concepts to solve problems.

Level 4 (*Extended Thinking*) requires complex reasoning, planning, developing, and thinking, most likely over an extended period of time. The extended time period is not a distinguishing factor if the required work is only repetitive and does not require applying significant conceptual understanding and higher-order thinking. For example, if a student has to take the water temperature from a river each day for a month and then construct a graph using the data gathered, this would be classified at Level 2. However, if the student is to conduct a river study that requires taking into consideration a number of variables, this would be at Level 4. At Level 4, the cognitive demands of the task should require analyzing and making inferences by considering a number of sources. Work could entail developing an original model for representing a problem with a number of facets. Students should be required to make several connections—to relate ideas within the content area or among content areas—and to select one approach among many alternatives on how the situation should be solved, in order to be at this highest level. Level 4 activities include developing and proving conjectures; designing and conducting experiments; making connections between a finding and related concepts and phenomena; combining and synthesizing ideas into new concepts; and critiquing experimental designs.

Depth-of-knowledge of an assessment task is different from the difficulty of that task. Difficulty of an item is a psychometric term that is represented by the percent of students who answer the item correctly. The difficulty of an item is related to depth-of-knowledge, but is also determined by other factors, such as the students’ opportunity to learn and repetitive actions. It is conceivable that an item at depth-of-knowledge (DOK) Level 1, such as “Recall the first 20 digits of pi,” is difficult, whereas an item that requires strategic reasoning, but of a type for which students have had extensive instruction, is actually less difficult.

Depth-of-knowledge is as relevant to classroom assessment as to large-scale assessment. For teachers to make judgments about students’ mathematical knowledge requires them to assure that the tasks used for students to demonstrate their knowledge is at an appropriate depth-of-knowledge level. Within a standards-based system, the appropriate level is defined by the content standards. In other systems, the appropriate level could be defined by the expected rigor,

or factors used to determine which students have the requirements for advancement or certification.

Range-of-Knowledge Correspondence

For assessments to be considered aligned with content standards, the assessments need to cover an adequate breadth of content, as expressed in the standards, in addition to addressing the categories and depth-of-knowledge. The range-of-knowledge correspondence criterion is used to judge whether a comparable span of knowledge expected of students by a standard is the same as, or corresponds to, the span of knowledge that students need in order to correctly answer the assessment items. The criterion for correspondence between span of knowledge for a content standard and an assessment considers the proportion of more refined delineations of content that are addressed on the assessment. Most content standards are further described by objectives or benchmarks. The range-of-knowledge correspondence criterion then considers the proportion of the objectives or benchmarks under a standard that has at least one corresponding item. In alignment studies, an assessment is considered to have attained a minimum acceptable level for range-of-knowledge correspondence if it has items corresponding to at least 50% of the objectives under a standard.

The concept of range applies to the relation between an assessment and other statements of content besides content standards. In any specification of a content domain of knowledge—e.g., operations with whole numbers, problems requiring proportional reasoning, or mathematical proof—the content can be further described by more specific content subdomains. For example, operations with whole numbers can be divided into addition with whole numbers, subtraction, multiplication, and division. The 2005 mathematics framework for the National Assessment of Educational Progress (NAEP) identifies five main content areas:

- number properties and operations,
- measurement,
- geometry,
- data analysis and probability, and
- algebra.

Each of these content areas is further defined by more refined content topics. For example, number properties and operations include:

- represent numbers,
- order numbers,
- compute with numbers,

- make estimates appropriate to given situations,
- use ratios and proportional reasoning, and
- apply number properties and operations to solve real-world and mathematical problems.

An acceptable range for NAEP would exist if the assessment included items from all six of these areas under Number Properties and Operations because the purpose of NAEP is to produce information on the performance of students in the states and the nation as a whole on the full spectrum of mathematical content. Creators of more limited assessments, constrained by the time available for testing and the number of possible items, have to make choices as to the number of the subtopics that can be addressed by an assessment. What proportion of the subtopics should be the minimal acceptable level for range will then depend on how student results will be reported, the purpose of the assessment, the number of subtopics, and the priorities assigned to topics. A test developed by a teacher, for example, to determine what students learned from a unit of instruction may only need to have items that related to number computations and estimation.

Range-of-knowledge correspondence is more difficult to attain if the content expectations are partitioned among a greater number of domains and a larger number of objectives. If a content standard or topic is further defined by a long list of very detailed and narrow content topics, then a greater number of items or more compound items (those that assess more than one mathematical idea) are needed to address an adequate proportion of these very detailed topics for an assessment to be considered as having sufficient range as expressed by the expectations. Very detailed statements of standards have been a problem for some states attempting to show that their assessments meet the alignment criteria of range. Because states are required to show that their assessments measure an adequate breadth of content as expressed by the state's content standards, this is a greater challenge if the five or six mathematics standards are delineated in a long list of objectives. Some states had developed standards in the mid 1990s to serve as curriculum guides for teachers. The intent of these curriculum standards was to give teachers as much detail as possible to help them make instructional decisions for students. The *No Child Left Behind* mandate, however, requires states to be held accountable for assuring that their assessments cover an adequate breadth of content, as expressed in their standards. As such, the curriculum standards are being used as accountability documents, not simply as curriculum guides. Standards serving as an accountability document do not require the detail in specifying

ing what content students should be able to know and do. As an accountability document, standards can articulate in general terms the expected knowledge and performance, while leaving the detailed specifications for other documents. Moreover, standards used for accountability should contain sufficient detail to guide policy development, assessment, and curriculum. Then the more detailed description of content, and precisely what students should know and do, can be explained at the required level of specificity in curriculum frameworks or assessment blueprints.

Balance of Representation

In addition to comparable depth and breadth of knowledge, aligned content standards and assessments require that knowledge be distributed equally in both. The range-of-knowledge correspondence criterion only considers the number of objectives or subtopics within a content standard or domain; it does not take into consideration how the assessment items or activities are distributed among these standards. The balance-of-representation criterion is used to indicate the degree to which one objective or subtopic is given more emphasis on the assessment than another and whether this is appropriate.

Not all mathematical content should be given the same emphasis across the grades. Number and operations generally would have greater emphasis than other topics in the earlier grades, while algebra and geometry would increase in emphasis in the higher grades. Also, within these broad mathematical areas, content topics will vary in importance, a factor that should be reflected on an assessment designed to measure students' mathematical knowledge as expected by curriculum standards or by other forms of expectations. Generally, assessment frameworks, such as NAEP, will specify how items should be distributed among the major top-

ics or content areas (see Table 30.2). Similar attention should be given to constructing a classroom assessment or evaluating a student's progress in learning. Not all mathematical content and ideas have the same degree of importance or the same number of attributes to warrant greater emphasis or a greater number of assessment tasks. Assessments should reflect these variations in importance and complexity and have appropriate balance of representation as reflected in the content standards and expectations.

Structure-of-Knowledge Comparability

The underlying conception of mathematics knowledge in expectations and assessments should be in agreement. For example, if standards indicate that students should understand mathematics "as an integrated whole" (National Council of Teachers of Mathematics, 1989), then the assessment activities should be directed toward the same ends. Both expectations and assessments should embody similar requirements for how students are to draw connections among ideas. Assessment of knowledge only as fragmented skills, for example, would not be consistent with the conception of how students should understand mathematics through the relationship among ideas as expressed in NCTM's *Standards* documents (1989, 1995, 2000).

The preliminary draft of the Illinois Academic Standards (Illinois State Board of Education, 1996) stated, "Solving problems is at the heart of mathematics. Mathematics is a collection of concepts and skills; it is also a means of investigation, reasoning, and communicating" (p. 25). One of five applications of learning in this document was "Making Academic Connections: Recognize and apply connections of important information and ideas within and among academic learning areas" (p. 26). The depiction of mathematics in these statements implies that students should learn a collection

Table 30.2 Distribution of Mathematics Questions as Specified by the NAEP 2005 Mathematics Framework (National Assessment Governing Board, 2004)

Content Areas	Grade 4 Target	Grade 8 Target	Grade 12 Target
Number properties and operations	40	20	10
Measurement	20	15	30*
Geometry	15	20	
Data analysis and probability	10	15	25
Algebra	15	30	35

* Measurement and geometry are combined for grade 12.

of concepts and skills along with learning how to solve problems and applying information to other fields. An assessment system aligned with a comparable structure of knowledge as represented by these expectations would have to gather information on students' understanding of specific concepts and skills, their ability to solve problems, and their application of mathematics to other academic learning areas.

Judging the structure of knowledge comparability, along with many of the other criteria, requires considering the full assessment system. Expectations and assessment are fully aligned according to this criterion if the structure of mathematical understanding expressed in expectations is represented in the assessment system. In the case of judging whether its assessment system is aligned with the Illinois Academic Standards, some part of the state system should gather information on how students are able to identify appropriate applications of mathematical ideas both within mathematics and in other fields. For example, students could be asked to develop a mathematical model of a physics experiment. There would be less alignment with the Illinois Academic Standards if little or no part of the assessment system asked students to demonstrate knowledge of how they are able to draw relationships among mathematical ideas and among mathematical ideas that have applications to other fields. The standards and assessments will show very low alignment in the structure of knowledge if students are only asked to demonstrate their knowledge of mathematics by recalling or applying isolated concepts, procedures, and skills.

As with the other alignment criteria, the structure-of-knowledge comparability applies to classroom assessments as well as to large-scale assessments. In determining what mathematics students know and can do, teachers need to consider the interaction among mathematical ideas and how students should make sense of these. In a standards-based system, the curriculum standards should communicate or imply how students are to understand the structure of mathematical ideas. When using other forms that indicate expectations, such as a curriculum document or textbook, teachers may need to probe deeper in order to understand the underlying connections among mathematical ideas and how students are to build a more sophisticated understanding of mathematics.

Structure-of-knowledge comparability also is related to the theory of learning that is implied in the curriculum and the expectations for learning. Such a theory of learning needs to clarify how students' knowledge of the content area develops, how their knowledge deepens through forming relationships among the concepts and skills in the content area, and the pathways students can use to achieve competence in

the content area (Webb, 2006). A fully aligned assessment will represent the underlying theory of learning and produce information that is comparable to this theory and how students are to know mathematics.

Beware of Using a Single Assessment for Multiple Purposes

The five alignment criteria I have discussed apply to all forms of assessments as discussed by de Lange, Wiliam, and Wilson (in this volume). Alignment is critical in a high-stakes standards-based system, as discussed by Wilson. In other high stakes environments such as selecting students for post-secondary education or in competitions, alignment of assessment content to standards or curriculum is less a concern than having reliable assessments with adequate predictive validity. Assessments used to select students should have psychometric qualities that are sufficient for use in verifying that the decisions made are accurate. For such tests, the criteria of range and depth of knowledge are of less concern than including items that will distinguish those students who have a greater probability of being successful in higher education from those who have a lower probability of being successful, or those who are experts from novices (Chi, Glaser, & Rees, 1982). The depth and breadth of items on the assessment is of concern only to the degree that these attributes predict what students will be required to know and do for future studies or to identify those who have specialized knowledge.

Some state assessment programs are confronting the issue of using one assessment for multiple purposes. State legislatures and policy makers (e.g., Michigan and Maine) are mandating that the high school graduation assessment be comparable to a college entry assessment, such as the College Board SAT or American College Testing ACT. Their rationale is that the K–12 curriculum in the state should prepare students sufficiently so they are ready for higher education. The high school graduation test should look forward to preparation for higher education rather than looking backwards to coverage of the curriculum. The preparation required for higher education is not always considered the same as what should constitute a quality K–12 education. A K–12 education should prepare students for higher education, but it must also prepare students to go into the work force and to become good citizens. For example, a strong background in algebraic rigor is important preparation for the study of mathematics in higher education, whereas a quality K–12 mathematics program may include data analysis, probability, and applying mathematical knowledge in context, as discussed by de Lange in this volume.

The state of Michigan conducted a study of the alignment of ACT and SAT assessments with its high school content standards to determine whether either of these assessments would be considered to be aligned with the content standards (Webb, 2005). If one of these assessments used to predict success in higher education was considered aligned as required by the *No Child Left Behind Act*, then the state could serve two purposes with the one assessment. The study indicated that the Michigan Educational Assessment Program (MEAP) assessment (51 items) had the best alignment with the state's challenging curriculum standards. Both the ACT (60 items) and the SAT (54 items) were found to overemphasize solution of algebraic equations, simplifying expressions, and determining geometric properties to a greater extent than is needed to sufficiently provide a measure of students' attainment of the stated expectations. Data analysis and probability were not considered sufficiently emphasized on the ACT and SAT in relationship to the Michigan content standards. Since both the ACT and the SAT have proven records in making higher education entry decisions, modifying either of these was not an option. The only other option was to supplement each of these assessments with items in the content areas with insufficient coverage in relationship to Michigan's content standards. Assessments are generally designed for specific purposes. When the results of an assessment are used for other than the intended purpose, there is an increased chance that invalid inferences will be made from the assessment results.

SUMMARY

In the age of assessment, gathering information to make decisions about students is pervasive in all aspects of the educational system. Critical to producing reliable and valid information about students' mathematical knowledge is having clear specifications about what students should know and be able to. Clear content specifications apply to all forms of assessments including large-scale assessments, high-stakes assessments, and formative assessments. In constructing or selecting an assessment, different criteria can be used to express the content students should know and be able to do, or how the content is measured by an assessment aligned to content standards or other statements of expectations. One criterion is that the assessment should include a sufficient number of tasks related to each topic so that reliable decisions can be made by the student's knowledge of each topic (*Categorical Concurrence*). The assessment also should

include tasks at a sufficient level of complexity (*Depth-of-Knowledge Consistency*) and that adequately cover the content (*Range-of-Knowledge Correspondence*). Because not all areas within a content domain have equal importance, the specification of content should include some indication of what emphasis should be given to different content areas within a content domain, and the distribution of items on the assessment should reflect this relative emphasis (*Balance of Representation*). In mathematics, rarely are students to learn only isolated facts, skills, and concepts; rather, they are expected to learn how mathematical ideas relate to each other and how to use these ideas in concert with each other. How the mathematical ideas within a content domain are structured and how students are to integrate these ideas with ideas in other content domains should be evident in expectations for student learning and in how assessments are constructed (*Structure-of-Knowledge Comparability*).

These criteria for content specifications are only a few factors that need to be considered in developing an assessment. Other factors include the underlying theory of learning, item type, length of the assessment, purpose, and how the results are to be reported. Another important consideration is that policy makers who mandate assessments, teachers, and consumers for assessment results all need to have some basic assessment literacy about what can and cannot be concluded from the scores of different types of assessments and what is necessary to develop a good assessment for a specific purpose. Advancement through the age of assessment requires taking responsibility for the appropriate use of assessment results, research on how assessments can be more effectively utilized by teachers to further student learning, and less invasive ways to gather information from students. Critical to all of this is a clear understanding and specification of the mathematics to be assessed.

Researchers have a special responsibility for attending to the specification of content on any assessment instrument they use. Reporting of research findings should include some indication of what content knowledge was assessed and how representative the content measured by an instrument is of the content from the targeted domain. This chapter has identified a number of dimensions for specifying content for assessment. In the age of assessment with the proliferation of tests, it is easy to find assessment results at all levels of the educational system. However, researchers who use results from existing assessments and their own instruments are challenged to insure that they are clear about what content knowledge and skills are being measured and that they have attended to the different dimension for specifying content.

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REFLECTIONS ON THE STATE OF AND TRENDS IN RESEARCH ON MATHEMATICS TEACHING AND LEARNING

From Here to Utopia

Mogens Niss

ROSKILDE UNIVERSITY, DENMARK

INTRODUCTION: IMAGINE UTOPIA

Why do we do research on the teaching and learning of mathematics? Although there are several different answers to this question depending on the degrees to which—and the ways in which—respondents are involved in the research enterprise, there is, I think, one ultimate answer that lies underneath all others: We do research on the teaching and learning of mathematics because there are far too many students of mathematics, from kindergarten to university, who get much less out of their mathematical education than would be desirable for them and for society. We believe that they could learn much more, and in much better ways, if the conditions and circumstances for teaching and learning were different. Simply put, at the end of the day research in mathematics education is done because we want to know why students do not get enough out of their mathematical education, and what we can do to remedy the situation.

Now, let us imagine that Utopia will have dawned on mathematics education research in, say, 2050, in the sense that the discipline will, by then, be able to

provide final, exhaustive and effective answers to all important questions in our field. What would this mean, and what would the imagined situation look like, more specifically? By referring to this situation as “Utopia,” it has already been admitted that it will never occur (and if it did, mathematics education researchers would be out of business, except—perhaps—as custodians of eternal treasures!). Nevertheless, pondering on the possible characteristics and potential implications of Utopia may be a means for reflection on where our field is today, and on where we are, or should be, heading in the future. Of course, the entire endeavour presupposes that by 2050 there will still be such a thing as explicit mathematics teaching and learning for at least some categories of students who need our services. In other words, it is a basic assumption for the following deliberations that the teaching and learning of mathematics will not have been integrated away into the teaching and learning of other subjects or been hidden in technological systems focusing on something else. Another basic assumption is that there will still be such a thing as institutions devoted to education (i.e., schools and universities and suchlike).

I submit that Utopia would mean that we would know and understand—and be able to document it—how to plan, orchestrate and carry out mathematics teaching so as to ensure that every student accomplishes an optimal learning of mathematics in accordance with his or her needs, wishes, capacity, and potential. On closer inspection, reaching Utopia would involve (at least) the following.

First of all, we would know and understand *why we should teach mathematics* to various categories of students in society, both from static and dynamic perspectives related to the development of society, acknowledging that the answer(s) to this question might be different for different student categories in different societies. This knowledge would be based on well founded and documented surveys of the location of mathematical insight and competencies in various quarters of society, thus providing convincing answers to the derived question: *who in society needs what mathematical insight and competencies, and for what purposes?*, where “society” is to be understood in a multi-faceted sense.

We would know and understand, in concrete terms, exactly what it means *to master mathematics* in all its dimensions and facets. More specifically, we would know, in an exhaustive and complete way, what the fundamental constituents of such mastery are, at an *overarching level* across age, grades, and institutions, and across mathematical topics. We would further know what it means to master different topics such as numbers, algebra, geometry, calculus, analysis, probability etc., as well as the interplay between topics. And we would know what it means and takes to be able to put different kinds of mathematics to use for extra-mathematical purposes in a variety of contexts and situations. We would further know what it means to be able to make mathematical inferences and to justify or assess mathematical statements and claims, as we would know what it is to pose and solve mathematical problems, and so forth and so on.

Then we would also have a complete knowledge and understanding of *how learning of mathematics takes place*, again in all the dimensions and facets of mathematics, for different categories of individuals and groups as a function of personal, social, societal, cultural, and contextual variables. This would also include knowing possible *limits* to mathematical learning for different sorts of individuals, for example as a function of biological, motivational and attitudinal factors. Moreover, we would know what paths mathematical learning can take, and what factors determine which paths are actually taken by different students under different circumstances.

We would know the life situations, interests, pre-occupations, and motivations of different categories of students so well that their *beliefs, affects, and attitudes to mathematics and its study*, and the impact of these

on their learning approach and behavior, would be known and understood by us. We would further know how we could influence (if desirable) these beliefs, affects, and attitudes in ways that pay due respect to students’ personal, social and cultural integrity.

Furthermore, we would have a complete knowledge and understanding of *how teaching of mathematics can assist* different individuals and groups of learners in accomplishing optimal learning of mathematics according to their needs and potentials. This would include knowing the ways, approaches, methods and techniques that help bringing about such learning, as well as the boundary conditions and limitations of their use and effect, as it would imply knowing the circumstances under which an approach or a method should not be put to use, as well as the effects of different approaches or methods under given circumstances. This would further include complete knowledge and understanding of what various kinds of learning materials and teaching aids (e.g., textbooks, ICT systems, concrete materials), and student activities can offer to the teaching and learning of mathematics, and how they can be utilized in optimal ways.

In addition, we would know and understand how the teaching and learning of mathematics can be viewed in relation to the teaching and learning of *other subjects*. What significant similarities (and differences) are there between mathematics and other educational subjects in these respects? This would also imply that we knew to what extent, and how, mathematics teaching and learning can gain from interaction with other subjects, both in terms of content and in terms of organized co-operation between subjects.

Assuming that live teachers will still be needed in Utopia as key players in the teaching of mathematics, we would have a complete overview and understanding of *what competencies different kinds of mathematics teachers, from pre-school to university, should possess* so as to be able to design, plan, orchestrate and carry out optimal teaching of mathematics. We would know *how to prepare* such teachers through pre-service education, and how to *maintain and develop their professional identity and competencies* by means of in-service activities of various sorts. We would further know and understand teachers’ beliefs, affects and attitudes towards mathematics and its teaching and learning, and the influence of these on students’ learning outcomes.

In Utopia we would have complete knowledge of the *ways in which students’ mathematical competencies and achievements can be assessed* in comprehensive and effective ways. This implies knowing the characteristics and qualities of a full array of assessment modes and instruments, including their administration and operation, again as a function of the boundary conditions, contexts and situations at issue. It would further

include knowing and understanding when to implement what modes and instruments for what aims.

Finally, we would know exactly *what policies*, conducted at different decisional and administrative levels, involving individuals, institutions, organizations, and national or international authorities, *would be efficient* so as to reach the objectives that follow from the above-mentioned points.

When it comes to the *meta-level*, we would have a complete overview and understanding of *mathematics education research as a discipline*, including its characteristic research questions, philosophical underpinnings, theoretical constructs, methodologies, techniques, and results, and including different schools of thought within, and approaches to, the field. We would know how, and why, this discipline resembles, and differs from, specific other disciplines, as we would know and understand the ways in which mathematics education research draws upon and is supported by other relevant disciplines, as well as the impact of this on the research enterprise.

THE NATURE OF THIS CHAPTER

The majority of the chapters in this handbook undertake the task of surveying and reviewing the development of mathematics education research, with regard to selected issues and topics, since the publication of the first Handbook, edited by Douglas Grouws, in 1992 (Grouws, 1992). The developments considered are to do with perspectives on, approaches to, and results of research, in order to portray what new land has been reclaimed by our field during the last decade and a half.

The nature of this, final, chapter is somewhat different. It might have been possible to concentrate on providing an overarching synthesis of the development of the field, a “synthesis of syntheses,” based on the “progress reports” constituted by the individual chapters. Although I shall, in fact, do something in that direction, I have chosen instead to position the current state and the “rate” of development of our field in relation to “Utopia.” How far away from Utopia are we, what progress has been made during the last fifteen years to move in that direction, and what—if anything—is missing in the course we have set to get there?

OBSERVATIONS CONCERNING TRENDS REFLECTED IN THE PREVIOUS CHAPTERS

This section presents and discusses four observations of a somewhat general nature made during my reading of the previous chapters of this Handbook.

First Observation: Three Kinds of Widening Perspectives of Research

The *first observation* that emerges from the previous chapters in this volume is the continual widening of the perspectives of research, within, by and large, any area of mathematics education.

Since the major take-off of mathematics education research in the late 1960's, our field has experienced a *widening of research perspectives* in three different respects.

Firstly, researchers have revealed, and continue to reveal, that the issues being discussed and the phenomena, objects and *problématiques* being investigated in our field are much more complex than we used to think. The set of factors that influence a given phenomenon or entity in significant ways, is much larger than acknowledged in earlier research. So, researchers insist, we have to include such additional factors in our research and pay due respect to their actual or potential roles. One example of this is the influence of students' metacognition and beliefs on their problem solving behavior (see the chapters by Schoenfeld, and Lesh & Zawojewski in this volume). Another example, represented in several chapters in this handbook (e.g., Cobb's and Artigue, Batanero, & Kent's, and Battista's), is the shift away from primarily taking into account the cognition of the individual learner towards studying him or her as situated in different kinds of socio-cultural surroundings and within different institutional practices: classroom, school or university, communities of practice, (sub)culture, society at large, or region. Moreover, the teaching and learning of mathematics is no longer seen as a function of student characteristics and teacher knowledge only. Teachers' beliefs and affect are important factors as well, as is reflected in Philipp's chapter on this very theme. The fact that mathematics teaching takes place in societal institutions under the influence of economic, political, ideological, and organizational forces, suggests that research should include national and local issues of policy and change. In this handbook two chapters (by Ferrini-Mundy & Floden, and by Tate & Rousseau) are devoted to such issues. Societies' focus on accountability and high-stakes testing, and on international comparisons such as TIMSS and PISA of student achievements in different countries, are now factors that have a direct impact on mathematics teaching and learning. Hence they should be studied by researchers, as depicted in the chapters by de Lange and Wilson.

One consequence of the growing complexity of our objects and subjects of study is that the empirical and theoretical frameworks and methodologies

adopted in research have to account for the resulting new and emergent factors and their interplay. This is what happens when Lesh and Zawojewski propose to include studies of the problem solving of people adapting mathematics for use in everyday environments and in work-places that require heavy use of mathematics. It is also what happens when Artigue, Batanero and Kent in their chapter talk about historico-epistemological and semiotic approaches to complement cognitive studies of tertiary mathematics education (e.g., focusing on linear algebra) and when they look at what neuroscience and embodied cognition have to offer to post-secondary mathematics education research. Thus, *the investigational frameworks themselves become more numerous and increasingly complex*. This development, in turn, gives way to the borrowing of constructs, frameworks and methodologies from an increasing number of other fields, such as neuroscience, psychology, education, philosophy, history, sociology, statistics, psychometrics, political science, economics, anthropology, ethnography, in addition, of course, to mathematics and all the disciplines and areas in which mathematics plays non-negligible parts.

In itself, the recognition that the teaching and learning of mathematics, and hence mathematics education research, are utterly complex is not new.

In the somewhat informal publication “Theory of Mathematics Education (TME)” (Steiner, et al, 1984; see also Steiner, 1985), which is a report partly of Topic Area 5 at ICME-5 in Adelaide, 1984, and partly of a subsequent mini-conference, the chief organiser Hans-Georg Steiner in his opening paper (pp. 16–32) made the following statement, which can almost be seen as a programmatic declaration for the establishment and development of a theory of mathematics education:

Mathematics education is a field whose domains of reference and actions are characterized by an *extreme complexity*: the complex phenomenon “mathematics” in its historical and actual development and its interrelation with other sciences, areas of practice, technology, and culture; the complex structure of teaching and schooling within our society; the highly differentiated conditions and factors in the learner’s individual cognitive and social development etc. In this connection the great variety of different groups of people involved in the total process plays an important role and represents another specific aspect to the given complexity.

Within the whole system several *sub-systems* have evolved. They do not always operate sufficiently well, especially they often lack mutual interconnection and cooperation. With respect to certain aspects and tasks mathematics education itself as a discipline and

professional field is one of these sub-systems. On the other hand, it also is the only scientific field to be concerned with the total system.

A *systems approach* with its self-referent tasks can be understood as an organizing *meta-paradigm* for mathematics education. It seems to be a necessity in order to cope with complexity at large, but also because of the fact that the *systems character* shows up in each particular problem in the field. (p. 16)

In his paper, and in his summary of it, Steiner (1984) warns against “. . . non-scientific *short-range pragmatism* “ as well as against “strong *reduction of complexity* favoring special aspects or research paradigms . . .” (i.e., against undue reduction of complexity) (p. 7). What we have seen in mathematics education research over the last couple of decades is that this complexity is now being taken seriously, not only at the rhetorical level but, more importantly, in actual practice. As indicated above, the chapters in this handbook demonstrate how this is taking place.

Secondly, along with the fact that mathematics teaching has developed in scope so as to address new categories of learners at all educational levels in increasingly varied environments and subject to more and more diverse conditions, *new phenomena, entities and problems* of mathematics education have been placed on the agenda of research. While mathematics education research in former times mainly focused on learners in primary and secondary school, and to some extent on their teachers, the scope has widened to such an extent that, today, pre-school children (as reflected in the chapter by Clements & Sarama), several sorts of tertiary students, also in fields to which mathematics is a service subject rather than a core subject, (as is the case with engineering, which is dealt with in a special section in the chapter on post-secondary mathematics education (by Artigue, Batanero, & Kent)) are now subjects of research. The same is true of adults returning to education, out-of school mathematics learning (see Presmeg’s chapter), children playing computer games, popularization of mathematics in informal settings including ICT, media and museums, and so on. As mathematics teaching is often delivered indirectly—in fields in which mathematics is integrated in ways that make it invisible or at least implicit—or is delivered by means of ICT systems in which it is hidden, new research items have been added to the agenda. Also, the teaching and learning of new mathematical topics are being studied. This Handbook has made an attempt at placing topics born to model features of the real world, such as probability and statistics under the aegis of mathematics education research. The chapters on statistics (by Shaugnessy) and probability (by Jones,

Langrall, & Mooney), and the statistics and probability section of the post-secondary education chapter are reflections of this. These are all instances of a different type of widening of research perspectives.

The addition of new items to the research agenda in combination with the endeavour to pay due respect to the complexity of the teaching and learning of mathematics, has led many mathematics educators to look at issues and problems that are actually outside of mathematics education proper but belong to general psychological, educational, or sociological spheres that have some bearing on mathematics education. Schoenfeld's chapter in large parts deals with the nature of empirical research in general, with examples drawn from pharmacological research as well as from mathematics education. Schoenfeld proposes a general schematization for conducting empirical research, starting from a real world situation, of which a conceptual model is constructed, and some of whose aspects are then captured in a representation system. Analyses are performed within the representation system, whose outcomes are interpreted within the conceptual model so as to allow for inferences to be made about the original real world situation. In his chapter, Cobb considers issues belonging to general epistemology, where he ends up making a plea for Putnam's pragmatic realism, and issues belonging to psychology, where he compares experimental psychology, cognitive psychology, socio-cultural theory and distributed cognition. The APOS theory (Action-Process-Object-Schema), by Dubinsky and colleagues, is actually a neo-Piagetian theory of general cognition, even though it is meant to be oriented towards mathematical cognition. Lots of research issues pertaining to curriculum reform and change (see the chapter by Ferrini-Mundy & Floden, and by Tate & Rousseau), to teacher's professional development (Sowder), to equity (see the Diversity in Mathematics Education chapter, and Bishop & Forgasz's chapter) are actually of a general nature, not specific to mathematics, as is also reflected in the literature reviewed in the respective chapters. In summary—and this is the third instance of widened perspectives to be mentioned here—we are witnessing that *phenomena, entities and problems beyond mathematics education* are now being dealt with by mathematics education researchers.

Second Observation: Constructing Theoretical Frameworks

While the previous observation included a widening of the theoretical and empirical investigational frameworks adopted in mathematics education research, the *second observation* focuses on mathematics

educators' actual struggle to establish such frameworks. It is evident from the chapters in this Handbook that although there is certainly no convergence to be seen in the frameworks employed, many of the authors attempt to establish one as far as their own chapter is concerned—often a framework in whose development they have (had) a key role themselves. Cobb argues for a deliberately permissive approach to investigational frameworks that he calls “theorizing as *bricolage*.” As mentioned above, Schoenfeld, in addition to revisiting his own construct “metacognition,” proposes a framework for empirical research that focuses on “Pasteur's quadrant,” which combines a quest for fundamental understanding with considerations of use. This framework is also meant to dissolve what he sees as an unfortunate dichotomy between quantitative and qualitative studies. Harel and Sowder base their chapter on the notion of “proof scheme,” a construct which they have developed over the last decade or so. Lesh and Zawojewski propose “a models and modelling perspective” on problem solving which differs from traditional perspectives by placing traditional problem solving as a subset of “applied problem solving as modelling activity,” rather than the other way round. In the chapter on post-secondary education, the authors outline Godino and Batanero's attempt, with colleagues, to create a framework incorporating ontological, anthropological and semiotic ideas while focusing on the relationship between students' personal meaning with respect to a mathematical concept and the intended institutional meaning.

This suggests that even for a particular theme or topic, like problem solving, or proof and proving, it is not a settled matter what framework to adopt. Once again an overarching, unifying framework, agreed upon by all researchers, is not in sight. In several chapters (see Cobb, Lesh & Zawojewski, and Artigue, Batanero, & Kent) considerable effort is spent on surveying and reviewing available frameworks, without necessarily advocating or constructing a particular one.

To this point I have talked of an investigational framework as a somewhat undefined notion. What does it actually mean? It is not a well-defined notion but I propose that we take it to consist of at least three components: (a) A perspective on the issues and questions to be investigated; (b) A set of theoretical constructs (e.g., concepts, notions, and assumptions) more or less sharply defined, coined to capture essential entities of significance to the issues and questions in focus of the framework; and (c) A set of preferred methods considered suitable for the investigation of the issues and questions of interest, with particular regard to the ways in which the theoretical constructs come into play. For a research approach to deserve

the name “investigational framework,” it is not necessary that the three components are explicitly introduced. The important thing is that they are actually present in the approach. I should make it clear at this point that to me the term “investigational framework” is not identical with the term “theory,” which in my view is a more elaborate construct to which I shall return later.

Third Observation: Political Pressure on Mathematics Education Research

My *third observation* is that several Handbook chapters (Cobb, Ferrini-Mundy & Floden, Schoenfeld) touch upon demands from political and administrative quarters for evidence-based research and accountability in education in general and in mathematics education in particular. In the USA this demand is very evident indeed, but in other countries, such as the UK and Denmark, similar developments may have been encountered and the movement is undoubtedly gaining momentum in a world dominated by neo-libertarian approaches to economy and politics. The thrust of the requirements inherent in this demand is that educational research should provide solid evidence, “proof,” of its claims, in particular of claims that lead to recommendations concerning subject matter content, processes, curricula, materials, assessment instruments, teaching approaches, and so forth. Unless such evidence is produced claims should be counted as unwarranted, and recommendations put forward on the basis of them should be discarded as “political views” disguised in scholarly or scientific gowns. In the U.S., the demands are accompanied by rather specific politically rooted requirements of the methodologies to be adopted by “acceptable” research, namely that such research should emulate clinical trials in pharmacology.

The vast majority of mathematics education researchers are likely to discard such requirements as being, in and of themselves, political in nature but disguised as a concern for scholarly health. The requirements are further discarded as expressions of a rather primitivistic reduction of the complexity of the practice and research of the teaching and learning of mathematics, a reduction which—as Schoenfeld attempts to demonstrate in his chapter on research method—also is too primitivistic to capture the pharmacological research, which by advocates of the evidence-based demand is meant to serve as a role model for educational research.

Although it is, on the one hand, annoying for seriously working mathematics education researchers to be met with potentially destructive requirements to their very enterprise, requirements that are even

given strength by being invoked in public funding of research, there are two other sides of the coin (you have to think of the coin as a rather thick one, so that the edge can be counted as a third side!). Firstly, the fact that politicians, administrators, and lobbyists find it worthwhile to take position vis-à-vis mathematics education research suggests that they take such research seriously, even if many of them counteract it. In a not too distant past it was almost out of the question that research was in play to underpin recommendations for practice. Many reactions against attempts to substantiate recommendations by reference to research seem to be based on dissatisfaction with the recommendations themselves rather than with the underpinning of them. However, it is not only politicians and lobbyists with particular perspectives and interests who are beginning to take mathematics education research and its outcomes seriously. Practising teachers, teacher trainers and others do the same but usually in more embracing ways, even though they often ask for more specific results that can guide them in their everyday work in concrete ways. All this being said—and here comes the third side of the coin—although it is indeed unreasonable to ask for evidence-based research in the pharmacological sense only, it should be admitted that the nature of our results are not always easy to understand and use, because they are multi-faceted and need substantial interpretation, and that we are much better at demonstrating what does not work and at providing explanations of why, than the opposite. Knowing what not to do does assist our delimitation of what to do, but not quite in ways that can produce operational guidelines for tomorrow’s practice.

Fourth Observation: The U.S. is the Focus of Attention

The *fourth observation* is probably one that can primarily be made by non-U.S. readers of the chapters in this Handbook. While it is certainly true that the bulk of the content of the previous chapters is completely international in scope and significance, it is also the case that some of the issues addressed, the perspectives adopted, and large parts of the literature reviewed, have distinct U.S. flavors to them. There are two reasons for this.

The first, which is obvious and natural, was also valid for the first Handbook, namely that a handbook crafted within the framework of the NCTM must pay particular attention to matters pertaining to and relevant for the education system(s) in the U.S., not the least so since important developments concerning the teaching and learning of mathematics in the U.S. and worldwide are spurred by the NCTM as an organisa-

tion and by quite a few of its members, including the authors of this volume. While the preoccupation with matters American is easy to explain, it is less obvious why many U.S. researchers are not so inclined to consider literature from other countries, even though there are hosts of non-American research publications available in English that provide insights which can shed light on numerous issues of interest to U.S. researchers. With noticeable exceptions this bias is also reflected in the chapters in this volume. Allow an outsider to note that there is a fair amount of truth in the final paragraph of Judith Sowder's chapter, when she writes "... we in the United States are too often insular and not open to answers found in other countries. Thus, more than one border needs to be crossed while we, researchers, administrators, policy makers, and teachers, seek to better educate teachers for the purpose of improving student learning." I believe that we would all benefit from the cross-fertilization that would arise if more U.S. researchers paid as much attention to non-U.S. research as non-U.S. researchers pay attention to research from the United States.

The other reason, which deserves some consideration, is of a different nature. Numerous chapters in this volume make explicit reference to the current politico-ideological situation in the U.S., and the ways in which it exerts an impact on mathematics education in general and on research in particular. As the "Math Wars" in California, Florida (see, for instance, the sections on these wars in the chapter by the Diversity in Mathematics Education Group) and elsewhere have shown, together with the present views and policies of the current national administration, mathematics education has become subject of a strongly politicized discourse. This seems to be rather unique in history. Although some of the heated debates and controversies over the so-called New Mathematics movement in the '50s and '60s made their way into the media and sometimes also reached national political levels in some countries (e.g., Norway) the discussion by and large remained an internal one amongst mathematics educators and mathematicians, accompanied by some spill-over into the public domain.

How can we interpret the entering of mathematics education into political agendas? Is this a singular phenomenon, local in time and space, related to the U.S. government 2001–2008? I don't think so. While governments in democracies come and go (and the latter is at least as important as the former), the political interest in the teaching and learning of mathematics goes beyond transient governments. It is deeply rooted in the societal underground. This is not the place to offer a more detailed analysis of the relationships between mathematics education, soci-

ety, and politics, but allow me to refer to a chapter, "Goals of Mathematics Teaching," that I wrote for another handbook, the *International Handbook of Mathematics Education* (Niss, 1996). In that chapter, I proposed that society provides mathematics education to its citizen for one or more of the following three fundamental reasons:

- in order to contribute to the *technological and socio-economic development* of society at large, either as such or in competition with other societies/countries;
- in order to contribute to *society's political, ideological and cultural maintenance and development*, either as such or in competition with other societies/countries;
- in order to *provide individuals with prerequisites which may help them to cope with life* in the various spheres in which they live: education or occupation; private life; social life; life as a citizen. (p. 13)

In countries that offer public mathematics education to all its children and youth, analyses suggest that the predominant reason is the first of these. The other two are important as well, but are subsidiary reasons adopted by some countries in certain eras. In principle, these three reasons are neither in concert nor in conflict with one another. However, in a given country at a given time the specific consequences drawn from them may be either in concert or in conflict. This happens if, for instance, the teaching time allocated to pursuing goals derived from the reasons favors one over the others, like if most activities were orchestrated to pursue goals related to the third reason.

I submit that what we are witnessing, as far as the political interest in mathematics education is concerned, is a struggle between the consequences of these reasons. The current U.S. administration seems to primarily insist on goals derived from the first reason, with some subsidiary attention being paid to the second one, while goals related to the third reason only deserve attention to the extent they are not in conflict with the primary ones. At the same time, many mathematics educators tend to emphasise goals related to the third reason, which is to do with empowering people to live in society, while maintaining that there is no intrinsic conflict between this and paying due attention to the other two (cf. the Handbook chapters by Bishop & Forgasz, and by the Diversity in Mathematics Education Group).

Potential conflicts between societal quarters advocating one reason over the others are not of a superficial nature. They are to do with fundamental views

and interests concerning the direction and ways in which society should move and develop. Let me mention just one point for illustration.

It is fair to claim that since World War II most societies in the world have actively sought to make mathematics education available to all citizens (for example, see Sowder, Bishop & Forgasz, and the chapter by the Diversity in Mathematics Education Group), first as regards primary education, then with respect to secondary education, and recently in more and more countries with respect to tertiary education also. What is important here is not the extent to which countries have actually been successful in this endeavor, but the very endeavor itself. The basic reason for this is, I submit, the technological and socio-economic reason. Now, in recent years we have been witnessing that the question has been raised, in some quarters in highly developed countries, whether it is really justified, from the point of view of societal needs and the needs of individuals in their everyday life, to subject everyone to a substantial dose of mathematics teaching. The fact that, today, so much mathematics of utilitarian societal relevance is embedded in technological systems in implicit ways, has been taken by some to suggest that perhaps not every citizen needs mathematics beyond a certain minimum, which might well be decreasing, for coping with his or her occupational and private life. Of course, any society needs a minority of high-level specialists with advanced mathematical education, but why, the argument goes, should this fact be used to bother the majority with painful mathematics beyond their needs and natural capacity, and why should those who are unhappy with the study of mathematics be allowed to prevent or jeopardize the development of front-line mathematical competencies with those who can and should benefit from demanding mathematics teaching? To counter this position, some mathematics educators might argue that although this line of reasoning might look seductive if a narrow utility argument based on the technological and socio-economic reason is adopted, a completely different conclusion would follow if focus were on equipping individuals with prerequisites for becoming knowledgeable, concerned, analytically inclined citizens with the capability to exert well-founded criticism of those in power in society. This would amount to pleading for goals related to the third of the fundamental reasons.

Such issues and their relatives will not disappear with transient governments but will remain with us in the future. I would even venture to predict that fights over such issues are likely to be intensified in the next couple of decades. Against this background, it will be interesting to see how a third NCTM research hand-

book will describe and analyse the “mathematico-political” situation in another decade and a half.

These deliberations suggest that USA, as far as political influence on mathematics education is concerned, is at present experiencing movements and developments that are going to spread to the rest of the world in the future, whence the preoccupation of several authors of this Handbook with this influence as it manifests itself in the U.S.

REFLECTIONS ON THE STATE OF AND PROSPECTS FOR MATHEMATICS EDUCATION RESEARCH

I am now ready to take a general look at how far we have come in approaching Utopia, if the distance is to be judged on the basis of the chapters in this Handbook. My overall conclusion is “not so far.” This being said, I hasten to add that on average a lot of progress has been made since the first Handbook appeared in 1992. Below, I shall try to outline the state of affairs as I interpret it.

Why Mathematics Education? For Whom?

To what extent do we fully know and understand *why we should teach mathematics to various categories of students in society*, and to what extent do we know and understand *who in society needs what mathematical insights and competencies, and for what purposes?* These issues are not key points in this Handbook, but aspects of them are briefly touched upon in some of the chapters, in particular the ones that deal with diversity and equity, in the engineering section of the chapter by Artigue, Batanero, and Kent, and in the chapter by Lesh and Zawojewski with its numerous remarks concerning problem-solving practices in professions that make heavy use of context-embedded mathematics.

It may seem pretty natural that a handbook on research on the teaching and learning of mathematics does not focus much on issues of “Why mathematics education? For whom?” because research tends to enter the stage only when mathematics education is actually being given/offered to certain groups of students. Nonetheless, well-founded answers to these questions are crucial to all aspects of practice of and research on the teaching and learning of mathematics. Most mathematics educators are convinced that it is essential to provide some mathematical education to everyone in society, although they are likely to disagree on what mathematics different kinds of people need and on how they need it. They would probably

subscribe to the arguments related to some or all of the three reasons indicated in the previous section, in addition, perhaps, to some others. Mathematics educators often have vivid images of where, how, and by whom mathematics is being used in society's work-places and in other disciplines and subjects, of what role mathematical competence plays for people as private and social individuals and as citizens, and of what role mathematics has in the development of society and culture from historical or contemporary perspectives. These images may well contain different sub-images with regard to different mathematical topics or processes.

Such images and convictions, which are often sound and strong, mostly rely on individual experience and impressions. Their soundness and strength notwithstanding, these images and convictions only rarely have a solid foundation and documentation in terms of research. This has at least three significant consequences. Firstly, a well-founded—I hesitate to write “evidence-based”—solid knowledge of the actual and potential nature, location, and distribution of mathematical knowledge and competence in different segments of different types of societies would help us orchestrate and tune teaching to different kinds of students much better than today. Secondly, we would be in a much better position to justify our claims about the relevance of mathematics education to various categories of learners, to people who have not accumulated similar piles of impressions and experiences to the ones that we, mathematics educators, have accumulated as individuals or as a community. Thirdly, in addition to providing better justification of our claims such well-documented knowledge would help us become better at communicating with others about the whole enterprise of mathematics teaching and learning.

Thus, I submit that we are lacking comprehensive overviews of mathematics in different kinds of societies and cultures and an insight into the educational consequences that may or should be drawn from such overviews. In other words, we are lacking a well-founded *socio-cultural ethnography of mathematics*. It is true that research within the ethno-mathematics movement (see Presmeg's chapter, and, for a specific example, Gerdes, 1996) has provided us with parts of such an ethnography as regards some “developing” countries and as regards particular cultural or ethnic groups in “developed” countries. It is also true that there are many illuminating studies of situated knowledge and communities of practice focusing on the discrepancy between in- and out-of-school contexts, like the famous study by Carraher, Carraher & Schliemann (1985) and Nuñez, Schliemann, & Carraher (1993) of

Brazilian street vendors. Studies of the mathematics of different professions, for example, banking and nursing (Hoyles, Noss, & Pozzi, 2001; Noss & Hoyles, 1996; Noss, Hoyles, & Pozzi, 2002) and engineering (Kent & Noss, 2003), have begun to appear, but are still rather sporadic and far from systematic across significant societal practices. Much more needs to be done in these directions if we are to come closer to Utopia, and the current mostly static, “snapshot,” studies ought to be complemented by dynamic, “moving picture,” studies of change and evolution.

In addition to studies of vocational and professional practices, and studies of mathematics in arts and crafts, studies are lacking that uncover the mathematical competencies involved in different kinds of citizenship. What does it take, for instance, to be able to understand, judge and take position towards specific civic and political issues, for example, concerning economy, environment, climate, planning, resource exploitation, epidemics, and demography? Many teaching materials contain examples referring to such issues, but they are usually tailored to fit into instructional sequences, and are normally not authentic in a large scale sense. What I am proposing is that studies within the socio-cultural ethnography of mathematics be extended to chart and survey aspects of citizenship on which mathematics has a bearing.

What Does It Mean to Master Mathematics?

The next issue to be addressed is to what extent we know and understand exactly *what it means to master mathematics* in all its dimensions and facets, *inside mathematics as a discipline and in extra-mathematical situations and contexts*, both at an overarching level (across age, grades, and institutions, and across mathematical topics), and with special regard to different topics. The general answer to this question is that we know and understand much more today than fifteen years ago, but that there is still a non-negligible distance to Utopia.

Marked progress has been made when it comes to understanding and describing what it means to master various mathematical topics. Essential parts of this progress are captured by the topics chapters in this Handbook, on whole number concepts and operations (by Verschaffel, Greer, & De Corte), on rational numbers and proportional reasoning (Lamon), on algebra learning (by Carraher & Schliemann, and Kieran, respectively), on geometric and spatial thinking (by Battista), on probability (by Jones, Langrall, & Mooney), and on statistics (by Shaughnessy and by Artigue, Batanero, & Kent at the tertiary level). Topics belonging to post-secondary levels such as calculus

and linear algebra are touched upon in the last-mentioned chapter, but are not dealt with as topics per se. One common feature of current portraits of the mastery of these topics is a much more sophisticated understanding of the interwoven variety of facets and components of each topic than was found one-and-a-half decade ago. Another common feature is that the mastery of one topic is conceptually and procedurally related to the mastery of other topics. Most importantly, perhaps, today mastery of a topic is portrayed as being intimately related to multi-faceted links to a variety of semantic environments that allow for interpretation and sense-making of concepts, rules, procedures, facts, and results.

Similar pictures can be drawn of the mastery of mathematical topics that are not so well represented in this Handbook such as discrete mathematics, analysis, and abstract algebra and geometry, which predominantly pertain to the tertiary level. Here the body of research is much more limited than for numbers, arithmetic, elementary algebra, and geometry, but the volume is growing, in particular as regards mathematical analysis and linear algebra.

What we are still largely lacking is an in-depth (profound rather than detailed) charting of how the mastery of a given topic is related to the mastery of other topics, and of how mastery of topics is related to mastery of problem solving, modelling, and proof and proving, respectively. Here, Lesh and Zawojewski in their chapter argue that there is no such thing as context- and situation-independent problem solving in mathematics, and that in general mathematical problem solving has not made any real progress during the last 10–15 years. Rather, they suggest, problem solving is developed along with the learning of mathematical concepts in rich environments that “ask for” or lend themselves to modelling. While it is pretty clear that there is not much evidence of subject-free problem solving, it seems to me that it remains an unsettled matter to what extent it is possible to identify and characterize generic problem solving within mathematics but across different domains.

There are certainly strong connections between problem solving and modelling. However, they are not, I submit, just two sides of the same coin, because much mathematical modelling concerning extra-mathematical domains is being performed for reasons different from solving problems in the usual sense of that word, and because a lot of intra-mathematical problem solving is being performed without regard to anything extra-mathematical. There is a fairly substantial research literature on the mastery of mathematical modelling that ought to be taken into consideration

in that context but which does not figure prominently in this Handbook.

During the last one or two decades mathematics education researchers have attempted to identify and characterize mastery of mathematics in terms that are not locked into particular mathematical topics and do not pertain to particular educational levels. One example of this endeavour can be found in the *Curriculum and Evaluation Standards* of the National Council of Teachers of Mathematics (NCTM, 1989) and in the *Principles and Standards for School Mathematics* (NCTM, 2000). The 1989 *Standards* listed—in addition to grade-specific mathematical topics—four characteristics of mathematics, meant to help foster students’ “mathematical empowerment”: “Mathematics as Problem Solving,” “Mathematics as Communication,” “Mathematics as Reasoning,” and “Mathematical Connections.” In the same spirit, the 2000 *Standards* provided a similar, but slightly expanded list: “Problem Solving,” “Reasoning and Proof,” “Communication,” “Connections,” and “Representation.” More recently, “Adding it Up” (National Research Council, 2001), edited by Kilpatrick, Swafford, and Findell, considered five strands of overarching *mathematical proficiency*: “conceptual understanding,” “procedural fluency,” “strategic competence,” “adaptive reasoning,” and “productive disposition.”

I and colleagues in Denmark have worked since the late 1990’s on developing the notion of a system of overarching *mathematical competencies* that are meant to capture what it means to master mathematics across topics and educational levels. I propose eight such competencies: “mathematical thinking competency,” “mathematical problem solving competency,” “mathematical modelling competency,” “mathematical reasoning competency,” “mathematical representations competency,” “symbols and formalisms competency,” “communications competency,” and “aids and tools competency.” (For a brief outline, see Niss, 2004a). This system of competencies also serves to underpin the mathematics framework of the OECD/PISA international comparisons of student achievements (OECD, 2003).

It appears that we have not yet come very far in establishing a comprehensive, consistent, and robust characterization of mathematical mastery in general terms, substantiated by a massive empirical corroboration. This is a point where lots of research is needed if we are ever to arrive in Utopia. But to quote the late British prime minister Winston Churchill, “This is not the end. It is not even the beginning of the end. But it is the end of the beginning.”

How Does Mathematics Learning Take Place?

To obtain complete knowledge and understanding of *how learning of mathematics takes place* is arguably the most formidable task in mathematics education research. How far have we come in accomplishing this task? It would be an impossible endeavour to attempt to answer this question in any kind of detail, at the end of a volume in which several hundred pages have been spent on surveying and reviewing the state of the art with regard to this very question. However, are there any aggregated conclusions to draw?

There are. On average a lot of progress has been made towards a deeper and more sophisticated insight into the mazes of mathematical learning, both in general and with respect to particular domains and processes. It is remarkable, however, that some of the chapters (e.g., Lamon's, and Lesh & Zawojewski's) talk with some pessimism about stagnation in research achievements over the last decade or so, although they do see promising new germs that are likely to open new avenues for coming to grips with how learning of certain topics or processes takes place or, rather, may take place. This is even more remarkable when contrasted with the much more optimistic tones found in other chapters (e.g., Verschaffel, Greer, & De Corte's and Harel & Sowder's) which report major progress since the last Handbook. The same is true of the learning of probability, where cognitive models of probabilistic reasoning have shed light on the learning of phenomena and concepts such as chance, random process, sample space, and probability measurement since the previous Handbook. Also the chapter by Zbiek, Heid, Blume, and Dick report on major progress in our understanding of how technology can facilitate learning in a multitude of ways, in particular when it comes to representation of mathematical objects, phenomena, and processes. Some chapters lie in between, for examples Battista's, when he reports on solid progress as regards the cognitive side of geometric and spatial reasoning—except for the role of what he calls technological enhancements offered by computer environments—but points to a lack of research that can account for the effects of affects and socio-cultural factors on students' learning.

This suggests that progress is not uniform across domains. Research can run out of steam within one domain while gaining momentum within another. It would be interesting to investigate the mechanisms behind such developments. It is certainly not the case that the domains in which research is reported to have stagnated have already reached Utopia. There is a long way to go for that to happen in all aspects of mathematics learning. Nevertheless, we have come

much closer to knowing and understanding how students' mathematics learning is construed, and the conditions under which it takes place. In one sentence: Mathematics learning is utterly complex, much more so than we thought. Through a multitude of studies over the last fifteen years research has been able to move beyond the mere recognition of complexity towards uncovering its scope and components. However, huge efforts remain in order to achieve major steps towards a justified reduction of complexity, the ultimate scientific goal of all scientific research.

What Are Students' Beliefs, Affects, and Attitudes to Mathematics?

In this Handbook, students' beliefs, affects, and attitudes to mathematics and its study have not been dealt with as an independent theme. Rather this theme has been parcelled out to be addressed, more or less sporadically or in passing (with some exceptions, e.g., Battista's, Lesh & Zawojewski's, and Schoenfeld's chapters), by the other chapters. This implies that there is not much of an overview to reflect on. Internationally, however, there is an amount of research worth considering—new research included (see, for instance, Leder, Pehkonen, & Törner, 2002)—but it may be that there has been some slowing-down in such research during the last decade.

One might hypothesize that the relative slowing down of research on beliefs, affects, attitudes, and motivation with regard to mathematics is a reflection of doubts being raised about how well-defined and well-delineated the basic notions are, and how clearly they can be disentangled from cognition in mathematics education. An attempt at defining key concepts is made by Philipp in his chapter on teachers' beliefs and affect. On the other hand, all mathematics educators, be they practitioners or researchers, seem to agree that there are affective factors that exercise a strong influence on students' learning. This suggests that there is a need for investing a considerable amount of effort on sorting things out, both conceptually and empirically, before Utopia is in sight in this respect.

How to Teach Mathematics So As to Ensure Optimal Learning Outcomes for Students?

One of my favorite quotations is from the Polish-American mathematician and mathematical physicist Mark Kac, who is claimed to have once said "I can't define my wife, but I can recognize her when I see her." This dictum (which may be apocryphal as I haven't been able to locate it anywhere) points to important limits to formalization of human experience. There

are many things that we know without being able to document them in a formalized way, while obeying certain rules and procedures. This is often the case when human beings' extraordinary pattern recognition is involved.

The Kac quotation, or to whomever it should be attributed, came to my mind when reading Hiebert and Grouws' chapter on the effects of classroom mathematics teaching on students' learning. One of their section headings bears a strong resemblance with the quotation: "A claim that appears obvious is strikingly difficult to specify." In several places in their chapter they return to statements like "teaching matters . . . but *how* is not easy to document," and "documenting the effects of teaching on learning is methodologically difficult." That goes for teaching in general and for mathematics education in particular. When broken down with regard to effective teaching for learning of particular topics or processes, these difficulties prevail, in some respects in an amplified form. Things get even more complicated when we want to investigate the specific influence of curricula and teaching materials, including textbooks, on students' learning, separating such influence from other instructional variables (see Stein, Remillard, & Smith's chapter).

Against this background, to review—here—our current knowledge and understanding of how teaching of mathematics can assist the optimal learning with different individuals and groups of learners is far beyond reach. Once again, we can ask whether there are aggregate conclusions to consider, across the chapters. And once again, there are.

Considerable progress has been made when it comes to devising, designing, orchestrating, and implementing mathematics teaching and student activities that seem to promote students' learning in forceful ways. But as indicated, it is difficult to document that the desirable learning outcomes result from teaching rather than from other variables in play. Characteristic of such teaching is that it pays due and articulate respect to the variety of factors that may influence student learning. In other words successful teaching makes a point of explicitly identifying, monitoring and controlling such factors as part of the teaching-learning environment. Moreover, research findings suggest that for students to obtain desirable learning outcomes it is important that they are engaged in activities where they have to "struggle" (in a productive sense of that word) with important mathematics, preferably in collaborative environments that challenge their work in constructive and supportive ways. Delicate balances are to be struck here, as "struggles" and "challenges" can so easily become obstacles rather than promot-

ers of learning, if they are in mismatch with students' situations and prerequisites. Striking such a balance is one objective in The Mathematical Task Framework, developed by Stein and her colleagues and referred to in several chapters in this Handbook.

Since the mid 1970s the role of information and communication technology (pocket calculators, computers, software systems) has been discussed intensively among mathematics educators. In the beginning the topic of the discussion was rather simplistic: Good or bad? And little solid evidence was provided for either position. Today we know and understand, much more than we did fifteen years ago, and in much more subtle ways, how technology can (and cannot) be put to use as a crucial instrument for teaching (see Zbiek, Heid, Blume, & Dick's chapter). This includes a deeper insight into the significance of the teaching and learning environments in which ICT is put to use and the conditions that govern them. One overall finding is that ICT as such can be both rather harmful and highly helpful to the teaching and learning of mathematics. The outcome is almost completely determined by the didactic and pedagogical design and implementation carried out by the teacher. Another finding (see Artigue, Batanero, & Kent's chapter) is that technology can no longer be seen as just a tool for mathematics education, an add-on. Rather we should consider technology as something that profoundly shapes and changes the very ways in which we work with and do mathematics.

We are not only making progress as regards understanding how teaching can contribute to learning. We have made considerable progress in characterizing and understanding what happens in the classroom, and in teaching practices. As regards American classrooms and teachers, the chapter by Franke, Kazemi, & Battey, is an eye-opener. Such research was only in its infancy fifteen years ago, and new panoramas are likely to be added to our knowledge in the years to come.

We are still far from being in a position where we can design and implement teaching that—guaranteed—leads to optimal learning outcomes for all the students we teach. The problem is not only to document the effects. Even though we have made remarkable progress in the design and implementation of successful and effective teaching for many categories of students, we have only limited successes in really assisting weak learners. So, here too, the way to Utopia is long.

How Should the Teaching and Learning of Mathematics and of Other Subjects Be Related?

On this issue the present Handbook is—as was the previous one—largely silent. It would of course

be unreasonable to criticize the book (i.e., the editor!) for this. The book has to limit itself to what is seen as its primary priorities. Besides, while there was a growing interest in the 1970s and 1980s in linking the teaching of mathematics to other subjects, mainly in order to provide avenues for import-export activities, which, from the point of view of mathematics, would provide fuel for work on mathematical applications and modelling, this interest was never really accompanied by research.

This is a pity. I submit that important insights would be gained and lessons to be learned, also for the benefit of mathematics education research, by studies that compare and contrast mathematics and other teaching subjects. We already know and understand, in the “folk wisdom sense” of Mark Kac, a lot of the differences and similarities in play between mathematics and other fields. But making them subject of research studies would make us much better at identifying and characterizing those features and factors that are tied to teaching and learning in general, or focused on certain types of subjects, and those that are specific to mathematics. Once such features and factors were identified and located we could invest special effort on uncovering the ways in which the features and factors, with respect to which mathematics differs from other subjects, influence the teaching and learning of mathematics. That, in turn, would help us understand when and why mathematics becomes difficult (or easy, for that matter) for students, and what we can do to assist them in overcoming these difficulties.

From the point of view of Utopia we can never claim to have arrived there until we are able to pinpoint the ways in which all important aspects of mathematics teaching and learning are similar to, respectively different from, the corresponding aspects of teaching and learning of other subjects. I therefore propose that such a comparative research programme be undertaken as one way to bring us closer to Utopia.

What Competencies Should Mathematics Teachers Possess and How Should They Be Developed?

There was a time, not so long ago, when the prevalent conception of teachers of mathematics could be condensed—and caricatured—as follows. A teacher of mathematics is a person who knows some mathematics, i.e. some concepts, facts, results, rules, methods, and procedures, on the one hand, and who knows how to teach students, in general terms, on the other hand. Thus the competencies a mathematics teacher should possess were of two kinds, subject matter knowledge, and general pedagogical skills. Satisfactory and effi-

cient teaching was expected to flow from the happy mating of these two kinds of competencies.

Today we know much better. We know that general pedagogy—or education—should not be kept separate from subject matter knowledge. We know that mathematical knowledge for teaching (see the chapter by Hill, Sleep, Lewis, & Ball) is an essential form of knowledge that is not identical to knowledge of mathematics as a theoretical edifice. We further know that teachers, in addition to possessing solid mathematical competencies, should also possess mathematics-specific “didactic competencies,” such as “curriculum competency,” “teaching competency,” “uncovering of learning competency,” “assessment competency,” “collaboration competency,” and “professional development competency” (see Niss, 2004a). We know that teachers’ beliefs and affects towards the nature and role of mathematics, mathematical activities, and the teaching and learning of mathematics, as well as towards students and their potentials for learning, are crucial elements in their luggage (see Philipp’s chapter). And we know that although teachers’ initial preparation does matter a lot for the ways in which they are initiated to their professional career, continual professional development focused on the teaching and learning of mathematics as an integrated entity, and the building of a professional identity in collaboration with different sorts of colleagues—peers, mentors, researchers—are even more critical to the way and extent to which they can develop as teachers who can help students learn mathematics to the best of their capacities.

An impressive amount of research over the last twenty years, on all aspects of teachers’ pre-service preparation, in-service development, and professional identity, in several countries, has brought us to where we are today. Sfard (2005) even talks of the present era in mathematics education research as the “era of the teacher.” Despite this progress, there are still many things that need to be done before we reach Utopia. For, there is no end to what it takes to be(come) a Utopian teacher, a fact that makes such a person more ideal than real. Against that background, it is important to identify the components that are particularly instrumental in the preparation and development of mathematics teachers, so as to reduce the infinite complexity of shaping the protagonists in the teaching and learning of mathematics.

How to Assess Students’ Mathematical Competencies and Achievements?

Assessment of students has always been a key element in mathematics education. Nevertheless research

has demonstrated that the assessment instruments typically employed in mathematics classrooms and in high-stakes testing are often so limited in scope that they can only assess a few aspects and components of students' knowledge and skills. And the ones that are left out tend to be the ones involved in aggregate and higher order competencies. Since "what you assess is what you see," this typically gives rise to a mismatch between the goals pursued in teaching and learning of mathematics and the assessment modes and instruments adopted. How far have we come in changing this picture so that we can become able to assess the full range of mathematical competencies in manners that are comprehensive, valid and reliable?

As is the case with teaching, there are several approaches to assessment that seem to work and are valid, if judged on an experiential basis. But it is difficult to convincingly demonstrate that they work, i.e. it is not always easy to achieve reliability.

Nevertheless, substantial progress has been made during the last decade and a half, as regards devising, designing and testing of assessment instruments and as regards research on assessment practices (see Wiliam's chapter in this Handbook). Along with such research, sharpened conceptual frameworks have been developed as tools to capture the multiple different purposes of assessment and the dimensions of the assessment modes and instruments. Research has also made progress when it comes to investigating the consequences and effects of high-stakes testing (see Wilson's chapter).

However, only very little progress has been made as regards the assessment of essential ingredients in mathematical competencies, such as asking questions, conjecturing, posing problems, constructing argument, including formal proofs, making use of and switching between representations, communication, and suchlike. Not only is research lacking, assessment instruments are largely lacking as well. Much assessment and testing is still focused on students' solving of already formulated problems. This shows that as far as assessment is concerned there is indeed a long way to Utopia.

What Policies Are Efficient to Achieve the Goals that Follow from Answers to the Previous Questions?

Issues of policy are, almost by definition, determined by local, including national, rather than global conditions and circumstances. It is therefore not so easy to draw many overarching lessons from the highly informative overviews of research made available to us by Ferrini-Mundy and Floden, and by Tate and

Rousseau in their chapters. Nor is it so clear, as in the other respect dealt with here, what constitutes Utopia with respect to policy. For in Utopia, the political atmospheres and environments prevalent in today's world, are not likely to be present. The reader might infer from this statement that I do not consider the current state of affairs, within which policies have to operate, ideal. There seems to me to be a non-negligible amount of room for political improvement in a number of places, but I shall refrain from specifying the directions in which such improvement should go.

Instead, I shall focus on one characteristic trait of policy in mathematics education in the U.S., a trait that cannot but impress a non-U.S. observer. I am thinking of the role of professional associations and organizations in mathematics education and mathematics, without which the situation would have been completely different, and, to be sure, worse, than is the case. The ways in which professional associations like NCTM, MAA, AMS, AAAS, ASI, among others, and committees created by them, have proactively worked to pave the way for reform in numerous aspects of the teaching and learning of mathematics, and have taken the lead in instigating research on them, is an inspiration for those of us who live in countries where change and reform is predominantly a top-down process, with political and administrative authorities holding the reins. In many of our countries, the role played by professional associations is often a reactive one, for instance limited to providing comments on reform proposals "sent down" from above. There is no reason why such associations should not take up similar activities as in the U.S. Although it is not so clear what exactly Utopia is, such activities would bring us closer.

What is the Nature of Mathematics Education Research as a Discipline?

Several chapters in this volume focus on fundamental aspects of mathematics education as a research discipline. This is true of the three "foundation" chapters by Cobb (on philosophy), Silver and Herbst (on theory), and Schoenfeld (on method), as well as of the present one. Moreover, the far majority of the other chapters actually contain sections in which the authors grapple with the nature of our field. This shows that authors feel it necessary to address this, fairly general issue, in order for them to portray and position their particular issue, topic, or theme within the larger research landscape.

In other words, researchers on the teaching and learning of mathematics are pretty much preoccupied with its foundations and characteristics. This would not be the case had the nature of our field been a

settled matter. Most research mathematicians, physicists, and historians do not include sections on foundations and scholarly identity in their publications on Lie groups, viscous liquids, or the French revolution. They work in fields with clear foundational identities, not necessarily in a philosophical, but in pragmatic and paradigmatic sense.

Mathematics education research does not have a clear identity. Well, it does when it comes to the objects of our investigations, the teaching and learning of mathematics in all its dimensions and facets. But when it comes to the goals of our research, opinions begin to differ, albeit not necessarily very much. Are we doing research exclusively in order to obtain intellectual elucidation of certain phenomena and processes, are we striving at ameliorating teaching and learning of mathematics for given groups of students, are we striving at empowering people to cope with life in a changing, complex world, or are we striving at improving society at large? When it comes to the nature of the activities we undertake to study our objects and pursue our goals, opinions really begin to differ.

Let us just take a look at one, “simple,” specific question that divides opinions among mathematics education researchers. The question is “Where does mathematics education research belong? To the social sciences, to the humanities, or to the sciences (including mathematics)? Somewhere else?” I do certainly not intend to answer this question here (well, I do: it belongs “somewhere else”), but the fact that researchers in our field come up with conflicting answers to it—although the majority would probably answer: “the social sciences,” and not just for pragmatic sociological or institutional reasons—suggests that there is a long way to go before we have a complete overview and understanding of the nature of our field in all its dimensions and facets, which is what Utopia is all about.

HOW DO WE GET CLOSER TO UTOPIA?

I shall finish by addressing three general issues that in my view deserve particular attention for research on mathematics teaching and learning to move substantially forward.

Concepts and Definitions in Research on the Teaching and Learning of Mathematics

The French mathematician Henri Poincaré (1908) in a well known article *L'avenir des Mathématiques* (reprinted as the chapter “The future of math-

ematics’ in “Science and Method”), made the often quoted statement that mathematics is the art of giving the same name to different things (“Je ne sais si j’ai déjà dit quelque part que la Mathématique est l’art de donner le même nom à des choses différentes”). This was meant to be opposed to the situation in poetry which may be characterised as the art of giving different names to the same thing. Anna Sfard has proposed, on several occasions (e.g., Sfard, 2005, p. 409f; see also the remarks by Silver and Herbst in this Handbook concerning international conversations about research in mathematics education), that there is a tendency for mathematics education research, too, to be the art of giving different names to the same thing. She makes a call for “conceptual accountability” (Sfard, 2005, p. 419). I share this view.

More generally, it seems to me that key concepts in research on the teaching and learning of mathematics are often used without being defined at all (e.g. “theory” or “theoretical,” see the next section), are vaguely defined (e.g. “problem,” “method,” “proof”), or are defined differently by different researchers (e.g., “modeling,” “reasoning,” “representation,” “belief”). Moreover, as Sfard has pointed out, even though definitions are being provided in scholarly publications, these definitions are seldom operational in the sense that there is an effective way to check whether a given entity actually falls under the definition or not.

Of course, as mathematics education research is usually reported in writing that makes heavy use of ordinary language, with technical terms interspersed, it is inconceivable that every single concept and term would be clearly and unambiguously defined, because such an endeavour would inevitably lead to an infinite regress. However, the fact that the task of defining key concepts, terms and notions as clearly as possible is extremely difficult should not be taken as an argument against trying our very best. Such a counter-argument would be similar to saying that “as a perfectly aseptic environment is impossible, one might as well conduct surgery in a sewer,” as the economist Robert Solow is claimed to have once said.

I would like to propose that committees of experts in our field be established to undertake the task of surveying and reviewing current usages attached to key concepts and terms relevant to their areas of expertise, in order to come up with draft definitions to be discussed in the mathematics education research community at large prior to revision, followed—hopefully—by agreed-upon definitions. If such work were coordinated across committees, preferably by the International Commission on Mathematical Instruction (ICMI), otherwise by other relevant bodies, we might achieve the mounting of a rather authoritative “Dictionary of concept and terms

in mathematics education research,” to be revised at regular intervals. Although coordination would eventually be needed, the initiation of such work would not necessarily have to await coordinated and concerted efforts by central bodies. Small-scale initiatives to pave the way would be valuable in their own right.

The Concept and Role of Theory in Mathematics Education Research

The notion of “theory” is crucial in any scholarly or scientific discipline, including research on the teaching and learning of mathematics. Schoenfeld, in his chapter, argues that any research methodology is fundamentally theory-laden, even though the theory may well be implicit. Although the words “theory” and “theoretical” come up in more and more research publications, it is neither clear what “theory” is actually supposed to mean, nor what foundations theories have and what parts they play in mathematics education. The absence of clear definitions of “theory” and “theoretical” in many publications that invoke these terms implies that they tend to become undefined notions in our discourses, thus laden with whatever connotations authors and readers bring to the interpretation of the research publications.

As is stated several times in this Handbook, there is no such thing as a well-established unified or unifying “theory of mathematics education” which is supported by a majority of mathematics education researchers. On the contrary, different groups of researchers represent different, if not opposing, schools of thought. This suggests that a lot of work needs to be done concerning the origins, foundations, developments, roles, and uses of theories in mathematics education research. We need work of a descriptive nature, focusing on uncovering and analysing the actual state of affairs, as well as work of a prescriptive, normative nature, focusing on what a theoretical underpinning of research in our field should look like. At the very least we should begin to discuss these issues. An excellent beginning has been made by Silver and Herbst in their chapter in this handbook. Their placing of “theory” in mathematics education as a mediator between the vertices of a triangle composed of “problems,” “research,” and “practice” seems very fruitful and worth further exploration.

I shall risk my skin by proposing the following definition of the concept of theory, not just within mathematics education research, but in general (see also Niss, 2006). The point is not (so much) to plea for acceptance of this particular definition but rather to stimulate further discussion of what a definition should look like.

A *theory* is a system of concepts and claims with certain properties, namely

- A theory consists of an *organized network of concepts* (including ideas, notions, distinctions, terms, etc.) *and claims* about some extensive domain, or a class of domains, consisting of objects, processes, situations, and phenomena.
- In a theory, the *concepts are linked in a connected hierarchy* (oftentimes, but not necessarily, of a logical or proto-logical nature), in which a certain set of concepts, taken to be basic, are used as building blocks in the formation of the other concepts.
- In a theory, the *claims are either* basic hypotheses, assumptions, or axioms, taken as *fundamental* (i.e., not subject to discussion within the boundaries of the theory itself), *or* statements obtained from the fundamental claims by means of *formal* (including deductive) *or material* (i.e., experiential or experimental with regard to the domain(s) of the theory) *derivation*.

Theories differ—greatly—with respect to the origin and nature of the *domains* (or classes of domains)—and the entities that populate them—to which they refer. They further differ with respect to the origin and nature of the *concepts* and conceptual *networks and hierarchies* involved, and regarding the fundamental and derived *claims* that are made. Theories also differ with respect to the kinds of methods by which claims are *derived* and *justified*. Finally, theories differ as far as their degrees of *stability*, *coherence*, and domain *coverage* are concerned.

In research, theories may serve different *purposes*. (A given theory, can, of course, serve several purposes at the same time.) One—crucial—purpose of a theory is to provide *explanation* of some observed phenomenon. Explaining a phenomenon by way of a theory means that the occurrence of the phenomenon can be obtained as a claim in the theory under specified conditions and circumstances, the fulfilment of which are substantiated as part of the explanation. A closely related purpose is to provide *predictions* of the (possible) occurrence of certain phenomena as a claim resulting from the (possible) fulfilment of the preconditions of their occurrence. Prediction will sometimes, but not necessarily always, rely on explanation of causes and mechanisms, while explanation will sometimes, but not necessarily always, give rise to predictions. Another purpose is to provide *guidance for action or behavior* by employing knowledge of substantiated claims within a theory as a means to plan

and implement action or behaviors so as to achieve desirable outcomes.

A fourth purpose of a theory, and perhaps the most important one as regards research on mathematics education, is to provide *a structured set of lenses*, through which aspects or parts of the world can be approached, observed, studied, analysed or interpreted. This takes place by selection of the elements to be considered important in the context, by focusing on certain features, issues, or problems; by adopting and utilising particular perspectives, and by providing a *methodology* for answering questions concerning the domain(s) considered.

A related, but not identical, purpose is to provide *a safeguard against unscientific approaches* to a problem, issue or theme, like, for example, haphazard and inconsistent choices with regard to terminology, research methodology, and interpretation of results. This purpose, which is a prominent one in mathematics education research, is pursued by articulating underlying assumptions and choices and making them subject to discussion; by situating one's research within some framework; and by explicating its characteristics vis-à-vis possible alternatives. A concise way of expressing this purpose is due to the late David Wheeler (personal communication) when he said "the purpose is simply to keep us honest."

Any mathematics educator has experienced criticism of our field from sceptical if not hostile colleagues in other fields (especially mathematics, psychology and general education) with regard to the foundations of our research and its results. Researchers in the humanities and the social sciences often have encountered similar criticism. This gives rise to the sixth and final purpose of theory in research: to provide (some) *protection against attacks from outside*.

In research on the teaching and learning of mathematics all these purposes seem to be in play, with varying emphases for different researchers. However, as the specific *role* of theory in a given piece of research is only seldom made explicit, let alone discussed, it is not so easy to detect this role empirically. This is, of course, the case with the rather numerous research papers which do not explicitly invoke or employ any theory at all. But it is also the case with publications that do invoke a theory without going beyond the mere invocation as such (Lerman & Tsatsaroni, 2003). This happens pretty often, when some theoretical framework is being referred to in the beginning or at the end of the publication without having any presence in between. In such cases one might be tempted to suppose that the main purpose in play is the last-mentioned one, unless it is just a matter of convention or ritual: there must be some mentioning of theory in a research pub-

lication. When a theory is actually employed in a piece of research, not just invoked, its role can be to serve as an overarching framework, to organise a set of observations and interpretations of singular but related phenomena into a coherent whole, to provide distinctions and terminology to the research, or to offer a research methodology. A number of other, more specific, roles have been identified in the chapter by Silver and Herbst. In mathematics education research, one role of theory seems to be predominant, namely that as a purveyor of sets of *hypotheses* put forward to account for observed phenomena. This relates to the overarching purpose of providing explanation. The fact that we usually encounter loosely organised sets of, typically a few, hypotheses rather than elaborate edifices of tightly organized networks of concepts and claims (as required in my definition of theory) is a reflection of the weak theoretical underpinning of our field.

It is characteristic that the majority of the theories actually invoked (but not necessarily employed) in mathematics education research are borrowed from other fields and disciplines. In addition to mathematics, we encounter statistics, psychometrics, philosophy, neuroscience, psychology, pedagogy, education, linguistics, semiotics, sociology, anthropology, political science, and history. There are also examples of what has been named "home-grown theories" developed more or less within the discipline of mathematics education research itself. Without engaging, here, in a discussion of the extent to which they are theories in the sense defined above, let me just mention, for example, APOS, Action-Process-Object-Schema (see, e.g. the chapter by Artigue, Batanero, & Kent), the distinction between concept definition and concept image, the process-object duality and its ramifications into encapsulation, reification, and procepts; the French theories of didactical situations (Brousseau), of didactical transposition (Chevallard), and conceptual fields (Vergnaud). American theories of socio-mathematical norms (Cobb and Yackel), teachers' mathematical knowledge for teaching (Ball and colleagues), and metacognition (Schoenfeld). All of these are treated in previous chapters.

The collection of theories put to use in mathematics education does not seem to be satisfactory or sufficient to serve as the foundation for research on the teaching and learning of mathematics. The theories imported from other fields are of too general a nature to be transposed to offer a sufficient pool of specific results and concrete methodologies, so as to provide complete guidelines for research in our field. The home-grown theories currently available are of too limited scopes to provide a comprehensive coverage of our field.

In Utopia, what would a full-fledged theory of mathematics education look like? I realise that Cobb, in the vein of his advocacy for “theorizing as *bricolage*” in his chapter in this Handbook, would probably deem this question at best irrelevant, at worst impossible. Similarly, Silver and Herbst, in their chapter, make a plea for what they call “an ecumenical approach,” while also seeing benefits in contemplating the possibility of a “grand theory of mathematics education.” Nevertheless, I think the question is worth taking seriously, in order to envision a potential role model for theory development. Without in any way pretending to be able to answer the question, a theory of mathematics education would, I would like to submit, contain at last the following sub-theories, somewhat similar to what Silver and Herbst call “middle-ranged theories,” linked together as a coherent and consistent system:

- a sub-theory of *mathematics as a discipline and a subject* in all its dimensions, including its nature, role, and exercise in *society and culture*;
- a sub-theory of *individuals’ and groups’ affective notions*, experiences, emotions, attitudes, and perspectives with regard to their actual and potential encounters with mathematics, and the outcomes thereof;
- a sub-theory of *individuals’ and groups’ cognitive notions*, beliefs, experiences, and perceptions with regard to their actual and potential encounters with mathematics, and the outcomes thereof;
- a sub-theory of *the teaching of mathematics* seen within all its institutional, societal, national, international, cultural and historical contexts;
- a sub-theory of *teachers of mathematics*, as individuals and as communities, including their personal and educational backgrounds and professional identities and development.

All sub-theories have to account for *situating* their objects, processes, situations, and phenomena *in all the contexts and environments that influence them*, be they scientific, biological, anthropological, linguistic, philosophical, economic, sociological, political, or ideological. Each sub-theory has to live up to the general requirements of a scientific/scholarly theory, for example, the ones outlined above, including accounting for the ways in which its claims are obtained and justified. The sub-theories *cannot just be juxtaposed*. They have to be integrated into a coherent and consistent whole, alone because mathematics is a constituent component in all of them.

For each of the five domains pointed to here it is probably possible to create several meaningful sub-

theories. When combined this gives rise to a potential of several competing “grand” theories of our field. So we shall presumably never have just one complete theory of mathematics education. However, based on our accomplishments in the past three decades we have the potential to really begin to undertake the task of moving towards such overarching theories.

Although it may be true that the search for a grand theory of mathematics education is futile, at least at the present state of development in our field, what would not be futile would be to undertake a meta-research programme that we might call *theory archaeology*. By this I mean a systematic effort to identify, uncover and analyse the alleged as well as the actual roles of “theory,” “theoretical framework,” and “theorizing” encountered in a wide selection of research publications in mathematics education research. Such a programme would help us transcend the rather unfortunate state of affairs in our field where terms derived from “theory” are ubiquitous without, in many cases, providing more coverage than did the emperor’s new clothes. At least such an endeavour would assist in keeping us honest, to reiterate, at a meta-level, David Wheeler’s dictum.

Past Insights Should Not Be Forgotten

From time to time researchers in our field deplore its apparent lack of accumulation. There are several causes for this deficiency. Here I shall concentrate on one of them.

We are not good enough at looking back at what our predecessors have accomplished and to take advantage of it in our current thinking. Our colleagues of the past are the ones that first broke what later became the roads on which we walk today. Their insights were no less deep than ours, but were shaped by experience, intuition and reflection in territories that were much more virgin than the territories at our disposal. These territories have later been cultivated step by step by means of research, much of which was foreseen and requested by our “ancestors.” But thirty to forty years ago, research, in the modern sense, was scarce. The approaches of our predecessors are sometimes pejoratively referred to as “opinionating.” Yet quite a few of their opinions were instrumental in pointing out what was worth investigating. We have a bad habit of discarding writings of the past, unless they are singular either because they are early instances of what later became mainstream developments, or because they can serve as especially articulate examples of the naïve and undeveloped past that we, thank God, have left long ago (was it yesterday?).

I plead for revisiting past accomplishments, not in order to honor our predecessors, although quite a few of them do indeed deserve it, but because it is in the best interest of our field to revisit the contributions of the past. We need a scholarly researched history of mathematics education research. There do exist excellent contributions to such a history (Kilpatrick, 1992, focusing primarily on the U.S.), but much more is needed, and of an international scope. Being aware of the history of mathematics education research would not only help us find forgotten, or even hidden, treasures, but would also make us less susceptible to the fads and fashion waves that haunt our field from time to time. The observations and comments made at the conclusion of my plenary lecture at ICME-9, Tokyo-Makuhari 2000 (Niss, 2004b), unfortunately do not seem to be outdated today, 6–7 years later. Allow me to quote myself :

As I see it, there is such a huge body of highly valuable outcomes of excellent research that need to be surveyed, structured, brought together and synthesized. It appears to be one of the weaknesses of our profession that many of us, myself included, tend to write and speak too much and read and contemplate too little. I would conjecture that if more systematic revisiting and synthesizing of research were carried out we would realise that “we know more than we know that we know.” (p. 53)

The other chapters of the present Handbook (re)present major steps forward to doing just that, identifying, digesting, surveying and synthesizing significant contributions to mathematics education research, so as to make us more conscious of what we know. Yet, the chapters also show that what we know is still highly diversified and complex, in the sense that it is not easy to produce brief and clear conclusive statements that are both correct, focused and concrete.

CONCLUSION

By its very nature, this chapter presents personal observations and reflections on the field of mathematics education as a research domain, based on all the previous chapters of this Handbook. What would Utopia—of course a point at infinity—in our field look like? Where are we today? What progress have we made during the last fifteen years? What could/should we do in order for us to make substantial new progress in our endeavour to approach Utopia? These are the questions that preoccupied me—and hopefully the readership as well—in the sections above.

Unavoidably, my observations and reflections are premised on my background, experiences, values, opinions, and taste. But I am profiting massively from the work of my colleagues, the authors of this Handbook. Needless to say, it has not been possible in this chapter to do justice to all the preceding ones. I want to conclude by expressing my gratitude to and admiration for the editor and the authors of this Handbook. By their efforts to identify, survey, and review fundamental parts and aspects of mathematics education research they have done invaluable service to our field and to our community, and not least to its newcomers, all of whom would be well advised to seriously study the relevant chapters of this volume at an early stage of their careers. In addition to bringing us up to date with significant accomplishments over the last couple of decades, the authors have succeeded in demonstrating that our field is not only alive and kicking but amidst a vibrant development of reclaiming promising new land. The construction of roads to Utopia is well under way.

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AUTHOR NOTE

I would like to extend my sincere thanks to the editor, Frank K. Lester, Jr., for having given me the unique opportunity to reflect on the chapters in this Handbook. Although this chapter has had no reviewers in the classical sense, its final version has greatly benefitted from his comments and suggestions.

About the Contributors

Michèle Artigue earned a Ph.D. in Mathematical Logic and began her career doing research in that area. She progressively moved toward doing research in mathematics education, thanks to her involvement in the activities of research, innovation and teacher training developed at the Institute of Research in Mathematics Education at the University Paris 7. In this field, beyond theoretical contributions on the relationships between epistemology and didactics, and on didactical engineering, her main research topics have been the teaching and learning of mathematics at university level, and especially the didactics of calculus and analysis, and the integration of computer technologies into mathematics education. She is currently full professor in the Mathematics Department at the University Paris 7 and director of the master program in didactics at this University. She has many editorial and scientific responsibilities, and since 1998 has been vice-president of the International Commission on Mathematics Instruction.

Deborah Loewenberg Ball is William H. Payne Collegiate Professor and Dean of the School of Education, at the University of Michigan. Ball's work draws on her many years of experience as an elementary classroom teacher. Her research focuses on mathematics instruction and on interventions designed to improve its quality and effectiveness. Her research groups study the nature of the mathematical knowledge needed for teaching and develop survey measures that have made possible analyses of the relations among teachers' mathematical knowledge, the quality of their teaching, and their students' performance. Of particular interest in this research is instructional practice that can intervene on significant patterns of educational inequality in mathematics education. Currently a member of the National Math Panel, Ball served on the Glenn Commission on Improving Mathematics Education for the 21st century, chaired the Rand Study Panel on programmatic research in mathematics education, co-chaired the international Study on the Professional Education of Teachers of Mathematics sponsored by the International Commission on Mathematical Instruction, and she is a trustee of the Mathematical Sciences Research Institute in Berkeley.

Carmen Batanero is Senior Lecturer of Mathematics Education at the University of Granada, Spain, where she teaches mathematics education and statistics education, and supervises doctoral dissertations on statistical education. Before joining this Department in 1988, she worked in the Department of Statistics where she taught probability and biostatistics courses for 11 years and completed her Ph.D. in Statistics (Stochastic Point Processes and Systems Reliability). Her primary research interest is statistics education, with special emphasis in the teaching of statistical inference. She is also interested in the theoretical and methodological bases for research on mathematics education and the education of mathematics teachers to teach statistics. Her publications include research papers, chapters in some international books, and books directed to teachers. She was co-editor of *Statistics Education Research Journal* (2002–2003), President of the International Association for Statistical Education (IASE) (2001–2003), and member of the International Commission for Mathematics Instruction (ICMI) Executive Committee (2003–2006). She also helped to establish the stochastic group at PME (Psychology of Mathematics Education). She has served on the International Programme Committee of several international conferences and is currently coordinating a Joint ICMI /IASE Study on Statistics Education.

Dan Battey is an Assistant Professor of Mathematics Education at Arizona State University, where he has been on the faculty since 2005. Previously, he was a post-doctoral fellow in Diversity in Mathematics Education at the UCLA Graduate School of Education & Information Studies. His work centers on engaging urban elementary school teachers in opportunities to learn within and from their practice of teaching mathematics in a way that sustains and generates change. Drawing on research of students' mathematical knowledge, he studies teacher knowledge, identity, and practice as well as equity issues within the context of urban schools.

Michael T. Battista is Professor of Mathematics Education in the College of Education at Michigan State University. Most of his research focuses on the learning and teaching of geometry and geometric measurement, and on the use of technology in mathematics teaching. He

has served on the editorial panel for the *Journal for Research in Mathematics Education*, is one of the authors of the geometry modules for the *Investigations in Number, Data, and Space* curriculum, is the geometry author for the grades 3-6 *Navigations* problem solving and reasoning books, is on the editorial panel for NCTM's 71st yearbook on geometry (2009), and is serving on NCTM's *Research Committee*. He has also served as a workshop co-leader in the Mathematical Association of America's *Preparing Mathematicians to Educate Teachers* program. Currently he is directing two NSF grants, the first to develop a *Cognition Based Assessment* system for elementary mathematics, and the second to investigate how elementary teachers learn, make sense of, and use research on students' mathematical thinking. At Michigan State, he is co-leader of the mathematics group for the Carnegie Foundation funded *Teachers for a New Era* project, and he is faculty leader for secondary mathematics in the secondary teacher education program.

Alan J. Bishop is Emeritus Professor of Education at Monash University, Melbourne, Australia where he has been since 1992. He took his PhD at Hull University in UK, then worked as Lecturer in mathematics education at Cambridge University, UK, from 1969–1992. His first area of research concerned visualisation, spatial ability and geometry learning, but in the 1970s he became more interested in socio-cultural aspects of mathematics education. He was Editor of the journal *Educational Studies in Mathematics* from 1978-90, has been an Advisory Editor to that journal since then, and is Managing Editor of the book series *Mathematics Education Library* published by Springer. He was Chief Editor of the two international handbooks of mathematics education published by Kluwer in 1996 and 2003, and has published numerous books, chapters and papers.

Glendon W. Blume is Professor of Education (Mathematics Education) at the Pennsylvania State University, where he has been a member of the faculty since 1986. His primary research interests focus on the areas of technology in mathematics teaching and learning and on secondary mathematics teachers' understanding of mathematics. He has been a co-Principal Investigator or Faculty Research Associate on grants funded by the National Science Foundation for over 15 years. He has served on the editorial panel for the 2005 Yearbook of the National Council of Teachers of Mathematics and has served on or chaired several committees of the Council. He has been a co-editor of the Yearbooks of the Pennsylvania Council of Teachers of Mathematics since 1989, and he has co-authored four book chapters that provide interpretive reports based on data from the National Assessment of Educational Progress from 1992 through 2003.

David W. Carraher is Senior Scientist at TERC, in Cambridge, MA, where he directs the TERC-Tufts *Early Algebra, Early Arithmetic Project* (www.earlyalgebra.terc.edu). He has been the Principal Investigator in NSF-funded longitudinal studies of the development, implementation, and evaluation of Early Algebra lessons in grades 2 to 5. He has published widely about learning mathematics in and out of school, rational number, and early algebra, and is the author of prize-winning mathematics education software. He is the director of research of the NSF-MSP project, *The Fulcrum Institute for Education in Science*, another collaboration between TERC and Tufts University. As Professor of Psychology in Brazil, he carried out influential research on everyday mathematics in and out of school. He chaired the scientific program committee of PME-95 (Psychology of Mathematics Education), edited, with R. Nemirovsky, a CD-based *JRME* Monograph about the *Uses of Video in Mathematics Education Research*, and co-edited, with Jim Kaput and Maria Blanton, *Algebra in the Early Grades*, published by Erlbaum in 2007. In 2006 he won the Premier Concours de Photos Numériques, Ste. Suzanne, Mayenne, France.

Douglas H. Clements is Chancellor's Professor of Education at University of Buffalo, SUNY. His primary research interests lie in the areas of the learning and teaching of geometry, computer applications in mathematics education, the early development of mathematical ideas, and the effects of social interactions on learning. His most recent interests are in creating, using, and evaluating a research for research-based curriculum and in taking successful curriculum to scale using technologies and learning trajectories. He has published over 90 refereed research studies, 6 books, 50 chapters, and 250 additional publications. He has directed several projects funded by the National Science Foundation and the U.S. Education Department's Institute of Educational Sciences, including curriculum development, investigations of children's development of geometric and spatial ideas, and experiments using large-scale randomized trials. His most recent projects include studies of the implementation of a research-based intervention including the *Building Blocks* preschool curriculum, which was developed with NSF funding. He was chair of the Editorial Panel of NCTM's research journal, the *Journal for Research in Mathematics Education*.

Paul Cobb is Professor of Mathematics Education at Vanderbilt University, where he has been a member of the faculty since 1992. He teaches courses in elementary mathematics methods, design research methodology, and the institutional setting of mathematics teaching and learning. His research interests focus on instructional design, the classroom microculture, and the broader institutional setting of mathematics teaching and learning. He is an

elected member of the National Academy of Education and received the Hans Freudenthal Medal for cumulative research program over the prior ten years from the International Commission on Mathematics Instruction (ICMI) in 2005. He an Invited Fellow of the Center for Advanced Studies in the Behavioral Sciences and received the award for outstanding article published in the *Journal for Research in Mathematics Education* in 1996. He currently serves on the editorial boards of seven journals in mathematics education and the learning sciences.

Erik De Corte is Emeritus Professor of Educational Psychology and member of the Center for Instructional Psychology & Technology at the University of Leuven, Belgium. His major research interest is to contribute to the development of theories of learning from instruction and the design of powerful learning environments, focusing thereby on learning, teaching, and assessment of thinking and problem solving, especially in mathematics. He was the first President (1985–1989) of the European Association for Research on Learning and Instruction (EARLI), and the founding editor of the EARLI journal *Learning and Instruction* (1990–1993). From 1987 until 2002 he was associate editor of the *International Journal of Educational Research*. He co-edited (with the late F.E. Weinert) the *International Encyclopedia of Developmental and Instructional Psychology* (1996). In 1997 he received the EARLI “Oeuvre Award for Outstanding Contributions to the Science of Learning and Instruction.” He was President of the International Academy of Education from 1998 until 2006. Respectively in 2000 and 2003 he was conferred the doctorate honoris causa of the Rand Afrikaans University, Johannesburg, and the University of the Free State, Bloemfontein, South Africa. During the academic year 2005–2006 he stayed as a Fellow at the Center for Advanced Study in the Behavioral Sciences at Stanford.

Jan de Lange is a Full Professor in Mathematics Education in the Department of Mathematics of the Faculty of Science at Utrecht University, The Netherlands. He has worked at the institute that is now known as the Freudenthal Institute since 1976 and has been leading this institute since. His interest in issues in mathematics education ranges from pre-primary to tertiary, from modeling and applications to assessment, from software to the web, and from educational design to implementation problems. Recently he has explored how cognitive psychology and educational neurosciences can, in some kind of complementarity, contribute to the theory of learning mathematics. He has been a member of the Mathematical Sciences Education Board at the National Research Council, Washington D.C., on the National Committee for TIMSS, on the international TIMSS-R committee, is Chair of the International Mathematics Experts Group of the PISA study of OECD

(Paris), a core member of the Numeracy and Literacy Commission of OECD, and member of a content validity commission of NAEP. He has been PI on many national and international projects, published several hundred articles, and delivered lectures in more than sixty countries. He is visiting Professor at the University of Wisconsin at Madison, and Director of FI-us at the University of Colorado at Boulder.

Thomas P. Dick is Professor of Mathematics and Coordinator of Collegiate Mathematics Education at Oregon State University, where he has been a member of the faculty since 1986 and directs both the Mathematics Learning Center and the Math Excel program. His primary research interests lie in the areas of advanced mathematical thinking, mathematical discourse, and the influences of technology on mathematics teaching and learning, especially calculus. He currently serves on the editorial panel for the *Journal for Research in Mathematics Education* (2004–2007) and as an associate editor for *School Science and Mathematics*. He is a former chair of the Committee for Research in Undergraduate Mathematics Education (American Mathematical Society/Mathematical Association of America) and a former chair of the College Board’s Advanced Placement Calculus Development Committee. He is an advisor on several projects involving the use of technological tools in mathematics instruction.

The Diversity in Mathematics Education Center for Learning and Teaching (DiME) is a consortium of three universities and three school districts that focuses on issues of equity in mathematics education. Consortium partners include the University of Wisconsin-Madison, the University of California at Los Angeles, and the University of California at Berkeley. About a dozen faculty and 30 graduate fellows and post doctoral fellows participate in DiME. DiME seeks to create new knowledge and research, new resources and tools, and a new generation of researchers and instructional leaders who are capable of making significant progress on issues of equity and diversity in mathematics education. Among the issues addressed by DiME research are (a) how opportunities for students from non-dominant cultures to learn mathematics with understanding are supported and how they are limited in schools and classrooms; (b) how teachers can be supported to engage in culturally relevant pedagogy, which builds on students’ cultural histories and engages them in using mathematics to address issues of social justice; and (c) how mathematics learning is mediated at multiple levels by issues of race and power. The chapter on Race, Culture and Mathematics Education is based on a center-wide initiative to address various explanations for differential mathematics achievement. Building on that work, an editorial panel was selected to take the lead in writing the handbook chapter. The panel

consisted of the following DiME members. **Vanessa R. Pitts Bannister** is an Assistant Professor of Mathematics Education at the Virginia Polytechnic Institute and State University located in Blacksburg, VA. She is interested in the relationships between teachers' pedagogy and knowledge of equity/diversity issues, curriculum, subject matter and student thinking. **Tonya Gau Bartell** is an Assistant Professor of Mathematics Education at the University of Delaware. Her primary research interest lies in examining how teachers learn to teach mathematics for social justice, including building understanding of how teachers might actually practice critical pedagogy and how teacher educators might help in-service teachers do so. **Dan Battley** is an Assistant Professor of Mathematics Education at Arizona State University. His research centers on teacher learning and equity issues in mathematics education, particularly in urban schools. **Victoria M. Hand** is an Assistant Professor of Mathematics Education at the University of Wisconsin at Madison. Her research investigates how particular pedagogical approaches, tools, and activity structures in mathematics classrooms afford and constrain students' negotiation of their sociocultural practices and identities. **Joi Spencer** is an Assistant Professor of Mathematics Education at the University of San Diego. Her research is interested in how discourse around race shapes the learning opportunities of African American children in mathematics. DiME Fellows also involved in writing the chapter include Filiberto Barajas, Rozy Brar, Kyndall Brown, Indigo Esmonde, Mary Q. Foote, Charles Hammond, Carolee Koehn, Mara G. Landers, Mariana Levin, Shiuli Mukhopadhyay, Ann Ryu, Marian Slaughter, Anita A. Wager. DiME directors Thomas Carpenter, Megan Franke, and Alan Schoenfeld provided oversight and guidance for the writing of the chapter.

Joan Ferrini-Mundy is a University Distinguished Professor in Mathematics Education at Michigan State University, where she has been Associate Dean and Director for Science and Mathematics Education in the College of Natural Science since 1999. Her experiences include two years as a Visiting Scientist at the National Science Foundation, and four years as the Director of the Mathematical Sciences Education Board at the National Research Council. For 18 years, Ferrini-Mundy was a member of the Mathematics Department at the University of New Hampshire. She has been active with the National Council of Teachers of Mathematics, serving on its Board of Directors from 1993 to 1996, and as chair of the Writing Group for the 2000 *Principles and Standards for School Mathematics*. Ferrini-Mundy has received numerous federal, state, and private foundation grants for projects in research and teacher education. Currently she is PI for an NSF Comprehensive Mathematics and Science Partnership project involving about 60 school districts in Michi-

gan and Ohio. She is also serving as Project Director for an initiative with the Michigan Department of Education to review and improve policies relative to teacher preparation. Her research interests include mathematics teacher knowledge and K–12 mathematics program improvement.

Robert E. Floden is Professor of Teacher Education, Measurement & Quantitative Methods, and Educational Psychology at the Michigan State University College of Education, where he has been a faculty member since 1977. His research has examined the effects of education policies on teaching and learning, with a special emphasis on roles of preservice teacher preparation and professional development. He is currently co-PI of Michigan State University's *Teachers for a New Era* initiative and co-PI on a project developing measures of teachers' mathematical knowledge for teaching algebra. From 1987 to 1989 Floden spent time at the Universität Tübingen as an Alexander von Humboldt Fellow. He was Features Editor of *Educational Researcher* from 1993 to 1995 and Editor of *Review of Research in Education* from 2003 to 2005. Floden served as president of the Philosophy of Education Society in 2004–5 and received the Margaret B. Lindsey Award for Distinguished Research in Teacher Education from the American Association of Colleges for Teacher Education in 2006. Floden's work has been published in the *Handbook of Research on Teaching*, the *Handbook of Research on Teacher Education*, and many journals.

Dr. Helen J. Forgasz is a Senior Lecturer in the Faculty of Education, Monash University (Australia), where she has been a member of the academic staff for four years. Her research interests in mathematics education include: gender equity issues, the affective domain, technology use for mathematics learning, and learning settings. From 2003–2006, Helen was the editor of the *Mathematics Education Research Journal* and is currently a member of the editorial boards of *Educational Studies in Mathematics* and the *Journal for Mathematics Teacher Education*. She is a member of the International Committee for the International Group for Psychology in Mathematics Education, and has served on the executive committee of the Mathematics Education Research Group of Australasia as Vice President (Research) 2001–2003 and Vice President (Conferences) 1999–2001. Helen has presented many papers at national and international conferences, and has published widely in scholarly and professional books and journals.

Megan Loef Franke is an associate professor in the Graduate School of Education and Information Studies at UCLA and the Director of Center X: Where Research and Practice Intersect for Urban School Professionals. Dr. Franke's research focuses on understanding teacher learning and professional development, particularly as it pertains to the teaching of mathematics. She uses her de-

veloping understanding of these issues to consider issues of urban school reform and social justice. Over the past 20 years Dr. Franke has worked with the Cognitively Guided Project to support and learn from teachers as they attempt to make use of the development of children's mathematical thinking. She has published in the *Journal for Research in Mathematics Education*, *Teaching and Teacher Education*, *American Educational Research Journal*, and *Journal for Mathematics Teacher Education*.

Brian Greer is currently an independent scholar, having held positions in the School of Psychology, Queen's University, Belfast, 1969–2000, and Department of Mathematics and Statistics, San Diego State University, 2000–2003. Major research and theoretical interests have included multiplicative reasoning, word problem solving, and the relationships between cognitive psychology and mathematics education. His current focus is on the cultural, historical, and political contexts of mathematics education

Douglas A. Grouws is Professor of Mathematics Education and William T. Kemper Fellow at the University of Missouri. He has a long history of research and scholarship in the mathematics education field. He was a member of the National Academy of Science committee that produced the recent report, *On Evaluating Curricular Effectiveness: Judging the Quality of K–12 Mathematics Evaluations*. He is author or co-author of three books including the editorship of the first *Handbook of Research on Mathematics Teaching and Learning*. A lifetime member of the National Council of Teachers of Mathematics, he currently serves on NCTM's task force on the interpretation of the results of the NAEP assessments in mathematics. He is co-Principal Investigator at the Center for the Study of Mathematics Curriculum (CSMC) and he currently directs two National Science Foundation funded projects: Mathematics Through Technology (MTT) and Comparing Options in Secondary Mathematics: Investigating Curriculum (COSMIC).

Guershon Harel is Professor at the Mathematics Department at the University of California, San Diego. Previously he served as Associate Editor of the *American Mathematical Monthly*, co-editor of the *Research in Collegiate Mathematics Education Series*, and Chair of the Editorial Board of the *Journal for Research in Mathematics Education*. Harel has research interest in cognition and epistemology of mathematics and their application in mathematics curricula and the education of mathematics teachers. Until the mid 1990s, Harel's research interest revolved around the notions of multiplicative conceptual field and advanced mathematical thinking, with particular attention to the concept of function, proof, and the learning and teaching of linear algebra. He is co-editor of two books in these areas: *The Development of Multiplicative Reasoning*

in the Learning of Mathematics and *The Concept of Function: Aspects of Epistemology and Pedagogy*. In the mid 1990s, he began to center his attention on the learning and teaching of proof, and since then he has been working collaboratively in this area with Larry Sowder.

M. Kathleen Heid is Distinguished Professor of Education and Professor of Mathematics Education at The Pennsylvania State University, where she has been a member of the faculty since 1984. Her research has centered on the impact of technology on the teaching and learning of mathematics, particularly algebra and calculus, and on mathematical knowledge of secondary teachers. She served on the Board of Directors of the National Council of Teachers of Mathematics from 2003 to 2006, served on the Board of Governors for the Mathematical Association of America from 1998 to 2003, and is currently Editor-Elect of the *Journal for Research in Mathematics Education*. She has co-directed several federally funded mathematics education projects centered on creating computer-intensive curricula for secondary schools and preparing teachers to teach mathematics in technology-intensive environments, she has served on the editorial boards of several research journals, and she is co-editor of *Research on Technology in the Teaching and Learning of Mathematics: Syntheses, Cases, and Perspectives*. She is currently co-PI for the NSF-funded Mid-Atlantic Center for Mathematics Teaching and Learning.

Patricio Herbst is an Associate Professor of Education at the University of Michigan, where he has been a member of the faculty since 1999. His primary research interest is on mathematics instruction in classrooms, particularly in relation to the teachers' management of students' reasoning and proving in high school geometry, and on the design and use of multimodal representations of teaching to elicit teachers' perspectives on different kinds of instruction. He has served on the Executive Board of the AERA Special Interest Group on Research in Mathematics Education and is on the editorial boards of *Journal of Mathematics Teacher Education*, *Educational Studies in Mathematics*, and *Recherches en Didactique des Mathématiques*.

James Hiebert is the Robert J. Barkley Professor of Education at the University of Delaware, where he teaches in programs of teacher preparation, professional development, and doctoral studies. His professional interests focus on mathematics teaching and learning in classrooms. He has edited books on students' mathematics learning and co-authored the books *Making Sense: Teaching and Learning Mathematics with Understanding* and *The Teaching Gap: Best Ideas from the World's Teachers for Improving Education in the Classroom*. He served on the National Research Council committee that produced *Adding It Up* and *Helping Children Learn Mathematics*, was

director of the mathematics portion of the TIMSS 1999 Video Study, and is a PI on the NSF-funded Mid-Atlantic Center for Teaching and Learning Mathematics. He has served on the editorial boards for *American Educational Research Journal*, *Cognition and Instruction*, *Elementary School Journal*, *Journal for Research in Mathematics Education*, *Journal of Educational Psychology*, *Journal of Mathematical Behavior*, and *Mathematics Thinking and Learning*.

Heather C. Hill is an assistant professor and associate research scientist at the University of Michigan. Her primary work focuses on developing measures of mathematical knowledge for teaching, and using these measures to evaluate public policies and programs intended to improve teachers' understanding of this mathematics. Her other interests include the measurement of instruction more broadly, instructional improvement efforts in mathematics, and the role that language plays in the implementation of public policy. She received a Ph.D. in political science from the University of Michigan in 2000 for work analyzing the implementation of public policies in law enforcement and education. She has served as section chair for AERA division L (politics and policy), and on the editorial board of *Journal for Research in Mathematics Education*. She is the co-author, with David K. Cohen, of *Learning Policy: When State Education Reform Works* (Yale Press, 2001).

Graham A. Jones, is an Emeritus Professor at both Illinois State University, where he was a faculty member from 1991–2002, and Griffith University, Australia, where he was Pro-Vice-Chancellor from 1986–1991. His major research activities have focused on the formulation and validation of cognitive models in probability, statistics, and number sense, and the use of these models in teaching and learning at the elementary and middle school levels. He has also investigated teacher knowledge in geometry, probability, and statistics during teacher enhancement projects. His publications include numerous research articles and books in these areas of interest and he has recently edited the book, *Exploring Probability in School: Challenges for Teaching and Learning* (Springer). From 1979 until 1984 he was the first president of the *Mathematics Education Research Group of Australasia* (1979–1984), and has served for varying periods of time on the editorial boards of the *Journal for Research in Mathematics Education*, *Mathematics Thinking and Learning*, *Mathematics Education Research Journal* (Australia) and *The Mathematics Educator* (Singapore). For the past 20 years he has regularly been involved in coordination roles, in teacher development and elementary education, at International Congresses in Mathematical Education.

Elham Kazemi is Associate Professor of Mathematics Education at the University of Washington, where she has

been on the faculty since 1999. Her research interests include studying the development of intellectual community among teachers and students in elementary mathematics. In particular, she has studied how teachers learn about children's reasoning in classroom practice and how teachers' collaborative inquiry creates new cultures of learning within schools. Her work with teachers is informed by her study of children's social and intellectual experiences in classrooms, and in recent years comparing children's experience across mathematics and literacy. Over the last six years, she has been an investigator in several federally-funded grants designing and studying professional education, especially for teacher leaders. Her works appears in places such as *Elementary School Journal*, *Theory into Practice*, *Journal of Mathematical Behavior*, and *Journal for Mathematics Teacher Education*.

Phillip Kent is a Research Officer in mathematics education at the School of Mathematics, Science and Technology, University of London Institute of Education. His research specializes in mathematics in workplaces and in professional practice, particularly in relation to the evolution of working practice due to information technology, and how new mathematical technologies may change the learning and use of mathematics in workplaces. He has done nationally-funded research projects and published widely in this area. He is a member of the London Knowledge Lab, an inter-disciplinary research institute of the University of London. Before joining the Institute of Education in 2001, he was at Imperial College, University of London, where he gained a PhD in Applied Mathematics in 1992, and worked for eight years in undergraduate curriculum development projects.

Carolyn Kieran is Professor of Mathematics Education at the Université du Québec à Montréal, where she has been a faculty member of the Department of Mathematics since 1983. Her primary research interest is the learning and teaching of algebra, with a particular focus on the roles played by computing technology. She served as President of the International Group for the Psychology of Mathematics Education from 1992 to 1995 and as a member of the Board of Directors of the National Council of Teachers of Mathematics from 2001 to 2004. Past contributions also include the Mathematics Learning Study committee and its project volume *Adding It Up*, the International Program Committee for the 12th ICMI Study on Algebra, the chair of the editorial panel of the *Journal for Research in Mathematics Education*, the editorial board of *A History of School Mathematics*, and the vice presidency of the Canadian Mathematics Education Study Group. Recent publications comprise chapters in the *Handbook of Research on the Psychology of Mathematics Education*, and *The Future of the Teaching*

and *Learning of Algebra*, and an article in the *International Journal of Computers for Mathematical Learning*.

Susan J. Lamon is a professor in the Department of Mathematics, Statistics, and Computer Science at Marquette University, where she has been on the faculty since 1989. Her research employs children's and adults' thinking to investigate the nature and development of proportional reasoning, its relationships to rational number knowledge, and its enhancement through the process of mathematical modelling.

Her research has led to the development and testing of classroom materials designed to facilitate rational number knowledge and proportional reasoning in elementary through university-level mathematics. In addition to serving on the executive committee of the International Committee on Teaching Mathematical Modelling and Applications (ICTMA), she reviews articles related to rational numbers and proportional reasoning for 10 major journals, serves on advisory boards for several federally-funded projects, and consults internationally with Ministries of Education to improve mathematics curricula.

Cynthia W. Langrall is a Professor in the Mathematics Department at Illinois State University. Her research interests are in the development of elementary and middle school students' probabilistic and statistical reasoning. Publications in these areas include several book chapters and articles in *Mathematical Thinking and Learning*, *Journal for Research in Mathematics Education*, and *Educational Studies in Mathematics*. She is also interested in research on teaching and teacher professional development and works with teachers in settings around the country to enhance the mathematics learning of all students. At Illinois State, her teaching is centered on theory and research courses for graduate students and mathematics content and methods courses for elementary and middle school prospective teachers.

Richard Lesh, Rudy Distinguished Professor of Learning Sciences at Indiana University, has been the Director of the Center for Research on Learning & Technology and is the Chair of the Learning Sciences Program in the Indiana University's School of Education. More than 250 journal articles and books reflect his international reputation concerning learning and problem solving in mathematics education, and research and assessment design in mathematics and science education. His projects, books, and software have been directed toward audiences ranging from kindergarten through college, as well as from teachers to researchers. He has been a Dean for Research at Northwestern University and Purdue University, a Principal Research Scientist at the Educational Testing Service in Princeton, and a Program Officer for the Research Division at the Education Directorate for the National Science

Foundation. Current or recent publications include: *The Handbook of Design Research in Mathematics & Science Education* (Kelly & Lesh, 2000), *Foundations for the Future in Mathematics Education* (Lesh, Hamilton & Kaput, in press), and *Beyond Constructivism: Models & Modeling Perspectives on Mathematics Problem Solving, Learning & Teaching* (Lesh & Doerr, 2003). He also is the associate editor for the *International Journal for Mathematical Thinking & Learning*.

Frank K. Lester, Jr. is Chancellor's Professor of Education, Martha Lea and Bill Armstrong Chair of Teacher Education, and Professor of Mathematics Education and Cognitive Science at Indiana University, where he has been a member of the faculty since 1972. His primary research interests lie in the areas of mathematical problem solving and metacognition, especially at the elementary and middle school levels. During the past 10 years he has become increasingly interested in the study of classroom assessment of problem solving and higher-order thinking in mathematics, as well as the theoretical and philosophical foundations for research on mathematics learning, teaching, and curriculum. From 1991 until 1996 he was editor of the *Journal for Research in Mathematics Education* and from 1999–2002 he was a member of the Board of Directors of the National Council of Teachers of Mathematics. He also served on the International Steering Committee for International Group for Psychology in Mathematics Education from 1988–1992. He has been on the advisory boards of several federally-funded projects and on the editorial boards of various journals, including *International Journal of Research on Mathematics, Science, and Technology Education*, *Journal of Educational Psychology*, *Mathematics Education Research Journal*, and *Mathematics Thinking and Learning*.

Jennifer Lewis is a doctoral student in mathematics education and teacher education at the University of Michigan, having been a teacher in elementary and middle schools. Her research focuses on teacher education efforts designed to improve mathematics instruction. She is also interested in research that examines education across the professions.

Edward S. Mooney is an associate professor of mathematics education at Illinois State University. His research has focused on middle school students' understanding of statistics. He has authored or coauthored articles in *Journal of Education for Students Placed At Risk*, *Mathematical Thinking and Learning*, and *Teaching Statistics*.

Mogens Niss is Professor of Mathematics and Mathematics Education. He was trained as a pure mathematician (specializing in topological measure theory) at the University of Copenhagen, where he spent the first years of his

academic career. In 1972 he joined the founding staff of Roskilde University, where he has remained ever since. This caused a reorientation of his research interests, which today include the teaching and learning of mathematical applications and modeling, the justification problem in mathematics education from societal perspectives, assessment in mathematics education, and the nature and role of mathematics education as an academic field. Within these fields he has published numerous papers and books. He is a member of the editorial board of several journals. During the years 1987–1998 he was a member of the Executive Committee of The International Commission on Mathematical Instruction, of which he was the Secretary General, 1991–1998. He has served on a multitude of national and international committees, commissions, and governing boards, often as a chair. He was the Chair of the International Programme Committee for ICME-10, held in Copenhagen in 2004. Also, he was the Director of the Danish KOM project, 2000–2002, and the Director of the Danish National Graduate School of Mathematics and Science Education.

Randy Philipp is professor of mathematics education in the School of Teacher Education at San Diego State University and a member and former associate director of the university's Center for Research in Mathematics and Science Education. His research focus is on the relationships among prospective and practicing mathematics teachers' knowledge, beliefs, and practices. He directed the federally-funded research and development project *Integrating Mathematics and Pedagogy* (IMAP). In addition to publications, video products, and a web-based beliefs survey, a Children's Mathematical Thinking course was developed and institutionalized through IMAP. He and Vicki Jacobs are currently co-directing federally-funded research to map a trajectory for the evolution of elementary school teachers engaged in sustained professional development. He coauthored *Middle-Grade Mathematics Teachers' Mathematical Knowledge and Its Relationship to Instruction* with Judy Sowder, Barbara Armstrong, and Bonnie Schappelle. He has served as associate editor of the *Journal for Research in Mathematics Education*, co-chair of the American Educational Research Association (AERA) Special Interest Group for Research in Mathematics Education, and chair of the AERA annual program for Division C, Section 2. In 2006 he became the first recipient of the Association of Mathematics Teacher Educators' Award for Excellence in Teaching.

Norma Presmeg has been a Professor in the Mathematics Department at Illinois State University since 2000. Prior to this appointment she served on the faculty of Florida State University for ten years, after emigrating from South Africa, where she served at the University of Durban-Westville

from 1986 to 1990. Following from her doctoral research at Cambridge University (1982–1985), her research interests over the last 25 years have involved the potential and pitfalls of visualization in the teaching and learning of mathematics, equity and the role of culture in this teaching and learning, the significance of mathematical prototypes, metaphor and metonymy, and more recently, the use of semiotics as a theoretical lens for research in mathematics education. She is currently an editor of *Educational Studies in Mathematics*, and book review editor of *Journal for Research in Mathematics Education*. She served as Co-Chair of the Special Interest Group: Research in Mathematics Education of the *American Educational Research Association* from 2000 to 2002, and she served on the International Steering Committee for the International Group for the Psychology of Mathematics Education (PME) from 1997 to 2001. She was a plenary speaker at the 30th Annual Meeting of PME in Prague, July 2006.

Janine Remillard is an Associate Professor at the University of Pennsylvania and chair of the Foundations and Practices of Education Division. Her research interests include teachers' use of and learning from mathematics curriculum materials and mathematics teaching and learning in urban schools and communities. She is currently a co-P.I. of *MetroMath: The Center for Mathematics in America's Cities*, an NSF-funded Center for Learning and Teaching and a research associate with the Center for the Study of Mathematics Curricula. Her work has been published in *Review of Educational Research*, *Curriculum Inquiry*, *Journal for Research in Mathematics Education*, *Elementary School Journal*, *Journal of Mathematics Teacher Education*, *the School Community Journal*, *Urban Review* and *Mathematics Thinking and Learning*. She co-editor of the forthcoming volume: *Teachers' Use of Mathematics curriculum Materials: Research Perspectives on the Relationship between Teachers and Curriculum Materials*.

Celia K. Rousseau is an assistant professor in the Department of Instruction and Curriculum Leadership at the University of Memphis. Her primary research interests lie in the areas of equity, critical race theory, and mathematics education in urban schools. She has authored or co-authored several articles and book chapters on these topics. She recently co-edited a book entitled: *Critical Race Theory in Education: All God's Children got a Song*. In addition, she has been involved in several projects in the Memphis area related to the professional development of secondary mathematics teachers.

Julie Sarama is Associate Professor of mathematics education at the University at Buffalo, SUNY. She conducts research on the implementation and effects of her own software environments in mathematics classrooms, young children's development of mathematical concepts and

competencies, implementation and scale-up of educational reform, and professional development. She is co-author of the award-winning *Turtle Math*, as well of over 40 refereed articles, 16 chapters, 7 units of the *Investigations in Number, Data, and Space* project, and the research-based *Building Blocks* curriculum. She also has co-authored 1 book, 20 software titles and many additional curriculum, journal, and magazine publications. She is current directing several projects funded by the National Science Foundation and the U.S. Education Department's Institute of Educational Sciences (IES), including two large-scale research projects investigating the implementation of educational reform and the professional development that is a necessary component of large-scale innovations. She has taught secondary mathematics and computer science, gifted math at the middle school level, preschool and kindergarten mathematics enrichment classes, and mathematics methods and content courses for elementary to secondary teachers.

Analúcia D. Schliemann is Professor of Education at Tufts University. She studied in Brazil, France, and England, where she obtained her Ph.D. in Developmental Psychology, and she was a Fulbright Scholar at LRDC, University of Pittsburgh. In Brazil, she worked for 20 years at the Federal University of Pernambuco. Her work on everyday mathematics has been published in English, Portuguese, Spanish, and Japanese. Some of her studies appear in *Street Mathematics and School Mathematics* (with T. Nunes and D. Carraher, New York: Cambridge University Press, 1993). At Tufts (since 1994), her research focuses on children's early algebraic reasoning and learning through classroom activities designed to explore the algebraic character of arithmetic in elementary school (see www.earlyalgebra.terc.edu). She co-authored, with D. Carraher and B. Brizuela, *Bringing Out the Algebraic Character of Arithmetic*, published by Erlbaum in 2007. She is equally at home in Theories of Learning and Development, Mathematics Education, Developmental Psychology, and Cross-Cultural Psychology.

Alan Schoenfeld holds the Elizabeth and Edward Conner Chair in Education and is Affiliated Professor of Mathematics at the University of California, Berkeley. Schoenfeld has served as president of the American Educational Research Association and as vice president of the National Academy of Education. He is a Fellow of the American Association for the Advancement of Science and a Laureate of Kappa Delta Pi. Schoenfeld obtained his Ph.D. in mathematics from Stanford University in 1973. Soon afterwards he began conducting research on mathematical thinking, teaching and learning. Major foci of his work have included problem solving, assessment, constructing analytic models of teaching, and issues of diversity; he has an ongoing concern with finding productive mechanisms

for systemic change and for deepening the connections between educational research and practice. Schoenfeld was one of the founding editors of *Research in Collegiate Mathematics Education*, and has served as associate editor of *Cognition and Instruction*. In all of his work, issues of research methods have been a central concern.

J. Michael Shaughnessy has taught and researched in mathematics and mathematics education for over 30 years. Much of his career has been devoted to the professional development of pre-service and in-service mathematics teachers. He is currently a Professor of Mathematics Education in the Department of Mathematics and Statistics at Portland State University, Portland, OR where he has worked for the last 15 years. His principal research interests lie in the areas of teaching and learning probability and statistics, and teaching and learning geometry. He was a member of the Board of Directors of the National Council of Teachers of Mathematics from 2001–2004, and served as co-chair of the Board for the Special Interest Group for Research in Mathematics Education of the American Educational Research Association from 2005–2007. He has served on the editorial boards of a number of journals, including the *Journal for Research in Mathematics Education*, *Mathematics Thinking and Learning*, and *Mathematics Education Research Journal*.

Edward A. Silver is the William A. Brownell Collegiate Professor of Education, and Professor of Mathematics at the University of Michigan. He also currently serves as Associate Dean for Academic Affairs in the University of Michigan's School of Education. His scholarly interests include the study of mathematical thinking, especially mathematical problem solving and problem posing; the design and analysis of intellectually engaging and equitable mathematics instruction for students; innovative methods of assessing and reporting mathematics achievement; and effective models for enhancing the knowledge of teachers of mathematics. He has published numerous articles, chapters, and books on these topics. For much of the 1990s Silver directed the QUASAR project, an ambitious design experiment, which stimulated and studied efforts to improve mathematics instruction in urban middle schools. He is a co-PI of the NSF-funded Center for Proficiency in Teaching Mathematics. He served as editor of the *Journal for Research in Mathematics Education* from 2000–2004. He has been on the advisory boards of several federally-funded projects and on the editorial boards of various journals, including the *American Educational Research Journal*, *Cognition and Instruction*, and *Mathematical Thinking and Learning*. He was the 2004 recipient of the Award for Outstanding Contributions of Educational Research to Practice from the American Educational Research Association.

Laurie Sleep is a doctoral student in mathematics education at the University of Michigan. Her research interests build upon her experience as an elementary teacher and include designing and studying ways to help prospective teachers develop mathematical knowledge for teaching. She teaches both mathematics methods and content courses for prospective teachers and was named a University of Michigan Outstanding Graduate Student Instructor in 2006. Sleep holds a B.A. in business-economics, a master's degree in education from UCLA, and a master's degree in mathematics from the University of Michigan.

Margaret Schwan Smith is an Associate Professor in the Department of Instruction and Learning in the School of Education and a Research Scientist at the Learning Research and Development Center, both at the University of Pittsburgh. She works with pre-service elementary, middle, and high school mathematics teachers at the University of Pittsburgh, with doctoral students in mathematics education who are interested in becoming teacher educators, and with practicing middle and high school mathematics teachers and coaches locally and nationally through several funded projects. Over the past decade she has been developing research-based materials for use in the professional development of mathematics teachers and studying what teachers learn from the professional development in which they engage. She is currently a member of the Board of Directors of the National Council of Teachers of Mathematics (2006–2009). In 2006 she was selected to receive the Chancellor's Distinguished Teaching Award given annually to honor outstanding faculty at the University of Pittsburgh.

Judith Sowder is Professor Emerita of Mathematics and Statistics at San Diego State University. For many years her research focused on children's development of number sense, but more recently on the instructional effects of teachers' mathematical knowledge at the elementary and middle school level. She served as editor of the *Journal for Research in Mathematics Education* from 1996 to 2000 and was elected to serve on the National Council of Teachers of Mathematics Board of Directors from 2000 to 2003. In 1996 she received the San Diego State University Alumni Association Award for Outstanding Teaching, Work and Contributions to the University and Community, and in 2000 the Lifetime Achievement Award from NCTM. She served on the International Committee (steering committee) for International Group for Psychology in Mathematics Education from 1992–1996. She has received research funding for several research projects from the National Science Foundation and the U. S. Department of Education, and a grant from QUALCOMM to develop a Professional Development Collaborative between San Diego State University and surrounding school districts. She

continues to serve on advisory boards for several projects, and is currently coauthoring a mathematics textbook for elementary and middle school teachers.

Larry Sowder is Professor Emeritus at San Diego State University, where he has been a faculty member since 1986. Earlier, he served as a high-school mathematics department chairman and physics teacher in northern Indiana before enjoying both graduate school at the University of Wisconsin and then a seventeen-year stint at Northern Illinois University. His research interests fall under a problem-solving rubric, ranging from an earlier focus on grades 4–8 students' work with applications of mathematics as embodied in the humble story problem, through the learning and teaching of mathematical proof devising. His collaborations with Guershon Harel have given him an enlightening and delightful professional development experience.

Mary Kay Stein holds a joint appointment as Professor in the School of Education and Senior Scientist at the Learning Research and Development Center, both at the University of Pittsburgh. Over the past decade, her research has transitioned from an exclusive focus on classroom-based mathematics teaching and learning to research that seeks to understand how institutional, interpersonal and policy contexts shape teachers' practice. Her work has been published in the *Journal for Research in Mathematics Education*, the *American Educational Research Journal*, *Teachers College Record*, *Urban Education*, and the *Harvard Educational Review*. She is the lead author of a widely used casebook for mathematics professional development, *Implementing Standards-Based Mathematics Instruction* and co-author of a book on educational reform in San Diego (*Reform as Learning*). Over the past several years, Dr. Stein has been a principal or co-principal investigator of a number of grants, including two each from the National Science Foundation, the Spencer Foundation, and the MacArthur Foundation. Dr. Stein has served on several national panels including the National Academy of Education's Panel on Strengthening the Capacity of Research to Impact Policy and Practice, and NCTM's Standards Impact Research Group.

William F. Tate is the Edward Mallinckrodt Distinguished University Professor in Arts & Sciences at Washington University. Professor Tate holds Arts and Sciences academic appointments in Education, American Culture Studies, Urban Studies, and Applied Statistics and Computation. He also is chair of the Department of the Education. Additionally, he is a participating faculty member in the School of Medicine's program in Audiology and Communication Sciences. Other responsibilities include serving as the principal investigator and project director of

the St. Louis Center for Inquiry in Science Teaching and Learning. The center is a multidisciplinary effort to build sustainable models of human resource development in the sciences and to integrate these models into the cultural resources of an urban community. He has authored scores of scholarly journal articles, book chapters, edited volumes, monographs, and textbooks focused on human resource development in mathematics, science, and technology education, urban studies, and race and American education. His most recent book project is a co-edited volume (with Gloria Ladson-Billings) entitled: *Education Research in the Public Interest: Social Justice, Action, and Policy*. He is president-elect of the American Educational Research Association.

Lieven Verschaffel has been a full professor in the faculty of Psychology and Educational Sciences of the University of Leuven since 2000. From 1979–2000, he held several research positions at the Fund for Scientific Research, Flanders (Belgium). His major research interest is psychology of (elementary) mathematics education. During the past 20 years he has been editor or co-editor of several scientific journals on research on learning and instruction in general and mathematics education in particular. He was editor of the major Dutch education journal *Pedagogische Studiën*, assistant editor of *Learning and Instruction*, and associate editor of *Research Dialogue in Learning and Instruction*. Currently, he is a member of the editorial board of *Learning and Instruction*, *Educational Research Review*, *Mathematical Thinking and Learning*, *Educational Studies in Mathematics* and *Adults Learning Mathematics*. Since 2005 he has been co-editor of the book series *New Directions in Mathematics and Science Education* (Sense Publishers). He is a much-sought-after speaker at international conferences and has written chapters for handbooks about his area of research. Currently, he is coordinator of the international scientific network “Design, development and implementation of powerful learning environments” sponsored by the Fund for Scientific Research–Flanders, and of the Concerted Research Action “Adaptive expertise in mathematics education,” sponsored by the Research Fund of the University of Leuven.

Norman L. Webb is a senior research scientist for the Wisconsin Center for Education Research of the University of Wisconsin-Madison. His current principle area of research is evaluation, assessment, and alignment. He has served as the principal investigator on a number of projects including the Adding Value project supporting the evaluation of NSF’s Mathematics Science Partnerships, the Study of Systemic Reform in Milwaukee Public Schools, and the Systemic Reform Study Team of the National Institute for Sciences. He is the leader of the evaluation team for the Center for Integrating Teaching and Learning and

an evaluator for SCALE. He has directed evaluations of curriculum and professional development projects. His academic training was in mathematics and mathematics education. He chaired the evaluation working group, one of four groups who wrote the Curriculum and Evaluation Standards for School Mathematics for the National Council of Teachers of Mathematics. He co-authored with Thomas Romberg, *Reforming Mathematics Education in America’s Cities*. He edited the NCTM 1993 yearbook on classroom assessment and wrote a chapter, “Assessment of Students’ Knowledge of Mathematics: Steps Toward a Theory,” for the 1992 *Handbook of Research on Mathematics Teaching and Learning*.

Dylan Wiliam is Deputy Director of the Institute of Education, University of London, where he is also Professor of Educational Assessment. After a first degree in mathematics and physics he taught in urban schools in London for 7 years, earning further degrees in mathematics and mathematics education. In 1984 he joined King’s College London where he led the mathematics teacher education program. Between 1989 and 1991 he coordinated the development of large-scale assessments for the national curriculum of England and Wales. After his return to King’s, he completed his PhD, dealing with the psychometric issues involved in creating systems of age-independent levels of attainment. From 1996 to 2001 he was the Dean of the School of Education at King’s College London, and from 2001 to 2003, he served as Assistant Principal of the College. Over the last 10 years, principally in collaboration with Paul Black, he has explored the potential for assessment to support learning (sometimes called assessment *for* learning, or formative assessment). From 2003 to 2006, he directed the Learning and Teaching Research Center at ETS, Princeton, NJ, where he led the development of a range of professional development materials for helping teachers use assessment to improve their teaching.

Linda Dager Wilson has worked in the field of mathematics education research since 1993, specializing in assessment issues, following 13 years as a high school mathematics teacher and a PhD in mathematics education from the University of Wisconsin. She was the assistant project director of the *Assessment Standards for School Mathematics*, published by the National Council of Teachers of Mathematics in 1995. While on the faculty at the University of Delaware, she served in Washington DC as a mathematics coordinator for the Voluntary National Test in Mathematics. Since coming to Washington in 1999, she has worked on several national projects, including the 2005 mathematics framework for the National Assessment of Educational Progress and more recent updates for grade twelve. While at the Center for the Study of Assessment Validity and Evaluation at the University of Maryland,

she developed methods for increasing access to mathematics assessment items for all learners. She recently chaired a task force on large scale assessment for NCTM. For the past three years she has worked as a consultant at Project 2061, taking primary responsibility for the development, scoring, and analysis of the student assessments in a research project on the teaching of middle grades mathematics.

Judith S. Zawojewski has been an Associate Professor of Mathematics Education at Illinois Institute of Technology since 2001. She has long been interested in the teaching and learning of problem solving and modeling in the context of classroom practice. Her commitment to link theory and practice is reflected in her publications, including co-authorship in a series of books for the National Council of Teachers of Mathematics: *Navigating through Problem Solving and Reasoning in Grade 3* (2004), *Grade 4* (2005), *Grade 5* (2006), and *Grade 6* (in press) and research chapters in *Beyond Constructivism: A Models and Modeling Perspective on Problem Solving, Learning and Instruction in Mathematics and Science Education* (2003), *Results from the Sixth Mathematics Assessment of the National Assessment of Educational Progress* (1997), and *Results from the Seventh Mathematics Assessment of the National Assessment of Educational Progress* (2000). Dr. Zawojewski has served as a reviewer for various research and practice-based journals, chaired the editorial panel for *Mathematics Teaching in the Middle School* from 1994–1997 and served on the editorial board of *School Science*

and *Mathematics* from 2001–2006. She has also held leadership positions and served on advisory boards for federally-funded research and development projects.

Rose Mary Zbiek is Associate Professor of Mathematics Education at The Pennsylvania State University. One strand of her research on technology addresses the impact of mathematical tools on teaching secondary school mathematics, particularly involving the use of computer algebra systems. Another strand of her work involves the use of technology in mathematical modeling, with a particular focus on developing understanding of curricular mathematics. Her current interests center on mathematical understandings of prospective secondary mathematics teachers and how these understandings synergistically relate to emerging classroom practice. She currently is Chair of the International Committee for Computer Algebra in Mathematics Education and recently completed her term as treasurer and board member of the American Education Research Association's Special Interest Group for Research in Mathematics Education. She served on the National Council of Teachers of Mathematics writing team for the *Curriculum Focal Points for Prekindergarten through Grade 8 Mathematics*. She is series editor of the National Council of Teachers of Mathematics' forthcoming Essential Understandings Project, which explicates key mathematical understandings needed by teachers of mathematics in Pre-Kindergarten through Grade 12.

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